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Crazy Merged

Condensed Books 1–17 + Papers 1–4, merged by section

One merged book: the condensed books and the Paper 1–4 answers combined, section by section, with each topic appearing once. Read each chapter top to bottom. Where a topic was answered in the papers, the full paper answer sits in its place (marked "Paper answer", with the papers it was asked in), followed by a dashed box of anything extra the book had. Paper answers not tied to a chapter are at the end of each section, then the section's Rapid Revision Sheet.

Section	Title
1	Basic Principles (General Urology)
2	Oncology, DSD, Gender & Ageing
3	Infections & Inflammation
4	Paediatric Urology 1
5	Paediatric Urology 2
6	Female Urology
7	Voiding Dysfunction
8	Nocturia & Male Incontinence
9	Andrology & Sexual Medicine
10	Testis, Penis & Urethra
11	Stones, Obstruction & Trauma
12	Bladder Cancer & Diversion
13	Kidney & UTUC
14	Adrenal & Retroperitoneum
15	Renal, Nephrology & Transplant
16	BPH
17	Prostate Cancer

SECTION 1

Basic Principles (General Urology)

1 Personal and Professional Development of the Urologist

1 · Professionalism

- **Unprofessional behaviour in medical school** is one of the strongest predictors of later disciplinary action by medical boards.
- Surgeons in the **highest quartile of patient complaints have more surgical complications**, so unprofessional behaviour is a measurable safety signal.
- The Patient Advocacy Reporting System (PARS) uses patient complaints, and the Coworker Reporting System (CORS) uses co-worker reports. After a CORS report, most individuals are never reported again.

2 · Burnout

- **Definition (triad)**: emotional exhaustion, depersonalisation and diminished personal accomplishment, in response to chronic interpersonal stressors. It is distinct from depression.
- Prevalence is about **40%** of physicians, and urologists approach **50%**. It causes more medical error.
- Physician suicide is **1.4×** the population rate in men and **2.7×** in women.
- **20% rule**: spending less than 20% of effort on the most meaningful activity raises burnout risk about **3×**.
- Drivers are loss of autonomy, purpose and community, electronic health record time and administrative burden.

Prevention strategy	Effect
Institutional awareness seminars	Minimally effective
Practice interventions (scribes)	More effective
Restoring autonomy over schedule	Significantly reduces burnout
Mindfulness, communication training, exercise	Evidence-based reduction

> **KEY POINT**
>
> Interventions that restore autonomy, purpose or community work; awareness alone does not.

3 · Four domains of learning

Domain	Framework	Assessment
Knowledge	Bloom's taxonomy (apex: create)	Easiest; certification exams
Psychomotor	Imitation, manipulation, precision, articulation, naturalisation (novice to master)	Proctorship (often needed for robotic and laser credentialing), simulation

Domain	Framework	Assessment
Communication	TeamSTEPPS	Often untaught
Affective	Krathwohl's taxonomy (apex: characterisation by a value set)	Formative

SAQ Operating room safety : the role of surgeon.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2021-Summer Paper 4, Q3a

Introduction

- About half of hospital adverse events involve surgery; half of these are preventable. The surgeon is **team leader** and ultimately responsible.
- **Swiss cheese model (Reason):** harm occurs when holes in several barriers line up, so safety needs layered defences.

WHO (World Health Organization) Surgical Safety Checklist (2008, 19 items)

Phase	Key checks
Sign in (before induction)	Identity, site, procedure, consent; site marked; allergy; airway; blood loss >500 mL
Time out (before incision)	Team introductions; confirm patient, site, procedure; surgeon states critical steps, duration, blood loss; antibiotic within 60 min; imaging displayed
Sign out (before leaving theatre)	Procedure recorded; counts correct; specimen labelled; equipment problems; recovery concerns

- Pilot in 8 hospitals: deaths fell 47%, major complications 36%.

Surgeon's responsibilities

1. **Universal Protocol** (Joint Commission, 2004): verification, **site marking** by the operating surgeon, patient awake, correct side only, time-out. Wrong-site surgery is a never event.
2. **Lead the time-out:** whole team engaged; repeat when procedure or surgeon changes.
3. **Retained surgical items** (sponges commonest): a count discrepancy raises risk about 100-fold; do not close until resolved; X-ray or RF (radiofrequency) tags.
4. **Fire safety:** the surgeon owns the **ignition source** (diathermy, light cables, laser). Let alcohol prep dry; FiO_2 (inspired oxygen fraction) ≤30% near the airway.
5. **Sharps:** double gloves, blunt needles, neutral zone (no hand-to-hand passing).
6. **Distractions:** "sterile cockpit" during critical steps (e.g. hilar clamping).
7. **Communication** (failure underlies up to 70% of events): Use **TeamSTEPPS** (Team Strategies and Tools to Enhance Performance and Patient Safety) tools: brief, debrief, **SBAR** (situation, background, assessment, recommendation), closed-loop check-back, **CUS** ("concerned, uncomfortable, safety issue"), two-challenge rule. Flatten hierarchy so juniors speak up.
8. **Self-monitoring: I'M SAFE** (illness, medication, stress, alcohol, fatigue, eating).

- **Also know:** positioning and padding; VTE (venous thromboembolism) prophylaxis; simulation.

Diagram to draw: Swiss cheese model with aligned holes.

Key points to remember:

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Also, from the condensed book (not in the paper answer):

- Communication errors contribute to about **70%** of medical errors and **27%** of malpractice claims.
- Situation awareness belongs to the individual, while a shared mental model belongs to the team.

KEY POINT

Certification tests the easiest domain, knowledge, but harm arises mainly in the communication, affective and psychomotor domains.

2 Evaluation of the Urologic Patient: History and Physical Examination

1 · Pain

- **Pain without tenderness is referred pain**: a non-tender painful testis may reflect an ipsilateral ureteric stone.

	Obstructive (colic)	Parenchymal
Cause	Distension of a hollow organ	Capsular stretch (pyelonephritis)
Pattern	Intermittent; restless, seeks a position	Constant; costovertebral angle tenderness; lies still

- Colic is mediated by prostaglandins and perceived in the **T11–L1** segments, which is why non-steroidal anti-inflammatory drugs (NSAIDs) work. A ureterovesical junction stone causes urgency and frequency.
- **Strangury** is stabbing pain at the end of micturition from an inflamed detrusor. An atonic bladder retains large volumes painlessly.

2 · Haematuria

SAQ Microscopic / asymptomatic haematuria and urinary biomarkers

Paper answer · P2 · P3 · merged from 2

Asked as: P3 SAQ "Asymptomatic haematuria and urinary biomarkers" (Asymptomatic Hematuria, Summer 2016 Paper III Q3d; Urinary biomarkers in diagnostic evaluation of hematuria, Summer 2021 Paper III Q3f) · P2 SAQ "Microscopic Hematuria" (Practice Set 1 Paper 2 Q3e; also 2016 Summer Paper 3 Q3d)

Definitions and epidemiology

- **MH (microscopic haematuria)**: **3 or more RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020).
- **A dipstick alone is not enough**: myoglobin, semen and bacterial peroxidases give false positives.
- **GH (gross haematuria)**: visible; exclude pigmenturia and menstrual blood.
- Prevalence about 6.5%. Malignancy in about 3% of MH (over 85% of these are bladder cancer) and in 14–26% of GH (DETECT I, IDENTIFY).
- Haematuria is evaluated by **cystoscopy and upper tract imaging**; urine tests are non-invasive **adjuncts** for bladder cancer.

Causes

- **Neoplasm**: bladder (over 85% of cancers found), UTUC (upper tract urothelial carcinoma) and RCC (renal cell carcinoma) about 1 in 1000 each, prostate.
- **Benign**: BPE (benign prostatic enlargement), stones, UTI (urinary tract infection), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal** (IgA nephropathy): dysmorphic RBCs, casts, proteinuria.
- **Others**: exercise, endometriosis, sickle cell.

- Anticoagulants do not explain haematuria; evaluate as usual.

Initial evaluation

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch syndrome), examination, **blood pressure, creatinine**, urine culture.
- Suspected UTI: culture, treat, then **repeat urinalysis**; persistent MH means full evaluation.
- Glomerular or renal picture (dysmorphic RBCs, casts or proteinuria suggest a glomerular source): nephrology **and** urological evaluation.

AUA/SUFU 2020 risk stratification

	Low (all)	**Intermediate (any)**	**High (any)**
Age	Women under 50, men under 40	Women 50–59, men 40–59	60 or over
Smoking (pack-years)	Never or under 10	10–30	Over 30
RBCs/HPF	3–10	11–25	Over 25
Gross haematuria history	No	No	**Yes**
Cancer risk	0–0.4%	1–2.5%	About 6%
Work-up	Shared decision: repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound)	Cystoscopy + RUS	Cystoscopy + **CT (computed tomography) urography**

- Additional risk factors (irritative symptoms, radiation, cyclophosphamide, occupational exposure) move a patient from low to intermediate.
- CT contraindicated: MR (magnetic resonance) urography, or retrograde pyelograms.
- Persistent MH after a negative ultrasound: CT or retrograde imaging.
- Hereditary RCC risk (VHL (von Hippel–Lindau)): upper tract imaging in any stratum.
- White-light cystoscopy. Cytology and urine markers are **not** used initially; cytology if CIS (carcinoma in situ) is suspected.
- **Gross haematuria**: cystoscopy, CT urography and cytology at any age.

Urine cytology

- **Sensitivity 31–62%, specificity 94–100%**: good for **high-grade tumours and CIS**, poor for low grade.
- **Paris System:** negative for HGUC (high-grade urothelial carcinoma), atypical, suspicious, HGUC (graded by nuclear-to-cytoplasmic ratio and nuclear features).
- Positive cytology with normal cystoscopy: random and prostatic urethral biopsies, upper tract washings.

Urinary biomarkers (sensitivity/specificity, Chou 2015 meta-analysis)

Marker	**Principle**	**Sens / spec**	**Note**
NMP22 (nuclear matrix protein 22)	Nuclear protein; lab or point-of-care	0.69/0.77	False positives: inflammation, stones

Marker	Principle	Sens / spec	Note
BTA (bladder tumour antigen) stat / TRAK	Complement factor H–related protein; dipstick or ELISA (enzyme-linked immunosorbent assay)	0.64/0.77	**False positive with blood** (factor H in serum)
UroVysion FISH (fluorescence in situ hybridisation)	Gain of chromosomes 3, 7, 17; loss of 9p21	0.63/0.87	After atypical cytology
ImmunoCyt	Antibodies to CEA (carcinoembryonic antigen) + two mucins	0.78/0.78	Highest sensitivity; operator dependent
Cxbladder	RT-PCR (reverse transcription polymerase chain reaction), 5 mRNAs	about 0.82/0.85	Needs validation

- **Also know:** Xpert Bladder (mRNA); AssureMDx (FGFR3, TERT, HRAS mutations + methylation).
- **Pattern:** more sensitive and less specific than cytology; they miss small low-grade tumours.

Role of markers in haematuria (guidelines)

- **AUA/SUFU 2020: no cytology or markers in the initial evaluation of microhaematuria**; cytology only after a negative work-up if CIS is suspected.
- **Gross haematuria:** cytology with cystoscopy and CT urography. A negative marker still leaves **≥10% cancer probability**, so cystoscopy is still needed.
- **No marker replaces cystoscopy** (AUA, EAU (European Association of Urology)); a negative test may one day help triage low-risk patients.
- **Recent advances:** urinary tumour DNA; methylation assays (Bladder EpiCheck).

Follow-up

- Negative evaluation: consider repeat urinalysis within 12 months; release if negative.
- Re-evaluate for gross haematuria, rising MH or new symptoms.

Diagram to draw: AUA/SUFU flowchart from MH to risk-based work-up (haematuria pathway), with markers shown as adjuncts.

> **Key points to remember:**
> - MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
> - Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.
> - Cytology: specific for high grade/CIS, insensitive for low grade.
> - BTA is falsely positive with blood.
> - No markers in initial microhaematuria work-up; none replaces cystoscopy.

Also, from the condensed book (not in the paper answer):

- The commonest cause of gross haematuria over 50 years is **bladder cancer**.

Clue	Suggests
Initial haematuria	Urethra or prostate

Clue	Suggests
Total haematuria	Bladder or above
Terminal haematuria	Bladder neck
Vermiform clots / cuboid clots	Upper tract / bladder
Clears with irrigation but recurs / clears once a catheter is placed	Upper tract / prostate

3 · Lower urinary tract symptoms

- A **bladder diary** (usually 48 h) gives functional capacity (normal **300–400 mL**), nocturia and polyuria. Frequency means more than 5–6 voids a day.
- Daytime frequency without nocturia suggests a psychogenic cause. Nocturia without daytime frequency suggests nocturnal polyuria.
- Irritative symptoms may be the only sign of carcinoma in situ (CIS), so keep a low threshold for cystoscopy in smokers over 50.

*See the paper answer **IPSS score** (Section 16, Ch 149).*

4 · Urinary incontinence

Type	Mechanism	Hallmarks
Continuous	Outlet bypassed (fistula, ectopic ureter)	Constant wetness
Overflow	Volume exceeds capacity	Often nocturnal; long-standing obstruction; cured by relieving it
Stress	Sphincter weakness	Cough or sneeze; multiparity; after prostatectomy
Urge	Leakage with urgency	Overactive bladder, cystitis, occult cancer
Mixed	Stress plus urge	Treating one alone may worsen symptoms
Functional	Cannot reach the toilet	Mobility, cognition

*See the paper answer **Embryology and management of ectopic ureter.** (Section 4, Ch 52).*

*See the paper answer **Nocturnal enuresis** (Section 5, Ch 47).*

Also, from the condensed book (not in the paper answer):

- Enuresis affects about **15%** at 5 years and **1%** at 15 years. Primary enuresis beyond 6 years needs urological evaluation, and secondary enuresis should raise the possibility of abuse or bullying.

5 · Sexual and other symptoms

- **Erectile dysfunction (ED)** (National Institutes of Health 1993) is inability to attain or maintain an erection sufficient for satisfactory intercourse. **ED heralds occult coronary artery disease.**
- With low libido and ED, measure early-morning testosterone on two occasions **≥2 weeks** apart.

*See the paper answer **Premature ejaculation** (Section 9, Ch 74).*

*See the paper answer **Haematospermia** (Section 9, Ch 74).*

Also, from the condensed book (not in the paper answer):

- The AUA/Sexual Medicine Society definition of premature ejaculation is within about 2 min.
- The commonest cause of **pneumaturia** is an enterovesical fistula (diverticulitis, Crohn disease). Bloody urethral discharge suggests urethral carcinoma.

6 · Drug history

Side effect	Drugs
ED	Thiazides, propranolol
Ejaculatory dysfunction	α-blockers, methyldopa, phenothiazines
Priapism	**Trazodone**, phenothiazines, hydralazine, prazosin
Retention	Anticholinergics, calcium-channel blockers, α-agonists (pseudoephedrine), antihistamines
Reduced spermatogenesis	Alkylating agents, marijuana
Gynaecomastia	Digoxin, cimetidine, verapamil

Antithrombotic	Action	Half-life
Warfarin	Vitamin K antagonist	36–42 h
Dabigatran	Direct thrombin inhibitor	8–17 h
Rivaroxaban / apixaban	Factor Xa inhibitors	5–12 h / 12 h
Clopidogrel / ticagrelor	P2Y12 blockade, irreversible / reversible	6 h / 6–8 h

- Half-life is not duration: **clopidogrel acts for the life of the platelet**.
- **Bleeding risk** is low for cystoscopy and ureteroscopy, intermediate for transurethral resection of the prostate (TURP) and of bladder tumour (TURBT), and **high for percutaneous nephrolithotomy (PCNL)**, artificial urinary sphincter and robotic nephrectomy.

7 · Physical examination

- **Bladder**: palpable above about **150 mL** and visible at about 500 mL. **Bimanual examination under anaesthesia after TURBT** is standard of care and independently predicts pT3.
- **Prostate**: a nodule as hard as a flexed knuckle needs biopsy.
- **Torsion**: horizontal lie, high testis and short cord with pain. An absent cremasteric reflex is highly sensitive in children; a present reflex argues against torsion but does not exclude it.
- Normal testis is **4–5 × 2–3 cm**. A **hard testicular mass is cancer until proven otherwise**, whereas an epididymal mass is almost always benign.
- **Female**: a Q-tip deflection **≥30°** on Valsalva means urethral hypermobility, but it is not required to diagnose stress incontinence. Document prolapse by POP-Q.
- **Neuro-urology**: the bulbocavernosus reflex tests **S2–S4**. Saddle sensory loss with an absent reflex localises a sacral root lesion such as cauda equina.

3 Indications and Interpretation of Laboratory Testing

1 · Urinalysis: collection and gross examination

- Examine urine within **2 h** or refrigerate it at 2–8 °C. Refrigeration precipitates crystals.
- In children, suprapubic aspiration has the lowest contamination and catheterisation is often preferred. A **bag specimen is never used for culture**.
- Red urine is not always blood (beetroot, rifampicin, phenothiazines).
- Cloudy urine is most often **phosphaturia**, which clears with acetic acid; pyuria does not.

2 · Dipstick chemistry

Interferent	False negative	False positive
Ascorbic acid	Glucose, bilirubin, blood, leucocyte esterase	–
Phenazopyridine / dyes	Leucocyte esterase	Nitrite, bilirubin, ketones
Alkaline urine	–	Protein
High specific gravity	Leucocyte esterase, glucose	–
Semen, menses, exercise	–	Blood
Mesna, levodopa	–	Ketones
Contamination (commonest)	–	Leucocyte esterase and nitrite

- **Specific gravity** is 1.001–1.035 and becomes **fixed at 1.010 in renal insufficiency**.
- pH is 4.5–8.0. A pH **>7.5** with infection means a urease-splitting organism (Proteus), which leads to struvite staghorn stones. Uric acid and cystine stones form in acid urine.

*See the paper answer **Renal tubular acidosis (distal and type 2)** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- The renal threshold for glucose is a serum level of about **180 mg/dL**, and SGLT2 inhibitors cause glycosuria. Nitroprusside detects acetoacetate but **not β-hydroxybutyrate**.

3 · Haematuria

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

Red urine	Sediment	Supernatant	Plasma
Haematuria	RBCs	Clear	–
Haemoglobinuria	No RBCs	Red, haem positive	Red
Myoglobinuria	No RBCs	Red, haem positive	Clear
Dyes, porphyria	No RBCs	Red, haem negative	–

Plasma is red in haemoglobinuria because haemoglobin is haptoglobin-bound; small unbound myoglobin is filtered at once.

	Glomerular	Non-glomerular medical	Urological
RBC shape	Dysmorphic	Round	Round
RBC casts	Present	Absent	Absent
Proteinuria	Significant	Often significant	Minimal, even with gross bleeding

- Proteinuria of 2+–3+ (100–300 mg/dL) is almost never urological, so refer to nephrology.

SAQ Berger's Disease.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2020-Summer Paper 1, Q3a

Berger's disease is IgA (immunoglobulin A) nephropathy (IgAN).

Definition

- **Commonest primary glomerulonephritis (GN) worldwide**, in children too. Dominant **mesangial IgA deposits** on immunofluorescence.
- Commonest cause of **glomerular haematuria** (about one-third). Second–third decades; male predominance.
- **IgA vasculitis** (Henoch–Schönlein purpura) is the systemic form.

Pathogenesis

- "Multi-hit" model: abnormal (galactose-deficient) IgA1 → autoantibodies → immune complexes → mesangial deposition, complement activation, proliferation. Mucosal (respiratory) infection triggers flares.

Clinical features

- **Recurrent painless macroscopic haematuria during or within days of an upper respiratory infection** ("synpharyngitic"), or after exercise. Post-streptococcal GN follows infection by weeks.
- Persistent microscopic haematuria, proteinuria, hypertension; occasional loin pain; rarely nephrotic syndrome or RPGN (rapidly progressive GN).

Evaluation (urologist's role)

- **Dysmorphic RBCs (red blood cells), RBC casts, proteinuria** → glomerular source. **Normal C3 and C4.**
- Without casts, complete the urological evaluation (imaging ± cystoscopy) to exclude a urinary tract cause.
- **Renal biopsy** confirms. Once diagnosed, repeat haematuria work-ups are not needed.

Oxford MEST-C score (biopsy)

Item	Grades
M: mesangial hypercellularity	M0, M1
E: endocapillary hypercellularity	E0, E1
S: segmental sclerosis	S0, S1
T: tubular atrophy/interstitial fibrosis	T0 ≤25%, T1 26–50%, T2 >50%
C: crescents	C0 none, C1 <25%, C2 ≥25% of glomeruli

Prognosis

- Often benign, but **20–30% reach ESKD (end-stage kidney disease)**.
- Poor: persistent proteinuria, hypertension, raised creatinine, older onset, high MEST-C.

Management

- No cure. **Supportive:** ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker), BP (blood pressure) control, low salt, stop smoking.
- **Newer drugs:** SGLT2 (sodium–glucose cotransporter 2) inhibitors; sparsentan (dual endothelin–angiotensin receptor blocker); targeted-release budesonide acting on gut mucosal lymphoid tissue.
- Persistent proteinuria above about 0.75–1 g/day despite supportive care: consider a 6-month steroid course (KDIGO (Kidney Disease: Improving Global Outcomes) 2021); weigh infection risk.
- Crescentic RPGN: steroids plus cyclophosphamide.
- **Transplant:** recurs histologically often, clinically rarely.

Diagram to draw: glomerulus with mesangial IgA deposits.

> **Key points to remember:**
> - Commonest primary GN; haematuria with upper respiratory infection.
> - Normal complement; biopsy shows mesangial IgA.
> - Oxford MEST-C; 20–30% ESKD.
> - ACE inhibitor or ARB is the mainstay.

Also, from the condensed book (not in the paper answer):

- Other clues are deafness with a family history (Alport syndrome), haemoptysis (Goodpasture syndrome), and a recent throat infection with low C3 (poststreptococcal glomerulonephritis).

4 · Proteinuria

SAQ Proteinuria

Paper answer · P1

Asked as:

- Proteinuria — *Summer 2017 Paper I Q3f (SAQ)*
- Proteinuria — *Summer 2021 Paper I Q3d (SAQ)*

Type: SAQ (10 marks)
(also answered in 2017-Summer Paper 1, Q3f)

Definition

- Normal protein excretion is **<150 mg/day** (another source: **80–150 mg/day**) (albumin <30 mg/day). The filter (glomerulus) blocks (holds back) albumin and the proximal tubule reabsorbs filtered small proteins.
- Normal mix: about 30% albumin, 30% globulins, 40% tissue proteins (Tamm–Horsfall).
- **Nephrotic range:** >3.5 g/day, with hypoalbuminaemia, oedema and hyperlipidaemia.

Detection and quantification (KDIGO, Kidney Disease: Improving Global Outcomes, categories by ACR)

Category	ACR (albumin-to-creatinine ratio)	PCR (protein-to-creatinine ratio)	24-h protein
A1 normal–mild	<30 mg/g (<3 mg/mmol)	<150 mg/g (<15 mg/mmol)	<150 mg
A2 moderate	30–300 mg/g	150–500 mg/g	150–500 mg
A3 severe	>300 mg/g (>30 mg/mmol)	>500 mg/g	>500 mg
- **Spot early-morning ACR is first line**; PCR if non-albumin protein suspected; 24-h urine with creatinine check. Alternatives: PCR, 24-h urine.			
- **Dipstick** (tetrabromophenol blue) detects albumin only (mainly albumin), from 20–30 mg/dL; misses dilute urine and **light chains**. **Sulfosalicylic acid** detects (precipitates) all proteins; dipstick negative but sulfosalicylic acid positive → suspect myeloma.			

Types

Type	Mechanism	Amount	Example
Glomerular (commonest)	Leaky filter	Often >2 g; albumin ≈70% (about 70%)	Diabetes, glomerulonephritis, FSGS (focal segmental glomerulosclerosis), amyloid
Tubular	Failed reabsorption of small proteins	<2 g (seldom >2–3 g); mainly globulins	Fanconi syndrome, heavy metals, drugs
Overflow	Excess plasma protein	Variable (0.3–2 g)	**Myeloma** (Bence Jones), myoglobin, haemoglobin

By timing	Features
Transient	Fever, exercise, heart failure; commonest in children
Orthostatic	Young men, upright only, <1 g; benign (normal first-morning sample). Test with a split (supine overnight vs upright day) collection
Persistent	Mostly glomerular; needs full work-up

Evaluation

Positive dipstick → repeat; exclude UTI (urinary tract infection), fever, exercise → persistent → quantify (ACR, PCR or 24-h protein) → creatinine/eGFR (estimated glomerular filtration rate), sediment (dysmorphic red cells / RBCs, casts), ultrasound → glomerular pattern (>2 g, albumin) → nephrology ± renal biopsy; C3, C4, ANA if systemic → globulins → electrophoresis, immunofixation (myeloma)

- UTI: urinary tract infection; eGFR: estimated glomerular filtration rate; RBC: red blood cell; ANA: antinuclear antibody.

Urological relevance

- **Haematuria work-up:** proteinuria, casts or dysmorphic cells suggest (suggests) medical renal disease; urological bleeding rarely raises protein much (adds little protein).
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy or partial nephrectomy:** >50% loss of mass causes hyperfiltration, proteinuria and FSGS. AUA (American Urological Association) 2021 counts proteinuria as a reason to prefer (favour) nephron sparing and refer to nephrology.
- **Living donors:** significant proteinuria excludes donation. **Also know:** mTOR (mammalian target of rapamycin) inhibitors cause it.

Management

- Treat the cause; blood pressure control; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

Diagram to draw: glomerulus and proximal tubule marking glomerular, tubular and overflow sites.

> **Key points to remember:**
> - Normal <150 mg/day; nephrotic >3.5 g/day.
> - ACR A2 30–300, A3 >300 mg/g.
> - Glomerular, tubular, overflow; orthostatic proteinuria is benign.
> - Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

5 · Infection: leucocyte esterase, nitrite and culture

	Leucocyte esterase	Nitrite
Detects	Host response (neutrophils)	Organism (gram-negative nitrate reductase)
Performance	Sensitive, non-specific	Specific, insensitive
Classic miss	Glycosuria, proteinuria	**Enterococcus**; low counts

- If only one is positive, do microscopy and culture before antibiotics. A dipstick never replaces culture.
- Culture thresholds are **≥10^5 CFU/mL** for clean catch and **≥10^3 CFU/mL** for a catheter specimen; **any growth on suprapubic aspiration is significant**.

6 · Sediment microscopy

- Squamous cells mean contamination, while **renal tubular cells always mean renal pathology**. Eosinophils suggest acute interstitial nephritis.
- Schistosoma haematobium ova have a **terminal spike**.

Cast	Meaning
Hyaline	May be normal (exercise, heat)
RBC	Glomerulonephritis

Cast	Meaning
WBC	Pyelonephritis, interstitial nephritis
Muddy-brown granular	Acute tubular necrosis
Fatty	Nephrotic syndrome

LAQ Urinary crystals

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 11 Paper 2, Q2

Introduction

- Crystals are precipitated urinary solutes seen on microscopy of the sediment. They occur in both normal and diseased states.
- Their value lies in suggesting stone type, metabolic disease, infection or drug toxicity. **Cystine is the one crystal that is never normal.**

How crystals form (physicochemistry)

- Urine is normally **supersaturated** with calcium oxalate and phosphate. Crystals form only when the concentration product passes a critical level.

Saturation zone	What happens
Undersaturated (below Ksp, solubility product)	No crystals; stones may dissolve
Metastable (Ksp to Kf, formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors effective
Unstable (above Kf)	Spontaneous nucleation, growth and aggregation

- **Nucleation** in vivo is mostly heterogeneous (on cell debris or other crystals). Crystals then grow, aggregate and are retained, for example on **Randall plaque**.
- **Inhibitors:** citrate, magnesium, pyrophosphate, Tamm–Horsfall protein, nephrocalcin, osteopontin. No inhibitor exists for uric acid or cystine, so these depend on pH and concentration.
- **Determinants of crystalluria:** urine volume, solute excretion, pH, temperature and inhibitors.

Examination of urine for crystals

- Fresh midstream specimen, examined within 2 hours at body or room temperature. **Refrigeration precipitates crystals** that may not redissolve.
- Dipstick pH (physiological range 4.5–8.0); centrifuged sediment; polarised light helps identify crystals.
- Phosphaturia (cloudy alkaline urine) **clears with acetic acid**; pyuria does not.
- Candida mimics calcium oxalate and red cells; budding and hyphae separate it.

Classification by urine pH and shape

Urine pH	Crystal	Shape	Clinical significance
Acidic	**Uric acid**	Rhomboid or lemon-shaped plates, rosettes; yellow-brown	Uric acid stones (pH <5.5; pKa 5.35), gout, tumour lysis
Acidic	Amorphous urates	Pink "brick-dust" granules	Usually normal (concentrated, cooled urine)
Acidic	**Cystine**	**Hexagonal** colourless plates	Always abnormal: cystinuria
Acidic or alkaline	**Calcium oxalate dihydrate**	**Envelope** (octahedral)	Common, often normal; hypercalciuria
Acidic or alkaline	Calcium oxalate monohydrate	Dumbbell or oval	Hyperoxaluria; ethylene glycol poisoning
Alkaline	**Triple phosphate (struvite)**	**Coffin-lid** prisms	Urease-producing infection (Proteus); staghorn stones
Alkaline	Amorphous phosphate	White granules	Benign phosphaturia
Alkaline	Calcium phosphate (brushite, apatite)	Needles or rosettes; amorphous	RTA (renal tubular acidosis), hyperparathyroidism
Alkaline	Ammonium biurate	"Thorny apple" spheres	Old specimens; ammonium urate stones (diarrhoea, laxative abuse, ileostomy)
Any	Cholesterol	Notched flat plates	Nephrotic syndrome, lipiduria

- **Rare metabolic crystals:** 2,8-dihydroxyadenine, brown spiculated spheres (APRT [adenine phosphoribosyltransferase] deficiency, radiolucent stones); xanthine (hereditary xanthinuria, high-dose allopurinol); tyrosine needles and leucine spheres (severe liver disease).

Drug crystals

- **Two routes:** the drug itself precipitates (direct), or it changes urine chemistry (indirect).

Drug	Crystal and features	Prevention
Indinavir (protease inhibitor)	Needles or fan-shaped plates; crystalluria in about two-thirds of new users; stones 4–13%; radiolucent, often invisible on CT	Hydration; more soluble below pH 5.5
Atazanavir	Highest stone risk of newer antiretrovirals	Hydration
Sulfonamides (sulfadiazine)	"Sheaves of wheat"	Hydration, alkalinise
Acyclovir (high-dose intravenous)	Birefringent needles; acute kidney injury	Slow infusion, hydration
Amoxicillin, ciprofloxacin (high dose)	Needles; worse in alkaline urine	Hydration
Methotrexate (high dose)	Crystal nephropathy	Alkalinise, hydration
Triamterene, ephedrine, guaifenesin, silicate antacids	Stones of drug or metabolite	Stop the drug

- **Indirect:** loop diuretics and steroids (hypercalciuria, calcium stones); acetazolamide and topiramate (alkaline urine, hypocitraturia, calcium phosphate); cytotoxics (uric acid); laxative abuse (ammonium urate).

Clinical significance

1. **Diagnosis of stone type** before stone analysis is available; crystals in a stone former help direct metabolic work-up.
2. **Recurrence risk:** persistent crystalluria in repeated early-morning samples, and large or aggregated crystals, suggest active stone disease.
3. **Infection:** struvite crystals with urine pH >7.2 signal a urease-producing organism.
4. **Acute kidney injury:** uric acid (tumour lysis), calcium oxalate monohydrate (ethylene glycol) and drug crystals cause intratubular obstruction.
5. **Monitoring therapy:** absence of cystine or uric acid crystals confirms adequate alkalinisation. - Most crystalluria in healthy people is transient and harmless.

Management principles

- Repeat on a fresh, warm specimen; confirm pH; 24-hour urine metabolic evaluation and stone analysis.
- **Fluids** for urine output above 2–2.5 L/day.
- **pH manipulation:** alkalinise (potassium citrate) for uric acid, cystine, methotrexate and sulfonamides; avoid alkali in calcium phosphate and struvite.
- Treat the cause: antibiotics and stone clearance for struvite; stop or change the drug; specific therapy (thiols for cystine, allopurinol for uric acid).

Diagram to draw: sketches of hexagon (cystine), envelope (calcium oxalate), coffin lid (struvite), rhomboid (uric acid), sheaf of wheat (sulfonamide).

> **Key points to remember:**
> - Cystine hexagons are always pathological.
> - Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
> - Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
> - Refrigerated urine precipitates crystals: examine fresh within 2 hours.

7 · Serum tests

- Check creatinine and estimated glomerular filtration rate (eGFR) whenever obstruction is suspected, because renal function sets the urgency of drainage.
- Creatinine rises only after about **50%** loss of GFR. CKD stages by eGFR (mL/min/1.73 m²) are G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29 and G5 <15.

8 · Tumour markers and cytology

SAQ Tumour markers in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3d

Definition and uses

- A substance made by a tumour, measured in blood, urine or tissue.
- Uses: screening, diagnosis, staging, prognosis, response, and detecting relapse. Ideal: specific, sensitive, proportional to bulk, short half-life.

Prostate

- **PSA (prostate-specific antigen):** a kallikrein serine protease (hK3) from luminal cells; **organ-specific, not cancer-specific** (BPH (benign prostatic hyperplasia), prostatitis, instrumentation raise it). Best marker for post-treatment follow-up.
- **Refinements:** density (PSA ÷ volume; cut-offs 0.08–0.15); %free PSA (lower in cancer; for PSA 4–10 ng/mL); velocity (little value); **PHI (Prostate Health Index) = (p2PSA ÷ free PSA) × √total PSA**; 4Kscore.
- **Urine:** PCA3 (prostate cancer antigen 3) after DRE (digital rectal examination); score = PCA3 mRNA ÷ KLK3 mRNA × 1000.
- **Also know:** tissue genomic classifiers.

Testis (germ cell tumours)

Marker	Half-life	Source	Never raised in
AFP (α-fetoprotein)	5–7 days	Embryonal, yolk sac	Pure seminoma, choriocarcinoma
β-hCG (human chorionic gonadotrophin)	24–36 hours	Choriocarcinoma, embryonal; seminoma in ~15%	Yolk sac tumour
LDH (lactate dehydrogenase)	24 hours	Non-specific; reflects bulk	—

- Measure before and after orchidectomy, on chemotherapy, in follow-up. Post-orchidectomy values give the **S stage** (S1: LDH <1.5 × normal, hCG <5000 mIU/mL, AFP <1000 ng/mL) and the **IGCCCG (International Germ Cell Cancer Collaborative Group)** risk group.
- Raised AFP with "pure seminoma" histology → treat as NSGCT (non-seminomatous germ cell tumour). False positives: liver disease, marijuana, LH (luteinising hormone) cross-reaction.
- **miR-371a-3p:** sensitive new marker; not raised by teratoma.

Bladder and upper tract

- **Urine cytology:** sensitive and specific for high-grade cancer, poor for low grade; for surveillance, not screening.
- NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), ImmunoCyt, UroVysion FISH (fluorescence in situ hybridisation): selected use only.

Others

- **Kidney and penis:** no validated serum marker.
- **Adrenal:** metanephrines, cortisol, aldosterone–renin ratio. **Neuroblastoma:** urinary VMA (vanillylmandelic acid)/HVA (homovanillic acid) raised in 90–95%.

Diagram to draw: marker decline curve after orchidectomy using half-lives.

Key points to remember:

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

SAQ Urine cytology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 1, Q3b

Definition and principle

- Microscopy of shed urothelial cells (Papanicolaou, 1945).
- **Principle:** high-grade cells are poorly cohesive, shed readily and look atypical; low-grade cells look almost normal. So cytology detects **high-grade tumours and CIS (carcinoma in situ)** but misses low grade.

Specimens

- Voided urine (not the first morning sample, whose cells have degenerated), at least 30 mL; bladder washing; upper tract washings.

Performance

- Sensitivity 31–62%, specificity 94–100%; strongly grade-dependent.
- **False positives:** instrumentation, stones, infection, radiotherapy, intravesical BCG (bacille Calmette–Guérin), polyomavirus ("decoy" cells), ileal conduit.
- **False negatives:** low grade, low cellularity, degeneration.

The Paris System

- Published 2016; **2nd edition 2022**. Aim: detect **HGUC (high-grade urothelial carcinoma)**. Key criterion: **N:C (nuclear-to-cytoplasmic) ratio**, plus hyperchromasia, coarse chromatin and irregular nuclear membrane.

Category	Criteria
Non-diagnostic	Inadequate
Negative for HGUC	N:C <0.5, or atypia explained by a benign cause
AUC (atypical urothelial cells)	N:C ≥0.5 and one feature
Suspicious for HGUC	N:C ≥0.7, at least two features, few cells (<5–10)
HGUC	N:C ≥0.7, at least two features, many cells
LGUN (low-grade urothelial neoplasm)	Fibrovascular cores; rarely diagnosable
Other malignancy	Squamous, adenocarcinoma, secondary

- It reduces equivocal "atypia" reports; cancer risk rises with each category.

Clinical uses

- **Not** for screening or initial microhaematuria evaluation (AUA/SUFU [American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2020).

- Gross haematuria; NMIBC (non-muscle-invasive bladder cancer) surveillance (mandatory in high risk); suspected CIS; upper tract tumours.
- **Positive cytology, normal cystoscopy:** image upper tracts, selective washings ± ureteroscopy, random bladder and prostatic urethral biopsies.
- **Also know:** FISH (fluorescence in situ hybridisation; chromosomes 3, 7, 17, 9p21), NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), CxBladder: more sensitive, less specific; none replaces cystoscopy.

Diagram to draw: normal urothelial cell beside an HGUC cell with high N:C ratio.

> **Key points to remember:**
> - High specificity, low sensitivity; good for high grade and CIS.
> - Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
> - Positive cytology, normal cystoscopy: upper tracts, random biopsies.
> - Not for screening or initial microhaematuria.

9 · Stone, adrenal and andrology tests

*See the paper answer **Metabolic evaluation in stone disease** (Section 11, Ch 99).*

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- Aldosterone **>15 ng/dL** with an aldosterone-to-renin ratio **>20** leads to adrenal vein sampling.

*See the paper answer **Semen analysis nomenclature.** (Section 9, Ch 69).*

Also, from the condensed book (not in the paper answer):

- Leucocytes **>1 million/mL** define leucocytospermia.

10 · Office tests

*See the paper answer **Uroflowmetry** (Section 7, Ch 107).*

Also, from the condensed book (not in the paper answer):

- Qmax **>15 mL/s** gives a low probability of obstruction and 10–15 mL/s is equivocal. A post-void residual **<100 mL** is generally acceptable.
- **No antibiotic prophylaxis** for simple cystoscopy or urodynamics without risk factors (age, steroids, immunodeficiency, anatomical anomaly, colonised material, long hospitalisation).

4 Basic Principles of Urologic Ultrasonography

1 · Physical principles

- Piezoelectric crystals convert alternating current into a longitudinal mechanical wave. Echo amplitude gives pixel brightness, and round-trip time gives depth. The urological range is **3.5–20 MHz**.
- Velocity equals frequency × wavelength. Speed of sound in soft tissue is **1540 m/s**, so frequency and wavelength vary inversely.

	Axial resolution	Lateral resolution
Separates	Objects one behind the other	Objects side by side
Improved by	Higher frequency	Narrow beam: focal zone at the depth of interest

> **KEY POINT**
> Absorption is proportional to frequency, so high frequency resolves finely but penetrates poorly: **7–18 MHz** for the testis and **3.5–5 MHz** for the adult kidney.

- **Reflection depends on the impedance difference at an interface, not on density.** Soft tissues cluster together (1.4–1.7 × 10^6 kg/m²·s), while air (0.0004) and bone (7.8) are extremes. Hence gel, bowel-gas obscuration and stone shadowing.

Artefacts

Artefact	Mechanism	Appearance
Acoustic shadowing	Strong reflection or attenuation	Dark band behind a stone; the stone looks crescentic, so size is inaccurate
Increased through-transmission	Less attenuation than surroundings	Bright tissue behind a simple cyst
Edging (critical angle)	Refraction along a curved surface	Shadow at the testicular pole
Reverberation	Bounce between two strong reflectors	Equally spaced echoes; the second at **twice the depth**
Comet tail	Many small reflectors (bowel gas)	Bright tail; may hide the lower renal pole
Twinkling	Rough reflective surface on colour Doppler	Colour mosaic behind a stone; detects small non-shadowing stones

2 · Modes

SAQ Doppler in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 1, Q3b

Principle

- **Doppler shift:** sound reflected from moving red cells changes frequency in proportion to velocity and the cosine of the insonation angle: $\Delta f = 2f_0 v \cos\theta / c$. Keep the angle **below**

60°.

- **RI (resistive index) = (PSV − EDV) / PSV** (peak systolic and end-diastolic velocity); reflects downstream resistance.

Modes

Mode	Shows	Use
Colour	Velocity and direction (red towards, blue away by convention)	Flow in kidney, testis, ureteric jets
Spectral (duplex)	Waveform at one point; velocities, RI	Penile and renal arteries
Power	Amplitude only; 3–5 times more sensitive, less angle-dependent	Low flow: torsion, prostate lesions

Applications

Organ	Findings
Scrotum	**Torsion:** absent intratesticular flow (compare with the other side); epididymitis: increased flow; **varicocele:** venous reflux on Valsalva; intratesticular RI >0.6 suggests poor spermatogenesis
Penis (after intracavernosal injection)	**PSV <25 cm/s = arterial ED (erectile dysfunction)**, 25–35 equivocal, >35 normal; **EDV >5 cm/s and RI <0.75 = veno-occlusive dysfunction**; Peyronie plaque assessment
Priapism	Ischaemic: absent cavernosal flow; non-ischaemic: high flow (PSV >50 cm/s), locates the fistula
Kidney	Normal RI 0.47–0.70; **raised RI in acute obstruction** (useful in pregnancy, where change in RI is more accurate); renal artery stenosis: high velocity and parvus-tardus waveform; arteriovenous fistula after biopsy or PCNL (percutaneous nephrolithotomy)
Transplant	Arterial thrombosis: absent signal; **venous thrombosis: reversed diastolic flow**; stenosis: high velocity; rejection: loss of diastolic flow, high RI
Bladder	Ureteric jets (absent on the obstructed side)
Tumours	Vascularity of renal and testicular masses; tumour thrombus; hypervascular prostate cancer foci

- **Also know:** intraoperative micro-Doppler in microsurgical varicocelectomy; contrast-enhanced ultrasound and elastography in multiparametric ultrasound.

Limitations

- Operator- and angle-dependent; poor in obesity and deep vessels; early torsion may show collateral flow, so **Doppler cannot exclude torsion**.

Diagram to draw: spectral waveform marking PSV and EDV with the RI formula.

> **Key points to remember:**
> - RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
> - Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
> - Absent intratesticular flow = torsion, but explore on clinical grounds.

Also, from the condensed book (not in the paper answer):

- **Elastography**: strain elastography (manual compression) is qualitative and operator-dependent. **Shear-wave elastography is quantitative** (shear waves 1–10 m/s; $E = 3\rho c^2$). Hard does not mean malignant.

3 · Contrast-enhanced ultrasound (CEUS)

LAQ Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy

Paper answer · P1 · merged from 3

Asked as: P1 LAQ "Iodinated and other contrast agents (use in urology) and their complications" · P1 SAQ "Non iodinated contrast materials" · P1 SAQ "Acetylcysteine and Contrast Induced Nephropathy (CIN)"

Introduction

- Iodine (atomic number 53, K-edge 33 keV) absorbs X-rays strongly. All iodinated agents are built on a **tri-iodinated benzene ring**.
- They are filtered by the glomerulus, not secreted or reabsorbed, so they concentrate in urine. Half-life is about 2 hours with normal function; biliary excretion rises in renal failure.
- Modern practice uses **non-ionic low-osmolar** (iohexol, about 600–850 mOsm/kg) or **iso-osmolar** (iodixanol, 290) agents. High-osmolar ionic agents caused far more reactions.
- Uses: CT (computed tomography) urography, IVU (intravenous urography), pyelography, cystourethrography, angiography.
- Non-iodinated agents are used when iodinated contrast is risky (allergy, renal impairment) or with MRI (magnetic resonance imaging) and ultrasound.

Classification of iodinated agents

Class	Example	Osmolality (mOsm/kg)	Iodine : particles
High-osmolar ionic monomer	Diatrizoate	~1500–2000 (5–8 × plasma)	3 : 2
Low-osmolar non-ionic monomer	Iohexol, iopamidol	~600–850	3 : 1
Low-osmolar ionic dimer	Ioxaglate	~600	6 : 2
Iso-osmolar non-ionic dimer	Iodixanol	~290	6 : 1

- **Why it matters:** a higher ratio gives more opacity per particle. Fewer particles mean less fluid shift, pain and reactions. So non-ionic low- or iso-osmolar agents are standard intravascularly.

Types of contrast agent and their uses in urology

Agent	Principle	Urological use
Iodinated: ionic high-osmolar (diatrizoate); non-ionic low-osmolar (iohexol); iso-osmolar dimer (iodixanol, 290 mOsm/kg)	Iodine absorbs X-rays; glomerular filtration	CT urography, angiography; retrograde pyelogram, urethrogram, cystogram
Gadolinium (GBCA, gadolinium-based contrast agent)	Shortens T1 relaxation; chelated	MRI of renal mass, prostate, MR urography

Agent	Principle	Urological use
Microbubbles	Gas in shell, intravascular	CEUS (contrast-enhanced ultrasound): cysts; voiding urosonography
Carbon dioxide	Negative contrast	Angiography in CKD (chronic kidney disease)

Complications of iodinated contrast

A. Hypersensitivity (allergic-like) reactions

- Mostly non-IgE (immunoglobulin E) direct mast-cell release: idiosyncratic, not dose-related. Severe reactions: 0.22% with ionic vs 0.04% with non-ionic agents (Katayama, 1990).

Type	Timing	Features
Immediate	Within about 1 hour (most within minutes)	Urticaria, angio-oedema, bronchospasm, hypotension; anaphylaxis if severe
Non-immediate	Hours to days	Maculopapular rash; T-cell mediated, milder

- Risk factors: previous reaction (strongest), asthma, atopy, severe cardiac disease, drug allergy, female sex. **Shellfish ("iodine") allergy is not a cross-reaction.**
- **Physiological reactions:** warmth, metallic taste, nausea, vasovagal hypotension with bradycardia, arrhythmia, seizures.
- Retrograde studies are safe in allergy.

B. Contrast-associated acute kidney injury (see the separate CIN section below)

C. Others

- **Metformin:** lactic acidosis only if AKI (acute kidney injury) develops. ACR (American College of Radiology): eGFR (estimated glomerular filtration rate) ≥30 and no AKI, continue. eGFR <30 or AKI, withhold 48 hours and restart when function (creatinine) is stable.
- **Thyroid:** iodine load can trigger hyperthyroidism/thyrotoxicosis (Jod-Basedow) in Graves' disease or nodular goitre. It blocks radio-iodine scans or therapy for several weeks.
- **Extravasation:** usually minor, but large power-injected volumes can rarely cause compartment syndrome or skin necrosis. Elevate the limb, apply compresses and observe. Get a surgical review if swelling increases, blisters form, or perfusion or sensation changes.
- **Also know:** iodide "mumps"; myasthenia flare; pregnancy (crosses placenta, not teratogenic); breastfeeding can continue; phaeochromocytoma safe with IV (intravenous) non-ionic agents; dialysis patients need no urgent dialysis.

Prevention and treatment of contrast reactions

Premedication for high-risk patients / prior reaction (ACR 2023)

Regimen	Drugs
1	Prednisone 50 mg orally 13, 7 and 1 hour before + diphenhydramine 50 mg 1 hour before
2	Methylprednisolone 32 mg orally 12 and 2 hours before + diphenhydramine 50 mg 1 hour before

- First consider ultrasound, MRI or non-contrast CT. Breakthrough reactions still occur, so keep resuscitation drugs ready. Retrograde pyelography is safe even after anaphylaxis, because

little contrast is absorbed.

Treating reactions `Stop injection → airway, oxygen, call for help Urticaria → observe ± H1-antihistamine (may progress) Bronchospasm → salbutamol inhaler 2 puffs (90 µg each), up to 3 times Anaphylaxis → intramuscular (IM) adrenaline 0.3–0.5 mg (1 mg/mL) every 5–15 min (repeat), max 1 mg; IV fluids Vasovagal → legs up, IV fluids, IV atropine Observe ≥30 min; later allergy testing; record the reaction`

Contrast-induced nephropathy (CIN) and contrast-associated AKI

Definitions

- **CIN (contrast-induced nephropathy)** (older term): creatinine rise ≥25% or ≥0.5 mg/dL within 48–72 hours of iodinated contrast, with no other cause.
- Current terms (ACR):
- **CA-AKI (contrast-associated acute kidney injury):** AKI within 48 hours of contrast, whatever the cause.
- **CI-AKI (contrast-induced AKI):** the subset actually caused by contrast.
- **KDIGO (Kidney Disease: Improving Global Outcomes) AKI:** creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for ≥6 hours.

Pathophysiology

- **Renal (afferent) vasoconstriction** causes **medullary hypoxia** and ischaemic tubular injury (ischaemic ATN, acute tubular necrosis), plus **direct tubular toxicity**. Usually transient; dialysis rarely needed.

Risk factors

- **Patient:** CKD (strongest), diabetes, age, dehydration, heart failure, anaemia, NSAIDs (non-steroidal anti-inflammatory drugs).
- **Procedure:** high contrast volume, repeated doses within 48–72 hours, intra-arterial route, high-osmolar agents.
- **eGFR thresholds:** real risk mainly at **eGFR <30 mL/min/1.73 m²** (IV contrast); ESUR (European Society of Urogenital Radiology) uses **<45** for intra-arterial first-pass exposure.
- ACR/NKF (National Kidney Foundation) 2020: older uncontrolled studies overstated the risk; it is rare but real.

Prevention

- Consider ultrasound or non-contrast CT instead.
- Lowest dose of a **low- or iso-osmolar** agent; weight-based dosing; a low tube voltage (80 kVp, kilovolt peak) needs less contrast. Avoid repeat doses within 48 hours; stop NSAIDs.
- **IV isotonic saline** (e.g. 1 mL/kg/h before and after) for high-risk patients (eGFR <30 not on dialysis): the only proven measure.
- **PRESERVE (2018):** sodium bicarbonate and **NAC (N-acetylcysteine)** gave no benefit over saline.
- **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59.
- Prophylactic dialysis (removing contrast by dialysis) does not reduce risk.
- **Metformin:** withhold for 48 hours only if eGFR <30 or AKI (see above).

Acetylcysteine

Introduction

- **NAC:** acetylated form of the amino acid cysteine; a **thiol (sulfhydryl, –SH) compound**. Given orally, intravenously or by nebuliser.

Mechanisms of action

- **Glutathione precursor:** supplies cysteine, the rate-limiting substrate for **glutathione** synthesis.
- **Antioxidant:** the free –SH group scavenges reactive oxygen species.
- **Mucolytic:** –SH breaks **disulfide bonds** in mucoproteins, lowering mucus viscosity.
- **Thiol–disulfide exchange:** in theory forms a soluble mixed disulfide with cystine.

Clinical uses

- **1. Paracetamol (acetaminophen) overdose: main indication.**
- Overdose saturates normal pathways; the toxic metabolite **NAPQI** (N-acetyl-p-benzoquinone imine) builds up once glutathione is exhausted, causing hepatic necrosis.
- NAC restores glutathione. It is most effective **within 8 hours**, but still worth giving later.
- **IV 21-hour regimen:** 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h. Shorter two-bag regimens are now used in some countries.
- **2. Mucolytic:** COPD (chronic obstructive pulmonary disease), cystic fibrosis.

Urological relevance

- **Contrast-induced AKI:** once widely used (early small trials, KDIGO 2012 suggestion). **PRESERVE (2018):** NAC gave no benefit over saline. It is **no longer recommended**; isotonic saline hydration is the only proven measure.
- **Cystinuria:** thiol action is theoretical; standard thiols are **tiopronin** and D-penicillamine, not NAC.
- **Cyclophosphamide cystitis: mesna** (another thiol) is the agent used.
- **Mucus after enterocystoplasty or neobladder:** sometimes used orally or in irrigation; evidence weak.
- **Male infertility:** antioxidant for oxidative stress (high ROS (reactive oxygen species), sperm DNA fragmentation); limited evidence.
- **Ischaemia–reperfusion** (renal surgery, transplant): antioxidant protection studied; not established.

Adverse effects

- **Anaphylactoid reactions** with IV loading (flushing, rash, bronchospasm): slow the infusion, give antihistamine.
- Nausea, vomiting; unpleasant sulphur smell; bronchospasm when nebulised in asthma.
- Safe in pregnancy and renal impairment for paracetamol poisoning; benefit outweighs risk.

Non-iodinated contrast agents

1. Gadolinium-based contrast agents (GBCAs)

- **Mechanism:** paramagnetic gadolinium **shortens T1 relaxation** of nearby water, giving bright enhancement. Free gadolinium is toxic, so it is **chelated** (linear or macrocyclic; macrocyclic is more stable). Usual dose 0.1 mmol/kg; renal excretion.

- **NSF (nephrogenic systemic fibrosis):** skin and organ fibrosis, joint contractures, no proven treatment. Risk: eGFR <30 mL/min/1.73 m², dialysis, AKI.

ACR group	Agents	NSF risk
I (linear)	Gadodiamide, gadopentetate, gadoversetamide	Highest
II	Gadobenate, gadoteridol, gadoterate, gadobutrol	**Little or none, even in ESRD (end-stage renal disease)**
III	Gadoxetate (liver-specific agent)	Few NSF data; recent ACR editions have moved some agents between groups, so check the current manual

- **Gadolinium deposition:** traces retained in brain (dentate nucleus) and bone, mainly with linear agents; no proven harm. Europe suspended most linear agents (2017).
- **Other:** allergic reactions rarer than with iodinated agents; not nephrotoxic at usual doses; avoid in pregnancy unless essential.
- **Uses:** renal mass, prostate mpMRI (multiparametric MRI), MR urography and angiography.

2. Ultrasound contrast (microbubbles)

- Gas core (perfluorocarbon or sulfur hexafluoride) in a lipid or albumin shell; strong harmonic echoes.
- Purely **intravascular**; gas exhaled by lungs. **No nephrotoxicity or radiation.** Rare cardiopulmonary reactions.
- **CEUS:** solid vs cystic renal mass (Bosniak), ablation monitoring, surveillance in CKD. Intravesical use: **voiding urosonography (ceVUS) for VUR (vesicoureteral reflux)**; ureteric patency after PCNL (percutaneous nephrolithotomy).

3. Other agents

- **CO_2 angiography:** negative contrast; not allergenic or nephrotoxic. Useful in CKD; **never above the diaphragm** (cerebral embolism).
- **Ferumoxytol** (iron oxide): MRI agent in renal failure. Lymphotropic iron nanoparticles help nodal staging (nodal metastases fail to darken).
- **Barium:** GI (gastrointestinal) studies only; contraindicated if perforation suspected.
- **Also know:** air as negative contrast (historical); radiotracers (^{99m}Tc, PET (positron emission tomography)) are not contrast agents; MRI safety (pacemakers, metal).

Diagrams to draw

- Tri-iodinated benzene ring, monomer versus dimer, on an osmolality scale.
- Chelated gadolinium (macrocyclic vs linear ring).
- Contrast, then vasoconstriction plus tubular toxicity, then medullary hypoxia, then AKI.
- Paracetamol to NAPQI to glutathione conjugation, with NAC replenishing glutathione.

Key points to remember:

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless; AMACING: no prophylaxis at eGFR 30–59.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic; paracetamol antidote, best within 8 h; cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux. CO_2 angiography: never above diaphragm.

Also, from the condensed book (not in the paper answer):

- Bubbles cannot cross the glomerulus and **never opacify the collecting system**.

4 · Safety and ALARA

	Mechanical	**Thermal**
Nature	Cavitation, streaming	Heating
Greatest	Near gas: lung, bowel	High frequency, focusing, long dwell, Doppler
Index	Mechanical index (MI): **≤0.7** low risk; limit time if >0.4 next to lung or bowel	Thermal index (TI) is the probability of a 1 °C rise; keep **<2** in urology

- **MI and TI are displayed estimates, not safety limits.**
- ALARA (as low as reasonably achievable) means low power, short examinations and adjusted focus. The **informed operator is the most important safety factor**.

5 · Renal ultrasound

- Normal kidney length is **10–12 cm** and width 4–5 cm, measured in the true midsagittal plane. Cortex is **>7 mm**, parenchyma **>15 mm** and intrarenal RI **0.47–0.70**.
- Cortex is slightly hypoechoic to liver; an iso- or hyperechoic cortex suggests medical renal disease.
- **Pyramids mimic calyces**: pyramids lie outside the hilar fat, and the collecting system lies inside it.
- Ultrasound-guided PCNL access is hard with a long skin-to-stone distance, a non-dilated system or a staghorn.
- After renal biopsy, complications rise with **severe hypertension, not the number of passes**. Haematuria appears within 6 h, so 6 h of observation equals admission.

*See the paper answer **Radiology for solid renal masses.** (Section 13, Ch 137).*

6 · Bladder and pelvic ultrasound

- Bladder volume = W × H × L × **0.625**.

*See the paper answer **Methods to estimate prostate size** (Section 16, Ch 149).*

Also, from the condensed book (not in the paper answer):

- **Bladder wall thickness predicts bladder outlet obstruction better than uroflowmetry**, post-void residual or prostate volume.
- **An automated bladder scanner is not an imaging study.** Fluid outside the bladder (ascites, urinoma, lymphocele) gives a wrong residual.

7 · Scrotal ultrasound

- Show both testes side by side, because **diffuse lymphoma or leukaemia is visible only by comparison**.
- Intratesticular PSV is **<10 cm/s**, and an RI **>0.6** suggests impaired spermatogenesis.

> **KEY POINT**
>
> In torsion, epididymal collaterals can appear within hours, and reactive hyperaemia can give a false-negative study. Ultrasound cannot diagnose torsion; only the surgeon can, so strong suspicion means exploration.

8 · Penile and urethral ultrasound

*See the paper answer **Doppler in urology** (Section 1, Ch 4).*

Also, from the condensed book (not in the paper answer):

- Arterial dilation **<75%** and acceleration time **>110 ms** also indicate arteriogenic ED. PSV is the most accurate measure, and arteriogenic ED flags systemic cardiovascular disease.

9 · Transrectal ultrasound (TRUS) and therapy

*See the paper answer **TRUS and prostate biopsy** (Section 17, Ch 155).*

*See the paper answer **HIFU** (Section 17, Ch 163).*

Also, from the condensed book (not in the paper answer):

- **Pregnancy**: ultrasound is first-line and safe in all trimesters, but grey-scale is poor for ureteric stones.

5 Principles of CT and Radiographic Imaging

1 · Radiation biology and safety

	Deterministic	Stochastic
Threshold	Yes	**None** (linear no-threshold model)
Dose increases	Severity	Probability (cumulative)
Mechanism	Cell loss	Mutation in an oncogene or tumour suppressor
Examples	Erythema, epilation, sterility, cataract	Cancer

- Absorbed dose is in gray (Gy). **Effective dose** (sievert, Sv) adds tissue weighting and sets exposure limits.

Study	Effective dose
Chest x-ray	**0.07 mSv**
Kidney–ureter–bladder (KUB) film / two-view abdomen	**0.7 / 1.4 mSv**
Ultra-low-dose CT abdomen–pelvis	About 1 mSv (1–2)
Low-dose CT	<3–4 mSv
Conventional CT abdomen–pelvis	**10–20 mSv**; 10 mSv carries about a 1 in 1000 cancer risk

> **KEY POINT**
>
> Stochastic risk has no threshold, so no dose is safe. Hence ALARA (as low as reasonably achievable): a risk–benefit decision before every ionising study, low-dose protocols and non-ionising alternatives.

2 · Radiographic studies

- KUB is used to follow radio-opaque stones, before shock-wave lithotripsy. CT detects more than **95%** of stones. An **upright film is most sensitive for free air**.

LAQ Describe IVU & its applications in Urology.

Paper answer · P1

Type: LAQ (25 marks)
Source: 2021-Summer Paper 1, Q2

Definition and principle

- IVU (intravenous urography; also IVP, intravenous pyelography or excretory urography) is a series of timed abdominal radiographs after intravenous iodinated contrast.
- Contrast is **freely filtered by the glomerulus and not reabsorbed**. As water is reabsorbed it is concentrated in the tubules (**nephrogram**) and then outlines the calyces, pelvis, ureters and bladder (**pyelogram**).
- So IVU shows **anatomy plus a rough, qualitative idea of function and drainage** of each kidney.
- **Current status:** largely replaced by CTU (CT [computed tomography] urography). It is still used where CT is unavailable (common in India) or when dynamic excretion and drainage

must be seen.

Contrast media

Type	Example	Feature
High-osmolar ionic	Diatrizoate	More reactions; obsolete for IV use
Low-osmolar non-ionic (standard)	Iohexol, iopamidol	Fewer reactions
Iso-osmolar non-ionic	Iodixanol	High-risk renal patients

Preparation

- Check creatinine or eGFR (estimated glomerular filtration rate), allergy, asthma, pregnancy, diabetes on metformin, myeloma.
- Laxative the night before (bowel gas hides the kidneys); short fast but **no dehydration**.
- Premedicate high-risk patients, e.g. prednisone 50 mg at 13, 7 and 1 h plus diphenhydramine 50 mg at 1 h.

Technique: film sequence

Film	Timing	Shows
Scout (control) KUB (kidney–ureter–bladder)	Before contrast	Calcification, bones, gas, technique
Contrast	Low-osmolar, about 1 mL/kg (usually 50–100 mL) IV	—
Nephrogram / nephrotomogram	About 1 min (60–90 s)	Renal size, outline, parenchyma
Pyelogram	5 min	Calyces; then apply **ureteric compression**
Compression film	10 min	Well-distended calyces and pelvis
Release (full-length) film	About 15 min	Whole ureters
Bladder and **post-void** films	20–30 min	Bladder outline, residue, lower ureters
Delayed films	Up to 24 h	Level of obstruction; **no excretion at 24 h = non-functioning kidney**
- **Also know:** films in expiration; prone film fills the ureters; oblique views; erect film for nephroptosis.		
- Compression is avoided in suspected obstruction or colic, aortic aneurysm, recent abdominal surgery and trauma.		

Contraindications

- Previous severe contrast reaction; significant renal impairment; pregnancy (relative); dehydration, myeloma.

Nephrogram patterns

Pattern	Causes
Absent	Renal artery occlusion, agenesis, avulsion
Delayed, increasingly dense	**Acute obstruction**, renal artery stenosis
Striated	Acute pyelonephritis, obstruction
Rim (cortical rim) sign	Renal infarction (capsular collaterals)
Bilateral persistent dense	Hypotension, ATN (acute tubular necrosis)

Applications in urology

Condition	IVU features
Calculi	Confirms an opacity is in the tract; **radiolucent stone = filling defect**; site, obstruction, calyceal anatomy before SWL (shock wave lithotripsy) or PCNL (percutaneous nephrolithotomy)
Acute obstruction	Delayed dense nephrogram, delayed excretion, dilatation down to the block, **standing column** of contrast, forniceal rupture
PUJ (pelviureteric junction) obstruction	Ballooned pelvis, delayed drainage, non-dilated ureter; crossing-vessel indentation
Duplex system	Upper moiety obstructed or non-functioning: lower moiety displaced, "**drooping lily**"
Ureterocele	"**Cobra head**" (spring onion) at the bladder
Horseshoe kidney	Low, medially rotated; lower calyces point medially
Retrocaval ureter	"Reverse J" or **fish-hook** at L3
Medullary sponge kidney	"**Paintbrush**" papillary striations
Papillary necrosis	**Ring sign**, "lobster claw" calyces
Genitourinary TB (tuberculosis)	**Moth-eaten** calyx, **phantom calyx**, Kerr's kink, **pipestem** or beaded ureter, contracted bladder, autonephrectomy
Urothelial tumour	Irregular filling defect, non-visualised calyx
Retroperitoneal fibrosis	Medial deviation and tapering of ureters with hydronephrosis
Pelvic lipomatosis	Pear-shaped bladder
Schistosomiasis	Ureteric striations, calcified bladder
- **Trauma: one-shot intraoperative IVP:** 2 mL/kg IV push, single film at 10 min. It confirms a functioning contralateral kidney before exploration in the unstable patient.	
- **Also know:** nephroptosis (descent >5 cm or two vertebral bodies supine to erect); urinary fistulae (ureterovaginal); renal masses (distortion); post-operative assessment; diverticula.	

Complications

- **Contrast reactions:** immediate (urticaria to anaphylaxis) or delayed (rash). Severe reaction: stop contrast, **IM (intramuscular) adrenaline 0.3–0.5 mg** every 5–15 min (maximum 1 mg), oxygen, β2-agonist. Observe 30 min.
- **CA-AKI (contrast-associated acute kidney injury):** prevent with hydration, low or iso-osmolar contrast, lowest dose.
- Radiation; metformin-related lactic acidosis in renal impairment.

Limitations and current role

- Two-dimensional; bowel gas; poor parenchymal detail; misses small masses; poor in renal failure.
- CTU is more sensitive for stones, masses and urothelial lesions, but gives a higher radiation dose than IVU. IVU remains useful in resource-limited settings.

Recent advances

- Split-bolus CTU (fewer phases); dual-energy CT (virtual unenhanced images); MR urography without radiation.

Diagram to draw: normal IVU with minor and major calyces, pelvis, ureter and the three narrowings; beside it a "drooping lily".

> **Key points to remember:**
> - Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
> - Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
> - Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
> - Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

P2 *Lower urinary tract trauma in pelvic fracture — plain and CT cystography (retrograde 350 mL), flame-shaped extraperitoneal versus intraperitoneal leak*

- Passive filling by excreted contrast misses ruptures. Extraperitoneal rupture can give a "molar tooth" pattern.
- A large extraperitoneal leak prevents distension and can **hide a coexisting intraperitoneal rupture**.

3 · Fluoroscopic studies

Setting	Peak kilovoltage (kVp)	Tube current (mA)
Changes	Photon energy	Photon number
Dose	**Rises exponentially**	Rises linearly
Penetration / contrast	Increases / decreases	Unchanged / increases

- **Retrograde pyelography** is chosen when intravenous contrast is hazardous (allergy, renal dysfunction). Use about 50% contrast, because dense contrast hides small filling defects. Over-pressure causes pyelovenous and pyelolymphatic backflow and sepsis.

SAQ **VCUG**

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 1, Q3c

Definition

- **VCUG (voiding cystourethrogram; MCU, micturating cystourethrogram):** fluoroscopic study of bladder filling and voiding after catheter instillation of contrast. It shows the bladder, **posterior urethra** and UVJ (ureterovesical junction) competence.
- **Gold standard for VUR (vesicoureteral reflux)** and PUV (posterior urethral valves).

Indications

- **Children:** febrile UTI (urinary tract infection) with abnormal ultrasound or recurrent infection (EAU [European Association of Urology]: also age <12 months); bilateral hydroureteronephrosis (PUV); dilated ureter; ureterocele; neurogenic bladder.
- **Adults:** stricture (with RGU [retrograde urethrogram]), diverticula, fistula, post-operative leak, voiding dysfunction.

Technique

- **Sterile urine** first. After UTI, wait until urine is sterile and the child is improving (about a week); the old 4–6-week delay is unnecessary.
- Scout film → catheter: **5 Fr feeding tube (infants), 6–8 Fr (children), 14–16 Fr (adults)** → confirm tip position.
- **Gravity fill** (≤70 cm H_2O) to expected capacity: <2 years weight (kg) × 7; 2–14 years (age × 30) + 30 mL; >14 years 300–500 mL. (Koff formula: (age + 2) × 30 mL.)
- **Oblique views** (UVJ, prostatic urethra) → remove catheter → **void under fluoroscopy** → **post-void film**.
- **Cyclic VCUG** (2–3 fill–void cycles) detects 12–20% more VUR.

Findings

- **VUR grade (International Reflux Study, 1981):**

Grade	Appearance
I	Ureter only, not dilated
II	Pelvis and calyces, no dilatation
III	Mild–moderate dilatation, minimal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation; papillary impressions lost

- Filling-phase reflux has a poorer resolution prognosis than voiding-only reflux.
- **PUV:** dilated, elongated posterior urethra, prominent bladder neck, trabeculation.
- Ureterocele, diverticula, residual urine; post-void pelvic retention suggests PUJ (pelviureteric junction) obstruction.

Limitations and complications

- Invasive, radiation, UTI, catheter trauma; snapshot (occult VUR); interobserver variability; autonomic dysreflexia (injury above T6).

Alternatives

- **Direct radionuclide cystography:** more sensitive, lower dose, poor anatomy.

- **CeVUS (contrast-enhanced voiding urosonography):** no radiation, accuracy close to VCUG.
- **PIC (positional instillation of contrast) cystography:** occult VUR with negative VCUG.

Diagram to draw: VUR grades I–V.

> **Key points to remember:**
> - Gold standard for VUR and PUV; sterile urine first.
> - Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
> - Grades I–II non-dilating, III–V dilating.

Also, from the condensed book (not in the paper answer):

- **Retrograde urethrography (RUG)** shows the anterior urethra. Use a 45° oblique, the balloon at about 2 mL in the fossa navicularis, and 20–30 mL of contrast. The prostatic urethra does not distend normally, which is not a stricture; the posterior urethra needs a combined VCUG.
- **Fluoroscopic renal access**: bull's-eye uses a C-arm tilted 15–30° toward the surgeon, with the needle end-on over the calyx.

4 · Fluoroscopy dose reduction

- Dose is proportional to fluoroscopy time. Recording it reduces it.
- Halving the collimator diameter cuts the exposed area **fourfold**. Low-dose mode halves mA and dose. Pulsed fluoroscopy at 8, 4 and 1 pulses/s saves **73%, 87% and 97%** against continuous (30 pulses/s).
- **Inverse square law**: doubling the distance gives ¼ of the dose.
- Place the source below the patient and the detector close to the patient. Wear a **0.25–0.5 mm** lead apron, a thyroid shield and glasses; the legs receive most scatter.

> **KEY POINT**
> Surgeon dose is mostly scatter from the patient, so everything that lowers patient dose protects the team.

5 · CT principles

SAQ Newer generation CT scan

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3a

Introduction

- Computed tomography (CT) is the reference imaging for stones, masses and trauma. Attenuation is expressed in Hounsfield units (HU): water 0, air −1000.

Evolution

Generation	Feature
Helical (~1989)	Continuous table movement, volumetric data

Generation	Feature
Multidetector (1998 → 320-row)	Sub-millimetre slices, 3D reconstruction
Dual-source/dual-energy (2005)	Two energies: material analysis
Photon-counting (US approval 2021)	Energy-sorts each photon: less noise, dose and metal artefact

Dual-energy CT (DECT)

- Two energies (e.g. 80 and 140 kVp) compared pixel by pixel: **uric acid vs non-uric-acid stones**, **virtual unenhanced** images (one phase fewer) and **iodine maps** (hyperdense cyst vs enhancing mass).

Low-dose CT

Protocol	Dose
Conventional abdomen/pelvis	10–20 mSv
Low-dose	<3–4 mSv
Ultra-low-dose	1–2 mSv
- Stone sensitivity stays ~95%; use standard dose if body mass index >30. EAU (European Association of Urology): ultrasound first; AUA (American Urological Association): CT first.	

Reconstruction

- Filtered back projection is noisy; iterative reconstruction halves dose; **deep learning reconstruction** gives good images at ~1 mSv.

Contrast phases and protocols

Phase	Timing	Use
Arterial	20–30 s	Renal arteries, donor anatomy, active bleeding
Corticomedullary	30–70 s	Hypervascular (clear cell) tumours, renal vein
Nephrographic	90–120 s	Most sensitive for renal masses
Excretory	7–10 min	Collecting system, ureter, urine leak
- **CT urography:** non-contrast, nephrographic and excretory phases (~20–30 mSv). **Split-bolus** merges phases, lowering dose but may miss subtle urothelial lesions.		
- **Renal mass:** enhancement ≥20 HU = solid, 10–20 equivocal, <10 cyst; Bosniak 2019 for cysts.		
- **Adrenal:** <10 HU unenhanced = lipid-rich adenoma; otherwise 15-minute washout (absolute >60% or relative >40% = adenoma).		

Phase	Timing	Use
- **Also know:** stone HU (uric acid 200–400, calcium oxalate 600–1200; indinavir invisible); bone window for fragments; partial volume causes pseudoenhancement of small cysts; ALARA (as low as reasonably achievable) dosing, since radiation risk has no threshold.		

Diagram to draw: Timeline: axial → helical → multidetector → dual-source → photon-counting.

> **Key points to remember:**
> - Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
> - DECT: uric acid stones, virtual unenhanced, iodine maps.
> - Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

Material	HU
Macroscopic fat (diagnostic of angiomyolipoma, AML)	≤ −10
Simple cyst	−9 to 20
Lipid-rich adrenal adenoma	<10
Hyperdense cyst (Bosniak II)	≥70 unenhanced

- Partial-volume **pseudoenhancement of cysts <10–15 mm** leads to overtreatment, so use thin slices and a central region of interest.

6 · Noncontrast CT (NCCT) for stones

- NCCT is the gold standard; all stones are visible except **protease-inhibitor (indinavir) stones**. Cutting tube current by up to 95% preserves accuracy, and ultra-low-dose CT gives less dose than a two-view KUB.
- **Avoid ultra-low dose** with BMI >30, very thin patients (phlebolith confusion), metal hardware or a suspected infected stone.
- CT, low-dose and ultra-low-dose CT are equally accurate (sensitivity about 95%). Ultrasound has sensitivity **84%** but specificity only **53%**; KUB sensitivity is **57%**.

Body	First imaging in suspected colic
AUA	CT in adults; standard dose if BMI >30; NCCT before PCNL or SWL
EAU	Ultrasound first, CT if inconclusive (randomised trial: less radiation, same outcomes)

7 · Contrast CT protocols

- The portal venous phase (70–90 s) shows liver and nodes for routine staging. The nephrographic phase (90–120 s) is **most sensitive for renal masses** and caval thrombus. Clear cell carcinoma enhances early and avidly; papillary enhances later.
- **Adrenal protocol** is noncontrast, portal venous and 15-min delay. Absolute washout = (PV − delayed)/(PV − unenhanced) × 100 >**60%**, and relative washout = (PV − delayed)/PV × 100 >**40%**, indicate adenoma.
- Nodal cut-offs are 10 mm for prostate cancer (poor sensitivity, so use PSMA PET-CT) and, in bladder cancer, a short axis >8 mm in the pelvis and >10 mm in the abdomen.

*See the paper answer **Radiology for solid renal masses.** (Section 13, Ch 137).*

8 · CT urography and CT cystography

- CT urography always includes an excretory phase. For poor distal ureteric opacification, **hydration is the simplest and most effective aid**, followed by a diuretic or prone position.
- Do **CT before TURBT**, because post-resection change mimics invasion. CT detects T3b/T4 but cannot separate muscle-invasive from non-invasive disease.

SAQ PET scans in urology.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2018-Summer Paper 1, Q3e

Principle

- **PET (positron emission tomography):** tracer emits a positron → annihilation with an electron → **two 511-keV photons at 180°** → coincidence detection. Fused with CT (computed tomography) or MRI.
- Quantitative (**SUV, standardised uptake value**); costly. Half-lives: ^{18}F 110 min, ^{68}Ga 68 min, ^{11}C 20 min (on-site cyclotron).

Tracers

Tracer	Mechanism	Main use
^{18}F-FDG (fluorodeoxyglucose)	Glucose analogue trapped after phosphorylation (Warburg effect)	Bladder, UTUC (upper tract urothelial carcinoma), penile, seminoma
PSMA (prostate-specific membrane antigen) ligands (^{68}Ga-PSMA-11, ^{18}F-DCFPyL, ^{18}F-PSMA-1007, rhPSMA-7.3)	Bind cell-surface PSMA	Prostate staging and recurrence
^{11}C/^{18}F-choline	Cell membrane synthesis	Prostate recurrence (older)
^{18}F-fluciclovine	Amino acid transport	Prostate recurrence
^{18}F-NaF (sodium fluoride)	Bone turnover	Bone metastases

Uses by organ

Prostate cancer

- FDG is poor (low glucose metabolism). PSMA has specificity >90%.
- **Primary staging (high risk): proPSMA (2020)**: PSMA PET/CT more accurate than CT plus bone scan (92% vs 65%). Nodal sensitivity only 30–40%, so a negative scan does not exclude micrometastases.
- **BCR (biochemical recurrence):** detection rises with PSA (prostate-specific antigen); about half positive below 0.5 ng/mL. PSMA beats fluciclovine (56% vs 26%). CONDOR: management changed in 63.9%.
- **Theranostics:** PSMA-positive mCRPC (metastatic castration-resistant prostate cancer) → **^{177}Lu-PSMA-617 (VISION 2021)**.
- NCCN (National Comprehensive Cancer Network): PSMA PET preferred; outcome benefit still unproven.

Testis

- **Seminoma residual mass >3 cm:** FDG-PET **≥6 weeks after chemotherapy**. Negative scan reliable (observe); positive → consider resection.
- **Not for NSGCT (non-seminomatous germ cell tumour):** teratoma is not FDG-avid.

Bladder and UTUC

- FDG for nodal and distant staging in MIBC (muscle-invasive bladder cancer): modest sensitivity, high specificity; urinary excretion hides the primary. Use when CT is equivocal.

Kidney

- FDG not routine (clear cell RCC (renal cell carcinoma) poorly avid). **^{89}Zr-girentuximab** (anti-CAIX, carbonic anhydrase IX) identifies clear cell RCC (ZIRCON).

Penis

- FDG confirms metastases in **palpable inguinal nodes**; poor for impalpable nodes.
- **Also know:** EMPIRE-1 (fluciclovine-guided salvage radiotherapy); false positives from inflammation; neuroendocrine cancer loses PSMA.

Diagram to draw: Positron annihilation with two 511-keV photons hitting a detector ring.

> **Key points to remember:**
> - proPSMA: PSMA PET 92% vs 65% accuracy.
> - Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
> - FDG poor in prostate and clear cell RCC.
> - VISION: ^{177}Lu-PSMA-617 in mCRPC.

SAQ PSMA PET Ct scan

Paper answer · P4

Type: SAQ (10 marks)
Source: 2019-Summer Paper 4, Q3a

Principle

- **PSMA (prostate-specific membrane antigen):** transmembrane glycoprotein over-expressed by prostate cancer, rising with grade.
- **PET/CT (positron emission tomography/computed tomography):** positron-labelled ligand binds PSMA; uptake as SUV (standardised uptake value).
- **Tracers:** ^{68}Ga-PSMA-11 (half-life 68 min); ^{18}F-DCFPyL, ^{18}F-rhPSMA-7.3, ^{18}F-PSMA-1007 (110 min, better images; PSMA-1007 little urinary excretion).
- **Specificity >90%**; beats choline and fluciclovine.

Indications

1. Primary staging (high risk):

- **proPSMA (2020):** accuracy **92% vs 65%** vs CT + bone scan.
- Nodal sensitivity only 30–40% against PLND (pelvic lymph node dissection): a negative scan does not exclude micrometastases.
- NCCN (National Comprehensive Cancer Network): may **replace** conventional imaging; AUA (American Urological Association): add-on.

2. BCR (biochemical recurrence):

- Detection rises with PSA (prostate-specific antigen): about **half positive below 0.5 ng/mL**, nearly all above 2.
- **Calais trial:** PSMA 56% vs fluciclovine 26%. **CONDOR:** management changed in 63.9%.
- Guides salvage radiotherapy (disease outside field in about 1 in 5).

3. Intraprostatic detection: PRIMARY trial: adding PSMA PET to MRI (magnetic resonance imaging) improved negative predictive value; 5-point PRIMARY score.

4. Theranostics: selects patients for **^{177}Lu-PSMA-617**. **VISION (2021):** mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: OS 15.3 vs 11.3 months.

Reporting

- **PROMISE miTNM** (molecular imaging TNM): miT, miN, miM staging plus expression score (vs blood pool, liver, parotid). PSMA-RADS also used.

Pitfalls

False positive	False negative
Coeliac/stellate ganglia, fractures, Paget's disease, fibrous dysplasia	Small nodes (<5 mm)
Other tumours (RCC (renal cell carcinoma), HCC (hepatocellular carcinoma))	PSMA-negative tumours (about 5–10%)
^{18}F-PSMA-1007 benign bone uptake	Neuroendocrine transformation; masking by bladder activity

- **Also know:** no proof yet that PSMA-guided treatment improves survival (stage migration).

Diagram to draw: PSMA ligand binding the cell-surface antigen with fused PET/CT image of a nodal lesion.

> **Key points to remember:**
> - proPSMA: accuracy 92% vs 65%.
> - BCR: about half positive below PSA 0.5 ng/mL.
> - Nodal sensitivity 30–40%: negative scan does not exclude nodes.
> - VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

9 · Iodinated contrast reactions and contrast-associated AKI

*See the paper answer **Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy** (Section 1, Ch 4).*

> **KEY POINT**
>
> Hypersensitivity is idiosyncratic (premedication, adrenaline), whereas nephrotoxicity is dose-related (hydration, less contrast). Prophylaxis for one does nothing for the other.

6 Magnetic Resonance Imaging

1 · Physics and sequences

- Magnetic resonance imaging (MRI) gives **better soft-tissue contrast and no ionising radiation**.
- **Fluid is dark on T1 and bright on T2.**
- **Diffusion-weighted imaging (DWI)**: high cellularity restricts water movement, so tumour is bright on DWI and dark on the apparent diffusion coefficient (ADC) map.
- **Chemical shift imaging**: a voxel holding both fat and water loses signal on the out-of-phase (OP) image, which means **microscopic fat**. Fat suppression nulls **macroscopic fat**.

2 · Indications, contrast and safety

*See the paper answer **Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy** (Section 1, Ch 4).*

SAQ MR urography

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 1, Q3a

Definition

- MRU (magnetic resonance urography): magnetic resonance imaging of kidneys, ureters and bladder giving anatomy and function **without ionising radiation**.

Techniques

	Static MRU	Excretory MRU
Sequence	Heavily T2-weighted; urine is bright	Gadolinium-enhanced T1
Contrast	None	GBCA (gadolinium-based contrast agent)
Depends on function	No	Yes; low-dose furosemide improves it
Best for	Dilated or obstructed system	Non-dilated system, excretion

- **Functional MRU (children):** hydration, bladder catheter, furosemide 1 mg/kg (maximum 20 mg), T2 images, then dynamic T1 after slow contrast injection. Patlak analysis gives differential function, GFR (glomerular filtration rate) and drainage.

Appearances

- Stones and calcification give **signal voids**; tumour, clot, gas and sloughed papillae are low-signal defects in bright urine.

Indications

- CT (computed tomography) urography contraindicated: contrast allergy, renal insufficiency, **pregnancy** (separates physiological from calculous dilatation; small stones missed).
- **Children:** complex anomalies, ectopic ureter in the incontinent girl, duplex systems, megaureter, crossing vessels in UPJ (ureteropelvic junction) obstruction, ectopic kidneys.
- Hydronephrosis: alternative to MAG3 (mercaptoacetyltriglycine) renography; equivalence not yet proven.

- Upper tract tumours and renal masses.

Advantages and limitations

Advantages	Limitations
No radiation; best anatomical detail	Cost, availability, long study (up to 1 hour)
Anatomy, function and drainage in one test	Sedation often needed under 6 years
As accurate as CTU (CT urography) for obstruction	Poor for stones; complex post-processing
Safe in iodinated contrast allergy	Pacemakers, metal implants, claustrophobia

Gadolinium safety

- **NSF (nephrogenic systemic fibrosis)** was the concern at GFR <30 mL/min/1.73 m². Group II agents (gadobenate, gadoteridol, gadoterate, gadobutrol) carry little or no risk even in end-stage disease.
- **Pregnancy:** gadolinium only if it clearly helps.
- **EAU (European Association of Urology):** CTU is preferred to MRU for upper tract urothelial cancer.

Diagram to draw: coronal static T2 MRU of a hydronephrotic kidney with a dark stone in the ureter.

> **Key points to remember:**
> - Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
> - Stones are signal voids; CTU preferred for urothelial cancer.
> - Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
> - NSF risk minimal with group II agents.

Also, from the condensed book (not in the paper answer):

- MRI is the modality of choice in **pregnancy, renal insufficiency and iodinated contrast allergy**.
- **Contraindications**: non-conditional pacemakers, ferromagnetic aneurysm clips, cochlear implants and orbital metal.

3 · Adrenal MRI

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- An adenoma shows an **OP signal drop** from intracytoplasmic lipid. Lipid-poor adenomas (**10–30%**) are indeterminate and need CT washout.
- **Myelolipoma** has macroscopic fat: bright on T1, it loses signal with fat suppression and shows the India ink artefact.
- **Traps**: metastases from clear cell renal cell carcinoma (RCC), hepatocellular carcinoma or liposarcoma contain lipid and mimic adenoma. One-third of phaeochromocytomas are T2-dark, so the "light bulb" sign is unreliable.

4 · Renal mass

*See the paper answer **Radiology for solid renal masses.** (Section 13, Ch 137).*

Also, from the condensed book (not in the paper answer):

- An **absent pseudocapsule** predicts **pT3a**.
- Calcium is a signal void, so MRI shows enhancing nodules in **calcified cysts** better than CT. Tumour thrombus enhances; bland thrombus does not.

5 · MR urography and bladder

*See the paper answer **MR urography** (Section 1, Ch 6).*

SAQ VI-RADS

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3b

Definition

- **VI-RADS (Vesical Imaging–Reporting and Data System):** a 5-point score for the likelihood of **muscle (detrusor) invasion** by bladder cancer on mpMRI (multiparametric magnetic resonance imaging). Panebianco et al., Eur Urol 2018.
- Standardises bladder MRI reporting, as PI-RADS (Prostate Imaging Reporting and Data System) does for the prostate.

Technique

- Multiplanar **T2W (T2-weighted)**, **DWI (diffusion-weighted imaging)** with ADC (apparent diffusion coefficient) map, and **DCE (dynamic contrast-enhanced)** T1.
- Done **before TURBT (transurethral resection of bladder tumour)**: resection oedema mimics invasion. Distend the bladder adequately.

Principle

- On T2W the detrusor is a **low-signal (dark) line**; urine, mucosa and submucosa are bright. The question is whether tumour **interrupts that dark line**.

Scoring

Score	Invasion	T2W	DCE	DWI
1	Highly unlikely	<1 cm, ± stalk; dark line intact	No early muscle enhancement	Muscle continuous
2	Unlikely	>1 cm with stalk and thick inner layer, or sessile; line intact	Inner layer enhances only	As 1
3	Equivocal	No category 2 features; line intact	No muscle interruption	As 1
4	Likely	**Line interrupted**	Enhancement extends into muscle	High signal into muscle
5	Highly likely	**Into perivesical fat**	Through wall into fat	High signal through wall and fat

- **Cut-off ≥3 = suspicious for MIBC (muscle-invasive bladder cancer)**; the "equivocal" category already counts as positive.

- Sensitivity and specificity roughly 80–90%; DWI is the main functional sequence.

Clinical uses

- Local staging before TURBT; earlier referral for neoadjuvant chemotherapy and cystectomy.
- Patients who cannot have iodinated contrast (renal impairment, allergy).
- **MRI-first pathway** (UK BladderPath trial) shortened time to correct treatment for MIBC.
- Response after neoadjuvant chemotherapy or bladder preservation: under study.

Limitations

- Needs an experienced radiologist; post-TURBT or BCG (bacille Calmette–Guérin) inflammation; small, bladder-neck or diverticular tumours; poor for CIS (carcinoma in situ) and nodes; does not replace histology.

Diagram to draw: bladder wall with tumour breaching the dark detrusor line (VI-RADS 4).

> **Key points to remember:**
> - Five points; tests the dark muscle line on T2W.
> - ≥3 suspicious for MIBC; 4 line broken, 5 fat.
> - Scan before TURBT.
> - Panebianco 2018; accuracy about 80–90%.

Also, from the condensed book (not in the paper answer):

- Upper tract urothelial carcinoma is homogeneous on DWI (ADC **≤451**), unlike central RCC.
- **T2 MRI is the best study for a urethral diverticulum**: it defines the neck, circumferential extent and any tumour or stone.

6 · Prostate MRI

*See the paper answer **Multiparametric MRI, PI-RADS and fusion biopsy** (Section 17, Ch 155).*

Also, from the condensed book (not in the paper answer):

- Prostate MRI has sensitivity **0.91** and specificity **0.37** for Grade Group ≥2, with negative predictive value **>90%**: it rules cancer out, not in.
- Dynamic contrast curves: type 1 persistent rise is normal, type 2 plateau is indeterminate, and **type 3 washout** is most suggestive.

Score	PI-QUAL (quality)	PRECISE (active surveillance)
1	All sequences non-diagnostic	Resolution
2	One sequence acceptable	Reduction
3	Rules in, not out	Stable
4	Rules in and out	Increase
5	All optimal	Stage progression

At PI-QUAL 1–2, give no PI-RADS score.

Trial	Design	Result	Take-home
PROMIS	MRI vs template biopsy	MRI highly sensitive	MRI before biopsy (level 1b)
PRECISION	Randomised: MRI ± targeted vs systematic biopsy	More significant, less insignificant cancer	MRI pathway standard
Göteborg-2	Screening trial	Targeted-only biopsy cut overdiagnosis	MRI-targeted screening

7 Genitourinary Tract Imaging: Basic Principles of Nuclear Medicine

1 · Principles and radionuclides

LAQ Radionuclide (isotope) renal scans: MAG3 and DMSA

Paper answer · P1 · P2 · merged from 2

Asked as: P1 LAQ "Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications" · P2 SAQ "Renal isotope scans and DMSA renal scan"

"Uro" is read as urology. "MAG 3" is taken to mean the Tc-99m MAG3 (mercaptoacetyltriglycine) renal scan: principle, technique, interpretation and uses.

Principles

- Nuclear medicine shows **function, not structure**. A radiotracer, given in a tiny (subpharmacological) dose, follows a physiological or molecular process; the patient is the radiation source.
- **Gamma camera** (sodium iodide crystal, photomultipliers, collimator): photon, collimator (admits photons from one direction), sodium iodide crystal (light), photomultiplier tubes, image. Planar (about 20 mm resolution) or **SPECT (single-photon emission computed tomography)**, 3D, with CT (about 10 mm).
- **PET (positron emission tomography):** the positron annihilates with an electron, giving two 511 keV photons 180° apart, detected in coincidence. This rejects noise and allows true quantitation (SUV, standardised uptake value); sharper (about 5 mm).

	Planar	SPECT	PET
Image	2D	3D, with CT (computed tomography)	3D, with CT or MRI (magnetic resonance imaging)
Resolution	About 20 mm	About 10 mm	About 5 mm
Quantitation	Semiquantitative	Semiquantitative	SUV
Main use	Renal function	Problem-solving	Cancer

- **^{99m}Tc (technetium-99m):** 6-hour half-life, single 140 keV photon: the workhorse. **^{18}F:** about 110 minutes, lowest positron energy, so the sharpest PET image. ^{68}Ga comes from a generator; ^{11}C (20 minutes) needs an on-site cyclotron.

Applications in urology (overview)

Area	Tracer	Use
Kidney	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid), -MAG3, -DMSA (dimercaptosuccinic acid)	GFR (glomerular filtration rate), split function, obstruction, scars, transplant
Bone	^{99m}Tc-MDP (methylene diphosphonate)	High-risk prostate cancer staging; low specificity

Area	Tracer	Use
Prostate	**PSMA (prostate-specific membrane antigen) PET**; choline; fluciclovine	PSMA preferred (NCCN [National Comprehensive Cancer Network]) for staging and recurrence; specificity >90%
Therapy	^{177}Lu-PSMA-617; radium-223	mCRPC (metastatic castration-resistant prostate cancer); VISION trial (2021)
FDG (fluorodeoxyglucose) PET	Glucose analogue trapped in cells	**Residual seminoma >3 cm**; penile cancer with palpable nodes; distant urothelial disease. Poor in prostate and clear cell RCC (renal cell carcinoma)
Renal mass	Sestamibi SPECT; girentuximab PET	Oncocytoma "hot" on sestamibi; clear cell RCC "hot" on girentuximab
Adrenal	^{123}I-MIBG (metaiodobenzylguanidine), ^{68}Ga-DOTATATE, FDG	Metastatic or extra-adrenal phaeochromocytoma; MIBG best in MEN-2 (multiple endocrine neoplasia type 2)
Sentinel node	^{99m}Tc-nanocolloid plus blue dye	Penile cancer, impalpable nodes
Scrotum	^{99m}Tc-pertechnetate	Torsion cold, epididymitis hot
Reflux	Direct radionuclide cystography	VUR (vesicoureteral reflux) follow-up: more sensitive, lower dose than VCUG (voiding cystourethrogram)
Infection	^{111}In-labelled leucocytes; ^{67}Ga	Occult abscess; not in transplant

Renal tracers

- Each tracer images the part of the nephron that handles it. Plasma clearance is about 20% filtration and 80% tubular secretion.

Feature	**^{99m}Tc-DTPA**	**^{99m}Tc-MAG3**	**^{99m}Tc-DMSA**
Handling	Glomerular filtration only	Protein-bound; tubular secretion, high extraction	Filtered, then retained in proximal tubules (megalin, cubilin); tubular cortical binding
Study	Dynamic	Dynamic	**Static cortical**, at 2 hours (P2: 2–4 hours)
Main use	**GFR**; captopril study; renovascular	**Split function, obstruction (first choice)**, drainage; transplant	**Pyelonephritis and scars**
Renal failure	Poor images in renal failure	Good images in renal failure	Delay imaging
Dose	5–10 mCi	5–10 mCi	5 mCi adult; 50 µCi/kg child

MAG3: principle

- **^{99m}Tc-MAG3:** protein-bound tracer cleared mainly by **proximal tubular secretion**. Extraction is nearly **60%** per pass, versus about 20% for DTPA.

- So it gives **good images even with poor GFR**, lower background and lower cost. **Agent of choice for split function and obstruction.**

Dynamic and diuretic renography (MAG3): method and interpretation

- **Preparation:** good hydration; bladder catheter for outlet obstruction or neurogenic bladder, and in children; cap a nephrostomy. Dose 185–370 MBq (5–10 mCi) intravenous bolus in adults.
- **Perfusion phase** (first minute): frames every 1–2 seconds; the kidney appears within seconds of the aorta (renal curve follows aortic upstroke).
- **Function (uptake) phase:** regions of interest with background subtraction. **Split function (DRF, differential renal function) = area under each curve at 1–3 minutes**; **normal 45–55%** (45:55 to 55:45). Normal MAG3 peak at 3–5 minutes; bladder by 10–15 minutes.
- **Drainage phase:** a rising curve needs furosemide. Upright or post-void images test gravity drainage.
- **GFR:** DTPA camera method (Gates) or plasma-sample clearance.
- **Diuretic renography ("well-tempered" diuretic renogram):** dilatation is not obstruction. Furosemide 0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children, given when the system is full; separates a patulous system from obstruction. Timing varies (F+20 classic, F+15, F0, F−15), so the report must always state when it was given.

T½ (half-clearance time) after furosemide	Meaning
<10 minutes	Not obstructed
10–20 minutes	Indeterminate
>20 minutes	Obstructed

- **Curve types (O'Reilly):** I normal; II rising (obstructed); IIIa dilated, not obstructed, washes out (falls) with diuretic; IIIb equivocal.

MAG3 uses

- PUJ (pelviureteric junction) obstruction and megaureter: **surgery if DRF <40% with poor drainage or falling DRF**; follow-up after pyeloplasty.
- Transplant: acute rejection (perfusion falls) versus delayed graft function (perfusion kept); urine leak.
- **Also know:** indirect cystography for VUR; split function before nephrectomy.

MAG3 pitfalls

- Poor function, dehydration, full bladder, wrong ROI (region of interest), mistimed diuretic, massive dilatation (capacious pelvis) and neonatal immaturity mimic obstruction. **Trend on serial scans** matters.

DMSA renal scan

Principle

- **^{99m}Tc-DMSA** is filtered and then taken up by **proximal tubular cells** (megalin and cubilin receptors). It is retained in the cortex and excreted very slowly.
- So it gives a **static map of functioning cortex**, not drainage.

Technique

- Intravenous dose: **50 µCi/kg in children, 5 mCi (185 MBq) in adults**.

- Images at **2–4 hours** (P1: 2 hours; later in renal failure): posterior and oblique views, **pinhole collimator** or **SPECT**, which is slightly more sensitive.
- Radiation about 1 mSv.

What it measures

- **DRF:** from geometric-mean counts; normal 45–55%. Better than MAG3 for ectopic, horseshoe or duplex kidneys and very poor function.
- **Cortical defects:** photopenic areas.

Indications

- **Acute pyelonephritis** (gold standard), especially in infants with febrile UTI (urinary tract infection).
- **Renal scarring** in VUR and recurrent UTI; top-down approach (DMSA first, VCUG only if abnormal).
- Function of a duplex upper moiety, ectopic or horseshoe kidney; MCDK (multicystic dysplastic kidney: no uptake).
- Deciding whether to reconstruct or remove a poorly functioning kidney.

Acute pyelonephritis versus scar

	Acute pyelonephritis	**Scar**
Timing	Within 1 week of fever (sensitivity falls by 1 month)	**≥6 months** after infection
Defect	Ill-defined, may be multiple	Sharp, with cortical thinning
Kidney size	Normal or swollen	Small or contracted (often small)
Repeat scan (at about 6 months)	Resolves (about 85%)	Persists

Goldraich scar grading

Type	**Finding**
1	No more than 2 scarred areas
2	More than 2 scars, normal tissue between
3	Generalised damage, small kidney
4	End-stage, shrunken kidney, little or no uptake

Limitations

- Cannot tell **congenital reflux dysplasia** (boys, grade IV–V VUR, no UTI) from acquired scar.
- Gross hydronephrosis gives false defects; no information on drainage; poor uptake in renal failure and proximal tubular disorders.
- **Also know:** MAG3 is best for obstruction; NICE (National Institute for Health and Care Excellence): late DMSA 4–6 months after atypical or recurrent UTI.

Other renal applications (special studies)

- **Captopril renography (DTPA):** the stenotic kidney keeps GFR by angiotensin II efferent constriction. Captopril removes this support, so uptake falls. Repeat off ACE (angiotensin-

converting enzyme) inhibitors: improvement **≥10%** off ACE inhibitors means renovascular hypertension is likely. Largely replaced by CT angiography.
- **Transplant (MAG3):** acute rejection: perfusion falls and worsens; delayed graft function: perfusion kept, improves. Shows urine leak without iodinated contrast.
- **Radionuclide (nuclear) cystography:** direct (tracer instilled into the bladder) or indirect (after MAG3). More sensitive and lower radiation dose than VCUG, but little anatomy. So VCUG first; nuclear cystography for follow-up of reflux.

Infection imaging

- **^{111}In-oxine leucocytes** (image at 24 hours): best for occult abdominal or genitourinary infection. ^{99m}Tc-HMPAO (hexamethylpropyleneamine oxime) leucocytes give better images, but urinary excretion obscures the tract. Neither in a transplant (leucocytes home to the graft).
- **Also know:** ^{67}Ga-citrate (non-specific, slow).

Oncology

- **FDG (^{18}F-fluorodeoxyglucose):** a glucose analogue phosphorylated and trapped (Warburg effect). Urinary excretion limits it to distant disease.
- **Seminoma residual mass >3 cm** after chemotherapy (EAU [European Association of Urology]). Not for NSGCT (non-seminomatous germ cell tumour), because teratoma is not FDG-avid.
- Penile cancer with palpable nodes; equivocal bladder or upper-tract metastases.
- Poor in prostate cancer and clear cell RCC.
- **Prostate: PSMA PET** is preferred (NCCN) for high-risk staging and biochemical recurrence. Specificity >90%; nodal sensitivity only 30–40%. Calais trial (2019): detection 56% (PSMA) versus 26% (fluciclovine). Bone scan (^{99m}Tc-MDP) remains conventional.
- **Renal mass: sestamibi SPECT** "hot" in oncocytoma; **girentuximab** (anti-CAIX, carbonic anhydrase IX) PET "hot" in clear cell RCC (ZIRCON trial).
- **Also know:** ^{123}I-MIBG and ^{68}Ga-DOTATATE for phaeochromocytoma; ^{99m}Tc-nanocolloid sentinel node biopsy in penile cancer; ^{99m}Tc-pertechnetate scrotal scan (torsion cold, epididymitis hot) when ultrasound is equivocal.

Therapy (theranostics)

- **^{177}Lu-PSMA-617:** beta emitter for mCRPC after ARPI (androgen receptor pathway inhibitor) and taxane: **VISION (2021)**. PSMA PET confirms the target first.
- **Radium-223:** alpha emitter for bone-only mCRPC.

Advantages and limitations

- Quantitative function, low radiation, safe in contrast allergy and pregnancy (MAG3 fetal dose <5 mGy).
- Poor anatomy; PET is costly; results depend on technique.

Diagrams to draw

- MAG3 renogram: perfusion spike, peak at 3–5 minutes, washout, 1–3 minute split window; renogram curve types I (normal), II (obstructed, rising), IIIa (dilated non-obstructed, falls after furosemide) and IIIb.
- Gamma camera chain: patient, collimator, crystal, photomultiplier, image.

- Posterior DMSA image: normal kidney and a scarred kidney with polar defects.

Key points to remember:

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 minutes obstructed, <10 minutes not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h; gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Also, from the condensed book (not in the paper answer):

- **Gallium-68 (^{68}Ga)** has a half-life of **68 min**, is generator-produced and has the **longest positron range**.
- Indium-111 (half-life **67 h**) and gallium-67 (**78 h**) suit delayed infection imaging. Gallium-67 citrate is transferrin-bound and non-specific, imaged at 48–72 h.
- Renal blood flow is about **1 L/min** (20% of cardiac output), and renal plasma flow about **600 mL/min**.

2 · Renal tracers and diuretic renography

*See the paper answer **Radionuclide studies and diuretic renography in PUJ obstruction** (Section 11, Ch 93).*

*See the paper answer **Radionuclide (isotope) renal scans: MAG3 and DMSA** (Section 1, Ch 7).*

Also, from the condensed book (not in the paper answer):

- Always check **whether and when furosemide was given**. With F−15, furosemide goes 15 min before the tracer, so diuresis is maximal when the tracer arrives.
- Captopril renography has largely been replaced by computed tomography angiography and Doppler.

ALGORITHM

1. Day 1: give oral captopril and perform a renogram; slow uptake and a low peak are abnormal.
2. If abnormal, repeat 1–2 days later off angiotensin-converting enzyme (ACE) inhibitors and calcium channel blockers.
3. Improvement of **≥10%** means a high probability of renovascular hypertension and likely benefit from revascularisation.

3 · Oncological PET, bone scan and PSMA

*See the paper answer **PET scans in urology.** (Section 1, Ch 5).*

*See the paper answer **PSMA PET Ct scan** (Section 1, Ch 5).*

Also, from the condensed book (not in the paper answer):

- **Bone scan** (^{99m}Tc-methylene diphosphonate) binds hydroxyapatite at osteoblastic activity. It is non-specific, shows a **flare** early in therapy, and a **superscan** is diffuse uptake with **absent renal activity**.
- Choline and fluciclovine are taken up by inflammation and benign prostatic hyperplasia, so they perform poorly in the prostate itself. PSMA shows normal uptake in the salivary and lacrimal glands.
- The National Comprehensive Cancer Network does not require conventional imaging before PSMA-PET.

Trial	Design	Result	Take-home
Calais	^{68}Ga-PSMA vs fluciclovine, biochemical recurrence	Detection 56% vs 26%	PSMA superior
CONDOR	^{18}F-DCFPyL, biochemical recurrence	Detection 61.5% at prostate-specific antigen (PSA) 0.8 ng/mL; management changed 64%	Detects at low PSA
EMPIRE-1	Randomised: fluciclovine-guided vs conventional salvage radiotherapy	12.5% absolute 3-year event-free gain	PET-guided planning helps
VISION	Randomised, metastatic castration-resistant, after androgen receptor inhibitor and taxane	^{177}Lu-PSMA-617 approved	Radioligand therapy

8 Evaluation and Management of Haematuria

1 · Definitions and causes

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

Also, from the condensed book (not in the paper answer):

- One positive urinalysis suffices because bleeding from cancer is intermittent. Specificity comes from **risk-stratifying the work-up, not from raising the threshold**.
- **Women** (male:female 4:1) present at a higher stage, because MH is put down to infection or gynaecological causes.
- The timing of gross haematuria (GH) in the stream points to the source:

Timing	Usual source
Initial	Urethra, distal to the external sphincter
Terminal	Trigone, bladder neck, prostate
Total	Bladder or upper tract
Cyclic (female)	Endometriosis

2 · Evaluation of microhaematuria (AUA/SUFU 2020)

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

Also, from the condensed book (not in the paper answer):

- **Renal ultrasound** detects cortical tumours but is poorly sensitive for upper tract urothelial carcinoma; this is why CT urography is reserved for high risk.
- **CT urography** phases: non-contrast for stones, nephrographic for renal masses, excretory for urothelium. The dose is **20–30 mSv**, which a split-bolus protocol reduces.
- UroVysion fluorescence in situ hybridisation targets chromosomes **3, 7, 17 and 9p21**.

3 · Gross haematuria

- The cancer yield is several-fold higher than in MH, so there is **no age threshold and no stratification**.
- GH without trauma or culture-proven infection needs **cytology, cystoscopy and CT urography**. With culture-proven infection, treat it and then repeat the urinalysis.
- In acute bleeding, assess haemodynamics, blood count, coagulation and post-void residual.
- **Clot retention**: pass a large-bore (22–24 Fr) three-way catheter, evacuate all clot manually, then run continuous bladder irrigation (CBI) until clear. Evacuation must be complete, because residual clot keeps the bladder distended so vessels cannot contract, and it fuels fibrinolysis.
- If this fails, perform cystoscopy under anaesthesia with clot evacuation, fulguration and biopsy of suspicious areas.
- Decide on stopping or reversing anticoagulation with the prescriber, weighing thrombotic risk. Never blame the anticoagulant without a full evaluation.

4 · Haemorrhagic cystitis

SAQ Haemorrhagic cystitis: diagnosis and management

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Diagnosis and management of hemorrhagic cystitis" · P3 SAQ "Management of Haemorrhagic cystitis."

Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after other sources are excluded (cystoscopy, cytology, upper tract imaging).

Causes and mechanism

Cause	Mechanism
Cyclophosphamide, ifosfamide	Liver forms **acrolein**, excreted in urine, which sloughs urothelium; onset within 48 h
Pelvic radiation (about 5%; 6 months to 10 years later)	**Obliterative endarteritis**, ischaemia, fibrosis, fragile neovessels
Viral	**BK polyomavirus**; adenovirus 11 and 35; after bone marrow or renal transplant
Others	Bacterial infection; busulfan, thiotepa; penicillins; amyloidosis

Diagnosis

- History: drugs, radiotherapy, transplant, anticoagulants. Severity: microscopic, gross, clots, clot retention.
- Blood count, coagulation, renal function, urine culture, viral PCR (polymerase chain reaction) if transplanted.
- **Exclude bladder cancer first:** cytology, cystoscopy with biopsy, upper tract imaging (CT urography). Cyclophosphamide and radiation are also risk factors for bladder cancer.

Prevention

- **Mesna** binds and inactivates acrolein; hyperhydration with forced diuresis, CBI (continuous bladder irrigation).

Management (escalate by severity; stability decides the branch)

`Resuscitate, transfuse, correct coagulopathy, treat infection → large-bore 3-way catheter + CBI → cystoscopy under anaesthesia: clot evacuation, fulguration, biopsy → intravesical alum 1% → bleeding after 48 h: hyperbaric oxygen (+ nephrostomies if bleeding continues) → formalin 1%, then 4% (cystogram first) → cystectomy + conduit diversion (Unstable at any stage: internal iliac embolisation)`

Intravesical agents

Agent	Mechanism and method	Caution / key point
Alum 1% (first line)	Protein precipitation, vasoconstriction; 200–300 mL/h; no anaesthesia	Avoid in renal failure (aluminium toxicity)

Agent	Mechanism and method	Caution / key point
Aminocaproic acid	Antifibrinolytic CBI, 200 mg/L	Only in a clot-free bladder; thromboembolism
Prostaglandins (carboprost), silver nitrate, hyaluronate (second-line options)	Vasoconstriction, coagulation, GAG (glycosaminoglycan) repair	Silver nitrate: cystogram
Formalin 1–4%	Protein fixation; **80–90% success**; under anaesthesia, instilled by gravity for 10–15 min	**Cystogram to exclude reflux** (occlude ureters if present); fibrosis, contracted (small) bladder, ureteric stricture

Other measures

- **HBOT** (hyperbaric oxygen therapy): 100% oxygen at 2–3 atmospheres, 30–40 sessions; raises tissue oxygen and drives neovascularisation; 80–90% response, but many relapse. **RICH-ART** trial: improved urinary symptom scores in radiation cystitis.
- **Nephrostomies:** divert urokinase-rich urine if bleeding continues.
- **Angioembolisation:** selective anterior division of internal iliac; spare the posterior division (gluteal pain); initial control about 90%. Used if unstable.
- **Cystectomy with conduit diversion:** last resort; remove the bladder (pyocystis risk).
- **Viral:** supportive (hydration, irrigation); cidofovir (intravenous or intravesical; evidence anecdotal).
- **Also know:** tranexamic acid and laser ablation are unproven; oral pentosan polysulfate in children.

Diagram to draw: stepwise escalation ladder.

> **Key points to remember:**
> - Acrolein (prevent with mesna); radiation endarteritis; BK virus.
> - Exclude bladder cancer.
> - Alum first; formalin only after cystogram.
> - Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

Also, from the condensed book (not in the paper answer):

- **Silver nitrate 0.5–1%** also needs a cystogram first, because it precipitates in the upper tract.
- **Aminocaproic acid** inhibits plasminogen activation (urokinase) and so blocks fibrinolysis; use it only in a clot-free bladder.
- **Formalin technique**: instil by gravity with the catheter **≤15 cm** above the pubis, up to **300 mL** or bladder capacity, and protect the urethra and skin.

5 · Prostatic haematuria

- It is a diagnosis of exclusion after a full GH work-up. **Benign prostatic hyperplasia is the commonest cause of GH in men over 60**, through increased microvessel density driven by vascular endothelial growth factor (VEGF).

- **5α-reductase inhibitors** (finasteride, dutasteride) lower VEGF and microvessel density. About **90%** improve, including anticoagulated men, with onset from 2 weeks to 9 months. Randomised trials show fewer recurrences than with watchful waiting.
- Prostate cancer bleeding usually means T4 invasion. Treat with androgen deprivation therapy with or without palliative radiotherapy.

ALGORITHM

1. Benign: start with a 5α-reductase inhibitor, then fulguration, then transurethral resection, laser vaporisation or enucleation, or simple prostatectomy, and finally prostatic artery embolisation (for poor-risk patients or bleeding after resection).
2. Malignant: start with androgen deprivation with or without radiotherapy, then fulguration or channel resection, then nephrostomy, and finally cystoprostatectomy.

6 · Urethral and upper tract bleeding

- **Urethrorrhagia** (blood at the meatus independent of voiding) means a lesion distal to the sphincter. Use retrograde urethrography and cystourethroscopy. With suspected trauma or penile fracture, **perform a urethrogram before any instrumentation**. In women, consider caruncle or diverticulum.
- **Renal arteriovenous malformation** is ~**75%** acquired (biopsy, partial nephrectomy, percutaneous nephrolithotomy, trauma). GH after a renal procedure needs prompt angiography and selective embolisation.
- **Ureteroiliac artery fistula** (radiation, vascular surgery, chronic stents) is treated by endovascular stenting.
- **Nutcracker syndrome**: the left renal vein is compressed between the aorta and the superior mesenteric artery.
- **Lateralising essential haematuria**: cystoscopy during bleeding shows the side, then ureteropyeloscopy inspects every calyx at low pressure for biopsy or fulguration.

Trial	Design	Result	Take-home
DETECT I	Prospective, UK, 3,556 referrals	Cancer in 13.8% with GH vs 3.1% with MH; 3.5% in GH under 45	Investigate GH at any age
RICH-ART	Randomised hyperbaric oxygen vs standard care, radiation cystitis	Improved urinary symptom score	Supports hyperbaric oxygen

9 Occupational Health in Urology

1 · Physical ergonomics

SAQ Operating room set-up

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3f (see also 2021-Summer Paper 4, Q3a for operating room safety)

Physical design

- **Zones:** protective (reception, changing) → clean (pre-op, recovery) → sterile (theatre, scrub) → disposal; one-way flow.
- **Ventilation:** positive pressure, about 20 air changes an hour, filtered air; laminar flow for implants.
- **Environment:** temperature about 20–24 °C, relative humidity about 20–60%. Patient warming: even 1–2 °C hypothermia raises SSI (surgical site infection) and blood loss (**Kurz 1996**: 3-fold wound infection).
- Power backup, piped gases, anaesthetic scavenging, radiolucent table.

Equipment layout by procedure

Setting	Arrangement
Endourology (TURP (transurethral resection of the prostate), URS (ureteroscopy), PCNL (percutaneous nephrolithotomy))	Lithotomy or prone; tower and monitor directly ahead; C-arm opposite surgeon; wires, baskets, dilators, spare scopes ready
Laparoscopic, upper abdomen	Surgeon and assistant face the abdomen; monitor opposite; scrub nurse near the foot
Laparoscopic, pelvic	30° Trendelenburg, split legs; surgeon on preferred side, assistant opposite
Robotic	Patient cart, surgeon console, vision cart; Si docks between legs, Xi from the side

- **Checklist before laparoscopic incision:** irrigation–suction, electrosurgical unit, full CO_2 cylinder plus spare, white-balanced camera, insufflator and Veress needle checked, **open tray ready**.
- Cables fixed; camera cable apart from diathermy; insufflator at eye level.

Positioning safety

- Pad bony prominences; axillary roll (lateral); avoid prolonged kidney rest and full flexion (rhabdomyolysis); no shoulder braces in steep Trendelenburg (brachial plexus).

Ergonomics and staff protection

- Monitor straight ahead, top at eye level; elbows 90–120°; table at the surgeon's pubic symphysis (laparoscopy).
- **Radiation:** ALARA (as low as reasonably achievable): time, distance, shielding; lead apron, thyroid shield, glasses; dosimeter.
- Smoke evacuation; sharps safety; vaccination.

Safety systems

- **WHO (World Health Organization) Surgical Safety Checklist** (sign in, time out, sign out); counts; fire-risk assessment (oxidiser, fuel, ignition).

Diagram to draw: theatre plan: table, tower, C-arm, anaesthetist, surgeon, scrub nurse.

> **Key points to remember:**
> - Zones with one-way flow; positive pressure, about 20 air changes an hour.
> - Monitor directly ahead at eye level; open tray ready for laparoscopy.
> - Hypothermia triples wound infection (Kurz).
> - ALARA: time, distance, shielding.

Also, from the condensed book (not in the paper answer):

- Work-related musculoskeletal disorders affect **46–90%** of urologists. They are a cumulative-load problem, so every fix either restores neutral posture or interrupts the load.
- Posture targets: in open surgery, work **5–10 cm** below the elbow with neck flexion **<25°**. In endourology, set the target between symphysis and umbilicus. At the robotic console, keep back flexion **<15°** and knees at 90–120°.
- Use steep-declination loupes (neck **<20°**) and two-piece lead that puts weight on the hips. A presbyopic surgeon should use workspace lenses, not progressives.

2 · Occupational exposures

*See the paper answer **Operating room set-up** (Section 1, Ch 9).*

*See the paper answer **Operating room safety : the role of surgeon.** (Section 1, Ch 1).*

Also, from the condensed book (not in the paper answer):

- **Fluoroscopy**: use pulsed screening at the lowest rate (1 pulse/s), collimate, and use last-image hold. Keep the source far from the patient and the image intensifier close. **Magnification increases dose.**
- **Pregnant surgeon**: wear a fetal dosimeter at the waist under the apron, use maternity or double-thickness lead, and minimise radiation cases.
- For CO_2 laser on condylomata, wear an N95 mask with the evacuator tip within **5 cm**. For intravesical chemotherapy, wear two pairs of chemotherapy gloves and use a closed-system transfer device.

3 · Physical and mental health

- **Atherosclerotic cardiovascular disease** is the most threatening modifiable cause of death in surgeons. Recommended sleep is **7–9 h**, and exercise **150 min** moderate weekly plus strength training twice.
- **Burnout** means emotional exhaustion, depersonalisation and low personal accomplishment, and affects about **50%** of urologists. Spending **≥20%** of time on meaningful work lowers it.
- A **second victim** develops guilt and post-traumatic stress after an adverse event and needs psychological first aid and peer support.

Stress zone	Features	Response
Ready (green)	Balanced, focused	Exercise, sleep, mindfulness
Reacting (yellow)	Irritable, poor concentration	Self-care, delegate
Injured (orange)	Anxiety, substance misuse, burnout	Support, less workload, time off
Ill (red)	Panic, severe depression, dependence	Mental-health professional, sick leave

Imposter syndrome is a cognitive distortion, commonest in trainees, and responds to cognitive behavioural therapy.

10 Ethics and Informed Consent

1 · Principles of medical ethics

SAQ Ethics and informed consent in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3e

Four principles (Beauchamp and Childress, 1979)

Principle	Meaning
Autonomy	Patient's deliberated self-rule; basis of consent and confidentiality
Beneficence	Act for the patient's benefit, judged by the patient's values
Non-maleficence	Avoid harm (*primum non nocere*); requires training and honest risk data
Justice	Fair distribution of resources, benefits and risks
- Hippocratic Oath gave beneficence, non-maleficence, confidentiality; autonomy and justice are modern additions (autonomy is bounded by justice).	
- **Jonsen's four-box method:** medical indications; patient preferences; quality of life; contextual features (family, cost, law).	

Informed consent: legal evolution

- From battery to negligence. **Schloendorff (1914):** right to bodily integrity. **Salgo (1957):** term "informed consent". **Canterbury v Spence (1972): reasonable-patient** disclosure standard.
- India: **Samira Kohli (2008)**: consent for one procedure does not cover another, except in emergency.

Elements

1. Diagnosis, proposed treatment and alternatives, including no treatment.
2. Risks, benefits and uncertainty compared with alternatives.
3. **Check understanding** (most often missed).
4. Voluntary decision by a patient with capacity; documented. - Material risks: no agreed cut-off (1% proposed); disclose rarer ones by severity; lay language. - Urology examples: incontinence and erectile dysfunction after prostatectomy; retrograde ejaculation after TURP (transurethral resection of the prostate); stent symptoms; conversion to open surgery.

Exceptions

- **Emergency** in an incapable patient (presumed consent); a conscious patient may still refuse.
- **No capacity:** surrogate using substituted judgement, then best interests.

- **Minors (<18 years):** parents consent; child gives assent.

Improving consent

- Decision aids, repeat-back, simple written material.
- **Shared decision making:** two-way exchange of options and patient values; vital when several options are reasonable (e.g. localised prostate cancer).

Special areas

- **End of life:** early goals-of-care talks, advance directives, DNR (do-not-resuscitate) orders.
- **Research:** written consent, ethics committee approval, extra protection for vulnerable groups.
- **Also know:** confidentiality; conflicts of interest; futility policies.

Diagram to draw: Jonsen four-box grid.

> **Key points to remember:**
> - Four principles: autonomy, beneficence, non-maleficence, justice.
> - Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
> - Commonest gap: not checking understanding.
> - Exceptions: emergency, incapacity (surrogate), minors.

Also, from the condensed book (not in the paper answer):

- All four principles are equal and **prima facie binding**; an ethical dilemma is a conflict between them that needs prioritising. Equal treatment is not always just treatment.
- In the four boxes, **quality of life draws on three principles** (autonomy, beneficence, non-maleficence) and **contextual features map to justice**.

2 · Informed consent

*See the paper answer **Ethics and informed consent in urology** (Section 1, Ch 10).*

Also, from the condensed book (not in the paper answer):

- The law moved from **battery to negligence to a patient-centred standard**. Mohr v Williams (1905) held that consulting a doctor is not consent.
- **Disclosing uncertainty improves trust.** Decision aids reduce decisional conflict; repeat-back works but is time-consuming.

3 · Capacity, exceptions and special situations

*See the paper answer **Ethics and informed consent in urology** (Section 1, Ch 10).*

Also, from the condensed book (not in the paper answer):

- If a surrogate does not follow the patient's wishes, go to the **ethics committee** and then a court-appointed conservator.
- The **Patient Self-Determination Act (1990)** requires institutions to tell patients of their right to an advance directive.
- For research, the institutional review board may **waive consent** for minimal-risk or de-identified data.

11 Perioperative Care in Urologic Surgery

1 · Preoperative evaluation

- Test selectively: an electrocardiogram at age **>40** or with cardiac history, clotting tests only with a bleeding or anticoagulant history, and a pregnancy test in every woman of childbearing age.

American Society of Anesthesiologists (ASA)	Definition
I / II	Healthy / mild systemic disease
III	Severe disease that limits activity, not incapacitating
IV	Incapacitating, a constant threat to life
V / VI	Moribund (death within 24 h) / brain-dead donor
E	Emergency suffix (never with VI)

2 · Cardiac and other risk

- Good reserve with low-stress (endoscopic) surgery needs no further testing, whatever the history. Capacity of **≥4 METs** (metabolic equivalents, about two flights of stairs) means no invasive work-up.
- The **Revised Cardiac Risk Index** has six unweighted predictors: ischaemic heart disease, heart failure, cerebrovascular disease, high-risk surgery, preoperative insulin and creatinine **>2 mg/dL**.
- After myocardial infarction (MI), waiting 4–6 weeks before elective surgery is reasonable. Frailty predicts complications independent of age.
- A forced expiratory volume in 1 s **<0.8 L** or <30% predicted is high pulmonary risk. Child class mortality is A 10%, B 30% and C 76–82%.
- Loss of >9 kg in 3 months is severe malnutrition; preoperative parenteral nutrition helps only these patients.

*See the paper answer **Stones in pregnancy** (Section 11, Ch 96).*

Also, from the condensed book (not in the paper answer):

- The fetal malformation threshold is **10 cGy**; computed tomography gives about 1 cGy.

3 · Optimising comorbidity

- Only β-blockade has improved cardiac outcome (Mangano atenolol trial).
- Stop smoking **≥8 weeks** before surgery; stopping later may raise pulmonary risk.
- Target glucose 80–250 mg/dL. Withhold metformin on the day of surgery and sodium-glucose cotransporter-2 inhibitors 3 days before (euglycaemic ketoacidosis).
- Presume adrenal suppression with prednisone >20 mg/day for >3 weeks or cushingoid features; ≤5 mg/day needs only the usual dose. Give hydrocortisone **50–100 mg** intravenously at induction, then 25–50 mg 8-hourly for 24–48 h.

4 · Antithrombotic management

- Stop warfarin 5 days before (international normalised ratio **<1.5**). Bridge moderate- and high-risk patients with therapeutic low-molecular-weight heparin (LMWH), last dose 24 h

before. Direct oral anticoagulants need no bridging.

Thrombotic risk	Mechanical valve	Atrial fibrillation ($CHADS_2$)	Venous thromboembolism (VTE)
High	Any mitral; caged-ball or tilting-disc aortic; stroke <6 months	5–6; stroke <3 months	<3 months; severe thrombophilia
Moderate	Bileaflet aortic plus risk factor	3–4	3–12 months; active cancer
Low	Bileaflet aortic alone	0–2	Single VTE >12 months

Drug	Stop before surgery
Aspirin, clopidogrel	7–10 days
Prasugrel / ticagrelor	7 / 5 days
Apixaban, rivaroxaban	48 h (72 h if high bleeding risk)
Dabigatran	48–96 h by creatinine clearance

KEY POINT

Stopping antiplatelets early after a coronary stent carries a 25–50% risk of MI. Delay elective surgery ≥30 days after a bare-metal stent and >1 year after a drug-eluting stent.

5 · Bowel preparation, temperature and positioning

*See the paper answer **Methods of preoperative bowel preparation.** (Section 12, Ch 133).*

SAQ Patient positioning in lap urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 1, Q3a (pneumoperitoneum physiology also answered in 2019-Summer Paper 1, Q2)

Principles

- Positioning uses **gravity** to move bowel away and gives port or robot access while keeping ventilation safe.
- Injury comes from **stretch, compression or ischaemia**, so position and duration act together; nerve injury occurs in about 2.8%. Shared team responsibility (ASA [American Society of Anesthesiologists] advisory, 2000).
- Secure IV (intravenous) access, oximetry and BP (blood pressure) cuff before final positioning: tucked arms become inaccessible.

Positions

Position	Used for	Set-up	Hazards
Steep Trendelenburg, low lithotomy or split legs	RARP (robot-assisted radical prostatectomy), cystectomy	Arms tucked and padded; chest strap over egg-crate; **no shoulder braces**; tilt about 22–30°	Brachial plexus; femoral nerve (excess hip flexion); facial and laryngeal oedema; corneal abrasion; tube into right bronchus

Position	Used for	Set-up	Hazards
Modified flank (30–45°)	Transperitoneal nephrectomy, pyeloplasty	Back on gel rolls; upper arm across chest; wide tape; tilt to test	Pressure points, slippage
Full lateral (90°), table flexed	Retroperitoneoscopy, adrenalectomy	**Axillary roll** 3–4 fingerbreadths below axilla; lower leg flexed 45°, upper straight on pillows	**Rhabdomyolysis**; sciatic stretch
Prone	Posterior adrenalectomy	Pad torso, elbows, hips	Airway, face and eyes

Rhabdomyolysis

- About 0.4% (1% retroperitoneal). Risk: men, BMI (body mass index) ≥25, **kidney rest**, full flexion, long cases. Downside hip pain, brown urine, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest (Campbell gives 20–30 minutes in one place, under 1 hour in another), avoid hypotension, keep cases under 5 hours.

Prevention and physiology

- Abduction ≤90°; pad ulnar groove and fibular head; recheck after every table move.
- **Head-down** raises cardiac output but splints the diaphragm; restrict fluids.
- Nerve injury on day 1: neurology review, conduction studies, physiotherapy.
- **Also know:** forearm compartment syndrome from infiltrated IV lines; Xi side-docking avoids stirrups.

Diagram to draw: full flank position with axillary roll and table break.

> **Key points to remember:**
> - Nerve injury about 2.8%; brachial plexus is commonest at risk.
> - Never shoulder braces in steep Trendelenburg.
> - Axillary roll below, not in, the axilla.
> - Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

*See the paper answer **Operating room set-up** (Section 1, Ch 9).*

Also, from the condensed book (not in the paper answer):

- Ileus beyond 72–96 h should prompt a search for obstruction or retroperitoneal haemorrhage.
- Femoral neuropathy comes from retractor blades on the psoas, not from hip position.

6 · Antibiotic prophylaxis

- Give it within **60 min** before incision, limit it to **≤24 h**, and redose after 4 h. Transrectal biopsy is a contaminated wound.

Procedure (American Urological Association, AUA)	First choice	Duration
Cystoscopy (sterile urine), shock-wave lithotripsy	Only with risk factors	Single

Procedure (American Urological Association, AUA)	First choice	Duration
Endourology, nephrectomy, prostatectomy, ileal conduit	Cefazolin	Single
Transrectal biopsy	Fluoroquinolone, or cephalosporin plus aminoglycoside	Single
Colon conduit	Cefazolin plus metronidazole	Single
Percutaneous renal surgery	First- or second-generation cephalosporin	≤24 h
Prosthesis	Aminoglycoside plus cephalosporin or vancomycin	≤24 h

7 · VTE prophylaxis

- VTE is the leading non-surgical cause of death after urological surgery, highest after cystectomy. Drug prophylaxis raises minor, not major, bleeding.
- Enoxaparin is **40 mg** daily (30 mg if creatinine clearance <30 mL/min). Unfractionated heparin (UFH) 5000 U is reversible and safe in renal failure.

AUA risk	Definition	Prophylaxis
Low	Minor surgery, <40 y	Early ambulation
Moderate	40–60 y, or minor surgery with risk factors	UFH 12-hourly or enoxaparin
High	>60 y, or 40–60 y with risk factors	UFH 8-hourly or enoxaparin
Highest	Cancer, prior VTE	Drug plus compression

A Caprini score **≥5** is highest risk. Give extended **28-day** LMWH after radical pelvic cancer surgery.

8 · Anaesthesia and transfusion

- Extubate only when the train-of-four ratio reaches **0.9**. Nitrous oxide expands closed gas spaces.
- Transfusion is often indicated at haematocrit **<21%** and rarely above 30%. Give platelets below 50,000 (100,000 for high-risk surgery).
- Transfusion-related acute lung injury is the commonest cause of transfusion death. Delirium may be the first sign of sepsis.

12 Fundamentals of Intraoperative Patient Safety

1 · Checklists, site verification and team safety

*See the paper answer **Operating room safety : the role of surgeon.** (Section 1, Ch 1).*

Also, from the condensed book (not in the paper answer):

- Wrong-site surgery occurs in about **1 in 112,000** cases, and up to 30% of root causes arise before admission.
- Mark only the correct side, never with an "X". For lower-tract endourology, mark the wristband, not the meatus. Use a professional interpreter, never family.
- Simulation-based team training changes behaviour; didactic teaching alone does not.

2 · Retained items and fire

- A retained surgical item is one found after skin closure, even in theatre. It occurs in **1 in 2800–10,000** cases; sponges are commonest, then drains, then needles.
- A correct count does not exclude a retained item, so obtain a radiograph when in doubt.
- The **Silverstein fire-risk score** gives one point each for open oxygen, an ignition source and a site above the xiphoid; **3** means high risk. Monopolar diathermy causes about 90% of fires.
- For buccal graft harvest, keep inspired oxygen ≤30%, stop nitrous oxide and warn anaesthesia before using diathermy.
- Place the diathermy return pad in full contact over muscle, away from metal implants, and prefer bipolar with pacemakers.

ALGORITHM

1. Fire: stop the procedure and pour saline or water.
2. Remove the drapes, then use an extinguisher.
3. Follow RACE: Rescue, Alert, Confine (gases off), Extinguish or evacuate.

3 · Positioning injuries (see Ch 11)

- In lithotomy, stirrups compress the common peroneal nerve at the fibular head and cause foot drop. Hip flexion with knee extension stretches the sciatic nerve.
- **Well-leg compartment syndrome** follows prolonged lithotomy (**>4 h**) with raised legs and Trendelenburg. Prevent it by limiting time or lowering the legs periodically. Suspect it with pain out of proportion and a raised creatine kinase, and treat by urgent fasciotomy.
- In the flank position, long cases and obesity risk rhabdomyolysis.

13 Incisions, Exposure and Drains

1 · Abdominal incisions

Incision	Indications	Notes
Lower midline	Prostatectomy, cystectomy, diversion, pelvic node dissection	**Commonest** urological incision
Upper or full midline	Retroperitoneal lymph node dissection (RPLND), trauma, bilateral nephrectomy	**Most versatile**; poor in obesity
Pfannenstiel	Pelvic surgery, specimen extraction	Rectus separated, not cut; **least hernia**
Gibson	Renal transplant, lower ureter	Extraperitoneal; hockey-stick extension adds hernias

- Close midline fascia with strong slowly absorbable suture, taking bites **1 cm** from the edge and 1 cm apart, with suture length to wound length **≥4:1**.
- No technique is superior for dehiscence or hernia, but avoid continuous rapidly absorbable suture.

2 · Approaches to the kidney

Incision	Indications	Trade-off
Midline transperitoneal	Trauma, inferior vena cava (IVC) thrombus, bilateral disease, horseshoe	Early vascular control; poor upper retroperitoneum
Subcostal or chevron	Radical nephrectomy, bilateral tumours, IVC thrombus	Early vascular control; liver, spleen and bowel at risk
Thoracoabdominal (9th–10th rib)	Level III thrombus; large upper-pole or adrenal mass	Best exposure; pain no worse than flank
Flank (12th supracostal or 11th transcostal)	Partial or simple nephrectomy, xanthogranulomatous pyelonephritis	Extraperitoneal; **pleura** at risk; transcostal causes flank bulge
Transverse upper abdominal	Paediatric Wilms tumour	–

KEY POINT

An anterior incision gives bilateral reach and early pedicle control at the cost of the bowel; a flank incision stays extraperitoneal at the cost of the pleura.

ALGORITHM

1. Pleural tear: pass a red rubber catheter through the tear and tie a purse-string around it.
2. Put the catheter end under water and give sustained inflation.
3. Withdraw it as bubbling stops. A large tear needs a chest tube.

- A routine chest radiograph after nephrectomy is unnecessary.

3 · Inguinal and scrotal incisions

- Radical orchidectomy uses an inguinal incision of 5–7 cm. **A scrotal incision is contraindicated in suspected testis cancer.**
- Subinguinal inguinal lymph node dissection risks flap necrosis, infection and lymphocele.

4 · Drains

- A Penrose drain is passive and suits the scrotum. Jackson-Pratt and Blake drains are active closed suction and suit most other surgery.
- Use a separate stab incision away from the inferior epigastric vessels, and secure the drain with non-absorbable suture.
- Routine drains are questioned after robotic prostatectomy, partial nephrectomy and cystectomy; drain when the findings demand it.

14 Sutures and Suturing

1 · Choosing a suture

- Braided suture knots securely but harbours bacteria; monofilament inflames less but has more memory.

Suture	Strength retained	Absorbed by
Plain / chromic gut	7–10 d / 21–28 d	70 / 90 d
Caprosyn	~10 d	56 d
Monocryl (poliglecaprone 25)	2 wk 20–40%	91–119 d
Vicryl (polyglactin 910, braided)	3 wk 40–50%	56–70 d
PDS (polydioxanone)	6 wk 35–60%	182–238 d
Maxon (polyglyconate)	6 wk 25%	180 d
Prolene, Mersilene (braided polyester)	Indefinite	Never

> **KEY POINT**
>
> Strength is lost long before absorption. Match the support needed: skin 1–2 weeks (Monocryl, gut), soft tissue 2–4 weeks (Vicryl), fascia ≥6 weeks (PDS, Maxon).

- Chromic gut is the most inflammatory absorbable suture; Caprosyn absorbs fastest.
- Silk marks the cord at orchidectomy before a possible retroperitoneal lymph node dissection.
- **Never leave non-absorbable suture or exposed metal staples in contact with urine**, because they form a stone nidus.
- V-Loc barbed suture is not for fascia; Stratafix Symmetric PDS is the only barbed suture strong enough for fascia.

2 · Needles and knots

- A taper point is commonest. A cutting needle has its edge on the inner curve; a **reverse cutting** needle has it on the outer curve and suits retention sutures.
- The smallest taper needle is RB (17 mm). A UR or GU 5/8-circle needle suits the deep pelvis.
- Prolene knots unravel, so add extra throws.

3 · Techniques

- A vertical mattress everts the edges; a figure-of-eight gives haemostasis.
- Run fascia only with slowly absorbable suture, and close port defects >8 mm.

Trial	Design	Result	Take-home
STITCH	Randomised, midline: 5 mm versus 1 cm bites	Hernia at 1 year 13% vs 21%	Small bites

15 Lower Urinary Tract Catheterisation

1 · Urethral catheters

SAQ Foley's (urethral) catheters, guidewires and DJ stents with their complications

Paper answer · P1

Asked as:

- Foley's catheter. — *Summer 2016 Paper I Q3e (SAQ)*
- Urethral catheters, guidewires and DJ stents — *Practice Set 18 SAQ Q3f (SAQ)* (also answered in part in 2016-Summer Paper 1, Q3e and 2015-Summer Paper 3, Q1)
- Complications if DJS — *Summer 2021 Paper I Q3c (SAQ) — Taken as complications of a DJ (double-J) ureteric stent.*

Type: SAQ (10 marks)

Foley's / urethral catheters

History

- **Frederic Foley**, a Boston urologist, developed the modern balloon catheter (Campbell: 1929; published in the 1930s) for **haemostasis after prostatectomy**. Reybard (1855) had made the first self-retaining one.

Design

- **Size: Fr (French) = 3 × diameter in mm.** Adults (adult) usually 14–16 Fr; clot retention 20–24 Fr three-way. Newborn 6 Fr.
- Normal calibre: (male) meatus 24 Fr (narrowest point), bladder neck 28 Fr, prostatic urethra 32 Fr, female urethra 22 Fr.
- **Balloon:** usually 10 mL; larger (30 mL) for traction. Use (inflate balloon with) **sterile water**, only after urine flows at the hub; volume is printed on the port.
- **Lumens:** one (intermittent, specimen); two-way (drainage + balloon) is routine; three-way adds irrigation for haematuria (stiff wall resists collapse). Each lumen reduces (shrinks) drainage size.
- **Tips:** straight; **coudé** (curved, for prostatic enlargement, tip up); **Councill** (open tip, over a wire).
- **Materials:** latex (cheap, allergy); silicone (long term, less urethritis and encrustation; better urethral tolerance); PVC (polyvinyl chloride); coatings (silver, antibiotic) reduce bacteriuria but not clinical infection.
- **Special:** Malecot (winged, no balloon); Lapides diagnostic (rings 1 cm apart); urodynamic.

Indications and contraindications

- **Indications:** retention, clot retention, output monitoring, long surgery, immobility, after urological surgery.
- **Absolute contraindication:** suspected urethral injury (blood at meatus, pelvic fracture). Do an RUG (retrograde urethrogram) first (**suspected urethral injury → RUG first**).
- **Relative:** AUS (artificial urinary sphincter): deactivate, then 12–14 Fr; stricture.

Technique

- Aseptic; 10–15 mL lignocaine gel. In men, advance **to the hub** and see urine before inflating. Replace the foreskin.
- **Difficult:** 18 Fr coudé → guidewire → cystoscopy, Councill over wire → suprapubic catheter.

Complications

- **CAUTI (catheter-associated urinary tract infection):** 75% of hospital UTIs; bacteriuria 5–10% per catheter-day; **biofilm** forms within minutes.
- Trauma (about 0.3% of inpatients): false passage, balloon inflated in the urethra. Also stricture, traumatic hypospadias, encrustation, paraphimosis, non-deflating balloon (puncture with a wire).

CAUTI prevention

- Review daily and remove early; closed dependent system; clean meatus with water; culture only if symptomatic. Treat only symptomatic infection and **change the catheter**.

Guidewires

Feature	Details
Uses	Access, safety wire, straightening ureter, guiding dilation and stents
Construction	Stainless-steel core with outer wrap; **nitinol** core is kink-resistant
Coating	PTFE (polytetrafluoroethylene): standard safety wire. **Hydrophilic** "glide": passes impacted stones, but too slippery to be a safety wire. Hybrid: hydrophilic tip, PTFE shaft
Size	0.018–0.038 inch (commonly 0.035/0.038); 80–260 cm, 145 cm most useful; floppy tip 1–3 cm
Tip and stiffness	Straight, angled, J; standard to super-stiff (straightens tortuous ureter)

DJ (double-J) stents: introduction

- The **double-J (double-pigtail)** stent (Finney 1978) is a self-retaining polymer tube with coils in the pelvis and bladder; adult usually 6 Fr, length by height. Consent should cover its symptoms and complications.
- **Materials:** polyurethane, silicone (hydrocoated silicone less painful); **metallic (Resonance)** for extrinsic malignant compression.
- **Indications:** obstruction (infected obstructed kidney, malignancy, pregnancy), after URS (ureteroscopy) with injury, pyeloplasty, transplant, leak.
- **Placement:** retrograde pyelogram → wire → stent over wire under fluoroscopy or vision.
- **Problems:** stent symptoms (bladder irritation and reflux) eased by α-blockers and antimuscarinics; migration; encrustation (forgotten stent); failure with extrinsic compression.

Classification of DJ stent complications

Timing	Complications
At insertion	**Misplacement**: proximal end below the obstruction; **submucosal** passage or perforation (suspect when force is needed)
Early	**Stent-related symptoms**; haematuria; infection and urosepsis; **vesicoureteric reflux** up the stent (flank pain on voiding)

Timing	Complications
Late	**Migration** (proximal or distal; short stent); **encrustation**; **stent failure** (obstruction); biofilm and colonisation; **fracture**; **forgotten stent**; rarely **ureteroarterial fistula**

Stent-related symptoms

- About **4 in 5 patients**: urgency, frequency, dysuria, haematuria, flank and groin pain; sexual and work problems.
- **Mechanism:** the **intravesical coil irritates the trigone**, and **reflux up the stent** distends the kidney on voiding.
- Measured by **USSQ** (Ureteral Stent Symptom Questionnaire).
- **Reduced by: α-blockers and antimuscarinics**; intraureteral placement; hydrocoated silicone; shortest dwell. Diameter does not matter.

Encrustation and forgotten stent

- Minerals and biofilm deposit on the stent. Risk rises with **dwell time**, infection, stone-forming urine and pregnancy. Leads to obstruction, sepsis, fracture, **ureteric avulsion on removal**, renal loss.
- **FECal grading (Forgotten, Encrusted, Calcified):** I minimal linear; II circular; III partial on one or both coils; IV complete on one coil; V both coils.
- **Management:** non-contrast CT (computed tomography); never pull by force; clear **bladder coil** (cystolithotripsy), then **ureter** (ureteroscopy), then **renal coil** (flexible ureteroscopy or PCNL [percutaneous nephrolithotomy]).

Stent failure

- Extrinsic compression (malignancy, radiation), debris, encrustation. Obstructive symptoms → image; change stent, **PCN** (percutaneous nephrostomy) or **metallic stent**.

Prevention

- Stent only when indicated; correct length; **stent register**; long-term stents changed every 3–6 months (local practice); **pregnancy: about every 4 weeks**.

Diagram to draw: cross-section of a three-way Foley; DJ stent in situ with bladder coil on the trigone and arrows showing reflux.

> **Key points to remember:**
>
> - Foley balloon catheter: designed for post-prostatectomy haemostasis.
> - Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
> - Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
> - CAUTI 5–10% per day; remove early; change the catheter when treating.
> - Hydrophilic wire to pass, PTFE wire to stay (safety).
> - DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
> - Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
> - Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Also, from the condensed book (not in the paper answer):

- Lapides showed that infection comes from stasis and overdistension, not bacteria alone, so **regular emptying, not sterility, prevents infection** in clean intermittent catheterisation.

Age	Straight (Fr)	Balloon (Fr)
Newborn	4–6	6
1–23 months	6–8	6–8
2–6 y	10	8–10
7–12 y	10–12	12–14
>12 y	14	14–16

2 · Difficult catheterisation

SAQ Outline algorithm for difficult catheter placement.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3b

Introduction

- **Normal calibre:** meatus 24 Fr (narrowest), bladder neck 28 Fr, prostatic urethra 32 Fr.
- **Causes:** BPH (benign prostatic hyperplasia, commonest), urethral stricture, false passage, meatal stenosis, bladder neck contracture, urethral trauma, abnormal anatomy.
- **Golden rule:** escalate **blind → wire → vision → suprapubic**; never repeat forceful blind passage.

Before starting

- History (surgery, radiotherapy, AUS (artificial urinary sphincter), reconstruction), DRE (digital rectal examination).
- **Absolute contraindication:** suspected urethral injury (pelvic fracture, blood at meatus, high-riding prostate) → **RUG (retrograde urethrogram)**; pelvic fracture urethral injury → suprapubic tube.
- **Technique:** 10–15 mL lubricant (lidocaine gel), penis stretched, steady pressure; inflate only after urine flows at the hub.

Algorithm (Campbell Fig. 15.6)

```
First attempt fails ├─ Prostatic obstruction suspected │ → 18-Fr coudé, tip (marker) pointing up │ → fails → flexible cystoscopy └─ Not suspected (stricture likely) → smaller catheter → fails → hydrophilic glidewire → fails → flexible cystoscopy At cystoscopy / with wire in bladder: ├─ Stricture → wire across → 5-Fr catheter, aspirate urine to confirm │ → super-stiff wire → serial dilatation (1 Fr above catheter size) │ → Councill (open-tip) catheter over wire └─ False passage → pass under vision (scope anterior) → wire → Councill over wire Wire cannot be passed / dense stricture / urethral disruption → suprapubic catheter (ultrasound-guided percutaneous, or open)
```

- **Why 18-Fr coudé:** stiffer, larger tip area, curved tip follows the bulbar-prostatic angle; 14–16 Fr usually buckle.
- **Blood on the tip** suggests false passage: stop blind attempts.

Special situations

Situation	Approach
AUS	Deactivate; smallest catheter 12–14 Fr; remove early
Obese or female	Exposure: Trendelenburg, retract pannus, two fingers in vagina to guide
Buried penis	Infrapubic pressure; flexible cystoscopy
Neobladder, continent pouch	Never force; scope + wire; suprapubic only with imaging guidance
PUV (posterior urethral valves) neonate	6–8 Fr, or 5-Fr open-ended over glidewire; else suprapubic or vesicostomy

Also know: filiforms, balloon dilators; complications: false passage, haematuria, stricture.

Diagram to draw: coudé catheter following the bulbar-prostatic curve with tip up.

> **Key points to remember:**
> - Blind → wire → vision → suprapubic.
> - BPH: 18-Fr coudé, tip up.
> - Councill catheter over wire after dilatation.
> - Suspected urethral injury: RUG first, no blind catheter.

Also, from the condensed book (not in the paper answer):

- Membranous strictures are the most prone to false passage. Pass the cystoscope with the camera angled anteriorly.
- With a neobladder, suprapubic puncture is hazardous (intraperitoneal), so use image guidance.

3 · Suprapubic catheterisation

- It is indicated when the urethra is inaccessible (trauma, stricture) and for long-term drainage.
- Percutaneous placement is contraindicated with a violated retropubic space, bowel over the bladder, or no ultrasound.

ALGORITHM

1. Confirm a full bladder with no overlying bowel on ultrasound, and tilt the patient head-down.
2. Make a 1–2 cm incision 1–2 fingerbreadths above the symphysis.
3. Aspirate urine with a spinal needle while infiltrating local anaesthetic.
4. Insert a trocar and sheath (16 Fr catheter) or use the Seldinger technique, without over-advancing.

- Bowel injury is the key complication. An experienced person should make the **first change at 4–6 weeks**.

4 · Complications

*See the paper answer **Catheter Associated Urinary Tract Infections (CAUTI)** (Section 3, Ch 26).*

Also, from the condensed book (not in the paper answer):

- Spinal cord injury raises bladder cancer risk, but surveillance cystoscopy is not recommended.

Retained catheter	Management
Balloon will not deflate	Cut the valve; pass a stiff wire end up the channel to puncture it, or use an ultrasound-guided needle; then cystoscopy to retrieve fragments
Knotted	Cystoscopic needle puncture of the balloon
Calcified	Cystoscopic laser lithotripsy

16 Fundamentals of Upper Urinary Tract Drainage

1 · Indications

- Obstruction is the commonest indication. Others are pre-stenting before retrograde intrarenal surgery, trauma (to prevent urinoma) and instillation of chemotherapy.
- In the **Whitaker test**, a renal pelvic minus bladder pressure **>12 cm H_2O** during constant-rate infusion means obstruction.

SAQ Interventional radiology in urology and therapeutic angioembolisation

Paper answer · P1

Asked as:

- Therapeutic angioembolization. — *Practice Set 7 Paper I Q3e (SAQ)*
- Interventional urological treatment in urology — *Practice Set 8 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

Assumed meaning: interventional radiology (image-guided percutaneous and endovascular) treatment in urology.

Principles

- Guidance by ultrasound, fluoroscopy or CT (computed tomography), usually under local anaesthesia. **Seldinger technique:** needle, guidewire, then devices over the wire.
- **Beforehand:** correct coagulopathy, manage anticoagulants, urine culture, antibiotic prophylaxis.
- **Requirements:** experienced team, resuscitation in the suite; unstable non-responders need surgery.

Non-vascular procedures

Procedure	Indications and points
PCN (percutaneous nephrostomy)	Obstructed infected kidney (as effective as retrograde stent), malignant ureteric obstruction, urinary leak or fistula, access for PCNL (percutaneous nephrolithotomy). Puncture a **posterior calyx** through the avascular Brödel plane. Transfusion 0.5–4%
Antegrade stenting	When retrograde stenting fails: malignant obstruction, ileal conduit, transplant ureter
Antegrade pyelography, Whitaker test	Define strictures; pressure–flow test for obstruction
Percutaneous drainage	Renal or perinephric abscess >3 cm, urinoma, lymphocele
Image-guided biopsy	Renal mass biopsy

Vascular (endovascular) procedures: therapeutic angioembolisation

Definition

- Deliberate **endovascular occlusion** of a target vessel, usually by transfemoral selective catheterisation under fluoroscopy, to stop bleeding, devascularise a tumour or close an

abnormal vessel.

- **Superselective** embolisation of a segmental branch preserves most parenchyma.

Embolic agents

Type	Examples	Use
Temporary	Autologous clot; gelatin sponge (2–6 weeks)	Priapism, prostate, bladder bleeding
Permanent particles	PVA (polyvinyl alcohol), microspheres	Tumours, PAE (prostate artery embolisation)
Mechanical	Microcoils, vascular plugs	Pseudoaneurysm, AVF (arteriovenous fistula), varicocele
Liquid	NBCA (N-butyl cyanoacrylate) glue, absolute ethanol	Tumour ablation, recurrent priapism

Urological indications

Procedure	Indications
Renal angioembolisation	Grade IV–V renal trauma; **pseudoaneurysm or arteriovenous fistula** after PCNL or partial nephrectomy (delayed bleed); bleeding AML (angiomyolipoma, first line); palliative embolisation of RCC (renal cell carcinoma)
Internal iliac (anterior division) embolisation	Intractable bladder haemorrhage; initial control about 90%. Spare the posterior division (gluteal ischaemia)
PAE (prostate artery embolisation)	LUTS (lower urinary tract symptoms) from BPH (benign prostatic hyperplasia); EAU (European Association of Urology) for men accepting lesser outcomes; AUA (American Urological Association) 2023 may offer
Varicocele embolisation	Coils or liquid sclerosants via femoral or jugular vein
High-flow priapism	Superselective cavernosal artery embolisation; resolution ≥75%; recurrence 30–40%
PTRA (percutaneous transluminal renal angioplasty)	Fibromuscular dysplasia; transplant renal artery stenosis

- **Renal trauma:** active contrast extravasation, pseudoaneurysm, AVF; success high in grade IV, lower in grade V. Failure predictors: instability, grade V, urinary extravasation. Repeat embolisation often succeeds.
- **Iatrogenic bleeding** after PCNL (percutaneous nephrolithotomy) or partial nephrectomy: **treatment of choice**; superselective coils nearly 100% technically successful.
- **AML (angiomyolipoma): first line for acute haemorrhage**; prophylactic for >6 cm, aneurysm >5 mm, growth or pregnancy plans. In TSC (tuberous sclerosis complex), embolise then give steroids.
- **RCC:** palliation of haematuria or pain in unfit patients; preoperative use no longer routine.
- **Intractable bladder haemorrhage** (radiation cystitis, cancer): bilateral **anterior division of internal iliac**; avoid the posterior division (gluteal pain).
- **Prostate:** PAE for BPH (benign prostatic hyperplasia) and refractory prostatic haematuria.
- **High-flow priapism:** superselective cavernosal artery embolisation; resolution ≥75%; recurrence → repeat with a non-resorbable agent.

- **Also know:** varicocele (venous coils); lymphatic embolisation for chylous ascites; ADPKD (polycystic kidney) pain.

Ablation

- **Cryoablation or RFA (radiofrequency ablation)**, percutaneous, for cT1a renal masses <3 cm; biopsy first. Higher local recurrence than partial nephrectomy, salvaged by repeat ablation.

Complications

- Contrast nephropathy, access-site haematoma, coil migration.
- **Post-embolisation syndrome:** fever, flank pain, nausea, leucocytosis; up to 72 hours; self-limiting.
- **Non-target embolisation:** bowel, bladder, rectum, gluteal muscle, penile skin.
- Renal infarction or abscess; ED (erectile dysfunction) after priapism embolisation.

Diagram to draw: prone PCN through a posterior calyx along the Brödel line; renal angiogram showing segmental pseudoaneurysm before and after coil occlusion.

> **Key points to remember:**
> - PCN equals retrograde stent for decompressing an obstructed infected kidney.
> - Superselective coils: treatment of choice for pseudoaneurysm and AVF.
> - Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
> - Bladder bleeding: internal iliac anterior division; spare posterior division.
> - Post-embolisation syndrome is common and self-limiting.

Also, from the condensed book (not in the paper answer):

- PCN avoids general anaesthesia in the very ill, while a stent avoids an external tube and can also be placed awake under local anaesthesia.

2 · Anatomy chooses the route

Situation	Preferred approach
Horseshoe kidney	Percutaneous via a more medial upper-pole calyx; computed tomography (CT) first, as retrorenal bowel is common
Pelvic or ectopic kidney	Supine anterior, image-guided or laparoscopy-assisted
Transplant	Anterior, ultrasound-guided
Ureteropelvic junction obstruction before pyeloplasty	Antegrade, to spare the ureter
Calyceal diverticulum, ileal ureter, transureteroureterostomy, urinary diversion	Antegrade

3 · Retrograde stenting and stents

*See the paper answer **Foley's (urethral) catheters, guidewires and DJ stents with their complications** (Section 1, Ch 15).*

Also, from the condensed book (not in the paper answer):

- Adult stents are typically 4.7–8 Fr and 22–30 cm; **6 Fr, 24–26 cm** is commonest.
- Under fluoroscopy alone, place the pusher marker at the mid-symphysis in men and the lower symphysis in women. This landmark is invalid after prostatectomy or with prolapse.
- The Ureteral Stent Symptom Questionnaire (USSQ) has **38 questions in six domains**. Stent length made no difference in a large randomised trial.

4 · Percutaneous nephrostomy

P1 *Pelvicalyceal anatomy and its relevance to PCNL; Prone vs supine PCNL — posterior calyx through the papilla along Brödel's avascular line, never the pelvis; calyx choice and pleural risk; prone versus supine*

- The American Urological Association (AUA) recommends antibiotic prophylaxis for all percutaneous renal surgery.
- A 21 G needle takes only a **0.018-inch** wire, which must be upsized; an 18 G needle takes a 0.035-inch wire. Dilate short, never beyond the target calyx.
- Balloon dilators are now commonest and bleed less but work poorly in scar, where metal Alken dilators are best.

Agent	Stop before access
Aspirin (primary prevention only)	1 week; continue if there is a cardiovascular indication
Warfarin	1 week (bridge if needed)
Clopidogrel	5 days
Non-steroidal anti-inflammatory drugs	3–5 days

Tube	Size	Notes
Foley or Councill	≤24 Fr	Fill balloon partly with saline, not contrast
Cope loop	6–14 Fr	Dislodges in obesity
Nephroureteral	8–12 Fr	Rarely dislodges
Malecot	16–30 Fr	Needs a stiffener

- Smaller tubes cause less pain and leak, with no more bleeding. Clamp before removal to test distal patency, and **exchange every 3 months**.

SAQ Tubeless PCNL

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Tubeless PCNL" · P4 SAQ "Tubeless PCNL"

Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

Selection

Suitable	Keep a nephrostomy
Complete clearance, single tract	Pyonephrosis or infected stone
Minimal bleeding	Significant bleeding
Intact collecting system	Significant perforation of (injury to) the collecting system
Supracostal access, bilateral cases (shown safe)	Residual stone needing a second look

Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent through the urethra, or a string.
3. Withdraw the sheath under vision; the **safety wire is removed last**.
4. Watch the tract; apply pressure (rib as backstop in thin patients). Fibrin glue in the tract did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

Evidence

- Meta-analyses of randomised trials: shorter stay and operative time, and less pain.
- The pain benefit is over large tubes (**>16 Fr**); there is **no benefit over small 8–9 Fr tubes**; small tubes (≤10 Fr) give similar comfort without more bleeding.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.
- Children: insufficient evidence.

Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

Current position

- Choose tubeless, small tube (≤10 Fr) or stent case by case; both stents and small tubes are acceptable (AUA, American Urological Association). Pairs well with mini-PCNL.

Diagram to draw: kidney with sealed tract and double-J stent.

Key points to remember:

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

5 · Complications

P2 *Complications of PCNL — immediate venous bleeding needs tamponade, bleeding at about 1 week is a pseudoaneurysm or arteriovenous fistula needing selective embolisation; pleural, colon and solid-organ injury*

- Major complications occur in about 10%; PCN transfusion is **0.5–4%**. Pleural injury needing intervention is about **5%** above the 12th rib and **25%** above the 11th.
- Colon injury occurs in about 0.1%; retrorenal colon is found in 10% prone versus 2% supine. Extraperitoneal injury is managed conservatively; peritonitis needs open repair.
- Air embolism shows as a sudden fall in end-tidal CO_2 and a mill-wheel murmur. Give 100% oxygen and position head down, right side up.

17 Principles of Urologic Endoscopy

1 · Evolution of the endoscope

LAQ Flexible endourological instruments, digital and current ureteroscopes

Paper answer · P4

Asked as:

- Briefly outline flexible endourological instrument and its role in management of urological disease — *Practice Set 1 Paper IV Q1 (LAQ)*
- Digital flexible ureteroscopes — *Summer 2019 Paper IV Q3c (SAQ)*
- Current ureter scopes. — *Summer 2022 Paper IV Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(flexible ureteroscopy in detail: see Practice Set 1 Paper 2, Q3b; digital scopes also in 2019-Summer Paper 4, Q3c)

Summer 2022 Q3d taken as "current ureteroscopes"; see also 2019-Summer Paper 4, Q3c.

Introduction

- A flexible endoscope has a **deflectable tip** and carries its image by a fibre bundle or a tip-mounted chip. It follows the natural curves of the urethra, ureter and collecting system.
- Physical basis: light travels along a curved glass fibre by **total internal reflection** (Tyndall).
- Marshall reported the first flexible URS (ureteroscopy) in 1964. It became therapeutic only when a **working channel** and **active tip deflection** were added in the early 1980s.
- **Fibreoptic flexible URS (ureteroscopy):** coherent glass-fibre bundle carries the image; fibres break → black spots; honeycomb (moiré) pattern.
- **Digital scopes** (first 2006): a sensor at the tip (**"chip-on-tip"**) converts light to an electrical signal, so the image does not degrade.

Classification of current ureteroscopes

Type	Key features
Rigid rod-lens	Larger calibre; largely historical
Semirigid	**Fibreoptic** image bundle; tip **≤7 Fr**; channel >3 Fr; often **two channels**; in-line or offset eyepiece
Flexible fibreoptic	Tip 4.5–7.5 Fr (smallest); image degrades as fibres break (black dots)
Flexible digital	"Chip-on-tip" CMOS (complementary metal-oxide semiconductor) or CCD (charge-coupled device) sensor plus LED (light-emitting diode); no camera head
Single-use flexible	Digital; LithoVue (2016) first; now several makers
Ureteral resectoscope	11.5 Fr; largely replaced by laser

Semirigid ureteroscope

- Uses **coherent fibre bundles** for the image (random fibres carry only light).
- **Two channels** are preferred: irrigation through one while an instrument is in the other, or a laser through one to break a stone held in a basket.
- **Offset eyepiece** gives a straight channel for rigid probes (pneumatic, biopsy).
- **Rule:** semirigid below the iliac vessels, flexible above.

Components

Component	Key features
Optics	**Fibreoptic:** a *coherent* image bundle carries the picture; a *random* light bundle carries illumination; cladding improves reflection. **Digital:** CCD (charge-coupled device) or CMOS (complementary metal oxide semiconductor) chip at the tip with LED (light-emitting diode) light; no camera head or light cable
Deflection	Lever → control wires through movable metal rings → wire fixed at the tip (deflection mechanism: lever pulls wires). "Intuitive" = tip moves with the lever. The reticle (notch) marks the plane of deflection
Working channel	Single channel shared by irrigation and instruments, so instruments reduce flow and deflection
Shaft	Tapered tip; passive (secondary) deflection segment behind the tip helps lower-pole access

- Flexible ureteroscope parts: optics, deflection mechanism (lever pulls wires), working channel.

Sensor technology (digital)

	CCD (charge-coupled device)	CMOS (complementary metal-oxide semiconductor)
Heat and power	More	Less (runs cooler)
Cost, speed	Higher, slower	Cheaper, faster
Status	Phased out	Current standard

- **LED (light-emitting diode)** at the tip; no camera head or light cable → better ergonomics.

Types

Instrument	Specifications	Main use
Flexible cystoscope	14–17 Fr; 37–40 cm; deflection 120–220°; channel 6.4–7.5 Fr; field about 120°	Office diagnosis and minor therapy
Flexible ureteroscope	Tip 4.5–7.5 Fr; deflection ≥270° both ways; channel **3.6 Fr**; dual-channel scope (2 × 3.3 Fr) exists	Kidney and upper ureter
Flexible nephroscope (or flexible cystoscope used through the tract)	Larger calibre; too wide for the UPJ (ureteropelvic junction)	Clearing calyces during PCNL (percutaneous nephrolithotomy)

Deflection and channel (flexible ureteroscope)

- Active deflection **≥270° both directions** (another source: **>270° both ways** in one plane); enhanced primary deflection eases lower-pole access / reaches the lower pole (ureteroinfundibular angle up to about 175°).
- Working channel **3.6 Fr**; dual-channel scopes exist.

Fibreoptic vs digital (digital versus fibreoptic)

Feature	Fibreoptic	Digital
Image	Honeycomb pattern; black dots as fibres break; degrades	Sharper, larger (magnified), stable image; no focusing or white balance
Size / Tip	Smallest tips, tapered	Larger (about 8.5 Fr), flatter; more access-sheath use
Outcome / Operative time	Reference; longer	Shorter operating time, equal (similar) stone-free rate (SFR)
Durability	Fragile; over-rotation damages fibres; repair about every 15 cases	**Not better** in later comparisons; not clearly better in trials
Processor	Platform-agnostic	Proprietary

- Surprise finding: in vitro, illumination was better with the fibreoptic cystoscope.
- Digital: sharper image, **shorter operating time, same stone-free rate**. But larger tip, so **more access sheaths and sheath complications**.
- Larger tip → more access-sheath use. Keep a fibreoptic scope for narrow UPJ (pelviureteric junction) or infundibulum.

Single-use scopes (single-use vs reusable)

- **LithoVue (2016):** first single-use digital ureteroscope; CMOS, LED, tip 7.7 Fr, shaft 9.5 Fr, 3.6 Fr channel, 270° each way. Single-use cystoscopes followed (2020).
- Rationale (reason): reusable ureteroscopes may need repair after only **3–5 cases**; reprocessing and repair add cost.
- Single-use advantages: maximal deflection each case, sterile, no reprocessing or repair; cost **lower below about 99 cases/year**.
- Best value in low-volume centres and for **large lower-pole stones**.
- Reusable: cheaper in high-volume centres; needs reprocessing (peracetic acid, glutaraldehyde, ethylene oxide).
- Cost advantage depends on case volume; environmental impact is similar to reusables. Most centres use a **hybrid fleet**.

Accessories

- **Guidewires:** 0.035/0.038 inch PTFE (polytetrafluoroethylene)-coated safety wire; hydrophilic wires to pass impacted stones (exchange, as too slippery for safety); nitinol-core and hybrid wires.
- **Access sheath** (11–16 Fr): repeated passes, lower intrarenal pressure; small risk of ureteric injury.
- **Laser fibres** 200–365 µm; **nitinol tipless baskets** ≤3 Fr; three-prong graspers; 3 Fr cup biopsy forceps; 3 Fr balloon dilating to 12 Fr; fulguration electrodes; pressure irrigation pumps.

- Flexible EHL (electrohydraulic lithotripsy) probes work; flexible pneumatic probes restrict deflection.

Role in urological disease

Area	Role
Lower tract	Haematuria work-up; bladder cancer surveillance, office biopsy and fulguration; NBI (narrow-band imaging) and blue light; stent removal or exchange; bladder stones, foreign bodies, strictures; botulinum toxin injection; diversions through a catheterisable channel; difficult ureteric access via a suprapubic tract
Stones	RIRS (retrograde intrarenal surgery): renal stones <20 mm, lower pole <10 mm; >2 cm when PCNL is unsafe
UTUC (upper tract urothelial carcinoma)	Diagnostic URS and biopsy (predicts grade in about 9 of 10); laser ablation of low-grade tumours; prefer to percutaneous route (seeding)
Obstruction	Endopyelotomy without crossing vessels; endoureterotomy for strictures <1.5 cm
Diagnosis	Positive cytology, filling defects, lateralising haematuria (haemangioma fulgurated), recurrent one-sided infection; foreign body retrieval
PCNL	Antegrade or retrograde flexible endoscopy raises stone-free rate; ECIRS (endoscopic combined intrarenal surgery)

Care and complications

- **Scope damage:** deflection wear; laser fibre scraping and perforating the channel. Pass fibres with the scope straight; **fire laser only when the fibre tip is seen touching the stone** (fire only with the fibre tip seen on the stone); leave the bladder empty to avoid buckling.
- **Sterilisation:** sterilise with ethylene oxide, glutaraldehyde or peracetic acid.
- **Patient:** ureteric injury, stricture, sepsis from high pressure. Use saline irrigation and antibiotic prophylaxis for all upper tract procedures.

Recent advances

- **Intrarenal pressure sensor** in the scope (LithoVue Elite): high pressure causes fever, SIRS (systemic inflammatory response syndrome), sepsis. Temperature sensing next.
- **AI (artificial intelligence):** stone recognition, laser setting adjustment, navigation.
- **Robotic flexible URS (ureteroscopy):** ergonomics, less radiation to surgeon.
- **Also know:** suction (access) sheaths, further miniaturisation.

Diagram to draw: cross-section of a flexible ureteroscope: chip, LED, working channel and deflection wires (scope tip cross-section showing CMOS chip, LEDs, 3.6 Fr channel and 270° deflection).

Key points to remember:

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.
- Pressure sensors and AI are the newest developments.
- Golden rule: never fire unless the fibre tip is seen on the stone.

SAQ Simulators in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3d

Rationale

- Fewer training cases and complex technology mean skills must be learnt **outside the patient**.
- Chapter 1: simulation is used to **acquire and maintain skills and to warm up** before operating.

Types

Type	Examples	Strengths and limits
Bench / box trainer	Peg transfer, cutting, suturing; chicken oesophagus anastomosis; 3D (three-dimensional)-printed perfused hydrogel kidney	Cheap, proven skill gains; limited realism
VR (virtual reality)	LapSim, LapMentor, RobotiX Mentor, SimNow (robotic prostatectomy); endourology trainers for cystoscopy, ureteroscopy, PCNL (percutaneous nephrolithotomy), TURP (transurethral resection of the prostate)	Records metrics (precision, economy of motion), builds curricula, sets expert benchmarks; costly; poor tissue feel
Live animal	Pig	Real bleeding; ethics, cost
Cadaver	Fresh-frozen	Real human anatomy; scarce
Team (non-technical)	High-fidelity theatre crises	Communication and teamwork

Validity (needed before a simulator is trusted)

- **Face:** looks real. **Content:** teaches the right skills. **Construct:** experts score better than novices. **Concurrent:** agrees with an established test. **Predictive:** simulator score forecasts operating performance.

Evidence (Chapter 18, Chapter 12)

- VR training **beats no training or box training** for novices and improves live performance and operating time.
- Robotic simulator scores correlate with live RARP (robot-assisted radical prostatectomy) performance.

- **FRS (Fundamentals of Robotic Surgery):** a multi-institutional RCT (randomised controlled trial) showed better speed and accuracy.
- Ureteroscopy team simulation taught non-technical skills better than lectures (randomised trial).

Structured curricula and assessment

- **EAU (European Association of Urology) E-BLUS** (European Basic Laparoscopic Urological Skills): online theory, then dry-model tasks, then examination.
- **Proficiency-based progression:** move on only after reaching expert benchmarks.
- Scoring: **GEARS (Global Evaluative Assessment of Robotic Skills)** and **OSATS (Objective Structured Assessment of Technical Skills)**.

Limitations

- Cost, poor tissue feel, limited proof of better patient outcomes.

Diagram to draw: pyramid: box trainer at the base, then VR, then animal or cadaver, then supervised surgery at the top.

> **Key points to remember:**
> - Bench, VR, animal, cadaver and team simulation.
> - Validity: face, content, construct, concurrent, predictive.
> - VR beats box or no training for novices.
> - E-BLUS and FRS; proficiency-based progression; GEARS.

Also, from the condensed book (not in the paper answer):

- **Desormeaux (1853)** excised a urethral papilloma, the **first therapeutic endoscopy**. Nitze (1877) placed the light source at the tip.

2 · Optics and enhanced imaging

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

*See the paper answer **CA bladder, etiology, newer investigations and treatment.** (Section 12, Ch 129).*

Also, from the condensed book (not in the paper answer):

- Narrow-band imaging (NBI) needs **no instillation**.
- Blue-light cystoscopy (Food and Drug Administration approved 2010): hexaminolevulinate is instilled **1 h** before, and blue light of 360–450 nm makes tumour fluoresce red while normal urothelium stays blue.

3 · Irrigation fluids and TUR syndrome

SAQ Irrigation fluids for TURP.

Paper answer · P2

Type: SAQ (10 marks)
Source: 2017-Summer Paper 2, Q3a

Ideal irrigant

- Transparent, **non-conducting** (monopolar), isotonic, non-toxic, non-haemolytic, cheap.
- **Why non-conducting?** In monopolar TURP (transurethral resection of the prostate), current flows loop → patient → pad; ionic saline would disperse it. **Bipolar** TURP completes the circuit in the instrument, so it uses **normal saline**.

Fluids

Fluid	Osmolality (approx.)	Points
Sterile water	0	Best vision, cheap; haemolysis, hyponatraemia; now avoided for TURP
Glycine 1.5%	~200 mOsm/kg	Commonest (UK, North America); glycine → ammonia, serine; transient blindness (retinal inhibitory neurotransmitter), encephalopathy
Sorbitol 3%	~165	Metabolised to fructose and glucose → hyperglycaemia; lactic acidosis
Mannitol 5%	~275	Near iso-osmolar; not metabolised; osmotic diuresis; volume overload in renal failure
Sorbitol 2.7% + mannitol 0.54%	~180	Continental Europe
Normal saline 0.9%	~308	Bipolar TURP, lasers (HoLEP, PVP); no TUR syndrome
- Plasma is about 290 mOsm/kg, so monopolar fluids are **hypo-osmolar**.		

TUR (transurethral resection) syndrome

- **Mechanism**: prostatic venous pressure is ~10 mmHg; irrigant at higher pressure enters open sinuses → volume overload, **dilutional hyponatraemia** and hypo-osmolality → cerebral oedema; plus glycine toxicity.
- **Features**: confusion, nausea, vomiting, visual disturbance, hypertension then hypotension, bradycardia, seizures, coma.
- **Risk factors**: gland >45 mL, resection >90 minutes (VA (Veterans Affairs) study), high irrigation pressure, open sinuses or capsular perforation.
- **Prevention**: bag height ~60 cm (each extra 10 cm more than doubles absorption), continuous-flow resectoscope, time limit, spinal anaesthesia (awake patient), bipolar TURP. Ethanol 1% tagging can monitor absorption.
- **Treatment**: stop surgery, check sodium; Na <120 mEq/L serious. Furosemide for overload; **3% saline** 100 mL boluses (max 300 mL) for neurological signs; raise sodium by 4–6 mEq/L initially, not over 12 mEq/L in 24 h (risk of central pontine myelinolysis).

Bipolar advantage

- Meta-analyses: no TUR syndrome, less transfusion and clot retention; NNT (number needed to treat) about 50 to prevent one TUR syndrome.

Diagram to draw: monopolar vs bipolar current pathways.

Key points to remember:

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

*See the paper answer **TUR syndrome** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- **Ureteroscopy always uses normal saline**, so that no hypotonic fluid is absorbed. Saline's only hazard is volume overload, without hyponatraemia.
- Serum sodium is usually **<125 mmol/L** in transurethral resection (TUR) syndrome. This chapter advises a correction of no more than 8–10 mmol/L in 24 h (the paper answer gives ≤12).

4 · Cystourethroscopy

SAQ Indications for outpatient cystourethroscopy.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3e

Introduction

- **Flexible cystourethroscopy** under local anaesthesia is the **commonest office procedure** in urology; no lithotomy needed.
- **Prerequisites:** consent; treat any UTI (urinary tract infection) first; lidocaine gel with **≥5 minutes dwell** (ideally 10–20); no antibiotic prophylaxis unless risk factors (age, diabetes, immunosuppression, catheters, prior UTI).

Diagnostic indications

Indication	Comment
Haematuria	All **gross haematuria** (with CT (computed tomography) urography, cytology). Microhaematuria by AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) risk: intermediate and high risk; shared decision in low risk
Bladder cancer surveillance	NMIBC (non-muscle-invasive bladder cancer): first check at 3 months; low risk then yearly; high risk 3–4-monthly for 2 years; also upper-tract follow-up of UTUC (upper tract urothelial carcinoma)
Recurrent UTI	Look for stone, diverticulum, stricture, fistula, foreign body
LUTS (lower urinary tract symptoms)	Not routine; before invasive BPH (benign prostatic hyperplasia) treatment (median lobe before urethral lift, bladder-neck contracture, stricture)
Pain syndromes	IC/BPS (interstitial cystitis/bladder pain syndrome): **Hunner lesion**; chronic pelvic pain with bladder pain or low flow
Women	Suspected urethral diverticulum, vesicovaginal fistula, mesh or suture erosion after sling, recurrent UTI after surgery

Indication	Comment
Others	Urethral stricture assessment and follow-up; abnormal urine cytology; radiation cystitis; foreign body

Therapeutic indications

- **Removal or exchange of ureteric stents** (exchange with fluoroscopy).
- **Biopsy and fulguration** of small (≤1 cm) low-grade recurrences; office laser ablation.
- **Intradetrusor botulinum toxin** injection (OAB (overactive bladder), neurogenic detrusor overactivity).
- Urethral bulking agents; stricture dilatation; small stones or foreign bodies; blue-light cystoscopy where available.

When not indicated or done in theatre

- Not needed for the index stress incontinence patient, initial OAB or uncomplicated LUTS.
- **No surveillance cystoscopy** in chronic indwelling catheters (AUA/SUFU 2021).
- Theatre if TURBT (transurethral resection of bladder tumour) needed, poor tolerance, children, or complex diversions.

Diagram to draw: haematuria risk-group flowchart.

> **Key points to remember:**
> - Haematuria and NMIBC surveillance are the two commonest indications.
> - Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
> - Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

Rigid instruments

- French (Charrière) gauge: **3 Fr = 1 mm** diameter. The diagnostic rigid sheath is 15–17 Fr. Choose the smallest scope that does the job.
- A rigid set has a lens, bridge, sheath and obturator. Use a 0° or 12° lens for the urethra, 25° or 30° for general work, and 70° or 120° for the dome, anterior wall and bladder neck.
- The **Albarran bridge** has a lever that deflects catheters and wires into an orifice behind a high bladder neck, a large median lobe or a reimplanted ureter.

Antibiotic prophylaxis: AUA versus EAU

Prophylaxis is driven by patient risk, not by the procedure.

	American Urological Association (AUA)	European Association of Urology (EAU)
Diagnostic cystoscopy	None unless patient risk factors are present	
TURP	Yes (<24 h)	Yes
TURBT	Yes	**Only if sepsis risk is high**
Ureteroscopy	**All patients**	According to local pathogens

	American Urological Association (AUA)	European Association of Urology (EAU)
Agents	Cefazolin or trimethoprim-sulfamethoxazole (ureteroscopy: TMP-SMX or a 1st/2nd-generation cephalosporin)	Local antibiogram
Added risk factors	Smoking, steroids, distant infection, prolonged stay, anatomical anomaly	Diabetes, weight extremes, prior urinary infection, previous instrumentation

Both share older age, malnutrition, immunodeficiency and catheters or colonised material. Treat any urinary tract infection before instrumentation.

Technique and special situations

- Prepare the genitalia with **aqueous povidone-iodine** at any age, because chlorhexidine and alcohol damage mucosa.
- **Orthotopic neobladder**: do not dilate or incise an anastomotic contracture without obstruction, as this risks incontinence. **Cutaneous pouch**: use a flexible scope through the channel for diagnosis, and treat percutaneously after imaging.

5 · Upper-tract endoscopy: indications

P2 *ESWL: mechanism, indications, factors affecting outcome — AUA/EAU stone size thresholds by site, unfavourable stone and anatomical factors*

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- The EAU prefers ureteroscopy (URS) for ureteric stones **>10 mm**.
- Diagnostic URS for upper tract urothelial carcinoma does not worsen survival but **raises intravesical recurrence**, so survey the bladder.
- Endoureterotomy also fails more with an allograft, an irradiated or ureteroenteric stricture, or a poorly functioning kidney. Balloon dilation alone is about 50% durable at 1 year.

6 · Ureteroscopes

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

*See the paper answer **Flexible ureteroscopy / RIRS and accessories** (Section 11, Ch 97).*

Also, from the condensed book (not in the paper answer):

- **Passive (secondary) deflection** off the upper pelvic wall lengthens the effective tip, but it **fails in gross hydronephrosis**, where there is no wall to push against.

7 · Accessories

*See the paper answer **Foley's (urethral) catheters, guidewires and DJ stents with their complications** (Section 1, Ch 15).*

LAQ LASERs in urology: holmium, thulium, KTP and newer laser technologies

Paper answer · P4 · merged from 4

Asked as: P4 LAQ "LASERs in urology: applications and newer generation laser technologies (new LASER for stone treatment)" · P4 SAQ "Holmium laser" · P4 SAQ "Thalium laser in urology" · P4 SAQ "KTP*** laser in urology"

*"Thalium" is taken to mean thulium (spelling error). "KTP***" is taken to mean the KTP (potassium titanyl phosphate) "GreenLight" laser; the asterisks are ignored.*

Principle

- LASER (light amplification by stimulated emission of radiation): a "pumped" gain medium releases identical photons. A mirror resonator amplifies them, and a partial mirror lets the beam out.
- Light is **monochromatic, coherent and collimated**, so it passes down thin fibres.
- **Chromophore** decides the effect. Water absorbs holmium and thulium light. Haemoglobin absorbs KTP light.
- Absorption sets **penetration**. Shallow penetration vaporises and cuts. Deep penetration coagulates and later sloughs.
- Coagulation occurs at about 60 °C; vaporisation at ≥100 °C.
- **Pulse duration** decides the stone mechanism. Short pulses (<10 µs) act photomechanically (shock waves). Long pulses (>250 µs: Ho:YAG [holmium:yttrium-aluminium-garnet], TFL [thulium fibre laser]) act **photothermally**.

Lasers used

Laser	Wavelength (nm)	Chromophore	Penetration	Main uses
Ho:YAG	2120 (Ch 19 quotes 2100; often quoted as 2100 nm)	Water	0.3–0.4 mm (other sources: 0.4 mm; 0.31 mm)	Stones, HoLEP, incisions, tumours
TFL	1940	Water (peak)	<0.1 mm (other sources: about 0.2 mm; 0.08 mm; chapter values 0.08–0.2 mm)	Dusting, enucleation
Tm:YAG	~2010 (other sources: 2013)	Water	0.2 mm	Vaporesection, enucleation
KTP/LBO	532	Haemoglobin	0.8 mm	PVP
Diode	940–1470	Water + haemoglobin	0.5–5 mm	Prostate vaporisation
Nd:YAG	1064	Neither strongly	Up to 10 mm	Coagulation of tumours
CO_2	10,600	Water	~0.1 mm	Skin and penile lesions

Holmium:YAG laser

Physics

- Solid-state, flashlamp-pumped, **pulsed** laser at **2120 nm** (often quoted as 2100 nm).

- Chromophore **water**; penetration about **0.4 mm**. It cuts and vaporises; heat spreads 2–3 mm and seals small vessels.
- Flexible quartz fibres; works in saline.

Mechanism on stones

- Long pulses (>250 µs) act **photothermally**, not by shock waves: long pulses heat the stone until it melts or decomposes.
- **Moses effect:** the start of each pulse makes a vapour channel; the rest of the pulse crosses it to the stone. So keep the fibre close and perpendicular.
- Breaks **all stone types**, including cystine and calcium oxalate monohydrate. It is the gold standard for stones and enucleation.

Settings

Mode	Settings	Effect
Fragmenting	0.6–1.0 J × 6–10 Hz	Pieces for basketing; more retropulsion
Dusting	0.2–0.3 J × 40–80 Hz	Fine dust; "popcorning" in a calyx
HoLEP enucleation	2.0 J × 40–50 Hz	Cutting along the capsule

- **Retropulsion** rises with pulse energy (not frequency) and falls with longer pulses or pulse modulation.
- Heat tracks power: ≤10 W in the ureter; good outflow.

Uses

- **Stones:** URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL (percutaneous nephrolithotomy), bladder stones.
- **BPH (benign prostatic hyperplasia):** HoLEP (holmium laser enucleation of the prostate).
- **Incisions:** strictures, bladder neck contracture, endopyelotomy, ureterocele, calyceal diverticulum neck.
- **Tumours:** UTUC (upper tract urothelial carcinoma) ablation; en bloc bladder tumour resection.
- Drawback: steep HoLEP learning curve.

Thulium lasers

Types and physics

	Tm:YAG (thulium:yttrium-aluminium-garnet)	TFL (thulium fibre laser)
Wavelength	2013 nm (P4 LAQ: ~2010 nm)	**1940 nm** (water absorption peak)
Source	Diode-pumped solid crystal	Diode-pumped thulium-doped silica fibre
Output	Continuous wave (now also pulsed)	Pulsed or continuous
Water absorption	High	About 4× holmium

- Optical penetration is the shallowest of urological lasers: about 0.2 mm (chapter values for TFL: 0.08–0.2 mm), against 0.31–0.4 mm for Ho:YAG.
- Shallow absorption gives precise vaporisation and clean incision, but **carbonisation**.

Thulium:YAG (2013 nm)

- Originally continuous wave (0.2 mm penetration, carbonisation a drawback): ThuVARP (vaporesection), ThuVAP (vaporisation), ThuVEP (vapoenucleation), **ThuLEP** (laser enucleation, 2010; mostly blunt dissection; size-independent like HoLEP).
- **Pulsed Tm:YAG:** newer pulsed solid-state version of the continuous-wave thulium laser for photothermal lithotripsy and enucleation.

TFL (1940 nm) for stones

- **Diode-pumped silica fibre** laser near the **water absorption peak** (about **4× holmium absorption**).
- **Low peak power, long pulses, very high frequency** (up to about 2000 Hz [P4 LAQ: above 2000 Hz]; holmium about 120 Hz); pulse energy as low as 0.025 J. This gives photothermal fine "dusting" and **minimal retropulsion**.
- **Thin fibres** (50–150 µm; P4 LAQ: down to about 150 µm) give better irrigation and scope deflection. Small, quiet, compact unit; no dedicated high-voltage outlet.
- Settings: start 0.6 J × 6 Hz; dusting 0.2 J × 40 Hz; avoid >40 Hz (pulses miss the stone); settings ≤40 Hz are often enough. "Paint, do not drill."
- **RCTs (randomised controlled trials):** TFL beat **low-power (30 W) holmium** (stone-free rate, time); it **equalled 120 W pulse-modulated holmium** in stone-free rate, with less retropulsion.
- Concerns: efficacy on all stone types; thermal injury at high power.

Thulium for prostate

- Techniques: ThuVARP, ThuVAP, ThuVEP, ThuLEP, ThuFLEP (thulium fibre laser enucleation).
- Settings: 70–90 W enucleation, 30 W coagulation; morcellation as for HoLEP.
- Outcomes: ThuLEP equals HoLEP; size-independent; safe on anticoagulants (AUA [American Urological Association]). Retrograde ejaculation about four-fifths.

Other thulium uses: en bloc bladder tumour resection; UTUC ablation; incision of strictures.

KTP/LBO (GreenLight) laser

Physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam passes through a **KTP or LBO (lithium triborate) crystal**. This doubles frequency and halves wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water. Optical penetration 0.8 mm; tissue effect 1–3 mm.
- Effect: **photoselective vaporisation** with a thin coagulation rim, so vaporisation is bloodless.

Platforms

Generation	Power
KTP	80 W
LBO HPS (High Performance System)	120 W
LBO XPS	**180 W**; 750 µm MoXy fibre (metal tip, fluid cooling)

- Side-firing fibre emits at 70°; 23–24 Fr cystoscope; saline irrigation.

PVP (photoselective vaporisation of prostate) technique

- Identify ureteric orifices; open the bladder neck first.
- Sweep 15–30° in 1–2 seconds at ≤2 mm working distance. Large bubbles mean efficient vaporisation.
- Pulsatile bleeding: coagulation mode (about 30 W). Energy >5 kJ/cc predicts complete vaporisation.

Indications

- AUA: prostates <30 cc and 30–80 cc; men at **higher bleeding risk** (anticoagulation continued). Not >80 cc (high retreatment).

GOLIATH trial

- RCT, 29 European centres, about 270 men, prostate ≤100 cc: **180 W PVP vs TURP** (transurethral resection of prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications at 6 months, 1 and 2 years.
- PVP: shorter catheter time and stay, fewer early reinterventions. TURP: faster. Retrograde ejaculation (about two-thirds) and 2-year reoperation equal.

Complications

- Dysuria and storage symptoms (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%; retrograde ejaculation.
- No TUR (transurethral resection) syndrome; transfusion very rare.
- **No tissue** for histology.

Other KTP uses: laparoscopic non-contact cutting (largely supplanted); KTP enucleation possible.

Applications by area

1. Stones

- **Ho:YAG** is the gold standard and breaks all stone types (see above). **TFL:** higher water absorption and low peak power give finer dust and less retropulsion.
- **Heat** depends on power, not laser type. Keep ≤10 W in the ureter; use good flow and an access sheath.

2. BPH

- **HoLEP:** endoscopic equivalent of open prostatectomy, **size-independent**. Enucleate at about 2 J × 40–50 Hz, then morcellate. The only endoscopic procedure superior to TURP. Less bleeding, shorter catheter time; safe on anticoagulants.
- **PVP (KTP/LBO 120–180 W):** haemoglobin absorption gives bloodless vaporisation, so anticoagulants can continue. **GOLIATH:** non-inferior to TURP. Avoid >80 cc (retreatment).
- **Thulium:** vaporisation, vaporesection, ThuLEP (equivalent to HoLEP). Diode vaporisation also used.
- Nd:YAG coagulation was abandoned (1 cm penetration, sloughing).
- AUA: HoLEP, ThuLEP or PVP for **high bleeding risk**.

3. Incisions

- Laser urethrotomy has no advantage over cold knife.

- Holmium incision of bladder neck contracture, ureteric stricture, endopyelotomy, ureterocele and calyceal diverticulum neck.

4. Urothelial tumours

- **NMIBC (non-muscle-invasive bladder cancer):** laser **en bloc resection** (Ho, Tm, KTP) gives detrusor in the specimen and no obturator jerk. Office ablation of small low-grade recurrences.
- **UTUC:** biopsy, then ablate by ureteroscopy:
- Ho:YAG: <0.5 mm, haemostatic.
- Nd:YAG: 5–6 mm coagulation, so stricture risk.
- Thulium: <0.2 mm.
- **PDT (photodynamic therapy):** a photosensitiser plus laser light makes singlet oxygen.

5. Penile and genital lesions

- PeIN (penile intraepithelial neoplasia) and small Ta/T1: **Nd:YAG preferred**, with base biopsies.
- CO_2 is too superficial for T1 (recurrence up to 50%). CO_2 or Nd:YAG for genital warts.

Newer generation laser technologies (new lasers for stone treatment)

- **Aims:** less retropulsion, finer dust, faster work.
- **1. High-power holmium (60–120 W; other sources: 100–120 W machines) with pulse modulation:** high frequency allows **dusting** (0.2–0.3 J × 40–80 Hz) and "popcorning"; long-pulse modes reduce retropulsion.
- **2. Pulse modulation (Moses technology and similar):** the first part of a pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the stone. Moses technology splits each pulse in two: the first opens the channel, the second strikes the stone. Less retropulsion; may shorten HoLEP.
- **3. TFL (1940 nm):** see above; ThuFLEP for prostate.
- **4. Pulsed Tm:YAG (2013 nm):** see above; it also does lithotripsy.
- **Others:** thin fibres; suction sheaths; AI (artificial intelligence) settings; 180 W GreenLight KTP/LBO (532 nm) vaporisation; diode lasers; burst wave lithotripsy (investigational).

Holmium versus TFL

	Ho:YAG	TFL
Wavelength	2120 nm	1940 nm
Water absorption	High	About 4× higher
Penetration	0.31 mm	0.08 mm
Peak power	High	Low
Retropulsion	More	Minimal
Fragments	Small	Smaller
Fibre	≥200 µm	≥150 µm

Principles of settings

- **Retropulsion rises with pulse energy**, not frequency; it falls with longer pulses.
- Keep the fibre close and perpendicular to the stone.
- TFL settings ≤40 Hz often enough.

Safety

- Wavelength-specific eyewear for all, including the patient. Door signage, covered windows, trained staff and a laser safety officer.
- Check fibres; never fire inside the scope. Guard against fire and thermal (urothelial) injury.
- **Heat follows total power, not laser type:** ≤10 W in the ureter, ≤25 W in the kidney; use low power, pauses, high flow, cool irrigation and an **access sheath**; urothelial damage begins above about 43 °C.

Diagrams to draw

- Water and haemoglobin absorption curves marking 532, 1064, 1940, 2013 and 2120 nm.
- Vapour bubble (Moses channel) between fibre tip and stone.
- Side-firing fibre at 70° vaporising lateral lobe.

> **Key points to remember:**
>
> - Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
> - Holmium: 2120 nm; water chromophore; pulsed; photothermal; Moses effect: fibre close and perpendicular.
> - Holmium settings: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
> - HoLEP size-independent, superior to TURP.
> - TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
> - TFL: low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
> - TFL beats low-power, equals high-power pulse-modulated holmium.
> - ThuLEP is size-independent and equal to HoLEP.
> - KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
> - GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
> - Nd:YAG for PeIN; eyewear for everyone.
> - Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

LAQ Energy sources in urology and laparoscopic surgery, including bipolar energy

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P3 LAQ "Therapeutic energy sources in urology (including laparoscopic energy devices)" · P4 LAQ "Basic energy modalities in urological surgery, including bipolar energy" · P2 SAQ "Energy sources in laparoscopic urology"

Introduction and principles

- Therapeutic energy is absorbed by tissue or stone and released as **heat, mechanical stress, cavitation or a chemical reaction**. Wavelength or frequency sets penetration; absorption sets the effect.
- All haemostatic devices achieve haemostasis (seal) by **denaturing protein**, by heat (current, light) or mechanical vibration.
- Heat effects: about **60 °C** coagulates protein; **100 °C** vaporises cell water; higher temperatures carbonise.

Classification

Group	Modalities
Electrical	Monopolar and bipolar electrosurgery, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration)
Electromagnetic	Lasers, MWA (microwave ablation), chemohyperthermia, PDT (photodynamic therapy), ionising radiation
Acoustic	Ultrasonic dissector, SWL (shock wave lithotripsy), intracorporeal ultrasonic and pneumatic lithotripsy, HIFU (high-intensity focused ultrasound), histotripsy, burst wave lithotripsy, Li-ESWT (low-intensity shock wave therapy)
Direct contact	Cryoablation, water vapour (Rezum), waterjet (Aquablation)

1. Electrosurgery: physics

- Alternating current at about **500 kHz** (radiofrequency). Mains current (50 Hz) depolarises nerve and muscle; at 500 kHz ions only oscillate and produce heat, too fast to stimulate nerve or muscle.
- **Ohm's law** ($V = I \times R$): tissue is the resistance. **Current density** decides the effect: a small tip concentrates current and heats (burns); a large pad disperses it.
- **Monopolar:** generator, active electrode, patient, dispersive pad, back to generator. Partial pad contact causes burns.
- **Bipolar:** current passes only between two electrodes (jaws) on the same instrument; less power, less spread, no dispersive pad (the patient is not in the circuit).
- **Electrocautery** (heated wire) differs: no current enters the patient.

Waveform	Character	Effect
Cut (50–80 W)	Continuous, low voltage	Cell water flash-boils; rapid boiling, clean incision, little char
Coagulation (30–50 W)	Interrupted (about 6% duty cycle), high voltage	Desiccation, deeper spread (up to 3 mm)
Fulguration (spray)	Higher voltage, arcs (sparks) across air	Non-contact surface haemostasis; surface charring of oozing beds
Blend	Cut with interruptions	Cutting plus haemostasis

Bipolar energy in detail

- Radiofrequency current (about 500 kHz). Less power and less lateral thermal spread (1–6 mm vs 5–7 mm monopolar). In endoscopy, **saline** irrigation can be used.

	Monopolar	Bipolar
Circuit	Loop to patient to pad	Active and return poles on the instrument
Irrigant	Glycine 1.5%, sorbitol–mannitol	Normal saline
TUR (transurethral resection) syndrome	Yes	Theoretically eliminated
Capacitive coupling, pad burns	Yes	No

- **Endoscopic uses:**

- **Bipolar TURP (transurethral resection of prostate):**
  - *Plasma system:* energy vaporises saline around the loop into plasma (orange glow of sodium); cuts at lower temperature while sealing vessels; platinum–iridium loop.
  - *TURis (transurethral resection in saline):* the sheath acts as return electrode; chapter notes report strictures and fistulae.
  - *True bipolar:* dual loops.
  - Outcomes: equal IPSS (International Prostate Symptom Score) and Qmax; less transfusion and clot retention; shorter catheter time and stay. NNT (number needed to treat) about 50 to prevent one TUR syndrome.
- **Bipolar TUVP (transurethral vaporisation):** PlasmaButton; AUA (American Urological Association) option for <30 and 30–80 cc.
- **Bipolar enucleation:** plasmakinetic energy can perform AEEP (anatomic endoscopic enucleation of prostate) satisfactorily; comparative data limited.
- **TURBT (transurethral resection of bladder tumour):** fewer obturator jerks and less bleeding in reviews; one RCT (randomised controlled trial) showed only less thermal artefact.
- **Laparoscopic and open uses:** bipolar forceps for pinpoint haemostasis; vessel sealers; prostatic pedicles, cystectomy.
- **Bipolar safety:** safer near pacemakers; touching tines complete the circuit without tissue. Obturator reflex reduced, not abolished.

Electrosurgery in endoscopic urology: present status

- **TURP:** still the reference BPH (benign prostatic hyperplasia) operation. Monopolar TURP needs non-conducting **1.5% glycine**, risking hyponatraemia (TUR syndrome); **bipolar** in saline matches monopolar efficacy with better safety and avoids TUR syndrome.
- **TURBT:** standard; obturator reflex risk on lateral wall. En bloc resection is increasing.
- **Obturator reflex:** lateral-wall TURBT stimulates the nerve, causing adduction and perforation. Prevent with paralysis, obturator block, partial filling, short bursts, bipolar.
- **EMDA–MMC (mitomycin C):** iontophoresis raises drug uptake in NMIBC (non-muscle-invasive bladder cancer); evidence low quality, not standard.

Laparoscopic energy devices

Device	Principle	Lateral spread	Notes
Monopolar (hook, scissors)	Tip to patient to return pad	**5–7 mm**	Versatile; stray-current risk
Bipolar	Current only between jaws	1–6 mm	No pad; no capacitive coupling
Vessel sealer (LigaSure, EnSeal, Caiman)	Bipolar current + jaw compression fuses collagen and elastin; impedance feedback stops output; built-in blade	1–3 mm	Seals vessels **≤7 mm** in one application only
Ultrasonic (Harmonic)	Blade vibrating at **25–55 kHz** (P4: about 55.5 kHz); friction and cavitation denature protein; no current through the patient	**≈1 mm**	Safe with pacemakers; jaws **>200 °C**, slow to cool, so do not touch bowel

Device	Principle	Lateral spread	Notes
Argon beam	Monopolar current through argon, non-contact	Superficial	Liver, spleen, raw surfaces; **gas embolism**

- Vessel sealers are routine in laparoscopic and robotic surgery.
- **Also know:** lasers (CO_2, KTP [potassium titanyl phosphate], holmium) now rarely used laparoscopically; laparoscopic RFA (target 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles) for small renal masses; robotic HotShears, bipolar Maryland, Vessel Sealer.

Electrosurgical injury and safety

Mechanism	Prevention
Direct (inadvertent) activation out of view	Activate only with tip in view
Direct coupling: active electrode touches another metal instrument	Whole metal tip visible
Capacitive coupling: current induced through intact insulation (metal cannula in plastic retainer)	Never mix metal and plastic ports (avoid hybrid trocars)
Insulation failure	Inspect insulation; **AEM (active electrode monitoring)**
Thermal spread to ureter, nerves or bowel	Bipolar near vital structures
Return pad burns; surgical fire (with oxygen)	Pad on muscle, away from implants; **REM (return electrode monitoring)** detects poor pad contact

- **AEM** detects insulation failure and capacitive coupling.
- Unseen bowel injuries present late (days) as perforation or peritonitis.
- Use lowest effective power (cut 50–80 W, coagulation 30–50 W), tip in view; bipolar near vital structures and in pacemaker patients.

2. Ultrasonic energy

- **Ultrasonic dissector:** see the device table; little lateral spread, but the blade stays hot.
- **Intracorporeal ultrasonic lithotripter:** vibrating probe with suction of fragments.

3. Lasers

- **Principle:** pumped gain medium in a mirrored cavity gives stimulated emission and a **monochromatic, coherent, collimated** beam.
- **Effect** depends on wavelength (chromophore: water, haemoglobin) and pulse duration.
- **Mechanisms:** photothermal (long pulses), photomechanical (short pulses, shock waves), photochemical (PDT).
- **Penetration predicts behaviour:** deep (Nd:YAG) coagulates widely with delayed sloughing; shallow (thulium) vaporises precisely with char.

Laser	nm	Penetration	Use and present status
KTP/LBO (potassium-titanyl-phosphate/lithium triborate; GreenLight)	532	0.8 mm	Prostate photovaporisation (haemoglobin); can continue anticoagulation

Laser	nm	Penetration	Use and present status
Nd:YAG (neodymium:yttrium-aluminium-garnet)	1064	10 mm	Deep coagulation (penile CIS [carcinoma in situ], warts); largely obsolete for BPH
TFL (thulium fibre laser)	1940	0.08 mm	Stones: beats low-power holmium, equals high-power; less retropulsion; enucleation (ThuLEP, thulium laser enucleation of prostate)
Tm:YAG (thulium:YAG)	2000	0.2 mm	Vaporesection, enucleation
Ho:YAG (holmium:YAG)	2120	0.31 mm	Gold standard lithotripsy; **HoLEP** (holmium laser enucleation of prostate) size-independent; strictures, UTUC (upper tract urothelial carcinoma) ablation

- Safety: eye protection for all; irrigant heating; fibre damage.

4. Acoustic energy and lithotripsy

- **SWL:** electrohydraulic, electromagnetic or piezoelectric sources. Shock waves break brittle stones by **spallation, shear and cavitation**. Slower rates (about 1 Hz) work better and injure less. **Status:** first line for renal stones ≤2 cm and favourable upper ureteric stones.
- **Intracorporeal:** pneumatic (ballistic), ultrasonic, combined, electrohydraulic, laser; pneumatic, ultrasonic and combined probes in PCNL (percutaneous nephrolithotomy); electrohydraulic now rare (ureteric injury).
- **HIFU:** focused beam (1500–10,000 W/cm^2) is absorbed and heats a rice-grain focus to necrosis; transrectal prostate. **Status:** focal or salvage prostate cancer (best for posterior tumours); investigational in guidelines.
- **Histotripsy** (cavitation liquefaction), **burst wave lithotripsy, ultrasonic propulsion:** investigational.
- **Li-ESWT:** for vasculogenic ED (erectile dysfunction) and Peyronie pain; evidence still limited.

5. Ablative energies (thermal ablation of tumours)

Modality	Mechanism	Key points and present status
Cryoablation	Argon expansion (Joule–Thomson); **below −40 °C, two freeze–thaw cycles**; visible ice ball	Small renal mass ≤3–4 cm in frail patients; focal or salvage prostate
RFA	375–900 kHz current; ionic agitation; target 60 °C; char at 150 °C	Small renal mass; oncological results similar to cryo; heat sink near vessels
MWA	915 MHz or 2.45 GHz; dielectric hysteresis	Renal mass; hotter, larger zones, less heat sink
HIFU	Focused ultrasound absorbed, causing necrosis	See acoustic energy
IRE	High-voltage microsecond DC (direct current) pulses open permanent membrane pores; non-thermal	Spares vessels and ducts; focal prostate; good continence and potency; trials ongoing

- **Cryo mechanism:** extracellular ice, then cell dehydration, then intracellular ice, then membrane rupture on thawing; ischaemia and apoptosis follow.
- **IRE mechanism:** electric pulses open pores in cell membranes that do not reseal, so cells die by apoptosis without heat; collagen scaffolds survive. It needs general anaesthesia with muscle paralysis and ECG (electrocardiogram)-synchronised pulses.

- **Guidelines:** AUA 2021: consider thermal ablation for **cT1a ≤3 cm**; biopsy before or at ablation. EAU (European Association of Urology): not routinely >3 cm (cryo >4 cm).

6. Light and chemical energy

- **PDT:** photosensitiser plus light gives singlet oxygen. **VTP (vascular-targeted photodynamic therapy, padeliporfin)** for low-risk prostate cancer: RCT (Azzouzi 2017) halved progression versus AS (active surveillance; 28% vs 58%); not widely adopted. **Photodynamic diagnosis** (blue light) aids TURBT.
- **Chemohyperthermia:** MMC with bladder wall heated to **42 °C** for NMIBC; used after BCG (bacillus Calmette-Guérin) failure or shortage; not standard. EMDA for mitomycin (see above).

7. Prostate (BPH): newer energies

- **Rezum:** steam (103 °C) releases latent heat by convection; no char, zone-confined; preserves ejaculation.
- **Aquablation:** robotic non-thermal saline waterjet; good for large glands; haemostasis needed.
- **Also know:** TUMT (transurethral microwave thermotherapy) and TUNA (transurethral needle ablation) largely abandoned.

8. Ionising radiation

- EBRT (external beam radiotherapy) and **brachytherapy** (LDR [low dose rate] or HDR [high dose rate]): standard for localised prostate cancer; trimodality bladder preservation; penile cancer (brachytherapy for small T1–T2 glans tumours).
- **Radioligand therapy** (^{177}Lu-PSMA [prostate-specific membrane antigen], VISION trial) and **radium-223** (ALSYMPCA) for mCRPC (metastatic castration-resistant prostate cancer).

Diagrams to draw

- Monopolar vs bipolar circuit (monopolar loop-to-pad beside bipolar loop with local return), with cut and coagulation waveforms.
- Laser penetration depths compared, with cryoprobe ice ball.

> **Key points to remember:**
>
> - Electrosurgery at about 500 kHz; current density decides heating.
> - Cut continuous low voltage; coagulation interrupted high voltage.
> - Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
> - LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
> - Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
> - Bipolar: no pad, saline irrigation, no TUR syndrome; bipolar TURP equals monopolar efficacy with fewer complications.
> - Holmium 2120 nm: gold standard for stones and HoLEP.
> - Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink; thermal ablation for cT1a ≤3 cm.
> - HIFU, IRE, VTP: focal prostate therapy, still evolving.

Also, from the condensed book (not in the paper answer):

- The need for dilation falls with scope size (**<1%** for a 7.4 Fr scope). A **zero-tip balloon** dilates right up to an impacted stone.
- **Electrohydraulic lithotripsy is not first line in the ureter** (AUA), because its poorly focused shock damages the ureter and scope. A rigid pneumatic probe needs a rigid scope.
- The **three-prong grasper is the safest retrieval device**, because it releases an oversized stone. Never basket an irretrievably large stone through a single-channel flexible scope.

8 · Ureteroscopy technique

*See the paper answer **Flexible ureteroscopy / RIRS and accessories** (Section 11, Ch 97).*

ALGORITHM

1. Perform cystoscopy and place a 0.038-inch PTFE safety wire.
2. For an impacted stone, pass an angled hydrophilic wire (with or without a torque catheter), then exchange it for a PTFE wire.
3. If obstruction with infection is suspected, aspirate the pelvis. If the aspirate is turbid, place a stent and postpone.
4. Drain the bladder and proceed: semirigid scope below the iliac vessels, flexible above.

- If the orifice will not admit the scope, the safest course is to **stent and return in 1–2 weeks**. **Never dilate a segment that contains a stone**, because it may perforate.
- Formal dilation is needed in about 8–25%. Fragment an impacted stone to dust of **≤2 mm**, or basket the fragments.
- Under-rotation is the commonest novice error, and over-rotation damages fibreoptic bundles. Use a 200 µm fibre when deflection is needed.
- The **T-tilt** (15° Trendelenburg plus 15° airplane away) helps at the lower pole. Give non-steroidal anti-inflammatory drugs, with or without an alpha-blocker, after surgery.

9 · Nephroscopy and complications

- The rigid nephroscope (about 24–26 Fr) passes through a 26–30 Fr Amplatz sheath, and a flexible nephroscope reaches calyces out of line.

*See the paper answer **Flexible ureteroscopy / RIRS and accessories** (Section 11, Ch 97).*

Also, from the condensed book (not in the paper answer):

- For mucosal injury or perforation, stop and stent for 2–4 weeks. Avulsion follows dragging an oversized basketed stone, and needs open or robotic reconstruction.

18 Fundamentals of Laparoscopic and Robotic Urologic Surgery

1 · Patient selection and contraindications

Absolute contraindications	Relative contraindications (caution, specific consent)
Uncorrectable coagulopathy; intestinal obstruction (unless being treated); abdominal wall infection; massive haemoperitoneum or haemoretroperitoneum; generalised peritonitis; suspected malignant ascites	Morbid obesity; extensive previous surgery; pelvic fibrosis; organomegaly; benign ascites; pregnancy; hernia; iliac or aortic aneurysm; surgeon inexperience

KEY POINT

Every relative contraindication is a problem of safe entry or of tolerating pneumoperitoneum. The answers recur: enter elsewhere (Palmer point, retroperitoneum), enter openly (Hasson), or insufflate less.

- Severe chronic obstructive pulmonary disease (COPD) needs arterial blood gases and lung function tests. Treat arrhythmias first, because hypercarbia and acidosis destabilise the myocardium.
- **Pregnancy**: enter cephalad to the fundus (Palmer point), use **8–12 mm Hg**, and operate in the **second trimester**; it is harder after 20 weeks.
- A diaphragmatic hernia lets carbon dioxide (CO_2) reach the mediastinum. Never insufflate through an abdominal-wall hernia.
- In morbid obesity, renal and adrenal laparoscopy still beats open surgery. Leave prostatectomy and cystectomy with body mass index (BMI) >30 until later in the learning curve.

2 · Preoperative preparation and set-up

*See the paper answer **Operating room set-up** (Section 1, Ch 9).*

*See the paper answer **Patient positioning in lap urology** (Section 1, Ch 11).*

Also, from the condensed book (not in the paper answer):

- **No mechanical bowel preparation** is needed unless bowel is used, and it shows no benefit even for cystectomy.
- Enhanced recovery after surgery (ERAS) allows solids until 6 h and a carbohydrate drink 2 h before surgery. It shortens stay by about **30%** and halves complications.
- Group and screen for nephrectomy, nephroureterectomy and prostatectomy. Crossmatch for partial nephrectomy, cystectomy and caval thrombectomy early in experience.

3 · Physiology of pneumoperitoneum

LAQ Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy

Paper answer · P1 · P4 · merged from 2

Asked as: P1 LAQ "Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy" · P4 SAQ "General complications related to laparoscopy."

"d/t" is read as "due to" and "lap" as laparoscopy.

Introduction to pneumoperitoneum

- **Pneumoperitoneum** is insufflation of gas, usually CO_2 (carbon dioxide), into the peritoneal cavity at 10–15 mmHg to create a working space.
- Its effects have **two drivers**: **pressure** (compression of the cava, renal veins, kidneys and diaphragm) and **absorbed CO_2** (hypercarbia, acidosis, sympathetic drive). Patient position and volume status modify both.
- **CO_2** is standard: non-combustible, cheap, highly blood-soluble, so embolism is less lethal. Helium or argon in severe lung disease; N_2O (nitrous oxide) supports combustion.

Choice of insufflant

Gas	Advantages	Disadvantages
CO_2 (standard)	Non-combustible; highly blood-soluble, so embolism is less lethal; cheap	Hypercarbia, acidosis, arrhythmia; COPD (chronic obstructive pulmonary disease) patients cannot clear it
Helium, argon	Inert; no hypercarbia	Poorly soluble, so gas embolism risk; costly. Helium used in lung disease or to salvage hypercarbia
N_2O (nitrous oxide)	Less peritoneal irritation	Supports combustion; no electrosurgery
Room air, oxygen	—	Embolism, combustion; abandoned

Effects by pressure (supine, Campbell Table 18.1)

Effect	5 mmHg	10 mmHg	20 mmHg	40 mmHg
Heart rate (HR)	↑	↑	↑	↓
MAP (mean arterial pressure), SVR (systemic vascular resistance)	↑	↑	↑	↑
Cardiac output	↔/↓	↔/↑	↔/↓	↓
Venous return, cardiac output (Summer 2019 source)	—	Variable	Variable	↓ sharply
GFR (glomerular filtration rate), urine output	↔	↓	↓↓	↓↓
Arterial PCO_2	↔	↑	↑	↑
Arterial pH	↔	↔/↓ (another source: ↔)	↓	↓

- Renal function is the first casualty (from 10 mmHg). Afterload rises at every pressure. Output collapses at 40 mmHg. Acidosis builds from 20 mmHg.

Cardiovascular (haemodynamic) changes

- **Afterload rises** at every pressure: aortic and splanchnic compression plus catecholamine and **vasopressin** release raise MAP and SVR.

- **Preload depends on volume status.** In normovolaemia or hypovolaemia, pressure up to 20 mmHg compresses the IVC (inferior vena cava) and venous return falls. In hypervolaemia the cava resists and venous return may rise. At ≥40 mmHg venous return falls sharply in everyone and output collapses.
- **Arrhythmias:** hypercapnia causes tachycardia and ventricular ectopics; peritoneal stretch causes **vagal bradycardia** or asystole. A new arrhythmia may herald pneumothorax, hypoxia or gas embolism.
- **CVP (central venous pressure) reads falsely high** because intrathoracic pressure rises; do not guide fluids by it.
- Leg venous stasis: theoretical DVT (deep vein thrombosis) risk.

Respiratory changes

- **Diaphragm splinted:** FRC (functional residual capacity) and compliance fall, peak airway pressure rises; atelectasis and V/Q (ventilation–perfusion) mismatch. Dead space is unchanged. PEEP (positive end-expiratory pressure) helps in lung disease.
- **Head-down** (steep Trendelenburg) adds a further fall in vital capacity, endobronchial tube slip and facial oedema.
- **Barotrauma** (pressure >15 mmHg for long): pneumothorax, pneumomediastinum.

Acid–base changes (CO_2 absorption)

- CO_2 is absorbed through the peritoneum and, extraperitoneally, through fat and disrupted lymphatics. Absorption rises over 30–60 minutes, then plateaus.
- **Hypercarbia and respiratory acidosis** with rising end-tidal CO_2. Managed by raising minute ventilation and lowering pressure.
- **COPD** patients cannot blow it off: arterial gases if insufflation exceeds 1 hour, continued in recovery; consider helium. **Hypercarbia peaks after extubation**, as stored CO_2 is released.

Renal changes

- **Oliguria** comes from **renal vein and parenchymal compression**, not ureteric compression or hormones. Cortical flow falls; medullary flow rises up to 20 mmHg, then falls. It reverses on desufflation.
- Keep pressure ≤10 mmHg (another source: ≤10–12 mmHg); avoid fluid overload. If needed give furosemide, mannitol 12.5–25 g or dopamine 2 µg/kg/min to keep urine >100 mL/hour, rather than fluid boluses (overload, oedema). Important in live donor nephrectomy.

Other systems

- **Splanchnic:** hepatic and mesenteric flow fall (rare mesenteric thrombosis); less ileus than open surgery; no rise in reflux.
- **ICP (intracranial pressure) and IOP (intraocular pressure)** rise with hypercarbia and head-down tilt: caution in head injury, shunts and glaucoma.
- **Hormonal:** catecholamines, cortisol and **vasopressin** rise; renin–angiotensin activation adds to afterload.
- **Stress and immune response:** cortisol, catecholamines and prolactin rise as in any surgery, but hyperglycaemia, insulin resistance, catabolism and IL-6 (interleukin-6) rise less than after open surgery, which may explain faster recovery; oncological benefit unproven.

- **Also know:** hypothermia from cold dry gas; shoulder-tip pain from diaphragmatic CO_2 irritation.

Effects of position and approach

Position	Heart rate	MAP	Cardiac output	Other effects
Head-up	↑	↓	↓	Output ↓, hypotension
Head-down (Trendelenburg)	↓	↑	↑ (but lungs worse)	Venous return and output ↑; lung volumes ↓, ICP and IOP ↑
Lateral flank, kidney rest	—	—	—	Lung V/Q mismatch; rhabdomyolysis risk with kidney rest

- **Summary:** head-down raises MAP and output but worsens lungs; head-up lowers output.
- **Retroperitoneal** insufflation causes smaller swings in output and venous pressures than transperitoneal; pelvic extraperitoneal insufflation raised arterial CO_2 more.

Safe practice

- Preoperative cardiopulmonary assessment; correct arrhythmias; optimise volume.
- **Practical rule: work at 10–12 mmHg** (12 mmHg keeps stroke volume unchanged); 20 mmHg only briefly (first trocar, venous bleeding); never exceed 20 mmHg for long.
- Monitor end-tidal CO_2, saturation, airway pressure and urine output.
- **Recent advances:** low-pressure (8 mmHg) surgery with valveless insufflators (AirSeal); helium (or argon) for lung disease (poorly soluble, so embolism risk).

General complications of laparoscopy

Incidence

- Historically about 4%, mortality 0.03–0.08%. Contemporary series 13–22% because procedures are complex. **Vascular, then adjacent-organ injuries are commonest.**
- Risk factors: complex procedures, operating time >4 h, creatinine >1.5 mg/dL, learning curve.

Complications by phase

Phase	Complication	Key points
Access (Veress, first trocar)	Preperitoneal insufflation; vessel or bowel puncture; **major vascular injury (0.11–2%)**	Confirm Veress tests; open access with adhesions. Major vessel: leave trocar in, laparotomy
Insufflation	**Gas embolism**; subcutaneous emphysema; pneumothorax (1.6–4%); pneumomediastinum, pneumopericardium; hypercarbia	Embolism: see below
Secondary trocars	Epigastric bleeding; crossing swords, rollover	Transilluminate; bladeless trocars; insert under vision
Dissection	Thermal or mechanical bowel injury; vascular injury; bladder or ureter injury	Four electrosurgical mechanisms below
Positioning	Nerve injury (about 2.8%: brachial plexus, femoral, sciatic); **rhabdomyolysis** (about 1% retroperitoneal)	Padding, kidney rest <1 h, cases <5 h
Exit	Bowel entrapment, late port bleeding	Inspect each site at 5 mmHg

Phase	Complication	Key points
Postoperative	Port-site hernia (0.3–5.4%), wound infection, shoulder-tip pain, DVT	Close fascia of large ports

Physiological complications: hypercarbia and acidosis; oliguria from renal vein compression; reduced venous return; raised intracranial pressure in head-down tilt (see above).

Gas embolism

- Gas enters an opened vein (renal vein, IVC or lumbar vein). Sudden collapse; end-tidal CO_2 rises abruptly, saturation falls, then end-tidal CO_2 falls; "mill-wheel" murmur.
- Treatment: stop insufflation, **left lateral, head down**, 100% oxygen, aspirate through a central line.

Electrosurgical (thermal) bowel injury: four mechanisms

1. **Inappropriate (direct) activation** out of view.
2. **Direct coupling** with another metal instrument.
3. **Capacitive coupling** (metal cannula in a plastic retainer).
4. **Insulation failure.** - Monopolar injury presents **late (up to 18 days)** with low-grade fever and pain; resect with margin.

Missed bowel injury (Bishoff)

- Not classic peritonitis: **pain at one trocar site, leucopenia, low fever, distension, diarrhoea** at 24–48 h.
- **CT (computed tomography) with oral contrast**; free gas is normal for up to 7–9 days.

Complications of retroperitoneoscopy

Introduction

- **Retroperitoneoscopy** is laparoscopic or robotic surgery in the retroperitoneum without entering the peritoneum. It gives direct hilar access and avoids bowel, but the space must be **created**.
- **Why its complications differ:** small space with no natural cavity; space made by finger and **balloon dissection** (commercial or Gaur glove balloon) behind Gerota's fascia and in front of the psoas; fewer landmarks; dissection close to the lens; gas tracks along tissue planes; CO_2 is absorbed through fat and disrupted lymphatics, so hypercarbia appears early if the port leaks.
- Minimally invasive renal surgery: about 15% complications, mostly postoperative. Learning curve about 20 cases.

Classification (by phase of surgery)

Phase	Complications
Access and space creation	Wrong plane, balloon rupture, peritoneal tear, muscle bleeding
Insufflation	Subcutaneous emphysema, hypercarbia, pneumothorax, gas embolism
Dissection	Vascular, visceral, nerve, thermal injury
Exit and recovery	Port bleeding, hernia, rhabdomyolysis, urinoma, lymphocele

1. Access and balloon-related

- **Balloon in muscle** (wrong plane, intramuscular) gives a postoperative **flank hernia** or bulge.

- **Too anterior** gives peritoneal breach or colon injury; **too posterior** gives bleeding from quadratus lumborum or psoas.
- **Balloon rupture:** latex glove balloon breaks into many fragments (silicone leaves one) that must be retrieved.
- **Peritoneal tear:** CO_2 enters the peritoneum and collapses the small working space. Fix by venting with a Veress needle or an extra port, enlarging the tear, or converting to transperitoneal.
- Disorientation: keep the **psoas horizontal** as the landmark.

2. Gas-related

- **Subcutaneous emphysema** (commonest; commoner than transperitoneal): crepitus over flank, chest and neck; pneumoscrotum. Cause: leak around the port. Prevent with a **balloon-tip (blunt-tip) trocar**, tight incisions and 12 rather than 15 mmHg.
- **Hypercarbia**, especially with emphysema; a sealed port removes the difference from transperitoneal surgery.
- **Pneumothorax (1.6–4%, commoner retroperitoneally):** pleura is near the upper pole and 12th rib. Signs: neck and chest emphysema; tension gives hypotension, absent breath sounds and high airway pressure. Treat with needle decompression (16-G needle in the second or third intercostal space, midclavicular line), then chest drain. Pneumomediastinum and pneumopericardium (about 0.8%) also occur.
- **Gas embolism** (renal vein injury, IVC or lumbar vein): managed as above.

3. Vascular injury

- Commonest complication of minimally invasive urology; major injury <1 in 1000 (P4: 0.11–2% at access) but fatal in 8–17%. Upper-tract work carries about five times the pelvic vascular risk.
- Sites: renal artery and vein, **lumbar and gonadal veins**, adrenal vein, IVC, aorta. In the small posterior space the IVC or aorta can be mistaken for hilar vessels.
- **Management of venous bleeding:** raise pressure temporarily to 25 mmHg, suction to see the site, compress or tamponade with a sponge, haemostatic sealant, extra 5-mm port for suturing or a Satinsky clamp, hand-assist, add ports, suture or clip, or convert if uncontrolled. Arterial: clip or staple; aortic injury needs conversion. Never clip blindly.

4. Visceral injury

- **Peritoneum and colon**; duodenum (right); spleen, pancreas (left); liver; pleura; ureter; adrenal; usually by thermal spread.
- Thermal bowel injury follows the four mechanisms above; it may present up to 18 days later; diagnose by CT with oral contrast.

5. Positioning and nerve injury

- **Rhabdomyolysis** (about 1% retroperitoneal; 0.4% overall): men, BMI (body mass index) ≥25, kidney rest, full flexion, long cases. Brown urine, hip pain, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest to <1 hour, cases <5 hours, avoid hypotension.
- **Nerves** (neuropraxia): brachial plexus (axillary roll), sciatic; in the field, **genitofemoral and lateral femoral cutaneous** on the psoas, ilioinguinal and iliohypogastric (thigh numbness, flank bulge).

6. Postoperative

- Bleeding or haematoma; port-site hernia or haematoma; wound infection.

- **Urinoma** (partial nephrectomy, pyeloplasty); **lymphocele** or chylous leak after hilar or node dissection.
- Delayed bowel injury; DVT.
- **Conversion** to transperitoneal or open surgery for bleeding, peritoneal tear or lost orientation.

Retroperitoneal versus transperitoneal

Retroperitoneal more	Retroperitoneal less
Emphysema, pneumothorax, hypercarbia, disorientation	Bowel injury, ileus, adhesions; peritoneum spared

Prevention

- Open (Hasson) access below the 12th rib tip; balloon wholly in the retroperitoneum; balloon-tip port; 12 mmHg; trocars under vision; respect the 12th rib and pleura; check haemostasis at low pressure.

Laparoscopic donor nephrectomy

Introduction

- First described by **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first** at every stage.

Donor evaluation `Questionnaire → physical and psychosocial review, blood group, HLA (human leukocyte antigen) typing → GFR by at least two methods, one measured → metabolic, serology, cancer screening → CT angiography for vessels → cardiac review if needed`

- Exclusions: kidney disease, malignancy, active infection, cardiovascular disease, diabetes, significant obesity; bilateral stones generally exclude.

Side selection

- **Left preferred:** longer renal vein, simpler dissection. **Right** if the left has multiple arteries or complex veins, or the right kidney is smaller: **the donor keeps the better kidney.**
- **Anatomy matters:** arterial variation in 25–40%; early (prehilar) branching and accessory arteries must be seen on CT, as they may contraindicate the laparoscopic route.

Techniques

Technique	Features
Pure transperitoneal	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds)	Hand port doubles as extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	No bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed instruments; single incision, less pain

Operative steps (left)

1. Reflect colon; divide **gonadal, adrenal and lumbar veins**.
2. Preserve periureteric tissue and the **gonadal–ureter triangle** (ureteric blood supply).
3. Keep pressure ≤10 mmHg; hydration and mannitol or furosemide to keep urine >100 mL/hour (counters pneumoperitoneum oliguria).

4. Divide ureter, then **artery first, then vein**. Vein stapled lateral to the adrenal vein; Endo-TA (non-cutting) stapler gives extra vein length.
5. Extract quickly; cold flush. Warm ischaemia a few minutes. - **Hem-o-lok:** manufacturer contraindicates it on the donor renal artery after stump bleeds. - **Open donor nephrectomy** is kept for complex anatomy or previous surgery, more often on the right.

Advantages versus open

- Less pain, shorter stay, faster return to work, better cosmesis; **more people volunteer to donate**; equal graft function.

Disadvantages and complications

- Longer operation, learning curve, shorter vessels (right vein), slightly longer warm ischaemia.
- Complications about **16%** (mostly gastrointestinal); Clavien ≥IV 2.5%; **mortality 3 per 10,000**; conversion 0.26–3.34%, mainly for vascular injury.

Long-term donor counselling

- Function 60–80% of pre-donation; lifetime ESRD (end-stage renal disease) **90 per 10,000** versus 14 in matched non-donors (Muzaale, 2014); more gestational hypertension.
- **Also know:** kidney paired donation for ABO-incompatible or crossmatch-positive pairs; living-donor grafts survive longer than deceased-donor grafts.

Diagrams to draw

- Abdomen under pressure with arrows to diaphragm (lungs), IVC (preload), aorta (afterload) and renal vein (oliguria), plus CO_2 absorption to blood.
- Abdominal wall showing capacitive coupling through a metal cannula in a plastic anchor.
- Flank cross-section showing balloon between psoas and Gerota's fascia, with peritoneum pushed medially and port positions for retroperitoneoscopy.
- Left renal hilum with gonadal, adrenal and lumbar veins, stapled artery and vein, and the gonadal–ureter triangle.

Key points to remember:

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it.
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Also, from the condensed book (not in the paper answer):

- Helium or argon can be switched to **to salvage intraoperative hypercarbia**.
- The commonest working pressure is **15 mm Hg**. A pressure of ≤10 mm Hg markedly reduces oliguria.

4 · Access

LAQ Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy

Paper answer · P1 · P4 · merged from 2

Asked as: P1 LAQ "Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases, and open vs laparoscopic vs robotic radical prostatectomy" · P4 SAQ "Different types of trocars in laparoscopy."

Access in laparoscopy

A. Transperitoneal: closed (Veress)

- Veress needle: 14-gauge, spring-loaded blunt stylet. It is an entry needle, not a trocar. Check flow and tip recoil first. Two "pops": fascia, then peritoneum.
- **Sites:** umbilicus (thinnest wall; aim towards pelvis in non-obese); lower quadrants; **Palmer point** (left subcostal, midclavicular line) with suspected adhesions; avoid scars.
- **Confirmation tests:** aspiration–injection–aspiration of saline; **drop test**; advancement test (1 cm free). Then insufflate: opening pressure <8–10 mmHg, set limit 15 mmHg.
- First trocar inserted blind, or under vision with an **optical trocar** (0° lens inside a clear tip).

B. Open (Hasson)

- 2-cm infra-umbilical incision; fascia and peritoneum opened under vision; stay sutures; blunt cannula.
- Safer with adhesions or previous surgery and early in training; more gas leak.

C. Retroperitoneal

- 2-cm incision below the tip of the 12th rib; finger dissection behind Gerota's fascia, in front of psoas, then **balloon dilatation** (commercial or **Gaur** glove-on-catheter balloon), then a balloon-tip port to prevent gas leak and emphysema.

D. Extraperitoneal (pelvic): infra-umbilical, behind rectus into the Retzius space with a balloon (used for extraperitoneal prostatectomy).

E. Others: hand-assisted (6.5–7.5 cm port); single-port/LESS (laparoendoscopic single-site); direct trocar entry without pneumoperitoneum.

- **Access complications:** preperitoneal insufflation, vessel or bowel puncture, gas embolism, subcutaneous emphysema.

Hasson cannula (Hassan's canula)

Assumed to mean the Hasson cannula, used for open laparoscopic access.

Introduction

- Described by **Harrith Hasson (1971)** for open ("direct vision") laparoscopy. The first port is placed under vision instead of a blind Veress needle and trocar.

- **Parts:** outer sleeve with a cone-shaped (olive) adapter, **blunt obturator**, struts to anchor fascial stay sutures, and a gas inlet with tap.

Indications

- **Previous abdominal surgery or expected adhesions** (main indication).
- Organomegaly, pregnancy, aortic aneurysm, children, thin patients.
- Retroperitoneoscopy and extraperitoneal access (open start, then balloon dilatation).

Technique (transperitoneal)

1. 2-cm semicircular infra-umbilical incision.
2. Open fascia and then peritoneum separately, under vision, finger-sized.
3. Confirm entry by sight and finger sweep (free adhesions).
4. Place two 0 silk fascial stay sutures.
5. Insert the cannula with its **blunt obturator**.
6. Seat the cone in the defect; wrap stay sutures round the struts to seal.
7. Remove obturator; insufflate at high flow (to 12–15 mmHg).
8. At closure, tie the stay sutures to close the fascia. - **Retroperitoneal:** 2–2.5 cm incision below the 12th rib tip, finger dissection onto psoas, balloon dilatation, then port.

Advantages over Veress

- No blind puncture: fewer major vessel and bowel injuries; no preperitoneal insufflation; no gas embolism from intravascular insufflation.
- Fast insufflation; easy fascial closure (less port-site hernia).

Disadvantages

- Larger incision; **gas leak** around the port; slightly longer in obese patients.
- Early retroperitoneal use caused subcutaneous emphysema and hypercarbia, now solved by balloon-cuffed blunt ports.

Complications

- Bowel injury if adherent under the wound (repair through the same incision); wound infection; port-site hernia; vascular injury is still possible.

Evidence

- Reviews show no single entry method is clearly safest for major injury; open entry has fewer failed entries.

Instrumentation for development of the extraperitoneal space

- Extraperitoneal laparoscopy works outside the peritoneum: the **flank retroperitoneum** (kidney, adrenal, ureter) or the **pelvic prevesical (Retzius) space** (extraperitoneal prostatectomy, lymphadenectomy).
- There is no natural cavity, so a working space must be **dissected before insufflation**: finger first, then balloon.

Instrument	Description and use
Open (Hasson) access set	Scalpel, S-retractors, artery forceps
Index finger	Blunt first pocket; feels psoas behind, Gerota's fascia in front

Instrument	Description and use
Commercial balloon dissector	Transparent silicone balloon on a stiff hollow shaft; inflated with room air by bulb (maximum 800 mL, 40 pumps). Oblong for the retroperitoneum, round for the pelvis. Laparoscope inside confirms position
Gaur balloon (1992)	Size 7 glove (or glove fingers, condom) tied with 0 silk on an 8-Fr red rubber catheter; filled with saline (600–800 mL). Cheap, but no rigid shaft or view; latex can rupture into fragments
Optical (visual) obturator trocar with 0° scope	Enters under vision; yellow retroperitoneal fat confirms the plane; scope sweeps the space
Blunt-tip balloon trocar	Internal balloon plus external foam cuff gives an airtight seal; prevents gas leak, emphysema and hypercarbia

Technique: retroperitoneal

1. Full flank, table flexed; 1.5–2.5-cm incision below the 12th rib tip.
2. Split muscles; open the lumbodorsal fascia; finger dissection between psoas and Gerota's fascia.
3. Balloon inflated cephalad then caudad, in front of psoas, behind Gerota's fascia.
4. Blunt-tip balloon trocar; insufflate; secondary ports under vision.

Technique: pelvic extraperitoneal

- Infraumbilical incision; open anterior rectus sheath, separate recti; finger tunnels behind the rectus to the pubis; pierce transversalis fascia; balloon midline, then each side.

Pitfalls

- Balloon in muscle causes a flank hernia. Too anterior causes peritoneal tear or colon injury; too posterior causes psoas or quadratus bleeding.
- Peritoneal tear shrinks the space; convert if needed. Extraperitoneal CO_2 needs more ventilation.

Trocars

Definition

- A **trocar** is a **cannula (port) plus an obturator**; 3–20 mm diameter, 5–20 cm long (Chapter 18).
- Parts: obturator tip, cannula, **valve** (keeps gas in), insufflation side-arm, retention mechanism.

Classification

Basis	Types
Obturator tip	Bladed (cutting) with spring safety shield; **bladeless (blunt dilating)**; **optical (visual)**; blunt (Hasson)
Material	Reusable metal; disposable plastic
Valve	Flap or trumpet valve (needs reducer); **multiseal** (5–12 mm, no reducer); **valveless (AirSeal)**
Retention	Smooth (suture-anchored); threaded sleeve; expandable arms; **balloon (Blunt Tip)**
Special	Hasson open cannula; radially expanding; hand ports; single-site multichannel ports; robotic 8-mm metal ports

Main types explained

- **Bladed trocars:** cut through tissue; more epigastric injury and port-site hernia. Now of historical interest.
- **Bladeless (blunt dilating) trocars:** spread muscle fibres; have replaced bladed ones. About **fivefold less inferior epigastric injury**; trocar-site bleeding 3% vs 9%. Switching cut port hernia from 1.83% to 0.19%. **Fascial closure only for midline ports >10 mm** or stretched sites (all ≥5 mm in children).
- **Optical trocars:** clear tip with a 0° scope inside; each layer is seen during entry.
- **Hasson cannula:** blunt obturator and cone, fixed with fascial stay sutures after open entry; safer with adhesions; leaks more gas.
- **Radially expanding trocar:** a sleeve over the Veress needle is dilated to 12 mm; less port-site bleeding and pain (Chapter 39).
- **Balloon (Blunt Tip) trocar:** internal balloon and external foam cuff seal the wall; prevents leak and **subcutaneous emphysema** in retroperitoneoscopy.
- **AirSeal valveless port:** stable pneumoperitoneum at 8–12 mmHg despite suction.
- **Single-site ports:** TriPort; da Vinci SP (single-port) 25–27-mm guide. **Hand ports:** GelPort.

Safety points

- **Never use a metal trocar with a plastic retaining ring**: capacitive coupling burns viscera out of view.
- Insert under vision; ports 8–10 cm apart.

Gases used

Gas	Advantages	Disadvantages
CO_2 (carbon dioxide; standard)	Non-combustible, cheap, **highly blood-soluble** (embolism less lethal), quickly absorbed	Absorbed, causing hypercarbia, acidosis, arrhythmia; dangerous in COPD (chronic obstructive pulmonary disease)
Nitrous oxide	Less peritoneal pain, fewer acid–base effects	**Supports combustion**: no electrosurgery
Helium	Inert, no hypercarbia; use in lung disease	Poorly soluble, so embolism risk; costly
Argon	Inert	Embolism risk
Air, oxygen	–	Embolism, combustion: abandoned

- **Physiology:** keep pressure 10–15 mmHg (12 mmHg more physiological). Pressure reduces venous return, raises afterload, causes **oliguria** (renal vein and parenchymal compression) and splints the diaphragm.
- CO_2 absorption plateaus at 30–60 minutes; higher with extraperitoneal insufflation. Hypercarbia peaks after extubation.
- **Gas embolism:** sudden collapse, millwheel murmur, end-tidal CO_2 rise then fall. Treat: stop insufflation, **left lateral, head-down**, 100% oxygen, aspirate via central line.

Comparison of open (RRP), laparoscopic (LRP) and robotic (RALP) radical prostatectomy

- **Background:** anatomical nerve-sparing open RRP (Walsh, 1982) was the standard. First LRP (Schuessler, 1997) was too slow; French centres standardised it in 2000, but laparoscopic

suturing of the anastomosis remained a barrier. The robot (da Vinci, 2000) removed that barrier, and RALP is now dominant in the USA.

Parameter	Open RRP	LRP	RALP
Vision	Direct, loupes	2D, unstable	**3D, 10× magnified**, stable
Operating time	Shortest	Longest early	Longer than open; falls with experience
Instruments	Hands, haptic feedback	Rigid, fulcrum effect	**Wristed** (7 degrees of freedom); no haptics
Learning curve	Moderate	**Steepest** (suturing)	Shorter
Blood loss	300–1000 mL	Less (pneumoperitoneum tamponade)	Least; **transfusion 1–2%**
Hospital stay, catheter	Longer; catheter ~2 weeks	Shorter	Usually 1 day; catheter ~1 week
Pain, return to work	More	Less	Less; normal activity 3–4 weeks
Positive margins	Similar	Similar	Similar (~15%; ~9% if pT2)
Continence (1 year)	~93% (expert)	Similar	Similar; possibly earlier
Potency	Surgeon-dependent	Similar	Similar
Bladder neck stricture	Up to 16% (older series)	Lower	**<2%** (mucosa-to-mucosa)
Unique risks	Wound infection, hernia	Conversion, CO_2 effects	Steep Trendelenburg (eye, nerve), robot failure (<0.5%)
Cost	Lowest	Moderate	**Highest**

Evidence

- **Yaxley 2016 (randomised, one open vs one robotic surgeon):** equal continence, potency and early margins at 3 months; less blood loss and shorter stay with robot.
- **Coughlin 2018:** no difference in urinary or sexual function at 24 months.
- **LAP-01 (robotic vs laparoscopic):** earlier continence with robot.
- **Concept:** the **surgeon, not the platform**, decides outcome; margins depend mainly on tumour (Gleason ≥7, stage) and surgeon volume. The robot's advantages are recovery and wider reach (obese patients, prior surgery).
- **Also know:** transperitoneal vs extraperitoneal approach (no consistent difference); Retzius-sparing RALP (earlier continence, more margins); single-port (SP) robot; new robotic platforms; telesurgery.

Diagrams to draw

- Veress needle sites (umbilicus, Palmer point) and trocar layout for RALP.
- Hasson cannula with cone, stay sutures and blunt obturator in the abdominal wall.
- Flank section with balloon between psoas and Gerota's fascia.
- Trocar with obturator tip types (bladed, bladeless, optical) and valve.

Key points to remember:

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision; main indication previous surgery or adhesions; avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon. Gaur balloon: glove on red rubber catheter. Balloon in muscle causes hernia.
- Trocar = cannula + obturator. Bladeless: fewer epigastric injuries and hernias. Balloon trocar seals and prevents retroperitoneal emphysema.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

Also, from the condensed book (not in the paper answer):

- The Veress needle should give flow with ≤2 mm Hg at 2 L/min. **Do not lift the fascia**, because this widens the gap between fascia and peritoneum rather than the abdominal space.
- At the umbilicus, angle towards the pelvis in thin patients and go perpendicular in obese patients, as the umbilicus migrates caudally.
- A **hand port placed first** is the safest entry for the inexperienced, as the intra-abdominal hand guides the secondary trocars.

	Transperitoneal	**Retroperitoneal**
Space	Large, familiar anatomy	Small; psoas is the key landmark; direct (artery-first) hilar access
Physiology	Larger haemodynamic swings	More emphysema and pneumothorax; CO_2 absorption similar with a sealed port
Best for	Large tumours, reconstruction	Previous transperitoneal surgery; posterior tumours

Vascular injury in upper-tract surgery is about **5×** the pelvic rate.

5 · Trocars and port placement

*See the paper answer **Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy** (Section 1, Ch 18).*

LAQ Laparoendoscopic single-site (single-port) surgery: LESS

Paper answer · P4

Asked as:

- Laproendoscopic single site surgery: LESS — *Winter 2023 Paper IV Q1 (LAQ)*
- Single port lap. Surgeries in urology — *Practice Set 2 Paper IV Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered as a long answer in 2023-Winter Paper 4, Q1)

Read as laparoendoscopic single-site surgery (LESS), i.e. single-port (single-site) laparoscopic and robotic surgery in urology. NOTES (natural orifice transluminal endoscopic surgery) is

covered in 2022-Summer Paper 4, Q1.

Definition and concept

- **LESS:** laparoscopic or robotic surgery in which all instruments and the camera enter (pass) through **one skin incision**, usually hidden in the umbilicus (umbilical) or below the waistline.
- Aim: further reduce access trauma, giving **better cosmesis, less pain and shorter stay**. Chapter 142 places it on a spectrum: multiport → hand-assisted → LESS → SP (single-port) robot → NOTES. Each step to fewer incisions buys cosmesis at the price of working space and triangulation.
- First urological single-port nephrectomies were reported in 2007; the first single-port robotic urological surgery followed in 2009.

Nomenclature (LESSCAR)

- Early names were many: SILS (single-incision laparoscopic surgery), SPA (single-port access), OPUS (one-port umbilical surgery), TUES (transumbilical endoscopic surgery), e-NOTES (embryonic NOTES).
- **LESSCAR (Laparoendoscopic Single-Site Surgery Consortium for Assessment and Research, 2008)** agreed the single umbrella term **LESS** and set standards for reporting, training and research.
- The term was agreed by LESSCAR (LESS Consortium for Assessment and Research, 2008).
- Sub-types (variants): transumbilical or extraumbilical; **pure LESS**; **"LESS + 1"** or needlescopic-assisted (an extra 2–5-mm port); **R-LESS** (robotic LESS); purpose-built **SP (single-port) robot**.

Access: ports and devices

Option	Description
Several low-profile trocars	Placed close together through one fascial incision (the future extraction incision)
Multichannel ports	TriPort 15 (two 5-mm + one 15-mm); TriPort+ (three 5-mm + one 10-mm); SILS port; GelPOINT (gel cap, trocars placed anywhere on it)
da Vinci SP entry guide	One 25–27-mm cannula carrying three articulating (wristed) 6-mm instruments + one flexible articulating camera

- Position: modified flank (transperitoneal) or full flank (retroperitoneal SP, 2.5–3-cm incision near the 12th rib tip, balloon dilatation).

The triangulation problem

- In standard laparoscopy ports are spread so instruments meet the target at an angle (the "diamond"). In LESS all shafts are **parallel and in-line** with the camera (instruments lie parallel to the camera).
- Results: **instrument clashing**, **"crossing swords"**, poor depth perception, poor retraction and hard suturing.
- Solutions:
- **Articulating instruments** (tip bends inside the abdomen) and **pre-bent or curved instruments**: they cross at the fascia, so the right hand works on the left side (**chiral manipulation**).

- **Flexible-tip or 5-mm low-profile scopes**, offset optics, extra-long instruments and instruments of **different lengths** to avoid handle clash.
- A 2-mm needlescopic instrument or extra 5-mm port (an extra needlescopic port) without hesitation.
- **Robotic platforms** restore wristed movement.

Robotic single-site surgery

Platform	Features
da Vinci Single-Site (Si/Xi)	Curved crossing cannulas; semi-rigid **non-wristed** instruments; software swaps hand control
da Vinci SP (FDA [Food and Drug Administration] approval 2018)	Purpose-built; one arm, no external collisions; instruments with elbow and wrist; 3D articulating camera; on-screen navigator shows instrument positions

- SP needs **10–25 cm between (from) cannula and (to) target** to deploy the joints; a GelPort lets instruments "float" outside the skin. Assistance through an **"SP + 1" port** or ROSI (remote-operated suction-irrigation). Limited by cost and availability.

Indications

- **Kidney:** simple and **donor nephrectomy**, radical nephrectomy, partial nephrectomy, **pyeloplasty**, nephroureterectomy, cyst decortication, cryoablation.
- **Adrenalectomy**; varicocelectomy; orchidopexy and pyeloplasty in children (paediatric).
- **Pelvis:** SP radical prostatectomy (transperitoneal, extraperitoneal); **SP transvesical simple prostatectomy** and bladder diverticulectomy.
- **Best case:** thin patient (patients), small or benign lesion (lesions), experienced surgeon.
- **Relative contraindications (avoid in):** obesity, large or hilar tumours, complex vascular anatomy, dense adhesions, unfit for pneumoperitoneum.

Outcomes versus conventional laparoscopy (Chapter 142)

- **Randomised trial (RCT, randomised controlled trial):** same operative time, blood loss, complications and stay; **less analgesia**.
- **Donor nephrectomy meta-analysis:** same stay, recovery, transfusion and conversion (otherwise equal); **lower pain scores**.
- **Renal mass:** same perioperative outcome and survival.
- **SP robotic partial nephrectomy vs multiport:** shorter stay, less opioid; same margins and complications; **longer warm ischaemia** in one review.
- An **extra port is often needed**; conversion is mostly to laparoscopy, rarely open.
- **Net effect:** gains are confined (limited) to **pain, cosmesis and stay**; oncological and functional outcomes are unchanged, but the operation is technically harder and longer.

Complications

- All laparoscopic complications (usual laparoscopic injuries: access injury, vascular and bowel injury, CO_2 effects).
- **Incisional and umbilical hernia** is higher (larger fascial defect): about 5% with da Vinci Xi single-site, lower with SP.
- Wound infection; conversion; longer operation; surgeon ergonomic strain. Prevent by careful fascial closure and case selection.

Recent advances

- **da Vinci SP** with same-day discharge pathways; **SP retroperitoneal** renal surgery.
- **Transvesical SP** prostatectomy; hidden incision endoscopic surgery (HIdES) in children.
- New single-port platforms; magnetic anchoring and guidance systems / instruments (restore triangulation).

Diagram to draw: umbilical multichannel port with articulating (crossed) instruments crossing at the fascia (chiral) versus a triangulated multiport diamond (layout).

> **Key points to remember:**
> - LESS = one incision; LESSCAR (2008) standardised the term.
> - Core problem: lost triangulation; solved by articulating or curved instruments and robots.
> - Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
> - da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Also, from the condensed book (not in the paper answer):

- **Triangulation**: place ports in a diamond around the target, with instruments at **<55°** to the horizontal and 25–45° between them. Place ports in the midline or at least 6 cm lateral.
- Before secondary trocars, inspect the whole abdomen for entry injury. Aim each port at the target, as redirecting widens the tract and causes emphysema.
- **Hand port**: the incision in centimetres equals the surgeon's glove size. Wound infection is about 9% and midline hernia 4–7%.

6 · Robotic systems and instruments

SAQ 3D printing in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3a

Definition

- **3D (three-dimensional) printing** is additive manufacturing: a solid object is built layer by layer from a digital model.
- In urology it turns a patient's CT (computed tomography) or MRI (magnetic resonance imaging) into a physical replica of kidney, tumour, vessels or prostate.

Workflow

`Thin-slice CT/MRI (DICOM files) → segmentation of kidney, tumour, vessels, calyces → STL (standard tessellation language) mesh → printer → post-processing`

- Takes hours to days; segmentation needs expertise.

Printing technologies

Method	Principle	Use
FDM (fused deposition modelling)	Melted plastic filament extruded	Cheap, rigid models

Method	Principle	Use
SLA (stereolithography)	Laser cures liquid resin	Fine detail, transparent models
Material jetting (PolyJet)	Jetted photopolymers, multi-colour	Tumour, vessels and calyces in different colours
SLS (selective laser sintering)	Laser fuses powder	Durable parts, devices
Bioprinting	Living cells in hydrogel "bio-ink"	Tissue engineering (see Practice Set 6 Paper 4, Q2)

Applications

1. **Partial nephrectomy planning:** complex hilar or endophytic tumours. The model shows the tumour against vessels and collecting system, so the surgeon plans the approach and rehearses steps. Promising.
2. **PCNL (percutaneous nephrolithotomy) planning:** staghorn stones, anomalous kidneys; choice of calyx. Early promise; 3D reformatting did not predict stone-free rate better.
3. **Prostate:** models with MRI lesions for margin and nerve-sparing planning and focal therapy.
4. **Patient education and consent:** patients understand their anatomy and the operation better.
5. **Training and simulation:** silicone or hydrogel phantoms for PCNL puncture, partial nephrectomy, pyeloplasty, anastomosis.
6. **Patient-specific devices:** prototype stents, catheters and surgical guides.

Limitations

- Cost, printing time and segmentation errors.
- Materials do not reproduce tissue feel or bleeding.
- Evidence mostly small series; no proven outcome benefit.

Future

- Virtual and augmented reality models on the robotic console, point-of-care printing, drug-eluting printed stents.

Diagram to draw: CT → segmentation → STL → printer → colour kidney model with tumour, artery and calyces.

> **Key points to remember:**
> - Additive manufacturing from CT/MRI via segmentation and an STL file.
> - Main uses: partial nephrectomy and PCNL planning, consent, simulation.
> - Limits: cost, time, unrealistic tissue, weak outcome evidence.

SAQ Hemostasis in laparoscopic surgery

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Hemostasis in laparoscopic surgery." · P4 SAQ "Hemostasis in lap surgery"

Principles

- Devices seal by **denaturing protein**. Plan from imaging.
- Keep a **haemorrhage tray** ready: laparoscopic Satinsky, 10-mm suction, vascular suture with Lapra-Ty, haemostatic agent.

Effect of pneumoperitoneum

- CO_2 at 12–15 mmHg **tamponades venous bleeding**, hiding it.
- Raise the pressure briefly for venous bleeding: 25 mmHg (P4: 20–25 mmHg).
- Before exit, **lower the pressure and inspect** the operative site and each port to unmask hidden bleeders and port-site bleeding: about 5 mmHg (P4: 5–10 mmHg).

Energy devices

Device	Lateral thermal spread	Note
Monopolar	5–7 mm	Capacitive coupling, insulation failure, direct coupling
Bipolar	1–6 mm	Current only between the jaws
Vessel sealer (LigaSure)	1–3 mm	Seals vessels **≤7 mm**; one application only
Ultrasonic (Harmonic)	≈1 mm	No current; blade >200 °C, cools slowly
Argon beam	Superficial	Raw surfaces; gas embolism risk

Clips

- **Titanium clips:** no diathermy near them (they slip).
- **Hem-o-lok** (locking polymer): circumferential dissection, see the tip beyond the vessel, feel the snap, at least two clips on the patient side with a 2-mm cuff.
- **FDA (Food and Drug Administration) warning:** after donor renal artery stump haemorrhages, the manufacturer **contraindicated Hem-o-lok on the living-donor renal artery**. A later series found it safe with correct technique, but use a stapler there.
- **Lapra-Ty** absorbable clips act as knots (renorrhaphy).

Staplers

- Endo GIA through a 12-mm port; **grey (2.0 mm) or white (2.5 mm)** loads for renal vessels; blue or green for thicker tissue.
- **Never fire across clips** (misfire). Reinforce a bleeding staple line with clips.

Sutures and haemostatic agents

- Intracorporeal or barbed sutures.
- Adjuncts: fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

Managing bleeding

- **Venous injury** (algorithm): 1. Raise pneumoperitoneum pressure (25 mmHg; P4: 20–25 mmHg). 2. Suction to identify the bleeding point. 3. Pledget tamponade. 4. Sealant. 5. Extra 5-mm port for suturing or Satinsky. 6. Hand-assist. 7. Convert.
- Keep the patient well filled (gas embolism).
- **Arterial:** minor bleeding, tamponade; renal artery, staple; aorta, convert.
- **Port-site bleeding:** cantilever the trocar to find it; diathermy or transfascial suture. Blunt trocars reduce it.

Diagram to draw: Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

Key points to remember:

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- EndoWrist instruments have about a **10-use** limit, and an Allen bolt opens jaws locked after failure.
- Robot camera ports are 8 mm (Xi) or 12 mm (Si). Place the assistant port caudal, not between the arms.
- The LapSac is the most tear-resistant bag for morcellation, and ring forceps puncture it least.

7 · Exiting the abdomen

ALGORITHM

1. Lower the pressure to 5 mm Hg to unmask bleeding.
2. Inspect the operative site and every port.
3. Place fascial sutures under vision and remove each port under vision, the scope last.
4. Close a hand port first, then reinsufflate.

Trocar	Adult	Child
Bladed	Close all 10–12 mm sites	Close all ≥5 mm sites
Blunt	Close midline >10 mm or stretched sites	

- With BMI >30, use a closure device. Transverse extraction incisions herniate less than vertical midline ones.

8 · Complications and management

*See the paper answer **Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy** (Section 1, Ch 18).*

Also, from the condensed book (not in the paper answer):

- **Bowel** is the commonest first-trocar injury. A through-and-through injury is seen only by viewing the primary port from a secondary port, so do this routinely. Leave the trocar in, gain access elsewhere and repair, with 4–5 L antibiotic irrigation. The commonest major vessels injured are the **aorta and common iliac artery**.
- **Thermal bowel injury**: resect a monopolar burn with **6 cm** margins, excise the visible area of a bipolar burn, and repair a mechanical injury laparoscopically.
- A Veress needle in a vessel is withdrawn without manipulation and re-sited. Always repair a bladder injury, and catheterise every major case.

9 · Training

*See the paper answer **Simulators in urology** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- Robotic surgeons with **>20 cases** have better outcomes, so mentor early cases.

19 Energy Modalities in Urology

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

1 · Physics essentials

Every modality depends on where the energy is absorbed.

- The five surgical effects are cutting, ablation, destruction, coagulation and fragmentation.
- Laser power equals pulse energy times frequency, so 0.5 J × 20 Hz gives 10 W. **10 bar = 1 MPa** (not 1 bar).
- High frequency penetrates shallowly and low frequency deeply, which is why a scrotal probe is linear and a renal probe curvilinear. Absorption usually rises with frequency.
- Photon energy is E = hf, so raising laser power sends more photons per second, not more energetic ones.
- Acoustic impedance is sound speed times density. A large mismatch (tissue–air, tissue–stone) reflects the wave, which explains gel coupling, gas and rib shadows and the stone acoustic shadow.
- Radiation force equals absorbed power divided by wave speed. It is negligible for light but useful for sound, and is the basis of ultrasonic stone propulsion.
- A shock is an abrupt pressure rise from nonlinear steepening, and is acoustic, not electrical. Cavitation is a vapour bubble torn open by negative pressure, whereas boiling is caused by a rise in temperature.

	Electromagnetic waves	**Elastic (acoustic) waves**
Devices	Lasers, electrosurgery, microwave, radiofrequency (RF)	Shock wave lithotripsy (SWL), ultrasound therapies, intracorporeal lithotripters
Medium and wavelength	Free space; microns (light)	Needs gas, liquid or solid; millimetres
Bioeffect	Absorption, giving heat	Heat, mechanical stress in solids, cavitation in fluids
Penetration	Short, so a fibre must reach the target	Centimetres, so transcutaneous delivery is possible

2 · Electrosurgery

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- **Pad placement**: use clean, well-perfused muscle and avoid bony prominences, scars, dense hair and metal implants. A **hip prosthesis is not a contraindication**; place the pad on the opposite side.
- **Pacemakers and implantable defibrillators**: prefer bipolar. If monopolar is needed, route the current path away from the device and use short bursts. Cardiology sets pacing to asynchronous and suspends anti-tachycardia therapy, an external defibrillator must be available, and the device is rechecked after surgery.
- **Surgical fire** needs fuel (drapes, alcohol prep), an oxidiser (oxygen-rich gas) and ignition. Electrosurgery is implicated in most theatre fires.

Endoscopic resection

*See the paper answer **TURP instruments, techniques and monopolar vs bipolar** (Section 16, Ch 151).*

*See the paper answer **TUR syndrome** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- Typical monopolar resection settings are cut **180 W** and coagulation **80–120 W**.
- Ions in the irrigant disperse current away from the loop, which is why monopolar irrigant must be ion-free, inert, non-toxic and transparent.
- Damaged sheath insulation causes loss of cutting or a urethral burn, so inspect the loop before mounting.

Device-assisted intravesical therapy for non-muscle-invasive bladder cancer (NMIBC)

	Electromotive drug administration (EMDA)	Chemohyperthermia
Energy	Electric field between an intravesical electrode and abdominal pads	915-MHz microwave via catheter (Synergo) with thermocouples
Mechanism	Iontophoresis, electro-osmosis and electroporation drive mitomycin C (MMC) into the lamina propria	Bladder wall at **42 °C** raises permeability and cancer-cell heat sensitivity
Evidence	Randomised controlled trial (RCT) of EMDA-MMC with bacille Calmette–Guérin (BCG): better recurrence- and progression-free survival. Meta-analysis: recurrence only. Cochrane: low quality	Not standard

3 · Lasers

*See the paper answer **LASERs in urology: holmium, thulium, KTP and newer laser technologies** (Section 1, Ch 17).*

*See the paper answer **Lasers in BPH, including green light** (Section 16, Ch 151).*

Laser lithotripsy: optical pulse duration decides the mechanism

Optical pulse	Mechanism	Bubble pressure	Fragments
Ultrashort <500 ns (Q-switched)	Plasma	Extremely high	Large
Short 0.5–10 µs (pulsed dye; frequency-doubled double-pulse neodymium:YAG [Nd:YAG], FREDDY)	Cavitation and shock (photomechanical)	>300 bar	Large
Long >250 µs (holmium:YAG [Ho:YAG], thulium fibre laser [TFL])	Photothermal	<20 bar	Small, dust
>20 ms or continuous wave	Photothermal with carbonisation	–	Adjacent tissue injured

- **Not in the paper answer:**
- Manufacturers quote the electrical (pump) pulse duration, but the **optical pulse duration** is the one that matters. Dosimetry, not the platform, sets the mechanism.
- **Moses effect**: the first ≈ **60 µs** boil a vapour channel. The fibre should be close and perpendicular to the stone (0° gives the most ablation; parallel gives none). Ho:YAG transients are only 8–20 bar at <1 J.
- The **pulsed dye laser (504 nm)** was safe because haemoglobin acts as an energy sink, but it failed on hard stones (cystine, brushite, calcium oxalate monohydrate).

Soft-tissue lasers

- The **CO_2 laser** (10,600 nm) is used for superficial penile lesions and cannot be used intracorporeally.
- The **diode laser** (940–1490 nm) is absorbed by water and haemoglobin variably, penetrates 0.5–5 mm and is used for prostate vaporisation.
- Nd:YAG forward scatter can injure bowel beyond the bladder wall. KTP is shallow because of haemoglobin absorption, not wavelength itself.
- **GreenLight** ablates best at a **4 mm/s** sweep and **≤2 mm** working distance. Gain plateaus beyond 180 W, but 180 W shortens operating time compared with 120 W.
- **Photodynamic therapy** treats the whole urothelium after one intravesical application of photosensitiser, by generating singlet oxygen.

4 · Acoustic therapy

Heating	Cavitation
Rises with frequency (absorption)	Rises as frequency falls (longer negative phase)
Dose-dependent	**Threshold effect**; nucleates at interfaces, stones and gas, so lung and bowel are vulnerable

- Do not perform SWL within **24 h** of ultrasound microbubble contrast. Doppler twinkling artefact is caused by microbubbles on the stone.
- Acoustic waves are non-ionising, so repeated safe exposures do not accumulate.

Shock wave lithotripsy

*See the paper answer **ESWL: mechanism, indications, factors affecting outcome, complications and current status** (Section 11, Ch 98).*

LAQ Energy sources for breaking stones in urology

Paper answer · P4

Asked as:

- Describe energy sources for breaking stones in Urology — *Practice Set 1 Paper IV Q2 (LAQ)*
- Describe energy sources for braking stones in uro — *Practice Set 5 Paper IV Q1 (LAQ)*

Type: LAQ (25 marks)
(also covered in 2025-Winter Paper 4, Q2 and Q3b; also answered in Practice Set 1 Paper 4, Q2)

Read as "energy sources for breaking stones in urology".

Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

Classification

Group	Sources
Extracorporeal	SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational)
Intracorporeal mechanical	Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic
Intracorporeal electrical	EHL (electrohydraulic lithotripsy)
Intracorporeal light	Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY

1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

Generator	Mechanism	Strength	Weakness
Electrohydraulic (original HM3, 1980)	Underwater spark at F1 of an ellipsoid reflector	Very effective	Electrode life <40,000 shocks; focus drifts
Electromagnetic (modern standard)	Coil pushes a metal membrane; lens or reflector focuses	Consistent; high energy at F2; long life	—
Piezoelectric	Many ceramic elements on a spherical dish	Long life; little pain	Small focal zone, low energy

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.

- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.
- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).
- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.
- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

	Ho:YAG	**TFL**
Wavelength	**2120 nm** (often quoted 2100)	1940 nm (water absorption peak)
Frequency	Up to about 120 Hz	Up to about 2000 Hz
Peak power	High	Low → less retropulsion
Fibre	≥200 µm	50–150 µm (more flow and deflection)
Product	Fragments and dust	Finer dust

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.
- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

Choice by site

Site	Usual energy
Ureter	Laser (semirigid or flexible); pneumatic distally; SWL
Kidney	SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser
Bladder	Laser, pneumatic, EHL, mechanical

Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

Diagram to draw: SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

> **Key points to remember:**
> - SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
> - Holmium 2120 nm, photothermal, breaks every composition.
> - TFL 1940 nm: low peak power, finer dust, less retropulsion.
> - EHL is poorly focused, so it is not first-line in the ureter.

Also, from the condensed book (not in the paper answer):

- Pulse parameters: peak negative pressure ≈ **10 MPa** lasting ≈ 5 μs, about 3 J per pulse, and about **2000 shocks** at 0.5–3 Hz.
- Each shock leaves bubbles that need ≈ **1 s** to dissolve, so faster firing builds a cavitation cloud that shields the stone and injures the kidney. Small air pockets in the coupling disproportionately reduce focal pressure.
- Piezoelectric sources focus at the centre of curvature of a spherical bowl and are the only type that produces both impulses and narrowband bursts.

Investigational acoustic stone therapies

	Burst wave lithotripsy (BWL)	Ultrasonic propulsion
Waveform	Narrowband, 5–100 cycles, 200–1000 kHz	About 1500 cycles in bursts of ≤3 s
Peak pressure and rate	5–8 MPa; ≥10 Hz	≤2.4 MPa; 50% duty cycle
Guidance and anaesthesia	Real-time ultrasound; none	Ultrasound; none
Action	Breaks stones into uniform fragments proportional to wavelength	Moves fragments by radiation force

KEY POINT

Acoustic therapy is management of the negative-pressure phase. Tension cracks brittle stones and spares ductile tissue, but the same tension nucleates bubbles that either destroy tissue deliberately (histotripsy) or shield the stone and injure the kidney (SWL fired too fast). Raise the frequency for heat and lower it for cavitation.

Soft-tissue ablation

*See the paper answer **HIFU** (Section 17, Ch 163).*

Also, from the condensed book (not in the paper answer):

- **HIFU** uses 1500–10,000 W/cm² in ~1-s pulses and is poorly visible on ultrasound, so gaps occur.
- **Cavitation histotripsy** uses >10,000 W/cm² to make a bubble cloud that liquefies tissue mechanically, and is **echogenic in real time**. **Boiling histotripsy** boils the focus within one burst (<100 ms). Both are investigational.

5 · Thermal and other ablation

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- In radiofrequency ablation the tissue is heated by current, so the probe itself is not hot. Early concerns about microwave ablation (collecting-system absorption, inconsistent zones) were not borne out, and its safety and efficacy equal radiofrequency ablation.
- **Rezum** injects 103 °C steam through an 18-G needle; each **9-s** injection brings tissue to ≥70 °C and stays within the zonal compartment.

KEY POINT

Current density explains electrosurgery, pulse duration explains laser lithotripsy, penetration depth (chromophore) explains soft-tissue lasers, and negative pressure (frequency and rate) explains SWL and histotripsy. In direct-contact modalities the tissue boundary, not absorption depth, limits the lesion.

6 · Landmark trials

Trial	Design	Result	Take-home
TFL vs low-power Ho:YAG	RCT, ureteroscopy for renal stones	TFL: higher stone-free rate, shorter time, less bleeding	TFL beats 30-W holmium
TFL vs high-power Ho:YAG	RCT vs 120-W pulse-modulated Ho:YAG	Stone-free equal; TFL less retropulsion; Ho:YAG less energy	The comparator explains discordant trials
Ultrasonic propulsion of residual fragments	RCT	Fewer relapses (growth, urgent visit, surgery) at 4 years	Clearing residual fragments matters

20 Complications of Urologic Surgery

1 · Classification and context

*See the paper answer **Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy** (Section 1, Ch 18).*

Also, from the condensed book (not in the paper answer):

- **Clavien–Dindo** is the recommended system and grades the **treatment required, not the event**. It omits intraoperative events and does not separate early from late.

Grade	Definition	Examples and traps
I	Deviation without drugs or intervention	Antiemetics, analgesics, electrolytes; wound opened at bedside
II	Other drugs	Blood transfusion, total parenteral nutrition
IIIa / IIIb	Surgical, endoscopic or radiological intervention	IIIa without, IIIb under general anaesthesia
IVa / IVb	Life-threatening, intensive care	IVa single organ (dialysis); IVb multiorgan. Central nervous system bleed or stroke counts; transient ischaemic attack does not
V	Death	–
Suffix "d"	Present at discharge	Flags follow-up; not a grade

2 · Positional and neuromuscular complications

*See the paper answer **Patient positioning in lap urology** (Section 1, Ch 11).*

Also, from the condensed book (not in the paper answer):

- Positional injury occurs in ≈ **1%** of laparoscopic and up to ≈ **6%** of robotic cases.
- In lithotomy the stirrup compresses the **common peroneal nerve at the fibular neck** (foot drop). Tucked arms in robotic surgery endanger the **ulnar nerve**.

Compartment syndrome and operative nerve injury

- **Lower-limb compartment syndrome** occurs in 1 in 3500 overall and up to **1 in 500** radical cystectomies. Perform fasciotomy if >**30 mmHg** and symptomatic. Pressure >50 mmHg for 3–4 h causes irreversible necrosis.
- **Genitofemoral** injury (psoas hitch, ≈ 5%) is avoided by placing sutures along the tendon. Repair an **obturator** transection with tension-free **7-0** non-absorbable epineural sutures.

3 · Urine leaks and urinary tract injury

Urine leak after partial nephrectomy

It presents at about day 13. Drain creatinine confirms it but does not locate it; computed tomography (CT) urography with delayed images shows the site.

ALGORITHM

1. Drainage and a Foley catheter.
2. Ureteric stent; drain off suction.
3. Radiological drainage of collections, then nephrostomy.
4. Nephrectomy as the last resort.

Ureteric injury

P3 *Ureteric injury: iatrogenic and upper ureteric; Uretero-vaginal fistula — danger sites, diagnosis, double-dye test, timing and repair by level*

- A partial tear is closed with 4-0/5-0 absorbable sutures over a stent for 4–8 weeks.
- Strictures are treated endoscopically if short, and surgically if **>2 cm** or after endoscopic failure.

Bladder injury and anastomotic leak

- Close a bladder injury in two layers and catheterise for 7–10 days.
- A **urethrovesical leak** after robotic prostatectomy occurs in up to ≈ 15% at days 7–10.

ALGORITHM

1. Keep the catheter patent and **never apply traction**.
2. Move the drain off suction; radiological drainage; then nephrostomy.
3. Formal repair of a major disruption **before day 7**, after which tissue will not hold sutures.

4 · Bleeding and vascular injury

P1 **P3** **P4** *Retroperitoneoscopy and laparoscopy complications; partial nephrectomy — leave the trocar in place, venous bleeding control; delayed haematuria means pseudoaneurysm, so angioembolise*

- Haemorrhage is the commonest intraoperative complication (≈ 2%).
- The commonest vessel injured at umbilical access is the **left common iliac vein**. Only the **internal iliac and inferior mesenteric** vessels may be ligated; all other major vessels are repaired.
- The superior mesenteric artery can be mistaken for the left renal artery in the lateral position, with mortality up to **75%**.

5 · Bowel injury

Most delayed injuries are thermal (monopolar spread >20 mm).

Injury	Management
Small tear without spillage	Primary repair
Colonic with faecal soiling	Colostomy
Thermal burn	**Always oversew, never conservative**; resect if extensive, because damage exceeds what is visible
Rectal (prostatectomy ≈ 0.5–1%; salvage ≈ 7%)	Multilayer closure, omental interposition; colostomy if irradiated or missed

Duodenal injuries perforate late even when <5 mm.

6 · Venous thromboembolism (VTE) and lymphocele

Symptomatic VTE occurs in ≈ **1%** after nephrectomy or prostatectomy and ≈ **5%** after radical cystectomy, mostly after discharge.

Caprini score: 1 point for age 41–60 or body mass index >25; 2 points for age 61–74, malignancy, or major or laparoscopic surgery >45 min; 3 points for age ≥75, personal or family VTE or thrombophilia; 5 points for stroke, arthroplasty or fracture. Risk is very low at 0, low at 1–2, moderate at 3–4 and **high at ≥5**.

Risk	Prophylaxis
Very low (transurethral, anti-incontinence)	Early ambulation
Low	Intermittent pneumatic compression (IPC)
Moderate	Low-molecular-weight heparin (LMWH), low-dose unfractionated heparin or IPC
High	Heparin plus IPC; IPC alone until haemostasis is secure if bleeding risk is high
High-risk cancer surgery	Add extended LMWH for ≈ **4 weeks** after discharge

Start pharmacological prophylaxis 2–12 h before surgery.

P1 *Lymphocele and postoperative lymphatic leak — incidence, observation, drainage, fenestration*

- Prevent it by clipping or sealing visible lymphatics and by a peritoneal flap after robotic prostatectomy.

7 · Landmark trials

Trial	Design	Result	Take-home
Preventer	Randomised, perioperative heparin after prostatectomy	No reduction in symptomatic VTE	Not mandatory in low-risk prostatectomy
Extended LMWH meta-analysis	Randomised trials, abdominopelvic cancer surgery	≥1 month of LMWH cut VTE without extra bleeding	Extend after cystectomy

21 Urologic Considerations in Pregnancy

1 · Physiological changes

System	Change	Urological relevance
Cardiovascular	Cardiac output up 30–50%	The uterus compresses the vena cava, so use a **15–20° left tilt** for every procedure
Haematological	Blood volume up 40–50%; hypercoagulable	Thromboembolism risk ~10×; white cell count up to 12,200, so it is unreliable for infection
Renal	Glomerular filtration rate up **40–65%**; creatinine ≈ **0.8 mg/dL**	A "normal" non-pregnant creatinine may mean impairment
Urine	Protein up to 300 mg/24 h; calciuria doubles	Persistent proteinuria: exclude pre-eclampsia, diabetes, renal disease and infection

Dipstick microscopic haematuria occurs in up to **20%** and is usually benign; pregnancy-specific causes are pre-eclampsia and placenta percreta.

2 · Physiological hydronephrosis

SAQ Bacteriuria in pregnancy and antibiotic policy

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Bacteriuria in pregnancy." · P3 SAQ "Baceriuria and use of antibiotic policy in pregnancy"

Read as: bacteriuria in pregnancy and the antibiotic policy for it.

Definition

- **ASB (asymptomatic bacteriuria):** $\geq 10^5$ CFU (colony-forming units)/mL of one uropathogen in a midstream sample without urinary symptoms ($\geq 10^2$ CFU/mL on a catheter sample).

Epidemiology

- Prevalence is **not raised** by pregnancy (about 2–10%; P3: about 2–7%), but ASB **persists and ascends**. Higher with low socioeconomic status, multiparity, sickle cell trait. Organisms as in non-pregnant women (mainly *E. coli*).

Why pregnancy matters (pathophysiology)

- **Progesterone** relaxes ureteric smooth muscle; the **uterus compresses the ureters at the pelvic brim** (right more). Dilatation from about 15 weeks; normal by 6 weeks postpartum.
- Result: **stasis and hydroureteronephrosis**; bacteriuria persists and ascends.
- GFR (glomerular filtration rate) rises 30–50%: creatinine and drug levels fall.
- **Untreated ASB leads to pyelonephritis in 15–45%** (historical), mostly in the third trimester (60–75% of cases). Treatment cuts this to **1–4%**.
- Pyelonephritis causes **preterm birth, low birth weight** and maternal sepsis. **Cochrane 2019:** treating ASB reduces pyelonephritis, preterm delivery and low birth weight.

Screening

- **Urine culture at 12–16 weeks** or the first visit (USPSTF [US Preventive Services Task Force], IDSA [Infectious Diseases Society of America]). Dipstick misses many cases.
- Treat all ASB found in pregnancy.

Antibiotic policy

Safe	Restricted or avoided
β-lactams (penicillins such as amoxicillin; cephalosporins such as cephalexin): throughout	**Nitrofurantoin:** stop near term (from 35 weeks): neonatal haemolysis (G6PD [glucose-6-phosphate dehydrogenase] deficiency); not for pyelonephritis
Fosfomycin: lower tract (cystitis) only	**TMP-SMX (trimethoprim-sulfamethoxazole):** avoid 1st trimester (folate, neural tube); caution near term (kernicterus)
Gentamicin: pyelonephritis	**Fluoroquinolones** (cartilage), **tetracyclines** (teeth, bone), chloramphenicol: avoid

- ACOG (American College of Obstetricians and Gynecologists) 2017: nitrofurantoin and sulfonamides first-line in 2nd–3rd trimester; 1st trimester only if no alternative.
- **ASB and cystitis:** culture-directed, **4–7 days** (cures more than a single dose); **repeat culture** after treatment.

Pyelonephritis

- **Admit**; IV (intravenous) β-lactam (ampicillin plus gentamicin, ceftriaxone or aztreonam) until afebrile, then oral (cephalexin) to complete ≥14 days (P3: 10–14 days).
- No improvement by **72 h: renal ultrasound** (stone, obstruction, abscess).

Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery, or close culture surveillance (Cochrane 2015: close surveillance equally effective).
- **Also know:** urinary diversion or augmentation, SCI (spinal cord injury): most treat ASB in pregnancy and consider prophylaxis.

Diagram to draw

- Right hydroureter compressed at the pelvic brim.

> **Key points to remember:**
> - Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
> - Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
> - β-lactams safe throughout; no fluoroquinolones or tetracyclines.
> - Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

Also, from the condensed book (not in the paper answer):

- It peaks in the second trimester and resolves by up to 12 weeks postpartum; its grade correlates poorly with pain.
- Symptomatic hydronephrosis without a stone is treated conservatively. In a randomised trial stenting relieved pain better but caused stent symptoms, so reserve drainage for infection, failing function or genuine obstruction.

3 · Urinary tract infection and drugs

*See the paper answer **Bacteriuria in pregnancy and antibiotic policy** (Section 1, Ch 21).*

Also, from the condensed book (not in the paper answer):

- Non-steroidal anti-inflammatory drugs are contraindicated (third-trimester ductus closure); treat colic with small frequent doses of morphine. Defer intravesical bacillus Calmette–Guérin and mitomycin C until after delivery.
- The **Pregnancy and Lactation Labeling Rule (2015)** abolished the A, B, C, D and X categories in favour of a narrative risk summary.

4 · Imaging

*See the paper answer **Stones in pregnancy** (Section 11, Ch 96).*

Also, from the condensed book (not in the paper answer):

- First-trimester exposure carries teratogenic risk; later exposure mainly raises childhood cancer risk. Needed CT or nuclear studies must not be withheld.
- Give gadolinium only if it clearly improves outcome. Iodinated contrast crosses the placenta but is not teratogenic in animals. A mercaptoacetyltriglycine (MAG3) scan gives a fetal dose of **<5 mGy**.

5 · Surgery and anaesthesia

- No anaesthetic agent is teratogenic at standard doses. **Never deny or delay necessary surgery in any trimester**, but postpone elective surgery. The **second trimester is safest**.
- Involve obstetrics, monitor the fetal heart, give steroids if the fetus is viable and preterm, and operate where caesarean is available. At laparoscopy, insufflate to **10–15 mmHg**.
- Perinatal loss is driven by the disease, not the operation.

6 · Urolithiasis

*See the paper answer **Stones in pregnancy** (Section 11, Ch 96).*

*See the paper answer **Bacteriuria in pregnancy and antibiotic policy** (Section 1, Ch 21).*

Also, from the condensed book (not in the paper answer):

- Nephrostomy carries more preterm labour than a stent. Early ureteroscopy is favoured because repeated exchanges add morbidity and still leave the stone.

7 · Malignancy

Urological cancer is rare (≈ **13 per million** pregnancies) and is diagnosed late.

- **Renal cell carcinoma** is the commonest. Operate as in non-pregnant women, timed to trimester; nephrectomy can be done at caesarean. Avoid targeted therapy and immunotherapy.
- **Angiomyolipoma** is prone to rupture: embolise if stable, operate if unstable.
- **Bladder cancer**: transurethral resection is safe at any gestation; defer intravesical therapy.

8 · Placenta accreta spectrum

Grade	Depth
Accreta	Attached to myometrium
Increta	Invades myometrium
Percreta	Through serosa into bladder, ureters or rectum

- The main risk factor is **previous caesarean**. Blood loss is 3–5 L and maternal mortality up to **7%**.
- MRI maps bladder and ureteric invasion. At cystoscopy, **do not biopsy**.
- Perform **caesarean hysterectomy at ≈ 34 weeks** with the placenta in situ and retrograde ureteric catheters. With detrusor invasion, do a partial cystectomy rather than dissect the plane. Genitourinary injury occurs in ≈ **20–29%**.

9 · Pregnancy after reconstruction

- After augmentation, pyelonephritis occurs in up to **60%**. Use **serum human chorionic gonadotrophin**, because urine tests are falsely positive after contact with bowel mucosa.

Reconstruction	Mode of delivery
Augmentation with intact bladder neck	Vaginal delivery not contraindicated
Bladder neck reconstruction or artificial sphincter	Caesarean preferred by most
Ureterosigmoidostomy	Vaginal delivery risks anal sphincter damage
Bladder exstrophy	Planned caesarean

> **KEY POINT**
>
> At caesarean after reconstruction, use a **high midline incision** with a urologist present.

- **Myelomeningocele**: latex allergy in up to 60%; folic acid **4–5 mg** daily from 3 months before conception to 12 weeks.
- **Renal transplant**: wait at least 1 year with stable function and stop mycophenolate before conception.

22 Intraoperative Consultations

1 · Principles and ethics

- The patient is unconscious and never consented to the change of plan, so the consultant protects the absent patient without blaming the colleague.
- Accepting the call creates a patient–physician relationship, so disclose the reason fully to the family.
- Four factors decide whether to act now or temporise: patient condition, procedure morbidity, long-term consequences and reversibility.

	Act now	**Temporise and defer**
Example	Low-grade ureteric injury, treated with a stent	Complex proximal high-grade injury
Morbidity / reversibility	Low / reversible	High / often irreversible
Action	Proceed and disclose afterwards	Await shared decision making

Repair complex injury at once only if deferral is the greater harm. If surgeons disagree, default to the primary surgeon unless that is harmful.

2 · Consultation time-out

ALGORITHM

1. Assess the field first, because the procedure cannot be named until it is seen.
2. Introductions, patient identity, condition, procedure, blood loss, added time and concerns.
3. Verbal agreement by all.

3 · Urethral injury

*See the paper answer **Outline algorithm for difficult catheter placement.** (Section 1, Ch 15).*

Also, from the condensed book (not in the paper answer):

- In women, repair a sling or dissection injury vaginally with an interposition flap (peritoneum, omentum or labial fat).

4 · Bladder injury

P2 *Lower urinary tract trauma in pelvic fracture — American Association for the Surgery of Trauma (AAST) bladder grades I–V*

- The bladder is the commonest organ injured in pelvic surgery. Retropubic tape injures it in **3–9%** (transobturator 0.5%), so cystoscopy is routine. **Transurethral resection of bladder tumour is the commonest source**: extraperitoneal perforation needs catheter drainage, and intraperitoneal perforation usually needs open repair.
- If injury is suspected but not seen, perform a cystotomy. Close in 1–2 absorbable layers, fill-test, and drain with a urethral catheter for **7–14 days**. No routine suprapubic tube is

needed.

> **KEY POINT**
>
> Ask three questions: is it closed, is it drained, and are the ureters intact? At the trigone, when in doubt, reimplant.

5 · Ureteric injury

*See the paper answer **Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury** (Section 11, Ch 94).*

Also, from the condensed book (not in the paper answer):

- Iatrogenic injury is **75–80%** of ureteric injuries and >90% are in the distal third. Avoid distal ureteroureterostomy (poor blood supply); use reimplant, then psoas hitch, then Boari flap.
- Thermal injury extends beyond the visible zone, so debride generously rather than simply stent.

6 · Renal, bowel and vascular injury

- **PCNL** (percutaneous nephrolithotomy) is the commonest cause of renovascular injury. Venous bleeding is tamponaded by clamping the nephrostomy; arterial bleeding needs selective angioembolisation, not exploration.
- In vascular injury, apply pressure and never clamp blindly.

7 · Placenta percreta and documentation

- The **posterior bladder** is the commonest site of invasion. Never biopsy at cystoscopy (exsanguinating haematuria). Caesarean hysterectomy at **34–38 weeks** is the gold standard; ureters are involved in up to 7%.
- Write a separate note of observed facts even if no injury is found. Never write "careless" or "avoidable".

23 Assessment of Urologic and Surgical Outcomes

1 · Framework

Donabedian	Structure	Process	Outcome
Definition	Context of care	What is done	Effect on patients
Examples	Volume, nurse:bed ratio	Antibiotic and thromboprophylaxis	Mortality, readmission, quality of life
Strength	Easy and cheap	Detects problems early; less affected by case mix	Most meaningful to patients
Weakness	Largely non-modifiable	Valid only if evidence links it to outcome	Confounded, so needs risk adjustment

> **KEY POINT**
>
> Without case-mix adjustment, surgeons who treat high-risk patients look worse, which discourages them from treating such patients.

2 · Long-term end points and bias

SAQ Clinical trial methodology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 1, Q3a

Definition

- A **clinical trial** is a prospective study that assigns people to an intervention to test its effect on health outcomes. The **RCT (randomised controlled trial)** is the gold standard because randomisation balances known and unknown confounders.

Phases

Phase	Purpose	Subjects
0	Microdose pharmacokinetics	Very few
I	Safety, dose, toxicity	20–80, often volunteers
II	Activity, efficacy signal	Patients, about 100–300
III	Comparative efficacy versus standard	Hundreds to thousands; basis of approval
IV	Post-marketing surveillance	General population

Design

- **Parallel**, **crossover** (own control; stable chronic disease), **factorial** (two questions at once), **cluster** or **adaptive**.
- Hypothesis: **superiority**, **non-inferiority** (pre-set margin) or equivalence.
- Surgical innovation follows the **IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study).

Randomisation and allocation

- Simple, **block** (equal numbers), **stratified** (by prognostic factors) or **minimisation**.
- **Allocation concealment** (central or web-based) stops selection bias.

Blinding

- Single (patient), double (plus assessor) or triple (plus analyst). Surgery: sham procedures or blinded assessors.

Endpoints

- One **primary endpoint** sets sample size; secondary endpoints are exploratory.
- **Hard** (overall survival) versus **surrogate** endpoints, valid only if they meet the **Prentice criteria**; in prostate cancer only MFS (metastasis-free survival) does.
- **PROs (patient-reported outcomes)** with validated tools; complications graded by **Clavien–Dindo** or CTCAE (Common Terminology Criteria for Adverse Events).

Sample size and analysis

- Set α (type I error, usually 0.05), power (80–90%) and expected effect.
- **ITT (intention-to-treat)** preserves randomisation; per-protocol supports non-inferiority claims.
- Kaplan–Meier curves for time-to-event data.

Ethics and reporting

- Declaration of Helsinki, ICH-GCP (International Council for Harmonisation–Good Clinical Practice), informed consent, ethics committee, DSMB (data safety monitoring board).
- **Registration** before enrolment (CTRI [Clinical Trials Registry–India]).
- **CONSORT** (Consolidated Standards of Reporting Trials): checklist plus **participant flow diagram** (25 items in the 2010 version; updated 2025).
- **Also know:** lead-time bias; Will Rogers phenomenon.

Diagram to draw: CONSORT flow diagram: enrolment → randomisation → allocation → follow-up → analysis.

> **Key points to remember:**
> - Phase III RCT with concealed allocation and blinding is the gold standard.
> - Primary endpoint drives sample size; surrogates need the Prentice criteria.
> - ITT analysis; CONSORT flow diagram for reporting.

SAQ Evidence-based medicine and critical appraisal

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Evidence-based medicine and critical appraisal of research papers" · P4 SAQ "Describe evidence based medicine."

"quaLliy" is taken to mean quality, i.e. critical appraisal.

Definition

- **EBM (evidence-based medicine):** "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.

- **Critical appraisal:** systematic assessment of a paper's **validity, results and applicability**; the "Appraise" step of EBM.

Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR [absolute risk reduction], **NNT [number needed to treat] = 1/ARR**), precision (95% CI [confidence interval]), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

Critical appraisal: three core questions

1. Is it valid? (internal validity)

- **PICO question**; suitable design.
- **RCTs (randomised controlled trials):** true randomisation, **allocation concealment**, comparable groups at baseline, **blinding**, complete follow-up (>80%), **ITT (intention-to-treat) analysis**.
- **Bias** checked:

Bias	Meaning
Selection	Groups differ at entry
Confounding	Third factor (comorbidity, frailty) linked to exposure and outcome; needs risk adjustment
Performance, detection, attrition	Unequal care, outcome assessment or dropouts
Lead-time, length-time; stage migration (Will Rogers)	Screening or better tests make survival look longer
Publication	Negative studies unpublished

- **Pitfalls in appraisal:** selection, confounding, lead-time, length-time, stage migration (Will Rogers) and attribution bias.

2. What are the results?

- Effect: RR (relative risk), **ARR, NNT = 1/ARR**.
- Precision: **95% CI**; p value; **sample size and power** (usually 80%; α 0.05).
- **Surrogate end points** must meet the **Prentice criteria** (prostate cancer: only metastasis-free survival does); validated outcome tools (validated PROs [patient-reported outcomes]); Clavien–Dindo grading.

3. Does it apply? (external validity)

- Similar patients and setting; benefit outweighs harm.

Levels of evidence (Oxford CEBM [Centre for Evidence-Based Medicine], as adapted by EAU [European Association of Urology])

Level	Type of evidence / design
1a	Systematic review or meta-analysis of RCTs
1b	At least one RCT

Level	Type of evidence / design
2a	Well-designed controlled study without randomisation
2b	Quasi-experimental (cohort) study
3	Non-experimental: case-control, correlation, case reports, case series
4	Expert committee reports or opinion

- **Grades:** A (consistent level 1), B (level 2–3), C (level 4 or consensus); D (some systems) = no recommendation possible.
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (now used by EAU).
- AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

Reporting checklists and quality tools

Design	Checklist
RCT	**CONSORT** (Consolidated Standards of Reporting Trials)
Systematic review / meta-analysis	**PRISMA** (Preferred Reporting Items for Systematic reviews and Meta-Analyses)
Observational	**STROBE** (Strengthening the Reporting of Observational studies in Epidemiology)
Diagnostic accuracy	**STARD** (Standards for Reporting Diagnostic accuracy studies)
Surgical innovation	**IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study)

- **Also know:** Cochrane RoB 2 (risk-of-bias) tool; trial registration; conflicts of interest; Clavien–Dindo; validated PROs; Donabedian model.

Limitations of EBM

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

Diagram to draw

- Evidence pyramid and the three overlapping circles of EBM.

> **Key points to remember:**
> - Sackett 1996: best evidence + clinical expertise + patient values.
> - 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
> - Level 1a = systematic review of RCTs; GRADE: strong or weak.
> - NNT = 1/ARR; beware lead-time, length-time and confounding bias.
> - Three questions: valid, results, applicable.
> - Randomisation, concealment, blinding, ITT, follow-up >80%.
> - CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Also, from the condensed book (not in the paper answer):

- **Attribution bias** (cause of death misassigned) is the greatest limitation of disease-specific mortality, because a known cancer is often written on the certificate. **Detection bias** arises when one arm is imaged more.
- The **ICECaP** analysis (Intermediate Clinical End Points in Cancer of the Prostate) found that prostate-specific antigen (PSA) end points did not track overall survival, whereas **metastasis-free survival** did.

SAQ Recist Criteria.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3a

Definition and evolution

- **RECIST (Response Evaluation Criteria In Solid Tumours):** standard imaging rules for tumour response and progression.
- **WHO (World Health Organization) 1981** (bidimensional) → **RECIST 2000** (unidimensional) → **RECIST 1.1 (2009)**, now standard. Used in trials for metastatic RCC (renal cell carcinoma), urothelial and prostate cancer.

Lesions at baseline (RECIST 1.1)

Category	Definition
Measurable lesion	Longest diameter **≥10 mm on CT (computed tomography)**, ≥20 mm on chest X-ray, ≥10 mm with callipers
Measurable node	**≥15 mm short axis**; 10–15 mm pathological but non-measurable; <10 mm normal
Non-measurable	Small lesions, blastic bone metastases, effusions, ascites
Target lesions	Up to **5 (maximum 2 per organ)**; largest, reproducible; **sum of diameters** recorded
Non-target	All other lesions, recorded as present or absent

Response categories

Response	Target lesions
CR (complete response)	All disappear; nodes <10 mm
PR (partial response)	**≥30% fall** in sum versus **baseline**
PD (progressive disease)	**≥20% rise** versus **nadir** and **≥5 mm absolute**, or any **new lesion**
SD (stable disease)	Neither PR nor PD

- **Overall response** combines target, non-target and new lesions; confirm response at ≥4 weeks in single-arm trials.
- **End points:** ORR (objective response rate = CR + PR), PFS (progression-free survival), duration of response.

Limitations

- Size only, ignoring necrosis (TKIs (tyrosine kinase inhibitors)); bone poorly assessed; observer variability.

Modifications

Criteria	Use
iRECIST (2017)	Immunotherapy: first progression is "unconfirmed", confirmed only on the next scan, allowing for **pseudoprogression**
Choi	Density fall counts as response
PCWG3 (Prostate Cancer Working Group 3)	Bone: 12-week scan is the new baseline (flare), then **≥2 new lesions** = progression; PSA (prostate-specific antigen) progression ≥25% and ≥2 ng/mL above nadir; soft tissue by RECIST
PERCIST	PET (positron emission tomography) metabolic response

Diagram to draw: number line of change in sum: −30% (PR), +20% (PD).

> **Key points to remember:**
> - RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
> - PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
> - iRECIST for immunotherapy; PCWG3 for prostate.

3 · Complications

A complication is any deviation from the expected postoperative course. The Common Terminology Criteria for Adverse Events (CTCAE) are clinician-graded and organ-based, for oncology drugs.

CTCAE grade	Definition
1	Mild; observation only
2	Moderate; minimal intervention; limits **instrumental** activities of daily living
3	Severe; hospitalisation; limits **self-care** activities of daily living
4	Life-threatening; urgent intervention
5	Death

Clavien–Dindo grades by the therapy needed, not perceived severity, which reduces subjectivity.

Grade	Definition	Example
I	No drug, surgical or radiological treatment; antiemetics, analgesics, diuretics allowed; wound opened at bedside	Ileus after cystectomy managed with fluids
II	Other drugs, **including blood transfusion**	Transfusion after nephrectomy
IIIa	Intervention **without** general anaesthesia	Percutaneous drainage of a pelvic collection
IIIb	Intervention **under** general anaesthesia	Unplanned return to theatre
IVa	Life-threatening, intensive care; single-organ dysfunction (including stroke, not transient ischaemic attack)	Myocardial infarction after nephrolithotomy
IVb	Multiorgan dysfunction	Renal failure with respiratory distress
V	Death	—

It does not capture multiple complications, and anaesthesia type varies by institution.

4 · Predicting complication risk

Functional status, comorbidity and frailty predict complications across surgery; adjust for them in every non-randomised comparison.

ECOG	Status	Karnofsky
0	Fully active	100
1	Restricted strenuous activity; light work	80–90
2	Self-care, no work; up >50% of waking hours	60–70
3	Limited self-care; bed or chair >50%	40–50
4	Completely disabled	10–30
5	Dead	0

ECOG is the Eastern Cooperative Oncology Group scale.

- **Comorbidity** drives treatment choice, so it confounds. Administrative data showed higher mortality after transurethral resection of the prostate than open prostatectomy, but adjusting for comorbidity removed the difference.
- **Charlson Comorbidity Index** (predicts 1-year mortality): weight 1 for myocardial infarction, heart failure, vascular, pulmonary disease and diabetes; 2 for hemiplegia, renal disease and any malignancy; 3 for moderate–severe liver disease; **6** for metastatic cancer and AIDS.
- **Fried frailty phenotype**: frail means **≥3 of 5** criteria (weak grip, exhaustion, slow walk, low activity, unintentional weight loss).

5 · Patient-reported outcomes

SAQ Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion

Paper answer · P1

Asked as:

- Health related quality of life instruments (HRQOL) in urology. — *Summer 2018 Paper I Q3c (SAQ)*
- Assessment of quality of life after urinary diversion — *Practice Set 3 Paper I Q3c (SAQ)*

Type: SAQ (10 marks)

Definition

- **HRQOL (HRQoL, health-related quality of life):** the parts of life affected by health: physical, emotional and social function and the patient's bother; the patient's view of how disease and treatment affect physical, emotional and social function, plus **bother**. It differs from a **symptom score**, which only counts symptoms.
- Must be **patient-reported**; doctors underestimate dysfunction (urinary, sexual and bowel dysfunction).

Types of instruments

Type	Measures	Examples	Limitation
Generic	Domains common to all diseases; allow comparison across diseases	**SF-36** (Short Form-36), SF-12, **EQ-5D** (EuroQol five-dimension; 5 items + VAS (visual analogue scale); gives utilities for QALYs (quality-adjusted life-years)), Sickness Impact Profile	Insensitive to diversion-specific issues
Cancer-specific	Cancer effects	**FACT-G** (Functional Assessment of Cancer Therapy–General), **EORTC QLQ-C30** (European Organisation for Research and Treatment of Cancer core questionnaire)	General cancer domains
Disease-specific	Domains of one disease; more responsive	See below	
Bladder-cancer-specific		**FACT-BI** (40 items), **EORTC QLQ-BLM30** (muscle-invasive module, 30 items), **BCI (Bladder Cancer Index**, 34 items: urinary, bowel, sexual domains), FACT-VCI (Vanderbilt Cystectomy Index)	Weight symptoms of one diversion, so biased in cross-diversion comparisons
Neobladder-specific		IONB-PRO (23 items)	Not usable across diversions

Urology-specific instruments

- **Prostate cancer: EPIC (Expanded Prostate Cancer Index Composite)**, EPIC-26, UCLA (University of California, Los Angeles) Prostate Cancer Index, FACT-P.
- **Bladder cancer:** FACT-BI, **Bladder Cancer Index**, EORTC QLQ-BLM30 and NMIBC-24, FACT-Vanderbilt Cystectomy Index.
- **Kidney cancer:** FKSI (FACT–Kidney Symptom Index).
- **LUTS (lower urinary tract symptoms): IPSS (International Prostate Symptom Score)** 7 items + 1 quality-of-life item; BPH Impact Index.
- **Incontinence and OAB (overactive bladder): ICIQ** (International Consultation on Incontinence Questionnaire), **OAB-q**, King's Health Questionnaire, I-QOL.
- **Sexual function: IIEF** (International Index of Erectile Function; 15 items), SHIM (5 items; no bother, so not true HRQOL), FSFI (Female Sexual Function Index).
- **Others:** NIH-CPSI (chronic prostatitis), O'Leary–Sant (interstitial cystitis), Wisconsin Stone QOL, USSQ (ureteric stent).
- **PROMIS (Patient-Reported Outcomes Measurement Information System):** free item banks, **computer-adaptive testing**, common T-score.

Psychometric properties (choosing a tool)

- A good tool is **reliable** (Cronbach α >0.7), **valid** in that population and **responsive** to change.

Property	Meaning	Threshold
Internal consistency	Items agree	Cronbach α >0.7 (>0.9 = redundant)

Property	Meaning	Threshold
Test–retest reliability	Stable over time	r >0.7
Content (face) validity	Right items present	Expert judgement
Criterion validity	Agrees with gold standard	r ≥0.7
Construct validity	Convergent and divergent	Correlations
Responsiveness	Detects real change	**MCID (minimal clinically important difference)**: AUA (American Urological Association) Symptom Index 3.1 points; otherwise ≥½ SD

- Validity is population-specific; translations need revalidation.
- Development: define domains → generate items → pilot → psychometric testing (Likert scales).

Quality of life after urinary diversion

Domains to assess

- Urinary: continence, catheterisation, stoma problems.
- **Body image**, sexual function, bowel function, social and emotional function, global HRQoL.

Evidence: ileal conduit versus neobladder

- **No diversion proven superior.** Studies are mostly cross-sectional and retrospective, without baseline data or randomisation.
- Verdict depends on instrument: generic tools mostly show **no difference** (largest group of studies); EORTC tools tend to **favour neobladder**; the urinary-focused BCI **penalises neobladder** for leakage.
- **Consistent signal:** body image favours neobladder; urinary function favours conduit.
- Large prospective cohort: global HRQoL exceeded baseline in all diversions by 6 months.
- Neobladder: daytime continence improves over 6–12 months; **night-time incontinence** (up to 50%) is the main bother.
- **Also know:** informed patients have less decision regret; most adapt to their diversion; PROMIS (Patient-Reported Outcomes Measurement Information System).

Practical approach

- Baseline and serial validated questionnaires (3, 6, 12 months), a disease-specific plus a generic tool, and realistic counselling.

Diagram to draw: Pyramid: symptoms → function → bother → overall HRQOL. Bar sketch: body image (neobladder better) versus urinary function (conduit better).

Key points to remember:

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-Bl, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

Also, from the condensed book (not in the paper answer):

- Clinicians consistently underestimate urinary, sexual and bowel dysfunction, so collect outcomes directly from patients. A symptom score counts what is wrong; an HRQOL instrument measures function plus bother.
- The International Prostate Symptom Score has 7 items scored **0–35**; the Female Sexual Function Index has **19 items in 6 domains**. Pad tests cannot separate stress from urgency incontinence.

★ Putting it together: everyday decisions

1 · Which imaging test first?

Question	First test
Renal colic	Low-dose non-contrast computed tomography (CT); ultrasound in pregnancy and children
High-risk haematuria	CT urography and cystoscopy
Renal mass	Multiphase CT or magnetic resonance imaging (MRI); enhancement >15–20 Hounsfield units is suspicious
Obstruction, split function	Mercaptoacetyltriglycine (MAG3) diuretic renogram
Prostate cancer before biopsy	Multiparametric MRI
Acute scrotum	Doppler ultrasound, never delaying exploration
Urethral stricture	Retrograde urethrogram

2 · Before theatre

ALGORITHM

1. Assess function, comorbidity and frailty.
2. Stop warfarin 5 days before (international normalised ratio <1.5), bridging only for high thrombotic risk; stop direct oral anticoagulants 48 h before without bridging; withhold metformin on the day.
3. Culture urine before instrumentation; give prophylactic antibiotics.
4. Give extended 28-day low-molecular-weight heparin after pelvic cancer surgery.
5. Complete the World Health Organization checklist.

3 · Irrigation and the infected obstructed kidney

*See the paper answer **TUR syndrome** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- Use saline for bipolar resection, laser and ureteroscopy; never use sterile water for resection, as it causes haemolysis and hyponatraemia.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Tissue engineering in urology: principles, status and recent advances

Paper answer · P1 · P3 · P4 · merged from 3

Asked as: P1 LAQ "Principles of tissue engineering and its use in urology" · P3 SAQ "Describe principles of tissue engineering and what its status in managemet of urological diseases is." · P4 LAQ "Tissue engineering in urology, including recent advances"

Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Regenerative medicine** is the wider term: cells, scaffolds, growth factors and gene therapy.
- **Why in urology:** bowel used for bladder and diversion causes mucus, stones, metabolic acidosis, infection and malignancy; grafts for long urethral defects are scarce. The aim is **to replace "wrong tissue" with "right tissue"**.

Principles: the triad plus bioreactor and blood supply

``` Cells + Scaffold + Signals (growth factors, mechanical cycling in a bioreactor) -> construct -> implanted -> vascularised and remodelled by the host

Biopsy -> cells expanded in culture -> seeded on scaffold (+ signals) -> conditioned in bioreactor -> implanted -> host vascularises and remodels ```

### Cell sources

| Source | Features, advantages and limitations |
|---|---|
| **Autologous differentiated cells** (urothelial and smooth muscle cells from a bladder biopsy) | Standard; no rejection; expand well in culture; unusable if the bladder is malignant or diseased |
| **Adult stem cells:** bone marrow or adipose mesenchymal, muscle-derived, **urine-derived** | Easy, non-invasive harvest; act mainly by paracrine (growth factor) effects |
| **Embryonic stem cells** | Pluripotent; ethical issues, teratoma risk |
| **iPSCs** (induced pluripotent stem cells) | Adult cells reprogrammed; patient-specific, no ethical issue; tumour risk |

- Gene-edited and gene-modified cells are being explored. Cell sheet engineering builds layers without a scaffold.

### Scaffolds (biomaterials)

- **Requirements:** biocompatible; biodegradable at the rate new tissue forms; porous for cell entry, attachment and vessel ingrowth; mechanically suited (a compliant bladder, a non-collapsing tube).

| Type | Examples | Feature |
|---|---|---|
| **Natural polymers** | Collagen, alginate, silk fibroin | Good cell recognition; weak |
```

Type	Examples	Feature
Acellular tissue matrices	Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis	Keep native architecture and stored growth factors
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer, poly(lactic-co-glycolic) acid)	Controlled shape and degradation; no bioactivity
Newer	Electrospun nanofibres, hydrogels, bio-ink; **"smart" scaffolds** releasing growth factors such as VEGF (vascular endothelial growth factor); silk fibroin	Mimic extracellular matrix

- **Seeding:** urothelial cells on the luminal side and smooth muscle cells on the outer side, copying native layers.
- **Unseeded (acellular)** matrices rely on host cells migrating in, so they work only for short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

Signals

- Growth factors (for example VEGF, for vessels) released from the scaffold, extracellular-matrix cues and mechanical stimuli direct cell growth and differentiation.

Bioreactors

- Culture chambers that perfuse nutrients and oxygen through thick constructs and apply **physiological conditioning**, such as cyclic stretch and filling to train bladder smooth muscle, so the tissue is functional before implantation.
- Types: static culture, spinner flask, rotating wall, perfusion and mechanical-stretch bioreactors. They also allow monitoring and sterile, standardised production.

Vascularisation and innervation (the main barrier)

- Oxygen diffuses only about **100–200 µm**, so thick constructs die centrally without vessels.
- Strategies: **omental wrap** (used by Atala), angiogenic factors, co-seeding endothelial cells, built-in (printed) vascular channels by bioprinting.
- Innervation and coordinated contraction cannot yet be recreated.

Clinical applications and status in urology

1. Bladder

- **Atala et al. (2006):** first seeded biodegradable bladder construct (autologous urothelial and muscle cells on collagen–PGA, wrapped in omentum) for augmentation in **7 myelomeningocele patients**; compliance improved.
- **Industry (Tengion) multi-institution phase II trial** in spina bifida: **no clinically or statistically significant improvement in compliance or capacity** (Chapter 48).
- Chapter 48 conclusion: a bladder with specialised urothelium and contractile muscle has **not** been achieved; it remains experimental.
- Chapter 111: an engineered bladder gives **storage only**; innervation for voluntary emptying cannot yet be recreated.

2. Urinary diversion: Neo-Urinary Conduit (Tengion, NCT01087687) (Chapter 135)

- Seeded biodegradable conduit to avoid bowel. Small enrolment; conduits survived about 250 days on average; **half developed stenosis or stricture.**
- Lessons: stiff, non-collapsible material; no cells formed around the connective-tissue sleeve; **no blood supply**.

3. Urethra

- Acellular bladder matrix onlay for hypospadias and short strictures: success in short onlay repairs.
- **Raya-Rivera et al. (2011, Lancet):** tubular PGA–PLGA scaffolds seeded with autologous cells in **5 boys** with posterior urethral defects; functional urethras at up to 6 years.
- Tissue-engineered oral mucosa (MukoCell) for urethroplasty, avoiding donor-site morbidity.

4. Other organs

- **Sphincter, detrusor:** autologous muscle-derived cell injection for SUI (stress urinary incontinence), investigational; for underactive detrusor, open-label studies improved voiding efficiency (Chapter 111). Biodegradable scaffold slings (Chapter 48).
- **Vagina:** engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59).
- **Penis:** engineered corporal tissue and tunica grafts (stem cell-seeded SIS in Peyronie disease), animal or experimental stage. ED (erectile dysfunction) stem cell therapy is investigational (Chapter 72).
- **Kidney:** decellularised–recellularised scaffolds (animal); renal assist devices; kidney organoids.
- **Testis:** spermatogonial stem cell banking for prepubertal cancer patients (future, Chapter 4).

Organ	Status
Bladder	**Atala (2006):** seeded collagen–PGA construct with omental wrap in **7 myelomeningocele patients**, compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Still experimental (Chapter 48)
Conduit	Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis or stricture**; stiff material, no blood supply (Chapter 135)
Urethra	Acellular matrix onlay for short strictures and hypospadias; **Raya-Rivera (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa avoids donor-site morbidity
Sphincter, detrusor	Autologous muscle-derived cells for SUI (investigational); biodegradable scaffold slings; underactive detrusor: open-label improvement in voiding efficiency (Chapter 111)
Vagina	Engineered vaginas in MRKH syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59)
Penis	Corporal tissue and tunica grafts: animal or experimental; stem cells for ED investigational
Kidney	Decellularised–recellularised scaffolds (animal), kidney organoids, renal assist devices: experimental
Testis	Spermatogonial stem cell banking in prepubertal cancer patients

- **Overall status:** still experimental except short urethral onlays.

Recent advances

3D bioprinting

- Layer-by-layer deposition of **"bio-ink"** (cells in hydrogel), with or without polymer, driven by patient CT (computed tomography) or MRI (magnetic resonance imaging) data.
- Advantages: exact patient-specific shape, several cell types placed precisely, built-in channels for vessels.
- Current urological use: 3D-printed (non-living) kidney and prostate models already aid planning and teaching; bioprinted urethra and bladder constructs are still at the animal stage.

Organoids

- 3D self-organising mini-organs grown from stem cells or tumour tissue.
- **Kidney organoids** from iPSCs: model development and disease, drug toxicity testing.
- **Tumour organoids** (bladder, prostate): personalised drug testing and biology (WNT/β-catenin work, Chapter 129).

Whole-organ decellularisation

- Kidneys are stripped of cells by detergent perfusion, leaving the matrix and vascular tree, then recellularised with the patient's cells; still at the animal stage. Renal assist devices are a parallel route.

Other

- Robotic bladder auto-transplantation and robotic retrieval of a **vascularised composite bladder allograft** (Chapter 135): a transplant, not engineered, route to "right tissue" that avoids bowel but needs immunosuppression.
- Smart (growth factor-releasing) scaffolds; urine-derived stem cells; gene-edited and gene-modified cells; cell sheet engineering.

Challenges

- **Vascularisation** of thick constructs (diffusion limit), **innervation** and contractility.
- Fibrosis and contracture; scale-up, cost, culture time; regulatory approval.
- Stem cell tumour risk. Promising pilots followed by disappointing controlled trials.

Future

- Vascularised bioprinted tissue, off-the-shelf cell-free smart scaffolds, combined gene and cell therapy.

Diagram to draw

- Biopsy, then cell culture, then seeding on scaffold, then bioreactor, then implant with omental wrap.

Key points to remember:

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
- Neo-urinary conduit: half stenosed; no blood supply.
- Raya-Rivera 2011: seeded tubular urethras in 5 boys.
- Still experimental except short urethral onlays.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

LAQ Robotics in urology (including robotic vs standard laparoscopy)

Paper answer · P4

Asked as:

- Role of robotics in urology — *Summer 2015 Paper IV Q1 (LAQ)*
- Discuss Robot and its role in urology. — *Winter 2017 Paper IV Q1 (LAQ)*
- Robots in urology — *Practice Set 4 Paper IV Q2 (LAQ)*
- Robotic vs std lap surgeries — *Practice Set 3 Paper IV Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2017-Winter Paper 4, Q1; 2015-Summer Paper 4, Q1) (robotics also answered in 2017-Winter Paper 4, Q1 and 2015-Summer Paper 4, Q1)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- Robotic surgery is **robot-assisted laparoscopy** with a **master–slave telemanipulator** (surgeon console, patient-side cart, vision cart). Access and physiology are the same; the interface differs.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled. RARP now dominates in the USA.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)); "Mona" first human trial (1997).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP (**Binder and Kramer**) in **2000**; robotic adrenalectomy 1999; first paediatric robotic procedure **2002**; robotic RPLND (retroperitoneal lymph node dissection) 2006.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) view magnified about 10 times, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; most have a 10-use limit; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
Standard (2000)	Three arms
S (2006) / Si (2009)	High definition, fourth arm / dual console for training
Xi (2014)	8-mm camera in any port, finer arms, side docking (docks from either side), no stirrups
SP (2018)	All instruments through one 2.5-cm cannula
da Vinci 5 (2024)	Force feedback

- Patents expired in 2019 (after patent expiry), so other platforms appeared (e.g. **Hugo, Versius** and the low-cost Indian **SSI Mantra**).
- **Also know:** ports 8–10 cm apart to avoid arm collisions; Xi docks from any side; retroperitoneal cases dock over the head.

Robotic vs standard laparoscopy: comparison

Feature	Standard laparoscopy	Robotic
Vision	Mostly 2D; assistant holds camera	**3D**, 10× magnified, surgeon controls camera
Instruments	Rigid, **4 DOF (degrees of freedom)**; fulcrum effect reverses movement	**EndoWrist, 7 DOF**, 90° articulation; intuitive movement
Precision	Tremor transmitted	**Tremor filtration, motion scaling**
Ergonomics	Standing, awkward	Seated, less fatigue
Suturing	Hard; steep learning curve	Easy; shorter learning curve
Haptic feedback	Present (limited)	Absent (force feedback only in newest system)
Port placement	Flexible	Ports 8–10 cm apart; docking time
Bedside	Surgeon at patient	Surgeon remote; emergency undocking delays
Cost	Low, reusable	High capital; instruments limited to about 10 uses
Failure	—	Robot failure about 0.4%

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost; cost-effective only at high volume
Tremor filtration, seated ergonomics	Limited haptic feedback
Less blood loss and shorter stay than open; fewer conversions than laparoscopy	Docking time, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, with less blood loss and shorter stay; **Coughlin 2018**: equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable. Bladder neck stricture under 2% vs up to 16% open.
- **Retzius-sparing** and **"hood"** techniques preserve continence structures and give earlier continence; ERAS (enhanced recovery after surgery) allows 23-hour or same-day discharge. Robotic surgery also widens candidacy (prior mesh, prior pelvic surgery) and allows salvage surgery.
- Versus laparoscopy: laparoscopic prostatectomy stalled on the vesicourethral anastomosis; the robot made it routine and dominant. A small RCT (randomised controlled trial; Porpiglia, 2013) favoured robotic continence and potency.

2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy (robotic vs laparoscopic gives): **fewer conversions**, **shorter warm ischaemia**, less (smaller) eGFR (estimated glomerular filtration rate) loss/fall and shorter stay; complications and margins similar. So more complex and hilar tumours are done minimally invasively.
- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.
- ICG (indocyanine green) guides selective clamping.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), with less blood loss and fewer transfusions (less transfusion).
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications. Continent diversion fell (11% in iROC).

4. Other procedures

Procedure	Role of the robot / why the robot helps
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in children and infants (small working space: about 1 L vs 6 L in adults); similar success to laparoscopy; easier suturing
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND (retroperitoneal lymph node dissection)	Template, nerve-sparing; short stay
Adrenalectomy, nephroureterectomy	Equivalent to laparoscopy; adrenalectomy suits the retroperitoneal approach
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, VVF (vesicovaginal fistula) / fistula repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

- **Radical nephrectomy, adrenalectomy:** no outcome advantage over laparoscopy; robot only adds cost.
- Outcomes depend more on the **surgeon** than on the platform.

Choosing robot vs laparoscopy

- Robot: complex excision plus reconstruction. Laparoscopy: extirpation, low-resource settings.

Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

Physiology and positioning

- **CO_2 pneumoperitoneum** (12–15 mm Hg, lower in children; another source: work at 10–12 mm Hg) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery increases venous return and cardiac output, and raises intracranial and intraocular pressure.
- The only strict contraindication to minimally invasive surgery is inability to tolerate pneumoperitoneum.

Complications specific to robotics

Problem	Details
Steep Trendelenburg (~22–25°)	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury (up to ~6%)	Brachial plexus (avoid shoulder braces); femoral, peroneal
Compartment syndrome, rhabdomyolysis	Long lithotomy or flank position; fasciotomy if >30 mmHg with symptoms
Thermal bowel injury / bowel burns	Insulation failure, capacitive coupling; presents late
Robot malfunction	About 0.4% (another source: <1%); half found before the patient enters; rarely forces conversion
Port-site hernia	Close larger port sites; transverse incisions cut it from 5.4% to 0.4% after RARP

- **Prevention**: secure inaccessible arms and lines before tilting, pad pressure points, limit tilt and lithotomy time, protect the eyes.

Training and credentialing

- Box trainer, **VR (virtual reality)** simulator (simulators), animal and cadaver labs, then proctored cases (proctoring). **FRS (Fundamentals of Robotic Surgery)** curriculum improved skills in an RCT. Five-day mini-residencies also help. Dual console and simulators for training. Proficiency comes after about 40 cases, but outcomes keep improving (outcomes improve) with volume.

Recent advances

- Single-port transvesical and extraperitoneal surgery; da Vinci SP (single port); force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency; latency limits it).
- New platforms (Hugo RAS, Versius, Indian SSI Mantra) may cut costs.

- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

Diagram to draw: console, vision cart and patient cart; RARP port placement; EndoWrist (7 DOF) beside a rigid instrument (4 DOF).

> **Key points to remember:**
> - da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
> - Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
> - RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
> - Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
> - Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
> - Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
> - Robot made prostatectomy routine; no benefit for simple extirpation.
> - Surgeon matters more than platform.

LAQ Natural orifice endoscopic surgery.

Paper answer · P4

Type: LAQ (25 marks)
Source: 2022-Summer Paper 4, Q1

Taken as NOTES (natural orifice transluminal endoscopic surgery) and its relatives (hybrid NOTES, LESS [laparoendoscopic single-site surgery]).

Definition and concept

- **NOTES:** entry into the peritoneal or retroperitoneal cavity through a deliberate incision in a hollow viscus reached through a natural orifice (mouth, vagina, bladder, rectum). The aim is **"scarless" surgery**: no abdominal wall wound, so no wound pain, infection or hernia.
- It differs from **endoluminal** surgery (e.g. ureteroscopy), which stays inside the lumen.
- **Pure NOTES:** all instruments through the orifice. **Hybrid NOTES:** natural orifice plus one or more small abdominal ports. Hybrid is what is actually used in humans.
- Chapter 142 places NOTES at the end of a spectrum: transperitoneal → retroperitoneal → hand-assisted → LESS → SP (single-port) robotic → NOTES. **Each step to fewer incisions buys cosmesis and less pain at the cost of working space and triangulation.**

History (milestones)

Year	Event
2002	Transvaginal laparoscopic nephrectomy in pigs (Gettman, Cadeddu)
2004	Transgastric peritoneoscopy in pigs (Kalloo)
2006	NOSCAR (Natural Orifice Surgery Consortium for Assessment and Research) white paper listed barriers
2008–2010	Hybrid, then pure, transvaginal nephrectomy in humans

Routes of access

Route	Features	Urological use
Transvaginal (posterior colpotomy)	Commonest; easy closure under vision; large specimens pass; women only	Nephrectomy, donor nephrectomy, specimen extraction
Transgastric	Upper abdomen reachable; gastrotomy closure and leak are the main risk	Experimental (peritoneoscopy, renal surgery in animals)
Transvesical	Sterile urine; bladder dome port; small calibre; closure difficult	Peritoneoscopy (animal, Lima), transvesical access to pelvis
Transurethral	Urologist's familiar route	Experimental transurethral prostatectomy in cadavers
Transcolonic/transanal	Faecal contamination risk	Rarely urological

Hybrid NOTES and natural orifice extraction

- Standard laparoscopy does the dissection; **the vagina is the access for a camera or instrument and the extraction route** (Chapter 142). This is the only NOTES form with real clinical urological use.
- **NOSE (natural orifice specimen extraction):** kidney removed through the vagina after laparoscopic nephrectomy, avoiding the extraction incision, which is the largest and most painful wound.
- **NOTES-assisted donor nephrectomy:** laparoscopic or robotic donor nephrectomy with **transvaginal kidney extraction**. Reported benefits: less pain, faster recovery, no extraction scar; warm ischaemia acceptable.

Steps of transvaginal (hybrid) nephrectomy

1. Lithotomy then modified flank; bowel and vaginal preparation; antibiotics.
2. Umbilical port (camera or instrument) plus one or two small ports.
3. **Posterior colpotomy** under laparoscopic vision; long port or flexible scope through the vagina.
4. Colon reflection, hilar dissection, ureter and vessels stapled.
5. Kidney in a bag, delivered through the colpotomy.
6. Colpotomy closed with absorbable sutures under vision.

LESS: the close relative

- All ports through **one skin incision** (umbilicus or below the waistline) using several low-profile trocars or a multichannel port (TriPort; da Vinci SP 25–27-mm entry guide).
- Problems: **no triangulation**, instrument clashing, in-line view. Solutions: articulating or bent instruments, flexible-tip scopes, instruments of different lengths.
- Evidence (Chapter 142): randomised trial versus laparoscopy showed the same time, blood loss and complications, with less analgesia. Donor meta-analysis: lower pain scores. Oncological outcome unchanged.
- Drawback (Chapter 20): **more incisional and umbilical hernias** because of the larger single incision.
- **SP robot (FDA [Food and Drug Administration] approval 2018)** has revived single-site surgery.

vNOTES (vaginal NOTES)

- A gynaecological development: a single-port device in the vagina gives access to the peritoneum.
- **HALON RCT (randomised controlled trial):** vNOTES hysterectomy was non-inferior to laparoscopic hysterectomy, with more same-day discharge and less pain.
- Urogynaecological uses: vNOTES apical prolapse repair (uterosacral suspension), adnexal surgery.

Challenges (NOSCAR list)

- **Safe access** and **secure closure** of the viscerotomy (leak, fistula); peritoneal **infection**.
- **Spatial orientation**: retroflexed, off-axis view.
- **Platform**: flexible scopes lack stiffness, traction and triangulation.
- Control of **haemorrhage** and other complications.
- Training and cost; dyspareunia and fertility concerns after colpotomy; transvaginal route unavailable in men.
- Chapter 142: adoption is limited by **no clear therapeutic benefit**, safety concerns and very selective patients; **pure NOTES has very few urological cases.**

Recent advances

- **Flexible robotic platforms** (Chapter 39): MASTER (NOTES, haptics), ViaCath.
- **Transvesical single-port** (LESS or da Vinci SP) surgery for diverticulectomy and foreign bodies (Chapter 124).
- Magnetic anchoring and guidance systems to restore triangulation.

Also know: LESSCAR consortium standardised the term LESS; "embryonic NOTES" (umbilicus as the natural scar); hidden-incision endoscopic surgery (HIdES) in children.

Diagram to draw: sagittal female pelvis showing posterior colpotomy with a transvaginal port and kidney extraction, plus umbilical port.

> **Key points to remember:**
> - NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
> - Transvaginal extraction removes the most painful wound.
> - LESS: one incision, lost triangulation, more hernias.
> - Barriers: closure, infection, orientation, platform; no proven benefit yet.

SAQ Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa

Paper answer · P1 · merged from 3

Asked as: P1 SAQ "Pioneers of urology: Hopkins, Huggins, Calne, Gaur and Asopa" · P1 SAQ "Andrew Novick." · P1 SAQ "Christian Chaussey"

Names read as Prof. Harold Hopkins ("Herald"), Dr Charles Huggins, Prof. Sir Roy Calne ("Caine"), Dr D. D. Gaur ("Gaue"), Hari Asopa, Andrew C. Novick and Christian Chaussy ("Chaussey"). The question is taken to ask for each person's contributions to urology. Novick is answered mainly through nephron-sparing surgery, and Chaussy through the history and principles of SWL.

Prof. Harold Hopkins (British physicist)

- **Fibreoptics (1954, with Kapany):** coherent bundles of glass fibres transmit an image by total internal reflection (Tyndall's principle). This made flexible endoscopes possible.
- **Rod-lens system (1966):** the old scope used thin glass lenses separated by long air spaces. Hopkins replaced them with **long glass rods separated by short air gaps**. Result: more light transmission, a brighter and wider image, and a **smaller calibre**. Developed commercially with Karl Storz.
- It underpins all modern rigid endoscopes and enabled the first rigid ureteroscope (1979).

Dr Charles B. Huggins (1901–1997)

- Canadian-born American urologist, University of Chicago.
- **1941, with Clarence Hodges:** castration (or oestrogen) relieved far-advanced disease and caused **regression of metastatic prostate cancer**, with falling serum acid phosphatase, pain relief and weight gain; injected androgens made it worse. This proved that **prostate cancer is hormone (androgen) dependent**.
- Used serum **acid phosphatase** to monitor response: the first tumour marker.
- **1945:** bilateral adrenalectomy for castration-resistant disease (relapse after castration, suspecting adrenal androgens); modest, brief responses; morbidity high.
- **Nobel Prize in Physiology or Medicine, 1966** (shared with Peyton Rous) for hormonal treatment of prostate cancer: the basis of hormonal cancer therapy.
- **Legacy:** basis of all ADT (androgen deprivation therapy): orchidectomy, oestrogens, LHRH (luteinising hormone-releasing hormone) agonists and antagonists, and later abiraterone (blocking adrenal androgens).
- **Also know:** canine prostate physiology; adrenalectomy for breast cancer.

Prof. Sir Roy Calne (Cambridge transplant surgeon)

- Showed in dogs that **6-mercaptopurine and then azathioprine** prolong renal allograft survival; azathioprine with steroids became standard immunosuppression.
- Performed the **first liver transplant in the UK (1968)**.
- **First clinical use of ciclosporin** in kidney transplantation (late 1970s), the calcineurin-inhibitor era; acute rejection fell markedly.

Dr D. D. Gaur (Mumbai, India)

- **1992: balloon dissection of the retroperitoneum.** A size 7 surgeon's glove tied on an 8-Fr red rubber catheter, inflated inside the retroperitoneum, creates a working space.
- This made **retroperitoneoscopy** practical and cheap; used for ureterolithotomy, nephrectomy and other renal surgery.

Hari S. Asopa (Indian urologist)

1. Asopa dorsal inlay urethroplasty (2001)

- Single-stage graft urethroplasty for anterior urethral stricture by a **ventral sagittal urethrotomy** approach.
- **Steps:** expose the urethra ventrally without circumferential mobilisation; open the urethra ventrally through the stricture; incise the **dorsal urethral plate** in the midline down to the corpora; **inlay** a free graft (buccal mucosa or penile/preputial skin) and quilt it to the tunica albuginea; close the urethra over a catheter.

- **Rationale:** the graft lies on the well-vascularised corporal bed; no mobilisation preserves spongiosal blood supply; suits bulbar and penile strictures.
- Ranks with **Barbagli dorsal onlay** and **Kulkarni one-sided** repair among dorsal graft techniques.

2. Asopa hypospadias repair (1970s)

- One-stage repair using a **vascularised inner preputial island flap** still attached to the outer preputial skin, transferred ventrally to form the neourethra; a forerunner of the Duckett transverse preputial island flap.

Andrew C. Novick

Who he was

- A **Cleveland Clinic** urologist and long-time chairman of its urology department. He was a world authority on renal surgery, an editor of *Campbell-Walsh Urology* (9th edition), and died in 2008.

Contributions

1. **Nephron-sparing surgery (NSS) / partial nephrectomy (PN):** he established its technique, indications and long-term cancer control. His group showed tumours ≤4 cm did as well with NSS as with radical nephrectomy, supporting the T1a/T1b split.
2. **Renovascular surgery:** aortorenal, splenorenal and hepatorenal bypass; **bench (ex vivo) surgery with autotransplantation** for complex lesions.
3. **Renal transplantation** surgery.

Indications for PN

Category	Situations
Imperative	Bilateral RCC (renal cell carcinoma); solitary functioning kidney
Relative	Contralateral kidney threatened (stones, stenosis, diabetes, hypertension); hereditary RCC
Elective	cT1a (≤4 cm) with a normal other kidney; now larger selected tumours

- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous thrombus.

Principles of NSS

1. Three-dimensional CT (computed tomography) maps tumour, vessels and collecting system.
2. **Early hilar control.** Keep warm ischaemia under about 25–30 minutes; if longer, use **ice-slush hypothermia** and mannitol.
3. Excise with a **negative margin** (width immaterial); intraoperative ultrasound for endophytic tumours.
4. **Watertight collecting-system closure**, vessel ligation, capsular renorrhaphy.
5. Drain for urine leak. - Function afterwards depends on parenchymal **quality and quantity preserved**; ischaemia matters less. At least 20–30% of one kidney is needed to avoid dialysis.

Legacy

- PN is now the AUA (American Urological Association) and EAU (European Association of Urology) standard for cT1a tumours.

Christian Chaussy (Munich): extracorporeal shock wave lithotripsy

Contribution and history

- Engineers at **Dornier** found by chance that shock waves from raindrop and micrometeorite research affected tissue.
- Chaussy, with Brendel and Schmiedt in **Munich**, developed the technique in animals and performed the **first human SWL (shock wave lithotripsy) on 7 February 1980 on the Dornier HM1**. They reported it in the *Lancet* in 1980.
- **HM3** (water bath, electrohydraulic spark, biplanar fluoroscopy): first installations Stuttgart 1983 and Indianapolis 1984. It remains the reference for efficacy.
- SWL made most open stone surgery obsolete.

Principles

- A shock wave is a short acoustic pulse: **peak positive 30–100 MPa, then a negative tail of about −10 MPa**.
- Generated at **F1**, focused onto the stone at **F2** through water or gel coupling (water and tissue have similar acoustic impedance); targeting by fluoroscopy or ultrasound.

Energy source	Feature
Electrohydraulic (HM1, HM3)	Underwater spark, ellipsoid reflector; most effective; electrode wear
Piezoelectric	Ceramic elements; long life, little pain; small focal zone, low energy
Electromagnetic (modern)	Coil and membrane with lens; consistent; dominant today

Mechanisms of fragmentation

- **Spallation:** reflected wave at the back surface creates tension.
- **Shear and tear** inside the stone.
- **Quasistatic squeezing** and **dynamic squeezing** (strongest force in initial breakage).
- **Cavitation:** collapsing bubbles erode the surface; excess bubbles shield the stone and injure tissue.

Practical points

- Better results at **60–90 shocks/min**, with power **ramping** and air-free coupling.
- Poor results: stone >1000 HU (Hounsfield units), skin-to-stone distance >10 cm, lower pole >1 cm.
- **Contraindications:** pregnancy, uncorrected coagulopathy, untreated UTI (urinary tract infection), distal obstruction, aortic aneurysm >4 cm.
- **Also know:** burst wave lithotripsy and ultrasonic propulsion are newer descendants.

Diagrams to draw

- Conventional lens scope versus Hopkins rod-lens scope (glass rods with short air gaps).
- Dorsal inlay: urethra opened ventrally, dorsal plate incised, graft on corpora (cross-section).
- PN steps: clamp, ice slush, excision, closure, renorrhaphy.
- Ellipsoid reflector with spark at F1 and stone at F2.

Key points to remember:

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966; 1945 adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery; PN indications imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer; negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1; HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

SAQ History of Laparoscopy

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3b

Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum; it grew from optics to robotic surgery.

Optics and access

Year	Pioneer	Milestone
1806	**Bozzini**	**Lichtleiter**, candle-lit light conductor
1877	Nitze	Distal electric light and lenses
1901	**Kelling**	Insufflated abdomen of a **dog**
1910	**Jacobaeus**	First human series; "laparoscopy"
1938	**Veress**	Spring-loaded needle
1966	**Hopkins**	**Rod-lens system**
1970s–80s	**Semm**	Automatic insufflator; operative laparoscopy
1970s	Hasson	Open access
1987	Mouret	Video cholecystectomy
- **Also know:** Desormeaux 1853, first therapeutic endoscopy; fibreoptics (Tyndall's principle) for flexible scopes; CO_2 became the standard insufflant.		

Urological milestones

Year	Pioneer	Milestone
1976	**Cortesi**	Impalpable testis, first urological use
1990–91	**Clayman**	**First laparoscopic nephrectomy**; nephroureterectomy 1991
1992	**Gaur (India)**	**Balloon retroperitoneal access**
1995	**Ratner**	Live-donor nephrectomy
1997	Schuessler	Laparoscopic radical prostatectomy (slow, not adopted)
2000	Guillonneau & Vallancien	Standardised prostatectomy
2000	**da Vinci**; **Binder**	Robot cleared in US; first robotic prostatectomy
2000s	**Menon**	Vattikuti Institute Prostatectomy
2007	Raman; Rane	Single-port nephrectomy

Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view.
- da Vinci generations: S 2006 (fourth arm), Si 2009 (dual console), Xi 2014 (flexible docking), SP 2018 (single port).

Indian contribution and recent advances

- Gaur's balloon made retroperitoneoscopy practical; indigenous robot SSI Mantra with telesurgery.
- New robots (Versius, Hugo, Hinotori), telesurgery, haptic feedback.

Diagram to draw: Timeline 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

> **Key points to remember:**
> - Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
> - Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
> - da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

SAQ Flow cytometry

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3d

Definition

- **Flow cytometry** measures physical and fluorescent properties of **single cells in suspension** flowing one at a time through a laser beam, thousands per second.

Principle

1. **Sample:** fresh cells, urine, bladder washings, or nuclei from paraffin blocks.
2. **Staining:** fluorochrome-labelled antibodies to CD (cluster of differentiation) antigens, or **DNA-binding dyes** (propidium iodide, DAPI [4′,6-diamidino-2-phenylindole]).
3. **Hydrodynamic focusing** lines the cells up in single file.

4. **Laser interrogation:** - **Forward scatter** = cell size. - **Side scatter** = internal complexity (granularity). - **Fluorescence** = amount of bound antibody or DNA.
5. Photomultiplier detectors → histograms and dot plots; **gating** selects populations. - **FACS (fluorescence-activated cell sorting)** physically separates selected cells.

DNA content analysis

- **DNA index** = tumour G0/G1 peak ÷ normal diploid peak. DI 1.0 = diploid; about 2.0 = tetraploid; any other = **aneuploid**.
- **S-phase fraction** estimates proliferation.
- Aneuploidy and high S-phase generally mean higher grade and worse prognosis.

Applications in urology

Area	Use
Bladder cancer	Ploidy of bladder washings: aneuploidy with high grade, CIS (carcinoma in situ), recurrence; replaced by **UroVysion FISH (fluorescence in situ hybridisation)**
Upper-tract urothelial cancer	Aneuploidy → poor long-term survival
Prostate cancer	Aneuploid tumours progress more after treatment than diploid
Neuroblastoma	**Hyperdiploid favourable**; diploid or tetraploid adverse; part of COG (Children's Oncology Group) risk
Urinalysis	Automated urine analysers count RBC (red blood cells), WBC (white blood cells) and bacteria
Transplantation	**Flow cytometric crossmatch**: more sensitive than CDC (complement-dependent cytotoxicity) crossmatch for donor-specific antibody; Luminex single-antigen beads; CD3 count to guide antithymocyte globulin
Male infertility	Sperm DNA fragmentation (SCSA [sperm chromatin structure assay]); haploid cells in testicular aspirates
Lymphoma	Immunophenotyping of testicular or renal lymphoma

Advantages and limitations

- **Advantages:** rapid, objective, quantitative, many parameters per cell.
- **Limitations:** architecture lost; near-diploid aneuploidy missed; normal cells dilute tumour peaks; ploidy now adds little independent value.

Diagram to draw: DNA histogram with diploid G0/G1 peak, S-phase and aneuploid peak.

> **Key points to remember:**
> - Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
> - DNA index 1.0 diploid; aneuploidy = worse prognosis.
> - Flow crossmatch is more sensitive than CDC crossmatch.

SAQ Haemostatic agents in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3e

Introduction

- Topical agents are **adjuncts** to surgical haemostasis (suture, clip, energy), used for diffuse ooze. No randomised study shows clear benefit of any single agent in partial nephrectomy.

Classification

Class	Examples	Mechanism / note
Passive (mechanical)	Gelatin sponge (Gelfoam), **oxidised regenerated cellulose (Surgicel)**, microfibrillar collagen, polysaccharide spheres	Matrix for platelets; needs intact clotting. Cellulose is acidic (antibacterial)
Active	Topical thrombin	Converts the patient's fibrinogen to fibrin
Flowable	Gelatin matrix + thrombin (**Floseal**)	Fills irregular beds, swells
Fibrin sealants	Fibrinogen + thrombin (Tisseel, Evicel)	Final clotting step; works despite coagulopathy; also seals leaks
Synthetic sealants and glues	PEG (polyethylene glycol) hydrogel (Coseal), cyanoacrylate, albumin–glutaraldehyde (BioGlue)	Seal; adhesive
Systemic	Tranexamic acid, aminocaproic acid, desmopressin	Antifibrinolytic; desmopressin for uraemic platelets

Urological uses

- **Partial nephrectomy:** renorrhaphy ± sealant or Surgicel bolster; sutureless repair in selected cases.
- **Laparoscopic venous injury:** pressure → sealant → suture.
- PCNL (percutaneous nephrolithotomy) tract sealing in tubeless cases; adrenal and spleen/liver ooze; prostatectomy and lymphadenectomy beds.
- **Fibrin glue:** ureteroscopic fulguration for upper-tract bleeding; fistula plugs.
- **Tranexamic acid:** lowers transfusion at radical cystectomy.

Intravesical agents (haemorrhagic cystitis)

Agent	Note
Alum 1% (50 g in 5 L)	First line; avoid in renal failure (aluminium toxicity)
Silver nitrate 0.5–1%	Chemical coagulation
Aminocaproic acid 200 mg/L irrigation	Thromboembolism and clot retention
Formalin 1–4%	Most effective; under anaesthesia; **cystogram first** to exclude reflux

Complications

- Swelling compresses nerves, ureter or spinal cord (gelatin, cellulose): use the least amount.
- Mimics abscess on imaging; foreign-body reaction; infection.
- Flowables entering open vessels → embolism or thrombosis.
- Bovine thrombin → antibodies and coagulopathy; human products → theoretical viral risk.
- **Also know:** energy adjuncts (bipolar, argon beam, vessel sealers); hyperbaric oxygen in radiation cystitis.

Diagram to draw: clotting cascade showing where passive agents, thrombin and fibrin sealant act.

> **Key points to remember:**
> - Agents are adjuncts, not substitutes for sutures.
> - Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
> - Surgicel: acidic, may mimic abscess.
> - Alum first line; formalin last, after cystogram.

SAQ Sterilisation of urological instruments

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3a

Definitions

- **Cleaning:** removes debris that shields organisms; always first.
- **Disinfection:** kills most microbes but not all spores. HLD (high-level disinfection) kills all except large numbers of spores.
- **Sterilisation:** kills all life, including spores.

Spaulding classification

Category	Contact	Requirement	Urological examples
Critical	Sterile tissue, vessels	Sterilisation	Resectoscope, ureteroscope, nephroscope, laparoscopic and robotic instruments
Semi-critical	Mucosa	HLD at least (sterilise if possible)	Flexible cystoscope, TRUS (transrectal ultrasound) probe
Non-critical	Intact skin	Low-level disinfection	Abdominal ultrasound probe

Reprocessing steps

1. Bedside pre-clean; leak test for flexible scopes.
2. Manual cleaning with enzymatic detergent; brush and flush channels.
3. Rinse → disinfect or sterilise → rinse → dry channels (alcohol, air) → store hanging; record tracking.

Methods

Method	Conditions	Use and drawbacks
Steam autoclave	121 °C for 15 min (15 psi) or 134 °C for 3–4 min	Cheapest, most reliable; metal instruments, autoclavable optics
Dry heat	160 °C for 2 hours	Glass; rarely used
Ethylene oxide	Low temperature	Heat-sensitive scopes, cables; toxic, carcinogenic, long aeration
Hydrogen peroxide gas plasma	Low temperature, short cycle	No toxic residue; not for cellulose; long narrow lumens limited

Method	Conditions	Use and drawbacks
Peracetic acid (Steris)	Automated liquid, low temperature	Sporicidal; ureteroscopes
Glutaraldehyde 2%	HLD in 20 min; sterilant after about 10 hours	Cheap; irritant, sensitiser; must rinse well
Gamma irradiation	Industrial	Single-use catheters, stents, sutures

Monitoring

- **Physical:** time, temperature, pressure records.
- **Chemical indicators:** autoclave tape; Bowie–Dick test (air removal).
- **Biological indicators:** *Geobacillus stearothermophilus* spores (steam, plasma); *Bacillus atrophaeus* (ethylene oxide, dry heat).

Urology-specific points

- A laser-perforated ureteroscope channel lets fluid in during sterilisation, destroying the scope.
- Poor channel cleaning causes outbreaks (Pseudomonas, mycobacteria).
- **Prions** resist routine methods: single-use or special protocols.
- **Single-use scopes** avoid reprocessing (about $107 per case for reusable ureteroscopes).
- **Also know:** laser fibre re-sterilisation; limited-life robotic instruments.

Diagram to draw: reprocessing cycle flowchart.

> **Key points to remember:**
> - Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
> - Autoclave 121 °C 15 min or 134 °C 3–4 min.
> - Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
> - Biological indicator: *G. stearothermophilus*.

SAQ Amyloidosis in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3c

Definition

- Extracellular deposits of misfolded protein in **β-pleated sheet fibrils**. **Congo red stain gives apple-green birefringence** under polarised light.

Classification

Type	Protein	Cause
AL (amyloid light-chain; primary)	Immunoglobulin light chain	Plasma cell dyscrasia, myeloma
AA (amyloid A; secondary)	Serum amyloid A	Chronic inflammation: TB (tuberculosis), rheumatoid arthritis, osteomyelitis, pressure sores, hidradenitis
ATTR (hereditary transthyretin)	Transthyretin	Autosomal dominant neuropathy

Type	Protein	Cause
Dialysis-related	β2-microglobulin	Long-term dialysis
Localised	Usually AL	Deposits in one organ only

Urological sites

- **Kidney (systemic):** nephrotic proteinuria, CKD (chronic kidney disease), enlarged kidneys, renal vein thrombosis, proximal tubular defects.
- **Bladder (usually localised): painless gross haematuria**; yellowish plaques or nodules that **mimic carcinoma**. Secondary bladder amyloid can bleed severely; Campbell lists amyloidosis among systemic causes of haemorrhagic cystitis.
- **Ureter and renal pelvis:** stricture or filling defect, often calcified; mimics urothelial cancer; obstruction.
- **Urethra:** rare; mass, obstructive voiding, haematuria.
- **Seminal vesicles:** senile deposits in older men; can mimic cancer invasion on MRI (magnetic resonance imaging).
- **Also know:** ATTR gives neurogenic bladder (underactive detrusor, LUTS [lower urinary tract symptoms] >80%) and erectile dysfunction; adrenal insufficiency; papillary necrosis.

Evaluation

- **Biopsy is diagnostic** (avoids unnecessary nephroureterectomy or cystectomy).
- Exclude systemic disease: serum and urine protein electrophoresis, immunofixation, free light chains, echocardiogram, renal function.

Management

- **Localised bladder:** TUR (transurethral resection) or fulguration; intravesical DMSO (dimethyl sulfoxide) reported; recurrence is common, so follow-up cystoscopy is needed.
- **Ureter:** ureteroscopic resection or segmental excision; stent for obstruction.
- **Urethra:** observe if mild; dilatation or urethroplasty if obstructive.
- **Refractory bleeding:** irrigation, embolisation, cystectomy.
- **Systemic:** AL: chemotherapy (daratumumab with bortezomib regimen, ANDROMEDA trial) or stem cell transplant; AA: treat inflammation (colchicine in familial Mediterranean fever); ATTR: tafamidis, patisiran. AA amyloid can recur in a renal transplant.

Diagram to draw: Congo red stained fibrils with apple-green birefringence.

> **Key points to remember:**
> - Congo red, apple-green birefringence.
> - Localised bladder amyloid mimics cancer; biopsy then TUR.
> - Always exclude systemic AL disease.

SAQ Collin's knife.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2018-Summer Paper 3, Q3b

*Taken to mean the **Collings knife**, the knife-shaped cutting electrode of the resectoscope (the reference chapters spell it both "Collins" and "Collings").*

Description

- A **hot (electrosurgical) knife electrode** fitted to the **resectoscope working element** in place of the loop.
- Thin **straight or angled (L- or hook-shaped) blade**; cuts a precise line with **cutting current** under irrigation and direct vision.
- **Monopolar** versions need **non-conducting irrigant** (glycine, sorbitol or mannitol); bipolar versions allow normal saline.
- Adult and **infant (9.5 Fr) resectoscope** sizes.
- **Principle:** it **incises rather than resects**; a deep, controlled cut releases a fibrous ring with little tissue loss.

Uses (from the chapters)

Use	Technique point
TUIP (transurethral incision of prostate)	Gland under 30 mL, no median lobe; incise at 5 and 7 o'clock from below the ureteric orifice to just proximal to the verumontanum, down to the capsule
Bladder neck contracture (after TURP (transurethral resection of prostate) or simple prostatectomy)	Incise until the scar springs open, to about 22 Fr; mitomycin C for recurrence
Ureterocele incision (older children)	Low transverse full-thickness incision near the bladder floor to reduce reflux
PUV (posterior urethral valve) ablation	Infant resectoscope only if urethral calibre is normal; incise at 5, 7 (± 12) o'clock; avoid hot-loop resection (stricture)
Bladder diverticulum neck	Incise the narrow neck so the diverticulum drains
PCNL (percutaneous nephrolithotomy) tract	Incise scarred fascia when rigid dilators fail

Alternatives

- **Cold knife (Sachse)** urethrotome: preferred for urethral stricture; hot knife unnecessary there.
- **Holmium or thulium laser**, Bugbee electrode, hook electrode, **Mohan's valvotome** (for valves).

Precautions and complications

- Stay away from the **ureteric orifices** and external sphincter; control depth (perforation, extravasation).
- TUR (transurethral resection) syndrome with monopolar irrigant if veins open.
- **Retrograde ejaculation** after bladder neck incision; stricture if the urethra is too small for the sheath.

Diagram to draw: resectoscope tip with L-shaped knife electrode, and the 5 and 7 o'clock TUIP incisions.

Key points to remember:

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

SAQ APOGEE

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 3, Q3c

Assumed meaning: the Apogee transvaginal mesh kit (American Medical Systems) for posterior vaginal wall and vault prolapse. It is not described in the chapter notes; facts below are general.

Definition

- A **prepackaged polypropylene mesh kit** placed transvaginally to support the **posterior compartment and vaginal apex** (DeLancey levels I and II). Anterior companion: **Perigee** (transobturator).
- Mesh: **type I macroporous monofilament polypropylene** (Amid classification), the only type that integrates into tissue.

Indications (historical)

- Symptomatic posterior and vault prolapse, mostly **recurrent** or with poor native tissue, avoiding abdominal surgery.

Technique (principle)

1. Midline posterior vaginal incision; dissect the rectovaginal space to the ischial spines.
2. Two small perianal incisions lateral and posterior to the anus.
3. Curved trocars pass through the **ischiorectal fossa**, piercing the iliococcygeus near the ischial spine, and deliver the mesh arms.
4. Mesh body fixed to the apex and laid over the rectovaginal septum **tension-free**; epithelium closed without trimming.

Complications

- **Mesh exposure** (IUGA/ICS [International Urogynecological Association/International Continence Society] term; "erosion" deprecated), extrusion, **perforation** into rectum or bladder.
- **Dyspareunia**, buttock and pelvic pain, mesh contraction, defecatory dysfunction, ischiorectal haematoma or abscess, rectal injury.
- **Management:** small exposure → vaginal oestrogen, then partial excision; pain or perforation → excision of mesh and arms.

Regulatory status

- **FDA (US Food and Drug Administration) 2011:** serious complications of transvaginal POP (pelvic organ prolapse) mesh "not rare", with no clear benefit over native repair. Reclassified high-risk in 2016.
- **2019:** FDA ordered remaining manufacturers to **stop selling transvaginal POP mesh**; Apogee-type kits are off the market. Synthetic mesh improves anatomy but complicates

more (Cochrane 2016).
- This does **not** apply to mid-urethral slings or **sacrocolpopexy mesh**.

Current alternatives

- Native posterior colporrhaphy or site-specific repair; uterosacral suspension or **sacrospinous fixation**; **sacrocolpopexy** (abdominal, laparoscopic, robotic) for apical prolapse; colpocleisis in the frail.

Diagram to draw: posterior mesh with two arms through the ischiorectal fossae.

> **Key points to remember:**
> - Apogee: transvaginal posterior/apical polypropylene mesh kit.
> - Exposure, dyspareunia and pain drove withdrawal.
> - FDA 2019 stopped transvaginal POP mesh sales.

SAQ Urethral stents and metallic stents in urology

Paper answer · P4

Asked as:

- Urethral stents — *Summer 2015 Paper IV Q3d (SAQ)*
- Metallic stents in urology — *Practice Set 3 Paper IV Q3c (SAQ)*
- Urethral stents. — *Practice Set 4 Paper IV Q3f (earlier version) (SAQ)*

Type: SAQ (10 marks)

(urethral stents also answered in 2015-Summer Paper 4, Q3d; ureteric stents in 2015-Summer Paper 3, Q1)

Definition and introduction

- A urethral stent is an endoscopically placed tube that keeps the **prostatic or bulbar urethra** open, mainly in men unfit for surgery.
- Polymer stents (polyurethane, silicone) fail early under **extrinsic compression** and need frequent changes. Metal stents **resist compression** and stay longer, but suffer **encrustation, tissue ingrowth and migration**.

Classification (urethral stents)

Type	Examples	Features
Temporary	**Memokath** (nitinol), **Allium**	Not epithelialised (designed to avoid epithelial incorporation); removable; left 6–12 months
Permanent, epithelialising	**UroLume**	Urothelium grows over it; **withdrawn**
Biodegradable	Experimental	Dissolve

Classification of metallic stents (by site)

Site	Type	Example	Features
Ureter	Full-length coiled metal, double pigtail	**Resonance** (2006)	Tightly coiled alloy; changed every 12 months

Site	Type	Example	Features
Ureter	Segmental self-expanding mesh, bare or covered	Wallstent-type; covered nitinol (Allium)	Placed across malignant stricture; bare mesh allows ingrowth
Ureter	Thermo-expandable nitinol coil	Memokath 051	Removable
Prostate	Temporary thermo-expandable nitinol coil	**Memokath**	Not epithelialised; removable
Prostate, bulbar urethra	Covered self-expanding, removable	Allium	Left 6–12 months
Bulbar urethra	Permanent epithelialising mesh	**UroLume**	Withdrawn

Ureteric metal stents

- **Indications:** malignant extrinsic obstruction; retroperitoneal fibrosis (first Resonance use); benign stricture in a patient unfit for repair; ureteroenteric stricture with limited life expectancy; failed tandem polymer stents.
- **Outcomes:** full-length metal stent: up to 3 years of drainage, pooled success about 72.5%, migration about 1%, but about one-third failed in one malignant series. Ureteroenteric strictures: patency >80% at 6–22 months.
- **Problems:** encrustation, difficult exchange, ingrowth, migration.

Urethral and prostatic stents

Indications

1. BPH (benign prostatic hyperplasia) retention in men unfit for TURP (transurethral resection of prostate) / unfit for surgery.
2. Recurrent short bulbar stricture in men unfit for urethroplasty.
3. **DSD (detrusor–sphincter dyssynergia)**: formerly (once) an alternative to sphincterotomy; now **abandoned** for complications.

Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed (with warm water) and softens when cooled (with cold water) for removal. A bulbar stent must **not cross the external sphincter**, or incontinence follows.

Complications

- **Migration** (temporary stents), **encrustation**, **hyperplastic ingrowth** with restenosis (UroLume), incontinence, pain and painful erections.

Results (outcomes)

- Prostatic stents free most unfit men from catheters short term, but many are later removed. Memokath variable. Removable bulbar stents succeed only in small series. UroLume worked early (good early results, then ingrowth) but failed long term from **hyperplastic ingrowth** and restenosis, so withdrawn, and many men still carry one.
- No stent removes **spongiofibrosis**, so none cures a stricture.

Guidelines

- **AUA (American Urological Association)**: **no (urethral) stent if urethroplasty is considered** (planned); stents make reconstruction harder.

Recent advances

- **Optilume** paclitaxel balloon (FDA (Food and Drug Administration) approval 2023 for anterior stricture): **ROBUST III**, after failed endoscopy, about 75% vs 27% stricture-free. Optilume BPH (EVEREST-1).
- **iTind** (temporary nitinol prostatic device): nitinol device left 5–7 days; AUA 2023 option for 25–75 mL glands without median lobe.
- **Also know:** biodegradable and drug-eluting stents (experimental).

Also know: a soft silicone "urethral stent" (splint) after hypospadias repair or urethroplasty is a different device; in complicated hypospadias repairs it reduces meatal stenosis and fistula.

Diagram to draw: sagittal urethra with prostatic stent and bulbar stent distal to sphincter; prostatic Memokath ending above the external sphincter.

> **Key points to remember:**
> - Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
> - Metal resists extrinsic compression; weakness is ingrowth and encrustation.
> - Resonance: coiled, full-length, changed every 12 months.
> - Memokath: thermo-expandable nitinol, removable.
> - Migration, encrustation, ingrowth; never cross the sphincter.
> - AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

SAQ Urological applications in nanotechnology.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2022-Summer Paper 4, Q3a

Taken as "applications of nanotechnology in urology".

Definition

- **Nanotechnology:** design and use of materials of about **1–100 nm**. At this scale, surface area and optical, magnetic and thermal properties change.
- **Why useful:** particles pass through leaky tumour vessels (**enhanced permeability and retention**), carry drugs or tracers, and can be targeted with ligands such as PSMA (prostate-specific membrane antigen) antibodies.

Applications

Area	Application
Imaging and staging	**USPIO (ultrasmall superparamagnetic iron oxide) lymphotropic MRI (magnetic resonance imaging):** normal nodes take up particles and lose T2 signal; metastatic nodes do not (Chapter 6; investigational in penile cancer, Chapter 84). Gold-nanoparticle CT (computed tomography) contrast, quantum dots
Sentinel node	Technetium-99m **nanocolloid** for dynamic sentinel node biopsy in penile cancer (Chapter 84)

Area	Application
Diagnosis	Nanowire and nanoparticle biosensors detecting PSA (prostate-specific antigen) and urinary markers at very low levels
Drug delivery	Nanoparticle (albumin-bound) paclitaxel in urothelial and prostate cancer (Chapter 164: tolerated, no pathological complete response); **NanoPac** intratumoral paclitaxel (Chapter 163); liposomal drugs; mucoadhesive intravesical carriers; siRNA (small interfering RNA) delivery
Focal ablation	**Gold nanoshells (AuroLase):** collect in tumour vessels and are heated by near-infrared laser; about three-quarters success on 12-month biopsy (Chapter 163). Magnetic nanoparticle hyperthermia
Irreversible electroporation	Electric pulses create **nanoscale membrane pores** ("NanoKnife"); spares matrix, vessels and ducts; investigational (Chapter 143)
Devices	Silver and other nanocoatings on catheters and stents against biofilm and encrustation
Tissue engineering	Electrospun nanofibre scaffolds that mimic extracellular matrix
Stone disease	"Calcifying nanoparticles" (nanobacteria) linked to Randall plaque, not proven causal (Chapter 95); nano-CT of stones and tissue

Limitations

- Toxicity and long-term tissue accumulation; uptake by liver and spleen; manufacturing cost; regulatory hurdles. Most uses remain experimental.

Diagram to draw: nanoparticle with drug core, polymer shell and surface PSMA ligand entering a leaky tumour vessel.

> **Key points to remember:**
> - 1–100 nm; enhanced permeability and retention allows tumour targeting.
> - USPIO MRI: benign nodes lose signal; metastatic nodes do not.
> - AuroLase gold nanoshells: photothermal focal therapy.

SAQ Morcellation

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3d

Definition

- Cutting a large specimen into small pieces so it leaves through a port, small incision or endoscope.
- **Manual:** ring forceps or Kelly clamp inside a bag. **Power (electromechanical):** rotating cylindrical blade (laparoscopy) or endoscopic tissue morcellator (prostate).

1. Endoscopic morcellation after enucleation

- Made **HoLEP (holmium laser enucleation of prostate)** practical; also ThuLEP (thulium laser enucleation).
- **Steps:** 1. Achieve good haemostasis first. 2. Pass a long nephroscope with a 5 mm channel. 3. Run two inflow lines fully open to keep the bladder distended. 4. Engage lobes with

suction; cut only in mid-bladder, away from the wall. 5. Collect fragments in a trap for histology.

- Devices: Piranha (rotating blades), VersaCut (oscillating).
- **Complications:** bladder mucosal injury (stop, release suction, disengage; mostly catheter only); intraperitoneal perforation needs laparotomy; **retained adenoma** (from malfunction or poor vision) needs repeat morcellation.

2. Laparoscopic morcellation (kidney, spleen)

- **LapSac** (nylon with polyurethane lining) is impermeable to tumour cells and bacteria.
- **Rules:** purpose-built bag; drapes; change gloves and instruments after; no deep forceps passes; keep pneumoperitoneum and watch the bag.

	Intact extraction	Morcellation
Incision	Larger	Small, better cosmesis
Pain, time, stay	Similar	Similar
Pathology	Full staging	Type and grade kept; size, capsule, vein invasion, and depth in UTUC (upper tract urothelial carcinoma) lost
Seeding	—	Rare port-site recurrence

- Splenectomy morcellation risks **splenosis** if the bag ruptures.

3. FDA 2014 warning (gynaecology)

- In 2014 the FDA (Food and Drug Administration) warned against laparoscopic power morcellation in hysterectomy or myomectomy for fibroids, because unsuspected uterine sarcoma (about 1 in 350) was spread and upstaged.
- A boxed warning followed; contained (in-bag) morcellation is advised.
- Lesson for urology: avoid uncontained power morcellation of possible malignancy.

Complications summary

- Bag rupture, seeding, lost staging, visceral or vascular injury, retained tissue.

Diagram to draw: enucleated lobes in a distended bladder with morcellator suction.

> **Key points to remember:**
> - Morcellator made HoLEP practical; distend bladder, haemostasis first.
> - Morcellation loses stage information; histology still obtained.
> - FDA 2014: uncontained power morcellation spreads occult sarcoma.

SAQ Stem cell therapy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3f (tissue engineering also in 2023-Winter Paper 1, Q1)

Definition

- Stem cells **self-renew** and **differentiate** into specialised cells. Therapy uses them to repair or replace damaged tissue.

Types

Type	Source	Comment
Embryonic	Blastocyst	Pluripotent; ethics, teratoma
Adult mesenchymal	Bone marrow, adipose (ADSC, adipose-derived stem cells), umbilical cord	Easy harvest; mainly paracrine
Muscle-derived (MDSC)	Skeletal muscle biopsy	Used for sphincter and detrusor
iPSCs (induced pluripotent stem cells)	Reprogrammed adult cells	Patient-specific; tumour risk
Urine-derived, spermatogonial	Urine; testis	Research

Mechanisms

- Differentiation into muscle, nerve or endothelium.
- **Paracrine action:** growth factors, anti-inflammatory, anti-fibrotic and neurotrophic signals.
- Homing to injured tissue; seeding of scaffolds.

Applications

- **SUI (stress urinary incontinence):**
- Autologous muscle-derived cells: thigh muscle biopsy → culture → transurethral or periurethral injection into the sphincter.
- Aim: regenerate the striated sphincter, not just bulk it.
- Early studies improved; placebo-controlled trials have not shown clear superiority, so it remains investigational.
- After childbirth injury, stem cell homing chemokines (CXCL12, CCL7) rise.
- **ED (erectile dysfunction):** intracavernosal bone-marrow, adipose or cord cells; animal and small uncontrolled human series only. SMSNA (Sexual Medicine Society of North America): experimental, IRB (institutional review board) protocols only.
- **Underactive detrusor:** muscle-derived cells injected into the bladder wall; two open-label studies improved voiding; research.
- **Tissue engineering:** cell-seeded scaffolds for bladder (Atala, 2006) and urethra (Raya-Rivera, 2011); adipose stem cell-seeded grafts for Peyronie disease (experimental).
- **Fertility:** spermatogonial stem cell banking in prepubertal cancer patients.
- **Kidney:** iPSC kidney organoids (research).

Challenges

- Poor cell survival and retention; tumour formation (embryonic, iPSC); standardisation, cost and regulation.
- Unregulated "stem cell clinics" offer unproven treatments outside trials, a growing ethical concern.
- **Also know:** cancer stem cells (a different concept) may drive tumour recurrence and resistance.

Diagram to draw: muscle biopsy → culture → periurethral injection into sphincter.

Key points to remember:

- Adult stem cells act mainly by paracrine signals.
- SUI muscle-derived cells and ED stem cells are investigational.
- Embryonic and iPSCs carry tumour risk.

SAQ Advances in lap surgeries

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 4, Q3a (related: LESS in 2023-Winter Paper 4, Q1; energy sources in 2015-Summer Paper 2, Q3d)

Introduction

- Since the first laparoscopic nephrectomy (Clayman, 1991), advances aim at **less access trauma, better vision, safer haemostasis, gentler physiology and faster training**.

1. Access and insufflation

- **Non-bladed dilating and optical trocars**: port-site hernia fell from 1.83% to 0.19%, wall bleeding from 0.83% to 0.16%.
- **Balloon dilators** for retroperitoneal and extraperitoneal space.
- **AirSeal** valveless system: stable pneumoperitoneum at **8–12 mmHg** despite suction, smoke evacuation, less CO_2 absorption.
- Lower pressure (12 rather than 15 mmHg): less pain and oliguria.

2. Fewer and smaller incisions

Technique	Gain	Limitation
LESS (laparoendoscopic single-site surgery)	Cosmesis, less analgesia	No triangulation, clashing
NOTES (natural orifice transluminal endoscopic surgery)	No abdominal scar (hybrid transvaginal extraction)	No proven benefit
Mini-laparoscopy (2–3 mm instruments)	Less pain, cosmesis	Instrument flexibility
Hand-assisted	Tactile feedback, fast bleeding control	Larger wound

3. Vision

- Chip-on-tip HD (high-definition) and **3D** laparoscopes; deflectable tips.
- **ICG (indocyanine green) near-infrared fluorescence**: vessel mapping for selective clamping in partial nephrectomy, ureteric identification, sentinel nodes.
- Laparoscopic ultrasound; 3D-printed planning models.

4. Haemostasis and suturing

- **Advanced bipolar sealers** (LigaSure: vessels ≤7 mm, 1–3 mm spread) and **ultrasonic shears** (about 1 mm spread, but very hot jaws).
- **Hem-o-lok** clips, **barbed sutures** (shorter anastomosis), Lapra-Ty clips, powered articulating staplers.

5. Perioperative care

- **ERAS (enhanced recovery after surgery)**: stay about 30% shorter, complications halved; no routine bowel preparation.

6. Training

- Box trainers, VR (virtual reality) simulators, EAU **E-BLUS** (European Basic Laparoscopic Urological Skills) certification, 5-day mini-residencies.

7. Robot-assisted laparoscopy

- Wristed 3D platforms (da Vinci Xi, SP 2018; Hugo, Versius, SSI Mantra) solved laparoscopic suturing (**see Q3b**).
- **Also know:** augmented reality, artificial intelligence, telesurgery.

Diagram to draw: LESS multichannel port in the umbilicus with articulating instruments.

> **Key points to remember:**
> - Blunt trocars cut port-site hernia to 0.19%.
> - AirSeal allows 8–12 mmHg working pressure.
> - ICG fluorescence guides selective clamping.
> - LESS: cosmesis at the cost of triangulation.

SAQ Artificial intelligence in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3d

Definitions

- **AI (artificial intelligence):** computer systems doing tasks that need human intelligence: recognising patterns, predicting, deciding.
- **ML (machine learning):** algorithms learn from data rather than fixed rules. **Supervised** (labelled data, e.g. tumour or not), **unsupervised** (finds clusters), reinforcement.
- **Deep learning:** multilayer **ANNs (artificial neural networks)**; **CNNs (convolutional neural networks)** are the commonest for images.
- **NLP (natural language processing)** and **LLMs (large language models)** handle text; **radiomics** extracts quantitative image features.

Development steps

`Data collection and labelling → training → internal validation → external validation → regulatory approval → use and monitoring`

Applications

Area	Examples
Imaging	CNNs detect stones on CT (computed tomography) with very high sensitivity; ANNs detect renal tumours; prostate MRI (magnetic resonance imaging) lesion detection; deep-learning reconstruction allows ultra-low-dose CT (about 1 mSv)
Pathology	Automated cancer detection and Gleason grading on digital slides; Paige Prostate was the first FDA (Food and Drug Administration)-authorised AI pathology tool (2021)

Area	Examples
Endoscopy	Real-time bladder-tumour detection at cystoscopy; future ureteroscopes setting laser energy, irrigation, pressure and temperature
Stones	Predict composition and stone-free outcome
Prognosis	Radiomics, pathomics and genomics combined for RCC (renal cell carcinoma) prognosis; AI on biopsy slides predicting benefit of ADT (androgen deprivation therapy) with radiotherapy
Robotic surgery	Video and kinematic data for skill assessment, outcome prediction, real-time guidance; future semi-autonomy
Paediatrics	Deep learning on ultrasound predicts CKD (chronic kidney disease) progression in valves; VUR (vesicoureteral reflux) infection-risk models
Others	Urodynamics interpretation; LLM chatbots for patient information

Advantages

- Fast big-data processing; consistency; decision support.

Limitations and ethics

- Needs large, high-quality labelled data; **bias** and poor generalisability (overfitting); "black box" decisions; few prospective or external validations.
- Privacy and data sharing; medico-legal liability; LLM "hallucinations"; regulation.
- AI assists, not replaces, the urologist or radiologist.

Diagram to draw: neural network: input layer (image pixels), hidden layers, output (stone or no stone).

> **Key points to remember:**
> - Deep learning is part of ML, which is part of AI; CNNs for images.
> - CNNs: very high sensitivity for CT stones.
> - Paige Prostate: first FDA-authorised AI pathology tool (2021).
> - Limits: data, bias, black box, validation, liability.

SAQ 3D laparoscopy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3b

Introduction

- Standard laparoscopy gives a **two-dimensional (2D)** image. Depth is judged from monocular cues (shadow, size, motion parallax, touch), which slows suturing.
- **Three-dimensional (3D) laparoscopy** restores **stereoscopic depth perception**, as in robotic surgery (Chapter 18: two-lens binocular systems give depth perception).

Principle

- Two lenses or two chip-on-tip sensors, a few millimetres apart, capture two images from slightly different angles, like the two eyes.
- The monitor shows both images; **polarised glasses** send one to each eye. The brain fuses them, and the **binocular disparity** is perceived as depth.

- **Also know:** active-shutter glasses; head-mounted displays; glasses-free (autostereoscopic) screens; 3D scopes at 0° and 30°, and with a deflectable tip; 3D 4K systems.

Advantages

- Better depth judgement and hand–eye coordination.
- **Shorter time for complex tasks**, especially intracorporeal suturing and knotting, with fewer errors.
- **Shorter learning curve**; novices gain more than experts.
- Gives robotic-type vision at much lower cost.

Disadvantages

- Visual strain, headache, dizziness or nausea in some users; glasses dim the image.
- Every team member needs glasses and a correct viewing distance and angle.
- Higher cost than 2D; 30° rotation needs care.
- No wristed instruments, tremor filtering or motion scaling: these come only with the robot.

Urological uses

- Reconstructive steps: **pyeloplasty**, vesicourethral anastomosis in laparoscopic radical prostatectomy, **partial nephrectomy** renorrhaphy, ureteric reimplantation, donor nephrectomy.

Evidence

- Randomised trials and meta-analyses in laboratory tasks and laparoscopic surgery show shorter operating and suturing times with 3D. Differences in complications and oncological outcomes are not established.

Feature	2D laparoscopy	3D laparoscopy	Robot
Depth perception	No	Yes	Yes (magnified)
Wristed instruments	No	No	Yes
Cost	Low	Moderate	High

Diagram to draw: two lenses at the scope tip, two images, polarised glasses, fused 3D image in the brain.

> **Key points to remember:**
> - Two lenses, two images, polarised glasses: binocular disparity gives depth.
> - Main gain: faster, more accurate suturing; shorter learning curve.
> - Drawbacks: eye strain, headache, cost, glasses for all.
> - Robotic vision at low cost, but without wristed instruments.

SAQ Telemedicine in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3c

Definition

- **Telemedicine** is delivery of health care at a distance using information and communication technology (video, phone, messaging, data transfer).

- **COVID-19 (coronavirus disease 2019)** distancing accelerated its adoption. Chapter 2: urologists' telehealth use rose sharply in 2019–2020.

Types

Type	Method	Urological example
Synchronous	Real-time video or phone consultation	Stone, PSA (prostate-specific antigen) and post-operative follow-up
Asynchronous (store-and-forward)	Reports, images, bladder diaries sent for later review	Scan reports, uroflow traces
Remote monitoring	Apps and home devices	Bladder diary apps, home uroflowmetry
Provider-to-provider	E-consults, telementoring, tele-tumour boards	Haematuria triage (Chapter 8)
Telesurgery	Remote robotic surgery	Experimental; limited by latency

Uses in urology

- **History-driven care:** LUTS (lower urinary tract symptoms) review, results, drug titration.
- ED (erectile dysfunction) evaluation by selected telemedicine (Chapter 72); men's health in large, sparse regions (Chapter 68).
- After mid-urethral sling, telemedicine follow-up is safe, and app-based follow-up gave higher satisfaction than office visits (Chapter 116).
- Internet-based pelvic floor training equalled a postal programme in an RCT (randomised controlled trial) (Chapter 114).

Benefits (Chapter 2)

- Access for rural and underserved patients; less travel; shorter waits; relatives join remotely; sustained satisfaction.

Limitations (Chapter 2)

- **No examination or in-office tests**: vital signs, cystoscopy, PVR (post-void residual).
- Needs connectivity and a secure, simple platform; hard for elderly or non-technical patients.
- No assistant to pre-gather results; more no-shows.
- **Medico-legal:** US licensure is state-based, and pandemic payment waivers have ended.

Indian framework

- **Telemedicine Practice Guidelines (2020)**: only a registered medical practitioner may consult; identity of patient and doctor confirmed; consent and records kept; narcotic and Schedule X drugs may not be prescribed.
- Government platform **eSanjeevani**; patient data covered by the Digital Personal Data Protection Act 2023.

Diagram to draw: patient (home or district hospital) linked by video and data to the urologist, with remote monitoring feeding in.

Key points to remember:

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

SAQ Virtual and augmented reality in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3e

Definitions

- **Virtual reality (VR):** the user is immersed in a fully computer-generated environment, usually through a head-mounted display.
- **Augmented reality (AR):** digital images are laid over the real view, for example a 3D (three-dimensional) tumour model on the robotic camera image.
- **Mixed reality:** virtual holograms are anchored in the room and can be handled.

Workflow

`CT (computed tomography) / MRI (magnetic resonance imaging) -> segmentation of organ, tumour, vessels, collecting system -> 3D virtual model -> viewed in VR, 3D-printed, or overlaid (AR) -> registration to the live surgical view`

Applications

Use	Examples
Pre-operative planning	Partial nephrectomy: tumour depth, arterial branches, selective clamping. **PCNL (percutaneous nephrolithotomy)**: calyx and tract choice (Chapter 98: immersive VR and 3D printing show early promise). Prostatectomy: MRI lesion model to plan nerve-sparing
Intra-operative navigation	AR overlay in the robot console to locate an endophytic tumour, hilar vessels, or a prostate lesion near the capsule
Education	VR anatomy, 360° operation videos; simulation, see Q3d
Patient counselling	Patients understand their anatomy and planned operation; better-informed consent
Paediatrics	VR distraction reduces anxiety before surgery and procedures such as catheterisation and VCUG (voiding cystourethrogram) (Chapters 35 and 38)

Limitations

- **Registration error:** organs move and deform with breathing, pneumoperitoneum and dissection, so a rigid overlay drifts; manual alignment is still common.
- Segmentation time and cost.
- Chapter 161: promising but **resource-intensive and hard to validate**.
- Chapter 98: 3D reformatting added nothing over orthogonal CT measurements for predicting stone-free rate.
- Little outcome evidence; cybersickness with headsets.

Future

- **AI (artificial intelligence)** for automatic segmentation and real-time tracking.
- **Digital surgery:** reconstruction of the patient's own anatomy allows rehearsal before the operation (Chapter 147).

Diagram to draw: CT kidney, segmented 3D model, and overlay of the model on the robotic view of a renal tumour.

> **Key points to remember:**
>
> - VR immerses; AR overlays on reality.
> - Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
> - Biggest problem: registration on moving, deforming organs.
> - Evidence for outcomes still limited.

Rapid Revision Sheet · Section 1

Operating room safety : the role of surgeon. (SAQ)

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Microscopic / asymptomatic haematuria and urinary biomarkers (SAQ)

- MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
- Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
- Any gross haematuria = high risk.
- Cytology: specific for high grade/CIS, insensitive for low grade.
- BTA is falsely positive with blood.
- No markers in initial microhaematuria work-up; none replaces cystoscopy.

Berger's Disease. (SAQ)

- Commonest primary GN; haematuria with upper respiratory infection.
- Normal complement; biopsy shows mesangial IgA.
- Oxford MEST-C; 20–30% ESKD.
- ACE inhibitor or ARB is the mainstay.

Proteinuria (SAQ)

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Glomerular, tubular, overflow; orthostatic proteinuria is benign.
- Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

Urinary crystals (LAQ)

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

Tumour markers in urology (SAQ)

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

Urine cytology (SAQ)

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

Doppler in urology (SAQ)

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy (LAQ)

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main CA-AKI risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic.
- NAC is the antidote for paracetamol overdose; best within 8 h.
- Cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
- CO_2 angiography: never above diaphragm.

Describe IVU & its applications in Urology. (LAQ)

- Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
- Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
- Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
- Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

VCUG (SAQ)

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

Newer generation CT scan (SAQ)

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

PET scans in urology. (SAQ)

- proPSMA: PSMA PET 92% vs 65% accuracy.
- Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
- FDG poor in prostate and clear cell RCC.
- VISION: ^{177}Lu-PSMA-617 in mCRPC.

PSMA PET Ct scan (SAQ)

- proPSMA: accuracy 92% vs 65%.
- BCR: about half positive below PSA 0.5 ng/mL.
- Nodal sensitivity 30–40%: negative scan does not exclude nodes.
- VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

MR urography (SAQ)

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

VI-RADS (SAQ)

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

Radionuclide (isotope) renal scans: MAG3 and DMSA (LAQ)

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 min obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h.
- DMSA is the gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Haemorrhagic cystitis: diagnosis and management (SAQ)

- Acrolein (prevent with mesna); radiation endarteritis; BK virus.
- Exclude bladder cancer.
- Alum first; formalin only after cystogram.
- Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

Operating room set-up (SAQ)

- Zones with one-way flow; positive pressure, about 20 air changes an hour.
- Monitor directly ahead at eye level; open tray ready for laparoscopy.
- Hypothermia triples wound infection (Kurz).
- ALARA: time, distance, shielding.

Ethics and informed consent in urology (SAQ)

- Four principles: autonomy, beneficence, non-maleficence, justice.

- Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
- Commonest gap: not checking understanding.
- Exceptions: emergency, incapacity (surrogate), minors.

Patient positioning in lap urology (SAQ)

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

Foley's (urethral) catheters, guidewires and DJ stents with their complications (SAQ)

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.
- Hydrophilic wire to pass, PTFE wire to stay (safety).
- DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Outline algorithm for difficult catheter placement. (SAQ)

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

Interventional radiology in urology and therapeutic angioembolisation (SAQ)

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: internal iliac anterior division; spare posterior division.
- Post-embolisation syndrome is common and self-limiting.

Tubeless PCNL (SAQ)

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

Flexible endourological instruments, digital and current ureteroscopes (LAQ)

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.

- Pressure sensors and AI are the newest developments.
- Golden rule: never fire unless the fibre tip is seen on the stone.

Simulators in urology (SAQ)

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

Irrigation fluids for TURP. (SAQ)

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

Indications for outpatient cystourethroscopy. (SAQ)

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

LASERs in urology: holmium, thulium, KTP and newer laser technologies (LAQ)

- Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
- Holmium: 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
- Holmium lithotripsy is photothermal; Moses effect: fibre close and perpendicular.
- Moses: two-part pulse through a vapour channel.
- Holmium: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
- HoLEP size-independent, superior to TURP; PVP for anticoagulated men (GOLIATH).
- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
- TFL: water peak, low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- ThuLEP is size-independent and equal to HoLEP.
- KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
- GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
- Nd:YAG for PeIN; eyewear for everyone.
- Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

Energy sources in urology and laparoscopic surgery, including bipolar energy (LAQ)

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
- Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.

- Holmium 2120 nm: gold standard for stones and HoLEP.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink.
- Thermal ablation for cT1a ≤3 cm.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy (LAQ)

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it (prevents leak).
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy (LAQ)

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision.
- Main indication for Hasson: previous surgery or adhesions.
- Hasson avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon in muscle causes hernia.
- Trocar = cannula + obturator.
- Bladeless trocars: fewer epigastric injuries and hernias.
- Balloon trocar seals and prevents retroperitoneal emphysema.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

Laparoendoscopic single-site (single-port) surgery: LESS (LAQ)

- LESS = one incision; LESSCAR (2008) standardised the term.
- Core problem: lost triangulation; solved by articulating or curved instruments and robots.
- Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
- da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

3D printing in urology (SAQ)

- Additive manufacturing from CT/MRI via segmentation and an STL file.
- Main uses: partial nephrectomy and PCNL planning, consent, simulation.
- Limits: cost, time, unrealistic tissue, weak outcome evidence. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Hemostasis in laparoscopic surgery (SAQ)

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

Energy sources for breaking stones in urology (LAQ)

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

Bacteriuria in pregnancy and antibiotic policy (SAQ)

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

Clinical trial methodology (SAQ)

- Phase III RCT with concealed allocation and blinding is the gold standard.
- Primary endpoint drives sample size; surrogates need the Prentice criteria.
- ITT analysis; CONSORT flow diagram for reporting.

Evidence-based medicine and critical appraisal (SAQ)

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.
- Three questions: valid, results, applicable.
- Randomisation, concealment, blinding, ITT, follow-up >80%.
- CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Recist Criteria. (SAQ)

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy; PCWG3 for prostate.

Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion (SAQ)

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-BI, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.

- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

Tissue engineering in urology: principles, status and recent advances (LAQ)

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
- Neo-urinary conduit (conduit trial): half stenosed; no blood supply.
- Raya-Rivera 2011: seeded tubular urethras in 5 boys.
- Still experimental except short urethral onlays.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

Robotics in urology (including robotic vs standard laparoscopy) (LAQ)

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

Natural orifice endoscopic surgery. (LAQ)

- NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
- Transvaginal extraction removes the most painful wound.
- LESS: one incision, lost triangulation, more hernias.
- Barriers: closure, infection, orientation, platform; no proven benefit yet.

Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa (SAQ)

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966.
- Huggins 1945: adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
- HM3 (1983) is still the efficacy benchmark.

- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

History of Laparoscopy (SAQ)

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

Flow cytometry (SAQ)

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

Haemostatic agents in urology (SAQ)

- Agents are adjuncts, not substitutes for sutures.
- Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
- Surgicel: acidic, may mimic abscess.
- Alum first line; formalin last, after cystogram.

Sterilisation of urological instruments (SAQ)

- Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
- Autoclave 121 °C 15 min or 134 °C 3–4 min.
- Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
- Biological indicator: *G. stearothermophilus*.

Amyloidosis in urology (SAQ)

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Collin's knife. (SAQ)

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

APOGEE (SAQ)

- Apogee: transvaginal posterior/apical polypropylene mesh kit.
- Exposure, dyspareunia and pain drove withdrawal.
- FDA 2019 stopped transvaginal POP mesh sales.

Urethral stents and metallic stents in urology (SAQ)

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

Urological applications in nanotechnology. (SAQ)

- 1–100 nm; enhanced permeability and retention allows tumour targeting.
- USPIO MRI: benign nodes lose signal; metastatic nodes do not.
- AuroLase gold nanoshells: photothermal focal therapy.

Morcellation (SAQ)

- Morcellator made HoLEP practical; distend bladder, haemostasis first.
- Morcellation loses stage information; histology still obtained.
- FDA 2014: uncontained power morcellation spreads occult sarcoma.

Stem cell therapy (SAQ)

- Adult stem cells act mainly by paracrine signals.
- SUI muscle-derived cells and ED stem cells are investigational.
- Embryonic and iPSCs carry tumour risk.

Advances in lap surgeries (SAQ)

- Blunt trocars cut port-site hernia to 0.19%.
- AirSeal allows 8–12 mmHg working pressure.
- ICG fluorescence guides selective clamping.
- LESS: cosmesis at the cost of triangulation.

Artificial intelligence in urology (SAQ)

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

3D laparoscopy (SAQ)

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

Telemedicine in urology (SAQ)

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

Virtual and augmented reality in urology (SAQ)

- VR immerses; AR overlays on reality.
- Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
- Biggest problem: registration on moving, deforming organs.
- Evidence for outcomes still limited.

SECTION 2

Oncology, DSD, Gender & Ageing

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

1 · Innate and adaptive immunity

LAQ Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T)

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Describe use and principles of immunology and Immunotherapy in Urological Oncology" · P4 LAQ "Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T."

Introduction

- Immunotherapy works by either **supplying a danger signal** that starts a Th1 response (BCG (bacille Calmette–Guérin), vaccine adjuvant) or **removing a brake** on an existing response (PD-1 (programmed death-1), CTLA-4 (cytotoxic T-lymphocyte antigen-4)).
- A **cancer vaccine** is a therapeutic agent that trains the patient's own T cells to recognise **tumour antigens**. It treats existing cancer; it does not prevent it.
- **BCG**, a tuberculosis vaccine, was among the first immune treatments for any cancer and is the original urological immunotherapy. It is non-specific; cancer vaccines aim for a **specific** anticancer response.

Principles of tumour immunology

Arm	Cells and role
Innate (rapid, pattern-based)	PAMPs (pathogen-associated molecular patterns) bind **TLRs (Toll-like receptors)**; macrophages, neutrophils; urothelium as sentinel (IFN (interferon), IL-8 (interleukin-8))
Bridge	**Dendritic cell (DC)**: takes up antigen, matures on danger signal, migrates to lymph node
Adaptive (specific, memory)	**CD8 killer** (sees class I MHC (major histocompatibility complex), ~9 amino acids); **CD4 helper** (class II); B cells
Brake	**Treg** (regulatory T cell, FoxP3+ CD4); MDSCs (myeloid-derived suppressor cells); M2 macrophages

- **Two signals**: TCR (T-cell receptor) meets MHC–peptide (signal 1) plus **CD28–B7** costimulation (signal 2). Without signal 2 the T cell becomes tolerant.
- **Th1 vs Th2**: IL-12 gives Th1 (IL-2, IFN-γ, TNF-α (tumour necrosis factor-α)), which is antitumour. IL-4 gives Th2 (antibody, chronic inflammation).
- **Checkpoints**: CTLA-4 (on Tregs) competes for B7; PD-1 on CD8 cells binds **PD-L1** (programmed death ligand-1); LAG-3 binds class II.

- **Adaptive immune resistance**: CD8 cells secrete IFN-γ, which induces PD-L1 on the tumour, which switches the same T cells off. So PD-L1 marks a suppressed ongoing response.

Immunoediting (three Es) and the rationale for immunotherapy in urology

- **Elimination** (small tumours destroyed), then **Equilibrium** (growth held), then **Escape** (clinical cancer).
- Escape is by loss of antigen or MHC, Treg expansion and checkpoint ligands. Vaccines try to restart the response.
- **Bladder:** inflammation-driven (schistosomiasis); CD8 infiltration is good; BCG success; high mutation load gives many neoantigens.
- **Kidney: CD8 paradox** (more CD8, worse outcome); rare spontaneous regression of metastases after nephrectomy; responses to IL-2 and interferon.
- **Prostate:** Treg-dominated and PD-L1 low, so checkpoint inhibitors fail; slow growth gives time to build a vaccine response; **organ-specific antigens** (PSA (prostate-specific antigen), PAP (prostatic acid phosphatase)).
- Vaccines have low toxicity and offer possible lasting memory.

BCG in NMIBC (non-muscle-invasive bladder cancer)

- Intravesical live attenuated mycobacteria.
- Mechanism (sequence): 1. Bacilli attach via **fibronectin**. 2. TLR2/4 activation → IL-8 and TNF-α. 3. Neutrophil wave → DC priming. 4. **Th1 CD4** response → memory.
- Th1 predicts success, Th2 failure. About half stay in remission for years.
- Used for intermediate/high-risk NMIBC and CIS (carcinoma in situ) with induction plus maintenance; standard for high-risk NMIBC.

Cytokines and gene/viral therapy

- **IFN-α** and **high-dose IL-2**: durable complete responses in a few mRCC (metastatic renal cell carcinoma) patients; capillary leak. Intravesical IFN-α: mixed results.
- **Nadofaragene firadenovec** (adenovirus carrying the IFN-α2b gene + Syn3): BCG-unresponsive CIS, CR (complete response) 53.4%; FDA (Food and Drug Administration) approved December 2022.
- **Cretostimogene**: oncolytic adenovirus with GM-CSF (granulocyte-macrophage colony-stimulating factor) gene.
- **N-803** (IL-15 superagonist) with BCG: FDA approved 2024 for BCG-unresponsive CIS.

Cancer vaccines: mechanism of action

- Two components: **antigen** (overexpressed in tumour) + **adjuvant** (a danger signal, often GM-CSF, which activates DCs).
- Sequence: `Intradermal antigen + adjuvant → immature DC takes up antigen; adjuvant matures it → DC migrates to lymph node → peptide on MHC I/II = signal 1; B7 costimulation = signal 2 → CD4 Th1 help (IL-2, interferon-γ) + CD8 cytotoxic T cells → T cells traffic to tumour → apoptosis of antigen-bearing cells → memory T cells`
- Without signal 2 the T cell becomes tolerant, hence the adjuvant.
- **Prime–boost:** the first dose primes, later doses boost.
- **Delayed effect:** survival may improve without tumour shrinkage or PSA fall.

- **Resistance:** PD-L1 on tumour, Tregs, MDSCs, MHC loss. This is the basis for combining vaccines with checkpoint inhibitors.

Vaccine platforms

Platform	Example
Whole-cell (autologous or allogeneic)	Reniale (RCC (renal cell carcinoma)); GVAX (prostate)
Peptide/protein	IMA901
Dendritic-cell (ex vivo loaded)	Sipuleucel-T; AGS-003 (rocapuldencel-T)
Viral vector (poxvirus)	PROSTVAC-VF; TroVax (MVA-5T4)
Heat-shock protein–peptide	Vitespen (HSPPC-96)
DNA/mRNA, personalised neoantigen	Phase I–II trials

Vaccines by organ

Renal cell carcinoma

- **IMA901:** 10 tumour-associated peptides + GM-CSF, with low-dose cyclophosphamide to deplete Tregs; HLA-A*02+ (human leucocyte antigen) metastatic RCC with sunitinib. **No OS (overall survival) benefit.**
- **AGS-003 (rocapuldencel-T):** autologous DCs loaded with tumour RNA, with sunitinib. **Terminated for lack of efficacy.**
- **TroVax** with standard first-line therapy: no OS benefit. **Vitespen** (adjuvant): no overall benefit.
- **Reniale** (adjuvant autologous tumour-cell vaccine): claimed benefit, not adopted.
- Checkpoint inhibitors, not vaccines, are today's RCC immunotherapy.

Prostate cancer

- **Sipuleucel-T:** the **only approved** vaccine (see note below).
- **PROSTVAC-VF:** vaccinia prime, fowlpox boost, encoding PSA + TRICOM (three costimulatory molecules). Phase III **closed for futility**.
- **GVAX:** allogeneic whole-cell lines secreting GM-CSF. Phase III trials **failed**.
- Also: DCVAC/PCa (phase III negative); DNA vaccines against PAP (trials).

Bladder cancer

- **BCG** (above); nadofaragene firadenovec, cretostimogene and N-803 (above).
- Peptide and neoantigen vaccines: experimental only.

Why most vaccines failed

- Tolerance to self-antigens; immunosuppressive tumour environment; dysfunctional patient DCs; bulky late disease; progression endpoints miss delayed effects.
- **Pattern:** IMA901, AGS-003, PROSTVAC and GVAX all failed; **only sipuleucel-T succeeded in phase III.**

Note on Sipuleucel-T

- **Nature:** autologous cellular immunotherapy: the patient's **CD54+ APCs (antigen-presenting cells)** activated with **PA2024**, a fusion protein of **PAP + GM-CSF**.
- **Manufacture and dosing:** 1. Leukapheresis collects PBMCs (peripheral blood mononuclear cells). 2. Cells incubated about 36 h with PA2024 at a central facility. 3. Infused

intravenously about 3 days after collection. 4. **Three doses at 2-week intervals** (one-month course).

- **Mechanism:** primes PAP-specific CD4 and CD8 T cells; later infusions boost.
- **IMPACT trial (Kantoff, NEJM 2010):** 512 men, 2:1 versus placebo, asymptomatic or minimally symptomatic mCRPC (metastatic castration-resistant prostate cancer).
- **OS 25.8 vs 21.7 months (HR (hazard ratio) 0.78)**, a gain of 4.1 months, despite placebo crossover.
- **No change in time to progression or PSA.**
- **FDA approval April 2010:** first therapeutic vaccine for a solid tumour.
- **Indication:** asymptomatic or minimally symptomatic mCRPC, good performance status, **no visceral metastases, no opioid need**; usually before chemotherapy.
- **Toxicity:** mild infusion reactions (fever, chills, fatigue); premedicate with paracetamol and antihistamine.
- **Limitations:** cost, logistics, no response marker; not available in India.

Checkpoint inhibitors

Site	Use (trial)
BCG-unresponsive CIS	Pembrolizumab if cystectomy declined/unfit (KEYNOTE-057: CR 41%)
Adjuvant MIBC (muscle-invasive bladder cancer) and UTUC (upper tract urothelial carcinoma)	Nivolumab after radical surgery (**CheckMate 274**: DFS (disease-free survival) 20.8 vs 10.8 months; included UTUC, benefit uncertain in that small subgroup). Atezolizumab (IMvigor010) negative
Metastatic UC (urothelial carcinoma)	Avelumab **maintenance** after platinum (JAVELIN Bladder 100, OS gain); pembrolizumab second line (**KEYNOTE-045**: OS 10.3 vs 7.4 months); **enfortumab vedotin (EV) + pembrolizumab**
RCC	Nivolumab + ipilimumab (CheckMate 214); IO (immuno-oncology) + TKI (tyrosine kinase inhibitor) combinations; adjuvant pembrolizumab (KEYNOTE-564, DFS HR 0.68)
Prostate	Monotherapy poor; pembrolizumab only for **MSI-H/dMMR** (microsatellite instability-high/mismatch repair-deficient) tumours

- VEGF (vascular endothelial growth factor) suppresses DCs and T-cell entry, so TKIs synergise with IO.
- **Biomarkers**: PD-L1 CPS (combined positive score) only for first-line single-agent pembrolizumab in cisplatin-ineligible UC; TMB (tumour mutational burden), MSI.

Adoptive cell therapy (brief)

- **CAR-T (chimeric antigen receptor T cells)**: T cells engineered against a surface antigen (e.g. PSMA (prostate-specific membrane antigen)). Investigational in GU (genitourinary) cancers; limited by poor trafficking, hostile microenvironment and cytokine release syndrome.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory

- Endocrinopathies need hormone replacement.
- BCG toxicity: cystitis, BCG sepsis.

Recent advances

- **EV-302**: enfortumab vedotin + pembrolizumab first line in metastatic UC, OS 31.5 vs 16.1 months versus platinum; now the preferred first line.
- **NIAGARA**: perioperative durvalumab with neoadjuvant cisplatin chemotherapy improved event-free and overall survival in MIBC.
- **Also know:** CheckMate 901 (nivolumab + gemcitabine–cisplatin); ADT (androgen deprivation) + IO trials; anti-CD38 and LAG-3 targets; DNA/mRNA personalised neoantigen vaccines (phase I–II).

Diagrams to draw:

- T cell–tumour synapse with TCR–MHC, CD28–B7, CTLA-4 and PD-1/PD-L1 blocked by antibodies.
- Sipuleucel-T flow: leukapheresis, PA2024 loading, reinfusion, T-cell priming.

Key points to remember:

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

Also, from the condensed book (not in the paper answer):

- Th1 cytokines (IL-2, TNF-α, IFN-γ) mean **BCG success**; Th2 cytokines (IL-4, IL-5, IL-10, IL-13) mean **BCG failure**.

ALGORITHM

1. Danger signal (TNF-α, TLR ligand) reaches a DC.
2. DC stops antigen uptake; raises MHC and B7-1/B7-2.
3. DC enters afferent lymphatics (CCL21 sensed by CCR7).
4. In the node it activates CD4, CD8 and B cells.

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; raised by IFN	Antigen-presenting cells (APCs)

	Class I MHC	Class II MHC
Peptide	~9 amino acids	13–25 amino acids

2 · Immune editing and escape

- Escape follows equilibrium, as antigen and MHC fall and Tregs rise.
- In adaptive immune resistance, CD8-derived IFN-γ induces tumour PD-L1, which switches off those PD-1-positive CD8 cells.

> **KEY POINT**
>
> PD-L1 is mostly induced by the immune attack, so it marks a suppressed response, not a fixed trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating
CTLA-4	CD80/CD86	Inhibitory; mainly on FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly on CD8 tumour-infiltrating lymphocytes (TILs)
LAG-3	MHC class II	Inhibitory

3 · Inflammation and genitourinary (GU) tumours

	Bladder	Kidney	Prostate
Does inflammation cause cancer?	Strong: schistosomiasis, chronic UTI, catheters, cyclophosphamide	Weak	Probable: proliferative inflammatory atrophy beside HGPIN
CD8 infiltration	Better outcome in MIBC	**Worse outcome — renal paradox**	May be suppressive
Dominant restraint	PD-L1, rising with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

4 · BCG in bladder cancer

> **ALGORITHM**
>
> 1. Instil 1–5 × 10^8 bacilli.
> 2. A fraction adheres via **fibronectin attachment protein**.
> 3. Urothelial TLR2/TLR4 release IL-8 and TNF-α.
> 4. Neutrophil wave (hours), then DC activation.
> 5. CD4 Th1 response in nodes (weeks), then memory.

- Memory explains durable remission in ~**50%** of NMIBC.
- **Nadofaragene firadenovec**, an adenovirus carrying the IFN-α2b gene with **Syn3**, is given 3-monthly for up to 12 months (CIS CR 53.4%). It was the FDA's **first gene therapy** (2022), for BCG-unresponsive NMIBC with CIS.

5 · Cancer vaccines

P4 *Rationale of using cancer vaccines in Urology — antigen + GM-CSF adjuvant; sipuleucel-T (PAP–GM-CSF, 3 doses 2-weekly, IMPACT); failed IMA901, AGS-003, PROSTVAC*

- Sipuleucel-T, the first FDA-approved solid-tumour vaccine, prolongs OS by **+4.1 months** with **no effect on time to progression**.

6 · Checkpoint blockade

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder

- First-line pembrolizumab alone (KEYNOTE-052) requires cisplatin-ineligibility with CPS **≥10**, or platinum-ineligibility; with CPS <10 it was worse than carboplatin.
- Atezolizumab and durvalumab were withdrawn, and adjuvant atezolizumab (IMvigor010) was negative.
- **Enfortumab vedotin (EV)**, an anti-nectin-4 antibody–drug conjugate, gave a 68% response rate with pembrolizumab when cisplatin-ineligible (EV-103).
- †In EV-302, EV plus pembrolizumab roughly doubled PFS and OS versus platinum and is now first line for all.

Kidney

- CTLA-4 on Tregs and PD-1 on CD8 TILs make dual blockade synergistic.
- VEGF blocks DC maturation and T-cell entry, so TKIs complement immunotherapy.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- In mCRPC, tumours are rarely PD-L1-positive; pembrolizumab gives a PSA response under 10%, and ipilimumab and IMbassador250 gave no OS benefit.
- The exception is **MSI-H/dMMR** tumours (1.2–12%), treated with pembrolizumab second line or later; its 2017 approval was the **first based on a biomarker, not a site**.
- †Immune-related adverse events of grade ≥2 (such as colitis or pneumonitis) need drug hold and corticosteroids; endocrinopathies need lifelong replacement.

> **KEY POINT**
>
> Successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, VEGF inhibition) or removes a brake (PD-1, CTLA-4); it fails where the brake is not limiting, as in Treg-dominated prostate cancer.

7 · Landmark trials

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only phase III-based second-line approval
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk MIBC	DFS 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, RCC after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS hazard ratio 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab for 1 year, high-risk clear-cell RCC	Recurrence or death reduced by 32%	Adjuvant standard

25 Molecular Genetics and Cancer Biology

1 · Hallmarks and basic molecular genetics

- **Local invasion** is the one hallmark that uniquely separates malignant from benign.
- **Alternative splicing** makes ligand-independent AR variants (AR-V7), causing castration resistance.
- **Somatic** testing finds targetable mutations; **germline** testing finds hereditary syndromes for surveillance and family cascade testing.

2 · Gene alterations: tumour suppressors and oncogenes

Oncogenes need one altered allele, suppressors usually two.

SAQ Oncogenes

Paper answer · P1

Asked as:

- Oncogenes — *Winter 2017 Paper I Q3f (SAQ)*
- Oncogene — *Practice Set 6 Paper I Q3f (SAQ)* (also answered in 2017-Winter Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **Proto-oncogene:** a normal gene that promotes cell growth and division (an "accelerator").
- **Oncogene:** its mutated or overactive form that drives (driving) uncontrolled proliferation. The change is **gain of function** and **dominant**: one altered allele is enough.
- The first human oncogene, **HRAS**, was isolated from (found in) the **T24 bladder cancer** cell line.

Mechanisms of activation

Mechanism	Effect	Example
Point (activating) mutation	Constitutively active protein (protein permanently "on")	RAS (bladder); FGFR3 (fibroblast growth factor receptor 3) in low-grade papillary bladder cancer; MET germline (germline MET) in hereditary papillary RCC (renal cell carcinoma)
Gene amplification	Increased copy number (more copies) → overexpression	MYC (prostate, bladder); HER2 (human epidermal growth factor receptor 2); AR (androgen receptor) in castration-resistant prostate cancer
Chromosomal translocation / fusion	Gene placed under a strong promoter, or fusion protein	**TMPRSS2–ERG** (androgen-driven ERG in about half of prostate cancers); BCR–ABL; TFE3 fusions (translocation RCC)
- **Also know:** promoter hypomethylation can switch oncogenes on.		

Functional classes

Class	Examples
Growth factors	PDGF (platelet-derived growth factor), sis

Class	Examples
Receptor tyrosine kinases	EGFR (epidermal growth factor receptor), HER2, FGFR3, MET, KIT (seminoma)
Signal transducers	**RAS** (binds GTP [guanosine triphosphate]; GTP-binding; mutation locks it "on"), BRAF, PI3K
Transcription factors	**MYC**, ERG, ETS family
Cell cycle / apoptosis regulators	Cyclin D1, BCL-2

Oncogene versus tumour suppressor gene

	Oncogene	Tumour suppressor
Normal role	Accelerator	Brake, DNA repair
Change	Gain of function	Loss of function
Alleles needed	One	Usually both (**Knudson two-hit**: retinoblastoma)
GU (genitourinary) examples	MYC, MET, FGFR3, ERG	RB1, TP53, VHL, PTEN

Urological significance

- **Bladder:** FGFR3 mutations in up to 80% of low-grade Ta tumours (10–20% of MIBC (muscle-invasive bladder cancer)); FGFR3 and TERT promoter mutations in >60% of low-grade papillary tumours; RAS (HRAS) mutually exclusive with FGFR3; HER2 in 9–12% of MIBC; MYC overexpressed in over half.
- **Kidney:** MET activation in papillary RCC.
- **Prostate:** TMPRSS2–ERG fusion, MYC (8q gain), AR amplification after (under) hormone therapy.
- **Testis:** KIT and KRAS mutations (seminoma).

Targeted therapy ("oncogene addiction")

- **Erdafitinib** (pan-FGFR inhibitor) for FGFR-altered (FGFR3/2-altered) advanced (metastatic) urothelial cancer; MET-active TKIs (tyrosine kinase inhibitors) in papillary RCC; AR-targeted drugs in prostate cancer; HER2-directed drugs (trastuzumab deruxtecan received a tumour-agnostic FDA approval in 2024 for HER2-positive solid tumours; disitamab vedotin is under study in urothelial cancer).

Diagram to draw: growth factor → receptor → RAS → RAF → MEK → ERK → MYC pathway.

> **Key points to remember:**
> - Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
> - HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
> - TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Also, from the condensed book (not in the paper answer):

- Two-hit exceptions are **dominant negative** mutation (mutant TP53) and **haploinsufficiency** (p27/CDKN1B).

3 · Cell cycle

- The cycle takes ~24 h, and cyclin-dependent kinases (CDKs) work only with their phase-specific **cyclin**.

Phase	Complex	Function
G1	Cyclin D–CDK4/6, cyclin E–CDK2	Phosphorylate RB1; cyclin D needs sustained mitogen (Ras/Raf/MEK)
S	Cyclin A–CDK2, then cyclin A–CDK1	S-phase genes
G2/M	Cyclin B–CDK1	Phosphorylates lamins, so the nuclear envelope breaks down

Checkpoint	Trigger	Mechanism
G1/S	Mitogens, nutrients, O_2, DNA damage	p21 (CDKN1A) and 14-3-3σ inactivate G1 CDKs; TP53 triggers apoptosis if damage is unrepaired
G2/M	DNA damage (radiation)	ATM activates CHEK1/CHEK2, which sequester Cdc25; TP53 induces p21 and GADD45
Spindle assembly	Unattached chromosome	Blocks anaphase-promoting complex; failure causes non-disjunction and **aneuploidy** (near-universal in solid tumours)

- CDK inhibitors are **Cip/Kip** (p21, p27, p57), acting broadly, and **INK4** (p15, p16, p18, p19), acting on CDK4/6.

RB1 — guardian of G1/S

ALGORITHM

1. Hypophosphorylated RB1 binds E2F and HDACs, silencing S-phase genes.
2. Cyclin D–CDK4/6 and cyclin E–CDK2 hyperphosphorylate RB1.
3. E2F is released and the restriction point is passed (irreversible).

- **TP53**, activated by DNA damage or oncogene stress, causes arrest, repair, senescence or apoptosis. **MDM2** is its negative regulator; TP53 is mutated in over half of cancers.

> **KEY POINT**
>
> Cancer must break both RB1 (entry) and TP53 (integrity), so both pathways are lost in almost every tumour.

4 · DNA damage and repair

Pathway	Lesion	Key proteins	Cancer link
Nucleotide excision repair (NER)	Bulky adducts, cross-links (UV, chemicals)	XPC, XPD, ERCC1/XPF	Xeroderma pigmentosum; **ERCC2 in bladder cancer**
Base excision repair (BER)	Single-base damage, single-strand breaks	MUTYH, OGG1, PARP1/2	OGG1 variants linked to RCC
Mismatch repair (MMR)	Mispairs, indel loops at repeats	MSH2/MSH6, MLH1/PMS2	Mutation rate raised ~100-fold; Lynch
Homologous recombination (HR)	Double-strand break (DSB)	ATM, RAD51, BRCA1, BRCA2	Breast/ovarian, prostate

Pathway	Lesion	Key proteins	Cancer link
Non-homologous end joining (NHEJ)	DSB	Ku70/Ku80, DNA-PKcs, ligase IV	Translocations

- Unrepaired DSBs (marked by **γH2AX** foci) cause apoptosis, the basis of radiotherapy. **HR** copies the sister chromatid and is precise; **NHEJ** is template-free and error-prone.

> **KEY POINT**
>
> Repair defects are targets: HR loss means PARP inhibitor sensitivity, MMR loss means immunotherapy, and ERCC2 loss means cisplatin sensitivity.

5 · Apoptosis

- In apoptosis, initiator caspases 8, 9, 10 activate executioners 3, 6, 7.

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Unrepaired DNA damage, lost survival signals	Fas ligand/TNF on Fas (CD95)/TNFR1
Sequence	Bax/Bak release cytochrome c; Apaf-1 apoptosome activates caspase-9	FADD recruits procaspase-8, giving caspase-8
Brakes	Bcl-2, Bcl-XL, Mcl-1; IAPs	FLIP
TP53	Dependent (transactivates Bax)	**Independent**, so a target when TP53 is mutated

- Bid links the two pathways. TP53 loss causes chemoradiotherapy resistance.

6 · Senescence, telomeres, autophagy

SAQ Telomerase

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 3, Q3b

Telomeres

- **Telomeres** are repetitive sequences (TTAGGG repeats) at chromosome ends. They protect ends from degradation and fusion.
- DNA polymerase cannot copy the very end of the lagging strand (end-replication problem). So about **100 base pairs are lost per division**: the "mitotic clock".
- Critical shortening triggers **replicative senescence** (Hayflick limit).

Telomerase: structure and function

- A specialised **reverse transcriptase** that adds telomeric repeats.
- Two core parts: **hTERT** (human telomerase reverse transcriptase, catalytic protein) and **hTR** (telomerase RNA, the template).
- **Normal expression:** only rapidly dividing tissues: germ cells (testis), intestinal epithelium, endometrium, activated lymphocytes. Most somatic cells lack it.

Role in cancer

```
Proliferation -> moderate shortening -> senescence/apoptosis (barrier 1) ->
checkpoint lost (p53) -> severe shortening -> chromosomal instability (barrier
```

2) -> telomerase activation -> immortal cancer cell

- **85–90% of human tumours** express high telomerase, giving unlimited replication.
- **ALT (alternative lengthening of telomeres):** telomerase-independent recombination; linked to ATRX or DAXX loss.
- **TERT promoter mutations** switch on hTERT.

Urological relevance

Organ	Finding
Bladder	TERT promoter mutation is the commonest mutation (over 60% even in PUNLMP, papillary urothelial neoplasm of low malignant potential). Urine markers: telomerase activity by **TRAP (telomeric repeat amplification protocol)**, a PCR (polymerase chain reaction)-based assay; hTERT messenger RNA; AssureMDx (FGFR3, TERT, HRAS mutations plus methylation). Inflammation gives false positives; not routine
Upper tract	hTR positive in almost all UTUC (upper tract urothelial carcinoma); rises with stage
Prostate	Short telomeres in cancer and premalignant lesions; short tumour telomeres predict recurrence; variable stromal telomeres predict cancer death

- **Also know:** senescent cells secrete SASP (senescence-associated secretory phenotype) cytokines in ageing and BPH (benign prostatic hyperplasia).

Therapy

- **Imetelstat** (telomerase inhibitor) is used in myelodysplastic syndrome.
- hTERT-based vaccines (UV1, GX301, INO-9012) are in early prostate, renal and bladder trials.

Diagram to draw: telomere-to-cancer flowchart with two barriers.

> **Key points to remember:**
> - Telomerase = hTERT (catalytic) + hTR (RNA template).
> - About 100 base pairs lost per division; 85–90% of tumours express telomerase.
> - TERT promoter mutation is the commonest mutation in bladder cancer.
> - TRAP assay detects urinary telomerase; inflammation gives false positives.

Also, from the condensed book (not in the paper answer):

- Telomerase is high in **85–90%** of tumours.
- **Senescent** cells are viable but permanently arrested; ADT induces senescence in some prostate cancers.
- **Autophagy** suppresses early tumours but sustains established ones.

7 · Epigenetics

- Epigenetic change alters expression without sequence change and is **reversible**. DNA methylation at CpG islands is maintained by DNMT1.
- Promoter hypermethylation silences suppressors as a "second hit": **GSTP1 in >90% of prostate cancer** (commonest somatic change), RASSF1A (up to 97% of bladder cancer) and p16. Global hypomethylation causes genomic instability.

- **H3K27me3 (written by EZH2)** represses genes; EZH2 is raised in mCRPC and predicts PSA recurrence.
- MicroRNA (18–25 nt) silences **many** mRNAs (**miR-371a-3p** in testis cancer) and siRNA **one**; long ncRNA HOTAIR drives castration resistance.

8 · Hypoxia, VHL and angiogenesis

ALGORITHM

1. In normoxia, HIF-α is prolyl-hydroxylated.
2. VHL (an E3 ubiquitin ligase) binds it for proteasomal degradation.
3. With hypoxia or VHL loss, HIF-α accumulates and dimerises with HIF-1β.
4. This switches on VEGF, angiopoietin and glycolytic genes.

- The **Warburg effect** (aerobic glycolysis) is best seen in clear cell RCC.

> **KEY POINT**
>
> VHL loss means permanent pseudohypoxia, driving angiogenesis, Warburg metabolism and immunosuppression; hence clear cell RCC responds to VEGF, HIF-2α and immune blockade.

- **Lineage plasticity** under driver targeting gives neuroendocrine prostate cancer.

9 · GU tumour genomics

Prostate

Gene	Significance
TMPRSS2–ERG	Commonest fusion (~half)
SPOP	Commonest mutated suppressor in primary tumours
TP53	Commonest alteration overall (more in advanced disease)
RB1, PTEN, TP53	Loss of ≥2 means aggressive variant, overlapping neuroendocrine
BRCA1/2	Germline plus somatic hit: give **PARP inhibitor**
ATM	Little PARP benefit
CDK12	Tandem duplications; poor response to hormones, chemo, PARP
MMR deficiency	Microsatellite instability (MSI): give immunotherapy
AR amplification	Late, under AR-targeted pressure; causes ADT resistance

Kidney

Subtype	Genomic hallmark	Therapy link
Clear cell (75–80%)	**3p loss** plus VHL, PBRM1, BAP1, SETD2	VEGF tyrosine kinase inhibitors (TKIs); belzutifan (HIF-2α)
Papillary (15–20%)	MET (7q); FH loss in subset	MET-active TKIs
Chromophobe	Mitochondrial DNA, mTOR pathway	mTOR-directed rational
Translocation	MiT/TFE fusions; young women	No targeted therapy

Subtype	Genomic hallmark	Therapy link
Sarcomatoid (any)	BAP1, CDKN2A loss	PD-L1-positive, so immunotherapy-responsive

- In clear cell RCC, VHL is mutated in 60–70%. **PBRM1** (~40%) means a better outcome and **BAP1** or **SETD2** (~10% each) a worse one.

Bladder (urothelial carcinoma, UC)

- UC has a high (APOBEC) mutation load, with TP53 most frequent.
- Of six consensus subtypes, **basal/squamous is largest (35%)**, luminal papillary 24%, neuroendocrine-like smallest 3%.
- **FGFR3** is altered in 15–20% of advanced UC and up to 37% of upper tract UC. These T-cell-poor tumours respond to **erdafitinib**, so test all advanced UC. HER2 is altered in 9–12% of MIBC.

Testis

- Testis cancer shows **12p gain** (†isochromosome 12p) and KIT/KRAS in seminoma; miR-371a-3p has >90% sensitivity and specificity.

Hereditary syndromes

SAQ Familial / hereditary urological malignancies

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Familial malignancies in urology" (Familial malignancies in Urology, Practice Set 2 Paper I Q3b; Familial malignancy in urology, Practice Set 8 Paper I Q3f) · P2 SAQ "Hereditary urological malignancies."

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: tumours are **early, bilateral and multifocal**.
- **4–6% of RCC (renal cell carcinoma)** is familial. All familial RCC syndromes are **autosomal dominant**; only HPRC (hereditary papillary renal carcinoma) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma, pancreatic cysts
HPRC	MET (7q31) oncogene	Multiple bilateral papillary (type 1, type 1-like)	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive**, often solitary papillary; collecting duct	Skin and uterine leiomyomas

Syndrome	Gene (chromosome)	Kidney tumour	Other features
BHD (Birt–Hogg–Dubé)	FLCN (folliculin, 17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13.3)	Multiple AMLs (angiomyolipomas); RCC 2–3% (P2: 1–3%)	Angiofibromas, cortical tubers

- **VHL type 1** (deletions, truncations): no phaeochromocytoma; **type 2** (missense): phaeochromocytoma.
- **Also know:** SDH (succinate dehydrogenase)-deficient RCC (paraganglioma); BAP1 (uveal melanoma, mesothelioma); Cowden (PTEN; breast, thyroid).

Management of familial kidney tumours

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save function for repeat operations.
- **Exception: HLRCC**: excise promptly, with wide margins.
- TSC AML: everolimus at 3–5 cm (P1; another source: if >3 cm); embolisation or excision (partial nephrectomy) if ≥5 cm.
- **Belzutifan** (HIF-2α (hypoxia-inducible factor 2α) inhibitor), FDA (Food and Drug Administration)-approved 2021 for VHL tumours.

Other urological cancers

- **Prostate: BRCA2**, BRCA1, ATM, CHEK2, PALB2, HOXB13, mismatch repair genes. Father and brother affected: 9-fold risk. BRCA2 gives aggressive prostate cancer. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): about 14-fold risk, about 75-fold with MSH2. Suspect if age <60 or a history of Lynch cancer; screen with MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.
- **Phaeochromocytoma:** about one-third hereditary: MEN-2 (multiple endocrine neoplasia type 2; RET), VHL, NF1 (neurofibromatosis type 1), SDHB (highest malignancy).
- **Also know:** Wilms tumour (WT1 gene: WAGR (Wilms, aniridia, genitourinary anomalies, retardation) and Denys–Drash; Beckwith–Wiedemann).

Approach

- Young age, multifocality or family history: genetic counselling, germline testing, screening of relatives.

Diagram to draw: VHL loss leading to HIF accumulation, VEGF (vascular endothelial growth factor) production and angiogenesis.

> **Key points to remember:**
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Also, from the condensed book (not in the paper answer):

- VHL carries a >70% lifetime risk of clear cell RCC (belzutifan). HLRCC gives papillary type 2 RCC, Lynch gives UC and prostate cancer, and HOXB13 carriers need earlier PSA screening.

KEY POINT

The germline syndrome names the somatic driver (VHL: clear cell; MET: papillary; FH: type 2; BRCA: prostate), and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss).

10 · Clinical NGS and biomarkers

- **Tumour-only** panels cannot separate somatic from germline variants.
- **Variant allele frequency (VAF)** depends on zygosity, copy number and purity; a heterozygous variant at 60% purity gives 30%. VAF of ~50% or ~100% suggests a **germline** variant, and single-digit VAF in cell-free DNA suggests **clonal haematopoiesis**.
- Circulating tumour DNA after treatment means minimal residual disease.
- A predictive marker needs a **biomarker–treatment interaction test** in a randomised trial; crossing lines mean predictive, parallel lines prognostic.
- Absolute benefit scales with baseline risk: a 50% relative reduction gives 15% absolute benefit at 30% risk but 1.5% at 3%.
- **HSD3B1 (1245A>C)** raises DHT synthesis, causing ADT resistance but abiraterone sensitivity.

11 · Landmark trials

Trial	Design	Result	Take-home
†PROfound	Olaparib vs enzalutamide/abiraterone, HR-altered mCRPC	Better radiographic PFS and overall survival, driven by BRCA1/2	Germline and somatic HR testing in metastatic prostate cancer
†THOR	Erdafitinib vs chemotherapy, FGFR-altered metastatic UC after immunotherapy	Longer overall survival	FGFR testing of all advanced UC

55 Management of Abnormalities of the External Genitalia in Males

1 · Normal development and penile size

- Male differentiation (weeks 9–13) needs Leydig testosterone, 5α-reductase conversion to DHT and a functional androgen receptor; any break causes undervirilisation.
- The genital tubercle forms the glans, urethral folds the shaft, and labioscrotal swellings the scrotum.
- **Stretched penile length (SPL)** runs from pubis (fat pad compressed) to glans tip. At term it is **3.5 ± 0.7 cm**; in adults 13.3 ± 1.6 cm.
- The micropenis cut-off is −2.5 SD: **1.9 cm** from term to 5 months, 1.5 cm at 30 weeks, 2.0 cm at 34 weeks, ≈ 3.7–3.9 cm at 5–11 years and 9.3 cm in adults. Compare with the cut-off, not the mean.
- Minipuberty (first 3 months) gives a second phase of penile growth. In Tanner staging, testicular change comes first (stage 2); semen analysis is done at Tanner 5.

2 · Prepuce: phimosis and paraphimosis

- **Physiological phimosis** is normal at birth: ≈ **90%** are **retractable by 3 years**, and < 1% remain phimotic at 17 years. Forceful retraction causes cicatricial phimosis.
- Treat persistent phimosis, balanitis, BXO or UTI.
- **Topical steroid**: betamethasone 0.05% twice daily for 30 days (or thrice daily for 21 days). About 1 in 6 fail.
- A retractile prepuce halves recurrent febrile UTI risk. The commonest acquired "penile lump" is **entrapped smegma**.
- **Paraphimosis** causes oedema within hours and may progress to gangrene.

ALGORITHM

1. Compress the glans with distal traction; this succeeds in most.
2. If it fails, reduce oedema: iced glove 5 min, granulated sugar 1–2 h, or punctures.
3. If still irreducible, do a dorsal slit, then circumcision.

3 · Circumcision

- The AAP (2012) found the benefits insufficient to recommend routine neonatal circumcision, but enough to justify access and payment.
- Before using a device (Gomco, Mogen, Plastibell), inspect the meatus to exclude hypospadias and megameatus. Older children need freehand surgery under GA.
- Analgesia is mandatory. **Dorsal penile nerve block is superior to EMLA**, whose prilocaine can cause methaemoglobinaemia.

Defer / avoid neonatal circumcision	Reason
Hypospadias, curvature, torsion, dorsal hood, buried or webbed penis	Prepuce needed for reconstruction
Large hydrocele / inguinal hernia	Skin pushed forward, causing secondary phimosis or a trapped penis

Defer / avoid neonatal circumcision	Reason
Coagulopathy (contraindication); hyperbilirubinaemia (some)	Haemorrhage
NICU setting	More complications

- **Benefits**: penile cancer occurs almost only in uncircumcised men. Febrile infant UTI falls, but **111** circumcisions prevent one UTI.
- There is ≈ **60%** protection against HIV acquisition in men, but no reduction in transmission to partners. HSV-2 and high-risk HPV are also lower, and no sexual dysfunction is proven.

Complications

SAQ Circumcision undone

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3d

"Circumcision undone" is read as an incomplete or complicated circumcision needing revision. Foreskin restoration is mentioned briefly at the end.

Introduction

- Circumcision complications: **0.2–5%**, rising **10- to 20-fold after infancy**.
- Most late problems concern **skin quantity** (too little or too much) or the **exposed meatus**.

Inadequate ("undone") circumcision

- **Too little skin removed** or asymmetrical excision: redundant prepuce covering the glans.
- Leads to **secondary (cicatricial) phimosis**, smegma retention, balanitis, adhesions and cosmetic distress.
- A large hydrocele or hernia pushes skin forward and raises this risk.

Other complications

Complication	Management
Bleeding (commonest, about 0.1%), usually frenular	Pressure, cautery, silver nitrate or suture
Glanular adhesions; **skin bridges** (can cause chordee, torsion)	Office lysis or excision; steroids do not help
Excess skin removal, degloving	Heals by secondary intention with ointment; severe: grafts
Trapped penis (cicatrix over glans)	Betamethasone 0.05% and retraction; else relaxing incision, flaps, Z-plasty
Meatal stenosis (almost only after circumcision)	Meatotomy or meatoplasty
Subcoronal fistula; glans amputation; necrosis (cautery on clamp)	Hypospadias-type repair; reattach glans within 8 hours

Evaluation

- Original procedure (age, device); examine skin quantity, meatus, curvature; exclude hypospadias, buried or webbed penis and BXO (balanitis xerotica obliterans).

Circumcision revision

- **Indications:** redundant or asymmetrical skin, phimosis, adhesions, cosmesis.
- **Needs general anaesthesia** in children, unlike neonatal circumcision.
- **Technique (sleeve):** mark inner and outer incisions (inner 5–10 mm from the corona), excise the excess collar, haemostasis, interrupted absorbable sutures.
- Lyse adhesions and skin bridges; correct torsion or chordee; meatotomy if needed; histology if BXO suspected.

Prevention

- Separate prepuce fully and inspect the meatus before using a device (Gomco, Mogen, Plastibell); defer with hypospadias, curvature, buried or webbed penis; trained operator.

Also know: foreskin restoration

- Traction tissue expansion or skin grafting, for men wanting a foreskin; results variable.

Diagram to draw: sleeve revision showing two circumferential incisions.

> **Key points to remember:**
> - Complications 0.2–5%; bleeding commonest.
> - Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
> - Meatal stenosis almost only in circumcised boys.
> - Revision needs general anaesthesia.

Also, from the condensed book (not in the paper answer):

- An amputated glans should be re-attached within **8 h**.
- Degloving heals by re-epithelialisation, with **no immediate suturing or graft**.
- A skin bridge can cause chordee or torsion, so excise it.

> **KEY POINT**
>
> Late sequelae are problems of skin quantity (too much removed: trapped penis, chordee, torsion; too little: redundancy, adhesions) or of the newly exposed meatus (stenosis).

Meatal stenosis and BXO

P2 P1 *Management of Urethral meatal stenosis / Lichen sclerosus (BXO) — normal calibre by age, post-circumcision meatitis, upward stream, ventral meatotomy; BXO features, steroid, circumcision, buccal mucosa*

- Meatal stenosis occurs almost exclusively in **circumcised** boys.
- BXO in boys is rare under 5 years. Steroids help only mild preputial disease, so **circumcision is preferred**, with or without meatoplasty.
- About 1 in 5 with meatal BXO need further meatal surgery.

4 · Inconspicuous penis versus micropenis

KEY POINT

Stretched length is the single discriminator. An inconspicuous penis (buried, webbed, trapped) has normal SPL hidden by skin, fat or scar, so treatment is anatomical. A micropenis is small but normally formed, so it needs endocrine work-up.

Buried penis type	Mechanism	Treatment
Congenital	Inelastic dartos; skin not anchored to deep fascia	Surgery
Obesity	Suprapubic fat hides shaft	Weight loss; liposuction only in adolescents
Trapped (post-circumcision)	Cicatrix embeds penis in fat	Dilation or betamethasone with retraction; then repair under GA at ≥ 6 months

- If retracting skin lateral to the shaft exposes the glans easily, the shaft is normal.
- **Repair** excises the dysgenic dartos bands, fixes dorsal subcutaneous tissue to pubic fascia and scrotal tissue to the ventral base, and covers with unfurled prepuce, Z-plasty, flap or graft.
- **Webbed penis** (penoscrotal fusion) is repaired by transverse incision with vertical closure, or Byars' flap. Avoid neonatal circumcision.

Micropenis

- Micropenis is normally formed but ≥ 2.5 SD below the mean (< **1.9 cm** at term). Testes are often small or undescended.
- A defect before week 12 (placental hCG) gives a small penis with hypospadias; gonadotrophin deficiency after 14 weeks gives a normally formed micropenis.

Cause	Examples	Hormones
Hypogonadotrophic hypogonadism (commonest)	Kallmann, Prader–Willi, Laurence–Moon–Biedl, CHARGE, hypopituitarism, midline brain defects	Low LH and FSH
Hypergonadotrophic	Rudimentary testes, gonadal dysgenesis, Robinow	High LH/FSH; no testosterone rise after hCG
Idiopathic	Delayed gonadotrophin timing	Normal axis
Partial androgen insensitivity (rare)	—	Ambiguity commoner

ALGORITHM

1. History, examination, SPL and **karyotype**.
2. Refer to a paediatric endocrinologist; measure testosterone before and after hCG, with or without a GnRH test.
3. Screen the anterior pituitary (glucose, Na/K, cortisol, thyroid); then MRI brain.
4. Test **end-organ response: intramuscular testosterone enanthate × 3 months**. Short courses spare height; prolonged ones advance bone age. Repeat at puberty.

- Most achieve male identity, standing voiding and satisfactory sexual function.

5 · Curvature (chordee without hypospadias) and torsion

SAQ Congenital Penile curvature..

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 10 Paper 1, Q3d

Definition and principle

- Curvature of the erect penis present from birth, with an **orthotopic meatus**.
- Rigidity needs **symmetric compliance** of the tunica albuginea; curvature means one aspect is less compliant.

Embryology

- DHT (dihydrotestosterone) drives fusion of the urethral folds and maturation of spongiosum, Buck's and dartos fascia. The urethra may form normally while the mesenchyme stays deficient → dysgenetic, inelastic ventral tissue.
- Normal fetal ventral curvature resolves by 20 weeks.

Devine and Horton classification (1973)

Type	Abnormal tissue
I	Corpus spongiosum deficient (chordee without hypospadias)
II	Dysgenetic Buck's and dartos fascia; normal spongiosum
III	Dartos only
IV	Normal layers; **asymmetric corporal compliance** (true congenital curvature)
V	Congenitally short urethra (very rare; doubted)

Two clinical groups

Feature	Chordee without hypospadias (I–III)	Congenital curvature (IV)
Direction	Ventral ± torsion	Ventral, lateral (often left), rarely dorsal
Erection size	In proportion to flaccid size	**Noticeably large**
Presentation	Childhood	18–30 years, noticed at puberty
- **Versus acquired** (Peyronie disease, fracture): congenital is painless, no plaque, no shortening.		

Evaluation

- Home photographs of erection; **artificial erection** (Gittes–McLaughlin saline test or prostaglandin) in theatre.
- Grade: mild <30°, moderate 30–45°, severe ≥45°.

Management

- Surgery if curvature hinders intercourse or distresses. 1. Deglove; excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum. 2. Residual curve: **midline ventral**

septotomy under artificial erection. 3. Still curved: **dorsal plication** contralateral to the curve: Nesbit (excise ellipses, close transversely), Yachia (Heineke–Mikulicz), 16-dot or TAP (tunica albuginea plication). **12 o'clock is nerve-free**, so midline plication is safe. 4. Severe: ventral **corporotomy with graft** (dermis, tunica vaginalis).

- **Rule:** if the meatus reaches the glans tip, do not divide the urethra; the corpora usually limit straightening.
- **Results:** single operation in almost all; recurrence 2–15%, mostly after plication.

Diagram to draw: penile cross-section: nerves at 11 and 1 o'clock, plication at 12.

> **Key points to remember:**
> - Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
> - Congenital curvature: young man, large erection, orthotopic meatus.
> - Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Also, from the condensed book (not in the paper answer):

- Repair if sexual dysfunction is expected, with concurrent penile surgery, or if growth will worsen it.

Type	Associations	Repair (stepwise)
Ventral (commonest)	Hypospadias; dorsal hood; a prominent frenulum causes distal chordee	Deglove; excise fibrous tissue superficial to Buck's; Byars' flaps; if still curved, dorsal plication, Nesbit or corporal rotation; if the urethra is short, reconstruct it; if frenular, frenulotomy
Dorsal	Isolated, epispadias	Ventral plication; protect dorsal neurovascular bundles
Lateral	Corporal asymmetry; straight when flaccid, noticed on erection	Plicate corpus **opposite** maximal curvature

- Work superficial to deep. An **artificial erection** locates the apex and confirms correction.
- **Penile torsion** is usually **counter-clockwise (to the left)**, < 60° and skin-level. If < 90°, deglove and rotate; if ≥ 90°, also divide basal bands and place non-absorbable anchoring sutures at the corporal base.

6 · Abnormal number, masses and lymphoedema

Feature	Aphallia	Diphallia
Incidence / defect	1 in 10–30 million; genital tubercle fails; 46,XY, normal scrotum and testes	1 in 5 million; from accessory to complete duplication
Associations	Cryptorchidism, VUR, renal agenesis, horseshoe kidney, imperforate anus, cardiopulmonary; ≈ 1/3 stillbirth/neonatal death	Hypospadias, bifid scrotum, bladder duplication, renal, pubic diastasis, anal, cardiac
Work-up / decision	Karyotype, anomaly screen, MRI; then neophallus or reassignment (reassigned may develop male identity)	Renal US, VCUG, penile US/MRI; individualised

- In aphallia the urethra most often opens **postsphincteric** (anal verge), with the best survival.

- Penile cysts are commonest and are excised; **juvenile xanthogranuloma is observed for 1 year**.
- Lymphoedema may be primary (Milroy, Meige, praecox) or acquired (anogenital Crohn disease); surgery is excisional, not drainage. Haemangiomas involute, while vascular malformations persist and are resected.

7 · Urethral anomalies

P2 *Megameatus intact prepuce / Anterior urethral valve — MIP: wide meatus under complete prepuce, found at circumcision, pyramid repair; AUV: bulbar, diverticulum lip, VCUG, endoscopic ablation*

- If MIP is recognised during circumcision, stop and preserve the prepuce.
- **Urethral duplication** is usually sagittal (≈ 94%), and the **ventral channel is the functional one**. The **Effmann** classification is type I incomplete (IA distal, IB proximal, blind), type II complete (IIA two meatuses, including IIA2 "Y-type" perineal; IIB one meatus), and type III with bladder duplication.
- Duplication presents with double stream, incontinence or UTI and is evaluated by VCUG, RUG and cystoscopy. **Never use the hypoplastic dorsal channel as the main urethra**.
- **Congenital urethral fistula** is subcoronal with a normal meatus and is closed in layers.
- **Cowper syringocele** (Maizels types: simple, perforated, imperforate, ruptured) needs endoscopic unroofing if symptomatic.
- **Megalourethra** is **scaphoid** (commoner; spongiosum deficient) or **fusiform** (rarer; cavernosa also deficient, often lethal). Both go with prune-belly and are repaired by reduction urethroplasty.

8 · Priapism

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

Type (children)	Cause	Treatment
Ischaemic	**Sickle cell** (commonest), leukaemia	Hydration, O_2, alkalinisation, analgesia, with or without transfusion; then aspiration with or without phenylephrine; then shunt
Non-ischaemic	Straddle injury causing an arteriocavernous fistula	Observe; if persistent, superselective embolisation
Stuttering	Sickle cell	Bedtime pseudoephedrine; oral β-agonist; long-term PDE5 inhibitor
Neonatal	Idiopathic, birth trauma, polycythaemia	None — resolves in 2–6 days

- Corporal blood gas decides urgency: ischaemic blood is dark, hypoxic and acidotic in a rigid, painful penis, while arterial blood is bright red in a non-rigid, painless one. Counsel about fibrosis and ED.

9 · Scrotal anomalies

- **Penoscrotal transposition** is failed inferomedial migration of the labioscrotal swellings. It goes with proximal hypospadias, and in the complete form **up to 75%** have upper-tract

anomalies, so do renal US and VCUG.

- Repair it with the hypospadias by **rotation-flap scrotoplasty**.

Anomaly	Feature	Association	Management
Bifid scrotum	Unfused folds	Proximal hypospadias	Scrotoplasty with hypospadias repair
Ectopic scrotum	Suprainguinal (commonest), infrainguinal, perineal	Ipsilateral renal agenesis/dysplasia, ectopic ureter; cryptorchidism; perineal lipoma usually isolated	Upper-tract US; scrotoplasty and orchidopexy at 6–12 months
Hypoplasia / agenesis	Under-developed / absent	Undescended testis (UDT); agenesis — all XY, bilateral UDT	—
Scrotoschisis	Rupture, extruded testis	Meconium peritonitis	Explore, orchidopexy, close

10 · Inguinal hernia and hydrocele

*All reflect incomplete closure of the **processus vaginalis**.*

Entity	Processus vaginalis	Plan
Indirect hernia	Patent; contents enter	Repair soon after diagnosis
Communicating hydrocele	Patent, fluid only	Repair if > 1–2 y, enlarging or fluctuating
Hydrocele of cord	Isolated patent segment	Rarely resolves; repair after 1 year
Scrotal (non-communicating)	Closed	Observe — most resolve by 1 y
Abdominoscrotal	Closed; through ring (≈ 1%; bilateral 30%)	Excise/drain; ring tightening unnecessary

- Hernia affects 1–5% of children and more with prematurity (≈ 30% if < 1 kg); right to left is 3:1. Bilateral risk is higher with prematurity, age under 1 year, cryptorchidism, ascites and VP shunts.
- An **intermittent bulge** means a patent processus, so operate. Incarceration needs immediate exploration, and reducible hernias elective repair within weeks; in preterm infants repair before discharge.

ALGORITHM

1. Skin-crease incision; open the external oblique, sparing the ilioinguinal nerve (not opened in young infants, Mitchell-Banks).
2. Split the cremaster and find the sac anteromedially; separate it from the vas and vessels.
3. Perform **high ligation** at the internal ring and tighten a wide ring.
4. Finally, **confirm the testis is scrotal**.

- Recurrence is 0.5–1% (3–6% after incarceration), from a missed or torn sac or a missed direct hernia. Other complications are iatrogenic cryptorchidism, atrophy and vas injury.
- Laparoscopic repair now matches open repair and closes the contralateral ring.
- Contralateral patency is ≈ 40% in infants, but a metachronous hernia develops in only ≈ **7%**, so routine contralateral exploration is not advised.

11 · Acute scrotum

Only cord torsion is time-critical.

Feature	Cord torsion	Appendage torsion	Epididymitis
Frequency	20–30%	**40–60% (commonest)**	5–15%
Peak age	Perinatal; 12–16 y	7–12 y	Infancy, puberty
Onset / vomiting	Acute severe; vomiting common	Sudden or insidious; uncommon	Insidious; rare
Cremasteric reflex	Usually absent	Present	Present
Signs	High-riding, horizontal lie	**Blue dot**, upper-pole nodule	Tender epididymis; pyuria 20–40%
Colour Doppler US (CDUS)	Absent/reduced flow; **whirlpool/snail sign**	Avascular upper-pole lesion, hyperaemia	Increased epididymal flow
Management	**Emergency exploration**	Ice, NSAIDs, rest	Supportive; antibiotics only if UTI

Intravaginal torsion

- The **bell-clapper** deformity is found in ≈ 12% at autopsy. Torsion affects ≈ 1 in 4000, is familial and is commoner if cryptorchid. Delay to presentation is the main cause of testis loss.
- A cremasteric reflex is present in up to 10% of torsions, so it does not exclude torsion.
- **TWIST score**: †swelling 2, hard testis 2, absent reflex 1, nausea/vomiting 1, high-riding 1. A score ≥ 5 is high risk, so go to theatre without imaging; ≤ 2 is low risk.
- On CDUS, absent flow is specific but only 63–90% sensitive. **Do not delay surgery for imaging when suspicion is high.**

Hours from onset	0–6	7–12	13–18	19–24	> 24	> 48
Orchidectomy risk	**≈ 5%**	20%	40%	60%	80%	90%

ALGORITHM

1. If torsion is suspected, explore the scrotum.
2. Detorse and apply warm packs. **Meanwhile fix the contralateral testis with non-absorbable suture**.
3. Reassess. If viable, then fix in a dartos pouch or with non-absorbable sutures. If non-viable, then do an orchidectomy; offer a prosthesis later.

- Manual detorsion fails in ≈ 1/3, and orchidopexy is still needed after success.
- Salvaged testes may still atrophy, with semen defects from ischaemia–reperfusion.
- **Intermittent torsion** precedes acute torsion in 30–50%, so offer **elective bilateral orchidopexy**.

Perinatal (extravaginal) torsion

- Cord and tunica twist together before scrotal fixation, giving a hard painless testis. Salvage is rare and it is bilateral in 5–22%; explore urgently if a normal postnatal exam was documented.

Appendage torsion, epididymitis, others

- Explore appendage torsion only if cord torsion is not excluded or pain is prolonged.
- Treat epididymitis with ice, NSAIDs and elevation, adding antibiotics if pyuria. A prepubertal boy with a positive culture needs renal US and VCUG.
- Idiopathic scrotal oedema shows the **fountain sign** and is treated with an antihistamine.

12 · Adolescent varicocele

P1 P3 *Varicocele – Pathophysiology / Varicocele: controversies — left-sided anatomy, nutcracker, grading, right/non-decompressing: image for mass; adolescent: observe, repair for asymmetry*

- Prevalence is ≈ **15%**; it appears after 10 years and peaks at Tanner 3.
- Measure testicular volume by **US (L × W × H × 0.71)**, serially. Peak retrograde flow > 38 cm/s correlates with asymmetry.
- Catch-up growth may be spontaneous, so **observation is the default**.

> **KEY POINT**
>
> **Indications for repair**: left testis > **20%** smaller (or bilateral hypotrophy), pain, abnormal semen (Tanner 5 / ≥ 18 y), or intratesticular varicocele with asymmetry. Total volume < 30 mL and low total motile count predict poor semen.

Technique	Recurrence	Hydrocele	Note
Open retroperitoneal (Palomo, mass ligation)	2–4%	Up to 30%	Lymphatics ligated
Laparoscopic, non-sparing	0–9%	11–32%	—
Laparoscopic artery/lymphatic-sparing	1–7%	0–4%	Lymphatic sparing cuts hydrocele
Microscopic subinguinal	0–10%	0–6%	Best balance; rare atrophy
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy (retrograde/antegrade)	6–35% / 4–12%	Occasional	Mainly for recurrence

- Hydroceles may appear 2 or more years after repair, and up to half need intervention.

13 · Epididymal and vasal anomalies

- Spermatoceles are postpubertal, contain sperm and are mostly observed.
- Bilateral absent vas suggests a **CFTR** mutation; unilateral absence suggests **renal anomalies**.

58 Management of Abnormalities of the Genitalia in Girls

1 · Embryology that explains the anomalies

- The **perineal body** splits the cloacal sphincter, so urogenital diaphragm and external anal sphincter are **both pudendal-innervated**.
- Müllerian ducts arise lateral to wolffian ducts, so a **müllerian anomaly travels with an ipsilateral renal anomaly**. Fused ducts form uterus, cervix and upper two-thirds of vagina; **sinovaginal bulbs** form the distal vagina.
- **Caudal regression** (weeks 4–5 insult) is linked to maternal diabetes; cranial extension gives VATER or MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).

2 · Evaluation and classification

- The **pull-down manoeuvre** (labia drawn caudally and laterally) shows the distal vagina. Discharge or bleeding needs vaginoscopy under anaesthesia.
- Ultrasonography (US) shows anatomy and upper tracts; **magnetic resonance imaging (MRI) is the gold standard** for müllerian anatomy.

Category	Mechanism	Examples
Hypoplasia/agenesis	Fails to form	Mayer–Rokitansky–Küster–Hauser (MRKH)
Vertical fusion	Faulty müllerian–sinus canalisation	Imperforate hymen, transverse septum, distal atresia
Lateral fusion	Failed midline fusion/resorption	Didelphys, Herlyn–Werner–Wunderlich syndrome (HWWS)

> **KEY POINT**
>
> Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery.

3 · Clitoris

- **Clitoromegaly** is commonest from **congenital adrenal hyperplasia (CAH)**, so check electrolytes, 17-hydroxyprogesterone and karyotype (Ch 59). A large hood with normal corpora is **pseudo-enlargement**; a **diminutive clitoris** suggests complete androgen insensitivity syndrome (CAIS).

4 · Vestibule: the interlabial mass

- **Urethral prolapse** affects prepubertal African-ancestry girls: a **doughnut with central meatus** needing no imaging. Sitz baths and topical steroid or oestrogen come first, then **circumferential excision**.
- A **Skene cyst** gives a **displaced meatus, eccentric stream** and usually ruptures.
- A **Gartner cyst / covered ectopic ureter** on the anteromedial vaginal wall is incised; upper-pole heminephrectomy follows if wetting or infection persists.
- A **prolapsed ureterocele** gives an infant retention and a pink-to-dusky mass; decompress, then reduce.

- An **ectopic ureter** below the sphincter causes **daytime wetting despite normal voiding**.

> **KEY POINT**
>
> Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Any cyst of urethral or vaginal origin needs a **renal US, always**.

- A **labial swelling** is commonest an inguinal hernia; **childhood asymmetric labium majus enlargement (CALME)** is physiological and is observed.

5 · Vagina: obstruction and absence

Block level and uterus presence decide the operation.

Entity	Level / uterus	Clue	Treatment
Imperforate hymen	Hymen; normal	Whitish bulge (neonate); bluish bulge and amenorrhoea (adolescent)	**Transverse** incision; not aspiration
Transverse septum	Upper/middle third; normal	Normal hymen, blind vagina	Excision, Z-plasty, balloon; hollow mould
Distal atresia	Sinus; normal	Shallow dimple; distended vagina per rectum	Pull-through with or without flaps
MRKH	Bulbs/plate; absent	Hymenal fringe and pouch; 46,XX	Dilation first
Cervical agenesis	Cervix; uterus present	Obstruction; MRI	Often hysterectomy and neovagina
HWWS	Lateral fusion; didelphys	Menstruates; lateral bluish bulge	Wide septal incision

- **Imperforate hymen** is the commonest obstructive anomaly; aspiration causes ascending infection.
- **Transverse septum** (~1 in 70,000): delay causes endometriosis, and **MRI distinguishes a high septum from cervical agenesis**. **Z-plasty** makes the scar longitudinal; the main complication is **stenosis**, and a **hollow mould** is used when a uterus is present.
- **Distal atresia**: the haematocolpos acts as a tissue expander; drain, then pull through.

MRKH (müllerian agenesis)

- MRKH (~1 in 5000) is 46,XX with **functioning ovaries**, the second commonest primary amenorrhoea after gonadal dysgenesis.
- Renal anomaly occurs in ≈ **one-third**, skeletal in 10–20%. **Type A** is symmetrical without extra-müllerian anomaly; **Type B** is asymmetrical with renal, Klippel–Feil or MURCS anomalies and needs full renal and skeletal work-up.

6 · Vaginal replacement

- **Dilation is first-line** (Frank; Ingram); surgery is no better. **Vecchietti** uses an olive with laparoscopic traction.
- **McIndoe** places a split-thickness graft over a stent; it stenoses, so dilation is mandatory. Alternatives include **buccal mucosa** and **robotic peritoneal vaginoplasty**.

- **Sigmoid** is the default bowel; excise ~3 cm of intervening bowel to avoid overlapping suture lines and fix it against prolapse. **Ileum** stenoses more and is kept for an irradiated pelvis or absent colon.
- Bowel risks discharge and **adenocarcinoma**; skin risks stenosis and **squamous carcinoma**. Every neovagina needs **annual examination**.

> **KEY POINT**
>
> **Cervical atresia**: a perineum-to-uterus anastomosis loses the endocervical barrier and risks life-threatening ascending infection. Hysterectomy with neovagina is often best; otherwise use endometrial suppression and IVF with transmyometrial transfer.

7 · Lateral fusion anomalies

HWWS girls menstruate, so obstruction is missed.

- **Didelphys** has a septate vagina in up to 75%; excise it only for symptoms. Incise a **septate uterus** only for recurrent pregnancy loss.
- **HWWS** is didelphys with a unilateral imperforate hemivagina: pain **without amenorrhoea**, a lateral bluish bulge and **ipsilateral renal agenesis**. Treat by wide septal incision.

8 · Acquired disorders

- **Vaginal rhabdomyosarcoma** is a grapelike bleeding mass under 2 years with the best prognosis of genital primaries. Chemotherapy is primary; IRS trials cut surgery from **100% to 13%**.
- **Labial adhesions** are the commonest interlabial abnormality (peak 13–23 months); **80% resolve in 1 year**, so treat only symptoms. Use topical oestrogen or betamethasone 0.05% before separation.

SAQ Female circumcision

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Female circumcision" · P2 SAQ "Female Circumcision"

Taken as female genital mutilation/cutting (Chapter 58 calls it "female circumcision (infibulation)"); the answer follows WHO (World Health Organization) definitions.

Definition and epidemiology

- **FGM (female genital mutilation):** all procedures that partly or totally remove the external female genitalia, or otherwise injure them, for non-medical reasons (WHO).
- About **200 million** living women and girls are affected; about 3 million African girls a year. Mainly Africa, the Middle East and Asia (Indonesia, Malaysia); practised in some Indian communities ("khatna").
- Usually done in childhood, at 4–10 years, without anaesthesia, as a rite of passage; it predates Islam. Pharaonic type = infibulation.

WHO classification

Type	Description
I (clitoridectomy)	Partial or total removal of the clitoral glans and/or prepuce
II (excision)	Removal of the clitoral glans and labia minora, with or without the labia majora

Type	Description
III (infibulation)	Narrowing of the vaginal opening (introitus) by cutting and joining the labia into a seal, with or without clitoral excision
IV (other)	Pricking, piercing, incising, scraping, cauterisation

- The older Toubia scheme numbers differ: infibulation is Type III in WHO.

Complications

- **Immediate:** pain, haemorrhage, shock, infection (tetanus), **acute urinary retention**, urethral or rectal injury, death.
- **Urological:** slow stream, dysuria, **recurrent UTI (urinary tract infection)**, meatal obstruction under the scar, urethral stenosis, incontinence, vaginal stone from urine stasis, **vesicovaginal fistula** after obstructed labour.
- **Gynaecological and sexual:** retained menses, dysmenorrhoea, inclusion cysts, keloid, dyspareunia, apareunia, infertility.
- **Obstetric:** obstructed labour, perineal tears, postpartum haemorrhage, neonatal death.
- **Psychological:** anxiety, depression, PTSD (post-traumatic stress disorder).

Evaluation

- Sensitive, culturally respectful history; examination to classify the type; drawings or photographs for consent.
- Flow rate and residual urine; cystoscopy if fistula is suspected.

Management

- **Deinfibulation (type III):** anterior **midline division of the scar** under local or regional anaesthesia to expose the urethral meatus and introitus; raw edges oversewn (subcuticular closure of each edge). High success.
- **Timing:** ideally before pregnancy, or antenatally, or in labour. Relieves retention, UTI and dyspareunia.
- **Never reinfibulate** (illegal in the UK).
- Excise cysts, treat stenosis, repair fistula; counselling.
- Clitoral reconstruction: less predictable, limited evidence and follow-up.

Legal and ethical aspects

- WHO and the UN (United Nations) condemn FGM as a human rights violation; WHO opposes its performance by health workers.
- Illegal in many countries (e.g. UK FGM Act 2003, with mandatory reporting in girls under 18). India has no specific law.

Diagram to draw: vulva showing types I–III and the deinfibulation incision line.

> **Key points to remember:**
> - WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
> - Urological: retention, recurrent UTI, stenosis, stones, fistula.
> - Midline deinfibulation for type III; never reinfibulate.

Also, from the condensed book (not in the paper answer):

- **Condylomata** suggest abuse, though perinatal transmission occurs; over 50% resolve.
- An **inguinal hernia** may present CAIS (~1–2.5%); the **round ligament is absent in genetic males**. If CAIS is found, preserve the testes.

59 Differences of Sexual Development

1 · Definition and nomenclature

- In a **difference of sex development (DSD)** chromosomal, gonadal or anatomical sex is atypical. The **2006 Chicago consensus** replaced "intersex" and "hermaphrodite" with karyotype-based terms.

Old term	2006 term
Male pseudohermaphrodite	46,XY DSD
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
XX male / XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis (not androgen insensitivity syndrome, AIS) MCQ

- Offer endocrine and genetic testing for **proximal hypospadias even with descended testes**.

2 · Normal sex development

- **SRY** upregulates **SOX9**, making Sertoli cells. Genes acting elsewhere explain syndromic DSD: **WT1** (Wilms tumour, nephropathy), **SF1** and **DAX1** (adrenal failure), SOX9 (campomelic dysplasia).
- **Sertoli anti-Müllerian hormone (AMH)** from 7–8 weeks makes Müllerian ducts regress by ≈ 10 weeks. **Leydig testosterone** from ≈ 9 weeks builds the Wolffian ducts.
- **5α-reductase 2 makes dihydrotestosterone (DHT)**, the active androgen for prostate, urogenital sinus and external genitalia.

> **KEY POINT**
>
> The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. A uterus in 46,XY means Sertoli (AMH) failure, that is dysgenesis, not a synthesis or receptor defect.

3 · Classification (2006 / Chicago)

Category	Conditions
Sex chromosome DSD	47,XXY Klinefelter; 45,X Turner; 45,X/46,XY mixed gonadal dysgenesis (MGD); 46,XX/46,XY chimerism
46,XX DSD — gonadal	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (SRY+, SOX9 duplication)
46,XX — androgen excess	Fetal: congenital adrenal hyperplasia (CAH: 21-hydroxylase, 11β-hydroxylase, 3β-HSD, P450 oxidoreductase); fetoplacental: aromatase deficiency; maternal: luteoma, drugs
46,XX — other	Cloacal anomalies; Mayer–Rokitansky–Küster–Hauser (MRKH); vaginal atresia

Category	Conditions
46,XY DSD — gonadal	Complete/partial gonadal dysgenesis (SRY, SOX9, SF1, WT1, DHH, MAP3K1); ovotesticular; testis regression
46,XY — androgen synthesis	LH receptor; Smith–Lemli–Opitz; StAR; 3β-HSD; 17α-hydroxylase; 17β-HSD3; 5α-reductase 2
46,XY — androgen action	Complete and partial AIS (CAIS, PAIS)
46,XY — other	Persistent Müllerian duct syndrome (PMDS); vanishing testis; isolated hypospadias; hypogonadotropic hypogonadism; cryptorchidism

- 3β-HSD and P450 oxidoreductase deficiency virilise 46,XX yet undervirilise 46,XY.

4 · 46,XX DSD

CAH is commonest and can kill a newborn through salt loss.

CAH — 21-hydroxylase deficiency

- CAH is the commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-hydroxylase deficiency (21-OHD) is **95%** of CAH; it is autosomal recessive, ≈ 1 in 15,000.
- 46,XX infants have normal ovaries and Müllerian structures, often with a **common urogenital sinus**; some look like a boy with **bilateral impalpable gonads**. Virilisation is graded **Prader I–V**.
- **17-hydroxyprogesterone (17-OHP) is raised**; measure it after **36 h**.
- A **salt-wasting crisis** (low sodium, high potassium, acidosis, shock) usually comes no earlier than **day 5–7**.
- Treat with glucocorticoid, mineralocorticoid and salt (hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone, NaCl in infancy; double-to-triple the dose in illness or surgery).
- Affected boys look normal, so **newborn screening** matters most.
- Males may get **testicular adrenal rest tumours**, which are benign and regress with more glucocorticoid but obstruct tubules, so bank sperm.
- Forms are **classic salt-wasting**, **simple virilising** and **nonclassic** (20–50% enzyme activity; hirsutism, oligomenorrhoea; may need no treatment).

Enzyme	Share	BP / K^+	46,XY	Signature
21-hydroxylase	95%	Salt wasting; low BP, high K^+	Normal	High 17-OHP
11β-hydroxylase	≈ 5%	Hypertension, low K^+	Normal	—
3β-HSD	<1%	Salt wasting	Undervirilised	High dehydroepiandrosterone (DHEA)
17α-hydroxylase	Rare	Hypertension, low K^+	Female/atypical; delayed puberty	Low androgens

Other 46,XX DSD

SAQ 46,XX male

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "46-XX Male" · P2 SAQ "46 XX Male"

Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.

Definition and classification

- **Testes and male genitalia in a person with a 46,XX karyotype.** Old term "XX sex reversal"; renamed **46,XX testicular DSD** (Chicago consensus 2006).
- First described by **de la Chapelle (1964)**. About **1 in 20,000** males.
- In the DSD classification it is a 46,XX DSD of gonadal development, together with ovotesticular DSD; **CAH (congenital adrenal hyperplasia)** is the androgen-excess type.

Genetics

Type	Mechanism	Phenotype
SRY-positive (about 80–90%)	**SRY** (sex-determining region Y) translocated from Yp to the X chromosome during paternal meiosis	Usually normal male genitalia
SRY-negative	Testis genes switched on without SRY: **SOX9** duplication or upregulation, SOX3, loss of RSPO1 or WNT4	More hypospadias, undescended testes, ovotestis

- SRY activates SOX9, so Sertoli cells form and secrete **AMH (anti-Müllerian hormone)**: no uterus. Leydig cell testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis always fails.

Clinical features

- Normal male identity; often found at **infertility work-up** for azoospermia, or prenatally (cell-free DNA 46,XX but male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia in some; progressive testicular regression.
- Height normal or **shorter than average** (no Y growth genes), unlike Klinefelter.

Feature	46,XX male	Klinefelter (47,XXY)
Frequency	1 in 20,000	1 in 500–1000
Height	Normal or short	Often tall
Genital anomalies	More common	Rare
Sperm retrieval	**Never (no AZF)**	Possible by micro-TESE (microdissection testicular sperm extraction)

Evaluation

- Semen analysis (azoospermia); FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY** confirms; DSD gene panel if SRY-negative; scrotal ultrasound.
- In an infant with atypical genitalia, exclude CAH (17-hydroxyprogesterone) and ovotesticular DSD.

Management

- **Testosterone replacement** when hypogonadal (libido, bone density, muscle).

- **Infertility:** testicular sperm extraction is futile; counsel for **donor sperm or adoption**.
- Repair hypospadias or undescended testis; reduction mammoplasty for persistent gynaecomastia.
- Genetic counselling, psychological support and bone density monitoring.

Diagram to draw: X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

> **Key points to remember:**
> - 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
> - Azoospermia always (AZF absent): no sperm retrieval.
> - Shorter than Klinefelter; FISH for SRY confirms.

Also, from the condensed book (not in the paper answer):

- Aromatase deficiency virilises mother and infant; maternal androgen exposure ends at birth, so no postnatal therapy is needed.

5 · 46,XY DSD

A uterus separates dysgenesis from synthesis or receptor defects.

Condition	Key features
Complete gonadal dysgenesis (Swyer)	Streaks, female genitalia, uterus; absent puberty; tumour 30–45%, so prophylactic gonadectomy; oestrogen; donor-ovum pregnancy
Partial gonadal dysgenesis	Dysgenetic testes, atypical genitalia, variable Müllerian structures. WT1: Denys–Drash (congenital nephrotic syndrome, Wilms 90%) or Frasier (late focal segmental glomerulosclerosis, gonadoblastoma 60%, no Wilms)
CAIS	X-linked; female genitalia, short blind vagina, no uterus; breasts by aromatisation; ≈ 50% have inguinal hernia; 1–2% of girls with hernia are CAIS
PAIS	Atypical genitalia; identity mostly male; pubertal breast growth
5α-reductase 2	Autosomal recessive; atypical/female at birth, blind pouch; **virilises at puberty**, fertility reported; testosterone:DHT **>20:1**; "guevedoces"
17β-HSD3	High androstenedione:testosterone ratio; virilises at puberty
LH receptor defect	Leydig hypoplasia; no hCG response
PMDS	AMH or AMH-receptor mutation (autosomal recessive); male genitalia with uterus and tubes; found at hernia repair or orchidopexy; 30–50% of transverse testicular ectopia
Vanishing testes	Loss after first trimester: male genitalia, impalpable testes, no Müllerian structures; undetectable testosterone/AMH, high gonadotropins at mini-puberty; testosterone from adolescence

- PAIS raised male may respond to high-dose testosterone; raised female, use a GnRH agonist or gonadectomy, then oestrogen. In CAIS, vaginal dilation usually suffices.
- **PMDS (Clarnette)**: type 1 is bilateral intra-abdominal (60–70%); type 2, testis in sac or scrotum with contralateral hernia (20–30%); type 3, transverse ectopia (10%).
- PMDS needs orchidopexy. The vasa hug the uterus, so Müllerian malignancy risk (3–8%) is weighed against careful removal sparing the vasa.

6 · Sex chromosome DSD and ovotesticular DSD

SAQ Mixed gonadal dysgenesis.

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 3, Q3e (DSD management in general in 2019-Summer Paper 3, Q3f)

Definition

- MGD (mixed gonadal dysgenesis): a **dysgenetic testis on one side and a streak gonad on the other**, with persistent Müllerian structures and variable virilisation.
- A sex chromosome DSD (difference of sex development), usually **45,X/46,XY mosaicism** (about 1 in 5,000 births). **Second commonest cause of neonatal atypical genitalia after CAH (congenital adrenal hyperplasia).**

Pathogenesis

- The Y-bearing line forms a testis; the 45,X line leaves a streak.
- AMH (anti-Müllerian hormone) and testosterone act **locally**, so anatomy is asymmetric: Wolffian duct on the **testis side**; tube and hemiuterus on the **streak side**.
- 45,X/46,XY spectrum: female (Turner + Y), atypical (MGD) or male with normal testes (about 85%).

Clinical features

- **Asymmetric genitalia:** phallus with chordee, perineal hypospadias, urogenital sinus; one testis in a hemiscrotum, the other side impalpable. Hernia may contain a uterus or tube.
- **Turner features:** short stature, coarctation, horseshoe kidney.

Evaluation (MDT, multidisciplinary team)

- **Karyotype** (many cells, for mosaicism); **17-OHP (17-hydroxyprogesterone)** to exclude CAH; LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, AMH.
- **Pelvic ultrasound** (uterus, asymmetric gonads); genitography, endoscopy; **laparoscopy with gonadal biopsy**. No gonadectomy during diagnosis.
- Echocardiography, renal ultrasound (Turner screening).

	MGD	Ovotesticular DSD
Karyotype	45,X/46,XY	Mostly 46,XX
Gonads	Dysgenetic testis + streak	Ovary + testis, or ovotestis
Tumour risk	15–35%	About 3%

Management

- **Sex of rearing:** individualised; more androgen exposure predicts male identity; dysphoria is commoner when raised female. Avoid irreversible infant surgery.
- **Gonads:** germ cell neoplasia 15–35% (gonadoblastoma, then dysgerminoma or seminoma) in streak and testis alike. **Remove streaks** (biopsy first if tissue uncertain).
- Raised male: keep a scrotal testis under surveillance; bring down an intra-abdominal one if possible.
- Raised female: remove or relocate the testis.

- **Surgery:** male: staged hypospadias repair, orchidopexy; female: feminising surgery deferred.
- **Hormones:** pubertal sex steroids; growth hormone. Infertility usual.

Diagram to draw: asymmetric anatomy: testis with vas; streak with tube and hemiuterus.

> **Key points to remember:**
> - 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
> - Second commonest cause of atypical genitalia after CAH.
> - Tumour risk 15–35%: remove streaks.
> - Screen for Turner features.

Also, from the condensed book (not in the paper answer):

- **Turner with Y** material: tumour 12–20%; the 2017 guideline advises gonadectomy at diagnosis, on low-level evidence.
- **Turner**: coarctation, bicuspid valve, horseshoe kidney (renal anomaly ⅓–½) and ovarian insufficiency ≈ 95%, so give lifelong cardiac and renal screening, growth hormone and oestrogen.
- **Ovotesticular DSD** has ovarian and testicular tissue; most are 46,XX, with a sinus and uterus. A **vas always accompanies a testis**, a tube an ovary.
- The ovary is usually normal and the testis dysgenetic; tumour risk is ≈ 3%. Biopsy both poles of an ovotestis and defer irreversible steps.

7 · Evaluation of the newborn with atypical genitalia

LAQ DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia

Paper answer · P3

Asked as:

- Management of DSD — *Summer 2019 Paper III Q3f (SAQ)*
- Evaluation and treatment of newborn with ambiguous genitalia — *Practice Set 4 Paper III Q2 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(DSD management also answered as an SAQ in 2019-Summer Paper 3, Q3f)

DSD read as: disorders/differences of sex development. Detrusor–sphincter dyssynergia is covered in 2016-Summer Paper 1, Q3a.

Introduction and definition

- Atypical ("ambiguous") genitalia point to a **DSD (difference/disorder of sex development)**: congenital conditions with atypical chromosomal, gonadal or anatomical sex (Chicago consensus 2006).
- It is a **medical emergency only for salt-wasting CAH (congenital adrenal hyperplasia)**; sex designation is not a "social emergency". Evaluate within 24–48 hours (neonatal evaluation, first 24–48 h).

Classification (by karyotype)

Group	Main causes (examples)
46,XX DSD	**CAH** (commonest; 21-hydroxylase 95%, 11β-hydroxylase about 5%, 3β-HSD (hydroxysteroid dehydrogenase) <1%); maternal androgens or aromatase deficiency; ovotesticular; testicular (SRY (sex-determining region Y) positive, SRY+)
46,XY DSD	Partial/complete gonadal dysgenesis; androgen synthesis defects (17β-HSD3, 5α-reductase); androgen insensitivity (CAIS/PAIS, complete/partial; PAIS = partial androgen insensitivity); LH (luteinising hormone) receptor defect
Sex chromosome DSD	45,X/46,XY MGD (mixed gonadal dysgenesis); 45,X Turner; 47,XXY Klinefelter; chimerism

History

- Consanguinity, **unexplained infant deaths** (CAH), infertility or amenorrhoea in relatives; maternal virilisation, progestogens or androgens; prenatal cfDNA (cell-free DNA) karyotype versus ultrasound discordance.

Examination (gender-neutral terms)

- Stretched phallic length, glans width, meatus, number of perineal openings, labioscrotal fusion and rugosity, pigmentation, anogenital distance. **Prader grade (stage) I–V** for virilisation.
- Dehydration, hypotension, hyperpigmentation (CAH); dysmorphism.
- **Gonads are the key sign.** Ovaries never descend, so **a palpable gonad means testicular tissue**. Bilateral impalpable gonads with atypical genitalia = CAH until proven otherwise (impalpable gonads suggest CAH). Asymmetry suggests MGD or ovotesticular DSD.
- Impalpable testis with hypospadias, or proximal hypospadias, also needs DSD work-up (Kaefer: DSD in about 30%, 50% if the testis is impalpable).

Investigations (tests)

- **Karyotype** at once (preliminary 2–3 days); **FISH (fluorescence in situ hybridisation) for X/Y/SRY** within hours.
- **CAH screen: 17-OHP (17-hydroxyprogesterone) at about 36 hours** (after 36 h; earlier values are falsely high); electrolytes and glucose daily. **Emergency:** exclude **salt-wasting CAH** (low sodium, high potassium, shock); **salt-wasting appears around day 5–7** (usually after day 5–7), so normal early electrolytes do not exclude CAH.
- **At about 1 week:**

Test	Meaning
AMH (anti-Müllerian hormone)	Present = functioning Sertoli cells
Testosterone, LH, FSH (follicle-stimulating hormone)	Testicular function (mini-puberty)
Androstenedione, DHEA-S (dehydroepiandrosterone sulfate)	Adrenal enzyme block
Testosterone:DHT (dihydrotestosterone) >20:1	5α-reductase deficiency
hCG (human chorionic gonadotropin) stimulation	No rise = no Leydig tissue or LH receptor defect

- **Pelvic ultrasound:** uterus (well seen neonatally), gonads, kidneys, cerebriform adrenals (CAH); then genitography, endoscopy, laparoscopy ± biopsy, genetic panels.
- **Genitography / endoscopy:** urogenital sinus, confluence level relative to bladder neck, cervix.
- If unclear: MRI (magnetic resonance imaging), **laparoscopy ± deep longitudinal gonadal biopsy**, gene panels (genetic panels), microarray, exome. **No gonadectomy during diagnosis (while diagnosing).**

Diagnostic algorithm

```
Palpable gonads: 0 -> CAH likely; 1 -> MGD, PAIS, ovotesticular; 2 -> 46,XY DSD (PAIS, 5α-reductase, synthesis defect) 46,XX + uterus + high 17-OHP -> CAH 46,XY, no uterus -> androgen synthesis/action defect 46,XY or 45,X/46,XY + uterus -> gonadal dysgenesis / MGD
```

- A precise diagnosis is reached in nearly all 46,XX but under 50% of 46,XY DSD.

Treatment

1. Emergency and medical management (CAH)

- Salt-wasting crisis: IV (intravenous) saline, dextrose, stress-dose hydrocortisone.
- Maintenance (Endocrine Society 2018): hydrocortisone 10–15 mg/m²/day in divided doses, fludrocortisone, salt supplements in infancy; stress dosing during illness and surgery.
- **Puberty:** oestrogen or testosterone induction if gonads fail.

2. Team and family

- **MDT (multidisciplinary team):** urologist, endocrinologist, geneticist, psychologist, neonatologist, ethicist; gender-neutral language. Congratulate parents; say "your baby", never "it"; open disclosure, peer support.

3. Sex of rearing (gender assignment)

- Weighs diagnosis, prenatal androgen exposure, likely gender identity, fertility, sexual function, family wishes and culture.
- Usually female: 46,XX CAH, CAIS (complete androgen insensitivity), complete gonadal dysgenesis. Trend male: PAIS, 5α-reductase and 17β-HSD3 deficiency.
- Micropenis (stretched length under 2 cm) alone is no reason for female rearing.

4. Surgery: timing and consent

- Irreversible surgery is ideally deferred until the child can participate (consent); parents decide with the team after balanced counselling.
- **Feminising** surgery (clitoroplasty, vaginoplasty, labioplasty) is irreversible, so deferral until the child can consent is favoured unless obstructed; severe CAH is debated.
- Clitoroplasty (ventral, nerve-sparing, glans preserved; Prader III–V); vaginoplasty by confluence (flap for low, partial/total urogenital mobilisation, pull-through for high); labioplasty. The vagina's position relative to the bladder neck, not common-channel length, decides the vaginoplasty. Stress steroids in CAH.
- **Masculinising:** proximal hypospadias repair, orchidopexy, often done in infancy (complication rates high).

5. Gonads (gonadal tumour risk)

Risk	Condition	Plan / management
High	Dysgenesis with Y, intra-abdominal (15–35%); PAIS non-scrotal (50%)	Gonadectomy
Intermediate	Turner with Y (12%)	Gonadectomy
Low	CAIS (2%); ovotesticular (3%)	Keep CAIS gonads until after puberty; ovotesticular: remove discordant tissue

Follow-up and recent advances

- Pubertal induction, psychology, fertility preservation (experimental gonadal tissue cryopreservation), transition to adult care.
- Prenatal cfDNA diagnosis; rapid exome/genome sequencing. Prenatal dexamethasone for CAH is experimental.

Diagram to draw: Prader stages I–V and the urogenital sinus with confluence level; karyotype-based DSD algorithm.

> **Key points to remember:**
> - Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
> - 17-OHP after 36 h; salt crisis around day 5–7.
> - Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
> - Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

Also, from the condensed book (not in the paper answer):

- Late presenters include CAIS, Swyer, MRKH, Turner with Y, Klinefelter and 5α-reductase deficiency; after puberty, high LH and testosterone with female genitalia means CAIS.

> **KEY POINT**
>
> **Emergency**: atypical genitalia with bilateral impalpable gonads is salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP, and watch for crisis from the end of week 1. Designation and surgery are not emergencies.

- A multidisciplinary team (MDT) reviews within **24–48 h**. A urologist without a team transfers early.
- Count palpable gonads (ovaries do not descend): none suggests CAH; one, MGD, ovotesticular DSD, PAIS or partial dysgenesis; two, 46,XY DSD.
- Hypospadias with undescended testis (Kaefer 1999): DSD ≈ 30%; 50% if testis impalpable; 65% if meatus proximal.

ALGORITHM

1. Send karyotype (≥45 cells, for mosaicism; FISH for X/Y in hours), electrolytes, and 17-OHP at ≈ 36 h.
2. Pelvic ultrasonography for uterus and gonads.
3. At ≈ 1 week: LH, FSH, testosterone, oestrogen, androstenedione, DHT, AMH, with or without hCG stimulation.
4. Genetic panel, microarray or exome.
5. If still unclear, then MRI, genitography, endoscopy, laparoscopy with or without biopsy (**no gonadectomy during diagnosis**).

- Normal AMH means functioning Sertoli (or granulosa) cells; high LH and FSH mean gonadal failure.
- A precise diagnosis is nearly always made in 46,XX DSD but in **<50%** of 46,XY.

8 · Gender assignment and MDT

- The **Money (1975)** model of early designation, early surgery and secrecy failed; shared decisions replaced it.
- Female trend: 46,XX CAH, CAIS, Swyer, LH receptor defect. Male trend: PAIS, 5α-reductase, 17β-HSD3, partial dysgenesis.
- Dysphoria is highest in 5α-reductase deficiency raised female (≈ half) and lowest in 46,XX CAH raised female.
- Micropenis (<2.5 SD; <2 cm at 0–5 months) needs an endocrine work-up; female reassignment is no longer recommended.

9 · Surgery

- **Feminising** surgery (Ch 60): clitoral surgery is irreversible and infant benefit is unproven. Ideally defer it until the individual can take part.
- **Masculinising** surgery (proximal hypospadias, orchidopexy) has high complications but is revisable, so infant repair is often chosen.
- In MRKH, dilation comes first. Uterine remnants are removed or suppressed, **never joined to a neovagina** (no cervix).

10 · Gonadal tumour risk

Risk rises with Y material (gonadoblastoma locus, TSPY), dysgenesis and intra-abdominal position, and falls with a scrotal gonad.

Risk	Condition	Germ cell tumour	Management
High	Dysgenesis with Y, intra-abdominal (Swyer, MGD)	15–35% (Swyer 30–45%)	Gonadectomy at diagnosis
	Frasier	60%	Gonadectomy
	Denys–Drash	40%	Gonadectomy
	PAIS, non-scrotal	50%	Gonadectomy
Intermediate	Turner + Y	12%	Gonadectomy (debated)
	17β-HSD3	28%	Monitor or gonadectomy

Risk	Condition	Germ cell tumour	Management
	PAIS / dysgenesis, scrotal	Unknown	Biopsy at puberty (≥30 tubules, OCT3/4)
Low	CAIS	2%	Keep until after breast development
	Ovotesticular DSD	3%	Remove discordant testicular tissue
?None	5α-reductase; Leydig hypoplasia	Unknown	Unresolved

11 · Ethics

- Respect autonomy, be transparent, and avoid irreversible procedures before consent unless necessary. Psychology is integral.

60 Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

1 · Principles

*See the paper answer **DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia** (Section 2, Ch 59).*

Also, from the condensed book (not in the paper answer):

- Endocrine Society 2018: families with experienced surgeons decide. Never use a flap for a high confluence. Introitoplasty is needed after puberty in 25–100%.

2 · Urogenital sinus and cloaca — classification

In a **urogenital sinus** the vagina still communicates with the urinary tract (usually the mid-distal urethra), and both exit as one **common channel**.

Entity	Distinguishing feature
Genital atypia states	Commonest; mostly congenital adrenal hyperplasia (CAH)
Pure urogenital sinus	Normal external genitalia; behaves like a cloaca
Persistent cloaca	Rectum also joins, so there is **one perineal opening**
Female exstrophy	Probably a form of urogenital sinus

- CAH causes half of all newborn atypical genitalia and **90%** of 46,XX cases. Salt wasting occurs in **67–75%**; get a karyotype within **48 h**.

> **KEY POINT**
>
> **Governing principle**: **true urethral length** (bladder neck to confluence) decides the operation far more than common-channel length, in sinus and cloaca. Confluence level is a continuum.

- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants). Virilisation lengthens the **vagina-to-meatus** distance, so a "high" CAH confluence is an illusion.

3 · Clinical evaluation

- The **first threat is metabolic**: monitor electrolytes for salt wasting before reconstruction. **Hypertension** suggests 11β-hydroxylase deficiency.
- A **suprapubic mass** (bladder, **hydrometrocolpos**, or both) is often the only early sign of a pure sinus.
- The **Huffman index** (glans width × length) is normally **<3.5 mm²**; **>10 mm²** is a concern.
- **Hydrocolpos in cloaca** forms because urine refluxes into poorly draining vagina(s). It is commoner with a common channel **>3 cm**, duplication and posterior cloaca.

Associated (cloaca)	Findings	Frequency
Renal	Solitary kidney, dysplasia, pelviureteric junction obstruction, ectopia	Majority

Associated (cloaca)	Findings	Frequency
Uterovaginal	Duplication/septation	**53–67%**
Spinal	Sacral anomalies; tethered cord ≈ one-third (mostly no skin stigmata)	Cord anomaly on magnetic resonance imaging (MRI) ≤ 43%
VACTERL; gut	Tracheo-oesophageal fistula; malrotation, duodenal atresia	Correct malrotation at colostomy

> **KEY POINT**
>
> Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). The number of perineal orifices tells you which world you are in.

4 · Imaging and endoscopy

CAH needs little imaging; a cloaca needs a full 3-D map.

Modality	CAH sinus	Pure sinus / cloaca
Ultrasonography (US)	Mandatory; **cerebriform adrenals**	Mandatory for kidneys; poor for Müllerian anatomy
Genitography	Least useful; often inaccurate	**Mandatory**, as a computed tomography (CT) or MRI cloacogram
MRI	Limited	Pelvis, spine, sphincter muscles
Endoscopy	At reconstruction	Early, separate: decompress and define

- On a **genitogram**, a **cervical impression** at the vaginal dome means female internal organs.
- Do a **gonadal biopsy** only if it would change the gender of rearing, using a **deep longitudinal incision** to reach testicular tissue within an ovotestis.
- **Prenatal cloaca** shows fetal ascites, bilobed pelvic cysts and hydronephrosis. **Always do echocardiography.**
- In duplicated vaginas the rectal fistula most often opens **within the vaginal septum**. At **early endoscopy**, **empty the vagina first**.

Common channel (Peña)	**<3 cm**	**>3 cm**
Anomalies; hydrocolpos	Fewer	More (nearly all have urological defects)
Abdominal approach	Rarely	More likely
Bladder	Better spontaneous voiding	Clean intermittent catheterisation (CIC) far more often

5 · DSD reconstruction — timing and consent

No trial compares early with late surgery.

- **For early surgery** gives pliable tissue; advocates seek a moratorium until consent.
- **Clitoroplasty** is only for severe hypertrophy (**Prader III–IV**) and is the most controversial component.

High-confluence vaginoplasty	Arguments
Early single-stage	Redundant clitoral/sinus tissue is usable; maternal oestrogen thickens tissue; a planned one-stage repair does better
After puberty	Infant repairs stenose; avoids dilation; allows consent; delay for the small vagina (**<3 cm**)
Hybrid	Clitorolabioplasty in infancy, vaginoplasty at puberty

> **KEY POINT**
>
> Without comparative data the decision is ethical, not evidential, and rests with parents and team. Endocrine Society CAH guideline (2018): families with experienced surgeons decide; the 2006 consensus limits surgery to Prader III–IV.

Body	Position
Lawson Wilkins and European paediatric endocrine societies' CAH group (LWPES/ESPE, 2002)	Surgery at 2–6 months; not between 12 months and adolescence
International intersex surgical society (ISHID) poll	Most prefer single-stage before 2 years
American Medical Association ethics council	Rejected a moratorium after 2 years' review
Advocacy groups; some US state bills	Moratorium until consent; bills not enacted

6 · Operative principles

Operate in this order: **clitoroplasty, then vaginoplasty, then labioplasty**.

Clitoral surgery	Technique	Verdict
Amputation; Lattimer recession	Excision; recession	Abandoned
Schmid (1961) MCQ	First corporeal excision preserving the neurovascular bundle	Basis of later techniques
Kogan (1983)	Subtunical excision, lateral Buck fascia incisions	—
Baskin (1999)	Defined neuroanatomy	**Ventral** incision spares the dorsal bundle
Pippi Salle (2007)	Corpora disassembled and buried in the labia	Reversible if male identity emerges

- The **Fortunoff (1964)** posterior perineal flap, now **omega-shaped**, underlies almost every repair. **Hendren and Crawford (1969)** showed that the **confluence level determines the type of vaginoplasty**.

Vaginoplasty	Indication	Principle	Pitfall
Cut-back	Labial fusion only	Incise sinus	Misses normal-calibre vagina
Flap	Low confluence	Posterior sinus opened, flap inset; confluence not moved	**Never for high confluence**: vaginal voiding, infection, incontinence

Vaginoplasty	Indication	Principle	Pitfall
Pull-through	Very high	Vagina separated; sinus becomes urethra	If the vagina falls short, add flaps
Replacement	Absent vagina	See Ch 58	—

KEY POINT

Ask: **must the confluence be moved?** A flap brings skin to a low vagina; a pull-through takes a high vagina off the urethra. A flap on a high confluence is the classic error.

7 · Low confluence repair

ALGORITHM

1. Ensure metabolic stability; give **stress-dose steroids in CAH** and antibiotics.
2. Do endoscopy; place a **Fogarty catheter in the vagina** and a Foley in the bladder.
3. Prepare the whole lower body for supine–prone rotation.

ALGORITHM

1. Place a glans traction suture.
2. Incise around the glans, keeping the inner prepuce for the hood (**second only to the glans in sensitivity**).
3. Raise the omega flap and make Y-incisions on the labia majora.
4. Deglove to the bifurcation, keeping the ventral plate intact.

- **Ventral subtunical excision**: incise Buck fascia on the **ventral-most** corpora and remove the erectile tissue, leaving the tunics and dorsal bundle untouched.

ALGORITHM

1. Flap vaginoplasty: raise the omega flap with its fat.
2. Free the posterior sinus off the rectum (**hardest step**).
3. Open the posterior sinus in the midline and **continue to normal-calibre vagina**.
4. Inset the flap.

Labioplasty: split clitoral skin forms the **labia minora**; the labia majora are moved down by **YV-plasty**.

KEY POINT

Two fixed points: never touch the dorsal neurovascular bundle; always open back to normal-calibre vagina.

8 · High confluence: pull-through

- Only ≈ **5%** of CAH cases are high; they are commoner in pure sinus.
- The **critical step** is separating the anterior vaginal wall from urethra and bladder neck, which risks sphincter and nerves.
- **ASTRA** (anterior sagittal transanorectal approach) is done prone, dividing the anterior rectal wall: excellent exposure, no colostomy, continence unaffected.

ALGORITHM

1. Position prone; incise in the midline between sinus and rectum.
2. Divide the sinus up to normal vagina.
3. Free the anterior wall off the urethra **under direct vision**.
4. Close the sinus as the urethra in 2–3 layers; mobilise the vagina.
5. Turn supine for flaps and labioplasty.

Err on the side of the vagina; if it is short, add skin flaps.

9 · Total and partial urogenital mobilisation

- **Total urogenital mobilisation (TUM, Peña 1997)** mobilises the whole sinus en bloc to the perineum without separation.
- **Never discard the sinus**: split **ventrally** it becomes the vestibule; split **dorsally** it becomes the anterior vaginal wall.

	TUM	**Partial (PUM, Rink)**
Extent	Retropubic and suprapubic	Stops at the **pubourethral ligament**
Use	Higher confluence; cloaca	Most sinus repairs
Risk	Sphincter/nerve injury, stress incontinence	Lower; convert to TUM if needed

10 · Outcomes of sinus/DSD surgery

- After infant vaginoplasty, **secondary introitoplasty** is needed after puberty in **25–100%**, because the orifice does not grow.
- **Azziz (1986)** MCQ reported poor coitus rates, but these reflect old **cut-backs**.
- **Confluence level, number of procedures and surgeon expertise** matter more than age; no data favour postpubertal surgery.
- **Elmasri meta-analysis (2018)**: female identity ≈ 89%; vaginal stenosis **27%**; only half had comfortable intercourse.

Two-procedure dilemma	**Infancy**	**After puberty**
Approach 1	Simple clitorolabioplasty	Extensive vaginoplasty
Approach 2	Extensive single-stage clitorolabiovaginoplasty	Simple introitoplasty

Both usually mean two operations, so the key factor is a **surgeon or team with a special interest**.

11 · Cloaca — initial management

- **Hendren's rule**: never do a separate rectal pull-through; repair rectum, vagina and urethra in one stage.

ALGORITHM

1. Decompress the bowel.
2. Decompress the genitourinary tract.
3. Correct nephron-destructive anomalies.
4. Then do the definitive repair.

Colostomy	Advantage	Drawback
Low loop	—	Bowel hard to obtain for vaginoplasty; interferes with pull-through
Right transverse, divided (Hendren)	Left colic supply intact; colon available for vaginal replacement	Absorbs urine, causing acidosis
Descending, divided (Levitt, Peña)	Less resorbing mucosa; enough distal colon; less prolapse	Leave a long sigmoid

- Urine absorbed by colon causes **hyperchloraemic acidosis**. **Preserve the appendix** for a later Malone antegrade continence enema (MACE).
- **Urinary decompression**: **CIC is first line**, then a cloacal cut-back; vaginostomy or vesicostomy only if CIC fails.
- **Kidneys**: structural anomalies occur in **60–83%**; chronic kidney disease (CKD) in **75%** and end-stage renal disease in **17%**. Urodynamics are abnormal in **90%**.

12 · Cloaca — definitive repair

Repair at **6–12 months**, after urinary evaluation and exclusion of hydrocolpos.

Common channel	Urethra	Procedure (Wood algorithm)
<3 cm	**>1.5 cm**	**TUM**
>3 cm	or <1.5 cm	Urogenital separation
Any	Any	Never transabdominal TUM (ischaemia)

KEY POINT

Why 1.5 cm? A urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags the bladder neck below the continence mechanism and causes leakage.

ALGORITHM

1. **PSARVUP (Peña; posterior sagittal anorectovaginourethroplasty)**: place catheters in each channel; position **prone jackknife**.
2. Mark the anal site with a muscle stimulator; incise in the midline.
3. Open the rectal fistula and mobilise the rectum.
4. Do TUM or vaginal separation.
5. Close the urethra in 2–3 layers with a **fat flap**; rebuild the perineal body; centre the rectum in the sphincter.

- It is **better to enter the vagina than the rectum or urethra**; separate it from the urethra by **sharp dissection, not cautery**.
- Aftercare: Foley for 2 weeks, rectal dilation from 2–3 weeks and colostomy closure at ≈ 3 months. Almost one-third empty poorly, so start CIC if in doubt.

13 · Cloaca — results and follow-up

- Outcome depends mostly on the **spine** and **common-channel length**.
- **Urinary**: about half void spontaneously. Peña: CIC in **28%** (channel <3 cm) vs **78%** (>3 cm). Review yearly for life.

- **Renal** function is preserved by aggressive bladder management. Faecal continence needs a bowel programme, then **MACE**.
- Vaginal stenosis or obstructed menstruation occurs in **36–41%**. Pregnancy is possible, with **caesarean preferred**.

ALGORITHM

1. Do US **6–9 months after thelarche**.
2. Repeat every 6–9 months until menses are normal.

65 Adolescent and Transitional Urology

1 · Definitions

- **Adolescent urology** bridges care for complex congenital conditions; the adult urologist lacks **whole-patient knowledge** (prior operations, altered anatomy, families).
- **Transition** (Got Transition) is the planned move from child/family-centred to adult/patient-centred care: **planning, transfer and integration**. Transfer is only one step.

> **KEY POINT**
>
> Renal function and HbA1c are expected to worsen, so **stability and retention in follow-up** are success.

2 · Who needs transition

Category	Examples	Implication
1. Complex congenital	Exstrophy, posterior urethral valves, complex hypospadias, spina bifida (SB)	Lifelong care
2. Treated, now quiescent	Vesicoureteric reflux	Early-adult review; may discharge with open return
3. Cured, with sequelae	Pelvic rhabdomyosarcoma	Late-effects surveillance
4. New adult concern	Fertility, pregnancy	Adult expertise

Failed transition (loss to follow-up, reliance on the emergency department) raises morbidity and mortality because **preventive surveillance is lost**.

3 · The process

"Three Ps"	Content	Success measure
Preparation	Independence from 12–14 years; Transition Readiness Assessment Questionnaire (TRAQ)	All families involved
Plan	Follow-up schedule; self-care (catheterisation, irrigation, medications)	Shared with patient and future team
Provider of the future	May or may not be a new team	Named provider

- The **paediatric provider must initiate**, and never during a crisis; **pregnancy** is the exception and needs expedited transfer with co-management.
- Parents fear losing control, so give the patient **private time** (sexual function) and then ask what may be shared.
- Begin at about **12 years** (AAP) and complete by the late teens or early 20s; age **26** is the latest (US parental insurance ends). If independence is unachievable, plan custodianship.

Got Transition core element	Age (years)
Policy	12–14
Tracking, readiness, planning	14–18

Got Transition core element	Age (years)
Transfer of care	18–21
Transition complete (integration confirmed)	18–23

> **KEY POINT**
>
> Transfer is not the endpoint: transition ends only when integration into adult care is confirmed.

4 · Routes and models of care

Patients arrive by transitioned referral (ideal), referral without transition, a new adolescent problem, or an **emergency after discharge**, which is the hardest and often the commonest.

Model	Strength	Drawback
1a Lifelong paediatric urologist	Continuity; suits limited life expectancy	Patient outlives the provider
1b Paediatric urologist in adult department	Continuity and adult expertise	Cost, privileges, geography
2 Transfer to an adolescent/adult specialist	Specialist coordinates the MDT	Cross-institution communication
3 Direct transfer, no plan	The default	Invisible until an emergency

Lifelong and multidisciplinary need are **independent**: exstrophy may need only a urologist, whereas SB needs the largest MDT. **Nursing** is the backbone.

5 · Development and consent

- Avoid genital surgery at **3–6 years**; the 1975 AAP panel favoured genital surgery at **<6 months**.
- Augmentation waits until the child can catheterise and wants dryness.
- **Emerging adults (18–25 years)** consent for themselves unless they lack capacity.
- Patients with **SB** usually have normal intelligence but weak **working memory and executive function**.
- **Medical PTSD** causes avoidance and procedural panic; CBT may help.

6 · Setting up a service

- The core MDT includes paediatric and adult urologists, specialist nurses, a care coordinator, nephrologist and psychologist.
- Review consent **with the family, then with the patient alone**; standing orders (for example, urine cultures) avoid ED visits.

Barrier	Solution
Patient/parent: attachment, fear	Address the child directly early; readiness questionnaires
Paediatric provider: family trust	Ask "Am I the best person?"
Adult provider: **unfamiliarity with congenital conditions (biggest)**; finance	Partnerships, networks, training

120 Aging and Geriatric Urology

1 · Concepts and biology

- **Chronological age is a poor descriptor**; physiological and functional age (function, frailty, geriatric syndromes) matter. The term "geriatrics" dates from Nascher (1909) MCQ .
- **Cellular senescence** drives ageing, either by telomere shortening (**Hayflick limit**, 1961) MCQ or by stress; its secretory phenotype (SASP) causes chronic inflammation.

2 · Clinical evaluation

- **Homeostenosis** is narrowed physiological reserve, so recovery after a stressor is poorer.

Tier	Content	Significance
Basic activities of daily living (ADLs)	Bathing, dressing, grooming, toileting, feeding, ambulation	**Bathing lost first**, feeding last; loss predicts dependency and postoperative outcome
Instrumental ADLs (IADLs)	Cooking, cleaning, transport, finances, telephone, medicines	Independent living

- A simple screen is **whether the patient can bathe independently**.
- Age is seldom an independent **surgical risk**; predictors are **longer operating time, poor ASA score and urgent surgery**. Optimise rather than "clear" the patient.

ALGORITHM

1. Screen for frailty before elective major surgery (SPAQI).
2. If positive, do a diagnostic frailty assessment.
3. Then do a comprehensive geriatric assessment (CGA).

- The **ACS NSQIP/AGS checklist** has 13 items, only 2 of which are organ systems; the geriatrician-led **CGA** adds physical performance, albumin, creatinine, BNP and FEV1.

3 · Geriatric syndromes

- Geriatric syndromes feed one another, so treating one helps others. **Frailty** is decreased reserve causing vulnerability, distinct from comorbidity and disability.

	Accumulated deficits index	Fried phenotype
Method	Count failing systems	(1) weight loss >4.5 kg or 5%/year; (2) weak grip; (3) slow gait; (4) low activity; (5) exhaustion
Cut-off	Continuous	**≥3 frail; 1–2 pre-frail**
Validity	Does not predict postoperative outcome	Predicts death, dependence, falls, postoperative complications

Polypharmacy — Beers criteria

Harm often comes via the **prescribing cascade**, in which a side effect is treated with another drug.

Drug	Concern and action
Antimuscarinics (in dementia/delirium), tricyclics, first-generation antihistamines	Anticholinergic burden: delirium, falls, retention. Avoid combinations; no tricyclics in BPH

Drug	Concern and action
NSAIDs	GI bleeding; renal and heart failure at CrCl <30 mL/min
Systemic oestrogen; testosterone	Cancer, cardiovascular harm. **Vaginal oestrogen safe**; testosterone only for confirmed symptomatic hypogonadism
Ciprofloxacin, macrolides (**not azithromycin**), co-trimoxazole	Warfarin potentiation: monitor INR; dose-reduce at CrCl <30
Benzodiazepines	Cognition, gait, delirium. Avoid with opioids or ≥2 other CNS drugs
Nitrofurantoin	Listed at CrCl <30 or chronic use; usable in moderate impairment
Desmopressin; tramadol	Hyponatraemia: treat nocturia's cause; check sodium

> **KEY POINT**
>
> Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance.

Genitourinary syndrome of menopause (GSM)

- GSM causes dryness, dyspareunia and UTI **plus urgency and urgency incontinence**. Sex-steroid loss drives OAB in both sexes, including men on ADT or 5-ARIs.
- Treat with **vaginal oestrogen**. Prior breast cancer is not an absolute bar.
- A **urethral caruncle** causes painless spotting, so exclude gynaecological cancer.

4 · Incontinence management

- Incontinence is common but not normal; behavioural therapy comes first (see Voiding Dysfunction book, Ch 110). †Transient causes (DIAPPERS): delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess output, restricted mobility, stool impaction.

*See the paper answer **Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management** (Section 7, Ch 103).*

Also, from the condensed book (not in the paper answer):

- Beers lists **oxybutynin and fesoterodine** as "avoid" in dementia, although the paper suggests fesoterodine for the elderly; trospium has the least central penetration.
- Fewer than half persist with **antimuscarinics** at 6 months; check the post-void residual after starting.

Nursing homes

- Catheter colonisation is **98%**, so treating asymptomatic bacteriuria (ASB) causes harm.

Loeb — catheter	Loeb — no catheter	NHSN
≥1 of: fever, new costovertebral tenderness, rigors, **acute mental status change**	Acute dysuria; or fever plus new urgency, frequency, suprapubic pain, haematuria or UI	Culture ≥10^5 CFU, no other source, and clinical signs

> **KEY POINT**
>
> A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used.

5 · UTI and asymptomatic bacteriuria

- **A causal link between confusion and UTI is not supported** (see Infections book, Ch 26).
- ASB affects up to 22% of older women and about 10% of older men and needs no antibiotics. USPSTF gives grade D against screening; IDSA advises treatment **only in pregnancy and before urological instrumentation**.

6 · Cancer, end of life

- **Prostate**: USPSTF grade D, so no PSA screening over 70. With life expectancy **<10 years**, plan non-curative care.
- **Bladder**: restaging TURBT for T1 is often omitted in the elderly yet improves survival. BCG is less effective (immunosenescence). Cystectomy at ≥75 gives comparable results with careful care.
- **Palliative care is not withdrawal**; it means aggressive symptom control after a goals-of-care discussion.
- The Age-Friendly "4 Ms" are Mobility, Medication, Mentation and what Matters; urologists must report **elder mistreatment**.

> **KEY POINT**
>
> Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide.

7 · Landmark trials

Trial	Design	Result	Take-home
ASPREE	RCT: aspirin vs placebo, healthy elderly	No disability-free survival gain; higher mortality	No primary-prevention aspirin
Testosterone Trials	RCTs, hypogonadal older men	No cognitive benefit; modest sexual-function gain	Not for cognition

Effective prevention includes vitamin D with calcium, antihypertensives, statins and exercise.

126 Management and Considerations in Transgender and Nonbinary Individuals

1 · Definitions and terminology

- **Transgender and gender-diverse (TGD)** people have a gender identity that differs from their sex assigned at birth. Assigned sex, gender identity, gender expression and sexual orientation are **independent axes**.

Term	Meaning
AFAB / AMAB	Assigned female/male at birth
Transgender man / woman	Male identity, AFAB / female identity, AMAB
Transmasculine / transfeminine	Direction of gender, not a binary endpoint
Nonbinary	Neither male nor female
Gender dysphoria	Distress from identity–assigned sex discrepancy
Gender incongruence	Diagnostic term: marked, persistent incompatibility

Not all TGD people have dysphoria; some need only social transition, others gender-affirming hormone therapy (GAHT) and/or surgery.

2 · Diagnosis and assessment

	DSM-5 (2013)	ICD-11 (2019)
Term	Gender dysphoria	**Gender incongruence** (replaced "gender identity disorder")
Criteria	Marked incongruence for ≥**6 months** , significant distress or impairment	Marked, persistent incongruence; desire for treatment of experienced gender
Chapter	Mental disorders	**Moved to Conditions Related to Sexual Health** MCQ

ICD-11 keeps a code for access but drops the mental-disorder label; **severe distress is not a prerequisite**.

WPATH Standards of Care 8 (SOC-8, 2022)

- Treat only marked, sustained incongruence, after excluding other causes and optimising coexisting conditions.
- Assess **capacity to consent**, and give **fertility counselling before any treatment**, not just surgery.
- **A single opinion** from a competent professional suffices for medical and surgical treatment.
- Before gonadectomy, **consider ≥6 months of hormones** unless not indicated.

> **KEY POINT**
>
> SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling stay firm because consequences are irreversible.

3 · Gender-affirming hormone therapy (GAHT)

GAHT aims for **physiological** levels of the affirmed gender in patients with persistent dysphoria, capacity, majority age and controlled comorbidities.

Direction	Agent	Dose
Transfeminine —	Oral oestradiol	2–6 mg/day
oestrogen	Transdermal oestradiol	0.025–0.2 mg/day
	Oestradiol valerate/cypionate IM	5–30 mg 2-weekly or 2–10 mg weekly
Transfeminine —	Spironolactone	100–300 mg/day
antiandrogen	Cyproterone acetate	25–50 mg/day
	GnRH agonist	3.75 mg monthly / 11.25 mg 3-monthly
Transmasculine —	Enanthate/cypionate	100–200 mg 2-weekly
testosterone	Undecanoate	1000 mg 12-weekly (rare in the US: oil embolism)
	Gel 1.6% / patch	50–100 mg/day / 2.5–7.5 mg/day

> **KEY POINT**
>
> **Why the asymmetry?** Oestrogen alone does not suppress testosterone to the female range, so transfeminine therapy needs an antiandrogen too; testosterone suppresses the ovarian axis on its own.

Transfeminine

- **Ethinyl oestradiol is not recommended** (about 20-fold thrombosis). Use 17β-oestradiol, target **100–200 pg/mL**, and **use transdermal if VTE risk** (age >45 or previous VTE).
- **Spironolactone** is commonest in the US and **cyproterone** in Europe; GnRH agonists and bicalutamide are alternatives.
- **Irreversible effects are breast development and reduced fertility.** Risks include VTE, CVD, hypertriglyceridaemia, liver dysfunction and osteoporosis.
- Monitor levels 3-monthly in year 1, then 1–2 times a year, aiming for testosterone **<50 ng/dL**; check potassium and creatinine on spironolactone.
- **After gonadectomy, reduce oestrogen by 50–75%** (titrate to LH, FSH and bone density) and **stop the antiandrogen**.

Transmasculine

- Contraindications are pregnancy, unstable coronary disease and untreated polycythaemia; oestrogen-sensitive cancer needs oncology input. **Contraception is essential** because testosterone is teratogenic.
- Target testosterone **300–1000 ng/dL**; persistent menses respond to a progestin or endometrial ablation.
- **Irreversible effects are clitoral growth, facial hair, voice change and male-pattern baldness.**
- The main risk is **erythrocytosis**, so monitor testosterone and **haematocrit** 3-monthly in year 1, then 1–2 times a year.
- **After gonadectomy, the testosterone dose is not reduced**. **Nonbinary** regimens are individualised (for example, **nandrolone** for muscle with little hair).

4 · Urological considerations

Lower urinary tract symptoms have five contributors: **natal anatomy, hormones, surgery, pelvic floor dysfunction and psychosocial factors**.

- **Testosterone-induced vaginal atrophy** (dryness, bleeding, urgency) is treated with topical vaginal oestrogen.
- **After vaginoplasty**, about a third report altered voiding (**nocturia is commonest**); meatal stenosis, stricture and fistula also occur (see Ch 127).
- **After phalloplasty**, post-void dribble comes from pooling at the native urethra–pars fixa junction; stricture and fistula are far commoner (see Ch 128).
- Treat **pelvic floor** dysfunction before surgery; this eases neovaginal dilation.
- **Tucking** causes stasis, prostatitis-like symptoms, epididymo-orchitis and UTI. Exclude atrophy, tucking and pelvic floor causes before blaming the operation.

5 · Fertility preservation

Counsel **before hormones and/or surgery**, as WPATH, ACOG, ASRM and the Endocrine Society advise.

	Transfeminine	Transmasculine
Standard	**Sperm cryopreservation**; hold GAHT for ≥**3 months** before collection	**Oocyte or embryo cryopreservation** after ovarian stimulation
Prepubertal, or cannot pause hormones	Testicular tissue cryopreservation (experimental)	Ovarian tissue; no hormone pause needed
Obstacles	Ejaculation distress: vibratory stimulation, electroejaculation or surgical retrieval	Bleeding returns; transvaginal scans cause distress

Off testosterone, menses return in 3–6 months, and most conceive within 6 months.

6 · Cancer screening — screen the organ, not the gender

- **Prostate (transfeminine)**: rare but may be aggressive. Offer shared decision-making on PSA at 55–69. **On GAHT, the upper limit is 1 ng/mL, or a rise of ≥0.3 ng/mL from nadir**, †because androgen suppression lowers PSA. After vaginoplasty, feel the prostate through the **anterior neovaginal wall**.
- **Testis**: self-examination only.
- **Cervix (transmasculine)**: cytology 3-yearly at 21–29 years; at 30–65, cytology 3-yearly or hrHPV testing 5-yearly; stop after 65. Use a small speculum and **vaginal oestradiol for 1–2 weeks before**.

7 · Affirming practice

- Keep a **detailed organ inventory** and use inclusive intake forms (name, pronouns).
- **Trauma-informed care** (SAMHSA) uses the "Four Rs": **Realise, Recognise, Respond, Resist retraumatisation**.

127 Male-to-Female Gender-Affirming Surgery: Vaginoplasty

1 · Aim and preparation

Many failures start with a patient who cannot dilate.

- **Mental health** input judges **surgical readiness**. Dysphoria-driven depression should **improve** after surgery, so it is no reason to defer; the key risk is **failure to dilate**.

Older criteria	WPATH Standards of Care 8 (SOC8)
Two assessments	**One** (inter-assessor agreement is high)
Mental health professional only	Any licensed clinician ≥ master's degree competent in transgender health
Hormones before gonadectomy	**6 months** suffices
Comorbidity "well managed"	Risks and benefits discussed
12 months' "real-life experience"	**Removed**: dangerous for those facing violence

Capacity is procedure-specific; if doubtful, a psychologist plans support.

> **KEY POINT**
>
> SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory.

- Stay within **1 hour's drive** of hospital for the first **3 weeks**.
- **BMI** mainly determines wound healing; no cut-off is proven, but many require **≤35 kg/m²**, and a higher BMI may suit vulvoplasty.
- **Permanent preoperative hair removal** of the scrotal–perineal graft is needed, because canal hair causes infection and hairballs.
- **Continue oestrogen perioperatively**; a 2022 meta-analysis found no added VTE risk.

Orchiectomy

- It may be the sole operation or a first stage, and afterwards spironolactone is stopped. Offer **fertility counselling and sperm banking to all**.
- Use a **vertical midline scrotal incision**; an inguinal incision may compromise the penile flap.

> **KEY POINT**
>
> Plan every orchiectomy as though vaginoplasty will follow: use a midline scrotal incision and leave cord fat for labial bulk.

2 · Penile inversion vaginoplasty (PIV)

SAQ Penile inversion vaginoplasty

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Penile inversion vaginoplasty technique." · P4 SAQ "Penile Inversion Vaginoplasty Technique"

Introduction

- The commonest gender-affirming genital operation for transgender women: a functional, aesthetic, sensate vulva and vagina.
- **Principle:** homologous penile and scrotal tissues transferred on their own blood supply.

Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care, version 8):** one assessment, **6 months of hormones** before gonadectomy.
- **Permanent hair removal** of the scrotal graft area (prevents intravaginal hair).
- BMI (body mass index) usually ≤35 kg/m²; stop smoking; sperm banking (fertility preservation); continue oestrogen.
- **Choose depth:** full-depth vs **vulvoplasty (zero-depth)**: a vulva without a canal, chosen for high surgical risk, prior prostatectomy, or no wish for penetrative intercourse.

Tissue transfer (homologous tissues)

Male structure	Becomes
Glans on dorsal neurovascular bundle	**Clitoris**
Preputial mucosal collar	Clitoral hood, inner labia minora
Dorsal bulbar urethral plate (spatulated urethra)	Vestibule
Penile shaft skin (inverted)	**Neovaginal lining**
Scrotal skin (full-thickness graft)	Extra (proximal) canal lining
Lateral scrotal skin + cord fat	Labia majora

Operative steps

`Lithotomy; tranexamic acid 1 g; catheter → harvest, defat and tubularise scrotal skin graft → bilateral orchidectomy (leave cord fat for labial bulk) → deglove penis between dartos and Buck's fascia (keep dartos with skin = blood supply) → separate dorsal neurovascular bundle + glans; excise ALL corporal erectile tissue → clitoroplasty; fix at level of adductor longus tendons → shorten and spatulate urethra; excise bulbospongiosus → canal dissection between prostate and rectum: central tendon, behind prostate, preserve Denonvilliers fascia, repeated rectal checks → invert penile skin tube + graft into canal; labiaplasty → pack canal; negative-pressure dressing`

- **Why excise all erectile tissue:** residual tissue engorges and obstructs (narrows) the canal.
- Depth depends on canal dissection, not penile length.
- **Robotic peritoneal flaps** add depth with less rectal injury; **intestinal vaginoplasty** if skin is inadequate or for revision.

Postoperative care

- Pack and catheter out on about day 5.
- **Lifelong dilation**: several times daily at first, e.g. 4 times daily for 6 weeks, tapering to daily after 1 year; pelvic floor physiotherapy; continue oestrogen.

Complications and outcomes

- **Wound dehiscence** (commonest, 12–33%); granulation tissue.

- **Rectovaginal fistula** (0.8% primary, 6.3% revision): the worst complication. Urethrovaginal fistula (0.8–3.9%).
- Urethral stricture, meatal stenosis or malposition (spraying).
- **Vaginal stenosis** (poor dilation); prolapse 4–10%.
- Most achieve orgasm (median about 6 months).

Diagram to draw: sagittal view of the lined neovaginal canal between prostate and rectum.

> **Key points to remember:**
> - Glans → clitoris; penile skin → canal; scrotal skin → graft.
> - Excise all erectile tissue; keep Denonvilliers fascia intact.
> - Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

Also, from the condensed book (not in the paper answer):

- Pad the calves in lithotomy (**common peroneal nerve**) and give **tranexamic acid 1 g IV at the start**, as bleeding is front-loaded.

ALGORITHM

1. Foley and drain; mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion).
2. Harvest scrotal graft and small posterior rhomboid flap (relieves introital tension); defat, tubularise.
3. Orchiectomy, leaving cord fat; penile disassembly; corporal excision and clitoroplasty.
4. Urethroplasty, then canal dissection; line canal, labiaplasty, pack.

- **Disassembly**: the neurovascular bundle is best **left on the dorsal tunica**, and **all erectile tissue is excised**.
- **Urethroplasty**: excise bulbospongiosus **only after canal dissection**, as it covers a rectal injury, and spatulate downward for a seated stream.
- **Canal**: use repeated digital rectal examination, **preserve Denonvilliers fascia** and close it if breached. **Packing** ensures graft take.
- Keep the labia majora incision **≥1 cm** from the groin crease to avoid dehiscence.

Vulvoplasty (zero-depth)

Vulvoplasty suits those not wanting receptive intercourse and avoids **lifelong dilation**, but is not freely reversible. **Prior radical prostatectomy favours vulvoplasty**.

3 · Abdominoperineal and intestinal options

- **Peritoneal (Davydov-type)**: robotic dissection cut rectal injury from **~7%** to about 2.3%. The **rectum may mimic seminal vesicles**, and peritoneal defects must be **closed** to prevent hernia.
- Flaps are either **two-flap** (bladder and sigmoid peritoneum) or **pull-through** (abdominal wall peritoneum).
- **Intestinal** vaginoplasty suits severe penoscrotal hypoplasia or revision stenosis; IBD, adhesions, irradiation and short gut contraindicate it. **Dilation is still required.**

KEY POINT

Lining ladder: penile and scrotal skin first, then peritoneum, then bowel. Each adds depth but imports donor pathology; none removes the duty to dilate.

4 · Complications

Complication	Cause / key fact	Management
Wound dehiscence (commonest minor)	Introitus, dilator tension	Local care; granulation: curette or cautery plus steroid
Rectovaginal fistula (most devastating)	**0.8%** primary, 6.3% revision	Intra-op: watertight closure and bulbospongiosus flap, with or without diversion. Established: stop dilating and revise later, **or** delay closure
Urethrovaginal fistula (0.8–3.9%)	Missed injury, distal stenosis	Distal: often left. Bladder-neck (incontinence): transvaginal repair
Urethral stricture (1–6%)	Spraying, UTI, retention	Exclude pelvic floor dysfunction; dilate if short, else meatal spatulation
Urethral bulge / upward meatus	Under-debulked spongiosum	Debulk, fix meatus downward
Vaginal stenosis	Poor dilation, scar, graft loss, pelvic floor dysfunction	Pelvic floor physiotherapy **first**; then scar excision and graft, or peritoneal/bowel augmentation
Prolapse (4–10%)	Poor graft take or fixation	Early repack; peritoneal fixation acts as a colpopexy

KEY POINT

Three failure modes: violated plane (fistulas), tissue not excised (bulge), graft not taken or maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient, so dilation is part of the operation.

5 · Outcomes and follow-up

- **Median time to orgasm is about 180 days**. Depth is 12–16 cm, and self-lubrication is not the norm.
- Stay is about **5 days**; on **day 5 the pack and catheter come out** and dilation is taught.
- Follow-up is weekly for 3 weeks, then at 6 weeks, 3, 6 and 12 months, then annually. †The prostate is retained; survey it via the anterior neovaginal wall.

Period	Dilation (15–20 min each)
0–6 weeks	4 × daily
6 weeks–3 months	3 × daily
3–12 months	2 × daily
>1 year	Daily; less if stable

KEY POINT

The body treats the canal as a defect to close, so stenosis is a lifelong risk. Any narrowing means more dilation, contacting the surgeon and physiotherapy.

128 Special Urologic Considerations in Transmasculine Individuals

SAQ Enlist the steps in female to male gender reassignment surgery.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2024-Winter Paper 4, Q3b

Preconditions

- Gender dysphoria diagnosed per **WPATH (World Professional Association for Transgender Health)** by an MDT (multidisciplinary team).
- **Testosterone:** chapter cites **at least 12 months** before genital surgery (clitoral growth plateaus at about 2 years). WPATH SOC8 (Standards of Care 8, 2022) relaxed this to about 6 months.
- **Urological screening:** uroflowmetry, PVR (post-void residual), renal ultrasound; treat LUTD (lower urinary tract dysfunction) first.

Stage 1: removal of female structures

1. **Subcutaneous mastectomy** (usually first).
2. **Hysterectomy + bilateral salpingo-oophorectomy**, preferably laparoscopic; remove whole cervix ± upper third of vagina.

Stage 2: genitoperineal transformation

3. **Vaginectomy** (needed if the urethra is lengthened): - Hydrodissection; circumferential introital incision. - Rectovaginal plane bluntly; urethrovaginal plane sharply, **no diathermy near the sphincter**. - Paravaginal haemostasis; levators closed in midline.
4. **Urethral lengthening, fixed (perineal) part:** vestibular mucosa and labia minora tubularised over 18 Fr; covered with bulbospongiosus.
5. **Phallus construction**, either: - **Metoidioplasty:** release clitoral chordee and urethral plate; lengthen urethra to the glans (preputial or labial flap ± buccal graft). Gives a 3–8 cm phallus with erection and sensation. - **Phalloplasty: RFFF (radial forearm free flap)** or ALT (anterolateral thigh) flap with **tube-in-tube urethra**; microvascular anastomosis; **clitoral nerve coapted** to a flap nerve.
6. **Urethral anastomosis** of phallic to fixed part; neomeatus at the tip.
7. **Scrotoplasty:** labia majora flaps rotated 180° in front (Ghent type). Monsplasty and perineal closure.
8. **SPC (suprapubic catheter)** + urethral catheter; urethral catheter out at 14–21 days.

Stage 3: prostheses

9. **Testicular prostheses** (17 mL) at 6 months or more.
10. **Erectile prosthesis** after 1 year, once protective sensation returns; single cylinder fixed to pubic bone (dedicated devices, e.g. ZSI 475 FTM [female-to-male]).

Complications

- **Fistula** (phallic base), **stricture** (fixed–phallic anastomosis), hair growth, dribbling, prosthesis infection or erosion, vaginectomy bleeding or rectal/bladder injury.

Diagram to draw: three parts of the neourethra (native, fixed, phallic).

> **Key points to remember:**
> - Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
> - No urethral lengthening without vaginectomy (fistula risk).
> - Commonest stricture: fixed–phallic anastomosis.

Also, from the condensed book (not in the paper answer):

- Haemorrhage is the commonest vaginectomy complication, and a two-stage repair is standard for long strictures.

1 · Scope

- Goals are a male appearance, **voiding standing**, sensation and penetration (see Ch 126).
- **Early puberty blockade has a surgical price**: the vagina is harder to remove and small labia majora give less scrotal tissue.

2 · Preoperative care and consent

- **Urological screening is mandatory** (uroflow, PVR, renal ultrasound), and voiding dysfunction, stones and hydronephrosis are treated first.
- SOC7 requires ≥**12 months** of testosterone before genital surgery (†SOC8: 6 months) for **clitoral hypertrophy**, which plateaus at about 2 years.

Agent	Helps	Hinders
Testosterone	Clitoral hypertrophy aids metoidioplasty	Vaginal atrophy: harder vaginectomy
GnRH agonist	Prevents female characteristics	Small labia majora: less scrotoplasty tissue

Consent covers fistula, stricture, vaginectomy bleeding and organ injury, voiding dysfunction, UTI and implant problems.

3 · Removal of female structures

- At hysterectomy the ureter lies beneath the uterine artery. †If the cervix is retained, screening continues.
- **Vaginectomy** risks haemorrhage, then bladder, then rectal injury. **Without vaginectomy there is no urethral lengthening** because of fistula risk.

ALGORITHM

1. Hydrodissect; circumferential introital incision.
2. Blunt rectovaginal plane; urethrovaginal plane **sharply, with no diathermy** (protects sphincter), then bladder plane.
3. If bladder injured: close, with or without cystoscopy.
4. Free levators, control paravaginal plexus; approximate levators.
5. Bulbospongiosus over urethra; layered closure.

Timing	Verdict
All-in-one	More major complications with phalloplasty; none extra with metoidioplasty
Early vaginectomy, phalloplasty later	Keeps blood loss away from free-flap perfusion
Two-stage (preferred)	Mastectomy, hysterectomy–oophorectomy; then vaginectomy and genital reconstruction

4 · Genital reconstruction

- The fixed (perineal) and phallic neourethra both lack spongiosum. One **dorsal clitoral nerve** is coapted for sensation.
- **Bulbospongiosus over the fixed part** reduces fistulas. Scrotoplasty flaps run on the **external pudendal artery**.

	Metoidioplasty	Phalloplasty
Phallus	Own clitoris, **3–8 cm**	Flap
Erection	Spontaneous	Needs prosthesis
Sensation	Largely kept	Depends on nerve coaptation
Penetration	Usually not	Principal goal
Donor scar	None	Visible

5 · Postoperative care and catheterisation

- The neourethral catheter comes out at **14–21 days**; the suprapubic catheter is clamped until voiding is normal.
- Catheterisation is hard because the curve into the native urethra is abrupt.

ALGORITHM

1. One gentle Coudé attempt, then one stylet-guided attempt.
2. If any difficulty, stop: flexible cystoscopy, guidewire, open-tip catheter.
3. In an emergency, place a **suprapubic catheter**.

6 · Urethral complications

- **Fistula** is commonest at the phallic base, and small ones close with diversion. If one persists >**3 months**, excise and close in layers with fat interposition.
- **Stricture** is commonest at the **fixed–phallic anastomosis**. A plateau uroflow prompts urethrography; treat after **≥6 months**.

Option	Use
Dilation	Low-grade meatal stenosis
Internal urethrotomy	Low success; never repeat
Meatotomy / Heineke–Mikulicz	Short meatal / short non-obliterative
Excision and primary anastomosis	Short, near-obliterative anastomotic
Free graft	Needs flap support or dorsal inlay
Two-stage (standard)	Long/complex; retubularise ≥6 months later

Option	Use
Flap / perineal urethrostomy	Recurrent, or no further reconstruction wanted

- **Hair** is prevented with a hairless flap or laser.
- **Post-void dribbling** is near-universal; if severe, exclude sacculation or a vaginal-remnant diverticulum.

> **KEY POINT**
>
> The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure.

7 · Prostheses

- Insert an erectile prosthesis ≥**1 year** after phalloplasty, once there is **protective sensation** and a stable urethra.
- Infection or erosion needs urgent explantation, with reimplantation ≥6 months later.

★ Putting it together

A · Newborn with atypical genitalia

ALGORITHM

1. **Emergency is adrenal:** CAH is commonest (half of all; 90% of 46,XX). Salt wasting (low Na, high K, shock) appears at day 5–7 (see Ch 59).
2. **Tests:** karyotype (SRY), 17-hydroxyprogesterone after 36 h, electrolytes, pelvic ultrasound.
3. **Team:** †defer sex assignment until MDT evaluation.
4. **Gonads:** dysgenetic with Y (30–45% tumour risk): prophylactic gonadectomy.

B · Urogenital sinus and cloaca

KEY POINT

True urethral length outweighs common-channel length. Low confluence needs a flap; high needs pull-through or urogenital mobilisation. Clitoroplasty only for Prader III–IV, sparing the dorsal bundle. Cloaca: colostomy, decompress hydrocolpos, imaging with endoscopy, then posterior sagittal repair; seek VACTERL and tethered cord (see Ch 60).

C · Absent vagina

Cause	Key feature	Management
Mayer–Rokitansky–Küster–Hauser	46,XX, no uterus; †image kidneys	Dilation first; else McIndoe, bowel, buccal or peritoneal
Complete androgen insensitivity	46,XY, testes, no uterus	Dilation usually suffices
Septum / imperforate hymen	Uterus present, haematocolpos	Hymen: incision, not aspiration; septum: excision

D · Immunotherapy and hereditary cancer

ALGORITHM

1. **Checkpoints:** PD-L1 silences PD-1-positive T cells; CTLA-4 acts at priming. †Immune adverse events grade ≥2: hold drug, give steroids; endocrinopathies need lifelong replacement (see Ch 24).
2. **Hereditary:** VHL (clear cell; belzutifan), HLRCC (aggressive papillary), Birt–Hogg–Dubé (hybrid oncocytic), Lynch (upper-tract urothelial).
3. BRCA prostate cancer: PARP inhibitor. FGFR-altered urothelial cancer: erdafitinib (see Ch 25).

E · Older patient before surgery

KEY POINT

Function, not age, predicts outcome. Screen frailty, then do a comprehensive geriatric assessment (cognition, delirium, nutrition, falls, polypharmacy). Avoid anticholinergics (prefer a β3-agonist). For new incontinence, seek transient causes first (†DIAPPERS) (see Ch 120).

F · Transgender and nonbinary patients

ALGORITHM

1. Screen the organs present: retained prostate (†PSA lower on oestrogen) and retained cervix (see Ch 126).
2. Vaginoplasty: stenosis and rectovaginal fistula (most devastating) (see Ch 127).
3. Phalloplasty: fistula and stricture, worse without vaginectomy (see Ch 128).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Discuss in brief, present concepts in various GU cancers

Paper answer · P4

Type: LAQ (25 marks)

Source: Practice Set 4 Paper 4, Q1

Read as: the current key concepts in diagnosis and management of each GU (genitourinary) cancer: kidney, upper tract, bladder, prostate, testis and penis.

Common themes

- **De-escalation:** surveillance and organ sparing for low-risk disease.
- **Precision:** molecular classification, germline testing and better imaging (mpMRI (multiparametric MRI, magnetic resonance imaging), PSMA (prostate-specific membrane antigen) PET (positron emission tomography)).
- **Perioperative systemic therapy:** immune checkpoint inhibitors (ICIs), ADCs (antibody–drug conjugates) and targeted drugs.
- **Robotic surgery** as the default approach.

1. Kidney: RCC (renal cell carcinoma)

- **WHO (World Health Organization) 2022:** new molecularly defined types (FH (fumarate hydratase)-deficient, SDH (succinate dehydrogenase)-deficient, TFE3-rearranged RCC).
- **Mechanism:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (vascular endothelial growth factor). This explains why VEGF TKIs (tyrosine kinase inhibitors) and the HIF-2α inhibitor **belzutifan** work.
- **Small renal mass:** active surveillance in the elderly or unfit; renal mass biopsy; **partial nephrectomy** for cT1 to avoid CKD (chronic kidney disease); thermal ablation for small tumours.
- **Adjuvant pembrolizumab** after nephrectomy for high-risk RCC (**KEYNOTE-564**: better disease-free and overall survival).
- **Metastatic:** IMDC (International Metastatic RCC Database Consortium) risk groups. First line is an **ICI combination**: nivolumab plus ipilimumab, or an ICI plus TKI (pembrolizumab–axitinib, nivolumab–cabozantinib, pembrolizumab–lenvatinib). **CARMENA (2018):** sunitinib alone was non-inferior to cytoreductive nephrectomy plus sunitinib in intermediate/poor risk, so cytoreduction is now selective. SBRT (stereotactic body radiotherapy) for oligometastases.

2. Upper tract urothelial carcinoma (UTUC)

- **Risk stratification (EAU (European Association of Urology)):** low risk (unifocal, <2 cm, low-grade, non-invasive on CT (computed tomography) urography) goes to **kidney-sparing** ureteroscopic ablation or segmental ureterectomy; UGN-101 (mitomycin gel) is an option. High risk goes to **RNU (radical nephroureterectomy) with bladder cuff** and template lymph node dissection.
- **Single postoperative intravesical mitomycin** after RNU cuts bladder recurrence (**ODMIT-C**: 28% to 16% at 1 year).

- **POUT:** adjuvant platinum chemotherapy improved disease-free survival in pT2–T4 or N+ disease.
- **Also know:** screen for **Lynch syndrome**; aristolochic acid (Balkan) nephropathy.

3. Bladder

- **NMIBC (non-muscle-invasive bladder cancer):** EAU risk groups; en bloc TURBT (transurethral resection of bladder tumour); blue-light or narrow-band cystoscopy; re-TURBT for T1; BCG (bacillus Calmette–Guérin) induction plus maintenance. **BCG-unresponsive** disease: radical cystectomy is standard; bladder-sparing options are pembrolizumab, nadofaragene firadenovec (gene therapy) and N-803 with BCG.
- **MIBC (muscle-invasive):** VI-RADS (Vesical Imaging Reporting and Data System) on MRI; **cisplatin-based neoadjuvant chemotherapy** then cystectomy; **trimodality** bladder preservation in selected patients; adjuvant **nivolumab** (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months); perioperative durvalumab with chemotherapy (**NIAGARA, 2024**) improved event-free and overall survival.
- **Metastatic: enfortumab vedotin plus pembrolizumab** (**EV-302**) replaced platinum first line; avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

4. Prostate

- **Diagnosis: MRI before biopsy** (**PRECISION**: more significant and fewer insignificant cancers); **transperineal biopsy** (less sepsis); **PSMA PET** staging (**proPSMA**: more accurate than CT plus bone scan); ISUP (International Society of Urological Pathology) grade groups; germline testing (BRCA2).
- **Localised: active surveillance** for low risk and selected favourable intermediate risk (**ProtecT, 15 years**: cancer mortality low and similar with surveillance, surgery or radiotherapy); robotic prostatectomy; **hypofractionated** radiotherapy; focal therapy in selected men.
- **Advanced:** mHSPC (metastatic hormone-sensitive) gets ADT (androgen deprivation therapy) plus an **ARPI** (androgen receptor pathway inhibitor); **triplet** with docetaxel (**PEACE-1, ARASENS**); prostate radiotherapy for low volume. EMBARK: enzalutamide for high-risk biochemical recurrence. mCRPC (metastatic castration-resistant): **PARP (poly-ADP ribose polymerase) inhibitors** for HRR (homologous recombination repair) mutations; **^{177}Lu-PSMA-617** (**VISION**: overall survival HR (hazard ratio) 0.62).

5. Testis

- **Stage I: surveillance** preferred for most seminomas and NSGCTs (non-seminomatous germ cell tumours); alternatives are one cycle of carboplatin (seminoma) or one BEP (bleomycin, etoposide, cisplatin) for high-risk NSGCT.
- **miR-371a-3p:** new serum marker for seminoma and NSGCT.
- **Low-volume stage II seminoma: primary RPLND (retroperitoneal lymph node dissection)** (SEMS, PRIMETEST) spares chemotherapy toxicity; robotic, nerve-sparing templates preserve ejaculation.
- **Also know:** IGCCCG (International Germ Cell Cancer Collaborative Group) 2021 update; survivorship care.

6. Penis

- **WHO 2022: HPV (human papillomavirus)-associated** vs **HPV-independent** squamous cell carcinoma; p16 staining; HPV vaccination for prevention.
- **Organ sparing:** topical treatment or laser for PeIN (penile intraepithelial neoplasia), glans resurfacing, wide excision, glansectomy; a negative margin of a few millimetres is enough.
- **Nodes:** for cN0 pT1b or higher, **DSNB (dynamic sentinel node biopsy)** or modified inguinal dissection. Bulky nodes: neoadjuvant TIP (paclitaxel, ifosfamide, cisplatin), then consolidative dissection.

Diagram to draw: six-row table: cancer, key diagnostic advance, localised treatment, advanced treatment.

> **Key points to remember:**
>
> - RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
> - UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
> - Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
> - Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

SAQ Prepucioplasty

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Prepucioplasty" · P4 SAQ "Prepucioplasty."

Read as: preputioplasty, a foreskin-preserving operation for phimosis.

Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring (forced retraction, balanoposthitis) or BXO (balanitis xerotica obliterans).

Indications

- Persistent symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks with gentle retraction). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance.
- Treat active infection first.

Techniques

Technique	Method
Dorsal slit with transverse closure	Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz principle)
Triple incision	Three longitudinal cuts over the ring, each closed transversely
Y–V plasty	V-flap of outer skin advanced into a Y-cut

- **Principle:** cutting the ring lengthwise and closing it crosswise widens its circumference.
- General anaesthesia or penile block; incise only the band; confirm full retraction; absorbable sutures.

Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.
- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.
- **Also know:** preputial reconstruction during hypospadias repair is also called preputioplasty; dehiscence is its risk.

Diagram to draw: dorsal incision through the ring, closed transversely.

> **Key points to remember:**
> - Steroid first; about 1 in 6 fail.
> - Cut lengthwise, close crosswise; daily retraction after.
> - BXO: circumcision preferred.

SAQ Secondary malignancies in urology

Paper answer · P1

Asked as:

- Secondary malignancies in Urology. — *Practice Set 3 Paper I Q3f (SAQ)*
- Secondary malignancies in Uro — *Practice Set 9 Paper I Q3c (SAQ)* (also answered in Practice Set 3 Paper 1, Q3f)

Type: SAQ (10 marks)

Interpreted (read) as second cancers caused by urological or oncological treatment (radiotherapy, chemotherapy, urinary diversion, immunosuppression).

Definition

- A **second (treatment-related) malignancy** is a new, histologically distinct cancer caused by treatment of a first disease. It appears after a latency of years, often **in the radiation field**.

Causes and risks

Cause	Second cancer	Key facts
Pelvic radiotherapy	**Bladder cancer**, rectal cancer, sarcoma (in-field)	Bladder RR (relative risk) 2–4 after cervical EBRT (external beam radiotherapy), 1.5–2 after prostate EBRT, smaller after brachytherapy, little with IMRT (intensity-modulated radiotherapy). Latency 15–30 years
Cyclophosphamide	**Bladder cancer**, UTUC (upper tract urothelial carcinoma)	Only chemotherapy proven to cause bladder cancer; metabolites phosphoramide mustard and acrolein. About 4.5-fold, dose-dependent (>20 g cumulative)
Urinary diversion / augmentation	Adenocarcinoma at the uro-intestinal junction	Ureterosigmoidostomy 6–29% after 10–20 years (registry 2.6%); cystoplasty 1.6%; ileal conduit 0.02%. Risk tracks colonic mucosa exposed to urine and faeces
Transplant immunosuppression	Skin cancer, PTLD (post-transplant lymphoproliferative disorder), RCC (renal cell carcinoma), bladder cancer	RCC mostly in native kidneys; bladder cancer about 3-fold and aggressive
Testicular GCT (germ cell tumour) treatment	Solid tumours, leukaemia	60–100% excess after cisplatin or radiotherapy, 200% (highest) after both; etoposide-related leukaemia
Paediatric tumours	In-field solid tumours, leukaemia	Wilms tumour 1% at 10 years

Mechanisms

- Radiation and alkylating (drugs) or topoisomerase-inhibitor chemotherapy: mutagenic DNA damage in normal tissue.
- Immunosuppression: loss of (lost) immune surveillance and oncogenic viruses (EBV [Epstein–Barr virus], HPV [human papillomavirus]).
- Diversion: urine–faeces contact at the anastomosis.

Prevention and surveillance

- IMRT, smaller fields; **mesna** with cyclophosphamide (reduces cystitis).
- **Haematuria after pelvic radiotherapy or cyclophosphamide: always cystoscopy.**
- Endoscopy from 5 years after ureterosigmoidostomy; augmentation surveillance from about 10 years; excise a defunctioned ureterosigmoid anastomosis.
- Testis cancer: fewer treatment modalities; low-dose CT (computed tomography) in surveillance.
- **Also know:** pioglitazone, aristolochic acid and phenacetin as drug-related urothelial carcinogens.

Diagram to draw: ureterosigmoid anastomosis showing (with) adenocarcinoma at the junction.

Key points to remember:

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

SAQ Genetic counselling

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 1, Q3d

Assumed meaning: genetic counselling in urological practice.

Definition

- A communication process that helps patients and families understand the **medical, psychological and family implications** of a genetic condition, so they can make informed choices. It is **non-directive, voluntary and confidential**.
- **Germline testing** (blood) finds inherited variants with family implications; **somatic testing** (tumour) guides treatment.

Steps

1. **Three-generation pedigree**.
2. **Risk assessment:** inheritance and recurrence risk (autosomal dominant 50%; autosomal recessive 25%).
3. **Pre-test counselling:** which test; possible results (pathogenic, **VUS [variant of uncertain significance]**, negative); insurance and psychological harms; consent.
4. **Post-test counselling:** interpret (do not act on a VUS); **cascade testing of relatives**; surveillance; targeted therapy.
5. **Reproductive options:** prenatal diagnosis, PGT (preimplantation genetic testing), donor gametes; psychosocial support.

Urological applications

Condition	Genes	Key point
Prostate cancer	BRCA2 (> BRCA1), ATM, CHEK2, PALB2, mismatch repair (Lynch), HOXB13	NCCN (National Comprehensive Cancer Network): test metastatic, high-risk, strong family history; BRCA2: aggressive disease, PARP (poly-ADP-ribose polymerase) inhibitors, earlier screening
Hereditary RCC (renal cell carcinoma)	VHL, MET, FH, FLCN, TSC1/2, SDH, BAP1	Counsel RCC at ≤46 years, bilateral or multifocal, family history
UTUC (upper tract urothelial carcinoma)	Lynch (MSH2 highest)	Screen if <60 years or Lynch-spectrum cancer history
ADPKD (autosomal dominant polycystic kidney disease)	PKD1 (chromosome 16), PKD2 (chromosome 4)	Testing selects living related donors

Condition	Genes	Key point
Cystinuria	SLC3A1, SLC7A9 (recessive)	Screen siblings
Male infertility	Karyotype, AZF (azoospermia factor) deletion, CFTR (cystic fibrosis transmembrane conductance regulator)	AZF deletions pass to sons by ICSI (intracytoplasmic sperm injection); absent vasa: test partner for CFTR
DSD (differences of sex development)	e.g. CAH (congenital adrenal hyperplasia, recessive), AIS (androgen insensitivity, X-linked)	Recurrence and gonadal tumour risk

Ethics and law

- Confidentiality; right not to know.
- **India:** PCPNDT Act (Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994) bans disclosure of fetal sex.

Diagram to draw: three-generation pedigree with standard symbols.

> **Key points to remember:**
> - Non-directive; pre-test and post-test counselling; cascade testing.
> - Prostate: BRCA2 most important; test all metastatic men.
> - RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

SAQ Noonan syndrome

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 22 SAQ Sheet, Q3a

Definition and genetics

- Noonan syndrome is an **autosomal dominant RASopathy** (disorder of the RAS–MAPK [mitogen-activated protein kinase] signalling pathway) with variable expression; many cases are de novo. Incidence about 1 in 1,000–2,500.
- **Karyotype is normal** (46,XY or 46,XX). The phenotype resembles Turner syndrome, hence the old name "male Turner".
- **Mechanism:** gain-of-function variants increase RAS–MAPK signalling, which disturbs growth and organ development.

Gene	Approximate share	Note
PTPN11 (encodes SHP2 phosphatase)	About 50%	Pulmonary stenosis common
SOS1	About 10%	Milder growth and learning effects
RAF1	About 5–10%	Hypertrophic cardiomyopathy
RIT1, KRAS, others	Smaller	KRAS: more learning difficulty

- Related RASopathies: Costello, cardiofaciocutaneous, NF-1 (neurofibromatosis type 1).

Clinical features

- Short stature; webbed neck; hypertelorism, ptosis, low-set posteriorly rotated ears; pectus deformity.

- **Heart:** pulmonary valve stenosis (commonest), HCM (hypertrophic cardiomyopathy), septal defects.
- **Bleeding tendency:** factor XI deficiency, platelet dysfunction.
- Lymphatic dysplasia: genital (penoscrotal) lymphoedema; mild learning difficulty.

Urological aspects

- **Cryptorchidism:** the main urological feature, in most affected boys (often bilateral).
- **Testicular dysfunction:** primary Sertoli cell dysfunction (low inhibin B, high FSH [follicle-stimulating hormone]) with poor spermatogenesis, even after orchidopexy. Leydig function is usually adequate; puberty is often delayed. Female fertility is usually normal.
- **Renal anomalies (about 10%):** mostly mild renal pelvic dilatation; also duplex system, solitary kidney, hypoplasia.
- Small excess of childhood cancers (JMML [juvenile myelomonocytic leukaemia], embryonal rhabdomyosarcoma).

Evaluation

- Clinical diagnosis (van der Burgt criteria), confirmed by gene panel; karyotype excludes Turner in girls.
- Before surgery: **echocardiography, coagulation screen and factor XI level**, anaesthetic review.
- Renal and bladder ultrasound at diagnosis.

Management

- **Orchidopexy by 18 months** (EAU [European Association of Urology] and AUA [American Urological Association]); laparoscopic for non-palpable testes.
- Follow-up: pubertal hormones, semen analysis, fertility counselling.
- **Also know:** growth hormone for short stature; multidisciplinary cardiac and haematology care.

Diagram to draw: RAS–MAPK cascade with SHP2 (PTPN11) acting upstream of RAS.

> **Key points to remember:**
> - Autosomal dominant, normal karyotype; PTPN11 in about half.
> - Cryptorchidism plus Sertoli dysfunction: male subfertility.
> - Check heart and clotting (factor XI) before orchidopexy.

SAQ Gene therapy in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3f (intravesical gene therapy also answered in 2022-Summer Paper 2, Q3a)

Definition

- Delivery of nucleic acid (DNA or RNA) into cells to treat disease: add, replace or silence a gene.

Vectors

Vector	Features
Adenovirus	Efficient, non-integrating, transient expression; replication-incompetent or oncolytic
Retrovirus, lentivirus	Integrate: long expression, risk of insertional mutagenesis
AAV (adeno-associated virus)	Low immunogenicity, long expression
Non-viral	Naked plasmid, liposomes, nanoparticles: safer, less efficient
Gene silencing	siRNA (small interfering RNA), antisense; CRISPR (clustered regularly interspaced short palindromic repeats) editing experimental

Strategies in urological cancer

1. **Immunogene (cytokine) therapy: nadofaragene firadenovec**: replication-incompetent adenovirus type 5 carrying the **interferon-α2b gene** with **Syn3** surfactant, which crosses the bladder's protective barrier. The urothelium keeps secreting interferon, overcoming the 1–2-hour dwell of protein interferon. 75 mL every 3 months; CIS (carcinoma in situ) complete response **53% at 3 months, 24% at 12 months**. **FDA (Food and Drug Administration) approved December 2022** for BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer).
2. **Oncolytic virus: cretostimogene (CG0070)**: replication-competent adenovirus; E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity.
3. **Suicide gene:** HSV-tk (herpes simplex virus thymidine kinase) turns ganciclovir or valacyclovir into a toxin; prostate cancer with radiotherapy (trials).
4. **Toxin gene under a tumour promoter:** BC-819 (diphtheria toxin A, H19 promoter).
5. **Gene-modified cells:** CAR-T (chimeric antigen receptor T cells) against PSMA (prostate-specific membrane antigen): early trials, toxic. Viral-vector vaccine PROSTVAC failed.

Benign conditions

- **OAB (overactive bladder) and ED (erectile dysfunction):** URO-902 (hMaxi-K) plasmid for the BK potassium channel relaxes smooth muscle; phase 1 only.
- **Primary hyperoxaluria type 1:** lumasiran (RNA interference against glycolate oxidase) lowers oxalate.
- Underactive detrusor: experimental.

Challenges

- Delivery barriers, vector immunity, short expression, cost, biosafety.

Diagram to draw: adenovirus → urothelial cell → interferon secretion → tumour apoptosis.

> **Key points to remember:**
> - Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
> - CIS CR 53% at 3 months, 24% at 12 months.
> - Cretostimogene: oncolytic adenovirus + GM-CSF.
> - Lumasiran: RNA interference for primary hyperoxaluria type 1.

SAQ Molecular targeted therapy in urological cancers

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3c

Concept

- **Targeted therapy** blocks a molecule the cancer depends on (receptor, kinase, transcription factor, DNA-repair enzyme, surface antigen), not all dividing cells.
- Classes: small-molecule TKIs (tyrosine kinase inhibitors, "-nib"), monoclonal antibodies ("-mab"), mTOR (mammalian target of rapamycin) inhibitors, HIF-2α (hypoxia-inducible factor-2α) inhibitor, PARP (poly-ADP-ribose polymerase) inhibitors, ADCs (antibody–drug conjugates), radioligands.

RCC (renal cell carcinoma)

- **Basis:** VHL (von Hippel–Lindau) loss → HIF accumulates even in normoxia → constant VEGF (vascular endothelial growth factor), PDGF (platelet-derived growth factor) and other targets.

Target	Drugs	Key evidence
VEGF receptors	Sunitinib, pazopanib, axitinib, cabozantinib (also MET, AXL), lenvatinib	COMPARZ: pazopanib noninferior to sunitinib; METEOR: cabozantinib beat everolimus
VEGF ligand	Bevacizumab	Modest PFS (progression-free survival) gain
mTOR	Temsirolimus (IV, poor risk), everolimus (oral)	Temsirolimus improved OS (overall survival) over interferon
HIF-2α	**Belzutifan**	VHL disease tumours; LITESPARK-005 beat everolimus after immunotherapy and TKI

- First line now: TKI **plus** checkpoint inhibitor. Everolimus also treats tuberous sclerosis angiomyolipoma.

Urothelial carcinoma

- **FGFR3 (fibroblast growth factor receptor 3)** alterations, typical of luminal tumours: **erdafitinib** (oral) after prior therapy.
- **ADCs: enfortumab vedotin** (Nectin-4) improved OS after platinum and immunotherapy; sacituzumab govitecan (Trop-2); HER2 (human epidermal growth factor receptor 2)-directed ADCs emerging.
- **EV-302 (2024):** enfortumab vedotin plus pembrolizumab, first line, roughly doubled OS versus platinum chemotherapy and is now standard (chapter notes predate it).

Prostate cancer (mCRPC, metastatic castration-resistant)

- **PARP inhibitors** exploit **synthetic lethality** in HRR (homologous recombination repair)-deficient tumours, especially BRCA2. **PROfound:** olaparib beat a second hormonal agent (imaging PFS 7.4 vs 3.6 months in BRCA/ATM).
- **PSMA (prostate-specific membrane antigen) radioligand: 177Lu-PSMA-617; VISION**: OS 15.3 vs 11.3 months.

Toxicity

- VEGF TKIs: hypertension, hand–foot syndrome, hypothyroidism. mTOR: mucositis, hyperglycaemia. Erdafitinib: hyperphosphataemia. Enfortumab: neuropathy, rash. PARP inhibitors: anaemia.

Limitations

- Resistance develops; cures rare; biomarker testing needed.

Diagram to draw: VHL–HIF–VEGF pathway with drug sites (TKI at receptor, bevacizumab at VEGF, mTOR and HIF-2α inhibitors).

> **Key points to remember:**
>
> - RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
> - Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
> - Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

Rapid Revision Sheet · Section 2

Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T) (LAQ)

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

Oncogenes (SAQ)

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Telomerase (SAQ)

- Telomerase = hTERT (catalytic) + hTR (RNA template).
- About 100 base pairs lost per division; 85–90% of tumours express telomerase.
- TERT promoter mutation is the commonest mutation in bladder cancer.
- TRAP assay detects urinary telomerase; inflammation gives false positives. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Familial / hereditary urological malignancies (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Circumcision undone (SAQ)

- Complications 0.2–5%; bleeding commonest.
- Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
- Meatal stenosis almost only in circumcised boys.
- Revision needs general anaesthesia.

Congenital Penile curvature.. (SAQ)

- Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
- Congenital curvature: young man, large erection, orthotopic meatus.
- Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Female circumcision (SAQ)

- WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, stones, fistula.
- Midline deinfibulation for type III; never reinfibulate.

46,XX male (SAQ)

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.

- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Mixed gonadal dysgenesis. (SAQ)

- 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
- Second commonest cause of atypical genitalia after CAH.
- Tumour risk 15–35%: remove streaks.
- Screen for Turner features.

DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia (LAQ)

- Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
- 17-OHP after 36 h; salt crisis around day 5–7.
- Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
- Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

Penile inversion vaginoplasty (SAQ)

- Glans → clitoris; penile skin → canal; scrotal skin → graft.
- Excise all erectile tissue; keep Denonvilliers fascia intact.
- Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

Enlist the steps in female to male gender reassignment surgery. (SAQ)

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Discuss in brief, present concepts in various GU cancers (LAQ)

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Prepucioplasty (SAQ)

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

Secondary malignancies in urology (SAQ)

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

Genetic counselling (SAQ)

- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

Noonan syndrome (SAQ)

- Autosomal dominant, normal karyotype; PTPN11 in about half.
- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Gene therapy in urology (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

Molecular targeted therapy in urological cancers (SAQ)

- RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
- Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
- Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 3

Infections & Inflammation

26 Infections of the Urinary Tract

1 · Definitions and classification

The label decides treatment length and whether to investigate.

- **Urinary tract infection (UTI)** is the urothelial inflammatory response to bacterial invasion, usually with bacteriuria and pyuria. Urine is **not sterile**.
- **Bacteriuria** and **pyuria** are not the same as UTI; bacteriuria without pyuria is colonisation. **Sterile pyuria**: look for tuberculosis (TB) see Ch 31, stone, urothelial injury and interstitial nephritis.
- **Cystitis** causes dysuria, frequency and urgency; mimics include urethritis, vaginitis, interstitial cystitis see Ch 28 and bladder carcinoma.
- **Acute pyelonephritis** is a clinical diagnosis: chills, fever and flank pain with bacteriuria and pyuria. Flank symptoms may be absent in spinal cord injury (SCI).
- **Chronic pyelonephritis** shows a focal coarse cortical scar over a **distorted calyx**.

	Uncomplicated UTI	**Complicated UTI**
Host	Healthy, non-pregnant woman	Compromised host and/or virulent or resistant organism
Tract	Structurally and functionally normal	Structural or functional abnormality, catheter
Organism	Eradicated by short oral course	Often multidrug-resistant (MDR), so culture is mandatory

- **Complicating factors** are an anatomical or functional abnormality (obstruction, neurogenic bladder [NGB], bladder outlet obstruction [BOO]), **male sex**, pregnancy, diabetes mellitus (DM), immunosuppression, catheter, hospital acquisition and symptoms >7 days. MCQ Male sex alone makes a UTI complicated.

Term	**Definition**	**Discriminator**
Isolated	First, or one remote UTI	—
Unresolved	Therapy failed; same organism	Never cleared
Reinfection	New inoculation (~ **70%** of recurrences, often different organism)	Cleared, then new event
Bacterial persistence (relapse)	Same organism from a focus (stone, prostate)	Focus remains; recurs quickly

- **Prophylaxis** prevents reinfection of a sterilised tract; **suppression** prevents growth from an ineradicable focus.
- **Catheter-associated UTI (CAUTI)** is the commonest cause of secondary bacteraemia and ~**80%** of nosocomial UTIs.
- Uncomplicated recurrent UTI does not cause scarring, hypertension or renal failure.

2 · Pathogenesis

Route	Frequency	Features
Ascending	Commonest	Bowel flora colonise the introitus and climb; reflux not required
Haematogenous	Uncommon	S. aureus, Candida; enhanced by obstruction
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Organisms

- **E. coli** causes ~**85%** of community and ~50% of hospital UTIs, but less than half of complicated UTI and urosepsis.
- **S. saprophyticus** causes ~10% of cystitis in young women; complicated UTIs involve Klebsiella, Pseudomonas and E. faecalis.
- **ST131 (O25b:H4)** is an MDR E. coli clone carrying extended-spectrum β-lactamase (ESBL), with fluoroquinolone resistance as its hallmark. MCQ Enterococcus is a Gram-positive coccus, not an Enterobacteriaceae.

Virulence — fimbriae

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Gene / adhesin	fim; **FimH** tip adhesin	pap operon on a pathogenicity island
Receptor	Mannosylated **uroplakins Ia/Ib**	Glycolipid (P blood-group) receptors in kidney
Tropism	Bladder, causing cystitis	Kidney, causing **pyelonephritis** (~**90%** of pyelonephritis strains)

- Adherence is the essential first step. In **phase variation** bacteria switch between piliated and non-piliated states.

Intracellular cycle (why cystitis recurs)

ALGORITHM

1. FimH binds uroplakin and the bacterium invades the umbrella cell (15 min–1 h).
2. It multiplies into an **intracellular bacterial community (IBC)**, which behaves like a biofilm (4–16 h).
3. Bacteria filament, escape and reinvade neighbouring cells (16–24 h).
4. Infected cells exfoliate, exposing deeper layers.
5. **Quiescent intracellular reservoirs (QIRs)** form (from ~2 weeks).

- QIRs are silent between attacks; urothelial disruption reactivates them, giving same-strain recurrence without a urological focus.

Host defences and susceptibility

- Lactobacilli are the introital barrier, disturbed by antimicrobials, spermicides and oestrogen loss.
- Bladder emptying is the key defence; **obstruction** makes infection life-threatening. In adults, vesicoureteral reflux (VUR) alone is harmless without high pressure or stasis.

- **DM** brings more abscess, emphysematous infection and papillary necrosis.

Renal papillary necrosis

- The main causes are **DM, obstruction with pyelonephritis, analgesic abuse and sickle cell disease**. A sloughed papilla obstructs the ureter; treat with antibiotics and early relief of obstruction.

3 · Evaluation and diagnosis

A fistula presents with pneumaturia or faecaluria, and an abscess with indolent fever and a flank mass.

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
Frequency/urgency	Prominent	Absent; no haematuria or suprapubic pain	Less marked; no fever
Clues / causes	E. coli	Discharge, odour, new partner; HSV, gonorrhoea, Chlamydia, Trichomonas, yeast, bacterial vaginosis	Discharge; initial-stream pyuria in men; N. gonorrhoeae, Chlamydia, HSV see Ch 29

- Dysuria with frequency and no vaginal discharge gives a ~**90%** probability of UTI; dysuria is the best single symptom.
- **Rule**: painless gross haematuria, or microhaematuria persisting after the infection clears, needs full upper and lower tract evaluation.
- Midstream urine is the default; a catheter sample is more accurate. With an indwelling catheter, replace it and sample the new one.

Urinalysis and culture

- **Leucocyte esterase** indicates pyuria. **Nitrite** is made by Enterobacteriaceae and is negative with Gram-positives and Pseudomonas; it is the best rule-in test. Dipsticks are better at **ruling out**.
- On microscopy, pyuria is ≥**5–10 WBC/high-power field (HPF)**, and squamous cells mean contamination. A positive culture without pyuria should be repeated on a catheter sample.

Threshold	Use
≥10^5 colony-forming units (CFU)/mL	Kass (originally pyelonephritis); ASB definition
≥**10^2 CFU/mL** of a known pathogen	Symptomatic women with cystitis; catheter specimens

- **Stone culture** predicts post-procedure sepsis better than bladder urine.

Imaging

- Imaging asks only whether there is obstruction, a collection to drain or a focus of persistence; most UTIs need none.
- **Indications in pyelonephritis**: possible obstruction, risk of papillary necrosis, prior reconstruction, **no response after 72 h**, DM, polycystic kidneys on dialysis, NGB and unusual organisms. Also image most men, compromised hosts and rapid recurrences.

Question	Best test
Stone	Non-contrast computed tomography (CT)

Question	Best test
Abscess, acute focal bacterial nephritis (AFBN), drainage planning	Contrast CT
Hydronephrosis, pyonephrosis, post-void residual (PVR)	Ultrasound (US)
Extrarenal extension	Magnetic resonance imaging (MRI)
Urethral diverticulum	Pelvic MRI with contrast
VUR; non-refluxing stump	Voiding cystourethrogram (VCUG)
Split function, scarring	Dimercaptosuccinic acid (DMSA) scan

Correctable causes of bacterial persistence (Box 26.6) include infection stones, chronic bacterial prostatitis, an infected atrophic kidney, ectopic ureter, foreign body, urethral diverticulum, an infected ureteric stump and fistula. Antibiotics cannot sterilise these, so **cure comes only by removal or drainage.**

4 · Antimicrobial principles

- In cystitis the **urinary** drug level matters; in parenchymal infection or bacteraemia, **serum and tissue** levels matter. Renal insufficiency and obstruction lower urine levels.
- **Collateral damage** means resistance, MDR colonisation, C. difficile and candidiasis, caused mainly by broad cephalosporins and quinolones.

SAQ Antimicrobial resistance

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3b (also answered in part in 2015-Summer Paper 1, Q3f)

Definition

- AMR (antimicrobial resistance): a microbe grows despite drug levels achievable in the patient. Reported as S (sensitive), I (intermediate), R (resistant) with the MIC (minimum inhibitory concentration).
- One of the most serious global health threats.

Types

Type	Mechanism	Relevance
Intrinsic	Species lacks the target	Predictable (enterococci and cephalosporins)
Acquired chromosomal	Mutants selected when urine level is close to MIC	Underdosing, poor compliance
Plasmid (extrachromosomal)	Horizontal transfer in bowel flora	**Commonest**; multidrug resistance after antibiotics

Mechanisms

- **Enzymes:** β-lactamases; aminoglycoside-modifying enzymes.
- **Altered target:** penicillin-binding protein; DNA gyrase mutation (quinolones).
- **Reduced entry** (porin loss) and **efflux pumps**.
- **Bypass:** enterococci scavenge folate (escape TMP-SMX (trimethoprim–sulfamethoxazole)).

Key resistant organisms

- **ESBL (extended-spectrum β-lactamase)** (plasmid, 1983): resists 3rd-generation cephalosporins; often co-resistant to fluoroquinolones and aminoglycosides. *E. coli* ST131 carries it.
- **AmpC; KPC (*Klebsiella pneumoniae* carbapenemase, 2001); NDM (New Delhi metallo-β-lactamase)**: carbapenem resistance.
- MRSA (methicillin-resistant *S. aureus*), VRE (vancomycin-resistant enterococci).

Drivers in urology

- Overuse of fluoroquinolones and cephalosporins; treating ASB (asymptomatic bacteriuria); long prophylaxis; catheters, stents, stones (biofilm).
- Transrectal biopsy: fluoroquinolone-resistant rectal *E. coli* causes sepsis.

Prevention: stewardship

1. Culture before treatment; follow the **local antibiogram**.
2. TMP-SMX only if local resistance ≤20%; oral fluoroquinolone for pyelonephritis only if ≤10%.
3. Prefer narrow, urine-confined drugs for cystitis (nitrofurantoin, fosfomycin, pivmecillinam: low resistance).
4. Shortest course; de-escalate to culture.
5. Do not treat ASB, except in pregnancy and before mucosa-breaching procedures.
6. Single-dose surgical prophylaxis; transperineal biopsy or rectal-swab-targeted prophylaxis.
7. Infection control and hand hygiene. - **Also know:** FDA (Food and Drug Administration) 2016 fluoroquinolone warning; *C. difficile*; ESBL: carbapenem, or fosfomycin/nitrofurantoin for cystitis.

Recent advances

- New β-lactam/β-lactamase inhibitor combinations (ceftazidime–avibactam, meropenem–vaborbactam); cefiderocol; gepotidacin.

Diagram to draw: Gram-negative cell showing β-lactamase, porin loss, efflux pump, altered target.

> **Key points to remember:**
> - Plasmid-mediated resistance is commonest.
> - ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
> - TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
> - Stewardship: culture, narrow, short, no ASB treatment.

SAQ ESBL producing enterobacterial UTI

Paper answer · P1

Asked as:

- ESBL producing enterobacterial UTI — *Summer 2019 Paper I Q3f (SAQ)*
- ESBL producing enterobacterial UTI. — *Practice Set 5 Paper I Q3c (SAQ)* (also answered in 2019-Summer Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **ESBLs (extended-spectrum β-lactamases): plasmid-encoded** enzymes that hydrolyse penicillins, **third-generation cephalosporins** and aztreonam. They are **inhibited by clavulanate** and spare **carbapenems** and cephamycins.
- Mainly *E. coli* and *Klebsiella* (Enterobacterales); first reported 1983.

Mechanism and spread

- Older types **TEM** and **SHV**; now **CTX-M** dominates worldwide. Spread through the community is linked to *E. coli* **ST131**, whose hallmark is fluoroquinolone resistance (fluoroquinolone resistance is its hallmark).
- Plasmids carry co-resistance genes: **fluoroquinolones, aminoglycosides, TMP-SMX (trimethoprim–sulfamethoxazole)**. So oral options are few.
- India has one of the highest prevalence rates.

Risk factors

- Prior antibiotics (especially cephalosporins, quinolones); recent hospital stay; **indwelling catheter**, urological instrumentation; recurrent UTI (urinary tract infection); diabetes; nursing-home residence; travel to high-prevalence areas; transrectal prostate biopsy.

Detection

- Screening: reduced susceptibility to ceftriaxone, cefotaxime or ceftazidime.
- **Phenotypic confirmation:** clavulanate synergy (**combination disc** or double-disc synergy test); automated systems.
- Genotypic PCR (polymerase chain reaction) for CTX-M.

Treatment

Infection	Options
Cystitis	**Nitrofurantoin** 5 days, **fosfomycin** 3 g single dose (E. coli), **pivmecillinam**; TMP-SMX or quinolone only if susceptible; single-dose aminoglycoside
Pyelonephritis, complicated UTI	Quinolone or TMP-SMX if susceptible; else (otherwise) **aminoglycoside**
Urosepsis, bacteraemia	**Carbapenem** (meropenem, ertapenem), then oral step-down by culture

- **MERINO trial (2018):** in ceftriaxone-resistant *E. coli*/*Klebsiella* bacteraemia, **piperacillin–tazobactam** had higher 30-day mortality than meropenem (12.3% vs 3.7%). Avoid pip-taz for serious ESBL infection.
- **Ceftolozane–tazobactam, ceftazidime–avibactam:** active but **reserved** for carbapenem-resistant organisms (stewardship).
- **Source control:** remove or change catheters; drain obstruction.
- Treat **asymptomatic bacteriuria** only before urological procedures or in pregnancy.
- **Febrile prostatitis after transrectal biopsy:** carbapenem is favoured.

Prevention

- **Antibiotic stewardship:** avoid broad-spectrum cephalosporins and quinolones; narrowest spectrum; shortest effective course; culture before treatment.
- Hand hygiene, contact precautions, catheter care.

- Pre-biopsy rectal swab culture for targeted prophylaxis; transperineal biopsy.

Also know: AmpC (not clavulanate-inhibited) and carbapenemases **KPC (*Klebsiella pneumoniae* carbapenemase)**, NDM (New Delhi metallo-β-lactamase) are wider resistance.

Diagram to draw: treatment flowchart from culture result to oral or IV (intravenous) choice.

> **Key points to remember:**
> - Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
> - Oral: nitrofurantoin, fosfomycin, pivmecillinam.
> - Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
> - Stewardship prevents spread.

Also, from the condensed book (not in the paper answer):

- Prior quinolone use is the strongest determinant of quinolone resistance, and plasmid resistance spreads among bowel flora.
- **Stewardship** means narrow spectrum, the shortest course, culture before treating recurrent or complicated UTI, and local antibiograms.

Agents

- Fosfomycin acts on the cell wall at a different step from β-lactams, so cross-resistance is rare. Enterococci scavenge folate, so TMP-SMX fails even when "sensitive".
- **Anti-pseudomonal** agents are antipseudomonal penicillins, **ceftazidime** (not ceftriaxone), aztreonam, aminoglycosides and fluoroquinolones.
- **Enterococcal** cover comes from aminopenicillins, piperacillin, nitrofurantoin and fosfomycin, and **never from cephalosporins**.
- Ertapenem misses Pseudomonas and Enterococcus.

Agent	Key points
Nitrofurantoin	Acts in urine only, so it is useless for pyelonephritis, prostatitis and bacteraemia. Toxicity includes pulmonary fibrosis and neuropathy. Beers 2015: avoid if creatinine clearance (CrCl) < **30 mL/min**; no long-term suppression in the elderly.
TMP-SMX	Good tissue levels; misses Enterococcus and Pseudomonas; potentiates warfarin.
Fosfomycin	3 g single dose; active against ESBL producers and vancomycin-resistant enterococci, so it is an oral option for MDR cystitis.
Fluoroquinolones	Broad, including Pseudomonas. Adjust at CrCl <30. FDA warnings include tendon rupture and aortic dissection; **2016: reserve for when there is no alternative in uncomplicated UTI**. Avoid in children and pregnancy.
Cephalosporins	Poor against enterococci; use in cystitis only if first-line agents cannot be used.
Carbapenems	Stable to ESBL and AmpC; penicillin cross-sensitivity <1%.
Aminoglycosides	Once daily 5–7 mg/kg; a nomogram level 6–14 h after the dose sets the interval at 24, 36 or 48 h. Monitor oto- and nephrotoxicity.
Aztreonam	Gram-negative spectrum without nephrotoxicity; useful in penicillin allergy and renal impairment.
Pivmecillinam	~2% E. coli resistance; Nordic first-line; not in USA.

KEY POINT

Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis; broad tissue-penetrating agents are kept for invasive infection. Resistance is made in the bowel, so breadth, duration and repetition matter more than MIC.

C. difficile infection (CDI)

- The usual culprits are cephalosporins, clindamycin, carbapenems, quinolones and TMP-SMX. Diagnosis needs ≥3 unformed stools in 24 h and toxigenic C. difficile or its toxin.

Episode (IDSA/SHEA 2021)	Treatment
Initial	**Fidaxomicin** 200 mg twice daily × 10 d; alternative oral vancomycin 125 mg four times daily × 10 d; metronidazole only if non-severe and others unavailable
First recurrence	Fidaxomicin or tapered vancomycin, with or without **bezlotoxumab** (anti-toxin B antibody)
≥2 recurrences	As above, or **faecal microbiota transplantation**
Fulminant	Oral/rectal vancomycin 500 mg four times daily and IV metronidazole

5 · Asymptomatic bacteriuria

Treating ASB usually harms; know the few groups who need it.

Group (Infectious Diseases Society of America [IDSA])	Specimen	Threshold
Women	**Two** consecutive voided, same strain	≥10^5 CFU/mL
Men	One voided	≥10^5 CFU/mL
Either sex	One catheterised	≥**10^2** CFU/mL

- Prevalence is 1–5% in premenopausal women and **100%** with a long-term catheter. Untreated ASB is harmless; treating it brings more symptomatic UTI and MDR.

Do NOT screen or treat (IDSA 2019)	Screen and treat
Premenopausal non-pregnant women; DM; elderly (community or institution); SCI; indwelling catheters; ASB with pyuria; children; renal transplant >1 month; other solid-organ transplant; **non-urological surgery (including arthroplasty and cardiac)**. No consensus: neutropenia.	**Pregnancy**; **urological procedures with anticipated mucosal bleeding** (transurethral resection [TUR], ureteroscopy, percutaneous nephrolithotomy [PCNL])

KEY POINT

ASB is treated only where bacteria are about to reach blood or tissue: pregnancy (hydroureter leading to pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective.

6 · Cystitis

Acute uncomplicated cystitis in women

- Diagnose on acute dysuria with a change in voiding. Fever, flank pain or no oral route should be treated as pyelonephritis.

Tier (IDSA/ESCMID)	Regimen	Caveat
First-line	**Nitrofurantoin** 100 mg twice daily × **5 d**	Not if pyelonephritis suspected
First-line	**TMP-SMX** 160/800 mg twice daily × **3 d**	Avoid if local resistance > **20%** or used in past 3 months
First-line	**Fosfomycin** 3 g single dose	Slightly lower efficacy
First-line	**Pivmecillinam** 400 mg twice daily × 3–7 d	Not in North America
Second-line	Fluoroquinolone × 3 d	Ecology and toxicity; last resort (FDA 2016)
Second-line	β-lactam (amoxicillin–clavulanate, cefpodoxime, cefadroxil) × 3–7 d	Lower efficacy; never amoxicillin alone

A **3-day** course beats a single dose and equals 7 days (7 if complicating factors). No post-treatment culture if asymptomatic.

Cystitis in men

- Male cystitis is complicated and about half have an abnormality, so evaluate and consider prostatitis see Ch 27.
- In afebrile men, **7 days is as good as 14 days** (Drekonja). The EAU suggests TMP-SMX 160/800 twice daily × 7 d, or a quinolone guided by local susceptibility.

Complicated UTI

Category (Canadian Urological Association [CUA])	Examples
Anatomical	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial, surgery, catheter
Obstruction	BOO, pelviureteric junction (PUJ) obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, VUR, incontinence, high PVR
Other	Pregnancy, stones, DM, immunosuppression

Always culture and correct the abnormality. Treat **10–14 days** by culture, IV to oral once afebrile.

Emphysematous cystitis

- Bladder-wall gas, typically in an elderly diabetic woman; fever occurs in only 30–50%. **CT confirms** it.
- ~**90%** settle with broad Gram-negative cover, catheter drainage and glucose control.

Unresolved bacteriuria

Cause (descending frequency)	On-treatment culture	Action
1. Resistant from outset (recent antibiotics)	Resistant	Change drug; no further work-up
2. Emergent resistance (underdosing)	Same species, now resistant	
3. Unsuspected second pathogen	New species present from start	
4. Rapid reinfection, new resistant species	New species; suspect enterovesical fistula	
5–7. Azotaemia; papillary necrosis; giant staghorn	Same species, still **sensitive**	Renal function and imaging
8. Non-compliance, Munchausen	—	—

7 · Recurrent UTI

Most recurrences are reinfections, so the work is prevention.

- Recurrent UTI means **2 in 6 months or ≥3 in 12 months**, each culture-documented. Reinfection is a different organism, or the same one >2 weeks after treatment.
- Risk factors include intercourse, spermicide, menopause, incontinence, cystocele and raised PVR.
- Evaluate with history, pelvic examination, **culture of each episode** and PVR. **Imaging and cystoscopy are not routine** (AUA, EAU, CUA).
- **Indications for imaging with or without cystoscopy (CUA)**: prior urinary surgery or trauma; stones; gross or persisting microhaematuria; obstructive symptoms or high PVR; urea-splitters; prior pelvic malignancy; persistence despite sensitivity-based therapy; DM or immunocompromise; pneumaturia, faecaluria or anaerobes; and repeated pyelonephritis. MCQ Recurrence frequency alone is **not** an indication.
- **Struvite** stones form when urease (Proteus mirabilis) produces magnesium ammonium phosphate, with bacteria inside the stone. The AUA advises complete removal in fit patients; unfit patients get acetohydroxamic acid and suppression.

Prevention (women, AUA 2019)

ALGORITHM

1. Confirm recurrent UTI with symptoms and culture.
2. If the same organism returns at short intervals, then evaluate for persistence (upper tract imaging and cystoscopy) and remove the nidus.
3. If it is reinfection, then start behavioural measures: ≥1.5 L water a day, frequent and postcoital voiding, stop spermicide.
4. Add cranberry ≥36 mg proanthocyanidin (PAC).
5. If UTIs follow intercourse, then give a **postcoital antibiotic**. If postmenopausal, then give **vaginal oestrogen**. If neither, then consider **methenamine hippurate** (normal renal function).
6. Use continuous antibiotic prophylaxis last.

Measure	Mechanism	Evidence / use
Water ≥1.5 L/day	Washout	RCT in premenopausal women: 1.7 vs 3.2 episodes/yr
Vaginal oestrogen	Restores lactobacilli, lowers pH	Strongest non-antibiotic evidence; ~90% reduction (Raz–Stamm). **Oral oestrogen is ineffective.**
Cranberry ≥36 mg PAC (twice daily)	Type A PAC blocks **P-fimbrial** adhesion	Cochrane 2023 favourable; safe with warfarin
Methenamine hippurate 1 g twice daily	**Formaldehyde in acid urine (pH <6)**	Non-inferior to daily prophylaxis; fails in NGB. Contraindicated if CrCl <50.
D-mannose	Inhibits **FimH**	Cochrane 2022 inconclusive
Immunostimulants (OM-89, Urovac, Uromune MV140)	Bacterial lysates	Some benefit; small biased trials
Probiotics, oral oestrogen, vitamin C	—	Not effective

Antibiotic strategy	Evidence	Position
Continuous low dose 6–12 months (nitrofurantoin, TMP, TMP-SMX, cephalexin)	Works only while taken; MDR	AUA "may prescribe"; EAU: last resort
Postcoital single dose	Effective; preferred for coital UTI	Acceptable
Self-start	Old data; overuse	Prefer analgesic, sample and observe, then culture-directed treatment

KEY POINT

Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigate only for features of persistence or complication; prevention targets flora, pH, adhesion and washout.

8 · Kidney infections

Acute pyelonephritis

- **White cell casts** prove a renal origin. Take blood cultures if toxic, admitted or pregnant; the usual cause is P-fimbriated E. coli.

Setting (IDSA)	Regimen	Days
Outpatient, no vomiting	Ciprofloxacin 500 mg twice daily if community quinolone resistance ≤**10%** ; levofloxacin 750 mg daily; TMP-SMX double-strength twice daily only if known sensitive. With or without one IV dose of ceftriaxone 1 g or an aminoglycoside.	Cipro 7; levo 5; TMP-SMX 14
Inpatient, severe/possible sepsis	Ampicillin and gentamicin; levofloxacin; ceftriaxone 1 g daily; carbapenem (ESBL risk). Gram-positive cocci: ampicillin–sulbactam with or without an aminoglycoside.	10–14 total
Pregnancy	Ampicillin and gentamicin, aztreonam or ceftriaxone, then oral cephalexin. **No quinolone.**	10–14

- **Obstruction** needs urgent decompression by **stent or percutaneous nephrostomy**.

ALGORITHM

1. If there is no sepsis or vomiting, then send a culture, with or without US, and treat as an outpatient with oral drugs for 7–10 d.
2. If there is sepsis, then take blood and urine cultures, image, and admit for IV treatment.
3. If improved by **72 h**, then step down to oral treatment.
4. If not improved, then repeat cultures and get a **CT**, and drain any obstruction or abscess.

KEY POINT

Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication.

Acute focal bacterial nephritis (AFBN)

- AFBN is a lobar infiltrate between pyelonephritis and abscess. On contrast CT it is wedge-shaped with no wall or liquefaction, whereas an abscess is round with a liquid centre and a **ring sign**. Give IV antibiotics for ≥7 days, then 7 days oral.

Emphysematous pyelonephritis (EPN)

SAQ Emphysematous pyelonephritis: prognostic factors and conservative management

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "EPN. Prognostic factors & conserv. mgt." · P3 SAQ "Prognostic indicators in emphysematous pyelonephritis."

Assumed meaning: EPN = emphysematous pyelonephritis.

Definition and pathogenesis

- EPN is an acute **necrotising** infection of the renal parenchyma and perirenal tissue by **gas-forming organisms**. It is a urological emergency.
- Organisms: *E. coli* (commonest), *Klebsiella*, *Proteus*.
- Most patients are **diabetic**; obstruction in about a quarter.
- **Mechanism (proposed):** high tissue glucose fermented to CO_2 in poorly perfused tissue.
- Reported mortality 19–43%.
- **CT (computed tomography)** is the imaging of choice.

Imaging-based prognostic classifications

1. Huang–Tseng CT classification (2000)

Class	Gas location
1	Collecting system only (emphysematous pyelitis)
2	Renal parenchyma only, no extrarenal extension
3A	Extension into the **perinephric** space
3B	Extension into the **pararenal** space
4	**Bilateral** EPN or EPN in a **solitary kidney**

2. Wan classification (CT pattern)

Type	CT	Mortality
I	Streaky or mottled gas, **no fluid**; rapid parenchymal destruction	50–60%
II	Renal or perirenal **fluid**, bubbly or loculated gas, or gas in collecting system	<20%

- **Why:** fluid means a walled-off (contained) collection that can be drained. Streaky gas without fluid means necrosis faster than the host can contain it.

Clinical and laboratory prognostic factors

- **Huang–Tseng risk factors: thrombocytopenia**, **AKI (acute kidney injury)/raised creatinine**, **altered sensorium**, **shock**.
- **Other adverse factors:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

Conservative management (now standard)

1. Resuscitation: fluids, inotropes, ICU (intensive care unit) if shocked; **tight glucose control**; correct electrolytes.

2. Antibiotics: broad-spectrum IV (intravenous), adjusted to culture; usually several weeks in total.

3. Drainage

- **Relieve obstruction:** DJ (double-J) stent or PCN (percutaneous nephrostomy).
- **PCD (percutaneous catheter drainage)** of gas and fluid collections under CT or ultrasound; multiple drains if needed; repeat CT to check.

How prognosis guides management (algorithm)

```
All: resuscitation + glucose control + IV antibiotics + relieve obstruction (stent/PCN) Class 1–2 -> antibiotics ± PCD (P3: antibiotics + PCD) Class 3–4, <2 risk factors -> antibiotics + PCD (usually succeeds) Class 3–4, >=2 risk factors, or PCD failure -> nephrectomy (emergency or delayed)
```

- Routine emergency nephrectomy carried 40–50% mortality, so drainage-first care replaced it.
- Delayed nephrectomy for a kidney that is non-functioning on renal scan.
- **Emphysematous pyelitis** (class 1, gas in the collecting system only) is often non-diabetic and often responds to antibiotics alone.

Diagram to draw: CT sketches of Huang–Tseng classes 1, 2, 3A/3B and 4.

> **Key points to remember:**
>
> - Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
> - Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
> - Wan type I (no fluid) 50–60% mortality; type II <20%.
> - Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

Also, from the condensed book (not in the paper answer):

- EPN is mostly seen in **diabetic women**; non-diabetics are usually obstructed, with less extensive disease.

CT pattern	Prognosis
Streaky/mottled gas, **no fluid**	50–60% mortality
Fluid with bubbly/loculated gas, or gas in collecting system	<20% mortality

Huang–Tseng class †	Extent of gas	Management †
1	Collecting system only (emphysematous pyelitis)	Antibiotics and percutaneous catheter drainage (PCD) or stent
2	Parenchyma, no extrarenal extension	Antibiotics and PCD
3A	Perinephric space	PCD if <2 risk factors; nephrectomy if ≥2 or PCD fails
3B	Pararenal space	
4	Bilateral or solitary kidney	PCD (nephron-sparing)

† The risk factors are thrombocytopenia, acute kidney injury (AKI), altered consciousness and shock.

> **KEY POINT**
>
> Fluid is a good sign and streaky gas a bad one: a contained collection can be drained; diffuse gas without fluid means parenchyma is necrosing faster than the host can wall it off.

Renal and perinephric abscess

SAQ Renal and perinephric abscess

Paper answer · P1

Asked as:

- Perinephric abscess — *Summer 2017 Paper I Q3c (SAQ)*
- Treatment of renal abscess — *Winter 2017 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Definition and types

- **Renal (cortical/corticomedullary) abscess:** pus confined to the renal parenchyma.
- **Cortical (renal carbuncle):** haematogenous *Staphylococcus aureus* from a skin source; formerly about 80% of cases.
- **Corticomedullary (now commonest):** ascending Gram-negative infection (*E. coli*, *Klebsiella*, *Proteus*) in a damaged or obstructed kidney (stones, reflux, diabetes).
- **Perinephric abscess:** pus **outside the renal capsule but within Gerota fascia**. Rupture through Gerota fascia gives a **paranephric** abscess (paranephric beyond it).

Aetiology (perinephric abscess)

- Rupture of a renal cortical abscess or pyonephrosis (commonest); extravasation of infected urine from obstruction (stone).
- **Haematogenous** spread in about a third (skin source, *S. aureus*).

- Infected perirenal haematoma or urinoma (trauma, surgery).
- Spread from bowel (Crohn disease, perforation), pancreas, spine.
- Risk factors: **diabetes (about a third)**, stones, obstruction, polycystic kidneys on dialysis.
- Organisms: *E. coli*, *Proteus*, *S. aureus*; anaerobes with bowel sources.

Clinical features

- Fever, flank pain, leucocytosis.
- Perinephric abscess is **insidious:** symptoms >5 days in most; more than a third afebrile.
- Flank pain, tenderness, flank mass (about half); psoas sign (flexed, externally rotated hip).
- **Suspect it when "pyelonephritis" persists:** pyelonephritis defervesces within 4–5 days; perinephric abscess does not.

	Pyelonephritis	**Perinephric abscess**
Symptoms before admission	≤5 days	>5 days
Fever on antibiotics	Never >4 days	≥5 days (median 7)

Investigations / diagnosis (needed to plan treatment)

- Leucocytosis, raised creatinine, pyuria; urine culture identifies the organism in only about a third; urine may be sterile (haematogenous or non-communicating abscess). Blood cultures positive in a quarter to a third.
- **CT (computed tomography): investigation of choice.** Renal abscess: round low-attenuation lesion with enhancing "ring sign". It shows the collection, its spread (psoas, flank), the cause (stone, obstruction) and plans drainage.
- Ultrasound: quick; guides drainage. Plain X-ray: loss of psoas shadow, scoliosis.
- AFBN (acute focal bacterial nephritis) is wedge-shaped without a wall and is treated medically.

Treatment

1. Antibiotics immediately (after blood and urine cultures)

Presumed source	**Empirical therapy**
Haematogenous (Gram-positive)	Penicillinase-resistant penicillin; vancomycin if allergic or MRSA (methicillin-resistant *S. aureus*)
Ascending (Gram-negative)	IV (intravenous) third-generation cephalosporin, antipseudomonal penicillin or aminoglycoside
Perinephric abscess	Aminoglycoside plus antistaphylococcal penicillin; cephalosporin or vancomycin if allergic. Adjust to culture
- Switch to oral once afebrile; continue until clinical and imaging resolution.	

2. Size-based drainage (renal abscess)

Size	**Management**
≤3 cm	Antibiotics alone (stable patient)
3–5 cm	Antibiotics alone or drainage; no consistent pathway
>5 cm	**Percutaneous drainage** (CT/ultrasound-guided) first line; may need several drains

Size	Management
Any size: immunocompromised or not responding	Percutaneous drainage

- **Perinephric abscess (if present) — lower threshold:** small (<3 cm) in an immunocompetent patient may resolve with antibiotics; drain if >3 cm. Larger or non-responding: **CT- or ultrasound-guided percutaneous drainage** (first line).
- Aspiration also distinguishes abscess from a cystic tumour; culture the pus.
- Serial ultrasound or CT until resolution. Worsening suggests perinephric spread, wrong diagnosis or resistant organism.

3. Treat the cause

- Relieve obstruction (stent or nephrostomy); treat stones once infection settles; control diabetes.

4. Surgery

- Open drainage for loculated or failed (percutaneous drainage) cases; **nephrectomy** for a non-functioning, destroyed or pyonephrotic kidney, at the same time or staged.

Outcome

- Perinephric abscess: mortality fell from 40–50% to about 12% with CT; delay in diagnosis causes most deaths.

Diagram to draw: coronal kidney showing renal, perinephric and paranephric spaces / abscess sites.

> **Key points to remember:**
> - Now mostly ascending Gram-negative abscess in a damaged kidney.
> - Perinephric: within Gerota fascia; diabetes in a third.
> - Fever beyond 4–5 days of antibiotics: get a CT.
> - CT is the investigation of choice; follow until resolution.
> - ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
> - Percutaneous drainage first line; nephrectomy if non-functioning.

Also, from the condensed book (not in the paper answer):

- A renal abscess ≤**3 cm** is treated with antibiotics; one >**5 cm** needs percutaneous drainage, as does any size in an immunocompromised or non-responding patient.
- A perinephric abscess has a lower drainage threshold (>3 cm); only a third are diagnosed at presentation, and delay causes nearly all deaths.

Infected hydronephrosis and pyonephrosis

- **Pyonephrosis** is infected hydronephrosis with suppurative destruction. Patients are very ill, **bacteriuria may be absent** if the ureter is blocked, and US shows a fluid–debris level.
- Give antibiotics with **immediate drainage** (stent or nephrostomy); nephrectomy if non-functioning.

SAQ Xanthogranulomatous pyelonephritis.

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 1, Q3a

Definition

- XGP (xanthogranulomatous pyelonephritis) is a rare, severe, **chronic** infection causing diffuse destruction of an enlarged, non-functioning kidney, with **lipid-laden foamy macrophages** (xanthoma cells).
- Up to a fifth of nephrectomies for pyelonephritis. Peak age 50–70 years; women; diabetes is a risk factor.

Pathogenesis

- **Triad: stone, obstruction, infection.** Stones in up to 80% (half staghorn).
- Low-virulence organisms (**Proteus** commonest, *E. coli*) in an obstructed, ischaemic kidney with an inadequate acute inflammatory response.

Pathology

- Diffuse (>80%) or segmental (focal).
- Gross: yellow nodules, pus-filled dilated calyces, a **contracted pelvis** encased by fibrosis, frequent **perinephric extension**.
- Microscopy: sheets of foamy histiocytes with lymphocytes, plasma cells and giant cells.
- **Malek staging:** I confined to kidney; II into perinephric fat; III into paranephric tissue or retroperitoneum.

Clinical features

- Flank pain, fever (about two-thirds), flank mass (about 60%), weight loss; persistent bacteriuria.
- Labs: anaemia, raised inflammatory markers, pyuria; liver dysfunction in up to half. One-third have **sterile urine** (prior antibiotics).

Imaging

- **CT (computed tomography), most useful:** enlarged kidney, little or no function, central calculus in a contracted pelvis; low-density cavities with rim enhancement (**"bear paw" sign**); extrarenal spread.
- Ultrasound: enlarged kidney with hypoechoic areas. DMSA (dimercaptosuccinic acid) scan quantifies lost function.
- **Differentials:** RCC (renal cell carcinoma), pyonephrosis, malakoplakia, lymphoma, tuberculosis.

Management

- Antibiotics to stabilise; drain if septic.
- **Nephrectomy** (with perinephric fat) for diffuse disease; partial nephrectomy for focal disease.
- Surgery is difficult: dense adhesions to psoas, great vessels, bowel and diaphragm. Remove the whole inflammatory mass, because simple drainage leads to **nephrocutaneous**

fistula.

- Laparoscopic nephrectomy is feasible; conversion rates are high.
- Frozen section can mimic clear cell carcinoma, and XGP can **coexist with RCC** or urothelial carcinoma. If cancer is not excluded, do nephrectomy.

Diagram to draw: CT sketch of an enlarged kidney with central staghorn and "bear paw" cavities.

> **Key points to remember:**
> - Stone, obstruction, Proteus infection → foamy macrophages.
> - CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
> - Mimics RCC; treat by complete nephrectomy, not drainage.

SAQ Malakoplakia

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3e

Definition

- "Soft plaque": a rare chronic granulomatous inflammation caused by **defective macrophage killing of bacteria**, mostly *E. coli*. Described by Michaelis and Gutmann (1902) and von Hansemann (1903).
- Commonest in the **bladder**; also ureter, kidney, prostate, testis, GI (gastrointestinal) tract, skin, lung.

Pathogenesis

- Macrophages ingest bacteria but cannot digest them (lysosomal defect, low intracellular cGMP [cyclic guanosine monophosphate]).
- Bacterial remnants become mineralised with calcium phosphate, forming laminated **Michaelis–Gutmann bodies**.
- Hosts are often immunocompromised, diabetic, chronically ill or **renal transplant recipients**.

Pathology

- Gross: soft yellow-brown plaques or nodules; renal tumour-like masses.
- Micro: sheets of large foamy histiocytes (**von Hansemann cells**) with **Michaelis–Gutmann bodies** (pathognomonic; "owl's eye" targetoid; stain with von Kossa, PAS [periodic acid–Schiff] and Prussian blue).
- MG bodies may be absent early; **α1-antitrypsin** immunostaining helps.

Clinical features

- Mostly >50 years; urinary tract **female:male 4:1**.
- Bladder: irritative LUTS (lower urinary tract symptoms), haematuria, recurrent *E. coli* UTI (urinary tract infection); plaques or masses mimicking cancer; distal ureteric stricture.
- Kidney: masses, fever, flank pain; renal vein or IVC thrombosis. Testis: epididymo-orchitis. Prostate: mimics carcinoma.

Investigations

- Urine culture (*E. coli*). Cystoscopy and **biopsy is diagnostic**.
- Imaging: enlarged kidneys with filling defects, **no stones, calcification or hydronephrosis** (unlike XGP).

	XGP (xanthogranulomatous pyelonephritis)	**Malakoplakia**
Cell	Lipid-laden foamy macrophages	Von Hansemann cells with MG bodies
Stone/obstruction	Usually (staghorn)	Absent
Laterality	Unilateral	Uni- or multifocal, bilateral
Treatment	Nephrectomy	Antibiotics; nephrectomy if unilateral, symptomatic

Management

- **Long-term antibiotics active inside macrophages:** fluoroquinolones (concentrate in macrophages), trimethoprim–sulfamethoxazole, rifampicin, doxycycline.
- **Bethanechol + ascorbic acid** raise intracellular cGMP and correct the degradative defect.
- Reduce immunosuppression where possible.
- Surgery if progression despite medical therapy: TUR (transurethral resection) of bladder lesions; **nephrectomy** for symptomatic unilateral renal disease.

Prognosis

- Bilateral or transplant-kidney disease: death often within 6 months; overall mortality can exceed 50%. Unilateral: long survival after nephrectomy.

Diagram to draw: histiocyte with Michaelis–Gutmann bodies.

> **Key points to remember:**
> - Defective macrophage digestion of *E. coli*.
> - Michaelis–Gutmann bodies are pathognomonic.
> - Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease.

Also, from the condensed book (not in the paper answer):

- In xanthogranulomatous pyelonephritis (XGP), urine is sterile in about a third but tissue culture is positive.

	XGP	**Malakoplakia**
Cell	Foamy lipid-laden macrophages	Von Hansemann cells, Michaelis–Gutmann bodies
Stones/obstruction	Usually	Absent
Organism	Proteus, E. coli	E. coli
Laterality	Unilateral	Uni-/multifocal, may be bilateral
Treatment	Nephrectomy	Intracellular antibiotics with or without nephrectomy

KEY POINT

Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics; pyonephrosis and large or obstructed abscesses need drainage; when the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy.

9 · Fournier gangrene and periurethral abscess

Skin hides the true spread; only early radical debridement saves lives.

- **Fournier gangrene** is necrotising fasciitis of the genitalia, perineum or perianal region. A source is found in ~**95%**, and the male:female ratio is ~10:1.
- Sources are **colorectal**, **urogenital** (stricture, instrumentation, catheter) and **dermal**. Risk factors are **DM**, obesity, alcoholism and immunosuppression.
- Synergistic organisms cause obliterative endarteritis, spreading along Buck's, dartos, Colles' and Scarpa's fascia.
- Pain out of proportion comes first, then crepitus and gangrene. **The skin underestimates fascial extent**, and CT must not delay surgery.

Score	Components	Use
Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC)	CRP, WBC, haemoglobin, sodium, creatinine, glucose	≥**6** suspicious; ≥**8** strongly predictive
Fournier's Gangrene Severity Index (FGSI) †	Nine vital signs and blood values, each scored 0–4	>9 predicts mortality

ALGORITHM

1. On suspicion, resuscitate and start broad IV antibiotics (piperacillin–tazobactam; or a carbapenem or vancomycin with clindamycin or metronidazole).
2. Perform **immediate radical debridement** back to bleeding fascia and leave the wound open.
3. Take the patient back for a **re-look at 24–48 h**.
4. If the source is urethral, then add suprapubic diversion; if rectal, then a colostomy.
5. Consider hyperbaric oxygen as an adjunct, then reconstruct.

- **Orchidectomy is almost never needed**, because the testicular blood supply is independent of the scrotal skin. Mortality is ~20–40%.
- A **periurethral abscess** needs drainage, suprapubic diversion and antibiotics.

10 · Urosepsis †

Term (Sepsis-3)	Definition
Sepsis	Organ dysfunction from a dysregulated response to infection: rise in Sequential Organ Failure Assessment (SOFA) score ≥**2**
Quick SOFA (qSOFA)	≥2 of: respiratory rate ≥22/min, altered mentation, systolic BP ≤100 mmHg
Septic shock	Vasopressors for mean arterial pressure (MAP) ≥65 mmHg **and** lactate >2 mmol/L despite fluids

ALGORITHM

1. Hour-1 bundle: measure lactate and take blood and urine cultures.
2. Give a broad IV antibiotic within 1 h.
3. If hypotensive or lactate ≥4, then give 30 mL/kg crystalloid.
4. If still hypotensive, then start noradrenaline for MAP ≥65.
5. Achieve **source control**: decompress with a stent or nephrostomy; no definitive stone surgery now.
6. De-escalate antibiotics by culture.

† Not in the source notes. Manipulating infected stones releases endotoxin, so drain first and operate later.

11 · Special populations

Pregnancy

*See the paper answer **Bacteriuria in pregnancy and antibiotic policy** (Section 1, Ch 21).*

Also, from the condensed book (not in the paper answer):

- GFR rises by 30–50%, so creatinine falls to about **0.5 mg/dL**, and a "normal" 0.8–0.9 may mean impairment.
- Severe or MDR pyelonephritis needs a carbapenem.

Elderly

- **Do not screen for or treat ASB**; bacteriuria with confusion is not UTI. The **absence** of pyuria **excludes** bacteriuria.
- The **revised McGeer criteria** require **localising** signs and ≥10^5 CFU/mL of ≤2 species in non-catheterised residents; catheterised residents may qualify on **systemic** signs.
- Treat for 3–6 days if uncomplicated and 10–14 if febrile.

Catheter-associated UTI

SAQ Catheter Associated Urinary Tract Infections (CAUTI)

Paper answer · P1

Type: SAQ (10 marks)
Source: 2017-Winter Paper 1, Q3c

Definition

- **CAUTI (catheter-associated urinary tract infection):** symptomatic UTI (urinary tract infection) with a catheter in place >2 days (or removed within 48 hours). Distinguish from **asymptomatic bacteriuria**, which needs no treatment.

Epidemiology and risk factors

- Commonest hospital-acquired infection (about 75–80% of hospital UTIs); commonest cause of secondary bacteraemia.
- Bacteriuria develops at about **5–10% per catheter-day**; 100% with long-term catheters. Intermittent catheterisation: 1–3% per catheterisation.

- **Risk factors:** duration (most important), female sex, older age, diabetes, breaks in the closed system, no systemic antibiotics.

Pathogenesis

- Entry at insertion, then **extraluminal** (along the outer surface; commoner in women) or **intraluminal** (from the drainage bag).
- **Biofilm:** bacteria in a matrix on the catheter surface, formed within minutes; it shields them from antibiotics. Urease-producers (*Proteus*) cause encrustation.
- Organisms: *E. coli*, *Pseudomonas*, *Proteus*, *Enterococcus*; long-term catheters polymicrobial.

Diagnosis

- Fever, rigors, flank or suprapubic pain, or delirium with no other source. Pyuria does not distinguish infection.
- Culture from a **freshly replaced catheter**, not the bag. **IDSA (Infectious Diseases Society of America): ≥10^3 CFU/mL** (colony-forming units) with symptoms; surveillance (CDC (Centers for Disease Control and Prevention)) definition uses ≥10^5.

Prevention (bundle)

- **Avoid unnecessary catheters;** review daily, remove early. Aseptic insertion.
- **Closed, dependent drainage**; bag below bladder; no routine irrigation.
- Meatal cleaning with water; no prophylactic antibiotics (delay bacteriuria only 3–4 days, select resistance).
- Alternatives: **clean intermittent catheterisation**, condom or suprapubic catheter.
- Silver-alloy catheters reduce bacteriuria in short-term use; no proven fall in symptomatic CAUTI.
- Do not treat asymptomatic bacteriuria (except pregnancy, before urological surgery).

Treatment

- **Replace or remove the catheter** before taking the culture and starting antibiotics, if in place >2 weeks (biofilm source).
- Empirical, then culture-directed antibiotics: **7 days** if prompt response; **10–14 days** if delayed.

Diagram to draw: catheter in bladder showing extraluminal and intraluminal routes and biofilm.

> **Key points to remember:**
> - Bacteriuria 5–10% per catheter-day; duration is the key risk.
> - Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
> - Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Also, from the condensed book (not in the paper answer):

- Catheters in place >30 days carry 4–5 species, and silicone is no better than latex.

Spinal cord injury

- All UTIs in SCI are complicated, and UTI is the commonest cause of fever. Symptoms are atypical: fever, spasticity, leakage, autonomic dysreflexia † and cloudy urine.

- **Clean intermittent catheterisation (CIC)** has the fewest long-term complications and an indwelling urethral catheter the most.

Drainage (NIDRR)	Significant bacteriuria
Indwelling urethral or suprapubic catheter	Any detectable count
Intermittent catheterisation	≥10^2 CFU/mL
Catheter-free men (clean voided)	≥10^4 CFU/mL

- Do not treat ASB. Treat afebrile UTI for 5–7 days; febrile UTI needs admission and **14 days** parenteral-first treatment. Change the catheter.
- Prophylaxis does not work in NGB; for recurrent UTI, get imaging and urodynamics (UDS) and lower storage pressure.
- A catheter for >**10 years** raises the risk of bladder cancer, so these patients need surveillance.

Diabetes, renal transplant

- In DM, do not treat ASB, but keep a low threshold to image pyelonephritis.
- After transplant, do not screen for ASB beyond 1–2 months. Treat uncomplicated UTI for 5–10 days and pyelonephritis for 14–21 days. Stents raise UTI by about half, so remove them within **4 weeks**.

12 · Fungal UTI

SAQ Fungal UTI

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3a

Introduction and organisms

- Fungal UTI (urinary tract infection) is rising with antibiotics, catheters and immunosuppression. **Candida** is commonest: *C. albicans*, then *C. glabrata* (often fluconazole-resistant); *C. krusei* is intrinsically resistant.
- Others in AIDS (acquired immunodeficiency syndrome) or transplant patients: Aspergillus, Cryptococcus, Mucorales, Histoplasma.

Risk factors and routes

- Antibiotics, indwelling catheters and stents, diabetes, steroids, neutropenia, prematurity, parenteral nutrition, obstruction.
- **Ascending** (catheter) causes most candiduria. **Haematogenous** spread from candidaemia: in disseminated candidiasis the kidney is the organ most often involved.

Clinical spectrum

- Asymptomatic candiduria (most) → cystitis → pyelonephritis → **fungus ball (bezoar)** causing obstruction → renal abscess or candidaemia.

Diagnosis

- Microscopy: budding yeasts and pseudohyphae (may mimic red cells). Repeat culture: contamination is common.

- Children: treat when repeated cultures exceed **10,000–15,000 CFU (colony-forming units)/mL**.
- Persistent funguria: ultrasound or CT (computed tomography) for bezoar; blood cultures if high risk.

Management

Situation	Treatment
Asymptomatic	Remove or change catheter, stop antibiotics; no antifungal
Treat asymptomatic only if	Neutropenia, very-low-birth-weight infant, before a urological procedure
Cystitis	Fluconazole 200 mg (3 mg/kg) daily for 2 weeks
Pyelonephritis	Fluconazole 200–400 mg (3–6 mg/kg) daily for 2 weeks
Fluconazole-resistant (*C. glabrata*, *C. krusei*)	Amphotericin B deoxycholate 0.3–0.6 mg/kg/day for 1–7 days ± flucytosine
Resistant cystitis	Amphotericin B bladder irrigation 50 mg/L
Fungus ball	PCN (percutaneous nephrostomy) drainage + systemic antifungal + local amphotericin irrigation; endoscopic or open removal if persistent

- Fluconazole concentrates in urine; **echinocandins and lipid amphotericin do not.**
- **Before procedures:** IDSA (Infectious Diseases Society of America) 2016 treats all candiduria (fluconazole 400 mg daily for several days before and after). AUA (American Urological Association) Best Practice Statement 2019 gives single-dose antifungal prophylaxis only for urothelium-breaching surgery. The two disagree.
- **Also know:** alkalinisation for bezoar; fluconazole caution under 6 months.

Diagram to draw: budding yeast with pseudohyphae; fungus ball in renal pelvis with nephrostomy.

> **Key points to remember:**
> - Candida commonest; *C. glabrata* often fluconazole-resistant.
> - Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
> - Fluconazole 2 weeks; echinocandins fail in urine.
> - Fungus ball: drain, antifungal, remove.

Also, from the condensed book (not in the paper answer):
- AUA BPS 2019 advises antifungal prophylaxis for high-pressure endoscopy (TURBT, TURP, therapeutic ureteroscopy) and open entry into the tract.

13 · Perioperative antimicrobial prophylaxis

One well-timed dose suffices; more only adds resistance.

Principle (AUA BPS 2019)	Rule
Indication	Any procedure breaching a tissue barrier, weighing host defence and consequences
Timing	Within **60 min** of incision; vancomycin and quinolones within 120 min

Principle (AUA BPS 2019)	Rule
Duration	**Single dose**; ≤**24 h** only for PCNL, prostheses, special cases
Agent	Local flora per antibiogram
Known infection	Sterilise urine before elective surgery (this is treatment); infected urine causes bacteraemia in ~half

- Host risk factors include age, smoking, steroids, immunodeficiency, hardware and infected stones.

Wound class	Examples	Antimicrobial prophylaxis (AP)
I clean	Scrotal/inguinal, retroperitoneal lymph node dissection (RPLND), nephrectomy without entering tract	Not needed (except groin/perineal incision, prosthesis)
II clean-contaminated	Any endoscopy; opening the urinary tract	Single dose
III contaminated	Infected stone PCNL, **transrectal biopsy**, bowel segments	Strongly recommended; anaerobic cover
IV dirty	Abscess, open trauma	Treatment, tailor to cultures

Procedure	AP?	Agent	Duration
Diagnostic cystoscopy / UDS (no risk factors)	No (EAU strongly against)	—	—
Cystoscopy with biopsy/fulguration	Host factors	TMP-SMX, amoxicillin–clavulanate, cephalosporin	Single
TURP / TURBT, laser prostate	All (TURBT: EAU only if high risk)	Cefazolin, TMP-SMX	Single (TURP short course with catheter acceptable)
Transrectal prostate biopsy	All	Fluoroquinolone or cephalosporin, with or without aminoglycoside; targeted by rectal swab	Single (≥1 day)
Ureteroscopy	All (diagnostic: uncertain)	TMP-SMX, 1st/2nd-gen cephalosporin	Single
PCNL	All	1st/2nd-gen cephalosporin; aminoglycoside and metronidazole	≤24 h
Shock-wave lithotripsy (SWL)	Only with risk factors (EAU: no)	TMP-SMX, cephalosporin	Single
Open/robotic entering tract	All	Cefazolin	Single
Colon segment	All	Cefazolin and metronidazole / cefoxitin / ertapenem, plus oral neomycin with erythromycin or metronidazole	Single

Procedure	AP?	Agent	Duration
Artificial urinary sphincter (AUS), penile prosthesis, sacral neuromodulation	All	Aminoglycoside and cephalosporin or vancomycin	≤24 h
Vaginal surgery incl. slings	All	Cefazolin/cefoxitin	Single
Penile (circumcision), clean scrotal	Likely none (scrotal: high-risk only)	Cefazolin	Single

- **Transrectal ultrasound (TRUS) biopsy** causes sepsis in 0.3–3.1%, mainly from quinolone-resistant E. coli. Use augmented or **targeted** prophylaxis; the AUA advises alternatives if local resistance is >20%.
- **UDS (SUFU 2017)**: give AP for neurogenic lower urinary tract dysfunction, raised PVR, ASB, immunosuppression, age >70, catheter or recent joint implant.
- **Catheter or stent removal**: give a single dose only if risk factors are present.
- **Endocarditis** (AHA): no AP solely to prevent it; with a high-risk valve and colonised urine, sterilise the urine first.
- **Orthopaedic hardware**: only a joint replaced <2 years ago or a high-risk host needs AP for high-bacteraemia procedures.

> **KEY POINT**
>
> Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit; otherwise give one well-timed, flora-matched dose. Anything longer is treatment.

14 · Landmark trials

Trial	Design	Result	Take-home
Raz & Stamm 1993	RCT, intravaginal oestriol vs placebo, postmenopausal recurrent UTI	0.5 vs 5.9 UTIs/patient-year; lactobacilli returned, pH fell	Vaginal oestrogen first-line after menopause
Hooton 2018	RCT, +1.5 L water/day vs none, premenopausal	Cystitis 1.7 vs 3.2/year	Hydration halves recurrences
Cai (ASB in recurrent UTI)	RCT, treat vs observe ASB in young women	Treatment increased recurrence and resistant E. coli	Do not treat ASB; E. faecalis may protect
ALTAR (methenamine)	Methenamine vs daily antibiotic prophylaxis, 12 months	Non-inferior	Non-antibiotic alternative
Drekonja 2021	RCT, afebrile male UTI, 7 vs 14 days ciprofloxacin/TMP-SMX	7 days non-inferior	7 days for afebrile men

27 Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

1 · Prostatitis: definitions and classification

Only categories I and II are infections; most men have category III.

- The Drach (1978) scheme, built on Meares–Stamey cultures, was replaced by the MCQ National Institutes of Health (**NIH, 1999**) classification.

NIH category	Entity	Defining feature
I	Acute bacterial prostatitis (ABP)	Acute prostatic urinary tract infection (UTI), often febrile
II	Chronic bacterial prostatitis (CBP)	Recurrent UTI, **same organism**; well between episodes
IIIA	Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS), inflammatory	Leucocytes in expressed prostatic secretion (EPS), post-massage urine (VB3) or semen
IIIB	CP/CPPS, non-inflammatory	No leucocytes
IV	Asymptomatic inflammatory	Incidental on biopsy or semen analysis

- I and II are **5–10%**; III is **90–95%**. **IIIA vs IIIB has no clinical significance**, and the prostate may not be the pain source.
- **Category III (NIH CPCRN)** needs pelvic pain for **≥3 of the previous 6 months**, excluding uropathogens, urogenital cancer, prior radiation, stricture and neurological bladder disease.
- Histological inflammation is near-universal (**>70%** at autopsy), so it cannot define disease; **Nickel grading** records its location, extent and grade.
- **Granulomatous prostatitis** classically follows **intravesical bacillus Calmette–Guérin (BCG)**, giving a **transient benign prostate-specific antigen (PSA) rise**.

2 · Category I: acute bacterial prostatitis

- Spread is mainly **ascending urethral**. Risks include catheters, instrumentation or biopsy, obstruction and insertive anal intercourse.
- **E. coli causes 65–80%**; Pseudomonas is commoner after instrumentation. After biopsy, organisms are often fluoroquinolone (FQ)-resistant or extended-spectrum β-lactamase (ESBL) producers.

> **KEY POINT**
>
> **Why young healthy men get ABP:** the organism, not the host — virulent uropathogenic E. coli that forms **biofilm** more often than cystitis strains; biofilm blocks antibiotics, so prostatic infection needs longer, well-penetrating therapy.

- Symptoms are frequency, dysuria, perineal pain, fever and rigors, with possible retention or sepsis. In febrile male UTI, **>90%** show raised PSA or prostate volume.

Evaluation

- First ask whether it was **spontaneous or after manipulation or biopsy**, as this decides empirical therapy.
- The prostate is tender and boggy; examine **gently and never massage**. Send midstream and **blood cultures**, and measure residual urine by **ultrasound (US)**.
- **No PSA** acutely; if raised, follow it to normal (up to **6 months**). Use transrectal ultrasound (TRUS) or CT only for suspected abscess, and **never biopsy**.

Treatment

- **Outpatient care only if** the man is not systemically ill, tolerates oral drugs and is not in retention.
- Systemic illness (**EAU**): amoxicillin or a 2nd-generation cephalosporin with an aminoglycoside, or a 3rd-generation cephalosporin. Ciprofloxacin only if local resistance **<10%** and fully oral outpatient care, or β-lactam anaphylaxis.
- **Febrile prostatitis after biopsy** (AUA White Paper): **no FQ or trimethoprim–sulfamethoxazole (TMP-SMX)**; use a **carbapenem**, amikacin or 2nd/3rd-generation cephalosporin.
- **Duration**: traditionally 4 weeks, but **2 weeks** of ciprofloxacin is equivalent; TB prostatitis needs ≥**6 months**. NSAIDs and α-blockers are useful adjuncts.
- **Retention**: brief urethral catheterisation is acceptable, but use a **suprapubic catheter for long-term** drainage.

Prostatic abscess

- Suspect it with high fever, immunosuppression or **no response after 48 h**. It is hypoechoic on TRUS, and **CT** shows spread.

ALGORITHM

1. If the abscess is <1–2 cm, then treat with antibiotics.
2. If it progresses and lies peripherally, then perform US-guided percutaneous drainage (recurrence 15–33%).
3. If it is large or drainage fails, then perform **transurethral unroofing**, using intraoperative US to locate it.
4. If it extends outside the prostate, then drain it openly (rare).

3 · Category II: chronic bacterial prostatitis

Recurrent same-organism UTI in men needs a long penetrating course.

- The prostate harbours the same organism and the man is **well between episodes**, whereas CP/CPPS pain is continuous.
- Organisms are Gram-negatives and E. faecalis; biofilm explains the poor response. The role of **Chlamydia** is debated.

Localisation test	Specimens	Interpretation
Meares–Stamey 4-glass (previous)	VB1 urethra; VB2 bladder (midstream); EPS; VB3 post-massage	Prostatic if EPS/VB3 counts exceed VB1/VB2
2-glass pre/post-massage (current)	Midstream pre-massage and first 10 mL post-massage	Predicts 4-glass in **>95%**

- There is no agreed cut-off, though some use a **10-fold rise**; **do not culture semen**.
- A residual volume **>180 mL** raises infection risk. Men **<45** with a first UTI have normal imaging, but check for **stricture**.

> **KEY POINT**
>
> **Why fluoroquinolones?** Prostatic entry is passive (lipid solubility, ionisation, protein binding, size). FQs are lipid-soluble **zwitterions** that concentrate in alkaline CBP fluid and cover Gram-negatives, Gram-positives and atypicals. Tetracyclines, macrolides and trimethoprim also penetrate.

Situation (EAU)	Regimen
First line	FQ (ciprofloxacin = levofloxacin) **4–6 weeks**
FQ-resistant, TMP-SMX-sensitive	TMP-SMX **3 months** (cure 30–50% vs 60–80% with FQ)
C. trachomatis	Azithromycin 500 mg daily 3 weeks, or doxycycline 100 mg twice daily 10 days; azithromycin is better than ciprofloxacin
T. vaginalis	Metronidazole 500 mg three times daily 14 days

- A macrolide can be added because it **penetrates biofilm**. Refractory cases need suppression or TURP.

4 · CP/CPPS — aetiology

An insult (infection, stress, trauma) in a susceptible man starts **neurogenic inflammation**, which other factors maintain, so single treatments fail.

Domain	Key evidence
Infection	Bacterial counts are equal in patients and controls; in mice, pain persists after clearance
Inflammation	Biopsy inflammation does not correlate with symptoms; symptoms track **cytokines**
Neurological	**Central sensitisation**
Pelvic floor	Tenderness and **failure to relax**; tender in **81%**
Psychosocial	**Catastrophising and helplessness predict** pain
Endocrine	Blunted adrenocorticotropic hormone (ACTH) response

5 · CP/CPPS — clinical features and evaluation

SAQ Chronic (non-bacterial) prostatitis: diagnosis, classification and management

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Chronic Prostatitis : diagnosis, classification and management." · P4 SAQ "Chronic non bacterial prostatitis."

Definition and classification: NIH (National Institutes of Health), 1999

Category	Description
I	Acute bacterial prostatitis

Category	Description
II	**Chronic bacterial prostatitis (CBP)**: recurrent UTI (urinary tract infection) with the **same organism**; well between episodes
IIIA	**CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, **inflammatory**: leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen
IIIB	CP/CPPS, **non-inflammatory**
IV	Asymptomatic inflammatory prostatitis (incidental)

- Chronic non-bacterial prostatitis is now called **CP/CPPS**, NIH category III: pelvic pain for **≥3 of the last 6 months** without uropathogenic infection. It accounts for 90–95% of prostatitis.
- The IIIA vs IIIB split has no treatment value.

Aetiology

- Interplay of past infection, neurogenic inflammation, **pelvic floor muscle dysfunction**, autoimmunity, psychological stress and **central sensitisation**.
- The prostate may not be the pain source.

Clinical features

- Perineal (commonest), testicular, suprapubic or penile pain; ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.

Evaluation (diagnosis of exclusion)

- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains (score 0–43).
- **DRE (digital rectal examination):** prostate tender in fewer than half; palpate the pelvic floor for **trigger points** (tender in about half).
- **Localisation tests:**
- **Meares–Stamey 4-glass:** VB1 (urethra), VB2 (bladder), EPS, VB3 (prostate).
- **2-glass** (pre- and post-massage urine): matches the 4-glass test in >95%.
- Urinalysis and culture, PVR (post-void residual), flow rate.
- Cytology if irritative symptoms or haematuria.
- Selected patients: cystoscopy (if low flow or bladder pain), urodynamics.
- **UPOINT phenotyping: U**rinary, **P**sychosocial, **O**rgan-specific, **I**nfection, **N**eurological/systemic, **T**enderness of muscles.

Management of category II (chronic bacterial prostatitis)

- Fluoroquinolone (lipid-soluble zwitterion, enters prostate) for **4–6 weeks** (EAU (European Association of Urology)).
- Chlamydia: azithromycin. Resistant organisms: TMP-SMX (trimethoprim-sulfamethoxazole) for 3 months. Refractory: suppression or TURP (transurethral resection of prostate).

Management of category III (CP/CPPS): multimodal, UPOINT-directed

Monotherapy trials fail.

Domain	Therapy
Urinary	**α-blockers** in recent-onset (<1 year), α-blocker-naive men; antimuscarinics

Domain	Therapy
Psychosocial	CBT (cognitive behavioural therapy), stress reduction
Organ (prostate)	**Quercetin**, **pollen extract** (small RCTs, randomised controlled trials); 5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia)
Infection	Quinolone or tetracycline for ≥6 weeks only if treatment-naive and <1 year (EAU), or positive VB3; no repeated courses
Neurological	Pregabalin, amitriptyline, duloxetine
Tenderness	**Pelvic floor physiotherapy, myofascial trigger-point release**, biofeedback

- **Others: acupuncture** (beat sham in a large RCT: 61.5% vs 38.3% response), aerobic exercise, PTNS (percutaneous tibial nerve stimulation), shockwave therapy; NSAIDs (non-steroidal anti-inflammatory drugs) only in combination.
- **Avoid (not recommended):** TURP or radical prostatectomy for pain.

Diagram to draw: UPOINT snowflake with a therapy for each domain.

> **Key points to remember:**
> - NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
> - CBP: same organism, fluoroquinolone 4–6 weeks.
> - CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
> - Multimodal therapy; α-blocker plus antibiotic only early.

Also, from the condensed book (not in the paper answer):

- The **NIH Chronic Prostatitis Symptom Index (NIH-CPSI)** † scores 0–43 (pain 0–21, urinary 0–10, quality of life 0–12), and a **≥6 points** fall is meaningful.
- **PSA is not indicated**, and a raised value must never be blamed on CP/CPPS.
- In men with bladder pain, about 42% meet interstitial cystitis criteria see Ch 28, and Hunner lesions respond to fulguration.

UPOINT phenotyping

Monotherapy fails because each man has a different mix of domains. UPOINT maps them for simultaneous treatment, and the domain count tracks severity.

Domain	Identify by	Therapy
Urinary	Lower urinary tract symptoms (LUTS), PVR	α-blocker; †antimuscarinic
Psychosocial	Depression, catastrophising	Psychology, stress reduction
Organ-specific	Prostate tenderness; bladder pain	†Quercetin/pollen; bladder therapy; 5α-reductase inhibitor (5-ARI) in BPH
Infection	Positive VB3 with negative VB2	Antibiotics
Neurologic/systemic	Pain beyond pelvis, irritable bowel syndrome (IBS), fibromyalgia	Tricyclics, pregabalin, acupuncture
Tenderness	Pelvic floor trigger points	Physiotherapy, myofascial release

6 · CP/CPPS — treatment

> **KEY POINT**
>
> **Governing fact:** no monotherapy has a positive definitive trial, because each tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team.

Modality	Evidence / role
Antibiotics	Quinolone trials negative. EAU: quinolone or tetracycline ≥**6 weeks** only if **treatment-naive and <1 year**. **No repeated courses without a positive culture**.
α-blockers	Trials negative; use in newly diagnosed men with voiding symptoms. Best combination: **antibiotic and α-blocker**.
Anti-inflammatories	Celecoxib helps pain only; steroids negative.
5-ARI	Dutasteride (REDUCE): ≥6-point NIH-CPSI fall in 49% vs 37%; for older men with BPH.
Neuropathic	Pregabalin missed its endpoint; amitriptyline needs ≥50 mg.
Conservative	Diet, **aerobic exercise** and cognitive therapy.
Acupuncture	Strongest conservative evidence: sham-controlled RCT, 61.5% vs 38.3% responders at 32 weeks.
Pelvic floor physiotherapy	Myofascial trigger-point release beats global massage (57% vs 21%); refer if the pelvic floor is tender.
Neuromodulation	Tibial or sacral nerve stimulation if frequency/urgency.
Surgery	No TURP or prostatectomy. **Bladder neck incision** only for proven obstruction.

7 · Acute orchitis

- **Bacterial** orchitis (age >35) is usually Gram-negative. **Mumps** is the commonest viral cause, following parotitis, with **atrophy in about half** but rare infertility. Autoimmune and vasculitic forms are treated with steroids.
- Ask about **pain duration**, because torsion is the key differential; get urine culture, STI tests and **scrotal US**. **Follow-up is mandatory** to avoid missing torsion or tumour.

8 · Acute epididymitis

Age and sexual history predict the organism.

- Epididymal pain and swelling for **<6 weeks**; infection is **ascending**.
- **In sexually active men <35, C. trachomatis and N. gonorrhoeae dominate**; **in men >35 and children, enteric organisms dominate**. Because >40% of men under 35 grow enterics, **screen all sexually active men for STI**.
- Send culture, first-void urine and STI NAAT; **Doppler US** excludes torsion.

Likely pathogen	Regimen
Chlamydia/gonorrhoea	Ceftriaxone **500 mg IM** (intramuscular) single dose (1 g if ≥150 kg) and doxycycline 100 mg twice daily **10 days**
STI and enteric (insertive anal sex)	Ceftriaxone 500 mg IM and levofloxacin 500 mg once daily 10 days

Likely pathogen	Regimen
Enteric only	Levofloxacin 500 mg once daily 10 days

- Give rest, elevation and NSAIDs, and treat partners. With **no improvement at 72 h**, repeat US for abscess or infarction.

9 · Chronic orchitis and epididymitis

- **Granulomatous orchitis** has infective causes (TB, brucellosis, syphilis) and an **idiopathic** form usually found after orchidectomy for suspected tumour. **Malakoplakia** shows **Michaelis–Gutmann bodies**.
- **Chronic epididymitis** (≥**3 months**) has a **Nickel classification**: inflammatory (infective, TB, **amiodarone**, idiopathic), obstructive (congenital, post-infective, post-vasectomy) and epididymalgia (normal epididymis).
- **TB epididymitis** is the commonest genital site, presenting as painless swelling with a **sinus**. Give four drugs for ≥**6 months**, and operate **only on anti-TB drugs**. **A mass not shrinking after 3 weeks needs inguinal exploration**.

10 · Chronic scrotal content pain (CSCP)

A cord block predicts who benefits from denervation surgery.

CSCP is intrascrotal pain for ≥**3 months** that interferes with life; there are no evidence-based guidelines.

- Post-procedural causes are **inguinal hernia repair** (commonest) and vasectomy (~**1 in 1000**, the consent figure). Others include varicocele, intermittent torsion and referred pain from a ureteric stone.
- **Wallerian degeneration** affects the **"trifecta nerve complex"**: cremaster, perivasal sheath and posterior lipomatous tissue. Pelvic floor dysfunction is common.
- **First decide** whether the pain is isolated or part of CP/CPPS, and get a **duplex US**.
- A **spermatic cord block** uses **20 mL 0.25% bupivacaine** at the **pubic tubercle**. **>50% relief** predicts success of microsurgical denervation of the spermatic cord (MDSC).
- Medical options are NSAIDs and **neuropathic agents**, without chronic opioids. Botulinum toxin equalled placebo in an RCT.

ALGORITHM

1. If there is an identifiable lesion, then do directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy, **microsurgical varicocelectomy**).
2. If the pain followed vasectomy, then offer **vasectomy reversal** (82–93%; wait ≥3 months).
3. If there is no lesion and the block is positive, then perform **MDSC**.
4. If the epididymis is focally affected, then perform epididymectomy.
5. Last resort: **inguinal orchidectomy**.

- **MDSC** is subinguinal and microscopic: divide the cord nerves, **preserve arteries and lymphatics** and ligate the internal spermatic veins. It cures 50–70%; consent for **testicular atrophy**. **Targeted trifecta MDSC** is equally effective and quicker.
- **Epididymectomy** succeeds in 50–75%. **Orchidectomy** is least reliable, and **inguinal beats scrotal** (75% vs 55%).

11 · Landmark trials

Trial	Design	Result	Take-home
NIH CPCRN ciprofloxacin/tamsulosin	RCT, 6 weeks, NIH-CPSI ≥15	No benefit vs placebo	Monotherapy fails in unselected CP/CPPS
REDUCE (CP/CPPS analysis)	Dutasteride vs placebo, 4 years	≥6-point fall 49% vs 37%	5-ARI for older men with BPH
MAPP myofascial physiotherapy	Feasibility vs global massage	57% vs 21% responders	Physiotherapy for tender pelvic floor
Acupuncture vs sham	Large RCT	61.5% vs 38.3% at 32 weeks	Evidence-based option
Botulinum toxin cord injection	Double-blind RCT	No better than local anaesthetic	Open-label series overestimate benefit

28 Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

1 · Definition and nomenclature

LAQ Interstitial cystitis / bladder pain syndrome

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management" (Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis, Winter 2024 Paper IV Q1 LAQ; Painful bladder syndrome management, Summer 2021 Paper IV Q3d SAQ) · P2 SAQ "Interstitial cystitis (Bladder pain syndrome)."

Read as: recent trends in the evaluation, diagnosis and management of IC/BPS (interstitial cystitis/bladder pain syndrome).

Definitions: how they have changed

- **NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987):** research criteria; missed about 60% of patients, so the definition was broadened.
- **AUA (American Urological Association), current:** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks**, without infection or another identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008):** chronic (**>6 months**) bladder-related pain with at least one other urinary symptom (persistent urge or frequency), after confusable diseases are excluded.
- **ICD-11 (International Classification of Diseases, 11th revision):** "chronic primary bladder pain syndrome".
- **Trend: splitting the syndrome.** "IC" is reserved for **Hunner lesion (HL) disease**, a true inflammatory disease; BPS without HL is a symptom complex. Both sit within UCPPS (urological chronic pelvic pain syndrome).

Phenotypes

- **Hunner lesion disease ("IC"):** about 20%, mainly over 50; true inflammation (dense lymphoplasmacytic infiltrate, mast cells).
- **Non-Hunner BPS (bladder pain syndrome):** bladder often histologically normal, or shows glomerulations on cystoscopy.
- Also pelvic-floor-dominant and widespread (centralised) pain phenotypes.

Pathophysiology (current view)

- Multifactorial: leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation; **central sensitisation**; stress-axis dysfunction; small-fibre neuropathy. No infection proven.
- **Associated conditions:** 75–80% have pain beyond the pelvis; IBS (irritable bowel syndrome, >30%), fibromyalgia, vulvodynia, endometriosis; tender pelvic floor in about 80%.

Recent trends in evaluation (AUA 2022), before treatment

- **Diagnosis of exclusion plus localisation:** tests aim to find the pain generator.
- **History:** pain worse on filling, relieved by voiding; "painful urgency" without urge incontinence, unlike OAB (overactive bladder). **Ask about ketamine.**
- **Examination: six-site pelvic-floor palpation** (anterior and posterior levator ani, obturator internus, each side); vulval swab test; DRE (digital rectal examination) in men.
- **Basic tests:** PVR (post-void residual), urinalysis, culture; cytology if risk factors; frequency–volume (voiding) chart.
- **Scores:** O'Leary–Sant ICSI/ICPI (Interstitial Cystitis Symptom/Problem Index), GUPI (Genitourinary Pain Index), pain VAS (visual analogue scale); **score pain and urinary symptoms separately.**
- **Body map:** widespread pain = 3 or more of 12 regions besides the pelvis.
- **Office cystoscopy if over 50 or HL suspected** (AUA 2022 change): HL is a stellate red area with radiating vessels, found only by looking.
- **Local anaesthetic challenge** (lidocaine with or without bupivacaine) localises pain to the bladder and selects intravesical therapy.
- **Dropped:** potassium sensitivity test (neither sensitive nor specific; obsolete); routine hydrodistension (glomerulations are non-specific); routine urodynamics (optional only).
- **Biopsy** only to exclude CIS (carcinoma in situ).

ESSIC classification

Cystoscopy with hydrodistension plus biopsy, e.g. "3C":

Cystoscopy	Biopsy
X not done; 1 normal; 2 glomerulations; 3 Hunner lesion	X not done; A normal; B inconclusive; C positive (inflammation, mastocytosis, fibrosis)

Differential diagnosis (exclude)

- UTI (urinary tract infection) including tuberculosis, CIS, stones, radiation or cyclophosphamide cystitis, **ketamine cystitis**, OAB, diverticulum, endometriosis.

Phenotype-based diagnosis and treatment (key trend)

Phenotype	Identified by	Treatment
Hunner lesion	Cystoscopy	Fulguration and/or triamcinolone; ciclosporin A if refractory
Anaesthetic-responsive bladder pain	Local anaesthetic challenge	Intravesical therapy
Pelvic-floor tenderness	Pelvic-floor examination	Myofascial physiotherapy
Widespread (centralised) pain	Body map, associated pain conditions	Systemic: amitriptyline, gabapentinoids, CBT (cognitive behavioural therapy), pain team

- **MAPP (Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain) reanalysis, 2023:** hydroxyzine/PPS (pentosan polysulfate) helped localised pain; amitriptyline helped widespread pain.

Management principles (AUA 2022)

- **Old AUA (2011/2014): six tiers**, from self-care to surgery.

- **AUA 2022 dropped the tiers:** grouped options, phenotype-guided, shared decisions; multimodal care allowed early; major surgery stays last.
- Stop treatments that fail; monitor with symptom scores.

Treatment options

Group	Options
Behavioural	Education; avoid trigger foods (coffee, citrus, spicy food, alcohol); stress management; flare plans
Physical	**Myofascial pelvic-floor physiotherapy** if tender (Standard); **avoid Kegel exercises**
Oral	**Amitriptyline** 10 mg at night, rising to about 50 mg; hydroxyzine; cimetidine; **PPS 100 mg three times daily** (**pigmentary maculopathy**: baseline and periodic retinal checks)
Intravesical	**DMSO (dimethyl sulfoxide) 50%, 50 mL, 15–20 min dwell**; heparin; alkalinised lidocaine (flares); also hyaluronic acid, chondroitin
Procedures	**HL: fulguration and/or triamcinolone injection** (75–90% improve); short (1–3 min) low-pressure (60–80 cmH_2O) hydrodistension; **onabotulinumtoxinA 100 U into the trigone** (must accept self-catheterisation; risk of retention); SNM (sacral neuromodulation)
Refractory HL	**Oral ciclosporin A** (watch kidney function and blood pressure)
Last resort (<10%)	Cystoplasty (augmentation) or diversion with or without cystectomy; pain may persist

- **Do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term oral (systemic) steroids.
- Pain management: gabapentinoids; opioids only via pain specialists.
- Most hydrodistension responses last under 2 months.
- Treat associated pain conditions (IBS, fibromyalgia) with other specialists.

Recent advances

- Urinary biomarkers and functional MRI (magnetic resonance imaging) for phenotyping (research); phenotype-stratified trials.
- **Also know:** tanezumab (anti-NGF (nerve growth factor)) not adopted; liposomal intravesical botulinum toxin no better than placebo; pudendal neuromodulation.

Diagram to draw: cystoscopic stellate Hunner lesion with radiating vessels, and the four-phenotype treatment chart.

> **Key points to remember:**
> - AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
> - AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
> - HL: fulguration or triamcinolone; ciclosporin if refractory.
> - Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

Also, from the condensed book (not in the paper answer):

- **IC/BPS** ranges from inflammatory disease with **Hunner lesions (HLs)** to pain with a normal bladder and many non-urological complaints.
- The **AUA** defines it as bladder-related pain, pressure or discomfort with lower urinary tract symptoms (LUTS) for **>6 weeks**, with no infection or other cause. † **ESSIC (2008)** requires **>6 months** and ≥1 urinary symptom; **type 3C** is classic HL disease.
- The **NIH/NIDDK (1987)** research criteria missed **~60%** when used clinically. The current **trend** is to call HL disease "IC" and non-HL disease "BPS".

2 · Epidemiology (headline)

- The RICE study found that **3.3–7.9 million US women** and 2.1–4.6 million men have IC/BPS-type symptoms.
- Female:male is **5:1** in clinics but **2–3:1** in populations, so men are under-diagnosed. **Ask men labelled CP/CPPS about bladder pain** see Ch 27.
- **HL prevalence** is **~20%**, rising with age.
- The course is chronic but **not progressive**, with flares, commonly after sexual activity.

> **KEY POINT**
>
> Uncontrolled series "work" but randomised controlled trials (RCTs) fail because of regression to the mean, placebo and extra attention. Expect improvement in the first 3–4 weeks of any treatment.

3 · Aetiology and pathophysiology

The cause is multifactorial, and each theory points to a treatment.

Theory	Mechanism / evidence	Treatment link
Glycosaminoglycan (GAG) layer / "leaky" urothelium	Urinary solutes such as potassium reach suburothelial nerves	Pentosan polysulfate (PPS), heparin, hyaluronic acid
Mast cells	Histamine release; marked in HL disease, but counts have **no diagnostic role**	Hydroxyzine, cimetidine, amitriptyline
Neurogenic	Neurogenic inflammation, pelvic cross-sensitisation and central sensitisation	Tricyclics, gabapentinoids, neuromodulation, CBT
Autoimmune	Excess of lupus, Sjögren syndrome and inflammatory bowel disease (IBD); lymphoplasmacytic infiltrate in HL	Ciclosporin A (CyA) for HL disease
Infection	No consistent organism; 18-week antibiotic RCT showed no benefit	No antibiotics without proven infection

> **KEY POINT**
>
> Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical.

4 · Pathology: Hunner lesion vs non-Hunner phenotype

- There is **no pathognomonic histology**. Biopsy only HLs or suspicious lesions, to exclude carcinoma in situ (CIS), tuberculous and ketamine cystitis.

Feature	HL disease ("IC")	Non-HL ("BPS")
Cystoscopy	Stellate erythema, radiating vessels, central scar; bleeds on filling ("waterfall"); rare at trigone	Normal, or glomerulations after distension
Histology	Ulceration, granulation, dense **lymphoplasmacytic** infiltrate, increased mast cells	Near-normal; sparse inflammation
Pain source	Bladder — end-organ disease	Often regional/widespread
Treatment	Fulguration / triamcinolone; CyA; dimethyl sulfoxide (DMSO) works better; cystectomy succeeds	Phenotype-directed; surgery often fails

5 · Associated conditions

- **75–80%** have pain beyond the pelvis. Among overlapping pain conditions, **irritable bowel syndrome (IBS)** is commonest (>30%), followed by **fibromyalgia**, chronic fatigue and migraine.
- **High-tone pelvic floor dysfunction (HTPFD)** occurs in **~80%**. **Autoimmune** disease shows the largest relative excess; in lupus, think of **lupus cystitis**.

> **KEY POINT**
>
> Commonest is not the same as most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Widespread pain predicts failure of bladder-directed therapy.

6 · Clinical features

Pain linked to filling separates IC/BPS from overactive bladder.

- Suprapubic pain is **worse with filling and relieved by voiding**, driving day and night frequency.

Feature	Overactive bladder (OAB)	IC/BPS	HTPFD
Urge	Sudden; fear of leakage	Persistent, mounting, pain-driven	Persistent; pelvic pressure
Pain	Absent	Present	Present
Incontinence	Urgency incontinence	Not a feature	—
Voiding	—	Frequency, low volumes	Obstructive

7 · Diagnosis — exclusion plus localisation

Basic evaluation (AUA 2022)

- Ask about the pain–filling link, triggers, pelvic irradiation, **cyclophosphamide and ketamine**.
- Examine for vaginal and urethral pathology, do a **Q-tip test** and DRE, and **palpate the pelvic floor**.

- Test post-void residual (PVR), urinalysis and culture, with cytology if at risk. Score pain and urinary symptoms separately, for example with the O'Leary–Sant indices (trial endpoint ≥30% fall).

Confusable disorders (ESSIC-modified)

Group	Disorder	Excluded by
Malignancy	Bladder cancer/CIS; prostate, gynae, colon	Cystoscopy with or without biopsy; prostate-specific antigen (PSA); pelvic exam
Infection	Bacterial, atypical, **TB**, herpes; prostatitis	Culture; **sterile pyuria needs TB culture**
Iatrogenic/drug	Radiation, cyclophosphamide, tiaprofenic acid, **ketamine**	History; ketamine causes upper-tract change
Outflow	Bladder neck obstruction, benign prostatic obstruction, retention	PVR, uroflow, (video)urodynamics
Structural/other	Stones, diverticulum; endometriosis, vulvodynia, atrophy; IBS, IBD	Imaging, cystoscopy, exam, laparoscopy, colonoscopy
Neuromuscular	Pudendal entrapment; pelvic-floor myalgia	Nerve block; pelvic-floor exam

- **Ketamine cystitis** causes HL-like inflammation, fibrosis and **hydronephrosis (~30%)**. Suspect it in the **young (<30) with visible inflammation**, and treat by **cessation**.
- **HTPFD** gives pelvic pressure, obstructive voiding and dyspareunia, with tender **myofascial trigger points**. Missed HTPFD causes poor response, so refer to a pelvic-floor physiotherapist.
- **Urodynamics** are not standard. Compliance is **normal**, and detrusor overactivity **does not exclude IC/BPS**.

Cystoscopy and bladder tests

- **AUA 2022** advises office cystoscopy when **HL is suspected, especially over 50**. Flexible cystoscopy shows most HLs **without hydrodistension**; biopsy **any trigonal lesion**.
- Relief after an **anaesthetic challenge** shows bladder-centred pain and selects intravesical therapy.
- The **potassium sensitivity test** (**0.4 M KCl** vs water) is **not recommended**, because it is painful and non-specific.
- **Hydrodistension** under anaesthesia fills at **80 cm H_2O for 1–3 min** to show **glomerulations**, which are **non-specific**. Biopsy only **after** distension and only suspicious lesions.

8 · Phenotype-directed management

*See the paper answer **Interstitial cystitis / bladder pain syndrome** (Section 3, Ch 28).*

> **KEY POINT**
>
> MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain**; amitriptyline helped **widespread pain**. A trial negative in all-comers can hide a positive result in the right phenotype.

9 · Treatment — AUA 2022 principles

The aim is coping and quality of life, not cure.

- There are **no tiers**, only grouped options: patients may move to any category **except major surgery** without failing "lower" ones. Stop ineffective treatment and seek other pain generators.
- The **Standard** is manual (myofascial) physiotherapy for pelvic-floor tenderness. Fulguration and/or triamcinolone for HLs is a **Recommendation**. Most drugs are **Options**. Chronic opioids are not routinely recommended.

ALGORITHM

1. If symptoms last ≥6 weeks, take a history, examine (pelvic floor, Q-tip, DRE), and check PVR, urinalysis and culture, with cytology if indicated.
2. Exclude confusable disease.
3. If HL is suspected (especially >50), then do cystoscopy. If HL is found, then fulgurate or inject triamcinolone; if refractory, give CyA.
4. If there is no HL, then match the phenotype: pelvic floor, physiotherapy; lidocaine responder, intravesical therapy; widespread pain, systemic therapy and CBT.
5. Combine behavioural, oral and intravesical treatment as needed, then BTX-A or sacral neuromodulation (SNM).
6. Only if bladder-centred and all else fails (rare), consider diversion with or without cystectomy.

10 · Treatment options (AUA 2022)

*See the paper answer **Interstitial cystitis / bladder pain syndrome** (Section 3, Ch 28).*

Also, from the condensed book (not in the paper answer):

- SNM is favoured over BTX-A for HTPFD or obstructive voiding.

10.1 Behavioural and pelvic-floor therapy

- Common **diet triggers** are caffeine, alcohol, fizzy drinks, spicy food and citrus; an elimination diet identifies the patient's own triggers.
- Education and behavioural modification alone improved 45% in the NIH amitriptyline RCT.
- In the myofascial physiotherapy RCT, 10 sessions beat global massage, **59% vs 26%**.

10.2 Oral therapy

- **Amitriptyline**: start **10 mg at night, adding 10 mg weekly to ~50 mg**. In the NIH RCT it gave 55% vs 45% (not significant), but **77% vs 53%** if ≥50 mg was tolerated. **Contraindications** are long QT, conduction block, recent myocardial infarction and heart failure.
- **Hydroxyzine** (H1 blocker) 25–50 mg at night had a negative RCT; **cimetidine** 400 mg twice daily has one positive RCT.
- **PPS 100 mg three times daily** is the only FDA-approved oral IC drug; only **3–6%** reaches urine and it acts in **3–6 months**. Its phase IV RCT was negative.
- **PPS maculopathy** is cumulative-dose related. **FDA (2020)**: retinal examination for all **within 6 months**, then periodically.

- **Ciclosporin A** is for HL refractory to fulguration or triamcinolone, at **1–3 mg/kg/day in two doses**. Its main toxicities are **nephrotoxicity and hypertension**, so monitor blood pressure, renal function and levels.
- **Antibiotics**: the AUA advises **none long-term**, and only one empirical course if never treated.

10.3 Intravesical therapy

Intravesical therapy is for **bladder-centred** pain; a poor lidocaine response predicts failure of every instillation. It is an AUA Option.

Agent	Regimen	Key points
DMSO 50%	50 mL, **15–20 min dwell**, every 1–2 weeks × 4–6	FDA-approved. Better in HL. Causes garlic breath and an initial flare; increases absorption of co-instilled lidocaine.
Heparin	10,000–40,000 U, 30–60 min, self-instilled	GAG mimic; benefit takes months.
Alkalinised lidocaine	200 mg with 8.4% bicarbonate daily × 5	Immediate effect, so useful for flares; tachyphylaxis.
Hyaluronic acid, chondroitin, PPS	Weekly then monthly	Weak/negative trials

- Cocktails are no better than DMSO alone.

10.4 Hunner lesion treatment

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone, with biopsy if never biopsied. **75–90%** improve, but repeat treatment is usually needed within 1–2 years.

ALGORITHM

1. Under general anaesthesia, map the HLs before distension causes bleeding.
2. Distend mildly and biopsy suspicious or trigonal lesions.
3. Fulgurate at **low volume** (Bugbee electrode), working from the **outer margin inwards**.
4. Inject triamcinolone 40 mg/mL (or 1:1 with bupivacaine/lidocaine) in 0.5-mL aliquots, 2–3 mm submucosally.
5. Fulgurate bleeders, empty the bladder and instil lidocaine–bupivacaine.

> **KEY POINT**
>
> Fulgurate from the margin inward: cautery erythema and oedema spread peripherally, so the true edge is lost if you start centrally.

10.5 Hydrodistension

- AUA Option: **short (1–3 min), low-pressure (60–80 cm H_2O)** distension, with benefit fading by 6 months.
- An anaesthetic capacity **<400 mL** marks a bladder-centred phenotype, often HL.

10.6 OnabotulinumtoxinA and neuromodulation

- **BTX-A** is an Option after failed conservative, oral and intravesical therapy: **100 U** into the bladder including the **trigone**, lasting about 5–9 months. The patient **must accept clean intermittent catheterisation (CIC)**.

- **SNM (S3)** is an Option and works best with **HTPFD or levator tenderness**, but IC/BPS has high explantation rates.

	BTX-A	**SNM**
Favoured when	Bladder-centred pain; can do CIC	Obstructive voiding, incomplete emptying, HTPFD, CIC impossible
Main risk	Retention needing CIC	Adverse events, explantation

10.7 Major surgery — last resort

- Surgery is only for **rare (<10%), selected, bladder-centred** patients failing all else, and pain may persist after cystectomy. If pain persists after a **differential spinal block** to T10, surgery will fail.

Predicts success	**Predicts failure**
HLs (refractory)	No HLs
Small fibrotic bladder; anaesthetic capacity <200–400 mL	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal/no response
Painful filling, pain on scope touch	Urethral/neuropathic pain; HTPFD; widespread pain

- Cystectomy with a conduit leaves about three-quarters pain-free. Prefer the **simplest diversion**, as a retained bladder risks pain and pyocystis.

11 · What NOT to do (AUA)

Do not offer	**Why**
Long-term oral antibiotics	RCT negative; adverse events 80% vs 40%
Intravesical bacillus Calmette–Guérin (BCG)	Ineffective; DMSO superior; † BCG sepsis
High-pressure, long hydrodistension†	No benefit; rupture, sepsis †
Long-term oral glucocorticoids	No benefit; toxicity
† Intravesical resiniferatoxin	Ineffective in RCT †
Kegel strengthening	Worsens HTPFD pain
Major surgery first / non-bladder-centred pain	Irreversible; pain persists

> **KEY POINT**
>
> Nothing is worse than irreversible surgery followed by continued pain — every predictor of surgical success marks bladder-generated pain.

12 · Landmark trials

Trial	**Design**	**Result**	**Take-home**
Myofascial PT RCT (FitzGerald 2012)	10 sessions myofascial vs global massage	59% vs 26%	Guideline Standard
NIH amitriptyline RCT (Foster 2010)	EBMP with or without amitriptyline	55% vs 45% (not significant); 77% vs 53% if ≥50 mg	Titrate to ≥50 mg; behavioural therapy works

Trial	Design	Result	Take-home
PPS phase IV RCT	PPS vs placebo	No difference (≈40% each)	Efficacy doubtful; remains Option
MAPP phenotype reanalysis (2023)	Two negative NIH RCTs by pain distribution	Hydroxyzine/PPS localised; amitriptyline widespread	Treat by phenotype

29 Sexually Transmitted Diseases

1 · Screening, prevention and partner notification

One sexually transmitted disease (STD) often hides another.

- The nucleic acid amplification test (**NAAT**) is the standard screen; men who have sex with men (MSM) need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia.
- **Partner notification** covers the previous **60 days** for urethritis, chlamydia and Mycoplasma genitalium; all partners for trichomoniasis; and the previous month for scabies or lice. Abstain for 7 days and until partners are treated. Test everyone at risk for HIV and syphilis.

2 · Urethritis

- **Gonococcal** urethritis makes up 5–20%. In non-gonococcal urethritis (**NGU**), C. trachomatis is commoner than M. genitalium, which is the leading cause of **persistent or recurrent** urethritis. Up to half have no pathogen.
- **Diagnosis** needs any one: discharge; Gram stain with **≥2 white blood cells (WBC) per oil-immersion field**; positive leucocyte esterase on first-void urine; or ≥10 WBC/hpf in first-void urine.
- Intracellular gram-negative diplococci in a symptomatic man confirm gonorrhoea; a negative Gram stain does not exclude it.
- **First-void urine NAAT** is preferred; **culture** is for treatment failure and children.

Pathogen	First line	Alternatives / notes	Follow-up
Gonorrhoea (CDC 2021)	**Ceftriaxone 500 mg IM once** (1 g if ≥150 kg)	Gentamicin 240 mg IM and azithromycin 2 g (β-lactam allergy); cefixime 800 mg. Europe: ceftriaxone 1 g and azithromycin 2 g. Add chlamydia cover unless excluded	Test of cure only for **pharyngeal** infection; retest at 3 months
Chlamydia / NGU	**Doxycycline 100 mg bd 7 days**	Azithromycin 1 g once	No test of cure; retest at 3 months
M. genitalium	Doxycycline 7 days, then **azithromycin** 1 g, then 500 mg daily × 3, if macrolide-sensitive; or **moxifloxacin** 400 mg × 7 days if resistant or untested	Single-dose azithromycin is inadequate	Abstain until cured
T. vaginalis (male)	Metronidazole 2 g once	Tinidazole 2 g once	Treat all partners

- Higher-dose **ceftriaxone monotherapy** reflects resistance; cephalosporins are safe unless a recent IgE-mediated penicillin reaction occurred. Treatment failure needs culture and susceptibility.
- **M. genitalium** has no cell wall, so Gram stain misses it; NAAT with a macrolide-resistance assay picks the second drug.
- There is no chlamydia test of cure, because nucleic acid persists ~3 weeks.

ALGORITHM

1. Persistent NGU: check adherence and look for an untreated partner.
2. Send NAAT for M. genitalium, with or without T. vaginalis.
3. If negative, then think of chronic pelvic pain syndrome.

KEY POINT

Men with gonorrhoea usually have symptoms; women are usually silent and present with pelvic inflammatory disease or infertility. Treating the man protects his partner.

3 · Epididymitis by age

Acute means <6 weeks; chronic (≥6 weeks) is mostly tuberculosis (see Ch 27). Always exclude **torsion**. Apply urethritis criteria and NAAT; add urine culture if enterics are likely.

Group	Organisms	Empiric regimen (10 days)
Young, sexually active	C. trachomatis, N. gonorrhoeae	Ceftriaxone 500 mg IM and doxycycline 100 mg bd
Insertive anal sex	Above, plus enterics	Ceftriaxone and levofloxacin 500 mg daily
>35 years, obstruction or instrumentation	E. coli	Levofloxacin alone

4 · Genital ulcer disease

Herpes is commonest, then syphilis. Appearance misleads, so test **every ulcer** with syphilis serology, HSV NAAT, an H. ducreyi test if endemic, and HIV. Biopsy non-healing ulcers, because carcinoma mimics them.

Disease (organism)	Lesion	Nodes	Test	Treatment
Primary syphilis (T. pallidum)	**Painless**, indurated, clean, single	Painless, rubbery, bilateral	Serology; darkfield	Benzathine penicillin G 2.4 MU IM once
Herpes (HSV-2 more than HSV-1)	**Painful** multiple vesicles that become shallow ulcers	Tender, bilateral	NAAT of unroofed base	Aciclovir and related drugs
Chancroid (H. ducreyi)	**Painful, ragged, undermined, purulent**	Tender **suppurative buboes**	Special culture; exclusion	Azithromycin 1 g once, or ceftriaxone 250 mg IM once, or ciprofloxacin 500 mg bd 3 days, or erythromycin 7 days
Lymphogranuloma venereum, LGV (C. trachomatis L1–L3)	Small, painless, transient	Painful, matted, fistulae; groove sign	Chlamydia NAAT	Doxycycline bd **21 days**
Donovanosis (Klebsiella granulomatis)	**Painless, beefy red**, bleeds easily	**Absent** (pseudobuboes)	**Donovan bodies** on crush preparation	Azithromycin 1 g weekly or 500 mg daily, ≥3 weeks **and until healed**

Genital herpes

- HSV lies latent in the **sacral ganglia** and reactivates with trauma, surgery or immunosuppression. Most transmission is from asymptomatic shedding. HSV-2 recurs more, so type it.
- Complications are aseptic meningitis and **urinary retention**; HSV-2 raises HIV risk 2–3-fold.

Drug	First episode (7–10 days)	Suppressive	Episodic
Aciclovir	400 mg tds	400 mg bd	800 mg bd 5 days
Valaciclovir	1 g bd	500 mg–1 g daily	500 mg bd 3 days
Famciclovir	250 mg tds	250 mg bd	1 g bd 1 day

- Treat every first episode. **Episodic** therapy must start within 1 day of lesions or in the prodrome.

Chancroid, donovanosis, LGV

- **Chancroid** needs a painful ulcer, a typical picture, and syphilis and HSV excluded. Aspirate or drain fluctuant buboes.
- **Donovanosis**: if antibiotics fail, biopsy, because **penile squamous cell carcinoma (SCC) mimics it**.

SAQ Lymphogranuloma venereum

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 1, Q3c

Definition

- **LGV (lymphogranuloma venereum)** is an invasive, lymphotropic STD (sexually transmitted disease) caused by ***Chlamydia trachomatis*** **serovars L1, L2 and L3**.
- Of 15 serovars, A–C cause trachoma and D–K anogenital infection. The L serovars invade lymphatics, causing lymphadenitis and scarring.

Epidemiology

- Endemic in the tropics, including India; increasing as **proctitis in MSM (men who have sex with men)**. Not more frequent or virulent in HIV (human immunodeficiency virus).

Clinical stages

Stage	Timing	Features
Primary	3–30 days	Small painless papule or ulcer, often unnoticed. Rectal inoculation → **proctitis** (bleeding, pain, tenesmus, discharge), now the commonest presentation
Secondary (inguinal syndrome)	2–6 weeks later	Fever, malaise; **painful, matted, unilateral inguinal or femoral buboes** that may form fistulae. **Groove sign**: nodes above and below the inguinal ligament
Tertiary	Untreated	Fibrosis and lymphatic block: **genital elephantiasis** (esthiomene in women), perirectal abscess, fistulae, **rectal stricture**

- Inguinal disease is commoner in men: the cervix and vagina drain to pelvic nodes.

Differential diagnosis of genital ulcer

- Syphilis (painless ulcer and nodes); herpes (painful vesicles); chancroid (painful ulcer, suppurative nodes); donovanosis (beefy ulcer, no nodes); **LGV: transient painless lesion, then painful matted nodes**.

Investigations

- **NAAT (nucleic acid amplification test)** of ulcer, rectal or node aspirate; LGV-specific genotyping confirms where available. Rectal chlamydia NAAT in all proctocolitis.
- Serology not routine. Screen for HIV, syphilis, gonorrhoea; offer PrEP (pre-exposure prophylaxis).

Treatment

- **Doxycycline 100 mg twice daily for 21 days** (first line).
- Alternatives: erythromycin 500 mg four times daily for 21 days, or azithromycin 1 g weekly for 3 weeks (pregnancy; consider test of cure by NAAT).
- Fluctuant buboes: aspirate through healthy skin to prevent sinus formation.
- **Late disease:** stricture dilatation or resection, fistula repair, reduction of elephantiasis.

Diagram to draw: groin with buboes above and below the inguinal ligament (groove sign).

> **Key points to remember:**
> - Serovars L1–L3; invasive and lymphotropic.
> - Proctitis is now the commonest presentation.
> - Groove sign; esthiomene; rectal stricture.
> - Doxycycline 100 mg twice daily for 21 days.

5 · Syphilis staging and serology

Stage	Features	Treatment (penicillin-allergic)
Primary (10–90 days)	Chancre and nodes; heals in 3–8 weeks	Benzathine penicillin **2.4 MU IM once** (doxycycline 14 days)
Secondary (3–5 months)	Rash on **palms and soles**, **condylomata lata**, alopecia, hepatitis, nephritis	As primary
Early latent (<1 year)	Positive serology only	As primary
Late latent / unknown	Positive serology only	Benzathine **weekly × 3** (doxycycline 28 days)
Tertiary	Gumma, aortitis, tabes dorsalis	Weekly × 3 if cerebrospinal fluid (CSF) normal
Neurosyphilis (any stage)	Central nervous system, eye, ear	**Aqueous penicillin G 18–24 MU/day IV 10–14 days**

- T. pallidum cannot be cultured, so **two serological tests** are used. The **nontreponemal test (NTT)** (VDRL, RPR) can be falsely positive; its titre falls with treatment, so it **monitors treatment**. The **treponemal test (TT)** (FTA-ABS, TP-PA, EIA) usually stays positive for life.
- A significant change is **fourfold (two dilutions)**. Use the same NTT each time.

ALGORITHM

1. **Reverse algorithm**
2. Start with a TT. If positive, then do an NTT.
3. If the NTT is negative, then do a second, different TT.
4. If that is positive and there is no past treatment, then treat as late latent syphilis.

- The **Jarisch–Herxheimer** reaction (fever and myalgia within 24 h) is treponemal lysis, not allergy; give NSAIDs.
- In primary or secondary syphilis, test for HIV and offer pre-exposure prophylaxis (PrEP) if negative.
- **Follow-up** serology is at 6 and 12 months (and 24 for latent disease). A fourfold rise means failure or reinfection, so retreat weekly × 3 and consider CSF.

6 · HPV and anogenital warts

- Human papillomavirus (HPV) **6 and 11** cause ~90% of warts. HPV **16 and 18** cause >70% of cervical cancer, and also penile and anal cancer.
- **Bowenoid papulosis** (HPV-16) is a high-grade squamous intraepithelial lesion. **Giant condyloma** (Buschke–Löwenstein) is a locally invasive verrucous carcinoma.
- Diagnosis is clinical. **Biopsy if** the diagnosis is uncertain, there is no response or worsening, the patient is immunocompromised, or the lesion is pigmented, indurated, fixed or ulcerated.

Treatment	Key points
Imiquimod 5% (patient)	3 nights/week for up to 16 weeks; wash off after 6–10 h. Weakens condoms
Podofilox 0.5% (patient)	bd for 3 days, then 4 days off. **Contraindicated in pregnancy**
Sinecatechins 15% (patient)	tds, not washed off. Weakens condoms. Not in HIV, herpes or pregnancy
Cryotherapy, trichloroacetic acid (TCA) 80–90% (provider)	Cryotherapy every 1–2 weeks; TCA weekly
Excision, curettage, CO_2 laser	For large warts; use smoke extraction

Change treatment if there is no response by 3 months; observation is acceptable.

- The **vaccine** is **nonavalent** (6, 11, 16, 18, 31, 33, 45, 52, 58). Give it at **11–12 years** (from 9), with catch-up to **26**, in both sexes. Starting <15 needs **2 doses**; ≥15 or immunocompromised needs **3 doses**.

7 · Molluscum and ectoparasites

- **Molluscum** contagiosum forms **umbilicated** papules that regress in 6–12 months; use cryotherapy, curettage or imiquimod. Extensive lesions suggest HIV.
- **Pubic lice**: use a **permethrin 1%** rinse for 10 min, or malathion 0.5% or ivermectin 250 µg/kg if resistant.
- **Scabies** causes nocturnal itch and burrows. Apply **permethrin 5%** to all skin, or give ivermectin 200 µg/kg repeated at 2 weeks. **Crusted** scabies (immunocompromised) needs topical treatment and repeated ivermectin, never lindane.

- Treat contacts from the past month and wash bedding.

8 · HIV for urologists

LAQ Describe clinical implication of AIDS on the Genito-Urinary Tract?

Paper answer · P1

Type: LAQ (25 marks)
Source: 2020-Summer Paper 1, Q1

Introduction

- AIDS (acquired immunodeficiency syndrome) is the late stage of HIV (human immunodeficiency virus) infection. HIV is a retrovirus that destroys CD4 T lymphocytes. It spreads by blood, semen, vaginal fluid and breast milk.

Stage	Features
1. Acute	2–4 weeks after infection; flu-like illness; high viral load; antibody tests may be negative
2. Clinical latency	Years; usually no symptoms
3. AIDS	**CD4 <200 cells/mm³** or an AIDS-defining condition

- **Concept:** genitourinary (GU) disease follows the falling CD4 count. Organisms become opportunistic, infections become chronic and the GU tract becomes a reservoir. With HAART (highly active antiretroviral therapy), the burden shifts to drug toxicity and diseases of ageing.

Diagnosis (urologist's role)

- Test anyone with, or at risk of, an STD (sexually transmitted disease), and every TB (tuberculosis) patient.
- Antigen–antibody immunoassay → antibody differentiation assay; if discordant, HIV RNA (acute infection).

Manifestations by site

Site	Manifestations
Genital skin and STDs	Ulcerative STDs (herpes, syphilis, chancroid) increase HIV acquisition and transmission; HSV-2 (herpes simplex virus 2) in 50–90%. Warts (HPV (human papillomavirus) 6, 11) in about 20%, may be giant; molluscum 5–18%. Perineal cellulitis (cover MRSA (methicillin-resistant *Staphylococcus aureus*)); Fournier gangrene commoner
UTI (urinary tract infection)	About fivefold commoner. Bacteriuria about 30% at CD4 <200, 11% at 200–500, none above 500. *E. coli* and other gram-negatives; *Pseudomonas* up to 30%. *Salmonella* UTI needs lifelong prophylaxis. Culture before treatment
Opportunistic UTI	No response, or CD4 <500: look for fungi (*Candida*, *Cryptococcus*, *Histoplasma*), TB, CMV (cytomegalovirus), adenovirus
Kidney	Opportunistic pyelonephritis (*Pneumocystis*, *Candida*, mycobacteria). AKI (acute kidney injury) from sepsis, dehydration, nephrotoxins (tenofovir, indinavir, amphotericin). Obstruction by stones, lymphoma, fungal balls

Site	Manifestations
GU TB	HIV raises active TB 15–20-fold. Disease is disseminated and bilateral, with **less fibrosis, fewer strictures and fewer contracted bladders**, because fibrosis needs an intact immune response. Mortality is higher. Same 6-month regimen; rifampicin lowers protease inhibitor levels
Prostate	Bacterial, TB and fungal prostatitis; prostatic abscess and urosepsis; relapse up to 70%. Cryptococcal prostatitis is a reservoir for relapsing meningitis, so give long-term antifungals
Testis and epididymis	Recurrent, chronic epididymo-orchitis (including TB). Testis harbours opportunists. Testicular atrophy and hypogonadism
Voiding	LUTS (lower urinary tract symptoms) worsen with disease stage. Retention may herald seroconversion. Neurogenic causes: HIV encephalopathy, toxoplasmosis, **CMV polyradiculopathy** (bladder and bowel dysfunction with sciatic pain: lumbar puncture, treat early), neuropathy
Sexual function	ED (erectile dysfunction) in about half of men under 50; protease inhibitors worsen it. **Protease inhibitors raise sildenafil levels 3–10-fold: start low.** Low libido, ejaculatory disorders

HIVAN (HIV-associated nephropathy)

- Nephrotic-range proteinuria and rising creatinine, often without oedema or hypertension. Typically CD4 <200 and viral load >1000 copies/mL; about 12 times commoner in Black patients.
- Biopsy: **collapsing FSGS (focal segmental glomerulosclerosis)** with tubular microcysts.
- Treatment: HAART plus ACE (angiotensin-converting enzyme) inhibitor. Without it, dialysis follows within about 10 months.

Stones

- **Indinavir** stones in up to 22% of treated patients. Indinavir precipitates in alkaline urine. Stones are radiolucent, and pure stones may be invisible even on CT (computed tomography).
- Stop the drug and hydrate: succeeds in up to 70%. Newer protease inhibitors rarely cause stones.

Malignancy

AIDS-defining	Non-AIDS-defining
Kaposi sarcoma (HHV-8 (human herpesvirus 8)), non-Hodgkin lymphoma, cervical cancer	Penile, anal, testicular, prostate cancers; Hodgkin lymphoma
Falls with HAART	Now about 70% of cancers in treated HIV

- **Kaposi sarcoma:** genital in about 3% of AIDS; glans lesions can block the meatus. Treat with ART (antiretroviral therapy) first. Local excision, laser or intralesional vinblastine; radiotherapy 15–30 Gy; systemic chemotherapy if disseminated.
- **Penile cancer:** HIV raises HPV infection; about fourfold risk of death.

Implications for urological practice

- **Universal precautions** for every patient: double gloves, no hand-to-hand passing of sharps.
- Occupational risk: percutaneous about 0.3%, mucous membrane about 0.09%.

- **PEP (post-exposure prophylaxis)** within 72 hours, for 28 days: tenofovir + emtricitabine plus raltegravir or dolutegravir. HIV test at 6 weeks, 3 and 6 months.
- CD4 <200: TMP-SMX (trimethoprim-sulfamethoxazole) *Pneumocystis* prophylaxis also lowers UTI.
- **Prevention:** male circumcision cuts heterosexual acquisition by 50–60% (three African RCTs (randomised controlled trials)); consistent condom use cuts risk about 80%; PrEP (pre-exposure prophylaxis).

Also know: IRIS (immune reconstitution inflammatory syndrome) on starting ART; serodiscordant couples: suppressed viral load with sperm washing and assisted reproduction; transplantation is possible with controlled HIV.

Diagram to draw: male GU tract outline labelled with the lesion at each site.

> **Key points to remember:**
> - AIDS = CD4 <200; UTI risk rises as CD4 falls.
> - HIV-TB: disseminated, less fibrosis, fewer strictures.
> - Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
> - PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Also, from the condensed book (not in the paper answer):

- Start post-exposure prophylaxis (PEP) within **72 h**, and only for mucosal, broken-skin or percutaneous contact with blood or genital fluid. Needlestick risk is **0.3%**.
- Circumcision cuts heterosexual acquisition by **50–60%** but does not protect partners. **After sexual assault** (men), give ceftriaxone 500 mg once, doxycycline 7 days, hepatitis B and HPV vaccines, and PEP within 72 h.
- PrEP is daily emtricitabine with tenofovir disoproxil fumarate, or tenofovir alafenamide in men only.

30 Cutaneous Diseases of the External Genitalia

1 · Approach to genital dermatoses

Genital rashes look alike, so biopsy early to avoid missing cancer.

- Genital skin has few reaction patterns, so mimics abound and the **biopsy threshold is low**. Ask about over-the-counter creams, and do a **full skin survey**, because extragenital lesions often give the diagnosis.
- Identify the **primary lesion**. A papule is solid and **≤0.5 cm**; larger solid lesions are nodules or plaques, and fluid-filled ones are vesicles (≤0.5 cm) or bullae. An **erosion** loses epidermis only and does not scar; an **ulcer** loses dermis too and scars.
- Bedside tests: KOH prep for hyphae or yeast, mineral oil prep for scabies, Tzanck smear for herpes, and Wood's lamp for erythrasma (coral-red) and vitiligo.

Biopsy: a 2–3 mm punch under local anaesthetic is often diagnostic; large or meatal lesions go to theatre.

2 · Principles of therapy

- Use **topical treatment first**; urologists overprescribe systemic antibiotics.
- Give **systemic antifungals** only for extensive dermatophytosis, Majocchi granuloma, severe immunocompromise or topical failure, and avoid systemic **ketoconazole** (hepatotoxicity, adrenal insufficiency).
- Give systemic prednisone for ≤3 weeks as a single morning dose.
- **Topical steroids** cause atrophy in thin genital skin, and the **prepuce acts as an occlusive dressing**. The rule is low potency and a short course; lichen sclerosus (LS) is the exception.
- Narrow-band UVB or the 308-nm excimer treats psoriasis and vitiligo. **PUVA causes genital squamous cell carcinoma (SCC)**, so it is avoided.

3 · Eczematous dermatoses

Exclude Candida, herpes, impetigo and drug reaction first.

Atopic dermatitis (AD)

- AD is relapsing, flexural and intensely pruritic; **90%** present before 5 years. Use emollients, a short topical steroid, calcineurin inhibitors and a sedating antihistamine; severe disease needs **dupilumab**.

Contact dermatitis

Feature	Irritant (ICD)	Allergic (ACD)
Mechanism	Direct cytotoxicity; dose-driven	Type IV hypersensitivity
Share	~**80%**	Minority; genital ACD is rare
Agents	Soaps, solvents, acids/alkalis	**Nickel** (commonest), neomycin, fragrances, latex, topical steroids
Picture	Burning, limited to the exposed zone	Demarcated, pruritic, weeping

Feature	Irritant (ICD)	Allergic (ACD)
Confirm	Exposure history	**Patch testing**
Treat	Avoidance, barrier	Remove allergen. If severe, a **3-week** prednisone taper (a short course rebounds)

Seborrhoeic dermatitis (SD)

- SD gives greasy-scaled plaques in sebaceous areas. It affects up to **83%** of patients with AIDS, so **extensive SD should prompt an HIV test**. A topical **antifungal** (ketoconazole) is the backbone.
- Lichen simplex chronicus (lichenification, in the LS differential) needs the itch–scratch cycle broken with a short potent steroid.

4 · Papulosquamous disorders

Tinea, erythrasma, secondary syphilis and neoplasia (Bowen disease, extramammary Paget disease (EPD), mycosis fungoides) mimic psoriasis, so "psoriasis" failing treatment needs KOH, syphilis serology and biopsy.

Psoriasis

- Psoriasis affects ~2% and forms sharp plaques with silvery scale; check nails, scalp, **umbilicus** and natal cleft. **Inverse psoriasis** loses its scale in folds, so use KOH to exclude tinea or Candida.

Step	Genital psoriasis treatment
First-line	Low-potency steroid (hydrocortisone butyrate 0.1%) continuously for **≤2 weeks**; triamcinolone 0.1% for thick plaque
Steroid-sparing	Calcipotriene, calcineurin inhibitors
Localised	308-nm excimer laser
Contraindicated	**PUVA** (dose-dependent genital SCC)
Extensive	Methotrexate, cyclosporine, apremilast, anti-TNF, ustekinumab, **ixekizumab (anti-IL-17), which has the best genital data**

Reactive arthritis (formerly Reiter)

- Reactive arthritis follows urethritis or enteric infection in **HLA-B27**-positive people. **Balanitis circinata** looks like psoriasis, so diagnose from **context**; it is self-limiting and needs only a low-potency steroid.

Lichen planus (LP)

- LP forms **polygonal, violaceous, flat-topped papules**; glans papules are white and reticular. Papular LP usually resolves within a year.
- **Erosive mucosal LP persists, scars and carries an SCC risk**, so review regularly and biopsy ulcers. Use an **ultrapotent** steroid (clobetasol 0.05%).

5 · Lichen sclerosus (LS) / BXO

SAQ Lichen sclerosus (BXO, balanitis xerotica obliterans)

Paper answer · P1

Asked as:

- Lichen sclerosis. — *Summer 2020 Paper I Q3c (SAQ)*
- BXO — *Practice Set 10 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)
(also answered in 2020-Summer Paper 1, Q3c)

Taken as lichen sclerosus (LS), formerly balanitis xerotica obliterans (BXO). BXO (balanitis xerotica obliterans) is taken as genital lichen sclerosus (LS) in the male.

Definition

- Chronic, progressive, **scarring inflammatory disease of genital skin**; possibly premalignant. BXO is its late stage in men.

Epidemiology

- 6–10 times commoner in females (prepubertal and menopausal).
- Men: peak 30–50 years in one chapter, older men in another; also children (boys with phimosis). Commonly found at circumcision for phimosis after infancy. 15–20% have extragenital disease.

Aetiology

- Cause unknown. Autoimmune links (thyroid disease, vitiligo, alopecia); familial clustering.
- **Chronic occlusion by urine** under the foreskin (dribbling, obesity) definitely increases risk.
- Infection and trauma are proposed but unproven.

Clinical features

- White, atrophic, sclerotic plaques with loss of architecture; purpura, fissures; itch, pain, painful erection, poor stream, spraying.
- Sequence: prepuce → **phimosis** → glans → **meatal stenosis** (sentinel site) → fossa navicularis and penile urethral stricture.
- **Spares the proximal (bulbar and posterior) urethra.**
- Buried penis from loss of shaft skin.
- **Mechanism of stricture:** voiding against meatal stenosis forces urine into the glands of Littre → inflammation → spongiofibrosis.

Evaluation

- Clinical diagnosis. **Biopsy** if doubtful or suspicious: hyperkeratosis, epidermal atrophy, basal vacuolation, dermal oedema with homogenised collagen, band-like lymphocytic infiltrate (lymphocytes).
- Uroflowmetry; **RUG (retrograde urethrogram) before meatal surgery**, because a proximal stricture often coexists (may coexist).
- **Differential:** lichen planus, Zoon balanitis, penile intraepithelial neoplasia.

Management

Situation	Treatment
Skin disease	**Clobetasol propionate 0.05%** daily for 1–3 months (another source: for about 3 months), then taper to twice-weekly maintenance; calcineurin inhibitors second line; retinoids for refractory disease
Phimosis	**Circumcision**, usually curative

Situation	Treatment
Meatal stenosis	Dilatation with clobetasol; extended meatotomy; BMG (buccal mucosal graft) meatoplasty
Anterior stricture	**Oral mucosal graft, never genital skin** (LS recurs in skin); staged or single stage
Panurethral or unfit	**Perineal urethrostomy** (easy because the proximal urethra is spared); self-dilatation

- Recurrence 40–50% after medical treatment (recurrence is common); restricture higher than in other aetiologies (restricture rates exceed other aetiologies); treatment follows the patient's goals.

Malignancy

- Penile SCC (squamous cell carcinoma) in **2.3–9%** on long follow-up; HPV (human papillomavirus)-independent type.
- **Lifelong follow-up**; biopsy any non-healing ulcer or nodule.

Diagram to draw: penis with LS at (showing) prepuce, glans, meatus and fossa involved, proximal urethra spared.

> **Key points to remember:**
> - Genital skin disease; meatus is the sentinel site.
> - Spares proximal urethra; perineal urethrostomy easy.
> - Clobetasol first; oral mucosa, never skin, for repair.
> - SCC in 2.3–9%: lifelong follow-up.

Also, from the condensed book (not in the paper answer):

- LS is 6–10× commoner in females, and affected men are usually >60; **15–20%** have extragenital disease, and in men the **perianal skin is spared**.
- Circumcision is usually curative, but LS may recur in the scar.

ALGORITHM

1. Biopsy.
2. Give a **potent steroid (clobetasol 0.05%) for 3 months**, then taper.
3. Continue twice-weekly maintenance (or mometasone 0.1% or a calcineurin inhibitor).
4. Lifelong surveillance of non-healing fissures, ulcers, nodules.

- Recurrence is **40–50%**, so maintain proactively. Meatal disease needs meatotomy or meatoplasty.

Disease	Steroid potency	Duration
AD / psoriasis / reactive arthritis	Low	Short (psoriasis ≤2 weeks)
Seborrhoeic dermatitis	Low	Initial severe phase only
Lichen planus	Ultrapotent	Until controlled
Lichen sclerosus	Potent	3 months, then maintenance

KEY POINT

Steroid strength tracks the scarring and malignant stakes of the disease. LS is the single exception to the short-course rule.

6 · Vesiculobullous and drug eruptions

Feature	Erythema multiforme (EM) minor	Stevens–Johnson syndrome (SJS)	Fixed drug eruption (FDE)
Trigger	**HSV**	**Drugs**: sulfonamides, NSAIDs, β-lactams, anticonvulsants	Sulfonamides, NSAIDs, tetracyclines, carbamazepine
Lesion	Symmetric **targets**; palms, soles, mouth	Blisters and necrosis; genitals involved in ~**60%**	Solitary violaceous plaque, often on the glans
Course	Self-limited, recurs	Mortality up to ~**30%**	**Recurs at the same site within 24 h** of re-exposure
Sequelae	None	Meatal stenosis, strictures	Brown pigmentation
Treatment	**Suppressive aciclovir prevents it**	See algorithm	Stop the drug

ALGORITHM

1. **SJS**
2. Stop the offending drug.
3. Give burn-type supportive care in an intensive care or burn unit (lowers mortality).
4. Limited data support cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis or IVIG; systemic steroids are controversial.

KEY POINT

One drug class can cause a trivial FDE or a lethal SJS; extent and systemic illness separate them. Genital FDE mimics herpes; "same spot each time" is the clue.

Autoimmune blistering disease needs perilesional biopsy with direct immunofluorescence.

7 · Noninfectious genital ulcers

A genital ulcer is infectious until proven otherwise (see Ch 29).

Behçet disease (BD)

- BD is an **HLA-B51**-linked vasculitis, **more severe in males**, with oral and genital ulcers, uveitis, epididymitis and **pulmonary artery aneurysm**. Oral ulcer is most sensitive; **genital ulcer is most specific**. No confirmatory test exists. Treat with colchicine, steroids or **anti-TNF**.

International Criteria for BD (2014) — ≥4 points	Points
Ocular lesions; oral aphthosis; genital aphthosis	2 each
Skin lesions; central nervous system; vascular; positive pathergy (optional)	1 each

Pyoderma gangrenosum (PG)

- PG accompanies inflammatory bowel disease and myeloproliferative disease as a painful ulcer with an **undermined violaceous border**. Biopsy the edge, treat with steroids, and avoid debridement.

8 · Infections

Balanitis and balanoposthitis

ALGORITHM

1. Remove irritants and improve hygiene.
2. Give a topical antibiotic or antifungal, then a short low-potency steroid.
3. If this fails, then **biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV).
4. If recurrent with phimosis, then **circumcision**.

Type	Clue	Management
Candidal	Satellite lesions; diabetes	Topical imidazole; check glucose
Irritant/allergic	Products, soaps	Avoidance
Circinate	Urethritis, HLA-B27	Low-potency steroid
Zoon (plasma cell)	Shiny red plaque, elderly uncircumcised	Biopsy, then circumcision
LS / BXO	White sclerotic glans, phimosis	Potent steroid; circumcision
Pseudoepitheliomatous keratotic micaceous balanitis (PEKMB)	Mica-like scaly plaque	Biopsy; excision

Soft-tissue infection

- Folliculitis needs hygiene. A furuncle needs incision and drainage with MRSA cover. Erysipelas has a sharp border and cellulitis an indistinct one; give antibiotics and **mark the border**.
- Fournier gangrene is a polymicrobial fascial infection with pain out of proportion, grey skin and crepitus; it needs emergency debridement (see Ch 26).

> **KEY POINT**
>
> Fournier gangrene is fascial, so skin shows a fraction of the necrosis; testes are spared. Imaging must never delay surgery. Debride to bleeding tissue, with a second look at 24–48 h.

- **Ecthyma gangrenosum** (P. aeruginosa bacteraemia in the immunocompromised) gives black eschar and needs immediate IV antipseudomonals.
- A **genital bite** needs amoxicillin–clavulanate 875/125 mg bd, irrigation and debridement.

Hidradenitis suppurativa (HS)

- HS is a **follicle-centred** disease of apocrine-bearing skin that causes abscesses, **sinus tracts** and scars, with fistulae and **SCC in chronic scars** as complications.

Hurley stage	Features	Mainstay
I	Abscess(es), no tracts or scarring	Topical clindamycin, deroofing
II	Recurrent abscesses, tracts, scarring; separated areas	Oral tetracycline, or clindamycin and rifampicin; deroofing; adalimumab if refractory
III	Diffuse interconnected tracts	Adalimumab and **wide excision**, graft or flap

- Advise stopping smoking and weight loss; **adalimumab** is given 40 mg weekly. **Repeated incision and drainage is discouraged** because the tract recurs.

Red groin rash

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Scrotum	**Involved**	**Spared**	—	May be involved
Look	**Satellite lesions**	Scaly advancing border, central clearing	Red-brown patch	Non-scaling plaque
Test	KOH: yeast	KOH: hyphae	Wood's lamp: **coral-red**	KOH negative
Treat	Topical imidazole **≥2 weeks**; test for diabetes	Topical azole or allylamine; terbinafine 250 mg/day if deep; no steroids	Clindamycin 1% or miconazole; oral erythromycin if severe	Low-potency steroid

> **KEY POINT**
>
> Azoles cover yeast and dermatophyte, so use them when unsure. Nystatin misses tinea; allylamines are weaker on Candida. A negative KOH from an untreated lesion means rethink the diagnosis.

Infestations

- Pubic lice and scabies both get permethrin 5%, **repeated at 1 week**; for scabies treat the whole body and the household. Ivermectin 200 µg/kg ×2 is the alternative, and lindane is disfavoured. Screen lice patients for other STDs.

9 · Premalignant and malignant lesions

SAQ Premalignant conditions in urology (including premalignant lesions of penis)

Paper answer · P1

Asked as:

- Premalignant conditions in urology — *Summer 2021 Paper I Q3a (SAQ)*
- Premalignant lesion of penis — *Practice Set 6 Paper I Q3a (SAQ)*

Type: SAQ (10 marks)

(premalignant conditions also answered in 2021-Summer Paper 1, Q3a)

Definition

- Altered tissue with a raised risk of invasive cancer, often **intraepithelial neoplasia with an intact basement membrane**. Treating it prevents cancer.
- Penis: a lesion with raised risk of invasive SCC (squamous cell carcinoma). The key entity is **PeIN (penile intraepithelial neoplasia)**: dysplastic squamous epithelium with an **intact basement membrane**.

Penis

Classification

1. PeIN (HPV-associated or HPV-independent)

- **HPV (human papillomavirus)-associated** (commonest; basaloid and warty) or differentiated (HPV-independent).
- **Mechanism (HPV):** viral E6 inactivates p53, E7 inactivates Rb (retinoblastoma protein); **p16** is the marker.
- **HPV-independent (differentiated):** arises in lichen sclerosus.
- **WHO (World Health Organization) 2022:** all PeIN is high grade; grades I–III abandoned.

Clinical entity / lesion	Site, appearance / key points	Risk of invasion
Erythroplasia of Queyrat	Glans or prepuce; red velvety (glans) plaque	**Progresses in 10–33%**
Bowen disease	Shaft; scaly red (shaft) plaque	About 5% progress
Bowenoid papulosis	Shaft papules, young men; HPV 16	Does not progress; may regress

2. Other lesions

- **Lichen sclerosus (BXO [balanitis xerotica obliterans]):** SCC (squamous cell carcinoma) in **2.3–9%** (HPV-independent type).
- **Buschke–Löwenstein tumour** (giant condyloma; verrucous carcinoma): HPV 6/11; locally invasive, locally destructive, rarely metastasises; may harbour SCC.
- **Also know:** cutaneous horn; PEKMB (pseudoepitheliomatous keratotic and micaceous balanitis, controversial); condyloma acuminatum; (chronic) phimosis; genital lichen planus; Kaposi sarcoma (HHV-8 [human herpesvirus 8], check HIV [human immunodeficiency virus]).

Evaluation

- Biopsy first: **multiple biopsies of adequate depth** to exclude invasion; p16 staining. Examine inguinal nodes; partner screening for cervical neoplasia.

Treatment (organ-preserving)

Situation	Treatment
Prepuce (preputial lesions)	Circumcision or excision with clear margin
Glans	**Topical 5-FU (5-fluorouracil) or 5% imiquimod**; **laser** (Nd:YAG [neodymium:yttrium-aluminium-garnet] preferred; CO_2 too superficial); **glans resurfacing** with skin graft
Shaft	Excision; Mohs surgery

Situation	Treatment
Lichen sclerosus	Clobetasol 0.05% for 3 months then maintenance; circumcision; lifelong follow-up
Buschke–Löwenstein	Wide excision; avoid radiotherapy (anaplastic change)

- Long-term follow-up (recurrence common after organ-preserving treatment). **Prevention:** neonatal circumcision, HPV vaccination, avoiding smoking.

Bladder

- **CIS** (carcinoma in situ): flat high-grade lesion, umbrella cells lost; untreated, **40–83%** invade muscle. Treat with BCG (bacillus Calmette–Guérin).
- **Urothelial dysplasia:** 14–19% progress; 60% to CIS with prior cancer.
- **Cystitis glandularis** (intestinal type): low risk of adenocarcinoma; resect and survey.
- **Leukoplakia** (keratinising squamous metaplasia): traditionally premalignant; Campbell now says **not** (no TP53 changes). Schistosomiasis and chronic catheters predispose to SCC.

Prostate

- **HGPIN** (high-grade prostatic intraepithelial neoplasia): precursor; isolated HGPIN no longer needs repeat biopsy. IDC-P (intraductal carcinoma of the prostate) usually means invasive cancer nearby.

Testis

- **GCNIS** (germ cell neoplasia in situ): precursor of all postpubertal GCTs (germ cell tumours) except spermatocytic tumour; **50% invasive within 5 years**. Treat by orchidectomy or **radiotherapy ≥20 Gy**.
- **Gonadoblastoma** in dysgenetic gonads with Y material; cryptorchidism.

Kidney and others

- **Also know:** papillary adenoma; acquired cystic kidney disease on dialysis; urethral diverticulum; bowel in augmentation.

Diagram to draw: normal urothelium vs dysplasia vs CIS (umbrella cells lost); penile epithelium with full-thickness atypia above an intact basement membrane.

> **Key points to remember:**
> - PeIN: intact basement membrane; WHO 2022 all high grade.
> - Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
> - Lichen sclerosus: SCC 2.3–9%.
> - Untreated bladder CIS: 40–83% invade.
> - GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

Also, from the condensed book (not in the paper answer):

- Penile intraepithelial neoplasia is HPV-related (type 16) or non-HPV (LS, chronic LP); biopsy **several areas**.
- Verrucous carcinoma needs local excision or Mohs surgery, not primary radiotherapy (anaplastic transformation).

Lesion	Treatment
SCC in situ (SCCis) — accessible (scrotum)	Excision with a **5-mm margin** (favoured)
SCCis — glans	Mohs surgery, laser, cryotherapy; topical 5-fluorouracil (5-FU) or imiquimod (higher recurrence)
Bowenoid papulosis	Observe in a reliable young man; or 5-FU, imiquimod, cryotherapy, laser
PEKMB	Biopsy (exclude verrucous carcinoma); excision or ablation

Tumour	Key features	Treatment
Basal cell carcinoma	Rare; pearly papule	Excision or Mohs
Kaposi sarcoma	HHV-8; classic, AIDS and transplant types; a glans lesion may block the meatus	Restore immunity; excision, laser or vinblastine if solitary; radiotherapy 15–30 Gy if regional; chemotherapy if disseminated
Melanoma	<1% of penile cancers	Wide excision or partial penectomy; immunotherapy if advanced
Extramammary Paget	Red plaque at apocrine sites; CEA and EMA positive; underlying carcinoma in **10–30%** (urethra, prostate, bladder, rectum)	**Mandatory search for internal malignancy**; excision or Mohs
Mycosis fungoides	Patches progress to plaques, then tumours; Sézary syndrome if in blood	Ultrapotent steroid, phototherapy

10 · Benign male genital lesions

Lesion	Site / look	Management
Pearly penile papules	Coronal rows of angiofibromas in 14–48% of young men, mistaken for condyloma	Reassure
Fordyce spots	Pin-sized shaft papules (ectopic sebaceous glands)	Reassure
Angiokeratoma of Fordyce	Red-purple scrotal papules; may bleed	Observe; laser if bleeding
Zoon balanitis	Shiny red glans plaque; elderly uncircumcised men	**Biopsy**, then **circumcision (curative)**
Median raphe cyst	Ventral, near the glans	Excision

> **KEY POINT**
>
> Zoon balanitis and erythroplasia of Queyrat are both red glans plaques in uncircumcised men: one benign and cured by circumcision, the other SCCis. A persistent red glans plaque always needs biopsy.

11 · Pigmentary disorders

- Vitiligo is accentuated by Wood's lamp; unlike LS, it has normal texture without atrophy or scarring.

31 Tuberculosis and Parasitic Infections of the Genitourinary Tract

1 · Urogenital tuberculosis — organism and pathogenesis

TB seeds the urinary tract late and heals by scarring.

LAQ Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder

Paper answer · P2 · P3 · merged from 2

Asked as: P3 LAQ "Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management" (Describe etiopathogenisis, clinical presentation, diagnosis and management of Renal TB?, Summer 2022 Paper III Q2 LAQ; Laboratory Diagnosis of Genitourinary Tuberculosis, Winter 2023 Paper III Q3f SAQ; Describe newer diagnostic modalities for Gu Kochs. Briefly outline role of all present diagnostic tests and give diagnostic algorithm, Practice Set 6 Paper III Q3d SAQ) · P2 LAQ "Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male."

Assumed meaning: "Gu Kochs" = genitourinary tuberculosis.

Introduction

- Caused by the MTBC (*Mycobacterium tuberculosis* complex); *M. bovis* (and BCG (bacille Calmette–Guérin)) is innately resistant to pyrazinamide.
- UG (urogenital) TB (tuberculosis) is 15–20% of pulmonary TB in developing countries; the second commonest extrapulmonary site after lymph nodes. **Kidney is involved in 80%.**
- Untreated, it destroys the kidney and contracts the bladder.

Etiopathogenesis

- **Primary infection:** inhaled bacilli spread to hilar nodes, then by lymph and blood to the well-perfused kidney and epididymis, before immunity develops (up to 12 weeks).
- **Granuloma:** macrophages, epithelioid cells and Langhans giant cells in a lymphocyte cuff, maintained by IFN-γ (interferon-γ). In 90–95% it contains the infection and it stays latent; viable bacilli stay dormant.
- **Reactivation** (lifetime 5–15%), even decades later: old age, diabetes, renal failure, HIV (human immunodeficiency virus), malnutrition, steroids or other immunosuppression. **Caseation** then releases bacilli into urine.

Route	Lesion
Haematogenous (principal)	Parenchymal disease: kidney, epididymis, prostate
Descending urinary	Luminal disease: ureteric stricture, cystitis, urethra
Ascending (intravesical BCG, about 1%)	Cystitis, prostatitis, epididymo-orchitis
Contiguous	From psoas, bowel

- **Renal sequence:** 1. Bacilli lodge in **periglomerular capillaries**. 2. Cortical granulomas coalesce. 3. **Caseation**; cavities open into calyces. 4. Papillary necrosis, abscess, calcification. 5. Bacilluria seeds ureter and bladder.
- **Autonephrectomy:** caseo-cavernous type, or fibrotic calcified "putty" kidney.

Site	Frequency	Pathology
Kidney	80%	Periglomerular granulomas, then caseating cavities, papillary necrosis, calcification; **autonephrectomy** ("putty kidney")
Ureter	—	Mural granulomas; stricture, commonest at the **VUJ (vesicoureteric junction)**; "beaded corkscrew" or "pipestem" ureter
Bladder	—	Cystitis starts around the ureteric orifices ("golf-hole" orifice); ulcers; after about a year of fibrosis it contracts (<100 mL); **thimble bladder <20 mL**
Epididymis	Commonest genital site	Starts in globus minor; beaded vas, posterior scrotal sinus
Prostate, seminal vesicles	22–49%	Calcification, abscess; infertility

- **Key concept:** healing occurs by fibrosis, and fibrosis obstructs. Infundibular and ureteric strictures and bladder contraction can **worsen during treatment**.
- **Also know:** granulomatous interstitial nephritis; miliary renal TB; amyloidosis; squamous metaplasia of the pelvis (squamous cell carcinoma risk); ESRD (end-stage renal disease) in about 7%.

Clinical presentation

- Renal TB is silent until it reaches the bladder; asymptomatic in about 8%.
- **Storage LUTS (lower urinary tract symptoms)** and dysuria (commonest); haematuria and flank pain (about one-third); colic <10% (sloughed papillae, clot, debris).
- Constitutional symptoms <20%. Recurrent "UTI (urinary tract infection)" not responding to antibiotics.
- Signs: renal angle tenderness, flank sinus, beaded vas, epididymal nodule; hypertension; renal failure if bilateral.

Diagnosis: principles and specimen

- GUTB (genitourinary tuberculosis) is paucibacillary with intermittent shedding, so **multiple samples** are needed.
- **Hierarchy of proof:** positive culture or DNA, then caseating granulomas on histology, then clinical picture with response to treatment.
- **Specimen:** early-morning first-void urine (EMU), **3–5 samples on consecutive days**, processed at once (acidic urine kills bacilli). Avoid prior fluoroquinolones.
- Also tissue, pus, FNAC (fine-needle aspiration cytology) or semen.

Laboratory tests

Test	Method	Comment / role
Urine routine	Microscopy	**Sterile (acid) pyuria** with or without microscopic haematuria in >90%
ZN (Ziehl–Neelsen) smear	Carbol fuchsin, acid-alcohol, methylene blue: red bacilli. Auramine fluorescence is more sensitive	Quick; sensitivity under 50%; saprophytic mycobacteria give false positives

Test	Method	Comment / role
Urine AFB (acid-fast bacilli) culture: gold standard	Decontaminate and concentrate. **LJ (Löwenstein–Jensen)** solid medium: 4–6 weeks. **MGIT (mycobacteria growth indicator tube)** liquid: about 10 days; use a liquid system as well	80–90% sensitivity with ideal technique; diagnosis; gives DST (drug-susceptibility testing); **the only test that monitors treatment**
NAAT (nucleic acid amplification test): GeneXpert MTB/RIF	Automated cartridge PCR (polymerase chain reaction); about 2 hours (result in 1–2 days)	Rapid diagnosis; detects MTB and **rifampicin resistance** (MDR (multidrug-resistant) TB surrogate). **Xpert Ultra** is more sensitive. Stays positive after cure, so cannot monitor treatment
TrueNat	Indian chip-based NAAT	NTEP (National TB Elimination Programme)
LPA (line probe assay)	DNA strip hybridisation	Rifampicin, isoniazid or second-line drug resistance
LF-LAM (lateral-flow lipoarabinomannan)	Urine antigen strip, under 30 minutes; point-of-care	HIV with symptoms, CD4 <200 or advanced disease (WHO (World Health Organization) 2019)
Histopathology	Biopsy (bladder, epididymis)	Caseating granulomas; NAAT on tissue raises yield
TST (tuberculin skin test)	5 TU (tuberculin units) PPD intradermal; read induration at 48–72 h	Proves infection only, not active disease; BCG false positive
IGRA (interferon-γ release assay)	ESAT-6/CFP-10 blood test (QuantiFERON, T-SPOT)	Unaffected by BCG; cannot separate latent and active; shows infection, not active disease

Newer diagnostic modalities

- **Liquid culture (MGIT):** about 10 days versus 4–6 weeks on LJ.
- **NAAT:** GeneXpert MTB/RIF (Xpert Ultra more sensitive), about 2 hours, detects rifampicin resistance; TrueNat; **LPA** for isoniazid resistance.
- **Urine LF-LAM:** strip, under 30 minutes; WHO 2019.
- **IGRA:** unaffected by BCG.
- **Imaging:** CT (computed tomography) urography; **MR (magnetic resonance) urography with DWI (diffusion-weighted imaging)** grades fibrosis.

Supportive tests

- ESR, blood count, renal function, HIV, blood sugar.
- Sputum and **chest radiograph in all** (mandatory; lung TB coexists in up to one-fifth).

Imaging and endoscopy

Imaging	Findings / role
KUB (kidney–ureter–bladder) film	Calcification (24–44%); lobar "putty" kidney

Imaging	Findings / role
IVU (intravenous urography) / CT urography	**Moth-eaten calyx**, **phantom calyx** (infundibular stenosis), cavities, hydrocalycosis, **Kerr's kink**, pipestem or beaded corkscrew ureter, ureteric strictures, calcification, small bladder; IVU replaced by CT
Ultrasound	Children, pregnancy, scrotum; follows hydronephrosis during treatment
MRU (magnetic resonance urography)	Children, pregnancy
Cystoscopy and biopsy	**"Golf-hole" ureteric orifice**, ulcers; biopsy if culture negative (yield 19–52%)
Chest radiograph	Mandatory (lung TB in up to one-fifth)

Diagnostic algorithm

```
Suspicion (sterile pyuria, storage symptoms, epididymal nodule) -> 3-5 EMU: ZN + culture (LJ + MGIT) + Xpert -> chest X-ray; CT urography Positive -> treat; DST guides regimen Negative -> cystoscopy/biopsy, aspiration + NAAT on tissue Still negative -> empirical trial if clinically compatible
```

Medical management (first step in all)

- **Multidrug therapy** because bacilli exist in different metabolic states and single drugs select resistance.
- Baseline liver and renal function, HIV, hepatitis, diabetes; DOT (directly observed therapy).
- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months, with **pyridoxine** against isoniazid neuropathy. India's NTEP uses 2HRZE/4HRE.
- Complicated disease: some treat 12–24 months, or ≥6 months after surgery (P2: some extend to 9–12 months). The new 4-month rifapentine–moxifloxacin regimen is not used for UG TB.
- Toxicity: hepatitis (all except ethambutol); optic neuritis (ethambutol).
- **Monitor with ultrasound** for worsening obstruction; stent or PCN (percutaneous nephrostomy) early. Steroids for strictures: anecdotal only.
- MDR-TB: second-line drugs (bedaquiline, pretomanid, linezolid).

Surgical management

- About half need surgery, **after at least 4–6 weeks of ATT (antituberculous therapy)**, when inflammation and bacillary load have settled.

Problem	Procedure
Obstruction	Early **DJ (double-J) stent** (succeeds in 41%) or **PCN**
Localised polar lesion not responding	Partial nephrectomy
Non-functioning kidney + persistent disease or hypertension	**Nephrectomy** (laparoscopic preferred; ligate artery and vein separately)
PUJ (pelviureteric junction) stricture	Pyeloplasty; ureterocalicostomy
Lower ureteric stricture	Reimplantation, psoas hitch, Boari flap; ileal ureter if long
Contracted bladder	Augmentation; substitution if <20 mL

- A stricture not improving after 6 weeks of drugs needs open repair.
- Contested: nephrectomy of a non-functioning kidney without active disease.

Thimble bladder in a 30-year-old man: assessment

Assess	How	Why
Disease activity	Repeat urine AFB culture; bladder biopsy	Do not reconstruct with active TB
Capacity and compliance	Cystogram, cystoscopy under anaesthesia, urodynamics	Size and type of reconstruction
Upper tracts	Ultrasound, CT urography, MCU (micturating cystourethrogram)	VUJ stricture or reflux needs reimplantation
Renal function	Creatinine, GFR (glomerular filtration rate), split function	Bowel reservoir needs creatinine <2 mg/dL, GFR >35–40 mL/min
Urethra, bladder neck, sphincter	Urethrogram, endoscopy	Orthotopic reservoir needs a patent, continent outlet
Patient	Dexterity, motivation, fertility wishes	May need CIC (clean intermittent catheterisation)

Thimble bladder: surgical options

Bladder state	Operation
Half the bladder diseased, capacity <100 mL	**Augmentation ileocystoplasty** ("clam" or onlay of detubularised ileum)
Severely contracted	Augmentation with **ileocaecum** or sigmoid
Thimble bladder (<20 mL)	**Substitution cystoplasty:** supratrigonal excision of the diseased bladder, keeping trigone and bladder neck, and an orthotopic detubularised ileal or ileocaecal reservoir

- **Principles:** detubularise and fold the bowel into a spherical, low-pressure reservoir (Laplace's law; interrupts peristalsis). Anastomose widely to the trigone.
- **Ureters:** if the VUJ is strictured or refluxing, excise to healthy ureter and **reimplant into the reservoir**: an antireflux tunnel under a colonic taenia, or an isoperistaltic afferent ileal limb. A Boari flap or psoas hitch is not possible in a contracted bladder.
- **Avoid orthotopic reconstruction** with poor renal function, a strictured or incompetent urethra, inability to perform CIC, active TB, or short bowel or inflammatory bowel disease. Then use a continent catheterisable pouch (Mitrofanoff) or an **ileal conduit**.
- **Also know:** laparoscopic and robotic augmentation; transurethral incision of bladder neck contracture.

Complications of bladder reconstruction

- **Mucus** and catheter blockage; **stones**; bacteriuria and UTI; **hyperchloraemic metabolic acidosis**; vitamin B12 deficiency if ileum is lost; **perforation** (needs regular emptying); incomplete emptying needing CIC; nocturnal incontinence; late tumour in the reservoir.

Follow-up

- Relapse 6.3–22% (mean about 5 years), so follow up with urine AFB culture and imaging for up to 10 years.
- After reconstruction: also creatinine, bicarbonate, B12 and ultrasound.

Diagrams to draw:

- Kidney showing caseous cavity, phantom calyx and VUJ stricture.
- Stepwise diagnostic flowchart (algorithm above) from sterile pyuria to culture and DST.
- Supratrigonal excision of a thimble bladder with an ileocaecal substitute and ureteric reimplantation.

Key points to remember:

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

Also, from the condensed book (not in the paper answer):

- UG TB is **15–20%** of pulmonary TB in developing countries. Infection stays latent in 90–95%; lifetime reactivation is **5–15%**, so UG TB often appears without active lung disease.
- **M. bovis (and bacille Calmette–Guérin, BCG) is innately pyrazinamide-resistant**.
- Isolated urogenital (UG) TB is generally **not infectious**, except during aerosol-generating procedures.
- Ascending spread: intravesical BCG causes UG TB in **~1%**.

KEY POINT

UG TB is blood-borne and lands in the two best-perfused UG organs, kidney and epididymis. Blood-borne seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction.

2 · Clinical features and pathology by site

- It is indolent, and patients are repeatedly treated for urinary tract infection (UTI). Constitutional symptoms occur in **<20%**. **Lower urinary tract symptoms dominate**; haematuria or flank pain occurs in ~⅓. **The hallmark is sterile pyuria** (>90%).

Site	Frequency / route	Signature lesion
Kidney	**80%**, haematogenous	Caseating cavities, then autonephrectomy (up to ⅓); "putty" kidney; end-stage renal failure ~7%
Ureter	Descending	Stricture at the vesicoureteric junction (VUJ); beaded corkscrew ureter. **Stricture obstruction is an important cause of renal failure**
Bladder	Descending	Starts at orifices; trigone spared. After ~1 year <100 mL; "thimble" <20 mL

Site	Frequency / route	Signature lesion
Epididymis	10–55%; second commonest; bilateral in ~⅓	Starts in the **globus minor (tail)**; posterior scrotal sinus in up to half
Vas	Contiguous	Thickened, beaded
Testis	From epididymis	Hard tumour-like mass
Prostate	22–49%	Calcified hard gland. "Prostatitis" responds partly to quinolones, then relapses
Seminal vesicles	Via vas or ejaculatory ducts	Low-volume ejaculate, azoospermia, haemospermia; infertility may be the first symptom
Urethra	1.9–4.5%	With prostatic disease; urethroscrotal fistula
Penis	Exceedingly rare	Ulcer; papulonecrotic tuberculid

3 · Diagnosis

*See the paper answer **Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder** (Section 3, Ch 31).*

Also, from the condensed book (not in the paper answer):

- KUB calcification progresses from punctate to globular to ring, and IVU may show cloverleaf calyces.

ALGORITHM

1. **Hierarchy of proof**
2. If culture or MTBC DNA is positive, then the diagnosis is made.
3. If negative, then caseating granulomas on tissue in the right context confirm it.
4. If still negative, then a compatible picture, exposure and response to empirical treatment suffice.
5. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.

- Culture sensitivity is 10–90% because shedding is intermittent. The Ziehl–Neelsen (ZN) smear detects **<50%**. Acidic urine inhibits mycobacteria, so culture promptly.
- **GeneXpert** detects rifampicin resistance, a surrogate for multidrug-resistant (MDR) TB. The WHO uses it in the initial work-up of all TB, but a negative result does not exclude TB, and it **cannot monitor treatment** because dead bacilli shed DNA.
- Fine-needle aspiration cytology of an epididymal nodule is diagnostic in ~⅔. At cystoscopy ulcers mimic cancer, and a **"golf-hole" orifice** means imaging the upper tract.
- Tuberculin skin test (**TST**) cut-offs are **≥5 mm** for contacts and the immunosuppressed, **≥10 mm** for immigrants, high-risk workers and drug injectors, and **≥15 mm** with no risk factors. The interferon-γ release assay (**IGRA**) has no BCG false positives and is preferred for people from endemic countries. Both show exposure only.

Modality	Signs / role
Plain film (KUB)	Calcification in 24–44%; **lobar "putty" kidney** is pathognomonic of end-stage disease; ureteric calcification is **intraluminal** (schistosomiasis: mural)

Modality	Signs / role
Intravenous urography (IVU)	**Moth-eaten calyx**; **phantom calyx**; **Kerr's kink**; **pipe-stem ureter**; contracted bladder. Findings are mostly **obstructive**
CT urography	Standard; granuloma may mimic renal cell carcinoma
Ultrasound (US)	**Serial monitoring of hydronephrosis during treatment**
MRI	Diffusion-weighted imaging separates pyonephrosis from hydronephrosis

KEY POINT

Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). So obstruction is the commonest finding and must be watched during treatment.

4 · Medical treatment

- Combine drugs to prevent resistance. Check baseline liver function, renal function and HIV status, and use directly observed therapy (DOT).

ALGORITHM

1. **Standard 6-month short course (2HRZE / 4HR)**
2. Intensive, 2 months: daily INH (H), rifampicin (R), pyrazinamide (Z) and ethambutol (E); stop E once fully susceptible.
3. Continuation, 4 months: HR daily (or thrice weekly by DOT).
4. Give pyridoxine throughout.

Drug	Daily adult dose	Signature toxicity	Monitor / note
INH	5 mg/kg (300 mg)	Neuropathy, hepatitis	Pyridoxine 25–50 mg
Rifampicin	10 mg/kg (600 mg)	Hepatitis, **drug interactions**	Antiretroviral and immunosuppressant levels; rifabutin with antiretrovirals
Pyrazinamide	Weight-banded 1000–2000 mg	Hyperuricaemia, hepatitis	M. bovis is resistant
Ethambutol	Weight-banded 800–1600 mg	Optic neuritis (red–green vision)	Check vision; the **only non-hepatotoxic** first-line drug

- Some give 12–24 months for complicated disease, as UG TB relapses more.
- **Monitor** liver function, vision, urine culture and **serial US**, because healing fibrosis can worsen strictures and bladder contraction.
- **Steroids** are anecdotal in UG TB; consider them if a stricture worsens on treatment.

5 · Surgical treatment

*About half need surgery, now mostly **reconstructive**.*

- The goal is organ preservation. Indications are obstruction or abscess, a non-functioning infected kidney, resistant hypertension and reconstruction.
- **Timing: 4–6 weeks after starting chemotherapy**, except in emergencies.

ALGORITHM

1. If uraemia or sepsis, then decompress urgently.
2. Try a retrograde double-J stent first (preferred; succeeds in < half, 41%).
3. If that fails, then try an antegrade stent.
4. If that fails, then percutaneous nephrostomy (PCN); loculated systems may need several.
5. Then repair definitively; if the kidney is unsalvageable, then nephrectomy.

- Removing a PCN without fixing the cause leaves a tuberculous cutaneous fistula.
- **Nephrectomy is established** for a non-functioning kidney with recalcitrant TB or resistant hypertension; **routine removal** of an asymptomatic non-functioning kidney is **contested**. Partial nephrectomy suits a polar lesion unresponsive to drugs.
- At nephrectomy, expect dense fibrosis and **ligate artery and vein separately** (late arteriovenous fistula).
- **Strictures** are commonest in the lower ureter, and endoscopic results are poor. Stent early for strictures arising on treatment and **image every stricture**; if worsening, give steroids, and **if there is no improvement after 6 weeks, operate**.

Site / gap	Procedure	Key point
Pelviureteric junction (PUJ), short	Dismembered pyeloplasty; flap if longer	Harder than congenital PUJ obstruction
PUJ unreconstructable	Ureterocalicostomy (lower pole)	Cover with capsule or omentum
Upper/mid ureter	Ureteroureterostomy; Davis intubated ureterotomy	—
Lower ureter	Excise, then ureteroneocystostomy	Bladder mobilisation gains 2–3 cm
Moderate gap	Psoas hitch	Up to **5 cm**; needs good capacity
Long gap	Boari flap, with or without hitch	**10–15 cm**; poor in a contracted bladder
Multiple / recurrent	**Ileal ureter**	Native ureter unusable

KEY POINT

Trap: in TB the bladder may itself be contracted, ruling out psoas hitch and Boari flap. Assess capacity first; ileal ureter, with or without augmentation, is then the option.

*See the paper answer **Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder** (Section 3, Ch 31).*

Also, from the condensed book (not in the paper answer):

- Contracted bladder: operate for intolerable symptoms or capacity **<100 mL**, once chemotherapy controls disease; a thimble bladder needs orthotopic substitution, and the spared trigone allows supratrigonal substitution.
- **Genital** surgery is only for failed medical therapy: **epididymectomy** if the testis is spared, otherwise orchidectomy. Obstructive azoospermia is often irreversible, so offer sperm retrieval.

6 · BCG-related infection, relapse, special situations

- **BCG UG TB** (~1%) causes granulomatous prostatitis, epididymo-orchitis, renal abscess or sepsis. For organ infection give INH and rifampicin for 3–6 months, since **pyrazinamide is useless**; for sepsis give triple therapy and steroids, and stop BCG permanently.
- **Relapse** occurs in **6.3–22%**, at a mean of ~5 years, so follow up for **10 years**.

Situation	Change	Duration
MDR-TB (INH and R resistant)	Second-line; BPaL (bedaquiline, pretomanid, linezolid), not yet approved for extrapulmonary TB	> 18 months; 6–9 months with BPaL
Extensively drug-resistant (XDR) TB	MDR plus fluoroquinolone and bedaquiline/linezolid resistance; 5–6 drugs	≥ 2 years
Pregnancy	HRE and pyridoxine; Z omitted in the USA (WHO includes it); breastfeeding allowed	9 months (without Z)
HIV on effective antiretroviral therapy (ART)	Standard daily regimen; rifabutin with protease inhibitors	6 months (9 if not on ART)
Renal transplant	Rifamycin with drug-level monitoring, or a rifamycin-free regimen	≥ 18 months if rifamycin-free

- **HIV** raises TB risk 15–20-fold, with more dissemination but **less fibrosis and fewer strictures**.

7 · Schistosomiasis (bilharzia)

Trapped eggs cause obstruction and squamous bladder cancer.

Species	Distribution	Target / venous home	Snail
S. haematobium	Africa, Arabian Peninsula	Vesical/pelvic plexus — UG tract	**Bulinus**
S. mansoni	Africa, Middle East, S America	Mesenteric — liver, gut	Biomphalaria
S. japonicum	China, SE Asia	Mesenteric — liver, gut	Oncomelania

ALGORITHM

1. **Life cycle**
2. Cercariae penetrate intact skin in fresh water and pass via the lungs to the liver, where they pair.
3. Paired adults move to the vesical plexus and live 3–5 years.
4. Eggs (**terminal spine**) cross the bladder wall into urine and hatch as miracidia.
5. Miracidia infect Bulinus, which releases cercariae.

- **Retained eggs cause the disease**; adults do not multiply in humans. Children aged 5–15 carry the heaviest burden.
- The **egg granuloma** forms polypoid masses (active), then fibroses and calcifies into **"sandy patches"**, leading to obstruction, metaplasia and cancer. Obstruction is **distal**; hydroureter may be **tonic** (reversible once relieved) or **atonic** (fibrotic).
- **Salmonella** shelters in the worm, causing chronic bacteriuria; treat both.
- **Bladder cancer** is rated **Group 1** by IARC. Patients are younger (40–50), and **SCC is 60–90%**; mass drug treatment in Egypt shifted SCC to urothelial cancer.

- Acute infection gives swimmer's itch and **Katayama fever**; chronic infection (from 8–12 weeks) gives **painless terminal haematuria**. Men may get haemospermia and infertility.

> **KEY POINT**
>
> Trap: a negative urine egg count does not exclude severe disease; the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients.

Test	Meaning	Limitation
Urine egg microscopy — **gold standard**	Active infection	Collect **09:00–15:00** (noon peak); negative early and in light or inactive disease
Rectal snip (squash prep)	Active or past infection; egg viability	Avoids bladder biopsy risks
Serology	Exposure	Stays positive, so it **never judges cure**
Circulating antigen	Live worms	Urine CCA is poor for S. haematobium
PCR	Most sensitive	Impractical in the field

- Imaging shows wall thickening, polyps and hydronephrosis on US, and a **calcified bladder ("fetal head in the pelvis")** with mural ureteric calcification on plain film.
- **Praziquantel (PZQ)** is the only WHO-listed drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart, curing 60–90%. PZQ is weak on immature worms, so repeat after several weeks.
- Active polypoid obstruction resolves with PZQ alone; late fibrotic stenosis needs surgery.

Complication	Surgery
Distal / intramural stenosis	Antireflux reimplantation; balloon dilatation (restenosis common); meatoplasty for meatal stenosis
Long / multifocal stenosis	Boari flap; ileal ureter; ileal conduit; long-term nephrostomy
Contracted bladder	Ileocystoplasty or diversion, always with PZQ
Chronic deep ulcer	Partial cystectomy (fulguration fails)

8 · Filariasis

In India, filariasis commonly causes hydrocele and chyluria.

- W. bancrofti causes >90% and is the **only cause of genital filariasis**; Brugia affects the leg. Microfilariae are **nocturnally periodic**, and doxycycline kills the Wolbachia endosymbiont.
- **Acute adenolymphangitis** is a **retrograde** lymphangitis. **Funiculoepididymitis** is a recurrent painful cord swelling that spares the testis and is often mistaken for tumour.
- **Hydrocele** is the commonest chronic lesion; elephantiasis comes late.
- Diagnosis uses a midnight smear, the **circulating antigen** card, and the US "**filarial dance sign**".

LAQ Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2024-Winter Paper 3, Q2 (also answered in 2018-Summer Paper 1, Q1, with lymphatic

anatomy)

Definition

- **Chyluria** is the passage of **chyle** (intestinal lymph rich in chylomicrons and protein) in the urine, making it milky.
- It needs an abnormal **lymphourinary fistula**, usually at a renal calyceal fornix (pyelolymphatic fistula).

Relevant lymphatic anatomy

- Intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction.
- Lumbar (para-aortic) trunks from the kidneys join the cisterna. Lymph flows cephalad and right to left.
- Valves normally stop backflow from the cisterna into renal hilar lymphatics.

Aetiology

Parasitic (commonest; India, China)	**Non-parasitic**
Wuchereria bancrofti (lymphatic filariasis; Culex mosquito in urban India)	Congenital: lymphangiectasia, renal lymphangioma
Rarely other parasites	Trauma or surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection])
	Retroperitoneal tumour, fibrosis, tuberculosis, pregnancy
- Genital and urinary filariasis is almost always *W. bancrofti*; Brugia species spare the genitalia.	

Pathogenesis

- **Life cycle:** a mosquito deposits infective larvae, which mature in central lymphatics over 6–9 months. Adults live 5–7 years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- **Obstructive theory:** adult worms and the host reaction cause **lymphangitis and fibrosis** of lymphatics between the intestinal trunks and thoracic duct.
- Lymphatics below the block **dilate**, their **valves fail**, and chyle **refluxes** into renal hilar and peripelvic lymphatics.
- Rising pressure ruptures the thin-walled lymphatics into a calyceal fornix, creating a **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- Chyluria appears **earlier than elephantiasis**, is usually **intermittent** and may remit spontaneously.

Clinical features

- **Milky urine**, worse after fatty meals and on standing; may clear after fasting.
- **Chylous clots:** colic, retention.
- **Haematochyluria** (blood-stained chylous urine).
- **Systemic effects** of fat and protein loss: weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- Endemic residence, filarial fever, hydrocele or lymphoedema.

- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Evaluation

1. Confirm chyle

- **Ether (chloroform) test:** milky urine shaken with ether clears, because the fat dissolves.
- **Urine triglycerides** positive; Sudan III staining shows fat globules; urine protein raised.
- Differentiate from **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial tests

- Night (midnight) blood smear: often negative by this stage.
- **Circulating filarial antigen** (card test), eosinophilia, PCR (polymerase chain reaction); ultrasound "filarial dance sign" in scrotal lymphatics.

3. Lateralise and localise the fistula

- **Cystoscopy after a fatty meal:** milky efflux from one ureteric orifice; bilateral in some.
- **Retrograde pyelography:** pyelolymphatic backflow.
- **Lymphoscintigraphy:** non-invasive; tracer seen in the kidney and bladder. MR lymphangiography or CT; pedal lymphangiography is the historical gold standard.

4. Baseline: albumin, haemoglobin, renal function.

Management

`Mild -> diet + antifilarial drugs Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection` **1. Conservative**

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids. MCTs are absorbed directly into portal blood, not lacteals, so less chyle forms.
- **DEC (diethylcarbamazine)** 2 mg/kg three times daily, traditionally for 12 days (a 1-day course is as effective). It kills microfilariae but only modestly kills adults.
- **Doxycycline** (anti-*Wolbachia*, 200 mg daily for 4–8 weeks); albendazole; ivermectin.

2. Endoscopic sclerotherapy (renal pelvic instillation)

- Ureteric catheter on the affected side, head-down position. Instil **silver nitrate (0.5–1%)** or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- Good initial response; recurrence common.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every hilar lymphatic. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy; optional **omental wrap** around the hilum. 5. On-table cystoscopy confirms clear efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.

- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open (shorter operation and stay, less blood loss, fewer complications).
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy obsolete.

4. Prevention

- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: retroperitoneal lymphatics with obstructed thoracic duct, dilated hilar lymphatics and pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
> - Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
> - Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
> - Chyluria precedes elephantiasis and is often intermittent.

Also, from the condensed book (not in the paper answer):

- Most chyluria remits on conservative treatment; instil sclerosant on the side identified at cystoscopy.

Drug	Action	Regimen / caution
Diethylcarbamazine (DEC)	Microfilaricidal, modest on adults	Drug of choice: 6 mg/kg/day; a single-day course equals a 12-day course. **Contraindicated if onchocerciasis or Loa loa is possible** (blindness, encephalopathy)
Albendazole	Micro- and macrofilaricidal	400 mg bd for 21 days; single dose in mass treatment
Ivermectin	Microfilaricidal	150–400 µg/kg once; caution with Loa
Doxycycline	Anti-Wolbachia; kills adults	200 mg daily for 4–8 weeks; improves lymphoedema and hydrocele; not in pregnancy or children

- Mass treatment uses albendazole with DEC in India, or with ivermectin where Loa or onchocerciasis is co-endemic.
- **Hydrocele** recurs after aspiration, so **excise the sac**. In funiculoepididymitis excise nodules and preserve the testis. Elephantiasis needs reduction scrotoplasty.
- **Chyluria** occurs when dilated retroperitoneal lymphatics rupture into the collecting system, giving milky urine with fat and protein loss. Exclude TB, tumour and trauma.

9 · Hydatid disease and other parasites

Never puncture a hydatid cyst blindly.

SAQ Hydatid disease of kidney

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3d

Introduction

- Cystic echinococcosis (CE) by *Echinococcus granulosus*, a cestode. Dog: definitive host; sheep and humans: intermediate.
- Ingested eggs → liver (commonest), lung, then **kidney (third site, about 2–3%)** via the systemic circulation. Usually single, unilateral, cortical.

Pathology

- Wall: **pericyst** (host fibrous layer), **laminated membrane** (ectocyst), **germinal layer** (endocyst) producing brood capsules, scolices and daughter cysts ("hydatid sand").
- **Closed** versus **open** cyst (ruptured into the collecting system → **hydatiduria**).

Clinical features

- Grows 1–2 cm a year; long asymptomatic; flank mass, dull pain.
- **Hydatiduria** ("grape-skin" membranes in urine; pathognomonic) with colic, in a minority.
- Infection; rupture (anaphylaxis). Renal function usually preserved.

Investigations

- **Serology** (ELISA (enzyme-linked immunosorbent assay)) is adjunctive only (sensitivity 60–90%); eosinophilia; Casoni test obsolete.
- **Ultrasound and CT (computed tomography):** thick-walled cyst, wall calcification, daughter cysts, detached membranes ("water-lily sign"). Stage guides treatment.

WHO (World Health Organization) stage	Appearance	Activity
CE1	Unilocular, double-line wall	Active
CE2	Multivesicular, daughter cysts	Active
CE3a / CE3b	Detached membranes / daughter cysts in solid matrix	Transitional
CE4 / CE5	Solid "ball of wool" / calcified wall	Inactive

- Retrograde pyelogram shows communication; image liver and lungs. **Never aspirate blindly** (anaphylaxis, seeding).

Management

- **Albendazole 400 mg twice daily**, 1–6 months: alone for small or inoperable cysts; around surgery to reduce seeding (± praziquantel).
- **Surgery is the mainstay** (open, laparoscopic or robotic), nephron-sparing when possible: 1. Pack the field with scolicide-soaked swabs (hypertonic saline, povidone-iodine). 2. Inject scolicide only if **not** communicating (chemical injury to the collecting system). 3. **Pericystectomy, partial nephrectomy or deroofing**; close communications; omentoplasty or drain. 4. **Nephrectomy** for a non-functioning kidney.
- **PAIR (puncture, aspiration, injection, re-aspiration):** chapter says liver only, not kidney; some series report percutaneous renal treatment.
- Calcified inactive cysts (CE4, CE5) may be observed.

Diagram to draw: layers of a hydatid cyst with daughter cysts and a communication into a calyx.

Key points to remember:

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating.

Also, from the condensed book (not in the paper answer):

- Pack the field with hypertonic (20%) saline swabs, and have the anaesthetist ready for anaphylaxis.
- Remove the cyst intact (pericystectomy) or do a partial nephrectomy; nephrectomy is only for a destroyed kidney.
- Enterobiasis, amoebiasis, trichomoniasis (see Ch 29) and fungal UTI (see Ch 26) are rarer genitourinary infections.

★ Putting it together: infection decisions

A · UTI: drug matches site

Situation	Choice	Principle
Uncomplicated cystitis	Nitrofurantoin 5 days, fosfomycin 3 g once, TMP–SMX 3 days (resistance <20%), pivmecillinam	Narrow, urine-confined
Pyelonephritis, complicated UTI, men	Quinolone or cephalosporin; carbapenem if resistant	Nitrofurantoin and fosfomycin do not reach tissue
ESBL producers	Fosfomycin or nitrofurantoin for cystitis; carbapenem if invasive	Co-resistance to quinolones
Asymptomatic bacteriuria	Treat only in pregnancy and before mucosa-breaching surgery	Otherwise harmful

B · Beyond antibiotics

ALGORITHM

1. **Image**: no response by 72 h, sepsis, obstruction, gas, febrile men.
2. **Drain** any obstructed infected kidney; treat the stone later.
3. Renal abscess: if ≤3 cm, then antibiotics; 3–5 cm, variable; if >5 cm, then percutaneous drainage.
4. **Emphysematous pyelonephritis**: gas within kidney: antibiotics and drainage; if beyond it, then nephrectomy (see Ch 26).
5. **Fournier: urgent repeated radical debridement**.

C · Recurrent UTI in women

ALGORITHM

1. Confirm each episode by culture.
2. Behaviour; vaginal oestrogen if postmenopausal.
3. Methenamine, cranberry, D-mannose.
4. Self-start or post-coital antibiotic; finally continuous prophylaxis.

D · Pelvic pain syndromes compared

	CP/CPPS	IC/BPS
Core	Male pelvic pain >3 months, no infection (NIH III)	Filling-related bladder pain, no other cause
Phenotyping	UPOINT; NIH-CPSI	Hunner vs non-Hunner
Treatment	By domain: α-blocker, pelvic-floor physiotherapy, neuromodulators, psychology	Behavioural, oral (amitriptyline, PPS), intravesical, Hunner fulguration, BTX-A; diversion last

E · GU tuberculosis in four lines

ALGORITHM

1. **Suspect:** sterile pyuria, contracted bladder, beaded vas.
2. **Confirm:** three early-morning urines for AFB, culture, GeneXpert.
3. **Treat:** 2HRZE, then 4HR; watch for ureteric stricture.
4. **Operate** after 4–6 weeks of drugs (see Ch 31).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management

Paper answer · P1

Asked as:

- Discuss general principles of genitourinary infections, characteristics of common organisms and management of bacteremia and septic shock. — *Summer 2016 Paper I Q2 (LAQ)*
- Describe Pathophysiology of urosepsis and its management. — *Summer 2017 Paper I Q2 (LAQ)*
- Describe pzthophysiology of urosepsis and its management. — *Practice Set 11 Paper I Q1 (LAQ)* (also answered in 2017-Summer Paper 1, Q2) — *"Pzthophysiology" is read as pathophysiology.*

Type: LAQ (25 marks)

Definitions

- **UTI (urinary tract infection):** urothelial inflammation from bacterial invasion, usually with bacteriuria and pyuria. **Sterile pyuria** needs evaluation (tuberculosis, stone, tumour).
- **Uncomplicated:** healthy non-pregnant woman, normal tract. **Complicated:** men, pregnancy, obstruction, stones, catheter, neurogenic bladder, diabetes, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months. Unresolved = never cleared; reinfection = new organism; **bacterial persistence** = same organism from a focus (stone, prostate).
- **Prophylaxis** assumes a sterile tract; **suppression** controls a focus that cannot be eradicated.
- **Urosepsis:** sepsis whose source is the urogenital tract (the urinary tract or male genital organs). It is usually Gram-negative and usually follows obstruction or instrumentation.

Pathogenesis of UTI

- **Routes:** ascending from bowel flora (commonest); haematogenous (*S. aureus*, *Candida*); lymphatic (rare).
- **Virulence (uropathogenic *E. coli*): type 1 pili** (mannose-sensitive) bind bladder uroplakins and cause cystitis. **P pili** (mannose-resistant) bind renal glycolipids and are found in about 90% of pyelonephritis strains. Haemolysin damages tissue. Bacteria invade umbrella cells and form intracellular communities; quiescent reservoirs cause recurrence.
- **Principle:** a strong host is infected only by virulent strains; a compromised host (obstruction, catheter) by almost any organism.
- **Host defences:** voiding washout, lactobacilli, urine osmolality and pH, uromodulin (Tamm–Horsfall protein), urothelial exfoliation.
- **Risk factors: obstruction** (stasis impairs defences and turns UTI life-threatening), VUR (vesicoureteral reflux; with infection it scars the growing kidney), stones, catheters, diabetes, pregnancy, spermicides.
- **Diagnosis:** dipstick (nitrite, leucocyte esterase) rules out better than it rules in. Culture is the gold standard: ≥10^5 CFU (colony-forming units)/mL (Kass); ≥10^2 in a symptomatic

woman or a catheter specimen. Image (ultrasound, CT (computed tomography)) in complicated infection or fever beyond 72 hours.

Characteristics of common organisms

Organism	Key features
E. coli	~85% of community and ~50% of hospital UTI; commonest cause of urosepsis; ST131 clone is ESBL (extended-spectrum β-lactamase) and quinolone resistant
Proteus	Urease splits urea into ammonia: alkaline urine (pH >7), **struvite stones**, persistence in stones
Klebsiella	Capsule; urease; ESBL and KPC (*K. pneumoniae* carbapenemase) producers
Pseudomonas	Nosocomial, catheter biofilm; needs ceftazidime, piperacillin, aminoglycoside, fluoroquinolone or meropenem
Enterococcus	**Always resistant to cephalosporins**; amoxicillin, nitrofurantoin; VRE (vancomycin-resistant)
S. saprophyticus	~10% of cystitis in young sexually active women
S. aureus	Haematogenous renal abscess; in urine, look for bacteraemia
Anaerobes, *Candida*	Abscesses, fistulae; *Candida* with catheters and diabetes (*Candida* in catheterised diabetics)

- **Urosepsis organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.
- **Also know:** emerging *Aerococcus urinae*; *Ureaplasma* and *Mycoplasma* with negative routine cultures; *Chlamydia* and tuberculosis do not grow on routine culture (sterile pyuria).
- **Antibiotic principles (see 2015-Summer Paper 1, Q3f):** urine levels matter in cystitis, tissue levels in pyelonephritis and bacteraemia. Use a narrow spectrum and short course; adjust for renal function; **drain obstruction**.

Bacteraemia, sepsis, septic shock and urosepsis

Sepsis-3 (2016)

Term (Sepsis-3, 2016)	Definition
Sepsis	Life-threatening organ dysfunction from a dysregulated host response to infection: acute rise in **SOFA (Sequential Organ Failure Assessment) score ≥2**
qSOFA (quick SOFA)	Respiratory rate ≥22/min, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk (of death) (prompt only)
Septic shock	Vasopressor needed for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite (adequate) fluids; mortality >40%
- SIRS (systemic inflammatory response syndrome) criteria were dropped from the definition as non-specific.	

Causes and risk factors (urological triggers)

- **Urological triggers:** infected obstructed kidney, PCNL (percutaneous nephrolithotomy), high-pressure ureteroscopy, prostate biopsy, TURP (transurethral resection of the prostate), infected stones, emphysematous pyelonephritis, abscess.
- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy, giving infected hydronephrosis or pyonephrosis. Pyonephrosis, emphysematous pyelonephritis, renal, perinephric and prostatic abscess.
- **Parenchymal infection:** emphysematous pyelonephritis, renal or perinephric abscess, prostatic abscess, Fournier gangrene.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy with high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy (sepsis 0.3–3.1%), catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.

Pathophysiology

- **Pathophysiology (summary):** endotoxin binds TLR4 (Toll-like receptor 4). Cytokines (TNF-α (tumour necrosis factor-α), interleukins IL-1 and IL-6) and nitric oxide cause vasodilatation, capillary leak, myocardial depression and DIC (disseminated intravascular coagulation), then organ failure.

1. Entry of bacteria (and endotoxin) into blood

- Obstruction causes stasis, impairs washout and raises pelvic pressure. Bacteria and endotoxin then enter the blood (circulation) by **pyelovenous and pyelolymphatic backflow**.
- High irrigation pressure at endoscopy does the same, and so does fragmenting infected stones, which release endotoxin (fragmenting infected stones releases endotoxin). Stone culture predicts postoperative sepsis better than bladder urine culture.
- An obstructed infected kidney is (behaves like) an **abscess with a collecting system**. Antibiotics barely reach it, and bladder urine may be sterile.

2. Host response `Endotoxin (LPS, lipopolysaccharide) / (or) Gram-positive wall products → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB) → cytokines TNF-α (tumour necrosis factor α), interleukins IL-1, IL-6, IL-8 → complement + coagulation activation + endothelial injury (activation)`

- **Vasodilatation:** inducible nitric oxide causes (lowers) low systemic vascular resistance ("warm" shock).
- **Capillary leak:** (endothelial) glycocalyx damage leads to hypovolaemia and tissue oedema.
- **Coagulopathy:** tissue factor release causes microthrombi, platelet consumption and DIC (disseminated intravascular coagulation).
- **Myocardial depression** and **mitochondrial dysfunction (failure):** cells cannot use oxygen, so lactate rises.
- **Immunoparalysis:** a later anti-inflammatory phase predisposes to (invites) secondary infection.

3. Organ dysfunction: AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver dysfunction (failure), DIC, multi-organ failure.

Clinical features

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often starts (begins) within hours of an endourological procedure (endourology).

Investigations

- **Lactate**, blood cultures ×2 (two blood cultures), urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, arterial (blood) gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas or abscess.

Management (Surviving Sepsis Campaign: hour-1 bundle 2018, guidelines 2021)

A. Resuscitation and antibiotics (SSC, Surviving Sepsis Campaign 2021) Recognise (SSC: do not use qSOFA alone for screening) → Lactate → blood cultures ×2 + urine culture → IV (intravenous) broad-spectrum antibiotics within 1 hour (shock or probable sepsis; within 3 h if sepsis only possible) → 30 mL/kg (balanced) crystalloid if hypotensive or lactate ≥4 mmol/L (within 3 h if hypotensive or hypoperfused) → noradrenaline first line for MAP ≥65 (target MAP 65); add vasopressin → SOURCE CONTROL: stent or nephrostomy (equally effective), drain abscess, change catheter; stone surgery later → refractory shock (persistent vasopressor need): hydrocortisone 200 mg/day; ICU (intensive care unit) → re-measure (repeat) lactate, capillary refill → de-escalate by culture at 48–72 h

- **Empirical antibiotics:** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

B. Source control: the key urological step

- **Drain the obstructed infected kidney urgently:** retrograde **double-J stent** or **percutaneous nephrostomy (PCN)**. Randomised trials show equal effect on sepsis (equally effective in a randomised trial (Pearle, 1998)).
- Stent: suits coagulopathy; avoid high-pressure contrast injection.
- PCN: better for (suits) large or impacted stones, pyonephrosis; allows (gives) pelvic urine (for) culture.
- **No definitive stone surgery during sepsis.** Treat the stone after (once) the infection clears.
- Other sources: drain renal or perinephric abscess; emphysematous pyelonephritis (percutaneous drainage, nephrectomy if diffuse gas without fluid); prostatic abscess drainage; change or remove (the) catheter; debride Fournier gangrene.

C. Supportive care (Support)

- Oxygen and (or) ventilation, glucose control (insulin if glucose ≥180 mg/dL), transfusion (transfuse) below haemoglobin 7 g/dL, thromboprophylaxis, stress-ulcer prophylaxis, renal replacement if needed. Watch for AKI (acute kidney injury), ARDS (acute respiratory distress syndrome) and DIC.

D. De-escalation and duration

- Narrow antibiotics by (to) culture at 48–72 h. Usual course (usually) 7–10 days (another source: 7–14 days); longer with an undrained focus or *S. aureus* bacteraemia; shorter courses suffice after good source control. Step down (switch) to oral therapy when (once afebrile and) stable.

Special situations

- **Emphysematous pyelonephritis** (diabetics): fluids, glucose control, antibiotics, drain; nephrectomy for diffuse gas without fluid.
- **Pregnancy:** stent or nephrostomy; avoid fluoroquinolones.
- **Fungal urosepsis:** remove catheter, antifungal, drain fungal balls.
- **Prognosis:** urosepsis does better than sepsis from other sites if the source is drained early; delay in drainage (delayed drainage) is the main avoidable cause of death.

Prevention

- Sterile urine before elective endourology; antibiotic prophylaxis; low irrigation pressure and access sheath; short operating time with infected stones; stage the procedure if pus is found.

Diagram to draw: pathway from *E. coli* adherence to bacteraemia and cytokine-driven shock; obstructed kidney leading to backflow, cytokine cascade and shock; obstructed kidney → pyelovenous backflow → cytokine cascade → shock and organ failure.

> **Key points to remember:**
>
> - *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
> - Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
> - Obstruction plus infection is the emergency; bladder urine may be sterile.
> - Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
> - Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
> - Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

SAQ Principles of antimicrobial therapy in UTI

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3f

Definitions

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract. **Complicated:** men, pregnancy, abnormal tract, catheter, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months.

Principles

- **Site decides the level:** urine in cystitis, tissue elsewhere; nitrofurantoin and fosfomycin lack tissue levels: **cystitis only**.

Killing pattern	Drugs	Dosing goal
Time-dependent (time above MIC [minimum inhibitory concentration])	β-lactams, carbapenems	Frequent doses or extended infusion
Concentration-dependent (peak:MIC)	Aminoglycosides, fluoroquinolones (FQs)	High once-daily dose

- **Resistance thresholds:** trimethoprim–sulfamethoxazole (TMP-SMX) if local resistance ≤20%; oral FQ for pyelonephritis if ≤10%.
- **Collateral damage:** FQs and cephalosporins select resistance and *C. difficile*; avoid them in cystitis.
- Culture all but simple cystitis; **drain** obstructed infected kidneys; shortest course.

Empirical choices

Scenario	Drug	Duration
Cystitis (women)	Nitrofurantoin 100 mg twice daily; fosfomycin 3 g; pivmecillinam 400 mg; TMP-SMX 160/800 mg twice daily	5 d; single dose; 3–5 d; 3 d
Cystitis (men)	TMP-SMX or per culture	7 d (= 14 d, Drekonja 2021)
Pyelonephritis (oral)	Ciprofloxacin 500 mg twice daily; levofloxacin 750 mg daily	7 d; 5 d
Pyelonephritis (intravenous)	Ceftriaxone 1 g daily; gentamicin 5–7 mg/kg once daily	10–14 d; oral when afebrile
Complicated / catheter UTI	Aminoglycoside or 3rd-generation cephalosporin per culture; change catheter	7–14 d
Urosepsis	Broad-spectrum **within 1 h**, source control	–

Asymptomatic bacteriuria

- Treat **only in pregnancy and before mucosa-breaching urological procedures**. Not in elderly, diabetic, catheterised or spinal cord injury patients.
- **Pregnancy:** avoid FQs and TMP-SMX; cephalexin is suitable.

Prophylaxis

- Recurrent: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics, ALTAR 2022), cranberry, post-coital antibiotics.
- Surgical: single dose within 1 hour; ≤24 h for percutaneous nephrolithotomy and prostheses.
- **Also know:** ESBL (extended-spectrum β-lactamase) producers; nitrofurantoin avoided if creatinine clearance <30 mL/min; gepotidacin (US 2025).

Diagram to draw: Concentration–time curve showing peak:MIC and time above MIC.

> **Key points to remember:**
> - Recurrent = ≥2 in 6 months or ≥3 in 12 months.
> - Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
> - Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

SAQ Ciprofloxacin

Paper answer · P1

Type: SAQ (10 marks)
Source: 2017-Summer Paper 1, Q3d

Class and mechanism

- Second-generation **fluoroquinolone**, derived from nalidixic acid.
- Inhibits **DNA gyrase** (topoisomerase II, main target in Gram-negatives) and **topoisomerase IV** (Gram-positives). This blocks DNA supercoiling and replication: **bactericidal**.

Spectrum

- Enterobacteriaceae, ***Pseudomonas aeruginosa***, *Haemophilus*, *S. saprophyticus*, *S. aureus* (methicillin-sensitive); *Legionella*, some mycobacteria.
- Weak against streptococci and enterococci; **no anaerobic cover**, so bowel and vaginal flora are partly spared.

Pharmacokinetics

- Oral bioavailability about 70%; IV (intravenous) and oral forms; half-life about 4 h.
- **Concentration-dependent killing:** AUC/MIC (area under curve to minimum inhibitory concentration) predicts effect; post-antibiotic effect.
- High urine and tissue levels; **good prostatic penetration**.
- Renal excretion: reduce dose if creatinine clearance <30 mL/min.

Doses

- Cystitis 250 mg twice daily × 3 days; pyelonephritis 500 mg twice daily × 7 days (or 1 g extended-release daily); IV 400 mg 12-hourly.
- Chronic bacterial prostatitis: 4–6 weeks.

Indications and restrictions

- Pyelonephritis and complicated UTI (urinary tract infection) **only where local resistance ≤10%**; prostatitis; prophylaxis for transrectal biopsy (falling, because of resistance).
- **Not first line for uncomplicated cystitis** (FDA (US Food and Drug Administration) 2016; European restrictions 2019): use only when no alternative.

Resistance

- *gyrA*/*parC* mutations, efflux pumps, reduced porins, plasmid *qnr* genes. Linked to ESBL (extended-spectrum β-lactamase) *E. coli* (ST131). Main cause of post-biopsy sepsis.

Adverse effects

- GI (gastrointestinal) upset (commonest), photosensitivity, *C. difficile*.
- **FDA boxed warnings:** tendinopathy and **Achilles rupture** (2008; risk higher with age, steroids, renal failure); peripheral neuropathy (2013); disabling effects (2016); psychiatric effects and dysglycaemia (2018); aortic aneurysm and dissection (2018).
- CNS (central nervous system): seizures, confusion; QT prolongation.

Interactions

- Antacids, iron, zinc, calcium block absorption (separate by ≥2 h). Raises theophylline and tizanidine levels; potentiates warfarin.

Pregnancy and children

- Avoid: **cartilage damage** in immature animals. Also avoid in breastfeeding.

> **Key points to remember:**
> - DNA gyrase and topoisomerase IV; concentration-dependent.
> - Use for pyelonephritis only if local resistance ≤10%.
> - Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

SAQ Complicated urinary tract infections.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2018-Summer Paper 1, Q3f

Definition

- **Complicated UTI (urinary tract infection):** infection in a **structurally or functionally abnormal tract** or a **compromised host**, or with a resistant organism; more likely to fail or recur.
- **FDA (US Food and Drug Administration) 2018:** pyuria + pathogen in urine or blood + fever, chills, flank pain, with an abnormality or catheter.
- **Uncomplicated:** healthy non-pregnant woman, normal tract.

Complicating factors

Category	Examples
Obstruction	BOO (bladder outlet obstruction), PUJ (pelviureteric junction) obstruction, stricture, stone, tumour
Voiding dysfunction	Neurogenic bladder, high PVR (post-void residual), VUR (vesicoureteral reflux)
Foreign body / iatrogenic	Catheter, stent, nephrostomy, recent instrumentation, hospital-acquired
Anatomical	Diverticulum, fistula, cystocele
Host	**Male sex**, pregnancy, diabetes, immunosuppression, renal failure, transplant, elderly
Other	Symptoms >7 days; recent antibiotics

- Each causes stasis, biofilm or weak defences.

Organisms

- ***E. coli*** under half; more *Klebsiella*, *Proteus*, *Pseudomonas*, *Enterococcus*, *Candida*. Often **MDR (multidrug-resistant)** or ESBL (extended-spectrum β-lactamase) producers.

Evaluation

- **Urine culture mandatory before antibiotics**; blood cultures if febrile.
- Renal function, glucose, lactate.
- **Imaging:** ultrasound first; **CT (computed tomography)** if no response in 72 hours, stones, abscess or emphysematous infection suspected.

Treatment

- **Empirical IV (intravenous) therapy** by local resistance (EAU (European Association of Urology)): amoxicillin + aminoglycoside, 2nd-generation cephalosporin + aminoglycoside, or IV 3rd-generation cephalosporin. Avoid fluoroquinolones if used in the last 6 months or resistance >10%. Carbapenem or newer agents for ESBL (ceftolozane–tazobactam, ceftazidime–avibactam).
- **Source control is essential:** drain obstruction (**stent or nephrostomy, equally effective**), change catheter, drain abscess, treat stones later.
- Switch to oral when afebrile and stable; de-escalate by culture.
- **Duration:** usually 7–14 days (14 days in men if prostatitis not excluded); 7 days may suffice once afebrile ≥48 hours.
- **Sepsis:** antibiotics within 1 hour, fluids, vasopressors, ICU (intensive care unit).
- Correct the underlying cause, or infection recurs.

Special forms

- **Also know:** emphysematous pyelonephritis and cystitis (diabetics); pyonephrosis; renal and perinephric abscess; XGP (xanthogranulomatous pyelonephritis); CAUTI (catheter-associated UTI); funguria; unresolved bacteriuria (resistance, staghorn "critical mass").

Diagram to draw: Flowchart: suspect cUTI → culture → imaging → empirical IV → drain obstruction → oral step-down.

> **Key points to remember:**
> - Complicated = abnormal tract or compromised host; male sex alone qualifies.
> - Culture mandatory; often MDR.
> - Drain obstruction: stent = nephrostomy.
> - 7–14 days; oral step-down when afebrile.

SAQ Drug Fosfomycin

Paper answer · P1

Type: SAQ (10 marks)
Source: 2022-Summer Paper 1, Q3c

Introduction

- Bactericidal **phosphonic acid derivative** antibiotic, chemically unrelated to other classes. Oral form: **fosfomycin trometamol**; intravenous form: fosfomycin disodium.

Mechanism of action

- Enters bacteria through glycerol-3-phosphate and glucose-6-phosphate transporters.
- Irreversibly inhibits **MurA** (enolpyruvyl transferase), blocking the **first step of peptidoglycan (cell wall) synthesis**. β-lactams and vancomycin act at later steps, so **cross-resistance is limited**.

Pharmacokinetics

- Oral bioavailability is modest, but the drug is excreted **unchanged in urine**. One 3 g dose keeps high urinary concentrations for about 1–2 days, which permits single-dose therapy. Low tissue levels.

- Concentrations fall in renal failure; avoid at very low creatinine clearance.

Spectrum

- **Escherichia coli**, including **ESBL (extended-spectrum β-lactamase)** producers; most Enterobacteriaceae (Klebsiella variable); *Enterococcus* including **VRE (vancomycin-resistant enterococcus)**; staphylococci.
- **Not reliable:** *Pseudomonas aeruginosa*, *Acinetobacter*.

Uses in urology

Indication	Dose
Uncomplicated cystitis in women (first-line, EAU (European Association of Urology) and IDSA (Infectious Diseases Society of America))	3 g oral sachet, single dose
Cystitis in pregnancy, asymptomatic bacteriuria (safe, category B)	3 g single dose
MDR (multidrug-resistant) lower UTI (urinary tract infection)	Oral option avoiding IV (intravenous) therapy; repeated doses used off-label
Prophylaxis for transrectal prostate biopsy	As effective as fluoroquinolone
Recurrent UTI prophylaxis	3 g every 10 days (EAU)

- **Not for pyelonephritis** or urosepsis orally (poor tissue levels). Emerging data suggest IV fosfomycin may help in bacteraemic MDR E. coli UTI.
- Not needed as prophylaxis for flexible cystoscopy.

Adverse effects

- Diarrhoea, GI (gastrointestinal) upset, headache, vaginitis; rare hypersensitivity. IV form: **sodium load**, hypokalaemia.

Resistance

- Transporter mutations, MurA changes, plasmid **fosA** enzymes. Resistance develops quickly in vitro, so systemic infections use combinations.

Evidence

- Huttner (2018) RCT (randomised controlled trial): nitrofurantoin 5 days gave higher clinical resolution than single-dose fosfomycin (70% vs 58%).

Diagram to draw: peptidoglycan synthesis showing fosfomycin (MurA), β-lactams and vancomycin sites.

> **Key points to remember:**
> - Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
> - 3 g single dose for cystitis; active against ESBL E. coli and VRE.
> - Not for pyelonephritis; poor Pseudomonas cover.

SAQ Papillary necrosis.

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 1, Q3a

Definition

- **RPN (renal papillary necrosis):** ischaemic necrosis of the renal papillae (inner medulla), which may slough into the collecting system.

Pathophysiology

- The papilla is vulnerable: it depends on the **vasa recta**, has **low oxygen tension** and a **hypertonic** environment. Almost every cause reduces medullary blood flow or oxygenation.
- **Sickle cell:** hypoxia and hypertonicity make cells sickle and block the vasa recta, causing microinfarcts.
- **Analgesics:** NSAIDs (non-steroidal anti-inflammatory drugs) block vasodilator prostaglandins; metabolites concentrate in the medulla.
- **Diabetes:** microangiopathy plus infection.

Causes ("POSTCARDS")

Letter	Cause
P	Pyelonephritis
O	Obstruction
S	Sickle cell disease or trait
T	Tuberculosis
C	Cirrhosis
A	Analgesic abuse
R	Renal transplant rejection; renal vein thrombosis
D	**Diabetes mellitus**
S	Systemic vasculitis
- **Also know:** infant dehydration and hypoxia; shock; renal candidiasis.	

Clinical features

- **Acute:** fever, flank pain, haematuria, **ureteric colic from a sloughed papilla**, sepsis; bilateral cases may cause AKI (acute kidney injury).
- **Chronic or incidental:** polyuria (poor concentration), sterile pyuria, recurrent UTI (urinary tract infection), CKD (chronic kidney disease). Retained papillae may **calcify** and harbour infection, including fungi.

Investigations

- Urine microscopy and culture; **strain urine** and send tissue for histology; creatinine, glucose, haemoglobin electrophoresis.
- **CT urography or IVU (intravenous urography):** early contrast tracking into the papilla ("**ball-on-tee**", "**lobster claw**"); "**ring sign**" (contrast around a detached papilla); clubbed calyces; filling defects; triangular ring-shaped calcification.
- Retrograde pyelography when renal function is poor.
- **Differential diagnosis:** tuberculosis, medullary sponge kidney, calyceal diverticulum, upper tract urothelial carcinoma.

Management

- **Treat the cause:** stop analgesics, control diabetes, hydrate, manage sickle crisis.
- **Culture-guided antibiotics.**
- **Obstruction by a sloughed papilla:** urgent drainage (stent or nephrostomy), then ureteroscopic extraction.
- Remove calcified retained papillae that cause persistent infection.
- **Follow-up:** renal function; analgesic abusers risk urothelial cancer.

Diagram to draw: calyx showing ring sign and lobster-claw appearance.

> **Key points to remember:**
> - Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
> - POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
> - Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

SAQ Cephalsporins

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 1, Q3c

Assumed meaning: cephalosporins (generations, spectrum, urological use).

Introduction and mechanism

- **β-lactam** antibiotics. They bind **PBPs (penicillin-binding proteins)** and block peptidoglycan cross-linking, so the cell wall fails: **bactericidal**, time-dependent killing.
- Mostly renal excretion (adjust dose in renal failure); **ceftriaxone** also has biliary excretion (no adjustment).
- Resistance: β-lactamases (**ESBL [extended-spectrum β-lactamase]**, AmpC), altered PBPs, reduced porin entry.

Generations

Generation	Examples	Spectrum	Urological use
First	Cefazolin (intravenous), cephalexin, cefadroxil (oral)	Gram-positive (not MRSA [methicillin-resistant *Staphylococcus aureus*]); *E. coli*, *Klebsiella*, *P. mirabilis*	**Surgical prophylaxis (cefazolin)**; cystitis in pregnancy; paediatric prophylaxis
Second	Cefuroxime, cefaclor; cefoxitin, cefotetan	Wider Gram-negative; cefoxitin and cefotetan add **anaerobes**	Prophylaxis for vaginal and bowel surgery; SWL (shock-wave lithotripsy)
Third	Ceftriaxone, cefotaxime, cefixime, cefpodoxime; **ceftazidime**	Most Enterobacterales; **only ceftazidime covers *Pseudomonas***	Pyelonephritis, urosepsis; **gonorrhoea (ceftriaxone 500 mg IM [intramuscular])**; prostate biopsy prophylaxis
Fourth	Cefepime	Gram-negatives incl. *Pseudomonas*, stable to AmpC; Gram-positive	Complicated, nosocomial UTI (urinary tract infection)
Fifth	Ceftaroline	MRSA + Gram-negatives	Limited urological role

- **Newer combinations** for complicated UTI with multidrug resistance: ceftazidime–avibactam, ceftolozane–tazobactam, cefiderocol.

Gaps in spectrum

- **Enterococci** (intrinsically resistant; never treat with cephalosporins), *Listeria*, MRSA (except ceftaroline), atypical organisms.
- ESBL and AmpC producers hydrolyse third-generation agents.

Adverse effects

- Hypersensitivity, less than penicillins. Cross-reactivity with third-generation agents is <1%; avoid after immediate (IgE [immunoglobulin E]-mediated) penicillin reaction.
- ***C. difficile* colitis**; "collateral damage" (selection of ESBL organisms).
- Positive Coombs test; bleeding with cefotetan and cefamandole.
- Ceftriaxone: biliary sludge; avoid in jaundiced neonates and with intravenous calcium.

Paediatric use

- Febrile UTI in neonates: **ampicillin + third-generation cephalosporin** (covers enterococcus). Oral cefixime 8 mg/kg/day. A first-generation agent is a good narrow-spectrum choice for many outpatient UTIs.

Diagram to draw: β-lactam ring binding a PBP and blocking peptidoglycan cross-links.

> **Key points to remember:**
> - Cefazolin = standard surgical prophylaxis.
> - Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
> - No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

SAQ Biofilm

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3c (also answered in part in 2015-Summer Paper 3, Q3d)

Definition

- A **structured microbial community in a self-made polymeric matrix**, adherent to a living or inert surface. The matrix contains exopolysaccharide, extracellular DNA and proteins.

Stages of formation

1. **Conditioning film:** host proteins coat the device within minutes.
2. **Attachment:** reversible, then irreversible (pili, adhesins, flagella).
3. **Matrix production and microcolonies.**
4. **Maturation:** 3-D (three-dimensional) structure with water channels; cell-to-cell signalling (quorum sensing).
5. **Dispersion:** free bacteria seed new infection.

Why antibiotics and immunity fail

- Poor drug penetration through the matrix.
- Slow-growing, dormant cells resist drugs that need active growth.

- Altered genes, envelope and targets.
- Survive **1000–1500 times** the usual killing concentration. Neutrophils are excluded.

Urological examples

Site	Point
Urethral catheter	Forms within minutes; **the catheter, not the bladder, is infected**. Proteus urease makes crystalline biofilm → encrustation and blockage
Ureteric stent, nephrostomy	Stent colonised more often than urine shows; nephrostomy changed every 3 months
Penile prosthesis, AUS (artificial urinary sphincter)	Revision of an "uninfected" device: about 10% infection
Infection stones	Bacteria inside struvite; antibiotics cannot sterilise
Chronic bacterial prostatitis	Biofilm commoner in prostatitis *E. coli*; needs longer, penetrating therapy
Recurrent cystitis	**Intracellular bacterial communities**: biofilm-like pods inside umbrella cells

- Common organisms: Pseudomonas, Proteus, *E. coli*, Enterococcus, coagulase-negative staphylococci, Candida.

Prevention and management

- Avoid devices; shortest dwell; aseptic insertion; closed dependent drainage.
- **Change the catheter** when treating symptomatic CAUTI (catheter-associated UTI (urinary tract infection)); remove infected stent or implant.
- Silver-alloy catheters reduce bacteriuria, not clinical UTI.
- Implants: antibiotic-coated devices, no-touch technique, **antiseptic washout** at revision (infection about 10% → 3%).
- Stones: complete clearance. Chemodissolution for catheter encrustation. Macrolides penetrate biofilm (prostatitis).
- **Also know (research):** quorum-sensing inhibitors, bacteriophages, new coatings.

Diagram to draw: five-stage biofilm cycle on a catheter surface.

> **Key points to remember:**
> - Biofilm: microbes in self-made matrix on a surface.
> - Survives 1000–1500 times normal antibiotic levels.
> - Treat CAUTI by changing the catheter.
> - Revision washout cuts prosthesis infection from about 10% to 3%.

SAQ Renal mucormycosis

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3c

Introduction

- Mucormycosis: an **angioinvasive** fungal infection by Mucorales (*Rhizopus* commonest; *Mucor*, *Lichtheimia*).

- Kidney involvement occurs in **disseminated** disease, or as **isolated renal mucormycosis**, reported particularly from India, sometimes in apparently immunocompetent young adults. Cases rose during the 2021 COVID-19 (coronavirus disease 2019) wave.

Risk factors

- Uncontrolled diabetes, especially **ketoacidosis**; corticosteroids; neutropenia and haematological cancer; transplantation; iron overload or deferoxamine; intravenous drug use; AIDS (acquired immunodeficiency syndrome).

Pathogenesis

- Spores inhaled or ingested → **haematogenous** seeding of the kidney.
- Hyphae invade arteries and veins → **thrombosis → infarction and haemorrhagic necrosis**.
- Acidosis and free iron favour growth (hence the ketoacidosis risk).

Clinical features

- Fever, flank pain, tender flank mass, haematuria, pyuria with sterile bacterial cultures.
- **Bilateral** disease → AKI (acute kidney injury) or anuria.
- Suspect when "pyelonephritis" fails antibiotics in a high-risk patient.

Investigations

- **CECT (contrast-enhanced computed tomography):** enlarged kidney(s), wedge-shaped or diffuse **non-enhancing** areas (infarcts), perinephric stranding or collection, renal vessel thrombosis.
- **Tissue diagnosis** (biopsy or nephrectomy specimen): **broad, ribbon-like, aseptate or pauciseptate hyphae branching at right angles**, invading vessels.
- KOH (potassium hydroxide) mount, calcofluor stain, fungal culture (often negative), PCR (polymerase chain reaction).
- Differential: *Aspergillus* has narrow septate hyphae with acute-angle branching.
- Image sinuses, lungs, brain.

Management (three pillars)

1. **Reverse predisposition:** control glucose, correct ketoacidosis, taper steroids and immunosuppression, stop deferoxamine.
2. **Early surgical debridement:** nephrectomy for a unilateral necrotic kidney; drain collections.
3. **Antifungal: liposomal amphotericin B 5 mg/kg/day** first line (ECMM/MSG ERC (European Confederation of Medical Mycology/Mycoses Study Group Education and Research Consortium) guideline, 2019). Step-down or salvage: **posaconazole or isavuconazole**. - **Fluconazole, voriconazole and echinocandins are inactive.** - Bilateral: antifungal plus dialysis.

Prognosis

- Mortality is high, especially with bilateral or disseminated disease and delayed diagnosis.

Diagram to draw: right-angle branching aseptate hypha invading a renal vessel.

Key points to remember:

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

SAQ Nitrofurantoin.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2020-Summer Paper 4, Q3e

Introduction

- A synthetic **nitrofuran**; **first-line for acute uncomplicated cystitis** (EAU (European Association of Urology)) because it concentrates in urine and spares bowel and vaginal flora.

Mechanism

- Reduced by bacterial flavoproteins (nitroreductases) to **reactive intermediates** that damage ribosomal proteins, DNA and several enzyme systems.
- **Bacteriostatic**, bactericidal at high urinary concentration.
- Multiple targets keep **acquired resistance low**.

Pharmacokinetics

- Well absorbed (better with food); **rapidly excreted in urine**. Serum and tissue levels are sub-therapeutic.
- Hence **useless for pyelonephritis, prostatitis, febrile or bloodstream infection**.
- Macrocrystals: slower absorption, less nausea.

Spectrum

Susceptible	Resistant
E. coli (most), ***Enterococcus faecalis***, *Staphylococcus saprophyticus*, *S. aureus* (not MRSA (methicillin-resistant *S. aureus*)), many *Klebsiella*, *Citrobacter*	***Proteus*** (intrinsic), *Pseudomonas*, *Serratia*, *Providencia*, *Morganella*, *Acinetobacter*

- Active against many ESBL (extended-spectrum β-lactamase)-producing *E. coli*.

Uses and doses

Indication	Regimen
Uncomplicated cystitis (women)	**100 mg twice daily × 5 days** (modified release)
Recurrent UTI (urinary tract infection) prophylaxis, adults	50–100 mg at night, or post-coital
Children: treatment / prophylaxis	3–5 mg/kg/day in 2 doses / 1–2 mg/kg at night (VUR (vesicoureteral reflux), after 3 months of age)
Pregnancy	Second and third trimesters; first trimester only if no alternative; **avoid near term**

Adverse effects

- **GI (gastrointestinal) upset** (commonest).
- **Pulmonary**: acute hypersensitivity (fever, cough, dyspnoea); **chronic interstitial fibrosis** with long-term use.
- **Peripheral neuropathy** (worse with renal impairment, diabetes, anaemia, B-vitamin deficiency).
- **Haemolysis in G6PD (glucose-6-phosphate dehydrogenase) deficiency**, including the neonate (hence avoid at term and <3 months of age).
- Hepatotoxicity (cholestatic or chronic hepatitis), rare.

Precautions

- **Renal impairment**: inadequate urine levels and more toxicity. Historically avoided if CrCl (creatinine clearance) <60; Beers Criteria (2015) allow short courses down to **CrCl 30 mL/min**; avoid long-term use in the elderly.
- Interactions: probenecid (blocks excretion), magnesium antacids, quinolones (antagonism).

Diagram to draw: spectrum table (susceptible versus resistant).

> **Key points to remember:**
>
> - Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
> - 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
> - Proteus intrinsically resistant; resistance in E. coli low.
> - Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

Rapid Revision Sheet · Section 3

Antimicrobial resistance (SAQ)

- Plasmid-mediated resistance is commonest.
- ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
- TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
- Stewardship: culture, narrow, short, no ASB treatment.

ESBL producing enterobacterial UTI (SAQ)

- Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
- Stewardship prevents spread.

Emphysematous pyelonephritis: prognostic factors and conservative management (SAQ)

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

Renal and perinephric abscess (SAQ)

- Now mostly ascending Gram-negative abscess in a damaged kidney.
- Perinephric: within Gerota fascia; diabetes in a third.
- Fever beyond 4–5 days of antibiotics: get a CT.
- CT is the investigation of choice; follow until resolution.
- ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
- Percutaneous drainage first line; nephrectomy if non-functioning.

Xanthogranulomatous pyelonephritis. (SAQ)

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

Malakoplakia (SAQ)

- Defective macrophage digestion of *E. coli*.
- Michaelis–Gutmann bodies are pathognomonic.
- Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Catheter Associated Urinary Tract Infections (CAUTI) (SAQ)

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + $\geq 10^3$ CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Fungal UTI (SAQ)

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

Chronic (non-bacterial) prostatitis: diagnosis, classification and management (SAQ)

- NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
- CBP: same organism, fluoroquinolone 4–6 weeks.
- CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
- Multimodal therapy; α-blocker plus antibiotic only early.

Interstitial cystitis / bladder pain syndrome (LAQ)

- AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
- AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
- HL: fulguration or triamcinolone; ciclosporin if refractory.
- Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

Lymphogranuloma venereum (SAQ)

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

Describe clinical implication of AIDS on the Genito-Urinary Tract? (LAQ)

- AIDS = CD4 <200; UTI risk rises as CD4 falls.
- HIV-TB: disseminated, less fibrosis, fewer strictures.
- Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
- PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Lichen sclerosus (BXO, balanitis xerotica obliterans) (SAQ)

- Genital skin disease; meatus is the sentinel site.
- Spares proximal urethra; perineal urethrostomy easy.
- Clobetasol first; oral mucosa, never skin, for repair.
- SCC in 2.3–9%: lifelong follow-up.

Premalignant conditions in urology (including premalignant lesions of penis) (SAQ)

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder (LAQ)

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria. (LAQ)

- Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
- Chyluria precedes elephantiasis and is often intermittent.

Hydatid disease of kidney (SAQ)

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management (LAQ)

- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
- Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

Principles of antimicrobial therapy in UTI (SAQ)

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

Ciprofloxacin (SAQ)

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

Complicated urinary tract infections. (SAQ)

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.

Drug Fosfomycin (SAQ)

- Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
- 3 g single dose for cystitis; active against ESBL E. coli and VRE.
- Not for pyelonephritis; poor Pseudomonas cover.

Papillary necrosis. (SAQ)

- Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
- POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
- Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

Cephalsporins (SAQ)

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

Biofilm (SAQ)

- Biofilm: microbes in self-made matrix on a surface.
- Survives 1000–1500 times normal antibiotic levels.
- Treat CAUTI by changing the catheter.
- Revision washout cuts prosthesis infection from about 10% to 3%.

Renal mucormycosis (SAQ)

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

Nitrofurantoin. (SAQ)

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 4

Paediatric Urology 1

41 Management Strategies for Vesicoureteral Reflux

1 · Definition and concept

We treat the infection risk, not the reflux.

- **Vesicoureteral reflux (VUR)** is retrograde urine flow to the upper tract, linked to **acquired scarring** and **congenital dysplasia**.
- **Infection, not reflux, is the injury**, so aim for sterile urine and good emptying.

2 · Embryology and antireflux mechanism

LAQ Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING

Paper answer · P1 · P3 · P4 · merged from 6

Asked as: P1 LAQ "Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management" (Describe embryological development of VU Junction and outline management of primary VUR?, Summer 2020 Paper I Q2; B/l congenital VUR - Pathophysiology and natural history, Practice Set 1 Paper I Q3f SAQ) · P3 LAQ "Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome" (Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR, Summer 2015 Paper III Q2; Discuss mgt. of b/l VUR, Practice Set 2 Paper III Q2; Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection, Practice Set 2 Paper III Q3d SAQ; Discuss the management of B/l VUR and discuss its long term outcome, Practice Set 6 Paper III Q1) · P4 LAQ "VUR: etiopathogenesis, recent trends in management and STING technique" (Describe etiopathogenesis of VUR and recent trends in management, Practice Set 2 Paper IV Q2; STING technique for VUR, Summer 2020 Paper IV Q3c and Practice Set 5 Paper IV Q3e SAQ) · P3 LAQ "Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?" · P3 SAQ "Endoscopic correction of VUR (STING procedure): indications and complications" (Endoscopic correction of VUR : indications and complications, Summer 2019 Paper III Q3e; Sting procedure, Winter 2024 Paper III Q3f) · P3 SAQ "Paquin and Gil-Vernet techniques of ureteric reimplantation"

"B/l congenital VUR" is read as bilateral primary (congenital) VUR (vesicoureteral reflux), often detected antenatally. STING is taken as subureteric Teflon injection, the endoscopic treatment of VUR.

Definition and epidemiology

- **VUR**: retrograde flow of urine from the bladder into the upper tract.
- **Primary**: congenitally deficient UVJ (ureterovesical junction) flap valve with a short intravesical tunnel and a normal bladder. **Secondary**: high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).

- Found in about **30% of children with UTI (urinary tract infection)**, about 30% of siblings and about 17% of children imaged for hydronephrosis. The younger the child, the likelier VUR.
- Infant boys predominate (higher voiding pressures, more UTI if uncircumcised); overall girls are affected twice as often. Less common in Black children.
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more.
- **Infection, not sterile reflux, injures the kidney.** Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia. So management targets infection risk, not the reflux itself.

Embryology of the VUJ (vesicoureteric junction)

1. **Ureteric bud (weeks 4–6)** arises from the caudal mesonephric (Wolffian) duct: ureter and duct form a "Y", and the common nephric (excretory) duct is the stem.
2. **Reciprocal induction:** the bud enters the metanephric blastema, branches into the collecting system and induces nephrons.
3. **Weeks 5–6:** the urorectal septum divides the cloaca into urogenital sinus (UGS) and anorectal canal.
4. **Absorption:** the common nephric duct is taken into the UGS (current view: removed by **apoptosis**), so ureter and duct open separately.
5. **Rotation (Mackie and Stephens):** the ureteric orifice moves **cranially and laterally** to the bladder base; the duct stays caudal as the ejaculatory duct, ending in the prostatic urethra (Gartner duct in females).
6. **Trigone:** classically ureteral; lineage tracing shows it is mostly bladder muscle. The **Chwalla membrane** over the orifice ruptures; its persistence explains ureterocele.
7. **Maturation:** ureteric muscle joins the detrusor as **Waldeyer's sheath**; the tunnel lengthens with growth, so VUR resolves with age.

Bud origin	Orifice	Result
Normal	Trigone corner	Tunnel length to diameter about **5:1** (Paquin); refluxing junctions about 1.4:1
Too low (early budding, early contact with UGS)	High, **lateral**	Short tunnel, so **primary VUR**; abnormal bud–metanephros contact gives **dysplasia**
Too high (late budding, late contact)	Caudal (distal), **medial**	**Ectopic ureter** (bladder neck, urethra, Wolffian structures), often obstructed
Two buds	Duplex	**Weigert–Meyer rule:** upper pole caudal-medial (ectopia, ureterocele); lower pole lateral (reflux)

- **Concept:** bud position fixes both orifice and kidney, so this single embryological fault (lateral orifice, short tunnel, dysplasia) explains why high-grade VUR and dysplasia coexist.

Clinical anatomy of the VUJ

- **Course:** juxtavesical ureter in **Waldeyer's sheath**, then an oblique intramural segment through the detrusor hiatus, then the submucosal tunnel, then the orifice (caudal and medial to the entry).
- **Intramural length 1.2–2.5 cm in adults** (0.5–0.8 cm in neonates). The **narrowest part** of the ureter (about 3–4 mm): the commonest stone impaction site.

- **Trigone:** superficial trigone (ureteric longitudinal muscle, running to the verumontanum); deep trigone (Waldeyer's sheath plus detrusor, to the bladder neck). Its base is the interureteric ridge. Waldeyer's sheath anchors the intramural ureter to the deep trigone.
- **Orifice (Lyon):** position A (medial trigone) to C (lateral); shape cone (normal), stadium, horseshoe, **golf-hole**. The more lateral and open, the more reflux.
- **Relations:** the vas (male) or uterine artery (female) crosses the ureter. **Pelvic plexus nerves lie dorsomedial to the VUJ**; wide extravesical dissection causes voiding dysfunction.
- **Clinical:** Hutch diverticulum at the hiatus; "fish-hook" ureter with BPH (benign prostatic hyperplasia).

Normal antireflux mechanism

- Balance of three factors: **ureteric peristalsis**, **UVJ anatomy** and **bladder dynamics** (low bladder pressure).
- **Passive flap valve:** the submucosal (intramural) ureter is compressed against firm detrusor backing as the bladder fills. This needs adequate tunnel length and fixation.
- **Paquin:** tunnel length to diameter about **5:1** normally versus 1.4:1 in refluxing units. Orifice shape also matters.
- **Active:** ureteric peristalsis opens the tunnel for each bolus, then it closes.
- **Bladder pressure:** high storage or voiding pressure overwhelms a normal valve.

Etiopathogenesis

1. Embryology (Mackie and Stephens): early budding gives a high, lateral orifice with a short tunnel (VUR) and dysplasia; late budding gives a distal, medial, ectopic and often obstructed ureter (see above).

2. Primary versus secondary

Primary	Secondary
Congenitally deficient UVJ: short tunnel, weak detrusor backing, lateral orifice	Normal UVJ overwhelmed by pressure: PUV, neurogenic bladder (VUR in up to 30% of spina bifida), dysfunctional voiding; or UVJ anomaly (ureterocele, duplication, paraureteral diverticulum)

- They overlap: "primary" VUR often has a treatable secondary component.

3. Bladder and bowel dysfunction (BBD): high voiding pressures and incomplete emptying distort the UVJ. Detrusor overactivity is the commonest urodynamic finding. Infant boys have high voiding pressures that settle with maturation.

4. Genetics: sibling prevalence about 30%, highest under 3 years; identical twins almost fully concordant; polygenic.

5. Infection: cystitis lowers the reflux threshold; *E. coli* endotoxin causes ureteric atony.

Pathogenesis of renal damage

	Reflux-associated dysplasia	Acquired scar
Cause	Abnormal bud (congenital)	Infected urine plus VUR
Typical patient	Boy, grade IV–V, antenatal hydronephrosis	Girl, recurrent febrile UTI, BBD
Preventable	No	Yes

- **Intrarenal reflux** through flat compound papillae at the poles needs only 2 mmHg in neonates, so scars are polar.
- **Sterile reflux rarely scars** unless pressures are high (Ransley and Risdon). Most scars follow early infections ("big bang", now debated).
- Sequelae: hypertension in **34–38%** of scarred children; CKD; poor growth; pre-eclampsia.
- **Also know:** massive bilateral VUR gives the megacystis–megaureter association.

International grading (International Reflux Study, 1981; VCUG (voiding cystourethrogram))

Grade	Findings
I	Non-dilated ureter only
II	Ureter, pelvis and calyces reached; no dilatation, sharp fornices
III	Mild–moderate dilatation of ureter, pelvis and calyces; minimal forniceal blunting
IV	Moderate dilatation and tortuosity; fornices obliterated but **papillary impressions kept**
V	Gross dilatation and tortuosity; **papillary impressions lost**; intrarenal reflux

- **Also know:** grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

Evaluation (children)

- History of febrile UTIs and BBD; urine culture; BP (blood pressure), growth, spine, urine protein, creatinine.
- **RBUS (renal and bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG** (gold standard): grade and urethra (excludes PUV). **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid) scan**: scars and split function; a defect persisting 6 months is a scar.
- **UDR** predicts resolution and breakthrough UTI better than grade.

Natural history

- Resolution comes from **elongation of the intravesical ureter** with growth.
- Resolution falls with grade and age: grade II 63–85%, grade III about 50%, **grades IV–V and bilateral grade III rarely** (International Reflux Study). Younger age resolves better; infants with grade IV–V may still resolve fairly well.
- **Bilaterality and older age delay resolution:** persistence is highest in bilateral grade IV (AUA (American Urological Association) 1997: about 10% resolve in 5 years), and in bilateral grade III at 6–10 years.
- **Consequences of bilateral damage:** adult hypertension in 34–38% of scarred children; CKD, with reflux nephropathy a leading paediatric transplant diagnosis; poor growth; pre-eclampsia (worse with bilateral scars). Correction cannot reverse dysplasia.

Risk stratification (Hidas)

Low risk	High risk
Male, >5 years	Female, <1 year (infant)
Unilateral, grade I–II	**Bilateral**, grade III–V

Low risk	High risk
No BBD, no scars	BBD, scars, multiple febrile UTIs

- Two-year breakthrough UTI on CAP (continuous antibiotic prophylaxis): 8.6% low, 26% intermediate, 62.5% high risk.

Recent trends in management

1. **Treat infection risk, not reflux:** risk stratification by sex, age, grade, laterality, BBD, scars and presentation (Hidas model).
2. **Fewer VCUGs:** ultrasound first (AAP (American Academy of Pediatrics) 2011, NICE (National Institute for Health and Care Excellence)); or "top-down" DMSA first. Contrast-enhanced voiding urosonography avoids radiation.
3. **Objective measures:** UDR predicts resolution and breakthrough UTI better than grade; machine-learning models.
4. **Selective CAP:** RIVUR halved recurrent UTI but did not reduce scarring; NNT (number needed to treat) 5 in high-risk and 18 in low-risk children. The Swedish Reflux Trial showed benefit in girls with dilating VUR.
5. **BBD first:** timed voiding, constipation treatment, anticholinergics. Topical steroid for phimosis in boys.
6. **Endoscopic injection:** Dx/HA (dextranomer/hyaluronic acid); HIT and double HIT (hydrodistension implantation technique) give about 90% resolution. Vantris works better but can obstruct late.
7. **RALUR-EV (robot-assisted extravesical reimplantation):** less pain and shorter stay; results approaching the open benchmark of **95.9%**.
8. **Less follow-up imaging:** VCUG may be omitted after surgery for low-grade VUR if ultrasound is normal.

Medical management

- **Aim:** prevent febrile UTI and scars while VUR resolves.
- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed. BBD raises breakthrough UTI and lowers resolution and endoscopic success.
- **CAP:** one low dose at night, a quarter of the treatment dose.

Agent	Age	Dose	Note
Trimethoprim or amoxicillin	<2 months (young infants)	—	Avoid sulfonamide (neonatal jaundice)
TMP-SMX (trimethoprim-sulfamethoxazole)	>2 months	2 mg/kg TMP	Avoid under 2 months (kernicterus); used in RIVUR
Nitrofurantoin	>2 months	1–2 mg/kg	Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung, liver toxicity

- **RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux, 2014)**, 607 children: recurrent UTI fell from **23.6% to 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD and in high risk.
- **Swedish Reflux Trial** (grade III–IV, age 1–2 years): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.

- **IRSC (International Reflux Study in Children) and Birmingham trials:** surgery equalled medical care for (new) scarring. **PRIVENT:** 6% absolute UTI reduction.
- **Circumcision** or topical steroid for phimosis: about 4 boys with VUR circumcised prevent one UTI (NNT about 4).
- After stopping CAP, 8–10% have a febrile UTI.
- Low-risk children: observation without CAP.
- **Guidelines:** AUA 2010/2017 recommends CAP for VUR with BBD and treating BBD first; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology) give high-risk children CAP and correct early after breakthrough infection.

Indications for correction

- Breakthrough febrile UTI on CAP despite good bladder and bowel habits.
- New scars or falling function.
- Persistent high-grade (IV–V) bilateral VUR (girls traditionally corrected before puberty; benefit unproven).
- Non-adherence or intolerance to CAP; parental choice.
- **Controversy:** asymptomatic bilateral high-grade VUR in an older boy with normal kidneys can be observed.

Case: 8-year-old boy with bilateral grade IV VUR

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars:** observation or CAP with BBD therapy. **Breakthrough UTI, scars or falling function:** correct surgically.
- **Choice here: bilateral Cohen.** `Exclude PUV/neurogenic bladder, treat BBD -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA) No -> observe or CAP; correct if UTI/scar`

Endoscopic correction: STING procedure

Definition and principle

- **STING (subureteric Teflon injection):** endoscopic injection of a **bulking agent** beneath the ureteric orifice. The mound supports and lengthens the intramural tunnel, restoring the **flap valve**.
- First done with Teflon by Matouschek (1981); popularised by **O'Donnell and Puri, 1986**. **Dx/HA (Deflux)** is now standard.

Bulking agents

Agent	Particle / property	Problem
Teflon (PTFE (polytetrafluoroethylene) paste)	90% <40 µm, permanent	**Migration**; never FDA (US Food and Drug Administration) approved
Dx/HA (Deflux)	Microspheres **80–250 µm**; gel resorbed	Volume loss about 23%; late recurrence; FDA approved 2001; standard agent
Polyacrylate (Vantris)	300 µm, permanent	Works better but late ureteric obstruction

- **Also know:** Macroplastique.

Indications

- Persistent primary VUR (grades II–IV) with **breakthrough febrile UTI** on prophylaxis, **new scars**, poor adherence or parental preference.
- Older girls; residual VUR after reimplantation; duplex or secondary VUR (lower success).
- **Relative contraindications:** grade V, **UVJ obstruction**, untreated BBD, large diverticulum.

Technique

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING:** needle enters **2–3 mm below the orifice at 6 o'clock**, bevel up, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.
4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding (the implant extrudes). Day case.

Technique	Needle entry	Result
STING	2–3 mm below the orifice, at 6 o'clock	Baseline
HIT (hydrodistension implantation technique)	Inside the hydrodistended orifice	Better than STING, all grades
Double HIT	Two intraureteric sites	About 90% resolution

Results

- Low-grade VUR about 90% after one injection.
- **Elder 2006 meta-analysis** (one injection): grade I–II about 79%, III 72%, **IV 63%**, V 51%; repeat injections raise success.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after initial success.
- Less effective in adults.

Complications

- **Failure or recurrence** (commonest): repeat injection once or twice, then reimplant (reimplantation is not harder after Dx/HA).
- **New contralateral VUR** after unilateral injection.
- **Ureteric obstruction:** early or **late-onset**, so long-term ultrasound.
- Febrile UTI despite a negative VCUG.
- **Calcified mound** (over half) mimics a ureteric stone. Teflon migration.

Follow-up

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound with or without VCUG at 3 months.

Ureteric reimplantation: principles and techniques

- **Principles:** exclude secondary VUR; gentle handling; tension-free; straight course; tunnel **5:1** (Paquin's rule).
- Open success **95.9%** overall (Elder 1997; AUA meta-analysis benchmark).

Technique	Method / tunnel	Strength / weakness
Cohen (intravesical)	Cross-trigonal	Commonest; success over 98%; one cystotomy for both sides; later ureteroscopy or retrograde access hard
Politano–Leadbetter (intravesical)	New superomedial hiatus above the old one	Long tunnel; blind hiatus risks bowel injury
Glenn–Anderson (intravesical)	Orifice advanced towards the bladder neck	Least kinking; short tunnel
Lich–Gregoir (extravesical)	Detrusorotomy, ureter laid in	No haematuria or spasm; no cystotomy; **retention up to 10% after bilateral repair** (pelvic plexus injury); stay close to the ureter
RALUR / RALUR-EV (robot-assisted laparoscopic ureteral reimplantation, extravesical)	Y-shaped detrusorotomy	Less pain, shorter stay; results approaching open
Paquin	Combined extra- and intravesical, suprahiatal	Long tunnel; for dilated ureters
Gil-Vernet	Intravesical trigonoplasty, no tunnel	Low-grade bilateral VUR

Paquin technique

- **Combined extravesical–intravesical** reimplant described by Paquin (1959), with the 5:1 rule.
- Steps: 1. Ureter mobilised **extravesically** and divided at the bladder. 2. Bladder opened; a **new hiatus** created higher on the posterior wall (**suprahiatal**). 3. Long **submucosal tunnel** from the new hiatus down to the original orifice site, length about 5 × diameter. 4. Ureter drawn through, spatulated, sutured to mucosa and anchored to detrusor. 5. Old hiatus closed.
- **Advantages:** long tunnel under direct vision; good for **dilated ureters and megaureter** (can combine with tapering).
- **Disadvantages:** kinking or obstruction at the new hiatus.

Gil-Vernet trigonoplasty

- **Intravesical, with no tunnelling and no ureteric mobilisation.**
- Steps: 1. Bladder opened. 2. **Transverse mucosal incision** across the trigone between the two orifices. 3. Intramural ureters with their muscle sheath **advanced medially** by one non-absorbable or absorbable stitch across the midline. 4. Mucosa closed.
- **Effect:** lengthens and medialises the intravesical ureters; corrects both sides at once.
- **Indications:** low to moderate grade (I–III), usually bilateral VUR, with non-dilated ureters.
- **Advantages:** quick, simple, low obstruction risk; easy later retrograde access.
- **Disadvantages:** unsuitable for high grade, dilated ureters or paraureteral diverticulum.

Feature	Paquin	Gil-Vernet
Approach	Extra- + intravesical	Intravesical
Ureter divided	Yes	No
Tunnel	New long submucosal	None (advancement)
Best for	High grade, dilated	Low grade, bilateral

Criteria for selecting the technique

Situation	Preferred option
Grades II–IV, parents want least invasive	Dx/HA injection (lower success at higher grade)
Bilateral VUR, small or trabeculated bladder	**Cohen** (both sides through one cystotomy)
Grade V, dilated ureter	Intravesical reimplant with or without tapering (Politano–Leadbetter or Cohen)
Unilateral, older child	Lich–Gregoir or RALUR-EV
Bilateral extravesical	Nerve-sparing dissection; warn about retention
Duplex system	Common sheath reimplant
Failed injection	Reimplant (not harder after Dx/HA)
Neurogenic or PUV bladder (secondary VUR)	Treat the bladder first (anticholinergics, CIC (clean intermittent catheterisation))

Complications and follow-up after reimplantation

- Ultrasound at 6–12 weeks.
- Complications: transient obstruction; persistent reflux (usually resolves by 1 year); contralateral VUR (about 1 in 10, usually resolves); leak; retention after bilateral extravesical repair.
- Obstruction early (oedema): stent or nephrostomy; late (ischaemia, kink): redo.
- Lifelong BP and renal checks if scarred; scarred women risk pre-eclampsia.

Long-term outcome

Outcome	Facts
Renal scarring	About 15% get a febrile UTI within 2 years; about 15% of those scar. Risk highest under 1 year. Much "scarring" in boys with high grade is **congenital dysplasia**, not preventable by any treatment
Hypertension	Reflux nephropathy may be the commonest cause of childhood hypertension after essential hypertension (renin–angiotensin). **34–38%** of scarred children are hypertensive as adults. Correcting VUR does not lower BP; nephrectomy of a small scarred kidney helps selected cases
CKD and ESKD (end-stage kidney disease)	Now rare: RRT (renal replacement therapy) from reflux nephropathy fell from 1.9 to 0.6 per million (USRDS (United States Renal Data System), 1996–2014). Bilateral scars cause hyperfiltration, proteinuria, concentrating defect, tubular acidosis
Pregnancy	VUR in late pregnancy: pyelonephritis about one-third versus under 5%. Scars: about **3× hypertension, 8× pre-eclampsia**; bilateral worse. Impaired function: fetal loss. Reimplanted women still get UTIs

Outcome	Facts
Growth; offspring	Catch-up growth after infection control; about 43% of screened offspring have VUR

- **Concept:** in adult life **the scar, not the reflux, carries the risk**, so correction does not remove it. Scarred patients need lifelong BP, proteinuria and creatinine checks.

Grade 3 VUR in an adult

- **Grade 3:** mild–moderate dilatation of ureter, pelvis and calyces, minimal forniceal blunting.
- **Adults rarely lose function from reflux alone**, unless high bladder pressure, stasis and UTI coexist. The **scar**, not the reflux, carries the risk. Spontaneous resolution is unlikely.
- **Causes:** persistent primary VUR; **secondary**: BOO (bladder outlet obstruction), neurogenic bladder, resection near the orifice, ureterocele incision, paraureteral diverticulum, transplant ureter, TB (tuberculosis).
- **Presentation:** recurrent febrile UTI, pyelonephritis (especially in pregnancy), loin pain on voiding, hypertension, proteinuria, CKD.
- **Evaluation:** VCUG (grade, bladder, diverticulum, outlet); ultrasound and **DMSA** for scars and split function; MAG3 (mercaptoacetyltriglycine) renogram if obstruction is suspected; **UDS (urodynamics)**: compliance, voiding pressure, residual; cystoscopy; blood pressure, creatinine, urine protein.
- **Treat secondary causes first** (relieve BOO; CIC and antimuscarinic for neurogenic bladder), then reassess.

Situation (primary grade 3)	Management
Incidental; no UTI, no scars	Observe: timed voiding, bowel care, prompt UTI treatment
Recurrent febrile UTI, new scars, falling function	**Correct the reflux**
Woman planning pregnancy with scars or prior pyelonephritis	Consider correction before pregnancy
Poorly functioning scarred kidney with infection or hypertension	Nephrectomy

- Low-dose prophylaxis selectively; adult evidence is limited.
- **Correction:** endoscopic Dx/HA injection (day case, repeatable, but less effective in adults); or **ureteroneocystostomy** (open, laparoscopic or robotic). Lich–Gregoir/RALUR-EV suits adults (no cystotomy; stay close to the ureter to protect nerves); Politano–Leadbetter (long tunnel); Cohen (over 98%; later retrograde access hard).
- **Psoas hitch** if length is needed; tapering seldom needed in grade 3.
- Follow-up: ultrasound at 6–12 weeks; lifelong blood pressure and renal checks if scarred.

Overall management algorithm

```
Treat BBD -> low risk: observe | high risk (bilateral): CAP Breakthrough UTI / new scar / persistent IV-V -> Dx/HA injection or reimplant (Cohen) Scars -> lifelong BP and proteinuria checks
```

Diagrams to draw:

- "Y" of ureteric bud and mesonephric duct with orifice migration (early-budding Y diagram).

- Long oblique tunnel versus short lateral refluxing tunnel; oblique intramural ureter in Waldeyer's sheath with submucosal tunnel and detrusor backing.
- Grades I–V.
- Cohen cross-trigonal reimplant; Gil-Vernet transverse trigonal incision and medial advancement stitch.
- Needle sites: STING below the orifice, HIT and double HIT inside the tunnel, with volcano mound.

Key points to remember:

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

Also, from the condensed book (not in the paper answer):

- Abnormal bud position also causes hypoplasia, so some "scars" are congenital.

3 · Primary vs secondary VUR

Secondary reflux needs the bladder treated first.

See the paper answer ***Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING*** *(Section 4, Ch 41).*

Also, from the condensed book (not in the paper answer):

- In spina bifida, VUR occurs in up to **30%**: start early clean intermittent catheterisation (CIC) and an anticholinergic or β3-agonist.
- **Reflux at low pressure or in early filling means poor resolution.** Treating detrusor overactivity improves resolution.

> **KEY POINT**
>
> Primary and secondary are not exclusive: "primary" VUR often has a treatable secondary component (overactivity, bladder and bowel dysfunction). Remove it and only true primary VUR remains.

4 · Epidemiology

- VUR is found in **~30%** of children with UTI and ~17% of those imaged for hydronephrosis. It is present in **38.6%** under 2 years versus 7.6% at 12–21 years.
- Girls are affected ~2× overall, but infant boys predominate.

5 · Grading — International Reflux Study (1981)

Grade	Ureter	Pelvis and calyces
I	Non-dilated, ureter only	Not reached
II	Not dilated	Reached, no dilatation
III	Mild–moderate dilatation	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate tortuosity	Moderate dilatation
V	Gross dilatation and tortuosity	Gross dilatation; **papillary impressions lost**

- Grades I–II are non-dilating and III–V dilating. Radionuclide cystography (RNC) uses 3 grades.
- **Ureteral diameter ratio (UDR)**, the largest distal ureteric diameter divided by the L1–L3 vertebral distance, predicts resolution and breakthrough UTI (BUTI) better than grade.

6 · Diagnosis

When to image after UTI

- **AAP 2011**, febrile UTI at 2–24 months: do **ultrasound (US) first**, and a voiding cystourethrogram (VCUG) only if US is abnormal or after a **second febrile UTI**.
- **NICE**: US, with VCUG mainly for atypical or recurrent UTI in younger children.
- **Top-down**: DMSA scan first, VCUG only if abnormal. A normal DMSA leaves a **<1%** chance of high-grade VUR but misses ~1/3 of (mostly low-grade) VUR.

Cystographic methods

*See the paper answer **VCUG** (Section 1, Ch 5).*

Also, from the condensed book (not in the paper answer):

- Direct RNC is more sensitive for grades II–V but **misses grade I**.
- Suspect coexisting **UVJ obstruction** with a distal aperistaltic segment, dilute contrast and delayed drainage. It is a **relative contraindication to endoscopic injection**.

Upper tract

- **Renal and bladder ultrasound (RBUS)** is the serial mainstay but cannot exclude VUR.
- **DMSA** is the gold standard for cortical defects (sensitivity 98%, specificity 92%); a defect that persists is a scar.

7 · Congenital dysplasia vs acquired scar

Only the acquired scar is preventable.

*See the paper answer **Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING** (Section 4, Ch 41).*

Also, from the condensed book (not in the paper answer):

- Hypertension develops in **34–38%** of scarred children. **Correcting VUR does not lower blood pressure**; removing a small kidney or segment may.

8 · Bladder and bowel dysfunction (BBD)

BBD is the main modifiable factor.

- **Treat constipation first**; it causes sphincter co-contraction.
- BBD brings more UTI on prophylaxis and after surgery, less resolution, and **lower endoscopic success, but not open surgical success** (AUA).
- **Phimosis**: in boys with VUR, circumcision cuts the number needed to treat (NNT) to prevent one UTI from ~111 to **~4**. Topical steroid is equivalent.

9 · Natural history and resolution

Grade	Spontaneous resolution
I–II	Most; grade II 63–85%
III	~50%; ~90% of resolutions within 4 years
IV–V, bilateral III	Few (infants with IV–V resolve well in year 1)

- Resolution favours **low grade, young age, unilateral reflux, no BBD, voiding-only reflux, higher bladder volume at onset and low UDR**.

10 · Medical management — continuous antibiotic prophylaxis (CAP)

CAP failure is the main trigger for surgery.

*See the paper answer **Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING** (Section 4, Ch 41).*

Also, from the condensed book (not in the paper answer):

- Give a low dose **at night**; avoid TMP-SMX in neonates (bilirubin displacement).
- **If febrile UTI breaks through CAP despite good bladder and bowel habits, stop medical management and correct the VUR.**

Risk stratification (Hidas)

Factor	Low risk	High risk
Sex / age	Male; >5 y	Female; <1 y
Laterality / grade	Unilateral; I–II	Bilateral; III–V
BBD / scars	No	Yes
Presentation	Prenatal hydronephrosis	Multiple febrile UTIs

- Two-year BUTI on CAP is **8.6% low, 26% intermediate, 62.5% high** risk. About two-thirds of children are low risk.

11 · Practical algorithm (AUA-based, risk-stratified)

ALGORITHM

1. Confirm true febrile UTIs; do RBUS. If multiple febrile UTIs or upper-tract findings, then VCUG.
2. In all, treat BBD (constipation first) with timed voiding.
3. If low risk (grade I–II, no BBD, male, older), then observe with or without CAP and treat febrile UTI promptly.
4. If high risk (grade ≥III, girl, BBD, scars, infant), then give CAP.
5. If breakthrough febrile UTI, new scars, or persistent high-grade VUR, then correct (endoscopic or reimplant).

- CAP helps little at grade ≤II without BBD, but helps at ≥III, at least in girls.

12 · Surgical correction

Surgery stops reflux but does not protect kidneys better.

- Open correction succeeds in **>95%**, but neither open nor robotic surgery beats medical therapy for scarring or growth. The adult UTI risk persists.
- **Endoscopic dextranomer/hyaluronic acid (Dx/HA)**: BBD lowers success; avoid it with UVJ obstruction. Calcified mounds can **mimic ureteric stones**.
- After a cross-trigonal reimplant (Ch 53), **retrograde ureteric access is difficult**.

13 · Associated conditions

- **UPJO with VUR**: **correct the obstruction first**; repairing both at once risks the ureteric blood supply.
- In a **duplex** kidney, VUR usually enters the **lower pole** (Weigert–Meyer).
- A **ureter entering a paraureteral diverticulum** will not resolve; a diverticulum beside the orifice resolves like primary VUR.
- Contralateral VUR with a multicystic dysplastic kidney usually resolves.
- **Megacystis–megaureter** (massive VUR recycling urine into a large thin bladder) and **obstructive refluxing megaureter** are treated by reimplant, not injection or outlet surgery.

14 · Sibling screening

- Sibling prevalence is ~**30%**, higher when younger.

- The target is **renal damage, not VUR**. AUA: **VCUG for siblings with a cortical abnormality or renal asymmetry on US, or a UTI history**; otherwise observe.

15 · Follow-up and adulthood

- If scarred, check BP, proteinuria and renal function for life.
- **Pregnancy**: scars bring ~3× hypertension and **~8× pre-eclampsia**. **Scarring, not VUR, drives morbidity.**

16 · Landmark trials

Trial	Design	Result	Take-home
IRSC	<9 y, high grade; CAP vs open surgery	UTI equal (38%); surgery fewer pyelonephritis; scars, 10-y function equal	Surgery ≈ medical for scarring
Birmingham Reflux Study	High grade; medical vs surgical, 5 y	New scars equal, all within 2 y	Equal; supports "big bang"
Swedish Reflux Trial	1–2 y, grade III–IV; CAP vs endoscopic vs surveillance	Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; boys no difference	Treat girls with dilating VUR
RIVUR	607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo	Recurrent UTI **12.9% vs 23.6%**; scarring unchanged	CAP prevents UTI, not scars
PRIVENT	576 children with UTI; CAP vs placebo	Absolute UTI reduction 6% (19% to 13%)	Modest benefit
French / low-risk trials	Grade I–III; CAP vs none	No significant difference	No benefit at low risk

> **KEY POINT**
>
> Trials disagree because of population risk: RIVUR was diluted by low-risk children; reanalysis gives NNT 5 high-risk vs 18 low-risk. Neither CAP nor surgery prevents scarring better.

49 Anomalies of the Kidney and Collecting System

1 · Embryology that explains everything

- The ureteric bud (UB) arises from the caudal wolffian duct (WD) at **5–7 wk**, invades the metanephric mesenchyme (MM) and branches; its tips induce nephrons (**reciprocal induction**).
- In weeks 6–9 the kidney ascends and rotates 90° ventromedially, so the pelvis faces medially.
- Renal anomalies travel with WD (male) and müllerian duct (MD, female) anomalies; the adrenal develops independently.

> **KEY POINT**
>
> Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet.

2 · Bilateral renal agenesis (BRA)

BRA is uniformly lethal, so recognise it and counsel.

- It occurs in **≈1 in 5000** births, **75% male**. First-degree relatives carry ≈1000× risk, so **scan parents and siblings by ultrasound (US)** of any infant with unilateral agenesis (URA), BRA or dysgenesis.
- The ureters are atretic and the bladder absent or hypoplastic. The flat adrenal gives the **"lying down" sign**.
- **Potter facies** and limb deformities are **deformations** from absent amniotic fluid, not malformations. Undescended testes (43%) and gastrointestinal anomalies (≈60%) are common.
- **Pulmonary hypoplasia** is the cause of death. **Peters two-step model**: a kidney-derived "renal factor" acts before 16 wk, then amniotic fluid in the canalicular phase (16–26 wk), so fluid alone only partly restores the lungs.
- **Diagnosis**: prenatally, severe oligohydramnios, no kidneys and a small chest. In a neonate, **anuria >24 h without bladder distension** means BRA until proved otherwise; confirm with US and power Doppler.
- **Prognosis**: ≈40% are stillborn and the rest die in 24–48 h. Amnioinfusion is experimental (RAFT, §9) and needs ethics oversight with full counselling.

> **KEY POINT**
>
> BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal end-stage renal disease (ESRD).

3 · Unilateral renal agenesis (URA) and the solitary functioning kidney (SFK)

The issues are genital anomalies and lifelong hyperfiltration.

- URA occurs in **1 in 1200**, male 1.8:1. Many apparent URAs are an **involuted multicystic dysplastic kidney (MCDK)**.
- Associations include VACTERL, MURCS, Turner, Kallmann, branchio-oto-renal and DiGeorge syndromes. **Maternal diabetes triples the risk**. An isolated preauricular pit or single umbilical artery needs no US; two or more anomalies do.
- **The ipsilateral ureter is absent in ≈60%**. The defect is in the UB, so the gonad is rarely affected.
- **Contralateral kidney**: vesicoureteral reflux (VUR) **≈30%**, pelviureteric junction (PUJ) obstruction ≈11%, vesicoureteric junction (VUJ) obstruction ≈7%.
- **Genital anomalies occur in 20–40%** (females 25–50%, males 10–15%).

	Male (WD derivatives)	Female (MD derivatives)
Always present	Testis, head of epididymis	Fimbriated end of tube
Commonest	Absent vas/epididymal body-tail/seminal vesicle (≈50%). **79% of men with absent vas lack the ipsilateral kidney**	Unicornuate/bicornuate, didelphys or septate uterus; vaginal atresia/duplication; Gartner duct cyst with blind ectopic ureter
Named syndrome	**Zinner**: seminal vesicle cyst and ipsilateral agenesis	**OHVIRA** (obstructed hemivagina and ipsilateral renal anomaly; Herlyn–Werner–Wunderlich): pubertal haematocolpos, pain
Trigger to image kidneys	Impalpable vas (infertility, herniotomy/orchidopexy)	Any uterine anomaly; pelvic US after thelarche, before menarche

- **Magee classification**: type I (before wk 4) gives a unicornuate uterus; type II (early wk 4) gives didelphys with obstructed hemivagina on the agenesis side; type III (after wk 4) gives isolated URA with normal genitalia.
- **MRKH** (46,XX müllerian agenesis): **type II has renal anomalies in 40–60%**; type I rarely does.
- **Work-up**: compensatory hypertrophy is seen from 20 wk in ≈90%. DMSA finds a missed ectopic kidney. A voiding cystourethrogram (VCUG) for contralateral VUR is **not routine**. Sport is not precluded.

Prognosis and surveillance of the SFK

- **Brenner hyperfiltration**: reduced nephron mass leads to glomerular hypertension, then albuminuria and sclerosis. Risk is highest when the SFK itself has a congenital anomaly (CAKUT), low birth weight or prematurity.
- In KIMONO, ≈⅓ had renal injury by ≈10 y, so follow-up is lifelong.

Situation	Blood pressure (BP) and first-morning urine albumin	Ultrasound	Creatinine
SFK without CAKUT	Annually	Every 5 y if glomerular filtration rate (GFR) >60 and no medication	Periodic, into adolescence
SFK with CAKUT	Twice yearly	As indicated	Periodic
GFR <60 or on antihypertensive/antiproteinuric	BP, albumin, creatinine 2–4×/year		

Hypertension or microalbuminuria (>30 mg/24 h) calls for an ACE inhibitor, salt limitation and no excess protein.

4 · Supernumerary kidney

- It has its **own capsule, collecting system and separate blood supply**, unlike a duplex kidney. About half are hydronephrotic, with obstruction, infection or stones in early adulthood.

5 · Ectopia (anomalies of ascent)

An ectopic kidney is easily mistaken for agenesis or tumour.

Simple ectopia

- It occurs in **1 in 900** at autopsy and is bilateral in ≈10%. **Pelvic is commonest**.
- The kidney is small and malrotated (pelvis anterior). **Hydronephrosis is common**, half from obstruction (mostly PUJ). **VUR occurs in 30–50%**, often into the orthotopic kidney.
- The blood supply is anomalous: **map it by CT or MR angiography (MRA) before surgery**.
- Genital anomalies occur in 15–45%. Continuous incontinence in a girl suggests an ectopic ureter from the ectopic kidney.
- **Never remove a "pelvic mass" without imaging**: solitary pelvic kidneys have been excised as tumours.

Cephalad and thoracic kidney

- Cephalad ectopia is seen with **omphalocele**. A thoracic kidney lies in the **foramen of Bochdalek**, is usually silent, and needs CT or MRU rather than thoracotomy.

6 · Fusion anomalies

Fused kidneys bring obstruction, stones and variable vessels.

Crossed renal ectopia

- The kidney lies opposite its ureteric orifice; **90% are fused**, making it the second commonest fusion anomaly. Left-to-right crossing is 3:1, and the ureteric orifice is orthotopic.
- **McDonald–McClellan types**: with fusion (90%), without fusion, solitary crossed and bilateral crossed (rarest).

Crossed fused form	Key feature
Inferior ectopia (≈⅔, commonest)	Crossed upper pole fused to normal lower pole; pelves anterior
Sigmoid (S-shaped)	Second commonest; pelves face opposite ways
Lump (cake)	Lobulated pelvic mass
L-shaped (tandem)	Crossed kidney transverse, anterior to L4
Disc (pancake)	Medial fusion, reniform outline kept
Superior ectopia	Least common; crossed lower pole to normal upper pole

- An **anterior pelvis means early fusion**; a medial pelvis means late fusion. VUR occurs in ≈20%.

- **A stone in the crossed ureter hurts on the anephric side**; pyelonephritis or PUJ obstruction hurts where the kidneys lie. Obtain MRU or MRA before major surgery.

Horseshoe kidney

- It is the **commonest fusion anomaly**: **1 in 400** (0.25%), male 2:1. **Lower-pole fusion occurs in 95%**, before rotation.
- The **isthmus** lies at L3/L4, **just below the IMA**, in front of the aorta and IVC.
- The calyces **point posteriorly**, and the ureter inserts high and bends over the isthmus. **The blood supply is highly variable**, so obtain MRA or CT before surgery.
- **≈60% of Turner patients have a horseshoe kidney**; it is also associated with neural tube, anorectal and VATER anomalies.
- **PUJ obstruction occurs in 13–34%**, but many hydronephroses are not obstructed. VUR occurs in 8–32%.
- **Stones affect ≈⅓ of adults**, so still do a metabolic work-up.
- **Wilms risk is 1.76–7.9×**; preoperative chemotherapy preserved ≈75% of parenchyma (NWTS). Renal cell carcinoma is not increased.
- At least 50% are asymptomatic. Pregnancy is unaffected and the kidney is transplantable.

> **KEY POINT**
>
> † PUJ obstruction in horseshoe kidney: dismembered pyeloplasty with dependent anterior anastomosis; symphysiotomy not routinely needed.

7 · Malrotation and vascular anomalies

- Malrotation occurs in ≈1 in 939. **Weyrauch** types: **ventral (no rotation, commonest)**, ventromedial, dorsal (rarest) and lateral (reverse). **The vessel course is the only clue** to direction.
- Hilar fibrosis can cause intermittent PUJ obstruction and stones; suspect malrotation when a stone lies in an unusual position.

SAQ Anomalies of renal artery.

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 2, Q3c

Normal pattern

- One renal artery from the aorta at L1–L2; anterior and posterior divisions; **five segmental end arteries** (apical, upper, middle, lower, posterior). The right artery passes behind the IVC (inferior vena cava).

Terminology

Term	Definition
Multiple renal arteries (preferred)	More than one vessel from the aorta to one kidney
Aberrant	Arises from a vessel other than aorta or main renal artery (iliac, SMA (superior mesenteric artery), gonadal)
Accessory	Two or more branches supplying the same segment

Incidence

- Single artery in **71–85%**; variants in 25–40%. Accessory arteries 25–28% (aorta T11–L4), usually to the **lower pole**. True aberrant vessels are rare except in **ectopic and horseshoe kidneys**.
- **Also know:** early prehilar branching; right artery anterior to IVC in about 5%.

Embryology

- The ascending kidney takes successive aortic branches; the lower ones normally degenerate. **Failure of degeneration** gives polar or multiple arteries.

Clinical significance

- **PUJ (pelviureteric junction) obstruction** from a lower-pole crossing vessel; but crossing vessels occur in a fifth to a quarter of normal PUJs.
- Compression of an infundibulum: hydrocalyx, pain, stones, UTI (urinary tract infection).
- **Surgery:** bleeding at endopyelotomy (incise posterolaterally); polar arteries are end arteries, so ligation infarcts that segment; **donor nephrectomy** needs CT (computed tomography) angiography; aortic surgery.
- Not linked to hypertension.

Renal artery aneurysm

- About **0.1%**. **Abeshouse types:** saccular (about 93%), fusiform, dissecting, arteriovenous.
- Mostly asymptomatic; hypertension, haematuria, pain; wreath-like calcification.
- **Treat** (endovascular or open) if rupture, **>2 cm**, expanding, **pregnancy or planned pregnancy**, symptoms or emboli.

Renal arteriovenous fistula

	Congenital (cirsoid AVM (arteriovenous malformation))	**Acquired**
Share	<25%	Majority (biopsy, PCNL (percutaneous nephrolithotomy), trauma)
Features	Bruit, haematuria, high-output failure	Often resolves
Treatment	Selective embolisation	Observe; embolise if bleeding

Diagram to draw: kidney with main artery, lower-pole accessory artery crossing the PUJ.

> **Key points to remember:**
> - Single artery in 71–85%; accessory arteries usually lower pole.
> - Crossing vessel can cause PUJ obstruction.
> - Treat aneurysm >2 cm or in pregnancy.

Also, from the condensed book (not in the paper answer):

- Treat an aneurysm for rupture, growth, pregnancy, size **>2 cm** or symptoms. An acquired AVF may close within months, so observe it.
- A congenital arteriovenous fistula presents with bruit, haematuria and high-output failure and rarely resolves.

8 · Collecting system anomalies

SAQ Pelvicalyceal anomalies other than PUJ obstruction

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3d

Introduction

- Congenital pelvicalyceal variants occur in about 4% of people.
- **Key question:** is there obstruction? Dilatation without obstruction needs no treatment; stasis causes stones and infection.

Anomalies

Anomaly	Features	Management
Calyceal diverticulum	Transitional-lined intrarenal cavity joined to a calyx (Type I, commoner, small) or pelvis (Type II, larger, symptomatic) by a narrow neck; 4.5 per 1000; congenital (persistent 3rd–4th generation ureteric branches) or acquired. **Stones in almost 40%**; milk of calcium	Asymptomatic: observe. Pain, infection, stones: ureteroscopic neck incision, **percutaneous ablation with fulguration**, laparoscopic marsupialisation; SWL (shock wave lithotripsy) ineffective
Hydrocalycosis	Dilated major calyx connected to the pelvis; infundibular obstruction by vessel, stenosis, scarring or achalasia	Symptomatic only: infundibulotomy or dilatation
Megacalycosis	**Non-obstructive**; malformed papillae; **12–20 calyces**; pelvis normal; male 6:1; normal diuretic renogram	**None**; may coexist with segmental megaureter
Infundibular / infundibulopelvic stenosis	Narrow infundibula ± small pelvis; usually bilateral, with VUR (vesicoureteric reflux), dysplasia; over a third reach renal failure (hyperfiltration)	Monitor creatinine and eGFR (estimated glomerular filtration rate); ACE (angiotensin-converting enzyme) inhibitors; surgery only if obstruction harms function: endoscopic, calicocalicostomy, ureterocalicostomy
Extrarenal pelvis	Large pelvis outside the sinus; mimics dilatation	None
Bifid pelvis	About 10%; normal variant	None
Unipapillary kidney	Single papilla and calyx; rare; usually hypoplastic kidney	None

- **Also know:** acquired infundibular stenosis (tuberculosis, post-PCNL (percutaneous nephrolithotomy)); Fraley syndrome (crossing artery on upper infundibulum).

Evaluation

- **Ultrasound:** cyst-like space; milk of calcium layers and shifts with position.
- **CT (computed tomography) urography or MRU (magnetic resonance urography): delayed contrast pooling** in a diverticulum.
- **Diuretic renogram (MAG3 (mercaptoacetyltriglycine)):** separates obstruction (hydrocalycosis) from non-obstruction (megacalycosis, extrarenal pelvis).

- **Retrograde pyelography:** maps the neck.

Differential diagnosis

- Obstructive hydronephrosis, papillary necrosis, pyelonephritic clubbing, tuberculosis, renal cyst.
- **Stones:** retrograde first (find, dilate or laser-incise the ostium); percutaneous for lower-pole diverticula.

Diagram to draw: kidney showing diverticulum with narrow neck, hydrocalyx and megacalyces.

> **Key points to remember:**
> - Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
> - Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
> - Infundibulopelvic stenosis: bilateral, renal failure risk.

Also, from the condensed book (not in the paper answer):

- Extrarenal pelvis: confirm with diuretic renography before calling it obstruction.
- Infundibulopelvic stenosis: monitor renal function yearly; if it falls, give an ACE inhibitor and restrict protein.

> **KEY POINT**
>
> Collecting-system anomalies split on one question: is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia.

9 · Landmark trials

Trial	Design	Result	Take-home
RAFT (Renal Anhydramnios Fetal Therapy)	Non-randomised multicentre; isolated BRA, serial amnioinfusion before 26 wk	94% live births; 82% survived ≥14 d with dialysis access; only 35% discharged on peritoneal dialysis; stopped early	Prevents lethal lung hypoplasia but leaves neonatal ESRD — research only

50 Renal Dysgenesis and Cystic Disease of the Kidney

1 · Genetics in one table

Gene	Renal phenotype
PAX2	Renal hypodysplasia or aplasia
EYA1	Branchio-oto-renal (BOR) syndrome
HNF1B	Renal cysts and diabetes syndrome
PKD1, PKD2 (primary cilium)	Polycystic kidney disease

2 · Dysgenesis: dysplasia vs hypoplasia

Dysplasia is the leading cause of childhood renal failure.

- **Dysgenesis** (abnormal size, shape or structure) includes dysplasia and hypoplasia.
- **Dysplasia** is a **histological** diagnosis of arrested metanephric differentiation. Its sine qua non is **primitive ducts** with fibromuscular collars; cartilage may also occur. Not all cysts are dysplastic, and not all dysplasia is cystic.
- Causes are primary UB failure (syndromes such as BOR and renal–coloboma) or fetal outflow obstruction. **It is the leading cause of childhood end-stage renal disease (ESRD).** Ultrasound (US) shows a small echogenic kidney with lost corticomedullary differentiation.
- **Hypoplasia** is reduced mass from too few cells; simple hypoplasia has a normal nephron number.

Entity	Defining feature	Course / management
Oligomeganephronia	Few, hypertrophied nephrons; usually bilateral	Polyuria, polydipsia, short stature; **proteinuria first sign**; ESRD in second decade. Fluids, correct salt loss/acidosis, protein 1.5 g/kg/d, ACE inhibitor; living related donor acceptable (not familial)
Ask-Upmark (segmental hypoplasia)	Mostly acquired reflux nephropathy; deep "slit-scars"	Girls ≥10 y, **severe hypertension**; partial/total nephrectomy cures blood pressure (BP); bilateral: medical, dialysis/transplant. Reflux correction does not cure hypertension
Hypodysplasia	With ectopic orifice, megaureter, ureterocele, posterior urethral valves (PUV), prune-belly	**Dysplasia is proportional to the degree of ureteric ectopia**: lateral orifice gives reflux-associated, medial/ureterocele gives obstruction-associated dysplasia

> **KEY POINT**
>
> The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together.

3 · Cystic disease: concepts and classification

Sort first by inheritance, then by the extrarenal clue.

- Cysts arise from tubules; the **primary cilium** is central.

- A **multicystic** kidney (MCDK) is dysplastic with no normal nephrons. A **polycystic** kidney starts normal, and its nephrons are progressively replaced by cysts.

Inheritable	Non-heritable
Autosomal recessive polycystic kidney disease (ARPKD)	Multicystic dysplastic kidney (MCDK)
Autosomal dominant PKD (ADPKD)	Benign multilocular cyst (cystic nephroma)
Juvenile nephronophthisis (recessive) / medullary cystic disease (dominant)	Simple cysts
Congenital nephrosis; familial hypoplastic glomerulocystic disease	Medullary sponge kidney; sporadic glomerulocystic disease
Syndromes: tuberous sclerosis, von Hippel–Lindau (VHL)	Acquired renal cystic disease; calyceal diverticulum

The extrarenal clue decides: hepatic fibrosis means ARPKD; berry aneurysm, ADPKD; retinitis pigmentosa, nephronophthisis; angiofibroma, tuberous sclerosis complex (TSC); haemangioblastoma, VHL.

4 · ARPKD

Huge echogenic kidneys and an always-fibrotic liver.

SAQ Autosomal recessive polycystic kidney disease.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2018-Summer Paper 3, Q3c

Definition and genetics

- **ARPKD (autosomal recessive polycystic kidney disease)**: bilateral, symmetrical renal enlargement from **dilated collecting ducts**, **always with congenital hepatic fibrosis**.
- Gene **PKHD1 on chromosome 6p**; protein **fibrocystin (polyductin)** on the **primary cilium** of tubular and bile-duct cells, so it is a **ciliopathy**.
- Incidence about 1 in 10,000–50,000 live births. Siblings have a 25% risk.
- "Infantile" is a misnomer: a **spectrum**; earlier onset means more severe disease.

Pathology

- Radially arranged dilated collecting ducts (**microcysts**).
- **Liver:** periportal fibrosis, biliary ectasia; **Caroli syndrome** if intrahepatic ducts dilate.

Clinical features

Age	Features
Fetus	Large echogenic kidneys, **oligohydramnios** → Potter facies, limb deformity, **pulmonary hypoplasia**
Neonate	Huge, smooth, hard flank masses; respiratory distress; rising creatinine; **30–50% die early**
Child	Hypertension, CKD (chronic kidney disease), growth failure, anaemia, osteodystrophy
Older child	**Portal hypertension**: varices, splenomegaly, cholangitis; liver function stays normal

- No association with renal tumours.

Investigations

- **Ultrasound:** bilateral very large, **homogeneously hyperechoic** kidneys (brighter than liver) from countless microcyst interfaces; discrete cysts appear with age.
- **Parents' ultrasound** normal (excludes ADPKD); **PKHD1 genetic testing**; liver ultrasound.
- **Differential:** neonatal ADPKD (autosomal dominant PKD), bilateral hydronephrosis, MCDK (multicystic dysplastic kidney), glomerulocystic disease, bilateral tumours, renal vein thrombosis.

Management (no cure)

- **Neonatal:** ventilation, fluid and electrolyte care, antihypertensives (ACE (angiotensin-converting enzyme) inhibitors).
- **CKD care:** nutrition, growth hormone, anaemia and bone treatment; dialysis and **kidney transplantation**.
- **Nephrectomy** (breathing, feeding, blood pressure, room for peritoneal dialysis): cautiously; **very early bilateral nephrectomy** (first 3 months) had the worst neurological outcomes (ARegPKD registry).
- **Liver:** variceal sclerotherapy, portosystemic shunt; **combined liver–kidney transplant** for severe liver disease.
- **Genetic counselling** and prenatal diagnosis.

ARPKD versus ADPKD

	ARPKD	ADPKD
Gene	PKHD1 (chromosome 6)	PKD1 (16), PKD2 (4)
Age	Perinatal	30–50 years
Cysts from	Collecting duct	Whole nephron
Ultrasound	Symmetric, homogeneous, echogenic	Large discrete cysts, asymmetric
Liver	Fibrosis (always)	Cysts
Other	None	Berry aneurysm, mitral prolapse

Diagram to draw: cut kidney with radial dilated collecting ducts.

> **Key points to remember:**
> - PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
> - Huge homogeneous echogenic kidneys with oligohydramnios.
> - Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

Also, from the condensed book (not in the paper answer):

- In later-onset disease, portal hypertension dominates.
- Nephrectomy costs nephron mass and brings dialysis earlier, so delay it where possible.

5 · ADPKD — paediatric focus

In children the key finding is hypertension with normal function.

*See the paper answer **ADPKD (Adult polycystic kidney disease)** (Section 13, Ch 136).*

Also, from the condensed book (not in the paper answer):

- **Contiguous gene syndrome**: deletion of PKD1 and the adjacent TSC2 gives TSC with severe early polycystic kidneys.
- A negative US in a young person does **not** exclude PKD2, so use genetic testing after counselling when certainty matters.
- Screen for aneurysm by MR angiography only for family history or high risk. Observe <7 mm; coil or clip if >7–10 mm or symptomatic.

> **KEY POINT**
>
> ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume.

6 · Juvenile nephronophthisis (NPH) / medullary cystic kidney disease (MCKD)

A salt-wasting cause of renal failure that ultrasound often misses.

- These ciliopathies give tubular atrophy, corticomedullary cysts and interstitial fibrosis in small-to-normal kidneys. **Polyuria and polydipsia occur in >80%**, with **salt wasting**. ESRD follows 5–10 y after presentation.

	Juvenile NPH	**MCKD**
Inheritance	Autosomal recessive	Autosomal dominant
ESRD	Mean 13 y; **10–20% of childhood renal failure**	Third–fourth decade
Hypertension	Absent	Present
Extrarenal	20%: **retinitis pigmentosa (Senior–Løken)**, Joubert, Cogan apraxia, hepatic fibrosis, Bardet–Biedl	None; type 2 — hyperuricaemia, gout in teenage males

- CT or MRI shows the cysts better than US. **Replace sodium, and avoid salt restriction and diuretics**. Transplant works, with no recurrence.

7 · Syndromes with renal cysts

Tuberous sclerosis complex (TSC)

- It is autosomal dominant: **TSC1 (chr 9), TSC2 (chr 16)**. Diagnosis needs 2 major features, or 1 major and 2 minor (angiomyolipoma is major).
- Kidney: cysts and **angiomyolipoma (AML) in 40–80%**, bilateral and multiple. AMLs never metastasise; the risk is haemorrhage. RCC occurs in ≈2%; suspect it in a growing fat-free lesion.
- **MRI is preferred**, because US misses 25–30% of fat-poor AMLs.
- **If an AML is >3 cm and growing, give everolimus (a mammalian target of rapamycin, mTOR, inhibitor) first line**. For an acute bleed, embolise and then give corticosteroids. Kidney-sparing surgery is second line; avoid nephrectomy. Image every 1–3 y, check BP, proteinuria and GFR yearly, and treat hypertension with an ACE inhibitor or ARB.

Von Hippel–Lindau disease

*See the paper answer **VHL gene and von Hippel–Lindau disease** (Section 13, Ch 137).*

Also, from the condensed book (not in the paper answer):

- Partial adrenalectomy for a small phaeochromocytoma avoids lifelong steroids.

8 · Multicystic dysplastic kidney (MCDK)

The MCDK is harmless; watch the opposite kidney.

- It has non-communicating cysts, **no normal parenchyma, no function** and an atretic ureter ("bunch of grapes").
- It is the **commonest renal cystic disease of the newborn and the second commonest neonatal abdominal mass**: 1 in 1000–4000. Bilateral disease is lethal.
- **The contralateral kidney** hypertrophies and has **vesicoureteral reflux (VUR) in 18–43% and pelviureteric junction (PUJ) obstruction in 3–12%**.
- About 40% involute (later imaged as "agenesis"). Hypertension and Wilms tumour are not increased.

US feature	MCDK	Severe PUJ obstruction
Cysts	Haphazard, varied, non-communicating	Calyces around central pelvis, communicating
Central large cyst	Absent	Present (pelvis)
Isotope uptake	None	Usually some

ALGORITHM

1. For a prenatally detected MCDK, do US on day 2–3 and at 1 month to assess the contralateral kidney and bladder.
2. If the diagnosis is in doubt, do a DMSA or MAG3 scan.
3. If there is a urinary tract infection (UTI) or an abnormal contralateral kidney or bladder, then do a voiding cystourethrogram (VCUG); otherwise do not.
4. Hand over to primary care to check BP, mass and UTI.

- **Nephrectomy** is only for mass effect, hypertension, cyst rupture or **increasing solid tissue**. Routine US surveillance has no proven benefit.

> **KEY POINT**
>
> MCDK removes itself; management looks to the solitary contralateral kidney and its reflux or obstruction.

9 · Benign multilocular cyst (cystic nephroma)

Imaging cannot exclude cystic Wilms tumour, so excise.

- **It is bimodal: boys <4 y (M 2:1, flank mass) and women >30 y (F 8:1, haematuria from herniation into the pelvis)**.
- **Excision is both diagnosis and treatment** (nephrectomy in children; partial if localised).

10 · Simple cysts in children and Bosniak

- **US criteria (all four)**: anechoic, thin smooth wall, posterior enhancement, round or oval. On CT a simple cyst is −10 to +20 HU without enhancement. A hyperdense cyst is benign if non-enhancing, homogeneous and ≤3 cm.
- **Children**: if a simple cyst is <15 mm and asymptomatic, repeat US at 2 y and discharge if stable. Continue follow-up if it is ≥15 mm or complex, or with a family history or genitourinary anomalies.

Bosniak	Features	Management
I	Simple	None
II	Fine septa, minimal calcification, hyperdense ≤3 cm	None
IIF	More septa/calcification, no measurable enhancement	Follow-up (≈11% progress)
III	Thick/irregular enhancing walls or septa	Excision (paediatric III–IV: >90% intermediate/malignant)
IV	Enhancing solid components	Excision

Treat a symptomatic benign cyst by aspiration with sclerosis or laparoscopic unroofing.

11 · Medullary sponge kidney, glomerulocystic and acquired cystic disease

SAQ Medullary sponge kidney

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 2, Q3e

Definition

- **MSK (medullary sponge kidney; precalyceal canalicular ectasia):** dilated (ectatic) distal **collecting ducts** with small cysts and diverticula **confined to the medullary pyramids**. Nephrons are initially normal.
- Usually **non-heritable** (sporadic); usually bilateral.

Pathogenesis of stones and nephrocalcinosis

- **Stasis and infection** in the ectatic ducts favour crystal deposition.
- **Tubular defects:**
- **Hypercalciuria** in one-third to one-half (mostly absorptive; sometimes renal leak).
- **Incomplete distal RTA (renal tubular acidosis)** in 30–40%, shown on acid loading. This causes **hypocitraturia** and alkaline urine.
- Impaired urinary concentration.
- Mutations in H^+-ATPase genes (ATP6V1B1, ATP6V0A4) support an MSK–distal RTA link.
- Result: calcium stones within dilated ducts and medullary **nephrocalcinosis**.

Clinical features

- Often silent for life; presents **after 20 years**.
- **Renal colic 50–60%**, UTI (urinary tract infection) 20–33%, gross haematuria 10–18%.

- Found in **2.6–21% of stone formers**, more often women.

Diagnosis

- **Urographic triad:** 1. Enlarged kidneys, with or without papillary calcification. 2. Elongated papillary tubules or cavities filling with contrast: **"bristles on a brush"** (paintbrush); when calcified, a **"bouquet of flowers"**. 3. Papillary blush and persistent medullary opacification.
- CT urography now replaces IVU (intravenous urography); plain CT shows medullary nephrocalcinosis.
- **Metabolic work-up:** 24-hour urine (calcium, citrate, pH), serum bicarbonate and potassium.
- **Differentials:** papillary necrosis, other medullary nephrocalcinosis (distal RTA, hyperparathyroidism), calyceal diverticulum.

Management

Agent	Target	Note
Potassium citrate	Hypocitraturia, hypercalciuria	**First choice**; reduces calciuria and recurrence
Thiazide	Hypercalciuria	Useful even without hypercalciuria
Inorganic phosphate	Hypercalciuria	If thiazide unsuitable; avoid with urease infection
Fluids, low-sodium diet	Volume, calciuria	All patients
- Treat infection; stones by SWL (shock-wave lithotripsy) or ureteroscopy; stones inside ducts may be unreachable.		
- **Prognosis:** usually benign; renal failure is rare.		

Diagram to draw: pyramid with ectatic ducts filled with contrast ("brush") and small calculi.

> **Key points to remember:**
> - Ectatic collecting ducts; sporadic; "bristles on a brush".
> - Hypercalciuria, incomplete distal RTA, hypocitraturia.
> - Potassium citrate is first-line prevention.

Also, from the condensed book (not in the paper answer):

- **Sporadic glomerulocystic disease** is a diagnosis of exclusion.

*See the paper answer **Acquired renal cystic disease** (Section 13, Ch 136).*

Also, from the condensed book (not in the paper answer):

- Acquired renal cystic disease affects 20–45% of children on dialysis, and a retroperitoneal bleed calls for nephrectomy because an occult RCC is likely.
- **Parapelvic cysts** mimic hydronephrosis on US; delayed contrast CT shows an undilated system, and management is conservative.

12 · Landmark trials

Trial	Design	Result	Take-home
HALT-PKD	Randomised: lisinopril and telmisartan vs lisinopril; rigorous vs standard BP	Dual blockade no benefit; low BP slowed total kidney volume (TKV) growth, lowered left ventricular mass and albuminuria, GFR unchanged	Single renin–angiotensin blocker; tight BP early
TEMPO / REPRISE	Randomised, tolvaptan vs placebo	Slower TKV growth and GFR decline	First disease-modifying drug; aquaresis, hepatotoxicity
ARegPKD registry	Observational ARPKD cohort	Bilateral nephrectomy <3 months gave the worst neurological outcome	Avoid early bilateral nephrectomy

51 Pathophysiology of Urinary Tract Obstruction

1 · Definition and classification

Function, not dilation, defines obstruction.

- **Obstruction** is any impediment to flow, calyces to meatus, that injures the kidney or lower tract: **acute or chronic, partial or complete, intrinsic or extrinsic, unilateral or bilateral**.
- It affects ≈ **1 in 500** live births; high-grade obstruction carries a lifetime end-stage renal disease (ESRD) risk of up to **30%**.

	Lower urinary tract obstruction (LUTO)	Upper tract
Commonest	Posterior urethral valves (PUV) and urethral atresia ≈ 50%	Ureteropelvic junction obstruction (UPJO), 1 in 1500; M:F 2:1
Others	Anterior urethral valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis, trisomy 18/21	Ureterovesical junction obstruction (UVJO), ectopic ureter, ureterocele, polyp, stones, crossing vessel
Outcome driver	ESRD and pulmonary hypoplasia	Unilateral function loss; contralateral kidney buffers

Presentation is prenatal hydronephrosis (mostly at 18–20 wk), or later infection, mass, pain or respiratory failure.

> **KEY POINT**
>
> Trap: **detrusor hypertrophy** causes UVJ obstruction by narrowing the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion.

2 · Haemodynamic changes

Rising pressure lowers GFR and renal blood flow (RBF), causing ischaemia. Injury begins within **2 h** of complete obstruction, worse if bilateral or solitary.

† Phase (complete)	Unilateral	Bilateral / solitary
I (0–1.5 h)	RBF rises, ureteric pressure rises (afferent dilatation: PGE2, nitric oxide)	Modest RBF rise
II (1.5–5 h)	RBF falls, pressure still high (efferent constriction)	RBF falls, pressure rises
III (> 5 h)	RBF and pressure fall (afferent constriction: angiotensin II, thromboxane)	Pressure stays high (retained solute, ANP), the basis of post-obstructive diuresis

- **Hypertension** is renin driven: bilateral ≈ **77%**, unilateral ≈ **20%**.

3 · Tubular function

- **Aquaporins and sodium transporters are downregulated**, causing a concentrating defect, salt wasting, polyuria and **distal renal tubular acidosis**.
- Potassium and other solutes are **wasted early and retained late**. FENa is < 1% in acute obstruction and > 1% with tubular injury.

4 · Pathology and fibrosis

- Fibrosis is the final common path, driven by stretch and **TGF-β/Smad**, amplified by angiotensin II.
- **Bladder** hypertrophies, compensates, then decompensates with poor compliance. Storage pressure > **40 cm H_2O** injures the kidneys.

5 · Recovery and compensatory growth

- Relief can reverse damage, but **incomplete recovery is common**.
- Compensatory growth is **glomerulomegaly, not new nephrons**.

6 · Post-obstructive diuresis (POD)

SAQ Post-obstructive diuresis

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3b

Definition

- POD (post-obstructive diuresis): marked polyuria after relief of obstruction, as the kidney sheds retained salt, water and urea.
- No agreed rate; a common working definition is **>200 mL/h for 2 consecutive hours or >3 L in 24 hours**. Up to 300 mL/h is common.
- **Physiological** (appropriate, self-limited) versus **pathological** (continues after euvolaemia, causing hypovolaemia and electrolyte loss).

Incidence and setting

- Unilateral obstruction: **about 2%** (the other kidney compensates). Bilateral or solitary kidney: **about 50%**.
- Commonest after catheterising **chronic retention with renal impairment**; also bilateral ureteric obstruction and PUV (posterior urethral valves).

Mechanisms

1. **Transporter dysregulation:** down-regulation of Na^+/K^+-ATPase and aquaporin-2 → concentrating defect (nephrogenic DI (diabetes insipidus)); loss of medullary tonicity.
2. **Volume expansion:** on relief, raised ANP (atrial natriuretic peptide), low aldosterone and sympathetic tone → salt and water loss.
3. **Osmotic diuresis** from retained urea.

Risk factors

- Bilateral obstruction, raised creatinine, oedema, heart failure, hypertension, confusion.
- After valve ablation: oligohydramnios, preterm birth, raised creatinine.

Clinical features

- Starts within hours of relief; most settle within 24–48 hours.
- Danger signs: postural hypotension, tachycardia, weight loss, hypernatraemia (free-water loss), low potassium, magnesium, phosphate.

Management

``` Relieve obstruction (catheter / nephrostomy)

|

Monitor hourly urine output, vitals, weight

|

Alert, can drink -> free oral fluids (most resolve)

|

Output persists / cannot drink / unstable -> admit; electrolytes every 4-6 h -> intravenous 0.45% saline: about 75% of urine output -> replace K, Mg, phosphate; adjust to urine sodium/osmolality ```

- **Do not overcorrect:** replacing every millilitre sustains the diuresis.
- **Gradual bladder decompression has no role**: it does not prevent haematuria or diuresis.
- PCN (percutaneous nephrostomy) for upper-tract obstruction; hydration alone for physiological POD.
- **Also know:** decompression haematuria; neonatal PUV concentrating defect; no approved drug therapy.

**Diagram to draw:** graph of urine output against time after catheterisation.

> **Key points to remember:**
> - Unilateral about 2%, bilateral about 50%.
> - Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
> - Replace about 75% of output with 0.45% saline; do not overcorrect.
> - Gradual decompression has no role.

**Also, from the condensed book (not in the paper answer):**

- Preterm infants have low aquaporin-2, hence their POD risk.

**ALGORITHM**

1. Monitor intake, output, and serum and urine electrolytes.
2. Replace losses and correct electrolytes.
3. **Do not over-replace**: it perpetuates POD.
4. Expect resolution within **24 h**; it seldom lasts > 48 h.

## 7 · Fetus and child

*See the paper answer **Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies** (Section 5, Ch 34).*
```

Also, from the condensed book (not in the paper answer):

- Second-trimester oligo-or anhydramnios carries **60–80%** mortality.
- In the Whitaker test, obstruction is a pelvic pressure > **12 cm H_2O** or a continuous rise.
- After birth, use ultrasound first, MAG3 for drainage, DMSA for scars and VCUG for outlet and reflux.

8 · Treatment principles

Drain to buy function, then reconstruct.

Option	Indication / key point
Clean intermittent catheterisation (CIC)	Neurogenic bladder; reduces UTI, stones, retention
Overnight catheter drainage	Valve bladder with nocturnal polyuria and progressive CKD
Vesicostomy	Infant with complex LUTO, unstable or unable to do CIC
Percutaneous nephrostomy	Upper tract; needs some hydronephrosis
Stent / cutaneous ureterostomy	Temporising; stent excellent for extrinsic obstruction
Pyeloplasty / reimplantation	Definitive; anastomose anterior to aberrant vessel; non-refluxing reimplant in children

- Use NSAIDs only if function is preserved. ESRD needs dialysis or transplant (Ch 61).

52 Ectopic Ureter, Ureterocele and Ureteral Anomalies

1 · Definitions and classification

- Ectopic ureter and ureterocele are one defect of **ureterotrigonal connection** on a continuum. The appearance changes the technique, not the strategy.
- An **ectopic ureter** is a single or duplex ureter that does not enter the trigone but enters **at or distal to the bladder neck**; merely opening near it is not ectopic.
- A **ureterocele** is **cystic dilation of the distal ureter** within the bladder or spanning bladder neck and urethra. It is never entirely urethral and never attached to wolffian structures.
- In a **duplex system, both lesions always involve the upper pole** ureter.

	Ectopic ureter	**Ureterocele**
Distal ureter	Dilated or not, opens ectopically	Cystic dilation
Position	At/distal to bladder neck; perineum, vagina, rectum (F) or wolffian ducts (M)	Bladder or spanning bladder neck and urethra
Wolffian attachment	Yes (male)	Never
Incontinence (untreated)	Yes, in females	No

Ureterocele classification

SAQ Ureterocele

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 3, Q3c (ectopic ureter also answered in 2018-Summer Paper 3, Q1)

Definition

- Cystic dilatation of the distal ureter, in the bladder or spanning bladder neck and urethra; **never wholly urethral**, never joined to Wolffian structures.
- Commoner in girls (about 4:1); about 80% drain the **upper pole of a duplex kidney** (Weigert–Meyer rule: upper pole orifice caudal-medial); about 10% bilateral.
- **Cause:** persistent **Chwalla membrane**; upper pole often dysplastic.

Classification

System	**Basis**	**Types**
Glassberg (most useful)	Position	**Intravesical** vs **ectopic (extravesical)**: part at bladder neck or urethra
Stephens	Orifice morphology	Stenotic, sphincteric, sphincterostenotic, **cecoureterocele** (cavity extends under urethra), blind, non-obstructed
Churchill	Function	Upper pole only / whole kidney / contralateral at risk

Clinical features

- Antenatal hydronephrosis (most); febrile UTI (urinary tract infection) or urosepsis; adult single-system type with stone.
- **Prolapse:** smooth interlabial mass in girls; BOO (bladder outlet obstruction).
- **No incontinence when untreated** (unlike female ectopic ureter).

Evaluation

- **Ultrasound:** thin-walled cyst in bladder, dilated upper pole and ureter.
- **VCUG (voiding cystourethrogram):** lower pole reflux ("drooping lily"), eversion (weak floor), prolapse.
- **DMSA (dimercaptosuccinic acid):** upper pole function; cystoscopy before incision.

Management

- **Goals:** preserve function; remove infection, obstruction, reflux; keep continence.
- **CAP (continuous antibiotic prophylaxis)** from birth if hydronephrotic.
- **Observation:** small, non-obstructing, low-grade reflux, non-functioning upper pole.

Option	Indication / result
TUI (transurethral incision)	Sepsis, neonates, intravesical type. Low transverse full-thickness incision; relieves obstruction in 78–97%; new reflux 0–50%. Intravesical without reflux: 70–80% cured; ectopic: secondary surgery up to 70%
Upper pole heminephrectomy	Non-functioning upper pole, no reflux; secondary surgery 40–50%
Ureteroureterostomy / pyeloureterostomy	Functioning upper pole
Excision + common sheath reimplant ± bladder neck repair	Persistent reflux, ectopic or cecoureterocele; persisting reflux 5–10%
Total reconstruction	Older child, non-functioning upper pole, lower pole reflux

- **Also know:** "watering-can" laser punctures (less reflux); pseudoureterocele; ureterocele disproportion.

Diagram to draw: duplex kidney with upper pole ureterocele and lower pole reflux.

> **Key points to remember:**
> - Upper pole of duplex; never wholly urethral; Chwalla membrane.
> - Intravesical vs ectopic predicts TUI success.
> - TUI relieves obstruction in 78–97%; new reflux 0–50%.

Also, from the condensed book (not in the paper answer):

- A **cecoureterocele** tunnels submucosally beyond the bladder neck into the urethra; it is often missed before surgery and risks incontinence.
- **Ureterocele disproportion** is a typical ureterocele with a small, dysplastic, non-dilated upper pole, so the kidney looks normal.

2 · Embryology essentials

Bud position explains upper pole obstruction and lower pole reflux.

- In a duplex, the upper pole bud arises **later and more cephalad** on the wolffian duct, so its orifice is carried **caudally** and becomes ectopic.
- **Weigert–Meyer rule**: **the upper pole orifice lies caudal (and medial) to the lower pole orifice**. So the upper pole gets ectopia or ureterocele (obstruction), and the lower pole gets a short tunnel and vesicoureteral reflux (VUR).
- Classically, a ureterocele results from failure of the **Chwalle membrane** to rupture. Renal dysplasia lies **downstream** of the connection defect.
- An ectopic ureter reaches the vagina, cervix or uterus only through the **Gartner duct**, never directly into müllerian structures.

3 · Ectopic ureter — sites and sex asymmetry

LAQ Embryology and management of ectopic ureter.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2018-Summer Paper 3, Q1

Definition

- An **ectopic ureter** is any ureter, single or duplex, that does **not open on the trigone**. True ectopia means an orifice **at or distal to the bladder neck**.
- Commoner in girls. In girls most ectopic ureters drain the **upper pole of a duplex kidney**; in boys most are **single-system**.

Embryology

- **Week 4–6:** the **ureteric bud** arises from the **mesonephric (wolffian) duct** and induces the metanephric blastema to form the kidney.
- The duct segment distal to the bud is absorbed into the **urogenital sinus**; the ureteric orifice then moves cranially and laterally, the wolffian orifice caudally to the verumontanum.
- **Mackie–Stephens bud theory:**

Bud position on wolffian duct	Absorption	Orifice	Result
Too caudal	Early	High and lateral, short tunnel	VUR (vesicoureteral reflux)
Too cranial	Late or never	Low and medial (bladder neck, urethra) or stays on wolffian duct	**Ectopic ureter**, often obstructed

- A bud from the wrong place meets poor blastema, so **renal dysplasia increases with the degree of ectopia**.
- **Duplex system:** the upper-pole bud is more cranial, absorbed later, and ends **lower and more medial** (**Weigert–Meyer law**): upper pole obstructs; lower pole refluxes.

Sites of ectopia (the key sex difference)

	Female	Male
Sites (commonest first)	Urethra, vestibule, vagina; rarely cervix, uterus, rectum (via **Gartner duct**)	Posterior urethra, seminal vesicle; also vas, ejaculatory duct
Relation to sphincter	Can open **below** the external sphincter	Always **above** the external sphincter

	Female	Male
Continence	**Continuous dribbling with normal voiding**	**Never incontinent**
Typical presentation	Constant dampness, perineal pus, UTI (urinary tract infection)	**Epididymo-orchitis**, UTI, flank pain, hydronephrosis

- **Zinner syndrome:** ectopic ureter into the seminal vesicle with ipsilateral renal agenesis or dysplasia and a seminal vesicle cyst.

Clinical features

- **Prenatal hydronephrosis**: most are detected this way.
- **Girl:** wet day and night despite normal voids; she is never dry for 30–60 minutes. Often misdiagnosed as voiding dysfunction.
- **Infection:** low-grade fever with spikes; urine culture may be negative because the infected moiety does not drain into the bladder.
- **Boy:** recurrent epididymitis in a young boy demands an upper-tract ultrasound.
- **Also know:** pain then perineal discharge; bilateral single-system ectopia with small bladder and wide bladder neck.

Evaluation

Test	What it shows
Ultrasound (first)	Dilated upper pole and ureter behind the bladder; parenchyma. Bladder view separates it from ureterocele
VCUG (voiding cystourethrogram)	Lower-pole or contralateral VUR ("drooping lily"), reflux into the ectopic ureter
DMSA (dimercaptosuccinic acid)	Gold standard for moiety function ("some" versus "none"); MAG3 (mercaptoacetyltriglycine) if drainage needed
MR (magnetic resonance) urography	Non-dilated upper pole, small dysplastic kidney, complex anatomy
Examination / cystovaginoscopy	Perineal orifice between urethra and vagina, cannulated for contrast

- **Also know:** clear urine leaking despite methylene blue in the bladder suggests an ectopic ureter; CT for a small unseen kidney.

Management

Goals: preserve renal function; remove infection, obstruction and reflux; achieve continence; with least morbidity. Start antibiotic prophylaxis in the neonate with significant dilation.

`Neonate with sepsis or massive dilation → end cutaneous ureterostomy (or nephrostomy); DMSA later DUPLEX system Upper pole non-functioning → upper-pole heminephroureterectomy Upper pole worth keeping → ureteroureterostomy (UU) / pyeloureterostomy + lower-pole VUR → common sheath reimplant (or lower-pole reimplant + distal UU) SINGLE system Functioning → ureteral reimplant (± tapering) Non-functioning → nephroureterectomy (laparoscopic) Bilateral → bladder neck reconstruction ± augmentation`

- **Upper-pole heminephroureterectomy** (open, laparoscopic or robotic): upper-pole ureter as a handle, passed behind the hilum; ligate polar vessels; excise the pole; close over fat.

Dissect on the ureter wall to protect the **lower-pole blood supply**. The stump is rarely a problem.

- **Ureteroureterostomy (UU):** end-to-side upper into lower pole ureter, distal (inguinal incision) or proximal (preferred with big size mismatch). **Stent the recipient lower-pole ureter** cystoscopically for identification. Now used even for non-functioning poles; feared hypertension has not been shown.
- **Common sheath reimplantation:** do not separate the two ureters (they share blood supply).

Complications

- Loss of lower-pole function (most serious); urinoma (up to 20% after minimally invasive partial nephrectomy, mostly harmless); stump infection; new reflux; persistent incontinence.

Recent advances

- Robotic UU and heminephrectomy; single-site surgery; non-dismembered side-to-side ureterovesicostomy in infants.

Diagram to draw: duplex kidney showing Weigert–Meyer orifices, and female versus male ectopic sites relative to the external sphincter.

> **Key points to remember:**
> - Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
> - Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
> - DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
> - Protect the lower-pole blood supply; stent the recipient ureter for UU.

Also, from the condensed book (not in the paper answer):

- A **single-system ectopic ureter** may mimic an "absent kidney" on ultrasound; use contrast CT or a renal function study.
- **Bilateral single-system ectopia** (rare) gives a hypoplastic bladder, poor bladder neck and bilateral dysplasia.

> **KEY POINT**
>
> Why does only the girl leak? The female ureter may open distal to the external sphincter and leaks continuously; the male ureter, confined to wolffian derivatives above it, presents as infection or obstruction.

- Male sites are the bladder neck, prostatic urethra, vas, seminal vesicle and ejaculatory duct.

4 · Clinical presentation

Prenatal

- Most are found as prenatal hydronephrosis. A fetal upper pole "cyst" is upper pole hydronephrosis until proven otherwise.
- **The diagnosis is made at the bladder**, because the upper tracts look identical; a large ureterocele can mask the bladder.

- Dilation of the ipsilateral lower pole or contralateral kidney suggests reflux. Dilation of the whole kidney and ureter suggests a single system.

Incidental and infection

- A "presumed ovarian cyst" may be a dilated ureter, so include it in the differential of a cystic pelvic mass.
- Do ultrasound in every child with urosepsis. If sepsis does not respond, do an **urgent transurethral incision (TUI)** of a ureterocele, or drain an ectopic ureter (usually end ureterostomy).
- An infected ectopic ureter gives low-grade spiking fever, and a **negative culture does not exclude infection**.
- **Any young male with bacterial epididymitis needs upper tract ultrasound**; it is often recurrent before diagnosis.

Incontinence

- Suspect it in a toilet-trained girl with **continuous low-volume dampness day and night** and normal voiding. Untreated ureteroceles do not cause incontinence.
- The key question is: **can she be dry for 30–60 minutes?** With an ectopic ureter she cannot.
- The leaking moiety may be non-dilated and missed on imaging, so suspicion rests on the history. Exclude voiding postponement and constipation.

Ureterocele prolapse

- It appears as a **smooth, congested, mucosa-covered interlabial mass**, often with voiding difficulty. A lobulated grape-like mass suggests **sarcoma botryoides** instead.
- Ultrasound confirms it; VCUG shows the prolapse, lower pole reflux and the "drooping lily".

Late presentation

- Teenagers and adults present with infection or pain; an adult single-system ureterocele often contains a **stone**. Upper pole heminephrectomy is the mainstay.

5 · Evaluation

- Examine the perineum of every continuously wet girl. The ectopic orifice lies **between the urethral meatus and the vagina**.

Imaging

Test	Answers	Note
Ultrasound	Anatomy; inferred function; bladder	First always; cannot separate ectopic ureter from ureterocele above bladder
Voiding cystourethrogram (VCUG)	Reflux (lower pole, contralateral, into ureterocele); bladder base, urethra	Optional if obstruction alone dictates incision; repeat after incision
Dimercaptosuccinic acid (DMSA)	Differential function — gold standard	Only "some vs none"
Mercaptoacetyltriglycine (MAG3) renogram	Function and drainage	Observation candidates; after 2 months

Test	Answers	Note
MR urography	Most detailed anatomy and function	Complex anatomy, non-dilated upper pole; needs sedation
Intravenous pyelogram (IVP)	Anatomy and qualitative function	Non-dilated ectopic ureter with incontinence

- A **ureterocele** is a thin-walled cyst within the bladder. A **pseudoureterocele** is a dilated ectopic ureter indenting the bladder; its wall is **bilaminar and thicker, and its lumen lies outside the bladder**.
- A full bladder can efface a ureterocele so it mimics an ectopic ureter, so scan through filling and voiding.
- On VCUG, the **"drooping lily"** is the lower pole pushed down and out by a dilated upper pole.
- † A single-system intravesical ureterocele gives a **"cobra head" (spring-onion)** sign on urography.
- **Preoperative function does not predict function after decompression**.

VCUG terms

Term	Meaning	Significance
Effacement	Flattens with filling	Mimics ectopic ureter
Prolapse	Moves into urethra	Outlet obstruction; predicts secondary surgery
Eversion	Diverticulum-like protrusion of bladder base	**Weak trigonal floor, so open repair likely**

Endoscopy

- **Choose the incision site** with little bladder filling, when the lowest part of the ureterocele is seen best.
- Detect any urethral extension (cecoureterocele) so the incision leaves no obstructing segment.
- A patulous bladder neck or cecoureterocele carries a later incontinence risk; consider bladder neck tightening.

6 · Management principles

- Goals: **preserve renal function; eliminate infection, obstruction and reflux; maintain continence; minimise total morbidity**.
- Start antibiotic prophylaxis early in significant hydronephrosis, even without reflux.
- Relieving obstruction may **create** reflux, or may let lower pole reflux resolve. This is the central trade-off, and decisions are empirical.
- **Fetal intervention**: none for ectopic ureter; for ureterocele only if it obstructs all units (experimental).

Neonatal algorithm

ALGORITHM

1. After prenatal hydronephrosis, do postnatal ultrasound.
2. If there is significant hydroureteronephrosis, start prophylaxis.
3. Do VCUG (optional if it will not change the decision to decompress), then a renal scan after 2 months.
4. If the ureterocele is small, with minimal hydronephrosis and no obstruction, observe.
5. If obstruction is significant and parenchyma is at risk, do TUI (or cutaneous distal ureterostomy if endoscopy is impossible).
6. If sepsis does not respond, do TUI (ureterocele) or nephrostomy or end ureterostomy (ectopic ureter); elective drainage is still needed later.

- **Observation** suits selected ureteroceles: no obstruction of the lower pole or contralateral kidney and lower pole reflux ≤ grade III, a non-functioning upper pole, or no obstruction on renogram. Counsel about unpredictable acute presentation.

7 · Operative options

Transurethral incision (TUI) of ureterocele

- TUI relieves obstruction in **78–97%**. It is the choice in **sepsis** and in the **infant**.
- Make a **low transverse full-thickness incision as distal and close to the bladder floor as possible** to reduce reflux. For an ectopic ureterocele, extend it into the urethral part so no obstructing flap remains.
- The **"watering-can"** technique (multiple holmium punctures) decompresses equally with less de novo reflux.
- It is a day-case procedure. Do ultrasound and VCUG at **4–6 weeks**.
- New reflux occurs in **0–50%**. **Intravesical** ureteroceles decompress without reflux in 70–80%; **extravesical** ones need secondary surgery in up to 70%. Lower pole reflux (risk ratio 1.74) and an extravesical site (2.78) predict reoperation.

> **KEY POINT**
>
> Why TUI first even if half need more surgery? It is outpatient, may be definitive, defers the second operation to an older child, and decompression often avoids tapering at reimplant. It is less appealing in an older child with a massive non-functioning upper pole, who needs definitive surgery at diagnosis.

Upper pole partial nephrectomy (heminephrectomy)

- It is for a non-functioning upper pole, by open, laparoscopic or robotic approach. The main concern is **not to damage the viable lower pole**.
- Use the upper pole ureter as a handle, passing it **behind the renal vessels**, ligate the upper pole vessels and resect at the demarcation.
- **Dissect on the wall of the upper pole ureter**, and leave its back wall on the lower pole ureter near the bladder, because they share a blood supply. A small distal stump is left; a large or refluxing one is excised low, sparing the bladder neck and vas.
- Outcomes: for ectopic ureter it is usually curative. For ureterocele, new reflux appears in **15–50%** and secondary surgery is needed in **40–50%**.

Ureteroureterostomy / pyeloureterostomy

- The upper pole is joined end-to-side into the lower pole system, for functioning and increasingly non-functioning upper poles; feared hypertension has not been substantiated.
- For a **distal** repair, **stent the lower pole ureter cystoscopically first**. A **proximal** repair is preferred for a dilated upper pole or size mismatch. Robotic repair matches open results, and size mismatch is no barrier.

Lower tract reconstruction

- **Ureterocele excision with common sheath reimplantation** relieves obstruction and reflux at once, and is the fallback for persistent reflux. It risks the bladder neck, sphincter and vagina.
- Mobilise **both ureters as a unit**, excise the ureterocele, **plicate the detrusor defect** and reimplant submucosally.
- A cecoureterocele remnant can act as an obstructing **urethral windsock** flap-valve.
- Reflux persists in **5–10%**, more often after tapering.
- **Total reconstruction** suits an **older child with a massive ureterocele, non-functioning upper pole and significant lower pole reflux**.

End ureterostomy for ectopic ureter

- It is for a neonate with sepsis or massive dilation. **Do not trim the redundant ureter**, because its length is needed for reimplant. Do DMSA after a few months, then reconstruct or remove.

8 · Decision summary

Upper pole function and lower pole reflux pick the operation.

Ectopic ureter (duplex)

ALGORITHM

1. If the upper pole is worth keeping and the lower pole refluxes, do a common sheath reimplant, or a lower pole reimplant with distal ureteroureterostomy.
2. If the upper pole is worth keeping and there is no lower pole reflux, do a proximal or distal ureteroureterostomy.
3. If function is unclear, or there is massive dilation or sepsis, do a temporary end ureterostomy, then DMSA, then reconstruct or remove.
4. If removal is chosen, do upper pole nephrectomy with ureterectomy.

Ureterocele (Table 52.1)

Procedure	Ideal indication	Advantage	Limitation
TUI	Infant; large ureterocele with VUR	Outpatient; effective; occasionally definitive	Reflux into ureterocele moiety needs bladder surgery
Upper pole nephrectomy	Older; large non-functioning upper pole; no VUR	Removes pathology; avoids bladder	Lower pole risk; may still need bladder surgery

Procedure	Ideal indication	Advantage	Limitation
Ureteroureterostomy / pyelostomy	Older; functioning upper pole; no VUR	Drains with little risk	Ureterocele left; may reflux
Excision and common sheath reimplant	VUR; functioning upper pole without major dilation	Removes obstruction and reflux; no renal risk	Complex; vagina/bladder neck risk; tapering

KEY POINT

VUR is the pivot: without VUR choose an upper tract solution; with VUR, a bladder-level solution, except in the infant, where TUI defers bladder surgery. Upper pole function then picks nephrectomy or ureteroureterostomy.

9 · Complications

- **Loss of lower pole function** after heminephrectomy is the most significant; prevent it by preserving the shared blood supply and using gentle traction.
- **Urinoma** occurs in up to 20% after laparoscopic or robotic cases; observe it if asymptomatic.
- **De novo reflux** after TUI or heminephrectomy is observed if there is no infection; otherwise do excision and reimplantation.
- **Voiding dysfunction** comes from poor trigonal support, which causes obstruction, poor emptying and recurrent VUR. Assess with VCUG **including lateral voiding views**, and treat by reconstruction, CIC or bulking.

10 · Other ureteral anomalies

Duplication

- It occurs in ~**1 in 125** (0.8%), F:M 1.6:1, and is mostly unilateral. **Look for contralateral duplication**: an occult second ectopic ureter explains persistent wetting.
- A **bifid ureter** is a partial duplication joining below the ureteropelvic junction (UPJ).
- **The lower pole gets single-system disease** (UPJ obstruction, VUR); **the upper pole gets ureteral-formation disease** (ectopia, ureterocele).
- Lower pole VUR is often high grade and resolves slowly; treat by common sheath reimplant, endoscopic injection or low ureteroureterostomy.

Triplication

- Smith types: 1, three separate ureters (**35%**, commonest); 2, incomplete with two orifices (21%); 3, trifid with one orifice (31%); 4, two ureters with three orifices (13%).

Valves, diverticulum, polyp

- † **Ureteral valves** are transverse mucosal folds with smooth muscle and proximal dilation; excise the segment. A ureteral **diverticulum** is usually a **blind-ending bifid ureter**.
- A **fibroepithelial polyp** is commonest at the **UPJ**, **89% male, 78% left**. At the UPJ, do pyeloplasty; elsewhere, **remove it ureteroscopically**.

Vascular position anomalies

LAQ Retrocaval ureter and IVC anomalous development

Paper answer · P1 · P3 · merged from 2

Asked as: P1 LAQ "IVC anomalous development and retrocaval ureter" · P3 SAQ "Retrocaval ureter"

Introduction

- The **IVC (inferior vena cava)** forms between the 6th and 8th weeks from three pairs of fetal veins by anastomosis, regression and persistence. The final IVC is **right-sided**, so long left veins must cross the aorta to reach it.
- Most anomalies are silent CT (computed tomography) findings. They **bleed at surgery, mimic lymph nodes, trap the ureter and alter IVC filter placement**.

Embryology of the IVC

Three paired venous systems

- Fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney: posterior cardinal and supracardinal veins dorsally, subcardinal veins ventrally.

Vein	Position	Adult derivatives
Posterior cardinal (first)	Dorsolateral	Iliac confluence; mostly regresses
Subcardinal	Ventromedial	Right: **suprarenal IVC**; intersubcardinal anastomosis: **left renal vein**; left: adrenal and gonadal veins
Supracardinal (last)	Dorsomedial	Right: **infrarenal IVC**; thoracic parts: **azygos and hemiazygos**

Segments of the definitive IVC (top to bottom)

1. **Hepatic**: right vitelline vein.
2. **Suprarenal**: right subcardinal vein (via the subcardinal–hepatic anastomosis).
3. **Renal**: subcardinal–supracardinal anastomosis.
4. **Infrarenal**: right supracardinal vein.
5. **Iliac confluence**: posterior cardinal veins.

Renal collar

- A venous ring encircles the aorta at renal level. The **ventral (pre-aortic) limb** persists as the left renal vein; the **dorsal (retro-aortic) limb** regresses.
- Normally the right **supracardinal** vein, which lies **dorsal** to the ureter, forms the infrarenal IVC, so the normal ureter lies in front of the IVC.

`Left-sided channels regress -> right-sided persist -> single right IVC + pre-aortic left renal vein`

Diagram to draw: three stages: paired cardinal veins, anastomoses with renal collar, definitive IVC.

Anomalies of IVC development

Anomaly	Embryological error	Key features
Retrocaval ureter ("preureteral vena cava")	Infrarenal IVC forms from a vein **ventral** to the ureter	Right only; 1 in 1500 autopsies; male 3:1; presents 3rd–4th decade. **Type I** low loop, fishhook, obstructs; **type II** high loop, sickle, mild

Anomaly	Embryological error	Key features
Circumaortic left renal vein	Dorsal limb persists with ventral	Commonest (2–16%); posterior limb runs lower into lumbar IVC
Retroaortic left renal vein	Ventral limb regresses, dorsal persists	Less common; may be compressed ("posterior nutcracker")
Left IVC	Left supracardinal persists, right regresses	Crosses to the right at renal level, in front of the aorta
Duplicated IVC	Both supracardinals persist	Left IVC joins left renal vein; double right cava can trap the ureter
Azygos continuation	Subcardinal–hepatic anastomosis fails	Large azygos; hepatic veins drain directly to right atrium; polysplenia, heterotaxy
Absent infrarenal IVC	Supracardinal and posterior cardinal failure	Deep vein thrombosis in the young

Retrocaval ureter: definition

- The **right ureter** passes **behind the IVC**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.
- In P3 the question was also asked as "Retrocecal ureter"; this is taken as a typo for retrocaval (circumcaval) ureter.

Retrocaval ureter: embryology

- If a vein **ventral** to the ureter persists as the infrarenal IVC, the ureter is trapped behind it.
- Campbell names a persistent **lumbar subcardinal** vein (the lumbar part of the ventral subcardinal vein; Chapter 52) or a **persistent posterior cardinal** vein (Chapter 93, and most texts); both mean that a ventral vein forms the IVC.
- Variant: double right IVC, with the ureter between the limbs.

Retrocaval ureter: incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; right side only.
- Presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis.
- Associations: horseshoe kidney, left renal anomalies; bilateral cases occur in situs inversus.

Retrocaval ureter: classification (Bateson and Atkinson)

	Type I (low loop), commoner	Type II (high loop)
Crossing	Lower, at L3–L4	Higher, at or near renal pelvis level
Shape	**J or "fishhook" / reverse-J**, S-curve	Smooth horizontal **sickle** curve
Obstruction	Yes, moderate–severe, at the iliopsoas edge	Mild or none (pelvis also crosses)

Retrocaval ureter: investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) is the alternative.
- IVU (intravenous urography): fishhook; ureter not seen beyond the J hook.
- Retrograde pyelography: **S-curve** or fishhook at L3–L4.
- **Diuretic renogram:** grades the degree of obstruction and gives split function.

- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**, so pyeloplasty fails.

Retrocaval ureter: management

- **Non-obstructed (usually type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it **anterior to the IVC**, and do a spatulated, tension-free anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; respect and preserve the ureteric blood supply.
- **Laparoscopic or robotic** repair (transperitoneal or retroperitoneal) is now preferred: equal success and faster recovery.
- Nephrectomy if the kidney is non-functioning.
- A pyeloplasty that misses the diagnosis fails.
- **Also know:** retroiliac ureter (L5–S1, with vasal anomalies); mimics of extrinsic ureteric crossing: spermatic / gonadal or lumbar vein, psoas (iliopsoas) tendon.

Diagram to draw: right ureter hooking behind the IVC with a fishhook pelvis.

Clinical implications of IVC anomalies

1. Diagnosis

- Venous-phase CT or MRI. A left or duplicated IVC mimics **para-aortic lymphadenopathy**; a large azygos mimics a mediastinal mass, risking needless biopsy.
- Retrocaval ureter: S-shaped or fishhook ureter at L3–L4 on retrograde pyelography; CT urography confirms; diuretic renography grades obstruction.

2. RPLND (retroperitoneal lymph node dissection)

- Preoperative CT must look for circumaortic or retroaortic veins and retrocaval ureter. A left or duplicated IVC lies in the left template; a retroaortic limb tears when lumbar vessels are divided.

3. Radical nephrectomy

- The hidden posterior limb of a circumaortic vein bleeds when the artery is approached from behind: ligate both limbs. Caval thrombus in a duplicated IVC needs rerouting of the left IVC.

4. Living donor nephrectomy

- The left kidney is preferred for its long vein. A circumaortic or retroaortic vein shortens it and adds bleeding risk; CT angiography is mandatory, and the right kidney may be chosen.

5. Retrocaval ureter repair

- As above: observe non-obstructed type II; for obstruction, divide, transpose in front of the IVC and re-anastomose over a stent, excising the aperistaltic segment.

6. Interventional and vascular

- **IVC filter:** with duplication, emboli bypass a single filter, so filter both cavae or place the filter suprarenally.
- Aortic surgery risks retroaortic vein injury. A retroaortic vein can cause haematuria or left varicocele.
- **Also know:** variable left adrenal vein entry (relevant to adrenal vein sampling); right gonadal vein drains into the right renal vein in up to 10%.

Key points to remember:

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

SAQ Retroileal Ureter

Paper answer · P2

Type: SAQ (10 marks)
Source: 2025-Winter Paper 2, Q3b

*"Retroileal" is taken as **retroiliac ureter**, the nearest real entity; retrocaval ureter is compared below (full answer: 2017-Winter Paper 3, Q3d).*

Definition

- The ureter passes **behind the common iliac artery** and is compressed there.
- Better called "**preureteral iliac artery**": the anomaly is presumed vascular; the ureter is trapped.

Embryology

- Normally the **ventral root of the umbilical artery** is replaced by a more **dorsal** branch, which lies behind the ureter.
- If the **ventral root persists** and the dorsal root fails, the ureter is trapped behind the artery.

Features

- **Very rare**; either side; **bilateral** cases reported.
- Obstruction at **L5–S1**.
- **Associated anomalies are common**, especially **vasal and mesonephric duct ectopia**: ureter and vas draining together into the posterior urethra; vasa ending in the ureters; ectopic vaginal ureter.

Presentation and diagnosis

- Flank pain, UTI (urinary tract infection) or hydroureteronephrosis down to the iliac crossing.
- **CT urography (computed tomography urography) with arterial phase** or MRI (magnetic resonance imaging) shows ureter and artery together; **diuretic renogram** for function and drainage.
- Caution: a vessel indenting an already dilated ureter may not be the cause.

Management

- Non-obstructed: observe.
- Obstructed: **divide the ureter, re-route it in front of the artery** and do a spatulated ureteroureterostomy (or reimplantation) over a double-J stent; open, laparoscopic or robotic.

Comparison with retrocaval ureter

	Retroiliac	Retrocaval (circumcaval)
Vessel	Common iliac artery	IVC (inferior vena cava)
Embryology	Persistent ventral umbilical artery root	Persistent subcardinal vein
Side	Either, may be bilateral	Right
Level	L5–S1	L3–L4, "fishhook"
Associations	Vasal/mesonephric ectopia	Horseshoe kidney, left renal anomalies
- **Also know:** distal ureter obstruction by uterine, umbilical, obturator or hypogastric vessels.		

Diagram to draw: ureter behind the common iliac artery at L5–S1, dilated above.

Key points to remember:

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

Also, from the condensed book (not in the paper answer):

- A retrocaval ureter is always right-sided and mimics UPJ obstruction, so a simple pyeloplasty fails.

53 Surgery of the Ureter in Children

1 · Pelviureteric junction (PUJ) obstruction — definition and aetiology

*See the paper answer **Congenital PUJ obstruction, failed pyeloplasty and salvage procedures** (Section 11, Ch 93).*

Also, from the condensed book (not in the paper answer):

- Paediatric hydronephrosis is a pelvic anteroposterior diameter (APD) of **12 mm**, but hydronephrosis is not obstruction.
- With a crossing vessel, **dismembering and transposing the ureter anterior to the vessel prevents recurrence**. For secondary PUJ obstruction from high-grade reflux, avoid reimplant and pyeloplasty at one sitting; stage them.
- For **lower-moiety PUJ obstruction in a duplex**, do ureteropyelostomy or pyeloplasty; if the moiety has ≤ **5–10%** function, do partial nephrectomy.

2 · Diagnosis and indications

Decide on function trend and dilatation, not on washout alone.

- Most are found on antenatal ultrasound (US) at ≈ 20 weeks.
- Neonatal renogram washout has false positives up to **44%**, so decide on the trend in differential renal function (DRF) and on dilatation, not on half-time (T½) alone.
- **Pyeloplasty indications**: increasing APD; low or falling DRF; breakthrough infection on prophylaxis; symptoms (pain; failure to thrive in infants; haematuria; rarely hypertension).

3 · Anderson–Hynes dismembered pyeloplasty

It is the **gold standard** for any access, with success ≈ **95%**. It preserves a crossing vessel, excises the adynamic segment and allows pelvic reduction.

ALGORITHM

1. Expose the PUJ and free the lower pole, looking for an accessory vessel.
2. Use bipolar diathermy sparingly and place stay sutures.
3. Decompress the pelvis with a 21-G needle, then dismember, with or without pelvic reduction.
4. **Spatulate laterally** into healthy ureter, keeping the stenotic segment as a "no-touch" handle.
5. Bring the apex of the spatulation to the inferior pelvis and make a tension-free, watertight 6-0/5-0 absorbable anastomosis.
6. Excise the stenotic segment last, irrigate out clot and cover with perinephric fat.

- For a long gap, mobilise the whole kidney. A drain is usually unnecessary.

Alternative	Best indication	Point
Foley Y-V-plasty	High insertion, **no vessel**	V base on dependent medial pelvis, apex at PUJ
Spiral flap	Long proximal stricture, dependent PUJ	Length:width ≤ 3:1
Vertical flap	Dependent PUJ, large box-shaped extrarenal pelvis	Flap length must match gap

Alternative	Best indication	Point
Vascular hitch (Hellström)	Anterior lower-pole vessel, **no intrinsic stenosis**: moderate hydronephrosis, no calyceal dilatation, good cortex and split function; peristalsing PUJ at surgery	Vessel fixed cranially in pelvic wall; no hypertension seen
Ureterocalicostomy	Failed repair, short scarred ureter, dilated lower calyx with thin cortex	Lower-pole nephrectomy exposes calyx; spatulated ureter joined over stent

Use flaps only when a tension-free primary anastomosis is impossible. Non-dismembered repair is inferior.

> **KEY POINT**
>
> Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)?

> **KEY POINT**
>
> One fixed principle, many routes: dismember, excise, spatulate, anastomose without tension. Failure is managed by repeating the principle.

4 · Access

- In **infants**, open surgery is preferred; **posterior lumbotomy** suits only infants and toddlers, while the flank approach suits any age.
- Laparoscopic results equal open, and robotic surgery has a shorter learning curve.

	Transperitoneal	Retroperitoneal
For	Easier landmarks; left transmesocolic	No bowel injury; leak extraperitoneal; shorter stay
Against	Bowel risk; urinary ascites causing ileus	Learning curve; small space
Note	Preferred for horseshoe kidney, retrocaval ureter	One randomised trial favoured it; table **not flexed**

- **Endopyelotomy** does poorly long-term under 18 years; its niche is **failed pyeloplasty** without a vessel, with a double-J stent for 6 weeks.

5 · Stents, outcomes, failure

Stent	Removal	Point
Internal double-J	Second anaesthetic unless string/magnet	Catheter 24–48 h; string gives more febrile UTI (girls)
Externalised nephroureteric	7–14 days, no anaesthesia	Fewer complications, shorter stay in infants
Stentless	—	Needs flawless technique

- Stent duration of 1–4 weeks does not change outcome.
- For emergency diversion before repair, use a **nephrostomy, not a retrograde stent**, because a stent inflames the PUJ.
- Success is 90–100%; the commonest reason for conversion is bleeding.

*See the paper answer **Congenital PUJ obstruction, failed pyeloplasty and salvage procedures** (Section 11, Ch 93).*

6 · Primary megaureter

Most resolve; the challenge is spotting the truly obstructed.

SAQ Primary obst. megaureter.

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 3, Q3f

Assumed meaning: primary obstructive megaureter (POM).

Definition

- **Megaureter:** ureter diameter **≥7 mm** (normal up to 16 years is 5.0–6.5 mm). It is a descriptive term.
- **Classification:** primary (intrinsic) or secondary (neurogenic bladder, outlet obstruction); and **obstructed, refluxing, non-refluxing non-obstructed, or refluxing and obstructed**.
- **POM:** an **adynamic juxtavesical segment** at the UVJ (ureterovesical junction) blocks peristalsis and causes functional obstruction above.

Pathogenesis

- Ureteric muscle matures cranially to caudally; the juxtavesical segment matures last, even after birth. This explains spontaneous resolution.
- Excess collagen, thick muscle sleeve, abnormal interstitial cells of Cajal.

Clinical features

- 10–23% of antenatal upper-tract dilatation. Boys, **left side**; **bilateral 25%**; contralateral dysplasia or obstruction 10–15%.
- Mostly asymptomatic; or UTI (urinary tract infection), pain, haematuria, stones.

Evaluation

- **Ultrasound:** hydroureteronephrosis.
- **VCUG (voiding cystourethrogram):** excludes reflux and PUV (posterior urethral valves).
- **MAG3 (mercaptoacetyltriglycine) renogram:** split function; washout T½ unreliable (false positive up to 44%).
- MRU (magnetic resonance urography) for anatomy.

Management

Conservative (most)

- CAP (continuous antibiotic prophylaxis) in infants; serial ultrasound and renograms; most resolve.
- Diameter <10 mm: virtually never needs surgery; ≥10 mm: a substantial minority do.
- Follow to early adulthood.

Indications for surgery

- Recurrent febrile UTI; progressive dilatation; **DRF (differential renal function) <40%**; **fall in DRF ≥5%**.

Infant (small bladder, huge ureter): temporise

- Cutaneous ureterostomy; refluxing reimplant; nephrostomy; **endoscopic balloon dilatation with double-J** for 2–6 months (70–80% success; complications up to a third).

Definitive repair (child)

- Excise stenotic segment → straighten → **taper** → **non-refluxing reimplant at 5:1** (Cohen or Politano–Leadbetter; extravesical for huge ureters). Stent afterwards.

Tailoring	Method	Use
Starr plication	Imbricate lateral wall	Moderate (<1.75 cm)
Kalicinski folding	Exclude and fold lumen	Moderate
Hendren excision	Excise lateral wall	Massive; risk of ischaemia

- Success over 90%.
- **Complications:** obstruction (stent first), VUR (vesicoureteral reflux; observe or inject), persistent dilatation.

Diagram to draw: dilated ureter above narrow adynamic segment; tapered reimplant.

Key points to remember:

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

Also, from the condensed book (not in the paper answer):

- In infancy the small bladder cannot take a safe tunnel for a massive ureter, so temporise with a **distal loop cutaneous ureterostomy** or a **refluxing reimplant**; balloon dilatation has weak evidence.
- Follow into adulthood, since some worsen at puberty.

KEY POINT

Why not rely on the renogram? Neonatal washout depends on immature function, hydration and a huge ureter's capacity, so serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures.

ALGORITHM

1. Excise the stenotic distal segment and straighten the ureter.
2. Taper if needed.
3. Reimplant non-refluxing at a **5:1** tunnel length:diameter.
4. Stent with a feeding tube for 7 days or a double-J for 4 weeks.

- Plication and folding preserve the blood supply; Hendren excisional tapering risks ischaemic stricture.
- Treat postoperative obstruction with a stent first; an ischaemic stricture needs redo.

7 · Surgery for vesicoureteral reflux

KEY POINT

Principles: exclude or treat bladder causes first; handle ureter and bladder gently; mobilise without tension; keep a tunnel length:diameter of **5:1** (Paquin); ensure a straight course with the bladder empty and full.

- Techniques are intravesical or extravesical, with the tunnel suprahiatal or infrahiatal.

Technique	Approach / tunnel	Strength	Weakness
Politano–Leadbetter	Intravesical; suprahiatal neo-hiatus, vertical tunnel	Long tunnel for high grade	Blind neo-hiatus (bowel, vas, vagina); kink if too lateral/anterior
Glenn–Anderson	Intravesical; infrahiatal toward bladder neck	Least kinking	Short tunnel
Cohen	Intravesical; cross-trigonal	Commonest; success **98%**; thick bladder; with bladder-neck reconstruction	Difficult later retrograde catheterisation

Technique	Approach / tunnel	Strength	Weakness
Lich–Gregoir	Extravesical detrusorotomy ≈ 5 cm	No haematuria/spasm; short catheter	Transient voiding dysfunction/retention up to **10%**, esp. bilateral
Robotic extravesical (RALUR-EV)	Y-shaped detrusorotomy 4–5 cm	Less pain, spasm, stay	Early series: leak, obstruction, ureteric injury

Cohen — key steps

ALGORITHM

1. Open the bladder vertically and put a feeding tube in the orifice as a handle.
2. Circumscribe the orifice and free the ureter, preserving its distal blood supply, until it reaches the contralateral wall without tension.
3. Make a submucosal tunnel across the trigone, with the new orifice above the contralateral one.
4. Suture mucosa to mucosa with 6-0, with a detrusor hitch at 6 o'clock.

Lich–Gregoir — key steps

ALGORITHM

1. With the bladder one-third full, divide the obliterated umbilical artery.
2. Free the ureter **hugging it**, only as far as the UVJ.
3. Distend the bladder and mark a 5-cm tunnel; incise the detrusor down to mucosa and raise wide flaps.
4. Lay the ureter in and close the detrusor caudocranially, catching adventitia at the apex.
5. Leave a Foley catheter for 12–48 h.

KEY POINT

Why voiding dysfunction? The pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ, so stay on the ureter and stop at the UVJ.

- In the American Urological Association (AUA) meta-analysis, open success is **95.9%**; minimally invasive results are converging.
- **Follow-up** is US at 6–12 weeks. A VCUG is needed for high grade, new hydronephrosis, UTI or a dysfunctional bladder.

Complications of reimplantation

	Early	Long-term
Reflux	Ipsilateral low grade usually resolves by 1 year. Contralateral ≈ 1 in 10 after unilateral repair, mostly resolves, so observe; intervene for pyelonephritis. Routine bilateral repair not justified	Short tunnel/untapered ureter, or **unrecognised secondary reflux** (neurogenic/valve bladder): treat the bladder first (anticholinergic and clean intermittent catheterisation)
Obstruction	Oedema, clot, haematoma, kink within 2 weeks: double-J or nephrostomy; most resolve	Ischaemia, angulation (hiatus too lateral/anterior), tight tunnel, stenosis: balloon and stent, then re-reimplant

- Contralateral reflux risk rises with age < 6 years, capacity < 50% of expected and a refluxing duplex.
- At **reoperation**, excise ischaemic ureter back to healthy tissue; for a short ureter use a **psoas hitch**, Boari flap or transureteroureterostomy.

KEY POINT

Early complications are tissue reactions, so drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder.

Endoscopic injection

*See the paper answer **Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING** (Section 4, Ch 41).*

Also, from the condensed book (not in the paper answer):

- After failure, repeat the injection once or twice, then reimplant; the encapsulated implant does not make reimplantation harder.

8 · Other ureteric lesions

- † **Retrocaval ureter** is right-sided with a fish-hook shape; re-anastomose it in front of the cava.
- A **congenital mid-ureteric stricture** is often misread as megaureter; excise it and do a ureteroureterostomy.
- A **fibroepithelial polyp** (upper third, male, left) causes pain and haematuria; remove it endoscopically.
- **Retroperitoneal fibrosis** is treated with steroids, then azathioprine, then a stent, and finally ureterolysis.

56 Hypospadias

1 · Definition, diagnosis and embryology

- The **triad** is an ectopic ventral meatus, ventral curvature and a dorsal hooded foreskin (the last two not always present). Incidence is ≈ **1 in 150–300** male births.
- **Megameatus with intact prepuce (MIP)**, ≈ 5%, hides under a normal foreskin, so retract before newborn circumcision. Avoid circumcision when curvature or ventral skin deficiency is present.

LAQ Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair

Paper answer · P2 · P3 · merged from 2

Asked as: P3 LAQ "Embryology of the penis and management of proximal hypospadias (including Koyanagi repair)" · P2 SAQ "Perineo-scrotal hypospadias"

Embryology of the penis

Indifferent stage (up to 8–9 weeks)

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

Hormonal control

- SRY (sex-determining region Y) drives the testis, whose **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

Primordium (germ layer)	Male	Female
Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce)	Glans, corpora	Clitoris
Urethral folds	Fuse to form penile urethra, median raphe	Labia minora
Labioscrotal swellings	Scrotum	Labia majora
Urethral plate (endoderm)	Whole urethra, including glanular	Vestibule

Urethral formation: the "double zipper"

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe. - The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated. - Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks. - **Prepuce** fuses dorsally first; ventral

fusion awaits urethral closure, hence the **dorsal hood** in hypospadias. - Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site). - **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

Diagram to draw: double-zipper urethral formation.

Definition of proximal (perineoscrotal) hypospadias

- Meatus on the proximal shaft, at the penoscrotal junction, scrotum or perineum (10–25% of hypospadias).
- Perineoscrotal hypospadias is a severe form, with severe ventral curvature, a short penis, bifid scrotum and often **penoscrotal transposition**.
- Cause: failed androgen-dependent urethral fold fusion (8–16 weeks).

Evaluation

- Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- DSD risk is 64% in proximal versus 7% in distal cases. A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- Prostatic utricle is commoner; cystoscopy in severe cases. No routine VCUG (voiding cystourethrogram) or renal ultrasound.

Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects are transient; evidence is mixed, but two RCTs (randomised controlled trials) showed fewer complications.

Curvature correction (decides one- versus two-stage)

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty ├─ residual <30° → dorsal midline plication (12 o'clock) └─ >30° → divide urethral plate └─ still >30° → ventral corporotomy ± graft → staged repair
```

	Dorsal plication (Nesbit, Baskin)	**Ventral lengthening**
Method	Shortens the convex dorsal side	1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft
Use	Mild (≤15–30°)	Moderate–severe
Recurrence	28%	9%

Diagram to draw: curvature algorithm.

Urethroplasty options

Approach	Technique	Comment
One-stage **proximal TIP (tubularised incised plate)**	Plate kept, incised and tubularised	Recurrent curvature, stricture: largely abandoned for severe cases
One-stage **onlay island flap**	Inner preputial flap on preserved plate	Diverticulum, fistula; higher complications
One-stage **TPIF (transverse preputial island flap, Duckett tube)**	Tube after plate division	High complications
One-stage **Koyanagi**	Meatus-based parameatal foreskin flap (see below)	No proximal circular anastomosis
Two-stage Bracka / STAG (staged tubularised autograft)	Stage 1: correct curvature, divide plate, quilt inner-prepuce or **buccal graft**. Stage 2 (≥6 months): tubularise, glansplasty, barrier layer	**Preferred for most proximal cases**; STAG lowest complications (about 23%)
Two-stage Byars	Dorsal preputial flaps moved ventrally, tubularised later	Supple flap bed; complications 43–81%

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; **scrotoplasty** for transposition.

Diagram to draw: two-stage graft repair: stage 1 graft bed, stage 2 tubularisation.

Koyanagi repair: definition and principle

- **One-stage repair of proximal hypospadias** with a **meatus-based parameatal foreskin flap** (Koyanagi). Skin strips round the meatus continue into the inner prepuce on both sides; they are swung ventrally and tubularised.
- **Principle:** the neourethra stays attached to the native meatus, so there is **no circular proximal anastomosis**, and the urethral plate can be divided to correct severe chordee.

Koyanagi repair: indications and contraindications

- **Indications:** proximal hypospadias (penoscrotal, scrotal, perineal) with **severe curvature** needing plate transection, and an **ample, unoperated prepuce**.
- **Contraindications:** prior circumcision or failed repair; small glans (give preoperative testosterone); curvature needing a ventral corporal graft (stage instead).

Koyanagi repair: procedure

`Stay suture; artificial erection → incise round the meatus, along both sides of the plate to the corona and round the inner prepuce (meatus-based "wings") → deglove; divide the plate; excise dysplastic dartos → residual curvature <30° → dorsal midline plication → split the dorsal prepuce; bring both wings ventrally on dartos pedicles → join inner edges, then outer edges, over a catheter → tube`

→ glanuloplasty, meatus at the tip → dartos barrier layer; skin cover ± scrotoplasty → stent 7–10 days; compressive dressing

- Modified and "neo-modified" versions add flap fixation and barrier layers to reduce fistula and diverticulum.

Diagram to draw (Koyanagi): meatus-based foreskin wings split dorsally and joined ventrally into a tube.

Koyanagi repair: complications

Complication	Cause
Fistula (commonest)	Long suture line; repair ≥6 months later
Meatal stenosis, stricture	Distal flap ischaemia
Diverticulum	No spongiosum; flap not fixed to the shaft
Glans dehiscence	Tension
Residual or recurrent curvature	Plication alone in severe cases
Flap necrosis, infection, poor cosmesis	Pedicle injury

- Because proximal repairs overall have **23–68%** complications, many now prefer **two-stage repair** (STAG, Bracka).

Complications and outcomes of proximal repair

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** is the commonest (3–45%; repair ≥6 months later; exclude distal obstruction).
- Others: glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, **diverticulum** (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans), hair-bearing urethra.
- A long, rigid neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.

Follow-up

- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

> **Key points to remember:**
> - DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
> - Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
> - <30° plication; >30° divide plate ± ventral graft and stage.
> - STAG/Bracka staged repair preferred; fistula commonest complication.
> - Koyanagi: one stage; meatus-based parameatal foreskin flap; plate divided; no proximal circular anastomosis.
> - Koyanagi: fistula commonest; diverticulum, meatal stenosis.

Also, from the condensed book (not in the paper answer):

- The critical window is **weeks 8–16**. Ventral prepuce fusion awaits urethral fusion, so the foreskin is incomplete in **95%**. MCQ

> **KEY POINT**
>
> Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close over an open urethra, and dysplastic ventral tissue tethers the shaft into curvature.

2 · Aetiology

It is **multifactorial**, but all causes mean too little androgen action in weeks 8–16.

- Placental insufficiency and growth restriction double the risk; assisted reproduction is an independent risk.
- Twin concordance is 9–27%, and chromosomal abnormality is found in up to 30%; about 200 syndromes include it. Endocrine disruptors are unproven in humans.
- It belongs to **testicular dysgenesis syndrome**, and a shorter anogenital distance (AGD) means more severe disease.

3 · Classification and severity

- Distal forms are **70–85%**. The meatus is the least reliable marker, so **classify after degloving** by meatus, curvature and plate.

LAQ Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula

Paper answer · P1 · P4 · merged from 2

Asked as: P4 LAQ "Hypospadias: present concepts in management, and hypospadias fistula" · P1 SAQ "Hypospadias repairs – single-stage versus staged techniques"

Introduction

- Read "present concepts" as the present (current) concepts in the management of hypospadias.
- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.
- The choice of repair depends on **curvature after degloving**, plate quality and glans size, not on meatal position alone.

Aetiology and associations

- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).
- Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.

Evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).

- No routine VCUG (voiding cystourethrogram) or renal ultrasound.

Classification: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

GMS (glans–meatus–shaft) score	Glans	Meatus	Shaft curvature
1	Good size, deep groove	Glanular	None
2	Adequate, grooved	Coronal	<30°
3	Small; narrow or flat plate	Mid or distal shaft	30–60°
4	Very small; flat plate	Proximal shaft, penoscrotal	>60°

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised plate).

Timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

Curvature decides the operation

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty (single-stage) ├─ residual <30° → dorsal midline plication (12 o'clock nerve-free zone), single-stage └─ >30° → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

Choice of technique

Situation	Current choice
Glanular meatus, no curvature	Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively
Distal or midshaft, good plate	**TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove
Narrow or shallow plate	TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification)
Proximal, curvature >30°	**Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar)
Proximal, curvature corrected, good tissue	One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned

Single-stage techniques

Group	Examples	Indication
Advancement	**MAGPI** (Duckett 1981); MIV	Glanular meatus, no curvature
Tubularisation	**TIP** (Snodgrass 1994); Thiersch–Duplay; GAP	Distal–midshaft; TIP in >90% of distal
Flap	Mathieu (flip-flap); **OIF** (onlay island flap)	Thin plate
Proximal one-stage	TPIF (Duckett tube); proximal OIF	Curvature corrected; good tissue
Graft	Dorsal buccal inlay	Narrow plate

- Pros: one anaesthetic.
- Proximal TIP is abandoned: devascularised plate, stricture, recurrent curvature.

Staged techniques

- Now the approach of choice for most proximal hypospadias.
- **Rationale (why stage):** a urethroplasty laid over a fresh corporal graft fails more often; stage 1 builds a supple, vascular bed. Stage 2 after **≥6 months**.

Type	Stage 1	Stage 2 (≥6 months)
Bracka / STAG (staged tubularised autograft)	Correct curvature, divide plate, quilt inner preputial or buccal graft	Tubularise, glansplasty, tunica vaginalis cover
Byars flaps	Split dorsal prepuce, transpose ventrally	Tubularise
Distal-first (Ulaanbaatar)	Curvature and distal urethra	Join proximal urethra

- Repeat the artificial erection before stage 2.

Single-stage versus staged: comparison of outcomes

Measure	Single-stage	Staged
Distal success	83–95%	–
Proximal complications	About 60% (long TIP, mixed)	STAG about 23% (lowest); Byars 43–81%
Recurrent curvature	Higher after plication	0–10%
Penile length, perception	Lower	Better
Extra procedures	Fewer planned	25–75% need more

Operative principles: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

Complications and outcomes (standardised)

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- Fistula is the commonest complication. A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.

- **Also know:** BXO (balanitis xerotica obliterans): excise and stage with buccal graft; redo repairs use buccal (lip, inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

Hypospadias fistula: definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus: the **commonest complication** of hypospadias repair, just under 10% at short-term follow-up.
- After TIP: 5.7% (**Pfistermuller 2015**); after proximal repair 3–45%. Usually at the original meatus or coronal margin.

Hypospadias fistula: aetiology

- **Distal obstruction** (meatal stenosis, stricture): raises voiding pressure and turbulence; the **key correctable cause**.
- **Ischaemia**, oedema, infection, haematoma, tension.
- **Technical:** overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- **Patient:** proximal meatus, glans <14 mm, redo surgery. A long, rigid neourethra raises resistance (Poiseuille's law).

Hypospadias fistula: prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

Hypospadias fistula: evaluation

- Split stream / multiple streams, usually within 2 years.
- Under anaesthesia: **bougie calibration** for distal narrowing, **urethroscopy** if irregular, and a **stress test** (compress proximally and distally, inject saline or methylene blue) for hidden tracts.

Hypospadias fistula: management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

Fistula	Repair
Small, shaft	Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier**
Large or thin tissue	Trap-door or island skin flap plus second layer
Coronal margin (glans dehiscence)	Implies distal narrowing or poor tissue: **redo urethroplasty and glansplasty**
With diverticulum	Excise, or reuse the redundant wall as a flap

- Recurrence about **20%**; reduce with interposed vascularised layers (dartos, tunica vaginalis, Cecil modification when tissue is scarce).
- Catheter: none for simple repairs, 7–14 days for complex.
- **Also know:** preputial fistula does not involve the urethra; congenital urethral fistula is closed the same way; follow through puberty.

```
UCF → wait 6–12 months → calibrate, urethroscopy, stress test ├─ distal obstruction → correct at same sitting ├─ small → excise + layered closure + barrier flap └─ large/coronal → flap repair or redo urethroplasty
```

Follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

Diagrams to draw: curvature algorithm; STAG stage 1 graft and stage 2 tubularisation; fistula repair showing neourethra, dartos barrier and skin with offset suture lines.

> **Key points to remember:**
>
> - Classify after degloving (GMS); curvature, not meatus, decides one versus two stages: curvature >30° after degloving means stage.
> - TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
> - Stage 2 after ≥6 months.
> - Hypospadias plus undescended testis: DSD work-up.
> - Follow through puberty; use patient-reported outcomes.
> - Fistula: commonest complication, just under 10%; TIP 5.7%.
> - Exclude distal obstruction; wait 6–12 months.
> - Layered closure with barrier flap; recurrence about 20%.

Also, from the condensed book (not in the paper answer):

- The glans–meatus–shaft (GMS) score (Merriman 2013) is scored in theatre after degloving.

4 · Associated anomalies and DSD work-up

- Hernia or hydrocele occurs in 9–16%, and undescended testis (UDT) in ≈ 7%.
- A **DSD work-up** is needed for UDT with proximal hypospadias, a perineal meatus or atypical genitalia. DSD risk is ≈ **50%** with a non-palpable vs 15% with a palpable testis, and 64% in proximal vs 7% in distal cases. Since 2006 severe hypospadias is classed as 46,XY DSD.
- Do ultrasound for Müllerian remnants, karyotype, testosterone, FSH, LH and AMH; a 46,XY karyotype does not exclude DSD (Ch 59).
- A large **prostatic utricle** causes stasis, UTI and difficult catheterisation. Routine VCUG and renal ultrasound are not needed.

> **KEY POINT**
>
> Hypospadias with cryptorchidism signals a systemic androgen problem, hence karyotype before surgery.

5 · Pre-operative assessment

- The goals are a straight erection, a urethra to the tip, good cosmesis and the fewest procedures. Exclude micropenis, and note that a deflected stream may justify surgery even when distal.
- **Timing**: the AAP advises 6–12 months, and the authors **6–18 months**, before genital awareness; younger repair has fewer complications.
- A glans width **≤14 mm** means more complications. Give testosterone cypionate **2 mg/kg intramuscularly at 5 and 2 weeks** before surgery if the glans is <15 mm, which gains about 4 mm. Two randomised trials showed fewer complications, but meta-analyses are inconclusive; side effects resolve within 6 months.

6 · Intra-operative principles

- Use a regional block, ketorolac 0.5 mg/kg, antibiotics, and a tourniquet or epinephrine. Cystoscopy is only for severe proximal or redo cases.
- Rapidly absorbed fine sutures beat polydioxanone, whose slow absorption favours fistulae. Close tension-free with non-overlapping lines, and **never use hair-bearing skin**.

7 · Curvature

Curvature decides one-stage or staged repair.

It comes from ventral skin tethering, dysplastic dartos, a short urethra and corporal disproportion.

ALGORITHM

1. Deglove to the penoscrotal junction and excise dysplastic dartos.
2. Make an **artificial erection**.
3. If straight, then proceed to urethroplasty.
4. If residual curvature is <30°, then do dorsal plication.
5. If >30°, then divide the urethral plate.
6. If still >30°, then do ventral corporal lengthening (usually staged).

- Mild is <30°, moderate 30–45° and severe ≥45°. Eyeballing underestimates curvature.
- Plication risks nerve injury, recurrence and shortening. Lengthening uses corporotomies, with or without a dermis or tunica vaginalis graft.
- Recurrence is 28% after plication vs 9% after grafting, and not dividing the plate is an independent risk, so correct fully the first time.

8 · Urethroplasty

Tubularised incised plate (TIP) repair is used for >90% of distal repairs.

Technique	Indication	Key point / pitfall
MAGPI	Glanular, mobile meatus, no curvature	Used more proximally gives a retrusive meatus
Urethromeatoplasty	Stenotic distal meatus	Heineke–Mikulicz
GAP (glans approximation)	Wide deep groove (MIP)	Two-layer neourethra

Technique	Indication	Key point / pitfall
Thiersch–Duplay	Wide plate	Tubularise over 12 Fr
TIP / Snodgrass	Distal, midshaft; narrow plate; mild curvature	Midline relaxing incision; stop 2–3 mm short; meatal stenosis; inlay graft if poor plate
Mathieu (flip-flap)	Distal, shallow plate	Narrow base causes ischaemia, fistula; fewer stenoses than TIP
Onlay island flap	Plate kept but narrow	Diverticulum if too wide; less stenosis
Duckett tube	Proximal; plate divided	Higher complications; diverticulum
Two-stage Bracka	Proximal severe curvature, redo, poor plate	Stage 1 quilted preputial/buccal graft; stage 2 (≥6 months) tubularise
Two-stage Byars	Proximal severe curvature	Dorsal prepuce flaps; highest staged complications

- Staged repair (STAG) has the lowest proximal complication rate (≈ 23%), because urethroplasty over a fresh corporotomy or graft fails. Repeat the artificial erection at stage 2. Proximal TIP has been abandoned because of devascularisation and recurrent curvature.

> **KEY POINT**
>
> Why proximal repairs fail (Poiseuille): resistance rises with length / radius⁴. A long, rigid tube ending in a non-distensible glans raises proximal pressure, causing fistula, diverticulum and stricture. 25–75% need extra procedures after planned two-stage repair.

- Glansplasty needs a tension-free **two-layer closure**, and a **barrier layer** (dartos or tunica vaginalis) reduces fistula.

9 · Post-operative care

- Leave a compressive dressing for ≈ **72 h**, and divert urine for **3–21 days** according to complexity. Short distal repairs need no stent. Give oxybutynin until the stent is out, with paracetamol and ibuprofen.

10 · Complications

Complication	Standard definition
Fistula	Urethral leak below meatus
Glans dehiscence	Wing separation; glans fusion <2 mm
Meatal stenosis	Obstructive symptoms and **<8 Fr** prepubertal / <12 Fr postpubertal
Diverticulum	Segmental sacculation on voiding
Recurrent curvature	**>30°** on erection

*See the paper answer **Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula** (Section 4, Ch 56).*

Also, from the condensed book (not in the paper answer):

- **Balanitis xerotica obliterans** needs complete excision and a staged graft. Recurrence is up to **40%**.
- **Meatal stenosis** is commoner after TIP than Mathieu and is treated by dilatation, then meatotomy or redo. A late **stricture** needs buccal urethroplasty, often staged.

> **KEY POINT**
>
> Nearly every complication traces to distal obstruction, ischaemia or tension, so repair begins by calibrating the distal urethra.

Reoperation

- Excise scar widely and transfer tissue rather than doing "small operations".
- **Buccal mucosa** is the graft of choice. Make it 10% larger, then tubularise ≥6 months later; lip mucosa suits the glans. Always do cystoscopy and an artificial erection.
- The **Cecil–Culp** procedure buries the penis in the scrotum when skin is short.

11 · Outcomes and follow-up

- A TIP meta-analysis (4,675 boys) found complications in 10.6% and reoperation in 4.5%. Distal success is 83–95%, while proximal complications are 23–68%.
- About 75% of complications occur by 3 years, so follow proximal cases **through puberty**. Recurrent UTI needs imaging for diverticulum, stricture or utricle.
- Proximal cases have worse adult sexual outcomes, and penile length is the chief complaint.

57 Etiology, Diagnosis and Management of the Undescended Testis

1 · Definitions

A retractile testis is watched; an undescended one is operated on.

LAQ Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods

Paper answer · P1 · P2 · P3 · merged from 4

Asked as: P1 LAQ "Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis" (Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child, Winter 2024 Paper I Q2; Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent, Practice Set 14 Paper I Q1) · P3 SAQ "Maldescended (undescended) testes and treatment algorithm" (Treatment algorithm of undescended testis, Summer 2018 Paper III Q3a; Maldescended testes, Practice Set 5 Paper III Q3b) · P2 SAQ "Undescended testis" · P1 SAQ "Methods of orchidopexy"

Maldescent is taken as any testis not in a stable scrotal position: undescended, ectopic, retractile and acquired.

Development of the testis

- **Indifferent gonad:** the gonadal (germinal) ridge forms at **4 weeks** on the medial urogenital ridge. It has four components: coelomic epithelium, underlying mesenchyme, adjacent mesonephros and **PGCs (primordial germ cells)**.
- PGCs arise in the **yolk sac wall in week 3** and migrate along the dorsal mesentery of the hindgut to the ridge (about 6 weeks).
- **SRY (sex-determining region Y)** on the short arm of Y, partly activated by SF1 (steroidogenic factor 1), switches on **SOX9** in pre-Sertoli cells. Sertoli cells then appear (about 40 days). Without SRY no Sertoli cells form and the gonad becomes an ovary.
- Sertoli cells enclose germ cells as **testicular cords (about 6 weeks)**, which canalise into seminiferous tubules.
- **Rete testis** forms with the cords and joins **5–12 mesonephric tubules**, which become efferent ductules. Full continuity (tubules, rete, efferent ductules, epididymis) is established by the **third month**.
- The testis turns round and a thick **tunica albuginea** forms. Testes become spherical after 7–8 weeks; ovaries stay oblong.

Hormone	Cell	Action	Male remnants
AMH (anti-Müllerian hormone)	Sertoli	Müllerian regression, weeks 8–10	Appendix testis, prostatic utricle
Testosterone	Leydig (from 8 weeks; driven by placental hCG (human chorionic gonadotrophin), later fetal LH (luteinising hormone))	Wolffian duct to epididymis, vas, seminal vesicle (weeks 8–12)	Appendix epididymis, paradidymis

Hormone	Cell	Action	Male remnants
DHT (dihydrotestosterone)	Formed locally by 5α-reductase type 2	External genitalia, prostate, scrotum (fused by 10–12 weeks)	–
INSL3 (insulin-like peptide 3)	Leydig	Gubernacular swelling (via RXFP2 receptor)	–

Gonadal descent: starting point

- At **7–8 weeks** the testis lies near the kidney (upper lumbar), held by the **cranial suspensory ligament (CSL)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks, attaches to the **epididymis** and runs to the future inguinal canal.
- **Female:** the CSL persists; the gubernaculum becomes the ovarian and round ligaments; the ovary descends only into the pelvis.

Phases of testicular descent (classic two-phase model)

	Transabdominal phase	Inguinoscrotal phase
Timing	Weeks 10–15	From the 7th month; inguinal passage rarely before 22 weeks; most testes scrotal after 27 weeks
Events	**CSL regresses**; **gubernaculum swells** (outgrowth); its cord shortens, holding the testis at the internal ring while the kidney and abdomen grow away (the kidney moves cranially)	Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; the bulb then regresses; the processus closes
Control	CSL regression by **androgen**; swelling by **INSL3** from Leydig cells, via its receptor RXFP2 (occurs even in androgen resistance)	**Androgen**, mainly indirectly via the **genitofemoral nerve (GFN)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure
Failure gives	Intra-abdominal testis	Canalicular or ectopic testis

- **Modern view:** the "transabdominal descent" is largely relative, not true migration. Barteczko and Jacob describe **five phases** (5 weeks to 28 weeks). The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis pass** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster growth (myogenesis).
- Spontaneous postnatal descent occurs mainly in the first 3–6 months.
- **Also know:** AMH defects (persistent Müllerian duct syndrome) cause cryptorchidism; Insl3-knockout mice are cryptorchid; the scrotum forms by 10–12 weeks under DHT.

Failed step	Result
CSL retained	Intra-abdominal testis
Abnormal or ectopic gubernacular attachment	Canalicular testis; ectopic (superficial inguinal pouch, perineal, femoral, prepenile)
Processus not closed	Hernia, hydrocele (patent in 87% of unilateral cases)

Failed step	Result
Fibrous remnant of processus / processus not fully involuted	Acquired (ascending) testis; acquired cryptorchidism

Abnormalities of descent: definitions and classification

Type	Description
Undescended (cryptorchid)	Arrested on the normal path: abdominal, "peeping" at the ring, canalicular, high scrotal/prescrotal
Ectopic	Off the normal path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, contralateral scrotum (transverse, crossed)
Retractile	Scrotal testis pulled up by the cremaster; stays down after sustained traction; review yearly (risk of ascent); some become acquired
Acquired (ascending)	Documented scrotal earlier, ascends later; diagnosed at 8–11 years; fibrous processus remnant
Recurrent, secondary, iatrogenic	Re-ascent after spontaneous descent; after hernia repair or orchidopexy
Vanishing vs agenesis	Lost after formation, in utero or perinatally (torsion; monorchia, anorchia) vs never formed

- Clinically: **palpable** (canalicular, high scrotal, ectopic) or **non-palpable** (intra-abdominal, "peeping", vanished/absent).

Epidemiology and aetiology

- 1–4% of term boys (P2: 2–4% at birth, 1–2% at 3 months); up to 45% of preterm boys. Most descend by 6 months.
- 75–80% palpable; 60–70% unilateral.
- Descent needs Leydig hormones: **INSL3** (transabdominal phase) and **androgen** (inguinoscrotal phase), with the genitofemoral nerve/CGRP guiding the gubernaculum.
- Risk factors: prematurity, low birth weight, syndromes (prune belly, cerebral palsy).

Associated anomalies and sequelae

- **Testicular: TTE (transverse testicular ectopia)**, polyorchidism, splenogonadal fusion; epididymal non-fusion.
- **PMDS (persistent Müllerian duct syndrome):** mutation of AMH or its receptor (AMHR2); bilateral cryptorchidism or TTE.
- **Genital and DSD (differences of sex development):** hypospadias (12–24%); bilateral non-palpable testes need karyotype and 17-OHP (17-hydroxyprogesterone) to exclude **CAH (congenital adrenal hyperplasia)**.
- **Syndromes:** prune belly (all), Klinefelter, androgen insensitivity, hypogonadotrophic hypogonadism, cerebral palsy, gastroschisis, omphalocele, myelomeningocele, PUV (posterior urethral valves).
- **Sequelae:** subfertility (worse if bilateral), **TGCT (testicular germ cell tumour) 2–5-fold**, torsion, hernia, trauma.

Evaluation

- **History:** prematurity, testes once scrotal (acquired), hernia, family history, salt-wasting illness.

- **Examination:** warm room; supine and frog-leg, then cross-legged or squatting. Sweep ("milk") from the anterior superior iliac spine down the canal to the scrotum to find the lowest position; check the superficial inguinal pouch.
- Sustained traction fatigues the cremaster: a **retractile** testis stays in the scrotum after gentle traction.
- Note penis size, meatus (hypospadias), scrotal development, syndromic features. Contralateral hypertrophy suggests a vanished testis.
- **No ultrasound or MRI (magnetic resonance imaging)**: inaccurate (ultrasound sensitivity about 45% for non-palpable testes); they never change the operation or remove the need for surgery, and delay surgery.

Bilateral undescended testis at 3 years

- This is **late presentation**; surgery is advised from 6 months, preferably by 12 months and by 18 months at the latest.
- **Bilateral palpable:** no investigations needed; plan orchidopexy, both sides usually in one sitting.
- **Bilateral non-palpable (or with hypospadias, small phallus or micropenis): treat as DSD until proven otherwise**, with urgent work-up:
- **Karyotype**; electrolytes and **17-OHP** to exclude **CAH** in a virilised 46,XX child.
- **Hormones:** LH, FSH (follicle-stimulating hormone), testosterone, **AMH and inhibin B**. Measurable AMH or inhibin B confirms testicular tissue; high gonadotrophins with absent AMH suggest **anorchia**. The **hCG stimulation test** (testosterone rise) is now less used.
- Pelvic ultrasound for Müllerian structures.
- **Both testes very high or of doubtful viability:** stage the sides, the second at 6 months, to avoid bilateral atrophy.
- **Biopsy** only if DSD or dysgenesis is suspected. **Bilateral anorchia:** testosterone at puberty, prostheses later.
- Adjuvant LHRH (buserelin) in bilateral cases is experimental.

Management principles (AUA (American Urological Association) 2014; EAU (European Association of Urology) similar)

- **Observe to 6 months** (corrected age) for spontaneous descent; refer if still undescended.
- **Orchidopexy from 6 months, preferably by 12 months, and by 18 months at the latest.** Reason: germ-cell and Leydig-cell loss starts in infancy; early surgery restores testicular growth and lowers cancer risk. **Kollin RCT (randomised controlled trial):** surgery at 9 months gave better testicular growth than at 3 years.
- **No hormonal therapy** (hCG or LHRH (luteinising hormone-releasing hormone)) to induce descent: descent in about 20% only; hCG may increase germ-cell apoptosis.
- **Goals of surgery:** tension-free **dependent scrotal** testis with intact blood supply; ligate the patent processus.

Treatment algorithm

```
Boy with empty scrotum ├─ Bilateral non-palpable ± abnormal phallus → DSD work-up (karyotype, hormones, electrolytes) └─ Otherwise → re-examine at 6 months corrected age Still undescended → refer → examination under anaesthesia (EUA) ├─ Retractile → yearly review (risk of ascent) ├─ Palpable (undescended or
```

ectopic) → inguinal or scrotal (Bianchi) orchidopexy └─ Non-palpable → diagnostic laparoscopy ├─ Testis seen → laparoscopic orchidopexy │ vessels short (testis not within 2–4 cm of ring) → Fowler–Stephens │ (clip testicular vessels; testis lives on vasal collaterals; │ one-stage, or two-stage with second stage at 6 months) or Shehata traction ├─ Vessels + vas enter ring → inguinal or scrotal exploration (excise nubbin) └─ Blind-ending vessels → stop (vanished testis)

- **Scrotal nubbin with contralateral hypertrophy** may start with scrotal exploration.
- **Postpubertal intra-abdominal testis with a normal contralateral testis: orchidectomy** (poor function, cancer risk); offer a prosthesis.

Methods of orchidopexy: palpable testis

- **Inguinal orchidopexy** (steps): 1. Groin incision. 2. Open external oblique (spare the ilioinguinal nerve). 3. Free the cord; divide the gubernaculum. 4. Separate and **ligate the sac (patent processus) at the internal ring**. 5. Retroperitoneal mobilisation; divide lateral bands. 6. **Prentiss manoeuvre** if needed (medial transposition under the inferior epigastric vessels) for extra length. 7. **Subdartos pouch.** Transalbugineal sutures are unnecessary.
- **Trans-scrotal (Bianchi) orchidopexy** for low testes: high scrotal incision; sac ligated through it; inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain.

Methods of orchidopexy: non-palpable testis

Examination under anaesthesia, then laparoscopy.

Method	Principle	Success
Primary laparoscopic	Mobilise vessels, preserve vasal collaterals, new hiatus medial to epigastric vessels	>90% (97%); primary orchidopexy, when possible, is most successful
Open transabdominal (Jones and Bagley)	Preperitoneal high mobilisation	95%
Fowler–Stephens (one- or two-stage)	Divide testicular vessels high; testis survives on **vasal and gubernacular collaterals**; second stage after 6 months	60–97% (one-stage 63%, two-stage 68%)
Shehata traction	Fix testis near the opposite anterior superior iliac spine; vessels lengthen; orchidopexy at 12 weeks	84–85%; atrophy 10%
Microvascular autotransplantation	Testicular vessels to inferior epigastric vessels	96%
Orchidectomy	Postpubertal, dysgenetic or atrophic testis	—

- Fowler–Stephens is for testes **not within 2–4 cm of the internal ring**.
- **Also know:** staged (two-stage) inguinal orchidopexy; Dessanti PTFE (polytetrafluoroethylene) sleeve traction; gubernaculum-sparing technique; redo approaches (Redman lateral, Cartwright fascial patch).

Complications of orchidopexy

- **Atrophy** (highest after Fowler–Stephens), retraction or re-ascent, vas or vessel injury, haematoma, infection; torsion very rare after subdartos fixation.

Follow-up and prognosis

- Check size and position at 6 months. Offer a prosthesis for an atrophic testis.
- **Fertility:** paternity about 80% (four-fifths) after unilateral and one-third after bilateral; worse when bilateral (oligospermia 42%, azoospermia 31% in one large review).
- **Cancer:** 2–5-fold TGCT risk (P1: about 3-fold), higher with late surgery; orchidopexy after 10–11 years carries relative risk 5.8 (Walsh).
- Counsel on fertility and cancer risk; teach **self-examination** after puberty.

Diagrams to draw:

- Sagittal fetus showing CSL, gubernaculum, processus vaginalis and both phases of descent.
- Line of descent with ectopic sites.
- Laparoscopic findings at the internal ring: testis, vessels entering the ring, blind-ending vessels.
- Fowler–Stephens collateral supply: testicular artery clipped, deferential and cremasteric arteries intact.

> **Key points to remember:**
>
> - SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
> - Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
> - Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
> - No imaging, no hormones.
> - Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
> - Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
> - Subdartos pouch; Prentiss manoeuvre gives extra length.
> - Fowler–Stephens relies on vasal collaterals.

Term	Key point (beyond the papers)
Undescended testis (UDT)	Not in the normal scrotal position (midpoint at or below mid-scrotum). It is palpable or non-palpable; non-palpable means undescended or absent.
High scrotal testis	Stable, above the scrotal midpoint; a mixed group, not routinely called undescended.
Retractile	One that rarely stays down or rests high may progress to acquired UDT.
Recurrent	Undescended at birth, descended spontaneously, then re-ascended (10–22%, at 1–5 years).
Secondary / testicular retraction	Suprascrotal after hernia repair / after orchidopexy (iatrogenic).
Agenesis	Never formed; comes with ipsilateral Müllerian persistence (extremely rare).

2 · Embryology of descent

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

Also, from the condensed book (not in the paper answer):

- The gubernaculum does not pull the testis; it swells, then regresses.

Phase	Timing	Event	Control
Transabdominal (relative)	10–15 weeks	Gubernacular swelling (outgrowth) from 12 weeks; cranial mesonephric ligament regresses by 13 weeks	INSL3 and androgen act together
Inguinoscrotal	**20–28 weeks**	The canal widens and the processus forms; rapid passage through the canal; the gubernaculum then regresses. Most are scrotal after 27 weeks.	Androgen via the genitofemoral nerve (**GFN**), which releases calcitonin gene-related peptide (**CGRP**)

> **KEY POINT**
>
> Outgrowth creates the canal; regression lets the testis follow, and a defect in either gives the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails.

Minipuberty is a hormonal surge peaking at 1–3 months, when gonocytes become **adult dark (Ad) spermatogonia**, the stem-cell pool. This is where a positional defect becomes a germ cell deficit.

3 · Epidemiology, aetiology and associations

Spontaneous descent ends by about 6 months.

- Prevalence is **2–4% at birth** (term), falling to **1–2% at 3 months**; up to 45% in preterm boys.
- About 88% of high scrotal testes descend spontaneously, but only about half of extrascrotal ones.
- 80–85% are isolated. Risk factors are **prematurity and low birth weight**, family history, maternal diabetes and smoking. Diethylstilbestrol doubles risk.
- It is a polygenic threshold trait, and part of **testicular dysgenesis syndrome** with hypospadias, infertility and testicular germ cell tumour (TGCT).
- No hormone predicts germ cell number, so hormones are not tested routinely outside DSD and anorchia.

Associated condition	Cryptorchidism	Implication
Prune-belly syndrome	All (bilateral abdominal)	Abdominal orchidopexy
Spigelian hernia	53–80%	Absent gubernaculum/canal
Cerebral palsy	25–54%	Often bilateral
Gastroschisis	5–25%	**50–55% descend spontaneously — observe**
Omphalocele, imperforate anus, myelomeningocele, posterior urethral valves	12–33%	Caudal-field anomalies are ipsilateral in 90%
Congenital diaphragmatic hernia	7–30%	Always ipsilateral

Associated condition	Cryptorchidism	Implication
Klinefelter (47,XXY)	> 50% of prepubertal Klinefelter boys	< 2% of non-syndromic UDT; more with hypospadias
Persistent Müllerian duct syndrome (PMDS)	Bilateral 59%; hernia uteri inguinalis; transverse ectopia 22%	AMH or AMHR2 mutation; adult infertility ≥ 80%, malignancy risk

Congenital vs acquired

Feature	Congenital	Acquired
Diagnosis	Birth/neonatal	Mean 8–11 years
Position	Often higher	Lower; often superficial inguinal pouch (SIP)
Processus vaginalis	Patent (87% unilateral)	Usually closed (patent 45–50%)
Epididymal anomalies	37–60%	11–31%

An acquired UDT is a never-fully-descended testis unmasked by growth, with semen results like congenital cases. Retractile testes share this pathology and ascend in about 2–3%, so they need **annual examination through childhood**.

4 · Histology, associated anomalies and fertility

Germ cells are lost from late infancy, so operate early.

- Germ cell loss begins **after infancy** and progresses; the contralateral testis is affected less.
- The UDT is smaller at birth, so the defect is intrinsic and worsened by position.
- Spermatogonia per tubule (S/T) is 2.5 at birth, with a nadir of 1.2 at 2–3 years.
- A **patent processus** is present in 87% of unilateral and 71% of bilateral cases. A UDT is at higher torsion risk.

Anomaly	Key features	Management
Polyorchidism	Usually left, scrotal in 75%; Kumar classification by drainage (A with vas, B without)	Observe a normal scrotal testis; orchidopexy if undescended with intact drainage
Splenogonadal fusion	Left side; the continuous form comes with limb defects and micrognathia	Recognise it to avoid a needless orchidectomy
Transverse testicular ectopia	Hernia with a contralateral non-palpable testis; PMDS in 20–50%	Laparoscopy; trans-septal placement if vasa fused

Trial	Design	Result	Take-home
Kollin et al.	Randomised trial: orchidopexy at 9 months vs 3 years	Early: more Sertoli and germ cells, better testicular growth (to 4 years)	Operate in first year; age at surgery, not hormones, predicts germ cells

Fertility outcome

Outcome	Former unilateral	Former bilateral
Oligospermia / azoospermia	31% / 14%	42% / 31%
Paternity	**≈ 90% of those attempting — as controls**	≈ 65% of those attempting (≈ 1/3 overall)

- In unilateral cases the normal testis carries fertility, so semen analysis overstates infertility; in bilateral cases it reflects true infertility.
- Ad spermatogonia at biopsy predict adult sperm counts; if they are absent, all counts are subnormal.
- Even after first-year surgery, 20–25% remain at infertility risk. Surgery **before 18 months** reduced assisted reproduction, and each 6-month delay added 5%.

5 · Malignancy

Orchidopexy lowers cancer risk only partly.

- TGCT risk is **2–5×** (pooled odds ratio (OR) ≈ 4). About 1% of cryptorchid men get TGCT, and 5–10% of TGCT patients had a UDT.
- In the **Swedish population study** the OR was about 2.8 if treated before puberty vs about 5.0 after. † Pettersson 2007: relative risk (RR) 2.23 if operated before 13 years vs 5.40 at 13 or older. A meta-analysis gave RR **5.8** for orchidopexy after 10–11 years.
- Untreated adults have OR ≈ 14.
- A persistently cryptorchid testis develops **seminoma** (74%), whereas tumours after orchidopexy are mostly non-seminoma (63%).
- Germ cell neoplasia in situ occurs in 2–3% of formerly cryptorchid men, so biopsy is for DSD and chromosomal cases. Biopsy at orchidopexy does not raise TGCT risk.
- Postpubertal UDT with a normal contralateral testis needs **orchidectomy** (up to age 50), with a prosthesis at the same sitting.
- **Microlithiasis** is not an independent TGCT risk, so routine ultrasound surveillance is not needed.

6 · Diagnosis

Examination decides; imaging adds nothing.

Examination — the gold standard

ALGORITHM

1. Warm room and hands. Examine supine, then cross-legged or squatting.
2. With lubricated fingers, sweep one hand from the anterior superior iliac spine (ASIS) along the canal and palpate the lowest position with the other.
3. Compress the canal and invaginate the scrotum to feel the SIP.
4. Apply sustained traction to fatigue the cremaster. If it **stays, it is retractile; if it springs back, it is undescended**.

- About **80% palpable**, and 60–70% are unilateral. At surgery a non-palpable testis is abdominal or peeping in 25–50%, vanished in 15–40% and a missed extra-abdominal testis in 10–30%.
- Examination under anaesthesia (EUA) is the final arbiter.

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

SAQ **Vanishing testes**

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3b

Definition

- A testis that **developed normally and was then lost**, usually from **torsion or a vascular accident in fetal life**, after genital masculinisation but before scrotal fixation.
- Unilateral loss gives **monorchia**; bilateral loss gives **anorchia** (testicular regression syndrome), which is rare.
- Excised "nubbins" contain haemosiderin.

Incidence

- **15–40% of nonpalpable testes** at surgery.

Clinical features

- **Contralateral compensatory hypertrophy:** volume >2 mL (sensitivity 72%, specificity 100%) or length about 19–20 mm suggests a solitary testis.
- No palpable intrascrotal "appendage" tissue; sometimes a scrotal nubbin.
- These aid counselling but **do not replace exploration**.
- **Anorchia:** micropenis in nearly half.

Evaluation

- **Imaging** (ultrasound, MRI [magnetic resonance imaging]) is not recommended: it does not change the operation.
- **Bilateral nonpalpable:** exclude DSD (difference of sex development) and CAH (congenital adrenal hyperplasia) first.
- Anorchia: high FSH (follicle-stimulating hormone) and LH (luteinising hormone); low AMH (anti-Müllerian hormone) and inhibin B; no rise after hCG (human chorionic gonadotrophin). Most reliable during mini-puberty.

Management: diagnostic laparoscopy (after examination under anaesthesia)

Laparoscopic finding	Action
Blind-ending vessels and vas above a closed ring	Diagnosis confirmed; stop
Vessels and vas entering the internal ring	Inguinal or scrotal exploration; excise the nubbin
Testis seen	Orchidopexy (single stage or Fowler–Stephens)
Neither vas nor vessels	Search up to the kidney (agenesis very rare)

- **Excise the remnant:** viable germ cells or tubules in 5–15%.
- **Scrotal-first exploration** is reasonable with a palpable nubbin and contralateral hypertrophy (length ≥1.8 cm); laparoscopy is then needed in about a quarter.
- **Contralateral fixation:** debated; not required by the AUA (American Urological Association) algorithm.
- **Prosthesis:** offer at least 6 months after scrotal surgery or after puberty, inguinal approach.
- **Anorchia:** testosterone from puberty. **Also know:** agenesis shows ipsilateral Müllerian persistence.

Diagram to draw: laparoscopic view of blind-ending vessels at a closed ring.

Key points to remember:

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

Imaging — not recommended

The AUA (2014) advises no ultrasound or MRI. Ultrasound sensitivity is only about 45%, and imaging merely delays surgery.

Bilateral non-palpable or UDT with hypospadias: DSD work-up

- A newborn with bilateral non-palpable testes needs an **urgent karyotype and hormones** to exclude **congenital adrenal hyperplasia (CAH)**, because of salt-wasting.
- Proximal hypospadias with UDT carries a chromosomal abnormality in 32–47% (see Ch 56).
- Micropenis with bilateral UDT suggests hypogonadotrophic hypogonadism or anorchia.

Marker	Anorchia	Bilateral abdominal UDT
FSH, LH	Raised	Normal
AMH, INHB (best prepubertal Sertoli markers)	Undetectable/low	Present
Basal testosterone	Low in first year only	Normal
Testosterone after hCG	No rise	Rises

hCG stimulation is no longer the test of choice, and **laparoscopy remains the confirmatory test**.

7 · Management principles and timing

- **Timing**: observe to **6 months** (corrected age), then refer; operate at **6–18 months**, preferably by 12.
- Retractile testes need no surgery but yearly examination. Acquired UDT should be operated on, especially if bilateral.
- A pubertal or postpubertal boy with a normal contralateral testis needs orchidectomy, with or without a prosthesis.

Hormonal therapy — not recommended (AUA)

- Intranasal LHRH gives descent in only about 20%, and hCG was never placebo-tested. hCG may increase germ cell apoptosis and reduce adult volume.
- Adjuvant LHRH for bilateral UDT with absent Ad spermatogonia (Hadziselimovic) has pilot data only.

KEY POINT

Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable. Descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement.

8 · Surgery for the palpable testis

The goal is a tension-free scrotal testis with intact vessels.

Inguinal orchidopexy — key steps

ALGORITHM

1. EUA, skin-crease incision, check the SIP.
2. Open the external oblique, sparing the ilioinguinal nerve.
3. Free the testis; divide the gubernaculum distal to the sac (protect a long-looping vas).
4. **Separate the patent processus from vas and vessels without skeletonising; high-ligate it at the internal ring** (main length gain).
5. If short, divide lateral spermatic bands and dissect retroperitoneally; then the **Prentiss manoeuvre** (medial to the inferior epigastric vessels).
6. Transverse scrotal incision and **subdartos pouch**; pass the testis untwisted.
7. Close dartos around the cord; transalbugineal sutures are unnecessary.

- If the testis cannot reach the scrotum, stage the procedure. Orchidectomy is only for an atrophic or abnormal testis, a postpubertal boy or an insufficient vas.
- Biopsy is not routine; it is indicated in DSD and considered in bilateral UDT or after 12–13 years.

Scrotal (Bianchi) orchidopexy

The testis is delivered through a high scrotal incision, the processus ligated and the testis fixed in a subdartos pouch. It is increasingly first choice for palpable, especially low, testes.

9 · The non-palpable testis

Laparoscopic findings at the ring decide the next step.

ALGORITHM

1. EUA: if now palpable, do an inguinal or scrotal orchidopexy.
2. If **still non-palpable, do diagnostic laparoscopy** (ring patency, vessels, vas, testis).
3. If **blind-ending vessels** (with or without vas) lie above a closed ring, the testis has vanished: stop (the only unequivocal endpoint).
4. If **vessels and vas enter the ring**, explore the groin/scrotum and excise the nubbin (viable tubules 5–15%).
5. If the testis is **intra-abdominal**, do a primary laparoscopic or Fowler–Stephens (one-or two-stage) orchidopexy; orchidectomy if postpubertal or dysgenetic.
6. If it is peeping at the ring, do a laparoscopic or inguinal orchidopexy.
7. If neither vas nor vessels are seen, search the perivesical area and retroperitoneum to the kidney.

- An open ring usually, but not always, means a viable distal testis, and apparently blind-ending vessels may coexist with an ectopic testis, so keep exploring if equivocal.
- The AUA does not require fixing the contralateral solitary testis.

Laparoscopic orchidopexy

ALGORITHM

1. Empty bladder and stomach. Open (Hasson) umbilical port; CO2 at 8–12 mmHg. Inspect both rings and add working ports.
2. Divide the gubernaculum distal to the ring and use it as a handle.
3. Incise peritoneum lateral to the vessels and distal to the vas; mobilise the vessels, **preserving the peritoneal triangle between vas and vessels** (keeps Fowler–Stephens possible).
4. Make a new hiatus medial to the inferior epigastric vessels.
5. Bring the testis through a trans-scrotal port to a subdartos pouch without tension.

- Ring closure is unnecessary, and the robot offers no clear advantage.

Fowler–Stephens (FS) and alternatives

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

Also, from the condensed book (not in the paper answer):

- Two-stage FS: clip the vessels high, wait about **6 months**, then mobilise on a wide peritoneal pedicle. Two-stage is slightly favoured.
- Keeping peritoneum over the vasal vessels and **sparing the gubernaculum** lowers atrophy. Avoid dividing vessels when possible, because it reduces germ cells.
- Operate on bilateral high testes about 6 months apart.

KEY POINT

Every abdominal technique answers one question: how to gain length without losing blood supply. FS sacrifices the testicular artery; traction gains length over time while keeping it, so vessel division is the last resort.

10 · Complications and follow-up

Complication	Cause / prevention	Management
Atrophy (≈ 2% modern; historically 24% abdominal/peeping vs 10% distal)	Vessel injury, skeletonisation, FS, tension	Offer a **prosthesis** ≥ 6 months after scrotal surgery or after puberty, inguinal route, fixed to dartos
Retraction / non-scrotal (≈ 2–10%)	Inadequate retroperitoneal mobilisation, unligated processus	Redo: a prescrotal testis by the scrotal approach; otherwise a lateral approach through unscarred tissue or a fascial patch on the cord. Results equal primary surgery.
Vas injury	Long-looping vas cut with the gubernaculum	Divide the gubernaculum distal to the sac; avoid skeletonising
Torsion after orchidopexy	Rare	Prevented by subdartos fixation
Laparoscopic	Bladder/vessel injury, hypercapnia, bowel obstruction	Hasson entry, bladder drainage

Follow-up: examine at ≥ **6 months** for position and size. Counsel on fertility and cancer, and teach self-examination from puberty.

61 Urologic Considerations in Pediatric Renal Transplantation

LAQ Urological evaluation and management of pediatric kidney transplant patient.

Paper answer · P1

Type: LAQ (25 marks)
Source: 2022-Summer Paper 1, Q2

Introduction

- Children needing KRT (kidney replacement therapy) often have CAKUT (congenital anomalies of the kidney and urinary tract). The urologist must ensure that the bladder that destroyed the native kidneys does not destroy the graft.
- **Five goals:** unobstructed drainage; a capacious, low-pressure reservoir with social continence; volitional emptying; no infection; the fewest operations.

Causes of ESKD (end-stage kidney disease) in children

- Commonest age at transplant 13–17 years.
- **Younger:** dysplasia, hypoplasia, aplasia and obstructive uropathy, mainly PUV (posterior urethral valves). **Older:** FSGS (focal segmental glomerulosclerosis) and neuropathic bladder.

Pretransplant evaluation

- **Every child:** urological history, renal ultrasound, uroflow with PVR (post-void residual) if toilet-trained.
- **VCUG (voiding cystourethrogram)** only with neuropathic bladder, PUV, VUR (vesicoureteral reflux), previous lower-tract surgery, recurrent febrile UTI (urinary tract infection) or unexplained hydronephrosis.
- **Urodynamics** for neuropathic bladder, PUV or large ureterocele, refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI. They measure capacity, compliance, detrusor activity, emptying and sphincter function, and are repeated.

```
History positive → ultrasound + VCUG → urodynamics Poor capacity/compliance → anticholinergic, CIC, cycling → still poor → augment Poor emptying → CIC ± catheterisable stoma Adequate → clear for transplant History negative → ultrasound + PVR → normal → clear
```

- An anuric child with no urological history needs no further tests.

Pretransplant management

1. Bladder optimisation (drugs → CIC → cycling → augmentation)

- Commonest problem: low capacity, detrusor overactivity, poor compliance. High pressure harms the kidney even without reflux.
- **Target:** age-expected capacity at detrusor pressure **<40 cm H_2O**.
- Antimuscarinics (oxybutynin titrated; solifenacin), mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage may worsen emptying, so add **CIC (clean intermittent catheterisation)**. Prove volitional emptying by repeated low PVRs.
- A family's adherence to CIC predicts adherence to immunosuppression.

2. Valve bladder versus neuropathic bladder

- PUV bladder moves from overactivity to **myogenic failure**: postpone augmentation. Neuropathic bladder may deteriorate under high post-transplant urine output: reconstruct earlier.

3. Defunctionalised bladder (anuria or diversion)

- **Bladder cycling:** catheter instillation of increasing saline volumes, with drugs, until age-expected capacity; confirm by urodynamics. **Never use the graft to test the bladder.**

4. Augmentation

- Indication: cannot reach **>75% expected capacity at <40 cm H_2O** with CIC every 3–4 hours despite maximal therapy. It does not clearly raise transplant complications.

Segment	Point
Dilated ureter (ureterocystoplasty)	Explore first: urothelium, no mucus or acidosis
Ileum, sigmoid	Most used; mucus, stones, infection, malignancy; lifelong follow-up
Stomach	Haematuria–dysuria syndrome in anuric patients; rarely used

- Continent stoma (Mitrofanoff) if the urethra is sensate or difficult; site it medial to the planned incision. Allow **6–12 weeks** healing before transplant.

5. Infection, nephrectomy and dialysis

- Treat stasis (double voiding, CIC); distinguish colonisation from UTI; consider circumcision in boys with valves.
- **Native nephrectomy** (default is to keep): uncontrolled hypertension, nephrotic syndrome, massive polyuria, recurrent pyelonephritis, large stones, malignancy risk, very large kidneys, massive VUR. Retroperitoneal route keeps PD (peritoneal dialysis).
- Keep a normal native ureter for later ureteroureterostomy. Intraperitoneal surgery on PD needs temporary haemodialysis.

At transplantation

- Retroperitoneal Gibson incision; children **<15 kg** may need aorta–caval anastomosis, sometimes transperitoneal.
- **Ureter:** extravesical non-refluxing ureteroneocystostomy (3–4 cm detrusorotomy) preferred; Barry tunnel; ureteroureterostomy to a normal native ureter.
- **Stent:** selective; double-J removed at 4–6 weeks, or early removal at 5–7 days.

Post-transplant complications

Complication	Rate	Management
Ureteric obstruction	6%	Usually distal ischaemic stenosis; >half within 100 days. Stent or nephrostomy; balloon dilatation; open repair (native ureter, bladder flap)
Urine leak	3%	Early; drain creatinine; catheter, nephrostomy, stent, then reimplantation
Graft VUR	2% (significant)	Febrile UTI → reimplant; asymptomatic → observe; injection works in one-third
Stones	2%	Treat if >1 cm and growing, obstructing or infected

- **Rising creatinine with hydronephrosis:** catheter drainage 1–2 weeks (bladder cause?) → biopsy if rejection is suspected → stent or nephrostomy. Obstruction and rejection can coexist.
- Also: lymphocele, **PTLD (post-transplant lymphoproliferative disorder)**, graft thrombosis (a leading cause of early graft loss), late infection reflecting bladder dysfunction.

Follow-up

- Lifelong ultrasound, PVR and bladder surveillance; recurrent UTIs lower graft GFR (glomerular filtration rate).

Diagram to draw: extravesical ureteroneocystostomy with detrusor flaps closed over the ureter.

> **Key points to remember:**
> - Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
> - PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
> - Never test a bladder with a graft; cycle the defunctionalised bladder.
> - Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

Also, from the condensed book (not in the paper answer):

- Graft reflux matters more than native reflux, because the kidney is solitary and immunosuppressed.

1 · Causes of ESKD in children

- The commonest cause of end-stage kidney disease (ESKD) is congenital anomalies of the kidney and urinary tract (**CAKUT**), then obstructive uropathy, mainly posterior urethral valves (**PUV**).
- Older children have FSGS or neuropathic bladder. The commonest age at transplant is **13–17 years**.

> **KEY POINT**
>
> **Five goals**: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it.

2 · Pre-transplant evaluation

- Every child needs a history and renal ultrasound, plus uroflow with post-void residual if toilet-trained.
- VCUG and urodynamics are selective: neuropathic bladder, PUV, reflux, prior surgery, febrile UTI or worsening hydronephrosis.

ALGORITHM

1. Positive history: US and VCUG. If dynamics are abnormal, urodynamics.
2. Treat with drugs and clean intermittent catheterisation (CIC), with or without bladder cycling, then reassess.
3. If still unsafe, augment; then clear for transplant.

3 · Bladder management

The bladder must be safe before the graft goes in.

- The commonest pattern is **low capacity, overactivity and poor compliance**, and high pressure harms the kidney even without reflux.
- The **target** is expected capacity at detrusor pressure **< 40 cm H_2O**, with social continence.
- Use antimuscarinics, mirabegron or botulinum toxin, with CIC as needed.
- Cycle a **defunctionalised bladder** with rising saline volumes, then repeat urodynamics. **Never use the allograft to test a bladder.**
- **Augment** if the bladder cannot reach > **75%** of expected capacity at < 40 cm H_2O with CIC no more often than every **3–4 h** despite maximal therapy. Postpone this in PUV, but augment earlier in neuropathic bladder.
- A dilated ureter is the best segment. Bowel augments need lifelong malignancy surveillance.
- Site a **Mitrofanoff medial to the planned transplant incision**.
- Reconstruct **before** transplant, allowing **6–12 weeks** to heal, except when transplanting before 2 years.

4 · Native nephrectomy

*See the paper answer **Pre transplant nephrectomy** (Section 15, Ch 91).*

Also, from the condensed book (not in the paper answer):

- Combine it with the transplant only to make room, as in a small child with huge kidneys.

5 · Transplant into the abnormal bladder

- A Gibson retroperitoneal incision is standard; a child < **15 kg** may need the aorta and vena cava through a transperitoneal approach.
- The preferred join is an **extravesical non-refluxing ureteroneocystostomy**. **Ureteroureterostomy** to the native ureter is the alternative and the salvage.
- Stent if difficult, removing the double-J at **4–6 weeks**.

6 · Urological complications

Complication	Rate	Timing / management
Ureteric stenosis	**6%**	Most within 100 days; distal, ischaemic. Focal: balloon and stent for 4–6 weeks; otherwise open repair (native ureter, bladder flap); do not delay.
Urine leak	**3%**	Early. Drain-fluid creatinine; check catheter, then US, then MAG3 scan or cystogram. Small: observe; large: drain or explore. Short necrosis: reimplant; long: native ureter, psoas hitch or flap.

Complication	Rate	Timing / management
Significant VUR	2%	See below
Stones	2%	Treat if > 1 cm and growing, obstructing or infected; one-stage modality

Rising creatinine with hydronephrosis

ALGORITHM

1. Indwelling catheter for 1–2 weeks: if creatinine falls, the bladder is the cause.
2. If not, stent and keep the catheter; biopsy if rejection is suspected.
3. If retrograde stenting fails, nephrostomy and antegrade study.

- The graft is unusually sensitive to obstruction, and obstruction and rejection coexist, so stent at a low threshold.
- An **allograft mass is post-transplant lymphoproliferative disorder (PTLD) until proven otherwise**.
- Recurrent febrile UTI from graft reflux with an optimised bladder needs an **open or robotic reimplant**; asymptomatic reflux is observed.

7 · Outcomes

- Urological complications rarely lose the graft, but **≥ 2 UTIs** lower graft GFR at 1 and 4 years.

63 Paediatric Urologic Oncology: Renal and Adrenal

1 · Wilms tumour (WT): epidemiology and genetics

Genetics decide who is screened and who needs kidney-sparing surgery.

LAQ Wilms tumour, including bilateral Wilms tumour

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Wilms tumour: classification, staging and management" · P3 SAQ "Bilateral Wilms tumour: treatment strategies"

Introduction

- Read as "Wilms tumour" (nephroblastoma); "Wilm's bladder" in one P2 wording is taken as a typo.
- WT (Wilms tumour) is an embryonal tumour of metanephric blastema. It forms 95% of kidney cancers under 15 years; median age 3.5 years; over 80% present before 5 years.

Genetics, syndromes and precursors

- **Genetics:** mostly sporadic; familial 1–2%. Genes: **WT1 (11p13)**, IGF2/11p15 (commonest change, about 70%), WTX, CTNNB1; **TP53 in anaplasia (in most anaplastic tumours)**.

Syndrome	WT risk
WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability / retardation; WT1 deletion)	50%
Denys–Drash (46,XY DSD [disorder of sex development], mesangial sclerosis; WT1 point mutation)	50%
BWS (Beckwith–Wiedemann syndrome), hemihypertrophy (11p15)	5–10%

- **Screening:** ultrasound every 3–4 months until 7 years when risk is >1%.
- **Nephrogenic rests** (precursors) occur in over one-third of WT kidneys and over 90% of bilateral cases.

Classification: histology

Group	Features	Significance
Favourable histology	**Triphasic:** blastema, epithelium (tubules), stroma; pseudocapsule	Survival >90%
Anaplasia (unfavourable)	Nuclei ≥3× neighbours (larger), hyperchromasia, abnormal mitoses; **focal** or **diffuse** (diffuse worse than focal)	Chemoresistant; poor even at stage I

- **SIOP (International Society of Paediatric Oncology) post-chemotherapy risk groups:**

Risk	Histology
Low	Completely necrotic
Intermediate	All others (regressive, mixed, epithelial, stromal, focal anaplasia)
High	**Blastemal-predominant**; diffuse anaplasia

- Residual blastema predicts relapse; stromal and epithelial tumours shrink little but are cured by excision.

Staging

COG (Children's Oncology Group): surgical-pathological, before chemotherapy

Stage	Description
I	Confined to kidney, completely resected, capsule intact, no renal sinus vessel invasion
II	Beyond kidney (renal sinus vessels, perirenal fat, vein thrombus) but completely resected
III	Residual abdominal disease: **positive nodes, any spill, prior biopsy**, positive margin, peritoneal implants
IV	Haematogenous metastases (lung commonest, liver, bone, brain)
V	Bilateral at diagnosis (each side staged separately)

- **SIOP staging** uses the same five stages but is applied **after preoperative chemotherapy**. Necrotic, chemotherapy-induced change in the sinus or perirenal fat does not upstage.
- **Biology:** combined **1p and 16q LOH (loss of heterozygosity)** predicts relapse; 1q gain is the next stratifier.

Clinical features

- Abdominal mass; haematuria about 20%; hypertension about 25% (renin).
- Look for aniridia, hemihypertrophy and varicocele. Varicocele or hepatomegaly suggests IVC (inferior vena cava) thrombus.
- Acquired von Willebrand disease in up to 8%.

Investigations

- Ultrasound with Doppler (IVC), then contrast CT (computed tomography) or MRI (magnetic resonance imaging) abdomen, then **chest CT** (lung is the commonest metastatic site; do it before anaesthesia).
- Blood count, renal and liver function, coagulation; syndrome review.

Management: two philosophies

	COG (North America)	SIOP (Europe)
Sequence	**Upfront transperitoneal radical nephrectomy**, then chemotherapy	**Preoperative VCR + AMD** (4 weeks; add DOX for 6 weeks if metastatic), no biopsy, then nephrectomy
Advantage	Untreated histology and biology for risk assignment	Tumour shrinks; spill almost zero
Risk basis	Stage + histology + LOH	Post-chemotherapy histology + stage

(VCR = vincristine; AMD = actinomycin D (dactinomycin); DOX = doxorubicin; RT = radiotherapy; NSS = nephron-sparing surgery.)

Surgical principles (radical nephrectomy)

- Transperitoneal approach; palpate renal vein and IVC before ligation; spare a tumour-free adrenal.
- **Avoid spill** (it makes the tumour stage III).

- **Sample at least 5 lymph nodes** (hilar, paracaval, para-aortic, interaortocaval); this is essential, formal dissection is not. Omitted sampling is the commonest protocol violation.
- IVC thrombus at or above the hepatic veins: preoperative chemotherapy to avoid bypass.

Postoperative therapy (COG, favourable histology)

Group	Treatment
Very low risk (<2 years, <550 g, stage I)	Nephrectomy alone (AREN0532)
Stage I–II	VCR + AMD
Stage III	VCR + AMD + DOX + flank RT
Stage IV	Three drugs + whole-lung RT; lung RT omitted if lung nodules clear by week 6 (AREN0533)
Anaplasia or 1p/16q LOH	Intensified regimens (regimen M adds cyclophosphamide + etoposide)

Bilateral WT (stage V): principles

- **Synchronous bilateral WT** is **5–7%**: **COG stage V**. Each kidney is staged separately.
- Linked to WAGR, Denys–Drash (about 50% risk) and BWS; **nephrogenic rests in over 90%**.
- Aim: **cure while preserving renal tissue (nephrons)**. Renal failure is about 15% at 15 years; ESRD (end-stage renal disease) mainly follows bilateral nephrectomy.
- **No upfront biopsy and no upfront nephrectomy.**

Bilateral WT: COG approach (AREN0534)

```
VCR + AMD + DOX × 6 weeks (no biopsy) → imaging at week 6 ├─ resectable / NSS feasible → bilateral NSS └─ not → 6 more weeks chemotherapy → surgery by week 12 poor response → bilateral OPEN biopsy → further therapy by each kidney's stage and histology ± flank RT
```

- **Why stop at 12 weeks:** in NWTS-4 (National Wilms Tumor Study 4), prolonged chemotherapy in poor responders delayed surgery and some tumours became anaplastic. A non-shrinking tumour is often differentiated and is cured by excision.
- **Why an open, bilateral biopsy:** anaplasia is discordant between kidneys in about 4 of 5.

Bilateral WT: surgery

- **Bilateral NSS** (partial nephrectomy or enucleation); frozen section; lymph node sampling; avoid spill.
- Compressed normal parenchyma is often underestimated, so try NSS in every case. Expert centres achieve bilateral NSS in over 90%.
- Enucleation: positive margins 23–36%, good outcome with flank radiotherapy.
- Unresectable both sides: bilateral nephrectomy and dialysis are rare and only for refractory tumour; transplant after **1–2 years** tumour-free.

Bilateral WT: adjuvant therapy and SIOP approach

- Adjuvant therapy is guided by the worst histology and margins: anaplasia or positive margin means intensified chemotherapy and flank radiotherapy.
- **SIOP:** same principle (no biopsy, NSS by week 12) but **two drugs, VCR + AMD**, if non-metastatic.

Unilateral nephron-sparing surgery

- For syndromic children and solitary kidney; SIOP UMBRELLA criteria are volume <300 mL, a substantial remnant and negative nodes.

Prognosis and outcomes

- Favourable histology: survival >90%.
- **AREN0534 (bilateral):** surgery by week 12 in 84%; bilateral NSS 39%; **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%**, versus 56% and 80.8% on NWTS-5.
- Salvage survival after relapse: 70–80% after VCR/AMD only; about 10% for anaplastic or blastemal relapse.

Late effects and follow-up

- **Late effects:** cardiotoxicity (DOX), second cancers (RT), infertility, renal failure (bilateral).
- **Follow-up (bilateral):** lifelong blood pressure, proteinuria, creatinine (renal function) and imaging, as late relapse occurs.

Diagrams to draw: COG staging diagram showing capsule, renal sinus, nodes and spill; bilateral tumours with planned partial nephrectomies; AREN0534 timeline (weeks 0, 6, 12).

> **Key points to remember:**
>
> - WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
> - Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
> - Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
> - COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
> - Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy; three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP uses two drugs.
> - AREN0534: 4-year OS about 95%; bilateral NSS to avoid ESRD.

Also, from the condensed book (not in the paper answer):

- WT is **6–7%** of childhood cancers and **95%** of kidney cancers under 15 y; median age **3.5 y**, younger if bilateral or syndromic. Refer for genetics if bilateral, under 2 y, with rests, syndromic features or family history.
- WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability) is a WT1 deletion; **Denys–Drash** syndrome (DDS) is a **dominant-negative** point mutation with early end-stage renal disease. Type I tumours (WT1) are stromal with intralobar rests; Type II (11p15, Beckwith–Wiedemann syndrome, BWS) are blastemal/epithelial with perilobar rests.
- Combined 1p and 16q loss of heterozygosity (~5%) earns the intensified **regimen M**; 1q gain (25–30%) is the next stratifier.

Syndrome risk	Examples
High (> 20%)	WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy
Moderate (5–10%)	BWS, isolated hemihypertrophy, Frasier (WT1), Simpson–Golabi–Behmel (GPC3)

Syndrome risk	Examples
Low	Li–Fraumeni, neurofibromatosis 1, Sotos, trisomy 18, Bloom

KEY POINT

Trap: BWS is the **commonest** predisposing syndrome but only moderate risk (21% bilateral); WAGR/DDS are rarer and highest.

ALGORITHM

1. If WT risk is > 1%, do a renal ultrasound (US) every **3–4 months** until **7 y**.
2. If a lesion looks suspicious, do computed tomography (CT) or magnetic resonance imaging (MRI).

- In BWS, hemihypertrophy and similar syndromes, add α-fetoprotein (AFP) 3-monthly to 4 y for hepatoblastoma.
- Screening lowers stage and allows nephron-sparing surgery (NSS), but survival benefit is unproven.

2 · Pathology

- Favourable histology is **triphasic**; the tumour is friable and spills easily.
- **Anaplasia** (chemoresistant): nuclei ≥ **3×** neighbours, hyperchromasia and abnormal mitoses; it is commoner in older children (~13% over 5 y).
- Nephrogenic rests are found in > 1/3 of WT kidneys and **> 90%** of bilateral cases; most regress. Perilobar rests are peripheral (BWS); intralobar rests lie anywhere (WAGR/DDS).

3 · Clinical features and work-up

- A varicocele persisting supine, or hepatomegaly, suggests inferior vena cava (IVC) thrombus. Gross haematuria needs cystoscopy at surgery.
- **Acquired von Willebrand disease** occurs in up to 8%; give desmopressin.
- Ascites beyond the cul-de-sac best predicts rupture; operate urgently only for bleeding. Imaging cannot judge nodes or separate WT from renal cell carcinoma (RCC) or clear cell sarcoma (CCSK).

4 · Staging and COG vs SIOP

COG operates first; SIOP gives chemotherapy first.

- Children's Oncology Group (COG) stages I–V are surgical–pathological. Stage III means **positive nodes, any spill**, implants, a positive margin or piecemeal removal.
- The International Society of Paediatric Oncology (SIOP) gives pre-operative vincristine (VCR) and dactinomycin (AMD) without biopsy.

KEY POINT

COG stage is surgical-pathological, so the surgeon, not the scan, decides chemotherapy intensity and irradiation: one spill converts stage I/II to III.

5 · Surgical principles

- Do a transperitoneal radical nephrectomy and inspect the liver, peritoneum and nodes.

- Sample nodes (COG **≥5 nodes**); omission is the commonest protocol violation and understages the tumour.
- Spill (~10%) causes local relapse; complications (~11%) are mainly haemorrhage and small-bowel obstruction.
- COG favourable histology: stage I–II get VCR and AMD; stage III–IV add doxorubicin (DOX) with flank/abdominal (and lung) radiotherapy (RT). Cyclophosphamide and etoposide are added in regimen M and for CCSK.

Bilateral WT (stage V) and NSS

*See the paper answer **Wilms tumour, including bilateral Wilms tumour** (Section 4, Ch 63).*

Also, from the condensed book (not in the paper answer):

- Use frozen section and a normal-kidney margin in NSS.

ALGORITHM

1. Synchronous bilateral WT: **no upfront nephrectomy, no biopsy**.
2. Give VCR, AMD and DOX for 6 weeks, then re-image.
3. If NSS is feasible, operate.
4. If not, give more chemotherapy, but do **surgery by week 12**.
5. If the tumour does not respond, do a **bilateral open biopsy**.

- Unilateral NSS suits syndromic children or a solitary kidney; UMBRELLA criteria are **<300 mL**, an adequate remnant and negative nodes.

IVC thrombus

- It occurs in ~6–10%. Below the hepatic veins, do upfront nephrectomy with **en bloc** thrombectomy (piecemeal means stage III).
- **At or above the hepatic veins, give pre-operative chemotherapy** to avoid bypass.

6 · Relapse and late effects

Relapse group	Definition	Salvage
Standard	Non-anaplastic, previously VCR/AMD only	70–80%
High	Non-anaplastic, after ≥ 3 drugs	40–50%
Very high	Anaplastic/blastemal	~10%

- Late effects track **RT and anthracycline**: heart failure follows DOX (rare below **150 mg/m²**). Renal failure after unilateral nephrectomy is only **0.6%** at 20 y.

7 · Other renal tumours

Tumour	Key facts	Management
Congenital mesoblastic nephroma (CMN)	Commonest **neonatal** renal tumour; ETV6–NTRK3 fusion in the cellular type	Radical nephrectomy alone; adjuvant only if incompletely resected
Clear cell sarcoma (CCSK)	3%; **bone** and brain metastases; late relapse; never bilateral	DOX-containing chemotherapy; RT even at stage I

Tumour	Key facts	Management
Rhabdoid tumour (RTK)	2%; **most lethal**; SMARCB1/INI1 loss; associated brain tumours	No standard; tazemetostat on trial
Cystic nephroma / cystic partially differentiated nephroblastoma (CPDN)	Identical on imaging; CPDN has blastema in the septa; DICER1	Excision cures
RCC	Outnumbers WT above 13–14 y. **Translocation RCC (TFE3/TFEB, Xp11.2)** is commonest.	Complete resection; resistant to chemotherapy and RT
Angiomyolipoma	Tuberous sclerosis; bleeding risk if > 3 cm and growing	Mammalian target of rapamycin (mTOR) inhibitor first; embolise for bleeding

KEY POINT

Age points to diagnosis: a neonate has CMN, a toddler WT or rhabdoid, an adolescent RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK).

8 · Neuroblastoma

Biology, not radical surgery, decides outcome.

- It is the commonest **extracranial solid tumour** and commonest infant malignancy; median age is 19 months, 50% are adrenal and ~70% are metastatic at diagnosis.
- Paraneoplastic syndromes include watery diarrhoea from vasoactive intestinal peptide and **opsoclonus–myoclonus–ataxia**.
- Urinary vanillylmandelic acid and homovanillic acid (**VMA/HVA**) are raised in **90–95%**. **Calcification and vessel encasement** favour neuroblastoma over WT.
- 123I-meta-iodobenzylguanidine (MIBG) scanning is required for staging; MRI is best for intraspinal extension. Screening caused **overdiagnosis** without fewer deaths.

International Neuroblastoma Risk Group (INRG) stage	Definition
L1	Localised, **no** image-defined risk factor (IDRF)
L2	Localised, ≥ 1 IDRF (vessel/nerve encasement)
M	Distant metastases
MS	Age **<18 months**; skin, liver, marrow (< 10%) only; no bone (INSS 4S: < 12 months)

- INRG is imaging-based, before treatment; the International Neuroblastoma Staging System (INSS) depends on resection and is surgeon-dependent.
- **MYCN amplification** (> 10 copies; ~20%) means high risk at any stage. M at ≥ 18 months is always high risk.
- Adverse features are 1p36 deletion, 11q LOH, 17q gain and **diploidy**; favourable features are **hyperdiploidy**, TrkA expression and age < 18 months.

KEY POINT

Trap: diploid is bad, hyperdiploid good. Stage 4S/MS has OS 80–88% with frequent regression; infants < 2 months may die of massive hepatomegaly or coagulopathy.

- **Low risk**: resection alone (stage 1 > 90% disease-free); chemotherapy only for symptoms, progression or adverse biology. Never remove the kidney or vital structures.
- **4S**: resecting the primary is not mandatory. **Intermediate**: chemotherapy, then resection.
- **High risk**: induction chemotherapy, delayed surgery after 3–4 courses, myeloablative consolidation with autologous stem-cell rescue, RT 15–30 Gy, and **13-cis-retinoic acid with anti-GD2 antibody**.
- In stage 4, radical resection improves local control but not survival.
- **Cord compression**: give chemotherapy first; laminectomy only for progressive deficit, as it causes scoliosis.

> **KEY POINT**
>
> Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection, so vital structures are never traded for margin.

9 · Adrenocortical tumours

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- Offer **TP53** testing to every child. Androgen excess causing **virilisation** is the commonest presentation.
- Work-up includes DHEA-S, testosterone, 17-hydroxyprogesterone, cortisol tests, a **urinary steroid profile**, metanephrines, CT/MRI and chest CT.
- Weight > 200 g and spill are adverse. Give **perioperative steroids** for cortisol-secreting tumours.

> **KEY POINT**
>
> Why open? Cure needs complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage.

10 · Landmark trials

Trial	Design	Result	Take-home
AREN0532/0533	1p/16q LOH treated with regimen M	Stage III–IV event-free survival (EFS) ~90% vs 61%	Biology-driven intensification
AREN0533 (lung)	Lung nodule response at week 6	RT omitted, EFS and overall survival (OS) preserved	Response-adapted RT omission
AREN0534	Bilateral WT: 3-drug, surgery by week 12	EFS 82%, OS 95% (vs 56%/81% in NWTS-5)	Fixed clock, maximise NSS
COG P9641	Low-risk neuroblastoma, surgery alone	EFS ~89%, OS ~97%	Chemo only if symptomatic
COG anti-GD2	Anti-GD2 + cytokines + retinoid vs retinoid	Antibody superior	Immunotherapy standard in high risk

Trial	Design	Result	Take-home
German NB97	Extent of resection, stage 4 > 18 months	No EFS/OS benefit	Radical resection not essential

64 Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

1 · Rhabdomyosarcoma (RMS): biology

Fusion status, not the microscope, now sets risk.

- RMS is the **commonest childhood sarcoma**; 20–30% are genitourinary (GU), at bladder/prostate (BP), paratesticular (PT) or vulvovaginal (VV) sites. GU-RMS 5-y survival is **>80%**.
- Syndromic cases (Li–Fraumeni, DICER1, neurofibromatosis 1, Costello) are usually embryonal; diagnosis under 3 y should prompt a germline search.

	Embryonal (ERMS)	**Alveolar (ARMS)**
Site	GU (commonest); the **botryoid** variant protrudes from bladder or vagina	Trunk, extremities, perineum
Age	Younger (~70% < 10 y)	Older, adolescents
Fusion	Usually negative	~80% **PAX3/7–FOXO1** [t(2;13), t(1;13)]
Prognosis	Better	Fusion-positive worse; fusion-negative ARMS behaves like ERMS

> **KEY POINT**
>
> Fusion status predicts outcome better than histology, so trials assign risk by **fusion status**.

2 · Presentation and work-up

- BP tumours cause haematuria or retention; PT tumours are a painless mass separate from the testis; VV tumours cause bloody discharge or a prolapsing mass.

ALGORITHM

1. Start with US.
2. CT or MRI of the primary and its nodal basin: BP, pelvic and retroperitoneal nodes up to the renal vessels; PT, retroperitoneal; VV, inguinal, pelvic and retroperitoneal.
3. **Chest CT** (lung is the commonest metastatic site).
4. Bone scan and bilateral marrow sampling; omit these in fusion-negative ERMS without nodal or lung disease.

3 · Stage, group and risk

The **stage** is pretreatment tumour–node–metastasis plus site. The **clinical group** records what the surgeon left. The **risk group** adds fusion status and age.

Children's Oncology Group (COG) stage	**Site**	**Size / nodes**
1	Favourable: **paratesticular, female GU**, orbit, head–neck	Any, M0
2	Unfavourable: **bladder/prostate**, extremity, parameningeal, trunk	≤ 5 cm, N0/NX

Children's Oncology Group (COG) stage	Site	Size / nodes
3	Unfavourable	≤ 5 cm N1, or > 5 cm any N
4	Any	M1

N1 means a node ≥ 1 cm in short axis. BP-RMS is never below stage 2.

Clinical group	Definition
I	Localised, completely resected, nodes negative
IIa / IIb / IIc	IIa: microscopic residual. IIb: involved nodes resected, most distal node negative. IIc: microscopic residual or most distal node positive.
III	Gross residual or **biopsy only**
IV	Distant metastases
Risk	Criteria
Low	Fusion-negative; stage 1 group I–II (III orbit only), or stage 2 group I–II
Intermediate	Fusion-negative stage 1–2 group III (non-orbit), stage 3 group I–II; any fusion-positive non-metastatic; stage 4 fusion-negative < 10 y
High	Stage 4 fusion-positive, or ≥ 10 y

> **KEY POINT**
>
> An organ-preserving **biopsy** places BP-RMS in group III, which means **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age < 1 y or > 10 y is independently adverse. Trap: most distal node positive = IIc, not IIb.

4 · Treatment principles

Chemotherapy does most of the work; surgery aims to save the organ.

- **Chemotherapy is universal**; local control is by radiotherapy (RT) and/or surgery. Complete resection (except PT) is done only if the organ is preserved.
- Primary exenteration fell from almost all cases in IRS-I to **13%** in IRS-IV, with overall survival unchanged.
- **Pretreatment re-excision** (before chemotherapy, without organ loss) improves GU survival and can change the group. **Delayed primary excision** cannot change the group and is done only if a negative-margin resection is likely.

Bladder/prostate RMS

- Biopsy is preferably **cystoscopic**; relieve obstruction with a **urethral catheter** first and avoid a suprapubic tube (seeding).
- An **end-of-therapy residual mass** is usually stroma or mature rhabdomyoblasts, so observe it if stable. Viable tumour after chemoradiation may need cystectomy.
- Only ~40% of preserved bladders are "normal", so do **urodynamics at 6 months** and renal US yearly for ≥ 10 y.

Paratesticular RMS

- Do a **radical inguinal orchidectomy**; after a scrotal approach, excise the remaining cord inguinally.
- Do an **ipsilateral nerve-sparing retroperitoneal lymph node dissection (RPLND)** before chemotherapy if **age ≥10 y** or nodes are radiologically N1.

> **KEY POINT**
>
> Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes.

Female GU, perineal, infants

- Vaginoscopy and biopsy; chemotherapy with or without brachytherapy controls most, and exenteration is reserved for unresponsive disease.
- Perineal tumours are mostly fusion-positive alveolar, with 5-y OS ~50%. RT causes most late effects, including bladder and sexual dysfunction, infertility and second cancers.

5 · Other bladder tumours

- **Urothelial** tumours in children are mostly low grade; treat by transurethral resection and follow with US. A recurrence should prompt a search for a syndrome (Lynch).
- **Spina bifida/exstrophy**: lifetime bladder cancer risk is **2–4%**, aggressive, and **not increased by augmentation**; evaluate haematuria promptly.

Benign lesion	Key point	Treatment
Inflammatory myofibroblastic tumour	ALK rearrangement; intermediate malignancy	Complete resection
Nephrogenic adenoma	Follows trauma, infection, surgery or transplant	TURBT and treat the cause; recurs
Eosinophilic cystitis	Biopsy essential; blood eosinophilia often absent	Steroids, antihistamines
Haemangioma	Severe haematuria; Klippel–Trenaunay, Sturge–Weber	Propranolol, laser; partial cystectomy

6 · Testicular tumours

Prepubertal tumours are mostly benign, so the testis can often be spared.

- Before puberty, **teratoma** is the commonest tumour (benign) and **yolk sac tumour (YST)** the commonest malignant one.
- Work-up is Doppler US with LDH, **α-fetoprotein (AFP)** and β-hCG; an **"onion-skin" appearance suggests epidermoid cyst**.

	AFP	β-hCG
Source	**YST invariably**	Embryonal carcinoma, mixed tumours; rare before puberty
Half-life	**5 days**	**24 h**
Caveat	Normally high in infancy; reaches adult level (< 10 ng/mL) by **6–9 months**	—

ALGORITHM

1. Use an inguinal approach.
2. If markers are normal and US looks benign, do a **testis-sparing partial orchidectomy** with frozen section.
3. If benign, close.
4. If frozen section shows germ cell malignancy, or AFP is raised, do a radical inguinal orchidectomy.

- Prepubertal teratoma needs partial orchidectomy and **no follow-up**; postpubertal teratoma is malignant and needs radical orchidectomy.
- **YST**: radical inguinal orchidectomy; **90%** are stage I, managed by **surveillance without RPLND or chemotherapy**. Stage III–IV disease gets cisplatin, etoposide and bleomycin.
- **Stromal** tumours (Leydig, Sertoli, juvenile granulosa) are treated by partial orchidectomy if markers are normal. Testicular adrenal rests in congenital adrenal hyperplasia respond to more steroid.

Disorders of sex development (DSD)

Tumour risk rises with Y material in dysgenetic gonads: high in gonadal dysgenesis, Frasier, Denys–Drash and partial androgen insensitivity (PAIS); intermediate in complete androgen insensitivity (CAIS); low without Y. Gonadoblastoma is premalignant.

> **KEY POINT**
>
> Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high.

7 · Other pelvic and genital tumours

- **Sacrococcygeal teratoma** causes neurogenic bladder in 36–51%, so follow with renal–bladder US and urodynamics for higher types. The **Currarino triad** also causes neurogenic bladder.
- A Sertoli–Leydig ovarian tumour should prompt **DICER1 screening**.
- **Urethral prolapse** mimics RMS in prepubertal girls; treat with topical oestrogen.

★ Putting it together: common paediatric decisions

A · VUR after febrile UTI

ALGORITHM

1. Do renal–bladder US; add a voiding cystourethrogram (VCUG) if UTIs recur or US is abnormal.
2. Treat bladder and bowel dysfunction (BBD) and constipation first.
3. Low risk (grade I–II, older boy) is observed; high risk (grade ≥ III, girl, BBD, scars, infant) gets continuous antibiotic prophylaxis.
4. Breakthrough UTI or new scars: inject or reimplant (see Ch 41, 53).

B · Dilated upper tract

Condition	Key test	Operate when	Operation
Pelviureteric junction (PUJ) obstruction	MAG3 diuretic renogram	Function falling or < 40%, symptoms	Anderson–Hynes pyeloplasty
Primary megaureter	US, MAG3, VCUG	Falling function, infection; most resolve	Tapered reimplant
Ureterocele / ectopic ureter	US, VCUG, functional scan	Obstruction, infection, incontinence	Incision, heminephrectomy or ureteroureterostomy (see Ch 52)
Multicystic dysplastic kidney	US and DMSA scan	Rarely; most involute	Observe

C · Undescended testis

ALGORITHM

1. **Palpable**: orchidopexy at **6–18 months**; no imaging.
2. **Non-palpable**: examination under anaesthesia, then laparoscopy to find and fix the testis or confirm it is absent.
3. **Bilateral non-palpable, or with hypospadias**: DSD work-up first (see Ch 59).

D · Acute scrotum

Never let a scan delay exploring torsion.

	Testicular torsion	Appendage torsion	Epididymitis
Typical	Sudden pain, high-riding testis	Upper-pole tenderness, "blue dot"	Gradual onset
Action	**Explore at once**; fix both sides, remove if non-viable	Analgesia, rest	Culture; antibiotics if infected

E · Newborn with atypical genitalia

> **KEY POINT**
>
> Bilateral impalpable gonads with atypical genitalia = salt-wasting congenital adrenal hyperplasia until proven otherwise. Check electrolytes, 17-hydroxyprogesterone and karyotype; crisis comes at days 5–7.

F · Wilms tumour vs neuroblastoma

	Wilms tumour	**Neuroblastoma**
Age	Median 3.5 years	Median ~18 months
Imaging	Intrarenal claw sign; IVC extension	Calcification, vessel encasement; crosses the midline
Markers	WT1, 11p15; WAGR, Denys–Drash	VMA/HVA raised in 90–95%; MYCN amplification = high risk
Strategy	COG: nephrectomy first; SIOP: chemotherapy first	By INRG risk group

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ TIP (Snodgrass) repair for hypospadias

Paper answer · P1 · P3 · merged from 2

Asked as: P3 SAQ "Snodgrass (TIP) repair for hypospadias, including extended indications" · P1 SAQ "TIP repair in hypospadias"

Introduction

- "Shodgrass" in one P3 wording is taken as the Snodgrass TIP repair, and "distal hypos." as distal hypospadias.
- **TIP (tubularised incised plate) urethroplasty** was described by Snodgrass in 1994, built on Rich's hinging incision (1989).
- Designed for distal hypospadias, it is now used for **over 90% of distal and about 80% of midshaft repairs**.
- Distal hypospadias forms 70–85% of cases.

Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate so that it can be tubularised **without tension**.
- The incised raw area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

Indications, prerequisites and assessment

- Distal and midshaft hypospadias, and redo cases with a healthy plate, when curvature is mild (**<30°** after degloving) and corrected by degloving or dorsal plication **without dividing the plate**.
- Supple, well-vascularised plate of adequate width and depth, and an adequate glans.
- **Glans width <14 mm** raises complications (glans width ≥14 mm is needed): give preoperative testosterone to enlarge it.
- **POST (plate objective scoring tool):** a low ratio (AB/BC ratio) of neomeatus length to glans fusion length means a poor plate, which favours a dorsal inlay graft.
- Timing: 6–12 months of age.
- Not for severe curvature needing plate division; proximal TIP is largely abandoned.

Extended indications

Indication	Rationale	Caveat
Midshaft hypospadias	Same principle	Good results
Proximal hypospadias with mild curvature, good plate	Single stage; no flaps	More fistula, stricture, recurrent curvature; Campbell-Walsh authors now prefer staging
Redo or failed repairs with preserved plate	Plate often survives	Worse than primary; inlay graft if scarred
MIP (megameatus intact prepuce)	Wide, deep plate	GAP (glans approximation procedure) alternative

Indication	Rationale	Caveat
Narrow or shallow plate: TIP + **dorsal inlay graft** ("Snodgraft")	Preputial or buccal graft fills the incision, so it cannot contract	Less meatal stenosis
Adults, circumcised boys	No preputial tissue needed	Success similar

- **Also know:** in redo cases with poor blood supply a dorsal buccal inlay beats onlay or tube; incise-and-graft is also used in adult urethral stricture (Asopa dorsal inlay) and female urethroplasty.

Contraindications

- Curvature >30° needing plate division; absent or scarred plate; penoscrotal transposition. Use staged repair instead (Bracka, STAG [staged tubularised autograft], Byars).

Operative steps

1. Glans stay (traction) suture; tourniquet.
2. U-incision around the plate and meatus; circumscribe and deglove; artificial erection.
3. Separate (mobilise) the glans wings deeply from the plate.
4. **Midline incision of the plate** from inside the meatus to the end of the plate, down to the corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, from distal to proximal, placing the first stitch about 3 mm (2–3 mm) short of the plate end to leave an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier layer.
7. Two-layer glansplasty; mucosal collar; skin closure with Byars flaps, or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin for bladder spasm.

Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, **meatal stenosis 3.6%**, stricture 1.3%.
- **Meatal stenosis is commoner than after Mathieu**; glans dehiscence also occurs.
- Distal repair success 83–95%.
- **Proximal and redo cases do significantly worse**; a dartos barrier layer improves results and reduces fistula.

Modifications

- **Dorsal inlay graft (Snodgraft)**: inner preputial skin graft in the incised bed for a shallow plate.
- Lateral plate incisions to 12 mm width.
- **Also know:** MAGPI (meatal advancement and glanuloplasty incorporated), Mathieu, Thiersch–Duplay, onlay flap.

Diagram to draw: cross-sections showing plate incision, inlay graft, tubularisation and dartos cover.

Key points to remember:

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate: dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

SAQ Glandular hypospadias

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 11 Paper 1, Q3b

"Glandular" is read as glanular hypospadias: meatus on the glans.

Definition

- The mildest **distal (anterior)** hypospadias: the meatus opens on the ventral glans, proximal to the normal tip.
- Hypospadias occurs in 1 in 150–300 male births; distal forms are 70–85%.
- Usual features: **dorsal hooded foreskin**, ventral glans tilt, little or no curvature. **MIP (megameatus intact prepuce)** variant (up to 5%): wide meatus with normal foreskin.

Embryology

- Urethral folds close from proximal to distal during weeks 8–16 under **DHT (dihydrotestosterone)**.
- Glanular urethra: ectodermal ingrowth (classic) or endodermal plate ("two-zipper", current view). Arrest at the last step gives a glanular meatus.

Evaluation

- Examine meatus, plate, glans size, curvature, torsion; retract the foreskin before any **newborn circumcision**.
- **GMS (glans–meatus–shaft) score** (each 1–4, total 3–12) predicts complications.
- No workup needed for isolated distal hypospadias; DSD (difference of sexual development) workup if a testis is undescended.

Indications and timing

- Deflected or spraying stream, meatal stenosis, curvature, cosmesis. A straight penis that voids standing may need no surgery; counsel parents.
- **Age 6–18 months** (AAP, American Academy of Pediatrics, 6–12 months).

Operations

Procedure	Suitable for	Key point
Urethromeatoplasty	Stenotic meatus, blind pit	Heineke–Mikulicz: vertical incision, horizontal closure

Procedure	Suitable for	Key point
MAGPI (meatal advancement and glanuloplasty, Duckett 1981)	Glanular meatus, mobile skin	Advance meatus, two-layer glanuloplasty; retracts if used more proximally
MIV (M inverted V) glansplasty	Glanular or coronal	Avoids MAGPI retraction
GAP (glans approximation)	Wide deep groove, fish-mouth meatus (MIP)	Tubularise and close glans wings
TIP (tubularised incised plate, Snodgrass 1994)	Narrow plate	Most used; slit-like meatus
Mathieu flip-flap; Thiersch–Duplay	Selected	Stenosis, fistula

- Preputioplasty or circumcision; dartos barrier layer.

Complications

- Meatal stenosis or retraction, fistula, **glans dehiscence**, residual curvature, poor cosmesis. Distal repairs succeed in about 80%.

Diagram to draw: MAGPI steps (meatal advancement, glans wings closed beneath).

> **Key points to remember:**
> - Glanular = most distal hypospadias; dorsal hood, minimal curvature.
> - MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
> - Repair at 6–18 months; may be left alone if function is normal.

SAQ Retractile testis?

Paper answer · P2

Type: SAQ (10 marks)
Source: 2020-Summer Paper 2, Q3b

Definition

- A **descended testis** that moves up out of the scrotum because of an active **cremasteric reflex**. It can be brought to the bottom of the scrotum by hand and **stays there**, at least for a while, after release.
- It is not undescended, but may be a mild form of the same disease and can later become an acquired undescended testis.

Epidemiology

- Common: suprascrotal in up to one-third of 4-year-olds; 11–15% of boys up to 11 years. Nearly all are scrotal after 12 years.

Examination (diagnosis is clinical)

- Warm room and hands, relaxed child, soap or gel; supine frog-leg (thigh abduction inhibits the cremasteric reflex), cross-legged or squatting.
- One hand sweeps from the internal ring down the canal; the other pulls the testis to the scrotal bottom. **Sustained gentle traction tires the cremaster.**
- Note size, mobility, contralateral testis and hernia.

Testis	Finding	Action
Retractile	Reaches scrotal bottom and stays	Observe yearly
Gliding (unstable high scrotal)	Pulled down but springs back	Treat as undescended
Undescended (palpable)	Cannot reach mid-scrotum	Orchidopexy by 18 months
Ascending (acquired)	Previously scrotal, now not	Orchidopexy

Natural history

- Ascent: 2.6% in a prospective birth cohort; 7–32% in specialist series.
- Operated retractile testes show small volume, patent processus vaginalis and epididymal anomalies, suggesting a continuum with cryptorchidism.

Management

- No ultrasound (adds nothing); no hormones (hCG [human chorionic gonadotrophin] neither separates nor treats).
- Reassure parents; **examine yearly** (AUA [American Urological Association] 2014) until puberty, because some testes ascend.
- **Orchidopexy** if it ascends, cannot be brought down, rests mostly high or becomes smaller. If unsure, examine under anaesthesia.

Diagram to draw: inguinoscrotal sketch: testis drawn by traction to the scrotal bottom.

> **Key points to remember:**
> - Retractile = comes down and stays down.
> - Gliding testis springs back: treat as undescended.
> - Yearly examination until puberty; operate if it ascends.

SAQ Megameatus intact prepuce

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 15 Paper 2, Q3f

Definition

- **MIP (megameatus intact prepuce)** is a hypospadias variant (up to 5%) with a **wide, "fish-mouth" glanular or coronal meatus**, a **completely formed prepuce** and **no significant curvature**. Described by Duckett and Keating (1989).

Embryology

- Normally the prepuce closes ventrally only after the urethral folds fuse (about 20 weeks), so 95% of hypospadias have a hooded, incomplete foreskin.
- In MIP the foreskin is complete, showing that **preputial and urethral development can dissociate**.

Clinical features

- Hidden under a normal foreskin; often found on **retraction before circumcision** or during circumcision.
- Wide, deep glans groove with a large, patulous meatus; well-developed spongiosum.
- Symptoms: spraying or deflected stream; cosmetic concern.

- **Advice:** retract the foreskin before every newborn circumcision. If found, the circumcision may be completed; prior circumcision does not harm later repair.

Evaluation

- Examine meatal site and size, glans and plate, curvature (artificial erection at surgery). GMS (glans–meatus–shaft) score.
- No routine upper-tract imaging for isolated distal hypospadias.

Management

- **Indication:** cosmesis and a straight stream; repair is optional. Usual timing 6–18 months.
- **Pyramid procedure (Duckett and Keating):** 1. Glans traction suture; retract prepuce. 2. Incise around the megameatus in a **pyramid shape**, apex on the ventral shaft. 3. Dissect the wide distal urethra free of the glans wings. 4. **Excise a ventral wedge** of redundant urethra to taper it. 5. Tubularise over a catheter; cover with dartos. 6. Approximate glans wings in two layers (glansplasty); preserve or reconstruct the prepuce.
- **Alternatives:** GAP (glans approximation procedure; pyramid = GAP with intact prepuce), TIP (tubularised incised plate) urethroplasty, Thiersch–Duplay tubularisation, Mathieu.

Complications

- Fistula, meatal stenosis or regression, glans dehiscence; generally low after distal repair.

Diagram to draw: fish-mouth glanular meatus with intact prepuce, and the pyramid incision.

> **Key points to remember:**
> - Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
> - Often found at circumcision; retract foreskin first.
> - Pyramid repair: dissect, taper and tubularise; glansplasty.

SAQ Techniques of chordee correction

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 17 Paper 2, Q3b (related: Practice Set 10 Paper 1, Q3d)

Definition and causes

- Chordee: **ventral penile curvature**, with or without hypospadias. It, not meatal position, decides one-stage versus staged repair.
- Causes: tethered ventral skin, fibrous dysplastic dartos and Buck's fascia, short urethral plate, **corporal disproportion**.
- Devine–Horton (1973) graded the dysplastic layers: I spongiosum, II Buck's and dartos, III dartos only, IV corporal disproportion.

Assessment

- Parents' photographs; **artificial erection** in theatre: Gittes–McLaughlin (tourniquet at the base plus saline), corporal compression, or prostaglandin. A tourniquet can hide basal curvature.
- Grade: mild <30°, moderate 30–45°, severe ≥45° (goniometer; eye underestimates).

Stepwise correction

```
Deglove to penoscrotal junction + excise dysplastic dartos → erection test ├─ Straight → urethroplasty ├─ Residual <30° → dorsal plication └─ >30° → divide urethral plate → still >30° → ventral lengthening (stage)
```

	Dorsal shortening (plication)	Ventral lengthening (corporotomy ± graft)
Methods	**Nesbit:** excise dorsal tunical ellipses, close transversely. **Baskin:** single midline plication in the **12 o'clock nerve-free zone**. Heineke–Mikulicz dorsal incision closed transversely; 16-dot plication	1–3 transverse **full-thickness corporotomies** (STAG, staged tubularised autograft); single **9-to-3 o'clock corporotomy** with graft: **dermis, tunica vaginalis**, SIS (small intestinal submucosa); dermal graft 20% larger than defect
Use	≤15–30°	Moderate–severe after plate division
Recurrence	28% (one series)	9%
Risk	Shortening; dorsal nerve injury off midline	Longer, staged; theoretical erectile dysfunction

- **Plate division** (transection) releases a short plate; not dividing it is an independent risk factor for recurrence. Repair is then staged.
- **Also know:** "fairy cuts" (superficial tunical incisions); lateral or dorsal curves need multipoint plication; correct **torsion** by dartos flap rotation.
- **Rule:** correct fully the first time and repeat the erection test before urethroplasty. Curvature may recur at puberty.

Diagram to draw: penile cross-section with dorsal nerves at 11 and 1 o'clock and midline plication at 12.

> **Key points to remember:**
> - Artificial erection after degloving decides the technique.
> - <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
> - Plication recurrence 28% vs grafting 9%.

SAQ Congenital ureteral anomalies

Paper answer · P3

Type: SAQ (10 marks)
Source: 2017-Summer Paper 3, Q3f

Embryology basis

- The **ureteric bud** arises from the **mesonephric (wolffian) duct** at about week 5 and induces the metanephric blastema.
- Anomalies arise from **abnormal budding**, **failed distal canalisation**, **abnormal muscle maturation** or **persistent embryonic vessels**.

Classification

Category	Anomaly	Key points
Number	**Duplication**: incomplete (bifid, Y-ureter) or complete	About 1 in 125; unilateral 6× bilateral. Lower pole: VUR (vesicoureteric reflux), PUJ (pelviureteric junction) obstruction. Upper pole: ectopia, ureterocele

Category	Anomaly	Key points
Termination (orifice)	**Ectopic ureter**	Opens at or below the bladder neck. **Females**: bladder neck to vagina, uterus; **continuous dribbling with normal voiding**. **Males**: always above external sphincter (posterior urethra, seminal vesicle, vas); **never incontinent**; epididymitis. Zinner syndrome
Structure	**Ureterocele**	Cystic dilation of distal submucosal ureter (Chwalle membrane). Intravesical vs ectopic (Glassberg); cecoureterocele (Stephens)
	PUJ obstruction	Intrinsic aperistaltic segment (infants) or crossing lower-pole vessel (older children)
	Megaureter (≥7 mm)	Obstructed, refluxing, non-refluxing non-obstructed, refluxing-obstructed; primary obstructive = adynamic distal segment
	Primary VUR	Short submucosal tunnel, lateral orifice
Position	**Preureteral (retrocaval) vena cava**	Persistent subcardinal vein traps the right ureter; fishhook (S) shape at L3–L4; type I obstructs, type II does not
	Retroiliac ureter; herniation	Rare

Weigert–Meyer law

- In complete duplication the **upper-pole ureter opens lower and more medial** (ectopic, obstructs, ureterocele); the **lower-pole ureter opens higher and lateral** (refluxes).
- Reason: the upper-pole bud arises more cephalad on the wolffian duct, is incorporated later and ends more caudally.
- **Also know:** blind-ending ureter, inverted-Y ureter, triplication, congenital stricture, ureteral valves, fibroepithelial polyp.

Evaluation

- Ultrasound, VCUG (voiding cystourethrogram), **DMSA (dimercaptosuccinic acid)** scan, MR (magnetic resonance) urography.

Management principles

- **Ureterocele:** endoscopic incision, upper-pole heminephrectomy or ureteroureterostomy.
- **Ectopic ureter:** heminephroureterectomy if non-functioning; otherwise ureteroureterostomy or reimplantation.
- **PUJ obstruction:** pyeloplasty. **Megaureter:** observe most; taper and reimplant (5:1) if function falls.
- **Retrocaval ureter:** divide, bring anterior to the cava, excise the aperistaltic segment, reanastomose.

Diagram to draw: complete duplication showing Weigert–Meyer orifice positions.

> **Key points to remember:**
> - Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
> - Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
> - Retrocaval ureter is a venous anomaly; always right-sided.

Rapid Revision Sheet · Section 4

Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING (LAQ)

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

Anomalies of renal artery. (SAQ)

- Single artery in 71–85%; accessory arteries usually lower pole.
- Crossing vessel can cause PUJ obstruction.
- Treat aneurysm >2 cm or in pregnancy.

Pelvicalyceal anomalies other than PUJ obstruction (SAQ)

- Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
- Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
- Infundibulopelvic stenosis: bilateral, renal failure risk.

Autosomal recessive polycystic kidney disease. (SAQ)

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

Medullary sponge kidney (SAQ)

- Ectatic collecting ducts; sporadic; "bristles on a brush".
- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

Post-obstructive diuresis (SAQ)

- Unilateral about 2%, bilateral about 50%.
- Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
- Replace about 75% of output with 0.45% saline; do not overcorrect.
- Gradual decompression has no role.

Ureterocele (SAQ)

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

Embryology and management of ectopic ureter. (LAQ)

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

Retrocaval ureter and IVC anomalous development (LAQ)

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; repair by anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

Retroileal Ureter (SAQ)

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

Primary obst. megaureter. (SAQ)

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair (LAQ)

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka staged repair preferred; fistula commonest complication.

- Koyanagi: one stage; meatus-based parameatal foreskin flap.
- Koyanagi: plate divided; no proximal circular anastomosis.
- Koyanagi: fistula commonest; diverticulum, meatal stenosis.

Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula (LAQ)

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- Curvature >30° after degloving: stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods (LAQ)

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
- Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
- Subdartos pouch; Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

Vanishing testes (SAQ)

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

Urological evaluation and management of pediatric kidney transplant patient. (LAQ)

- Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

Wilms tumour, including bilateral Wilms tumour (LAQ)

- WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.

- COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
- Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy.
- Bilateral: COG three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP two drugs.
- AREN0534: 4-year OS about 95%.
- Bilateral NSS to avoid ESRD.

TIP (Snodgrass) repair for hypospadias (SAQ)

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate: dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

Glandular hypospadias (SAQ)

- Glanular = most distal hypospadias; dorsal hood, minimal curvature.
- MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
- Repair at 6–18 months; may be left alone if function is normal.

Retractile testis? (SAQ)

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

Megameatus intact prepuce (SAQ)

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.

Techniques of chordee correction (SAQ)

- Artificial erection after degloving decides the technique.
- <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
- Plication recurrence 28% vs grafting 9%.

Congenital ureteral anomalies (SAQ)

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.

SECTION 5

Paediatric Urology 2

32 Embryology of the Human Genitourinary Tract

1 · Overview

- Development is mostly complete by the second trimester. The **indifferent stage** lasts to about **8–9 wk**; then androgens drive the sexes apart.

2 · Kidney and ureter

Bud position decides kidney quality and orifice site.

Three successive kidneys

Kidney	Features
†Pronephros	Cervical, week 4; non-functional and regresses
Mesonephros	†Transient, but its **Wolffian duct** persists; tubules become efferent ductules (male) or epoöphoron (female)
Metanephros (permanent)	†Week 5; ureteric bud and metanephric blastema form the nephrons

Reciprocal induction †

- The **ureteric bud** grows from the Wolffian duct into the metanephric mesenchyme and forms the ureter and collecting system.
- Mesenchymal **GDNF** binds **RET** on the bud to drive branching. **WT1** makes the mesenchyme competent, and loss of **PAX2** causes agenesis or hypoplasia (renal-coloboma syndrome).
- Nephrogenesis ends at about 34–36 wk; no nephrons form after birth.

> **KEY POINT**
>
> One defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder.

Ascent and rotation †

- The kidney ascends in weeks 6–9 and its hilum rotates medially; failure gives a pelvic or malrotated kidney. Fused lower poles make a **horseshoe kidney**, trapped under the inferior mesenteric artery.

3 · Cloaca, bladder, trigone and urethra

LAQ Bladder embryology and exstrophy-epispadias complex

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 LAQ "Bladder and ureter development, exstrophy–epispadias complex and its management" · P1 LAQ "Describe embryology of bladder. What are the proposed theories for

extrophy epispadias complex? Describe skeletal abnormalities." · P2 SAQ "Embryogenesis of urinary bladder"

Introduction

- Assumed meanings: "ectopia vesicae" means classic bladder exstrophy (CBE); "single stage repair of exstrophy of UB" means single-stage (complete primary) repair of exstrophy of the urinary bladder, mainly CPRE (complete primary repair of exstrophy, Mitchell).
- **EEC (exstrophy–epispadias complex)** is one spectrum: glanular epispadias, then **CBE (classic bladder exstrophy)** (over 50%), then cloacal exstrophy.
- CBE incidence about 1 in 46,000; male:female 2.3:1. Recurrence in offspring of an affected parent: 1 in 70.

Development of the ureter

- **Weeks 4–6:** the **ureteric bud** arises from the distal **Wolffian (mesonephric) duct**. GDNF (glial cell line-derived neurotrophic factor) from the metanephric mesenchyme induces a single bud (extra buds cause duplication).
- **Reciprocal induction:** the bud invades the metanephric blastema. The bud stalk forms the ureter; its branches form the pelvis, calyces and collecting ducts; the blastema forms the nephrons.
- **Wall:** urothelium, lamina propria, smooth muscle and serosa, the same plan as the bladder.
- **Chwalla membrane** at the future orifice normally ruptures; failure is the classic explanation of ureterocele.

Development of the bladder: division of the cloaca

- The **cloacal membrane** (bilaminar ectoderm–endoderm) occupies the infraumbilical wall. **Mesenchyme grows in** between its layers and forms the lower abdominal muscles and pelvic bones.
- **Cloacal division (weeks 5–6; P2: weeks 4–6):** the **urorectal septum** (Tourneux and Rathke folds) grows caudally and divides the endodermal cloaca into a dorsal **anorectal canal** (rectum, posterior) and a ventral **UGS (urogenital sinus)** (bladder, anterior). It meets the cloacal membrane, which perforates to form the urogenital and anal openings.
- Paired genital tubercles fuse **cephalad** to the membrane.
- **Wolffian (mesonephric) duct entry** is the landmark that divides the UGS:
- **Cranial vesicourethral canal** (continuous with the allantois) forms the **bladder and pelvic urethra**.
- **Caudal (definitive) UGS** forms the phallic urethra in males, and the vestibule and distal vagina in females.

Development of the bladder: wall layers

- **Urothelium:** UGS endoderm.
- **Lamina propria, detrusor, adventitia / serosa:** surrounding splanchnic mesenchyme, by epithelial–mesenchymal signalling (epithelium and mesenchyme induce each other).

Trigone and ureterovesical junction (weeks 5–6)

- The **common excretory duct** (Wolffian duct distal to the bud) dilates and is absorbed into the UGS; **apoptosis** separates the ureter from the Wolffian duct.

- The arms rotate (**Mackie–Stephens**): the ureteric orifice moves **cranially and laterally**; the Wolffian duct ends caudally as the ejaculatory duct at the verumontanum.
- **Trigone muscle:** classic view, ureteric extension; current view (mouse lineage tracing), mostly bladder smooth muscle.

Bud origin	Orifice	Result
Too low (early contact)	High, lateral, short tunnel	VUR (vesicoureteral reflux)
Too high (late contact)	Caudal, medial	Ectopic ureter, often obstructed
Either	Abnormal bud–blastema contact	Dysplasia (VUR and dysplasia coexist)

- **Weigert–Meyer:** in duplex systems the upper-pole orifice is caudal and medial.

Allantois, urachus and fetal bladder function

- **Months 4–5:** the bladder descends into the pelvis; the allantoic duct elongates and obliterates as the **urachus** (median umbilical ligament).
- Urine production starts at about **week 10**. The fetal bladder is seen in about half at 10 weeks and almost all by 13 weeks; it empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

Bladder anomalies explained by embryology

Anomaly	Basis
Exstrophy	Cloacal membrane not reinforced by infraumbilical mesoderm
Urachal (patent urachus, sinus, cyst, diverticulum)	Failed obliteration at different levels
Agenesis, duplication	UGS or trigone failure; septation
Congenital (Hutch) diverticulum	Detrusor deficiency at the ureteric hiatus
VUR, ectopic ureter	Abnormal bud position

Developmental basis of EEC: core defect

- The **cloacal membrane is not reinforced by mesodermal ingrowth**. This mesoderm normally forms the lower abdominal muscles and pelvic bones, so the thin membrane ruptures (**Marshall–Muecke** overdevelopment theory).
- **Timing and extent of rupture decide the variant:** before septum descent gives cloacal exstrophy; after gives CBE; limited rupture gives epispadias.

Developmental basis of EEC: theories

Theory	Proposed defect	Result
Cloacal membrane overdevelopment (Marshall–Muecke)	Abnormally large membrane in week 4 blocks medial mesenchymal migration	Rupture before septum descent: cloacal exstrophy; after: bladder exstrophy; limited: epispadias
Lateral body-wall fold failure	Folds fail to meet in midline	Abdomen and pelvis: cloacal; pelvis alone: classic
Caudal genital hillocks	Genital tubercles fuse **below** rather than above the membrane	Dorsal urethral defect (epispadias)

Theory	Proposed defect	Result
Abnormal body-stalk insertion	Caudal insertion prevents mesenchyme interposition	Superficial unstable membrane disintegrates
Bony pelvis first	Pelvic ring primordium fails to rotate	Pelvic structures cannot meet, so the bladder herniates

- All theories converge on **absent infraumbilical midline mesoderm**.

Aetiology of EEC

- Multifactorial (inconsistent twin concordance). **22q11.2 duplication** is the commonest variant; **ISL1** susceptibility gene; p63 under-expressed in exstrophic urothelium.
- **IVF (in vitro fertilisation)** raises risk about 7.5-fold and early high-dose progesterone about tenfold; maternal smoking is linked to cloacal exstrophy; folate to the mildest form.

Resulting anatomy in classic exstrophy

- Exposed (open) bladder plate; epispadias; low umbilicus; anterior anus; wide levator hiatus; inguinal hernias.
- **Male genitalia:** anterior corpora about 50% shorter; short, upturned penis with dorsal chordee.
- **Female:** bifid clitoris, short vagina, anterior stenotic introitus; later prolapse.
- **Bladder:** immature, not neuropathic (raised type III collagen falls after closure). Ureters enter with no obliquity, so **VUR (vesicoureteric reflux) occurs in 100%** after closure.

Skeletal abnormalities: classic bladder exstrophy

3D (three-dimensional) CT (computed tomography) studies

- **Rotational:** posterior pelvis externally rotated (mean 12° each side); anterior segment externally rotated (mean 18° each side); sacroiliac joint rotated; acetabular and femoral retroversion.
- **Dimensional: pubic diastasis** (mean 4.8 cm); **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%; posterior segment length normal; larger sacrum.
- Bone histology is normal, so growth potential is normal.
- **Clinical effects:** short pendular penis (anterior segment short and splayed); wide hips, waddling gait and out-toeing in early childhood. These cause little disability and improve with growth.
- **Pelvic floor:** levator ani displaced posteriorly and flattened; anterior anus; later pelvic organ prolapse.
- **Spine:** minor variants (spina bifida occulta, lumbarisation, sacralisation) fairly common; scoliosis and dysraphism uncommon.

Skeletal abnormalities: cloacal exstrophy and isolated epispadias

- **Cloacal exstrophy:** widest, **asymmetric** diastasis (about twice CBE), sacroiliac malformation, hip dislocation.
- **Spinal anomalies in 85–100%** (lumbar myelodysplasia commonest; lipomeningocele); MRI (magnetic resonance imaging) for all.
- Limb defects: **clubfoot** (commonest), limb shortening, tibial and fibular deformity.

- **Isolated epispadias:** male symphysis widened (less than CBE); female symphysis usually closed or narrow.

Management: prenatal and at birth

- **Prenatal:** prenatal diagnosis in only about 25%. Ultrasound shows no bladder filling, low umbilicus, wide pubic rami, small genitalia and an enlarging lower abdominal mass. Counsel; deliver at an exstrophy centre.
- **At birth:** tie the cord with silk (no clamp); cover the plate with plastic wrap; saline irrigation at nappy change. Renal ultrasound; multidisciplinary team.
- **Assess:** template size and polyps (**≥3 × 3 cm** suits newborn closure), diastasis and pelvic malleability under anaesthesia, urethral plate and penile length, associated anomalies.
- Small template, penoscrotal duplication, hypoplastic bladder: **delayed closure at 6–9 months** (delay closure rather than do a bad early closure).

Pelvic osteotomy and immobilisation

- **Purpose:** closing the pelvic ring places the bladder neck deep in the pelvis, reduces tension and helps continence.
- **Indications:** closure **after 72 hours**, diastasis **>4 cm**, stiff (poorly malleable) pelvis, failed closure, or cloacal exstrophy (always).
- **Combined bilateral anterior innominate + vertical iliac osteotomy** (most used) with external fixator and Buck traction for 4–6 weeks.
- Under 72 hours with malleable pubes: closure without osteotomy, with modified Bryant traction for 4 weeks.
- Spica casting fails more often (failure with external fixation plus traction 7.5% versus spica 47%).

Modern staged repair of exstrophy (MSRE)

- Aim of closure: convert exstrophy into incontinent epispadias, with enough outlet resistance for bladder growth but not enough to harm the kidneys.

Stage	Age	Procedure	Purpose
1	Newborn	Bladder, posterior urethra and abdominal wall closure; ± osteotomy	Converts to complete epispadias; outlet resistance stimulates bladder growth
2	6–12 months	Epispadias repair (**modified Cantwell-Ransley**), after testosterone	Raises capacity; straightens penis
3	5–7 years (another source: 4–7 years)	BNR (bladder neck reconstruction): modified **Young-Dees-Leadbetter (YDL)** + bilateral **Cohen** reimplantation	Continence; needs capacity **≥100 mL** and a motivated child

- **Closure keys: radical division of the urogenital diaphragm fibres** so the vesicourethral unit sits deep in the pelvis (missing this is the commonest error); ureteric stents; suprapubic tube 4 weeks; pubic nylon mattress suture.
- **Alternatives:** CPRE (Mitchell) with penile disassembly; Kelly radical soft-tissue mobilisation; Erlangen total repair (single stage); Mainz ureterosigmoidostomy. Most still need an outlet procedure.

Single-stage repair (CPRE and others)

- Single-stage repairs aim for fewer operations and continence without bladder neck repair.

Repair	Features
CPRE (Mitchell)	Newborn closure + penile disassembly + bladder neck tailoring
Erlangen ("total")	All in one at about 8 weeks, no osteotomy
Combined closure + epispadias repair	Delayed or failed closures, boys >6 months

- **Selection:** template ≥3 × 3 cm, elastic, few polyps; experienced team. Not for penoscrotal duplication, hypoplastic bladder or bilateral hydronephrosis.
- **CPRE technique:** 1. Combined general and epidural anaesthesia; **osteotomy if >72 hours** (3 days) old, diastasis >4 cm or stiff pelvis. 2. Incise around the plate; free the bladder; **divide the urogenital diaphragm fibres** to the levator hiatus so the unit sits deep. 3. **Penile disassembly:** right and left hemicorporoglanular bodies plus a urethral wedge on spongiosum (now often left attached to the glans). 4. Urethra placed ventrally between the corpora; corpora joined; meatus usually **hypospadiac**. 5. Bladder and neck closed and tapered; ± bilateral ureteric reimplantation (BUR-CPRE). 6. Pubic No. 2 nylon mattress suture; suprapubic catheter 4 weeks, ureteric stents. 7. Immobilise: external fixation plus traction (failure 7.5% versus spica 47%).
- **Complications:** same frequency as MSRE but **more serious**: dehiscence, prolapse, vesicocutaneous fistula, **glans or corporal loss** (ischaemia), early reimplantation for VUR or hydronephrosis; retention in girls.

Follow-up, failures and salvage

- **Follow-up:** prophylaxis; ultrasound; yearly cystogram and cystoscopy under anaesthesia.
- Dehiscence or prolapse: wait 4–6 months, then reclose with osteotomy.
- Failed BNR (no 3-hour dry interval by 2 years): bulking, redo YDL, or **augmentation + bladder neck closure + Mitrofanoff** (continent catheterisable stoma).

Cloacal exstrophy (OEIS) management

- OEIS: omphalocele, exstrophy, imperforate anus, spinal defects.
- Stabilise; neurosurgery (spinal defects 85–100%); separate bowel from hemibladders; colostomy; osteotomy always; gender congruent with karyotype; continence mostly by augmentation with CIC (clean intermittent catheterisation).

Outcomes and long-term issues

- Daytime dryness after MSRE about **70–74%** (another source: 70–75%), versus about 56% after CPRE with YDL; most still need bladder neck surgery after CPRE.
- **Bladder capacity at neck repair is the main predictor**; each failed closure lowers it. **Best continence predictors: template size at birth and a successful first closure.**
- **Also know:** variants (superior vesical fissure, duplicate exstrophy, pseudoexstrophy); adenocarcinoma risk (cytology, cystoscopy; colonoscopy after ureterosigmoidostomy); reduced male fertility and uterine prolapse in adults; female epispadias; psychosocial support.

Diagrams to draw: cloaca divided by the urorectal septum (vesicourethral canal, UGS, allantois) with failed mesodermal ingrowth of the cloacal membrane; Wolffian duct with

ureteric bud and its absorption into the UGS; exstrophy pelvis with diastasis; MSRE timeline; penile disassembly into two hemicorpora and urethral wedge.

Key points to remember:

- Bud from Wolffian duct; position decides VUR or ectopia.
- Urorectal septum weeks 5–6 (P2: 4–6); mesonephric duct entry is the landmark; vesicourethral canal forms the bladder; urachus from allantois.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
- Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
- MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
- Osteotomy if closure after 72 h or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL, template size and a successful first closure predict continence.

Also, from the condensed book (not in the paper answer):

- Anorectal malformations arise from the **cloacal membrane**, not the urorectal septum (weeks 5–6).

Weigert–Meyer rule (duplex kidneys)

SAQ Posterior urethral valves: embryology, management and valve bladder

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 SAQ "Embryology and Management of P.U. valve patient." · P1 SAQ "Embryology of Posterior Urethral Valves." · P2 SAQ "Valve bladder: management"

Introduction

- PUV (posterior urethral valves): the commonest cause of bladder outlet obstruction in boys (1.4–2.2 per 10,000 births).
- Morgagni described them (1769); Hugh Hampton Young named and classified them (1919).

Normal development

- **Weeks 5–6:** the **urorectal septum** divides the cloaca into the anorectal canal (rectum) and the **urogenital sinus**.
- The **Wolffian (mesonephric) ducts** join the cloaca just before division. Their orifices normally migrate to the **verumontanum**, leaving small folds (plicae colliculi).
- The **urogenital membrane** normally disappears completely.
- Fetal urine starts at 10–14 weeks; after 16 weeks amniotic fluid is mainly fetal urine. So obstruction in this period damages bladder, kidneys and lungs.

Embryology of the valves: Young types and their origin

Type	Anatomy	Embryology
I (about 95%)	Leaflets from the verumontanum fuse anteriorly just proximal to the external sphincter	Abnormal anterior / anterolateral insertion of the Wolffian duct ends into the cloacal wall: hypertrophied inferior urethral crest
II	Folds from verumontanum to bladder neck	Not real (Stephens 1955): secondary bladder-neck hypertrophy
III (5–10%)	Annular (ring) diaphragm with a central hole	**Persistent urogenital membrane**

Other theories and genetics

- Hypertrophied mucosal folds (oldest); cloacal remnants; Wolffian–Müllerian maldevelopment.
- **COPUM (congenital obstructing posterior urethral membrane, Dewan):** one oblique membrane, torn by the first catheter into apparent type I leaflets. Plausible but unproven.
- **Cobb's collar (Moormann's ring):** a congenital ring-like narrowing of the proximal bulbar urethra, just distal to the external sphincter; some authors group it with type III valves.
- **Genetics:** mostly sporadic; chromosome 11 duplications; copy number variants (worse renal outcome); TBX5, PTK7.

Diagram to draw: posterior urethra with type I leaflets versus a type III membrane, with keyhole VCUG.

Consequences of obstruction

- Thick bladder, dilated posterior urethra (**keyhole sign**), hydroureteronephrosis, **renal dysplasia**, oligohydramnios, pulmonary hypoplasia.
- Pop-offs (VUR (vesicoureteral reflux), urinoma, ascites) relieve pressure.
- Dysplasia is fixed in utero; ablation cannot reverse it.

Antenatal management

- Bilateral hydroureteronephrosis, thick bladder, **keyhole sign**, oligohydramnios.
- **Vesicoamniotic shunt** helps survival (lungs), not kidneys (PLUTO trial, 2013).

Neonatal management

1. **Drain the bladder:** 5–7 Fr feeding tube (coudé if it curls); minimal balloon inflation.
2. **Stabilise:** ventilation; correct **post-obstructive diuresis**, hyponatraemia, acidosis, hyperkalaemia; antibiotics.
3. **Confirm:** VCUG (voiding cystourethrogram), renal ultrasound, MAG3 (mercaptoacetyltriglycine) at 2–3 months.
4. **Primary endoscopic valve ablation** (preferred): laser, Bugbee or hook at 5 and 7 o'clock ± 12; VCUG within 1 month.
5. **Vesicostomy** if the urethra is too small or drainage fails. **Upper-tract diversion** rarely: no proven renal benefit.
6. **Circumcision:** cuts febrile UTI (urinary tract infection) sharply. - **VUR** (50–80%) is secondary; treat the bladder, not the ureter.

Prognosis

Nadir creatinine (first year)	ESRD (end-stage renal disease) risk
< 0.4 mg/dL	Nil
0.4–0.69	2%
0.7–0.99	27%
> 1.0	Up to 100%

- Lifetime ESRD risk 20–50%.

Valve bladder: definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV ablation, without residual obstruction, because of bladder dysfunction.

Valve bladder: pathophysiology (vicious cycle)

```` Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual

- polyuria (nephrogenic diabetes insipidus) → bladder never empty → pressure transmitted to dilated upper tracts → they refill the bladder → renal damage ```
- Three processes: **polyuria, poor sensation, residual urine**; together they keep the upper tracts dilated.
- Age pattern: hyperreflexia in infancy, better compliance in childhood, then a large atonic bladder (myogenic failure) after puberty.

**Diagram to draw:** vicious-cycle diagram of the valve bladder.

### Valve bladder: clinical features and evaluation

- Late continence (about 5.5 years), incontinence, UTI, worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**
- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

### Valve bladder: management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin 0.1 mg/kg twice daily) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** (catheter for 7–10 hours) for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones). - Reflux is treated through the bladder; reimplantation is rarely needed.

### Long-term care and transplantation

- **CKD (chronic kidney disease)** care.
````

- ESRD in 20–50%. Before **transplantation**, optimise the bladder (videourodynamics, CIC or overnight drainage); graft survival is then similar.

Key points to remember:

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: polyuria, poor sensation, residual urine; overnight drainage and CIC break the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

Also, from the condensed book (not in the paper answer):

- Male ectopia stays above the external sphincter, so there is no continuous wetting; †female ectopia below it causes continuous dribbling with normal voids.
- With an **ectopic vas**, the vas joins the ureter and both drain into the prostatic urethra.

4 · Gonads

Two testicular hormones explain every disorder of sex development (DSD).

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

Also, from the condensed book (not in the paper answer):

- **SRY** makes Sertoli cells enclose primordial germ cells in cords at about **6 wk**; Leydig cells appear from 8 wk.
- **Müllerian-inhibiting substance (MIS, AMH)** from Sertoli cells makes the Müllerian ducts regress at **8–10 wk**; without it, the uterus and tubes persist.
- **Testosterone** from Leydig cells turns the Wolffian duct into epididymis, vas and seminal vesicle.

Ovary — the default

- Without SRY there is no MIS or androgen, so development is female.

KEY POINT

Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every DSD is a combination of the two.

5 · Gonadal descent

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

6 · Genital ducts

Male accessory glands

- The **prostate** buds at **10 wk** from urogenital sinus (UGS) epithelium, mostly lateral to the **verumontanum**.
- Androgen acts via the **androgen receptor in mesenchyme**; stromal type 2 5α-reductase makes **dihydrotestosterone (DHT), which has about 10× the AR affinity** of testosterone. Its deficiency therefore gives a rudimentary prostate and severe hypospadias.

Female tract

- The Müllerian ducts use the **Wolffian duct as a guidewire**, which explains combined Müllerian and renal anomalies.
- The **entire vaginal epithelium and hymen are UGS-derived** (Bulmer view). Female Wolffian remnants include the **Gartner duct**.

7 · External genitalia

*See the paper answer **Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair** (Section 4, Ch 56).*

Also, from the condensed book (not in the paper answer):

- The tunica is thickest at the 12 o'clock nerve-free zone, the dorsal plication site for mild to moderate curvature.
- The shaft is fused by **16 wk**. In females the folds do not fuse, leaving an open vestibule.

8 · Developmental step and resulting anomaly

Step that fails	Anomaly
†No bud or no induction (GDNF–RET, PAX2)	Renal agenesis (absent hemitrigone)
†Abnormal bud–blastema contact	Dysplasia, hypoplasia, MCDK
†Early bifurcation or two buds	Bifid ureter or complete duplication
Bud too low / too high	VUR / ectopic ureter
†Ascent or rotation fails	Pelvic, malrotated or horseshoe kidney
Cloacal membrane or septation	Anorectal malformation; †cloacal anomalies, exstrophy
Allantois persists	Urachal remnants
Fold fusion arrests	Hypospadias
Urethral plate splits	Urethral duplication
Androgen excess in 46,XX	Virilisation (CAH)
Cranial suspensory ligament retained	Intra-abdominal testis
Gubernacular attachment abnormal	Canalicular or ectopic testis
Gubernacular migration fails	Congenital cryptorchidism
Processus fails to close / to involute	Hydrocele, hernia / ascending testis
One Müllerian duct absent	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys

Step that fails	Anomaly
Septum persists	Uterine obstruction or duplication
Vaginal plate fails to canalise	Transverse vaginal septum
Inferior end fails to open	Imperforate hymen, the commonest vaginal obstruction
Müllerian and Wolffian both abnormal	OHVIRA

9 · Clinical correlations

- **Hypospadias** (about 1 in 250) is an **arrest, not a deformation**.
- **CAH** may make a 46,XX infant look fully male. Suspect it with **impalpable gonads and hyperpigmentation**, since it mimics hypospadias with bilateral impalpable testes.
- A **streak gonad** makes no MIS or testosterone, so the child is female; give hormones and **remove the streak if 46,XY**, because of gonadoblastoma risk.
- **Persistent Müllerian duct syndrome** is 46,XY with uterus and tubes, usually from an MIS-receptor defect, found at exploration for bilateral abdominal testes.
- **OHVIRA** presents after menarche with cyclic pain or pyocolpos. The kidney is often **dysplastic with an ectopic ureter into the obstructed hemivagina**; if missed, leakage persists after septum resection.

> **KEY POINT**
>
> Homologues: Littre and Skene; Cowper and Bartholin; spongiosum and vestibular bulb; gubernaculum and the ovarian plus round ligaments; labioscrotal swellings and labia majora. Traps: the vagina is UGS-derived; the appendix testis is Müllerian and the appendix epididymis Wolffian.

33 Urologic Aspects of Pediatric Nephrology

1 · Renal function in the child

- CAKUT is the **commonest cause of paediatric end-stage kidney disease (ESKD)**.
- GFR reaches the adult level by **2 years**. Creatinine in the first 48 h is **maternal**; by day 7 it is the neonate's own (**<0.5 mg/dL** at term).
- †Schwartz: estimated GFR = 0.413 × height (cm) / creatinine (mg/dL).

> **KEY POINT**
>
> The neonatal kidney is "hormonally deaf": maternal creatinine, salt wasting, hyperkalaemia, low bicarbonate and poor concentration reflect immaturity, not disease.

2 · Haematuria

- Microscopic haematuria is **>5 red blood cells (RBCs)/high-power field**; it affects about 1% of school children and mostly resolves.
- **Fresh microscopy is critical**: red urine is non-glomerular, while brown urine with RBC casts is glomerular. A positive dipstick without RBCs means haemoglobin or myoglobin.
- **UTI is the commonest cause of gross haematuria**. Trauma, pain or clots need urological work-up, often CT; suspected GN needs complement, ASO and nephrology referral.
- In isolated microscopic haematuria, **VCUG and cystoscopy are not indicated**.

3 · Proteinuria

- Normal urine protein-to-creatinine ratio (UPCR) is **<0.2 mg/mg** (<0.7 in newborns).
- **Transient** (fever, exercise) and **orthostatic** proteinuria are benign; **fixed** proteinuria needs full evaluation with or without biopsy.

4 · Glomerular disease

Under 1 year causes are genetic; at 1–10 years think of minimal change, post-streptococcal GN, HSP and HUS; from 7 years, lupus, IgA, MPGN and Alport.

Nephrotic syndrome

- Proteinuria **>40 mg/m²/h or UPCR >2.0** with hypoalbuminaemia and hyperlipidaemia.
- In primary disease give **steroids first, without biopsy**, since minimal change is 90% under 10 years and mostly steroid-sensitive.

Category	Definition	Action
Steroid-sensitive	Remission within 8 weeks	~50% relapse
Frequently relapsing	≥2 relapses in 6 months or ≥4 in 12	Steroid-sparing: cyclophosphamide, mycophenolate, rituximab, calcineurin inhibitors
Steroid-dependent	2 consecutive relapses on steroids	
Steroid-resistant	No remission after 6–8 weeks	Biopsy; usually FSGS, which recurs after transplant

Nephritic syndrome

Complement sorts the causes:

C3	C4	Diagnoses
Low	Normal	Post-infectious GN, C3 glomerulopathy
Low	Low	Lupus nephritis, MPGN
Normal	Normal	IgA, HSP, Alport, ANCA GN, anti-GBM

- In **post-streptococcal GN**, C3 normalises by **8–12 weeks**; if it stays low, biopsy.
- **IgA nephropathy** gives painless gross haematuria **during** a respiratory infection. **Alport** adds sensorineural deafness.
- **HUS** is the commonest cause of severe AKI in a young child; atypical (complement) HUS gets eculizumab.

5 · Tubulopathies

*See the paper answer **Renal tubular acidosis (distal and type 2)** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- Type 4 RTA (**hyperkalaemia**) from obstruction improves after urological correction.

6 · Cystic disease

- **MCDK** has no function, and bilateral disease is lethal. In MCDK, VUR or PUJ obstruction occurs in up to 25%, so do a VCUG, especially with hydroureteronephrosis. Follow with US and BP until involution, then yearly.
- **ARPKD** causes enlarged kidneys and hepatic fibrosis, with >75% ESKD by 5 years. Nephronophthisis causes polyuria and ESKD by adolescence.

7 · Hypertension

- Hypertension is repeated BP **>95th percentile** for age, sex and height.
- Infants have coarctation or renovascular disease, young children Wilms or neuroblastoma, school-age children **renal parenchymal disease including reflux nephropathy**, and obese adolescents essential hypertension.
- Drugs are for stage 2, symptoms, end-organ damage or a secondary cause. Target <90th percentile, or **<50th** in CKD.

8 · Acute kidney injury

KDIGO stage	Creatinine	Urine output
I	Rise ≥0.3 mg/dL in 48 h or 1.5–1.9× baseline	<0.5 mL/kg/h for 6–12 h
II	2.0–2.9×	<0.5 for ≥12 h
III	≥3× or ≥4.0 mg/dL or clearance <35 (<18 years)	<0.5 for ≥24 h or <0.3 for 12 h

- Creatinine lags GFR by 24–72 h. FeNa **<1%** means prerenal and >3% intrinsic.
- Dialysis is for diuretic-resistant overload, refractory hyperkalaemia or acidosis, or uraemia.

9 · Chronic kidney disease

- CKD stages by GFR are I ≥90, II 60–89, III 30–59, IV 15–29 and V **<15** or dialysis.
- Complications include growth failure from growth hormone resistance, bone disease, anaemia and acidosis.
- **Do not restrict protein**; strict BP control is the only proven way to slow progression.

10 · Dialysis and transplantation

*See the paper answer **Urological evaluation and management of pediatric kidney transplant patient.** (Section 4, Ch 61).*

Also, from the condensed book (not in the paper answer):

- Peritoneal dialysis is first choice if **<9 years / <20 kg**; omphalocele, gastroschisis, diaphragmatic hernia and exstrophy contraindicate it.
- Living donors give less delayed graft function (5% vs 15%) and a longer graft half-life (20 vs 12.5 years).
- Many centres await **≥10 kg and 65 cm** before transplant because of graft thrombosis.

34 Perinatal Urology

1 · Fetal renal development and prenatal imaging

Amniotic fluid is fetal urine, so obstruction threatens the lungs.

LAQ Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies

Paper answer · P1 · P3 · merged from 2

Asked as: P3 LAQ "Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO)" · P1 SAQ "Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation"

Introduction and terminology

- Assumed meaning: HN (hydronephrosis) and ANH (antenatal hydronephrosis) are now called **UTD (urinary tract dilation)**.
- Major congenital anomalies occur in about **2–3% of births**. Urinary tract anomalies are among the commonest detected prenatally (commonly quoted as 20–30% of all prenatal anomalies).
- **UTD is the second commonest prenatal anomaly after cardiac defects, in 1–3% of pregnancies.**
- Mostly UTD on the 20-week scan, usually mild and transient: mild 57–88%, moderate 10–30%, severe 1.5–13% of UTD. The serious group is **LUTO (lower urinary tract obstruction)**: PUV (posterior urethral valves), urethral atresia, PBS (prune-belly syndrome).

The prenatal anomaly scan (16–20 weeks)

- The **anomaly scan is done at 18–20 weeks** (ACOG (American College of Obstetricians and Gynecologists)); by 20 weeks most urological anomalies are visible.
- Screening US (ultrasound) at 18–20 weeks detects dilation well (sensitivity about 90%) but predicts the diagnosis poorly.
- **Why this window:** fetal urine production begins at about 14 weeks, and after about 16 weeks amniotic fluid is mainly fetal urine, so **oligohydramnios reflects renal failure or outlet obstruction** and causes **pulmonary hypoplasia**. The bladder should be seen filling and emptying.
- Survey kidneys, collecting system, ureters, bladder, genitalia, amniotic fluid, abdominal wall and spine. AFI (amniotic fluid index): normal 8–18 cm; oligohydramnios <5–6 cm.

Anomalies detected and their signs

Anomaly	Signs
PUV (posterior urethral valves)	Male; thick bladder; **keyhole sign**; bilateral hydroureteronephrosis; oligohydramnios
Bilateral renal agenesis	**Anhydramnios after 16 weeks**, no bladder, no renal arteries: lethal Potter sequence
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, no central pelvis
ARPKD (autosomal recessive polycystic kidney disease)	Large, homogeneously **bright** kidneys; oligohydramnios

Anomaly	Signs
Bladder exstrophy	**Bladder never seen**, low cord insertion, lower abdominal mass, wide pubis
Prune belly	Thin-walled large bladder, empty scrotum
Myelomeningocele	Spinal defect, lemon and banana signs

- **Also know:** duplex kidney with ureterocele (intravesical cyst); cloacal exstrophy (adds omphalocele, spinal defects); hydrocolpos; ambiguous genitalia; renal ectopia.

Diagram to draw: fetal keyhole sign.

Causes of antenatal UTD (SFU [Society for Fetal Urology] consensus)

Cause	Incidence
Transient/physiological dilation	50–70%
UPJ (ureteropelvic junction; PUJ, pelviureteric junction) obstruction pattern	10–30%
VUR (vesicoureteral reflux)	10–40%
UVJ (ureterovesical junction) obstruction/megaureter	5–15%
MCDK	2–5%
Duplex system, ureterocele, ectopic ureter	5–7%
LUTO (PUV, prune-belly)	0.1–0.5%

- The chance of postnatal pathology rises with grade: about 12% (mild) to 88% (severe). UPJ obstruction and PUV track grade; **VUR is not predicted by grade**.

Grading systems

APD (anteroposterior renal pelvic diameter, APRPD): transverse image at the hilum. APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.

Grade	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

SFU (postnatal only): 1 pelvis barely split; 2 pelvis and major calyces; 3 minor calyces uniformly dilated; 4 plus parenchymal thinning.

UTD consensus 2014 (risk stratification): six parameters: APD, central/peripheral calyces, parenchymal thickness, parenchymal appearance, ureter, bladder (plus amniotic fluid prenatally).

Prenatal parameter	A1 low risk	A2–3 increased risk (any one)
APD 16–27 weeks	4 to <7 mm	≥7 mm
APD ≥28 weeks	7 to <10 mm	≥10 mm
Calyces	Central or none	Peripheral
Parenchyma, ureter, bladder	All normal	Abnormal
Amniotic fluid	Normal	Unexplained oligohydramnios

- **A1:** one repeat US after 32 weeks. **A2–3:** US every 4 weeks to delivery. Both: postnatal US after 48 hours.

Prenatal algorithm (unilateral UTD, male fetus)

```
Anomaly scan: unilateral UTD → Detailed scan: contralateral kidney, ureter, bladder wall and emptying, posterior urethra (keyhole), amniotic fluid, other anomalies ├─ Bilateral / thick bladder / keyhole / oligohydramnios ─► LUTO pathway │ (fetal medicine; karyotype; Ruano staging; possible shunt) ├─ A1 ─► repeat US after 32 weeks └─ A2–3 ─► US every 4 weeks to delivery → Normal delivery at local centre; no fetal intervention for unilateral UTD → Postnatal US after 48 hours (earlier if solitary kidney or suspected LUTO)
```

- Repeat scans because the lesion may progress and a second anomaly is more likely. **Fetal MRI** (magnetic resonance imaging) only for complex anatomy.
- Counsel the parents: most cases resolve; prognosis with a normal opposite kidney is excellent.

Antenatal management of fetal uropathy (suspected LUTO): evaluation

- Male fetus, thick-walled "keyhole" bladder, bilateral hydroureteronephrosis, oligohydramnios: LUTO until proved otherwise.
- Anomaly scan, **fetal echocardiogram, karyotype** (one-fifth of megacystis is aneuploid); fetal MRI if unclear.
- **Serial fetal urine** (3 samples, 48–72 hours apart). Favourable: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <6 mg/L; not validated.
- **Ruano staging:** I mild, normal fluid: observe. **II severe, function preserved** (oligohydramnios, favourable urine): intervene. III cysts, dysplasia, unfavourable urine: intervention for the lungs only.

Antenatal management of fetal uropathy: options

Option	Notes
Observation, serial scans	Most cases
VAS (vesicoamniotic shunt)	Dislodgement, blockage, preterm labour
Fetal cystoscopy + laser valve ablation	Diagnoses and treats; fistula up to 10%
Serial amnioinfusion	Lungs in anhydramnios; under trial
Early delivery / termination	Mature lungs, falling fluid / poor prognosis

- Shunt or fetal cystoscopy is done at 18–24 weeks in severe LUTO.
- **PLUTO (2013):** shunt versus conservative (31 fetuses): better perinatal survival, **no renal benefit**.

Antenatal management of fetal uropathy: counselling and delivery

- Multidisciplinary team: fetal medicine, paediatric urology, nephrology, neonatology.
- Intervention improves survival, not renal function; complications **over 40%**.
- Deliver where the bladder can be decompressed at once; prophylaxis, then VCUG (voiding cystourethrogram).

Postnatal algorithm

- **First US after 48 hours**, because neonatal dehydration under-reads dilation. Postnatal normal: APD <10 mm, no calyceal or ureteral dilation.

	P1 (SFU 1–2)	P2 (SFU 3)	P3 (SFU 4)
Findings	APD 10–15 mm, central calyces	APD ≥15 mm, peripheral calyces or dilated ureter	Abnormal parenchyma or bladder
Repeat US	3–6 months	Individualised	1 month
Antibiotic prophylaxis	No	Discretion; **yes if ureter >7 mm**	Yes
VCUG	No	Discretion	Yes
MAG3 (mercaptoacetyltriglycine) renogram	No	Discretion	If persistent

- **Male-specific points:** VCUG confirms a normal posterior urethra. **Circumcision** protects (NNT [number needed to treat] 10). UTI (urinary tract infection) risk: female sex, uncircumcised boys, high grade, dilated ureter.
- Prophylaxis: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin and trimethoprim only after 8 weeks.
- MAG3: DRF (differential renal function) <40%, T½ >20 minutes, or delayed retention suggest obstruction. MRU (magnetic resonance urography) for anatomy; DMSA (dimercaptosuccinic acid) for MCDK or duplex function.
- **Surgery indications (UPJ):** rising APD, DRF <40% with poor drainage, falling DRF, symptoms (UTI, stone, pain). Megaureter surgery: DRF <40%, febrile UTI; defer reimplant beyond 1 year.
- Outcome: SFU 1 resolves in nearly all by 2.5 years; SFU 2 about 95% by 4 years.

Bilateral dilation and PUV: how they differ

- Bilateral hydroureteronephrosis, thick bladder, keyhole sign and oligohydramnios mean LUTO until proved otherwise.
- At birth: 5 Fr feeding tube, prophylaxis, creatinine, early VCUG; prenatal intervention improves survival but not renal function (PLUTO).

Diagram to draw: UTD A1/A2–3 prenatal and P1–P3 postnatal flowchart; keyhole bladder with bilateral hydroureteronephrosis and shunt.

Key points to remember:

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- ANH = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- UTD A2–3 = APRPD ≥7 mm (16–27 weeks) or any abnormal feature.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

Also, from the condensed book (not in the paper answer):

- The metanephric kidney appears at **day 28** and urine production starts at **14 weeks**. After **~16 weeks** amniotic fluid is mainly urine, so obstruction causes oligohydramnios and pulmonary hypoplasia.
- The amniotic fluid index (AFI) is normally 8–18 cm; oligohydramnios is **<5–6 cm**.
- Ultrasound severity tracks PUJ obstruction and posterior urethral valves (PUV), but not VUR or duplex. Fetal MRI is used only for anatomy, such as cloaca or exstrophy.

KEY POINT

Prenatal detection is for triage and counselling; there is no proof it improves outcome.

2 · Grading systems

- **Anteroposterior pelvic diameter (APD)** is measured transversely at the hilum and is **meaningless without its gestational age**. Severe is >10 mm in the 2nd trimester and >15 mm in the 3rd.
- Postnatal pathology rises from about 12% (mild) to 88% (severe), but **VUR shows no trend**.
- **SFU** grading (postnatal): 1, sinus barely split; 2, major calyces; 3, minor calyces with parenchyma spared; 4, parenchymal thinning. All SFU 3 and about 70% of SFU 4 kidneys keep DRF >40%.
- Pyramidal thickness **≤3 mm** predicts pyeloplasty and reduced function.
- The **urinary tract dilation (UTD)** classification uses six parameters plus oligohydramnios and covers the whole tract, pre- and postnatally.

3 · Prenatal follow-up

ALGORITHM

1. Normal APD ceiling, all else normal: **<4 mm (16–27 weeks) → <7 mm (≥28 weeks) → <10 mm (postnatal >48 h)**.
2. If A1 (APD 4 to <7 mm at 16–27 wk, or 7 to <10 mm at ≥28 wk), repeat US once after 32 weeks.
3. If A2–3 (any abnormal parameter), scan every 4 weeks until delivery.
4. In both, do a postnatal US after 48 h.

- Exstrophy, cloaca, ARPKD, outlet obstruction and myelomeningocele need a multidisciplinary prenatal consult.

4 · Fetal LUTO and oligohydramnios

Fetal intervention saves lungs, not kidneys.

- The picture is a male fetus with bilateral hydroureteronephrosis, a non-cycling bladder, the **"keyhole" sign** and oligohydramnios. Only fetal cystoscopy can separate PUV, urethral atresia and prune-belly syndrome (PBS).
- Prognosis is poor with oligohydramnios **before 27 weeks**, cortical cysts or echogenicity.
- Fetal urine biomarkers are **not validated** (European consensus 2022). †Classic favourable values were Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L and β2-microglobulin <4 mg/L.

Ruano stage	Fluid / kidneys	Urine biochemistry	Intervention
I — mild	Normal fluid and echogenicity	Favourable	Not indicated
II — severe, function preserved	Oligo/anhydramnios; hyperechoic; no cysts	Favourable within 3 samples	**Indicated** — for lungs and kidneys
III — severe, function abnormal	Anhydramnios; hyperechoic, with or without cysts and dysplasia	Unfavourable after 3 samples	Maybe — for lungs only

Options

- Options are early delivery, vesicoamniotic shunt (VAS), fetal cystoscopy and serial amnioinfusion; the window is **18–24 weeks**.
- Complications occur in about **44%**, most often shunt failure, followed by preterm labour and membrane rupture.

Feature	VAS	Fetal cystoscopy
Diagnoses cause	No	Yes
Mechanism	Bladder-to-amniotic bypass	Laser valve ablation
Specific complications	Blockage, migration, bowel herniation	Urethral fistula (≤10%)
Survival / renal function (PUV)	Improved / not improved	Improved / short-term only

- After VAS, over half have renal insufficiency and a third need renal replacement therapy.

KEY POINT

Counsel: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent renal replacement, because dysplasia is often established first.

5 · Postnatal imaging

- Do the **first US after 48 h**, because neonatal dehydration under-reads dilation.
- Postnatal P1 (APD 10 to <15 mm, central calyces), P2 (≥15 mm, peripheral calyces or abnormal ureter) and P3 (abnormal parenchyma or bladder) roughly equal SFU 1–2, 3 and 4. **Classify by the highest-risk feature**.
- **VCUG** finds VUR in up to 31%; order it by **UTI risk** (females, uncircumcised males, high-grade dilation, dilated ureter, abnormal bladder).
- Contrast-enhanced voiding urosonography avoids radiation but shows the urethra poorly.
- **MAG3** is preferred over DTPA. Consider surgery if **DRF <40%**, T½ >20 min or retention persists upright. Pitfalls are dehydration and poorly timed furosemide.
- **DMSA** shows scars, dysplasia and duplex upper-pole function; **MRU** shows ectopic ureters and crossing vessels but needs sedation.

6 · Postnatal management, UTI and prophylaxis

- **P1** needs only a US at 3–6 months; **P2** needs CAP **if the ureter is >7 mm**; **P3** needs CAP, VCUG and MAG3 if dilation persists.
- In P3 without VUR, give CAP until pyeloplasty or **1 year**, as about 90% of UTIs occur in year 1.
- UTI occurs in 8–22%; agreed risks are **female sex, high-grade dilation and intact foreskin**. Circumcision has a number needed to treat of **10**.
- Evidence for CAP is weak. Use amoxicillin or a cephalosporin from birth; nitrofurantoin or trimethoprim only from **≥8 weeks**.

ALGORITHM

1. US after 48 h: if normal, the dilation was transient.
2. If P1, repeat US at 3–6 months.
3. If P2, give CAP if the ureter is >7 mm; VCUG and MAG3 at discretion.
4. If P3, give CAP, do a VCUG and repeat US at about 1 month; if dilation persists, do a MAG3.
5. If bilateral severe dilation, treat as outlet obstruction: decompress, CAP, urgent evaluation.

7 · Specific antenatal diagnoses

- **Transient** dilation (50–70%) is commonest, and mild dilation resolves in 90–98%, mostly by 2–4 years.

PUJ obstruction pattern

- It is the commonest **surgical** cause. Third-trimester APD **15 mm** has about 80% sensitivity and specificity for pyeloplasty.

- Serial US suffices if P2, APD <20 mm and pyramidal thickness >3 mm, since DRF >40% is likely.
- Operate for rising dilation, DRF <40% with poor drainage, falling DRF, upright retention or symptoms.

Vesicoureteral reflux

- Prenatally detected VUR is typically male, high-grade and bilateral, with congenital dysplasia; VUR after febrile UTI is typically female, lower-grade and acquired. Screen girls with high-grade UTD and uncircumcised boys.

Megaureter / ureterovesical junction obstruction

- Most resolve; surgery is predicted by a ureter **>13 mm** and DRF <30%.
- BAPU: a prenatal ureter **>7 mm after 30 weeks** needs postnatal US, CAP and VCUG.
- Operate for DRF <40%, febrile UTI or pain. **Under 1 year, temporise** with ureterostomy or a refluxing reimplant, and reimplant after 1 year.

> **KEY POINT**
>
> In PUJ, VUR and megaureter, dilation alone is not an indication: operate for function loss, infection or progression; grade sets surveillance.

LUTO conditions

- **PUV** is the commonest LUTO. **PBS** combines megacystis, a deficient abdominal wall and impalpable testes.
- **Urethral atresia** means a catheter will not pass; it is managed by vesicostomy, then progressive dilation of the anterior urethra (PADUA). Massive reflux and MMIHS also cause megacystis.

> **KEY POINT**
>
> Bilateral dilation with normal fluid means urine still leaves the fetus (reflux, megacystis); with low fluid it means true outlet obstruction.

Neonatal management of LUTO

ALGORITHM

1. Have the team at delivery and resuscitate the lungs first.
2. Pass a 5 Fr feeding tube, with a finger in the rectum to guide it.
3. Confirm position on a plain film (KUB).
4. If it fails, place a US-guided suprapubic catheter.

- **Do not use a balloon catheter**, which obstructs the posterior urethra and ureteric orifices. Judge decompression by the **creatinine nadir**, and warn of post-obstructive diuresis.
- Oliguria after decompression suggests a thick bladder obstructing the ureters; treat with antimuscarinics.

Multicystic dysplastic kidney

- VUR occurs in about 20%. There is **no Wilms association**, so no routine nephrectomy; follow with US and blood pressure.

Duplex, ureterocele, ectopic ureter

- By **Meyer–Weigert**, the upper-pole orifice is inferomedial and obstructs, while the lower-pole orifice is superolateral and refluxes.
- Suspect a duplex when the ureter is more dilated than the pelvis.
- Endoscopic puncture of a ureterocele is increasingly first-line, and mandatory for outlet obstruction or sepsis. High-grade VUR usually needs ureterocele excision and common-sheath reimplant.

Exstrophy

- Both forms show a non-visualised bladder, low umbilicus and pubic diastasis. **Cloacal exstrophy (OEIS)** adds omphalocele, imperforate anus and spinal anomalies.

Cloaca, urogenital sinus, CAH

- A **cloaca** shows a single channel, a pelvic cyst and calcified meconium; a **urogenital sinus** has no bowel obstruction and normal external genitalia.
- **CAH** (21-hydroxylase, autosomal recessive) is 75% salt-wasting. Only **1 in 8** at-risk pregnancies is an affected girl, so prenatal treatment is controversial.

Myelomeningocele

Fetal closure before 26 weeks reduces shunting and improves motor outcome, but gives **no bladder benefit**.

8 · Neonatal urological emergencies

Urinary ascites and retention

- Urine causes up to 30% of neonatal ascites, and **PUV is the commonest cause**.
- Spontaneous bladder rupture heals with a catheter, but injury from an umbilical catheter needs repair.
- Retention >24 h suggests PUV or atresia in boys, and urogenital sinus or ureterocele in girls.

Abdominal masses

- **Congenital mesoblastic nephroma** is the commonest solid renal mass under 6 months.
- **Adrenal haemorrhage** has no Doppler flow, shrinks on serial US and calcifies like an eggshell, whereas **neuroblastoma** is vascular and persists. Haemorrhage may track down the processus and mimic an acute scrotum.

Anorectal malformation and ambiguous genitalia

- Anorectal malformation has genitourinary or spinal anomalies in up to **75%**, so get renal and spinal US.
- In ambiguous genitalia, defer gender assignment and **exclude salt-wasting first**; the commonest cause is CAH.

Interlabial masses

Mass	Position	Management
Imperforate hymen	Midline bulging membrane	Hymenal incision
Urethral prolapse	Midline collar at meatus	Conservative
Prolapsed ureterocele	Off-midline from meatus; hymen intact	US and VCUG

Mass	Position	Management
Paraurethral cyst	Off-midline, displaces meatus	Most resolve

Gross haematuria

First exclude an oestrogen-withdrawal bleed and urate crystals, then get a Doppler US.

Feature	Renal vein thrombosis	Renal artery thrombosis	Cortical necrosis
Risk	Maternal diabetes, prematurity, umbilical venous catheter, polycythaemia	Umbilical artery catheter	Hypoperfusion
Presentation	**Triad**: haematuria, flank mass, thrombocytopenia	Hypertension, heart failure	Renal failure
Treatment	Hydration, with or without anticoagulation	Remove catheter, with or without thrombolysis	Supportive

Scrotal masses and torsion

- **Prenatal torsion** gives a hard, non-tender testis at birth with almost no salvage, so do urgent contralateral orchidopexy.
- **Postnatal torsion** (first 30 days) has salvage up to 44%, so explore immediately and fix the other side.

9 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT, VAS vs conservative in fetal LUTO; 31 randomised, stopped early	Fewer perinatal deaths; no renal benefit; 44% complications	Shunting saves lungs, not kidneys
Great Ormond Street pyeloplasty RCT	APD ≥15 mm, DRF >40%: pyeloplasty vs observation	Most observed kidneys stable or improved; a minority (APD 20–40 mm) lost function	Observation safe with monitoring
MOMS†	RCT, prenatal vs postnatal myelomeningocele closure	Less shunting, better motor outcome; no urodynamic benefit	Fetal closure is not for the bladder

35 Urologic Evaluation of the Child

1 · Triage by acuity

Triage asks: **what is lost by waiting** — an organ, a life, or nothing?

Tier	Conditions
Emergent	Acute scrotum; acute abdomen; AKI; obstruction with UTI and/or AKI; retention; incarcerated hernia; paraphimosis; priapism; trauma; newborn ambiguous genitalia, exstrophy, renal vein thrombosis
Urgent (24 h)	Gross haematuria **without infection**; newborn bilateral or solitary-kidney hydronephrosis; renal or testicular tumour; spina bifida
Semiurgent (1–3 wk)	After treated febrile UTI; postnatal dilation (no outlet obstruction); reducible hernia; interlabial mass; amenorrhoea
Routine	Hydrocele; circumcision; incontinence or enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testis; varicocele

KEY POINT

Context shifts the tier: a hernia is emergent if incarcerated but semiurgent if reducible; hydronephrosis is urgent if bilateral or in a solitary kidney.

2 · History

- For hydronephrosis, ask about prenatal history, progression and **amniotic fluid**, a proxy for fetal renal function.
- Obtain **consent from the guardian and assent from the child**; adolescents get a private history.

3 · Examination

Children hide shock until late.

- Minimum systolic BP is **≥60** in neonates, ≥70 at 1 month–1 year, **70 + 2×age** at 1–10 years and ≥90 above 10 years.
- **Shock shows first as tachycardia and tachypnoea; hypotension means decompensation.**

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

Also, from the condensed book (not in the paper answer):

- **Micropenis** is a stretched length below mean − 2.5 SD, which is **1.9 cm** at term.
- In **Tanner staging**, the first male sign is testicular enlargement (stage II), not hair.
- A **simple sacral dimple** (<5 mm, within the cleft, <25 mm above the anus) needs no imaging; image for a lipoma, tag, deep dimple or deviated cleft.

4 · Urine and blood tests

- In febrile children under 24 months, use a **catheter or suprapubic aspirate**; **bag specimens have high false-positive rates** and help only if negative.

- The AAP defines UTI as pyuria and/or bacteriuria with **≥50,000 colony-forming units (CFU)/mL** (catheter or aspirate) in a symptomatic child.

Test	Threshold
Microscopic haematuria	**>5 red cells (RBC)/high-power field** (adult ≥3), on >2 occasions
Calcium/creatinine ratio (hypercalciuria)	>0.4 at age 2–5 years; >0.2 above 5 years
Normal protein/creatinine ratio	<0.7 in the first year; <0.4 up to 4 years; <0.2 after 5 years; >3.0 is nephrotic
Specific gravity	<1.010 dilute (water excess, diabetes insipidus); >1.020 dehydration

- Always **confirm dipstick blood by microscopy**, and test proteinuria on a **first-morning** sample.

5 · Imaging

Only US and MRI avoid radiation; keep doses as low as reasonably achievable.

Modality	Answers
US	Kidney, bladder, scrotum, torsion, spine
VCUG	Reflux, urethra, genitogram
CT	Severe trauma, stones
MRI	Tumours, ectopia, duplication, Müllerian or sinus anatomy
MAG3 renogram	**Obstruction** and differential function
DMSA scan	**Scarring** or pyelonephritis, and differential function

6 · Emergent presentations

Acute scrotum

- The commonest cause is **torsion of the appendix testis**, with point tenderness in a prepubertal boy; treat it supportively.
- **Cord torsion** (mostly over 10 years) gives severe pain, vomiting, a high-riding testis and an absent cremasteric reflex.
- It is a **clinical diagnosis**: explore without imaging and fix the other side.

TWIST item	Points	Score and action
Testicular swelling	2	0: no US
Hard testis	2	1–5: scrotal US
Absent cremasteric reflex; nausea or vomiting; high-riding testis	1 each	**≥6**: explore without US

Other emergencies

- In **ambiguous genitalia**, exclude **salt-wasting CAH** (21-hydroxylase in about 95%). Crisis comes on day **4–15**, so do not discharge early; check **17-hydroxyprogesterone and electrolytes**.
- In **exstrophy**, tie the cord with silk (no clamp), cover the plate with cling film and transfer.

- **Infected obstruction** needs prompt drainage by stent or nephrostomy.
- In **paraphimosis**, exclude a **hair tourniquet**, then compress the glans under analgesia; a dorsal slit is the fallback.
- Ischaemic **priapism** is usually from **sickle cell disease**; give oxygen, fluids and analgesia.
- In **trauma**, the **mechanism** decides CT. After straddle injury consider **sexual abuse**; meatal blood needs a retrograde urethrogram, not a blind catheter.
- In **retention**, expected capacity (mL) is **(age + 2) × 30**. Try analgesia and disimpaction before a catheter.

7 · Common referral presentations

Antenatal hydronephrosis

- Scan **before discharge** if bilateral, in a solitary kidney or with suspected outlet obstruction; otherwise after **48 h**.
- A boy with bilateral dilation and a thick bladder needs a **VCUG before discharge**, catheterised meanwhile.

Haematuria

- Glomerular haematuria is tea-coloured with casts and protein; urological haematuria is bright red with clots.
- Investigate all gross haematuria with urinalysis, culture and US. Isolated microhaematuria is usually benign, so repeat it over 2–3 weeks.

Bladder bowel dysfunction and enuresis

- **Always examine the back**, and treat stepwise, starting with behavioural therapy.

Scrotal, inguinal and genital

- A communicating **hydrocele** is smaller in the morning and is treated by high ligation at the internal ring.
- An **undescended testis** with hypospadias or micropenis suggests a DSD.
- For **varicocele**, the Lambert volume is L×W×H×**0.71**; a non-decompressing or right-sided one needs a retroperitoneal US.
- Prepubertal **testicular tumours** are mostly benign, and tumour markers guide testis-sparing surgery.
- **Meatal stenosis** is treated by meatotomy under EMLA, and symptomatic **labial adhesions** by oestrogen or steroid cream.

> **KEY POINT**
>
> Red flags: absent cremasteric reflex with acute pain; ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension; abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft.

8 · Child comfort

- Distraction, midazolam, intranasal ketamine and EMLA cream can avoid general anaesthesia.

36 Pediatric Urogenital Imaging

1 · Principles — ALARA and the three harms

- Under **ALARA**, **every child is treated as radiation sensitive**. **Sedation risk usually exceeds contrast risk**, and ultrasonography (US) is **safe at all ages and gestations**.

> **KEY POINT**
>
> Every choice trades **radiation, sedation and contrast** against yield; the long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply.

2 · Ultrasonography

US comes first, but shows anatomy, not function.

SAQ Ultrasonography in neonates

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "USG in neonates" · P3 SAQ "HSG in neonates"

Introduction

- Assumed meaning: ultrasonography (USG) of the neonatal urinary tract, including antenatal hydronephrosis follow-up. "HSG" (hysterosalpingography) in the P3 wording is not done in neonates, so it is taken as USG.
- First-line imaging, used for antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies), febrile UTI (urinary tract infection), mass and poor stream.

Why ultrasound first

- No radiation or sedation; repeatable. Shows **anatomy, not function**.

Timing and technique

- **First scan after 48 hours**: earlier, physiological (relative) dehydration under-reads dilation.
- **Scan early if outlet obstruction (PUV, posterior urethral valves) is suspected or the kidney is solitary**; catheterise in suspected obstruction.
- Higher-frequency probes than in adults; kidneys supine and prone; length against age nomograms; bladder full and after voiding; Doppler.
- Measure **APD (anteroposterior pelvic diameter)** on a transverse mid-kidney view; assess calyces, parenchyma, ureters and bladder wall (thick if over 3 mm when full).

Normal neonatal kidney (pitfalls)

- Cortex **iso- or hyperechoic** to liver.
- **Prominent hypoechoic pyramids** are normal and mimic dilated calyces or cysts (not hydronephrosis).
- Normal postnatal **APD <10 mm**, with no calyceal or ureteric dilation.

Grading

- **SFU (Society for Fetal Urology) grades:** 0 none; 1 pelvis only; 2 major calyces; 3 major and minor calyces; 4 grade 3 plus parenchymal thinning.

- **UTD 2014, prenatal:** A1 (APD 4 to <7 mm before 28 weeks, 7 to <10 mm after) versus A2–3 (larger, or any other abnormality).

Antenatal hydronephrosis follow-up (UTD 2014, postnatal)

UTD risk	Findings	Action
P1 (SFU 1–2)	APD 10 to <15 mm, central calyces	Repeat US at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis
P2 (SFU 3)	APD ≥15 mm, peripheral calyces or dilated ureter	Repeat US; VCUG and prophylaxis at discretion / optional (give prophylaxis if ureter >7 mm)
P3 (SFU 4)	Abnormal parenchyma or bladder	Repeat US at 1 month; VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan

Diagram to draw: SFU grades 0–4 as kidney outlines.

Specific neonatal findings

Condition	Ultrasound finding
PUJO (pelviureteric junction obstruction)	Dilated pelvis, normal ureter; pyramids under 3 mm predict pyeloplasty
VUR (vesicoureteral reflux)	Normal first scan in up to 25%
PUV	Thick bladder, dilated posterior urethra (**keyhole sign**), bilateral hydroureteronephrosis; VCUG before discharge
Duplex anomalies	Ureterocele (thin-walled intravesical cyst), ectopic ureter, megaureter
MCDK (multicystic dysplastic kidney)	Non-communicating cysts of varying size, no central pelvis, scant parenchyma
ARPKD (autosomal recessive polycystic kidney disease)	Bilateral large, bright (echogenic) kidneys
Renal vein thrombosis	Large echogenic kidney, no venous flow; haematuria, mass, thrombocytopenia, hypertension
Adrenal haemorrhage	Avascular, evolving suprarenal mass (neuroblastoma is vascular)
Other masses	Mesoblastic nephroma
Spine	Occult dysraphism before arch ossification (up to 6 months)
DSD (differences of sex development)	Müllerian structures, gonads

- **Also know:** hydrometrocolpos, scrotal masses, twinkling artefact (stones).

Limitations

- Normal US does not exclude VUR. Function needs MAG3 renography (after renal maturation) or DMSA (dimercaptosuccinic acid).

Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

Key points to remember:

- First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

Also, from the condensed book (not in the paper answer):

- Fetal urine is made from **10–12 weeks**; by **20 weeks** US detects most anomalies.
- SFU grade 4 is defined by parenchymal thinning, not more dilatation.
- **Bilateral hydroureteronephrosis with a thick, poorly emptying bladder needs a VCUG** for posterior urethral valves (PUV); megaureter or ureterocele needs diuretic renography or MRU.

Cysts, masses, stones

Lesion	Side	US hallmark
MCDK (multicystic dysplastic)	Unilateral	Non-communicating random cysts, scant echogenic parenchyma
ARPKD (recessive polycystic)	Bilateral	Large homogeneously **bright** kidneys, not discrete cysts
ADPKD (dominant)	Bilateral	Parenchyma almost replaced by cysts
Multilocular cystic nephroma	Unilateral	Like MCDK, more parenchyma; infancy
Cystic Wilms	Unilateral	Overlaps; 2–4 years

- A neonatal solid mass is usually **mesoblastic nephroma**; a toddler's is Wilms. Neonatal enlargement with acute kidney injury and hypertension suggests **renal vein thrombosis**.
- **Stones** are echogenic **with** shadowing (fat is without), and Doppler **twinkling artefact** confirms them.

Infection and trauma

- In UTI, US is not routine at the start; **do US if there is failure to improve on antibiotics**. It misses low-grade VUR and scars, where DMSA is better. **Normal US neither risk-stratifies APN nor predicts VUR.**

Contrast-enhanced voiding urosonography (CeVUS)

- Microbubbles grade VUR on five grades with **no radiation**. Area under curve is **0.97**, so it is recommended for VUR.

Bladder/urethra

- The **keyhole sign** (thick bladder and dilated posterior urethra) means PUV.
- To tell clot from tumour, scan supine then prone, because a **clot moves**.

Scrotum and genitalia

- **Never delay exploration for suspected torsion**.
- The **whirlpool sign** gives torsion sensitivity ~96% vs ~76% for colour Doppler, with specificity ~99%.

- Appendage torsion is an avascular nodule with surrounding hyperaemia. **Teratoma** is the commonest prepubertal tumour; epidermoid shows **onion rings**.
- **Do not image a routine undescended testis** (AUA Choosing Wisely) except in an obese child.

3 · Voiding cystourethrography

VCUG is the reference test for reflux and urethral anatomy.

- **After UTI**, do it once the child is improving on antibiotics; a 4–6-week wait is unnecessary, though very early VCUG may overgrade VUR.
- Indications are hydronephrosis with **dilated ureter(s)**, febrile or recurrent UTI, suspected PUV, epididymitis in boys, and UTI in a circumcised boy.
- **Occult VUR**: VCUG finds VUR in only ~40% of DMSA-proven APN. **PIC cystography** (positional instillation of contrast at the orifice) is used when VCUG is negative but febrile UTI recurs.
- **Trauma**: with meatal blood, do a **retrograde urethrogram (RUG) before any catheter**. Capacity (mL) = **(age + 2) × 30**, max 400. A "pie-in-the-sky" bladder means pelvic fracture with extraperitoneal injury.

4 · Nuclear medicine

DMSA

- DMSA is taken up by proximal tubules, so APN shows photon-deficient areas.

Feature	Acute pyelonephritis	Scar
Contour	Preserved	Volume loss, concavity
Timing	Within days; resolves over ~5 months	Lesion at **≥6 months** = scar

- A third to a half of acute lesions scar.

MAG3 diuretic renography

*See the paper answer **Radionuclide studies and diuretic renography in PUJ obstruction** (Section 11, Ch 93).*

Also, from the condensed book (not in the paper answer):

- Give furosemide **1 mg/kg IV** at maximal filling; if given too early, drainage is measured before the system is full.
- There is **no validated paediatric T½ cut-off**, so decide on age, function, curve, upright drainage and serial studies. A **biphasic curve** and poor upright drainage predict progression.

Radionuclide cystography and testicular scan

- **Direct** cystography is more sensitive and lower-dose than VCUG but grades coarsely, so it suits follow-up. **Indirect** cystography is insensitive and not routine.

5 · MR urography

- MRU gives anatomy and function without radiation and suits complex anomalies.

- Protocol: hydration and catheter, then **furosemide 1 mg/kg (max 20 mg)**, T2, gadolinium and dynamic T1; function is measured by **Patlak** analysis.
- It needs sedation under 6 years and is best for **bladder rhabdomyosarcoma** and **local staging of renal tumours**.

6 · CT and radiography

- CT is chosen for speed in trauma and staging, not for hydronephrosis.
- **In trauma, arterial-phase images miss collecting-system injury, so delayed excretory images are mandatory.** CT cystography separates intra- from extraperitoneal rupture.
- CT rarely changes stone care. IVP is obsolete.

7 · Choice of test by clinical question

Question	First line	Next	Avoid
Hydronephrosis — obstruction?	US	MAG3; MRU	CT, IVP
Hydroureteronephrosis, with or without thick bladder	US	VCUG	—
VUR	VCUG	CeVUS; direct cystogram; PIC if occult	Indirect cystogram
APN / scar	DMSA	—	Grey-scale US; MRI
Stones	US	Low-dose CT; KUB	Reflex CT
Renal mass	US	Contrast CT; MRI	—
Bladder mass	US (full bladder)	MRI	—
Trauma	CT with delayed phase; US	RUG; cystography	Catheter before RUG if meatal blood
Acute scrotum	Clinical, with or without Doppler/whirlpool	Scintigraphy if equivocal	Imaging delaying exploration
Undescended testis	Examination	US only if obese	Routine US, MRI
DSD	US and genitography	MRI	CT

37 Infection and Inflammation of the Paediatric Genitourinary Tract

1 · Overview and the paradigm shift

- Urinary tract infection (UTI) affects ~**1% of boys, 3–5% of girls**, and 30–50% recur.
- Congenital dysplasia now explains more scarring than reflux does, so the approach is **selective imaging, prompt treatment and prophylaxis only for the high-risk child**.

2 · Pathogenesis and virulence

- Infection is **ascending**: gut bacteria colonise the perineum, enter the bladder, adhere to umbrella cells and may ascend. Uropathogenic E. coli (UPEC) causes most UTIs.

Factor	Mechanism	Association
Type 1 fimbriae (mannose-sensitive)	FimH binds uroplakins, allowing entry	Cystitis, asymptomatic bacteriuria (ASB)
P fimbriae (mannose-resistant)	Bind Galα1–4Galβ (P blood group)	**Pyelonephritis**
Aerobactin	Siderophore	Growth in iron-poor urine
α-Haemolysin, CNF1	Cell lysis, cytoskeletal damage	Inflammation, invasion
Capsule (K antigen)	Antiphagocytic, anticomplementary	Pyelonephritis
Biofilm, flagella	Shield; motility	Recurrence; ascent

- **Intracellular bacterial communities** (IBCs) form a **quiescent reservoir** that reseeds the lumen for months, causing same-strain recurrence despite adequate antibiotics.

> **KEY POINT**
>
> IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations.

3 · Host risk factors

- In the first year UTI affects boys **3.7%** vs girls 2%; by 6 years, boys 2% and girls 7%.
- **Circumcision** lowers UTI in the first 6 months **~10-fold**. The number needed to circumcise is **111** normal boys, 11 with recurrent UTI and **4** with high-grade VUR.
- The **faecal reservoir** recycles the same strain, and antibiotics such as TMP-SMX destroy protective flora.
- **VUR** is found in 1–2% of newborns and **25–40%** after a first UTI. Most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change.
- Treating **bladder and bowel dysfunction (BBD)** cuts recurrence and aids VUR resolution.
- In **neurogenic bladder**, 40–80% on CIC are bacteriuric; **ASB without VUR needs no antibiotics**, and frequent catheterisation prevents UTI.
- **Catheter-associated UTI** is the commonest nosocomial infection, so remove catheters early. Early infancy (IgG nadir) is the highest-risk period.

KEY POINT

Host factors either raise inoculum/dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lower defence (immature immunity, receptor-bearing blood groups); prevention reverses whichever applies.

4 · Classification

- UTI is **complicated** (anomaly, dysfunction, foreign body; all neonatal UTI) or uncomplicated; upper or lower; first or recurrent.
- Febrile UTI is presumed pyelonephritis, but only **50–66%** of febrile children show acute changes on DMSA scan, the gold standard.

Recurrence	Between episodes	Cause	Remedy
Unresolved bacteriuria	Never sterile	Non-compliance, malabsorption, resistance	Culture-directed retreatment
Bacterial persistence	Sterile; same organism	Nidus: stone, anomaly, papilla, catheter, stent	**Remove nidus**
Reinfection	Sterile; new organism	New ascent	Prevention

Asymptomatic bacteriuria

- Kass definition: two cultures of **≥10^5 colony-forming units (CFU)/mL** of the same organism, without symptoms.
- Infants with ASB are treated and scanned. In school-age girls without VUR or anomaly, **do not screen or treat**, because this selects resistance.

5 · Diagnosis

Clinical

- Fever is ≥38 °C. A toxic look, age <30 days and ≥39.4 °C predict bacteraemia.
- Under 2 years, risk factors are fever >40 °C, previous UTI, suprapubic tenderness and an **uncircumcised penis**. **Another fever source, such as otitis, only halves UTI risk.**

Urine collection

Method	Contamination	Role
Bag	50–60% (false-positive culture up to 75%)	**Reliable only when negative**; never culture
Clean catch	~26%	Toilet-trained; preferred by NICE
Catheter	10–15%	Sensitivity 95%, specificity 99%; **standard <2 years non-toilet-trained**
Suprapubic aspiration (SPA)	<1%	Reference; US guidance gives 80–90% success

- Collect **before antibiotics** unless delay is dangerous. For 2–24 months the AAP accepts bag urinalysis; if positive, culture a catheter or SPA sample.

Test	Sensitivity	Specificity	Use
Leukocyte esterase (LE)	~79%	~87%	Negative helps **rule out**
Nitrite	~50%	**~98%**	Positive **rules in**
LE or nitrite or microscopy	99.8%	70%	Best rule-out
Pyuria ≥5 WBC/HPF	~74%	~86%	No advantage over LE
Gram stain, fresh urine	~91%	92–96%	Most reliable rapid test

- **Nitrite is falsely negative** with short bladder dwell (infants), dilute urine and gram-positives (Enterococcus).
- UTI without pyuria is rare, which separates it from ASB. A threshold of **≥10 WBC/HPF** works better under 2 years.
- **Culture is essential** unless urinalysis is entirely normal.

Guideline	Specimen	Significant count (CFU/mL)
Kass	Voided	$\geq 10^5$
AAP 2011 (2–24 months)	Catheter/SPA	**≥50,000 + pyuria**
EAU	Midstream/catheter	10^3–10^4 monoculture
RCH Melbourne	SPA / catheter	Any gram-negative / $>10^3$
ISPN (Italy)	Catheter / midstream	$>10^4$ / $>10^5$

- **The cleaner the specimen, the lower the accepted count.** Mixed growth and skin flora suggest contamination.
- CRP is sensitive, not specific. **Procalcitonin ≥0.5 ng/mL** best predicts acute pyelonephritis (APN) and later scarring.

6 · Imaging after UTI

- Routine imaging has not been shown to prevent scarring or renal failure, so **order a test only if it changes management**.
- US is abnormal in ~15% after a first febrile UTI, but only 1–2% need action. It is **urgent if the child is unwell or not responding at 48 h**.
- VCUG is the gold standard for VUR, done with sterile urine in a well child; CeVUS is an alternative.
- DMSA shows APN best within **1 week**; a defect at **≥6 months** is a scar, which cannot be told from dysplasia. CT is only for occult stones or mass vs inflammation.

Guideline	Ultrasound	VCUG	Late DMSA
AAP (2–24 months)	Yes, after first febrile UTI	If US abnormal (or recurrence)	No
NICE <6 months	Yes	Atypical/recurrent	Atypical/recurrent
NICE >6 months	Atypical/recurrent	No	Atypical/recurrent
Top-down	No	If acute DMSA abnormal	If acute DMSA abnormal
RCH Melbourne	If <3 months or seriously ill	No	No

- **NICE "atypical" UTI** means seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, no response within **48 h**, or a non-E. coli organism. Late DMSA is at 4–6 months.
- NICE and AAP miss about half of grade III–V VUR but spare most children a VCUG, because high-grade VUR declares itself at recurrence.

7 · Treatment

- **Delay matters**: new-scar odds double if antibiotics start after 72 h rather than within 48 h, so start empirically after urine collection.
- **Oral therapy equals intravenous** for non-toxic children over 2 months who tolerate oral drugs.
- **Admit** if under 1 month, toxic, dehydrated, vomiting or of doubtful compliance. Neonates need a septic work-up and IV therapy, switching to oral after ≥24 h afebrile.
- 90% are afebrile by 48 h. **If not improving at 48 h, do RBUS and broaden cover**.

Infection	Duration
Febrile UTI	**7–14 days** (shorter is inferior)
Afebrile cystitis	**2–4 days** (beats single dose; equals 7–14 days)
Lobar nephronia	≥3 weeks
Renal abscess	Antibiotics; drain if no response
Pyonephrosis	Antibiotics and drainage

- Choose by the **local antibiogram**. TMP-SMX and amoxicillin are poor empirical agents; prefer **cephalexin**, and nitrofurantoin for cystitis. **Proteus is a boys' organism**.
- **Neonates** need Enterococcus cover, so give **ampicillin with a third-generation cephalosporin or an aminoglycoside**.
- Use broad cover after previous UTI, recent antibiotics or admission, or with an anomaly. Fluoroquinolones are reserved for complicated UTI.

Agent	Dose (mg/kg/day)	Restriction
Cephalexin	50–100 in 4	—
Cefixime	8 once daily	—
Amoxicillin–clavulanate	20–40 in 3	—
TMP-SMX	6–12 (TMP) in 2	**<6 weeks/premature**: kernicterus
Nitrofurantoin	3–5 in 2	**<3 months**, GFR <50%, G6PD deficiency; **never for febrile UTI**
Ceftriaxone (IV)	75 once daily	—
Gentamicin (IV)	7.5 (once daily acceptable)	Nephro-/ototoxicity

No repeat culture is needed if the organism was sensitive.

8 · Renal scarring and sequelae

- Scars favour the **poles**, where compound papillae allow intrarenal reflux, and risk rises with VUR grade.

Pyelonephritic episode	1st	2nd	3rd	4th	5th
Scar risk	5%	10%	20%	40%	60%

- Childhood UTI causes **<1% of CKD**. Scars lead to hypertension and pre-eclampsia, so bilateral scars need lifelong checks of BP, function and proteinuria.

9 · Prevention and prophylaxis

- Recurrence is **10–30%**, highest in the first 3–6 months.
- **Continuous antibiotic prophylaxis (CAP)** is not given after a first UTI (AAP, NICE). It is for the **high-risk** child: girls with dilating VUR (≥III), BBD or recurrent febrile UTI.
- Give about **¼ of the therapeutic** dose at bedtime, using a different agent from treatment. Options are TMP-SMX (not under 6 weeks), nitrofurantoin and a cephalosporin in young infants.
- With VUR and BBD, recurrence is ~45% vs 15%; treat BBD with timed voiding and laxatives, then anticholinergics, biofeedback or CIC.
- **Circumcision** helps high-risk boys (recurrent febrile UTI, high-grade VUR, hydroureteronephrosis). Cranberry and probiotics are at most adjuncts.

10 · Special infections

*See the paper answer **Xanthogranulomatous pyelonephritis.** (Section 3, Ch 26).*

Also, from the condensed book (not in the paper answer):

- In children, XGP has calculi in only 38–70% and presents with failure to thrive and anaemia.
- **Lobar nephronia** mimics a mass on CT and needs antibiotics for ≥3 weeks. **Pyonephrosis** must be **drained**.
- **Viral haemorrhagic cystitis** is due to adenovirus 11; **BK virus** after marrow transplant is treated with **cidofovir**.
- **Prepubertal epididymitis** is usually sterile; if culture-positive or recurrent, look for ectopic ureter or utricle. **Balanoposthitis** needs circumcision only if recurrent.

11 · Landmark trials

Trial	Design	Result	Take-home
RIVUR	607 children, VUR I–IV, TMP-SMX vs placebo, 2 years	Recurrent UTI ~halved (number needed to treat 8); **no difference in scarring**	Prevents recurrence, not scars; most benefit with BBD, febrile index UTI
Swedish Reflux Trial	Randomised, dilating VUR, 1–2 years	Prophylaxis reduced recurrence in girls	CAP for girls with dilating VUR

38 Core Principles of Perioperative Management in Children

1 · Perioperative surgical home and enhanced recovery

- **Enhanced recovery after surgery (ERAS)** combines counselling, shorter fasting, judicious IV fluid, no routine tubes or drains, opioid-sparing analgesia and early feeding and mobilisation. It shortens stay with no more complications.

2 · Preoperative assessment

- Risk is raised by upper respiratory tract infection (URTI), asthma exacerbation, dehydration, uncontrolled diabetes, prematurity and fever.
- **Preoperative anxiety** causes more pain and emergence delirium; explanation and play help.
- **Investigations are indication-driven**, with no routine bloods, chest X-ray or ECG. Check urine hCG in females of menstrual age; it can be falsely positive after augmentation cystoplasty.
- Send a **urine culture** before any procedure opening the urinary tract and reschedule until sterile. Treat asymptomatic bacteriuria before endourology.
- **URTI** leaves the airway hyperreactive for **6–8 weeks**, risking laryngospasm and bronchospasm. Postpone elective surgery at least 4, ideally 6 weeks.
- **Sepsis**: give antibiotics within **1 h** of septic shock and avoid neuraxial blocks. In **trauma**, assume a full stomach.

3 · Fasting (NPO, nil by mouth)

Minimum fast	Material
2 h	Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h)
4 h	Breast milk
6 h	Infant formula, non-human milk, light meal, tube feeds
8 h	Regular meal

- Non-human milk counts as a light meal. List the youngest first, because they dehydrate faster.
- **Diabetes**: list the child first, stop metformin 24 h before and halve the long-acting insulin the night before; a long case needs IV insulin.
- Omit ACE inhibitors that morning, stop warfarin and aspirin 7–10 days before, and never stop clonidine abruptly.

4 · Anaesthesia considerations

Neonate and preterm infant

- **Postoperative apnoea** peaks at 12–23 h, so give 23-h monitoring if postconceptional age is **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term). Defer elective surgery beyond these ages.

Anaesthetic neurotoxicity

- The FDA warning (2016) says repeated or lengthy (**> 3 h**) anaesthesia at **< 3 years** may affect brain development. Human trials of a single short exposure are negative (§11), so do not delay necessary surgery.

Comorbidities

- **Asthma**: give a β2-agonist before induction. **Chronic kidney disease**: avoid surgery within 4–6 h of haemodialysis and give no NSAIDs in CKD 3–4.
- **Sickle cell disease**: above all avoid dehydration, and also hypoxia, cold and tourniquets; acute chest syndrome is the main cause of death.
- **Phaeochromocytoma**: give α-blockade before β-blockade, with volume loading, and expect hypoglycaemia after resection.
- **Congenital adrenal hyperplasia**: give IV stress-dose hydrocortisone with glucose-containing fluid.

Spina bifida

- Confirm VP shunt function. **Latex allergy** can cause anaphylaxis, so care should be latex-free from birth.
- **Autonomic dysreflexia** (lesion at or above **T6**): bladder distension causes severe hypertension and bradycardia. Stop the stimulus, drain the bladder, sit the child up and give a rapid antihypertensive.

5 · Fluid and electrolyte management

Hypotonic fluids can cause fatal hyponatraemia.

- **Holliday–Segar 4-2-1 rule**: **4 mL/kg/h** for the first 10 kg, plus **2 mL/kg/h** for the next 10 kg, plus **1 mL/kg/h** per kg above 20. For 22 kg this is 62 mL/h.
- Add dextrose under 2–4 months; give no potassium if dialysed.

Maintenance fluid	Historical	Current
Tonicity	Hypotonic (0.45% saline)	Isotonic (0.9% saline or lactated Ringer) with dextrose
Potassium	10 mEq/L if < 10 kg; 20 mEq/L if > 10 kg	

> **KEY POINT**
>
> **Why isotonic?** Surgical stress, pain, nausea and hypoxia trigger non-osmotic ADH release, so hypotonic fluid adds free water and causes hyponatraemia (< 135 mmol/L). A child's large brain-to-skull ratio makes the resulting encephalopathy potentially fatal.

6 · Blood loss and transfusion

- Estimated blood volume (EBV) is 80–90 mL/kg in a term neonate and 70–75 mL/kg in a child. Maximum allowable blood loss = EBV × (starting − target haematocrit) / starting haematocrit.
- Transfuse at haemoglobin **< 7 g/dL** in stable children; 10–15 mL/kg raises haemoglobin by about 2–3 g/dL.

7 · Temperature

- Infants cool quickly, causing acidosis, coagulopathy and apnoea, so use a warm theatre, forced-air warming and warmed fluids.

8 · Antibiotic prophylaxis and intraoperative crises

- **Timing**: give within **60 min** of incision; vancomycin and fluoroquinolones within **120 min**.
- Doses: cefazolin **30 mg/kg** (redose at 4 h); ampicillin 50 mg/kg (redose at 2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg and ciprofloxacin 10 mg/kg (not redosed).

	Anaphylaxis	**Malignant hyperthermia (MH)**
Triggers	Neuromuscular blockers, β-lactams, latex	Succinylcholine, volatile agents (not nitrous oxide)
First sign	Collapse, bronchospasm, hives minutes after induction	Unexplained hypercarbia, tachycardia, masseter spasm; late — acidosis, hyperkalaemia, rigidity
Treatment	**Adrenaline**, 100% oxygen, fluids; send tryptase	Stop triggers, **dantrolene**, clean circuit; admit 24–48 h (recrudescence)

9 · Analgesia

- Under 3 years, use the FLACC pain score.
- **Regional blocks under general anaesthesia** are safe and opioid-sparing (penile, **caudal**, epidural). Caudal 0.25% bupivacaine is 0.5–1 mL/kg; maximum bupivacaine is 2.5 mg/kg.
- Give **multimodal, scheduled** analgesia: acetaminophen 15 mg/kg 6-hourly and ibuprofen 10 mg/kg 8-hourly. Ketorolac is limited to 72–96 h, and no NSAIDs are given under 6 months.
- **Codeine is contraindicated**, because ultrarapid metabolisers make lethal amounts of morphine. Limit discharge opioids to 3–5 days.

10 · Postoperative care, VTE and day-case discharge

- **Emergence delirium** occurs at 2–8 years, within 30 min of waking; the child does not recognise parents, unlike a child in pain.
- **Venous thromboembolism** is rare before puberty; at-risk adolescents need mobilisation and mechanical prophylaxis, with LMWH if several factors are present.
- **Day-case** surgery suits ASA I–II children outside the apnoea-risk ages having a short, low-blood-loss procedure.

11 · Landmark trials

Trial	**Design**	**Result**	**Take-home**
GAS	Randomised; infant hernia, awake regional vs sevoflurane < 1 h	No cognitive difference at 2 and 5 years	Single short anaesthetic safe
PANDA	Sibling-matched; single exposure < 3 years	No IQ difference at 8–15 years	Repeated exposure untested
MASK	Cohort; exposure < 3 years	Negligible deficit after single exposure	Reassuring

39 Principles of Laparoscopic and Robotic Surgery in Children

1 · Evolution and indications

SAQ Discuss diagnostics and operative pediatric laparoscopy in urology

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3a

Evolution

- Diagnostic (non-palpable testis, 1976) → nephrectomy (1994) → pyeloplasty (1995) → robotic (2002).

Diagnostic laparoscopy

- **Non-palpable testis** (procedure of choice, after examination under anaesthesia): note vessels, vas, testis and internal ring.

Finding	Meaning
Blind-ending vessels and vas, closed ring	Vanishing testis: stop
Vessels and vas entering ring	Explore groin; excise remnant
Intra-abdominal testis	Orchidopexy (primary or Fowler-Stephens)
- **DSD (differences of sex development):** gonadal inspection and biopsy, Müllerian structures.	

Operative laparoscopy

- **Testis:** primary orchidopexy (>90% success); Fowler-Stephens one- or two-stage; gonadectomy in DSD.
- **Kidney:** nephrectomy, heminephrectomy, **pyeloplasty** (commonest MIS (minimally invasive surgery) procedure).
- **Ureter and bladder:** extravesical or vesicoscopic (cross-trigonal) reimplantation, Mitrofanoff, augmentation, bladder neck reconstruction, utricle and urogenital sinus.
- **Also know:** RPLND (retroperitoneal lymph node dissection) for paratesticular rhabdomyosarcoma.

Technical principles

- **Access:** open Hasson or Veress at umbilicus (Palmer's point if scarred); equal safety.
- 3–5 mm ports; robots use 8 mm. Transperitoneal (most space), retroperitoneal (spares peritoneum, useful in peritoneal dialysis), vesicoscopic.
- **Pressures:** CO_2 absorption, reduced venous return, raised airway pressure; reversible oliguria (no fluid bolus).

Age	0–1 month	2–12 months	1–2 years	2–8 years
Max IAP (intra-abdominal pressure)	≤6 mmHg	≤8	≤10	≤12

Age	0–1 month	2–12 months	1–2 years	2–8 years
- **Positioning:** modified flank for kidney; supine steep Trendelenburg for pelvis. Infants: minimise inhalational anaesthetic, rectal tube to reduce bowel distension.				
- **Only absolute contraindication:** inability to tolerate pneumoperitoneum.				

Complications

- Right common iliac vessel injury (behind umbilicus; commonest major), inferior epigastric (commonest minor), bowel injury, subcutaneous emphysema, **port-site hernia 0.5–3%** (close fascia).

Pros and cons

- Shorter stay, better cosmesis, deep pelvic access; pain benefit unproven. Robot: cost, learning curve; infant working space only about 1 L.

Diagram to draw: laparoscopic view of internal ring with blind-ending vessels.

> **Key points to remember:**
> - Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
> - Pressure ceiling 6–12 mmHg by age.
> - Right common iliac at Veress entry; close port fascia.

Also, from the condensed book (not in the paper answer):

- The first paediatric urological laparoscopy was for non-palpable testis in **1976**. The wristed robot removed the suturing bottleneck, so **reconstruction** grew, with pyeloplasty the commonest.
- The only strict contraindication is **inability to tolerate pneumoperitoneum**; age and size are not contraindications.

2 · Physiology of pneumoperitoneum

P1 *Pneumoperitoneum physiology; laparoscopic access and gases — pressure vs CO2 effects on heart, lungs, kidney, ICP; vagal bradycardia; CO2 choice; gas embolism signs and treatment (left lateral head-down, oxygen, aspirate)*

- Effects depend on pressure and reverse on desufflation, so use the **lowest age-appropriate pressure**: intra-abdominal pressure (IAP) < 12 mm Hg, and about 7 mm Hg is feasible.
- Infant oliguria is transient and is not a guide to resuscitation, so give **no fluid boluses**.

Age	0–1 month	2–12 months	1–2 years	2–8 years
Maximum IAP	**≤ 6**	**≤ 8**	**≤ 10**	**≤ 12 mm Hg**

- **Cardiovascular**: IAP > 10 mm Hg lowers venous return and cardiac index. Insufflation can trigger a **vagal reflex**, so desufflate and reinsufflate slowly.
- **Respiratory**: infants have a compliant chest wall and low FRC, so tidal volume falls, worse with Trendelenburg, and CO2 absorption affects them most.

- Laparoscopy is generally safe with **VP shunts**. Infants cool fast, so use warmed humidified gas.

3 · Access

- Open (Hasson) and closed (Veress) entry do not differ in injury rates (Cochrane); confirming intraperitoneal position is what protects.
- After scarring or failed umbilical entry, use **Palmer's point** (left midclavicular line, about 3 fingerbreadths below the costal margin).

4 · Ports, instruments and placement

- **3- and 5-mm** laparoscopic ports suit infants, whereas robots need **8 mm**. Robotic wristed instruments have 7 degrees of freedom against 4 for laparoscopic ones.
- Triangulation is essential in laparoscopy, while the robot tolerates linear placement. **HIdES** puts ports below the Pfannenstiel line.

5 · Approaches

	Transperitoneal	**Retroperitoneal**	**Vesicoscopic**
Space	Largest; familiar	Limited	Limited; small bladder predicts complications
Strength	Complex reconstruction	Avoids peritoneum; preserves peritoneal dialysis	Intravesical access
Limitation	Colon mobilised or mesenteric window	Unfamiliar view; peritoneal tear collapses space	Learning curve; port-site closure
Uses	Testis, kidney, ureter, bladder	Pyeloplasty, (hemi)nephrectomy, lymph node dissection	Cross-trigonal reimplantation

- In **left pyeloplasty**, a transmesenteric window avoids colonic mobilisation.

6 · Robotics in children

- **Working space** is about **1 L at 1 year vs 6 L in adults**, so arms collide; wide port spacing and bowel preparation help.
- **Limitations**: there are no 5-mm robotic ports, and single-port needs a 25-mm cannula and about **10 cm** to the target, which is prohibitive in small children.
- **Benefits** are shorter stay and less narcotic use; incisions under 15 mm contract with growth.

7 · Complications

- The major vascular injury is to the **right common iliac** artery at Veress entry; the minor one is the **inferior epigastric**, so insert ports under vision.
- Port-site hernia peaks around 1 year, so **close fascia** at every port.
- Experience is the strongest predictor, with fewer complications above 12 cases a year.

KEY POINT

Ergonomics: set the table at 70–80% of elbow height and the monitor at eye level. The robot offloads the arm but fixes the neck, so use the clutch to keep the elbows on the armrest.

40 Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

1 · Principles and terminology

- Non-invasive assessment excludes anatomical or neurological causes, then defines **storage vs emptying** dysfunction (ICCS 2014).
- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older child = **(age in years + 2) × 30 mL**. Small is <65% EBC; large is >150% (some use >130%).

2 · History

- **Secondary enuresis** is wetting after **≥6 months** dry, so look for stressors.
- In girls, poor **toilet posture** activates the pelvic floor and causes incomplete emptying; a footstool helps.
- **Holding manoeuvres** (potty dance, squatting) suppress detrusor overactivity (DO).
- Bowel dysfunction may be the sole cause of urinary symptoms. Target a painless **Bristol 3–4 stool daily**.
- **Rome IV functional constipation** (developmental age ≥4 y, ≥1 month) needs **≥2 of**: ≤2 stools/week; ≥1 soiling/week; retentive posturing; painful or hard stools; large rectal mass; stools blocking the toilet.
- **Neurological red flags** are perineal numbness, or gait or limb change with new LUTS.

ALGORITHM

1. **Bowel and bladder dysfunction (BBD)**
2. Apply Rome IV; obtain a 7-day diary, urinalysis and pelvic US.
3. **Treat the bowel first.**
4. If LUTS persist, do uroflow with EMG and start urotherapy.
5. If that fails, add an anticholinergic, then a β3-agonist; then botulinum toxin, biofeedback or neuromodulation.

- The **48-h frequency–volume chart** shows capacity and polyuria and is **mandatory** when first-line therapy fails.
- Scores: **DVSS** best tracks treatment, and **DVISS** is best for diagnosis.

3 · Physical examination

- Measure height, weight and BP at **every visit**.
- **Neurological**: include perineal sensation, anal tone and sacral reflexes (anal wink, bulbocavernosus); their absence means the S2–S4 arc is interrupted.
- **Spine**: a simple sacral dimple (<2.5 cm from anus) is low risk and gets spinal US up to 6 months. High-risk signs (hair tuft, lipoma, vascular malformation, multiple stigmata, VACTERL) need **MRI**.
- **Genitalia**: **continuous incontinence in a girl suggests an ectopic ureteric orifice.**

4 · Laboratory tests

- Do a **urinalysis for every child with LUTS**; **culture is not routine.** On CIC, culture only with fever, pain, new incontinence or spasm.

5 · Non-invasive urodynamics

*See the paper answer **Uroflowmetry** (Section 7, Ch 107).*

Also, from the condensed book (not in the paper answer):

- Uroflow is valid only if toilet-trained and voiding **≥50% EBC** (replacing the adult ≥150 mL rule), with more than one curve.
- A staccato curve never reaches zero, whereas an interrupted one does.

> **KEY POINT**
>
> The curve reflects detrusor against outlet. Too much detrusor gives a tower; too much outlet gives a plateau; intermittent outlet activity gives staccato; intermittent detrusor effort gives an interrupted curve. Patch EMG tells which side is at fault.

- **Pelvic US**: the bladder wall is abnormal if **>3 mm full / >5 mm empty**. A rectal diameter **>30 mm** without urge means faecal loading.
- **PVR** is abnormal at 4–6 y if **>30 mL or >21%** of capacity and at 7–12 y if **>20 mL or >15%**. It is valid only **<5 min** after voiding with the bladder at 50–115% EBC, so repeat it.

6 · Invasive urodynamics

Filling pressure, not wetting, decides renal risk.

Indications

- **Neurogenic**: spinal dysraphism, tethered cord, sacral agenesis, transverse myelitis or spinal cord injury (SCI).
- **Severe obstruction**: posterior urethral valves (PUV) or urethral atresia.
- Non-neurogenic LUTS: study only after failed 2nd/3rd-line therapy, and only if it will change management.

Technique

- **Avoid sedation or general anaesthesia**; if unavoidable, use ketamine rather than midazolam.
- Routine prophylaxis is not needed, because post-study UTI is about **1.4%**.
- Use a **6 Fr dual-lumen** urethral and a rectal catheter, and **fill at 5–10% of EBC per minute** with warm fluid.
- **Stop filling** at strong urge, discomfort, voiding, leak greater than fill rate, Pdet persistently **>40 cmH_2O**, or volume **>150% EBC**.
- Report the first fill, which gives the highest Pdet. **Videourodynamics** adds trabeculation, VUR with its onset volume and pressure, and bladder-neck shape.

Storage phase

- **Pdet = Pves − Pabd**, and normal filling is flat. **DO** means phasic involuntary rises.

- **Compliance = ΔV/ΔPdet**: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better guide.

SAQ Leak point pressures and ALPP

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications" · P2 SAQ "ALPP"

Definition

- **LPP (leak point pressure):** the pressure at which leakage is first seen, spontaneous or provoked. There are two independent types:

Feature	ALPP (abdominal LPP), incl. VLPP (Valsalva LPP) and CLPP (cough LPP)	DLPP (detrusor LPP)
Definition	Total Pves (vesical pressure) at leak during raised abdominal pressure, **no detrusor contraction**	Lowest Pdet (detrusor pressure) at leak, **without contraction or straining**
Measures	Sphincter strength against abdominal pressure	Bladder pressure against outlet resistance
Bad result	**Low = bad** (weak sphincter)	**High = bad**: >40 cmH_2O means upper-tract risk; hydronephrosis or reflux in 85% (McGuire 1981)
Used in	SUI (stress urinary incontinence)	Neurogenic or low-compliance bladder

- **ALPP (abdominal leak point pressure):** the **intravesical pressure** at which urine leaks because of raised abdominal pressure, **without a detrusor contraction**. It measures the urethra's ability to resist abdominal pressure: **lower ALPP = weaker sphincter**.
- Provoked by Valsalva (**VLPP**, the intravesical pressure at which urine leaks during Valsalva) or cough (**CLPP**).
- Seen only in SUI; there is **no normal value** (continent women never leak).
- Newer term: "urodynamic stress test" (ICS, International Continence Society).

Technique (VLPP/ALPP)

1. Small dual-lumen urethral catheter plus rectal catheter; zero to atmosphere.
2. Patient **standing or sitting**; fill to **150 mL** (original method; some use 200 mL).
3. Ask for a **graded Valsalva** or cough; record the **lowest total Pves at visible leak** (include baseline, not the change).
4. No leak: repeat every 50 mL to capacity; then, at capacity, **remove the urethral catheter** and read pressure from the rectal line.
5. Report volume, catheter size and provocation (Valsalva or cough).

Interpretation (McGuire 1993)

VLPP / ALPP (cmH_2O)	Inference
<60	**ISD (intrinsic sphincter deficiency)**
60–90	Equivocal; mixed
≥90 (>90)	**Urethral hypermobility**, little ISD

Clinical implications and uses of ALPP

- Separates ISD from hypermobility; a low VLPP/ALPP (ISD) traditionally favours a sling or bulking agent over colposuspension.
- Post-prostatectomy incontinence: many men leak only once the urethral catheter is removed.

Limitations

- Not standardised: falls with **larger catheters** and **higher volumes**.
- **VLPP is lower than CLPP** in the same woman (VLPP < CLPP), so state the method.
- Invalid during DO (detrusor overactivity) or poor compliance; prolapse raises it or masks the leak (reduce it).
- ISD and hypermobility coexist; VLPP/ALPP alone **does not predict surgical outcome** well.
- **VALUE trial (Nager 2012):** for uncomplicated SUI, office evaluation was non-inferior to urodynamics before surgery. AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023 SUI guideline: urodynamics optional in the index patient.
- **EAU (European Association of Urology):** do not use leak point pressure to grade severity.

DLPP: clinical implications

- **McGuire (1981):** DLPP **>40 cmH_2O** in myelodysplastic children leads to hydronephrosis or VUR (vesicoureteral reflux) in **85%**.
- A strong outlet keeps the patient dry but transmits pressure to the kidneys; a weak outlet "pops off" safely.
- **Hostile bladder** (≥40 cmH_2O): CIC (clean intermittent catheterisation) plus antimuscarinic; botulinum toxin, augmentation or vesicostomy if needed.
- **Caveat:** 40 is not a validated "safe" value; 20 cmH_2O is more sensitive in children.

Also know

- DO leak point pressure (ICS 2017); MUCP (maximum urethral closure pressure) ≤20 cmH_2O suggests ISD.

Diagram to draw

- Pves, Pabd and Pdet traces marking ALPP (leak on Valsalva with flat Pdet) and DLPP.

> **Key points to remember:**
> - VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
> - ALPP low = weak sphincter; DLPP high = upper-tract danger.
> - VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
> - VLPP < CLPP; catheter size and volume alter it.
> - DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
> - Lower DLPP is safer; no validated safe cut-off.

Also, from the condensed book (not in the paper answer):

- A DLPP **>40 cmH_2O**, or no leak at high pressure, signals danger; the adult ALPP cut-off (<60 cmH_2O) has no paediatric norms.

KEY POINT

McGuire (1981), myelodysplasia: with leak pressure >40 cmH_2O, hydronephrosis or secondary VUR developed in **85%**. Aim for the lowest achievable pressure.

Voiding phase

- **High Pdet with low Qmax means obstruction.**
- Anatomical obstruction has a **quiet EMG**; functional obstruction (DSD) has an **active EMG**.
- Normal Pdet at Qmax is about 66 cmH_2O in boys and 57 in girls; infant boys reach 118 physiologically MCQ.

KEY POINT

Upper-tract threats: compliance <20 (especially <10); DLPP >40 cmH_2O; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow.

- **UMPIRE** (spina bifida) studies at about **3 months**, then at 1, 2 and 3 years, then yearly to 5 y if VUR or a hostile bladder. Hydronephrosis is a late sign, so re-study on any change and before reconstruction.

7 · Evaluation in specific conditions

Condition	Typical findings	When to study
Non-neurogenic LUT dysfunction / enuresis	DO, dysfunctional voiding	Only if refractory
Spinal dysraphism	UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high**	Baseline, serial, pre-surgery
Sacral agenesis	Areflexia with weak sphincter, or DO with DSD; level does not predict	If voiding dysfunction
Anorectal malformation	Tethered cord common	Baseline about 1 y after repair
PUV	DO, then poor compliance, then myogenic failure ("valve bladder" about 15%)	Serial
Cerebral palsy	Upper motor neuron; DO, DSD rare	Only if conservative therapy fails
SCI	Spinal shock causes retention; higher lesions have more DSD	Video-urodynamics 3–6 months after injury
Transverse myelitis	DO most often; motor recovery does not predict bladder	Early, with CIC

KEY POINT

UMPIRE dropped DSD from high risk because reviewers disagreed on who had it, not because DSD is benign. Neurogenic bladder goals are normal capacity, low storage pressure, complete emptying and upper-tract protection.

8 · Landmark trials

Trial	Design	Result	Take-home
McGuire 1981	Myelodysplasia urodynamic cohort	DLPP >40 cmH_2O led to upper-tract damage in 85%	40 cmH_2O = safety threshold

42 Bladder Anomalies in Children

1 · Classification and embryology

LAQ Embryological development of urinary bladder and management of Hutch Diverticulum?

Paper answer · P3

Type: LAQ (25 marks)
Source: 2021-Summer Paper 3, Q1 (bladder diverticulum also answered in 2016-Summer Paper 2, Q3c)

A. Embryology of the bladder

1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. The cloacal membrane perforates to form the urogenital and anal openings.
- The **Wolffian (mesonephric) duct entry** is the landmark:
- **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
- **Caudal (definitive) UGS** → phallic urethra in males; distal vagina and vestibule in females. The whole female urethra comes from the pelvic part.

2. Wall layers

- **Urothelium:** endoderm of the UGS.
- **Lamina propria, detrusor and adventitia:** surrounding splanchnic mesenchyme. Epithelium and mesenchyme induce each other (epithelial–mesenchymal signalling).

3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the mesonephric ducts.
- Ureteric orifices migrate **cranially and laterally**; the Wolffian ducts end caudally as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric (Wolffian) tissue. Current view (mouse lineage tracing): **mostly bladder smooth muscle**, minor ureteric contribution.
- **Bud position (Weigert–Meyer):** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis. The allantoic duct elongates and obliterates into the **urachus** (median umbilical ligament).

5. Function

- Urine production starts at **week 10**. Fetal bladder is seen from 10 weeks, almost always by 13 weeks, and empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

B. Anomalies explained by embryology

Anomaly	Embryological basis
Agenesis	Cranial UGS atrophy, or trigone not formed; survival only with ectopic ureters
Exstrophy	Cloacal membrane not reinforced by mesoderm
Duplication	Complete or incomplete; sagittal or coronal
Urachal (patent urachus, sinus, cyst, vesicourachal diverticulum)	Urachus not obliterated at different levels
Congenital diverticulum	Focal detrusor deficiency, often at the ureteric hiatus

C. Hutch diverticulum: definition

- **Paraureteral diverticulum**: mucosa herniates through the **ureteric hiatus**, between the intravesical ureter and the roof of the hiatus, superolateral to the orifice, sparing the trigone. **Hutch** divided them into primary and secondary.

	Primary (congenital)	Secondary (acquired)
Cause	Congenitally deficient detrusor at the hiatus	High pressure: PUV (posterior urethral valves), BOO (bladder outlet obstruction), neurogenic bladder (described in paraplegics), dysfunctional voiding, prior surgery
Bladder	Smooth; single, often intermittent	Trabeculated; one of many

D. Pathophysiology

- No muscle backing → **VUR**; no muscle in the sac → cannot empty → stasis, UTI (urinary tract infection), stones; tumour in adults.
- An enlarging sac damages **Waldeyer's sheath** and pulls the ureter out → UVJ (ureterovesical junction) dysfunction, reflux, later **obstruction** and renal damage.
- Large sacs compress the bladder neck → retention (vicious circle).

E. Evaluation

- **US (ultrasound):** sac, residual urine, hydronephrosis.
- **VCUG (voiding cystourethrogram), gold standard:** oblique and post-void films show size, VUR and whether the ureter enters the sac. Repeat if suspicion persists (sac is dynamic).
- **DMSA (dimercaptosuccinic acid):** scars; **MAG3 (mercaptoacetyltriglycine) renogram** if obstruction.
- **Urodynamics** if secondary; cystoscopy (orifice, neck); in adults enter the sac, cytology.

F. Management

1. Treat the cause first (secondary): valve ablation, BOO surgery, CIC (clean intermittent catheterisation) and antimuscarinics. Remodelling may make excision unnecessary.

2. Primary diverticulum:

Situation	Management
Small, asymptomatic, no VUR	Observe

Situation	Management
VUR, ureter beside the sac	As primary VUR (resolves like VUR of the same grade); many excise if it persists. Dx/HA (dextranomer/hyaluronic acid) injection possible
Ureter enters the sac	VUR will not resolve → surgery
Breakthrough febrile UTI, new scars, large sac with residual, stones, obstruction, retention	**Diverticulectomy + ipsilateral reimplantation**

3. Operative steps (open, laparoscopic or robotic): `Cystoscopy ± ureteric catheter → cystotomy → evert sac or dissect it from its pseudocapsule (0.5 cm mucosal cuff round the neck) → close detrusor defect in two layers (firm backing) → reimplant ureter: Cohen cross-trigonal or Politano–Leadbetter (tunnel 5:1) → catheter ± suprapubic tube, drain`

- Extravesical or combined approaches suit selected cases. In adults, endoscopic outlet surgery is done first.

G. Complications and follow-up

- Ureteric injury, leak, persistent VUR, obstruction, recurrence (syndromic patients).
- US at 6–12 weeks; VCUG for febrile UTI; urodynamics if voiding is abnormal.

Also know: posterolateral multiple diverticula (Ehlers–Danlos, Williams, Menkes); hypoplasia; megacystis.

Diagram to draw: cloacal division by the urorectal septum; Hutch sac beside the ureteric orifice.

> **Key points to remember:**
> - Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
> - Hutch: herniation through the ureteric hiatus → VUR.
> - Ureter entering the sac: VUR will not resolve.
> - Secondary: treat the outlet first; else diverticulectomy + reimplantation.

Also, from the condensed book (not in the paper answer):

- Anomalies are **structural** (exstrophy, duplication, agenesis, urachal, diverticula) or **functional**; prune-belly syndrome (PBS) is both.

2 · Fetal bladder on ultrasound

- The fetal bladder empties every **15–20 min**, so if unseen, **repeat in the same sitting**. Amniotic fluid is mainly fetal urine after **16 weeks**.
- In exstrophy there is no filling with **normal amniotic fluid**.

	Agenesis	Hypoplasia	Exstrophy
US	Absent after repeat scan	Small, never reaches capacity	No filling; bladder template on abdominal wall
Core defect	Cranial UGS atrophy or no trigone	Inadequate filling or storage	Open bladder plate

	Agenesis	Hypoplasia	Exstrophy
Outcome	Survives only with ectopic ureteric drainage	May grow; augmentation often	Reconstruction (see Ch 43)

- **Agenesis** is very rare (about 1:600,000, mostly female); ureters drain into müllerian structures or the rectum, and it is treated by ureterosigmoidostomy or stoma.

3 · Megacystis

- In classic congenital megacystis, **massive VUR causes the dilation**; today the term means any prenatally dilated bladder.

Gestation	Normal	Abnormal
10–14 weeks	Length ≤6 mm; <10% of CRL	**≥8 mm**; ≥13% of CRL
15–40 weeks	Length about GA − 5 mm	> GA + 12 mm
Later (functional)	Empties within 45 min	No emptying in 45 min

- **Causes**: LUTO, with PUV in about 50%, urethral atresia or anterior valves; non-obstructive causes are PBS, MMIHS and cloacal anomalies. Male:female is 8:1.
- **Aneuploidy is about 1/5** (trisomy 13/18), so karyotype.
- In euploid fetuses, 7–15 mm mostly resolves, while **>15 mm** never does. **Persistent megacystis with oligohydramnios means LUTO.**

MMIHS

- MMIHS is a visceral myopathy (**ACTG2**), mostly female, with megacystis and normal or increased fluid. The bladder is acontractile and most need **lifelong CIC**; TPN complications are the leading cause of death.

> **KEY POINT**
>
> A female fetus with megacystis cannot have PUV, so think of MMIHS, cloacal anomaly or urethral atresia.

4 · Bladder duplication

- **Complete sagittal duplication is commonest**, with external genital duplication in up to 90% and lower GI duplication in up to 42%.

ALGORITHM

1. Early: relieve obstruction and prevent infection.
2. Later, if both sphincters are competent, then join the distal urethras.
3. If one is incompetent, then close that bladder neck and excise its urethra.
4. Join duplicated vaginas, with vulvoplasty.

- **Sphincter competence, not anatomy, decides reconstruction.**

5 · Bladder diverticula

- A primary (Hutch) diverticulum is observed if small and excised if there is VUR. For a secondary one, **treat the outlet first**.

- **Syndromic** diverticula (**Ehlers–Danlos, Williams, Menkes**) are multiple and giant; resect only for UTI or stones, as recurrence is common.

6 · Urachal anomalies

	Patent urachus	Sinus	Cyst	Vesicourachal diverticulum
Open	Whole canal	Umbilical end	Isolated segment	Bladder apex
Presents	Urine from the umbilicus in a neonate	Draining umbilicus	Infection (S. aureus), painful mass	Incidental
Imaging	US; fistulogram or VCUG	Sinugram	US; CT/MRI if extensive	Incidental
Treat	Excise with cuff	Complete excision	Drain, then excise	Only if a narrow neck causes stones or UTI

- The differential is a patent **omphalomesenteric duct**. Incidental remnants are increasingly excised only if symptomatic.

ALGORITHM

1. If infected, drain and give antibiotics first.
2. Infraumbilical incision; circumscribe the tract, preserving the umbilicus.
3. Dissect extraperitoneally and excise with a **bladder cuff**.
4. Two-layer closure; catheter overnight.

7 · Acquired lesions

Lesion	Key features	Treatment
Haemangioma	Klippel–Trénaunay, Sturge–Weber; haematuria	Fulguration or Nd:YAG laser
Fibroepithelial polyp	2nd commonest paediatric bladder tumour	Endoscopic resection
Nephrogenic adenoma	Female 5:1; after infection, surgery, augmentation	TUR; **80% recur**, so surveillance cystoscopy
Cystitis cystica	Almost all female; mimics tumour	Long-term antibiotic prophylaxis
Eosinophilic cystitis	Male	**Biopsy mandatory**; steroids and antihistamines
Haemorrhagic cystitis	Cyclophosphamide or ifosfamide; BK virus, adenovirus	Mesna prevents; pentosan polysulfate

43 Exstrophy–Epispadias Complex

1 · Spectrum, epidemiology and embryology

Rupture of the cloacal membrane explains the whole spectrum.

*See the paper answer **Bladder embryology and exstrophy-epispadias complex** (Section 5, Ch 32).*

Also, from the condensed book (not in the paper answer):

- Spica cast fails in 66% without osteotomy, and diastasis >6 cm needs staged closure.
- Rupture timing explains the phenotypes: early, extensive rupture opens bladder **and** hindgut (cloacal exstrophy); later rupture opens the bladder only; rupture at the genital tubercle gives epispadias.

2 · Anatomy of classic bladder exstrophy

- **Pelvis**: the acetabula are retroverted and the posterior segment is normal. The resulting waddling gait and out-toeing improve with growth.
- **Pelvic floor**: the levator ani lies posterior and flattened. Continent patients have the shortest intersymphyseal distance and deepest bladder neck, hence **deep placement, osteotomy and pelvic fixation**.
- The **intrasymphyseal band** tethers the posterior vesicourethral unit to the pubic rami. It contains **no striated sphincter**, so radical incision down to the levator hiatus is essential.
- **Indirect inguinal hernia** is common, often bilateral in males, so repair it at closure.
- **Anorectum**: the anus is anterior. **Rectal prolapse before closure** resolves with closure, but prolapse **after closure** suggests outlet obstruction and needs urgent cystoscopy.
- **Male**: the **anterior corpora are about 50% shorter** and 30% wider. The prostate is never circumferential. Cavernous nerves lie laterally, so potency is retained, though retrograde ejaculation may follow BNR.
- **Female**: the vagina is short with a stenotic anterior orifice, the **cervix sits in the anterior vaginal wall** and the clitoris is bifid. A wide levator hiatus causes **uterine prolapse**, even in nulliparous women, and **osteotomy does not prevent it**.
- **Bladder**: it is **immature, not neuropathic**. The exposed plate develops polyps, squamous metaplasia and cystitis glandularis, which carry an **adenocarcinoma risk**. Compliance falls after BNR, so follow-up is lifelong.
- The ureters enter laterally without obliquity, so **vesicoureteral reflux (VUR) occurs in 100%** of closed bladders and is corrected by reimplantation at BNR.

3 · Variants

Except superior vesical fissure, variants are managed like CBE.

Variant	Feature	Management
Skin-covered (split symphysis), the commonest	Musculoskeletal defect; urinary tract intact	Formal closure at birth
Duplicate exstrophy	Everted mucosal patch plus a closed pelvic bladder	Excise patch, close wall

Variant	Feature	Management
Superior vesical fissure	Rupture only at the top; sphincter intact	Toilet-trains normally; no BNR
Pseudoexstrophy	Musculoskeletal defect only; low umbilicus	—
Epispadias with bladder prolapse	Prolapse may show only on Valsalva	Repair, with or without osteotomy

4 · Prenatal diagnosis

- Only about 25% are diagnosed prenatally. Signs are **absent bladder filling**, a low umbilicus, wide pubic rami, small genitalia and an **enlarging lower abdominal mass**. Plan **delivery at an exstrophy centre**.

5 · Newborn care and selection

A good first closure best predicts continence.

ALGORITHM

1. Tie the cord with 2-0 silk (no clamp, which traumatises mucosa).
2. Cover the plate with non-adherent plastic wrap; irrigate with saline at each nappy change.
3. Assess heart and lungs; obtain a renal US.
4. Transfer to an exstrophy centre for MDT counselling.

- **Assess** the template (≥**3 × 3 cm** is suitable), diastasis, pelvic malleability under anaesthesia and penile length. A plate that **bulges on crying** is adequate.
- A plate <3 cm, polyp-covered or fibrotic is **inadequate**. Penoscrotal duplication, hypoplastic bladder and bilateral hydronephrosis preclude primary closure.
- A small plate is closed **later, at 6–9 months, with osteotomy**, because a bad closure ruins continence. If it has not grown by 6–12 months, excise it and divert. After delayed closure a **catheterisable channel is needed far more often**.

6 · Surgical philosophies

Repair	Stages / feature	Weakness
Modern staged repair (MSRE)	Closure at birth; epispadias repair at 6–12 months; BNR with reimplant at 4–7 years	Multiple operations
Complete primary repair (CPRE, Mitchell)	Newborn closure with penile disassembly and neck tailoring	Hypospadias; glans loss; early reimplant
Kelly	Closure, then radical soft-tissue mobilisation at 3–6 months, then hypospadias repair; avoids osteotomy	Fistula, stenosis
Erlangen	Total repair at about 8 weeks without osteotomy	—
Mainz	Closure, then sigma pouch at about 2 years	Colonic malignancy

KEY POINT

Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all still need an outlet procedure. Compressing the reconstruction adds ischaemic and outlet risk without removing that need.

7 · Primary closure (MSRE stage 1)

Closure creates an incontinent epispadias with enough outlet for growth.

ALGORITHM

1. General anaesthesia with **epidural** (catheter to T10–T12).
2. Outline a 2-cm mucosal strip from the trigone to mid-shaft (boy) or vaginal orifice (girl); incise the bladder–skin junction.
3. Free the bladder from the rectus and strip peritoneum off the dome.
4. **Divide the urogenital diaphragm fibres sharply down to the levator hiatus**; free the corpora from the rami.
5. Close bladder, neck and posterior urethra; **taper over a 12-Fr sound**.
6. Place a suprapubic Malecot (4 weeks) and ureteric stents; leave the **urethra unstented**.
7. Close the pubis with a **No. 2 nylon horizontal mattress** suture, knot away from the neourethra; fashion a neoumbilicus.

- **Tapering** gives enough resistance for bladder growth and against prolapse, but not enough to harm the kidneys.
- **Ureteric stents** stay 10–14 days because oedema can obstruct the ureters; a urethral stent causes necrosis.
- The **commonest error** is inadequate urogenital diaphragm dissection, after which the unit is pulled forward and prolapses.

8 · Osteotomy and immobilisation

Osteotomy and rigid immobilisation cut closure failure several-fold.

- In a **newborn <72 h** whose malleable pubes meet easily, osteotomy is not needed. **If in doubt, osteotomise.**
- Osteotomise if closure is **>72 h** after birth, diastasis is **>4 cm**, the pelvis is stiff, the template is large, after **failed closure**, and in **cloacal exstrophy**. Diastasis >6 cm needs staged closure with a fixator.

Type	Position	Features
Combined bilateral anterior innominate and vertical iliac (preferred)	Supine	Best malleability; least recurrence
Bilateral posterior iliac (original)	Prone	Worse approximation, more recurrence
Anterior innominate alone	Supine	Less mobility

- Aftercare is **modified Buck traction for about 4 weeks with an external fixator for 4–6 weeks**. Without osteotomy, use **modified Bryant traction** (hips 90°) for 4 weeks.

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
Spica cast	47%	30%	**66%**

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
External fixation with traction	**7.5%**	8%	— (few)

- Failure falls with traction plus fixator for **>4 weeks**; osteotomy type and epidural make no difference. Complications are pin-site inflammation, transient femoral nerve palsy and delayed union.

9 · After primary closure

- At 4 weeks calibrate the outlet, scan and check the residual. Give **continuous antibiotic prophylaxis**, because all have VUR.
- Perform **yearly cystoscopy and cystogram under anaesthesia** to measure capacity. If capacity is **<30 mL at 1–2 years**, a continence procedure may not be feasible.
- If the **outlet is too tight**, escalate from dilation to incision and revision. Recurrent UTI with a distended bladder suggests **intrapubic stitch erosion**.

10 · Epispadias repair (stage 2)

Urethroplasty raises outlet resistance, so the bladder grows.

- It is done at 6–12 months, before BNR, because urethroplasty **increases capacity**. Give **testosterone enanthate 2 mg/kg intramuscularly (IM) at 5 and 2 weeks** before every urethroplasty.
- **Modified Cantwell–Ransley** (preferred): the plate is freed but its **distal 1 cm stays on the glans**; it is tubularised over 8 Fr and the **corpora are rotated over the neourethra**, which then lies ventral.
- **Chordee** is corrected by dorsomedial corporotomy with cavernocavernostomy; skin cover uses ventral preputial flaps.
- Early fistula occurs in about 1 in 6, usually at the base; **coronal fistula is rare**.
- **Mitchell–Bägli disassembly** risks plate ischaemia, and most end **hypospadiac**. Glans or corporal loss is linked to closure **without osteotomy**, so if ischaemia is seen, osteotomise immediately.
- **CPRE**: about half need reimplant in year 1, 65–80% need an outlet procedure, and most end on CIC.
- **Kelly**: unaided voided continence is under 1 in 5, with glans loss reported.
- **Combined closure and epispadias repair** suits failed or delayed closures and boys >6 months, but risks ischaemic penile loss.
- **Female closure**: bladder, urethra and vagina move deep as one unit, with **monsplasty and joining of the clitoral halves**.

11 · Bladder neck reconstruction (stage 3)

Capacity at BNR best predicts dryness.

- **Prerequisites**: capacity **≥100 mL** under anaesthesia, stable urodynamics, a motivated child of 5–7 years and a voiding programme. Reimplant earlier for pyelonephritis or worsening hydronephrosis.

ALGORITHM

1. **Modified YDL**
2. Open the bladder neck transversely.
3. Perform **Cohen cross-trigonal reimplantation**.
4. Outline a posterior mucosal strip **15 × 30 mm**; denude the lateral muscle triangles.
5. Tubularise the strip and **double-breast the muscle flaps** over it.
6. Suspend to the rectus fascia; suprapubic tube for 3 weeks, then clamp trials.

- After MSRE, about **70–75%** are dry by day (intervals >3 h); CPRE with YDL is lowest (about 56%).
- **Predictors** are template size, **successful primary closure** and, most importantly, **capacity at BNR**. Closure within 72 h or with osteotomy makes continence likely, whereas delayed closure without osteotomy gives about 10%. After two closures, fewer than 1 in 5 void continently.
- Day dryness comes within about 12 months and night dryness takes 2–3 years; **not dry by 2 years means failure**.

Category (after BNR)	Definition
Continent	Dry day and night, interval ≥3 h
Socially continent	Dry 3 h by day, occasional wet nights
Wet	Dry <3 h by day, wet at night

12 · Failures and complications

Each failed closure shrinks the bladder.

- **Failed closure** means dehiscence, **bladder prolapse**, outlet stricture or vesicocutaneous fistula. Causes are incomplete pelvic-diaphragm mobilisation, inadequate immobilisation, infection and tube malfunction.
- In expert series, major dehiscence or prolapse occurs in about 3% and femoral palsy in about 2%.
- **Reclosure** waits at least 4–6 months and uses **osteotomy**, rigid immobilisation and radical urogenital diaphragm division. After secondary closure, voided continence is only about 1 in 6.
- Outlet stricture presents with UTI, an enlarging bladder, stones or rectal prolapse; obstruction for more than 6 months usually means augmentation.
- **Failed BNR** means no 3-h dry interval by 2 years. **Redo YDL** only if the neck is patulous, capacity >100 mL and urodynamics stable; otherwise do **bladder neck transection, augmentation and a Mitrofanoff channel**.
- **Genital**: urethrocutaneous fistula is commonest; glans or corporal loss may need **radial forearm phalloplasty**.

13 · Diversion and augmentation

- **Ureterosigmoidostomy** needs anal continence. It risks pyelonephritis, hyperchloraemic acidosis and **colonic adenocarcinoma about three decades later**, so **yearly colonoscopy** is required.

- If the bladder is too small, use a **non-refluxing colon conduit**. After failed BNR, do **augmentation with an appendiceal channel, with or without neck closure**; stones are the commonest complication.
- **Continent diversion** is indicated for closures that do not grow, inadequate capacity or wetness after BNR, and upper-tract decompensation.

14 · Long-term issues

- **Malignancy**: augmented exstrophy bladders develop adenocarcinoma, so survey endoscopically from 5 years after reconstruction.
- **Male**: libido, erection and orgasm are preserved. Fertility is impaired by retrograde ejaculation and oligospermia; about 1/4 achieve paternity, half via ICSI.
- **Female**: a stenotic introitus needs **perineal flap vaginoplasty**, because a cutback worsens prolapse. **Uterine prolapse** is treated by sacral suspension. **Pregnancy** ends in a **planned caesarean with a urologist present**.
- Psychosocial problems need lifelong support and transition care.

15 · Epispadias

LAQ Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias)

Paper answer · P3

Asked as:

- Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias — *Practice Set 6 Paper III Q2 (LAQ)*
- Isolated penopubic Epispadias. — *Summer 2022 Paper III Q3f (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(isolated penopubic epispadias also answered in 2022-Summer Paper 3, Q3f)

Development of the urethra

1. Division of the cloaca (weeks 5–6)

- The urorectal septum divides the cloaca into the dorsal anorectal canal and the ventral **UGS (urogenital sinus)**.
- Wolffian duct entry is the landmark. Above it, the **vesicourethral canal** forms the bladder and pelvic urethra. Below it, the **definitive UGS** forms the phallic urethra in males, and the urethra and vestibule in females.
- UGS also forms the prostate (buds at 10 weeks) and Cowper and Littre glands; the **utricle** is Müllerian.
- The epithelium of the **entire male urethra is endodermal** (UGS origin).

2. Penile urethra: the "double zipper" (androgen-dependent)

- Indifferent genital tubercle until 8–9 weeks. Testosterone (Leydig cells, 8 weeks) is converted to DHT (dihydrotestosterone) by **type 2 5α-reductase**.
- The **urethral plate** (endoderm) forms in the tubercle.
- **Opening zipper:** the plate canalises proximal to distal into a urethral groove bounded by the urethral folds.

- **Closing zipper:** the folds fuse in the ventral midline, forming the tubular urethra and median raphe.
- **Glanular urethra:** formed by direct canalisation of the plate inside the glans (current endodermal view; Glenister's ectodermal ingrowth theory is outdated).
- **Timetable:** coronal sulcus 11–12 weeks; fusion complete 16 weeks; urethra complete 18 weeks; fetal curvature gone 20 weeks.
- The prepuce forms with the glanular urethra, so hypospadias has a hooded foreskin.
- **Female:** the vestibular plate canalises but the folds do not fuse, giving an open vestibule and labia minora.

Classification of congenital urethral anomalies

Mechanism	Anomaly
Failed ventral fold fusion	**Hypospadias** (1 in 250 males; commonest at shaft–glans junction); congenital urethral fistula
Dorsal defect	**Epispadias** (exstrophy–epispadias complex)
Obstruction	**PUV (posterior urethral valves)**, Young type 1 (95%) and type 3; **anterior urethral valves** with diverticulum; urethral atresia or hypoplasia; meatal stenosis
Duplication	**Urethral duplication**: sagittal; ventral channel is functional; Effmann type I blind, II complete, III with caudal duplication
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); prune-belly association
Cystic or remnant	Cowper's syringocele; enlarged utricle (proximal hypospadias); parameatal cyst

Epispadias (including isolated penopubic epispadias): definition, embryology, classification and clinical features

- Part of the **exstrophy–epispadias complex**: a **dorsal urethral defect** (defect of the dorsal urethral wall): the roof is replaced by a broad (dorsal) mucosal strip (urethral plate). Cause (embryology): absent mesodermal reinforcement of the cloacal membrane; the genital tubercles fuse **caudal** to it, giving a dorsal urethral defect.
- **Male (1 in 117,000; incidence of isolated male epispadias about 1 in 117,000), classification by meatus:** glanular, penile, **penopubic** (subsymphyseal) (about 70% of male cases are complete (penopubic), with incontinence). **Female (1 in 484,000), Davis:** vesicular (patulous), subsymphyseal, retrosymphyseal (incontinent).
- **Features (clinical):** penopubic meatus; outlet may admit a finger → **total incontinence**. **Short, broad, upturned penis with dorsal chordee**, like exstrophy; pubic diastasis (less than exstrophy); small bladder from no outlet resistance; **VUR (vesicoureteral/vesicoureteric reflux) 30–40% (male), 30–75% (female)**. Female (Davis retrosymphyseal type): bifid clitoris, depressed mons, incontinence.

Evaluation

- (Renal) ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy and closure) if pubic diastasis is over 3 cm, the intrasymphyseal bar is poor, sphincter is absent or capacity is tiny (tiny bladder).

Management of epispadias

- **Aims:** continence, safe upper tracts, a straight functional penis of good length.
- **Preoperative testosterone** 2 mg/kg intramuscularly (intramuscular), 5 and 2 weeks before urethroplasty.

Stage 1: penile reconstruction and urethroplasty (6–12 months)

- Release chordee: divide suspensory ligaments and scar; free corpora from the inferior pubic rami.
- **Modified Cantwell–Ransley:** outline an 18-mm plate to the tip; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; free the plate from the corpora (distal 1 cm left on the glans); tubularise over 8 Fr; transpose it (the tubularised plate is moved) **ventral to the corpora**, which are rotated medially ± cavernocavernostomy; close glans wings; Z-plasty at the base; stent 10–12 days.
- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.
- Outlet resistance rises, so capacity grows.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **YDL (Young–Dees–Leadbetter)** bladder neck repair + Marshall–Marchetti–Krantz suspension + **ureteric reimplantation**. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

Female epispadias: at 6–12 months excise the urethral roof, close over 10 Fr, unite clitoral halves and labia minora, monsplasty; YDL at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula up to (about) one-fifth (at the penile base); long-term fistula and stricture about 5% (stricture rare); erections and paternity (usually) preserved.
- Better than exstrophy (bladder never exposed or scarred). Failures (failure): bulking injection (injectables), augmentation, continent stoma.

Diagram to draw: double zipper (canalisation then fusion); penopubic epispadias (dorsal penopubic meatus and plate).

> **Key points to remember:**
> - Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
> - About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
> - Diastasis over 3 cm (>3 cm) → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
> - Continence after YDL: males 47–82%, females 70–87%.

Also, from the condensed book (not in the paper answer):

- Glanular epispadias is never incontinent.

	Male	Female
Incidence	1 in 117,000	1 in 484,000
Classification	Glanular, penile, **penopubic** (about 70%, incontinent)	Davis: vesicular, subsymphyseal, retrosymphyseal (incontinent), in rising severity
VUR	30–40%	30–75%
Presentation	At birth	May present only with incontinence

ALGORITHM

1. **Female repair (6–12 months)**
2. Incise vertically through the urethral wall at 9 and 3 o'clock.
3. Excise the urethral roof (dorsal wedge taper) and close over a 10-Fr Foley.
4. Unite the clitoral halves and labia minora; bring mons fat over the suture line.
5. Catheter 5–7 days; BNR at 4–5 years.

- Treat as exstrophy (osteotomy and closure) if diastasis is >3 cm, the sphincter is scant or capacity is small.
- In **males** with a small bladder, do urethroplasty first, then BNR at about 100 mL (4–5 years).
- Female continence after BNR is about **87%**, as good as in CBE girls, because the bladder was never exposed.

16 · Cloacal exstrophy

A multisystem defect; neurological level sets disability.

- Incidence is 1 in 200,000–400,000. **OEIS** stands for Omphalocele, Exstrophy, Imperforate anus and Spinal defects. **Two hemibladders flank exstrophied ileocaecal bowel**, with a rudimentary hindgut and separated hemiphalluses.

Feature	CBE	Cloacal
Spinal defects	Uncommon	**85–100%** (lipomeningocele commonest)
Omphalocele	Rare	Near-universal
Upper-tract anomalies	Uncommon	41–60%; solitary kidney commonest
Osteotomy	Selective	Always
Continence	Mostly voiding	Mostly augmentation with CIC

- Only about 1 in 5 has a sacral skin sign, so all need spinal MRI. A 46,XY child may have an **intravesical phallus**, so biopsy before calling it aphallia.
- Prenatally, the **"elephant trunk" sign** is prolapsed ileum. **Gender** follows **karyotype**.
- **Bowel**: never discard hindgut; tubularise the caecal plate with an end colostomy.

ALGORITHM

1. Stage 1 (newborn): close the omphalocele, separate the caecal plate, join the hemibladders and form a colostomy.
2. Stage 2: bladder closure with **combined osteotomy** and fixation for 6–8 weeks.
3. At 5–7 years: YDL only if capacity >100 mL (rare); usually augmentation with a continent stoma.

- Combined osteotomy gives about 90% closure success but no continence gain; staged closure beats single-stage.
- Primary closure fails in about 31–43%. Continence relies on augmentation with CIC, and the **neurological level** determines disability.

44 Prune-Belly Syndrome

LAQ Prune belly syndrome

Paper answer · P1 · P3 · merged from 2

Asked as: P1 LAQ "Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management." · P3 SAQ "Prune belly syndrome"

Definition

- **PBS (prune belly syndrome)**, also called Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **bilateral intra-abdominal testes** and a **dilated, poorly contracting urinary tract**.
- 1 in 29,000–44,000 live births; **95% male**.
- **Survival depends on renal dysplasia and lung hypoplasia, not on the abdominal wall.**

Antenatal diagnosis

- **Ultrasound** may suggest PBS from 11–14 weeks, but the classic signs are not consistent even at 30 weeks.
- **Findings:** huge bladder filling the abdomen; bilateral hydroureteronephrosis; urachal diverticulum; dilated posterior urethra; echogenic kidneys with cortical cysts (dysplasia); irregular abdominal circumference; fetal ascites; oligohydramnios when severe.
- **Differential diagnosis:** PUV (posterior urethral valves), urethral atresia, megacystis–megaureter.
- Fetal dilatation does **not** predict postnatal renal function.
- **Prenatal management:** intervention has no proven renal benefit. Vesicoamniotic shunting is justified only for **urethral atresia with progressive oligohydramnios**, or to decompress the abdomen and avoid dystocia. Termination is hard to justify except with early, severe oligohydramnios. Plan delivery in a centre with neonatal intensive care.

Diagram to draw: fetal ultrasound sketch: huge bladder, dilated posterior urethra, bilateral hydroureteronephrosis.

Pathogenesis (theories, none proven)

Theory	Mechanism
Early (fetal) urethral obstruction	Back-pressure dilates the tract and stretches the wall; but most have no obstruction after birth
Lateral plate mesoderm defect	This mesoderm forms ureters, bladder, prostate, urethra and gubernaculum, so one defect explains the triad
Intrinsic urinary tract (ureteric) defect	Primary ureteric dilatation and fetal ascites
Yolk sac defect	Primary defect of the yolk sac

- **Genetics:** mostly sporadic, normal karyotype; smooth-muscle genes **MYOCD (myocardin)** and **FLNA (filamin A)** in multiplex families.

Pathophysiology and features by organ

- **Kidneys:** dysplasia in about 50% (Potter type II from mesenchymal defect; type IV cystic from obstruction).
- **Ureters:** long and tortuous, worse distally. Collagen replaces smooth muscle, so walls do not coapt and peristalsis fails: **a failed pump, not a blocked pipe**. The result is stasis and infection. **VUR (vesicoureteric reflux) in about 75%.** The proximal ureter has better muscle.
- **Bladder:** large, smooth, collagen-rich; urachal pseudodiverticulum (urachus patent in 25–30%); splayed trigone. Normal compliance but poor contractility, so voiding is "unbalanced", with residual urine.
- **Prostate and urethra:** prostatic hypoplasia dilates the posterior urethra; distal obstruction in about 20%; urethral atresia; **megalourethra** (scaphoid: spongiosum deficient; fusiform: all erectile bodies, more severe).
- **Testes:** intra-abdominal; azoospermia usual; paternity only by sperm retrieval and ICSI (intracytoplasmic sperm injection).
- **Extra-genitourinary (about 75%):** pulmonary hypoplasia; cardiac about 10%; gastrointestinal at least 30% (malrotation); orthopaedic up to 65% (talipes, hip dysplasia).

Classification (Woodard)

Category	Features	Outlook	Management
I	Oligohydramnios, **pulmonary hypoplasia**, dysplasia, urethral atresia	Perinatal death	Supportive; bladder drainage only
II	Full triad, moderate or unilateral dysplasia / moderate uropathy, **no pulmonary hypoplasia**	Risk of CKD (chronic kidney disease)	Individualised (controversial, debated group)
III	Mild or incomplete; **stable function**	Good; majority	Surveillance

- **Also know:** RUBACE score (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism, Extra-genitourinary) grades morbidity / severity in survivors; "pseudo-prune" presents late with renal failure.

Postnatal management: neonatal

Team: neonatology, nephrology, urology, cardiology.

1. Heart and lungs first: **chest radiograph** for pneumothorax and hypoplasia.
2. Creatinine, BUN (blood urea nitrogen), electrolytes; follow the **trend**, because the first value reflects the mother.
3. Renal and bladder ultrasound once stable.
4. Antibiotic prophylaxis; **circumcision**; strict asepsis.
5. **VCUG (voiding cystourethrogram)** only with renal insufficiency or suspected outlet obstruction.
6. **DMSA (dimercaptosuccinic acid) at 4–6 weeks** for split function; MAG3 (mercaptoacetyltriglycine) for drainage.
7. Outlet obstruction: catheter or suprapubic tube.

Diagram to draw: VCUG showing large bladder and dilated posterior urethra.

Postnatal management: urinary tract

- **Conservative** for category III: surveillance, treat infection.
- **Surgery for:** recurrent upper tract infection, sepsis, progressive dilatation or rising creatinine / renal decline, high-grade VUR, true obstruction.
- **Temporary diversion:** wide **cutaneous vesicostomy (Blocksom)**. Avoid high ureterostomy, which wastes the good proximal ureter.
- **Reconstruction after 3–6 months** (lung maturity): excise the distal ureter, taper and reimplant the healthier proximal ureter; reduction cystoplasty gives little lasting benefit (capacity recurs). **CIC (clean intermittent catheterisation)** or Mitrofanoff for poor emptying.

Orchidopexy

- **Bilateral transabdominal at about 6 months**, often with other surgery; later, **staged Fowler–Stephens** (better than single-stage) or laparoscopic.

Abdominoplasty

- Aims: cosmesis; better cough, voiding / emptying and defecation.

Technique	Key feature
Randolph	Transverse; excises the worst lower wall
Ehrlich	Vertical; umbilicus kept; vest-over-pants overlap
Monfort	Vertical; umbilicus kept; central fascial plate with lateral flaps overlapped; transperitoneal access, good exposure for combined surgery

- **Also know:** Furness (extraperitoneal plication); Dénes; corset.

Prognosis and follow-up

- About 20% die perinatally; **up to 30% develop chronic renal failure**.
- **Nadir creatinine <0.7 mg/dL at 1 year** predicts stable function.
- Lifelong timed voiding, cultures and imaging; transplant when needed.

> **Key points to remember:**
> - Triad; 95% male; dysplasia and lungs decide survival.
> - Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
> - Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
> - Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

Also, from the condensed book (not in the paper answer):

- Renal failure is driven by pyelonephritis, dysplasia and polyuria; transplant outcomes are equal.
- Female PBS is ~5%, with normal gonads.

1 · Definition and aetiology

- The **triad** (Eagle–Barrett) is deficient abdominal wall muscle, bilateral intra-abdominal testes and a dilated urinary tract. Incidence is ~**1 in 29,000–44,000** and **95% male**; it is commoner in twins and African ancestry.

- **Survival follows the kidneys and lungs, not the belly**; the wall is least prognostic.
- There is no inheritance pattern; multiplex families show the smooth-muscle genes **MYOCD** and **FLNA** MCQ .
- Theories are early urethral obstruction, a lateral plate mesoderm defect, an intrinsic urinary tract defect and a yolk sac defect.

> **KEY POINT**
>
> Weakness of the obstruction theory: most have **no obstruction after birth**. The tract is **dilated, collagen-rich and largely non-obstructed**, and this drives management.

2 · Genitourinary pathology

- **Kidneys**: dysplasia in ~**50%**, worst with outlet obstruction without urachal decompression. Dilation does not correlate with dysplasia; **infection, not obstruction, is the main threat**.
- **Ureters**: the distal ureter is worse, so the **proximal ureter is preserved** in reconstruction. Vesicoureteric reflux (VUR) occurs in ~**75%**. The ureter is a **failed pump, not a blocked pipe**.
- **Bladder**: massive, thick but smooth, with a urachal pseudodiverticulum; the urachus is patent at birth in **25–30%**. It has normal compliance, large capacity and poor sensation; about half void normally but may deteriorate.
- **Prostatic hypoplasia** dilates the posterior urethra and causes ejaculatory failure. **Urethral atresia** is lethal unless the urachus is patent. **Megalourethra** is scaphoid (spongiosum only, milder) or fusiform (cavernosa too, more severe).
- **Testes** lie over the iliac vessels. Infant germ cells are normal, so the abdominal environment destroys later potential, hence early orchidopexy.

3 · Extragenitourinary anomalies

About **75%** have them, and nearly **half premature**. Pulmonary problems (>50% of survivors) include hypoplasia, pneumothorax and pneumonia; cardiac (~10%), gastrointestinal (≥30%, malrotation) and orthopaedic (up to 65%, talipes, hip dysplasia, knee dimples) anomalies are common.

> **KEY POINT**
>
> Two mechanisms: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation).

4 · Presentation and spectrum

- **Prenatal** appearance mimics posterior urethral valves: massive bladder and hydroureteronephrosis. Dilation does **not** predict function.

Woodard category	Kidneys / lungs	Course	Urological management
I	Dysplasia; **pulmonary hypoplasia**; oligohydramnios, urethral atresia	Stillbirth or early death (~20% perinatal mortality overall)	Supportive; simple bladder drainage

Woodard category	Kidneys / lungs	Course	Urological management
II	Full triad; minimal or unilateral dysplasia; no lung hypoplasia	Risk of progressive renal insufficiency	Individualised — **controversial group**
III	Incomplete or mild; no dysplasia; stable	Majority; good	Surveillance; surgery for UTI or deterioration; orchidopexy in year 1

- The abdominal wall is not a sorting criterion. **Pseudo-prune** (partial flaccidity, unilateral undescended testis) is diagnosed late and has higher late renal failure.

5 · Evaluation

ALGORITHM

1. Team; **heart and lungs first** (chest X-ray).
2. Creatinine trend; the day-1 value is maternal.
3. Renal–bladder ultrasound.
4. Prophylaxis and circumcision; then selective imaging.

- A voiding cystourethrogram is done only for renal insufficiency or suspected outlet obstruction, under antibiotic cover. A **DMSA** scan is done at **4–6 weeks**; MAG3 renograms are unreliable with massive dilation. Outlet obstruction is drained by a percutaneous suprapubic tube.

6 · Management philosophy

	Conservative	Early reconstruction
Premise	Dilation is non-obstructive and may improve	Stasis, reflux and infection drive renal loss
Operate when	Proven obstruction, intractable infection	Progressive dilation, high-grade VUR, recurrent upper tract infection, declining function
Best for	Category III	Selected category II

- Delay major reconstruction to **≥3–6 months** for the lungs. Single-stage comprehensive surgery adds no morbidity.

7 · Urinary tract surgery

- **Blocksom vesicostomy** (sepsis, renal failure, atresia): make the stoma larger than usual, as it tends to stenose. Avoid proximal ureterostomy.
- **Ureteroplasty**: excise the distal ureter, taper the upper part and reimplant. Reduction cystoplasty is reserved for a large urachal diverticulum, as capacity recurs; clean intermittent catheterisation empties better. Circumcision reduces UTI.

8 · Orchidopexy

- A broad mesorchium defeats the inguinal approach. Do a **bilateral transabdominal orchidopexy at ~6 months**, often with other surgery. Later, a **staged Fowler–Stephens** procedure is preferred.

9 · Abdominoplasty

- Cosmetic gain is certain; functional gains (cough, defecation, emptying) are plausible. Operate from ~6 months with other surgery.

Technique	Incision / umbilicus	Key step / note
Randolph	Transverse, 12th rib to 12th rib	Excision of the lower wall; lateral bulge persists
Ehrlich	Vertical; umbilicus on a pedicle	Vest-over-pants advancement
Monfort	Vertical ellipse; umbilicus in situ	Central fascial plate with lateral flaps; **transperitoneal**, so ideal with reconstruction
Dénes	Fusiform	Modified Ehrlich
Furness	Midline, limited	Plication; peritoneum not opened

ALGORITHM

1. **Monfort**
2. Skin ellipse, umbilicus spared; incise the central plate at the lateral rectus borders.
3. Do the genitourinary surgery.
4. Suture plate edges to scored lateral peritoneum.
5. Overlap lateral flaps over drains.

10 · Long-term outcome

- Creatinine **<0.7 mg/dL at 1 year** predicts stable function. Up to **30%** develop chronic renal failure, driven by pyelonephritis, dysplasia and polyuria. Transplant outcomes are equal.
- **Bladder**: timed and double voiding; if these fail, use CIC.
- **Fertility**: puberty and erections are normal, but **azoospermia means no spontaneous paternity**; sperm retrieval with ICSI is used.

> **KEY POINT**
>
> Long-term care is a contest against **stasis**: lowering residual urine and infection protects the kidneys, while dysplasia and polyuria set a ceiling surgery cannot raise.

45 Posterior Urethral Valves

1 · Definition and classification

- **Posterior urethral valves (PUV)** are the commonest cause of bladder outlet obstruction in infant boys. Ablation relieves obstruction but **not the embryological injury** to bladder and kidney, so care is lifelong.
- Lower urinary tract obstruction (LUTO) affects ~2.2/10,000 births; PUV is commonest (~1.4/10,000), then urethral atresia and prune-belly.

*See the paper answer **Posterior urethral valves: embryology, management and valve bladder** (Section 5, Ch 32).*

Also, from the condensed book (not in the paper answer):

- Type III resembles a congenital stricture; it is contentious but accepted.

Young type	Anatomy	Status
I (95%)	Leaflets from verumontanum fuse anteriorly just proximal to external sphincter	Obstructing; the lesion treated
II	Folds from verumontanum to bladder neck	Non-obstructing; probably not real
III (5–10%)	Annular membrane, central aperture (persistent urogenital membrane)	Obstructing

- Copy number variant burden predicts higher nadir creatinine and CKD stage 5.

2 · Pathophysiology

- Obstruction causes detrusor hypertrophy, then **fibrosis replaces smooth muscle**. The bladder is the **driver**: renal dysfunction, reflux (VUR) and voiding dysfunction act through it.
- The voiding cystourethrogram (VCUG) triad is a **dilated posterior urethra, hypertrophied bladder neck and flattened verumontanum**.
- Ureters dilate from **pressure transmission, VUR (up to 70%) and polyuria**, then lose peristalsis, causing stasis and infection.
- Renal injury is **obstructive uropathy** (partly reversible) plus **dysplasia** (fixed at birth, which sets the ceiling). Tubular injury causes **nephrogenic polyuria** and post-obstructive diuresis.
- Angiotensin II induces TGF-β1 and fibrosis; ACE inhibitors may help selected children.

Organ / lesion	Effect	Natural history
Lung — hypoplasia	Respiratory failure	Commonest cause of neonatal death
Kidney — obstructive uropathy	Reversible insufficiency	Improves; may recur with bladder dysfunction
Kidney — dysplasia	Irreversible insufficiency	Progressive failure, hypertension
Kidney — tubular injury	Salt and water wasting	Nephrogenic diabetes insipidus; worsens with age

Organ / lesion	Effect	Natural history
Bladder	Poor sensation, hypercontractility, low compliance; later myogenic failure	Lifelong, changes with age
Ureters	Poor coaptation, stasis	Most retain hydronephrosis — **not** a sign of residual obstruction

"Pop-off" mechanisms

- **VURD** (unilateral high-grade VUR into a non-functioning dysplastic kidney) was thought protective. This is **refuted**: most with normal creatinine at 2 years lose it by 8–10 years, and urodynamics are worse. Urinoma, ascites and diverticula are debated but managed the same way.

> **KEY POINT**
>
> Every pop-off is false reassurance: normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child.

3 · Diagnosis

Antenatal

- Ultrasound shows a **thick distended bladder, bilateral hydroureteronephrosis, "keyhole sign" and oligohydramnios**; an echogenic or cystic cortex means dysplasia. It is only ~50% accurate for the cause.
- About half are diagnosed antenatally, but this has **not** improved renal outcome. A male fetus with these findings needs postnatal US and a **VCUG before discharge**.

Postnatal

- **US** comes first: the wall is thick if **>3 mm distended / >5 mm empty**.
- **VCUG** is definitive, showing a dilated elongated posterior urethra with **abrupt funnelling** at the valve; about half have high-grade VUR. Use a 5–8 Fr feeding tube and **withdraw the catheter during voiding**.
- A **MAG3** renogram is done at **2–3 months**. Creatinine in the first 48 h is maternal; follow the trend to the **nadir** in year 1.

4 · Fetal intervention

- Decisions hinge on **oligohydramnios, hydronephrosis degree and laterality, and gestational age**. Favourable features are oligo- (not an-)hydramnios, normal karyotype, no cortical cysts and favourable urine.

Fetal urine (after 20 weeks)	Favourable
Sodium / chloride	**<100 / <90 mEq/L**
Osmolality	**<200 mOsm/L**
β2-microglobulin	**<6 mg/L** (most sensitive)

- **Triple serial sampling** 48–72 h apart beats one sample, though prediction was poor on systematic review.
- A **vesicoamniotic shunt** may prevent **pulmonary hypoplasia** but gives **no renal benefit**. Fetal cystoscopic laser ablation is no better. Both carry **40–50%** pregnancy loss.

KEY POINT

Fetal intervention can rescue the lung, not the kidney: fluid deficiency is still in progress, but dysplasia is already established.

5 · Neonatal presentation and initial management

- **Pulmonary hypoplasia** is the commonest early death and the usual cause of delayed surgery.

ALGORITHM

1. Stabilise the airway.
2. Pass a **5–7 Fr feeding tube**. If it curls: coudé, curved stylet, or rectal finger.
3. Confirm on US; minimal balloon fill.
4. Creatinine, electrolytes, fluids.
5. US, then VCUG; antibiotic prophylaxis.
6. Ablate when stable.

- **Post-obstructive diuresis** needs monitoring of output, weight and electrolytes with IV replacement. **Hyponatraemia** is transient pseudohypoaldosteronism, treated with sodium.
- **Urinoma or ascites** (3–10%) resolves with bladder drainage.
- **Delayed presentation** (UTI, incontinence, renal failure) is common: over 40% present after 6 months, and late does not mean milder.

6 · Surgery

Primary valve ablation (preferred)

- Ablation restores flow and **normal bladder cycling**, superior to passive diversion. Use a **7.5–9 Fr cystoscope** with holmium laser or Bugbee.
- Incise at **5 and 7 o'clock, with or without 12 o'clock**. Valves are thin and avascular, so **incise, don't resect**.
- For a small urethra, use vesicostomy or **PADUA** (progressive urethral dilation with up-sized tubes). Leave a catheter ≥24 h and **repeat VCUG within 1 month**.

Complication	Cause / prevention	Management
Urethral stricture	Over-resection, hot loop, oversized instrument	Dilation / urethrotomy; urethroplasty if dense
Meatal injury	Resectoscope through small meatus	Smaller scope / PADUA
Residual valve	Incomplete incision	Persistent funnel on VCUG: re-ablate
Sphincter injury	Ablating distal to valves	Incontinence — precise incision

- **Bladder neck incision** gives no benefit and risks retrograde ejaculation; it is not recommended.

Vesicostomy

- It is used for a very small or premature infant, failed decompression, or persistent deterioration. It **does not defunctionalise** the bladder and matches ablation in final outcome.

SAQ Vesicostomy.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2016-Summer Paper 4, Q3f

Definition

- **Cutaneous vesicostomy**: the bladder dome is brought to the lower abdominal skin as an incontinent stoma draining into the nappy. It "vents" the bladder, protecting the upper tracts; usually a **temporary** bridge to reconstruction.

Indications

- **PUV (posterior urethral valves)**: infant too small for a resectoscope; upper tract deterioration despite drainage or ablation.
- **Prune-belly syndrome**: acute renal failure, urosepsis, urethral atresia. A larger stoma is needed because stenosis is common.
- **Neurogenic bladder**: high pressure despite CIC (clean intermittent catheterisation) and drugs, or CIC impossible; worsening hydronephrosis; febrile UTI (urinary tract infection) with high-grade reflux.
- **Also know:** urethral atresia; MMIHS (megacystis-microcolon-intestinal hypoperistalsis syndrome); urogenital sinus anomalies.

Blocksom technique (Duckett modification)

1. Transverse 2 cm incision midway between umbilicus and pubis. (In prune-belly, excise any urachal diverticulum.)
2. Incise the rectus sheath; separate the recti.
3. Fill the bladder; mobilise the **dome** up to the urachus, sweeping peritoneum off.
4. Bring the **dome** (not the anterior wall) to the wound. This reduces prolapse of the posterior wall.
5. Anchor detrusor to rectus fascia; stoma should take about 24 Fr.
6. Open the dome; suture mucosa to skin. No catheter needed.

Advantages

- Simple; no catheter or appliance.
- The bladder still fills and empties through the stoma at low leak pressure, so compliance is **not** lost; it remains usable for reconstruction or transplantation.

Complications

- **Stomal stenosis**, mucosal **prolapse**, peristomal dermatitis, stones, UTI; leakage harder to manage as the child grows.

Closure

- When the cause is treated (valve ablation, urethral growth) or at definitive reconstruction; check bladder function first (cystogram or urodynamics).
- Excise the stoma, free the bladder from fascia, close in two layers, catheter drainage for a few days.

Alternatives

Option	Use
Cutaneous ureterostomy or pyelostomy	Rare; persistent upper tract obstruction
Ileovesicostomy	Unable to catheterise; 10–15 cm ileum
Appendicovesicostomy (Mitrofanoff, 1980)	Continent catheterisable channel

Diagram to draw: sagittal view of bladder dome sutured to fascia and skin.

> **Key points to remember:**
> - Blocksom: dome-based stoma midway between umbilicus and pubis.
> - Bladder keeps cycling; compliance preserved.
> - Stenosis and prolapse are the main complications.

Also, from the condensed book (not in the paper answer):
- The urachus is excised with a small dome cuff; using the dome keeps the posterior wall taut.

Upper tract diversion, circumcision and nephrectomy

- Upper tract diversion gives **no renal protection**; use it only for deterioration **despite full lower tract drainage**.
- UTI risk is 50–60% in PUV; circumcision cuts it by over 80%, so offer it to all.
- VURD nephroureterectomy is rare; **keep the ureter** for augmentation.

7 · VUR in PUV

- VUR occurs in 50–80% and is **secondary**. It resolves after ablation in 25–40% and does not predict renal outcome.
- Treat the **bladder**. Reimplant only for recurrent febrile UTI despite circumcision and optimal bladder care.

> **KEY POINT**
> VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading.

8 · Bladder dysfunction and valve bladder syndrome

*See the paper answer **Posterior urethral valves: embryology, management and valve bladder** (Section 5, Ch 32).*

Also, from the condensed book (not in the paper answer):
- Stop oxybutynin if residual and capacity rise, as myogenic failure can follow.

ALGORITHM
1. High-pressure voiding causes fibrosis and weaker contraction.
2. Post-void residual (PVR) rises; nephrogenic polyuria adds volume.
3. Upper tracts store urine and refill the bladder, so it is never empty.
4. Pressure is back-transmitted; renal function declines, causing more polyuria.

- Continence comes at ~5.5 vs 2.9 years. Do **uroflow and PVR** in every toilet-trained boy; new incontinence after dryness is a red flag.
- An α-blocker suits early bladder-neck obstruction and an antimuscarinic the overactive wall; both harm once the bladder is hypocontractile. Do **videourodynamics** serially.

9 · Prognosis for renal function

- Lifetime end-stage renal disease (ESRD) is **20–50%**. Risks are dysplasia, **year-1 nadir creatinine**, recurrent UTI and bladder dysfunction.

Nadir creatinine (1 year)	ESRD risk
<0.8 mg/dL	Minimal (best single predictor)
>1.2 mg/dL	High
<0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL	**0% / 2% / 27% / 100%**

- **0.7 mg/dL is the watershed**, where risk leaps. On US, echogenicity, cysts and a smaller renal parenchymal area predict a worse outcome.

10 · CKD, ESRD and transplantation see Ch 61

- Do **videourodynamics in every candidate** and optimise with CIC or overnight drainage.
- Graft and patient survival are equal if the bladder is managed; late graft loss follows bladder dysfunction. Augmentation can be done before or after transplant.

11 · Long-term follow-up, sexual function and fertility

- Transition to adult urologists who know valves. Adult LUTS are 2–3 times commoner; erectile function and fertility are similar overall but worse with incontinence or renal insufficiency.
- Lifelong checks are blood pressure, creatinine, urine protein, US, and uroflow with PVR.

12 · Related urethral lesions

SAQ Anterior urethral valve

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 17 Paper 2, Q3c

Definition and epidemiology

- AUV (anterior urethral valve): a congenital obstructing fold or cusp in the anterior urethra (bulbar, penoscrotal or penile). It is the **commonest congenital obstructive lesion of the anterior urethra** but **25–30 times less common than PUV (posterior urethral valves)**.
- Often linked to an **anterior urethral diverticulum**: the distal lip of the diverticulum fills and acts as a valve, or a semilunar fold hangs from the roof.

Embryology (unclear)

- Hypoplastic corpus spongiosum over the segment suggests a place on the **hypospadias spectrum**, or faulty union of urethral mucosa and fossa navicularis epithelium; a ruptured dilated Cowper (bulbourethral) gland duct is also proposed.

Clinical features (vary with severity and age)

- Antenatal hydronephrosis, LUTO (lower urinary tract obstruction) in severe cases.
- **Ventral penile bulge** during voiding, with urine expressed by compressing the shaft; post-void dribbling, weak stream, incontinence.
- Palpable bladder, UTI (urinary tract infection), renal insufficiency.

Evaluation

- **VCUG (voiding cystourethrogram):** dilated anterior urethra proximal to the valve, diverticulum, trabeculated bladder, bladder diverticula, often VUR (vesicoureteric reflux).
- Ultrasound, creatinine; urethroscopy confirms.
- **Differential:** Cowper syringocele, megalourethra, acquired diverticulum, stricture.

Management

- **Premature or small infant:** vesicostomy until the urethra takes a cystoscope.
- **Most cases: endoscopic valve ablation** (Bugbee electrode or laser), usually definitive.
- **Large diverticulum:** open excision and urethroplasty over a catheter.
- Then manage VUR, infection and bladder dysfunction.

AUV versus PUV

Feature	AUV	PUV
Frequency	25–30 times rarer	Commonest infant outlet obstruction
Site	Bulbar, penoscrotal, penile	From the verumontanum
Sign	Penile bulge	Weak stream, palpable bladder
Bladder dysfunction	Up to 80%	Near universal
Renal outcome	78% normal function	ESRD (end-stage renal disease) 20–50%

Prognosis

- Depends on pre-treatment creatinine and GFR (glomerular filtration rate). Azotaemia, VUR and UTI together raise the risk of poor renal outcome 25-fold.

Diagram to draw: VCUG sketch: dilated bulbar urethra with diverticulum ending at a valve lip.

> **Key points to remember:**
> - 25–30 times rarer than PUV; diverticulum common.
> - Penile bulge on voiding; diagnosis by VCUG.
> - Endoscopic ablation usually definitive; 78% keep normal renal function.

Also, from the condensed book (not in the paper answer):

- **Urethral atresia** needs decompression to survive; treat with vesicostomy, then PADUA.
- **Urethral duplication** (Effmann I–III): the **ventral** channel has the sphincter, so it is the one moved to the glans.
- **Urethrorrhagia** in peripubertal boys resolves in up to 92%; treat voiding dysfunction and reserve cystoscopy for atypical cases.
- Anorectal malformation often has a rectourinary fistula; catheterise before PSARP.

13 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT: vesicoamniotic shunt vs conservative in LUTO	Trend to better 28-day survival; poor long-term survival and renal function in both	Shunt may help lungs, not kidneys
Circumcision RCT	Circumcision and prophylaxis vs prophylaxis	Febrile UTI 3% vs 49% over 2 years	Circumcise boys with PUV
Oxybutynin RCT	Oxybutynin to toilet training vs observation post-ablation	Better hydronephrosis/VUR; no change in creatinine, UTI, continence	Selective, monitored use

46 Neuromuscular Dysfunction of the Lower Urinary Tract in Children

1 · Definition and causes

- **Neurogenic lower urinary tract (LUT) dysfunction** is bladder and/or sphincter dysfunction from aberrant innervation.
- The commonest cause is **neural tube defect (NTD)**, mainly spina bifida (SB). Survival is about 90% to 10 years, so **renal failure** is a leading late morbidity; **10%** of SB deaths are urological. Clean intermittent catheterisation (CIC, Lapides 1971) was the pivotal advance.

Congenital	Acquired
Myelomeningocele; occult NTD (lipomeningocele, tethered cord); sacral agenesis; anorectal malformation (ARM)	Brain: cerebral palsy (CP), tumours, infarcts. Cord: trauma, transverse myelitis. Pelvis: extensive surgery

2 · Myelodysplasia (NTD)

SAQ Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty

Paper answer · P1 · P3 · merged from 2

Asked as: P3 SAQ "Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty" · P1 SAQ "Neurogenic bladder in myelodysplasia"

Introduction

- Myelodysplasia (NTD [neural tube defect], mainly **myelomeningocele**, an open NTD) is the commonest cause of neurogenic LUT (lower urinary tract) dysfunction in children; other causes are occult dysraphism, sacral agenesis and anorectal malformation.
- Lesions: lumbosacral 47%, lumbar 26%, sacral 20%.
- **Bony or motor level does not predict bladder function**, and the lesion can progress (re-tethering), so every newborn needs urodynamic testing.
- About 10% of deaths in SB (spina bifida) are urological. Folic acid 400 µg daily before conception prevents 50–70% of NTDs.

Patterns of dysfunction

- Detrusor: overactive (NDO, neurogenic detrusor overactivity) or underactive. Outlet: competent, incompetent or dyssynergic.
- Newborns: most have detrusor contractions; over a third are acontractile. Needle EMG (electromyography): intact sacral arc about 40%, partial denervation 25%, complete 35%.

Prenatal assessment

- **Maternal serum AFP (alpha-fetoprotein)** is raised in open defects; amniotic AFP and acetylcholinesterase confirm.
- **Ultrasound:** spinal defect; "lemon" and "banana" signs of Chiari II; ventriculomegaly; hydronephrosis, thick bladder. Fetal MRI (magnetic resonance imaging) for level.
- Multidisciplinary counselling at a fetal centre.

- **MOMS (Management of Myelomeningocele Study) 2011:** closure **before 26 weeks** reduced shunt need (about 40% versus 82%) and improved motor and cognitive scores, but caused prematurity (34 versus 37 weeks) and uterine thinning. **No proven bladder benefit**: CIC (clean intermittent catheterisation) and reconstruction rates unchanged.

Neonatal assessment (baseline; EAU [European Association of Urology] 2022)

Test	Timing / purpose
Examination	Spine, legs, anal tone, perineal sensation, bladder
Renal and bladder ultrasound	Before discharge, then 6-monthly for 2 years (serially); hydronephrosis, wall thickness
Residual urine	Start CIC if not emptying
Creatinine	Within 3 months (cystatin C later; better with low muscle mass)
VCUG (voiding cystourethrogram) or video-UDS	Reflux; dilating VUR (vesicoureteric reflux) in about 7%
UDS (urodynamics) with EMG	Within 3 months, once stable after closure
DMSA (dimercaptosuccinic acid)	Baseline scarring

Urodynamic risk (McGuire 1981; UMPIRE [Urologic and Renal Protocol for the Newborn and Young Child with Spina Bifida])

Risk	DLPP (detrusor leak point pressure) or end-filling pressure
High (hostile)	40 cm H_2O or more
Intermediate	25–39 cm H_2O, NDO or poor compliance
Low (safe)	Under 25 cm H_2O

- Poor compliance and DSD (detrusor–sphincter dyssynergia) also predict upper-tract damage.
- Repeat studies: the lesion can change (re-tethering).

Diagram to draw: neonatal UDS trace with DLPP over 40 cm H_2O.

Management: kidneys first, then continence

- **Early CIC (Lapides 1971) plus oxybutynin** for high-risk (hostile) bladders: hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) falls from about 30% to under 2% (P1: 0–1.6%).
- Other drugs: solifenacin, mirabegron; **botulinum toxin A** about 10 U/kg (maximum 360 U), continence 65–87%.
- Surgery: vesicostomy (bridge); **augmentation cystoplasty**; Mitrofanoff channel; bladder-neck sling, reconstruction or artificial sphincter.
- VUR: CIC and antimuscarinic first; avoid Credé; reimplant during augmentation if grade ≥III.
- **Also know:** latex allergy; bowel programme; sexual dysfunction (about 75% of men have ED [erectile dysfunction]; sildenafil helps); precocious puberty.

Bladder myoplasty: definition

- The P3 wording "bladder myeloplsty" is assumed to mean "bladder myoplasty" (a muscle procedure on the bladder wall).

- **Detrusor myoplasty:** wrapping a **vascularised, innervated skeletal muscle flap** round the bladder so that voluntary contraction of that muscle empties an **acontractile bladder**.

Bladder myoplasty: indications

- **Acontractile or underactive detrusor** with catheter dependence: lower motor neuron lesions (spina bifida, cauda equina injury), myogenic failure.
- Needs good bladder capacity and compliance, and a competent but not obstructing outlet.

Latissimus dorsi detrusor myoplasty (LDDM; Stenzl)

- **Free latissimus dorsi flap:** artery and vein joined to the **inferior epigastric vessels**; motor nerve joined to an **intercostal nerve**.
- Muscle wrapped **spirally over more than three-quarters of the bladder** and anchored to pelvic floor fascia and ligaments.
- The patient voids by contracting the transferred muscle (abdominal effort).
- **Results:** most voided with low residual; bladder contractility index rose to the strong range.
- **Complications** in about one-third: thromboembolism, abscess, wound infection; none long term.
- **Rectus abdominis detrusor myoplasty:** simpler, less morbid; small spinal cord injury series with spontaneous voiding.
- **Status: investigational.** Standard care stays **clean intermittent catheterisation**.

Diagram to draw: latissimus flap spiralled round the bladder with vascular and nerve anastomoses.

Other "myoplasty": detrusor myotomy/myectomy (autoaugmentation)

- Purpose is the opposite: **increase capacity and compliance** in a small, poorly compliant or overactive bladder.
- **Technique (Cartwright and Snow, 1989):** strip and excise the detrusor over the dome, leave the mucosa intact, and anchor the detrusor edges (psoas hitch) so the mucosa bulges as a **wide-mouthed diverticulum**.
- **Pros:** urothelial lining, no mucus, no metabolic change, no bowel.
- **Cons:** seldom adds significant capacity; compliance gain may not last; **not recommended for adult detrusor overactivity**.
- **Also know:** seromuscular enterocystoplasty covers the bulge with demucosalised sigmoid.

Alternatives

- **Also know:** sacral neuromodulation, sacral anterior root stimulation, reduction cystoplasty (abandoned), muscle-derived stem cells (research).

Key points to remember:

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm H_2O or more = hostile (McGuire 1981); early CIC + antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

Also, from the condensed book (not in the paper answer):

- Trospium (a quaternary amine) has few central effects.

Classification and prevention

- **Myelomeningocele** is open, commonest and most severe; **SB occulta** is closed and usually benign.
- The neural tube closes by **day 35**. **Folic acid** cuts NTDs by 50–70%: all women of child-bearing age take **400 µg/day** from 4 weeks before conception, and women with SB take **4–5 mg/day**.
- Recurrence is **5%** after one affected child. Almost all open NTDs have **Chiari II**, and most have hydrocephalus.
- **LUT function is not predictable from bony level or the legs**, so urodynamics defines every bladder.

KEY POINT

Management of Myelomeningocele Study (**MOMS**): prenatal repair (<26 weeks) reduced death or shunt need (RR 0.70) and improved motor and cognitive scores at 30 months, at the cost of prematurity, uterine-scar thinning or dehiscence, and more re-tethering.

Pathophysiology

- In **detrusor–sphincter dyssynergia (DSD)**, the sphincter fails to relax during detrusor contraction; it is diagnosed by electromyography.

KEY POINT

Every pattern is detrusor (over/under) × outlet (competent/incompetent/dyssynergic). Dangerous: **overactive or poorly compliant detrusor against a closed or dyssynergic outlet**, where storage pressure rises and causes reflux and hydroureteronephrosis. An incompetent outlet "pops off": the child is wet but the kidneys are safe.

Newborn evaluation

ALGORITHM

1. Before discharge: renal–bladder US and post-void residual (need for CIC?).
2. After week 1: serum creatinine.
3. Once stable: urodynamics, **safe from ≈3 months**.
4. Cystography: ideally videourodynamics, otherwise VCUG.
5. Optional baseline DMSA scan (EAU).
6. US at 6, 12, 18 and 24 months, then yearly.

- Guidelines come from ICCS 2012, BAPU 2016 (most selective), SBA 2018 and EAU 2022. In the UMPIRE newborn cohort, dilating VUR was **7%**.
- **Cystatin C** beats creatinine, because low muscle mass hides a falling GFR. High-grade VUR acts as a pop-off, so urodynamic pressures underestimate risk.

Urodynamic risk factors

Category	Definition
Hostile / high	Detrusor leak point pressure (DLPP) or end-filling pressure **≥40 cmH_2O**; NDO with DSD (DSD removed in UMPIRE revision)
Intermediate	NDO, reduced compliance, 25–39 cmH_2O
Abnormal but safe	<25 cmH_2O
Normal	Normal capacity, compliance >15 mL/cmH_2O, no NDO/DSD, minimal residual

- Poor compliance independently predicts renal injury.

Tethered cord

- Growth stretches a low-lying cord, so gait, sensation and bladder deteriorate. Suspect it with new wetting, rising pressure, new UTIs, worsening scoliosis or back pain. Do urodynamics **before untethering and 3–4 months after**.

3 · Proactive (early) management

Priorities in order are **kidneys, then continence, then UTI avoidance, then sexual function and fertility**.

- **Early CIC and an antimuscarinic** (Bauer) treat hostile dynamics. All paediatric societies endorse this; there is **no benefit if the child empties spontaneously**. If hydronephrosis persists, add overnight drainage.
- Give **no** antibiotic prophylaxis on CIC; treat bacteriuria only if symptomatic.

Outcome	Expectant	Early CIC + antimuscarinic
Hydroureteronephrosis	18–80%	0–15%
Renal scarring	27–50%	4–12%
End-stage renal disease (ESRD)	≈30%	0–1.6%
Augmentation	—	18–24% fewer

KEY POINT

Proactive won because it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure).

4 · Management ladder

ALGORITHM

1. CIC and antimuscarinic, with or without a β3-agonist.
2. If that fails: intradetrusor botulinum toxin.
3. If CIC is impossible: vesicostomy as a bridge.
4. Then augmentation, with or without a catheterisable channel and an outlet procedure see Ch 48.

Pharmacotherapy

- **Antimuscarinics** are first-line. FDA-licensed drugs are **oxybutynin** (≥5 y; 0.1–0.6 mg/kg/day, max 5 mg/dose; oral, patch or intravesical), **solifenacin** (≥2 y), **fesoterodine** (≥6 y, >25 kg) and **mirabegron** (≥3 y; may cause hypertension).
- α-Sympathomimetics for low-outlet incontinence are unlicensed and help a minority.

Botulinum toxin A

- Under general anaesthesia, inject **≈10 U/kg (max 360 U)** into about 30 sites, **sparing the trigone**. Continence is 65–87%, lasting 6–10 months, and repeats stay effective. Poor pre-existing compliance predicts poor response.

Surgery

- **Vesicostomy** is a temporary bridge when CIC is unmanageable. **Augmentation** (ileum preferred) treats the refractory bladder, with a channel or outlet procedure as needed see Ch 48. After ileum, check B12 yearly.

VUR in the neurogenic bladder

- VUR is secondary, and treating the bladder resolves 30–55%. **Avoid Credé** with a reactive sphincter, because it worsens VUR.
- Antireflux surgery is for breakthrough febrile UTI despite low pressure. At augmentation, **reimplant grade ≥III**.

5 · Bowel management and latex allergy

- Neurogenic bowel is managed stepwise with diet, laxatives and transanal irrigation, then a **Malone antegrade continence enema (ACE)**, often at augmentation.
- Latex sensitisation risks intraoperative anaphylaxis, so care is latex-free **from birth**.

6 · Sexual function and transition

- Function is **inversely proportional to neurological level**. Central precocious puberty affects about half of girls.
- **Males** have erectile dysfunction in about 75%; **sildenafil** works in about 80%. **Females** conceive in 70–80%; avoid the combined pill and IUD. Offer a structured transition.

7 · Other causes

- **Sacral agenesis** is linked to maternal diabetes; look for flat buttocks and a short gluteal cleft. Level does not predict the bladder, and about a third present late.
- **Spinal cord injury**: CIC during spinal shock; a **suprasacral lesion later gives NDO with DSD**. **Autonomic dysreflexia** (lesion **≥T6**) causes severe hypertension with bladder filling, so monitor blood pressure during urodynamics.
- **Transverse myelitis** causes acute retention in over 95%; **late CIC leads to poorer compliance**, so do early urodynamics and CIC.
- **Cerebral palsy** gives coordinated overactivity; upper-tract risk is **<5%**, so urodynamics is only for failed conservative care or recurrent UTI.
- **ARM** has a cord anomaly in 18–50%, so do spinal US before 3 months; neuropathy comes from the cord, not the PSARP. Pelvic surgery can injure pelvic nerves.

> **KEY POINT**
>
> Lesion level predicts pattern. **Suprapontine** (CP) gives coordinated overactivity and low risk. **Suprasacral cord** (SCI, myelitis, tethering) gives NDO with DSD and high risk. **Sacral/peripheral** gives underactivity. Myelodysplasia is mixed and unpredictable.

8 · Landmark trials

Trial	Design	Result	Take-home
MOMS	RCT, prenatal vs postnatal repair	Less shunting, better motor score; more prematurity, uterine dehiscence	Fetal repair in selected centres
McGuire 1981	Myelodysplasia cohort	DLPP >40 cmH_2O led to upper-tract damage	40 cmH_2O = safety line
Sildenafil in SB	Double-blind placebo RCT	≈80% improved erections	First-line for ED

47 Functional Disorders of the Bladder and Bowel in Children

1 · Normal bladder and bowel function

LAQ Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency

Paper answer · P1 · P3 · merged from 2

Asked as: P1 LAQ "Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children" · P3 SAQ "DD of frequency and urgency in 3 yr male child and treatment"

Introduction

- The LUT (lower urinary tract) has two opposite functions: **storage** at low pressure with a closed outlet, and **emptying** by a coordinated detrusor contraction with a relaxed outlet. Bladder and sphincter act reciprocally.

Neurophysiology: peripheral innervation

	Parasympathetic	Sympathetic	Somatic
Origin	**S2–S4** (sacral parasympathetic nucleus)	Thoracolumbar (T10–L2)	**Onuf nucleus** (S2–S4 ventral horn)
Nerve	Pelvic	Hypogastric	Pudendal
Bladder	Contracts (ACh [acetylcholine] on **M3**; ATP)	Relaxes (noradrenaline on **β3**)	–
Outlet	Relaxes urethra (**NO**, nitric oxide)	Contracts bladder neck (**α1**)	Contracts EUS (external urethral sphincter; nicotinic)
Role	Voiding	Storage	Voluntary closure

- All three nerves carry afferents. **Aδ fibres** (tension receptors in muscle) signal normal fullness. **Unmyelinated C fibres** are "silent" normally; after inflammation, SCI (spinal cord injury) or obstruction they are recruited and drive overactivity (the basis of capsaicin and resiniferatoxin).
- Detrusor has more M2 receptors, but **M3 mediates contraction**.

Neurophysiology: reflex and central control

- **Storage (spinal reflexes):** low afferent activity triggers sympathetic outflow (detrusor inhibition, bladder-neck closure) and the **guarding reflex** (pudendal; sphincter activity rises with filling).
- **Voiding (spinobulbospinal reflex):** high afferent activity reaches the **PAG (periaqueductal grey)** and then the **PMC (pontine micturition centre, Barrington nucleus)**; the sphincter relaxes first, then the detrusor contracts. The PMC is an on–off switch.
- **Forebrain decides:** the insula senses fullness; the **medial prefrontal cortex** tonically holds the switch "off" until voiding is socially appropriate.
- Lesion patterns: above the pons, overactivity with a coordinated sphincter; between pons and sacral cord, overactivity with **DSD (detrusor–sphincter dyssynergia)**; sacral or

peripheral, underactive bladder.

- **Central transmitters:** glutamate excitatory; GABA (γ-aminobutyric acid) and glycine inhibitory; serotonin inhibits voiding; dopamine tonically inhibits the PMC (lost in Parkinson disease).

```
Filling → Aδ afferents → sacral cord → PAG low traffic: spinal storage reflexes (sympathetic + guarding) high traffic + cortex "permits": PMC → EUS relaxes → detrusor contracts
```

Diagram to draw: bladder with pelvic, hypogastric and pudendal nerves, plus brainstem loop (PAG–PMC).

Maturation in children

- Bladder visible on fetal ultrasound by 14 weeks. Infants void reflexly but **wake before voiding** (cortical arousal gates voiding).
- Infant voiding is often interrupted (transient dyscoordination), with detrusor pressure often **>100 cmH2O**; frequent small voids largely disappear by **2 years**.
- Capacity rises: **EBC (expected bladder capacity) = (age + 2) × 30 mL** (infants: weight in kg × 7 mL).
- Toilet training: bowel control, then daytime bladder, then night-time bladder. Persisting dyscoordination after toddlerhood is dysfunction, not immaturity.

Definitions: frequency and urgency

- **Frequency:** ≥8 voids a day (ICCS, International Children's Continence Society). **Urgency:** sudden, compelling need to void.
- At 3 years toilet training may still be ongoing, so decide first whether the pattern is abnormal.

Types of voiding dysfunction: non-neurogenic (functional) LUT dysfunction (ICCS)

Type	Key feature
OAB (overactive bladder)	Urgency ± frequency (≥8 voids a day), urgency incontinence; tower flow
Voiding postponement	Habitual holding; holding manoeuvres; infrequent voids (≤3 a day)
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction during voiding in a neurologically normal child; **staccato** flow
Underactive bladder	Needs abdominal straining to void; interrupted flow
Giggle incontinence	Complete emptying with laughter
Vaginal reflux	Small leak just after voiding in girls
Enuresis	Wetting in sleep, ≥5 years; monosymptomatic or non-monosymptomatic

- **BBD (bladder and bowel dysfunction):** LUT dysfunction plus constipation; rectal loading worsens symptoms and UTI (urinary tract infection).
- **Hinman (non-neurogenic neurogenic bladder):** the severe end of dysfunctional voiding, with trabeculated bladder, VUR (vesicoureteric reflux) and renal damage.

Types of voiding dysfunction: neurogenic LUT dysfunction

- Causes: **myelomeningocele** (commonest), occult dysraphism / **tethered cord**, sacral agenesis, anorectal malformation, cerebral palsy, SCI, transverse myelitis, pelvic tumour surgery (sacrococcygeal teratoma).
- Pattern = detrusor (overactive or underactive) × outlet (competent, incompetent or dyssynergic). Bony level does not predict bladder function.
- **Hostile bladder: DLPP (detrusor leak point pressure) ≥40 cmH2O** (McGuire 1981), poor compliance, or NDO (neurogenic detrusor overactivity) with DSD.

Types of voiding dysfunction: anatomical

- **PUV (posterior urethral valves)** "valve bladder": overactivity, then poor compliance, then myogenic failure; ectopic ureter.

Differential diagnosis of frequency and urgency (3-year-old boy)

Group	Causes and clues
Infection	**UTI**: fever, dysuria; balanoposthitis; viral haemorrhagic cystitis
Functional	**BBD**/OAB; voiding postponement; **extraordinary daytime urinary frequency** (sudden very frequent small voids, no nocturia or wetting, often after stress; self-limiting)
Bowel	**Constipation**: rectal mass presses on bladder; commonest associated cause
Obstruction	**PUV**, mild late form; **meatal stenosis** (circumcised boys; thin deflected stream); severe phimosis
Neurogenic	Occult spinal dysraphism, tethered cord (sacral dimple, hair tuft, gait or foot abnormality)
Bladder irritants	Stone, foreign body, hypercalciuria, pinworms, rarely **rhabdomyosarcoma**
Polyuria	Diabetes mellitus, diabetes insipidus, chronic kidney disease
Behavioural	Anxiety, family stress, developmental disorders

Evaluation

- **History:** onset, nocturia, wetting, stream, fever, **bowel habit (Bristol stool chart)**, thirst, stressors.
- **Examination:** abdomen, meatus and foreskin, **lumbosacral spine**, lower-limb neurology.
- **Urinalysis and culture**; glucose; specific gravity >1.005 makes diabetes insipidus unlikely.
- **Bladder and bowel diary.**
- **Ultrasound** of kidneys and bladder: wall thickness, residual, dilation, stones.
- **Uroflowmetry with PVR (post-void residual) and patch EMG (electromyography)** if the child cooperates: bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive/DSD), plateau (obstruction).
- **Selective:** VCUG (voiding cystourethrogram) for suspected PUV or febrile UTI with abnormal ultrasound; spinal MRI (magnetic resonance imaging) for neurological signs; urine calcium/creatinine ratio.
- Invasive (video)urodynamics only for neurogenic, anatomical or refractory cases.

Diagram to draw: flowchart: urinalysis, then ultrasound with residual, then targeted tests.

Management

- **Treat constipation first:** disimpaction, then maintenance **polyethylene glycol**, diet and toilet routine.
- **Standard urotherapy** for 4 weeks: timed voiding every 2–3 hours, relaxed posture, adequate fluids, avoid caffeine.
- **UTI:** antibiotics per culture.
- **OAB persisting after urotherapy: oxybutynin 0.3–0.6 mg/kg/day** in 2–3 doses (maximum 15 mg/day; watch for constipation); mirabegron; botulinum toxin; parasacral TENS (transcutaneous electrical nerve stimulation).
- **Dysfunctional voiding: biofeedback**; α-blocker; sphincter botulinum toxin if refractory.
- **Extraordinary daytime frequency:** reassurance; resolves in weeks to months.
- **Enuresis:** alarm or desmopressin.
- **Meatal stenosis:** meatotomy. **PUV:** endoscopic valve ablation. **Tethered cord:** neurosurgical release.
- **Neurogenic: CIC (clean intermittent catheterisation; Lapides 1971) + antimuscarinic**, then intradetrusor botulinum toxin, then augmentation.

Key points to remember:

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile.
- Always ask about bowels; treat constipation first.
- Frequency/urgency: rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

Also, from the condensed book (not in the paper answer):

- The International Children's Continence Society (ICCS) now uses expected bladder capacity = (age + 1) × 30 mL; the paper's (age + 2) is the older formula†.
- In **infancy**, voiding pressures are often **>100 cm H2O** and interrupted voiding is normal; both are gone by **~2 yr**. Dyscoordination later is **dysfunction, not immaturity**.
- Late toilet training is linked to incontinence, enuresis and urinary tract infection (UTI).

KEY POINT

Bladder and bowel are one unit, sharing pelvic space and afferents. A loaded rectum causes overactivity and poor emptying, so treating constipation improves lower urinary tract symptoms (LUTS).

2 · Definitions — ICCS

Bladder and bowel dysfunction (BBD) is lower urinary tract (LUT) dysfunction plus constipation and/or faecal incontinence in a neurologically normal child.

Group	Symptom	ICCS definition
Storage	Increased frequency	**≥8 voids/day**

Group	Symptom	ICCS definition
	Decreased frequency	**≤3 voids/day**
	Urgency	Sudden immediate need to void
	Nocturia	Waking at night to void
Voiding	Hesitancy	Difficulty initiating when ready
	Straining	Valsalva to initiate and maintain
	Intermittency	Non-continuous stream
	Retention	Cannot void despite effort; distended bladder
Other	Holding manoeuvres; incomplete emptying; recurrent UTI	Incomplete-emptying sensation counts even with minimal post-void residual (PVR); recurrent UTI is itself a BBD symptom

- **Daytime urinary incontinence (DUI)** is leakage while awake at age **≥5 yr**. **Enuresis** is intermittent leakage while asleep at ≥5 yr: **monosymptomatic (MNE)** has no other LUTS, **non-monosymptomatic (NMNE)** has some.
- **Faecal incontinence** is mostly **retentive** (overflow); ~1 in 5 is non-retentive.
- † **Expected bladder capacity (EBC)** = (age in years + 1) × 30 mL (ICCS, to ~12 yr).

Daytime LUT conditions

Condition	Definition / mechanism	Key features
Overactive bladder (OAB)	Urgency, with or without frequency, nocturia, incontinence; no UTI or other pathology	Urgency incontinence, holding manoeuvres
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction **during voiding**, neurologically intact; staccato flow	Learned; high PVR, UTI, secondary vesicoureteral reflux (VUR)
Underactive bladder	Raises abdominal pressure to initiate, maintain or complete voiding	Infrequent, interrupted flow, PVR
Voiding postponement	† Habitual deferral with holding until urgency incontinence	† Low frequency; oppositional defiant disorder
Giggle incontinence	Extensive emptying during or after laughing; LUT otherwise normal	Distinguish from laugh-provoked OAB leak and vaginal reflux
Vaginal reflux	Vagina fills retrograde during voiding	Prepubertal girl; small leak just after voiding
Extraordinary daytime frequency	† ≥ hourly small voids, no incontinence or nocturia	† Benign, self-limiting; exclude UTI, diabetes
Hinman syndrome (non-neurogenic neurogenic bladder)	End stage of dysfunctional voiding	Trabeculation, VUR, hydronephrosis, renal damage, hypertension

KEY POINT

Trap: dysfunctional voiding (neurologically intact) is not detrusor–sphincter dyssynergia (DSD), which needs a neurological lesion. Urgency incontinence is not always OAB: a voiding postponer also leaks with urge, and anticholinergics help briefly, then worsen emptying.

3 · Clinical presentations and associations

- **Acute retention** (peak 2–4 yr) follows genital inflammation, instrumentation or **faecal impaction**; try **analgesia, sitz bath and/or enema before a catheter**.
- **Functional constipation** affects ~14% and is driven by a painful-stool **withholding cycle**. Exclude organic causes such as anorectal malformation (ARM), Hirschsprung disease and spinal causes.
- **ADHD, autism and intellectual disability** have the highest BBD rates and respond worse.

4 · Evaluation

*Evaluation mainly **excludes organic disease**; avoid over-investigation.*

Warning signs

Red flags are rectal bleeding, weight loss or vomiting (organic disease); poor growth or hypertension (chronic kidney disease, CKD); a lumbosacral skin lesion or abnormal leg neurology (spinal MRI); an abnormal anus (ARM); low specific gravity (SG) or glycosuria (diabetes insipidus, DI, or diabetes mellitus); and constant dribbling in a girl (ectopic ureter).

- Ask about bowels in **every** child with LUTS, and use a **3-day bladder diary** and a **bowel diary**. Bristol types 1–2 indicate constipation.
- A **gush on separating the labia** means vaginal reflux. Routine rectal examination is **not** needed.

Investigations (stepwise, non-invasive)

- **ICCS response**: complete 100%, partial 50–99%, none <50%.
- **Urinalysis**: SG >1.005 makes DI unlikely. Persistent proteinuria needs a first-morning protein:creatinine ratio (>**0.2 mg/mg** is abnormal).
- Measure **PVR** within 5 min of voiding at 50–115% EBC.
- **Uroflow** (3 flows, >50% EBC): bell is normal, tower suggests OAB, staccato dysfunctional voiding, interrupted underactive bladder, and plateau obstruction.
- Pelvic floor **EMG** active during voiding means dysfunctional voiding; an **EMG lag time <2 s** means detrusor overactivity. **Video urodynamics** is only for treatment failure or suspected obstruction or upper-tract risk.

Bowel — Rome IV functional constipation (child ≥4 yr)

ALGORITHM

1. ≥2 of, ≥1/week for ≥1 month, irritable bowel syndrome (IBS) criteria not met:
2. ≤2 defecations in toilet/week; ≥1 faecal incontinence/week; retentive posturing/withholding; painful or hard stools; large rectal faecal mass; large-diameter stools that may block the toilet.

- Pain is **not** a Rome IV criterion.
- No routine labs or **plain radiograph**. A rectal diameter **≥30 mm** on US suggests constipation (European Association of Urology/European Society for Paediatric Urology, EAU/ESPU).
- If refractory, use a transit study, contrast enema and **anorectal manometry** (absent RAIR needs rectal biopsy for Hirschsprung).

5 · Management of daytime incontinence

ALGORITHM

1. History, exam, urinalysis, PVR, with or without uroflow.
2. Exclude neurogenic causes, diabetes, CKD and anatomical causes.
3. Optimise neurodevelopmental disorders; **treat constipation first**.
4. If wetting is only at night, treat as MNE.
5. Otherwise, **standard urotherapy** (≥4 wk); re-evaluate.
6. Then **specific urotherapy** by diagnosis.
7. If refractory: botulinum toxin or neuromodulation.

Standard urotherapy

- Give information, fluid and diet advice and diaries. Sit with hips and knees at ~90° and feet supported, and use **timed voiding** 6–7 times a day. **Reward adherence, not dryness**. Urotherapy cures DUI in more than half.

Dysfunctional voiding

- Diagnose by **staccato flow with active pelvic floor EMG**.

Renal risk	Management
High (hydronephrosis, VUR, rising creatinine)	Video urodynamics; lower storage pressure; clean intermittent catheterisation (CIC) past the sphincter
Low	Urotherapy, double voiding, treat constipation, short-course prophylaxis for recurrent UTI

- **Biofeedback is the first-line specific therapy**. **α-blockers** (doxazosin) lower PVR. If refractory, give **intrasphincteric onabotulinumtoxinA** 50–300 units.

Overactive bladder

Use drugs only after urotherapy and bowel management.

Agent	Dose / point
Oxybutynin (non-selective)	Immediate release **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); extended release (ER) 5–20 mg/day; patch 3.9 mg/day. Only US-approved drug for paediatric OAB; causes dry mouth and constipation.
Tolterodine	1–2 mg twice daily or ER 2–4 mg; ER randomised controlled trial (RCT) negative
Solifenacin (M3-selective)	2.5–10 mg daily; positive phase III RCTs (LION, LEOPARD)
Trospium (quaternary amine)	Does not cross the blood–brain barrier

Agent	Dose / point
Mirabegron (β3-agonist)	25–50 mg if >35 kg; approved ≥3 yr for **neurogenic** detrusor overactivity only; safe add-on

- Check PVR and stools before and during antimuscarinics.
- **Intradetrusor onabotulinumtoxinA** (refractory): **5 units/kg, max 100 units**; it lasts ~12 months.

Neuromodulation options are parasacral TENS over S2–S3 (beat oxybutynin, but a sham RCT was negative), the weaker tibial nerve stimulation, and off-label sacral nerve stimulation (SNS), which helps bladder and bowel.

Other conditions

- **Underactive bladder**: timed and double voiding and biofeedback; CIC if PVR is high.
- **Postponement**: timed voiding and treat comorbidity; † avoid anticholinergics.
- **Giggle**: **methylphenidate** or **biofeedback**.
- **Vaginal reflux**: void with legs abducted (sit backwards on the toilet). **Extraordinary frequency**: † reassure.

6 · Nocturnal enuresis

SAQ Nocturnal enuresis

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Nocturnal enuresis" · P3 SAQ "Enuresis"

Definition (ICCS [International Children's Continence Society])

- Intermittent urinary incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

Epidemiology

- About **5–10% at 7 years**; falls with age, with spontaneous resolution about 15% a year; about 2% of adults; familial.

Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (poor or reduced night-time ADH [antidiuretic hormone] rise).
2. **Small (reduced) nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep. - **Why sleep apnoea matters:** obstructed breathing makes arousal difficult and releases atrial natriuretic peptide, which raises urine output. - Comorbidities: constipation, sleep-disordered breathing, ADHD (attention deficit hyperactivity disorder).

Diagram to draw: triad: nocturnal polyuria, small bladder capacity, arousal failure.

Evaluation

- **History:** day symptoms, bowels, snoring, fluids, family history.
- **Examination:** spine, genitalia, abdomen for faecal loading; urinalysis.

- **Bladder diary:** daytime voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL in the chapter (ICCS uses [age + 1] × 30 mL); night urine output by weighing nappies.
- Uroflowmetry, PVR (post-void residual) and ultrasound only for NMNE or before adding anticholinergics. No invasive tests in MNE.

Management: general

- **Treat daytime LUTS and constipation first** (in NMNE).
- Education removes blame; standard urotherapy. Waking, fluid restriction alone and punishment do not work.

Management: first line

	Alarm	**Desmopressin**
Target	Arousal (conditioning)	Night urine output
Use	Nightly for months; parent wakes child; continue to 14 dry nights	0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed
Response	50–70%; durable	Rule of thirds; immediate but relapses off the drug
Caution	Needs motivated family	Restrict / limit evening fluids (hyponatraemia)

- Overlearning (extra evening fluid once dry) reduces alarm relapse.

Management: failure and resistant cases

- **Failure:** switch or combine.
- **Second line:** add an anticholinergic (oxybutynin) after checking PVR and bowels.
- **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists; treat sleep apnoea (adenotonsillectomy).

Adult enuresis

- New enuresis in an adult may signal high-pressure chronic retention: catheterise and treat the obstruction.

> **Key points to remember:**
> - ≥5 years; secondary = after 6 months dry.
> - Alarm (durable) and desmopressin (rule of thirds) are first line.
> - Treat daytime symptoms and constipation first.

Also, from the condensed book (not in the paper answer):

- Prevalence is **5–10% at 7 yr**; ~15% resolve each year, and ~2% persist into adulthood.
- The **triad** is nocturnal polyuria (relative antidiuretic hormone deficiency), reduced nocturnal capacity or overactivity, and failure to arouse.
- In **NMNE**, treat bothersome daytime LUTS and constipation first. In **MNE**, treat from **~6 yr**.

	Enuresis alarm	Desmopressin
Use	Nightly; the parent wakes the child; continue until **14 consecutive dry nights**	**0.2–0.4 mg** tablet or 120–240 µg melt ~1 h before bed; stop if no effect after ~1 week
Response	50–70%; slow	Immediate; rule of thirds; best with nocturnal polyuria
Durability	**Durable**; overlearning lowers relapse	**Not durable**; drug holidays or special nights
Safety	Nil	Hyponatraemia: † no fluids from 1 h before dose

ALGORITHM

1. MNE at ≥6 yr: start the alarm or desmopressin.
2. If it fails, check use, then switch or **combine**.
3. Re-evaluate: hidden daytime LUTS, constipation, sleep-disordered breathing, neurodevelopmental disorder; uroflow and PVR.
4. Then **add an anticholinergic** to desmopressin.
5. If refractory: imipramine, a β3-agonist or specialist drugs.

- **Imipramine** 25–50 mg is **cardiotoxic** (overdose, long QT), so do an electrocardiogram if there is syncope, palpitations or a family history of sudden death.

7 · Functional constipation — treatment

SAQ Management of Defecation Disorders resulting into LUTS

Paper answer · P4

Type: SAQ (10 marks)
Source: 2025-Winter Paper 4, Q3f

Assumed meaning: bladder and bowel dysfunction, where constipation or defecation disorders cause LUTS (lower urinary tract symptoms).

Why the bowel causes LUTS

- **Shared space:** a loaded rectum compresses the bladder, reducing capacity and obstructing outflow.
- **Shared nerves:** bladder and bowel afferents converge on the same spinal segments; cross-organ sensitisation causes urgency and detrusor overactivity.
- **Shared muscle:** a pelvic floor that contracts during defecation (**dyssynergic defecation**) also contracts during voiding (**dysfunctional voiding**) → high PVR (post-void residual), UTI (urinary tract infection), reflux.
- Result: urgency, daytime incontinence, enuresis, retention, recurrent UTI.

Evaluation

- Stool frequency, pain, **Bristol chart**; 3-day bladder and bowel diary.
- **Rome IV (child):** at least 2, weekly for 1 month: ≤2 defecations/week; faecal incontinence weekly; retentive posturing; painful or hard stools; large rectal mass; large-diameter stools.
- **Rome IV (adult), for 3 months:** ≥2 of straining, hard stools, incomplete evacuation, blockage, manual manoeuvres (each in >25% of defecations), <3 stools/week.
- Examine abdomen, spine and perineum; exclude organic causes (Hirschsprung disease, hypothyroidism, neurogenic).

- **Also know:** rectal diameter ≥30 mm on ultrasound; no routine X-ray; manometry if refractory.

Management

Rule: treat the bowel first (EAU, European Association of Urology/ESPU, European Society for Paediatric Urology).

1. **Disimpaction:** oral PEG (polyethylene glycol) 3350, 1–1.5 g/kg/day for 3–6 days; enemas if needed.
2. **Maintenance:** PEG 0.4 g/kg/day (range 0.2–0.8), at least 2 months; lactulose alternative; stimulants (senna) for expulsion.
3. **Urotherapy:** toileting after meals (gastrocolic reflex); footstool posture; normal fluid and fibre.
4. **Biofeedback, pelvic floor physiotherapy** for dyssynergia.
5. **Refractory: transanal irrigation** (water via rectal cone; a quarter to a third stop); **ACE (antegrade continence enema, Malone)**; botulinum toxin to the sphincter; sacral or tibial neuromodulation. - **Adults:** laxatives, prucalopride or linaclotide, biofeedback; rectocele repair; irrigation in neurogenic bowel. - **Caution:** anticholinergics worsen constipation; LUTS return.

Diagram to draw: pelvic cross-section showing loaded rectum compressing bladder.

> **Key points to remember:**
> - Treat the bowel first.
> - Rome IV: ≥2 criteria for 1 month (child).
> - PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

Also, from the condensed book (not in the paper answer):

- Daily enemas for 3–6 days are an alternative to PEG.

ALGORITHM

1. If impacted, **disimpact** with oral polyethylene glycol (PEG) or enemas.
2. Give education, a diary, toilet sitting and **maintenance PEG**.
3. Review at 2 wk; if failing, check adherence, dose, impaction, mental health.
4. Continue for **≥2 months**. Wean when ≥1 month symptom-free, with ≥3 stools/week and Rome IV not met.
5. Relapse: reinstate, or refer to gastroenterology.

Laxative	Dose
PEG 3350 — disimpaction	**1–1.5 g/kg/day × 3–6 days** (>3 yr)
PEG 3350 — maintenance	**0.2–0.8 g/kg/day**, start 0.4 (>6 months)
Lactulose — maintenance	1–2 g/kg once or twice daily (safe in infants)

- **Titrate** PEG: increase it if there is no stool in 24 h, and add a **stimulant** (senna, bisacodyl) if watery accidents occur. There is **no dependence**.
- Refractory cases may need biofeedback, transanal irrigation or SNS. Before an antegrade continence enema (**ACE**), do **colonic manometry**, because diffuse dysmotility may need

resection (see Ch 48).

8 · Recurrent UTI and BBD

- In RIVUR/CUTIE, recurrence was **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**.
- Secondary VUR often resolves once BBD is treated.
- **EAU/ESPU**: exclude BBD in every toilet-trained child with febrile or recurrent UTI, and give prophylaxis during BBD treatment. The **AUA** advises prophylaxis for BBD with VUR.

9 · Landmark trials

Trial	Design	Result	Take-home
RIVUR + CUTIE (BBD analysis)	Prophylaxis vs placebo, <6 yr after febrile UTI, with or without VUR	BBD + VUR highest recurrence; BBD alone > VUR alone	Treat BBD; prophylaxis for BBD + VUR
LION / LEOPARD	Phase III placebo RCTs, solifenacin	Improved all efficacy measures; mild–moderate side effects	Best paediatric antimuscarinic evidence

48 Lower Urinary Tract Reconstruction in Children

1 · Goals and principles

- The priorities, in order, are: **protect the kidneys, then prevent urinary tract infection (UTI), then achieve continence**.
- The indication is a hostile bladder (poor compliance, high pressure, poor emptying) despite medical care, mostly in spina bifida; also exstrophy, posterior urethral valves and bilateral ectopic ureters.
- **Clean intermittent catheterisation (CIC)** is the key enabler, and the main determinant of outcome is **patient and family commitment**.

> **KEY POINT**
>
> Every reconstructive step trades storage for emptying. A continent reservoir that is not catheterised causes hydronephrosis, sepsis and perforation, so incontinence can be "protective". CIC and motivation are prerequisites.

2 · Patient selection and preoperative evaluation

- **Storage**: a chronic filling pressure **>40 cm H2O** causes hydroureteronephrosis, a falling glomerular filtration rate (GFR) and acquired vesicoureteral reflux (VUR). Start anticholinergics and CIC; persistent poor compliance needs **augmentation**.
- **Outlet**: resistance must exceed bladder pressure during all activity; a deficient outlet needs a procedure.
- **Emptying**: nothing predicts voiding after augmentation, so every patient practises CIC beforehand.

Urodynamic finding	What it decides
Capacity and fluoroscopic bladder shape	Length of bowel segment
Detrusor leak point pressure (DLPP), end-filling pressure, open bladder neck	Need for a bladder neck procedure
VUR with a history of UTI	Reimplantation
Sphincter electromyography (EMG)	Dyssynergia

- Fill at **≤10% of predicted capacity per minute**, because fast filling falsely lowers compliance.
- Every child has ultrasound (US). Hydronephrosis needs a renogram **with a bladder catheter** and voiding cystography or videourodynamics.
- Check electrolytes, blood urea nitrogen (BUN) and creatinine. If raised, involve nephrology and measure 24-h urine **volume**: the reservoir must hold **3–4 h** of output.
- In **renal insufficiency**, acidosis **always worsens** with ileum or colon, so consider stomach. In **liver failure**, ammonia reabsorption is dangerous.
- Low-grade secondary **reflux** usually resolves after augmentation; correct high-grade reflux at reconstruction.
- In short gut, diarrhoea or faecal incontinence, spare the ileocaecal valve; add an antegrade continence enema (ACE) if needed.

- A sensate urethra, wheelchair use or prior bladder neck repair favour an abdominal stoma. Do all steps in **one operation** where possible.
- **Sterile urine** is mandatory; bowel preparation is optional. With a ventriculoperitoneal shunt, infection is rare but shunt failure increases.

3 · Managing the ureters

- Reimplant into the native bladder, with or without a **psoas hitch** (avoid the genitofemoral and femoral nerves). A short ureter needs transureteroureterostomy.
- Into bowel: **colon** under a taenia; **stomach** submucosally; **ileum** needs a **nipple valve** (hardest).

4 · Bladder augmentation (enterocystoplasty)

*See the paper answer **Intestinal segments in urology and orthotopic neobladder** (Section 12, Ch 133).*

Also, from the condensed book (not in the paper answer):

- Base deficit is the most sensitive marker; height can fall even without frank acidosis.
- Mucus: colon more than ileum more than stomach; stomach forms uric acid stones.

Principles

- **Bivalve the native bladder widely**; a narrow anastomosis makes a diverticulum.
- **Always detubularise** and **reconfigure into a sphere**. Intact bowel contracts at **60–100 cm H_2O**, and a sphere gives the most volume per surface area.
- Length: ileum **20–40 cm**, colon **~20 cm**; err on the large side.
- Use **no permanent sutures or staples**, because they seed stones.

Segments

- **Ileocystoplasty**: take ileum at least **15–20 cm proximal to ileocaecal valve** (to protect B12 absorption) and open it antimesenterically into a U or W.
- **Gastrocystoplasty** suits **renal insufficiency with acidosis, short gut and irradiated bowel**. **Jejunum** is avoided (salt-losing acidosis).

Segment	Advantages	Disadvantages
Ileum (default)	Least pressure activity; plentiful, mobile mesentery	Hyperchloraemic acidosis; B12 loss; mucus; hard to make antireflux
Ileocaecal	Taenial reimplant; appendix; imbricated valve	Diarrhoea in neurogenic bowel; more mucus
Sigmoid	Taenial reimplant; close to bladder	Strongest contractions; most mucus; more reaugmentation
Stomach	No acidosis; least mucus; fewer struvite stones; usable in renal failure	Hypokalaemic hypochloraemic alkalosis; haematuria–dysuria syndrome; tumours
Ureter	No mucus, no metabolic effect	Needs a massively dilated ureter of a non-functioning kidney; limited volume
Autoaugmentation (detrusorectomy)	Extraperitoneal; no bowel	Seldom adds capacity; compliance may not last

- Seromuscular and tissue-engineered augments are experimental. Technique matters more than segment, so use ileum unless there is a reason not to.

> **KEY POINT**
>
> Augmentation is needed less now, thanks to early anticholinergics, CIC, botulinum toxin and neuromodulation (immediate vs expectant treatment: 17% vs 41% augmented). This is not trial-proven, but it is the rationale for treating at diagnosis.

Incontinent alternatives

- **Vesicostomy** is an infant bridge for worsening hydronephrosis, febrile UTI with high-grade VUR, or high pressure. **Ileovesicostomy** suits those who cannot catheterise.

5 · Continent catheterisable channels

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

Also, from the condensed book (not in the paper answer):

- In a **flap valve**, a narrow tube tunnelled into a **low-pressure** reservoir is **compressed** as pressure rises; the appendix needs only **~2 cm** of tunnel. Nipple valves (Kock, Mainz) are **effaced** by pressure (about 85% vs more than 90–95% continent). A flap valve has **no pop-off**.

ALGORITHM

1. Detach the appendix with a caecal cuff and preserve the mesoappendix.
2. Tunnel the **distal end** submucosally.
3. Bring the caecal end to skin without tension or twist, with a **skin flap** in the stoma.
4. Keep the channel **short and straight**; fix the bladder to the abdominal wall.
5. Test catheterisation after each step.
6. Stent with 6–12 French (Fr) for 10–14 days; the surgeon does the first catheterisation.
7. Then CIC at least 4-hourly by day.

- Site the stoma at the umbilicus or right lower quadrant. The appendix can be **split** for both an ACE and a channel.
- **Yang–Monti**: a 1–2 cm ileal ring is retubularised **transversely** over 12 Fr; it needs about twice the revisions of the appendix. The **Casale** bladder flap spares bowel but has high stenosis.
- **Stomal stenosis** is the commonest complication and occurs early. Stomal incontinence (rare) means a short tunnel or high reservoir pressure.
- **Ureterosigmoidostomy and Mainz II** are largely abandoned because of acidosis, pyelonephritis and cancer.

6 · Bladder outlet procedures

Outlet procedures sacrifice voiding and pop-off, so neurogenic repairs need **concurrent augmentation** and surveillance for **≥5 yr**.

Procedure	Mechanism	Voiding	Main problem / place
Fascial sling	Coaptation and elevation	Rarely adequate, so CIC	Better in girls; only predictor of success is augmentation; AUS still possible later
Artificial urinary sphincter (AUS)	Inflatable bladder-neck cuff	**Preserved**	Infection, erosion, malfunction; best if voiding
Bulking injection	Submucosal	Preserved	Poor durability
Young–Dees–Leadbetter	Tubularised trigone	Possible	Exstrophy (see Ch 43); poor in neurogenic
Kropp	Anterior bladder tube under trigone (flap valve)	Abolished	Highest continence; hard urethral CIC (boys); new VUR
Pippi Salle	Anterior flap onlaid on trigonal strip	Abolished	Easier CIC; lower night continence; fistula
Bladder neck closure	Complete separation and continent stoma	Abolished	After failed repairs; no pop-off

- The **sling** suits a marginal bladder neck in an augmented child on CIC.
- The **AUS** cuff goes at the bladder neck in girls and prepubertal boys and is left **deactivated ~6 weeks**; most neurogenic children still need CIC. New hydronephrosis means unmasked bladder hostility.
- **Kropp** and **Salle** repairs often need later augmentation. Bladder neck **closure** with **interposed omentum** keeps fistula low, and suits reliable catheterisers only.

> **KEY POINT**
>
> Choosing the outlet: a child who voids gets an AUS (the only option preserving voiding). A child on CIC who is being augmented gets a sling (girls) or flap-valve lengthening with a channel. After failed repairs, do bladder neck closure with a Mitrofanoff. Always re-study the bladder afterwards.

7 · Malone antegrade continence enema (MACE)

- † An appendicocaecostomy gives scheduled antegrade washouts and faecal continence in spina bifida, anorectal malformation and refractory constipation. Plan it with the urinary reconstruction.

8 · Postoperative care

- Irrigate the suprapubic tube **≥3 times daily**. Do a **cystogram at ~3 weeks**, then start CIC 2–3-hourly by day, **never beyond 4–5 h**.

9 · Complications of augmentation

*Up to ⅓ need further surgery: **bowel is not bladder**.*

Segment	Derangement	Management
Ileum, colon	**Hyperchloraemic metabolic acidosis** (overt mainly with renal impairment)	Bicarbonate; **base deficit** is most sensitive

Segment	Derangement	Management
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis	Avoid
Stomach	**Hypokalaemic hypochloraemic alkalosis** with vomiting; **haematuria–dysuria**	Fluids, H2 blockers or proton pump inhibitor
Stomach and bowel	Near neutral	Complex patients

- Acid load demineralises bone and lowers height, so treat base deficit early. Check B12 long term.
- **Bacteriuria** is expected with CIC; treat only symptomatic infection and **urea-splitters**.
- **Stones** are mostly **struvite**; prevent them with **daily irrigation ≥240 mL** saline, which also clears mucus.

Delayed spontaneous perforation

- Counsel a **~9%** risk, mostly in myelodysplasia. **Outlet procedures double the risk**, as do missed CIC and substance abuse.
- Signs are pain, distension, fever, sepsis and shoulder-tip pain; poor sensation means **late presentation** and deaths. Do a **cystogram** with a post-drain film, then a CT cystogram if negative. Treat by **immediate laparotomy and repair**; manage conservatively only if stable, sterile and contained.

Malignancy

*See the paper answer **Malignancy in augmented bladder and after urinary diversion** (Section 12, Ch 133).*

Also, from the condensed book (not in the paper answer):

- After ureterosigmoidostomy, adenocarcinoma arises at the anastomosis after **~26 yr**, from N-nitroso compounds.
- After augmentation it is reported from **4 yr**, is highest after **gastrocystoplasty**, and is aggressive. Bladder cancer in spina bifida is ~2–4%. There is no consensus on annual cystoscopy; consider it after gastrocystoplasty, in exstrophy and after transplant.

Pregnancy

- Vaginal delivery is possible; caesarean after major bladder neck repair or AUS, with a urologist present.

10 · Follow-up and transition

When	Tests
3, 6, 12 months	US; electrolytes, BUN, creatinine, urine culture (with or without urodynamics by 6 months)
Yearly, lifelong	US, chemistry (with or without blood gas, B12), CIC and irrigation compliance; urodynamics if new hydronephrosis, incontinence or VUR
Symptomatic	† Haematuria needs cystoscopy; pain or sepsis needs perforation excluded

- Late chronic kidney disease after augmentation is driven by **CIC non-compliance**, so adult urology must police CIC and the upper tracts.

11 · Landmark trials

Trial	Design	Result	Take-home
Early vs expectant neurogenic bladder care	Cohort, treated at diagnosis vs expectantly	Augmentation 17% vs 41%	Treat hostile bladder early
Daily bladder irrigation	Augmented cohort, ≥240 mL saline daily	Marked fall in stones	Routine daily irrigation
Tissue-engineered bladder	Industry multicentre trial	No gain in compliance or capacity	Bowel remains standard

★ Putting it together: common paediatric pathways

A · Antenatal urinary tract dilation (UTD)

ALGORITHM

1. **Prenatal:** A1 before 28 weeks: ultrasound (US) after 32 weeks. A2–3: US 4-weekly. Male, thick bladder, bilateral: suspect posterior urethral valves (PUV).
2. **Postnatal:** first US **after 48 hours**, except suspected PUV or bilateral severe dilation.
3. Normal (<10 mm): discharge; mild: repeat US; moderate–severe or dilated ureter: voiding cystourethrogram (VCUG), MAG3, consider prophylaxis (→ Ch 34).

B · First febrile urinary tract infection (UTI)

ALGORITHM

1. **Diagnose:** catheter or suprapubic sample if not toilet-trained; bag only rules out.
2. **Treat** within 48 h (>72 h doubles new scarring); oral if well.
3. **Image:** US all; VCUG if US abnormal, atypical or recurrent; DMSA at 4–6 months if needed.
4. **Prevent:** treat bladder and bowel dysfunction and constipation; prophylaxis if high risk (→ Ch 37, 41).

C · Newborn with PUV

ALGORITHM

1. Stabilise breathing → 5–7 Fr feeding tube → creatinine, electrolytes, fluids.
2. US, then VCUG; prophylaxis.
3. When stable, **endoscopic ablation**; if the urethra is small, vesicostomy.
4. Prognosis: first-year nadir creatinine (<0.8 mg/dL good); lifelong renal follow-up (→ Ch 45).

D · Neurogenic bladder in spina bifida

Step	What	Why
Birth	US, post-void residual, creatinine; latex-free care	Baseline
≈3 months	Urodynamics	Hostile: leak point or end-filling ≥40 cmH_2O, poor compliance, dyssynergia
Hostile bladder	Early clean intermittent catheterisation (CIC) and antimuscarinic	Kidney protection
Failing	Botulinum toxin, vesicostomy (infant), then augmentation	Safe storage (see Ch 46, 48)
Always	Bowel programme; watch for re-tethering	Safety

E · Daytime wetting and bedwetting

ALGORITHM

1. Urinalysis, residual with or without uroflow; exclude organic causes.
2. **Constipation first**, then urotherapy ≥4 weeks.
3. Antimuscarinic for overactive bladder, biofeedback for dysfunctional voiding; toxin or neuromodulation if refractory.
4. **Monosymptomatic enuresis:** alarm or desmopressin; combine if partial (→ Ch 47).

F · Augmentation segment

Segment	Advantages	Disadvantages
Ileum	Default; compliant	Mucus, hyperchloraemic acidosis, B12 loss
Ileocaecum / colon	Appendix for channel	Diarrhoea in neurogenic bowel; acidosis
Stomach	Suits renal failure; little mucus	Haematuria–dysuria, alkalosis
Ureter	No mucus or metabolic effect	Needs a massive ureter

KEY POINT

Reconstruction trades storage for emptying: an unreliably catheterised reservoir causes sepsis and perforation, so CIC commitment comes first.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ CSIC.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2018-Summer Paper 3, Q3e

*Taken to mean **CISC – clean intermittent self-catheterisation**.*

Definition and principle

- Regular, **clean (not sterile)** passage of a catheter by the patient (or carer) to empty the bladder, then removal.
- **Lapides (1971–72)**: infection comes from **stasis and overdistension** (poor bladder-wall blood flow), not from bacteria alone. Regular complete emptying protects the bladder wall and upper tracts, even though bacteriuria develops.
- Gold standard for chronic emptying failure.

Indications

- **Neurogenic bladder:** spina bifida, spinal cord injury, multiple sclerosis, cauda equina, diabetic cystopathy.
- **Underactive or acontractile detrusor**; chronic retention; post-operative retention.
- **Detrusor–sphincter dyssynergia**: CISC bypasses the obstructed sphincter.
- After **augmentation or continent diversion**; **valve bladder**; high residual after botulinum toxin.
- **Contraindications:** poor hand function or cognition without a carer, urethral false passage or stricture, unwillingness.

Technique

- Wash hands → clean meatus with water → lubricated catheter (or pre-lubricated **hydrophilic** catheter) → pass until urine flows → drain fully → withdraw slowly.
- Sizes: adults about 12–14 Fr; children by age (8–10 Fr).
- **Frequency:** 4–6 times a day, keeping each volume **below about 400–500 mL** (child: below expected bladder capacity).
- Combine with **antimuscarinics** for overactivity; bladder diary.

Catheters

Type	Note
Plain PVC (polyvinyl chloride), reusable after washing	Cheap; common in India
Hydrophilic-coated single-use	Less urethral trauma; longer time to first UTI (urinary tract infection)
Compact or closed-system	Convenience, outdoor use

Children

- Start **early (infancy)** in spina bifida with hostile bladder; families adapt best; child involved from 3–5 years.

- **Mitrofanoff appendicovesicostomy** (1980) if urethral catheterisation is difficult (wheelchair, sensate urethra, girls).

Complications

- **Asymptomatic bacteriuria** (common): **do not treat or give prophylaxis**; treat only symptomatic UTI.
- Urethral trauma, bleeding, **false passage**, stricture, epididymitis, bladder stones (hair, mucus), rarely lost catheter.

Outcome

- Less pyelonephritis, epididymo-orchitis and urosepsis than indwelling catheters; preserves the upper tracts; with drugs, continence in many.

Diagram to draw: neurogenic bladder management ladder: CISC + antimuscarinic → botulinum toxin → augmentation + channel.

> **Key points to remember:**
> - Lapides: overdistension, not bacteria, causes infection; clean is enough.
> - 4–6 times daily, under about 500 mL each.
> - Do not treat asymptomatic bacteriuria.
> - Mitrofanoff when urethral access is hard.

Rapid Revision Sheet · Section 5

Bladder embryology and exstrophy-epispadias complex (LAQ)

- Bud from Wolffian duct; position decides VUR or ectopia.
- Urorectal septum weeks 5–6 (P2: weeks 4–6); mesonephric duct entry is the landmark.
- Vesicourethral canal forms the bladder; urachus from allantois.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
- Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
- MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
- Osteotomy if closure after 72 h or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL predicts continence; template size and first closure success predict continence.

Posterior urethral valves: embryology, management and valve bladder (SAQ)

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: polyuria, poor sensation, residual urine.
- Valve bladder: overnight drainage and CIC; overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies (LAQ)

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.

- ARPKD large bright kidneys; MCDK non-communicating cysts.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

Ultrasonography in neonates (SAQ)

- First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

Discuss diagnostics and operative pediatric laparoscopy in urology (SAQ)

- Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
- Pressure ceiling 6–12 mmHg by age.
- Right common iliac at Veress entry; close port fascia. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Leak point pressures and ALPP (SAQ)

- VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

Embryological development of urinary bladder and management of Hutch Diverticulum? (LAQ)

- Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
- Hutch: herniation through the ureteric hiatus → VUR.
- Ureter entering the sac: VUR will not resolve.
- Secondary: treat the outlet first; else diverticulectomy + reimplantation.

Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias) (LAQ)

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
- Diastasis over 3 cm (>3 cm) → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

Prune belly syndrome (LAQ)

- Triad; 95% male; dysplasia and lungs decide survival.
- Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
- Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
- Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

Vesicostomy. (SAQ)

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.

- Stenosis and prolapse are the main complications.

Anterior urethral valve (SAQ)

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty (SAQ)

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm H_2O or more = hostile (McGuire 1981); early CIC + antimuscarinic.
- Early CIC plus antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency (LAQ)

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.
- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

Nocturnal enuresis (SAQ)

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Management of Defecation Disorders resulting into LUTS (SAQ)

- Treat the bowel first.
- Rome IV: ≥2 criteria for 1 month (child).
- PEG disimpaction 1–1.5 g/kg/day for 3–6 days. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

CSIC. (SAQ)

- Lapides: overdistension, not bacteria, causes infection; clean is enough.
- 4–6 times daily, under about 500 mL each.
- Do not treat asymptomatic bacteriuria.
- Mitrofanoff when urethral access is hard.

SECTION 6

Female Urology

101 Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

1 · Perineum and external genitalia

- The meatus lies **1 cm** anterior to the vaginal opening.
- **Skene's (paraurethral) glands** sit beside the meatus and are the prostate homologue; obstructed periurethral glands form a diverticulum or cyst.
- **Bartholin's gland** lies under the posterior end of each vestibular bulb and may form a cyst or abscess.
- The **deep perineal space** holds the urethral sphincter, urethrovaginal sphincter, compressor urethrae and deep transverse perineal; the **superficial space** (under Colles fascia) holds the superficial perineal muscles, clitoris, vestibular bulbs and Bartholin's glands.
- The perineal muscles and external anal sphincter converge at the **perineal body**, so episiotomy is angled laterally to spare the sphincter.
- The **labia majora have a dual supply** (superficial external pudendal from above, posterior labial from below), so the **Martius flap** is reliable.

2 · Bony pelvis and ligaments

- The **ischial spine** marks apical sutures.

Fixation point	Used for	Adjacent danger
Sacrospinous ligament (SSL): ischial spine to lower sacrum/coccyx	SSL fixation (apical prolapse)	Pudendal nerve/vessels near the spine; sacral plexus just posterior
Cooper's ligament (medial superior pubic ramus)	Burch colposuspension for stress urinary incontinence (SUI)	Obturator vessels; aberrant obturator
Obturator foramen; canal superolateral	Transobturator tape (TOT) trocar	Obturator nerve/vessels, causing groin pain or bleeding
Anterior longitudinal ligament (ALL) over S1	Sacrocolpopexy mesh	Presacral veins, middle sacral vessels, left common iliac vein, right ureter
Uterosacral ligament	Uterosacral (apical) suspension	Ureter (lateral, closest at the cervix); sacral nerve entrapment

- The pudendal nerve leaves through the **greater** sciatic foramen and re-enters through the **lesser** to reach Alcock's canal.

3 · Pelvic floor muscles

- The pelvic diaphragm is the **levator ani** and coccygeus; the levator is supplied directly by **S3–S5 (mainly S4)**.

LAQ Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair

Paper answer · P1 · P2 · merged from 3

Asked as: P1 LAQ "Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia" · P1 SAQ "Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options." · P2 SAQ "Vaginal prolapse repair."

Introduction and definitions

- Assumed meanings: "procedentia" is read as procidentia (complete uterine prolapse); "fertility sparing" / "fertility preserving" means uterus-preserving treatment for a woman who wants future pregnancy; "pelic floor prolapse" and "vaginal prolapse repair" mean pelvic organ prolapse (POP) in women and the surgical repair of vaginal wall and vault prolapse.
- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing.
- **POP (pelvic organ prolapse):** descent of the anterior wall, posterior wall, uterus/cervix or vault. Anterior is commoner than posterior, which is commoner than apical. It is a disease only if symptomatic: a quality-of-life condition.
- Pelvic organs are supported by the **levator ani**, the **endopelvic fascia and its ligaments** and the **perineal membrane and perineal body**, anchored to the **bony pelvis**. Failure causes POP and SUI. Every SUI operation uses one of these anchors.

Bony pelvis and surgical anchors

- **Bony landmarks** and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).
- **Sacrospinous ligament (SSL)**, 7–8 cm, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine, the sacral plexus just posterior.
- **Sacral promontory:** left common iliac vein and right ureter each about 2.7 cm away.

Pelvic floor muscles: pelvic diaphragm (levator ani + coccygeus)

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops (as a sling) behind the anorectal junction	Anorectal angle; anorectal sling
Pubococcygeus (pubovaginal part)	Posterior pubis and arcus tendineus to viscera (vagina, perineal body, anus) and anococcygeal raphe	Slings urethra and vagina; closes hiatus, lifts urethra
Iliococcygeus	ATLA (arcus tendineus levator ani), ischial spine to raphe, coccyx	Horizontal **levator plate**
Coccygeus (not levator; separate)	Ischial spine to sacrum, coccyx	Lies on sacrospinous ligament

- Levator tone (tonic contraction) closes the **urogenital hiatus**, through which all POP passes; the uterus and upper vagina rest on the levator plate.

- Levator is palpable through the lateral vaginal walls for PFMT (pelvic floor muscle training) assessment.
- **Muscle fails first, fascia second:** once injury (birth injury, avulsion) opens the hiatus, ligaments carry the load and fail; the fascia then carries the whole load and stretches.
- Levator defects in up to **20% of primiparous women**; avulsion predicts advanced prolapse.

Endopelvic fascia and ligaments

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix): the **cardinal–uterosacral complex** suspends the apex.
- The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix.
- **Pubocervical (pubovesical) fascia** attaches laterally to the **ATFP (arcus tendineus fasciae pelvis)**, the "white line" from the lower pubis to just above the ischial spine.
- **Rectovaginal fascia** supports the posterior wall.
- **Pubourethral ligaments** suspend the urethra from the pubis and anchor the mid-urethra; **urethropelvic ligaments** anchor it to the ATFP.

Perineal membrane, perineal body and urethra

- The perineal membrane spans the pubic arch and is pierced by the urethra and vagina; above it lies the striated urogenital sphincter. The perineal body anchors the distal vagina.
- **Striated urogenital sphincter:** sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**.
- Urethra 4–5 cm; the continence zone is the **proximal and mid-urethra**. The **oestrogen-sensitive vascular submucosa** gives up to about 30% of closure force.

DeLancey's levels of support (1992)

Level	Region	Structures	Loss produces
I – suspension	Cervix/upper third	Cardinal–uterosacral complex	**Uterine or vault prolapse**, enterocele
II – attachment	Middle third	Paravaginal attachment to ATFP, levator fascia	Cystocele (central or paravaginal), rectocele; anterior or posterior wall prolapse
III – fusion	Distal third	Urethra, perineal membrane, perineal body	**Urethral hypermobility**, perineal descent, distal rectocele

- **Central cystocele:** midline weakness, rugae lost. **Paravaginal cystocele:** ATFP detachment (Richardson 1981), rugae preserved.
- Level I is the keystone: always correct the apex.

Diagram to draw: sagittal female pelvis with levator plate and DeLancey levels I–III, with sacrospinous and sacrocolpopexy fixation points; hammock cross-section.

Innervation

- **Pudendal nerve (S2–S4):** sphincters and perineal muscles; runs through Alcock's canal near the ischial spine; stretched in delivery.
- **Nerve to levator ani (S3–S5):** runs on the muscle's pelvic surface, separate from the pudendal; neuropathy in about a quarter of primiparous women.

- Autonomic pelvic splanchnic and hypogastric nerves: injured in radical pelvic surgery.

Theories of continence and the operations they produced

Theory	Mechanism	Surgical answer
Pressure transmission (Enhorning 1961)	Hypermobile urethra drops out of the abdominal pressure zone	Restore position: Burch, MMK (Marshall–Marchetti–Krantz)
Hammock (DeLancey 1994)	Urethra lies on anterior vaginal wall and endopelvic fascia attached to ATFP and levator; abdominal pressure compresses it against this firm backing; repositioning not essential	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament laxity at the **mid-urethra**	Tension-free mid-urethral sling
Intrinsic sphincter deficiency (ISD) (McGuire 1978)	Urethra fails to close	Sling, bulking, AUS (artificial urinary sphincter)

- ISD: VLPP (Valsalva leak point pressure) <60 cmH_2O or maximal urethral closure pressure <20 cmH_2O. Most SUI includes some ISD. So slings work in hypermobility, but Burch does poorly in severe ISD with a fixed urethra.

Why prolapse occurs: risk factors (Bump & Norton)

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens and the ligaments carry the whole load and stretch.
- **Predisposing:** genetics, collagen disorders (Ehlers–Danlos, Marfan).
- **Inciting: vaginal delivery** (strongest; forceps worse), parity, hysterectomy.
- **Promoting:** obesity, constipation, cough.
- **Decompensating:** ageing.

Clinical presentation of prolapse

- **Vaginal bulge** (most specific), pressure, heaviness, worse by evening.
- **Splinting/digitation** to void or defecate (high-grade POP).
- Urinary: voiding difficulty (urethral kinking), OAB (overactive bladder); SUI in about 40%, which may be **occult**.
- Bowel (rectocele): obstructed defecation, stool trapping.
- Sexual dysfunction, discharge, ulceration; procidentia may cause hydronephrosis.

Examination

- Lithotomy (supine) and **standing** with Valsalva / strain; Sims or split speculum for each compartment; lateral sulci (paravaginal defect); rectovaginal examination.
- **Cough stress test**, repeated with prolapse reduced (**occult SUI**); **Q-tip** angle change >30° = hypermobility; levator strength and avulsion.

POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996)

- Nine points in cm relative to the **hymen (0)**: inside (above) negative, outside positive; measured on maximal strain, recorded on a 3 × 3 grid.

Point	Definition
Aa / Ap	Anterior / posterior wall 3 cm inside hymen (−3 to +3)
Ba / Bp	Most distal point of anterior / posterior wall (−3 if no prolapse)
C	Cervix or vault cuff
D	Posterior fornix (omitted after hysterectomy)
gh, pb, tvl	Genital hiatus, perineal body, total vaginal length (at rest)

Stage	Most distal point (leading edge)
0	No prolapse
I	>1 cm above hymen (< −1 cm)
II	Within ±1 cm of hymen (−1 to +1 cm)
III	>1 cm below hymen (> +1 cm) but < (tvl − 2) cm
IV	Complete eversion (≥ tvl − 2)

- **Also know:** Baden–Walker grades 0–4; stage I–II often normal (POSST study).

Investigations

- Urinalysis, PVR (post-void residual), bladder diary.
- **Cough test or urodynamics with prolapse reduced** for occult SUI; urodynamics also if ISD or mixed symptoms.
- Renal ultrasound in procidentia (hydronephrosis).
- Dynamic MRI or translabial ultrasound if complex or recurrent.
- A pessary trial confirms that the POP causes the symptoms.

Principles of prolapse repair

- Operate for bother, not anatomy. The first operation gives the best chance.
- Identify each compartment by POP-Q. **The apex is the cornerstone:** failing to support it greatly raises recurrence; uncorrected level I loss causes recurrent prolapse after SUI repair.
- **Three categories: restorative** (native tissue), **compensatory** (graft or mesh, for weak tissue or failed repair), **obliterative**.
- Vaginal route: fewer complications, quicker recovery. Abdominal route: more durable.
- Screen for **occult SUI** with prolapse reduced and consider a concomitant sling (OPUS, CARE trials). Finish with **cystoscopy** to confirm ureteric efflux.

Surgical options by compartment

Compartment	Operations	Key points
Anterior (cystocele)	**Anterior colporrhaphy** (midline plication of pubocervical fascia); **paravaginal repair** (fascia reattached to the ATFP)	Central versus lateral defect; does not support the apex
Apical, vaginal (uterus, vault)	**Uterosacral suspension** (sutures; ureter injury 1–11%; cystoscopy); **sacrospinous fixation** (suture 1.5–3 cm medial to the ischial spine, avoiding the pudendal bundle; gluteal pain 15%); iliococcygeus suspension; McCall culdoplasty; vaginal hysterectomy; uterus-sparing hysteropexy; Manchester	OPTIMAL trial: uterosacral = sacrospinous at 2 years

Compartment	Operations	Key points
Apical, abdominal (compensatory)	**Sacrocolpopexy** (open, laparoscopic, robotic): macroporous polypropylene mesh from vaginal apex to the anterior longitudinal ligament over S1 / sacral promontory	Most durable (>90%); mesh exposure about 3–10% (P2: about 10% long term, CARE)
Posterior (rectocele)	**Midline fascial plication** without levatorplasty; site-specific repair; perineorrhaphy (levels II–III)	Levator plication causes dyspareunia; constipation may persist
Obliterative	**Le Fort** (partial, uterus in situ, lateral channels) or total colpocleisis	Elderly, not sexually active; success >90%; coital regret ≤13%

- Cochrane 2016: vaginal apical repair has more recurrence and dyspareunia than sacrocolpopexy, but suits many older women.
- **April 2019:** the US FDA (Food and Drug Administration) ordered transvaginal synthetic mesh for (anterior) POP withdrawn; slings unaffected.
- Anterior colporrhaphy is not an SUI operation.

Importance of anatomy in SUI surgery

Operation	Anatomy used	Anatomical hazards
Burch colposuspension	Space of Retzius (avascular: symphysis in front, bladder behind, endopelvic fascia below). Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's (iliopectineal) ligament**	Bleeding from Cooper's ligament and perivesical veins; ureteric kinking (cystoscopy); needs a mobile vagina
Paravaginal repair / VOS (vagino-obturator shelf)	Lateral vaginal sulcus to **ATFP** or obturator fascia (level II)	Obturator vessels and nerve
Retropubic MUS (mid-urethral sling), TVT (tension-free vaginal tape; Ulmsten 1995)	Tape under mid-urethra (integral theory), through **endopelvic fascia** and Retzius; suprapubic exits about 2 cm from midline	Bladder perforation (4.5%), vessels, bowel
TOT (transobturator tape; Delorme 2001)	Through obturator foramen / membrane, **obturator internus and externus**; exit at adductor longus tendon notch	Groin pain (6.4%), obturator neurovascular bundle; bladder injury rare (0.6%)
SIMS (single-incision mini-sling)	Anchors in obturator internus or endopelvic fascia	Works by restricting mobility
Autologous PVS (pubovaginal sling)	Rectus fascia at the **bladder neck** (proximal urethra), arms to rectus	Voiding dysfunction; best for ISD (compression)
Bulking agents	**Submucosa** coaptation	Repeated injections

- **Why mid-urethra:** it is the zone of maximal closure pressure and pubourethral support. A loose tape lets the posterior urethral wall be compressed and kinked during stress.
- **TOMUS (2010):** retropubic 80.8% versus transobturator 77.7% objective success, not different.
- **SISTEr:** autologous sling more successful than Burch at 2 years, with more voiding dysfunction.

- **Also know:** needle suspensions abandoned (sutures pulled through).

Diagram to draw: cross-section of urethra on the vaginal hammock (DeLancey) with Burch, bladder-neck sling and mid-urethral sling sites.

Procidentia: definition and assessment

- **Procidentia:** complete uterine (or vault) prolapse with the uterus wholly outside the hymen, **POP-Q stage IV**.
- Assess all compartments, occult SUI with prolapse reduced, PVR, hydronephrosis (ureteric kinking at the hiatus), cervical length and ulceration.
- Abnormal bleeding or cervical pathology needs gynaecological evaluation first, because it may argue against keeping the uterus.

Fertility-sparing treatment of procidentia

Principle: restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

1. Conservative (first line)

- PFMT, weight loss, treat constipation.
- **Pessary:** space-filling type (Gellhorn, cube) for procidentia; removed for conception; review about 6-monthly for discharge and erosion.

2. Uterus-preserving surgery

Operation	Technique	Fertility points
Sacrospinous hysteropexy (vaginal)	Cervix sutured into the SSL **1.5–3 cm medial to the ischial spine**	Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); later data suggest superiority. Buttock pain, pudendal injury
Uterosacral hysteropexy (vaginal or laparoscopic)	Shorten and reattach uterosacral ligaments to cervix, sutures lateral to medial	No mesh; cystoscopy for ureteric kinking
Sacrohysteropexy (open, laparoscopic, robotic)	Mesh from posterior cervix to anterior longitudinal ligament over the promontory	Most durable apical support; later delivery usually by caesarean
Manchester (Fothergill)	Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy	Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted

- **Also know:** Indian abdominal (cervicopexy) slings for young nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

3. Choosing `Procidentia, wants pregnancy -> PFMT + pessary -> fails -> sacrospinous hysteropexy or laparoscopic sacrohysteropexy (avoid Manchester) -> recurrence after childbearing -> hysterectomy + vault suspension or sacrocolpopexy`

Complications and follow-up

- Bladder, ureter or rectal injury; retention; de novo SUI; dyspareunia.

- Recurrence (mostly posterior overall; anterior after sacrospinous fixation).
- Mesh exposure: 3.4–5.4% after sacrocolpopexy in meta-analyses (P1 SAQ: about 3–10%; P2: about 10% long term, CARE); ureteric kinking.
- The retained uterus needs cervical screening.

Key points to remember:

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone: always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch to Cooper's ligament; TOT through obturator foramen; PVS at bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

Also, from the condensed book (not in the paper answer):

- The **levator hiatus** transmits the urethra, vagina and rectum, and the **levator plate** is a horizontal visceral shelf; injury leaves the hiatus lax, the plate tilts and **prolapse** follows.

4 · Endopelvic fascia and visceral ligaments

- The **intermediate** fascial stratum (parametrium, paracolpium, pubocervical fascia) **gives most organ support**.
- The **pubourethral ligaments** (puboprostatic homologue) and anterior vaginal wall form a mid-urethral hammock; weakness causes SUI.
- A vesicopelvic fascia defect at the arcus tendineus gives a paravaginal cystocele.
- The **cardinal (Mackenrodt's) ligament** carries the pelvic plexus, so dividing it at hysterectomy can cause bladder dysfunction.
- Distinguish the **infundibulopelvic ligament** (ovarian vessels, ureter beneath) from the **ovarian ligament** (ovary to uterus) MCQ.

DeLancey levels of vaginal support †

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

KEY POINT

Support ligaments are connective tissue condensed around vessels and nerves, so the best anchors (uterosacral, cardinal) are exactly where sutures injure the ureter, pelvic plexus and sacral nerves. See Ch 108

5 · Urethra and continence anatomy

- The urethra is **4.7–5.1 cm** long and hangs from the posterior pubourethral ligaments on the anterior vaginal wall.
- **The mucosa and spongy vascular submucosa generate most closure pressure and are oestrogen-dependent**, which is the rationale for vaginal oestrogen.
- The **internal sphincter** is smooth and autonomic at the bladder neck; the **external sphincter** (sphincter urethrae, **compressor urethrae**, **urethrovaginal sphincter**) is striated, voluntary (pudendal) and covers the distal two-thirds.
- **Hypermobility** is one cause of SUI. An autologous pubovaginal sling sits at the **bladder neck**; a synthetic mid-urethral sling (MUS) sits at the **mid-urethra**.

6 · Bladder and ureter

The ureter is the structure most often injured.

- The **dome** is peritonealised, so intraperitoneal injury needs formal repair. Injury to the **base** causes vesicovaginal or vesicouterine fistula.
- The **pelvic ureter** crosses the iliac bifurcation, passes behind the ovary, runs **under the uterine artery** ("water under the bridge"), then crosses the anterior fornix to the bladder.

ALGORITHM

1. **Ureteric danger points**:
2. Pelvic brim, at the infundibulopelvic ligament.
3. Under the uterine artery (hysterectomy).
4. Beside the cardinal/uterosacral ligaments near the cervix.
5. Anterior vaginal fornix (vaginal surgery).
6. Sacral promontory: the right ureter (sacrocolpopexy).

7 · Nerves at surgical risk

- The **pudendal nerve (S2–S4)** crosses the SSL **close to the ischial spine** into Alcock's canal; pudendal block is given transvaginally just behind the spine.
- In SSL fixation, place sutures about 2 cm medial to the spine, through (not around) the ligament †.

Nerve	Procedure / mechanism	Deficit
Common peroneal	Lithotomy: compression at the fibular head	Foot drop
Sacral plexus (L4–S4)	SSL fixation (plexus exits just posterior to the SSL)	Posterior leg motor/sensory loss
Pudendal	SSL fixation near the spine	Perineal pain, sensory loss, sphincter dysfunction
Sacral trunks S1–S4	Uterosacral suspension (deep sutures)	Buttock/posterior thigh pain
Ilioinguinal (L1), genitofemoral (L1–L2)	Hernia repair, caesarean, hysterectomy	Mons/labial numbness or pain

Nerve	Procedure / mechanism	Deficit
Obturator (L2–L4) †	TOT trocar, obturator lymphadenectomy	Groin pain, adductor weakness
Posterior femoral cutaneous (S1–S3)	Gluteal pressure, pelvic tumour	Perineal sensory loss

KEY POINT

Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Leg pain and swelling after long lithotomy suggest **compartment syndrome**, an emergency († fasciotomy).

8 · Vessels and lymphatics

Internal iliac	Anterior division	Posterior division
Visceral	Obliterated umbilical, superior vesical, uterine, vaginal, middle rectal	None
Parietal	Obturator, internal pudendal, inferior gluteal	Iliolumbar, lateral sacral, superior gluteal

- The **internal iliac** has rich collaterals, so ligation or embolisation is tolerated; ligating the **external iliac** threatens the limb.
- The **presacral plexus** is valveless and under high pressure, and a 1–2 mm vein can bleed more than **1 L/min**.

ALGORITHM

1. **Sacrocolpopexy**
2. Expose the ALL at the promontory over the L5/S1 disc.
3. Open the peritoneum distally.
4. Suture into the ALL over **S1** (strongest; less discitis).
5. Avoid the left common iliac vein, middle sacral vessels, presacral plexus and right ureter (each about 2.7 cm from the mid-promontory).

Organ	First nodes
Distal ⅓ urethra, lower ⅓ vagina, vulva	Inguinal, then pelvic
Proximal ⅔ urethra, upper ⅔ vagina	Iliac, obturator (pelvic)
Bladder, uterus	Perivisceral, then pelvic, presacral and great-vessel nodes
Ovary MCQ	Along the gonadal vessels to para-aortic nodes (bypasses the pelvis)

9 · Endoscopic anatomy

ALGORITHM

1. Distal urethra: glands, diverticular ostia, mesh, stones.
2. Mid-urethral coaptation (external sphincter).
3. Bladder neck coaptation.
4. Trigone and orifices (30° lens).
5. Survey the whole mucosa.

- Squamous metaplasia and cystitis cystica are benign, but biopsy if in doubt.

10 · Imaging

- **Translabial ultrasound** shows bladder-neck mobility, prolapse, and **mesh, MUS and bulking agents**.
- On **endoanal ultrasound**, an internal sphincter defect is a bright break in dark muscle; an external sphincter defect is a dark lesion in bright muscle.
- **T2-weighted MRI is best for urethral diverticulum** and separates other cysts such as Gartner, Skene and Bartholin cysts.
- **MR defecography** grades descent below the **pubococcygeal line** and is used for defaecatory dysfunction or symptoms out of proportion to examination.

108 Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

1 · Definitions and terminology

The type of UI decides treatment.

- **Urinary incontinence (UI)** is complaint of involuntary urine loss (ICS/IUGA, 2010).
- Symptoms are non-specific, so combine them with signs and observations (diary, urodynamics [UDS]). **Bother drives treatment**: without quality-of-life impact, none may be needed.

Type	Complaint of loss…	Urodynamic correlate / trap
Stress UI (SUI)	On effort, cough, sneeze	Urodynamic SUI: leak with abdominal pressure rise **and no detrusor contraction**
Urgency UI (UUI)	With urgency	**Detrusor overactivity (DO) not required**; DO incontinence (DOI) is a separate UDS diagnosis
Mixed UI (MUI)	With urgency and effort	Urge- or stress-predominant, or equal
Overflow UI	Frequent small-volume loss with retention	Underactivity or obstruction; high post-void residual (PVR)
Functional UI †	Impaired mobility/cognition, normal lower urinary tract	Frail elderly, long-term care
Continuous UI	Day and night, few voids	Suspect fistula
Nocturnal enuresis	During sleep	Differs from nocturnal UUI (wakes, cannot reach toilet)

- **Urgency** is a sudden compelling desire that is hard to defer (pathological), whereas **urge** is normal desire.
- **Overactive bladder (OAB)** is urgency with or without UUI, usually with frequency and nocturia; it is **defined by urgency**. Frequency ≥ **8 voids/24 h** is abnormal once polyuria is excluded.

2 · Epidemiology of female UI

- Remission reaches about 12%/year and is **lowest for MUI**.
- **Pregnancy** is the commonest cause of new UI in young women (about **40%**, mostly SUI). UI at 3 months postpartum usually persists, and pelvic floor muscle training (PFMT) during and after pregnancy reduces it.
- **Hysterectomy** can cause or cure UI; vaginal and laparoscopic routes carry more risk than open.

3 · Risk factors for female UI

Factor	Key point
Ageing	Strongest independent non-modifiable factor. SUI peaks at 40–59 years; UUI/MUI dominate after 60. Over 40% in long-term care at admission

Factor	Key point
Vaginal delivery	Causes later **SUI, not UUI**. Caesarean section (CS) protects only if **every** delivery is by CS; number needed to treat (NNT) is **13** to prevent one moderate–severe UI. Elective CS protects more than CS in labour. Baby ≥4 kg: OR about 1.5
Parity	First delivery OR about 1.5, then modest rise; strongest for SUI
Obesity	Body mass index (BMI) ≥30 more than doubles risk (SUI, MUI); **reversible** with weight loss. Little effect on SUI surgery success
Menopause / oestrogen	**Systemic oestrogen increases UI** (new UI nearly doubled vs placebo). Vaginal oestrogen is not linked to SUI and treats atrophy and recurrent UTI
Race	SUI commoner in White than Black/Asian women (higher closure pressure in Black women)
Smoking	Urgency, frequency, UUI (dose-dependent)
Caffeine	UUI, MUI, OAB, but not SUI
Diabetes	Up to 2×; neuropathy and macroalbuminuria are independent factors

> **KEY POINT**
>
> Obstetric injury and loss of support cause **stress** UI; ageing, neurological, metabolic and lifestyle factors cause **urgency**. So the dominant type moves from stress to urge with age.

Male UI (contrast)

- In men **UUI is commoner than MUI, which is commoner than SUI**. Male SUI is nearly always iatrogenic (radical prostatectomy [RP]) or neurological, and remission reaches **40%**.
- Post-RP UI is 8–60% depending on definition; after TURP it is about 1%.

4 · Physiology of continence

- During filling, **pelvic** parasympathetic outflow is suppressed, **hypogastric** sympathetic β-action relaxes the detrusor and α-action closes the bladder neck, and the **pudendal** nerve (Onuf's nucleus) keeps the sphincter closed.
- In **women**, the continence zone is the **proximal and mid-urethra**; the bladder neck is often incompetent in continent women. The oestrogen-sensitive spongy submucosa gives up to **30%** of closure.
- **Support**: the anterior vaginal wall is the firm backing. Lax vaginal support with intact pubourethral ligaments causes **shear** in the continence zone.
- **Male vs female**: men tolerate losing one outlet unit (TURP) because the bladder neck is powerful; in women the bladder neck is weak, so external sphincter deficiency causes UI.

> **KEY POINT**
>
> The female outlet relies on a **single** mid-urethral sphincter against a mobile backing, so continence needs support, intrinsic urethral properties and muscle together.

5 · General pathophysiology: bladder vs outlet

Mechanism	Causes	Result
Loss of suprapontine inhibition	Stroke, multiple sclerosis (MS), Parkinson; early diabetes mellitus (DM)	Neurogenic DO, causing UUI
Outlet obstruction	Women: after anti-incontinence surgery; men: bladder outlet obstruction (BOO)	DO with new (de novo) UUI
Detrusor underactivity	Conus/cauda lesion, neuropathy (DM, alcohol), radical hysterectomy, abdominoperineal resection (APR), opioids including tramadol	Retention, then **overflow UI**
Poor compliance	Pelvic radiotherapy	High detrusor leak point pressure (LPP), causing UI
Sphincter deficiency	RP, urethral/anti-incontinence surgery, cord injury, α-blockers, muscle relaxants	Intrinsic sphincter deficiency (ISD), causing SUI

- In **DM**, early polyuria and UUI give way to lost sensation and contractility, with poor emptying, UTI and overflow.
- In multiple system atrophy ≥20% leak after TURP.
- In post-RP UI, **ISD predominates (>70%)**, and DO or poor compliance contributes in >30%.
- Each failed operation adds fibrosis and ISD; **labour** causes fascial injury and **pudendal denervation**.

6 · Pathophysiology of female SUI

LAQ Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery

Paper answer · P3 · merged from 2

Asked as: P3 LAQ "Female pelvic floor, continence mechanism, classification and management of SUI" · P3 LAQ "SUI: evaluation in females, role of UDS, and surgical therapy in males"

Definition and mechanism

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, exertion, coughing or sneezing (ICS, International Continence Society).
- Continence needs **urethral pressure greater than bladder pressure** at rest and during stress.
- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.
- Two mechanisms: **urethral hypermobility** (loss of the suburethral support, the DeLancey "hammock") and **ISD (intrinsic sphincter deficiency)**. Most women have some ISD.

Anatomy of the female pelvic floor

1. Pelvic diaphragm

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction	Anorectal angle
Pubococcygeus (pubovaginal)	Pubis, ATFP (arcus tendineus fasciae pelvis) to viscera, raphe	Slings urethra and vagina
Iliococcygeus	Arcus tendineus levator ani, ischial spine to coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine to sacrum, coccyx	Lies on sacrospinous ligament
- Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, then fascia carries the load.		

2. Perineal membrane and perineal body

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

3. Endopelvic fascia and ligaments

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

4. DeLancey levels (1992)

Level	Structures	Defect
I suspension	Cardinal–uterosacral	Apical prolapse, enterocele
II attachment	ATFP, levator fascia	Cystocele (central or paravaginal), rectocele
III fusion	Perineal membrane, perineal body	**Urethral hypermobility**, perineal descent

Continence mechanism in the female

1. Intrinsic urethral factors (smooth and striated sphincters, vascular submucosa)

- **Mucosal seal**: folded urothelium coapts. The **oestrogen-dependent submucosal vascular plexus** provides up to **30%** of closure force.
- **Smooth muscle** (internal sphincter, bladder neck and proximal urethra): involuntary tone. The female bladder neck is weak and often open in continent women.
- **Striated urogenital sphincter**: the rhabdosphincter, mostly **slow-twitch** fibres, gives tonic mid-urethral closure. Distal parts are the **compressor urethrae** and **urethrovaginal sphincter**.
- The **continence zone is the proximal and mid-urethra.**

2. Support (extrinsic)

- The urethra lies on the **anterior vaginal wall and endopelvic fascia**, attached laterally to the **ATFP** and levator ani.

- **Pubourethral ligaments** anchor the mid-urethra; **levator ani** (pubococcygeus) tone keeps the backing firm.

3. Neural

- **Pudendal nerve (S2–4)** drives the sphincter (**guarding reflex**). Urethral pressure rises **just before** a cough reaches it, so closure is active, not just transmitted.

4. Theories of continence and SUI

Theory	Mechanism	Surgical implication
Pressure transmission (Enhorning 1961)	Hypermobile urethra drops out of the abdominal pressure zone	Restore position (Burch)
Hammock (DeLancey 1994)	Abdominal pressure compresses the urethra against a firm suburethral layer; a lax layer fails	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament laxity at the mid-urethra	Tension-free MUS (mid-urethral sling)
ISD (McGuire & Lytton 1978)	Urethra itself fails to close (VLPP [Valsalva leak point pressure] <60, MUCP [maximum urethral closure pressure] <20 cmH_2O)	Sling, bulking, AUS (artificial urinary sphincter)
- Most SUI has some ISD. **Hypermobility alone never causes SUI.**		

Classification of SUI

1. Blaivas & Olsson (1988), on video-urodynamics

Type	Bladder neck at rest	On stress
0	Closed, above lower border of symphysis	Descends and opens, but no leak seen (history of SUI)
I	Closed, above symphysis	Descends **<2 cm**; leaks; little or no cystocele
IIA	Closed, above symphysis	Descends **>2 cm** with cystourethrocele; leaks
IIB	Closed but **at or below** lower border of symphysis	Proximal urethra opens; leaks with or without further descent
III	**Open at rest** without detrusor contraction	Leaks with minimal stress = **ISD**

2. Functional: hypermobility versus ISD

	Urethral hypermobility	ISD
Q-tip test	Angle change **>30°**	Often fixed urethra
VLPP / ALPP	**≥90 cmH_2O**	**<60 cmH_2O** (60–90 equivocal)
MUCP	Normal	**≤20 cmH_2O**
Cause	Childbirth, ageing, obesity	Previous surgery, radiation, neuropathy, ageing

3. Other

- **Severity (Stamey):** grade 1 leak on cough or sneeze; grade 2 on minimal effort (walking, rising); grade 3 at rest.
- **Index (uncomplicated)** versus **complicated** SUI (recurrent, prolapse, neurogenic, high PVR [post-void residual], radiation); pure versus mixed incontinence.

Evaluation of SUI in women

History

- Leakage with cough or activity means SUI; after urgency, UUI (urgency urinary incontinence); both, MUI (mixed). A positive SUI history predicts urodynamic SUI in about three-quarters.
- Severity (pads/day), bother; voiding symptoms, prolapse, obstetric history, prior surgery or radiation, neurological disease, drugs.
- **Questionnaires:** ICIQ-UI SF (International Consultation on Incontinence Questionnaire–UI Short Form); UDI-6 (Urogenital Distress Inventory: bother); IIQ-7 (Incontinence Impact Questionnaire: impact). **Bladder diary** (3 days).

Examination (comfortably full bladder)

- Neurological (S2–S4 sensation, anal tone, bulbocavernosus reflex); atrophy; prolapse by **POP-Q (Pelvic Organ Prolapse Quantification)**.
- **Cough stress test:** cough supine, then standing if negative. Leakage synchronous with cough confirms SUI. Reduce prolapse to unmask **occult SUI**.
- **Q-tip test:** swab to bladder neck, Valsalva; rotation **>30°** = hypermobility. Visual mobility examination is now routine.

Tests

Test	Method and use
Urinalysis	Required; exclude infection, haematuria
PVR	Required before surgery; ultrasound; repeat if high (no agreed cut-off)
1-hour pad test (ICS)	500 mL fluid, standard exercises; >1.4 g positive; 1–10 g mild, 11–50 g moderate, >50 g severe
24-hour pad test	Normal activities; more reproducible
Cystoscopy	Haematuria, irritative symptoms, previous mesh

Role of urodynamics (UDS) in SUI

- UDS (urodynamic studies) assess storage and voiding. The key question in SUI is: **will the result change the operation?** Pure, demonstrable SUI gains little. **UDS is not routine.**

Parameter	Method and meaning
USI (urodynamic stress incontinence)	Leak with raised abdominal pressure, no detrusor contraction
ALPP/VLPP (abdominal/Valsalva leak point pressure)	Standing, from 150 mL, Valsalva or cough; **<60 cmH_2O = ISD**, 60–90 equivocal, ≥90 = little ISD (McGuire 1993)
MUCP	Urethral pressure profile; a low MUCP (≤20 cmH_2O) also suggests ISD; does not diagnose or grade SUI
Filling cystometry	DO (detrusor overactivity), compliance, capacity; cough-induced DO mimics SUI
Pressure–flow study	Voiding dysfunction, poor contractility (retention risk after sling)
Prolapse reduction	Occult SUI
- If no leak with the catheter in place, repeat with it removed.	

Evidence

- **VALUE (ValUE) trial (2012), 630 women** with uncomplicated SUI: office evaluation alone was **non-inferior** to office evaluation plus UDS before surgery (success 77.2% vs 76.9%).
- **Dutch VUE study:** immediate midurethral sling was non-inferior to UDS-tailored treatment.
- **SISTEr:** UDS did not predict outcome.

Guidelines

- **AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023:** UDS may be omitted in the index patient with demonstrable SUI.
- **UDS advised in non-index patients:** unclear diagnosis or SUI not demonstrated; mismatch between symptoms and findings; urgency-predominant MUI; neurogenic bladder; raised PVR or voiding dysfunction; prolapse stage ≥3 without demonstrable SUI; failed previous anti-incontinence or prolapse surgery.
- **EAU (European Association of Urology):** no routine UDS for uncomplicated SUI; do not use leak point or urethral pressure to grade severity.
- **Men:** before AUS or sling for post-prostatectomy incontinence, UDS confirms USI and finds DO, poor compliance or weak contractility. The MASTER trial recruited men with USI.

Conservative management (first line)

- Weight loss, fluid and caffeine adjustment.
- **Supervised PFMT (pelvic floor muscle training)** for at least 3 months (Cochrane: more cure than no treatment).
- **Pessary**; vaginal oestrogen for atrophy.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): increases Onuf's nucleus drive to the sphincter; approved in Europe, not the USA.

Surgical options for female SUI: principles

Operation	Principle (how it works)	Best for	Key facts
MUS: retropubic (TVT, tension-free vaginal tape, Ulmsten 1995), transobturator (TOT; Delorme 2001; inside-out de Leval), single-incision	Tension-free type I polypropylene tape under the **mid-urethra** acts as a new pubourethral ligament; the urethra kinks against it during stress	First-line index SUI	**First line** (AUA/SUFU strong recommendation). **TOMUS**: retropubic 80.8% vs transobturator 77.7% cure (equivalent); retropubic often preferred in ISD. Retropubic: more bladder perforation; transobturator: more groin pain
SIMS (single-incision mini-sling)	Short tape, vaginal only	Option	Non-inferior to TOT for subjective cure in most studies; conditional recommendation (2023)

Operation	Principle (how it works)	Best for	Key facts
Autologous pubovaginal sling (PVS) (rectus fascia or fascia lata)	At the **bladder neck**, minimal tension; restores the hammock, giving bladder-neck support and compression	**All types, including ISD**, recurrent SUI, neurogenic SUI, failed MUS, after mesh removal, mesh-averse	Gold standard. **SISTEr**: sling beat Burch (47% vs 38%) but more voiding dysfunction and UTI (urinary tract infection)
Burch colposuspension (open, laparoscopic or robotic)	Paravaginal fascia beside the mid-urethra and bladder neck sutured to **Cooper's ligament**; lifts the bladder neck back into the pressure zone and gives a stable backing	Hypermobility; women declining mesh; at abdominal surgery	85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD
Paravaginal repair, VOS (vagino-obturator shelf)	Vagina to ATFP or obturator fascia		Flatter, less obstructive
MMK (Marshall–Marchetti–Krantz)	Bladder neck to symphysis periosteum	Historical	Osteitis pubis 0.7–2.5%; abandoned
Urethral bulking agents	Submucosal injection at bladder neck or proximal urethra improves coaptation	ISD with fixed urethra; frail patients; minimal surgery	Less durable; repeat injections
AUS	Bladder-neck cuff; constant circumferential closure	Severe refractory ISD, fixed urethra	Second-line; erosion, infection, revision
- **Also know:** needle suspensions abandoned (sutures pull through); anterior colporrhaphy is not an SUI operation; CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension).			

Burch steps (algorithm): Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

Retropubic vs transobturator MUS

	Retropubic	Transobturator
Bladder perforation	4.5%	0.6%
Voiding dysfunction (TOMUS surgery)	More (2.7% vs 0%)	Less
Groin pain	1.3%	6.4%
ISD, long-term durability	Often preferred	Possibly more repeat surgery

Complications of female SUI surgery

- **MUS complications:** perforation, retention, de novo urgency, mesh exposure (about 2%), pain.
- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal): urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS [International Urogynecological Association/International Continence Society] terms): exposure, extrusion, perforation into bladder or urethra, pain; manage by excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.

Male SUI: surgical therapy

- **Aetiology:** commonest after **radical prostatectomy (RP)**, which removes the internal sphincter, leaving continence to the rhabdosphincter. Also TURP (transurethral resection of the prostate), radiation, pelvic fracture.
- **Work-up:** 24-h pad weight (mild <150 g, moderate <400 g, severe above); standing cough test; **cystoscopy** (anastomotic stricture, sphincter coaptation); UDS if mixed or poor contractility.
- **Timing:** PFMT first; no surgery for 6–12 months, as continence keeps improving (other P3 answer: continence improves for up to 2 years, but do not delay surgery beyond 6 months for severe leak).

Device	Mechanism	Use and results
Transobturator sling (AdVance XP)	Repositions the bulb 3–4 cm proximally; needs residual sphincter	Mild–moderate leakage, no radiotherapy; cure about 60%
Quadratic sling (Virtue)	Compression plus elevation	Mild; limited data
Adjustable sling (ATOMS, adjustable transobturator male system)	Cushion adjusted via scrotal port	Mild–moderate; about two-thirds dry
ProACT balloons	Bladder-neck compression	More erosion
Bulking agents	Submucosal bulk	Small leaks, unfit men; rarely durable; not advised
AUS	Circumferential compression	**Gold standard**; all severities, moderate–severe, after radiation; about 80% continent

AMS 800 AUS

- **Components:** urethral **cuff** (bulbar, usually 4.0–4.5 cm); **PRB (pressure-regulating balloon) 61–70 cmH_2O** (51–60 high risk; 71–80 bladder neck) in a preperitoneal pocket; scrotal **pump**. Filled with 23 mL.
- **Technique:** perineal incision; cuff on the proximal bulbar urethra; **deactivate for 4–6 weeks**.
- **Needs** dexterity and cognition. Social continence (0–1 pad) 55–77%; satisfaction about 90%; lifespan 7–10 years.

- **Complications:** infection 1–3% (explant; reimplant after 3–6 months); **erosion** 2–15% (explant, catheter about 3 weeks); urethral **atrophy** (late recurrent leak); mechanical failure; retention.
- **Erosion risks:** radiation, prior erosion, urethroplasty; use a **transcorporal cuff** (tunica adds bulk) or lower PRB. Atrophy: smaller or tandem cuff.

Algorithm (AUA/SUFU 2019) `Severe SUI or any radiotherapy -> AUS Moderate, no radiotherapy -> AUS or male sling Mild, no radiotherapy -> sling (or PFMT, AUS) Sling failure -> AUS`

- **MASTER RCT (randomised controlled trial):** sling was **non-inferior** to AUS for continence at 12 months; AUS remains preferred for heavy leak (>250 g/day) or radiation.

Diagrams to draw

- DeLancey levels and hammock cross-section.
- Sites of Burch sutures, bladder-neck sling and mid-urethral tape; sagittal pelvis showing retropubic and transobturator tape paths.
- Cystometrogram trace with abdominal pressure spike and leak (ALPP).
- AMS 800 (cuff, pump, PRB) and male transobturator sling path.

> **Key points to remember:**
>
> - Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
> - DeLancey: I suspension, II attachment (ATFP), III fusion.
> - Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
> - Blaivas types 0, I, IIA, IIB, III; type III = ISD.
> - Q-tip >30° = hypermobility; VLPP / ALPP <60 cmH_2O = ISD; MUCP ≤20 cmH_2O.
> - Burch corrects position; slings and AUS correct function.
> - VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
> - Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
> - PFMT first; MUS first-line surgery.
> - SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
> - PVS for ISD; urethral injury → no mesh.
> - Men: AUS gold standard; sling for mild, non-irradiated.
> - AUS PRB 61–70 cmH_2O; deactivate 4–6 weeks.
> - Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

Also, from the condensed book (not in the paper answer):

- Support does not explain everything: some women with marked hypermobility are dry, and some with a fixed urethra leak severely.

Hypermobility vs ISD

- ISD equals **type III SUI** (Blaivas) MCQ and is usually seen after previous surgery.
- A **"pipestem" urethra** is fixed and non-coapting; subtler ISD coexists with hypermobility in most SUI.
- **Valsalva leak point pressure (VLPP)** is the abdominal pressure that causes leakage. **VLPP < 60 cm H_2O means ISD**, historically treated by bulking or pubovaginal sling.

KEY POINT

Slings work in hypermobility but Burch fails in ISD. Almost all SUI includes some ISD, so sphincter-directed procedures (pubovaginal sling, mid-urethral sling [MUS], bulking) work in hypermobile women. Burch corrects only hypermobility, so it is weak with severe ISD and a fixed urethra.

7 · OAB pathophysiology (brief)

- Neurogenic causes (§5) and outlet obstruction (BOO, post-sling) both lead to DO and de novo urgency.
- With pelvic organ prolapse (POP), urgency is mainly due to obstruction from kinking. **OAB improves after POP repair in 60–80%**, more after anterior/apical repair, with no reliable predictor.

8 · Insensible incontinence

- It is usually ISD or DOI; if treatment fails, seek rarer causes. Post-void dribble suggests diverticulum (pelvic MRI), continuous loss with normal voids suggests ectopic ureter (upper-tract imaging), and frequent small losses suggest overflow (UDS and PVR).

9 · Pelvic organ prolapse: definition and staging

- **POP** is descent of the uterus/vault and/or vaginal walls from loss of support. **Name the vaginal segment** (anterior, apical, posterior), not the presumed organ.
- The most specific symptom is a **visible or palpable bulge**, which correlates best once past the hymen.
- Never attribute bleeding or pain to POP; relief with a **pessary trial** predicts benefit from surgery.

POP-Q stage	Most distal point on maximal strain
0	Aa, Ba, Ap, Bp = −3; C/D ≤ −(tvl − 2)
1	> 1 cm above hymen (< −1)
2	Within 1 cm of hymen (−1 to +1)
3	> +1 but < +(tvl − 2)
4	Complete eversion: ≥ +(tvl − 2)

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

Also, from the condensed book (not in the paper answer):

- The stage is set by the single most distal point, and measuring right after pessary removal underestimates it.
- POP-Q is the IUGA/ICS-recommended system; Baden–Walker grades have poor interobserver reliability.

10 · POP epidemiology and risk factors

- Loss of support is seen in 43–90%, and **asymptomatic stage 1–2 is normal**; symptomatic POP affects 3–12%.

- **Anterior POP is commoner than posterior, which is commoner than apical**. Apical descent accompanies most significant anterior POP and is universal at stage 4. **So when planning cystocele repair, assess the apex and consider vault suspension.**
- POP progresses slowly; progression predictors are age >60, Ba ≥4 cm and genital hiatus (gh) ≥5 cm.

Risk factor	POP	UI (contrast)
Vaginal childbirth	Strongest; rises steeply with parity; forceps worse. Elective CS risk is about nulliparous; CS in labour about vaginal	SUI, not UUI
Age	About 40% rise per decade; surgery peaks in the 7th decade	Shift to UUI/MUI
Obesity	About 1.4–2.5-fold	>2-fold, reversible
Hysterectomy	Raises risk, most if done for POP (no vault suspension)	Cause or cure
Race	Hispanic highest, Black lowest	White highest
Oestrogen / menopause	Not consistently associated	Oral oestrogen increases UI
Other	Family history (about 5× in siblings), connective-tissue disease, chronic constipation (2.5×)	—

- Recurrence after surgery is predicted by preoperative stage 3–4.

11 · Pathophysiology of POP

- The **levator ani** is the primary support; the uterosacral–cardinal ligaments hold the organs in position over it.

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

Also, from the condensed book (not in the paper answer):

- The commonest cystocele defect at surgery is detachment of the arcus tendineus fasciae pelvis (ATFP) from the ischial spine.

ALGORITHM

1. **POP sequence**
2. Labour causes levator trauma or denervation.
3. The muscle loses tone and the urogenital hiatus widens.
4. Fascia and ligaments now carry the whole load.
5. They stretch and fail (ageing, straining, hysterectomy): POP.
6. POP widens the hiatus further.

- MRI shows levator defects in up to **20%** of primiparas and none in nulliparas; **avulsion** predicts advanced, multicompartment POP.
- Levator neuropathy follows vaginal delivery or CS in labour but almost never elective CS, so **labour itself is the injury**.
- An altered collagen type I:III ratio and Ehlers–Danlos or Marfan syndrome increase risk.

12 · POP and UI interaction

Prolapse can hide SUI until repair.

- 40–55% of women with SUI have significant anterior POP, as do about 20% with OAB.
- **Occult SUI** is SUI seen only with the prolapse reduced (up to a quarter of advanced POP).

ALGORITHM

1. A cystocele kinks the urethra, and the obstruction masks SUI.
2. POP repair relieves the kink, so SUI is unmasked.
3. Detect it by UDS or stress test **with the prolapse reduced** (pessary or speculum).

- De novo SUI after POP surgery is predicted by age >66, DM and low urethral closure pressure.
- Examine for POP in every UI evaluation, because POP missed at sling surgery means later prolapse surgery.

13 · Landmark trials

Trial	Design	Result	Take-home
CARE	Stress-continent women, abdominal sacrocolpopexy with or without Burch	Burch reduced postoperative SUI; SUI on UDS with prolapse reduced predicted it	Prophylactic Burch at sacrocolpopexy prevents de novo SUI
OPUS	Stress-continent women, vaginal POP repair with prophylactic MUS vs sham	UI at 1 year 27% vs 43% (NNT about 6); more bladder perforation, UTI, bleeding, retention	Fewer de novo SUI, more adverse events, so counsel

105 Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

1 · Definitions and risk factors

Anatomical prolapse is common; disease means symptoms.

- Risk factors for **urinary incontinence (UI)** are age, obesity, smoking, comorbidity and postmenopausal hormones; vaginal delivery causes stress UI (SUI).
- **Pelvic organ prolapse (POP)** is commonest anteriorly, then posteriorly, then apically. Half of parous women have anatomical POP, but disease means **symptoms attributable to it**. A **vaginal bulge** is the most specific symptom.

Category	POP risk factors
Primary only	Older age, high body mass index (BMI), high birth weight, levator defects, low education, poverty
Recurrence only	**Younger age**, preoperative Pelvic Organ Prolapse Quantification (POP-Q) stage 3–4
Both	Vaginal delivery, parity, levator hiatal ballooning
Protective (primary)	Caesarean delivery; **smoking** MCQ

- **ICS UI terms** follow the trigger: SUI, urgency UI (UUI) and mixed UI (MUI). **Insensible** UI occurs without awareness, and **occult SUI** appears only after POP reduction.

2 · History

- Ask directly and exclude reversible causes with **DIAPPERS**: Delirium, Infection, Atrophic vaginitis, Psychological, Pharmacological, Excess urine output, Restricted mobility, Stool impaction.
- **SUI is urethral failure, UUI is bladder failure, and MUI is both**. In MUI, treat the more **bothersome** component first.
- **POP** gives a bulge and pressure that worsen later in the day. **Splinting** suggests high-grade POP. Anterior POP kinks the urethra and causes voiding difficulty; posterior POP obstructs defecation.
- SUI coexists in about **40%** of women with POP, and many expect repair to cure leakage, so counsel them.
- **Occult SUI** is found in about ⅓ before vaginal POP repair, and **de novo SUI** can appear after POP surgery; both need a shared decision on concurrent versus staged surgery.
- Ask about neurological disease, diabetes, radiation, pelvic or mesh surgery and obstetric trauma. **Local** oestrogen may improve UI and **systemic** oestrogen may worsen it.
- **Drugs**: **α-blockers worsen SUI in women**, and α-agonists, anticholinergics and opioids impair emptying. Diuretics, SGLT2 inhibitors and lithium cause polyuria. **ACE inhibitors cause cough, which worsens SUI**.

3 · Physical examination

- Atrophic (oestrogen-deficient) tissue is pale, flat, dry and without rugae.

SUI testing

- AUA/SUFU 2023 says: **objectively demonstrate SUI with a comfortably full bladder before surgery**.

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

Also, from the condensed book (not in the paper answer):

- The Q-tip test is now mostly for teaching and trials; **visual urethral mobility** (meatal rotation < or **>30°**) is routine.

POP examination

- Examine in lithotomy **and standing**, while straining, to see the worst prolapse.
- Use a **split Graves speculum** to isolate each compartment in turn. If the blade is expelled with posterior foreshortening, there is **level I loss, meaning vault prolapse**; if it stays, there is an isolated rectocele or enterocele.

POP-Q system

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

Also, from the condensed book (not in the paper answer):

- Total vaginal length (tvl) is measured with the prolapse reduced, and the stage is set by the single most distal point.

Stage	POP-Q criteria	Baden–Walker grade (by eye)
0	Aa, Ba, Ap, Bp −3; C or D ≤ −(tvl − 2)	0: normal
I	Leading edge < −1 cm	1: upper vagina to midplane
II	Leading edge −1 to +1 cm	2: midplane to hymen
III	> +1 cm but < +(tvl − 2)	3: through/beyond hymen
IV	≥ +(tvl − 2): complete eversion	4: complete eversion

> **KEY POINT**
>
> Stage II is a 2-cm window around the hymen, and the III/IV boundary depends on tvl. Example: tvl 8, Bp +5 lies between +1 and +6, so **stage III**.

Other elements

- The **bulbocavernosus reflex** (S2–S4) is present in only about **70%** of normal women, so its absence alone is not pathological.
- For **prior mesh**, IUGA/ICS terms distinguish extrusion (through vaginal epithelium) from perforation (into bladder, urethra or another viscus).

4 · Supplemental evaluation

The uncomplicated patient needs few tests.

Urinalysis (UA) and post-void residual (PVR) are universal. For the AUA/SUFU 2023 index patient having SUI surgery, **UA, PVR and objective SUI demonstration** are required;

cystoscopy, urodynamics (UDS) and imaging are not routine. In non-neurogenic OAB only UA is required (PVR optional).

See the paper answer ***Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery*** *(Section 6, Ch 108).*

Also, from the condensed book (not in the paper answer):

- Unexplained haematuria or pyuria also makes a patient non-index.

Symptom quantification

- A 3-day **bladder diary** is both diagnostic and therapeutic (see Ch 114), and its largest void equals functional capacity.
- Use the ICIQ-UI SF for UI and the **PFDI / PFIQ** for POP.
- **Pad tests** are mainly for research. In the ICS 1-hour test, 500 mL is drunk in <15 min, standard activities follow, and the pad is weighed.

	1-hour test	**24-hour test**
Normal	<1.4 g	<4.4 g
Mild	1–10 g	4–20 g
Moderate	11–50 g	21–74 g
Severe	>50 g	≥75 g

- In the **double-dye tampon test** (oral phenazopyridine and intravesical methylene blue), orange staining means a ureterovaginal and blue a vesicovaginal fistula.
- Measure **PVR** by bladder scan and repeat it. There is no agreed cut-off, but asymptomatic women are almost always <100 mL; a raised PVR changes counselling before SUI surgery or antimuscarinics.
- **UDS** are not needed in the index SUI patient (see Ch 107).
- **Cystoscopy** is not routine; use it for haematuria, urgency or irritative symptoms, recurrent UTI, prior sling or mesh, and pelvic radiation.

Imaging

Modality	**Main indication**
Voiding cystourethrogram	Reflux, diverticula, cystocele
Renal ultrasound	Hydronephrosis: chronic retention, **severe POP (ureteric kinking)**, fistula, neurogenic bladder
Translabial ultrasound	**Modality of choice for mesh** (CT and MRI poor); levator defects
MRI / dynamic MRI	**Urethral diverticulum (standard of care)**; complex or recurrent POP; symptoms out of proportion to findings

- A diverticulum gives the **3 Ds**: dysuria, postvoid dribbling and dyspareunia (see Ch 122).

5 · Management pathway

ALGORITHM

1. History, examination, UA and PVR.
2. Exclude DIAPPERS causes, UTI, obstruction and tumour.
3. Classify the UI and grade the bother.
4. If there are red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation or pelvic surgery, prolapse beyond the introitus, fistula), refer for specialist work-up.
5. If not, start initial management and reassess.
6. Add tests only if they change management; then surgery.

Diagnosis	Initial management
SUI	Lifestyle and behavioural therapy including pelvic floor muscle training (PFMT) (see Ch 114); † pessary; surgery if these fail
UUI / OAB	Lifestyle, behavioural therapy; antimuscarinic trial (see Ch 113)
Significant PVR	Treat constipation, review drugs; catheter drainage if PVR 200–500 mL, then reassess
Symptomatic POP	† Observation, PFMT, pessary or surgery, chosen by bother and goals

6 · Landmark trials

Trial	Design	Result	Take-home
† VALUE	Randomised controlled trial, uncomplicated demonstrable SUI: office evaluation with or without UDS	Office evaluation alone non-inferior	Preoperative UDS not needed in index SUI

114 Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

1 · Definition and principles

- **Conservative treatment** is any therapy that is neither surgery nor drugs.
- **Behavioural treatment** changes behaviour or environment, or teaches skills. It is effective and **essentially risk free**, so it is first-line in all guidelines for stress (SUI), urgency (UUI) and mixed (MUI) urinary incontinence, overactive bladder (OAB) and nocturia.
- Programmes are multicomponent and individualised, combining education, pelvic floor muscle training (PFMT), bladder training (BT), biofeedback (BF), electrical stimulation (E-stim) and lifestyle advice.
- Outcome is unpredictable, so offer it to anyone willing; dementia patients benefit from caregiver-led programmes.

	Outlet-directed	**Bladder-directed**
Target	Urethral closure	Bladder function via voiding habits
Examples	PFMT; behavioural training with urge suppression	BT; delayed voiding
	Patient-dependent	**Caregiver-dependent**
Examples	BT, PFMT, behavioural training	Timed voiding, habit training, prompted voiding
Needs	Intact function, learning, motivation	Knowledgeable, motivated caregiver
Suits	Cognitively and functionally intact	Impaired, nursing home

> **KEY POINT**
>
> Why first-line if surgery cures SUI more often? Conservative therapy is risk free and at least as effective as drugs. This asymmetry of harm drives every recommendation.

2 · Assessment

- A **bladder diary for 3–7 days** records intake, voids and volumes, urgency, leaks and pads. It estimates **functional capacity**, and its **self-monitoring effect** makes it assessment, outcome measure and treatment at once.
- **Education** comes first: anatomy and the mechanism of the patient's own leakage. Improvement is gradual and needs daily practice.

3 · Lifestyle modifications

Cheap adjuncts; evidence is weak except for weight loss.

Fluid management

- **Inadequate intake is more common** than excess; self-restriction causes dehydration, urinary tract infection (UTI), constipation and concentrated irritant urine.
- Excess intake worsens frequency, so reduce it if 24-h output is >~**2100 mL**.

- **Target** a **normal range, not restriction**: **30 mL/kg/day**, minimum **1500 mL**, at set times.
- Change the **timing**, not the total: stop fluids **3–4 h before bed** for nocturia. For oedema-driven nocturia, use stockings or leg elevation in the late afternoon, or a mid-to-late afternoon loop diuretic.

Caffeine and irritants

- Caffeine is a **diuretic and irritant** that raises filling detrusor pressure and is a risk factor for detrusor overactivity (DO).
- About **250 mg/day** (≈2 cups) is associated with moderate–severe UI in men.
- Only a subset is caffeine-sensitive. Try a 3–5-day reduction, tapering with decaf to avoid withdrawal headache.

Bowel function

- Constipation and straining are linked to lower urinary tract symptoms (LUTS) through shared innervation, cross-sensitisation and straining-induced **pudendal denervation**. **Faecal impaction** can cause OAB or overflow.
- Advise fibre **with adequate fluid**, exercise, and regular evacuation after meals.

Obesity, smoking, diet

- Obesity roughly doubles UI risk. **Loss of 5–10% body weight** sustained for 12 months substantially reduces UI, mainly SUI; bariatric surgery resolves UI in morbid obesity.
- Behavioural weight loss in overweight women is the **only grade A lifestyle recommendation**.
- For smoking, no UI recommendation is possible (grade D); advise cessation for general health. A fruit, vegetable and whole-grain diet helps prevent UI after menopause (B).

4 · Pelvic floor muscle training (PFMT)

Most failures come from a wrong contraction.

- PFMT, introduced by **Arnold Kegel in 1948**, is the cornerstone for SUI, UUI, MUI, urgency and frequency.
- **Mechanism**: contraction lifts the urethra and compresses it against the symphysis. Training causes striated muscle **hypertrophy**, which raises urethral pressure and improves bladder-neck support. There is also a **skill** route (the knack).

Assessment of PFM function

- A correct contraction gives **perineal elevation**; perineal descent means the patient is straining.
- On **digital vaginal exam**, levator ani lies just above the hymen at **4 and 8 o'clock**. Assess tone, symmetry, hold and quick flicks, and note **overactivity** (hypertonic, cannot relax).
- Scales: **Modified Oxford** is the most used; **Brink** scores pressure, duration and displacement.

Teaching

- Say: "Tighten as if stopping gas, pull up and in", without using buttocks or abdomen. For men: "move the penis up and down".

- The commonest errors are **co-contraction** (rectus, gluteals) and **Valsalva substitution**; counting aloud prevents breath-holding.
- **Verify correct contraction before prescribing.** Failure to isolate the muscle is probably the commonest cause of poor outcome; a pamphlet does not ensure it.

Regimen

No regimen is ideal; good results come with **45–60** contraction–relaxation pairs per day.

ALGORITHM

1. Empty the bladder.
2. **Lying**: 5 quick (1–2 s) and 5 sustained (5, building to 10 s) contractions.
3. Repeat **sitting**, then **standing** (total 30).
4. Rest as long as each contraction (work:rest **1:1 to 1:2**).
5. Do this twice daily = **60/day**.

- **The knack**: contract **just before and during** a cough, sneeze or lift. It reduces leakage immediately.

Evidence

- PFMT beats no treatment for SUI, UUI and MUI, short and long term; pure SUI does best.
- **Supervised training beats unsupervised** (grade A), and **age is no deterrent**.
- Antenatal PFMT in continent women prevents late-pregnancy and postpartum UI; give supervised postpartum PFMT if UI persists ~3 months.
- **After radical prostatectomy (RP)**, PFMT hastens continence in the first 3 months, and behavioural training helped men leaking years after RP. Adding E-stim or BF does not help.
- In **neurological OAB** (multiple sclerosis, incomplete spinal cord injury, Parkinson disease, stroke), PFMT is safe and first-line (B). PFMT adds to tolterodine for UUI.

Cones and devices

- **Weighted vaginal cones** (2 × 20 min/day) work by prompting a reflex contraction, but they **cannot be retained with POP or a wide introitus** and are no better than PFMT. Motion-sensing, app-guided devices can distinguish lift from strain and modestly improve results.

5 · Urge suppression

- A well-timed PFM contraction **reflexly inhibits the detrusor**, whereas rushing raises bladder pressure.

ALGORITHM

1. When the urge comes, do not rush.
2. Stand still or sit, and relax.
3. Do several quick PFM squeezes, then distract yourself.
4. The urge subsides like a wave; then walk normally to void.
5. Once mastered, wait 5 min after the urge, then extend.

- Used alone for UUI, it gives a mean **60–80%** reduction in incontinence, and also reduces urgency, frequency and nocturia.

- For **nocturia**, suppress the urge on waking; void if it has not passed in 1–2 min.
- It is at least as good as antimuscarinics and adds to drugs. In men on an α-blocker, adding behavioural therapy reduced nocturia more than adding an antimuscarinic.

6 · Biofeedback (BF)

- BF is a **teaching technique**, not a treatment. It shows PFM activity in real time by manometry or surface electromyography (EMG); EMG separates PFM from abdominal activity, whereas pressure BF cannot separate lift from strain.
- BF **adds no clear benefit to supervised PFMT**. Use it for very weak PFMs, poor proprioception, an uncertain contraction, or motivation.

7 · Electrical stimulation (E-stim)

- Non-implanted stimulation (vaginal or anal probe, or surface electrodes) produces **PFM contraction** and **detrusor inhibition**.

	Long-term (chronic)	**Maximal**
Intensity	Below sensory threshold	Just below pain threshold
Aim	Detrusor inhibition via pudendal afferents	Urethral closure (direct and reflex striated contraction)
Dose	6–12 h/day for months	15–30 min, several times a week to twice daily

Indication	**Frequency**
UUI	**5–20 Hz** (low)
SUI (including post-prostatectomy)	**20–50 Hz** (higher)
MUI	~20 Hz or alternating
All	Pulse 200–1000 µs, rectangular, biphasic

- E-stim beats sham but **adds nothing to PFMT**, so PFMT is first-line (A). Its **niche** is women who **cannot contract voluntarily**.
- **Tibial nerve stimulation** for UUI or OAB: transcutaneous equals percutaneous in efficacy, with fewer adverse effects (see Ch 115).

> **KEY POINT**
>
> E-stim is dosed by goal. Sub-sensory, low-frequency, prolonged stimulation inhibits the detrusor; near-painful, higher-frequency, brief stimulation contracts the sphincter. Volitional PFMT does both, so E-stim earns its place only where volitional contraction is absent.

8 · Bladder training and scheduled voiding

P1 *Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management — BT breaks urgency, frequent voiding, reduced capacity cycle; void by clock not urge (dissociates voiding from urgency)*

- BT also works for SUI and MUI; it is grade A first-line in women, but evidence in men is insufficient.
- Adding behavioural training to drugs gives benefit that persists after the drug stops (UITN).

ALGORITHM

1. Start with a diary.
2. Set an initial interval of ~1 h while awake (30 min if needed).
3. Void on waking, at each interval and at bedtime, but not between.
4. If leaks and urgency are controlled, increase by **15–30 min/week**.
5. Target an interval of **3–4 h**; review weekly with reinforcement.

- BT beats no treatment for UUI, SUI and MUI, and the Agency for Healthcare Research and Quality (AHRQ) rates it equal to PFMT.
- **Delayed voiding** has no schedule: at each urge, suppress it, **wait 5 min**, and extend gradually. BT is triggered by the clock; delayed voiding by the urge plus a wait.

Caregiver-dependent toileting: elderly, nursing homes

Programme	Patient role	Key features
Timed voiding	Passive	Fixed 2–4 h by day; dryness goal. Also for impaired sensation and infrequent voiders. ~20% of frail elders become dry, 30–50% improve; a 2–3-day trial predicts success
Habit training	Passive	Schedule individualised from the diary, shorter than the usual interval, to **pre-empt** expected leaks; suits diuretic or time-patterned UI
Prompted voiding (PV)	Responsive	**Monitoring**, then **prompting**, then **praise**; every 2–2.5 h. 25–40% of nursing-home residents respond (daytime leaks ≤1); a 3-day trial predicts
Functional incidental training	Responsive	PV plus mobility and strength exercise
Bladder training	Active	Lengthens interval; needs ability and motivation

- **Only BT lengthens the interval**; caregiver programmes aim at dryness, and PV success depends more on the caregiver than the patient.
- First exclude reversible causes (DIAPPERS) and check post-void residual (PVR) (see Ch 105).

9 · High-tone PFM dysfunction

A tight pelvic floor needs relaxing, not strengthening.

- Raised tone causes pelvic pain, hesitancy and incomplete emptying. Treat by **down-training**: contract, then relax, with work:rest **1:4 or longer**; BF is useful.

10 · Mechanical devices

Devices are **intravaginal** (support) or **intraurethral/meatal** (occlude); no device is proven superior.

Continence pessaries

- **Rationale**: the pessary **stabilises the bladder neck, lengthens the urethra and compresses it against the pubis**, like a sling. The knob of the ring sits at the urethrovesical junction.
- **Types** for SUI: incontinence **ring** (with knob), **dish** and **Hodge**. For pelvic organ prolapse (POP), support pessaries (ring, Gehrung) suit milder prolapse and space-filling ones

(Gellhorn, cube, donut) suit advanced POP.

- **Fitting**: choose the largest comfortable size, one finger-breadth from the wall. The patient walks, coughs, strains and voids before leaving, with no expulsion, pain or obstruction.
- **Contraindications**: active pelvic infection, severe ulceration, silicone or latex allergy, **likely non-compliance with follow-up**.
- **Complications**: discharge and odour. Neglected use causes **erosion, impaction, vesicovaginal or rectovaginal fistula**, and bleeding.
- **Care**: follow-up every **24 weeks** is acceptable (versus 12; grade C), inspecting for erosion each time.
- About half improve. The International Consultation on Incontinence (ICI) says pessary or PFMT may be offered; pessary with PFMT beats PFMT alone for POP symptoms. See **ATLAS** under trials.

Other devices

- **Uresta** (reusable bell-shaped pessary) and **Impressa** (disposable tampon-like device) give suburethral support. Urethral inserts are balloon plugs for pure SUI, with risks of UTI, haematuria and migration.
- Use pads when treatment fails or is declined, and prevent **incontinence-associated dermatitis**.
- **An indwelling catheter is the last resort**: unmanageable retention, urine-contaminated sacral or perineal wounds, terminal care, or preference. Clean intermittent self-catheterisation (CISC) is preferred for retention.

11 · Vaginal oestrogen

- For SUI with atrophy, PFMT with vaginal oestrogen beats oestrogen alone (C). Local oestrogen improves urgency, frequency, recurrent UTI and pessary tolerance (see Ch 113).

12 · Adherence, prevention and delivery

- **Adherence caps efficacy** and declines over time; realistic expectations, follow-up, reminders and linking exercise to daily cues help.
- **Prevention in childbearing**: supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (A). Offer a postnatal programme after forceps or a baby ≥**4000 g**. Antenatal PFMT does **not** prevent postnatal prolapse.
- **Prevention at prostatectomy**: PFMT before surgery, with or without early postoperative PFMT, shortens time to continence.

13 · Evidence grading: key ICI recommendations

Grade (Oxford)	Meaning
A	Consistent level 1 evidence; effectively mandatory
B	Consistent level 2/3 studies or majority of randomised controlled trial (RCT) evidence
C	Level 4, or Delphi-processed expert opinion
D	**No recommendation possible** (not "weak")

ICI 2023 recommendation	Grade
Supervised PFMT first-line for UI in women of all ages	A
PFMT over E-stim as first-line	A
BT first-line for women with UI	A
Behavioural weight loss for overweight women with UI	A
Antenatal supervised PFMT in continent women to prevent UI	A
PFMT reduces POP symptoms (stage reduction uncertain)	A; B
PFMT over drugs first-line (fewer adverse effects, less dropout)	B
PFMT before surgery (surgery more effective but riskier)	B
Adding BF, cones or E-stim to PFMT: not routine	B/C
Fluid adequacy and caffeine reduction	B
Smoking cessation for UI	D

- AHRQ (2018): behavioural therapy is **more likely than anticholinergics to cure UUI** (high strength).

14 · Landmark trials

Trial	Design	Result	Take-home
First behavioural vs drug RCT	Older women, UUI: behavioural vs titrated drug	Behavioural fewer leaks, more satisfaction	Behavioural at least equal to drugs for UUI
MOTIVE	Men with OAB, no obstruction: behavioural vs oxybutynin extended release (ER)	Equivalent reduction in voids	Behavioural therapy works in men
UITN (BE-DRI)	Urgency-predominant UI: tolterodine ER with or without behavioural	Combination better satisfaction, bother; persisted after drug stopped	Add behavioural training to drugs
ATLAS	Stress-predominant UI: pessary vs behavioural vs both	3 mo: behavioural better satisfaction; combination not better; no difference at 12 mo	Pessary reasonable alternative; behavioural first
Weight-loss RCT (PRIDE)	Overweight women: 6-month intensive weight loss vs education	Greater UI reduction	Recommend 5–10% weight loss (A)
GBT	Women with UI: one 2-h bladder-health class vs none	Modestly better ICIQ-SF score to 12 months	Group classes cheap, scalable

116 Retropubic Suspension Surgery for Incontinence in Females

1 · Mechanism

- Stress urinary incontinence (SUI) arises from **urethral hypermobility** and **intrinsic sphincter deficiency (ISD)**, alone or combined.
- Suspension restores **pressure transmission** and a firm backstop for compression.

> **KEY POINT**
>
> Suspension corrects **hypermobility, not the sphincter**: it lifts and stabilises bladder neck and proximal urethra, so presupposes a mobile urethra. A fixed ISD urethra needs compression (sling, artificial urinary sphincter (AUS)).

2 · Patient selection

- Do a stress test supine, then standing if negative; a Q-tip angle >30° means hypermobility.
- AUA/SUFU minimum: **post-void residual (PVR) and urinalysis**; urodynamics only if leak is not shown. Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cm H_2O suggests ISD.
- **Indications** (Burch remains an AUA/SUFU option with midurethral sling (MUS), autologous sling and bulking): SUI with hypermobility, or the stress part of mixed incontinence (MUI); **concomitant abdominal surgery** (sacrocolpopexy); **difficult vaginal access**; **mesh aversion**.

Contraindications	Raise failure risk
Immobile vagina (cannot reach Cooper's)	Baseline urgency, preoperative anticholinergic
Immobile urethra (negative Q-tip ≈ **5-fold** failure): use sling or AUS	Obesity, menopause, age >60
ISD alone; "lead-pipe" (fixed, open) urethra	Prior hysterectomy or SUI surgery; advanced POP; MUI

3 · Anatomy

- **Retzius** is an avascular preperitoneal space between symphysis and bladder. **Cooper's ligament** lies on the superior pubic ramus.
- The **arcus tendineus fascia pelvis (ATFP)** runs from symphysis to ischial spine.

4 · Burch colposuspension

Paravaginal tissue is sutured to Cooper's; sutures elevate, and **fibrosis maintains** the repair.

ALGORITHM

1. Low lithotomy, prep including vagina, Foley.
2. Pfannenstiel or low midline incision.
3. Enter Retzius; sweep fat laterally.
4. Find bladder neck by Foley balloon and vaginal finger.
5. Dissect bladder off vagina **laterally**; expose Cooper's.
6. Place sutures, then cystoscopy.

SAQ Burch colposuspension

Paper answer · P1

Type: SAQ (10 marks)
Source: 2022-Summer Paper 1, Q3b (related: 2017-Winter Paper 3, Q2)

Definition

- Retropubic suspension (**Burch, 1961**) of the paraurethral and para-bladder-neck vaginal wall to **Cooper's (iliopectineal) ligament**; Tanagho (1976) modified it.

Principle

- SUI (stress urinary incontinence) from **urethral hypermobility**: the urethra drops out of the retropubic position. Elevation restores **pressure transmission** and a backstop; scarring holds it.
- It treats hypermobility, not ISD (intrinsic sphincter deficiency).

Indications

- SUI or stress-predominant MUI (mixed urinary incontinence) with hypermobility, especially with **concomitant abdominal surgery** (sacrocolpopexy), **mesh refusal**, or poor vaginal access.
- **Contraindications:** immobile urethra or vagina, ISD alone, pure urgency incontinence.

Work-up

- Cough stress test, PVR (post-void residual), urinalysis; urodynamics only if SUI is not demonstrated.

Steps (open)

1. Low lithotomy, Foley; Pfannenstiel incision; enter the **space of Retzius**.
2. Vaginal finger elevates the vagina; sweep bladder medially. **Avoid midline dissection** over urethra.
3. Two to four 0 or 1-0 delayed-absorbable or non-absorbable sutures each side through vaginal wall (not lumen): distal at mid-urethra, proximal at bladder neck, placed laterally.
4. Pass through Cooper's ligament; tie with vaginal elevation, without tension; leave about **two fingerbreadths** between bladder neck and pubis.
5. **Cystoscopy:** ureteric efflux, no suture in bladder. - Laparoscopic and robotic versions match open short-term results.

Outcomes

- Cure 85–90% at 1 year, about 70% at 5 years (Cochrane).

- **SISTEr (2007):** autologous fascial sling more effective but more voiding dysfunction and UTI.
- **Ward and Hilton:** TVT (tension-free vaginal tape) equivalent to colposuspension; meta-analyses favour midurethral slings.
- **CARE (2006):** prophylactic Burch at sacrocolpopexy reduced postoperative SUI.

Complications

- Voiding dysfunction 2–27%, de novo storage symptoms 8–27%, **de novo prolapse** 2.5–27%; bladder or ureteric injury; bleeding.
- **Also know:** MMK (Marshall–Marchetti–Krantz; osteitis pubis 0.7–2.5%), paravaginal repair, VOS (vagino-obturator shelf); failure predictors: BMI (body mass index) >25, age >60, prior surgery.

Diagram to draw: sagittal pelvis with sutures from paravaginal fascia to Cooper's ligament.

> **Key points to remember:**
> - Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
> - Cure 85–90% at 1 year, ~70% at 5 years.
> - Niche: concomitant abdominal surgery, mesh avoidance.

Also, from the condensed book (not in the paper answer):

- Tie **distal first** and **loosely** (Tanagho), leaving **≈ 2 fingerbreadths** between bladder neck and pubis; over-tightening obstructs.

> **KEY POINT**
>
> **Danger points**: never dissect midline over urethra or bladder neck (Santorini bleeding, injury, obstruction if sutures medial). Passing through Cooper's may bleed suddenly: use pressure, clips, sutures.

Approach	Bottom line
Open (Pfannenstiel)	Cure **85–90% at 1 y, ≈70% at 5 y**
Laparoscopic / robotic	Subjective cure equivalent to open to 18 months
Needle suspension (Pereyra, Stamey, Raz)	Abandoned: sutures tore through

5 · Other retropubic procedures

A higher anchor means more obstruction.

Technique	Anchor	Holds by	Obstruction
Burch	Cooper's ligament	Sutures; high V-shaped angle	Greater
Marshall–Marchetti–Krantz (MMK)	Symphyseal periosteum	Sutures; high	Greater
Paravaginal (Richardson)	ATFP	Scarring; flatter	Less
Vagino-obturator shelf (VOS)	Obturator fascia (or iliopectineal)	Scarring; flatter	Less

> **KEY POINT**
>
> AUS in women is second-line for ISD (recurrent or neurogenic). Place it at the bladder neck, preferably **abdominally**; transvaginal placement raises infection and erosion.

6 · Perioperative care

- **Cystoscopy is mandatory**: no suture in bladder or urethra, and bilateral efflux. If efflux is absent, release sutures.
- For incomplete emptying, use clean intermittent catheterisation (CIC), preferred to suprapubic, until PVR is consistently **<150 mL**.

7 · Outcomes and complications

- Burch roughly equals retropubic MUS but is slower and causes more voiding dysfunction.
- Pubovaginal sling (PVS) is more effective but more obstructive (SISTEr).
- Add prophylactic Burch at sacrocolpopexy in stress-continent women (CARE); with existing SUI, MUS beats Burch.
- Voiding dysfunction (2–27%) is treated by CIC, then urethrolysis (Ch 118). De novo overactive bladder (OAB, 8–27%) needs obstruction excluded.
- **Posterior prolapse** (2.5–27%) follows because anterior elevation opens the cul-de-sac.
- **Osteitis pubis** is unique to MMK (0.7–2.5%) and causes suprapubic pain at 2–12 weeks; treat with rest and analgesia.

8 · Landmark trials

Trial	Design	Result	Take-home
SISTEr	Randomised trial (RCT), 655: Burch vs autologous fascial PVS	PVS higher cure at 2 and 5 y; more urinary infection, voiding dysfunction, de novo UUI	Sling more effective, more morbid
Ward–Hilton (TVT vs colposuspension)	First RCT, 348	No difference to 5 y; TVT more perforation; Burch more voiding dysfunction, later POP	MUS equals Burch, less invasive
CARE	RCT, 322 stress-continent, sacrocolpopexy with or without Burch	Less postoperative SUI with Burch, sustained 2 y	Add prophylactic Burch at sacrocolpopexy

118 Slings: Autologous, Biologic, Synthetic and Mid-Urethral

1 · How slings work — theories of continence

The theory decides where the sling sits and how tight.

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

Also, from the condensed book (not in the paper answer):

- Supporting integral theory, incontinent women have ≈**40% less collagen** in skin and round ligament MCQ .
- **Urethral hypermobility is a SIGN** of damaged support, **not the cause** of stress urinary incontinence (SUI).
- A pubovaginal sling (PVS) sits under **minimal tension at the bladder neck** and rebuilds the hammock. A mid-urethral sling (MUS) lies **loose under the mid-urethra** and arrests the posterior wall. Neither should obstruct at rest.

Historical milestones MCQ

- McGuire reintroduced the fascial PVS (1978). The tension-free vaginal tape (**TVT**, 1995) was the first retropubic MUS and defined the two complications **unique to MUS**: urinary tract mesh penetration and vaginal exposure. The transobturator route followed in 2001 and the single-incision mini-sling (SIMS) in 2006.

Guideline status (AUA/SUFU)

- MUS for the index patient is a **strong recommendation**, grade A. Either route is conditional, grade A; SIMS is conditional, grade B.
- Mesh-specific risks **and** non-mesh alternatives must be discussed before a synthetic MUS (Clinical Principle).

2 · Preoperative assessment and patient selection

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

Also, from the condensed book (not in the paper answer):

- Q-tip movement **>30°** on straining defines hypermobility.
- Further evaluation, including urodynamic studies (UDS), is needed for unclear diagnosis, neurogenic bladder, urgency-predominant mixed urinary incontinence (MUI), high post-void residual (PVR), POP-Q stage ≥3 without SUI on reduction, or prior surgery. Cystoscopy is not part of the index evaluation.

Urodynamic studies — the controversy

- UDS may be **omitted** in the index patient with demonstrable SUI (conditional, grade B), based on **ValUE**.
- A maximal urethral closure pressure (MUCP) **<20 cm H_2O** defines type 3 SUI, that is intrinsic sphincter deficiency (**ISD**). A Valsalva leak point pressure (VLPP) **<60 cm H_2O** also

means ISD; a value **>90** means little or no ISD.

- ISD is a **matter of degree** and now largely a clinical diagnosis; it matters most with a **hypomobile urethra**.
- **Detrusor overactivity (DO)** is the most consequential bladder finding and may improve or worsen, so consider treating it first. Detrusor underactivity predicts voiding difficulty but should **not exclude** surgery.
- Offer non-surgical options to all: weight loss, **pelvic floor muscle training (PFMT)** before surgery for most, and pessaries.

3 · Choosing the sling

Situation	Preferred option	Reasoning
Index SUI, mobile urethra	MUS (retropubic or transobturator); SIMS acceptable	Equal efficacy, least morbidity
ISD with mobile urethra	Retropubic preferred over transobturator	Meta-analyses favour retropubic
Fixed, immobile urethra	Retropubic MUS, autologous PVS, bulking, adjustable sling	AUA Expert Opinion; mobility predicts MUS success
Recurrent SUI after failed MUS	Retropubic MUS or autologous PVS	Retropubic better than transobturator for recurrence
Mesh complication, urethral injury, fistula, diverticulum, reconstruction	**Autologous PVS**	No mesh near a damaged urethra; adds tissue
Declines mesh	Autologous PVS or Burch see Ch 116	—
Neurogenic SUI on clean intermittent catheterisation (CIC)	Autologous PVS (may be tight)	Only once UDS confirm safe capacity and compliance
Concomitant pelvic organ prolapse (POP) surgery	MUS, PVS or Burch	Conditional, grade C

4 · Pubovaginal sling

- **Indications:** primary or recurrent SUI from **hypermobility and ISD**, stress-predominant MUI, anteroapical POP, neurogenic SUI, **urethral reconstruction** (trauma, mesh perforation, interposition at fistula or diverticulum repair), and failed Burch or MUS.
- **Autologous fascial PVS is the gold standard for all forms of SUI.** It gives **dynamic compression without obstruction**; too much tension causes retention and erosion.
- Width **<2 cm** compresses without a constricting band.

Material	Behaviour	Status
Autologous — rectus fascia, fascia lata	Best ingrowth and neovascularisation, minimal reaction; negligible erosion	**Gold standard**; cost is harvest pain and wound complications
Allograft	Processing weakens it; variable, less durable	Mostly withdrawn
Xenograft (porcine, small intestinal submucosa (SIS), bovine)	Weaker, encapsulated; cure falls with time; porcine inferior in trials	Mostly withdrawn

Material	Behaviour	Status
Synthetic at bladder neck	Urethral perforation **15×** and vaginal exposure **14×** other materials (especially woven polyester)	**No longer used at bladder neck**
Vaginal wall sling	Weak; inclusion cysts, vaginal shortening	Historical

KEY POINT

Fibrosis paradox: more fibrosis should mean better support, but the inflammation that drives it also degrades the sling and causes erosion. Autologous fascia remodels with minimal inflammation, the ideal balance.

- **Rectus or fascia lata:** strength and cure are equal. Fascia lata avoids abdominal wound problems but causes thigh pain; use it after **ventral hernia repair**.

5 · Autologous PVS — operative procedure

- **Counsel** about voiding dysfunction and de novo urgency, and teach CIC in selected women. Give one dose of cefazolin and mechanical thromboprophylaxis.

ALGORITHM

1. Through a 6–7 cm Pfannenstiel incision, mark and excise a **2 × 8 cm** rectus graft.
2. Close the fascia with No. 1 polydioxanone (PDS).
3. Clean the graft, stitch No. 1 PDS to each end and keep it in saline.

- Keep the transverse incision **≥2 cm above the symphysis** for tension-free closure. A fascia lata graft is taken over the iliotibial band, and the fascia lata is not closed.
- **Vaginal part:** make an **inverted-U incision**. With the **bladder empty**, perforate the **endopelvic fascia** and finger-dissect the retropubic space.

ALGORITHM

1. With the bladder empty, pass Stamey needles **from above**, on the back of the pubis, into the vaginal incision.
2. Perform **cystoscopy (70° lens)**. If there is an injury, repass **more laterally** and re-scope after each pass.
3. Confirm ureteric efflux, centre the graft and tag its ends above.

- Do any POP repair **before tensioning** and remove the weighted speculum.
- **Tension:** tie the sutures over the rectus with **two fingerbreadths** of slack, deliberately **loose**. Tie tighter only if lifelong CIC is planned.
- Postoperatively, do a voiding trial on day 1; if it fails, use Foley or CIC and retry within 5 days. No heavy lifting and no intercourse for **6 weeks**.

6 · PVS outcomes

PVS rarely fails for stress leakage; "failure" is usually urgency.

- Results are durable even in complex patients (radiation, spinal cord injury, trauma). Cure stays high even with MUCP <4 cm H_2O, so **severe ISD is not a contraindication**.
- **Recurrent SUI:** a repeat PVS, or a PVS after failed MUS, succeeds in most.

- The commonest "failure" is **storage symptoms or urgency urinary incontinence (UUI)**; de novo UUI occurs in 2–22%, predicted by prior POP surgery.
- **Cochrane, PVS versus MUS:** PVS is probably no better, maybe less effective; MUS is shorter, with fewer complications (except bladder perforation) and less voiding and storage dysfunction.
- A damaged urethra does worst, yet PVS still dries most. In **MUI**, PVS cure equals that of pure SUI but the urgency outcome is unpredictable.

7 · PVS complications and management

Voiding dysfunction and bladder outlet obstruction (BOO)

- Incidence is 2.5–35%, **more than Burch** (SISTEr: all outlet procedures were in the PVS arm). Retention >4 weeks is ≈8%; **permanent retention is ≤5%**.
- The usual cause is **excess tension**; poor contractility and **straining to void** add to it.
- It presents as weak stream, retention, **storage symptoms only**, recurrent urinary tract infection (UTI) or pain, even with a low PVR. Exclude new POP kinking the urethra.
- Prediction is unreliable, so a low detrusor pressure means **counselling about CIC**, not refusing surgery. Evaluate by examination, PVR, cystoscopy and selective video-UDS; a high-pressure, low-flow pattern is often **absent in women**.

> **KEY POINT**
>
> **The key diagnostic criterion is the temporal relationship** between surgery and symptoms, not a urodynamic number. Preoperative UDS does not predict obstruction (SISTEr).

ALGORITHM

1. Most transient retention resolves within **10 days**: use catheter or CIC, timed voiding, PFMT, anticholinergic.
2. Autologous obstruction often improves, so wait up to **3 months**.
3. If retention is complete, act earlier: at **<6 weeks**, loosen the sling under anaesthesia by downward urethral traction.
4. If **>6 weeks** or conservative care fails, then do sling incision or urethrolysis.

- Downward loosening is **not for synthetic slings**. Dilation risks fibrosis, and **transurethral bladder-neck incision fails** because the sling is extraluminal.

Procedure	Features
Sling incision (transvaginal)	84–100% success; least morbid, so **first choice**; ≈20% recurrent SUI, not prevented by interposition
Transvaginal urethrolysis	Least effective for PVS: cannot relieve suburethral compression
Suprameatal urethrolysis	Better access to the lateral wings
Retropubic urethrolysis	Best results; after failed transvaginal attempt

- Urethrolysis succeeds in 65–93%, with recurrent SUI in 0–19%. Failure is usually **inadequate lysis**, and a repeat relieves ≈90%.
- **Storage symptoms persist** in over half: use an antimuscarinic or β3-agonist, **sacral neuromodulation (SNM)**, or onabotulinumtoxinA 100 U.

- Recurrent SUI after lysis is treated with bulking, repeat PVS or MUS.

Perforation, exposure and other complications

- Perforation is rare with fascia. Urethral perforation needs **sling excision and repair** (with or without Martius flap); bladder erosion is removed endoscopically.
- An exposed synthetic PVS is removed, and recurrent SUI is treated with a second, **autologous** PVS.
- Mortality is ≈**3 per 10,000** anti-incontinence procedures; harvest-site **incisional hernia** occurs in ≈1%.

8 · Mid-urethral sling — mechanism and materials

- The MUS **secures the posterior urethral wall**, with dynamic **kinking**. Mesh is fixed along its **whole length**, so 50–93% stay dry after midline incision.
- Preoperative **immobility predicts failure**; the ideal is a loose sling with a mobile urethra. **Exception, SIMS:** success needs restricted mobility, so it is set tighter.

> **KEY POINT**
>
> **Why does a loose tape work?** At rest it does not touch the urethra, so voiding is unobstructed. On coughing the mobile urethra descends onto the tape; the posterior wall is arrested, the urethra kinks and closes. A fixed urethra never meets the tape, and tightening the tape to compensate obstructs.

- The material is **type 1 (Amid) macroporous monofilament polypropylene** with pores **>75 µm**, which admits macrophages and fibrovascular ingrowth. Microporous tapes (<50 µm) had high exposure and were withdrawn.

9 · MUS operative procedures

Each route has its own danger structures.

SAQ Mid-urethral slings: TOT and TVT

Paper answer · P1 · P2 · merged from 2

Asked as: P2 SAQ "Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT" · P1 SAQ "TOT in SUI"

Definition and history

- Both are polypropylene **MUS (mid-urethral slings)** for female SUI (stress urinary incontinence), based on the **integral theory** (Petros and Ulmsten, 1990): a loose mid-urethral tape replaces the pubourethral ligaments as a backboard on straining.
- **TVT (tension-free vaginal tape, retropubic):** retropubic polypropylene MUS; Ulmsten, 1995–1998.
- **TOT (transobturator tape):** a synthetic MUS of macroporous monofilament polypropylene, passed through the obturator foramina instead of the retropubic space. Delorme (2001) designed it to avoid bladder, bowel and vessel injury.

Mechanism

- **Integral theory:** pubourethral ligaments, the suburethral vaginal hammock and pubococcygeus maintain continence at the mid-urethra. Hypermobility is a sign of their damage, not the cause.

- The loose tape at the mid-urethra supports the **posterior urethral wall**, which is compressed against it on straining (dynamic kinking); it is not tight at rest.
- TOT anchors in the obturator muscles (**obturator internus and externus**).

Material

- **Type 1 (Amid) macroporous monofilament polypropylene**, pores >75 µm: allows macrophage entry and fibrous ingrowth, so it resists infection.

Indications and evaluation

- SUI or stress-predominant mixed incontinence after failed PFMT (pelvic floor muscle training); index SUI with urethral hypermobility.
- TVT preferred over TOT in **ISD (intrinsic sphincter deficiency)** or recurrent SUI.
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated.
- Counsel about permanent mesh, mesh risks and alternatives (colposuspension, autologous sling, bulking) (AUA/SUFU [American Urological Association/Society of Urodynamics and Female Urology]).

Technique: differences

	TVT (retropubic)	**TOT (transobturator)**
Path	Vagina to retropubic space to suprapubic skin, hugging the pubis; cystoscopy essential	Labiocrural fold to obturator membrane to vagina (outside-in), or reverse (inside-out)
Structures at risk	Bladder, bowel, iliac and epigastric vessels	Obturator nerves and vessels, vaginal fornix

TVT technique

1. Local, spinal or general anaesthesia; lithotomy.
2. Two suprapubic stab incisions 2 cm lateral to midline; 1.5 cm midline vaginal incision 1.5 cm from the meatus; minimal dissection, endopelvic fascia not opened.
3. **Bottom-up** 5 mm trocar hugging the back of the pubis, through the retropubic space to the skin (catheter guide deflects the bladder).
4. **Cystoscopy with a 70° lens, full bladder**, after each pass; if perforated, re-pass more laterally.
5. Tension **loosely** with a clamp between tape and urethra; remove sheath.

TOT technique (outside-in)

1. Lithotomy with hips hyperflexed; 1.5 cm midline vaginal incision 1.5 cm below the meatus; dissect laterally to the ischiopubic ramus.
2. Groin stab incision in the labiocrural fold, at the notch between inferior pubic ramus and **adductor longus tendon**.
3. Trocar perforates the obturator membrane (felt as a give), turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal fornices (sulci).
5. Tension-free: a clamp or scissors pass easily between tape and urethra. - **Inside-out** (de Leval; vagina to thigh): fewer vaginal perforations, more voiding dysfunction.

Outcomes

- TVT: Ulmsten (1998): 91% cured at 1 year. **Ward–Hilton:** equal to colposuspension at 5 years. Long-term cure varies 48–97% by definition.
- **TOMUS (Richter, 2010):** 597 women; 12-month objective cure **80.8% retropubic vs 77.7% transobturator** (not significantly different, equivalent); voiding dysfunction needing surgery 2.7% vs 0%.
- TOMUS at 5 years: 51.3% vs 43.4%; equivalence not shown; satisfaction similar.
- **Cochrane review (Ford, 2017):** cure similar up to and beyond 5 years; repeat surgery possibly commoner after TOT.
- TOT is poorer with **ISD** or a **fixed urethra**: retropubic sling preferred.

Complications: comparison (Cochrane, 2017)

Complication	TVT (retropubic)	TOT
Bladder perforation	**4.5%**	**0.6%**
Groin pain	1.3%	**6.4%**
Voiding dysfunction, bleeding, vascular injury	More	Less
Mesh exposure	About 2%	About 2%
Long-term repeat surgery	Less	Possibly more

TVT complications

Complication	Notes
Bladder perforation	4.5% (TOT 0.6%); re-pass and continue
Urethral injury	Abandon mesh; staged repair
Bleeding/haematoma	2–3%; rarely bowel or vascular injury
Voiding dysfunction	Surgery needed 2.7%; incise tape in midline within 3 months
Mesh exposure	About 2%
Others	De novo urgency, suprapubic pain (groin pain less than TOT)

TOT complications

- Groin or thigh pain (usually temporary; NSAIDs [non-steroidal anti-inflammatory drugs]), vaginal mesh exposure, voiding dysfunction (incise sling within 4 weeks), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- Urethral injury during either sling: **do not place mesh** (AUA guideline).

Choosing between TVT and TOT

- **TVT preferred: ISD** (ALPP [abdominal leak point pressure] <60 cmH_2O or MUCP [maximum urethral closure pressure] <20 cmH_2O), fixed urethra, recurrent SUI. Failure is 12 times likelier after TOT with low leak point pressure (O'Connor, 2006).
- **TOT preferred:** previous retropubic surgery, obesity, avoiding bladder injury.
- AUA: either may be offered to the index patient.
- **Also know:** SIMS (single-incision mini-slings; non-inferior to TOT); mesh warnings (FDA [Food and Drug Administration] 2010; UK pause 2018): counsel about alternatives (Burch colposuspension, autologous fascial sling, bulking agents).

Diagram to draw

- Coronal pelvis: U-shaped retropubic TVT at mid-urethra versus flat TOT, the tape under the mid-urethra through both obturator foramina.

Key points to remember:

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
- TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

Also, from the condensed book (not in the paper answer):

- Document counselling about the **permanent implant**, mesh-specific complications and non-mesh alternatives.
- Bottom-up beats top-down, with better subjective cure and fewer perforations and exposures.

ALGORITHM

1. Infiltrate local; make two suprapubic stabs ≈2 cm from midline.
2. Make a 1.5 cm midline vaginal incision 1.5 cm from the meatus.
3. Make minimal lateral tunnels to the endopelvic fascia, **not perforating** it (unlike PVS).
4. The catheter guide deflects the bladder to the opposite side.
5. Pass the trocar through Retzius and rectus, **hugging the back of the pubis**; repeat on the other side.
6. Perform cystoscopy.
7. Tension loosely (clamp between tape and urethra), remove sheath, trim.

- Keep the trocar in **metal-to-bone contact** with the back of the pubis, so it stays extravesical and extraperitoneal.
- **Transobturator, outside-in (TOT):** hips **hyperflexed (120°)**; puncture the labiocrural fold at the adductor longus notch, pierce the obturator membrane and turn around the ramus onto the finger. Check the vaginal sulcus for a button-hole.
- **Inside-out (TVT-O)** has the same cure, with more voiding dysfunction and fewer vaginal perforations.
- **SIMS** uses only a vaginal incision and anchors in **obturator internus**; set it **tighter**, close to the urethra.

Feature	Retropubic (TVT)	Transobturator	SIMS
Path / anchor	Retzius, rectus fascia; U-shape	Obturator foramen; flatter hammock	Obturator internus only
Cure (index SUI)	Equal to transobturator up to 5 y; best long-term data (>17 y)	Equal; slightly more repeat surgery	Non-inferior to transobturator (TVT-Secur excluded)

Feature	Retropubic (TVT)	Transobturator	SIMS
ISD / recurrent SUI	**Preferred**	Worse with low VLPP	Limited data
Bladder perforation	**4.5%**	**0.6%**	Rare
Vascular/visceral injury, bleeding, voiding dysfunction	More	Less	Less
Pain	Suprapubic 2.9%	**Groin 6.4%** vs 1.3% retropubic	Least
Vaginal exposure	About 2%	About 2%	Similar (excluding TVT-Secur)
Cystoscopy	Mandatory (70°)	Recommended; check the urethra	Essential (30° suffices)

10 · Intraoperative cystoscopy

SAQ Management of perforated mid urethral sling

Paper answer · P1

Type: SAQ (10 marks)
Source: 2023-Winter Paper 1, Q3a

Terminology (IUGA/ICS, International Urogynecological Association / International Continence Society)

- **Perforation:** mesh in a hollow organ (bladder, urethra). **Exposure:** mesh seen through unhealed vaginal epithelium. **Extrusion:** passage out of a tissue. "Erosion" is no longer used.
- Two settings: **intraoperative trocar injury** and **late mesh perforation**.

A. Intraoperative trocar injury

- Bladder injury: 4.5% retropubic versus **0.6% transobturator**; registries 2.7–3.8%.
- **Detection:** cystoscopy with a **70° lens and full bladder distension** after every trocar pass.

Injury	Management
Bladder	Withdraw trocar, **repass more laterally**, confirm on cystoscopy, complete the sling; catheter overnight to 72 h. Benign; no later mesh perforation.
Urethra	**Do not place mesh** (AUA/SUFU, American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction). Repair, catheter drainage; **staged** anti-incontinence surgery after healing.

B. Late perforation of bladder or urethra

- **Presentation:** recurrent UTI (urinary tract infection), haematuria, urgency, pain, stones on mesh, voiding dysfunction, fistula. Often months later.
- **Evaluation:** examination, **cystourethroscopy**, translabial ultrasound (best imaging for mesh); exclude fistula.

Management

- **Bladder:** endoscopic excision (scissors, resection or holmium laser) flush with the wall; or transvesical laparoscopic or robotic excision; open cystotomy if large.
- **Urethra:** transvaginal excision of mesh in the field and layered urethral repair; **Martius flap** if complex or fistula.
- **Continence after excision:** SUI recurs in many (44–100% after synthetic sling revision). Options: concomitant **autologous pubovaginal sling** (Shah 2013: good results) or staged. A damaged urethra predicts poor continence.
- Avoid urethral dilation after mesh slings; it can cause perforation.

Prevention

- Empty bladder; trocar hugging the pubis; bottom-up retropubic route; transobturator route when suitable.

Also know: Amid type I macroporous polypropylene integrates best; chronic pain or infection may need excision from both sides of the bone.

Diagram to draw: retropubic and transobturator trocar paths with bladder at risk.

> **Key points to remember:**
> - Bladder trocar injury: repass laterally, continue.
> - Urethral injury: abandon mesh; stage the repair.
> - Late perforation: excise mesh, repair, consider autologous sling.

Also, from the condensed book (not in the paper answer):

- Inspect the urethra too after transobturator and SIMS passes. A bladder injury is benign: leave a catheter overnight to 72 h.

11 · MUS outcomes

These are the figures quoted when counselling.

- Retropubic slings are the most studied (17–20 years); objective cure stays high and subjective cure is lower. Cochrane (81 trials): retropubic and transobturator cure are equal from short to long term.
- TVT equals colposuspension at 5 years, with less de novo urgency and less later POP.
- In TOMUS at 5 years, satisfaction stayed high (79% retropubic, 85% transobturator) despite falling objective success.
- Long term, **de novo urgency rises** (≈⅓ at 10–20 years); retention is 0–7%.
- In Cochrane, SIMS did worse than retropubic, driven by **TVT-Secur (withdrawn)**; others about equal transobturator, with less pain. The key long-term endpoint is **reoperation**.

Group	Key points
MUI	Urgency resolves in ≈60–75% (best when stress-predominant). De novo urgency risk rises with age, obesity, nocturia, DO and low capacity. Persistent DO: transobturator < retropubic < PVS. New storage lower urinary tract symptoms (LUTS) may mean BOO
ISD	About 50% cure with urodynamic ISD. An immobile urethra predicts failure more strongly than a low VLPP, and low VLPP is not a contraindication to retropubic MUS. Transobturator: 12× odds of persistent SUI with low VLPP

Group	Key points
Recurrent SUI	No level 1 evidence; repeat MUS is commonest, retropubic better than transobturator. Cure ≈81–90% but below primary surgery (≈62% vs 86%); more urgency, and more bladder perforation after prior retropubic surgery
Concomitant POP	A concomitant sling lowers postoperative SUI and reoperation but raises voiding dysfunction. About 30% of SUI resolves after POP repair alone. A prophylactic sling only after counselling (OPUS)
Elderly	Safe; similar or lower cure, more urgency and voiding dysfunction. Q-tip >30°: 90% continent; <30°: 42%; <10°: 80% stay incontinent. Cure ≈64% at ≥75 y vs 89% at 55–64 y. **Frailty, not age**, predicts readmission. Do not exclude by age (AUA Expert Opinion)
Obese	Success >80% but lower (≈87% vs 66% at 5 y), worse with BMI >35; more wound infection, fewer bladder punctures

12 · MUS complications and management

> **KEY POINT**
>
> **Concurrent or staged sling at POP repair**. Concurrent: one operation and less SUI (≈39% falls to 8–19%), but more obstruction, storage symptoms and mesh events. Staged: avoids an unnecessary sling but commits a third to a second operation. Decide after testing for occult SUI with the prolapse reduced.

- Bladder trocar injury is ≈2–4%, almost all retropubic; risk rises with prior retropubic surgery and thin habitus. Vascular (≈0.07%) and bowel (≈0.04%) injuries are rare, but deaths have been reported.
- †Unexplained hypotension, falling haemoglobin, pain, ileus or sepsis after a retropubic sling needs urgent imaging or exploration.
- Bleeding occurs in 2–3%, more with retropubic slings. A retropubic haematoma is usually **asymptomatic, so observe**.
- **UTI is the commonest complication** (3–4% at 30 days). It is less after transobturator slings and marks a worse course.
- Vaginal wound infection is 0.7–2%; rarely there is abscess or **adductor myositis**, which may present years later.
- **Groin pain** after transobturator slings affects 16–26% early and ≈4% at 12 months. Treat with NSAIDs, physiotherapy and trigger-point injection; if refractory, excise, and **complete removal** relieves more than partial.

Voiding dysfunction and retention

- The tape is **too tight** or **too proximal**, or POP kinks the urethra. Obstruction is commoner after retropubic slings (TOMUS: 2.7% vs 0%). Evaluate by **temporal relation**, examination, PVR, cystoscopy and selective UDS.

ALGORITHM

1. Retention or high PVR: CIC or catheter; most resolve within 2 weeks.
2. If obstruction persists, perform a **transvaginal midline sling incision at 2–4 weeks**.
3. Beyond 3 months the tape is fixed, so excise its lateral portion.
4. Formal urethrolysis is rarely needed.

- **Urethral dilation is harmful; do not perform it**. After incision 70–100% stay continent. **Early** release gives better voiding and **less** recurrent SUI.

KEY POINT

Why is the timing different from PVS? Fascial obstruction often settles, so one waits up to 3 months. Mesh is fixed by ingrowth and rarely loosens; late release needs wider dissection and causes more recurrent SUI, so release within ≈4 weeks.

Storage symptoms, exposure and sexual function

- De novo overactive bladder (OAB) is ≈6–12%, equal for all MUS types. A clear temporal link means **partial BOO first**, and **cystoscopy is mandatory**. Treat with behavioural measures and drugs, then sling revision, SNM, percutaneous tibial nerve stimulation (PTNS) or botulinum toxin.
- Vaginal exposure is ≈2%; †it causes discharge, bleeding, dyspareunia or partner discomfort. †Small exposures get topical oestrogen with or without excision; larger ones are excised. Urinary tract mesh needs excision and repair, with no immediate new mesh see Ch 119.
- De novo dyspareunia is ≈3–14%, and sling removal may relieve it. Many women improve as **coital incontinence** stops.

13 · Adjustable slings

- **Remeex** (mesh on sutures to a suprapubic varitensor, adjusted after surgery) is a niche option for **complex SUI**. Cure is ≈85–90%, but complications reach 32–51% and there is no level 1 evidence.

14 · Assessing the failed or troublesome sling

ALGORITHM

1. Define the symptom: SUI, storage, voiding, or pain/discharge.
2. Establish the **timeline** in relation to surgery.
3. Examine for exposure, POP, urethral mobility and tenderness over the tape.
4. Check PVR and urine culture, perform **cystoscopy**, and add UDS selectively.
5. Tailor treatment to the cause.

- If SUI recurs with a mobile urethra, repeat a retropubic MUS or do a PVS; if the urethra is fixed or there is ISD, use PVS, bulking or an adjustable sling.
- If storage symptoms have a temporal link, treat obstruction first. Pain or exposure needs excision (complete for pain) see Ch 119.

15 · Mesh controversy and regulatory status

- The **2010 FDA** safety communication on mesh for POP and SUI was based on **MAUDE** database reports. Professional bodies still endorse MUS with documented counselling.
- †The 2019 FDA order stopping transvaginal mesh concerned **POP**, not MUS. The UK (2018) paused MUS use pending safeguards see Ch 119.

16 · Landmark trials

Trial	Design	Result	Take-home
ValUE	RCT, non-inferiority, 630, uncomplicated SUI: UDS vs office evaluation	Office evaluation non-inferior at 1 y	UDS may be omitted in the index patient
SISTEr	RCT, 655: rectus PVS vs Burch	PVS higher cure (47% vs 38% at 2 y); more voiding dysfunction, UTI and UUI; all BOO surgery in PVS arm; persisted at 5 y	PVS more effective but more obstructive
Ward–Hilton	RCT: TVT vs open colposuspension	Equal to 5 y; TVT more bladder injury; Burch more morbidity and POP	MUS replaced Burch
TOMUS	RCT, 597: retropubic vs transobturator	80.8% vs 77.7% (1 y, equivalent); surgery for voiding dysfunction 2.7% vs 0%; †more groin and neurological symptoms with transobturator; 5 y 51% vs 43% (equivalence not met)	Either; retropubic slightly more durable, more voiding dysfunction
OPUS	RCT (327): POP repair with or without prophylactic retropubic MUS in stress-continent women	Incontinence at 12 months 27% vs 43%; NNT 6.3; more perforation, UTI and bleeding	Reduces SUI but adds risk, so counsel

117 Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

1 · Goals and principles

Prolapse is a quality-of-life problem, so operate for bother.

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

Also, from the condensed book (not in the paper answer):

- **Success means absence of symptoms** (no bulge), which drives satisfaction more than anatomy does.
- Obliterative surgery is contraindicated if intercourse is desired. Open and laparoscopic routes have equal complications, but laparoscopic recovery is faster.
- **The first operation is the best chance**, because planes are then lost and recurrence rises with each re-attempt. Preoperative education improves satisfaction, and declining surgery is a legitimate choice.

2 · Choice of procedure by compartment

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

Also, from the condensed book (not in the paper answer):

- Enterocele (point D) is closed by culdoplasty: **McCall** vaginally, **Halban** or **Moschcowitz** abdominally.

Compartment	POP-Q point	Organ
Anterior	Aa / Ba	Urethrocele / cystocele
Middle	C / D	Cervix or cuff / enterocele
Posterior	Ap / Bp	Rectocele / enterocele; perineal body

Site	Vaginal	Abdominal
Aa — urethra	Anterior repair; bladder neck suspension; sling	Retropubic urethropexy
Ba — bladder	Anterior repair; paravaginal; colpocleisis; wedge colpectomy	Paravaginal; sacrocolpopexy
C — cervix/cuff	Uterosacral ligament suspension (USLS); iliococcygeus; sacrospinous ligament fixation (SSLF); Manchester; hysteropexy; vaginal hysterectomy; colpocleisis	Hysterectomy; USLS; sacrocolpopexy; uterine suspension
D — cul-de-sac	**McCall** culdoplasty	**Halban**, **Moschcowitz** culdoplasty
Ap — posterior	Posterior repair; site-specific repair	Colpoperineopexy

KEY POINT

Only paravaginal repair and uterosacral suspension are done by both routes. McCall is vaginal; Halban (linear sutures) and Moschcowitz (purse-string, which may angulate the ureter) are abdominal.

3 · Anterior compartment

Anterior repair has the highest recurrence.

- A defect may be **central** (midline thinning of the pubocervical fascia) or **lateral/paravaginal** (detached from the arcus tendineus fasciae pelvis, ATFP); most are combined. **Anterior repair does not suspend the apex.**

ALGORITHM

1. In lithotomy, catheterise, hydrodissect and incise in the midline from apex to bladder neck.
2. Dissect the epithelium off the pubocervical fascia laterally to the ATFP and reduce the bladder.
3. Plicate with interrupted 2-0/3-0 delayed-absorbable sutures from bladder neck to apex.
4. Trim judiciously and close; give intravenous (IV) dye, do **cystoscopy** and pack.

- Deep sutures may enter or kink the bladder or ureter, so completion cystoscopy is mandatory.
- Anterior repair is abandoned for stress urinary incontinence (SUI): AUA meta-analysis failure approached **40%**.
- Recurrence depends on the **definition**: 30–46% for "POP-Q ≥2" versus 9–23% for "beyond hymen".
- Polypropylene lowers recurrence versus native tissue but causes far more graft complications without better QoL. Hence the **withdrawal of transvaginal synthetic POP mesh** (US 2019) see Ch 119.
- A concomitant sling **halves anterior recurrence**, more than any graft.

Complication	Point / management
De novo SUI	Rises with stage: stage 2 **41%** , early 3 **53%** , advanced 3–4 **66%** ; point Ba predicts it
Detrusor overactivity	De novo in 5–7%; pre-existing overactivity resolves in up to 63%
Retention	Usually from the sling: clean intermittent catheterisation (CIC); if persistent, sling incision or urethrolysis
Bleeding (>350 mL)	Bleeding means the wrong plane; worse at redo
Bladder injury	Drain the bladder before perforating endopelvic fascia; two-layer closure; Martius if irradiated
Others	Ureteric obstruction, vesicovaginal fistula (VVF), vaginal shortening, dyspareunia

KEY POINT

Confirming ureteric efflux: indigo carmine (blue, IV), fluorescein 10% (green, IV) or phenazopyridine (orange, oral). If there is no efflux, pass a ureteric catheter or remove plication sutures. Never close without proving patency.

Paravaginal repair

- Reattach the pubocervical fascia to the ATFP with non-absorbable sutures 1 cm apart (**5–7** vaginal, **4–6** abdominal), then do cystoscopy.
- Lateral recurrence is 2–8%, but central recurrence is ~**22%**, so these are independent failures. The abdominal route is more durable.
- **Neither colporrhaphy nor paravaginal repair treats SUI.** The vaginal route has more serious complications (transfusion 12%).

4 · Occult SUI and the concomitant sling

- High-stage anterior POP kinks the urethra and masks SUI; SUI on prolapse reduction occurs in 8–66% of continent women.
- Test **with prolapse reduced** (pack, speculum or pessary); sensitivity is low and specificity high.
- **Staged** argument: sling obstruction needing intervention (8.5%) equals intervention for SUI without a sling (8.3%).
- The **OPUS calculator** uses 7 predictors (accuracy 0.72 vs 0.54 for the stress test). In **CUPIDO**, urodynamics added nothing in asymptomatic women.

KEY POINT

A sling also helps the prolapse: distal anterior support makes the cystocele repair last. **Why not everyone?** In OPUS it caused more bladder perforation, UTI, bleeding and incomplete emptying. Weigh per patient.

ALGORITHM

1. In advanced anterior POP, do a reduction stress test, with or without urodynamic studies (UDS).
2. If SUI is symptomatic, occult or urodynamic, then offer a concomitant procedure.
3. If she is continent and the test is negative, then weigh individual risk and benefit (OPUS calculator).
4. The sling is **selective, not routine**.

5 · Apical compartment

The apex is the cornerstone of prolapse repair.

Uterosacral ligament suspension (USLS)

- **The ureter lies 4 cm from the ligament at the sacrum but only 0.9 cm at the cervix**. Injury is usually **kinking**, not ligation. Sutures near the sacrum can entrap S1–S4 roots, causing buttock and posterior thigh pain.
- The **safe zone** is the intermediate portion, ~1 cm posterior, at the level of the ischial spine; pass needles **lateral to medial**.

ALGORITHM

1. After vaginal hysterectomy, place cuff stays at 3 and 9 o'clock, excise the enterocele sac, pack bowel and palpate the spines.
2. Place Allis clamps at 5 and 7 o'clock and pass **2–4** non-absorbable sutures per side.
3. Do any anterior repair or sling now.
4. Bring the sutures through the apical pubocervical and rectovaginal fascia and tie.
5. Perform **cystoscopy after tying, before trimming**. If there is no efflux, remove the most lateral ipsilateral suture first.

- Compared with SSLF, USLS keeps the midline axis, closes the cul-de-sac and risks the pudendal vessels less.
- Apical success is ~98% and anterior ~81%; ureteric kinking or injury occurs in **1–11%**.
- **OPTIMAL**: USLS equals SSLF (~65% at 2 years on a strict composite), and perioperative pelvic floor muscle training (PFMT) adds nothing.

Sacrospinous ligament fixation (SSLF)

- SSLF is extraperitoneal and preserves length. The ligament lies under coccygeus; the pudendal bundle runs round the spine and the inferior gluteal vessels lie behind.
- The **safe zone** is **1.5–3.0 cm medial to the ischial spine**, at limited depth. Too lateral risks the pudendal bundle; too medial hits weak ligament and sacral nerves.

ALGORITHM

1. Mark the apex, incise in the midline, enter the perirectal space (rectum medial) and palpate spine and ligament.
2. Place retractors and pass 2 sutures (Capio or Miya hook).
3. Complete other repairs, then attach the sutures to the apex (pulley stitch if permanent).
4. Close the vagina halfway and tie the apex flush, with **no suture bridge**.
5. Do cystoscopy and close.

- SSLF is usually right-sided, with 2 sutures 2 cm apart. The **pulley stitch** keeps the knot out of the lumen.
- Cure is ~85% and apical recurrence ~5%; stage 4 predicts recurrence.
- **Drawback:** the axis is deflected posteriorly and to the right, causing **anterior recurrence** (~18%).
- Gluteal pain occurs in **15%** and resolves in 2–3 months. Pudendal injury, rectal injury and haemorrhage can also occur.

> **KEY POINT**
>
> Both ligaments are safest in their **middle third** and dangerous at their bony ends: the intermediate portion for USLS, and 1.5–3 cm medial to the spine for SSLF.

Iliococcygeus suspension is for deficient uterosacrals, with sutures **1 cm distal to the ischial spine**; anterior recurrence is commonest.

Sacrocolpopexy (SCP)

- **Indications:** other defects (including SUI) repairable at one sitting, failed vaginal repair, young sexually active women, isolated vault prolapse, or wish for the most durable repair. It

maximises vaginal length without distortion.

- Use **type I macroporous monofilament polypropylene**; fascia lata recurs in ~1/3 versus <1/10 with polypropylene.

ALGORITHM

1. Low lithotomy with a vaginal stent or EEA (end-to-end anastomosis) sizer; sigmoid to the left.
2. Incise the peritoneum over the promontory and along the **right** of the rectum (watch the right ureter) down to the anterior longitudinal ligament.
3. If there is a large enterocele, then add a Halban culdoplasty.
4. Dissect the bladder off the anterior vagina (6–8 cm) and the rectum off the posterior wall.
5. Attach a Y-mesh to both walls **without entering the lumen**, tension it, and fix it with 2–3 permanent sutures at **S1**.
6. **Retroperitonealise** the mesh, add an anti-incontinence procedure if indicated, and perform cystoscopy.

- Fixation too low (S3–4) risks the presacral venous plexus; too high risks the middle sacral vessels and the **left common iliac vein**.
- Avoid hypersuspension, which causes pain and de novo SUI. Sutures or tacks into the promontory can cause spondylodiscitis.
- Open, laparoscopic and robotic SCP succeed in 87–100%; the robot saves little blood but costs more.
- Failures are mainly **posterior or distal**, with almost no apical retreatment; ~20% later need vaginal repair, so counsel.

SCP complication	Rate / management
Haemorrhage (presacral)	~4%; †pressure, sutures, bone wax, sterile thumbtacks
Cystotomy / enterotomy / ureter	3% / 1.6% / 1%; bladder is the commonest robotic visceral injury
Ileus / small bowel obstruction	3.6% / 1.1%
Mesh exposure	**~10% by 7 years** (E-CARE); more with microporous mesh; excision see Ch 119

- **Hysterectomy at SCP:** total hysterectomy raises exposure, so go **supracervical** or close the cuff meticulously.
- Cochrane, vaginal apical repair vs SCP: the vaginal route gives more prolapse awareness (relative risk 2.1), repeat surgery (2.3), recurrence (1.9), SUI (1.9) and dyspareunia (2.5), yet it suits many older women.

ALGORITHM

1. If she is elderly or comorbid and not sexually active, then offer colpocleisis.
2. If she is fit, has primary prolapse and prefers the vaginal route, then offer USLS or SSLF, with or without vaginal hysterectomy or hysteropexy.
3. If young or sexually active, or with recurrence, short vagina or high stage, then offer SCP (minimally invasive).

6 · Hysterectomy vs uterine preservation

- †Hysterectomy alone does not cure prolapse, so always resuspend the apex.
- Sacrospinous hysteropexy is non-inferior to vaginal hysterectomy with USLS.
- †Preservation needs a normal uterus: exclude abnormal bleeding, endometrial or cervical disease and Lynch syndrome, and keep cervical screening up to date.

7 · Obliterative surgery — colpocleisis

For frail older women it is quick, low-risk and effective.

- **Total colpocleisis** (after hysterectomy) denudes and inverts the whole vagina. **Le Fort** (uterus in place) excises anterior and posterior rectangles and leaves **lateral channels** for drainage.
- Both need **high perineorrhaphy and levator myorrhaphy** and cystoscopy. If the uterus stays, evaluate the endometrium; hysterectomy removes cancer and **pyometra** risk but raises complications.
- Spare ≥**1.5 cm** of distal anterior wall, because urethral traction causes SUI. Add a sling for SUI or occult SUI.
- Success is ~90–98% and satisfaction >90%, but regret reaches **13%**.

8 · Posterior compartment

Bowel symptoms often persist after rectocele repair.

- Symptoms are bulge, splinting or digitation, trapping (**not faecal incontinence**) and dyspareunia. First exclude slow transit and **anismus**.
- **Benefit is predicted** by bulge, splinting and a large rectocele, but not by constipation. **Constipation often persists**.
- **The transvaginal route beats transanal** (recurrence 7% versus 66%).
- **Midline fascial plication**, with or without perineal body repair, is the International Consultation on Incontinence (ICI) procedure of choice. In **site-specific** repair the commonest defect is **transverse**.
- Perineorrhaphy should leave a smooth contour, because ridges cause dyspareunia.

ALGORITHM

1. Repair anterior and apical compartments first; hydrodissect and incise in the midline.
2. Dissect the rectovaginal fascia off the epithelium; if the hernia is large, place a purse-string at its base.
3. Plicate in the midline (2-0 absorbable), incorporate the perineal body, trim and close.

- **Levator plication has been abandoned** because dyspareunia reaches 50%. **Grafts** are not supported (complications RR 1.8).

9 · Complications across POP surgery — summary

Problem	Prevention / management
Ureteric injury (USLS highest, 1–11%)	Bites lateral to medial; cystoscopy after tying; remove the most lateral suture; stent; reimplant
Bladder/rectal injury	Correct plane, drained bladder, rectal finger; layered closure

Problem	Prevention / management
Mesh exposure	Type I mesh, full-thickness flaps, supracervical hysterectomy; oestrogen or excision
Recurrence	Anterior after USLS/SSLF, posterior after SCP; most are asymptomatic
Buttock pain	Middle-third placement; observe up to 3 months; nerve injection
De novo SUI / retention	Reduction testing; selective sling; CIC; sling incision

10 · Landmark trials

Trial	Design	Result	Take-home
CARE	RCT: SCP with or without Burch, continent women, stage 2–4	SUI 24% vs 44% at 3 months	Prophylactic Burch reduces de novo SUI
E-CARE	7-year extension	SUI 62% vs 77%; mesh exposure ~10%; composite failure 34–48%	The benefit is a delay; exposure rises with time
OPUS	RCT: vaginal POP repair with MUS vs sham	27% vs 43% at 12 months; more perforation, UTI, bleeding, retention	Selective sling
OPTIMAL	RCT: USLS vs SSLF, with or without PFMT	Equal (~65%) at 2 y; PFMT no effect	Native apical repairs are equivalent
Bilateral SSLF vs abdominal SCP	RCT	Earlier failure and 3× cystocele reoperation after SSLF	Abdominal more durable
Transvaginal vs transanal rectocele	RCT	Recurrence 7% vs 66%	Repair vaginally

119 Genitourinary Complications of Mesh and Their Repair

1 · Mesh, terminology and regulation

- Unlike the abdominal wall, the pelvic floor is mobile, contaminated and close to viscera. Mesh harms are pain, exposure, extrusion and injury to nearby structures.

Amid type	Pore	Example	Behaviour
I	Macroporous >75 µm	Knitted monofilament polypropylene	**Only recommended mesh**
II	Microporous ~10 µm	Polytetrafluoroethylene (PTFE, Gore-Tex)	Encapsulated; >30% full-length slings excised
III	Macropores in microporous base	Mersilene, Mycromesh	Extrusion ~20%
IV	Submicronic	Silicone-coated	Extrusion/sinus up to 40%

KEY POINT

Why pore size matters: bacteria (~1 µm) enter pores that macrophages (~50 µm) cannot, so infection is sealed from immunity. Large pores (~1500 µm) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. Mesh adds no cancer or autoimmunity.

*See the paper answer **Management of perforated mid urethral sling** (Section 6, Ch 118).*

Also, from the condensed book (not in the paper answer):

- IUGA/ICS **CTS (Category–Time–Site) coding**: Category 1–3 vaginal (1 no separation; 2 exposure ≤1 cm; 3 >1 cm or extrusion), 4 urinary tract, 5 bowel, 6 skin/musculoskeletal, 7 systemic; suffix A asymptomatic, B symptomatic, C infection, D abscess. Time T1 ≤48 h, T2 48 h–2 months, T3 2–12 months, T4 >12 months. Site S1 vaginal suture line, S2 vagina away from it, S3 trocar path, S4 other skin/musculoskeletal, S5 intra-abdominal.
- The FDA classed transvaginal POP mesh as **class III (2016)** and it stopped selling in **2019**; slings and sacrocolpopexy (**transabdominal POP mesh, class II**) are exempt and proven safe, whereas mini-slings are not.

2 · Risk factors and prevention

- **Contraindications to vaginal mesh** are simultaneous urethrovaginal fistula repair, urethral injury or mesh perforation, and diverticulectomy. With **intraoperative urethral injury, abandon mesh** and place a sling later.
- Relative risks are prior fistula or diverticulum surgery, radiation, oestrogen-deficient tissue, smoking, obesity and age extremes; in men, radiotherapy and urethrotomy.
- POP mesh risk rises with high mesh volume, **shallow vaginal dissection**, hysterectomy, diabetes and sexual activity.

3 · Evaluation

ALGORITHM

1. History and **prior operative notes**.
2. Speculum and pelvic exam: exposure, bands, point tenderness, levator hypertonicity.
3. If haematuria, recurrent UTI, storage LUTS or voiding pain, then cystourethroscopy.
4. If voiding dysfunction, then urodynamics (UDS).
5. Translabial ultrasound for a sling; **MRI** for sacrocolpopexy mesh (invisible on CT); CT for haematoma.

Time never excludes a complication; slings can expose decades later.

4 · Midurethral sling complications

Vaginal exposure

- It follows infection, haematoma, excess tension or superficial placement, and presents with bleeding, discharge, dyspareunia, partner pain or UTI.
- A small early exposure gets **topical oestrogen** (1 g cream nightly for 2 weeks, then 2–3 times weekly; or estradiol 10 µg tablet).
- **Excision**: raise a flap, expose the full width of mesh, pass an instrument behind, divide, excise at least **1 cm** around the defect and close. Avoid blind lateral dissection, which causes vulval haematoma after transobturator (TO) slings.

Pain

- Causes are shrinkage, tension, obturator neuralgia and pelvic floor hypertonia.
- **Immediate severe pain**: exclude haematoma and consider **early removal** before ingrowth.
- Delayed focal pain gets pelvic floor physical therapy, trigger point injection and oestrogen, then segment excision (60–90% improve). Diffuse late pain reflects central sensitisation, and excision may fail.
- Remove **as much vaginal mesh as possible** (81% pain-free). For TO **groin pain**, add a thigh incision to obturator externus.

Bladder perforation

- Cystoscopy with a **full bladder** detects trocar puncture. A missed one causes storage LUTS, haematuria, UTI, stones or vesicovaginal fistula (VVF).
- **Endoscopic** removal has low morbidity, but mesh stays in the wall and re-extrudes. **Open or robotic transvesical** removal excises 2 cm beyond the defect with two-layer closure and a catheter for 2–3 weeks; re-extrusion is minimal.

Urethral perforation

- Causes are trocar injury, over-tensioning and **urethral dilation** (avoid it; divide an obstructing sling). It presents with voiding pain, retention or haematuria.
- Repair: inverted-U flap, peel the mesh off, minimal urethrotomy, layered closure with or without a **Martius** flap, and non-overlapping suture lines. A stricture needs buccal urethroplasty.
- After excision, SUI occurs in **42%**, fewer with a simultaneous fascial sling.

Voiding dysfunction

- Midline **sling division** resolves retention in 95% and symptoms in 75%. Urgency incontinence persists in up to 76% and is de novo in up to 45%. Recurrent SUI (20–74%) is treated by bulking, repeat MUS or fascial sling.
- Persistent symptoms need UDS, then wider excision and urethrolysis.

> **KEY POINT**
>
> **How much to remove**: retention needs midline division; exposure at least 1 cm each side; pain maximal vaginal removal, rarely arms. **Never remove asymptomatic mesh**; more removal means more bleeding and more POP and SUI recurrence.

5 · Transvaginal POP mesh

- In the largest randomised trial (n = 1358), native and mesh repair were equal at 6 years (reoperation 12% vs 10%); 8% of the mesh arm needed excision.
- Exposure is 10–12% and removal 5–8%. **Contraction** (~12%) is unique to POP mesh and causes tightness and dyspareunia. Bladder perforation is 0.7–3.5%, rectal 0.15% and fistula 0.2%.
- Treat **conservatively** (observe, oestrogen) if <1 cm, mildly symptomatic and uninfected; only **size** predicts success. After office trimming, 73% later need theatre.
- **Excision**: raise thick flaps, then clamp behind the mesh and divide, which is the **most dangerous step**. Cut arms under tension rather than chasing them, then repair natively.
- Afterwards, cystocele recurs in 19%, 20% need repeat excision and 3% develop VVF.

6 · Sacrocolpopexy mesh

- Exposure is ~**10.5% at 7 years**, higher with total hysterectomy (23% vs 5%) and stage 3–4.
- Features are discharge, granulation, dyspareunia, cuff abscess, bladder or bowel perforation and **discitis**; image cross-sectionally.
- Cut an **exposed suture** below the knot in the office. Oestrogen for small exposures and vaginal excision often fail; partial colpocleisis succeeds in ~50% but is unsuitable if sexually active.
- **Abdominal removal** suits failed vaginal repair, pain, infection, discitis or ureteric complication. **Remove the entire infected mesh**, including sacral sutures, because residual mesh causes sinus, abscess and fistula; bowel injury occurs in up to 29%.

7 · Male slings

- **Advance XP** causes retention in 14–21%, usually for days; incise one arm if it persists. **Virtue** causes perineal or scrotal pain in 14–17%, chronic in 0–7%.
- **Urethral extrusion** usually follows a perforation missed without intraoperative cystoscopy. Excise lateral to the urethra and repair primarily or with buccal graft, avoiding transection.

121 Urinary Tract Fistulae

1 · Definition and universal principles

- A **urinary tract fistula** is an abnormal epithelial- or mesothelial-lined connection from the urinary tract to another cavity (bowel, vessel, genital tract) or skin.
- Causes are iatrogenic (commonest in the developed world), malignancy, inflammation, infection, ischaemia, trauma and childbirth.
- It persists because **urine keeps flowing through a tract that cannot heal**, so it epithelialises. Every principle diverts flow or restores healing, and the **first repair has the best chance**.

Management (whole course) — Table 121.1	Operation (any route) — Table 121.2
Optimise nutrition	Evaluate concomitant injury; rule out malignancy
Control infection	Adequate exposure; remove all foreign material
Continuous drainage (catheter/stent)	Tension-free closure; vascularised edges
Eliminate obstruction	Multiple, watertight, **non-overlapping** layers
Consider malignancy — **always biopsy** if any history	Consider interposition; drain after

> **KEY POINT**
>
> Trap: a fistula may be the only sign of recurrent cancer, so biopsy the tract after prior malignancy or radiation.

2 · Vesicovaginal fistula (VVF) — aetiology

VVF is the commonest acquired urinary fistula (**>75%** of genitourinary fistulae).

	Developing world — obstetric (>90%)	Developed world — iatrogenic
Cause	Obstructed labour; young, small pelvis, poor nutrition	Unrecognised surgical injury; **>75%** after hysterectomy
Mechanism	Fetal head impacted for days causes pressure necrosis and a large defect	Dissection, cautery, clamp crush, bladder caught in cuff closure
Extent	Trigone, urethra, ureter often involved; "obstructed labour injury complex" (renal, rectal, orthopaedic, neurological, psychosocial)	Solitary, **supratrigonal at vaginal cuff**

- **Hysterectomy** causes VVF in ≈ 0.8/1000, lowest after vaginal and highest after laparoscopic surgery. Distorted anatomy and concomitant prolapse, incontinence or mesh surgery raise risk; immediate recognition and repair prevent fistula.
- **Universal cystoscopy at hysterectomy** raises intraoperative detection but has an inconsistent effect on delayed injury. AUGS advises it with any prolapse or anti-incontinence procedure, AAGL after laparoscopic or robotic hysterectomy, and ACOG makes no recommendation after benign hysterectomy.
- **Radiation** causes obliterative endarteritis, fibrosis and necrosis. VVF follows 0.6–2% of cervical cancer radiotherapy, typically **1.5–2 years** later (up to 30); risk is dose-dependent and worse in smokers.

- Other causes are mesh, **bevacizumab** (anti-VEGF) and cancer, and rarely pessary, intrauterine device, infection or endometriosis.

3 · Classification of VVF

Of >25 systems, the WHO uses simple vs complex.

Axis	Goh	Waaldijk
Basis	Distance of distal edge from external urethral meatus	Closing (continence) mechanism and urethral involvement
Types	1: >3.5 cm · 2: 2.5–3.5 · 3: 1.5–2.5 · 4: <1.5 cm	I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement (A/B = without/with circumferential defect) · III: miscellaneous (ureteric)
Size	a <1.5 · b 1.5–3 · c >3 cm	Small <2, medium 2–3, large 4–5, extensive ≥6 cm
Tissue	i none/mild fibrosis · ii moderate/severe · iii special (radiation, ureteric, circumferential, previous repair)	—

KEY POINT

Trap: Goh type 1 is farthest from the meatus (best) and type 4 closest (worst); closer means worse closure and continence. Goh predicts closure better. Always describe size, location (trigonal vs supratrigonal), tissue loss, and ureteric and urethral involvement.

4 · Diagnosis of VVF

- The classic symptom is **continuous leakage day and night** in any position; a small tract may leak only positionally.
- **Timing points to mechanism**: a missed cystotomy leaks in the first days, thermal or crush injury at **7–10 days**, and radiation after years. Fever, flank pain or ileus suggest urinoma or ureteric injury.
- On speculum the defect is usually at the vaginal apex after hysterectomy, and vaginal fluid creatinine approximates urine creatinine. Biopsy friable tissue after cancer or radiation, and optimise oestrogen, smoking, glucose and nutrition.
- **Single dye test**: backfill ~300 mL methylene blue, insert a tampon and walk 15–60 min. Proximal stain means VVF, distal stain means urethrovaginal fistula or SUI, and dye from the cervical os means vesicouterine fistula.
- **Double dye test**: oral phenazopyridine 100–200 mg plus blue in the bladder, then a tampon. **Orange means ureterovaginal fistula (UVF)**, blue means VVF, and both means both.
- **Cystoscopy is integral** (sensitivity ~92%) and shows size, number, orifices, bladder neck, mesh and capacity, and allows biopsy. Vaginoscopy shows the vaginal side. A cystogram (lateral view) or T2 MRI helps difficult cases.
- **Upper tract imaging is mandatory**, because ureteric injury coexists in up to **12%**. Use CT urography with a delayed phase, or MR urography if contrast allergy or renal failure. **Partial ureteric obstruction with VVF means UVF until proven otherwise.** If unclear, do retrograde pyelogram and stent, or percutaneous nephrostomy (PCN).

- Urodynamics helps planning: 10–15% have SUI needing an autologous sling, and radiation may reduce capacity and compliance.

> **KEY POINT**
>
> Work-up order: is it urine (creatinine), where is it from (dye tests, endoscopy), and what else is damaged (hence compulsory upper tract imaging). A preoperative PCN also allows antegrade ureteroscopy to find the ureter in scar.

5 · VVF — timing, conservative care, repair

LAQ Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 LAQ "Vesicovaginal fistula (VVF): management options and early vs delayed repair" · P2 SAQ "VVF repair: early versus delayed timing, and surgery for recurrent VVF" · P1 SAQ "Eponymous techniques of VVF repair"

Definition and aetiology

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract (communication) causing continuous leakage; the commonest acquired urinary fistula (>75% of genitourinary fistulae).
- **Developing world (low-resource settings):** obstetric (obstructed labour) in >90%; the fetal head impacted for days causes pressure necrosis.
- **Developed world (high-resource settings):** iatrogenic in >75%, mainly after hysterectomy, usually supratrigonal at the vaginal cuff.
- Other causes: radiation, malignancy, mesh, foreign body.
- Common mechanism: **ischaemic necrosis**; continuous urine flow keeps the tract open until it epithelialises with time.
- Time to presentation: cystotomy, days; thermal injury, 7–10 days; radiation, months to years.

Classification

Goh (distance from meatus)	Size	Tissue
1: >3.5 cm; 2: 2.5–3.5; 3: 1.5–2.5; 4: <1.5 cm	A <1.5 cm; B 1.5–3 cm; C >3 cm	i mild fibrosis; ii moderate/severe; iii radiation, ureteric, circumferential, previous repair
- **Also know:** Waaldijk (closing mechanism involvement); WHO (World Health Organization) simple vs complex.		

Evaluation (guides choice)

- Speculum; fluid creatinine.
- **Dye tests:** single (blue backfill, tampon); **double** (oral phenazopyridine plus blue): orange = UVF (ureterovaginal fistula), blue = VVF.
- **Cystoscopy and vaginoscopy:** number, size, relation to ureteric orifices and bladder neck, mesh; **biopsy the tract if any history of cancer** (the fistula may signal recurrence).

- **Upper tract imaging (CTU, computed tomography urography) is mandatory:** to exclude a coexisting ureteric (ureterovaginal) fistula (up to 12–25%).
- Urodynamics: capacity, compliance, SUI (stress urinary incontinence) in 10–15%.
- Prepare: treat UTI (urinary tract infection), optimise nutrition, vaginal oestrogen for atrophy; counsel about residual incontinence, dyspareunia and possible reoperation.

Principles of management

- Optimise nutrition, treat infection, drain continuously, relieve obstruction, exclude malignancy.
- Surgery: excise scar, wide mobilisation, tension-free, multilayer, **non-overlapping**, watertight closure; remove foreign material; consider interposition; drain afterwards. **The first repair has the best chance.**
- At repair: treat infection; ureteric stents if within 1 cm of a ureter; vaginal or abdominal route (open, laparoscopic, robotic); **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter drainage for about 2 weeks (≥10–14 days).

Option 1: Conservative and endoscopic (conservative trial while waiting)

- **Catheter drainage** cures about 15%. Best if small (<1 cm, ideally <3 mm), fresh, started (catheterised) within **3 weeks**, and non-irradiated. Once epithelialised (≥6 weeks), closure is unlikely. EAU (European Association of Urology): if tried, at least 12 weeks.
- **Fulguration** plus 2 weeks' drainage for tiny tracts <3.5 mm.
- **Sealants** (fibrin glue, cyanoacrylate, collagen): tiny series; not recommended by EAU.

Timing: early versus delayed repair (the debate)

	Early repair	**Delayed repair**
Definition (varies)	Days, <6 weeks or <3 months	Traditionally ≥3 months
Rationale	Clean iatrogenic tissue heals well; ends wetness and distress early	Inflammation settles, necrotic tissue sloughs, margins become clear
Evidence	**Comparable closure** in most series; Waaldijk: immediate obstetric repair gave very high first-attempt closure	One small series: more recurrence when repaired <6 weeks
Drawback	Friable tissue if inflamed	Prolonged distress, no better outcome

- Some authors avoid the **4–12 week** window (active healing phase).
- **EAU consensus:** no minimum delay if there is **no active infection and no recent irradiation**.
- **Timing by cause:**
- **Clean iatrogenic / post-hysterectomy:** early repair.
- **Obstetric with necrosis (necrotic edges):** catheter trial, then repair at about 3 months.
- **Radiation:** wait **≥12 months** after radiotherapy; biopsy to exclude recurrence; interposition flap; consider diversion if the bladder is poor.
- **Recurrent (redo) fistula:** wait until inflammation settles (commonly about 3 months); radiation ≥12 months after radiotherapy; failed radiation repairs were redone after **≥6 months**.

Option 2: Surgical route

Vaginal	Abdominal (open, laparoscopic, robotic)
Low, distal or trigonal, accessible apex	Supratrigonal, narrow vagina, no descent
Ureters distant	Ureter involved, needs reimplantation
Less pain, blood loss, shorter stay	Augmentation, **Tanagho bladder flap** or diversion possible
Flap-splitting, offset flaps	Large (≥4 cm primary, **≥2 cm recurrent**), radiation
Failed **Latzko** can be cured by a second Latzko	**O'Conor** bivalving finds the scarred plane
Risk: vaginal shortening	
- First-attempt closure about **90% by either route**; choose the route the surgeon knows best.	

Eponymous and named techniques: transvaginal

- **Sims:** first successful systematic vaginal repair (silver wire).
- **Flap-splitting (Raz):**
- Ureteric stents if within 1 cm; Foley through the tract for traction.
- Inverted J or U incision with a 1–3 mm cuff; raise anterior and posterior vaginal flaps.
- Layer 1: bladder (3-0 or 4-0), including the tract edge. Layer 2: perpendicular perivesical layer. Fill test 200–300 mL.
- Optional Martius or peritoneal flap; advance the opposite vaginal flap ≥2 cm beyond. Closure ≥90%.
- **Latzko (1942), high partial colpocleisis:**
- Denude vaginal epithelium 1–2 cm around the tract; the bladder is **not opened or dissected**.
- Close the vaginal walls over the tract in layers.
- ≥90% success in non-irradiated post-hysterectomy apical VVF; favoured for radiation VVF and tracts near the ureters. Drawback: **vaginal shortening and dyspareunia**.
- **Also know:** vaginal cuff excision (all apical scar excised as an inverted cone).

Eponymous and named techniques: abdominal

- **O'Conor and Sokol (1951), suprapubic extraperitoneal:**
- Midline cystotomy at the dome, **bivalved down** to the fistula; ureteric catheters.
- Excise the tract; separate bladder from vagina 2–3 cm beyond.
- Close the vagina; **omental or peritoneal flap** between; close the bladder in layers; suprapubic catheter.
- Modifications: **transvesical** (limited cystotomy) and **extravesical** (no cystotomy; fewer LUTS [lower urinary tract symptoms]).
- **Laparoscopic (1994) and robotic (2005):** equal closure (85–100%) with less blood loss, shorter stay and less morbidity; vaginal stent, backfill, ICG (indocyanine green) perfusion check. Robotic repair of recurrences gives high closure.

Interposition flaps (radiation, recurrent, ischaemic or large VVF)

Flap	Blood supply	Route	Use
Martius (1928) (labial fibrofatty pad)	Internal pudendal (inferior), external pudendal (superior)	Vaginal	Mid–distal VVF, bladder neck; does not reach the apex
Peritoneal (vaginal)	Local; pouch of Douglas fold	Vaginal or abdominal	High apical post-hysterectomy VVF
Omentum (J-flap)	Gastroepiploic arteries	Abdominal	Radiation, recurrent, large
Gracilis, Singapore, rectus	Myocutaneous	Vaginal/perineal	Refractory, large tissue loss
- RCT (randomised controlled trial): Martius added nothing in simple VVF, with donor-site pain in nearly a third.			
- **Interposition is widely used in recurrent fistula** (accepted, though not proven superior).			
- **Also know:** rectus abdominis flap (last resort).			

Recurrent VVF

- Persists after a failed repair; redo tissue is scarred and poorly vascularised, so the first repair has the best chance.
- Failure causes: tension, ischaemia, overlapping suture lines, infection, poor drainage, **malignancy, radiation**, missed second tract.
- Before redo: biopsy, cystoscopy and vaginoscopy, CT urography and dye test to exclude a **ureterovaginal fistula**, urodynamics; treat infection, optimise nutrition.
- Route: abdominal for recurrent ≥2 cm; a failed Latzko can be redone vaginally; robotic repair gives high closure.
- Interposition flap; catheter ≥10–14 days, then cystogram.
- More than one repair predicts residual incontinence.

Postoperative care

- Catheter ≥10–14 days (urethral and/or suprapubic); antimuscarinic or β3-agonist for spasm; cystogram before removal; stents out at 4–6 weeks.

Option 3: Urinary diversion (when repair is unwise)

- Multiple failures, severe radiation damage, massive tissue loss: conduit or continent reservoir/diversion.
- Unfit: ureteric occlusion with nephrostomies.
- Resource-poor: ureterosigmoidostomy as a last resort.

Outcomes

- Residual urgency or incontinence despite closure (Goh 3–4 predicts; more than one repair predicts); retention (vertical bladder closure); dyspareunia (narrow vagina, fistula ≥3 cm). ICS (International Continence Society): pelvic floor physiotherapy.

Diagrams to draw

- Flap-splitting repair: perpendicular bladder layers, Martius flap, advanced vaginal flap.
- Latzko denuded area versus flap-splitting layers.
- Timing timeline: early window, 4–12 week zone, 3 months (obstetric), 12 months (radiation).
- Martius flap between bladder and vaginal closures.

Key points to remember:

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

Also, from the condensed book (not in the paper answer):

- Size >3 cm predicts failure, and radiation strongly predicts failure with ureteric stricture, poor compliance and sphincter deficiency, so consider diversion.
- Apical VVF (8–10 cm from introitus) lies beyond Martius reach, so use peritoneum or omentum.

Setting	Timing of repair	Rationale
Obstetric (traditional)	≥3 months, catheter meanwhile	Inflammation settles, necrosis sloughs
Iatrogenic, clean tissue	Early; European Association of Urology (EAU): no minimum delay if no infection or recent radiation	Comparable closure; less distress
Radiation-induced	≥**12 months** after radiation ends	Late injury still evolving

- **Catheter** drainage is favourable for tracts <1 cm catheterised within **3 weeks**, and unfavourable at ≥6 weeks or with radiation or malignancy. The EAU trial lasts ≥**12 weeks**. Sealants are possible for <1 cm in poor surgical candidates.
- **Location**: within **1 cm** of a ureter, stent it; if the ureter is involved, reimplant abdominally. Bladder neck involvement needs reconstruction, plus a pubovaginal sling if needed.
- **Route**: there are no RCTs, and first-attempt closure is ≈ **90%** either way. Transvaginal repair has less blood loss, pain and stay.
- Abdominal repair suits large fistulae (≥4 cm primary, ≥2 cm recurrent), urethral, bladder-neck or ureteric involvement, a non-capacious vagina, radiation, or no lithotomy. Vaginal repair risks shortening and dyspareunia, whereas abdominal repair spares sexual function.

ALGORITHM

1. **Transvaginal flap-splitting (Table 121.5)**
2. Cystoscopy: stent close ureters; wire through the tract.
3. Suprapubic catheter (SPC) and urethral catheter; episiotomy if narrow.
4. Foley ≥8 Fr through the fistula for traction; hydrodissect.
5. **J- or U-incision** with a 1–3 mm cuff; raise wide anterior and posterior flaps.
6. Layer 1: 3-0/4-0 delayed absorbable. Layer 2: **perpendicular** to layer 1.
7. Leak test 200–300 mL; Martius or peritoneal flap if needed.
8. Advance the opposite flap ≥2 cm beyond (layer 3); cystoscopy for efflux.

- **Postoperative drainage is the most important part**: urethral catheter, often with SPC, for ≥10–14 days. Give an anticholinergic or β3-agonist for spasms, remove stents at 4–6 weeks, and advise pelvic rest for ~6 weeks.
- **Latzko** leaves the bladder unopened and the tract unexcised: denude 1–2 cm and close in two or more layers. It succeeds in ≥90% of non-irradiated cases and suits close orifices, but shortens the vagina.

ALGORITHM

1. **O'Conor (Table 121.7)**
2. Ureteric catheters; low midline extraperitoneal approach.
3. Sagittal dome cystotomy; **bivalve** down around the fistula; excise the tract.
4. Separate bladder from vagina ≥2–3 cm beyond; close the vagina.
5. Place **omentum or peritoneum** beyond the vaginal suture line.
6. Close bladder in 1–2 layers; SPC with or without urethral catheter; fill test.
7. Drain 10–14 days, then cystogram.

- O'Conor causes more de novo urgency and LUTS from long suture lines. Transvesical repair suits low-lying fistulae, and extravesical repair suits supratrigonal fistulae. Robotic O'Conor uses omentum or peritoneum, with indocyanine green to check edge vascularity.

6 · Interposition flaps

- Flaps reinforce, fill dead space and import blood supply. Use them for **complex VVF** (radiation, recurrent, ischaemic obstetric, poor tissue); an RCT showed **no benefit of Martius** in simple VVF.
- **Martius** is supplied by the internal pudendal (inferior), external pudendal (superior) and obturator (lateral) vessels. Keep the **superior** pedicle for VVF (inferior for rectovaginal). It reaches mid-distal VVF, bladder neck and urethra, but **not the apex**.
- A vaginal peritoneal flap suits the apex, and **omentum** suits radiation, recurrent or large fistulae. Rectus, gracilis and Singapore flaps are a last resort.
- After repair, residual incontinence (predicted by Goh 3–4 and more than one repair) drives quality of life, and ICS advises physiotherapy for all obstetric fistula. Divert if the fistula is unrepairable.

7 · Ureterovaginal fistula (UVF)

LAQ Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula

Paper answer · P3 · merged from 2

Asked as: P3 LAQ "Urological complications in OBS-GYN and uretero-vaginal fistula" · P3 LAQ "Urological complications of OBGY surgery and urethrovaginal fistula"

Assumed meaning: OBS-GYN / OBGY = obstetric and gynaecological practice and surgery.

Part A: Urological complications in OBS-GYN

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding, prolapse or mesh surgery.

By setting

Setting	Complications
Obstetric	**Obstructed labour** (fetal head impacted for days): pressure necrosis, so large VVF (vesicovaginal fistula), urethral loss, ureteric involvement; over 90% of VVF in the developing world. **Caesarean**: bladder injury, ureteric injury, **vesicouterine fistula** (Youssef syndrome: cyclical haematuria, amenorrhoea, continence preserved). Post-partum or post-epidural **retention** (overdistension)
Gynaecological surgery	**Bladder injury** (commonest organ). **Ureteric injury** (over 90% distal third; ligation commonest). **Fistulae**: VVF (over 75% follow hysterectomy in the developed world), UVF (ureterovaginal fistula), urethrovaginal fistula
Radical hysterectomy	Pelvic plexus injury: retention, poor compliance; manage with CIC (clean intermittent catheterisation)
Prolapse and continence surgery	Ureteric kinking at apical suspension; mesh or sling exposure, perforation, outlet obstruction, pain
Radiotherapy	Radiation cystitis, VVF, ureteric stricture

By complication: key facts

Complication	Key facts
Bladder injury (commonest organ injured)	Caesarean 0.27–0.3%; hysterectomy about 0.3–0.75% (laparoscopic highest); retropubic tape 3–9%. Repair: debride, 2-layer closure, fill test, catheter 7–14 days; check ureters if trigonal
Ureteric injury	Gynaecology causes over half of iatrogenic injuries; over 90% distal third. Hysterectomy overall 0.2–2.4%: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery 1.7–4.3%. **Ligation** is the commonest gynaecological mechanism
Fistulae	VVF about 0.8 per 1000 hysterectomies; UVF; vesicouterine fistula after caesarean (Youssef syndrome); urethrovaginal fistula
Voiding dysfunction	Radical hysterectomy (pelvic plexus injury) 16–80% historically: retention, poor compliance; CIC and wait. Post-partum or post-epidural overdistension
Mesh and sling	Exposure, erosion into bladder or urethra, outlet obstruction, pain, dyspareunia

Complication	Key facts
Radiation	Radiation cystitis; VVF in 0.6–2% after cervical cancer radiotherapy; ureteric stricture

- **Ureteric danger sites (four):** pelvic brim (near the infundibulopelvic ligament); **under the uterine artery** in the cardinal ligament; lateral to (beside) the uterosacral ligament; anterolateral vaginal fornix near the UVJ (ureterovesical junction). Most injuries are in the **distal 6 cm**.
- **Prevention and recognition:** haemostasis and anatomy; know the danger sites; prophylactic stents only selectively, if high risk (they aid recognition but do not prevent injury); cystoscopy with dye (IV indigo carmine or oral phenazopyridine) after hysterectomy where indicated, minimally invasive hysterectomy, and prolapse or continence surgery.
- Only about one-third of ureteric injuries are recognised at open surgery. A missed injury presents with fever, ileus, flank pain, urinoma or leakage; do CTU (computed tomography urography) with delayed films, then retrograde stenting. Distal repair: ureteroneocystostomy ± psoas hitch or Boari flap (see Part B).

Part B: Ureterovaginal fistula (UVF)

Definition

- Abnormal communication between the ureter and the vagina, causing continuous leakage.

Aetiology and pathogenesis

- Usually an **unrecognised injury to the distal third of the ureter** during pelvic surgery. About **three-quarters follow an often uneventful abdominal hysterectomy**.
- **Mechanisms:** ligation, crush, laceration, thermal injury, **devascularisation** or ischaemia. The injured segment necroses and leaks into the vaginal cuff, which is the weakest point.
- Diagnosis is delayed in 18–88%; **delayed recognition raises fistula risk about fourfold**.
- Other causes: obstetric, cancer, radiation, endometriosis, foreign body.

Clinical features

- **Continuous leakage with normal voiding** (the other kidney fills the bladder), usually appearing **within 4 weeks**, often with fever, flank pain or ileus, as thermal or ischaemic injury necroses.

	VVF	UVF
Voiding	Reduced	Normal
Upper tract	Normal	Hydronephrosis common
First line	Catheter, repair	Stent

Investigations (confirm before treatment)

Test	Method and reading
Double dye test	Oral phenazopyridine 100–200 mg (orange urine); after 1 hour, blue dye instilled into the bladder; tampon; walk about. **Orange = UVF**; blue = VVF; both = both
Three-swab test	Blue dye in the bladder; three swabs stacked in the vagina. Blue upper swab = VVF; **wet, unstained swab = UVF**

Test	Method and reading
CT urography, delayed phase	Hydronephrosis, extravasation; MR (magnetic resonance) urography if contrast is contraindicated
Cystoscopy + RGP (retrograde pyelogram)	Abrupt cut-off or contrast leak; stent at the same sitting
- **Exclude a coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury.	

Management

- **Goals:** dryness and preservation of renal function. Management has moved from nephrectomy, through reimplantation, to endourology.
- **1. Endourological (first line): stent first**
- **Retrograde double-J stent as soon as possible** if a wire passes. Stenting **within 2 weeks** of detection works best, succeeding far more often than at 2–6 weeks, because the ureter heals over the stent before fibrosis.
- Wire will not pass: **PCN (percutaneous nephrostomy)** with antegrade stent, or rendezvous.
- Stent for **4–6 weeks**, then remove and re-image. Most stentable fistulae close; a stricture may follow.
- **2. Surgical repair (stent fails or impossible)**
- **Timing:** no consensus. Early repair equals delayed: early repair took less time and blood loss with equal success; a systematic review found equal success before and after 6 weeks. Delay only if tissue is still inflamed.
- **Ureteroneocystostomy** (distal third, so usually feasible):
  1. Transperitoneal or extraperitoneal approach; identify the ureter above the fistula.
  2. Divide above the diseased segment; dissect as distally as possible for length; **preserve periureteric adventitia**, especially laterally where vessels enter.
  3. Mobilise the bladder; spatulate; tension-free mucosa-to-mucosa anastomosis over a double-J stent; refluxing or tunnelled.
  4. Drain; catheter 7–14 days.
- **Adding length:**

Technique	Extra reach
Psoas hitch (sutures parallel to psoas fibres, spare genitofemoral nerve)	Up to about 5 cm
Boari flap (base ≥4 cm, length:base ≤3:1)	10–15 cm
Downward nephropexy; ileal ureter	Longer defects
- Open, **laparoscopic and robotic repair give equal results**; vaginal reimplantation reported. **Success ≥90%.**	
- The vaginal opening is left to heal; no separate vaginal repair is needed.	

Algorithm `Leak + normal voiding -> exclude VVF (dye test, cystoscopy) -> CT urography / RGP -> wire/stent passes? Yes -> double-J 4-6 wk -> re-image ->`

healed / persistent No -> PCN + antegrade stent -> fails -> ureteroneocystostomy + psoas hitch / Boari flap

Complications and follow-up

- Hydronephrosis, infection, ischaemic stricture.
- Ultrasound 1–2 weeks after stent removal; some add a renal scan at 6 weeks. Stricture usually appears within a year but may appear **up to 30 months** later, so follow beyond this.

Part C: Urethrovaginal fistula

Aetiology

- Developing world: **obstructed labour**, often with a VVF; also trauma, radiation, catheter injury.
- Developed world (iatrogenic): **urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of urethrovaginal fistulae), slings, bulking agents, anterior repair or mesh, indwelling catheters (pressure necrosis), pessaries, radiation, trauma; also gender-affirming and paediatric genital surgery.

Clinical features

- Depend on site relative to the sphincter. **Distal** fistula: often asymptomatic, split stream, vaginal voiding, post-void dribble. **Proximal** (bladder neck): continuous leakage or SUI (stress urinary incontinence).
- Dyspareunia, recurrent UTI (urinary tract infection), perineal irritation; partner discomfort with exposed mesh.

Evaluation

- **Speculum examination** usually suffices; **methylene blue test** if rugae hide it (distal stain on tampon). Assess prolapse, SUI, atrophy.
- **VCUG (voiding cystourethrogram), lateral view;** VUDS (videourodynamics) for associated SUI or detrusor overactivity.
- **Cystourethroscopy** (flexible is better: outflow at the lens distends the urethra): size, site, bladder neck, exposed mesh, other fistulae. **About 20% have a separate VVF.** Image the upper tract.
- Goh location (distance from external meatus): type 1 >3.5 cm, 2: 2.5–3.5, 3: 1.5–2.5, 4: <1.5 cm. Closer means worse closure and continence.
- Optimise: treat UTI, vaginal oestrogen, nutrition; biopsy if previous cancer.

Management

- Asymptomatic distal fistula: may need no repair. Timing: early if tissue is healthy; otherwise after inflammation settles.
- **Standard: transvaginal rotational-flap repair:**

Step	Details
1. Exposure	Lithotomy (prone gives better anterior exposure); cystourethroscopy, cannulate tract; suprapubic catheter (Lowsley tractor), urethral catheter
2. Incision	Hydrodissect; circumscribe tract with 1–2 mm cuff, minimal excision; inverted **J- or U-shaped** vaginal flap incision
3. Flaps	Raise vaginal flap; mobilise periurethral tissue widely to bladder neck; open retropubic space if sling planned

Step	Details
4. Closure	**Close urethra** over a urethral sound: **4-0 absorbable** sutures, transverse or longitudinal; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test
5. Interposition	**Martius flap** (labial fat pad, external pudendal pedicle) most often; omentum, gracilis, rectus fascia occasionally
6. Cover	Advance vaginal flap beyond closure (2-0)

- **Autologous pubovaginal sling** for SUI or bladder-neck involvement; **never synthetic** (erodes the ischaemic urethra). Remove any mesh in the field.
- **Near-total urethral loss:** tubularised anterior bladder flap neourethra. Irreparable cases: diversion.
- **After repair:** urethral catheter out at 2–3 days; suprapubic catheter ≥14 days; lateral VCUG before removal; nothing per vagina for 6 weeks.

Outcomes and complications

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI in up to half. Sling then or staged later.
- Urethral stricture, recurrence, dyspareunia.

Diagrams to draw

- Ureter passing under the uterine artery; double dye test; ureteroneocystostomy with psoas hitch.
- Urethrovaginal repair: inverted-U vaginal flap, two perpendicular urethral layers, Martius flap.

Key points to remember:

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Ureteric injury is mostly distal and missed.
- UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
- Orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
- Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

Also, from the condensed book (not in the paper answer):

- Up to **25%** of VVF have a coexisting UVF. Prophylactic stents are selective (e.g. caesarean hysterectomy for placenta accreta).

ALGORITHM

1. Suspect UVF; exclude VVF; confirm with CTU or retrograde pyelogram.
2. Stent; remove at 4–6 weeks and reimage.
3. If resolved, stop. If persistent or stenting fails, then do ureteroneocystostomy.

KEY POINT

UVF: wet but voids normally. The ureter heals over a stent if reached early, so the detection-to-stent interval largely decides whether surgery is needed.

8 · Vesicouterine fistula (VUF)

- It follows **low transverse caesarean** and joins the uterine body above the internal os to the posterior bladder; a competent cervix **preserves continence**.
- **Youssef syndrome** is menouria, amenorrhoea, first-trimester miscarriage and continence. Bladder endometriosis is the differential of menouria.
- Dye appears from the cervical os, and cystoscopy shows a posterior supratrigonal defect; add cystogram, hysterosalpingography or MRI.
- Spontaneous closure is ~5%. After 3–4 months, separate transabdominally and close in layers with omentum or peritoneum, or do hysterectomy if the family is complete. Deliver later pregnancies by caesarean.

9 · Urethrovaginal fistula

*See the paper answer **Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula** (Section 6, Ch 121).*

10 · Rectourethral fistula (RUF)

LAQ Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management

Paper answer · P3

Asked as:

- Etio-pathogenesis, diagnosis and management of Recto-urethral fistula? — *Summer 2020 Paper III Q1 (LAQ)*
- Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula — *Winter 2025 Paper III Q1 (LAQ)* (also answered in 2020-Summer Paper 3, Q1, as recto-urethral fistula)

Type: LAQ (25 marks)

Assumed meaning (recto-prostatic): rectourethral fistula at the level of the prostatic urethra or the vesicourethral anastomosis.

Definition

- **RUF (rectourethral fistula):** a communication between the posterior urethra / prostatic urethra (or the vesicourethral anastomosis after prostatectomy) and the distal rectum.
- Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

Aetiology

Group	Causes
Congenital	Anorectal malformation in boys (rectobulbar or rectoprostatic fistula)
Surgical (commonest acquired)	**RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate) and open (simple) prostatectomy
Radiation and energy	**EBRT (external beam radiotherapy) plus brachytherapy** (commonest), then brachytherapy alone, then pelvic radiotherapy; cryotherapy; HIFU (high-intensity focused ultrasound): low after primary, up to about 9% after salvage HIFU
Others	Trauma (penetrating injury, pelvic fracture); Crohn disease; advanced rectal or prostate cancer

Pathogenesis

- **Surgical:** the rectum is torn during apical or posterior dissection behind Denonvilliers fascia, especially with small glands and salvage surgery. An unrecognised tear (near the apex) leaks urine (urine leaks into it), an abscess forms and the tract epithelialises.
- Rectal injury at RP: laparoscopic about 1%, open and robotic under 0.5%, salvage RP about 7%. (Another source: rectal injury is commonest with retropubic and laparoscopic RP, lowest with robotic RP, and was over 15% in historical open salvage series.)
- **Radiation or ablation:** obliterative endarteritis makes the tissue hypoxic and fibrotic. Necrosis follows months to years later. These fistulae are large and heal poorly.
- **Congenital:** abnormal cloacal development leaves the rectum ending in the posterior urethra.

Classification: Martini (after prostate cancer treatment)

Axis	Category	Definition
Stage (size on CT (computed tomography) urography or cystography)	I / II / III	≤1.5 cm / >1.5 cm / any size with sphincter damage
Grade (tissue)	0	Surgical injury, no prior non-surgical treatment
	1	Prior radiation, brachytherapy, HIFU or other energy
	2	Salvage RP or salvage ablation
- **Tissue quality (grade), not size, chiefly decides the repair.**		

Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding.
- Recurrent UTI (urinary tract infection), dysuria, urosepsis; coexisting SUI (stress urinary incontinence).

Diagnosis

- **DRE (digital rectal examination):** may feel the defect on the anterior rectal wall.
- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.

- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; gives size and site for planning.
- Upper-tract imaging to exclude ureteric injury; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: fistula repair alone will not cure severe SUI; an AUS (artificial urinary sphincter) may be needed later; a later AUS erodes more often.

Prevention

- At RP: a recognised tear without gross spillage → two-layer (multilayer) closure, air test, **omental or peritoneal interposition**, drain, antibiotics; colostomy if (previously) irradiated.
- Radiotherapy: a **hydrogel rectal spacer** lowers rectal dose and late rectal toxicity (randomised trial: late rectal toxicity 2.0% vs 7.0%).

Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close urethra and rectum separately in **tension-free, non-overlapping layers with vascularised (interposition) tissue interposed**.
- **Conservative:** no sepsis and no faecaluria → catheter (urethral or suprapubic), low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; choice depends on site, radiation and prior repair. **A diverting colostomy usually comes first.**

Approach	Essentials	Strengths	Weaknesses
Perineal (Young; Goodwin)	Separate and close urethra and rectum separately; **gracilis flap** between (or dartos; buccal graft)	High closure; most stomas reversed	Urinary incontinence common
Transsphincteric (York-Mason)	Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure	Avoids scarred plane; spares sexual function	Poor membranous exposure; no flap; faecal incontinence up to 20%
Abdominal (transvesical, extravesical, O'Conor-like) / robotic	Omental flap	Large or irradiated fistulae, extensive resection	Morbid; poor deep-pelvic view; small robotic series
Laparoscopic / robotic	Open pouch of Douglas; close rectum in two layers; omental or peritoneal flap; can combine with perineal gracilis	Deep-pelvic vision	Small series only
Salvage	Cystoprostatectomy, ileal conduit ± permanent colostomy	Unreconstructable irradiated field	Major surgery

`RUF → sepsis control, catheter ± colostomy, biopsy small, non-irradiated, no faecaluria → conservative ~12 weeks persistent, grade 0 → perineal (gracilis) or York-Mason irradiated or large → abdominal/robotic (omentum) or perineal (gracilis) unreconstructable → urinary + faecal diversion Congenital → newborn colostomy → posterior sagittal anorectoplasty`

- **PSARP (posterior sagittal anorectoplasty)** separates the rectum from the urethra and places it within (in) the sphincter complex.

Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, urethral stricture, ED (erectile dysfunction), wound infection, permanent stoma.

Diagram to draw: sagittal pelvis showing (with) the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy to exclude cancer.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Martini	Category
Stage (size)	I ≤1.5 cm · II >1.5 cm · III any size with sphincter damage
Grade (tissue)	0 surgical injury only · 1 prior radiation/energy · 2 salvage radical prostatectomy (RP) or salvage ablation

11 · Vesicoenteric fistula (VEF)

- Colovesical is commonest. M : F is **3 : 1** because the uterus shields the bladder.
- Causes in order are **diverticulitis**, **Crohn disease** (ileovesical) and colorectal cancer (10–15% of colovesical fistulae), then bladder or prostate cancer, radiation and appendicitis.
- It causes **pneumaturia** (50–70%), faecaluria and polymicrobial UTI. **Gouverneur syndrome** is suprapubic pain, frequency, dysuria and tenesmus.

Test	Role
Poppy seed test	Highest detection of presence; cheap; no location
CT (preferred)	Air in bladder, wall thickening, adherent bowel; shows cause
Cystoscopy	Nonspecific erythema/oedema
Colonoscopy	Poor for tract; **excludes cancer**
MRI	Crohn, anatomy

- Treat conservatively if symptoms are minimal and disease benign (especially Crohn); spontaneous closure is <2%.
- **Resect the diseased bowel with the fistula** with single-stage anastomosis; use Hartmann for abscess, contamination, advanced cancer or radiation. Leave or oversew a small bladder defect, do **partial cystectomy** if malignant, and interpose omentum.

> **KEY POINT**
>
> VEF is a bowel disease presenting in the bladder: remove the diseased bowel, and the bladder usually heals itself.

Renocolic fistula: spontaneous ones are mostly renal (xanthogranulomatous pyelonephritis [XGP]) and need nephrectomy; small iatrogenic ones after PCNL settle with antibiotics, drainage and nil by mouth.

12 · Urocutaneous fistulae

- **Vesicocutaneous** fistula follows long-term SPC, open prostatectomy or radiation, and persists with infection, foreign body, tumour or bladder outlet obstruction (**BOO**). Drain and relieve BOO; if unhealed, do fistulectomy or partial cystectomy.
- **Urethrocutaneous** fistula is the commonest hypospadias complication (up to 20%). Wait ≥**6 months** and exclude distal stricture. Close <1 cm locally; ≥1 cm or redo needs vascularised flaps (**tunica vaginalis** better than dartos).
- **Nephrocutaneous** fistula follows stones, XGP, prior surgery or tuberculosis. The kidney is non-functioning in ~90%, so do **nephrectomy**.

13 · Pubovesical fistula (PVF)

- A tract from bladder, prostate or urethra to the pubic symphysis causes **osteomyelitis**. It follows prostate radiotherapy plus repeated outlet procedures, median ~10 years after radiation, and also artificial urinary sphincter surgery.
- It presents with **pubic pain on walking**, abscess, urosepsis or sinus. **MRI is most sensitive and specific**; bone biopsy guides antibiotics because urine culture is unreliable.
- Conservative care usually fails. Debride bone, excise the fistula, fill with a rectus or omental flap, and give 4–6 weeks of antibiotics; if bladder or sphincter function is poor, do **cystoprostatectomy and diversion**.

14 · Urovascular fistulae

- **Arteriocalyceal** fistula follows PCNL, transplant or trauma; treat by **angioembolisation**.
- **Ureteroarterial fistula (UAF)** is an emergency. Most (85%) are secondary to long-term stents, pelvic surgery, radiation, cancer or vascular grafts. It presents with **intermittent haematuria**, and ~20% are unstable.
- Diagnose with **angiography**, provoking by stent removal only with endovascular treatment ready; a negative CT angiogram does not exclude it.
- Treat with an **endovascular stent graft** with or without internal iliac embolisation; use open surgery for infection, abscess or enteric involvement.

15 · Renal urine leak and anastomotic leak

- After **trauma**, a leak occurs in 1–7%. Use CT with a 10-min delayed phase; a non-opacified distal ureter suggests ureteropelvic junction avulsion. Most resolve, and persistent leaks need stent or nephrostomy.
- After **partial nephrectomy** (0.5–20%), a central tumour predicts leak. Drain, adding stent or nephrostomy if needed.
- **Vesicourethral anastomotic leak after RP** (Patil): I extraperitoneal <6 cm, II extraperitoneal ≥6 cm, III intraperitoneal. Prolong the catheter, then withdraw the drain, then place stents or nephrostomies; reoperate last.

16 · Fistula after gender-affirming surgery

- After phalloplasty or metoidioplasty, urethrocutaneous fistula and stricture are common; **primary colpectomy** roughly halves fistula. Small ones close with a catheter. After transfeminine surgery, neovaginal fistula is rare and is repaired with interposition after diversion.

17 · Landmark trials

Trial	Design	Result	Take-home
Martius flap in simple obstetric VVF	RCT, repair ± Martius	No difference in closure; ~⅓ donor-site pain/numbness	Reserve Martius for complex fistula
Prophylactic ureteric stents in gynaecological surgery	RCT (+ meta-analysis)	No/small reduction in injury (number needed to treat 224)	Selective, not routine, stenting

122 Bladder and Female Urethral Diverticula

1 · Bladder diverticula — definition and pathology

*See the paper answer **Bladder diverticulum.** (Section 12, Ch 124).*

Also, from the condensed book (not in the paper answer):

- Diverticular cancer: radical cystectomy gives the same stage-for-stage survival; upstaging is common.
- A diverticulum is urothelium herniating through the muscularis propria; an outer pseudocapsule gives the surgical plane.
- With **no functional muscle, the sac cannot contract**, so post-void residual (PVR) is high and stasis leads to infection, stones and malignancy.
- About 90% occur in adults; M : F ≈ **9 : 1**.

Feature	Congenital	Acquired
Age / number	<10 years; solitary	Men >60; often multiple
Site	Posterolateral to ureteric orifice (~90% near ureterovesical junction [UVJ]); vesicoureteral reflux (VUR) common	Ureterovesical hiatus commonest; anywhere
Cause	Detrusor weakness at UVJ	Bladder outlet obstruction (**BOO**) or neurogenic — almost always
Wall / presentation	Smooth; acute urinary tract infection (UTI)	Trabeculated; incidental (lower urinary tract symptoms [LUTS], haematuria, UTI)
Malignancy	Not increased	Increased

- Up to 60% of congenital cases have a syndrome, neuropathic bladder or BOO, e.g. **Menkes**, **Williams**, **Ehlers-Danlos**, prune-belly or posterior urethral valves.
- Acquired cases are due to benign prostatic hyperplasia in ~**70%**; in women, causes include dysfunctional voiding, prolapse, bladder neck obstruction and post-sling obstruction.
- A **Hutch diverticulum** is paraureteric, in a neuropathic bladder with VUR.

> **KEY POINT**
>
> Congenital: weak wall, normal pressure. Acquired: normal wall, high pressure, so an acquired diverticulum sends the search back to the outlet.

2 · Bladder diverticula — evaluation

- Symptoms are nonspecific (incomplete emptying, double voiding).
- Do rectal examination, PSA, urinalysis, culture, **cytology**, endoscopy and upper and lower tract imaging, and seek BOO or occult neuropathy.
- Voiding cystourethrogram (**VCUG**) shows size, VUR and emptying. **Paradoxic enlargement** means the sac fills on voiding and refills the bladder afterwards.
- CT or MRI shows number, masses and relation to ureters and rectum.
- Hydroureteronephrosis (HUN) occurs in ~7% of adults, from compression, underlying dysfunction or VUR; the pelvic ureter deviates **medially**.

- Video-urodynamics (**VUDS**) shows a **pressure-sink effect**: contractility looks falsely low, and a raised PVR may be diverticular urine.
- **Cystoscopy** inspects the whole sac with cytology from it. Biopsy with extreme care, as perforation seeds tumour.
- With VUR and a congenital diverticulum, reimplant **selectively**. Treating the cause may make excision unnecessary.

> **KEY POINT**
>
> Two jobs: find the cause (outlet or neurogenic) and exclude cancer in a sac each test sees imperfectly, so tests are combined.

3 · Malignancy in bladder diverticula

- Prevalence is 0.8–10%, driven by stasis and inflammation. Urothelial carcinoma is **70–80%** and squamous cell carcinoma (SCC) 20–25%.
- With **no muscularis propria**, spread is early and extravesical, transurethral resection (TUR) risks perforation, and staging is inaccurate.
- **Radical cystectomy** gives the same stage-for-stage survival. Partial cystectomy or diverticulectomy suits selected low-grade, low-stage tumours.

> **KEY POINT**
>
> Treat a diverticular tumour as potentially invasive from the outset; missing muscle explains early spread, perforation risk and inaccurate staging.

4 · Bladder diverticula — management

Treat the outlet first; operate for complications, not size.

ALGORITHM

1. Assess with history, cystoscopy, cytology and VCUG.
2. If a lesion is suspicious, then biopsy and give cancer therapy.
3. If not, image the upper tract. If HUN or symptoms, then do urodynamics and treat BOO, then reassess.
4. If a problem from the diverticulum persists: fit → diverticulectomy; unfit → clean intermittent catheterisation (CIC).
5. If asymptomatic and uncomplicated, then survey.

- **Diverticulectomy indications**: refractory LUTS, relapsing UTI, **stones**, **carcinoma or premalignancy**, upper tract deterioration (obstruction, VUR). **Size alone is not an indication.**
- **Treat the outlet first or simultaneously**; a small sac may then empty, and concomitant prostatectomy is common.
- If observing, counsel about cancer and survey with symptoms, cytology and cystoscopy. The unfit with poor emptying need CIC or a catheter.
- **Endoscopic** treatment for the unfit resects the neck to muscle, but large sacs do poorly.
- Diverticulectomy is open, laparoscopic or robotic: use an ipsilateral ureteric catheter, dissect in the pseudocapsule, close the neck in layers, and reimplant a ureter entering the sac.

5 · Female urethral diverticulum (UD) — anatomy and aetiology

SAQ Female urethral diverticula

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3e

Definition

- **UD (urethral diverticulum):** a urine-filled periurethral sac within the periurethral fascia, connected to the urethra by an **ostium**.
- Third to seventh decades; prevalence up to 1–6%.

Aetiology

- Mostly **acquired**. **Routh theory** (most accepted): infection of **periurethral glands** leads to duct obstruction, suburethral abscess, rupture into the lumen and an epithelialised sac.
- Hence the ostium is **posterolateral at mid or distal urethra** in ≥90%.
- **Also know:** childbirth trauma theory (but 20–30% are nulliparous).

Clinical features

- "Three Ds" (dysuria, dyspareunia, post-void dribbling) in only about 5%.
- Commonest: frequency, urgency, pain, **recurrent UTI (urinary tract infection)**.
- Tender anterior vaginal wall mass; about one-third incontinent.
- **Complications:** stones (4–10%); 10% show metaplasia, dysplasia or carcinoma (**adenocarcinoma** commonest).

Evaluation

- Urine culture; **cystourethroscopy** (ostium seen in under half).
- **Imaging sensitivity: MRI (magnetic resonance imaging) 97%**, double-balloon urethrography 95%, ultrasound 82%, VCUG (voiding cystourethrogram) 74%.
- MRI shows saddlebag or circumferential extension.
- **Videourodynamics:** true SUI (stress urinary incontinence) versus dribbling from the sac.
- **Differential diagnosis:** Skene cyst (distal, distorts meatus, no communication), Gartner cyst, leiomyoma, caruncle, bulking agent.

Classification

- **L/N/S/C3 (Leach):** location, number, size, configuration, communication, continence.
- **Also know:** Leng–McGuire (absent fascia = pseudodiverticulum).

Management

- **Asymptomatic or unfit:** suppressive antibiotics, post-void stripping; counsel about cancer.
- **Symptomatic: transvaginal diverticulectomy.**
- **Principles:** 1. Sterile urine; topical oestrogen if atrophic. 2. **Inverted-U** anterior vaginal wall flap. 3. **Preserve periurethral fascia** as a separate layer. 4. Excise the **entire sac** and close the ostium. 5. **Watertight** urethral closure over a catheter (as small as 12 Fr). 6. **Multilayered, non-overlapping** closure; obliterate dead space. 7. **Martius flap** if poor tissue.
- **Bothersome SUI:** concomitant **autologous pubovaginal sling**; **never synthetic mesh**.

- **Also know:** Spence–Duckett marsupialisation for distal lesions only.
- Pericatheter VCUG at 14–21 days.

Complications

- Recurrence 10–22%; de novo SUI 7–16%; urethrovaginal fistula 1–8%; stricture; recurrent UTI.

Diagram to draw: sagittal section of a mid-urethral sac and its ostium.

> **Key points to remember:**
> - Periurethral gland infection (Routh); posterolateral ostium.
> - MRI most sensitive (97%).
> - Preserve fascia; non-overlapping layers; Martius flap.
> - No synthetic mesh at diverticulectomy.

Also, from the condensed book (not in the paper answer):

- Configuration is simple in ~73%, saddlebag in ~17% and circumferential in ~10%; ~50% are mid-urethral and 6–19% multiple.
- Carcinoma in UD is **adenocarcinoma** > urothelial > SCC. Local excision alone recurs, so use anterior exenteration or cystourethrectomy with or without radiotherapy.

- **Periurethral glands** lie posterolaterally, with ducts draining into the **distal third**; UD lie between the leaves of the urethropelvic ligament. The striated sphincter is mid-urethral, so plan repair for continence.
- In the gland theory, a gland abscess herniates into the fascia, usually **ventrally**, then ruptures into the lumen; about a third are proximal, however.

6 · UD — presentation and evaluation

- All three Ds occur in only ~**5%**. Features are irritative LUTS, pain, recurrent UTI, mass (~⅓), incontinence (~⅓), haematuria and discharge; 10–20% are asymptomatic. **Size does not track symptoms**, and UD is often misdiagnosed for years.
- In pregnancy, give antibiotics or aspirate, and repair after delivery.
- A mass lies 1–3 cm inside the introitus, and a **hard mass suggests stone or cancer**. Check atrophy, stress test and prolapse.
- **Cystourethroscopy** shows the ostium at **4 and 8 o'clock** mid-urethra in 15–89%; not seeing it must not stop work-up.
- **VUDS** shows SUI in up to 50% and separates true SUI from dribbling from the sac.
- **MRI is best** (T2 high, maps extent) but may miss small cancers. Ultrasound shows an anechoic sac, and VCUG underestimates size.

> **KEY POINT**
>
> UD symptoms mimic common LUTS and do not track size, so suspicion and examination make the diagnosis. Imaging confirms and **maps** it, especially extension towards the bladder neck (bladder or ureteric injury, postoperative incontinence).

7 · Differential diagnosis of periurethral masses

Lesion	Clue	Treatment
UD	Mid/proximal; communicates	Excision, reconstruction
Leiomyoma	Firm, mobile, solid; oestrogen-dependent	Transvaginal enucleation
Skene cyst/abscess	Lateral to and distorts meatus; non-communicating	Aspiration, marsupialisation, excision
Gartner duct cyst	Anterolateral; may receive ectopic ureter — image upper tract	Observe, sclerotherapy, excision
Vaginal wall cyst	Müllerian commonest	Excise if symptomatic
Urethral prolapse	Red "doughnut"; prepubertal, postmenopausal	Oestrogen; excision
Caruncle	Red, posterior meatal lip; postmenopausal	Oestrogen; excise if atypical
Bulking agent	Injection history	Excise if symptomatic

Skene gland adenocarcinoma may raise PSA.

> **KEY POINT**
>
> Two questions sort most periurethral masses: does it communicate (only UD), and where is it relative to the meatus (Skene, prolapse and caruncle lie at the meatus)? A firm, mobile, solid mass is a leiomyoma.

8 · Classification and SUI

System	Basis	Use
L/N/S/C3 (Leach)	Location, Number, Size, Configuration, Communication, Continence	Standard; not validated
Ginsburg–Genadry	Location	Distal: marsupialisation; proximal: excision
Leng–McGuire	Periurethral fascia present?	Absent = **pseudodiverticulum**, needing flap/graft

- Significant de novo stress urinary incontinence (SUI) occurs in **7–16%**; risks are size ≥30 mm, a proximal or circumferential sac and a short urethra. Most settles within 12 months.
- About 50% of preoperative SUI improves after excision, so operate only for **bothersome** SUI, using an **autologous pubovaginal sling, never synthetic**.

9 · Surgical repair of UD

- Operate for symptoms, stones or suspected cancer; treat the asymptomatic or unfit with suppressive antibiotics and postvoid stripping.
- **Spence-Duckett marsupialisation** suits only distal-third UD in women who are not sexually active.
- Principles (Box 122.2): vascularised flap, separate fascia, excise ostium and entire sac, watertight multilayer non-overlapping closure, close dead space, and **preserve or create continence**.

ALGORITHM

1. Lithotomy and Foley; suprapubic catheter (**SPC) at the start**, never the end (it would cross the repair).
2. Make an **inverted-U incision** and raise the vaginal flap off the fascia.
3. Incise the fascia **transversely** and raise fascial flaps.
4. Excise the entire sac to the ostium, including the inflamed urethral edge.
5. Close the urethra watertight over a catheter (as small as 12 Fr).
6. Close fascia **perpendicular** to it, obliterating dead space.
7. If tissue is poor or fascia attenuated, then add a **Martius** flap.
8. Advance the vaginal flap (3 layers; 4 with Martius); antibiotic pack.

- A pseudodiverticulum needs a Martius flap or fascial graft. Do a **pericatheter VCUG at 14–21 days**, repeated weekly if leaking.

10 · Complications

Complication	Incidence	Notes
Recurrent UD	**10–22%** (pooled ~10%)	Incomplete excision, dead space; circumferential, complex, proximal, multiple
Incontinence	1.7–16%	Conservative 12 months, then autologous sling
Urethrovaginal fistula	0.9–8.3%	Distal: may need no repair; proximal: repair ± Martius
Stricture	0–5%	—
Recurrent UTI	up to 31%	Investigate for recurrence
Others	—	Distal urethral necrosis, bladder/ureteric injury, dyspareunia

Redo excision gives less relief and more fistula, so image persistent symptoms for recurrence or stricture.

> **KEY POINT**
>
> Complete sac removal, watertight closure, off-axis fascia and obliterated dead space prevent the three failures: recurrence, fistula, incontinence.

Feature	Bladder diverticulum	Female UD
Origin	Herniation through detrusor	Infected periurethral gland ruptured into lumen
Cause to treat	BOO / neurogenic	Infection; coexisting SUI
Commonest cancer	Urothelial (then SCC)	Adenocarcinoma
Key study	Cystoscopy + cytology; VCUG	MRI
Operate for	Complications, not size	Symptoms or complications
Operation	Diverticulectomy + outlet treatment	Transvaginal excision, layered reconstruction

77 Sexual Function and Dysfunction in the Female

SAQ Female sexual dysfunction

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 3, Q3f

Definition and prevalence

- FSD (female sexual dysfunction): persistent desire, arousal, orgasm or pain problems causing **personal distress**.
- About **40%** of women in their lifetime; **HSDD (hypoactive sexual desire disorder) commonest**. Almost half of women with UI (urinary incontinence) have FSD.

Physiology and response models

- Arousal: pudendal afferents, pelvic parasympathetic efferents → **NO (nitric oxide)–cGMP (cyclic guanosine monophosphate)** → engorgement and lubrication. Oestrogen maintains tissues; testosterone drives desire.
- **Masters and Johnson** (linear): excitement, plateau, orgasm, resolution; Kaplan added desire. **Basson** (circular): desire is often responsive.

Classification

DSM-5 (Diagnostic and Statistical Manual, 2013)	ICSM (International Consultation on Sexual Medicine)/ISSWSH (International Society for the Study of Women's Sexual Health)
Female sexual interest/arousal disorder (≥3 of 6 criteria)	HSDD and arousal disorder kept separate (cognitive vs genital arousal)
Female orgasmic disorder	Orgasm disorders
Genito-pelvic pain/penetration disorder	Genital-pelvic pain dysfunction
–	PGAD (persistent genital arousal disorder)
- ≥6 months with distress; lifelong or acquired; generalised or situational.	

Causes

- Hormonal (menopause, GSM [genitourinary syndrome of menopause]), drugs (SSRIs [selective serotonin reuptake inhibitors], antihypertensives), diabetes, depression, relationship factors.
- Urological: incontinence, prolapse, sling dyspareunia, cystectomy nerve or clitoral injury.

Evaluation

- Clinician must ask. **PLISSIT** (permission, limited information, specific suggestions, intensive therapy); BSSC-W (Brief Sexual Symptom Checklist for Women) screen; **FSFI (Female Sexual Function Index, 19 items, six domains)**.
- Examination: atrophy, lichen sclerosus, cotton-swab test, pelvic floor tone. Laboratory tests rarely help.

Management

- Treat causes; education; sex therapy (sensate focus, mindfulness).
- **HSDD (premenopausal, FDA [US Food and Drug Administration] approved): flibanserin** 100 mg at bedtime (5-HT1A agonist/5-HT2A antagonist; alcohol caution); **bremelanotide** 1.75 mg subcutaneously on demand (melanocortin agonist). Postmenopausal: transdermal testosterone (off-label).
- **Arousal:** testosterone; PDE5 (phosphodiesterase type 5) inhibitors only with neurovascular disease.
- **Orgasm:** education, directed masturbation.
- **Pain/vestibulodynia:** pelvic floor physiotherapy, dilators, biofeedback, topical lidocaine, vaginal oestrogen; vestibulectomy if refractory.
- **Also know:** ospemifene, intravaginal prasterone (menopausal dyspareunia).

Diagram to draw: Basson circular sexual response model.

> **Key points to remember:**
> - About 40% prevalence; HSDD commonest; distress required.
> - DSM-5 merges desire and arousal, dyspareunia and vaginismus.
> - Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

Also, from the condensed book (not in the paper answer):

- Flibanserin: about half respond by 8 weeks; REMS. Bremelanotide: ≤8 doses/month; hyperpigmentation. Testosterone: stop at 6 months if no benefit.
- Genital-sparing cystectomy allows sex at about 6 vs 14 weeks; combined anterior and posterior repair increases dyspareunia; TOT spares the dorsal clitoral nerve.

1 · Functional anatomy

The **distal vagina, clitoris and urethra** share blood supply and nerves and respond as one unit, the **clitoral complex**, so urethral or vaginal surgery can change sexual function.

- The **clitoris** is the penile homologue, **separate from the urethra**; its two corpora continue as crura on the ischiopubic rami.
- The glans is very sensitive but **not erectile**. The dorsal nerves lie **superficially on the dorsum**, outside the tunica, so hood or clitoral surgery can injure them.
- **Erectile** tissue is the clitoral body, crura and vestibular bulbs; labia minora, glans, periurethral tissue and vaginal walls are vascular but do not engorge.
- **Skene** glands are the prostate homologue (PSA positive) and **Bartholin** glands the bulbourethral homologue. MCQ The G-spot has no agreed anatomical basis.
- Nerve supply of the labium majus and the glans draining **directly to deep inguinal nodes** are minor exam points.

2 · Physiology of the sexual response

ALGORITHM

1. Touch on the clitoral complex travels in pudendal (somatic) afferents to **S2–S4**.
2. Parasympathetic efferents (nervi erigentes, then cavernous nerves) dilate arterioles, causing engorgement and lubrication.

The reflex has **somatic afferents and visceral efferents**. **Orgasm** is cerebral, with **sympathetic** output (T12–L1) contracting the uterus and pudendal reflexes contracting pelvic striated muscle.

KEY POINT

Pudendal injury removes the **trigger**; pelvic plexus or cavernous nerve injury removes the **response**.

ALGORITHM

1. Neuronal nitric oxide synthase (nNOS) makes nitric oxide (NO), which activates guanylate cyclase to make cyclic GMP (cGMP).
2. cGMP lowers Ca^{2+}, so smooth muscle relaxes and tissue engorges.
3. Phosphodiesterase type 5 (PDE5) degrades cGMP; vasoactive intestinal polypeptide (VIP) works with NO.

Desire balances **excitation** (dopamine, noradrenaline, melanocortins, oxytocin, testosterone, oestrogen) against **inhibition** (serotonin, opioids, prolactin, progesterone); licensed HSDD drugs act on this balance.

	Oestrogen	**Testosterone**
Role	Maintains the **substrate**: epithelium, collagen, lubrication, acidic pH, NOS/VIP	Drives **desire**; inactive when bound to sex hormone–binding globulin (SHBG)
Deficiency	Thin, dry, fibrotic vagina causing dyspareunia; risk rises at oestradiol **≤50 pg/mL**	Lower libido, arousal, sensation and orgasm
Treatment risk	—	Virilisation: early reversible (acne, hirsutism); late irreversible (voice, alopecia, clitoromegaly)

Model	**Stages**	**Key idea**
Masters & Johnson (1966), linear	Excitement, plateau, orgasm, resolution	**No desire phase**
Kaplan (1977), linear	Desire, arousal, orgasm	Adds desire
Whipple & Brash-McGreer, circular	Seduction, sensations, surrender, reflection	Reflection feeds next seduction
Basson (2002), circular	Neutrality, receptivity, arousal, **responsive desire**, satisfaction	Spontaneous desire **not needed**; intimacy is the motive

KEY POINT

Under Basson, lack of spontaneous desire (common after menopause) is not a dysfunction, which is why DSM-5 merged desire and arousal.

3 · Classification

DSM-5 requires **≥6 months** and **clinically significant distress**; low desire without distress is not a disorder.

Domain	DSM-IV-TR	DSM-5
Desire and arousal	HSDD; female sexual arousal disorder (FSAD); sexual aversion disorder	**Female sexual interest/arousal disorder (FSIAD)**, **≥3 of 6**: less interest; fewer fantasies; less initiation/receptivity; less pleasure, response to cues or sensation (in 75–100% of encounters). Aversion dropped
Orgasm	Female orgasmic disorder (FOD) after normal excitement	FOD: delay, infrequency, absence **or lower intensity** in **75–100%**; normal-excitement requirement removed
Pain	Dyspareunia; vaginismus	**Genito-pelvic pain/penetration disorder (GPPPD)**, ≥1 of: penetration difficulty; pain; fear of pain; pelvic floor muscle (PFM) tensing

- **ICSM 2016** keeps HSDD and FSAD **separate** and adds **persistent genital arousal disorder (PGAD)** and orgasmic illness syndrome; **ISSWSH** adds a urological **genital arousal disorder**.
- About **40%** have a lifetime sexual problem but only about **8–12%** a distressing disorder. **HSDD is commonest**, FOD second.

4 · Causes (biopsychosocial)

- **Biological**: diabetes, hypertension, coronary disease, thyroid disease, hyperprolactinaemia, renal failure, spinal cord injury (SCI), multiple sclerosis (MS), vulval dermatoses, incontinence, pregnancy, breastfeeding and menopause.
- **Psychological**: depression (bidirectional, so screen every HSDD), anxiety, body image, abuse, conflict and the **partner's erectile dysfunction or premature ejaculation**.
- **Drugs**: SSRIs, SNRIs, antipsychotics, antihypertensives, antiandrogens, GnRH agonists, opiates and alcohol. MCQ Combined hormonal contraceptives and oral oestrogens raise SHBG.
- **Genitourinary syndrome of menopause (GSM)** causes dryness, burning, dyspareunia (often **vestibular**), urgency, dysuria and recurrent UTI. Dryness causes pain, which lowers desire, so **treat the tissue first**.
- **Cancer treatment** harms by castration, anti-oestrogens (aromatase inhibitors cause severe GSM), pelvic radiotherapy (stenosis) and nerve injury.

5 · Evaluation

Women rarely volunteer sexual problems, so the **clinician must ask**, gender-neutrally.

ALGORITHM

1. **P**ermission.
2. **L**imited **I**nformation: cycle, ageing, drugs.
3. **S**pecific **S**uggestions: lubricants, **vaginal oestrogen**, positions, vibrators.
4. **I**ntensive **T**herapy means **referral** (PLISSIT).

- **DSDS** (Decreased Sexual Desire Screener) is a **validated** 5-item screen; yes to items 1–4 with no contributing cause means **generalised acquired HSDD**. The ACOG BSSC-W is not validated.
- **FSFI** (19 items, 6 domains) is most used; FSDS-R measures **distress**; **PISQ-12/IR** is the **only validated** tool for prolapse and incontinence.
- History: lifelong or acquired, generalised or situational, the **order problems appeared**, drugs, surgery, LUTS, prolapse and the **partner's** function.
- Examination is essential with pain. **pH ≥4.5** without infection or semen means atrophy. Do a **Q-tip test** of the vestibule and **retract the hood** for adhesions.
- Lichen sclerosus leads to introital stenosis; lichen planus (**Wickham striae**) to vaginal stenosis.
- **Overactive** PFM causes dyspareunia and **underactive** PFM causes incontinence and prolapse; fornix tenderness suggests endometriosis.
- Labs are rarely needed. Do **not** measure a **random testosterone**: there is no female range and no level predicts HSDD.

6 · Hypoactive sexual desire disorder

- ISSWSH defines HSDD as ≥6 months of lost desire **with distress**, not due to drug, illness, pain, abuse or partner. **Exclude relationship conflict first.**
- HSDD peaks in **midlife** and is defined by distress, not hormone level.

ALGORITHM

1. Screen (DSDS), then history and examination.
2. **Correct modifiable factors**: drugs, depression, GSM, pain, partner's ED.
3. Education; sex or couples therapy.
4. If **generalised acquired** HSDD persists, then drug; review at 8 weeks.
5. **Refer** if situational HSDD fails counselling, or psychiatric illness or sexual trauma.

	Flibanserin	**Bremelanotide**
Mechanism	Central **5-HT1A agonist / 5-HT2A antagonist**: lowers serotonin, raises dopamine/noradrenaline (removes the brake)	Central **melanocortin agonist** releasing dopamine (presses the accelerator)
Licence	FDA: only **premenopausal** acquired generalised HSDD	
Dose	**100 mg at bedtime** daily; about half respond by 8 weeks	**1.75 mg subcutaneous** ≥45 min before sex; ≤1/24 h, **≤8/month**
Side effects	Dizziness, somnolence, nausea	Nausea, flushing, headache, **hyperpigmentation**
Cautions	**Alcohol** (hypotension, syncope); strong CYP3A4 inhibitors; REMS restricted prescribing	Uncontrolled hypertension or cardiovascular disease

- **Transdermal testosterone** (off-label) improves desire, arousal, orgasm and satisfaction in **postmenopausal HSDD**, with no cardiovascular signal but unknown long-term safety.
- The 2019 Global Consensus targets **premenopausal physiological levels**, checks levels at baseline and 3–6 weeks, and says **stop at 6 months** if there is no benefit. Bupropion (off-label) is an alternative.

7 · Arousal disorder and PGAD

- **Subjective and genital arousal correlate poorly**, so vasodilators raise flow without felt arousal. Use mindfulness-based sex-focused CBT.
- **Transdermal testosterone** has the best evidence. PDE5 inhibitors help only with neurovascular impairment (type 1 diabetes, SCI, MS, SSRI-induced). **Bupropion 150 mg twice daily** treats SSRI-induced dysfunction.
- **PGAD** is distressing genital arousal for hours to days, **not relieved by orgasm** and without desire. Causes include Tarlov cysts, pelvic varices and **stopping an SSRI**; treat with CBT, duloxetine or pregabalin.

8 · Female orgasmic disorder

- FOD is a **diagnosis of exclusion requiring distress**: exclude drugs, disease, stress and **inadequate stimulation** (including the partner's ED/PE).
- **No orgasm at coitus without clitoral stimulation is normal variation.**
- **Education is the cornerstone**, with directed masturbation, sensate focus and mindfulness. If arousal is intact, increase stimulation (vibrators; MCQ EROS clitoral vacuum is the only FDA-cleared FSD device).

ALGORITHM

1. **SSRI-induced dysfunction**: was it present before treatment?
2. If not, wait for tolerance (~4 months).
3. If it persists at 3–6 months, then lower the dose, or switch to or add **bupropion** or mirtazapine; consider sildenafil.

9 · Genito-pelvic pain / penetration disorder

- **Dyspareunia** is superficial or deep (endometriosis, prolapse, pelvic floor). **Vaginismus** is involuntary contraction preventing penetration.
- **Vulvodynia** is vulval burning pain for **≥3 months** without a cause; it is localised or generalised, provoked or unprovoked.
- **Provoked vestibulodynia (PVD)** is commonest (about **10–12%** of women). Primary PVD starts at first penetration; secondary PVD (about two-thirds) follows pain-free sex.
- The mechanism is vestibular nerve proliferation with a **hypertonic pelvic floor** and fear-avoidance.
- **Clitorodynia**: hood adhesions trap keratin and form painful pseudocysts, so **examine the glans routinely**.

ALGORITHM

1. Education and vulval care.
2. Lidocaine 5% or EMLA **10 min before intercourse**.
3. **Multimodal pelvic floor physical therapy** (myofascial release, down-training, biofeedback, graded **dilators**) with psychological/couples therapy.
4. If hypo-oestrogenic, then vestibular oestrogen.
5. If localised PVD is refractory, then **vestibulectomy**.

- **Multimodal physical therapy is standard of care** for PVD; EMG biofeedback is best if the pelvic floor is overactive.
- Oestrogen deficiency **increases** vulval nerve density, so **topical beats systemic and vestibular beats vaginal**.
- Second-line options are amitriptyline, gabapentin and intralevator botulinum toxin. **Vestibulectomy** beats non-surgical therapy in localised neuroproliferative disease.

Treatment of GSM

ALGORITHM

1. Non-hormonal lubricants/moisturisers and regular activity (NAMS first line).
2. Then **low-dose vaginal oestrogen** (minimal absorption, no progestogen) or vaginal DHEA (prasterone 6.5 mg nightly).
3. If vaginal products are impossible, then ospemifene 60 mg orally (SERM, vaginal agonist; VTE risk).
4. If vasomotor symptoms, then systemic hormone therapy.

KEY POINT

After **breast cancer**, use non-hormonal measures first; vaginal oestrogen only after oncology discussion (aromatase inhibitors).

10 · Pelvic surgery and sexual function

Surgery harms by three routes: **nerves** (lateral vaginal wall, pelvic plexus), **clitoral blood supply** (distal urethra) and **vaginal shape**. Counsel **before and after**.

- **Radical cystectomy**: loss of desire is commonest, and about half can have intercourse. Preserve function by saving the bundles on the **lateral vaginal walls**, dividing the urethra at the **bladder neck**, and keeping vaginal dimensions.
- **Genital-sparing** cystectomy (if oncologically safe) lets sex resume at **~6 vs 14 weeks**; diversion type is irrelevant. **Never trade oncological control.** See Ch 101
- **Hysterectomy** for benign disease leaves function unchanged or better; after **rectal surgery** dysfunction is common, and nearly universal after abdominoperineal resection.
- **Prolapse repair**: bother and body image, not stage, predict function. Dyspareunia is commoner after vaginal than abdominal repair, and after anterior mesh than **sacrocolpopexy**.
- **Combined anterior and posterior repair increases dyspareunia**, so stage it in sexually active women. **Keep length, avoid excess skin excision, preserve the introitus.** See Ch 119
- **SUI surgery**: about a third improve and most are unchanged; **curing coital incontinence** predicts improvement. Retropubic tape (**TVT**) may denervate the vaginal wall, while transobturator tape (**TOT**) spares the dorsal clitoral nerve. Dyspareunia is a strong reason for sling removal.
- **Coital incontinence** (about 1 in 10 incontinent women): leakage at penetration is stress; at orgasm, detrusor overactivity.
- **Bladder pain syndrome causes the worst FSD**, through pain, PFM overactivity and vulvodynia. **Sacral neuromodulation** improves sexual function whatever the cause.

- **Genital cosmetic surgery** has poor evidence; screen for **body dysmorphic disorder**.

11 · Special populations

Ask about orientation and gender identity; offer fertility preservation before transition. Oestrogen risks thromboembolism; testosterone risks erythrocytosis (haematocrit **>50%**), so check haematocrit and testosterone 3-monthly in year 1, then 1–2 times a year. SCI impairs sensation, lubrication and orgasm. See Ch 126

12 · Landmark trials

Trial	Design	Result	Take-home
Flibanserin phase III	RCTs vs placebo, premenopausal	More satisfying events and desire, less distress	First FDA-approved drug for HSDD
RECONNECT (bremelanotide)	2 phase III RCTs	More desire, less distress	On-demand option for premenopausal HSDD
Bupropion for SSRI-induced dysfunction (Cochrane)	Meta-analysis	Benefit at 150 mg twice daily	Add or switch to bupropion

★ Putting it together: female pelvic decisions

A · Incontinent woman

ALGORITHM

1. **Classify** (history, diary): stress (SUI), urgency (UUI), mixed, overflow, functional. Check stress test, prolapse (POP), residual, urinalysis (Ch 105).
2. **Urodynamics** are not needed for uncomplicated demonstrable SUI (VALUE). Use them if mixed, voiding dysfunction, prior surgery, neurological or high-grade POP.
3. **Conservative first:** weight, fluids, supervised pelvic floor and bladder training, pessary, vaginal oestrogen (Ch 114).
4. If this fails, then **SUI surgery** (Ch 116, 118); for UUI, drugs, botulinum toxin, neuromodulation.

B · Choosing an SUI operation

Operation	Best for	Drawbacks
Retropubic mid-urethral sling (MUS)	Standard; intrinsic sphincter deficiency	Perforation (cystoscopy mandatory), voiding dysfunction
Transobturator MUS	Avoiding the retropubic space	Groin/thigh pain; weaker in severe SUI
Autologous pubovaginal sling	Recurrent SUI, urethral repair, mesh declined	More voiding dysfunction
Burch	Mesh declined; concurrent abdominal surgery	Posterior prolapse, de novo urgency
Bulking agent	Frail, minimal surgery	Lower, short-lived cure

C · Prolapse: which repair?

ALGORITHM

1. **POP-Q** stage II is a leading edge within 1 cm of the hymen (Ch 105).
2. Observe if asymptomatic; offer pessary or surgery if symptomatic.
3. **Support the apex**: uterosacral/sacrospinous suspension or sacrocolpopexy (most durable) (Ch 117).
4. If **frail, not sexually active**, then colpocleisis.
5. If **occult SUI**, then discuss a concomitant sling (OPUS: fewer leaks, more complications).

D · Vesicovaginal fistula

KEY POINT

The first repair has the best chance. Do a double-dye test, upper tract imaging and, if irradiated, biopsy. Post-hysterectomy fistulae are supratrigonal and mostly repaired vaginally; go abdominal for ureteric involvement, large or high fistulae, or failed vaginal repair. Use tension-free, non-overlapping layers, a vascularised flap (Martius or omentum) for poor tissue, then a cystogram (Ch 121).

E · Mesh complications

ALGORITHM

1. If suspected, examine and cystoscope.
2. If exposure is small, then oestrogen with or without trimming; otherwise excise and close.
3. If mesh is in bladder or urethra, then remove and repair without new mesh.
4. Remove only enough to solve the problem, never asymptomatic mesh (Ch 119).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ Endoscopic procedure of SUI

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3f

Read as endoscopic injection of urethral bulking agents.

Introduction

- **SUI (stress urinary incontinence)** results from urethral hypermobility and **ISD (intrinsic sphincter deficiency)**.
- **Bulking agents**, injected submucosally under cystoscopic vision, are the least invasive surgical option.

Mechanism

- Bulk at the proximal urethra improves **mucosal coaptation** and resistance.
- It does not correct hypermobility, but most SUI includes some ISD.

Indications

- ISD with a **fixed, immobile urethra** (low VLPP [Valsalva leak point pressure] <60 cmH_2O, historically).
- Women wanting a **low-risk** procedure; elderly or unfit; future pregnancy.
- Persistent leakage after a sling; neurogenic SUI (safer for bladder compliance than a sling).
- **Avoid in:** active UTI (urinary tract infection), untreated detrusor overactivity.

Agents

Agent	Note
Polyacrylamide hydrogel (Bulkamid)	Non-particulate
Polydimethylsiloxane (Macroplastique)	Silicone particles
Calcium hydroxylapatite; carbon-coated zirconium beads	Particulate
Bovine collagen (skin test); PTFE (polytetrafluoroethylene); autologous fat	Withdrawn or abandoned: allergy, migration, granuloma, embolism

Technique

1. Sterile urine; local anaesthesia; cystoscope (0–30° lens) in the urethra.
2. **Transurethral:** needle into the submucosa about 1–1.5 cm distal to the bladder neck; or **periurethral:** needle alongside the urethra, watched cystoscopically.
3. Inject at 2–4 sites until the mucosal cushions **coapt in the midline**.
4. Avoid an indwelling catheter (it moulds the bulk); use CISC (clean intermittent self-catheterisation) for retention.

Outcomes

- Early improvement is common, but cure is lower than with slings and **declines with time**; repeat injections are often needed.

- **Cochrane review (Kirchin 2017):** limited evidence of benefit.
- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction):** an option, with counselling on durability.
- **Men:** EAU (European Association of Urology): **do not offer** for post-prostatectomy incontinence.

Complications

- Transient retention, UTI, dysuria, haematuria, urgency.
- Sterile abscess, migration, erosion; may mimic a urethral diverticulum on imaging.

Diagram to draw: cystoscopic view of coapted mucosal cushions after injection.

> **Key points to remember:**
> - Bulking improves coaptation; does not correct hypermobility.
> - Best for fixed urethra with ISD, or unfit patients.
> - Less durable than slings; repeat injections common.
> - Not recommended for post-prostatectomy incontinence.

SAQ Endometriosis in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 22 SAQ Sheet, Q3d

Definition and epidemiology

- Endometriosis is endometrial glands and stroma outside the uterus. Urinary tract involvement occurs in about 1–5% of women with endometriosis, mostly with deep disease.
- Sites: **bladder** (commonest, about 80%) > ureter (about 15%) > kidney, urethra (rare).

Pathogenesis

- Retrograde menstruation (Sampson), coelomic metaplasia, lymphovascular spread, and **iatrogenic implantation** at pelvic surgery (e.g. caesarean section).
- Bladder lesions start on the **serosa** (vesicouterine pouch) and invade the detrusor.
- Ureteric disease: **extrinsic** (periureteric fibrosis, commoner) or intrinsic (muscularis or mucosa); usually distal, left more often.

Clinical features

- **Bladder:** cyclical haematuria or frequency, urgency and dysuria with menses; suprapubic pain. Differential: BPS (bladder pain syndrome), vesicouterine fistula (menouria).
- **Ureter:** often **silent**; hydronephrosis and loss of renal function may go unnoticed.

Evaluation

- Transvaginal and transabdominal ultrasound gauge size and depth; **renal ultrasound in every woman with deep endometriosis**.
- **MRI (magnetic resonance imaging):** extent; high T1 signal from blood products.
- **Cystoscopy:** limited value (disease is extravesical); may show a bluish submucosal lesion. Biopsy excludes carcinoma.

- CTU (computed tomography urography) and MAG3 (mercaptoacetyltriglycine) renogram for obstruction and split function.
- Staging: rASRM (revised American Society for Reproductive Medicine) I–IV; #Enzian records bladder and ureter.

Management (with gynaecologist)

- **Medical:** progestins (dienogest), combined pill, GnRH (gonadotrophin-releasing hormone) analogues. Symptomatic relief; not for obstructive ureteric disease.
- **Bladder: partial cystectomy** (full-thickness excision with margin), laparoscopic or robotic; stent ureters if near. TUR (transurethral resection) is incomplete, with recurrence and perforation risk.
- **Ureter:**
- Extrinsic, mild: ureterolysis.
- Intrinsic or stricture: segmental resection with ureteroureterostomy, or **reimplantation with psoas hitch or Boari flap**.
- Non-functioning kidney: nephrectomy.
- Treat all pelvic disease at the same operation to lower recurrence.
- **Also know:** risk factor for VVF (vesicovaginal fistula) and ureteric injury at hysterectomy; rare malignant change (clear-cell, endometrioid).

Diagram to draw: endometriotic nodule in the vesicouterine pouch invading the detrusor.

> **Key points to remember:**
> - Bladder commonest; cyclical haematuria.
> - Ureteric disease silent: screen kidneys.
> - Partial cystectomy, not TUR.

SAQ Urethral caruncle.

Paper answer · P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3f

Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

Differential diagnosis

Lesion	Distinguishing feature
Urethral mucosal prolapse	**Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women
Urethral carcinoma	Firm, ulcerated, indurated, fixed; bleeding mass; biopsy
Urethral diverticulum	Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging)
Skene gland cyst or abscess	Lateral to meatus; distorts it; does not communicate
Others	Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus
- Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied.	

Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions: 1. Grasp the tip and apply traction to expose the base. 2. Excise at the base. 3. Reapproximate mucosa with absorbable sutures if the defect is large. 4. Foley catheter for a few days.
- Complications: bleeding, recurrence, meatal stenosis.

Diagram to draw: meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

SAQ Acute urinary retention in females.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2017-Summer Paper 4, Q3f

Introduction

- AUR (acute urinary retention) is sudden inability to void. It is uncommon in women and is often non-obstructive, so search for the cause.

Causes

Group	Examples
Obstructive	Large cystocele or prolapse; tight sling after incontinence surgery; urethral diverticulum, stricture; pelvic mass, **retroverted gravid uterus**, haematocolpos; cancer
Functional	**Fowler syndrome**; dysfunctional voiding; primary bladder neck obstruction (rare)
Neurogenic	MS (multiple sclerosis; may present first with retention), cauda equina, transverse myelitis, diabetes
Inflammatory	Cystitis, vulvovaginitis, **genital herpes**, Bartholin abscess
Drugs	Anticholinergics, opioids, tricyclics, calcium-channel blockers
Postoperative/postpartum; psychogenic	Epidural, pain; diagnosis of exclusion

Fowler syndrome

- Young women (20–35) with painless retention and bladder capacity over 1 litre.
- **Urethral sphincter EMG (electromyography):** complex repetitive discharges, so the sphincter fails to relax. Urethral closure pressure >100 cm H_2O.

Evaluation

- **History:** drugs, surgery, delivery, neurological and bowel symptoms, genital lesions.
- **Examination:** pelvic (prolapse, mass); neurological (perineal sensation, anal tone).
- **Tests:** drained volume; urine culture; creatinine; pregnancy test; ultrasound.
- MRI (magnetic resonance imaging) spine if neurological signs; cystoscopy.
- **Later:** video-urodynamics (female obstruction, e.g. Qmax ≤12 mL/s with Pdet ≥20 cm H_2O); sphincter EMG.

Management

- **Immediate:** urethral catheter; suprapubic if urethral access fails.
- **Treat the cause:** stop drugs; treat infection or herpes; pessary or prolapse repair; reposition the uterus; sling division; diverticulectomy.
- **TWOC (trial without catheter)** after correction.
- **Persistent retention:**
- **CISC (clean intermittent self-catheterisation)** rather than an indwelling catheter.
- **SNM (sacral neuromodulation):** about 70% success in Fowler syndrome.
- Bethanechol not recommended. Bladder neck incision only with caution (incontinence).

Diagram to draw: flowchart: AUR → catheterise → find cause → treat → TWOC → CISC or SNM.

Key points to remember:

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention.

Rapid Revision Sheet · Section 6

Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone.
- Always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch: Cooper's ligament; TOT: obturator foramen; PVS: bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 cmH_2O = ISD; MUCP ≤20 cmH_2O.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 cmH_2O; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

Burch colposuspension (SAQ)

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

Mid-urethral slings: TOT and TVT (SAQ)

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.

- TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

Management of perforated mid urethral sling (SAQ)

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF (LAQ)

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula (LAQ)

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Ureteric injury is mostly distal and missed.
- UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
- Orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
- Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management (LAQ)

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy to exclude cancer.

- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Female urethral diverticula (SAQ)

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

Female sexual dysfunction (SAQ)

- About 40% prevalence; HSDD commonest; distress required.
- DSM-5 merges desire and arousal, dyspareunia and vaginismus.
- Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

Endoscopic procedure of SUI (SAQ)

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.

Endometriosis in urology (SAQ)

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Urethral caruncle. (SAQ)

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.

Acute urinary retention in females. (SAQ)

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 7

Voiding Dysfunction

102 Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis

1 · Bony pelvis

- The sacrum and two innominate bones (**ilium, ischium, pubis**, fused at the acetabulum) meet at the sacroiliac (SI) joints behind and the symphysis in front.
- The **sacrospinous** ligament divides the sciatic foramen into **greater and lesser**.
- The pelvic brim is **50–60°** to horizontal. The **arcuate line** separates false from true pelvis. The iliac crest is at **L4**.

2 · Lower abdominal wall

Fascia and extravasation

- The umbilicus is the **T10** dermatome, at L3. **Scarpa's** fascia fuses with fascia lata about 1 cm below the inguinal ligament and continues as **Colles fascia**, then dartos.
- Colles is fused to the ischiopubic rami and perineal membrane, so **anterior urethral injury** fills penis and scrotum with blood and urine as a **"butterfly" perineal haematoma**. It spares the thigh but can spread **upward** deep to Scarpa's to the flank.
- †With Buck's fascia intact the haematoma stays on the shaft; a butterfly means Buck's is breached.

Rectus sheath

	Above arcuate line	Below arcuate line
Anterior	External oblique and part of internal oblique	All three aponeuroses
Posterior	Rest of internal oblique and transversus	**Absent**: transversalis fascia and peritoneum only

- Tendinous intersections tether rectus to the anterior sheath, so it can be **divided transversely without retracting**.
- Segmental nerves enter **laterally**, so a paramedian incision lateral to rectus causes atrophy and ventral hernia.

Inguinal canal

- It is **4 cm** long and carries the **spermatic cord and ilioinguinal nerve**. The deep ring is midway between the anterior superior iliac spine and pubic tubercle, **lateral to the inferior epigastric vessels**.
- The internal oblique and transversus roof closes onto the floor as a **shutter mechanism**.
- Direct hernias are **medial** and indirect hernias **lateral** to the inferior epigastrics. At **radical orchidectomy** the cord is ligated at the internal ring, so the stump retracts retroperitoneally.

Peritoneal folds (laparoscopic map)

Fold (medial to lateral)	Contains	Significance
Median	Urachus	Bladder apex to umbilicus
Medial (paired)	Obliterated umbilical arteries	Incise **lateral** to them to reach obturator fossa and retropubic space
Lateral (paired)	Inferior epigastric vessels	Mark deep ring and hernia type

3 · Pelvic floor and fascia

- Obturator internus fascia thickens into the **tendinous arch of levator ani**, the origin of the pelvic diaphragm (levator ani and coccygeus).
- Levator ani comprises **puborectalis** (a sling behind the anorectal junction for **faecal continence**), pubococcygeus and iliococcygeus. It is tonically contracted, relaxes to void, and **Kegel exercises** increase its tone.
- **Parietal fascia** lines the walls and **visceral fascia** sheaths the organs; their lateral attachments, the **endopelvic fascia**, carry pelvic vessels and are the gateway to the prostate.
- The **arcus tendineus fascia pelvis** marks the parietal–visceral junction and anchors urethral and bladder supports.

4 · Perineum

- The **perineal membrane** spans the pubic arch, is pierced by the urethra and supports the sphincter.
- The **perineal body** is the midline keystone, blending with Denonvilliers; injury at perineal prostatectomy causes urinary and faecal incontinence.
- Perineal prostatectomy MCQ : **Young** divides the central tendon, leaving the external sphincter on the rectum; **Belt** passes **under** it.
- The **internal pudendal artery** runs in **Alcock's canal** and becomes the dorsal penile artery.
- The **pudendal nerve (S2–S4)** supplies both external sphincters, levator ani, ischio- and bulbospongiosus, ending as the dorsal penile nerve.

5 · Pelvic vessels and lymphatics

- The **external iliac** gives the inferior epigastric and deep circumflex iliac. **Accessory obturator vessels** from the inferior epigastric cross the obturator fossa and are a PLND hazard.
- **Internal iliac**: the posterior trunk is parietal (superior gluteal, ascending lumbar, lateral sacral); the anterior trunk is visceral (umbilical, obturator, middle rectal, inferior vesical, internal pudendal, inferior gluteal).
- The patent umbilical gives the **superior vesical** and artery of the vas. The **inferior vesical** (not internal pudendal) supplies the neurovascular bundle (NVB).
- The iliac veins join **behind the internal iliac artery**, a site of major PLND bleeding.

- The **dorsal venous complex (DVC)** enters under the symphysis in front of the striated sphincter; ligating it at prostatectomy can damage the sphincter.
- Lymph often **crosses sides**. External iliac nodes drain bladder and urachus; internal iliac nodes drain prostate and bladder. The **node of Cloquet** is the **distal limit of PLND**.

6 · Pelvic nerves

Nerve	Roots	Key point
Ilioinguinal	L1	Through inguinal canal to anterior scrotum
Genitofemoral	L1–L2	On psoas; **lateral limit of PLND**; genital branch supplies cremaster
Femoral	L2–L4	Injured by psoas hitch, retractor blades, inguinal dissection
Obturator	L2–L4	Iliac bifurcation to obturator canal; PLND injury gives **weak adduction**
Pudendal	S2–S4	Perineal sensation; levator, urinary and anal sphincters
Nervi erigentes	S2–S4	Parasympathetic to bladder, rectum

> **KEY POINT**
>
> Anterior scrotum is L1 (ilioinguinal, genitofemoral); posterior scrotum is sacral (pudendal, posterior femoral cutaneous).

*See the paper answer **Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate** (Section 16, Ch 148).*

Also, from the condensed book (not in the paper answer):

- The NVB lies at **5 and 7 o'clock** at the apex; injury at prostatectomy or rectal resection causes erectile dysfunction.

7 · Pelvic viscera

Rectum

- The rectum begins at **S3**. Opening the **rectovesical pouch** exposes the seminal vesicles in the posterior robotic approach. **Denonvilliers fascia** separates rectum from prostate.
- The distal rectum near the apex is the **commonest rectal injury site** at prostatectomy.

Bladder

- The bladder is extraperitoneal except the dome, which leads to the **urachus** (adenocarcinoma, fistula, cyst).
- Circular bladder-neck fibres form the **internal sphincter**; these **α1-adrenergic** fibres maintain continence and **antegrade ejaculation**, and their damage adds to post-prostatectomy incontinence.
- Lymph goes mainly to the **external iliac** nodes, and also to internal iliac and obturator nodes, bilaterally.

Ureterovesical junction

- The intramural ureter runs **1.5–2 cm** obliquely in **Waldeyer's sheath**; being narrow, it traps stones.
- It is a **flap valve** in which detrusor backing compresses the ureter. Weak backing causes **vesicoureteral reflux (VUR)**, and a weak hiatus allows a **Hutch diverticulum**.

- The pelvic ureter narrows at the iliac crossing and passes **behind the vas**. Its blood supply enters **laterally**, so stripping the lateral attachments causes stricture.

8 · Endoscopic anatomy

ALGORITHM

1. Under local anaesthetic, inspect bladder, orifices and trigone with a flexible cystoscope.
2. Retroflex at the neck: intravesical prostatic protrusion, hidden tumours.
3. Find the verumontanum and ejaculatory ducts; the external sphincter coapts just distal, then the bulbar urethra.

KEY POINT

The verumontanum is the sentinel at transurethral resection of the prostate (TURP): resection distal to it injures the external sphincter and causes incontinence.

9 · Radiographic anatomy

- A **phlebolith** mimics a distal ureteric or bladder stone.
- On a **cystogram**, absent lateral distension suggests an extraperitoneal pelvic haematoma or post-prostatectomy lymphocele.
- Voiding cystourethrography (**VCUG**) best shows the posterior urethra, as the sphincters relax. On retrograde urethrography (**RUG**), steady pressure overcomes the external sphincter.
- The membranous urethra is the **shortest, narrowest** segment and is not a stricture.
- Computed tomography (**CT**) needs contrast except for **stones**. In **extraperitoneal rupture**, urine lies in the retropubic and perivesical spaces.
- Malignant nodes are rounded and heterogeneous; size is most specific at ≥**1 cm** short axis.
- **CT urogram** excretory images are taken at **7–10 min**.
- Magnetic resonance imaging (**MRI**) has the best soft-tissue contrast. A distended bladder wall over **3 mm** is thickened.

103 Physiology and Pharmacology of the Bladder and Urethra

1 · Overview

- The lower urinary tract (LUT) has two functions, **storage** (continence) and **emptying** (micturition), integrating the central nervous system (CNS), peripheral nerves, detrusor smooth muscle (DSM), urothelium, urethra, pelvic floor and external urethral sphincter (EUS).

> **KEY POINT**
>
> **Why single-organ therapy fails:** bladder and urethra act reciprocally, so boosting contractility in detrusor underactivity fails unless the urethra also relaxes. Afferent modulation acts on both (reduce it for overactivity, augment it for underactivity), making it the most promising therapeutic lever.

2 · Functional anatomy

Bladder

- The **male bladder neck** is a complete smooth muscle ring with dense sympathetic supply; it closes in ejaculation, ensuring **antegrade ejaculation**. Women lack this collar.
- The **lamina propria** is the "functional centre", containing the afferent plexus, interstitial myofibroblasts and capillaries.

Stroma

- The wall is about **50% collagen, 2% elastin**, mainly types I, III and IV.
- **Poorly compliant bladders** have more connective tissue relative to muscle and a higher **type III : type I collagen** ratio, so capacity falls.

Urethra and sphincters

- The urethra has smooth muscle (lissosphincter) and an **intramural striated rhabdosphincter (EUS)**, separate from the periurethral pelvic floor.
- The male EUS is a crescent in front of the prostate, becoming a horseshoe distally. The female EUS is a horseshoe over the ventral urethra, and a subepithelial vascular plexus forms a **"urethral seal"**.

Feature	Slow-twitch	Fast-twitch
Role	Sustained resting tone	Rapid extra squeeze on coughing
Fatigue / metabolism	Slow; oxidative	Rapid; anaerobic
Where	Intramural rhabdosphincter, giving **~50% of static urethral resistance**	Periurethral pelvic floor (mixed)
Rhabdosphincter mix	Male **65% slow / 35% fast**; female **87% slow / 13% fast**	

3 · Urothelium: barrier and sensor

The urothelium senses stretch; its damage drives urgency and pain.

Barrier

- **Umbrella cells** carry **uroplakin** plaques, which form the barrier and are the binding site of **type 1-fimbriated E. coli**. **Tight junctions** block paracellular flow.
- Loss of umbrella cells (infective or radiation cystitis) lets solutes leak in, so symptoms worsen with distension.

Sensor–transducer function

Urothelial cells behave like sensory neurons, carrying purinergic, adrenergic, muscarinic and TRP receptors. On stretch they release mediators onto afferents, directly or via myofibroblasts.

Mediator	Released by / when	Effect
ATP	Stretch; more with injury, inflammation, ageing	Excites suburothelial **P2X3** afferents, giving fullness and pain; P2X3-null mice are hyporeflexic
Nitric oxide (NO)	Especially inflammation	Blocks afferent Ca^{2+} channels: **afferent inhibition**
Acetylcholine (ACh), non-neuronal	Basal; more with stretch, ageing	Modulates afferents and DSM, so antimuscarinics may act partly here
Prostaglandins (PGs)	Urothelium is the main source	Detrusor regulation, cytoprotection
Inhibitory factor (unknown)	Basal	Reduces muscarinic contraction; not NO or PG

- The afferent threshold is set by the **ATP (excitatory) : NO (inhibitory) balance**. ATP is raised in detrusor overactivity (DO) and in bladder pain syndrome/interstitial cystitis (BPS/IC).
- **Suburothelial myofibroblasts** are linked by connexin 43 (**Cx43**) gap junctions and respond to ATP and ACh; their M2/M3 expression correlates with urgency in idiopathic DO.

> **KEY POINT**
>
> Damaged urothelium causes symptoms twice over: by **solute leak** into the lamina propria and by **altered signalling** (more ATP, less NO) to afferents.

4 · Detrusor smooth muscle physiology

Each contraction step is a drug target.

- Not every detrusor cell is innervated, so **gap junctions** spread excitation. Ca^{2+}–calmodulin drives **myosin phosphorylation**, and activity is mostly tonic.

ALGORITHM

1. ACh binds **M3** receptors (Gq).
2. Intracellular Ca^{2+} rises via L-type entry and phospholipase C (PLC), whose IP3 releases sarcoplasmic reticulum (SR) Ca^{2+}.
3. Ca^{2+}–calmodulin activates myosin light chain kinase (MLCK), which phosphorylates MLC20.
4. Cross-bridges cycle and the muscle contracts.
5. As Ca^{2+} falls, myosin light chain phosphatase (MLCP) dephosphorylates MLC20 and the muscle relaxes.

- Tone depends on Ca^{2+} and the **MLCK : MLCP balance**. In **calcium sensitisation**, RhoA-activated **Rho kinase** and protein kinase C (PKC) inhibit MLCP, so force rises without more Ca^{2+}.
- **L-type voltage-dependent Ca^{2+} channels (VDCC)** are the main trigger for contraction (blocked by nifedipine, diltiazem, verapamil).
- K^+ channels repolarise the cell, and **BK is the key brake**: Ca^{2+} sparks open BK, hyperpolarising the cell. BK knockout gives overactive bladder (OAB), and BK is reduced in neurogenic DO (NDO).

G protein	Receptors	Effect
Gq	M1, M3, M5 ("odd"); α1; AT II; H1; 5-HT2	IP3 and DAG release Ca^{2+} and activate PKC: contraction
Gi/o	M2, M4 ("even")	Lower cAMP
Gs	β3	Raise cAMP, activating PKA: relaxation

- **Purinergic (P2X1) contraction** is minimal normally but appears in obstructed, OAB and ageing bladders. It suppresses muscarinic coupling, explaining detrusor hyperactivity with impaired contractility (**DHIC**): urgency with incomplete emptying.
- The detrusor is deliberately **poorly coupled**, protecting filling. More Cx43 in bladder outlet obstruction (BOO) and NDO, induced by cytokines and nerve growth factor (NGF), allows whole-bladder myogenic contraction, which is DO.

KEY POINT

Motor–sensory coupling: spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and so reduce sensory input; efferent and afferent functions are inseparable.

5 · Receptors and drug targets

Receptor	Location	Function	Drug target
M3 (Gq)	Detrusor; urothelium; salivary glands	**Mediates contraction**	Antimuscarinics
M2 (Gi)	Detrusor (**numerically predominant**)	Amplifies M3 by opposing β-cAMP relaxation; the balance shifts from M3 to M2 in neurogenic and obstructed bladders	Non-selective antimuscarinics

Receptor	Location	Function	Drug target
β3 (Gs)	Detrusor (**most expressed adrenoceptor**); afferents	Relaxes during storage; increases compliance	Mirabegron, †vibegron
α1A	Urethra, prostate	Urethral closure; female resting tone	α-blockers (benign prostatic hyperplasia, BPH); α-agonists increase closure
α1D	Bladder (predominant α); spinal cord	Storage symptoms of obstruction	Tamsulosin improves storage lower urinary tract symptoms (LUTS)
α2	Prejunctional urethra; ganglia	Reduces noradrenaline release	—
P2X3	Suburothelial afferents	Senses ATP; increased in BPS/IC	AF-219 (proof-of-concept)
P2X1	Detrusor	Purinergic contraction; increased in BOO, ageing	Potential for DHIC
TRPV1	C fibres; urothelium	Senses heat >43 °C, protons, capsaicin; increased in NDO	Capsaicin and resiniferatoxin (RTX) desensitise
TRPM8	Afferents	Senses cold (bladder cooling reflex)	Experimental
NK1	Spinal cord	Afferent transmission	Aprepitant reduced urgency (trial)
Nicotinic	EUS	Sphincter contraction	—

LAQ Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management

Paper answer · P1 · P2 · P3 · merged from 4

Asked as: P1 LAQ "Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management" · P2 SAQ "Overactive bladder and its drug therapy" · P3 SAQ "OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron" · P1 SAQ "Compare and contrast various detrusor relaxants in management of OAB"

Definition

- **OAB (overactive bladder), ICS (International Continence Society): urgency**, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), in the absence of UTI (urinary tract infection) or other obvious pathology.
- **OAB dry** versus **OAB wet** (with UUI).
- OAB is a **symptom** diagnosis. DO (detrusor overactivity) is a **urodynamic** finding; only 44–69% of patients with urgency show DO.

Pathophysiology

All mechanisms converge on **more afferent signal reaching awareness**: the bladder turns the signal up, or the CNS (central nervous system) fails to turn it down.

Mechanism	Explanation
Afferent (sensory)	Aδ fibres normally sense filling. Unmyelinated **C fibres** are "silent", but become spontaneously active and hypersensitive in pathological states
Urothelial	Urothelium acts as a transducer: mechanical, inflammatory or chemical stimuli release ATP (adenosine triphosphate), acetylcholine and other mediators onto suburothelial afferents
Myogenic	Patchy denervation causes receptor upregulation and spontaneous contraction; exaggerated **micromotions** cause urgency, and coordinated spread across the wall gives DO
Neurogenic / central	Forebrain and PAG–PMC (periaqueductal grey–pontine micturition centre) normally inhibit voiding; loss of this inhibition lets "afferent noise" be felt as urgency
Other	Bladder–bowel crosstalk (constipation), ischaemia–reperfusion, chronic inflammation, altered urinary microbiome

- **Risk factors and associations:** ageing, female sex, BOO (bladder outlet obstruction: an animal model, but only an indirect cause in humans; DO persists in up to 60% after TURP [transurethral resection of the prostate]), anxiety and depression, metabolic syndrome, poor toileting habits.

Evaluation

- **Minimum:** history, examination, **urinalysis**.
- Bladder diary (3–7 days; P2: 3 days), questionnaires, PVR (post-void residual) when indicated.
- Urodynamics and cystoscopy only for complicated or refractory cases, or if urodynamics will change an invasive decision.

Outline of management (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]; not a compulsory sequence)

`Education + behavioural therapy (first line) ↓ (4–12 weeks) Antimuscarinic or β3-agonist (mirabegron, vibegron); combinations ↓ PTNS / onabotulinumtoxinA / SNM (third line) ↓ Augmentation cystoplasty or urinary diversion`

Line	Treatment
Behavioural	Education; reduce fluids by about 25%; cut caffeine; weight loss; **bladder training** (timed voids, interval increased by 15–30 minutes); **PFMT (pelvic floor muscle training)** with urge suppression: 50–80% fewer leaks
Drugs	**Antimuscarinics** (oxybutynin, solifenacin, extended-release trospium, darifenacin, fesoterodine, tolterodine); **β3-agonists** (mirabegron, vibegron); combinations
Third line	**PTNS (percutaneous tibial nerve stimulation)** weekly × 12; **onabotulinumtoxinA 100 U** intradetrusor; **SNM (sacral neuromodulation)** at S3 (success about 70–80%; revision >30% at 5 years)
Last resort	Augmentation cystoplasty or urinary diversion

Non-pharmacological management

1. Education and lifestyle

- Explain normal bladder function; set realistic goals.

- Reduce fluid intake by about 25% if excessive; cut caffeine; treat constipation.
- **Weight loss:** about 8 kg loss in obese women reduced UUI episodes.

2. Bladder training (BT) (breaks the cycle of urgency, frequent voiding and smaller capacity)

1. Keep a bladder diary.
2. Start a fixed interval (often 1 hour while awake).
3. Void by the clock, not by urge.
4. Increase by 15–30 minutes a week as control improves.
5. Target interval 3–4 hours. - Works by **uncoupling the void from the urge**.

3. PFMT and urge suppression

- Kegel exercises: quick 1–2 s flicks and 5–10 s holds, twice daily.
- A pelvic floor contraction **reflexly inhibits the detrusor**. Urge suppression: stop, stay still, squeeze quickly several times, relax, let the urge pass, then walk calmly to the toilet.
- PFMT and BT reduce incontinence episodes by **50–80%**.
- **Also know:** biofeedback helps patients who cannot find the muscle; electrical stimulation at 5–20 Hz inhibits the detrusor; timed and prompted voiding for cognitively impaired patients.

4. PTNS

- 34-gauge needle 3–4 cm above the medial malleolus; 20 Hz, 200 µs; confirm great-toe flexion or sole tingling. **30 min weekly for 12 weeks**, then maintenance.
- The posterior tibial nerve (L4–S3) shares sacral roots with the bladder. **SUmiT trial:** 55% vs 21% (sham) responders.

5. SNM

- Mechanism: stimulates **afferents** that modulate brainstem and forebrain control.
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or **staged tined lead** at **S3** (anal "bellows" and great-toe flexion). ≥50% improvement on diary leads to implanting the pulse generator in the buttock.
- Success about 70–80%. **InSite trial:** 61% vs 42% (standard medical therapy). Revision >30% at 5 years; MRI (magnetic resonance imaging)-conditional rechargeable devices now available.

6. Intradetrusor onabotulinumtoxinA

- Cleaves SNAP-25 (synaptosomal-associated protein 25), blocking acetylcholine release; also reduces afferent signalling.
- **100 U in 10 mL at 20 sites** cystoscopically; effect lasts about 6–9 months.
- About 25% become completely dry. Risks: UTI about 20%, retention needing CISC (clean intermittent self-catheterisation). **ROSETTA:** greater UUI reduction than SNM at 6 months, but more UTI.

7. Last resort: augmentation cystoplasty or urinary diversion.

Drug therapy: place and mechanisms

- Drugs are second line, after behavioural therapy, as first-line pharmacotherapy (antimuscarinics and β3 agonists alongside each other; AUA/SUFU). Two main classes: **antimuscarinics** and **β3-adrenoceptor agonists**.

- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent urgency signalling.
- Muscarinic side effects by subtype: **M1** (brain) cognition; **M2** heart rate; **M3** (glands, gut) dry mouth, constipation.
- Selectivity is by **receptor subtype** (M3) or **tissue** (bladder over salivary gland); tissue selectivity matters more.
- **β3 agonists:** β3 is the commonest (most highly expressed) adrenoceptor in human detrusor. Activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage, sparing the voiding contraction.

Agents

Drug	Distinguishing feature	Usual dose
Oxybutynin	Non-selective; direct relaxant; metabolite causes most dry mouth; patch, gel	5 mg two to three times daily; ER (extended-release)
Tolterodine	Bladder (over salivary) selective; QT prolongation at high dose	2 mg twice daily (IR, immediate-release); ER 4 mg daily
Fesoterodine	Prodrug of tolterodine's active metabolite	4–8 mg daily
Solifenacin	10-fold M3 > M2; long half-life	5–10 mg daily
Darifenacin	About 30-fold M3-selective	7.5–15 mg daily
Trospium	Quaternary; does not cross the blood–brain barrier (little brain entry)	20 mg twice daily
Mirabegron (β3)	Inhibits CYP2D6 (a liver enzyme): metoprolol interaction; small blood-pressure rise	25–50 mg daily
Vibegron (β3)	No CYP2D6 interaction	75 mg daily

Darifenacin vs solifenacin vs tolterodine

Feature	Darifenacin	Solifenacin	Tolterodine
Selectivity	**Most M3-selective (about 30-fold)**; not tissue-selective	M3 about 10-fold over M2; bladder-selective	Non-subtype-selective; **bladder-selective**
Dose	7.5–15 mg daily (ER)	5–10 mg daily	2 mg twice daily (IR); 4 mg daily (ER)
Half-life	Moderate (ER once daily)	**Long** (about 2–3 days): once daily	Short; active metabolite 5-HMT (5-hydroxymethyl tolterodine)
Metabolism	CYP3A4 (a liver enzyme) and CYP2D6	CYP3A4	CYP2D6 (and 3A4)
Dry mouth	Dose-related	Dose-related	Lower with ER
Constipation	**Highest** (gut M3)	Moderate–high	Lower
CNS	Spares M1: **preferred in elderly**	Better persistence	High dose: QT prolongation
Dose limit	Lower dose with CYP3A4 inhibitors or liver disease	5 mg in severe renal or moderate liver impairment	Halve with renal or liver impairment

- **Efficacy is similar**; response is individual, so switch agent or dose.

- **STAR trial:** flexible solifenacin beat tolterodine ER for urgency and incontinence.

Mirabegron (β3 agonist)

- First **β3-adrenoceptor agonist** for OAB; FDA (Food and Drug Administration) approval 2012.
- **Mechanism:** activation of adenylyl cyclase raises **cAMP**, relaxing the detrusor during **storage**. Reduces non-voiding contractions and afferent firing. The voiding contraction (muscarinic M3) is spared, so PVR barely rises.
- **Dose:** oral, extended release, once daily: start **25 mg**, increase to **50 mg** after 4–8 weeks. **25 mg maximum** in severe renal impairment (eGFR [estimated glomerular filtration rate] 15–29) or moderate hepatic impairment; avoid in end-stage renal disease and severe hepatic impairment.
- **Moderate CYP2D6 inhibitor:** raises levels of metoprolol, desipramine, flecainide; monitor digoxin.
- **Efficacy:** phase III trials (SCORPIO, ARIES, CAPRICORN): fewer incontinence episodes and micturitions than placebo. **TAURUS** (12 months): efficacy similar to tolterodine, with less dry mouth.
- **Adverse effects:** hypertension, tachycardia, headache, nasopharyngitis, UTI. **Dry mouth and constipation near placebo rates**, far below antimuscarinics.
- **Contraindicated:** severe uncontrolled hypertension (≥180/110 mmHg). Caution with arrhythmia, and with BOO when combined with antimuscarinics (retention).
- Other uses: add-on to α-blocker in men with BPH (benign prostatic hyperplasia); neurogenic detrusor overactivity.
- **Also know:** vibegron 75 mg (no CYP2D6 interaction).

Antimuscarinics versus β3 agonists

Feature	Antimuscarinics	β3 agonists
Efficacy	Similar	Similar (TAURUS: mirabegron ≈ tolterodine)
Dry mouth, constipation, blurred vision	Common; main reason for stopping	Near placebo
Cognition	Central M1 blockade; **dementia risk 10–65% higher with use >3 months**	No anticholinergic burden
Retention	Up to 3% in men	Low
Contraindications	Urinary or gastric retention, uncontrolled narrow-angle glaucoma	Uncontrolled (severe) hypertension (≥180/110 mmHg)

Choice of drug and special groups

- **Elderly and cognitive impairment:** antimuscarinic use >3 months is linked to 10–65% more new-onset dementia (SUFU white paper). Prefer a **β3 agonist** (no anticholinergic burden), or ER trospium, darifenacin or fesoterodine. Beers criteria: avoid antimuscarinics in dementia.
- ER forms cause less dry mouth. Poor persistence is the main problem; switch dose or drug before third-line therapy.

- **Combination:** solifenacin plus mirabegron beats each alone (**SYNERGY, SYNERGY II**), without extra cardiovascular effects; in non-responders it equals solifenacin 10 mg with less dry mouth (**BESIDE**).
- **Men:** α-blocker first; add an antimuscarinic (if PVR is low) or β3 agonist; retention up to 3%, so check PVR.
- **Also know:** vaginal oestrogen in atrophy; desmopressin for nocturnal polyuria; imipramine (children, mixed incontinence); intravesical oxybutynin; refractory OAB: onabotulinumtoxinA 100 U and neuromodulation.

Diagrams to draw

- Bladder showing urothelial afferents, C fibres and micromotions, with brainstem and forebrain (cortical) inhibitory pathways.
- Detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.
- Muscarinic subtypes M1 (brain), M2 (heart), M3 (bladder, gland, gut) with each drug's selectivity.

Key points to remember:

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

Also, from the condensed book (not in the paper answer):

- **Oxybutynin**: its metabolite **desethyloxybutynin** causes most of the dry mouth, so the controlled-release form is better tolerated.
- CNS-penetrant agents block M1, causing delirium and cognitive decline that is reversible within weeks of stopping.
- Outlet drugs are **α-agonists** (ephedrine, midodrine); systemic oestrogen may worsen urinary incontinence (UI).

SAQ Duloxetin

Paper answer · P3

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 3, Q3e

Read as: duloxetine, mainly its role in stress urinary incontinence.

Introduction

- **Duloxetine** is an SNRI (serotonin–noradrenaline reuptake inhibitor), also an antidepressant.
- Licensed in Europe (not USA) for **moderate to severe SUI (stress urinary incontinence) in women**; off-label in men.

Mechanism

- During storage, bladder afferents drive glutamate excitation of pudendal motor neurons in **Onuf's nucleus (S2–S4)**, which keep the striated urethral sphincter closed.
- Serotonin (5-HT2) and noradrenaline (α1) amplify this glutamate drive. Duloxetine blocks their reuptake, so **sphincter tone and the sneeze-induced continence reflex increase**.
- In voiding, glutamate drive switches off, so retention does not occur.
- Central effects also reduce bladder activity.

Dose

- **40 mg twice daily**; starting at 20 mg twice daily for 2 weeks reduces nausea. Taper on stopping.

Evidence

- Phase III RCTs (randomised controlled trials) in women: about 50% median fall in incontinence episodes versus about 30–40% with placebo; better quality of life; **cure rare**.
- Benefit is temporary and lasts only while taken.
- Male PPI (post-prostatectomy incontinence): small, short-term benefit.

Guideline position

- **EAU (European Association of Urology):** may be offered to women for **temporary** improvement, not cure; warn about side effects. Men: offer for SUI (weak), and **inform of off-label use and adverse effects** (strong).
- **NICE (National Institute for Health and Care Excellence):** not first line; not routinely second line; only if surgery is unwanted or unsuitable.
- Adding PFMT (pelvic floor muscle training): evidence conflicting.

Adverse effects

- **Nausea** (commonest, early), dry mouth, dizziness, insomnia, constipation; **high discontinuation**.
- Raised blood pressure, hyponatraemia (elderly), suicidal ideation, discontinuation syndrome.

Contraindications and interactions

- MAOIs (monoamine oxidase inhibitors), including **methylene blue** (serotonin toxicity); CYP1A2 inhibitors (fluvoxamine, ciprofloxacin); hepatic impairment; severe renal impairment; uncontrolled hypertension or narrow-angle glaucoma.

Diagram to draw: Onuf's nucleus with 5-HT and noradrenaline terminals driving the pudendal nerve.

Key points to remember:

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA.

6 · Peripheral innervation

*See the paper answer **Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency** (Section 5, Ch 47).*

Also, from the condensed book (not in the paper answer):

- Parasympathetic outflow is **S2–S4**; sympathetic is †**T10–L2**.
- Some parasympathetic postganglionic neurons are **intramural**, so cauda equina or pelvic plexus injury **decentralises rather than denervates** the bladder.
- C fibres are mucosal, mostly **silent** and capsaicin/RTX-sensitive, and they drive a new **spinal reflex** after spinal cord injury (SCI), inflammation, BOO or ageing.
- Afferent sensitisation by ATP, NGF and inflammatory mediators leads to central sensitisation, and **cross-organ sensitisation** explains BPS/IC overlapping irritable bowel syndrome. μ-opioids raise sphincter tone, causing postoperative retention.

7 · Central control

Lesion level predicts the bladder pattern.

- The pontine micturition centre (**PMC**) is **Barrington's nucleus** ("M region"); lesions (stroke, multiple sclerosis, tumour) cause **retention**. It does **not** project to Onuf's nucleus but relaxes the sphincter **indirectly** via GABA/glycine interneurons.
- The **L-region** (ventrolateral pons) is the storage centre and projects to Onuf's nucleus.
- The periaqueductal grey (**PAG**) relays afferents to the PMC, forming a binary "on–off" switch; without higher control the bladder empties at a critical volume, as in infants.
- In the **forebrain**, the **insula** registers desire and the **prefrontal cortex** decides, keeping the reflex under **tonic suppression**. Frontal white-matter lesions cause **incontinence**.

KEY POINT

Lesion localisation: a forebrain lesion removes tonic inhibition, giving DO with preserved coordination. A pontine lesion gives retention. A suprasacral cord lesion gives DO with detrusor–sphincter dyssynergia (**DSD**). A sacral or cauda lesion gives a decentralised, acontractile bladder (see Ch 104).

8 · Storage and emptying phases

- Storage pressure stays low through intrinsic wall properties, quiet parasympathetic outflow and a **sympathetic storage reflex**, which is weak in humans.
- **Guarding reflex**: filling recruits pudendal motoneurons, raising EUS activity, and on coughing urethral pressure rises **before** pressure transmission. Its suppression in voiding is supraspinal, so SCI causes **DSD**.
- **Sphincter-to-bladder reflex**: pudendal afferents inhibit bladder preganglionic neurons, the basis of "holding on" and neuromodulation.

ALGORITHM

1. Tension-receptor (Aδ) firing exceeds threshold.
2. If the cortex permits, the PAG/PMC switch flips.
3. Sympathetic and pudendal outflow switch off; the **urethral sphincter relaxes first**, and urethral smooth muscle relaxes (NO).
4. The detrusor contracts (parasympathetic) and flow begins.
5. Urethral afferents reinforce the contraction until the bladder is empty.

- **Urethra-to-bladder reflexes**: urethral flow facilitates contraction, linking stress urinary incontinence (SUI) with detrusor instability.
- After SCI, weeks of spinal shock (retention) precede an NGF-driven **C-fibre spinal reflex**, giving NDO with DSD.
- **Pdet = Pves − Pabd**. Detrusor power (Pdet × flow, Q) is roughly constant, so voiding **Pdet reflects outlet resistance, not contractility**.
- Women with SUI void at Pdet at Qmax ≈ **19 cmH_2O**, healthy men at ≈ **60–70 cmH_2O**. Low pressure in women does not mean impaired contractility, and male nomograms are invalid in women.

9 · Urethral closure mechanisms

Each SUI treatment targets one closure force.

- α-blockade lowers urethral pressure by about **one third**, so smooth muscle (α1A) tone matters.
- **Mucosal seal**: lamina propria vessels fill to seal the lumen, and oestrogen increases this flow.
- **Passive pressure transmission** reaches the proximal urethra; in the **hammock hypothesis (DeLancey)** it is compressed against a firm posterior fascial layer.
- **Active** guarding adds a fast-twitch reflex squeeze on coughing.

> **KEY POINT**
>
> Continence = **active muscle tone + passive anatomical coaptation**. The integral theory led to the mid-urethral sling (MUS), the SUI standard, yet DeLancey's data suggest intrinsic sphincter function matters more than support (see Ch 109).

10 · Compliance

SAQ Urinary bladder compliance

Paper answer · P3

Type: SAQ (10 marks)
Source: 2017-Summer Paper 3, Q3e

Definition

- **Compliance** is the change in volume for a change in detrusor pressure: **C = ΔV / ΔPdet**, in **mL/cmH_2O**.
- It reflects **low-pressure storage** during filling.

Physiological basis

- **Viscoelastic wall**: smooth muscle, collagen (about 50% of the wall) and elastin (about 2%). Coiled collagen straightens as the bladder fills.
- More collagen or less elastin makes the wall stiff.

Measurement (filling cystometry)

- ICS (International Continence Society) points: Pdet at **start of filling** and at **cystometric capacity** (or before a leak-causing contraction), **excluding detrusor contractions**.
- **No agreed normal value**; compliance depends on capacity. Below **20 mL/cmH_2O** (some use 10–20) suggests poor accommodation. In children, normal is often taken as **>15 mL/cmH_2O**.

Pitfalls

Pitfall	Effect	Fix
Fast filling	False low compliance	Slow or stop filling; pressure falls
Sustained low-amplitude detrusor contraction	Mimics low compliance	Stop filling: pressure returns to baseline
Pop-off: VUR (vesicoureteric reflux), diverticulum, leaking outlet	Falsely good compliance	Video-urodynamics; occlude outlet
- A small bladder can give a low number without danger (**compliance paradox**). **Absolute storage pressure matters more than the ratio.**		

Causes of low compliance

- **Neurogenic**: spina bifida, spinal cord injury, lower motor neuron and decentralisation injury (radical pelvic surgery).
- **Outlet obstruction**: PUV (posterior urethral valves), long-standing BPO (benign prostatic obstruction).
- **Structural**: radiation cystitis, tuberculosis, schistosomiasis, multiple bladder operations.

Consequences and DLPP

- **DLPP (detrusor leak point pressure)**: lowest Pdet at which leakage occurs without contraction or strain.
- **McGuire (1981)**: DLPP **>40 cmH_2O** in myelodysplastic children led to hydronephrosis or VUR in 85%. Not a validated safe limit (20 cmH_2O is more sensitive); aim as low as possible.
- A strong outlet keeps the patient dry but transmits high pressure to the kidneys.

Management

- **Clean intermittent catheterisation + antimuscarinics** (first line).
- **Intradetrusor botulinum toxin**.
- **Augmentation cystoplasty** if refractory; vesicostomy in infants.
- Repeat urodynamics to confirm safety.

Diagram to draw: cystometrogram: normal flat versus steep rising filling curve.

Key points to remember:

- C = ΔV/ΔPdet; <20 mL/cmH_2O suggests poor compliance.
- McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

Also, from the condensed book (not in the paper answer):

- Normal filling keeps pressure **<10 cmH_2O**. Determinants are viscoelasticity, active tone and geometry.
- †A detrusor leak point pressure (DLPP) ≥ **40 cmH_2O** in a low-compliance neurogenic bladder predicts upper tract damage (see Ch 104).

11 · Pathophysiology links

- **Idiopathic OAB**: DO is found in a minority (11% in one cohort), so symptoms reflect dysregulated sensation: more ATP/TRPV1 and Cx43, less BK, and NGF plasticity.
- **BOO** causes hypertrophy and partial denervation, which lead to **muscarinic supersensitivity** and more P2X1 and Cx43. NGF stays raised, which explains **persistent storage symptoms after prostate surgery**.
- **BPS/IC** shows more NGF, ATP/P2X3, TRPV1 and mast cells; tanezumab and intravesical RTX failed (see Ch 113).
- **Underactive bladder (UAB)** is a symptom complex, whereas detrusor underactivity is the urodynamic finding.
- A post-void residual (PVR) ≥ **300 mL** warrants evaluation, as do bothersome LUTS, hydronephrosis, renal insufficiency or culture-proven urinary tract infection (UTI); PVR <300 mL does not prove normal contractility.

12 · Ageing

- Ageing causes **hypofunction**: contractility, sensation and closure pressure fall. Capacity does not fall per se, so small capacity means coexisting DO.
- **α-contraction increases (α1D) and β-relaxation decreases**, so noradrenaline contracts the aged bladder.
- The aged CNS is prone to overactivity, giving urgency with poor emptying (DHIC). †Prefer β3 agonists or low-CNS-penetrant agents in the elderly.

13 · Neuromodulation and botulinum toxin — mechanisms

- Sacral neuromodulation (**SNM**), pudendal stimulation and posterior tibial nerve stimulation (**PTNS**) treat refractory OAB **and** non-obstructive retention by acting on **afferent axons**.
- In OAB, somatic input inhibits preganglionics and transfer to the PMC; in retention, it switches off an overactive **guarding reflex**.
- **OnabotulinumtoxinA (OBTX)**: the light chain cleaves **SNAP-25**, reducing ACh release. It also lowers sensory receptors, neuropeptides and urothelial ATP.
- RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin.

14 · Landmark trials

Trial	Design	Result	Take-home
ABC	RCT: intradetrusor OBTX vs oral antimuscarinic, idiopathic OAB with urgency urinary incontinence (UUI)	Equal fall in daily UUI; OBTX gave complete dryness more often	Antimuscarinic: dry mouth; OBTX: catheterisation, UTI
ROSETTA	RCT: OBTX 200 U vs SNM, women with refractory non-neurogenic OAB, 2 years	Comparable efficacy; OBTX more often dry	OBTX more UTIs (note 200 U dose)

104 Pathophysiology and Classification of Lower Urinary Tract Dysfunction

1 · Normal function: two phases

Storage requires	Voiding requires
Low detrusor pressure (normal compliance), appropriate sensation	Detrusor contraction of adequate magnitude and duration
Outlet closed at rest and on straining	Fall in smooth and striated sphincter resistance
No detrusor overactivity (DO)	No anatomic obstruction

- **Storage is mostly passive**. More collagen (inflammation, bladder outlet obstruction [BOO], decentralisation) gives **drug-unresponsive** low compliance.
- Storage relies on α1 bladder-neck tone, β3 detrusor relaxation and the spinal **guarding reflex**; voiding uses pelvic parasympathetic outflow coordinated by the **pons**.
- **Stress continence** needs equal pressure transmission to the proximal urethra, active reflex closure and a **mucosal seal**.

2 · Pathophysiology

Every failure is of the **bladder, outlet, or both**: the basis of the **Wein functional classification**.

Storage failure

- **Bladder overactivity** means involuntary contractions, low compliance, or both.
- **Overactive bladder (OAB)** (International Continence Society, ICS) is urgency with or without urgency urinary incontinence (UUI), usually with frequency and nocturia. With UUI assume DO, as urodynamic studies often miss it.
- **Hypersensitivity** without overactivity occurs in interstitial cystitis/bladder pain syndrome (IC/BPS). **Outlet underactivity** follows denervation, surgery, trauma, ageing or loss of female support.

	Hypermobility (genuine stress urinary incontinence, SUI)	Intrinsic sphincter deficiency (ISD)
At rest	Outlet competent	Bladder neck poorly functional or open
Cause	Poor pelvic support	Prior surgery, scarring, neurological
Treatment	Correct hypermobility	Coaptation or compression

> **KEY POINT**
>
> Urinary incontinence can occur from ISD alone, but **never from hypermobility alone**. In **DeLancey's hammock**, a firm suburethral layer lets abdominal pressure compress the urethra.

Emptying failure

- **Bladder underactivity** is neurogenic, myogenic (overdistension, drugs) or psychogenic.

- **BOO** (ICS) is low flow and/or raised post-void residual (PVR) **with high detrusor pressure**. Male causes are prostate, bladder neck contracture and stricture; female causes are prolapse and slings.
- **Functional** obstruction includes striated sphincter dyssynergia (commonest), smooth sphincter dyssynergia (**autonomic dysreflexia**), primary bladder neck obstruction and dysfunctional voiding (**non-neurogenic**).

3 · Functional classification (Wein) — 2 × 2 table

	Because of the bladder	**Because of the outlet**
Failure to store	DO; low compliance; hypersensitivity; underactivity with overflow	Hypermobility; ISD. (Plus **fistula**)
Treatment	Bladder training; antimuscarinics, β3-agonists, onabotulinumtoxinA; neuromodulation; augmentation	Pelvic floor training; α-agonists, duloxetine; bulking; slings; artificial urinary sphincter (AUS)
Failure to empty	Underactivity: neurogenic, myogenic, psychogenic	Anatomic obstruction; sphincter dyssynergia; bladder neck dysfunction
Treatment	Valsalva/Credé; parasympathomimetics; electrical stimulation	α-blockers, 5α-reductase inhibitors (5ARIs); prostatectomy, incision; sphincter onabotulinumtoxinA, sphincterotomy
Circumvent	Pads or collecting devices, antidiuretics, catheterisation, diversion	

> **KEY POINT**
>
> Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ An α-agonist tightens the outlet (storage); an α-blocker relaxes it (emptying). OnabotulinumtoxinA in the detrusor treats storage; in the sphincter it treats emptying.

Combined deficits (T10 paraplegia: DO and dyssynergia) are managed by external collection, or by **abolishing DO** plus clean intermittent catheterisation (CIC).

4 · Urodynamics by phase and site

Phase	**Bladder**	**Outlet**
Storage	Filling cystometry; detrusor leak point pressure (DLPP)	Urethral pressure profile; Valsalva leak point pressure (VLPP)
Voiding	Voiding cystometry (Pdet)	Fluoroscopy; sphincter electromyography (EMG)

Flowmetry and PVR assess both phases but **cannot localise** the fault.

5 · Other classifications

LAQ Pathophysiology, classification and management of LUTS.

Paper answer · P2

Type: LAQ (25 marks)
Source: 2024-Winter Paper 2, Q1 (medical and surgical management of BPH also answered in 2015-Summer Paper 2, Q1)

Definition

- **LUTS (lower urinary tract symptoms):** symptoms of the storage, voiding and post-micturition phases; not disease-specific; up to half of older men.
- **BPH (benign prostatic hyperplasia)** is histology; BPE is enlargement; BPO is prostatic obstruction; **BOO (bladder outlet obstruction)** has any cause. Not every man with LUTS has BPO.

Classification

1. By phase (ICS, International Continence Society)

Storage	Voiding	Post-micturition
Frequency, urgency, **nocturia** (waking ≥1 time to void), urgency incontinence	Weak or split stream, hesitancy, intermittency, straining, terminal dribble	Feeling of incomplete emptying, post-micturition dribble

- Storage symptoms form **OAB (overactive bladder)**: urgency ± urgency incontinence, usually with frequency and nocturia; most bothersome.

2. Functional (Wein)

	Because of the bladder	Because of the outlet
Failure to store	DO (detrusor overactivity), low compliance, hypersensitivity	Sphincter weakness (SUI, stress urinary incontinence)
Failure to empty	DUA (detrusor underactivity)	BPO, stricture, bladder neck contracture, sphincter dyssynergia

3. Severity: IPSS (International Prostate Symptom Score), 7 questions scored 0–5 plus a quality-of-life question: mild 0–7, moderate 8–19, severe 20–35.

4. By cause: prostate, bladder, urethra, neurological, polyuria, infection, stone, tumour.

Pathophysiology

- **Prostate:** BPH is an increase in cell number of stroma and glands in the **transition zone** and periurethral region. **Type 2 5α-reductase** in stromal fibroblasts makes **DHT (dihydrotestosterone)**; androgens are permissive, not mitogenic. Ageing, inflammation and metabolic syndrome contribute.
- **Static component:** gland bulk; the **capsule** transmits expansion to the urethra (TUIP, transurethral incision of the prostate, relieves obstruction without removing tissue).
- **Dynamic component: α1A-adrenergic** smooth-muscle tone; the basis of α-blockers.
- Size does not correlate with symptoms or obstruction.
- **Bladder response:** hypertrophy → compensation → decompensation. Collagen causes trabeculation and low compliance; DO causes storage symptoms; late detrusor failure causes retention. One-third keep storage symptoms after surgery.
- **Also know:** nocturnal polyuria (night output >33% of 24 h over age 65); BPO complications: AUR (acute urinary retention), UTI (urinary tract infection), stones, haematuria, renal failure.

Evaluation (EAU, European Association of Urology, men ≥40)

- **Basic:** history (comorbidity, drugs, sexual function), IPSS, DRE (digital rectal examination) and neurological check, **urinalysis (mandatory)**, PVR (post-void residual), PSA (prostate-specific antigen) if life expectancy >10 years.

- **Frequency–volume chart, 3 days:** for storage symptoms or nocturia.
- **Before treatment:** uroflowmetry (Qmax, maximum flow rate, <10 mL/s suggests BOO); ultrasound for volume and median lobe.
- **Selected:** creatinine and renal ultrasound if large PVR; cystoscopy for haematuria, stricture or before PUL (prostatic urethral lift).
- **Pressure–flow study** (separates BOO from DUA) before surgery if voided volume <150 mL, PVR >300 mL, age <50 or >80, or failed surgery.

Management: EAU algorithm

```
Bothersome LUTS? No -> watchful waiting, lifestyle advice Absolute surgical indication? Yes -> surgery Otherwise: Nocturnal polyuria -> lifestyle +/- desmopressin Storage predominant -> antimuscarinic or beta3-agonist Prostate >40 mL -> 5ARI (5-alpha-reductase inhibitor) +/- alpha1-blocker Others -> alpha1-blocker or PDE5 inhibitor Residual storage -> add antimuscarinic or beta3-agonist Failure or complication -> surgery
```

- **Watchful waiting:** education, fluid 1.5–2 L/day (none 2 h before bed), less caffeine and alcohol, double voiding, bladder training; yearly review.

Drug	Example	Mechanism and notes
α1-blocker	Tamsulosin 0.4 mg, silodosin 8 mg, alfuzosin 10 mg	Relax prostatic smooth muscle; acts in days; dizziness, ejaculatory dysfunction, IFIS (intraoperative floppy iris syndrome)
5ARI	Finasteride 5 mg, dutasteride 0.5 mg	Lower DHT; volume falls about 20% over 6–12 months; **PSA halves**; reduces AUR and surgery
Antimuscarinic	Solifenacin, tolterodine	Block detrusor M3 receptors; for storage symptoms if PVR <200 mL
β3-agonist	Mirabegron 50 mg, vibegron	Relax detrusor during storage
PDE5I (phosphodiesterase-5 inhibitor)	Tadalafil 5 mg daily	Smooth-muscle relaxation; useful with ED (erectile dysfunction)

- **MTOPS:** progression cut by doxazosin 39%, finasteride 34%, combination 66%; **CombAT** confirmed. Combine when volume >30–40 mL and PSA >1.5 ng/mL.

Surgery

- **Indications:** refractory retention, recurrent UTI, bladder stones, recurrent haematuria, BPO-related renal insufficiency; failed medical therapy.

Prostate volume	Options
<30 mL	TUIP, TURP (transurethral resection of the prostate)
30–80 mL	**TURP (reference standard)**, bipolar TURP, PVP (photoselective vaporisation of the prostate), enucleation, PUL, Rezum (water vapour), Aquablation
>80 mL	**HoLEP (holmium laser enucleation)** or thulium enucleation, open or robotic simple prostatectomy

- Ejaculation preserved: PUL, Rezum. Anticoagulated: PVP, HoLEP. PAE (prostatic artery embolisation) in selected men.

- **TURP complications:** bleeding, **TUR syndrome** (hyponatraemia with glycine; avoided by bipolar saline), retrograde ejaculation, stricture, incontinence.
- **Also know:** iTind; female LUTS follow OAB and SUI guidelines.

Diagram to draw: BPH → BPE → BPO → BOO → bladder changes → LUTS.

> **Key points to remember:**
> - ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
> - Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
> - 5ARI for >40 mL; MTOPS combination cut progression 66%.
> - TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

Also, from the condensed book (not in the paper answer):
- Chronic obstruction causes hypertrophy, collagen deposition, diverticula and DO, so intervene promptly.

ICS urodynamic classification (most precise)

Domain	Storage	Voiding
Detrusor	Normal; DO (neurogenic or idiopathic)	Normal; underactive; acontractile
Sensation, capacity, compliance	Normal, high/increased, low/reduced, absent	—
Urethra	Normal or incompetent closure	Normal; BOO, dysfunctional voiding, detrusor–sphincter dyssynergia (DSD)

- **Maximum cystometric capacity is when one can no longer delay** voiding.
- DSD is detrusor contraction with involuntary striated sphincter contraction. † **Lapides** and **Bors–Comarr** are older neurological classifications. MCQ

6 · Nocturia and neurogenic dysfunction

- **Polyuria** is >**40 mL/kg/24 h**. **Nocturnal polyuria** is a night-to-24-h volume ratio >**20%** in the **young, >33% elderly**; causes include loss of the nocturnal vasopressin rise and sleep apnoea.
- In **IC/BPS**, normally silent **C fibres** become hyperactive through central sensitisation.

Lesion	Detrusor	Sphincter
Suprapontine	DO	Synergic
Suprasacral spinal	DO	DSD, raised PVR
Sacral	Acontractile	Impaired

> **KEY POINT**
>
> DSD needs an intact sacral arc **disconnected from the pons**, so it is suprasacral, never suprapontine.

106 Evaluation and Management of Males With Urinary Incontinence

1 · Definition and determinants

- **Urinary incontinence (UI)** (International Continence Society, ICS) is any involuntary loss of urine. In men the **external urethral sphincter (EUS) is most important**.
- Risk factors include **age (major, independent)**, lower urinary tract symptoms (LUTS), infection, cognitive or neurological impairment, obesity, diabetes, prostatectomy and drugs.
- Reversible causes, **DIAPPERS**: Delirium, Infection, Atrophic urethritis, Pharmacological, Psychiatric, Endocrine, Reduced mobility, Stool impaction.
- Male:female is about 1:2. **Urgency UI (UUI) predominates**, then stress UI (SUI), then mixed UI (MUI).

2 · Types

Type (ICS)	Mechanism	Male context
SUI: on effort, cough	Sphincter deficiency	Almost only **after prostatectomy**
UUI: with urgency	Detrusor overactivity (DO) overcomes a normal outlet	~**90%** have DO on urodynamics; it is "wet" overactive bladder (OAB)
MUI	DO and sphincter deficiency	Uncommon; may follow prostatectomy
Nocturnal enuresis	Wetting in sleep	New in an adult man: suspect **high-pressure chronic retention** and renal failure
Continuous	Fistula, exstrophy/epispadias, ectopic ureter	Rare
With chronic retention	Residual ≥**300 mL**	"Overflow" is no longer an ICS term; spinal cord injury (SCI) at or above T6 risks autonomic dysreflexia

3 · Evaluation

- The history covers timing, triggers, prior prostate or pelvic surgery, neurology, bowel and sexual function, drugs and the **most bothersome symptom**.
- Examine the abdomen, genitalia, prostate (digital rectal examination, DRE), anal tone, neurology and cognition, and do a **cough test supine then standing**.
- Record **body mass index (BMI)**: weight loss may help SUI, and high abdominal pressure can overcome an artificial urinary sphincter (AUS) cuff.
- **Dipstick** all men; haematuria needs cytology, cystoscopy and upper-tract imaging.
- In a **bladder diary**, **3 days equal 7 days**; include a weekend.
- The International Prostate Symptom Score (IPSS) does **not measure incontinence**, so use the **ICIQ-UI Short Form** (0–21). A **pad test** is not routine and cannot separate UUI from SUI.

Severity	Pads/day	Pad weight/day
Mild	1	<200 g

Severity	Pads/day	Pad weight/day
Moderate	2–4	200–400 g
Severe	>4	>400 g

- **Flow and post-void residual (PVR)**: flow cannot separate bladder outflow obstruction (BOO) from underactivity.
- Check prostate-specific antigen (PSA) only if life expectancy is >10 years and it would change management.
- Do urodynamic studies (**UDS) only if they alter management**, i.e. before invasive treatment (ICI, EAU and AUA guidance) (see Ch 107).
- **Urodynamic stress incontinence (USI)** is leakage on effort without detrusor contraction. **Cough-associated DO incontinence** mimics SUI.

> **KEY POINT**
>
> Every man: history, examination, BMI, dipstick, diary, questionnaire, flow, PVR. Optional: PSA, bloods, imaging, endoscopy, invasive UDS, usually just before surgery.

4 · Urgency urinary incontinence

> **ALGORITHM**
>
> 1. Lifestyle, bladder training and pelvic floor muscle training (PFMT) ≥6 weeks (ideally 3 months).
> 2. Antimuscarinic or β3-agonist, or both.
> 3. Urodynamics optional; then onabotulinumtoxinA, sacral neuromodulation (SNM) or posterior tibial nerve stimulation (PTNS).
> 4. Last resort: augmentation or ileal conduit.

- Cut fluid by **25%** if intake is >1 L/day, and avoid caffeine.
- **Antimuscarinics** have similar efficacy, and extended-release forms are preferred. **Anticholinergic burden** for ≥3 months raises dementia risk by about 50%.
- **β3-agonists** are co-first-line with less dry mouth; with mirabegron, **monitor blood pressure**. In **cognitive impairment, use a β3-agonist or early surgery**.
- Mirabegron plus solifenacin beats escalating solifenacin. Tadalafil is the only phosphodiesterase-5 inhibitor for LUTS with erectile dysfunction.

	OnabotulinumtoxinA	SNM	PTNS
Delivery	Cystoscopic, **100 U** (200 U neurogenic DO)	S3 lead, implanted generator	Needle behind medial malleolus, 30 min weekly × 12
Strength	Repeat every 6–9 months	Most effective, durable >10 years	Least invasive; better for dry OAB
Drawback	Retention in ~**10%**, so the man must accept self-catheterisation	Most invasive, costliest	Frequent visits

- If the SNM test gives ≥**50%** improvement, a permanent generator is implanted (see Ch 115).

5 · Stress urinary incontinence

ALGORITHM

1. Supervised PFMT ≥3 months, with or without biofeedback and electrical stimulation (better started pre-op); consider duloxetine.
2. Exclude anastomotic or urethral stricture and bladder dysfunction (UDS).
3. Choose surgery by severity.

*See the paper answer **Post-prostatectomy incontinence** (Section 8, Ch 123).*

Also, from the condensed book (not in the paper answer):

- After radical prostatectomy wait **6–12 months** before surgery.
- The **AUS is the gold standard**, with about 80% continence (Scott 1973 MCQ), and is activated at about 6 weeks; †deactivate it before catheterisation or endoscopy to prevent erosion.
- Radiotherapy, severe leakage and prior transurethral resection worsen fixed-sling results, so choose an AUS. **ProACT** is **contraindicated after radiotherapy**.

EAU 2023 recommendation (post-prostatectomy incontinence, PPI)	Grade
Offer AUS for moderate–severe PPI	Strong
Warn of AUS/ProACT complications, mechanical failure, explantation	Strong
Inform of duloxetine side effects, off-label use	Strong
Fixed transobturator sling for mild–moderate PPI	Weak
Do not offer bulking agents; no ProACT after radiotherapy	Weak

6 · Mixed and other types

- In **MUI**, treat the most bothersome component first; DO may mask SUI, so treat DO and repeat UDS.
- In high-pressure chronic retention with enuresis, **catheterise first**, then relieve BOO.
- Enuresis with a low residual is usually OAB and is treated with an antimuscarinic or β3-agonist, with or without desmopressin.
- Postmicturition dribble is treated with **urethral milking**.
- **Red flags** needing a specialist are haematuria, pain, recurrent infection, pelvic radiotherapy, abnormal DRE and high PVR.

7 · Landmark trials

Trial	Design	Result	Take-home
MASTER	Non-inferiority randomised controlled trial (RCT), male sling vs AUS for urodynamic SUI after prostatectomy	Equivalent at 12 months; secondary outcomes favoured AUS; few completely dry	Sling is an acceptable alternative; counsel realistically

107 Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

1 · Definition and role

- **Urodynamics (UDS)** is the dynamic study of urine transport, storage and evacuation. Use it when simple tests fail, to **diagnose the cause of symptoms** and to assess **diseases that silently damage the upper tracts** such as spinal cord injury (SCI), multiple sclerosis (MS) and radiation cystitis.

Wein functional classification — the frame for every study

Site	Overactive	Underactive
Bladder	Failure to store	Failure to empty
Outlet	Failure to empty	Failure to store

- Set the questions first and aim to **reproduce the symptom**; an unrecorded abnormality is not excluded.

> **KEY POINT**
>
> **Box 107.1 — Urodynamic risk factors** (treat even if asymptomatic): impaired compliance; detrusor–external sphincter dyssynergia (DESD); detrusor–internal sphincter dyssynergia (DISD); high-pressure detrusor overactivity (DO) throughout filling; detrusor leak point pressure (DLPP) **≥ 40 cmH_2O**; poor emptying with high storage pressure.

2 · Preparation

- **Decide medication status in advance**: test on regular dosing to assess drug effect, or stop the drug first to uncover aetiology.
- **ICS pre-invasive work-up**: symptom and bother score, **3-day frequency–volume chart**, free flow and postvoid residual (PVR).
- **Urinalysis before every invasive study**; postpone for symptomatic urinary tract infection (UTI).

Antibiotic prophylaxis (AUA / SUFU Best Practice Policy)

The "index" patient needs **no routine prophylaxis** (level 1). Non-index patients get a **single dose within 1 h**: oral trimethoprim–sulfamethoxazole first; alternatives include a cephalosporin, amoxicillin–clavulanate, an aminoglycoside or a fluoroquinolone.

Prophylaxis YES	Prophylaxis NO
Neurogenic LUT dysfunction (NLUTD); bladder outlet obstruction (BOO) or raised PVR; age > 70; **asymptomatic bacteriuria**; immunosuppression (especially renal transplant); chronic catheters including clean intermittent catheterisation (CIC); high-risk orthopaedic implant	Diabetes; **recurrent UTI history**; recent hospitalisation; obesity; menopause; gender; prior urological surgery; cardiac valve disease (not routinely)

Trap: asymptomatic bacteriuria is treated here, unlike in most of urology.

3 · Non-invasive components

SAQ Uroflowmetry

Paper answer · P2 · P3 · merged from 2

Asked as: P2 SAQ "Uroflowmetry." · P3 SAQ "Uroflowmetry"

Definition

- A non-invasive urodynamic test that records the **volume of urine voided per unit time** (mL/s). It reflects detrusor contraction and outlet resistance together.

Method

- Patient comes with a **comfortably full bladder** (normal desire) and voids in private, in the usual (preferred) position.
- **Flowmeters:** weight transducer (load cell), rotating disc, capacitance dipstick.
- **Voided volume ≥150 mL** (Qmax is unreliable below 125–150 mL). Repeat 2–3 times; ask whether the void was typical and compare with the frequency–volume chart.
- Measure **PVR** (post-void residual) by ultrasound afterwards.

Parameters

Parameter	Meaning
Qmax (maximum flow rate)	Most useful (key) value; normal men >15 mL/s, women >20 mL/s; read from the smoothed curve, ignoring spikes
Qave (average flow rate)	Volume ÷ flow time; less useful and less accurate for obstruction
Voided volume (VV)	Must be ≥150 mL
Flow time vs voiding time	Voiding time includes interruptions
Time to Qmax	Normally early, in the first third of the void
Voiding efficiency	VV ÷ (VV + PVR) × 100

Curve patterns (ICCS, International Children's Continence Society)

Pattern	Suggests
Bell-shaped	Normal
Tower (high, short)	Overactive bladder
Plateau (flattened, box, prolonged)	Fixed obstruction: **urethral stricture**, PUV (posterior urethral valves)
Prolonged, low, skewed	BPO (benign prostatic obstruction) or detrusor underactivity
Staccato (never reaches zero)	Sphincter overactivity, dysfunctional voiding
Intermittent / interrupted (reaches zero), straining	Detrusor underactivity with abdominal straining, obstruction, or DSD (detrusor–sphincter dyssynergia)

Interpretation

- **Qmax <10 mL/s:** obstruction likely (high specificity). **Qmax <15 mL/s:** more sensitive, less specific.

- **Nomograms / volume correction: Siroky** (men), **Liverpool** (men and women, centiles). Corrected Qmax = Qmax ÷ √VV.
- **Artefacts:** "wag" (moving stream), knocking the flowmeter, straining spikes. Read Qmax from the smoothed curve, not single spikes.
- Children: valid if voided volume ≥50% of expected capacity; accept Qmax only if the peak lasts ≥2 s.

Limitations

- **Cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**; only a pressure–flow study can.
- Varies with volume, position, anxiety and between days.

Uses

- BPH (benign prostatic hyperplasia): evaluation before active treatment and follow-up after medical or surgical treatment.
- Follow-up after **urethroplasty** or optical urethrotomy (stricture recurrence).
- Children: dysfunctional voiding, BBD (bladder and bowel dysfunction), after posterior urethral valve ablation; with patch EMG (electromyography).
- Women: voiding dysfunction, including before stress incontinence surgery.
- Pressure–flow study before BPH surgery if Qmax >10 mL/s (P2: EAU [European Association of Urology]; P3: AUA [American Urological Association], "may be done").

Diagram to draw

- Flow curves: normal bell, tower, plateau (stricture), prolonged low (BPO), interrupted/intermittent.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
> - Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
> - Cannot distinguish BOO from DUA.

Also, from the condensed book (not in the paper answer):

- Minimum voided volume is **VV 150 mL**, because maximum flow (Qmax) is linear above it and hyperbolic below.
- Normal Qmax is > 15 mL/s in men (< 10 suggests obstruction) and > 20 mL/s in women.

4 · Equipment, transducers and quality control

- The minimum is vesical (Pves), abdominal (Pabd) and subtracted detrusor pressure (Pdet) with flow. Infuse saline or contrast, **not CO_2**.

Feature	External water-filled	Air-charged	Microtip
Reference level	**Upper border of symphysis pubis**; re-level on position change	None (no hydrostatic effect)	—

Feature	External water-filled	Air-charged	Microtip
Problems	Air bubbles; motion and hydrostatic artefact; underdamped	Overdamped (signals > 3 Hz attenuated), so coughs are blunted	Expensive, reusable

- Systems are **not interchangeable** (Pdet may differ by up to 10 cmH_2O); water-filled systems underpin most nomograms.

> **KEY POINT**
>
> **Quality control** (ICS Good Urodynamic Practice): zero to atmosphere, flush air, and keep resting Pdet near 0. Do a **cough test** before filling, at intervals and before voiding; a biphasic Pdet spike means re-flush or reposition. Rectal contractions cause an unexplained fall in Pdet.

5 · Filling cystometry (CMG)

CMG shows whether storage is at safe, low pressure.

- Filling to "permission to void" gives the **maximum cystometric capacity (MCC)**. **Pdet = Pves − Pabd**; rectal Pabd is preferred.
- Standard adult filling is **30–100 mL/min**; the ICS physiological rate is < body weight (kg)/4 mL/min. Neurogenic bladders need a physiological rate and body-warm saline (EAU). DO is commoner upright.
- A **normal ("stable")** bladder keeps Pdet near zero with no involuntary contractions. **Sensation** (first sensation, first desire, strong desire, urgency) is the only subjective parameter.

Detrusor overactivity (DO)

- DO means involuntary detrusor contractions (IDC) during filling, either **idiopathic** or **neurogenic**. The label comes from the history, as tracings look identical.
- Absence does not exclude it: **up to 50%** of women with urgency urinary incontinence (UUI) show no DO, and some asymptomatic volunteers do show it.
- In BOO with overactive bladder (OAB), a **single terminal IDC** predicts storage-symptom resolution after transurethral resection of the prostate (TURP). In neurogenic DO, higher Pdetmax and longer IDCs predict upper-tract deterioration.
- **Cough-associated DO** mimics stress urinary incontinence (SUI) but is detrusor in origin.

Compliance

*See the paper answer **Urinary bladder compliance** (Section 7, Ch 103).*

Also, from the condensed book (not in the paper answer):

- Storage pressure **> 40 cmH_2O** itself leads to upper-tract damage.
- Outlet pop-off hides poor compliance, which may appear only after a sling or artificial urinary sphincter (AUS), so occlude the outlet.

6 · Leak point pressures

*See the paper answer **Leak point pressures and ALPP** (Section 5, Ch 40).*

Abdominal leak point pressure, ALPP (cmH_2O)	Inference (McGuire)
< 60	ISD

Abdominal leak point pressure, ALPP (cmH_2O)	Inference (McGuire)
60–90	Equivocal
≥ 90	Little or no ISD (hypermobility)

- Intrinsic sphincter deficiency (ISD) and hypermobility coexist; ALPP poorly predicts surgical success.
- **Occult SUI** is leakage only on prolapse reduction in a continent woman with pelvic organ prolapse (POP).
- **DLPP > 40 cmH_2O** is not a validated safe cut-off (20 is more sensitive in children), so aim **as low as achievable** by lowering bladder pressure or the outlet (e.g. sphincterotomy for DESD), then repeat the UDS.

7 · Urethral pressure profilometry (UPP)

- **Maximum urethral closure pressure (MUCP)** is maximum urethral minus intravesical pressure (normal 40–60 cmH_2O in women).
- **MUCP ≤ 20 cmH_2O** means ISD ("type III" SUI) and predicts sling or suspension failure.
- The EAU advises against using UPP or leak point pressure to grade severity.

8 · Pressure–flow studies (PFS)

Only PFS tells obstruction from a weak bladder.

Pdet	Flow	Interpretation
Low/normal	High/normal	Unobstructed
High	Low	**BOO**
Low	Low	Detrusor underactivity (obstruction may be hidden)

- ICS **detrusor underactivity** is reduced contraction strength or duration, so emptying is prolonged or incomplete. **Acontractile detrusor** replaces "areflexia".

Men — nomograms and indices

Index	Formula	Cut-offs
Bladder outlet obstruction index (BOOI)	PdetQmax − 2Qmax	**> 40 obstructed · 20–40 equivocal · < 20 unobstructed**
Bladder contractility index (BCI)	PdetQmax + 5Qmax	**> 150 strong · 100–150 normal · < 100 weak**

- BOOI ≥ 40 needs Pdet ≥ 40 even at zero flow, so a decompensated detrusor masks obstruction.
- **Stop tests** estimate isovolumetric pressure (Piso); in men Piso < 50 cmH_2O suggests underactivity.
- About 2/3 of men with benign prostatic obstruction (BPO) have DO; half to two-thirds resolve after relief.
- MCQ In the bethanechol supersensitivity test (Lapides), 2.5 mg is given subcutaneously; a rise **≥ 15 cmH_2O** at 100 mL is positive (neurogenic), but the test is unreliable.

Women

- Women void at low pressure, so the **BOOI and ICS nomogram cannot be used**. Suspect BOO after anti-incontinence surgery or with prolapse.

Method	Criterion	Comment
Cut-points	e.g. Qmax ≤ 12 mL/s and PdetQmax ≥ 25 cmH_2O (Defreitas); Chassagne ≤ 15 / ≥ 20	Anatomic obstruction only; overlap
Videourodynamics, VUDS (Nitti)	Radiographic obstruction between bladder neck and meatus with a sustained contraction of any magnitude	Localises; works despite weak detrusor
Blaivas–Groutz	Free Qmax vs Pdetmax; none/mild/moderate/severe	Two separate voids; **overdiagnoses**
Solomon–Greenwell	PdetQmax − 2.2Qmax: < 0 unlikely; > 18 almost certain	Includes functional obstruction

9 · EMG and sphincter coordination

- Surface perineal electrodes are standard. Normally EMG quietens and urethral pressure falls **before** Pdet rises.
- The **guarding reflex** (EMG rise during an IDC) is normal, not dyssynergia.

	DESD	Dysfunctional voiding
Cause	**Suprasacral cord lesion** (the higher, the likelier)	Learned; no neurological lesion
Behaviour	Involuntary sphincter contraction during DO or voiding	Intermittent sphincter contraction during voiding
Significance	High pressure causes poor compliance and upper-tract risk	LUT symptoms; fluoroscopy improves accuracy

- **DISD** needs a suprasacral lesion above the T10–L1 sympathetic outflow and is seen on **VUDS only**. **Primary bladder neck obstruction (PBNO)** is also invisible to EMG and needs fluoroscopy.

10 · Videourodynamics

SAQ Urodynamic studies, including video urodynamics

Paper answer · P2 · P4 · merged from 3

Asked as: P2 SAQ "Uro Dynamic Studies." · P2 SAQ "Vesicourodynamics" · P4 SAQ "Video Urodynamics"

"Vesicourodynamics" is taken as videourodynamics (VUDS): urodynamics with simultaneous fluoroscopic imaging of the bladder and urethra.

Definition and principles

- UDS (urodynamic studies) test bladder and urethral function in storage and voiding.
- ICS (International Continence Society): **decide the question first** and **reproduce the symptom**; a normal study does not exclude disease.
- Before invasive UDS: symptom score, 3-day **frequency–volume chart**, free uroflow and PVR (post-void residual); treat UTI (urinary tract infection) first.

Components of conventional UDS

Test	Method	Main values
Uroflowmetry	Voided volume ≥150 mL	Bell curve normal; Qmax <10 mL/s (men) suggests obstruction or weak detrusor; plateau = stricture
PVR	Bladder scan or catheter	Emptying
Filling cystometry	Bladder and rectal lines, fill with saline 30–100 mL/min (adults); **Pdet = Pves − Pabd**; zero at atmosphere, transducer at symphysis	Sensation, capacity, **DO (detrusor overactivity)**, **compliance** = ΔV/ΔPdet (<20 mL/cmH_2O impaired)
Leak point pressures	ALPP (abdominal LPP): cough or Valsalva at 150 mL	Low ALPP = weak sphincter (ISD, intrinsic sphincter deficiency)
	DLPP (detrusor LPP)	**>40 cmH_2O** risks upper tracts (McGuire)
Pressure–flow study	Void with lines in	High pressure–low flow = obstruction; low–low = underactivity
EMG (electromyography)	Perineal surface electrodes	DESD (detrusor–external sphincter dyssynergia)
UPP (urethral pressure profile)	Catheter withdrawn	Limited value
Videourodynamics	Above with contrast and fluoroscopy	VUR (vesicoureteric reflux), bladder neck, site of obstruction

Indices (men)

- **BOOI (bladder outlet obstruction index)** = PdetQmax − 2Qmax: **>40 obstructed**, 20–40 equivocal, <20 unobstructed.
- **BCI (bladder contractility index)** = PdetQmax + 5Qmax: **>150 strong**, 100–150 normal, **<100 weak**.
- Women: no validated nomogram (Nitti video criteria).

Indications for UDS

- Male LUTS (lower urinary tract symptoms) before surgery when unclear (young men, equivocal flow, failed TURP [transurethral resection of the prostate]); complex, mixed or recurrent incontinence.
- **Neurogenic bladder** (spinal injury, MS [multiple sclerosis], spina bifida), radiation cystitis, before augmentation.
- Children: neurogenic bladder, PUV (posterior urethral valves); fill at **5–10% of expected capacity per minute**; expected capacity = (age + 2) × 30 mL.

Video urodynamics (VUDS): definition and set-up

- **VUDS:** multichannel urodynamics (pressure–flow, with or without EMG) recorded at the same time as fluoroscopy, using **contrast as the filling fluid**.
- Vesical and rectal catheters (Pdet [detrusor pressure] = Pves − Pabd), flowmeter, EMG; C-arm (most flexible) or a tilting table.

- Position for the question: **SUI (stress urinary incontinence) views standing**; voiding in the usual position (women sitting, men standing).
- **Screen only at key moments** (provocation, pressure rise, voiding) to limit radiation.

What fluoroscopy adds

Situation	VUDS finding
NLUTD (neurogenic lower urinary tract dysfunction)	**Gold standard** (EAU); DESD and **DISD** (detrusor–internal sphincter dyssynergia, seen only on VUDS); trabeculation, diverticula
VUR	Pressure, volume and grade at onset; reflux "pops off" pressure, so compliance looks falsely safe
Bladder neck	Only way to see and assess it; **PBNO** (primary bladder neck obstruction) in young men and women
Female obstruction	**Nitti criteria:** radiographic narrowing between bladder neck and meatus during voiding, with a sustained detrusor contraction
Dysfunctional voiding	"Spinning-top" urethra at the external sphincter
SUI	Leakage with ALPP; open bladder neck at rest suggests ISD
Male obstruction	Localises prostatic obstruction, bladder neck contracture or stricture

- **PBNO types (young men):** I high pressure, low flow; II normal pressure, low flow with bladder-neck narrowing; III delayed bladder-neck opening.

Indications for VUDS

- Known or suspected NLUTD; unexplained retention in women; young men with LUTS; complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant.
- **Children:** PUV, spinal dysraphism; contrast-enhanced ultrasound VUDS avoids radiation.

Urodynamic risk factors (upper-tract danger)

- Impaired compliance; DESD; DISD; high-pressure DO throughout filling; DLPP >40 cmH_2O; poor emptying with high storage pressure.

Limitations of VUDS

- Radiation, cost, space, availability, contrast allergy.
- EAU (European Association of Urology): optimum test in NLUTD. AUA (American Urological Association) guideline: lowest radiation dose that answers the question.

Also know

- Ambulatory UDS; bethanechol test; antibiotic prophylaxis for risk groups.

Diagrams to draw

- Normal trace: Pves, Pabd, Pdet, flow, EMG.
- VUDS trace beside a voiding cystourethrogram image.

Key points to remember:

- Pdet = Pves − Pabd.
- Ask a question first; reproduce the symptom.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 cmH_2O: upper-tract risk.
- VUDS = urodynamics + fluoroscopy with contrast.
- VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

Also, from the condensed book (not in the paper answer):

- If VUDS is unavailable in NLUTD, do CMG with PFS (EAU).
- Fluoroscopy is critical for PBNO types II and III.

11 · Ambulatory urodynamics

- Natural filling shows more DO and better compliance than pump filling; use it for symptoms unexplained by conventional UDS.

12 · Indications by condition (AUA/SUFU, EAU)

Order UDS when the result will change treatment.

Female SUI

- In uncomplicated demonstrated SUI (the "index" patient) UDS may be omitted (AUA/SUFU 2023; EAU: not routinely).
- Do UDS in non-index women: prior anti-incontinence or POP surgery, symptom–examination mismatch, voiding dysfunction, significant UUI, raised PVR, unconfirmed SUI or NLUTD.

Male LUTS

- No UDS is needed to start an α-blocker for lower urinary tract symptoms (LUTS).
- AUA/SUFU: do **PFS when obstruction matters, before invasive, morbid or irreversible treatment**, because without obstruction surgery does worse.
- EAU PFS triggers before invasive treatment are prior failed treatment, inability to void > 150 mL, **Qmax > 10 mL/s**, PVR > 300 mL, and age > 80 or < 50.
- **Impaired compliance is the only absolute urodynamic indication to treat BOO.**

Women with voiding symptoms; NLUTD

- PFS separates obstruction from impaired contractility; functional obstruction (PBNO, dysfunctional voiding) needs VUDS.
- The NLUTD goal is **low-pressure filling**, even at the cost of catheterisation; only UDS separates DO from poor compliance.
- UDS is essential before treatment and in follow-up in SCI and myelomeningocele.

13 · Landmark trials

Trial	Design	Result	Take-home
ValUE	RCT, 630 women, uncomplicated SUI: office evaluation with or without UDS	Success ~77% both arms	Office evaluation alone non-inferior before SUI surgery
Dutch embedded RCT	Discordant UDS randomised: immediate mid-urethral sling vs UDS-tailored	Equal cure	UDS discordance need not change surgery
SISTEr	Burch vs pubovaginal sling	ALPP and DO did not predict outcome	UDS adds little in stress-predominant SUI

109 Neuromuscular Dysfunction of the Lower Urinary Tract

1 · Principles and classification

The lesion level predicts the bladder and sphincter pattern.

- The International Continence Society (ICS) uses **five parameters**: sensation; detrusor (normal, overactive, areflexic, impaired); compliance; smooth sphincter (synergic or dyssynergic); striated sphincter (synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone).

Level	Typical disorders	Detrusor	Smooth sphincter	Striated sphincter
Suprapontine	Stroke, tumour, Parkinson disease (PD), dementia, cerebral palsy	Overactive	Synergic	Synergic (with or without impaired voluntary control; PD bradykinesia)
Multiple system atrophy (MSA)	—	Overactive, later impaired	**Open**	Denervated
Suprasacral cord	Spinal cord injury (SCI), multiple sclerosis (MS), transverse myelitis	Overactive	Synergic (dyssynergic if above T7)	**Dyssynergic**
Sacral cord / cauda equina	Sacral SCI, disc, myelodysplasia	Areflexic	Competent non-relaxing; may open	Fixed tone
Peripheral	Radical pelvic surgery, diabetes, tabes	Impaired / areflexic	Open (pelvic surgery)	Fixed tone / synergic

> **KEY POINT**
>
> Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**. An open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**.

2 · Lesions at or above the brainstem

Stroke

- **Acutely** there is retention ("cerebral shock"). **Chronically** there is detrusor overactivity (DO) with **synergic sphincters**.
- **True detrusor–sphincter dyssynergia (DSD) does not occur. Pseudodyssynergia** is an electromyographic (EMG) flare from voluntary attempts to inhibit a contraction.
- In men with lower urinary tract symptoms (LUTS), do a full urodynamic study (UDS) before outlet surgery.

Dementia, brain injury, tumour, others

- **Dementia**: anticholinergics worsen cognition, so a **β3 agonist is strongly preferred** (SUFU).

- **Brain injury** above the pontine micturition centre (PMC) gives DO with synergic sphincters; brainstem injury below the PMC can give DSD. Frontal **tumours** can cause retention.
- **Normal-pressure hydrocephalus** gives dementia, gait ataxia and incontinence, improved by a shunt.
- **Cerebral palsy** gives DO with synergic sphincters, but about half of adults with LUTS have poor compliance or a detrusor leak point pressure (DLPP) ≥40 cm H_2O, so evaluate fully.

Parkinson disease vs multiple system atrophy

- **PD** LUTS occur in 35–70% and are mainly storage symptoms. Sphincter **bradykinesia** mimics DSD.
- Treat with overactive bladder (OAB) drugs with low central effect, such as mirabegron or trospium; onabotulinumtoxinA, tibial nerve stimulation (PTNS) and sacral neuromodulation (SNM) also work.
- **Transurethral resection of the prostate (TURP) in PD is not contraindicated.** The old "post-TURP incontinence in PD" cases were MSA, so confirm PD, exclude MSA and do videourodynamics first.
- **MSA** affects the intermediolateral columns and **Onuf nucleus**, causing early severe LUTS and erectile dysfunction (ED).
- In MSA, DO comes early and poor emptying late. Treating DO worsens emptying, so add clean intermittent catheterisation (CIC); give desmopressin for nocturia. **Prostatectomy is hazardous.**

Feature	MSA	PD
Urinary symptoms / ED	Precede or accompany parkinsonism; ED near universal	Years after diagnosis; ED ~⅓
Dominant problem; PVR	UI; high postvoid residual (PVR)	Frequency/urgency; PVR normal
Bladder neck	**Open**	Synergic
Striated sphincter EMG	**Denervation**	Normal (bradykinesia)
Detrusor	DO, then areflexia	DO; areflexia uncommon
Outlet surgery	Worsens	Acceptable after work-up

3 · Spinal cord disease

Cord lesions add dyssynergia and silent upper-tract damage.

Multiple sclerosis

- LUTS occur in 50–90% and are the presenting complaint in up to 15%. On UDS, DO is found in ~⅔, DSD in 30–65% of those with DO, and areflexia in ~¼.
- Symptoms do not track the disease, so **baseline UDS is essential**.
- Keep management **reversible**, because the pattern changes: antimuscarinic with CIC, onabotulinumtoxinA, or SNM in selected patients. Credé and Valsalva voiding are contraindicated, and an indwelling catheter is the last resort.

LAQ Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 1 Paper 2, Q2

Assumed meaning: cervical spinal cord injury (traumatic, operated, or from the surgery itself), giving a suprasacral lesion above T6.

Introduction

- A cervical cord lesion lies **between the PMC (pontine micturition centre) and the sacral cord**. It causes tetraplegia and the classic suprasacral pattern.
- Complete transection is rare; incomplete lesions are commonest. Manage by **urodynamics, not neurology**.

Pathophysiology

- The PMC normally coordinates detrusor contraction with sphincter relaxation. Cutting the descending pathway leaves a **spinal reflex**.
- **Why NDO (neurogenic detrusor overactivity) appears:** unmyelinated C-fibre afferents become sensitised (NGF (nerve growth factor)), sprout and drive a new sacral reflex. Bulbospinal inhibition is lost.

Phase	Detrusor	Outlet	Clinical
Spinal shock (usually 6–12 weeks; up to 1–2 years)	Areflexic, smooth bladder	Bladder neck closed; reduced closure pressure	**Retention**; overflow only
Recovery (suprasacral pattern)	**NDO**, no sensation below lesion	**Striated DSD (detrusor–sphincter dyssynergia)**; lesion above T7–T8 adds **smooth sphincter dyssynergia**	Reflex leakage, incomplete emptying, high pressure

- **Blaivas DSD types:** type 1, sphincter relaxes at peak pressure (incomplete lesions); type 2, sporadic contractions; type 3, crescendo–decrescendo obstruction (types 2–3 with complete lesions).
- **Consequences:** high storage and voiding pressure, VUR (vesicoureteric reflux) in 17–25%, hydronephrosis, stones (about 7% in 10 years), UTI (urinary tract infection). Untreated, about half of men with DSD deteriorate (worse if complete, continuous DSD).
- Upper tract risk also rises with low compliance (<12.5 mL/cm H_2O) and NDO for over a third of filling.
- **VUR:** treat the bladder first (lower the pressure); reflux surgery in a thick bladder is difficult.
- **Also know:** cervical myelopathy may instead give an underactive detrusor.

Autonomic dysreflexia

- Occurs with lesions **above T6** (cervical 60%), after spinal shock. Bladder or bowel distension triggers unopposed spinal sympathetic outflow, as medullary inhibition is lost.
- **Features:** pounding headache, severe hypertension (systolic rise ≥20 mmHg defines it at urodynamics), bradycardia, sweating and flushing above the lesion. It can cause stroke or seizures.

- **Treatment:** sit up, **drain the bladder** (check catheter), then **nitroglycerin 2% paste** above the lesion; intravenous nitroprusside or labetalol if persistent. Prophylaxis with terazosin or prazosin. Monitor blood pressure during cystoscopy and urodynamics; use spinal or general anaesthesia.

Evaluation

- History, examination, **hand function** (decides self-catheterisation), transfers, carers.
- Urinalysis, creatinine, **renal ultrasound**. Men with good hand function can self-catheterise; others need carers or a suprapubic catheter.
- **VUDS (videourodynamics) after spinal shock**, with blood pressure monitoring: compliance, NDO, DSD, VUR, **DLPP (detrusor leak point pressure) ≥40 cm H_2O** = upper tract risk.
- AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2021: suprasacral lesions are "unknown risk" until tested; DSD or poor compliance = high risk.

Management

Goals: protect kidneys, low-pressure storage, low-pressure emptying, no infection, continence, social acceptability. Start with the simplest reversible option.

- **Spinal shock: CIC (clean intermittent catheterisation)** preferred; indwelling catheter early is acceptable. Defer irreversible procedures.

Aim	Options
Emptying	**CIC: gold standard**, volumes <500 mL; carer CIC in tetraplegics. **Suprapubic catheter** better than long-term urethral catheter. Credé/Valsalva not recommended
Storage (lower pressure)	**Antimuscarinics** ± mirabegron (β3 agonist); **intradetrusor onabotulinumtoxinA 200 U** if refractory (repeat when the effect wears off)
Refractory small bladder	**Augmentation ileocystoplasty** ± continent catheterisable channel (Mitrofanoff)
Men unable to do CIC	**External sphincterotomy** or intrasphincteric botulinum toxin → reflex voiding into condom catheter. High failure; DLPP ≥40 cm H_2O afterwards = failure. Urethral stents abandoned
Neuromodulation	**SARS (sacral anterior root stimulator, Brindley) + posterior sacral rhizotomy**: abolishes NDO and dysreflexia, allows voiding on demand
Last resort	Ileovesicostomy; bladder neck closure + suprapubic catheter; ileal conduit

- **Also know:** sacral neuromodulation is used for neurogenic bladder without SCI (spinal cord injury); hydrophilic catheters delay first UTI.

Complications and follow-up

- **UTI:** treat only symptomatic bacteriuria (5–14 days); no routine prophylaxis.
- Stones, VUR (treat the bladder first), bladder cancer (no screening cystoscopy, AUA/SUFU 2021).
- Lifelong: yearly review and renal function, upper tract imaging every 1–2 years, repeat urodynamics if anything changes.

`Cervical SCI → spinal shock (retention): CIC → VUDS at 3 months: NDO + DSD → antimuscarinic + CIC → Botox → augmentation → cannot CIC: sphincterotomy + condom / SARS / ileovesicostomy`

Diagram to draw: neural axis: PMC, lesion at cervical cord, sympathetic T10–L2, sacral S2–S4, with the resulting NDO + DSD.

> **Key points to remember:**
> - Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
> - Autonomic dysreflexia above T6: drain bladder first, nitrates next.
> - CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
> - DLPP ≥40 cm H_2O threatens the kidneys.

Also, from the condensed book (not in the paper answer):

- Spinal shock usually lasts **6–12 weeks**; bladder reflexes return with the lower-limb deep tendon reflexes.
- Smooth sphincter dyssynergia occurs with cord lesions at **T7–T8** (column T6) or above.
- The sacral cord begins at vertebral column T12–L1, so state column and cord level separately.

	Suprasacral (complete)	**Sacral (complete)**
Detrusor	Overactive	Areflexic
Compliance	Normal initially	Normal or high; may fall
Smooth sphincter	Synergic (dyssynergic above T7–T8)	Competent non-relaxing; may later open
Striated sphincter	Dyssynergic	Fixed tone, no voluntary control
Somatic signs	Spasticity, hyperreflexia	Flaccidity, reduced reflexes
Management	Lower pressure (drugs or toxin) and CIC	CIC and keep compliance safe

- **Neurology does not equal urodynamics**, and thoracolumbar (T10–L2 column) lesions are least predictable, so **manage by UDS, not by level**.
- Upper-tract risks are detrusor fibrosis, leak point pressure ≥75 cm H_2O and DO for >⅓ of filling; DLPP >40 cm H_2O is the classic threshold. Damage can occur **without vesicoureteral reflux (VUR)**.

Autonomic dysreflexia (AD)

- AD occurs in SCI **above T6** and may appear years later. Triggers include bladder or rectal distension, a blocked catheter, instrumentation, stones, sores and fracture.
- Risk during UDS is highest with complete cervical lesions, DSD and poor compliance.

ALGORITHM

1. Sit the patient upright and loosen clothing.
2. **Remove the stimulus**: drain the bladder, unblock the catheter, empty the rectum.
3. If blood pressure stays high, apply nitroglycerin 2% paste above the lesion (first line).
4. If it still persists, give labetalol or intravenous nitroprusside.
5. For prophylaxis, give terazosin or prazosin (α1-blockers).
6. If intractable, consider sacral rhizotomy, sympathectomy or cordectomy.

- Nifedipine (10–20 mg sublingual) is now banned in many centres because of uncontrolled hypotension.

VUR and women with SCI

- A **chronic urethral catheter is worst** for the upper tracts. VUR artefactually enlarges capacity and lowers pressures on UDS.
- Women paraplegics do CIC, while tetraplegics often need indwelling catheters.

Other cord disorders

- **Transverse myelitis** causes retention, then DO with DSD, and is managed as SCI. **Cervical myelopathy** gives DO with or without DSD, or underactivity, so UDS guides treatment.
- In **myelomeningocele**, level correlates poorly with UDS. The typical adult has an **areflexic bladder and open bladder neck** with a fixed sphincter, so leaks at a fixed pressure; DSD occurs in 10–15%.
- Poor compliance, high pressure and DO predict renal decline. For stress UI, bulking is safer than a sling or artificial urinary sphincter (AUS) because it does not reduce compliance.
- **Tethered cord** presents after growth spurts, usually with DO. A sensory neurogenic bladder (tabes, B12) has a large capacity.

4 · Lesions distal to the cord

Root and nerve lesions give a weak bladder behind a fixed outlet.

- **Disc disease** (L4–L5, L5–S1) most often gives a **normally compliant areflexic bladder**. Do **pre-laminectomy UDS** for clinical and medicolegal reasons.
- **Cauda equina syndrome** causes saddle anaesthesia and an acontractile detrusor. **Decompression within 48 h** gives better urinary and rectal recovery.
- **Radical pelvic surgery** (abdominoperineal resection, radical hysterectomy) gives impaired contractility, fixed striated tone, an **open non-functional bladder neck** and lower compliance, so the patient leaks or retains.
- After pelvic surgery, **use CIC and wait 6–12 months**; do **no prostatectomy unless BOO is proven**.
- Herpes zoster and herpes simplex (**Elsberg syndrome**) cause transient retention.

SAQ Bladder dysfunction non genitourinary systemic disease

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 2, Q3a

Taken as lower urinary tract dysfunction caused by systemic (non-genitourinary) diseases.

Principle

- The **site of the lesion**, not the disease name, predicts the bladder:
- Brain above the pons: **DO (detrusor overactivity), no DSD (detrusor–sphincter dyssynergia)**.
- Spinal cord: DO **with DSD**.
- Peripheral nerves or roots: **areflexia, reduced sensation, high PVR (post-void residual)**.
- **Urodynamics** before treatment, because symptoms do not predict findings.

Diabetes mellitus (most important)

- **Diabetic cystopathy:** sensory then motor autonomic neuropathy. Impaired sensation leads to large capacity, poor contractility, low flow and later high PVR and UTI (urinary tract infection).
- **DO is in fact the commonest urodynamic finding** (early disease or coexisting cerebrovascular disease); hypocontractility comes late.
- Mechanisms: glycation end products, sorbitol pathway, oxidative stress, polyuria.
- Differentiate from BOO (bladder outlet obstruction) by pressure–flow study. Treatment: **timed voiding**, CIC (clean intermittent catheterisation), cautious antimuscarinics.

Other systemic diseases

Group	Disease	Usual finding
Neurological	Stroke	Acute retention, then DO; no true DSD
	Parkinson disease	DO; pseudodyssynergia; TURP (transurethral resection of prostate) possible after excluding MSA (multiple system atrophy)
	Multiple sclerosis	DO about 38%, DO with DSD 29%, areflexia 26%; changes over time
Infective	HIV/AIDS	DO, DSD; areflexia with CMV (cytomegalovirus)
	Herpes zoster/simplex, Guillain–Barré	Retention, areflexia; reversible
	Spinal tuberculosis, Lyme, HTLV-1	DO with DSD or areflexia
Autoimmune and connective tissue	SLE (systemic lupus erythematosus)	DO; lupus cystitis
	Systemic sclerosis	Areflexia, low compliance
Infiltrative	Amyloidosis (transthyretin)	Low sensation, underactivity, SUI (stress urinary incontinence)
Nutritional	B12 deficiency, alcohol (thiamine)	Mixed; reversible with replacement
Endocrine	Hyperthyroidism	Poor emptying; resolves with treatment

Also know: myasthenia gravis; schizophrenia; gastroparesis.

Management principles

- Treat the disease; reversible therapy.

- Storage: antimuscarinics (trospium, or mirabegron in elderly brains), botulinum toxin, neuromodulation.
- Emptying: CIC; avoid Credé with DSD.

Diagram to draw: neuraxis with lesion levels and the bladder pattern at each.

> **Key points to remember:**
> - Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
> - Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
> - Urodynamics before treating; reversible treatment first.

Also, from the condensed book (not in the paper answer):

- **Guillain–Barré** causes retention; use CIC until recovery.

> **KEY POINT**
>
> Distal lesions hit the peripheral reflex arc, giving areflexia with a non-relaxing or fixed outlet. Compliance falls only when preganglionic parasympathetic fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery.

5 · Specific functional and miscellaneous conditions

Detrusor–sphincter dyssynergia

SAQ Detrusor–external sphincter dyssynergia (DESD / DSD)

Paper answer · P1

Asked as:

- External Sphincter Dyssynergia — *Summer 2016 Paper I Q3a (SAQ)*
- Detrusor sphincter dyssynergia — *Practice Set 10 Paper I Q3f (SAQ)* (also answered in 2016-Summer Paper 1, Q3a)

Type: SAQ (10 marks)

Definition

- **DESD / DSD (detrusor–external sphincter dyssynergia):** the striated sphincter contracts involuntarily (involuntary contraction) during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **It requires a lesion between the pons and the sacral cord:** suprasacral spinal cord injury (SCI), MS (multiple sclerosis), transverse myelitis, myelomeningocele. Brain lesions (stroke, Parkinson's) do not cause it (**never with brain lesions**).
- Lesions (at or) above T6 may add bladder-neck (**smooth sphincter**) dyssynergia and autonomic dysreflexia.

Blaivas classification

Type	EMG (electromyography) pattern	Lesion
1	Activity rises with detrusor pressure, then relaxes suddenly at peak	Incomplete

Type	EMG (electromyography) pattern	Lesion
2	Sporadic contractions throughout voiding	Complete
3	Crescendo–decrescendo contraction; obstruction throughout	Complete
- Alternative: intermittent versus continuous (continuous means a complete lesion). Type has no relation to lesion level (type is unrelated to level).		

Diagnosis

- **Pressure–flow study with EMG:** EMG rises during the detrusor contraction, with high pressure and interrupted flow. Needle EMG is more reliable than surface.
- **VUDS (videourodynamics):** narrowed (narrow) membranous urethra with dilated posterior urethra; shows VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion; Hinman syndrome in children).

Significance

- Worse when complete, continuous and **in men**: about half of untreated men get VUR, hydronephrosis, stones or urosepsis. Other risk factors: high filling pressure, indwelling catheter.
- **Management logic:** DSD is both a storage and an emptying failure. Lower bladder pressure, then empty it.
- DLPP (detrusor leak point pressure) ≥40 cmH_2O endangers the kidneys.

Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin, with condom drainage. The goal is DLPP <40; DLPP ≥40 afterwards means failure; repeat procedures are common.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents are abandoned; oral drugs disappoint. Credé and Valsalva voiding contraindicated.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

Diagram to draw: urodynamic trace with rising detrusor pressure (Pdet), EMG bursts and interrupted flow.

> **Key points to remember:**
> - Lesion between pons and sacral cord; never brain lesions.
> - Blaivas 1 incomplete; 2 and 3 complete.
> - Worst in men; DLPP ≥40 cmH_2O hostile.
> - CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

Also, from the condensed book (not in the paper answer):

- Blaivas type 1 (incomplete) can be observed unless there is VUR, hydronephrosis or AD; types 2–3 need treatment.
- Without a neurological lesion, doubt DSD (except Hinman syndrome in children).

Functional outlet disorders

- **Dysfunctional voiding** (Hinman) is striated sphincter contraction during voiding without neurological disease, proven on pressure–flow–EMG.
- **Non-neurogenic chronic retention** is PVR ≥**300 mL** persisting ≥6 months on ≥2 occasions. High-risk patients need drainage, preferably suprapubic.
- **Primary bladder neck obstruction** affects young and middle-aged men with a normal prostate, and videourodynamics localises it. α-blockers help some, but **bladder neck incision is definitive**.
- **Female BOO** is rare. Groutz criteria are free maximum flow (Qmax) ≤12 mL/s and detrusor pressure at Qmax ≥20 cm H_2O.
- **Fowler syndrome** is retention in young women with complex repetitive discharges on needle EMG and urethral closure pressure >100 cm H_2O. **SNM is the treatment** (~70% succeed).

Other conditions

- Rarer causes include myasthenia (more incontinence after prostatectomy), Isaacs syndrome, infective myelopathies and lupus cystitis (treated with steroids). After **radiation**, early storage symptoms recover by 6 months.

6 · Evaluation and risk stratification

- Under AUA/SUFU 2021 and EAU 2022 neurogenic LUT dysfunction (NLUTD) guidelines, **the primary goal is to prevent renal damage.**
- Everyone gets history, examination (cognition, **hand dexterity**, spasticity), urinalysis, PVR and diaries.
- **Defer full assessment during spinal shock or acute brain injury**, and give no irreversible therapy then.

Risk (AUA/SUFU)	Typical	Defining findings
Low (all)	Suprapontine or uncomplicated peripheral	Spontaneous voiding, low PVR, stable; normal imaging and renal function
Unknown	Infrapontine–suprasacral; sacral/infrasacral	Storage and voiding symptoms, raised PVR, recurrent UTI; not yet assessed, so do UDS, upper-tract imaging and renal function
Moderate	—	Neurogenic retention; DO with incomplete emptying; normal tracts
High (any one)	—	Poor compliance, raised storage pressure, DO with DSD, VUR; hydroureteronephrosis, scarring, large stones; abnormal renal function

KEY POINT

Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); high risk means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not by diagnosis.

7 · Management

Start with the simplest, most reversible step.

ALGORITHM

1. Aim for low-pressure storage with regular emptying: CIC with an antimuscarinic or β3 agonist.
2. If refractory, give intradetrusor onabotulinumtoxinA.
3. In selected patients, try neuromodulation.
4. If these fail, do augmentation or continent diversion.
5. As a last resort, do bladder neck closure with a suprapubic catheter, ileovesicostomy or incontinent diversion.

- **CIC is the gold standard**; keep volumes <**500 mL**. Credé and Valsalva are not recommended.
- CIC beats an indwelling urethral catheter, and for a chronic catheter **suprapubic is better than urethral**.
- For refractory neurogenic DO, use **intradetrusor onabotulinumtoxinA** (~60% continue long term) and warn about CIC. Intrasphincteric toxin is not routine.
- The Brindley stimulator is used in complete SCI (see Ch 115). PTNS suits patients who void and have OAB; SNM suits NLUTD without SCI or spina bifida.
- **UTI**: treat bacteriuria only if symptomatic, correct stones and residuals, and give **no prophylaxis** for CIC or indwelling catheters.
- **Bladder cancer** incidence is similar to the general population, but it is often muscle-invasive at diagnosis. **AUA/SUFU 2021 advise no routine surveillance cystoscopy**; scope for haematuria.

8 · Follow-up

Follow-up is lifelong, because upper-tract damage is silent.

Body	Schedule
AUA/SUFU 2021 (risk-based)	Low risk: none routine. Moderate/high: yearly history, examination and renal function; upper-tract imaging every 1–2 years; repeat UDS for change; after bowel reconstruction, yearly electrolytes, acid–base and imaging
EAU	Specialist review every 1–2 years; ultrasound and residual 6-monthly; dipstick 2-monthly; yearly examination and bloods
American Paraplegia Society (SCI)	Annual for 5–10 years, then alternate years; renogram: a >20% fall in renal plasma flow means investigate; annual cystoscopy with indwelling catheter (**not endorsed** by AUA/SUFU)

110 Overactive Bladder

1 · Definitions and terminology

OAB is a symptom diagnosis built on urgency.

*See the paper answer **Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management** (Section 7, Ch 103).*

Also, from the condensed book (not in the paper answer):

- In overactive bladder (OAB), **urgency is the defining symptom**, hence "urgency" rather than "urge" incontinence.
- OAB is a symptom syndrome, whereas detrusor overactivity (DO) is a urodynamic observation that may be asymptomatic or artefact.

Term	ICS definition (all patient-reported)	
Increased daytime frequency	Patient considers they void too often by day; upper normal	**7** voids
Nocturia	Waking ≥1 time to void; each void preceded and followed by sleep	
Urgency	Sudden, compelling desire to void, difficult to defer	
UUI	Involuntary leakage associated with urgency	

- **Urgency urinary incontinence (UUI) is the most bothersome lower urinary tract symptom (LUTS) in both sexes**. Frequency bothers at ≥**8** daytime voids and nocturia at ≥**2**.
- **Isolated nocturia** usually means nocturnal polyuria (night urine >33% of 24-h output in older adults) or poor sleep.
- **Mixed urinary incontinence (MUI)** is stress UI (SUI) with UUI; SUI with "OAB dry" is **not** MUI.

OAB versus bladder pain syndrome (BPS)

BPS requires bladder pain, pressure or discomfort with ≥1 urinary symptom; pain is excluded from OAB, though overlap is large.

Feature	OAB	BPS
Primary fear	**Leakage**	**Pain**
Filling sensation	Phasic urgency waves	Steady, accelerating pain
Voided volumes	Variable	More consistent
Deferral	Limited, because of leak risk	Possible, at cost of pain
Site	Perineum/penile base, urethra/vagina	Suprapubic, with or without perineal
UUI	Yes	No

2 · Epidemiology and burden

- Adult OAB affects ≈ **12–29%**, depending on the definition (EpiLUTS highest, EPIC lowest).
- Women have UUI 2–3× as often as men, so the sex gap is in **wet** OAB. Prevalence rises with age, to ~40% of men over 75.

- Fewer than half tell a clinician. About **92%** stop the first antimuscarinic within 2 years, mainly because of **unmet expectations**, then side effects.
- OAB dry progresses to UUI at ~1% a year. Among third-line options, onabotulinumtoxinA 100 U is the most cost-effective and sacral nerve stimulation (SNS) the costliest.

3 · Pathophysiology

- **DO is no longer the sine qua non**: only about half with DO have urgency, and **44–69%** with urgency show DO.
- The periaqueductal grey and pontine micturition centre are the main regulatory level. Afferent, urothelial, myogenic (exaggerated micromotions) and central mechanisms all contribute (see pointer).

> **KEY POINT**
>
> Final common pathway: more afferent signal reaches awareness. The bladder or a neighbour turns it up, or the CNS fails to turn it down. Hence DO and obstruction correlate only loosely with OAB.

4 · Aetiology and risk factors

- Risks are **ageing** (clearest), female sex, metabolic syndrome, bowel and hormonal factors, depression and anxiety.
- **Bladder outlet obstruction (BOO)** is only an **indirect** cause in humans, since OAB is as common in women, in whom BOO is rare.

5 · Clinical assessment

- The **minimum** work-up (AUA/SUFU and EAU) is **history, examination and urinalysis**, followed by empirical treatment; the EAU adds questionnaires and a diary.
- **History** covers bother, SUI, voiding symptoms, haematuria, fluids and caffeine, occult neurology, pelvic surgery or radiotherapy, bowel function (faecal incontinence in ~**50%** of women with OAB), drugs and causes of polyuria.

Drug class	Relative contraindications
Antimuscarinics	Uncontrolled closed-angle glaucoma; cognitive impairment; retention; impaired gastric emptying
β3-adrenergic agonists	Uncontrolled hypertension; arrhythmia

- **Examination**: frailty, cognition and oedema, but the **pelvic and genital examination is most important** (prolapse, atrophy, stress test; digital rectal examination in men; perineal sensation).
- **Urinalysis is mandatory**; microhaematuria needs full evaluation.
- **Postvoid residual (PVR)** is not needed before conservative or oral therapy if uncomplicated; measure it with retention risk, voiding symptoms, prior outlet surgery or neurology.
- Use a **diary** for 3–7 days; **maximum voided volume is the functional capacity.** Questionnaires include ICIQ-OAB and OAB-q; MCQ OAB-V8 is a screening tool, not a severity score.

6 · Urodynamics (UDS)

- UDS is **not** in the initial work-up. It is **indicated** when the result decides invasive or irreversible treatment, after multiple failures, or with suspected BOO.
- DO is found in OAB wet in **>90% of men** but is absent in 30–40% of women. It correlates **best with UUI and worst with nocturia**.
- Stop antimuscarinics to show DO, and continue them to explain refractory symptoms.
- **Assess emptying**: each **10 mL** of PVR raises the risk of catheterisation after onabotulinumtoxinA by **27%**. DO persists in up to **60%** after transurethral resection of the prostate (TURP).

7 · Management

Treatment follows the patient's goals.

- OAB is benign, so **no treatment is acceptable**. Set **realistic expectations** of improvement, not cure. AUA/SUFU lines are **not a compulsory sequence**; allow **4–12 weeks** per therapy.

ALGORITHM

1. Confirm OAB and exclude mimics.
2. Educate, then start conservative measures.
3. If not enough, add drugs.
4. If drugs fail, offer tibial nerve stimulation, sacral neuromodulation or onabotulinumtoxinA.
5. Rarely, do augmentation or diversion.
6. Reassess at every step.

- Reducing fluid by ~**25%** helps, and cutting caffeine improves urgency and frequency.
- Pelvic floor muscle training with bladder training reduces leaks by **50–80%**. **Weight loss** reduces SUI and UUI (PRIDE trial).
- Drugs (see Ch 113): response is idiosyncratic, so switch or combine before invasive work-up.
- In **dementia**, antimuscarinic use >**3 months** raises risk by 10–65%, so prefer a β3 agonist.
- In benign prostatic hyperplasia with OAB, an α1-blocker plus antimuscarinic works; retention occurs in ~3%, so take care if PVR is raised.

Third-line (see Ch 115)

Frailty predicts functional decline and 1-year morbidity of up to 50%, even after minor procedures.

	Tibial nerve stimulation	**OnabotulinumtoxinA**	**SNS**
Efficacy	60–80% respond	UUI reduced 59%; ~¼ dry	5-year success 70–80%
Drawback	12 weekly visits; response by weeks 8–10	Urinary tract infection (UTI) ~20%; catheterisation 0–12% in trials, real-world ~20% women, ≤35% men	Revision >30% at 5 years; battery
Risk	Lowest	Moderate	Highest

- **ICI pathways**: in men, treat the outlet with the bladder; in women, treat the most bothersome MUI component first; in the frail elderly, put comorbidity and carer goals first.

> **KEY POINT**
>
> Escalate by invasiveness and reversibility, not severity alone. OAB is benign, so patient goals, frailty and PVR decide how far to climb.

111 Underactive Detrusor

SAQ Detrusor underactivity and its management

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Management of detrusor underactivity" · P4 SAQ "Detrussor underactivity."

Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span."
- **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** the symptom complex of slow stream, hesitancy and straining, with or without incomplete emptying.
- DUA is common in elderly men and women having urodynamics: found in 40–48% of men over 70.

Pathophysiology and causes

- Failure may be **myogenic** (myocyte loss, fibrosis), **efferent** (poor nerve activation) or **afferent** (reduced sensation cannot sustain the voiding reflex).

Type	Examples
Neurogenic	Cauda equina, disc prolapse, spinal injury, diabetic neuropathy, multiple system atrophy, early stroke, radical pelvic surgery
Myogenic	Long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing
Drugs	Anticholinergics, opioids
Functional	Fowler syndrome, dysfunctional voiding

Evaluation

- Bladder diary, flow rate, **PVR (post-void residual)**, sacral neurological examination, spinal MRI if indicated (deficit), drug review.
- **Pressure-flow study is the only accepted (diagnostic) test:** low detrusor pressure with low Qmax.
- **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 × Qmax:** <100 weak, 100–150 normal, >150 strong. Also PIP (projected isovolumetric pressure), Watts factor, stop test.
- Ambulatory urodynamics if no contraction is seen on the standard study (the "bashful bladder").

Management

- **Goals:** protect kidneys, prevent UTI (urinary tract infection) and stones, improve symptoms. No drug reliably improves contraction.

Aim / option	Details
Conservative drainage	Timed and double voiding; treat constipation; stop causative drugs
CISC (clean intermittent self-catheterisation)	**Mainstay** for high PVR; fewer infections than an indwelling catheter. Suprapubic catheter if CISC impossible

Aim / option	Details
Lower outlet resistance	α-blockers (modest benefit); pelvic-floor biofeedback
Outlet surgery (TURP, transurethral resection of the prostate)	In selected obstructed men; little help in low-PVR men; may free catheter-dependent fit men
SNM (sacral neuromodulation)	For non-obstructive retention, especially Fowler syndrome; needs some detrusor function; intact sensation predicts response
Sacral anterior root stimulator (Brindley)	Complete spinal cord injury, with posterior rhizotomy
Parasympathomimetics (bethanechol, distigmine)	**Not recommended / not routine**: weak evidence; systemic side effects (bronchospasm, rarely cardiac arrest)
Intrasphincteric botulinum toxin	Experimental
Diversion	Mitrofanoff when urethral CISC fails; rarely ileal conduit

- Avoid Credé or Valsalva voiding (reflux risk).
- SNM and parasympathomimetics need some intact detrusor and sensory function; a fibrosed detrusor responds to nothing.

Investigational

- Intravesical electrotherapy, intravesical prostaglandins, TAC-302, latissimus dorsi detrusor myoplasty, muscle-derived stem cells.

Diagram to draw

- Pressure-flow plot on the Schäfer nomogram with BCI lines at 100 and 150.

> **Key points to remember:**
> - BCI = Pdet@Qmax + 5Qmax; <100 weak.
> - CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
> - SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

Also, from the condensed book (not in the paper answer):

- Poor surgical outcome: voiding pressure under 45 cm H_2O, age over 80, PVR over 1500 mL.
- Significant PVR proposal (over 40% of capacity); multisystem atrophy 52–95%.
- Sphincter toxin 50–100 U; no bladder neck incision in women.

1 · Definitions and symptoms

Drugs and surgery lower outlet resistance but cannot strengthen the detrusor.

- **Detrusor underactivity (DUA)** (International Continence Society [ICS] 2002, urodynamic) is a contraction of **reduced strength and/or duration**, causing prolonged or incomplete emptying. **No thresholds** are given. It is managed mainly by intermittent self-catheterisation (ISC).
- A **"bashful bladder"** fails to void only during the urodynamic study (UDS), unlike a truly **acontractile detrusor**.

- **Underactive bladder (UAB)** (ICS 2018) is the symptom set: slow stream, hesitancy and straining, with or without incomplete emptying. It has **no pathognomonic symptom** and overlaps with overactive bladder (OAB) and bladder outlet obstruction (BOO).
- A significant postvoid residual (PVR) is not agreed; one proposal is over **40%** of functional capacity (voided plus PVR).

Phase	Symptoms
Voiding	Weak stream, intermittency, hesitancy, straining
Post-micturition	Incomplete emptying
Storage	Frequency and nocturia, **or** reduced desire with infrequent voiding
Very large PVR	Incontinence, especially in sleep

> **KEY POINT**
>
> **Sensation decides the picture.** Absent sensation makes an infrequent voider, with overflow worse at night. Preserved sensation makes a frequent voider with incomplete emptying and nocturia. DUA may be asymptomatic.

2 · Epidemiology

- BOO is rare in women, so low flow with high PVR in a woman suggests DUA.
- UDS series find DUA in men under 50 in **9–28%**, men over 70 in ~**48%** and women in 12–45%.
- It peaks in institutionalised elderly people, often as **detrusor hyperactivity with impaired contractility (DHIC)**: an involuntary contraction emptying under half the bladder.
- Elderly patients often also have detrusor overactivity (DO), BOO or stress incontinence, so do **UDS before lower urinary tract symptom (LUTS) surgery in the elderly**.
- Non-neurogenic male DUA shows **minimal progression** over 10 years.

3 · Aetiopathogenesis

Confirmed causes are neurological disease and diabetes; probable ones are BOO, ageing and atherosclerosis.

Mechanism	Site	Result
Myogenic	Myocyte, extracellular matrix (ECM)	Weak contraction despite normal drive
Efferent	Sacral cord, roots, pelvic nerves	Impaired activation
Afferent	Peripheral afferents, spinal columns	**Early termination** of the voiding reflex
Brain	Pontine micturition centre (PMC), periaqueductal grey (PAG), limbic, prefrontal	Failed integration

- Ageing contributes through ECM fibrosis and falling afferent sensation; in the **"stiff balloon" hypothesis**, afferent drive fades after a small void, so the contraction stops early and leaves a PVR.
- **Drug causes are reversible**: anticholinergics (including antihistamines, antipsychotics, tricyclics, antiparkinsonian drugs) and **opioids**.

- †**Fowler syndrome** (young women, non-relaxing sphincter) responds to sacral neuromodulation.
- **BOO** causes **decompensation** in animals, but most men in acute retention have preserved detrusor function.
- **Diabetes**: early diuresis causes hypertrophy and storage symptoms; late damage impairs sensation and emptying.

Condition	DUA	Key point
Stroke	~Half retain acutely	"Cerebral shock"; chronic sequel is DO
Parkinson disease	<20%	DO commoner; DUA drug-related
Multisystem atrophy	**52–95%**	Efferent parasympathetic atrophy; mistaken for Parkinson
Multiple sclerosis	~20%	Lumbosacral plaques
Radical pelvic surgery	Hysterectomy: ~72% LUT dysfunction	Nerve-sparing reduces it; many recover by 1 year
Guillain–Barré, zoster	—	Reversible

4 · Diagnosis

Only a pressure-flow study proves a weak detrusor.

- **Invasive UDS is the only accepted method**. There are no consensus criteria, and nearly all measure strength only.
- Typically maximum flow (Qmax) and detrusor pressure at Qmax (Pdet@Qmax) are both low; isovolumetric estimates correct for their inverse relation.

Measure	Formula	Thresholds / note
Bladder contractility index (BCI) (Abrams)	Pdet@Qmax + 5Qmax	>150 strong; 100–150 normal; **<100 weak** ; male-derived
Projected isovolumetric pressure (PIP) (Schäfer)	Pdet@Qmax + K·Qmax (K 5 in men, 1 in older women)	<50 very weak; 50–100 weak
Detrusor contraction coefficient (DECO)	BCI/100	<1 weak

- †The **BOO index** is Pdet@Qmax − 2Qmax: over 40 is obstructed and under 20 unobstructed.
- If the patient fails to void on UDS, do **ambulatory UDS**: over four-fifths of apparently acontractile patients contract.

5 · Management

Safe drainage protects the kidneys; no drug reliably helps.

ALGORITHM

1. Evaluate: diary, digital rectal examination, urinalysis, flow, PVR.
2. If there is a neurological deficit, then do spinal MRI.
3. Stop offending drugs; treat constipation.
4. Timed or double voiding.
5. Lower outlet resistance where relevant.
6. If PVR is high, then ISC; if ISC is impossible, then a suprapubic catheter.
7. Selected patients: sacral neuromodulation (SNM), outlet surgery, diversion.

Aim	Routine	Experimental	Not recommended
Drainage	Timed/double voiding; catheterisation; diversion	—	Credé, Valsalva, reflex voiding
Lower outlet resistance	Pelvic floor biofeedback; α-blockers; outlet surgery	Sphincter botulinum toxin	—
Improve detrusor function	SNM; anterior root stimulator	Intravesical prostanoids; intravesical electrotherapy; myoplasty	**Muscarinic agonists; anticholinesterases**

- **Credé or Valsalva** is used only with an incompetent sphincter, as otherwise it causes reflux.
- **ISC** causes urethral bleeding in ~⅓; keep volumes below ~500 mL. Suprapubic is the best long-term indwelling route.
- **Bethanechol and distigmine** cause bronchospasm, diarrhoea, salivation and **rare fatal cardiac arrest**, and are useless with fibrosis.
- **SNM** works best with a sphincter or pelvic floor disorder and may help DHIC, where treating DO worsens DUA and vice versa (see Ch 115).
- The **Brindley** stimulator, combined with rhizotomy, is for complete spinal cord injury. Intravesical electrotherapy had a negative sham-controlled trial.

KEY POINT

Every therapy needs a surviving link: Brindley needs efferents and muscle; neuromodulation and electrotherapy need afferents and muscle. **Preserved sensation predicts response** to bethanechol, toxin and electrotherapy.

- **Sphincter toxin** is given as 50–100 U cystoscopically (men) or periurethrally (women). About half of idiopathic cases improve; it is short-lived and unlicensed.

	DUA, low PVR, symptomatic	DUA, catheter-dependent
Reasoning	Emptying at low Pdet, so BOO is unlikely to matter	Lowering resistance may allow voiding
Evidence	Transurethral resection of the prostate (TURP) at ~11 years: no benefit over conservative care	Flow rarely normalises, but many void
Verdict	**Surgery unlikely to help**	Consider in young, fit patients

- Outcome in retention is poor with voiding pressure under **45** cm H_2O, age over **80** or PVR over **1500 mL**.
- In women, avoid bladder neck incision because of incontinence and stenosis.
- **Diversion**: a continent catheterisable stoma suits young women with idiopathic retention refractory to SNM; an ileal conduit treats renal deterioration from intramural ureteric obstruction. Reduction cystoplasty is abandoned.

113 Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

1 · Framework

Drugs follow the Wein scheme (see Ch 104). Most lower urinary tract (LUT) drugs fail on **selectivity**, because the same receptor in heart and vessels caps the dose.

2 · Antimuscarinics and β3-agonists

First-line overactive bladder (OAB) drugs; side effects and age drive choice.

*See the paper answer **Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management** (Section 7, Ch 103).*

Also, from the condensed book (not in the paper answer):

- Beers avoidance of oxybutynin; myasthenia and severe colitis as contraindications.
- MILAI II add-on to mirabegron; combination doubles constipation.
- †Antimuscarinics act mainly in **storage**, so retention is uncommon.
- **Oxybutynin** causes the most dry mouth and has the most central nervous system (CNS) penetration, so avoid it in the elderly (Beers).
- Tolterodine is 2 mg bd or extended-release (ER) 4 mg od. **Darifenacin** is most M3-selective.
- **Trospium** is a renally excreted quaternary amine. Other doses: propiverine 20 mg, imidafenacin 0.1–0.2 mg bd.
- †**Side effects by receptor**: dry mouth, constipation and blurred vision (M3); tachycardia (M2); cognition (**M1**).
- †**Contraindications**: **uncontrolled narrow-angle glaucoma**, urinary or gastric retention, myasthenia gravis, and severe ulcerative colitis or toxic megacolon.
- †In the elderly, cumulative **anticholinergic burden** favours a β3-agonist or non-CNS-penetrant agent. Men with bladder outlet obstruction are safe if postvoid residual (PVR) is low.
- Mirabegron goes from 25 to 50 mg od; avoid it in severe uncontrolled hypertension (≥**180/110** mm Hg). Vibegron needs no titration and had no blood pressure (BP) effect.

ALGORITHM

1. †OAB: behavioural therapy, then an antimuscarinic or a β3-agonist; if elderly or at cognitive risk, then choose the β3-agonist first.
2. If the response is poor, then switch or combine.
3. If refractory, then onabotA 100 U, sacral neuromodulation (SNM) or percutaneous tibial nerve stimulation (PTNS).
4. Last resort: augmentation cystoplasty.

3 · Combination therapy

Direction	Evidence	Agents
Mirabegron added to failing antimuscarinic	SYNERGY, SYNERGY II, BESIDE, MILAI	**Solifenacin only**
Antimuscarinic added to failing mirabegron	MILAI II (52 weeks)	Solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine ER 4 mg
Vibegron plus antimuscarinic	Phase IIb (vibegron 100 plus tolterodine 4 mg)	Extrapolated; no trial after monotherapy failure

> **KEY POINT**
>
> In non-responders, combination **equals** solifenacin 10 mg with less dry mouth: it gains **tolerability, not efficacy**. Heart rate, BP, QT and PVR are unchanged; constipation roughly doubles.

- An **α-blocker plus desmopressin** suits refractory nocturia in benign prostatic hyperplasia (BPH), on uncontrolled evidence.
- Adding **vaginal oestrogen** is **not established**; †systemic oestrogen worsens urinary incontinence (UI).

4 · Intradetrusor onabotulinumtoxinA (onabotA)

The main third-line option.

SAQ Botulinum toxin in urology

Paper answer · P1 · P2 · P3 · P4 · merged from 4

Asked as: P1 SAQ "Botulinum toxin in urology: mechanism, indications and doses" · P2 SAQ "Botulinum neurotoxin and its role in urology" · P3 SAQ "Botulinum toxin in urology" · P4 SAQ "Botulinum toxin in urology"

"Botulimum" / "Botulinim" in the questions is read as botulinum toxin.

Introduction

- Neurotoxin of *Clostridium botulinum*, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**, the only licensed formulation; type B is short-acting.
- 150 kDa: a heavy chain (100 kDa; binding and entry) and a zinc-dependent light chain (50 kDa).

Mechanism of action

1. The heavy chain binds the cholinergic nerve terminal and the toxin is internalised.
2. The light chain (a zinc protease) enters the cytosol.
3. It **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE (vesicle-docking) protein. Vesicles cannot fuse, so **acetylcholine release stops**: chemical denervation. - **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) sensory receptors on C fibres, so urgency and pain fall. - The effect lasts about 6–9 months and wanes as nerve terminals sprout, so injections are repeated; repeat injections stay effective and response is maintained.

Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor at **about 20 sites (100 U, in 10 mL) or 30 sites (200 U, in 30 mL)**, usually sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

Urological uses: indications and doses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK): incontinence down about 59%, about 25% dry. **ABC** trial: equal to antimuscarinic, more complete dryness. **ROSETTA**: comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection) (P2, P3: 200 U used, equal to or ≥ SNM; another source: slightly better than SNM)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY trials): less incontinence, larger capacity, lower detrusor pressure; works with or without antimuscarinics
Paediatric NDO (spina bifida)	About **10 U/kg (maximum 360 U)**, about 30 sites, in published (Campbell) series; FDA (Food and Drug Administration) label (2021, age ≥5) 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation
DSD (detrusor–sphincter dyssynergia), underactive detrusor	Intrasphincteric injection 50–100 U, mainly in men unable to do CISC (clean intermittent self-catheterisation)
IC/BPS (interstitial cystitis/bladder pain syndrome)	100 U intradetrusor, often trigonal; AUA (American Urological Association) option, off-label
BPH (benign prostatic hyperplasia)	100–300 U intraprostatic: an early small trial was positive, but large placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo (saline/sham)**; not recommended

- **Also know:** pelvic floor spasm; paediatric dysfunctional voiding (sphincter); chronic pelvic pain; chronic orchialgia (RCT negative); ED (erectile dysfunction) trials. Units of different toxin products are not interchangeable.

Complications

- **Retention needing CISC** (the main risk): 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; commoner in NDO. Each 10 mL rise in baseline PVR raises the risk about 27%. **The patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.

Contraindications

- Active UTI, myasthenia gravis, aminoglycosides, pregnancy, unwillingness to self-catheterise.

Recent advances

- Paediatric NDO: FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.

- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

Diagram to draw

- Cholinergic synapse (nerve terminal) showing SNAP-25 cleavage by the light chain.

> **Key points to remember:**
> - Cleaves SNAP-25; acts on efferent and afferent nerves.
> - 100 U idiopathic OAB; 200 U NDO; spare trigone.
> - Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
> - Main risks: retention needing CISC, UTI.
> - Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
> - Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

Also, from the condensed book (not in the paper answer):

- Repeat ≥12 weeks apart; aminoglycoside caution.
- The dose is **100 U** for idiopathic OAB and **200 U** for neurogenic detrusor overactivity (NDO).
- †Patients must accept clean intermittent catheterisation (CIC) first, especially in non-catheterising NDO.
- The commonest adverse event (AE) is urinary tract infection (UTI). The effect lasts ≈6–9 months.

5 · Nocturia: desmopressin †

*See the paper answer **Desmopressin (DDAVP)** (Section 8, Ch 112).*

Also, from the condensed book (not in the paper answer):

- Low-dose sublingual desmopressin is 25 µg in women and 50 µg in men; hyponatraemia risk rises at age ≥**65**.

6 · Increasing outlet resistance (SUI)

No drug is standard for stress urinary incontinence (SUI).

†Pelvic floor muscle training and surgery are first-line.

Drug	Mechanism	Status
Duloxetine †	Serotonin–noradrenaline reuptake inhibitor at **Onuf's nucleus**; raises pudendal tone	40 mg bd; nausea causes discontinuation; Europe, not USA
Phenylpropanolamine, midodrine, norepinephrine	Systemic α-agonists	Weak evidence; BP rises first; phenylpropanolamine banned (stroke)

Drug	Mechanism	Status
Imipramine †	Anticholinergic plus α-like outlet effect	Mixed UI, enuresis; orthostatic hypotension, arrhythmia; caution in elderly
Clenbuterol, vaginal phenylephrine gel	β-agonist; local α-agonist	Abandoned

7 · Facilitating emptying

- †**Bethanechol** has **no convincing controlled evidence** in detrusor underactivity and causes cholinergic AEs; avoid in asthma, peptic ulcer, bradycardia and obstruction.
- †**α-blockers** help in BPH, primary bladder neck obstruction and neurogenic LUT dysfunction, and aid catheter trials.
- †Sphincteric onabotA is an option for the striated sphincter.
- †**Drugs that cause retention** include anticholinergics, pseudoephedrine, opioids, benzodiazepines, anaesthesia and epidurals; stopping them helps more than adding any drug.

8 · Failed drugs

Agent	Indication	Reason for failure
Vitamin D3, elocalcitol; calcium antagonists	Urgency UI, OAB	No urodynamic effect
K^+ channel openers; URO-902 gene therapy	OAB	Dose capped by heart; phase 1 only
RimabotulinumtoxinB	Toxin A-refractory DO	Short duration (≤3 months)
Intraprostatic onabotA, PRX302, NX-1207	BPH lower urinary tract symptoms	No better than sham injection

9 · Landmark trials

Trial	Design	Result	Take-home
SYNERGY / II	Phase III, 12 weeks / 12 months	Solifenacin 5 plus mirabegron 50 better than monotherapy for UI	Effective, safe long-term
BESIDE	Solifenacin 5 non-responders	Add-on better than solifenacin 5, equal to solifenacin 10	Alternative to dose escalation
MILAI II	52 weeks, mirabegron non-responders	Four antimuscarinic add-ons safe, effective	Add antimuscarinic to mirabegron
DIGNITY	Pooled RCTs, NDO	OnabotA 200 U equal in anticholinergic users/non-users	Efficacy independent of anticholinergic
EMBARK	RCTs, refractory OAB	OnabotA 100 U effective	Standard idiopathic OAB dose

115 Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

LAQ Neuromodulation for voiding dysfunction, including sacral neuromodulation

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P4 LAQ "Neuromodulation for voiding dysfunction, including sacral neuromodulation" · P2 SAQ "Neuromodulation" · P3 SAQ "Sacral neuromodulation"

Taken as neuromodulation for lower urinary tract dysfunction.

Definition and history

- **Neuromodulation:** electrical stimulation of a peripheral nerve that alters the activity of central (reflex) pathways controlling the bladder. It acts on the nervous system (**afferents**), not directly on muscle.
- Direct muscle stimulation failed (painful, local). Stimulating **afferents** needs less current and affects the whole organ.
- **SNM (sacral neuromodulation)** (Tanagho and Schmidt, about 1985): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator). PTNS (percutaneous tibial nerve stimulation): FDA (Food and Drug Administration) approval 2000.
- **Rule:** exclude anatomical causes first (obstruction, stones, cancer, fistula). Stimulation cannot correct anatomy or restore muscle.

Mechanism

- Sacral roots are mixed nerves. Current excites **afferents to the spinal cord and forebrain** (areas of awareness and attention), altering PMC (pontine micturition centre) output. It acts on **afferent** nerves, not motor nerves.
- **OAB (overactive bladder):** heightened central control (afferent input) damps PMC firing, so there are fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling (sensation) and pelvic floor tone, so the PMC switches on and the **EUS (external urethral sphincter) relaxes**. Hence one therapy treats **both** opposite disorders.
- **Tibial:** the nerve carries **S2–S3** fibres; inhibition of OAB-type (C-fibre) reflexes is **opioid-mediated** (naloxone blocks it); long-term cortical plasticity.

Modalities

Modality	Target and method	Main use
SNM	Tined lead in **S3** foramen + implanted generator	Refractory OAB, NOUR (non-obstructive urinary retention), faecal incontinence
PTNS	34-gauge needle above medial malleolus	Refractory OAB
TTNS (transcutaneous tibial)	Surface patch, home use	OAB (less effective)
Implantable tibial	Small subcutaneous device over tibial nerve	OAB, avoids clinic visits

Modality	Target and method	Main use
Pudendal	Tined lead in Alcock's canal	IDO (idiopathic detrusor overactivity), NDO (neurogenic DO), SNM failure (off-label)
DGNS (dorsal genital nerve stimulation)	Dorsal penile/clitoral nerve	Experimental; NDO
IVES (intravesical electrical stimulation)	Intravesical cathode	Underactive bladder in children
SARS (sacral anterior root stimulator)	Anterior S2–S4 roots + dorsal rhizotomy	Complete SCI (spinal cord injury)

A. Sacral neuromodulation: indications

- **Third line** for OAB after behavioural and drug therapy (AUA [American Urological Association] and EAU [European Association of Urology] agree).

FDA approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997	Bladder pain syndrome
Urgency–frequency 1999	Neurogenic bladder (small studies positive; low-level evidence)
Non-obstructive urinary retention (e.g. Fowler's syndrome) 1999	Pelvic pain
Faecal incontinence 2011	

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography) activity; responds well to SNM.
- Retention needs some detrusor function; success falls with age.

SNM: technique

Test phase (with a bladder diary)

Test phase	PNE (percutaneous nerve evaluation)	Staged tined lead
Lead	Temporary, office, local anaesthesia (dislodges)	Permanent tined lead, theatre, fluoroscopy
Duration	5–7 days	Up to 4 weeks
Use	OAB	More reliable; higher conversion; preferred for retention (NOUR)

- **Landmark:** prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

Root	Pelvic response	Foot response
S2	—	Calf and foot (external) rotation
S3 (target)	**Bellows**, perineal sensation	**Great-toe flexion**
S4	Bellows/perineal only	None

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral stimulation no better.

SNM: outcomes

- **InSite:** success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years (another source: about half keep benefit at ≥5 years).
- **ROSETTA:** onabotulinumtoxinA 200 U reduced UUI slightly more (−3.9 vs −3.3 episodes/day) with higher satisfaction, but caused more UTI (urinary tract infection) (24% vs 10%) and CIC (clean intermittent catheterisation, self-catheterisation); SNM removal 9% at 2 years.
- NOUR: most stop catheterising.

SNM: complications and contraindications

- Pain at the IPG site, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costly (costliest option); battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

B. Tibial nerve stimulation

- **PTNS technique:** 34-gauge needle **3–4 cm (three fingerbreadths) above the medial malleolus**, behind the tibia; ground pad on the same leg; 20 Hz, 0–10 mA. Confirm by **toe flexion/fanning** or sole tingling. **30 min weekly × 12 weeks**, then maintenance.
- **SUmiT** (n = 220, validated sham): response **55% vs 21%**. Extension: 77% relapse-free at 3 years with tapering maintenance.
- **OrBIT:** PTNS comparable to extended-release tolterodine, with fewer side effects.
- **TTNS:** home self-treatment; skin impedance limits amplitude, so less effective.
- **Implantable devices** remove clinic visits: **BlueWind** (miniature, external control; benefit sustained at 3 years); **eCoin** (coin-sized, battery-powered, office insertion under local anaesthesia).

C. Other targets

- **Pudendal nerve stimulation:** tined lead via a posterior approach to Alcock's canal. Better than SNM in a blinded crossover; **salvage after failed SNM**. Off-label.
- **DGNS:** purely sensory; acute DO suppression in SCI; **conditional (on-demand)** stimulation possible. Electrode fixation is difficult.
- **IVES (Katona):** intravesical cathode in saline activates Aδ afferents, producing detrusor contraction. Needs some afferent and muscle function; daily sessions; used in children; the sham-controlled trial was negative.
- **Brindley SARS:** stimulates anterior S2–S4 roots for voiding, defecation and erection. **Dorsal rhizotomy** abolishes reflex incontinence and DSD (detrusor–sphincter dyssynergia). The sphincter relaxes faster than the detrusor between bursts, giving **post-stimulus voiding**. Needs intact efferents.
- **Also know:** intravaginal stimulation (SUI, stress urinary incontinence); sacral dermatome stimulation; closed-loop SNM.

Comparison of third-line OAB options

	PTNS	OnabotulinumtoxinA 100 U	SNM
Invasiveness	Lowest	Moderate	Implant
Drawback	12 weekly visits	UTI, retention/CIC	Revision, cost
Retention	No role	Worsens	**Treats**

Recent advances

- Rechargeable stimulators (FDA 2019; ≥15 years); non-rechargeable devices lasting ≥10 years (2021).
- **Full-body (whole-body) MRI (magnetic resonance imaging)-conditional** systems at 1.5 T and 3 T.
- **Implantable tibial nerve stimulators** (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives; PTNS and pudendal stimulation as alternatives to SNM.
- Standardised tined-lead placement improves outcomes and lowers revision.
- Bilateral stimulation no better.

Diagram to draw

- Lead in the S3 foramen with IPG in the buttock, and PTNS needle above the medial malleolus.

> **Key points to remember:**
> - Afferent modulation explains SNM benefit in both OAB and retention.
> - S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
> - SNM revision >30% at 5 years.
> - SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
> - ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
> - Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
> - Full-body MRI-conditional devices at 1.5 T and 3 T.
> - Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
> - Standardised tined-lead placement improves outcomes and lowers revision.

Also, from the condensed book (not in the paper answer):

- Urodynamic DO does not predict success; PNE gives false negatives.

1 · Evolution and mechanism

One device treats both overactivity and retention.

- Direct detrusor stimulation failed, so stimulation now uses an **indirect route via afferents**.
- Milestones MCQ : sacral neuromodulation (SNM) ~1985 (Tanagho, Schmidt); percutaneous tibial nerve stimulation (PTNS) derives from acupuncture point **SP6**.
- **Current theory**: sacral afferents reach forebrain awareness areas, which reduces pontine micturition centre (PMC) firing in overactive bladder (OAB); in underactive bladder, raised filling awareness lets the PMC switch on.
- **Paradox**: in retention it works mainly by **striated sphincter relaxation**, not detrusor contraction.

2 · Sacral neuromodulation

- **Third-line** for idiopathic OAB (American Urological Association/Society of Urodynamics [AUA/SUFU] and European Association of Urology [EAU] agree).
- First get a frequency–volume diary and exclude anatomical causes, urinary tract infection (UTI) and a significant postvoid residual (PVR). Urodynamic detrusor overactivity (DO) **does not predict** success.
- In **non-obstructive urinary retention (NOUR)**, PVR is the only objective outcome. Fowler syndrome predicts response; age lowers the odds.

Food and Drug Administration (FDA) approved	Not approved
Urgency urinary incontinence (UUI)	Bladder pain syndrome
Urgency–frequency	Pelvic pain syndrome
Chronic NOUR	Neurogenic bladder

- †Faecal incontinence is also approved.
- Test success is ≥**50%** improvement in ≥1 diary parameter against a 3–5-day baseline; in retention, catheterisation falls ≥50%, ideally to none.
- The **S3 target** is ≈9 cm above the coccyx tip, 1–2 cm lateral. S3 gives anal "bellows" and **great-toe flexion**; S2 gives calf or foot rotation; S4 is perineal only.
- Percutaneous nerve evaluation (PNE) is an office test with a temporary lead for 5–7 days; lead migration lowers conversion, and it is not used in NOUR.
- The staged approach places a permanent tined quadripolar lead under fluoroscopy, tested for up to 4 weeks, with the generator in the buttock at stage 2.
- Programming is 210 µs, 10–16 Hz, at sensory threshold. Cycling equals continuous, and **bilateral is not superior** (randomised controlled trial, RCT).
- Battery life is ≥**10 years** (non-rechargeable) or ≥15 years (rechargeable). Devices are whole-body **MRI**-conditional at **1.5 and 3 T**.
- Complications: revision ~3% and explantation ~9% at 2 years (ROSETTA). †Site or leg pain is reprogrammed, then revised; infection needs explant.

3 · Tibial nerve stimulation

ALGORITHM

1. Supine; insert a 34-gauge needle ≈3–4 cm above the medial malleolus, behind the tibia, at 60–90°; ground pad on the same leg.
2. Stimulate at 20 Hz, 200 µs, 0–10 mA.
3. Confirm **great-toe flexion/toe fanning** or sole tingling.
4. Give **12 weekly sessions × 30 min**, then taper to maintenance.

- Indications are drug-refractory or intolerant OAB; it is safe in neurogenic OAB and in children.
- **Transcutaneous tibial nerve stimulation (TTNS)** uses home surface electrodes; it is cheaper but less effective than PTNS, and equals tolterodine with fewer side effects.
- Success is a ≥50% fall in urgency or UUI, and **relapse is likely without maintenance**. Results are better without DO.

4 · Other targets

- **Pudendal** stimulation (posterior approach) beat SNM in a blinded crossover and is salvage after failed SNM; it is not FDA approved.
- Dorsal genital nerve, sacral dermatome and intravaginal stimulation have limited roles and low-quality evidence.
- †**Intravesical stimulation** suits neurogenic underactive bladder with incomplete lesions, since afferents must be intact.

5 · Sacral anterior root stimulation (Brindley)

It restores voiding in complete spinal cord injury.

- Anterior root stimulation gives voiding, defecation and erection. It is combined with **posterior sacral rhizotomy**, which abolishes reflex incontinence and autonomic dysreflexia, raises capacity and compliance, and protects the kidneys.
- †Voiding is "post-stimulus" because the sphincter relaxes faster than the detrusor. †Rhizotomy costs reflex erection, reflex ejaculation and residual sacral sensation.

6 · Landmark trials

Trial	Design	Result	Take-home
SUmiT	RCT, PTNS vs validated sham, 12 weeks, n = 220	Responders 55% vs 21%	First sham-proven PTNS efficacy; maintenance keeps 77% relapse-free at 3 years
InSite	RCT, SNM vs standard medical therapy, mild–moderate OAB	Success 61% vs 42% at 6 months	SNM beats continued drugs
ROSETTA	RCT, onabotulinumtoxinA 200 U vs SNM, refractory UUI women	Toxin slightly fewer leaks at 6 months; equal at 2 years; more UTI, some clean intermittent catheterisation (CIC)	Both valid third-line; registered OAB toxin dose is 100 U

★ Putting it together: voiding dysfunction decisions

A · Functional classification as treatment map

	Bladder	Outlet
Failure to store	Overactivity, low compliance: drugs, onabotulinumtoxinA, neuromodulation, augmentation	Sphincter weakness: pelvic floor training, duloxetine, sling, artificial sphincter, bulking
Failure to empty	Underactive, acontractile: clean intermittent catheterisation (CIC); sacral neuromodulation (SNM) for non-obstructive retention	Obstruction or dyssynergia: α-blocker, outlet surgery, sphincterotomy, CIC

If neither is fixable, use a catheter or diversion (see Ch 104).

B · Lesion level and urodynamics

Lesion	Detrusor	Striated sphincter	Key risk
Above pons (stroke, Parkinson disease, dementia)	Overactive	Synergic	Urgency incontinence; low upper-tract risk
Pons–sacral cord (spinal cord injury [SCI], multiple sclerosis)	Overactive	**Dyssynergic**	High pressure damages upper tracts; autonomic dysreflexia (AD) if above T6
Sacral, cauda equina, peripheral (diabetes, pelvic surgery)	Areflexic, reduced sensation	Fixed tone	Retention, overflow

KEY POINT

A leak point or storage pressure ≥40 cmH_2O, or poor compliance, threatens the kidneys and must be treated (see Ch 107, 109).

C · OAB ladder

ALGORITHM

1. Behavioural measures, bladder training and pelvic floor training.
2. Antimuscarinic or β3-agonist (β3 in the elderly). If partial response, then combine (see Ch 113).
3. If refractory, then onabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn about CIC), tibial nerve stimulation or SNM (see Ch 115).
4. Last resort: augmentation or diversion.

D · SCI bladder

ALGORITHM

1. **Spinal shock**: catheter until reflexes return.
2. **Then** urodynamics; CIC plus antimuscarinic or toxin for high-pressure overactivity.
3. If CIC is impossible, then sphincterotomy or diversion; survey upper tracts lifelong.
4. **AD** (above T6): hypertension, bradycardia, headache, flushing. Sit up, remove the stimulus and give a short-acting antihypertensive (see Ch 109).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ EMG

Paper answer · P1

Type: SAQ (10 marks)

Source: Practice Set 9 Paper 1, Q3b

EMG is taken as urodynamic electromyography of the striated urethral/anal sphincter and pelvic floor.

Definition and principle

- EMG (electromyography) records the electrical potentials of depolarising striated muscle membranes, here the **external urethral sphincter and pelvic floor**, during filling and voiding.
- It shows **coordination** between detrusor and sphincter, not sphincter strength. The bladder neck (smooth muscle) is invisible to EMG.
- External anal sphincter activity mirrors the urethral sphincter, except in some neurological diseases.

Electrodes

Surface (patch)	Concentric needle
Perianal patches at 3 and 9 o'clock (ICS [International Continence Society] standard: two skin electrodes plus reference)	Needle into the sphincter (girls beside the meatus; boys transperineal)
Comfortable; poorer signal; movement artefact	Best signal; painful; dislodges
Coordination only	Also motor unit analysis: acute versus chronic denervation

Normal pattern

- Filling: activity **rises gradually** (recruitment, guarding reflex).
- Voiding: EMG **falls silent before** detrusor pressure rises and stays silent until flow ends.
- **Lag time** (uroflow with patch EMG in children): relaxation to flow 2–6 s is normal; <2 s suggests DO (detrusor overactivity); >6 s suggests primary bladder neck dysfunction.

Abnormal patterns

Finding	Meaning
DSD (detrusor–sphincter dyssynergia)	Sphincter contracts during detrusor contraction; needs a **suprasacral** cord lesion (SCI [spinal cord injury], MS [multiple sclerosis]). Blaivas types 1–3
Dysfunctional voiding	Same tracing without neurological lesion (learned)
Pseudodyssynergia	EMG rise from straining or trying to suppress a contraction
Denervation potentials	Sacral cord or cauda equina lesion; MSA (multiple system atrophy)
Complex repetitive discharges, decelerating bursts	**Fowler syndrome** (retention in young women); predicts SNM (sacral neuromodulation) response

Clinical uses

- Neurogenic bladder (spina bifida, SCI); voiding dysfunction in children; female retention; before sphincter surgery.
- **Limitation:** surface EMG misdiagnoses some dysfunctional voiders; **VUDS (videourodynamics)** is more accurate.

Diagram to draw: tracing of Pdet, flow and EMG: EMG silent during normal voiding versus active in DSD.

> **Key points to remember:**
> - EMG judges coordination, not strength.
> - True DSD needs a suprasacral lesion.
> - Lag time 2–6 s normal.
> - Fowler syndrome: complex repetitive discharges on needle EMG.

Rapid Revision Sheet · Section 7

Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management (LAQ)

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

Duloxetin (SAQ)

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Urinary bladder compliance (SAQ)

- C = ΔV/ΔPdet; <20 mL/cmH_2O suggests poor compliance.
- McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

Pathophysiology, classification and management of LUTS. (LAQ)

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

Uroflowmetry (SAQ)

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
- Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
- Cannot distinguish BOO from DUA.

Urodynamic studies, including video urodynamics (SAQ)

- Pdet = Pves − Pabd.

- Ask a question first; reproduce the symptom.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 cmH_2O: upper-tract risk.
- VUDS = urodynamics + fluoroscopy with contrast.
- VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male (LAQ)

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm H_2O threatens the kidneys.

Bladder dysfunction non genitourinary systemic disease (SAQ)

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

Detrusor–external sphincter dyssynergia (DESD / DSD) (SAQ)

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 cmH_2O hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

Detrusor underactivity and its management (SAQ)

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
- SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

Botulinum toxin in urology (SAQ)

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

Neuromodulation for voiding dysfunction, including sacral neuromodulation (LAQ)

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
- SNM revision >30% at 5 years.
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.

- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

EMG (SAQ)

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

SECTION 8

Nocturia & Male Incontinence

112 Nocturia

1 · Definition and terminology

- **Nocturia** (International Continence Society, ICS, 2019) is the number of voids during the **main sleep period**, from intending to sleep until intending to rise. The period follows the **individual's own schedule**, so shift workers are included.
- The definition does **not exclude 1 void per night** and does **not incorporate bother**.
- ICS 2019 dropped "wake **to void**"; non-urological causes are **leverage for interdisciplinary care, not grounds to exclude**.
- Guidance is from the European Association of Urology (EAU) and ICS; the AUA overactive bladder (OAB) guideline calls **isolated nocturia a separate entity**.

Key diary terms

Term	Definition / formula	Meaning
Actual nightly voids (ANV)	Voids in the main sleep period	Severity count
First uninterrupted sleep period (FUSP)	Hours from sleep onset to the first night void	Key sleep measure
Nocturnal urine volume (NUV)	Night voids **plus the first morning void**	Volume made during sleep
Nocturnal polyuria index (NPi)	NUV / 24-h volume	Proportion made at night
Nocturnal urine production (NUP)	NUV / hours of sleep	Hourly rate
Maximum voided volume (MVV); nocturnal MVV (NMVV)	Largest void in 24 h; largest in the sleep period	Proxies for functional capacity
Nocturia index (Ni)	NUV / MVV	>1: night output exceeds capacity
Modified Ni (MNi)	NUV / NMVV	Uses night data only
Predicted nightly voids (PNV)	Ni − 1	Voids expected if the bladder filled fully each time
Nocturnal bladder capacity index (NBCi)	ANV − PNV	>0: patient voids below capacity

Trap: the first morning void is **split**. Its volume goes into NUV, but it counts as daytime frequency, not nocturia.

2 · Why nocturia matters

- **It is common**: up to **3 in 5** people >70 years void ≥2 times nightly; age is the most robust correlate.
- **Bother** is reported by most people with ≥**2** voids, but **any nocturia that brings a patient to a urologist is clinically significant**.

- **Sleep**: nocturia is the **commonest cause of adult sleep disruption**. Most first voids occur within **2–3 h** of sleep onset, breaking early slow-wave (N3) sleep and causing daytime fatigue.
- **Falls and hip fracture** are linked to nocturia more than to other lower urinary tract symptoms (LUTS).
- **Systemic disease**: nocturia may herald cardiometabolic (strongest), renal, endocrine or neurological disease; ≥2 voids is linked to higher mortality.

> **KEY POINT**
>
> The first void breaks restorative slow-wave sleep; the symptom may be the visible edge of systemic disease.

3 · Pathophysiology — production, capacity and sleep

Remember **"Sleep CALM"**: **S**leep disorders, **C**omorbidities, **A**ctions (behaviour, lifestyle), **L**ower urinary tract dysfunction (LUTD) and **M**edications.

- Every factor upsets **night production**, **functional capacity** or **sleep**; the sleep link is **bidirectional** ("convenience" voids).

Nocturnal polyuria (NP) and global polyuria

Criterion	Threshold	Comment
NPi (ICS)	> **20%** in young adults to > **33%** in older adults	Most used, but discriminates poorly
NUP (ICS alternative)	> **90 mL/h**	More specific for production-driven nocturia
Global polyuria	> **40 mL/kg/24 h** (>2.8 L at 70 kg)	Uncommon but serious; **always investigate**

- **Krimpen study**: NPi >33% was found in 92% of older men with ≥2 voids, but also in 70% of men **without** nocturia.
- Excess night production contributes to **most adult nocturia**.

Functional capacity and the mismatch

- Normal capacity is usually >300–400 mL; **150–300 mL** suggests diminished capacity, which may be **24-hour** (all voids small) or **night-specific** (partly convenience voids).
- **Ni is the strongest diary determinant of severity and of treatment response**.

> **ALGORITHM**
>
> 1. Worked example:
> 2. NUV is 800 mL and MVV is 200 mL, so Ni = 4.
> 3. PNV = Ni − 1 = 3.
> 4. The diary shows 6 night voids, so NBCi = 6 − 3 = **3**: the patient voids well below capacity.

Investigate diminished night capacity when NBCi is >**1.3**; a value >**2** is strongly associated with severe nocturia.

4 · The bladder diary

EAU (strong): use a diary for female LUTS, and for male LUTS with storage symptoms or nocturia.

ALGORITHM

1. Give the patient a measuring container.
2. Record time and volume of every void and every drink.
3. Mark bedtime (intending to sleep) and rising time.
4. Grade urgency and note leakage for each void.

- International Consultation on Incontinence (ICI) urgency codes: **0** no need (social void), **1** normal desire, **2** urgency that passed, **3** urgency but dry, **4** urgency with leak.
- **Diary ANV is the gold standard for severity**; questionnaires overestimate it. **FUSP is the most important sleep measure**.
- Duration: EAU advises **3 days** (Male guideline: at least 3); because of "diary fatigue", fewer than half complete it.
- A **nocturnal-only diary** lets MNi substitute for Ni but **cannot diagnose global polyuria**; EAU does not recommend it for women.

KEY POINT

Every cause works through production against capacity, gated by sleep; their ratio (Ni) predicts severity better than either alone.

5 · Evaluation — the Sleep CALM work-up

Do a basic work-up in all, then tailored tests and referral. The **TANGO** questionnaire screens all domains.

Factor	History / examination	Tests (basic, then ancillary)
Sleep	Habits, apnoea, restless legs, pain, mood	STOP-BANG and sleep diary; then polysomnography, sleep specialist
Comorbidities	Blood pressure, volume overload, neurological signs	Metabolic panel; then HbA1c, ECG, echo, natriuretic peptide, urine albumin, calcium, thyroid, water deprivation
Actions	Fluid, alcohol, caffeine, sodium; body mass index (BMI)	Fluid intake diary
LUTD	LUTS, prior surgery; abdominal, genital, rectal examination	Urinalysis, post-void residual (PVR); then prostate-specific antigen (PSA), uroflowmetry, cystoscopy
Medications	Reconcile drugs and **dose timing**	—

- **Trap**: urology is one row of five; first-line is to **optimise every Sleep CALM abnormality**.

6 · Sleep disorders

Obstructive sleep apnoea (OSA) is the best-described link; frequent nocturia predicts severe OSA.

Arm	OSA mechanism	Effect of continuous positive airway pressure (CPAP)
Production (main)	Hypoxia causes pulmonary vasoconstriction, raising right atrial transmural pressure. This releases **atrial natriuretic peptide (ANP)**, causing natriuresis.	Lowers night production
Capacity	Intermittent hypoxia causes oxidative stress and detrusor instability	Less detrusor overactivity
Sleep	More light sleep, less slow-wave sleep	Increases slow-wave sleep

- CPAP reduces nocturia by up to ≈**2 voids** in moderate–severe OSA, among the **largest effects of any intervention**. EAU Female (strong): refer suspected OSA to a sleep clinic.
- **STOP-BANG** scores Snoring, Tired, Observed apnoea, Pressure, BMI >**35**, Age >**50**, Neck >**40 cm** and Gender male. A score ≥**3** means high risk; the tool is highly sensitive.
- Ask the **primary motive for getting up** (urge, or already awake). Melatonin and insomnia therapy may help.

> **KEY POINT**
>
> OSA raises production, lowers capacity and fragments sleep — hence one of the largest nocturia responses to treatment.

7 · Comorbidities

- Comorbidities act mainly through **excess night production**. Nocturia may be the **first sign** of hypertension, metabolic syndrome, coronary, diabetic or early kidney disease.
- EAU: consider systemic causes in all; even **non-bothersome nocturia merits a frequency–volume chart**.
- **Volume overload** (e.g. heart failure): leg fluid is mobilised on lying down and excreted at night.
- **Hypertension**: blood pressure normally dips **10–20%** in sleep. **Non-dipping**, from renin–angiotensin–aldosterone dysregulation, causes **nocturnal natriuresis**.

Disorders of water homeostasis

Arginine vasopressin (AVP) acts on collecting-duct **V2 receptors**; defects give **24-hour polyuria**.

Condition	Defect	Causes
Central diabetes insipidus (DI)	Reduced AVP secretion	Brain tumour, haemorrhage, trauma, autoimmune
Nephrogenic DI	Kidney insensitive to AVP	Drugs, electrolytes, renal failure
Primary polydipsia	AVP suppressed by high intake	Idiopathic; psychiatric

Arginine-stimulated **copeptin** appears superior to water deprivation. Always evaluate global polyuria.

> **KEY POINT**
>
> Comorbidity acts through the kidney (third-spaced fluid, nocturnal natriuresis, failed antidiuresis). Global polyuria and non-bothersome nocturia both warrant looking beyond urology.

8 · Actions — lifestyle and behaviour

- **Timing matters more than amount**. Water causes diuresis within ≈1 h; caffeine and alcohol within 2–3 h, and they disturb sleep.
- Restrict fluid, caffeine and alcohol from **3–4 h** before sleep; older adults should drink more by day.
- Also: **sodium** restriction, weight loss, exercise, **compression stockings**, bladder training, pelvic floor muscle training (PFMT), treat constipation.

9 · Lower urinary tract dysfunction

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), underactivity, neurogenic bladder and menopause syndrome See Voiding Dysfunction book, Ch 110.
- **Isolated nocturia is not a LUTS**, so it is not expected to respond to bladder or prostate drugs.
- Do urinalysis in all, and **PVR in both sexes**. Check PSA only if it changes management. †Urodynamics are not routine See Voiding Dysfunction book, Ch 107.

Benign prostatic obstruction

- In benign prostatic hyperplasia (BPH) trials, **α-blockers** carry most of the nocturia effect; 5α-reductase inhibitors (5ARIs) alone are weakest. Gains are fractions of a void with a large placebo response: **statistical, not clinical, significance** See Voiding Dysfunction book, Ch 113.
- **Surgery**: TURP beat microwave therapy, α-blockers and watchful waiting; **counsel** that nocturia may respond sub-optimally.

Overactive bladder

- Tolterodine was negative for night voids but reduced urge-related voids; **fesoterodine** (nocturnal urgency endpoint) was positive.
- Patients with a **raised NBCi** respond best to an evening dose. β3-agonists give a marginal benefit (≈0.1–0.2 voids over placebo).
- **Behavioural therapy** comes first and matches or beats drugs. Antimuscarinics are on the **Beers Criteria** for people ≥65 years, the core nocturia population.

Therapy	EAU Male LUTS	EAU Female LUTS
α1-blocker	Offer for nocturia with LUTS (weak)	—
5ARI	Offer with LUTS and prostate > **40 mL** (weak)	—
Phosphodiesterase-5 (PDE5) inhibitor	**Do not offer**	—
Antimuscarinic	Offer with OAB (weak)	Offer after counselling (**strong**)

Therapy	EAU Male LUTS	EAU Female LUTS
PFMT; vaginal oestrogen	—	Offer PFMT (**strong**); offer oestrogen (weak)

KEY POINT

LUT drugs act on capacity and outflow, but most nocturia is production-driven — effects are statistically significant, clinically small. Treat the mechanism, not the organ.

10 · Medications causing nocturia

Mechanism	Drugs
Increased urine output	Diuretics; SSRIs (block antidiuretic hormone); **calcium channel blockers** (oedema, raised ANP); tetracycline; lithium (reduces aquaporin-2); **SGLT-2 inhibitors** (glycosuria)
Insomnia / central nervous system effects	Stimulants; α- and β-blockers; salbutamol, theophylline; corticosteroids, thyroxine; SSRIs; phenytoin
Bladder toxicity	Ketamine; tiaprofenic acid; cyclophosphamide

Adjust dose timing first, or substitute another drug.

11 · Management algorithm and refractory nocturia

ALGORITHM

1. Take a history, examine, and obtain a diary and basic tests.
2. Identify the Sleep CALM factors.
3. Optimise each factor, with multidisciplinary help.
4. Add conservative measures.
5. If nocturia stays refractory, then consider an antidiuretic or a timed diuretic in selected patients.
6. Repeat the diary at every step.

- Treat interacting causes together (EAU **"Shared Care Pathway"**). **Target night volume even when capacity is small**.

Desmopressin

SAQ Desmopressin (DDAVP)

Paper answer · P3

Asked as:

- Desmopressin. — *Winter 2023 Paper III Q3b (SAQ)*
- DDAVP — *Practice Set 3 Paper III Q3c (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 3, Q3b)

Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**, desmopressin), a synthetic analogue of AVP (arginine vasopressin/antidiuretic hormone).

- Deamination and D-arginine substitution prolong action (give longer action) and make it a **selective V2 agonist (selective V2 agonism) without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells raises cAMP (cyclic adenosine monophosphate), inserting **aquaporin-2**; water is reabsorbed, so urine is less and concentrated (V2 receptor → cAMP → aquaporin-2 insertion → water reabsorption → less, concentrated urine).
- Endothelial V2 action releases von Willebrand factor and factor VIII (haemostatic use).

Formulations

Form	Dose
Tablet	0.2–0.4 mg at bedtime, about 1 h before bed (enuresis)
Oral lyophilisate (melt)	120–240 µg, about 1 h before bed
Low-dose nasal spray (nocturia)	0.83 or 1.66 µg, 30 min before bed
Sublingual ODT (orally disintegrating tablet)	27.7 µg women, 55.3 µg men
IV / subcutaneous	0.3 µg/kg (bleeding)

Uses (in urology)

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. **Rule of thirds**: one-third full, one-third partial, one-third no response. Acts (works) at once but relapses on stopping. Restrict fluids in the evening (avoid evening fluid); add oxybutynin if partial.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer below (under) 65; low dose above (over) 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia** (multiple sclerosis, MSA [multiple system atrophy]).
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired, in Wilms tumour). Tachyphylaxis after about 3 (repeated) doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, baseline sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women are more sensitive.
- Most events occur in the **first 2–3 weeks**.

Monitoring and contraindications

- Sodium at baseline, then within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

Diagram to draw: V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

Key points to remember:

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early.

Also, from the condensed book (not in the paper answer):

- Desmopressin is the **only US-approved nocturia drug**, for **≥2 voids per night due to NP** (level 1b); some NP patients have a **blunted nocturnal AVP rise**.
- †Also contraindicated with systemic or inhaled glucocorticoids and estimated glomerular filtration rate (eGFR) <50 mL/min.

	Nasal spray	**Orally disintegrating tablet (ODT)**
Dose	1.66 mcg if <65 with no risk factors; **0.83 mcg** if ≥65 or at risk	**Sex-specific**: women **25 mcg**; men **50 mcg** (75 mcg added hyponatraemia, not efficacy)
Timing	≈30 min before bed	Sublingual, ≈60 min before bed

- **Women are more sensitive**, because the V2 receptor gene is on the **X chromosome**.
- †Advise minimal fluid from 1 h before until 8 h after the dose.
- Hyponatraemia is defined as sodium <**130 mmol/L**; most cases occur within **3 weeks**.
- **Sodium monitoring** (EAU): baseline, **day 3**, day 7, 1 month, then periodically (US label: baseline, within 7 days and ≈30 days). EAU Female: avoid if baseline sodium is low.
- A **low dose** still works and avoids antidiuresis outlasting sleep. **Age 65 is the pivot.**

Timed diuretics

- Furosemide acts for 6–8 h. Evening dosing **causes** NP, whereas **morning or early-afternoon** dosing treats it (level 1b).
- Small trials (bumetanide 4–6 h, or furosemide 40 mg 6 h before bed) cut voids; monitor sodium.
- **Trap**: afternoon furosemide with bedtime desmopressin worked in a trial, but loop diuretics are a **contraindication** to desmopressin in US labelling.

KEY POINT

Refractory nocturia: move urine production out of the night. The price is sodium balance — dose, age and monitoring define safe use.

12 · Landmark trials

Trial	**Design**	**Result**	**Take-home**
MTOPS	Doxazosin, finasteride, combination, placebo	Doxazosin and combination beat placebo; finasteride did not	α-blocker drives benefit
VA Cooperative	Terazosin, finasteride, combination, placebo	Only terazosin separated from placebo	5ARI alone does not help
CombAT	Tamsulosin, dutasteride, combination	Combination −0.5 vs −0.3/−0.4 voids	Significant but small

Trial	Design	Result	Take-home
Fesoterodine RCT	Nocturnal urgency primary endpoint	Positive	Helps urgency-driven night voids
Nasal spray phase III	0.83/1.66 mcg vs placebo	Both beat placebo	Basis of approval
ODT sex-specific RCTs	Women 25; men 50/75 mcg	Positive; 75 mcg caused more hyponatraemia	Sex-specific dosing

123 Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

1 · Pathophysiology and device mechanisms

- Male **sphincteric (stress) urinary incontinence (SUI)** needs failure of **both** sphincters. Radical prostatectomy (RP), the commonest cause, removes the internal sphincter, leaving the rhabdosphincter.
- Other causes: radiotherapy (RT), pelvic fracture, neuropathy, vesicourethral anastomotic stenosis (VUAS) treatment. See Voiding Dysfunction book, Ch 106.

Device	Mechanism	Site	Adjustable	Needs residual sphincter
Artificial urinary sphincter (AUS, AMS 800)	Circumferential compression, cycled by the patient	Bulbar urethra or bladder neck	No	No
Advance / Advance XP	**Repositions** and lengthens the urethra; no compression	Bulbar, transobturator	No	**Yes**
Virtue (quadratic)	Prepubic compression and elevation	Bulbar, four arms	No	Less dependent
ATOMS (adjustable transobturator male system)	Cushion compression	Bulbar, transobturator	Yes, via scrotal port	Less dependent
ProACT	Two para-urethral balloons	Bladder neck	Yes, via scrotal ports	—

- The **AUS** has a cuff, a scrotal pump and a pressure-regulating balloon (PRB) that sets cuff pressure. The cuff refills passively over minutes.
- A transobturator sling raises leak point pressure **without obstruction** by elevating the bulb and rhabdosphincter.

> **KEY POINT**
>
> Every device compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but compress — only the Advance depends absolutely on a coapting residual sphincter.

2 · Indications and device selection

- Indication: **irreversible sphincter incompetence** with bothersome incontinence (UI) after conservative care. Do not delay surgery beyond **6 months** for severe SUI.
- 24-hour pad weight grades severity: mild **<150 g**, moderate **<400 g**. AUS is preferred if >250 g or after RT (MASTER trial See Ch 106).

Stratum (AUA/SUFU)	Recommendation
Severe UI or **any RT**	AUS first-line; sling only with counselling
Moderate, no RT	AUS or male sling; adjustable balloons in specialised centres

Stratum (AUA/SUFU)	Recommendation
Mild, no RT	Continue pelvic floor muscle training (PFMT), sling, possibly AUS

- Counsel: ~1 thin pad/day, AUS lifespan **7–10 years**; failed slings mostly go to AUS.
- **A sling needs** mild–moderate SUI, a contractile bladder (if hypocontractile, choose AUS), a non-irradiated urethra and a coapting sphincter.
- **An AUS needs** manual dexterity and cognition, a healthy urethra and adequate bladder compliance.

Absolute — need an alternative	Relative
Low compliance threatening the kidneys; low capacity; urethra too poor (alternatives: clamp, catheter, bladder neck closure, diversion)	Frequent transurethral work (bladder cancer, refractory VUAS, radiation cystitis), because it risks erosion and infection

- Radiation cystitis needs hyperbaric oxygen and **6 months** free of bleeding before AUS.

> **KEY POINT**
>
> Each device's contraindication mirrors the other's requirement.

3 · The high-risk ("fragile") urethra

- Strong erosion risk factors are **prior RT, prior erosion or infection, and urethroplasty** (worse than RT, as transection divides the bulbourethral arteries).
- After a prior erosion, 5-year repeat erosion approaches **50%**.
- Countermeasures: optimise medically, use a **transcorporal (TC) cuff**, a PRB of **51–60 cm H2O**, a cuff 0.5 cm larger, and nightly deactivation.

> **KEY POINT**
>
> Erosion = spongiosal ischaemia under constant pressure. Countermeasures add tissue (TC tunica) or lower pressure — never squeeze harder.

4 · Operative technique

- Sterile urine; give IV antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg plus a first/second-generation cephalosporin or vancomycin.
- Place the cuff on the **proximal bulbar urethra, proximal to corporal fusion**, where spongiosum is thickest.

ALGORITHM

1. Midline perineal incision; divide bulbospongiosus.
2. Incise Buck fascia and make a 2-cm tunnel around the urethra.
3. Measure and place the cuff (usually 4.0–4.5 cm).
4. Place the PRB preperitoneally and fill it with **23 mL**.
5. Place the pump subdartos in the anterior scrotum.
6. Cycle the device; do urethroscopy.
7. Leave it **deactivated for 4–6 weeks**.

- Less tight cuffs continent better; a **3.5-cm cuff** raises erosion after RT. †Deactivate the cuff before any later catheterisation or endoscopy.

PRB (cm H2O)	Use
51–60	High-risk bulbar (RT, prior erosion, atrophy); trades continence for safety
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

- **Tandem cuff**: a second cuff ≥1.5–2 cm distal; not first-line, because the distal cuff erodes more.
- **Transcorporal cuff** (Webster 2002): ventral tunica stays on the dorsal urethra and adds bulk. Use it for a urethra <3.5 cm, revisions, prior erosion or urethroplasty; erosion benefit is unproven.
- **Bladder neck AUS** suits an intact prostate (neuropathic bladder) and is **contraindicated after RP**; cuff ≥8 cm, PRB 71–80.

ALGORITHM

1. **Advance XP**:
2. Perineal incision; open bulbospongiosus.
3. Divide the perineal body to release the bulb.
4. Pass the helical trocar outside-in below the adductor longus tendon.
5. Fix the mesh to the spongiosum.
6. Tension under cystoscopy until the mucosa coapts (bulb moves 3–4 cm proximally).

- **ATOMS** leaves bulbospongiosus **intact** and is adjusted in clinic via the scrotal port; **ProACT** balloons are placed at the bladder neck under fluoroscopy.

5 · AUS complications and management

ALGORITHM

1. **Retention**:
2. Confirm the cuff is deactivated.
3. Insert a small catheter for 24–48 h.
4. If the second trial fails, then place a **suprapubic cystostomy**, avoiding the PRB.
5. If retention lasts weeks, then the cuff is undersized: replace it.

- **Infection** occurs in **1–3%** (up to 10% if high-risk), usually staphylococci; **scrotal pain is the key early sign**. If antibiotics fail, explant all; salvage only if not eroded, septic or immunosuppressed.
- **Erosion** occurs in **2–15%** (up to 35% with several risk factors). Remove **all components**. If <50% of the circumference is lost, leave a catheter ~3 weeks; if >50%, do in situ repair or anastomosis.
- Reimplant after **3–6 months** at a **new site**, using a TC cuff with or without a low-pressure PRB.
- **Atrophy** causes a **gradual** recurrent leak. Downsize, relocate (TC cuff distally) or add a tandem cuff; **never simply raise PRB pressure**.
- **Mechanical failure**: within 3 years, replace the failed part; **after 3 years, replace the entire device**.

Recurrent leak after AUS	Clue, then how to confirm
Deactivated	Sudden; reactivate
Fluid leak	Sudden; PRB shrinks on imaging
Atrophy	Gradual; PRB volume normal (>20 mL)
Erosion / detrusor overactivity	Cystoscopy / urodynamics

> **KEY POINT**
>
> Sudden = device, gradual = urethra. Manage every complication to keep a salvageable urethra for the next cuff.

6 · Sling complications and failure

Device	Commonest	Erosion / explant
Transobturator	Transient groin pain; retention or urgency 4–10%	0–0.4%
ATOMS	More complications	Erosion/infection ~10%; explant 10–16%
ProACT	**Perforation** (~5%); migration; revision 22%	~4%

- Cystoscope every worsening sling. After a failed sling the **AUS is the gold standard**, with results as good as primary AUS.

7 · Special circumstances

- A stricture at the cuff may herald erosion.
- With a penile prosthesis, synchronous and staged placement do equally well; if staged, place the AUS first. **Climacturia** responds to a sling or AUS, adding a **mini-jupette** graft if there is erectile dysfunction.

8 · Outcomes

SAQ Post-prostatectomy incontinence

Paper answer · P2

Asked as:

- Post prostatectomy incontinence. — *Summer 2021 Paper II Q3a (SAQ)*
- Prostatectomy incontinence — *Practice Set 11 Paper II Q3f (SAQ)*

Type: SAQ (10 marks)

(Practice Set 11 Paper II Q3f: also answered in 2021-Summer Paper 2, Q3a)

Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI. Rates vary with definition ("no pad" vs "safety pad") and reporting; expert no-pad continence is in the mid-90s % at 1 year; continence improves for up to 2 years.

Pathophysiology

- RP removes the internal sphincter (bladder neck, prostatic urethra), so continence depends (entirely) on the external **rhabdosphincter**.
- Causes: apical injury to rhabdosphincter or pudendal branches (sphincter or nerve injury at the apex), short membranous urethra, afferent denervation; also detrusor overactivity, low compliance, **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction (Rocco stitch) give **earlier, not better final** continence.

Evaluation

- History (type, severity, radiotherapy, cancer status), bladder diary, **24-hour pad weight**: mild <200 g, moderate 200–400 g, severe >400 g (another source: mild <150 g, moderate <400 g, severe more; thresholds vary).
- **Standing cough test**, grade (grades) 0 to 4 (0–4: leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream).
- Urinalysis, creatinine, PVR (post-void residual), PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and judges (shows) residual sphincter coaptation; urodynamics when needed (if mixed symptoms).

Management

1. **Wait 6–12 months** (improves up to 2 years). Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally preoperative (started before surgery); pads, condom catheter or (penile) clamp.
2. **Duloxetine** (off-label). **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019, EAU [European Association of Urology]):

Situation	Choice
Mild–moderate, no radiotherapy	**Male sling**: AdVance / AdVance XP (repositions the bulbar urethra; needs a coapting sphincter) or adjustable systems (ATOMS [adjustable transobturator male system])
Severe, or any radiotherapy	**AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff pressure 61–70 cm H_2O
ProACT balloons	Expert centres; avoid after radiotherapy

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence (0–1 pad) 55–77%; revision 10–29% at 3 years; lasts (lifespan) 8–10 years. Erosion risks: radiotherapy (risk higher after radiotherapy), prior erosion, urethroplasty (use a **transcorporal cuff** for fragile urethra).
- **Also know:** climacturia; treat VUAS first.

Diagram to draw: AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

Key points to remember:

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months.

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

Also, from the condensed book (not in the paper answer):

- Cure clusters at 55–65% for all devices (ProACT ~60–67%); Advance XP results are poorer after RT and in severe UI.

★ Putting it together

A · Nocturia: three diary questions, by danger

ALGORITHM

1. **Global polyuria?** If >40 mL/kg/24 h, then investigate for diabetes, diabetes insipidus or polydipsia.
2. **Nocturnal polyuria?** If night fraction >20% (young) or >33% (older), then seek heart failure, sleep apnoea, oedema or evening fluids; consider desmopressin with sodium checks.
3. **Reduced night capacity?** If NBCi >1.3, then treat overactive bladder, obstruction or residual (See Ch 112).

B · Post-prostatectomy incontinence

KEY POINT

Pelvic floor training first; operate on severe leakage by ~6 months. Pad weight: mild <150 g, moderate <400 g. A sling suits mild–moderate leakage, a contractile detrusor and no radiotherapy. Leakage >250 g, radiotherapy or a failed sling needs a sphincter. For erosion risk, add tissue (transcorporal cuff) or lower pressure; never tighten. Remove eroded or infected devices (See Ch 123).

Rapid Revision Sheet · Section 8

Desmopressin (DDAVP) (SAQ)

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Post-prostatectomy incontinence (SAQ)

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 9

Andrology & Sexual Medicine

66 Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

1 · Scrotum

Each layer is a surgical plane from the abdominal wall.

Layer (superficial to deep)	Continuous with	Note
Tunica dartos	Scarpa, Colles, penile dartos	Smooth muscle
External spermatic fascia	External oblique	—
Cremasteric fascia	**Internal oblique**	**Striated** cremaster; thermoregulation
Internal spermatic fascia	Transversalis fascia	—
Tunica vaginalis (parietal, visceral)	Peritoneum	Hydrocele collects between its layers; visceral layer fused to albuginea

- The anterior wall carries the external pudendal vessels and the ilioinguinal and genitofemoral nerves; the posterior wall carries the perineal vessels and nerve. Nothing crosses the raphe.

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

> **KEY POINT**
>
> Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum. †So scrotal violation in testis cancer adds the inguinal nodes to the at-risk field.

2 · Testis

Structure

- Volume is **15–25 mL**, length 4.5–5.1 cm. The capsule is visceral tunica vaginalis, **tunica albuginea** (with smooth muscle) and tunica vasculosa.
- The **mediastinum testis** is the posterior echogenic band on ultrasound (US). The appendix testis is a **müllerian** remnant.
- About 250 lobules hold **600–1200 seminiferous tubules**, which drain to the rete testis.
- The interstitium is **20–30%** of volume and holds the **Leydig cells**. **Sertoli** tight junctions form the blood–testis barrier, aided by endothelium and peritubular myoid cells (see Ch 67).

Arterial supply — three sources

Artery	Origin	Point
Testicular (internal spermatic)	Aorta below renal artery	Main supply; larger than the other two combined

Artery	Origin	Point
Deferential	Superior (or inferior) vesical	Vas, cauda; key collateral, lost after vasectomy
Cremasteric	Inferior epigastric	Tunica vaginalis; anastomoses at mediastinum

- Collaterals may save the testis after testicular artery ligation, but atrophy or azoospermia can occur.

> **KEY POINT**
>
> **Surgical rules**: no traction suture through the lower pole. Biopsy the **medial or lateral upper pole**; the midportion has fewest vessels. After vasectomy the deferential collateral may be gone, so preserve the testicular artery in later cord surgery and counsel about devascularisation.

Veins — pampiniform plexus

The plexus cools the artery by **countercurrent heat exchange** to **2–4 °C below rectal** temperature. A varicocele disrupts this.

Group	Accompanies	Drains to
Anterior	Testicular artery	Inferior vena cava (IVC) on the right; left renal vein on the left
Middle	Vas	Pelvic veins
Posterior	Posterior cord	Epigastric and pudendal veins

*See the paper answer **Varicocele: pathophysiology, controversies and surgical management** (Section 10, Ch 87).*

Also, from the condensed book (not in the paper answer):

- **High (retroperitoneal) ligation** can take artery and vein, since collaterals suffice; **inguinal or subinguinal** ligation takes veins only, because arterial ligation there risks devascularisation.

Lymphatics and nerves

- Dividing lymphatics at varicocelectomy causes a **hydrocele**; microsurgery preserves them.
- Supply is autonomic, with **no somatic innervation**; the genital branch of the genitofemoral nerve supplies the tunica vaginalis.
- The **"trifecta" nerve complex** (perivasal, posterior periarterial/lipomatous, intracremasteric) is the target of microsurgical cord denervation for chronic orchialgia.

Imaging

Modality	Role
Ultrasonography (US), 7.5–10 MHz with Doppler	**First line**: pain, mass, swelling, epididymo-orchitis, recurrent varicocele
Computed tomography (CT)	Not primary; cord or inguinal disease; air in Fournier gangrene
Magnetic resonance imaging (MRI)	Second line: indeterminate US, intra- versus extratesticular; nonpalpable testis (MR angiography)

- If torsion is suspected, **explore**; Doppler must not delay surgery.

- Varicocele is a **clinical diagnosis**. Use US (standing, Valsalva) only if examination is equivocal.

3 · Epididymis

- Caput, corpus and cauda lie on the posterolateral testis. **Principal** cells form the blood–epididymis barrier, and **clear** cells acidify the lumen to keep stored sperm quiescent.
- The testicular artery supplies caput and corpus, and the deferential artery the cauda. Nerves and smooth muscle increase toward the cauda, which shifts from storage to propulsion.
- Wolffian (mesonephric) remnants include the **appendix epididymis**; müllerian (paramesonephric) remnants are the **appendix testis** and prostatic utricle.

4 · Vas deferens

- It runs **posterior** to the cord vessels through the inguinal canal, crosses **anterior to the ureter**, forms the ampulla and joins the seminal vesicle.
- It is **30–35 cm** long with a **0.2–0.7 mm** lumen and has **three muscle layers** (longitudinal, circular, longitudinal).
- About half the cord nerves surround the vas: sympathetic fibres drive emission, sensory fibres cause chronic scrotal pain.
- A **vasogram** can cause obstruction, so do it **only at reconstruction**.

5 · Seminal vesicles and ejaculatory ducts

- Behind the bladder base, they join the ampulla as ejaculatory ducts opening on the verumontanum at **5 and 7 o'clock**.
- They bud from the mesonephric duct alongside the ureteric bud, so **renal anomalies coexist**.
- The fluid is ~70% of semen and alkaline. It carries **fructose**, prostaglandins and **semenogelin**, which forms the coagulum later liquefied by prostate-specific antigen (PSA).
- On **transrectal ultrasound (TRUS)**, a post-ejaculation anteroposterior (AP) diameter **>16 mm** or volume **>6 mL** means obstruction. **MRI** is the modality of choice; CT is for stones.

> **KEY POINT**
>
> A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction.

6 · Prostate

*See the paper answer **Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate** (Section 16, Ch 148).*

Also, from the condensed book (not in the paper answer):

- TRUS volume is length × width × height × **0.52**, which underestimates. The normal gland is **~25 mL**.
- **Multiparametric MRI** is scored by PI-RADS v2.1; targeted biopsy finds less insignificant cancer.

7 · Urethra

LAQ Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias

Paper answer · P1

Asked as:

- Describe the embryology and anatomy of male Urethra. Describe different endoscopic management techniques of treatment of urethral strictures. — *Summer 2019 Paper I Q1 (LAQ)*
- Discuss anatomy of urethra. Classify its congenital anomalies and discuss management of epispadias. — *Practice Set 5 Paper I Q1 (LAQ)* (male urethral anatomy also answered in 2019-Summer Paper 1, Q1; isolated epispadias in 2022-Summer Paper 3, Q3f)

Type: LAQ (25 marks)

Embryology

- **Cloaca** is divided by the **urorectal septum** (weeks 4–7) into the anterior **UGS (urogenital sinus)** and the posterior anorectal canal.
- The UGS has three parts: **vesical** (bladder), **pelvic** (prostatic and membranous urethra) and **phallic** (urethral plate of the penis).
- **Posterior urethra:** from the pelvic UGS. Prostatic buds appear at 10 weeks; the **utricle** (Müllerian remnant) and ejaculatory ducts (Wolffian) open on the verumontanum.
- **Anterior urethra:** Leydig cells make testosterone from about 8 weeks; type 2 5α-reductase converts it to DHT (dihydrotestosterone), which masculinises the genital tubercle. Germ layers: ectoderm gives skin and prepuce, mesoderm the corpora, endoderm the **urethral plate**.
- **Penile urethra: "double zipper"** 1. **Opening zipper:** the urethral plate canalises proximal to distal into a urethral groove between the urethral folds. 2. **Closing zipper:** the folds fuse in the ventral midline, forming the tube and the median raphe. Folds fused by 16 weeks; urethra complete by about 18 weeks.

Glanular urethra	Ectodermal ingrowth theory (Glenister)	Endodermal theory (current)
Formed by	Surface ectoderm grows into the glans to meet the endodermal urethra	Direct canalisation of the urethral plate inside the glans
Epithelium	Mixed ectoderm and endoderm	Endoderm (UGS) along the whole urethra

- **Clinical link:** the shaft–glans junction (change of mechanism) is the commonest hypospadias site. **Also know:** urethral duplication; PUV (posterior urethral valves); utricular cyst.

Anatomy of the male urethra

- About 18–20 cm long in adults; normal diameter 8–9 mm (wide). **Posterior urethra** runs from the bladder neck to the **perineal membrane**; **anterior urethra** from there (perineal membrane) to the meatus, inside the corpus spongiosum.

Segment	Length / features	Lining
Prostatic	Length of prostate; urethral crest, **verumontanum** with utricle (Müllerian remnant) and ejaculatory ducts; angle divides preprostatic and prostatic parts	Urothelium
Membranous	2–2.5 cm; shortest and narrowest; striated sphincter	Pseudostratified columnar
Bulbar	Widens as the bulb; **Cowper ducts at 3 and 9 o'clock**; lies **eccentric, close to (nearer) the dorsum** of spongiosum	Pseudostratified columnar
Penile	About 15 cm; central, uniform calibre; glands of Littré	Pseudostratified columnar
Fossa navicularis	Glanular; lacuna magna on the roof	Stratified squamous

- **Sphincters:** (1) **internal (preprostatic) smooth muscle sphincter** at the bladder neck, autonomic, gives continence at rest; (2) **external striated rhabdosphincter**, horseshoe-shaped around the membranous urethra, supplied by the pudendal nerve (mainly S3, some S2; Onuf nucleus), plus its intrinsic smooth muscle.
- **Blood supply (arteries):** internal pudendal → common penile artery → artery to the bulb (proximal; bulbourethral branches) and **retrograde flow via the dorsal arteries through the glans**. This **dual supply** allows bulbar transection in anastomotic urethroplasty.
- **Venous** (veins) to pudendal and prostatic plexuses; **lymph** to internal and external iliac nodes.
- **Female urethra:** about 4 cm; smooth muscle plus striated sphincter over the distal two-thirds; **Skene glands** (prostate homologue) near the meatus; oestrogen-dependent mucosal seal.

Endoscopic management of stricture

Principle: healing after incision or dilation is by **secondary intention**, a race between epithelialisation and wound contraction. No endoscopic method removes **spongiofibrosis**, so cure is limited to short, superficial bulbar strictures.

Favourable	Unfavourable
Bulbar, **<1.5–2 cm**	Penile, fossa, longer
No dense spongiofibrosis	Straddle injury, LS (lichen sclerosus)
First procedure	Repeat; luminal narrowing >74%

1. Dilation

- Stretches the scar without tearing it (bleeding means tearing).
- **Safest: balloon dilation over a guidewire** under vision; blind filiforms condemned. Sounds or self-dilation for palliation.

2. DVIU (direct vision internal urethrotomy)

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define site and length.
2. Guidewire across; **normal saline** irrigation.
3. **Sachse cold knife** at 12 o'clock, through scar to healthy tissue; avoid deep cuts (corpora: ED (erectile dysfunction), fistula).

4. Catheter **3–7 days**; longer adds nothing. - **Laser urethrotomy** (holmium, others): **no advantage over cold knife**. - Results: **Steenkamp and Heyns** (randomised): dilation equals urethrotomy; a second attempt rarely and a third never succeeds. **Santucci (2010):** about 8% stricture-free long term.

3. Adjuncts against recurrence

- **CISC (clean intermittent self-catheterisation)** tapering over 3–6 months: delays recurrence, which returns on stopping.
- **Mitomycin C** injection reduces recurrence; steroid injection, colchicine: weak data.

4. Urethral stents

- **UroLume** (permanent): **withdrawn** (ingrowth). **Memokath**, removable stents: small series only.

5. Optilume (paclitaxel-coated balloon)

- FDA (Food and Drug Administration) 2023. **ROBUST III:** 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy.

Complications: recurrence (commonest), bleeding with erections, irrigant extravasation, UTI (urinary tract infection) and sepsis, ED, spongio-cavernous fistula, incontinence.

Guideline (AUA (American Urological Association) 2016): urethrotomy or dilation for short (<2 cm) bulbar strictures; **urethroplasty, not repeat endoscopy**, after failure. The reconstructive ladder is abandoned.

Classification of congenital urethral anomalies

Group	Anomalies
Abnormal meatal position	**Hypospadias** (ventral); **epispadias** (dorsal)
Obstructive	**PUV (posterior urethral valves)**, Young types I–III; **anterior urethral valves** (with diverticulum); urethral atresia or hypoplasia (congenital stricture); congenital meatal stenosis
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); anterior urethral diverticulum; Cowper duct cyst (syringocele); enlarged prostatic utricle
Duplication	**Effmann classification**: type I blind incomplete; type II complete (IIA2 "Y-type"); type III with caudal duplication. The ventral channel is the functional one
Accessory openings	Congenital urethrocutaneous fistula; parameatal urethral cyst
Female	Epispadias; urethral prolapse; persistent urogenital sinus

Epispadias: definition and classification

- A **dorsal urethral wall defect**, replaced by a broad dorsal mucosal strip (urethral plate); part of the **exstrophy–epispadias complex**, mostly seen with exstrophy.
- Embryology: an overdeveloped cloacal membrane blocks mesenchymal ingrowth; genital tubercles fuse caudal to it.

	Male	Female
Incidence	About 1 in 117,000	About 1 in 484,000

	Male	Female
Types	Glanular, penile, **penopubic** (about 70%, incontinent)	**Davis:** vesicular (patulous meatus), subsymphyseal, retrosymphyseal (incontinent)
Genitalia	Short, upturned penis with **dorsal chordee**	**Bifid clitoris**, flat glabrous mons
VUR (vesicoureteric reflux)	30–40%	30–75%

- The more proximal the meatus, the worse the continence (glanular never incontinent). Pubic diastasis is less than in exstrophy.

Evaluation

- Renal ultrasound; VCUG (voiding cystourethrogram) for reflux; cystoscopy and **bladder capacity** under anaesthesia.
- **Key decision:** use the exstrophy pathway (osteotomy and closure) if pubic diastasis >3 cm, poor intrasymphyseal bar, absent sphincter tissue or tiny bladder.

Management

Aims: continence with safe upper tracts, and a straight penis of adequate length that works for intercourse.

- **Preoperative testosterone** (2 mg/kg intramuscular, 5 and 2 weeks before).

Stage 1: urethroplasty and penile reconstruction (6–12 months)

1. Release dorsal chordee: divide suspensory ligaments; free corpora from the inferior pubic rami.
2. **Modified Cantwell–Ransley:** an 18-mm plate strip is outlined; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; the plate is tubularised over an 8-Fr stent and moved **ventral to the corpora**, which are rotated medially over it (diamond corporotomies if curvature persists).
3. Glans wings closed; skin flaps resurface the dorsum; base Z-plasty; stent 10–12 days. - **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias. - Urethroplasty adds outlet resistance, so the bladder grows before stage 2.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **Young–Dees–Leadbetter** repair with Marshall–Marchetti–Krantz suspension and **ureteric reimplantation**. Osteotomy is rarely needed.
- **Good capacity at presentation:** single-stage repair is possible.
- **Female:** urethroplasty and genitoplasty (clitoral halves and labia joined, mons reconstructed) at 6–12 months; bladder neck repair at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula about 1 in 6 (penile base); stricture rare; neourethra catheterisable; erections and paternity preserved.
- Better than exstrophy: the bladder was never exposed or scarred. Failures: injectables, augmentation.

Diagram to draw: sagittal male urethra with segments, sphincters and a 12 o'clock urethrotomy; Cantwell–Ransley with plate transposed under the corpora.

> **Key points to remember:**
> - Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
> - Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
> - DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
> - Optilume ROBUST III 74.6% vs 26.8%.
> - Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
> - About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

8 · Penis

Structure

- There are paired **corpora cavernosa** (crura on the pubic arch) and a **corpus spongiosum** (bulb, urethra, glans).
- The septum is incomplete distally, so the corpora act as **one unit** (†one-sided injection fills both).
- The **tunica albuginea** (2–3 mm) has an inner **circular** and an outer **longitudinal** layer; it is thin ventrally.
- **Buck fascia** encloses the corpora and splits around the spongiosum. From midline outward it holds the **deep dorsal vein, then arteries, then nerves**.
- Sinusoidal smooth muscle is **~45%** of volume. Endothelial nitric oxide synthase (NOS) sustains erection.

Arteries

The internal pudendal artery runs through Alcock canal to become the **common penile artery**:

Branch	Supplies
Bulbourethral	Spongiosum, urethra, glans
Cavernosal	Enters at hilum, runs centrally; helicine arteries fill the sinusoids (erection)
Dorsal	Under Buck fascia to the glans; circumflex branches

Veins, nerves

- The superficial dorsal vein drains skin; the **deep dorsal vein** (under Buck fascia) drains to the periprostatic plexus.
- In the distal two-thirds, sinusoids drain via the subtunical plexus and **emissary veins**; the proximal third drains via cavernous and crural veins.
- The **dorsal nerve** (pudendal, **S2–S4**) gives sensation; the **cavernous nerves** (pelvic plexus) give erection.

> **KEY POINT**
>
> Erection is veno-occlusive: filled sinusoids compress subtunical and emissary veins against the inelastic tunica. †Tunical or smooth-muscle damage defeats this and causes venous leak.

Penile duplex (after intracavernosal injection)

Parameter	Threshold	Meaning
Peak systolic velocity	**<30 cm/s**	Arterial insufficiency
End-diastolic velocity	**>5 cm/s**	Veno-occlusive dysfunction
Resistive index	**<0.80**	Veno-occlusive dysfunction

MRI shows fracture, prosthesis malposition and tumour.

67 Male Reproductive Physiology

1 · Hypothalamic–pituitary–gonadal axis

Every hormonal work-up reads this axis.

	Peptide hormones	Steroid hormones
Examples	Luteinising hormone (LH), follicle-stimulating hormone (FSH)	Testosterone, oestradiol
Storage	Secretory granules	Not stored; secretion equals production
Receptor	Cell surface, acting via cyclic AMP (cAMP)	Intracellular, altering transcription; carried bound in plasma

Hypothalamus

- Gonadotrophin-releasing hormone (GnRH) is a decapeptide whose only function is LH/FSH release. Half-life is **5–7 min**.
- Its rhythms are pulsatile (every **90–120 min**), circadian (testosterone peaks early morning) and seasonal.
- **Pulsatility is essential**: a continuous GnRH agonist (leuprolide) desensitises the pituitary, giving castrate testosterone.
- In **Kallmann syndrome** GnRH neurones fail to migrate, which causes hypogonadotrophic hypogonadism. †Anosmia coexists because the neurones migrate with the olfactory placode.

GnRH modulation	Factors
Negative	Opioids (β-endorphin), FSH, LH, testosterone
Positive	Prostaglandin E2, insulin, **kisspeptin** (puberty), **leptin**
Variable	Dopamine (catecholamines)

Anterior pituitary

- LH and FSH share an **α subunit** with TSH and hCG and have a specific **β subunit**.
- **Hyperprolactinaemia abolishes gonadotrophin pulsatility**, so testosterone falls. Low prolactin is not pathological.

	LH	FSH
Pulses	8–16/24 h; follow GnRH	Smaller; partly GnRH-independent (inhibin, activin)
Target	**Leydig cell**	**Sertoli cell** (and spermatogonia)
Action	Steroidogenesis: cholesterol to pregnenolone to testosterone	Tubule growth; **initiates spermatogenesis at puberty**; maintains quantitative output
Main feedback	Testosterone (hypothalamus)	Oestradiol (pituitary) and **inhibin B**

Testicular hormones and feedback

- Testosterone output is ~**5 mg/day**. 5α-reductase converts it to the more potent **dihydrotestosterone (DHT**), needed in most peripheral tissues; aromatase converts it to **oestradiol**, which feeds back especially on FSH.
- FSH makes Sertoli cells produce androgen-binding protein (ABP), transferrin, lactate and inhibin. **Inhibin** suppresses FSH; **activin** stimulates it.

> **KEY POINT**
>
> There are two loops: LH with Leydig cells (testosterone), and FSH with Sertoli cells (inhibin, oestradiol). So **LH reports Leydig function, while FSH and inhibin B report the tubules**. High FSH with low inhibin B means spermatogenic failure.

Development and ageing

- **SRY** (Yp) starts the testis. **DAX1** and **WNT4** are "antitestis" genes; the female pathway is not a passive default.

Fetal cell	Product	Effect
Leydig	Testosterone	Wolffian duct forms internal genitalia
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testicular descent
Target tissue	DHT	External genitalia
Sertoli	Müllerian-inhibiting substance (MIS/AMH)	Müllerian regression; keeps germ cells out of meiosis

- Puberty (~12 years) is triggered by kisspeptin, leptin and melatonin.
- **Ageing**: testosterone is ~one-third lower by the 7th decade and FSH ~threefold higher. SHBG rises and feedback is blunted, so failure is both peripheral and central.

2 · Testis

Architecture and blood flow

- It has **600–1200 tubules** and ~**700 million Leydig cells** in young adults.
- The cord holds one artery in ~half of men, so at microsurgical varicocelectomy ligating a single artery can cause atrophy.
- **Countercurrent exchange** cools arterial blood **2–4 °C** below core and recycles testosterone. It is lost in varicocele and cryptorchidism.

Leydig cells and testosterone synthesis

- **Rate-limiting step**: LH raises cAMP, and **steroidogenic acute regulatory (StAR) protein** moves cholesterol into the inner mitochondrial membrane to become pregnenolone.
- Further enzymes (3β-HSD, 17α-hydroxylase/17,20-lyase, 17β-HSD) make testosterone; their mutations cause 46,XY disorders of androgen biosynthesis.
- †In blood, ~44% of testosterone is bound tightly to SHBG, ~54% loosely to albumin and ~2% is free. Free plus albumin-bound is "bioavailable".

Testosterone peak	Timing	Role
Fetal	12–18 weeks	Builds reproductive tract
Neonatal	~2 months	Imprints target tissues
Pubertal	12–17 years	Masculinisation
Adult	Peak in 2nd–3rd decade, then decline	Maintains androgen-dependent organs

Sertoli cells and blood–testis barrier

LAQ Spermatogenesis, azoospermia and non-obstructive azoospermia

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P1 LAQ "Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest" · P2 SAQ "Azoospermia and non-obstructive azoospermia" · P3 SAQ "Spermatogenesis"

Maturation arrest is taken as spermatogenic maturation arrest on testicular histology.

Spermatogenesis: definition

- Production of haploid spermatozoa from diploid spermatogonia in the seminiferous tubules: about **64 days** (42–76); about 123 million sperm a day.

Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- The hypothalamus releases **GnRH (gonadotropin-releasing hormone)** in pulses; pulsatility is essential (constant agonist exposure shuts the axis down). Kisspeptin triggers puberty.
- **LH (luteinising hormone)** acts on Leydig cells to make testosterone (about 5 mg/day). **FSH (follicle-stimulating hormone)** acts on Sertoli cells.
- Feedback: testosterone (and oestradiol) on LH; **inhibin B** from Sertoli cells on FSH. So FSH and inhibin B report the tubules, and LH reports the Leydig cells.
- **Intratesticular testosterone** is about 50-fold higher than serum (held by **ABP, androgen-binding protein**) and is essential for postmeiotic maturation. Exogenous testosterone suppresses LH and so stops spermatogenesis.
- The testis is kept 2–4 °C below core temperature by the pampiniform countercurrent exchange.

Sertoli cells and the blood–testis barrier

- Nurse cells: nourish germ cells; secrete ABP, inhibin and fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** divide the tubule into a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). Post-meiotic cells are "foreign" to the immune system; the barrier and local tolerance protect them.

Phases of spermatogenesis

Phase	Events
1. Proliferation (spermatocytogenesis)	**Dark type A (Ad)** reserve stem cells self-renew (GDNF [glial cell line–derived neurotrophic factor] niche); **pale type A (Ap)** divide every 16 days into type B spermatogonia, then primary spermatocytes. Daughter cells stay joined by cytoplasmic bridges
2. Meiosis	Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; synapsis and crossing-over) gives secondary spermatocytes; meiosis II gives 4 haploid round spermatids
3. Spermiogenesis	No division: acrosome from Golgi, flagellum from centriole, nucleus condenses to about 10%, mitochondria form the midpiece, cytoplasm shed
4. Spermiation	Sertoli cell releases sperm into the lumen

- **Cycle, stage, wave:** six stages in humans; cycles are arranged as a helical wave, so production is continuous.
- Defective meiotic recombination causes maturation arrest and sperm aneuploidy.

Epididymal maturation and transport

- Testicular sperm are immotile. During 2–12 days of epididymal transit they gain **progressive motility** and fertilising ability; they are stored in the cauda.
- **Capacitation** and the acrosome reaction occur in the female tract.

Clinical relevance of spermatogenesis

- A semen sample reflects events about 3 months earlier; reassess 3 months after treatment.
- Disruptors: heat (varicocele, cryptorchidism), exogenous testosterone, chemotherapy, radiation.

Azoospermia: definition

- No sperm in the **centrifuged** ejaculate on **at least two** semen analyses; 10–15% of infertile men.
- **NOA (non-obstructive azoospermia):** azoospermia due to **failure of spermatogenesis** (testicular failure); about 60% of azoospermic men.

Classification

Type	Causes	Clues
Pre-testicular (hypogonadotropic)	Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone or anabolic steroids**	Low FSH, LH, testosterone
Testicular (NOA, spermatogenic failure)	Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy or radiation, idiopathic	Small soft testes, **FSH >7.6 IU/L**, long axis ≤4.6 cm
Post-testicular (OA, obstructive azoospermia)	CBAVD (congenital bilateral absence of the vas deferens), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction)	Normal testes, **FSH ≤7.6**, dilated epididymis
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Causes of NOA

- **Genetic: Klinefelter (47,XXY)**, the commonest; **Y microdeletion** (AZF [azoospermia factor] a, b, c); other karyotype anomalies.
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic:** commonest overall.
- *Note:* hypogonadotropic hypogonadism is pre-testicular and treatable medically (hCG [human chorionic gonadotropin] ± FSH).

Evaluation

- History (mumps, surgery, chemotherapy, drugs, testosterone use); examination (testis size, **vasa**, epididymis, varicocele).
- **Hormones:** FSH (reflects germ-cell mass), LH, testosterone, oestradiol, prolactin.

- NOA: small soft testes, palpable vasa, normal ejaculate volume. **FSH >7.6 IU/L and testis long axis ≤4.6 cm** point to NOA; FSH ≤7.6 with normal testes (>4.6 cm) suggests obstruction.

```
Azoospermia (2 centrifuged samples) ├─ Vasa absent → CBAVD → CFTR testing, both partners; renal scan/US ├─ Volume <1 mL → post-ejaculate urine: sperm = retrograde; none → TRUS for EDO └─ Normal volume: FSH ≤7.6 IU/L + testis long axis >4.6 cm → obstructive FSH >7.6 + testis ≤4.6 cm (small) → spermatogenic failure
```

- **Genetics (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine): karyotype** and **Y microdeletion** in all azoospermic men with NOA, with counselling. **AZFa or AZFb complete deletion: no sperm, do not operate**; AZFc: sperm often found. **CFTR (cystic fibrosis transmembrane conductance regulator)** testing in CBAVD.
- **TRUS (transrectal ultrasound):** seminal vesicle >1.5 cm wide or a midline cyst suggests EDO.
- **Testis biopsy:** combined with sperm extraction and freezing.

Histology patterns in NOA

Pattern	Histology / meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages present, fewer cells	Best
Maturation arrest	Halts at one germ-cell stage	About 40%
SCO (Sertoli cell–only)	No germ cells	About 30%

Maturation arrest

Definition

- A histological pattern of **spermatogenic failure** in which germ cells develop normally up to one stage and **the sequence halts** at a premeiotic, meiotic or postmeiotic step. No mature sperm, or very few, are made.

Classification

- **Early:** arrest at spermatogonia or **primary spermatocyte** (meiotic failure).
- **Late:** arrest at the **spermatid** stage.
- **Complete** (all tubules) versus **incomplete** (foci progress to sperm): incomplete gives better retrieval.

Causes

- **Genetic:** complete **AZFb deletion** causes arrest; meiotic recombination defects in about half of men with maturation arrest; karyotype anomalies.
- **Acquired:** heat (varicocele, cryptorchidism), chemotherapy, radiation, drugs, orchitis; often idiopathic.

Clinical features and diagnosis

- Azoospermia or severe oligospermia, normal semen volume, palpable vasa.
- **Testis size and FSH are often normal**, so it can mimic obstruction (FSH ≤7.6 IU/L, long axis >4.6 cm).
- **Diagnosis needs testicular biopsy** (Bouin's fixative, never formalin), usually done as sperm extraction.

- Karyotype and Y microdeletion testing in all azoospermic men (AUA/ASRM).

Management

- Optimise: treat varicocele, stop exogenous testosterone; off-label hormonal stimulation (clomiphene, hCG) unproven.
- **Micro-TESE** with ICSI: retrieval **about 40%** (versus 30% in SCO).
- **Complete AZFb deletion: retrieval highly unlikely; do not operate.**
- No sperm: donor sperm or adoption. Round spermatid injection is experimental.
- Genetic counselling: sons inherit Y deletions.

Management of azoospermia

1. Pre-testicular (hypogonadotropic)

- Stop exogenous testosterone. **hCG 1500–5000 IU 2–3 times weekly ± FSH**; pulsatile GnRH in hypothalamic failure; dopamine agonist for prolactinoma.

2. Obstructive

- Microsurgical **vasovasostomy** or **vasoepididymostomy** (clear watery fluid or sperm parts in vasal fluid: vasovasostomy; thick toothpaste-like fluid without sperm: vasoepididymostomy).
- **TURED (transurethral resection of the ejaculatory ducts)** for EDO.
- CBAVD or unreconstructable: sperm retrieval by **MESA (microsurgical epididymal sperm aspiration)**, PESA (percutaneous epididymal sperm aspiration) or TESA + **ICSI (intracytoplasmic sperm injection)**; freeze sperm at reconstruction.

3. Non-obstructive

- **Optimisation (before retrieval):** stop exogenous testosterone; repair a clinical varicocele (evidence weak; no definitive evidence in NOA); off-label hormonal therapy (clomiphene, anastrozole, hCG) for low testosterone.
- **Micro-TESE (microdissection testicular sperm extraction) + ICSI:**
- **Principle:** spermatogenesis in NOA is **focal**. Under ×20–25 magnification, **larger, dilated, opaque tubules** are chosen; the least tissue is removed and the blood supply is preserved.
- Sperm found in **about 50%**; better than conventional TESE (Klinefelter 68%, Sertoli cell–only 30%).

Predictor	Retrieval (%)
Post-orchidopexy	74
AZFc deletion	70
Klinefelter	68
Post-chemotherapy	53
AZFa or AZFb complete deletion	**0: do not operate**

- FSH and testis size **do not** reliably predict retrieval.
- **ICSI:** fresh or frozen testicular sperm. Live birth per cycle is modest; miscarriage is higher (sperm DNA damage). Male offspring of AZF deletion carriers inherit the deletion.

4. Ejaculatory

- Retrograde: sympathomimetics, or sperm from alkalinised urine. Anejaculation: vibratory or electroejaculation.

5. Failure (no sperm): donor sperm or adoption; genetic counselling.

Also know

- Spermatogonial stem cell renewal by GDNF; in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagrams to draw

- Cross-section of a seminiferous tubule showing basal and adluminal compartments with germ cells maturing towards the lumen.
- Hypothalamic–pituitary–testis axis with the three levels of failure.
- Opened testis under the microscope: plump tubules versus thin Sertoli cell–only tubules.
- Maturation arrest: seminiferous tubule showing germ cells up to spermatocytes with no spermatids or sperm.

> **Key points to remember:**
>
> - Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
> - Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
> - FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
> - Confirm azoospermia on two centrifuged samples.
> - FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
> - Karyotype and Y microdeletion in all.
> - AZFa/b complete deletion: do not operate (no retrieval).
> - NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
> - Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
> - FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
> - Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

SAQ Male reproductive immunology and immunobead test

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Male reproductive immunology" · P3 SAQ "Immunobead test for semen"

Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

Immune privilege

Anatomical barrier: blood–testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids). If the barrier is breached, sperm antigens meet the immune system.

Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.

- Sertoli and local mediators: galectin-1, TGF-β (transforming growth factor-β), interferon-γ, soluble Fc receptor, **androgens**.
- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest); torsion, trauma, biopsy; infection (orchitis, prostatitis, STI [sexually transmitted infection]); heat (varicocele, cryptorchidism); genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross (poor penetration of) cervical mucus, failure to bind the zona and impaired fertilisation; head-bound antibody matters most.
- **Indications to test:** sperm agglutination or isolated asthenozoospermia; unexplained infertility; after vasectomy reversal.

Tests for ASA: immunobead test (IBT) and MAR

- The IBT detects ASA bound to sperm (direct) or free in body fluids (indirect).
- **Principle:** polyacrylamide beads coated with rabbit anti-human **IgG, IgA or IgM**.

Test	Method	Use
Direct IBT	Patient's motile sperm washed free of seminal plasma; mixed with beads coated with anti-human IgG/IgA; motile sperm with attached beads counted	Sperm-bound antibody: % bound, class and site (head/tail)
Indirect IBT	Antibody-free donor sperm incubated with heat-inactivated patient serum, seminal plasma or female serum/cervical mucus; then direct IBT done	Free antibody; patient sperm too few or immotile; female partner
MAR (mixed antiglobulin reaction)	Fresh unwashed semen + immunoglobulin (Ig) G (or IgA)-coated latex beads + antiglobulin (anti-IgG); beads stick to antibody-coated motile sperm; mixed sperm–bead clumps counted	Quick, cheap screen; IgG/IgA only

- Report: **% of motile sperm with beads**, immunoglobulin class, and **site** (head, midpiece, tail). Tail-tip binding is ignored.
- **WHO (World Health Organization) 2010 (5th edition) threshold: ≥50% of motile sperm bound is clinically significant.**
- **Head binding** and **IgA** matter most (IgA is produced locally and blocks mucus penetration; IgM is too large to enter semen).
- Both direct tests need motile sperm.
- **Also know:** ELISA (enzyme-linked immunosorbent assay) and flow cytometry; serum tests correlate poorly with sperm-bound antibody.

Management of ASA

Treatment	Comment
Condoms	Lack good evidence
Corticosteroids	Two controlled trials conflict; risk of avascular necrosis; not recommended

Treatment	Comment
Sperm washing + intrauterine insemination (IUI)	Bypasses cervical mucus; for mild binding; modest benefit, lacks good evidence (washing removes unbound, not bound, antibody)
IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Preferred, usual treatment; fertilisation largely unaffected, bypassing antibody effects

Infection and inflammation

- **Leukocytospermia:** >1 × 10^6 peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- **Orchitis:** mumps (post-pubertal, atrophy); bacterial (*Chlamydia*/gonococcus if young, *E. coli* if older); tuberculous; **autoimmune**, primary (antibodies with azoospermia) or secondary to vasculitis.
- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; seminal plasma suppresses female tract immunity.

Diagrams to draw

- Seminiferous tubule with Sertoli tight junctions and compartments.
- Motile sperm with beads attached at head and tail.

Key points to remember:

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% bound significant; head-bound IgA matters most.
- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- Treatment of choice: IVF/ICSI.
- Leukocytospermia >1 million/mL.

Also, from the condensed book (not in the paper answer):

- **Inhibin B** marks Sertoli function; Sertoli tight junctions are the body's strongest, yet open to let spermatocytes pass.

3 · Spermatogenesis

A semen test reflects events about two months earlier.

*See the paper answer **Spermatogenesis, azoospermia and non-obstructive azoospermia** (Section 9, Ch 67).*

Also, from the condensed book (not in the paper answer):

- Output is ~ **123 million/day**, about 1000 per heartbeat.
- †Spermatogenesis needs high intratesticular testosterone, so exogenous testosterone suppresses it.

Stem cells

- Failed migration of primordial germ cells causes extragonadal germ cell tumours or Sertoli cell–only testes.

- "Selfish spermatogonia" (FGFR2/FGFR3 mutations) cause paternal-age disorders and spermatocytic seminoma.

Meiosis

	Mitosis	Meiosis
Divisions / daughters	1 / 2 identical	2 / 4 varied, haploid
Synapsis, crossover	None	Prophase I (synaptonemal complex, chiasmata)
Centromere division	Anaphase	**Anaphase II**

- **Defective recombination** occurs in ~half of maturation arrest. Altered crossover causes aneuploidy, so IVF/ICSI offspring have more chromosomal abnormalities.

Genetics

Y microdeletions of the azoospermia factor (AZF) region occur in **15%** of azoospermic and 6–8% of severely oligospermic men, the commonest defined molecular cause.

Deletion	Histology	Sperm retrieval
Complete AZFa	Sertoli cell–only	Highly unlikely
Complete AZFb	Maturation arrest	Highly unlikely
AZFc	Hypospermatogenesis / focal spermatogenesis	**Often found**
Partial AZFa/b	—	Ejaculated sperm possible

Paternal age matters because spermatogonia divide ~23 times a year, raising de novo dominant mutations (achondroplasia, Apert, Marfan), autism, schizophrenia and **DNA fragmentation**.

4 · Epididymis

- The 3–4 m duct runs from caput through corpus to cauda. Testicular sperm are **immotile and cannot fertilise**, except by ICSI.
- **Transport** takes **2–12 days**, driven mainly by spontaneous rhythmic contraction.
- **Storage**: the cauda holds about half the sperm; prolonged storage increases DNA fragmentation.
- **Maturation** gives progressive motility and zona binding. **Fertility is reached mostly by the late corpus or early cauda**.

> **KEY POINT**
>
> Pregnancy occurs even after vasoepididymostomy at the efferent ductules, but **the more distal the anastomosis, the better the fertility**. After vasectomy for >15 years, motility is lower after reversal because of epididymal dysfunction.

- The epididymis is **androgen dependent**, mainly on DHT, which explains impaired semen on 5α-reductase inhibitors.
- **Heat** abolishes storage, and **denervation** (spinal cord injury, diabetes) impairs transport.

5 · Vas deferens, seminal vesicles, ejaculatory ducts

Semen chemistry shows whether these organs are present and open.

- With a **muscle:lumen ratio of ~10:1** the vas is the most muscular hollow viscus.

- The **seminal vesicle** gives ~**80%** of seminal volume in the late fractions. Its fluid is alkaline and carries **fructose** and prostaglandins; **semenogelin I** forms the coagulum.
- The **ejaculatory duct** has no sphincter; reflux is prevented by its **acute insertion angle**.
- Obstruction may be **physical** or **functional** (vesicle myopathy from diabetes or drugs), and static TRUS may not tell them apart.
- An **ejaculate with pH <7.2, low volume and no fructose** means absent or blocked seminal vesicles, as in congenital bilateral absence of the vas (CBAVD) or ejaculatory duct obstruction.

6 · Spermatozoon, capacitation and fertilisation

Part	Contents
Head (4.5 × 3 µm)	Compacted nucleus; acrosome (enzymes)
Neck	Proximal centriole; axoneme starts
Midpiece	~75 mitochondria in a helix, outer dense fibres; ends at annulus
Principal piece / endpiece	Axoneme with fibrous sheath

- The axoneme is **"9 + 2"**, and **dynein** slides the doublets. Inner dynein arm mutants have no motility; outer arm mutants have reduced motility.
- **Paternal mitochondria are degraded** in the egg, so mitochondrial inheritance is maternal.
- **Capacitation** in the female tract gives hyperactivated motility and readiness for the **acrosome reaction**.
- Binding to zona **ZP3** triggers the acrosome reaction. Sperm **phospholipase C zeta (PLCζ)** triggers calcium oscillations, which activate the oocyte.

Sperm defects

- Morphology is graded by strict (Kruger) criteria. Head defects rarely reflect DNA quality, except **pinhead** and **multiple-tailed** sperm.
- In **globozoospermia** (round head, no acrosome) fertilisation fails even with ICSI, because of PLCζ deficiency.

	Ciliary dyskinesia (immotile cilia)	MMAF (multiple morphological abnormalities of the flagella; fibrous sheath dysplasia)
Motility	<10%	Severely reduced
Morphology	Normal on light microscopy; axonemal defect on electron microscopy (EM)	Short, bent, coiled or absent flagella
Cause	Recessive; respiratory disease, dextrocardia (Kartagener)	Genetic; also varicocele, reactive oxygen species (ROS)
Treatment	IVF-ICSI	Correct cause; else IVF-ICSI

7 · Ejaculation physiology †

Each cause of ejaculatory failure hits a specific phase.

ALGORITHM

1. **Emission** (sympathetic, T10–L2 via the hypogastric nerve; α1): vas, seminal vesicle and prostate contract and deliver semen to the posterior urethra.
2. The **bladder neck closes**. If it fails to close, then ejaculation is retrograde.
3. **Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus and pelvic floor contractions, with a relaxed external sphincter, give an antegrade ejaculate.

SAQ Retrograde ejaculation

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3f

Definition

- RE (retrograde ejaculation): semen passes backwards into the bladder at orgasm because the **bladder neck fails to close**; orgasm is preserved, ejaculate absent or low volume.

Physiology

Phase	Event	Control
Emission	Vas, seminal vesicles, prostate deliver semen to posterior urethra	Sympathetic T10–L2
Ejection	**Bladder neck closes**, external sphincter relaxes, bulbospongiosus contracts	Somatic S2–S4 + sympathetic reflex
Orgasm	Cerebral perception	Cortical

Causes

- **Anatomical: TURP (transurethral resection of prostate)** 42–100%, TUIP (incision) 5–45%; most other resections and enucleations; previous bladder neck reconstruction. Ejaculation-sparing techniques, PUL (prostatic urethral lift) and Aquablation preserve ejaculation.
- **Neurogenic: diabetic autonomic neuropathy** (commonest medical), RPLND (retroperitoneal lymph node dissection; less with nerve sparing), colorectal surgery, sympathectomy, aortic surgery, spinal cord injury, multiple sclerosis.
- **Drugs:** α-blockers (silodosin, tamsulosin), antipsychotics, antidepressants, thiazides.
- **Congenital:** Müllerian duct cyst, Wolffian duct anomaly, prune belly syndrome.

Evaluation

- Low-volume or absent ejaculate with orgasm; infertility.
- **Post-ejaculatory urinalysis:** void, ejaculate, void again; centrifuge. **>5–10 sperm per HPF (high-power field)** confirms RE; significant if urinary sperm approach or exceed antegrade sperm.
- Sperm and fructose in urine separate RE from **failure of emission**; TRUS (transrectal ultrasound) if EDO (ejaculatory duct obstruction) suspected.

Treatment

- Reassure if fertility not desired; stop offending drug.

- **Sympathomimetics** (close bladder neck): **pseudoephedrine** 60–120 mg (120 mg 6-hourly the day before and twice on the day); midodrine, ephedrine, phenylpropanolamine. Side effects: dry mouth, hypertension.
- **Imipramine** 25 mg twice daily; antihistamines (brompheniramine) add to sympathomimetics.
- **Bladder neck reconstruction:** poor results; collagen injection experimental.
- **Sperm retrieval:** alkalinise urine (sodium bicarbonate), void or catheterise after orgasm, wash sperm in buffer → IUI (intrauterine insemination), IVF (in vitro fertilisation) or ICSI (intracytoplasmic sperm injection).
- **Prevention:** nerve-sparing RPLND, ejaculation-preserving procedures for BPH (benign prostatic hyperplasia).

Diagram to draw: bladder neck open at ejaculation with semen entering the bladder.

> **Key points to remember:**
> - TURP and diabetic neuropathy are the commonest causes.
> - Post-ejaculatory urine: sperm and fructose confirm RE.
> - Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

68 Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

1 · The men's health centre

The urologist is often the only doctor a man sees.

- Men avoid preventive care but seek help for erectile dysfunction (ED), infertility and voiding, so the urologist is the **front door** to a multidisciplinary centre and triages the whole man.

Age band	Dominant need	Key partners
18–39	Sexual, reproductive, testicular ("rule out hypogonadism"); preventive	Sports medicine, physiotherapy, mental health
40–64	Prostate and colon screening; disease management	Gastroenterology (GI), cardiology, endocrinology, bariatrics, pulmonology
65+	Falls, neurodegeneration, hearing/vision, nutrition	Geriatrics, neurology, audiology, ophthalmology

2 · The longevity gap

- Male life expectancy was **76.1** versus 81.1 years for females (2016).
- Males have higher mortality in **12 of 15** leading causes; the largest gap is **suicide (>3-fold)**.
- Prostate cancer is the only US malignancy with rising stage IV at diagnosis, so discuss screening.

3 · Urologic screening

- Screen sexual and voiding complaints together, with PSA and urinalysis per guidelines.

Questionnaire	Screens
Sexual Health Inventory for Men (SHIM)	Erectile function
Androgen Deficiency in the Aging Male (ADAM)	Androgen-deficiency symptoms
International Prostate Symptom Score (IPSS)	Lower urinary tract symptoms (LUTS)

4 · Urologic disorders of focus

Testosterone deficiency (TD)

SAQ Male menopause (andropause, late-onset hypogonadism)

Paper answer · P4

Asked as:

- Male menopause. — *Summer 2017 Paper IV Q3e (SAQ)*
- Andropause — *Winter 2017 Paper IV Q3e (SAQ)*
- Male menopause — *Summer 2019 Paper IV Q3f (SAQ)*

Type: SAQ (10 marks)

Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause"). Same topic as 2017-Summer Paper 4, Q3e and 2017-Winter Paper 4, Q3e; full answer given here.

Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Menopause" (or "Andropause") is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

Pathophysiology

- Testosterone falls slowly from middle age.
- Testosterone in the 70s is about one-third lower than in young men. (Another source: mean testosterone in the seventh decade is about **one-third lower** than in young men.)
- **Testis:** fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so free testosterone (bioavailable [free + albumin-bound] testosterone) falls more.
- Worse with obesity, diabetes and **opioids**.

Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

Diagnosis

- **Morning (7–11 am) total testosterone, twice (repeated).** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L). EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low (Low LH → **prolactin**); pituitary MRI (magnetic resonance imaging) if testosterone <150 ng/dL or prolactin high.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

Treatment

- Lifestyle first: weight loss, exercise, treat comorbidities.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months (another source: at about 3 months).
- **Contraindications:**
- Prostate or breast cancer.
- **Wish for fertility** (TRT stops sperm production [suppresses spermatogenesis]; use hCG [human chorionic gonadotrophin] or clomiphene).
- Haematocrit >50% (raised haematocrit).
- Untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop/hold if >54%), PSA.

Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

Diagram to draw: hypothalamic–pituitary–testis axis with ageing defects.

> **Key points to remember:**
> - Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
> - SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
> - TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Also, from the condensed book (not in the paper answer):

- For haematocrit, the AUA withholds TRT at >50% and the Endocrine Society at >54%; then reduce the dose or phlebotomise.
- **Randomised trials show TRT benefit** for obesity, metabolic syndrome, diabetes, muscle and bone, but **TRT is not a treatment for depression**.
- For men on chronic opioids, the Endocrine Society advises screening for TD and considering TRT. Exclude secondary causes before starting TRT.

> **KEY POINT**
>
> † **Fertility caution**: exogenous T suppresses LH/FSH and intratesticular T, which causes oligo- or azoospermia. Men wanting children should avoid TRT and use clomiphene, human chorionic gonadotrophin (hCG) or aromatase inhibitors (see Ch 69).

† TRT	Points
Contraindications	Near-term fertility wish; breast cancer; untreated prostate cancer; haematocrit >50%; untreated severe obstructive sleep apnoea (OSA); uncontrolled heart failure; MI/stroke within 3–6 months
Target	Mid-normal T (≈450–600 ng/dL)
Formulations	Gels, injections, pellets, each with its own monitoring plan

Erectile dysfunction

- ED affects >50% of men aged 40–70.
- **Every man with ED, of any age, is at cardiovascular disease (CVD) risk**. †Penile arteries are small, so ED precedes coronary events by ~2–5 years.
- Treatment steps from lifestyle to PDE5 inhibitors (never with nitrates), then injection, intraurethral or vacuum therapy, then prosthesis, which has the highest satisfaction. Platelet-rich plasma and shock wave therapy remain in trials.

BPH and infertility

- Refer BPH when medication fails, is not tolerated or is not wanted.
- Infertility is no pregnancy after 1 year; a male factor is present in ~half (see Ch 69).

5 · Cardiovascular prevention

- Only **lowering low-density lipoprotein cholesterol (LDL-C)** benefits primary prevention; raising HDL or lowering triglycerides does not. Screen lipids from age 20.

Statin decision (ACC/AHA)	Action
LDL-C ≥190 mg/dL	High-intensity statin with or without a nonstatin
LDL-C 70–189, 10-year risk ≥**7.5%**	Moderate–high intensity statin
Risk 5–7.5%	Moderate intensity may be started
Statin-intolerant, risk >20%	Ezetimibe or PCSK9 inhibitor

- USPSTF, NICE and ESC start statins at ≥10% risk.
- A **coronary artery calcium (CAC) score** is most useful at **intermediate risk**; zero means <1% atherosclerosis.

Blood pressure (BP) (ACC/AHA 2017)	Systolic	Diastolic
Normal	<120	and <80
Elevated	120–129	and <80
Stage 1	130–139	or 80–89
Stage 2	≥140	or ≥90

Elevated or stage 1 BP gets lifestyle change. Drugs are for stage 2, or stage 1 with age ≥65, T2DM, CVD, chronic kidney disease or 10-year risk ≥10%.

6 · Obesity and metabolic syndrome

Fat lowers testosterone, and low testosterone adds fat.

- Body mass index (BMI): overweight 25–29.9; obesity I 30–34.9, II 35–39.9, III ≥40.
- Advise weight loss at BMI ≥30, or 25–29.9 with CVD risk, OSA or osteoarthritis; GLP-1 agonists are approved.
- †Metabolic syndrome (ATP III) needs ≥3 of: waist >102 cm, triglycerides ≥150 mg/dL, HDL <40 mg/dL, BP ≥130/85, glucose ≥100 mg/dL. Adipose aromatase raises oestradiol, which lowers LH, so low T and metabolic syndrome drive each other.

7 · GI, skin and mental health

- Evaluate **constipation** in every man with UTI or retention. Colorectal screening runs from **45–75** years.
- Iron deficiency anaemia carries a >30-fold GI malignancy risk and needs investigation.
- Psoriasis is linked to MI, and suicide is much higher in men.

69 Male Infertility

1 · Definition and overview

Half of infertility involves the man, so always examine him.

- Infertility is no pregnancy after **1 year** of unprotected intercourse. It affects ~15% of couples; male and female factors contribute equally.
- Most conceptions occur in the first 3–6 cycles, so evaluation may start after 6 months.
- Idiopathic infertility is the largest group. **Varicocele is the commonest identifiable cause**, then obstruction.

† Classification	Examples
Pre-testicular (endocrine)	Hypogonadotropic hypogonadism (HH), Kallmann, prolactinoma, exogenous androgens, opioids, obesity, congenital adrenal hyperplasia (CAH)
Testicular	Klinefelter, Y microdeletion, cryptorchidism, torsion, orchitis, varicocele, chemotherapy/radiation, Leydig failure
Post-testicular	Obstruction (epididymal, vasal, ejaculatory duct), congenital bilateral absence of the vas (CBAVD), ejaculatory dysfunction, antisperm antibodies, structural sperm defects

2 · History

LAQ Discuss the role of urologist in reproductive endocrinology and male infertility practice.

Paper answer · P2

Type: LAQ (25 marks)
Source: 2022-Summer Paper 2, Q1

Introduction

- **Infertility:** no conception after 12 months of regular unprotected intercourse; about **15% of couples** (WHO, World Health Organization). The male contributes about half, yet many men are never examined.
- The urologist (andrologist) has five roles: **diagnose** the male factor, **treat endocrine causes**, **correct surgical causes**, **retrieve sperm for ART (assisted reproductive technology)**, and protect **general health and future fertility**.
- Commonest diagnoses: idiopathic, **varicocele** (commonest correctable), obstruction.

1. Diagnostic role

- **History:** duration, coital timing (semen is best after 1–2 days' abstinence), partner's age (fecundity falls after 35), cryptorchidism, mesh herniorrhaphy, RPLND (retroperitoneal lymph node dissection), diabetes, spinal injury, smoking, cannabis, opioids, **exogenous testosterone or anabolic steroids**.
- **Examination:** androgenisation; **testis size** (reflects spermatogenic mass); epididymis; **both vasa present?**; varicocele (standing, Valsalva).
- **Semen analysis**, at least two. WHO 6th edition (2021) 5th centiles: volume 1.4 mL, concentration 16 million/mL, total count 39 million, progressive motility 30%, normal forms 4%. These describe fertile men, not fertility cut-offs.

- **Genetics:** karyotype and **Y-chromosome microdeletion** (AZF, azoospermia factor) in non-obstructive azoospermia or severe oligospermia; **CFTR (cystic fibrosis transmembrane conductance regulator)** testing of both partners in CBAVD (congenital bilateral absence of the vas deferens).
- **Imaging:** scrotal US (ultrasound); **TRUS (transrectal US)** if volume <1 mL (ejaculatory duct obstruction, EDO), after post-ejaculate urine excludes retrograde ejaculation; renal US in vasal agenesis.

2. Endocrine evaluation (the urologist's reproductive endocrinology)

- Spermatogenesis needs a **high intratesticular testosterone (T)**. Morning total T + SHBG (sex hormone-binding globulin) + albumin gives bioavailable T; low is below about **280–300 ng/dL**; repeat borderline values (T is pulsatile and circadian).

Test	What it tells
LH (luteinising hormone)	Localises low T: high → testicular (Leydig) failure; low/normal → pituitary or hypothalamus
FSH (follicle-stimulating hormone) / inhibin B	Germ-cell mass. Azoospermia with **FSH >7.6 IU/L and testis long axis ≤4.6 cm → spermatogenic failure**; FSH ≤7.6 and >4.6 cm → obstruction
Oestradiol (E2)	**T (ng/dL) : E2 (pg/mL) <10** → aromatase excess (obesity)
Prolactin	Only with low T, low libido/ED (erectile dysfunction), headache or field defect
TSH (thyroid-stimulating hormone), cortisol, 17-hydroxyprogesterone	Only if signs; CAH (congenital adrenal hyperplasia)

3. Medical (endocrine) treatment

Condition	Mechanism	Treatment
Hypogonadotropic hypogonadism (Kallmann, pituitary)	Testis normal, signal missing: most treatable cause	**hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly**, add hMG (human menopausal gonadotropin) 75 IU or recombinant FSH; pulsatile GnRH (gonadotropin-releasing hormone) pump for hypothalamic failure. Sperm may take 1–2 years
Anabolic steroids / exogenous T	Negative feedback suppresses LH → intratesticular T falls → azoospermia	Stop; hCG ± clomiphene to speed recovery. **Never give T to a man wanting fertility** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine)
Hyperprolactinaemia	Prolactin inhibits GnRH	Repeat level; pituitary MRI (magnetic resonance imaging) if high; **cabergoline** (fewer side effects than bromocriptine)
Partial hypogonadism, obesity	Aromatisation of T to E2 suppresses LH	Weight loss; **clomiphene 25 mg daily** (off-label), **anastrozole 1 mg daily** or letrozole if T:E2 <10
CAH	Adrenal androgens suppress gonadotropins; testicular adrenal rest tumours	Glucocorticoids

Condition	Mechanism	Treatment
Primary testicular failure (Klinefelter 47,XXY)	Germ cells absent or few	No drug restores sperm; exogenous T worsens it; **micro-TESE** (microdissection testicular sperm extraction)

4. Surgical role

Problem	Operation
Clinical varicocele + abnormal semen	**Microsurgical subinguinal varicocelectomy** or embolisation: improves semen, DNA fragmentation and pregnancy. Not for US-only varicocele
Vasal or epididymal obstruction	**Vasovasostomy / vasoepididymostomy** (microsurgical intussusception)
EDO	**TURED** (transurethral resection of ejaculatory ducts)
Non-obstructive azoospermia	**Micro-TESE**, best technique; no operation for **AZFa/AZFb** deletions (no sperm found)
Unreconstructable obstruction	PESA/MESA (percutaneous/microsurgical epididymal sperm aspiration), TESA (testicular sperm aspiration)
Retrograde ejaculation; anejaculation	Sympathomimetics, urine sperm retrieval; **penile vibratory stimulation**, electroejaculation

5. Partner of the ART team, counselling and men's health

- Choose the least invasive route: natural conception, IUI (intrauterine insemination), IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection). **Cryopreserve sperm** at retrieval.
- **Fertility preservation:** sperm banking before chemotherapy, radiotherapy or RPLND; **onco-TESE** if azoospermic.
- **Genetic counselling:** AZFc deletion passes to sons; CFTR risk to offspring; paternal age ≥40 years raises offspring risk.
- **Infertility is a marker of general health:** higher risk of **testis cancer** (scan every azoospermic man), later comorbidity and mortality.

Recent advances: sperm DNA fragmentation testing; microfluidic sperm selection; genomic panels; prepubertal testicular tissue cryopreservation (experimental).

Also know: antisperm antibodies; globozoospermia and ciliary dyskinesia (ICSI); androgen receptor defects (high T and LH).

Diagram to draw: hypothalamic–pituitary–testis axis with feedback loops (T and E2 on LH; inhibin B on FSH).

> **Key points to remember:**
> - FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
> - Exogenous testosterone is a contraceptive: never give it for infertility.
> - Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
> - Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Also, from the condensed book (not in the paper answer):

- The **female partner's age is the most important determinant**; fecundity falls after 35.
- Paternal age ≥40 needs counselling about offspring bipolar disorder, schizophrenia, autism and ADHD (ASRM/AUA 2021).
- Semen quality peaks after **1–2 days' abstinence**, so advise daily intercourse around ovulation; most lubricants impair sperm.

Gonadotoxins and drugs

> **KEY POINT**
>
> **Exogenous testosterone (T) is a contraceptive**. It is aromatised to oestradiol (E2), which feeds back on the pituitary and lowers LH. Intratesticular T then falls, and spermatogenesis falls with it. Recovery usually takes ≥4 months after stopping (after ≤2 years' use); after longer use it is unknown.

- Other harmful drugs include antiandrogens, spironolactone, corticosteroids, anabolic steroids and some antiretrovirals. Cannabis, alcohol and smoking also impair semen.

Infection and childhood conditions

- Infection injures either **anatomically** (duct scarring) or **functionally** (leucocytes, reactive oxygen species [ROS]).
- After torsion, about half have later spermatogenic effects.

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Paternity	Near normal	Markedly reduced
Inhibin B	~⅔ of controls	~⅓ of controls

- **Laterality is the key prognostic factor.**
- The American Urological Association (AUA) advises orchidopexy within **18 months**, because germ cells are lost from 15–18 months. Later surgery before 10 years still improves sperm retrieval.

3 · Physical examination

Testis size and the vasa sort most azoospermic men.

- Examine the man standing. An engorged epididymis suggests obstruction, and testis size reflects spermatogenic mass.

Testis size	Abnormal
Seager caliper (long axis)	**≤4.6 cm**
Prader orchidometer (volume)	≤20 mL
Scrotal ultrasound (US)	Most accurate; added value unclear

- Unilateral vasal absence needs renal US; bilateral absence needs CFTR testing and renal US.
- Palpable seminal vesicles suggest ejaculatory duct obstruction (EDO).

Varicocele grade	Definition
Subclinical	Seen only on Doppler; **do not treat**

Varicocele grade	Definition
I	Palpable only with Valsalva, standing
II	Palpable without Valsalva, not visible
III	Visible through scrotal skin

4 · Endocrine evaluation

ALGORITHM

1. Take a morning sample for total T, sex hormone-binding globulin (SHBG) and albumin, and calculate bioavailable T.
2. Measure LH and follicle-stimulating hormone (FSH), then E2.
3. Measure prolactin only if indicated.

- Adequate total T is ≥**280–300 ng/dL**. Low T is common even with density >20 million/mL, so assess every infertile man.
- Low SHBG (obesity, diabetes) makes total T under-read; high SHBG (ageing, cirrhosis) makes it over-read.
- A T (ng/dL) to E2 (pg/mL) ratio **<10:1** means aromatase excess.
- Prolactin of 20–50 µg/L needs no further work-up; a significant rise needs pituitary MRI.

5 · Semen analysis

SAQ Semen analysis nomenclature.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3e

Collection (WHO, World Health Organization, 2021, 6th edition)

- Masturbation into a clean wide-mouthed non-toxic container (special condom if needed); complete sample.
- Abstinence **2–7 days** (evidence: 1–2 days best). Keep at 20–37 °C; analyse within **1 hour**, after liquefaction (~30 min).
- At least **two samples, 2–3 weeks apart**, because of variability.

Lower reference limits (5th centile, fertile men)

Parameter	Lower limit
Volume	**1.4 mL**
Concentration	**16 million/mL**
Total count	**39 million/ejaculate**
Total motility	**42%**
Progressive motility	**30%**
Normal forms (strict)	**4%**
Vitality	**54%**
Peroxidase-positive leucocytes	<1 million/mL

- These are reference values, not fertility cut-offs. The 6th edition grades motility as progressive, non-progressive or immotile.

Nomenclature

Term	Meaning
Normozoospermia	All parameters at or above reference
Oligozoospermia	Total count (or concentration) below reference
Asthenozoospermia	Progressive motility below reference
Teratozoospermia	Normal forms below reference
OAT (oligoasthenoteratozoospermia)	All three abnormal
Azoospermia	No sperm, **including centrifuged pellet**
Cryptozoospermia	Sperm absent in fresh smear, present only after centrifugation
Aspermia	No ejaculate (dry orgasm)
Hypospermia	Volume below reference
Necrozoospermia	Low vitality (many dead sperm)
Leucocytospermia	Peroxidase-positive white cells above 1 million/mL
Haemospermia	Blood in ejaculate
Globozoospermia	Round-headed sperm without acrosome

Notes

- **Low volume or aspermia:** post-ejaculatory urine for retrograde ejaculation; TRUS (transrectal ultrasound) for EDO (ejaculatory duct obstruction).
- **Vitality:** eosin–nigrosin, or **HOS (hypo-osmotic swelling)** test, which keeps sperm usable for ICSI (intracytoplasmic sperm injection).
- Only azoospermia is definitive; values overlap between fertile and infertile men.
- **Also know:** Kruger strict morphology; DNA fragmentation; antisperm antibodies (MAR, mixed antiglobulin reaction); semen pH no longer recommended.

Diagram to draw: normal sperm (head, midpiece, tail) beside a WHO reference box.

> **Key points to remember:**
> - WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
> - Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
> - Two samples 2–3 weeks apart; abstinence 2–7 days.

Also, from the condensed book (not in the paper answer):

- **Low volume (<1.0 mL)** suggests EDO, CBAVD, hypoandrogenism or retrograde ejaculation. Do a **postejaculatory urinalysis**; if there are no sperm in urine, do transrectal ultrasound (TRUS).
- †Sperm DNA fragmentation testing is not routine; use it for recurrent pregnancy loss or failed assisted reproduction (ART).

6 · Genetic testing

ALGORITHM

1. If non-obstructive azoospermia (NOA) or density **<5 million/mL**, then do karyotype and Y-microdeletion testing (ASRM/AUA 2021).
2. If CBAVD (or idiopathic EDO), then test CFTR in the man **and his partner**.

SAQ Genetic causes of male infertility. Enumerate and classify.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2017-Summer Paper 3, Q3d

Introduction

- Genetic defects underlie many cases of **azoospermia** and **severe oligozoospermia**. They decide **sperm retrieval odds** and **transmission to offspring** through ICSI (intracytoplasmic sperm injection).

Classification

Group	Condition	Key features
1. Chromosomal, numerical	**Klinefelter syndrome (47,XXY)**	**Commonest genetic cause**; 1 in 500–1000; small testes, high FSH (follicle-stimulating hormone), low testosterone, usually azoospermia. Non-mosaic in 80–90%. Testicular sperm in about 50% (micro-TESE, microdissection testicular sperm extraction)
	47,XYY; **46,XX male** (SRY gene translocated to X)	XX male: azoospermia, no sperm retrievable
2. Chromosomal, structural	**Robertsonian and reciprocal translocations**, inversions, ring Y	Oligozoospermia; recurrent miscarriage
3. Y chromosome microdeletions (Yq, AZF (azoospermia factor))	**AZFa** (USP9Y, DDX3Y), **AZFb**, **AZFc** (DAZ genes)	Complete AZFa or AZFb deletion: **no sperm at retrieval**. AZFc: sperm often in ejaculate or testis; sons inherit the deletion
4. Single-gene: excurrent ducts	**CFTR (cystic fibrosis transmembrane conductance regulator) mutations**: CBAVD (congenital bilateral absence of vas deferens); idiopathic ejaculatory duct obstruction	Low-volume azoospermia, normal spermatogenesis; **test both partners** (ΔF508)
5. Single-gene: hypothalamic–pituitary	**Kallmann syndrome** (KAL1, anosmia); GnRH (gonadotropin-releasing hormone) receptor mutations	Low LH (luteinising hormone) and FSH; treat with hCG ± FSH
6. Androgen action	**Androgen receptor** mutation (long CAG repeat); 5α-reductase type 2 deficiency	High testosterone and LH, poor sperm

Group	Condition	Key features
7. Sperm structure and motility	**Globozoospermia** (SPATA16, DPY19L2: no acrosome); macrocephalic sperm (AURKC); **PCD (primary ciliary dyskinesia)**/Kartagener (DNAH5, DNAI1: immotile sperm)	ICSI; macrocephalic sperm: ICSI not advised (aneuploidy)

Genetic testing indications

- **Karyotype and Y microdeletion:** non-obstructive azoospermia and severe oligozoospermia (**<5 million/mL**, ASRM/AUA (American Society for Reproductive Medicine/American Urological Association) 2021; EAU (European Association of Urology) uses <10 million/mL for karyotype).
- **CFTR:** CBAVD, idiopathic obstructive azoospermia; screen the partner; counsel before ICSI.
- **Also know:** Noonan syndrome, myotonic dystrophy.

Diagram to draw: Y chromosome long arm showing AZFa, b, c.

> **Key points to remember:**
> - Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
> - AZFa or AZFb complete deletion: do not attempt retrieval.
> - CBAVD = genital form of cystic fibrosis; test both partners.

Also, from the condensed book (not in the paper answer):

- AZFc deletion with oligozoospermia declines progressively, so **cryopreserve sperm** early.

7 · Imaging

- Do not screen, because incidental microlithiasis causes harm. Varicocele is clinical.

SAQ Ejacultory duct obstruction

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 3, Q3b

Assumed meaning: ejaculatory duct obstruction (EDO).

Anatomy

- Intraprostatic continuations of the seminal vesicles, joined by the vasal ampullae; they lack muscle and open at the **verumontanum**. Seminal vesicles give about 60% of the ejaculate and its fructose.

Aetiology

- **Congenital (commonest):** midline (Müllerian/utricular) cyst, Wolffian duct cyst, atresia. Idiopathic EDO carries CFTR (cystic fibrosis transmembrane conductance regulator) mutations in about half.
- **Acquired:** infection, instrumentation, duct stones, compression.
- **Zinner syndrome:** seminal vesicle cyst, ipsilateral renal agenesis, EDO.

- Complete bilateral EDO causes under 5% of azoospermia. Partial EDO gives oligoasthenospermia.

Clinical features

- Infertility, painful ejaculation, haematospermia, perineal pain.

Diagnosis

Semen (complete bilateral EDO)

- Volume **<1.0 mL**, **acidic pH**, low or absent **fructose**, azoospermia.
- Palpable vas, normal FSH (follicle-stimulating hormone) and testes.
- Exclude retrograde ejaculation and CBAVD (congenital bilateral absence of the vas deferens).

Investigations

Test	Method / finding
TRUS (transrectal ultrasound)	Seminal vesicle AP diameter **>1.5 cm**, duct >2.3 mm, midline cyst. Low specificity
Seminal vesicle aspiration	**≥3 sperm per high-power field** confirms obstruction
Vesiculography	Contrast into vesicles; invasive
Chromotubation	Indigo carmine into vesicles; dye at the veru on cystoscopy

Management

TURED (transurethral resection of the ejaculatory ducts)

- For azoospermic or severely oligoasthenospermic men, after the couple declines IVF (in vitro fertilisation).
- **Steps:** TRUS-guided aspiration (cryopreserve sperm) → instil indigo carmine with contrast → 24-Fr resectoscope, finger in rectum → resect the veru until dye effluxes. Spare bladder neck, sphincter and rectum. Catheter overnight.
- **Results:** volume rises in most; sperm appear in about half. Cold-knife incision alone re-obstructs.

Complications

- **Urine reflux** into the ducts, **epididymitis**, retrograde ejaculation, rectal injury, re-obstruction.

Alternatives

- **Seminal vesiculoscopy** (4–6 Fr via utricle): avoids reflux; restenosis unknown. Sperm retrieval with ICSI (intracytoplasmic sperm injection).

Diagram to draw: sagittal prostate with midline cyst compressing the ducts at the veru.

> **Key points to remember:**
> - Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
> - TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
> - TURED: sperm return in about half; reflux and epididymitis are its price.

8 · Azoospermia: obstructive (OA) vs non-obstructive (NOA)

This split decides between reconstruction and sperm extraction.

	OA	NOA (spermatogenic dysfunction)
FSH	**≤7.6 IU/L**	>7.6 IU/L
Testis long axis	>4.6 cm	≤4.6 cm
Management	Reconstruction (see Ch 70) or sperm retrieval	Genetic tests; microdissection testicular sperm extraction (micro-TESE) with intracytoplasmic sperm injection (ICSI)

ALGORITHM

1. If the vasa are absent, then test CFTR in both partners.
2. If the vasa are present and volume is <1 mL, then do postejaculatory urinalysis. Sperm in urine means retrograde ejaculation; no sperm means TRUS for EDO.
3. If volume is normal, then use FSH and testis size to separate OA from NOA.

9 · Specific causes and treatment

Klinefelter syndrome (47,XXY)

SAQ Klinefelter's syndrome.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3b

Definition and genetics

- **47,XXY**: commonest sex chromosome aneuploidy and **commonest genetic cause of male infertility**; 1 in 500–1000 male births.
- Non-mosaic in **80–90%**; the rest mosaic **46,XY/47,XXY** (milder, more sperm) or 48,XXXY variants. Extra X from non-disjunction (maternal age).
- Underdiagnosed: most never diagnosed; increasingly detected on NIPT (non-invasive prenatal testing).

Pathophysiology

- Seminiferous tubules **hyalinise** progressively from puberty; germ cells are lost. Leydig cells fail, so testosterone falls. Puberty starts on time, then stalls.

Clinical features

Age	Features
Infant	Usually normal; micropenis, cryptorchidism, hypospadias
Adolescent/adult	**Small firm testes (<3.5 cm, 1–4 mL)**, gynaecomastia, tall stature with long legs (only ~30%), sparse hair, female fat distribution
Neurodevelopment	Language, learning and attention difficulties

Hormones and tests

- **High FSH (follicle-stimulating hormone) and LH (luteinising hormone), low or low-normal testosterone, high oestradiol** (raised oestradiol:testosterone ratio). Inhibin B

low.
- Semen: **azoospermia** in most (~8% have ejaculated sperm).
- **Karyotype** confirms (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine, guideline: karyotype all men with non-obstructive azoospermia or severe oligozoospermia).

Management

- **Fertility: micro-TESE** (microdissection testicular sperm extraction) + ICSI (intracytoplasmic sperm injection). Sperm found in **~40–50%** (68% in one series); offspring are usually chromosomally normal.
- Spermatogenesis declines with age, so retrieve early (late adolescence or young adulthood). Some give clomiphene, anastrozole or hCG (human chorionic gonadotropin) beforehand.
- **TRT (testosterone replacement therapy):** for hypogonadism (bone, muscle, mood, libido). It **suppresses spermatogenesis**, so offer sperm retrieval or banking **before** starting.
- Gynaecomastia: reduction mammoplasty if persistent. Speech, educational and psychological support.

Associated risks

- Osteoporosis, metabolic syndrome, diabetes, VTE (venous thromboembolism), autoimmune disease.
- **Mediastinal (extragonadal) germ cell tumours**; male breast cancer (~1% lifetime).

Diagram to draw: hypothalamic–pituitary–testis axis showing high FSH and LH, low testosterone.

> **Key points to remember:**
> - 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
> - Small firm testes, high FSH and LH, azoospermia.
> - Micro-TESE ~50% success; retrieve sperm before testosterone.
> - Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

Also, from the condensed book (not in the paper answer):

- **Trap**: exogenous T suppresses residual spermatogenesis, so extract sperm before starting T. If fertility is wanted, use aromatase inhibitors or human chorionic gonadotrophin (hCG).

Hypogonadotropic hypogonadism — most treatable cause

State	Hormones	Treatment
Kallmann (anosmia; KAL1, GNRHR)/complete HH	Low LH, low FSH	hCG with or without recombinant FSH (rFSH) or human menopausal gonadotrophin (hMG)
Isolated LH deficiency ("fertile eunuch")	Low LH, normal FSH	hCG
Isolated FSH deficiency	Low FSH	FSH
Hypothalamic GnRH deficiency	Low GnRH	Pulsatile subcutaneous GnRH every 2 h, for ≥6 months

State	Hormones	Treatment
Partial HH (common)	LH inappropriately low for T	Selective oestrogen receptor modulator (SERM) or aromatase inhibitor

- Give hCG 1500–5000 IU 2–3 times weekly (max 10,000/week), titrated to T; add hMG or rFSH 75 IU 2–3 times weekly. After androgen abuse, allow 1–2 years.
- Treat prolactinoma with cabergoline, and CAH with corticosteroids.

Empirical medical therapy

Drug	Dose	Notes
Clomiphene (SERM)	25 mg daily or 50 mg alternate days; max 100 mg	Blocks E2 feedback, so LH and FSH rise; off-label; no change in prostate-specific antigen (PSA) or haematocrit; target T 600–800 ng/dL
Anastrozole / letrozole	Anastrozole 1 mg daily	Raises T and lowers E2; use when T:E2 <10:1 (obesity); caution with bone density

Primary failure and androgen resistance

- High LH with low T has no accepted therapy; exogenous T is contraindicated, so offer sperm extraction.
- Androgen receptor defects give high T, LH and E2 and are treated with ICSI.

CBAVD

SAQ CBAVD

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 3, Q3f

Definition

- **CBAVD (congenital bilateral absence of the vas deferens):** both vasa absent, causing **obstructive azoospermia** with normal spermatogenesis.
- Often a **genital form of CF (cystic fibrosis)**: nearly all men with CF lack both vasa.

Genetics

- **CFTR (cystic fibrosis transmembrane conductance regulator)**: chloride channel gene; more than 1600 mutations.
- Severe mutation on both alleles: CF. **Mild mutation on one or both alleles: CBAVD.** Commonest severe mutation **ΔF508**; the **5T variant** (intron 8) is a common mild allele.
- A minority have other mesonephric duct gene defects instead, with more **renal agenesis**.

Clinical features

- **Vasa not palpable** (Meacham: a presumed vas slipping away three times is absent).
- Caput epididymis present and distended; body and tail often absent.
- Normal testis size, FSH (follicle-stimulating hormone) and testosterone.
- Seminal vesicles absent or hypoplastic: **semen volume <1.0 mL, acidic, fructose-negative**, azoospermic.
- Sinus or chest symptoms may reveal mild CF.

Evaluation

- Semen analysis; FSH; **TRUS (transrectal ultrasound)** for seminal vesicles.
- **Renal ultrasound** (renal agenesis in about 1 in 10).
- **CFTR testing of the man and his female partner** (carrier rate about 1 in 20 in northern Europeans). Panels cover 25–40 mutations, so a negative panel leaves residual risk.
- Testis biopsy unnecessary when FSH and testis size are normal.

Management

- **Genetic counselling**: if the partner carries a mutation, offspring may have CF; PGT (preimplantation genetic testing) or donor sperm can be offered.
- Reconstruction is impossible, so **surgical sperm retrieval + ICSI (intracytoplasmic sperm injection)**:

Technique	Points
MESA (microsurgical epididymal sperm aspiration)	Best: many motile sperm; cryopreserve several vials
PESA (percutaneous epididymal aspiration)	Local anaesthetic; fewer sperm
TESA/TESE (testicular aspiration/extraction)	If epididymal retrieval fails
- ICSI results are good; one study reported more miscarriage and stillbirth.	
- **Also know:** CUAVD (unilateral absence) is a Wolffian duct defect with ipsilateral renal agenesis.	

Diagram to draw: absent vas and seminal vesicle with intact testis and caput epididymis.

> **Key points to remember:**
> - CBAVD = mild CFTR mutation; test both partners.
> - Low-volume, acidic, fructose-negative azoospermia; normal FSH.
> - Renal ultrasound; MESA + ICSI.

Varicocele

*See the paper answer **Varicocele: pathophysiology, controversies and surgical management** (Section 10, Ch 87).*

Also, from the condensed book (not in the paper answer):

- Varicocele affects ~15–20% of all men versus 35–50% of infertile men (see Ch 70).

Antisperm antibodies and structural defects

- For antisperm antibodies, washing fails and steroid trials conflict, so use **ICSI**.
- Globozoospermia needs ICSI with or without oocyte activation, and ICSI is not recommended for macrocephalic multiflagellate sperm (AURKC).

10 · Lifestyle and obesity

- Obesity raises E2 and lowers LH and T, and worsens ART outcomes, so advise weight loss. Stop smoking, cannabis, excess alcohol and anabolic steroids, and avoid heat.

11 · Assisted reproduction

Sperm count decides the technique.

- †Choose by total motile sperm count (TMSC): IUI if processed TMSC is ≳5–10 million; IVF below that or after failed IUI; ICSI for severe oligozoospermia, retrieved sperm, antisperm antibodies or prior fertilisation failure.
- †Retrieval in OA is by percutaneous or microsurgical epididymal sperm aspiration (PESA/MESA) or testicular aspiration; in NOA it is micro-TESE.

70 Surgical Management of Male Infertility

1 · Principles

- In vitro fertilisation (IVF) with intracytoplasmic sperm injection (ICSI) treats even non-obstructive azoospermia (NOA), but it is costly, burdens the female partner and bypasses natural genetic barriers.
- Treating the man (reconstruction for obstructive azoospermia (OA), varicocelectomy) is safest and cheapest, and moves the couple **down the ladder**: donor sperm, then IVF/ICSI with testicular sperm, then IVF/ICSI with ejaculated sperm or intrauterine insemination (IUI), then natural conception.

2 · Surgical anatomy

Structure	Arteries (origin)
Testis	**Testicular** (aorta) — main; deferential (internal iliac/superior vesical); cremasteric (inferior epigastric; mainly tunica vaginalis)
Epididymis	Superior/medial epididymal (from testicular a.) supply the caput; inferior epididymal (from deferential a.) supplies the cauda; interconnected
Vas	Abdominal end: deferential only. Testicular end: deferential and inferior epididymal. **No supply from cremasteric or cord vessels**

- The testicular artery **exceeds deferential and cremasteric combined**; after vasectomy, preserve it at any later surgery.
- The epididymis can be dissected off the testis if the **superior epididymal artery is intact**; this gains **4–6 cm**.
- If the vas is blocked and devascularised at two sites, the segment between fibroses. So **never do two simultaneous vasovasostomies on one vas**; excise it and do one tension-free anastomosis.
- **Excurrent ducts**: 7–11 efferent ducts enter the caput, but a **single tubule** runs from the caput–corpus junction to the vas. So injury distal to the caput (even **one stitch**) obstructs the whole side.

3 · Diagnostic testis biopsy

LAQ Azoospermia: testicular biopsy and surgical sperm retrieval techniques

Paper answer · P1 · P2 · P3 · P4 · merged from 4

Asked as: P4 LAQ "Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival." · P2 SAQ "Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction)" (asked as "Sperm extraction in Obstructive azoospermia", "Testicular aspiration of sperms and assisted reproduction" and "Sperm Retrival techniques") · P1 SAQ "Testicular biopsy and TESA (testicular sperm aspiration)" (asked as "TESA" and "Testicular biopsy") · P3 SAQ "Sperm retrieval techniques"

Definitions

- **Azoospermia:** no sperm in the **centrifuged pellet** of the ejaculate on **at least two** semen analyses. Found in about 1% of men and 10–15% of infertile men.
- Not **cryptozoospermia** (rare sperm in pellet) or **aspermia** (no ejaculate).
- **OA (obstructive azoospermia):** no sperm in the centrifuged ejaculate despite **normal spermatogenesis**, from blockage of the ducts.
- **TESA (testicular sperm aspiration):** percutaneous needle aspiration of seminiferous tubules under local anaesthesia to retrieve sperm for **ICSI (intracytoplasmic sperm injection)**, which needs only one sperm per oocyte.

Classification

Type	Causes	Clues
Pre-testicular	Kallmann syndrome, pituitary disease, **exogenous testosterone**	Low FSH (follicle-stimulating hormone), LH (luteinising hormone)
Testicular: **NOA (non-obstructive azoospermia)**	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy, mumps	Small testes, high FSH
Post-testicular: **OA**	CBAVD (congenital bilateral absence of the vas deferens), vasectomy, epididymal or ejaculatory duct obstruction, hernia-repair injury	Normal testes and FSH
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Evaluation (brief)

- Absent vasa (CBAVD): CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners. Volume <1 mL: post-ejaculate urine, then TRUS (transrectal ultrasound).
- **FSH ≤7.6 IU/L and testis long axis >4.6 cm: obstruction likely;** FSH >7.6 and axis ≤4.6 cm: spermatogenic failure.
- **Karyotype and Y microdeletion** in NOA. **AZF (azoospermia factor) a or b complete deletion: no sperm, do not operate.**

Testicular biopsy

Indications

- Azoospermia with **normal testicular size, palpable vasa and normal FSH**: separates OA from NOA (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Since ICSI needs one sperm, biopsy is now **diagnostic and therapeutic**: plan it as TESE (testicular sperm extraction) and **cryopreserve** sperm.
- **Not needed:** CBAVD with normal FSH (spermatogenesis always present); high antisperm antibodies after iatrogenic vasal injury. Do not operate with complete AZFa or AZFb deletion.

Types

Type	Method	Use	Limits
Open biopsy	Small incision in the albuginea, pea-sized tissue	**Gold standard** for histology and retrieval	Open procedure
Percutaneous core biopsy	14 G biopsy gun under cord block	Fresh retrieval in OA; core of tissue	Blind; may injure epididymis or testicular artery; avoid in scarred scrotum
FNA (fine-needle aspiration), i.e. TESA	23 G needle, suction, several passes	OA retrieval; backup to epididymal methods	Few tubules, poor architecture; low yield in NOA
Microdissection biopsy (micro-TESE)	Operating microscope, plump tubules	NOA	Long operation

- **Also know:** multi-site FNA "mapping" in NOA.

Open biopsy technique

1. Local anaesthesia with **spermatic cord block** (or general/spinal); choose the larger testis.
2. 2 cm transverse scrotal incision; open tunica vaginalis.
3. Stay sutures; 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch prep** at once, then wet squash prep for sperm; if negative, contralateral biopsy.
7. Close albuginea and tunica vaginalis separately; no antibiotics.

Histology patterns

Pattern	Meaning
Normal spermatogenesis	Suggests obstruction
Hypospermatogenesis	All stages, fewer cells
Maturation arrest	Stops at a germ-cell stage
SCO (Sertoli cell–only)	No germ cells
Tubular sclerosis/hyalinisation	End-stage (e.g. Klinefelter)

Surgical sperm retrieval: indications

- **OA with normal spermatogenesis:** CBAVD, unreconstructable obstruction, failed or declined vasectomy reversal, back-up at reconstruction (vasoepididymostomy; frozen for IVF [in vitro fertilisation]), or failed epididymal aspiration.
- **NOA (testicular failure):** only testicular retrieval works; TESA gives poor yield; **micro-TESE (microdissection testicular sperm extraction)** preferred.
- **Anejaculation** (spinal cord injury): vibratory stimulation or electroejaculation first; retrieval after these fail.
- All retrieved sperm need **ICSI**, because sperm from obstructed systems have poor motility; cryopreserve spare sperm.

Techniques of sperm retrieval

Technique	Method / site	Best for	Merits, limits and notes
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput epididymis, local anaesthesia	OA	Simple; motile but few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened under microscope at 16–25×; micropipette by **capillary action**, never suction	**OA (best)**	Many (most) motile sperm, multiple frozen vials; needs microsurgical skill
TESA	Testis, 23 G needle with suction, many passes	OA	Least invasive; immature, immotile sperm, small yield; poor in NOA
Percutaneous core biopsy	14 G biopsy gun	OA	Fresh retrieval; core of tissue; blind; vessel injury
Conventional TESE	Open biopsy, one or more sites	OA, NOA	When epididymis is absent or empty; gold standard for tissue diagnosis; blind sampling
Micro-TESE (Schlegel 1999)	Testis bivalved (opened in an avascular plane); **larger, opaque tubules** picked under the microscope	**NOA (best)**	Most sperm, least tissue; sperm in about half; long operation
Vasal aspiration	Vasotomy, testicular end flushed	At reconstruction	

MESA technique

1. Testis delivered; epididymis inspected at 16–25×.
2. Dilated tubule opened with a microknife; fluid checked at once.
3. No motile sperm: close and **move proximally** (start distally), up to the efferent ductules (proximal sperm are more motile; the most motile are often in the efferent ductules).
4. Collect by **capillary action** into micropipettes, never suction; 4–12 pipettes.
5. **Cryopreserve** in several aliquots; close tubule with 10-0 suture.

TESA technique

1. Spermatic cord block with local anaesthetic.
2. Testis fixed; **23 G needle** (or angiocatheter) on a high-suction syringe.
3. Multiple passes in different directions draw tubules into the needle (often 10–20 passes needed; P1 table: 3–4 passes).
4. Immediate wet examination for sperm; repeat until sperm are found. **Cryopreserve** spare sperm.

TESA: advantages and limitations

- **Advantages:** office procedure, no microsurgery, cheap, repeatable.
- **Limitations:** few tubules with poor architecture (not diagnostic histology); small numbers of immature, immotile sperm; often too few to freeze; blind.

Micro-TESE logic

- Spermatogenesis in NOA is **focal**. Plump, larger tubules hold sperm; thin white ones are Sertoli cell–only.
- Embryologist examines tissue in theatre; map positive sites; if negative, explore the other testis.
- It removes the least tissue for the most sperm, and is better than conventional TESE.

Choice: epididymal sperm first (MESA) in OA; testicular (TESA/TESE) if epididymal sperm are absent or retrieval fails; micro-TESE in NOA.

Results

- OA: retrieval almost always succeeds; good pregnancy rates.
- NOA: micro-TESE finds sperm in **about half**; fine-needle aspiration yields far fewer sperm than micro-TESE. Retrieval depends on the cause:

Condition	Micro-TESE retrieval
Post-orchidopexy	74%
AZFc deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
SCO	30%
AZFa/AZFb deletion	**0%: do not operate**

- **Before retrieval in NOA:** stop testosterone; repair clinical varicocele; off-label clomiphene, anastrozole or hCG (human chorionic gonadotropin).

Assisted reproduction and use of retrieved sperm

- **IUI (intrauterine insemination):** washed sperm into the uterus; mild male factor.
- **IVF:** oocytes and sperm combined in the laboratory.
- **ICSI:** one sperm injected into each oocyte; essential for surgically retrieved (testicular) sperm (poor motility).
- Frozen retrieved sperm equal fresh, except possibly in spinal cord injury.
- NOA pregnancies: more miscarriage (sperm aneuploidy, DNA damage); genetic counselling. No sperm: donor sperm or adoption.
- ICSI bypasses natural selection: risk of transmitting genetic defects. Offer karyotype, Y-microdeletion and CFTR testing as indicated before ICSI.
- Female burden: ovarian stimulation, multiple pregnancy.

Complications

- **Haematoma** (commonest), pain, infection (rare), testicular vascular injury (TESA, core biopsy).
- **Inadvertent epididymal biopsy** (most serious: causes obstruction); epididymal obstruction (PESA).
- Testicular atrophy or transient fall in testosterone (TESE).

Also know

- Testicular sperm may beat ejaculated sperm when DNA fragmentation is high.

- Post-mortem retrieval needs prior consent.
- Microfluidic sperm sorting.

Diagrams to draw

- Testis and epididymis showing PESA, MESA and TESA sites; testis with 23 G needle on a suction syringe and multiple passes; flow to ICSI.
- Open biopsy: small albugineal window with extruding tubules snipped off.
- Micro-TESE: bivalved testis with plump (dilated) spermatogenic tubules beside thin Sertoli cell–only tubules.

Key points to remember:

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
- OA: normal FSH and testes; MESA is best (capillary action, never suction; move proximally): many motile sperm, multiple vials.
- TESA: 23 G needle, suction, local anaesthesia; for OA; TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best (sperm in about half); TESA yield poor.
- AZFa/AZFb deletion: 0% retrieval, do not operate.
- All surgically retrieved sperm need ICSI; cryopreserve extras.
- Haematoma commonest; epididymal biopsy most serious.

Also, from the condensed book (not in the paper answer):

- Biopsy is usually bilateral and planned as testicular sperm extraction (TESE), with an andrology laboratory on site for wet examination and freezing.

ALGORITHM

1. Give a cord block; choose the larger testis (opposite a varicocele).
2. Make a 2 cm incision, open the tunica vaginalis, place stay sutures.
3. Incise the albuginea 3–4 mm, using bipolar on crossing vessels.
4. Take a pea-sized sample with iris scissors (no forceps).
5. Do a touch prep, then a wet prep. If negative, then biopsy the other side.
6. Close the tunica vaginalis with continuous 5-0; no antibiotics.

- A non-absorbable suture marks a positive site for future TESE.

Complication	Point
Haematoma — most common	Prevent with magnification, bipolar and continuous tunica vaginalis closure
Epididymal biopsy — most serious	If that tubule held sperm, obstruction is now certain

4 · Vasography

- **Absolute indications (all three)**: azoospermia, complete spermatogenesis on biopsy and at least one palpable vas. Relative: severe oligospermia with normal biopsy, high sperm-bound antibodies, and low volume with poor motility (partial ejaculatory duct obstruction (EDO)).
- Do it **only at planned reconstruction**, because of scar and stricture. It cannot diagnose epididymal obstruction, the commonest non-vasectomy site.

ALGORITHM

1. High scrotal incision; find the vas at the straight–convoluted junction; hemivasotomy at 25×.
2. **Examine vasal fluid first.** If motile sperm, then barbotage and cryopreserve **before any dye**.
3. Pass a 24-G sheath into the abdominal end and inject 1 mL saline. If flow is easy, then it is patent.
4. Rarely, give dilute indigo carmine; blue urine confirms.
5. Use formal contrast only for obstruction above the internal ring.
6. Close with 10-0 mucosa and 8-0 muscularis.

Vasal fluid / vasogram	Meaning and action
No sperm, abdominal end patent	Epididymal obstruction: vasoepididymostomy
Copious fluid, many sperm	Vasal obstruction or EDO: locate it (vasogram, or 2-0 Prolene threaded down)
Block at ejaculatory ducts; both vasa fill from one side	EDO or midline cyst: transurethral resection of the ejaculatory ducts (TURED)
Vas ends blindly	Partial vasal agenesis: cystic fibrosis gene (CFTR) testing; retrieve and freeze
Inguinal block	Inguinal or crossed vasovasostomy
Many sperm, normal vasogram	Retrograde ejaculation, emission failure, aperistalsis

- Use **indigo carmine, not methylene blue**, which kills sperm (use it only after banking).
- Complications are **stricture** (repeated punctures, poor closure, non-water-soluble contrast), devascularisation, haematoma and granuloma.
- **Transrectal seminal vesiculography** aspirates dilated vesicles under transrectal ultrasound (TRUS); sperm there show a patent vas and epididymis, allowing TURED without scrotal exploration.

5 · Vasovasostomy (vasectomy reversal)

Evaluation

- Prior fertility proves spermatogenesis. Check follicle-stimulating hormone (FSH) if testes are small or soft; raised FSH means poorer prognosis.
- An indurated epididymis predicts vasoepididymostomy. A **sperm granuloma at the testicular end gives a better prognosis**, because it vents pressure, whatever the interval.

- Operate only if able to do vasoepididymostomy.

Approach and preparation

- The approach is scrotal for most and inguinal for hernia or orchidopexy obstruction. **Never strip the perivasal sheath**.
- **Length**, in escalating steps: dissect to the external then internal ring; free the convoluted vas (4–6 cm; **do not unravel**); reroute beneath the inguinal floor; take the epididymis off the testis. This bridges gaps up to **10 cm**.
- A healthy cut end has a white mucosal ring, soft muscularis and a bleeding edge. **If the muscularis is gritty or bleeding poor, recut.**
- At the testicular end **aspirate, never flush**. Flush saline into the abdominal end to prove patency.

Intraoperative fluid grading — the decision

Vasal fluid	Microscopy	Operation
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin	Sperm with tails	Vasovasostomy
Copious, creamy yellow	Many sperm heads	Vasovasostomy
Scant, granuloma present	Barbotage: sperm	Vasovasostomy
Thick white **toothpaste**	No sperm/heads	**Vasoepididymostomy**
Scant white thin, or dry without granuloma	No sperm/heads	**Vasoepididymostomy**
Any	Grape-like clusters of heads	**Vasoepididymostomy**

> **KEY POINT**
>
> Clear watery fluid still gets vasovasostomy: the epididymis has not blown out, and sperm usually return. Toothpaste fluid means secondary epididymal blow-out; a discrete level (dilated above, collapsed below) confirms it. When in doubt or inexperienced, do vasovasostomy.

Anastomotic principles

Principle	Reason
Accurate mucosa-to-mucosa	Testicular lumen is 2–5× the abdominal one
Leakproof	Sperm are antigenic; fluid lacks clotting factors, so the seal depends on mucosal sutures
Tension-free	Tension gives early sperm, then falling counts, then azoospermia
Good blood supply; healthy mucosa/muscularis; atraumatic	Recut until bleeding and soft; redo if back-walled or torn

- **Multilayer microdot**: six mapped dots guide **six 10-0 mucosal sutures**, then 9-0 deep muscularis, 9-0 adventitia and 8-0 sheath. The two-layer alternative is 10-0 mucosa and 8-0 full thickness.
- Reinforcing vasal sheath sutures (ReVas) relieve tension and raise patency.
- **Crossed vasovasostomy** suits unilateral inguinal obstruction with a contralateral atrophic testis, or one unreconstructable side.

- **Varicocele with reversal**: vasal vessels are compromised, so stage varicocelectomy ≥6 months later.

Post-operative care and outcomes

- Check semen at **6 weeks**, then 6 months if no pregnancy. Poor motility gets a 4-week prednisone course. Azoospermia at 6 months means redo.
- An anastomotic granuloma heralds obstruction. **Late stricture** shows as falling motility, then count. **Bank sperm once motile sperm appear.**
- With sperm in vasal fluid on at least one side, sperm return in virtually all. Pregnancy is about half, depending on vasectomy interval and female age.

6 · Vasoepididymostomy

- The epididymis is a single tubule ~200 µm wide and 12–15 feet long, with its thinnest wall at the caput. So a **more distal anastomosis is easier and gives better pregnancy**.
- **Indications**: reversal by the fluid criteria; or non-vasectomy OA with proven spermatogenesis, no vasal sperm and a patent vas and ejaculatory ducts.

ALGORITHM

1. Hemitransect the vas at the straight–convoluted junction; sample fluid; confirm patency.
2. Open the tunica vaginalis; choose a site **above** the obstruction where tubules are dilated. If unclear, then puncture distal to proximal until sperm appear.
3. Make a 3–4 mm tunic window.
4. Bring the vas through the tunica vaginalis, **fixed with 5-0 Prolene to the albuginea (tension relief)**.
5. Place posterior tunic–vas 8-0, then intussusception sutures, then outer tunic–vas adventitia.

Technique	For	Against
Longitudinal intussusception (LIVE) — used for all	Two 10-0 longitudinal sutures placed in the **still-distended** tubule; incise between; tubule pulled into vas; bloodless, leakproof	Fluid not checked before sutures set
End-to-side	Bloodless; fluid checked first	Suturing a collapsed tubule
End-to-end	Rapid level finding; bridges gap	Haemostasis hard; inferior epididymal supply lost

- **No simultaneous varicocelectomy**; stage it after 6 months.
- Late shutdown occurs, so **bank sperm during surgery and once they appear**. If it fails, redo or retrieve sperm for ICSI.

7 · TURED

*See the paper answer **Ejacultory duct obstruction** (Section 9, Ch 69).*

Also, from the condensed book (not in the paper answer):

- For retrograde ejaculation give pseudoephedrine 60 mg 90 min before, or use alkalinised urinary sperm.

ALGORITHM

1. Use a 24-Fr resectoscope with a finger in the rectum.
2. Resect the **verumontanum** to open the dilated orifices or cyst, sparing bladder neck, sphincter and rectum.
3. **Blue efflux confirms** adequacy; coagulate minimally.
4. Catheter overnight; 7 days of antibiotics.

8 · Sperm retrieval

*See the paper answer **Azoospermia: testicular biopsy and surgical sperm retrieval techniques** (Section 9, Ch 70).*

Also, from the condensed book (not in the paper answer):

- Microsurgical epididymal sperm aspiration (MESA) gives most motile sperm and multiple vials. **Microdissection TESE (micro-TESE) is best in NOA**.

KEY POINT

Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved.

ALGORITHM

1. Micro-TESE: general or regional anaesthesia.
2. At 10×, find the avascular plane on the anterior albuginea; make a wide equatorial incision and bivalve the testis.
3. Excise larger tubules; an andrologist in theatre examines them; map positives.
4. If negative, then explore the other testis.
5. Close the albuginea with 5-0 and the tunica vaginalis with 4-0.

9 · Cryopreservation

- Freeze at diagnostic biopsy, at every reconstruction (before any dye), and from the ejaculate once motile sperm return after reversal.

10 · Varicocelectomy

*See the paper answer **Varicocele: pathophysiology, controversies and surgical management** (Section 10, Ch 87).*

Also, from the condensed book (not in the paper answer):

- Conventional inguinal repair has hydrocele 3–15% and recurrence 5–15%, because it misses scrotal and periarterial veins.
- Ligate vasal veins only if >2.5 mm, keeping one set: **one intact deferential set guarantees venous return**.

ALGORITHM

1. Local anaesthesia with sedation; 2–3 cm incision over the external ring.
2. Deliver the cord on a Penrose, excluding the ilioinguinal nerve and genital branch.
3. **Map arteries with micro-Doppler before handling.**
4. Open the spermatic fasciae; work at 10–25× with 1% papaverine; loop the artery.
5. Ligate **all veins except vasal**; preserve lymphatics.
6. Deliver the testis; ligate external spermatic and gubernacular veins.
7. Do the impulse test. If there is no impulse, then a vein was missed.

- **End point**: only testicular and cremasteric arteries, lymphatics, and the vas with its vessels remain.

KEY POINT

Recurrence means a vein was missed; hydrocele, a lymphatic tied; atrophy, an artery tied. Microsurgery with testicular delivery addresses all three.

11 · Landmark trials

Trial	Design	Result	Take-home
Madgar (1995)	Randomised controlled trial (RCT): varicocelectomy vs no surgery in infertile men	1-year pregnancy 44% vs 10%	Repair of clinical varicocele with abnormal semen improves fertility

71 Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

1 · Definition and key concepts

- **Erectile dysfunction (ED)** is the consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction (American Urological Association (AUA) 2018).
- Erection needs **sinusoidal (trabecular) smooth muscle relaxation, arterial dilation and venous compression (veno-occlusion)**. Rigidity comes from trapping blood, not from inflow alone.
- Milestones MCQ: nitric oxide (NO) was identified in the 1990s, and oral phosphodiesterase type 5 inhibitors (PDE5Is) arrived in 1998.

2 · Functional anatomy

Corpora

- There are paired dorsal **corpora cavernosa** and a ventral **corpus spongiosum**, which surrounds the urethra and expands as the glans.
- The septum is **incomplete** in humans, so one injection fills both corpora.
- Sinusoidal oxygen tension (Po2) is venous when flaccid (**~35 mm Hg**) and arterial when erect (**~90 mm Hg**).

Tunica albuginea

- The **inner circular layer** gives off **intracavernous pillars** (absent at 5–7 o'clock). The **outer longitudinal layer** is thicker dorsally and absent in the ventral groove.
- The ventral tunica is ~0.8 mm against ~2.2 mm dorsally. So 5–7 o'clock is the weak zone for **penile fracture** and implant crossover or perforation.
- **Veno-occlusion**: emissary veins run obliquely between the layers and are compressed as the sinusoids stretch the tunica. Cavernous arteries escape because they lie in a periarterial sheath.
- The spongiosal tunica is thin, single-layered and absent in the glans, so spongiosum and glans are **low-pressure arteriovenous shunts** and never become rigid.

Arteries

- The internal iliac gives the **internal pudendal**, which becomes the common penile artery with three branches.
- The **cavernous** artery is the main corporal supply. Its **helicine arteries** are tortuous when flaccid and straight when erect, and open into sinusoids without capillaries.
- The **dorsal** artery fills the glans, and the **bulbourethral** artery supplies the bulb and spongiosum.
- **Accessory pudendal arteries** occur in ~one-third and may be dominant. Preserving them at radical prostatectomy speeds erectile recovery.

Veins

Sinusoids drain into the **subtunical venous plexus**, which exits through **emissary veins** in the tunica.

System	Drains	Destination
Deep dorsal vein (with or without circumflex and periurethral veins)	Glans and distal two-thirds of corpora	Periprostatic (Santorini) plexus
Cavernous and crural veins	Proximal (infrapubic) penis	Internal pudendal veins
Superficial dorsal vein	Skin, subcutaneous tissue	Saphenous veins

> **KEY POINT**
>
> Vascular anatomy varies widely between men, so the same "nerve-sparing" operation gives different erectile outcomes.

3 · Haemodynamics: phases of erection and detumescence

LAQ Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED

Paper answer · P1

Asked as:

- Write in detail about physiology of erection. How will you evaluate a diabetic with ED? — *Practice Set 5 Paper I Q2 (LAQ)*
- Neurotransmitters in erectile function — *Practice Set 12 Paper I Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(erection physiology also in Practice Set 5 Paper 1, Q2)

Introduction

- Erection is **cavernous smooth-muscle relaxation**: low cytosolic Ca^{2+} dilates helicine arteries and sinusoids; expanding sinusoids compress emissary veins (veno-occlusion).
- Pathways: parasympathetic **S2–S4** cavernous nerves (erection), sympathetic **T11–L2** (detumescence), pudendal (sensation, ischiocavernosus rigidity).

Functional anatomy

- **Corpora cavernosa:** sinusoids lined by endothelium, separated by **trabecular smooth muscle**. Flaccid tone depends on smooth-muscle contraction.
- **Tunica albuginea:** inner circular and outer longitudinal layers. **Emissary veins** pass obliquely between them and are compressed when the corpora expand: the **veno-occlusive mechanism**.
- **Arteries:** internal pudendal → common penile → **cavernous (helicine branches)**, dorsal and bulbourethral arteries. Accessory pudendal arteries in about one-third.
- Spongiosum and glans have a thin or absent tunica, so no veno-occlusion; they stay low-pressure.

Haemodynamic phases

Phase	Events	Intracavernous pressure
Flaccid	Smooth muscle contracted; low inflow	Low
Latent (filling)	Arterial dilation; **inflow peaks**	Barely rises
Tumescence	Sinusoids fill; venous outflow compressed	Rising

Phase	Events	Intracavernous pressure
Full erection	Maximal veno-occlusion	About 100 mmHg
Rigid	**Ischiocavernosus** contraction (pudendal nerve)	Several hundred mmHg
Detumescence	Initial, slow, fast phases	Falls to baseline

Molecular mechanism and peripheral neurotransmitters

- **Relaxation (erection):** NO (nitric oxide) from **nNOS (neuronal nitric oxide synthase)** in cavernous nerves initiates erection; endothelial NOS (eNOS) sustains it.
- NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → protein kinase G (PKG) → K^+ channels open, Ca^{2+} channels close, Ca^{2+} sequestered → **low cytosolic Ca^{2+}** → relaxation.
- Second route: PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide) → **cAMP (cyclic adenosine monophosphate)** → protein kinase A.
- **PDE5 (phosphodiesterase type 5)** degrades (breaks down) cGMP and ends relaxation (ends / terminates erection); PDE5 inhibitors block it.

Peripheral pro-erectile transmitters

Transmitter	Source / mechanism
NO (nitric oxide)	Principal. **nNOS (neuronal NOS)** initiates; **eNOS (endothelial)** sustains. NO → soluble guanylyl cyclase → **cGMP** → PKG → K^+ channels open, Ca^{2+} channels close → relaxation
ACh (acetylcholine)	Indirect: endothelial NO release, inhibits adrenergic nerves; atropine does not block erection
PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide), CGRP (calcitonin gene-related peptide)	**cAMP** → PKA (protein kinase A); VIP role minimal in man
CO, H2S, EDHF (endothelium-derived hyperpolarising factor)	Gaseous/endothelial relaxants via cGMP or K^+ channels

- O2 is a NOS substrate: low flaccid Po2 (~35 mmHg) limits NO; erect Po2 ~90 mmHg.
- **Also know:** carbon monoxide and hydrogen sulfide also raise cGMP; acetylcholine acts indirectly through endothelial NO (atropine does not block erection).

Anti-erectile transmitters (flaccidity and detumescence)

- **Contraction (flaccidity): noradrenaline (α1)** (principal), endothelin-1, angiotensin II → raised Ca^{2+} → calmodulin → myosin light-chain kinase → contraction.
- **Noradrenaline (α1)**: principal; α2 prejunctional reduces noradrenaline and NO release.
- **Endothelin-1** (ET-A), angiotensin II (AT1), PGF2α, thromboxane A2: Gq → IP3 → Ca^{2+} rise → Ca^{2+}–calmodulin → MLC (myosin light-chain) kinase → contraction.
- **Rho-kinase** inhibits myosin (MLC) phosphatase (calcium sensitisation) and maintains (sustaining) tone.
- **Detumescence:** NO release stops, PDE5 degrades cGMP and sympathetic discharge contracts the smooth muscle, so emissary veins reopen.
- Gap junctions (connexin-43) synchronise the smooth muscle.

Neural control

Pathway	Origin	Role
Parasympathetic	**S2–S4** → pelvic plexus → cavernous nerves	Erection
Sympathetic	**T11–L2** → hypogastric nerves	Detumescence, emission
Somatic	Pudendal nerve (Onuf nucleus)	Penile sensation; ischiocavernosus rigidity

- **Three types: psychogenic** (brain, chiefly by reducing sympathetic tone), **reflexogenic** (genital touch, sacral arc; preserved in upper cord injury), **nocturnal** (REM [rapid eye movement] sleep).
- **Central:** MPOA (medial preoptic area) and **PVN (paraventricular nucleus)**. Dopamine and oxytocin promote; serotonin, prolactin, GABA (γ-aminobutyric acid) and opioids inhibit.

Central neurotransmitters (PVN [paraventricular nucleus], MPOA [medial preoptic area])

Promote	Inhibit
Dopamine (D2 in PVN; apomorphine)	**Serotonin** (5-HT1A, 5-HT2; but 5-HT2C pro-erectile)
Oxytocin	GABA (γ-aminobutyric acid)
Central NO	Opioids (high dose), cannabinoids
Central noradrenaline; melanocortins (MC4R)	**Prolactin**

- Dopamine and oxytocin converge on NO in the PVN. Noradrenaline is pro-sexual centrally but anti-erectile peripherally.

Clinical relevance

- PDE5 inhibitors, intracavernosal alprostadil (cAMP), papaverine, phenylephrine for priapism (α1), SSRIs (selective serotonin reuptake inhibitors) cause dysfunction, hyperprolactinaemia lowers libido.

Diagram to draw: NO–cGMP pathway in the cavernous smooth-muscle cell.

Evaluation of a diabetic with ED (erectile dysfunction)

Why diabetics: ED is commoner, earlier and more severe in DM (diabetes mellitus). Mechanisms: endothelial NO loss, autonomic neuropathy, arteriopathy, smooth-muscle loss and fibrosis (veno-occlusive failure), AGEs (advanced glycation end products) and low testosterone. ED is a **sentinel of cardiovascular disease**.

1. History

- Onset, duration (≥3 months), severity, morning erections, libido, ejaculation; partner.
- **IIEF-5 (International Index of Erectile Function, SHIM):** 5–7 severe, 8–11 moderate, 12–16 mild-to-moderate, 17–21 mild, 22–25 no ED.
- Diabetes: type, duration, **glycaemic control**, complications (retinopathy, nephropathy, neuropathy, foot disease), autonomic symptoms (postural giddiness, gastroparesis, bladder dysfunction).
- Cardiovascular: angina, exercise tolerance, smoking, hypertension, dyslipidaemia, nitrates.
- Drugs: antihypertensives, antidepressants. Depression screen.

2. Examination

- BMI, waist, blood pressure, peripheral pulses.

- Genitalia: Peyronie plaques, testicular size, gynaecomastia.
- Neurological: genital sensation, **bulbocavernosus reflex**, foot sensation. Penile neuropathy can exist without foot neuropathy.

3. Laboratory

- **HbA1c (glycated haemoglobin)** and fasting glucose, **lipid profile**, creatinine/eGFR (estimated glomerular filtration rate), urine albumin.
- **Morning total testosterone**; if low, repeat with free testosterone, LH (luteinising hormone), prolactin. TSH (thyroid-stimulating hormone) if indicated.

4. Cardiovascular risk stratification (Princeton Consensus)

- **High risk** (unstable angina, recent infarction, uncontrolled hypertension, some arrhythmias): defer sexual activity and treatment; cardiology referral. **Indeterminate:** cardiac assessment (e.g. stress test), then restratify. **Low:** start treatment.

5. Specialised tests (selected cases)

- **CIS (combined intracavernous injection and stimulation):** alprostadil 10–20 µg; a rigid erection excludes major vascular disease.
- **Penile duplex Doppler:** PSV (peak systolic velocity) **<25 cm/s = arterial insufficiency** (>35 normal); EDV (end-diastolic velocity) **>5 cm/s** and RI (resistive index) **<0.75 = veno-occlusive dysfunction**.
- **NPTR (nocturnal penile tumescence and rigidity), RigiScan:** separates psychogenic from organic ED.
- **Neurophysiological:** bulbocavernosus reflex latency, dorsal nerve conduction velocity, biothesiometry.
- DICC (dynamic infusion cavernosometry and cavernosography) and angiography: rarely, as diffuse diabetic vascular disease is not suited to revascularisation.

```
Diabetic with ED → history + IIEF-5 + examination → HbA1c, lipids,
testosterone, renal function → Princeton cardiac risk → treat risk factors →
PDE5 inhibitor trial → failure → duplex/CIS → ICI or prosthesis
```

> **Key points to remember:**
> - nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
> - Noradrenaline α1 is the main anti-erectile transmitter.
> - Dopamine/oxytocin promote; serotonin/prolactin inhibit.
> - Rigid phase: ischiocavernosus, suprasystolic pressure.
> - Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
> - Diabetic ED: check HbA1c, testosterone and cardiac risk.

Also, from the condensed book (not in the paper answer):

- Full-erection intracavernous pressure (ICP) plateaus at **~100 mm Hg**; the rigid phase is suprasystolic, driven by the pudendal nerve via ischiocavernosus.
- **Trap**: arterial inflow peaks in the latent phase, not at rigidity. This is why peak systolic velocity (PSV) on Doppler is measured early.
- Detumescence has three phases: initial (transient ICP rise), slow (veins gradually reopen) and fast (outflow restored).

4 · Neuroanatomy

Peripheral and spinal

	Sympathetic	Parasympathetic	Somatic
Origin	**T11–L2** (often T10–T12)	**S2–S4** intermediolateral columns	S2–S4; motor from **Onuf nucleus**
Route	Sympathetic chain, superior hypogastric plexus, hypogastric nerves, then pelvic plexus	Pelvic nerves (nervi erigentes) to the pelvic plexus	Pudendal nerve; afferent via dorsal nerve of penis
Role	**Detumescence**; emission	**Erection** (cavernous nerves)	Sensation; ischiocavernosus (rigidity), bulbocavernosus (ejaculation)

- **Cavernous nerves** arise from the pelvic plexus, carry both autonomic types and run posterolateral to the prostate. Pelvic ganglia lie along the bladder–prostate junction and seminal vesicles, so they are vulnerable in radical pelvic surgery.
- The **bulbocavernosus reflex** (glans squeeze contracts bulbocavernosus) has limited value in ED work-up.

Three types of erection

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Psychogenic	Audiovisual, fantasy	Brain modulates spinal centres, chiefly by suppressing sympathetic tone	May persist with sacral lesions; absent with lesions above T9
Reflexogenic	Genital touch	Sacral reflex arc to the cavernous nerves	Preserved in upper cord lesions
Nocturnal	Rapid eye movement (REM) sleep	Cholinergic pontine tegmentum on; locus coeruleus (noradrenergic) and raphe (serotonergic) silent	—

Supraspinal (central)

- The **medial preoptic area (MPOA)** is dopaminergic and integrates hormonal and sensory cues. The **paraventricular nucleus (PVN)** sends pro-erectile **oxytocin** neurons to the lumbosacral nuclei, and the nucleus paragigantocellularis is the **serotonergic brake**.

Central neurotransmitters

*See the paper answer **Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED** (Section 9, Ch 71).*

Also, from the condensed book (not in the paper answer):

- Apomorphine is a D1/D2 agonist acting in the PVN, which releases oxytocin and then NO. Melanocortins act on the MC4 receptor.
- Serotonin 5-HT1A and 5-HT2 receptors inhibit erection but facilitate ejaculation.

> **KEY POINT**
>
> NO synthase (NOS) inhibitors block both apomorphine- and oxytocin-induced erections, so NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally.

5 · Molecular mechanisms of smooth muscle tone

Every erectogenic drug acts on one of these pathways.

Contraction (flaccid state)

- Corporal smooth muscle is **contracted at rest** through myogenic tone, adrenergic tone and endothelial constrictors.
- **Calcium pathway**: noradrenaline (α1), endothelin-1 (ETA), angiotensin II (AT1), PGF2α and thromboxane A2 act via Gq and phospholipase C to form inositol trisphosphate (IP3), raising Ca^{2+}. Ca^{2+}–calmodulin then activates **myosin light chain (MLC) kinase**, which phosphorylates MLC20.
- **Ca^{2+} sensitisation by RhoA/Rho-kinase**: it inhibits **MLC phosphatase (MLCP)**, giving sustained contraction without a further Ca^{2+} rise.
- Rho-kinase is raised in diabetes mellitus (DM), hypertension, hypercholesterolaemia, hypogonadism, ageing and nerve injury, and it also drives fibrosis.

Relaxation (erection)

- **NO–cGMP**: NOS converts L-arginine and O2 into NO and citrulline. **Neuronal NOS initiates and endothelial NOS (eNOS) sustains** erection.
- NO activates soluble guanylyl cyclase to make cyclic GMP (cGMP), which activates **protein kinase G (PKG-I)**. PKG lowers intracellular Ca^{2+}, inhibits Rho-kinase and activates MLCP, so the muscle relaxes. PKG-I knockout mice cannot erect, so PKG is the bottleneck.
- **cAMP pathway**: PGE1, vasoactive intestinal peptide (VIP) and calcitonin gene-related peptide (CGRP) act via Gs to raise cyclic AMP (cAMP), which activates protein kinase A (PKA). It is **NO-independent**, which is why intracavernosal alprostadil works when nerves or endothelium fail.
- **PDE5 (cGMP-specific) is the principal cavernous PDE**. PDE6 is retinal (visual side effects); PDE2, 3 and 4 degrade cAMP. Papaverine is nonspecific.
- **Gap junctions (connexin-43)** synchronise relaxation, so a local input gives a uniform erection.
- Oxygen acts as a switch: erect Po2 >50–60 mm Hg permits full NOS activity, while low flaccid Po2 suppresses it.
- Excess **adenosine** with increased A2B signalling causes prolonged erection in adenosine deaminase-deficient and sickle cell mice, the **priapism models**.
- Acetylcholine acts indirectly (endothelial NO, presynaptic adrenergic inhibition), so **atropine does not abolish** neural erection MCQ.

Second messenger	Activators	Kinase	Degraded by	Drug
cGMP	NO, CO, H2S, natriuretic peptides	PKG-I	PDE5	PDE5Is

Second messenger	Activators	Kinase	Degraded by	Drug
cAMP	PGE1, VIP, CGRP, adenosine (A2)	PKA	PDE2, 3, 4	Alprostadil; papaverine (both)

Detumescence

NO release stops and PDE5 hydrolyses cGMP. Sympathetic noradrenaline (α1; the principal transmitter of flaccidity), endothelin-1 and Rho-kinase then contract the trabeculae, so the veins reopen.

> **KEY POINT**
>
> Corporal tone is a tug-of-war over one phosphorylation site (MLC20): Ca^{2+}–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" means the phosphatase.

6 · Classification of ED

Approach	Categories
Neurovascular mechanism	Failure to **initiate** (neurogenic, psychogenic, endocrine) · to **fill** (arterial) · to **store** (veno-occlusive)
Aetiological	Psychogenic · neurogenic · hormonal · arteriogenic · cavernosal (venogenic) · drug-induced · systemic disease · mixed (commonest)

Most organic ED has a psychogenic component.

7 · Risk factors and endothelial dysfunction

ED often signals vascular disease before the heart does.

- **Age** is the strongest demographic predictor, but DM, heart disease and hypertension carry more risk (Massachusetts Male Aging Study (MMAS)).
- **Endothelial dysfunction**: oxidative stress causes **NOS uncoupling**, while arginase and asymmetric dimethylarginine (ADMA) further lower NO.
- **Artery-size hypothesis**: smaller penile arteries become symptomatic first. ED preceded coronary artery disease (CAD) by a mean of **~2 years**, so ED is a **sentinel** and needs cardiac risk stratification (see Ch 72).
- The common end-path is raised Rho-kinase and transforming growth factor-β (TGF-β), which lower the smooth muscle-to-collagen ratio and cause veno-occlusive failure.

8 · Pathophysiology by type

Psychogenic

- The brain inhibits the spinal centres, and excess sympathetic tone contracts the muscle. Causes include performance anxiety, partner conflict, depression and post-traumatic stress disorder (PTSD).

Neurogenic

- **Brain** causes include Parkinson disease, stroke, tumours and dementia. **Cord** causes include SCI (commonest), spina bifida, disc disease and multiple sclerosis (MS).

Complete lesion	Reflexogenic erection	Psychogenic erection
Upper cord (above sacral centre)	Preserved in **~95%**; brief, need continuous stimulation	Lost
Lower cord (sacral)	Preserved in only **~25%**	May occur (thoracolumbar pathway); marginal tumescence

- **Iatrogenic**: radical prostatectomy, cystoprostatectomy, abdominoperineal resection (APR) and pelvic radiation. Most men have at least a transient decline despite nerve sparing, worse with baseline ED and age >60.
- Cavernous nerve injury causes apoptosis and fibrosis, leading to venous leak. This is the rationale for early penile rehabilitation.
- **Pelvic fracture** injures the cavernous nerves or pudendal artery. With posterior urethral injury, early realignment preserves potency better than delayed anastomosis.

Hormonal

- **Testosterone (T)** governs **libido** more than mechanics, but it maintains NOS, PDE5 and smooth muscle. Low libido is the commonest sexual symptom of low T.
- AUA: measure **morning, fasting** T in all men with ED. Low T is **<300 ng/dL**; treat only with ≥2 low levels and symptoms. Nocturnal erections need ~200 ng/dL.
- Nocturnal erections are androgen-dependent, but erections to visual stimuli are largely independent MCQ. Dihydrotestosterone (DHT) is the key penile androgen.
- **Hyperprolactinaemia** (prolactinoma, antipsychotics) suppresses gonadotrophin-releasing hormone (GnRH), causing low T and low libido.
- A low T:oestradiol ratio (obesity), hyperthyroidism and hypothyroidism also cause ED.

Vasculogenic – arteriogenic

- Atherosclerotic or traumatic occlusion of the hypogastric–pudendal–cavernous tree lowers perfusion pressure, giving a slower, softer erection.
- Risks are hypertension, hyperlipidaemia, smoking, DM, obesity, pelvic irradiation and blunt perineal trauma. Trauma causes focal stenosis in young men, who are revascularisation candidates.
- Chronic hypoxia raises TGF-β1 and collagen, so "inflow failure becomes storage failure".
- In hypertension, angiotensin receptor blockers (ARBs) and nebivolol protect penile tissue.

Vasculogenic – veno-occlusive (venous leak)

- Causes are **tunical** (ageing, DM, Peyronie disease, penile fracture), **fibroelastic trabecular change** preventing sinusoidal expansion, **insufficient relaxation** (anxiety, adrenergic tone, endothelial dysfunction), and **acquired shunts** after priapism shunt surgery. Congenital ectopic or crural veins cause primary ED.

> **KEY POINT**
>
> Veno-occlusion is an emergent property of healthy tissue (elastic tunica and fully relaxed trabeculae). So most "venous leak" is really cavernosal smooth muscle disease, which is why venous ligation surgery fails and is investigational.

Metabolic and systemic disease

- **DM** causes the earliest, most severe ED through arteriopathy, nitrergic neuropathy, endothelial dysfunction, advanced glycation end products (AGEs), raised Rho-kinase and hypogonadism. **Key**: the NO source is damaged but the response to exogenous NO is preserved, so PDE5Is work until fibrosis sets in.
- **Metabolic syndrome and obesity** improve with weight loss, Mediterranean diet, exercise and bariatric surgery. In **hyperlipidaemia**, oxidised low-density lipoprotein (LDL) inhibits NO, and statins are linked to better International Index of Erectile Function (IIEF) scores.
- **Chronic kidney disease**: up to **80%** of men with end-stage renal disease have ED, and it persists in ~half after transplant.
- Other associations include HIV, cirrhosis, chronic obstructive pulmonary disease, obstructive sleep apnoea and lower urinary tract symptoms.

Drug-induced

Class	Cause ED	Mechanism / note	Alternative
Antihypertensives	**Thiazides**, **nonselective β-blockers**, clonidine	Thiazide – unknown; β-blocker – prejunctional α2	ACE inhibitors, ARBs, nebivolol, α1-blockers (ejaculatory dysfunction), calcium channel blockers
Antidepressants	Selective serotonin reuptake inhibitors (SSRIs; paroxetine worst), tricyclics	Spinal 5-HT2/5-HT3 inhibition, less dopamine and NO; post-SSRI dysfunction may persist	Bupropion, mirtazapine, nefazodone, buspirone; add a PDE5I
Antipsychotics	Typical > atypical	Dopamine block causes hyperprolactinaemia	Newer agents
Antiandrogens	LHRH agonists/antagonists, androgen receptor antagonists, cyproterone, **5α-reductase inhibitors (5ARIs)**	5ARI: least T effect yet persistent ED after stopping	None
Recreational	Tobacco (dose–response, ≥20 pack-years), heavy alcohol, opioids	Endothelial damage, neuropathy, hypogonadism	Stop or moderate
Others	Cimetidine, spironolactone, carvedilol	—	—

Trazodone causes priapism, and among anticonvulsants carbamazepine and phenytoin are worse than lamotrigine.

Primary (lifelong) ED

- It is usually psychogenic, but exclude hypoplastic cavernous arteries, congenital venous leak, corporal atresia and micropenis.

KEY POINT

Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction and TGF-β-driven fibrosis.

72 Evaluation and Management of Erectile Dysfunction

1 · Definition and significance

- **Erectile dysfunction (ED)** (American Urological Association, AUA): a consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction. Minimum **3 months**, except after trauma or surgery.
- About **20%** of adult men have ED (up to 15% at 40–49 y; up to 40% at 60–69 y).
- **ED predicts cardiovascular disease (CVD)** through shared endothelial dysfunction, as strongly as smoking or a family history of myocardial infarction (MI). Younger onset means greater cardiovascular (CV) risk.

Independent association (multivariate)	Risk factors
Strongest	Antidepressant use, then antihypertensive use
Strong	Diabetes mellitus (DM), the strongest disease; obstructive urinary symptoms
Moderate	Hypertension, prostate enlargement, smoking, body mass index (BMI) ≥30, inactivity
Weak/null after adjustment	CVD, hypercholesterolaemia

- Drugs cause up to 25% of ED (see Ch 71); opioids double the risk.

2 · Management principles

- **Shared decision making** (AUA 2018): all non-contraindicated options are offered **in parallel, not ranked**, and the patient may choose any first. They are phosphodiesterase type 5 inhibitors (PDE5Is), the vacuum erection device (VED), intraurethral (IU) alprostadil, intracavernosal injection (ICI) and prosthesis.
- In testosterone (T) deficiency (symptoms and total T below 300 ng/dL), restoring T probably improves the other ED treatments.

Cardiac risk assessment – Princeton Consensus

*See the paper answer **Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED** (Section 9, Ch 71).*

Also, from the condensed book (not in the paper answer):

- Every man with ED is at potential CV risk; sex equals moderate exertion. **Refer** after failed therapy, for a young trauma patient, for deformity, in medicolegal cases, and for complex endocrine, psychiatric or vascular problems.

3 · Basic evaluation

History

- Ask about onset, pattern, morning erections, libido, ejaculation, curvature and prior treatment.
- Ask about DM, CVD, neurological disease, pelvic surgery or radiation, trauma and drugs. Screen for depression.

Psychogenic	Organic
Sudden onset	Gradual onset
Complete, immediate loss	Incremental progression
Situational	Global
Waking/nocturnal erections present	Poor or absent

> **KEY POINT**
>
> Preserved nocturnal/morning erections prove an intact neurovascular axis, the strongest historical pointer to psychogenic ED.

Questionnaires

- The **International Index of Erectile Function (IIEF)** has 15 items in 5 domains: erectile function, orgasm, desire, intercourse satisfaction and overall satisfaction.

*See the paper answer **Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED** (Section 9, Ch 71).*

Also, from the condensed book (not in the paper answer):

- The Sexual Health Inventory for Men (SHIM) has 5 items scored 1–5 over the past 6 months. Questionnaires grade severity and response but do **not** identify the cause.

Physical examination

- Look for hypogonadism (testes, gynaecomastia), vascular disease (blood pressure (BP), obesity, pulses), neuropathy (genital sensation, bulbocavernosus reflex) and structural problems (plaque, micropenis, chordee).

Laboratory tests

- **Recommended**: fasting glucose or glycated haemoglobin (HbA1c), lipids, serum chemistry, full blood count and **morning (7–11 am) total testosterone**.
- Repeat a low T, because about one-third of mildly low values normalise, and add luteinising hormone (LH) and prolactin. Check prostate-specific antigen (PSA) if prostate disease is suspected or before T therapy.

> **KEY POINT**
>
> The basic work-up finds the disease behind the ED (undiagnosed DM, dyslipidaemia, hypogonadism, CV risk); it does not prove ED.

4 · Hormonal evaluation

*See the paper answer **Male menopause (andropause, late-onset hypogonadism)** (Section 9, Ch 68).*

Also, from the condensed book (not in the paper answer):

- The AUA threshold is **300 ng/dL (10.4 nmol/L)**; above about 230 ng/dL, sexual function correlates only weakly with T. Order a pituitary magnetic resonance imaging (MRI) scan if T is **<150 ng/dL**.
- Hyperprolactinaemia is caused by antipsychotics, tricyclics, opiates, prolactinoma, hypothyroidism and renal failure.

5 · Specialised testing

These tests are for young trauma patients, primary ED, surgical candidates, medicolegal cases and failed therapy.

Test	Line	Answers	Indication
Combined intracavernosal injection and stimulation (CIS)	1st	Can the vessels produce a rigid erection?	Screening
Penile duplex Doppler	2nd	Arterial or veno-occlusive? Asymmetry?	Most reliable, least invasive
DICC (dynamic infusion cavernosometry/cavernosography)	3rd	Is there a venous leak, and where?	Trauma, lifelong ED, before venous surgery
Selective internal pudendal arteriography	3rd	Site of arterial lesion; is the inferior epigastric donor usable?	Young trauma patient before revascularisation

CIS test

- Inject alprostadil 10–20 µg or Bimix/Trimix 0.3 mL into the lateral base, then compress, stimulate and rate rigidity. A rigid, sustained erection means normal haemodynamics; anxiety or a low dose gives false negatives, so redose.
- If there is no detumescence by about 1 h, inject diluted phenylephrine (500 µg/mL) every 3–5 min.

Penile duplex Doppler

- Measure cavernous artery peak systolic velocity (PSV) and end-diastolic velocity (EDV) at the base every 5 min up to 20 min after ICI.

Parameter	Arterial insufficiency	Veno-occlusive dysfunction
PSV	**<25 cm/s** (25–35 borderline; >35 normal)	Stays >25 cm/s
EDV	—	**>5 cm/s**
Resistive index (RI) = (PSV − EDV)/PSV	—	**<0.75** (normal >0.9)
Other	Acceleration time >122 ms; side-to-side PSV difference >10 cm/s; artery dilation <75%	Rapid detumescence
When assessed	Within 5 min	15–20 min (needs full relaxation)

> **KEY POINT**
>
> Venous leak can be diagnosed only with adequate inflow and full smooth muscle relaxation; an anxious patient mimics veno-occlusive dysfunction, so redose first. Anatomical variants can falsely suggest arteriogenic ED.

DICC, NPTR and neurological tests

- DICC measures flow-to-maintain, pressure decay and cavernous artery systolic occlusion pressure after a vasodilator; cavernosography localises the leak.
- Nocturnal penile tumescence and rigidity (NPTR) testing in a sleep lab is the standard. The home **RigiScan** measures **radial** rigidity; normal is 4–5 erections a night, ≥60% rigid for

≥10 min. It suits obscure cause, medicolegal cases and suspected psychogenic ED.

6 · Treatment

ALGORITHM

1. Treat reversible causes (lifestyle, drugs, hormones, psychosocial) and stratify CV risk.
2. Shared decision: PDE5I, VED, IU alprostadil or ICI. Low-intensity extracorporeal shock wave therapy (Li-ESWT) is investigational.
3. If the response is inadequate, then optimise, combine or switch.
4. If still inadequate, then offer a penile prosthesis (see Ch 75).

Lifestyle and risk-factor modification

- Stopping smoking, exercise, weight loss and a Mediterranean diet improve function; control DM, lipids and BP.
- **Change medication**: replace thiazides or β-blockers with calcium channel blockers, angiotensin-converting enzyme inhibitors or angiotensin receptor blockers. For a selective serotonin reuptake inhibitor (SSRI), switch to bupropion or mirtazapine, or add a PDE5I.
- **Psychosexual therapy** works best combined with medical therapy.

Hormonal therapy

- Give T only for confirmed deficiency, as a ~3-month trial; it improves libido more than erection and is synergistic with PDE5Is.
- Monitor digital rectal examination, PSA, haematocrit, liver tests and lipids at baseline, 3–6 months and yearly. The main adverse effect (AE) is erythrocytosis; randomised trials show no causal CV link.
- Preparations include gels (transfer risk), intramuscular enanthate/cypionate (200–250 mg every 2–3 weeks) and undecanoate depot (750 mg at 0 and 4 weeks, then every 10 weeks). Avoid methyltestosterone because it is hepatotoxic.
- T does not correct hyperprolactinaemia: stop the drug, give bromocriptine for prolactinoma, and operate for failure or visual compromise.

PDE5 inhibitors (first-line oral)

They block cyclic guanosine monophosphate (cGMP) breakdown, so they **enhance, not initiate**, erection; sexual stimulation is obligatory.

	Sildenafil	**Vardenafil**	**Tadalafil**	**Avanafil**
Doses (mg)	25, 50, 100	5, 10, 20	5, 10, 20 (2.5/5 daily)	50, 100, 200
Time to peak, T (h) max	0.8 (~1 h before)	0.7–0.9 (~1 h)	**2** (~2 h)	**0.3–0.5** (~30 min)
Onset (min)	15–60	15–60	15–120	15–60
Half-life (h)	3–5	4–5	**17.5**	3–5
Fatty food	Lowers absorption	Lowers absorption	**No effect**	Lowers absorption

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Characteristic AE	Visual (blue tinge; PDE6), flushing	Visual, flushing; **QT prolongation, so avoid with class 1A/3 antiarrhythmics** and congenital long QT	**Back pain/myalgia** (PDE11)	Fewest visual effects
Nitrates	**Contraindicated (all)**			

- Headache is the commonest AE. Efficacy is equal across agents: about 70% achieve intercourse, and 40–50% in DM and after prostatectomy.
- **Nitrates** in any form, including amyl nitrite "poppers", cause profound hypotension. If angina occurs, give no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil). There is no antidote; treat hypotension with head-down tilt, fluids and phenylephrine.
- Use caution with α-blockers (stabilise first, start low) and aortic stenosis; CYP3A4 inhibitors need a lower dose.
- They are not studied after MI or stroke within 6 months, with BP <90/50 or >170/100, or in retinitis pigmentosa. For sudden visual loss, stop the drug and seek care.
- **Before labelling failure**, check stimulation, timing, food, maximum dose and **up to 9–10 attempts**; correct hypogonadism, glycaemia and lipids; then switch agent or try daily tadalafil.
- **Penile rehabilitation** after prostatectomy is unproven, since REINVENT and other nightly-dosing trials were negative.

Other oral agents (yohimbine, apomorphine, trazodone) have little or no proven benefit.

Vacuum erection device

- Negative pressure fills the corpora and a base ring traps the blood; **remove it within 30 min**. The erection is hinged at the base, cool and cyanotic.
- Up to 90% get an erection but only 30–70% are satisfied; it suits older men in stable relationships. Take care with anticoagulants.

Intraurethral alprostadil (MUSE)

- A prostaglandin E1 (PGE1) pellet of 125–1000 µg is inserted into the urethra after voiding. About 50% respond, so it is a niche option: PDE5I failure after nerve injury, add-on to a PDE5I, or soft glans after prosthesis.
- Urethral pain affects about one-third; use a condom if the partner is pregnant.

Intracavernosal injection

ICI acts directly on smooth muscle, **independent of nerves and nitric oxide (NO)**, so it works when PDE5Is fail (neurogenic, post-prostatectomy, DM).

Agent	Mechanism	Notes
Alprostadil (PGE1) 5–40 µg	Adenylate cyclase raises cyclic AMP (cAMP)	Only FDA-approved ICI; ~70% efficacy; metabolised locally; **penile pain** commonest AE; least priapism and fibrosis
Papaverine	Nonspecific PDE inhibitor; Ca^{2+} channel block	Cheap; raises liver enzymes, causes priapism and fibrosis, so not used alone

Agent	Mechanism	Notes
Phentolamine	α-antagonist	Adjunct only; hypotension, reflex tachycardia
Trimix (alprostadil 10 µg, papaverine 30 mg and phentolamine 1 mg per mL)	Synergistic	≈90%; rescues monotherapy failures; less pain
Bimix	Papaverine with phentolamine, or alprostadil with phentolamine	≈90%

- Titrate in the office from a low dose, since neurogenic ED is very sensitive, and inject the lateral shaft.
- **Priapism** is highest with papaverine mixes and in neurogenic or young men; the patient must seek care beyond 4 h. Long-term use causes fibrosis and curvature.
- Contraindications: psychological instability, priapism risk (sickle cell), severe coagulopathy, unstable CVD, poor dexterity (the partner may inject), and monoamine oxidase (MAO) inhibitors, because phenylephrine reversal risks a hypertensive crisis.

Topical agents (alprostadil cream, the over-the-counter MD3000 gel) give modest benefit.

Restorative and investigational

Therapy	Proposed mechanism	Evidence/status
Li-ESWT	New vessels; nerve, endothelial, smooth muscle regeneration	RCT meta-analysis: IIEF better than sham in **mild–moderate vasculogenic** ED, the best evidence; still investigational (Sexual Medicine Society of North America: research only)
Platelet-rich plasma	Growth factors	Conflicting RCTs
Stem cells	Regeneration	Uncontrolled series only
Botulinum toxin ICI	Blocks sympathetic noradrenaline	Short-term benefit in refractory ED

Combination therapy

Monotherapy fails in nearly 40%. Useful pairs are daily tadalafil with on-demand sildenafil, a PDE5I with T, IU, ICI or VED, and a prosthesis with IU for a soft glans. PDE5I with ICI is off-label, so test it in the office.

Surgery

- A **penile prosthesis** suits men whose other therapy is contraindicated, ineffective or unwanted, and those with corporal fibrosis or Peyronie disease (see Ch 75).

LAQ Surgical management of ED and penile prostheses (types, biofilms)

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P2 LAQ "Surgical management of erectile dysfunction and penile prosthesis" (asked as "Discuss surgical management of erectile dysfunction", "Penile Prosthesis" and "Describe various penile prosthesis in ED") · P3 SAQ "Types of penile prosthesis. What are biofilms in penile prosthesis?" · P4 SAQ "Penile implants."

Introduction

- Surgery for ED (erectile dysfunction) means **penile prosthesis** (main operation) and, rarely, **penile vascular surgery**.
- Prosthesis: most effective, but last-line, treatment of ED; **satisfaction 80–90%**. First devices: Scott inflatable (1972), Small–Carrion rod (1978).
- Lue's goal-directed approach: surgery is for men who fail, cannot use or decline oral PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injection, intraurethral alprostadil or a VED (vacuum erection device).
- Detailed vascular testing is needed only before vascular surgery.

Indications and contraindications for a prosthesis

- **Indications:** ED not responding to, contraindicated for, or unacceptable with (refusal of) oral PDE5Is, injections or vacuum devices; common after prostatectomy, pelvic radiotherapy, diabetes.
- Corporal (cavernosal) fibrosis: after ischaemic priapism (**>36 h**), infection or implant removal; early implant preserves length.
- **Peyronie disease with ED**; penile or pelvic trauma.
- **Contraindications:** active skin, urinary (UTI [urinary tract infection]) or systemic infection; uncontrolled diabetes; unrealistic expectations; reversible ED.
- In insensate men (spinal cord injury) use an inflatable device, because rods erode.

A. Penile vascular surgery

	Arterial revascularisation	**Venous surgery**
Defect	Arteriogenic (poor inflow)	Veno-occlusive (venous leak)
Work-up	Duplex; selective pudendal angiography (also checks the inferior epigastric donor)	DICC (dynamic infusion cavernosometry and cavernosography) locates the leak
Technique	Inferior epigastric artery to dorsal artery (revascularisation) or to deep dorsal vein (arterialisation)	Ligation or embolisation of deep dorsal, superficial dorsal or crural veins; crural plication
Selection	Age <55, non-smoker, non-diabetic, no venous leak, focal internal pudendal stenosis; best in young men after pelvic or perineal trauma	Not defined
Status	Strictly selected men	Investigational
Complications	Glans hyperaemia (commonest), shunt thrombosis, inguinal hernia	Glans numbness, shortening, curvature, skin necrosis

B. Penile prosthesis: types

Type	**Devices / examples**	**Features**	**Best for**
Malleable (semirigid) rods	Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel)	Two bendable rods; simple, cheap; almost never fail (mechanical failure exceedingly rare; P3: no mechanical failure); always firm; erode more in insensate men	Poor manual dexterity; cost
Mechanical rods	Spectra (discontinued)	Articulating segments	–

Type	Devices / examples	Features	Best for
2-piece inflatable	Ambicor (discontinued)	Cylinders + scrotal pump; no reservoir; poor flaccidity	–
3-piece inflatable	AMS 700 CX/LGX/CXR; Coloplast Titan, Titan Touch, Titan Narrow; Rigicon Infla 10	Two cylinders, scrotal pump, abdominal reservoir; **best rigidity and true flaccidity**; most popular	Most men

Three-piece inflatable details

- **Expansion:** girth only (CX, Titan) or girth and length (LGX, up to 25%; Infla 10 AX). **Narrow** cylinders (CXR, Titan Narrow) for fibrotic corpora.
- **Material:** AMS and Rigicon use silicone–fabric–silicone layers; Titan uses polyurethane (Bioflex).
- **Coatings:** InhibiZone (rifampicin + minocycline) on AMS; hydrophilic coating on Titan, dipped in antibiotic.
- **Reservoir:** space of Retzius, or **ectopic submuscular** after pelvic surgery or cystectomy; lockout valve prevents autoinflation.

Consent

- The erection may be shorter (pseudocapsule scar); the aim is a firm penis for intercourse; sensation and ejaculation may reduce; infection or mechanical failure may need reoperation.

Preoperative preparation

- Treat phimosis and skin, fungal and urine infection; sterile urine; stop anticoagulants and aspirin.
- Blood glucose **<200 mg/dL on the day of surgery** (more important than HbA1c [glycated haemoglobin]).
- Shave in theatre; chlorhexidine–alcohol skin preparation (about 40% fewer wound infections than povidone-iodine); MRSA (methicillin-resistant *Staphylococcus aureus*) decolonisation if a carrier.
- IV vancomycin or a cephalosporin plus gentamicin, 1 hour before incision, for up to 24 hours.

Operative steps (3-piece, penoscrotal approach)

`Catheter → upper scrotal incision → expose corpora lateral to urethra → stay sutures → corporotomy → dilate proximally to ischial tuberosity, distally under glans (instruments dorsolateral; 7→13 mm) → goalpost test for crossover → measure → cylinders equal to or 1 cm shorter → test inflation → close corporotomies → pump in dependent subdartos (dartos) pouch → reservoir in space of Retzius → connect tubing → test inflation → irrigate → close in layers`

- Dilate with Hegar or Brooks dilators, instruments pointing dorsolaterally; check for crossover or perforation; cylinders sized to corporal length (equal to or 1 cm shorter than measured length).
- **Approaches:** penoscrotal (commonest; secures the pump; urethra at risk); infrapubic (reservoir placed under direct vision; dorsal nerve at risk); subcoronal or ventral penile for rods.

- Malleable rods are placed slightly shorter: 0.5–1 cm shorter than the measured length; girth chosen with the two-dilator test.
- **Surgical principles:** antibiotics, glucose control, no-touch technique.
- **Intraoperative problems:** crossover (redilate laterally); crural perforation (suture sling); distal urethral perforation or urethral injury (abandon that side for 5–6 months); fibrotic corpora (cavernotomes, narrow cylinders); curvature (modelling for Peyronie's disease); floppy glans (glans fixation).
- **Postoperative:** mummy wrap with glans visible; cylinders partly inflated; drain out by 24 hours; activation at 4–6 weeks.

Biofilm

- **Definition:** a structured microbial community in a self-made **polysaccharide matrix**, attached to the implant.
- **Stages:** 1. Conditioning film of host proteins coats the device within minutes. 2. Reversible, then irreversible bacterial adhesion. 3. Microcolonies. 4. Maturation into a 3D structure with water channels and quorum sensing. 5. Dispersal of free bacteria, causing clinical infection.
- **Organisms:** coagulase-negative staphylococci commonest; also *S. aureus*, Gram-negatives, *Candida*.
- **Why antibiotics fail:** the matrix blocks penetration, bacteria are slow-growing (persisters), and they survive up to **1000–1500 times** normal killing levels.
- **Relevance:** organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

Complications

- **Mechanical failure** (commonest; inflatables): repair in about **15% at 5 years, 30% at 10 years**.
- **Infection (1–3%)**: biofilm on the device.
- **Prevention:** glucose <200 mg/dL; treat skin and urine; decolonise MRSA carriers; intravenous vancomycin or cephalosporin plus gentamicin (up to 24 h); chlorhexidine–alcohol preparation; **antibiotic-coated devices halve infection** (InhibiZone rifampin–minocycline; hydrophilic Titan dipped in antibiotic); **no-touch technique** (under 0.5%).
- **Infected implant:** pus, fixed pump or exposed parts. Remove **all** components.
- **Mulcahy salvage:** remove all parts; wash with dilute povidone-iodine (3 minutes) then saline or antibiotic; change gloves, drapes and instruments; insert a new device at once. Success **84%**; fails with early onset, aggressive organisms, cellulitis. Delayed reimplantation loses length.
- **Intraoperative:** crural perforation, crossover, urethral injury.
- **Others:** erosion or extrusion (catheterisation, rods in insensate men); reservoir erosion into bladder or bowel; reservoir herniation; pump migration; autoinflation; penile shortening; **glans necrosis** (diabetes, smoking, radiation, previous implant); glans hypermobility; cylinder aneurysm.

Special situations (also know)

- Ischaemic priapism >36 hours: early implantation preserves length.
- Transplant recipients: no extra infection.
- After phalloplasty: single cylinder, high erosion.

- Peyronie's disease: modelling.

Outcomes

- Patient and partner satisfaction **80–90%**, the highest of any ED treatment; the main complaint is shorter length.

Recent advances

- Minimal-dilation technique; flat ectopic reservoirs; penile transplantation. Regenerative therapies (Li-ESWT [low-intensity shock wave therapy], stem cells) remain experimental.

Diagrams to draw

- 3-piece inflatable prosthesis: two cylinders, scrotal pump, reservoir in the space of Retzius.
- Biofilm life cycle.

> **Key points to remember:**
>
> - Prosthesis is the last line but gives 80–90% satisfaction.
> - 3-piece inflatable gives the best rigidity and flaccidity.
> - Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
> - Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
> - Biofilm: microbes in a matrix; antibiotics fail.
> - Infection 1–3%; infected implant: remove all parts; Mulcahy salvage 84%.
> - Repairs 15% at 5 years.
> - Arterial revascularisation only for young men with post-traumatic focal stenosis.

7 · Landmark trials

Trial	Design	Result	Take-home
Prostate Cancer Prevention Trial (placebo arm)	Prospective cohort, men ≥55	ED gave 45% more cardiac events at 5 years	ED is a CV sentinel; stratify risk
REINVENT	RCT: nightly vs on-demand vardenafil vs placebo after nerve-sparing prostatectomy	No difference in spontaneous recovery	Scheduled "rehabilitation" PDE5I is unproven

73 Priapism

1 · Definition and classification

- **Priapism** is a full or partial erection lasting **≥4 h** beyond, or unrelated to, sexual stimulation. Untreated ischaemic priapism causes fibrosis and permanent erectile dysfunction (ED).
- It is classified by **oxygenation, not duration**: failed outflow causes time-critical ischaemia, while excess arterial inflow gives an oxygenated, painless erection that can wait.

Feature	Ischaemic (low-flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high-flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Failed outflow, like a compartment syndrome	Repeated self-limiting episodes, often nocturnal	Fistula between cavernosal artery and sinusoids
Typical cause	Drugs, sickle cell disease (SCD), blood disorders	SCD	Blunt perineal/penile trauma, needle injury
Pain / rigidity	Painful (within 6–8 h); fully rigid, tender	Painful; rigid during episode	Painless; partially rigid
Blood gas / Doppler	Low oxygen, high CO_2, acidotic; absent or low flow	Ischaemic in each episode	Arterial gas; high flow, fistula or blush
Management	Emergency decompression	Treat episode as ischaemic, plus prophylaxis	Observe, then embolise
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

> **KEY POINT**
>
> Trap: stuttering priapism is **ischaemic**, so any episode lasting ≥4 h needs the same first-line treatment.

2 · Aetiology

LAQ Priapism: diagnosis and management

Paper answer · P2 · P3 · merged from 2

Asked as: P3 LAQ "Priapism: Diagnosis and Management" (also asked as P3 SAQ "Priapism") · P2 SAQ "Management of priapism"

Definition

- Full or partial erection lasting **≥4 hours** beyond sexual stimulation, or unrelated to it (AUA/SMSNA [American Urological Association/Sexual Medicine Society of North America] 2021–2022).

Classification (types)

	Ischaemic (low flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high flow)
Share	**95%**	Subset of ischaemic	<5%

	Ischaemic (low flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high flow)
Lesion	Veno-occlusion	Repeated (recurrent) ischaemic episodes	Cavernosal artery–sinusoid (arterio-sinusoidal) fistula after trauma
Pain, rigidity	Painful, fully rigid	Painful, recurrent, often nocturnal	Painless, partial
Typical cause	ICI (intracavernosal injection), drugs, SCD (sickle cell disease)	SCD	Perineal or needle trauma
Urgency	**Emergency**	Emergency per episode + prophylaxis	Not an emergency

Aetiology

- **Ischaemic:** ICI (papaverine, phentolamine, PGE1 [prostaglandin E1]), including recreational use; α-blockers, trazodone, antipsychotics, cocaine; heparin or warfarin withdrawal; **SCD** (29–42% of affected men in their lifetime; 63% of paediatric ischaemic priapism); leukaemia (1–5%, high white cell count); penile metastases; scorpion and spider venom.
- **Non-ischaemic:** straddle or coital injury, pelvic fracture, needle injury. Onset is often delayed, when the clot over the arterial tear lyses.

Pathophysiology

- **Ischaemic:** failed outflow acts like a compartment syndrome. Hypoxia, hypercarbia and acidosis cause sinusoidal thrombosis, smooth-muscle necrosis and TGF-β (transforming growth factor beta)-driven fibrosis, so ED (erectile dysfunction) follows. Irreversible damage starts within hours.
- **Stuttering (molecular):** low basal NO (nitric oxide) from eNOS (endothelial NO synthase) dysfunction downregulates PDE5 (phosphodiesterase 5). When an erection starts, cGMP (cyclic guanosine monophosphate) is not broken down, so relaxation is unopposed. Low Rho-kinase and high adenosine add to this. This explains why daily PDE5 inhibitors prevent episodes (they reset PDE5 regulation).
- **Non-ischaemic:** unregulated, well-oxygenated arterial inflow, so fibrosis is unusual.

Diagnosis

- **History:** duration, pain, drugs, trauma, SCD or blood disease, prior episodes, baseline erectile function.
- **Examination:** rigid tender corpora with a soft glans and spongiosum (ischaemic); partial, non-tender (high flow); perineal bruising; vital signs before phenylephrine.
- **Corporal blood gas** (cornerstone; decides the type); high flow resembles arterial blood:

	Colour	pO2 (mmHg)	pCO2 (mmHg)	pH
Ischaemic	Dark red	**<30**	**>60**	**<7.25**
Flaccid (mixed venous)	Medium red	40	50	7.35
Non-ischaemic (arterial)	Bright red	>90	<40	7.40

- **PDUS (penile duplex ultrasound, colour Doppler)** when the gas is equivocal: ischaemic = absent cavernosal flow or PSV (peak systolic velocity) <50 cm/s with no diastolic flow;

high flow = PSV >50 cm/s, may show the fistula. Angiography only for embolisation.
- Blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology: in parallel, never delaying treatment.

Management of ischaemic priapism (AUA/SMSNA 2021)

``` Counsel on ED risk → dorsal penile block <36 h: aspiration (19 G, penoscrotal junction, 3 or 9 o'clock) ± saline irrigation + phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h (monitor blood pressure, pulse: reflex bradycardia) → fails → distal shunt (Winter, T-shunt, Al-Ghorab) ± tunnelling → proximal shunt (Quackels) or penoscrotal decompression

> 36 h or refractory → early (immediate) penile prosthesis (or analgesia and counselling) ```

- Aspiration alone succeeds in under 1 in 4. Cold (10 °C) saline irrigation was better than body temperature in an RCT (randomised controlled trial); avoid chilled saline in SCD.
- Ice, oral agents, hydration and IV (intravenous) fluids are adjuncts only; in SCD, intracavernosal treatment still comes first, with concurrent haematology care.

| Shunt | Method |
|---|---|
| **Winter** (distal) | Tru-Cut (biopsy) needle through glans; least effective |
| **Ebbehoj / T-shunt** (distal) | Scalpel through glans; T-shunt rotates blade 90° |
| **Al-Ghorab** (distal, open) | Dorsal glans incision; excise corporal (tunica) tips |
| **Snake tunnelling** | 7/8 Hegar dilator passed proximally through distal shunt |
| **Quackels / Sacher** (proximal) | Cavernosum to spongiosum at bulb; stagger ≥1 cm |
| **Grayhack / Barry** (proximal) | Cavernosum to saphenous / dorsal vein |
| **Penoscrotal decompression** | Proximal corporotomy, clot suction (evacuation); allows prosthesis |

- **ED by duration** (Bennett and Mulhall, SCD series): ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- **Early prosthesis** (Zacharakis 2014): less shortening and fewer revisions than delayed insertion: shortening 3% vs 40%, revision 9% vs 27%, satisfaction 96% vs 60%.

### Stuttering priapism

- An episode lasting ≥4 hours is treated as ischaemic; home self-injection of phenylephrine.
- **Prophylaxis:** daily PDE5 inhibitor (Burnett 2014 RCT: primary endpoint missed, but promising), pseudoephedrine, etilefrine; hormonal agents (GnRH [gonadotropin-releasing hormone] agonists, antiandrogens, finasteride) at the cost of libido. **Also know:** digoxin; baclofen.

### Non-ischaemic priapism (AUA/SMSNA 2022)

- Observation, perineal compression and ice; about 60% resolve. Reassess at 4 weeks.
- Persistent: PDUS, then **selective cavernosal artery embolisation** (resolution ≥75%, recurrence 30–40%). For recurrence, repeat embolisation with non-resorbable PVA (polyvinyl alcohol), not surgery. Surgical ligation is a last resort (ED about 50%).

### Complications

- ED, corporal fibrosis, penile shortening; shunt-related oedema, haematoma, urethral injury or fistula, skin necrosis, infection.
```

Diagram to draw: distal (glans) and proximal shunt sites on a penile outline.

> **Key points to remember:**
> - Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
> - Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
> - ED risk 0% ≤12 h, 100% ≥36 h; early prosthesis beyond 36 h or if refractory.
> - High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

Also, from the condensed book (not in the paper answer):
- Metastases obstruct venous outflow.

Ischaemic – extra points

- Many other drugs, including oral phosphodiesterase type 5 inhibitors (PDE5Is), can cause it, so take a full drug history.
- In leukaemia, give cytotoxics and leukapheresis alongside penile treatment.
- **Recurrence after a shunt** may mean shunt thrombosis. Give **aspirin 325 mg + heparin 5000 U subcutaneously** before surgery, then aspirin 81 mg daily for 2 weeks.

Sickle cell disease

- Deoxygenated haemoglobin S polymerises, causing haemolysis and low nitric oxide (NO), which obstruct outflow.
- Lifetime risk is **29–42%**; SCD causes 23% of adult and **63% of paediatric** ischaemic priapism. Stuttering is commoner than major episodes.

Non-ischaemic

- Onset is often delayed ≥1 day after trauma, because the clot over the arterial tear disperses in a nocturnal erection.
- Transient post-ischaemic hyperaemia after treatment mimics it; colour Doppler distinguishes them.

3 · Molecular pathophysiology

- Low basal NO reduces cyclic guanosine monophosphate (cGMP) feedback, so **PDE5 is down-regulated**. A neural NO surge then produces cGMP that cannot be broken down, so relaxation is unopposed.
- Paradox: **a daily PDE5I restores PDE5 and prevents** recurrent episodes.

4 · Evaluation

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

*See the paper answer **Doppler in urology** (Section 1, Ch 4).*

Also, from the condensed book (not in the paper answer):
- Record baseline erectile function, because it is needed for prosthesis counselling.
- The glans and spongiosum stay soft in both types; their involvement suggests another disease.

ALGORITHM

1. History, examination and corporal blood gas.
2. If ischaemic, then acute ischaemic algorithm.
3. If indeterminate, then penile/perineal PDUS.
4. If non-ischaemic, then expectant algorithm.

5 · Acute ischaemic priapism (AUA/SMSNA)

- Irreversible damage begins 4–12 h after presentation, so counsel about ED risk first. Conservative measures are **adjuncts only; never delay first-line treatment**. Oral terbutaline or pseudoephedrine is not recommended.

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

ALGORITHM

1. If <36 h, then aspirate ± irrigate and inject intracavernosal phenylephrine (BP and cardiac monitoring).
2. If that fails, then distal shunt ± tunnelling; tunnel if not done.
3. If still erect, then proximal shunt or penoscrotal decompression; if refractory, prosthesis.
4. If >36 h, then counsel high ED risk with analgesia/follow-up, OR early penile prosthesis.

Aspiration / irrigation

- After a penile block, insert a **19-G needle** at the penoscrotal junction at **3 or 9 o'clock**; one side usually suffices. Aspirate, repeat and irrigate with saline.
- Aspiration alone works in fewer than 1 in 4. Cold (10 °C) irrigation gave 96% vs 60% in a randomised controlled trial (RCT), but **avoid chilled saline in SCD**.

Phenylephrine

- It is a selective α1-agonist with the least cardiac effect.
- Dilute to **100–500 µg/mL** and give **1-mL aliquots every 3–5 min** for up to **1 h** before declaring failure; the maximum is about 1 mg in 1 h (European Association of Urology).
- Toxicity includes hypertension, headache, arrhythmia and **reflex bradycardia**. Monitor, especially with cardiovascular disease or monoamine oxidase (MAO) inhibitors.

Shunts

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

Also, from the condensed book (not in the paper answer):

- AUA: proximal shunts only after a failed distal shunt with tunnelling. Complications include urethral injury or fistula, necrosis, infection and pulmonary embolism.

Early penile prosthesis

- It is used for priapism **>36 h** or failed shunts, because oedematous corpora dilate easily, while delayed surgery meets dense fibrosis.
- Early vs delayed implantation: shortening 3% vs 40% and satisfaction 96% vs 60%.
- Prior distal shunts raise the risk of distal perforation (see Ch 75).

6 · Non-ischaemic priapism

*It is **not an emergency**.*

- About 60% resolve with observation, compression and ice.

ALGORITHM

1. Observe and reassess at 4 weeks.
2. If it persists, then PDUS (fistula site, size), counsel, and selectively embolise the internal pudendal/cavernosal artery.
3. If it recurs, then repeat embolisation with a non-resorbable agent.
4. Only after repeated failure, ligate surgically.

Agent	Type	Feature
Autologous clot	Temporary	Children; early recurrence
Gelatin sponge	Temporary (2–6 weeks)	Reabsorbed
Microcoils	Permanent	Theoretical ED risk
Polyvinyl alcohol particles	Permanent	**Preferred for repeat** (AUA)
N-butyl cyanoacrylate	Permanent	Second line for recurrence

- Embolisation succeeds in ≥75%, with 30–40% recurrence. Open ligation causes ED in about 50%.

7 · Stuttering priapism — prevention

- An episode ≥4 h is treated as acute ischaemic priapism. Short episodes are managed with fluids, walking and analgesia, and home self-injected phenylephrine for breakthrough.
- No prophylactic drug has robust evidence.

Agent	Dose	Note
PDE5I	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Most promising; start when flaccid; the only RCT missed its primary endpoint
Pseudoephedrine	60–120 mg at night	Off-label first-line α-agonist
Etilefrine	50–100 mg daily	α1-agonist; Cochrane: insufficient evidence
GnRH agonists, antiandrogens (bicalutamide, flutamide, cyproterone)	Depot / oral	Suppress libido and erections; avoid before puberty is complete
Finasteride	5 mg, tapered to 3 mg then 1 mg	One SCD series
Digoxin	0.25–0.5 mg	Inhibits Na^+/K^+-ATPase
Baclofen	Oral 40 mg at night; intrathecal	Neurogenic/reflex erections

SCD specifics

- First-line aspiration and phenylephrine are the same; give IV hydration and analgesia **concurrently, never instead**. Exchange transfusion is not a primary treatment (AUA); involve haematology and avoid chilled irrigation.

8 · Landmark trials

Trial	Design	Result	Take-home
Ateyah 2005	RCT, ICI-induced priapism, irrigation temperature	10 °C saline 96% vs 60% at 37 °C	Cold irrigation (not in SCD)
Burnett 2014	RCT, SCD, sildenafil vs placebo	Primary endpoint missed; open-label benefit	PDE5I prophylaxis plausible, unproven
Zacharakis 2014	Early vs delayed prosthesis	Shortening 3% vs 40%; satisfaction 96% vs 60%	Implant early when ED is inevitable

74 Disorders of Male Orgasm and Ejaculation

1 · Spectrum

- Ejaculatory dysfunction (EjD) is the **most common male sexual disorder**. It includes premature ejaculation (PE), delayed ejaculation (DE), anejaculation and retrograde ejaculation (RE). Painful ejaculation, anhedonia and postorgasmic illness syndrome (POIS) are rarer.
- Orgasm is a separate **cortical** event, so ejaculation and orgasm can dissociate.

2 · Physiology of ejaculation

- Lumbar interneurons at **L3–L5** form the **spinal ejaculation generator**.

Phase	Events	Mediation
Emission	Vas, seminal vesicles and prostate deliver semen to the posterior urethra	Autonomic, **T10–L2**
Ejection	Bulbospongiosus, ischiocavernosus, levator ani contract rhythmically; **bladder neck closes**; external sphincter relaxes	Somatic **S2–S4 (pudendal)** and a sympathetic reflex
Orgasm	Cerebral processing of pudendal input	Cortical

- **Dopamine (D2) promotes** and **serotonin (5-HT) inhibits** ejaculation.
- 5-HT2C stimulation delays and 5-HT1A shortens ejaculation, so lifelong PE is thought to be **5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity**.

> **KEY POINT**
>
> Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic.

3 · Premature ejaculation — definition and classification

Short latency alone is not PE; loss of control and distress are needed.

SAQ Premature ejaculation

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Premature Ejaculation" · P3 SAQ "Premature ejaculation."

Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed: 1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a clinically significant, bothersome fall in latency, often to **≤3 minutes** (acquired). 2. Inability to delay ejaculation on all or nearly all penetrations. 3. Negative personal consequences: distress, frustration, avoidance of intimacy.
- **AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2020:** lifelong PE (premature ejaculation) within about **2 minutes**.
- **Types:** lifelong, acquired, variable and subjective (normal latency).

Physiology and aetiology

- Ejaculation = **emission** (sympathetic T10–L2) then **expulsion** (somatic S2–S4, pudendal). **Serotonin brakes and dopamine accelerates**.
- **Lifelong PE:** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.
- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

Evaluation

- Diagnosis is by **history**: latency, control, bother; lifelong or acquired.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials. Population median 5.4 minutes.
- **PEDT (Premature Ejaculation Diagnostic Tool):** 5 items; score ≥11 suggests PE, 9–10 probable PE. Also PEP (Premature Ejaculation Profile) and IPE (Index of Premature Ejaculation).
- IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

Office algorithm (diagram to draw):

1. Confirm PE.
2. Secondary to ED? → treat the cause.
3. Otherwise → classify lifelong or acquired.

Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Acquired PE: treat the cause first (ED, prostatitis, thyroid).

Drug	Dose and timing
Dapoxetine (short-acting SSRI [selective serotonin reuptake inhibitor]; only licensed oral drug)	30–60 mg, 1–3 h before intercourse
Paroxetine (strongest delay) 10–40 mg, sertraline, fluoxetine, citalopram	Daily, off-label; full effect 2–3 weeks; do not stop abruptly
Clomipramine	12.5–50 mg daily or 3–4 h before
Topical lidocaine–prilocaine spray (licensed in Europe 2014)	5–30 min before (P1: at least 5 min before); condom avoids partner numbness
PDE5Is (phosphodiesterase type 5 inhibitors)	With ED, alone or with SSRI
Tramadol	Low evidence; addiction risk

- SSRIs raise synaptic serotonin, which delays ejaculation. Avoid in bipolar disorder.
- **Also know:** high dropout from drug therapy.

> **Key points to remember:**
> - ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
> - Lifelong is neurobiological; acquired is usually secondary.
> - Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Also, from the condensed book (not in the paper answer):

- DSM-5 duration (≥6 months, 75–100% of occasions) and severity bands are in the table below.

	Lifelong	Acquired	Variable	Subjective
Onset	From first encounters, every partner	After previously normal latency	—	—
Latency	≤ ~1 min (mostly 30–60 s)	Markedly reduced, often ≤3 min	Occasionally short	Normal or long
Basis	Neurobiological, genetic	Medical/psychological	Normal variation	Psychological/cultural
Disorder?	Yes	Yes	No	No

- The International Society for Sexual Medicine (ISSM) requires ejaculation within **about 1 min** (lifelong) or a bothersome fall, often to **≤ about 3 min** (acquired). It is the first **evidence-based** definition, for vaginal intercourse only. **Latency alone never suffices**.

	ISSM	DSM-5 (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ ~1 min	≈1 min	≈**2 min** since sexual debut
Acquired	Bothersome fall, often ≤3 min	Specifier only	Marked fall (2–3 min or ≥50%)
Duration/frequency	Always or nearly always	≥6 months, 75–100% of occasions	—
Severity	—	Mild 30–60 s; moderate 15–30 s; severe <15 s	—
Basis	Evidence	Same data	Expert opinion

- The AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2-min line trades specificity for sensitivity.
- Community stopwatch intravaginal ejaculatory latency time (IELT) is positively skewed, median **5.4 min**; ≤1 min is statistically abnormal.

4 · Premature ejaculation — aetiology and evaluation

- Lifelong PE affects younger men with few comorbidities; acquired PE affects older men with metabolic syndrome or diabetes.
- **PE with erectile dysfunction (ED)** affects up to half of men with ED, who rush before a partial erection is lost.
- PE is the main sexual symptom in **chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS)**, and it improves when **hyperthyroidism** is treated, though routine thyroid tests are not indicated.
- Examination is desirable in lifelong and **mandatory in acquired PE**, with digital rectal examination (DRE) for prostatitis.

5 · Premature ejaculation — treatment

ALGORITHM

1. Confirm PE.
2. If it is secondary to ED or another dysfunction, then treat that primary cause.
3. If lifelong, then SSRI ± behavioural therapy (patient preference).
4. If acquired, then treat the cause ± behavioural therapy/SSRI; try withdrawal after 6–8 weeks.
5. If variable or subjective, then reassurance, education, psychotherapy.

- Behavioural therapy is mainly an adjunct.

Drug	Dose / timing	Notes
Dapoxetine	30–60 mg, 1–3 h before	**Only licensed oral PE drug** (>50 countries, not FDA); short half-life; works from the first dose; no PDE5I interaction; many stop it
Paroxetine (daily)	10–40 mg	**Strongest delay**; off-label
Sertraline / fluoxetine / citalopram	50–200 / 20–40 / 20–40 mg daily	Off-label
Clomipramine (a tricyclic antidepressant, TCA)	12.5–50 mg daily or 3–4 h before	Off-label
Topical lidocaine/prilocaine	20–30 min before (spray ≥5 min)	Condom advised; level 1a evidence
Phosphodiesterase type 5 inhibitors (PDE5Is)	30–50 min before	Best with coexisting ED; add-on to SSRI
Tramadol	25–50 mg, 3–4 h before	Low evidence; addiction risk

- **Daily selective serotonin reuptake inhibitors (SSRIs)** act in 5–10 days, fully at **2–3 weeks**.
- Use caution in bipolar disorder, with aspirin or NSAIDs, and under 18 years. **Never stop them abruptly.**
- Meta-analyses show that a PDE5I beats placebo and that a PDE5I with an SSRI beats an SSRI alone.

On-demand dapoxetine	Daily off-label SSRI
Licensed; fewer side effects; little withdrawal; infrequent intercourse	Greater delay; full effect 2–3 weeks; must taper; established relationships

KEY POINT

The real limiter in PE is adherence, not efficacy: most stop SSRIs and topicals (cost, lost spontaneity, "antidepressant" stigma).

6 · Delayed ejaculation, anejaculation and anorgasmia

- DE (DSM-5) is a marked delay or absence of ejaculation despite adequate stimulation, on ~75–100% of occasions, with distress. There is **no validated cut-off**, but >25–30 min with distress suggests DE.
- **Anorgasmia** is absent perceived orgasm. DE is mostly acquired and rises with age; its hallmark is normal erections with **low subjective arousal**.

Category	Causes
Afferent	Ageing (penile afferents, Pacinian corpuscles)
Psychogenic	Inhibited ejaculation; performance anxiety; religious orthodoxy; **idiosyncratic vigorous masturbation**
Endocrine	Hypogonadism, **hypothyroidism** (DE; hyperthyroidism causes PE), hyperprolactinaemia
Neurogenic disease	Spinal cord injury (SCI), then diabetic autonomic neuropathy, then multiple sclerosis (MS)
Neurogenic surgery	Radical prostatectomy, rectal surgery, sympathectomy, aortic aneurysm repair, retroperitoneal lymph node dissection (RPLND)
Anatomical (causes RE)	Transurethral resection of prostate (TURP), bladder neck incision
Congenital / infective	Müllerian cyst, Wolffian anomaly, prune belly; urethritis, tuberculosis (TB), schistosomiasis
Drugs	**SSRIs** (~7-fold DE risk), TCAs, antipsychotics, α-methyldopa, thiazides, alcohol

- **Superior hypogastric plexus** injury at RPLND causes dry ejaculation, which **nerve-sparing RPLND** preserves in most men.
- Ask whether masturbation works and whether he has orgasm without semen (RE or obstruction). Examine the vasa and do a DRE; check morning testosterone (T), HbA1c and lipids.

ALGORITHM

1. Orgasm? If never or sometimes, then inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique).
2. If always but no ejaculate, then look for **sperm in post-orgasm urine**; if present, RE.
3. If absent (aspermia), then failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO).

	Bilateral EDO	RE	Failure of emission
Antegrade volume	Very low	Low or absent	Absent
pH, fructose	Low (seminal vesicle fluid, ~80% of volume, is missing)	—	—
Post-orgasm urine	No sperm	**Sperm and fructose**	No sperm

Treatment

- Treat the cause, and counsel about anejaculation and sperm retrieval before pelvic surgery. Refer to a psychosexual therapist once organic and drug causes are excluded (AUA).
- Normalise T, change the offending drug and change masturbatory style. Drugs are all off-label, with about 40% success.

Drug	Action	Dose
Cabergoline	Central D2 agonist	0.5–2 mg every 3 days; **annual echocardiography** (valvular fibrosis)
Bupropion	Dopaminergic	150 mg once or twice daily
Pseudoephedrine	α1 agonist	120 mg 1–2 h before
Reboxetine	Noradrenaline reuptake inhibitor	4–8 mg daily
Cyproheptadine	5-HT antagonist	4–12 mg 3–4 h before (SSRI-induced DE); sedation
Amantadine, buspirone	—	Anecdotal
Intranasal oxytocin	Oxytocinergic	Both RCTs negative

7 · Retrograde ejaculation

*See the paper answer **Retrograde ejaculation** (Section 9, Ch 67).*

Also, from the condensed book (not in the paper answer):

- Diagnosis: **>5–10 sperm/high-power field** in urine passed after masturbation confirms RE.
- Aquablation preserves ejaculation better than TURP in randomised controlled trials.

8 · Sperm retrieval in SCI

- Reflex erection needs the sacral arc, but ejaculation needs thoracolumbar outflow, so it is rare (<5%) with complete upper motor neuron lesions; RE is common.

ALGORITHM

1. **Penile vibratory stimulation (PVS)** is first line, with up to ~70% success; it needs an intact reflex at the T12–S1 level.
2. If PVS fails, then **electroejaculation (EEJ)**: a rectal probe stimulates hypogastric sympathetic efferents; effective and safe.
3. If EEJ fails, then prostate massage, then surgical retrieval (TESE/TESA/MESA/PESA); testicular sperm extraction succeeds in most.

- Lesions above T6 risk **autonomic dysreflexia**: pretreat with **nifedipine** and monitor BP.

9 · Other disorders

Disorder	Key points
Painful ejaculation	Prostatitis/CPPS, urethritis, benign prostatic hyperplasia (BPH), seminal vesicle stones, EDO, α-blockers (least with alfuzosin); often none. Surgery (TURP, transurethral resection of ejaculatory ducts (TURED), pudendal neurolysis) weakly supported
Haematospermia	Usually benign (see pointer below). Seminal vesiculoscopy if refractory (see Ch 70); evaluate if >40 y or persistent
Climacturia	Leak at orgasm after radical prostatectomy. Steps: void first, condom, pelvic floor training, constriction ring, male sling
Ejaculatory anhedonia	Orgasm without pleasure; depression, hypogonadism, hyperprolactinaemia, SSRIs
Post-SSRI sexual dysfunction	Genital numbness, low desire, ED, anhedonia persisting months to years
POIS	Myalgia, fatigue, flu-like state within 30 min of orgasm
Orgasmic headache	Dull bilateral ache rising with excitement; sudden, intense at orgasm

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SAQ Haematospermia

Paper answer · P1

Asked as:

- Hematospermia — *Winter 2025 Paper I Q3c (SAQ)*
- Hematospermia — *Practice Set 9 Paper I Q3d (SAQ)* (also answered in 2025-Winter Paper 1, Q3c)

Type: SAQ (10 marks)

Definition

- Blood in the ejaculate. It alarms patients but is usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.
- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified.
- **Pseudo-haematospermia:** blood from the partner (partner's blood; condom test) or haematuria.

Causes

Group	Examples
Iatrogenic (commonest identifiable)	Prostate biopsy (about 10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy, BPH (benign prostatic hyperplasia) procedures
Inflammation and infection	Prostatitis, vesiculitis, urethritis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis
Obstruction and cysts	Ejaculatory duct obstruction or calculi; seminal vesicle cyst (Zinner syndrome) or utricle cyst
Tumours	**Prostate cancer**; rarely urethral, bladder or seminal vesicle
Vascular	Prostatic varices, BPH vessels, AV (arteriovenous) malformation

Group	Examples
Systemic	Hypertension, bleeding disorders, anticoagulants, liver disease; also abstinence, trauma

Evaluation

- **History:** episodes, recent biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants.
- **Examination:** blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, semen culture, STI screen, **PSA (prostate-specific antigen)** in men over 40, coagulation if indicated.

Who needs full work-up (full work-up if): age **≥40**, persistent beyond several weeks or recurrent, haematuria, abnormal DRE or PSA, risk factors for TB or cancer, or a distressed patient.

- **TRUS (transrectal ultrasound):** cysts, calculi, dilated seminal vesicles (>1.5 cm) or ejaculatory ducts (>2.3 mm) suggest obstruction.
- **MRI (magnetic resonance imaging) pelvis:** best for seminal vesicle haemorrhage, cysts and prostate cancer.
- **Cystoscopy and urine cytology**; early-morning urine for acid-fast bacilli.

Management

- **Young, single episode, normal tests: reassure**.
- **Treat the cause:** antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- **Persistent prostatic bleeding:** 5-ARI (5-alpha-reductase inhibitor), e.g. finasteride.
- **Refractory or obstructive:** TURED (transurethral resection of ejaculatory ducts), cyst unroofing or aspiration, **seminal vesiculoscopy** (4–6 Fr small scope through the ejaculatory duct or utricle, irrigation, laser for stones).

Diagram to draw: algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

> **Key points to remember:**
> - Usually benign; prostate biopsy is the commonest identifiable cause.
> - Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
> - Exclude prostate cancer and genitourinary tuberculosis.
> - TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

75 Surgery for Erectile Dysfunction

1 · Place of surgery and devices

A prosthesis is the last step for erectile dysfunction (ED).

*See the paper answer **Surgical management of ED and penile prostheses (types, biofilms)** (Section 9, Ch 72).*

Also, from the condensed book (not in the paper answer):

- Coloplast cylinders are polyurethane (Bioflex), so keep cautery **<35 W**; Boston and Rigicon silicone cylinders tolerate cautery.

2 · Selection and counselling

- **Consent is expectation management**: the device restores rigidity, not the former erection.
- The commonest disappointment is a **shorter erection**, because the pseudocapsule limits expansion (worse in Peyronie disease). Document it.
- With **phimosis**, circumcise first and implant about 6 weeks later.
- Raised glucose on the day needs intravenous (IV) insulin. Nasal methicillin-resistant Staphylococcus aureus (MRSA) carriers get chlorhexidine bathing and mupirocin 2% for 5 days.

3 · Surgical approaches

Approach	Advantages	Cautions
Penoscrotal	Majority choice; secure subdartos pump; good crural exposure	Urethra at risk, so Foley first; reservoir via inguinal canal
Infrapubic (Scott)	Reservoir under vision	Dorsal neurovascular bundle: stay sutures at 10 and 2 o'clock; patient pulls pump down daily
Subcoronal	Rods; allows straightening	Circumcise at the same time, or lymphoedema follows
Ventral penile	Rods without bending; foreskin kept	—

4 · Operative technique (three-piece)

ALGORITHM

1. Catheter; expose the tunica.
2. Two 2-0 polydioxanone (PDS) stay sutures per corpus; longitudinal corporotomy.
3. Dilate proximal, then distal (Metzenbaum, Hegar/Brooks 7→13 mm).
4. Goalpost test; measure (Furlow).
5. Cylinders in; test inflation 1. Close corporotomies; test inflation 2.
6. Reservoir, then pump; connect; test inflation 3.
7. Irrigate and close.

- Dilators point **dorsolaterally**, away from urethra and septum. The **goalpost test** (equal height and symmetry) detects crossover.

- The cylinder is equal to or 1 cm shorter than measured, because oversizing causes buckling and S-deformity; rear tip extenders adjust length.
- The pump sits in a dependent subdartos pouch, anterolateral to the testis.

Reservoir placement

- The **Retzius** space is entered with the bladder empty; the iliac vessels lie about 1 inch lateral.
- In a hostile pelvis (radiation, cystectomy, robotic prostatectomy, transplant) use an **ectopic** site, made safe by the **lockout valve**.
- **Submuscular suprafascial** is the commonest ectopic site. Fill it nearly full, because underfilling leads to migration.

5 · Intra-operative problems

- **Fibrotic corpora** follow priapism or explant (pelvic radiation does not add difficulty). Use cavernotomes or scar excision and **narrow cylinders**, always two.
- **Crossover** gives a lopsided erection and unequal measurements. Leave a large dilator in the common cavity and redilate the other side laterally; the Foley should lie midline between cylinders.
- **Crural perforation** (one dilator deeper, lengths differ by >1 cm) is repaired with a **suture sling** (2-0 Prolene through the rear tip, over the corporotomy).
- **Urethral perforation** at the corporotomy is closed and the operation proceeds. If distal (fossa navicularis, irrigant at the meatus), **abandon that cylinder**, place the contralateral one, leave no catheter and replace at 5–6 months.
- A **hypermobile glans** drooping ventrally is the supersonic transporter (**SST**) deformity. It is fixed with glans sutures (3-0 non-absorbable).
- **Residual curvature** is treated with high-rigidity cylinders (AMS 700 CX, Titan, Infla 10 X) and **modelling** (clamp tubing, bend 90° away for 90 s), then plication or incision with or without graft. Less than 20° is acceptable (see Ch 76).

> **KEY POINT**
>
> No catheter after distal urethral perforation: the urinary stream draws fluid into the urethra (Bernoulli), so urine does not enter the defect. Wait 5–6 months for scar to obliterate the false tract. Most intra-operative errors are **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley).

> **KEY POINT**
>
> After robotic prostatectomy the peritoneum is not replaced and bowel lies near the internal ring, so a blind inguinal Retzius reservoir can injure bowel. Choose an ectopic site or place it under vision.

6 · Complications

Infection usually costs the whole device.

Infection (1–3%)

- Risks include hyperglycaemia, revision, MRSA and scarred corpora.

- American Urological Association (AUA) prophylaxis is IV vancomycin, or a 3rd-generation cephalosporin plus gentamicin, 1 h before incision and for ≤24 h. Extended oral courses are unproven; in high-risk men add an antifungal, with or without piperacillin–tazobactam.
- Coatings (InhibiZone rifampin–minocycline; hydrophilic Titan/Infla) halve infection, and a **no-touch** technique gives <0.5%.
- Fever and erythema suggest infection; pus, an exposed device, progressive pain or a **pump fixed to the scrotal wall** are definitive. **Remove ALL components**, as retained parts harbour biofilm.

Option	Technique	Caveats
Explant, delayed replacement	Antibiotic irrigation drains for 3 days; reimplant at 2–6 months	Fibrosis, loss of several cm; vacuum erection device (VED) meanwhile
Mulcahy salvage	Remove all, antiseptic washes, new drapes, immediate new device	~84%; fails with onset <2 months, aggressive organisms, purulence with cellulitis
Delayed salvage	IV vancomycin and gentamicin, salvage at 3 days	Aggressive infection
Malleable salvage	Rods instead	Quick; convert later

- Washes use dilute povidone-iodine (0.35–3.5%) for 3 min, then saline or antibiotic; hydrogen peroxide is dropped as cytotoxic.
- An uninfected revision carries ~10% infection, falling to ~3% with antiseptic washout of all cavities.

Mechanical failure

- This is the **commonest complication**: leak, pump malfunction, autoinflation, kinking or cylinder aneurysm.
- If the device is >3 years old, replace **all** parts; check any retained part with the **ohmmeter test**. A rod that loses rigidity has a core fracture and is replaced.

Extrusion and erosion

Feature	Extrusion	Erosion
Position	Tip through tunica, still under skin/glans	Through skin or into urethra
Cause	Forceful dilation, oversizing, aggressive use	Infection, crossover, catheterisation; rods in insensate men (spinal cord injury)
Management	**Distal corporoplasty** (new dorsomedial plane behind pseudocapsule)	Remove all, washout, contralateral spacer, replace at 5–6 months

- **Reservoir erosion** into the bladder needs reservoir exchange and repair; into bowel, remove the whole device.
- **Penile necrosis** is linked to diabetes, vascular disease, smoking, prior implant and radiation. Avoid circumferential incisions and tight wraps, and remove cylinders promptly if it occurs.
- An early basal bulge is corporotomy disruption (repair); a late one is cylinder aneurysm (replace).

7 · Special situations

- After **priapism** >36 h, an **immediate implant** preserves length and relieves pain; avoid distally expanding cylinders (see Ch 73).
- In **Peyronie disease** length loss is greater, so use non-expanding high-rigidity cylinders with modelling.
- Transplant and HIV-positive men (adequate CD4) have no excess infection.

8 · Post-operative care

- It is day surgery. Leave cylinders **partially inflated** so the pseudocapsule forms large, remove the drain within 24 h and keep the penis upward for 4–6 weeks.
- Activation is at **4–6 weeks**, then the device is cycled daily. Dissatisfaction is mainly about length and rigidity.

9 · Penile revascularisation †

- †Only young men (<55) with a focal post-traumatic pudendal occlusion qualify (inferior epigastric to dorsal artery). It is not recommended for generalised atherosclerosis or venous leak.

76 Diagnosis and Management of Peyronie Disease

1 · Definition

- **PD** (American Urological Association, AUA 2015) is an acquired penile abnormality with **fibrosis of the tunica albuginea**, with or without pain, deformity, erectile dysfunction (ED) and distress.

> **KEY POINT**
>
> Myths corrected: PD is **not rare** (3–20%; higher in diabetes and ED). **Spontaneous resolution is rare** and up to half worsen untreated, so "wait 6–12 months" is a myth. **Calcification is not a marker of chronic disease**.

2 · Anatomy and pathophysiology

- The outer longitudinal tunica layer is **absent at 5–7 o'clock**, so the dorsum buckles; **60–70% of plaques are dorsal** with septal involvement.
- **Microtrauma** from bending during intercourse causes **delamination at the septal insertions**.
- **Transforming growth factor-β1 (TGF-β1)** is central. Collagen becomes disorganised and elastin decreases, so the penis **curves toward the plaque**.
- A **susceptible host** is needed, and PD is associated with **Dupuytren and Ledderhose** disease; only 16–40% recall trauma.

3 · Clinical features and phases

- Features are pain, deformity (curvature, indentation, **hourglass**, hinge, shortening), plaque and ED. Hourglass or hinge gives an **unstable penis**.
- Kelami grading: ≤30° mild, 31–60° moderate, >60° severe. Ventral and lateral curves are harder to manage than dorsal.

Feature	Active (acute)	Stable (chronic)
Pain	Present (flaccid, erection, intercourse)	Usually resolved
Deformity	Progressive	No progression
Aim	Prevent progression; non-steroidal anti-inflammatory drugs (NSAIDs) for pain; traction	Correct deformity (injection, surgery)
Surgery	Not appropriate	When **≥12 months from onset and stable ≥6 months**

- The differential includes congenital curvature, chordee, post-traumatic fibrosis and tumours (epithelioid sarcoma mimics PD).

4 · Evaluation

- The **key question** is: "If straight with current rigidity, would it be adequate for penetration?" If **no**, offer a prosthesis with straightening; the answer also predicts post-operative erectile function.
- Measure **stretched penile length (SPL)** from pubis to corona dorsally; this is essential, as patients overestimate loss.

- Extensive **calcification** on ultrasound predicts poor intralesional response and more surgery.

Levine calcification grade	Size	Surgery likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm, or multiple ≥1 cm	Increased

- **Intracavernosal injection (ICI) with or without penile duplex Doppler ultrasonography (PDDUS)** is required by the AUA **before any invasive treatment**. It measures curvature and gives peak systolic velocity, end-diastolic velocity and resistive index (RI).

5 · Non-surgical management

SAQ Management of Peyronie's disease

Paper answer · P1

Type: SAQ (10 marks)
Source: 2017-Winter Paper 1, Q3a

Definition

- **PD (Peyronie's disease)** (AUA (American Urological Association) 2015): acquired fibrosis of the tunica albuginea, with pain, deformity, ED (erectile dysfunction) and/or distress. Prevalence 3–20%; up to half worsen untreated.

Phases

- **Acute (active, first 12–18 months):** pain, worsening deformity; aim to stop progression. **Stable:** pain gone, deformity unchanged for ≥3–6 months; aim to straighten.

Evaluation

- History: pain, deformity, Dupuytren disease, ability to penetrate. **Key question:** "If your penis were straight, would your current rigidity allow intercourse?" (predicts postoperative function).
- Plaque, SPL (stretched penile length); **office intracavernosal injection ± duplex Doppler ultrasound** before any invasive treatment (curvature by goniometer, calcification, vascular status).
- **Kelami grading:** ≤30° mild, 31–60° moderate, >60° severe; note hourglass or hinge. PDQ (Peyronie's Disease Questionnaire).

Non-surgical treatment

Option	Evidence/position
NSAIDs (non-steroidal anti-inflammatory drugs)	For pain (only oral agent AUA supports)
Oral vitamin E, tamoxifen, potaba, pentoxifylline	No reliable curvature benefit; not recommended
CCH (collagenase *Clostridium histolyticum*)	Stable disease, curve 30–90°, intact erections. **IMPRESS I/II:** 34% improvement vs 18% placebo. Up to 4 cycles (2 injections of 0.58 mg) with modelling. Risk: corporal fracture

Option	Evidence/position
Verapamil, interferon α-2b (intralesional)	May offer; stabilise and modestly improve
Traction therapy	Gains length, reduces curvature; useful in acute phase
ESWT (extracorporeal shock wave therapy)	Pain only; no effect on curvature

Surgery

- **Indications:** stable disease (≥12 months, stable ≥6 months), deformity preventing intercourse, failed conservative therapy.

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Erection adequate (± PDE5I (phosphodiesterase-5 inhibitor))? ├─ No → penile prosthesis (inflatable) + manual modelling; │ residual >30° → tunical incision; graft if defect >2 cm └─ Yes → curve <60–70°, no hourglass/hinge → plication (Nesbit, 16-dot, tunica albuginea plication) curve >60–70°, hourglass, hinge, short penis → plaque incision/partial excision + graft (pericardium, small intestinal submucosa)
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- **Plication** shortens the convex side: simple, low ED risk, but loses length.
- **Grafting** restores length and girth but carries higher ED risk (5–54%).

Diagram to draw: Nesbit wedge excised opposite the plaque.

> **Key points to remember:**
> - Ask the "straight penis" question; duplex before surgery.
> - CCH: IMPRESS, stable 30–90° curve, intact erections.
> - Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Also, from the condensed book (not in the paper answer):

- **Oral**: the AUA "should not offer" any except **NSAIDs for pain**; vitamin E, tamoxifen, colchicine and carnitine give no benefit.
- **Extracorporeal shock wave therapy (ESWT)** helps pain, not curvature; radiotherapy is harmful.

Collagenase Clostridium histolyticum (CCH, Xiaflex)

- It is the only PD drug approved by the Food and Drug Administration (FDA, 2013). It degrades collagen **types I and III** and spares elastin, vessels and nerves.
- **Protocol (IMPRESS)**: 2 injections of **0.58 mg** 24–72 h apart, then **model** the penis 24–72 h after the second. Give up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling is essential.
- **AUA criteria** are stable disease, curvature **>30° and <90°** and intact erections (with or without medication). IMPRESS excluded ventral plaques, extensive calcification and disease <12 months.
- Response is poor with calcification, curve ≥60°, severe indentation or hinge.
- Bruising and pain are common; the serious risk is **corporal rupture** (0–4.9%), so there is no intercourse until modelling ends.

Other intralesional agents

Agent	Mechanism	Evidence	AUA
Verapamil	Calcium channel blocker (CCB): reduces fibroblasts and extracellular matrix, increases collagenase	Randomised trials conflicting; stabilises; inferior to CCH; cheap	May offer
Interferon α-2b	Reduces fibroblasts and collagen, increases collagenase	Randomised controlled trial (RCT) vs saline: curvature reduced 27% vs 9%; flu-like symptoms (pre-treat with NSAIDs)	May administer
Nicardipine	CCB	Reduces pain and plaque; no curvature benefit	—

- Verapamil is given as 10 mg diluted (10–20 mL).

Mechanical therapy

- **Penile traction therapy (PTT)** in the acute phase reduces curvature, pain and the need for surgery, and gains length; it needs at least 3 h/day.
- The AUA makes no recommendation for traction, vacuum or combinations.

6 · Surgery

Indications

- Disease is **≥12 months from onset, with deformity stable ≥6 months**.
- Deformity or poor rigidity prevents intercourse, and conservative therapy has failed.

Consent

- The goal is a **functionally straight** penis (≤20°; European Association of Urology, EAU, ≤15°). Residual or recurrent curvature occurs in up to 16%.
- **Length loss** follows every procedure (plication more than grafting); 70–80% have already lost length, so document SPL.
- **New ED** is commoner after grafting than plication (~30% if pre-op rigidity is suboptimal); pre-op erection quality predicts it best.

Choice of procedure — algorithm

ALGORITHM

1. Ask: is the erection adequate (with or without a PDE5I)?
2. If **no**, place an inflatable prosthesis with modelling. If the curve persists, incise with or without a graft (graft if defect >2 cm).
3. If **yes**, do PDDUS. If the curve is simple and <60–70°, with no hourglass or hinge and acceptable length loss, **plicate** (if length loss is unacceptable, graft).
4. If **yes** but the curve is complex/multiplanar >60–70°, or there is severe hourglass, hinge or a short penis (<9 cm stretched), do **plaque incision/partial excision with graft** if the ED risk is accepted (otherwise prosthesis).

Plication (shortening)

- It shortens the **convex side opposite the plaque**. It is simple, with the least new ED, but does not fix hourglass, hinge or shortening.

Technique	Method	Point
Nesbit (1965)	Transverse elliptical full-thickness excision opposite the curve, closed transversely	Original; for a ventral curve, lift Buck fascia and take dorsal wedges
Yachia	Vertical 0.5–1.5 cm incision closed transversely (Heineke–Mikulicz)	No tunica removed; long incisions narrow the shaft
Essed–Schroeder	Non-absorbable figure-of-eight, buried knots	First incisionless plication
16-dot	Permanent extended Lembert, 4 dots per plication	Popular; no incision
Tunica albuginea plication (TAP, Levine)	Paired partial-thickness transverse incisions (outer layer only); buried permanent suture and 3-0 polydioxanone (PDS)	Cavernosal tissue undisturbed

Plaque incision/partial excision and graft (lengthening)

- It suits complex deformity with **strong erections** and no veno-occlusive dysfunction, and restores calibre and 0.5–3 cm of length.
- ED predictors are age >55, veno-occlusive dysfunction (RI <0.80), a large graft, a ventral curve and a curve >60°.
- **Key steps**: elevate Buck fascia with the neurovascular bundle, make an H/double-Y incision **crossing the septum**, and sew in the graft (pericardium or small intestinal submucosa) with running 4-0 PDS.

Prosthesis with straightening

- PD with PDE5I-refractory ED is treated in one operation; an **inflatable** device is preferred.
- **Manual modelling** (Wilson–Delk): inflate fully, clamp the tubing and bend opposite the curve for 60–90 s, with the hand on the shaft, not the glans. It straightens 86–100%.
- Residual curvature >30° needs a relaxing incision, with a graft if the defect is >2 cm. Length loss is the commonest complaint (see Ch 75).

7 · Landmark trials

Trial	Design	Result	Take-home
IMPRESS I and II	Phase III RCT, CCH vs placebo 2:1, curvature 30–90°	Curvature reduced 34% (17°) vs 18% (9.3°); bother improved	Basis of FDA approval; modelling essential
Interferon α-2b multicentre RCT	Intralesional interferon (IFN) vs saline	Curvature reduced 27% vs 9%	First RCT of intralesional therapy

★ Putting it together: andrology decisions

A · Infertile man: work-up

ALGORITHM

1. **History and examination**: testosterone use, mumps, cryptorchidism; testis volume, vasa, varicocele (see Ch 69).
2. **Two semen analyses** (abstinence 2–7 days). Low volume with acid pH suggests duct obstruction or congenital bilateral absent vas (CBAVD).
3. **Hormones**: follicle-stimulating hormone (FSH) and testosterone; if testosterone is low, add luteinising hormone (LH) and prolactin.
4. **Genetics** (azoospermia, severe oligozoospermia): karyotype, Y microdeletions; CFTR in CBAVD.

B · Azoospermia: obstructive or not

	Obstructive (OA)	Non-obstructive (NOA)
Testis	Normal volume	Small, soft
FSH	Normal (≤7.6 IU/L)	Raised (>7.6 IU/L)
Clues	Absent vasa, low volume, vasectomy or hernia repair	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy
Treatment	Vasovasostomy (VV) or vasoepididymostomy (VE), resection of ejaculatory ducts, or sperm retrieval for ICSI (see Ch 70)	Micro-TESE with ICSI; fix a clinical varicocele first; no retrieval if complete AZFa/b deletion

KEY POINT

† Reversal suits a younger partner wanting >1 child; ICSI suits an older partner or female factor. Copious vas fluid or sperm means VV; toothpaste-like fluid or none means VE (see Ch 70).

C · Erectile dysfunction ladder

ALGORITHM

1. **Assess:** IIEF-5 score, HbA1c, lipids, morning testosterone. ED is a cardiovascular sentinel, so stratify cardiac risk (see Ch 72).
2. **Modify** risk factors; treating hypogonadism rescues PDE5 inhibitor (PDE5I) failures.
3. **Shared decision:** PDE5I (never with nitrates), vacuum, intraurethral alprostadil or intracavernosal injection.
4. **Penile prosthesis** (inflatable three-piece) if others fail; highest satisfaction. (see Ch 75).
5. **Young, focal post-traumatic arterial occlusion** → revascularisation.

D · Priapism: ischaemic vs non-ischaemic

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Clinical	Rigid, painful	Semi-rigid, painless; after perineal trauma
Cavernous blood gas	pO_2 <30, pCO_2 >60 mm Hg, pH <7.25	Like arterial blood: pO_2 >90, pCO_2 <40
Doppler	Little or no flow	High flow, fistula or pseudoaneurysm
Treatment	Aspirate with or without irrigation, then phenylephrine (100–500 µg every 3–5 min, up to 1 h), then distal shunt with or without tunnelling, then proximal shunt. >36 h or refractory: early prosthesis	Observe; selective embolisation if persistent

KEY POINT

Sickle cell: treat as above without waiting for transfusion. Stuttering priapism: daily PDE5I, nightly pseudoephedrine, or hormonal suppression after puberty (see Ch 73).

E · Peyronie disease

ALGORITHM

1. **Active**: analgesia, traction. Collagenase (CCH): stable dorsal/lateral 30–90°, intact erections; else verapamil, interferon α-2b.
2. **Surgery**: ≥12 months from onset, stable ≥6 months (see Ch 76).
3. If **erections are good**: simple <60–70°, no hourglass/hinge, adequate length → plication; complex, hourglass/hinge or short → incision/partial excision + graft (more ED).
4. If **erections are poor** despite PDE5I → inflatable prosthesis with modelling.

F · Premature ejaculation (PE)

KEY POINT

Diagnosis needs short latency (lifelong ≈1 min or less), poor control and distress. Treat ED first. Options: stop–start or squeeze, lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral drug outside the US), daily SSRI (paroxetine best), tramadol (off-label, addiction risk) (see Ch 74).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ Oligospermia and oligoasthenozoospermia

Paper answer · P2 · P3 · merged from 2

Asked as: P3 SAQ "Oligospermia and oligoasthenozoospermia" (asked as "Oligoanaesthospermia" and "Oligospermia") · P2 SAQ "Oligospermia."

Assumed meaning (oligoanaesthospermia): oligoasthenozoospermia (low sperm count with poor motility).

Definition (WHO [World Health Organization] 2021, 6th edition, 5th centiles)

- **Oligozoospermia:** sperm concentration below the lower reference limit: **WHO 2021 (6th edition) 16 million/mL** (5th centile; total count 39 million per ejaculate). Earlier thresholds were 20 million/mL (1999, the legacy cut-off) and 15 million/mL (2010).
- **Severity:** mild to moderate 5–16 million/mL; **severe oligozoospermia <5 million/mL**; cryptozoospermia: sperm only in a centrifuged pellet.

Parameter (WHO 2021 5th centiles)	Lower reference value
Concentration	16 million/mL (total 39 million)
Volume	1.4 mL
Progressive motility	30%
Total motility	42%
Normal forms (strict)	4%
Vitality	54%

- These centiles come from fertile men; they are a reference, not pass/fail fertility cut-offs. Define oligospermia in context: read a single value with testis size, FSH (follicle-stimulating hormone) and repeated values.

Causes

- **Varicocele** (commonest correctable; heat and oxidative stress) and **idiopathic** (largest group).
- **Pre-testicular (hormonal):** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, obesity (raised oestradiol), hyperprolactinaemia, opioids.
- **Testicular: varicocele**, cryptorchidism, mumps orchitis, torsion, chemotherapy or radiotherapy, gonadotoxins; genetic (Klinefelter, Klinefelter mosaic, **AZFc [azoospermia factor c] deletion**).
- **Post-testicular (motility):** infection or **leucocytospermia** (over 1 million/mL), antisperm antibodies, partial ejaculatory duct obstruction (EDO; low volume), PCD (primary ciliary dyskinesia).
- **Lifestyle:** smoking, alcohol, cannabis, heat, drugs, gonadotoxins.

Evaluation

- **At least two semen analyses, 2–3 weeks apart**, same abstinence (WHO 2–7 days of abstinence; parameters peak at 1–2 days).
- History and examination: testis size/volume, varicocele, vasa, secondary sex characters.

- Morning **testosterone, LH (luteinising hormone), FSH**; oestradiol and prolactin if testosterone is low.
- **Under 5 million/mL** (especially with raised FSH or small testes): **karyotype and Y-microdeletion** testing (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine] 2020 guideline).
- Selected: vitality test (eosin or hypo-osmotic swelling), antisperm antibodies, semen culture, scrotal Doppler, TRUS (transrectal ultrasonography) if volume is low (<1.0 mL), DNA fragmentation.

Management

- **General:** stop smoking, weight loss, avoid heat; **stop exogenous testosterone** and gonadotoxins; treat infection.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen; it improves semen parameters and pregnancy rates. Never treat a subclinical one.
- **Medical:**
- Hypogonadotropic: hCG (human chorionic gonadotropin) ± FSH.
- Idiopathic with low testosterone: off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if the testosterone:oestradiol ratio is low (<10).
- Antioxidants: weak evidence.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe or failed IUI: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc carriers because counts may fall.
- Spermatogenesis takes about 64 days, so judge response (reassess) after 3–6 months.

Diagram to draw: hypothalamic–pituitary–testis axis with sites of treatment.

> **Key points to remember:**
> - WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
> - Karyotype and Y-microdeletion if under 5 million/mL.
> - Repair palpable varicoceles only; never give testosterone.

SAQ Avanafil

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3e

Introduction

- Second-generation oral **PDE5I (phosphodiesterase type 5 inhibitor)**; **FDA (US Food and Drug Administration) approved 2012**; Europe 2013. Structure differs most from the other PDE5Is.

Mechanism

- Sexual stimulation releases NO (nitric oxide), which raises cGMP (cyclic guanosine monophosphate) and relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; avanafil blocks this.
- **Enhances, does not initiate, erection**: stimulation is needed.

- **More selective for PDE5** over PDE6 (retina) and PDE1 than sildenafil, so **no visual disturbance** in the comparison table.

Pharmacokinetics (Campbell Table 72.7)

Feature	Avanafil	Sildenafil	Tadalafil
Time to peak	**0.3–0.5 h (fastest)**	0.8 h	2 h
Half-life	3–5 h	3–5 h	17.5 h
Fatty food	Reduces absorption	Reduces	No effect
Doses	**50, 100, 200 mg**	25–100 mg	5–20 mg
Blue vision	No	Yes	Rare

- Taken about 15–30 minutes before sex; start 100 mg; maximum once daily. Metabolised by CYP3A4 (cytochrome P450 3A4).

Efficacy

- Effective in general ED (erectile dysfunction), **diabetes** and **after nerve-sparing radical prostatectomy** (REVIVE trial programme): improves IIEF (International Index of Erectile Function) erectile scores and successful-intercourse rates, some within 15 minutes. Efficacy equivalent across PDE5Is; no head-to-head superiority.

Adverse effects

- Headache, flushing, nasal congestion, dyspepsia; back pain rare.

Contraindications and interactions

- **Nitrates in any form**: absolute; no nitrate within 12 hours of a dose.
- Avoid strong CYP3A4 inhibitors (ketoconazole, ritonavir); maximum 50 mg with moderate inhibitors.
- **α-blockers:** start 50 mg once stable (additive hypotension).
- Caution: aortic stenosis, recent myocardial infarction or stroke, hypotension, retinitis pigmentosa. Not recommended in severe renal impairment, dialysis or severe hepatic impairment.
- **Also know:** proposed prophylaxis for recurrent ischaemic priapism; can cause priapism rarely.

Advantages

- **Fastest onset**, fewer visual and muscle effects; suits on-demand spontaneous use.

Diagram to draw: NO–cGMP pathway in cavernosal smooth muscle with the PDE5 block.

> **Key points to remember:**
> - Peak 0.3–0.5 h; doses 50, 100, 200 mg.
> - Highly PDE5-selective: no blue vision.
> - Nitrates absolutely contraindicated.

SAQ CRE & BEP

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3f

"CRE" is read as carbapenem-resistant Enterobacterales, and BEP as bleomycin, etoposide and cisplatin chemotherapy for germ cell tumours; the two are answered separately.

Part A: CRE (carbapenem-resistant Enterobacterales)

- **Definition:** Enterobacterales (Klebsiella pneumoniae, Escherichia coli, Enterobacter) resistant to at least one carbapenem, or producing a carbapenemase.

Mechanism	Example
Class A carbapenemase	KPC (Klebsiella pneumoniae carbapenemase)
Class B metallo-β-lactamase	**NDM (New Delhi metallo-β-lactamase)**: common in India
Class D	OXA-48
Non-enzymatic	ESBL (extended-spectrum β-lactamase) or AmpC plus porin loss

- **Urological risk:** catheters, stents, prolonged stay, prior carbapenem or fluoroquinolone use, transrectal biopsy, ICU (intensive care unit) care.
- **Treatment** (by culture and carbapenemase type; ID (infectious disease) input):
- KPC, OXA-48: ceftazidime–avibactam; meropenem–vaborbactam or imipenem–relebactam for KPC.
- **NDM:** ceftazidime–avibactam plus aztreonam, or cefiderocol.
- Cystitis: single-dose aminoglycoside, fosfomycin; polymyxins as last resort. Avoid tigecycline for UTI (urinary tract infection) (low urine levels).
- **Prevention:** antibiotic stewardship, contact precautions, rectal screening of high-risk patients, early catheter removal, transperineal rather than transrectal biopsy.

Part B: BEP (bleomycin, etoposide, cisplatin)

- Standard first-line chemotherapy for metastatic **GCT (germ cell tumour)**; replaced PVB (less neuromuscular toxicity).
- **Usual 21-day cycle:** cisplatin 20 mg/m² days 1–5, etoposide 100 mg/m² days 1–5, bleomycin 30 units days 1, 8, 15.

IGCCCG (International Germ Cell Cancer Collaborative Group) risk	Regimen
Good	BEP ×3 or EP ×4
Intermediate/poor	BEP ×4 (VIP ×4 if lungs poor)
Stage I NSGCT (non-seminomatous GCT), high risk	BEP ×1 adjuvant

- **Toxicity:** bleomycin lung fibrosis (keep FiO_2 low at later RPLND (retroperitoneal lymph node dissection)) and Raynaud phenomenon; cisplatin nephrotoxicity, ototoxicity, neuropathy; myelosuppression; infertility (bank sperm); late cardiovascular disease and second cancers.

Diagram to draw: BEP 21-day cycle timeline.

Key points to remember:

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low FiO_2 and restricted fluids at surgery.

SAQ Tadalafil

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3b

Pharmacology

- Selective **PDE5 (phosphodiesterase type 5) inhibitor**; FDA (Food and Drug Administration) approved 2003.
- **Mechanism:** sexual stimulation releases NO (nitric oxide), raising cGMP (cyclic guanosine monophosphate), which relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; tadalafil blocks this. It **augments, not initiates,** erection.
- Structurally distinct from sildenafil; cross-reacts with **PDE11**, little with PDE6 (retina).

Property	Tadalafil (vs sildenafil)
Time to peak	2 h (0.8 h)
Half-life	**17.5 h** (3–5 h)
Fatty food	No effect (reduces absorption)
Window	Widest (up to 36 h)

Uses and doses

1. **ED (erectile dysfunction):** 10–20 mg on demand, or 2.5–5 mg once daily. Intercourse success about 70%; 40–50% in diabetes and after prostatectomy.
2. **BPH (benign prostatic hyperplasia) with LUTS (lower urinary tract symptoms), with or without ED: 5 mg daily**, the only PDE5 inhibitor licensed. IPSS (International Prostate Symptom Score) falls about 6 points (placebo 3.6) **without Qmax (peak flow rate) change**; mechanism is NO–cGMP relaxation, perfusion and reduced afferent activity, not de-obstruction. Equal to tamsulosin for IPSS. EAU (European Association of Urology) 2022 and AUA (American Urological Association) 2021: option for moderate–severe LUTS.
3. **Also know:** fixed finasteride–tadalafil (FDA 2021); penile rehabilitation (benefit unproven); prevention of stuttering priapism (PDE5I paradox); pulmonary arterial hypertension.

Adverse effects

- Headache, dyspepsia, **back pain and myalgia** (characteristic), flushing, nasal congestion. Visual effects rare.
- NAION (non-arteritic anterior ischaemic optic neuropathy): no proven link; stop if sudden visual loss.

Contraindications and interactions

- **Nitrates in any form: absolute.** After tadalafil, no nitrates for **48 h** (24 h for sildenafil).

- Caution: α-blockers (start lowest dose of the added drug), aortic stenosis, hypotension. CYP3A4 inhibitors (ketoconazole, ritonavir) raise levels.
- Not studied: MI (myocardial infarction) or stroke within 6 months, unstable angina, BP <90/50 or >170/100 mmHg, retinitis pigmentosa.

Diagram to draw: NO–cGMP–PDE5 pathway in cavernosal smooth muscle.

> **Key points to remember:**
> - Half-life 17.5 h; food-independent; back pain.
> - 5 mg daily for LUTS: IPSS improves, Qmax does not.
> - No nitrates for 48 h after a dose.

SAQ Sertoli cell only syndrome

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3a (non-obstructive azoospermia also answered in 2025-Winter Paper 2, Q3a)

Definition

- SCOS (Sertoli cell only syndrome), **germ cell aplasia** (Del Castillo syndrome): seminiferous tubules lined only by Sertoli cells, with **complete loss of spermatogonia**. Leydig cells usually normal.
- The worst histological pattern of NOA (non-obstructive azoospermia).

Types

Type	Feature
Pure (complete, congenital)	All tubules lack germ cells
Mixed (focal, acquired)	Some tubules keep foci of spermatogenesis, the basis of sperm retrieval

Aetiology

- **Developmental:** primordial germ cells fail to migrate into the testis.
- **Genetic: complete AZFa (azoospermia factor a) deletion** gives pure SCOS; **AZFc** may give SCOS with foci; Klinefelter (47,XXY).
- **Acquired:** cryptorchidism, chemotherapy or radiation, mumps orchitis, torsion; idiopathic.

Clinical features

- Normal virilisation; **azoospermia** with normal ejaculate volume; small soft testes.

Evaluation

- Semen: azoospermia on two centrifuged samples.
- **FSH (follicle-stimulating hormone) high**, **inhibin B low** (Sertoli marker); LH (luteinising hormone) normal or high; testosterone normal or low.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- **Karyotype and Y microdeletion** before surgery.
- **Histology** confirms, usually at sperm retrieval.

Pattern	Meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages, fewer	Best
Maturation arrest	Stops at one stage	About 40%
SCOS	No germ cells	**About 30%**

Management

- **Counsel** on genetic findings.
- **Complete AZFa or AZFb deletion:** retrieval essentially nil; do not operate; donor sperm or adoption.
- **Micro-TESE (microdissection testicular sperm extraction):** at ×20–25, **larger, opaque, yellow tubules** (foci of spermatogenesis) are selected; SCOS tubules are thin, white and stringy. Live births are reported even in SCOS.
- Sperm used for **ICSI (intracytoplasmic sperm injection)**.
- Optimise first: stop exogenous testosterone; hormonal stimulation evidence weak.
- **Also know:** spermatogonial stem cell transplantation and in-vitro spermatogenesis are experimental.

Diagram to draw: tubule cross-section: normal layered epithelium versus Sertoli cells only.

> **Key points to remember:**
> - Complete loss of spermatogonia; high FSH, low inhibin B.
> - Complete AZFa deletion: pure SCOS, no retrieval.
> - Micro-TESE retrieval about 30%: worst histology, not hopeless.

SAQ Oxidative stress and male infertility

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3b

Definition

- **Oxidative stress (OS):** imbalance where ROS (reactive oxygen species) exceed the antioxidant capacity of semen. The term **MOSI (male oxidative stress infertility)** has been proposed.
- Low ROS levels are **physiological**: needed for capacitation, hyperactivation, acrosome reaction and sperm–oocyte fusion.

Sources of ROS

Source	Examples
Leucocytes (main)	Infection, prostatitis, epididymitis, leucocytospermia
Immature sperm	Excess residual cytoplasm
Varicocele	Heat and hypoxia
Lifestyle	**Smoking**, alcohol, obesity, heat
Others	Ageing, diabetes, pollutants, radiation

Why sperm are vulnerable

- Membrane rich in **polyunsaturated fatty acids** → lipid peroxidation.
- Little cytoplasm, so few antioxidant enzymes; limited DNA repair.
- **Seminal plasma antioxidants:** superoxide dismutase, catalase, glutathione peroxidase; taurine, vitamins C and E, zinc.

Effects

- Lipid peroxidation → loss of **motility** and membrane fluidity.
- **Sperm DNA fragmentation** → poor embryo quality, miscarriage, lower ART (assisted reproductive technology) success.
- Mitochondrial damage and apoptosis.

Evaluation

- Semen analysis: peroxidase-positive WBC (white blood cells) **>1 million/mL** = leucocytospermia.
- **Direct ROS:** chemiluminescence (luminol).
- **Indirect:** total antioxidant capacity, malondialdehyde (lipid peroxidation), ORP (oxidation-reduction potential).
- **DNA fragmentation tests:** TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labelling), SCSA (sperm chromatin structure assay), SCD (sperm chromatin dispersion), Comet.
- Use selectively (unexplained infertility, recurrent miscarriage, failed ART); not routine.

Management

- **Lifestyle:** stop smoking, reduce weight and alcohol, avoid heat.
- **Treat cause:** antibiotics for infection; **varicocelectomy** for clinical varicocele (reduces DNA fragmentation).
- **Antioxidants** (vitamins C and E, CoQ10 [coenzyme Q10], zinc, selenium, carnitine): Cochrane review, low-quality evidence; **MOXI trial (2020)** showed no gain in semen parameters, DNA fragmentation or live birth. Avoid excess: **reductive stress** also harms.
- High DNA fragmentation with failed ICSI (intracytoplasmic sperm injection): **testicular sperm** has less fragmentation.

Diagram to draw: balance of ROS sources against seminal antioxidants, with effects on sperm.

> **Key points to remember:**
> - Leucocytes and immature sperm are main ROS sources.
> - Lipid peroxidation and DNA fragmentation are the key injuries.
> - Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

SAQ Dhat syndrome

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3c

Definition

- **Dhat syndrome:** a **culture-bound syndrome** of South Asia. The man believes he is losing **semen ("dhat")** in urine or nocturnal emissions, and attributes weakness, fatigue and sexual problems to it.
- Term coined by **Wig** (India); listed in ICD-10 (International Classification of Diseases, 10th revision) and in the DSM-5 (Diagnostic and Statistical Manual, 5th edition) glossary of cultural concepts of distress.
- Related syndromes: **Shen-kui** (China), **Koro** (genital retraction).

Cultural basis

- Traditional belief that semen is a **precious vital fluid** made from many drops of blood; its loss is felt to drain body and mind.

Epidemiology

- Young men, often unmarried or newly married, rural, lower education; common in urology and sexual medicine clinics.

Clinical features

- **Somatic:** fatigue, weakness, aches, poor appetite.
- **Psychological:** anxiety, low mood, guilt (about masturbation), poor concentration, sleep loss.
- **Sexual:** fear of ED (erectile dysfunction) or PE (premature ejaculation), worry about penis size and fertility.
- Whitish urine is usually phosphaturia, oxaluria or normal secretions.
- **Comorbidity: depression** and **anxiety** in many; somatisation.

Evaluation

- Non-judgmental sexual history and belief system.
- Examination of genitalia; urine microscopy (phosphates, oxalates, sperm, pus cells).
- Exclude organic disease: **UTI (urinary tract infection), prostatitis, STI (sexually transmitted infection), diabetes**, true spermatorrhoea.
- Screen for depression and anxiety.

Management

1. Counselling (main)

- **Sex education:** semen is continuously produced; nocturnal emission and masturbation are normal and harmless.
- Reassure without dismissing.

2. Psychological therapy: CBT (cognitive behavioural therapy) for misbeliefs and anxiety.

3. Drugs

- **SSRIs (selective serotonin reuptake inhibitors)** for depression, anxiety or associated PE; short anxiolytic course if needed.
- Treat genuine ED or PE.

4. Avoid harm

- Avoid unnecessary "tonics", antibiotics and repeated tests; they reinforce the belief.
- Refer to psychiatry when depression is significant.

Prognosis

- Good with counselling; drop-out is common.

Diagram to draw: flowchart: complaint of semen loss → exclude organic cause → counsel → treat depression or anxiety.

> **Key points to remember:**
>
> - Culture-bound semen-loss anxiety of South Asia.
> - Depression and anxiety common; exclude UTI, prostatitis, STI.
> - Counselling and sex education first; SSRIs for comorbidity.

Rapid Revision Sheet · Section 9

Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias (LAQ)

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

Spermatogenesis, azoospermia and non-obstructive azoospermia (LAQ)

- Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
- Confirm azoospermia on two centrifuged samples.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b complete deletion: do not operate (no retrieval).
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

Male reproductive immunology and immunobead test (SAQ)

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- Treatment of choice: IVF/ICSI.
- Leukocytospermia >1 million/mL.

Retrograde ejaculation (SAQ)

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

Male menopause (andropause, late-onset hypogonadism) (SAQ)

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Discuss the role of urologist in reproductive endocrinology and male infertility practice. (LAQ)

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Semen analysis nomenclature. (SAQ)

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.

Genetic causes of male infertility. Enumerate and classify. (SAQ)

- Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
- AZFa or AZFb complete deletion: do not attempt retrieval.
- CBAVD = genital form of cystic fibrosis; test both partners.

Ejacultory duct obstruction (SAQ)

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

Klinefelter's syndrome. (SAQ)

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

CBAVD (SAQ)

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

Azoospermia: testicular biopsy and surgical sperm retrieval techniques (LAQ)

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
- OA: normal FSH and testes; ICSI essential.
- OA: MESA (capillary action); MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA: 23 G needle, suction, local anaesthesia; for OA with normal spermatogenesis.
- TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best; finds sperm in about half of NOA; TESA yield poor.
- AZFa/AZFb deletion: 0% retrieval (no sperm), do not operate.
- All surgically retrieved (testicular) sperm need ICSI; cryopreserve extras.
- Haematoma commonest; epididymal biopsy most serious.

Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED (LAQ)

- nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

Surgical management of ED and penile prostheses (types, biofilms) (LAQ)

- Prosthesis is the last line but gives 80–90% satisfaction.
- 3-piece inflatable gives the best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; coated devices and no-touch technique; infected implant: remove all parts; Mulcahy salvage 84%.
- Satisfaction 80–90%; repairs 15% at 5 years.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

Priapism: diagnosis and management (LAQ)

- Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
- ED risk 0% ≤12 h, 100% ≥36 h; consider early prosthesis beyond 36 h.
- Beyond 36 h or refractory: early penile prosthesis (or analgesia and counselling).
- High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

Premature ejaculation (SAQ)

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Haematospermia (SAQ)

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

Management of Peyronie's disease (SAQ)

- Ask the "straight penis" question; duplex before surgery.
- CCH: IMPRESS, stable 30–90° curve, intact erections.
- Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Oligospermia and oligoasthenozoospermia (SAQ)

- WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
- Karyotype and Y-microdeletion if under 5 million/mL.

- Repair palpable varicoceles only; never give testosterone.

Avanafil (SAQ)

- Peak 0.3–0.5 h; doses 50, 100, 200 mg.
- Highly PDE5-selective: no blue vision.
- Nitrates absolutely contraindicated. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

CRE & BEP (SAQ)

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low FiO_2 and restricted fluids at surgery. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Tadalafil (SAQ)

- Half-life 17.5 h; food-independent; back pain.
- 5 mg daily for LUTS: IPSS improves, Qmax does not.
- No nitrates for 48 h after a dose.

Sertoli cell only syndrome (SAQ)

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

Oxidative stress and male infertility (SAQ)

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

Dhat syndrome (SAQ)

- Culture-bound semen-loss anxiety of South Asia.
- Depression and anxiety common; exclude UTI, prostatitis, STI.
- Counselling and sex education first; SSRIs for comorbidity. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 10

Testis, Penis & Urethra

78 Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

1 · Definition, boundaries and organs

LAQ **Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.**

Paper answer · P1

Type: LAQ (25 marks)
Source: 2024-Winter Paper 1, Q1 (related: 2019-Summer Paper 1, Q2; 2020-Summer Paper 1, Q3b)

Anatomy: definition and boundaries

- The space behind the parietal peritoneum. **Anterior:** posterior peritoneal reflection; **posterior:** posterior abdominal wall; **cranial:** diaphragm; **caudal:** extraperitoneal pelvis.

Group	Contents
Primarily retroperitoneal (formed there)	Kidneys, ureters, adrenals, aorta, IVC (inferior vena cava)
Secondarily retroperitoneal (mesentery fused to the wall)	Ascending and descending colon, duodenum D2–D3, pancreas (not the tail)
Vessels, lymphatics, nerves	Lumbar nodes, cisterna chyli, sympathetic trunks, autonomic plexuses, lumbosacral plexus

Posterior abdominal wall

- **Flank layers:** external oblique, internal oblique, transversus abdominis, transversalis fascia. **Posterior:** psoas, quadratus lumborum, erector spinae.
- **Lumbodorsal (thoracolumbar) fascia:** three layers that fuse into one at the **tip of the 12th rib**, the standard entry point for retroperitoneoscopy.
- **Ribs:** access above the 12th rib risks the pleura; the intercostal bundle runs under the rib above.

Fasciae and spaces (three embryological strata)

Space	Boundaries	Contents
Anterior pararenal	Posterior peritoneum and anterior renal fascia	Colon, duodenum, pancreas
Perirenal	Anterior (Toldt) and posterior (Zuckerkandl) laminae of Gerota's fascia	Kidney, adrenal, perirenal fat, hilum, upper ureter, gonadal vessels
Posterior pararenal	Posterior lamina and transversalis fascia	Fat only

- The laminae fuse laterally as the **lateroconal fascia**. The perirenal cone is **open inferiorly**, so urine, blood or pus track into the pelvis.
- Perirenal fat is fine and pale; pararenal fat is coarse and yellow-orange.

Vessels, lymphatics and nerves

- **Aorta** enters at T12. Unpaired anterior branches supply the gut (coeliac, SMA (superior mesenteric artery), IMA (inferior mesenteric artery)); paired lateral branches supply adrenals, kidneys (L1–L2) and gonads; lumbar arteries go to the wall.
- At the hilum the **renal vein lies anterior to the artery**. The **left renal vein is long** (gonadal, adrenal and lumbar tributaries); the right is short.
- Lymph flows right to left; **cisterna chyli** at L1–L2.
- **Nerves at risk:** sympathetic trunk (medial to psoas), genitofemoral (on psoas), subcostal, iliohypogastric, ilioinguinal (on quadratus); hypogastric plexus (ejaculation).

Principles of laparoscopic retroperitoneal surgery

1. **No natural cavity:** the working space must be created (finger and balloon dissection) before insufflation.
2. **Psoas is the key landmark:** it forms the floor. Keep it horizontal on screen; the kidney in Gerota's fascia lies anterior, and the peritoneum is pushed medially.
3. **Direct posterior route to the hilum:** the renal artery, posterior to the vein, is met first, allowing early arterial control.
4. **Peritoneum stays intact:** no bowel mobilisation or adhesiolysis; no spill of tumour, urine or pus into the peritoneum.
5. **Small space:** closely spaced ports, limited triangulation, lens smudging.
6. **Gas behaviour:** CO_2 (carbon dioxide) absorbs through fat, so hypercarbia and subcutaneous emphysema are commoner; haemodynamic swings are smaller than transperitoneal.

Preferred for	Limitations
Adhesions, peritoneal dialysis, morbid obesity, pregnancy; posterior tumours, pyeloplasty, cyst surgery, renal biopsy	Small space, unfamiliar landmarks, harder suturing, learning curve

- **Evidence:** randomised trials and meta-analysis: blood loss, complications and oncology similar to transperitoneal; the consistent advantage is **shorter operative time**.

Salient steps: retroperitoneal laparoscopic nephrectomy

1. **Position:** full flank (90°), table flexed, axillary roll; surgeon behind the patient.
2. **Access (open Hasson):** 1.5–2.5-cm incision just below the 12th rib tip; split muscle, open the lumbodorsal fascia; finger confirms psoas behind and Gerota-covered lower pole in front.
3. **Space creation:** balloon dilator (commercial, or Gaur glove balloon with 600–800 mL saline) anterior to psoas, posterior to Gerota's fascia; blunt-tip balloon-cuff trocar seals the port.
4. **Ports:** two more trocars under vision, one off the 12th rib tip, one posterior and superior to the camera.
5. **Dissection:** along psoas, find the ureter and gonadal vein medially; lift ureter and lower pole to put the hilum on stretch; see arterial pulsation.
6. **Hilum:** clip or staple the **artery first**, then the vein; on the left a **lumbar vein** usually needs division.

7. **Mobilise:** upper pole (adrenal preserved or taken), then lateral and anterior attachments; radical = outside Gerota's fascia.
8. **Extract:** clip ureter, bag the specimen, remove via an extended port; close. - **Problems:** peritoneal tear (space collapses), emphysema, pneumothorax, vascular injury, flank hernia (intramuscular balloon), rhabdomyolysis (about 1%). **Bail-out:** open the peritoneum and convert.

Diagram to draw: transverse section showing the three spaces, Gerota's laminae and the balloon anterior to psoas.

> **Key points to remember:**
> - Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
> - Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
> - Psoas is the landmark; artery met first.
> - Main advantage: shorter operative time, peritoneum untouched.

Also, from the condensed book (not in the paper answer):

- Primary organs lie in the perirenal space; secondary organs lie in the anterior pararenal space.
- Secondary organs exclude the first part of the duodenum (**D1**) and the pancreatic tail; the **anal canal** is primarily retroperitoneal.

> **KEY POINT**
>
> Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney.

2 · Posterior abdominal wall and surface landmarks

- The upper pole is at the 11th rib (left) and 12th (right, lower). The hilum is at **L1**; the lower pole at L3–L4.
- **Psoas major** (T12–L5 to lesser trochanter) flexes the hip with iliacus. **Quadratus lumborum** lies posterolateral to psoas and stabilises the 12th rib; psoas minor is inconstant.
- Dividing the **costovertebral (lumbocostal) ligament**, which tethers the 12th rib to L1–L2, frees the rib in posterior approaches.
- The **intercostal bundle** lies under the rib above (vein, artery, nerve from top down), so the nerve is least protected. Supracostal access may cross the pleura and has more complications than subcostal.
- **Dorsal lumbotomy** incises the lumbodorsal fascia vertically, lateral to erector spinae and quadratus, so **no muscle is cut**.

3 · Retroperitoneal fasciae and spaces

Collections spread along these spaces.

Three fetal strata form three fasciae and spaces: outer (transversalis; posterior pararenal), intermediate (Gerota; perirenal, genitourinary) and inner (fusion fascia; anterior pararenal,

gastrointestinal).

- The posterior Gerota lamina fuses medially with transversalis; this **must be incised to reach the hilum**.
- The Toldt (anterior) lamina is thinner than the Zuckerkandl (posterior) lamina.
- The perirenal cone may cross the midline below the hilum. Urinoma, haematoma and abscess are contained laterally but **track caudally**.
- The **anterior pararenal** space holds colon, pancreas and D2–D3, and is entered from the **white line of Toldt**.

4 · Gastrointestinal relations (operative hazards)

- **Right**: the pancreatic head lies in front of the inferior vena cava (IVC), and D2 lies on the right hilum, hence the **Kocher manoeuvre**. D3 crosses between the superior mesenteric artery (SMA) and the aorta.
- **Left**: pancreatic tail, spleen and stomach lie near the upper pole, and splenic vessels are at risk when the pancreas is lifted off renal fascia.

5 · Great vessels

Aorta and branches

The aorta enters at **T12**, left of the IVC, and bifurcates at **L4**.

Branch	Type	Level	Supplies
Inferior phrenic	Paired lateral	First, below hiatus	Diaphragm; superior adrenal
Celiac	Anterior	T12	Foregut
Middle adrenal	Paired lateral	SMA level	Adrenal
SMA	Anterior	L1–L2	Midgut
Renal	Paired lateral	L1–L2, below SMA	Kidney; inferior adrenal
Gonadal	Paired **anterolateral**	Below renals	Testis/ovary
Lumbar	Paired posterior	Four pairs, L1–L4	Wall, spine, cord
Inferior mesenteric (IMA)	Anterior	L3–L4	Hindgut

- Supernumerary renal arteries occur in ≥ ¼ (see Ch 84).
- **Ureteric supply**: proximal from renal; middle from aorta, common iliac and gonadal; distal from internal iliac. The abdominal ureter is supplied **medially**, the pelvic ureter **laterally**.
- IMA ligation is safe because the marginal artery of **Drummond** feeds the colon. **SMA ligation is not safe**: it causes severe bowel ischaemia.

IVC and veins

- The IVC passes the diaphragm at **T8**.
- The **left renal vein** crosses in front of the aorta below the SMA. It receives gonadal (right-angle entry), adrenal and ≥ 1 **lumbar vein** (avulsion risk). The **right** is short, without tributaries, and duplicated in one-sixth. Both enter the IVC at L1.
- The **right gonadal vein** enters the anterolateral IVC; the short **right adrenal vein** enters the posterolateral IVC.

- Left varicocele predominates (longer vein, perpendicular entry). A **new right varicocele** needs imaging for a retroperitoneal mass, e.g. renal cell carcinoma with venous thrombus.
- Dissect the right ureter **lateral** to the gonadal vein; medial dissection risks IVC injury and gonadal avulsion.
- The inferior mesenteric vein (**IMV**) lies in the left mesocolon and at reoperation is mistaken for the gonadal vein.

Variants (essential)

*See the paper answer **Retrocaval ureter and IVC anomalous development** (Section 4, Ch 52).*

Also, from the condensed book (not in the paper answer):

- † A retroaortic left renal vein (≈ 2–3%) is injured when lumbars are divided or the aorta rolled in retroperitoneal lymph node dissection (RPLND).
- † A circumaortic vein (≈ 5–7%) has a posterior limb that mimics a lumbar vein, so check the pre-operative CT.
- † Retrocaval ureter is a **venous** anomaly with a "reverse-J" on pyelogram, repaired by dismembered pyeloplasty in front of the IVC.

6 · Lymphatics

LAQ Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma

Paper answer · P1 · P2 · merged from 2

Asked as: P1 LAQ "Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications" (asked as "Lymphatic drainage of testis and principles for designing templates for RPLND" and "Anatomy of lymphatic drainage of testis and scrotum. Describe management implications of penile malignancies") · P2 LAQ "Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis."

The P1 second part (Summer 2021) is taken as the management implications of lymphatic drainage in penile cancer, i.e. management of the inguinal and pelvic nodes.

Embryological basis

- **Why retroperitoneal:** the testis develops at L1–L2 beside the mesonephros (near the kidney) and descends, carrying its vessels and lymphatics; its artery, vein and lymphatics stay retroperitoneal. So its lymph returns to where it began: the **retroperitoneal (lumbar) nodes near the renal hilum**, not the groin. The scrotum is skin and drains to the **groin**.

Lymphatic drainage of the testis

- **Route:** lymphatics of the testis and the caput and corpus of the epididymis run up the **spermatic cord** with the testicular (gonadal) vessels, through the internal ring and along the gonadal vessels to the lumbar nodes near the renal hilum (**para-aortic and interaortocaval nodes**). Lymph capillaries do not enter seminiferous tubules.
- **Onward:** lumbar trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta, behind the right crus; hence retrocrural nodes), then the **thoracic duct** to the left innominate vein. This explains **left supraclavicular** node metastases.
- Lymph flows **cephalad and right to left**.

- **Epididymis:** caput and corpus drain with the testis; **cauda epididymis** drains with the vas to the **external iliac** nodes.

Retroperitoneal nodal regions

Region	Extent
Right paracaval (with precaval, retrocaval)	IVC (inferior vena cava) midline to right ureter
Interaortocaval	IVC midline to aortic midline (between IVC and aorta)
Left para-aortic (with pre-aortic, retroaortic)	Aortic midline to left ureter

- **Also know:** ten mapped regions include right and left common iliac, interiliac and right and left suprahilar.

Primary landing zones

	Right testis	Left testis
Primary (landing zone)	**Interaortocaval** (below renal vessels)	**Left para-aortic** (and left hilar)
Secondary / next echelon	Right paracaval, precaval, then para-aortic	Interaortocaval; pre-aortic
Crossover / contralateral	Small but real to left para-aortic (another source: common; right to left occurs)	None significant; rare to the right, bulky disease only (another source: rare; implies bulk)

- **Crossover is only right to left**, matching the overall lymph flow.
- **Donohue mapping (1982):** contralateral and suprahilar spread is rare but rises with bulk; interiliac spread is very rare.
- **Pelvic, iliac and inguinal nodes** are involved only when drainage is altered (or with retrograde spread from bulky disease): previous scrotal or inguinal surgery (orchidopexy, hernia repair), scrotal violation, or tumour invading the tunica vaginalis or scrotum.
- The retroperitoneum is the first metastatic site in 70–80% (choriocarcinoma spreads by blood). CT (computed tomography) understages 25–35% of stage I NSGCT (non-seminomatous germ cell tumour); 5–9 mm landing-zone nodes are suspicious.

Lymphatic drainage of the scrotum

- Scrotal skin drains to the **ipsilateral superficial inguinal nodes**. Lymphatics, like vessels and nerves, **do not cross the median raphe**.

Clinical implications: testis and scrotum

- **Trans-scrotal orchidectomy or biopsy is contraindicated.** It leaves cord behind and opens scrotal skin lymphatic routes, so tumour can reach inguinal and pelvic nodes, causing local and inguinal or pelvic recurrence. Use **radical inguinal orchidectomy** with high cord clamping.
- Inguinal or pelvic spread from the testis means altered drainage: scrotal violation, previous orchidopexy or hernia repair, scrotal invasion.
- RPLND (retroperitoneal lymph node dissection) templates and para-aortic or "dog-leg" radiotherapy fields follow the landing zones.
- Scrotal skin cancer is staged at the ipsilateral groin.

Principles of designing RPLND templates

1. Map-based design: follow the side-specific landing zones to remove likely disease and spare the rest.

- Historical dissection: bilateral infrahilar + suprahilar + interiliac: most men anejaculatory.
- **Ray (1974)** first templated landing zones; **Donohue (1982)** mapping confirmed side-specific spread; **Weissbach and Boedefeld (1987)** reduced left template with frozen section.
- So a template must include the primary landing zone, the next echelon and the known crossover route.

2. Fixed boundaries

- **Superior:** renal hila (renal vessels). **Lateral:** ureters. **Inferior:** ureter crossing the common iliac artery. Always remove the **gonadal vein and cord stump** to the internal ring.

3. Omit suprahilar and interiliac regions in non-bulky disease, as spread there is rare. This reduces chylous ascites, renal, pancreatic and duodenal injury, and protects the hypogastric plexus.

4. Unilateral modified templates leave contralateral sympathetic fibres untouched. Because of right-to-left crossover, both templates take the para-aortic nodes **above the IMA (inferior mesenteric artery)**.

Template	Content
Right modified	Paracaval, precaval, retrocaval, interaortocaval, right common iliac; para-aortic **above IMA**; omit para-aortic below IMA
Left modified	Left para-aortic, preaortic, retroaortic, left common iliac; interaortocaval above IMA; omit paracaval/precaval/retrocaval
Full bilateral infrahilar	All three regions both sides (paracaval, precaval, retrocaval, interaortocaval, pre-aortic, para-aortic, retroaortic, both common iliacs), between renal vessels, ureters and ureteric crossing of iliacs
Historical extended	Plus suprahilar and interiliac (thoracoabdominal); most anejaculatory

- Omitting para-aortic above the IMA (right) or interaortocaval (left) is controversial.

5. Nerve sparing to preserve ejaculation: antegrade ejaculation (emission and bladder-neck closure) depends on **L1–L4 postganglionic sympathetic fibres** (about three each side) joining the **superior hypogastric plexus at or below the IMA** and running over the sacral promontory. Identify them prospectively before dividing tissue.

- **Unilateral templates** leave contralateral fibres untouched (antegrade ejaculation 75–97%). **Prospective nerve sparing** (Jewett 1988) identifies fibres: 90–100%.

6. Completeness ("duplicate open RPLND"): split-and-roll over the great vessels, **divide lumbar vessels** to clear retrocaval and retroaortic tissue, clip lymphatics at the cephalad limit (left renal vein) to prevent chylous ascites. Incomplete RPLND or mass excision gives worse outcomes (GETUG T93BP; 33% local relapse after post-chemotherapy mass-only resection).

7. Extend if needed: enlarged nodes at operation call for the full bilateral template (frozen section, Weissbach 1987). Convert to bilateral if nodes are suspicious at surgery or the primary has somatic-type malignancy.

Template by clinical setting

Setting	Template
Primary RPLND (high-risk stage I or low-volume stage II NSGCT, normal markers)	Default: modified unilateral nerve-sparing **or** full bilateral infrahilar nerve-sparing; no randomised trial. Modified allowed: low-volume, clinically negative nodes. Main risk: out-of-template nodes 3–23% (right > left)
PC-RPLND (post-chemotherapy), residual mass >1 cm, normal markers	**Full bilateral infrahilar + residual mass** (standard). Main risk: crossover with bulk; viable tissue unrecognisable
Modified PC template allowed only if	Mass ≤5 cm in landing zone, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good or intermediate risk

- Against unilateral templates: 3–23% of node-positive men have nodes outside the modified template, more with right tumours.
- *Sources differ:* Campbell testis-neoplasm chapter calls full bilateral nerve-sparing the standard; the surgery chapter accepts either.

RPLND outcomes and complications

- Primary RPLND: pN+ about 20–30%; cancer-specific survival (CSS) about 98.8–100%; retroperitoneal recurrence <2% after full template.
- **Antegrade ejaculation >90%** with templates plus nerve sparing (modified template alone: early 75–87%, refined 97%).
- Complications: ileus (commonest), wound infection, chylous ascites (0.2–2.1% primary; 2–12% post-chemotherapy), pulmonary (bleomycin lung toxicity; low FiO_2), ejaculatory loss.
- **Also know:** nephrectomy needed at PC-RPLND (left-sided mass >10 cm); GETUG T93BP showed incomplete RPLND predicts relapse.

Stage I seminoma: definition, staging and work-up

- Disease confined to the testis: any pT, N0, M0 on imaging. Stage IA = pT1; IB = pT2–4; IS = markers still raised after orchidectomy.
- **Work-up:** AFP (alpha-fetoprotein), β-hCG (human chorionic gonadotropin) and LDH (lactate dehydrogenase) before and after orchidectomy. CT abdomen and pelvis; chest CT or radiograph. Baseline testosterone; **offer sperm banking**. A raised AFP means non-seminoma.
- **Radical inguinal orchidectomy:** early cord control at the internal ring. It cures 80–85% of stage I seminoma.
- **Risk factors for relapse: tumour >4 cm** and **rete testis invasion**. Five-year relapse is 12%, 16% and 32% with 0, 1 and 2 factors. LVI (lymphovascular invasion) predicts relapse in non-seminoma, not seminoma.

Stage I seminoma: options after orchidectomy (cancer-specific survival about 100% with each)

	Surveillance	Carboplatin	Radiotherapy
Method	Clinical review, markers, CT	**One cycle, AUC (area under the curve) 7**	**20 Gy** in 10 fractions to the para-aortic strip
Relapse	14–20%	About 5%	3–5%

	Surveillance	Carboplatin	Radiotherapy
Relapse site	Retroperitoneum (84–100%)	Retroperitoneum	Pelvis, chest, left supraclavicular
Drawbacks	Late relapse, many scans, compliance	Underdosing; late effects unknown	Second cancers, GI (gastrointestinal) toxicity, oligospermia
Status	**Preferred**	If unwilling or unsuitable for surveillance; risk factors	If unwilling or unsuitable for surveillance

- **Carboplatin (Calvert):** 7 × (GFR [glomerular filtration rate] + 25) mg, using isotope GFR.
- **Trials: MRC TE18** (Medical Research Council): 20 Gy equals 30 Gy, with less toxicity. **MRC TE19/EORTC (European Organisation for Research and Treatment of Cancer) 30982:** one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE10: para-aortic field equals dog-leg, with 2% pelvic relapse.
- **Risk-adapted (EAU [European Association of Urology]):** surveillance for low risk; with risk factors discuss surveillance or carboplatin (treating all would overtreat two-thirds).

```
Inguinal orchidectomy + markers + CT │ Stage I seminoma │ Surveillance (preferred) ── unwilling / unreliable ──► Carboplatin AUC7 ×1(–2) │ (or radiotherapy 20 Gy) Relapse (mostly retroperitoneal) ├─ small nodes ► dog-leg radiotherapy with boost, or chemotherapy └─ bulky/distant ► BEP ×3 or EP ×4
```

- BEP: bleomycin, etoposide, cisplatin; EP: etoposide, cisplatin.

Stage I seminoma: relapse, IGCCCG and follow-up

- Relapses are almost always cured; median 14 months, but 10–20% occur after 4 years.
- **IGCCCG:** seminoma is good risk unless there are non-pulmonary visceral metastases (intermediate). **There is no poor-risk seminoma**, and markers do not change the risk group.
- **Surveillance follow-up:** examination and markers every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly; CT every 4–6 months (years 1–3), then 6-monthly to year 7, then yearly. CT finds >85% of relapses.

Lymphatic drainage of the penis

Superficial system	Deep system
Prepuce and shaft skin	Glans, frenulum, corpora
Dorsal network divides at the base and drains to both superficial inguinal groups	Trunks circle the corona and run under Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes

- **Two rules:** drainage is **bilateral and unpredictable in side** (presymphyseal crossover), but **orderly in sequence**: superficial inguinal, then deep inguinal, then external iliac and obturator. **No skip metastases.**
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**; sentinel nodes are mostly **superomedial**, then central. **Deep nodes** (1–3) lie medial to the femoral vein; **Cloquet's node** is the highest.

Management implications in penile cancer

1. Prognosis and timing

- Nodal status is the **strongest prognostic factor**. Penile SCC (squamous cell carcinoma) has a long **locoregional phase**, so the groin is a **curable station**.
- **Kroon (2005):** early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about **84% vs 35%** when delayed until palpable.
- Manage both groins **within 3 months**; remove **at least 8 nodes per side**.

2. Bilateral management

- Crossover means **both groins** are staged or treated; with one palpable groin, the other holds micrometastasis in about 30%.

3. Impalpable groin (cN0): risk-based

Risk	EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance if compliant, or surgical staging
High	**pT1b or higher**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if DSNB unavailable

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk and is staged surgically (ILND first).
- **DSNB:** ultrasound ± FNAC (fine-needle aspiration cytology) first, because a tumour-replaced node diverts flow (false negative). Then technetium-99m nanocolloid, blue dye and gamma probe.
- **No-skip spread and superomedial sentinels** justify **modified ILND (Catalona)**: smaller field, saphenous vein preserved; a positive frozen section leads to complete radical ILND.

4. Palpable nodes (cN+)

- About half are reactive, but do not delay with antibiotics. Ultrasound-guided **FNAC**; if positive, **radical ILND** on that side and stage the other.
- **Bulky (≥4 cm), fixed or bilateral nodes:** neoadjuvant **TIP** (paclitaxel, ifosfamide, cisplatin) × 4, then surgery in responders.

5. Pelvic nodes

- Stepwise spread means a negative groin needs no pelvic dissection.
- **PLND (pelvic lymph node dissection)**, ipsilateral, if **more than 2 inguinal nodes or ENE (extranodal extension)**. ENE in any node is pN3.

6. Complications of ILND

- Wound infection, **flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis). Reduced by modified templates, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.
- **Also know:** adjuvant therapy for pN2–3; Cabanas sentinel biopsy abandoned.

Recent advances

- **Robotic RPLND** for low-volume disease (same templates, same rules).
- RPLND for small-volume stage II seminoma (**SEMS, PRIMETEST**).
- Serum **miR-371a-3p** microRNA for selection, and it detects relapse better than standard markers.

- MRI (magnetic resonance imaging) instead of CT for seminoma surveillance (TRISST trial) to reduce radiation.

Diagrams to draw

- Retroperitoneum showing great vessels with the three nodal regions, right (interaortocaval) and left (para-aortic) landing zones and the right-to-left crossover arrow; outline the right and left modified templates, IMA level and hypogastric plexus.
- Beside it, the femoral triangle with Daseler's zones and Cloquet's node.

Key points to remember:

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila, ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND: full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum (ipsilateral superficial inguinal) and penis drain to inguinal nodes; never operate trans-scrotally.
- Seminoma relapse risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

Also, from the condensed book (not in the paper answer):

- Lumbar nodes are the primary drainage of kidney, adrenal, ureter and gonad.

KEY POINT

The testis is embryologically retroperitoneal, so it drains to where it began. Crossover is one-way (right to left), so right templates include interaortocaval and some para-aortic tissue. See Ch 80, 81.

7 · Nerves

Autonomic

- **Parasympathetic** outflow is craniosacral (III, VII, IX, X, S2–S4). **Sympathetic** outflow is thoracolumbar, **T1–L2**.
- **Sympathetic trunks** lie medial to psoas behind the great vessels, and are met when lumbar vessels are divided.
- **Ejaculation**: † postganglionic fibres from the **T12–L3** ganglia cross in front of the aorta and IVC and converge in the hypogastric plexus below the IMA. They drive **emission** and **bladder neck closure**, so extensive dissection causes anejaculation or retrograde ejaculation, while erection is spared. See Ch 81.

Lumbar (lumbosacral) plexus

Nerve	Roots	Course / motor	Sensory
Subcostal	T12	Below 12th rib	Skin over hip
Iliohypogastric	L1	Over quadratus; internal oblique, transversus	Gluteal, pubic
Ilioinguinal	L1	As above	Upper medial thigh, penile root, anterior scrotum
Genitofemoral	L1–L2	**On anterior psoas**; genital branch to cremaster	Anterior scrotum; upper anterior thigh
Lateral cutaneous thigh	L2–L3	Sensory only	Lateral thigh
Obturator	L2–L4	Medial to psoas, to obturator canal; adductors	Medial thigh
Femoral	L2–L4	Lateral to psoas; quadriceps	Anterior thigh, medial leg

Nerve	Setting	Result of injury
Genitofemoral	Psoas hitch, laparoscopic varicocelectomy, RPLND	Anterior scrotal/thigh numbness
Obturator	Lateral-wall TURBT (bladder tumour resection); pelvic node dissection	Adductor jerk causing bladder perforation
Femoral	Retractor blade on psoas; lithotomy	Loss of knee extension
Hypogastric plexus	Extensive retroperitoneal dissection	Anejaculation/retrograde ejaculation

8 · Imaging anatomy

- On axial CT the **duodenum lies against the right hilum** and the left renal vein in front of the aorta.
- † Before RPLND or nephrectomy, check accessory renal arteries, retro-/circumaortic left renal vein and nodal size by zone.

79 Neoplasms of the Testis

1 · Epidemiology and risk factors

- Over 95% are **germ cell tumours (GCTs)**: seminoma or non-seminomatous GCT (NSGCT); 5% are extragonadal. It is the commonest solid malignancy in men aged **20–40 y**.
- Cisplatin raised metastatic cure to **80–90%**, so the aim is **minimum toxicity without losing curability**.
- Bilateral in **2%**, mostly metachronous.
- **Risk factors**: White race, cryptorchidism, family history, personal history, germ cell neoplasia in situ (GCNIS) and infertility. Undescended testis (UDT) has a relative risk (RR) of **4–6**, falling to 2–3 with orchidopexy before puberty. A previous GCT gives a 12-fold contralateral risk, yet the 15-y cumulative incidence is only 2%.
- **GCNIS** is an arrested fetal gonocyte formed **before birth**. It lies beside invasive GCT in 80–90% and invades in **50% at 5 y** and 70% at 7 y. Contralateral GCNIS is found in 5–9%, and in ≈36% with atrophy, UDT or age <40.
- Microlithiasis alone needs no work-up; it matters only after a previous GCT.

> **KEY POINT**
>
> GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes risk.

2 · Pathology and classification

LAQ Classify testicular tumors and describe management of seminoma

Paper answer · P3

Type: LAQ (25 marks)
Source: Practice Set 1 Paper 3, Q1 (stage I seminoma also answered in 2018-Summer Paper 2, Q2)

Introduction

- Over 95% of testicular tumours are GCTs (germ cell tumours). Seminoma is 52–56% of GCNIS (germ cell neoplasia in situ)-derived GCTs, peaks in the fourth–fifth decade.
- **Risk factors:** cryptorchidism, previous contralateral GCT (twelvefold), family history, GCNIS, White race, infertility.
- All postpubertal GCTs except spermatocytic tumour carry **isochromosome 12p**.

WHO (World Health Organization) classification (2016, retained 2022)

Group	Tumours
1. GCNIS-derived GCTs	GCNIS; **seminoma**; **NSGCT (non-seminomatous GCT)**: embryonal carcinoma, postpubertal yolk sac tumour, choriocarcinoma and other trophoblastic tumours, postpubertal teratoma, teratoma with somatic-type malignancy; mixed GCT
2. Non-GCNIS-derived GCTs	**Spermatocytic tumour** (formerly spermatocytic seminoma); prepubertal teratoma, dermoid and epidermoid cyst; prepubertal yolk sac tumour

Group	Tumours
3. Sex cord–stromal tumours (<5%)	Leydig cell, Sertoli cell (including large cell calcifying), granulosa cell (adult, juvenile), fibroma–thecoma, mixed
4. Mixed germ cell–sex cord stromal	Gonadoblastoma (DSD [difference of sex development])
5. Others	Lymphoma (commonest testicular tumour in older men), rete testis and collecting duct tumours, adnexal and paratesticular tumours, metastases

- Any NSGCT element, however small, makes the tumour an NSGCT and is treated as such.
- **Seminoma pathology:** sheets of clear cells with lymphocyte-rich fibrovascular septa. Immunohistochemistry: CD117 and PLAP (placental alkaline phosphatase) positive, CD30 negative. Syncytiotrophoblasts (hCG [human chorionic gonadotropin]) in about 15%. "Anaplastic seminoma" is no longer recognised.
- **Spermatocytic tumour** (<1%): older men, no GCNIS, no i(12p), benign unless sarcomatous.

Evaluation and staging

- Scrotal ultrasound; AFP (alpha-fetoprotein), β-hCG and LDH (lactate dehydrogenase) before and after orchidectomy. **A raised AFP means NSGCT**, whatever the histology.
- **Radical inguinal orchidectomy** with early cord control at the internal ring; never trans-scrotal. Offer sperm banking and prosthesis.
- CT (computed tomography) of chest, abdomen and pelvis; TNMS (tumour, node, metastasis, serum markers) staging.
- About 80% of seminomas are clinical stage (CS) I; 15–20% CS II (nodes IIA ≤2 cm, IIB 2–5 cm, IIC >5 cm).

Management of seminoma by stage

CS I (cure by orchidectomy alone 80–85%; cancer-specific survival about 100% with any option)

	Surveillance (preferred)	**Carboplatin**	**Radiotherapy**
Method	Examination, markers, CT for at least 5 years	1 cycle **AUC (area under the curve) 7**, dose = 7 × (GFR [glomerular filtration rate] + 25) mg	**20 Gy** para-aortic strip
Relapse	14–20%	About 5% (1 cycle)	3–5%
Drawback	Late relapses (10–20% after 4 years)	Underdosing; late effects unknown	Second cancers, GI (gastrointestinal) toxicity

- **Relapse risk factors:** tumour **>4 cm** and **rete testis invasion** (relapse 12%, 16%, 32% with 0, 1, 2 factors). LVI (lymphovascular invasion) is not a factor in seminoma.
- **Trials:** MRC TE19/EORTC (Medical Research Council/European Organisation for Research and Treatment of Cancer) 30982: one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE18: 20 Gy equals 30 Gy.
- Surveillance relapses are 84–100% retroperitoneal and almost all salvaged.

CS IIA–B

- **Non-bulky (≤3 cm):** dog-leg radiotherapy with nodal boost (25–30 Gy in the chapter; EAU [European Association of Urology] uses 30 Gy for IIA and 36 Gy for IIB), or chemotherapy, or **primary RPLND (retroperitoneal lymph node dissection)** (SEMS trial).
- **Bulky (>3 cm) or multiple nodes:** BEP (bleomycin, etoposide, cisplatin) ×3 or EP (etoposide, cisplatin) ×4.

CS IIC–III: chemotherapy by IGCCCG (International Germ Cell Cancer Collaborative Group)

Risk	Seminoma definition	5-year survival	Regimen
Good (90%)	Any primary, **no NPVM (non-pulmonary visceral metastases)**, normal AFP, any hCG/LDH	86%	BEP ×3 or EP ×4
Intermediate (10%)	NPVM present	72%	BEP ×4
Poor	**None for seminoma**	–	–
- Carboplatin alone is inferior in advanced disease.			

Post-chemotherapy residual mass

```
Residual mass after chemotherapy ├─ <3 cm ──────────► observe (necrosis about 90%)
                                 └─ >3 cm ─► FDG-PET (fluorodeoxyglucose positron emission tomography) at ≥6 weeks
                                            ├─ negative ► observe
                                            └─ positive ► resection (full template PC-RPLND) or salvage
```

- Surgery is difficult because of desmoplasia; complete resection only 58–74%.

Relapse and follow-up

- After surveillance or radiotherapy: radiotherapy for small retroperitoneal relapse, otherwise first-line BEP. After chemotherapy: VeIP (vinblastine, ifosfamide, cisplatin) or high-dose chemotherapy.
- Surveillance schedule: review every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly. CT detects >85% of relapses.

Recent advances

- Serum **miR-371a-3p** microRNA: sensitive marker of residual disease and relapse.
- MRI (magnetic resonance imaging) for surveillance to cut radiation; primary RPLND for low-volume stage II.

Diagram to draw: WHO classification tree (GCNIS-derived vs non-GCNIS-derived vs sex cord–stromal) and the residual-mass flowchart.

> **Key points to remember:**
> - WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
> - CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
> - Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
> - No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Also, from the condensed book (not in the paper answer):

- Isochromosome 12p, i(12p), proves GCT origin in a carcinoma of unknown primary and in somatic-type malignancy from teratoma.

Subtype	Markers	IHC / histology	Hallmark
Seminoma	Human chorionic gonadotrophin (hCG) ≈15%; **never alpha-fetoprotein (AFP)**	CD117+, CD30−, placental alkaline phosphatase (PLAP)+	4th–5th decade; can transform to NSGCT
Spermatocytic	None	OCT3/4−, PLAP−; three cell types	6th decade; never mixed, not with UDT; **benign** unless sarcomatous/anaplastic
Embryonal carcinoma (EC)	AFP/hCG or normal	AE1/AE3+, OCT3/4+, CD30+†, c-KIT−	Totipotent; spreads with normal markers; predicts occult spread
Yolk sac	AFP, not hCG	Schiller–Duval bodies (≈50%), hyaline globules	Pure form: children, mediastinum
Choriocarcinoma	Very high hCG	Syncytio- and cytotrophoblasts	Spreads by blood (lung, liver, brain); bleeds; poor risk
Teratoma	Normal/mild AFP	≥2 germ layers; maturity irrelevant after puberty	Benign histology, malignant genetics; **chemoresistant**

- Teratoma is present in ≈half of mixed GCTs. Unresected, it causes growing teratoma, somatic transformation or late relapse, so surgery is integral to cure.
- hCG shares its α-subunit with TSH and LH, so it can cause hyperthyroidism and gynaecomastia.

> **KEY POINT**
>
> Management follows the most chemoresistant element: EC drives occult metastasis, choriocarcinoma haemorrhage, teratoma surgery.

3 · Presentation

- A **painless mass** is commonest. Metastases are present at diagnosis in ≈2/3 of NSGCT and 15% of seminoma; 10–20% present with metastatic symptoms such as back pain or a left supraclavicular node.
- About 1/3 are first misdiagnosed, so re-examine epididymo-orchitis 2–4 weeks after antibiotics.

4 · Diagnosis and initial management

Scrotal ultrasound (US)

- GCT is **hypoechoic**: homogeneous suggests seminoma, heterogeneous suggests NSGCT.
- In metastatic GCT with normal palpation, US seeks a **burned-out primary**, which still needs orchidectomy; normal testes mean extragonadal GCT.

- An impalpable <10 mm lesion with normal markers is GCT in 20–50%: options are orchidectomy, inguinal exploration with frozen section, or serial US.

Serum tumour markers

*See the paper answer **Tumour markers in urology** (Section 1, Ch 3).*

Also, from the condensed book (not in the paper answer):

- Half-lives: AFP **5–7 days**, hCG **24–36 h**, LDH 24 h.
- After orchidectomy, a slow decline with negative imaging is monitored, but **rising markers mean chemotherapy**.
- An unstable patient with very high AFP/hCG gets chemotherapy first and orchidectomy later.

False positives (hepatic disease, hypogonadal LH cross-reactivity, marijuana) occur, and miR-371a-3p is a GCT-specific marker not raised by teratoma.

Radical inguinal orchidectomy

- Remove testis and cord to the internal ring within 1–2 weeks. **Trans-scrotal approach or biopsy is contraindicated**. Orchidectomy alone cures 80–85% of clinical stage (CS) I seminoma and 70–80% of CS I NSGCT.
- **Testis-sparing surgery** suits synchronous bilateral or solitary testis (<2–3 cm, adequate androgen) or a suspected benign lesion, with frozen section; GCNIS then gets ≥20 Gy.
- Biopsy the contralateral testis if atrophy, UDT, age <40 or suspicious US.

5 · Staging

Spread and imaging

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- CT abdomen–pelvis for all misses disease in **25–35% of CS I NSGCT**, so landing-zone nodes of 5–9 mm are suspicious. Equivocal nodes with normal markers are re-imaged in 6–8 weeks.
- Brain imaging if hCG >10,000 IU/L; no routine bone scan or FDG-PET at diagnosis.
- Chest CT if markers raised, abdominal disease, abnormal X-ray, or CS I NSGCT with LVI/EC.

Base decisions on CT and markers within 4 weeks of treatment, and offer **sperm cryopreservation** before RPLND, chemotherapy or radiotherapy.

TNM staging — the only TNM with a serum (S) category

Category	Definition
pTis	GCNIS
pT1	Testis/epididymis, **no LVI**; may invade tunica albuginea
pT2	LVI, or tunica vaginalis
pT3 / pT4	Spermatic cord / scrotum

Category	Definition
N1 / N2 / N3	≤2 cm / >2–5 cm / >5 cm (pN1 ≤5 nodes, pN2 >5 or extranodal; pN invalid after chemotherapy)
M1a / M1b	Non-regional nodes or lung / other viscera
S1	LDH <1.5×N and hCG <5000 mIU/mL and AFP <1000 ng/mL
S2	LDH 1.5–10×N or hCG 5000–50,000 or AFP 1000–10,000
S3	LDH >10×N or hCG >50,000 or AFP >10,000

Stage	Criteria
IA / IB	pT1 / pT2–4, N0 M0 S0
IS	Any pT, N0 M0, **S1–3** (raised markers, no visible disease)
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	M1a, S0–1
IIIB	N1–3 or M1a with S2
IIIC	N1–3 or M1a with S3, or **any M1b**

IGCCCG risk — metastatic disease

Risk	NSGCT	Seminoma
Good	Testis/retroperitoneal primary, no NPVM, AFP <1000, hCG <5000, LDH <1.5×N (56%; 5-y survival 92%)	Any primary, no NPVM, any hCG/LDH, normal AFP (90%; 86%)
Intermediate	Any of AFP 1000–10,000, hCG 5000–50,000, LDH 1.5–10×N (28%; 80%)	**NPVM** (10%; 72%)
Poor	**Mediastinal primary**, NPVM, AFP >10,000, hCG >50,000 or LDH >10×N (16%; 48%, now ≈71%)	**None**

NPVM = non-pulmonary visceral metastases.

> **KEY POINT**
>
> IGCCCG uses markers at the start of chemotherapy, not pre-orchidectomy values, and is invalid at relapse. Seminoma risk depends on NPVM alone; mediastinal NSGCT is always poor risk.

6 · Treatment principles

- Centralise care, give full-dose cisplatin and resect all residual disease.

	Seminoma	NSGCT
CS I / II / III	85 / 10 / 5%	≈1/3 each
Occult metastasis CS I	10–15%	25–35%
Radiosensitive	Exquisitely	No (except brain)
Metastatic teratoma	Rare	Frequent, so surgery needed
Markers	Rarely useful	Central

- Seminoma relapses contain NSGCT in 10–15%, so biopsy non-responders.

GCNIS management

- Options are orchidectomy, radiotherapy **≥20 Gy** (lower doses allow recurrence) or observation. Up to 40% later need testosterone after radiotherapy.
- With a normal contralateral testis and a wish for paternity, orchidectomy is preferred because scatter harms the other testis. Cisplatin only partly clears GCNIS.

7 · NSGCT management

Every CS I option cures ≈100%; the choice is about toxicity.

CS I NSGCT

- Adjuvant therapy overtreats 70–80%.
- **Risk factors** are **LVI** (strongest) and **EC predominance**. High-risk tumours relapse in ≈50%, low-risk in <20%.
- **Surveillance**: relapse ≈28%, >90% within 2 y and mostly retroperitoneal; relapses are chemo-naive and curable.
- **Primary RPLND** removes teratoma and finds pathological stage II in 19–28%, mostly cured by surgery alone. pN1 is observed; pN2–3 (relapse >50%) gets **2 cycles adjuvant** (relapse ≤1%) or observation.
- **Bleomycin–etoposide–cisplatin (BEP) ×1** gives the lowest relapse (≈2–3%) but leaves teratoma, exposes all to late toxicity, and relapses are chemoresistant.
- Full bilateral nerve-sparing template (ejaculation >90%) is standard; a modified template is acceptable for clinically negative nodes.

ALGORITHM

1. Assess CS I NSGCT for LVI and EC predominance.
2. If low risk (neither), then surveillance.
3. If high risk, then surveillance, BEP×1 or expert nerve-sparing RPLND, chosen by preference and toxicity.

KEY POINT

Surveillance maximises quality-adjusted survival when relapse risk is <≈1/3. All options cure ≈100%; they differ in overtreatment versus difficulty of salvage.

CS IS, IIA–B

- **CS IS** is treated with induction chemotherapy.
- **CS IIA–B** with raised markers or nodes **>3 cm** gets chemotherapy, with or without post-chemotherapy RPLND (PC-RPLND). With normal markers and nodes <3 cm, **primary RPLND** is used: 13–35% are pN0, ≈30% have teratoma, and half avoid chemotherapy.

CS IIC–III: induction by IGCCCG

P2 *CRE & BEP — BEP×3/EP×4 good, BEP×4 (VIP if lungs poor) intermediate/poor, toxicities*

- BEP×3 equals BEP×4; etoposide–cisplatin (EP) ×3 and carboplatin are inferior.
- Second line is TIP×4 (paclitaxel, ifosfamide, cisplatin), VeIP×4 or VIP; TIP is best for late relapse.
- High-dose chemotherapy (HDCT) failed in three randomised trials, so it is for specialised centres.

Post-chemotherapy residual masses (NSGCT)

- With normal markers and a residual mass **>1 cm**, do **PC-RPLND** with a full bilateral template. Histology is necrosis in ≈40%, teratoma in ≈45% and viable tumour in ≈15%.
- No model, imaging or **FDG-PET** predicts necrosis (teratoma is not avid), so always resect.
- After a complete response (≤1 cm), observe, or resect if pre-chemotherapy nodes were >3 cm (controversial).
- With multiple sites, resect all, retroperitoneum first, because its necrosis predicts necrosis elsewhere. Somatic transformation (6–8%) needs complete resection.
- After viable tumour, adverse factors are intermediate/poor risk, incomplete resection and >10% viable cells.

Relapsed NSGCT

SAQ Growing teratoma syndrome

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3a

Definition

- GTS (growing teratoma syndrome; Logothetis, 1982): metastatic masses of an NSGCT (non-seminomatous germ cell tumour) **enlarge during or after chemotherapy while AFP (alpha-fetoprotein) and hCG (human chorionic gonadotrophin) are normal or falling**. Resection shows **mature teratoma only**.
- Uncommon: about 2–8% of metastatic NSGCT.

Pathogenesis

- Teratoma is **histologically benign but genetically malignant** (i(12p), aneuploidy) and **chemoresistant**.
- Chemotherapy kills the sensitive elements (embryonal, yolk sac), so markers fall; the teratoma keeps growing, mostly by enlarging cysts. Chemotherapy-induced differentiation is a second theory.
- Main risk factor: teratoma in the orchidectomy specimen.

Clinical features

- Retroperitoneum commonest; also lung, mediastinum, supraclavicular nodes.
- Often silent; or compression: ureteric obstruction, IVC (inferior vena cava) compression or thrombosis, bowel obstruction, back pain.

Evaluation

- Markers before every cycle; **rising markers mean viable cancer, not GTS**.
- CT (computed tomography): enlarging, well-defined, **cystic** mass, sometimes with fat or calcification.
- **FDG-PET (fluorodeoxyglucose positron emission tomography) has no role**: teratoma is not FDG-avid.
- **Differential:** progressing viable GCT (germ cell tumour; markers rise); malignant transformation (somatic sarcoma or carcinoma); necrosis or fibrosis (mass shrinks).

Management

- **Stop chemotherapy**; do not escalate to salvage chemotherapy (ineffective).
- **Prompt complete surgical resection**: PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection), with resection of every other site; adjacent vessel, kidney or bowel resection may be needed. Delay lets the mass become unresectable.
- Radiotherapy is ineffective.
- **Outcome:** favourable after complete resection (long-term survival 75–90% after resection of residual teratoma). Incomplete resection leads to regrowth, **malignant transformation** (rhabdomyosarcoma commonest) or **late relapse**.
- **Also know:** CDK4/6 (cyclin-dependent kinase 4/6) inhibitors studied for unresectable teratoma; lifelong CT and marker follow-up.

Diagram to draw: graph of mass size rising while marker curve falls during chemotherapy cycles.

> **Key points to remember:**
> - Enlarging mass plus normal or falling markers equals GTS.
> - Teratoma is chemoresistant and not FDG-avid.
> - Treatment is complete surgical resection, not more chemotherapy.

Also, from the condensed book (not in the paper answer):

- **Chemo-naive relapse** gets induction by IGCCCG, with cure >95%.
- **Early relapse (≤2 y) after chemotherapy** gets TIP/VeIP/VIP×4 or HDCT, then post-salvage surgery for residual masses.
- **Desperation surgery** is for rising markers with a single resectable site; survival is 33–57%.
- **Late relapse (>2 y)** occurs in ≈3%. Yolk sac tumour is commonest and chemoresistant, so **complete resection** is the treatment.

8 · Seminoma management

*See the paper answer **Classify testicular tumors and describe management of seminoma** (Section 10, Ch 79).*

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- CS I risk factors are tumour **>4 cm** and rete testis invasion (relapse 12/16/32% with 0/1/2), but they are not validated, so surveillance is preferred for all.
- IIA–B nonbulky (≤3 cm): dog-leg radiotherapy 25–30 Gy with a 5–10 Gy boost, BEP×3/EP×4, or primary RPLND (SEMS, COTRIMS). Bulky or multiple nodes get chemotherapy.
- Carboplatin ×2 gives relapse ≈1.5%. Seminoma relapses late (median 14 months), and radiotherapy carries second malignancy risk.

Residual seminoma masses are necrosis in ≈90%, and many regress over 12–18 months.

ALGORITHM

1. If the residual seminoma mass is <3 cm, then observe.
2. If >3 cm, then FDG-PET ≥6 weeks after chemotherapy (earlier scans give false positives).
3. If PET-negative, then observe.
4. If PET-positive, then template PC-RPLND (or repeat PET/observe). Mass-only excision gives 33% local relapse.

- Relapse after radiotherapy gets first-line chemotherapy; after chemotherapy, VeIP×4 or HDCT.

9 · Brain metastases

- Present in ≈1% at diagnosis, with **choriocarcinoma or very high hCG**; they can bleed fatally. Give BEP×4, then resect residuals.

10 · Treatment sequelae and follow-up

- RPLND can cause ejaculatory loss (>80% after full template without nerve sparing). Orchidectomy alone causes hypogonadism in 10–20%.
- Chemotherapy toxic death is 0–2.4% (good risk) and 3–4.4% (intermediate/poor). Azoospermia is near universal; ≈half recover by 2 y, 80% by 5 y.
- Late effects: Raynaud 20–45%, neuropathy 14–43%, ototoxicity, hypogonadism 15–40%, **cardiovascular disease** and **second malignancy** (60–100% excess after cisplatin or radiotherapy, 200% after both).
- Follow-up is most intensive in years 1–2 (NSGCT) or 1–3 (seminoma).

KEY POINT

Because cure is expected, late cardiovascular disease and second cancers, which rise with each added modality, decide between equally curative options.

11 · Non-germ-cell tumours†

- †**Leydig cell**: the commonest sex cord–stromal tumour, causing precocious puberty or gynaecomastia. ≈10% are malignant; these resist chemo- and radiotherapy, so nodes need RPLND. A small marker-negative lesion suits testis-sparing excision.
- †**Sertoli**: mostly benign; the large-cell calcifying variant occurs with Carney complex and Peutz–Jeghers.
- †**Lymphoma**: the commonest testicular tumour in men **>50–60 y** and the commonest bilateral one. It is treated by orchidectomy, then R-CHOP with CNS prophylaxis and contralateral testis radiotherapy.

12 · Landmark trials

Trial	Design	Result	Take-home
IGCCCG 1997	Pooled 5202 NSGCT, 660 seminoma	Markers, NPVM, mediastinal primary predict outcome	Risk groups select regimen
PVB vs BEP	RCT, advanced	Equal survival, less neurotoxicity	BEP standard

Trial	Design	Result	Take-home
BEP×3 vs ×4	RCT, good risk	Equal survival	BEP×3
VIP vs BEP×4	RCT, intermediate/poor	Equal, more toxicity	VIP only if lungs at risk
MRC TE19/EORTC 30982	Carboplatin ×1 vs radiotherapy, CS I seminoma	5-y RFS 94.7% vs 96%	Equivalent
MRC 20 vs 30 Gy	RCT, CS I seminoma	Equal relapse, less toxicity	20 Gy PA
Albers (German)	BEP×1 vs unilateral RPLND, CS I NSGCT	2-y PFS 99% vs 92%	Favours BEP×1
Williams	Adjuvant ×2 vs observation, pathological stage II	Relapse 6% vs 49%; same survival	Both acceptable
MRC surveillance	2 vs 5 CTs, CS I NSGCT	No difference	Fewer scans safe

RCT = randomised controlled trial; RFS = relapse-free survival; PFS = progression-free survival; PA = para-aortic.

80 Surgery of Testicular Tumours

1 · Evaluation of the testis mass

It is urgent, because delay escalates curative therapy.

- Examine supraclavicular nodes, breasts, abdomen and scrotum. Scrotal ultrasound (US) mainly checks the **contralateral testis** (bilateral synchronous ≈ **1%**). Markers (AFP, β-hCG, LDH) give diagnosis, stage and post-orchidectomy baseline.
- Discuss a prosthesis and † offer sperm banking.

2 · Radical inguinal orchidectomy

Treatment of choice: **early cord control** before handling the testis, removing testis, epididymis and cord to the internal ring. † A scrotal approach breaches a second (inguinal) lymphatic field and leads to local recurrence.

ALGORITHM

1. Make a 3–5 cm incision over the canal and open the external oblique.
2. **Preserve the ilioinguinal nerve**.
3. Mobilise the cord at the pubic tubercle and apply a **Penrose early clamp**.
4. Deliver the testis and divide the gubernaculum.
5. Dissect the cord to the internal ring.

- **High ligation**: ligate vas and gonadal vessels **separately** with **non-absorbable** suture, leaving a **1–2 cm** tail to mark the stump for excision at retroperitoneal lymph node dissection (RPLND).

KEY POINT

† **Scrotal violation**: inguinal exploration, high cord ligation and excision of the scrotal scar. Routine hemiscrotectomy is unnecessary; chemotherapy-era outcomes are not compromised.

Partial orchidectomy (testis-sparing surgery)

SAQ **Testis sparing surgeries for CA testis**

Paper answer · P4

Type: SAQ (10 marks)
Source: 2016-Summer Paper 4, Q3b

Definition and rationale

- **Partial orchidectomy**: inguinal excision of the tumour, preserving the rest of the testis. The standard remains **radical inguinal orchidectomy**; sparing aims to keep androgen production and avoid lifelong testosterone.

Indications

Suspected germ cell tumour (GCT), all of:

- Synchronous bilateral tumours or tumour in a **solitary testis**.
- Tumour **≤2 cm** (up to 2–3 cm, under 30% of testicular volume), preferably polar.

- Adequate preoperative testosterone.
- Not with a normal contralateral testis.

Suspected benign or indeterminate lesions:

- Small impalpable lesion (<10 mm) with normal markers: most benign (cyst, infarct, Leydig nodule), but 20–50% are small seminomas. Lesions <5 mm are rarely malignant.
- **Prepubertal**: teratoma and epidermoid cyst with normal AFP (alpha-fetoprotein): partial orchidectomy is the standard.

Preoperative work-up

- Ultrasound of both testes; AFP, beta-hCG (human chorionic gonadotropin), LDH (lactate dehydrogenase); testosterone, LH (luteinising hormone); **sperm banking**; consent for radical orchidectomy.

Technique

1. Inguinal incision; early cord control with a soft clamp.
2. Deliver the testis; isolate with towels.
3. **Intraoperative ultrasound** localises the lesion.
4. Cold ischaemia optional if resection <30 minutes.
5. Incise the tunica parallel to the segmental vessels; excise the mass with a rim of normal tubules.
6. **Frozen section** (benign versus malignant) and biopsies of adjacent parenchyma for GCNIS (germ cell neoplasia in situ).
7. Close the tunica; fix the testis in the scrotum.

Management of GCNIS

- **Radiotherapy about 18–20 Gy** to the remnant: prevents invasive GCT, keeps Leydig function, but causes sterility; up to 40% later need testosterone.
- German Testicular Cancer Study Group: no local recurrences with radiotherapy; most men who omitted it recurred.
- Radiotherapy may be deferred for paternity under close follow-up. Local recurrence → completion orchidectomy.

Complications and follow-up

- Haematoma, atrophy, hypogonadism, local recurrence. Follow with self-examination, ultrasound, markers and testosterone.

Diagram to draw: testis with polar tumour excised with a rim of normal tissue.

> **Key points to remember:**
> - Solitary or bilateral tumour, ≤2 cm, normal testosterone.
> - Frozen section plus adjacent biopsies for GCNIS.
> - GCNIS: 18–20 Gy keeps Leydig function but sterilises.

Also, from the condensed book (not in the paper answer):

- Needs **all three**: polar, **≤2 cm**, contralateral testis compromised/absent. **Never elective with a normal contralateral testis.**
- Masses **<5 mm** with negative markers have low malignancy, so selective surveillance is an option. Cold ischaemia can be omitted if resection takes **<30 min**.
- GCNIS options are completion orchidectomy, **18–20 Gy** or observation (local recurrence ≈ 7.5%). **Local recurrence always needs completion orchidectomy**.

Delayed orchidectomy

Life-threatening metastatic germ cell tumour (GCT) gets chemotherapy first, then **delayed orchidectomy in all**, because the testis is a sanctuary with **discordant response**. An apparent extragonadal retroperitoneal primary shows intratubular neoplasia in up to 42%, so lateralising nodes mean orchidectomy on that side.

3 · RPLND — terminology and indications

Term	Indication
Primary RPLND	High-risk clinical stage (CS) I or low-volume stage II non-seminomatous GCT (NSGCT), **normal markers**
Post-chemotherapy (PC-RPLND)	After induction; residual **>1 cm**, normal markers
Salvage PC-RPLND	After second-line chemotherapy
Desperation	Markers still elevated; resectable
Redo	Retroperitoneal recurrence after prior RPLND
Late relapse	**>24 months** after complete response

Each step down means more chemoresistant disease and harder surgery.

Preoperative planning

- CT within **6 weeks** and markers within **7–10 days**; note vascular variants; no bowel prep.
- After **bleomycin**, test lung function and ask anaesthesia to use **low FiO_2, restrictive fluids**.
- Total IVC occlusion by thrombus needs inferior vena cava (IVC) resection; partial needs reconstruction.

4 · Templates

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- The bilateral template is bounded by renal vessels, ureters and common iliac arteries. A modified post-chemotherapy template needs a lesion **≤5 cm**.

5 · Open RPLND technique

- The approach is **transperitoneal** or **extraperitoneal** (less ileus).

- Incise from the caecum to the inferior mesenteric vein (IMV) and along the line of Toldt to the foramen of Winslow, freeing duodenum and pancreas. Use **liberal clips** to prevent chyle leak.
- **Split and roll**: split at 12 o'clock on the aorta from the left renal vein down to the inferior mesenteric artery (IMA), roll packets off and ligate lumbars.
- Lumbar arteries are usually 3 pairs, with the **2nd pair at IMA level**. The **right sympathetic trunk**, posterolateral to the IVC, is mistaken for lymphatic tissue.

Nerve sparing

- Left fibres run along the lateral aortic border near the IMA. Right fibres emerge **beneath the IVC** and cross the aorta obliquely, so they are at risk when lumbar veins are ligated.

> **KEY POINT**
>
> Identify fibres **prospectively, before dividing anything**; loop them and pass the packet beneath. Avoid traction, which causes neuropraxia and transient anejaculation.

Postoperative care

- No routine drain; if one is used, remove it at **<100 mL/24 h** on diet. No nasogastric (NG) tube, low-fat diet early, and home on days 3–5.

6 · Auxiliary procedures (PC-RPLND 23–45%)

- **Nephrectomy** is commonest, usually **left**. The strongest predictor is a residual mass **>10 cm**.
- **Vascular** repair is planned in every PC-RPLND, using a patch or graft.
- **IVC**: narrowing **<25%** is repaired by venorrhaphy, >25% by patch or graft; ligation is acceptable. **The aorta is never ligated**.
- **Liver**: histology is discordant in ≈half; resect, or ablate if **<3 cm** and unresectable. **Thorax**: do RPLND first, then resect lesions >1 cm unless the retroperitoneum shows fibrosis only (thoracic necrosis then in ≈89%).

7 · Decision making

- **Complete remission (≈70%)**: generally observe, though ≤1 cm residuals hold teratoma in ≈ 1/5 and viable tumour in ≈ 1/20.
- **Necrosis is predicted by** no teratoma in the primary, normal pre-chemotherapy AFP/β-hCG, raised LDH, >50% shrinkage and residual <2 cm.
- **After primary RPLND**: pN0, teratoma only or pN1 are observed; pN2 gets 2 cycles or observation. Observation cures ≈70%.

8 · Histology of residual mass after chemotherapy

Finding	Frequency	Outcome/action
Fibrosis/necrosis	**≈40%**	Relapse-free (RFS) and cancer-specific survival (CSS) ≈95%; no adjuvant
Teratoma	**≈45%**	RFS 80–90%; must be resected
Viable GCT	**≈15%**	Survival 50–70%; consider 2 adjuvant cycles

> **KEY POINT**
>
> **Resect every NSGCT residual >1 cm**: no imaging (including PET †) or marker separates necrosis from teratoma, and teratoma is chemoresistant, grows, relapses late and transforms. Surgery is both staging and the only treatment.

- **Fizazi** model for viable GCT: incomplete resection, ≥10% viable cells and IGCCCG intermediate/poor risk. With 0, 1 or ≥2 factors, 5-y overall survival (OS) is 100/83/51%.
- Adjuvant chemotherapy: NCCN for incomplete resection or intermediate/poor risk; EAU for incomplete resection only. Not after salvage chemotherapy.

9 · High-risk PC-RPLND

- **Salvage**: viable GCT is at least doubled; OS 60–75%.
- **Desperation**: OS ≈54%. Operate if disease is confined and completely resectable, with markers declining or plateaued, or chemotherapy exhausted. Do **not** operate with rapidly rising markers or multisite progression. **Postoperative marker normalisation** is the only OS predictor.
- **Redo** is mostly for technical failure (retroaortic/retrocaval tissue), so do **complete great-vessel mobilisation the first time**.
- **Late relapse** (yolk sac, somatic-type malignancy) is treated by **surgery first**.

10 · Unique situations

- **Seminoma residual**: dense desmoplasia, viable in ≈10%. If **<3 cm**, observe; if **>3 cm**, do FDG-PET (≥6 weeks †). If avid, choose selected PC-RPLND, salvage chemotherapy or HDCT.
- **Sex cord stromal**: malignant features include >4 cm, necrosis, mitoses and lymphovascular invasion; RPLND is of unproven benefit.

11 · Outcomes, fertility and complications

- **Ejaculation**: template or nerve sparing preserves antegrade ejaculation in **>90%** (primary) at no oncological cost. † If lost, options include sympathomimetics and sperm retrieval.
- Complications: primary 10.6–24%, PC up to 56%. Mortality: primary ≈0, PC <1%.

Complication	Key facts / management
Pulmonary (bleomycin)	PC 3–5%, acute respiratory distress syndrome. Low FiO_2, restrictive fluids; accept SpO_2 **88–90%**
Ileus	Up to 26% (commonest); extraperitoneal approach, no NG tube, early feeding
Chylous ascites	Primary 0.2–2.1%, PC 2–12%; linked to suprahilar dissection (cisterna chyli L1–L2)
Lymphocele	Symptomatic ≤1.7%; drain if infected
Venous thromboembolism (VTE)	Primary <1%, PC 0.1–3.1%; drug prophylaxis if prior chemotherapy, VTE or obesity
Neurological	Femoral neuropraxia; paraplegia with bulky disease

*See the paper answer **Lymphocele and postoperative lymphatic leak (chylous ascites)** (Section 15, Ch 91).*

ALGORITHM

1. Paracentesis; drain if it reaccumulates.
2. Very-low-fat, high-protein diet with medium-chain triglycerides (MCT) and octreotide.
3. If no response, nil by mouth with total parenteral nutrition (TPN).
4. If refractory, lymphangiography with or without embolisation, peritoneovenous shunt or exploration.
5. Continue a low-fat diet ≥**4 weeks** after resolution.

12 · Landmark trials

Trial	Design	Result	Take-home
Williams (Testicular Intergroup)	RCT, pathological stage II: 2 adjuvant cycles vs observation	Adjuvant near-abolished relapse; equal survival	Observe pN1; individualise pN2
SEMPET	FDG-PET in post-chemotherapy seminoma residuals	>3 cm: specificity 100%, NPV 96%; ≤3 cm PPV 37%	PET for >3 cm only

81 Minimally Invasive Retroperitoneal Lymph Node Dissection

SAQ Minimally invasive and robotic RPLND

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Minimal invasive retroperitoneal lymph node dissection" · P4 SAQ "Robotic RPLND"

Introduction

- **MI-RPLND (minimally invasive retroperitoneal lymph node dissection)** for testicular GCT (germ cell tumour): laparoscopic (1992) or robotic (2006). **It must duplicate open RPLND exactly.**

Indications

- **Primary:** marker-negative stage I NSGCT (non-seminomatous GCT), especially high-risk IB; small-volume stage II NSGCT or seminoma (1–2 nodes ≤3 cm in the landing zone).
- **Postchemotherapy:** only a small solitary residual mass with little desmoplasia; **open** for large masses or vessel involvement.
- **Expanding indication:** primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy (SEMS, PRIMETEST).
- **Patient selection:** serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND.

Templates and nerve sparing

- Boundaries: renal hila, ureters, ureter crossing the common iliacs; plus spermatic cord.
- **Primary:** extended **unilateral modified template** with nerve sparing (right: paracaval, precaval, interaortocaval; left: para-aortic, preaortic; both take nodes above the IMA (inferior mesenteric artery)). If nodes are enlarged, the dissection becomes bilateral.
- **Postchemotherapy:** bilateral full template.
- **Nerve sparing:** identify the sympathetic chains and L1–L4 postganglionic fibres to the hypogastric plexus below the IMA.
- **Diagram to draw:** right and left modified templates.

Steps (robotic)

1. Positioning: **modified lateral decubitus** (unilateral) or **supine steep Trendelenburg with peritoneal hammock** (any template).
2. Mobilise colon; Kocherise duodenum (right); divide gonadal vein.
3. **Split-and-roll:** split over the great vessel from common iliac to renal vein; roll packets off.
4. **Divide lumbar vessels** to clear retrocaval and retroaortic tissue.
5. Clip lymphatics (left renal vein); bag each packet.

Feature	Laparoscopic	Robotic
Retraction	Assistant	Fourth arm
Instruments	Straight	Wristed
Suturing, vessel control	Difficult	Easier

Outcomes

- Stay 1–3 days, blood loss mostly ≤200 mL; conversion about 5%; **antegrade ejaculation ≥90%** (robotic nerve sparing after primary surgery); chylous ascites as open. Node yields approach open (about 30).
- **Major bleeding:** pressure, raise pneumoperitoneum, convert if uncontrolled (open set ready).

Concerns and guidelines

- Concerns: staging-only dissections, frequent adjuvant chemotherapy, unindicated surgery, catastrophic bleeding, follow-up <5 years; no randomised trials.
- **Guidelines:** AUA (American Urological Association): experienced surgeons may offer it; EAU (European Association of Urology): specialist centres only.

Recent advances

- Serum miR-371a-3p for selection of stage I patients with occult nodes.
- RPLND for small-volume stage II seminoma (SEMS, PRIMETEST) to avoid chemotherapy and radiotherapy.

> **Key points to remember:**
> - Same template, split-and-roll, lumbar division and nerve sparing as open.
> - Best for chemotherapy-naive low-volume disease.
> - Conversion about 5%; ejaculation preserved ≥90%.
> - Expanding indication: primary RPLND for small-volume stage II seminoma.
> - Serum miR-371a-3p may select stage I patients with occult nodes.

Also, from the condensed book (not in the paper answer):

- The benchmark recurrence-free rate is ≈**85%** for limited pN+ without adjuvant chemotherapy; low MI recurrence is **confounded** by high pN0 and adjuvant rates.

1 · Indications and rationale

- Goals in early-stage disease are to preserve cure, avoid **dual therapy** (surgery and chemotherapy) and minimise lifelong toxicity.
- **Primary** RPLND (retroperitoneal lymph node dissection) is for marker-negative clinical stage (CS) I non-seminomatous germ cell tumour (NSGCT), and small-volume stage II NSGCT and seminoma. It spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0.
- **Post-chemotherapy (PC-RPLND)** is for NSGCT residual **>1 cm** and select seminomas.
- The robot adds fourth-arm retraction and **circumferential great-vessel dissection with lumbar control**.

> **KEY POINT**
>
> Much recovery gain came from Enhanced Recovery After Surgery (ERAS) pathways, which help every approach, so the benefit of access alone is smaller than it looks.

2 · Guidelines and controversy

- The AUA (early stage) says it **may be offered** by a surgeon expert in germ cell tumours and minimally invasive surgery (MIS), with the patient told long-term data are lacking. The EAU restricts it to **specialist centres** expert in open and MIS.
- With no noninferiority evidence, it is highly controversial. Concerns are staging-only dissection without lumbar division, excess adjuvant chemotherapy, operating on the wrong patients, conversion in **≈5%**, and short follow-up with reported peritoneal carcinomatosis.

3 · Patient selection

- There are no absolute contraindications; relative ones are extensive prior surgery and morbid obesity.
- **Suited to MIS**: chemo-naive high-risk IB; small-volume II (1–2 landing-zone nodes, none **>3 cm**); a post-chemotherapy solitary landing-zone mass with little desmoplasia.
- **Favour open**: most PC-RPLND, a large mass, large chemotherapy shrinkage, possible vessel reconstruction, somatic malignancy or retrocrural disease.

4 · Unchanging operative principles

- Imaging within **≈4 weeks** and markers within **1–2 weeks**.
- **MI-RPLND must completely duplicate open RPLND**: template, completeness and nerve-sparing are unchanged.

ALGORITHM

1. Safe entry and exposure.
2. **Split-and-roll** nodal removal.
3. Prospective nerve sparing.
4. **Lumbar vessel division**.
5. Spermatic cord removal, then closure.

- Chemo-naive patients may have an extended unilateral nerve-sparing template, converted to bilateral if enlarged nodes are found. Post-chemotherapy, the **bilateral full template** is standard.
- Incomplete RPLND or mass-only excision worsens survival.

5 · Preparation, positioning and ports

- Plan for haemorrhage with an **open set in theatre**; after bleomycin, manage fluids and oxygen carefully. A high-fat meal the day before highlights lymphatics.
- Positioning is lateral decubitus, laparoscopic supine, or robotic steep Trendelenburg with a **peritoneal hammock** that allows any template.

6 · Operative steps

- **Right**: incise the Toldt line to the foramen of Winslow and **Kocherise**; divide the gonadal vein at the inferior vena cava (IVC); split over the IVC; clip lymphatics; roll the paracaval packet, spare fibres and divide lumbars. **Do not ligate the right renal artery** in the interaortocaval space.

- **Left**: divide the gonadal vein at the left renal vein and control the **lumbar vein entering the left renal vein**.
- Lower the pneumoperitoneum to check haemostasis; **no routine drain**; discharge at **1–2 days**.

7 · Complications and management

ALGORITHM

1. Major vascular injury: apply pressure and raise the pneumoperitoneum.
2. Inform anaesthesia, get blood and call vascular help.
3. If uncontrolled, keep pressure, undock and convert.

- For MIS repair, **identify the injury before acting: no blind clipping**.
- Prevent chylous ascites with liberal clips rather than energy devices; rates match open, and it is treated by low-fat diet, paracentesis, octreotide and parenteral nutrition.
- Antegrade ejaculation is kept in **≥90%** after primary nerve-sparing.

8 · Outcomes vs open

- There is **no randomised trial**. MIS gives shorter stay and less blood loss but longer operative time; brisk bleeding is the unique risk.
- Robotic node yields approach the open benchmark of **≈30**, but adjuvant chemotherapy is still too frequent. Post-chemotherapy data are too small.

9 · Landmark trials

Trial	Design	Result	Take-home
GETUG T93BP	Good-risk NSGCT: BEP×3 vs EP×4	Most relapses/deaths without or with incomplete RPLND	Complete RPLND matters

82 Tumours of the Penis

1 · Overview

- Squamous cell carcinoma (SCC) is **~95%** of penile malignancy, and **0.4–0.6%** of male cancers in North America and Europe, but up to **10%** in parts of Asia, Africa and South America.
- Prognosis rests on **primary stage and histology** and on **nodal metastasis**.

2 · Premalignant lesions: penile intraepithelial neoplasia (PeIN)

*See the paper answer **Premalignant conditions in urology (including premalignant lesions of penis)** (Section 3, Ch 30).*

Also, from the condensed book (not in the paper answer):

- Erythroplasia of Queyrat (EQ) is a red velvety glans lesion in uncircumcised men that mimics balanitis; it progresses in **10–33%**.
- Bowen disease is a scaly shaft plaque mimicking psoriasis or eczema; it progresses in **~5%**. Bowenoid papulosis (HPV-linked shaft papules in young men) **does not progress**.

3 · Aetiology and risk factors of SCC

Factor	Key point
No neonatal circumcision	Neonatal circumcision virtually eliminates invasive cancer; it protects against invasive SCC, **not CIS** (carcinoma in situ). Adult circumcision gives little or no protection
Phimosis	Present in **25–75%**; **7–10×** risk. It mediates the non-circumcision risk, as smegma causes inflammation and fibrosis
HPV	Low risk 6, 11; high risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest**. Present in **~50%**; wives have ~3× cervical cancer risk
Human immunodeficiency virus (HIV)	~4× risk of penile cancer death; PeIN progresses ~6 years earlier
Tobacco	All forms; dose-dependent and independent of phimosis
Lichen sclerosus (LS)	Cancer in **2.3–9%**; HPV-**independent** SCC
Trauma, obesity	Scarred shaft; obesity causes buried penis and functional phimosis
†**Psoralen and ultraviolet A (PUVA)**	† Genital exposure markedly increases SCC, so shield the genitalia

- **Prevention** is by neonatal circumcision (AAP 2012: benefits outweigh risks), hygiene, no tobacco and HPV vaccination (nonavalent adds 31/33/45/52/58).

> **KEY POINT**
>
> Two routes converge on the preputial sac: HPV-driven (E6/E7, p16-positive, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision prevents almost none.

4 · Natural history and spread

- Most arise on the glans or prepuce; **flat ulcerative tumours metastasise earlier**.
- **Buck fascia** is a temporary barrier; once it and the tunica are breached, tumour enters the corpora and spreads by blood.
- Distant metastases are uncommon (**1–10%**) and late. Untreated, death comes within **2 years** from femoral erosion haemorrhage or sepsis.

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- Pelvic stations are external iliac, internal iliac and obturator; distant sites (lung, liver, bone, brain) are rare without nodal disease.

5 · Presentation and diagnosis

- It presents as induration, a warty growth or an ulcer, often hidden by phimosis. Examine the lesion and **palpate both groins**.
- **Biopsy is mandatory before definitive therapy**, because depth and grade predict occult nodes and superficial biopsies mislead.
- **Hypercalcaemia without bone metastases** comes from PTH-related substances, tracks bulky inguinal disease and may resolve after node excision. Treat with saline, then bisphosphonate.

6 · Histology

Subtype (WHO 2022)		HPV	Features	Nodal risk
Usual (keratinising)	**45–65%**	Independent	Usually grade II; lymphovascular (LVI) or perineural invasion (PNI) in ~⅓	25–40%
Warty		Associated	Cauliflower, fibrovascular cores; slow	17–25%
Basaloid		Associated	Small cells, necrosis, deep	**>½ node positive**
Papillary		Independent	Low grade, no koilocytosis	≤12%
Verrucous		Independent	**Broad pushing border**	**No metastasis**; recurs if under-excised
Sarcomatoid		Independent	Spindle cells	**Most aggressive**
Rare: pseudoglandular, adenosquamous; clear cell, lymphoepithelioma-like (HPV)		—	Deep invasion	Pseudoglandular, clear cell worst

- **Any anaplastic cells make it grade III**. **LVI** predicts nodes in every study, and **PNI** does so strongly.
- **High-risk HPV or p16 positivity means better disease-specific survival (DSS)**, not better nodal status.

7 · Imaging

A negative scan never clears the groin.

- **Primary**: examination is reliable; **MRI** (with or without artificial erection) helps if equivocal and organ sparing is planned.
- **cN0 groin**: **no imaging detects micrometastases**, so stage surgically; use CT only if obese or after prior groin surgery.
- **cN+ or T3–T4**: CT chest, abdomen and pelvis. **FDG-PET/CT** is best for pelvic and distant disease.

8 · Staging: TNM 8th edition (2017)

T1a versus T1b decides who needs groin staging.

T	Definition
Tis	PeIN
Ta	Non-invasive localised SCC (incl. pure verrucous without destructive invasion)
T1	Glans: lamina propria · Foreskin: dermis/lamina propria/dartos · Shaft: tissue between epidermis and corpora
T1a	**No LVI, no PNI, not high grade**
T1b	**LVI and/or PNI, or high grade (grade III/sarcomatoid)**
T2	**Corpus spongiosum**, with or without urethra
T3	**Corpora cavernosa** (incl. tunica albuginea), with or without urethra
T4	Scrotum, prostate, pubic bone

N	Clinical (cN)	Pathological (pN)
1	Palpable mobile unilateral node	**≤2 unilateral**, no extranodal extension (ENE)
2	Mobile ≥2 unilateral or bilateral	**≥3** unilateral or bilateral
3	Fixed mass or pelvic nodes	**ENE or pelvic nodes**

Stage	TNM
I	T1a N0
IIA	T1b, T2 N0
IIB	T3 N0
IIIA / IIIB	T1–3 N1 / N2
IV	T4, any N3, any M1

- In the 8th edition, **urethral invasion no longer upstages**.

> **KEY POINT**
>
> Traps: T1b and T2 share stage IIA; any N3 or T4 is stage IV even without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). The primary is staged by tissue, the groin by surgery.

9 · Management of the primary tumour

- **Penectomy** is the oncological gold standard (local recurrence **0–8%**), but over half of patients are ≤60, so preserve where safe.
- **Organ preservation** suits PeIN, Ta, T1 and grade I–II in **compliant** men. It is **contraindicated** by size **≥4 cm**, grade III, deep cavernosal invasion, or unfitness for salvage.
- Local recurrence is higher, but **early recurrence is salvaged without survival cost**, except in pT3 or high grade.
- **Margins**: microscopic spread is **~5 mm** for grade I–II and **~10 mm** for grade III. Only margins **<1 mm** raised recurrence, so a **negative frozen-section margin (>1 mm) is enough**; the 2-cm dogma is abandoned.

Stage / site	Treatment
PeIN, glans	Laser, resurfacing; topical alternative
Ta / PeIN — foreskin, shaft skin	Excision to negative margin; laser; topical (Tis only)
Ta, T1 glans	Excision, resurfacing, glansectomy, radiotherapy (RT; not Ta)
T2 glans, no gross cavernosal involvement	**Total glansectomy** with or without distal corporal transection; partial penectomy; RT
T2 with corporal invasion, T3	**Partial or total penectomy**
T4	Neoadjuvant chemotherapy, then surgery in responders
Recurrence after conservation	Re-excision or penectomy; superficial low-grade re-treated conservatively

- **Glans resurfacing** strips the epithelium and adds a split-skin graft, keeping erections.
- **Laser**: CO_2 reaches only ~0.1 mm (recurrence up to 50%), so **Nd:YAG** (up to 6 mm) is preferred, with base biopsies; recurrence is <10% for CIS and 10–25% for T1.
- † Partial penectomy leaves a stump for standing micturition; total penectomy needs a **perineal urethrostomy**.

> **KEY POINT**
>
> Margin width depends on grade, not a fixed 2 cm. Low-grade tumours spread millimetres and their recurrences are salvageable, so preservation is safe; in high-grade or cavernosal disease local relapse is lethal and amputation regains its place.

10 · Inguinal nodes (see Ch 84)

LAQ Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P2 LAQ "Importance and management of Inguinal lymph nodes in CA penis." · P3 SAQ "Inguinal Block Dissection?" · P4 SAQ "Videoendoscopic inguinal lymphadenectomy"

"Inguinal block dissection" is taken as ILND (inguinal lymph node dissection), mainly for penile cancer.

Importance

- Presence and extent of inguinal metastasis is the **strongest prognostic factor** in penile SCC (squamous cell carcinoma).
- A long **locoregional phase** precedes distant spread, so the groin is a **curable station**; lymphadenectomy alone can cure.
- **Kroon (2005)**: early ILND for impalpable nodes gave 3-year disease-specific survival about 84% vs about 35% when delayed until nodes were palpable (ENE [extranodal extension] about 20% vs 95%).
- Manage both groins **within 3 months** of diagnosis; aim for **at least 8 nodes per side**.
- About 80% 5-year survival with 2 or fewer unilateral nodes, no ENE, no pelvic nodes.

Lymphatic drainage and surgical anatomy

- **Superficial system** (prepuce, shaft skin) drains to superficial inguinal nodes. **Deep system** (glans, frenulum) drains to the corona, then within Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes.
- Presymphyseal crossover makes drainage **bilateral** and unpredictable in side; with one palpable groin, the other has micrometastasis in about 30%.
- **Stepwise, no skip**: superficial inguinal, then deep inguinal, then external iliac; so superficial status gates deep dissection and inguinal status gates pelvic dissection.
- **Femoral triangle:** superior **inguinal ligament**, lateral **sartorius**, medial **adductor longus**; floor fascia lata; apex where the muscles cross.
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**:

Zone	Related vessel
Central	Saphenofemoral junction
Superolateral	Superficial circumflex iliac vein
Inferolateral	Lateral femoral cutaneous and superficial circumflex veins
Superomedial (sentinel nodes mostly here)	Superficial external pudendal and superficial epigastric veins
Inferomedial	Great saphenous vein

- **Deep nodes** (1–3) medial to the femoral vein; **Cloquet's node** is the highest.

Staging (TNM [tumour–node–metastasis] 8th edition)

N	Clinical	Pathological
1	One mobile unilateral node	2 or fewer unilateral, no ENE
2	Mobile, 2 or more unilateral or bilateral	3 or more (unilateral or bilateral)
3	Fixed mass or pelvic nodes	ENE or pelvic nodes

- **T1a**: no LVI (lymphovascular invasion), no PNI (perineural invasion), not high grade. **T1b**: LVI or PNI or high grade.

Evaluation

- **Careful bilateral palpation**: cN0 or cN+ (number, side, mobile or fixed). Imaging (US [ultrasound], CT [computed tomography], MRI) only if examination is unreliable (obesity, previous groin surgery).

- **cN+ or high risk**: FDG-PET/CT (fluorodeoxyglucose positron emission tomography–CT) or CT chest–abdomen–pelvis.
- About half of palpable nodes are reactive, yet an antibiotic trial is **no longer recommended** (delays treatment). US-guided **FNAC (fine-needle aspiration cytology)** is specific; a negative result never ends the pathway.

Management of the impalpable groin (cN0): risk-based

Risk (EAU-ASCO [European Association of Urology–American Society of Clinical Oncology] 2023)	Primary	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance (compliant) or surgical staging
High	**pT1b or more**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if unavailable or DSNB positive

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk, staged surgically (ILND first).
- **DSNB**: US ± FNAC first (a tumour-replaced node diverts flow and causes a false negative); then 99mTc (technetium-99m) nanocolloid, blue dye and gamma probe; serial sectioning. False-negative about 7% at expert centres.
- **Also know:** Cabanas anatomical sentinel biopsy abandoned.

Management of palpable nodes (cN+)

`Palpable node → US-guided FNAC positive → radical ILND same side + stage other side (superficial ILND, FS-ILND or DSNB) negative → excisional biopsy → negative: observe Bulky (≥4 cm), fixed, bilateral or pelvic nodes → neoadjuvant TIP × 4 → responders: ilioinguinal resection → cisplatin-unfit: chemoradiation`

- **PLND (pelvic lymph node dissection)**: if **more than 2 inguinal nodes or ENE** (pN2–3); ipsilateral; external iliac commonest, then obturator.
- **Neoadjuvant TIP** (paclitaxel, ifosfamide, cisplatin), **Pagliaro (2010)**: objective response 50%; survival mainly in responders. InPACT (International Penile Advanced Cancer Trial) is testing the sequence.
- **Adjuvant**: chemotherapy for pN2–3 (weak evidence); radiotherapy for multiple nodes or ENE (at least 50 Gy; 54 Gy for ENE).

Inguinal block dissection (ILND): indications

- Penile SCC: **proven cN+**; **cN0 high risk** (pT1b or more, G3) if DSNB is positive or unavailable. Also vulval cancer, melanoma.
- Within 3 months; at least 8 nodes per side. Bulky or fixed nodes: chemotherapy first.

ILND templates

Template	Extent
Radical (cN2 or more)	Superficial + deep; saphenous often ligated; sartorius transposed

Template	Extent
Fascia-sparing ILND (FS-ILND)	May be used for cN1
Modified (Catalona)	Smaller field: nothing lateral to femoral artery or below fossa ovalis; **saphenous spared**; no sartorius transposition; positive frozen section: convert to complete (radical) ILND
Superficial	Above fascia lata only

Technique: open radical ILND

1. Frog-leg position; incision **2 cm below and parallel** to the inguinal ligament.
2. Flaps **deep to Camper's fascia** (preserves the skin capillaries).
3. Boundaries: inguinal ligament, sartorius, adductor longus. Divide saphenous at apex; open fascia lata; clear along the femoral vessels; **never lateral to the femoral artery** (femoral nerve).
4. **Sartorius transposition** over the vessels; suction drains until <30 mL/day.
5. Pelvic dissection if more than 2 inguinal nodes or ENE.

Videoendoscopic inguinal lymphadenectomy (VEIL)

Definition and aim

- **VEIL:** ILND through thigh ports, beneath Scarpa's fascia; laparoscopic (Tobias-Machado) or robotic (**RVEIL**).
- **Aim:** open-surgery template without the groin incision that causes flap necrosis.

Indications

- Penile SCC, **impalpable nodes**, intermediate/high risk (pT1b+, G3, LVI).
- **Mobile, non-bulky palpable nodes.**
- **Not for** bulky, fixed (cN3) or skin-involving nodes: these need chemotherapy, then open surgery.
- Also vulval cancer, melanoma.
- DSNB still comes first for impalpable nodes (EAU-ASCO).

Technique

- Mark an inverted triangle on the skin **before insufflation**, because gas distorts landmarks. `Position: supine frog-leg / low lithotomy → 2 cm camera incision below the femoral triangle apex → finger dissection beneath Scarpa's fascia → two lateral and medial working ports (+ assistant port) → CO2 at 15 mmHg (higher briefly to create space) → develop space up to the inguinal ligament → remove nodal packet off fascia lata, apex to ligament → preserve saphenous vein if possible; clip tributaries → frozen section → deep dissection medial to femoral vein if positive → bag retrieval, suction drain in dependent part`
- **Template:** same as open: superficial ± deep; Daseler's five zones; Cloquet's node is the highest deep node. **Never dissect lateral to the femoral artery** (femoral nerve).

Outcomes vs open

	Open ILND	VEIL/RVEIL
Wound infection, necrosis, dehiscence	Higher	**Lower**

	Open ILND	VEIL/RVEIL
Hospital stay, drain time	Longer	Shorter
Operative time	Shorter	About 2 h (slightly longer per side)
Nodal yield, recurrence	Reference	**Similar**
Evidence	Retrospective	Retrospective, selection-biased

- Commonest complication of VEIL: **lymphocele**, then lymphoedema and cellulitis.

Robotic VEIL

- 3D vision, wristed instruments, ICG (indocyanine green) for lymphatics; adequate dissection on open re-exploration (phase I). No VEIL vs RVEIL comparison.

Complications of ILND

- Wound infection, **flap necrosis**, dehiscence, seroma, **lymphocele**, **lymphoedema**, DVT/PE (deep vein thrombosis/pulmonary embolism).
- Reduced by transverse incision, correct flap plane, modified template, saphenous sparing, drain removal by volume (<30 mL/day), thromboprophylaxis and minimally invasive access (VEIL); sartorius transposition increases them.

Follow-up

- Examination 3-monthly for 2 years, then 6-monthly to 5 years (recurrence mostly within 2 years); then self-examination.

Diagrams to draw

- Femoral triangle with Daseler's zones and Cloquet's node.
- Femoral triangle with VEIL port sites below the apex.

> **Key points to remember:**
> - Nodal status is the strongest prognostic factor; ILND can cure.
> - pT1b or higher: stage groins surgically (DSNB preferred) within 3 months; at least 8 nodes per side.
> - Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
> - PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT testing the sequence).
> - Femoral triangle: ligament, sartorius, adductor longus; flaps deep to Camper's; stay medial to femoral artery.
> - VEIL: same template, fewer wound complications, equal nodal yield; not for bulky, fixed or skin-involving nodes.

SAQ Prognostic index score in carcinoma penis

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3e

Assumed meaning: prognostic factors, risk-group systems and nomograms in penile SCC (squamous cell carcinoma). No single universally used "prognostic index score" exists.

Introduction

- Prognosis rests on primary histology and stage and, above all, **inguinal node status**.

1. Nodal status (most important factor)

Nodal finding	5-year survival
Node negative	About 73%
≤2 nodes, unilateral, no ENE (extranodal extension)	About 80%
>2 nodes	0–50%
ENE or **pelvic nodes**	Poor (pelvic about 14%)

- TNM (tumour-node-metastasis) 8th edition: pN1 (1–2 unilateral nodes) about 90% 3-year DSS (disease-specific survival); pN3 (ENE or pelvic) about one-third.

2. Primary tumour factors (predict occult nodes in cN0)

- **Depth:** corporal invasion → about 60% node positive.
- **Grade:** any grade III component = grade III.
- **LVI (lymphovascular invasion)** and **PNI (perineural invasion)**: strongest predictors; define pT1b.
- **Subtype:** verrucous and PeIN (penile intraepithelial neoplasia) do not metastasise; basaloid and sarcomatoid aggressive.

3. Risk-group systems

System	Low	Intermediate	High
Earlier EAU (European Association of Urology), after Solsona	Tis, Ta, T1 G1	T1 G2	T2 or more, or G3
Current EAU–ASCO (American Society of Clinical Oncology) / TNM	PeIN, Ta, **T1a** (nodes 0–18%)	—	**T1b or more** (one-third to over half)

- Low risk → groin surveillance; high risk → **DSNB (dynamic sentinel node biopsy)** or ILND (inguinal lymph node dissection).

4. Nomograms and indices

- **Ficarra nomogram:** predicts nodal involvement (thickness, growth pattern, grade, LVI, corporal invasion, clinical node status).
- **Kattan nomogram:** predicts cancer-specific survival.
- A pathological **prognostic index** (grade + anatomical level of invasion + PNI; Cubilla group) has been proposed.
- **Status:** none externally validated; not used routinely.

5. Molecular markers

- **HPV (human papillomavirus)/p16 positive:** better DSS (not node status).
- Adverse: p53, PD-L1 (programmed death-ligand 1), Ki-67, high **NLR** (neutrophil-to-lymphocyte ratio), raised CRP (C-reactive protein).

6. Other

- Inguinal recurrence after observation: only about one-third survive 5 years. Size >5 cm worse.

Diagram to draw: flowchart: primary pathology → low or high risk → surveillance or DSNB/ILND.

> **Key points to remember:**
> - Node status is the strongest prognostic factor.
> - LVI, PNI, grade III define high-risk T1b.
> - Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

Also, from the condensed book (not in the paper answer):

- Cure is ~**80%** with ≤2 unilateral nodes and no ENE or pelvic nodes; with positive pelvic nodes, 5-year survival after surgery alone is ~**14%**.
- Occult node rates are **0–18%** for PeIN/Ta/T1a, one-third to over half for T1b and above, and ~60% with corporal invasion.

ALGORITHM

1. **cN0, low risk**: surveillance. If a node appears, do fine-needle aspiration cytology (FNAC); if positive, inguinal lymph node dissection (ILND) and contralateral staging.
2. **cN0, high risk (T1b+)**: bilateral dynamic sentinel node biopsy (**DSNB**) or superficial/fascia-sparing/minimally invasive ILND. If positive, completion ILND.
3. **cN1–2, mobile, <4 cm**: biopsy, then ILND and contralateral staging. If >2 nodes or ENE, add pelvic lymph node dissection (PLND).
4. **Fixed, ≥4 cm or pelvic**: neoadjuvant TIP (cisplatin-eligible) or chemoradiation, then resect responders.
5. **Adjuvant therapy** if >2 nodes, bilateral nodes, ENE or pelvic nodes.

- DSNB refinements (pre-operative ultrasound with or without FNAC, serial sectioning) cut false negatives from ~18% to ~7%. Morbidity is greatest with radical ILND, then superficial or fascia-sparing, then minimally invasive, and least with DSNB.
- No pelvic skip metastases occur, so PLND is mainly staging.

11 · Radiotherapy

- **Circumcise first** to expose the lesion and prevent oedema and phimosis. Surgical salvage remains curative.
- **External-beam RT (EBRT)**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks** with **bolus**; ~55% are recurrence-free at 5 years.
- **Brachytherapy** (Ir-192) suits **T1–T2 glans tumours <4 cm**: ~60 Gy over ~5 days gives ~79% relapse-free survival and ~85% preservation at 5 years, with better erections than EBRT.
- **Late effects** are **meatal stenosis** (10–45%, at 18–24 months) and ulceration; **1 in 5 local recurrences occur after 5 years**.
- For the groins, give **no prophylactic RT** for cN0. Adjuvant RT for **≥2 nodes or ENE** is ≥50 Gy, with **54 Gy for ENE and 57–60 Gy for gross residual**. Chemoradiation (cisplatin if GFR >50 mL/min) treats inoperable nodes.

12 · Systemic therapy

- No randomised trials are completed. **TIP** (paclitaxel, ifosfamide, cisplatin) is most active; **Bleomycin regimens are discouraged**.
- **Neoadjuvant TIP ×4** for bulky N2–N3 or pelvic nodes gives ~50% response, and ~⅔ of responders stay disease-free, so **response selects for surgery**.
- Adjuvant chemotherapy has no proven survival gain but may be considered for pN3 or pelvic nodes.
- Palliative platinum is first line; checkpoint inhibitors help ~1 in 6 in trials.

13 · Follow-up

Groin relapse after observation is usually fatal.

- After organ preservation, use close follow-up and self-examination.
- Observed groins are checked 2–3-monthly for 2 years, then less often. Only ~⅓ of cN0 men with groin relapse survive 5 years.

14 · Non-squamous penile tumours (~5%)

Tumour	Key features / treatment
BCC	Proximal shaft; almost never metastasises. Treat by excision or Mohs
Melanoma	<1%; glans; spreads by blood and lymph; AJCC cutaneous staging. Bilateral modified ILND if Breslow ≥1 mm, ulceration or Clark IV–V (DSNB reasonable); adjuvant nivolumab for stage IIIB–IV
Sarcoma	Malignant are **proximal**; vascular commonest; **local recurrence characteristic**. Superficial: wide excision or partial; deep corporal: total. No ILND unless palpable
Extramammary Paget	Penoscrotal eczematoid plaque; CK7+, CEA+, S-100−; ~½ have another cancer, so screen. Wide excision (up to 3 cm) guided by **mapping biopsies**
Lymphoma	Usually systemic: R-CHOP even if isolated
Metastases	From **bladder, prostate, rectum**; **priapism** is the commonest sign; most die within a year; palliate

15 · Landmark trials

Trial	Design	Result	Take-home
Neoadjuvant TIP (MD Anderson)	Phase II; TIP ×4, cN2–N3 M0	Response 50%; median OS 17.1 months; durable mainly in responders	Standard before surgery for bulky/pelvic nodes
Netherlands Cancer Institute early vs delayed	Early DSNB vs dissection after surveillance relapse	3-y survival 84% vs 35%; ENE near-universal if delayed	Stage high-risk cN0 groins early

84 Inguinal Node Dissection

1 · Why the groin matters in penile cancer

Groin surgery alone can cure node-positive disease.

- Penile squamous cell carcinoma (SCC) has a **locoregional phase**, so the groin is a **curable station**.
- **Nodal status is the strongest prognostic factor**. Extranodal extension (ENE), even in one node, makes it pN3.
- Early versus delayed inguinal lymph node dissection (ILND) for impalpable nodes gave ENE ≈20% vs ≈95% and 3-year disease-specific survival ≈84% vs ≈35%. So manage the groins within **3 months** of diagnosis.
- An adequate dissection removes ≥**8 nodes per side**.

2 · Indications and risk groups

	Low risk	Intermediate	High risk
NCCN	pTis, pTa, pT1a	pT1b G1–2	≥pT2, or G3/G4
EAU-ASCO	pTa, pTis, G1 pT1a	G2 pT1a	≥pT1b
cN0 — NCCN	Surveillance	Modified/radical bilateral ILND or DSNB	Same
cN0 — EAU-ASCO	Surveillance	Surveillance (compliant) vs DSNB	DSNB; ILND if unavailable

NCCN = National Comprehensive Cancer Network; EAU-ASCO = European Association of Urology–American Society of Clinical Oncology; DSNB = dynamic sentinel node biopsy; G = grade. **Trap**: pT1b G2 is intermediate (NCCN) but high (EAU-ASCO).

- For the impalpable high-risk groin, **NCCN leads with ILND** and **EAU-ASCO leads with DSNB**. Select on **primary pathology only**, not HPV or nomograms.
- **cN+**: about half of palpable nodes are metastatic. Do fine-needle aspiration cytology (FNAC), then excisional biopsy if negative. If any is positive, do **bilateral radical ILND**. Give no antibiotic trial.
- With unilateral palpable nodes, the other side has micrometastasis in ≈**30%**, so stage it.
- **Bulky or fixed (cN3)**: give cisplatin–taxane chemotherapy first, then surgery. Resect upfront only if chemotherapy-ineligible or vessel invasion is imminent; involved skin needs a myocutaneous flap.

3 · Staging work-up

- Palpable nodes or high risk need 18FDG-PET/CT or CT chest–abdomen–pelvis. Imaging misses deposits <1 cm, so it has no role in the impalpable groin.
- **FNAC** is specific but insensitive: if positive, do ILND; if negative, do DSNB or ILND and never discharge.
- **DSNB**: do ultrasound, with or without FNAC, first, because a **tumour-replaced node** reroutes tracer and gives a false negative. Then use ^{99m}Tc nanocolloid and blue dye with serial sectioning. Positive: ILND †; negative: surveillance.

KEY POINT

Every groin test fails toward **false negatives**; a negative result never ends the pathway.

4 · Lymphatic anatomy

*See the paper answer **Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy** (Section 10, Ch 82).*

Also, from the condensed book (not in the paper answer):

- The **femoral triangle** is bounded by the inguinal ligament, **adductor longus** medially and **sartorius** laterally.
- Raise flaps deep to **Camper's fascia**, which holds the capillary network; too thin causes necrosis and too thick leaves nodes.
- In the fossa ovalis the order, lateral to medial, is **nerve, artery, vein**, so **never dissect lateral to the femoral artery**.

5 · Templates

Template	Extent	Features
Radical	Whole femoral triangle, superficial and deep	Fascia lata opened; saphenous ligated; sartorius transposed
Modified (Catalona)	Excludes lateral to femoral artery and caudal to fossa ovalis	Shorter incision, saphenous preserved, thicker flaps, no transposition
Superficial	Above fascia lata only	Fascia lata preserved

Modified ILND nodes go to frozen section; if **positive, convert to radical** †.

6 · Open radical ILND

ALGORITHM

1. Frog-leg position; incision ≈2 cm below and parallel to the inguinal ligament.
2. Raise flaps 3 cm above the ligament to the apex; divide the saphenous at the apex (spare if disease minimal).
3. Open fascia lata over sartorius and adductor longus; dissect cephalad on the femoral vessels.
4. Divide the saphenous at the junction; clear deep nodes up to the femoral canal.
5. **Sartorius transposition**; close the femoral canal.
6. Suction drains; tack the flaps.

7 · Pelvic lymph node dissection (PLND)

- If the inguinal frozen section is positive, do PLND at the same sitting; otherwise stage it.
- † EAU: with ≥2 positive inguinal nodes or ENE (pN2–3), do ipsilateral PLND, open or minimally invasive.

8 · Video-endoscopic ILND (VEIL)

*See the paper answer **Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy** (Section 10, Ch 82).*

Also, from the condensed book (not in the paper answer):

- The commonest robotic complication is lymphocele.

9 · Complications and post-op care

Fear of complications must not cause omission.

Complication	Cause	Prevention / management
Wound infection, breakdown (commonest)	Skin flora, dead space	Asepsis, tack flaps, incisional negative pressure
Flap necrosis	Breach of Camper's capillary network	Correct plane, modified template, VEIL; debride, then flap or negative pressure †
Lymphocele / seroma	Unsealed lymphatics	Sealing devices; drain until **<30 mL/day**; drain symptomatic or infected collections †
Lymphoedema	Lymphatic interruption	Saphenous sparing, modified template, compression hose ≥1 month
Others	Phlebitis, neurapraxia, pulmonary embolism (PE), femoral bleeding	Thromboprophylaxis; stay medial to artery

Mobilise early. NCCN advises antibiotic prophylaxis, though the authors do not use it routinely.

10 · Follow-up

Most groin relapses occur within 2–3 years (median ≈1 year). EAU-ASCO advises examination 3-monthly for 2 years, then 6-monthly to 5 years, with imaging only if node-positive; NCCN images everyone.

83 Tumours of the Urethra

1 · Benign urethral tumours

Lesion	Who / where	Clue	Treatment
Leiomyoma	Reproductive-age women; posterior wall	Firm suburothelial mass; grows in pregnancy	Excision; may recur
Haemangioma	Men, any age	Bluish sessile lesion, painless bleeding	Fulguration/laser; open excision if extensive
Fibroepithelial polyp	Boys; bulbar/posterior	Polyp on a stalk; filling defect	Transurethral resection

2 · Anatomy, histology and lymphatics

Tumour site predicts its type and its nodal basin.

Feature	Male (~20 cm)	Female (~4 cm)
Segments	Anterior: fossa navicularis, pendulous, bulbar. Posterior: membranous, prostatic	Anterior = distal ⅓; posterior = proximal ⅔
Epithelium	Urothelium in the prostatic urethra, columnar in the anterior urethra, squamous in the fossa navicularis	Proximal ⅓ urothelium; distal ⅔ stratified squamous
Histology	Urothelial carcinoma (UC) 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%. **Anterior: up to 80% SCC; posterior: mostly UC**	UC 28–45%, SCC 19–29%, **adenocarcinoma 28–38%** (near-equal)
Lymphatics (both sexes)	Anterior drains to **inguinal** nodes (bulbar also external iliac); posterior drains to **pelvic** nodes	

> **KEY POINT**
>
> Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women it is also the **continence boundary**: the distal third can be excised without incontinence.

3 · Staging: AJCC 8th edition

T	Penile urethra and female urethra	Prostatic urethra
Ta / Tis	Non-invasive papillary / carcinoma in situ (CIS)	Tis also includes prostatic duct involvement **without stromal invasion**
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (bladder)	Bladder, rectum
N	N1 single, N2 multiple regional nodes (inguinal, pelvic, presacral): **number, not size**	

- **Evaluation** is by examination under anaesthesia with cystourethroscopy, bimanual palpation and biopsy, plus **MRI** for local extent and CT chest if invasive.

- **Palpable inguinal nodes (20–30%) are almost always metastatic**, so, unlike penile cancer, no antibiotic trial is given.

4 · Male primary urethral cancer (PUC)

Distal tumours are curable by surgery; bulbar tumours need multimodal therapy.

- Patients are >60 years. **Urethral stricture** is the commonest risk factor (up to 50%); others are lichen sclerosus, pelvic radiotherapy (RT) and HPV 16.
- † Biopsy a stricture that bleeds or recurs rapidly.
- **Prognosis** depends on stage and location: distal tumours do 2–3-fold better, and bulbar survival is as low as 20–30%.

Pendulous / fossa navicularis (distal)

- Surgery alone controls low-stage disease, ranging from transurethral resection to **distal urethrectomy with or without partial penectomy**, to total urethrectomy with perineal urethrostomy, with or without total penectomy.
- **Penile preservation** is endorsed by EAU and NCCN, using options such as topical 5-fluorouracil (5-FU) for distal CIS, distal urethrectomy, or glansectomy. A positive margin needs re-excision or adjuvant RT.
- A **margin of 5 mm** is now standard (1 cm traditionally).
- **Primary RT** is only for early anterior lesions in men refusing surgery, and it is ineffective for bulbar PUC.

Bulbar (proximal anterior)

- The few low-stage tumours are treated by transurethral resection, or segmental excision and anastomosis.
- **Advanced** disease needs radical cystoprostatectomy, pelvic lymphadenectomy, total urethrectomy and penectomy. Surgery alone gives ~¼ 5-year survival, so use **multimodal therapy**.

Inguinal nodes (both sexes)

- There is **no proven benefit of prophylactic inguinal dissection in cN0**. For cN1, give neoadjuvant chemotherapy or chemoradiotherapy, then lymphadenectomy.

5 · Male posterior (prostatic) urethral cancer

Only stromal invasion needs cystectomy.

- It is usually **secondary to bladder UC**, found in 20–40% at radical cystectomy (RC), so always evaluate the bladder and upper tracts.
- Diagnose by TURP biopsy; it is often CIS. BCG for high-grade superficial disease gives ≥70% response.

Involvement	Treatment	5-year survival
Mucosal	TURP + BCG	67–74%
Ductal / acinar (still Tis)	TURP + BCG, or RC with or without urethrectomy	67–74%
Stromal invasion	**RC + urethrectomy + pelvic lymphadenectomy**, with or without neoadjuvant chemotherapy	~36%; follows bladder stage

> **KEY POINT**
>
> The only question is "stroma breached?" TURP before BCG debulks, improves contact and stages. Ductal disease is not T2, so do not jump to cystectomy.

6 · Female urethral cancer

- It is ~⅓ of PUC, in women >60 years. Risks include chronic irritation, HPV and **urethral diverticulum (6–10% harbour malignancy**, mostly adenocarcinoma).
- Distal tumours are SCC; proximal ones are UC or adenocarcinoma, which is **mucinous** or **clear cell** (in diverticula).
- **Location decides continence-sparing excision**; stage with MRI pelvis.
- **Stage dominates prognosis**: distal tumours do better than proximal, which do better than pan-urethral.

Treatment

Situation	Management
Small non-invasive	Endoscopic resection or laser, with or without BCG
Low-stage distal	**Distal urethrectomy** (transvaginal) **or RT** (brachytherapy with or without external beam, ~60 Gy): equal, ≥70% cured. Stress incontinence (>40%), meatal stenosis; ~1 in 5 recur locally. RT complications 20–50% (stricture, necrosis, fistula)
Proximal / pan-urethral	**Anterior exenteration**: RC, total urethrectomy, extended pelvic lymphadenectomy and anterior vaginal wall, with or without hysterectomy and pubic rami. Alone, most die, so use **multimodal** therapy
Palpable groin nodes	Chemoradiotherapy, chemotherapy or lymphadenectomy; often curable. Pelvic nodes are poor

- RT local control depends on the **length of urethra involved**, not histology or size.

7 · Multimodal therapy (both sexes)

- For advanced or proximal disease, give **cisplatin-based neoadjuvant chemotherapy** (EAU, NCCN) or **chemoradiotherapy**, then consolidative surgery. Response predicts survival.
- **Chemoradiotherapy** is 45–55 Gy to the genitalia, perineum and groins with 5-FU and mitomycin C. It is preferred for advanced SCC, though every responder develops a stricture.

Histology	Regimen
UC	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin); gemcitabine–cisplatin
UC or SCC	Paclitaxel, ifosfamide, cisplatin (TIP); cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gemcitabine, 5-FU, leucovorin, cisplatin

Stage (NCCN, anterior male / female)	First line
Tis, Ta, T1	Repeat transurethral resection, with or without intraurethral BCG or chemotherapy
Male T2 pendulous	Partial urethrectomy with or without partial penectomy; if margin positive, chemoradiotherapy, surgery or RT

Stage (NCCN, anterior male / female)	First line
Male T2 bulbar	Urethrectomy with or without RC; if ≥pT3/N1, adjuvant chemotherapy or chemoradiotherapy
Female T2	RC + urethrectomy, or distal urethrectomy by location, or chemoradiotherapy
≥T3 N0	Chemoradiotherapy with or without surgery; or neoadjuvant chemotherapy, then surgery or RT
N1	Chemoradiotherapy (preferred for SCC) or systemic therapy, with or without consolidative surgery

- Recurrence after resection of invasive disease is ~50%, so survey the urethra, groins and body.

8 · Urethral recurrence (UR) after radical cystectomy

Urethral risk decides whether a neobladder is safe.

Male

- Risk is **1–15%** (~5%), mostly within 2 years.
- **Risk factors** are **prostatic urethral involvement** (15% vs 5%), **positive urethral margin**, cutaneous diversion, CIS and multifocality. An orthotopic neobladder (ONB) is protective (~5% vs 9%).
- Routine prophylactic urethrectomy is abandoned, because 64–100% of specimens are negative.
- **Before ONB**, confirm a negative intraoperative frozen section of the urethral margin.
- **Urethrectomy** is absolute for a positive margin or gross urethral tumour, and relative for diffuse CIS, prostatic urethral involvement or multifocal disease. Delayed urethrectomy is as good as simultaneous.

Female

- The main risk is **bladder neck involvement**, but only 30–50% of these have urethral tumour, so it is not an absolute contraindication.
- ONB is allowed if the bladder neck, urethra and anterior vaginal wall are uninvolved and frozen section is negative; UR after ONB is <5%.

Surveillance and treatment

- Urethral wash cytology has no proven survival benefit; if positive, do urethroscopy and biopsy.
- **Total urethrectomy is treatment of choice**, including the meatus; after ONB, convert to cutaneous diversion. **Intraurethral BCG treats CIS** (not papillary or invasive). Invasive UR has median survival ~17 months.

85 Benign Disorders of the Penis and Urethra

1 · Reconstructive surgical technique

- **Cautery**: use **bipolar on the urethra** to limit spongiosal damage, and monopolar on subcutaneous tissue and muscle.
- A **bougie à boule** calibrates. **Goodwin sounds** dilate over a guidewire, so they are safer than van Buren sounds.
- **Sutures** are always **absorbable**, because permanent ones form a stone nidus; monofilament polydioxanone (PDS) is preferred. Support is needed for **30–60 days**, so avoid chromic, which lasts 14 days. Use **5-0** for most urethral work and **6-0** for a single-layer mucosal anastomosis (non-transecting excision and primary anastomosis, EPA).

2 · Tissue transfer: grafts and flaps

LAQ Grafts in urology and buccal mucosa

Paper answer · P1 · P3 · merged from 2

Asked as: P1 LAQ "Enlist various grafts used in urology and role of BMG in urology" · P3 SAQ "Uses of buccal mucosa in urology"

Graft versus flap

	Graft	Flap
Blood supply	Separated; gets a new supply from the bed	Moved with its own pedicle
Depends on	A well-vascularised host bed	Its pedicle
Example	BMG (buccal mucosal graft)	Penile fasciocutaneous island flap

Graft take (about 96 hours)

Phase	Duration	Mechanism
Imbibition	about 48 h	Graft "drinks" nutrients from the bed; graft is cooler than core
Inosculation	about 48 h	Bed and graft vessels connect; true microcirculation returns

- A graft lives on its bed, so it does best on thick vascular beds. Take fails with poor beds (scar, radiation), haematoma, seroma, infection or shear. Hence **quilting or pie-crusting** and **immobilisation for 5–10 days** (bolster or vacuum dressing).

Grafts used in urology

Group	Graft	Main uses
Skin	**STSG** (split-thickness skin graft), meshed	Staged urethroplasty only; penile and scrotal cover (Fournier gangrene, buried penis)
	FTSG (full-thickness skin graft)	Fossa navicularis, penile urethra (avoid in lichen sclerosus)
	Posterior auricular (Wolfe) FTSG	Staged hypospadias repair
Mucosa	**Buccal, lingual, labial (oral mucosa)**	Urethra, hypospadias, ureter
	Bladder epithelium (mucosa)	Urethra, hypospadias; meatal prolapse

Group	Graft	Main uses
	Rectal (colonic) mucosa	Panurethral stricture when oral mucosa is unavailable
Other autologous	Dermis	Tunica albuginea in Peyronie disease
	Tunica vaginalis	Peyronie disease, urethral cover; easy harvest
	Saphenous vein	Peyronie defects; vascular repair
	Rectus fascia, fascia lata	Pubovaginal sling
Allograft / xenograft	Pericardium, porcine SIS, porcine dermis, cadaveric fascia	Peyronie grafting ("off the shelf"); slings
Synthetic	Polypropylene (macroporous type I mesh)	Mid-urethral sling, prolapse
	Dacron, PTFE (polytetrafluoroethylene)	Vascular grafts; not advised for Peyronie (infection, fibrosis)
Newer	Collagen fleece (TachoSil); acellular matrix, tissue-engineered oral mucosa	Peyronie; experimental urethra

- Vein is not a true graft: it survives by direct endothelial perfusion and its vasa vasorum.
- **Peyronie grafts:** pericardium (thin, strong, does not contract) and SIS (cut about 25% oversize; contracts) are commonest; AUA gives no preference.
- **STSG versus FTSG:** STSG carries little dermis, takes easily, can be meshed (up to sixfold) but contracts. FTSG carries full dermis, contracts less, looks better but is more "fastidious" in take.

Buccal mucosa: history and why it works

- **BMG** is the workhorse graft for urethral reconstruction. **Humby 1941** (hypospadias); **Bürger 1992** and **El-Kasaby 1993** (stricture).
- Thick, **non-keratinised**, elastin-rich, hairless epithelium used to a **wet environment**; resists infection.
- Thin, **highly vascular lamina propria** with a panlaminar plexus, so imbibition and inosculation are rapid and take is excellent.
- Little contraction; handles and holds sutures well; low donor morbidity.

Harvest

1. Nasotracheal intubation; mouth gag; mark the graft (about 4–5 × 2 cm per cheek; P3: 4–6 × 2–2.5 cm).
2. Stay **1 cm from the commissure** and avoid **Stensen's (parotid) duct** opposite the upper second molar.
3. Infiltrate, incise, raise the mucosa off buccinator; haemostasis.
4. Donor site: leave open or close; both acceptable.
5. Back table: **defat** the graft, cut about 10% oversize, fenestrate with a needle; quilt to the bed. - **Lingual** (under the tongue) and **labial (lip)** grafts are alternatives. Buccal and lingual give equal urethroplasty success; lip mucosa is thinner, suits the glans.

Uses in urethroplasty

Technique	Principle	Best for
Dorsal onlay (Barbagli, 1995)	Urethra rotated, dorsal urethrotomy; graft spread-fixed on corpora	Bulbar; reliable bed
Ventral onlay	Ventral urethrotomy; graft covered by spongioplasty	Bulbar with healthy spongiosum
Dorsal inlay (Asopa)	Ventral approach (urethrotomy); dorsal plate incised; graft inlaid	Penile and bulbar, no mobilisation
Augmented anastomosis	Excise narrowest part, floor anastomosis, graft roof	Long, dense bulbar stricture
Kulkarni (2012)	One-sided dissection keeps ventral blood supply; penile invagination; dorsal graft (onlay)	Panurethral, one stage
Staged (Johanson, Bracka)	Stage 1 lay open and graft; stage 2 tubularise after 6 months or more	**Lichen sclerosus**, failed hypospadias
BMG perineal urethrostomy	Graft completes dorsal wall to skin	Panurethral or failed repairs

- Dorsal and ventral results are similar (dorsal equals ventral); graft equals flap; onlay beats tubes.
- **Skin is avoided in lichen sclerosus** because the disease recurs in skin; use staged BMG.

Other uses of BMG

- **Ureter:** BMG onlay ureteroplasty (first human report 1999), now robotic with an **omental wrap**; for long proximal or mid strictures in dense scar. CORRUS (Collaborative of Reconstructive Robotic Ureteral Surgery) failure about 15–25% by site.
- Redo and staged hypospadias; urethral closure in rectourethral fistula; bladder neck contracture; female urethral stricture; Peyronie grafting (small defects only).

Complications

- **Donor:** pain, swelling (first month), numbness (up to 6 months), mouth tightness, salivary duct injury.
- **Recipient:** graft failure, restricture, sacculation, fistula, ED.

Outcomes

- Bulbar BMG urethroplasty about 85–90% success; most failures within 12 months.
- Results fall with time and in penile, redo and lichen sclerosus cases.

Recent advances

- Tissue-engineered oral mucosa; acellular matrix; robotic BMG ureteroplasty.

Diagram to draw: bulbar cross-section showing dorsal onlay, ventral onlay and dorsal inlay (Asopa) positions.

Key points to remember:

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, thin vascular lamina propria (excellent take), wet-adapted.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.
- Bulbar success about 85–90%.

Also, from the condensed book (not in the paper answer):

- **Extragenital full-thickness skin graft (FTSG)** fails in the urethra because its greater mass makes take ischaemic. The exception is the **posterior auricular (Wolfe) graft**, which behaves like a split-thickness skin graft (STSG).
- Oral mucosa graft (OMG) has a **panlaminar plexus** and sacculates less than skin. Buccal onlay failures occur in the **first 12 months**.

KEY POINT

Graft trade-off: a dense fine plexus and low mass favour take, while carried dermis or lamina propria gives durability and resists contraction. This explains STSG vs FTSG, the failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal.

Flaps

- Flaps are classified by **composition** (fasciocutaneous, fascial, musculocutaneous, muscle or visceral), **blood supply** (axial on a named vessel, or random as in a Z-plasty), **movement** (advancement, rotation or transposition) and **location** (local, distant pedicled or free).

3 · Surgical anatomy

Fascial layers decide flap planes and how blood spreads.

- Layers, deep to superficial, are **tunica albuginea, Buck's fascia, dartos, skin**. Dartos is fat-free, carries the skin vessels and fuses with Colles fascia.
- **Skin supply**: the superficial external pudendal artery feeds a subdermal plexus, which is the basis of the McAninch circular flap.
- **Deep supply**: the internal pudendal artery continues as the common penile artery, giving the **bulbourethral, dorsal and cavernosal** arteries.

Situation	Spread
Buck's intact (penile fracture)	Confined to the shaft, giving an **eggplant deformity**
Buck's ruptured (straddle)	Within Colles–Scarpa: scrotum, perineum, abdominal wall; limited on the thigh by fascia lata
Fournier gangrene	From perineum and scrotum up the abdominal wall via Scarpa

KEY POINT

Two envelopes, two vascular worlds: dartos carries the skin supply (so skin flaps are raised on dartos); Buck's encloses the erectile bodies fed by the common penile artery.

4 · Lichen sclerosus (LS)

*See the paper answer **Lichen sclerosus (BXO, balanitis xerotica obliterans)** (Section 3, Ch 30).*

Also, from the condensed book (not in the paper answer):

- Malignant transformation (mostly SCC) reaches up to **8%**.
- **Steroid**: clobetasol propionate or betamethasone 0.05% once or twice daily for 1–2 months, then on alternate days for 3–4 months.
- Meatal stenosis is treated by extended meatotomy or BMG onlay, and early repair may avert a panurethral stricture.

> **KEY POINT**
>
> LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, retrograde urethrography (RUG) before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy.

5 · Other urethral lesions

Lesion	Key features	Management
Urethral haemangioma	Bloody discharge; often extends beyond view, so image with **MRI**; benign	Observe; small: laser or triamcinolone; large: excision, staged repair
Reactive arthritis (Reiter)	Arthritis, conjunctivitis, urethritis after Chlamydia/Shigella/Salmonella; HLA-B27 in ~2/3; **circinate balanitis**	Urethritis is self-limiting
Amyloidosis	Urethral mass, obstruction, haematuria; needs biopsy	Observe; dilate or urethroplasty if obstructive
Urethrocutaneous fistula	Mostly iatrogenic (hypospadias, phalloplasty)	Treat the **underlying process** as well as the tract
Male urethral diverticulum	Congenital (all layers) or acquired (no smooth muscle; behind stricture, catheter, trauma); dribbling, UTI, stone	Cystourethroscopy and RUG; repair and treat the cause
Enlarged utricle	Müllerian remnant, with proximal hypospadias	A catheter may enter the utricle; UTI, post-void dribble

6 · Prepuce and meatus

- **Paraphimosis (emergency)**: a trapped foreskin causes oedema and can lead to glans necrosis. Reduce by steady compression, with the thumbs pushing the glans and the fingers pulling the foreskin. If irreducible, do an emergency **dorsal slit**; offer elective circumcision for recurrence.
- **Phimosis** is physiological in infants (resolving at 1–8 years); in adults it leads to balanitis and cancer. Treat with steroid, dorsal slit or circumcision.
- **Balanitis**: **Candida** is commonest. Treat with hygiene and antifungal, and **check blood glucose**.

SAQ Meatal stenosis and submeatal stricture in males

Paper answer · P1 · P2 · merged from 2

Asked as: P2 SAQ "Management of Urethral meatal stenosis." · P1 SAQ "Treatment of submeatal stricture urethra in males"

Definition

- **Meatal stenosis:** narrowing of the external urethral meatus below the normal calibre for age.
- **"Submeatal stricture"** is read as a stricture of the meatus and fossa navicularis (glanular urethra). In adults, meatal stenosis is usually the tip of a **fossa navicularis stricture**.

Age	Normal calibre
Under 4 months	6–10 Fr
1–12 months	10–12 Fr
1–10 years	12–14 Fr
Over 10 years	14–18 Fr

Causes

- **Children:** almost always after **neonatal circumcision**: wet nappies irritate the exposed meatus (ammoniacal meatitis) and the ventral lips fuse into a thin membrane. Rarely congenital with hypospadias.
- **LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]):** main inflammatory cause in adults today; begins at the glans and meatus and progresses proximally.
- **Iatrogenic:** catheter or instrument trauma, TURP (transurethral resection of prostate).
- **Failed hypospadias repair.**

Clinical features

- Thin, forceful stream deflected **upwards**; spraying; straining, frequency, incomplete emptying; pain at the start of voiding.

Evaluation

- Child: watch the child void; calibrate with a bougie à boule. Uroflowmetry. Urethrography only for UTI (urinary tract infection) or suspected proximal lesion.
- Adult: uroflowmetry; **RUG (retrograde urethrogram) before any meatal surgery**, as a proximal stricture often coexists, especially in LS.
- Biopsy suspicious or atypical lesions: LS turns malignant (squamous cell carcinoma) in up to 8%.

Treatment

Option	Method	Indication / comment
Topical steroid	Clobetasol 0.05% once or twice daily for 1–2 months, then alternate days; gentle dilation and nappy hygiene in children	Meatitis and early narrowing in a child; early LS
Dilation or self-calibration	Meatal dilator lubricated with clobetasol	Elderly, unfit; seldom durable

Option	Method	Indication / comment
Ventral meatotomy (child)	EMLA (eutectic mixture of local anaesthetics) cream in clinic, or general anaesthesia; parents spread the edges with a dilator tip 3 times daily for 7–10 days	Symptomatic stenosis in a child
Ventral meatotomy (adult)	Crush and incise ventrally; **three sutures** (apex and each side), mucosa to glans skin; dilate afterwards	Simple stenosis; in LS extend it with an inverted-V relaxing incision (extended meatotomy)
Dorsal meatotomy (Y-V plasty)	Excise the scar ridge, advance a flap	Scarred or recurrent; effective but cosmesis poorer
Fossa navicularis reconstruction (flaps)	**Jordan flap:** ventral urethrotomy through the stenosis; **transverse ventral penile skin island on dartos** transposed and inverted into the defect. **De Sy:** ventral longitudinal island advanced	Fossa navicularis involved, non-LS; excellent function and cosmesis; not in LS
Graft	**BMG (buccal mucosal graft)** inlay or onlay, one or two stages; glans wings developed	**LS: oral mucosa, never genital skin**
Perineal urethrostomy	Bypass	Pan-anterior LS, failed repairs, elderly

- LS affects genital skin, so skin flaps recur; it spares the proximal urethra, so perineal urethrostomy works.

Complications

- Bleeding, restenosis (commoner in LS), spraying, poor cosmesis.

Diagrams to draw

- Ventral meatotomy: crush, incise ventrally, suture mucosa to glans at apex and both sides.
- Jordan flap: transverse ventral skin island on dartos inverted into the ventral meatotomy.

> **Key points to remember:**
> - Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
> - Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
> - Non-LS: meatotomy or Jordan/De Sy skin island flap.
> - LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Adult circumcision (sleeve)

ALGORITHM

1. Penile block. Foreskin retracted: mark the cuff incision across the frenular base; incise to Buck's.
2. Foreskin advanced: mark the coronal incision.
3. Excise the sleeve using **bipolar**, because adult vessels do not seal by pressure.
4. Quadrant dartos sutures, then skin closure.

- **Human immunodeficiency virus (HIV)**: the foreskin has many CD4+ cells and microtears. Circumcision lowers female-to-male HIV by ~**60%**, and also lowers herpes simplex and HPV.

7 · Penile curvature

*See the paper answer **Congenital Penile curvature..** (Section 2, Ch 55).*

Also, from the condensed book (not in the paper answer):

- Congenital curvature is a smooth whole-shaft curve, whereas chordee without hypospadias shows a hooded prepuce and ventral inelasticity.

Devine–Horton type	Abnormal tissue
I	Spongiosum deficient (chordee without hypospadias, I–III)
II	Dysgenetic Buck's and dartos; urethra/spongiosum normal
III	Dartos only
IV	All normal; asymmetric tunica = **congenital curvature**
V	Congenital short urethra; its existence is doubted

ALGORITHM

1. Excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum.
2. Artificial erection; ventral midline septotomy.
3. If still curved, do dorsal tunical ellipse excision or plication under the elevated neurovascular bundle.

- **Acquired non-Peyronie curvature** (see Ch 76) follows deep urethrotomy, neglected or **subclinical fracture**, or a septal tear. Erections are normal, with no shortening. Treat by **scar excision and graft**, not contralateral plication.

8 · Penile transplantation

A composite allotransplant restores appearance, erection and voiding after penile loss, but needs lifelong immunosuppression; the alternative is phalloplasty.

9 · Adult acquired buried penis (AABP)

It is self-perpetuating and usually needs surgery.

- Risks are **obesity**, aggressive circumcision, scrotal lymphoedema and LS.
- It is a **vicious cycle**: the escutcheon tethers the penis, and trapped urine causes inflammation, LS and scarring, which trap it further. So **weight loss alone rarely cures**.
- The **Pariser** classification grades the repair needed: I skin flap, II graft, III scrotoplasty, IV escutcheonectomy, V panniculectomy.
- **Evaluate the urethra**, because a severe stricture needs urethroplasty **before** grafting.

ALGORITHM

1. Slit the phimotic ring and place glans traction.
2. Circumcising incision; excise non-viable skin.
3. Do escutcheonectomy (harvest the graft from it; keep 2–3 cm of skin at the base).
4. Tack Scarpa to the rectus fascia or pubis; place drains.
5. Cover with STSG (0.012–0.020 inch), fibrin glue and a bolster for 5–7 days.

- **Scrotal lymphoedema** may be congenital, infective (**Wuchereria bancrofti**, lymphogranuloma venereum), malignant or after node dissection or radiation. Resect **all** involved tissue and cover with spared perineal or lateral scrotal skin.
- **Foreign body injection** (silicone, paraffin) causes fibrosis, necrosis and abscess. Excise it all and graft, and screen for **body dysmorphic disorder**.

10 · Landmark trials

Trial	Design	Result	Take-home
African circumcision trials (South Africa, Kenya, Uganda)	Three randomised controlled trials (RCTs), adult male circumcision vs none	~60% reduction in female-to-male HIV acquisition	Circumcision is an HIV-prevention strategy

86 Urethral Stricture Disease

1 · Definitions, anatomy, aetiology

LAQ Urethral stricture: etiology, classification, DVIU, management and failed VIU

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Urethral stricture disease: etiology, presentation and management (including recent advances)" (asked as "Describe etiology, clinical presentation and management of urethral strictures" and "Describe etiology, clinical presentation and recent advances in management of Urethral Structure Disease") · P3 LAQ "Urethral stricture: classification, DVIU and surgery after failed VIU" (asked as "Classify urethral strictures. Discuss in brief, surgical management of stricture after failed VIU" and SAQ "DVIU (Direct Visual Internal Urethrotomy)")

"Structure" is taken to mean urethral stricture disease.

Definition

- A **urethral stricture** is a fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar, often with variable **spongiofibrosis**.
- Posterior narrowing is named differently (WHO [World Health Organization] 2004, ICUD [International Consultation on Urological Diseases] 2010): after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery or radiation, **stenosis or contracture**.
- Area falls with the square of diameter: halving calibre (30 to 15 Fr) loses about three-quarters of the lumen area, so symptoms appear late.

Anatomy relevant to surgery

- **Bulbar urethra lies eccentric (dorsal)** in the spongiosum; the penile urethra is central.
- **Dual blood supply** (bulbar arteries plus retrograde flow from the glans) lets the bulb be divided and mobilised.

Aetiology

Category	Examples and mechanism
Idiopathic (commonest, about equal to iatrogenic)	No cause found
Iatrogenic	Catheters (very common in LMICs [low- and middle-income countries]), TURP (transurethral resection of prostate), TURBT (transurethral resection of bladder tumour), cystoscopy, hypospadias repair, radiotherapy. Mechanism: mucosal injury, pressure ischaemia, infection
Traumatic	**Straddle injury**: bulb crushed against the pubis; pelvic fracture (PFUI) gives a posterior obliterative defect
Inflammatory	Gonorrhoea (historically commonest, now rare); **lichen sclerosus (LS)**. Chlamydia: no proven link
Congenital	Rarest; infants only (short anterior stricture)

- **LS**: chronic skin disease starting at the glans and meatus. High-pressure voiding forces urine into Littre glands, causing inflammation and deep spongiofibrosis. It **recurs in any genital skin** used to repair it.

Classification

1. By site (WHO 2004 / ICUD 2010 terminology)

Segment	Term
Anterior: meatus, fossa navicularis, penile (pendulous), bulbar	**Stricture**
Membranous urethra after pelvic fracture	**PFUI** / distraction defect
Posterior urethra after prostatectomy or radiotherapy; bladder neck	**Stenosis or contracture**

2. By aetiology: idiopathic, iatrogenic, traumatic, inflammatory, congenital (table above).

3. By length: short (up to about 2 cm), long, multifocal, **panurethral** (whole anterior urethra).

4. By depth (Jordan's spectrum): (A) mucosal fold, (B) iris constriction, (C) full-thickness with minimal spongiofibrosis, (D) full-thickness (deep) spongiofibrosis, (E) fibrosis beyond the spongiosum (periurethral fibrosis), (F) complex stricture with fistula or abscess.

5. By complexity: simple (short, bulbar, first treatment) versus complex (long, multiple, LS, radiation, failed previous treatment, fistula).

- **PFUI** (Ch 125): major distraction, minor distraction, partial injury.
- **Also know: LSE system** (Lumen calibre, Segment involved, Etiology) for standard reporting.

Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- **Retention** after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

Evaluation (define the 4 Ls: location, length, depth, density)

Test	Role
Uroflowmetry and PVR (post-void residual)	Low, flat plateau-shaped flow curve; screen and follow-up
Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)	Site and length; steep oblique views; done by the surgeon
Flexible urethroscopy	Calibre, appearance; paediatric scope sees beyond
Sonourethrography	Length and depth of spongiofibrosis
MRI (magnetic resonance imaging)	PFUI defect alignment only; not routine

- **Urethral rest** (suprapubic catheter 6–8 weeks) shows the true length. Avoid blind dilation and instrumentation before surgery. In retention, place a **suprapubic catheter** rather than force a dilation.
- Before urethroplasty: treat infection; look for LS at the meatus; examine the mouth; record erectile function.

Management principles

- First agree the aim: **palliation or cure**.
- The old "reconstructive ladder" (repeat endoscopy before surgery) is abandoned (archaic), because repeated endoscopy makes later urethroplasty harder.

1. Dilation

- Soft or balloon dilation over a guidewire. It stretches, not tears, the scar. Cures only epithelial strictures without spongiofibrosis.

2. DVIU (direct vision internal urethrotomy, VIU)

Principle

- Incision of the stricture scar into healthy tissue under vision (Sachse cold-knife urethrotome), so the lumen heals enlarged by **secondary intention**: a **race between epithelialisation and wound contraction**. The stricture recurs when contraction wins, which is usual when spongiofibrosis is deep. Deep spongiofibrosis cannot be cured endoscopically.

Indications

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis, any non-bulbar
Short (<1.5–2 cm)	Longer
Superficial, no dense spongiofibrosis	Dense scar (straddle, LS)
First procedure	Repeat; luminal narrowing >74%

Technique

1. RUG and VCUG define length and site.
2. Saline irrigation; **guidewire** through the stricture.
3. **Cold-knife incision traditionally at 12 o'clock** through scar to healthy spongiosum. Campbell notes the spongiosum is thinnest from 10 to 2 o'clock; deep cuts enter the corpora (ED [erectile dysfunction], fistula).
4. Catheter **3–7 days**; longer catheterisation adds no benefit. - **Laser** (holmium, Nd:YAG [neodymium:yttrium–aluminium–garnet], others): **no advantage over cold knife** (laser = cold knife).

Adjuncts

- **CISC (clean intermittent self-catheterisation)/self-dilation**, tapering over 3–6 months: delays recurrence, which returns when stopped.
- Mitomycin C injection (reduces recurrence), steroid injection (limited data); removable stents; permanent UroLume stents withdrawn.

Complications

- **Recurrence** (commonest); bleeding (often with erections); **irrigant extravasation**; UTI, sepsis; ED from deep cuts; spongio-cavernous fistula; incontinence if sphincter incised.

Why repeat VIU is futile after failure

- VIU succeeds only in **bulbar strictures <1.5 cm without dense spongiofibrosis, at the first attempt**. Recurrence shows deep spongiofibrosis, which an incision cannot remove.

- **Steenkamp 1997, Heyns 1998 (randomised trials):** dilation = urethrotomy; a second urethrotomy helps few, a third none. **A second urethrotomy adds virtually nothing.**
- **Santucci & Eisenberg (2010):** about **8% stricture-free** long term after a first DVIU; repeat procedures almost always fail.
- Each repeat dilation or urethrotomy lengthens the scar and lowers later urethroplasty success; urethroplasty is more **cost-effective** after one failure.
- **AUA (American Urological Association) guideline (2016)** and EAU (European Association of Urology): DVIU or dilation may be offered for **short (<2 cm) bulbar** strictures; offer **early urethroplasty, not repeated endoscopy**, after one failed dilation or DVIU; no permanent stents (no stent if urethroplasty is possible).

3. Urethroplasty (including surgery after failed VIU)

Stricture	Operation	Principle and key points
Bulbar, short (1–2 cm, up to 3–5 cm proximally)	**EPA (excision and primary anastomosis)**: gold standard, about 90–95% success	Excise all scar, spatulate both ends widely, tension-free anastomosis. **Non-transecting** or **vessel-sparing** variants keep bulbar arteries
Bulbar, longer	**BMG (buccal mucosal graft) onlay**: dorsal (**Barbagli**), ventral, dorsal inlay (**Asopa**)	Graft spread-fixed on a well-vascular bed (corpora or spongiosum)
Bulbar with an obliterated segment; long dense	**Augmented anastomosis**	Excise the narrowest part, anastomose the floor, graft the roof
Penile	One-stage BMG inlay or onlay; **penile skin flap** (Orandi, Quartey, circular)	Skin island carried on dartos; not in LS; avoid EPA here (chordee)
LS, failed hypospadias, infection	**Staged repair** (Johanson): stage 1 open and graft; tubularise after about 6 months or more	Oral mucosa, not genital skin, in LS
Panurethral	Kulkarni one-sided, one-stage dorsal graft, or staged	—
Elderly, unfit, complex, multiple failures	**Perineal urethrostomy** (BMG-augmented)	Simple, durable bypass
Meatal stenosis; fossa navicularis	Meatotomy; flap or graft	Topical clobetasol for early LS meatal disease
PFUI (including after failed VIU)	Suprapubic catheter, then delayed **perineal anastomotic urethroplasty** after 3 months or more	Progressive steps: bulbar mobilisation, intracrural separation, inferior pubectomy, rerouting of the urethra

- **Onlay is better than tube** repairs. Graft and flap results are equal; grafts are simpler.
- **Avoid genital skin in LS** (use oral mucosa).
- **Not fit or declines surgery: Optilume** paclitaxel-coated balloon; intermittent self-dilation; long-term suprapubic catheter.
- **Urethral stents** (UroLume) are withdrawn.

Recent advances in management

1. **Ladder abandoned**: repeated endoscopy lowers later urethroplasty success, so **early urethroplasty** is offered after one failed DVIU (AUA and EAU guidelines).
2. **Optilume paclitaxel-coated balloon** (FDA [Food and Drug Administration] 2023): paclitaxel inhibits fibroblasts. **ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after (repeated) failed endoscopy; IPSS (International Prostate Symptom Score) 9 vs 19.9.
3. **Anti-scarring adjuncts after DVIU**: mitomycin C injection reduces recurrence; intermittent self-dilation; removable stents. Permanent UroLume stents withdrawn.
4. **Vessel-sparing and non-transecting EPA**: preserve bulbar arteries, theoretically less ED.
5. **Augmented anastomosis**: excise the narrowest segment, graft the rest.
6. **Kulkarni one-sided dorsal BMG** through a perineal incision for panurethral strictures.
7. **Staged oral mucosa repair for LS**; avoid skin. **BMG perineal urethrostomy** reduces neomeatal stenosis.
8. **Robotic reconstruction** of bladder neck contracture and posterior stenosis (Y-V plasty, BMG).
9. **Tissue-engineered oral mucosa** (investigational). **Prevention**: silicone and hydrophilic catheters, smaller endoscopes.
10. **PROMs (patient-reported outcome measures)**: USS-PROM (Urethral Stricture Surgery PROM) is increasingly used to judge success. - **Also know:** dartos-island fossa repair; lingual graft; clobetasol for early LS.

Complications of urethroplasty

- Recurrence, ED (overall similar to circumcision; higher with long repairs), chordee, post-void dribble, graft-site pain, numbness.

Follow-up

- Voiding cystourethrogram at catheter removal (about 3 weeks); then uroflow, PVR and symptom score; RUG, cystoscopy or urethroscopy if flow falls.
- Most EPA recurrences occur in the first 6 months; graft results decline with time.

Diagrams to draw

- RUG of a bulbar stricture.
- Cross-section of dorsal onlay versus ventral onlay.
- EPA with spatulated ends.
- Urethrotome cutting at 12 o'clock over a guidewire.

> **Key points to remember:**
>
> - Idiopathic and iatrogenic are commonest; LS recurs in skin.
> - Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
> - DVIU only once, for bulbar strictures <1.5–2 cm at first attempt; do not repeat it.
> - Santucci: about 8% long-term success; repeat DVIU futile.
> - AUA: urethroplasty after failure; laser = cold knife; ladder abandoned.
> - EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
> - Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
> - ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

Also, from the condensed book (not in the paper answer):

- Straddle strictures often present late; prostate cancer treatment causes posterior stenosis.

Site	Correct term
Anterior urethra	Stricture (epithelial scar with or without spongiofibrosis)
Membranous urethra after pelvic fracture	Distraction defect; pelvic fracture urethral injury (PFUI)
Posterior urethra after trauma/radical prostatectomy	Posterior urethral stenosis
Prostatic urethra, bladder neck	Contracture or stenosis

- The common penile artery supplies the corpus spongiosum. The bulbar urethra lies dorsally, so a deep 12 o'clock urethrotomy in the bulb is dangerous.
- In lichen sclerosus (LS), microabscesses form in the glands of Littre; antibiotics relieve symptoms, not the stricture.

> **KEY POINT**
>
> Retrograde glans flow allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation causes ischaemia and PFUI repair failure.

> **KEY POINT**
>
> Aetiology predicts biology: iatrogenic and straddle strictures are focal scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it.

2 · Presentation and evaluation

- **Acute retention**: if a catheter fails, do a retrograde urethrogram (RUG) and place a **suprapubic cystostomy**, not blind dilation. Pass any guidewire under flexible endoscopic vision.
- Four determinants guide treatment: **location, length, depth, density**.
- The treating surgeon does a dynamic RUG and voiding cystourethrogram (VCUG). Do not dilate at first urethroscopy. MRI is for tumours and PFUI only.
- **Hydraulic trap**: voiding pressure holds the urethra above the stricture open, so a segment that cones down is scarred.
- **Urethral rest** with a suprapubic tube for **6–8 weeks** reveals the true length.

3 · Decision making

- First agree on **palliation or cure**. The **reconstructive ladder is archaic**: go to the procedure most likely to cure.

ALGORITHM

1. If bulbar, <1.5 cm, without dense spongiofibrosis and untreated, then dilate or do one internal urethrotomy; if it recurs, then urethroplasty.
2. If bulbar 1–2 cm (up to 3–5 cm proximally), then excision and primary anastomosis (EPA).
3. If longer bulbar, then oral mucosa graft (OMG) onlay, with or without augmented anastomosis.
4. If penile, then onlay (graft or flap); if hostile bed, LS or failed hypospadias, then staged OMG.
5. If panurethral, options exhausted or elderly, then perineal urethrostomy.

4 · Dilation and internal urethrotomy (DVIU)

*See the paper answer **Urethral stricture: etiology, classification, DVIU, management and failed VIU** (Section 10, Ch 86).*

Also, from the condensed book (not in the paper answer):

- DVIU does worse when luminal narrowing exceeds **74%**; the safest dilation is a **balloon** over an endoscopically placed guidewire.

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis
<1.5 cm	Longer
No dense deep spongiofibrosis	Dense (e.g. straddle)
First procedure	Repeat

- Self-catheterisation tapers over **3–6 months**; the stricture recurs when it stops. A 6-week catheter is no better than 3–7 days.

KEY POINT

No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; each repetition spends the patient's reconstructive capital.

5 · Excision and primary anastomosis (EPA)

- EPA is the **gold standard** for short bulbar strictures; recurrence is rare. Rules: **excise all fibrosis, spatulate widely, keep it tension-free**.
- Length comes from spongiosal mobilisation, freeing Buck's fascia and the intracrural space, and detaching bulbospongiosus from the perineal body. **The more proximal the stricture, the longer the excisable segment.**

	Transecting	Vessel-sparing transecting	Non-transecting
Bulbar arteries	Divided	Preserved	Preserved
Steps	Excise, spatulate, anastomose	Divide at membranous–bulbar junction, dorsal spatulation	Dorsal stricturotomy, excise scar from within, close transversely
Use	Most	Proximal bulbar	Proximal

Vessel-sparing protects men with poor retrograde supply (hypospadias, trauma, prior surgery); its other benefits are unproven.

6 · Substitution urethroplasty

- **Buccal mucosa graft (BMG)** is the workhorse, one-stage or staged. Skin grafts are not used in LS. Grafts do best in the **bulb**, which has a thick vascular bed.
- Graft position may be ventral (with spongioplasty), lateral, or **dorsal (Barbagli)**, spread-fixed on the corpora.
- **Augmented anastomosis** often beats pure onlay. **Kulkarni** is a one-stage panurethral dorsal onlay via the perineum with **penile invagination**.
- **Staged**: stage 1 lays the urethra open and quilts the graft to dartos. Stage 2 tubularises it to about 28 Fr after at least 6 months. It is used for LS and failed hypospadias.

> **KEY POINT**
>
> A graft lives on its bed: it succeeds where the bed is thick and vascular (bulb, dorsal corpora), and is staged when the bed, or skin disease, cannot be trusted in one step.

- **Flaps** use hairless genital skin on dartos, such as the **Orandi** ventral longitudinal flap or a **circular island** flap up to **15 cm**; fossa navicularis strictures use a Jordan or De Sy island.
- Grafts equal flaps; flaps suit irradiated or multiply operated urethras. **Onlay beats tube**.
- **LS**: skin repairs fail, so use **oral mucosa only**, often staged.
- **Perineal urethrostomy** opens the urethra dorsally to healthy mucosa, sutures it to skin, and can add a BMG to complete the dorsal wall.

7 · Meatal stenosis

- **Child**: post-circumcision meatitis needs ventral meatotomy and parental dilation; early meatitis responds to 0.05% clobetasol.
- **Adult**: the fossa is usually involved, so evaluate it as a stricture before ventral meatotomy or dorsal Y-V plasty.

8 · Pelvic fracture urethral injury (PFUI)

*See the paper answer **Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management** (Section 11, Ch 125).*

Also, from the condensed book (not in the paper answer):

- PFUI occurs in about **10%** of pelvic fractures. Repair at **3–6 months**; endoscopic core-through is not curative.
- Failure is mostly **ischaemic stenosis** of the mobilised bulb; internal pudendal status decides the order of surgery (table below).

ALGORITHM

1. On-table endoscopy; exaggerated lithotomy; perineal incision.
2. Mobilise the bulb; pass a sound via the suprapubic tract; excise fibrosis.
3. Spatulate to **≥30 Fr**; place 8–10 tension-free 4-0 PDS sutures over a silicone stent.
4. If tension remains, then add steps in order: mobilisation, intracrural separation, infrapubectomy, corporal rerouting.

- Success is in the high **90s %**.

Internal pudendal status	Potency	Action
Intact ≥1 side	Often potent	Reconstruct
Reconstituted only	Never potent	Reconstruct; revascularise for ED later
Bilateral occlusion	—	**Revascularise first** (ischaemic stenosis risk)

Screen at-risk men with **duplex ultrasound, then angiography if abnormal**.

9 · Vesicourethral stenosis and fistula

- Stenosis after prostate cancer treatment is incised deep at 3 and 9 o'clock, adding mitomycin C if recalcitrant. Obliteration needs robotic or perineal repair, often followed by an artificial urinary sphincter.
- **Rectourethral fistula**: a simple one (<2 cm, non-irradiated) is repaired perineally or by York-Mason. A complex (irradiated) one needs faecal diversion and **gracilis interposition, with or without BMG**.

10 · Complications and follow-up

- Recurrence is early after EPA and within 12 months after grafts. ED is uncommon, higher with long repairs and straddle injuries, and often improves.
- Follow up with symptoms, uroflow and post-void residual, and urethroscopy if suspicious.

11 · Landmark trials

Trial	Design	Result	Take-home
ROBUST III	Phase III RCT, Optilume (paclitaxel balloon) vs endoscopic control after multiple failed endoscopic treatments	1-year stricture-free 74.6% vs 26.8%	Drug-coated balloon beats repeat endoscopy; role as first-line undefined

87 Surgery of the Scrotum and Seminal Vesicles

1 · Surgical anatomy of the scrotum

- Lax, rugated skin gives **poor tamponade**, so haematomas keep enlarging and haemostasis must be meticulous.
- Dartos keeps necrotising fasciitis off the testes. **Transdermal absorption** is about **40×** that of other skin, and the skin is rich in 5α-reductase.

Territory	Artery	Parent
Scrotum, anterior	Superficial and deep external pudendal	Femoral
Scrotum, posterior	Posterior scrotal	Perineal (internal pudendal)
Testis, primary	Testicular	Aorta
Testis, secondary	Deferential	Superior vesical, from internal iliac
Testis, secondary	Cremasteric	Inferior epigastric, from external iliac

KEY POINT

Three trunks (aorta, internal iliac, external iliac) anastomose in the cord and at the cauda. So the testicular artery can be ligated in Fowler–Stephens orchidopexy, and the superior epididymal artery sacrificed in vasectomy reversal.

2 · Peri-operative principles and access

- Shave with a **razor on the day of surgery**. The American Urological Association (AUA) advises a single IV cephalosporin or clindamycin **only with risk factors**.
- Benign disease uses a scrotal incision. **Possible malignancy** needs an inguinal incision, with the cord clamped before the testis is handled.
- **Always close the dartos**, because most haematomas come from inadequate dartos closure.

3 · Scrotal wall: excision, scrotectomy, reconstruction

- **Fournier gangrene** is the commonest indication for partial scrotectomy. Give fluids and broad-spectrum antibiotics, and **debride urgently**, because it spreads **2–3 cm/h**; mortality is about **20%**. Re-debride every **24–48 h**.
- Scrotal violation in non-seminomatous germ cell tumour (NSGCT) calls for partial scrotectomy of the scar.
- If more than **50–60%** of skin is lost, do not reconstruct at once; protect the testes in thigh pockets or under a vacuum dressing (VAC).

ALGORITHM

1. If <50% and tension-free, then close primarily.
2. If <50% with tension, then local scrotal advancement or secondary intention.
3. If >50%, then VAC, then stretched scrotal advancement or delayed primary closure.
4. If no local tissue, then fasciocutaneous thigh flap; meshed split-thickness skin graft (STSG) last.

KEY POINT

The scrotum works because it is lax. Graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor.

4 · Hydrocele

- **Congenital** hydrocele comes from a patent processus; about **80%** resolve by 18 months. **Acquired** hydrocele reflects a production–absorption imbalance.
- **Colour Doppler ultrasound first** excludes tumour and decides the scrotal or inguinal route.

Technique	Steps	Best for
Lord plication	Small opening; radial plicating sutures; no dissection	Thin-walled, smaller; least haematoma, no drain
Jaboulay (Winkelman)	Excise part of parietal layer, evert remainder behind testis	Thin, floppy sacs
Bottleneck	Trim to 2-cm cuff, tack edges	Excisional
Excision	Excise sac, oversew edges	Thick, large, multiloculated; lowest recurrence

KEY POINT

The more tunica excised, the lower the recurrence but the more haematoma. Recurrence after any successful repair is **0–6%**.

- **Aspiration and sclerotherapy** suit the unfit; aspiration alone recurs, so add a sclerosant such as doxycycline. With **retreatment**, resolution is **85–96%**, but results are poor if multiloculated or more than **750 mL**.

5 · Spermatocele and epididymal surgery

- Operate on a spermatocele only for symptoms, and **counsel about obstructive azoospermia**.
- A suspected malignant epididymal tumour is approached inguinally; a proven benign one by scrotal microsurgical excision.
- **Epididymectomy** gives **poor results for chronic epididymitis** (>6 weeks), so it is a last resort; it works better with a **palpable lesion**.

6 · Varicocele repair

SAQ Varicocele: pathophysiology, controversies and surgical management

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 SAQ "Varicocele: controversies and present status of treatment" (asked as "Controversies in management of varicocele" and "Present status of varicocele treatment") · P1 SAQ "Varicocele – Pathophysiology" · P2 SAQ "Surgical management of bilateral varicocele"

Definition and epidemiology

- Abnormal dilatation and tortuosity of the **pampiniform plexus** of the internal spermatic veins.
- About **15% of men, 35% with primary and 75–81% with secondary infertility** (P3: up to 80%). Appears after age 10, peaks at Tanner 3. About 90% left-sided.
- With sensitive tests most are bilateral; the clinical significance of subclinical ones is uncertain.
- **Venous anatomy:** anterior (internal spermatic), middle (deferential) and posterior (cremasteric) groups; the last two give return after repair and cause recurrence.

Aetiology: why it develops

1. **Left-sided anatomy:** the left vein enters the left renal vein at a right angle, 8–10 cm higher than the right enters the IVC (inferior vena cava). The taller column overloads its valves. The right vein enters obliquely, which acts as a valve.
2. **Valve incompetence** or absence; continuous reflux predicts progression.
3. **Nutcracker:** left renal vein compressed between aorta and SMA (superior mesenteric artery).
4. **Collateral pathways**; **genetic** (4–8 times commoner in first-degree relatives); tall, thin habitus. - **Secondary:** an isolated **right-sided or non-decompressing** varicocele (one that does not decompress lying down) suggests a retroperitoneal or renal mass (tumour), so image the abdomen.

Pathophysiology: how it damages the testis

Mechanism	Explanation
Heat (primary)	Lost countercurrent heat exchange; the testis normally runs 2–4 °C below core
Oxidative stress	Raised reactive oxygen species cause sperm DNA fragmentation
Hypoperfusion and stasis	Hypoxia, toxin accumulation
Apoptosis	Germ cell loss
Leydig dysfunction	Lower testosterone than controls

- The injury is **progressive and duration-dependent** (atrophy), and may affect **both** testes.

Consequences

- **Testicular hypotrophy** (asymmetry above about 20% guides adolescent repair); poor semen; hypogonadism (lower testosterone); ache on standing (2–11%).
- Repair improves semen, DNA fragmentation, testosterone and pregnancy.

Grading

Grade	Finding
Subclinical	Doppler only (not treated)
I	Palpable with Valsalva only
II	Palpable, not visible
III	Visible

Controversies and present status of treatment (indications)

Question / situation	Evidence	Current position
Does repair improve fertility? Infertile couple, **palpable** varicocele, abnormal semen	Madgar RCT (randomised controlled trial, 1995): pregnancy 44% versus 10% at 1 year; meta-analyses mixed	**Repair clinical** varicocele with **abnormal semen** in an infertile couple with normal female partner (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine])
Subclinical varicocele?	No benefit for semen or pregnancy	**Do not treat**
Normal semen?	No proven benefit	Not for fertility
Adolescent varicocele	Testis volume is a surrogate; link to adult fertility unproven	Observe; repair for **testis asymmetry (hypotrophy) above 20%**, pain, or abnormal semen (Tanner 5 or 18 years)
NOA (non-obstructive azoospermia)	Small series: sperm returns to ejaculate in some	No definitive evidence; may be considered before micro-TESE (microsurgical testicular sperm extraction)
Before IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Repair may improve live-birth rate or allow less invasive ART (assisted reproductive technology)	Discuss, especially with oligozoospermia
High DNA fragmentation	Repair lowers fragmentation	Reasonable indication
Hypogonadism (low testosterone)	Repair raises testosterone in hypogonadal men	Emerging indication
Pain	Dull ache on standing; relief in most after microsurgery	Repair after other causes excluded

- **Bilateral disease: do not repair subclinical** sides; repair each palpable side.

Choice of technique

Technique	Recurrence	Hydrocele	Comment / bilateral use
Microsurgical subinguinal/inguinal	Lowest, **0.5–1%**	**About 0**	**Best, preferred**: artery and lymphatics preserved; two small incisions, local or regional anaesthesia
Retroperitoneal (Palomo)	High with artery spared (15–25%)	High with mass ligation (about 7%)	Children (artery ligation acceptable)
Laparoscopic	3–15%	Up to 12%	Both sides through same ports; general anaesthesia; serious morbidity possible
Embolisation / sclerotherapy	4–25% (P2: 15–25%)	Nil	Outpatient; femoral (left) or jugular (right) access; technical failure

Microsurgical steps (each side)

1. Subinguinal incision over the external ring; deliver the cord.
2. Microscope and **micro-Doppler** to map the testicular artery; papaverine for spasm.
3. Ligate all internal and external spermatic veins (all veins except vasal); preserve **artery, lymphatics, vas and one set of vasal veins**.
4. Deliver the testis to ligate gubernacular and external spermatic veins (lower recurrence).
5. **Impulse test** confirms no missed vein.

Bilateral varicocele: specific points

- **Never ligate both testicular arteries**: in adults this has caused azoospermia and atrophy. So artery-sparing microsurgery is safest bilaterally.
- Image for a renal or retroperitoneal mass if a varicocele is isolated right-sided or non-decompressing.
- Repair any coexisting hydrocele.

Complications and outcome

- **Hydrocele** (lymphatic ligation; commonest after non-microsurgical repair); testicular artery injury (atrophy); recurrence (diagnose by upright colour Doppler showing reflux).
- **Predictors of poor result:** recurrence, artery ligation, hydrocele. Earlier repair recovers more.

Diagrams to draw

- Left and right testicular veins draining into renal vein and IVC.
- Spermatic cord cross-section at microsurgery: artery, lymphatics and vas preserved, veins ligated.

> **Key points to remember:**
>
> - Left vein: right-angled entry, 8–10 cm higher column; 90% left.
> - Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage; injury is progressive.
> - Right or non-decompressing varicocele: image for a renal mass.
> - Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
> - Madgar 1995: 44% versus 10% pregnancy.
> - Adolescents: observe; repair for asymmetry above 20%.
> - Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
> - Never ligate both testicular arteries.

Also, from the condensed book (not in the paper answer):

- †Embolisation suits recurrence after surgery. For orchialgia, **microsurgical repair** gives more complete relief.

7 · Vasectomy

- **Counsel**: the man is not sterile at once, so contraception continues until post-vasectomy semen analysis (PVSA). Repeat vasectomy is needed in <**1%**; pregnancy occurs in about **1**

in 2000 even after an azoospermic PVSA; chronic pain affects about **1–2%**.

- Give **no routine antibiotics**. **Isolation** determines morbidity; **occlusion** determines efficacy.

Feature	Conventional (CV)	Minimally invasive (MIV)	No-scalpel (NSV)
Skin opening	1.5–3 cm	≤**10 mm**	Puncture <10 mm
Status	Difficult cases	**Recommended**: less pain, bleeding, infection	Original MIV (Li Shunqiang, China, 1974)

- Open on the **straight vas**, because the convoluted vas is harder now and at reversal.

AUA-recommended occlusion (failure ≤1%)	Note
Mucosal cautery (MC) and fascial interposition (FI)	FI = internal spermatic fascia over one end
MC alone (no clips/ligatures)	Cautery of lumen 0.5–1.5 cm
Open-ended: abdominal end MC and FI, testicular end open	Less congestive epididymal pain; granuloma linked to better reversal

> **KEY POINT**
>
> Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division is required; segment excision is optional (**1 cm** is adequate).

ALGORITHM

1. Do PVSA at **8–16 weeks**.
2. If azoospermia or rare non-motile sperm (RNMS, ≤**100,000/mL**), then stop contraception.
3. If motile sperm, then repeat at 6 months; if still motile, then it is a failure, so repeat the vasectomy.

- **Early failure** is technical, such as a missed vas; **late failure** is recanalisation after a successful PVSA.
- **Sperm granuloma** is expected rather than a complication. Post-vasectomy pain affects about **2%**; if a cord block helps, then denervation or reversal follows.

8 · Orchidectomy and prosthesis

- **Bilateral simple orchidectomy** treats hormone-sensitive metastatic prostate cancer. It is cheaper than androgen deprivation therapy but has no proven survival benefit.
- **Unilateral** orchidectomy is for trauma or non-salvageable torsion, and for orchialgia only after all conservative measures. **Epididymis-sparing** orchidectomy gives a better contour.
- **For orchialgia, use the inguinal route**, because it removes more of the proximal ilioinguinal, iliohypogastric and genitofemoral nerves.
- **Prosthesis**: offer it to every man lacking a testis, before cancer orchidectomy. Pierce the fill port ≤**5 times**, never after insertion. A **high scrotal subdartos** pouch has the least infection and erosion.

9 · Testicular torsion and fixation

- †**Acute torsion**: if suspicion is high, explore without waiting for imaging. Detorse and assess, then fix or remove; **always fix the contralateral testis**.
- Intermittent torsion (pain <2 h) and postpubertal retractile testis need elective orchidopexy.

Fixation	Method	Note
Suture fixation	Albuginea to dartos (3-0 absorbable)	Re-torsion still possible; transparenchymal sutures damage spermatogenesis
Dartos pouch	Subdartos pouch with a purse-string at the cremaster exit	No transparenchymal suture, so **preferred when fertility matters**

10 · Chronic scrotal pain (>3 months)

ALGORITHM

1. History, examination, duplex ultrasound; if a cause is found, then treat it.
2. If none, then exclude extrascrotal causes (abdomen, pelvis, spine, hip, prostatitis, pelvic floor).
3. If non-specific, then give support, NSAIDs, tricyclics or gabapentin for ≥3 months.
4. If epididymal, then epididymectomy.
5. If testicular or post-vasectomy, then do a cord block: if relief >**50%**, then microsurgical denervation (or reversal); if <50%, then pain clinic, with inguinal orchidectomy last.
6. If post-hernia, then ilioinguinal block; if relief >**90%**, then neurolysis.

- **Microsurgical denervation of the spermatic cord (MDSC)** is done through a subinguinal incision. A micro-Doppler helps **preserve testicular and cremasteric arteries, lymphatics, and the vas with its vessels**; everything else, including the trifecta nerve complex, is divided.

11 · Complications of scrotal surgery

- **Haematoma** is commonest; drain it if large, infected or enlarging. A falling haematocrit after orchidectomy suggests a slipped cord ligature with a **retroperitoneal bleed**, which needs re-ligation through a retroperitoneal extension.

12 · Seminal vesicles

- The seminal vesicles (SV) are Wolffian derivatives that bud at about **12 weeks** and make about **60%** of seminal fluid. The **neurovascular bundles lie lateral to their tips**.
- **MRI is most accurate**. On transrectal ultrasound (TRUS), suspect obstruction if the SV is >**1.5 cm** or the ejaculatory duct >**2.3 mm**; confirm by **aspiration showing sperm**.
- **Zinner syndrome** is an SV cyst with **ipsilateral renal agenesis** and ejaculatory duct obstruction (EDO), often with an ectopic ureter into the cyst.
- Observe or aspirate small cysts. **Transurethral unroofing** is preferred for cysts, EDO and stones; large symptomatic cysts need laparoscopic or robotic excision.

Benign	Malignant
Cystadenoma, papillary adenoma, hydatid cyst, amyloid	Adenocarcinoma, mixed epithelial–stromal, phyllodes, sarcoma, carcinoid, seminoma, yolk sac

- Image tumours with MRI and biopsy by TRUS if inconclusive. Secondary invasion is far commoner. Adenocarcinoma needs radical excision and lymphadenectomy.

13 · Surgical approaches to the seminal vesicles

Approach	Use	Main risk
Transvesical transtrigonal	Large benign cyst; ectopic ureter into cyst	Urinary leak, ureter
Perivesical	Children, large cyst with or without nephroureterectomy	Ureter, vas
Transperitoneal (laparoscopic/robotic)	Bilateral lesions; dominant, because it opens neither organ	Access injuries
Perineal	At perineal prostatectomy	Neurovascular bundle
Transcoccygeal	Hostile abdomen/perineum	Rectum

- A rectal injury needs two-layer repair; a ureteric injury needs reimplantation over a stent.

★ Putting it together

A · Testicular mass: the first week

ALGORITHM

1. Ultrasound.
2. Tumour markers AFP, hCG and LDH **before** orchidectomy; sperm bank.
3. **Radical inguinal orchidectomy** (never trans-scrotal).
4. Repeat markers (AFP half-life 5–7 days, hCG 1–1.5 days); CT chest, abdomen and pelvis.

B · Germ cell tumour by stage

Stage	Seminoma	Non-seminoma (NSGCT)
CS I	Surveillance (occult spread 10–15%); single-dose carboplatin	Low risk: surveillance. High risk (lymphovascular invasion, embryonal-predominant): surveillance, BEP ×1 or nerve-sparing RPLND
II	≤3 cm: radiotherapy, chemo or RPLND; >3 cm: chemo	Normal markers and small nodes: RPLND; otherwise chemo
Advanced (IGCCCG)	Full-dose cisplatin. Good: BEP ×3 or EP ×4. Intermediate/poor: BEP ×4 (VIP ×4 if lung concern).	
Residual mass	>3 cm: FDG-PET (≥6 weeks post-chemotherapy)	Any >1 cm: post-chemotherapy RPLND (teratoma)

C · Penile cancer: primary and groin

ALGORITHM

1. **Primary:** if PeIN, then topical 5-FU/imiquimod, laser or glans resurfacing. If invasive, then organ-sparing surgery, glansectomy, or partial/total penectomy; margins <1 cm are acceptable.
2. **cN0:** if low risk (pTis/pTa/pT1a), then surveillance; if intermediate/high, then DSNB or modified ILND; if positive, then radical.
3. **cN+:** if FNA positive, then bilateral radical ILND; if cN3, then neoadjuvant TIP first. Add pelvic dissection if ≥2 inguinal nodes or extranodal extension.

D · Anterior urethral stricture

Stricture	Treatment
Bulbar <1.5 cm, first time	Dilation or DVIU once; if it recurs, urethroplasty
Bulbar 1–2 cm (3–5 cm proximally)	EPA
Longer bulbar	BMG onlay or augmented anastomosis
Penile	Onlay graft/flap; staged if hostile bed, LS, failed hypospadias
Panurethral, frail	Perineal urethrostomy
Pelvic fracture urethral injury	Suprapubic catheter, then delayed perineal anastomosis

In retention, place a suprapubic catheter; never blind-dilate.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ Surgical management of primary scrotal cancer

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Surgical management of primary scrotal cancer" · P4 SAQ "Surgical management of primary scrotal cancer."

Introduction and aetiology

- Rare. **SCC (squamous cell carcinoma)** is commonest. **Percivall Pott (1775)** linked it to chimney soot in chimney sweeps: the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus), chronic inflammation.
- **Other types:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Lymph drains to **superficial inguinal nodes, crossing the midline**: both groins at risk.

Evaluation

- Painless nodule or ulcer on the anterolateral scrotum, often presenting late; **biopsy** confirms.
- Examine both groins (lymphatics cross the midline); FNAC (fine-needle aspiration cytology) of palpable nodes; CT (computed tomography) of abdomen and pelvis for staging.

Staging (Lowe modification of Ray and Whitmore)

Stage	Extent
A1	Confined to scrotum
A2	Invades testis, cord, penis, pubis or perineum
B	Inguinal nodes only
C	Pelvic nodes, no distant spread
D	Distant metastases

Management of the primary

1. **Wide local excision with a 2 cm margin**, full thickness including dartos; frozen sections for margins.
2. Testes and cord preserved unless invaded; hemi- or total scrotectomy for large tumours.
3. Stage A2: en bloc removal of involved testis, penis, perineum or pubis.
4. Positive margin: re-excise.

Reconstruction (Chapter 87)

- Scrotal laxity usually allows primary closure.

Defect	Management
<50%, no tension	Primary closure
<50%, tension	Scrotal advancement flap or secondary intention

Defect	Management
>50%	Testes in thigh pockets (pouches); local or thigh flaps; later STSG (split-thickness skin graft) or flap

Management of nodes (extrapolated from penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.
- **Impalpable:** surveillance; dynamic sentinel node biopsy or modified ILND for high-grade/invasive tumours.
- **Bulky or fixed:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.
- Routine prophylactic dissection is not advised.

Other histologies

- **EMPD:** wide excision (up to 3 cm) guided by **mapping biopsies**; screen for associated GU (genitourinary) and other cancers.
- **Also know:** BCC, excision or Mohs surgery; melanoma, excision by Breslow depth ± sentinel node; sarcoma, wide excision.

Diagram to draw: elliptical wide excision with 2 cm margin and bilateral inguinal node drainage.

> **Key points to remember:**
> - Pott 1775: chimney sweeps, first occupational cancer.
> - Wide local excision, 2 cm margin; testes spared unless invaded.
> - Nodes: managed as in penile cancer; nodes decide prognosis.

SAQ Perineal urethrostomy and eponymous urethroplasty techniques

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 SAQ "Eponymous urethroplasty techniques and perineal urethrostomy" · P1 SAQ "Perineal urethrostomy." · P2 SAQ "Perineal urethrostomy."

Introduction: urethroplasty options

- Options: **EPA (excision and primary anastomosis)**, the gold standard for short bulbar strictures; **graft or flap onlay**; **staged repair**.
- **Principles:** grafts need a vascular bed; onlay beats tube; graft equals flap.

Eponymous urethroplasty techniques

Eponym	Technique	Use
Johanson	Two-stage: stricture laid open, edges sutured to skin (urethrostomy); tubularised later	LS (lichen sclerosus), failed hypospadias, infection
Blandy	Inverted-U scrotal or perineal skin flap into a ventral urethrotomy	Bulbar strictures; classic perineal urethrostomy (PU)
Turner-Warwick	Two-stage scrotal inlay; transpubic approach, omental wrap	Complex, posterior
Webster	Progressive perineal approach: mobilisation, crural separation, inferior pubectomy, rerouting	PFUI (pelvic fracture urethral injury)

Eponym	Technique	Use
Barbagli	**Dorsal onlay** BMG (buccal mucosal graft) spread-fixed to corpora	Bulbar
Asopa	**Dorsal inlay** graft through a ventral urethrotomy, no mobilisation	Penile and bulbar
Kulkarni	One-sided dorsolateral onlay; panurethral repair via perineum with penile invagination	Panurethral (LS)
Orandi	Ventral longitudinal penile skin island flap on dartos	Pendulous
Quartey	Showed skin island is a passenger on a fascial (dartos) flap	Basis of all genital skin flaps
McAninch	Circular fasciocutaneous penile skin flap	Long anterior strictures (up to 15 cm)
Jordan	Vessel-sparing EPA; ventral transverse island for fossa	Proximal bulbar; fossa navicularis

- **Also know:** Monseur dorsal urethrotomy; Andrich–Mundy non-transecting anastomosis; Bracka two-stage BMG.
- **Diagram to draw:** bulbar cross-section showing dorsal onlay (Barbagli), dorsal inlay (Asopa) and ventral onlay.

Perineal urethrostomy: definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses**, rather than reconstructs, a diseased anterior urethra.

Indications

- **Complex anterior strictures:** LS panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk men** unwilling or unfit for staged reconstruction.
- **First stage** of a staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for penile or urethral cancer.
- Temporary: stricture after phalloplasty; Fournier gangrene; resectoscope access in morbid obesity.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU suits LS and is easy in LS.

Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS raise restricture.

Techniques

1. Classical (Blandy) PU: inverted-U perineal skin flap (a random-pattern advancement flap) advanced into a ventral bulbar urethrotomy (the opened bulbar urethra).

2. BMG PU, steps:

1. High lithotomy; inverted-U incision; split (divide) bulbospongiosus in the midline.
2. Free the spongiosum and **transect** it; **dorsal urethrotomy** into healthy mucosa.

3. Bring the healthy ventral urethra to perineal skin; calibrate proximally to **30 Fr**; close the distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin. - The graft prevents the circumferential scar that narrows a deep PU.

Diagram to draw: sagittal perineum with the inverted-U flap sutured into the opened bulbar urethra.

Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or periodic dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Compared with staged reconstruction it is a shorter, single operation with less morbidity.
- High mid-term success and satisfaction in selected patients.

> **Key points to remember:**
> - Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
> - Orandi and McAninch flaps; Quartey flap anatomy.
> - Johanson staged; Blandy inverted-U PU.
> - PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
> - Ideal for LS, failed repairs and elderly patients.
> - BMG PU prevents circumferential scarring; calibrate to 30 Fr.

SAQ Testicular microlithiasis

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3a

Definition

- **TM (testicular microlithiasis):** multiple small (<3 mm) bright echogenic foci **without acoustic shadowing** scattered in the testicular parenchyma on ultrasound.
- **Classic TM:** ≥5 microliths per testis (or per image); **limited TM:** <5, of no clinical significance.

Pathology

- Laminated calcified bodies **inside seminiferous tubules**, formed on degenerated tubular cells that Sertoli cells fail to clear. Affected tubules are often obliterated.

Prevalence

Population	Prevalence
General population	5–10% (population-based 4.2%, rising with age)
Multicentre review, >37,000 males	2.9%
Boys with cryptorchidism	3.5%
Previously cryptorchid testis vs contralateral	14% vs 19% (no difference)

Associations

- Cryptorchidism, subfertility, **GCNIS (germ cell neoplasia in situ)** and TGCT (testicular germ cell tumour), Klinefelter syndrome, Down syndrome.

Cancer significance

- Tumours were found in 4.64% with TM vs 0.33% without, but cryptorchidism was not accounted for.
- **TM is not clearly an independent risk factor** for TGCT. An EAU (European Association of Urology) paediatric panel review of several hundred children found only one cancer through puberty.
- **Exception:** contralateral TM in a man with previous GCT (germ cell tumour) signals a higher GCNIS risk.

Management

- **Isolated TM, no risk factors:** reassure; **no further evaluation, no routine ultrasound follow-up**.
- **TM with risk factors** (previous GCT, cryptorchidism, infertility, atrophic testis): teach **monthly self-examination from puberty**; ultrasound only for a palpable change.
- **Biopsy:** not for TM alone. Contralateral biopsy at orchidectomy is considered when GCNIS risk is high: atrophic testis (<12 mL), cryptorchidism, age <40.
- Counsel carefully: incidental TM causes anxiety and unnecessary tests.

Differential diagnosis on ultrasound

- Coarse calcification, "burned-out" tumour scar, stellate scar (these need orchidectomy if intratesticular), scrotal pearls, rete testis ectasia.

Also know: routine screening ultrasound for infertility is discouraged because of incidental TM; follow-up schedules remain undefined.

Diagram to draw: ultrasound sketch of a testis with scattered non-shadowing echogenic dots.

> **Key points to remember:**
> - Classic TM ≥5 microliths; limited <5.
> - Not an independent cancer risk factor in isolation.
> - Self-examination only with risk factors; no routine imaging.

SAQ Muscle assisted urethroplasty

Paper answer · P4

Asked as:

- Muscle assisted urethroplasty — *Winter 2023 Paper IV Q3b (SAQ)*
- Muscle assisted urethroplasty — *Practice Set 5 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3b)

Read as muscle-flap-assisted graft urethroplasty: a vascularised muscle flap used as bed or cover for a graft. "Muscle-sparing" bulbar urethroplasty is noted at the end.

Rationale

- A free graft (e.g. **BMG [buccal mucosal graft]**) has no blood supply. It survives by **imbibition** (about 48 h, nutrients from the bed), then **inosculation** (new microcirculation). So **"a graft lives on its bed"**.
- **Poor beds:** radiation, multiple failed repairs, fistula, pelvic-fracture scar, infection. Here a **muscle flap brings blood supply**, fills dead space and separates suture lines.

Muscle flaps used

Flap	Supply	Use
Gracilis (muscle-only, axial, multiple pedicles)	Medial circumflex femoral branch, proximal thigh	Irradiated posterior urethra, rectourethral fistula, complex perineal repair
Bulbospongiosus	Perineal branches	Covers a ventral graft; support in bulbar repair
Rectus abdominis (VRAM, vertical rectus abdominis myocutaneous)	Inferior epigastric	Large pelvic or perineal defects

Technique (gracilis-assisted BMG)

1. Lithotomy; perineal exposure; excise scar or fistula.
2. Close rectal side; urethra opened.
3. Gracilis harvested through a medial thigh incision; pedicle preserved; tunnelled to perineum.
4. Muscle interposed between rectum and urethra, or used as the graft bed.
5. **BMG quilted onto the muscle** as onlay; urethra closed over a catheter.
6. Catheter drainage; urethrogram before removal.

Results (Chapter 86)

- Gracilis with or without BMG for rectourethral fistula: **non-irradiated, all closed**, no later strictures. **Irradiated:** most closed in one stage, but more strictures and permanent bowel diversion.
- Muscle interposition is integral; BMG is invaluable for urethral closure.

Complications

- Thigh wound problems; graft contraction, restricture, fistula recurrence.

"Muscle-sparing" bulbar urethroplasty (Barbagli)

- The **bulbospongiosus muscle, central perineal tendon and perineal nerves** are left intact; dorsal BMG onlay through a lateral approach.
- Aim: less post-void dribbling and ejaculatory dysfunction.

Diagram to draw: gracilis flap tunnelled to the perineum, lying between rectum and urethra with BMG on it.

Key points to remember:

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

SAQ Gracilis myocutaneous flap in Uro.

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 4, Q3d (gracilis-assisted urethroplasty also in 2023-Winter Paper 4, Q3b)

Definition

- A pedicled **axial** flap of gracilis muscle, with or without skin, rotated from the medial thigh to bring **vascularised bulk** to the perineum, groin or vagina.

Anatomy

Feature	Detail
Origin–insertion	Inferior pubic ramus and body of pubis → medial proximal tibia (pes anserinus)
Blood supply	**Mathes–Nahai type II**: dominant pedicle from the **medial circumflex femoral artery** (profunda femoris), entering about 8–10 cm below the pubic tubercle; minor distal pedicles from the superficial femoral artery
Nerve	Anterior branch of the **obturator nerve**
Skin paddle	Reliable over the proximal two-thirds only
Function	Expendable (other adductors compensate)

Why it is used

- Brings blood supply into **irradiated or scarred** fields, fills dead space and **separates suture lines**. Near the perineum, easy to tunnel.

Uses in urology

1. **Rectourethral fistula** (perineal): non-irradiated all closed; irradiated mostly closed, but more strictures and permanent colostomy.
2. **Complex or irradiated VVF (vesicovaginal fistula)** with tissue loss.
3. **Urethroplasty** in a poor bed: muscle as bed for BMG (buccal mucosal graft).
4. Extensive perineal or scrotal loss after **Fournier gangrene**.
5. **Groin cover** of femoral vessels after inguinal node dissection with skin excision.
6. Vaginal reconstruction after exenteration.

Technique

1. Lithotomy; line from pubic tubercle to medial femoral condyle; gracilis lies just behind it.
2. Medial thigh incision; skin paddle over the proximal muscle if needed.
3. Divide the distal tendon; ligate minor pedicles; **preserve the dominant pedicle**.

4. Tunnel subcutaneously to the perineum; rotate without twisting.
5. Interpose and fix; drain the donor site.

Complications

- Donor seroma, dehiscence, **distal skin-paddle necrosis**, thigh numbness; flap loss rare.
- **Also know:** alternatives: Martius, omentum, VRAM (vertical rectus abdominis myocutaneous), Singapore flap.

Diagram to draw: gracilis tunnelled on its proximal pedicle between urethra and rectum.

> **Key points to remember:**
> - Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
> - Skin paddle unreliable over the distal third.
> - Main uses: rectourethral fistula, complex VVF, Fournier, groin cover.

Rapid Revision Sheet · Section 10

Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. (LAQ)

- Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
- Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
- Psoas is the landmark; artery met first.
- Main advantage: shorter operative time, peritoneum untouched.

Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma (LAQ)

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila (renal vessels), ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND (post-chemotherapy): full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum to ipsilateral superficial inguinal nodes; scrotum and penis drain to inguinal nodes; never operate trans-scrotally.
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.
- Seminoma risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

Classify testicular tumors and describe management of seminoma (LAQ)

- WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
- CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
- Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
- No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Growing teratoma syndrome (SAQ)

- Enlarging mass plus normal or falling markers equals GTS.
- Teratoma is chemoresistant and not FDG-avid.
- Treatment is complete surgical resection, not more chemotherapy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Testis sparing surgeries for CA testis (SAQ)

- Solitary or bilateral tumour, ≤2 cm, normal testosterone.
- Frozen section plus adjacent biopsies for GCNIS.
- GCNIS: 18–20 Gy keeps Leydig function but sterilises.

Minimally invasive and robotic RPLND (SAQ)

- Same template, split-and-roll, lumbar division and nerve sparing as open.

- Best for chemotherapy-naive low-volume disease.
- Conversion about 5%; ejaculation preserved ≥90%.
- Expanding indication: primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy.
- Robotic nerve sparing preserves antegrade ejaculation in ≥90% after primary surgery.
- Serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND (retroperitoneal lymph node dissection).

Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy (LAQ)

- Nodal status is the strongest prognostic factor; ILND can cure.
- pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
- Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
- PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT [International Penile Advanced Cancer Trial] testing the sequence).
- Femoral triangle: ligament, sartorius, adductor longus.
- Flaps deep to Camper's; stay medial to femoral artery.
- Within 3 months; at least 8 nodes per side.
- VEIL: same template as open; fewer wound complications; equal nodal yield.
- VEIL not for bulky, fixed or skin-involving nodes.

Prognostic index score in carcinoma penis (SAQ)

- Node status is the strongest prognostic factor.
- LVI, PNI, grade III define high-risk T1b.
- Nomograms (Ficarra, Kattan) unvalidated; p16 favourable. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Grafts in urology and buccal mucosa (LAQ)

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, wet-adapted; thin vascular lamina propria gives excellent take.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.
- Bulbar success about 85–90%.

Meatal stenosis and submeatal stricture in males (SAQ)

- Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
- Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: steroid (clobetasol), buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Urethral stricture: etiology, classification, DVIU, management and failed VIU (LAQ)

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU (VIU) only once, for short bulbar strictures (<1.5–2 cm) at first attempt; do not repeat it.
- Santucci: about 8% long-term success; repeat DVIU futile.

- AUA: urethroplasty after failure; laser = cold knife; early urethroplasty, ladder abandoned.
- EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
- Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
- Optilume (ROBUST III): 74.6% vs 26.8% stricture-free at 1 year.

Varicocele: pathophysiology, controversies and surgical management (SAQ)

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.
- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.
- Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
- Never ligate both testicular arteries.

Surgical management of primary scrotal cancer (SAQ)

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

Perineal urethrostomy and eponymous urethroplasty techniques (SAQ)

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU.
- PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly patients.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Testicular microlithiasis (SAQ)

- Classic TM ≥5 microliths; limited <5.
- Not an independent cancer risk factor in isolation.
- Self-examination only with risk factors; no routine imaging.

Muscle assisted urethroplasty (SAQ)

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

Gracilis myocutaneous flap in Uro. (SAQ)

- Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
- Skin paddle unreliable over the distal third.
- Main uses: rectourethral fistula, complex VVF, Fournier, groin cover. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 11

Stones, Obstruction & Trauma

93 Management of Upper Urinary Tract Obstruction

1 · Ureteropelvic junction obstruction (UPJO): presentation

- Trauma: see Ch 94, Ch 125; stones: see Ch 95–100.
- UPJO is usually congenital. **Neonates** show prenatal hydronephrosis on ultrasonography, with azotaemia if bilateral or solitary.
- **Older children and adults** have intermittent flank pain with vomiting, haematuria after minor trauma, urinary tract infection (UTI) and, rarely, hypertension.

2 · Diagnosis

Prove impaired drainage before operating.

> **KEY POINT**
>
> **Dilation is not obstruction.** Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows where; functional tests (renography, Whitaker) show whether it matters. Combine one of each.

Modality	Key point
Ultrasonography (first in neonates)	Large medial pelvis, rounded calyces joined by dilated infundibula (unlike multicystic kidney); ureter not seen
Non-contrast CT (acute pain)	Hydronephrosis to UPJ without stone; excludes radiolucent stone, tumour
Contrast CT	Shows **crossing vessels**; can substitute for renography
Diuretic renography	Differential function, drainage; best follow-up study
Retrograde pyelography	**At the time of repair** (avoids introducing infection)
Nephrostomy and antegrade study	When retrograde fails; tests recoverability; permits Whitaker

SAQ Radionuclide studies and diuretic renography in PUJ obstruction

Paper answer · P1

Asked as:

- Radionuclide studies for PUJ obstruction. — *Summer 2021 Paper I Q3e (SAQ)*
- Radionuclear studies in PUJ obstruction — *Practice Set 4 Paper I Q3e (SAQ)*
- Diureteric renography — *Practice Set 2 Paper I Q3f (SAQ)*

Type: SAQ (10 marks)

(Practice Set 4: also answered in 2021-Summer Paper 1, Q3e. Practice Set 2: PUJ obstruction renography also in 2021-Summer Paper 1, Q3e; tracers: see Practice Set 2 Paper 1, Q3e)

"Diureteric" is read as diuretic renography.

Introduction / principle

- **Dilatation is not obstruction.** PUJ (pelviureteric junction) obstruction is impaired drainage that damages the kidney. **Diuretic renography** is the functional standard (functional gold standard for PUJ obstruction) and best follow-up test (best follow-up study).
- A capacious pelvis retains tracer passively and furosemide washes it out; an obstructed system cannot empty even under diuresis.

Tracers

Tracer	Handling	Use
^{99m}Tc-**MAG3** (technetium-99m mercaptoacetyltriglycine)	Tubular secretion	**Agent of choice** (tracer of choice): split function and drainage; good images (good curves) even with poor function
^{99m}Tc-**DTPA** (diethylenetriaminepentaacetic acid)	Glomerular filtration	GFR (glomerular filtration rate); poor in renal impairment; an alternative to MAG3
^{99m}Tc-**DMSA** (dimercaptosuccinic acid)	Cortical retention	Static split function and scars; no drainage data

"Well-tempered" diuretic renogram (standard protocol)

Element	How	Why
Hydration	Oral, or IV (intravenous) fluid in children (about 15 mL/kg)	Dehydration mimics obstruction
Bladder catheter	Urethral catheter	Removes full-bladder (back-pressure) and reflux (VUR, vesicoureteral reflux) effects
Furosemide	0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children	Maximal diuresis
Timing	**F+20** (classic; 20 min after tracer, when the pelvis is full; Campbell authors use F+15), **F0** (with tracer) or **F−15** (15 min before, so the system is already in diuresis)	Diuretic before filling (early diuretic into an unfilled system) misleads; always state timing
Regions of interest	Regions of interest over cortex (for split) and pelvis (dilated pelvis for drainage); then upright images	Wrong placement gives spurious values

- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- First study after about 4–6 weeks of age (immature function).

Outputs

- **DRF (differential renal function)** from counts at 1–3 min (1–3 minute counts); normal 45–55%.
- **Drainage half-time (T½, half-clearance time)** after furosemide, from maximal filling, not from injection:

T½ after furosemide	Interpretation
<10 min (minutes)	Not obstructed
10–20 min (minutes)	Equivocal (indeterminate)
>20 min (minutes)	Obstructed

- **O'Reilly curves:** I normal; II obstructed (keeps rising); IIIa dilated, non-obstructed (falls after furosemide); IIIb equivocal.
- **Also know:** P40 (activity remaining / left at 40 min) raises sensitivity; output efficiency; NORA (normalised residual activity); a **biphasic curve** and poor upright (post-void) drainage predict progression.

Indications

- PUJ (pelviureteric junction) obstruction, antenatal hydronephrosis, megaureter, after pyeloplasty, transplant ureter.

Indications for pyeloplasty from renography

- Consider surgery for:
- **DRF <40%**, **fall in DRF** on serial comparable scans (**>10%** on serial scans; ≥5% in children; 10% in older series), **T½ >20 min** (minutes) or upright retention; also (poor drainage with) symptoms, infection, increasing dilatation.
- Adult DRF <15%: likely non-salvageable; nephrostomy and reassess, or nephrectomy.

Pitfalls

- Dehydration, early diuretic (mistimed diuretic), full bladder, wrong ROI (region of interest), very poor function, very capacious pelvis.
- One Campbell chapter calls T½ >40 min obstructed; children have no agreed (paediatric T½) cut-off (false positives up to 44%; neonatal false-positive rates reach 44%); one Campbell chapter uses **T½ >40 min** in adults, so **serial trends and curve shape** matter / decide.

Follow-up and alternatives

- After pyeloplasty: renogram 1 month after stent removal; spontaneous drainage means success.
- Equivocal (cases) → **Whitaker test** (>15–20 cm H_2O obstructed) or MRU (magnetic resonance urography).

Diagram to draw: normal, obstructed and dilated non-obstructed renogram curves (O'Reilly curves I, II, IIIa and IIIb) with furosemide arrow.

> **Key points to remember:**
> - MAG3 is the tracer of choice; hydrate and catheterise.
> - Well-tempered: hydrate, catheter, standard diuretic dose and timing.
> - Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
> - DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
> - Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

Also, from the condensed book (not in the paper answer):

- **Renography**: hydrate, then give furosemide **0.3–0.5 mg/kg** at **20 min**; mercaptoacetyltriglycine (MAG3) is preferred.
- **Obstruction is a half-time (T½) >40 min**. After successful pyeloplasty, drainage occurs even before furosemide.

ALGORITHM

1. **Whitaker test**
2. Place a nephrostomy and a bladder catheter; perfuse at **10 mL/min**.
3. If pressure stays ≤12–15 cm H2O, there is no obstruction; if it exceeds **15–20 cm H2O**, there is.
4. It is invasive, so use it only when other tests are non-diagnostic.

3 · Indications and choice of intervention

- **Indications**: symptoms; impaired or falling function; stones or infection; rarely hypertension.
- **Neonates** are selectively observed; operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults** with indeterminate results have serial renography.
- The **procedure of choice** is dismembered pyeloplasty, now robotic at most major centres; endopyelotomy fails more.
- Drain before repair only for infection, azotaemia (solitary or bilateral) or unrelenting pain.

Endopyelotomy subgroup (landmark series)	Success
Crossing vessel present vs absent	**42% vs 86%**
High- vs low-grade hydronephrosis	60% vs 81%
Both excluded	95% (≈ pyeloplasty)

- A crossing vessel is **prognostic, not a contraindication**, but it pushes primary UPJO towards pyeloplasty.
- **Secondary UPJO: cross over.** Failed endopyelotomy is treated by pyeloplasty; failed pyeloplasty by endourology first.
- **Nephrectomy** suits a thin shell with a normal contralateral kidney, stones with chronic infection, or multiple failed repairs. Adults with <**15%** differential function are non-salvageable; if uncertain, stent or drain and re-scan.

4 · Endopyelotomy

- **Principle**: a full-thickness **lateral** incision into peripelvic fat heals over a stent; lateral is **free of crossing vessels**.
- Pyeloplasty (≈**95%**) wins because it makes a wide dependent funnel; endourology wins only on recovery.
- **Contraindications**: segment >**2 cm**, active infection, uncorrected coagulopathy. Urine must be sterile first.

Percutaneous antegrade

- Best for **UPJO with pelvicalyceal stones**. Puncture the **mid-posterior or superolateral calyx**, not the lower pole, whose acute infundibular angle blocks passage.
- Stent with a 14/7-Fr endopyelotomy stent, traditionally for **6 weeks**; 2 and 4 weeks are equal.
- **Clear stones before incising**, because fragments migrate periureterally and cause fibrosis.

Retrograde ureteroscopic (standard)

SAQ Retrograde endopyelotomy

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 1, Q3f

Definition and principle

- Endoscopic **full-thickness incision of the UPJ (ureteropelvic junction)** from inside the ureter, through to peripelvic fat, followed by stenting so the ureter heals with a wider lumen (Davis "intubated ureterotomy" concept, 1943).
- **Retrograde ureteroscopic holmium laser endopyelotomy** is now the standard endourological route: direct vision without a percutaneous tract.

Indications and contraindications

Favourable	Unfavourable / contraindicated
Functionally significant UPJO (UPJ obstruction); short segment	**Long stenotic segment**
Secondary UPJO (failed pyeloplasty)	**Upper-tract stones** (treat percutaneously instead)
Moderate hydronephrosis, fair function	**Crossing vessel**, **gross hydronephrosis**, poor function
- Pre-operative CT (computed tomography) angiography or Doppler to look for crossing vessels.	

Technique

1. General anaesthesia; retrograde pyelogram; hydrophilic safety wire into the pelvis.
2. Access sheath; flexible ureteroscope to the UPJ (semirigid in some women). Dilate a narrow intramural ureter or pre-stent.
3. Holmium laser, 200-µm fibre, 0.8–1.2 J, 10–15 Hz.
4. **Lateral incision** (no crossing vessels laterally), deepened into fat and extended into normal ureter and pelvis.
5. Optional balloon dilation to 24 Fr; DJ (double-J) stent and Foley catheter.
6. Stent out after some weeks; **diuretic renogram 4 weeks later**; follow up for 24–36 months.

Results

- Holmium series 70–80% at 5 years (range 44–89%). Excluding crossing vessels >4 mm raises success to about 90%.
- Crossing vessel present versus absent: 42% versus 86% (antegrade series).

- Still below pyeloplasty (about 95%), but most cost-effective endourological option.

Complications

- Urine leak, stent migration, infection, bleeding (lateral cut spares vessels), stricture (rare with small scopes); ureteroarterial fistula reported.
- **Also know:** antegrade (percutaneous) endopyelotomy for UPJO with stones; cautery wire balloon (more haemorrhage); balloon dilation alone (about 42% success); failure then needs pyeloplasty.

Diagram to draw: UPJ with lateral full-thickness incision and crossing vessel lying anterior/posterior.

> **Key points to remember:**
> - Lateral full-thickness incision over a stent.
> - Crossing vessel and gross hydronephrosis predict failure.
> - Success 70–80%, below pyeloplasty (about 95%).

Also, from the condensed book (not in the paper answer):
- Control bleeding with haemostatic laser settings or 10-min balloon tamponade.

ALGORITHM
1. General anaesthesia; working and safety wires; flexible ureteroscope via a sheath.
2. With a 200-µm holmium fibre, incise **laterally** into fat and into normal ureter.
3. Leave a stent and Foley; do a renogram 4 weeks after stent removal.

- Percutaneous endopyeloplasty (endoscopic Heineke–Mikulicz) fails in secondary UPJO.

Complications and follow-up

ALGORITHM
1. **Percutaneous bleeding**
2. Bed rest and transfusion.
3. **Do not irrigate** the nephrostomy, so clot tamponades.
4. If bleeding persists, do **selective embolisation**.

- Renogram at 1 and 6 months, then yearly for **2–3 years**. Pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years.
- After failure, get CT angiography; a significant crossing vessel needs **pyeloplasty, not repeat incision**.

5 · Pyeloplasty

KEY POINT

Principles (open = minimally invasive): widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** transition.

LAQ Laparoscopic pyeloplasty.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2016-Summer Paper 3, Q2

Introduction

- **PUJO (pelviureteric junction obstruction)** is impaired flow from pelvis to ureter that, if untreated, causes progressive renal damage (Koff).
- Causes: **intrinsic** adynamic segment (abnormal circular muscle), **extrinsic** crossing lower-pole vessel, high insertion; secondary (failed repair, stone).
- **Anderson–Hynes (1949) dismembered pyeloplasty** is the gold standard by any route. First laparoscopic pyeloplasty **1993 (Schuessler)**; robotic **2002**. Success **over 90–95%**, equal to open, with less morbidity.

Indications

- Symptoms (pain, Dietl's crisis), infection, stones, hypertension (rare).
- Impaired split function or **fall of 10% or more** on serial renography; rising APD (anteroposterior diameter); obstructed solitary kidney or bilateral disease.
- Children: most antenatal hydronephrosis resolves; about one-third need surgery.
- **Contraindications**: uncorrected coagulopathy, untreated UTI (urinary tract infection), cardiopulmonary disease barring pneumoperitoneum.
- **Nephrectomy instead** if split function under 15% in adults with a normal other kidney.

Preoperative evaluation

- **USG (ultrasonography)**: large pelvis, small calyces, ureter not seen.
- **MAG3 (mercaptoacetyltriglycine) diuretic renogram**: split function and an obstructive curve that does not wash out after furosemide. Usual teaching: washout half-time under 10 minutes normal, 10–20 minutes equivocal, over 20 minutes obstructed (Ch 93 quotes over 40 minutes). Split function under 40% supports repair.
- **CT (computed tomography) angiography**: anatomy, **crossing vessels**, stones.
- **Retrograde pyelogram** at surgery: confirms the level, excludes ureteric stricture.
- **Whitaker test** if equivocal: perfusion 10 mL/min; pelvic pressure over 15–20 cm H2O means obstruction.
- Sterile urine. Drain beforehand (stent or nephrostomy) only for infection, azotaemia or severe pain.

Approaches

	Transperitoneal	**Retroperitoneal**
Space	Large, familiar landmarks	Small, created by balloon
Bowel	Colon reflected (left: **transmesocolic** window)	Not entered; leak stays extraperitoneal
Use	Commonest; easier suturing	Avoids bowel; harder orientation
- One paediatric randomised trial favoured retroperitoneal (shorter stay); overall outcomes similar.		

Steps (transperitoneal Anderson–Hynes)

1. Cystoscopy, retrograde pyelogram, catheter.

2. 45–60° flank; pneumoperitoneum 12–15 mmHg in adults (under 12 mmHg in children); 3–4 ports, camera at umbilicus.
3. Reflect colon medially, or open the mesocolon over the pelvis.
4. Find the ureter; trace it to the PUJ with its periureteric tissue; minimal diathermy. Look for a **crossing vessel**.
5. Hitch stitch on the pelvis; transect the pelvis above the PUJ; **reduce** a redundant pelvis if needed.
6. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a handle (no-touch).
7. **Transpose** the PUJ in front of any crossing vessel.
8. Anastomose the spatulation apex to the most dependent pelvis with 4-0 absorbable (5-0/6-0 in infants): back wall, stent, front wall, then close the pelvis. Watertight, tension-free, funnel-shaped.
9. **Antegrade double-J** over a wire; clamp the catheter: urine from the stent side holes confirms the bladder coil.
10. Excise the narrow segment; drain near (not on) the suture line.

Crossing vessel

- It lowers endopyelotomy success (42% vs 86%), so pyeloplasty is preferred. **Never divide** a lower-pole vessel (segmental infarction).
- **Vascular hitch (Hellström)**: vessel moved up onto the pelvis, only if the PUJ is intrinsically normal and peristalsing.

Stenting and postoperative care

- Double-J for **4–6 weeks** (children 1–4 weeks: no difference). Externalised stent in infants avoids a second anaesthetic.
- Foley for 24–48 hours (bladder pressure refluxes up the stent); drain out when dry. If drain output rises after Foley removal, replace the Foley.
- **MAG3 renogram** after stent removal, then ultrasound; follow for years.

Robotic pyeloplasty

- 3D vision, wristed suturing, tremor filtration, **shorter learning curve**. Now first-line at major centres.
- Success 94.7–100%, the same as laparoscopy; advantage is ergonomic, at higher cost.

Outcomes

- Laparoscopic: operating time 2–4 hours; complications 2–16%; conversion 0–5.5%; stay about 2.7 vs 4.2 days (open); success 87–99%.
- **Up to 30% of failures appear after 2 years**, so follow-up must be long.

Complications

- Urine leak and urinoma, stent symptoms or migration, UTI, haematuria, ileus, bowel or vessel injury, port-site hernia, recurrent obstruction.

Management of failure

```
Early (clot, oedema) -> nephrostomy, wait Persistent obstruction -> CT
angiography (missed vessel?) short, no vessel -> endopyelotomy (about 70%)
otherwise -> redo laparoscopic/robotic pyeloplasty (about 89% after failed
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open; buccal graft onlay) small intrarenal pelvis -> ureterocalicostomy poor function -> nephrectomy

Also know: Y-V plasty (high insertion); Culp spiral flap (long narrowing); stones removed through the pyelotomy with a flexible scope; safe in horseshoe and pelvic kidneys.

Recent advances

- Single-port robot (2018); indocyanine green fluorescence for redo cases; buccal mucosa ureteroplasty.

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition over the crossing vessel, dependent anastomosis.

> **Key points to remember:**
> - Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
> - Transpose over a crossing vessel; never divide it.
> - Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
> - Robotic equals laparoscopic in outcome, with easier suturing.

LAQ Congenital PUJ obstruction, failed pyeloplasty and salvage procedures

Paper answer · P1 · P2 · merged from 3

Asked as: P1 LAQ "Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management." · P1 SAQ "Management of failed pyeloplasty" · P2 SAQ "Salvage procedures for Complicated PUJO."

Definition

- PUJO (pelviureteric junction obstruction) is impaired flow of urine from the pelvis into the ureter. **Koff:** obstruction is "any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage". **Dilatation is not obstruction.**
- About 1 in 1500 live births; commonest cause of antenatal hydronephrosis. Male and left predominance; **bilateral in 10–40%**.

Pathogenesis

Type	Lesion	Typical patient
Intrinsic (commonest)	Short adynamic segment: interrupted circular muscle, excess collagen, fewer pacemaker (interstitial Cajal-like) cells. The peristaltic wave cannot cross, so the pelvis empties poorly	Neonate or infant, antenatal finding
Extrinsic	**Lower-pole crossing vessel** anterior to the PUJ; the ureter kinks over it	Older child or adult with intermittent pain
Secondary	Kinking by a tortuous ureter in high-grade VUR (vesicoureteral reflux); fibroepithelial polyp; high insertion	Any age

- **Crossing vessels** are persistent embryonic renal arteries. They occur at a fifth to a quarter of **normal** PUJs, so a vessel alone does not prove cause.
- **Also know:** proposed incomplete ureteric recanalisation; horseshoe kidney and duplex lower moiety associations.

Effects of obstruction (pathophysiology)

- Pelvic pressure rises; filtration and perfusion fall.
- **Immature kidney:** apoptosis and impaired nephrogenesis. **Mature kidney:** hypertrophy, then fibrosis (TGF-β [transforming growth factor-beta] signalling). Fibrosis, not dilatation, causes loss of function.
- **Also know:** concentrating defect, salt wasting; rarely renin-mediated hypertension.

Clinical features

- **Antenatal ultrasound** (commonest); infants: UTI (urinary tract infection), flank mass, failure to thrive.
- Older children and adults: **intermittent loin pain with vomiting (Dietl crisis)**, haematuria after minor trauma, stones, infection.
- **Associated anomalies (up to 50%):** contralateral PUJO, dysplasia or MCDK (multicystic dysplastic kidney), VUR (up to 40%).

Diagnosis

1. Antenatal grading (APD, anteroposterior renal pelvic diameter)

	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

- **UTD (urinary tract dilation) 2014 consensus:** six parameters (APD, central/peripheral calyces, parenchymal thickness and appearance, ureter, bladder). Prenatal A1 (low) or A2–3; postnatal **P1** (APD 10–<15 mm, central calyces), **P2** (≥15 mm or peripheral calyces or ureter), **P3** (abnormal parenchyma).
- **SFU (Society for Fetal Urology) grades 0–4:** 1 pelvis just seen; 2 pelvis plus major calyces; 3 all calyces, normal parenchyma; **4 parenchymal thinning**.

2. Postnatal ultrasound after **48 hours** (neonatal dehydration under-reads dilatation): large pelvis, rounded calyces, ureter not seen.

3. MAG3 (technetium-99m mercaptoacetyltriglycine) diuretic renogram ("well-tempered": hydrated, catheterised, furosemide)

- **DRF (differential renal function)** (normal 45–55%); drainage half-time <10 min normal, 10–20 equivocal, **>20 min obstructed** (adult chapter: >40 min).

4. Others: VCUG (voiding cystourethrogram) by UTI risk; MRU (magnetic resonance urography) or CT (computed tomography) angiography for vessels; **Whitaker test** (10 mL/min; >15–20 cm H_2O obstructed) if equivocal.

Primary management

Conservative: most resolve, about **one-third need surgery**; serial ultrasound and renogram.

Indications for pyeloplasty

- **DRF <40%** or a **fall of >10%** on serial scans.
- **Increasing APD** or SFU 4 / P3 with poor drainage.
- **Breakthrough UTI**, symptoms, stones, hypertension.

- Bilateral or solitary kidney.

Anderson–Hynes dismembered pyeloplasty: gold standard (~95% success)

1. Expose the PUJ; free the lower pole and look for a crossing vessel.
2. Stay sutures; transect the pelvis; reduce a redundant pelvis.
3. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a "no-touch" handle.
4. **Transpose** the PUJ anterior to any vessel (never divide it).
5. Watertight, tension-free, dependent anastomosis (5-0/6-0 absorbable); excise the segment.
6. Double-J or externalised stent (1–4 weeks equivalent); Foley 24–48 h with a double-J.

- **Route:** open preferred in infants; laparoscopic equals open with less pain; **robotic** (easier suturing, shorter learning curve) is now primary at most major centres.
- **Alternatives:** Foley Y-V plasty (high insertion); Culp spiral or vertical flap (long segment); **vascular hitch** (crossing vessel with normal PUJ); **endopyelotomy** (73% overall; **42% with vs 86% without a crossing vessel**), mainly for secondary PUJO.
- **Nephrectomy:** DRF **<15% in adults** (children: no agreed cut-off).

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition anterior to the crossing vessel, dependent anastomosis.

Complications after pyeloplasty

- Urine leak, stent problems, UTI.

Failed pyeloplasty: definition

- Persistent pain, UTI or stones; **obstructive diuretic renogram** (T½ >20 minutes); unchanged or increasing hydronephrosis; falling split function.
- Dilatation settles slowly; judge by renography.
- About 1 in 9 children need a secondary procedure. Most failures occur within 1–2 years (P2: within 2 years), but up to 30% occur later.

Causes of failure

- **Ischaemia** (over-stripping) and **tension** at the anastomosis.
- **Missed crossing vessel**; inadequate excision of the narrow segment; non-dependent anastomosis; kinking.
- **Urine leak** or infection causing periureteric fibrosis.
- Missed distal stricture or high-grade VUR.

Early failure (first days or weeks)

- Pain, fever, increasing dilatation: usually **oedema or clot**; temporary nephrostomy or stent, and most settle.

Evaluation of late failure

- **MAG3 diuretic renogram:** obstruction (drainage) and split function.
- **CT angiography:** crossing vessel.
- Antegrade or retrograde pyelography (stricture length); **Whitaker test** if equivocal.
- Drain first (stent or nephrostomy) if infected or function is uncertain, then reassess.

Salvage options

Option	When	Results/notes
Endopyelotomy (antegrade or retrograde ureteroscopic, laser)	Short stricture, no significant crossing vessel, moderate hydronephrosis; first choice after failed open repair	About 70% after failed pyeloplasty; lateral incision; stent 4–6 weeks
Redo pyeloplasty (open, laparoscopic, robotic)	Most failures; long or dense stricture, crossing vessel; failed endopyelotomy	About 86–89% (P2: about 83–89%); best choice in most children; wide mobilisation of kidney and ureter; transperitoneal (scarred retroperitoneum)
Ureterocalicostomy	Scarred, small intrarenal pelvis; dilated lower-pole calyx	About 70% in redo series; **resect** lower-pole parenchyma (not nephrotomy)
Buccal mucosa onlay graft	Long scarred PUJ/UPJ (ureteropelvic junction) or proximal ureter, dense scar	Robotic with omental wrap; failure about 15–25% (CORRUS)
Appendiceal onlay flap	Right-sided proximal stricture	Pedicled on appendiceal artery
Flaps (spiral, vertical), downward nephropexy	Gain length	Combine with redo repair
Ileal ureter	Long defect / long ureteric loss	Contraindicated if creatinine >2 mg/dL, bladder dysfunction, IBD (inflammatory bowel disease), radiation
Autotransplantation	Solitary or poor contralateral kidney; other repairs impossible; long ureteric loss	About 90% long term
Nephrectomy	Split function <15% with normal contralateral kidney; multiple failures	Definitive; laparoscopic

- Rule of thumb: failed pyeloplasty: try endourology first if anatomy suits; failed endopyelotomy: redo pyeloplasty. Redo pyeloplasty is the most reliable.
- **Also know:** robotic near-infrared fluorescence (indocyanine green, ICG) to identify the ureter and perfusion; Davis intubated ureterotomy (historical).

Principles of redo surgery

- Transperitoneal route in children (retroperitoneum is scarred).
- Ureteric catheter; wide mobilisation of kidney and ureter; **excise all scar**; spatulate; **tension-free, watertight** anastomosis over a stent; wrap with omentum or perirenal fat.
- **Also know:** near-infrared fluorescence shows perfusion; downward nephropexy.

Diagram to draw: ureterocalicostomy: lower-pole parenchyma excised, spatulated ureter sewn to lower calyx.

Follow-up

- After primary pyeloplasty: renogram and ultrasound after stent removal.
- After redo: stent out at 4–6 weeks; renograms for 2–3 years (late failures).

Recent advances

- Robotic redo with buccal graft and ICG fluorescence; pyramidal thickness ≤3 mm predicts surgery; urinary biomarkers.

Key points to remember:

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

Also, from the condensed book (not in the paper answer):

- Dismembered (Anderson–Hynes 1949) repair suits any insertion, allows pelvic reduction and **transposition over crossing vessels**, and alone excises the abnormal UPJ.
- Open approach is flank extraperitoneal by default; anterior for horseshoe or pelvic kidney.

ALGORITHM

1. Dissect the ureter with periureteral tissue; place a lateral marking stitch.
2. Excise the UPJ; spatulate **laterally**; bring the apex to the inferior pelvis, anterior to any crossing vessel.
3. Do a watertight absorbable anastomosis, with or without reduction; leave a stent and drain.

- Remove the **stent** at **4–6 weeks**. Reserve an added nephrostomy for secondary UPJO.
- Laparoscopic and robotic success is >**90–95%**, equal to open; the robot's advantage is ergonomic.

ALGORITHM

1. Remove the Foley at 24–36 h, then the drain if output is negligible.
2. If output rises after Foley removal (reflux along the stent), **replace the Foley for 7 days**.
3. Remove the stent at 4–6 weeks; then renography.

6 · Other UPJ reconstructions

> **KEY POINT**
>
> Every UPJ repair asks **what tissue bridges the gap while keeping blood supply**. Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small.

Procedure	Indication	Note
Foley Y-V-plasty	**High insertion**	Not with crossing vessels or need for reduction
Culp–DeWeerd spiral flap	Large pelvis, long proximal narrowing	Longest flap; length:width **≤3:1**
Scardino–Prince vertical flap	Dependent UPJ, box-shaped pelvis	Shorter flap
Davis intubated ureterotomy	Long strictures, with spiral flap	Heals over stent (secondary intention)
Ureterocalicostomy	Small intrarenal pelvis, horseshoe, failed pyeloplasty	Ureter to lower-pole calyx

- Outline flaps longer than needed; trim the apex, never the base.

SAQ Ureterocalycostomy

Paper answer · P2 · P3 · merged from 2

Asked as: P3 SAQ "Indication, procedure and complications of Uretero-Calycostomy?" · P2 SAQ "Uretero-calycostomy."

Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

Procedure (open, laparoscopic or robotic)

`Flank position (45–60°); pre-placed double-J stent → mobilise ureter with generous periureteral tissue → mobilise kidney; partial lower-pole nephrectomy to expose the calyx widely (not simple nephrotomy) → close (ligate) the pelvic outlet; spatulate ureter laterally → apex stitch to lateral calyx wall; second at 180° → interrupted mucosa-to-mucosa sutures, all placed then tied → capsule closed over cut surface, away from anastomosis → cover with perinephric fat, peritoneum or omentum; drain`

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.

Postoperative care and follow-up

- Stent out at 4–6 weeks; urogram about 1 month later, then renogram.
- Later endoscopy needs **antegrade upper-pole access**.

Complications and outcome

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, loss of the renal unit.
- Success about **70%** in the largest (mostly redo) series; better in small paediatric (robotic) series. The access route does not affect outcome.

Alternatives

- Endopyelotomy, redo pyeloplasty, flap repair, Davis intubated ureterotomy, ileal ureter, buccal graft, autotransplantation, nephrectomy (poor kidney); calicovesicostomy for a low-lying kidney.

Diagram to draw: kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
> - Stricture is the main failure; success about 70% in redo series.

7 · Retrocaval ureter

*See the paper answer **Retrocaval ureter and IVC anomalous development** (Section 4, Ch 52).*

8 · Ureteral stricture — evaluation and endourology

Exclude cancer; endoscopy suits only short, well-vascularised strictures.

- **Causes**: malignancy, stone, radiation, surgical ischaemia, periureteral fibrosis, endoscopy, infection (tuberculosis, schistosomiasis).
- **Exclude malignancy first** with CT, with or without ureteroscopy and biopsy.
- Stricture follows ureteroscopy (URS) in ≈**3%** vs 1.5% after shock-wave lithotripsy; an impacted stone >2 months gives ≈1 in 4.
- Endourology needs ipsilateral function ≥**25%** (the algorithm uses 20%).
- **Stents** suit intrinsic strictures; extrinsic compression needs tandem or metallic stents. Chronic stenting suits unfit or cancer patients.
- **Balloon dilation** is inflated until the **waist disappears**, then stented 2–4 weeks. Contraindications: infection, stricture >**2 cm**. It works best (≈85%) in short, recent, non-anastomotic, well-vascularised strictures; anastomotic ≈50%.
- **Endoureterotomy** (retrograde holmium) is first line under 2 cm. Cut full thickness into **2–3 mm of normal ureter** each side, over a safety wire.

Level	Incision	Avoids
Distal/mid ureter	**Anteromedial**	Iliac vessels

Level	Incision	Avoids
Proximal ureter	**Lateral/posterolateral**	Great vessels

- Complete obliteration in high-risk patients can be crossed by **cutting toward the light**, then stenting 8–10 weeks.

ALGORITHM

1. If <2 cm and function >20%, then endoscopic incision.
2. If >2 cm and function >20%, then reconstruction.
3. If function <20%, then nephrectomy (children may still merit repair).
4. Transplant ureter: consider balloon. Follow with renography for 2 years.

KEY POINT

Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across.

9 · Ureteral stricture — reconstruction ladder

Defect length and level choose the repair.

Technique	Defect bridged	Level
Ureteroureterostomy	**2–3 cm**	Upper/mid
Ureteroneocystostomy	4–5 cm	Distal
Psoas hitch	**6–10 cm**	Below pelvic brim
Boari flap	**10–15 cm**	Lower/mid; spiral may reach pelvis
Renal descensus	5–8 cm	Upper; adds to any repair

- **Ureteroureterostomy** suits short defects only, because tension causes stricture. Preserve adventitia, spatulate 180° apart, stent and cover with omentum; success is >90%.

ALGORITHM

1. **Persistent leak**
2. X-ray the stent position.
3. Pull the drain off the anastomosis and stop suction.
4. Place a Foley and give an anticholinergic; if it persists, place a nephrostomy.

SAQ Boari flap

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3d

Definition

- A **tubularised flap of anterior bladder wall**, based posterolaterally on the side of the lesion, swung up to bridge a distal or mid-ureteric gap. Described in dogs 1894, humans 1947; laparoscopic 2001, robotic 2008.

Reconstructive ladder (distal ureter)

Option	Gap bridged
Ureteroneocystostomy	Distal 4–5 cm
+ Psoas hitch	Adds up to 5 cm (total 6–10 cm)
Boari flap (+ hitch)	**10–15 cm**; spiral flap may reach pelvis (right)
Downward nephropexy	Adds up to 5–8 cm
- Beyond: TUU (transureteroureterostomy), ileal ureter, autotransplantation.	

Indications

- Long distal or mid-ureteric loss: gynaecological injury (commonest), stricture, failed reimplantation, distal TCC (transitional cell carcinoma) ureter, tuberculosis.

Contraindications

- **Small, contracted or poorly compliant bladder** (radiation, tuberculosis); uncorrected outlet obstruction or neurogenic bladder. Not usually for acute unstable trauma.
- Preoperative: urodynamics or cystogram for capacity; antegrade and retrograde studies for length.

Technique

1. Mobilise bladder; divide contralateral superior vesical pedicle if needed.
2. Mark flap on the **non-distended** bladder: **base ≥4 cm, apex ≥3 cm, length:base ≤3:1**; length = gap + 3 cm if tunnelled.
3. Raise flap; **psoas hitch** its base (sutures superficial, parallel to fibres: spare genitofemoral and femoral nerves).
4. Anastomose spatulated ureter: **non-refluxing** 3 cm submucosal tunnel, or **refluxing** direct (apex about 3 times ureteric diameter).
5. Stent (double-J); tubularise in two layers 2-0 absorbable; drain, catheter 7–14 days. - Laparoscopic and robotic: transperitoneal, refluxing anastomosis, results excellent.

Complications

- **Recurrent stricture** (commonest; ischaemia or tension), reduced capacity with frequency and urgency, leak, nerve palsy, iliac vessel injury, pseudodiverticulum.
- eGFR (estimated glomerular filtration rate) fell in 44% within 2 years after reimplantation: counsel.
- **Alternative:** bladder advancement flap, similar reach and simpler.

Diagram to draw: bladder outline with oblique flap (4 cm base, 3 cm apex), hitched and tubularised to ureter.

> **Key points to remember:**
>
> - Bridges 10–15 cm; combine with psoas hitch.
> - Base ≥4 cm, length:base ≤3:1, measure bladder empty.
> - Contraindicated in small contracted bladder.

SAQ Trans Uretero Ureteric anastomoses

Paper answer · P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3a

Definition

- **TUU (transureteroureterostomy)**: the injured (donor) ureter is brought across the midline and joined **end-to-side** to the opposite (recipient) ureter. First clinical use in 1934.

Indications

- Lower ureteric loss that reimplantation, psoas hitch or Boari flap cannot bridge: small, scarred or irradiated bladder; long ureteric loss; pelvic phlegmon; associated rectal, vascular or extensive bladder injury.
- One ureter short while the other is reimplanted with a psoas hitch (hitch only one side).
- Undiversion and reconstruction in children (Hendren).
- Usually a **delayed** repair, not acute.

Contraindications

Absolute	**Relative (disease that may involve both ureters)**
Donor ureter too short to reach without tension	Stone disease, **urothelial carcinoma**
Diseased recipient ureter	Retroperitoneal fibrosis, pelvic radiation
	Chronic pyelonephritis, genitourinary tuberculosis
- Recipient reflux must be corrected at the same time, so do a preoperative VCUG (voiding cystourethrogram). Less favoured in children (long life, both kidneys at risk).	

Technique (steps)

1. Midline transperitoneal incision; reflect the colon.
2. Mobilise the **donor ureter with its adventitia** (blood supply); divide it just above the diseased segment.
3. Expose about 5 cm of recipient ureter with minimal mobilisation.
4. Tunnel the donor ureter **behind the sigmoid mesentery, above the IMA (inferior mesenteric artery)** to avoid kinking.
5. **Anteromedial ureterotomy** of the recipient (at least 2 cm); **spatulated, tension-free end-to-side** anastomosis with fine absorbable interrupted sutures.
6. **Double-J stent** from donor kidney to bladder; drain. (Laparoscopic and robotic routes are feasible.)

Complications

- Urine leak (commonest), urinoma, stricture or obstruction, kinking under the IMA.
- **Main risk: one unilateral problem becomes bilateral obstruction.**
- Later endoscopic access is difficult; use the antegrade route.

Results

- Success 90–97% in adults; complications (mostly leak) in about one-quarter of complex cases.

Diagram to draw: donor ureter crossing above the IMA to an end-to-side anastomosis, with stent.

> **Key points to remember:**
> - End-to-side, spatulated, tension-free; cross above the IMA.
> - Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
> - Risk: unilateral injury becomes bilateral.

Also, from the condensed book (not in the paper answer):

- In adults, refluxing and non-refluxing reimplants are equal. Treat outlet obstruction first.
- **Renal descensus** rotates the kidney inferomedially and fixes its lower pole; the renal vein limits descent.
- **Ileal ureter** is isoperistaltic. Contraindications: creatinine >**2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis. Start endoscopic surveillance from year 3.
- **Buccal mucosa onlay** suits dense scar. **Autotransplantation** suits an absent or poor contralateral kidney, or when other repairs are impossible.

> **KEY POINT**
>
> Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung.

10 · Ureteroenteric anastomotic stricture

- Strictures occur in 4–9% of conduits, so **upper-tract surveillance is lifelong**.
- **The cause is ischaemia**. The **left is commoner**, because it needs extra length, passes under the sigmoid mesentery and angulates at the inferior mesenteric artery.
- Risks: **non-refluxing anastomosis** (≈13% vs 2–3%), carcinoma in situ at the margin, obesity and major complications.
- Conduit dilation is often non-obstructive, so confirm with renography; rapid progression with a mass suggests recurrent cancer.
- **Open revision is the gold standard** (≈91% vs 26% endoscopic); robotic is equal. Endourology is **antegrade**. Endoscopic success is ≈50% under 1 cm and better on the right.

ALGORITHM

1. If <2 cm and function >20%: right side, endoscopic incision; **left side, formal repair** (sigmoid mesentery bleeding).
2. If >2 cm, then formal repair.
3. If function <20%, then nephrectomy or observation.

11 · Retroperitoneal fibrosis

LAQ Retroperitoneal fibrosis: diagnosis and (medical) management

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 LAQ "Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis." · P1 SAQ "Retroperitoneal fibrosis" · P2 SAQ "Retro peritoneal fibrosis medical management"

Introduction

- RPF (retroperitoneal fibrosis) is a rare inflammatory fibrosis (fibrous plaque) encasing the aorta, IVC (inferior vena cava) and ureters, centred on the **distal aorta at L4–L5**.
- Incidence about 1 per 200,000–500,000 a year; age 40–60; male:female 2–3:1.

Etiology

Idiopathic (about 70%)	Secondary (about 30%)
Part of **chronic periaortitis**, a large-vessel vasculitis	**Drugs** (commonest identifiable cause): methysergide and other ergots, β-blockers, phenacetin
Proposed antigen: **ceroid** (oxidised lipid in plaque)	**Malignancy:** lymphoma (commonest), myeloma, carcinoid, pancreas, prostate, sarcoma
Probably autoimmune: membranous glomerulonephritis, ankylosing spondylitis, granulomatosis with polyangiitis; ANA (antinuclear antibody) positivity predicts relapse	Radiotherapy; asbestos
Part of multifocal fibrosclerosis: sclerosing mediastinitis and cholangitis, orbital pseudotumour, Riedel thyroiditis	Infection: tuberculosis, actinomyces, gonorrhoea, schistosomiasis

- Many idiopathic cases are now classed as **IgG4 (immunoglobulin G4)-related disease** (raised serum IgG4, IgG4-positive plasma cells, storiform fibrosis).

Pathology

- **Gross:** flat, tan, dense plaque centred on the **distal aorta at L4–L5**, wrapping the ureters; may invade ureter or psoas.
- **Histology changes with stage:**
- Early (active): collagen with capillary proliferation and lymphocytes, plasma cells and fibroblasts (steroid-responsive).
- Late (burnt-out): acellular, avascular sheets of collagen.
- **Trap:** malignant RPF looks identical except for small islands of tumour cells, so take **deep, multiple biopsies**.
- **Mechanism of obstruction:** extrinsic compression and loss of ureteric peristalsis; the lumen stays patent.

Clinical features

- **Dull lower back or flank pain** (commonest); weight loss, anorexia, nausea, malaise, fever, hypertension.
- About half present with **uraemia** or oliguria/anuria; symptoms last 4–6 months before diagnosis.

- **Extension:** IVC (leg oedema, DVT [deep venous thrombosis]); renal vein (gross haematuria); spermatic cord and scrotum. Examination is usually unrevealing.

Diagnostic evaluation

Laboratory

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein) in half to two-thirds; leucocytosis; anaemia; raised creatinine (renal failure); ANA; serum IgG4.

Imaging

Test	Findings and role
CT (computed tomography) / CT urography	Homogeneous soft-tissue mass enveloping aorta, IVC and ureters; **medial deviation** of proximal and mid-ureters with smooth tapering
Retrograde pyelography	When renal function is poor
MRI (magnetic resonance imaging)	Low T1; **high T2 signal means active disease**; enhancement falls with treatment
PET-CT (positron emission tomography)	**Most sensitive test of activity**; guides biopsy and follow-up
Renography	Split function (repair or nephrectomy)

- **Biopsy** (CT-guided or at ureterolysis) if atypical or malignancy is suspected; may be omitted with no malignancy history, no lymphadenopathy and typical imaging.

Treatment algorithm

```
RPF with hydronephrosis -> Relieve obstruction (stents; nephrostomy if anuric or septic) -> Treat post-obstructive diuresis -> Stop culprit drug; exclude malignancy -> Active disease: steroids (+/- second-line drugs) -> Failure or burnt-out fibrosis: ureterolysis
```

Initial management

- Emergent **stents or PCN (percutaneous nephrostomy)**; nephrostomy if critically ill, uraemic and anuric with little urine. Replace fluid during **post-obstructive diuresis**.
- **Stop the culprit drug;** treat any secondary cause.

Medical management

Agent	Role and regimen
Prednisone (first line)	About 60 mg daily (about 1 mg/kg), tapered to 5 mg, total course up to 2 years. About 80% respond; relapse on tapering 25–50%
Mycophenolate mofetil (with steroids), azathioprine, cyclophosphamide, ciclosporin, colchicine	Second line or steroid-sparing
Tamoxifen; medroxyprogesterone	Thought to inhibit fibroblasts
Rituximab	Refractory or IgG4-related disease

- Steroids work on the cellular, active phase: pain and symptoms settle in days, ESR falls and the mass shrinks. Active inflammation predicts response; burnt-out fibrosis responds poorly.
- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen, so steroids remain first line.

- Drainage plus medical therapy clears obstruction in about two-thirds; only about **15% need ureterolysis**.

Surgical ureterolysis

- For failed medical therapy, burnt-out (fibrotic) disease or tissue diagnosis. **Bilateral**, even if hydronephrosis is unilateral; open, laparoscopic or robotic.
- Steps: 1. Pre-place stents to identify the ureters. 2. Mobilise the colon; take deep biopsies for frozen section. 3. Start at the distal, non-dilated ureter. 4. "Split and roll": right-angle clamp between ureter and plaque; incise the fibrosis over it. 5. Close any ureterotomy with absorbable suture. 6. **Separate the ureter from the fibrotic bed so it cannot be re-encased.**

Protection method	How
Intraperitonealisation	Ureter lateralised; peritoneum closed behind it
Lateral retroperitoneal placement (lateralisation)	Ureter displaced laterally (equivalent results)
Omental wrap	Omentum split on the gastroepiploic arteries, wrapped round each ureter; adds blood supply; best for extensive disease

- Postoperative steroids may reduce recurrence.
- **Outcomes:** open about 94% stent-free at 1 year; laparoscopic about 90%; robotic with omental wrap about 86%.

Other options

- Long-term metallic stents (unfit); autotransplantation (rarely needed) if ureterolysis is impossible; nephrectomy for an unrecoverable (dead) kidney.

Follow-up and monitoring

- Symptoms, ESR/CRP, creatinine, ultrasound; periodic CT/MRI or PET-CT for activity; remove stents once the ureters drain.
- Watch for relapse and steroid side effects (diabetes, osteoporosis).

Diagram to draw: periaortic plaque at L4–L5 drawing both ureters medially, with an omental wrap.

> **Key points to remember:**
> - Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
> - Medially deviated ureters; PET-CT best shows activity; biopsy deep.
> - Drain first, then prednisone (about 80% respond).
> - Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
> - Rituximab for IgG4-related or refractory disease.
> - About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

ALGORITHM

1. If uraemic, decompress with a **nephrostomy** or stents and manage the **post-obstructive diuresis**.
2. Stop any culprit drug; biopsy.
3. Give steroids; if they fail, do ureterolysis.

- Steroids work while the lesion is cellular; second line is immunosuppressants or **tamoxifen**. Ureterolysis with an omental wrap is most definitive.

94 Upper Urinary Tract Trauma

LAQ Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management

Paper answer · P2

Asked as:

- Traumatic renal vascular parenchymal injuries : diagnosis and management. — *Summer 2017 Paper II Q2 (LAQ)*
- Diagnosis and Treatment of renal trauma — *Practice Set 9 Paper II Q1 (LAQ)*
- Grade 4 Renal trauma. — *Practice Set 3 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 9 Paper II Q1: also answered in 2017-Summer Paper 2, Q2; grade IV in Practice Set 3 Paper 2, Q3c) (Practice Set 3 Paper II Q3c: renal trauma also answered in 2017-Summer Paper 2, Q2)

Assumed meaning: diagnosis and management of traumatic renal parenchymal and renal vascular injuries.

Introduction

- Kidney is the **commonest urological organ injured** (up to 5% of all trauma); **blunt injury causes 80–90%**; firearms cause most penetrating injuries.
- **Rapid deceleration** stretches the kidney at its fixed points (hilum and UPJ, ureteropelvic junction): arterial intimal tear and thrombosis, vein disruption, pedicle or UPJ avulsion.
- Main renal artery injury is rare (about 0.05% of blunt trauma).

Classification: AAST (American Association for the Surgery of Trauma) 2018

Grade	Injury
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration ≤1 cm, no extravasation
III	Laceration >1 cm without collecting-system injury; **any vascular injury or active bleeding contained within Gerota's fascia**
IV	Laceration into collecting system with extravasation; renal pelvis laceration and/or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction (thrombosis)
V	**Main renal artery or vein laceration or hilar avulsion**; devascularised kidney with active bleeding; shattered kidney
- Advance one grade for multiple injuries, up to grade III. Change from 1989: main vessel injury moved from IV to V; pseudoaneurysm and AVF (arteriovenous fistula) now grade IV.	

Definition: AAST 2018, grade IV in detail

- Laceration into the **collecting system with urinary extravasation**.
- **Renal pelvis laceration and/or complete UPJ (ureteropelvic junction) disruption.**
- **Segmental artery or vein injury** (pseudoaneurysm, AVF [arteriovenous fistula]).
- **Active bleeding beyond Gerota's fascia.**
- **Segmental or complete infarction** from thrombosis without active bleeding.
- Change from 1989: main-vessel injury moved to grade V; segmental vascular and UPJ injuries added to grade IV.

Clinical diagnosis

History and examination

- Mechanism is the most important (key) history; mechanism matters. Signs: flank bruise or tenderness, **fractured ribs 9–12**, flank or low chest wound.

Haematuria

- Test the **first urine sample**. Significant: gross haematuria, or microscopic (microhaematuria) (>5 RBC [red blood cells] per high-power field, hpf) with **shock (SBP [systolic blood pressure] <90 mmHg)**.
- Degree does **not** match severity; it may be **absent in pedicle injury** and UPJ avulsion (in about half of UPJ injuries), and in grade IV.

Indications for imaging (AUA (American Urological Association) and EAU (European Association of Urology))

1. Blunt trauma with gross haematuria.
2. Blunt trauma with microhaematuria and **SBP (systolic blood pressure) <90 mmHg** at any time.
3. Rapid deceleration (high-speed collision, fall), even without haematuria.
4. Penetrating trauma with likely renal injury, any haematuria.
5. Children: image liberally (hypotension appears late). - Microhaematuria without shock or deceleration (microhaematuria alone without shock): observe without imaging.

Imaging

- **Contrast-enhanced CT (computed tomography)** is the gold standard. Arterial/nephrographic phase (~80 s) shows bleeding; **delayed excretory phase** (5–10 min / minutes) shows urinary extravasation (urine leak).
- **CT red flags:**

Finding	Suggests
Medial haematoma	Pedicle (vascular) injury
Medial extravasation, no contrast in distal ureter (without distal ureteric contrast)	Renal pelvis or UPJ avulsion
Absent enhancement, cortical rim sign	Main renal artery occlusion
Active contrast blush (vascular contrast blush)	Arterial bleeding needing AE (angioembolisation)

Finding	Suggests
Haematoma rim >4 cm (another source: >3.5–4 cm), medial laceration	High risk of intervention; bleeding likely to need intervention (≥2 of rim, blush, medial laceration predict intervention)
- **One-shot IVP (intravenous pyelography)** on table (2 mL/kg, one film at 10 min) in the unstable patient: confirms a **functioning contralateral kidney**. FAST (focused assessment with sonography for trauma) is non-specific. Renal vein injury is poorly seen on CT.	

Management

`Unstable despite resuscitation → laparotomy (± damage control) (or angioembolisation / AE if team ready, else laparotomy) Stable → stage with CT Grade I–III → non-operative Grade IV–V → non-operative in selected; AE for active bleeding Active blush / pseudoaneurysm / AVF → superselective AE Extravasation → observe (90% heal); stent ± nephrostomy if urinoma enlarges, fever, obstruction, or UPJ/pelvis injury`

- **Non-operative**: bed rest until gross haematuria clears, serial haemoglobin (**every 4–6 h**); success >90% in blunt injury and selected penetrating ones. **Non-operative success** about 85% for grade IV–V even in penetrating series; most failures need only AE, stent or nephrostomy.
- Failure: instability, or falling haemoglobin despite 2–3 units of blood.
- Urinary extravasation heals spontaneously in **76–90%** (another source: up to 90%).
- **Angioembolisation**: superselective coils for active bleeding, pseudoaneurysm, AVF or delayed bleeding; repeat AE often succeeds.
- **Reimaging** (reimage / repeat CT at ~48 hours) for grade IV–V with deep laceration (deep lacerations), or new fever, pain or falling haematocrit (routine reimaging debated).

Indications for exploration

Absolute	Relative
Haemodynamic instability, no or transient response	Extravasation with devitalised parenchyma (>25%; lowers the threshold)
Expanding or pulsatile retroperitoneal haematoma	Arterial thrombosis
Suspected pedicle avulsion	Associated colon or pancreatic injury
UPJ avulsion	Incomplete staging

Operative technique (operative steps)

- Midline transabdominal (laparotomy); life-threatening injuries first.
- **Vascular control before opening Gerota's fascia**: lift small bowel; incise retroperitoneum over the aorta above the inferior mesenteric artery → find the left renal vein crossing the aorta → loop vein and both arteries; then reflect colon, open Gerota's fascia.
- **Renorrhaphy**: debride → suture-ligate vessels → watertight collecting-system closure → close defect over bolster → omental cover → drain. Polar injury: partial nephrectomy.

- **Nephrectomy** if unreconstructable or unstable. **Damage control**: pack, resuscitate in ICU (intensive care unit), relook (return) at 24–48 h.

Renovascular injury

- **Main artery thrombosis** (intimal tear): unilateral with normal other kidney → observe (revascularisation rarely succeeds). **Solitary or bilateral → revascularise** (warm ischaemia ideally <120 min); endovascular stent for (or stent an) intimal flap.
- **Vein**: left renal vein can be ligated near the IVC (inferior vena cava) (gonadal, adrenal collaterals); right cannot. Repair with 5-0 vascular suture; segmental veins may be ligated.

Complications

- **Delayed bleeding** (within 3 weeks; usually within 21 days): AE (embolise).
- Urinoma or abscess: stent, nephrostomy, drain.
- **Hypertension**: Goldblatt (arterial stenosis), **Page kidney** (haematoma compression), segmental ischaemia, AVF.
- Hydronephrosis, loss of function; **also know:** AKI (acute kidney injury).

Follow-up

- Blood pressure, urinalysis, creatinine; EAU: DMSA (dimercaptosuccinic acid) scan for salvaged grade III–V kidneys; imaging if new symptoms.

Diagram to draw: left renal vein crossing the aorta, with vessel loops on the vein and artery. For grade IV: kidney with laceration into a calyx, urinoma and segmental pseudoaneurysm.

> **Key points to remember:**
> - AAST 2018: segmental vessel injury grade IV; main vessel grade V.
> - Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
> - Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
> - Delayed CT phase is essential; blush means AE.
> - Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
> - Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
> - Revascularise only a solitary kidney or bilateral injuries.

Also, from the condensed book (not in the paper answer):

- Polar injury is treated by partial nephrectomy with omental cover.

1 · Renal injury — mechanism and presentation

- The kidney is the commonest urological organ injured. Blunt trauma causes 80–90%; most are grade I–III and heal conservatively.
- **Deceleration** injures the fixation points (hilum and ureteropelvic junction, UPJ), causing artery thrombosis, vein disruption, and pedicle or UPJ avulsion.
- **Penetrating** injury causes more bleeding, nephrectomy and complications; high-velocity missiles cause delayed necrosis.
- Signs: flank ecchymosis and **ipsilateral rib 9–12 fractures** (strongest association).

2 · Haematuria

- Significant haematuria is **gross**, or **microhaematuria (>5 red blood cells per high-power field) with systolic blood pressure (SBP) <u>≤90 mm Hg</u>**.
- Degree does not match severity; haematuria is absent in ~⅓ of blunt renovascular injuries and **~½ of UPJ injuries**. Test the first aliquot.

> **KEY POINT**
>
> Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight.

3 · AAST kidney injury scale (2018)

The AAST grade, set on CT, predicts intervention and nephrectomy.

Grade	AAST 2018
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration **<u>≤1 cm</u>**, no extravasation
III	Laceration >1 cm, no collecting-system rupture; **any vascular injury or active bleeding contained within Gerota's**
IV	Collecting system laceration with extravasation; pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's; segmental/complete infarction without active bleeding
V	**Main renal artery or vein laceration** or hilar avulsion; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for bilateral or multiple injuries, up to III; grade by the highest finding. MCQ Main-vessel injury was grade IV in 1989 and is now V; segmental pseudoaneurysm or arteriovenous fistula (AVF) is now IV.

4 · Indications for imaging (AUA = EAU)

- **Penetrating** trauma with **any haematuria** needs CT, if the patient is stable.
- **Blunt** trauma needs CT with rapid deceleration, gross haematuria, or microhaematuria with SBP <90 at any time; otherwise blunt microhaematuria is **observed without imaging**.
- **Children** hold blood pressure until ~50% blood loss, so image liberally.

5 · Imaging studies

Without a delayed phase, urinary extravasation is missed.

- Gold standard: **contrast CT with arterial/nephrographic and delayed (~<u>10 min</u>) phases**. A medial haematoma with normal arteries suggests vein injury.
- Focused assessment with sonography for trauma (FAST) cannot separate haematoma from urine.

CT finding	Suggests
Medial haematoma	Vascular pedicle injury
Medial extravasation; no contrast in distal ureter	Renal pelvis/UPJ avulsion

CT finding	Suggests
Absent global enhancement (cortical rim sign)	Main renal artery occlusion
Contrast blush	Brisk bleeding; strongly predicts need for angioembolisation (AE)
≥2 of: rim distance >4 cm, medial laceration, blush	High-risk grade IV

- The **MiGUTS nomogram** predicts bleeding intervention; its largest contributor is **haematoma rim distance**.

ALGORITHM

1. **One-shot intravenous pyelogram (IVP)** (unstable, no CT): **2 mL/kg contrast → single film at 10 min**. It confirms a working contralateral kidney.

Perinephric collection	Hounsfield units (HU)	Feature
Urinoma	0–20	Contrast pooling on delayed phase
Haematoma	>30	—
Abscess	>20	Rim enhancement

6 · Nonoperative management

- Stable, staged **grade I–IV and selected V** injuries are observed, whatever the mechanism. Success is >90% in blunt injury, even with extravasation; early mobilisation is safe.
- **Failure triggers**: instability; transfusion of **2–3 units**; new pain or guarding; enlarging urinoma with fever. Most failures need only stent, nephrostomy or AE.
- **Penetrating** injury breaches Gerota's fascia, losing tamponade, so more fail; stabs do better than gunshots.
- **Reimaging** (AUA): CT at ~48 h for **grade IV–V deep lacerations** or deterioration, not for uncomplicated I–III.

7 · Urinary extravasation

- Parenchymal leaks **heal in 76–90%**. **Drain early** when urine cannot pass antegrade: pelvis or UPJ avulsion, ureteric clot, UPJ obstruction.
- A symptomatic or enlarging urinoma gets a retrograde stent and catheter; if that is impossible, a nephrostomy.

> **KEY POINT**
>
> Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained.

8 · Intervention

Needless exploration releases tamponade and costs kidneys.

Absolute indications	Relative (increasingly observed)
Instability, no or transient response	Extravasation with significant devascularisation

Absolute indications	Relative (increasingly observed)
Expanding or pulsatile haematoma	Colon or pancreatic injury
Suspected pedicle avulsion; UPJ avulsion	Arterial thrombosis; parenchymal extravasation

- **AE** is superselective, raises salvage and can be repeated. **Failure predictors**: instability, grade V, urinary extravasation, gross haematuria.
- At laparotomy, exploration is **not mandatory** for staged grade I–III. The only absolute indication is a **pulsatile or expanding haematoma**.

ALGORITHM

1. **Vascular control first**
2. Eviscerate the small bowel; incise peritoneum over the aorta above the inferior mesenteric artery (IMA), up to the ligament of Treitz.
3. Follow the aorta to the left renal vein; loop and retract it; loop both renal arteries.
4. Reflect the colon and open Gerota's fascia; occlude the vessels if bleeding.

- **Damage control** (hypothermia, acidosis, coagulopathy): pack, close temporarily, resuscitate and return at 24–48 h.

ALGORITHM

1. Expose and gain vascular control; debride only devitalised tissue.
2. Suture-ligate vessels; close the collecting system **watertight**.
3. Approximate parenchyma over a bolster; cover with Gerota's fascia or **pedicled omentum**; drain.

- Colon or pancreatic injury no longer mandates nephrectomy.
- **Revascularise** only a solitary kidney or bilateral injury, with warm ischaemia **<120 min**; unilateral repair salvages only ~⅓.
- The **left renal vein may be ligated; the right cannot**.
- Hypertension follows thrombosis repair in ≥43%, so monitor blood pressure for a year.

9 · Complications and follow-up

- Delayed bleeding comes within 21 days and is treated by AE.
- **Page kidney** (subcapsular compression) is treated with an ACE inhibitor or receptor blocker, then drainage if refractory; post-traumatic AVF needs AE.
- EAU: no imaging for grade I–II; salvaged grade III–V get a dimercaptosuccinic acid (DMSA) scan.
- Bleeding risk peaks at **2–6 weeks**; a solitary kidney does not bar contact sport.

10 · Ureteral injury

Often silent; late diagnosis means fistula or kidney loss.

LAQ Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury

Paper answer · P3

Asked as:

- Iatrogenic ureteric injury — *Practice Set 2 Paper III Q1 (LAQ)* (upper ureteric injury also answered in 2023-Winter Paper 3, Q2)
- Discuss management of ureteric injuries occurring during gynec surgeries — *Practice Set 4 Paper III Q1 (LAQ)* (also answered in Practice Set 2 Paper 3, Q1)
- Discuss Etiology, Diagnosis and Management of Upper ureteric injury. — *Winter 2023 Paper III Q2 (LAQ)*

Type: LAQ (25 marks)

Assumed scope: surgical (iatrogenic) injury at any level of the ureter, in gynaecological, colorectal, vascular and urological surgery. Assumed meaning (upper ureteric injury): injury to the upper (proximal) ureter, from the PUJ (pelviureteric junction) to the pelvic brim, from external trauma or surgery.

Introduction

- Ureteric trauma is uncommon: 1–2.5% of genitourinary trauma. **Iatrogenic injury is commoner overall** than external trauma.
- **The proximal ureter is the commonest site of external injury. The distal ureter is the commonest (usual) iatrogenic site** in non-urological surgery.
- Gynaecological surgery has historically caused about half of all iatrogenic ureteric injuries; hysterectomy is the main culprit.
- It is mainly a problem of **recognition** (the core problem): thermal, ligation and devascularisation injuries are invisible at operation and declare themselves days to weeks later. Injury is often **missed** and declares itself later as urinoma, sepsis, stricture or fistula.
- Late diagnosis brings urinoma, sepsis, AKI (acute kidney injury), fistula and renal loss, and needs nephrostomy about ten times more often.

Aetiology and incidence (EAU (European Association of Urology) trauma guideline data)

1. External trauma

- **Penetrating** (most): gunshot wounds up to about 90%. Blast damages the intramural blood supply up to 2 cm from the wound.
- **Blunt** (up to about 40%): rapid deceleration avulses the ureter at the **PUJ**; linked to lumbar transverse process fractures.
- **Over 90% have associated injuries** (small bowel commonest, colon, vessels, kidney 10–28%). Mortality 6–9% reflects these.

2. Iatrogenic

Source	Key facts (upper-ureter relevance)
Gynaecological	Historically about half. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery highest (1.7–4.3%). Endometriosis and placenta accreta raise risk
Colorectal	0.15–10%; laparoscopic equals open
Colorectal, spinal, retroperitoneal surgery	Ligation, transection, cautery

Source	Key facts (upper-ureter relevance)
Urological / Ureteroscopy	Ureteroscopy is the commonest (iatrogenic) cause in some series: abrasion, perforation 0.2–2%, avulsion under 0.5% (another source: <0.3%) (proximal third most at risk); stricture about 3% (another source: 0.3–3%). Risk: proximal stone, impacted stone, long operation, large sheath, inexperience. Also radical prostatectomy, partial nephrectomy or ablation (late proximal stricture)
Partial nephrectomy, thermal ablation	Proximal stricture months later (lower-pole and hilar tumours)
Vascular (aortic) surgery	Aorto-iliac bypass: transient hydronephrosis 12–20%; up to 85% unrecognised (missed initially). Devascularisation. Rare ureteroarterial fistula (chronic stent plus pelvic surgery or radiation)

Gynaecological incidence (EAU trauma guideline data)

Procedure	Ureteric injury (%)
Caesarean section (emergency)	0.01–0.06
Vaginal hysterectomy	0.02–0.5
Abdominal hysterectomy	0.03–2.0
Minimally invasive hysterectomy	0.13–6.0 (laparoscopic = robotic, about 0.9)
Prolapse surgery	**1.7–4.3 (highest)**

- **Danger sites in the pelvis (applied anatomy):** pelvic brim near (at) the infundibulopelvic ligament; **under the uterine artery** ("water under the bridge"); beside the uterosacral ligament; vaginal fornix / vaginal cuff near the UVJ (ureterovesical junction).
- Up to a third are recognised during open surgery, fewer during laparoscopy; the uterosacral region is hardest to see.
- **Mechanisms:** ligation, crush, laceration, transection, avulsion, thermal injury, **devascularisation**. About 10% of women get a large share of distal ureteric blood from the uterine artery (the uterine artery supplies much of the distal ureter), so some (ischaemic) injuries are unavoidable.
- **Risk factors:** previous surgery, adhesions, malignancy, radiotherapy, heavy bleeding, long operation, learning curve; endometriosis (encasement, adhesions, medial deviation), placenta accreta.
- **URS (ureteroscopy) risks:** long operation, 12/14-Fr sheath, impacted or proximal stone, inexperience.

Classification: AAST (American Association for the Surgery of Trauma) ureter scale

Grade	Injury (description)
I	Haematoma or contusion, no devascularisation (without devascularisation)
II	Laceration <50% of circumference
III	Laceration ≥50%
IV	Complete transection, <2 cm devascularised (with <2 cm devascularisation)
V	Avulsion, >2 cm devascularised (with >2 cm devascularisation)

- Advance one grade for bilateral injury, up to grade III.

Prevention

- See (identify) the ureter at the **pelvic brim** (crossing the common iliac artery) and trace it before ligating pedicles. Good haemostasis and exposure.
- **Prophylactic stents** (or lighted stents) help recognition in high-risk (complex) cases, but do not clearly reduce (prevent) injury. **ICG (indocyanine green)** instilled into the ureter (intraureteric) makes it fluoresce (fluoresces) for hours.
- **Cystoscopy** after minimally invasive hysterectomy and prolapse surgery, with oral phenazopyridine or IV (intravenous) dye to show efflux.
- URS: safety wire; ≤14-Fr sheath in an unstented ureter; fragment before basketing; "stop, stent, stage".

Diagnosis and recognition

Suspicion (trauma)

- Penetrating flank or abdominal wound; high-energy deceleration.
- **Haematuria is absent in 25–45%**, so its absence does not exclude injury.

Intraoperative (best outcome)

- Inspect the ureter (at laparotomy); IV (intravenous) indigo carmine or methylene blue; cystoscopy for efflux (jets), with retrograde (on-table) pyelogram or stent. Check the other ureter and bladder. About 11% are missed at trauma laparotomy.
- Methylene blue is an MAO (monoamine oxidase) inhibitor: avoid with SSRIs/SNRIs (selective serotonin / serotonin–noradrenaline reuptake inhibitors; serotonin toxicity), G6PD (glucose-6-phosphate dehydrogenase) deficiency and pregnancy.

Delayed presentation (most cases) / postoperative

- Mean diagnosis about day 10: fever, ileus, flank pain, leucocytosis, rising creatinine, rising drain output, drain or wound urine leak, urinoma, anuria (bilateral), or **ureterovaginal fistula** (UVF): normal voiding plus constant leak, because the other kidney fills the bladder, usually within 4 weeks.

Imaging and tests

Test	Role
CT (computed tomography) urography with 10-minute delayed images (both ureters)	Best test in stable patients. Signs: extravasation, non-opacified distal ureter, delayed nephrogram, urinoma
Retrograde pyelography	Most sensitive; shows site; allows stenting at once
Antegrade nephrostogram	When retrograde fails or gap is wide
One-shot IVU (intravenous urography)	Non-diagnostic in 33–100%; never rely on it

- **Drain-fluid (or vaginal fluid) creatinine** above (far above) serum creatinine confirms urine.
- **Double-dye test** separates VVF (vesicovaginal fistula) from ureterovaginal fistula: oral phenazopyridine plus intravesical methylene blue. Orange pad = UVF; blue = VVF; both = both.

Management

Principles

- **Stability decides timing; level and length decide the operation.**

- Repair: mobilise sparing (preserve) adventitia; debride to bleeding tissue (edges); **spatulated, tension-free, watertight, stented** repair (anastomosis) with fine absorbable suture; drain; retroperitonealise; omentum (omental wrap) for thermal, radiation, blast or vascular injury; non-tunnelled reimplant.
- **No bowel interposition or autotransplantation acutely.**

By timing ``` Recognised at surgery -> repair at once (stable patient) Unstable -> ligate + later nephrostomy, or exteriorised stent Recognised late -> try retrograde stent (or nephrostomy + antegrade stent)

|-- succeeds -> stent 4-8 weeks, then image |-- fails -> nephrostomy; early repair if <=7 days, otherwise delayed repair after about 6 weeks Uncontrolled sepsis / collections -> drain; repair regardless of timing Ureteric injury suspected

|-- Unstable -> damage control: ligate + later PCN, | or exteriorised stent / cutaneous ureterostomy |-- Stable, found at surgery -> immediate repair over stent |-- Found late / postoperatively -> Attempt retrograde stent (+/- PCN, drain) -> Fails -> PCN; delayed definitive repair Suspected late injury -> retrograde stent (or nephrostomy + antegrade stent)

```
|-- succeeds -> stent 4-8 weeks, then image
|-- fails -> nephrostomy
     |-- found <=7 days -> early repair
     |-- later -> repair after about 6 weeks
```

Sepsis / uncontrolled collections -> drain; repair regardless of timing ```

At surgery, by injury

Injury	Action
Suture through wall, minor thermal	Remove suture; stent
Partial tear	Close transversely (transverse closure, 4-0/5-0) over stent, 4–8 weeks
Ligation	Remove ligature (deligate); stent if viable, excise and repair if not
Transection, lower ureter (usual)	**Ureteroneocystostomy ± psoas hitch** (95–100% success)
Long distal gap	**Boari flap** + hitch (gaps up to 12 cm)
Transection above pelvic brim	Ureteroureterostomy
Unstable patient	Ligate + later nephrostomy, or exteriorised stent

Repair by grade

- **Grade I–II (contusion, partial):** double-J stent for about 6 weeks; or close a laceration transversely over a stent. Excise non-viable contused ureter.
- **Grade III–V:** formal reconstruction.

By level (complete injury)

Level	Repair
Upper	Ureteroureterostomy; ureteropyelostomy (PUJ (pelviureteric junction) avulsion); ureterocalicostomy
Mid	Ureteroureterostomy ± downward nephropexy; TUU (transureteroureterostomy) when delayed

Level	Repair
Lower (commonest iatrogenic)	**Ureteroneocystostomy, non-tunnelled ± psoas hitch** (95–100% success); **Boari flap** for gaps up to 12 cm
Long loss (delayed only)	Ileal ureter or Yang–Monti; buccal graft ureteroplasty; autotransplantation (last resort)

Options for the upper ureter

Procedure	Key points
Ureteroureterostomy	Commonest acute repair; short defects (2–3 cm); spatulate 5–6 mm, 180° apart; 5-0 absorbable over double-J; up to 90% success; leak 10–24%, stricture up to 12%
Ureteropyelostomy (reimplant into pelvis)	PUJ avulsion; spatulated
Ureterocalicostomy	Pelvis or PUJ destroyed; lower-pole parenchyma resected
Renal descensus	Mobilise kidney downwards; gains up to 8 cm
TUU (transureteroureterostomy)	Delayed; long loss when the bladder cannot reach; 90–97% success. Avoid with stones, RPF (retroperitoneal fibrosis), TB (tuberculosis), urothelial cancer or radiation
Ileal ureter / Yang–Monti	Long or panureteric loss; 81–100% success. Avoid if creatinine >2 mg/dL, bladder dysfunction, inflammatory bowel disease, radiation enteritis. Acidosis about 20%
Appendiceal interposition	Right side; about 92% short-term success
Buccal mucosa graft ureteroplasty	Delayed long strictures; onlay with omental wrap, often robotic; about 87% success
Autotransplantation	Last resort before nephrectomy; 5–10% renal units lost
Nephrectomy	Rarely; unstable with unreconstructable kidney, or poor function

Iatrogenic endoscopic injury

- **URS perforation:** stop the procedure and place a **stent** about 4 weeks (± nephrostomy), image later; later reconstruction if stricture develops. **Avulsion:** nephrostomy, delayed reconstruction; avoid nephrectomy.

Fistulae

- **Ureterovaginal fistula (UVF):** stent first (best / ideally within 2 weeks; most heal); if stenting fails, reimplant with psoas hitch or Boari flap (success ≥90%). Early and delayed repair do equally well.
- **Ureteroarterial fistula:** endovascular covered stent.
- **Also know:** TUU (transureteroureterostomy), ileal ureter, buccal graft ureteroplasty and autotransplantation for long delayed defects; robotic reconstruction (repair) for delayed repair matches open; avoid nephrectomy.

Complications and follow-up

- Urinoma, abscess, leak, fistula, stricture, loss of renal function.
- Persistent leak: check stent position, take drain off suction, catheterise the bladder, then nephrostomy.

- **Follow-up:** ultrasound or diuretic renogram at 4–12 weeks and (at) 1 year; strictures (late stenosis) can be silent and may appear up to 30 months.

Medicolegal

- Disclose and document; malpractice claims most often follow gynaecological surgery.

Diagram to draw: psoas hitch with non-tunnelled reimplant; ureter's relation to the uterine artery (ureter passing under the uterine artery); spatulated ureteroureterostomy over a double-J stent.

> **Key points to remember:**
> - Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
> - Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
> - AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
> - CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
> - Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
> - Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
> - Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Also, from the condensed book (not in the paper answer):

- Delayed upper-ureter repair may use appendix (right) or buccal graft.

AAST ureter grade	Description
I	Contusion/haematoma without devascularisation
II	Laceration **<50%** transection
III	Laceration ≥50% transection
IV	Complete transection, **<2 cm** devascularisation
V	Avulsion, >2 cm devascularisation

- Iatrogenic injury is commoner (distal); external is mostly gunshot (proximal). **Haematuria is absent in 25–45%**.
- The **distal ureter** is the commonest iatrogenic site; prolapse surgery is highest-risk, and minimally invasive hysterectomy is riskier than open.
- **Ureteroscopy**: stricture ~3%, avulsion <1%. At a tear, stop basketing and stent.
- Preoperative stents aid **recognition**, not proven prevention. Check efflux at cystoscopy after minimally invasive hysterectomy and prolapse surgery. **Methylene blue** is a monoamine oxidase inhibitor; avoid it in pregnancy and G6PD deficiency.

Modality	Role
CT urography, **10-min delays**	Best in a stable patient; trace both ureters
Retrograde ureterography	**Most sensitive**; allows stenting
Antegrade study	When retrograde fails
One-shot IVP	Often non-diagnostic; never rely on it

- CT signs: **no contrast distal to the injury**, medial extravasation (UPJ avulsion), delayed nephrogram, urinoma.

> **KEY POINT**
>
> Ureteral injury is diagnosed by absent contrast beyond it, so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative.

ALGORITHM

1. If stable at laparotomy, repair now; if unstable, do damage control and repair later.
2. If contusion, stent, or excise and repair.
3. If found late or partial, stent; if stenting fails, place a nephrostomy and repair later.
4. If ureterovaginal fistula, try a stent first.

- **Damage control**: ligate the proximal stump with long silk and place a nephrostomy once stable, or exteriorise a single-J stent.
- Principles: spare adventitia; debride to bleeding edges; make it **spatulated, tension-free, stented and watertight**; drain. **No bowel or autotransplantation acutely.**
- In trauma, reimplant **non-tunnelled**, because tunnelled reimplants stricture more.
- Psoas hitch beats distal ureteroureterostomy, because the distal stump's supply is tenuous; spare the genitofemoral and femoral nerves.

95 Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

1 · Overview and epidemiology

- **Removal is surgical, but recurrence is metabolic**. Prophylaxis: see Ch 99.
- Lifetime prevalence is **1–15%**; in the United States ≈ **10%** and rising.
- **Sex**: males outnumber females 2–3 : 1, but the gap is narrowing. **Oestrogen is protective**, hence a second female peak at menopause.
- **Race**: highest in White, lowest in Asian people. **Age**: peak in the 4th–6th decades; paediatric stones are rising.
- **Climate and occupation**: heat lowers urine volume and pH, sunlight causes vitamin D-mediated hypercalciuria, and sedentary work causes hypercalciuria.
- **Obesity**, **metabolic syndrome** and **diabetes** raise risk. **Insulin resistance** unifies these: it lowers urine pH, causing uric acid stones, and hyperinsulinaemia causes hypercalciuria.
- Hypertension, cardiovascular disease and chronic kidney disease (CKD) are associated. Fluid intake is inversely related (proven by randomised controlled trial, RCT).

2 · Physicochemistry and stone formation

Prevention lowers supersaturation or aids inhibition.

LAQ Urolithiasis: etiopathogenesis, medical management and prevention of recurrence

Paper answer · P1 · P2 · P4 · merged from 3

Asked as: P4 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" / "Describe medical management of recurrent stone disease and discuss recent advances in instrumentations of RIRS" · P2 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" (also P2 SAQ "Describe medical management of renal calculi") · P1 LAQ "Discuss pathophysiology of renal stones and also describe how to prevent its recurrence"

Introduction

- Urolithiasis is a **chronic metabolic disease** punctuated by stone episodes; recurrence is up to **50% within 5 years** (another source: about half recur within 5–10 years). Removal is surgical; recurrence is metabolic. Recurrent formers need a **metabolic evaluation** and targeted therapy.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20% (calcium oxalate with or without apatite about 80% in all), uric acid and struvite about 7% each, cystine 1–3%.
- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy inside the kidney.

Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass,

sarcoidosis.

- **Anatomical (stasis)**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

Physicochemistry: state of saturation

State	Concentration product (CP)	What happens
Undersaturated	Below Ksp (solubility product)	No crystals; stones may dissolve
Metastable	Between Ksp and Kf (formation product)	No spontaneous nucleation, but growth on existing crystals; inhibitors act here. **Zone of therapy**
Unstable	Above Kf	Spontaneous nucleation, growth, aggregation

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

Steps of stone formation

1. **Supersaturation:** low volume; high calcium, oxalate or uric acid; low citrate.
2. **Nucleation:** homogeneous, or **heterogeneous** on debris or other crystals (needs less energy).
3. **Growth and aggregation:** aggregates, not single crystals, can occlude a tubule within the 5–7 minute nephron transit (**free-particle theory**).
4. **Retention (fixed-particle theory): Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) point to a secondary cause: enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria.

Inhibitors, promoters and matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm–Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

Pathophysiology by urinary abnormality

Abnormality (definition)	Mechanism
Hypercalciuria (commonest; >200 mg/day, or >4 mg/kg/day)	Absorptive (gut), renal leak (secondary high PTH [parathyroid hormone]), resorptive (primary hyperparathyroidism): Pak classification
Hyperoxaluria (>40 mg/day)	**Primary:** hepatic enzyme defect (PH1, AGXT gene); **enteric:** malabsorbed fat binds calcium, freeing oxalate; dietary
Hyperuricosuria (>600 mg/day)	Urate promotes calcium oxalate nucleation
Hypocitraturia (<320 mg/day)	Distal RTA, diarrhoea, hypokalaemia, animal protein (acid load, intracellular acidosis)
Uric acid stones	**Low urine pH (<5.5)** more than excess uric acid; pKa 5.5

Abnormality (definition)	Mechanism
Cystine	Autosomal recessive defect of tubular cystine reabsorption
Struvite	**Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate and carbonate apatite; biofilm

Medical management: aims

- Relieve pain, dissolve suitable stones and **prevent recurrence** (up to 50% at 5 years).

Acute colic and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory. Urgent drainage if infection with obstruction.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm** after migration from the kidney: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

Dissolution (chemolysis)

- **Uric acid** stones (radiolucent) dissolve with oral potassium citrate to alkalinise (EAU [European Association of Urology]: pH 7.0–7.2 for dissolution); allopurinol if hyperuricosuric.
- **Cystine:** alkalinise to pH 7.0–7.5 with thiol drugs. **Struvite:** acetohydroxamic acid occasionally dissolves.

Prevention of recurrence

1. Screen every first-time former (AUA [American Urological Association], EAU)

- History, serum electrolytes, creatinine, calcium, uric acid; urine pH and culture; **stone analysis**; imaging. This sorts patients into high and low risk.

2. General measures (every former)

- **Fluid for urine ≥2.5 L/day** (drink about 3 L). **Borghi 1996**: urine >2 L cut 5-year recurrence (12.1% vs 27%; recurrence halved).
- **Normal calcium**, low sodium (≤2.3 g/day), less animal protein, more fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension]-type diet). **Borghi 2002**: this diet halved recurrence compared with a low-calcium diet. Restricting calcium raises oxalate absorption.
- Limit high-oxalate foods (spinach, nuts, chocolate); weight loss.
- Citrus juices (lemonade, orange) are adjuncts only.

3. Metabolic evaluation that guides therapy (high-risk or recurrent formers, children) (see 2015-Summer Paper 2, Q2)

- **Stone analysis** (infrared spectroscopy or X-ray diffraction); serum chemistry: serum calcium, uric acid, creatinine, electrolytes, PTH if calcium is high-normal.
- One or **two 24-hour urines** on usual diet, at least 4 weeks after an episode: volume, pH, calcium, oxalate, citrate, uric acid, sodium, potassium, creatinine, cystine. Therapy targeted to these results reduces stone events.
- Targets: volume >2.5 L, calcium <250 mg (women) or <300 mg (men), oxalate <40 mg, citrate >320 mg, uric acid <600 mg.

4. Selective drug therapy

Abnormality	Drug and dose	Rationale and notes
Hypercalciuria	**Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption plus volume contraction. Add potassium citrate: thiazide lowers potassium, and hypokalaemia causes hypocitraturia
Primary hyperparathyroidism	Parathyroidectomy	Cures resorptive hypercalciuria
Hypocitraturia, distal RTA, chronic diarrhoea	**Potassium citrate 30–60 mEq/day**	Alkali load reduces tubular (proximal) citrate reabsorption; avoid urine pH ≥7
Hyperuricosuria / hyperuricosuric calcium oxalate stones	Allopurinol 300 mg/day	Watch hypersensitivity; reduce dose in renal impairment
Uric acid stones	Potassium citrate to **pH 6.0–6.5** (avoid >7.0: calcium phosphate); allopurinol only for hyperuricosuria or failed alkalinisation	A pH disease: pKa 5.5; dissolution possible
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; liquid potassium citrate; magnesium; cholestyramine	Calcium binds oxalate in the gut
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA [ribonucleic acid] interference against glycolate oxidase)	ILLUMINATE-A: 84% vs 0% reached near-normal urine oxalate
Cystinuria	Urine ≥3 L/day (another source: >3 L); low sodium and protein; potassium citrate to pH **7.0–7.5**; then **tiopronin** (or D-penicillamine)	Aim cystine <250 mg/L; thiols form soluble mixed disulfides
Struvite	**Complete clearance**; culture-specific antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily (another source: two to three times daily) if surgery impossible	Urease inhibitor; risk of thromboembolism

- **Also know:** recurrent calcium stones with a normal work-up: thiazide or potassium citrate (AUA: even without abnormality); citrus juices are adjuncts.
- **Trials**: Ettinger 1988 (chlorthalidone) and Barcelo 1993 (potassium citrate) reduced new stones. **NOSTONE (2023)**: hydrochlorothiazide 12.5–50 mg did not reduce recurrence, which questions the older trials.

5. Monitoring and follow-up

- Repeat 24-hour urine within **6 months** of starting therapy, then yearly; reanalyse any new stone.
- Serum potassium 1–2 weeks after starting a thiazide or citrate; imaging for new stones.

Diagram to draw: saturation zones (Ksp, Kf) and Randall plaque on a renal papilla.

Diagram to draw: flowchart from stone analysis and 24-hour urine to abnormality to selective drug therapy.

Recent advances in prevention

- **ROKS (Recurrence of Kidney Stone) nomogram** predicts individual risk.
- **Also know:** NOSTONE (2023) negative; citrus juices are adjuncts; *Oxalobacter formigenes* loss; lumasiran for PH1.

Recent advances in RIRS instrumentation

Advance	Principle	Benefit or limitation
Digital flexible ureteroscopes	CMOS (complementary metal-oxide semiconductor) chip at the tip; LED (light-emitting diode) light	Sharper image that does not degrade; tip larger (~8.5 Fr); no more durable
Single-use scopes (LithoVue, 2016)	7.7 Fr tip, ≥270° deflection, new each case	No repair or reprocessing; cheaper below ~99 cases a year; for lower pole and difficult cases
Thulium fibre laser (about 1940 nm)	Long pulse, very high frequency	Fine dust, minimal retropulsion; faster than Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) at low settings
Pulse-modulated (Moses) Ho:YAG	Pulse shaped through a vapour bubble	Less retropulsion, faster dusting
Access sheaths (12/14 Fr)	Outflow channel	Lower intrarenal pressure and temperature; severe ureteric injury in ~1 in 8
Suction sheaths and FANS (flexible and navigable suction access sheath)	Vacuum at the sheath; flexible tip follows the scope into calyces	Aspirates dust; lower pressure; better stone-free rates in early studies
Intrarenal pressure monitoring	Sensor in the wire or scope (LithoVue Elite)	High pressure causes backflow, SIRS (systemic inflammatory response syndrome) and sepsis; safe limits not yet defined
Robotic flexible ureteroscopy (Avicenna Roboflex)	Seated console controls the scope	Ergonomics, less radiation to surgeon

- **Laser safety**: tissue damage above 43 °C, so keep power ≤10 W in the ureter and ≤25 W in the kidney, with continuous cool irrigation.
- **Also know:** dusting versus fragment-and-basket; nitinol tipless baskets; anti-retropulsion devices; Monarch robotic platform for PCNL (percutaneous nephrolithotomy) access.

Key points to remember:

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

Also, from the condensed book (not in the paper answer):

- Free-particle theory (crystals trapped as plugs) vs fixed-particle theory (anchored on plaque).
- Stones form by **supersaturation, then nucleation, growth and aggregation, then retention**. The concentration product (CP) multiplies ion concentrations; the solubility product (Ksp) is CP at saturation; the formation product (Kf) is CP at spontaneous crystallisation.

State	CP	What happens
Undersaturated	< Ksp	No crystals; stones may dissolve
Metastable	Ksp–Kf (most urinary salts)	No homogeneous (spontaneous) nucleation; growth and heterogeneous nucleation possible; **inhibitors effective, so this is the zone of therapy**
Unstable	> Kf	Spontaneous nucleation; inhibitors ineffective

- The **relative saturation ratio (RSR)** is CP ÷ Ksp. Below 1, stones dissolve; above 1, they grow.
- Changing pH lowers the saturation of **phosphate, uric acid and cystine, but not oxalate**. **Calcium and oxalate contribute equally** to calcium oxalate saturation.
- Urine holds calcium oxalate at ≈ 4× its solubility in water; with inhibitors it precipitates only at **7–11×**.
- Nephron transit is too short for one crystal to block a tubule, so aggregation occludes.
- **Randall plaque** (MCQ 1937) is **calcium apatite**. It starts in the **thin-limb basement membrane**, spreads through the interstitium and erodes the papilla; calcium oxalate then grows on it.

Stone former	Randall plaque	Ductal plugs
Idiopathic calcium oxalate	Present, as the anchor	Uncommon, mild
Intestinal bypass	Absent	Apatite plugs in inner medullary collecting ducts; injury, fibrosis
Brushite	Present	Apatite plugs; interstitial fibrosis

Stone former	Randall plaque	Ductal plugs
Distal renal tubular acidosis (RTA), hyperparathyroidism, cystinuria	—	Present

> **KEY POINT**
>
> **Plaque vs plug**: a plaque is an idiopathic interstitial apatite anchor; a plug is intratubular obstruction with injury, caused by systemic disease. **Ductal plugs should prompt a search for a secondary cause.** Infection or shock-wave lithotripsy may convert a calcium oxalate former into a brushite former.

3 · Inhibitors and promoters

- Inhibitors raise the supersaturation needed for nucleation. **Uric acid and cystine have no inhibitor**; only pH and concentration matter. Promoters: low volume and citrate; high calcium, oxalate, phosphate, uric acid.

Inhibitor	Action	Key feature
Citrate	Complexes calcium; inhibits precipitation, agglomeration and urate-induced nucleation	Most important small-molecule inhibitor
Magnesium	Complexes oxalate	Synergy with citrate, phytate
Glycosaminoglycans	Bind crystal surface	Lower in stone formers
Nephrocalcin	Inhibits calcium oxalate monohydrate nucleation, growth, aggregation	Formers excrete isoforms lacking γ-carboxyglutamic acid
Tamm-Horsfall protein	Inhibits aggregation, not growth	Most abundant urinary protein, made in the thick ascending limb (TAL); **a promoter in acid urine**
Osteopontin	Inhibits nucleation, growth, aggregation, epithelial binding	Inducible

- **Matrix** is ≈ **2.5%** of stone weight.

4 · Mineral metabolism

- **Calcium**: 30–40% is absorbed in the small bowel, transcellularly (calcitriol-dependent) at low intake and paracellularly at high intake.
- **Oxalate-rich food lowers calcium absorption, and luminal calcium lowers oxalate absorption**.
- **Calcitriol**, made in the proximal tubule under **parathyroid hormone (PTH) and hypophosphataemia**, is the most potent stimulator of gut calcium absorption.
- **PTH** causes bone resorption, raises renal calcium reabsorption and calcitriol, and lowers phosphate reabsorption; it has **no direct gut effect**.

Segment	Filtered calcium reabsorbed	Route
Proximal tubule	**60–65%**	Paracellular; claudin-2
TAL	25–30%	Paracellular, lumen-positive; claudin-16/19; the calcium-sensing receptor (CaSR) acts through claudin-14 to block it

Segment	Filtered calcium reabsorbed	Route
Distal convoluted tubule (DCT)	8–10% (fine control)	Transcellular: entry by TRPV5 channel, then carried by calbindin-D28k

- **Oxalate**: 6–14% is absorbed, unregulated, and excreted almost entirely in urine; half is dietary, half hepatic. **Ascorbic acid** is the largest endogenous source; Oxalobacter formigenes degrades oxalate.

5 · Classification

Stone	Frequency
Calcium oxalate (pure or with apatite)	**60%**
Hydroxyapatite	20%
Brushite	2%
Uric acid	≈ 7% (8–10% US, rising)
Struvite	≈ 7% (falling)
Cystine	1–3% (up to 10% children)
Triamterene, silica, 2,8-dihydroxyadenine (2,8-DHA)	< 1% each

Condition	Defect	% of formers
Absorptive hypercalciuria (incl. phosphate leak)	Increased gut calcium absorption	20–40
Renal hypercalciuria	Renal calcium leak	5–8
Resorptive hypercalciuria	Primary hyperparathyroidism (PHPT)	3–5
Hyperuricosuric calcium	Purine excess	10–40
Hypocitraturic calcium	Idiopathic, diarrhoea, distal RTA, thiazide	10–50
Hyperoxaluric calcium	Primary, enteric, dietary	2–15
Hypomagnesiuric calcium	Reduced gut magnesium absorption	5–10
Idiopathic low urine pH	Low pH	15–30
Low urine volume	Inadequate intake	10–50
Infection; cystinuria	Urease; cystine transport	1–5; < 1
No abnormality	—	< 3

> **KEY POINT**
>
> Classify by the **urinary abnormality, not the stone**: composition starts the work-up. Fewer than 3% have no abnormality. Work-up: see Ch 99.

6 · Hypercalciuria

Fasting urine and PTH reveal the source.

- It is the commonest abnormality (35–65%).

- **Definition**: > **200 mg/day** after 1 week on a 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day.
- The **Pak classification** (gut, kidney, bone) is an exam framework, not a therapy guide.

Feature	Absorptive	Renal	Resorptive
Defect	Increased gut absorption	Renal leak	PHPT
Serum calcium	Normal	Normal	**High**
PTH	**Suppressed**/normal	**High** (secondary)	High
Fasting urinary calcium	**Normal** (< 0.11 mg/dL glomerular filtrate)	**High**	High
Serum phosphate	Normal (low in phosphate leak)	Normal	Low
Sequence	Hyperabsorption suppresses PTH and raises the filtered load	Renal loss raises PTH, which raises calcitriol	PTH drives bone resorption and calcitriol, causing hypercalcaemia

- Absorptive **type I** persists on calcium restriction; **type II** normalises.
- **CYP24A1** loss stops calcitriol inactivation, causing hypercalciuria and nephrocalcinosis. In **renal phosphate leak**, hypophosphataemia raises calcitriol and gut calcium absorption.

Disorder	Gene	Features
Dent disease (X-linked)	ClC-5	Hypercalciuria, **low-molecular-weight proteinuria**, nephrocalcinosis, renal failure
Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC)	Claudin-16, -19	TAL magnesium/calcium loss
Bartter	NKCC2, ROMK, ClC-Kb, barttin; type V activating CaSR	Salt wasting, hypokalaemic alkalosis, hypercalciuria

KEY POINT

Fasting urine and PTH separate gut from kidney: in renal leak, fasting calcium and PTH are high; in absorptive hypercalciuria, fasting calcium is normal and PTH suppressed.

- **PHPT** causes ≈ 5% of stones, and stones occur in < 5% of PHPT. Suspect it if serum calcium is > **10.1 mg/dL**; repeat with intact PTH, and use ionised calcium for normocalcaemic PHPT. A **thiazide challenge** unmasks it.

Cause	Mechanism	PTH	Clue
PHPT	Autonomous PTH	High/inappropriately normal	Commonest in outpatients
Sarcoidosis	Macrophage 1α-hydroxylase	Suppressed	Corrects with **corticosteroids**
Malignancy	PTH-related protein (PTHrP), osteolysis (RANKL)	Suppressed	Commonest in inpatients

Cause	Mechanism	PTH	Clue
Glucocorticoid excess	Increased bone resorption	Raised	Cushing syndrome

7 · Hyperoxaluria

- **Definition**: > **40 mg/day**. Types: primary, enteric, dietary and idiopathic.

SAQ Primary Hyperoxaluria

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 10 Paper 2, Q3f

Definition

- Rare **autosomal recessive** errors of **hepatic glyoxylate metabolism**: glyoxylate is not converted to glycine and is oxidised to oxalate, an end product cleared only by the kidney.
- Hyperoxaluria = urinary oxalate >40 mg/day; in PH (primary hyperoxaluria) often **>100 mg/day**.

Types

	PH1 (~70%)	PH2 (~10%)	PH3 (~10%)
Enzyme (gene)	**AGT** (alanine-glyoxylate aminotransferase; **AGXT**), liver peroxisome, **pyridoxine (B6) cofactor**	GRHPR (glyoxylate reductase/hydroxypyruvate reductase)	HOGA1 (4-hydroxy-2-oxoglutarate aldolase), mitochondria
Urine marker	**Glycolate**	**L-glycerate**	4-hydroxy-2-oxoglutarate
Course	**Most severe**; ESKD (end-stage kidney disease) in 2nd–3rd decade	Milder, can reach ESKD	Childhood stones; rarely ESKD

Pathophysiology

- High urinary oxalate → calcium oxalate crystals in tubules → **nephrocalcinosis** and recurrent stones.
- **Vicious circle:** falling GFR (glomerular filtration rate) reduces oxalate clearance → plasma oxalate rises → **systemic oxalosis** (bone, retina, heart, vessels, skin).

Clinical features

- Spectrum from **infantile oxalosis with renal failure** to occasional adult stones; some diagnosed only after a transplant fails.
- Clues: childhood stones, nephrocalcinosis, family history, **calcium oxalate monohydrate** stones.

Diagnosis

`24-h urinary oxalate ↑ (child: spot oxalate:creatinine ratio) → urinary glycolate / L-glycerate / 4-hydroxy-2-oxoglutarate → genetic testing (AGXT, GRHPR, HOGA1) all normal → secondary (enteric, dietary) hyperoxaluria`

- Plasma oxalate when GFR is low; screen siblings and related living donors.

Management

- **Hyperhydration** (about 3 L/m²/day), **potassium citrate**, normal calcium intake.
- **Pyridoxine** trial in all PH1: responsive mainly with the Gly170Arg mutation; it restores AGT activity and peroxisomal targeting.
- **Lumasiran** (RNA interference against glycolate oxidase, which cuts glyoxylate supply): **ILLUMINATE-A**: 84% vs 0% reached urinary oxalate ≤1.5 × upper limit of normal. **Also know:** nedosiran.
- Stones: ureteroscopy or PCNL (percutaneous nephrolithotomy); limit SWL (shock wave lithotripsy).
- **ESKD:** intensive dialysis, then **combined liver–kidney transplant for PH1** (only the liver replaces the enzyme; a lone kidney would be re-damaged). Kidney alone usually suffices for PH2.

Diagram to draw: hepatic glyoxylate pathway showing the AGT block (PH1) and the lumasiran target (glycolate oxidase).

> **Key points to remember:**
> - PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
> - Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
> - Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

ALGORITHM

1. **Enteric hyperoxaluria**
2. Fat malabsorption leaves fatty acids in the gut lumen.
3. Fatty acids saponify calcium and magnesium, so less calcium binds oxalate and more oxalate is free.
4. Free oxalate undergoes **colonic hyperabsorption**, helped by bile salts.

- **Enteric** is the commonest acquired form: small-bowel resection or Crohn disease, jejunoileal bypass and malabsorptive bariatric surgery (not band or sleeve).
- It **needs an intact colon**; with an ileostomy, uric acid or ammonium urate stones form instead. Diarrhoea also causes hypocitraturia, hypomagnesiuria and low urine pH.
- **Dietary**: nuts, chocolate, tea, spinach, rhubarb and vitamin C. **Severe calcium restriction is counter-productive**, as less calcium binds gut oxalate.

8 · Hyperuricosuria, hypocitraturia, hypomagnesiuria

- **Hyperuricosuria** is > **600 mg/day**. Urate promotes calcium oxalate stones by **adsorbing inhibitors, "salting out" and epitaxy**; allopurinol cut recurrence (trial below).
- **Hypocitraturia** is < **320 mg/day**; **acid–base status is the main determinant**. Causes: distal RTA, chronic diarrhoea, animal protein, thiazides (through hypokalaemia), ACE inhibitors.

> **KEY POINT**
>
> Citrate excretion reads the proximal-tubular **intracellular pH**: acidosis, hypokalaemia or an acid diet make the tubule reclaim citrate. Hence potassium citrate is the treatment (see Ch 99).

- Hypomagnesiuria: **two RCTs of magnesium showed no benefit**. **Low urine pH** (< 5.5) causes uric acid and calcium oxalate stones.

9 · Renal tubular acidosis

SAQ Renal tubular acidosis (distal and type 2)

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P2 SAQ "Renal tubular acidosis" · P1 SAQ "Distal Renal tubular acidosis." · P3 SAQ "Type 2 Renal tubular acidosis?"

Definition

- **RTA (renal tubular acidosis)**: hyperchloraemic, normal AG (anion gap) metabolic acidosis from defective tubular H^+ secretion or HCO_3^- reabsorption.
- **Distal (type 1) RTA:** the collecting duct cannot secrete H^+ despite systemic acidosis. Result: **hypokalaemic, hyperchloraemic, normal anion gap metabolic acidosis** with **urine pH >6**: the only acidosis with alkaline urine.
- **Proximal (type 2) RTA:** failure of the proximal tubule to reclaim bicarbonate, causing hyperchloraemic, normal anion gap metabolic acidosis.

Comparison of types

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	H^+ secretion (α-intercalated cell)	Proximal HCO_3^- reabsorption	Aldosterone deficiency or resistance
Urine pH in acidosis	**Always >6**	<5.5 at steady state	Low
Potassium	Low	Low	**High**
Serum HCO_3^-	Low, may be very low	15–18 mEq/L (P3: 14–18 mEq/L)	Mildly low
Urine calcium	High	Normal or mildly high	Low
Urine citrate	**Profoundly low**	Near normal	–
Stones	**Common: up to 70%; calcium phosphate, nephrocalcinosis**	Uncommon	Uncommon
Causes	Sjögren, lupus, inherited	Fanconi	Diabetes, obstruction
Treatment	**Potassium citrate** 1–2 mEq/kg/day	Alkali 10–15 mEq/kg/day plus potassium	Fludrocortisone, low-potassium diet, stop offending drugs

Distal (type 1) RTA

Normal acidification and the defect

- **α-intercalated cells** secrete H^+ by apical H^+-ATPase; bicarbonate exits by the basolateral Cl^-/HCO_3^- exchanger (AE1). Normally urine pH can fall to 4.5.
- In distal RTA, potassium is secreted in place of H^+, causing hypokalaemia.

Causes

Inherited	Acquired (secondary)
SLC4A1 (AE1): mostly dominant	Sjögren syndrome, SLE (systemic lupus erythematosus)
ATP6V1B1, ATP6V0A4 (H^+-ATPase): recessive, **sensorineural deafness**	Obstructive uropathy, pyelonephritis, medullary sponge kidney
CA2: mixed proximal–distal (carbonic anhydrase II deficiency)	Hypercalciuria, hyperparathyroidism, amphotericin B

Why stones form (urological importance)

- **Stones in up to 70%**: **calcium phosphate** stones and **nephrocalcinosis**. Stones lead to diagnosis in up to half.
- **Profound hypocitraturia** (most important): acidosis increases proximal citrate reabsorption.
- **Hypercalciuria:** bone buffering releases calcium. **Fixed alkaline (high) urine pH** precipitates calcium phosphate.

Clinical features

- Stones, nephrocalcinosis, weakness (hypokalaemia), osteomalacia or rickets; children fail to thrive (growth failure).
- **Incomplete distal RTA:** normal bicarbonate, but urine pH cannot fall below 5.5 after an acid load.

Proximal (type 2) RTA

Pathophysiology

- The proximal tubule reclaims **85–90% of filtered bicarbonate** (luminal carbonic anhydrase; exit by the basolateral Na–HCO_3 cotransporter).
- **Self-limiting:** bicarbonate wasting lowers plasma HCO_3^- until the filtered load drops below the lowered threshold; the intact distal tubule then copes, giving a **steady state at about 14–18 mEq/L (P2: 15–18 mEq/L) with urine pH <5.5**.
- **Hypokalaemia** from distal sodium bicarbonate delivery; worsens on alkali.

Causes

Isolated	Generalised (Fanconi syndrome)
Carbonic anhydrase inhibitors (acetazolamide, **topiramate**)	**Cystinosis** (commonest in children), Lowe, Wilson, galactosaemia
Carbonic anhydrase II deficiency (mixed)	**Ifosfamide, cisplatin**, aminoglycosides, valproate
	Myeloma, amyloidosis, lead

- **Fanconi:** glycosuria, aminoaciduria, phosphaturia, uricosuria.

Clinical features

- Children: growth failure, **rickets**; adults: osteomalacia, weakness.
- **Stones uncommon**: urinary citrate stays near normal and urine is acid at steady state. Topiramate and acetazolamide are exceptions (alkaline urine, calcium phosphate stones).

Type 4 (hyperkalaemic) RTA

- Hyperkalaemia, low urine pH; diabetes, obstructive uropathy. Amiloride and trimethoprim cause hyperkalaemic distal RTA.

Diagnosis

- Bloods: low bicarbonate, high chloride, low potassium (types 1 and 2); urine pH >6 on a fasting morning sample in distal RTA; urine can acidify below 5.5 in proximal RTA.
- **UAG (urine anion gap) = (Na^+ + K^+) − Cl^-** reflects urinary ammonium: **positive** (low ammonium excretion) in renal causes; negative in GI (gastrointestinal) loss such as diarrhoea.
- **Ammonium chloride load** (0.1 g/kg orally, hourly urine pH for about 6 h): **failure to fall below 5.5** confirms distal RTA. The **furosemide–fludrocortisone test** is a better-tolerated alternative.
- **Bicarbonate loading:** urine pH rises; **fractional bicarbonate excretion >15%** indicates proximal RTA. Look for Fanconi features; drug history.
- 24-hour urine: low citrate, high calcium (distal); ultrasound for nephrocalcinosis.

Treatment

- **Distal RTA: potassium citrate** (about 1–2 mEq/kg/day in divided doses): corrects acidosis, hypokalaemia and hypocitraturia, reduces calcium excretion and stone formation. Sodium alkali is less suitable because sodium raises urinary calcium. Treat the cause; manage stones endourologically.
- **Proximal RTA:** treat the cause; **large alkali doses (about 10–15 mEq/kg/day)** as potassium citrate or mixed citrate, replacing potassium; thiazide; phosphate and vitamin D for rickets.
- **Type 4:** fludrocortisone, low-potassium diet, stop offending drugs.

Urological relevance

- Obstruction causes hyperkalaemic distal RTA, reversible by relief.
- Topiramate and acetazolamide mimic RTA (alkaline urine, low citrate, calcium phosphate stones).
- Potassium citrate beats sodium alkali, which raises calcium excretion.
- **Also know:** CKD (chronic kidney disease) acidosis; amiloride and trimethoprim (hyperkalaemic distal RTA); mixed type (carbonic anhydrase II deficiency); medullary sponge kidney; laxative abuse (ammonium urate stones); inherited forms (recessive with deafness); growth failure in children.

Diagram to draw: nephron with defect sites; α-intercalated cell with H^+-ATPase, AE1 and the stone mechanism; proximal tubular cell with carbonic anhydrase.

> **Key points to remember:**
>
> - Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
> - NH_4Cl load: pH fails to fall below 5.5 in distal RTA.
> - Potassium citrate is the treatment for distal RTA.
> - Type 2 (proximal): bicarbonate wasting; HCO_3^- 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
> - Fractional HCO_3^- excretion >15%; stones uncommon in type 2.
> - Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
> - Positive UAG = renal cause.

Also, from the condensed book (not in the paper answer):

- Genetics: SLC4A1 (dominant); ATP6V1B1/ATP6V0A4 (recessive, with **sensorineural hearing loss**).

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4
Defect	Distal H^+ secretion	Proximal HCO_3^- reabsorption	Aldosterone deficiency/resistance
Serum K^+	Low	Low	**High**
Urine pH	**Always > 6.0**	< 5.5 at steady state	Low
Urinary citrate	**Profoundly low**	Near normal	—
Stones	**Up to 70%**: calcium phosphate, nephrocalcinosis	Uncommon	Uncommon
Associations	Sjögren, lupus, obstruction; genetic	Fanconi	Diabetic nephropathy

- **Incomplete distal RTA**: electrolytes are normal, but urine pH cannot fall below 5.5 after an ammonium chloride load.
- In distal RTA, **profound hypocitraturia** is probably the most important stone factor.

> **KEY POINT**
>
> Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria, the recipe for **calcium phosphate** stones.

10 · Uric acid stones

LAQ Uric acid calculi: etiopathogenesis, medical management and allopurinol

Paper answer · P2

Asked as:

- Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management, — *Practice Set 9 Paper II Q2 (LAQ)*
- Allopurinol — *Practice Set 10 Paper II Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Uric acid stones form **8–10%** of stones in the USA and are increasing with obesity and diabetes. They are **radiolucent** and can be **dissolved medically**.

Etiopathogenesis

1. Why humans form uric acid stones

- Humans and apes lack hepatic **uricase**, which converts uric acid to allantoin (10–100 times more soluble). Blood and urine uric acid are therefore high.

2. Physical chemistry: pH is the key

- Uric acid is a weak acid, **pKa about 5.35–5.5**. At pH 5.5 half is free uric acid and half is urate.

- Free uric acid is poorly soluble (about 96–100 mg/L); sodium urate is about 20 times more soluble. At **pH 6.5, over 90% is ionised**.
- **No natural inhibitor** of uric acid crystallisation is known, so stone formation depends almost entirely on supersaturation.

3. The three determinants

Determinant	Causes	Importance
Low urine pH (<5.5)	Insulin resistance, obesity, type 2 diabetes, gout, high animal-protein diet, chronic diarrhoea, ileostomy	**Most important**; most formers have normal uric acid excretion
Low urine volume	Hot climate or occupation, diarrhoea, dehydration	Concentrates uric acid
Hyperuricosuria (>600 mg/day)	Purine-rich diet, gout, myeloproliferative disorders, tumour lysis, uricosuric drugs, inherited defects	Least often the driver in idiopathic cases

4. Why urine is too acid: insulin resistance

- Insulin normally stimulates **ammonia production** from glutamine in the proximal tubule and the Na^+/H^+ exchanger NHE3.
- Insulin resistance reduces ammonium excretion. Acid is then buffered as titratable acid, so urine pH falls.
- **Renal lipotoxicity:** fat accumulates in the proximal tubule, which then burns fatty acids instead of glutamine, so ammoniagenesis falls further.
- Endogenous acid production is also higher. Uric acid stones are thus a **renal manifestation of the metabolic syndrome**; about a third of stones in diabetics are uric acid.

5. Aetiological categories

- **Congenital:** HGPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency (Lesch–Nyhan), PRPP (phosphoribosylpyrophosphate) synthetase overactivity, hereditary renal hypouricaemia (URAT1, the urate transporter).
- **Idiopathic:** low urine pH without gout ("gouty diathesis").

6. Uric acid and calcium stones

- In **hyperuricosuric calcium oxalate stone disease**, urate promotes calcium oxalate by removing inhibitors, by "salting out" and by **epitaxy** (urate crystals act as a nidus).
- **Also know:** ammonium acid urate stones (laxative abuse, diarrhoea, ileostomy); theobromine (cocoa) inhibits uric acid crystallisation in vitro.

Diagnosis in brief

- Radiolucent on X-ray; CT **200–400 HU (Hounsfield units)**; **dual-energy CT** identifies uric acid.
- Urine pH persistently <5.5; serum uric acid; 24-hour urine; screen for diabetes.

Rationale of medical management

Principle: uric acid stones are a **"pH disease" more than a uric acid disease**. Raising pH turns insoluble uric acid into soluble urate. Diluting urine lowers supersaturation. Lowering uric acid production helps only when excretion is high.

1. Fluids

- Urine output **at least 2.5 L/day** lowers uric acid concentration.

2. Diet and lifestyle

- Limit non-dairy **animal protein** (it gives purines and an acid load; dietary purines provide about 30% of urinary uric acid).
- More fruit and vegetables (alkali load). Weight loss, exercise and glucose control correct the underlying insulin resistance.

3. Urinary alkalinisation (mainstay)

- **Potassium citrate 30–60 mEq/day** in two or three doses. Citrate is metabolised to bicarbonate, raising urine pH.
- **Target pH 6.0–6.5** for prevention (EAU [European Association of Urology]: 6.2–6.8); **avoid pH >7.0**, which risks calcium phosphate stones.
- **Why potassium, not sodium:** sodium alkali raises urinary calcium and can cause calcium stones. Sodium bicarbonate or sodium citrate is used if potassium is unsafe (renal impairment, hyperkalaemia).
- Patients **monitor urine pH with dipsticks**.

4. Dissolution (oral chemolysis)

- Pure uric acid stones dissolve over weeks to months. **EAU: pH 7.0–7.2**; stent or nephrostomy if obstructed or infected; monitor by imaging.

5. Xanthine oxidase inhibitors

- **Allopurinol 300 mg/day** lowers uric acid production. It is **not first-line**, because low pH, not hyperuricosuria, is the usual cause.
- Indications: hyperuricosuria with hyperuricaemia (e.g. gout), recurrence despite alkalinisation (AUA, American Urological Association), hyperuricosuric calcium oxalate stones.
- Reduce the dose in renal impairment. Risks: rash; **allopurinol hypersensitivity syndrome** (2–8% have reactions; may be fatal); Stevens–Johnson syndrome.
- **Febuxostat** is an alternative. **Rasburicase** (recombinant uricase) prevents uric acid nephropathy in tumour lysis.

Landmark trial: Ettinger (1986), randomised placebo-controlled trial: allopurinol more than halved recurrence in hyperuricosuric, normocalciuric calcium oxalate stone formers.

Allopurinol (in detail)

Introduction and mechanism

- A **purine (hypoxanthine) analogue** that inhibits **xanthine oxidase (xanthine dehydrogenase)**. This blocks hypoxanthine → xanthine → uric acid, lowering serum and urinary uric acid.
- Metabolised in the liver to **oxypurinol**, the active metabolite. It has a long half-life and is cleared by the kidney, so the dose is cut in renal impairment.

Uses in urology

Indication	Rationale
Recurrent calcium oxalate stones with hyperuricosuria (>600 mg/day) and normal urinary calcium	Urate "seeds" calcium oxalate (heterogeneous nucleation, salting out, inhibitor adsorption). **Ettinger (1986) RCT (randomised controlled trial): recurrence more than halved**; AUA (American Urological Association) recommends it
Uric acid stones	**Not first line:** low urine pH is the main cause, so alkalinise with potassium citrate to pH 6.0–6.5. Allopurinol is added for hyperuricosuria, gout or failure of alkalinisation
2,8-DHA (dihydroxyadenine) stones (APRT (adenine phosphoribosyltransferase) deficiency)	Blocks adenine → DHA; febuxostat more effective
Tumour lysis prophylaxis; gout; HPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency, PRPS (phosphoribosylpyrophosphate synthetase) overactivity	Prevents urate nephropathy

Dose

- Adult **300 mg/day** (start 100 mg, titrate); child 4–10 mg/kg/day (maximum 300 mg). Reduce for low creatinine clearance.

Adverse effects

- Rash (commonest), myalgia, gastrointestinal upset; gout flare on starting; interstitial nephritis.
- **Hypersensitivity in 2–8%**, from skin reactions to **allopurinol hypersensitivity syndrome** (fatal in up to 27%) and Stevens-Johnson syndrome. Risk is higher with high dose, renal failure, diuretics and statins. **HLA-B*58:01** carriers (some Asian populations) have higher risk.
- **Xanthine stones** (radiolucent): rare, mainly with high doses in Lesch-Nyhan syndrome.

Drug interactions

- **Azathioprine and 6-mercaptopurine:** xanthine oxidase metabolises them, so toxicity (marrow suppression) rises. Reduce their dose sharply (important after transplant).

Alternatives

- **Febuxostat:** non-purine xanthine oxidase inhibitor, no dose change in mild to moderate renal impairment. Carries a cardiovascular-death warning.
- **Also know:** rasburicase (urate oxidase) for tumour lysis.

Diagram to draw:

- Sigmoid curve of uric acid solubility against urine pH, marking pKa 5.5 and target 6.0–6.5.
- Purine pathway (hypoxanthine → xanthine → uric acid) showing the xanthine oxidase block.

Key points to remember:

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.
- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Also, from the condensed book (not in the paper answer):

- Uric acid forms about one-third of stones in type 2 diabetes.

Determinant	Causes	Importance
Low urine pH	Insulin resistance/obesity, animal protein, diarrhoea, gout	**Single most important**
Low volume	Diarrhoea, heat	Concentrates uric acid
Hyperuricosuria	Purine diet, gout, myeloproliferative disease, uricosurics	Least often the driver

- Humans lack hepatic **uricase**, so uric acid is not converted to soluble allantoin.
- The pKa of uric acid is **5.35**. Undissociated uric acid dissolves only to ≈ 100 mg/L, but at pH 6.5 > 1200 mg/L stays soluble.
- Stones are radiolucent on plain film but visible on non-contrast CT.

ALGORITHM

1. **Why the urine is acid in insulin resistance**
2. Insulin resistance and lipotoxicity lower proximal ammoniagenesis and sodium–hydrogen exchanger 3 (NHE3) activity.
3. H^+ is excreted as titratable acid instead of ammonium.
4. Urine pH stays **persistently low**, uric acid stays undissociated, and stones form.

KEY POINT

Uric acid stones are a disease of **urine pH, not uric acid**. No inhibitor exists, so raising pH above the pKa removes the driver; alkalinisation can dissolve stones (see Ch 99).

11 · Cystine and struvite stones

Both recur aggressively without lifelong prevention or complete clearance.

SAQ Cystinuria

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3a

Definition and genetics

- **Autosomal recessive** defect of proximal tubular (and jejunal) reabsorption of the dibasic amino acids **COLA: cystine, ornithine, lysine, arginine**. Only cystine is insoluble, so only

cystine forms stones.

- Transporter subunits and ICC (International Cystinuria Consortium) types:

Gene	Subunit	Chromosome	Type
SLC3A1	rBAT (heavy)	2p	A
SLC7A9	$b^{0,+}$AT (light)	19q	B
Both	—	2 and 19	AB

- Fractional excretion of cystine rises from about 0.4% to up to 100%. Incidence 1 in 1000 to 1 in 17,000; **1–3% of adult stones, up to 10% of paediatric stones**; males more severe.

Pathophysiology

- **Pure supersaturation disease**: no specific inhibitor. Solubility is pH dependent: about 300 mg/L at pH 5, 400 mg/L at pH 7, 1000 mg/L at pH 9.
- Coexisting hypocitraturia (commonest), hyperuricosuria or hypercalciuria cause mixed calcium–cystine stones.

Clinical features

- First stone at mean about 12 years; recurrent, bilateral, staghorn stones; CKD (chronic kidney disease) from repeated surgery (about a quarter of adults have eGFR [estimated glomerular filtration rate] <60).

Diagnosis

- **Hexagonal crystals** in urine (always abnormal); cyanide–nitroprusside screen; 24-hour urinary cystine (normal <60 mg/day/1.73 m^2; homozygotes often >400).
- Stone: yellow, waxy, faintly radiopaque (sulphur); **resists SWL (shock wave lithotripsy)**.

Management

- **Fluids:** urine output **≥3 L/day**, including at night.
- **Diet:** sodium ≤2300 mg/day; limit animal protein (not in children).
- **Alkalinise to pH 7.0–7.5** with potassium citrate 30–60 mEq/day; avoid >7.5 (calcium phosphate stones) and sodium alkali (raises cystine excretion).
- **Thiols** if cystine stays >250 mg/L: thiol–cystine mixed disulfide is soluble.
- **Tiopronin** (choice): 300 mg twice daily, up to 1200 mg/day.
- **D-penicillamine:** 250 mg/day, titrated; rash, proteinuria, nephrotic syndrome, pancytopenia.
- Captopril: weak evidence.
- Monitor blood count, liver tests, urine protein.
- **Surgery:** ureteroscopy or PCNL (percutaneous nephrolithotomy) with laser; nephron-sparing approach.

Diagram to draw: proximal tubule cell with rBAT/$b^{0,+}$AT transporter and hexagonal crystal.

> **Key points to remember:**
>
> - Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
> - Hexagonal crystals; target cystine <250 mg/L.
> - Urine ≥3 L, pH 7.0–7.5, then tiopronin.

SAQ Struvite stones.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3a

Definition and epidemiology

- **Infection stones:** magnesium ammonium phosphate hexahydrate ($MgNH_4PO_4 \cdot 6H_2O$), usually mixed with **carbonate apatite**.
- 5–15% of stones (falling); female:male ~2:1; highest in India and Pakistan. Risk: neurogenic bladder, spinal injury, diversion, catheters, obstruction, diabetes, elderly.

Pathogenesis

```
Urease-producing bacteria Urea + H2O -> 2NH3 + CO2 NH3 -> NH4+ (urine pH > 7.2) CO2 -> carbonate; phosphate -> PO4(3-) + Mg -> struvite; + Ca -> carbonate apatite
```

- Sterile urine has no urease, so infection is essential. Ammonia damages the GAG (glycosaminoglycan) layer; bacteria live inside the stone in a **biofilm**, so antibiotics cannot sterilise a retained stone.
- **Organisms: Proteus mirabilis** (commonest), Klebsiella, Pseudomonas, Providencia, Staphylococcus, **Ureaplasma** (easily missed). E. coli is rarely urease-positive.

Clinical features

- Recurrent UTI (urinary tract infection), fever, flank pain, haematuria; often few symptoms. Grow fast into **staghorn** calculi. Untreated: pyonephrosis, XGP (xanthogranulomatous pyelonephritis), sepsis, renal loss.

Evaluation

- Urine pH >7; culture (urine and stone cultures often differ). NCCT (non-contrast computed tomography): burden and anatomy; DMSA (dimercaptosuccinic acid) or renogram for split function. Stone analysis; 24-h urine (many have hypercalciuria or hypocitraturia).

Management

- **Complete stone clearance** is the goal: each fragment is an infected nidus.
- Culture-specific antibiotics before surgery; drain an obstructed infected kidney first.
- **PCNL (percutaneous nephrolithotomy): first line** for staghorns (AUA, American Urological Association). Add flexible nephroscopy, ECIRS (endoscopic combined intrarenal surgery) or second look. SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery) for small stones only.
- Anatrophic nephrolithotomy rarely; nephrectomy for a poorly functioning kidney.
- **Prevention:** surveillance cultures, suppressive antibiotics, fluids, correct stasis or metabolic defects.
- **AHA (acetohydroxamic acid)** 250 mg three times daily (urease inhibitor), only when surgery is exhausted. Side effects in up to 30% (haemolytic anaemia, DVT (deep vein thrombosis)); avoid if creatinine >2.5 mg/dL or pregnancy.
- **Also know:** hemiacidrin irrigation (historical); matrix stones (radiolucent, Proteus).

Diagram to draw: urease cascade; staghorn with PCNL tract.

Key points to remember:

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

Also, from the condensed book (not in the paper answer):

- **Cystinuria genes**: SLC3A1 (type A), SLC7A9 (type B), or both (type AB). It is autosomal recessive, and with **no inhibitor** cystine is soluble only at alkaline pH (≈ 1000 mg/L at pH 9).
- **Struvite**: **Proteus mirabilis** is commonest; **Ureaplasma** is easily missed. **Less than half of staghorns are infection stones**, and bacteria within the stone demand complete clearance.

ALGORITHM

1. **Urease cascade**
2. Urease splits urea into ammonia (NH_3) and CO_2.
3. Ammonia forms NH_4^+ and OH^-, raising urine pH to 7.2–8.0.
4. With magnesium, phosphate forms **struvite**; with calcium, carbonate apatite.

12 · Miscellaneous and drug-related stones

	Xanthine	2,8-DHA
Enzyme deficient	Xanthine dehydrogenase (XDH)	Adenine phosphoribosyltransferase (APRT)
Radiology	Both radiolucent, so mistaken for uric acid	
Treatment	—	Allopurinol or febuxostat (more effective)

- **Ammonium acid urate** stones are radiolucent and follow diarrhoea, ileostomy, **laxative abuse** and infection.

Route	Drug	Mechanism / stone
Direct	**Indinavir**	Insoluble at physiological pH; stones 4–13%; radiolucent even on CT
Direct	Triamterene; guaifenesin/ephedrine; silicates	Drug stones
Indirect	Acetazolamide, **topiramate**, zonisamide	Mimic distal RTA, causing calcium phosphate stones
Indirect	Loop diuretics	Hypercalciuria (up to 64% of low-birth-weight infants on furosemide)
Indirect	Steroids, vitamin D, thiazides, laxatives, cytotoxics	Hypercalciuria, hypocitraturia, urate

13 · Anatomic predisposition and pregnancy

Anomaly	Stones	Metabolic findings
Ureteropelvic junction obstruction (UPJO)	≈ 20%; children ~70× risk	Common (hypercalciuria first); recurrence after repair, often contralateral

Anomaly	Stones	Metabolic findings
Horseshoe kidney	≈ 36% adults	All tested abnormal; **hypocitraturia**; calcium stones ≈ 89%
Calyceal diverticulum	Up to 40%	Abnormal in half; UTI up to 40%
Medullary sponge kidney	Frequent, with nephrocalcinosis	Hypercalciuria, hypocitraturia; distal RTA on acid loading

KEY POINT

Anatomy sets the stage but metabolism writes the stone: anatomical stone formers still need **metabolic evaluation**, and pyeloplasty alone does not prevent recurrence (see Ch 99).

- **Pregnancy**: stones occur in the 2nd–3rd trimester. Physiological hydronephrosis affects up to 90%, **right more than left**.
- Raised GFR and placental calcitriol promote stones, while raised citrate and magnesium inhibit them; the **net risk is unchanged**, but stones skew to **calcium phosphate**.

14 · Landmark trials

Trial	Design	Result	Take-home
Borghi water trial	RCT, high fluid vs no advice, first-time idiopathic calcium formers	Recurrence 12% vs 27%	Urine volume is the one proven universal modifier (see Ch 99)
Ettinger allopurinol trial	RCT, allopurinol vs placebo, hyperuricosuric normocalciuric calcium oxalate formers	> 2-fold fewer recurrences	Allopurinol for hyperuricosuric calcium oxalate stones (see Ch 99)

96 Acute Care of Urinary Lithiasis

1 · Presentation and diagnosis

Fever with obstruction is an emergency.

- **Renal colic** is flank-to-groin pain; a UVJ stone causes penile-tip pain and frequency.
- Haematuria occurs in ~**90%**, so it is absent in >10%.
- **Fever means an infected obstructed system**, needing urgent intervention.
- Send bloods, urinalysis and a **pregnancy test** in women of reproductive age. Urine >50 WBC/hpf or nitrites suggest infection.

Modality	Role	Limitation
Non-contrast CT (NCCT)	Gold standard (sens 97%, spec 95%); shows alternative diagnoses	8–16 mSv; misses matrix and indinavir stones
Low-dose NCCT (≤**3 mSv**)	Equal for stones ≥3 mm	Weak <3 mm and in obesity
Ultrasound (US)	First line in pregnancy, children, known formers	Up to 40% still need CT
Plain X-ray (KUB)	**Avoid** for acute diagnosis	Lowest accuracy
MRI	Second line in pregnancy and children	Poor stone detection

> **KEY POINT**
>
> Unknown diagnosis: NCCT answers everything at once. Known stone: US showing hydronephrosis suffices.

2 · Pain management

- **NSAIDs** block prostaglandin release and match opioids with less vomiting.

Agent	Place	Note
Ketorolac 30 mg	**First line** (EAU, NICE)	Avoid in renal failure, GI bleed, pregnancy
IV paracetamol	Second line	NSAID barred or insufficient
Opioids	Intractable pain	**Continue the NSAID**: morphine plus ketorolac beats either
Antispasmodics; forced fluids	**Ineffective**	Fluids only for vomiting

3 · Medical expulsive therapy (MET)

SAQ Medical expulsive therapy (MET) in urinary calculi

Paper answer · P2

Asked as:

- Medical expulsive therapy in urinary calculi. — *Summer 2021 Paper II Q3d (SAQ)*
- Medical expulsive Rx for urinary calculi. — *Practice Set 11 Paper II Q3e (SAQ)*

Type: SAQ (10 marks)
(also answered in 2021-Summer Paper 2, Q3d)

Definition

- MET (medical expulsive therapy): drugs that help a ureteric stone pass into the bladder. **α-blockers** are the main agents.

Rationale: natural history

- Spontaneous passage depends on **width**, size and site:

Width	Spontaneous passage	Location	Spontaneous passage
0–3 mm	98%	Vesicoureteric junction (small stones)	92%
5 mm	65%	Distal ureter	75%
>6.5 mm	9%	Proximal ureter	48%

Mechanism

- **α1 receptors** (mainly **α1A and α1D**) are abundant in (dominate in) the distal ureter and VUJ (vesicoureteric junction). Blockade reduces spasm and the force and frequency of peristalsis (peristaltic force and frequency), so a larger urine bolus pushes the stone / builds higher pressure above the stone.
- Others: calcium-channel blockers (nifedipine); PDE5 (phosphodiesterase-5) inhibitors under study. **NSAIDs (non-steroidal anti-inflammatory drugs) relieve pain but do not aid passage.**

Drugs

- **Tamsulosin 0.4 mg daily** (most studied); **silodosin 8 mg daily** (better expulsion in meta-analyses but twice the retrograde ejaculation); alfuzosin, doxazosin.

Evidence

Study	Result
SUSPEND (Pickard, 2015): 1136 patients; tamsulosin, nifedipine or placebo	No difference in need for intervention at 4 weeks (19% vs 20%)
STONE (Meltzer, 2018): stones <9 mm (up to 9 mm)	Tamsulosin no better than placebo
Meta-analyses and a large distal-stone RCT (randomised controlled trial)	Benefit for **distal stones >5 mm**

- Stratified, the trials agree: benefit lies in **distal stones of 5–10 mm** (smaller ones pass anyway; larger rarely pass).

Indications and guideline position (AUA [American Urological Association] and EAU [European Association of Urology])

- AUA and EAU: offer MET for **ureteric stones <10 mm**, especially distal, when immediate intervention is not needed (EAU: distal >5 mm).
- **After SWL (shock wave lithotripsy)** for faster fragment clearance (to clear fragments). Children: off-label but effective.

Contraindications (need urgent drainage instead; drain urgently)

- Infection with obstruction or sepsis, solitary or bilateral obstruction, renal deterioration (impairment), refractory pain or vomiting.

Protocol

- NSAID analgesia; counsel on off-label use, dizziness, hypotension, retrograde ejaculation.
- Periodic KUB (kidney–ureter–bladder radiograph) or ultrasound follow-up; **intervene after 4–6 weeks** if not passed.

Diagram to draw: ureter with α1 receptors concentrated distally (at the distal ureter and VUJ [vesicoureteric junction]) and a stone above it (above the vesicoureteric junction).

> **Key points to remember:**
> - Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
> - SUSPEND 2015: no benefit in an unselected population.
> - Trial of passage 4–6 weeks, then intervene.

Also, from the condensed book (not in the paper answer):

- Measure **transverse width**, not length; a 6 mm wide stone passes in ≈ 33%.
- Trial of passage, with or without MET, lasts **4–6 weeks**.

4 · Urgent intervention

- **AUA indications**: obstruction with infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Stent and nephrostomy (PCN)** are equal for sepsis; stents cause dysuria and haematuria, PCN a longer stay.
- **Stent failure** is predicted by P-CUT (ureteric diameter at the stone minus stone width) ≥**7.5 mm**, stone ≥9.5 mm and multiple stones; prefer PCN then.
- Acute SWL (proximal) or URS (distal <10 mm) is done **only without infection**.

ALGORITHM

1. Give analgesia; get NCCT.
2. If infection or an AUA criterion, then drain only (stent/PCN); treat the stone later.
3. If not and stone <10 mm, then trial with or without MET for 4–6 weeks.
4. If it fails, then URS or SWL.

5 · Special populations

SAQ Stones in pregnancy

Paper answer · P2 · P3 · merged from 2

Asked as: P2 SAQ "Management of Stones in pregnancy." / "Urolithiasis in pregnancy" · P3 SAQ "Management of lower ureteric stone of size less than 1 cm in pregnancy?"

Background

- 1 in 200 to 1 in 1500 pregnancies; renal colic is the **commonest non-obstetric cause of admission**; mostly second or third trimester.
- Linked with preterm delivery, pre-eclampsia and premature rupture of membranes.
- **Why calcium phosphate stones:** raised GFR (glomerular filtration rate) and placental calcitriol cause hypercalciuria; urine is alkaline. Raised citrate, magnesium and urine flow protect, so stone rates barely change.

- **Physiological hydronephrosis** (up to 90% by third trimester, right-sided) mimics obstruction.

Evaluation

- Urinalysis, **urine culture always**, blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; **transvaginal US** for the distal ureter; a rise in Doppler **RI (resistive index)** suggests acute obstruction.
- Second line: **MRI (magnetic resonance imaging)** (stone as a filling defect) or **low-dose CT (computed tomography)**. Keep total fetal dose **≤50 mGy** (under 50 mGy).

Management algorithm

```
Symptomatic stone in pregnancy (including lower ureteric stone <1 cm) ├─ sepsis, solitary kidney, renal failure, uncontrolled pain │ → urgent stent or nephrostomy (+ antibiotics) ├─ pain controlled, no infection → observe (50–80% pass) └─ failed observation → ureteroscopy + holmium laser or temporary stent/nephrostomy until delivery
```

Conservative management

- Hydration, paracetamol, small doses of morphine (opioids).
- **NSAIDs (non-steroidal anti-inflammatory drugs) are contraindicated** (ductus arteriosus closure in the third trimester).
- **MET (medical expulsive therapy)** with tamsulosin is off-label.

Ureteroscopy

- **URS (ureteroscopy) with holmium laser:** first-line intervention and definitive treatment for a non-complex stone in any trimester; ideal for a distal (lower ureteric) stone; second trimester is the safest window.
- Neuraxial anaesthesia, lateral tilt, minimal fluoroscopy (lead shield beneath the patient, US guidance), fetal monitoring.
- Third-trimester URS may raise preterm labour risk.

Temporising drainage

- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid: change stents about **every 4 weeks** (P3: every 4–6 weeks; 4–6 weeks in series), nephrostomy about every 3 weeks; definitive treatment after delivery.

Other modalities

- **Asymptomatic renal stones:** observe until after delivery.
- **SWL (shock wave lithotripsy) contraindicated. PCNL (percutaneous nephrolithotomy) deferred** until after delivery except in expert centres.

After delivery

- Clear residual stones; **metabolic evaluation** postpartum.

Diagram to draw: the management flowchart above.

Key points to remember:

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

Also, from the condensed book (not in the paper answer):

- Resistive index difference ≥**0.06** means obstruction.
- About 30% need intervention; change stent/PCN every **4–6 weeks**. URS is safe in any trimester (ACOG 2019).

Population	Pitfall	Key point
Elderly	Atypical pain; first stone after 60 unusual	Always CT to exclude aortic/cardiac mimics; drain early
Transplant kidney	Denervated, so **painless**	Fever or renal failure; endoscopic, else percutaneous

6 · Landmark trials

Trial	Design	Result	Take-home
SUSPEND	RCT, tamsulosin vs placebo, any ureteral stone, n = 1136	Intervention at 4 weeks 19% vs 20%	Unstratified; does not abolish MET for distal 5–10 mm

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LAQ Management of upper ureteric stones

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 2 Paper 2, Q2 (SWL also answered in 2019-Summer Paper 2, Q2 and 2021-Summer Paper 1, Q1; flexible URS in 2019-Summer Paper 2, Q3b)

Introduction

- **Upper (proximal) ureter:** UPJ (ureteropelvic junction) to the upper border of the sacroiliac joint.
- Proximal stones pass least often, so many need intervention. Dangers: renal damage and sepsis.

Evaluation

- **NCCT (non-contrast CT)**: gold standard; gives size, density (HU, Hounsfield units), SSD (skin-to-stone distance) and impaction. KUB (kidney–ureter–bladder radiograph) for SWL follow-up; creatinine, urine culture, coagulation.

1. Emergency: drain first

- **AUA (American Urological Association) indications for urgent intervention:** infection with obstruction, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Obstructed infected kidney: drain only** (double-J stent or PCN, percutaneous nephrostomy; equal for sepsis); definitive treatment later. PCN suits large stones (stent fails if transverse diameter ≥9.5 mm).

2. Observation and MET (medical expulsive therapy): limits

Spontaneous passage	Rate
By size	1–4 mm 78%; 5–7 mm 60%; 8–10 mm 39%
By site	UVJ (ureterovesical junction) 92%; distal 75%; mid 60%; **proximal 48%**

- **Observation** suits small (<5–6 mm) uncomplicated stones: NSAIDs (non-steroidal anti-inflammatory drugs) first.
- **MET (alpha-blockers, tamsulosin or silodosin):** reduce spasm so a larger urine bolus pushes the stone. Benefit is clear only for **distal stones 5–10 mm**; meta-analysis shows **no benefit for proximal stones**. SUSPEND (2015) showed no overall benefit.
- Trial of passage is limited to **4–6 weeks**, with imaging.

3. Choice of active treatment

- Factors: stone (size, density, impaction), patient (habitus, anticoagulation, preference), surgeon and equipment.

EAU (European Association of Urology) recommendation for proximal ureteric stones

Size	First choice	Alternative
<10 mm	SWL or URS (ureteroscopy)	–
>10 mm	URS (retrograde or antegrade)	SWL

Size	First choice	Alternative
Large (>15–20 mm), impacted	PCNL with antegrade URS	Laparoscopic ureterolithotomy

SWL versus URS

	SWL (shock wave lithotripsy)	URS
Stone-free rate (AUA/EAU meta-analysis, approximate)	<10 mm about 90%; >10 mm about 70%	<10 mm about 80%; >10 mm about 80%
Sessions	More retreatment	Single session; higher SFR at 2 weeks, equal at 3 months
Anaesthesia	Sedation or none	Usually general
Complications	Fewer	Perforation, stent pain
Anticoagulation	Must stop	Laser URS can continue

- **SWL fails** with size >10–15 mm, impaction (CT "ring" sign), HU >1000, SSD >10 cm, cystine or COM (calcium oxalate monohydrate) stones.
- **SWL technique:** supine; 60–90 shocks/min; ramp energy; no routine pre-stent. Contraindicated in pregnancy, bleeding disorder, untreated UTI (urinary tract infection), distal obstruction, uncontrolled hypertension.

4. Ureteroscopy for proximal stones

- **Flexible URS** preferred: it follows a stone that migrates into the kidney.
- Safety wire; access sheath (12/14 Fr); tight ureter: pre-stent and return after 2–4 weeks.
- **Laser:** holmium:YAG (2120 nm) or thulium fibre laser; dusting or fragmenting; keep **≤10 W in the ureter** to avoid thermal injury.
- **Retropulsion** into the kidney: reduced by low-energy settings and antiretropulsion devices (Stone Cone, NTrap); pneumatic lithotripsy works poorly above the brim.
- **Push-back:** stone deliberately pushed into the renal pelvis with a catheter, then treated by SWL, RIRS (retrograde intrarenal surgery) or PCNL; now largely replaced by in-situ flexible URS.
- Stent only for perforation, impaction, infection or a planned second stage.

5. Antegrade approach (PCNL with antegrade URS)

- Indications: large (>15–20 mm), hard or **impacted proximal stones**, failed retrograde access, urinary diversion, transplant, concomitant renal stones.
- **Upper or mid-pole posterior calyx** gives a straight line to the UPJ; flexible nephroscope or antegrade URS down the dilated ureter; high SFR.

6. Laparoscopic (or robotic) ureterolithotomy

- Indications: large impacted stones, failed SWL/URS, where endourology is unavailable, or with concomitant UPJ obstruction (pyeloplasty).
- Steps: retroperitoneal or transperitoneal access; identify the bulge; **clamp the ureter above the stone** to prevent migration; longitudinal ureterotomy; remove; double-J stent; close with 4-0 absorbable sutures; drain.
- High single-stage SFR; risk of urine leak. **Open ureterolithotomy** is now rare.

Complications of treatment

Complication	Points
Perforation	0–4%; stop, stent about 4 weeks
Avulsion	0.06–0.5%; **proximal third most at risk**; nephrostomy and delayed reconstruction
Stricture	3–6%; image all after URS
Sepsis	About 5% after URS
SWL	Steinstrasse, haematoma, colic

Also know: pregnancy: conservative, then stent or URS (no SWL).

Recent advances

- Single-use digital scopes; pressure sensors; thulium fibre laser; robotic URS; **BWL (burst wave lithotripsy)**.

Diagram to draw: algorithm for proximal stone: drain if infected; <10 mm SWL or URS; >10 mm URS; large impacted antegrade or laparoscopic.

> **Key points to remember:**
> - Proximal stones pass in only 48%; MET does not help proximal stones.
> - <10 mm: SWL equals URS; >10 mm: URS first (EAU).
> - Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
> - Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Also, from the condensed book (not in the paper answer):

- SWL **within 48 h** of an acute obstructing stone beats delayed SWL.
- Little is gained beyond 2 SWL sessions; **forced diuresis is useless**.

1 · Background and indications

- Non-contrast CT (NCCT) is the gold standard; ultrasound misses mid-ureteric stones. US with a plain film (KUB) suits follow-up.
- Medical expulsive therapy is controversial after SUSPEND; intervene after **4–6 weeks**.
- The ladder is SWL, then ureteroscopy (URS), then percutaneous nephrolithotomy (PCNL) with antegrade URS.

Tier	Trigger
Elective	Failed trial of passage
Urgent	Unrelenting pain; obstruction-related renal dysfunction
Extremely urgent	Obstructive pyelonephritis or sepsis: **drain only**, never manipulate the stone

- Work-up: creatinine and **urine culture**; exclude coagulopathy and untreated UTI.

Modality	Antiplatelet/anticoagulant	Bridging
PCNL	Stop if feasible	Short-acting heparin if high risk
SWL	Stop if feasible	Not recommended; choose another modality
Laser URS	**Need not stop**	None; more haematuria and stent symptoms

KEY POINT

Only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen; laser URS works inside the lumen under vision.

2 · Choosing treatment

- Upper ureter ≤2 cm and distal ureter: stone-free rate (SFR) is similar, but SWL needs more sessions.
- **Impaction** (thick wall around the stone on NCCT) defeats SWL, so treat endoscopically.
- A large hard proximal stone needs PCNL with antegrade URS; if unfit, staged URS.

Modality	Size guidance	
SWL	Historical limit 2 cm; SFR falls above	**10–15 mm**
URS	No definite limit	
PCNL with or without antegrade URS	>15–20 mm, large hard proximal	

KEY POINT

Constant trade-off: SWL means fewer complications but more sessions; URS means one session with slightly more morbidity.

3 · Ureteroscopes

	Semirigid	Fibreoptic flexible	Digital flexible	Single-use digital
Image	Stable unless bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered	Smallest tip	~8.5 Fr	7.7 Fr
Cost	Cheap	Repair ~every **15** cases		Low volume

- Semirigid scopes suit the distal and mid ureter; flexible scopes suit proximal stones.
- Digital scopes are **not more durable**; keep a fibreoptic scope for a narrow ureter.

4 · Access

Most injuries happen at access, so be safe before fast.

SAQ Flexible ureteroscopy / RIRS and accessories

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P2 SAQ "RIRS (Retrograde intrarenal surgery)" / "RIRS" / "Flexible Ureteroscopy." / "Flexible URS" · P1 SAQ "Flexible Ureterorenoscopy" · P3 SAQ "Accessories for RIRS"

Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureterorenoscopy (fURS; flexible ureteroscopy) inside the kidney; it inspects and treats the whole upper tract retrogradely by the natural route.
- Marshall first reported it in 1964; a working channel and active deflection made it therapeutic. Active deflection, miniaturised digital scopes and the holmium laser made it practical.

Instrument

Feature	Details
Tip	4.5–7.5 Fr, tapered (shaft larger) (digital tips up to about 8.5 Fr)
Working channel	**3.6 Fr** (laser fibre, basket), shared by accessories and irrigation
Deflection	**≥270° both ways** (reaches lower pole); lower pole angle averages about 140° (up to 175°)
Fibreoptic	Smallest tip; image degrades as fibres break (black spots); repair about every 15 cases
Digital ("chip-on-tip", CCD [charge-coupled device] or CMOS [complementary metal-oxide semiconductor] chip)	Better image; larger tip (about 8.5 Fr); not more durable
Single-use (LithoVue, 2016)	7.7 Fr; no repair, reprocessing or sterilisation cost; cheaper below about 99 cases a year

Indications

- **Stones:** renal stones **<2 cm**; lower pole <1 cm (or 1–2 cm / larger as an alternative to PCNL [percutaneous nephrolithotomy]); patients who must stay on anticoagulants (first choice when anticoagulants cannot be stopped); failed SWL (shock wave lithotripsy); obesity; bleeding disorders; staged treatment of larger stones.
- **Others:** diagnosis and laser ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; short strictures and endopyelotomy; calyceal diverticulum; infundibular stenosis; essential haematuria; foreign bodies.

Technique

1. Sterile urine; single-dose antibiotic; α-blocker or **pre-stenting** if the ureter is tight.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **Ureteral access sheath (UAS)** (usually 12/14 Fr) over a stiff wire under fluoroscopy: lowers intrarenal pressure and temperature and allows repeated passes, but causes ureteric injury in nearly half (severe in about 1 in 8).
4. Low-pressure irrigation. **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) or thulium fibre laser**: dusting (low energy, high frequency) or fragment-and-basket. Keep power ≤10 W in the ureter and ≤25 W in the kidney (heat above 43 °C damages tissue).
5. Lower pole stones: relocate the stone to an upper calyx with a basket.
6. Inspect the ureter on withdrawal; stent unless uncomplicated. **Omit stent** if the case was uncomplicated with normal anatomy, normal renal function, no ureteric injury and no second stage (AUA [American Urological Association]). - **Difficult (unfavourable) lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum (EAU [European Association of Urology] factors against SWL; they also hinder fURS).

Accessories

Guidewires (0.018–0.038 inch, 145–260 cm)

Type	Construction	Use	Weakness
Hydrophilic	Nitinol core, slippery when wet	Bypassing impacted stones, tortuous ureter	Displaces easily

Type	Construction	Use	Weakness
PTFE (polytetrafluoroethylene)-coated	Steel or nitinol	Safety and working wire	Damages scope channel on backloading
Hybrid	Floppy hydrophilic tip, stiff PTFE shaft	Both roles	—

- **Safety wire** keeps access and allows stenting after injury (AUA, EAU).
- **Also know:** open-ended or dual-lumen catheter (pyelogram, second wire); stiff wire for the sheath; balloon dilators.

UAS (ureteral access sheath)

- Hydrophilic sheath with tapered obturator; usually **12/14 Fr** (inner/outer); tip below the PUJ (pelviureteric junction).
- **Pros:** lower intrarenal pressure and temperature; repeated passes; less scope wear.
- **Cons:** ureteric injury in nearly half, severe in about **1 in 8** (less if pre-stented); no proven fall in infection.
- **Suction sheaths and FANS (flexible and navigable suction access sheath):** aspirate dust and keep pressure low.

Retrieval devices

- **Nitinol tipless baskets** (about 1.5–2.4 Fr): kink-resistant, open flush in calyces; relocate lower-pole stones. Small baskets allow more flow but are fragile. Graspers release easily.

Laser and fibres

- Ho:YAG or thulium fibre laser; small-core fibres (about 200 µm) keep deflection and flow; **≤10 W in the ureter**.

Irrigation and antiretropulsion

- Gravity, pressure bag, hand or foot pumps (hand pumps: more fever); pressure sensors emerging.
- Stone Cone, NTrap, BackStop gel: mainly ureteric stones.

Stents

- **Double-J**; AUA: may omit after uncomplicated URS (ureteroscopy).

Outcomes

- Stone-free rate is higher than SWL for renal stones <2 cm but falls with size and lower pole location. Larger stones may need staged procedures.
- Reported rates depend on the imaging used (CT strictest).

Complications

- Stent symptoms (commonest); **sepsis about 5%**; stricture about 3%; perforation (a few per cent); bleeding or subcapsular haematoma <1%; **avulsion <1%** (nephrostomy, delayed reconstruction); scope damage.

Recent advances

- Single-use digital scopes; **thulium fibre laser** and pulse-modulated (Moses) holmium (less retropulsion, finer dust); **suction access sheaths and FANS**; intrarenal pressure sensors

(pressure-sensing scopes); robotic flexible ureteroscopy; AI (artificial intelligence) navigation.

Diagram to draw: flexible scope through an access sheath over a wire, deflected into the lower pole calyx, with a tipless basket in the lower calyx.

> **Key points to remember:**
> - 3.6 Fr channel; ≥270° deflection.
> - 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
> - Laser power ≤10 W ureter, ≤25 W kidney.
> - RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
> - Sepsis about 5%, stricture about 3%.
> - Hydrophilic wire passes; PTFE wire stays.
> - Nitinol tipless basket for calyces.

LAQ Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2015-Summer Paper 3, Q1

Introduction

- A stent keeps the upper tract draining past obstruction, injury or an anastomosis. **Zimskind 1967** placed the first indwelling splint; **Finney 1978** introduced the self-retaining **double-J (double-pigtail)** stent.

Types and materials

Feature	Details
Size	3–8.5 Fr (adult usually 6 Fr); adults 20–30 cm by height, children about age + 10 cm
Polymers	Polyurethane, silicone (least encrusting, soft); hydrocoated silicone causes less pain
Metallic (Resonance)	Resists extrinsic compression: malignant obstruction, retroperitoneal fibrosis
Others	Hybrid metal-reinforced, thermo-expandable nitinol (Memokath), biodegradable (experimental)
Designs	Single-pigtail diversion stent; open-ended catheter to identify the ureter; modified bladder ends (Polaris loop: no benefit)

Also know: anatomy: three narrowings (PUJ (pelviureteric junction), iliac vessels, intramural ureter, the narrowest at 3–4 mm). Placement: retrograde pyelogram first, stent over a wire under fluoroscopy; antegrade route if retrograde fails.

Diagram to draw: double-J stent in situ with both coils and side holes.

Indications

- Obstruction: infected obstructed kidney (stent equals PCN (percutaneous nephrostomy)), malignancy, stricture, pregnancy.
- After URS (ureteroscopy), pyeloplasty, reimplantation, transplant; injury, leak, fistula; pre-stenting a tight ureter.

Stent symptoms and complications

- About **4 in 5** have urgency, frequency, haematuria or flank pain, measured by the **USSQ (Ureteral Stent Symptom Questionnaire)**. Cause: **intravesical material irritating the bladder and reflux up the stent**. Diameter and length matter little.
- Relief: **alpha-blockers and antimuscarinics**; intraureteral placement; shortest dwell; opioid-free ERAS (enhanced recovery after surgery) analgesia.
- **Misplacement** (submucosal) and **migration** (too short).
- **Stent failure** from compression, debris or encrustation: worst with malignant or irradiated ureter, bilateral obstruction; then PCN.
- **Encrustation** rises with dwell time, infection and stone-forming urine. **Forgotten stent** causes sepsis and renal loss; keep a register.
- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear encrustation; II circular encrustation; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.
- **Encrusted stent**: never pull by force; NCCT (non-contrast computed tomography); clear the **bladder coil**, then **ureter**, then **renal coil** (flexible URS or PCNL (percutaneous nephrolithotomy)).

Ureteroscopy

	Semirigid	Flexible
Tip	7 Fr or less	4.5–7.5 Fr; 3.6 Fr channel
Use	Distal and mid ureter	Upper ureter, kidney
Deflection	None	Over 270°

- Segura's dictum: always a safety wire. Tight ureter: stent and return after **2–4 weeks**; routine pre-stenting is unnecessary.
- **UAS (ureteral access sheath)**, usually 12/14 Fr: lowers intrarenal pressure and eases repeated passes, but **severe injury in about 1 in 8**; less after pre-stenting.
- **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm)** or **TFL (thulium fibre laser, about 1940 nm)**. **Dusting** (low energy, high frequency) is faster but leaves fines; **fragment and basket** confirms stone clearance and gives stone for analysis. Keep pressure and temperature low with outflow.

Steps of URS: (1) sterile urine, antibiotic dose; (2) cystoscopy and **safety guidewire**; (3) semirigid inspection, dilate only if needed; (4) access sheath over a working wire under fluoroscopy (for flexible URS); (5) low-pressure irrigation, laser, basket; (6) inspect the ureter on withdrawal; (7) decide on stent.

- **Also know:** anti-retropulsion devices; digital and single-use scopes; URS suits patients who cannot stop anticoagulants.
- Complications: sepsis about 5% (treat positive urine culture first), stricture about 3%, bleeding under 1%, perforation (stop and stent), avulsion under 1% (PCN, delayed repair).

Stent after URS?

- **Denstedt 2001** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: stent may be omitted when anatomy and renal function are normal, there is no ureteric injury and no second stage is planned. **EAU**

(European Association of Urology) agrees.

- **Stent needed**: perforation, stricture or tight ureter, infection, planned second stage, solitary kidney, impacted stone.
- Removal at 3 days gave more unplanned visits than at 7 days; a string allows home removal. Stenting at SWL (shock wave lithotripsy) does not improve stone passage.

```
Uncomplicated URS -> no stent (or short string stent) Complicated -> double-J:
injury 2-6 wk; second stage until re-look
```

Recent advances

- Suction sheaths, intrarenal pressure sensors, biodegradable stents.

> **Key points to remember:**
> - Finney 1978 double-J; symptoms from intravesical stent and reflux.
> - Encrusted stent: bladder coil, ureter, then kidney.
> - Denstedt 2001: no routine stent after uncomplicated URS.

Also, from the condensed book (not in the paper answer):

- **α-Blockers** ~1 week pre-URS give easier access, higher SFR and less stent pain.
- Reserve the sheath for stones **above the iliac vessels**.
- For a difficult ureter, place a stent and return after **2–4 weeks**.

ALGORITHM

1. If the wire will not pass, place an open-ended catheter 1–2 mm short of the stone.
2. Do a retrograde pyelogram.
3. Use contrast hydrodissection with a hydrophilic wire.
4. Advance the catheter past the stone, then exchange for a stiff wire.

- If the catheter abuts the stone, the wire exits at 90° and causes **perforation**. If the ureter is inaccessible, **stent and stage**.
- If the scope will not enter the orifice, use **railroading** between two wires, a dual-lumen catheter or scope rotation.

Dilation method	Best for
Indwelling stent	Whole ureter narrow; gentlest, but commits to a second procedure
Serial/coaxial dilators	Long narrow segments
Balloon	Focal stenosis; inflate until the waist resolves, hold 1–2 min
Semirigid scope	Distal ureter

- **Pre-stenting is not needed** for ureteral stones; it helps only renal stones.
- The access **sheath** helps with large burden, dusting and high power, lowering intrarenal pressure (IRP) and heat.

5 · Lithotripsy

- **Basketing**: keep the stone **5–10 mm from the tip**, **never basket blind**, and when in doubt **fragment first**.

Source	Key points
Electrohydraulic	High ureteral trauma; abandoned
Pneumatic	~12 Hz; **rigid/semirigid only**; best below brim; retropulsion
Holmium laser (Ho:YAG)	Short pulse fragments; long or modulated pulse lowers retropulsion; dusting is low energy, high frequency
Thulium fibre laser (TFL)	Low peak power and long pulse give fine dust; minimal retropulsion; >40 Hz misses

*See the paper answer **LASERs in urology: holmium, thulium, KTP and newer laser technologies** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- Impacted stone: target the centre first, so the sides fall away from the wall.

Setting	Ho:YAG	TFL
Fragmentation	0.6–1.0 J × 6–10 Hz	0.6 J × 6 Hz short pulse (3.6 W)
Dusting	0.2–0.3 J × 40–80 Hz	0.2 J × 40 Hz (8 W)

- **Settings, not only the laser, decide outcome**.
- **Thermal** damage begins above **43 °C**, and **heat tracks power, not laser type**.
- Cap at **≤10 W in ureter** and ≤25 W in kidney, with continuous irrigation, because stopping it causes thermal stricture.

6 · After URS

Most need neither long antibiotics nor a stent.

- Give a **single perioperative intravenous antibiotic dose**; extend only after urosepsis this episode.
- Stents cause symptoms in ~4/5; stentless uncomplicated URS is safe.
- Stent for perforation, stricture, tight ureter, infection or a second stage.

7 · SWL for ureteral calculi

- SWL is an option at all levels.

Absolute contraindications	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot stop (for treatment and 48 h); distal obstruction; untreated UTI; uncontrolled hypertension	Skeletal deformity or obesity preventing focusing; adjacent aneurysm

- **Predictors**: size matters most, and URS is better ≥10 mm. Impaction (CT "ring" sign) is possibly the strongest.
- Cystine, brushite and calcium oxalate monohydrate resist SWL. Density >1000 HU and skin-to-stone distance >10 cm are proven for renal stones only.
- **No routine pre-stenting**. Hold aspirin, NSAIDs and anticoagulants for treatment and 48 h. Give antibiotics only with stents, struvite or recent UTI.
- Position the upper ureter supine, the mid-ureter **prone** and the distal ureter supine; **60–90 shocks/min** beats 120. α-Blockers raise SFR.

8 · Special situations

- **Pregnancy**: treat conservatively first; **SWL is contraindicated**; URS is safe with shielded fluoroscopy. Stents encrust, so change them **every 4 weeks**.
- **Diversion or reimplant**: use SWL (prone) or **antegrade flexible URS**. Transplant: both routes are safe.

9 · Complications

- **Perforation** (0–4%): stop, stent for ~4 weeks and drain any urinoma.
- **Avulsion** (0.06–0.5%) follows forceful extraction, mostly in the **proximal third**; place a **nephrostomy and repair later**.
- **Stricture** occurs in 3–6% and is silent, so image every patient.
- **Sepsis** (~5%) is linked to positive culture, pre-stenting, diabetes and long surgery.

> **KEY POINT**
>
> Nearly every serious URS complication is force without vision: blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage.

10 · Landmark trials

Trial	Design	Result	Take-home
Stentless URS	RCT, stent vs none after uncomplicated URS	Safe without stent	Stent only for named indications
TFL vs Ho:YAG	RCT, low TFL settings	TFL faster; equal ureteral SFR	Low settings optimal for TFL

98 Surgical Management of Upper Urinary Tract Calculi

1 · Choosing the approach

- The framework is the 2016 AUA/Endourological Society guideline, built on index patients and shared decision making.

Domain	Factors
Stone	Size, number, location, composition
Renal anatomy	Obstruction or stasis, hydronephrosis, ureteropelvic junction obstruction (UPJO), calyceal diverticulum, horseshoe or ectopic kidney, lower pole
Patient	Infection, obesity, habitus deformity, coagulopathy, age, hypertension, renal failure or transplant, solitary kidney, diversion, pregnancy

> **KEY POINT**
>
> Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude a modality, not surgery.

Asymptomatic renal stones

- The natural history is not benign: **~50%** have a stone event within **5 years**. **Size predicts intervention**; risk rises above 5 mm and above 10 mm.
- With **recurrent UTI and a non-obstructing stone**, remove it whatever its composition, and **culture the stone**.
- Treat residual fragments **>4 mm** after ureteroscopy (URS).
- **Treat small asymptomatic contralateral stones** during surgery on the symptomatic side.

Preoperative evaluation

- AUA: non-contrast CT (NCCT) **should** precede percutaneous nephrolithotomy (PCNL), and **may** guide the choice of shock wave lithotripsy (SWL) versus URS.
- SWL works best when density is **<900 HU** and skin-to-stone distance (SSD) is **<10 cm**.
- Everyone needs urine culture within 1 month.

Modality selection

Scenario (AUA)	Offer	Caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	SWL has fewer complications; URS has higher stone-free rate (SFR)
Total burden >2 cm	**PCNL**; staged URS an option	SWL not first-line
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line, because fragments clear poorly
Lower pole <1 cm	SWL or URS	—
Staghorn	PCNL	—
Cannot stop antiplatelet/anticoagulant	**URS**	Avoid SWL and PCNL

- EAU: for >20 mm use PCNL, then URS or SWL; for **10–20 mm, SWL and endourology are in equipoise**; for <10 mm use SWL or URS, then PCNL.

Stone morphometry scores

SAQ Stone complexity scores

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3b

Purpose

- Grade stone and kidney complexity, chiefly to **predict stone-free rate (SFR) after PCNL (percutaneous nephrolithotomy)**. Used for counselling, choosing the surgeon, auditing and comparing series.

Main scores

Score	Variables and scoring	Predicts
Guy's Stone Score (Thomas, 2011)	**Grade I:** single stone in mid or lower pole, or pelvis, simple anatomy. **II:** single upper-pole stone, or multiple stones with simple anatomy, or single stone with abnormal anatomy. **III:** multiple stones with abnormal anatomy, calyceal diverticulum stone, or partial staghorn. **IV:** complete staghorn, or any stone in spina bifida or spinal injury	SFR on KUB (kidney, ureter, bladder radiograph); not complications
S.T.O.N.E. nephrolithometry	**S**ize (area), **T**ract length (≤100 or >100 mm), **O**bstruction (hydronephrosis), **N**umber of involved calyces, **E**ssence (density, ≤950 or >950 HU [Hounsfield units]); total 5–13	SFR, blood loss, operative time, stay; needs CT (computed tomography)
CROES nomogram (Clinical Research Office of the Endourological Society)	Stone burden, location, staghorn, number, prior treatment, **annual centre volume**	SFR; only multicentre derivation, but SFR on KUB
S-ReSC (Seoul National University Renal Stone Complexity)	Number of **9 sites** involved on CT: pelvis, upper and lower major calyces, anterior and posterior minor calyces of the three groups; score 1–9	SFR; not complications
RUSS (Resorlu–Unsal Stone Score)	1 point each: stone >20 mm; lower-pole stone with infundibulopelvic angle <45°; stones in more than one calyx; abnormal anatomy; score 0–4	SFR after **RIRS** (retrograde intrarenal surgery)

Limitations

- Each score added what the previous lacked: anatomy, then CT depth and density, then centre factors, then calyceal distribution.
- Mostly predict clearance, **not complications**. Meta-analysis: heterogeneous; no single generalisable score.
- **Also know:** Triple D score predicts SWL (shock wave lithotripsy) success, not PCNL.

Diagram to draw: kidney outline showing the 9 S-ReSC sites.

Key points to remember:

- Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
- S.T.O.N.E.: five variables, score 5–13.
- CROES includes centre volume; RUSS is for RIRS.

2 · Shock wave lithotripsy

LAQ ESWL: mechanism, indications, factors affecting outcome, complications and current status

Paper answer · P2

Asked as:

- Describe indication, mechanism of stone fragmentation and current status of ESWL? — *Summer 2019 Paper II Q2 (LAQ)*
- Describe factors affecting outcome of ESWL. Discuss complications of ESWL and strategies to avoid them. — *Summer 2021 Paper II Q1 (LAQ)* (Mechanism, indications and current status are answered in 2019-Summer Paper 2, Q2.)
- Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them. — *Practice Set 13 Paper II Q2 (LAQ)* (also answered in 2021-Summer Paper 2, Q1; mechanism and indications in 2019-Summer Paper 2, Q2)

Type: LAQ (25 marks)

Introduction

- ESWL (extracorporeal shock wave lithotripsy), now SWL, breaks stones with focused acoustic shock waves generated outside the body (generated at F1 and focused on the stone at F2). First human use 1980 (Dornier HM1).
- A shock wave is a short pulse: peak positive pressure 30–100 MPa, then a negative tail of about −10 MPa.
- Stones break by spallation, shear, squeezing and **cavitation**. Cavitation also causes most tissue injury, so steps that improve outcome usually reduce complications too.
- Outcome = SFR (stone-free rate) and need for retreatment.

Components of a lithotripter

- Energy source at F1 (first focus); focusing system; coupling (water or gel); fluoroscopy and/or ultrasound to place the stone at F2. A focal zone wider than the stone improves breakage.

Energy source	Principle	Advantages	Disadvantages
Electrohydraulic (HM3)	Underwater spark; ellipsoid reflector	Most effective	Electrode wear; variable shocks
Piezoelectric	Ceramic elements on a spherical dish	Long life; little pain	Low energy; tiny focal zone
Electromagnetic (modern machines)	Coil moves a metal membrane; lens focuses	Consistent; high energy at F2	—

Mechanisms of stone fragmentation

- Each shock creates stress inside the stone. Above a critical value, cracks form; repeated shocks make them grow and join (**fatigue**).

Mechanism	How it works
Spallation	Wave reflects and inverts at the back surface of the stone; tension breaks off the distal face
Shear and tear (superfocusing)	Waves inside the stone reflect and focus; cracks form where shear peaks
Quasistatic squeezing	Wave travels faster in stone than fluid; a ring of pressure compresses the stone
Dynamic squeezing	Shear waves at the stone edges combine; strongest force in initial breakage
Cavitation	Negative phase forms bubbles; collapse erodes the surface. Excess bubbles shield the stone and injure tissue

Indications

- **Renal (AUA/EAU):** <2 cm outside the lower pole and lower pole <1 cm (SWL or URS [ureteroscopy]); EAU: 10–20 mm equal to endourology; >20 mm only after PCNL (percutaneous nephrolithotomy).
- **Ureteric:** any level, best <10 mm; emergency SWL within 48 hours works well. Patients avoiding anaesthesia and stents; children.
- **Poor choices:** >2 cm or staghorn; lower pole >1 cm (**Lower Pole I**: stone-free 21% SWL vs 91% PCNL); cystine, brushite, calcium oxalate monohydrate; infection stones.

Contraindications

- **Absolute:** pregnancy; uncorrected coagulopathy or (ongoing) antithrombotics; untreated UTI (urinary tract infection); obstruction distal to the stone; nearby arterial aneurysm (aneurysm near the stone; aortic >4 cm); severe uncontrolled (untreated) hypertension.
- **Relative:** obesity or skeletal deformity preventing targeting; pacemaker.

Factors affecting (influencing) outcome / predictors of success

1. Stone factors

Factor	Favourable	Unfavourable
Size (most important)	<10 mm	Renal >20 mm (PCNL [percutaneous nephrolithotomy] first); ureteric ≥10 mm favours URS (ureteroscopy)
Density on NCCT (non-contrast computed tomography / non-contrast CT [computed tomography])	<900 HU (Hounsfield units) clears best	>1000 HU predicts failure
Composition	Calcium oxalate dihydrate, uric acid	Cystine, brushite, calcium oxalate monohydrate (dense, homogeneous)
Impaction (ureter)	Recent, mobile	Chronic impaction (soft-tissue ring of thick ureteral wall around the stone on CT, unchanged position): may be the strongest predictor

Factor	Favourable	Unfavourable
Burden and site	Single, renal pelvis or upper calyx	Multiple sites; lower pole >1 cm

- Infection stones and stones in several sites: prefer a more efficient modality.

2. Renal anatomical factors

- **Lower pole (lower pole anatomy):** fragments drain against gravity. **Lower Pole I (Albala, 2001):** stones >1 cm, SFR **21% SWL vs 91% PCNL**. A steep infundibulopelvic angle and a long (>10 mm), narrow (<5 mm) infundibulum reduce clearance.
- Obstruction distal to the stone (UPJO [ureteropelvic junction obstruction], infundibular stenosis, calyceal diverticulum) traps fragments. Horseshoe and ectopic kidneys drain poorly.

3. Patient factors

- **SSD (skin-to-stone distance) >10 cm** (obesity) predicts failure. Skeletal deformity can prevent targeting.
- Respiratory movement of 10–15 mm moves the stone out of F2.
- **Triple D score:** stone density, stone volume and skin-to-stone distance together predict success.

4. Machine and technique factors (technique to optimise results)

1. **Energy source:** electrohydraulic (HM3) breaks best; piezoelectric has low energy and a tiny focal zone; modern machines are electromagnetic. A focal zone wider than the stone breaks better.
2. Treat infection; stop antithrombotics; control blood pressure. No routine stent: it does not improve clearance.
3. **Analgesia:** good analgesia, GA (general anaesthesia) or sedation keeps the stone in focus, because stone movement reduces breakage.
4. **Coupling:** gel as a mound, free of air; air pockets in the gel cut fragmentation by up to two-fifths.
5. **Rate 60–90 shocks/min** beats 120/min, because the bubble cloud clears between shocks (bubbles clear between shocks), so less shielding and renal injury.
6. **Ramping:** start at low energy and step up (ramping energy up also helps); this causes protective vasoconstriction.
7. Positioning (prone for mid-ureteric stones over the sacrum), continuous imaging (monitor with imaging), experienced team. Little gain beyond two sessions (more than two sessions add little).
8. **Adjuncts:** α-blockers (faster fragment clearance); potassium citrate; percussion, diuresis and inversion for lower pole fragments.

Complications

Group	Complication	Comment
Fragment-related	Renal colic; **steinstrasse** ("stone street" of fragments in the ureter); residual fragments	Steinstrasse rises several-fold with burden >2 cm^2
Infective	UTI (urinary tract infection), pyelonephritis, urosepsis (sepsis)	Infection stones, stents

Group	Complication	Comment
Renal tissue injury	Haematuria (transient, common); **subcapsular or perirenal haematoma** (minority; rarely embolisation)	Risk: age, hypertension, obesity, diabetes, coronary disease, thrombocytopenia, anticoagulation, high energy, many shocks, narrow-focus machines
Other organs	Liver, spleen, pancreas, colon / bowel, lung (children), skeletal muscle	Very rare
Cardiac	Arrhythmia during treatment	Pause or ECG (electrocardiogram) gating
Skin	Petechiae, bruising	Minor

- **Refuted long-term effects** (**not** proven): new hypertension, diabetes, decline in (loss of) renal function and infertility.

Strategies to avoid complications

1. **Select.** Absolute contraindications as above. Send large, dense, lower pole or impacted stones to URS or PCNL.
2. **Prepare.** Urine culture; treat bacteriuria. Prophylaxis only with risk factors (stent, nephrostomy, struvite, recent UTI, bowel reconstruction). Stop antithrombotics before and for the day and (about) 48 hours after. Control blood pressure; reprogramme / check pacemakers.
3. **Protect the kidney.** Slow rate; **slow ramping** (ramp over at least 4 minutes) (induces protective renal vasoconstriction and lowers haematoma risk); fewest shocks and lowest effective energy.
4. **Prevent steinstrasse.** PCNL or URS for large burdens. Routine stenting does not improve SFR, though it may reduce steinstrasse with large stones. α-blockers after SWL.
5. **Treat early.** Steinstrasse: MET (medical expulsive therapy), SWL to the lead fragment or URS; **nephrostomy if fever or obstruction**. Haematoma: bed rest, transfusion, embolisation if bleeding continues.
6. **Follow up** with KUB (kidney-ureter-bladder radiograph) and ultrasound.

Current status

- **Declining use worldwide.** URS has grown with small digital scopes and lasers, giving higher single-session stone-free rates. Newer lithotripters are also less effective than the HM3.
- **SWL vs URS:** SWL has fewer complications, no stent or general anaesthesia, lower cost, but more retreatment. Guidelines still list it first-line for small renal and ureteric stones.

Recent advances

- **Burst wave lithotripsy:** bursts of 5–100 ultrasound cycles at 4–8 MPa; less cavitation; done awake (no anaesthesia); uniform fragments.
- **Ultrasonic propulsion** moves fragments to help them pass.

Diagram to draw: ellipsoid reflector with F1 (spark) and F2 (stone), stone and gel coupling, with spallation and squeezing arrows on the stone; box of failure predictors.

Key points to remember:

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

Also, from the condensed book (not in the paper answer):

- **Absolute contraindications** are pregnancy, uncorrected coagulopathy or antithrombotics, untreated UTI, severe untreated hypertension, aortic or renal artery aneurysm (aorta >**4 cm**), obstruction distal to the stone, and inability to target the stone.
- No prophylactic antibiotic is needed if urinalysis is negative and there are no risk factors (AUA, EAU).
- The electrohydraulic HM3 (Dornier, 1980) remains the efficacy benchmark MCQ.

3 · Ureteroscopy (retrograde intrarenal surgery, RIRS)

URS is the workhorse for small stones and antithrombotic patients.

- Pre-treat bacteriuria and give a single antibiotic dose.
- **Antithrombotics**: URS is the modality for those who must continue them; antiplatelets increase admissions.

Instruments and principles

- **Safety wire**: AUA and EAU recommend it, although a multicentre study found no outcome difference. Never force a wire past an impacted stone.
- The **ureteral access sheath (UAS)** lowers intrarenal pressure but gives **no clear fall in infection**. It injures the ureter in nearly half, less if **pre-stented**.
- **Anti-retropulsion** devices reduce stone migration.
- AUA allows **stent omission** only when all four apply: normal anatomy, normal renal function, no ureteral injury and no planned second stage.

Success

"Clinically insignificant fragments" cause reintervention in up to a fifth, so report zero-fragment rates.

Complications

Complication	Rate	Risk factors / management
Sepsis	~5%	Pre-stent, positive culture, ischaemic heart disease, age, long surgery, diabetes; treat the urine first
Subcapsular/perirenal haematoma	<1%	Hydronephrosis, thin cortex, long surgery, hypertension, UTI

Complication	Rate	Risk factors / management
Stricture	~3%	UAS injury, impacted stone; ultrasound surveillance (rate underestimated)
Perforation	Few %	Stop and stent
Avulsion	<1%	Proximal ureter; immediate nephrostomy, delayed reconstruction

4 · Percutaneous nephrolithotomy

PCNL clears the most stone; safe access is the key skill.

- Landmarks MCQ: Fernström and Johansson performed the first percutaneous stone removal in 1976.
- For 1–2 cm stones PCNL gives higher SFR than SWL or URS. It is the treatment of choice for **staghorn** and most stones >2 cm.

PCNL first-line	Applications
Burden	Stones >2 cm, staghorn, matrix stones
Lower pole	>1 cm; lower pole diverticulum or infundibular stenosis
Poor retrograde route	Urinary diversion, inaccessible lower tract, prior ureteral stricture, injury or reconstruction
Coexisting pathology	UPJO; infected stones in non-ambulatory patients

Preoperative

- Stop anticoagulants with the prescriber (bridge if high risk); **low-dose aspirin is safe**.
- **NCCT is the gold standard**, showing calyceal anatomy, SSD, HU and the surrounding organs.
- For a poorly functioning kidney, get a renogram; below 10–20% function, nephrectomy may suit.
- Give single-dose, culture-directed prophylaxis. **Tranexamic acid** reduces transfusion without more thromboembolism.

Positioning

SAQ PCNL patient positioning: prone vs supine

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each." · P2 SAQ "Describe various pt positions in PCNL"

Introduction

- PCNL (percutaneous nephrolithotomy) position decides calyceal access, working space, airway safety and simultaneous retrograde access.

Indications for PCNL

- Renal stones **>2 cm**, staghorn, lower pole >1 cm, calyceal diverticulum, failed retrograde route or urinary diversion, UPJ (ureteropelvic junction) obstruction.

Positions

Position	Set-up	Advantages	Disadvantages
Prone (Goodwin, 1955)	Lithotomy for ureteric catheter, then turned prone; chest and pelvic bolsters; horizontal bolsters at nipple line and ASIS (anterior superior iliac spine) bring the kidney down more than vertical ones; arms ≤90°, face and knees padded	**Posterior calyx and upper pole** access; large working area; multiple tracts; familiar	Turning (repositioning); airway hard to reach; pressure and positioning injuries; retrorenal colon commoner prone; needs split-leg table for ECIRS
Prone split-leg / prone-flexed	Legs apart for flexible cystoscopy; hip flexion opens the costovertebral space	ECIRS (endoscopic combined intrarenal surgery) prone	Needs split-leg table; prone cystoscopy learning curve
Supine (Valdivia-Uría, 1987)	Ipsilateral flank at table edge on a lumbar bolster; arm across chest	**Airway retained**; no repositioning; single position, shorter time; **gravity drains fragments**, low intrarenal pressure; safer anaesthesia	Smaller working area; collapsed system; anterior calyces; longer tract; mobile kidney; harder upper pole
Galdakao-modified supine Valdivia (Scoffone)	Supine flank tilt with ipsilateral leg extended and other leg in lithotomy	**Simultaneous retrograde access (ECIRS)** in one position	Longer, more horizontal tract
Bart's flank-free modified supine (Bach)	Bolster under the thorax, flank free	More flank space	Fewer data
Lateral decubitus (flank)	Kidney-up flank position	Obese, kyphoscoliosis, poor respiratory reserve	Unfamiliar fluoroscopic orientation

Technique: prone vs supine

Feature	Prone	Supine
Puncture and tract	Posterior calyx puncture under fluoroscopy or ultrasound; tract more vertical; large working area	Puncture in posterior axillary line; tract more horizontal and longer; kidney more mobile; air outlines anterior calyces
Drainage	Against gravity; higher pelvic pressure	Gravity drains fragments, low intrarenal pressure
Anaesthesia	Airway access difficult; turning needed	Airway retained; no repositioning

Positioning safety (prone)

- Bolster shins so toes float; pad knees; arms ≤90° to the torso (brachial plexus); axillary and pectoral rolls; padded face and eyes.

Specific complications

- **Prone:** ocular injury, neuropraxia (brachial plexus), pressure injury, endotracheal tube dislodgement while turning.
- **Supine: colon injury** (if puncture too anterior), access failure to upper pole, kidney displacement during dilatation.
- **Both:** bleeding (transfusion now about 1%), pleural injury above the 12th rib, sepsis, bowel, spleen or liver injury.

Choosing a position

- **Prone:** standard anatomy, upper pole or multiple posterior tracts.
- **Supine variants:** obesity, cardiopulmonary disease, kyphosis, ECIRS, ileal conduit or continent channel, combined ureteric stone.
- **Spinal deformity:** individualise; pad every pressure point.
- **Evidence:** stone-free rates and complications similar; supine often shortens operating time; choice is mainly surgeon preference.

Diagram to draw: axial CT cross-section showing prone (vertical) and supine (horizontal) tracts.

> **Key points to remember:**
> - Prone: posterior and upper pole access, large working area.
> - Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
> - Horizontal bolsters drop the kidney more.
> - Similar outcomes; choose by anatomy and experience.

Access: the most important step

- The **ideal** target is a posterior calyx with a short, wide infundibulum.
- **Supracostal (above the 12th rib)** upper pole access reaches most of the system but raises chest complications. Lower pole access risks colonic injury.

Step	Eye-of-the-needle (bulls-eye)	Triangulation
Align	C-arm rotated 15–30° toward surgeon; needle seen end-on	Mediolateral in the oblique view, cephalocaudal in the anteroposterior (AP) view
Depth	Rotate back past AP	Short advances, alternating views

- Confirm entry by urine return, not contrast.
- **Ultrasound access** avoids radiation and shows adjacent organs.
- In **endoscopic combined intrarenal surgery (ECIRS)**, a ureteroscope watches the puncture, giving fewer tracts and higher clearance.

Dilatation and tract size

- Dilate with Alken metal, Amplatz serial or **balloon** dilators (one pass, 20–30 atm), keeping a safety wire.
- Multiple tracts raise bleeding and reduce function, especially in chronic kidney disease.

SAQ Mini-PCNL and newer PCNL techniques

Paper answer · P3 · P4 · merged from 2

Asked as: P4 SAQ "Miniperc" / "Newer PCNL techniques and nomenclature of miniaturised PCNL" · P3 SAQ "Mini Perc."

Definition and why miniaturise

- PCNL (percutaneous nephrolithotomy) through a **smaller tract** than standard, aiming to reduce bleeding and renal trauma. First "mini-perc" by **Jackman (1998)**, 13 Fr, in children.
- In PCNL, bleeding rises with tract size. Smaller tracts mean less bleeding and pain, at the cost of longer operating time and poorer vision.

Nomenclature (by tract or outer sheath size)

Name	Tract size	Features
Standard	24–34 Fr	Ultrasonic or pneumatic lithotripsy; graspers
Mini	15–22 Fr	First "mini-perc": Jackman 1998, 13 Fr; irrigation washes fragments out ("vacuum-cleaner" effect)
Super-mini	About 10–14 Fr	Sheath with **active suction** of fragments
Ultra-mini	11–14 Fr	3 Fr inline telescope in 11–13 Fr sheath
Micro-perc	<10 Fr	Single-step "all-seeing needle" with optics; laser only, fragments pass spontaneously

- **Problem:** names overlap and cut-offs are contested (paediatric chapter: mini <20–24 Fr), so many authors report the **actual sheath size in Fr**.
- The **working channel**, not sheath size, decides which instruments pass: the smallest systems take only a laser fibre.

Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); laser (holmium/thulium) or small pneumatic probes.
- **Fragment removal**: irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Ultrasound or fluoroscopic access; supine or prone; ECIRS (endoscopic combined intrarenal surgery) possible.
- Exit: often **tubeless** or small nephrostomy (≤10 Fr: less pain).

Newer PCNL techniques

1. **Supine PCNL and ECIRS:** Galdakao-modified supine; simultaneous retrograde flexible ureteroscopy watches puncture and dilatation; fewer tracts.
2. **Ultrasound-guided access:** less radiation; shows pleura and viscera.
3. **Single-step or balloon dilatation.**
4. **Suction sheaths:** lower pressure, faster clearance.
5. **Lasers:** high-power and pulse-modulated holmium, TFL (thulium fibre laser) for dusting through small tracts.
6. **Flexible nephroscopy** after bulk removal: higher stone-free rate.
7. **Tubeless or small (≤10 Fr) nephrostomy:** less pain.

- **Also know:** robotic or artificial intelligence-assisted puncture under study.

Indications and contraindications

- Renal stones **1–2 cm**, **lower pole stones** >1 cm (alternative to RIRS [retrograde intrarenal surgery]), failed SWL (shock wave lithotripsy) or RIRS, **paediatric stones**, calyceal diverticular stones; selected larger stones or extra staghorn tracts.
- **Contraindications**: as for PCNL (untreated infection, uncorrected coagulopathy, pregnancy).

Advantages over standard PCNL

- Less bleeding and haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage; stone-free rates non-inferior to standard PCNL.
- Children: a 24–30 Fr tract in a child equals about a 72 Fr tract in an adult, so miniaturisation matters most here.

Disadvantages

- **Longer operative time**, poorer vision, slower fragment clearance.
- **Intrarenal pressure**: high-pressure irrigation through a tight sheath, especially when the scope nearly fills it, raises pelvic pressure and causes pyelovenous backflow, **fever and sepsis**. Keep outflow free, use low-pressure irrigation, or suction sheaths.

Evidence

- Mini vs standard (meta-analysis): **non-inferior stone-free rate**, less bleeding (fewer transfusions), shorter stay, longer operation; same complications.
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears against a 24 Fr tract**.
- Mini-PCNL versus RIRS, large matched study: similar stone-free and retreatment rates; RIRS shorter stay.

Diagram to draw: scale bar of tract sizes compared (micro, ultra-mini, mini, standard).

> **Key points to remember:**
>
> - Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).
> - Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
> - Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
> - Working channel decides instruments; suction lowers pressure.

Completion and exit

- Finish with **flexible nephroscopy or URS**.

ALGORITHM

1. Confirm the collecting system and ureter are clear.
2. Withdraw the sheath under vision, with the safety wire out last.
3. Compress the tract.
4. If bleeding persists, re-inflate the balloon or place a **Kaye tamponade catheter**, and alert the anaesthetist.
5. If it still persists, arrange **selective angioembolisation**.

- Drain with a nephrostomy (≤10 Fr means less pain) or a double-J stent, or go tubeless.

Complications

- Cochrane (42 trials): **PCNL equals RIRS for major complications**, with fewer secondary procedures.
- Complications include haemorrhage (delayed bleeding from arteriovenous fistula or pseudoaneurysm needs angioembolisation), visceral injury, hydrothorax or pneumothorax, and sepsis.

5 · Special anatomy

Anatomy	Key adaptation
Horseshoe (stones in ~1 in 5)	SWL: long SSD. PCNL: access is medial and caudal; the upper pole posterior calyx is usually subcostal; more colonic injury; UPJO is frequent
Calyceal diverticulum/infundibular stenosis	Treat only if pain, UTI or haematuria. URS for upper or interpolar (laser-incise or dilate the ostium); lower pole: PCNL. Fulguration controversial
Pelvic kidney	SWL or URS <2 cm; PCNL with ultrasound, with or without CT or laparoscopic guidance
Urinary diversion	Antegrade wire access most reliable
Transplant	Always remove stones. The denervated graft presents with fever, oliguria and haematuria, not colic. Ultrasound-guided PCNL
Spinal deformity	SWL poor; URS good; PCNL equal clearance but more infection

- **Laparoscopic or robotic** surgery is only for when SWL, URS and PCNL fail or are expected to.

6 · Landmark trials

Trial	Design	Result	Take-home
Lower Pole I	Randomised controlled trial (RCT), PCNL vs SWL, lower pole >1 cm	SFR 91% vs 21%	SWL not first-line for lower pole >1 cm
Contralateral stone RCT (2022)	Remove vs observe small asymptomatic stones during surgery	Relapse 16% vs 63%	Treat contralateral stones

Trial	Design	Result	Take-home
Cochrane PCNL vs RIRS (2023)	42 trials	No difference in major complications; fewer secondary procedures	PCNL no longer "more morbid"

99 Prevention of Urinary Stone Disease

1 · Stone disease as a chronic condition

Recurrence is the disease, so prevention is lifelong.

- Up to **50%** recur within 5–10 years. The **ROKS** nomogram predicts a second stone.
- **Low urine volume** is the single most important risk; others are young age, male sex, obesity, family history and low citrate.

2 · Screening and metabolic evaluation

LAQ Metabolic evaluation in stone disease

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Discuss metabolic evaluation in urinary calculus disease." / "Describe metabolic evaluation in stone disease" / "Discuss evaluation for urolithiasis and management of bilateral renal staghorn calculi" · P1 SAQ "Metabolic Evaluation in Stone Disease"

Introduction and rationale

- Stone disease is a chronic metabolic disorder: first-time formers have about a **50% risk of recurrence within 10 years** (another source: up to 50% recur within 5–10 years).
- Evaluation finds a treatable cause and directs targeted therapy; therapy matched to 24-hour urine results reduces stone events. The **ROKS (Recurrence of Kidney Stone) nomogram** estimates individual risk.

Physicochemical basis (why we measure these)

- Stones form when urine is **supersaturated**: nucleation, growth, aggregation and retention, often on **Randall plaque**.
- In the **metastable zone** (between solubility and formation products) crystals grow on existing nuclei but do not form spontaneously; above it nucleation is spontaneous.
- **Promoters** (calcium, oxalate, uric acid, low volume) raise supersaturation; **inhibitors** (citrate, magnesium, Tamm–Horsfall protein) raise the supersaturation needed to crystallise. The 24-hour urine measures both.

Who needs evaluation

- **Screening evaluation** for every first-time former (AUA [American Urological Association], EAU [European Association of Urology]).
- **24-hour urine for high-risk and recurrent formers:** recurrent formers, **children**, solitary kidney, bilateral or large burden, GI (gastrointestinal) disease, bariatric surgery or bypass, obesity, metabolic syndrome, diabetes, primary hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, family history; uric acid, cystine or infection stones. Bilateral staghorns are high risk by definition.

Step 1: Screening evaluation (every first-time former)

Component	What is assessed
History	Colic, UTI (urinary tract infection) and recurrent UTI, previous stones, fluid and diet; GI disease or bariatric surgery, gout, sarcoidosis, family history; lithogenic drugs: calcium, vitamins C and D, acetazolamide, topiramate, steroids, triamterene, indinavir

Component	What is assessed
Blood	Creatinine, sodium, potassium (electrolytes), calcium, uric acid, bicarbonate (if RTA suspected); **PTH (parathyroid hormone)** if calcium is high or high-normal
Urine	Urinalysis: pH, crystals (hexagonal = cystine), pyuria; culture; cyanide–nitroprusside cystine screen, confirmed by quantitative 24-hour cystine
Stone analysis	Infrared spectroscopy or X-ray diffraction: uric acid points to metabolic syndrome, cystine to a genetic cause, struvite to infection
Imaging	NCCT: burden predicts future events; uric acid usually <500 HU (Hounsfield units)

- Screening is a **triage**: low risk gets general advice; high risk goes on to 24-hour urine.

Imaging in evaluation for urolithiasis

Imaging	Role
NCCT (non-contrast CT)	Gold standard: sensitivity 97%, specificity 95%. Size, site, burden, density (uric acid 200–400 HU; calcium oxalate 600–1200 HU), skin-to-stone distance
Low-dose NCCT (≤3 mSv)	Equal accuracy for stones ≥3 mm
Ultrasound	First line in pregnancy and children
KUB (plain X-ray)	Follow-up of radio-opaque stones only
CT urography	Anatomy before PCNL (percutaneous nephrolithotomy)
MAG3 (mercaptoacetyltriglycine) renogram or DMSA (dimercaptosuccinic acid) scan	Split function

Step 2: Extended metabolic evaluation (24-hour urine)

- **One or two 24-hour urines** on the usual diet, at least 4 weeks after the episode, stone-free and infection-free. Creatinine checks completeness (men about 20–25, women about 15–20 mg/kg/day).
- Measure volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine (cystine if suspected).

Parameter	Desirable per day	Abnormality and meaning
Volume	**>2.5 L**	Low volume: commonest, easiest to fix
Calcium	<250 mg women, <300 mg men (or <4 mg/kg)	**Hypercalciuria**: absorptive (PTH suppressed), renal leak (PTH high, serum calcium normal), resorptive (primary hyperparathyroidism)
Oxalate	**<40 mg**	Dietary; **enteric** (fat malabsorption, gastric bypass: fat binds calcium, freeing oxalate; often 60–70 mg); **primary** (hepatic, often >100 mg, childhood; type 1 (AGXT gene) ~70%, type 2 (GRHPR), type 3 (HOGA1))
Citrate	**>320 mg**	Hypocitraturia: distal RTA, diarrhoea, hypokalaemia, high animal protein
Uric acid	<600 mg (labs: <750 women, <800 men)	Hyperuricosuria promotes calcium oxalate stones
pH	5.8–6.2	<5.5 uric acid (metabolic syndrome); >6.5–7 calcium phosphate, RTA, infection

Parameter	Desirable per day	Abnormality and meaning
Sodium	<100 mmol	High sodium raises calcium excretion
Cystine	Negative screen	Cystinuria; keep <250 mg/L

- **Uric acid stones** depend mainly on pH: below the pKa (5.5) uric acid is undissociated and insoluble.
- **Infection stones**: urease-splitting organisms, pH >7, struvite.
- **Also know:** magnesium, potassium, phosphate; hypomagnesiuria; RSS (relative supersaturation) is calculated from these and guides therapy; medullary sponge kidney and calyceal diverticulum (stasis plus metabolic defects); 2,8-dihydroxyadenine stones (APRT [adenine phosphoribosyltransferase] deficiency).

Interpreting hypercalciuria (Pak classification)

Type	Defect	Serum calcium	PTH	Fasting urine calcium
Absorptive (20–40%)	Increased gut absorption	Normal	Normal or suppressed	Normal
Renal leak (5–8%)	Tubular calcium loss	Normal	High (secondary)	High
Resorptive (3–5%)	Primary hyperparathyroidism	High	High	High

- Fast and calcium-load subtyping is no longer advised, because it does not change treatment.

Special tests

- **Distal RTA:** urine pH >6, low citrate, calcium phosphate stones. Incomplete forms: oral **ammonium chloride load** (0.1 g/kg) or furosemide–fludrocortisone test fails to lower urine pH below 5.5.
- **Cystinuria:** autosomal recessive transport defect (cystine, ornithine, lysine, arginine); quantitative 24-hour cystine; solubility about 300 mg/L at pH 5 and 400 mg/L at pH 7, so aim for <250 mg/L.
- **Primary hyperoxaluria:** genetic testing (type 1, AGXT gene, about 70%).

Evaluation algorithm

`Stone episode → screening evaluation Low risk → general advice High risk → 24-h urine ×1–2 → targeted therapy → repeat within 6 months → yearly`

Diagram to draw: supersaturation zones (undersaturated, metastable, unstable) and the evaluation flowchart (first stone, screening, low risk with general advice or high risk with 24-hour urine and targeted therapy).

Treatment linked to abnormality

- **General**: urine >2.5 L, sodium ≤2.3 g/day, normal calcium, less animal protein, more fruit and vegetables.

Abnormality	Treatment
Low volume	Fluid for urine >2.5 L/day
Hypercalciuria	**Thiazide** (hydrochlorothiazide 25 mg twice daily, chlorthalidone, indapamide) plus potassium citrate; plus sodium restriction

Abnormality	Treatment
Primary hyperparathyroidism	**Parathyroidectomy**
Hypocitraturia, distal RTA	**Potassium citrate** 30–60 mEq/day
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; magnesium
Primary hyperoxaluria type 1	Pyridoxine, **lumasiran** (RNAi [ribonucleic acid interference]); liver ± kidney transplant
Hyperuricosuria	**Allopurinol** 300 mg/day
Uric acid stones	**Potassium citrate** to urine pH 6.0–6.5
Cystinuria	Urine ≥3 L/day; pH **7.0–7.5**; **tiopronin**
Struvite	Complete clearance; AHA (acetohydroxamic acid) if surgery impossible

- **Borghi 1996**: urine >2 L/day cut 5-year recurrence (12% vs 27%). **Borghi 2002**: low-salt, low-protein, normal-calcium diet halved recurrence versus low calcium. **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence.

Special groups and follow-up

- **Children:** all evaluated (spot ratios if not toilet-trained; consider Dent disease, monogenic causes). **Pregnancy:** defer 24-hour urine (physiological hypercalciuria).
- **Follow-up (AUA):** repeat 24-hour urine **within 6 months** of starting therapy, then yearly; blood tests for drug side effects (potassium on thiazide or citrate); re-analyse every new stone. Adherence is poor; citrus juices are only adjuncts.

Recent advances

- **Lumasiran** (RNA interference) for primary hyperoxaluria type 1; **NOSTONE (2023):** hydrochlorothiazide did not reduce recurrence; ROKS nomogram.

Staghorn calculus (bilateral staghorn question)

- **Definition:** a branched stone filling the renal pelvis and extending into the calyces. **Partial:** pelvis and some calyces; **complete:** pelvis and all major calyces.
- **Composition:** classically struvite (magnesium ammonium phosphate) with carbonate apatite. Bacterial **urease** (Proteus, Klebsiella, Ureaplasma) splits urea into ammonia, so urine pH rises above 7.2 and struvite crystallises inside a bacterial biofilm. Fewer than half of modern staghorns are infection stones; calcium phosphate, cystine and uric acid also occur.
- **Why treat:** untreated staghorns destroy the kidney and cause sepsis, XGP (xanthogranulomatous pyelonephritis) and death.

Staghorn: work-up before surgery

- NCCT with coronal and 3D views: burden, calyceal anatomy, colon and spleen.
- **Renal function:** eGFR (estimated glomerular filtration rate) and split function, because both kidneys are at risk.
- **Infection:** culture-specific antibiotics before surgery; drain an obstructed or infected kidney (nephrostomy through a calyx usable later for PCNL). Culture the stone.
- **Guy's stone score (2011)** predicts stone-free rate:

Grade	Stone
I	Single stone in mid or lower pole or pelvis, simple anatomy
II	Single upper-pole stone; multiple stones in simple anatomy; or single stone in abnormal anatomy
III	Multiple stones in abnormal anatomy; calyceal diverticulum; partial staghorn
IV	Staghorn; or any stone in spina bifida or spinal injury

- **Also know:** STONE score, CROES nomogram, S-ReSC score.

Staghorn: treatment options (AUA, EAU)

- **Goal: complete clearance**, because each fragment is an infected nidus; also clear infection and preserve function.

Option	Role
PCNL	**First line.** Multiple tracts if needed; upper-pole access reaches pelvis, lower pole and ureteropelvic junction. Flexible nephroscopy raises clearance; second-look for residue
ECIRS (endoscopic combined intrarenal surgery)	PCNL (usually supine) plus retrograde flexible ureteroscopy: fewer tracts, less bleeding, higher clearance
Staged RIRS (retrograde intrarenal surgery)	Selected partial staghorns; patients who cannot stop anticoagulants
SWL (shock wave lithotripsy)	Not as monotherapy; adjunct for residual fragments
Anatrophic nephrolithotomy (Boyce–Smith)	Incision along Brödel's avascular line, artery clamped, cold ischaemia, infundibula reconstructed. Now rare; when endoscopy cannot clear the stone in a reasonable number of sessions
Nephrectomy	Poorly functioning kidney (split below about 10–20%) that is infected or painful; XGP

Bilateral staghorns: sequence

``` Sepsis or obstruction on either side? Yes -> drain + antibiotics -> treat that side once sterile No -> treat the better-functioning (or symptomatic) kidney first

|

NCCT: stone-free? -> second look if residue

|

Other kidney after recovery (selected: same-session bilateral PCNL, or PCNL + contralateral RIRS) ```

- **Why the better kidney first:** it secures renal function early; some start with the simpler side. No trial defines the order.
- **Same-session bilateral surgery:** feasible at experienced centres if the first side was short and uncomplicated; saves an anaesthetic but risks bilateral bleeding or obstruction.
- Check creatinine after each side. Remove a non-functioning kidney only after the other side is cleared.
```

Staghorn: complications and follow-up

- PCNL risks: bleeding (transfusion about 1%; angioembolisation), sepsis, pleural, colon, spleen or liver injury.
- NCCT for stone-free status; treat residual fragments >4 mm. Surveillance cultures.
- 24-hour urine even in struvite formers, then targeted therapy; urine ≥2.5 L/day.
- **AHA** 250 mg three times daily, a urease inhibitor, only when surgery is exhausted. Avoid if creatinine >2.5 mg/dL or in pregnancy; causes DVT (deep vein thrombosis), haemolytic anaemia.

Diagram to draw: staghorn with an upper-pole PCNL tract and a retrograde flexible scope (ECIRS).

> **Key points to remember:**
> - Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
> - Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
> - Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
> - Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
> - NCCT gold standard (97%/95%).
> - Staghorn: PCNL first line; complete clearance is the goal.
> - Bilateral: drain an infected side first, then the better kidney.
> - AHA only when surgery is not possible.

Also, from the condensed book (not in the paper answer):

- A 24-hour urine is **not** part of screening.
- Lithogenic drugs include calcium, vitamins C and D, acetazolamide and steroids.

3 · General measures (all formers)

- **Fluid** is the cornerstone: aim for urine ≥**2.5 L/day**.
- **Limit nondairy animal protein and sodium**, which raise urinary calcium and lower citrate.
- Keep **calcium intake normal**, because restriction increases stones.

> **KEY POINT**
>
> Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption.

4 · Hypercalciuria: thiazides

- The AUA makes thiazides the **preferred drug** for hypercalciuria with recurrent calcium stones, **and** for recurrent calcium stones with a normal evaluation.
- **Hypokalaemia causes intracellular acidosis, which leads to hypocitraturia**. So add **potassium citrate**, and check potassium 1–2 weeks after starting or changing the dose.
- With amiloride, **never add potassium**. Avoid triamterene, as it forms stones.
- Side effects include gout, diabetes and sexual dysfunction.

Target	Drug and dose	Key side effects
Calcium	Hydrochlorothiazide 25 mg twice daily; chlorthalidone 25–50 mg; indapamide 2.5 mg	Hypokalaemia, hyperuricosuria, hypocitraturia
Citrate	Potassium citrate 30–60 mEq/day	Gastrointestinal (GI) upset, hyperkalaemia
Uric acid	Allopurinol 300 mg	Rash, myalgia, hypersensitivity
Oxalate	Magnesium; pyridoxine 100 mg	Diarrhoea (magnesium); ataxia, numbness (pyridoxine)
Cystine	Tiopronin; D-penicillamine 250 mg titrated	See thiol table
Urease	Acetohydroxamic acid 250 mg three times daily	Heaviest burden (below)

5 · Hyperoxaluria

- Stone risk rises continuously with urinary oxalate. Avoid large portions of spinach, nuts, chocolate and tea.
- **Enteric** hyperoxaluria follows malabsorption or Roux-en-Y gastric bypass; urine oxalate is >40 mg/day. Unabsorbed fat binds calcium, which frees oxalate for colonic absorption.
- Treat with **calcium up to 1 g/day with meals**, very high fluid, **liquid** potassium citrate, magnesium and a low-oxalate diet.

*See the paper answer **Primary Hyperoxaluria** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- Lumasiran is approved for PH1.

6 · Hypocitraturia

- AUA offers **potassium citrate** 30–60 mEq/day for recurrent calcium stones with low citrate, **and** for those with no abnormality.
- Sodium citrate does not lower urinary calcium. Use caution in renal impairment.
- Over-alkalinising to **pH ≥7** creates **calcium phosphate** risk.
- **Alkali citrate, not total citrate, matters**.

7 · Uric acid stones: a pH disease

*See the paper answer **Uric acid calculi: etiopathogenesis, medical management and allopurinol** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- Give allopurinol only for hyperuricaemia with hyperuricosuria, or failed alkalinisation.

8 · Cystine stones

ALGORITHM

1. **Dilute**: urine ≥**3 L/day**, including at night; sodium ≤2300 mg/day; moderate protein (not in children).
2. Aim for cystine <**250 mg/L**.
3. If not enough, **alkalinise** with potassium citrate to pH **7.0–7.5**.
4. If still forming stones, add a **thiol** (tiopronin).

- pH >7.5 risks calcium phosphate, and sodium alkali raises cystine excretion.
- Thiols form a soluble cystine–drug disulfide.

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol; largely replaced	**Thiol of choice**; fewer stop it	Weak evidence
Dose	250 mg/day, titrate	300 mg twice daily, up to 1200 mg/day	—
Toxicity	Rash, fever, arthralgia, pancytopenia, nephrotic syndrome	Asthenia, GI upset, rash, joint aches	—

9 · Infection (struvite) stones

- **Clear the stone completely**, because every fragment harbours bacteria.
- **Acetohydroxamic acid (AHA)** is a urease inhibitor. AUA: use it only for residual or recurrent struvite after surgical options are exhausted.
- AHA toxicity is ~30%, with **deep venous thrombosis** the most severe. It is contraindicated if creatinine is >**2.5 mg/dL** or in pregnancy.
- Most struvite formers are also hypercalciuric or hypocitraturic, so do a 24-hour urine.

10 · Monitoring

ALGORITHM

1. Start therapy.
2. Repeat the 24-hour urine within **6 months**.
3. Then check urine and blood (for side effects) at least yearly.
4. If a new stone forms, **re-analyse** it.
5. Image periodically.

11 · Landmark trials

Trial	Design	Result	Take-home
Borghi 1996	Randomised controlled trial (RCT), high fluid, calcium oxalate formers	5-year recurrence 12% vs 27%	Urine >2 L/day prevents stones
Borghi 2002	RCT, hypercalciuric formers	Low protein and salt with normal calcium halved events vs low calcium	Never restrict calcium

Trial	Design	Result	Take-home
Ettinger 1988	Double-blind RCT, chlorthalidone	~90% fewer stones	Thiazides prevent calcium stones
Barcelo 1993	RCT, potassium citrate	Fewer new stones at 3 years	Citrate prevents recurrence
ILLUMINATE-A	Phase 3 RCT, lumasiran, PH1	84% vs 0% reached near-normal urine oxalate	RNA interference for PH1

100 Lower Urinary Tract Calculi

1 · Bladder calculi

- Bladder stones are ~5% of urinary stones. Types are **migrant**, **primary** (endemic) and **secondary**.

	Primary (endemic)	Secondary
Patient	Children; peak **2–4 years**; boys far more than girls	Men >60 with outlet obstruction
Cause	Cereal diet low in animal protein and phosphate	Bladder outlet obstruction (BOO), foreign body, catheter, infection
Composition	Ammonium acid urate, calcium oxalate, uric acid	**Struvite**
Recurrence	Solitary; rarely recurs	Recurs unless the cause is removed

- Foreign bodies act as a nidus: a migrated intrauterine device, sutures, balloon fragments or hair during clean intermittent catheterisation (CIC).

> **KEY POINT**
>
> Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child.

- Adults present with **terminal haematuria** and lower urinary tract symptoms.

2 · Management of bladder stones

The route, not the stone, dictates modality.

Option	Indication / key point
Chemodissolution	Adjunct only; for catheter encrustation
Extracorporeal shock wave lithotripsy (ESWL)	Artificial sphincter, penile prosthesis, neobladder, high-risk patient; works poorly in continent reservoirs
Transurethral cystolithotripsy	Most popular; holmium laser of choice; risks urethral stricture
Percutaneous cystolithotomy	Difficult urethra (bladder neck reconstruction or closure); spares the urethra
Open cystolithotomy	Small-capacity bladder, stricture, no safe endoscopic route

3 · Special situations

Treat the cause along with the stone.

- **BOO** is present in 45–79%. **A bladder stone secondary to BOO is an absolute indication for BPH surgery**, but **prove BOO** first.
- **Diversion or augmentation** stones arise from acidosis-driven hypercalciuria, enteric hyperoxaluria, urease producers and **mucus**.
- A **continent pouch: go percutaneous**, which protects continence. A conduit with stomal stenosis needs stone clearance and stomal revision.
- **Spinal cord injury**: risk peaks **3 months** after injury; **CIC beats indwelling catheters**. Stones ≤4 cm get transurethral laser; larger ones percutaneous or open.

- **Transplant**: **remove all exposed suture**. **Children**: use percutaneous cystolithotripsy for large stones.

4 · Urethral calculi

- Urethral stones are 0.3–1% of stones, in males. Most are **migratory**, and the commonest site is the posterior urethra.
- They cause **acute painful retention**; anterior stones give a palpable mass. ~98% are radio-opaque.

Site	Treatment
Posterior	**Push back** into the bladder, then laser; else treat in situ
Anterior	**No pushback**; in situ holmium laser; urethrotomy for large stones
Fossa navicularis	Forceps; ventral meatotomy
Diverticular / stricture	Extraction with diverticulectomy or stricture repair, same sitting or staged

5 · Unusual sites

- **Preputial** stones follow phimosis; treat by circumcision and send the **excised skin to histopathology**.
- **Prostatic** stones are linked to BPH, **not cancer**.
- **Seminal vesicle** stones cause haematospermia; **TRUS is most accurate**.

54 Management of Paediatric Kidney Stone Disease

1 · Epidemiology

Children carry stone disease for life.

- Early onset means stones before **24 years**. About 50% recur within 3–5 years.

Feature	Children	Adults
Sex ratio	Female predominance	Male (historically 3 : 1)
Calcium oxalate (CaOx)	≈ **80%**, mostly idiopathic, on Randall's plaques	75–80%
Calcium phosphate (CaP) / uric acid	CaP commoner and uric acid rarer, because urine is more alkaline	≈ 5% each
Lifetime stakes	Chronic kidney disease (CKD), hypertension, low bone mineral density (BMD); cumulative radiation	Shorter

- **High-risk** groups include bladder augmentation (≈ 1%/year) and malabsorption.

> **KEY POINT**
> Childhood is the bone-accrual window, so seek low BMD early.

2 · Diagnosis and imaging

ALGORITHM

1. If a stone is suspected, do grey-scale **ultrasound (US)** with colour Doppler.
2. If US is negative but suspicion is high, or there is hydronephrosis, then do **low-dose non-contrast computed tomography (NCCT)**.
3. If US is negative and suspicion is low, stop.

- US shows the **twinkle artefact** (sensitivity > 70%, specificity > 95%) but overestimates size by 1–2 mm. NCCT is ≈ 100% accurate.

3 · Metabolic evaluation: every child

An abnormality raises recurrence up to **fivefold**, so do a **full evaluation after the first stone**.

- **History**: ketogenic diet, vitamin D, furosemide, topiramate, prematurity.
- **Examination clues**: dysmorphism suggests Williams syndrome, rickets suggests Dent disease or HHRH, and tetany suggests FHHNC.
- Do **stone analysis** first: cystine, struvite, 2,8-dihydroxyadenine and xanthine are each diagnostic.
- With hypercalcaemia and hypercalciuria, check for **CYP24A1** loss.

Metabolite	Normal (child)
Calcium	< **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y)
Oxalate	< 40–50 mg/1.73 m^2/day

Metabolite	Normal (child)
Citrate (lower limit)	Hypocitraturia is < 128 mg/g creatinine (male), < 300 (female)
Cystine	< 50–60 mg/1.73 m²/day
Uric acid	< 815 mg/1.73 m²/day; spot < 0.56 mg/dL glomerular filtration rate

KEY POINT

Never use adult limits: infants excrete more calcium, so the school-age 0.21 ratio overcalls them. Citrate's abnormality is too little.

4 · Urinary abnormalities

Hypercalciuria: commonest (30–50%)

Serum calcium	Causes
High	Hyperparathyroidism, CYP24A1, hypervitaminosis D, immobilisation, Williams
Low	Hypoparathyroidism; autosomal dominant hypocalcaemic hypercalciuria (calcium-sensing receptor gain)
Normal	Prematurity, loop diuretics, topiramate, ketogenic diet, Dent, Bartter, FHHNC, distal renal tubular acidosis (dRTA), HHRH; if none, **idiopathic**

Hyperoxaluria

*See the paper answer **Primary Hyperoxaluria** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- Only 10–30% of primary hyperoxaluria type 1 (PH1) respond to pyridoxine.
- Thiols and lumasiran are the only drugs approved for paediatric stone prevention.

Others

- **Hypocitraturia**: incomplete dRTA needs an acid-load test. **Cystinuria** may present in infancy with a staghorn.

Monogenic disease	Inheritance	Signature
Dent (CLCN5)	X-linked	Low-molecular-weight proteinuria, nephrocalcinosis, CKD
Bartter	Recessive	Salt wasting, hypokalaemic **alkalosis**, hypercalciuria
FHHNC (CLDN16/19)	Recessive	Hypomagnesaemia, tetany, nephrocalcinosis
dRTA	Dominant/recessive	Hypokalaemic **acidosis**, hypocitraturia, nephrocalcinosis

5 · Management principles

- Tamsulosin is off-label (≤ 6 weeks).
- The residual fragment threshold is **4 mm on US**: 2 mm on CT plus the ≈ 2 mm US overestimate.
- Give antibiotic prophylaxis for all URS and PCNL, but for SWL only with infection risk; limit it to ≤ 24 h.

Scenario (AUA/Endourological Society 2016)	Recommendation
Uncomplicated ureteric ≤ **10 mm**	Observation with or without an α-blocker
Ureteric, failed or unlikely to pass	URS or SWL (strong)
Before URS	No routine pre-stenting
Renal ≤ **20 mm**	SWL or URS
Renal > 20 mm	PCNL or SWL (with stent or nephrostomy); authors prefer PCNL
Before PCNL	Low-dose CT (strong)
Asymptomatic non-obstructing renal	Surveillance with US
Open/laparoscopic/robotic	Only with anatomic abnormality

6 · Ureteroscopy

The small ureter is the main limit, so never force access.

- Clearance is ≈ 90%, but worse under 6 years and in the lower pole.

ALGORITHM

1. Confirm a sterile culture; place in lithotomy; do cystoscopy.
2. Do a retrograde pyelogram and place a safety wire.
3. Calibrate the orifice with an 8/10 Fr coaxial dilator.
4. **If it will not pass, then place a 4.6–4.8 Fr stent for ≥ 14 days and return later.**
5. If it passes, do URS.
6. Stent if the case was long, needed many passes, or caused trauma or oedema.

- **No balloon or active orifice dilatation**; rely on passive stent dilatation.

7 · Shockwave lithotripsy

- SWL equals URS for small stones.
- Favourable: < 10 mm, non-lower-pole, < 1000 HU. Unfavourable: cystine, brushite, infundibulopelvic angle > 45°.
- Use < 20 kV and ≤ 2000–3000 shocks (< 2000 in infants); **80/min is better than 120/min**.
- Absolute contraindications are bleeding diathesis, untreated UTI and pregnancy.

8 · Percutaneous nephrolithotomy

- PCNL is first-line for stones > 20 mm. A 24–30 Fr tract in a child equals ≈ 72 Fr in an adult, so **miniaturise**.

ALGORITHM

1. Prevent hypothermia; place a ureteric catheter.
2. Position prone (latex precautions in myelomeningocele).
3. Puncture a posterior calyx under US or fluoroscopy.
4. Dilate serially to an 18 Fr sheath (24 Fr for staghorn).
5. Fragment and check with a flexible scope.
6. Place a nephrostomy (7 Fr infant to 16 Fr adolescent).

9 · Secondary prevention

- **Fluid**: at least maintenance; > 2–2.5 L/day in adolescents; 1.5–2 L/m² in cystinuria or PH.
- **Sodium** < 2–3 mEq/kg/day. **Calcium is not restricted**, because it binds gut oxalate.
- Target urine pH > 6.5 for uric acid and > 7 for cystine; avoid > 6.5 in calcium formers.

Drug	Indication	Child dose / caution
Hydrochlorothiazide	Hypercalciuria despite salt restriction	1–2 mg/kg/day; for hypokalaemia add potassium citrate
Potassium citrate	Hypocitraturia, uric acid, cystine	2–4 mEq/kg/day; CaP risk if pH > 6.5
Tiopronin	Cystinuria failing fluid and alkali	Fewer adverse effects than penicillamine
Allopurinol	HPRT, PRPS, adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx	4–10 mg/kg/day; can cause xanthine stones
Pyridoxine	PH1 (trial in all)	10–30% respond
Lumasiran	PH1	RNA interference against glycolate oxidase

PH1 with end-stage disease needs liver–kidney transplantation.

10 · Landmark trials

Trial	Design	Result	Take-home
ILLUMINATE-A	Randomised, lumasiran vs placebo, PH1	Urinary oxalate lowered by 65%	RNA interference therapy for PH1

125 Genital and Lower Urinary Tract Trauma

1 · Principles

- Key decision: **immediate reconstruction vs deliberate delay**. Testicular rupture is repaired at once; pelvic fracture urethral injury (PFUI) gets a suprapubic tube now and urethroplasty later.
- Imaging: ultrasonography for scrotum and equivocal penile fracture; computed tomography (CT) cystography for bladder; retrograde urethrography (RUG) for urethra.

2 · Penile fracture

SAQ Fracture penis

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 15 Paper 2, Q3c

Definition and mechanism

- **Penile fracture** is a tear of the **tunica albuginea** with rupture of the corpus cavernosum, in the **erect** penis.
- Abnormal bending of the erect penis causes an abrupt rise in intracavernosal pressure beyond tunical strength → **transverse tear**.
- The tunica (inner circular, outer longitudinal layers) is **thinnest ventrolaterally**, so most tears are ventral or lateral, proximal to mid shaft.
- **Causes:** intercourse (commonest; rear-entry), masturbation, rolling over; self-inflicted "taqaandan" (Middle East); collagenase injection for Peyronie disease.
- **Bilateral tears in about 10%**; urethral injury about 20% in the West vs about 3% in Asia and the Middle East.

Clinical features

- **Cracking or popping sound → pain → rapid detumescence → swelling and discoloration.**
- **"Eggplant deformity"** when Buck's fascia is intact; if torn, haematoma spreads to scrotum, perineum and suprapubic area.
- Penis deviates **away from the tear**; tunical defect may be palpable.
- **Urethral injury:** blood at meatus, haematuria, retention.

Investigations

- **Diagnosis is clinical** (AUA [American Urological Association] Urotrauma Guideline).
- **Ultrasound** for equivocal cases (localises the tear); **MRI** (magnetic resonance imaging) if still equivocal (100% negative predictive value). Cavernosography is discouraged.
- Urethra: **intraoperative flexible cystoscopy** (AUA and EAU [European Association of Urology] preference) or RUG (retrograde urethrogram).
- **"False fracture":** dorsal vein or artery rupture.

Management

- **Immediate surgical repair** (less curvature, ED [erectile dysfunction] and nodules than conservative care). 1. Incision: ventral penoscrotal or lateral over the defect; subcoronal degloving if site unclear (preputial ischaemia risk). 2. Evacuate haematoma; find the tear (artificial erection helps). 3. Close the tunica with **interrupted 2-0/3-0 absorbable sutures**; avoid debriding erectile tissue. 4. Urethra: partial → close over catheter; complete → tension-free repair; catheter 2–3 weeks.
- Abstinence about 1 month. Delay up to 7 days does not worsen outcome. After collagenase injection without urethral injury, conservative care is acceptable.

Complications

- About 20%: nodule, curvature, ED, painful erection, infection, stricture or fistula.

Diagram to draw: cross-section of penis showing tunical tear, Buck's fascia and haematoma spread.

> **Key points to remember:**
> - Snap, detumescence, eggplant deformity.
> - Urethral injury about 20% (West); suspect if bilateral.
> - Immediate exploration and interrupted absorbable tunical closure.

Also, from the condensed book (not in the paper answer):

- Bilateral tears (~10%) carry more urethral injury, and absence of blood at the meatus does not exclude it.
- Collagenase ruptures mostly occur after the 14-day observation window; manage them conservatively unless the urethra is injured.

ALGORITHM

1. Explore promptly. Through a ventral penoscrotal (or small lateral) incision, evacuate haematoma and find the tear (artificial erection).
2. Irrigate; close the tunica with interrupted **2-0/3-0 absorbable** sutures.
3. If the urethra is injured, repair it.

- A breached Buck's fascia gives a butterfly haematoma, and a negative MRI reliably excludes a fracture. Avoid deep corporal ligation and excessive debridement.

3 · Penetrating, bite, amputation, zipper and strangulation

- **Gunshot**: explore immediately without waiting for imaging. Debride, give antibiotics and close the tunica.
- Urethral injury occurs in **15–50%** of penile gunshots, so check with RUG or intraoperative dye. Viable tissue is repaired primarily; high-velocity tissue loss needs suprapubic diversion and staged repair.
- **Dog bites**: irrigate, debride and close **primarily**, with amoxicillin–clavulanate and tetanus and rabies cover. **Human bites** present late and infected, so **leave them open** and give amoxicillin–clavulanate or moxifloxacin.

Penile amputation

- It is usually self-mutilation, so **always get psychiatric consultation**. **Microsurgical replantation** is preferred; otherwise repair urethra and corpora macroscopically.
- Transport the part **"double bagged"**: saline gauze in a sterile bag, inside an outer bag of ice slush.
- Erections return in >50% with either technique, but microvascular repair gives less stricture and sensory loss. If the part is lost, formalise the stump; total loss needs phalloplasty.

ALGORITHM

1. Suprapubic cystostomy; debride.
2. Tunica with 2-0 absorbable; urethra over a catheter with 4-0.
3. Anastomose dorsal artery with **11-0**, dorsal vein with **9-0** and dorsal nerve with **10-0** nylon.
4. Provide skin cover.

- **Zipper** injury (children): under penile block, try mineral oil once; if that fails, cut the slider's **median bar** with bone cutters.
- **Strangulation**: in children think of hair tourniquet and **child abuse**. Escalate from lubrication to the **string technique**, then to cutting tools with saline cooling in theatre.

4 · Testicular injury

- About 80% of penetrating scrotal wounds also involve neighbouring structures, including the **femoral vessels**.
- Consider rupture in every blunt injury; haematoma size does not predict it. **Ultrasonography** sensitivity is only ~64–75%, so an **equivocal scan must not delay exploration**.
- A non-palpable testis after trauma suggests **dislocation**, commonest to the superficial inguinal pouch (50%). Trauma does not exclude torsion.

AAST testis grade	Injury
I	Contusion/haematoma
II	Subclinical laceration of tunica albuginea
III	Tunical laceration with <50% parenchymal loss
IV	Major laceration with ≥50% parenchymal loss
V	Total destruction or avulsion

ALGORITHM

1. Explore; remove necrotic and extruded tubules.
2. Close the tunica with small absorbable sutures.
3. If the defect is large, use a tunica vaginalis flap or graft (never synthetic); drain.

- Explore a **significant haematocele** (up to 80% hide rupture) and an **intratesticular haematoma** (delayed orchidectomy 15%).
- Always explore penetrating injury; ligate an injured vas and reconstruct later.
- Salvage is **>90%** if repaired within **72 h**; non-operative care leads to infection, atrophy and late orchidectomy.

5 · Genital skin loss

- Fournier gangrene debridement is the commonest cause. Treat it with antibiotics, resuscitation and **repeated debridement**, then negative pressure wound therapy; hyperbaric oxygen is unproven.

	Penis	**Scrotum**
Graft	**Thick, non-meshed** split-thickness skin graft (STSG) 0.012–0.015 inch, because meshed graft contracts	**Meshed** STSG after local flaps (exudate escapes, so better take)
Key step	Excise residual subcoronal skin, which otherwise causes lymphoedema; never reuse avulsed skin	Suture the testes together to avoid a bifid scrotum; ≤50% loss closes primarily
Note	Graft is insensate, but sexual function is preserved	Thigh pouches not recommended

- Graft 5–7 days after the last debridement, once the wound is granulating, under a bolster or vacuum dressing.

6 · Bladder injury

Intraperitoneal tears need surgery; most extraperitoneal tears heal on a catheter.

LAQ Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Classification and treatment of lower track trauma in # pelvis" (also P2 SAQ "Pelvic # assessment and management" and P2 SAQ "Early surgical intervention in urethral injuries") · P3 LAQ "Describe early and late management of PFUD injuries"

PFUD is taken as pelvic fracture urethral distraction defect, i.e. PFUI (pelvic fracture urethral injury). Lower urinary tract trauma means bladder and urethral injury in pelvic fracture.

Introduction

- Pelvic fracture accompanies **83–95% of bladder ruptures**, yet only **5–10% of pelvic fractures** injure the bladder. Urethral injury (PFUI) occurs in about **10% of male** and up to 6% of female pelvic fractures.
- 80–94% have other major (associated) injuries, which cause the deaths; long-term disability comes from wrong management.

Mechanism

- **EPR (extraperitoneal rupture):** pelvic disruption (shearing) tears the bladder at its fascial attachments, or bone spicules lacerate it.
- **IPR (intraperitoneal rupture):** a blow on a **full bladder** bursts the dome, the weakest part, covered by peritoneum.
- **PFUI:** the posterior urethra is fixed to the pubis (puboprostatic ligaments, perineal membrane), so displacement shears it, usually at the **bulbomembranous junction**. In prepubertal boys the tear may extend into the prostate and bladder neck.
- Highest risk: straddle fracture of all four pubic rami, vertically unstable fractures, symphyseal diastasis.
- The sphincter is usually avulsed but stays functional, so most men stay continent after repair.
- The gap fills with scar: an obliterative **distraction defect**, not a true stricture.

Classification

Bladder: AAST (American Association for the Surgery of Trauma) grades I–V

Grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal wall laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Practical: contusion, EPR (simple or complex), IPR, combined.

Urethra: Goldman I–V

Type	Injury
I	Posterior urethra stretched, intact
II	Disruption above the urogenital diaphragm (perineal membrane)
III	Disruption of membranous urethra through the urogenital diaphragm, extending below the membrane into the bulb (commonest)
IV	Bladder neck injury extending into urethra (IVa: bladder-base injury mimicking IV)
V	Pure anterior urethral injury

- **Campbell-Walsh practical grading: partial injury; minor distraction** (a cystoscope passes retrogradely into the bladder); **major distraction** (a cystoscope cannot pass into the bladder). Practically partial vs complete disruption. Stricture risk and need for urethroplasty rise from partial to major.
- Anterior urethra: contusion, incomplete or complete.

Assessment and clinical features

- **ATLS (Advanced Trauma Life Support)** resuscitation first; associated injuries take priority; pelvic binder and stabilisation.
- Bladder: **gross haematuria (nearly all)**, suprapubic pain, inability to void, free fluid.
- Urethra: **triad: blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate and butterfly perineal haematoma are unreliable.
- **Vaginal and rectal examination**; vaginal examination in every female.

Investigations

- **Blood at the meatus mandates RUG (retrograde urethrogram) before any catheter** (AUA [American Urological Association]). Method: 16 Fr catheter tip in fossa navicularis, 3 mL balloon, penile traction, oblique position, 20 mL contrast, film during injection. Extravasation with bladder filling means partial injury; without filling, complete. Females: urethroscopy.
- **Cystography:** absolute indication: **gross haematuria + pelvic fracture** (rupture in ~29%). Relative: gross haematuria alone; microscopic haematuria with fracture.
- **CT cystography:** retrograde fill with **350 mL** of 2–4% contrast; no drainage film needed.
- Plain cystogram: scout, full-bladder and **drainage films** (posterior leaks otherwise missed).
- Clamping the catheter after IV contrast is inadequate.

- **EPR:** flame-shaped pelvic contrast. **IPR:** contrast outlines bowel loops.

Bladder injury: management

- **Uncomplicated EPR:** 22 Fr urethral catheter 2–3 weeks; cystogram before removal; repair if leak persists beyond 4 weeks.
- **Immediate repair:** IPR; penetrating injury; bladder neck injury; bone fragments in the bladder; rectal or vaginal injury; poor drainage or clots; open pelvic fixation or laparotomy for other reasons.
- IPR: open or laparoscopic two-layer closure; inspect ureteric orifices; cystogram at 7–10 days. At laparotomy, repair EPR from inside the bladder, leaving the pelvic haematoma undisturbed.
- Suprapubic tube adds nothing over urethral catheter (AUA). Antibiotics only around IPR repair.
- Rectal or vaginal injury: separate the walls, no overlapping suture lines, interpose tissue.

Posterior urethra (PFUI): early management

- Key decision: **immediate repair versus deliberate delay**. Clean, localised injuries are repaired at once; PFUI is drained and reconstructed later.

```
Blood at meatus → RUG Partial → one gentle catheter attempt (confirm position) Complete → SPC (suprapubic catheter): percutaneous 16 Fr or open (if bladder repair needed) ± brief primary realignment if stable (flexible cystoscopy, antegrade + retrograde) Bladder neck/rectal injury → immediate repair Delayed (2–3 months) → up-and-down-o-gram → perineal excision and primary anastomosis (corporal split → pubectomy → rerouting)
```

1. ATLS resuscitation; associated injuries take priority.
2. **SPC (suprapubic cystostomy): the standard of care.** Percutaneous (peel-away sheath, 16 Fr Foley; avoid small "punch" tubes) or open through a lower midline incision, which allows bladder inspection (needed if bladder repair is required). - SPC is safe with pelvic plating (AUA Urotrauma guideline); place and tunnel it high, away from hardware.
3. **Primary realignment (PR)** (AUA Urotrauma option): a brief attempt in a stable patient, acutely or within days, by combined antegrade and retrograde flexible cystoscopy over a guidewire (avoid prolonged attempts). Catheter out at 4–6 weeks; keep the SPC and remove it 7–14 days later if voiding well.
4. Partial tear: one gentle catheter attempt allowed.
5. **Immediate open exploration/repair** only for **bladder neck or rectal injury**, or when laparotomy is needed anyway. - **Also know:** female urethral injuries are usually repaired early; in boys tears may extend into the bladder neck.

SPC versus primary realignment

	SPC + delayed urethroplasty	**Primary realignment**
Advantage	Simple; haematoma untouched	Partial tears may heal; aligned ends ease repair
Drawback	Complete injuries need urethroplasty	Most still form a 1–2 cm stenosis
Evidence	Prospective data: **no significant difference** in need for urethroplasty	Modern PR does not add ED (erectile dysfunction) or incontinence

Anterior urethra: early management

Injury	Early management
Contusion, incomplete straddle tear	Urethral catheter alone
Major straddle injury	SPC (standard); primary realignment for lesser injuries; delayed anastomotic urethroplasty
Low-velocity gunshot or stab	**Immediate primary repair:** debride minimally, spatulate, tension-free anastomosis over a catheter
High-velocity or close-range shotgun	SPC, staged repair
Penile fracture with urethral tear	**Immediate repair:** partial tear oversewn; complete tear debrided and anastomosed; catheter 2–3 weeks

- Penile fracture: urethral injury in about 20% of Western and 3% of Asian series; **intraoperative flexible cystoscopy** before catheterisation. Limit spongiosal debridement.

Late management of PFUI

Timing: delayed repair at 2–3 months (P3: after **3 months**), once the scar is stable and the patient is ambulant.

Evaluation

- **"Up-and-down-o-gram":** cystogram plus RUG while the patient tries to void, to measure the gap length; antegrade flexible cystoscopy via the SPC tract if the bladder neck stays closed / does not open. Bladder neck appearance does **not** predict its later function.
- **MRI (magnetic resonance imaging)** for complex cases: gap length, prostatic displacement (cephalad displacement predicts pubectomy).

Endoscopic: DVIU (direct vision internal urethrotomy) only for short, incomplete stenosis. "Cut-to-the-light" or repeated DVIU for gaps ≥1 cm fails (false passages).

Perineal excision and primary anastomosis (EPA; delayed anastomotic urethroplasty): treatment of choice

1. Exaggerated lithotomy (under 5 hours); sounds via meatus and SPC tract.
2. Midline or lambda incision; mobilise the bulbar urethra up to the penoscrotal junction (not beyond: chordee).
3. Divide the urethra at the scar; excise **all** fibrosis until healthy prostatic mucosa appears over the sound (identify the prostatic apex).
4. Spatulate both ends to 30 Fr; 8–10 interrupted 4-0 absorbable sutures; tension-free mucosa-to-mucosa anastomosis over a silicone catheter.

Progressive perineal approach (each step shortens the path to bridge a longer gap):

Step	Manoeuvre
1	Bulbar urethral mobilisation (enough for most)
2	Separation of the corporal bodies (corporal splitting, intracrural space)
3	Inferior wedge pubectomy (Kerrison rongeur)
4	Supracrural rerouting of the urethra around one corpus

- **Abdominoperineal or transpubic approach:** urethrorectal fistula, failed repair, bladder neck injury, children.
- **Postoperative:** voiding study at 21–28 days; cystoscopy at about 6 months.

Outcomes and complications

- Urethroplasty success above 95%. Restenosis 5–15%; often DVIU-treatable because the scar is already excised.
- **ED** up to 82%, mostly from the injury (cavernous nerves, pudendal arteries; nerve, arterial, venous). **Incontinence <4%** (uncommon; bladder neck).
- Missed IPR: ascites, acidosis, azotaemia (peritoneal reabsorption).
- Fistula, abscess, neurogenic voiding dysfunction; follow up ≥1 year.
- **Also know:** chordee, penile shortening, ejaculatory dysfunction, lithotomy neuropathy, penile Doppler before repair in impotent men.

Diagram to draw: pelvis with EPR (flame-shaped) vs IPR on cystogram; RUG showing partial vs complete posterior urethral disruption; sagittal pelvis showing the bulbomembranous gap and the four progressive perineal steps.

> **Key points to remember:**
> - Blood at meatus: RUG before catheter.
> - Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
> - EPR: catheter 2–3 weeks; IPR: immediate repair.
> - PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
> - Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
> - Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
> - Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
> - ED mostly from the injury; incontinence <4%.

Also, from the condensed book (not in the paper answer):

- Underfilling the bladder gives false-negative cystograms.
- Bone spicules in the wall prevent healing, so repair them; a ureteric orifice injury needs a stented reimplant.

AAST bladder grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal laceration ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Intraperitoneal leakage causes chemical peritonitis and, once reabsorbed, **acidosis and azotaemia**.

KEY POINT

Immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, laparotomy for other reasons.

7 · Posterior urethral injury (PFUI)

Drain now and reconstruct later.

*See the paper answer **Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management** (Section 11, Ch 125).*

Also, from the condensed book (not in the paper answer):

- PFUI complicates ~10% of male and up to 6% of female pelvic fractures, mostly **straddle (four-rami)** and unstable fractures.
- ED is predicted by symphyseal diastasis, lateral displacement and a long gap. Endoscopic "cut-to-the-light" fails for gaps ≥1 cm.

AAST urethra grade	Injury
I	Contusion — blood at meatus, normal urethrogram
II	Stretch — elongation without extravasation
III	Partial disruption — extravasation, contrast reaches bladder
IV	Complete disruption, <2 cm separation
V	Complete, >2 cm separation or extension into prostate/vagina

ALGORITHM

1. If there is blood at the meatus, do RUG before any catheter.
2. Place a 16-Fr catheter in the fossa navicularis and inflate the balloon with 3 mL.
3. Apply penile traction, position oblique, inject 20 mL contrast and image during injection.

- In females, examine the vagina in every pelvic fracture. A suprapubic tube is safe with pelvic hardware if tunnelled high in the midline.

8 · Anterior urethral injury

- Most are **straddle injuries** crushing the bulbar urethra against the pubis. Contusion or incomplete tear needs a catheter; major injury a suprapubic tube.
- Later, the 1.5–2 cm obliterated bulbar segment is treated by **excision and primary anastomosis** (>95% success). Urethrotomy through an obliteration is futile.

9 · Landmark trials

Trial	Design	Result	Take-home
Multi-institutional PFUI study	Prospective, suprapubic tube vs early endoscopic realignment	No difference in stenosis or urethroplasty need	Suprapubic tube remains standard; realignment optional, brief

62 Paediatric Genitourinary Trauma

1 · Renal injury: how the child differs

- The kidney is the commonest site of paediatric genitourinary (GU) trauma. Most injuries are **blunt** and grade 1–3; half have other injuries.
- Congenital anomalies such as ureteropelvic junction (UPJ) obstruction are commoner than in adults.

Feature	Child	Adult
Protection	Less perirenal fat, weaker wall, kidney below the ribs	Better cushioned
Blood pressure	Maintained until ≈ **25–30% volume loss** ; hypotension is late	Falls earlier
Haematuria	May be absent (up to 2/3 of grade ≥ 2 injuries)	Classic trigger
Imaging decision	**Mechanism-led**; keep radiation as low as reasonably achievable (ALARA)	Led by haematuria and hypotension

2 · Indications for imaging

Setting	Indication (American Urological Association [AUA])
Stable blunt	Gross haematuria
Stable blunt	Microscopic haematuria with systolic blood pressure (SBP) < 90 mm Hg
Mechanism / examination	Rapid deceleration; significant flank blow; rib fracture; flank ecchymosis
Penetrating	Any injury to abdomen, flank or lower chest, even without haematuria

- Triggers: fall > **10 feet (3 m)**, high-speed collision, flank blow; rib, spine, femur or pelvic fracture. All four markers together (haematuria, hypotension, associated injury, mechanism) give > **98% sensitivity**.
- Paediatric hypotension (1–10 years) is SBP < 70 + (2 × age) mm Hg.

> **KEY POINT**
>
> Trap: "no haematuria, so no imaging". In children both adult markers fail, so mechanism drives the decision.

3 · Imaging studies

- Renal ultrasound (US) detects grade 3–5 without radiation and suits screening, but **contrast computed tomography (CT)** is preferred. Add a **delayed (excretory) phase** if a leak is suspected.

4 · Grading

American Association for the Surgery of Trauma (AAST): grades 1–3 are **low** and 4–5 are **high**.

Grade	1989 AAST	2011 revision
1	Contusion or non-expanding subcapsular haematoma	Subcapsular haematoma/contusion

Grade	1989 AAST	2011 revision
2	Perirenal haematoma; cortical laceration < 1 cm	Same, within Gerota fascia
3	Laceration > 1 cm, no collecting-system injury	Same, into medulla
4	Laceration into collecting system; main artery/vein injury with contained haemorrhage	Collecting system with extravasation; segmental vessel injury; pelvis laceration or **UPJ disruption**; shattered kidney
5	**Shattered kidney**; hilar avulsion	Main artery/vein laceration, avulsion or thrombosis

KEY POINT

Grades 1–3 differ only in depth (1 cm) and never involve the collecting system; entry makes grade 4. Shattered kidney is grade 5 (1989) but grade 4 (2011).

High-risk CT criteria (grade 4–5)

Criterion	Predicts
Perirenal haematoma > 3.5 cm adult, > **2.2 cm child**	Bleeding needing angioembolisation
Vascular contrast extravasation; medial laceration	Bleeding
> 25% parenchyma devitalised	Bleeding and open surgery
Medial urinary extravasation	UPJ disruption
Extravasation with no contrast in distal ureter	Urine leak needing intervention

With 0–1 factor, < 10% need intervention; with 2–3, > 50% do (grade 4a vs 4b).

5 · Non-operative management (NOM)

- NOM is the **default at every grade**, including grade 4–5 and stable penetrating injury, provided the child is stable.

ALGORITHM

1. Confirm stability; grade the injury.
2. Bed rest, with serial haemoglobin and abdominal examination.
3. When gross haematuria resolves and counts are stable, mobilise.

- Grade 1–2 needs only **23-hour observation**. Repeat imaging: AUA CT at 48 h (grade 4–5), EAU US at 2–3 days; but **symptoms, not imaging, predict complications**.
- Grade 3 (devitalised) to 5: ≈ **25%** need embolisation, ≈ **15%** urinoma intervention, ≈ 5% exploration.

6 · Complications during NOM

- **Urine leaks** occur only in grade 4–5, and up to **85%** resolve. A urinoma shows **flank pain, ileus and low-grade fever**.
- If no contrast reaches the distal ureter, do retrograde pyelography (RPG). Escalate from **stent and Foley catheter** (AUA first line, ≈ 90%; out at 4–6 weeks) to percutaneous drain, then nephrostomy, then open repair.

Bleeding	Persistent	Delayed
Mechanism	Vessels fail to tamponade	Fistula or pseudoaneurysm ruptures as the haematoma liquefies
Timing	From injury	Typically **1–2 weeks** (up to 1 month)
Stable	Allow 2–3 transfusions, then embolise	Early embolisation unless mild
Unstable	Urgent angioembolisation	

- Superselective embolisation succeeds in > 80%.

7 · Operative management

- Operate for **instability from renal bleeding**, pulsatile or expanding haematoma, or failed embolisation. A stable non-pulsatile haematoma found at laparotomy is **left alone**.
- A **one-shot pyelogram** (2 mL/kg, film at 10–15 min) confirms a **contralateral kidney**.

*See the paper answer **Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management** (Section 11, Ch 94).*

Also, from the condensed book (not in the paper answer):

- Early vascular control did not lower nephrectomy in the one randomised trial.

ALGORITHM

1. Confirm a contralateral kidney.
2. Open Gerota fascia; control the vessels; debride; close the collecting system, then parenchyma over a bolster.
3. Drain. If the injury is penetrating, inspect the ureter.

- **UPJ disruption** follows deceleration, often without haematuria; CT shows **medial extravasation, empty distal ureter and often no laceration**. Repair early if stable; otherwise nephrostomy and repair at 2–3 months.

8 · Follow-up

- Scarring: grade 1–2 none, grade 3 ≈ 50–60%, grade 4–5 ≈ 100%, so do **US at 2–4 months**.
- **Hypertension** is rare and renin-mediated, so use renin–angiotensin blockade. Check blood pressure yearly after grade 3–5.

9 · Ureteral trauma

- Most are penetrating. **Haematuria is absent in 25–50%** and ≈ 10% are missed at laparotomy, so diagnose by **delayed-phase CT** or RPG.

AAST grade	Injury
I	Contusion/haematoma, no devascularisation
II / III	< 50% / ≥ **50%** transection
IV	Complete transection, < 2 cm devascularisation
V	Avulsion, > **2 cm** devascularisation

- Found within 1 week: repair if stable; ligate and place a nephrostomy if unstable. Found after 5–7 days: stent first; if that fails, nephrostomy and reconstruction at 6–12 weeks.

*See the paper answer **Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury** (Section 11, Ch 94).*

Also, from the condensed book (not in the paper answer):

- **Avoid transureteroureterostomy in children**, because a long life puts the other kidney at risk.

10 · Bladder injury

Feature	Child	Adult
Bladder position	**More intraperitoneal**, so isolated injury is commoner (≈ half have no fracture)	Pelvic organ
Pelvic fracture	Half as frequent, so it implies higher energy and more urethral injury	Commoner
Posterior urethral injury	Prostatic; **bladder neck injury commoner**	Bulbomembranous

- **Gross haematuria or retention** calls for **retrograde cystography** with retrograde urethrography (RUG); delayed-phase CT may miss a rupture.
- AAST: I contusion; II extraperitoneal < 2 cm; III extraperitoneal ≥ 2 cm or intraperitoneal < 2 cm; IV intraperitoneal ≥ 2 cm; V bladder neck or trigone.
- Extraperitoneal tears get a catheter, but are repaired for bone fragments, hardware, rectal or vaginal tear, or bladder neck injury. Intraperitoneal tears are **always repaired**. Keep the catheter 1–2 weeks, then do a cystogram.

> **KEY POINT**
>
> Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor and many need secondary procedures.

11 · Urethral injury

Grade/type	AAST	Goldman
I	Contusion: meatal blood, normal RUG	Posterior urethra stretched, intact
II	Stretch, no extravasation	Rupture above intact urogenital diaphragm
III	Partial: extravasation, contrast reaches bladder	Rupture through diaphragm into bulbar urethra
IV	Complete, no bladder filling, < 2 cm gap	Bladder rupture extending into proximal urethra
V	≥ 2 cm gap or into prostate/vagina	Anterior urethral injury

- Do **RUG before any catheter**; if a catheter will not pass, place a **suprapubic tube**.
- Realignment is a brief attempt only; EAU advises against immediate urethroplasty. **Bladder neck injury is repaired at once.**

ALGORITHM

1. Divert with a suprapubic tube and wait several months.
2. Assess with cystogram and RUG, with or without endoscopy.
3. Do transperineal excision and primary anastomosis; if there is tension, add pubectomy.
4. Catheter for 2–3 weeks; cystogram; remove the suprapubic tube.

- Restricture ≈ 10%, incontinence ≤ 20%, and erectile dysfunction up to half (gap > 2.5 cm).

12 · Genital injury

KEY POINT

Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Involve social services before discharge.

- Penis AAST: I skin; II Buck fascia; III avulsion, glans/meatus laceration, defect < 2 cm; IV partial penectomy or ≥ 2 cm; V total penectomy.
- **Circumcision** is the commonest cause; the **Mogen clamp** causes the worst (glans amputation).
- Testis AAST: I contusion; II subclinical tunica tear; III < 50% loss; IV ≥ 50% loss; V destruction. If US shows tunica discontinuity, haematocele or focal flow loss, **explore and close the tunica**.
- Hymenal, vaginal or perianal injury means **penetration: evaluate for abuse**; a **normal examination does not exclude abuse**.

★ Putting it together

A · Renal colic

ALGORITHM

1. Give a non-steroidal anti-inflammatory drug (NSAID; ketorolac 30 mg) unless there is renal failure, bleeding or pregnancy. Confirm with non-contrast CT, or ultrasound in pregnancy and children.
2. With infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration or refractory pain, **decompress now** (stent or nephrostomy; equal). Never lithotripsy while infected.
3. **Otherwise**, if the stone is <10 mm, try passage with or without an α-blocker (distal 5–10 mm) for 4–6 weeks; if that fails, do ureteroscopy (URS) or shock wave lithotripsy (SWL).

B · Stone operation

Stone	First choice	Notes
Ureteric, any level	URS or SWL	URS clears more, SWL has fewer complications; hard proximal >15–20 mm: consider percutaneous nephrolithotomy (PCNL)
Renal <2 cm, not lower pole	SWL or URS	SWL works best <900 Hounsfield units and skin-to-stone <10 cm
Lower pole >1 cm	PCNL or URS	SWL not first line (Lower Pole I)
Burden >2 cm, staghorn	PCNL	Staged URS an option

KEY POINT

SWL absolute contraindications: pregnancy, coagulopathy or antithrombotics, untreated infection, uncontrolled hypertension, arterial aneurysm, and obstruction distal to the stone. No routine stent after SWL or uncomplicated URS.

C · Prevention

ALGORITHM

1. **All patients:** urine ≥2.5 L/day, normal calcium, low sodium and less animal protein.
2. **If high risk**, give therapy guided by 24-hour urines, repeated within 6 months.

Abnormality	Treatment
Hypercalciuria	Thiazide and potassium citrate (hypokalaemia lowers citrate)
Hypocitraturia	Potassium citrate
Uric acid	Alkalinise to pH 6.5–7; allopurinol only for hyperuricosuria/gout or failed alkali
Cystine	≥3 L/day fluid, pH 7.0–7.5, then tiopronin
Struvite	Complete clearance; check metabolism

D · Renal trauma

ALGORITHM

1. **Image** with delayed-phase CT for gross haematuria, microhaematuria with shock, deceleration or penetrating injury.
2. If stable with grade I–IV (or selected V), manage **non-operatively** (>90% succeed). Embolise a contrast blush; stent a symptomatic leak.
3. If unstable or expanding, **explore**; a one-shot pyelogram (2 mL/kg, 10 min) confirms the other kidney.

E · Bladder and urethra

Injury	Management
Extraperitoneal bladder rupture	22 Fr catheter, cystogram at 2–3 weeks; repair if at laparotomy, bone spicule, or unhealed at 4 weeks
Intraperitoneal bladder rupture	Surgical repair
Pelvic fracture urethral injury	Suprapubic tube (or brief realignment) now, delayed urethroplasty later
Penile fracture	Prompt tunical repair; check the urethra
Testicular rupture	Immediate repair

Retrograde cystography must fill to ≥350 mL.

F · Ureteric defects

Defect	Repair
2–3 cm, upper/mid	Spatulated ureteroureterostomy
Lower ureter, 6–10 cm	Psoas hitch reimplant
10–15 cm	Boari flap
Longer	Transureteroureterostomy, ileal ureter (creatinine <2 mg/dL), autotransplant, nephrectomy

G · Children

KEY POINT

Stones: ultrasound first; full metabolic evaluation after one stone. Renal ≤20 mm: SWL or URS; >20 mm: PCNL. **Trauma:** image by mechanism (fall >3 m, deceleration); a perirenal haematoma >2.2 cm predicts embolisation.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ Renal autotransplantation

Paper answer · P1 · P3 · merged from 2

Asked as: P3 SAQ "Auto-transplantation of Kidney?" / "Role of auto transplant of kidney in Urology." / "Renal Autotransplantation" · P1 SAQ "Autotransplantation"

Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in **1963** for proximal ureteric injury.
- **Principle:** bringing the kidney to the bladder replaces the ureter without bowel; the bench gives a cold, bloodless field.

Role (indications)

Group	Examples
Ureteric loss (commonest)	Long-segment loss or stricture; long iatrogenic or traumatic loss; repeated failed repairs (**delayed setting only, never acute**)
RPF (retroperitoneal fibrosis)	When ureterolysis fails or is impossible and function is satisfactory
Tumour	Total ureterectomy for UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy in a solitary kidney; left RCC (renal cell carcinoma) needing suprarenal caval resection (new drainage for the right kidney)
Vascular	Complex branch renal artery aneurysm, stenosis or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome
Pain	Selected loin pain–haematuria syndrome

- **Preferred over ileal ureter** when the contralateral kidney is absent or poor, or bowel is unsuitable (creatinine >2 mg/dL, inflammatory bowel disease, radiation enteritis).
- **Contraindications:** poor renal function, severe iliac atherosclerosis, unfit patient.

Procedure

`CT (computed tomography) angiography, renogram, ureteric study → harvest as for live donor nephrectomy (open/laparoscopic/robotic): maximal vessel length, preserve ureteric blood supply → cold perfusion on bench ± ex vivo repair → Gibson incision, extraperitoneal iliac fossa → renal vein to external iliac vein (end-to-side); renal artery to external iliac (end-to-side) or internal iliac (end-to-end) → drainage: pyelovesicostomy (± Boari flap), ureteroneocystostomy, or native ureter left in continuity → stent, catheter, drain`

- Laparoscopic harvest plus Gibson implantation: less analgesia, faster recovery; **robotic autotransplantation** emerging.

Complications and outcome

- Arterial or venous **thrombosis**, bleeding, urine leak, anastomotic stricture, lymphocele, infection.
- **Renal unit loss 5–10%** (lower in transplant centres); major (Clavien–Dindo ≥III) complications about 15–20%.
- **Long-term success about 90%.**

Follow-up

- Doppler ultrasound early; renogram and blood pressure later.

Alternatives

- Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti, buccal ureteroplasty, renal descensus.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

> **Key points to remember:**
> - For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
> - Delayed setting only; bench surgery for complex vascular or tumour cases.
> - Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
> - Unit loss 5–10%; success about 90%.

SAQ Newer methods of stone analysis

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Newer methods of stone analysis" / "Newer techniques of stone analysis" · P4 SAQ "Newer methods of stone analysis and their significance in management of calculus disease."

Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former (every first stone); **repeat if stones recur on treatment** (evolving metabolic condition), or on early recurrence after clearance.
- Composition points to the cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.
- Composition can change over time (e.g. calcium oxalate converting to brushite), so re-analyse recurrent stones.

Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**, so it cannot separate COM (calcium oxalate monohydrate, whewellite) from COD (dihydrate, weddellite), or apatite from brushite.
- High error rates; misses rare stones (2,8-DHA, drugs); needs a large sample.

Newer methods

Method	Principle	Comment
FTIR (Fourier-transform infrared) spectroscopy	Each compound absorbs a characteristic infrared spectrum	Commonest reference method; tiny sample; detects proteins and drugs
XRD (X-ray diffraction)	Crystal lattice diffracts X-rays in a unique pattern	Gold standard for crystals; misses amorphous material
Polarising microscopy	Crystal optics (birefringence)	Cheap; needs an experienced observer (observer-dependent)
Morpho-constitutional analysis (Daudon)	Stereomicroscopy of surface and section, plus FTIR	Subtypes point to cause (e.g. type Ia COM: low urine volume; Ic: primary hyperoxaluria)
SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)	Surface structure and elements	Research
Micro-CT (micro-computed tomography)	3-D non-destructive structure	Research
Raman spectroscopy	Scattered light spectrum	Research

In vivo prediction

- **DECT (dual-energy computed tomography):** two tube energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid stones** with near-perfect accuracy; it estimates composition in vivo, mainly uric acid versus calcium stones.
- **Hounsfield units:** SWL (shock wave lithotripsy) works best below about 900–1000 HU; >1000 HU predicts SWL failure; cystine and brushite are dense.
- **Endoscopic recognition** of morphology (lost when stones are dusted). Dusting loses the specimen: collect fragments.

Significance in management

- **Uric acid:** oral dissolution by urinary alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite, cystine:** hard, SWL-resistant; prefer endoscopy.
- **Brushite:** can convert from calcium oxalate after SWL or infection; avoid over-alkalinisation.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency; treat with allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

Diagram to draw: FTIR spectrum outline and COM (dumbbell) versus COD (envelope) crystal shapes.

Key points to remember:

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
- DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.
- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

SAQ What are nanobacteria? Describe their role in urolithiasis.

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 1, Q3e

Definition

- **Nanobacteria**, now called **CNPs (calcifying nanoparticles)**, are spherical particles far smaller than bacteria with a **calcium apatite shell**, described in bovine and human serum by **Kajander and Çiftçioglu (1998)** as *Nanobacterium sanguineum*.
- **Their living nature is disputed.** Proposed as the smallest self-replicating bacteria, but later work showed "replication" is mineral nucleation, and that they are **mineralo-protein complexes** (apatite with serum proteins such as fetuin-A). Their bacterial DNA proved to be contamination. Most now regard them as **non-living**.

Claimed properties

Property	Claim
Size	Pass through bacteria-retaining filters
"Growth"	Very slow, in serum-containing medium
Shell	Apatite; resistant to heat and gamma irradiation
Detection	Electron microscopy, von Kossa and Hoechst staining, specific antibodies, culture
Susceptibility	Tetracycline and EDTA (ethylenediaminetetraacetic acid) inhibit them

Proposed role in urolithiasis

- **Nidus theory:** CNPs in blood or urine are filtered or excreted, deposit apatite in renal tubular cells and interstitium, and act as a **nucleating nidus** for crystal growth.
- **Link to Randall plaque:** idiopathic calcium oxalate stones grow on **Randall plaque**, an interstitial **apatite** deposit that starts in the basement membrane of the thin loop of Henle and erodes through the papilla. CNPs are **associated with**, but **not proven to cause**, Randall plaque (Campbell, Chapter 95).
- Early studies found apatite nanoparticles in a large majority of stones; rat studies reported renal calcification after injection.
- Also implicated in atherosclerotic calcification, polycystic kidney cysts and chronic prostatitis.

Therapeutic implications (unproven)

- Tetracycline-based regimens with EDTA were tried in stones and chronic prostatitis/chronic pelvic pain syndrome; no RCT (randomised controlled trial) evidence supports routine use.

Current position

- Not in stone guidelines; pathogenesis is explained by supersaturation, inhibitors, Randall plaque and ductal plugs.

Diagram to draw: renal papilla with interstitial apatite (Randall plaque) eroding through urothelium, CNPs as nidus, and calcium oxalate stone attached.

> **Key points to remember:**
> - Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
> - Proposed nidus for apatite and Randall plaque; association, not causation.
> - Tetracycline/EDTA therapy unproven.

SAQ Tear drop bladder

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3a

Definition

- A **tear-drop (pear-shaped or inverted-pear) bladder** is a bladder that is symmetrically **compressed from both sides** and elongated upwards on cystogram, IVU (intravenous urogram) or CT (computed tomography).
- It is a radiological sign of a **bilateral perivesical mass or collection**, not a disease itself.

Causes

Group	Causes
Collections	**Pelvic haematoma** (pelvic fracture: commonest in trauma); urinoma (extraperitoneal leak); bilateral lymphocele (after lymphadenectomy or transplant); pelvic abscess
Fat	**Pelvic lipomatosis**
Muscle	**Iliopsoas hypertrophy** (muscular young men)
Vascular	Dilated pelvic venous collaterals (IVC [inferior vena cava] occlusion); bilateral iliac artery aneurysms
Nodes and tumour	Pelvic lymphadenopathy (lymphoma, metastases); pelvic retroperitoneal fibrosis; pelvic tumours

Pelvic lipomatosis (the classic cause)

- Overgrowth of benign, non-encapsulated fat in the perivesical and perirectal spaces; mainly in men, more in Black men.
- **Presentation:** LUTS (lower urinary tract symptoms), suprapubic pain, constipation; often incidental.
- **Imaging:**
- Plain film: pelvic lucency. IVU: elevated, elongated pear-shaped bladder; **medial deviation of both ureters**; hydronephrosis.
- Barium enema: straightened, elongated rectosigmoid.

- **CT or MRI (magnetic resonance imaging):** fat density is diagnostic.
- **Cystoscopy:** high bladder neck, difficult passage; often **cystitis glandularis**, which can progress to adenocarcinoma. Biopsy and surveillance are needed.
- **Management:** observe if no obstruction. Upper-tract obstruction → stent or nephrostomy, ureteric reimplantation or urinary diversion. Weight loss rarely helps; surgical excision of fat is difficult.

Evaluation of a tear-drop bladder

- History: trauma, pelvic surgery or transplant, malignancy.
- **CT** shows the cause (blood, fat, fluid, muscle, nodes, vessels). Ultrasound for collections; aspiration (creatinine for urinoma, lymph for lymphocele).

Management

- Treat the cause: pelvic haematoma usually observed; lymphocele drained or marsupialised (laparoscopic fenestration); urinoma drained and leak repaired.

Diagram to draw: cystogram outline of a bladder squeezed into a tear drop by bilateral pelvic haematoma.

> **Key points to remember:**
> - Tear-drop bladder = bilateral extrinsic compression.
> - Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
> - Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

SAQ Calyceal diverticulum – classification and management

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3c

Definition

- Cystic cavity in the renal cortex, **lined by transitional epithelium**, connected to a calyx (or pelvis) by a **narrow neck**. Fills passively with urine.
- About 4.5 per 1000; equal in children and adults; no side or sex preference.

Classification

	Type I	Type II
Frequency	Commoner	Less common
Connects to	Minor calyx (usually upper pole)	Renal pelvis or major calyx
Size, symptoms	Small, usually silent	Larger, often symptomatic

Aetiology

- **Congenital** (favoured): persistent third- and fourth-generation ureteric bud branches.
- **Acquired:** cortical abscess draining into a calyx, infundibular stenosis, trauma, reflux.

Clinical features

- Stasis: **stones in up to 40%**, milk of calcium, infection, pain, haematuria.

Evaluation

- Ultrasound mimics a simple cyst. **CT (computed tomography) urography with delayed images shows contrast pooling**; milk of calcium layers and shifts with position.
- Retrograde pyelography maps the neck before intervention.
- **Differential:** hydrocalycosis (dilated major calyx from infundibular obstruction), megacalycosis, simple cyst.

Management

- **Asymptomatic:** observe with ultrasound.
- **Treat** pain, recurrent infection, haematuria, abscess, symptomatic stones.

Approach	Technique	Best for
Ureteroscopy	Find the ostium, laser-incise or balloon-dilate the neck, remove stones	Upper or mid-pole, small stones
PCNL (percutaneous nephrolithotomy)	Direct puncture onto stone, clear stones, fulgurate lining, dilate or create neck	Posterior, lower-pole, larger stones; long-term results good
Laparoscopic or robotic	Unroof, remove stones, close neck, obliterate cavity, pack with fat	Anterior, thin-parenchyma, failed endoscopy
Partial nephrectomy	Excision	Rare; destroyed segment

- Endoscopy keeps the cavity draining by widening the neck; laparoscopy abolishes it by closing the neck.
- **Also know:** SWL (shock wave lithotripsy) gives symptom relief but low stone clearance; fulguration of lining is debated.

Diagram to draw: kidney with upper-pole diverticulum, narrow neck to calyx, stone inside.

> **Key points to remember:**
> - Transitional lined, narrow neck; type I calyceal, type II pelvic.
> - Stones in up to 40%; delayed CT shows pooling.
> - Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

SAQ Pelvic lipomatosis

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3e

Definition

- Rare benign **overgrowth of non-encapsulated mature fat** in the perivesical and perirectal spaces, compressing the bladder, ureters, rectum and pelvic veins.
- Mostly men aged 20–60; associated with obesity and hypertension. Cause unknown; obesity alone does not explain it (weight loss rarely shrinks the fat). Chronic inflammation and venous obstruction are proposed.

Clinical features

- LUTS (lower urinary tract symptoms), haematuria, recurrent UTI (urinary tract infection), suprapubic or perineal pain, constipation.
- Leg oedema or venous thrombosis from venous compression.

- **Hydronephrosis and renal failure** from ureteric compression in a substantial minority.
- **Proliferative cystitis (cystitis glandularis)** is common; it can transform to **adenocarcinoma**.

Evaluation

- **IVU (intravenous urography) or cystogram:** elevated, elongated **"pear-" or "teardrop-shaped" bladder**; medially displaced pelvic ureters; high prostate.
- **Barium enema:** elongated, narrowed rectosigmoid.
- **Ultrasound:** echogenic pelvic fat; hydronephrosis.
- Medial deviation of **both** pelvic ureters also occurs in retroperitoneal fibrosis and after abdominoperineal surgery.
- **CT (computed tomography) or MRI (magnetic resonance imaging): diagnostic**; homogeneous fat density surrounding pelvic organs, separates it from tumour.
- **Cystoscopy:** elevated trigone, long prostatic urethra (rigid scope may be difficult); biopsy for cystitis glandularis.
- **Differential of pear-shaped bladder:** pelvic haematoma, lymphadenopathy, liposarcoma, iliopsoas hypertrophy, venous collaterals, pelvic fibrosis.

Management

- **Surveillance** (often progressive): ultrasound for hydronephrosis, renal function, periodic cystoscopy for cystitis glandularis.
- Weight loss, antibiotics and steroids give limited or no benefit; COX-2 (cyclo-oxygenase-2) inhibitors for cystitis glandularis are case reports only.
- **Ureteric obstruction:** stents or nephrostomy; ureteric reimplantation (difficult, dense fat).
- **Advanced or failed cases:** urinary diversion; cystectomy if adenocarcinoma or unmanageable bladder.
- Surgical excision of fat is difficult: no tissue planes, heavy bleeding.
- Resect cystitis glandularis transurethrally and treat obstruction or inflammation.

Diagram to draw: cystogram outline of pear-shaped bladder with medially deviated ureters.

> **Key points to remember:**
> - Pear-shaped bladder with medial ureters; CT shows fat.
> - Cystitis glandularis common; adenocarcinoma risk.
> - Watch for hydronephrosis; diversion if obstruction progresses.

SAQ Physiology of UPJ in relation to urine transport

Paper answer · P2

Type: SAQ (10 marks)
Source: 2023-Winter Paper 2, Q3f

UPJ = ureteropelvic (pelviureteric) junction.

Anatomy relevant to function

- The UPJ is the apex of a funnel-shaped pelvis; its spiral smooth muscle is continuous with calyces and ureter.

Pacemaker and electrical activity

- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves carry pain and only modulate.
- **Primary pacemaker: atypical smooth muscle cells** at the pelvicalyceal border; they fire spontaneously (HCN, hyperpolarisation-activated cyclic nucleotide-gated, channels).
- **ICC-like (interstitial cells of Cajal-like) cells** (c-Kit positive) in the pelvis and UPJ relay the signal to typical smooth muscle and act as back-up pacemakers.
- Signal spreads through **gap junctions**. Calcium entry gives a plateau action potential; calcium–calmodulin activates myosin light-chain kinase, causing contraction.

Bolus formation and transport

1. Urine fills the pelvis at low pressure; the UPJ is closed.
2. Pelvic contraction pushes urine through the opening UPJ into the upper ureter.
3. The UPJ closes behind it, forming a **bolus**.
4. Ureteric walls **coapt** behind the bolus; the wave (2–6 per minute, 20–80 cm H_2O) carries it to the bladder. - At **low flow**, the UPJ acts as a filter: several pelvic contractions occur for each ureteric contraction. - As **flow rises**, ureteric frequency reaches 1:1, then bolus volume rises. - At **very high flow**, boluses merge into a continuous column and the ureter dilates **without obstruction**.

Effect of obstruction

- **Acute:** stronger, more frequent contractions.
- **Chronic:** dilatation, walls fail to coapt, peristalsis becomes ineffective; raised pelvic pressure lowers GFR (glomerular filtration rate).
- **Intrinsic UPJO (UPJ obstruction):** abnormal muscle, excess collagen, fewer ICC-like cells, so an **aperistaltic segment** cannot transmit the wave.
- **Also know:** endotoxin and progesterone reduce peristalsis. Diuretic renogram and Whitaker test (chapter: pressure difference >12 cm H_2O suggests obstruction; classic teaching: <15 unobstructed, >22 obstructed) assess transport.

Diagram to draw: pelvis, UPJ and ureter showing bolus formation and coaptation.

> **Key points to remember:**
> - Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
> - Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
> - Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

SAQ Complications of PCNL, including bleeding and persistent haematuria after PCNL

Paper answer · P2

Asked as:

- Complications of PCNL — *Practice Set 2 Paper II Q3e (SAQ)*
- What are the causes of bleeding in PCNL? How will you manage persistent hematuria 6 days after PCNL!? — *Practice Set 6 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)
(mini-PCNL complications in 2022-Summer Paper 1, Q1) (PCNL complications also in Practice Set 2 Paper 2, Q3e)

Introduction

- PCNL (percutaneous nephrolithotomy) is the standard for renal stones >2 cm. Most complications arise from **access and tract dilatation**.
- Graded by the **Clavien–Dindo** system (I: no intervention; II: drugs or transfusion; III: intervention; IV: ICU [intensive care unit]; V: death).
- Bleeding is the commonest serious complication of PCNL. Transfusion was historically ~7%, now nearer 1% with smaller tracts. Angioembolisation is needed in <1%.

Incidence (systematic review, Seitz 2012, quoted by EAU [European Association of Urology])

Complication	Rate
Fever	10.8%
Transfusion	7% (nearer 1% with modern and mini tracts)
Thoracic	1.5%
Sepsis	0.5%
Organ injury	0.4%
Angioembolisation	0.4%
Death	0.05%

Main complications

Complication	Cause	Prevention and management
Acute bleeding	Infundibular or segmental vessel injury	Fornix puncture; place sheath, tamponade with balloon or **Kaye catheter**, clamp nephrostomy; transfuse
Delayed bleeding	Pseudoaneurysm, AV (arteriovenous) fistula	**Selective angioembolisation**
Infection, SIRS (systemic inflammatory response syndrome), sepsis	Infected stone, high intrarenal pressure	Sterile urine, culture-specific antibiotics, low-pressure irrigation; stop and drain if pus
Pleural (pneumothorax, hydrothorax)	Supracostal access, highest above 11th rib	Subcostal where possible, puncture in expiration; chest drain
Colon	Retrorenal colon, left lower pole	CT review; extraperitoneal: double-J stent and withdraw nephrostomy into colon (colostomy tube)
Spleen, liver	Upper-pole access, organomegaly	Ultrasound guidance; conservative or surgery
Collecting system	Perforation, extravasation	Stop, drain; later infundibular stenosis
Fluid absorption	Long operation, perforation	**Normal saline**; hyponatraemia, overload

Also know: venous air embolism, DVT (deep vein thrombosis) and pulmonary embolism, neuropraxia and eye injury (prone), residual fragments, loss of renal function (multiple tracts), urinoma, nephrocutaneous fistula, retained hardware.

Risk factors

- Staghorn, multiple tracts, supracostal access, large tract, long operation, CKD (chronic kidney disease), diabetes, positive urine culture, inexperience.
- **Bleeding risk factors:** infundibular or upper-pole puncture, multiple or large tracts, staghorn stone, long operation, antithrombotic drugs, coagulopathy, diabetes.

Reducing risk

- **Tranexamic acid** reduces transfusion (Cochrane).
- Ultrasound-guided access, mini-PCNL, ECIRS (endoscopic combined intrarenal surgery).
- **Bleeding prevention:** puncture through the **calyceal fornix** along a posterior calyx (Sampaio: no arterial injury); correct clotting; mini-tracts.

Bleeding in PCNL

Causes

Timing	Cause
Intraoperative	Parenchymal or venous tract bleeding; infundibular artery injury; through-and-through puncture; posterior segmental artery injury (upper-pole access); torn infundibulum or pelvis
Early postoperative	Tract bleeding after sheath removal; clot retention
Delayed (about 1 week)	**Pseudoaneurysm**, **AV (arteriovenous) fistula**

Mechanism of delayed bleeding

- **Pseudoaneurysm:** clot seals an arterial tear, then ruptures intermittently.
- **AV fistula:** artery and vein heal together; the thin vein cannot withstand arterial pressure.

Management at day 6 (persistent haematuria)

- At day 6, assume an **arterial lesion** until proved otherwise. `Resuscitate: IV access, crossmatch, transfuse; Hb, clotting → Stop anticoagulants; catheter, irrigate clots → Stable, mild: bed rest, clamp nephrostomy, serial Hb → Persistent or recurrent bleeding or Hb fall → CT angiography (arterial blush) or direct angiography → SUPERSELECTIVE ANGIOEMBOLISATION (coils) → Fails or unstable after embolisation → open exploration, partial or total nephrectomy`
- **Angiography** diagnoses >90% and treats in the same sitting. It may miss intermittent bleeds; non-selective embolisation loses renal tissue.
- Clot colic or obstruction: double-J stent or nephrostomy.
- Follow-up: Hb and creatinine; check for infection and abscess.

Intraoperative ladder (for completeness)

- Pressure on tract → nephrostomy clamped (tamponade) → balloon reinflation or **Kaye tamponade catheter** → angioembolisation.

Diagram to draw: kidney cross-section with posterior calyx, Brödel line, pleura and retrorenal colon; angiogram sketch: pseudoaneurysm before and after coiling.

> **Key points to remember:**
> - Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
> - Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
> - Early bleeding venous: tamponade.
> - Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
> - Treatment: superselective angioembolisation.
> - Supracostal access increases pleural injury; colon injury at left lower pole.

SAQ Steinstresse.

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 2, Q3e

Assumed meaning: steinstrasse.

Definition

- **Steinstrasse** (German, "stone street"): a column of stone fragments in the ureter that fails to pass and may obstruct. Typically after **SWL (shock wave lithotripsy)**; occasionally after PCNL (percutaneous nephrolithotomy) or URS (ureteroscopy).
- Incidence is low overall (about 4–7% after SWL, EAU [European Association of Urology]) but rises several-fold with stone burden **>2 cm²**.

Risk factors

- Large burden (>2 cm), renal pelvic stones, hard stones that break into large fragments, high energy, impaired ureteric drainage.

Classification (Coptcoat)

Type	Composition
I	Small fragments only
II	Larger lead fragment (4–5 mm) followed by small fragments
III	Large fragments throughout

Clinical features

- Asymptomatic (silent obstruction), colic, nausea; fever and sepsis if infected; anuria and AKI (acute kidney injury) in a solitary kidney. Most lie in the distal ureter.

Evaluation

- KUB (kidney, ureter, bladder) X-ray, ultrasound, or non-contrast CT (computed tomography); creatinine, urine culture, blood counts.

Management

```
Steinstrasse ├─ infection / sepsis → urgent PCN (percutaneous nephrostomy) + antibiotics ├─ asymptomatic, no obstruction → MET + analgesia, imaging every 1–2 weeks └─ failure, pain, obstruction: ├─ lead fragment (type II/III) → SWL to lead fragment └─ URS with laser lithotripsy (most definitive)
```

- **MET (medical expulsive therapy):** α-blockers (tamsulosin) improve fragment clearance and reduce pain.
- **PCN** is preferred in infected obstruction: drains without manipulating the ureter, and fragments often pass once the kidney is decompressed. A stent is an alternative.
- Retained fragments with poor progress: URS clears the column.

Prevention

- Choose PCNL or RIRS (retrograde intrarenal surgery) for large burdens.
- Energy ramping, slower shock rate; MET after SWL.
- **Stent before SWL:** may reduce steinstrasse with large stones but does not improve stone-free rate and adds stent symptoms. The chapters disagree: Chapter 97 says it may reduce steinstrasse; Chapter 98 advises against stenting at SWL.

Diagram to draw: ureter with lead fragment and trailing column (Coptcoat types I–III).

> **Key points to remember:**
> - Risk rises several-fold with burden >2 cm².
> - Coptcoat I–III by fragment size.
> - Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

SAQ Treatment of inferior calyceal calculus

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 2, Q3b

Introduction

- Lower pole is the **commonest site** of renal stones. Gravity-dependent position makes fragments hard to clear, so **SWL (shock wave lithotripsy) works poorly** here.

Factors in decision

- **Stone:** size, density (<900 HU [Hounsfield units] favours SWL), composition.
- **Anatomy unfavourable for SWL clearance:** acute **infundibulopelvic angle**, long and narrow infundibulum, skin-to-stone distance >10 cm.
- **Patient:** obesity, anticoagulants (favour URS [ureteroscopy]), pregnancy, preference.
- Evaluation: NCCT (non-contrast CT), urine culture, renal function.

Treatment by size

Size	Options
Asymptomatic, small	Observation acceptable; about half have an event in 5 years. Treat if growing, symptomatic, infected or occupational
<10 mm	SWL or **RIRS** (retrograde intrarenal surgery). AUA (American Urological Association): either
10–20 mm	EAU (European Association of Urology): SWL only if favourable anatomy; otherwise **RIRS or PCNL** (percutaneous nephrolithotomy)
>10 mm (AUA)	**PCNL** (highest SFR [stone-free rate]) or URS; SWL not first-line
>20 mm	PCNL first; staged RIRS an alternative

Evidence

- **Lower Pole I (Albala 2001):** stones >1 cm: SFR 91% PCNL vs 21% SWL.
- Lower Pole II: stones ≤1 cm, URS vs SWL: no significant SFR difference.

Technique points

- **RIRS:** flexible ureteroscope (deflection up to 270°); relocate the stone to an upper calyx with a basket before laser (holmium or thulium fibre) lithotripsy; access sheath.
- **PCNL:** direct lower pole posterior calyx puncture; mini-PCNL reduces bleeding (transfusion near 1%).
- **SWL adjuncts:** percussion, diuresis, inversion therapy and alpha-blockers improve fragment clearance.
- **Also know:** lower pole calyceal diverticulum or infundibular stenosis favours PCNL; residual fragments >4 mm after URS need treatment.

Follow-up

- Imaging for residual fragments; metabolic evaluation if recurrent.

Diagram to draw: infundibulopelvic angle and infundibular length/width on an IVU (intravenous urogram) tracing.

> **Key points to remember:**
> - Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
> - <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
> - Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
> - Reposition stone during RIRS.

SAQ Ovarian vein syndromes

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 15 Paper 2, Q3e

Definition

- **Ovarian vein syndrome** (Clark, 1964) is extrinsic compression of the **right ureter** by a dilated or thrombosed **right ovarian vein**, causing intermittent hydroureteronephrosis. It is a **controversial** entity; many cases are now attributed to other causes.
- The plural also covers ovarian vein thrombosis and pelvic congestion.

Anatomy (basis)

- Ovarian veins run in the infundibulopelvic ligaments. The **right** enters the IVC (inferior vena cava) obliquely; the **left** enters the left renal vein at a right angle.
- Gonadal vessels **cross anterior to the ureter**; ovarian veins dilate greatly in pregnancy.

Types

Syndrome	Mechanism	Features
Ovarian vein (ureteric) syndrome	Dilated right ovarian vein compresses the ureter at the crossing	Multiparous women; right flank pain (premenstrual or in pregnancy); recurrent UTI (urinary tract infection)

Syndrome	Mechanism	Features
Ovarian vein thrombosis	Postpartum hypercoagulability and venous stasis; mostly right-sided	Fever and right lower-quadrant or flank pain after delivery; may obstruct ureter; risk of pulmonary embolism
Pelvic congestion syndrome	Incompetent, refluxing ovarian veins with pelvic varices; sometimes secondary to nutcracker (left renal vein compression)	Chronic pelvic pain worse on standing, dyspareunia, post-coital ache

Controversy

- Right-sided dilatation in pregnancy is mostly **physiological hydronephrosis** (progesterone plus uterine compression; stops at the pelvic brim). Venous compression is rarely proven, so diagnosis is by **exclusion**.

Investigations

- Ultrasound; **CT urography** or IVU (intravenous urography): right hydroureter to a smooth oblique extrinsic indentation, normal ureter below.
- **Diuretic renogram** to prove obstruction; retrograde pyelography.
- CT or MR (magnetic resonance) venography, Doppler; **gonadal venography** (gold standard for reflux).
- **Exclude:** stone, retrocaval ureter, iliac vessel crossing, retroperitoneal fibrosis, endometriosis.

Management

- **Asymptomatic:** observe. **Obstruction:** stent or nephrostomy if infected or pregnant; definitive **ureterolysis with ligation and division of the ovarian vein** (open or laparoscopic).
- **Thrombosis:** anticoagulation and antibiotics.
- **Pelvic congestion:** progestogens or GnRH (gonadotropin-releasing hormone) analogues; **coil or sclerotherapy embolisation**; ligation.

Diagram to draw: right ovarian vein crossing anterior to the right ureter, with proximal hydroureteronephrosis.

> **Key points to remember:**
> - Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
> - Postpartum ovarian vein thrombosis: right side, anticoagulate.
> - Treatment: ureterolysis with vein division, or embolisation.

SAQ Post traumatic bladder neck injury: acute and delayed management?

Paper answer · P3

Type: SAQ (10 marks)
Source: 2021-Summer Paper 3, Q3c

Introduction

- The bladder neck is the **internal sphincter**. It is torn mainly in **pelvic fracture**, often with prostatic urethral, rectal or vaginal injury; **commoner in children**.
- Missed injury causes **incontinence, fistula and stricture**.

Diagnosis

- Pelvic fracture with gross haematuria, retention or blood at the meatus; **vaginal examination** in females.
- **RUG (retrograde urethrogram)**, then **retrograde CT or plain cystography** (350 mL).
- Grading: AAST (American Association for the Surgery of Trauma) bladder **grade V** = laceration into the bladder neck or ureteric orifice; Goldman type IV = bladder rupture extending into the proximal urethra.

Acute management

- Resuscitate; if unstable, damage control (drainage), repair later.
- **Bladder neck injury needs immediate repair** (AUA (American Urological Association) and EAU (European Association of Urology)), unlike isolated posterior urethral distraction (suprapubic tube, delayed repair). `Lower midline incision → anterior cystotomy → identify ureteric orifices (stent if near) → minimal debridement; re-approximate neck to prostatic urethra, layered absorbable sutures over a 22 Fr catheter → rectal or vaginal tear: separate layers, no overlapping suture lines, interpose omentum (colostomy for rectum) → remove bone spicules; drain ± suprapubic tube`
- Catheter 2–3 weeks; cystogram before removal.

Delayed management (3–6 months)

Problem	Management
Associated urethral distraction	Cystogram + RUG + endoscopy; **anastomotic urethroplasty first**, not combined with bladder neck surgery
Bladder neck contracture	Endoscopic incision or resection; open Y-V plasty if refractory
Incontinence (both sphincters damaged)	Urodynamics; **AUS (artificial urinary sphincter)**; bladder neck reconstruction (Young–Dees–Leadbetter) or fascial sling; last resort **bladder neck closure + Mitrofanoff** ± augmentation
Fistula	Layered repair with tissue interposition
- Bladder neck appearance on imaging **does not predict** continence; restore continuity first, then treat incontinence.	
- Continence is poor whenever the bladder neck is involved (paediatric series: incontinence up to 20% overall).	

Diagram to draw: pelvic fracture with bladder neck tear and repair over a catheter.

> **Key points to remember:**
> - Bladder neck injury = immediate repair.
> - Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
> - Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

SAQ Role of open and laparoscopic stone surgeries in modern era of endourology.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2021-Summer Paper 4, Q3f

Introduction

- SWL (shock wave lithotripsy), URS (ureteroscopy) and PCNL (percutaneous nephrolithotomy) treat almost all stones. Open surgery is now a small minority in centres with full endourology, but remains in resource-limited settings.
- **AUA (American Urological Association)/Endourological Society 2016:** consider laparoscopic, robotic or open surgery **only when SWL, URS and PCNL fail or are expected to fail**; not first line.

Indications

Category	Examples
Endourology failed or unlikely	Very complex staghorn needing many PCNL sessions; failed multiple procedures
Anatomical problem treated together	**PUJ (pelviureteric junction) obstruction + stone:** pyeloplasty with pyelolithotomy; anterior calyceal diverticulum; infundibular stenosis
Difficult access	Pelvic or ectopic kidney (laparoscopy-assisted PCNL); morbid obesity; skeletal deformity
Poor renal function	**Nephrectomy** for a non-functioning kidney; partial nephrectomy for a non-functioning pole
Large impacted ureteric stone	Laparoscopic ureterolithotomy after failed URS/SWL
Concomitant surgery	Other open or laparoscopic procedure
Bladder	Very large stone; with open prostatectomy, diverticulectomy or augmentation; small bladder; urethral stricture

Procedures

- **Pyelolithotomy:** simple, extended (Gil-Vernet, sinus dissection) or coagulum.
- **Anatrophic nephrolithotomy (Smith and Boyce):** for complex staghorn. Clamp renal artery; cool with ice slush; give mannitol; incise along **Brödel's avascular line**; remove stone; repair infundibula; close.
- Radial nephrotomy, ureterolithotomy, cystolithotomy.

Laparoscopic and robotic surgery

- Transperitoneal or retroperitoneal pyelolithotomy and ureterolithotomy; robotic anatrophic nephrolithotomy reported.
- Best for large pelvic stone in an extrarenal pelvis: high single-stage clearance, less bleeding than PCNL. Drawbacks: longer operation, calyceal stones need a flexible scope.

Feature	Open	Laparoscopic/robotic
Morbidity	Large incision, pain, long stay	Less pain, shorter stay
Kidney	Ischaemic loss of function (anatrophic)	Similar repair, less wound morbidity

Feature	Open	Laparoscopic/robotic
Later surgery	Adhesions make redo difficult	Fewer

Complications

- Bleeding, urinary leak, infection, loss of renal function, residual fragments.

Diagram to draw: Brödel's line incision for anatrophic nephrolithotomy.

> **Key points to remember:**
> - Open/laparoscopic only when endourology fails or is unsuitable.
> - Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
> - Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

SAQ Diet in stone disease

Paper answer · P4

Asked as:

- Diet in stone disease. — *Winter 2023 Paper IV Q3f (SAQ)*
- Diet in stone disease. — *Practice Set 5 Paper IV Q3f (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3f)

Principle

- Diet lowers urinary **supersaturation** by **dilution**, by reducing promoters (calcium, oxalate, uric acid, sodium) and by raising inhibitors (citrate).
- General advice for all; recurrent formers tailored to **24-hour urine**.

General measures

Measure	Target	Mechanism and evidence
Fluid	Urine **≥2.5 L/day** (about 3 L intake), spread through the day	Most important single factor. **Borghi 1996 RCT (randomised controlled trial):** recurrence 12.1% vs 27% at 5 years
Normal calcium	1000–1200 mg/day, with meals	Binds oxalate in the gut. **Restriction increases stones**. **Borghi 2002:** normal-calcium, low-salt, low-protein diet halved stone events vs low-calcium diet
Low sodium	Salt about 4–5 g/day	Sodium increases urinary calcium, lowers citrate
Limit animal protein	0.8–1.0 g/kg/day	Acid and purine load: hypercalciuria, hypocitraturia, hyperuricosuria
Fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension] diet)	Daily	Alkali and citrate; DASH linked to fewer stones
Weight	BMI (body mass index) 18–25	Obesity: low urine pH, more oxalate and uric acid

- **Citrus:** lemonade (most evidence) and orange juice raise citrate; orange juice adds alkali. **Volume matters more than drink type.**

- Avoid sugar-sweetened drinks and **high-dose vitamin C** (converted to oxalate).

Stone-specific diet

Stone or abnormality	Diet
Hyperoxaluria	Avoid large portions of **spinach, beets, chocolate, nuts, tea**; calcium with meals
Enteric hyperoxaluria	Strict low oxalate and **low fat**; **calcium up to 1 g/day with meals**; fluid; magnesium
Uric acid	Less animal protein (purines give about 30% of urinary uric acid); fruit and vegetables; weight loss
Cystine	Fluid **3–5 L/day** including night; sodium **≤2300 mg/day**; less animal protein (not in children)
Struvite	Infection control; diet has little role

Adherence

- Poor: only half reach 2.5 L at 6 months; apps and smart bottles help.

Diagram to draw: "stone-prevention plate": water, fruit and vegetables, normal dairy, little meat and salt.

> **Key points to remember:**
> - Urine ≥2.5 L/day (Borghi 1996).
> - Never restrict calcium (Borghi 2002).
> - Low salt, less animal protein, more fruit and vegetables.
> - Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

SAQ ECIRS (endoscopic combined intrarenal surgery)

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3b

Definition

- **ECIRS:** PCNL (percutaneous nephrolithotomy) and retrograde flexible ureteroscopy performed **at the same time, in one position**, by two surgeons.
- Term and technique popularised by **Scoffone (2008)** in the **GMSV (Galdakao-modified supine Valdivia) position**. Combined access dates back to the 1980s (Lawson's retrograde needle placement, 1983).

Rationale

- Each route covers the other's blind spots: the retrograde scope guides and watches the puncture, and reaches calyces the rigid nephroscope cannot.
- Result: one tract, better clearance.

Positioning

Position	Features
GMSV (preferred)	Supine; flank on a bolster at the table edge; ipsilateral leg extended, contralateral leg in lithotomy
Bart's modified supine	Variant of supine
Prone split-leg	Easier posterior calyces; needs a split-leg table

- **Supine advantages:** airway access, gravity drainage of fragments, no turning, simultaneous retrograde access.

Technique

1. Cystoscopy; UAS (ureteral access sheath) placed; flexible ureteroscope maps the stones and chooses the target calyx.
2. Ultrasound or fluoroscopic puncture **watched endoscopically** (double bull's-eye technique).
3. Wire grasped and brought out: through-and-through access.
4. Tract dilated under vision (balloon) and sheath placed.
5. Antegrade lithotripsy and suction; the retrograde scope **relocates stones** from inaccessible calyces into the nephroscope's view, or lasers them.
6. Final check with both scopes; stent or tubeless exit. - Mini-ECIRS uses a mini-PCNL tract.

Indications

- Staghorn and complex multi-calyceal stones, combined renal and ureteric stones, anomalous kidneys, failed previous treatment, urinary diversion, obese or cardiopulmonary patients (supine).

Advantages (meta-analysis)

- Single position, **fewer tracts, less bleeding**, fewer ancillary procedures, **higher stone-free rate**; less fluoroscopy; the UAS keeps intrarenal pressure low.

Disadvantages

- Two surgeons and two sets of equipment; cost; longer set-up; ureteric injury from the sheath; less working space in supine.

Diagram to draw: GMSV position with nephroscope from the flank and flexible ureteroscope from below in the same calyx.

> **Key points to remember:**
> - Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
> - Retrograde scope guides the puncture and relocates stones.
> - Fewer tracts, less bleeding, higher stone-free rate.

Rapid Revision Sheet · Section 11

Radionuclide studies and diuretic renography in PUJ obstruction (SAQ)

- MAG3 is the tracer of choice; hydrate and catheterise.
- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
- Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

Retrograde endopyelotomy (SAQ)

- Lateral full-thickness incision over a stent.
- Crossing vessel and gross hydronephrosis predict failure.
- Success 70–80%, below pyeloplasty (about 95%).

Laparoscopic pyeloplasty. (LAQ)

- Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
- Transpose over a crossing vessel; never divide it.
- Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
- Robotic equals laparoscopic in outcome, with easier suturing.

Congenital PUJ obstruction, failed pyeloplasty and salvage procedures (LAQ)

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

Ureterocalycostomy (SAQ)

- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
- Stricture is the main failure; success about 70% in redo series.

Boari flap (SAQ)

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

Trans Uretero Ureteric anastomoses (SAQ)

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.

- Risk: unilateral injury becomes bilateral.

Retroperitoneal fibrosis: diagnosis and (medical) management (LAQ)

- Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
- Medially deviated ureters; PET-CT best shows activity; biopsy deep.
- Drain first, then prednisone (about 80% respond).
- Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
- Rituximab for IgG4-related or refractory disease.
- About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management (LAQ)

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
- Delayed CT phase is essential; blush means AE.
- Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
- Revascularise only a solitary kidney or bilateral injuries.

Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury (LAQ)

- Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
- Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
- AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
- Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
- Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Urolithiasis: etiopathogenesis, medical management and prevention of recurrence (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

Primary Hyperoxaluria (SAQ)

- PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.

- Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
- Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

Renal tubular acidosis (distal and type 2) (SAQ)

- Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
- NH_4Cl load: pH fails to fall below 5.5 in distal RTA.
- Potassium citrate is the treatment for distal RTA.
- Type 2 (proximal): bicarbonate wasting; HCO_3^- 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
- Fractional HCO_3^- excretion >15%; stones uncommon in type 2.
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause.

Uric acid calculi: etiopathogenesis, medical management and allopurinol (LAQ)

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.
- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Cystinuria (SAQ)

- Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Struvite stones. (SAQ)

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

Medical expulsive therapy (MET) in urinary calculi (SAQ)

- Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
- SUSPEND 2015: no benefit in an unselected population.
- Trial of passage 4–6 weeks, then intervene.

Stones in pregnancy (SAQ)

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

Management of upper ureteric stones (LAQ)

- Proximal stones pass in only 48%; MET does not help proximal stones.

- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Flexible ureteroscopy / RIRS and accessories (SAQ)

- 3.6 Fr channel; ≥270° deflection.
- 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
- Sepsis about 5%, stricture about 3%.
- Hydrophilic wire passes; PTFE wire stays.
- Nitinol tipless basket for calyces.

Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

Stone complexity scores (SAQ)

- Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
- S.T.O.N.E.: five variables, score 5–13.
- CROES includes centre volume; RUSS is for RIRS.

ESWL: mechanism, indications, factors affecting outcome, complications and current status (LAQ)

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

PCNL patient positioning: prone vs supine (SAQ)

- Prone: posterior and upper pole access, large working area.
- Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
- Horizontal bolsters drop the kidney more.
- Similar outcomes; choose by anatomy and experience.

Mini-PCNL and newer PCNL techniques (SAQ)

- Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).
- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.

- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
- Working channel decides instruments; suction lowers pressure.

Metabolic evaluation in stone disease (LAQ)

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
- NCCT gold standard (97%/95%).
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

Fracture penis (SAQ)

- Snap, detumescence, eggplant deformity.
- Urethral injury about 20% (West); suspect if bilateral.
- Immediate exploration and interrupted absorbable tunical closure.

Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management (LAQ)

- Blood at meatus: RUG before catheter.
- Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
- EPR: catheter 2–3 weeks; IPR: immediate repair.
- PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
- Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
- ED mostly from the injury; incontinence <4%.

Renal autotransplantation (SAQ)

- For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
- Unit loss 5–10%; success about 90%.

Newer methods of stone analysis (SAQ)

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
- DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.

- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

What are nanobacteria? Describe their role in urolithiasis. (SAQ)

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

Tear drop bladder (SAQ)

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

Calyceal diverticulum – classification and management (SAQ)

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

Pelvic lipomatosis (SAQ)

- Pear-shaped bladder with medial ureters; CT shows fat.
- Cystitis glandularis common; adenocarcinoma risk.
- Watch for hydronephrosis; diversion if obstruction progresses.

Physiology of UPJ in relation to urine transport (SAQ)

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

Complications of PCNL, including bleeding and persistent haematuria after PCNL (SAQ)

- Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
- Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.
- Supracostal access increases pleural injury; colon injury at left lower pole.

Steinstresse. (SAQ)

- Risk rises several-fold with burden >2 cm^2.
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

Treatment of inferior calyceal calculus (SAQ)

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

Ovarian vein syndromes (SAQ)

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

Post traumatic bladder neck injury: acute and delayed management? (SAQ)

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

Role of open and laparoscopic stone surgeries in modern era of endourology. (SAQ)

- Open/laparoscopic only when endourology fails or is unsuitable.
- Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
- Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

Diet in stone disease (SAQ)

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

ECIRS (endoscopic combined intrarenal surgery) (SAQ)

- Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
- Retrograde scope guides the puncture and relocates stones.
- Fewer tracts, less bleeding, higher stone-free rate. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 12

Bladder Cancer & Diversion

129 Tumours of the Urinary Bladder

1 · Epidemiology (headline)

- It is the **fourth most prevalent** cancer and the **costliest per patient over a lifetime**.
- The median age at diagnosis is **73**, and the **male:female ratio is about 3:1**.
- **Women present later and do worse stage for stage**, as haematuria is mistaken for infection.

2 · Risk factors

The exposure predicts the histology.

LAQ CA bladder, etiology, newer investigations and treatment.

Paper answer · P2

Type: LAQ (25 marks)
Source: 2016-Summer Paper 2, Q1

Introduction

- Commonest urinary tract cancer; over 90% **UC (urothelial carcinoma)**. Men 3 times commoner; women present later. About 70% are **NMIBC (non-muscle-invasive bladder cancer)**; it recurs often.

Aetiology (risk factors)

Factor	Key facts
Smoking	30–40% of UC; risk 2–3 times. Aromatic amines form DNA adducts; **NAT2 (N-acetyltransferase 2) slow acetylators** at higher risk. Risk nears never-smokers >15 years after quitting
Occupation (5–10%)	Aromatic amines (dye, rubber, leather, painters, hairdressers); polycyclic hydrocarbons (aluminium, chimney sweeps)
Environment	**Arsenic** in water (West Bengal, Bangladesh, Taiwan)
Radiation	Pelvic radiotherapy 1.5–4 times; latency 15–30 years
Drugs	**Cyclophosphamide** (dose-dependent, above 20 g); pioglitazone (conflicting)
Chronic inflammation	*Schistosoma haematobium*, long-term catheter, stones → **SCC (squamous cell carcinoma)**; exstrophy → adenocarcinoma
Genetic	First-degree relatives 2 times; **Lynch syndrome** (MSH2); GSTM1 deletion
- **Also know:** obesity raises risk; diet has no proven role.	

Pathology and staging

- **WHO (World Health Organization) 2004 grading**: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. Grade predicts progression.
- **Papillary pathway** (FGFR3; recurs, rarely progresses) versus **flat CIS (carcinoma in situ)** (TP53, RB; progresses).

TNM (tumour, node, metastasis), 2017	Extent
Ta / Tis	Papillary non-invasive / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	One pelvic node / several / common iliac

Clinical features and standard work-up

- **Painless gross haematuria (85%)**; irritative symptoms suggest CIS.
- **Cystoscopy and biopsy (gold standard)**; **CT (computed tomography) urography**; **urine cytology** (sensitivity 31–62%, specificity 94–100%, detects high grade; **Paris System** reporting).

Newer investigations

Test	Principle	Value
Blue-light cystoscopy (PDD, photodynamic diagnosis) with HAL (hexaminolevulinate)	Porphyrins in tumour fluoresce red under blue light	About one-third more CIS; recurrence benefit inconsistent
NBI (narrow-band imaging)	415 and 540 nm light absorbed by haemoglobin highlights vessels	More tumours; many false positives
Urine biomarkers	NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), **UroVysion FISH (fluorescence in situ hybridisation)** (chromosomes 3, 7, 17, 9p21), ImmunoCyt, Cxbladder	More sensitive, less specific than cytology; **none replaces cystoscopy**
mpMRI (multiparametric MRI) with VI-RADS (Vesical Imaging-Reporting and Data System)	Muscle-line integrity scored 1–5	**≥3 suspicious for muscle invasion**; about 80–90% accurate; before TURBT
En bloc TURBT (transurethral resection of bladder tumour)	Tumour removed in one piece with detrusor	Better staging and muscle sampling, fewer perforations; recurrence similar
- **Also know:** optical coherence tomography; urinary tumour DNA; PET-CT (positron emission tomography) for doubtful metastases.		

Treatment

1. TURBT: complete, with muscle in the specimen. **Re-TURBT within 6 weeks** for T1, incomplete resection or no muscle. **Single chemotherapy instillation within 6 hours** (mitomycin C, epirubicin or gemcitabine) for low-risk tumours; never BCG (bacillus Calmette–Guérin) at this time.

2. NMIBC by risk (AUA (American Urological Association))

Risk	Definition	Treatment
Low	Solitary low-grade Ta ≤3 cm; PUNLMP	Single instillation; surveillance
Intermediate	Low-grade Ta recurrent, >3 cm or multifocal; low-grade T1; high-grade Ta ≤3 cm	Chemotherapy or BCG; maintenance 1 year
High	High-grade T1; large, multifocal or recurrent high-grade Ta; any CIS; variant histology; lymphovascular invasion	**BCG induction + 3-year maintenance**; early cystectomy if very high risk
- **BCG** induces a Th1 immune response; six weekly doses from 2–4 weeks after TURBT, then **SWOG (Southwest Oncology Group) maintenance** (recurrence-free survival 76.8 vs 35.7 months). Only agent that reduces progression.		
- **BCG-unresponsive**: radical cystectomy; if unfit, **pembrolizumab** (KEYNOTE-057), **nadofaragene firadenovec**, gemcitabine–docetaxel, TAR-200.		

3. MIBC (muscle-invasive bladder cancer), cT2–T4a N0M0

- **Cisplatin-based NAC (neoadjuvant chemotherapy)** (dose-dense MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) or gemcitabine–cisplatin) → **radical cystectomy + pelvic lymphadenectomy**. NAC adds 5–8% absolute survival (SWOG 8710: pT0 38% vs 15%). Cisplatin-ineligible: GFR (glomerular filtration rate) <60 mL/min, poor performance status, hearing loss, neuropathy, heart failure.
- **Diversion**: ileal conduit or orthotopic neobladder. Robotic equals open (**RAZOR**).
- **Trimodality therapy**: complete TURBT + chemoradiation (5-FU (fluorouracil) + mitomycin C, **BC2001**). For solitary T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- **Adjuvant nivolumab** for ≥pT3 or N+ (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months). Partial cystectomy for selected solitary dome tumours.

4. Metastatic: **enfortumab vedotin + pembrolizumab** (**EV-302**: nearly doubled survival versus platinum). Otherwise platinum then **avelumab maintenance**; **erdafitinib** for FGFR3 alterations.

Follow-up and recent advances

- Cystoscopy at 3 months, then by risk; upper-tract imaging in high risk.
- Molecular subtypes (basal tumours respond to cisplatin); **NIAGARA** (2024): perioperative durvalumab with NAC improved survival.

Diagram to draw: bladder wall layers with Ta, Tis, T1–T4 marked.

> **Key points to remember:**
> - Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
> - Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
> - High-risk NMIBC: BCG with 3-year maintenance.
> - MIBC: cisplatin NAC + cystectomy; trimodality in selected.

Also, from the condensed book (not in the paper answer):

- **Cyclophosphamide** is the only proven chemotherapeutic carcinogen, via **phosphoramide mustard** (acrolein causes haemorrhagic cystitis, not cancer). Risk is ~4.5× above 20 g, with latency peaking at 10–14 years.
- **Radiation**: external beam radiotherapy for cervical cancer raises risk 2–4×, and for prostate cancer 1.5–2×.
- **Exstrophy** carries a malignancy risk of 3.3–7.5%, and **>90% is adenocarcinoma**.
- **Smoking** causes **30–40%** of urothelial carcinoma (UC) and raises risk **2–3×**.
- **N-acetyltransferase 2 (NAT2) slow acetylators** carry the highest risk, because acetylation detoxifies aromatic amines. Risk falls to near that of never-smokers **>15 years** after quitting.
- **Schistosoma haematobium** causes chronic inflammation and fibrosis, which leads to **squamous cell carcinoma (SCC)**. Long-term catheters in spinal cord injury and stones also cause SCC.
- Germline variants (MSH2, BRCA1/2) occur in 13–24% of UC, so test selected high-grade patients.

3 · Pathology

Benign lesions (reactive mimics)

Lesion	Hallmark	Potential / action
Squamous/glandular metaplasia	Trigone; squamous in up to 40% of females	None; no biopsy
Inverted papilloma	Trigone; endophytic nests, no atypia; <1% of tumours	Benign, ~1% recur; resect
Urothelial papilloma	Delicate stalks; FGFR3 mutant, TP53/RB wild type	None; resect
Nephrogenic adenoma	Tubule-like glands after injury	None; re-resect if it recurs
Leukoplakia	Keratinising squamous metaplasia; floating white flakes	**Not premalignant**, so no follow-up
Cystitis cystica/glandularis	From proliferating von Brunn nests	Glandularis: low adenocarcinoma risk, so resect, treat cause and survey

Flat (precursor) lesions

Loss of polarity means dysplasia; loss of umbrella cells means carcinoma in situ (CIS).

Lesion	Key features	Significance
Hyperplasia	≥10 layers, no atypia	May precede low-grade papillary tumour (FGFR3)
Reactive atypia	Atypia proportionate to inflammation; polarity kept	Coexists with dysplasia/CIS, so investigate
Atypia of unknown significance	Atypia disproportionate to inflammation	Limited utility
Dysplasia	Loss of polarity, umbrella cells kept; CK20, p53	14–19% progress; **60%** progress to CIS if there was prior cancer
CIS	Flat, high grade, noncohesive pleomorphic cells; umbrella cells lost	Precursor of invasive high-grade disease

UC grade (2004 WHO) – noninvasive spectrum

Non-muscle-invasive bladder cancer (NMIBC) makes up about **70%** of UC and includes Ta, CIS and T1.

	PUNLMP*	Low-grade Ta	High-grade papillary	CIS
Morphology	Papillary >7 cells, minimal atypia, solitary, trigone	Branching stalks, some atypia	Fused papillae, many mitoses	Flat, severe atypia
Genetics	TERT promoter, FGFR3	Chromosome 9 loss; FGFR3, PI3K, RAS	TP53/RB	RB, TP53, PTEN; **no FGFR3**
Behaviour	Recur ~20% at 5 y; progress <1%	Recur 48–71%; progress 2–12%	>80% invade if untreated	With invasive tumour: 45–65% 5-y mortality

*Papillary urothelial neoplasm of low malignant potential. **Grade, not recurrence, predicts danger.**

Histological variants of UC

About 80% have mixed (mostly squamous) differentiation. **Altered growth patterns behave worse than their stage.**

Variant	Hallmark	Chemosensitive?	Management
Micropapillary (0.7–2.2%)	Tight clusters with retraction artefact; **any amount counts**; ERBB2 amplification	No	**Early radical cystectomy (RC) even for NMIBC**
Sarcomatoid (0.3–0.6%)	Large infiltrative mass	Poor	Immediate RC
Plasmacytoid (<1%; includes signet ring)	Eccentric nuclei; **intraperitoneal spread**; CDH1 truncation	Yes, but relapse is common	Neoadjuvant chemotherapy (NAC) whenever possible
Nested (M:F 6:1)	Mimics von Brunn nests and inverted papilloma	No	Often upstaged; stage-matched outcome as UC
Squamous (16–22%, up to 60%)	Basal subtype	NAC active in mixed tumours	Linked to recurrence and progression
Glandular (~10%)	Gland-like spaces	–	Outcome as conventional UC

Non-urothelial malignancy

	Small cell	SCC	Adenocarcinoma	Urachal adenocarcinoma
Setting	Rare; dedifferentiated UC	2–5% in West; majority where schistosomiasis is endemic	**Secondary is commoner than primary** (colon, prostate, uterus, cervix); exstrophy	~1/3 of bladder adenocarcinomas; from the allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic syndromes	Locally advanced, fewer metastases	Cribriform/glandular; **colonoscopy** to exclude a primary	Dome/anterior wall; Sheldon staging
Treatment	**Cisplatin–etoposide first**, then RC or radiation	RC; preoperative radiation worsened survival	RC; 5-fluorouracil–cisplatin by analogy	**En bloc dome with urachal ligament and umbilicus**

> **KEY POINT**
>
> Small cell is systemic disease: primary cystectomy gives poor disease-free survival that adjuvant chemotherapy does not rescue, whereas neoadjuvant chemotherapy markedly improves survival. SCC in spinal cord injury fell after the switch to intermittent self-catheterisation, showing that inflammation drives it.

Urachal carcinoma (2016 WHO criteria): the epicentre is in the dome or anterior wall, there is no widespread cystitis glandularis, and a secondary source is excluded. TNM staging is unsuitable.

Sheldon stage	Extent
I	Urachal mucosa
II	Confined to urachus
IIIA / IIIB / IIIC	Bladder / abdominal wall / peritoneum
IV	Metastatic

4 · Molecular biology

	Papillary pathway	Flat pathway
Origin and product	Hyperplasia gives low-grade papillary tumour	Dysplasia gives CIS, then high-grade invasive tumour
Genetics	**FGFR3** (up to 80% of low-grade Ta vs 10–20% of MIBC), chromosome 9 deletion, PIK3CA	**TP53** (up to 76% of MIBC), RB1, CDKN2A, PTEN loss
Behaviour	Recurs, rarely progresses	Progresses

- HRAS/KRAS mutations are **mutually exclusive with FGFR3**, because they act in the same pathway.

The Cancer Genome Atlas (TCGA) subtype	Luminal	Basal/squamous
Markers	FGFR3, GATA3, KRT20, PPARG, FOXA1	CK5, CK6, CK14; low GATA3 and FOXA1

The Cancer Genome Atlas (TCGA) subtype	Luminal	Basal/squamous
Behaviour	Less aggressive; papillary	Aggressive; squamous; women; T3/T4
Targets	FGFR3 inhibitors, ERBB2	Cisplatin NAC; checkpoint inhibitors; EGFR

- The **consensus classification (2020) has six classes**. Luminal papillary has the best survival, **basal/squamous (35%) is the commonest**, and neuroendocrine-like (3%) is the worst. Luminal-infiltrated tumours respond poorly to cisplatin NAC.

5 · Clinical presentation and evaluation

About one in five visible haematurias hides a cancer.

- **Painless gross haematuria occurs in 85%**. Irritative symptoms suggest CIS.
- Evaluate every gross haematuria, even on anticoagulants. Check prostate-specific antigen, because ~10% of recurrent gross haematuria is prostate cancer.
- After a negative work-up, repeat it at 3–5 years only if haematuria persists.

	Gross haematuria	Asymptomatic microhaematuria
Malignancy	~20%	2.6–5.2%
Cystoscopy	All adults	Age ≥35 years
Upper tract	Computed tomography urogram (CTU)	CTU (AUA 2012); ultrasound may replace it
Cytology/markers	Reasonable	Avoid routinely

The dipstick has sensitivity 91–100% but specificity only ~65%, so **confirm with microscopy**.

Cystoscopy and enhanced imaging

Cystoscopy with biopsy is the gold standard; flexible equals rigid. White light misses small tumours and CIS.

	Blue light (HAL–BLC)	Narrow-band imaging (NBI)
Principle	Hexaminolevulinate (HAL) makes porphyrins build up in tumour, which fluoresce red under blue light	No dye; 415 and 540 nm light is absorbed by haemoglobin, so vessels stand out
Benefit	Finds more Ta and CIS; CIS is seen only on BLC in ~¼	Better CIS detection; lower 12-month recurrence in low risk
Status	Adjunct; only HAL is FDA approved	Adjunct

Urine cytology and biomarkers

*See the paper answer **Urine cytology** (Section 1, Ch 3).*

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

ALGORITHM

1. Cytology shows high-grade cells but cystoscopy is negative.
2. Take random bladder biopsies and prostatic urethral biopsies.
3. If these are negative, take selective upper-tract washings for cytology, with or without ureteroscopy.

6 · Staging and risk groups

Stage	TNM, American Joint Committee on Cancer (AJCC) 2017
0a / 0is	Ta (noninvasive papillary) / Tis (CIS)
I	T1 – lamina propria (subepithelial connective tissue)
II	T2a inner / T2b outer half of muscularis propria, N0
IIIA	T3a (microscopic perivesical) / T3b (macroscopic) / T4a (prostatic stroma, uterus, vagina) N0; or T1–T4a N1 (single true pelvic node)
IIIB	T1–T4a N2 (≥2 true pelvic nodes) or **N3 (common iliac)**
IVA / IVB	T4b (pelvic/abdominal wall) or M1a (distant nodes) / M1b (organs)

- **The 2017 change**: regional nodes without distant metastasis are now stage III (previously IV). MCQ Common iliac nodes are N3, not M1a.
- **T1a/T1b substaging** is recommended, because ~50% of T1 are upstaged to ≥T2 at cystectomy.
- **Prostatic urethra**: mucosal CIS is not adverse in itself but escapes intravesical therapy. Only **prostatic stromal invasion is T4a** (5-year survival historically <25%).

NMIBC risk group	Definition
Low	Primary, solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta that is multiple and/or recurrent and/or >3 cm
High	Any high grade, CIS or T1

130 Management Strategies for Non–Muscle-Invasive Bladder Cancer

1 · Definition and natural history

- **Non-muscle-invasive bladder cancer (NMIBC)** is urothelial carcinoma that has not invaded the detrusor: Ta, T1 and carcinoma in situ (CIS). It makes up **70–80%** of new bladder cancers.
- Recurrence occurs in **40–60%**; progression ranges from ~6% in low-grade (LG) Ta to 17% in high-grade (HG) T1.
- **Grade, not stage, is the strongest predictor of progression**, and CIS comes second. Primary CIS progresses in **>50%**.
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in **one-third to one-half**.

> **KEY POINT**
>
> NMIBC is two diseases. LG disease is a **recurrence** problem; HG disease is a **progression** problem. Chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) treats progression, and cystectomy is for progression that cannot be controlled.

2 · Risk stratification

*Risk is **dynamic**, so reassess it at every recurrence.*

American Urological Association/Society of Urologic Oncology (AUA/SUO) 2016 (updated)

Risk	Criteria	Intervention
Low	LG solitary Ta **≤3 cm**; papillary urothelial neoplasm of low malignant potential (PUNLMP)	Single postoperative chemotherapy instillation
Intermediate	LG Ta recurring within 1 yr; LG Ta solitary >3 cm or multifocal; **LG T1**; **HG Ta ≤3 cm**	6-week induction of chemotherapy or BCG; maintenance to 1 yr
High	HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement	Induction BCG and maintenance to **3 yr**. If BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene or gemcitabine–docetaxel

EORTC and CUETO scoring models

Factor	EORTC	CUETO
Number, prior recurrence rate, T stage, concurrent CIS, grade (WHO 1973)	✓	✓
Tumour diameter	✓	—
Age, sex	—	✓

- EORTC **overestimates** risk, because its patients had little maintenance BCG. CUETO, derived in BCG-treated patients, fits better. MCQ Both use **WHO 1973** grading.

3 · Endoscopic management

TURBT: principles

- Transurethral resection of bladder tumour (TURBT) quality depends on **complete resection** and **detrusor in the specimen**; missing muscle means higher mortality.
- Image the upper tracts at diagnosis. Upper tract tumours occur in **<5%**, more often with trigonal tumour, CIS and high risk.
- **Diverticular tumour**: there is no detrusor, so invasion beyond the lamina propria is **T3a**. For HG, strongly consider partial cystectomy or RC.
- **En bloc resection** gives fewer perforations, more detrusor and better staging, but the **same recurrence**.

Repeat (second-look) TURBT

SAQ Re-look (second-look) TUR bladder tumour

Paper answer · P4

Asked as:

- Re-look TUR bladder tumour. — *Summer 2017 Paper IV Q3d (SAQ)*
- Re-look TUR bladder tumour. — *Practice Set 5 Paper IV Q3d (earlier version) (SAQ)*

Type: SAQ (10 marks)

Read as second-look (repeat or restaging) TURBT.

Definition

- A **planned second (restaging) TURBT (transurethral resection of bladder tumour)** of the primary site, including detrusor muscle, a few weeks after the first. It completes resection and corrects staging. It is a quality indicator of NMIBC (non-muscle-invasive bladder cancer) care.

Rationale

- **TURBT is a staging operation first.** Completeness and muscle in the specimen are the quality markers.
- **Incomplete resection** — residual tumour is common:
- At the resection site in **≥40%** when re-resected within weeks; tumour at first check cystoscopy in up to 45%.
- **HG (high-grade) Ta:** residual up to 50%, upstaging up to 15%.
- **Understaging:** in **HG T1**, re-resection shows worse disease or upstaging to muscle invasion in **25–30%**. This rises to about half if the first specimen had **no muscle**.
- Understaging is about twice as likely without muscle.

Indications (AUA [American Urological Association] / EAU [European Association of Urology])

Recommended	Consider	Not needed
Incomplete first (initial) resection	HG Ta, large or multifocal (AUA) (residual up to 50%, upstaging up to 15%)	Low-grade Ta

Recommended	Consider	Not needed
All T1 tumours	Outside resection with no slides for review	Primary CIS (carcinoma in situ)
No detrusor muscle in specimen (except low-grade Ta and primary CIS)		Clear muscle invasion
		Cystectomy already planned

Timing

- **2–6 weeks** after the first TURBT (EAU); **within 6 weeks** (AUA), before BCG (bacille Calmette–Guérin). This allows healing yet avoids delaying intravesical BCG.

Technique

- Resect the **primary-site scar and base deep into detrusor**, plus any residual tumour; send the base / specimens separately. Use blue light or NBI (narrow-band imaging) if available; it helps.

Benefits / evidence

- Corrects stage and **changes treatment in about one-third**.
- **Divrik RCT (randomised controlled trial, 2010)** in T1: re-TUR improved recurrence-free and progression-free survival (lower recurrence and progression).
- Better BCG response with minimal residual tumour / when BCG follows re-TUR.

Result-based plan (decision after re-TUR)

- **Residual T1 at re-resection** means (predicts) progression in up to **80%**: consider early cystectomy. `No tumour (T0) / Ta → BCG induction + maintenance Residual HG T1 → early cystectomy (or BCG with close review) Muscle invasion (≥T2) → treat as muscle-invasive: neoadjuvant chemotherapy + cystectomy`

Recent advances

- **En bloc resection** gives muscle (more detrusor) more often, less residual tumour, and may reduce re-TUR. It is not yet a proven substitute.
- Expert pathology review changes management in about one-third.
- **Also know:** maximal TURBT before trimodality therapy.

Diagram to draw: flowchart: T1 or no muscle → re-TUR 2–6 weeks → T2: cystectomy pathway / residual T1: consider cystectomy / T0–Ta: BCG; and bladder wall showing first resection scar and deep second resection into detrusor.

> **Key points to remember:**
>
> - Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
> - Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
> - Residual T1 → progression (progresses) in up to 80%.

Recommended (within 6 weeks, include muscularis propria)	Consider
Incomplete resection; all T1 (especially without muscle); no muscle in specimen	High-risk HG Ta (multifocal and/or large)

Biopsies and office management

- Random biopsies are **not** needed in low risk; consider them with positive cytology or high risk. Biopsy the prostatic urethra if there is CIS or a bladder neck/trigone tumour.
- Small (**≤1 cm**) LG recurrences can be fulgurated in the office, but this needs a **prior LG diagnosis and negative cytology**. **Active surveillance** also suits selected patients.

Enhanced cystoscopy

	Blue-light (hexaminolevulinate, HAL)	Narrow-band imaging (NBI)
Principle	Porphyrins fluoresce red under blue light; needs intravesical instillation	415 and 540 nm light is absorbed by haemoglobin; no agent
Gain	~**one-third more CIS**; smaller papillary tumours	More papillary tumours and CIS
Weakness	Cost; no 3-yr recurrence benefit in a pragmatic trial	False positives ~60%

4 · Single post-operative instillation

LAQ Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P4 LAQ "Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy" · P2 SAQ "Intravesical therapy for TCC of bladder" · P3 SAQ "Immediate post TURBT intravesical Therapy for TCC"

Introduction

- NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) is 70–80% of bladder cancers. "Urothelial carcinoma" now replaces "transitional cell carcinoma" (TCC).
- Two problems: **recurrence** (40–60%) and **progression** (about 6% in low-grade Ta to 17% in high-grade T1). Intravesical therapy follows complete TURBT (transurethral resection of bladder tumour).
- **Rule:** chemotherapy reduces recurrence only. BCG (bacille Calmette-Guérin) is the only agent that also reduces progression.

Risk-adapted use (AUA [American Urological Association]/SUO)

Risk	Criteria	Therapy
Low	LG (low-grade) solitary (primary) Ta ≤3 cm; PUNLMP	Single post-op instillation only
Intermediate	LG Ta recurrent, >3 cm or multifocal; LG T1; HG (high-grade) Ta ≤3 cm	6-week chemotherapy or BCG induction; maintenance 1 year
High	HG T1; HG Ta large, multifocal or recurrent; any CIS; LVI (lymphovascular invasion); variant histology	BCG induction + maintenance 3 years

- EAU (European Association of Urology) adds a "very high" group; EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables.

1. Single immediate instillation after TURBT

Rationale

- After TURBT, NMIBC recurs in 40–60%. Many early recurrences come from **floating cells implanting** on the raw resection bed. A single chemotherapy instillation (SI) kills them

before implantation.

Timing

- Implantation occurs within hours, so instil **as early as possible: within 6 hours** (P3: ideally within 2–6 h, in theatre or recovery). Guidelines allow up to 24 h, but MMC given at 24 h lost its benefit (no benefit at 24 hours).

Agents

Agent	Dose	Note
MMC (mitomycin C)	40 mg in 20–40 mL	Alkylating; skin rash
Gemcitabine	2 g in 100 mL, 1 h	Standard in USA; mild toxicity
Epirubicin	50 mg	Chemical cystitis
- **Also know:** pirarubicin works; thiotepa does not.		

Technique: exclude perforation; instil via catheter; retain 1–2 h; drain.

Evidence

- **Sylvester 2016 individual-patient meta-analysis:** about 35% relative reduction in recurrence; **no benefit** with >1 recurrence/year or EORTC recurrence score ≥5.
- **SWOG (Southwest Oncology Group) S0337:** gemcitabine 2 g versus saline in suspected low-grade NMIBC: 4-year recurrence 35% vs 47%; no excess toxicity.
- No effect on progression.

Who benefits

- **Low risk** (primary solitary low-grade Ta ≤3 cm): SI alone; greatest benefit.
- Intermediate risk: SI then adjuvant course. High risk: BCG follows.

Contraindications

- **Suspected perforation**, extensive resection, bleeding needing irrigation: withhold.
- **Never BCG immediately**: intravasation of live bacteria causes sepsis and death.

Complications

- Irritative symptoms (commonest), chemical cystitis, skin desquamation (MMC), contracted bladder; serious harm mainly after perforation.
- **Also know:** continuous saline irrigation may also reduce recurrence.

2. Adjuvant chemotherapy

- **MMC** (alkylating), 40 mg in 20 mL weekly ×6. Optimise it (Au, 2001): empty bladder, fluid restriction, sodium bicarbonate (alkalinised urine), concentrated dose. Skin rash.
- **Gemcitabine** (nucleoside analogue): minimal absorption, well tolerated. **Anthracyclines** (epirubicin, doxorubicin) cause chemical cystitis.
- Intermediate risk: 6-week induction, then monthly maintenance for 1 year.
- **Gem/Doce:** gemcitabine 1 g, drain, then docetaxel 37.5 mg; weekly ×6, then monthly for 2 years. Used in BCG-unresponsive disease and during BCG shortage (details under combination therapy).

3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin**. Cytokines follow (IFN-γ [interferon-gamma], IL [interleukin]-2, IL-12, TNF-α [tumour necrosis factor-alpha]). A **Th1 (T-helper 1)** response

gives cell-mediated tumour kill; TRAIL release is important.

- **Technique:** 1 vial in 50 mL saline. Start **2–4 weeks** after TURBT (re-epithelialisation). Urinalysis first; retain 1–2 hours.
- **SWOG schedule (Lamm, 2000):** 6 weekly induction doses, then 3 weekly doses at 3 and 6 months, then every 6 months to 3 years. Median recurrence-free survival 76.8 vs 35.7 months.
- **EORTC 30962:** full dose; 1 year for intermediate, 3 years for high risk. **NIMBUS:** reduced frequency was inferior.
- CIS complete response up to 84%; BCG beats MMC only with maintenance. Only maintenance BCG reduces progression.
- **Contraindications:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk intravasation); previous BCG sepsis; total incontinence; immunosuppression (now relative). Relative: active UTI (urinary tract infection), liver disease, previous TB (tuberculosis).

BCG toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Anticholinergics, analgesics, NSAIDs (non-steroidal anti-inflammatory drugs)
2	Severe or >48 h; persistent fever	Culture; isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; isoniazid + rifampicin + ethambutol (triple therapy); steroids only with cover

- BCG is resistant to **pyrazinamide**. BCG sepsis 0.5%.

BCG failure

- **Adequate BCG:** ≥5 of 6 induction + ≥2 of 3 maintenance doses.
- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite adequate BCG, or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Also know:** refractory, intolerant, exposed, delayed relapse.

BCG-unresponsive options

- **Timely radical cystectomy** is the standard (EAU, AUA).
- If unfit or refusing:
- **Pembrolizumab** (KEYNOTE-057): IV every 3 weeks; CIS complete response 41%.
- **Nadofaragene firadenovec:** adenovirus with interferon-α2b gene, every 3 months; CIS complete response 53%.
- **N-803:** IL-15 superagonist with BCG; CIS complete response 71%.
- Gem/Doce; valrubicin; **TAR-200** (gemcitabine-releasing device); cretostimogene (trials).

Device-assisted therapy

- **EMDA (electromotive drug administration):** current increases MMC tissue delivery 4–7-fold. Sequential BCG + EMDA-MMC improved outcomes in T1.
- **Chemohyperthermia:** RITE (radiofrequency, 42 °C); HIVEC (heated recirculating MMC).
- **Also know:** UGN-102 MMC gel; interferon with low-dose BCG.

Combination intravesical therapy

Rationale

- Single agents fail: chemotherapy reduces **recurrence** only; BCG reduces **progression** but fails in about a third.
- Combinations add **different mechanisms**, overcome resistance, and are useful during **BCG shortage** and in **BCG-unresponsive** NMIBC.

1. Sequential chemotherapy (doublets)

Regimen	Method	Results
Gem/Doce (gemcitabine–docetaxel)	Gemcitabine 1 g in 50 mL (60–90 min), drain, then docetaxel 37.5 mg (60–120 min); weekly ×6, then monthly for 2 years	High-grade recurrence-free survival about 65% at 1 year, 52% at 2 years; CIS complete response 60% at 1 year; **84% avoid cystectomy**
Gem/MMC	Gemcitabine 1 g, then MMC 40 mg	Durable recurrence-free about 38%

- **Why it works:** gemcitabine (antimetabolite) and docetaxel (microtubule stabiliser) act on different cell-cycle phases.
- Oral sodium bicarbonate and antiemetics aid tolerance. No level 1 evidence; phase III BRIDGE compares it with BCG in BCG-naive patients.

2. Chemotherapy with BCG

- Sequential MMC–BCG: **no benefit** over BCG alone.
- **Exception: BCG alternating with EMDA-MMC** in T1: lower recurrence and progression than BCG alone. A later trial was non-significant.
- Chemohyperthermia (RITE, radiofrequency-induced thermochemotherapeutic effect) potentiates BCG.

3. BCG with immunomodulators

- **BCG + interferon-α (α2b):** no gain over BCG; allows BCG dose reduction to one-third.
- **N-803 (nogapendekin alfa)** = an IL-15 (interleukin-15) superagonist activating CD8 T and NK (natural killer) cells. With BCG in BCG-unresponsive CIS: complete response 71%, 45% at 12 months. FDA (Food and Drug Administration)-approved in 2024 (chapter: "under review").

4. BCG with checkpoint inhibitors (BCG-naive high risk)

- **CREST** (subcutaneous sasanlimab + BCG) and **POTOMAC** (durvalumab + BCG): improved event-free or disease-free survival over BCG alone.
- **ALBAN** (atezolizumab + BCG): negative. KEYNOTE-676 (pembrolizumab) ongoing.

5. Device-assisted and novel combinations

- **Chemohyperthermia** (MMC at 42 °C: RITE; HIVEC, hyperthermic intravesical chemotherapy) increases uptake.
- **TAR-200:** gemcitabine-releasing intravesical device (± cetrelimab); very high early complete response in BCG-unresponsive CIS (SunRISe-1).
- **Cretostimogene** (oncolytic adenovirus) + pembrolizumab: about two-thirds complete response at 12 months.

Toxicity of combinations

- Additive cystitis; immune-related events with checkpoint inhibitors.

Diagrams to draw: BCG mechanism (fibronectin, cytokines, Th1, kill) with SWOG schedule timeline; Gem/Doce sequence timeline (instil, dwell, drain, instil); floating cells implanting on the resection bed and killed by instillation.

Key points to remember:

- Single instillation within 6 h (at most 24 h); never BCG immediately; withhold if perforation; best for low-risk solitary tumours.
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Only maintenance BCG reduces progression; SWOG 3-year schedule; NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

Also, from the condensed book (not in the paper answer):

- Give mitomycin C (MMC) within **6 h** (level 1 evidence). After perforation it risks chemical cystitis, contracture and rare deaths.

5 · Intravesical chemotherapy

*See the paper answer **Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation** (Section 12, Ch 130).*

Also, from the condensed book (not in the paper answer):

- Chemotherapy reduces **recurrence**, with **no proven effect on progression**. In intermediate risk, MMC or gemcitabine equals BCG. Give a 6-week induction, then monthly for 1 yr.
- A quinolone for 24–48 h after each BCG dose reduces cystitis without loss of efficacy.

Agent	Class	Dose	Signature toxicity
Mitomycin C	Alkylating	40 mg/20 mL weekly ×6	Rash, severe skin reaction, contracted bladder 5%
Gemcitabine	Nucleoside analogue	2 g/100 mL	Mild; minimal absorption
Doxorubicin / epirubicin	Anthracycline	50 mg/50 mL	Chemical cystitis, contracted bladder
Valrubicin	Anthracycline	800 mg/75 mL	FDA approved for BCG-refractory CIS; complete response (CR) ~1/5
Thiotepa	Alkylating	60 mg/30 mL	Myelosuppression
Docetaxel	Taxane	75 mg/100 mL	Myelosuppression

- **Gemcitabine–docetaxel (Gem/Doce)** is increasingly used in BCG-naive HG disease.

6 · Bacille Calmette-Guérin (BCG)

BCG prevents progression, but only with maintenance.

Mechanism

BCG binds fibronectin, releasing cytokines that recruit immune cells and produce a **T-helper 1 (Th1)** cytotoxic response. No strain is superior.

Administration

ALGORITHM

1. Wait **2–4 weeks** after TURBT (re-epithelialisation).
2. Check urinalysis before each dose to exclude infection and gross bleeding.
3. Instil 1 vial in 50 mL saline by catheter and retain it for 1–2 h, restricting fluids beforehand.
4. If catheterisation is traumatic, delay several days.

Contraindications

Absolute	Relative
Immediately after TUR; gross haematuria; traumatic catheterisation; previous BCG sepsis; immunosuppression*; total incontinence	Urinary tract infection; liver disease (isoniazid cannot then be given for sepsis); previous tuberculosis; anti-TNF drugs; poor performance status; advanced age

- Most absolute contraindications mean a **breached urothelium, allowing intravasation**. *Transplant and steroid patients may receive it, with weaker response.

Efficacy

- BCG is the **only intravesical agent that reduces progression**, and only with **maintenance**.
- In CIS, the initial CR is up to **84%**, and about half are durable.
- A second induction for persistent CIS gives a further 30–50% CR, but beyond 2 courses progression risk rises to 20–50%. **The response at 6 months predicts prognosis.**

Schedule

ALGORITHM

1. **SWOG (Southwest Oncology Group; Lamm 2000)**: give induction weekly ×6.
2. Then give 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**.
3. If high risk, give full dose for 3 yr. If intermediate risk, give 1 yr (EORTC 30962).

Toxicity management

Grade	Features	Management
1	Cystitis, mild haematuria, fever <38.5°C, lasting <48 h	Symptomatic; consider culture
2	Severe symptoms or >48 h	Culture, chest X-ray, liver tests; isoniazid, rifampicin and pyridoxine; resume at ½–⅓ dose

Grade	Features	Management
3	Haemodynamic change, persistent fever; allergic (arthralgia, rash); organ involvement (liver, lung, kidney)	**Stop BCG**; isoniazid, rifampicin and ethambutol; prednisone only **with** antimycobacterial cover; infectious disease input

- MCQ BCG is resistant to **pyrazinamide**. Asymptomatic granulomatous prostatitis needs no treatment.

BCG shortage

ALGORITHM

1. Prioritise high-risk patients.
2. Reduce the dose, especially in maintenance, and split vials.
3. Use chemotherapy: single agent for intermediate, Gem/Doce for high risk. Early RC for the highest risk.

7 · Recurrence after BCG

SAQ BCG therapy and BCG failure

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "BCG therapy in Bladder Cancer" · P2 SAQ "BCG failure"

Introduction

- BCG (bacille Calmette-Guérin), a live attenuated mycobacterium (Morales, 1976), is standard for high-risk NMIBC (non-muscle-invasive bladder cancer) and the only intravesical agent that reduces both **recurrence and progression**.

Mechanism

- BCG binds **fibronectin** on the urothelium; cytokines (IFN-γ [interferon-gamma], IL-2 [interleukin-2] and TNF-α [tumour necrosis factor-alpha]) and immune cells follow.
- A **Th1 (T-helper 1)** response gives cell-mediated tumour kill. It needs repeated stimulation, which is why **maintenance** is essential.

Indications

- **High-risk NMIBC:** HG (high-grade) T1, large/multifocal/recurrent HG Ta, **CIS (carcinoma in situ)** (initial complete response up to 84%).
- **Intermediate risk:** option. Not for low risk.

Technique and schedule

- Start **2–4 weeks after TURBT (transurethral resection of bladder tumour)**; urinalysis first; atraumatic catheter; retain **1–2 hours**.
- **Induction:** weekly × 6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** weekly × 3 at 3 and 6 months, then every 6 months to **3 years** (recurrence-free survival 76.8 vs 35.7 months).
- **EORTC (European Organisation for Research and Treatment of Cancer) 30962:** intermediate risk, full dose for 1 year; high risk, full dose for 3 years. **NIMBUS:** reduced frequency was inferior.

Contraindications

- **Absolute:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk **intravasation**); previous BCG sepsis; total incontinence; immunosuppression (now debated).
- **Relative:** UTI (urinary tract infection), liver disease, prior TB (tuberculosis).

Toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Symptomatic
2	Severe or >48 h	Isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; add ethambutol; steroids only with cover
- BCG is resistant to **pyrazinamide**.		

BCG failure: definitions

- "Failure" was poorly defined; consensus categories now guide treatment.

Category	Definition	Implication
Adequate BCG	≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a second induction; for T1, ≥5 of 6 induction alone	Threshold for "unresponsive"
BCG-unresponsive	Persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within **6 months (Ta/T1) or 12 months (CIS)** of last dose; **HG T1 at first 3-month check**	More BCG futile; cystectomy
BCG-resistant	CIS or HG Ta at 3 months after induction	Reinduction: 30–50% respond
Delayed relapse	HG relapse beyond these windows	Rechallenge may work
BCG-intolerant	Toxicity prevented adequate BCG	Salvage chemotherapy

- **Why BCG fails:** inadequate dosing; PD-L1 (programmed death-ligand 1) upregulation on tumour cells.

Evaluation of BCG failure

- Cystoscopy, cytology, **re-TURBT** and biopsies to exclude invasion.
- **Extravesical disease:** prostatic urethral biopsy (24–30%) and CT urography (upper tract 0.8–25%).

Management of BCG failure

1. **Timely radical cystectomy** is standard (EAU [European Association of Urology], AUA [American Urological Association]): HG T1 at 3 months progresses in 82%; CIS failure in about 50%.
2. **Bladder sparing** (unfit or unwilling):

Option	Key result (CIS complete response, CR)
Pembrolizumab IV (KEYNOTE-057)	41% at 3 months, 19% at 12 months; grade 3–4 toxicity 13%
Nadofaragene firadenovec (adenovirus carrying IFN-α2b [interferon alpha-2b] gene) every 3 months	53% at 3 months, 24% at 12 months
Gemcitabine–docetaxel sequential	60% at 1 year; 84% avoid cystectomy (retrospective)
N-803 + BCG (IL-15 [interleukin-15] superagonist)	45% at 12 months

- Salvage CR rates halve by 12 months.
- **Also know:** BCG-refractory (older term); gemcitabine, gemcitabine–mitomycin, valrubicin, chemohyperthermia, EMDA (electromotive drug administration), chemoradiation; TAR-200; cretostimogene.

Recent advances

- PD-1/PD-L1 (programmed death-1/ligand-1) inhibitor + BCG in BCG-naive high risk improved event-free survival (CREST, POTOMAC).
- TAR-200 gemcitabine-releasing device.

Diagrams to draw: SWOG schedule timeline; flowchart: HG recurrence after BCG, was BCG adequate?, unresponsive, then cystectomy or bladder sparing.

> **Key points to remember:**
> - Only BCG reduces progression; maintenance is essential.
> - Never immediately after TURBT; start at 2–4 weeks.
> - Adequate BCG = 5/6 induction + 2/3 maintenance.
> - Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
> - Timely cystectomy is standard; salvage CR rates halve by 12 months.

Also, from the condensed book (not in the paper answer):

- After failure, check the prostatic urethra and upper tracts, because recurrence there develops in about half on long follow-up.

> **KEY POINT**
>
> **Trap:** HG T1 at 3 months after induction alone counts as unresponsive. CIS or HG Ta at 3 months is "resistant", so give more BCG.

Bladder-sparing options (unfit for or declining RC)

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
Pembrolizumab (KEYNOTE-057)*	Anti-PD-1; 200 mg intravenously every 3 weeks up to 2 yr	41%, then 19%	63%; grade 3–4 toxicity 13% (immune)
Nadofaragene firadenovec*	Adenoviral interferon-α2b gene; intravesical 3-monthly	53%, then 24%	71%; grade 3–4 4%

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
N-803 + BCG	Interleukin-15 superagonist	71% (any time), then 45%	~85%
Gem/Doce	Weekly ×6, monthly ×2 yr	—, then 60% (43% at 2 yr)	84%; no level 1 evidence

*FDA-approved. CR roughly halves by 12 months, and none matches cystectomy.

8 · Early radical cystectomy in high-risk NMIBC

- T1 failing BCG at 3 months progresses in **82%**, and CIS in ~50%.
- Immediate RC for clinical T1 gives 10-yr disease-free survival of **92%** (64% if understaged). Even a 12-week delay worsens survival.
- **Patients who progress to muscle invasion under surveillance do worse than those with de novo muscle-invasive disease.**

ALGORITHM

1. **Strongly consider RC** if: BCG-unresponsive HG; persistent HG T1 on re-resection; T1 with CIS, LVI or variant histology; deep lamina propria invasion; diffuse CIS.
2. Also if: diverticular HG tumour; distal ureteric or prostatic ductal involvement; incompletely resectable tumour; patient preference.

9 · Surveillance

Cystoscopy with cytology is the gold standard. **Tumour at the 3-month cystoscopy** and number at first resection best predict recurrence.

Risk	Cystoscopy	Cytology	Upper tract imaging
Low	At 3 months, then annually from 12 months; consider stopping at **≥5 yr**	Consider	Only if haematuria
Intermediate	3–6-monthly for 1–2 yr, then 6–12-monthly for 2 yr, then annually	Consider	Consider every 1–2 yr
High	**3–4-monthly for 2 yr**, then 6-monthly for 2 yr, then annually **for life**	Same schedule	Annually for 2 yr, then lengthen

- Restart the clock at each recurrence, using the risk of the **latest resected** tumour.
- Positive cytology with negative cystoscopy needs biopsies and upper-tract work-up.
- Urinary markers (UroVysion FISH is the most specific) do not replace cytology.
- **Extravesical recurrence**: upper tract UC occurs in <1% at low risk and ~10% at high risk. The prostatic urethra is involved in 20–40% of high-risk patients at 10 yr; HG ductal disease needs cystoprostatectomy.

10 · Prevention

- **Advise stopping smoking** at every visit, although its effect on recurrence is uncertain.

11 · Landmark trials

Trial	Design	Result	Take-home
SWOG S0337	Immediate gemcitabine 2 g vs saline, suspected LG	4-yr recurrence 35% vs 47% (LG 34% vs 54%); no extra toxicity	Gemcitabine is the standard single instillation
SWOG maintenance (Lamm 2000)	Induction with or without maintenance BCG, recurrent Ta/T1/CIS	Median RFS 76.8 vs 35.7 months; only 16% completed	Maintenance is essential
EORTC 30962	Full vs ⅓ dose; 1 vs 3 yr	Toxicity stops ~7–8%	Intermediate: full dose 1 yr; high: full dose 3 yr
NIMBUS	Reduced-frequency (9 doses) vs standard, HG	Recurrence doubled; stopped early	Do not reduce frequency
KEYNOTE-057	Single-arm pembrolizumab, BCG-unresponsive CIS	CR 41% (3 mo), 19% (12 mo)	First systemic therapy approved for NMIBC

131 Management of Muscle-Invasive and Metastatic Bladder Cancer

1 · Natural history and histology

MIBC kills through micrometastases, so cure needs systemic treatment too.

- **20–30%** of new bladder cancers are muscle-invasive (MIBC), and **~12%** are node-positive or metastatic.
- **Secondary** MIBC (**~20%**), which has progressed from non-muscle-invasive disease, does worse than **primary** MIBC, with more understaging and less response to neoadjuvant chemotherapy (NAC).
- Most deaths after local therapy occur within **2 years**, from micrometastases present at diagnosis.
- Urothelial carcinoma (UC) is 90%. **Radical cystectomy (RC) is standard whatever the histology**; subtype changes only the use of NAC and radiation.

Variant	Key points	Treatment
Squamous (5%)	Schistosomiasis, chronic cystitis	Poor chemo- and radiosensitivity, so up-front RC with lymphadenectomy
Adenocarcinoma (2%)	Commonest in exstrophy; urachal; exclude colorectal or breast metastasis	RC (urachal: partial cystectomy)
Small cell	High stage; ectopic ACTH, hypercalcaemia	**Chemosensitive**, so NAC with etoposide and cisplatin, then RC
Micropapillary	Frequent lymphovascular invasion (LVI)	If predominant, up-front RC
Mixed UC + variant	Quantify the variant	UC-predominant: NAC; variant-predominant: RC

2 · Clinical staging

Stage by resection depth, examination under anaesthesia (EUA), computed tomography (CT) with excretory phase, then chest imaging. In cT2, pathology upstages **40–54%**; LVI, variant histology, female sex and secondary MIBC predict upstaging.

TURBT

- Resect all visible tumour if safe, with a separate deep specimen; this is needed for bladder preservation, and pT0 at RC improves survival.
- If high-grade T1 or no muscle, repeat resection within **6 weeks**.
- **Prostatic urethral biopsy** (trigone or bladder-neck tumour, carcinoma in situ (CIS), multifocal disease) is a full-thickness loop from mid-prostate to verumontanum at **5 and 7 o'clock**.
- **Frozen urethral margin at cystoprostatectomy** is mandatory if the prostatic biopsy was positive, and advisable whenever a neobladder is planned; a neobladder is made only if it is negative.

Bimanual EUA

Examine before and after resection with the bladder empty; EUA more often understages than overstages.

cT	Bimanual finding
T2a	Nonpalpable
T2b	Induration, no three-dimensional mass
T3a	Mobile three-dimensional mass
T4a	Invades prostate, vagina, rectum
T4b	Fixed to pelvic sidewall

Imaging and bloods

- **Image before resection**, as post-resection inflammation mimics T3. CT nodal sensitivity is only **30–50%**; hydronephrosis suggests extravesical disease.
- Magnetic resonance imaging stages T best. The **Vesical Imaging-Reporting and Data System (VI-RADS)** grades the muscularis line: intact in 1–3, interrupted in 4, into fat in 5. A score **≥3** is suspicious.
- FDG PET/CT is reserved for equivocal imaging.
- Chest CT is needed for all; bone or brain imaging only with symptoms or raised alkaline phosphatase. Renal function decides NAC and orthotopic diversion.

3 · Pathological staging (AJCC 8th) and prognostic markers

Category (American Joint Committee on Cancer)	Definition
T2a / T2b	Inner / outer half of muscularis propria
T3a / T3b	Perivesical fat microscopic / macroscopic
T4a / T4b	Prostatic stroma, seminal vesicles, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	Single true-pelvis node (perivesical included) / multiple / **common iliac**
M1a / M1b	Nodes beyond common iliacs / non-nodal
Stage II / IIIA / IIIB	T2N0 / T3–T4a N0 or T1–T4a N1 / N2–N3
Stage IVA / IVB	T4b any N, or M1a / M1b

- Traps: common iliac is **N3, not M1**; T4b N0 is stage IVA. pT4a needs **stromal** invasion, since urethral or ductal CIS does not upstage.
- **Subtypes**: luminal survives best, **basal gains most from NAC**, and p53-like is chemoresistant.
- **Circulating tumour DNA**: over a third stay positive after RC and do badly; in IMvigor010, adjuvant atezolizumab helped only them.

4 · Radical cystectomy: principles and extent

- **RC with bilateral pelvic lymph node dissection (PLND) is standard for resectable MIBC**, ideally after cisplatin NAC. Delay >**12 weeks** worsens survival.
- **Men**: bladder, perivesical tissue, prostate and seminal vesicles are removed. **Women** (anterior exenteration): bladder, urethra, uterus, cervix, ovaries, tubes and anterior vaginal wall.

- **Male organ-sparing** needs no bladder-neck disease and negative prostatic biopsies; incidental prostate cancer is found in ¼–½.
- **Female organ-sparing** needs no base or trigone tumour. Keeping the uterus and anterior vagina supports a neobladder; keep ovaries premenopause unless there is breast or ovarian cancer risk.

> **KEY POINT**
>
> Spare an organ only where it is oncologically silent. The gatekeepers are, in men, the prostatic urethra and bladder neck; in women, the bladder neck, base and anterior vaginal wall.

5 · Pelvic lymph node dissection

Nodal status is the strongest predictor of survival.

LAQ Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types" · P4 LAQ "Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer?"

Introduction

- MIBC (muscle-invasive bladder cancer; TCC, transitional cell carcinoma, now called urothelial carcinoma) is cT2–T4a disease.
- Nearly half of patients treated by RC (radical cystectomy) alone develop metastases from occult micrometastases, so cure needs local plus systemic therapy, planned by an MDT (multidisciplinary team).
- Cure improves with **systemic therapy before surgery** and a **meticulous lymph node dissection** (accurate staging, possible therapeutic value).
- UTUC (upper tract urothelial carcinoma) behaves alike, but RNU (radical nephroureterectomy) removes a kidney, changing chemotherapy timing.

Staging

- Complete TURBT / Re-TURBT (transurethral resection of bladder tumour) with detrusor muscle; **MRI (magnetic resonance imaging) with VI-RADS** (Vesical Imaging-Reporting and Data System; ≥3 suggests invasion); CT (computed tomography) urogram and chest CT; bone scan if symptomatic.
- Assess fitness and cisplatin eligibility.

Management algorithm

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cT2–T4a N0 M0 ├─ Cisplatin-eligible → cisplatin NAC (neoadjuvant chemotherapy) │ → RC + bilateral PLND (pelvic lymph node dissection) ├─ Cisplatin-ineligible → RC + PLND directly │ (pT3–4 or N+ after either → adjuvant nivolumab or chemotherapy) ├─ Selected, motivated → trimodality therapy (TMT) └─ Unfit → chemoradiation / palliative radiotherapy cN+ → induction chemotherapy → consolidation RC in responders M1 → systemic therapy
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1. Cystectomy pathway: radical cystectomy (standard of care for cT2–T4a N0 M0)

- **Cisplatin-based NAC:** ddMVAC (dose-dense methotrexate, vinblastine, doxorubicin, cisplatin) or GC (gemcitabine–cisplatin) ×4 (details below).
- **Cisplatin-ineligible:** direct cystectomy.
- **RC + standard bilateral PLND:** obturator, external and internal iliac nodes. **SWOG S1011:** extended PLND gave no survival benefit (details below).
- Men: cystoprostatectomy. Women: anterior exenteration; organ-sparing in selected cases.
- Do not delay beyond **12 weeks** from diagnosis (worse survival). Diversion: ileal conduit, orthotopic neobladder or continent pouch.
- **Adjuvant:** cisplatin chemotherapy if no NAC; **nivolumab** for pT3–4 or N+ (details below).

2. Bladder preservation: TMT (trimodality therapy)

- Maximal (visibly complete) TURBT + RT (radiotherapy) (64 Gy, or 55 Gy in 20 fractions) + radiosensitiser. **BC2001:** mitomycin C + 5-FU (fluorouracil) improved 2-year locoregional disease-free survival (67% vs 48%).
- Best candidates: cT2, solitary tumour, **no CIS (carcinoma in situ), no hydronephrosis**, good bladder function.
- 5-year cancer-specific survival 50–84%; salvage cystectomy 10–30%.
- **Partial cystectomy:** solitary dome or urachal tumour, no CIS.
- **Also know:** chemotherapy alone or radical TURBT alone is not recommended.

3. Metastatic disease

- First line: **EV (enfortumab vedotin) + pembrolizumab** (EV-302: about doubled overall survival (OS) vs platinum). Alternatives: platinum chemotherapy then **avelumab maintenance** (JAVELIN Bladder 100); erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

Neoadjuvant chemotherapy (NAC) in MIBC

Rationale

- Treats micrometastases early, while tumour burden is low; better tolerated before surgery, so more patients complete it.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

Eligibility: cT2–T4a N0 M0 and cisplatin-fit. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, grade ≥2 hearing loss, grade ≥2 neuropathy, NYHA (New York Heart Association) class III–IV failure.

Trial	Regimen	Result
SWOG 8710 (Southwest Oncology Group)	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC vs RC	pT0 38% vs 15%; median survival 77 vs 46 months (P4: 5-year OS 57% vs 43%); **P = 0.06**
BA06 30894 (EORTC/MRC, European Organisation for Research and Treatment of Cancer/Medical Research Council)	CMV (cisplatin, methotrexate, vinblastine) ×3	16% relative reduction in mortality; 10-year OS 36% vs 30%
ABC meta-analysis (Advanced Bladder Cancer)	Cisplatin combinations	OS HR (hazard ratio) 0.86: **5% absolute 5-year survival gain**

Trial	Regimen	Result
VESPER	ddMVAC ×6 vs GC ×4	ddMVAC better than GC: organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity
NIAGARA	Perioperative durvalumab + GC, then adjuvant durvalumab, vs GC	Better event-free survival (HR 0.68) and OS (HR 0.75)

- **Guidelines (EAU, European Association of Urology; AUA, American Urological Association; NCCN, National Comprehensive Cancer Network):** cisplatin-based NAC for all eligible patients; ddMVAC 3–6 cycles or GC 4 cycles.
- **No survival benefit:** single-agent cisplatin, carboplatin regimens, neoadjuvant radiotherapy.
- **Cisplatin-ineligible:** proceed straight to RC; neoadjuvant pembrolizumab (PURE-01) and atezolizumab (ABACUS) are investigational.
- The chapter calls neoadjuvant immunotherapy unproven; NIAGARA has since made perioperative durvalumab + GC an option.

Adjuvant therapy in MIBC

	Neoadjuvant	Adjuvant
Selection	Clinical stage (understaging risk)	True pathological stage, but still overtreats
Delivery	Better tolerated	Post-RC morbidity; a quarter never receive it
Evidence	Level 1	Weaker: small, early-closed trials

Adjuvant chemotherapy

- Trials were small or closed early. **ABC 2022 update:** cisplatin-based adjuvant chemotherapy improves 5-year OS by about 6%. Benefit greatest in node-positive and extravesical disease.
- Use: cisplatin-naive patients (no NAC) with **pT3–4 or N+**: ddMVAC or GC.

Adjuvant immunotherapy

- **CheckMate 274:** nivolumab for 1 year after RC (pT3–4a or N+, with or without prior NAC): median disease-free survival **20.8 vs 10.8 months**. It also included UTUC.
- **IMvigor010** (atezolizumab): negative overall (included pT2); benefit in ctDNA (circulating tumour DNA)-positive patients.
- Use nivolumab if cisplatin-ineligible or declining chemotherapy, or with ypT3 or worse after NAC.
- **Also know:** adjuvant pelvic radiotherapy (Cairo trials) improves pelvic control in pT3–4; ctDNA-guided therapy is emerging.

Lymphadenectomy in MIBC

- **Primary drainage:** obturator, internal and external iliac, presacral. **Secondary:** common iliac, para-aortic, paracaval.
- Node positivity: pT2 8–26%, pT3 33–50%, pT4 36–80%.

Template	Extent
Standard	Bilateral external and internal iliac and obturator nodes; genitofemoral nerve laterally, node of Cloquet caudally, up to the common iliac bifurcation
Extended	Adds common iliac and presacral nodes, up to the aortic bifurcation
Super-extended	Up to the inferior mesenteric artery

- **Evidence:** German phase III (limited vs extended): no survival benefit. **SWOG S1011** (most had NAC): median nodes 24 vs 39, **no difference** in disease-free survival (HR 1.10) or OS; more grade ≥3 events. So **standard bilateral PLND** is the standard of care.
- **Node count:** fewer than 10 nodes doubled the risk of death (SWOG 8710 reanalysis). **Also know:** lymph node density, extranodal extension (prognostic).
- Unexpected nodes at RC: frozen section, proceed if resectable within the pelvis. Clinically node-positive: chemotherapy first, then RC in responders.

Neoadjuvant and adjuvant therapy in UTUC

- **Why neoadjuvant:** CKD (chronic kidney disease) is common and GFR falls after RNU, losing cisplatin eligibility. Give platinum **while both kidneys work**.
- Biopsy under-stages, so select by high-risk features (high grade, invasion on imaging, hydronephrosis).
- **EA8141** (phase II): accelerated MVAC, or gemcitabine–carboplatin if CrCl (creatinine clearance) 30–49; complete response 14%, below ypT2N0 60%.
- **AUA/SUO (Society of Urologic Oncology):** neoadjuvant platinum is a standard option for high-risk UTUC, especially if postoperative CrCl will fall below 50.
- **Adjuvant: POUT** (phase III, pT2–T4 or N+ after RNU): gemcitabine–platinum vs surveillance; disease-free survival HR 0.45, 3-year 71% vs 46%. Weaker with carboplatin, favouring neoadjuvant cisplatin. Adjuvant nivolumab (CheckMate 274) also included UTUC.

Lymphadenectomy in UTUC

- Node positivity 12–25%. Lymph drains **right to left and top to bottom**, so the template depends on side and level.

Primary site	Right	Left
Renal pelvis, upper two-thirds ureter	Hilar, paracaval, retrocaval, **interaortocaval**	Hilar, para-aortic
Lower third ureter	Ipsilateral common, external, internal iliac, obturator, presacral	Same

- Value: pN0 better than pNx; ≥8 nodes improved disease-specific survival above pT1. Indicated for high-grade or invasive disease; robotic yield equals open. Risk: chylous leak.

Urinary diversion options

- Ileal conduit (simplest, commonest), continent cutaneous pouch, or **orthotopic neobladder**. No diversion is proven superior for QoL (quality of life).

Neobladder: principles

1. **Reliable outlet:** intact rhabdosphincter, tumour-free urethra (frozen section).

2. **Low pressure:** complete **detubularisation** and cross-folding abolish peristalsis; a **sphere** gives most volume (Laplace).
3. **Capacity:** 44 cm of folded ileum holds <200 mL at first, **500–600 mL by 1 year**. - **Contraindications:** positive urethral margin, stricture, eGFR (estimated GFR) <35–40 mL/min, liver failure, inability to self-catheterise. Age, obesity and nodes are not bars.

Critical evaluation of neobladder types

Type	Bowel; ureteric technique	Strengths	Weaknesses
Studer	40–44 cm ileum + isoperistaltic afferent limb; refluxing	Simple, no staples, takes short ureters; stricture <3%; good long-term function	Night incontinence; reflux on cystogram (harmless)
Hautmann W	About 70 cm ileum; refluxing since 1997	Largest capacity, early continence; stricture fell 9.5% → 1%	More retention, acidosis, B12 loss
T pouch (Stein–Skinner)	44 cm + tunnelled T-limb flap valve	Antireflux; accepts dilated ureters	Complex; more secondary procedures (USC [University of Southern California] RCT)
Abol-Enein serous tunnel	40 cm W, serosal troughs	Antireflux; stricture about 3.8%	Long ureters needed; no dilated ureters
Hemi-Kock	Intussuscepted nipple	Good early results	Late nipple stenosis, stones; abandoned
Camey II; Padua (VIP)	65 cm or 42 cm ileum; Le Duc tunnel	Early designs	Le Duc stricture 7–29%
Mainz III; right colon; sigmoid (Reddy)	Colon ± ileum	When ileum short or diseased	High-pressure contractions, poor compliance, more leakage

Critical points

- **Reflux vs obstruction:** Valsalva voiding pressurises the whole system equally, so reflux is benign; every antireflux valve can obstruct. **USC RCT** (484 patients): Studer = T pouch for renal function and infection; T pouch needed more secondary procedures. Hence **Studer and Hautmann** are the most popular.
- **Segment:** ileum is most compliant; colon and stomach give high pressures.
- **Capacity trade-off:** a larger pouch gives earlier continence but more retention and metabolic load.
- **Robotic intracorporeal neobladder:** comparable outcomes; long-term data awaited.

Neobladder outcomes, complications and follow-up

- **Day continence 80–90%; night 20–50%.** Retention: men 4–10%, **women 20–30%**.
- **Late:** hyperchloraemic metabolic acidosis, B12 deficiency, mucus, stones, ureteroileal stricture, pouch rupture, urethral recurrence (men about 7% at 5 years).
- **Follow-up:** blood gas, B12, upper-tract imaging, urethral cytology if CIS.

Diagrams to draw: bladder wall T-staging (T2a, T2b, T3, T4) beside the standard PLND template; standard vs extended pelvic templates, and right vs left UTUC templates; Studer pouch with afferent limb and urethral anastomosis.

Key points to remember:

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

Also, from the condensed book (not in the paper answer):

- About **25%** are node-positive at RC. Recurrence is **70–80%** if node-positive vs ~30% if node-negative.
- Standard PLND reaches the common iliac bifurcation; extended reaches the aortic bifurcation. About 10 nodes is the usual cut-off for pelvic recurrence risk.

6 · Intraoperative decision making

- If **clinically node-positive**, give cisplatin chemotherapy first, then **consolidative RC** in responders.
- **Unexpected nodes**: proceed only if completely resectable within the true pelvis. **Abort RC** if nodes are unresectable, the bladder is fixed to the sidewall or the rectosigmoid is invaded.
- **Ureteric margin** is involved in 6–8% and raises upper-tract recurrence (4–10%, mostly at 2–4 years). Frozen section is not mandated, but send one for gross tumour.
- **Male urethral recurrence** is 1–8% and highest with **prostatic stromal invasion** (~30%), so consider urethrectomy then. A positive pre-op prostatic biopsy does **not** bar a neobladder; the **frozen urethral margin decides**.
- **Female urethra** is involved in 2–12%. Urethrectomy is mandatory only for T4 disease involving urethra or vagina; send a proximal urethral frozen section in every neobladder candidate.

7 · Outcomes after RC

Stage	5-year disease-specific survival
≤pT1 / pT2 / pT3 / pT4	~90% / 70–81% / 40–52% / 16–44%
N0 / N+	62–80% / 21–35%

- **~50%** ultimately die of bladder cancer, mostly from metastases within 2 years.
- **Pelvic recurrence**: low risk (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 with positive margins or <10 nodes) **39–46%**.

8 · Neoadjuvant therapy

Cisplatin NAC gives a level 1 survival gain.

*See the paper answer **Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy** (Section 12, Ch 131).*

Also, from the condensed book (not in the paper answer):

- About half of patients are cisplatin-ineligible, and they go straight to local therapy.
- Neoadjuvant immunotherapy (IO) phase II trials: **ABACUS** (atezolizumab) complete response 31%; **PURE-01** (pembrolizumab) pT0 42%.

Criterion	Cisplatin-ineligible (any one; ~50%)	Platinum-ineligible
Renal	Glomerular filtration rate **<60 mL/min**	Creatinine clearance <30 mL/min
ECOG performance status	≥2	≥2 (text) / ≥3 (table)
Hearing loss / neuropathy	Grade ≥2 / ≥2	— / grade ≥3
Heart failure	NYHA III–IV	NYHA III–IV; multiple high-grade comorbidities

- The NCCN prefers dose-dense methotrexate, vinblastine, doxorubicin and cisplatin (ddMVAC).
- **Neoadjuvant IO** blocks the PD-1/PD-L1 checkpoint and was not yet recommended in the source. †NIAGARA has since shown that perioperative durvalumab with gemcitabine–cisplatin (GC) improves event-free and overall survival (OS).
- Neoadjuvant radiation downstages but gives no survival gain.

9 · Adjuvant therapy

- The target is **pT3–4 or N+**. Post-RC morbidity prevents delivery in ~¼, so neoadjuvant treatment is preferred.
- **Chemotherapy** is not standard, but discuss ddMVAC or GC in **cisplatin-naive** pT3–4/N+ patients.
- **Nivolumab for 12 months** is given for ≥pT3/N+ after NAC, or if cisplatin-ineligible or declining. IMvigor010 (atezolizumab) was negative.
- **Radiation** may be considered for pT3/N+, where local failure is 20–40%.

10 · Bladder preservation

Selected patients keep their bladder with similar survival.

Trimodality therapy (TMT)

LAQ Bladder-preserving treatment of TCC

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Classification of bladder cancer and bladder-preserving treatment of TCC" · P4 LAQ "Discuss bladder preservation surgery for TCC of urinary bladder."

Read as: classification of bladder cancer, and bladder-preserving treatment of TCC (transitional cell carcinoma, urothelial carcinoma), including TURBT (transurethral resection of bladder tumour), radical TURBT, partial cystectomy and trimodality therapy.

Classification of bladder cancer

1. Histological

Type	Share	Note
Urothelial carcinoma	~90%	Variants: micropapillary, plasmacytoid, sarcomatoid, nested; squamous or glandular differentiation
Squamous cell carcinoma	~5%	Schistosomiasis, catheter, stones
Adenocarcinoma	~1–2%	Urachal (dome), exstrophy
Small cell / others	<1%	Neuroendocrine; sarcoma

2. Grade (WHO 2004/2016, World Health Organization)

- PUNLMP (papillary urothelial neoplasm of low malignant potential), low-grade and high-grade carcinoma. WHO 1973 (G1–G3) is still reported by some.
- Low-grade (FGFR3 pathway) recurs but rarely progresses. High-grade and CIS (carcinoma in situ; TP53/RB pathway) progress.

3. Stage: TNM 2017

T	Definition
Ta / Tis	Non-invasive papillary / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall

- N1 one true-pelvic node; N2 ≥2 pelvic nodes; N3 common iliac; M1a distant nodes; M1b organs.

4. Clinical groups

- **NMIBC (non-muscle-invasive, ~70%)**: Ta, Tis, T1. Risk groups: low (solitary primary low-grade Ta <3 cm), intermediate, high (high-grade, T1 or CIS).
- **MIBC (muscle-invasive)**: T2–T4a. Metastatic.
- **Molecular:** luminal papillary versus basal/squamous subtypes.

Why bladder preservation (introduction)

- Standard for MIBC (muscle-invasive bladder cancer) is cisplatin NAC (neoadjuvant chemotherapy) plus RC (radical cystectomy) with PLND (pelvic lymph node dissection).
- RC is morbid and patients are often elderly. Bladder preservation suits **(1) unfit patients** and **(2) fit, motivated patients with limited disease**.
- **Why single modalities fail:** they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%; occult nodes about 25%).

Options

Option	Use / status
Radical TURBT alone	Not a primary treatment
Partial cystectomy + PLND	Highly selected solitary tumour **<3 cm** in mobile bladder (dome), **no CIS**, 1–2 cm margin, frozen sections; **urachal adenocarcinoma**
RT (radiotherapy) alone	Inferior; quasi-palliative
Chemotherapy alone	Inferior; not recommended
TMT (trimodality therapy)	**Best-studied, guideline-accepted**

TURBT for NMIBC: the standard

- Complete resection with **detrusor muscle in the specimen**; separate base biopsies; cold-cup biopsy of suspicious areas.
- **En bloc resection** (loop or laser) keeps architecture, gives more detrusor and fewer perforations.
- **Repeat TURBT within 6 weeks** for all T1, incomplete resection or no muscle. Residual tumour in ≥40%.
- Then intravesical chemotherapy or BCG (bacillus Calmette–Guérin) by risk group.

Radical TURBT for MIBC

- Rationale: about 10% are pT0 at cystectomy. Only for tumour limited to superficial muscle, with a negative re-resection.
- About a third later progress to MIBC on long follow-up; selected series report 5-year CSS (cancer-specific survival) about 82%. Not a primary treatment.

Partial cystectomy

Ideal candidate	Contraindication
Small (**<3 cm**), **solitary** tumour in a mobile part of bladder (dome)	**CIS**, multifocal tumour
No CIS, no hydronephrosis	Trigone or bladder neck tumour
1–2 cm margin achievable away from the ureteric orifice and trigone; adequate residual capacity	Small bladder; prostatic urethral involvement
Urachal adenocarcinoma (treatment of choice: en bloc dome, urachus and umbilicus)	

Technique (steps):

1. NAC (neoadjuvant cisplatin chemotherapy) if cisplatin-eligible.
2. PLND (always).
3. Cystotomy away from tumour.
4. Full-thickness excision with perivesical fat, **frozen-section margins**.
5. Ureteric reimplantation if needed.
6. Watertight closure; irrigation (prevents seeding); drain.
7. Cystogram before catheter removal.

Outcomes: 5-year OS (overall survival) 67–70%; intact bladder 65–74%; salvage RC up to a quarter. Late recurrence, so lifelong follow-up.

Trimodality therapy (TMT)

Components: (1) **maximal, visibly complete TURBT**; (2) **RT** to bladder ± pelvic nodes; (3) concurrent **radiosensitising chemotherapy**.

Selection criteria:

- cT2 (selected cT3–4a), small solitary tumour.
- **No hydronephrosis, no extensive CIS (no CIS)**, visibly complete TURBT.
- Good bladder function; motivated, compliant patient.
- Only 6–19% of MIBC meet strict criteria.

Radiosensitising regimens:

Regimen	Evidence
Mitomycin C + 5-FU (fluorouracil)	**BC2001**: 2-year locoregional DFS (disease-free survival) **67% vs 48%** vs RT alone, no excess late toxicity
Cisplatin (± 5-FU or paclitaxel)	MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols
Gemcitabine	Cisplatin-ineligible
Carbogen + nicotinamide (BCON)	Hypoxia modification

Radiation dose and schedules: 64–66 Gy in 32 fractions or **55 Gy in 20** (non-inferior).
`Split course (MGH/RTOG): TURBT → chemoRT ~40 Gy → cystoscopy + biopsy CR → consolidation to 60–66 Gy Residual tumour → immediate cystectomy Continuous (current): TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr → cystoscopy + biopsy and imaging at 3–4 months ├─ Complete response → surveillance (NMIBC recurrence: TURBT + BCG) └─ Persistent / recurrent MIBC (invasive) → salvage cystectomy (10–30%)`

Response assessment: cystoscopy with biopsy of tumour site, cytology, CT (computed tomography)/MRI (magnetic resonance imaging). **CR (complete response)** leads to surveillance. Non-response leads to salvage RC.

Salvage cystectomy:

- Needed in **10–30%** for persistent or recurrent muscle-invasive disease; it is the designed safety net.
- NMIBC (non-muscle-invasive) recurrence: TURBT and intravesical BCG.

Outcomes:

- 5-year CSS 50–84%, OS 36–74%.
- MGH/RTOG: CR 69%; 5-year DSS (disease-specific survival) 71%; 80% of 5-year survivors keep the bladder.
- Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and similar CSS (81% vs 84%).
- Low late toxicity; good quality of life.

Comparison with radical cystectomy

	Radical cystectomy	Bladder preservation (TMT)
Staging	Pathological	Clinical only

	Radical cystectomy	Bladder preservation (TMT)
Nodes	Removed	Irradiated or not treated
Function	Diversion; sexual dysfunction	Native bladder; RT (bowel/bladder) toxicity
Follow-up	Upper tract, urethra	**Intensive cystoscopy: 3-monthly for 2 years, to 10 years**, plus CT
Evidence	Standard	No completed RCT (SPARE closed for poor accrual)

Follow-up after preservation

- Cystoscopy every 3–4 months for 2 years, 6-monthly in years 3–4, then yearly to 10 years.
- CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5 years; cytology.

Recent advances

- **Immunotherapy + chemoradiation:** PD-1/PD-L1 (programmed death-1 / ligand-1) inhibitors with TMT under trial (e.g. SWOG/NRG S1806 with atezolizumab; KEYNOTE-992 with pembrolizumab); results awaited.
- Hypofractionation 55 Gy/20 non-inferior (BC2001–BCON meta-analysis: same toxicity).
- Bladder sparing after complete response to NAC (± checkpoint inhibitor) with biomarker selection (DNA repair gene mutations, circulating tumour DNA); RETAIN: about half recurred.
- Robotic partial cystectomy; molecular subtypes.
- **Also know:** RT alone is quasi-palliative; chemotherapy alone not recommended.

Diagram to draw: TNM T-stage diagram of the bladder wall, with the TMT flowchart from TURBT to chemoradiation, response check, surveillance or salvage cystectomy.

Key points to remember:

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.

Also, from the condensed book (not in the paper answer):

- **Dose**: a continuous course to **60–66 Gy** with restaging at 3–4 months is commonest; **55 Gy in 20 fractions** is non-inferior.
- Concurrent chemotherapy beats radiation alone in two randomised trials.

KEY POINT

TMT contains cystectomy rather than replacing it. Salvage RC is the built-in safety net, so selection and lifelong cystoscopy decide success. Complete resection cuts the need for cystectomy (~11% vs 42%).

Single modalities (none is primary therapy)

Modality	Use	Why it fails
Chemotherapy alone	Complete responders declining RC	Early intravesical recurrence (~half by 5 months)
Radiation alone (64–66 Gy/32 or 55 Gy/20 fractions)	Unfit for surgery and chemoradiation; palliation	Inferior to chemoradiation; local control 27% in T3
Radical TUR	Superficial muscle invasion only	Understaging; ~25% have occult nodes
Partial cystectomy with PLND	<3 cm solitary, no CIS, margin away from orifices; **urachal adenocarcinoma (treatment of choice)**	Late recurrence, so lifelong follow-up; convert to RC if the frozen margin is positive

11 · Metastatic urothelial carcinoma

First-line choice hinges on platinum eligibility.

- Bajorin adverse factors are Karnofsky status <80% and visceral metastasis.
- **Cisplatin-eligible**: GC or ddMVAC. **GC equals MVAC** and is less toxic (von der Maase), and dose-dense beat standard MVAC (EORTC 30924).
- **Cisplatin-ineligible, platinum-eligible**: gemcitabine–carboplatin (EORTC 30986) or enfortumab vedotin (EV) with pembrolizumab. †EV-302 later showed EV with pembrolizumab roughly doubled OS vs platinum, making it the preferred first line.
- **Platinum-ineligible**: pembrolizumab (KEYNOTE-052) or atezolizumab if PD-L1-positive. Trap: **first-line IO alone is only for platinum-ineligible patients.**
- **Maintenance avelumab** is given if there is no progression after platinum (JAVELIN Bladder 100). IO combined with chemotherapy is not recommended.
- **Second line** is **pembrolizumab** (KEYNOTE-045, regardless of PD-L1); atezolizumab failed (IMvigor211). Platinum can be rechallenged if the last dose was ≥12 months earlier.
- **Antibody–drug conjugates**: **EV** (Nectin-4; neuropathy, rash) beat chemotherapy after platinum and IO (EV-301). **Sacituzumab govitecan** (Trop-2; neutropenia) is third line.
- **Erdafitinib**, an oral FGFR inhibitor, treats FGFR3/FGFR2 alterations after platinum, with 40% response; it causes **hyperphosphataemia**.
- **IO toxicity** includes rash, colitis, hepatitis, pneumonitis, myocarditis and endocrinopathies, which are often **irreversible**.

ALGORITHM

1. If platinum-eligible, give GC or ddMVAC (or gemcitabine–carboplatin, or EV with pembrolizumab).
2. If there is response or stable disease, give avelumab maintenance.
3. On progression, give pembrolizumab, or erdafitinib if FGFR-altered.
4. Then give EV, then sacituzumab govitecan.
5. If platinum-ineligible, give pembrolizumab, then EV.

12 · Follow-up

- **After RC**: CT with excretory phase and chest CT 6-monthly for 2–3 years, then yearly to 5 (EAU, AUA). Check renal function at each visit and vitamin B12 yearly from year 2.
- Upper-tract recurrence (4–10%) is the commonest **late** one; pelvic and distant recurrences come mostly within 2 years.
- **After bladder preservation**: cystoscopy 3–4-monthly in years 1–2, 6-monthly in years 3–4, then yearly to **10 years**, with CT urogram and chest CT 3–6-monthly for 2 years, then yearly to 5.

13 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8710	MVAC ×3, then RC, vs RC	Median OS 77 vs 46 months; pT0 38% vs 15%	pT0 is a surrogate; OS P = .06
BA06 30894 (EORTC/MRC)	Cisplatin, methotrexate, vinblastine (CMV) ×3, then RC or radiotherapy	10-year OS 36% vs 30%	Largest positive NAC trial
ABC meta-analysis	11 randomised NAC trials	OS hazard ratio (HR) 0.86; +5% at 5 years	Combination platinum only
VESPER	ddMVAC ×6 vs GC ×4	Organ-confined 77% vs 63%; 3-year progression-free survival 66% vs 56%	ddMVAC better but more toxic
SWOG S1011	Standard vs extended PLND	No difference in disease-free survival (DFS) or OS	Standard PLND suffices
CheckMate 274	Adjuvant nivolumab vs placebo, pT3–4/N+	DFS 20.8 vs 10.8 months	Adjuvant standard
BC2001	Radiation with or without mitomycin C/5-fluorouracil	Locoregional DFS 67% vs 48%	Sensitise radiation
von der Maase	GC vs MVAC, metastatic	OS 14.0 vs 15.2 months (not significant)	Equal efficacy, GC less toxic
EORTC 30986	Gemcitabine–carboplatin vs MCAVI, cisplatin-ineligible	Equal OS, less toxicity	Gem–carbo standard
JAVELIN Bladder 100	Avelumab maintenance vs none after platinum	OS 21.4 vs 14.3 months (HR 0.69)	Maintenance standard
KEYNOTE-045	Pembrolizumab vs chemotherapy, second line	OS 10.3 vs 7.4 months	Second-line standard
EV-301	EV vs chemotherapy after platinum and IO	OS 12.9 vs 9.0 months	ADC third line

132 Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

1 · Framing

- Transurethral resection (TUR) stages every patient and cures low-grade non-muscle-invasive bladder cancer (NMIBC); muscle-invasive disease (MIBC) needs radical cystectomy (RC) and pelvic lymph node dissection (PLND), after cisplatin neoadjuvant chemotherapy (NAC) if eligible.

2 · Transurethral resection of bladder tumour (TURBT)

Complete resection with muscle in the specimen makes staging correct.

Pre-operative evaluation

- **Continue aspirin**; hold other anticoagulants and P2Y12 inhibitors by half-life. **Treat active infection before resection**.
- Image the upper tracts by CT urogram, MRI, or ultrasound with retrograde study. Iodinated contrast needs a glomerular filtration rate (GFR) **≥60** mL/min; gadolinium needs GFR **>30**.

Set-up and inspection

ALGORITHM

1. Antibiotic within 1 h (avoid fluoroquinolones).
2. Inspect with 30° and 70° lenses (120° if incomplete).
3. Map tumours; add blue-light or narrow-band imaging if available (more tumours and carcinoma in situ (CIS) found, less recurrence).

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% saline
Electrolyte disturbance	Higher risk	Lower risk
Obturator reflex	Present	Reduced

Resection principles

- Remove all visible tumour with **detrusor in the specimen** and a **~2 cm** margin of normal tissue where possible.
- Then examine under anaesthesia (EUA) with the bladder empty: a **mass palpable after resection is cT3b**.
- **En bloc resection** keeps orientation and detrusor, and improves T1 staging.
- Maximal resection is therapeutic in NMIBC, but before RC for MIBC it gives no survival gain; residual tumour is only prognostic.

Difficult locations

Site	Hazard	Countermeasure
Lateral wall	**Obturator jerk**: nerve stimulation makes the leg adduct and the loop perforates	Low distension; bipolar; muscle relaxant (general anaesthesia)

Site	Hazard	Countermeasure
Ureteric orifice	Coagulation causes stricture and hydronephrosis	**Pure cutting current**; routine stent not needed; short-term stent after large resection or heavy cautery is reasonable
Dome	Intraperitoneal perforation; heat injury to bowel	Low distension; suprapubic pressure; straight loop; minimal cautery

Bladder perforation

	Extraperitoneal	Intraperitoneal
Site / frequency	Away from dome; commoner	Dome, posterior wall; rare, large ones morbid
Clues	—	Bladder will not distend; defect seen; abdomen distends
Confirm	Cystogram at same sitting	
Management	**Foley drainage and observation**	**Laparotomy**, inspect bowel, repair; catheter and drain

> **KEY POINT**
>
> Perforation can follow adequate resection of advanced tumour. It **rules out immediate intravesical chemotherapy**, because the drug would leak outside the bladder.

Post-operative intravesical agents

*See the paper answer **Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation** (Section 12, Ch 130).*

Also, from the condensed book (not in the paper answer):

- Single mitomycin C or epirubicin **within 24 h** gives a ~**13%** absolute fall in recurrence (meta-analysis); gemcitabine (SWOG S0337) cut recurrence by 12% at 4 years. They reduce recurrence, not progression.
- **BCG (bacille Calmette–Guérin) is never given immediately**: wait **2–4 weeks** with no gross haematuria, because of the risk of systemic absorption.

3 · Radical cystectomy: preparation

- **Mark a stoma for every patient** (sitting and standing) and consent for a conduit.
- **Avoid routine mechanical bowel prep** and oral antibiotics; omitting them speeds recovery and cuts wound complications.
- **Alvimopan** (peripheral μ-opioid antagonist), given 30–90 min pre-op and then twice daily, lessens ileus and shortens stay.
- Give an IV antibiotic within 1 h covering skin and bowel flora, e.g. **cefoxitin**.
- Start **venous thromboembolism (VTE) prophylaxis** (compression and drug) **before induction** and extend it to **4 weeks**: 28 rather than 8 days gives fewer major VTE without extra bleeding. Enoxaparin is standard; apixaban is equal, with better compliance.

4 · Open radical cystectomy: technique

ALGORITHM

1. Lower midline; Retzius space; divide urachus and vas or round ligament.
2. Mobilise caecum and sigmoid; make the **retrosigmoid window** (left ureter).
3. Free and divide ureters with adventitia.
4. Retractor (check vena cava); PLND; staple, clip or seal vesical pedicles.

- **Distal ureteric frozen section** is controversial: a positive margin predicts upper-tract recurrence, but clearing it has no proven benefit.

Male

ALGORITHM

1. Incise peritoneum over seminal vesicles; separate rectum; incise Denonvilliers fascia.
2. Control posterior pedicles; open endopelvic fascia.
3. Ligate dorsal venous complex (DVC); divide urethra.

- **Rectal injury is repaired primarily**, with or without faecal diversion.
- **Urethral frozen section** before a neobladder has a very high negative predictive value. **A positive result rules out orthotopic neobladder**, so warn the patient beforehand.
- **Nerve sparing** leaves few men potent at 1 year, and local recurrence occurs even in pT2. Avoid it in advanced disease or pre-existing erectile dysfunction.
- **Prostate sparing** improves potency and continence, but occult prostate cancer is common. It needs normal DRE and prostate-specific antigen (PSA), a TURP and a negative frozen section.

Female

- The standard is **anterior exenteration**, essential if EUA is positive or cT4a. Discuss **ovarian sparing** if disease is not locally advanced.

ALGORITHM

1. Ligate ovarian vessels; incise posterior fornix; control pedicles; excise urethra.
2. Watertight **clamshell vaginal closure** (keeps girth, loses length).

- **Vagina and urethra sparing for neobladder** needs ≤cT2, an uninvolved bladder neck and a negative urethral frozen section; preserve the striated sphincter.

5 · Pelvic lymphadenectomy

Template	Boundaries	Evidence
Standard	Lateral genitofemoral nerve; medial internal iliac artery; distal Cooper ligament; proximal ureter crossing common iliac	Reference
Extended	Standard plus common iliac and presacral	Better staging; possible survival gain
Superextended	Up to inferior mesenteric artery (preaortic)	No benefit over extended

- **≥10 nodes** removed goes with better survival (NMIBC at RC: >20).
- A randomised trial found **extended not superior to limited** for recurrence-free or cancer-specific survival, but its staging value is certain.

6 · Partial cystectomy

- Indications are a **solitary small MIBC without CIS** and **urachal adenocarcinoma** (en bloc umbilicus, urachus and dome with frozen margins).
- Oncology is as for RC: PLND, NAC if eligible, and consent for RC. Muscle-invasive recurrence needs salvage RC.

ALGORITHM

1. PLND; open the bladder away from tumour.
2. Full-thickness excision with fat and a 1–2 cm frozen-section margin; reimplant ureter if needed.
3. Watertight closure; warm water irrigation (against seeding); drain.
4. Cystogram day 7, then remove catheter.

7 · Robotic radical cystectomy (RARC)

KEY POINT

No effective salvage exists: urine spillage, poor PLND or a positive soft-tissue margin leads to near-uniformly fatal recurrence, so robotics is judged on oncology first.

- Indications match open surgery; prior surgery and obesity are relative contraindications. A **15-mm left assistant port** allows stapling and bagging.

ALGORITHM

1. Release sigmoid; develop presacral plane; open cul-de-sac to prostate apex.
2. Open Retzius; control anterior pedicle; free ureters (**never grasp**).
3. **Extended PLND with ureters intact**.
4. Clip ureters, send margin, pass left ureter under sigmoid.
5. Posterior pedicles, DVC, urethra (clip proximally only if neobladder).

- PLND before ureteric division keeps urine output as a resuscitation guide. Bag each packet whole; the obturator nerve is **not** the posterior limit, so go down to the hypogastric vein.

8 · Complications and post-operative care

- Gastrointestinal complications lead, then infection. For **ileus**, use alvimopan, early feeding and opioid-sparing analgesia, with **no routine nasogastric tube**.
- Ureteric devascularisation causes ischaemic stricture (keep adventitia).

9 · Landmark trials

Trial	Design	Result	Take-home
Grossman 2003	NAC and RC vs RC	Median survival 46 vs 77 months	Offer cisplatin NAC

Trial	Design	Result	Take-home
SWOG S0337	Single gemcitabine vs saline after TUR	12% fewer recurrences at 4 years	Immediate gemcitabine for low grade
RAZOR (2018)	Non-inferiority RARC vs open; open diversion	2-year progression-free survival 72.3% vs 71.6%; less blood loss, shorter stay	Robotic oncologically equivalent
iROC	RARC with intracorporeal diversion vs open	More days alive and out of hospital at 90 days; fewer VTE, wound complications	Recovery advantage

133 Use of Intestinal Segments in Urinary Diversion

1 · Surgical anatomy

A diversion depends on its pedicle's blood supply.

- **Stomach** is supplied by the right gastroepiploic (gastroduodenal, from hepatic) and left gastroepiploic (from splenic) along the greater curve. The graft is antrum or a fundal wedge, which must **never reach the pylorus or lesser curve**.
- **Small bowel** is ~22 ft: the proximal 2/5 is jejunum (wider, single arcades) and the distal 3/5 ileum (multiple arcades). Bowel >**8 cm** from a straight vessel may not survive, so clear ≤8 cm of mesentery at a segment end.
- **Avoid pelvic small bowel after radiotherapy**, including the fixed terminal ileum. A right colic arising from the ileocolic lets the ascending colon reach the pelvis.
- **Colon** is supplied by the superior mesenteric artery (SMA: ileocolic, right colic, middle colic) and inferior mesenteric artery (IMA: left colic, sigmoid and superior haemorrhoidal). The internal iliac gives the middle haemorrhoidal; all link through the **marginal artery of Drummond**.

Watershed	Site	Between
Griffiths point	Splenic flexure	Middle colic and ascending left colic
Sudeck's point	Rectosigmoid	Last sigmoid and superior haemorrhoidal

> **KEY POINT**
>
> Watershed zones get the smallest end branches of two arteries, so they go ischaemic first in hypotension or embolism; site anastomoses to one side.

2 · Segment selection

LAQ Intestinal segments in urology and orthotopic neobladder

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Use of intestinal segments in urology" / "Various uses of intestines in Uro" / "Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder." (also P2 SAQ "Neobladder") · P1 SAQ "Instestinal segments in urology"

Introduction

- Bowel is used to divert urine (conduit, continent pouch, neobladder), store urine, carry it to the skin, enlarge the bladder (augmentation), replace the ureter, make catheterisable channels and line a neovagina.
- Choice depends on renal function, previous surgery or irradiation, bowel disease, mesenteric length and the reconstruction needed.

Uses at a glance

Use	Segment and operation
Incontinent diversion	**Ileal conduit** (10–15 cm ileum, 10–15 cm from the ileocaecal valve); colon conduit (transverse) after pelvic radiation; jejunal conduit only if nothing else (avoid)

Use	Segment and operation
Continent cutaneous pouch	Indiana (right colon, plicated ileum as channel), Mainz I, Kock (ileal nipple valves); with appendix (Mitrofanoff) or tapered ileum channel
Orthotopic neobladder	Studer (40–44 cm ileum plus afferent limb), Hautmann (~70 cm ileum, W shape); about 50–70 cm detubularised ileum joined to the urethra
Rectal reservoir	Ureterosigmoidostomy, Mainz II: now rare (acidosis, cancer)
Augmentation cystoplasty	"Clam" ileocystoplasty (20–40 cm ileum; ~25 cm ileum), sigmoid, caecum; gastrocystoplasty in renal failure. For refractory neurogenic or small contracted bladder; needs CIC (clean intermittent catheterisation)
Ureteric replacement	**Ileal ureter**, isoperistaltic ileum, for long defects; Yang–Monti tubes
Catheterisable channels	**Mitrofanoff** appendicovesicostomy; **Yang–Monti** ileal tube (1–2 cm ileum retubularised transversely) when the appendix is absent; MACE (Malone antegrade continence enema) for bowel
Neovagina	**Sigmoid** preferred (wide lumen, mobile mesentery, less stenosis); ileum if pelvis irradiated; MRKH (Mayer–Rokitansky–Küster–Hauser), cloaca, gender-affirming vaginoplasty

- Ileal ureter contraindications: creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease or radiation enteritis.

Choice of segment

Segment	Electrolyte disturbance	Advantages / use	Drawbacks
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Secretes H^+ and Cl^- (acid); less mucus; usable for renal failure, short gut, dense adhesions	**Haematuria–dysuria syndrome**, ulcers
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic acidosis**	Outside radiation field	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Default: mobile, compliant, constant (reliable) blood supply; default conduit, ureteric replacement	B12 and bile-salt loss; obstruction ~10%
Colon	Hyperchloraemic metabolic acidosis, hypokalaemia	Easy antireflux tunnel; transverse colon outside (spared by) the pelvic radiation field	Diarrhoea if ileocaecal valve used; obstruction ~5%
Appendix	—	Mitrofanoff catheterisable channel	—

Technical principles of using bowel

- **Preparation:** ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed. Assess renal function. Mark the stoma site before surgery when a stoma is possible, through the rectus.
- **Mesentery:** transilluminate, preserve a major mesenteric arcade, wide base; clear ≤8 cm of mesentery from a segment end; avoid watershed areas (Griffiths point at splenic flexure, Sudeck point at rectosigmoid).
- **Bowel anastomosis:** well-vascularised, no tension, no spillage, close the mesenteric window.

- **Detubularisation** (Kock): opening the bowel on its antimesenteric border and refolding it **abolishes coordinated (peristaltic) contractions** and gives more volume for the same surface area; a **sphere** gives the largest volume for its surface area; by Laplace's law (pressure ∝ wall tension/radius) a larger radius means lower pressure, so pressure stays low.

Ureterointestinal anastomoses

Technique	Type	Stricture
Bricker	Refluxing, end to side	4–22% (mean 6%)
Wallace	Refluxing, ureters joined end to end	~3% (lowest)
Leadbetter–Clarke, Goodwin	Nonrefluxing (antireflux) tunnel in colon	8–14%

- Principles: spatulate, preserve periureteric tissue, tension-free, mucosa to mucosa, soft stents (leak 3–5% without). Bring the bowel to the ureter, not the ureter to the bowel.
- Refluxing (Bricker, Wallace; stricture about 3–6%) versus antireflux tunnels (more strictures).
- Fix and retroperitonealise the anastomosis. The left ureter passes behind the sigmoid mesentery; avoid angulation under the IMA (inferior mesenteric artery), a common stricture site.
- Reflux with normal unobstructed ureters has not been shown to harm the adult kidney; antireflux tunnels stricture more. Obstruction, not low-pressure reflux, damages kidneys.

Stomas

- **Protruding (nipple, "rosebud") stoma** for appliance wearers: less stenosis, better appliance fit. **Flush stoma** for continent pouches that are catheterised.
- **Loop-end (Turnbull) stoma** for obese patients with a short mesentery. Site is marked before surgery, through the rectus muscle.

Selecting conduit versus continent diversion

- **Renal function needed for continent diversion:** creatinine <2 mg/dL or GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; urine osmolality ≥600 mOsm/kg after water deprivation; minimal proteinuria; normal liver function; ability to self-catheterise.
- **Contraindications to continent diversion:** poor renal or hepatic function, inability or unwillingness to self-catheterise, inflammatory bowel disease or short gut. A neobladder also needs a tumour-free urethra and a competent sphincter.
- A conduit is simplest and fastest; continent options improve body image but carry more reoperations and metabolic problems.

Complications of bowel in the urinary tract

Metabolic (worse with longer segment, longer contact time, poorer renal function)

- **Hyperchloraemic acidosis (ileum, colon):** NH_4^+ (ammonium) replaces Na^+ on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate (potassium citrate if hypokalaemic); chlorpromazine or nicotinic acid limit chloride transport.
- **Hypokalaemia:** worse with colon; replace potassium before correcting acidosis (paralysis risk).
- **Jejunal syndrome:** treat with sodium chloride and thiazide. **Gastric alkalosis:** H2 blockers or PPI (proton-pump inhibitor).

- **Altered sensorium:** hyperammonaemia (drain the pouch/diversion, lactulose), hypomagnesaemia.
- **Drug reabsorption:** phenytoin, methotrexate; drain pouches during chemotherapy.
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Nutrition:** B12 deficiency (give parenteral B12 after 5–7 years of ileal diversion), bile-salt diarrhoea, short-gut effects.

Other

- Infection (bacteriuria in about three-quarters of conduits; treat only symptoms or urea-splitting organisms, or pure Proteus or Pseudomonas).
- Mucus.
- **Stones:** colon conduit 3–4%, ileal conduit 10–12%, continent (caecal) reservoirs ~20% (mucus, staples).
- **Augment perforation (pouch rupture):** surgical emergency; regular CIC prevents it.
- Ureterointestinal stricture (lifelong risk), renal deterioration, bowel obstruction, stomal stenosis and parastomal hernia.
- **Secondary malignancy:** ureterosigmoidostomy 6–29% after 10–20 years (endoscopy from 5 years); augments (endoscopy from 10 years). Neovagina: stenosis, mucus discharge, adenocarcinoma reported.

Orthotopic neobladder: definition and principles

- An **orthotopic** detubularised bowel (usually ileal) reservoir joined to the native urethra after radical cystectomy, when the urethra is tumour-free, to allow **volitional voiding** per urethra without a stoma. 80–90% of men and about 75% of women are medically eligible.
- **Principles:** 1. **Reliable outlet:** intact rhabdosphincter (pudendal nerve), healthy, tumour-free urethra. 2. **Low pressure:** complete **detubularisation** abolishes coordinated contractions; folding into a **sphere** gives most volume for surface area. 3. **Adequate capacity:** ≥500 mL when mature (44 cm double-folded ileum: <200 mL early, **500–600 mL** by 1 year). 4. Unobstructed upper tracts.

Neobladder: selection and contraindications

- **Absolute contraindications:** tumour at the **urethral margin** (positive urethral margin on frozen section), urethral stricture, poor renal function (eGFR [estimated glomerular filtration rate] <35–40 mL/min), severe liver disease, inability to self-catheterise.
- **Relative:** prostatic stromal invasion (men); bladder-neck or anterior vaginal tumour (women); poor sphincter; prior radiation (irradiated pelvis); frailty. Age, obesity and CIS (carcinoma in situ) alone are not bars.

Neobladder techniques compared

Design	Segment and shape	Ureters	Advantage	Drawback
Studer	40–44 cm ileum, double-folded U + afferent limb	Refluxing, isoperistaltic limb	Simple, no staples, short ureters; stricture <3%	Reflux on cystogram (benign)
Hautmann W	About 70 cm ileum, W	Refluxing (since 1997)	Largest early capacity, early continence	More late retention, more absorption
T pouch	44 cm + 15 cm T limb	Antireflux flap valve	Accepts dilated or short ureters	More reoperations than Studer

Design	Segment and shape	Ureters	Advantage	Drawback
Abol-Enein	40 cm ileum, W	Serous-lined tunnel	Durable antireflux	Needs long, non-dilated ureters
Camey II	65 cm ileum, U	Le Duc antireflux tunnel	Early detubularised design	Le Duc strictures 7–29%
Hemi-Kock	61 cm ileum	Intussuscepted nipple	Good early results	Nipple stenosis, staple stones
Padua (VIP)	About 42 cm ileum, funnel	Le Duc	Anatomical funnel; robotic	—
Mainz III	10–15 cm caecum + 20–30 cm ileum (ileocaecal)	Submucosal tunnel	Good compliance	Loses ileocaecal valve
Sigmoid (Reddy)	35 cm sigmoid colon, U	Tunnelled	Spares ileum (short bowel)	Poor compliance, urgency

- **Hautmann vs Studer:** larger initial capacity gives earlier continence but more late retention and absorption.
- **Studer steps (brief):** isolate ileum 15–20 cm proximal to the ileocaecal valve; open distal 40–44 cm antimesenterically; sew into a U and fold into a sphere; stented ureters end-to-side to the afferent limb; dependent point to urethra.
- **USC (University of Southern California) RCT (randomised controlled trial):** Studer vs T pouch, no difference in renal function at 3 years; T pouch needed more secondary procedures.

Neobladder outcomes

Outcome	Result
Day continence	80–90% (3–6 months)
Night incontinence	20–50% (no sphincter reflex, mucosal water secretion)
Retention needing CIC	Men 4–10%; women 20–30%: men leak, women fail to empty
Urethral recurrence (men)	About 7% at 5 years

Neobladder complications

- **Early:** urine leak, ileus, infection, mucus plugging.
- **Late: hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, ureteroileal stricture (<3%), stones, pouch rupture, incisional hernia, urethral recurrence.

Care and follow-up

- Neobladder: mucus irrigation, bicarbonate; voiding training (sit, relax, strain), timed voiding, night alarm. Annual metabolic panel, B12, upper tracts; urethral cytology if CIS.
- Lifelong for all bowel reconstructions: electrolytes, bicarbonate, renal function, B12, upper-tract imaging (ultrasound or CT) for stricture and stones; endoscopy of colonic or augmented segments as above.

Recent advances

- Robotic intracorporeal diversion (complications similar to open in experienced hands); robotic intracorporeal neobladder comparable to open; tissue-engineered bladder

(experimental) and alternatives to bowel (ureterocystoplasty, autoaugmentation) remain limited.

- **Also know:** neobladder uptake is low despite 80–90% of men being eligible.

Diagram to draw: ileal conduit (loop) with mesenteric arcade and Wallace anastomosis.

Diagram to draw: Studer U-pouch with afferent limb, ureters and urethral anastomosis.

Key points to remember:

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Also, from the condensed book (not in the paper answer):

- Losing **ileum** causes B12, bile-salt and fat malabsorption; losing the **ileocaecal valve** causes bacterial overgrowth, diarrhoea and bicarbonate loss. **B12** liver stores last **3–5 yr**, so give empirical parenteral B12 after 5–7 yr of ileal diversion.
- **Stomach** suits renal failure, short gut and dense adhesions, but causes the **haematuria–dysuria syndrome** (treat with H2 blockers). Use **jejunum** only if nothing else works.

3 · Bowel preparation and enhanced recovery

SAQ Methods of preoperative bowel preparation.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2022-Summer Paper 3, Q3d

Rationale

- Bowel used in urinary diversion is opened into a sterile urinary tract. Faecal spillage causes wound infection, abscess, leak and sepsis.
- Flora: aerobes (*E. coli*, *Enterococcus faecalis*); anaerobes (*Bacteroides*, *Clostridium*).

Segment	Organisms per gram
Jejunum	$10–10^5$
Distal ileum	$10^5–10^7$
Ascending colon	$10^6–10^8$
Descending colon	$10^{10}–10^{12}$

Methods

1. Dietary: clear liquids the day before; clear drinks until 2 hours before surgery.

2. Mechanical bowel preparation (MBP) reduces faecal **amount**, so total bacteria, not concentration.

- **PEG (polyethylene glycol)–electrolyte** lavage: non-absorbable; preferred in elderly, renal or heart failure, cirrhosis.
- Sodium phosphate, magnesium citrate; enemas.
- Historical: 3-day purgation, whole-gut irrigation by nasogastric tube (9–12 L saline), mannitol (feeds bacteria).
- Harms: dehydration, electrolyte loss; liquid stool may spill more.

3. Oral antibiotic preparation reduces bacterial **concentration**.

- **Neomycin + erythromycin base** (Nichols–Condon, 1977: septic complications 43% with MBP alone vs 9% with added antibiotics); neomycin + metronidazole; kanamycin.
- Harms: diarrhoea, *C. difficile* colitis, candidiasis; useless in obstruction.

4. Intravenous prophylaxis (most important): anaerobic plus aerobic cover (e.g. cephalosporin + metronidazole) within 1–2 hours before incision; it must be in tissue before contamination.

5. ERAS (enhanced recovery after surgery): no routine MBP; prehabilitation, multimodal opioid-sparing analgesia, chewing gum, **alvimopan** (peripheral μ-opioid antagonist, 12 mg before surgery), early feeding and mobilisation.

Evidence

- Colorectal meta-analyses and Cochrane: MBP does not reduce leak or abscess; may raise SSI (surgical site infection).
- Oral antibiotics added to MBP reduce SSI and leaks in large colectomy series.
- Small cystectomy trials: no difference without preparation.

Who still needs full preparation

- Salvage cystectomy after radiation, or cystectomy for benign disease (radiation cystitis, fistula, neurogenic bladder); immunosuppression.

Also know: wound protectors reduce SSI; enema before transrectal biopsy.

Diagram to draw: bacterial load along the gut.

> **Key points to remember:**
> - Mechanical lowers amount; antibiotics lower concentration.
> - Nichols–Condon: neomycin + erythromycin.
> - Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

Also, from the condensed book (not in the paper answer):

- The descending colon holds **$10^{10–12}$/g** bacteria; oral antibiotics cut this to ~10^2/g. The **IV antibiotic**, given 1–2 h before incision and covering anaerobes, matters most.
- **Alvimopan** (peripheral μ-opioid antagonist) is given as 12 mg 30 min–5 h pre-op, then twice daily, limited to 15 doses for inpatients.
- **Enhanced recovery after surgery (ERAS)** uses **no routine bowel prep**; postoperative ileus is the commonest cystectomy complication.
- Salvage or benign cystectomy (radiation, fistula, neurogenic) still needs full mechanical and antibiotic prep.

4 · Intestinal anastomoses

1. **Get exposure** and wall off the field; check the **blood supply** (transilluminate, bleeding edges, no tension).
2. **Prevent spillage** with non-crushing clamps, and irrigate the segment.
3. Join **serosa to serosa, inverted** and watertight, with **no strangulating sutures** and the **mesentery aligned**.

- Breakdown follows ischaemia, faecal sepsis, a **drain on the anastomosis** or irradiated bowel (most lethal). Sutured and stapled are equal. Leak presents late (2nd week); a neobladder–vaginal fistula causes continuous incontinence.
- **Obstruction** occurs in ~10% with ileum or stomach and ~5% with colon, mostly from adhesions. Prevent it with unirradiated bowel, closed mesenteric apertures, omentum and **reperitonealisation**.
- **Ogilvie syndrome** is a dilated caecum; exclude obstruction with a water-soluble enema. If the caecum is **>12 cm** or white count rises, give neostigmine or decompress by colonoscopy; operate for ischaemia or peritonitis.

5 · Ureterointestinal anastomoses

- Mobilise minimally, keep the periadventitia, spatulate, and sew mucosa to mucosa. **Bring the bowel to the ureter**, retroperitonealise, and use soft stents (they cut leak from 3–5% to near zero).

Technique	Type	Key feature	Stricture	Leak
Leadbetter–Clarke	Nonrefluxing	Colon; tunnel under taenia (2.5–3 cm)	8–14%	2.5%
Goodwin	Nonrefluxing	Transcolonic tunnel made from inside (3–4 cm)	No reliable data	
Bricker	Refluxing end-to-side	Any segment; simple	4–22% (mean 6%)	~3%
Wallace	Refluxing end-to-end	Ureters joined together (side-by-side, Y, head-to-tail)	**~3% (lowest)**	~2%

KEY POINT

Conduit pressure is damped by ureteric peristalsis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop colonisation, so a stented refluxing anastomosis is the default; reflux also allows a loopogram.

- **Stricture** follows ischaemia, leak, radiation or infection, so surveillance is lifelong. The classic site is the **left ureter under the IMA**.
- Open reanastomosis is most durable; endoincision works in ~50–60% at 2 yr. Endourology fails more if the stricture appears <1 yr post-op, is **≥1.5 cm**, or is left-sided.

6 · Conduit diversion

- The **ileal conduit** is simplest, with fewest early complications; avoid it in short bowel, inflammatory small-bowel disease and irradiated ileum.

ALGORITHM

1. Take **10–15 cm** of ileum, 10–15 cm from the ileocaecal valve, keeping a major arcade.
2. Restore continuity by ileoileostomy; close the mesenteric window; flush the segment.
3. Pass the left ureter behind the sigmoid mesentery; make stented anastomoses.
4. Fix the base retroperitoneally in the right lower quadrant (RLQ); make the stoma.

- Intractable stomal bleeding suggests portal-hypertensive varices, treated by TIPS or embolisation.
- A **jejunal conduit** (15–25 cm from Treitz, left upper quadrant stoma) is used only if ileum and colon are unusable.

	Transverse	**Sigmoid**	**Ileocaecal**
Best for	**Extensive pelvic irradiation**; intestinal pyelostomy	Exenteration with colostomy	Long ureteric replacement
Supply	Middle colic	Sigmoid vessels	Ileocolic
Avoid	—	Sigmoid disease; ligated hypogastrics with rectum in situ; pelvic irradiation	—

- Avoid colon conduits in inflammatory bowel disease and chronic diarrhoea. An **ileal vesicostomy** suits neurogenic bladder and is reversible.
- Remove stents at days 4–6; drain retroperitoneally, **never intraperitoneally**.

Complication (%)	**Ileal conduit**	**Colon conduit**	**Continent**
Bowel obstruction	10	4	0.8
Ureterointestinal obstruction	8	Antireflux 20	4
Acidosis needing treatment	16	5	Ileum 8; colon 27
Renal deterioration	18	15	—

A gastric conduit causes no acidosis (0%).

7 · Stomas

- Mark sitting and supine; go **through the rectus** (lateral stomas herniate); stay ≥5 cm from the incision and avoid creases, scars and irradiated skin.
- **Keep the subcutaneous fat**, cut the fascia in a cross, and make the aperture admit two fingers.
- A **nipple (rosebud)** stoma suits an appliance (2–3 cm spout, less stenosis); a **flush** stoma suits a continent catheterisable channel.

	End stoma	**Loop (Turnbull)**
Stenosis	Higher	Lower
Parastomal hernia	1–4%	4–20%
Best for	Standard	**Obese** (thick, short mesentery)

- Parastomal hernia occurs in 2–6.6% of ileal conduits. Stomal stenosis occurs in 20–24% of ileal and 10–20% of colon conduits, and in >50% of paediatric catheterisable stomas.

8 · Renal function

- Renal deterioration after conduits is 10–74% (ileal conduit 18%), and ~6% die of renal failure. After ureterosigmoidostomy it is the leading cause of death beyond 15 yr.
- Continent diversions demand more of the kidney than conduits. **Tubular acidification matters as much as glomerular filtration rate (GFR).**

Test	**Threshold for continent diversion**
Creatinine and urine protein	**<2.0 mg/dL**, no proteinuria
GFR	**>35 mL/min**
Ammonium chloride load	Urine pH **≤5.8**
Water deprivation	Osmolality **≥600 mOsm/kg**

- **Bowel invalidates** urine osmolality and pH and serum urea and creatinine, because they are reabsorbed.

9 · Metabolic complications

Absorption depends on the **segment, surface area, dwell time, solute concentration, renal function and pH**.

Segment	**Na**	**K**	**Cl**	**pH**	**Associated**	**Treatment**
Stomach	—	Low	Low	High	Raised aldosterone; lethargy, seizures, arrhythmia	H2 blocker, PPI; arginine HCl; remove segment
Jejunum	Low	High	Low	Low	Raised renin, angiotensin; vomiting, dehydration	IV saline, bicarbonate, thiazide
Ileum/colon	—	Low	High	Low	Total-body K depletion; hypocalcaemia	Alkali; chlorpromazine, nicotinic acid

> **KEY POINT**
>
> Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na and Cl, high K); ileum and colon absorb NH4Cl (high Cl). **Only jejunum is hyperkalaemic**.

- **Gastric alkalosis** is usually trivial, but severe with renal failure or dehydration. Above a **gastrin** of **120 ng/L**, small gastrin rises cause large bicarbonate rises. Treat stepwise: rehydration and H2 blocker, arginine HCl if life-threatening, omeprazole, then segment removal.
- In the **jejunal syndrome**, NaCl loss causes hypovolaemia and activates renin–aldosterone, so the kidney keeps losing K-rich urine. It is worse with proximal segments, can be fatal, and needs long-term oral salt and thiazide.
- **Hyperchloraemic acidosis** is worst with **ureterosigmoidostomy** and colonic pouches; $NH4^+$ also enters via K^+ channels. Mucosal villous atrophy occurs, but transporters persist, so the risk persists.
- Alkali options are sodium bicarbonate, citrate solutions, or potassium citrate when K is needed. **Chlorpromazine 25–50 mg tds** or **nicotinic acid 400 mg 3–4×/day** inhibit cyclic AMP; they limit, but do not correct, acidosis.
- **Hypokalaemia** is commoner with colon. **Trap**: correcting acidosis alone causes profound hypokalaemia and flaccid paralysis, so give K with bicarbonate. Mg and Ca are also low.
- **Altered sensorium** is usually **ammonia** (liver dysfunction plus ureterosigmoidostomy). Treat by draining the diversion, oral neomycin, protein restriction, IV arginine glutamate and lactulose.
- **Drugs are reabsorbed**, notably **phenytoin, some antibiotics and methotrexate**, so **drain continent pouches during chemotherapy**.
- **Osteomalacia** is commonest with ureterosigmoidostomy, from acidosis, vitamin D resistance and renal Ca loss. Alkaline phosphatase is raised. Correcting a base deficit >2.5 mEq/L prevents bone loss; treat with alkali and calcium, then active vitamin D.
- **Infection**: ~3/4 of conduit urines are bacteriuric; **treat pure Proteus or Pseudomonas** and observe asymptomatic mixed cultures.
- **Stones** (struvite, Ca phosphate) follow acidosis, urease organisms, **staples** and mucus: colon conduit 3–4%, ileal conduit 10–12%, continent caecal reservoir ~20%.

Cancer

SAQ Malignancy in augmented bladder and after urinary diversion

Paper answer · P1 · P4 · merged from 2

Asked as: P4 SAQ "Secondary tumours after urinary diversion" · P1 SAQ "Cancer in augmented bladder"

Definition

- New cancers arising in or next to bowel used for urinary diversion or bladder augmentation, usually after a **long latency** (10–20 years or more; usually decades). Distinct from recurrent urothelial cancer.
- Bladder augmentation (enterocystoplasty) is mainly for neurogenic bladder. **Earliest reported tumour 4 years after surgery**; tumours after augmentation are often **aggressive and advanced at diagnosis**.

Risk by diversion (German registry, Kälble 2011)

Diversion	Risk	Surveillance
Ureterosigmoidostomy	2.6% (6–29% in long-term series); mean latency about 26 years	Regular colonoscopy **from 5 years**
Augmentation cystoplasty	1.6%	Endoscopy, often from 10 years (see surveillance below)
Colonic neobladder	1.3%	—
Ileocaecal pouch	0.14%	If symptomatic
Ileal neobladder	0.05%	No routine
Ileal conduit	0.02%	No routine

- **Pattern:** risk is highest where urothelium meets colon and is bathed in faeces; purely ileal diversions are lowest.

Risk factors

- **Gastrocystoplasty** (highest among augments), **exstrophy**, immunosuppression after transplant, smoking, chronic infection, long latency.
- The underlying disease may itself carry risk: bladder cancer in spina bifida is about 2–4%, mostly stage III or IV, and non-gastric augmentation did not appear to increase that risk.

Pathogenesis

- **Urothelium–colonic junction** exposed to urine and faeces; tumours arise at the ureterocolic anastomosis. Ureterosigmoidostomy is the extreme model: urothelium beside colonic mucosa bathed in faeces.
- **N-nitrosamines (N-nitroso compounds)** from bacterial action on urinary nitrates in stagnant, infected urine (probably enhancers, not sole cause).
- Chronic inflammation at the anastomosis releases growth factors (EGF, epidermal growth factor) that drive proliferation; ornithine decarboxylase; abnormal mucin; perianastomotic urothelium is genetically unstable.
- Removing the urothelium prevents tumours experimentally.
- **Defunctioned anastomoses still form cancer:** when converting a ureterosigmoidostomy, excise the old anastomosis.

Pathology

- Mostly **adenocarcinoma**, usually at the anastomosis (ureterocolic or **enterovesical**); also adenomatous polyps, urothelial carcinoma (aggressive, often metastatic), signet-ring and other types, sarcoma.
- **Nephrogenic adenoma:** benign; probably the commonest tumour after cystoplasty.

Clinical features

- **Gross haematuria**, rectal bleeding, change in bowel habit, ureteric obstruction (hydronephrosis), recurrent UTI (urinary tract infection), pelvic pain, new incontinence, mass.
- Neurogenic patients may notice symptoms late.

Surveillance and prevention

- **Ureterosigmoidostomy:** regular colonoscopy from year 5.

- **Augmentation:** chapters differ: Campbell bladder-surgery chapter advises **endoscopy yearly from 10 years**; paediatric chapter: no consensus, not shown cost-effective (Mayo group: unnecessary). Higher-risk groups benefit most.
- Investigate any haematuria promptly: cystoscopy with biopsy, cytology, CT (computed tomography) urography; biopsy and staging.
- **Prevention:** prefer ileal segments; counsel patients; regular clean intermittent catheterisation, daily bladder irrigation (less stasis, mucus and stones), treat urea-splitting infection, stop smoking, lifelong (transitional) follow-up.

Management

- **Ureterosigmoidostomy:** resection of the segment with the anastomosis; conversion to conduit or continent diversion.
- **Augmentation:** partial or total (radical) cystectomy with **excision of the augment**, lymphadenectomy and new urinary diversion (conduit or continent pouch).
- Chemotherapy as for histology; prognosis poor because of late stage.
- **Also know:** after cystectomy, recurrent urothelial cancer may arise in the upper tracts or a retained urethra (urethral recurrence about 7–9%); needs separate surveillance. Stones, perforation and metabolic acidosis also occur after augmentation.

Diagrams to draw: ureterosigmoid anastomosis with tumour at the urothelial–colonic junction; clam ileocystoplasty with tumour at the enterovesical suture line.

> **Key points to remember:**
> - Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
> - Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
> - Excise defunctioned anastomoses.
> - Gastric segment, exstrophy, immunosuppression: highest risk in augments.
> - Haematuria needs cystoscopy; treatment is cystectomy.

Also, from the condensed book (not in the paper answer):

- After **ureterosigmoidostomy**, cancer develops in 6–29% after **10–20 yr**. Defunctioned anastomoses still develop cancer, so **excise them, do not ligate**.

10 · Neuromechanical aspects

- Detubularising bowel into a sphere disrupts coordinated peristalsis and, by Laplace's law, increases volume and lowers pressure; this is the basis of reservoirs.

11 · Landmark trials

Trial	Design	Result	Take-home
Alvimopan (Lee 2014)	Randomised double-blind vs placebo, radical cystectomy	Bowel recovery 5.5 vs 6.8 days; stay 7.4 vs 10.1 days; less POI	Use with opioid-based analgesia

134 Continent Urinary Diversion

1 · Definition and basic principles

LAQ Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion" · P2 LAQ "Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction."

Read as: principles of continence in continent diversion, orthotopic neobladder versus non-orthotopic (heterotopic) urinary diversion after radical cystectomy, and the Mitrofanoff principle in urinary and GI (gastrointestinal, bowel) reconstruction, i.e. antegrade continence enema channels.

Introduction, definition and types

- A continent diversion stores urine in a bowel reservoir and empties it **at will**: by voiding through the native urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous pouch**).
- **Neobladder definition**: an **orthotopic** reservoir of detubularised bowel (usually ileum) joined to the native urethra after radical cystectomy, allowing **volitional voiding** without a stoma.
- Evolution: ileal conduit (Bricker, 1950), then continent cutaneous (Kock, 1982), then orthotopic (Camey, 1979; Hautmann, 1988; Studer, 1989).
- **Ileal conduit** (Bricker, 1950) remains the commonest diversion worldwide; **ureterosigmoidostomy** is the oldest continent diversion.
- AUA (American Urological Association)/SUO (Society of Urologic Oncology) and EAU (European Association of Urology): counsel every patient on all options. **80–90% of men and ~75% of women** are medically eligible for a neobladder, but far fewer receive one.

Group	Examples	How urine is held and emptied
Orthotopic	Studer, Hautmann W, T pouch, Padua, Mainz III	Native rhabdosphincter; voids by relaxing and straining
Incontinent cutaneous	Ileal conduit; transverse colon, sigmoid or jejunal conduit	Stoma and appliance
Continent cutaneous	Indiana pouch; right colon pouch with appendix (Mitrofanoff) or Yang–Monti channel; Kock (historical)	Self-catheterisation through a valved channel
Rectal	Ureterosigmoidostomy, Mainz II (sigma-rectum pouch)	Anal sphincter; urine mixed with stool

Principles of continence

- **Continence is a pressure balance:** low reservoir pressure must stay below adequate outlet resistance. Failure of either side causes leakage.

A. Low-pressure, adequate-capacity reservoir

- **Detubularisation** (opening bowel on its antimesenteric border) and cross-folding abolish coordinated peristaltic contractions.
- **Spherical shape:** greatest volume for a given surface area. By **Laplace's law** (P = 2T/R; wall tension = pressure × radius), a larger radius gives lower pressure for the same wall tension. Result: a low-pressure reservoir.
- **Capacity ≥500 mL** when mature. Bowel stretches with time (ileum more than colon); 44 cm of double-folded ileum holds <200 mL at first and 500–600 mL within a year.
- Ileum is most compliant; colon has pressure waves.

B. Reliable outlet

- **Orthotopic:** intact **rhabdosphincter** (pudendal nerve), healthy urethra, careful apical dissection, nerve sparing. Voiding is by Valsalva and pelvic floor relaxation. Night leakage (20–50%) reflects loss of the guarding reflex.
- **Continent cutaneous:** a stable efferent channel that is easy to catheterise and has a durable, dependable continence mechanism (see continent cutaneous mechanisms below).

C. Protect the upper tracts and metabolism

- **Unobstructed ureterointestinal anastomosis** to protect the kidneys.
- **Enough renal function** (eGFR, estimated glomerular filtration rate, about ≥35–40 mL/min) to clear the acid load reabsorbed by bowel mucosa.
- Ability to self-catheterise.

Studer as an example of the principles: orthotopic low-pressure ileal reservoir (**Studer, 1989**): detubularised ileal reservoir with a long **isoperistaltic afferent limb**, joined to the urethra. **Detubularisation** stops coordinated contractions, so pressure stays low; a **sphere** (Laplace) gives most volume for surface area. The **isoperistaltic afferent limb** limits reflux (damps reflux during Valsalva voiding) and accepts **short ureters**. No staples in the reservoir. Continence depends on the **rhabdosphincter**.

Choice of bowel segment

Segment	Points
Ileum (default)	Most compliant, lowest pressure; mesentery reaches the urethra; mucosa atrophies, so absorption falls
Ileum + caecum (Mainz)	Near-ileal compliance; loses ileocaecal valve
Colon (right, sigmoid)	Less distensible, pressure waves, so more night leakage; used when ileum is short or diseased
Stomach	Secretes acid: helps in renal impairment; haematuria–dysuria; rarely used

Reflux versus obstruction

- Voiding by Valsalva raises pressure equally throughout, so reflux into a low-pressure pouch is benign, while **every antireflux valve can obstruct** (Le Duc tunnel stricture 7–29%; late Kock nipple stenosis).
- Direct refluxing anastomosis (Wallace, Leadbetter): stricture about 3–6%.
- **USC (University of Southern California) RCT (randomised controlled trial)**, 484 patients: Studer vs T pouch, no difference in renal function or infection at 3 years; the antireflux T pouch gave no renal benefit and needed more secondary procedures (reoperations).

- Studer: reflux on cystogram, yet kidneys preserved in about 95%.

Continent cutaneous mechanisms

Mechanism	Example	How it works	Weakness
Nipple valve (intussuscepted)	Kock (historical), Mainz I	Reservoir pressure compresses the intussuscepted segment	Same pressure effaces it; failure ≥15%, reoperation about one-third; late nipple stenosis
Flap valve: narrow tube (appendix) tunnelled submucosally or in taenia	**Mitrofanoff** (Penn, right colon pouch), Yang–Monti ileal tube if no appendix, Kropp	Narrow tube in a submucosal tunnel is coapted as pressure rises	No pop-off; stomal stenosis
Imbricated ileocaecal valve + tapered ileum	**Indiana** pouch	Continence zone lies at the plicated valve	Bowel frequency, B12 loss
- Used when continence is wanted but the urethra is unusable; umbilical stoma preferred.			
- **Also know** (native outlet): Kropp, Pippi Salle, slings, AUS (artificial urinary sphincter), bulking, bladder neck closure.			

The Mitrofanoff principle

- **Mitrofanoff (1980):** any supple, narrow tube implanted in a **submucosal tunnel** of a low-pressure reservoir, with its other end at skin, gives a **continent catheterisable channel**.
- As the reservoir fills, pressure transmitted through the mucosa flattens the tunnelled segment against the muscle backing: a **flap valve**. Continence lies in the tunnel, not in the conduit.
- Appendix: only a **2 cm tunnel** suffices because of its small calibre.

Conduits

Conduit	Points
Appendix (first choice)	Natural tube, minimal harvest morbidity; needs length ≥5–6 cm and ≥12 Fr; caecal cuff or **Mitchell caecal extension** adds length
Yang–Monti	2–3 cm ileum opened on the antimesenteric border and retubularised transversely over 12 Fr; double or spiral Monti for obese; twice the revisions
Tapered ileum	Long enough; kinks if redundant
Ureter, fallopian tube	More stenosis
Casale bladder flap	No bowel; consumes bladder capacity

Technique (appendicovesicostomy)

1. Mobilise caecum; detach appendix with a caecal cuff, preserving mesoappendix.

2. Spatulate tip; tunnel submucosally (posterolateral or anterior bladder wall), intra- or extravesically.
3. Bring the base to the umbilicus or right iliac fossa with skin flaps, without tension or twist; keep it short and straight; fix the bladder to the abdominal wall.
4. Catheterise after each step. Channel catheter 2–4 weeks; then CIC every 4 hours by day.

Mitrofanoff applications in urinary reconstruction

- **Neurogenic bladder** (spina bifida, spinal injury), often with augmentation, when urethral CIC is difficult (wheelchair-bound, sensate urethra).
- After bladder neck reconstruction or closure; exstrophy; valve bladder.
- **Continent cutaneous diversion** after cystectomy: Penn and right colon pouch (appendix in taenia; up to 100% continence); Yang–Monti if no appendix.
- Unusable urethra: complex stricture, urethrectomy.

Mitrofanoff applications in GI reconstruction: MACE

- **MACE (Malone antegrade continence enema; Malone, 1990):** appendix brought to skin as an appendicocaecostomy with an antireflux tunnel (flap valve) so stool does not leak.
- Antegrade washouts empty the colon, giving **faecal continence** in neurogenic bowel (spina bifida), anorectal malformations, cloaca and refractory constipation.
- Options: Monti-MACE; **split appendix** for both MACE and urinary channel; caecostomy button. Yang–Monti for the urinary channel when the appendix is absent or used for MACE.

Results and complications of continent catheterisable channels

- **Continence >95%** for flap valves; nipple valves about 85%.
- **Stomal stenosis is the commonest problem**; also false passage, channel polyps or granulation, pouch stones, revision (suprafascial or subfascial).
- Missed CIC: no pop-off, so **reservoir rupture** or upper tract damage.

Patient selection

	Orthotopic only	All continent diversions
Absolute contraindications	Tumour at the urethral margin (frozen section); significant urethral stricture	Severe renal (eGFR <40 mL/min) or hepatic dysfunction (liver failure); physical or mental inability to self-catheterise
Relative	Poor external sphincter; bladder-neck or anterior vaginal wall tumour (women); prostatic stromal invasion (men); irradiated sphincter (irradiated pelvis)	Frailty, poor cognition, social problems, prior radiation or bowel resection

- **Renal function**: creatinine <1.7–2.2 mg/dL or eGFR **>35–40 mL/min** (below about 35–40 mL/min = severe renal impairment; another source: <40 mL/min), because the kidney must excrete acid reabsorbed from stored urine. Formal tests: urine pH ≤5.8 after ammonium chloride, osmolality ≥600 mOsm/kg after water deprivation.
- **Urethral recurrence**: in men the strongest predictor is prostatic stromal invasion (prostatic urethral tumour raises urethral recurrence risk); in women, bladder-neck or anterior vaginal wall tumour. **Intraoperative frozen section** of the urethral margin decides. CIS (carcinoma in situ) or multifocality alone do not exclude a neobladder.

- **Age and obesity alone are not contraindications**; comorbidity, cognition and support decide. A conduit suits the frail or unsupported patient.
- Continent cutaneous pouch: continence wanted but urethra unusable.
- Mark a conduit stoma site in everyone: up to ~10% change plan intraoperatively.

Orthotopic versus non-orthotopic: comparison

Feature	Orthotopic neobladder	Ileal conduit	Continent cutaneous	Ureterosigmoidostomy
Appliance	None	Bag	None; small stoma	None
Body image	Best	Worst	Good	Good
Complexity	High	Lowest	Highest	Moderate
Continence	Day 80–90%; night incontinence 20–50%	Not applicable	High (Indiana, Mitrofanoff)	Depends on anal sphincter
Catheterisation	10–20% of men, ~30% of women	None	Always	None
Bowel used	Long ileal segment (Studer 54–56 cm)	10–15 cm ileum	Right colon + terminal ileum	None isolated
Typical late problems	Night leakage, retention, acidosis	Stomal stenosis (20–24%), parastomal hernia, renal deterioration	Stomal stenosis, stones, difficult catheterisation	Severe acidosis, pyelonephritis, cancer at the anastomosis

Neobladder types

Design	Segment	Ureters
Studer	40–44 cm ileum + afferent limb	Refluxing, isoperistaltic limb
Hautmann W	About 70 cm ileum	Refluxing; largest capacity
T pouch	44 cm + T limb	Antireflux flap valve
Camey II; Padua	Ileum	Antireflux (Le Duc)
Mainz III; sigmoid	Ileocaecal; colon	Tunnelled

	Studer	Hautmann
Ileum	54–56 cm, U + afferent limb	~70 cm folded in a W
Initial capacity	Smaller early capacity	Larger: earlier continence
Late	Less late retention	More retention, reabsorption

Studer technique (operative)

- Brief: isolate ileum 15–25 cm from the ileocaecal valve; open the distal part on the antimesenteric border, sew into a U, fold into a sphere; end-to-side ureteric anastomoses to the afferent limb; anastomose the dependent point to the urethra. 1. **Isolate 54–56 cm of terminal ileum**, ending **15–25 cm** above the ICV (ileocaecal valve) (another source: **15–20**

cm proximal to the ICV); restore bowel continuity. 2. **Reservoir: distal 40–44 cm** (two 20–22-cm limbs) opened on the antimesenteric border; sew the two limbs (medial edges) into a **U** (back wall). 3. **Afferent limb:** leave the proximal **10–15 cm** tubular as the afferent limb (another source: **about 15 cm**; original 20 cm; 10–12 cm in later figures); close its end. 4. **Fold the U in the opposite direction** (across) and close the anterior wall into a **sphere**. 5. **Ureters:** spatulated **end-to-side, refluxing** ureteroileal anastomoses to the afferent limb, interrupted 4-0, **stented**. 6. **Urethral anastomosis** at the most dependent point with pre-placed sutures; catheter and suprapubic tube.

- Capacity rises from <200 to **500–600 mL** in a year.
- **Intracorporeal robotic Studer** follows the same steps.

Postoperative care, voiding training and follow-up

- Mucus irrigation; oral bicarbonate; cystogram before catheter removal.
- **Voiding training**: void **sitting**, relax the pelvic floor, strain gently; **timed voiding** every 2 hours, extended to 3–4 hours as capacity grows (under 200 mL to 500–600 mL in a year). **Night alarm.**
- Lifelong follow-up: annual metabolic panel, B12, upper-tract imaging; urethral wash cytology if CIS (carcinoma in situ).

Current status of orthotopic neobladder

1. Preferred where suitable: closest to a native bladder. AUA/SUO and EAU: counsel every cystectomy patient on all options and refer to an experienced surgeon.

2. Under-used. 80–90% of men and about 75% of women are eligible, yet only about 9% of US patients receive any continent diversion, and this is falling; continent diversion in RAZOR 21%, iROC 11%. Causes: comorbid patients, surgeon training, intracorporeal robotics.

3. Designs in use. Studer (40–44 cm U plus isoperistaltic afferent limb) and **Hautmann W** (about 70 cm; refluxing since 1997) dominate; T pouch or serous-lined tunnel when antireflux is wanted.

4. Oncological safety.

- Men: urethral recurrence about 7% at 5 years; main risk is prostatic stromal involvement. The neobladder seems protective (meta-analysis HR (hazard ratio) 0.44).
- Women: risk lies with bladder-neck tumour; no skip lesions, so preserve the urethra. **Frozen section** of the urethral margin decides in both sexes.
- Local recurrence <10%, rarely disabling the pouch; nodal disease is not a bar. Selected **post-radiation** patients do well.

5. Function (outcomes)

Outcome	Result
Day continence	80–90% (most by 3–6 months); another source (Studer): 90–98%; about 90%
Night incontinence	20–50% (up to half leak): no sphincter (guarding) reflex, mucosal water secretion, nocturnal diuresis
Retention needing CIC	Men 4–10%; **women 20–30%** (pouch angulation): men leak, women retain
Self-catheterisation	**10–20% of men, ~30% of women**

6. Complications. Early morbidity equals conduit. Late: hyperchloraemic metabolic acidosis, vitamin B12 deficiency, mucus, ureteroileal stricture, stones, pouch rupture, pouch–vaginal fistula (5–10%), hernia.

7. Robotic intracorporeal neobladder: comparable to open; long-term data lacking.

8. Quality of life: no diversion proven superior; body image favours neobladder.

- **Also know:** nerve sparing and uterine preservation aid continence.

Metabolic complications (bowel in the urinary tract)

Segment	Disturbance
Ileum, colon	**Hyperchloraemic metabolic acidosis** (ammonium chloride absorbed); hypokalaemia, worse with colon
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis
Stomach	Hypochloraemic, hypokalaemic alkalosis

- Acidosis is severe in most ureterosigmoidostomies; most colonic (Mainz) pouches need alkali. Treat with sodium bicarbonate or citrate, and replace potassium as well.
- **Also know:** osteomalacia; vitamin B12 deficiency and bile-salt diarrhoea after ileal loss (liver B12 stores last 3–5 years); stones (ileal conduit 10–12%, continent caecal reservoirs ~20%); bacteriuria; drug reabsorption (methotrexate, phenytoin); ammoniagenic encephalopathy; growth delay in children.
- **Secondary cancer**: ureterosigmoidostomy 6–29% after 10–20 years (yearly endoscopy from 5 years); ileal conduit and ileal neobladder lowest.

Surgical complications

- **Early (≤90 days)**: ileus, urine leak (long pouch suture lines; drain percutaneously), infection, mucus retention; not more than with a conduit.
- **Late**: ureteroileal stricture ~3–6% (Wallace or direct end-to-side) (Studer/neobladder: **<3%**); neobladder retention, urethral stricture, rare pouch rupture; ventral and parastomal hernia; **hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, stones, pyelonephritis (commonest late, Studer), urethral recurrence.
- Continent cutaneous channels: stomal stenosis commonest (see results above).

Quality of life

- No randomised trial. Body image favours the neobladder, urinary function the conduit; no diversion is superior on HRQoL (health-related quality of life).

Recent advances

- RARC (robot-assisted radical cystectomy): **RAZOR (2018)** non-inferior 2-year progression-free survival (open diversion in both arms); **iROC**: intracorporeal diversion gave more days alive and out of hospital at 90 days.
- Intracorporeal neobladders and nerve- or organ-sparing cystectomy are increasing.

Diagrams to draw:

- Studer neobladder and ileal conduit side by side; Studer U-pouch with afferent limb, ureters and urethral anastomosis; ileal segment marked (afferent limb plus 40–44-cm U), then the folded spherical pouch with afferent limb.
- Flap valve (tunnelled appendix collapsing as pressure rises) beside a nipple valve.

Key points to remember:

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL; 500–600 mL by 1 year) against a reliable outlet.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel; Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

Also, from the condensed book (not in the paper answer):

- Orthotopic is now preferred (natural voiding, no appliance, fewer revisions); cutaneous suits pre-existing incontinence or urethral disease.

KEY POINT

Continence is a pressure balance: low reservoir pressure (from detubularisation) against adequate outlet resistance (sphincter or efferent valve). If either fails, the patient leaks. Camey I (tubular ileum) failed because its pressure was too high.

2 · Patient selection

Selection means excluding a bad urethra, kidneys or hands.

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

Also, from the condensed book (not in the paper answer):

- **Locally advanced or node-positive (N+) disease found at surgery is not a contraindication**: pelvic recurrence is ~7% (13% if extravesical or N+) and rarely disables the neobladder. The exception is a grossly positive margin at pubis, rectum or side wall.
- eGFR 40–59 is safe, as decline is no faster than with a conduit.
- A short or fatty mesentery is the commonest technical reason to abandon orthotopic diversion.

ALGORITHM

1. Estimated glomerular filtration rate (eGFR) >40.
2. Intact external sphincter and adequate bowel.
3. Not frail; no severe liver, cardiac or psychological disease.
4. Men: no distal prostatic urethral cancer. Women: no palpable anterior vaginal tumour or cervical invasion.
5. If all are met and the patient will self-catheterise, offer an orthotopic neobladder.

- Mark a conduit site in everyone, through mid-rectus and away from creases, umbilicus and scars.

Urethral recurrence

Feature	Men	Women
Dominant risk	Prostatic urethral involvement; worst with **stromal invasion** (usually N+, so give neoadjuvant chemotherapy)	**Bladder-neck tumour**; anterior vaginal wall invasion
Pre-op	Transurethral resection (TUR) biopsy at 5 and 7 o'clock beside the verumontanum — optional	Examination under anaesthesia (EUA); bladder-neck biopsy
Definitive test	Intraoperative **frozen section of urethral margin** (more predictive than TUR biopsy; the American Urological Association (AUA) requires a negative margin)	
Bladder carcinoma in situ (CIS)	Not predictive	
Skip lesions	—	None: a negative bladder neck means a negative urethra

- Risk rises with depth: none < CIS/mucosal/ductal < stromal. **The neobladder itself is protective** (hazard ratio (HR) 0.44).
- Prostatic involvement needs urethral surveillance (wash cytology, urethroscopy).
- In women, keep the urethra if bladder-neck biopsy and proximal urethral frozen section are negative.

Patient-related factors

- **Renal function** is the key contraindication. Proceed if creatinine <**1.7–2.2 mg/dL** or eGFR >**35–40 mL/min**, with hydration and oral bicarbonate and no nephrotoxins.
- **Self-catheterisation** is needed by 10–20% of men and ~30% of women with a neobladder, and by all with cutaneous pouches. The need may appear years later, so all must be willing; multiple sclerosis, quadriplegia or frailty favour a conduit.
- **Sphincter weakness** is managed with an artificial urinary sphincter (AUS) in men, and Burch or sling in women (who will likely catheterise).
- **Prior radiation** is acceptable without incontinence, stricture, fistula or severe bowel damage; the final decision is made intraoperatively.
- After **prior prostatectomy**, pre-op continence predicts outcome. After **prior bowel resection** with short bowel, use a sigmoid neobladder.

3 · Preserving continence at cystectomy

The **rhabdosphincter** (pudendal nerve) is critical. It lies **anterior and lateral** to the urethra in both sexes.

Men

- Never dissect posteriorly beyond the apex. Incise endopelvic fascia and puboprostatic ligaments minimally; control the dorsal venous complex (DVC) with **absorbable** suture, as clips or staples erode into the anastomosis.

ALGORITHM

1. Open the endopelvic fascia; incise the puboprostatic ligaments slightly.
2. Ligate the DVC close to the apex.
3. Incise the anterior urethra and place sutures.
4. Place posterior sutures through the rectourethralis, then divide the posterior urethra.
5. Tag the sutures until the ureteroenteric anastomosis is done.

- Place sutures **before** full division, as the male urethra retracts. Apical frozen section with tumour means urethrectomy and cutaneous diversion or conduit; atypia or focal CIS means a second distal margin.
- Nerve sparing has a controversial continence benefit; it may help night continence by keeping urethral sensation.

Women

- Sparing uterus and anterior vagina in selected low-stage disease avoids fistula, improves sexual function and reduces retention. A palpable bladder-neck or vaginal tumour needs en bloc urethrectomy and cutaneous diversion.

ALGORITHM

1. Divide the posterior pedicles and develop the plane off the anterior vagina.
2. Use the Foley balloon to mark the bladder–urethra junction.
3. Free the bladder neck and the first 0.5 cm of urethra.
4. Clamp the bladder neck, divide just distal to it, and send a frozen section.

- Leave endopelvic fascia untouched; the bladder neck is not needed for female continence.
- Always interpose an **omental flap** between vagina and pouch, and keep vaginal and urethral suture lines apart.

4 · Choice of bowel and ureteric reflux

Ileum gives lowest pressure; obstruction, not reflux, loses kidneys.

- **Ileum is the default**, being most distensible; its mucosa atrophies, lowering mucus and absorption. Colon has pressure waves; stomach has poor compliance and haematuria–dysuria. Close with absorbable suture, as staples cause stones.
- **Antireflux is unnecessary**: Valsalva raises pressure equally throughout, and every valve can obstruct. A direct (Wallace or Leadbetter) anastomosis strictures in **3–6%**, against 7–29% for Le Duc. **Obstruction, not reflux, destroys kidneys.** Kock nipple failures are late (>10 yr) and silent.

5 · Orthotopic neobladder designs

*See the paper answer **Intestinal segments in urology and orthotopic neobladder** (Section 12, Ch 133).*

Also, from the condensed book (not in the paper answer):

- Hautmann became refluxing in 1997, cutting stenosis from 9.5% to 1%. Studer is the most popular design; the sigmoid neobladder has the worst night continence.

ALGORITHM

1. **Studer:**
2. Isolate ileum ending 15–20 cm proximal to the ileocaecal valve; keep a 40–44 cm reservoir (two 20–22 cm limbs) and an afferent limb.
3. Open the reservoir, oversew the back wall, and fold it the opposite way.
4. Make stented end-to-side ureteroileal anastomoses.
5. Join the dependent point to the urethra.

KEY POINT

Hautmann vs Studer is a capacity trade-off: a bigger initial pouch is continent sooner but empties worse and absorbs more later.

6 · Continent cutaneous reservoirs

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

Also, from the condensed book (not in the paper answer):

- **Indiana** uses the right colon and ~10 cm of terminal ileum, with a double-imbricated ileocaecal valve and tapered ileum; the whole colon is opened to abolish caecal peristalsis.
- An **umbilical stoma** has less stenosis and is easiest for paraplegics; the appendix takes only 14–16 Fr catheters.

7 · Perioperative care

- **Enhanced recovery after surgery (ERAS)** means prehabilitation, carbohydrate loading, restrictive fluids, no nasogastric (NG) tube, early diet and opioid-sparing analgesia. **Alvimopan** cut ileus from ~23% to ~7%; ileus is the commonest gastrointestinal complication.
- **Drainage**: single-J stents and a 24 Fr urethral catheter irrigated for mucus; remove the catheter at **3 weeks**. Then use timed voids, lengthened progressively, with pelvic floor exercises.

8 · Results and complications

Men tend to leak, women to retain.

Early (<90 days) morbidity is no higher than with an ileal conduit.

Complication	Key facts	Management
Urine leak	Commoner (long suture lines)	Catheter, stents; if undrained, percutaneous drain with or without nephrostomies; **avoid early open repair**
Bacteriuria	Very common	Treat only if symptomatic or Proteus/Pseudomonas; in febrile urinary tract infection (UTI), look for obstruction, stones, residual; if recurrent, nitrofurantoin or methenamine
Ureteroileal stricture	Can appear after 10 yr, often silent	Suspect on CT or ultrasound (US); confirm by pouchogram or antegrade study; endoscopy (retreatment >50%) inferior to open
Pouch stones	Kock and stapled pouches	Endoscopic
Pouch perforation	Rare; radiation, chronic retention	Cystogram; catheter with or without drains if stable
Pouch–vaginal fistula	5–10%; vaginal excision, radiation	Lateral cystogram before catheter removal; beyond the first weeks it won't heal, so repair transvaginally or transabdominally, or convert to cutaneous
Stomal stenosis (cutaneous)	Up to ~20%; after 90 days, radiation	Usually open revision
Efferent limb necrosis	Rare, ischaemic	Endoscopy for level; explore if below fascia
Incisional hernia	Common; impairs emptying	Repair

Continence

- Day continence comes over **3–6 months**, ultimately **80–90%**. Night continence lags into year 2, and nocturnal incontinence persists in **20–50%**. Risks are age >65, colon, diabetes, no nerve sparing and hysterectomy.
- Night leakage reflects loss of the sphincter–detrusor reflex; advise voiding at bedtime and an alarm 2–3 times a night.
- Evaluate persistent leakage after the 12–18-month plateau, excluding fistula and using urodynamics.

Stress incontinence	Men	Women
Bulking agent	Marginal	Mild only; <50%, poorly durable
Definitive	AUS or male sling	Pubovaginal sling — most then catheterise
Safety rule		No blind retropubic needle passage; use bone anchors or the prepubic route

Retention

- It presents as infection, overflow, palpable pouch or new hydronephrosis. It affects men in 4–10% (~21% by 10 yr) and **women in 20–30%**.
- In women the cause is **mechanical**: the pouch falls back and angulates the pouch–urethral junction. Prevent with omentum behind the pouch, vaginal suspension, sacrocolpopexy and uterine preservation.

- Treat with clean intermittent self-catheterisation (CISC), as drugs are useless; exclude stricture or recurrence.

> **KEY POINT**
>
> Men fail by **leaking** (sphincter without reflexes), so augment the sphincter. Women fail by **not emptying** (unsupported pouch kinks), so support the pouch.

9 · Follow-up

- Stricture occurs in the first 4 months, recurrence from 4 months to 3 yr, and stones, late obstruction and urethral or upper-tract tumour after 3 yr.
- Review 4-monthly in year 1, 6-monthly to 3 yr, then annually (examination, metabolic panel, blood count), with CT and chest imaging to 5 yr.
- Check B12 yearly. Do urethral wash cytology in men, and bimanual examination and voided cytology in women. After 5 yr continue lifelong upper-tract US, metabolic panel and blood count.
- No diversion has proven superior quality of life; the neobladder gives better body image and the conduit better urinary function.

10 · Landmark trials

Trial	Design	Result	Take-home
USC Studer vs T pouch	Randomised, 484 radical cystectomy (RC) patients	No difference in renal function or infection at 3 yr; T pouch more secondary procedures	Antireflux adds cost without benefit
RAZOR	Open vs robotic RC, extracorporeal diversion	Continent diversion only 21%	Low uptake even with expert surgeons
iROC	Open vs robotic RC, intracorporeal diversion	Continent diversion 11%	Intracorporeal approach further reduces continent diversion

135 Minimally Invasive Urinary Diversion

1 · Context

- Robot-assisted radical cystectomy (RARC) equals open oncologically in trials (RAZOR), but with **extracorporeal** diversion (ECUD); intracorporeal diversion (ICUD) is untested.
- In the International Robotic Cystectomy Consortium (IRCC), ICUD rose ≈**11%/year**, driven by the **ileal conduit**, the commonest robotic diversion.

2 · Patient selection

A bar to one step is not a bar to all.

Obesity, age, advanced disease and prior surgery no longer exclude.

Decision	Contraindication
Minimally invasive route	Poor cardiopulmonary tolerance of steep Trendelenburg (relative; eased by Xi and AirSeal)
Use of ileum	Short bowel; severe inflammatory bowel disease; radiation enteritis
Neobladder (absolute)	Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function (as for open)

ALGORITHM

1. Pasadena Consensus learning pathway:
2. RARC with extracorporeal diversion.
3. Intracorporeal conduit (≈30–100 cases).
4. Continent diversion (≈25–50 cases). Target time: **≤4 h conduit, ≤5 h neobladder**.

3 · Perioperative bundle

Enhanced recovery (ERAS) uses alvimopan, goal-directed fluids, immediate oral diet and opioid-sparing analgesia, giving less ileus and a shorter stay. Caution: a stay **<5 days** about doubles 90-day major complications.

Element	Recommendation
Fasting	Solids to 6 h, clear fluids to 2 h
Bowel preparation	**None**, and no oral antibiotics (electrolyte upset, liquid spillage, no benefit)
Thromboprophylaxis	Pneumatic compression and stockings; low-molecular-weight heparin before and up to **4 weeks** after
Antibiotics	Broad-spectrum intravenous within **1 h**

4 · Position and ports

- Use lithotomy and steep Trendelenburg, with Hasson entry after prior surgery.
- **Six to seven ports sit more cephalad** to reach over small bowel and take nodes to the aortic bifurcation.
- A neobladder adds a left stapling port; a suprapubic 12-mm port serves bowel reanastomosis and extends for specimen extraction.

5 · Nerve sparing

One plexus serves sexual function and neobladder continence.

	Male	Female
Danger points	Presacral and medial common iliac; high pedicle ligation; seminal vesicles and Denonvilliers fascia; prostatic pedicles and apex	Lateral vaginal wall and bladder base (inferior hypogastric plexus)
Manoeuvre	Athermal, intrafascial, traction-free pedicle control	Preserve tissue around proximal urethra and anterior vaginal wall; extract via Pfannenstiel, not transvaginally
Sphincter	One-third overlapped by dorsal venous complex; dorsal penile nerve branches within **3 mm** of apex	Mainly mid–distal third, so continence survives loss of bladder neck

6 · Intracorporeal ileal conduit

ALGORITHM

1. Pass the left ureter under the sigmoid mesentery to the right.
2. Choose 12 cm ileum, 15–20 cm from the ileocaecal valve; place the marionette stitch.
3. Isolate with a 45-mm stapler.
4. Do ureteroileal anastomoses (left first) and stent.
5. Reanastomose bowel, then form the stoma.

- The **marionette stitch** (1-0 silk through the stoma site and conduit end) is held **untied**; it replaces the missing hand and positions the conduit.
- Avoid a redundant segment (See Ch 133), and tag cut ends to prevent malrotation.
- The **enterotomy** is an athermal cold-scissors buttonhole, because thermal slits gave more strictures.
- Single-J stents are used, and the **guidewire comes out only after anchoring**. Use a Bricker (separate end-to-side) or Wallace (Y, one anastomosis) technique.

7 · Intracorporeal neobladder

The key change is **early neobladder–urethral anastomosis** (Karolinska): dependent ileum is anchored to the urethra first, then the pouch is built in place. Design principles: See Ch 134.

Design	Ileum	Signature
Karolinska (modified Studer)	50 cm	10-cm afferent limb; Penrose sling
USC (Studer)	65 cm (longest)	44+16 chimney+5 discarded
UCLH Pyramid	50 cm	Koie-derived cross-fold
Saint-Augustin modified Y	40 cm (shortest)	Refluxing, uncrossed ureters
Padua	42 cm	Only stapled pouch; split-nipple antireflux
FloRIN	50 cm	Asymmetrical U; L-shaped neo-trigone; antireflux
Roswell Park (Hautmann W)	45–50 cm	One trough at a time; chimneys

ALGORITHM

1. Urethral anastomosis (Van Velthoven barbed suture).
2. Detubularise distal 40 cm (cold scissors); form the posterior plate.
3. Join ureters to afferent limb or chimney, stented.
4. Close the anterior wall once the 22-Fr catheter is confirmed.

- A **tension-free urethral anastomosis** lacks tactile feedback. It is helped by urethral length, less Trendelenburg, perineal pressure and Penrose traction. Incising mesenteric peritoneum risks pouch ischaemia.

Intracorporeal Indiana pouch

ALGORITHM

1. Mobilise the right colon.
2. Divide ileum 15 cm from the ileocaecal junction and transverse colon proximal to the middle colic; restore continuity.
3. Detubularise along the anterior taenia and fold.
4. Bring ureters through the posterior plate on double-J; place a caecostomy.
5. Taper ileum over 14 Fr, imbricate the valve, and form an umbilical stoma.

KEY POINT

Capacity comes from detubularised right colon; continence from tapered ileum and the imbricated ileocaecal valve.

8 · Postoperative care

Avoid overhydration; there is no routine nasogastric tube (Cochrane). Irrigate the neobladder **8-hourly** for mucus.

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Day 1
Neobladder catheter	**3–4 weeks** , after leak-free pouchogram
Externalised stents / double-J	≈1 week / cystoscopically at 3–4 weeks
Feeding-tube stents tied to Foley	With the catheter

9 · Outcomes

Most problems come from the diversion, and many appear late.

- **Operative time** depends most on diversion type (conduit ≈70 min shorter); ICUD and ECUD take similar time. Proficiency needs ≥**66 RARCs**.
- **Complications** are mostly diversion-related, infection commonest. Two-thirds of reoperations occur after 3 months, so 90-day reporting underestimates them.
- **Parastomal hernia** reaches ≈one-third by 3 years (risks: long operation, fascial defect ≥30 mm, low eGFR).
- ICUD versus ECUD (IRCC) is neutral for high-grade complications and readmission after adjustment, with less transfusion.

- **Reoperation** reaches ≈half by 5 years; predictors are neoadjuvant chemotherapy and any high-grade complication.
- **Uretero-enteric stricture** rises over 5 years; robotic revision equals open.

> **KEY POINT**
>
> Night continence is worse because the urethrosphincteric guarding reflex is inactive in deep sleep, so continence depends on reservoir capacity. Hence detubularisation, nerve and sphincter preservation, a night alarm and limiting night output. Assess at 6–12 months.

- Good ECOG status and retained filling sensation predict continence. Up to 25% of men need clean intermittent catheterisation (CIC) after open neobladder.
- Intracorporeal series report day continence 68–100%, night 55–85%, and CIC up to 11%.

10 · Landmark trials

Trial	Design	Result	Take-home
RAZOR	RCT RARC vs open, extracorporeal diversion	Oncological equivalence	Says nothing about ICUD
iROC	Multicentre RCT, intracorporeal vs open	Open: worse QoL at 5 weeks, more disability at 5 and 12 weeks; no difference after 12 weeks; 11% continent	ICUD gains are early and recovery-related

124 Bladder Surgery for Benign Disease

1 · Surgical anatomy essentials

- The bladder is an extraperitoneal organ (mucosa, lamina propria, detrusor, perivesical fat) derived from the endodermal urogenital sinus.
- **Anteriorly** lies the space of Retzius, the usual access. **Posteriorly**, peritoneum reflects near the prostatorectal junction (men) or vesicouterine space (women), and Denonvilliers fascia separates bladder and prostate from rectum. **Laterally** lie the pelvic sidewall, obturator space, and vas or round ligament.
- The **medial umbilical ligaments** are the obliterated umbilical arteries; the **median umbilical ligament** is the urachus, which can persist as a patent urachus, sinus or cyst.
- The **uterine artery** crosses over the ureter near the internal os, and the **vas** crosses it near the seminal vesicle tip. The distal ureter runs medial to the medial umbilical ligament.
- Superior and inferior vesical arteries form the posterolateral pedicles; lymph drains mainly to external iliac nodes. Nerves are sympathetic T10–L2 and parasympathetic S2–S4; the pudendal nerve (Onuf's nucleus) supplies the sphincter.

> **KEY POINT**
>
> Three landmarks organise every benign bladder operation: Retzius for access, the peritoneal reflection for the posterior bladder, and the distal ureter for safety. They decide whether surgery stays extraperitoneal.

> **KEY POINT**
>
> Common aftercare: watertight **two-layer 2-0 polyglactin** closure (mucosa, then seromuscular), leak test, large-bore Foley for about **1 week**, drain out at about **48 h** if output is low, and a **cystogram before catheter removal**. With high drain output, drain creatinine above serum means a urine leak.

2 · Bladder diverticula

A sac without muscle cannot empty: stasis, infection, stones, cancer.

SAQ Bladder diverticulum.

Paper answer · P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3c

Definition

- Herniation of bladder mucosa through the **muscularis propria**, forming a sac with a neck. With no functioning muscle it **cannot empty** (stasis).
- Male:female about 9:1.

Classification

Feature	Congenital (primary)	Acquired (secondary)
Age	Children (<10 years)	Men >60 years

Feature	Congenital (primary)	Acquired (secondary)
Cause	Weak detrusor at the ureteric hiatus	**BOO (bladder outlet obstruction)**: BPH (benign prostatic hyperplasia) in about 70%, stricture; neurogenic bladder
Number, site	Solitary; **paraureteral (Hutch)**	Often multiple, anywhere
Bladder wall	Smooth	Trabeculated
Associations	VUR (vesicoureteral reflux); Ehlers–Danlos, Williams, Menkes syndromes	Stones, cancer
- **Also know:** iatrogenic (after cystotomy or reimplantation); cellule, saccule and diverticulum form a continuum.		

Clinical features

- Usually incidental while investigating LUTS (lower urinary tract symptoms), haematuria or UTI (urinary tract infection).
- **Double voiding**, recurrent UTI, stones, retention, ureteric obstruction.
- **Malignancy** in 0.8–10% (urothelial 70–80%, squamous 20–25%). With no muscle layer, tumours invade the fat early (beyond lamina propria = **T3a**).

Investigations

- **VCUG (voiding cystourethrogram)**: size, VUR, emptying, **paradoxical filling** during voiding.
- **Cystoscopy**: enter and inspect the sac (flexible scope); cytology; biopsy carefully (perforation risk).
- **CT (computed tomography) urography**: tumour, stones, ureteric deviation, hydronephrosis.
- **Urodynamics**: BOO or neurogenic dysfunction (urine "sinks" into the sac, so contractility looks falsely low).

Management

- **Treat the cause first** (BOO); a small diverticulum may then empty and need only surveillance.
- **Observation** (cystoscopy, cytology) if asymptomatic; **CIC (clean intermittent catheterisation)** if unfit with poor emptying.
- **Endoscopic**: neck incision or resection ± fulguration in frail patients.
- **Diverticulectomy indications**: refractory symptoms, recurrent UTI, stones, upper-tract damage or VUR, premalignant change.
- **Tumour inside**: partial or radical cystectomy.

Diverticulectomy technique

- Open, laparoscopic or robotic; endoscopic prostate surgery first, same session.
- Steps: ureteric catheter if near the ureter → cystotomy → evert or dissect the sac from its pseudocapsule → close in two layers → catheter about 1 week, cystogram before removal.

- Complications: ureteric injury, leak.

Diagram to draw: paraureteral (Hutch) diverticulum with narrow neck and ureter.

> **Key points to remember:**
> - No muscle → stasis, UTI, stones, cancer.
> - Acquired type: treat the BOO first.
> - Tumour invading beyond lamina propria is T3a.

Also, from the condensed book (not in the paper answer):

- Voiding cystourethrogram (VCUG) shows hidden diverticula and the outlet obstruction, and decides between a preoperative stent and simultaneous reimplantation.
- A fibrous **pseudocapsule** gives the surgical plane. Most acquired diverticula are asymptomatic and incidental.
- CT urogram shows hydronephrosis, ureteric displacement and stones. Send cytology in all, add flow rate or urodynamics, and obtain a **negative urine culture** before surgery.
- At cystoscopy, enter every sac and note neck size and distance from the orifice; treat stricture or bladder neck contracture before or with the diverticulectomy.

Situation	Management
Small, asymptomatic, uncomplicated	Observe, or endoscopic fulguration/resection
Large with residual urine, recurrent urinary tract infection (UTI), stones, pain	Diverticulectomy
VUR or ureteric obstruction	Diverticulectomy with or without reimplantation
Cancer in diverticulum	Partial or radical cystectomy, not diverticulectomy

Operative principles

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**, and augmentation for neurogenic bladder. Do **endoscopic outlet surgery first**, because later irrigation disrupts fresh suture lines.
- Localise the sac by selective filling or flexible cystoscopic transillumination, and place a ureteral catheter if it lies near the ureter.
- Open extraperitoneal surgery remains the gold standard, even for posterior sacs; laparoscopic and robotic surgery is reasonable in experienced hands.

ALGORITHM

1. **Open intravesical diverticulectomy**
2. Fill the bladder with about 200 mL via an 18 Fr Foley.
3. Lower midline incision; split recti; incise transversalis fascia.
4. Sweep peritoneum off the dome and open a vertical cystotomy.
5. Identify the orifices (ureteral catheter if concerned) and excise the sac.
6. Place a 20–22 Fr Foley, close in two layers, leak-test, and drain.

Step	Intravesical	Combined intra-/extravesical
Delivery	Sac everted or dissected from the pseudocapsule from inside; mucosal incision about **0.5 cm** round the neck	A finger in the sac pushes it out; the neck is incised from outside
Hazard	Ureteric orifice injury	**Excess bladder wall resection** if the inner and outer incisions do not meet
Closure	Two layers 2-0 polyglactin with a secure seromuscular layer, which prevents recurrence	

Laparoscopic / robotic

- Use steep Trendelenburg and cystoscopy first, with pneumoperitoneum at **15 mm Hg**. Place the assistant port opposite the sac, circumscribe the neck, and excise and bag the sac with the ureter in view; staplers risk stones forming on the staples.
- If the sac sits at the ureteral hiatus, amputate the neck, make a new hiatus and reimplant over a double-J. Extraperitoneal, single-site and single-port variants exist.

Complications

- **Urinary extravasation is commonest**. It shows as high drain output, infected urinoma, or ileus and urinary ascites after transperitoneal surgery. Check drain creatinine and do CT cystography.

ALGORITHM

1. **Managing a leak**
2. If asymptomatic, keep Foley and drain.
3. If a symptomatic collection forms, drain it percutaneously.
4. Do a cystogram before removal; if the leak persists, choose nephrostomies or exploration and closure.

- Other complications are UTI, fistula, bladder spasm, wound infection and bowel injury.
- Unrecognised ureteric injury presents with acute kidney injury or flank pain; try a retrograde stent, and if it fails, place a nephrostomy and repair later.

KEY POINT

Diverticulectomy is a dissection problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely, and fix the outlet first.

3 · Bladder stones, foreign bodies and fistulae

A bladder stone signals stasis, so treat the cause too.

- The cause is **stasis**: bladder outlet obstruction, neurogenic bladder, or augmented and substituted bladders. Foreign bodies are mostly iatrogenic (mesh, migrated intrauterine device).
- **Non-contrast CT** is best for stone burden, prostate volume and hydronephrosis. Cystoscopy is needed only for haematuria, suspected cancer or prostate assessment.
- Treat the stone **and** its cause; the concomitant operation often picks the route (transvesical prostatectomy, diverticulectomy or enterocystoplasty).

Approach	Best for	Advantage	Limitation
Transurethral cystolitholapaxy	Most stones	No incision	Prolonged instrumentation risks urethral stricture
Percutaneous cystolithotripsy	Large stones	Nephroscope, short stay, no stricture	Contraindicated after prior surgery, diversion or augmentation (bowel injury)
Open cystolithotomy	Very large burden; with prostatectomy, diverticulectomy, augmentation	Historic gold standard	Incision, longer catheter
Laparoscopic/robotic	Large burden with other procedures; foreign bodies, eroded mesh	Minimally invasive	Technically demanding

- **Percutaneous** access two fingerbreadths above the symphysis is dilated to a **30 Fr** sheath for a 24–26 Fr nephroscope. Stones under 1 cm are grasped whole and larger ones fragmented; the suprapubic catheter is clamped at 24–48 h. After prior surgery, image first to exclude interposed bowel.
- **Transvesical laparoscopy** uses a pneumovesicum of **8–12 mm Hg**, avoids the peritoneum and suits eroded mesh. Transperitoneal laparoscopic or robotic cystolithotomy allows simultaneous prostatectomy or diverticulectomy.
- **Open** cystolithotomy opens the bladder vertically, removes the stones, inspects for tumour and intravesical lobe, and closes in two layers with a drain.
- Refractory fistulae (colovesical, vesicovaginal) may need partial cystectomy, and a refractory urinary fistula may justify simple cystectomy.

4 · Partial cystectomy for benign disease and urachal surgery

- Indications are leiomyoma, paraganglioma, endometriosis, urachal anomalies and fistulae; most benign tumours need only transurethral resection (TUR).
- **Avoid it with small capacity or severe dysfunctional voiding.** Exclude malignancy first, because cancer needs wide margins and lymphadenectomy. Lesions near the orifice need reimplanting.

Lesion	Clue	Investigation	Treatment / trap
Leiomyoma	Women of childbearing age; endovesical, intramural or extravesical	Transvaginal ultrasound (US), MRI	Small endovesical: TUR; large or intramural: wide local resection
Paraganglioma	Episodic hypertension, hypotension or syncope on **micturition or sex**; or painless haematuria	Plasma metanephrines; MIBG scan; yellow raised mucosa	**Alpha-blockade before any instrumentation**; partial cystectomy preferred over TUR (recurrence)
Endometriosis	Cyclical haematuria with menses	US, MRI (bright on T1); cystoscopy limited	Partial cystectomy (disease is extravesical)
Urachal anomaly	Umbilical discharge, mucinuria, infected mass; malignant risk	US/CT; VCUG or sinogram; cystoscopy	Culture and antibiotics, with or without drainage; if recurrent, excise urachus with a dome cuff

- Dome and anterior lesions are approached extraperitoneally, posterior ones transperitoneally, with no node dissection. **Aspirate before opening to avoid spillage**.
- Excise full thickness with about a **2-cm margin** if capacity allows, with frozen-section margins, and reimplant if the orifice is involved.
- **Urachal excision** takes the whole tract with peritoneal wings and a small dome cuff, with **no wide margin or frozen section**; the umbilicus may be kept if benign. Open excision uses a periumbilical midline incision down to a transmural dome cuff.
- Complications are extravasation (commonest), ureteric injury and reduced capacity.

> **KEY POINT**
>
> Leiomyoma, paraganglioma and endometriosis infiltrate and recur if incompletely excised, so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, a margin). Urachal surgery is the exception: the whole tract matters, not a wide margin.

5 · Simple (supratrigonal) cystectomy

A last resort that helps only if the bladder is the source.

- It treats end-stage refractory dysfunction: radiation cystitis, neurogenic bladder, refractory incontinence, fistula, bladder neck contracture, refractory anastomotic stricture, and interstitial cystitis/bladder pain syndrome (IC/BPS).
- Prove the bladder is the source, since non-vesical pain persists. In IC/BPS, success is predicted by a **small fibrotic bladder, small capacity under anaesthesia and Hunner lesions**.
- Diversion alone risks pyocystis, bleeding, pain and secondary cancer.

Feature	Simple (benign)	Radical (cancer)
Prostate / anterior vagina	Preserved unless indicated	Removed
Lymphadenectomy	None	Performed
Bladder opened	Often, to find the trigone and bladder neck	Avoided
Excision	Bivalve and amputate near the trigone; stump oversewn and residual mucosa cauterised	En bloc

Steps are to divide the ureters, bivalve the bladder, divide the pedicles, amputate each half and cover the stump with posterior peritoneum. An ileal conduit usually follows, and robotic surgery may reduce complications. Drains stay 5–7 days, until drain creatinine equals serum, and complications are common.

6 · Ureteral reimplantation in adults

Adults need reimplantation mainly for injury and stricture.

- Indications are distal ureteral trauma, iatrogenic injury, **stricture** and late-presenting injury, and reimplantation alongside diverticulectomy or partial cystectomy. Primary VUR is rare.
- In thermal or gunshot injury, debride even normal-looking ureter; assess partial injury with on-table retrograde pyelogram or methylene blue.
- Strictures follow impacted stone, radiation or chronic inflammation. An atrophic kidney needs decompression, then a renogram.

- Measure the gap with antegrade and retrograde imaging and exclude malignancy. A **poorly functioning kidney needs nephrectomy, not reimplantation**. **Persistent extravasation despite adequate stenting** is an indication.

> **KEY POINT**
>
> Refluxing reimplantation is acceptable in adults. Refluxing anastomoses show no faster renal decline, more stenosis or more infection than tunnelled ones, while needing less ureter and less bladder mobilisation.

Feature	Nonrefluxing (tunnelled)	Refluxing (direct)
Tunnel	**3–5 × ureteric diameter** (5:1 robotically)	None
Ureter length / mobilisation	More	Less; reaches the posterior dome
Use	High-grade reflux with renal risk	Generally sufficient in adults

Technique	Access	Neo-hiatus	Signature step
Modified **Lich-Gregoir** (nonrefluxing)	Extravesical	Posterior wall, superior-medial to native orifice	Detrusor trough over intact mucosa; detrusor closed over ureter
Modified **Politano-Leadbetter** (nonrefluxing)	Transvesical	Cephalad-medial to native hiatus	Submucosal tunnel; right-angle clamp draws ureter through
Refluxing extravesical	Extravesical	Posterior dome, lateralised	Direct spatulated anastomosis
Refluxing transvesical	Transvesical	Posterior dome over a Kelly clamp	Clamp pulls ureter in

- Divide the ureter proximal to disease, with a frozen-section margin in stricture. Mobilise the bladder off peritoneum, dividing the ipsilateral medial umbilical ligament. Test reach on the **decompressed** dome.
- **Lich-Gregoir**: a small cystotomy forms the neo-orifice, detrusor is incised toward the bladder neck sparing mucosa, a spatulated 4-0 anastomosis is stented, and detrusor is closed over the ureter.
- **Politano-Leadbetter**: through an anterior cystotomy, a neo-orifice about 1.5 cm superior-medial to the native one leads into a submucosal tunnel.
- For a poorly vascularised ureter, use a non-transecting side-to-side refluxing anastomosis.
- Preserve periureteric adventitia, spatulate, and keep knots outside the lumen. The stent is a **6 Fr × 26–28 cm double-J**.
- Primary reimplantation suits defects up to **4 cm**; longer gaps need a psoas hitch or Boari flap, and renal descent adds up to **5 cm**. Further reach comes from dividing the contralateral medial umbilical ligament and pedicle.
- Enter **superior-medially, toward the bladder neck**, because lateral sites kink on filling.
- Open surgery is preferred for trauma; laparoscopic and robotic surgery give less blood loss and shorter stay with equal patency.
- Aftercare is a Foley for 1 week with a cystogram and a stent for **≈6 weeks**. A **leak** is managed by draining the urinoma and keeping the Foley and stent; if refractory, place a nephrostomy.

- **Late obstruction** usually follows stent removal (ischaemia, tension, narrow neo-orifice, poor spatulation, radiation, lateral site); short strictures are treated endoscopically and high-grade ones by redo.

> **KEY POINT**
>
> Reimplantation fails from ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other.

7 · Bridging distal ureteral defects

*See the paper answer **Boari flap** (Section 11, Ch 93).*

Also, from the condensed book (not in the paper answer):

- Nonrefluxing Boari length is the defect plus about 3 cm; longer flaps need a wider base.
- Beyond reach, options are transureteroureterostomy, ileal ureter and autotransplantation.

Option	Reach	Rule
Primary reimplant	≤4 cm	Toward bladder neck
Psoas hitch	Gains about 5 cm	Sutures superficial, parallel to the fibres; reimplant before tying
Boari flap	Up to **15 cm**	Base ≥**4 cm** , apex ≥**3 cm** , length:base ≤**3:1**

- Hitch plus flap shortens the flap and preserves capacity. The hitch sutures go into the psoas minor tendon, avoiding the **genitofemoral** (surface) and **femoral** (deep) nerves; reimplant on the superolateral dome.
- The Boari flap is based on the diseased side, anchored to psoas and tubularised in two layers.
- A bladder advancement flap gives comparable reach and is simpler. A **small-capacity, poorly compliant bladder** contraindicates both hitch and flap.
- Complications are femoral or genitofemoral neurapraxia, iliac vessel injury, ischaemic stricture of a narrow-based flap, and reduced capacity.

> **KEY POINT**
>
> The bladder is the donor tissue: hitch first, flap second. Every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and bladder capacity.

8 · Enterocystoplasty (augmentation)

Augmentation protects the kidneys from high bladder pressure.

- Indication: neurogenic bladder with poor compliance **despite** clean intermittent catheterisation (CIC), anticholinergics, botulinum toxin and sacral neuromodulation, especially with upper tract deterioration or incontinence with skin breakdown.
- Bowel lowers pressure only if **detubularised**, which abolishes peristalsis.
- Work-up includes video-urodynamics, serum chemistry, B12 and bowel history; **confirm the patient can catheterise**. Metabolic effects: See Ch 133.

- Add a sling, artificial urinary sphincter, bladder neck closure, catheterisable stoma or nonrefluxing reimplant as needed.

Segment	Note
Ileum (commonest)	25 cm taken **≈20 cm** proximal to the ileocaecal valve (spares B12 and bile-salt absorption); acidosis risk
Ileocaecal	When a catheterisable channel is wanted
Sigmoid	Colonoscopy first
Stomach	Possible; needs a higher incision

ALGORITHM

1. Isolate ileum with a broad mesenteric base and check reach to the dome.
2. Restore bowel side-to-side; close the window.
3. **Detubularise** on the antimesenteric border and fold into a U.
4. **Bivalve the bladder widely** and make a wide-mouthed anastomosis.
5. Suprapubic tube with or without Foley; leak-test; drain.

- A narrow mouth creates a poorly emptying pseudodiverticulum.
- Early complications are extravasation (preserve the arcade), UTI, bowel injury, fistula and intra-abdominal infection.
- **Perforation** follows missed catheterisation and is often masked by sensory loss. Diagnose with CT cystogram and proceed to **emergency exploration**.
- Stones are the commonest long-term complication. Neoplasia risk means surveillance endoscopy from **10 years**, with routine metabolic and vitamin bloods.
- If the urethra is not catheterisable, do a continent ileocaecocystoplasty; if the patient cannot catheterise, an incontinent ileovesicostomy.

KEY POINT

Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem. Detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, bloods and endoscopic surveillance are the price.

9 · Continent catheterisable channels

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

Also, from the condensed book (not in the paper answer):

- Channels suit adequate-capacity bladders after failed urethral CIC; augment if compliance is poor.
- The bladder is drained for 2–4 weeks before channel catheterisation starts. Always have a fallback if the appendix is absent or short.
- A Monti uses 2–4 cm of ileum; a double Monti risks channel diverticula; vas, colon and stomach channels do poorly.

ALGORITHM

1. **Minimally invasive appendicovesicostomy**
2. Detach the appendix with a caecal cuff and spatulate it.
3. Make a ≈4-cm submucosal detrusor tunnel and anastomose.
4. Bring the stoma to the right lower quadrant or umbilicus.
5. Drain the bladder for **2–4 weeks**. Continence is **>95%**.

KEY POINT

Use the appendix when present and long enough; reconfigure ileum when not. Continence comes from tunnel length through detrusor, and missed catheterisation endangers the bladder, not just the channel.

10 · Vesicovaginal fistula (VVF) †

P3 P2 *VVF: management options, early vs delayed repair — causes, double-dye test, CT urography, catheter trial, timing, vaginal (Latzko, flap-split) vs abdominal (O'Conor) routes, interposition flaps*

- Leakage usually starts 1–3 weeks after hysterectomy, and a coexisting ureteric injury is present in about 10%.
- A ureterovaginal fistula is treated with a ureteric stent first, and by ureteroneocystostomy if that fails.

★ Putting it together

A · NMIBC: risk and treatment

AUA/SUO risk is assigned after complete TURBT.

Risk	Features	Adjuvant	Cystoscopy
Low	Low-grade (LG) solitary Ta ≤3 cm; PUNLMP	Single post-op instillation	3 mo, then yearly
Intermediate	LG Ta recurring <1 yr/>3 cm/multifocal; LG T1; high-grade (HG) Ta ≤3 cm	Induction MMC/gemcitabine or BCG; maintain 1 yr	3–6-monthly for 1–2 yr
High	HG T1; HG Ta >3 cm/multifocal/recurrent; CIS; variants; LVI; HG prostatic urethra	Re-TURBT (T1); BCG with **maintenance 3 yr** (SWOG); early RC if highest-risk	3–4-monthly for 2 yr, 6-monthly for 2 yr, then yearly; upper tracts yearly for 2 yr
BCG-unresponsive	HG despite adequate BCG (≥5/6 induction + ≥2/3 maintenance)	RC; if unfit: pembrolizumab (CIS), nadofaragene, gem–docetaxel	—

B · MIBC: cystectomy vs preservation See Ch 131, 132

	NAC + RC + pelvic node dissection	Trimodality therapy
Who	Default if fit; NAC (gem–cis or ddMVAC) if cisplatin-eligible	Solitary T2, complete TURBT, **no CIS/hydronephrosis**, good function; or unfit
Cisplatin-ineligible	GFR <60, ECOG ≥2, grade ≥2 hearing loss or neuropathy, or NYHA III–IV: local therapy	
After	Adjuvant nivolumab (CheckMate 274): ≥pT3/N+ after NAC, or no NAC	Lifelong cystoscopy; salvage RC 10–30%

C · Choosing the diversion See Ch 133–135

	Ileal conduit	Orthotopic neobladder	Continent cutaneous
Best for	Frail, poor renal function, cannot catheterise	Default continent option, either sex	Unusable urethra or sphincter
Must have	Usable (non-irradiated) ileum	eGFR **>35–40** ; intact sphincter; negative urethral frozen section; will catheterise	eGFR >35–40; dexterity; efferent limb
Main problems	Stoma problems, ureteroileal stricture	Night leakage; late retention (women); acidosis	Stomal stenosis, catheterisation difficulty, stones

D · Metabolic complications by segment

Segment	Disturbance	Treatment
Ileum / colon	**Hyperchloraemic metabolic acidosis**	Bicarbonate, potassium citrate

Segment	Disturbance	Treatment
Jejunum	**Hyponatraemic hypochloraemic hyperkalaemic acidosis**	Sodium chloride; avoid jejunum
Stomach	**Hypochloraemic hypokalaemic alkalosis**; haematuria–dysuria syndrome	H2 blocker/PPI

Losing terminal ileum lowers B12 and causes bile-salt diarrhoea; buffering acidosis in bone causes osteomalacia. See Ch 133

E · Metastatic urothelial carcinoma

ALGORITHM

1. **Fitness**: cisplatin-, platinum-only or platinum-ineligible.
2. **First line**: enfortumab vedotin (EV) with pembrolizumab (EV-302); otherwise gem–cis (gem–carbo if cisplatin-ineligible).
3. **If no progression on platinum**, give avelumab maintenance (JAVELIN Bladder 100).
4. **Later lines**: pembrolizumab; erdafitinib (FGFR-altered); EV; sacituzumab govitecan. See Ch 131

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman

Paper answer · P2

Asked as:

- Describe in detail the management of T1G3 TCC of urinary bladder? — *Summer 2019 Paper II Q1 (LAQ)*
- Discuss management of T1G3 bladder tumor — *Practice Set 7 Paper II Q3a (SAQ)*
- Discuss treatment of 25 yr female with bladder TCC T1NoMo and C13 — *Practice Set 4 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 7 Paper II Q3a: also answered in 2019-Summer Paper 2, Q1) (Practice Set 4 Paper II Q1: T1G3 management also answered in 2019-Summer Paper 2, Q1)

Practice Set 4 Paper II Q1: "C13" is read as CIS (carcinoma in situ): T1 N0 M0 urothelial carcinoma with concurrent CIS in a 25-year-old woman.

Introduction and risk category

- T1G3 is high-grade (HG) urothelial carcinoma (the current name for TCC, transitional cell carcinoma) that invades (invading) the lamina propria but not the muscularis propria (not muscle). TCC is now urothelial carcinoma. CIS is flat and always high grade.
- T1 is 20–30% of NMIBC (non-muscle-invasive bladder cancer) and its most dangerous form: recurrence >80%, progression to muscle invasion 30–50%.
- Grade, not stage, drives progression. So every HG T1 is **high risk** (AUA [American Urological Association]/SUO [Society of Urologic Oncology] and EAU [European Association of Urology]).
- **T1 with CIS is high-risk NMIBC** (AUA and EAU). CIS is the second strongest progression factor after grade. Untreated CIS progresses to muscle invasion in 40–83%.
- **EAU 2021 very high risk:** T1 high grade (HG) with CIS plus at least one of age >70, multiple papillary tumours, or diameter ≥3 cm. In this woman (25-year-old), age does not count, so number and size decide. LVI (lymphovascular invasion) and variant histology also make it very high risk.
- **Young woman:** long life expectancy, fertility, pregnancy and sexual function shape every choice.

Risk factors for progression

- **Adverse factors:** concurrent CIS (carcinoma in situ), LVI (lymphovascular invasion), variant histology, size >3 cm, multifocality, prostatic urethral involvement, residual T1 at re-resection.

Factor	Significance
Concurrent CIS (carcinoma in situ)	Second most important factor after grade
Depth of invasion (substaging)	Invasion deep to the muscularis mucosae is worse; T1a/T1b proposed, no agreed method

Factor	Significance
LVI (lymphovascular invasion)	Suggests occult spread
Variant histology	Micropapillary (any amount), plasmacytoid, sarcomatoid, nested
Size >3 cm, multifocal, recurrent	More residual tumour and progression
Prostatic urethra, diverticulum	Poor BCG contact; diverticulum lacks muscle
Residual T1 at re-TURBT	Progression up to 80%
HG T1 at 3 months after BCG	Progression about 82%
- **Also know:** EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables; hydronephrosis suggests muscle invasion.	

Evaluation

- Cystoscopy, urine cytology (usually positive in CIS), CT (computed tomography) urography (upper tract tumour <5%, more with CIS), bimanual examination under anaesthesia (including vagina).
- Smoking and occupational history.
- **Pregnancy test** before CT and BCG (bacille Calmette-Guérin); expert pathology review.

Step 1: Complete, good-quality TURBT (transurethral resection of bladder tumour)

- Resect all visible tumour **with detrusor muscle** in the specimen; send the base separately.
- Blue-light cystoscopy or NBI (narrow-band imaging) finds more CIS (**blue-light cystoscopy** finds about one-third more CIS). **En bloc resection** (where possible) keeps orientation for substaging.
- Biopsy the prostatic urethra if tumour is at the bladder neck or trigone, or with CIS.
- Biopsy red velvety areas and the **bladder neck** (relevant to a later neobladder).

Step 2: Re-TURBT (restaging) in all T1

- Within **2–6 weeks** (within 6 weeks), before BCG (bacille Calmette-Guérin).
- **Why:** residual tumour in ≥40%; upstaging to muscle invasion in 25–30% (about half if no muscle in the first specimen). **Residual T1: progression up to 80%.** Re-TURBT improves recurrence- and progression-free survival: Divrik's RCT (randomised controlled trial) showed better recurrence-free and progression-free survival.

```
HG T1 on TURBT → re-TURBT at 2–6 weeks ├─ ≥T2 → neoadjuvant chemotherapy + cystectomy ├─ residual T1 / very high risk → offer early radical cystectomy └─ T0, Ta or CIS → BCG induction + 3-year maintenance
```

- Practice Set 7 version: HG T1 → re-TURBT; T0, Ta, CIS, T1 → BCG (bacille Calmette-Guérin); very high risk → offer early radical cystectomy.
- Practice Set 4 version: T1 + CIS → re-TURBT within 6 weeks; T0/Ta/CIS → BCG induction + 3-year maintenance.

Step 3a: BCG – standard for most patients (bladder preservation)

- **Mechanism:** BCG attaches to (binds) **fibronectin**, triggers cytokines (IFN-γ [interferon-gamma], IL [interleukin]-2) and a **Th1 (T-helper 1)** immune (cytokine) response that kills

tumour. It is the **only intravesical agent that reduces progression**.

- **Schedule:** start 2–4 weeks after TURBT, never immediately (intravasation and sepsis); induction weekly ×6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** 3 weekly doses at 3 and 6 months, then every 6 months (6-monthly) to **3 years**. Median recurrence-free survival 76.8 vs 35.7 months. **EORTC 30962:** full dose for 3 years in high risk.
- **CIS:** tumour-free up to 84%, about half durable; persistent CIS at 3 months: second course gives a further 30–50% complete response.
- BCG is live: avoid pregnancy and breastfeeding during treatment.
- **Contraindications (avoid with):** gross haematuria, traumatic catheterisation, symptomatic UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis common; BCG sepsis needs isoniazid, rifampicin, ethambutol and steroids (antituberculous drugs and steroids).

Step 3b: Early (immediate) radical cystectomy

- **Rationale:** clinical T1 is understaged in one-third to one-half. Patients who progress under surveillance do worse than those presenting with muscle invasion (progressors do worse than primary muscle-invasive disease). Immediate cystectomy gives about **92% 10-year disease-free survival**.
- **AUA:** offer to fit patients with persistent HG T1 on re-resection, or T1 with CIS, LVI or variant histology.
- **EAU very-high-risk group (2021 model; risk factors: age >70, multiple papillary tumours, diameter ≥3 cm):** T1 HG with CIS plus ≥1 risk factor; T1 HG without CIS but with all three; also LVI, variant histology, CIS of the prostatic urethra (prostatic urethral CIS), persistent T1 at re-TURBT. Offer immediate cystectomy; BCG if declined or unfit.
- **Offer for (Practice Set 4):** very high risk, residual HG T1, LVI, variants, diffuse CIS, or patient preference.
- Also: diverticular tumour, tumour not controllable endoscopically.

Step 3c: Cystectomy tailored to a young woman

- Standard female operation is anterior exenteration (bladder, urethra, anterior vagina, uterus, cervix, ovaries).
- **Organ-sparing** (≤cT2, bladder neck uninvolved): keep the **uterus and ovaries** (fertility, hormones, avoids surgical menopause; ovaries also make testosterone). Keep the **vagina and neurovascular tissue** (vaginal dissection no more dorsal than 2 and 10 o'clock) for sexual function and continence. Oncological safety of sparing is not fully known.
- **Orthotopic neobladder** suits her: needs a negative bladder-neck biopsy and **urethral frozen section** (negative bladder neck means negative urethra; CIS does not predict urethral involvement). Day continence 80–90%; **retention 20–30% in women**, so she must accept CIC (clean intermittent self-catheterisation). See Practice Set 4 Paper 2, Q3e.
- **Fertility:** if uterus or ovaries must go, offer oocyte or embryo cryopreservation first. Pregnancy after a neobladder is possible; caesarean uses a high midline incision to avoid the pouch.

Step 4: BCG failure (BCG-unresponsive disease)

- **Adequate BCG for T1:** ≥5 of 6 induction doses.

- **BCG-unresponsive:** HG T1 at the 3-month check after induction; persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Standard:** timely radical cystectomy (EAU, AUA).
- **Unfit or refusing (declined):** pembrolizumab (KEYNOTE-057: CIS complete response 41%); nadofaragene firadenovec (53%); N-803 with BCG; gemcitabine–docetaxel; TAR-200; trials.
- **Also know:** BCG-intolerant; delayed relapse may respond to BCG rechallenge.

Follow-up (AUA/SUO high risk)

- Bladder preserved: cystoscopy and cytology every 3–4 months (3-monthly) for 2 years, 6-monthly for 2 years, then yearly for life.
- Upper-tract imaging yearly for 2 years (regular upper-tract imaging).
- Positive cytology with normal cystoscopy: check prostatic urethra and upper tracts.
- After neobladder: B12, metabolic panel, upper tracts.

Recent advances

- Checkpoint inhibitors with BCG in BCG-naive disease (trials); urine markers predicting BCG response; TAR-200.

Diagram to draw: bladder wall layers showing CIS, Ta, T1a/T1b (above and below the muscularis mucosae) and T2.

> **Key points to remember:**
> - Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
> - BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
> - T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
> - Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
> - BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
> - Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

SAQ Optical coherence tomography in carcinoma bladder

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "Optical coherence tomography in carcinoma bladder" · P2 SAQ "What is optical coherence tomography. What is its potential role in management of CA bladder?"

Introduction

- **WLC (white-light cystoscopy)** is the gold standard but subjective: many biopsies are negative and small lesions are missed. WLC misses flat lesions and cannot show depth.
- **OCT (optical coherence tomography)** is an **"optical biopsy"** technique listed among enhanced cystoscopic methods for NMIBC (non-muscle-invasive bladder cancer): real-time cross-sectional imaging of the bladder wall. It works like ultrasound but uses light.

What is OCT: principle

- OCT is the **optical analogue of B-mode ultrasound**. **Near-infrared light** (about 1300 nm) is shone on tissue, and the echo delay of back-scattered light is measured instead of sound, using **low-coherence interferometry** (back-scattered light measured against a reference beam).
- The delay of reflected light gives depth. Scanning across builds a cross-sectional image.
- **Resolution about 10–20 µm** (near histology). **Depth about 2–3 mm** (P2: only **1–2 mm**): urothelium, lamina propria and superficial muscle.
- First described in 1991 (ophthalmology).

Technique

- A fibre-optic probe passes through the cystoscope's working channel.
- WLC or blue light finds the lesion; the probe is placed **in contact** with it and read at once (real time).

Image interpretation

Tissue	OCT appearance
Normal	Three layers: thin, dark, low-scatter urothelium; bright lamina propria with clear boundary; darker muscularis visible below
Inflammation	Layers preserved but thickened
Papillary tumour (Ta)	Thickened urothelium, papillary pattern; lamina propria boundary intact
Invasive (T1)	Layers lost; **loss (break) of the urothelium–lamina propria boundary** = invasion

Potential role in carcinoma bladder

1. **Diagnosis:** tell tumour from inflammation at suspicious areas; separate benign from malignant flat lesions in real time; fewer unnecessary biopsies.
2. **Staging aid: Ta vs T1** (lamina propria invasion) during TURBT (transurethral resection of bladder tumour); helps in diverticula and thin walls.
3. **Resection margins:** checks margins and completeness of TURBT.
4. **Surveillance** of NMIBC, including after BCG (bacille Calmette-Guérin).
5. **Upper tract:** through the ureteroscope, OCT correlates well with final grade and stage of UTUC (upper tract urothelial carcinoma).

Evidence and limitations

- Small studies show **high sensitivity but lower (moderate) specificity**. Inflammation, scar, recent TURBT and post-BCG change mimic tumour (false positives). Adding OCT to fluorescence (blue-light) cystoscopy improves specificity.
- Tiny field of view: needs WLC or PDD (photodynamic diagnosis) to guide it.
- Too shallow to judge (deep) muscle invasion; cannot grade (no cellular detail); cost; learning curve.
- Still **investigational**; not recommended routinely by guidelines.

Other enhanced cystoscopy techniques

Technique	Principle
PDD (blue light, HAL, hexaminolevulinate)	Tumour porphyrins fluoresce red; detects about one-third more CIS (carcinoma in situ)
NBI (narrow-band imaging)	415/540 nm light absorbed by haemoglobin; shows vessels; many false positives
CLE (confocal laser endomicroscopy)	Cell-level surface images after fluorescein
- **Also know:** Raman spectroscopy.	

Diagram to draw: OCT image showing the three normal bladder layers beside a tumour with loss of layering.

> **Key points to remember:**
> - Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (P2: 1–2 mm).
> - Loss of layering suggests tumour; boundary loss suggests T1.
> - Role: Ta vs T1; cannot assess deep muscle.
> - High sensitivity, moderate specificity, false positives from inflammation; investigational.

SAQ Intravesical gene therapy.

Paper answer · P2

Type: SAQ (10 marks)
Source: 2022-Summer Paper 2, Q3a

Concept

- A gene is instilled into the bladder so that urothelial and tumour cells make an anticancer product, or the vector itself kills them.
- **Why the bladder suits it:** direct contact, little systemic exposure, response checked by cystoscopy and cytology.
- **Barrier:** the glycosaminoglycan layer resists vectors → **transduction enhancers** (Syn3; DDM, n-dodecyl-β-D-maltoside).
- **Indication:** high-risk **BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer)** in patients unfit for or declining radical cystectomy. BCG-unresponsive: persistent or recurrent CIS (carcinoma in situ) or high-grade Ta/T1 after adequate BCG, or relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.

Agents

Agent	Vector, mechanism	Key data
Nadofaragene firadenovec	**Replication-incompetent adenovirus type 5** carrying the **interferon-α2b gene**, with **Syn3** surfactant. Urothelium keeps secreting interferon → apoptosis (TRAIL, TNF-related apoptosis-inducing ligand) and immunity, overcoming the short dwell of interferon protein	75 mL every **3 months**, 1-hour dwell. CIS: **complete response 53% at 3 months, 24% at 12 months**; papillary high-grade RFS (recurrence-free survival) 44% at 12 months; serious events 2%. **FDA (Food and Drug Administration) approved December 2022**, first bladder gene therapy
Cretostimogene grenadenorepvec (CG0070)	**Replication-competent oncolytic adenovirus**: E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity	Two 6-week inductions + maintenance; single-arm phase III BOND-003 in BCG-unresponsive CIS: high CR (complete response) rate; combination with pembrolizumab under study
Inodiftagene vixteplasmid (BC-819)	Plasmid: **diphtheria toxin A** gene under tumour-active **H19** promoter	Investigational

Administration and toxicity

- Catheter instillation, patient turned during dwell; biosafety precautions.
- Mostly grade 1–2: **bladder spasm, urgency, discharge around catheter, fatigue**.

Place in practice

- Alternatives: radical cystectomy (standard), **pembrolizumab** (KEYNOTE-057), gemcitabine–docetaxel, **N-803** (interleukin-15 superagonist) + BCG.
- Limits: responses wane (median CR duration about 10 months); single-arm trials; cost.

Also know: non-viral IL-12 (interleukin-12) plasmid (EG-70); early adenoviral attempts failed through poor transduction.

Diagram to draw: adenovirus transducing a urothelial cell → interferon secretion → tumour apoptosis.

> **Key points to remember:**
> - Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
> - CIS CR 53% at 3 months, 24% at 12 months.
> - Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

SAQ Von Bruns Nest

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 10 Paper 2, Q3e

Read as: von Brunn nests of the urothelium.

Definition

- **Von Brunn nests** are solid nests of benign urothelial cells lying in the **lamina propria**. They form when the basal layer of the surface urothelium buds downward and the buds pinch off.
- A common normal or reactive finding in most adult bladders, especially at the **trigone and bladder neck**; also in the ureter and renal pelvis.

Pathogenesis: the proliferative cystitis spectrum

`Chronic irritation (infection, stones, catheter, outlet obstruction) → urothelial buds → von Brunn nests (solid) → central degeneration and fluid → cystitis cystica (cysts) → glandular (columnar / goblet cell) metaplasia → cystitis glandularis`

Lesion	Histology	Malignant potential
Von Brunn nests	Solid urothelial nests, lamina propria	None
Cystitis cystica	Nests with central cyst, urothelial lining	None
Cystitis glandularis (typical)	Cuboidal or columnar lining	Low
Cystitis glandularis, **intestinal type**	Goblet cells, mucin	Possible link to **adenocarcinoma**

Clinical features

- Usually incidental. When florid: irritative symptoms, haematuria, infection; cystoscopy shows submucosal nodules or cysts that can mimic tumour.
- **Associations:** chronic inflammation, outlet obstruction, **pelvic lipomatosis** (florid cystitis glandularis), bladder exstrophy.

Differential diagnosis (pathologist's trap)

- **Nested variant of urothelial carcinoma:** deceptively bland but aggressive (male:female 6:1). Clues: deep infiltrative nests reaching muscle, nuclear atypia.
- **Inverted papilloma:** endophytic nests with trabeculae, usually at the trigone.
- Nephrogenic adenoma; urothelial carcinoma with glandular differentiation.

Management

- None for isolated nests; treat the cause (infection, stone, catheter, obstruction).
- **TURBT (transurethral resection) or biopsy** if a mass is present, to exclude carcinoma.
- **Cystitis glandularis:** remove the irritant and arrange **endoscopic surveillance**, because rare change to adenocarcinoma is reported (especially the intestinal type and in exstrophy).
- Upper tract: benign, but coexisting upper-tract urothelial cancer cannot be excluded, so follow up.

Diagram to draw: urothelium budding into lamina propria, forming a solid nest, then a cyst, then glandular metaplasia.

Key points to remember:

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

SAQ Mitomycin

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 11 Paper 2, Q3b

Pharmacology

- MMC (mitomycin C) is an antitumour antibiotic from *Streptomyces caespitosus*.
- **Mechanism:** after reductive activation in the cell it acts as an **alkylating agent**, cross-linking DNA and stopping DNA synthesis; cell-cycle non-specific.
- Molecular weight 334 Da: **little systemic absorption** from the bladder.

Uses in NMIBC (non-muscle-invasive bladder cancer)

1. **Single immediate post-TURBT (transurethral resection of bladder tumour) instillation:** 40 mg. Given **within 6 hours** it reduces recurrence; at 24 hours it does not, because it works by preventing implantation of floating cells. Best for a solitary low-grade papillary tumour. Avoid if perforation or extensive resection is suspected.
2. **Adjuvant induction:** 40 mg weekly for 6 weeks (± maintenance) in low- and intermediate-risk disease. Reduces recurrence but **not progression**; BCG (bacille Calmette–Guérin) with maintenance is superior in high risk.
3. **Optimised delivery (Au, 2001):** empty bladder, overnight fluid restriction, sodium bicarbonate (less drug breakdown), 40 mg in 20 mL; retain 1–2 hours.
4. **Device-assisted:** EMDA (electromotive drug administration) increases tissue uptake; chemohyperthermia at 42 °C (RITE [radiofrequency-induced thermochemotherapeutic effect], HIVEC [hyperthermic intravesical chemotherapy]).
5. **Sequential gemcitabine then MMC** for BCG-exposed disease.

Other urological uses

- **Upper tract:** single bladder dose after nephroureterectomy (ODMIT-C: bladder recurrence 28% to 16% at 1 year). **UGN-101 (mitomycin gel)**, FDA (Food and Drug Administration) approved 2020 for low-grade UTUC (upper tract urothelial carcinoma): complete response 59%, ureteric stricture about 44%. UGN-102 for low-grade intermediate-risk bladder tumours.
- **Bladder-sparing chemoradiation:** 5-FU (5-fluorouracil) + MMC (BC2001: better locoregional control than radiation alone).
- **Antifibrotic:** injected after incision of recurrent bladder neck contracture or urethral stricture.

Side effects

- Chemical cystitis (30–40%); **skin rash, palmar or genital dermatitis** (signature toxicity; wash after voiding); contracted bladder (about 5%); encrustation cystitis; myelosuppression

rare.
- Extravasation through a perforation can cause pelvic necrosis.

Diagram to draw: timeline: TURBT, single MMC within 6 hours, then weekly induction ×6.

> **Key points to remember:**
> - Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
> - Reduces recurrence, not progression.
> - Signature toxicity: skin rash; contracted bladder about 5%.

SAQ Mullerionosis of UB

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3b

"Mullerionosis of UB" is read as müllerianosis of the urinary bladder.

Definition

- Benign lesions of **müllerian-type epithelium** in the bladder wall, in three forms:

Lesion	Tissue
Endometriosis (commonest)	Endometrial glands **and stroma**
Endocervicosis	Endocervical mucinous glands
Endosalpingiosis	Ciliated tubal epithelium

- **Müllerianosis** strictly means two or more of these together.
- Women of reproductive age, often after **caesarean section**. Usually **posterior wall or dome** (vesicouterine pouch).
- Bladder is the commonest urinary tract site of endometriosis, ahead of the ureter.

Pathogenesis

- **Implantation:** retrograde menstruation (Sampson) into the vesicouterine pouch.
- **Iatrogenic implantation** at caesarean or pelvic surgery (primary vs iatrogenic forms).
- **Metaplasia** of coelomic or subperitoneal müllerian remnants (the "secondary müllerian system").
- Lymphatic or vascular spread; direct extension of anterior uterine adenomyosis.

Clinical features

- **Cyclical haematuria** or irritative LUTS (lower urinary tract symptoms) with menses; suprapubic pain, dysmenorrhoea, dyspareunia.
- Endocervicosis: non-cyclical mass or LUTS.
- **Also know:** menouria differential includes vesicouterine fistula (Youssef syndrome).

Evaluation

- **Ultrasound** (transabdominal plus transvaginal): size and depth of a hypoechoic nodule.
- **MRI (magnetic resonance imaging):** extent; haemorrhagic foci **bright on T1**, fibrotic thickening dark on T2.
- **Cystoscopy:** bluish submucosal nodule with neovascularity, best seen during menses. Of **limited value**, because disease is mainly **extravesical** (from serosa inwards).

- **Biopsy:** excludes adenocarcinoma, especially with endocervicosis. Immunohistochemistry: ER/PR (oestrogen/progesterone receptor) positive; CD10 marks endometrial stroma.
- Ultrasound of upper tracts for ureteric involvement.
- **Differential:** adenocarcinoma, cystitis glandularis, nephrogenic adenoma, urachal remnant.

Management

- **Hormonal suppression** (combined pill, progestins, GnRH [gonadotropin-releasing hormone] agonists): symptom relief; recurrence after stopping.
- **TUR (transurethral resection):** incomplete for transmural disease; perforation risk; recurs.
- **Partial cystectomy** (laparoscopic or robotic) with a margin: **primary treatment**. Ureteric stent or reimplantation if the orifice is involved.
- Excise other pelvic endometriosis with the gynaecologist.
- Malignant change (endometrioid or clear cell adenocarcinoma) is rare.

Diagram to draw: sagittal pelvis with a dome nodule growing from the vesicouterine pouch into the detrusor.

> **Key points to remember:**
> - Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
> - Cyclical haematuria; MRI bright on T1.
> - Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

SAQ Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2023-Winter Paper 3, Q3e

Definition

- **CIS (carcinoma in situ, Tis):** flat, non-invasive, **high-grade** urothelial carcinoma. It is not "premalignant".
- Histology: severe atypia, loss of polarity, non-cohesive pleomorphic cells, **loss of umbrella cells** (dysplasia keeps them).
- **Types:** primary (isolated, about 10%), secondary or concurrent (with papillary tumour, about 90%).

Pathogenesis

- **Flat pathway:** dysplasia → CIS → high-grade invasive cancer.
- Driven by **TP53, RB and PTEN** loss; **no FGFR3** mutation (unlike low-grade papillary tumours).
- **Field change:** multifocal; spreads to prostatic urethra, distal ureters and upper tract.

Clinical importance

- Untreated, **40–83% progress** to muscle invasion. Primary CIS progresses in >50%.
- "CIS alone" hides invasion at cystectomy in up to 20%.

- **Second strongest prognostic factor after grade**; any CIS makes NMIBC (non-muscle-invasive bladder cancer) **high risk**.
- About 80% have **irritative symptoms**; it mimics cystitis.

Diagnosis

- **Urine cytology:** most useful in CIS, because the cells are poorly cohesive. Positive cytology with normal cystoscopy needs mapping.
- Cystoscopy: red velvety patch or normal-looking mucosa; **blue-light cystoscopy** (hexaminolevulinate) detects about one-third more CIS; NBI (narrow-band imaging).
- **Biopsies:** cold-cup from suspicious and random sites; prostatic urethra; upper tract imaging.

Management

- **BCG (bacillus Calmette–Guérin) is first line:**
- Induction: 6 weekly instillations.
- **Maintenance** (SWOG [Southwest Oncology Group] 2000: 3 weekly doses at 3 and 6 months, then 6-monthly for 3 years): median recurrence-free survival 76.8 vs 35.7 months.
- Complete response up to 84%; durable in about half.
- **Persistent CIS at 3 months:** second BCG course; 30–50% more respond.
- **BCG-unresponsive** (persistent or recurrent CIS within 12 months of adequate BCG): **radical cystectomy**.
- Unfit or unwilling: **pembrolizumab** (KEYNOTE-057: complete response 41%), nadofaragene firadenovec, gemcitabine–docetaxel.
- Prostatic urethral CIS: TURP plus BCG; ductal involvement needs cystoprostatectomy.

Follow-up

- Cystoscopy and cytology 3-monthly for 2 years, then 6-monthly; upper-tract imaging.

Diagram to draw: flat versus papillary pathways of bladder cancer.

> **Key points to remember:**
> - Flat, high-grade; TP53/RB; no FGFR3.
> - Untreated 40–83% progress; always high risk.
> - BCG with maintenance; cystectomy if BCG-unresponsive.

SAQ Virtual cystoscopy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3a

Definition and principle

- **Virtual cystoscopy (VC)** is a computer-generated, three-dimensional (3D) endoluminal view of the bladder. It is reconstructed from thin-section **CT (computed tomography)** or **MRI (magnetic resonance imaging)** data of a distended bladder.
- Software "flies through" the lumen using **surface or volume rendering**; multiplanar images also show wall thickness and extravesical spread.

Technique

Step	CT virtual cystoscopy	MR virtual cystoscopy
Distension	Air or CO2 through a catheter (negative contrast), or excretory-phase contrast (positive contrast, no catheter)	Urine-filled bladder; urine bright on T2
Acquisition	Thin slices, supine and prone (moves dependent urine off the posterior wall)	3D T2 or T1 contrast sequences
Drawback	Radiation; catheter for gas	Cost, time, motion artefact

Advantages

- Non-invasive; no anaesthesia.
- Sees where a scope cannot: through a **tight urethral stricture**, inside **diverticula**, beyond a large median lobe.
- With **CTU (CT urography)**, upper tracts and staging are imaged in one study (Chapter 5: high accuracy for bladder tumours if well distended).

Limitations

- Detects polypoid tumours ≥5 mm well; misses **small and flat lesions, including CIS (carcinoma in situ)**.
- No colour or mucosal detail, **no biopsy and no treatment**.
- False positives: trabeculation, clot, debris, folds of an underdistended bladder, prostatic impression.

Indications (selective)

- Unfit for, or refusing, cystoscopy; urethral stricture; diverticulum; bladder evaluation during CTU for haematuria.
- **Guideline position:** EAU (European Association of Urology) and AUA (American Urological Association) do not recommend VC in place of cystoscopy. Flexible cystoscopy remains the standard.

Related advances

- **Also know:** MRI **VI-RADS (Vesical Imaging–Reporting and Data System)** scores muscle invasion (Chapter 6); blue-light and narrow-band cystoscopy improve CIS detection (Chapter 8); virtual ureteroscopy uses the same principle.

Diagram to draw: thin-slice CT of a gas-distended bladder, then a 3D endoluminal fly-through showing a polypoid tumour.

> **Key points to remember:**
> - 3D endoluminal reconstruction from thin-section CT or MRI.
> - Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
> - No biopsy; does not replace cystoscopy.
> - Useful beyond strictures and inside diverticula.

SAQ Bladder transplantation

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3f

Rationale

- After cystectomy, or when the bladder cannot be reconstructed, the usual substitute is **bowel**. Bowel in the urinary tract causes mucus, metabolic acidosis, stones, infection and malignancy risk.
- Chapter 135: two frontiers aim to **remove bowel from urinary reconstruction**: tissue engineering and bladder transplantation.

Tissue engineering (why it has not succeeded)

- **Atala et al. (2006):** first seeded biodegradable construct for neurogenic augmentation. A later multi-institution trial showed **no significant gain in compliance or capacity** (Chapter 48).
- Engineered conduit trial: **half developed stenosis or stricture**, because of stiff material, poor cell coverage and **no blood supply** (Chapter 135).
- Engineered bladder: storage only (Chapter 111). See 2022-Summer Paper 4, Q2.

Bladder transplantation: development (Chapter 135)

- A **VCA (vascularised composite allograft)**: the donor bladder is transplanted with its own blood vessels.
- Stepwise preclinical work at USC (University of Southern California): pigs, perfused cadavers, then brain-dead human donors.

Technique (robotic)

1. **Robotic retrieval** of the bladder on its vesical vascular pedicles from the donor.
2. **Back-table vascular reconstruction**.
3. **Robotic implantation**: vessels to recipient iliac vessels; ureters joined to the graft.
4. **ICG (indocyanine green) fluorescence** confirms perfusion. Vascularity and mucosal integrity held for over 12 hours.

First clinical case

- Reported in **2025** from Los Angeles: a combined kidney and bladder transplant from a deceased donor, in a dialysis patient who had lost most of his bladder.

Problems

- **Lifelong immunosuppression**: infection, malignancy, drug nephrotoxicity. This is the major limitation (Chapter 135), especially after cancer.
- **Denervated graft**: stores urine; voiding may need straining or CIC (clean intermittent catheterisation).
- Rejection monitoring, donor supply, unknown long-term function.

Likely candidates

- Already immunosuppressed patients (kidney transplant recipients) with an absent or non-functioning bladder.

Diagram to draw: donor bladder on its vascular pedicles anastomosed to recipient iliac vessels, ureters entering the graft.

Key points to remember:

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

SAQ Artificial urinary bladder

Paper answer · P4

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 4, Q3d

Read as a man-made bladder substitute (prosthetic, engineered or transplanted). Bowel neobladder: see Practice Set 4 Paper 2, Q3e.

Need

- Bowel substitutes absorb urine and secrete mucus, causing acidosis, mucus, stones, infection, perforation and cancer. The aim is a substitute that **is bladder-like, not bowel**.

Requirements of an ideal substitute

- Low-pressure storage of adequate volume; continence.
- Complete, voluntary emptying; no reflux or upper-tract damage.
- Biocompatible: no encrustation, infection, rejection or cancer risk; durable.

Approaches

Approach	Principle	Status
Alloplastic (prosthetic) bladder	Synthetic reservoir (silicone, polymers) joined to ureters and urethra	Failed: encrustation, infection, leak at the tissue–prosthesis junction, migration, fibrosis
Tissue-engineered bladder	Autologous urothelial and muscle cells seeded on a biodegradable scaffold, omental wrap	Atala (2006): 7 patients improved; phase II trial showed no significant gain in capacity or compliance; experimental
Engineered conduit	Seeded scaffold as urinary conduit	Neo-Urinary Conduit trial: half developed stenosis or stricture; stiff, no blood supply
Bladder transplantation	Robotic retrieval and transplant of a vascularised composite bladder allograft	Feasible in models and brain-dead donors; first human case reported in 2025; needs immunosuppression

Why progress is slow

- **Vascularity:** thick constructs die centrally.
- **Innervation and contraction:** an engineered bladder stores urine but cannot yet empty voluntarily.
- **Urine–material interface:** foreign material encrusts and becomes infected.
- Fibrosis and contracture.

Alternatives in clinical use

- Augmentation (bowel, ureterocystoplasty, autoaugmentation); continent or orthotopic diversion; ileal conduit.

- **Also know:** neuroprosthetic emptying (sacral anterior root stimulator), artificial urinary sphincter, intraurethral valve-pump devices for underactive bladders.

Future

- Bioprinted vascularised constructs, stem cells (urine-derived, iPSCs, induced pluripotent stem cells), smart scaffolds.

Diagram to draw: biopsy → cell culture → seeding on bladder-shaped scaffold → omental wrap → anastomosis to bladder base.

> **Key points to remember:**
> - Prosthetic bladders failed (encrustation, infection, leak).
> - Atala 2006 promising; later trial no gain; still experimental.
> - Barriers: vascularity, innervation, urine–material interface.

Rapid Revision Sheet · Section 12

CA bladder, etiology, newer investigations and treatment. (LAQ)

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

Re-look (second-look) TUR bladder tumour (SAQ)

- Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
- Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
- Residual T1 → progression (progresses) in up to 80%.

Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation (LAQ)

- Single instillation within 6 h (at most 24 h); never BCG immediately (SWOG S0337 gemcitabine).
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Withhold if perforation; best for low-risk solitary tumours.
- Only maintenance BCG reduces progression; SWOG 3-year schedule.
- NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

BCG therapy and BCG failure (SAQ)

- Only BCG reduces progression; maintenance is essential.
- Never immediately after TURBT; start at 2–4 weeks.
- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy (LAQ)

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab (adds durvalumab) improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.

- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

Bladder-preserving treatment of TCC (LAQ)

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.

Intestinal segments in urology and orthotopic neobladder (LAQ)

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Methods of preoperative bowel preparation. (SAQ)

- Mechanical lowers amount; antibiotics lower concentration.
- Nichols–Condon: neomycin + erythromycin.
- Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

Malignancy in augmented bladder and after urinary diversion (SAQ)

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
- Excise defunctioned anastomoses.
- Gastric segment, exstrophy, immunosuppression: highest risk.
- Haematuria needs cystoscopy; treatment is cystectomy.

Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic (LAQ)

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL) against a reliable outlet; 500–600 mL by 1 year.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR (estimated glomerular filtration rate) >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.

- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; refluxing ureteric anastomosis is acceptable (USC (University of Southern California) trial).
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel.
- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

Bladder diverticulum. (SAQ)

- No muscle → stasis, UTI, stones, cancer.
- Acquired type: treat the BOO first.
- Tumour invading beyond lamina propria is T3a.

Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman (LAQ)

- Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
- BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
- T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
- Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
- BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
- Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

Optical coherence tomography in carcinoma bladder (SAQ)

- Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (P2: 1–2 mm).
- Loss of layering suggests tumour; boundary loss suggests T1.
- Role: Ta vs T1; cannot assess deep muscle.
- High sensitivity, moderate specificity, false positives from inflammation; investigational.

Intravesical gene therapy. (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

Von Bruns Nest (SAQ)

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

Mitomycin (SAQ)

- Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.

- Reduces recurrence, not progression.
- Signature toxicity: skin rash; contracted bladder about 5%.

Mullerionosis of UB (SAQ)

- Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
- Cyclical haematuria; MRI bright on T1.
- Disease is extravesical: cystoscopy limited, partial cystectomy preferred. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder. (SAQ)

- Flat, high-grade; TP53/RB; no FGFR3.
- Untreated 40–83% progress; always high risk.
- BCG with maintenance; cystectomy if BCG-unresponsive. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Virtual cystoscopy (SAQ)

- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

Bladder transplantation (SAQ)

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

Artificial urinary bladder (SAQ)

- Prosthetic bladders failed (encrustation, infection, leak).
- Atala 2006 promising; later trial no gain; still experimental.
- Barriers: vascularity, innervation, urine–material interface. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 13

Kidney & UTUC

136 Benign Renal Tumours

1 · The benign small renal mass

Many resected "cancers" are benign.

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Only the **simple cyst** and **fat-containing angiomyolipoma (AML)** are imaging diagnoses; size, sex and age are unreliable.
- Management follows the risk of renal cell carcinoma (RCC); the SRM algorithm is in Ch 137.

2 · Renal cysts and the Bosniak classification

- A **simple cyst** (any size, homogeneous, **≤20 Hounsfield units (HU)**) needs no further imaging. Otherwise use Bosniak.

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
I	Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (**3 mm**) enhancing wall or septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI	Likely benign	Active surveillance up to **5 yr**
III	Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall or septum	**~50%**	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	**~90%**	

KEY POINT

Ladder: wall ≤2 mm, 3 mm, ≥4 mm, irregular, then nodule. Traps: any calcification fits II, a class I wall may enhance, and growth of I/II is not malignancy.

KEY POINT

Why surveil small III/IV? Cystic RCC is indolent and mostly fluid, so size predicts poorly (proposed **pT1c**) and the number needed to treat is very high.

- European Association of Urology (EAU): treat Bosniak IV as a solid mass, and biopsy a solid component only if it changes management.

Diffuse cystic disease

SAQ ADPKD (Adult polycystic kidney disease)

Paper answer · P2

Type: SAQ (10 marks)
Source: 2017-Winter Paper 2, Q3d

Genetics

- ADPKD (autosomal dominant polycystic kidney disease): commonest inherited cystic kidney disease (about 1 in 400–1000); 7–15% of patients on haemodialysis.

Gene	Chromosome	Share	ESRD (end-stage renal disease)
PKD1 (polycystin-1)	16	~85%	50s
PKD2 (polycystin-2)	4	~15%	70s

- Polycystins (primary cilium) restrain proliferation.

Clinical features (30–50 years)

- **Pain** (commonest): mass, cyst haemorrhage, infection, stones. **Haematuria** about 50%; **stones** 20–30%.
- **Hypertension:** about 50% at 20–35 years with normal function; stretched vessels release renin.
- **Renal failure:** tracks kidney volume; worse with PKD1, male sex.
- **Extrarenal:** liver cysts (commonest; up to 90% by 50); **intracranial aneurysm 10–30%**; seminal vesicle (40%), pancreatic, arachnoid cysts; mitral valve prolapse; diverticulosis.
- RCC (renal cell carcinoma) is not commoner, but is more often bilateral and sarcomatoid.

Diagnosis (ultrasound in at-risk relatives)

Age	Ravine (PKD1 families)	Pei unified (2009)
15–29	≥2 cysts, one or both kidneys	≥3 cysts in total (15–39)
30–59	≥2 cysts in each kidney	≥2 in each kidney (40–59)
≥60	≥4 cysts in each kidney	≥4 in each kidney

- CT or MRI shows TKV (total kidney volume), which predicts progression. Genetic testing if equivocal or for a living donor.

Management

- **BP <130/80 mmHg** with ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker); HALT-PKD: tighter control slowed TKV growth.
- **Tolvaptan** (V2-receptor antagonist; lowers cyclic adenosine monophosphate): **TEMPO 3:4 (2012)** slowed TKV growth and function loss; **REPRISE (2017)** slowed eGFR (estimated glomerular filtration rate) decline in later disease. Causes polyuria and liver toxicity.
- **Pain:** exclude infection, stone, tumour; avoid NSAIDs (non-steroidal anti-inflammatory drugs); aspiration, then laparoscopic unroofing.

- **Infected cyst:** lipid-soluble antibiotics (trimethoprim–sulfamethoxazole, fluoroquinolones); drain if no response.
- **Haemorrhage:** usually settles with rest. **Aneurysm screening** (MR angiography) if family history.
- **ESRD:** dialysis or transplant (no recurrence). **Native nephrectomy** before or at transplant for recurrent infection, bleeding, pain, suspected tumour or no room for the graft.

Diagram to draw: large cystic kidneys, liver cysts and berry aneurysm.

> **Key points to remember:**
> - PKD1: chromosome 16, 85%, earlier ESRD.
> - Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
> - Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

SAQ Acquired renal cystic disease

Paper answer · P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3b

Definition

- **ARCD (acquired renal cystic disease)**: bilateral cysts in kidneys of patients with **ESRD (end-stage renal disease)** not due to inherited cystic disease. Diagnosis: **≥3 cysts in each kidney**.
- Described in 1977; occurs with haemodialysis and peritoneal dialysis.

Epidemiology

Duration of renal failure	Prevalence
Before dialysis	About 10%
3 years of dialysis	44%
5 years	60%
≥10 years	>90%
- Male:female 3:1; independent of the cause of renal failure.	

Aetiology

- **Uraemia, not dialysis**, is the cause; dialysis only gives cysts time to form.
- Tubular obstruction (fibrosis, oxalate crystals); uraemic toxins; growth factors (EGF (epidermal growth factor)); compensatory hyperplasia.
- Cysts may regress after transplant (data conflict); cancer risk persists.

Clinical features

- Mostly asymptomatic.
- **Bleeding into cysts (up to 50%)**: loin pain, haematuria, retroperitoneal haematoma; worsened by uraemic coagulopathy and heparin.
- Cyst infection, stones, polycythaemia (erythropoietin).
- **Neoplasia**: adenomas in 20–25%. **RCC (renal cell carcinoma)** risk about 3 times that without ARCD. **Acquired cystic disease-associated RCC** is a distinct subtype (oxalate

crystals, good prognosis); papillary RCC is also common.

Evaluation

- **Ultrasound**: small echogenic kidneys, many small cysts.
- **CT (computed tomography) with and without contrast**: best for tumours. MRI (magnetic resonance imaging) for small lesions; gadolinium risks NSF (nephrogenic systemic fibrosis).

Feature	ARCD	ADPKD (autosomal dominant polycystic kidney disease)
Kidney size	Small (<300 g)	Large (>800 g)
Cysts	Small	Large
Extrarenal	None	Liver cysts, aneurysms

Management

- **Bleeding**: conservative; persistent → **embolisation or nephrectomy**; peritoneal dialysis avoids heparin.
- Painful cyst: aspiration; infected cyst: drainage, nephrectomy if it fails.
- **Tumour >3 cm: nephrectomy**. <3 cm: nephrectomy in fit patients, or yearly CT and excise if growing.
- **Screening** (debated): younger fit patients after 3 years of dialysis, and before transplant.
- After transplant, native-kidney RCC is more aggressive; >70% of transplant RCCs arise there.

Diagram to draw: small kidney with multiple cysts and a small solid tumour.

> **Key points to remember:**
> - ≥3 cysts per kidney in ESRD; >90% after 10 years.
> - Uraemia causes it; transplant may regress cysts, not risk.
> - RCC risk 3 times; image dialysis patients with haematuria.

Also, from the condensed book (not in the paper answer):

- Acquired cystic kidney disease (ACKD) affects **80%** by 10 yr of dialysis, with RCC risk **≥10×** (multifocal, bilateral, often papillary).
- PKD1 relatives aged 15–29 need **≥3 cysts** for diagnosis; a normal scan excludes disease only from age 40.

3 · Oncocytoma

Biopsy cannot reliably separate it from chromophobe RCC.

- It is the **commonest benign enhancing renal mass**, up to half of benign SRMs <4 cm.
- **Birt–Hogg–Dubé** (folliculin, chr 17) gives fibrofolliculomas, lung cysts with pneumothorax, oncocytoma and chromophobe RCC (chRCC).
- A **stellate scar** is non-specific, whereas **segmental enhancement inversion** favours oncocytoma.

	Oncocytoma	Chromophobe RCC
Cytokeratin 7 (CK7)	Rare, single-cell; KIT positive	Diffuse positive

	Oncocytoma	Chromophobe RCC
Cytogenetics	Normal, −1, −Y, 11q13 rearrangement	Extensive abnormalities

> **KEY POINT**
>
> **Why can a core not clear oncocytoma?** chRCC has oncocytic areas, so some biopsy "oncocytomas" are RCC. EAU: no definitive core diagnosis.

4 · Angiomyolipoma

AML can bleed catastrophically, so treatment follows bleeding risk.

SAQ Renal angiomyolipoma and its management

Paper answer · P2 · P3 · merged from 2

Asked as: P2 SAQ "Renal aangiomyolipoma" (Practice Set 7 Paper 2 Q3f) · P3 SAQ "Management of AML." (2021-Summer Paper 3 Q3d)

Assumed meaning: AML = renal angiomyolipoma (not acute myeloid leukaemia).

Introduction and pathogenesis

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; AML in 55–90%: younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34) leads to unchecked **mTOR (mammalian target of rapamycin)** signalling, which drives growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis); treat as RCC (renal cell carcinoma).

Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock).
- CKD (chronic kidney disease) in TSC.

Diagnosis

- **CT (computed tomography): macroscopic fat (below about −15 to −20 HU [Hounsfield units]) on non-contrast images is diagnostic.**
- US (ultrasound): hyperechoic (so are some RCCs).
- MRI (magnetic resonance imaging): signal loss on fat suppression, India-ink artefact.
- Fat-poor lesion: MRI ± biopsy (HMB-45 [human melanoma black-45], Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

Management algorithm (Restrepo 2022)

Situation	Management
Acute haemorrhage	Resuscitate, then **SAE (selective arterial embolisation)**; nephrectomy if SAE fails or is unavailable

Situation	Management
Symptomatic (pain, haematuria) or poor access to care	SAE, NSS (nephron-sparing surgery), ablation in selected cases
Asymptomatic sporadic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy**	Active treatment
None of these	**Surveillance**: imaging (US or CT) at 6 and 12 months, then yearly
TSC, asymptomatic, **>3 cm**	**Everolimus**; MRI every 1–3 years

- The old **4 cm** cut-off overtreats and should not be a general trigger; EAU (European Association of Urology) 2023 favours surveillance for most.

Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis) is common; abscess, infarction.
- **Partial nephrectomy:** preferred surgery; essential in TSC (multifocal, recurrent).
- **Thermal ablation** (radiofrequency, cryo): limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%** (placebo); side effects stomatitis, acne, hyperlipidaemia.

Diagram to draw

- Management algorithm by bleeding, symptoms, size and TSC.

> **Key points to remember:**
> - Macroscopic fat on non-contrast CT is diagnostic.
> - Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
> - SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Also, from the condensed book (not in the paper answer):

- An observed AML ≥4 cm needs CT angiography to look for an aneurysm.
- EAU (sporadic AML): use a mammalian target of rapamycin (mTOR) inhibitor only if unfit for embolisation or surgery.

ALGORITHM

1. **Spontaneous retroperitoneal haemorrhage** (Wunderlich): resuscitate, then selective arterial embolisation (**SAE**). Nephrectomy only if SAE is unavailable or unsuitable.
2. **Symptomatic** (pain, haematuria, barriers to care): SAE, then nephron-sparing surgery (NSS) or RN; ablation in selected cases.
3. **Asymptomatic sporadic**: if >**6 cm**, aneurysm **≥5 mm**, growth >**0.25 cm/yr**, or childbearing age/pregnancy, then treat; otherwise surveil.
4. **Tuberous sclerosis (TSC)**: if >**3 cm** or growing, then everolimus first line; SAE or kidney-sparing resection second.

> **KEY POINT**
>
> **Why abandon 4 cm?** It was poorly specific and caused overtreatment. "Barriers to care" counts because surveillance needs access to imaging and emergency care.

5 · Other benign epithelial and mesenchymal tumours

- **Papillary adenoma** is **≤15 mm** and cannot metastasise, so it is acceptable in a donor kidney.
- **Metanephric adenoma** (women in their 50s, polycythaemia) is **WT1 positive, CK7 negative**; excise it by NSS.
- **Mixed epithelial and stromal tumour and adult cystic nephroma** form one family and are managed by Bosniak.
- **Leiomyoma** is often really a lipid-poor AML; **Melan-A** is positive only in AML.

Tumour	Key trait
Haemangioma	Sturge–Weber, Klippel–Trénaunay; usually excised
Haemangioblastoma	Von Hippel–Lindau (VHL); mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria
Reninoma (juxtaglomerular cell tumour)	Young women; refractory hypertension, hypokalaemia, high renin; excision cures
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · Landmark trial

Trial	Design	Result		Take-home
EXIST-2	Phase III randomised trial, everolimus vs placebo, AML in TSC or lymphangioleiomyomatosis	≥50% volume reduction in **42% vs 0%**; no bleeding needing embolisation		Approved; mTOR inhibition first line in TSC

137 Malignant Renal Tumours

1 · Overview, epidemiology & aetiology

Most kidney cancers are incidental, so overtreatment is the modern risk.

- Renal cell carcinoma (RCC) makes up 2–3% of adult cancers and is the most lethal common urological cancer. Male to female ratio is **2:1**, peak age 55–75 y, and **4–6% are familial**.
- Over 60% are incidental and ~70% are organ-confined.
- Risk factors: **tobacco** (best accepted; relative risk ~1.4, dose-dependent); **obesity** (the "obesity paradox" of higher BMI but lower stage); **hypertension** (the disease, not its drugs); chronic kidney disease (CKD) and end-stage renal disease (ESRD); family history (2–4×).
- In children and young adults the commonest RCC is **Xp11/TFE3 translocation RCC**. These patients do well with surgery even when node-positive, so do a formal lymph node dissection (LND) at radical nephrectomy (RN).

2 · Classification & pathology

WHO 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC not otherwise specified (NOS)
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; fumarate hydratase (FH)-deficient; succinate dehydrogenase (SDH)-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; **epithelioid angiomyolipoma**; metastases

> **KEY POINT**
>
> Trap: epithelioid angiomyolipoma (AML) is malignant, whereas classic AML, papillary adenoma and oncocytoma are benign. The papillary type 1/2 split was **abandoned** because "type 2" was several diseases (MET-altered, MiT translocation, FH-deficient), so WHO moved to molecular definitions.

Gross features

- RCC has a **pseudocapsule** of compressed parenchyma and is not infiltrative, except collecting duct and sarcomatoid tumours. Cystic degeneration occurs in 10–25% (better prognosis).
- Venous involvement occurs in ~10% and can reach the right atrium.
- Sporadic RCC is bilateral in 2–4% and multicentric in 10–20%; occult satellites are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Hypervascular; clear cytoplasm; carbonic anhydrase IX (CA-IX) positive; sarcomatoid 5–8%	Most venous invasion; worse when localised, better when metastatic
Papillary (10–15%)	**Trisomy 7, 17**, loss of Y; MET altered 81%	Hypovascular; CK7, AMACR; multicentric up to 40%	Good (5-y cancer-specific survival (CSS) 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in Birt–Hogg–Dubé (BHD)	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	From intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; **cup-like CA-IX**	Seen in ESRD and VHL; now benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8 positive; diagnosis of exclusion	Poor; urothelial-like, so may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	**Sickle cell trait**	Young Black patients; infiltrative	Dismal, median 8–13 mo
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells with papillae; FISH confirms	40% of paediatric RCC; late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test ALK before calling a tumour "unclassified"	Responds to ALK inhibitors in some
Unclassified (1–5%)	—	Shrinking catch-all	Poor

- **Indolent types:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC and clear cell papillary. **Lethal types:** medullary, FH-deficient, collecting duct, unclassified and sarcomatoid.
- Because chromophobe and oncocytoma are hard to separate, WHO 2022 added "oncocytic renal neoplasm of low malignant potential".
- **Sarcomatoid** change (1–5%) is dedifferentiation of any subtype, with median survival <1 y. Sarcomatoid or rhabdoid features mean automatic grade 4 and must be reported in every RCC.

> **KEY POINT**
>
> Localised clear cell RCC does worse than papillary or chromophobe because it is intrinsically aggressive. Yet metastatic clear cell does better, because VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs).

WHO/ISUP grading: replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100
3	Nucleoli conspicuous, eosinophilic at **×100**
4	Extreme pleomorphism, giant cells, **rhabdoid/sarcomatoid**

Nuclear grade is an independent prognostic factor.

3 · Molecular biology & hereditary syndromes

The VHL pathway explains clear cell vascularity and drug targets.

SAQ VHL gene and von Hippel–Lindau disease

Paper answer · P2

Asked as:

- VHL gene. — *Practice Set 2 Paper II Q3f (SAQ)*
- Von Hippel Lindau disease — *Practice Set 8 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

(Practice Set 8 Paper II Q3c: VHL gene also answered in Practice Set 2 Paper 2, Q3f)

Introduction

- **VHL (von Hippel–Lindau)** is a **tumour suppressor gene on chromosome 3p25**, identified in 1993; three exons, 213-amino-acid protein.
- Germline mutation causes **VHL disease**: an **autosomal dominant** familial cancer syndrome (about 1 in 36,000), penetrance >95% by 65 years.
- It is the familial form of **clear cell RCC** (renal cell carcinoma). Somatic VHL loss occurs in most sporadic clear cell RCC; chromosome 3p is altered in over 90%.

Function / genetics and pathogenesis: the oxygen-sensing pathway

- VHL protein binds elongin B, elongin C, cullin-2 and RBX1 to form (VHL protein forms) an **E3 ubiquitin ligase** that degrades **HIF (hypoxia-inducible factor)** in normoxia.
- **Normoxia:** hydroxylated **HIF -1α and -2α** are tagged for proteasomal destruction.
- **VHL loss (or hypoxia):** loss of VHL stabilises HIF, which (chiefly HIF-2α) accumulates, joins ARNT (aryl hydrocarbon receptor nuclear translocator) and switches on **VEGF (vascular endothelial growth factor)**, PDGF (platelet-derived growth factor), TGF-α (transforming growth factor-α), GLUT1, erythropoietin and CA-IX (carbonic anhydrase IX).
- Hence the **hypervascularity** (**highly vascular** tumours) of clear cell RCC.
- **Knudson two-hit:** one germline plus one somatic hit, so VHL tumours are **early, bilateral and multifocal**.

VHL disease manifestations

Lesion	Frequency
Craniospinal haemangioblastoma (cerebellum commonest)	60–80%
Retinal angioma (earliest)	25–60%

Lesion	Frequency
Clear cell RCC (mean age 39; commonest cause of death)	25–75%
Renal cysts (commonest renal lesion)	Up to 76%
Phaeochromocytoma	10–25%
Pancreatic cysts / neuroendocrine tumours	45–75% / 11–17%
Epididymal cystadenoma	25–60% of men
- Also endolymphatic sac tumour.	
- **RCC is now the commonest cause of death.**	

Genotype–phenotype

Type	Mutation	Phaeochromocytoma	RCC
1	Deletions, truncations	Low	High
2A	Missense	High	Low
2B	Missense	High	High
2C	Missense	Phaeochromocytoma only	Absent

Diagnosis

- Family history plus one lesion; otherwise ≥2 haemangioblastomas, or 1 plus a visceral lesion. Genetic testing of relatives.

Surveillance (VHL Alliance 2020)

- Eye check (examination) yearly from infancy; metanephrines yearly from 5 years; CNS (brain and spine) MRI every 2 years from 11; **abdominal MRI every 2 years from 15**.

Clinical implications / urological management

- **3 cm rule:** observe solid renal tumours <3 cm; intervene when largest reaches 3 cm; then nephron-sparing surgery (enucleation) or ablation, after excluding phaeochromocytoma.
- **Nephron-sparing surgery** (enucleation) is standard; not curative: local recurrence 75–85%, yet 10-year disease-specific survival 81–94%. Ablation for repeated small tumours.
- Exclude phaeochromocytoma before any surgery (normetanephrine raised; α-blockade). **Partial (cortical-sparing) adrenalectomy.**
- **Targeted therapy:** VEGF TKIs (tyrosine kinase inhibitors), bevacizumab, mTOR (mammalian target of rapamycin) inhibitors.
- **Belzutifan** (HIF-2α inhibitor): FDA (US Food and Drug Administration) approved 2021 for VHL disease; renal response 30–50% (another source: responses in about half of renal tumours); anaemia.
- **Also know:** PBRM1, BAP1, SETD2 on 3p.

Diagram to draw: VHL–HIF pathway in normoxia versus (vs) VHL loss, with drug targets.

> **Key points to remember:**
>
> - AD; 3p25 tumour suppressor; VHL protein degrades HIF.
> - VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
> - Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
> - Belzutifan (HIF-2α inhibitor) approved 2021.

*See the paper answer **Familial / hereditary urological malignancies** (Section 2, Ch 25).*

Also, from the condensed book (not in the paper answer):

- BAP1 mutation (up to 14% of sporadic clear cell) gives high grade and rapid growth, so these tumours are unsuitable for active surveillance (AS).
- Loss of **CDKN2A** (9p) carries a poor prognosis in all subtypes.

ALGORITHM

1. Normoxia: the VHL protein joins elongin B/C, cullin-2 and RBX1 to form an E3 ligase, which degrades hypoxia-inducible factor (HIF)-1α/2α.
2. If VHL is lost (or in hypoxia), **HIF-2α** accumulates and pairs with ARNT. This raises VEGF, PDGF, TGF-α, GLUT1, erythropoietin and CA-IX → hypervascular clear cell RCC.

Drug targets: VEGF ligand (bevacizumab); VEGF-receptor TKIs (sunitinib, pazopanib, axitinib, cabozantinib); mTOR (temsirolimus, everolimus); HIF-2α (**belzutifan**, FDA-approved in 2021 for VHL, ~50% response); CTLA-4 (ipilimumab); PD-1 (nivolumab, pembrolizumab); PD-L1 (atezolizumab, avelumab).

Syndrome (beyond topics)	Gene (chr)	Renal tumour	Other / management
Hereditary leiomyomatosis and RCC (HLRCC)	FH (1q42–43)	Aggressive, often **solitary**	Uterine leiomyosarcoma; **prompt excision at any size, wide margins**; PN only if a wide margin is feasible; FDG-PET finds small metastases
SDH-RCC	SDHA/B/C/D, SDHAF2	Onset ≤40 y	Paraganglioma/phaeochromocytoma, GIST; excision, PN preferred
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast, thyroid; 3-cm rule
Tuberous sclerosis	TSC1/TSC2	AMLs; clear cell 1–3%	AML: surveil <3 cm, everolimus 3–5 cm, embolise or excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma; nephron-sparing excision
MiTF-associated	MiTF p.E318K	RCC >5×	Melanoma; management undefined

4 · Clinical presentation

- The **classic triad** (flank pain, gross haematuria, mass) is rare and means advanced disease. Other clues are weight loss, bone pain, a **non-reducing or right-sided varicocele**, and leg oedema from inferior vena cava (IVC) obstruction.
- Over 50% of spontaneous perirenal haematomas are due to tumour, so repeat CT later.

Paraneoplastic syndromes (10–20%)

Syndrome	Frequency / mechanism	Key point
Raised erythrocyte sedimentation rate (ESR)	Commonest	Marker of advanced disease

Syndrome	Frequency / mechanism	Key point
Hypercalcaemia	Up to 13%; parathyroid hormone-related protein (PTHrP) or osteolytic metastases	**The only one managed medically**: hydration, then furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy with or without metastasectomy
Hypertension	Renin, artery compression, arteriovenous fistula	—
Polycythaemia	Erythropoietin	—
Stauffer syndrome	3–20%; interleukin-6	Non-metastatic liver dysfunction (ALP raised almost always); exclude liver metastases; normalises after nephrectomy in 60–70%; persistence means viable tumour and poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders

- Paraneoplastic syndromes resolve only with excision or systemic therapy, so persistence after nephrectomy means residual disease.

5 · Diagnosis

Imaging sorts a mass into cyst, fat or enhancing tumour.

LAQ Radiology for solid renal masses.

Paper answer · P4

Type: LAQ (25 marks)
Source: 2018-Summer Paper 4, Q1

Introduction

- Most renal masses are now found incidentally. Imaging is the **strongest predictor of malignancy**; size and **enhancement** are the two key variables.
- About 20% of solid enhancing cT1 masses are benign (oncocytoma, fat-poor AML (angiomyolipoma)).
- **Goals:** solid or cystic? enhancing? fat? subtype? stage and surgical anatomy.

A. CT (computed tomography): the standard test

Renal mass protocol (dedicated, thin slices):

Phase	Timing	Use
Non-contrast	–	Baseline HU (Hounsfield units), fat, calcification, haemorrhage
Corticomedullary	30–70 s	Hypervascular masses; arteries; clear cell vs other subtypes
Nephrographic	90–120 s	**Most sensitive** for detection (tumour enhances less than parenchyma); renal vein and IVC (inferior vena cava) thrombus
Excretory	7–10 min	Collecting system relation

Enhancement thresholds (rise from unenhanced value):

Rise	Meaning
≥20 HU	Enhancing: solid tumour
10–20 HU	Equivocal (MRI, magnetic resonance imaging, or CEUS)
<10 HU	Non-enhancing: cyst

- **Pitfall:** pseudoenhancement of small (<10–15 mm) cysts; dual-energy CT iodine maps help.
- **Macroscopic fat = AML:** fat ≤ −10 to −20 HU on unenhanced CT is diagnostic. Fat **with calcification** suggests RCC (renal cell carcinoma).
- **Fat-poor AML:** hyperdense on unenhanced CT, homogeneous, avid early enhancement and washout.
- **Subtype clues:** clear cell RCC enhances early, avidly and heterogeneously (necrosis); papillary RCC is homogeneous and **hypoenhancing**; chromophobe and oncocytoma intermediate, with a central scar. Overlap is large.

B. MRI

- **Indications:** contrast allergy, CKD (chronic kidney disease), equivocal CT enhancement, young patients, thrombus. Enhancement = signal rise ≥15–20%. Macrocyclic gadolinium carries very low NSF (nephrogenic systemic fibrosis) risk.

Feature	Clear cell	Papillary	Chromophobe / oncocytoma	AML
T2 signal	High, heterogeneous	**Low**, homogeneous (haemosiderin)	Central T2-bright scar	Fat signal
Enhancement	Avid (>200%)	Weak	Intermediate / avid	Avid
Diffusion (ADC, apparent diffusion coefficient)	High	Lowest	Intermediate	–
Chemical shift	**Signal drop** (intracellular lipid, ~59%)	No drop	–	**India-ink** artefact; loss on fat saturation

- **ccLS (clear cell likelihood score, 1–5):** structured MRI algorithm for solid cT1a masses using T2 signal, corticomedullary enhancement and microscopic fat, plus diffusion and **segmental enhancement inversion** (favours oncocytoma). Score 4–5 suggests clear cell RCC; 1–2 a non-clear-cell tumour. It guides biopsy versus surveillance.

C. Ultrasound and CEUS (contrast-enhanced ultrasound)

- Grey-scale: cyst versus solid; poor below 2 cm. AML is hyperechoic, but so are some small RCCs.
- **CEUS** (microbubbles, not nephrotoxic): shows enhancement; used in CKD or allergy, indeterminate lesions, ablation and surveillance.

D. Cystic masses: Bosniak 2019 (CT and MRI)

Class	Key feature	Malignancy / action
I	Simple, thin wall ≤2 mm, no septa	Benign; none
II	1–3 thin septa; homogeneous hyperdense (≥70 HU unenhanced)	Benign; none

Class	Key feature	Malignancy / action
IIF	Wall or septa 3 mm, or ≥4 thin septa	<10%; follow 6, 12 months, then yearly for 5 years
III	Wall or septa ≥4 mm, or irregular	~50%; surgery or surveillance
IV	**Enhancing nodule**	~90%; treat

E. Functional and molecular imaging

- **99mTc-sestamibi SPECT/CT** (single-photon emission CT): mitochondria-rich **oncocytoma** and hybrid oncocytic or chromophobe tumours take it up; clear cell RCC does not.
- **Girentuximab PET (positron emission tomography)** targets CA-IX (carbonic anhydrase IX) on clear cell RCC. **ZIRCON** (^{89}Zr-girentuximab, indeterminate cT1 masses): about 85% sensitivity and 87% specificity for clear cell RCC.
- **FDG-PET** (fluorodeoxyglucose PET): not recommended (low sensitivity). PSMA (prostate-specific membrane antigen) PET: experimental.

F. Renal mass biopsy

- **Indications:** result may change management; suspected lymphoma, metastasis or infection; before ablation or surveillance; metastatic disease. Not needed if surgery is planned regardless.
- Cores: PPV (positive predictive value) 98.8%; subtype accuracy ≥90%, grade 60–80%; non-diagnostic ~14%. Seeding almost nil. Oncocytoma cannot be confirmed on cores.

G. Staging and surgical planning

- **Venous thrombus:** CT sensitivity 78% (renal vein), 96% (IVC). MRI or multiplanar CT shows the cephalad tip; enhancing thrombus is tumour, non-enhancing is bland.

Mayo level	Extent	T stage
0	Renal vein	T3a
I	IVC ≤2 cm above renal vein	T3b
II	IVC below hepatic veins	T3b
III	Intrahepatic IVC, below diaphragm	T3b
IV	Above diaphragm or atrium	T3c

- Nodes ≥2 cm usually malignant. Chest CT; bone scan only if symptomatic.
- **Nephrometry:** R.E.N.A.L. (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location), PADUA, C-index: predict complexity and complications; guide partial versus radical nephrectomy.
- **After ablation:** persistent or new enhancement = residual or recurrence.
- **Recent:** radiomics and AI (artificial intelligence).

Diagram to draw: algorithm: incidental mass → renal-protocol CT/MRI → fat (AML) / cyst (Bosniak) / enhancing solid → biopsy or treat.

Key points to remember:

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

Also, from the condensed book (not in the paper answer):

- The strongest predictors of malignancy are **male sex (~3×)** and **size**: 20–30% of masses ≤4 cm and ~40% of ≤2 cm are benign, while few ≥7 cm are benign.
- A cT1 mass is ~20% benign, ~60% indolent and ~20% potentially aggressive.
- Renal mass biopsy (RMB): **multiple cores over fine-needle aspiration**; haematoma ~5%, specificity 94%. A non-malignant result does not prove the mass is benign.

- **Multiphase CT** is the single most important test: enhancement **>15–20 Hounsfield units (HU)** suggests RCC, and fat **<−20 HU** means AML. MRI is best for IVC thrombus and organ invasion.

ALGORITHM

1. If simple cyst, then observe.
2. If complex or solid: eGFR >45 → CT or MRI with/without contrast; eGFR 30–45 → hydrate, then the same; eGFR <30 → non-contrast CT or gadolinium MRI; contrast allergy → ultrasound (USG), contrast USG or MRI.
3. Then: fat → AML; complex cyst → Bosniak; solid enhancing → management options, with or without RMB.

- **Bosniak** (see Ch 136): malignancy is 3–10% in IIF, ~50% in III and 75–90% in IV. The key discriminator is unequivocal **measurable enhancement**. Avoid ablation and biopsy of cystic masses; AS is acceptable for III/IV.
- Do RMB when the result may change management, or when lymphoma, metastasis or infection is suspected. Do not biopsy patients who will be managed the same way regardless. Biopsy before ablation.
- **Screening:** there is none for the population or for autosomal dominant polycystic kidney disease. In ESRD, screen with USG or CT from the 3rd year of dialysis if fit, and continue after transplant.
- Offer genetic counselling for RCC at **≤46 y**, multifocal or bilateral tumours, or a suggestive history or pathology.

6 · Staging & prognosis

Pathological stage is the strongest predictor of outcome.

TNM (AJCC 8th)	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%

TNM (AJCC 8th)	Definition	5-y survival
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%
T3a	Renal vein/segmental branches, **pelvicalyceal system**, perirenal/sinus fat, within Gerota's	45–75%
T3b	IVC below diaphragm	40–55%
T3c	IVC above diaphragm or **caval wall invasion**	20–45%
T4	Beyond Gerota's, incl. **contiguous ipsilateral adrenal**	0–35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0–30%

Stage groups: I is T1N0; II is T2N0; III is T1–2N1 or T3 any N; IV is T4 or M1. The 8th edition moved pelvicalyceal invasion to T3a.

> **KEY POINT**
>
> TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus a supradiaphragmatic thrombus) are lost. Record all of them and use integrated nomograms.

- **Work-up:** abdominal CT and chest X-ray (chest CT if symptoms or high risk). Bone scan only with raised ALP, bone pain or poor performance status (PS); no PET. Treat an abnormal adrenal as involved.
- **Pathological stage** is the most important factor, then grade, size, subtype, necrosis and sarcomatoid change. Node-positive 5-y survival is 5–30%, and with a thrombus but otherwise confined disease it is 45–69%.
- Venous wall invasion is as important as thrombus level, and collecting-system invasion is adverse.
- Nomograms: **UISS** uses TNM, grade and ECOG PS; **SSIGN** (Mayo) uses Stage, Size, Grade and Necrosis; **Leibovich** predicts recurrence in clear cell RCC.

7 · Management of localised RCC

Every option trades cancer control against kidney function.

LAQ Nephron-sparing surgery for renal masses: indications, approaches and recent advances

Paper answer · P1 · P2 · P3 · P4 · merged from 4

Asked as: P1 LAQ "Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy" · P2 LAQ "Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches" · P3 LAQ "Nephron-sparing surgery for renal masses: methods, recent advances and neoadjuvant therapy" · P4 SAQ "Nephron Saving Surgery (NSS)"

"Nephron sparing nephrectomy" and "nephron saving surgery" are read as nephron-sparing surgery (NSS): partial nephrectomy and ablation. "Nepron" is read as nephron.

Embryology of the kidney

- Three successive kidneys form from **intermediate mesoderm**, in cranial-to-caudal order:

Kidney	Time and site	Fate
Pronephros	Week 3–4, cervical	Non-functional; regresses
Mesonephros	Week 4, thoracolumbar	Functions briefly; tubules become efferent ductules; **mesonephric (Wolffian) duct** persists
Metanephros	Week 5, sacral	**Definitive kidney**

- **Reciprocal induction:** the **ureteric bud (UB)** grows from the caudal Wolffian duct (about week 5) into the **metanephric blastema**.
- UB branches into ureter, pelvis, calyces and collecting ducts.
- Blastema, induced by UB tips, forms nephrons (glomerulus to distal tubule).
- **Molecular control:** GDNF (glial cell line-derived neurotrophic factor) from the blastema acts on RET on the UB; SLIT2/ROBO2 restricts GDNF to one bud. Also WT1, PAX2.
- Urine production is established by 14 weeks; nephrogenesis ends before birth.
- **Ascent and rotation (weeks 6–9):** the kidney rises from the pelvis to the lumbar region and rotates 90° so the hilum faces medially. It takes successively higher arteries; persistent lower ones become accessory arteries.
- **Clinical correlation:** no UB gives agenesis; abnormal induction gives MCDK (multicystic dysplastic kidney); failed ascent gives ectopia; fused lower poles trapped by the IMA (inferior mesenteric artery) give horseshoe kidney; also malrotation, duplication.

Surgical anatomy of the kidney

- **Size and position:** about 11 cm, 125–170 g; T12–L3, lies on the cone-shaped psoas, so the upper pole is more medial and posterior, and the kidney is angled 30–50° behind the coronal plane. The right kidney is 1–2 cm lower (liver).
- **Relations:**

Aspect	Right	Left
Superior	Liver, adrenal, 12th rib	Spleen, stomach, adrenal, 11th–12th ribs
Medial/anterior	Duodenum (2nd part), hepatic flexure	Pancreatic tail with splenic vessels, splenic flexure, jejunum
Posterior (both)	Diaphragm, psoas, quadratus lumborum, subcostal nerve	Same

- **Coverings (inside out):** true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below. So it contains tumour and abscess (collections), yet fluid tracks into the pelvis.
- **Hilum (front to back):** renal vein, artery, pelvis, then the posterior segmental artery.
- **Vascular anatomy (the basis of partial nephrectomy)**
- **Arteries:** renal artery from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava).
- **Five segmental end arteries**: posterior (first branch, behind the pelvis) plus four from the anterior division (apical, upper, middle, lower). With no anastomoses, occlusion causes segmental infarction, but each segment can be clamped or resected separately. Then interlobar (columns of Bertin), arcuate, interlobular arteries.
- **Accessory arteries** in 25–28%, end arteries too.

- **Brödel line:** an avascular plane between anterior (two-thirds) and posterior (one-third) territories, just behind the lateral border. Incision here bleeds least (anatrophic nephrolithotomy).
- **Veins anastomose freely**, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Parenchyma / internal structure:** cortex and 8–18 pyramids; 0.4–1.2 million nephrons per kidney.
- **Collecting system:** papilla, minor calyces (7–9), major calyces, pelvis. Puncture through the **calyceal fornix**, not the infundibulum (Sampaio).
- **Also know:** lymph drains to para-aortic (left) or paracaval/interaortocaval nodes (right); sympathetic T8–L1.

Small renal mass (SRM)

- **Definition:** an enhancing renal mass **≤4 cm (cT1a)**, mostly incidental.
- **Biology:** 20–30% of masses ≤4 cm and about 40% ≤2 cm are benign (oncocytoma, fat-poor angiomyolipoma) (P2: about **20%** of enhancing cT1 masses are benign); most cancers are indolent (<2% of the cancers metastasise). About 20% benign, 60% indolent, 20% potentially aggressive.

Definition and rationale of NSS

- NSS (nephron-sparing surgery) is complete local removal of the tumour with **maximal preservation of functioning parenchyma**. PN (partial nephrectomy) is the standard form; thermal ablation is the non-excisional form.
- **Why spare nephrons:** RN (radical nephrectomy) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN gives equal cancer control for small masses (cT1a) with better function and keeps about **90%** of global function (RN 65–70%).
- **CKD from surgery alone** is stable, with near-normal survival if the new GFR (glomerular filtration rate) stays >45; **CKD from medical disease** (diabetes, hypertension) progresses. So NSS matters most when medical renal disease coexists.
- **Concept:** function after PN depends mainly on the **parenchyma preserved** (preserved parenchymal volume), less on warm ischaemia time.

Indications for nephron-sparing surgery

Category	Situations
Imperative (absolute)	Solitary (functioning) kidney; bilateral tumours (bilateral RCC, renal cell carcinoma); RN would mean dialysis
Relative	Contralateral kidney threatened: stones, renal artery stenosis, hydronephrosis, diabetes, (severe) hypertension, proteinuria; hereditary RCC
Elective	Localised tumour with a normal contralateral kidney; **cT1a (≤4 cm) with normal contralateral kidney: standard of care**; cT1b–T2 when feasible
Benign	Symptomatic AML (angiomyolipoma); focal atrophy in duplex kidney; infected calyceal diverticulum

- Now extended (increasingly used) for selected cT1b–T2, complex or hilar tumours when function matters (better function).
- **Relative contraindications: <20% of nephron mass retained**; pedicle encasement; collecting-system invasion; **venous tumour thrombus**; cT4; cN1 (nodes).

Role of NSS by clinical setting

Setting	Role of NSS
cT1a (≤4 cm)	PN is the **standard of care** when intervention is needed; RN is overtreatment
cT1b–T2	PN when feasible and function matters; otherwise debatable
Hereditary RCC	Observe until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision)
Bilateral tumours, solitary kidney	Imperative PN, staged if bilateral; avoids dialysis
Frail, comorbid, cT1a <3 cm	Thermal ablation or active surveillance
Locally advanced, thrombus, cN1, metastatic	Generally RN or cytoreductive surgery

Choosing the option for a small renal mass (AUA, American Urological Association, 2021)

Option	Role
Active surveillance	<2 cm or predominantly cystic; elderly, comorbid (frail patients), life expectancy <5 years. Imaging every 3–6 months. Intervene if >3 cm or growth >5 mm a year (DISSRM registry data: almost no metastases)
Thermal ablation (cryoablation, radiofrequency)	cT1a **<3 cm**, percutaneous, biopsy first. Higher local recurrence than PN, salvaged by repeat ablation
Partial nephrectomy	**Prioritise PN for cT1a** needing treatment. Imperative: solitary kidney, bilateral tumours, CKD or proteinuria; also familial RCC, young patients
Radical nephrectomy	Preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new (post-RN) eGFR likely **>45**. Otherwise PN

- **PN principles:** negative margin (width does not matter); maximise preserved parenchyma; avoid long warm ischaemia; enucleation for familial or multifocal tumours.

Preoperative evaluation and planning

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging) with 3D (three-dimensional) reconstruction; chest imaging; eGFR (estimated glomerular filtration rate) and proteinuria (CKD stage).
- **Renal mass biopsy (RMB)** when it may change management, and before (preferred) or at ablation (non-diagnostic about 14%). Genetic counselling if age ≤46, multifocal or bilateral.
- **Nephrometry scores** predict complexity, complications, ischaemia time and conversion:

R.E.N.A.L. component	Scored as
Radius	≤4, 4–7, ≥7 cm
Exophytic/endophytic	≥50% exophytic, <50%, entirely endophytic
Nearness to collecting system	≥7, 4–7, ≤4 mm
Anterior/posterior	Descriptor

R.E.N.A.L. component	Scored as
Location to polar lines	Polar, crossing, central
- Total: **low 4–6, moderate 7–9, high 10–12** complexity. **PADUA** (Preoperative Aspects and Dimensions Used for an Anatomical) and C-index are alternatives. 3D models help (shorten ischaemia in a randomised trial).	

Methods of NSS

Group	Methods
Surgical PN	Open, laparoscopic, robot-assisted (RAPN)
Excision technique	Standard (wedge) resection with a rim; enucleation; polar resection or heminephrectomy (≥30% of parenchyma; large polar tumours)
Thermal ablation	Cryoablation, RFA (radiofrequency ablation), MWA (microwave ablation)
Developmental	SBRT (stereotactic body radiotherapy), HIFU (high-intensity focused ultrasound), irreversible electroporation

Surgical approaches for PN

Approach	Features
Open, extraperitoneal flank (supracostal 11th/12th rib, 8 cm mini-flank)	Posterior exposure; solitary kidney, complex or central tumours, cold ischaemia
Open transperitoneal (subcostal, chevron, midline)	Bilateral or large tumours, vascular reconstruction
Laparoscopic transperitoneal	Large space (more space), easier suturing; anterior tumours
Laparoscopic retroperitoneal	No bowel mobilisation, direct hilum; posterior tumours
Robot-assisted (RAPN)	Easier suturing, shorter warm ischaemia, fewer conversions; transperitoneal for anterior and retroperitoneal for posterior tumours; equal oncology

- Open PN suits complex tumours in a solitary kidney.
- First laparoscopic PN 1993 (Winfield).
- **Robotic vs laparoscopic:** shorter warm ischaemia, fewer conversions. **Robotic vs open:** less blood loss, shorter stay, fewer complications; same margins and eGFR.
- Haemostasis: sliding-clip renorrhaphy, barbed sutures, sealants.

Operative steps of PN

1. Flank extraperitoneal or transperitoneal (open), laparoscopic or robotic approach.
2. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
3. Expose and defat the kidney; **intraoperative ultrasound** maps the tumour borders and depth.
4. Vascular control: clamp the artery (± vein); selective or off-clamp for small tumours. Use **ice slush** (cold ischaemia) if ischaemia is expected to be **≥25–30 minutes** (solitary kidney, CKD).
5. Excise with a thin rim of normal parenchyma, or **enucleate** along the pseudocapsule (hereditary, familial, multifocal). **A negative margin is the priority; its width does not**

matter.

6. **Two-layer renorrhaphy** (sliding clip): inner layer closes vessels and collecting system; outer capsular layer over a haemostatic bolster.
7. Unclamp; check haemostasis; close Gerota fascia; drain.

Enucleation vs standard resection

	Standard (wedge) resection	Enucleation
Plane	Through normal parenchyma, thin rim	Along the tumour pseudocapsule
Advantage	Fewer positive margins	Less bleeding, maximal parenchyma, eases zero ischaemia
Use	Default	Hereditary RCC (e.g. VHL [von Hippel–Lindau]), familial, multifocal, solitary kidney, severe CKD

Vascular control and ischaemia

Strategy	Verdict
En bloc or artery-only clamp	Equivalent; traditional warm ischaemia <25–30 minutes; ice slush if longer is expected
Early unclamping (after inner layer)	May reduce pseudoaneurysm
Selective or super-selective segmental clamping	Avoids global ischaemia; no functional gain in a randomised trial
Off-clamp (zero ischaemia)	Small exophytic tumours; more blood loss; no long-term eGFR benefit in randomised trials
Parenchymal compression	Peripheral tumours
Hypothermia, mannitol	No benefit with a normal contralateral kidney

- Function depends mainly on the **quantity of parenchyma preserved**; ischaemia is secondary within accepted limits.
- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes. RAPN: shorter ischaemia than laparoscopy, less blood loss than open.

Thermal ablation (ablative NSS)

- **Cryoablation** (double freeze–thaw), **RFA**, **MWA**; mostly **percutaneous** (CT or ultrasound guided).
- AUA 2021: alternative for **cT1a <3 cm**, especially elderly, comorbid, solitary kidney or VHL; biopsy before (preferred) or at ablation.
- **Cryoablation:** cell death at ≤−40 °C; iceball 5–10 mm beyond the tumour. **RFA/MWA:** heat causes coagulative necrosis.
- Success = loss of enhancement on CT/MRI. Local recurrence is higher than PN but treatable by repeat ablation (equal after repeat ablation).
- **Also know:** SBRT for non-surgical candidates; active surveillance for frail patients.

Special situations

- **Hereditary RCC (VHL):** observe until the largest tumour is **3 cm**, then enucleate (except HLRCC: wide excision).
- **Bilateral tumours:** bilateral PN, usually staged; if one side needs RN, do the PN first.

- **Solitary kidney:** temporary dialysis about 3.5%; local recurrence 3–5%.
- **Also know:** bilateral Wilms tumour; kidney-sparing upper-tract urothelial cancer surgery.

Role of neoadjuvant therapy

- **Aim:** downsize a tumour so that PN becomes possible, mainly an **imperative indication** (solitary kidney, bilateral tumours) where radical nephrectomy means dialysis.
- **TKIs (tyrosine kinase inhibitors):** pazopanib, sunitinib, axitinib. Tumour volume falls **10–30%**; resection becomes possible in **40–45%**, including PN candidates. Best in clear cell RCC.
- **Toxicity:** serious adverse events about 20%, mainly **wound healing**. Stop TKIs several half-lives before and after surgery.
- **Checkpoint inhibitors:** nivolumab is safe but shrinks little; trials ongoing.
- **Status:** not standard (trials only); **upfront surgery whenever feasible**. Adjuvant pembrolizumab (KEYNOTE-564) suits high-risk resected disease.

Complications of PN

- **Urine leak** (drain, then stent); **haemorrhage, pseudoaneurysm or AV (arteriovenous) fistula** (**delayed haematuria**: angioembolisation); AKI (acute kidney injury), renal insufficiency; temporary dialysis in a solitary kidney.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** when >50% of a solitary kidney is lost: proteinuria, FSGS (focal segmental glomerulosclerosis); give an ACE (angiotensin-converting enzyme) inhibitor.

Follow-up

- AUA risk-based imaging; eGFR and urine protein at every visit.

Outcomes and evidence

- Elective PN: **local recurrence 1–2%**, cancer-free survival >90%; imperative 3–5%. Margin width does not matter if negative.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN better function, similar oncology (cancer control); overall survival slightly favoured RN (10-year overall survival 81% RN vs 76% PN), but the trial is flawed.
- **DISSRM (Delayed Intervention and Surveillance for Small Renal Masses) registry:** surveillance almost never led to metastasis; about 24% had delayed intervention.

Recent advances

Area	Advance and evidence
Planning	R.E.N.A.L. and PADUA scores; **3D models**: randomised trial showed shorter operation and ischaemia, less blood loss
Platform	**RAPN:** shorter warm ischaemia than laparoscopic; less blood loss than open. **Single-port robot:** shorter stay, less opioid
Vascular control	Off-clamp, segmental, selective or super-selective clamping (guided by **NIRF (near-infrared fluorescence)** with ICG, indocyanine green), early unclamping. **Randomised trials: no long-term eGFR gain**
Excision	**Enucleation** along the pseudocapsule: less blood loss; hereditary, multifocal, solitary kidney

Area	Advance and evidence
Reconstruction	Sliding-clip and **barbed suture** renorrhaphy; single-layer gives better function, fewer pseudoaneurysms; sutureless with sealants in selected
Neoadjuvant	TKI downsizing in imperative cases (see above)
Outcome reporting	**Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes

Diagrams to draw: segmental arteries with the Brödel line on a transverse kidney section; clamped renal artery, tumour excised with a rim, two-layer renorrhaphy; RAPN with selective segmental clamp and renorrhaphy.

Key points to remember:

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm (after biopsy), surveillance <2 cm or frail.
- Imperative, relative, elective; elective PN is standard for cT1a.
- AUA: RN only if complex + no CKD + new GFR likely >45.
- R.E.N.A.L. 4–6 / 7–9 / 10–12.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Enucleation for hereditary/multifocal.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials; 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

Also, from the condensed book (not in the paper answer):

- AUA 2021: refer to nephrology if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30. Do LND for clinically positive nodes and adrenalectomy only if suspicious.
- CKD follows RN in 65% vs PN in 20%; surgical CKD progresses ~0.7%/y vs medical CKD ~4.7%/y. **Predicted GFR after RN ≈ global GFR × contralateral split function × 1.25**.
- **Triggers to intervene on AS**: >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms.

- On AS, small masses grow 0.1–0.3 cm/y and metastasise in <2%; image every 3–6 mo.
- PN function depends first on parenchymal quality, then on the **amount of vascularised parenchyma preserved (the main modifiable factor)**, and only secondarily on ischaemia.
- Leak and bleeding are commoner after PN than RN (steps: Ch 142).

- In a solitary kidney at least 20–30% of the kidney is needed. If bilateral tumours make PN impossible, options are RN with dialysis or neoadjuvant TKI.
- Enucleation is controversial in sporadic RCC because 40–50% infiltrate the pseudocapsule.

Radical nephrectomy

- Robson's principles (early vascular ligation, removal outside Gerota's, adrenalectomy, extended LND) are now modified: perifascial dissection is still essential (~25% of cT1b/T2 invade fat), but there is **no routine adrenalectomy** (unless enlarged on imaging or extensive tumour) and **no routine extended LND**.
- Laparoscopic RN is standard without gross invasion, major venous involvement or bulky nodes; open surgery uses a subcostal or thoracoabdominal approach (see Ch 141).

Thermal ablation (see Ch 143)

- Thermal ablation (TA) works best <2.5–3 cm and gives no pathological staging. Cryoablation is lethal at ≤−20 °C with a double freeze–thaw; neither cryoablation nor radiofrequency is superior.
- Recurrence is mostly salvaged by repeat TA, and nodular enhancement means recurrence.

Familial RCC: 3-cm rule (National Cancer Institute)

- VHL tumours <3 cm had no metastases at ~5 y, while ~25% metastasise when larger. **HLRCC is the exception**.
- Recurrence is near-universal, so surveillance is lifelong. **Exclude phaeochromocytoma** and central nervous system haemangioblastoma before surgery.

8 · Locally advanced RCC

Even with a caval thrombus, complete surgery can cure.

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Clamp infrarenal IVC, then contralateral renal vein, then suprarenal IVC; occlude lumbars
III	Intrahepatic, below diaphragm	T3b	Mobilise liver and milk thrombus below hepatic veins; or suprahepatic clamp with Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass with or without hypothermic arrest; avoid opening the heart if thrombus can be brought below the atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins; graft/reconstruct

- Image close to surgery, because the thrombus grows. **No routine renal artery embolisation** (more deaths). Do coronary angiography before bypass (levels III–IV) and use intraoperative transoesophageal echocardiography.
- Level and wall invasion predict survival, and level IV without metastasis is curable. Mortality is 5–10% for supradiaphragmatic thrombus.

Locally invasive (T4)

- Only ~40% of suspected invasion is confirmed. En bloc resection is the only cure, but prognosis is poor.

Lymph node dissection

- Only 30–43% of enlarged nodes are malignant. With ≥2 risk factors (≥10 cm, cT3/T4, grade 3–4, sarcomatoid change, necrosis), >40% are node-positive.
- LND gave no survival benefit (EORTC 30881), so it is done for suspicious nodes, mainly for staging.

9 · Recurrence, adjuvant therapy & follow-up

A third of recurrences come late; one adjuvant drug works.

Local recurrence	After RN	After PN	After TA
Rate	2–4%	1.4–10%	Cryo 3–10%, radiofrequency ablation 5–20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone
Management	Only 20–40% isolated, so do a full work-up; en bloc resection gives 30–50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- A positive PN margin is usually surveilled; completion nephrectomy is only for gross involvement.
- About 20–35% metastasise after "curative" surgery, but cytokines, vaccines and most TKIs failed as adjuvants. **Adjuvant pembrolizumab** is FDA-approved for high risk.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1–2	X-ray
Intermediate	pT1 grade 3–4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging starts at 3–6 mo and intensifies with risk; very high risk is followed for up to 10 y. A microscopic positive margin moves the patient up one level. **30% of recurrences occur after 5 y**.
- Benign pathology needs no imaging. After TA, do contrast imaging within 6 mo, then follow the intermediate schedule.

10 · Other renal malignancies

Lymphoma and metastases need biopsy and drugs, not nephrectomy.

Tumour	Clues	Management
Sarcoma (1–2%)	**Leiomyosarcoma commonest** (50–60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins; the first resection is the best chance; margin and grade give prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%); massive adenopathy, splenomegaly; prior lymphoma means recurrence until proven otherwise	**Biopsy, avoid nephrectomy**; R-CHOP with or without radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung primary commonest; multifocal, 5–30 HU	Biopsy; systemic therapy; nephrectomy only for refractory bleeding
Carcinoid	**Horseshoe kidney (82×)**	Excision; good prognosis
Small cell	Usually advanced	Surgery and platinum
Primitive neuroectodermal tumour	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than in children

- In a suspected sarcoma, any RCC feature excludes it (the differential is sarcomatoid RCC).

> **KEY POINT**
>
> Non-RCC malignancies are almost all **hypovascular**. The key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis: biopsy, don't resect). RCC nodes are interaortocaval (right) or para-aortic (left), so nodes elsewhere suggest lymphoma.

11 · Landmark trials

Trial	Design	Result	Take-home
EORTC 30904	Randomised trial (RCT), RN vs elective PN, ≤5 cm, normal contralateral kidney	PN better function; 10-y overall survival (OS) better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed
EORTC 30881	RCT, RN with or without LND	No survival benefit	Underpowered, low-risk (4% node-positive); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	Disease-free survival (DFS) 6.8 vs 5.6 y; no OS gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	DFS hazard ratio 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry, AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

1 · Epidemiology & risk factors

UTUC is often invasive at diagnosis, and its causes act as a field effect.

LAQ Upper tract urothelial carcinoma

Paper answer · P2 · P3 · merged from 2

Asked as: P3 LAQ "Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management" · P2 LAQ "Etiopathogenesis and management of Upper tract TCC"

TCC (transitional cell carcinoma), including TCC of the ureter, is taken as UTUC (upper tract urothelial carcinoma).

Introduction and epidemiology

- **UTUC** is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Incidence is rising, **driven by ureteric tumours**.
- Peak age 70–90; male:female 2:1.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall. Concurrent bladder cancer in 17%.
- Bladder recurrence follows in 22–47%, mostly from downstream seeding.
- **Site (ureter): distal ureter 70%, mid 25%, proximal 5%**, explained by downstream implantation.
- Standard is **RNU (radical nephroureterectomy) with bladder cuff**; kidney-sparing versus radical surgery is chosen by **risk**.
- URS (ureteroscopy) biopsy is small and shallow, so it **cannot stage**. Stage is inferred from grade, endoscopic appearance, imaging and cytology together.

Aetiology (etiopathogenesis)

Factor	Key facts and mechanism
Tobacco	Commonest and most important modifiable factor; risk 2.5–7-fold. Aromatic amines are activated to mutagenic N-hydroxylamines
Aristolochic acid	Balkan endemic nephropathy (UTUC 40% of renal tumours there) and Chinese herb nephropathy. Aristolactam–DNA adducts cause A:T to T:A transversions in TP53. Familial but **not inherited**; often bilateral
Lynch syndrome (HNPCC, hereditary nonpolyposis colorectal cancer)	Germline MMR (mismatch repair) mutation: MLH1, MSH2, MSH6, PMS2. Risk ~14-fold, highest with MSH2; found in 5–9% of screened UTUC
Occupation	"Amino tumours": aniline dyes, benzidine, β-naphthylamine (dyes, rubber, petrochemicals); latency up to 20 years
Analgesic abuse	Phenacetin: papillary necrosis; capillarosclerosis is pathognomonic; history in 11% of ureteric tumours
Arsenic	Well water in south-west Taiwan (blackfoot disease)

- **Also know:** cyclophosphamide (acrolein); chronic infection and stones (squamous carcinoma); alcohol.
- **Lynch screening:** suspect if young (age <60), or a personal or family history of Lynch-spectrum cancer (colon, endometrium). Test the tumour for MSI (microsatellite instability) or MMR immunohistochemistry, then germline testing.

Pathogenesis

- Two theories both operate. **Intraluminal seeding** (one clone spreading downstream) explains distal predominance and frequent bladder recurrence (22–47%). **Field effect / field change** (carcinogen in urine, independent clones) explains multifocality.
- Sequence: hyperplasia, dysplasia, CIS (carcinoma in situ). Variant histology in ~25%. Molecular subtypes include FGFR3-mutant (low grade), TP53/MDM2-mutant and hypermutated (Lynch).

Pathology and spread

- 85% papillary, 15% sessile; WHO (World Health Organization) 2004/2016 low or high grade. Ureteric tumours are low grade and low stage in 55–75%.
- **Why ureteric tumours do worse:** the renal parenchyma is a barrier for pelvic tumours, but periureteric fat holds a rich lymphovascular plexus. At equal stage, ureteric tumours have more local and distant failure.
- Spread: direct, lymphatic (**mid and distal ureter to pelvic nodes**; upper ureter to hilar and retroperitoneal), blood (lung, liver, bone).

Clinical features

- **Haematuria** (56–98%), **flank pain** (20–32%, dull from hydronephrosis/obstruction or colic from clots); lumbar mass about 10%; incidental about 15%.
- Weight loss, bone pain or a mass suggest advanced disease.

Diagnosis (investigations)

1. Cystoscopy: always, to exclude coexisting bladder cancer (concurrent in 17%). Positive cytology with a normal bladder and urethra means the upper tracts must be examined.

2. Imaging

Test	Role and key points
CTU (computed tomography urography)	First line; best test; highest accuracy (sensitivity ~92%, specificity ~95%). Radiolucent **filling defect**, hydronephrosis, ureteric wall thickening, non-visualisation. Tumour ~46 HU (Hounsfield units) versus uric acid stone ~100 HU. Misses flat lesions (CIS)
MRU (magnetic resonance urography)	If contrast or radiation is contraindicated; sensitivity ~75% for tumours <2 cm
Retrograde pyelography	Poorly filled calyces; accuracy ~75%; done with ureteroscopy. **Goblet (champagne-glass) sign**, the ureter dilated just below the tumour; **Bergman's sign**, the catheter coiling below it
FDG-PET/CT (fluorodeoxyglucose positron emission tomography)	Nodes and metastases when contrast is contraindicated
Renogram	Split function; contralateral kidney must be assessed

Test	Role and key points
- **Hydronephrosis** predicts invasive disease (invasive ureteric tumour in up to 80%).	

3. Urine cytology

- Sensitivity ~65%, grade-dependent (grade 1 ~20%, grade 3 ~75%). **Selective** ureteric washings are more accurate than voided urine. Take it before contrast. FISH (fluorescence in situ hybridisation) is optional for surveillance.

4. Flexible URS with biopsy: the key test

- Adds accuracy (75–85% to ~90%). Biopsy gives **grade correctly in ~90%**, but **stage cannot be assessed** (shallow samples; a Ta biopsy is often T3 at surgery).
- Indications: diagnostic doubt, planned kidney-sparing treatment, solitary kidney.
- Diagnostic URS before RNU raises bladder recurrence, without changing survival.

Staging (TNM [tumour, node, metastasis] 8th edition, 2017)

T	Renal pelvis	Ureter
Ta / Tis	Non-invasive papillary / carcinoma in situ	Papillary non-invasive / carcinoma in situ
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Muscularis	Muscularis
T3	**Peripelvic fat or renal parenchyma**	Periureteric fat
T4	Adjacent organs, or **through the kidney into perinephric fat**	Adjacent organs
- **N1** single node ≤2 cm; **N2** node >2 cm or multiple. **M1** distant metastasis.		
- **AJCC (American Joint Committee on Cancer) grouping:** stage I = T1, II = T2, III = T3; **any N+, T4 or M1 = stage IV**.		
- **Grade:** WHO 2004/2016: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade.		
- Regional nodes: pelvis and upper ureter to hilar and retroperitoneal; lower ureter to pelvic.		

Risk stratification

EAU (European Association of Urology)

Low risk (all needed)	High risk (any one)
Unifocal	Multifocal
Under 2 cm (<2 cm)	2 cm or more (P2: >2 cm)
Low-grade cytology and biopsy	High-grade cytology or biopsy

Low risk (all needed)	High risk (any one)
No invasion on CTU	Hydronephrosis; invasion on imaging
—	Previous radical cystectomy; variant histology

- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023** splits each group into favourable and unfavourable (obstruction, multifocality, lower-tract involvement). Ablation is **preferred** in low-risk favourable, **offered** in low-risk unfavourable, **rare** in high-risk favourable, **palliative** in high-risk unfavourable. Endoscopic therapy suits tumours **under 1.5 cm**.
- **Prognostic factors:** stage (strongest), grade, LVI (lymphovascular invasion), nodes, sessile architecture, necrosis >10%, ureteric site, surgical delay >12 weeks.

Management A: kidney-sparing (minimally invasive and endoscopic) treatment

Indications

- **Elective**: low-risk disease, whatever the other kidney.
- **Imperative**: solitary kidney, bilateral tumours, CKD (chronic kidney disease), unfit for RNU, after shared decision.
- The patient must accept frequent recurrence and **lifelong endoscopic surveillance**.

Ureteroscopic (retrograde) management

- Closed system, lowest morbidity; for small ureteric and renal tumours.
- **Steps**: cystoscopy, then selective cytology and saline barbotage, then no-touch semirigid then flexible URS, map all lesions, biopsy (3 Fr forceps or basket) into fixative, ablate the base with laser (holmium or thulium, after biopsy), stent optional.

Laser	Penetration	Note
Ho:YAG (holmium:yttrium-aluminium-garnet)	Under 0.5 mm	Safe in thin ureter
Nd:YAG (neodymium:YAG)	Up to 5–6 mm	Coagulative necrosis; transmural injury risk
Thulium	Under 0.2 mm	Strong haemostasis

- Avoid circumferential fulguration or ablation (stricture); stage large tumours.
- Results: upper-tract recurrence 23–90% (P2: about half), bladder recurrence 15–45%, progression 0–20%. **Grade is the strongest predictor.**

Percutaneous (antegrade) management

- Percutaneous resection for large pelvic tumours, lower-pole or inaccessible calyces and urinary diversions.
- **Steps**: retrograde catheter, puncture a calyx away from the tumour (upper or middle calyx for pelvic tumours), dilate to 30 Fr, cold-cup, bipolar resection or laser with the base sent separately, nephrostomy left, **second-look nephroscopy**.
- Removes more tumour and can **stage**; the tract allows topical therapy. But it is an **open system**: tract seeding about 0.3%, bleeding.

Segmental ureterectomy

- **Distal ureterectomy with bladder cuff and reimplantation** (psoas hitch or Boari flap) for distal tumours, including high-grade ones when the kidney must be saved. It gives a

staging specimen; add pelvic lymphadenectomy if high grade.

- First examine the whole tract above for multifocality. Lower-ureteric tumours with unfavourable features need resection, not endoscopy.

Site	Operation
Distal ureter	Distal ureterectomy with bladder cuff and reimplantation; psoas hitch (~5 cm extra) or Boari flap (10–15 cm)
Mid or proximal	Excision with 1–2 cm margins (frozen section) and ureteroureterostomy (rare)
Whole ureter	Ileal ureter or autotransplantation
- Refluxing reimplant eases surveillance; non-refluxing limits seeding (debated).	
- Laparoscopic or robotic routes are feasible; outcomes equal RNU in selected patients.	
- Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); UGN-101 mitomycin gel for low grade (see topical therapy). Lifelong endoscopic surveillance.	

Management B: RNU with bladder cuff (standard for high risk)

- **Principles**: kidney outside Gerota's fascia; clip the ureter early; never open the collecting system; remove the **whole intramural ureter and cuff** (stump recurrence 30–75% if left); bag extraction intact; watertight bladder closure; template lymphadenectomy for high risk.
- **Laparoscopic and robotic RNU steps**: modified flank, 3–4 ports; reflect colon; clip ureter; artery then vein; kidney outside Gerota's fascia (adrenal spared); ureter traced under the superior vesical artery; cuff excised (extravesical or open); two-layer closure.
- Abandoned (higher recurrence): transurethral (endoscopic) "pluck", stripping, laparoscopic stapling.
- Open, laparoscopic and robotic routes are oncologically equivalent: laparoscopic **equivalent to open** (AUA/SUO), with less blood loss and shorter stay; less favoured in bulky T3/T4. Robotic yields more nodes.
- **Template lymphadenectomy** for high risk: hilar plus paracaval (right) or para-aortic (left); pelvic nodes for lower ureteric tumours.

Management C: adjuvant topical therapy

- **Single postoperative intravesical chemotherapy** (AUA/SUO strong): **ODMIT-C (2011)** MMC 40 mg cut bladder recurrence 28% to 16% at 1 year; **Ito (2013)** pirarubicin, 42% to 17% at 2 years.
- **Upper-tract instillation** via nephrostomy (low pressure, 25 cm H2O), ureteric catheter or reflux up a double-J. **BCG** (or MMC): 6-week course for upper-tract CIS, and after complete ablation of high-risk favourable Ta/T1. Risks: sepsis, stricture.
- **UGN-101 (Jelmyto)**: mitomycin reverse-thermal gel, **FDA (Food and Drug Administration) 2020** for low-grade UTUC not manageable endoscopically. **OLYMPUS**: complete response 59%, 56% disease-free at 12 months; ureteric stricture about 44%.

Management D: perioperative (systemic) chemotherapy

- **POUT (2020)**: adjuvant platinum after RNU for pT2–T4 or N+: 3-year DFS (disease-free survival) **71% vs 46%**. Gemcitabine–cisplatin; carboplatin if creatinine clearance 30–49.
- **Neoadjuvant** platinum is preferred when renal function is borderline, i.e. when post-RNU renal function would rule out cisplatin (RNU removes cisplatin eligibility).
- **Also know:** adjuvant nivolumab (CheckMate 274) if platinum-ineligible, but no clear benefit in the UTUC subgroup.

Management E: metastatic disease

- First line: **enfortumab vedotin + pembrolizumab** (EV-302, 2023): overall survival ~31 vs ~16 months versus platinum chemotherapy. Alternative: gemcitabine–platinum, then avelumab maintenance.
- Later: pembrolizumab (KEYNOTE-045), erdafitinib for FGFR3 alterations. Node-positive: chemotherapy first, then consolidative surgery.

Prognosis

- 5-year disease-specific survival: localised ~89%, regional ~63%, distant ~17%. Stage is the strongest factor.

Surveillance and follow-up (AUA/SUO)

- **Kidney-sparing**: cystoscopy and URS within 3 months, URS again at 6 and 12 months; imaging 3–6-monthly (high risk) or 6–9-monthly (low risk), then yearly to 5 years; URS and CTU at closer intervals, for life.
- **After RNU**: cystoscopy and cytology from 3 months; CTU yearly (more often if pT2 or more).

Recent advances

- Lynch syndrome screening; thulium fibre laser; single-use digital ureteroscopes.

Diagrams to draw: ureteroscopic laser ablation of a pelvic tumour beside RNU with bladder cuff; RNU specimen with bladder cuff beside lymph node templates; TNM levels of a renal pelvic tumour (T3 parenchyma versus T4 perinephric fat), beside the diagnostic flowchart (CTU, cystoscopy, URS biopsy, risk group); distal ureterectomy with psoas-hitch reimplantation, beside the goblet sign.

Key points to remember:

- Tobacco, aristolochic acid (A:T to T:A in TP53) and Lynch (MSH2) are the key causes.
- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

Also, from the condensed book (not in the paper answer):

- Upper tract urothelial carcinoma (UTUC) is **5–10%** of urothelial cancers, but **40%** of renal neoplasms in the Balkans. Concurrent bladder cancer occurs in **17%**, ~60% are invasive at diagnosis, and multifocality is **10–20%**.
- After UTUC, the bladder recurs in **22–47%**.
- **EAU: screen for Lynch syndrome (LS) at the first interview** if age <60, a personal LS-spectrum cancer, a first-degree relative (FDR) <50 with LS cancer, or two FDRs with LS cancer.

KEY POINT

Why UTUC presents invasive: the muscle layer is thin, so tumours reach muscle, parenchyma or periureteric fat earlier than bladder tumours.

- **Smoking** carries a relative risk (RR) of **2.5–7**. Chronic infection or stones (RR 1.5–2) favour squamous cell carcinoma (SCC). Cyclophosphamide, long-term anthranoid laxatives and pelvic radiotherapy are iatrogenic causes.

KEY POINT

Aristolochic acid, arsenic, phenacetin and Lynch all behave as a **field disease** (multiple, bilateral recurrences). This favours kidney-sparing treatment, but Balkan endemic nephropathy patients often have renal insufficiency.

- **Lynch**: UTUC risk is ≈**14×**, and ≈**75×** in MSH2 carriers. If mismatch repair proteins are lost on immunohistochemistry, do germline testing.

2 · Pathology & molecular biology

- Over 90% are urothelial carcinoma. Variant histology (≈**25%**: squamous, glandular, sarcomatoid, micropapillary, neuroendocrine) is aggressive, and **micropapillary** is independently adverse.
- Ureteric inverted papilloma harbours malignancy in ≈**18%**, so follow it for ≥2 yr.

- SCC (0.7–7%) is commoner in the pelvis and invasive at presentation; adenocarcinoma is rare and linked to stones.
- Microsatellite instability distinguishes UTUC from bladder cancer, is prognostic and flags LS; the **RAS-mutated** subtype does worst.

3 · Diagnosis

Biopsy gives grade but not stage.

- Haematuria is the commonest presentation, then flank pain. **Always do cystoscopy**; positive cytology with normal cystoscopy (including urethra) means the upper tracts must be evaluated.
- **CT urography (CTU)** is first line (EAU prefers it to MR urography): sensitivity ≈**92%**, specificity ≈95%, but flat lesions are missed. Hydronephrosis means an invasive ureteric tumour in up to 80%.
- For MR urography, avoid gadolinium if creatinine clearance <30 mL/min. Do a split-function scan before radical surgery.
- Cytology detects ≈65% overall (G1 20%, G3 75%); selective washing beats voided urine, and it should be **collected before contrast**.
- Flexible ureteroscopy (URS) with biopsy raises accuracy to ≈90%; biopsy grade is ≈90% accurate, but visual grading only ≈70%.

> **KEY POINT**
>
> **Why URS biopsy cannot stage:** samples are small and shallow, so a Ta biopsy is often T3 at resection. Stage is **inferred** from grade, appearance, hydronephrosis and cytology, which is why kidney-sparing patients need intensive follow-up.

- **Diagnostic URS indications:** uncertainty, planned kidney-sparing treatment, solitary kidney. **Pitfall:** URS before radical nephroureterectomy (RNU) increases intravesical recurrence without changing survival, so avoid it when CTU and cytology already justify RNU.

4 · Staging & grading

Pelvic T3 vs T4 is a classic trap.

TNM 8th	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or **renal parenchyma**; ureter: periureteric fat. pT3a = microscopic parenchymal; pT3b = macroscopic or peripelvic fat (worse)
T4	Adjacent organs, or through kidney into perinephric fat
N1 / N2	Single node ≤2 cm / single >2 cm or multiple

- AJCC stages: I = T1; II = T2; III = T3; **IV = T4, any N+ or M1**.
- Pelvic and upper ureteric tumours drain to hilar and retroperitoneal nodes, and lower ureteric tumours to pelvic nodes. WHO grades are PUNLMP (managed as low grade), low and

high grade.

5 · Prognosis & risk stratification

The risk group decides between kidney-sparing treatment and RNU.

- **Stage matters most**, then grade. SEER 5-yr cancer-specific survival after RNU: ≤T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- **Ureteric and multifocal tumours do worse** than pelvic ones at the same stage, because there is no parenchymal barrier and the periureteric lymphovascular plexus is rich.
- Age alone is never a reason to deny RNU. Do **RNU within 12 wk** for invasive disease. Size **2 cm** is the best cut-off for ≥pT2.
- Post-op adverse factors include nodal disease, lymphovascular invasion (never seen on biopsy, so report it on every specimen), positive margins, necrosis, sessile growth, CIS and variant histology.

Low risk — ALL	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
<2 cm	>2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

> **KEY POINT**
>
> Every pre-op factor is a stand-in for invasion, so one adverse feature means high risk. **Trap:** a small, low-grade, unifocal tumour **with hydronephrosis is high risk**.

6 · Recurrence patterns

- Bladder recurrence after UTUC is 15–75% at 5 yr, usually within 2 yr.
- Upper tract tumours after bladder cancer occur in 2–4%, late (median >3 yr). About **30%** of bacillus Calmette–Guérin (BCG)-treated bladder CIS (or cystectomy for BCG-refractory disease) develop UTUC.

> **KEY POINT**
>
> **Asymmetric recurrence:** antegrade seeding makes bladder recurrences common, early and monoclonal, while the field effect gives multifocal tumours of different clones. Both operate.

- Intravesical recurrence after RNU is linked to male sex, prior bladder cancer, ureteric or multifocal tumours, laparoscopic or extravesical cuff technique, positive margins and diagnostic URS.
- **Pan-urothelial disease** (bladder plus ≥2 extravesical sites) has no evidence-based algorithm; note that cystectomy before RNU may lose cisplatin eligibility.

7 · Management overview & follow-up

ALGORITHM

1. Work-up: cystoscopy and CTU, with or without URS biopsy and cytology; assess contralateral function; screen for LS; then assign the risk group.
2. If **low risk**, then kidney-sparing treatment (endoscopic or segmental) with intensive surveillance.
3. If **high risk**, then RNU with bladder cuff, with or without LND, within 12 wk.
4. If **cN+**, then systemic chemotherapy first, and surgery only if there is a response.
5. After RNU, give a single intravesical chemotherapy dose; all patients have cystoscopic follow-up.

- Pre-op nomograms (≥pT2 or non-organ-confined) select patients for neoadjuvant chemotherapy, given before RNU because nephrectomy often removes cisplatin eligibility.
- Lymph node dissection (LND) gives the best staging, but its curative role is debated (technique and POUT: Ch 139).
- Follow-up: cystoscopy for all. After kidney-sparing treatment, add URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory), do yearly upper-tract CT. Surveillance is prolonged because the risk is constant.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Epidemiology, risk factors, pathology, staging and imaging: Ch 138

1 · Principles & patient selection

Tumour risk and renal reserve together decide between removing and saving the kidney.

*See the paper answer **Upper tract urothelial carcinoma** (Section 13, Ch 138).*

Also, from the condensed book (not in the paper answer):

- Endoscopy carries ~3× more reinterventions than RNU; ipsilateral recurrence after segmental resection is 33–55%; no routine adrenalectomy.
- **Standard** treatment of upper tract urothelial carcinoma (UTUC) is radical nephroureterectomy (RNU) with bladder cuff, if the other kidney is normal and the patient fit.
- UTUC is **unforgiving of spillage**, so every approach keeps the collecting system closed.
- Work-up is CT or MR urogram, cystoscopy, then ureteroscopy (URS) with biopsy and selective cytology; URS raises accuracy from ≈75% to 85–90%.
- Keep diagnostic endoscopy brief and low-pressure to avoid seeding.
- Low-grade (LG) tumours are mostly Ta/T1; high-grade (HG) tumours are mostly invasive.

*AUA/SUO risk group	Features	Ablation	Systemic therapy
Low – favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	**Preferred**	No
Low – unfavourable	LG with obstruction, multifocality or prior lower-tract tumour	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High – unfavourable	HG cytology, invasion on imaging, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

*American Urological Association / Society of Urologic Oncology.

KEY POINT

Low grade does not mean automatic ablation: obstruction or multifocality makes an LG tumour unfavourable, where RNU keeps a role.

KEY POINT

Two axes: tumour risk and renal reserve. High risk with adequate reserve means extirpation; low risk means preservation. Imperative indications are where they conflict.

ALGORITHM

1. Imaging suspicion: CT or MR urogram.
2. Endoscopy with biopsy and cytology; assign risk group; shared decision.
3. If high risk with adequate glomerular filtration rate (GFR), then RNU with or without neoadjuvant chemotherapy.
4. If low risk or poor GFR, then kidney-sparing management (KSM).

- Imperative KSM must not significantly compromise cancer control; dialysis risk is weighed against oncological risk.
- Unfavourable lower ureteric tumours need distal ureterectomy, not endoscopy.
- **Expectant management** (AUA/SUO option) is only for major comorbidity or high risk of end-stage renal disease (ESRD); untreated UTUC progresses.

2 · Endoscopic (kidney-sparing) management

- **Grade** is the key selector and the strongest predictor of recurrence. AUA/SUO: tumours **<1.5 cm**; caution at ≥1.5 cm.
- For a bulky or suspected HG tumour, use a **staged approach**: biopsy and partial ablation only, avoiding perforation, and complete it once pathology confirms suitability.
- Versus RNU, survival is similar but there are **~3× more reinterventions** and lifelong surveillance.

	Retrograde ureteroscopic	**Antegrade percutaneous**
System	**Closed**	**Open** – tract seeding ≈0.3%, all HG
Best for	Low-volume ureteric, small renal	Large pelvic/upper ureteric, lower-pole calyx, diversion
Staging	Not possible	Possible – base sent separately
Morbidity	Low	Higher (≈25%), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical therapy	Repeat URS; catheter or stent reflux	Via retained nephrostomy

- **URS:** saline barbotage improves cytology, and a tight ureter is pre-stented rather than forcefully dilated. The usual method is a **3-Fr biopsy, then laser**; resection never goes beyond the lamina propria.
- A tumour protruding from the orifice needs ablation or resection of the whole intramural ureter.
- **Never fulgurate circumferentially**, because a stricture follows.
- Lasers: holmium:YAG <0.5 mm, ideal for the ureter; neodymium:YAG 5–6 mm, with transmural risk; thulium <0.2 mm, with strong coagulation.
- **Percutaneous:** puncture the calyx distal to a peripheral tumour, never through it, and dilate to 30 Fr. Resect with cold-cup forceps or a **bipolar** resectoscope, since monopolar risks hypo-osmotic absorption.
- A **nephrostomy is always left** for second look and topical therapy. HG or invasive histology means RNU.
- In a diversion, use flexible or antegrade endoscopy.

KEY POINT

Does URS spread tumour? Prior ureteroscopy increases bladder recurrence (seeding), but pre-RNU URS has not increased metastasis. A single post-op intravesical instillation mitigates this.

3 · Segmental ureterectomy

- It saves nephrons **and** gives a staging specimen; in selected patients it equals RNU.
- **URS of the entire tract above is mandatory**. For HG, add ipsilateral pelvic lymph node dissection (LND).
- Ipsilateral recurrence is **33–55%**, mostly distal and lifelong.
- Segmental resection with uretero-ureterostomy is rare and needs 1–2 cm margins with frozen section.
- Distal ureterectomy with reimplant suits LG low-stage tumours, or selected HG distal ones when the kidney must be kept.
- **Boari flap:** contraindicated by a small or irradiated bladder. The base is ≥2 cm wider than the apex, and the width ≥3× the ureteric diameter.
- **Ileal ureter:** 20–25 cm of ileum ≥15 cm from the ileocaecal valve, isoperistaltic.

Manoeuvre	Length gained	Condition
Psoas hitch	≈5 cm	Mobile bladder; avoid nerve entrapment
Boari flap with hitch	Further 10–15 cm	Adequate capacity
Renal descensus with nephropexy	Up to 8–10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 · Positive cytology & carcinoma in situ (CIS)

Trace a positive cytology before removing any kidney.

ALGORITHM

1. Positive voided cytology: repeat it.
2. CT urogram and bladder biopsies. If the bladder is positive, then treat the bladder.
3. If still positive with a negative bladder (or after BCG "6+1"), then selective upper-tract cytology, retrograde pyelography (RPG), prostatic urethral biopsy (men) and URS.

- Upper-tract CIS is a diagnosis of exclusion: persistently positive selective cytology with normal URS and imaging.
- **No RNU for cytology alone**, because of false positives and later contralateral disease.
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy; RNU if more than surface disease is proven.
- CIS with HG papillary UTUC needs radical or multimodal treatment.

5 · Topical therapy in the upper tract

- Indications are solitary kidney, bilateral UTUC and ESRD risk. Routes are nephrostomy (≤25 cm H_2O), ureteric catheter or stent reflux.
- Always use **low pressure and sterile urine**, as sepsis deaths are reported.

- Adjuvant instillation after papillary ablation has **no proven benefit**. BCG is acceptable after complete ablation of HG favourable Ta/T1; its commonest complication is renal granulomas.
- **UGN-101 (Jelmyto)** is mitomycin C (MMC) in a reverse-thermal gel, FDA-approved in 2020 for LG UTUC **not endoscopically manageable**. Stricture occurs in ≈44%.

6 · Radical nephroureterectomy

Remove the whole ureter and cuff without spillage.

- **Indications:** non-metastatic high-risk pelvic or proximal ureteric tumour (HG, obstruction, parenchymal invasion), with cuff and LND; or LG with unfavourable features or failed or infeasible endoscopy.
- **Kidney first, cuff last. Clip the ureter early**, control the hilum before manipulation and stay outside Gerota's fascia. **Adrenalectomy is not routine.**
- Remove the **entire intramural ureter and orifice**: a retained stump recurs in **30–75%** and cannot be surveyed. After bowel diversion, excise the ureteroenteric anastomosis.
- Extravesical or transvesical excision is acceptable; lasso, pluck, stripping and stapling are abandoned because of spillage and positive margins. Close in two layers.
- Laparoscopic or robotic: wide perihilar margin and extract **intact in a bag**.
- Ureteric blood supply is anteromedial in the upper third, medial in the middle and **lateral** in the lower third.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

> **KEY POINT**
>
> Principle beats approach: port-site seeding and carcinomatosis follow spillage, not the approach.

Lymphadenectomy

*See the paper answer **Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy** (Section 12, Ch 131).*

Also, from the condensed book (not in the paper answer):

- Outcome ranks pN0 better than pNx, better than pN+; the prognostic cliff is T2 to T3.
- LND has value in **T2–T4**, especially for pelvis and proximal ureter, and the surgeon, not the approach, sets the yield.
- Caudal limits: renal pelvis to the inferior mesenteric artery; upper two-thirds of ureter to the aortic bifurcation; lower third to the pelvis.

7 · Perioperative adjuncts

- Bladder tumours follow in 30–50% after RNU or endoscopy, mostly within 2 yr. **AUA/SUO strongly recommend** a single dose after RNU (optional after endoscopy).

- Post-op (day 3–7), do a cystogram, then give MMC or gemcitabine with a 1–2 h dwell. An **intraoperative** instillation, clamped until the ureter is clipped, needs no cystogram.
- **Renal function is the hinge**, so give platinum **before** nephrectomy. Neoadjuvant is favoured if post-op creatinine clearance (CrCl) is expected **<u><50</u>** mL/min.
- In EA8141, patients with CrCl >50 received accelerated MVAC ×4.
- Adjuvant (POUT): gemcitabine–cisplatin (CrCl >50) or gemcitabine–carboplatin (30–49). cN+ disease receives chemotherapy, then consolidative surgery.

8 · Complications

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; no circumferential fulguration
Percutaneous	Bleeding, irrigant absorption, perforation, secondary ureteropelvic junction obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 · Follow-up

- Every visit includes **cystoscopy**, cytology and a metabolic panel; after KSM add ipsilateral URS.
- Use CT or MR urography (RPG if no contrast) and chest imaging.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6–9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3–6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	With cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3–6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3–6-monthly ×2 yr; chest CT 6–12-monthly ×5 yr

10 · Landmark trials

Trial	Design	Result	Take-home
ODMIT-C	Randomised controlled trial (RCT), MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% vs 16% at 1 yr	Single-dose instillation standard
Pirarubicin trial (Ito)	RCT, single pirarubicin dose within 5 days of RNU	Bladder recurrence 42% vs 17% at 2 yr	Confirms single post-RNU instillation
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable

Trial	Design	Result	Take-home
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option
POUT	Phase III adjuvant platinum vs surveillance, pT2–4/N+	Stopped early; disease-free survival (DFS) hazard ratio **0.45**; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

141 Contemporary Open Surgery of the Kidney

1 · Role & preoperative work-up

- Open surgery offers **exposure and early great-vessel control**. It suits renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours and a hostile abdomen (minimally invasive surgery: Ch 142).
- **Renal function** decides partial (PN) versus radical nephrectomy (RN). Estimate glomerular filtration rate (GFR) with the **CKD-EPI 2021 race-free** equation, which beats MDRD.
- If GFR is **<60** or there is significant proteinuria, ask for a perioperative nephrology consult.
- In advanced RCC, check blood count and metabolic panel (MSKCC and IMDC risk models), liver tests (**Stauffer syndrome**) and cardiopulmonary fitness.
- Contrast is generally safe if GFR is **>30**. If thrombus is suspected, MR or CT venography must be **≤1 month old**, because thrombus propagates rapidly.
- Give cefazolin ≤1 h pre-incision, pneumatic compression and subcutaneous heparin 5000 IU; no routine bowel preparation.
- **Renal artery embolisation** lets the vein be tied first when the hilum is hidden, but it is no longer routine because of postinfarction syndrome and harder surgery if delayed >48 h.

2 · Surgical anatomy & approaches

Most disasters come from misread anatomy.

- At risk on the **right** are liver, duodenum and colon; on the **left**, spleen, pancreatic tail, colon and superior mesenteric artery (SMA).
- The **right renal artery** lies behind the renal vein and IVC and can be reached in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- The **SMA** leaves the front of the aorta **1–2 cm above LRV** and veers left, so it can be mistaken for the left renal artery.
- The right gonadal and adrenal veins drain into the IVC; the left ones drain into the LRV.
- Anterior exposure reflects the colon along the **white line of Toldt**. On the right, add a **Kocher** manoeuvre without cautery, because duodenal injury is devastating.
- **Cattell-Braasch** (right colon and midgut mobilisation) and partial **Mattox** (left medial visceral rotation) expose the great vessels.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour or IVC thrombus (early aortic access); bilateral	Bowel mobilisation needed
Anterior subcostal	Unilateral transperitoneal	Muscle division
Chevron	Renovascular; RN with thrombectomy; both pedicles	Extensive muscle division
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN or PN; 8-cm mini-flank above the 11th	Pleura
Transcostal (11th, rib resected)	Retroperitoneum at an old rib tip	Rib division

Incision	Best for	Limitations
Thoracoabdominal, dorsal lumbotomy	Largely historical; thoracoabdominal is still an option for level III–IV thrombus	

- The **flank** position lowers venous return and lung function, so it is poor in cardiopulmonary disease; pad against rhabdomyolysis.
- In the **supracostal** approach, push the pleura up and divide the intercostals close to the **upper border of the rib** to spare the bundle. Spare the **subcostal nerve**.

3 · Simple nephrectomy

- Stay **within Gerota's fascia**, on the capsule, for a benign nonfunctioning kidney (obstruction, infection, stones, reflux), and spare the adrenal.
- Benign PN suits focal atrophy, a duplex pole or a symptomatic angiomyolipoma.

4 · Radical nephrectomy

- Robson's operation included adrenal and extended nodes; now the **normal adrenal is spared** and lymph node dissection (LND) is selective.
- **Indications:** tumours not amenable to PN, such as a nonfunctioning kidney, tumour replacing the kidney, nodes, cytoreduction or venous thrombus.
- Staging is abdominal CT or MRI and **chest CT**; a bone scan only for symptoms or raised alkaline phosphatase.
- Biopsy only if it changes the plan, such as an infiltrative mass or before systemic therapy.
- T4 invasion means altered enhancement of the organ, not just contact.

ALGORITHM

1. **Right:** survey the abdomen; open Toldt's line down to the iliac vessels; take down the hepatic flexure; Kocher; expose the IVC.
2. Loop and palpate the renal vein; find the artery behind it; tie **artery, then vein**; remove the kidney outside Gerota's.
3. **Difficult hilum:** ligate the artery in the interaortocaval space, so the kidney decompresses; then double-ligate the flaccid vein.
4. Alternative: start at the lower pole and lift the kidney from posterior to anterior to show the artery.
5. **Left:** mobilise the splenic flexure down to the iliacs; divide the renocolic ligament (pad the pancreas).
6. Follow the aorta or gonadal vein to the LRV. Tie the artery above and behind it, confirm occlusion, then the vein.

KEY POINT

Why artery first? Tying the vein first leaves the kidney engorged, so collaterals bleed. Arterial ligation shrinks the kidney and softens the vein, which can then be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or the SMA has been tied.

- **Lumbar veins:** use silk ties and **no clips on the IVC side**, because clips block emergency caval clamping.

- **Lost pedicle control:** tell the anaesthetist, compress, use two suckers and **never clamp blindly**.
- **Adrenalectomy** only for an upper-pole tumour with a lost plane, an adrenal mass or diffuse involvement. Metastasis occurs in **<5%**, and contralateral metastasis is as common as ipsilateral.
- **LND:** nodes are positive in ≈5%, and the EORTC RCT showed no overall survival benefit, though it included many low-stage tumours.
- **Do LND if nodes are enlarged**: a short axis **≥1.5 cm ≈ 50% positive**. Consider it in high risk (>10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- The template runs from crus to bifurcation. **Clip lymphatics above the renal vein** to prevent chylous ascites.
- **Closure:** fill with saline and give a big breath; bubbles mean a pleural tear, which is sutured over a 14-Fr catheter. Take a **chest X-ray in all**.

5 · IVC tumour thrombectomy

Surgery is the only cure; thrombus level decides the plan.

- RCC is the commonest cause (**5–10%** of RCC). Untreated, most die within 1 yr, so surgery (RN and LND, with or without systemic therapy) is the backbone.
- **Tumour thrombus** differs from **bland thrombus** (clot below the obstruction). **MR venography** best shows extent, wall invasion and bland thrombus.
- Anticoagulate for venous thromboembolism, bland thrombus or IVC occlusion. Use **no suprarenal filters**, because thrombus grows onto the device.
- Use **transoesophageal echo (TEE)** for levels II–IV, and involve a **cardiothoracic surgeon for III–IV**.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC **≤2 cm** above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At or above major hepatic veins, below diaphragm	Liver mobilisation with or without Pringle; occlusive thrombus: suprahepatic clamp; nonocclusive: venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) with deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip or staple); do not extract
D	Mixed with tumour	Resect infrarenal IVC; keep lumbars as collaterals

ALGORITHM

1. **Level I (right):** control the IVC before handling the tumour; ligate the right artery interaortocavally.
2. Loop the LRV and the suprarenal and infrarenal IVC with **Rummel tourniquets**.
3. Milk the thrombus toward the ostium; place a Satinsky round the ostium with the thrombus inside the jaws.
4. Do an ostial cavotomy keeping ≥half the IVC width, and extract.
5. **Level I–II:** ligate the renal artery early. Tighten tourniquets in order: **infrarenal IVC, contralateral renal vein, suprarenal IVC**.
6. Open the cava, extract, flush and inspect the wall. Close with 4-0 Prolene.
7. Release the **caudal clamp first**, flush on a held breath, then release the cranial clamp.

- A **left tumour (level II)** is harder: tie the left artery first, then Kocher, **split-and-roll** the IVC and excise the LRV ostium en bloc. Do not clamp the right renal artery.
- **Level III:** keep bypass on standby and mobilise the liver as in transplantation. The **major hepatic veins are never divided**, and **Budd–Chiari syndrome is a contraindication**.
- The **Pringle** manoeuvre is needed if clamping above the hepatic veins. Limit it to ≤60 min, ideally **≤20 min**.
- **Level III sequence:** clamp infrarenal IVC, contralateral renal vein, porta hepatis, then suprahepatic IVC; open and extract. Unscrapable wall invasion needs IVC resection.
- Do a 2–5 min **clamp trial**. If cardiac output falls by >50% or mean arterial pressure by >30%, use bypass. Occlusive thrombus tolerates clamping because of azygos collaterals; free-floating thrombus does not.
- Incomplete tumour removal is universally fatal, so never compromise IVC control to save time.

	VVB	CPB with DHCA
Access	Femoral to jugular, percutaneous	Sternotomy; cool to 16–18 °C and drain 95% of blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar and intercostal back-bleeding	Cardiac and neurological events
Use	Least invasive; not with atrial or iliac bland thrombus, Budd–Chiari	Level IV; bloodless field; ≤60 min (<40 better)

- In **VVB**, place cannula tips far from the thrombus, because dislodgement causes massive pulmonary embolism (PE).

LAQ Vascular grafts and vascular surgery in urology

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 LAQ "Vascular grafts and vascular surgery in urology" (asked as "Vascular Grafts in Urology", Summer 2021 Paper III Q2 LAQ; "Vascular Surgery in Urology", Winter 2024 Paper III Q3a SAQ) · P1 SAQ "Vascular grafts in urology" (Practice Set 5 Paper 1 Q3e) · P2 SAQ "Role of vascular grafts in urology" (2015-Summer Paper 2 Q3e)

Assumed meaning: blood-vessel grafts and patches (conduits) used in urological surgery: transplantation, renal and IVC (inferior vena cava) tumour surgery, RPLND (retroperitoneal

lymph node dissection), sarcoma, renovascular repair and autotransplantation, dialysis access and penile surgery. Tissue grafts (buccal mucosa) are not included.

Assumed meaning (vascular surgery): vascular procedures performed or met by the urologist (control and repair of vessels, caval surgery, transplant and renovascular surgery, penile vascular surgery and access), not vascular surgery as a specialty.

Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would fail or leave tension or narrowing.
- **Patch:** widens a partly resected wall. **Interposition graft:** replaces a segment. **Bypass:** routes flow around a lesion.

Classification of graft materials

Type	Examples	Advantages	Disadvantages
Autologous vein	Great saphenous, gonadal, superficial femoral, donor external iliac vein	Resists infection; suits small vessels and contaminated fields	Limited length; valves; harvest wound
Autologous artery	Internal iliac (hypogastric), **Carrel aortic patch**	Best for renal artery repair	Limited supply
Allograft (homograft)	Cadaveric vein or artery	Off the shelf; avoids xenograft	Availability; degeneration
Xenograft (biological)	**Bovine or porcine pericardium**	Pliable, strong patch, low infection	Patch only; cost
Synthetic	**PTFE (polytetrafluoroethylene, Gore-Tex)**, ringed PTFE, Dacron (polyethylene terephthalate), collagen-impregnated Dacron	Off-the-shelf, large calibre, any length	Infection needs removal; venous grafts thrombose

Principles

- **Control first:** proximal and distal control with vessel loops, Rummel tourniquets and atraumatic clamps (bulldog, Satinsky, DeBakey). Artery before vein.
- **Choice of graft: autologous** vein or artery for small vessels and contaminated fields; **synthetic** (PTFE or Dacron) for large, high-flow vessels with no bowel contamination.
- **Venous (low-pressure) grafts thrombose** more than arterial ones, so use the **widest graft** possible.
- Patch: **smooth side luminal**, rough thrombogenic side outward; edges must not invert.
- Systemic heparin before clamping; flush air and debris before tying; tension-free, everting, spatulated anastomosis with fine monofilament polypropylene (Prolene).
- **Sutures:** polypropylene (Prolene), double-armed. IVC or aorta 3-0/4-0; renal vessels 4-0/5-0; transplant anastomoses 5-0/6-0.
- Antiplatelet or anticoagulant afterwards as indicated.
- Wrap synthetic grafts in **omentum** or retroperitoneal fat, away from bowel and urine.

Application 1: renal transplantation

- Renal vein and artery end-to-side to external iliac vessels; Carrel patch.

- **Short right renal vein:** extend with donor IVC, free donor external iliac vein, or **spiral gonadal vein graft**.
- **Multiple arteries (18–43%):** Carrel patch; on the back table, conjoined side-to-side or end-to-side (bench) reconstruction. **Preserve the lower-pole artery** (it supplies the ureter).
- **Atherosclerotic recipient iliac artery:** mount the graft artery on a **Gore-Tex or bovine pericardial patch**.
- **Children under 15 kg:** aorta and IVC anastomosis, transperitoneal.
- **Transplant renal artery stenosis (1–23%):** angioplasty or stent first; surgical interposition graft or bypass if it fails, which risks graft loss.

Application 2: RCC (renal cell carcinoma) with IVC tumour thrombus

- **Mayo levels 0–IV:** level I milking and partial caval clamp; II full caval control; III liver mobilisation ± venovenous bypass; IV cardiopulmonary bypass ± deep hypothermic arrest.

Situation	Repair
Residual lumen ≥50%	Primary closure
Residual lumen <50%	**Patch cavoplasty**: bovine or autologous pericardium, PTFE, Dacron or saphenous vein
Circumferential wall invasion	**Caval replacement: PTFE 16–20 mm**; contralateral renal vein reimplanted into the graft
Infrarenal bland thrombus	Filter or caval interruption, sparing lumbar collaterals

- **Leibovich–Bower patch width** = normal caval diameter × % diameter resected × 3.
- Left tumour with suprarenal caval resection: the right kidney has no collaterals, so give it new drainage (autotransplantation, or **saphenous vein graft** to the splenic, portal or inferior mesenteric vein).
- After caval replacement: low-dose heparin, then **lifelong aspirin 81 mg**.

Application 3: PC-RPLND (post-chemotherapy RPLND)

- Vascular procedures in **9–15%**; aorta most often.
- **IVC:** narrowing <25%: primary repair; >25%: patch or interposition graft. **Ligation is tolerated** (slow occlusion builds collaterals).
- **Aorta:** never ligated; must be repaired or grafted. Small injury: repair or pericardial patch; resection: **PTFE or Dacron** (about 1% need replacement). Graft also if the duodenum is injured, to prevent aortoduodenal fistula.

Application 4: retroperitoneal sarcoma

- Caval resection ± reconstruction (Dacron, PTFE, homograft); grafts **thrombose in 10–40%**; anticoagulate at least 3 months.

Application 5: renovascular repair and autotransplantation

- Aortorenal bypass (saphenous vein, internal iliac artery, PTFE); splenorenal or hepatorenal bypass.
- Ex vivo bench repair of branch aneurysm or stenosis (or ureteric loss) with internal iliac artery or saphenous vein, then **autotransplantation** (see Q3e). Angioplasty or stent first for fibromuscular dysplasia.

- Traumatic renal artery injury: endovascular stent for the main renal artery; attempt open repair only for solitary kidney or bilateral injury (warm ischaemia <120 minutes).

Application 6: dialysis access

- Arteriovenous fistula first; **AV (arteriovenous) graft (PTFE)** when no native fistula is possible (poor veins): usable in 2–3 weeks; more stenosis, thrombosis and infection than AVF (arteriovenous fistula).

Application 7: iatrogenic or traumatic vascular injury

- Interposition graft.
- SMA (superior mesenteric artery): repair or graft bypass. SMV (superior mesenteric vein): autologous vein graft; Gore-Tex only if no vein.

Application 8: penile surgery

- Priapism: **Grayhack** saphenous vein–cavernosal shunt.
- Vein grafts for Peyronie tunical defects (survive by endothelial perfusion); synthetic grafts not recommended.
- **Arterial revascularisation** (inferior epigastric artery to dorsal artery) for young men with post-traumatic pudendal stenosis; venous ligation investigational.

Application 9: endovascular

- Angioembolisation for trauma, AML (angiomyolipoma) bleeding, pseudoaneurysm and AVF after partial nephrectomy.

Complications

- **Thrombosis** (caval grafts 10–40%), anastomotic stenosis (intimal hyperplasia), bleeding, pseudoaneurysm.
- **Graft infection** (synthetic grafts usually need removal); graft–enteric or urinary fistula.
- **Ureteric:** transient hydronephrosis after aorto-iliac grafts (12–20%), symptomatic stricture 1–2%, **ureteroarterial fistula** (treat with endovascular covered stent).

Recent advances

- Endovascular covered stent grafts; bioengineered acellular vessels (under study); robotic autotransplantation.

Diagrams to draw

- IVC patch cavoplasty and PTFE caval replacement with left renal vein reimplanted.
- Mayo IVC thrombus levels with the matching vascular control.
- Bench arterial reconstruction.

> **Key points to remember:**
>
> - Proximal and distal control, heparin, flush, everting Prolene anastomosis.
> - Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
> - Mayo levels decide caval control and bypass.
> - IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
> - PC-RPLND: IVC may be ligated; aorta must be grafted.
> - Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
> - Bench surgery plus autotransplantation for complex renovascular lesions.

Also, from the condensed book (not in the paper answer):

- Suprarenal IVC ligation with a right tumour is acceptable if the LRV is tied caval to its tributaries, which act as collaterals.

KEY POINT

"Shortcuts" that become "longcuts": opening the IVC before all inflow is controlled; side-biting a level I; clips on the IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping the test clamp.

6 · Partial nephrectomy

*See the paper answer **Nephron-sparing surgery for renal masses: indications, approaches and recent advances** (Section 13, Ch 137).*

Also, from the condensed book (not in the paper answer):

- Hyperfiltration injury (focal segmental glomerulosclerosis, proteinuria) is rare unless nephron mass falls by >80%.
- Multifocality is ≈2% in clear cell and ≈10% in papillary RCC, so inspect the whole kidney with intraoperative ultrasound.
- PN is size-independent if it is safe; collecting-system invasion is a relative contraindication.

Question	Evidence
Warm ischaemia	Aim **<30 min**; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss and stay

KEY POINT

PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes. Hypothermia and mannitol are optional.

- Always **control the hilum first**, even if not clamping, so a clamp can go on at once.
- **Enucleation** follows the pseudocapsule plane; **resection** takes a 5–10 mm rim and is usually clamped.
- **Renorrhaphy:** the inner layer closes vessels and collecting system with absorbable sutures (permanent sutures become a stone nidus); the outer layer is capsular over a bolster.

7 · Complications & management

Injury	Management
SMA	Injured when mistaken for the left renal artery or by Cattell-Braasch traction. Potentially lethal, so call a vascular surgeon for primary repair or graft

Injury	Management
Inferior mesenteric artery or vein (IMA/IMV)	Ligation safe if the marginal artery of Drummond is patent
Superior mesenteric vein (SMV)	Ligation causes bowel ischaemia, splanchnic hypertension and hypotension. **Never ligate**; repair with 6-0 Prolene venorrhaphy. If transected, reanastomose or use autologous vein; leave the abdomen open
Duodenum	Haematoma narrowing the lumen: incise serosa and muscularis, drain. Tear: debride, 2-layer closure, omentum, drain, nasogastric tube
Pancreas	Glue and drain (check amylase); ductal injury: general surgeon; late collection: percutaneous drain, nil by mouth, octreotide
Pleura	Purse-string over a 14-Fr catheter, evacuate on a positive-pressure breath; large tear: chest tube

- In thrombectomy, air embolism is prevented by releasing the caudal clamp first. Use **no cell saver**, because it spreads tumour.
- A **urine leak** mostly appears within 72 h. If drain output is >50 mL/24 h, a drain creatinine above serum means urine; a persisting urinoma needs a double-J stent and Foley, and most resolve in 4–6 wk.
- A significant **pseudoaneurysm** (flank pain, haematuria or shock) needs **angiography** with **selective embolisation**, because CT angiography can miss it and cannot treat it.
- An **arteriovenous fistula** is observed, then coil-embolised, with surgery if complex.
- **Renal insufficiency** is usually mild and transient; exclude obstruction, drug toxicity and thrombosis.

142 Laparoscopic and Robotic Surgery of the Kidney

1 · Principles & patient selection

Fitness here means fitness for the gas.

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ; first laparoscopic partial nephrectomy (PN): Winfield, 1993.
- Minimally invasive surgery (MIS) gives less pain, blood loss and convalescence than open, with **equivalent long-term outcomes** and the same indications.
- **General anaesthesia is mandatory.** The CO_2 pneumoperitoneum impairs ventilation, venous return and cardiac output, so risk is high in chronic obstructive pulmonary disease (COPD) and cardiac disease.
- In severe COPD with hypercarbia, lower the pressure, then use helium, then convert.
- With the robot, **docking and undocking delay emergency access**, so the team must communicate closely.
- In obesity, **visceral adipose tissue**, not body mass index (BMI), predicts difficulty. Prior surgery is **not a contraindication**.

> **KEY POINT**
>
> The patient must tolerate the **pneumoperitoneum**, not the renal operation: cardiopulmonary reserve, not age or BMI, decides fitness.

2 · Access & port placement

Routes are transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted (HAL) and single-site (LESS).

	Transperitoneal	Retroperitoneal
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	Reached after colon reflection	Direct, especially the posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- **TP** uses a 30–45° modified flank position with 3–5 trocars.
- **Robotic** arms are spaced **8 cm (Si) / 6 cm (Xi)**, forming a **broad-based triangle** with the camera at the apex.
- **RP** uses the full flank position; the space is made by finger or balloon below the 12th rib in the posterior axillary line.

> **KEY POINT**
>
> RP pitfalls: a balloon inflated within muscle causes a flank hernia; too anterior injures the peritoneum or colon; too posterior makes the quadratus or psoas bleed. Bail-out: open the peritoneum and continue transperitoneally.

- **HAL** gives tactile feedback and helps novices, donors and scarred kidneys; a hand through the port can compress a bleeding vessel.
- **LESS and the single-port robot** improve cosmesis but lose triangulation, with unchanged outcomes.

3 · Nephrectomy — operative steps

ALGORITHM

1. Reflect colon; find ureter and gonadal vessels on the psoas.
2. Lift kidney to expose the hilum; artery, then vein.
3. Upper pole (adrenal spared or taken); ureter and lateral attachments; bag; extract.

- Keep the **lateral attachments** until late, or the kidney rolls over the hilum. On the right, dissect cold near the duodenum and do a **Kocher**.
- The **ureter** lies just behind the gonadal vessels and is a useful handle.
- **Hilum:** divide the **artery first, then the vein**, with an endo-GIA stapler. **Never leave clips in the staple line.**
- **En bloc hilar stapling** is safe (no arteriovenous fistula, AVF) and saves ~45 min.
- **Extraction:** intact in a bag, or morcellated in an impermeable **LapSac**, which keeps type and grade but loses size and stage.

4 · Simple vs radical nephrectomy

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple** nephrectomy is for benign nonfunctioning kidneys, such as chronic pyelonephritis, xanthogranulomatous pyelonephritis, obstruction, reflux or tuberculosis; after pyonephrosis or stones, use hand assistance or a subcapsular approach.
- **Radical nephrectomy (RN):** MIS equals open oncologically at 5 and 10 yr. In RCTs of RP versus TP, the only consistent RP advantage is **shorter operative time**.
- **Lymph node dissection (LND)** has staging value only, with no overall survival benefit; the AUA advises removing clinically positive nodes.
- Tumours >10 cm are feasible, though landmarks are lost; conversion is higher laparoscopic than robotic.
- Selected **inferior vena cava (IVC) thrombus** cases can be done by experts, with early arterial ligation that may retract the thrombus (Ch 141). Laparoscopic **cytoreductive** nephrectomy allows earlier systemic therapy.

KEY POINT

Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery.

5 · Partial nephrectomy

*See the paper answer **Nephron-sparing surgery for renal masses: indications, approaches and recent advances** (Section 13, Ch 137).*

SAQ Role of renal nephrometry in management of RCC.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2020-Summer Paper 1, Q3f

Definition

- **Nephrometry scores** grade the anatomical complexity of a renal mass on CT (computed tomography) or MRI (magnetic resonance imaging), to standardise RCC (renal cell carcinoma) reporting and plan treatment.

R.E.N.A.L. score (Kutikov and Uzzo, 2009)

Component	1 point	2 points	3 points
Radius (maximum diameter)	≤4 cm	>4 to <7 cm	≥7 cm
Exophytic/endophytic	≥50% exophytic	<50% exophytic	Entirely endophytic
Nearness to collecting system or sinus	≥7 mm	>4 to <7 mm	≤4 mm
Anterior/posterior	Suffix a, p or x (no points)		
Location to polar lines	Entirely above or below	Crosses a polar line	>50% across, crosses midline, or entirely between
- Suffix "h": touches main renal vessels. **Total 4–12:** low 4–6, moderate 7–9, high 10–12.			

PADUA (Preoperative Aspects and Dimensions Used for an Anatomical classification) score (Ficarra, 2009)

- Polar location, exophytic rate, renal rim (lateral or medial), renal sinus, collecting system and size, plus anterior/posterior face. **Total 6–14:** 6–7 low, 8–9 intermediate, ≥10 high risk.

C-index (centrality index)

- Distance from tumour centre to kidney centre ÷ tumour radius. **Lower value = more central, more complex.**

Role in management

Decision	Use
Partial vs radical nephrectomy	Feasibility and risk of PN (partial nephrectomy) vs RN (radical nephrectomy)
Approach	Open vs minimally invasive; hilar clamping
Predict outcomes	Warm ischaemia, blood loss, complications, conversion, collecting-system entry; urine leak (R and N components)
Function	Parenchymal loss, eGFR (estimated glomerular filtration rate) decline

Decision	Use
Ablation	Cryoablation complications rise with complexity
Pathology	Higher scores: malignancy, higher grade

- **AUA (American Urological Association) 2021:** with high oncological potential, prefer RN only if PN would be challenging even for experts, there is no CKD (chronic kidney disease) or proteinuria, and new baseline eGFR will likely exceed 45.

Limitations

- Interobserver variation; ignores patient factors and surgeon experience.

Also know: ABC (arterial based complexity), DAP (diameter–axial–polar) scores; **MAP (Mayo Adhesive Probability) score** for adherent perinephric fat.

Diagram to draw: kidney with polar lines and the five R.E.N.A.L. components.

> **Key points to remember:**
> - RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
> - PADUA 6–14; C-index lower = more central.
> - Predicts PN feasibility, ischaemia time and complications.
> - MAP score: adherent perinephric fat.

Also, from the condensed book (not in the paper answer):

- In cT1b/T2 tumours, PN gives better estimated GFR with equal oncology, but **tumours >5 cm upstaged to pT3a recur more after PN**.
- The collecting system is entered in about 60%, so always drain.

ALGORITHM

1. Intraoperative ultrasound (size, depth); open Gerota's away from the lesion; score the capsule.
2. Clamp the hilum (Satinsky en bloc or bulldogs); cold-scissor excision.
3. Close vessels and collecting system; renorrhaphy; unclamp; drain.

- **Trifecta:** negative margin, no major complication within 30 days, and **WIT ≤25 min**.
- **Enucleation** gives less blood and function loss but more positive margins, with the same survival. It suits familial, multifocal, solitary-kidney and severe chronic kidney disease (CKD) cases.
- **Renorrhaphy:** sliding-clip and barbed sutures shorten warm ischaemia time (WIT), and **single-layer** closure gives better function and fewer pseudoaneurysms.
- The **collecting system** is entered in **~60%**; repair it with absorbable suture and **always drain**.
- **Ischaemia:** the historical warm limit is **30 min**, but **preserved parenchymal volume predicts function better than WIT**. Use cold ischaemia for a solitary kidney, CKD or a long expected clamp time.

Hilar strategy	Best for	Verdict
Artery-only vs artery and vein	Any	Equivalent

Hilar strategy	Best for	Verdict
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (near-infrared fluorescence, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

> **KEY POINT**
>
> No clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped.

- Robotic PN has fewer conversions and shorter WIT than laparoscopic, and less blood loss than open.
- **Positive margins (0–7%)** matter only in high-risk tumours, so the answer is **surveillance, not reoperation**.
- **Port-site recurrence is ~0.1%** and follows spillage, not the gas.
- A **hilar** tumour touches the renal artery or vein; consider enucleation with superficial renorrhaphy bites.
- Salvage surgery after PN or ablation often ends in RN, because the planes are lost.

6 · Other renal procedures

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first proves the cyst causes the pain; excise the wall, biopsy, fulgurate the base
Laparoscopic renal biopsy	Failed percutaneous biopsy, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex
Nephropexy	Descent >5 cm or 2 vertebrae from supine to erect, with pain, reduced flow or obstruction	Usually right side, slim young women; 20% asymptomatic. Fix the capsule to psoas or quadratus fascia
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Suture the neck, obliterate the cavity with fat
Nephrolysis	Chyluria, loin pain–haematuria	Lateralise by cystoscopy; hilar lymphatic ligation with or without omental wrap
Ablation	≤3 cm, comorbid, solitary kidney, hereditary RCC	Percutaneous preferred. Cryo: two freeze–thaw cycles, iceball 0.5–1 cm beyond tumour. Radiofrequency ablation (RFA): 60–100°C (>105°C vaporises), 1 cm margin, no real-time imaging; more retreatment

7 · Complications & management

Dangerous complications happen out of view, so seek them.

*See the paper answer **Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy** (Section 1, Ch 18).*

Also, from the condensed book (not in the paper answer):

- The overall complication rate is **~15%** (PN > RN > simple), with a learning curve of about 20 nephrectomies.
- **In left RN, the superior mesenteric artery (SMA) can be mistaken for the renal artery**, leading to bowel necrosis.

ALGORITHM

1. Major bleed: locate; temporary control (grasper or pressure); inform the anaesthetist (blood, open set).
2. Clear the field: raise pneumoperitoneum or add a trocar; then definitive clip or suture.
3. If that fails, then convert and call a vascular surgeon.

- Vascular injury happens at **access** or at the hilum. **Never clip or suture blindly.**
- At the end of the case, **lower the insufflation pressure** to unmask venous bleeding.
- **Haematuria at ~day 12 after PN** suggests a pseudoaneurysm or AVF and is treated by selective embolisation.
- A **urine leak** after PN (0.8–10%) is drained percutaneously, with a stent if it persists.
- **Bowel injury** (<1%, mortality ~3%) is thermal in about half. It presents with **pain at a single trocar site**, distension, diarrhoea and a **normal or low white count**.
- Pancreatic injury is insidious, so inspect at the end. A diaphragm injury shows as a sudden rise in airway pressure and is closed with a purse-string, followed by a chest X-ray.
- Bowel injury is repaired laparoscopically if stable, or by laparotomy if unstable or with gross spillage.
- **Neuromuscular** injury (2.7%) includes flank rhabdomyolysis, treated by hydration.

> **KEY POINT**
>
> MIS complications are the open ones in altered form and timing: the lethal ones (access vessel injury, thermal bowel injury) happen out of view and present late or atypically.

8 · Laparoscopic live donor nephrectomy †

P4 P1 *Laparoscopic donor nephrectomy — left preferred (longer vein), donor keeps better kidney; gonadal–ureter packet; artery then vein; no Hem-o-lok on donor artery; mortality 3/10 000; lifetime ESRD risk*

- † Donors need lifelong annual blood pressure and renal function checks.

143 Focal Therapy for Renal Tumours

1 · Rationale & indications

Ablation suits patients whom surgery harms more than the tumour.

- Small renal cell carcinoma (RCC) rarely kills; chronic kidney disease (CKD) and comorbidity do (Ch 137).
- **Partial nephrectomy (PN) is the gold standard** for cT1a. Thermal ablation (TA) is the alternative: cryoablation (CA), radiofrequency ablation (RFA) or microwave ablation (MWA).
- **Indications:** high surgical risk, solitary kidney, bilateral tumours, or hereditary syndromes (von Hippel–Lindau, VHL) needing repeat treatment. Recovery is faster and renal function equal or better than PN.

> **KEY POINT**
>
> **Trap:** TA is never the default for a fit patient.

Guideline	Position on TA
American Urological Association (AUA) 2021	Consider for cT1a **≤3 cm**, especially comorbid; PN prioritised; percutaneous preferred
European Association of Urology (EAU) 2022	For frail/comorbid; TA not routinely **>3 cm**; CA not **>4 cm**
National Comprehensive Cancer Network (NCCN) 2022	**Only one allowing cT1b** (non-surgical candidates); may need multiple treatments
American Society of Clinical Oncology (ASCO) 2017	Option if complete ablation achievable

2 · Selection & work-up

The tumour stays in place, so biopsy gives the only histology.

ALGORITHM

1. Counsel: age, comorbidity, CKD risk, size and location.
2. Biopsy, then re-counsel with histology.
3. Choose PN, surveillance or percutaneous ablation.

- **The AUA advises biopsy of all**, ideally at a **separate earlier session**. Post-ablation, biopsy only a new enhancing lesion.
- Treat urinary infection first; colonised urine (ileal conduit) risks abscess.
- Complication predictors are the RENAL score (low 4–6, moderate 7–9, high 10–12), American Society of Anesthesiologists (ASA) score >3 (about 3×) and $(MC)^2$ (CA only). Day-case suits ≤3 cm, low complexity and no anticoagulation.

3 · Mechanisms

*See the paper answer **Cryotherapy for CA prostate** (Section 17, Ch 163).*

Also, from the condensed book (not in the paper answer):

- Cryo target is **≤−40 °C**; the iceball extends **5–10 mm beyond tumour**, pushing the −40 °C isotherm past the tumour edge.
- Heat: **≥60 °C** causes coagulative necrosis; ≥100 °C causes vaporisation and charring. Lethal: 45 °C for 60 min, 55 °C for 5 min, **70 °C for 1 min**.
- **Heat sink:** large vessels carry heat or cold away, so perivascular tumour survives. Counter with colder or longer freezes, more probes, or MWA.

4 · Choice of energy

- RFA electrodes vary: the cool-tip gives the largest zone, the umbrella the most spherical, and a single monopolar tip suits only ≤2 cm.

	Cryoablation	**RFA**	**MWA**
Physics	Argon cools (**Joule–Thomson**); helium warms	450–1200 kHz current causes ionic friction	915 MHz/2.45 GHz causes **dielectric hysteresis**
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	**Double freeze–thaw** (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	**Yes**	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	**Least**
Zone visible	**Iceball**	**No**; inferred from impedance and temperature	Gas bubbles on ultrasound (US)
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

> **KEY POINT**
>
> **Why RFA is self-limiting:** charring raises impedance and halts current (cap 105 °C, electrodes cooled). MWA radiates energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- In CA the second freeze enlarges necrosis. MWA for cT1 uses 45–65 W for 5–15 min.

5 · Technique

- **Percutaneous is standard.** Laparoscopic is kept for prior scarring or failed hydrodisplacement. General anaesthesia with jet ventilation reduces renal motion.

ALGORITHM

1. Give intravenous (IV) antibiotics.
2. Position prone or flank on the computed tomography (CT) gantry.
3. Place probes under image guidance; confirm on non-contrast CT.
4. Ablate, then contrast CT for completeness and haematoma.

- **Hydrodisplacement** pushes organs away (colon >1 cm) with saline or 5% dextrose (**D5W for RFA**, as saline conducts current). Use caution in cardiac failure.
- **Pyeloperfusion** (central tumours) irrigates a ureteral catheter at the **opposite temperature**: cold for RFA/MWA, warm for CA.
- **CA:** cryoprobes sit **1–2 cm apart** and come out only after full thaw. Bleeding rises with size and probe number (>4 cm). **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; overlap ablations for larger lesions.
- **MWA:** antennas go tangentially, never into the collecting system: 1 for ≤2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm.

6 · Follow-up, success & salvage

With no specimen, imaging alone shows success.

- **Success is complete loss of contrast enhancement** on CT/MRI.
- **Incomplete ablation** enhances on the first scan and is re-ablated. **Local recurrence** later enlarges or enhances and needs biopsy. **Enlargement after any modality is recurrence until proved otherwise.**
- CA zones shrink >50% in year 1; RFA zones barely shrink (**fibrotic halo**).
- **AUA:** CT/MRI with and without contrast plus chest imaging at **6, 12, 24, 36, 48, 60 months**, then shared decision. US is not routine unless a contrast-enhanced (CEUS) protocol exists.
- Salvage is re-ablation, PN, radical nephrectomy (RN), or stereotactic body radiotherapy (SBRT) if comorbid.

7 · Oncological outcomes vs PN

- There is no randomised trial. Local recurrence is commoner after TA but **equal once re-ablation is counted**.
- **Metastasis-free and cancer-specific survival are equal**; overall survival is lower in these older, comorbid patients.
- **Size predicts failure:** RFA does worse at ≥3 cm, and T1b worse than T1a. No energy is superior.

KEY POINT

Why comparisons are biased: benign and under-graded tumours flatter TA; occult pT3a penalises it; non-surgical candidates die of other causes first. Few RCC deaths occur, so trials need huge cohorts.

8 · Complications & management

- Major complications (Clavien ≥3) are **≤5%**, fewer than PN; percutaneous beats laparoscopic, and MWA is lowest.

Complication	Prevention	Management
Haemorrhage – commonest; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Ablate (RFA/MWA coagulate); otherwise selective embolisation
Urinoma, fistula, pelviureteric junction (PUJ) or ureteric stricture	Pyeloperfusion; CA for central tumours	Conservative or stent; drain a large urinoma
Bowel or other organ injury (tumour <1 cm away)	Hydrodisplacement, positioning	Fistula: stent, low-residue diet; diversion if persistent
Pneumothorax (access above 12th rib)	Recognise during procedure	Aspirate; chest tube rarely
Nerve injury / RFA pad burn	Pads level on posterior thighs	Neuralgia is self-limiting
Infection <1%, may be lethal; abscess at 1–6 months	Urine culture, prophylaxis	†Drain and antibiotics

9 · Newer modalities (investigational)

- **High-intensity focused ultrasound (HIFU)** ablates incompletely; **histotripsy** (non-thermal) has no clinical data yet.
- **SBRT** needs no anaesthesia but leaves viable tumour on biopsy; it may have a salvage role.
- **Irreversible electroporation** spares vessels but controls cancer poorly.

144 Treatment of Advanced Renal Cell Carcinoma

1 · Overview

In metastatic RCC, drug choice rests on risk group and mechanism.

LAQ Metastatic RCC: assessment and management

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Metastatic RCC (mRCC): assessment and management, including recent concepts" · P3 SAQ "Discuss the management of Metastatic Renal Cell Carcinoma."

Introduction

- About one-third of RCC (renal cell carcinoma) patients have metastases at diagnosis; 20–40% of localised tumours metastasise later.
- Five-year survival is about 12%, but median survival now approaches 3–5 years.
- Treatment moved from cytokines to VEGF (vascular endothelial growth factor)-targeted TKIs (tyrosine kinase inhibitors), and now to ICI (immune checkpoint inhibitor) combinations. Surgery has become selective.

Assessment (before treatment)

- **Clinical:** symptoms of metastases (cough, bone pain, neurological deficit), weight loss; paraneoplastic syndromes (hypercalcaemia, polycythaemia, hypertension, Stauffer syndrome); performance status (Karnofsky or ECOG [Eastern Cooperative Oncology Group]); comorbidity.
- **Laboratory:** haemoglobin, corrected calcium, neutrophils, platelets, LDH (lactate dehydrogenase), ALP (alkaline phosphatase), creatinine, liver tests.
- **Imaging:** contrast CT (computed tomography) of chest, abdomen and pelvis (or MRI, also for IVC [inferior vena cava] thrombus). Bone scan only with bone pain, raised ALP or poor performance status; brain MRI (magnetic resonance imaging) with neurological symptoms; that is, bone scan or brain MRI only when symptoms or tests suggest spread; PET not routine.
- **Histology:** nephrectomy specimen or **percutaneous biopsy** before systemic therapy. It separates clear cell from non-clear cell RCC and shows sarcomatoid change, which alters drug choice.

Prognostic risk groups (decide first-line therapy)

Adverse factor	MSKCC (Memorial Sloan Kettering), cytokine era	IMDC (International Metastatic RCC Database Consortium), TKI era
Karnofsky <80%	Yes	Yes
Haemoglobin below lower limit of normal	Yes	Yes
Corrected calcium raised	>10 mg/dL	> upper limit of normal (ULN)
LDH >1.5 × ULN	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to systemic therapy

Adverse factor	MSKCC (Memorial Sloan Kettering), cytokine era	IMDC (International Metastatic RCC Database Consortium), TKI era
Neutrophils, platelets > ULN	No	Yes (two factors)

- IMDC 6 factors; MSKCC model: 5 factors including LDH.
- Favourable 0 factors, intermediate 1–2, poor ≥3. IMDC median survival: 43.2, 22.5 and 7.8 months.

Management: principles

- Multidisciplinary care: systemic therapy is the backbone; surgery and radiotherapy are added selectively.

1. First-line systemic therapy, clear cell RCC (NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

IMDC risk	Preferred	Alternatives (if IO, immuno-oncology, not possible)
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib or pazopanib; active surveillance for slow, low-volume (indolent) disease
Intermediate/poor	The same three, **or nivolumab + ipilimumab**	Cabozantinib; sunitinib; pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (the TKI target). ICIs block PD-1 (programmed death-1) or CTLA-4 (cytotoxic T-lymphocyte antigen-4), releasing T-cell brakes and giving durable responses.
- **Trials (all versus sunitinib):** CheckMate 214 (nivolumab + ipilimumab: OS [overall survival] HR [hazard ratio] 0.63; overall survival benefit in intermediate/poor risk); KEYNOTE-426 (pembrolizumab + axitinib); CheckMate 9ER (nivolumab + cabozantinib); CLEAR (pembrolizumab + lenvatinib: PFS [progression-free survival] 23.9 vs 9.2 months). Benefit is independent of PD-L1 (programmed death ligand-1) status, so PD-L1 does not select patients.

2. Later lines: switch mechanism

- After an IO combination: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor). IO rechallenge is not advised (CONTACT-03 negative).
- After both IO and TKI: **belzutifan**; LITESPARK-005 showed better PFS than everolimus.
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR), both with OS gain.

3. Cytoreductive nephrectomy (CN): from routine to selective

- Cytokine era: SWOG 8949 and EORTC (European Organisation for Research and Treatment of Cancer) trials showed longer survival with CN + interferon-α.
- **CARMENA (2018):** sunitinib alone was non-inferior to CN + sunitinib in MSKCC intermediate/poor risk (median OS 18.4 vs 13.9 months).
- **SURTIME:** deferred CN after sunitinib was safe; OS favoured deferral.
- **Now:** immediate (upfront) CN only for fit patients with low metastatic burden who do not need immediate systemic therapy; **deferred CN** in responders. Avoid in poor risk, poor performance status or brain metastases.

- Palliative nephrectomy or angioembolisation for haematuria (bleeding) or pain.

4. Metastasectomy and local therapy (radiotherapy)

- Complete resection of solitary or few metastases gives 5-year survival of 35–50%. Better with complete resection, solitary lung lesion, disease-free interval >1 year, metachronous disease.
- **SBRT (stereotactic body radiotherapy)** or ablation for oligometastases; radiosurgery for brain metastases; palliative radiotherapy or surgery for cord compression and weight-bearing bones; palliative radiotherapy for bone pain.

5. Adjuvant therapy after complete resection

- **KEYNOTE-564:** pembrolizumab for 1 year after surgery in high-risk clear cell RCC, including **M1 NED** (no evidence of disease after complete metastasectomy): disease-free survival HR 0.63; FDA (Food and Drug Administration) approval 2021; a later report showed OS benefit.

6. Supportive care and special groups (non-clear cell RCC)

- Zoledronic acid or denosumab for bone metastases. Immune-related adverse events (colitis, hepatitis, hypophysitis, thyroiditis) need steroids and drug interruption.
- Non-clear cell RCC: clinical trials preferred; cabozantinib or sunitinib; gemcitabine + carboplatin (platinum) for collecting duct carcinoma.
- **Also know:** high-dose IL-2 (interleukin-2) gives durable complete response in 7–9% (another source: in a few percent); temsirolimus for poor risk; chemotherapy and hormones are ineffective.

Follow-up

- CT every 8–12 weeks on therapy (response by RECIST [Response Evaluation Criteria in Solid Tumours]); monitor thyroid, liver, blood pressure and cortisol.

Recent advances

- Belzutifan, also approved for VHL-associated tumours; neoadjuvant ICI trials; biomarkers to choose IO/IO or IO/TKI.

Algorithm

```
Metastatic RCC -> biopsy + IMDC risk -> IO/TKI (all risks) or nivo + ipi (intermediate/poor) -> deferred CN / metastasectomy / SBRT in responders -> progression: switch class (TKI, belzutifan)
```

Diagram to draw: the flowchart above (metastatic RCC, biopsy and IMDC risk, IO combination ± deferred CN in responders, metastasectomy or SBRT, switch drug class on progression).

> **Key points to remember:**
>
> - IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
> - IO/TKI combinations for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
> - CARMENA and SURTIME: no routine upfront CN.
> - Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
> - Belzutifan after IO and TKI.

Also, from the condensed book (not in the paper answer):

- About **1/3** present with metastases; a further **20–40%** of localised tumours metastasise later. mRCC 5-yr survival is about **12%**; 10-yr <5%.

Standard care is now **immune checkpoint (immuno-oncology, IO) combinations**.

2 · Prognostic models

- **MSKCC (Motzer)** came from the cytokine era; **IMDC (Heng)** came from first-line VEGF-targeted therapy.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal (LLN)	< LLN
Corrected calcium	**>10 mg/dL**	> upper limit of normal (ULN)
Lactate dehydrogenase (LDH)	**>1.5 × ULN**	—
Time factor	No prior nephrectomy	**<1 yr** from diagnosis to systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median overall survival (OS)	20 mo	43.2 mo
Intermediate (1–2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

> **KEY POINT**
>
> Three factors are shared. IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and short interval. The IMDC risk group **decides the first-line regimen**.

3 · Biology and drug targets

- The **VHL** protein (3p) tags **hypoxia-inducible factor (HIF)-α** for degradation; when VHL is lost, HIF drives VEGF, PDGF, TGF-α, GLUT-1 and erythropoietin. Drugs block the ligand (bevacizumab), receptors (tyrosine kinase inhibitors, TKIs), HIF translation (mTOR inhibitors) or HIF-2α (belzutifan). MET/AXL/FGFR escape explains broader later TKIs.
- **CTLA-4** and **PD-1/PD-L1** switch off T cells. Blocking them gives autoimmune toxicity, and benefit shows as OS and durability rather than progression-free survival (PFS).
- VEGF inhibitors are immunomodulatory, the rationale for IO plus TKI. Clear cell RCC (ccRCC) is chemoresistant.

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A; IV	Bleeding, proteinuria, hypertension (HTN)
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, hand–foot syndrome (HFS); rarely used
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off**	HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism**

Agent	Target / route	Signature toxicity / note
Pazopanib	VEGFR, PDGFR, KIT	**Hepatotoxicity (raised ALT)**; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, **MET, AXL**	Fatigue, diarrhoea, HFS; frequent dose reduction
Lenvatinib	VEGFR1–3, **FGFR1–4**, PDGFR-α, RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; **IV weekly**	Mucositis, hyperglycaemia, hyperlipidaemia, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	**HIF-2α inhibitor**; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis, hypophysitis, vitiligo
Avelumab	**Anti-PD-L1**; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · Surgery in metastatic disease

Surgery now serves systemic therapy; the question is who and when.

- **Cytoreductive nephrectomy (CN) rationale:** spontaneous regression of metastases is <1%, and a bulky primary suppresses T cells. CN helps only as a platform for systemic therapy.
- Fewer than 1/4 reach systemic therapy with ECOG 2, multiple sites, or brain or liver metastases.
- **CARMENA caveats:** about 45% were poor risk, about 25% of the CN arm were off-protocol, and it stopped early underpowered. It does **not** show harm in fit, low-burden, favourable patients.

ALGORITHM

1. Assess mRCC: histology, IMDC risk, PS and metastatic burden.
2. If fit, with low burden, favourable features and a resectable primary, then consider upfront CN followed by systemic therapy.
3. If intermediate/poor risk or high burden, then give systemic therapy first and offer deferred CN to responders.

KEY POINT

CN helps only patients fit enough, with disease slow enough, to reach systemic therapy; operating first on poor-risk patients delays it.

- **Metastasectomy** does best with complete resection, a solitary lesion, age <60, disease-free interval >1 yr, lung site, size <4 cm and metachronous disease, all markers of indolent biology.

- **Palliative** nephrectomy treats pain and paraneoplastic syndromes; try **angioembolisation for haematuria** first. Brain and spinal metastases get surgery and radiation.

5 · Systemic therapy for ccRCC

First line follows IMDC risk; later lines switch mechanism.

	IFN-α	High-dose IL-2
Response	10–15%	15–20%
Durable complete response (CR)	<2%	**7–9%**; >60% of CRs stay disease-free
Regimen	Schedule/route immaterial	**600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	**Vascular leak**: hypotension, multiorgan failure; early mortality 2–5%

- IL-2 is now kept for carefully selected fit patients with clear cell histology and prior nephrectomy; IFN-α is not used first line.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	**Active surveillance**, axitinib, high-dose IL-2
Intermediate / poor	Same three **+ ipilimumab + nivolumab**	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, **temsirolimus**

KEY POINT

Choosing: IO/TKI gives higher response and PFS and is better tolerated; IO/IO gives deeper durable responses (CR 9%) with autoimmune toxicity; never compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk (favourable did better on sunitinib in CheckMate 214). PD-L1 does not select. Use a TKI or temsirolimus if IO is contraindicated.

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	**Nivolumab or cabozantinib** (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, **belzutifan**, cabozantinib, lenvatinib + everolimus, **tivozanib**

- Tivozanib is used after ≥2 prior therapies. TKI + mTOR and nivolumab + sunitinib/pazopanib combinations failed.
- **Ipilimumab + nivolumab toxicity:** grade 3–4 events **46%**, high-dose steroids 35%, discontinuation 22%. Dose: nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4.
- † For immune-related events, hold IO and give steroids for grade ≥2 (prednisolone 1–2 mg/kg for grade 3–4), tapering over ≥4 weeks. Grade 4 means permanent stopping.
- TKI toxicity is managed by dose reduction or interruption; mTOR inhibitors need glucose, lipid and phosphate monitoring.

6 · Non-clear cell RCC

- Non-clear cell RCC (nccRCC) is 15–25%. No agent is proven, so offer a **trial to all**.
- Papillary tumours respond to sunitinib in only 5–10%; temsirolimus beat IFN in the nccRCC subgroup.
- **Hereditary papillary RCC (HPRC)** carries a germline activating **MET** mutation (chromosome 7) and gives type 1 papillary RCC. Sporadic papillary has **trisomy 7 in >2/3**. MET inhibitors (savolitinib, crizotinib) are active here.
- **Hereditary leiomyomatosis and RCC (HLRCC)**: fumarate hydratase loss gives type 2 papillary RCC. The cells depend on glycolysis, so **bevacizumab + erlotinib** restricts glucose (partial response about 2/3).
- **Collecting duct** carcinoma is treated with **gemcitabine + carboplatin**; **sarcomatoid** tumours with gemcitabine-based therapy.

7 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs **18.4** mo, hazard ratio (HR) 0.89	No upfront CN in intermediate/poor
SURTIME	Immediate vs deferred CN	OS 15.0 vs **32.4** mo, HR 0.57	Systemic therapy first is safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Later comparator
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; patients preferred pazopanib (PISCES)
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS better, OS not	Little used
Temsirolimus phase III	vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer	Lenvatinib + everolimus
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF	PFS HR 0.58; **OS HR 0.70**	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib
TIVO-3	Tivozanib vs sorafenib, after 2–3 lines	PFS HR 0.73; OS same	Third-line tivozanib

Trial	Design	Result	Take-home
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, **HR 0.73**; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS better; response 60 vs 40%	All risk groups
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS better; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, **HR 0.39**; response 71%	Largest PFS/response effect

★ Putting it together

A · Small renal mass

	Active surveillance (AS)	Ablation (Ch 143)	Partial nephrectomy (PN)	Radical nephrectomy (RN)
Best for	Frail, life expectancy <5 y; <2 cm or cystic	cT1a **≤3 cm**, comorbid; biopsy first	cT1a standard; imperative if solitary, bilateral, hereditary or CKD	Only if high complexity, no CKD/proteinuria, and new eGFR **>45**
Numbers	Growth 0.1–0.3 cm/y; treat if >3 cm or >5 mm/y	Cryo: 2 freeze–thaw cycles; RFA/MWA >60–70 °C	Trifecta: R0, no complication, warm ischaemia ≤25 min	New GFR ≈ GFR × contralateral split × 1.25
Weakness	Imaging adherence	More local recurrence; no pathology	Leak, bleeding	CKD 65% vs 20% after PN

B · Renal lesions

Lesion	Imaging	Genetics / IHC	Action
Simple cyst	≤20 HU, no enhancement	—	Nothing
Complex cyst	Bosniak v2019	—	≤2 cm surveil; >4 cm excise
Angiomyolipoma (AML)	Fat ≤−20 HU	TSC2/TSC1; HMB-45, Melan-A+	Embolise if >6 cm, aneurysm >5 mm or bleeding; everolimus in TSC
Oncocytoma	Central scar	KIT+, CK7 single cells; Birt–Hogg–Dubé	Surveil or PN (chromophobe mimic)
Clear cell RCC	Hypervascular	3p/VHL loss activates HIF-2α; CA-IX	PN/RN/ablation/AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple losses; diffuse CK7	As RCC; good prognosis
Lymphoma/metastasis	Infiltrative	—	**Biopsy; don't operate**
UTUC	Filling defect, ~46 HU, cytology+	MSI suggests Lynch	Risk-stratify, then KSM or RNU

C · UTUC: kidney-sparing (KSM) vs radical nephroureterectomy (RNU)

	KSM (endoscopic or segmental)	RNU + bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low grade, no invasion on CT urogram; or imperative (solitary, CKD, bilateral)	ANY of multifocal, >2 cm, high grade, hydronephrosis, prior cystectomy, variant

	KSM (endoscopic or segmental)	RNU + bladder cuff
Extras	Surveillance ureteroscopy; UGN-101 mitomycin gel	≤12 wk; template node dissection; single post-op intravesical chemo
Systemic	—	Neoadjuvant platinum if post-op CrCl <50; adjuvant (POUT) ≥pT2/N+

D · Open vs minimally invasive

- **Open (Ch 141)** is kept for IVC thrombus needing bypass, complex anatomy, prior surgery and trauma.
- **Laparoscopic/robotic (Ch 142)** is standard for organ-confined RN and most PN, with faster recovery and the same oncology.

E · Metastatic RCC

ALGORITHM

1. **Score IMDC (6 factors, Ch 144)**: 0 is favourable, 1–2 intermediate, ≥3 poor.
2. **First line**: IO + TKI (pembro + axitinib, nivo + cabozantinib, pembro + lenvatinib) at any risk; ipi + nivo only if intermediate/poor.
3. **Cytoreductive nephrectomy**: upfront only if fit and low burden; otherwise deferred (CARMENA, SURTIME).
4. **Later lines**: switch mechanism; belzutifan after IO + TKI.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2019-Summer Paper 3, Q1

Definition and biology

- **SRM (small renal mass):** an enhancing solid or complex cystic renal mass **≤4 cm (cT1a)**, usually found incidentally.
- **Benign share falls with size:** 20–30% of masses ≤4 cm and about 40% of masses ≤2 cm are benign. At ≥7 cm few are benign and about half are high grade.
- Male sex triples the risk of malignancy. Under 2% of small cancers metastasise.

Differential diagnosis

Group	Entities	Pointers
RCC (renal cell carcinoma)	**Clear cell** (commonest), papillary (10–15%), chromophobe (3–5%), cystic RCC, rare molecular subtypes	Clear cell enhances strongly; papillary enhances weakly
Other malignant	Urothelial carcinoma, lymphoma, metastasis, sarcoma, epithelioid AML (angiomyolipoma)	Infiltrative growth
Benign solid	Oncocytoma, fat-poor AML, papillary adenoma, metanephric adenoma	See below
Cystic	Haemorrhagic or complex cyst, cystic nephroma	Bosniak category
Inflammatory, vascular	Abscess, focal pyelonephritis, tuberculosis; aneurysm, AVM (arteriovenous malformation), infarct, pseudotumour	Clinical picture

- **Oncocytoma:** commonest benign enhancing SRM (up to half of benign SRMs). Central stellate scar and segmental enhancement inversion suggest it but are not specific. A core cannot separate it from chromophobe RCC, which has oncocytic areas. Sestamibi SPECT (single-photon emission computed tomography) is under evaluation.
- **Fat-poor AML:** no visible fat (fat below −20 HU [Hounsfield units] diagnoses classic AML); homogeneous, T2-dark, mimics RCC; biopsy HMB-45 and Melan-A positive.
- **Also know:** metanephric adenoma (hypovascular, polycythaemia, WT1 positive); papillary adenoma (≤15 mm).

Narrowing the differential

- **Multiphase CT (computed tomography):** enhancement over 15–20 HU means solid tissue. MRI (magnetic resonance imaging) helps below 2 cm.
- **RMB (renal mass biopsy)** if it will change management (AUA [American Urological Association]); seeding almost unreported; non-diagnostic about 14%.
- Then surveillance, ablation or **PN (partial nephrectomy)**, preferred for cT1a.

Haemostasis in LPN (laparoscopic partial nephrectomy)

Principle: a clear field for excision, then secure closure of the bed. Plan with R.E.N.A.L. score and 3D (three-dimensional) CT; intraoperative ultrasound.

1. Vascular control

Technique	Method	Comment
Main artery ± vein clamp	Bulldog clamps, or laparoscopic Satinsky en bloc	Artery-only equals artery–vein. Aim warm ischaemia ≤25 min (trifecta)
Selective or super-selective	Clamp the tumour's segmental branch; 3D CT, NIRF (near-infrared fluorescence)	Avoids global ischaemia; no functional gain in an RCT (randomised controlled trial)
Off-clamp ("zero ischaemia")	No clamping; small exophytic tumours	More blood loss; same long-term eGFR (estimated glomerular filtration rate)
Early unclamping	Release after the inner suture layer	Shorter ischaemia; possibly fewer pseudoaneurysms
Cold ischaemia	Ice slush, cold saline via ureteric catheter, cold arterial perfusion	Solitary kidney, CKD (chronic kidney disease), long repair; benefit unproven

- Function depends mainly on parenchyma preserved. Mannitol: no benefit in an RCT. **Also know:** parenchymal clamp for peripheral tumours.

2. Excision: score the capsule, cut with cold scissors, clip visible vessels. Enucleation along the pseudocapsule bleeds less.

3. Renorrhaphy (most reliable step)

- **Inner layer:** running 3-0 or 2-0 absorbable suture closes cut vessels and collecting system (opened in about 60%).
- **Outer layer, sliding-clip technique:** mattress sutures anchored by a locking clip; clips slid down to the capsule give tension without knots. Pledgets prevent capsular tears; oxidised cellulose bolster fills the defect.
- **Barbed suture:** holds tension without knots, so ischaemia is shorter.
- Single layer gave better function and fewer pseudoaneurysms than two layers.

4. Energy: argon beam or bipolar coagulation of the bed. Argon can destroy collecting-system sutures and adds gas, so vent the ports.

5. Haemostatic agents (adjuncts, no proven benefit in RCTs): gelatin–thrombin matrix; fibrin sealant; oxidised cellulose; PEG (polyethylene glycol) hydrogel; cyanoacrylate. Sutureless closure only for selected small tumours.

6. Final check: unclamp, inspect, **lower pneumoperitoneum** to unmask venous bleeding; drain.

Postoperative bleeding

- Early (hypotension, falling haemoglobin): resuscitate, CT, angioembolisation.
- **Delayed haematuria (about day 12):** pseudoaneurysm or arteriovenous fistula; selective angioembolisation preserves function. Endophytic and posterior tumours carry higher risk.

Diagram to draw: sliding-clip renorrhaphy over a bolster, with bulldog clamp on the artery.

Key points to remember:

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

SAQ Immunotherapy / immunomodulation in RCC

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Immunotherapy in (metastatic) RCC" (2024-Winter Paper II Q3e; Practice Set 4 Paper II Q3f) · P4 SAQ "Immuno modulation in Renal, cell carcinoma" (2025-Winter Paper 4 Q3a)

Assumed meaning: immunomodulation (immunotherapy) in renal cell carcinoma. Overall management of metastatic RCC: see 2024-Winter Paper 3, Q3b.

Rationale

- RCC (renal cell carcinoma) is immunogenic and chemoresistant: rare (<1%) spontaneous regression after nephrectomy, and tumour-infiltrating T cells.
- **Checkpoints:** PD-1 (programmed death-1) on CD8 T cells binds tumour PD-L1 and switches the T cell off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) on regulatory T cells damps priming. Blockade **releases the brake** on existing anti-tumour immunity.
- VEGF (vascular endothelial growth factor) is immunosuppressive, so TKIs (tyrosine kinase inhibitors) synergise with checkpoint inhibitors.

Cytokine era (non-specific stimulation)

	IFN-α (interferon-alpha)	High-dose IL-2 (interleukin-2)
Response	10–15%	15–20%
Durable complete response	<2%	**7–9%**
Toxicity	Flu-like	**Capillary leak**; needs intensive care
Role	Obsolete	Selected fit patients only

- High-dose IL-2 dose: 600,000–720,000 IU/kg intravenously every 8 hours, maximum 15 doses; over 60% of complete responders stay disease-free.

Checkpoint inhibitors: trials (all clear cell)

Trial	Regimen vs comparator	Result
CheckMate 025	Nivolumab vs everolimus, second line after TKI	OS (overall survival) 25.0 vs 19.6 months
CheckMate 214	Nivolumab + ipilimumab vs sunitinib	Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9%
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	Better PFS (progression-free survival) and OS
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	Better PFS and OS; PFS 16.6 vs 8.3 months
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2 months; response 71%

Trial	Regimen vs comparator	Result
JAVELIN	Avelumab + axitinib	No OS benefit

Treatment sequencing

- **First line:** an IO/TKI (immuno-oncology plus TKI) combination for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor risk only.
- **After IO:** switch to a VEGF-targeted TKI. IO rechallenge is not recommended: it failed in CONTACT-03 and TiNivo-2. Belzutifan after IO and TKI.

Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC, including resected metastases (M1 NED, no evidence of disease). DFS (disease-free survival) HR 0.68 (P4 and another source: HR 0.63); later OS benefit (HR 0.62).
- Atezolizumab and nivolumab-based adjuvant trials were negative.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; high-dose steroids (methylprednisolone 1–2 mg/kg); infliximab or mycophenolate if refractory

- Endocrinopathies: hormone replacement.

Biomarkers and also know

- **PD-L1 does not select patients.** Sarcomatoid tumours respond well. Gene signatures are investigational; antibiotics may reduce benefit.
- Non-clear cell data are scant.
- Vaccines and allogeneic transplant failed.

Diagram to draw

- T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

> **Key points to remember:**
> - IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
> - High-dose IL-2: durable complete response 7–9%.
> - CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
> - KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
> - irAE grade ≥2: hold drug and give steroids.

SAQ Biomarkers for patients with advanced RCC

Paper answer · P2

Type: SAQ (10 marks)
Source: 2023-Winter Paper 2, Q3d

Introduction

- A **prognostic** biomarker predicts outcome regardless of treatment; a **predictive** biomarker predicts benefit from a specific drug.
- In advanced RCC (renal cell carcinoma), only clinical risk models are used routinely. No molecular biomarker is yet validated to select treatment (EAU, European Association of Urology; NCCN, National Comprehensive Cancer Network).

Clinical and laboratory: risk models

Factor	MSKCC (Memorial Sloan-Kettering Cancer Center)	IMDC (International Metastatic RCC Database Consortium)
Karnofsky performance status <80%	Yes	Yes
Haemoglobin < lower limit of normal	Yes	Yes
Corrected calcium high	>10 mg/dL	> upper limit of normal
LDH (lactate dehydrogenase) >1.5 × normal	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to treatment
Neutrophils, platelets > normal	No	Yes

- IMDC: favourable 0 factors (median survival 43 months), intermediate 1–2 (22.5 months), poor ≥3 (7.8 months).
- IMDC group guides first-line choice (nivolumab–ipilimumab not for favourable risk).
- **Also know:** neutrophil-to-lymphocyte ratio, CRP (C-reactive protein), liver or bone metastases (poor prognosis).

Tissue biomarkers

- **PD-L1 (programmed death-ligand 1):** benefit of checkpoint combinations is **independent of PD-L1**, so it is not used to select patients.
- **Sarcomatoid features:** respond well to checkpoint inhibitors.
- **Gene expression signatures:** IMmotion150 showed high T-effector signature tumours do better with atezolizumab–bevacizumab; high angiogenic signature tumours do well with sunitinib. Investigational.
- **Genomic:** BAP1 loss and CDKN2A loss indicate poor prognosis; PBRM1 data conflicting. MET alterations in papillary RCC (cabozantinib); FH (fumarate hydratase) deficiency. TMB (tumour mutational burden) is low and not predictive.

Circulating and other biomarkers

- **KIM-1 (kidney injury molecule-1):** high serum levels predict recurrence and benefit from adjuvant immunotherapy (investigational).
- **ctDNA (circulating tumour DNA):** RCC sheds little DNA; methylation assays are promising.
- **Also know:** antibiotics during checkpoint therapy reduced survival (microbiome); sunitinib-induced hypertension is linked to better outcome.

Diagram to draw: IMDC risk groups with survival curves.

Key points to remember:

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

SAQ Tivozanib

Paper answer · P2

Type: SAQ (10 marks)
Source: 2024-Winter Paper 2, Q3b

Introduction

- Oral **VEGFR TKI (vascular endothelial growth factor receptor tyrosine kinase inhibitor)** used in advanced RCC (renal cell carcinoma).
- Targets **VEGFR-1, -2 and -3** (also c-KIT and PDGFR-β, platelet-derived growth factor receptor-β); highly potent and selective.

Mechanism and rationale

- In clear cell RCC, **VHL (von Hippel–Lindau) loss** lets HIF (hypoxia-inducible factor) accumulate, driving VEGF and angiogenesis.
- Tivozanib blocks VEGFR signalling, starving the tumour of new vessels.
- High selectivity means fewer off-target effects (less hand-foot skin reaction and diarrhoea than sorafenib).

Dose and pharmacology

- **1.34 mg orally once daily, 21 days of a 28-day cycle.**
- Long half-life (about 4–5 days); dose reduction for toxicity or hepatic impairment.

Key trials

Trial	Setting	Result
TIVO-1 (phase III)	0–1 prior therapy, versus sorafenib	PFS (progression-free survival) 11.9 vs 9.1 months; response 33% vs 23%; OS (overall survival) not better (28.8 vs 29.3 months)
TIVO-3 (phase III)	After 2–3 prior systemic therapies including a VEGFR TKI, versus sorafenib	PFS 5.6 vs 3.9 months (HR, hazard ratio, 0.73); response 18% vs 8%; OS not different (HR 0.97); benefit also after prior checkpoint inhibitor
TiNivo-2 (phase III)	After checkpoint inhibitor; tivozanib ± nivolumab	**Negative:** adding nivolumab did not improve PFS (no immunotherapy re-challenge)

Approval and place

- FDA (Food and Drug Administration) declined in 2013 because of the OS concern in TIVO-1; EMA (European Medicines Agency) approved first-line use in 2017.
- **FDA March 2021:** relapsed or refractory advanced RCC **after ≥2 prior systemic therapies**.
- NCCN (National Comprehensive Cancer Network): later-line option, mainly after immunotherapy combinations.

Adverse effects

- **Hypertension** (commonest; control before starting), dysphonia, fatigue, diarrhoea, proteinuria, hypothyroidism.
- Bleeding, arterial thrombosis, cardiac failure, **impaired wound healing** (stop before surgery), RPLS (reversible posterior leukoencephalopathy syndrome).
- **Monitoring:** blood pressure, thyroid tests, urine protein, liver tests.

Diagram to draw: VHL loss → HIF → VEGF → VEGFR, blocked by tivozanib.

> **Key points to remember:**
> - Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
> - TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
> - FDA 2021: after ≥2 prior systemic therapies.
> - TiNivo-2 negative; main toxicity hypertension.

SAQ Zero-ischaemia partial nephrectomy (laparoscopic and robotic)

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 4, Q3e (clamping strategies and laparoscopic partial nephrectomy haemostasis also in 2019-Summer Paper 3, Q1)

Definition and rationale

- **Zero-ischaemia PN (partial nephrectomy):** excision and renorrhaphy **without clamping the main renal artery**, so the kidney stays perfused (Gill and colleagues, 2011).
- **Why:** warm ischaemia injures nephrons (historical limit 30 minutes; trifecta ≤25). Most relevant in a **solitary kidney or CKD (chronic kidney disease)**.

Techniques

Technique	Method
Off-clamp	Hilum looped for rescue but not clamped; cold-scissor excision with strong suction
Anatomical (super-selective)	Microdissect the higher-order arterial branches entering the tumour; occlude only these with small bulldogs
Selective segmental clamping	Only the tumour's segment is ischaemic ("minimal ischaemia")
Parenchymal compression	Parenchymal clamp across a polar tumour
Enucleation	Pseudocapsule plane bleeds less, so suits off-clamp work

Aids

- **3D (three-dimensional) CT (computed tomography) reconstruction** maps segmental arteries; intraoperative Doppler ultrasound.
- **NIRF (near-infrared fluorescence) with ICG (indocyanine green)** on robotic consoles: after selective clamping, perfused kidney glows and the ischaemic tumour zone stays dark.
- **Robot:** wristed instruments and fast suturing ease off-clamp work.

- **Haemostasis:** clip or suture bed vessels as met; sliding-clip or barbed renorrhaphy; clamp if bleeding hides the margin.

Selection

- **Ideal:** small (cT1a), exophytic, peripheral tumour; experienced surgeon.
- **Caution:** endophytic, hilar or central tumours.

Evidence

- Off- versus on-clamp: **better early eGFR (estimated glomerular filtration rate), more blood loss**; **no difference beyond the early period**.
- **Robotic RCTs (randomised controlled trials, e.g. CLOCK):** no difference in eGFR or split function up to 2 years.
- RCT of super-selective clamping with NIRF: no 6-month gain; stopped for futility.
- **Concept:** parenchymal volume preserved, not ischaemia time within accepted limits, governs function.
- **Also know:** early unclamping; hypothermia for long ischaemia; mannitol no benefit.

Diagram to draw: hilum with a micro-bulldog on a tertiary branch; tumour zone dark on NIRF.

> **Key points to remember:**
> - Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
> - NIRF with ICG confirms the right branch.
> - RCTs: no long-term eGFR gain; more blood loss.
> - Parenchyma preserved matters more than clamp time.

Rapid Revision Sheet · Section 13

ADPKD (Adult polycystic kidney disease) (SAQ)

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

Acquired renal cystic disease (SAQ)

- ≥3 cysts per kidney in ESRD; >90% after 10 years.
- Uraemia causes it; transplant may regress cysts, not risk.
- RCC risk 3 times; image dialysis patients with haematuria.

Renal angiomyolipoma and its management (SAQ)

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

VHL gene and von Hippel–Lindau disease (SAQ)

- AD; 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
- Belzutifan (HIF-2α inhibitor) approved 2021.

Radiology for solid renal masses. (LAQ)

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

Nephron-sparing surgery for renal masses: indications, approaches and recent advances (LAQ)

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm, surveillance <2 cm.
- Imperative, relative, elective; elective PN is standard for cT1a; ablation or surveillance for frail patients with tumours <3 cm.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- AUA: RN only if complex + no CKD + new eGFR/GFR likely >45.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function.
- Enucleation for hereditary/multifocal; ablation for cT1a <3 cm after biopsy.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
- 3D models: randomised benefit.

- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

Upper tract urothelial carcinoma (LAQ)

- Tobacco, aristolochic acid (A:T → T:A in TP53) and Lynch (MSH2) are the key causes.
- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

Vascular grafts and vascular surgery in urology (LAQ)

- Proximal and distal control, heparin, flush, everting Prolene anastomosis.
- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- Mayo levels decide caval control and bypass.
- IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
- PC-RPLND: IVC may be ligated; aorta must be grafted.
- Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
- Bench surgery plus autotransplantation for complex renovascular lesions.

Role of renal nephrometry in management of RCC. (SAQ)

- RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
- PADUA 6–14; C-index lower = more central.
- Predicts PN feasibility, ischaemia time and complications.
- MAP score: adherent perinephric fat. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Metastatic RCC: assessment and management (LAQ)

- IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI (IO combinations) for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
- Belzutifan after IO and TKI.

Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy. (LAQ)

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.

- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

Immunotherapy / immunomodulation in RCC (SAQ)

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
- irAE grade ≥2: hold drug and give steroids.

Biomarkers for patients with advanced RCC (SAQ)

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

Tivozanib (SAQ)

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

Zero-ischaemia partial nephrectomy (laparoscopic and robotic) (SAQ)

- Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
- NIRF with ICG confirms the right branch.
- RCTs: no long-term eGFR gain; more blood loss.
- Parenchyma preserved matters more than clamp time. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 14

Adrenal & Retroperitoneum

145 Surgical and Radiographic Anatomy of the Adrenals

LAQ Phaeochromocytoma: adrenal anatomy, hormones and perioperative management

Paper answer · P1 · P2 · merged from 2

Asked as: P1 LAQ "Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma" (Practice Set 6 Paper 1 Q1) · P2 SAQ "Phaeochromocytoma: overview, pre-operative and intra-operative management" (asked as "Pre operative management of pheochromocytoma", Summer 2020 Paper II Q3c; "Intra op mgt of pheo", Practice Set 7 Paper II Q3c; "Pheochromocytoma", Practice Set 1 Paper II Q3d)

Assumed meaning: "arenal" = adrenal glands.

Adrenal anatomy

Gross

- Paired retroperitoneal glands on the upper renal poles (11th–12th ribs), inside Gerota fascia, separated from the kidney by thin connective tissue. Yellow-orange, 4–6 × 2–3 cm, about 5 g each.

Feature	Right	Left
Shape, level	Triangular; higher	Crescentic; lower (to L1)
Relations	Liver anteriorly, duodenum anteromedially, IVC (inferior vena cava) medially, diaphragmatic crus behind	Pancreas body and splenic vessels anteriorly, aorta medially, crus behind
Vein	**Short (<1 cm), into posterior IVC**: tears easily	**Longer, into left renal vein**, often joined by inferior phrenic vein

Blood supply

- **"Many arteries in, one vein out."** Superior adrenal (from inferior phrenic), middle (aorta) and inferior (renal artery) arteries break into 10–50 small branches.
- Cortical sinusoids drain into the medulla, so cortisol-rich blood reaches the chromaffin cells. **Cortisol induces PNMT (phenylethanolamine-N-methyltransferase)**, which is why the medulla makes adrenaline.

Nerves: preganglionic sympathetic fibres (T11–L2, via splanchnic nerves, no synapse in the coeliac ganglion) end directly on chromaffin cells. The medulla is a modified sympathetic ganglion.

Embryology: cortex from coelomic mesoderm (adrenogonadal ridge, weeks 3–4); medulla from neural crest. Hence adrenals stay orthotopic in renal agenesis, and adrenal rests occur along the path of gonadal descent.

Histology (zones from outside in: glomerulosa, fasciculata, reticularis = "salt, sugar, sex")

Zone	Share	Product	Main control
Glomerulosa	~15% of cortex	Aldosterone	Angiotensin II, potassium
Fasciculata	~80%	Cortisol (up to 20 mg/day)	ACTH (adrenocorticotropic hormone)
Reticularis	5–7%	DHEA (dehydroepiandrosterone), DHEA-S, androstenedione	ACTH
Medulla	~10% of gland	Adrenaline 80%, noradrenaline 19%, dopamine 1%	Sympathetic nerves

Steps in adrenal hormone production

1. Steroids (cortex)

- Cholesterol (mainly from LDL [low-density lipoprotein]) is carried into mitochondria by **StAR (steroidogenic acute regulatory protein)**: the rate-limiting step, driven by ACTH.
- **Side-chain cleavage enzyme (CYP11A1)** converts it to **pregnenolone**. The enzymes present in each zone then decide the product: `Pregnenolone -3βHSD-> Progesterone -21OH-> 11-deoxycorticosterone -11βOH-> Corticosterone -aldosterone synthase (CYP11B2)-> ALDOSTERONE (glomerulosa only) Pregnenolone -17αOH-> 17-OH pregnenolone/progesterone -21OH-> 11-deoxycortisol -11βOH-> CORTISOL (fasciculata) 17-OH pregnenolone -17,20-lyase-> DHEA -> androstenedione (reticularis)`
- 3βHSD (3β-hydroxysteroid dehydrogenase). Steroids are not stored; they diffuse out and act on intracellular receptors.
- **Regulation:** CRH (corticotropin-releasing hormone) drives ACTH, which drives cortisol, with negative feedback and a morning peak; RAAS (renin–angiotensin–aldosterone system) and potassium control aldosterone.
- **Clinical link:** 21-hydroxylase deficiency is the commonest cause of CAH (congenital adrenal hyperplasia).

2. Catecholamines (medulla) `Tyrosine -tyrosine hydroxylase (rate-limiting)-> L-DOPA -decarboxylase-> Dopamine -dopamine β-hydroxylase-> Noradrenaline -PNMT (cortisol-induced)-> Adrenaline`

- Stored in vesicles, released by exocytosis. **COMT (catechol-O-methyltransferase)** inside chromaffin cells forms **metanephrine and normetanephrine** continuously, which is why metanephrines are the best test. MAO (monoamine oxidase) and the liver form VMA (vanillylmandelic acid).

Phaeochromocytoma: definition and epidemiology

- Catecholamine-secreting tumour of adrenal medullary chromaffin cells; **paraganglioma** is the extra-adrenal form (organ of Zuckerkandl, bladder).
- 1–2 per 100,000 a year; about 5% of adrenal incidentalomas.
- **"Rule of 10" is outdated:** extra-adrenal up to 25%, **hereditary about one-third**, bilateral commoner in hereditary disease.

Genetics

- **MEN2 (multiple endocrine neoplasia type 2)**, RET: about 50% risk, adrenal, low malignancy.
- **VHL (von Hippel–Lindau)**: normetanephrine only.
- **NF1 (neurofibromatosis type 1)**.
- **SDHB (succinate dehydrogenase subunit B):** extra-adrenal, **30–50% metastatic**. (SDHB/SDHD = succinate dehydrogenase subunits.)
- Offer genetic testing to all: about one-third hereditary (MEN2, VHL, NF1, SDHB/SDHD).

Clinical features

- Triad: **headache, sweating, palpitations**. Sustained hypertension is commoner than paroxysmal; pallor, hyperglycaemia; >20% asymptomatic. Epinephrine-dominant tumours can cause hypotensive episodes.

Why preparation is needed

- Handling the tumour releases catecholamines: hypertensive crisis, arrhythmia, MI (myocardial infarction), stroke. After vein ligation, sudden withdrawal causes hypotension and hypoglycaemia.
- Phaeochromocytoma surgery is a **two-phase haemodynamic problem**: catecholamine **surge** while the tumour is handled, then catecholamine **withdrawal** once the vein is tied.
- Surges occur at intubation, positioning, pneumoperitoneum and tumour manipulation.
- Mortality fell from up to 50% to <3% with blockade, now advised for **all** proven tumours, even normotensive.

Pre-operative step 1: confirm, localise, test genes

- **Biochemistry first:** plasma free or 24-hour urinary fractionated **metanephrines**. Tumours convert catecholamines continuously, so metanephrines are more sensitive than episodic catecholamines. Chromogranin A; clonidine suppression for borderline results.
- **Imaging:** CT (computed tomography): >10 HU (Hounsfield units) unenhanced, no rapid washout; MRI (magnetic resonance imaging).
- **Functional imaging** if extra-adrenal or hereditary: ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); FDG (fluorodeoxyglucose) PET (best for SDHB); ^{123}I-MIBG (best in MEN2).
- **Never biopsy** (hypertensive crisis). Iodinated contrast is safe.
- Genetic testing (about one-third hereditary).

Pre-operative step 2: cardiac assessment

- ECG (electrocardiogram) and echocardiography for catecholamine cardiomyopathy; joint plan with the anaesthetist.

Pre-operative step 3: drug blockade (7–14 days)

Drug	Action	Points
Phenoxybenzamine (first line)	Irreversible non-selective α-blocker	10 mg twice daily, increased every 2–3 days (titrated); about 1 mg/kg total
Doxazosin, prazosin	Selective α1-blockers	Fewer side effects

Drug	Action	Points
Calcium channel blocker	Smooth muscle relaxation	Alternative strategy or add-on
β1-blocker (atenolol, metoprolol)	Controls tachycardia	**Only after adequate α-blockade**
Metyrosine	Blocks tyrosine hydroxylase	Add-on; sedation, extrapyramidal effects

- **α-blockade for all**, 7–14 days.
- **Never β before α:** β-blockade removes β2 vasodilatation, leaving α-vasoconstriction unopposed, which causes crisis.
- **Target:** seated BP (blood pressure) 120–130/80 mmHg; a mild postural drop (systolic >90 mmHg) shows adequate blockade.

Pre-operative step 4: volume expansion

- High-salt diet and fluids once blockade starts; 1–2 L IV (intravenous) fluid the night before. This prevents post-excision hypotension.

Pre-operative step 5: day of surgery

- Last phenoxybenzamine dose the night before surgery; omit the morning dose.

Intra-operative management

Prerequisites

- α-blockade 7–14 days (phenoxybenzamine or doxazosin), β-blocker only after α, volume expansion; ECG and echocardiography; joint plan with the anaesthetist.

Monitoring

- **Arterial line before induction**, central venous line, ECG, urine output, frequent glucose; cardiac output or transoesophageal echo if cardiomyopathy.

Anaesthesia

- Deep, smooth induction (propofol, opioid such as fentanyl or remifentanil); attenuate the intubation response.
- **Avoid triggers:** histamine-releasing drugs (morphine, atracurium), ketamine, ephedrine, metoclopramide, droperidol, desflurane.
- Sevoflurane or isoflurane; vecuronium or rocuronium; magnesium sulphate as an adjunct (reduces catecholamine release and arrhythmia).

Haemodynamic control

Event	Action
Hypertensive surge	**Pause manipulation**; nitroprusside, nicardipine or clevidipine, nitroglycerin, phentolamine
Tachyarrhythmia	**Esmolol**, lidocaine; amiodarone if needed
Hypotension after vein ligation	**Aggressive volume**, then α-agonist (phenylephrine, noradrenaline); vasopressin if refractory
Hypoglycaemia (insulin rebound)	Dextrose infusion, glucose checks

Surgical principles

- **Laparoscopic adrenalectomy** (transperitoneal or posterior retroperitoneoscopic) is standard; open for large or invasive tumours.
- **Minimal handling**; "no-touch" dissection away from the tumour.
- **Early adrenal vein ligation** where possible, after warning the anaesthetist.
- Remove intact in a bag (avoid capsule rupture and seeding).
- **Cortical-sparing** (partial) adrenalectomy in MEN2, VHL and bilateral tumours.
- Steroid cover if bilateral adrenalectomy.

Post-operative management

- HDU (high dependency unit) overnight: blood pressure, heart rate, glucose.
- Fluids ± vasopressors for **hypotension**.
- Glucose monitoring for **rebound hypoglycaemia** (insulin no longer suppressed).
- Steroid cover after bilateral adrenalectomy.

Follow-up and metastatic disease

- Metanephrines after surgery, then **yearly for life** (recurrence up to 16%).
- Malignancy is defined only by **metastases**. Treat with metastasectomy, ^{131}I-MIBG, CVD (cyclophosphamide, vincristine, dacarbazine).

Diagrams to draw

- Posterior view of both adrenals with the three arteries and the two veins.
- Flowchart: metanephrines, imaging, α-blockade, β-blockade, salt and fluids, surgery.
- Intra-operative BP curve: surges during handling, fall after vein ligation, with drugs marked.
- Catecholamine pathway to metanephrines (COMT).

> **Key points to remember:**
> - Right vein short into IVC; left vein into left renal vein.
> - StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
> - Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
> - Metanephrines first; never biopsy.
> - α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
> - Volume expansion prevents hypotension; watch glucose.
> - Arterial line before induction; avoid histamine releasers and ephedrine.
> - Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
> - After vein ligation: fluids, vasopressors, glucose.
> - One-third hereditary; SDHB most often metastatic.

Also, from the condensed book (not in the paper answer):

- Neonatal gland weight and adrenal rests (§1).

1 · Embryology

- The cortex and gonad share the adrenogonadal ridge, which explains adrenal rests; neural crest forms the medulla.
- At birth the gland is **2×** adult weight. The **fetal zone involutes in year 1**.
- In **renal agenesis** the adrenal is **discoid**.

- **Adrenal rests** occur in ≈1% of adults and in 7.5–15% of neonates along gonadal descent; in congenital adrenal hyperplasia (CAH) they form **testicular masses**.

2 · Zones and hormones

Cortex is ≈90% and medulla ≈10%. †Outside inwards: **GFR = salt, sugar, sex**.

- Glomerulosa ~15% makes aldosterone; fasciculata **~80%** makes cortisol (lipid-laden "clear cells"); reticularis 5–7% makes DHEAS; the medulla (~10%) holds chromaffin cells.
- The medulla is a **modified sympathetic ganglion** of axonless postganglionic neurones.

3 · Relations and blood supply

- Each gland weighs ≈5 g. The right is **retrocaval**, directly above the upper pole; the left lies medial to the upper pole.
- **Exposure**: on the right, liver above, inferior vena cava (IVC) medial, duodenum and kidney below. On the left, pancreas and spleen above, splenic flexure medial.
- Three arteries enter via many twigs, but **one vein drains**. Cortical blood drains into the medulla, so cortisol drives **epinephrine synthesis**.
- The **right vein** is short and enters the **posterior IVC**, so it can avulse. The **left** enters the left renal vein, joined by the inferior phrenic.

> **KEY POINT**
>
> The right side is riskier: high, retrocaval, with a short vein whose avulsion tears the cava.

5 · Imaging appearance

- **CT** shows an inverted **V/Y** with limbs of 3–6 mm. Unenhanced **≤10 HU** means lipid-rich.
- **MRI** has the best contrast; on **T2 the gland resembles fat**. Ultrasound separates solid from cystic.
- Adrenal venous sampling (AVS) is the only **functional** study; cavography shows intraluminal tumour.

146 Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

1 · Applied anatomy and physiology

*See the paper answer **Phaeochromocytoma: adrenal anatomy, hormones and perioperative management** (Section 14, Ch 145).*

Also, from the condensed book (not in the paper answer):

- Anomalous venous drainage occurs in up to **9%**, which matters for adrenal vein sampling (AVS).
- **Pituitary failure or exogenous steroids spare aldosterone**, because the glomerulosa obeys angiotensin II and potassium; the inner zones atrophy without ACTH.

2 · Evaluation of an adrenal mass – principles

KEY POINT

Two questions: **(1) Is it malignant?** (imaging). **(2) Is it hormonally active?** (biochemistry). Yes to either means treat, usually by resection. Imaging cannot identify functional adenomas, so biochemistry is never omitted.

Category	Indications for adrenalectomy (Box 146.1)
Functional	Autonomous cortisol secretion; phaeochromocytoma; lateralised aldosterone excess
Oncological	>4 cm (except myelolipoma); suspicious imaging (lipid-poor, heterogeneous, irregular, infiltrating); growth >1 cm on follow-up; isolated metastasis (multidisciplinary)
Myelolipoma	Very large and/or symptomatic
At nephrectomy for renal cell carcinoma (RCC)	Adrenal abnormal or not visualised; thrombus to adrenal vein level
Bilateral	Failed pituitary surgery for Cushing disease; selected ectopic ACTH; AIMAH; PPNAD (§7)

3 · Imaging

LAQ Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management" (asked as "Diagnosis and management for Adrenal Carcinoma", Summer 2018 Paper IV Q2 LAQ; "Approach to evaluation of adrenal mass/s", Summer 2022 Paper IV Q3f SAQ) · P2 SAQ "Algorithm for management of an incidentaloma" (Practice Set 3 Paper 2 Q3d)

Assumed meaning (P2): adrenal incidentaloma.

Definition and epidemiology

- **Adrenal incidentaloma:** mass **≥1 cm** found on imaging done for another reason.

- Two questions: **Is it malignant? (imaging)** and **Is it functioning? (biochemistry).**
- ACC (adrenocortical carcinoma): rare (0.7–2 per million per year), aggressive cortical tumour. **Bimodal** (children <10 years; adults 40–50 years); slight female excess.
- **ACC syndromes:** Li–Fraumeni (TP53), Beckwith–Wiedemann (IGF-2, insulin-like growth factor 2), Lynch, MEN-1 (multiple endocrine neoplasia 1), Carney complex, FAP (familial adenomatous polyposis). Surgery is the only cure.

Differential diagnosis of an adrenal mass

- Non-functioning adenoma (commonest); cortisol-secreting adenoma 5.3%; phaeochromocytoma 5.1%; aldosteronoma 1%; ACC 4.7%; metastasis 2.5%; myelolipoma, cyst, haemorrhage.
- **Bilateral:** metastases, lymphoma, **tuberculosis** and histoplasmosis, haemorrhage, CAH (congenital adrenal hyperplasia), macronodular hyperplasia, bilateral phaeochromocytoma (MEN-2 [multiple endocrine neoplasia type 2], VHL [von Hippel–Lindau]).

Clinical features of ACC

- **Functional in 50–79% of adults** (90% of children):
- Cortisol (commonest): rapidly progressive Cushing syndrome, often with virilisation.
- Androgens: virilisation in women. **Oestrogen in a man (feminisation) suggests carcinoma.**
- Aldosterone: rare.
- Non-functional: pain, fullness, weight loss, or incidental.

Incidentaloma algorithm

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Adrenal mass ≥1 cm ├─ IMAGING: unenhanced CT (computed tomography) │ ≤10 HU (Hounsfield units), homogeneous → benign (lipid-rich) adenoma │ (or MRI chemical-shift signal drop-out → lipid-rich adenoma) │ >10 HU → washout CT (15 min) or chemical-shift MRI │ APW >60% or RPW >40% → adenoma │ indeterminate → MDT (multidisciplinary team): repeat imaging 6-12 months, FDG-PET or surgery │ Macroscopic fat → myelolipoma ├─ HORMONES (all patients) │ 1-mg overnight DST (dexamethasone suppression test) │ plasma or urine metanephrines │ ARR (aldosterone-renin ratio) if hypertension or low potassium │ sex steroids only if ACC suspected └─ DECISION functioning, suspicious, >4 cm, or growing → adrenalectomy benign, non-functioning → discharge
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Hormonal work-up (all lesions; ESE–ENSAT, European Society of Endocrinology–European Network for the Study of Adrenal Tumours)

Axis / hormone	Tests
Glucocorticoid (cortisol)	**1-mg overnight DST** (or 24-h urine or late-night salivary cortisol): ≤1.8 µg/dL (50 nmol/L) excludes; 1.8–5.0 µg/dL = possible autonomous secretion (**MACS**, mild autonomous cortisol secretion); >5.0 µg/dL autonomous; ACTH (adrenocorticotropic hormone)
Sex steroids and precursors	DHEA-S (dehydroepiandrosterone sulphate), androstenedione, testosterone, 17-OH-progesterone, 11-deoxycortisol; oestradiol in men and postmenopausal women (only if ACC suspected)
Mineralocorticoid (aldosterone)	**ARR** (aldosterone-to-renin ratio) if hypertension or hypokalaemia

Axis / hormone	Tests
Catecholamines	**Plasma free or 24-h urinary fractionated metanephrines**: must exclude phaeochromocytoma
Bilateral	17-hydroxyprogesterone (CAH); exclude adrenal insufficiency

- Why: confirms cortical origin, plans steroid cover, gives **tumour markers** for follow-up.
- **Never biopsy or operate before excluding phaeochromocytoma.** Biopsy only for suspected metastasis in known cancer.

Imaging

- Abbreviations: CT, HU, MRI (chemical-shift magnetic resonance imaging), MDT, FDG-PET (fluorodeoxyglucose positron emission tomography).
- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.
- Malignant features: heterogeneous, irregular, invasion, >4 cm. Macroscopic fat = myelolipoma.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5% (another source: carcinoma in ~2% <4 cm, ~6% at 4–6 cm, ~25% >6 cm). Benign imaging overrides size.
- **ACC size:** mean 10–12 cm; >90% exceed 5 cm.
- **CT in ACC:** irregular, heterogeneous, necrosis, calcification; **unenhanced >10 HU** (mean ~39 HU vs ~8 HU in adenoma); **slow washout** (absolute <60%, relative <40% at 15 min). Growth ~1.9 cm/year.
- **MRI in ACC:** no chemical-shift signal loss; best for IVC (inferior vena cava) thrombus and invasion.
- **FDG-PET/CT:** ACC is avid, but PET cannot separate ACC from metastasis.
- **Staging:** CT chest, abdomen, pelvis; bone or brain only if symptomatic.
- **No biopsy** if ACC is suspected (seeding, poor accuracy); only for unresectable or metastatic disease, after excluding phaeochromocytoma.

Management of incidentaloma (ESE–ENSAT 2023)

- **Surgery:** cortisol excess, phaeochromocytoma (after alpha-blockade), aldosteronoma (after AVS, adrenal vein sampling), malignant features or suspicious imaging, >4 cm in fit patients, significant growth (≥5 mm and >20%). Laparoscopic for benign; open for suspected ACC.
- **MACS:** screen for hypertension, diabetes and osteoporosis; surgery individualised.
- **Follow-up:** homogeneous **≤10 HU, non-functioning: benign, no further imaging whatever the size**; metanephrines may be omitted. Indeterminate non-resected: re-image at 6–12 months; growth >20% and ≥5 mm is concerning.
- **Bilateral:** metastases, lymphoma, tuberculosis, haemorrhage, CAH (17-hydroxyprogesterone); exclude adrenal insufficiency.

ACC pathology: Weiss score (≥3 of 9 = malignant)

Domain	Criteria
Nuclear / cytology	High nuclear grade; >5 mitoses per 50 high-power fields; atypical mitoses
Architecture	≤25% clear cells; diffuse architecture; necrosis
Invasion	Venous; sinusoidal; capsular

- SF-1 (steroidogenic factor 1) confirms cortical origin. **Ki-67** and mitotic rate grade the tumour (low vs high grade).

ENSAT staging

Stage	Definition	5-year disease-specific survival (ENSAT 2009)
I	T1 (≤5 cm), N0 M0	~80%
II	T2 (>5 cm), N0 M0	~60%
III	T3 (periadrenal fat) or T4 (adjacent organ, venous thrombus) or N1	~50%
IV	Distant metastasis only	~13%

- ENSAT differs from older UICC (Union for International Cancer Control) staging by moving T4 and N1 without metastasis into stage III.

Management of ACC

1. Surgery (stages I–III)

- **Open transperitoneal adrenalectomy** for suspected ACC, especially >6 cm or invasive. Laparoscopy risks capsule rupture and peritoneal carcinomatosis; only for small organ-confined tumours by experts.
- **R0, intact capsule:** rupture gives dismal results. **En bloc** removal of involved kidney, liver, spleen, pancreas or IVC; IVC thrombectomy (bypass if needed).
- **Locoregional lymphadenectomy** (periadrenal fat, renal hilum, para-aortic or paracaval) improves staging and may reduce recurrence.
- High-volume surgeon (ESE–ENSAT: >6, preferably ≥20 adrenalectomies a year). Hydrocortisone cover if cortisol-secreting.

2. Adjuvant therapy

- **Mitotane** (adrenolytic DDT [dichlorodiphenyltrichloroethane] derivative) for **high risk**: stage III, R1/Rx, or Ki-67 >10% (ESE–ENSAT 2018), for at least 2 years.
- Target blood level **14–20 mg/L**; needs **glucocorticoid replacement** (it destroys the cortex and raises cortisol-binding globulin).
- Toxicity: gastrointestinal, neurological, hepatic, hypothyroidism, gynaecomastia.
- **ADIUVO** (RCT, randomised controlled trial; R0, stage I–III, Ki-67 ≤10%): no recurrence-free survival benefit, so observe low-risk patients.
- **Adjuvant radiotherapy** (tumour bed): R1 or Rx, or stage III. It cuts local recurrence but not overall survival.

3. Advanced or metastatic (stage IV)

- **EDP-M** (etoposide, doxorubicin, cisplatin + mitotane) is first line. **FIRM-ACT** (2012): versus streptozotocin–mitotane, response 23% vs 9% and PFS (progression-free survival) 5.0 vs 2.1 months; overall survival not significantly better.
- Mitotane alone for low-volume, slow disease.
- **Surgery:** resect oligometastases or late recurrence; debulk only if >90% removable.
- **Hormone control:** metyrapone, ketoconazole, mifepristone, etomidate. Radiotherapy for bone and brain metastases.

Follow-up of ACC

- CT chest, abdomen, pelvis and baseline-raised hormones **every 3 months for 2 years**, then 6-monthly to 5 years.

Prognosis of ACC

- Overall 5-year survival **20–47%**. Bad factors: stage, R1, high Ki-67 or mitoses, cortisol excess, age, size >12 cm.

Recent advances

- Molecular classes (CIMP, CpG island methylator phenotype) predict prognosis.
- Checkpoint inhibitors give modest responses; IGF-1R (IGF-1 receptor) inhibitor linsitinib failed (GALACCTIC).
- ADIUVO-2 tests mitotane ± platinum–etoposide in high risk.

Diagrams to draw

- The incidentaloma imaging and biochemistry algorithm above.
- ACC algorithm: 1. Adrenal mass: hormones + CT (HU, washout). 2. Suspected ACC: open R0 en bloc resection + LND (lymph node dissection). 3. Risk-based adjuvant mitotane or radiotherapy.

Key points to remember:

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST: >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

Also, from the condensed book (not in the paper answer):

- Up to **7%** of metastases are <10 HU, so every lesion still needs biochemistry; RCC and hepatoma metastases can also wash out.
- **Unenhanced CT**: **≤10 Hounsfield units (HU)** means a lipid-rich adenoma, which covers >70% of adenomas. Chemical-shift MRI drop-out is about as accurate.
- **CT washout** is the gold standard for lipid-poor lesions, with phases at 0, 60 s and 15 min. It is an adenoma if the absolute percentage washout (APW) is **>60%** or the relative washout (RPW) **>40%**.

ALGORITHM

1. Unenhanced CT: if ≤10 HU or MRI drop-out, then adenoma.
2. If >10 HU, then do a 15-min washout CT.
3. If APW >60% / RPW >40%, then it is an adenoma; if below, manage as indeterminate.

Size	Malignancy (large unselected series)	Surgical series
<4 cm	0.1%	≈2–5% adrenocortical carcinoma (ACC)
4–6 cm	2.4%	≈6–10%
>6 cm	19.5%	≈25%

- **4 cm** is the consensus threshold (sensitive, poorly specific); 6 cm is more specific.
- **Benign imaging overrides size**: no <10 HU lesion was malignant in the largest series.
- Heterogeneity, irregular margins and invasion are suspicious. **Macroscopic fat** (<−30 HU) means myelolipoma. **Calcification is not reassuring**, as it occurs in up to 30% of ACC.
- Classic follow-up re-images at 6 and 12 months; growth of ≥1 cm means consider resection.

Functional imaging

SAQ MIBG scan

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 2, Q3c

Principle

- MIBG (meta-iodobenzylguanidine) is a guanethidine analogue that resembles **noradrenaline**.
- It enters chromaffin and sympathetic cells through the **noradrenaline transporter** and is stored in **adrenergic secretory vesicles**. So it maps catecholamine-handling tissue: function, not just anatomy.

Isotopes

	^{123}I-MIBG	^{131}I-MIBG
Use	**Diagnostic** (preferred)	Older diagnostic; mainly **therapeutic**
Image quality	Better; allows SPECT-CT (single-photon emission computed tomography)	Poorer
Emission	Gamma	Beta (therapy) plus gamma

Technique

- **Thyroid blockade** with potassium iodide before and after injection (free iodine).
- **Stop interfering drugs:** labetalol, tricyclic antidepressants, sympathomimetics, reserpine, some calcium channel blockers.
- Intravenous injection; planar and SPECT-CT images at about 24 hours.
- **Normal uptake:** salivary glands, liver, heart, bladder, thyroid if unblocked; normal adrenals faint.

Uses

1. Phaeochromocytoma and paraganglioma

- **Extra-adrenal, multiple, metastatic or recurrent disease**; biochemistry positive but imaging negative.
- **Isolated normetanephrine excess** may indicate an extra-adrenal tumour; MIBG finds it.

- Tumour >5 cm: MIBG or FDG (fluorodeoxyglucose) PET (positron emission tomography) to exclude metastases before surgery.
- **Best in MEN-2 (multiple endocrine neoplasia type 2).** Raised metanephrines after surgery: MIBG can unmask metastases.

2. Neuroblastoma

- ^{123}I-MIBG is **required by COG (Children's Oncology Group) for staging**, replacing skeletal survey and bone scan. More sensitive for relapse. Non-avid tumours: PET.

3. Therapy (^{131}I-MIBG)

- Confirm uptake first. Metastatic phaeochromocytoma: symptom response in up to two-thirds; complete response <5%. High-specific-activity form FDA (Food and Drug Administration)-approved 2018.
- Relapsed or refractory neuroblastoma: response about 28–39%.

Limitations

- **Cold scans in SDHB (succinate dehydrogenase subunit B) and VHL (von Hippel–Lindau)**; aggressive metastases lose uptake.
- **FDG PET is now gold standard** for staging, and **^{68}Ga-DOTATATE PET/CT** is replacing it at some centres.
- **Also know:** carcinoid and medullary thyroid cancer imaging; cardiac sympathetic imaging.

Diagram to draw: whole-body MIBG showing an adrenal hot spot with normal liver, heart and bladder uptake.

> **Key points to remember:**
> - Noradrenaline analogue; block thyroid; stop labetalol.
> - ^{123}I diagnostic, ^{131}I therapeutic.
> - Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

Also, from the condensed book (not in the paper answer):

- For phaeochromocytoma >5 cm, do MIBG or FDG PET (best in SDHB/VHL) pre-op to exclude metastases. Iodinated contrast does **not** precipitate crisis.

4 · Hormonal work-up (all lesions >1 cm)

> **KEY POINT**
>
> Every incidentaloma gets a **1-mg dexamethasone suppression test (DST) and metanephrines**. Add an **aldosterone-to-renin ratio (ARR)** if there is hypertension or unexplained hypokalaemia. **Never biopsy, sample or excise before excluding phaeochromocytoma**.

Incidentalomas comprise cortisol adenoma 5%, phaeochromocytoma 5%, aldosteronoma 1%, ACC 5% and metastasis 2.5%, so **≈18%** are potentially surgical.

Cortisol

SAQ **Cushing syndrome and the dexamethasone suppression test**

Paper answer · P2

Asked as:

- Cushing Syndrome — *Practice Set 2 Paper II Q3c (SAQ)*
- Dexamethasone suppression test — *Practice Set 5 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

Definitions

- **Cushing syndrome:** symptomatic hypercortisolism. **Cushing disease:** due to a pituitary ACTH (adrenocorticotropic hormone)-secreting adenoma.
- **MACS (mild autonomous cortisol secretion):** autonomous cortisol without clinical signs (replaces "subclinical Cushing"); raises cardiovascular, bone and diabetes risk.

Causes

Group	Cause	Share
Exogenous	Glucocorticoids (oral, inhaled, topical)	Commonest overall
ACTH-dependent (80–85% of endogenous)	Pituitary adenoma (Cushing disease)	About 70%
	Ectopic ACTH (bronchial carcinoid, small cell lung cancer, phaeochromocytoma)	10–15%
ACTH-independent	Adrenal adenoma, ACC (adrenocortical carcinoma), AIMAH (macronodular hyperplasia), PPNAD (pigmented nodular disease; Carney complex)	About 10%

Clinical features

- Central obesity, moon face, plethora (90–100%); purple striae, hypertension, hirsutism, ED (70–90%).
- **Discriminating:** proximal weakness, bruising.
- **Urological:** nephrolithiasis in up to 50%; hypogonadotropic hypogonadism (low libido, ED).

Diagnosis

Step 1: exclude exogenous steroids (including creams and sprays) first. Step 2: confirm cortisol excess (two tests)

Test	Method	Abnormal
Overnight 1-mg DST (dexamethasone suppression test)	1 mg at 11 pm; 8 am cortisol	>1.8 µg/dL (50 nmol/L)
Late-night salivary cortisol	11 pm, twice	Raised
24-h UFC (urinary free cortisol)	Two collections	Raised; used in pregnancy

Step 3: locate the source

- **Low ACTH:** adrenal; adrenal CT or MRI.
- **High ACTH:** pituitary MRI; **IPSS (inferior petrosal sinus sampling) with CRH (corticotropin-releasing hormone)**: central gradient = pituitary; none = ectopic (chest CT).
- Pitfalls: pseudo-Cushing (depression, alcohol), oestrogens, enzyme inducers.

Dexamethasone suppression test

Principle

- **Dexamethasone** is a potent synthetic glucocorticoid that is **not detected by the cortisol assay**. In a normal axis it suppresses pituitary ACTH (adrenocorticotropic hormone) by negative feedback, so morning cortisol falls.
- **Autonomous** cortisol (adrenal adenoma) or excess ACTH drive (pituitary, ectopic) fails to suppress.
- "Low-dose" 1 mg is still 3–4 times physiological glucocorticoid.

Indications

- Suspected Cushing syndrome; **every adrenal incidentaloma** (5–8% secrete cortisol); suspected ACC (adrenocortical carcinoma).

Types

Test	Method	Abnormal
Overnight 1-mg LD-DST (low-dose DST; preferred screen)	Dexamethasone 1 mg at 11 pm–midnight; serum cortisol 8–9 am	See below
48-hour 2-mg LD-DST	0.5 mg 6-hourly for 48 hours; cortisol at 48 hours	>1.8 µg/dL; improves accuracy
High-dose DST	8 mg overnight or 2 mg 6-hourly for 48 hours	Over 50% suppression suggests pituitary source; **no longer routine** (some ectopic tumours also suppress)

Interpreting the 1-mg test

Morning cortisol	Meaning
≤1.8 µg/dL (50 nmol/L)	Normal; >90% sensitive, rules out
1.8–5.0 µg/dL	Possible autonomous cortisol secretion
>5.0 µg/dL (140 nmol/L)	Autonomous secretion; >95% specific
- ESE (European Society of Endocrinology) 2023 incidentaloma guideline: cortisol >1.8 µg/dL without overt signs = **MACS (mild autonomous cortisol secretion)**.	
- If abnormal: repeat testing and measure **morning ACTH** (low = adrenal; high = pituitary or ectopic, IPSS [inferior petrosal sinus sampling]).	

Pitfalls

- **False positive:** oral contraceptives (raise cortisol-binding globulin; up to 50%); CYP3A4 inducers (phenytoin, carbamazepine, rifampicin); missed dose; pseudo-Cushing states (depression, alcohol, obesity, stress).
- **False negative:** CYP3A4 inhibitors (itraconazole, ritonavir, diltiazem).
- **Choice of test:** pregnancy → UFC (urinary free cortisol); epilepsy → LNSC (late-night salivary cortisol); renal failure → DST (not affected by GFR [glomerular filtration rate]).

Management

Cause	Treatment
Cushing disease	Trans-sphenoidal surgery (cure 60–80%); radiotherapy
Adrenal adenoma	**Laparoscopic adrenalectomy**
ACC	Open radical adrenalectomy, mitotane
Ectopic	Resect source; otherwise bilateral adrenalectomy
AIMAH, PPNAD	Bilateral adrenalectomy

- **Bilateral adrenalectomy:** for failed pituitary surgery or life-threatening hypercortisolism. Cost: lifelong replacement, **Nelson syndrome** (8–29%).
- **Medical (bridge):** metyrapone, ketoconazole, etomidate, mitotane, mifepristone.

Perioperative care (urological relevance)

- **Contralateral adrenal is suppressed:** autonomous cortisol suppresses the opposite adrenal; adrenal insufficiency in about 99.7% after surgery for overt Cushing.
- Give **perioperative steroid cover** and taper after adrenalectomy: **Hydrocortisone 100 mg IV** on the day of surgery, then slow oral taper; recovery takes a median 11 months.

Diagram to draw: HPA (hypothalamic–pituitary–adrenal) axis with dexamethasone feedback and with ACTH-dependent and ACTH-independent sites.

> **Key points to remember:**
> - Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
> - 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
> - Oral contraceptives and enzyme inducers cause false positives.
> - ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
> - High-dose DST no longer routine.
> - Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

Also, from the condensed book (not in the paper answer):

- 1-mg DST: **≤1.8 µg/dL (50 nmol/L)** excludes autonomy; 1.8–5.0 means possible autonomy; **>5.0 µg/dL (140 nmol/L)** is >95% specific but misses up to 18% of Cushing syndrome.
- Urinary free cortisol (UFC) is insensitive for mild excess and unreliable if GFR is <60 mL/min.

Catecholamines

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Sensitivity 97–99% – best rule-out	>97% by tandem mass spectrometry
Conditions	**Supine after 20 min rest**, fasting	Normal renal function
Highly indicative	Normetanephrine >2.2 nmol/L and/or metanephrine >1.2 nmol/L (≈4× upper limit of normal)	—

- **False positives** come from paracetamol, mesalazine, sulfasalazine (stop ≥5 days), tricyclics, **phenoxybenzamine**, severe illness and seated sampling.
- If equivocal, repeat supine off β-blockers, check chromogranin A, or do a **clonidine suppression** test.

5 · Biopsy and incidentaloma management

- Biopsy answers only "is this a metastasis?", because cytology cannot separate adenoma from carcinoma. ESE–ENSAT advises **no biopsy without a history of malignancy**; complications occur in 3–12%, with seeding in ACC.

ALGORITHM

1. If imaging characterises the lesion, then do not biopsy.
2. If it is functional, then do not biopsy.
3. If ACC is suspected, then do not biopsy.
4. If the result will change management (e.g., known primary) and justify risk, then biopsy, **after excluding phaeochromocytoma**.

- Incidentalomas appear on ≈5% of scans; ≈93% are inactive and need no treatment.

6 · Primary aldosteronism (PA)

LAQ How will you evaluate and manage 38 yr male with uncontrolled hypertension

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 7 Paper 2, Q1 (renovascular hypertension also answered in Practice Set 2 Paper 2, Q1; phaeochromocytoma in Practice Set 6 Paper 1, Q1)

Assumed focus: secondary (urological, renal and adrenal) causes of hypertension in a young adult.

Introduction

- **Resistant hypertension:** BP (blood pressure) above goal despite 3 drugs of different classes, including a diuretic, at optimal doses; or needing 4 or more drugs.
- Poor control below 40 years is a classic trigger to search for a **secondary cause**; many are curable.

Causes

Group	Causes	Clues
Renovascular	FMD (fibromuscular dysplasia), atherosclerotic RAS (renal artery stenosis), Takayasu arteritis, reninoma	Bruit; creatinine rise >30% on ACE (angiotensin-converting enzyme) inhibitor; flash pulmonary oedema; kidneys differing >1.5 cm

Group	Causes	Clues
Renal parenchymal	CKD (chronic kidney disease), ADPKD (autosomal dominant polycystic kidney disease), reflux nephropathy, glomerulonephritis, Page kidney	Raised creatinine, proteinuria, haematuria
Obstructive uropathy	Bilateral obstruction (about 77% hypertensive), unilateral (about 20%), chronic retention	LUTS (lower urinary tract symptoms), palpable bladder, hydronephrosis
Primary aldosteronism (PA)	Adenoma (Conn), bilateral hyperplasia	Hypokalaemia (present in <40% only)
Phaeochromocytoma, paraganglioma	Sporadic or hereditary	Paroxysmal headache, sweating, palpitations
Cushing syndrome	Exogenous steroid, adrenal adenoma, pituitary	Central obesity, striae, diabetes
- **Also know:** sleep apnoea; drugs (NSAIDs [non-steroidal anti-inflammatory drugs], steroids, decongestants, liquorice); alcohol; thyroid and parathyroid disease; coarctation.		

Evaluation

Step 1: Exclude pseudo-resistance

- Correct cuff technique; home or ambulatory monitoring to exclude white-coat effect; check adherence, doses, interfering drugs, salt and alcohol.

Step 2: History and examination

- Paroxysms, weakness or polyuria (hypokalaemia), stones, UTI (urinary tract infection) or childhood reflux, snoring, family history.
- BP both arms and radio-femoral delay; abdominal bruit; palpable kidneys or bladder; cushingoid habitus; café-au-lait spots (NF1 [neurofibromatosis type 1]); fundus.

Step 3: Baseline tests and target-organ damage

- Creatinine, eGFR (estimated glomerular filtration rate), electrolytes, urinalysis with albumin–creatinine ratio, glucose, lipids, ECG (electrocardiogram), echocardiography.
- **Ultrasound KUB (kidneys, ureters, bladder):** renal size and asymmetry, scars, cysts, masses, hydronephrosis, residual urine.

Step 4: Targeted screening (method and interpretation)

Cause	Screen	Confirm or localise
PA	Morning **ARR (aldosterone-to-renin ratio)** >20–30 with aldosterone ≥15–20 ng/dL; stop spironolactone first; correct potassium	**Intravenous saline 2 L over 4 h** (aldosterone stays >5 ng/dL) or oral salt loading → adrenal CT (computed tomography) → **AVS (adrenal vein sampling)**; skip AVS if age <40 with a clear unilateral adenoma >1 cm

Cause	Screen	Confirm or localise
Phaeochromocytoma	**Plasma free or 24-h urinary fractionated metanephrines**	CT or MRI (magnetic resonance imaging); ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); genetic testing
Cushing	**1-mg overnight dexamethasone suppression** (cortisol ≤1.8 µg/dL excludes); late-night salivary cortisol; 24-h urinary free cortisol	ACTH (adrenocorticotropic hormone): low = adrenal, high = pituitary or ectopic
Renovascular	**Duplex Doppler**: peak systolic velocity >180–200 cm/s, renal/aortic ratio >3.5	CT or MR angiography; catheter angiography with pressure gradient if intervention planned
Parenchymal, obstructive	Ultrasound, urine tests	Renogram (DTPA [diethylenetriaminepentaacetic acid] or MAG3 [mercaptoacetyltriglycine]) for split function and drainage; CT urography

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Uncontrolled BP, age 38 → exclude pseudo-resistance → baseline tests + ultrasound KUB → ARR, metanephrines, dexamethasone test, renal Doppler ├─ positive → confirm → localise → treat cause └─ all negative → essential: optimise drugs
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Management

1. General (all patients)

- Lifestyle: salt restriction, weight loss, exercise, limit alcohol.
- Optimise **ACE inhibitor or ARB (angiotensin receptor blocker) + calcium channel blocker + thiazide-like diuretic**. Fourth drug: **spironolactone** (PATHWAY-2, 2015: best add-on in resistant hypertension).

2. Cause-specific

Cause	Treatment
PA, lateralised	**Laparoscopic adrenalectomy**: biochemical cure about 94%, but only 37–52% need no antihypertensives
PA, bilateral or unfit	Spironolactone 25–50 mg/day (up to 400 mg) or eplerenone
Phaeochromocytoma	**α-blockade 7–14 days**, β-blocker only after α, volume expansion → laparoscopic adrenalectomy; metanephrines yearly for life
Adrenal Cushing	Adrenalectomy with steroid cover; pituitary disease → trans-sphenoidal surgery
FMD	**Angioplasty alone** (cure common in the young)
Atherosclerotic RAS	Medical therapy (**ASTRAL 2009, CORAL 2014**: stenting no benefit); stent for flash pulmonary oedema or falling function
Branch disease, aneurysm	Surgical bypass or ex vivo repair with autotransplantation
Reninoma	Partial nephrectomy (cures most)
Small non-functioning kidney (reflux, segmental hypoplasia)	**Nephrectomy** if renal vein renin lateralises (ratio >1.5)
Page kidney	Drain or decapsulate

Cause	Treatment
Obstruction	**Relieve it** (catheter, nephrostomy, pyeloplasty, prostate surgery); watch post-obstructive diuresis
CKD	ACE inhibitor or ARB for proteinuria; scheduled diuretics

3. Follow-up: home BP, creatinine, potassium; duplex after angioplasty.

Recent advances

- **Catheter renal denervation** (radiofrequency or ultrasound), approved in the USA in 2023 for selected resistant hypertension; SPARTACUS trial questioned routine AVS.

Diagram to draw: stepwise algorithm from pseudo-resistance to cause-specific treatment.

> **Key points to remember:**
> - Hypertension below 40 years: screen for secondary causes.
> - ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
> - Metanephrines for phaeochromocytoma; α before β.
> - FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Also, from the condensed book (not in the paper answer):

- PA causes 3–13% of hypertension; most patients (63–91%) are **normokalaemic**.

Subtype	%	Surgically curable?
Bilateral (idiopathic) hyperplasia	60	No – mineralocorticoid receptor antagonist (MRA)
Aldosterone-producing adenoma	35	Yes
Unilateral hyperplasia	2	Yes
Carcinoma, ectopic (ovarian), familial hyperaldosteronism (FH)	<1 each	Carcinoma/ectopic yes; FH no

- **FH-I** (glucocorticoid-remediable) results from a chimeric gene that puts aldosterone under ACTH control; it causes early severe hypertension and is treated with glucocorticoid.
- **Screen (Endocrine Society 2016)**: resistant hypertension (≥3 drugs including a diuretic) or control on ≥4 drugs; BP ≥150/100 on three days; hypertension and hypokalaemia; **incidentaloma with hypertension**; sleep apnoea; onset <20 y or stroke <50 y; family history of PA; disproportionate target-organ damage.
- **ARR** is measured in the morning, seated, as plasma aldosterone concentration (PAC, ng/dL) divided by plasma renin activity (PRA, ng/mL/h). It is positive if **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2 and correct K^+ first.

Drug	Effect on ARR	Action
Spironolactone, eplerenone, amiloride	Profound	Stop ≥4–6 weeks; use α1-blocker or calcium channel blocker (CCB)
β-blockers	Lower renin, so **false positive**	May continue (or stop)
ACE inhibitors / angiotensin receptor blockers	Raise renin, so false negative	Continue; undetectable renin is suspicious

Drug	Effect on ARR	Action
Calcium channel blockers	Neutral	Continue

Confirmatory test	Protocol	Positive
Oral sodium loading	3 days high salt; 24-h urine Na >200 mmol	Urine aldosterone >12 µg/day
IV saline infusion	2 L 0.9% saline over 4 h, recumbent	PAC >5 ng/dL (>10 suggests adenoma)
Captopril challenge	25–50 mg, when salt load is contraindicated	PAC stays >15 ng/dL

- **Skip confirmation** if there is spontaneous hypokalaemia, undetectable renin and PAC >20 ng/dL.
- CT all surgical candidates to exclude ACC, but ≈20% of aldosteronomas are <1 cm, so CT alone misdirects ≈25%.
- **AVS** (sensitivity 95%, specificity 100% if done well) is done in the morning with K^+ corrected, sampling both adrenal veins and the IVC for aldosterone and cortisol.
- **Selectivity** proves the catheter is in the vein (adrenal-to-IVC cortisol >1.1–5:1). **Lateralisation**: a cortisol-corrected aldosterone ratio **>2–4:1** indicates ipsilateral adrenalectomy.
- **Omit AVS** if age <40 with a clear unilateral adenoma >1 cm and a normal contralateral gland, or if ACC is suspected.

ALGORITHM

1. If there is a screening indication, measure the ARR.
2. If the ARR is positive, confirm with a suppression test. If there is a family history or onset <20 y, test for FH-I.
3. If not a surgical candidate, then MRA.
4. If a surgical candidate, do CT. If age <40 with a unilateral >1 cm nodule, then adrenalectomy.
5. Otherwise do AVS: if lateralised, laparoscopic adrenalectomy; if not lateralised, MRA.

- After surgery, check PAC and ARR; † stop K^+ and MRA, and watch for **hyperkalaemia** for weeks because the contralateral glomerulosa is suppressed. BP settles over 6 months; residual hypertension is likelier in older, obese men with long-standing disease.
- MRA: spironolactone 25–50 mg, up to a maximum of 400 mg (gynaecomastia, erectile dysfunction); eplerenone 25 to 100 mg (selective).

7 · Cushing syndrome and cortisol excess

*See the paper answer **Cushing syndrome and the dexamethasone suppression test** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- Post-op insufficiency follows ≈65% of MACS resections; the HPA axis recovers in a median of 6.5 (MACS) to 11 (Cushing) months.
- **Urological**: hypogonadotropic hypogonadism and **urolithiasis in up to 50%**, so test cushingoid stone formers.

ALGORITHM

1. Exclude exogenous steroid.
2. Screen with 1-mg DST, late-night salivary cortisol (LNSC) ×2 or UFC ×2.
3. If abnormal, endocrinologist confirms; measure **morning ACTH**.
4. If ACTH low: adrenal, do CT.
5. If normal/high: pituitary MRI and **inferior petrosal sinus sampling after CRH**. A central-to-peripheral gradient means Cushing disease; no gradient means ectopic.

- **Bilateral adrenalectomy** is for failed pituitary treatment, life-threatening hypercortisolism, or unresectable or occult ectopic ACTH. The cost is lifelong replacement, crisis risk and **Nelson syndrome** (chiasmal compression, oculomotor palsies).

	ACTH-independent macronodular adrenal hyperplasia (AIMAH)	**Primary pigmented nodular adrenocortical disease (PPNAD)**
Glands	Massive, nodules up to 4 cm, >60 g each	Normal size; small black/brown nodules in atrophic cortex
Association	McCune–Albright	Carney complex (≈half)
Clue	Looks ACTH-driven – check ACTH	Paradoxical rise in UFC after dexamethasone
Treatment	Bilateral adrenalectomy; unilateral removal of the larger gland has worked in AIMAH	

- **Mild autonomous cortisol secretion (MACS)**: individualise surgery for hypertension, obesity, dysglycaemia or osteopenia; progression to overt Cushing is very unlikely.
- **Post-op adrenal insufficiency is expected** after overt Cushing (≈99.7%), because the contralateral gland is atrophic.
- Give hydrocortisone 100 mg IV on the day of surgery, then an oral taper (25–50 mg/day). Alternatively give dexamethasone 0.5 mg 6-hourly, where a morning cortisol <50 nmol/L confirms cure. Maintenance is 10–12 mg/m²/day in 2–3 doses.

8 · Phaeochromocytoma and paraganglioma (PPGL)

These can kill on the table, so block before surgery.

*See the paper answer **Phaeochromocytoma: adrenal anatomy, hormones and perioperative management** (Section 14, Ch 145).*

Also, from the condensed book (not in the paper answer):

- ≈24% of apparently sporadic cases carry germline mutations.
- 75–90% are adrenal; paraganglioma is extra-adrenal. Headache occurs in 60–90%.

KEY POINT

The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014).

Syndrome	Gene	Features	PPGL risk / malignancy
MEN-2A	RET 10q11.2	Medullary thyroid cancer (MTC), hyperparathyroidism	50% / ≈3%; adrenal, bilateral
MEN-2B	RET	MTC, mucosal neuromas, marfanoid	50% / ≈3%
VHL type 2	VHL 3p25	RCC, haemangioblastoma, epididymal cystadenoma	10–20% / ≈5%; normetanephrine only
Neurofibromatosis 1 (NF1)	NF1 17q11	Neurofibromas, café-au-lait	1% / ≈11%
PGL-4	**SDHB** 1p36	Abdominal/thoracic paragangliomas	≈20% / **30–50%**
PGL-1	SDHD 11q23 (**maternal imprinting** – paternally transmitted)	Head and neck, multifocal	≈20% / <3%

All are autosomal dominant. VHL type 1 has no phaeochromocytoma, whereas type 2 does.

- On CT the mean is ≈35 HU; the T2 "light bulb" sign is unreliable.
- **Block all with abnormal biochemistry, even if normotensive.** Phenoxybenzamine causes hypotension after resection; α1-selective drugs do so less. CCBs can be sole therapy in mild cases.
- **Metyrosine** 250 mg 2–3 times a day, increased to 1.5–2 g, is added to α-blockade for large or metastatic tumours; it causes sedation and **extrapyramidal** effects.

KEY POINT

Why never β before α? β-blockade removes β2-mediated arteriolar dilatation. Catecholamines then act unopposed on α1 receptors and cause a hypertensive crisis. Hence use selective β1 agents.

- **Targets**: seated BP 120–130/80; standing systolic >80–90 mmHg; † heart rate 60–70/min. **Volume repletion** is key.
- In **MEN-2 and VHL**, do **cortical-sparing partial adrenalectomy**.
- Malignancy is defined **only by metastases**; risk factors are SDHB, extra-adrenal site, >5 cm and dopamine secretion.
- Check metanephrines at ≈6 months, then annually for life, because recurrence occurs even >15 years later.

9 · Non-functional benign lesions

Lesion	Key features	Management
Myelolipoma	Fat and haematopoietic tissue; macroscopic fat diagnostic; may exceed 10 cm	Exception to 4-cm rule; still do hormonal work-up. Excise if symptomatic or large; embolise a bleed

Lesion	Key features	Management
Cysts	Endothelial 45%, pseudocyst 39% (harbours malignancy ≈7%); >30 HU means haemorrhage	Endocrine work-up. **Excise if >5 cm, thick-walled, heterogeneous or symptomatic**
Ganglioneuroma	Young; encases vessels without narrowing them	Benign; diagnosed at resection
Oncocytoma	Rare; large; ≈10% functional	Resect; >30% malignant (unlike renal)

10 · Adrenocortical carcinoma (ACC)

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- Overall 5-year survival is 20–47% (stage IV <5%), and 60–80% recur after resection.

ENSAT stage	Definition	At diagnosis
I	T1 (≤5 cm) N0 M0	3–4%
II	T2 (>5 cm) N0 M0	29–46%
III	T3 (periadrenal fat) or T4 (adjacent organ) or N1 or venous thrombus	11–19%
IV	Distant metastasis only	39–49%

- Resect recurrences if low grade with a long disease-free interval.
- † Mitotane often needs higher glucocorticoid doses, because it raises binding globulins and speeds clearance.
- **Paediatric ACC** is >90% functional (virilisation), warrants TP53 testing and has a better prognosis.

11 · Adrenal metastases

- With known malignancy, **>50%** of new adrenal lesions are metastases. RCC and hepatoma deposits may wash out, so growth and biopsy decide. **Still exclude phaeochromocytoma**.
- Resect an isolated metastasis in selected patients: non-small cell lung cancer (level 1 evidence) and melanoma, and RCC as part of cytoreduction.

12 · Adrenal insufficiency

	Primary (Addison)	Secondary (pituitary/hypothalamic)
Causes	Autoimmune (anti-21-hydroxylase), **tuberculosis**, bilateral haemorrhage, infiltration, **bilateral adrenalectomy**, drugs (ketoconazole, abiraterone)	**Exogenous glucocorticoid suppression** (commonest for surgeons), pituitary tumour/apoplexy, Sheehan, critical illness
Aldosterone	Lost	Preserved
Skin	Hyperpigmentation (high ACTH)	Pallor
Electrolytes	Low Na^+, **high K^+**; salt craving	Low Na^+, normal K^+
Replacement	Hydrocortisone + fludrocortisone	Hydrocortisone only

- A **crisis** mimics an acute abdomen. Those at risk have a cortisol-secreting tumour, bilateral adrenalectomy or a **previous contralateral nephrectomy/adrenalectomy**, so review old notes.
- Diagnose with morning cortisol and ACTH, then ACTH stimulation. Give ½–⅔ of the hydrocortisone dose on waking.

> **KEY POINT**
>
> † **Adrenal crisis**: do not wait for results. Draw cortisol and ACTH, give **hydrocortisone 100 mg IV bolus**, then 200 mg/24 h (infusion or 50 mg 6-hourly). Give 0.9% saline (1 L in the first hour) with dextrose, treat the precipitant, and taper over 1–3 days. Fludrocortisone is unnecessary while hydrocortisone is >50 mg/day. Teach sick-day rules and issue a steroid card.

Surgical stress (chronic steroids)	Example	Cover
Minor	Inguinal hernia, <1 h	Hydrocortisone 25 mg
Moderate	Joint replacement	Usual dose + hydrocortisone 50–75 mg intra-op
Major	Oesophagectomy, bypass	Usual dose + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Glucocorticoid	Potency	Replacement (mg/day)	Half-life (h)
Hydrocortisone	1	20	8–12
Prednisone / prednisolone	4	5	18–36
Dexamethasone	25–50	0.5	36–54

13 · Congenital adrenal hyperplasia (CAH)

- CAH is autosomal recessive, and **21-hydroxylase deficiency is >95%**; low cortisol raises ACTH, causing hyperplasia and androgen excess (see Ch 59).
- Non-classic CAH may present as an adult mass, and testicular rests mimic tumour, so consider them before orchidectomy.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTACUS	RCT: CT- vs AVS-guided PA treatment	Same antihypertensive intensity at 1 y; CT and AVS discordant in half; persistent aldosteronism 20% (CT) vs 10% (AVS)	AVS remains standard for biochemical cure
FIRM-ACT	RCT: mitotane + etoposide, doxorubicin, cisplatin (EDP) vs mitotane + streptozotocin, advanced ACC	Better response and progression-free survival; overall survival similar	M-EDP first-line
ADJUVO	RCT: adjuvant mitotane vs observation, low/intermediate-risk R0 stage I–III ACC	No difference in recurrence	No routine adjuvant mitotane in low risk

15 · MCQ one-liners

MCQ ONE-LINERS

1. Commonest ectopic ACTH source: bronchial carcinoid.

147 Surgery of the Adrenal Glands

SAQ Adrenal Sx.

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3d (preoperative phaeochromocytoma management also in 2020-Summer Paper 2, Q3c)

Assumed meaning: surgery of the adrenal gland (adrenalectomy).

Surgical anatomy

- Right: pyramidal; **short vein into posterior IVC (inferior vena cava)**, so it is the riskier side. Left: crescentic; long vein into left renal vein. Three arteries (inferior phrenic, aorta, renal), one vein.

Indications (AAES [American Association of Endocrine Surgeons] 2022)

- **Functional, any size:** aldosteronoma (Conn; after AVS [adrenal venous sampling]), cortisol-secreting adenoma, phaeochromocytoma, sex-hormone tumour, failed pituitary surgery for Cushing disease.
- **Malignant potential:** ≥4 cm; growth; >10 HU (Hounsfield units) unenhanced; washout <50%; suspected ACC (adrenocortical carcinoma); solitary metastasis.
- Myelolipoma >7 cm or symptomatic; cysts.
- **Contraindications (correct first):** uncontrolled hypertension, coagulopathy, peritonitis (transperitoneal); pregnancy relative.

Preoperative preparation

- Biochemistry: metanephrines in **every** case before surgery.
- **Phaeochromocytoma:** α-blockade 2 weeks (phenoxybenzamine or doxazosin), β-blocker only after α, salt and fluids.
- Cushing: hydrocortisone 100 mg IV on the day of surgery. Conn: correct potassium.

Approaches

Approach	Use
Laparoscopic transperitoneal (gold standard since 1992)	Familiar view, larger tumours
Retroperitoneoscopic (lateral or prone)	Prior abdominal surgery, small tumours
Robotic	Either route
Open (subcostal, chevron, thoracoabdominal, flank)	**ACC**, large or invasive tumours

Steps (laparoscopic transperitoneal)

- **Right:** elevate liver → Kocher manoeuvre → follow IVC → **control adrenal vein early** (warn anaesthetist) → free gland to psoas → bag intact.
- **Left:** reflect colon and spleen → follow left renal vein to adrenal vein → clip → free from pancreas and splenic vessels.
- Kidney attachment last (handle). ACC: open en bloc, no rupture.
- **Partial adrenalectomy:** bilateral or hereditary phaeochromocytoma, solitary gland.

Complications

- Bleeding (adrenal vein avulsion, IVC); spleen, pancreatic tail, liver, bowel injury; pneumothorax.
- **Phaeochromocytoma:** surges during handling, **hypotension after vein ligation**, rebound hypoglycaemia.
- **Adrenal insufficiency** after Cushing surgery (~99.7%): steroid taper.
- Conn: **hyperkalaemia** for weeks (high-salt diet).

Diagram to draw: right and left adrenal veins and relations.

> **Key points to remember:**
> - Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
> - Control the vein early; right vein short into IVC.
> - Open en bloc surgery for ACC.

Also, from the condensed book (not in the paper answer):

- Size and imaging criteria for incidentaloma (§2).

1 · Surgical anatomy (essentials)

- The **right** gland lies behind the bare area of the liver. Its vein is **<1 cm** long and enters the **posterior IVC at 45°**. The **left** gland is more mobile.

2 · Indications

Two gates: any functional tumour, or malignant potential.

Category	Indication
Functional (any size)	Aldosteronoma; cortisol adenoma; Cushing disease after failed trans-sphenoidal surgery; phaeochromocytoma (PCC); virilising/feminising tumour
Non-functional	Adrenocortical carcinoma (ACC); symptomatic cyst/myelolipoma; solitary metastasis
Size	**≥4 cm** ; >2 cm growing ≥1 cm in 12 months
Imaging	Non-contrast CT **>10 HU**; washout <50% with delayed >35 HU; chemical-shift MRI not adenoma; FDG-PET uptake equal to or above liver

> **KEY POINT**
>
> Trap: bilateral hyperplasia causing aldosteronism is **not** surgical, so adrenal venous sampling (AVS) comes before adrenalectomy for aldosteronism.

Incidentaloma work-up

- ACC is found in ~2% and autonomous secretion in up to 20%. Measure sex steroids if ≥4 cm or virilised.
- All get a 1-mg dexamethasone suppression test (never spot cortisol) and metanephrines (>2× upper limit is significant).
- If hypertensive, check the aldosterone-to-renin ratio (ARR >20 with aldosterone >20 ng/dL), then salt loading, then AVS; up to 40% are normokalaemic.
- Inert, ≤4 cm, ≤10 HU lesions get CT at 12 months. Operate on myelolipoma (≤ −20 HU) only if **>7 cm**, symptomatic or atypical.

3 · Contraindications

- Correct physiology first. Low-dose aspirin may continue; necessary surgery in pregnancy can be done in any trimester. Confirm that a prior nephrectomy has not removed the ipsilateral adrenal before contralateral adrenalectomy.

4 · Pre-operative preparation

Pathology	Preparation	Why
PCC	α-blockade for **2 weeks** (phenoxybenzamine; doxazosin, calcium channel blocker, metyrosine alternatives); β **only after α**; volume expansion	Contracted patients collapse once vein tied
Cushing	†Hydrocortisone 100 mg at induction, tapered over months; thromboprophylaxis	Contralateral gland is suppressed
Aldosteronoma	†Correct K^+; control BP	†Hypokalaemia causes arrhythmia

5 · Approaches

- **Minimally invasive surgery is standard** (since 1992).
- The **retroperitoneoscopic** route (lateral or **prone**) avoids bowel, so it suits prior abdominal surgery; bilateral small tumours can be done prone without repositioning.
- **Open** surgery is for suspected **ACC**, large tumours, or invasion of the IVC or organs.

6 · Operative steps

Control the vein, then free the gland.

ALGORITHM

1. **Right**:
2. Elevate the liver; Kocher the duodenum; expose the IVC.
3. Dissect up from the renal vein to the short adrenal vein.
4. **Warn anaesthesia**, then clip early.
5. Dissect superiorly and laterally down to psoas; bag the gland intact.

ALGORITHM

1. **Left**:
2. Reflect the colon at Toldt, outside Gerota; lift spleen and pancreas.
3. Follow the left renal vein to the adrenal vein; clip after warning.
4. Protect pancreas and splenic artery; divide superomedial tissue; bag.

- **Retroperitoneoscopic**: enter below the 12th rib tip with a balloon. Along psoas the **renal artery is met first**; the right vein enters the posterolateral IVC above the hilum.

> **KEY POINT**
>
> Ligate the vein early in PCC to limit the catecholamine surge; hypotension follows, so tell anaesthesia first.

7 · Special pathologies

- **PCC intra-op**: treat a surge with nitroprusside, nitroglycerin or phentolamine, and arrhythmia with esmolol.
- **ACC** needs a high-volume surgeon (>6, ideally ≥20/year) and an **open, intact, en bloc** resection.
- **Metastasis** (two-thirds from lung or kidney) is resected when the adrenal is the only disease; renal origin and <5 cm do best.
- Avoid ablation of PCC or ACC outside trials.

8 · Partial (cortical-sparing) adrenalectomy

- **Indications**: bilateral or hereditary PCC, familial syndromes, prior contralateral adrenalectomy, children, Conn adenoma and cysts.
- Arteries enter superiorly, medially and inferiorly, so **lateral dissection is safe**; the main vein may be divided. †Retain about one-third of a gland, guided by intra-op ultrasound.

9 · Complications

- For bleeding, apply pressure, raise the pneumoperitoneum, side-bite the IVC, and convert if uncontrolled.
- †Check glucose hourly for 24–48 h after PCC. Treat insufficiency with IV hydrocortisone 100 mg and saline. For hyperkalaemia after aldosteronoma, stop K^+ and spironolactone, occasionally adding fludrocortisone.

10 · Outcomes

- PCC recurs in up to **16%** over life, so check BP and metanephrines annually.
- †Aldosteronoma surgery corrects hypokalaemia in nearly all and cures hypertension in about half. Advanced-stage ACC has a 5-year survival **<15%**.

140 Retroperitoneal Tumours

1 · Anatomy of the retroperitoneum

See the paper answer ***Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.*** *(Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- **Depth of psoas invasion predicts nerve injury**: genitofemoral injury causes thigh paraesthesia, while femoral injury weakens hip flexion and knee extension.

KEY POINT

The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement.

2 · Classification

The group decides first treatment.

Group	Entities	First treatment
Germ cell / gonadal	Seminoma, nonseminoma, sex cord stromal	Platinum chemotherapy (germ cell tumour, GCT); RP lymph node dissection, RPLND (sex cord stromal)
Mesodermal — malignant	Liposarcoma, leiomyosarcoma, undifferentiated pleomorphic sarcoma (UPS), synovial, Ewing, desmoplastic small round cell tumour (DSRCT), unclassified	Surgery; chemotherapy (Ewing, DSRCT)
Mesodermal — benign/intermediate	Lipoma, leiomyoma, solitary fibrous tumour (SFT), PEComa, desmoid, myelolipoma, GIST	Excision / observation
Neurogenic (~10%)	Schwannoma, neurofibroma, malignant peripheral nerve sheath tumour (MPNST), ganglioneuroma, neuroblastoma, paraganglioma	Surgery
Haematological (~1/3 of primary RP)	Lymphoma, plasmacytoma	Chemotherapy — biopsy only
Cystic	Lymphangioma, mucinous cystadenoma, cystic mesothelioma, cystic teratoma	Complete excision
Unknown primary	Poorly differentiated carcinoma encasing vessels	Histology-led

3 · Clinical evaluation

Tissue only if it changes treatment.

- Most are painless or incidental. The sequence is **imaging, then serology, then tissue**.

- **CT chest/abdomen/pelvis** comes first (NCCN). **Negative HU (fat)** suggests liposarcoma, angiomyolipoma or myelolipoma, but CT cannot tell origin or malignancy. **MRI** shows vascular involvement; PET-CT is selective.
- In a young man check **AFP and hCG**; in an older man, PSA. Add blood count and electrophoresis; † metanephrines if paraganglioma is possible.
- **Biopsy** only when it **changes first treatment**: lymphoma, GCT, Ewing, or high-grade sarcoma for neoadjuvant therapy. Use a **core (14–16 G coaxial) rather than FNA**; † take a retroperitoneal route, with the tract inside the future resection field.
- Avoid biopsy of purely cystic lesions, † an unblocked paraganglioma, or a young man with raised AFP/hCG.

4 · Mesenchymal tumours (sarcoma)

- Soft-tissue sarcoma is <1% of adult cancers; **15–20%** arise in the RP.

Entity	Key features	Treatment
Well-/de-differentiated (WD/DD) liposarcoma (commonest, **~60%**)	Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+	Radical compartmental resection
Leiomyosarcoma (**~20%**)	From vessel media (IVC); no fat; SMA, desmin, h-caldesmon+	Negative margins + adjuvant
UPS (ex-MFH)	5th–6th decade; no fat, peripheral nodular enhancement	Wide resection
Unclassified (≤20%)	Exclusion; prior RT in a quarter	Screen MDM2
Synovial	Calcification; TLE1; **t(X;18)**	Chemosensitive
SFT	**NAB2-STAT6**; Doege-Potter hypoglycaemia (IGF)	En bloc; radiosensitive
Ewing	EWSR1-FLI1 t(11;22); infiltrative	Chemotherapy-based
DSRCT	Young men; peritoneal implants; WT1-EWSR1 t(11;22)	Multimodal; poor
PEComa	HMB-45 + muscle markers; TSC2	Excision; mTOR inhibitors if unresectable
Desmoid	Nuclear β-catenin; APC/CTNNB1; never metastasises; recurs in up to half	Surgery or observation

- **Liposarcoma**: **dense nodules in a fatty mass mean de-differentiation** and are the biopsy target. WD recurs locally in up to 40% but rarely metastasises; DD spreads distantly in up to 20%. Mean time to recurrence is ~**2 years**, so surveillance is lifelong.
- If unresectable, anthracycline or trabectedin offers modest benefit. **Pleomorphic liposarcoma** is MDM2-negative and metastasises early to lung.
- **Leiomyosarcoma** obstructs veins, urine and bowel; biopsy underestimates its aggressiveness.

> **KEY POINT**
>
> MDM2 amplified (WD/DD) means local disease, won or lost at the margin; MDM2 absent (pleomorphic) means systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy because no margin is achievable.

- † **FNCLCC grade** adds differentiation (1–3), mitoses per 10 high-power fields (1: 0–9, 2: 10–19, 3: ≥20) and necrosis (0 none, 1 <50%, 2 ≥50%): **2–3 = G1, 4–5 = G2, 6–8 = G3**. Grade and histology, not size, predict distant spread.

5 · Neurogenic tumours

Tumour	Key features	Management
Schwannoma	Paravertebral; encapsulated; S100+; slow	Local resection curative (often enucleable)
Neurofibroma	NF-1 or sporadic; malignant change up to 20%	Surgical
MPNST	<10% of RP sarcoma; half NF-1; H3K27me3 loss; spreads along roots	Resect with **parent nerve**; 5-year ~70% R0 vs ~20% R1
Ganglioneuroma	3rd–4th decade; calcification up to 30%	Excision
Neuroblastoma	MYCN adverse; MIBG staging	Paediatric chemotherapy with or without surgery
Paraganglioma	Commonest at organ of Zuckerkandl (below IMA); T2-bright; secretory ~40%; metastases 20–50%; NF-1, MEN-2A, SDHB	Surgery after α-blockade† (see Ch 145)

6 · Germ cell, haematological, cystic, unknown primary

- **GCT**: an RP mass in a young man is GCT until proven otherwise; extragonadal disease is often a **burnt-out testicular primary**. Give platinum first, then resect all nonseminomatous residual masses, because teratoma is chemoresistant and marker-silent (see Ch 79–80).
- **Sex cord stromal** tumours resist chemo and radiotherapy, so do early RPLND in high-risk node-negative primaries.
- **Lymphoma** forms confluent nodal masses: biopsy, never resect.
- **Cystic** lesions (lymphangioma, mucinous cystadenoma, cystic mesothelioma) need **complete excision**, because aspiration or marsupialisation recur.
- **Unknown primary**: if midline or i(12p)-positive, give platinum; a single deposit is excised or irradiated.

> **KEY POINT**
>
> Histology decides first treatment: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning sarcoma resection.

7 · Surgery for RP sarcoma

The first operation decides cure.

- **Complete en bloc (R0) resection at the first operation** in an experienced centre is the best chance.

- **Compartmental resection** removes adherent uninvolved neighbours (kidney, colon, psoas fascia), which lowers local recurrence. Preserve duodenum, pancreatic head, major vessels and femoral nerve unless invaded. † **Never enucleate**, because the pseudocapsule contains tumour.
- Nephrectomy is needed in up to **55%** but rarely leads to dialysis; † get a split renogram first.
- After **bleomycin**, keep FiO_2 low and fluids restrictive.
- Through a midline laparotomy, **split and roll** over the great vessels and ligate lumbar vessels and retrocrural lymphatics to prevent chylous ascites.
- The **aorta is always reconstructed**. A chronically occluded **IVC** tolerates ligation; IVC grafts thrombose in 10–40%, so anticoagulate for ≥3 months.

8 · RPLND in germ cell tumours

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- Primary RPLND gives survival ~100% with either template. In post-chemotherapy RPLND, a full bilateral infrahilar template is standard; unilateral only if the mass is **≤5 cm** in the landing zone, markers are normal, and IGCCCG risk is good or intermediate.
- Sparing the L1–L4 fibres and the superior hypogastric plexus gives antegrade ejaculation in **90–100%**.

9 · Complications

*See the paper answer **Lymphocele and postoperative lymphatic leak (chylous ascites)** (Section 15, Ch 91).*

Also, from the condensed book (not in the paper answer):

- Chylous ascites occurs in 2–7% after chemotherapy. Other risks are pancreatic leak (check drain amylase), VTE after caval reconstruction and femoral neuropraxia.

ALGORITHM

1. **Chylous ascites**:
2. Paracentesis and a medium-chain triglyceride (MCT) diet, with or without octreotide.
3. If it reaccumulates, place a drain.
4. Then nil by mouth and total parenteral nutrition.
5. If output stays >100 mL/day, then peritoneovenous (LeVeen) shunt or ligation.
6. Continue the MCT diet for 1–3 months after resolution.

10 · Adjuvant and neoadjuvant therapy

Neither is routine; surgery is the mainstay.

- **Radiotherapy**: pre-op is better tolerated than post-op. **STRASS** showed no gain, so it is **not standard**; † an exploratory benefit in low-grade liposarcoma keeps it an option.
- **Chemotherapy**: WD liposarcoma and leiomyosarcoma are largely resistant; myxoid liposarcoma, synovial sarcoma and Ewing respond. **Standard neoadjuvant chemotherapy is only for Ewing**.

11 · Recurrence and follow-up

- † Local recurrence kills in liposarcoma, whereas leiomyosarcoma fails distantly (lung, liver).
- † CT every 3–6 months for 2–3 years, 6-monthly to 5 years, then annually; lifelong for liposarcoma.
- Re-resect slow-growing resectable recurrence (WD liposarcoma).

12 · Landmark trials

Trial	Design	Result	Take-home
EORTC-62092 STRASS	Phase III, 266 patients: pre-op RT + surgery vs surgery	No abdominal recurrence-free survival difference (HR 1.01); more complications	Pre-op RT not standard
STRASS II (ongoing)	Neoadjuvant doxorubicin + ifosfamide/dacarbazine + surgery vs surgery	Awaited	Neoadjuvant chemotherapy unproven

★ Putting it together: adrenal and retroperitoneal decisions

A · Adrenal incidentaloma: two questions

ALGORITHM

1. **Malignant?** If unenhanced CT is ≤10 HU or MRI shows drop-out, then adenoma. If >10 HU, do washout CT (APW >60% / RPW >40% = adenoma). If >4 cm, suspicious or growing, consider resection.
2. **Functioning?** Test all with 1-mg DST and metanephrines, plus ARR if hypertensive or hypokalaemic.
3. If **silent, <4 cm and benign**, then no follow-up.

B · Functional tumours compared

	Aldosteronism	**Cushing**	**Phaeo**
Screen	ARR	1-mg DST, salivary or urinary cortisol	Metanephrines
Next	Confirm, then CT, then AVS	ACTH: low = adrenal; normal/high = pituitary/ectopic	CT/MRI, functional imaging; genetics
Pre-op	Correct K^+	Steroid cover	α then β; fluids, salt
Post-op	Stop K^+/spironolactone; hyperkalaemia	Steroids until axis recovers	Hypotension, hypoglycaemia
Medical	Mineralocorticoid antagonist (bilateral)	Steroidogenesis inhibitors	Long-term α-blockade

C · Adrenalectomy approach

KEY POINT

Use laparoscopy for benign tumours. Operate open for large suspected carcinoma or invasion, because rupture causes recurrence. On the right, control the short caval vein carefully. Partial adrenalectomy suits bilateral or hereditary disease.

D · Retroperitoneal mass

ALGORITHM

1. **Image**, then **exclude** paraganglioma (metanephrines), germ cell tumour (testis, markers) and lymphoma.
2. If it changes treatment, do a **core biopsy** by the retroperitoneal route.
3. **Sarcoma:** en bloc R0 resection at the first operation in a specialist centre; never enucleate. Follow liposarcoma with lifelong CT.

Rapid Revision Sheet · Section 14

Phaeochromocytoma: adrenal anatomy, hormones and perioperative management (LAQ)

- Right vein short into IVC; left vein into left renal vein.
- StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
- Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
- Metanephrines first; never biopsy.
- α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
- Volume expansion prevents hypotension; watch glucose.
- Arterial line before induction; avoid histamine releasers and ephedrine.
- Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
- After vein ligation: fluids, vasopressors, glucose.
- One-third hereditary; SDHB most often metastatic.

Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma (LAQ)

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST: >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

MIBG scan (SAQ)

- Noradrenaline analogue; block thyroid; stop labetalol.
- ^{123}I diagnostic, ^{131}I therapeutic.
- Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

Cushing syndrome and the dexamethasone suppression test (SAQ)

- Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
- High-dose DST no longer routine.
- Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

How will you evaluate and manage 38 yr male with uncontrolled hypertension (LAQ)

- Hypertension below 40 years: screen for secondary causes.
- ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
- Metanephrines for phaeochromocytoma; α before β.
- FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Adrenal Sx. (SAQ)

- Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
- Control the vein early; right vein short into IVC.
- Open en bloc surgery for ACC. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 15

Renal, Nephrology & Transplant

88 Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

1 · Kidney: position and relations

The kidney's neighbours are what you injure at surgery.

- The kidney is ≈**11 cm** long, 125–170 g (male), at T12–L3. The **right lies 1–2 cm lower**; both descend ≈3 cm on inspiration or standing.

P1 *Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal mass — relations, coverings, Gerota open below, hilum order*

- Traction on the hepatorenal or splenorenal ligament tears the liver or splenic capsule.

LAQ Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL

Paper answer · P1

Asked as:

- Describe anatomy of pelvicalyceal system in detail and its application in endourological surgeries — *Winter 2017 Paper I Q1 (LAQ)*
- Anatomy of renal collecting system and its relevance to PCNL — *Practice Set 1 Paper I Q3d (SAQ)* (also answered in 2022-Summer Paper 1, Q1 and 2017-Winter Paper 1, Q1)
- Describe anatomy of kidney in relation to PCNL and describe indication, contraindication, procedure and complications of mini-PCNL. — *Summer 2022 Paper I Q1 (LAQ)* (also answered in 2017-Winter Paper 1, Q1 (pelvicalyceal anatomy) and 2017-Winter Paper 4, Q3d (Miniperc))

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Papilla → minor calyx → infundibulum → major calyx → renal pelvis → UPJ (ureteropelvic junction); lined by urothelium. Anatomy is very variable, so map it (CT (computed tomography) urogram or pyelogram) before every endourological procedure.
- Safe PCNL (percutaneous nephrolithotomy) access depends on the kidney's position, neighbours, vessels and calyces.

Surgical anatomy relevant to PCNL: position and orientation

- Lies T12–L3 (right 1–2 cm lower); descends about 3 cm on inspiration.
- It lies on the cone-shaped psoas, so the upper pole is medial and posterior and the lower pole lateral and anterior.
- Tract: skin, flank muscles, thoracolumbar fascia, paranephric fat, Gerota fascia, perirenal fat, cortex, then **pyramid and papilla** into the calyx.

Anatomy of the calyces (collecting system)

- **Minor calyx:** a cup around one papilla; its two side projections are the **fornices**; it drains by an infundibulum to a major calyx and pelvis. **Compound calyces** (2–3 papillae) are commonest at (cluster at) the poles. Average 7–9 calyces per kidney (range 4–19 or more).
- **Distribution:** upper pole usually 3 calyces, interpolar 3–4, lower pole 2–3.
- **Drainage pattern:** upper pole by a single midline infundibulum; lower pole by a single midline infundibulum or paired calyces; hilar region (hilar ones) by **anterior and posterior rows** of paired calyces.
- **Orientation:** the kidney lies on psoas, rotated 30–50° behind the coronal plane (posteriorly), so **posterior calyces point posterolaterally** (towards the prone surgeon). On a pyelogram anterior and posterior calyces (rows) overlap.
- **Brödel vs Hodson pattern (Kaye and Reinke):** in a Brödel-type kidney the posterior calyces lie lateral and the anterior calyces medial; in a Hodson-type kidney the reverse. Brödel type is commoner on the right, Hodson type on the left, so confirm calyx orientation with contrast/air or ultrasound before puncture.

Renal pelvis

- Pyramidal, apex to the ureter; capacity 3–10 mL. **Intrarenal** (in the sinus), **extrarenal** (larger, longer infundibula, mimics dilatation) or mixed.
- **Duplex (divided) pelvis:** upper and lower parts join at the hilum, with no direct connection between upper and lower calyces (does not link upper and lower calyces). Suspect it when one pole cannot be reached from a puncture.

Sampaio classification (Sampaio types; 140 resin endocasts)

Type	Frequency	Midzone (hilar) drainage
A-I	45%	Midzone calyces drain into the superior or inferior calyceal group (via a polar group)
A-II	17.2%	**Crossed calyces**: one to the superior and one to the inferior group (to both groups)
B-I	21.4%	Independent major calyceal group (major calyx) from the midzone
B-II	16.4%	1–4 minor calyces enter the pelvis directly (straight into the pelvis)

- **Perpendicular minor calyx (11.4%):** drains straight into the pelvis, hidden on a pyelogram (pyelography); narrow neck (≤4 mm) hinders clearance.
- **Crossed calyces:** the limb to the inferior group is **ventral in 87.5%**.

Vascular relations (the basis of safe access)

- Hilum, anterior to posterior: vein, artery, pelvis, **posterior segmental artery** (the only branch passing behind the pelvis).
- Five segmental **end arteries**; interlobar arteries run in the **columns of Bertin**, hugging the anterior and posterior faces of the infundibula (beside the infundibula).
- **Brödel line:** avascular plane between anterior and posterior arterial territories, just posterior to (behind) the lateral border, along the posterior calyces. Posterior calyces along it are the target.

Puncture route / site (Sampaio)	Vascular injury
Upper-pole infundibulum	67% (arterial 26%); posterior segmental artery
Mid-pole infundibulum	Arterial 23%

Puncture route / site (Sampaio)	Vascular injury
Lower-pole infundibulum	Infundibular artery (present) in 38%
Calyceal fornix	Venous <8%, **no arterial injury**

- So the needle passes **through the papilla along the calyx axis**, never into an infundibulum or the pelvis.

Relations (the hazards of access)

- **Relations:** pleura reaches the 12th rib posteriorly, lung the 10th intercostal space; liver and spleen descend in inspiration; **retrorenal colon** is commoner at the lower pole and when prone.

Structure	Relation	Practical point
Pleura, lung	Pleura reaches the 12th rib; lung the 10th intercostal space	Prefer subcostal or 11th-space access; risk rises above the 12th rib, highest above the 11th; puncture in expiration
Liver, spleen	Over the upper poles; descend in inspiration	Risk with supracostal access, inspiration, organomegaly
Colon	Retrorenal colon, mainly at the lower pole, commoner prone and on the left	Review CT; ultrasound shows bowel

Applications in endourology

1. PCNL (percutaneous nephrolithotomy) access

- Ideal target: a **posterior calyx with a short, wide infundibulum**, punctured (entered) **through the papilla and fornix** along the calyx axis, never into (through) the infundibulum or pelvis; identified by injected air, C-arm rotation or ultrasound. This passes through a pyramid and avoids interlobar arteries.
- **Upper-pole posterior access** gives the straightest route (best reach) to the pelvis, UPJ and most calyces (staghorn; staghorn branches). Chest risk (pneumothorax, hydrothorax) rises above the 12th rib, highest above the 11th. Puncture in expiration.
- **Lower-pole access** is chest-safe but risks (retrorenal) colon (left) and reaches the UPJ poorly (poor UPJ reach).
- Identifying (identify) posterior calyces: rainbowing the C-arm towards the surgeon shortens them (they shorten when the C-arm rotates towards the surgeon); injected air fills non-dependent (posterior, when prone) calyces (rises posteriorly when prone); ultrasound also shows bowel and pleura.
- **Also know:** traps: perpendicular or crossed calyces, duplex pelvis.

2. RIRS (retrograde intrarenal surgery)

- Entering the lower-pole infundibulum needs about **140° deflection** (104–175°); ≥270° scopes see it, but working there is harder.
- **Unfavourable lower-pole anatomy:** steep infundibulopelvic angle, long calyx (>10 mm), narrow infundibulum (<5 mm). Relocate stones to an upper calyx with a basket.
- Lower-pole stone >1 cm: PCNL preferred (**Lower Pole I, 2001:** stone-free 91% PCNL vs 21% SWL (shock wave lithotripsy)).

3. Endopyelotomy

- Incise the UPJ **laterally**, where crossing vessels are absent. A crossing vessel lowers success (42% vs 86%).
- Antegrade route: mid-posterior or superolateral calyx; not lower pole (acute angle to the UPJ).

4. **Infundibulotomy and calyceal diverticulum**

- Incise infundibular stenosis **superiorly or inferiorly**; never anteriorly.
- **Calyceal diverticulum:** cortical cavity joined to the fornix by a narrow neck; treat if symptomatic. Upper or interpolar: ureteroscopy, incise or dilate the neck. Lower pole: direct puncture onto the stone, mini-PCNL; fulguration optional.

5. **Open/anatrophic nephrolithotomy:** incision along the Brödel line.

- **Also know:** horseshoe kidney (medial access, upper pole usually subcostal); pelvic kidney (ultrasound or laparoscopic guidance); transplant kidneys; megacalycosis mimics obstruction; AV (arteriovenous) fistula or pseudoaneurysm → angioembolisation.

Mini-PCNL: definition

Term	Tract size
Standard	24–34 Fr
Mini	15–22 Fr
Ultra-mini	11–14 Fr
Micro-perc	<10 Fr

- First mini-perc: Jackman, 1998 (13 Fr).

Indications

- Renal stones 1–2 cm; lower-pole stones >1 cm; failed SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery).
- Calyceal diverticulum (lower pole), children, poor retrograde access (urinary diversion, ureteric stricture), selected stones >2 cm or extra tracts for staghorns.

Contraindications

- Untreated UTI (urinary tract infection), uncorrected coagulopathy, pregnancy (EAU, European Association of Urology).
- Tumour in the access tract or suspected malignant renal tumour; no safe window; unfit for anaesthesia.

Procedure

1. Urine culture, blood count, coagulation, creatinine; **non-contrast CT (computed tomography)**; antibiotic prophylaxis.
2. Lithotomy: ureteric catheter and retrograde pyelogram.
3. Prone, or supine (Galdakao-modified) for single-position retrograde access.
4. 18 G puncture of a posterior calyx under fluoroscopy (bull's-eye or triangulation) or ultrasound; urine return; guidewire and safety wire.
5. One-step, serial or balloon dilatation to 15–22 Fr; place the sheath.
6. Mini-nephroscope; holmium/thulium laser or pneumatic lithotripsy. Fragments wash out by the irrigation **"vacuum-cleaner" effect** or a **suction sheath**.
7. Flexible nephroscopy to clear calyces.

8. Exit: tubeless, small nephrostomy (≤10 Fr: less pain) or double-J stent.

Advantages and limitations

- Meta-analysis: **non-inferior stone-free rate**, smaller haemoglobin drop, **fewer transfusions**, shorter stay.
- Longer operation, poorer vision. A tight sheath raises **intrarenal pressure**, causing pyelovenous backflow and **fever or sepsis**. Advantages disappear against 24 Fr tracts.

Complications (grade by Clavien–Dindo)

Type	Points
Bleeding	Acute tract bleeding: tamponade sheath, balloon or Kaye catheter. Delayed: pseudoaneurysm or AV (arteriovenous) fistula, treated by angioembolisation
Infection	Fever, SIRS (systemic inflammatory response syndrome), urosepsis
Thoracic	Pneumothorax, hydrothorax: chest drain
Visceral	Colon (stent plus nephrostomy withdrawn into the colon), spleen, liver, duodenum
Collecting system	Perforation, extravasation, later infundibular stenosis; residual stones
Others	Irrigation absorption, venous gas embolism, DVT (deep vein thrombosis), neuropraxia, renal function loss, death

Diagram to draw: transverse section of kidney showing anterior and posterior calyces (posterior calyx), Brödel line and the needle through the posterior calyceal fornix.

> **Key points to remember:**
> - Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
> - Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
> - Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
> - Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
> - Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
> - Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
> - Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
> - High intrarenal pressure in mini-PCNL causes fever and sepsis.

Also, from the condensed book (not in the paper answer):

- Upper-pole excision through the diaphragm risks pneumothorax.

2 · Coverings and compartments

- Each perirenal space is separate, so extravasation rarely crosses the midline. The **anterior pararenal space** (colon, duodenum, pancreas) does cross it; the posterior pararenal space holds fat only.
- Transperitoneally, open the **white line of Toldt**. Reach the right hilum by a **Kocher manoeuvre**; on the left, reflect the pancreatic tail, spleen and splenic vessels medially.

3 · Parenchyma and its imaging

- Cortex is ≈1 cm. **Columns of Bertin** carry interlobar arteries, so the safe track is **through pyramid and papilla**.
- Glomerular filtration rate (GFR) is ≈**125 mL/min**; ≈99% of filtrate is reabsorbed.

Modality	Normal kidney
Ultrasound	Cortex iso-/hypoechoic to liver (hyperechoic in newborn)
Computed tomography (CT)	Unenhanced 30–60 Hounsfield units (HU), cortex = medulla; enhanced 80–120 HU
Magnetic resonance (MRI)	T1 cortex brighter; T2 cortex slightly darker

CT phase	Timing	Use
Arterial	20–30 s	Vessels
Corticomedullary	30–70 s	Cortex enhances
Nephrographic	**80–120 s**	**Best for renal masses**
Excretory	>3 min	Collecting system, ureter

- On intravenous urography, no excretion at 24 h means a non-functioning kidney.
- **Corticomedullary differentiation** is lost in chronic kidney disease and acute kidney injury and returns with recovery.

Congenital anomalies

- **Malrotation** shows a central pelvis with calyces medial to it. In **ectopia**, poor drainage causes stasis, infection and stones.
- **Horseshoe**: lower poles fuse and ascent is arrested by the inferior mesenteric artery (**IMA**); it brings ureteropelvic junction obstruction (UPJO), stones, infection and multiple arteries.

4 · Renal vessels, lymphatics and nerves

End arteries make partial nephrectomy possible.

LAQ Renovascular hypertension: pathophysiology, evaluation and treatment

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment" (asked as "Pathophysiology of renovascular hypertension (RVH) and iscemic nephropathy. Describe treatment of RVH in detail", Summer 2022 Paper II Q2; "Discuss evaluation of & management of reno-vascular hypertension", Practice Set 2 Paper II Q1; "Anat. of Renal Vasculature + Dx modalities & Rx optns for Renovasc. HTN.", Practice Set 4 Paper II Q2) · P1 LAQ "Pathophysiology of Renovascular hypertension" (Practice Set 8 Paper 1 Q2)

"Iscemic" is read as ischaemic nephropathy.

Anatomy of the renal vasculature

Renal arteries

- Arise from the aorta at the **L1–L2 disc**. The longer **right artery passes behind the IVC (inferior vena cava)**. Branches supply the adrenal, renal pelvis and proximal ureter.

- **Hilum, anterior to posterior:** renal vein, renal artery, pelvis; the posterior segmental artery lies behind the pelvis.
- **Anterior division** gives four segmental branches (apical, upper, middle, lower). **Posterior division** gives the posterior segmental artery: the first, most constant branch, passing behind the pelvis. If it crosses in front of the ureter, it can cause UPJO (ureteropelvic junction obstruction).
- **Five segmental arteries are end arteries** with no significant anastomosis. Occlusion causes segmental infarction; each segment can be resected separately.
- **Sequence of branching:** 1. Segmental 2. Lobar 3. Interlobar (columns of Bertin) 4. Arcuate (corticomedullary junction) 5. Interlobular 6. Afferent arteriole, glomerulus, efferent arteriole 7. Peritubular capillaries or vasa recta
- **Brödel line:** avascular plane between anterior and posterior territories, just behind the lateral border. Used for anatrophic nephrolithotomy and posterior calyceal access.
- **Variations (25–40%):** accessory arteries in 25–28% (T11–L4, rarely iliac or SMA [superior mesenteric artery]). They are end arteries, usually to the lower pole. Relevant to donor nephrectomy and endopyelotomy.

Renal veins

- Veins **anastomose freely** (venous collars, subcapsular plexus), so segmental vein ligation is safe. 3–5 segmental veins form the renal vein, anterior to the artery.

	Right renal vein	**Left renal vein**
Length	2–4 cm	6–10 cm
Tributaries	None	Left adrenal, left gonadal, lumbar
Course	Direct to IVC	Between SMA and aorta (nutcracker syndrome)
Anomalies	Accessory veins commoner	Circumaortic (2–16%); retroaortic

Relevance to RVH (renovascular hypertension): atheroma affects the **ostium and proximal third**; **FMD (fibromuscular dysplasia)** affects the **mid-distal artery and branches**; a stenosed accessory artery can cause segmental RVH.

Definitions

- **RVH:** hypertension caused by RAS (renal artery stenosis) that is relieved by revascularisation or nephrectomy. RAS is common in older (vascular) adults; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from haemodynamically significant RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

Causes

	Atherosclerotic RAS	**FMD**
Share	Most (about 90%)	Most of the rest
Patient	Older, smokers, diabetic, vascular disease	Young women
Site	**Ostium and proximal third** (aortic plaque)	**Mid and distal artery, branches**; "string of beads" (medial fibroplasia, commonest type)

	Atherosclerotic RAS	FMD
Course	Progresses to occlusion; renal function lost	Rarely occludes
Treatment	Medical first	**Balloon angioplasty without stent**; good BP (blood pressure) response

Also know: Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), renal artery aneurysm, dissection, embolism, **Page kidney** (perinephric compression, e.g. subcapsular haematoma), transplant RAS.

The RAAS (renin–angiotensin–aldosterone system)

- **Renin** is released by juxtaglomerular cells of the afferent arteriole in response to: 1. low afferent arteriolar pressure (intrarenal baroreceptor); 2. low sodium chloride delivery to the **macula densa**; 3. sympathetic **β1** stimulation.
- Renin cleaves hepatic **angiotensinogen** to angiotensin I. **ACE (angiotensin-converting enzyme)**, mainly in lung endothelium, converts it to **angiotensin II**.
- **Angiotensin II (AT1 receptor) actions:** systemic arteriolar vasoconstriction; **aldosterone** release (sodium retention, potassium loss); proximal tubular sodium reabsorption; thirst and ADH (antidiuretic hormone); sympathetic facilitation; **efferent arteriolar constriction**, which maintains GFR; cardiac and vascular hypertrophy and fibrosis.
- Normally, rising BP and sodium suppress renin (negative feedback). In RVH the stenosed kidney "sees" low pressure, so this feedback fails.

Pathophysiology of RVH: Goldblatt models (basis of the tests)

- **Goldblatt (1934)** clipped the canine renal artery and produced sustained hypertension, reversed by early clip removal. Loop: low perfusion pressure releases juxtaglomerular renin, which produces angiotensin II (vasoconstriction) and aldosterone (sodium retention).

	Two-kidney, one-clip (2K1C)	One-kidney, one-clip (1K1C)
Human equivalent	Unilateral RAS, normal opposite kidney	Bilateral RAS or stenosed solitary kidney
Sodium	Normal contralateral kidney excretes the extra sodium (**pressure natriuresis**)	No normal kidney: sodium and water retained
Renin	**High** from stenotic side; contralateral suppressed	Rises, then returns to normal as volume expands
Type	**Renin (angiotensin)-dependent**	**Volume-dependent**
ACE inhibitor effect	BP falls; captopril test positive	Less BP effect; may cause **acute renal failure / acute kidney injury**

Phases of 2K1C hypertension

1. **Acute phase:** renin-dependent; ACE inhibitor or clip removal reverses it.
2. **Transition:** sodium retention and secondary hyperaldosteronism; renin falls towards normal.
3. **Chronic phase:** hypertension damages the contralateral kidney (nephrosclerosis) and resistance vessels. BP no longer depends on the stenosis, so **revascularisation cures less** (cure falls with duration). This explains poorer results in long-standing hypertension.

Other contributing mechanisms

- Renal afferent nerve signals raise **sympathetic activity**.
- Angiotensin II generates **oxidative stress** (reactive oxygen species), lowers **nitric oxide** and raises **endothelin**, worsening vasoconstriction.
- Inflammation and fibrosis; left ventricular hypertrophy and diastolic dysfunction.

Pathophysiology of ischaemic nephropathy

- The kidney receives about **25% of cardiac output** (P1: 20–25%), far more than its metabolic need. **Autoregulation** holds GFR until stenosis is severe (about **70% or more**, or a significant translesional pressure gradient).
- Beyond this, the stenotic kidney keeps its GFR by angiotensin II-mediated **efferent constriction**. Blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR: hence the **creatinine rise >30%** after these drugs.
- Chronic hypoperfusion causes hypoxia, inflammation, microvascular rarefaction, **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive injury add damage.
- Early disease is reversible ("hibernating" kidney); established fibrosis is not.
- **Flash pulmonary oedema:** bilateral RAS causes volume retention plus angiotensin II-driven diastolic dysfunction.

Evaluation: clinical clues (who to test)

- Onset <30 years (FMD) or severe hypertension after 55 years; **resistant** (≥3 drugs) or accelerated hypertension; sudden worsening of controlled BP.
- **Creatinine rise >30% after ACE inhibitor/ARB**; **flash pulmonary oedema**; kidneys differing in size by >1.5 cm; abdominal bruit; **hypokalaemia** (secondary hyperaldosteronism).

Evaluation: basic tests

- Creatinine and eGFR (estimated GFR), potassium, urinalysis and proteinuria, ECG; exclude primary aldosteronism and phaeochromocytoma.

Evaluation: imaging and functional tests

Test	Method and finding	Comment
Duplex Doppler US (ultrasound)	High PSV (peak systolic velocity), above about 180–200 cm/s; renal/aortic velocity ratio >3.5; **RI (resistive index) = (PSV − EDV)/PSV**	First line, no contrast; operator dependent. **RI ≥0.8 (>0.8) means parenchymal disease and predicts poor response**
CT angiography	Stenosis, ostial plaque, beads	Best non-invasive anatomy; contrast nephropathy
MR (magnetic resonance) angiography	Gadolinium-enhanced	Caution if eGFR <30 mL/min

Test	Method and finding	Comment
Captopril renography (DTPA, diethylenetriaminepentaacetic acid)	Day 1 after oral captopril: delayed uptake, low peak; repeat off ACE inhibitor and calcium channel blocker. Improvement of ≥10% = high probability RVH	Removing angiotensin II (efferent constriction) drops GFR in the stenosed kidney, proving renin-dependence; DTPA is cleared purely by filtration; use declining
Catheter angiography	Gold standard; **translesional (pressure) gradient** (systolic ≥20 or mean ≥10 mmHg = significant)	Done when intervention is planned
Renal vein renin	Ratio >1.5 lateralises	Before nephrectomy

Algorithm `Clinical clue -> Duplex US (or CTA/MRA if eGFR allows) -> no RAS: treat as essential hypertension -> RAS: FMD -> angioplasty Atherosclerotic -> optimal medical therapy -> high-risk features + viable kidney -> angiography + gradient -> stent`

Treatment 1: medical (first line for atherosclerotic RAS)

- **ACE inhibitor or ARB** controls renin-dependent BP; check creatinine and potassium at 1–2 weeks. Calcium channel blockers, diuretics.
- **Statin, antiplatelet, smoking cessation, diabetes control**: most deaths in these patients are cardiovascular.

Treatment 2: landmark trials (atherosclerotic RAS)

- **ASTRAL (2009):** stenting plus medical therapy versus medical therapy alone gave **no benefit** in renal function, BP, events or mortality; procedural complications.
- **CORAL (2014):** stenting versus optimal medical therapy gave **no reduction** in cardiovascular or renal events; only a small BP fall.
- Hence **no routine revascularisation** of atherosclerotic RAS (ESC [European Society of Cardiology] 2017).

Treatment 3: indications for revascularisation

- **FMD with hypertension** (first choice).
- Atherosclerotic RAS with **flash pulmonary oedema** or recurrent unexplained heart failure, **rapidly declining function** with bilateral or solitary-kidney RAS, or truly resistant hypertension.
- Kidney must be viable (not small and atrophic; RI <0.8).

Treatment 4: endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.
- **Ostial atherosclerotic lesion: angioplasty plus stent**, because a balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.
- Complications: contrast nephropathy, **atheroembolism**, dissection, rupture, thrombosis, access haematoma.

Treatment 5: surgical revascularisation

- Indications: failed endovascular therapy, **branch disease or aneurysm**, children, coexisting aortic surgery.

Technique	Use
Aortorenal bypass (saphenous vein; internal iliac artery in children; PTFE [polytetrafluoroethylene])	Standard
Transaortic endarterectomy	Bilateral ostial plaque
Splenorenal (left) / hepatorenal (right) bypass	Hostile (diseased) aorta
Ex vivo bench repair + autotransplantation	Branch FMD, aneurysms
Nephrectomy (partial or total)	Small non-functioning kidney with lateralising renin

Transplant RAS

- Raised velocity on Doppler; contrast imaging confirms; angioplasty or stent first, surgery if this fails.

Follow-up

- BP, creatinine, duplex at intervals for restenosis; cure is commoner in FMD than atherosclerosis.

Diagrams to draw

- 2K1C versus 1K1C with the renin–angiotensin loop (low pressure, renin, angiotensin II, aldosterone, hypertension).
- RAAS cascade from renin release to angiotensin II actions.
- Renal artery dividing into anterior (four segments) and posterior divisions, with the Brödel line.

> **Key points to remember:**
>
> - Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
> - Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
> - Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
> - Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
> - Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
> - RVH is proved only by cure; duplex first, angiography with gradient before intervention.
> - ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
> - FMD gets balloon angioplasty without stent; ostial atheroma: stent.
> - Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

Also, from the condensed book (not in the paper answer):

- **CT angiography** is the gold standard for vascular anatomy but misses vessels <2 mm.
- Right-sided lymph drains to interaortocaval/paracaval nodes but **may cross to left para-aortic** nodes.

5 · Percutaneous puncture and vessels

*See the paper answer **Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL** (Section 15, Ch 88).*

Also, from the condensed book (not in the paper answer):

- A through-and-through puncture hits anterior segmental vessels and causes marked haemorrhage.

> **KEY POINT**
>
> Why the fornix? Arteries run on both faces of the infundibula, and a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Vascular injury causes haemorrhage, arteriovenous fistula, pseudoaneurysm and loss of function.

6 · Pelvicalyceal system

*See the paper answer **Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL** (Section 15, Ch 88).*

Also, from the condensed book (not in the paper answer):

- Dilatation is mimicked by an extrarenal pelvis, **megacalycosis** and a calyceal diverticulum.

7 · Ureter: course and relations

Injuries occur at crossings; stones stick at narrowings.

- The adult ureter is 22–30 cm. Stones impact at three **narrowings**: ureteropelvic junction (UPJ), iliac crossing and ureterovesical junction (UVJ).
- It lies on psoas and crosses the **common iliac bifurcation** (landmark), with gonadal vessels in front. The **pelvic brim** is a commonest injury site.
- In females it passes under the **uterine artery 1.5–2 cm** from the cervix ("water under the bridge"). In males the vas loops over it.
- The **intramural** segment is 1.2–2.5 cm long. Bladder pressure closes this oblique tunnel, which prevents reflux.

8 · Radiologic anatomy of the ureter

- Because of **peristalsis**, the whole ureter is rarely opacified; a continuous column suggests obstruction or atony.
- A stricture needs **proximal dilatation**; crossing vessels and iliac indentation can mimic it.
- Incomplete duplication (one bud, one orifice) is commoner than complete duplication (two buds, two orifices).

KEY POINT

Weigert–Meyer rule: the upper-moiety orifice is **inferior and medial** and obstructs (ureterocele); the lower moiety refluxes (vesicoureteral reflux, VUR).

- Bilateral medial displacement suggests retroperitoneal fibrosis or pelvic lipomatosis.

9 · Ureteric blood supply, nerves, histology

Segment	Artery	Enters	Endoureterotomy
Proximal	Renal	Medially	**Lateral**
Middle	Common iliac	Posteriorly	**Anterior**
Distal	Superior vesical	Laterally	**Medial**

- The longitudinal anastomosis is fragile, so preserve adventitia and avoid lateral retraction to prevent ischaemic stricture.
- Peristalsis is **myogenic**, so denervation does not cause reflux. Pain (T11–L1) refers to flank, groin, scrotum or labia.
- In **UPJO**, the junction has excess longitudinal muscle and collagen.

10 · Endoscopic anatomy

- The trigone is the bladder's most vascular part. Orifices lie 1–2 cm from the midline and move laterally on filling.

Lyon axis	Grades
Position	A medial trigone (normal) · B · C lateral
Configuration	0 volcano (normal) · 1 stadium · 2 horseshoe · 3 golf-hole

- A higher grade means more reflux. The **intramural ureter is narrowest** (3–4 mm) and may need dilatation.
- The lower-pole infundibulum needs ≈**140°** deflection.

ALGORITHM

1. Infundibulotomy: incise superior, then inferior, then posterior quadrants.
2. **Never anterior**.

90 Fundamentals of Nephrology

1 · Why a urologist needs nephrology

- Each kidney has ≈**1 million nephrons**, fixed at birth. When nephron mass falls (nephrectomy, agenesis), the rest **hyperfilter**.
- Interstitial fibroblasts make **erythropoietin**; the proximal tubule makes **calcitriol** by 1α-hydroxylation.

> **KEY POINT**
>
> The glomerulus is a pressure filter set by arteriolar tone; the tubule is energy-hungry and near-hypoxic. Most disorders fail one or the other.

2 · Measuring kidney function

Serum creatinine (SCr) often misleads in surgical patients.

SAQ GFR estimation in prospective renal donor

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "GFR estimation in a prospective renal donor and creatinine clearance" (asked as "Describe methods to estimate GFR in a prospective renal donor", Summer 2019 Paper I Q3a; "Creatinine clearance", Practice Set 4 Paper I Q3d) · P2 SAQ "GFR estimation: clinical assessment and in the prospective transplant donor" (asked as "Methods for GFR estimation in prospective tx donor", Practice Set 9 Paper II Q3a; "GFR clinical assessment", Winter 2023 Paper II Q3a)

Assumed meaning: tx = transplant (prospective living kidney donor).

Definition

- GFR (glomerular filtration rate) is the plasma volume filtered by both kidneys per minute; normal about 120 mL/min/1.73 m². It is the standard index of kidney function.
- The ideal marker is freely filtered and neither secreted nor reabsorbed.

Why it matters in the donor

- The donor loses about half the nephron mass; function settles at 60–80% of the pre-donation level.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** sets GFR thresholds and asks for **measurement, not just estimation**, at a decision point: GFR is assessed by at least two methods, **including a measured one**.

Method 1: serum creatinine and eGFR (estimated GFR)

- Creatinine comes from muscle and is freely filtered (not reabsorbed), but generation depends on **muscle mass**, diet (cooked meat, creatine supplements) and age. It is also secreted, so it **overestimates GFR**, increasingly as function falls. Trimethoprim blocks secretion.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021 creatinine equation:** age and sex, **no race term**; currently recommended; used for staging. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate above 60 mL/min, where donors lie: a screening test only.

Method 2: cystatin C

- 13-kDa protein made by all nucleated cells; freely filtered, then fully reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- The **combined creatinine–cystatin C equation** is more accurate than either alone; it is the most accurate estimate.

Method 3: measured 24-hour creatinine clearance

- **Definition:** creatinine clearance (CrCl) is the volume of plasma cleared of creatinine per minute. It estimates GFR because creatinine is freely filtered and not reabsorbed.
- Normal about 120 mL/min/1.73 m^2 in young men, slightly lower in women; falls with age.
- **CrCl = (U × V) / P**: U = urine creatinine, V = urine flow (mL/min), P = plasma (serum) creatinine.
- **Method:** discard the first morning void, collect all urine for 24 hours including the next morning's first void; take serum creatinine during the collection.
- **Completeness check:** creatinine excretion should be **20–25 mg/kg/day in men, 15–20 mg/kg/day in women**.

Limitations of creatinine clearance

- **Tubular secretion** of creatinine is small in health but rises as function falls, so CrCl **overestimates GFR** in CKD (chronic kidney disease).
- Collection errors are common. Creatinine generation depends on muscle mass, diet and age.
- Drugs blocking secretion (**trimethoprim, cimetidine**) raise serum creatinine without real injury.
- Invalid in AKI (acute kidney injury).
- **Also know:** cimetidine pretreatment blocks secretion to improve accuracy; averaging urea and creatinine clearance helps in advanced CKD.

Estimated clearance: Cockcroft–Gault

- **CrCl = (140 − age) × weight (kg) / (72 × serum creatinine mg/dL)**, × 0.85 in women.
- Used for **drug dosing**; not normalised to body surface area; inaccurate in obesity.

Method 4: mGFR (measured GFR), the reference standard

Marker	Method / comment
Inulin	Urinary clearance; gold standard, research only
Iohexol	Plasma clearance; safe and accurate (another source: plasma or urinary clearance)
Iothalamate	Urinary or plasma clearance
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling, or gamma camera (Gates method); purely filtered; also gives **split function**

Method 5: split function

- DTPA renogram gives each kidney's share (differential function); MAG3 or DMSA (dimercaptosuccinic acid) are alternatives.
- A difference >10% (beyond 45:55) matters: **leave the better kidney with the donor**.

KDIGO 2017 donor decision

- Initial **eGFRcr**; confirm with **mGFR, measured creatinine clearance or eGFRcr-cys** (creatinine–cystatin C eGFR).

GFR (mL/min/1.73 m²)	Decision
≥90	Accept
60–89	Individualise (age, lifetime risk)
<60	Decline

- Indian practice: **DTPA GFR** with split function plus CT (computed tomography) angiography is standard.

Clinical uses of GFR and creatinine clearance

- **CKD staging (KDIGO):** G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m², always with albuminuria (ACR, albumin-to-creatinine ratio).
- **Living donor evaluation:** KDIGO 2017 table above; confirm with measured GFR.
- **Urology:** drug and contrast dosing; **cisplatin eligibility** (CrCl <60 mL/min usually unfit); continent diversion (>35 mL/min); partial versus radical nephrectomy.
- Extremes of body size, amputees and pregnancy, where eGFR is unreliable: measure creatinine clearance.
- **Acute change:** eGFR is invalid when function is unstable; AKI uses creatinine change and urine output. NGAL (neutrophil gelatinase-associated lipocalin) is emerging.

Diagrams to draw

- Plasma tracer clearance curve after a single injection.
- Nephron showing filtration plus a small secreted fraction of creatinine.

Key points to remember:

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

Also, from the condensed book (not in the paper answer):

- SCr is falsely low with small muscle mass, and falsely raised by **trimethoprim and cimetidine**, which block secretion. Cystatin C is raised by hyperthyroidism, steroids and inflammation.

KEY POINT

Trap: after cystectomy or bladder repair, a rising SCr with ascites may be an **intraperitoneal urine leak** reabsorbed across the peritoneum, not kidney injury.

*See the paper answer **Proteinuria** (Section 1, Ch 3).*

Also, from the condensed book (not in the paper answer):

- **eGFR is invalid when function is changing**, so acute kidney injury (AKI) is defined by SCr change and urine output.

3 · Acute kidney injury

Glomerular haemodynamics

- Net filtration pressure ≈**10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32.

Arteriolar change	Renal blood flow	GFR
Afferent constriction	Falls	Falls
Afferent dilatation	Rises	Rises
Efferent constriction	Falls	Rises
Efferent dilatation	Rises	Falls

Angiotensin II blockade therefore drops GFR when perfusion is low.

Definition and staging (KDIGO)

Stage	Serum creatinine	Urine output
1	Rise ≥**0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†)	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or SCr ≥4 mg/dL with acute rise ≥0.5 (†or dialysis started)	<0.3 mL/kg/h for 24 h, or anuria 12 h

LAQ Acute renal failure (AKI) and acute tubular necrosis

Paper answer · P2

Asked as:

- Etiology, pathogenesis and management of acute renal failure. — *Winter 2024 Paper II Q2 (LAQ)* (acute tubular necrosis also answered in 2022-Summer Paper 2, Q3c)
- Acute tubular necrosis. — *Summer 2022 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Acute renal failure is now called AKI (acute kidney injury).

Definition and staging: KDIGO (Kidney Disease: Improving Global Outcomes)

- AKI (per KDIGO) = rise in serum creatinine **or** fall in urine output; either alone is enough.

Stage	Serum creatinine	Urine output
1	Rise ≥0.3 mg/dL in 48 h, or ×1.5 baseline in 7 days	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or ≥4 mg/dL, or RRT (renal replacement therapy)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- One-fifth to one-third of inpatients; mortality about 25%, about 50% (about half) if RRT is needed.

- **ATN definition:** ATN is the commonest **intrinsic** cause of AKI (acute kidney injury): tubular injury from **ischaemia or toxins**. A misnomer: cell death is mainly **apoptosis**.

Etiology

Type	Causes
Prerenal	Hypovolaemia (bleeding, vomiting, burns), heart failure, sepsis, hepatorenal syndrome; worsened by NSAIDs (non-steroidal anti-inflammatory drugs) and ACE (angiotensin-converting enzyme) inhibitors
Intrinsic	**ATN (acute tubular necrosis):** ischaemia, contrast, aminoglycosides, cisplatin, myoglobin. **AIN (acute interstitial nephritis):** β-lactams, NSAIDs, PPIs (proton pump inhibitors), checkpoint inhibitors. Glomerulonephritis; vascular (renal artery or vein thrombosis, atheroemboli)
Postrenal	BOO (bladder outlet obstruction), bilateral ureteric obstruction or obstructed single kidney: stones, tumours, retroperitoneal fibrosis, ureteric injury at surgery

- Prerenal AKI and ATN cause about three-quarters of hospital AKI.

Causes of ATN

Ischaemic	Toxic
Prolonged prerenal state: shock, sepsis, haemorrhage, burns, cardiac surgery, intraoperative hypotension, renal artery clamping, **transplant (delayed graft function)**	**Exogenous:** aminoglycosides, amphotericin, cisplatin, iodinated contrast. **Endogenous:** myoglobin (rhabdomyolysis, e.g. prolonged flank position), haemoglobin, light chains

Pathogenesis (pathophysiology)

- **Autoregulation** holds GFR (glomerular filtration rate) by afferent dilatation (prostaglandins, myogenic reflex) and efferent constriction (angiotensin II, tubuloglomerular feedback).
- **Prerenal:** low perfusion lowers glomerular pressure. Tubules are intact, so they **conserve sodium and water**. NSAIDs block afferent dilatation; ACE inhibitors block efferent constriction. Prolonged prerenal state becomes ATN.
- **Ischaemic ATN:**
- **Why tubules:** the medulla works near hypoxia (medullary oxygen tension is only 10–20 mmHg), while the **S3 proximal tubule (segment)** and **medullary thick ascending limb** consume much oxygen and are most vulnerable.
- **Loss of polarity:** Na^+/K^+-ATPase loses its basolateral position (moves off the basolateral membrane) → sodium reabsorption fails → high sodium at macula densa → **tubuloglomerular feedback** constricts afferent arteriole (afferent constriction) → GFR falls (protective).
- Cells die (mostly apoptosis) and slough (dead cells slough) → **muddy-brown casts** block (obstruct) tubules; filtrate leaks back; endothelial injury and inflammation worsen hypoxia; autoregulation is lost.
- Phases (clinical course): initiation → **maintenance** (oliguria, rising creatinine) → **recovery** (polyuria: tubules regenerate before concentrating ability returns).
- **Toxic ATN:** vasoconstriction plus direct tubular injury. **Contrast:** risk with age, diabetes, CKD (chronic kidney disease). **Rhabdomyolysis** (long lateral positioning): myoglobin causes toxicity, cast obstruction and hypoperfusion.

- **Postrenal:** raised tubular pressure, then afferent vasoconstriction; recovery depends on duration (dog studies of complete obstruction: full at 1 week, 70% at 2 weeks, none at 6 weeks).

Evaluation (diagnosis: ATN versus prerenal)

- History (fluid loss, hypotension, drugs, contrast), volume status, bladder scan.
- **Ultrasound early** in unexplained AKI to exclude obstruction.

Index	Prerenal	ATN
Urine sodium	<20 mmol/L	>20 mmol/L
FeNa (fractional excretion of sodium)	<1%	**>1%**
FeUrea (on diuretics)	<35%	>35%
Specific gravity	>1.015	<1.010
Sediment	Bland	**Muddy-brown granular casts**, tubular cells
Fluid challenge (response to fluids)	Quick (rapid) recovery	Lag of days to weeks

- Creatine kinase; white cells (AIN); red cell casts (glomerulonephritis); biopsy only if recovery lags (fails).

Management

General

- Treat (remove) the cause; **avoid hypotension** (MAP, mean arterial pressure, >65 mmHg or near baseline) and nephrotoxins; stop NSAIDs and ACE inhibitors or ARBs (angiotensin receptor blockers) (RAAS (renin–angiotensin–aldosterone system) blockers); adjust drug doses.
- Isotonic crystalloid (fluids) for hypovolaemia; aggressive fluids in rhabdomyolysis; intravenous fluid before contrast (pre-contrast hydration).
- **Postrenal:** urethral or suprapubic catheter; **double-J stent or PCN (percutaneous nephrostomy)**; infected obstructed kidney is an emergency.
- AIN: stop drug ± steroids; glomerulonephritis: nephrology and biopsy.

Complications

- **Hyperkalaemia:** calcium gluconate 10 mL of 10% (protects heart) → insulin 10 U with 50 mL 50% dextrose, salbutamol nebuliser (shift) → binders, diuretics or dialysis (remove).
- **Volume overload:** loop diuretics in escalating doses. **Diuretics (only for volume overload) do not treat ATN.**
- Acidosis: bicarbonate. Uraemic bleeding: desmopressin 0.3 µg/kg.

Dialysis indications (AEIOU), refractory to medical therapy

	Indication
A	Acidosis
E	Electrolytes, mainly hyperkalaemia
I	Intoxications / ingestions (lithium, salicylate, alcohols)
O	Overload with respiratory distress

	Indication
U	Uraemia: encephalopathy, pericarditis, bleeding

- Intermittent haemodialysis via 10–12 Fr jugular or femoral catheter; **CRRT (continuous RRT)** if haemodynamically unstable. **STARRT-AKI (2020):** early start gave no survival benefit.

Post-obstructive diuresis

- Causes: retained urea (osmotic diuresis), volume expansion with ANP (atrial natriuretic peptide), and down-regulated aquaporin-2 and sodium transport.
- Up to 300 mL/h for about 2 days. Free oral fluids; if unable to drink, replace about 75% of output with 0.45% saline; electrolytes every 4–6 hours. Gradual bladder decompression has no role.

Outcome

- Full or partial recovery, or dialysis dependence; persistence ≥90 days = CKD. ATN recovery takes days to weeks.
- **Also know:** (early) biomarkers NGAL (neutrophil gelatinase-associated lipocalin), KIM-1 (kidney injury molecule-1); AKD (acute kidney disease, under 3 months); gadolinium (groups I and III) avoided in AKI.

Diagram to draw: tubular cell losing polarity → sloughing → casts and back-leak.

> **Key points to remember:**
> - KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
> - FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
> - S3 segment and thick ascending limb are most vulnerable.
> - Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
> - Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Also, from the condensed book (not in the paper answer):

- **3 months** separates acute kidney disease (AKD) from CKD.

Prerenal AKI

- Urine is bland, fractional excretion of sodium (FeNa) is low, and function **recovers immediately** on correction.
- **Autoregulation**: the **myogenic reflex** dilates the afferent arteriole via prostaglandins, so it is **disabled by NSAIDs** (non-steroidal anti-inflammatory drugs).
- In **tubuloglomerular feedback (TGF)**, angiotensin II constricts the efferent arteriole, so it is **disabled by renin–angiotensin–aldosterone system (RAAS) blockade**.
- **Modifiable** factors are hypotension, anaemia, sepsis, nephrotoxins, rhabdomyolysis and hypercalcaemia.

*See the paper answer **Acute renal failure (AKI) and acute tubular necrosis** (Section 15, Ch 90).*

Also, from the condensed book (not in the paper answer):

- High FeNa after diuretics does not prove ATN; use fractional excretion of urea (FeUrea). **Response to correction is the single most important test.**
- Keep mean arterial pressure (MAP) >**65 mm Hg**. **Fluids** (mOsm/L): plasma and 0.9% saline 308; Plasmalyte 294; Ringer's lactate 273; 0.45% saline 154; 5% dextrose (D5) 0, so D5 is free water.

Intrinsic AKI: the cast localises the compartment

	Acute tubular necrosis (ATN)	Acute glomerulonephritis (GN)	Acute interstitial nephritis (AIN)
Cause	Ischaemia, toxins	Immune complex, anti-glomerular basement membrane (anti-GBM), cell-mediated	Drug, infection, systemic
Urine	Granular casts	Dysmorphic red cells, **red cell casts**, proteinuria	White cells, **white cell casts**, with or without eosinophils

Acute tubular necrosis

- **Rhabdomyolysis** (obese patient, long flexed lateral robotic case) needs **early aggressive isotonic fluid**; alkalinisation is unproven. †Prevent it with padding and limited flexion time.
- **Give diuretics only for symptomatic overload; they do not treat ATN.**

Postcontrast AKI

- This is an SCr rise within **48 h** of intravenous (IV) iodinated contrast. Risks include CKD, diabetes and the arterial route.
- **IV fluid before contrast is the only effective prevention** (†for eGFR <30 or AKI); use the lowest dose.
- **Gadolinium** groups I and III cause nephrogenic systemic fibrosis (**NSF**), so avoid them in AKI, eGFR <**30** and dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) is allowed. **Group III (gadoxetate) is treated as group I.**

Acute interstitial nephritis

- **Drugs cause most**, notably β-lactams, NSAIDs, proton pump inhibitors and **checkpoint inhibitors**.
- Suspect it in unexplained hospital AKI with **sterile pyuria**. Urine eosinophils are insensitive and non-specific MCQ.
- **Stop the drug**; early steroids may help, and about half recover fully.

Glomerulonephritis

- In microhaematuria, a rising SCr, proteinuria, red cell casts, hypertension or purpura call for **urgent nephrology and biopsy**, not cystoscopy first.

Complications of AKI

- **Overload** is urgent only with respiratory distress. Give a high-dose IV loop diuretic and add metolazone or acetazolamide (**sequential blockade**).
- For uraemic bleeding, give desmopressin (**DDAVP**) **0.3 µg/kg** IV/IM 30–60 min before the procedure.

Renal replacement therapy

- AEIOU (acidosis, electrolytes, ingestions, overload, uraemia) means failed medical management, not a biochemical threshold. **Early dialysis gives no benefit.** Continuous therapy is not superior.
- Non-recovery for ≥**90 days** means CKD or end-stage kidney disease (ESKD).

4 · Drug dosing in reduced kidney function

- In AKI, eGFR overestimates function, so reassess doses daily.
- †**Cockcroft–Gault** = (140 − age) × weight (kg) / (72 × SCr mg/dL), × 0.85 if female.
- Avoid NSAIDs; †withhold RAAS blockers in dehydration or perioperative hypotension ("sick-day" rule).
- †Avoid nitrofurantoin if eGFR <30 (UK guidance allows short courses at 30–44). †Stop metformin at eGFR <30, and withhold it before iodinated contrast in AKI or eGFR <30.

5 · Chronic kidney disease

- **KDIGO** (Kidney Disease: Improving Global Outcomes) defines CKD as abnormal structure or function for ≥**3 months**. Classify by **cause, GFR and albuminuria**.

GFR (mL/min/1.73 m²)	A1 albumin-to-creatinine ratio (ACR) <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G1 ≥90	Low*	Moderate	High
G2 60–89	Low*	Moderate	High
G3a 45–59	Moderate	High	Very high
G3b 30–44	High	Very high	Very high
G4 15–29	Very high	Very high	Very high
G5 <15 (kidney failure)	Very high	Very high	Very high

*G1–G2 count as CKD only with albuminuria or structural damage. Albuminuria can shift risk two bands: eGFR 43 with ACR 1100 mg/g is **G3bA3, very high**.

- **Each perioperative AKI is a modifiable progression factor**.

SAQ Metabolic abnormalities in CKD

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 2, Q3e

Definition

- **CKD (chronic kidney disease), KDIGO (Kidney Disease: Improving Global Outcomes):** abnormal kidney structure or function for **≥3 months**. Staged by GFR (glomerular filtration rate) G1–G5 (G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m²) and albuminuria A1–A3.
- Each abnormality is a lost kidney function: excretion, acid–base balance, hormone production.

Abnormalities

Abnormality	Mechanism	Management
Sodium and water	Fixed urine concentration; sodium retention → overload, hypertension	Salt restriction; diuretics (chlorthalidone works even in advanced CKD)
Hyperkalaemia	Less distal sodium delivery; RAAS (renin–angiotensin–aldosterone system) blockers lower aldosterone	Potassium binders and diuretics, so RAAS blockers can continue
Metabolic acidosis	Less ammonium excretion; retained sulphate and phosphate (anion gap rises late); bicarbonate loss	Sodium bicarbonate or citrate to keep bicarbonate **≥22 mEq/L**; slows progression, protects bone and muscle
CKD-MBD (mineral and bone disorder)	Phosphate retention → FGF-23 (fibroblast growth factor 23) rises → less **calcitriol** → hypocalcaemia → **secondary hyperparathyroidism**	Phosphate restriction, binders with meals, active vitamin D; calcimimetic; parathyroidectomy if tertiary
Anaemia	Erythropoietin falls, mainly below GFR 30	Iron first, then erythropoietin; Hb (haemoglobin) target 10–11.5 g/dL (harm above 13)
Hyperuricaemia	Reduced excretion	Treat gout; asymptomatic lowering unproven (CKD-FIX)
Lipids	High triglycerides, low HDL (high-density lipoprotein)	Statin ± ezetimibe if ≥50 years, not on dialysis (SHARP)
Glucose	Insulin resistance, but slower insulin clearance → hypoglycaemia	Lower insulin doses

Bone disease (renal osteodystrophy)

- **High turnover:** osteitis fibrosa (PTH [parathyroid hormone] excess).
- **Low turnover:** adynamic bone (oversuppressed PTH), osteomalacia.
- **Tertiary hyperparathyroidism:** autonomous gland, hypercalcaemia; persists after transplant.
- **Vascular calcification** adds cardiovascular death.

Other endocrine and uraemic effects

- **Also know:** hypogonadism, hyperprolactinaemia, erectile dysfunction (better after transplant); growth failure in children; protein-energy wasting; dialysis amyloid; uraemic platelet dysfunction (desmopressin).
- **Urological relevance:** dose drugs by GFR; avoid gadolinium if eGFR (estimated GFR) <30; each AKI (acute kidney injury) speeds progression.

Diagram to draw: CKD-MBD cascade: phosphate ↑ → FGF-23 ↑ → calcitriol ↓ → calcium ↓ → PTH ↑ → bone disease.

> **Key points to remember:**
> - FGF-23 rises first; secondary then tertiary hyperparathyroidism.
> - Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
> - Hyperkalaemia: binders rather than stopping RAAS blockade.

Also, from the condensed book (not in the paper answer):

- Tertiary hyperparathyroidism (HPT) is treated with a calcimimetic, then subtotal parathyroidectomy.

Slowing progression

- Target glycated haemoglobin <7.5% and blood pressure <**130/80**; the agent matters only in proteinuric disease (RAAS blockade).
- **Sodium–glucose cotransporter-2 (SGLT2) inhibitors** protect kidney and heart and cut mortality, with or without diabetes. Add **finerenone** in CKD with type 2 diabetes.

6 · Haematuria and proteinuria: when to involve a nephrologist

- †Microhaematuria is ≥**3** red blood cells per high-power field (RBC/HPF) on microscopy; a positive dipstick needs confirmation.
- **Glomerular clues** are dysmorphic red cells, red cell casts, raised ACR or protein-to-creatinine ratio (PCR), rising SCr, hypertension and systemic disease.

ALGORITHM

1. Haematuria with or without proteinuria: send urine microscopy, ACR, SCr/eGFR.
2. If there are casts, dysmorphic cells, AKI or systemic features, refer for **urgent nephrology**.
3. If isolated non-glomerular haematuria, do a urological evaluation.
4. If isolated albuminuria, stage CKD and refer per criteria.

- †KDIGO referral criteria: AKI or an abrupt GFR fall; GFR <30; ACR ≥300 mg/g (≥30 mg/mmol); a fall >5 mL/min/1.73 m²/year; red cell casts or haematuria >20 RBC/HPF; BP refractory to ≥4 drugs; persistent potassium abnormality; recurrent stones; hereditary disease.

7 · Electrolyte and acid–base disorders

Water and sodium

- **Sodium disorders are water disorders**. **Volume depletion overrides hypo-osmolality** in driving antidiuretic hormone (ADH).
- Of 1 L, D5 leaves **83 mL** in the circulation, 0.9% saline 250 mL and 5% albumin 1000 mL.

Hyponatraemia

- †Transurethral resection (TUR) syndrome is absorption of hypotonic glycine or sorbitol during monopolar resection.

Volume	Urine	Causes
Hypovolaemic	Na >20	Diuretics, mineralocorticoid deficiency, cerebral salt wasting
Hypovolaemic	Na <20	Vomiting, diarrhoea
Euvolaemic	Osm > serum	Syndrome of inappropriate ADH (SIADH), cortisol deficiency, hypothyroidism, pain, drugs
Euvolaemic	Osm low	Polydipsia, low solute ("tea and toast", beer potomania)
Hypervolaemic	Na >20	Kidney failure

Volume	Urine	Causes
Hypervolaemic	Na <20	Heart failure, cirrhosis, nephrotic

- Treat SIADH with water restriction, then vaptans, salt with furosemide, or urea.
- **Rate**: in chronic hyponatraemia, raise sodium ≤**8–12 mEq/L per 24 h**, or **osmotic demyelination** follows. Treat overcorrection with DDAVP and free water.
- With severe symptoms, give **3% saline to raise sodium 4–6 mEq/L**, then slow down.

Hypernatraemia

- Nephrogenic diabetes insipidus (DI) follows lithium, hypercalcaemia, hypokalaemia and **relief of obstruction**.
- Give free water. In chronic cases, lower sodium ≤**0.5 mEq/L/h**, or cerebral oedema follows.

Potassium

SAQ Management of acute hyperkalaemia

Paper answer · P1

Asked as:

- Management of Acute Hyperkalemia — *Winter 2025 Paper I Q3e (SAQ)*
- Mgt. of acute hyperkalemia. — *Practice Set 6 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

(also answered in 2025-Winter Paper 1, Q3e)

Definition and physiology

- Serum potassium **>5.5 mmol/L**; mild 5.5–5.9, moderate 6.0–6.4, **severe ≥6.5** (UK Kidney Association). Above 6.0 is generally urgent.
- About 98% of potassium is intracellular; the ratio sets the resting membrane potential, so a high level (high levels) causes **arrhythmia and cardiac arrest**.
- Defence: rapid shift into cells (insulin, β_2-agonists via Na^+/K^+-ATPase), then renal excretion (needs distal sodium delivery, urine flow, aldosterone).

Causes

Mechanism	Examples
Pseudohyperkalaemia	Haemolysis, tourniquet, fist clenching, thrombocytosis
Reduced excretion	AKI (acute kidney injury), CKD (chronic kidney disease), **obstructive uropathy** (type 4 RTA [renal tubular acidosis]), adrenal insufficiency; drugs: ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers), spironolactone, NSAIDs (non-steroidal anti-inflammatory drugs), tacrolimus
Shift out of cells	Acidosis, insulin deficiency, β-blockers, suxamethonium
Cell release	**Rhabdomyolysis** (prolonged lithotomy or flank position), tumour lysis, crush
Urological	**Jejunal conduit** (reabsorbs urinary potassium)

Assessment

- Repeat the sample to **exclude pseudohyperkalaemia** if the ECG (electrocardiogram) is normal.

- **ECG changes:** peaked T (earliest) → prolonged PR, wide QRS → loss of P → sine wave → ventricular fibrillation (VF). Ascending weakness.
- Creatinine, glucose, blood gas, CK (creatine kinase); stop potassium sources.

Emergency treatment (in order)

Step	Drug and dose	Onset and effect
1. Protect the heart	**Calcium gluconate 10%, 10 mL IV (intravenous)** over 5–10 min; repeat if ECG unchanged	1–3 min; lasts 30–60 min; does not lower potassium
2. Shift into cells	**Insulin 10 units regular IV with 50 mL of 50% dextrose**	15–30 min; lowers about 0.6–1 mmol/L; check (monitor) glucose (hypoglycaemia)
	Salbutamol 20 mg nebulised	Additive to insulin
	Sodium bicarbonate	Only with metabolic acidosis
3. Remove	Loop diuretic if passing urine; **binders**: sodium zirconium cyclosilicate (about 1 h), patiromer (about 7 h), sodium polystyrene sulfonate (slow; colonic necrosis)	Hours
	Haemodialysis	Refractory or oliguric, ESKD (end-stage kidney disease)

- **Relieve obstruction** (catheter, nephrostomy); watch for post-obstructive diuresis.
- Monitor ECG, potassium and glucose; treat the cause.

Diagram to draw: flowchart: ECG change or K^+ ≥6.5 → calcium → shift → remove.

> **Key points to remember:**
> - Calcium stabilises the membrane but does not lower potassium.
> - Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
> - Binders, diuretics and dialysis remove potassium; relieve obstruction.

ALGORITHM

1. **Stabilise** if there is any ECG change: calcium gluconate 10%, 10 mL IV.
2. **Shift**: insulin 10 U with 50 mL of 50% dextrose; nebulised salbutamol 20 mg.
3. **Remove**: haemodialysis if oliguric or ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; a binder.
4. Stop culprit drugs.

- **Sodium polystyrene sulfonate (SPS)** is slow and causes **colonic necrosis**, so it is not first-line.
- **Hypokalaemia** from **vomiting** is caused by kidney loss, not gastric loss. Replace potassium chloride, **oral preferred**, 20–40 mEq repeated.

Calcium

- Saline and loop diuretics raise urine calcium; **thiazides lower it**.
- **Hypercalcaemia in a stone former**: non-suppressed PTH means primary HPT; if suppressed, check PTH-related peptide (renal, bladder, squamous cancer) and 1,25-vitamin D (sarcoid, lymphoma). Low urine calcium suggests familial hypocalciuric hypercalcaemia.

- Treat if symptomatic or ≥**14 mg/dL**: **isotonic saline** 200 mL/h to urine 100–150 mL/h; loop diuretic only if overloaded; calcitonin; IV bisphosphonate; denosumab.

Acid–base

- Anion gap (AG) = Na − (Cl + HCO_3); it is raised if >12. If the AG rises less than the bicarbonate falls, a non-AG acidosis coexists.
- **Urine AG** = (Na + K) − Cl. Negative means ammonium excretion, so gut loss; positive means a renal cause (renal tubular acidosis, RTA).

*See the paper answer **Renal tubular acidosis (distal and type 2)** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- **Topiramate and zonisamide** cause calcium phosphate stones. MCQ Laxative abuse leads to **ammonium acid urate stones**.
- †Bowel conduits and neobladders absorb ammonium chloride, causing hyperchloraemic non-AG acidosis.

8 · Urinary tract obstruction and post-obstructive diuresis

Relief of obstruction can cause dangerous fluid loss.

Term	Meaning
Hydronephrosis	Pelvicalyceal dilatation, with or without obstruction
Obstructive uropathy (OU)	Obstruction at any level
Obstructive nephropathy	Irreversible parenchymal loss

- OU is **curable**, so exclude it (†by ultrasound) in any unexplained AKI or CKD.

*See the paper answer **Post-obstructive diuresis** (Section 4, Ch 51).*

ALGORITHM

1. Relieve the obstruction, then allow free oral fluids; do not give routine IV fluid.
2. If output stays high, or there is hypovolaemia, or the patient is obtunded or nil by mouth, then check a panel every **4–6 h** (Na, K, phosphate, Ca, Mg).
3. Replace ≈**75%** of the output with **0.45% saline** plus electrolytes.

KEY POINT

Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement sustains it.

Recovery after obstruction

Canine study	Model	Recovery
Vaughan and Gillenwater	Complete unilateral ureteral obstruction	100% at 7 days; 70% at 14 days; 30% at 4 weeks; **none at 6 weeks**

- Duration, not dilatation, decides the injury. A small stone with low-grade obstruction can be observed safely for **2–4 weeks**; beyond 4 weeks the outcome is unpredictable.
- Keep a **low threshold for a stent or nephrostomy**. †Infection with obstruction needs urgent drainage.

9 · Landmark trials

Trial	Design	Result	Take-home
†PRESERVE	Bicarbonate vs saline; N-acetylcysteine (NAC) vs placebo	No difference	Saline; no NAC
†STARRT-AKI	Accelerated vs standard renal replacement therapy	No survival benefit; more dialysis dependence	Dialyse for AEIOU, not early
†DAPA-CKD / EMPA-KIDNEY	SGLT2 inhibitor, CKD with or without diabetes	Less progression	Standard in CKD
†FIDELIO-DKD	Finerenone, CKD with type 2 diabetes	Fewer kidney and cardiovascular events	Add to RAAS blockade
†CHOIR / TREAT	High vs lower haemoglobin target	Harm with high target	Avoid Hb >13 g/dL

91 Renal Transplantation

1 · End-stage disease and why transplant

Transplant beats dialysis, so plan it early.

- KDIGO stage **G5 (eGFR <15 mL/min/1.73 m²)** is end-stage renal disease (ESRD).
- **Transplant is optimal therapy**, and **pre-emptive** transplant (before dialysis) is best. It may triple life expectancy, but death risk is **higher for the first 3–6 months**, then much lower (Wolfe).

2 · Recipient evaluation

> **KEY POINT**
>
> Two questions: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)?

- Deceased-donor listing is allowed at eGFR ≤**20**.

Recurrence risk	Diseases	Implication
Very low	Autosomal dominant polycystic kidney disease (ADPKD), dysplasia, Alport, **genetic** focal segmental glomerulosclerosis (FSGS)	None
Higher	**Primary FSGS**, membranoproliferative GN, membranous	Counsel; monitor
Histological often, clinical rarely	IgA nephropathy	—
Metabolic	Primary hyperoxaluria	Combined liver–kidney

Disease coded in native-kidney genes cannot recur in the graft; disease from circulating or immune factors can.

- **Cardiac**: coronary artery disease is the main cause of death; screen high-risk patients.
- **Vascular**: heavy iliac calcification may prevent anastomosis (non-contrast CT). ADPKD with a family history of intracranial bleed needs an aneurysm screen.
- **Obesity** is a relative contraindication that increases wound complications, delayed graft function (DGF) and rejection.
- **Infection**: active tuberculosis contraindicates until treated. Screen for hepatitis B, C and HIV, and **vaccinate before** transplant, because live vaccines are barred afterwards.

Malignancy	Transplant
Prostate Gleason ≤6; superficial non-melanoma skin cancer; incidental renal tumour <1 cm	Without prior treatment
Basal/squamous skin, melanoma in situ, renal cell carcinoma <3 cm, small low-grade thyroid, superficial bladder cancer	No wait after cure
All other	After treatment, individualised wait

SAQ Pre transplant nephrectomy

Paper answer · P2

Type: SAQ (10 marks)
Source: 2022-Summer Paper 2, Q3e

Assumed meaning: removal of the native kidney(s) before or at kidney transplantation.

Principle

- **Default: leave native kidneys.** They give residual urine (less fluid restriction), erythropoietin and vitamin D activation; nephrectomy adds morbidity.
- Remove them only when they will threaten the patient or graft under **immunosuppression**.

Indications

Group	Indication	Reason
Infection	Chronic pyelonephritis, infected stones, recurrent upper UTI (urinary tract infection) with **high-grade VUR (vesicoureteral reflux)** or obstruction, XGP (xanthogranulomatous pyelonephritis)	Sepsis under immunosuppression
ADPKD (autosomal dominant polycystic kidney disease)	No room for graft (kidney below iliac crest), recurrent cyst infection or bleeding, pain, early satiety	Space and symptoms
Malignancy	Suspicious mass or complex cyst (acquired cystic disease), high tumour risk (e.g. Denys–Drash)	Most RCC (renal cell carcinoma) in recipients arises in native kidneys
Hypertension	Uncontrolled despite multiple drugs	Renin-driven
Heavy proteinuria	Nephrotic syndrome with malnutrition or thrombosis (e.g. congenital nephrotic syndrome)	Hides graft proteinuria, hypercoagulability
Others	Massive polyuria, large stone burden	Graft fluid balance, infection

Timing

- **Before transplantation:** infection or tumour; reconstruction 6–12 weeks earlier.
- **Simultaneous:** ADPKD needing space, ipsilateral through the transplant incision; avoided in small children.
- **After transplantation:** symptomatic ADPKD once the graft works.

Technique

- **Laparoscopic** (transperitoneal or retroperitoneal), open, or **embolisation** in high-risk patients.
- **PD (peritoneal dialysis) patients:** retroperitoneal route keeps PD going; transperitoneal surgery needs temporary haemodialysis.
- Bilateral: prone approach for small kidneys; morcellation or open for large ADPKD kidneys.
- **Ureter:** transect high and leave a normal ureter (for later ureteroureterostomy); **remove a dilated refluxing ureter completely**; keep it as a possible continent stoma if catheterisation may be needed.

Complications

- Bleeding, transfusion (sensitisation), infection, bowel injury, hypotension after bilateral removal, loss of residual function.

Also know: pre-transplant urodynamics; bladder augmentation if capacity <75% expected or pressure >40 cmH_2O.

Diagram to draw: enlarged ADPKD kidney reaching the iliac fossa beside the planned graft site.

> **Key points to remember:**
> - Default: leave native kidneys.
> - Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
> - Retroperitoneal route for PD; leave a normal ureter.

3 · Immunological compatibility

LAQ Discuss the role of tissue typing and cross match in live and cadaveric

Paper answer · P3

Type: LAQ (25 marks)
Source: 2018-Summer Paper 3, Q2

The question is incomplete; it is taken to mean "... live and cadaveric (deceased-donor) renal transplantation".

Introduction

- Two immune barriers decide compatibility: **preformed antibodies** (anti-A/B isoagglutinins and anti-HLA (human leukocyte antigen) antibodies) and **T-cell responses** to mismatched HLA.
- **Tissue typing** measures how well donor and recipient HLA match. **Antibody screening and crossmatch** detect preformed antibodies before the kidney meets them. Together they prevent **hyperacute rejection**, now practically eliminated.

The HLA system

- HLA is the human MHC (major histocompatibility complex), a gene cluster on **chromosome 6 (short arm)**. Genes are inherited as two **haplotypes** (one from each parent) and expressed codominantly.

	Class I	**Class II**
Loci	HLA-A, -B, -C	HLA-DR, -DQ, -DP
Expression	All nucleated cells	Antigen-presenting cells, endothelium
Presents to	CD8 (cytotoxic) T cells	CD4 (helper) T cells

- **Sibling** chance: 25% two-haplotype identical, 50% one-haplotype, 25% none. A parent or child always shares one haplotype.

Tissue typing methods

Method	Principle	Note
Serological (microlymphocytotoxicity)	Lymphocytes + typed antisera + complement; dead cells take up dye	Low resolution; largely replaced
Molecular (DNA-based)	PCR (polymerase chain reaction) with SSP (sequence-specific primers), PCR-SSO (sequence-specific oligonucleotide probes), sequence-based typing, **NGS** (next-generation sequencing)	High (allele-level) resolution; now standard

- **Matching:** usually reported as mismatches at **A, B and DR** (0–6). **DR matters most**, then B, then A; DQ increasingly typed. Fewer mismatches mean better graft survival and less sensitisation for a future transplant (Collaborative Transplant Study, Opelz). The effect is larger in deceased-donor grafts.

ABO compatibility

- Same rules as transfusion: **O universal donor, AB universal recipient**.
- **A2 (non-A1) donors** can go to O or B recipients with low anti-A titre.
- **ABO-incompatible** living transplant is possible: lower titres (plasmapheresis or immunoadsorption, rituximab, IVIG (intravenous immunoglobulin)) to a centre target before surgery.

Sensitisation and antibody screening

- Anti-HLA antibodies come from **pregnancy, blood transfusion, previous transplant** (and infection).
- **PRA (panel reactive antibody):** recipient serum tested against a panel of lymphocytes; result as % of panel killed.
- **Solid-phase (Luminex) single-antigen beads:** identify each antibody and its strength (MFI, mean fluorescence intensity). Antibodies against a donor are **DSA (donor-specific antibodies)**.
- **cPRA (calculated PRA):** % of donors the patient would be incompatible with, calculated from listed "unacceptable antigens". It replaced cell-panel PRA for allocation; very high cPRA gets allocation priority.

Crossmatch

Type	Method	Meaning if positive
CDC (complement-dependent cytotoxicity)	Donor T and B lymphocytes + recipient serum + complement; cell death scored. **DTT** (dithiothreitol) removes IgM autoantibody; AHG (antihuman globulin) increases sensitivity	**Positive T-cell CDC with IgG DSA = absolute contraindication** (hyperacute rejection; Patel and Terasaki, 1969)
Flow cytometry crossmatch	Antibody binding on donor cells detected by fluorescent anti-IgG	More sensitive; CDC-negative/flow-positive = higher risk of AMR (antibody-mediated rejection); desensitise or exchange
Virtual crossmatch	Compare recipient's bead-defined antibodies with donor HLA type	Speeds deceased-donor allocation; confirm by cell-based crossmatch

- **B-cell positive** alone: class II antibody or non-specific binding; interpret with bead results.

- Crossmatch uses a **recent serum** plus historical peak sera.

Living versus deceased donor

	Living donor	Deceased (cadaveric) donor
Typing	ABO and HLA early in donor work-up; choose best of several donors	ABO and HLA at donor referral
Crossmatch	At work-up and a **final crossmatch shortly before surgery**	At the time of offer, often virtual first; done against the list rapidly (cold ischaemia)
Role of matching	Less critical: even unrelated living donors do better than well-matched deceased donors (Terasaki, 1995)	Allocation uses HLA match, sensitisation (cPRA), waiting time and donor quality (KDPI, Kidney Donor Profile Index)
If incompatible	**Paired (swap) exchange**, desensitisation, ABO-incompatible protocol	Patient waits; acceptable-mismatch programmes

- **India:** swap (paired) donation is legal under the Transplantation of Human Organs (THOA) amendment, 2011.

Managing the incompatible or sensitised recipient

- **Kidney paired donation**, including chains with altruistic donors.
- **Desensitisation:** plasmapheresis + low-dose IVIG, or high-dose IVIG, ± rituximab; **imlifidase** (cleaves IgG) for highly sensitised deceased-donor recipients (conditional EU approval). Desensitised HLA-incompatible living transplant gave better survival than waiting on dialysis (Montgomery, 2011).

After transplant

- Monitor for **de novo DSA** (especially class II) with dd-cfDNA (donor-derived cell-free DNA) and biopsy; de novo DSA predicts graft loss.

Recent advances

- **Eplet (epitope) matching**, high-resolution NGS typing, virtual crossmatch, imlifidase.

Diagram to draw: CDC crossmatch: donor lymphocytes + recipient serum + complement, dead cells take up dye (positive).

> **Key points to remember:**
> - HLA on chromosome 6; match A, B, DR; DR matters most.
> - Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
> - Flow and bead assays are more sensitive; cPRA guides allocation.
> - Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

4 · Deceased donation

LAQ Deceased donor kidney: selection, maintenance and retrieval techniques

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval" (asked as "Describe selection criteria for cadaver renal donor. Describe in details kidney retrieval surgery", Summer 2021 Paper II Q2; "Describe procedure of

identification, maintenance, distribution and retrieval of kidney in a 25 yr old brain dead pt", Practice Set 3 Paper II Q1) · P1 SAQ "Kidney retrieval techniques for organ donation" (2023-Winter Paper 1 Q3c)

Assumed meaning: the Indian deceased-donor pathway: identification and brain-stem death certification, donor maintenance, allocation and retrieval. Marginal donor: see 2016-Summer Paper 3, Q3f; brain death certification: 2018-Summer Paper 4, Q3c; also answered in 2022-Summer Paper 4, Q3b.

Types of donor

- **Deceased (cadaver): DBD** (donation after brain death: circulation maintained until cold perfusion) or **DCD** (donation after circulatory death: warm ischaemia).
- **Living:** related or unrelated (THOA, Transplantation of Human Organs Act, in India).
- A 25-year-old brain-dead patient is an ideal **standard-criteria DBD** donor: circulation continues until cold perfusion, so warm ischaemia is minimal.
- India: **THOA 1994**, amended 2011, **Rules 2014**, makes brain-stem death legal death; **NOTTO** (National Organ and Tissue Transplant Organisation) allocates organs.

Identification and selection criteria

Potential donor: ventilated patient with catastrophic brain injury (head injury, intracranial bleed, hypoxia). ICU (intensive care unit) informs the **transplant coordinator**; **mandatory request** to the family.

Donor evaluation

- Cause of death, medical and social (high-risk behaviour) history; kidney: terminal creatinine, urine output, hypertension, diabetes, ultrasound.
- Laboratory: ABO and HLA (human leucocyte antigen) typing; serology and NAT (nucleic acid testing) for HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), EBV (Epstein–Barr virus), syphilis, toxoplasma.

Absolute contraindications

- Active or metastatic cancer, haematological malignancy, past melanoma, untreatable viral infection.
- Treated bacterial infection (even bacterial meningitis) is acceptable; viral encephalitis needs care.

Standard versus marginal donor

Category	Definition
ECD (expanded criteria donor), UNOS (United Network for Organ Sharing)	Age ≥60, or 50–59 with 2 of: hypertension, creatinine >1.5 mg/dL, death from CVA (cerebrovascular accident)
KDPI (Kidney Donor Profile Index)	Replaced ECD in the US (2014); >85% = marginal; longevity matching with EPTS (Estimated Post-Transplant Survival)
Paediatric <15 kg	En bloc transplant

- **Remuzzi biopsy score** for marginal kidneys: 0–3 single, 4–6 dual, ≥7 discard.
- **Also know:** HCV (hepatitis C virus)-viraemic donors with antivirals.

Legal death and consent: certification of brain-stem death (THOA)

- **DBD preconditions:** apnoeic coma with a known irreversible cause; exclude drugs (sedatives, relaxants), hypothermia, metabolic causes and shock.

- **Board of four (4-member Board):** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon (else a nominated surgeon, physician, anaesthetist or intensivist); treating doctor. **None from the transplant team.**

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI
Oculovestibular (caloric)	Ice-cold water into each ear (drum seen): no eye movement	VIII, III, VI
Cranial motor response	Supraorbital pressure: no grimace	V, VII
Gag and cough	Touch pharynx; suction catheter in trachea: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**; **Form 10**. Ancillary tests (angiography, perfusion scan) only if examination is impossible.
- **DCD (Maastricht):** I dead on arrival; II failed resuscitation (both uncontrolled); **III arrest after withdrawal of support (controlled, commonest)**; IV arrest in a brain-dead donor; V euthanasia.
- **Consent:** next-of-kin (near relative) consent after a **mandatory request** by ICU staff; in medico-legal cases inform police.

Donor maintenance

- **Why:** herniation causes a **catecholamine storm**, then loss of sympathetic tone (hypotension), **diabetes insipidus (DI)**, hypothermia, DIC (disseminated intravascular coagulation), low thyroid hormone and cortisol, and pulmonary oedema.
- **"Rule of 100s":** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Measures:** fluids guided by CVP (central venous pressure), no starch colloids (nephrotoxic); **vasopressin** first (pressor and treats DI), then noradrenaline; methylprednisolone 15 mg/kg, thyroid hormone, insulin; lung-protective ventilation; warming; correct sodium, potassium and clotting.

Distribution (allocation)

- **NOTTO**, New Delhi, with **ROTTOs** (regional) and **SOTTOs** (state organisations), keeps registries and waiting lists.
- **Sequence:** hospital and state list, then region, then national.
- **Kidney criteria:** ABO compatibility, waiting time on dialysis, HLA match, priority for children and previous living donors, **negative crossmatch** (donor spleen or nodes). Many states let the retrieving hospital keep one kidney.
- 2023 policy removed the domicile requirement and upper age limit for registration.

Deceased multi-organ kidney retrieval (DBD)

1. **Checks:** death certificate, consent, blood group; coordinate thoracic, liver, pancreas and kidney teams.
2. **Incision:** midline xiphoid-to-pubis laparotomy with median sternotomy.
3. **Exposure:** mobilise right colon and small-bowel mesentery off the retroperitoneum (Cattell–Braasch manoeuvre); expose aorta and IVC (inferior vena cava).
4. **Prepare cannulation:** encircle the distal aorta above the bifurcation; ligate the inferior mesenteric artery; prepare the supracoeliac aorta for clamping. **Look for and preserve aberrant (accessory) renal arteries**; keep accessory arteries from the iliacs.
5. **Cold perfusion:** systemic heparin; cannulate the distal aorta; clamp the supracoeliac aorta; **vent** the IVC (or right atrium); flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; crushed (surface) ice; open Gerota's fascia laterally to cool the kidneys.
6. **Remove heart, lungs, liver and pancreas first.**
7. **Kidneys en bloc (last):** lift both with aorta, IVC, perinephric fat and ureters off the posterior wall; keep periureteric tissue; divide ureters low. Divide aorta and IVC (upper limit set by the liver team, lower at the bifurcation); lift everything anteriorly and divide prevertebral tissue.
8. **Extras:** iliac vessels for reconstruction; spleen and lymph nodes for crossmatch.

DCD: after death and a no-touch period, **super-rapid** laparotomy, direct aortic cannulation and IVC venting limit warm ischaemia; kidneys usually go on machine perfusion.

Back-bench (back-table) preparation

- Separate the kidneys: open the aorta posteriorly to see the renal ostia and divide it between them, keeping an aortic **Carrel patch** for each.
- Keep the IVC with the right kidney to lengthen the short right renal vein.
- Ligate tributaries and lymphatics (prevents lymphocele); trim fat without decapsulation; spare the lower pole–ureter triangle.
- Document anatomy and injuries; biopsy marginal kidneys; triple-bag in cold solution on ice at 4 °C; label.

Living donor nephrectomy

- **Principles:** donor safety first; maximal vessel length; keep periureteric tissue and the lower pole–ureter triangle; minimal handling; short warm ischaemia.
- **Left kidney preferred** (longer vein).

Technique	Features
Open (flank or mini-incision)	Complex anatomy, more often right side
Laparoscopic (Ratner, 1995)	Preferred; less pain, faster recovery
Hand-assisted	About two-thirds of US cases; tactile control
Retroperitoneoscopic, LESS (laparoendoscopic single-site), robotic	Selected centres

- **Sequence:** 1. Hilar dissection. 2. Ureter divided. 3. Artery then vein stapled or clipped. 4. Rapid extraction. 5. Cold flush.

- **Risks:** complications about 16%; mortality 3 per 10,000; conversion 0.26–3.34%.

Preservation

- Cooling to 4 °C slows metabolism. **Hypothermic machine perfusion** reduces DGF (delayed graft function) versus static cold storage (Moers, 2009), but not long-term graft loss.
- Keep cold ischaemia short: cold ischaemia >24 hours raises thrombosis risk.
- Normothermic perfusion is experimental.

Pitfalls

- Polar artery or intimal injury, short right vein, ureteric devascularisation, capsular tears.

Diagram to draw

- Aorta and IVC with distal aortic cannula, supracoeliac clamp, IVC vent and en bloc kidneys.

Key points to remember:

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Also, from the condensed book (not in the paper answer):

- En bloc kidneys go to an adult with body mass index <30 and low immunological risk; thrombosis is 10–15%.

	Donation after brain death (DBD)	**Donation after circulatory death (DCD)**
Warm ischaemia	Minimal	Present, so rapid aortic cannulation and cooling are needed
Preservation	Static cold storage	Usually machine perfusion (reduces DGF)

ALGORITHM

1. Midline incision and sternotomy; mobilise bowel off the retroperitoneum.
2. Control the aorta and inferior vena cava (IVC).
3. Find aberrant or accessory arteries **before cannulation**.
4. Perfuse; ice via opened Gerota fascia.
5. Remove kidneys en bloc with **periureteric tissue preserved**.
6. Separate on the back table; store cold.

5 · Living donation

SAQ Evaluation and selection criteria of the living renal donor

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P2 SAQ "Pre-operative evaluation of the living (related and unrelated) renal donor" (asked as "Pre operative investigations for renal donor", Summer 2017 Paper II Q3b; "Pre-op assessment of unrelated donor", Practice Set 8 Paper II Q3a) · P3 SAQ "Assessment criteria for kidney donor." (2022-Summer Paper 3 Q3a) · P1 SAQ "Extended Criteria for Selection of Live Renal Donor" (2025-Winter Paper 1 Q3a)

Assumed meaning: evaluation and acceptance criteria for a living kidney donor; for the unrelated question, a living unrelated kidney donor (Indian context); "extended criteria" read as the expanded (marginal, "medically complex") living donor. Related: 2023-Winter Paper 4, Q3e; marginal donor in 2024-Winter Paper 4, Q3e.

Principles

- **Donor safety first. KDIGO (Kidney Disease: Improving Global Outcomes) 2017** replaced fixed cut-offs with a **risk-based approach**: the donor's estimated lifetime ESKD (end-stage kidney disease) risk is compared with the centre's acceptable threshold.
- Multidisciplinary team with an **independent donor advocate**.
- For an unrelated donor the medical work-up equals that of a related donor; the extra step is **legal and ethical**: proof of genuine, unpaid motive.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011** (Rules 2014): near relatives may donate; others need authorisation committee approval; swap donation allowed.

Legal assessment of the unrelated donor (THOA)

- A donor "other than near relative" donates for **affection or attachment**: **Authorisation Committee approval is mandatory**.
- Committee: hospital-based if >25 transplants a year, otherwise district or state; members are not from the transplant team.
- Process: joint application; **video-recorded interview** of donor and recipient; verify identity, documents, financial status and long association; exclude middlemen and payment; confirm free, informed consent.
- Foreign recipient with unrelated Indian donor not approved; appeal within 30 days.

Medical assessment sequence

1. Compatibility (immunological)

- ABO group, HLA (human leukocyte antigen) typing, **crossmatch** (CDC [complement-dependent cytotoxicity] and flow), DSA (donor-specific antibody). Incompatible pairs: paired exchange.
- Unrelated donor: more HLA mismatch is expected, yet living unrelated grafts outperform deceased-donor grafts.

2. Kidney function and urine

- GFR (glomerular filtration rate) by two methods, one measured (e.g. DTPA [diethylenetriamine pentaacetic acid] GFR or creatinine clearance).

- Urinalysis, culture; persistent microscopic haematuria: cystoscopy, imaging, ± kidney biopsy.

3. Medical and metabolic

- Blood pressure (ambulatory if borderline), fasting glucose, **HbA1c (glycated haemoglobin)**, OGTT (oral glucose tolerance test) if at risk; lipids, liver tests, coagulation, BMI (body mass index).
- Active cancer, chronic infection and cardiovascular disease exclude.

4. Infection screen

- HIV (human immunodeficiency virus), HBsAg (hepatitis B surface antigen), anti-HBc, HCV (hepatitis C virus), CMV (cytomegalovirus), syphilis; **tuberculosis** screen.

5. Cardiopulmonary: ECG (electrocardiogram), chest X-ray, echo; stress test if indicated.

6. Age-appropriate cancer screening: Pap smear, mammogram, PSA (prostate-specific antigen), stool test or colonoscopy.

7. Urological: stones (metabolic work-up), haematuria, malignancy, recurrent UTI (urinary tract infection).

8. Anatomy

- **CT (computed tomography) angiography:** number of vessels, early branching, stones, masses.
- **Split function** (DTPA or DMSA [dimercaptosuccinic acid]) if asymmetric: the **better kidney stays with the donor**; left preferred (longer vein).

9. Psychosocial: voluntary, no coercion or payment; independent donor advocate.

Acceptance criteria: standard, extended and exclusion

Factor	Standard (accept)	Extended (individualise)	Exclude (decline)
Age	Adult	Older donors (often >60) acceptable; no upper limit	Under legal age
GFR, mL/min/1.73 m²	≥90	**60–89**	<60
Albuminuria	<30 mg/g ACR (P1: <30 mg/day)	**30–100** mg/g ACR (P1: mg/day)	>100 mg/g ACR (P1: mg/day)
Hypertension	None	Controlled on 1–2 drugs, no target-organ damage	Uncontrolled, organ damage
Obesity	BMI <30	**30–35**; individualised risk	Many centres: >35
Glucose	Normal	Prediabetes; selected type 2 diabetes	Type 1 diabetes
Stones	None	Single past stone, low recurrence risk; small stone removed ex vivo	Bilateral or recurrent; cystinuria, primary hyperoxaluria
Anatomy	Single vessels	Multiple arteries, simple cyst, mild FMD (fibromuscular dysplasia)	Major anomaly

- ACR = albumin–creatinine ratio.

- P2 (unrelated-donor answer) states that diabetes, significant obesity and uncontrolled hypertension exclude.

Extended criteria (marginal, medically complex living donor)

- A living donor with a risk factor that would once have excluded donation; used because of organ shortage.
- **Why older donors are acceptable:** their remaining lifespan is short, so lifetime ESKD risk is low. Young donors carry the highest lifetime risk. Healthy older donors are acceptable.
- **Also know:** isolated microscopic haematuria after negative work-up; treated low-risk cancer; APOL1 (apolipoprotein L1) high-risk genotype (test in African ancestry); future pregnancy (more pre-eclampsia in later pregnancy).
- **Safeguards:** measured GFR and split function (better kidney stays with the donor); CT angiography; laparoscopic retrieval; THOA authorisation; independent donor advocate.
- **Outcome:** grafts from older living donors do about as well as standard deceased-donor kidneys.

Counselling figures (Muzaale 2014)

- Function settles at 60–80% of baseline.
- Lifetime ESKD (ESRD, end-stage renal disease) 90 per 10,000 donors vs 14 in matched healthy non-donors; mortality 3 per 10,000.

Diagrams to draw

- Donor work-up flowchart: 1. Legal approval. 2. Compatibility. 3. Medical. 4. Anatomy. 5. Surgery.
- Risk-threshold scale (accept, individualise, decline).

> **Key points to remember:**
> - KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
> - GFR 60–89 and albuminuria 30–100 are individualised.
> - Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
> - ABO and crossmatch first; CT angiography for anatomy.
> - Better kidney stays with the donor.
> - THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
> - Interview video-recorded; no commerce.

Also, from the condensed book (not in the paper answer):

- Donor mortality is **3/10,000**. An independent donor advocate is mandatory.
- **Laparoscopic donor nephrectomy** is standard.

6 · The operation

- On the back table, ligate hilar lymphatics (lymphocele). **Never dissect the triangle between lower pole and ureter.** Extend a short right vein with donor IVC.

- **Multiple arteries** (18–43%) mean more DGF but similar survival; use a Carrel patch, conjoining or separate anastomoses.
- **Always keep the lower pole artery**, because it supplies the ureter; upper pole branches ≤2 mm may be ligated.
- The graft goes in the retroperitoneal iliac fossa. **Contralateral placement** brings the pelvis and ureter anterior.

	Gibson	Hockey stick
Course	Oblique, 2 cm above pubis to near the anterior superior iliac spine (ASIS)	Midline suprapubic, up the rectus edge
Access	Iliac vessels	Better to common iliac, IVC
Hernia	Lower	Higher; rectus denervation

ALGORITHM

1. **Vein first**: end-to-side to the external iliac vein, 5-0/6-0 polypropylene.
2. Artery end-to-side to the external iliac artery (punch arteriotomy; avoid calcified clamp sites).
3. Reperfuse, then position the kidney without kink or stretch.

- A flaccid graft suggests poor inflow and a tense, engorged graft poor outflow; check with intraoperative ultrasound.

ALGORITHM

1. **Lich–Gregoir** extravesical: fill the bladder; incise the detrusor down to mucosa.
2. Open the mucosa distally and spatulate the ureter.
3. Suture mucosa to mucosa with 5-0/6-0 absorbable, with or without a stent.
4. Close the detrusor loosely as a short tunnel.

- Document stent removal, because forgotten stents encrust.

7 · Surgical complications

Early vascular problems lose grafts within hours.

SAQ Vascular complications after renal transplantation, including transplant renal artery stenosis

Paper answer · P1

Asked as:

- Vascular Complications after Renal Transplantation in Recipient — *Winter 2025 Paper I Q3d (SAQ)*
- Transplant renal artery stenosis — *Practice Set 20 SAQ Q3d (SAQ)*

Type: SAQ (10 marks)

(vascular complications of transplantation also answered in 2025-Winter Paper 1, Q3d)

Introduction

- Uncommon but a major cause of **early graft loss**. Doppler ultrasound is the first test.

- **Concept:** no inflow gives anuria; blocked outflow gives a swollen, tender graft with haematuria; reduced inflow gives hypertension and fluid retention.

Main complications

	Arterial thrombosis	Venous thrombosis	TRAS (transplant renal artery stenosis)
Incidence	0.2–5%	0.1–8.2%	1–23% (commonest)
Timing	First days	First week	Months (usually 3 months–2 years)
Presentation	Sudden anuria	Painful swollen graft, haematuria, oliguria	Refractory hypertension, rising creatinine, flash pulmonary oedema
Doppler	No arterial flow	**Reversed diastolic flow**, clot in vein	High velocity at stenosis; **parvus-tardus** waveform distally
Treatment	Immediate exploration; thrombectomy (catheter thrombolysis described); nephrectomy if non-viable	Immediate exploration, thrombectomy; nephrectomy if necrotic	**Angioplasty or stenting** (first choice); surgical revision if that fails

- **Thrombosis risks:** extreme donor or recipient age, **cold ischaemia >24 hours**, multiple arteries, technical error (kink, intimal flap), hypovolaemia, rejection, thrombophilia (factor V Leiden, antiphospholipid antibodies).
- **TRAS causes:** anastomotic technique, clamp injury, recipient atherosclerosis (iliac inflow stenosis mimics it). Contrast angiography confirms.

Transplant renal artery stenosis (TRAS) in detail

Definition and incidence

- TRAS (transplant renal artery stenosis): narrowing of the graft artery or its inflow; **commonest vascular complication (1–23%)**, usually 3 months to 2 years after transplant.

Types and causes

- **Anastomotic:** suture technique, end-to-end anastomosis to the internal iliac artery, no aortic (Carrel) patch.
- **Post-anastomotic (distal):** clamp or perfusion-cannula intimal injury, kinking, rejection.
- **Pre-anastomotic ("pseudo-TRAS"):** recipient iliac atherosclerosis.
- **Also know:** CMV (cytomegalovirus) infection, delayed graft function, deceased donor.

Pathophysiology and clinical features

- Reduced graft perfusion activates the **renin–angiotensin–aldosterone system**.
- **Refractory hypertension**, rising creatinine, fluid retention, **flash pulmonary oedema**; graft bruit.
- Clue: creatinine rises sharply after an ACE (angiotensin-converting enzyme) inhibitor.

Evaluation

- **Doppler ultrasound first:** high peak velocity at stenosis (about >200–250 cm/s), and distal **parvus-tardus** waveform with low resistive index.

- MRA (magnetic resonance angiography) or CO_2 angiography to spare contrast.
- **Contrast angiography is the gold standard** and allows treatment at once. Iodinated contrast risks nephrotoxicity; gadolinium is avoided at low GFR (glomerular filtration rate).
- **Differential of graft dysfunction with hypertension:** rejection, calcineurin-inhibitor toxicity, ureteric obstruction.

Management

- **Medical:** for mild stenosis with stable function: blood pressure control (calcium-channel blockers), statin, antiplatelet; avoid or monitor ACE inhibitors.
- **Percutaneous angioplasty with or without stent: treatment of choice** for significant stenosis (usually >50–60%). High technical success; reverses hypertension and graft dysfunction. Stents reduce restenosis.
- **Surgical revision** (resection and re-anastomosis, interposition graft, bypass) if angioplasty fails or anatomy unsuitable; risks graft loss.
- Follow-up Doppler after angioplasty detects restenosis; untreated TRAS ends in graft loss.

Other vascular complications

- **Haemorrhage:** hilar vessels or anastomosis; re-explore if unstable.
- **Pseudoaneurysm (<1%):** technical or **mycotic**; rupture risk; endovascular or open repair, often graft nephrectomy, prolonged antibiotics.
- **Post-biopsy AV (arteriovenous) fistula:** persistent haematuria; usually closes; large or with steal: **coil embolisation**. Subcapsular haematoma can cause a **Page kidney**.
- **Graft torsion:** intraperitoneal grafts; prevent with nephropexy.
- **DVT (deep vein thrombosis)** of the iliac vein; **lymphocele** compressing the iliac or renal vein.
- **Limb ischaemia** from iliac dissection; internal iliac use: gluteal claudication, erectile dysfunction.

Prevention

- Anastomose away from atheroma; check the lie for kinking; avoid clamp injury and kinking; use a Carrel patch where possible.
- Extend a short right renal vein with donor IVC (inferior vena cava).
- Short cold ischaemia; anticoagulate thrombophilic patients; avoid hypovolaemia.
- At reperfusion, a soft kidney suggests poor inflow; a tense, engorged kidney suggests outflow block.

Diagram to draw: Doppler waveforms: normal, parvus-tardus (TRAS; distal to stenosis) and reversed diastolic flow (venous thrombosis).

> **Key points to remember:**
>
> - TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
> - Refractory hypertension plus rising creatinine: Doppler first.
> - Venous thrombosis: reversed diastolic flow, swollen tender graft.
> - Arterial thrombosis: sudden anuria; explore immediately.

Also, from the condensed book (not in the paper answer):

- Bleeding (brisk drain, falling haematocrit) needs transfusion and re-exploration.

SAQ Lymphocele and postoperative lymphatic leak (chylous ascites)

Paper answer · P1

Asked as:

- Lymphocele — *Practice Set 8 Paper I Q3a (SAQ)*
- Postoperative lymphatic leak — *Practice Set 14 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Postoperative lymphatic leak taken as lymphocele and chylous ascites after RPLND (retroperitoneal lymph node dissection), nephrectomy, pelvic node dissection and transplantation.

Definition and types

- **Lymphocele:** a collection of lymph **without an epithelial lining** (a pseudocyst with a fibrous wall), caused by divided lymphatics that keep leaking. It is the **commonest complication of PLND (pelvic lymph node dissection)**.
- **Chylous ascites:** lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks.

Settings and incidence

- **PLND** (radical prostatectomy, cystectomy): radiological in up to 60%, clinically significant 2–5%. An extended template roughly doubles the rate.
- **Renal transplantation:** from lymphatics around the recipient iliac vessels or donor hilum. Radiological 0.6–33.9% (symptomatic 0.03–26%); most appear within 6 months, peaking at 6 weeks.
- Also after RPLND (retroperitoneal lymph node dissection; lymphocele ≤1.7%), inguinal dissection, nephrectomy.
- **Chylous ascites:** RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection.

Type	Definition	Setting and frequency
Lymphocele	Lymph collection **without epithelial lining**	**PLND (pelvic lymph node dissection):** radiological up to 60%, significant 2–5%. **Transplant:** symptomatic 0.03–26%, peak 6 weeks. **RPLND:** ≤1.7%
Chylous ascites	Lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks	RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection

- **Risk factors:** extensive dissection, extended templates, suprahilar dissection near the cisterna chyli (L1–L2), post-chemotherapy surgery, obesity, heparin prophylaxis, mTOR (mammalian target of rapamycin) inhibitors, acute rejection, retransplantation (re-transplant).

Clinical features

- **Lymphocele:** most are silent (usually silent). Large ones compress:
- iliac vein: leg oedema, **DVT (deep vein thrombosis)**;
- bladder: frequency, urgency, pelvic pressure;

- ureter or graft: hydronephrosis, rising creatinine (graft dysfunction).
- Infection (may become infected): fever, rigors, sepsis.
- **Chylous ascites:** rising girth, early satiety, dyspnoea; milky drain fluid.

Diagnosis

- **Ultrasound:** anechoic, often septated, thin-walled collection. **CT (computed tomography)** maps it before intervention. Air or rim enhancement means infection.
- **Aspirate:** lymph has creatinine equal to (equals) serum (urinoma: much higher), lymphocytes; culture excludes abscess. Chyle: **triglyceride 2–8 × serum**, protein >3 g/dL.

Management

Lymphocele (treat by symptoms, not size alone)

Situation	Treatment
Asymptomatic, <5 cm	Observe; most resolve
Symptomatic	**Percutaneous drain** ± sclerotherapy (povidone-iodine, doxycycline)
Recurrent or thick-walled	**Laparoscopic fenestration (marsupialisation)** into a peritoneal window (best)
Infected	Antibiotics + percutaneous drainage; exclude urine leak
Laparoscopy unsafe	Open drainage

`Asymptomatic, <5 cm → observe Symptomatic → percutaneous drain (± sclerotherapy) Recurrent / thick-walled → laparoscopic peritoneal fenestration (best) Infected → drainage + antibiotics; exclude urine leak`

- **Success:** laparoscopic fenestration best, open next, drain or sclerotherapy intermediate, **aspiration alone worst** (reaccumulates, recurs, risks infection).
- Sclerosants (povidone-iodine, doxycycline) may hinder later fenestration.
- Fenestration lets the peritoneum absorb lymph (internal drainage): the definitive cure.
- **Lymphangiography with embolisation:** high success; long-term data awaited.

Chylous ascites (stepwise)

1. Paracentesis; drain if it reaccumulates.
2. **Very low-fat, high-protein diet with MCT (medium-chain triglycerides)**, which are absorbed directly into portal blood, bypassing lacteals; plus **octreotide**.
3. Nil by mouth with **TPN (total parenteral nutrition)**.
4. Refractory: lymphangiography with embolisation, peritoneovenous shunt or surgical ligation. - Continue a low-fat diet for **≥4 weeks** after resolution.

Prevention

- **Clip or ligate visible lymphatics** (clips often preferred; no robust data) (cephalad at the renal hilum in RPLND; perivascular at transplant). Spare lymphatics over the external iliac artery (leg lymph). Avoid unneeded suprahilar dissection.
- **Peritoneal interposition flap** fixed to the bladder (peritoneal flap after robotic PLND) reduces clinically significant lymphocele. Transperitoneal access is **not** protective (fluid loculates within 48–72 h).
- Drain after transplantation and major nodal dissections.

Diagram to draw: pelvic lymphocele compressing the external iliac vein and bladder, with a laparoscopic peritoneal window.

> **Key points to remember:**
> - Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
> - Lymph creatinine = serum; urinoma creatinine higher.
> - Laparoscopic fenestration best (is best); aspiration alone worst.
> - Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
> - Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Transplant nephrectomy

- Indications are **graft intolerance syndrome** (pain, swelling, haematuria, fever; steroids first), infection and tumour, not an asymptomatic failed graft.
- **Early (<6–8 weeks), go extracapsular**; later, go **subcapsular**, except for tumour.

8 · Rejection

Chronic antibody injury now limits graft survival.

- The usual regimen is induction, then tacrolimus and mycophenolate, with or without steroids.

LAQ Allograft rejection and immunobiology of renal transplantation

Paper answer · P1 · P4 · merged from 2

Asked as: P4 LAQ "Allograft rejection, immunobiology of renal transplant, marginal kidney donor and newer immunosuppressant drugs" (asked as "Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant", Winter 2017 Paper IV Q2; "Describe immunobiology of renal transplant and newer immunosuppressant drugs", Summer 2020 Paper IV Q1; "Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs", Practice Set 3 Paper IV Q1; "Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation", Summer 2015 Paper IV Q2) · P1 SAQ "Renal allograft rejection." (2020-Summer Paper 1 Q3d)

Definition and introduction

- Rejection is the recipient's immune attack on donor (allo) antigens, which the recipient's immune system recognises as foreign: **HLA (human leucocyte antigen)** class I (A, B, C) and II (DR, DQ, DP), ABO and non-HLA antigens.
- Immunosuppression has cut acute rejection from over 60% (pre-ciclosporin era) to **below 10%**. Slow, subclinical **chronic antibody-mediated injury** is now the main problem and the main cause of late graft loss.
- Organ shortage drives use of **marginal donors**. Their kidneys carry more DGF (delayed graft function) and shorter graft survival, but transplantation still beats dialysis.

Transplant (target) antigens

Antigen	Features
HLA = MHC (major histocompatibility complex)	Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells. Class II mismatch is most immunogenic

Antigen	Features
ABO blood group	On endothelium; incompatibility causes hyperacute rejection unless desensitised
Minor and non-HLA antigens	Rejection despite HLA identity

- **Sensitisation:** transfusion, pregnancy or previous transplant creates preformed anti-HLA antibodies.
- **Testing:** CDC (complement-dependent cytotoxicity) and flow crossmatch (donor lymphocytes with recipient serum); **Luminex single-antigen beads** define each recipient's antibodies and allow a **virtual crossmatch**; cPRA (calculated panel-reactive antibody) expresses sensitisation.

Innate immunity: the first trigger

- **Ischaemia–reperfusion injury** releases danger signals (DAMPs, damage-associated molecular patterns). These activate **Toll-like receptors**, complement and macrophages, and up-regulate MHC on the graft. Hence DGF raises rejection risk.
- **Dendritic cells** bridge innate and adaptive immunity: activated, they up-regulate MHC and costimulatory B7 and migrate to lymph nodes.

Allorecognition (how T cells see the graft)

Pathway	Mechanism	Main role
Direct	Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells (donor antigen-presenting cells)	Very strong (high T-cell frequency); early acute TCMR (T-cell-mediated rejection)
Indirect	Recipient APCs (antigen-presenting cells) present **processed donor peptides on self-MHC**	Lifelong; helps B cells make DSA (donor-specific antibody), so alloantibody and chronic rejection
Semi-direct	Recipient APCs acquire **intact donor MHC** (membrane transfer) and present it	Links both pathways

Three-signal model of T-cell activation

1. **Signal 1 (antigen):** TCR (T-cell receptor) binds MHC–peptide; **calcineurin** dephosphorylates NFAT (nuclear factor of activated T cells), which drives IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation): CD28 binds CD80/86 (B7)**. Signal 1 without signal 2 causes **anergy**. CTLA-4 (cytotoxic T-lymphocyte antigen-4) is the natural brake.
3. **Signal 3:** IL-2 binds **CD25** (IL-2 receptor) and signals through **mTOR (mammalian target of rapamycin)**, moving the cell cycle from G1 to S; clonal proliferation follows, which needs de novo purine synthesis.

Effector mechanisms

- **Cellular:** CD4 Th1 cells (IL-2, IFN-γ [interferon-γ]) recruit macrophages; CD8 cytotoxic T cells kill by perforin and granzyme, causing **tubulitis and endarteritis (arteritis)**.
- **Humoral:** B cells with T-cell help become plasma cells that make **DSA**. DSA binds endothelium and fixes complement, leaving **C4d** on peritubular capillaries, with **microvascular inflammation** (glomerulitis, capillaritis). NK (natural killer) cell injury is complement-independent, so AMR (antibody-mediated rejection) may be C4d-negative.

- **Regulation: Tregs** (regulatory T cells, FoxP3+) suppress responses; the basis of tolerance research.

Types of rejection

Type	Timing	Mechanism and features
Hyperacute	Minutes–hours	Preformed ABO/HLA antibody; complement, thrombosis; graft turns blue; **prevented by crossmatch**
Accelerated acute	2–5 days	Memory (anamnestic) response in sensitised recipients
Acute TCMR	Days–months (mostly first 6 months)	Interstitial infiltrate, tubulitis, arteritis
Acute AMR	Any time	DSA, C4d, microvascular inflammation (capillaritis)
Chronic active AMR	Months–years	**Transplant glomerulopathy** (GBM [glomerular basement membrane] double contours), arterial intimal fibrosis; commonest (main) cause of late graft loss
Subclinical	—	Histological rejection, stable function

- **Risk factors:** HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF, **non-adherence**, retransplant, previous rejection.

Diagnosis and monitoring

- **Clinical:** rising creatinine, oliguria; fever and graft tenderness are now rare.
- **Exclude:** obstruction, urine leak, lymphocele, renal artery stenosis or thrombosis (ultrasound with Doppler), CNI (calcineurin inhibitor) toxicity, BK virus nephropathy, recurrent disease.
- **Monitoring: de novo DSA** predicts graft loss; **dd-cfDNA (donor-derived cell-free DNA)** has negative predictive value >90% and avoids unnecessary biopsy; gene-expression panels.
- **Biopsy is the gold standard;** Banff grades the findings and classifies TCMR and AMR.

Banff classification

Category	Key criteria
Borderline	Foci of tubulitis with mild interstitial inflammation
TCMR IA / IB	Interstitial inflammation >25% with moderate (IA) or severe (IB) tubulitis
TCMR IIA / IIB	Mild-moderate (IIA) or severe (IIB) intimal arteritis
TCMR III	Transmural arteritis or fibrinoid necrosis
Active AMR	All three: (1) tissue injury (microvascular inflammation, arteritis, TMA [thrombotic microangiopathy]); (2) antibody–endothelium interaction (C4d, microvascular inflammation or gene transcripts); (3) **circulating DSA**

Treatment of rejection

- **TCMR:** IV (intravenous) pulse methylprednisolone; **rATG (rabbit antithymocyte globulin) / ATG** for Banff II–III or steroid-resistant rejection.

- **AMR** (not standardised): steroids, **plasmapheresis, IVIG (intravenous immunoglobulin)**, rituximab; eculizumab in selected patients; anti-IL-6 (interleukin-6) for chronic active AMR.
- Optimise maintenance (tacrolimus, mycophenolate, steroids); address adherence.

Marginal kidney donor: categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002: age **≥60**, or **50–59** with **≥2** of **hypertension**, **creatinine >1.5 mg/dL**, **death from CVA (cerebrovascular accident)**. About 70% higher risk of graft loss.

2. KDPI (Kidney Donor Profile Index): percentile of a 10-factor risk index. **KDPI >85%** replaced ECD in the US (December 2014). Low-KDPI kidneys go to recipients with the longest EPTS (Estimated Post-Transplant Survival): **longevity matching**.

3. DCD (donation after circulatory death): Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of therapy	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled

- Warm ischaemia needs rapid cannulation and cooling, and DCD kidneys usually go on machine perfusion. DGF is **higher (roughly 40–50%)**, but long-term survival is similar to DBD (donation after brain death).

4. Others: AKI (acute kidney injury) donors; **HCV (hepatitis C virus)-viraemic** donors into negative recipients with DAAs (direct-acting antivirals); **paediatric donors <15 kg** (en bloc); older or hypertensive living donors (KDIGO [Kidney Disease: Improving Global Outcomes] 2017 risk-based approach).

- **Absolute exclusions:** metastatic or haematological cancer, melanoma, untreatable viral infection. Treated bacterial infection is acceptable; screen by serology and nucleic acid testing.

Marginal kidney donor: assessment and optimisation

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, arteries, each 0–3): **0–3 single**, **4–6 dual kidney transplant**, **≥7 discard**.
- **Cold storage** at 4 °C in UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution slows metabolism.
- **HMP (hypothermic machine perfusion)** reduces DGF (**Moers, NEJM 2009**: about 21% vs 27%; greatest benefit in ECD kidneys); Campbell notes no long-term survival difference. Normothermic perfusion allows viability testing (limited experience).
- **ESP (Eurotransplant Senior Program):** donors **≥65** to recipients **≥65**, prioritising short cold ischaemia over HLA matching. Informed consent is essential.
- **Ojo (JASN 2001):** ECD transplantation improves survival over dialysis, although graft survival is lower than with standard kidneys.
- With DGF, give antibody induction and delay nephrotoxic CNI therapy.

Standard immunosuppression (current practice)

- **Induction** (over 90% receive it) plus **tacrolimus + mycophenolate** ± steroids (mycophenolate: GI [gastrointestinal] upset, teratogenic).
- Acute rejection is below 10%; chronic AMR is now the main problem. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

Newer immunosuppressive drugs

Drug	Mechanism (site)	Key points
Belatacept	**CTLA4-Ig**: binds CD80/86, blocks **CD28 costimulation (signal 2)**	FDA (Food and Drug Administration) 2011; IV 10 mg/kg induction, then **5 mg/kg 4-weekly** (monthly); CNI-free. **BENEFIT**: better GFR (glomerular filtration rate), more early acute rejection; 7-year better patient and graft survival. **Contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD [post-transplant lymphoproliferative disorder])
Extended-release tacrolimus	CNI: binds FKBP12 and inhibits calcineurin (**signal 1**), so less IL-2 transcription	Once daily, better adherence, steadier levels; nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor
Mycophenolate	Inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis	Standard partner of tacrolimus
mTOR inhibitors (sirolimus, everolimus)	Bind FKBP12 and block mTOR (**signal 3**), arresting G1 to S	Not nephrotoxic; fewer CMV (cytomegalovirus)/BK infections; antitumour effect. **TRANSFORM (2018)**: everolimus + reduced (low-dose) CNI non-inferior. Poor wound healing, **lymphocele**, proteinuria, dyslipidaemia
Basiliximab	Anti-CD25 (IL-2 receptor), non-depleting (**signal 3**)	Low-risk induction, **20 mg IV days 0 and 4**
rATG; **alemtuzumab**	T-cell depletion; anti-CD52	High-risk induction, DGF; infection (CMV), PTLD
Rituximab; bortezomib	Anti-CD20 (B cells); proteasome inhibitor (plasma cells)	AMR, desensitisation
Eculizumab	Anti-C5 (complement)	Selected AMR; aHUS (atypical haemolytic uraemic syndrome) recurrence
Imlifidase	Enzyme that **cleaves IgG**	**EMA (European Medicines Agency) 2020**: crossmatch-positive, highly sensitised patients
Anti-IL-6 therapy (tocilizumab, clazakizumab)	Blocks IL-6 pathway	Chronic active AMR

- **Also know:** anti-CD40 costimulation blockade is under trial. **Voclosporin** is approved for lupus nephritis, **not transplantation**.

Recent advances

- **Tolerance induction** by mixed chimerism (combined kidney and donor stem-cell transplant) and Treg therapy; normothermic repair; **pig-kidney xenotransplant** into a living human (March 2024).

Diagrams to draw

- Three-signal T-cell activation (APC–T-cell synapse showing signals 1–3) with drug sites: CNI at signal 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis.
- Banff biopsy features: tubulitis, arteritis, peritubular capillary C4d.

Key points to remember:

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

Also, from the condensed book (not in the paper answer):

- De novo donor-specific antibody (DSA) is linked to class II mismatch and under-immunosuppression.
- Hyperacute rejection is †irreversible, so the graft is removed.

9 · Landmark trials

Trial	Design	Result	Take-home
Wolfe 1999	Registry: transplant vs wait-listed dialysis	Risk 2.84 early; equal at 106 days; long-term 0.32	Early peril, durable survival gain

★ Putting it together: kidney function and transplant decisions

A · Rising creatinine after urological surgery

ALGORITHM

1. **Postrenal first:** check the catheter and scan. After cystectomy, a rise with ascites may be a urine leak (drain creatinine ≫ serum).
2. **Prerenal vs intrinsic:** FeNa <1% with bland urine means volume depletion; >1% with granular casts means acute tubular necrosis. Stop nephrotoxins, RAAS blockers and metformin.
3. **Dialysis** only for refractory "AEIOU": acidosis, electrolytes (K), intoxication, overload, uraemia.

B · Post-obstructive diuresis

KEY POINT

After bilateral or solitary-kidney relief, monitor output and electrolytes. Diuresis is mostly physiological; †it is pathological if it persists after euvolaemia. †Do not replace every millilitre; if the patient cannot drink, replace ≈75% with 0.45% saline.

C · Transplant ureteric complications

Problem	Clue	Management
Urine leak (early)	High drain output; fluid creatinine ≫ serum	Foley and nephrostomy ± stent; explore if large or persistent
Ureteric stricture	Falling function ± mild hydronephrosis	Nephrostomy, then dilation or endoureterotomy; open repair is most durable
Lymphocele	Fluid creatinine ≈ serum; may obstruct ureter or vessels	Treat only if symptomatic (<15%); laparoscopic window is best
Stent infection	Colonised stent	Remove at 14–21 days (earlier risks leak; later risks infection)

Prevent them with a stent and by sparing the lower pole–ureter triangle.

D · Rejection at a glance

Type	Cause	Point
Hyperacute	Preformed donor-specific antibody (DSA)	Crossmatch prevents it; †irreversible, so remove graft
Acute	T-cell-mediated (TCMR) or antibody-mediated (AMR), Banff	Biopsy. TCMR: IV steroids, then antithymocyte globulin (ATG). AMR: steroids, IV immunoglobulin, plasmapheresis, rituximab
Chronic	Mainly antibody-mediated	Main late loss; subclinical

E · Child going for transplant

Fix the bladder before transplant.

ALGORITHM

1. Most have congenital anomalies (CAKUT), chiefly valves.
2. Make the bladder safe: age-appropriate capacity at detrusor pressure <40 cm H_2O (antimuscarinics, catheterisation, augmentation).
3. Remove refluxing ureters or kidneys causing infection, polyuria or hypertension.
4. A poor bladder destroys the graft.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Vascular anastomosis and vascular access for dialysis

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Principles of vascular anastomosis and vascular access surgery for HD" (asked as "Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD", Practice Set 5 Paper II Q2 LAQ and Practice Set 12 Paper II Q3a SAQ) · P4 SAQ "Vascular access for dialysis." (2020-Summer Paper 4 Q3b)

Assumed meaning: HD = haemodialysis. Vascular access also answered in 2025-Winter Paper 2, Q3c.

Introduction

- A vascular anastomosis joins two vessels so that flow is smooth, leak-free and lasting (transplantation, autotransplantation, caval repair, dialysis access).
- **Types:** end-to-end, **end-to-side** (commonest: transplant, fistula), side-to-side, and patch (Carrel aortic patch).
- Haemodialysis needs a blood flow of about **250–350 mL/min**; access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Principles of vascular anastomosis

Principle	How and why
Exposure and control	Proximal and distal control (vessel loops, Rummel tourniquets, bulldog or Satinsky clamps)
Anticoagulation	Systemic heparin before clamping; heparinised saline flush; prevents thrombosis in the static column
Atraumatic handling	Hold the adventitia only, never the intima; clamp clear of atheroma, because clamp injury causes intimal dissection and stenosis
Healthy ends	Trim diseased ends; resect venous valve cusps at the suture line; prevents stenosis and leak
No tension, twist or kink	Adequate length; trial the final position before suturing
Intima-to-intima eversion	Smooth endothelium faces the lumen; adventitia and rough surfaces stay outside, as they are thrombogenic
Wide opening	Spatulate or bevel small vessels, or use end-to-side with an elliptical arteriotomy (aortic punch), to avoid stenosis (stricture)
Suture	**Non-absorbable monofilament polypropylene**, double-armed: smooth, low drag, low thrombogenicity. IVC (inferior vena cava) or aorta 3-0/4-0; transplant vessels 5-0/6-0; wrist fistula about 6-0/7-0
Needle direction	Equal, small, close bites; pass **inside-out on the artery** so plaque and intima are tacked down, not lifted into a flap
Continuous or interrupted	Continuous is quick and haemostatic; **interrupted** for small vessels and children, allowing growth
Technique	**Carrel triangulation** (three stay sutures keep the back wall clear); **parachute** (sew loosely, then draw up; one knot) or tack both apices and run to the midpoint

Principle	How and why
Flush and release	Before the last knot, briefly release clamps to flush air, clot and debris; leave slack ("growth factor") to avoid purse-stringing; release venous/distal clamp first, then arterial
Confirm	Haemostasis, pulse, thrill, Doppler

- **Failure:** early from technical error (thrombosis, leak, stenosis); late from **intimal hyperplasia**.
- **Also know:** vascular staplers; grafts (autologous vein for small or contaminated fields; PTFE [polytetrafluoroethylene] for large vessels).

Options for vascular access surgery for HD

Option	Construction / description	Ready	Comment
1. Non-tunnelled CVC (central venous catheter)	Dual-lumen 10–12 Fr in internal jugular or femoral vein	At once	AKI (acute kidney injury) and emergencies; acute dialysis, days only
2. Tunnelled cuffed CVC	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; infection, central stenosis
3. Native AVF (arteriovenous fistula)	Artery joined to a superficial vein, usually vein end to artery side	6–8 weeks (matures)	**First choice:** best patency, fewest infections
a. Snuffbox or **radiocephalic (Brescia–Cimino, 1966)**	Radial artery to cephalic vein at wrist	6–8 weeks	Most distal; preserves proximal sites; higher failure to mature (most fail to mature)
b. Brachiocephalic	Brachial artery to cephalic vein at elbow	6–8 weeks	Higher flow; more steal
c. Brachiobasilic transposition	Deep basilic vein mobilised and moved superficially	Longer, often two stages	When the cephalic vein is unusable
4. AVG (arteriovenous graft)	PTFE loop or straight: forearm loop, upper-arm or thigh	2–3 weeks (usable)	Poor veins; more stenosis (venous anastomosis), thrombosis, infection
5. HeRO (Haemodialysis Reliable Outflow) graft	Graft outflow bypasses occluded central veins into the right atrium		Central vein occlusion
6. Endovascular AVF (WavelinQ, Ellipsys)	Percutaneous radiofrequency or thermal anastomosis between adjacent forearm vessels		No incision; selected anatomy

- **Also know:** lower-limb femoral vein transposition when arms are exhausted; peritoneal dialysis needs no vascular access.

Planning

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**; individualised (KDOQI [Kidney Disease Outcomes Quality Initiative] 2019 "ESKD Life-Plan").
- **Timing:** refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.

- **Vein preservation:** preserve arm veins; no venepuncture, cannula or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications of access

Complication	Treatment
Failure to mature (juxta-anastomotic stenosis)	Angioplasty (PTA), ligate side branches
Stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then treat the stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft, remove catheter

- **Also know:** venous hypertension; high-output cardiac failure.

Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

Recent advances

- **Endovascular AVF** (WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels.

Diagram to draw

- Radiocephalic end-to-side fistula at the wrist with Carrel triangulation stay sutures.

> **Key points to remember:**
> - Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
> - AVF first, distal, non-dominant arm, about 6 months ahead.
> - Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
> - Thrombosis sits on a stenosis; steal: banding or DRIL.

LAQ Describe in details basic patho-physiology of urine production and transport

Paper answer · P1

Type: LAQ (25 marks)
Source: 2015-Summer Paper 1, Q1

Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled to a low-pressure bladder.

A. Functional anatomy

- Glomerular barrier (fenestrated endothelium, basement membrane, podocytes) restricts by size and negative charge, so albumin is excluded.
- **Juxtamedullary nephrons** (long loops) build the medullary gradient.

B. Renal blood flow (RBF) and autoregulation

- RBF ~25% of cardiac output. The medulla is hypoxic (10–20 mmHg), endangering the thick ascending limb and S3 segment.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure 80–180 mmHg:
- **Myogenic:** the afferent arteriole constricts when stretched and dilates (prostaglandins) when pressure falls; NSAIDs (non-steroidal anti-inflammatory drugs) abolish this.
- **Tubuloglomerular feedback:** low NaCl at the macula densa releases renin; angiotensin II constricts the efferent arteriole. Angiotensin blockade abolishes this.

C. Glomerular filtration

- GFR ~125 mL/min (180 L/day); filtration fraction ~20%.
- **Net filtration pressure** ≈ glomerular hydrostatic 60 – Bowman's space 18 – oncotic 32 = **~10 mmHg**.
- Efferent constriction lowers RBF but **raises** GFR; afferent constriction lowers both. Obstruction raises Bowman's space pressure, so GFR falls.
- **Also know:** GFR estimated from creatinine or cystatin C (race-free 2021 equations).

D. Tubular handling

Segment	Key transporter	Handling	Drug/disease link
Proximal tubule	Na^+/H^+ exchanger, SGLT2 (sodium–glucose cotransporter 2)	~**65%** Na^+ and water (iso-osmotic); all glucose; most bicarbonate and phosphate; secretes drugs	SGLT2 inhibitors
Thin descending limb	Aquaporin-1	Water out, solute stays: fluid concentrates	–
Thick ascending limb	NKCC2 (Na^+-K^+-$2Cl^-$)	~**25%** NaCl; water-impermeable "diluting segment"	Loop diuretics
Distal convoluted tubule	NCC (Na^+-Cl^- cotransporter)	5–7% Na^+; Ca^{2+} (parathyroid hormone)	Thiazides lower urinary Ca^{2+}
Collecting duct	ENaC (epithelial Na^+ channel), AQP2	Principal cells: Na^+/K^+ (aldosterone), water (ADH); intercalated cells: acid	Amiloride, spironolactone; distal RTA (renal tubular acidosis)

E. Concentration

- **Countercurrent multiplication:** ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling adds about half. Vasa recta exchange without washing out the gradient.
- **Antidiuretic hormone (ADH)** is released by high osmolality or low volume. Signal chain: V2 receptor → cAMP (cyclic adenosine monophosphate) → protein kinase A → aquaporin-2 (AQP2) inserted into the apical membrane → water reabsorbed. Concentration needs both a hypertonic medulla and AQP2; urine osmolality ranges ~50–1200 mOsm/kg.

Hormone	Site	Effect
Angiotensin II, aldosterone	Efferent arteriole, proximal tubule; collecting duct	Na^+ retention, K^+/H^+ loss, GFR maintained
Atrial natriuretic peptide	Afferent arteriole, collecting duct	Natriuresis, lowers renin
Parathyroid hormone	Proximal and distal tubule	Phosphaturia, Ca^{2+} retention

F. Ureteric transport

- **Pacemaker:** atypical smooth muscle cells at the minor calyces/pelvicalyceal border.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves (T11–L1, S2–4) carry only pain.
- Calcium entry triggers action potentials spread by gap junctions; calcium–calmodulin then activates myosin light-chain kinase.
- Walls coapt behind each bolus: 2–6 waves/min at 20–80 cmH_2O. Very high flow makes boluses coalesce, dilating the ureter **without obstruction**.
- Ureterovesical tunnel length:diameter 5:1 prevents reflux; bladder pressure >40 cmH_2O stops efflux.

Factor	Effect on transport
Obstruction	Acute: stronger contractions; chronic: dilatation, lost coaptation
Infection	Gram-negative endotoxin inhibits peristalsis (dilatation without obstruction)
Pregnancy	Right > left dilatation: compression at pelvic brim, progesterone
Drugs	α1-blockers and calcium-channel blockers relax (medical expulsive therapy); NSAIDs lower pelvic pressure in colic

G. Bladder storage and voiding

- **Storage** below ~10 cmH_2O: compliance, sympathetic β3 detrusor relaxation, α1 bladder-neck and pudendal sphincter tone.
- **Voiding:** pelvic parasympathetic (S2–4) acetylcholine on M3 receptors contracts the detrusor; nitric oxide relaxes the urethra.

H. Pathophysiology of obstruction (unilateral: triphasic)

`Phase 1 (0–1.5 h): RBF up, pressure up (afferent dilatation: prostaglandins, nitric oxide) Phase 2 (1.5–5 h): RBF down, pressure up (efferent constriction) Phase 3 (>5 h): RBF down, pressure down (afferent constriction: angiotensin II, thromboxane A2, endothelin)`

- Ischaemic injury starts **within 2 hours** of complete block.
- **Tubules:** down-regulated aquaporins and Na^+ transporters cause concentrating defect, salt wasting and distal RTA.
- **Fibrosis** (final path to failure): stretch, inflammation, TGF-β (transforming growth factor-β), angiotensin II, microRNAs.
- **Bilateral obstruction or solitary kidney:** pressure stays high longer and natriuretic peptides and urea accumulate, so **post-obstructive diuresis** follows relief in ~50% (vs ~2% unilateral).
- **Clinical:** hypertension (~77% bilateral, ~20% unilateral); contralateral hypertrophy; pyelorenal backflow acts as a pop-off; forniceal rupture causes urinoma.

- **Also know:** immature kidneys fail to develop; outlet obstruction causes detrusor hypertrophy then decompensation; diuretic MAG3 (mercaptoacetyltriglycine) renogram (half-time >40 min = obstructed); Whitaker test (>15–20 cmH_2O = obstructed); tolvaptan in polycystic disease.

Diagram to draw: Nephron with transporters; triphasic graph.

> **Key points to remember:**
> - RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
> - Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
> - Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

LAQ Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management)

Paper answer · P1

Asked as:

- Describe anatomy of lymphatic drainage of genitourinary system, etiopathogenesis and management of Chyluria — *Summer 2018 Paper I Q1 (LAQ)*
- Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria. — *Practice Set 8 Paper I Q1 (LAQ)*
- Management of chyluria — *Practice Set 10 Paper I Q1 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2018-Summer Paper 1, Q1 and 2024-Winter Paper 3, Q2)

General plan of lymphatic drainage

- Lymph flows **cephalad and right → left** and returns to the left innominate (brachiocephalic) vein.
- Pelvic nodes (external iliac, internal iliac, obturator, presacral) → common iliac → lumbar (para-aortic, interaortocaval, paracaval) nodes → **lumbar trunks**. These join the intestinal trunk at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). The **thoracic duct** rises behind the aorta to the left innominate vein.
- Pelvic lymph crosses freely between right and left (the two sides). Organs supplied by lateral aortic branches (kidney, adrenal, ureter, gonad) drain to lumbar nodes.
- **Lumbar groups:** left lumbar (preaortic, left para-aortic, retroaortic), interaortocaval, and right lumbar (precaval, paracaval, retrocaval). They drain organs supplied by lateral aortic branches: kidneys, adrenals, ureters and gonads.
- Preaortic nodes (coeliac, superior and inferior mesenteric) form the **intestinal trunk**, which carries **chyle** from the intestinal lacteals.
- Lumbar and intestinal trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). It continues as the **thoracic duct**, which rises behind the aorta to the left innominate vein.
- **Lymphatic anatomy (summary):** intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction. Renal lumbar lymphatics join the cisterna; valves normally prevent backflow.

Drainage by organ

Organ	Primary nodes	Notes
Left kidney	Left lateral para-aortic	Capsular and deep hilar networks drain to renal vein nodes; may go retrocrural
Right kidney	Right paracaval, interaortocaval	May cross to left para-aortic
Ureter	Abdominal: para-aortic (left), paracaval/interaortocaval (right); middle: common iliac; pelvic: common, external, internal iliac	Plexus in muscle and adventitia
Bladder	Mainly **external iliac**; also internal iliac, obturator	Begins in lamina propria; bilateral cross-drainage
Prostate	**Obturator and internal iliac**; presacral	Three routes: superolateral (side wall), posterior (presacral), apical (internal pudendal)
Testis	Left: left para-aortic; right: interaortocaval, then right paracaval	Follows gonadal vessels; crossover only right → left. Epididymal cauda and vas → external iliac
Penis	Skin → superficial inguinal (both groins); glans and corpora → superficial and deep inguinal, then pelvic	Shaft lymphatics converge dorsally
Scrotum	Ipsilateral superficial inguinal	Does not cross the raphe
Urethra (male)	Penile → inguinal; bulbomembranous and prostatic → internal/external iliac, obturator	Female: proximal → pelvic; distal → inguinal

Lymphatic drainage of the urinary tract (intrinsic lymphatics and regional nodes)

Organ	Intrinsic lymphatics	Regional nodes
Kidney	Superficial capsular and deep hilar networks along the interlobular, arcuate and interlobar arteries, to nodes at the renal vein	**Left:** left lateral para-aortic (sometimes retrocrural, or thoracic duct above the diaphragm). **Right:** interaortocaval and right paracaval (may cross to left para-aortic)
Ureter	Plexuses in the muscle and adventitia	Abdominal: left para-aortic (left); paracaval and interaortocaval (right). Middle: common iliac. Pelvic: common, external and internal iliac
Bladder	Begins in the lamina propria; a few perivesical nodes	Mainly **external iliac**; also internal iliac and obturator; cross-drainage to both sides
Prostate	Periprostatic plexus	Obturator and internal iliac; presacral; external iliac
Urethra	Submucosal plexus	Male prostatic and membranous: internal iliac, obturator; penile: inguinal, then external iliac. Female: proximal to pelvic, distal to inguinal nodes

- **Applied:** upper tract cancers spread to lumbar nodes; bladder and prostate cancers to pelvic nodes. Renal drainage is asymmetric: right renal lymph can cross left; left stays left.
- Valves in the renal hilar lymphatics normally stop backflow from the cisterna chyli. **Failure of these valves is the basis of chyluria.**

Chyluria: definition

- Passage of **chyle** (intestinal lymph rich in chylomicrons, triglyceride and protein) in urine, giving milky urine (making it milky; urine looks milky). Caused by an abnormal **lymphourinary (chylourinary) fistula**, usually at the renal fornices / a renal calyceal fornix (**pyelolymphatic fistula**).

Etiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; *Culex* mosquito)	Congenital lymphangiectasia, renal lymphangioma
Brugia (lymphatic filariasis that spares the genitals; rarely urological)	Trauma, surgery (e.g. partial nephrectomy, RPLND (retroperitoneal lymph node dissection))
	Retroperitoneal tumour or fibrosis, tuberculosis, pregnancy

Pathogenesis (obstructive theory)

- **Life cycle:** mosquito-borne larvae mature into adult worms that live for years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- Adult worms live in retroperitoneal lymphatics and cause lymphangitis and fibrosis. This **obstructs** flow between the intestinal lymph trunks and the thoracic duct.
- Proximal lymphatics dilate and their valves fail, so chyle **refluxes** into renal hilar lymphatics.
- Raised pressure ruptures them into a calyceal fornix → **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- **Obstructive (reflux) theory:** `Adult worms + host immune reaction -> lymphangitis and fibrosis -> obstruction between intestinal trunks/cisterna chyli and thoracic duct -> dilatation of retroperitoneal lymphatics, valve incompetence -> reflux of chyle into renal hilar and peripelvic lymphatics -> raised pressure ruptures thin-walled lymphatics into a calyceal fornix -> pyelolymphatic fistula -> chyluria`
- **Pathogenesis (filarial):** adult *Wuchereria bancrofti* worms and the host reaction cause **lymphangitis and fibrosis** between the intestinal trunks, cisterna chyli and thoracic duct. Lymphatics below the block dilate and their valves fail. Chyle refluxes into renal hilar lymphatics, which rupture into a calyx.
- The fistula is usually renal (hilar lymphatics lie against thin fornices); rarely ureteric or vesical.
- Chyluria appears **earlier than genital elephantiasis**, is usually **intermittent** and may remit spontaneously. It worsens after fatty meals, when lacteal flow is highest. So every treatment is judged against natural remission.
- **Non-parasitic causes** work the same way: lymphatic obstruction (tumour, fibrosis), direct disruption (surgery, trauma) or congenital lymphangiectasia.
- **Effects:** loss of fat, protein and lymphocytes causes weight loss, hypoproteinaemia and lowered immunity. Fibrin in chyle forms clots.

Clinical features

- Milky urine, worse after fatty meals; **chylous clots** (retention, colic); **haematochyluria**.

- Loss of fat and protein → weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Grade	Features
Mild	Intermittent; no systemic effect
Moderate	Persistent; clots
Severe	Haematochyluria, clot retention, weight loss, hypoalbuminaemia

Evaluation (before treatment)

1. Confirm chyle

- **Ether (chloroform) test**: milky urine shaken with ether clears because fat dissolves. **Urine triglycerides** (and protein) positive/raised; Sudan III stains fat globules.
- Distinguish from / exclude **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial status (filarial tests)

- Night blood smear (often negative at this stage / late), **circulating filarial antigen** (card) test, eosinophilia; PCR (polymerase chain reaction).

3. Lateralise the side and localise the fistula

- **Cystoscopy after a fatty meal** shows milky efflux from the ureteric orifice (one, sometimes both).
- **Retrograde pyelography** shows **pyelolymphatic backflow**.
- **Define fistula: lymphoscintigraphy** (non-invasive, shows tracer in kidney and bladder), pedal lymphangiography (historical gold standard), MR (magnetic resonance) lymphangiography or CT.

4. Baseline: serum albumin, haemoglobin, renal function; urine culture before instillation.

Management (outline / algorithm)

`Mild → diet + DEC Persistent / moderate → add sclerotherapy (can repeat) Severe or failed sclerotherapy → renal hilar lymphatic disconnection Mild -> low-fat, high-protein diet + MCT oil + DEC Moderate/persistent -> endoscopic sclerotherapy (silver nitrate 0.5-1%, povidone-iodine) Severe or failed -> renal hilar lymphatic disconnection (open, laparoscopic, robotic) Confirm chyle -> grade severity Mild -> diet + antifilarial drugs; observe Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection`

1. Conservative (all patients)

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids.
- **Why MCT:** MCTs (medium-chain fats) are absorbed straight/directly into the portal blood, not into lacteals, so chyle formation falls (less chyle is formed).
- **DEC (diethylcarbamazine)** 6 mg/kg/day for 12 days kills microfilariae (another source: 2 mg/kg orally three times daily, traditionally 12 days; a 1-day course is as effective). It has only modest effect on adults.

- **Doxycycline** 200 mg daily for 4–8 weeks kills *Wolbachia* (endosymbiont; anti-*Wolbachia*) and so sterilises or shortens the life of adult worms. Albendazole and ivermectin are alternatives.
- Correct anaemia and protein deficit. Most mild cases settle on this, and many remit spontaneously.

2. Endoscopic sclerotherapy (renal pelvic instillation sclerotherapy)

- Ureteric catheter on the affected (leaking) side, head-down position. Instil **silver nitrate** (0.5–1%) or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- **Contraindications:** active UTI (urinary tract infection), bilateral simultaneous instillation (risk of anuria).
- Good initial success/response; recurrence is common, but it is repeatable and avoids surgery.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every/all hilar lymphatics. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy ± optional **omental wrap** around the hilum as a barrier. 5. **On-table cystoscopy** confirms clear urine/efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open in meta-analysis: shorter operation and stay, less blood loss, faster bowel recovery, fewer complications.
- **Complications:** lymphocele, renal vein injury, recurrence.
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy (now) obsolete; prevent by clipping lymphatics during renal and nodal surgery.

4. Follow-up and prevention

- Urine triglycerides and weight; recurrence needs repeat sclerotherapy or surgery.
- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: Retroperitoneal lymphatics with cisterna chyli, obstruction (a block above it / obstructed thoracic duct), dilated hilar lymphatics, and reflux to a pyelolymphatic fistula at a calyceal fornix.

Key points to remember:

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

LAQ Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD)

Paper answer · P2

Asked as:

- Discuss principles of hemodialysis and vasular access in details. — *Winter 2017 Paper II Q1 (LAQ)*
- Vascular access for hemodialysis — *Winter 2025 Paper II Q3c (SAQ)*
- Hemodialysis access. — *Practice Set 10 Paper II Q3c (SAQ)*
- Access techniques in dialysis — *Practice Set 12 Paper II Q3f (SAQ)*
- Discuss techniques of dialysis — *Practice Set 5 Paper II Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(Winter 2025 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b) (Practice Set 10 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c) (Practice Set 12 Paper II Q3f: CAPD also answered in Practice Set 1 Paper 2, Q3a) (Practice Set 5 Paper II Q3e: HD principles also answered in 2017-Winter Paper 2, Q1; CAPD in 2018-Summer Paper 2, Q3e)

Assumed meaning (Winter 2017): "vasular" is a typo for "vascular" access for haemodialysis. Assumed meaning (Practice Set 12): access for PD (peritoneal dialysis) and HD (haemodialysis); emphasis on PD catheter insertion, since HD vascular access is answered in full in Practice Set 12 Paper 2, Q3a. Assumed meaning (Practice Set 5): principles and techniques of HD (haemodialysis) and PD (peritoneal dialysis), with continuous therapies.

Introduction

- Haemodialysis (HD) removes solutes and water from blood across a semipermeable membrane in an extracorporeal circuit. It is the commonest KRT (kidney replacement therapy) for ESKD (end-stage kidney disease) and severe AKI (acute kidney injury). It needs an access giving about **250–350 mL/min** of blood flow.
- Access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Principles of haemodialysis (physical principles)

Process	Driving force	What it removes
Diffusion	Concentration gradient across the membrane	Small solutes (urea, creatinine, potassium); bicarbonate enters blood
Convection	Transmembrane pressure; solute is dragged with water	Middle molecules (e.g. β2-microglobulin)
Ultrafiltration	Transmembrane (hydrostatic) pressure (HD) or osmotic (PD) gradient	Excess water, down to "dry weight"

- **Countercurrent flow:** dialysate runs opposite to blood (**countercurrent**), keeping (to keep) a gradient along the whole dialyser. Clearance rises with flows, membrane area and time.
- **Dialyser:** blood (250–350 mL/min) passes through hollow fibres (hollow-fibre) of **synthetic membrane**; high-flux membranes clear more middle molecules.
- **Dialysate:** ultrapure (reverse-osmosis) water with sodium about 140 mmol/L, potassium about 2 mmol/L, bicarbonate and calcium, adjusted to the patient.
- **Anticoagulation:** heparin; heparin-free or regional citrate if bleeding risk is high (e.g. after surgery).
- **Schedule:** about 4 h (hours) three times (3 times) a week. **Also know:** haemofiltration, haemodiafiltration (HDF), CRRT (continuous renal replacement therapy).
- **Access:** AVF (arteriovenous fistula), AVG (arteriovenous graft) or CVC (central venous catheter).

Adequacy

- **Kt/V** (dialyser urea clearance × time ÷ urea distribution volume): target 1.4, minimum 1.2 (Kt/V ≥1.2) per thrice-weekly session (KDOQI [Kidney Disease Outcomes Quality Initiative] 2015). **URR (urea reduction ratio) ≥65%.**

Indications

- **Acute ("AEIOU"), refractory to medical treatment:** Acidosis; Electrolytes (hyperkalaemia); Ingestions (lithium, salicylates, alcohols); Overload with respiratory distress; Uraemia (encephalopathy, pericarditis, bleeding).
- **Chronic:** symptomatic ESKD, usually at eGFR (estimated glomerular filtration rate) 5–10; IDEAL (2010): early start gave no benefit.

Complications of haemodialysis

- **During dialysis:** hypotension (commonest), cramps, arrhythmia, air embolism, bleeding. **Disequilibrium syndrome:** rapid urea removal in early sessions shifts water into the brain (cerebral oedema); prevent with short, gentle first sessions.
- **Long term:** cardiovascular disease (main cause of death), mineral–bone disorder, calciphylaxis, β2-microglobulin amyloidosis (carpal tunnel), infection, hepatitis B and C; acquired cystic kidney disease (about 80% in ESKD; RCC [renal cell carcinoma] in 1–2% of these).
- **Access-related:** access infection, thrombosis, steal.

Peritoneal dialysis

- The **peritoneum** is the membrane: diffusion for solutes; **glucose or icodextrin** draws water osmotically.
- **Tenckhoff catheter** (two Dacron cuffs), placed open, laparoscopically (omentopexy) or percutaneously (see PD access below).
- **CAPD (continuous ambulatory PD):** about 4 exchanges of 2 L daily. **APD (automated PD):** night cycler.
- **Adequacy:** weekly Kt/V ≥1.7; PET (peritoneal equilibration test) classifies transport.
- **Complications:** peritonitis, exit-site infection, catheter malfunction, hernia, hyperglycaemia, peritoneal fibrosis.

Continuous and hybrid therapies

- **CRRT (continuous renal replacement therapy):** CVVH (continuous veno-venous haemofiltration; convection), CVVHD (haemodialysis; diffusion), CVVHDF (both). For haemodynamically unstable AKI (acute kidney injury).
- **Also know:** SLED (sustained low-efficiency dialysis); HDF (haemodiafiltration: CONVINCE 2023 lowered mortality versus high-flux HD); home and nocturnal HD.

HD versus PD

	HD	PD
Setting	Centre or home	Home, continuous
Haemodynamics	Rapid shifts	Gentle; preserves residual function
Favoured in	Abdominal surgery, failed PD	Children, cardiac instability, no vascular access

Types of vascular access

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino, 1966)	Radial artery to cephalic vein at the wrist, end-to-side (now usually vein end to artery side)	≥6–8 weeks	**First choice**: best patency, fewest infections; highest failure to mature
Brachiocephalic AVF	Brachial artery to cephalic vein at the elbow	≥6–8 weeks	Higher flow; more steal
Brachiobasilic (transposition) AVF	Deep basilic vein transposed superficially	Longer	When the cephalic vein is unusable
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene), forearm loop or upper-arm straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; highest infection and central stenosis

- Non-tunnelled 10–12 Fr jugular or femoral catheters are for acute dialysis only.

Planning the access

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first** (US **Fistula First**, 2003). **KDOQI 2019** asks for (favours) an individual "ESKD Life-Plan": the right access for the right patient at the right time.

- **Timing:** refer at eGFR <30; create an AVF at least (about) **6 months** before dialysis is expected.
- **Vein preservation:** no venepuncture, cannula or PICC (peripherally inserted central catheter) in the non-dominant arm (that arm); avoid subclavian catheters (lines).
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm), venography if central stenosis is suspected.
- **Maturity, "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, by (at) 6 weeks.

Access complications

Complication	Key point	Treatment
Failure to mature	Juxta-anastomotic stenosis; side veins	Angioplasty; ligate (side) branches
Stenosis	Neointimal hyperplasia; AVG at venous anastomosis	PTA (percutaneous transluminal angioplasty), stent graft, revision
Thrombosis	Usually on an underlying stenosis	Thrombectomy, then treat (fix) the stenosis
Steal syndrome	Hand ischaemia; diabetes, brachial-based access	Banding, DRIL (distal revascularisation–interval ligation), ligation
Aneurysm	Repeated needling at one site	Repair if skin is thin
Infection	CVC > AVG > AVF	Antibiotics; excise graft; remove catheter

- **Also know:** brachiobasilic transposition AVF; venous hypertension (central stenosis); high-output cardiac failure (banding); ischaemic monomelic neuropathy; graft seroma.

Monitoring

- Each session: thrill and bruit, arm-elevation test (a healthy fistula collapses), prolonged bleeding after needling, rising venous pressure, falling Kt/V. Duplex scan when abnormal; treat a stenosis causing clinical change. Rotate needle sites ("rope-ladder").

Peritoneal dialysis access

Catheter: Tenckhoff silicone catheter with **two Dacron cuffs**. The deep cuff sits in the rectus muscle; the superficial cuff about 2–4 cm from the exit. The tip lies in the **pelvis (rectovesical or rectouterine pouch)**; the exit site points **downwards** to reduce infection.

Insertion techniques

Technique	Steps	Comment
Open (mini-laparotomy)	Paramedian incision → split rectus → open posterior sheath and peritoneum → catheter to pelvis on a stylet → purse-string round deep cuff → tunnel → exit site	Simple; general or local anaesthesia
Laparoscopic (advanced)	Rectus sheath tunnel; **omentopexy**, adhesiolysis, hernia repair as needed	Best function; fixes causes of outflow failure
Percutaneous (Seldinger)	Needle, guidewire, dilator and peel-away sheath under local anaesthesia, ± ultrasound or fluoroscopy	Bedside; avoid after major abdominal surgery

Technique	Steps	Comment
Peritoneoscopic	Small-scope guided placement	Variant of percutaneous
- **Break-in:** delay full-volume exchanges about 2 weeks to prevent leak; urgent start uses small supine volumes.		
- **Complications:** outflow failure (constipation, omental wrap, migration), leak, hernia, exit-site or tunnel infection, peritonitis, bowel perforation.		
- **Contraindications:** peritoneal scarring, uncorrectable hernia, inability to perform exchanges.		

Haemodialysis catheter access (brief, insertion)

- **Non-tunnelled CVC (central venous catheter):** dual-lumen 10–12 Fr, internal jugular or femoral vein by **ultrasound-guided Seldinger**; for AKI (acute kidney injury); avoid subclavian (central stenosis).
- **Tunnelled cuffed CVC:** right internal jugular, tip at the cavoatrial junction under fluoroscopy; bridge to AVF.
- **AVF (arteriovenous fistula)** first choice, then **AVG (arteriovenous graft)**; HeRO and endovascular AVF in selected patients (see Practice Set 12 Paper 2, Q3a).

Recent advances

- **Endovascular AVF** (percutaneous creation: WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels; home and nocturnal HD.
- **CONVINCE (2023):** high-dose HDF lowered all-cause mortality compared with high-flux HD.

Diagram to draw: dialyser with countercurrent blood and dialysate (beside a PD catheter in the pelvis), and a radiocephalic end-to-side fistula at the wrist; Tenckhoff catheter with two cuffs, tunnel, downward exit and pelvic tip.

> **Key points to remember:**
> - Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
> - HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
> - AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
> - Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
> - Tenckhoff: two cuffs, pelvic tip, downward exit.
> - Laparoscopic placement with omentopexy gives best function.
> - Avoid subclavian lines; AVF first for long-term HD.
> - CRRT for unstable AKI; PD preferred in children.

SAQ Brain-stem death: pathophysiology, guidelines and declaration

Paper answer · P1 · P2 · P4 · merged from 3

Asked as: P2 SAQ "Brainstem death: biology and clinical implications" (asked as "Brain stem Death: Biology and clinical implications", Winter 2024 Paper II Q3d; "Brainstem death", Practice

Set 13 Paper II Q3c) · P1 SAQ "Patho-physiology of Brain stem death" (2015-Summer Paper 1 Q3d) · P4 SAQ "Brain-stem death: guidelines for declaration and the Board of Medical Experts (brain dead committee)" (asked as "Brain dead committee", Summer 2018 Paper IV Q3c; "Guidelines for declaration of brain stem death", Summer 2022 Paper IV Q3b)

Assumed meaning (P4): the Board of Medical Experts that certifies brain-stem death under Indian law.

Definition and legal basis

- **Brain-stem death:** irreversible loss of the capacity for consciousness and of the capacity to breathe, from permanent brainstem failure; equivalent to death of the person (legally death).
- **India: THOA (Transplantation of Human Organs Act) 1994** legalised brain-stem death; **amended 2011** (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Internationally: Harvard criteria (1968), AAN (American Academy of Neurology) criteria (updated 2023).
- Donation after brain death (DBD) keeps the circulation until procurement, so warm ischaemia is minimal.

Biology (pathophysiology)

Sequence

1. Brain injury causes oedema and raised ICP (intracranial pressure).
2. Cerebral perfusion falls, leading to herniation ("coning").
3. Pons: Cushing reflex (hypertension, bradycardia).
4. Medulla: catecholamine storm (hypertension, tachyarrhythmia).
5. Loss of sympathetic tone: vasodilatation, hypotension.

System	Consequence
Heart and vessels	Storm causes myocardial injury and arrhythmia, then **hypotension**
Lungs	Neurogenic pulmonary oedema
Pituitary	**Diabetes insipidus (DI)** (no ADH, antidiuretic hormone): polyuria, hypernatraemia; low thyroid hormone and cortisol
Hypothalamus	**Hypothermia** (poikilothermia)
Blood	DIC (disseminated intravascular coagulation) from brain thromboplastin
Whole body	Cytokine injury to organs; hyperglycaemia

Composition of the Board of Medical Experts (India; Rules 2014)

Member	Role
1. Registered medical practitioner **in charge of the hospital**	Hospital authority
2. **Independent specialist**, nominated by the in-charge from a panel approved by the Appropriate Authority	Independent check
3. **Neurologist or neurosurgeon** from the approved panel (if unavailable: surgeon, physician, anaesthetist or intensivist)	Expert
4. Medical practitioner **treating the patient**	Clinical history

- **No member may belong to the transplant team.**

Certification: preconditions

- Apnoeic coma on a ventilator with a **known irreversible cause**.
- Exclude drugs (**sedatives, relaxants**), hypothermia (35–36 °C by protocol), metabolic or endocrine causes, shock.

Certification: brainstem reflexes (all must be absent)

Brainstem reflex	Method	Cranial nerves
Pupillary light reflex	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI
Oculovestibular (caloric)	~50 mL ice-cold water into each ear (drum visible): no eye movement	VIII, III, VI
Motor response in cranial nerve area	Central painful stimulus (supraorbital pressure): no facial grimace	V, VII
Gag and cough	Touch pharynx; pass suction catheter down trachea: no response	IX, X

Certification: apnoea test and timing

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; watch for breathing effort; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg** (or ≥20 mmHg above baseline) is a positive test. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**, recorded on **Form 10**.
- **Ancillary tests** are not mandatory; used when clinical examination cannot be completed (e.g. facial trauma): EEG (electroencephalogram), cerebral angiography, perfusion scan, transcranial Doppler.

Clinical implications: linked duties after declaration

- **Mandatory request:** ICU (intensive care unit) staff must inform relatives of the option to donate; a **transplant coordinator** counsels; next of kin give consent.
- **NOTTO** (National Organ and Tissue Transplant Organisation), with regional and state bodies (ROTTO, SOTTO: regional and state organ and tissue transplant organisations), allocates organs and keeps the registry.

Clinical implications: donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (pressor, also treats DI); thyroid hormone, methylprednisolone, insulin; warming; protective ventilation.
- **Also know:** cold storage at 4 °C (University of Wisconsin solution); donor exclusions (metastatic cancer, melanoma, untreatable viral infection).

Recent advances

- Hypothermic machine perfusion reduces delayed graft function.

Diagrams to draw

- Herniation sequence leading to storm, DI, hypothermia, DIC.
- Certification flowchart: 1. Coma. 2. Preconditions. 3. Test 1. 4. 6 hours. 5. Test 2. 6. Form 10. 7. Consent. 8. NOTTO.

> **Key points to remember:**
>
> - Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
> - Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
> - THOA: four-member Board; no transplant-team member.
> - Two tests ≥6 h apart; Form 10; Rule of 100s.

SAQ Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens

Paper answer · P1 · P2 · P4 · merged from 3

Asked as: P1 SAQ "Immunosuppressants in renal transplantation (classification, protocols, ciclosporin)" (asked as "Immunosuppressants", Summer 2021 Paper I Q3f; "Classify immunosuppresants. Describe few common protocols", Practice Set 3 Paper I Q3b; "Cyclosporins", Practice Set 4 Paper I Q3f) · P2 SAQ "Pre transplant Immunosupression treatment options" (2023-Winter Paper 2 Q3e) · P4 SAQ "Steroid-free immunosuppression in renal transplantation" (Practice Set 3 Paper 4 Q3f)

Taken as immunosuppressant drugs in renal transplantation; "Cyclosporins" taken as ciclosporin (cyclosporin A); "pre-transplant" taken as induction immunosuppression and desensitisation given before or at transplantation. Also answered in 2020-Summer Paper 4, Q1; tacrolimus in 2017-Winter Paper 3, Q3c.

Principle

- T-cell activation needs three signals: **signal 1** (TCR [T-cell receptor] binds donor antigen; calcineurin then drives IL-2 [interleukin-2] transcription), **signal 2** (costimulation, CD28–CD80/86) and **signal 3** (IL-2 receptor signalling through mTOR [mammalian target of rapamycin], driving proliferation). Each drug blocks a step.
- Rejection risk is highest in the first weeks.

Phases

- **Induction** (peri-transplant intense immunosuppression), **maintenance** (lifelong), **rescue** (treatment of rejection).
- **Desensitisation** removes preformed antibody in HLA (human leucocyte antigen)- or ABO-incompatible recipients.

Classification

Class	Drug(s)	Mechanism / action	Main toxicity
CNI (calcineurin inhibitors)	**Tacrolimus**, ciclosporin	Block calcineurin (signal 1)	**Nephrotoxicity**, diabetes, tremor, hypertension; ciclosporin: gum hypertrophy, hirsutism

Class	Drug(s)	Mechanism / action	Main toxicity
Antimetabolites (antiproliferatives)	**Mycophenolate**, azathioprine	Block purine synthesis (IMPDH [inosine monophosphate dehydrogenase])	Diarrhoea, leucopenia
mTOR inhibitors	Sirolimus, everolimus	Block signal 3	**Lymphocele, poor wound healing**, proteinuria, dyslipidaemia; not nephrotoxic, antitumour
Corticosteroids	Prednisolone, methylprednisolone	Block cytokine transcription (IL-1, IL-2) and redistribute lymphocytes	Diabetes, bone loss, cataract
Costimulation blocker	**Belatacept**	CTLA4-Ig binds CD80/86 (signal 2)	**PTLD** (post-transplant lymphoproliferative disorder); avoid if EBV (Epstein–Barr virus)-negative
Depleting antibodies	rATG (rabbit antithymocyte globulin), alemtuzumab	T-cell depletion	Infection, PTLD
Non-depleting antibody	**Basiliximab**	Anti-CD25 (IL-2 receptor)	Few
Antibody / B-cell agents	Rituximab, IVIG (intravenous immunoglobulin), plasmapheresis, bortezomib, eculizumab, imlifidase	Antibody removal or blockade; antibody-mediated rejection, desensitisation	—

Pre-transplant assessment of immunological risk

- cPRA (calculated panel-reactive antibody), DSA (donor-specific antibody) on Luminex beads, crossmatch (CDC [complement-dependent cytotoxicity] and flow).

Induction agents

Agent	Mechanism	Use and drawbacks
Basiliximab	Anti-CD25 (IL-2 receptor); non-depleting	Low-risk recipients; 20 mg IV (intravenous) on days 0 and 4; few side effects
rATG	Polyclonal T-cell depletion	High-risk (sensitised, retransplant, delayed graft function); less rejection than basiliximab; cytokine release, leucopenia, infection, PTLD
Alemtuzumab	Anti-CD52; profound depletion	Allows steroid avoidance; off-label
IV methylprednisolone	Cytokine suppression	Given intraoperatively with all regimens

- **Over 90%** receive induction, then **tacrolimus + mycophenolate** ± steroids; steroids in more than two-thirds. Acute rejection has fallen from ≥60% to <10%.

- In living-donor transplants, tacrolimus and mycophenolate often start days before surgery.

Common protocols

1. **Standard-risk (most recipients):** basiliximab 20 mg IV on days 0 and 4 + intraoperative methylprednisolone; then **tacrolimus + mycophenolate + tapering prednisolone**.
2. **High-immunological-risk** (sensitised, retransplant, DGF [delayed graft function] expected): **rATG** induction, then triple therapy.
3. **Steroid-avoidance or early withdrawal:** low-risk recipients, usually after depleting induction (see below).
4. **CNI-sparing:** belatacept- or mTOR-based regimens to spare nephrotoxicity; TRANSFORM: everolimus with reduced CNI non-inferior.
5. **ABO-incompatible or HLA-sensitised:** desensitisation with rituximab, plasmapheresis and IVIG; imlifidase for crossmatch-positive (see below).
6. **Rejection:** T-cell-mediated: pulse IV methylprednisolone, then rATG (for Banff grade II–III); antibody-mediated: steroids, plasmapheresis, IVIG, rituximab.

- **Monitoring: tacrolimus trough levels** (higher early, lower later); CYP3A4 (cytochrome P450 3A4) interactions (azoles, macrolides raise levels; rifampicin lowers).

Desensitisation: HLA-incompatible

- **Plasmapheresis** (removes antibody) plus **low-dose IVIG** after each session; or **high-dose IVIG** (immunomodulation).
- **Rituximab** (anti-CD20) reduces B cells that renew antibody. Bortezomib (plasma cells) and eculizumab (anti-C5) in selected cases.
- **Imlifidase:** bacterial enzyme that **cleaves all IgG** within hours, turning a positive crossmatch negative; EMA (European Medicines Agency) conditional approval 2020 for highly sensitised patients.
- Desensitised live-donor transplant gives better survival than staying on dialysis (Montgomery, 2011).

Desensitisation: ABO-incompatible

- **Rituximab** 2–4 weeks before; **plasmapheresis or immunoadsorption** to lower anti-A/B titres to a centre target (commonly ≤1:8); IVIG; tacrolimus–mycophenolate started before surgery. Splenectomy is now obsolete.
- Post-transplant titre monitoring; A2 donors carry lower risk.
- **Alternatives: kidney paired exchange** avoids desensitisation; acceptable-mismatch programmes.

Steroid-free immunosuppression

Why avoid steroids

- **Metabolic:** PTDM (post-transplant diabetes mellitus), hypertension, dyslipidaemia, weight gain, hence **cardiovascular death** (leading cause of death with a functioning graft).
- **Bone:** osteoporosis, avascular necrosis; **growth failure** in children.
- Cataract, infection, poor healing.

Strategies

Strategy	Timing
Steroid avoidance	No steroids, or perioperative doses only

Strategy	Timing
Early (rapid) withdrawal	Stopped within the first week; most studied
Late withdrawal	Months later; more rejection, less favoured

Regimen

- **Induction is essential:** depleting rATG or alemtuzumab; basiliximab in low risk.
- **Maintenance:** tacrolimus at target levels plus MMF (mycophenolate mofetil).

Selection

- **Suitable:** low immunological risk: first transplant, non-sensitised, no DSA, prompt graft function, good adherence.
- **Unsuitable:** highly sensitised, DSA, retransplant after immunological loss, DGF, early rejection, **recurring steroid-responsive disease** (IgA nephropathy, lupus), steroids needed for other illness.

Evidence

- **Woodle et al. (2008):** withdrawal after day 7 versus low-dose maintenance: similar 5-year outcomes; **more mild, steroid-sensitive rejection**.
- **HARMONY (Lancet 2016):** steroids stopped on day 8: similar rejection and graft survival, **less PTDM**.
- **Cochrane review (2016):** more acute rejection, but no difference in patient or graft survival up to 5 years.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2009:** low-risk recipients with induction may stop steroids within the first week; if continued beyond it, continue rather than withdraw.
- **Practice:** over two-thirds still take maintenance steroids (Chapter 91); steroid-free maintenance is increasing in low-risk children.

Risks and safeguards

- Early rejection; possibly more chronic injury and recurrence. Safeguards: tacrolimus levels, adherence, DSA monitoring, biopsy for rising creatinine.

Ciclosporin (cyclosporin A)

Introduction

- Cyclic peptide of 11 amino acids from the fungus *Tolypocladium inflatum*; the first **CNI**.
- Its introduction in the early 1980s cut acute rejection from at least 60% (pre-ciclosporin era) towards today's <10%. Tacrolimus has now largely replaced it.

Mechanism

1. Ciclosporin binds cyclophilin in the cytoplasm.
2. The complex inhibits calcineurin.
3. NFAT (nuclear factor of activated T cells) stays phosphorylated.
4. No IL-2 transcription, so T-cell activation is blocked (signal 1).

Pharmacokinetics and monitoring

- Variable oral absorption; the **microemulsion form** absorbs more predictably.
- Metabolised by **CYP3A4** and P-glycoprotein; narrow therapeutic index, so whole-blood **trough (C0) or 2-hour (C2) levels** guide dosing.

Uses

- **Transplantation:** maintenance with an antimetabolite and steroids, when tacrolimus is not tolerated.
- **IC/BPS (interstitial cystitis/bladder pain syndrome):** refractory **Hunner lesions**, 1–3 mg/kg/day; better than pentosan polysulfate.
- **Others:** retroperitoneal fibrosis after steroid failure; frequently relapsing nephrotic syndrome; severe psoriasis and other dermatoses.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent arteriolar vasoconstriction (reversible); chronic striped fibrosis, arteriolar hyalinosis; TMA (thrombotic microangiopathy)
Metabolic	Hypertension, hyperkalaemia, hypomagnesaemia, **hyperuricaemia** (gout), dyslipidaemia
Cosmetic	**Gingival hyperplasia, hirsutism**
Neurological	Tremor, PRES (posterior reversible encephalopathy syndrome)
Immune	Infection, skin cancer, PTLD

Interactions

- **Levels rise:** azoles, erythromycin and clarithromycin, diltiazem, verapamil, grapefruit.
- **Levels fall:** rifampicin, phenytoin, carbamazepine, St John's wort.
- More nephrotoxicity with NSAIDs (non-steroidal anti-inflammatory drugs) and aminoglycosides.

Versus tacrolimus

- Ciclosporin: more hirsutism, gum hypertrophy, hypertension and hyperlipidaemia; less diabetes and tremor.
- **ELITE-Symphony (2007):** low-dose tacrolimus gave less rejection and better GFR (glomerular filtration rate) than ciclosporin regimens.

Urological relevance

- **Infection:** UTI (urinary tract infection), **BK virus** (ureteric stricture, nephropathy), CMV (cytomegalovirus).
- **Malignancy:** skin cancer, PTLD, RCC (renal cell carcinoma, mostly native kidneys), **bladder cancer about 3-fold** and more aggressive.
- **Surgery:** lymphocele and wound problems with mTOR inhibitors; stones (CNI, steroids): CNIs promote transplant stones (hyperuricaemia, hyperoxaluria, hypocitraturia).
- **Also know:** belatacept (BENEFIT trial: better GFR), extended-release tacrolimus, TRANSFORM (everolimus with reduced CNI non-inferior).

Diagrams to draw

- APC (antigen-presenting cell)–T cell synapse with signals 1–3 and the drug acting at each.
- T cell with the cyclophilin–calcineurin–NFAT–IL-2 pathway.
- Antibody titre falling with plasmapheresis sessions to target before transplant.
- Timeline: induction, steroids stopped by day 7, tacrolimus–MMF maintenance.

Key points to remember:

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
- Basiliximab for low risk; rATG for high immunological risk.
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.
- Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
- Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
- Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

SAQ CAPD and CAPD peritonitis

Paper answer · P1 · P2 · merged from 2

Asked as: P2 SAQ "CAPD (Continuous Ambulatory Peritoneal Dialysis)" (Summer 2018 Paper II Q3e; Practice Set 1 Paper II Q3a) · P1 SAQ "Management of CAPD peritonitis" (Practice Set 21 SAQ Sheet Q3d)

Definition and principle

- CAPD (continuous ambulatory peritoneal dialysis): home dialysis using the **peritoneum** (about 1–2 m^2) as the membrane; dialysate dwells in the abdomen.
- **Diffusion:** urea, creatinine and potassium move into dialysate down a concentration gradient.
- **Ultrafiltration:** an osmotic agent draws water by osmosis: **glucose** (1.5%, 2.5%, 4.25%) or **icodextrin** (for the long dwell).
- **CAPD regimen:** about **4 exchanges of 2 L daily**, 4–6-hour dwells, overnight long dwell. **APD** (automated peritoneal dialysis) uses a night cycler.

Catheter

- **Tenckhoff catheter:** silicone with **two Dacron cuffs** (deep cuff in rectus muscle; superficial cuff about 2 cm from the exit), exit site pointing downward, tip in the pelvis.
- **Insertion:** open; **laparoscopic** (rectus sheath tunnelling, omentopexy, adhesiolysis); percutaneous Seldinger. Delay full volumes about 2 weeks to prevent leaks.

Adequacy

- Weekly **Kt/V urea ≥1.7** (peritoneal plus residual renal urea clearance × time ÷ urea distribution volume), with clinical well-being.

- **PET** (peritoneal equilibration test) identifies high transporters, who do better on short dwells (APD).

Advantages over haemodialysis

- Home-based, no vascular access, **haemodynamically gentle**, **preserves residual renal function**. Favoured in children, poor vascular access and unstable heart disease.

Contraindications

- **Absolute:** peritoneal adhesions or fibrosis (scarring), loss of membrane function (membrane failure), uncorrectable hernia, nobody able to do exchanges.
- **Relative:** recent abdominal surgery, stoma, abdominal wall infection, massive polycystic kidneys, morbid obesity, severe malnutrition.

Complications (overview)

- **Peritonitis** (see below).
- **Exit-site and tunnel infection:** prevent with topical mupirocin or gentamicin.
- **Mechanical:** outflow failure (constipation, omental wrap, migration), pericatheter **leak**, **hernias** (raised abdominal pressure), hydrothorax, scrotal oedema (patent processus vaginalis).
- **Metabolic:** hyperglycaemia, weight gain, hypokalaemia, protein loss.
- **Long term:** ultrafiltration failure; **EPS (encapsulating peritoneal sclerosis)**: rare, presents as bowel obstruction.
- **Also know:** the PD (peritoneal dialysis) catheter is removed after a successful transplant; retroperitoneal laparoscopy protects the peritoneum for PD patients.

CAPD peritonitis: introduction

- The leading cause of technique failure and transfer to haemodialysis. Commonest organism: **coagulase-negative *Staphylococcus*** (touch contamination).
- ISPD (International Society for Peritoneal Dialysis) 2022 target: centre rate **≤0.40 episodes per patient-year**.

CAPD peritonitis: diagnosis (any 2 of 3)

1. Abdominal pain and/or **cloudy effluent**.
2. Effluent **WCC (white cell count) >100/µL with >50% neutrophils** (dwell ≥2 hours).
3. Positive effluent culture.

CAPD peritonitis: initial steps

- Effluent cell count, Gram stain, **culture in blood-culture bottles** (culture-negative rate should be <15%).
- Examine exit site and tunnel; analgesia; heparin in bags if fibrin.

CAPD peritonitis: empirical antibiotics

- **Intraperitoneal route preferred** (intravenous if septic); start immediately after samples.
- Cover **Gram-positives** (vancomycin or first-generation cephalosporin, e.g. cefazolin) **and Gram-negatives** (third-generation cephalosporin, e.g. ceftazidime, or aminoglycoside). Choice follows centre sensitivities.
- Intermittent (once daily in the long dwell) or continuous dosing; avoid prolonged aminoglycoside (residual function, ototoxicity).

- **Antifungal prophylaxis** (nystatin or fluconazole) during antibiotics.

CAPD peritonitis: organism-specific duration

Organism	Duration
Coagulase-negative *Staphylococcus*, streptococci, culture-negative	2 weeks
S. aureus, enterococcus, Gram-negatives	3 weeks
Pseudomonas	3 weeks, two antipseudomonal drugs
Polymicrobial enteric	Add metronidazole; surgical review (bowel perforation)
Fungal	**Immediate catheter removal** + antifungal

CAPD peritonitis: catheter removal

- **Refractory** (effluent not clear after 5 days of the right antibiotics), relapsing, fungal, refractory exit-site or tunnel infection.
- Haemodialysis meanwhile; new catheter after at least 2 weeks.

CAPD peritonitis: definitions

- **Relapse:** same organism (or sterile) within 4 weeks of finishing therapy.
- **Recurrent:** different organism within 4 weeks.
- **Repeat:** same organism after 4 weeks.

CAPD peritonitis: prevention

- Training, exit-site mupirocin or gentamicin, prophylaxis before colonoscopy or gynaecological procedures.

Diagrams to draw

- Tenckhoff catheter: two cuffs (deep and superficial), tunnel, downward exit site, pelvic tip.
- Peritonitis management flowchart from cloudy bag to catheter removal.

> **Key points to remember:**
> - Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
> - Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
> - Peritonitis: two of cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
> - Intraperitoneal Gram-positive plus Gram-negative cover.
> - Remove catheter for fungal or refractory (5 days) peritonitis.

SAQ THOA/THOTA legislation and hospital authorization committee

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "THOA 1994 and its 2011 amendment (THOTA) and 2014 Rules" (Amendments to THOA: 2011; HOTA and its modification) · P4 SAQ "Hospital authorization committee for live related kidney donation."

Indian law; not covered in the chapter notes. Assumed meaning (Practice Set 11): India's Transplantation of Human Organs Act (THOA) 1994, its 2011 amendment and 2014 Rules.

Legal framework and background: THOA 1994 aims

- **THOA (Transplantation of Human Organs Act) 1994**; amended 2011 as **THOTA (Transplantation of Human Organs and Tissues Act)**; **Rules 2014**.
- THOA 1994 regulates removal, storage and transplantation of organs, **prevents commercial dealing** (banned commercial dealing) and **recognised brain-stem death as legal death**, allowing deceased donation.
- Purpose: **prevent organ trade** while allowing genuine donation.
- Organ trade continued and deceased donation stayed low. The **Transplantation of Human Organs (Amendment) Act 2011** responded; **Rules followed in 2014**.

Main provisions of THOA 1994

- **Living donors: near relatives** (spouse, parents, children, siblings) after verification. Others, donating "by reason of affection or attachment", need **Authorisation Committee** approval.
- **Appropriate Authority** registers and inspects hospitals; only registered centres may transplant.
- **Brain-stem death certification by a Board of four:** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon; treating doctor. **None from the transplant team.** Two sets of tests ≥6 hours apart.
- **Organ sale is a punishable offence.**

2011 amendment (THOTA)

Area	Change	Purpose
Scope	**Tissues** added; renamed the **Transplantation of Human Organs and Tissues Act (THOTA)**; **tissue banks** registered	Regulates tissue banks
Near relatives	**Grandparents and grandchildren** added to spouse, children, parents and siblings	Wider related donor pool
Swap donation	**Paired exchange** (swap donation) between near-relative pairs allowed, with Authorisation Committee approval	Overcomes ABO or crossmatch incompatibility
Retrieval centres	Hospitals with ICU (intensive care unit) facilities can register as **non-transplant organ retrieval centres**	
Transplant coordinator	Mandatory in every registered transplant hospital	Organises donation
Mandatory ("required") request	ICU doctor or staff must ask families of a potential brain-dead donor about donation and inform the retrieval centre	More deceased donors
Brain-death board	If a neurologist or neurosurgeon is unavailable, a surgeon, physician, anaesthetist or intensivist nominated by the hospital can serve	Smaller hospitals can certify
Eye donation	Trained technicians may remove corneas	
Registry and network	National human organ and tissue removal and storage network and **national registry** of donors and recipients (now run by NOTTO, the National Organ and Tissue Transplant Organisation)	Fair allocation

Area	Change	Purpose
Penalties	**Higher penalties** for organ trade (up to 10 years' imprisonment; fine up to ₹1 crore) (see below)	Deterrence

Penalties under the amendment

- **Removal of organs without authority:** imprisonment up to **10 years** and fine up to ₹20 lakh.
- **Commercial dealing in organs:** imprisonment **5–10 years** and fine **₹20 lakh to ₹1 crore** (P4: imprisonment up to 10 years and fine up to ₹1 crore).
- Registered medical practitioner: name removed from the register (doctors' registration removed) for **3 years** for a first offence and **permanently** for a later offence.

Rules 2014 (THOT Rules)

- Standard forms for consent and certification (**Form 10** for brain-stem death); brainstem reflexes and apnoea test set out.
- Authorisation Committee interviews donor and recipient to exclude coercion or payment.
- **Also know:** regional and state networks (ROTTO, SOTTO); 2023 guidelines removed the domicile rule and age limit for deceased-donor listing.

Living donor categories and approval

Donor	Approval
Near relative: spouse, son, daughter, father, mother, brother, sister, grandparents, grandchildren	Verified by the hospital's competent authority
Other than near relative (affection or attachment)	**Authorisation Committee approval mandatory**
Swap (paired) donation between near-relative pairs (2011)	Authorisation Committee
Foreign donor or recipient	Authorisation Committee; embassy certifies relationship

Hospital authorisation committee: constitution

- **Hospital-based** if a centre does more than **25 transplants a year**; otherwise **district or state committee**.
- **Members:** medical director or superintendent (chair); two senior doctors of the hospital **not in the transplant team**; two persons of high integrity and social standing (preferably one woman), nominated by the State; state committees add health-department nominees.

Hospital authorisation committee: functions and procedure

1. Receive the **joint donor–recipient application** with consent forms, fitness certificates and identity and relationship documents.
2. **Personal interview** of donor, recipient and families; proceedings **video-recorded**.
3. Verify **relationship** (documents; **HLA [human leukocyte antigen] or DNA testing** if doubtful).
4. Exclude **commercial dealing**: financial status, reasons, long association, middlemen.
5. Confirm donor understands risks and consents freely, without coercion.
6. Check **domicile** (other-state donors).

7. Decide promptly; decision recorded and displayed; reasons given if refused. **Appeal** to the State Government within **30 days**.

Safeguards and restrictions

- Unrelated donors and foreigners need **Authorisation Committee** approval.
- **Minors and mentally ill persons** cannot be living donors except in exceptional, approved cases.
- **Foreign recipient with unrelated Indian donor** not approved.

Diagrams to draw

- Deceased donation flowchart: potential donor (brain-dead patient) in ICU → required request → transplant coordinator → brain-stem death board (Board of four) → family consent → retrieval → national registry and allocation (NOTTO allocation).
- Living donor approval flowchart: application → near relative (competent authority) or other (committee) → interview, verify → decision → appeal.

> **Key points to remember:**
>
> - THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
> - 2011: tissues added; grandparents and grandchildren became near relatives.
> - Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
> - Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
> - Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
> - Hospital-based committee if >25 transplants a year; no transplant team member.
> - Non-relatives, swap and foreigners need committee approval.
> - Interview video-recorded; appeal within 30 days.

SAQ Marginal kidney donor and Remuzzi score

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P4 SAQ "Marginal donor and Remuzzi score for pre-implantation biopsy" (Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys.) · P2 SAQ "Marginal kidney donor" · P3 SAQ "Marginal Donor"

"Preemptive" is read as pre-implantation (procurement) biopsy.

Definition

- A **marginal (extended) donor** is one whose kidney carries a **higher risk of DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- **Ojo (2001):** ECD transplantation still gives better survival than staying on dialysis.

Deceased marginal donors

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002

- Age **60 or over**, or
- Age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**.

- Graft-loss risk about 1.7 times that of a standard donor (about 70% higher).

2. KDPI (Kidney Donor Profile Index) >85%

- Replaced ECD in the US (2014); 10 donor factors (age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV [hepatitis C virus], DCD).
- **Longevity matching:** lowest-KDPI kidneys go to recipients with the best EPTS (Estimated Post-Transplant Survival).

3. DCD (donation after circulatory death), Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of support	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled
- Warm ischaemia raises DGF (roughly 40–50%); long-term survival is similar to DBD (donation after brain death).		

4. Others

- Donor AKI (acute kidney injury).
- **HCV-positive (HCV-viraemic)** donors with antiviral treatment.
- **Paediatric donors under 15 kg (en bloc).**
- **Excluded:** metastatic or haematological cancer, melanoma, untreatable viral infection.

Living marginal donors

- Older donors, controlled hypertension (older or hypertensive living donors), obesity, small stones, multiple renal arteries. Acceptable after careful evaluation and consent.

Remuzzi score (Karpinski/Remuzzi)

- Pre-implantation biopsy (ideally at least 25 glomeruli), read on rapid sections.
- **Four components (glomeruli, tubules, interstitium, vessels), each 0–3: total 0–12.**

Component	0	1	2	3
Glomerular global sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial and arteriolar narrowing	None	Wall thickness < lumen diameter	Wall = or slightly > lumen	Wall far > lumen, severe narrowing or occlusion

Allocation by score

Total	Grade	Use
0–3	Mild	**Single** kidney transplant
4–6	Moderate	**Dual** kidney transplant

Total	Grade	Use
7–12	Severe	**Discard**

- Remuzzi (NEJM 2006): biopsy-guided older-donor kidneys, single or dual, had graft survival similar to young-donor kidneys.
- **Limitations:** sampling error; wedge biopsy overestimates sclerosis; interobserver variation.

Optimising use of marginal kidneys

- Short cold ischaemia.
- **HMP (hypothermic machine perfusion)** reduces DGF versus cold storage (Moers, 2009), without changing long-term survival; normothermic perfusion tests viability.
- **Old-for-old** matching: **ESP (Eurotransplant Senior Program)**, donors 65 or over to recipients 65 or over, with short cold ischaemia.
- Recipient: informed consent; induction; delayed CNI (calcineurin inhibitor) if DGF.

Urological relevance

- Older and ECD kidneys have more **urine leak and ureteric stricture** (ischaemic distal ureter), as do multiple arteries and DGF.

Diagrams to draw

- Remuzzi grid with four compartments scored 0–3.
- Maastricht categories on a timeline of circulatory arrest.

> **Key points to remember:**
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
> - KDPI >85% replaced ECD (2014).
> - Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
> - 0–3 single, 4–6 dual, ≥7 discard.
> - HMP reduces DGF; Maastricht III is the commonest DCD.

SAQ Renin-angiotensin system

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3c

Definition

- Hormonal cascade controlling **blood pressure, extracellular volume, Na^+/K^+ balance and GFR (glomerular filtration rate)**.

Cascade

```
Angiotensinogen (liver) ↓ renin (juxtaglomerular cells) — rate-limiting
Angiotensin I ↓ ACE (angiotensin-converting enzyme), mainly lung Angiotensin II
→ AT1 / AT2 receptors ↓ ACE2 Angiotensin (1-7) → Mas receptor (protective)
```

Juxtaglomerular apparatus and renin release

- Granular cells in the afferent arteriole secrete renin and act as baroreceptors; the **macula densa** senses tubular NaCl.

Renin stimulated by	Renin inhibited by
Low renal perfusion pressure (baroreceptor)	High perfusion pressure
Low NaCl at macula densa	High NaCl at macula densa
Sympathetic β1 activity	Angiotensin II (negative feedback)
Prostaglandins, hypovolaemia	Atrial natriuretic peptide, β-blockers, NSAIDs (non-steroidal anti-inflammatory drugs)

Actions of angiotensin II

- **AT1 receptor:** vasoconstriction; **aldosterone** (distal Na^+ retention, K^+ loss); direct proximal Na^+ reabsorption; thirst and ADH (antidiuretic hormone); **fibrosis**.
- **AT2 receptor:** vasodilatation; fetal urinary tract development.
- **Intrarenal:** efferent constriction maintains GFR when perfusion falls, so ACE inhibitors and ARBs (angiotensin receptor blockers) drop GFR in bilateral or solitary-kidney renal artery stenosis.

Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis (older men, ostial) or fibromuscular dysplasia (young women). Diagnose by duplex ultrasound, computed tomography or magnetic resonance angiography, captopril renography. **Renal vein renin ratio >1.5** with contralateral suppression predicts cure by revascularisation or nephrectomy.
- **Page kidney:** subcapsular haematoma compresses parenchyma; ACE inhibitor/ARB, then drainage.
- **Reninoma (juxtaglomerular cell tumour):** young women, ≤3 cm mass, refractory hypertension, **hypokalaemia**; excision cures most.
- **Obstruction:** hypertension (~77% bilateral, ~20% unilateral); angiotensin II drives fibrosis.
- **Also know:** post-traumatic hypertension (arteriovenous fistula, scarring); reflux nephropathy (renin-guided segmental resection); hyperkalaemia with angiotensin blockade in CKD (chronic kidney disease).
- **Drugs:** ACE inhibitors (cough, angio-oedema from bradykinin), ARBs, aliskiren (direct renin inhibitor), aldosterone antagonists; all contraindicated in pregnancy.

Diagram to draw: Juxtaglomerular apparatus and cascade.

Recent advances

- Renal denervation (US approval 2023) for resistant hypertension.

> **Key points to remember:**
> - Renin is rate-limiting; released by low pressure, low NaCl, β1.
> - Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
> - Renin ratio >1.5 lateralises; reninoma excision cures most.

SAQ Drugs associated with Acute Renal Failure.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2016-Summer Paper 1, Q3c

Introduction

- AKI (acute kidney injury), KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL in 48 hours or ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

Classification by mechanism

Mechanism	Drugs	How
Haemodynamic (prerenal)	**NSAIDs** (non-steroidal anti-inflammatory drugs)	Block prostaglandin afferent dilatation
	ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers)	Remove efferent constriction that holds GFR (glomerular filtration rate)
	Ciclosporin, tacrolimus, contrast; diuretics	Afferent vasoconstriction; volume loss
ATN (acute tubular necrosis)	**Aminoglycosides**, amphotericin B, vancomycin, cisplatin, ifosfamide, contrast	Direct tubular toxicity
AIN (acute interstitial nephritis)	**β-lactams**, sulfonamides, quinolones, rifampicin, NSAIDs, **PPIs (proton pump inhibitors)**, allopurinol, **checkpoint inhibitors**	Cell-mediated hypersensitivity
Crystals	Aciclovir, methotrexate, indinavir, sulfadiazine; tumour lysis	Intratubular obstruction
Microangiopathy	Gemcitabine, VEGF (vascular endothelial growth factor) inhibitors	Endothelial injury
Rhabdomyolysis	Statins (with fibrates)	Myoglobin toxicity, precipitation

- **"Triple whammy":** NSAID + ACE inhibitor/ARB + diuretic in a dehydrated patient.
- **False creatinine rise:** trimethoprim and cimetidine block creatinine secretion.

Recognition

Type	Urine
Prerenal	FeNa (fractional excretion of sodium) <1%, bland
ATN	FeNa >1%, muddy-brown casts
AIN	Sterile pyuria, white cell casts, ± rash, fever, eosinophilia
- AIN appears days to weeks after the drug starts; it is the commonest biopsy lesion with checkpoint inhibitors. Urine eosinophils are unreliable.	

Prevention and management

- Identify risk (CKD (chronic kidney disease), age, diabetes). Dose by eGFR (estimated GFR); monitor drug levels; once-daily aminoglycoside.
- Hydrate before contrast or cisplatin; alkalinise for methotrexate.

- **Stop the culprit**, withhold NSAIDs and ACE inhibitors, restore volume. **Early steroids for AIN** (only about half of antibiotic AIN recovers fully).
- Dialysis for refractory AEIOU: acidosis, electrolytes, intoxication, overload, uraemia.

Diagram to draw: glomerulus with afferent (prostaglandin) and efferent (angiotensin II) control.

> **Key points to remember:**
> - NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
> - ATN: aminoglycosides, amphotericin, cisplatin, contrast.
> - AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
> - Crystals: aciclovir, methotrexate, indinavir.

SAQ Hypokalemia

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3f

Definition and physiology

- **Hypokalaemia:** serum potassium **<3.5 mEq/L**; severe <2.5 mEq/L.
- About **98% of body potassium is intracellular**; the ratio sets the resting membrane potential.
- Two defences: **shift into cells** (insulin, β2-adrenergic stimulation, alkalosis) and **renal excretion** (distal nephron). Distal secretion rises with high urine flow, distal sodium delivery, **aldosterone** and alkalosis.

Causes

Mechanism	Causes
Low intake	Malnutrition
Shift into cells	Insulin, catecholamines (stress, β-agonists), alkalosis, refeeding, periodic paralysis
GI (gastrointestinal) loss	Diarrhoea, villous adenoma, **ureterosigmoidostomy and colonic diversions** (colon secretes potassium, ileum absorbs it)
Renal loss, high BP (blood pressure)	**Primary hyperaldosteronism (Conn)**, renin-secreting juxtaglomerular tumour, Cushing, Liddle, **abiraterone** (mineralocorticoid excess)
Renal loss, normal BP	Loop and thiazide diuretics, **postobstructive diuresis**, osmotic diuresis, **RTA (renal tubular acidosis) types 1 and 2**, Bartter, Gitelman, hypomagnesaemia, amphotericin, cisplatin

- **Vomiting:** potassium is lost mainly in urine (alkalosis → kaliuresis), not in gastric juice.

Clinical features

- Muscle weakness, cramps, ileus, **flaccid paralysis**, rhabdomyolysis.
- Arrhythmias; potentiates digoxin toxicity.
- Polyuria (nephrogenic diabetes insipidus); **hypocitraturia** → calcium stones.
- **ECG (electrocardiogram):** flat T wave, **prominent U wave**, ST depression, prolonged QU; ventricular arrhythmias.

Evaluation

- Repeat potassium; magnesium, bicarbonate, glucose.
- **24-h urine potassium:** <20 mmol/day suggests extrarenal loss; higher suggests renal loss.
- Renal loss: check blood pressure and acid–base; **aldosterone–renin ratio** if hypertensive (only a minority of Conn patients are hypokalaemic).

Management

- Treat the cause; stop offending drugs.
- **Potassium chloride** preferred (potassium citrate if acidotic, phosphate if hypophosphataemic). **Oral** 20–40 mEq repeated when possible.
- **Intravenous** for severe or symptomatic cases: usually ≤10 mEq/h peripherally; faster only centrally with ECG monitoring. Never as a bolus.
- **Correct magnesium**, or potassium will not correct.
- **Urinary diversion:** replace potassium **before or with bicarbonate**; correcting acidosis alone drives potassium into cells → paralysis.
- Correct before abiraterone and adrenalectomy for Conn syndrome (spironolactone preoperatively).

Diagram to draw: ECG strip showing flat T and U wave.

> **Key points to remember:**
> - Shift or loss; urine potassium separates renal from extrarenal.
> - ECG: flat T, U wave.
> - Diversion acidosis: give potassium with or before bicarbonate.
> - Correct magnesium first.

SAQ Hyperfiltration injury in AKI and CKD

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3c

Definition

- **Hyperfiltration:** a rise in **SNGFR (single-nephron glomerular filtration rate)** in surviving nephrons when nephron mass falls; whole-kidney GFR (glomerular filtration rate) may look normal.
- **Brenner hypothesis (1982):** the adaptation itself damages glomeruli.

Physiology

- Net filtration pressure ≈ 10 mmHg (hydrostatic 60 − Bowman's 18 − oncotic 32).
- Afferent dilatation raises flow and filtration; **efferent constriction (angiotensin II) raises filtration pressure**.

Mechanism

Nephron loss → afferent dilatation + efferent constriction → raised glomerular capillary pressure and flow (hyperfiltration) → glomerular hypertrophy; podocytes cannot divide → stretch, detachment → proteinuria → FSGS (focal

segmental glomerulosclerosis) → tubulointerstitial inflammation, fibrosis → more nephron loss (vicious cycle)

In AKI (acute kidney injury)

- Incomplete repair leaves fewer nephrons, which hyperfilter; creatinine may normalise while **renal reserve is lost**.
- This drives the **AKI-to-CKD (chronic kidney disease) transition**; AKI persisting ≥90 days becomes CKD.
- Each perioperative AKI episode is a **modifiable risk factor** for CKD progression.

In CKD

- Final common pathway of progression; marked by **albuminuria (KDIGO (Kidney Disease: Improving Global Outcomes) A stages)**, hypertension, falling eGFR (estimated GFR).

Urological settings

Setting	Point
Nephrectomy or donation	Compensatory hypertrophy 20–30%; donor function 60–80% of pre-donation
Partial nephrectomy, solitary kidney	>50% parenchymal loss → proteinuria, FSGS; marked when total nephron mass falls >80%; ESRD (end-stage renal disease) if <20–30% of one kidney remains
Congenital solitary kidney, MCDK (multicystic dysplastic kidney)	Monitor BP (blood pressure), proteinuria, yearly ultrasound

- **Also know:** reflux nephropathy, infundibulopelvic stenosis, diabetes, obesity, pregnancy, low birth weight (low nephron endowment).

Prevention and treatment

- Nephron sparing; avoid NSAIDs (non-steroidal anti-inflammatory drugs) and nephrotoxins.
- **BP <130/80 mmHg**; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)** in proteinuric disease: efferent dilatation lowers glomerular pressure.
- **SGLT2 (sodium–glucose cotransporter-2) inhibitors:** restore tubuloglomerular feedback, causing afferent constriction. **DAPA-CKD** and **EMPA-KIDNEY** slowed progression with or without diabetes.
- Finerenone (type 2 diabetic CKD); dietary protein restriction.

Diagram to draw: glomerulus with dilated afferent and constricted efferent arterioles.

> **Key points to remember:**
> - Hyperfiltration = raised SNGFR; Brenner 1982.
> - Leads to FSGS, proteinuria and the AKI-to-CKD transition.
> - ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

SAQ UW solution and organ preservation solutions

Paper answer · P2

Asked as:

- UW solution. — *Summer 2017 Paper II Q3c (SAQ)*

- Organ preservation solution — *Practice Set 7 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)

Assumed context: preservation of the kidney for transplantation.

Introduction

- **University of Wisconsin solution** (Belzer, late 1980s): the **gold standard** cold-storage solution for kidney, liver and pancreas.

Principles of cold preservation

- Cooling / **Hypothermia** to **4 °C** cuts (slows) metabolism to roughly a tenth, but does not stop it, so injury accumulates with time.
- Ischaemia causes ATP (adenosine triphosphate) loss (depletion) → **Na^+/K^+-ATPase pump failure** → **cell swelling**; anaerobic glycolysis → **acidosis**; calcium influx. On reperfusion, hypoxanthine is converted by xanthine oxidase to **oxygen free radicals** → ischaemia–reperfusion injury, seen clinically as (clinically this appears as) **DGF (delayed graft function)**.
- A solution must therefore prevent swelling and oedema, buffer acid, scavenge radicals and supply ATP precursors. It therefore needs:
- **Impermeants** (lactobionate, raffinose, mannitol, glucose) to prevent cell swelling.
- **Buffers** (phosphate, histidine) against acidosis.
- **Colloid** (hydroxyethyl starch, polyethylene glycol) against interstitial oedema.
- **Antioxidants and energy precursors** (glutathione, allopurinol, adenosine).
- Electrolytes: "intracellular" (high K^+, low Na^+) or "extracellular" (low K^+, high Na^+) type.

UW composition (per litre)

Component	Amount	Purpose
Lactobionate (K^+ salt)	100 mmol	Impermeant anion; prevents cell swelling; chelates calcium
Raffinose	30 mmol	Impermeant trisaccharide
Hydroxyethyl starch	50 g	Colloid; prevents interstitial oedema
KH_2PO_4	25 mmol	Buffer
$MgSO_4$	5 mmol	Membrane stability
Adenosine	5 mmol	ATP precursor
Glutathione	3 mmol	Antioxidant
Allopurinol	1 mmol	Xanthine oxidase inhibitor; fewer free radicals
Insulin, dexamethasone, antibiotic	Small	Additives
- **Intracellular-type**: K^+ about 125 mmol/L, Na^+ about 30 mmol/L; osmolality ~320 mOsm/kg; pH 7.4.		

UW drawbacks

- **Viscous** (starch): slow flush. **High K^+**: flush before reperfusion (hyperkalaemia, vasospasm). May crystallise (filter).

Comparison of solutions

Feature	UW	HTK (Custodiol)	Euro-Collins
Type	Intracellular (high K^+)	Low K^+, low Na^+	Intracellular
Impermeant	Lactobionate, raffinose	Mannitol	Glucose (enters cells; weaker)
Buffer	Phosphate	**Histidine** (strong)	Phosphate
Colloid	HES (hydroxyethyl starch)	None (low viscosity)	None
Others	Antioxidants, adenosine	Tryptophan, ketoglutarate	—

Solution	Type and key components	Points
UW (University of Wisconsin, Belzer)	Intracellular; lactobionate, raffinose, hydroxyethyl starch, glutathione, adenosine, allopurinol	Standard for static storage; viscous; flush out high K^+ before reperfusion
HTK (histidine–tryptophan–ketoglutarate, Custodiol)	Low K^+; histidine buffer, tryptophan (membrane stabiliser), ketoglutarate (substrate), mannitol	Low viscosity, fast cooling; large volume; outcomes similar to UW
Euro-Collins	Intracellular; glucose impermeant, phosphate	First generation; largely abandoned
Celsior	Extracellular; lactobionate, mannitol, glutathione, histidine	Designed for heart; also abdominal organs
IGL-1 (Institut Georges Lopez)	UW-like, but polyethylene glycol and reversed Na^+/K^+	Extracellular; less viscous
- Kidney outcomes with UW and HTK (histidine–tryptophan–ketoglutarate) are similar; Euro-Collins largely abandoned. **Also know:** Celsior, IGL-1, Marshall's (hypertonic) citrate.		

Practical use and methods

- Living donor, very short cold ischaemia: cold lactated Ringer's suffices; deceased donor or exchange: UW or HTK.
- **Static cold storage:** in situ aortic flush, IVC (inferior vena cava) venting, ice slush; storage at 4 °C. Keep (aim) cold ischaemia short (ideally <24 h / hours): DGF and thrombosis rise with time (beyond).
- **Hypothermic machine perfusion** (Belzer MPS (machine perfusion solution), gluconate-based): pulsatile cold perfusion; flow and resistance help assess viability. Reduces DGF vs static storage; **Moers (2009)**: DGF 20.8% vs 26.5%, 1-year graft survival 94% vs 90%, without long-term survival gain (the chapter notes no difference in long-term survival). Used especially for DCD (donation after circulatory death) kidneys.
- **Normothermic perfusion:** experimental; allows assessment and repair; limited experience.

Diagram to draw: ischaemia(–reperfusion) cascade with the UW / solution component acting at (that blocks) each step.

Key points to remember:

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).
- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
- Machine perfusion reduces DGF.

SAQ Post transplant malignancy

Paper answer · P2

Type: SAQ (10 marks)
Source: 2023-Winter Paper 2, Q3c

Introduction

- Cancer risk after kidney transplantation is about 2–3 times the general population; a leading cause of late death with a functioning graft.

Causes

- **Lost immune surveillance** from immunosuppression (dose and duration matter; depleting antibodies add risk).
- **Oncogenic viruses:** EBV (Epstein–Barr virus) → PTLD (post-transplant lymphoproliferative disorder); HHV-8 (human herpesvirus-8) → Kaposi sarcoma; HPV (human papillomavirus) → skin and anogenital cancer; hepatitis B/C → liver cancer.
- **Direct drug effects:** azathioprine sensitises skin to UV (ultraviolet) light; calcineurin inhibitors promote tumour growth. mTOR (mammalian target of rapamycin) inhibitors are **antitumour**.
- **Pre-existing risk:** ESRD (end-stage renal disease), acquired cystic disease, analgesic nephropathy; rarely donor-transmitted.

Common cancers

Cancer	Key facts
Non-melanoma skin	Commonest; squamous exceeds basal cell (reverse of general population); multiple, aggressive
PTLD	Mostly B-cell, EBV-driven; highest in first year, children and **EBV-negative recipient of EBV-positive donor**; may present as graft mass
Kaposi sarcoma	HHV-8; regresses with less immunosuppression and switch to sirolimus
RCC (renal cell carcinoma)	4.6% of post-transplant cancers; **90% in native kidneys**; graft tumours mostly papillary, low grade
Bladder cancer	About **3-fold** risk; more aggressive
Prostate	Same incidence as general population

Management

- **Reduce immunosuppression** and switch to an mTOR inhibitor; sirolimus conversion reduced new skin squamous cancers (Euvrard, 2012).
- **PTLD:** reduce immunosuppression, rituximab, then chemotherapy (R-CHOP).

- RCC, prostate and bladder: treat as in the general population; protect graft vessels and ureter at surgery. BCG (bacillus Calmette–Guérin) is usable in selected bladder cancer.

Prevention and screening

- Pre-transplant cancer screening and waiting periods (none after curative treatment of small RCC <3 cm, Gleason 6 prostate cancer, superficial bladder cancer).
- Sun protection, yearly skin examination, age-appropriate screening.
- KDIGO (Kidney Disease: Improving Global Outcomes) does **not** endorse routine native-kidney screening; screen only high-risk recipients.

Diagram to draw: pie chart of post-transplant cancers (skin largest).

> **Key points to remember:**
> - Skin cancer commonest; squamous more than basal cell.
> - PTLD: EBV-negative recipient of EBV-positive donor, first year.
> - RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
> - Reduce immunosuppression and switch to mTOR inhibitor.

SAQ Mycophenolate Mofetil.

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 2, Q3b (immunosuppressants also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)

Introduction

- **MMF (mycophenolate mofetil):** an antimetabolite immunosuppressant; morpholinoethyl ester **prodrug** of MPA (mycophenolic acid). With tacrolimus it forms maintenance therapy in **over 90%** of kidney recipients.

Mechanism

- MPA is a reversible, non-competitive inhibitor of **IMPDH (inosine monophosphate dehydrogenase)**, the key enzyme of **de novo guanosine nucleotide synthesis**.
- **Why lymphocyte-selective:** T and B lymphocytes depend on the de novo pathway; other cells use the salvage pathway. MPA also inhibits the type II isoform found in activated lymphocytes.
- Result: arrest of T- and B-cell proliferation and reduced antibody formation.

Pharmacology

Feature	Detail
Dose (kidney)	**1 g twice daily** oral or IV; EC-MPS (enteric-coated mycophenolate sodium) 720 mg twice daily is equivalent
Metabolism	Hydrolysed to MPA; liver glucuronidation to inactive MPAG (MPA glucuronide); **enterohepatic recirculation**
Excretion	MPAG in urine; accumulates in renal failure
Interactions	**Ciclosporin lowers MPA** (blocks enterohepatic recirculation), tacrolimus does not; antacids, cholestyramine reduce absorption; never with azathioprine

Uses

- Kidney transplant maintenance (tacrolimus + MMF ± steroids); other solid organs.
- Lupus nephritis, vasculitis; steroid-refractory immune-related adverse events of checkpoint inhibitors.

Adverse effects

- **GI (gastrointestinal):** diarrhoea, nausea, colitis (commonest; dose split or EC-MPS).
- **Marrow:** leucopenia, anaemia, thrombocytopenia: blood count weekly in month 1.
- **Infection:** CMV, BK virus (ureteric stricture), rarely PML (progressive multifocal leucoencephalopathy).
- **Malignancy:** skin cancer, PTLD (post-transplant lymphoproliferative disorder).
- **Teratogenic:** miscarriage, ear and facial malformations; contraception mandatory; switch to azathioprine before conception.
- Advantage: **not nephrotoxic**; no hypertension, diabetes or dyslipidaemia.

Evidence

- Three mid-1990s randomised trials (US, European, Tricontinental) showed fewer early acute rejections than azathioprine or placebo, so MMF replaced azathioprine.
- **ELITE-Symphony (2007):** low-dose tacrolimus + MMF + daclizumab + steroids gave the lowest rejection and best GFR.

Diagram to draw: purine synthesis pathway showing IMPDH block (de novo) and the salvage pathway spared.

> **Key points to remember:**
> - Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
> - 1 g twice daily; ciclosporin lowers MPA levels.
> - Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

SAQ Renal transplant in neurogenic bladder

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 2, Q3a

Why the bladder matters

- A neurogenic bladder with **poor compliance, low capacity or detrusor overactivity** transmits high pressure to the graft even without reflux. Poor emptying adds infection. Recurrent pyelonephritis, not augmentation, causes graft loss.
- **Five goals:** unimpeded drainage, low-pressure capacious storage, effective emptying, no infection, fewest procedures.

Pretransplant evaluation

- Renal ultrasound, VCUG (voiding cystourethrogram) and **urodynamics** (capacity, compliance, overactivity, emptying, sphincter). Repeat them; the bladder changes over time.
- **Target:** storage to expected capacity at detrusor pressure **<40 cm H_2O**.

- **Defunctionalised bladder** (anuria or diversion): **bladder cycling** (saline fills of increasing volume via catheter), then urodynamics.

Pretransplant optimisation (stepwise)

`Antimuscarinic (oxybutynin), mirabegron, botulinum toxin → CIC (clean intermittent catheterisation) ± overnight drainage → bladder cycling if defunctionalised → still >40 cm H₂O, <75% expected capacity or CIC ≤3-hourly → augmentation`

- **Augmentation:** ileum or sigmoid most used; dilated ureter (ureterocystoplasty) preferred when available (no mucus or metabolic effects). **Stomach avoided** (haematuria–dysuria syndrome in anuric patients). Augmentation does not clearly raise transplant complications.
- **Mitrofanoff** channel if urethral CIC is difficult; site it **medial** to the planned transplant incision.
- Do reconstruction **before** transplant; allow **6–12 weeks** to heal. Neuropathic bladders may deteriorate under high graft output, so earlier reconstruction is justified.
- **Native nephrectomy** only for named reasons: recurrent upper-tract infection, massive reflux, stones, uncontrolled hypertension.

At transplant

- Extravesical non-refluxing ureteroneocystostomy into bladder or augment (know the augment pedicle). Ureteroureterostomy to a native ureter is an option.
- Ileal conduit drainage if the bladder is unusable.

Post-transplant care

- Continue CIC; colonisation is expected, so treat only symptomatic UTI (urinary tract infection).
- **Creatinine rise with hydronephrosis:** catheter drainage for 1–2 weeks first. If it improves, the bladder is the cause; otherwise stent or nephrostomy and consider rejection.
- Febrile UTI with graft reflux: reimplant after optimising the bladder.
- Lifelong surveillance of augments (malignancy, stones).

Diagram to draw: stepwise bladder optimisation ladder.

> **Key points to remember:**
> - Target <40 cm H_2O storage before transplant.
> - Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
> - Pyelonephritis, not augmentation, loses grafts.

SAQ Pre-emptive renal transplant

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3a

Definition

- Kidney transplantation **before starting maintenance dialysis** in a patient with advanced CKD (chronic kidney disease).

Rationale

- Every year on dialysis adds cardiovascular disease, vascular calcification and bone–mineral damage that the graft later inherits. Pre-emptive transplantation **avoids this exposure**.
- Transplantation beats dialysis: Wolfe (1999) showed early risk is higher, equal at 106 days, then 68% lower long term.

Advantages

- Better patient and graft survival than transplantation after dialysis; longer dialysis time worsens outcome (registry data).
- No dialysis access surgery, access infection or catheter complications.
- Fewer transfusions, so less HLA (human leucocyte antigen) sensitisation.
- Better quality of life, employment; lower cost.
- Children: better growth and development.

Frequency

- 28.7% of US transplants in 2021 (living donor 51%, deceased donor 21%, children 56%). Commonest with a **living donor**.

Timing and evaluation

- Evaluate early, ideally before dialysis. KDIGO (Kidney Disease: Improving Global Outcomes) 2020: refer when eGFR (estimated glomerular filtration rate) is <30 mL/min/1.73 m², 6–12 months before expected dialysis. Living-donor transplant is usually done at eGFR <20 with progressive decline.
- Deceased-donor listing (US) at eGFR ≤20.
- Full recipient work-up: cardiac, vascular (iliac calcification), infection (TB [tuberculosis], hepatitis B and C, HIV [human immunodeficiency virus]), malignancy, psychosocial; recurrence risk (primary FSGS [focal segmental glomerulosclerosis]).
- **Urological:** bladder function and lower tract (valves, neurogenic bladder); native nephrectomy only for specific indications.
- India: mainly living related donors under THOA (Transplantation of Human Organs Act).

Limitations

- Needs early referral and a ready donor.
- Transplant may come earlier than needed, using up dialysis-free native function; adherence untested.
- In advanced CKD, bowel augmentation may speed decline.

Also know: ABO-incompatible or paired exchange programmes widen pre-emptive options.

Diagram to draw: Wolfe curve: relative risk of death after transplant vs dialysis over time.

> **Key points to remember:**
> - Transplant before dialysis = best outcome.
> - Refer at eGFR <30; transplant usually at eGFR <20.
> - Commonest with living donors and children.

SAQ Tacrolimus.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2017-Winter Paper 3, Q3c

Introduction

- **Tacrolimus (FK506)**: macrolide CNI (calcineurin inhibitor) from *Streptomyces tsukubaensis*; about 10–100 times more potent than ciclosporin. With mycophenolate it is maintenance in **over 90%** of kidney recipients.

Mechanism

`Tacrolimus + FKBP12 (FK506-binding protein 12) → complex inhibits calcineurin (Ca²⁺/calmodulin phosphatase) → NFAT (nuclear factor of activated T cells) stays phosphorylated, cannot enter nucleus → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)`

- Ciclosporin binds **cyclophilin** instead; same end result.

Pharmacokinetics

- Poor, variable oral bioavailability; reduced by food. Binds red cells, so **whole-blood trough (C0)** levels are monitored (narrow therapeutic index).
- Metabolised by **CYP3A4/3A5**; effluxed by **P-gp (P-glycoprotein)**. CYP3A5 expressers need higher doses.

Formulations

- Immediate-release (twice daily); prolonged-release (Advagraf, once daily); LCP-tacrolimus (Envarsus, once daily, higher bioavailability); intravenous; topical. Once-daily forms aid adherence.

Dosing and monitoring

- Started about 0.1–0.2 mg/kg/day in two doses, then adjusted to trough. Targets vary by centre and risk; **ELITE-Symphony low-dose arm: 3–7 ng/mL**.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent vasoconstriction (reversible); chronic arteriolar hyalinosis, striped fibrosis; TMA (thrombotic microangiopathy)
Metabolic	**NODAT/PTDM (new-onset/post-transplant diabetes)**, more than ciclosporin; **hyperkalaemia**, **hypomagnesaemia**, hyperuricaemia
Neurological	**Tremor**, headache, paraesthesia, seizures, PRES (posterior reversible encephalopathy syndrome)
Others	Hypertension, GI (gastrointestinal) upset, alopecia, infection (CMV (cytomegalovirus), BK polyomavirus), malignancy (skin, PTLD (post-transplant lymphoproliferative disorder))

Drug interactions

- **Raise levels (CYP3A4 inhibitors):** azoles, clarithromycin/erythromycin, diltiazem, verapamil, protease inhibitors, grapefruit; diarrhoea.
- **Lower levels (inducers):** rifampicin, phenytoin, carbamazepine, St John's wort.
- Additive nephrotoxicity: NSAIDs (non-steroidal anti-inflammatory drugs), aminoglycosides; ACE (angiotensin-converting enzyme) inhibitors add to hyperkalaemia.

Versus ciclosporin

	Tacrolimus	Ciclosporin
Acute rejection	Lower	Higher
Diabetes, tremor, alopecia	More	Less
Hirsutism, gum hypertrophy, hyperlipidaemia, hypertension	Less	More
- **ELITE-Symphony (2007):** low-dose tacrolimus + mycophenolate + daclizumab + steroids gave the **lowest rejection, best GFR (glomerular filtration rate) and graft survival** at 1 year versus ciclosporin or sirolimus regimens.		
- **Also know:** CNI minimisation with belatacept or everolimus; topical tacrolimus in genital dermatoses (lichen planus, Zoon balanitis).		

Diagram to draw: T-cell showing FKBP12–calcineurin–NFAT–IL-2 pathway.

> **Key points to remember:**
> - FKBP12 → calcineurin → NFAT → IL-2 blocked.
> - CYP3A4/P-gp; trough-guided; many interactions.
> - Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
> - ELITE-Symphony: low-dose tacrolimus best.

SAQ Urological Complications post Kidney Transplantation.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2023-Winter Paper 3, Q3d

Introduction

- Urological complications are the **commonest surgical complications** (8–9.5%). Most arise from the **ureter and ureterovesical anastomosis**.
- The distal donor ureter depends on a single renal-artery blood supply, so ischaemia is the root cause. **Denervation** masks pain.
- A **routine stent** reduces leak and stricture; remove at **14–21 days** (earlier increases leak; later increases UTI [urinary tract infection]).

Complications

Complication	Key facts	Management
Urine leak (1.2–8.9%; commonest early)	Distal ureteric necrosis; lost lower-pole artery; technical	Fluid creatinine far above serum. Foley + PCN (percutaneous nephrostomy) + antegrade stent; early reimplant or native ureteroureterostomy if large
Ureteric stricture (0.5–9%; commonest beyond 3 months)	Ischaemia; long cold ischaemia, rejection, BK (BK polyomavirus) and CMV (cytomegalovirus)	Nephrostomy first; balloon or laser endoureterotomy; **open reimplant, native ureteroureterostomy or Boari flap best long-term**
Lymphocele (symptomatic 0.03–26%; peak 6 weeks)	Divided lymphatics; mTOR (mammalian target of rapamycin) inhibitors	Treat only if symptomatic; **laparoscopic peritoneal fenestration** best

Complication	Key facts	Management
VUR (vesicoureteric reflux)	50–86% radiological; <1% significant	Treat cause; endoscopic injection low grade; reimplant
Haematuria	Anastomotic; persistent after biopsy = AV fistula	Self-limiting; selective embolisation
Stones (about 1%)	Hyperparathyroidism; calcineurin inhibitors	As native kidney; watch silent obstruction
UTI (commonest infection)	Stent, reflux, steroids	Treat 14–21 days if pyelonephritis

- **Obstruction:** early is technical, oedema or compression (haematoma, lymphocele); late is stricture, stone or lymphocele.
- **Encrusted stent:** CT, nephrostomy, then PCNL (percutaneous nephrolithotomy), RIRS (retrograde intrarenal surgery) or ESWL (shock-wave lithotripsy).
- **Lower tract:** bladder capacity recovers; BPH (benign prostatic hyperplasia) treated by TURP (transurethral resection of prostate) after 3 weeks, never with a stent in place.
- **Also know:** ED (erectile dysfunction) persists in 20–50%; bladder cancer risk about threefold; RCC (renal cell carcinoma) mostly in native kidneys; renal artery stenosis and thrombosis.

Diagram to draw: transplant kidney in iliac fossa showing leak, stricture and lymphocele sites.

> **Key points to remember:**
> - Ureter is the weak point; ischaemia is the cause.
> - Leak early, stricture late; open repair best.
> - Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

SAQ Female kidney donors: special precautions

Paper answer · P4

Asked as:

- Female kidney donors: enumerate special precautions — *Winter 2023 Paper IV Q3d (SAQ)*
- Female kidney donors: enumerate special precautions — *Practice Set 4 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3d)

Introduction

- Most living donors, especially in India, are **women** (wives, mothers). Guidance: **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** living donor guideline.

Before donation

1. **Free, uncoerced consent:** interview the donor alone; independent donor advocate; family pressure and gender inequity screened. In India the THOA (Transplantation of Human Organs Act) authorisation committee verifies this.
2. **Pregnancy:** pregnancy test before contrast CT (computed tomography) and surgery; **reliable contraception** during work-up. Pregnant women do not donate.

3. **Obstetric history:** previous **pre-eclampsia, gestational hypertension or gestational diabetes** signals higher future hypertension, diabetes and kidney risk; evaluate and counsel.
4. **Future pregnancy counselling (KDIGO):** donation **increases gestational hypertension and pre-eclampsia** (Garg et al., NEJM 2015: about 11% vs 5% in non-donors). Maternal and fetal outcomes are otherwise similar (Chapter 91).
5. **Oestrogen therapy:** combined OCPs (oral contraceptive pills) and HRT (hormone replacement therapy) raise **VTE (venous thromboembolism)** risk; stop them some weeks before surgery and give barrier or progestogen-only contraception; VTE prophylaxis.
6. **Gynaecological and cancer screening:** cervical cytology, breast examination or mammography (age-appropriate); pelvic ultrasound if indicated.
7. **Urine tests:** repeat urinalysis **outside menstruation** to avoid false haematuria; exclude recurrent UTI (urinary tract infection).
8. **GFR (glomerular filtration rate):** low muscle mass makes creatinine-based estimates unreliable; **measured GFR** preferred. Note nephron mismatch with a large recipient.
9. Correct **iron-deficiency anaemia**.

Operative

- Laparoscopic donor nephrectomy; **Pfannenstiel** extraction for cosmesis; transvaginal extraction possible.

After donation

- **Defer pregnancy** until recovery and stable function; many programmes advise at least 6–12 months (another source: often 6–12 months).
- In pregnancy: inform obstetrician; **high-risk antenatal care**; monitor BP (blood pressure), proteinuria and creatinine; low-dose aspirin may be considered.
- **Lifelong annual BP, creatinine and proteinuria**; avoid NSAIDs (non-steroidal anti-inflammatory drugs).

Diagram to draw: flowchart: consent → pregnancy test → hormone review → GFR → surgery → pregnancy follow-up.

> **Key points to remember:**
> - Ensure free consent (Indian gender gap).
> - Pregnancy test; contraception; stop oestrogens.
> - Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
> - Defer pregnancy; high-risk antenatal follow-up.

SAQ Laparoscopic donor nephrectomy (including pros and cons)

Paper answer · P4

Asked as:

- Laparoscopic donor nephrectomy — *Practice Set 2 Paper IV Q3a (SAQ)*
- Pros and cons of lap donor nephrectomy — *Practice Set 2 Paper IV Q3d (SAQ)*

Type: SAQ (10 marks)
(living donation also in 2023-Winter Paper 1, Q3c; the pros and cons are asked again in Practice Set 2 Paper 4, Q3d) **See Practice Set 2 Paper 4, Q3a (full answer).**

Introduction

- **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first.**
- Work-up includes measured GFR (glomerular filtration rate) and CT (computed tomography) angiography.

Side selection

- **Left preferred:** longer renal vein, simpler dissection.
- **Right** if the left has multiple arteries or complex veins, or the right is smaller or abnormal: the donor keeps the better kidney. The right vein is short.

Techniques

Technique	Features
Pure transperitoneal laparoscopic	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds of US cases)	Hand port is also the extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	Balloon-made space; no bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed dissection; single umbilical incision; cost

Key steps

1. Reflect colon; divide gonadal, adrenal and lumbar veins (left).
2. Keep periureteric tissue and the gonadal–ureter triangle (ureteric blood supply).
3. Hydration and diuretic before vessel division.
4. Divide ureter, artery, then vein (stapler; see Q3f on Hem-o-lok).
5. Extract quickly; cold flush.

Warm ischaemia

- Arterial occlusion to cold flush: a few minutes, slightly longer than open, without effect on long-term graft function.

Pros and cons (versus open)

Pros	Cons
Less pain, shorter stay, faster (earlier) return to work	Longer operation; learning curve
Better cosmesis; **more people volunteer to donate** (more willing donors)	Slightly longer warm ischaemia
Magnified view of hilum	Shorter vessels, especially right vein (on the right)
Equal graft function and survival (graft function equals open)	Pneumoperitoneum lowers renal blood flow; cost

- **Risks (safety):** complications about 16% (mostly gastrointestinal); mortality **3 per 10,000**; conversion 0.26–3.34%. Lifetime ESRD (end-stage renal disease) 90 per 10,000 versus 14 in matched non-donors.

Diagram to draw: left hilum with gonadal, adrenal and lumbar veins and the stapled artery and vein.

> **Key points to remember (including pros and cons angle):**
> - Ratner 1995; left kidney preferred (longer vein).
> - Hand-assisted in about two-thirds of US cases.
> - Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
> - Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
> - Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
> - Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

SAQ Artificial kidney

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3e

Read as devices that replace kidney function: the dialyser (classic "artificial kidney") and newer wearable, implantable and bioartificial kidneys.

Definition

- An **artificial kidney** removes solutes and water from blood across a semipermeable membrane. The haemodialysis dialyser is the standard example.
- Pioneered by **Kolff (1943)**; long-term use followed the Scribner shunt (1960) and Brescia–Cimino AV (arteriovenous) fistula (1966).

Principles

Process	Mechanism	Removes
Diffusion	Concentration gradient across membrane	Small solutes (urea, potassium)
Convection (haemofiltration)	Solute dragged with water	Middle molecules
Ultrafiltration	Hydrostatic pressure gradient	Water
Adsorption	Binding to membrane or sorbent	Some proteins, toxins

- **Hollow-fibre dialyser**; blood and dialysate flow **countercurrent** to keep the gradient.
- **Schedule:** blood flow 250–350 mL/min, 3–5 hours, 3–5 times a week; adequacy by Kt/V (urea clearance × time ÷ volume), at least 1.2 per session.
- **Peritoneal dialysis** uses the peritoneum as membrane (glucose or icodextrin draws water).

Limitations of current dialysis

- Intermittent: fluid and solute swings cause cramps.
- No endocrine (erythropoietin, vitamin D activation) or tubular metabolic function.
- Access thrombosis, infection, steal; cardiovascular disease, vascular calcification.

- **Transplantation remains the best renal replacement therapy.**

Newer artificial kidneys

Device	Principle	Status
WAK (wearable artificial kidney)	Belt-worn; sorbent regenerates a small volume of dialysate	Early human trials
Wearable or portable peritoneal dialysis	Continuous sorbent-regenerated peritoneal fluid	Development
Implantable bioartificial kidney	Silicon nanopore haemofilter plus bioreactor of renal tubule cells; driven by blood pressure, no pump	Preclinical
RAD (renal assist device)	Human proximal tubule cells in a hollow-fibre cartridge with continuous dialysis	Phase II in acute kidney injury; not adopted
Decellularised or organoid kidneys; pig kidney xenotransplant	Regenerative and xenograft routes	Experimental (see Practice Set 6 Paper 4, Q2)

Urologist's role

- AV fistula and graft work-up, laparoscopic PD (peritoneal dialysis) catheter placement with omental fixation, transplant.

Diagram to draw: hollow-fibre dialyser with countercurrent blood and dialysate flow.

> **Key points to remember:**
> - Diffusion, convection, ultrafiltration across a semipermeable membrane.
> - Dialysis lacks endocrine and tubular function; transplant is best.
> - Wearable, implantable bioartificial and RAD devices are investigational.

SAQ Robotic kidney transplantation

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3f

Introduction

- **RAKT (robot-assisted kidney transplantation)**: the graft is implanted intracorporeally on the iliac vessels using the robot.
- Milestones: Hoznek (2002, robot-assisted anastomosis); Giulianotti (2010, obese recipient); **Menon and Ahlawat (Vattikuti–Medanta, 2014)** standardised the transperitoneal technique with **regional hypothermia**, assessed by the IDEAL (idea, development, exploration, assessment, long-term study) framework.

Indications and selection

- Living-donor grafts mainly; increasingly deceased donors.
- **Obese recipients** gain most, as open transplant carries more wound complications.
- **Contraindications (relative):** heavy iliac calcification on non-contrast CT (computed tomography), multiple graft vessels early in experience, previous transplant on that side, dense adhesions.

Technique (Vattikuti–Medanta)

1. Lithotomy, steep Trendelenburg; transperitoneal ports.
2. Periumbilical **GelPOINT** incision for graft and ice insertion.
3. Peritoneal flap; expose the **external iliac vessels**.
4. Graft wrapped in gauze with ice slush; **ice slush in the iliac fossa** for regional hypothermia.
5. **Vein first:** renal vein end-to-side to external iliac vein, to shorten arterial ischaemia.
6. Renal artery end-to-side to external iliac artery; unclamp and reperfuse.
7. Extravesical **Lich–Gregoir** ureteroneocystostomy over a double-J stent.
8. Graft retroperitonealised with the peritoneal flap to prevent torsion.

Advantages

- Small incision: **fewer wound infections, less lymphocele**, less pain, faster recovery, better cosmesis.
- Magnified 3D (three-dimensional) view and precise suturing in deep pelvis or obese patients.

Disadvantages

- **Longer rewarming (anastomosis) time**; cost; learning curve; pneumoperitoneum and Trendelenburg effects.
- No tactile feedback for calcified vessels.

Outcomes

- Multicentre and ERUS (European Association of Urology Robotic Urology Section) series: graft function and survival comparable with open transplant, with lower surgical-site complications.

Complications

- Graft thrombosis, haemorrhage, urine leak, ureteric stricture, lymphocele (less), port-site hernia; DGF (delayed graft function).

Diagram to draw: pelvis with graft on external iliac vessels and ureter to bladder dome; ice slush via GelPOINT.

> **Key points to remember:**
>
> - Vattikuti–Medanta technique with regional hypothermia (2014).
> - Vein first, then artery, then Lich–Gregoir.
> - Best for obese recipients; graft function equals open.

Rapid Revision Sheet · Section 15

Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL (LAQ)

- Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
- Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
- Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
- Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
- Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
- High intrarenal pressure in mini-PCNL causes fever and sepsis.

Renovascular hypertension: pathophysiology, evaluation and treatment (LAQ)

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

GFR estimation in prospective renal donor (SAQ)

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

Acute renal failure (AKI) and acute tubular necrosis (LAQ)

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Metabolic abnormalities in CKD (SAQ)

- FGF-23 rises first; secondary then tertiary hyperparathyroidism.
- Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
- Hyperkalaemia: binders rather than stopping RAAS blockade.

Management of acute hyperkalaemia (SAQ)

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

Pre transplant nephrectomy (SAQ)

- Default: leave native kidneys.
- Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
- Retroperitoneal route for PD; leave a normal ureter.

Discuss the role of tissue typing and cross match in live and cadaveric (LAQ)

- HLA on chromosome 6; match A, B, DR; DR matters most.
- Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
- Flow and bead assays are more sensitive; cPRA guides allocation.
- Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

Deceased donor kidney: selection, maintenance and retrieval techniques (LAQ)

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Evaluation and selection criteria of the living renal donor (SAQ)

- KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
- GFR 60–89 and albuminuria 30–100 are individualised.
- Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
- ABO and crossmatch first; CT angiography for anatomy.
- Better kidney stays with the donor.
- THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
- Interview video-recorded; no commerce.

Vascular complications after renal transplantation, including transplant renal artery stenosis (SAQ)

- TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
- Refractory hypertension plus rising creatinine: Doppler first.
- Venous thrombosis: reversed diastolic flow, swollen tender graft.
- Arterial thrombosis: sudden anuria; explore immediately.

Lymphocele and postoperative lymphatic leak (chylous ascites) (SAQ)

- Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
- Lymph creatinine = serum; urinoma creatinine higher.
- Laparoscopic fenestration best (is best); aspiration alone worst.
- Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
- Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Allograft rejection and immunobiology of renal transplantation (LAQ)

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

Vascular anastomosis and vascular access for dialysis (LAQ)

- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal: banding or DRIL.

Describe in details basic patho-physiology of urine production and transport (LAQ)

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management) (LAQ)

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.

- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD) (LAQ)

- Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
- HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
- AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
- Tenckhoff: two cuffs, pelvic tip, downward exit.
- Laparoscopic placement with omentopexy gives best function.
- Avoid subclavian lines; AVF first for long-term HD.
- CRRT for unstable AKI; PD preferred in children.

Brain-stem death: pathophysiology, guidelines and declaration (SAQ)

- Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
- Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens (SAQ)

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
- Basiliximab for low risk; rATG for high immunological risk.
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.
- Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
- Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
- Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

CAPD and CAPD peritonitis (SAQ)

- Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.

- Peritonitis: two of cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
- Intraperitoneal Gram-positive plus Gram-negative cover.
- Remove catheter for fungal or refractory (5 days) peritonitis.

THOA/THOTA legislation and hospital authorization committee (SAQ)

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
- 2011: tissues added; grandparents and grandchildren became near relatives.
- Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
- Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
- Hospital-based committee if >25 transplants a year; no transplant team member.
- Non-relatives, swap and foreigners need committee approval.
- Interview video-recorded; appeal within 30 days.

Marginal kidney donor and Remuzzi score (SAQ)

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard.
- HMP reduces DGF; Maastricht III is the commonest DCD.

Renin-angiotensin system (SAQ)

- Renin is rate-limiting; released by low pressure, low NaCl, β1.
- Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
- Renin ratio >1.5 lateralises; reninoma excision cures most.

Drugs associated with Acute Renal Failure. (SAQ)

- NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
- ATN: aminoglycosides, amphotericin, cisplatin, contrast.
- AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
- Crystals: aciclovir, methotrexate, indinavir.

Hypokalemia (SAQ)

- Shift or loss; urine potassium separates renal from extrarenal.
- ECG: flat T, U wave.
- Diversion acidosis: give potassium with or before bicarbonate.
- Correct magnesium first.

Hyperfiltration injury in AKI and CKD (SAQ)

- Hyperfiltration = raised SNGFR; Brenner 1982.
- Leads to FSGS, proteinuria and the AKI-to-CKD transition.
- ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

UW solution and organ preservation solutions (SAQ)

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).

- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
- Machine perfusion reduces DGF.

Post transplant malignancy (SAQ)

- Skin cancer commonest; squamous more than basal cell.
- PTLD: EBV-negative recipient of EBV-positive donor, first year.
- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

Mycophenolate Mofetil. (SAQ)

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

Renal transplant in neurogenic bladder (SAQ)

- Target <40 cm H_2O storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

Pre-emptive renal transplant (SAQ)

- Transplant before dialysis = best outcome.
- Refer at eGFR <30; transplant usually at eGFR <20.
- Commonest with living donors and children.

Tacrolimus. (SAQ)

- FKBP12 → calcineurin → NFAT → IL-2 blocked.
- CYP3A4/P-gp; trough-guided; many interactions.
- Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
- ELITE-Symphony: low-dose tacrolimus best.

Urological Complications post Kidney Transplantation. (SAQ)

- Ureter is the weak point; ischaemia is the cause.
- Leak early, stricture late; open repair best.
- Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Female kidney donors: special precautions (SAQ)

- Ensure free consent (Indian gender gap).
- Pregnancy test; contraception; stop oestrogens.
- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

Laparoscopic donor nephrectomy (including pros and cons) (SAQ)

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.

- Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
- Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

Artificial kidney (SAQ)

- Diffusion, convection, ultrafiltration across a semipermeable membrane.
- Dialysis lacks endocrine and tubular function; transplant is best.
- Wearable, implantable bioartificial and RAD devices are investigational.

Robotic kidney transplantation (SAQ)

- Vattikuti–Medanta technique with regional hypothermia (2014).
- Vein first, then artery, then Lich–Gregoir.
- Best for obese recipients; graft function equals open. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 16

BPH

148 Anatomy and Development of the Prostate

1 · Development (embryology)

The androgen pathways that build the prostate also drive BPH.

- **Origin**: the urogenital sinus (UGS), endoderm in mesoderm. Incomplete cloacal division (imperforate anus, urorectal fistula) impairs development.
- **Dihydrotestosterone (DHT)**: †5α-reductase type 2 in UGS mesenchyme converts testosterone to DHT, the more potent androgen receptor (AR) ligand. **SRD5A2** loss leaves the prostate incomplete; castration or inactive AR leaves it absent. †The same enzyme is the target of 5α-reductase inhibitors (5-ARIs) in BPH.
- **Androgen is not rate-limiting**: testosterone is present at week 9 but budding starts at week 10, gated by **AR demethylation** in the UGS.

Week	Event
6–7	Cloaca divides into the ventral UGS and the dorsal anorectum
≈9	Testosterone activates **mesenchymal** AR, whose andromedins instruct the epithelium
10	Epithelial buds invade the mesenchyme; this fixes the adult ductal pattern

Mesenchyme instructs, epithelium responds

- Budding and branching need **mesenchymal (stromal)** AR, not epithelial AR; UGS mesenchyme can even turn bladder epithelium into prostate (Cunha).
- **Andromedins** are androgen-induced stromal signals, chiefly **fibroblast growth factor 10 (FGF10)**; Nkx3-1 specifies the epithelium.

> **KEY POINT**
>
> McNeal: **re-awakening of embryonic induction** in ageing transition-zone stroma may initiate new glands, one theory of BPH.

- Gestational diethylstilboestrol, phthalates and dioxin disrupt development; dioxin-induced hyperinnervation may explain variable α-blocker response.

2 · Zonal anatomy (McNeal)

LAQ Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate

Paper answer · P1 · P3 · P4 · merged from 3

Asked as: P1 LAQ "Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer" (Anatomy of prostate and its neurovascular bundles with its clinical details; Describe in detail, anatomy of prostate with special reference to concepts in surgical management if localised CA prostate) · P4 LAQ "Prostate anatomy, newer surgical concepts in CA prostate and neurovascular bundle preservation in radical prostatectomy"

(Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate; also SAQ: Neurovascular bundle preservation in Radical prostatectomy: recent techniques) · P3 SAQ "Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate"

"if localised" is read as "of localised" prostate cancer.

Introduction

- A fibromuscular, glandular organ of about 20 g around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**), in front of the rectum.
- Buds from the urogenital sinus at week 10, driven by androgen acting on the **mesenchymal** AR (androgen receptor).
- RP (radical prostatectomy) aims at the "trifecta" of cancer control, continence and potency. Modern RP is anatomical: each anatomical step or new technique either secures a clear margin or saves a named structure (nerves, sphincter, anterior support).

Relations

- **Anterior:** retropubic space (of Retzius), puboprostatic ligaments, DVC (dorsal venous complex).
- **Posterior:** Denonvilliers fascia, then rectal ampulla, hence DRE (digital rectal examination) and TRUS (transrectal ultrasound).
- **Lateral:** levator ani with its fascia; NVB (neurovascular bundle) posterolaterally.
- **Base:** bladder neck; seminal vesicles behind. **Inside:** the urethra bends about 35° at the **verumontanum**, where the ejaculatory ducts open beside the utricle.

Zonal anatomy (McNeal 1981)

Zone	Share of glands	Position	Clinical / surgical relevance
Peripheral (PZ)	About 70%	Posterior and lateral; longest ducts	**70% of cancers**; felt on DRE; biopsy target; posterolateral, so ECE (extracapsular extension) runs along the nerves
Central (CZ)	About 25%	Cone around the ejaculatory ducts at the base	Few cancers; mimics HGPIN (high-grade prostatic intraepithelial neoplasia)
Transition (TZ)	About 5%	Two lobes beside the preprostatic sphincter; shortest ducts	**BPH (benign prostatic hyperplasia)**; anterior cancers (about 15%) missed by DRE
Periurethral glands	Small	Within the preprostatic sphincter	Middle-lobe BPH
Anterior fibromuscular stroma (AFMS)	No glands	Anterior	Anterior tumours reach it; anterior margin risk in posterior approaches

- **Capsule:** there is **no true capsule**, so cancer breaching the gland shows "extraprostatic extension". In BPH the compressed PZ forms the **surgical capsule**, the plane of enucleation. The capsule transmits growth pressure to the urethra, so TUIP (transurethral incision) relieves obstruction.

Fascial investments and dissection planes

Layer	Position	Surgical rule
Prostatic fascia	On the gland; carries DVC tributaries anteriorly	Stays on the prostate in nerve sparing
Levator fascia	Covers levator; fuses with prostatic fascia as the lateral pelvic fascia	Incised to reach the NVB
Denonvilliers fascia	Between prostate and rectum; thickest at base and vesicles	Excised completely for the posterior margin

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral (posterior) ECE is suspected.

Nerve-sparing plane	Lies	Trade-off
Intrafascial	Under the prostatic fascia (between capsule and prostatic fascia)	Most nerves kept; more positive margins
Interfascial	Between prostatic and levator fascia	**Standard nerve-sparing plane** for organ-confined disease
Extrafascial	Outside levator and Denonvilliers fascia	Widest margin; ED (erectile dysfunction) likely

- Chapter 160 (Walsh) calls intrafascial "never used"; Chapter 161 accepts it in selected men.

Blood supply and lymphatics

- **Inferior vesical artery** ends as: **urethral group** (enters at the vesicoprostatic junction, 5 and 7 o'clock; supplies the adenoma) and **capsular group** (posterolateral, in the lateral pelvic fascia, marking the cavernous nerves).
- **Accessory pudendal arteries** run over the anterolateral prostate; dividing them reduces penile inflow. Preserve them; they may be the main penile supply.
- **Veins:** the deep dorsal vein passes under the pubis into a superficial branch and lateral (**Santorini**) plexuses, draining to the internal iliac vein. They anastomose freely, so tears bleed heavily.
- DVC: CO_2 pneumoperitoneum tamponades it, so robotic blood loss is low.
- **Lymph:** obturator, internal and external iliac, common iliac, presacral nodes.

Neurovascular bundle

- **Pelvic plexus:** parasympathetic S2–S4 plus sympathetic (T10–L2) via the hypogastric nerves; beside the rectum, centred on the seminal vesicle tip. It gives the **cavernous nerves** (parasympathetic).
- **NVB of Walsh (Walsh and Donker, 1982):** capsular vessels plus **cavernous nerves**. The nerves are invisible; the capsular vessels mark them.
- It runs posterolaterally in the lateral pelvic fascia **between prostatic and levator fascia**, **3–5 mm from the capsule**, outside Denonvilliers fascia.
- At the apex they lie at **5 and 7 o'clock**, posterolateral to the sphincter. The nerves then swing to about **3 and 9 o'clock** at the mid-urethra and enter the corpora behind the dorsal penile artery. The apex is where most nerve injuries occur.

- Distribution varies (is diffuse): main posterolateral bundle, posterior "hammock", anterolateral fibres (the basis of high anterior release).
- **Injury mechanisms:** incision, entrapment (clip or suture), **heat** (thermal), **traction**; so dissect athermally without traction.
- **Striated sphincter:** a vertical tube over the apex (Oelrich), pudendal nerve; passive continence.

Continence anatomy

Passive	Active
Smooth muscle of bladder neck and urethra; anterior support / suspensory system (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC)	Horseshoe striated sphincter at the apex, supplied by pudendal nerve branches that enter it at the apex

Clinical details

- **BPH surgery:** TURP (transurethral resection of the prostate) stays proximal to the verumontanum (sphincter lies just distal); urethral-group vessels at 5 and 7 o'clock bleed.
- **Cancer:** PZ cancer exits posterolaterally along nerves; anterior TZ tumours are missed by DRE and biopsy, so MRI (magnetic resonance imaging) helps.
- **Radical prostatectomy (RP) steps:** 1. Incise endopelvic fascia; spare the pubourethral ligament. 2. Ligate the DVC over the urethra before division. 3. Preserve accessory pudendal arteries. 4. Excise Denonvilliers fascia; the apex is the commonest positive-margin site.
- Excise an NVB for palpable extracapsular disease; margins are equal with or without nerve sparing.
- **Also know:** rectal injury at the ampulla; PDE5 (phosphodiesterase type 5) inhibitor rehabilitation.

Concepts and newer concepts in surgical management of localised cancer

1. **Selection:** RP suits localised disease (mainly intermediate and selected high risk) with a life expectancy of more than 10 years; low risk is usually offered active surveillance.
2. **RARP (robot-assisted RP):** 3D, 10× magnified view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal early function and early oncology versus open; less blood loss (300 versus 1400 mL) and shorter stay. Surgeon matters more than platform.
3. **Risk-graded (risk-stratified) nerve sparing:** each side is graded from full (intrafascial) through partial to none (for example the Tewari four-grade scale), using grade, positive cores, mpMRI (multiparametric magnetic resonance imaging) capsular contact and nomograms, side by side.
4. **NeuroSAFE:** intraoperative frozen section of each (the whole) posterolateral surface; if positive, that NVB is resected. It aims to allow more nerve sparing without more positive margins; the UK NeuroSAFE PROOF RCT tested its effect on erectile function.
5. **Anterior-support preservation:** - **Retzius-sparing** (Bocciardi, 2010): fully posterior through the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins (anteriorly), so avoid for anterior tumours. - **Hood technique** (Tewari, 2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC; early data. - Retzius-sparing and Hood are aimed at continence; potency equal at 12 months.

6. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length, watch for an apical lip, divide sharply without cautery (cold division); control the DVC distally or late. Posterior (Rocco) reconstruction may aid early continence.
7. **PLND (pelvic lymph node dissection) templates:**

Template	Nodes	Drainage captured
Limited	Obturator	Least
Standard	+ External iliac	About 38%
Extended	+ Internal iliac	About 63%
+ Common iliac	+ Common iliac	About 75%

- Extended PLND when nodal risk exceeds 5% (EAU [European Association of Urology]). It stages but does not improve survival (staging, not therapeutic); lymphocele is the commonest complication. 8. **Single-port robot (da Vinci SP, 2018):** one 2.5-cm cannula; extraperitoneal or transvesical; same-day discharge. 9. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy or TULSA (transurethral ultrasound ablation). Spare sphincter, NVB and rectum.
- **Also know:** PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; salvage robotic RP; new platforms; ERAS (enhanced recovery after surgery).

Nerve-sparing techniques (NVB preservation): recent techniques

- **Walsh anatomic (open):** release apex to base from the rectal surface; small clips; **no thermal energy**.
- **High anterior release** (open) and **Veil of Aphrodite** (robotic): lateral prostatic fascia lifted high anteriorly as a "veil". Select by **Tsuzuki criteria**: ≥2 of PSA (prostate-specific antigen) >10 ng/mL, Gleason >6, abnormal DRE, perineural invasion, high core involvement → >10% risk, so avoid.
- **Antegrade release (robotic):** incise levator fascia, sweep NVB posterolaterally, clip pedicle at the "groove".
- **Athermal, traction-free:** clips or bulldogs, cold scissors.
- Graded nerve sparing, NeuroSAFE, Retzius-sparing and Hood: see above.

Adjuncts

- **Penile rehabilitation:** PDE5i (phosphodiesterase type 5 inhibitor), then injections.
- **Also know:** sural nerve grafts and nerve stimulation tests (abandoned); amniotic membrane wraps, fluorescence nerve imaging (experimental).

Diagram to draw

- Axial prostate showing the three fasciae, the NVBs at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

Key points to remember:

- PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
- Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

Also, from the condensed book (not in the paper answer):

- The urethra bends **≈35°** at the verumontanum, where 13–15 ducts open. This is the distal limit of transurethral resection (TURP) and enucleation.
- The transition zone (TZ) has the shortest ducts and gives †≈20% of cancers; the peripheral zone (PZ) has the longest.

KEY POINT

Trap: the central zone (CZ), around the ejaculatory ducts, is not the TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). The enlarging TZ compresses PZ into the **surgical (pseudo)capsule**, the enucleation plane (see Ch 152).

3 · Fascia, vessels †

- †The capsule is incomplete anteriorly and apically. The neurovascular bundles lie outside it, so enucleation spares them.
- †Prostatic veins join the vertebral (**Batson**) plexus, a route to spinal metastases.

4 · Histology and cell types

Feature	Luminal (secretory)	Basal
Keratins	8, 18	5, 14
Marker	Prostate-specific antigen (PSA), NKX3-1	KRT5/14
AR	High	Low
Secretes	PSA, prostatic acid phosphatase (PAP)	Almost none

- The urethra and proximal ducts carry club, hillock and neuroendocrine cells; BPH stroma contains prominent lymphocytes.

5 · Secretions and clinical markers

- Prostatic fluid is **37–44%** of semen and acidic (pH ≈6.3); seminal vesicle fluid is the alkaline majority.
- Kallikreins liquefy semen within **20–30 min**; incomplete liquefaction at 60 min is nonliquefaction.

- **Ion trapping**: macrolides concentrate in acidic fluid; inflammation alkalinises it.
- PSA, PSMA and PAP all occur outside the prostate, so "prostate-specific" is a misnomer.

Product	Key facts
PSA (KLK3)	Serine protease (19q13.4); liquefies semen
PSMA	Folate hydrolase; rarely deleted in cancer, hence PSMA-PET and ^{177}Lu-PSMA
PAP	First tumour marker (Huggins and Hodges, 1941) MCQ
Zinc	Highest of any tissue; inhibits mitochondrial aconitase, so citrate is secreted

KEY POINT

Benign cells: high zinc blocks aconitase, so citrate is exported. Cancer: **zinc is lost** and citrate is oxidised, so citrate is low (†the basis of MR spectroscopy).

149 Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

1 · Definitions and the four overlapping entities

Histology, size, obstruction and symptoms differ; treatment depends on which.

*See the paper answer **Pathophysiology, classification and management of LUTS.** (Section 7, Ch 104).*

Also, from the condensed book (not in the paper answer):

- †BOOI = PdetQmax − 2Qmax: **>40 obstructed, 20–40 equivocal, <20 unobstructed**.
- "Hypertrophy" is wrong because BPH raises cell **number**, not size; "prostatism" is obsolete (Abrams coined "LUTS") MCQ .

Term	How measured
Benign prostatic hyperplasia (BPH): a histological increase in epithelial and stromal cell number in the periurethral/transition zone (TZ)	Histology
Benign prostatic enlargement (BPE): enlargement caused by BPH	Digital rectal examination (DRE), transrectal ultrasound (TRUS), MRI
Bladder outlet obstruction (BOO): obstruction from any cause (BPH, stricture, cancer, bladder neck sclerosis)	Pressure–flow (best); flow rate (indirect)
Benign prostatic obstruction (BPO): BOO caused by BPE	Pressure–flow
Lower urinary tract symptoms (LUTS): storage, voiding and post-micturition; not prostate-specific	International Prostate Symptom Score (IPSS)

- The four entities overlap only **partially**; BOO may exist without BPE (for example, a stricture). Detrusor ageing, polyuria, sleep disorders and impaired contractility can all mimic BPH.

> **KEY POINT**
>
> Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are separate; obstruction is diagnosed only by pressure–flow.

2 · Epidemiology

- At autopsy there is **no BPH under 30**; prevalence is ≈ **50% in the 50s**† and **88% in the 80s**. It is **similar across all ethnic and geographic groups**.
- No clinical definition is globally accepted. BPH is described by **symptoms** (IPSS), **enlargement** and **obstruction**, plus bother. The proposed volume cut-off is **>30 mL**; a community composite is IPSS >7, maximum flow rate (Qmax) <15 mL/s and volume >20 mL.

SAQ IPSS score

Paper answer · P3

Type: SAQ (10 marks)
Source: 2022-Summer Paper 3, Q3e

Definition

- **IPSS (International Prostate Symptom Score):** validated, self-administered questionnaire grading male LUTS (lower urinary tract symptoms) over the past month.
- Based on the **AUA-SI (American Urological Association Symptom Index, 1992)**; adopted by WHO (World Health Organization) in 1993 by adding one quality-of-life question.

Items

No.	Symptom	Phase
1	Incomplete emptying	Voiding
2	Frequency (again within 2 hours)	Storage
3	Intermittency	Voiding
4	Urgency	Storage
5	Weak stream	Voiding
6	Straining	Voiding
7	Nocturia	Storage

- **Scoring:** items 0–5. 0 not at all; 1 less than 1 time in 5; 2 less than half the time; 3 about half; 4 more than half; 5 almost always. **Nocturia scored by number of times** (0 to ≥5).
- **Total 0–35:** voiding subscore 4 items, storage 3 (not validated).

Severity	Score
Mild	0–7
Moderate	8–19
Severe	20–35

- **QoL (quality of life)/bother question:** "If you were to spend the rest of your life with your urinary condition as it is now, how would you feel?" Scored 0 (delighted) to 6 (terrible); reported separately, outside the 35.

Uses

- First step in LUTS work-up (AUA and EAU [European Association of Urology]); grades severity and bother.
- Guides treatment: mild, non-bothersome → watchful waiting; bothersome → medical therapy or surgery.
- Monitors response: **a 3-point fall is the minimum perceptible change**.
- Standard end point in trials.

Limitations

- **Not disease-specific:** women score similarly; does not diagnose BOO (bladder outlet obstruction) or BPH (benign prostatic hyperplasia); correlates poorly with urodynamics, prostate size and flow.
- No incontinence, pain or post-micturition items.
- Needs literacy.

Alternatives

- **VPSS (Visual Prostate Symptom Score):** pictures, for low literacy.
- BPH Impact Index; ICIQ-MLUTS (International Consultation on Incontinence Questionnaire–Male LUTS); DAN-PSS (Danish Prostate Symptom Score); OAB-q (overactive bladder questionnaire); LURN (Lower Urinary Tract Research Network) SI-29 adds incontinence and pain.

Diagram to draw: IPSS chart with scoring columns.

> **Key points to remember:**
> - 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
> - QoL 0–6 is separate; IPSS = AUA-SI + QoL.
> - 3-point change is noticeable; IPSS does not diagnose BOO.

SAQ Methods to estimate prostate size

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3f (TRUS also answered in 2017-Winter Paper 1, Q3e)

Why measure

- Chooses treatment: 5ARI (5α-reductase inhibitor) if >30 mL (AUA (American Urological Association)) or >40 mL (EAU (European Association of Urology)); type of BPH (benign prostatic hyperplasia) surgery.
- **PSAD (PSA density)** = PSA (prostate-specific antigen) ÷ volume; >0.15 suggests cancer. Also AUR (acute urinary retention) risk, brachytherapy planning.
- Normal adult gland 20–25 g; about 48 g at 60–70 years. **1 cm³ ≈ 1 g**.

Methods

Method	Technique	Accuracy and use
DRE (digital rectal examination)	Estimate size, consistency	**Underestimates** above 30 mL (lateral lobes impalpable); flat gland predicts <30 mL
PSA as surrogate	Age-specific cut-offs for **>40 mL:** >1.6 ng/mL (50s), >2.0 (60s), >2.3 (70s)	AUC (area under curve) 0.76–0.78; within ±20% of TRUS in most
TAUS (transabdominal ultrasound)	Full bladder; transverse width and height, sagittal length	Less precise than TRUS; also PVR (post-void residual) and **IPP (intravesical prostatic protrusion)**
TRUS (transrectal ultrasound)	Ellipsoid formula; planimetry	**Method of choice**
MRI (magnetic resonance imaging)	T2-weighted ellipsoid or segmentation (manual or automated)	Most accurate and reproducible
CT (computed tomography)	Axial images	Overestimates; poor capsule definition
Cystoscopy	Urethral length, lobes, median lobe	Rough; guides procedure choice

TRUS formulas

Shape assumed	Formula
Ellipsoid (standard)	**π/6 (0.52) × transverse × AP (anteroposterior) × length**
Sphere	π/6 × transverse³
Prolate spheroid	π/6 × transverse² × AP
Planimetry	Sum of serial transverse areas

- Transverse and AP at the widest axial section; length sagittally. All formulas correlate >0.90 with specimen weight.
- **Planimetry:** probe on a stepper, images every 3–5 mm, areas summed; most precise; used for brachytherapy.

IPP grading (suprapubic, sagittal, bladder 100–200 mL)

- Grade I <5 mm, II 5–10 mm, III >10 mm; ≥10 mm predicts BOO (bladder outlet obstruction) and failed trial without catheter.

Diagram to draw: TRUS transverse and sagittal views with three measured diameters.

> **Key points to remember:**
> - Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
> - DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
> - Planimetry most precise; MRI most reproducible.

Also, from the condensed book (not in the paper answer):

- Probability of obstruction by Qmax: <10 mL/s high, 10–15 intermediate, >15 low. Qmax falls almost linearly with age (Olmsted median 20.3 mL/s at 40–44 versus 11.3 at 75–79).
- DRE underestimates size by 25% to ≥50%.

ALGORITHM

1. Total prostate volume (TPV) ≈ **25 mL in the 30s → 35–45 mL in the 70s**.
2. Transition zone volume (TZV) ≈ 15 → 25 mL.

3 · Aetiology

> **KEY POINT**
>
> BPH is an **accumulation** disease: volume = proliferation minus death. Established BPH shows equal or reduced replication and raised bcl-2. Androgens act chiefly by **suppressing cell death**, so withdrawal shrinks the gland.

- New glands otherwise form only in the fetus, hence the theory of **embryonic reawakening** of stromal induction.

3.1 Androgens: permissive, not causal

- Prepubertal castrates and men with androgen defects **never develop BPH**; withdrawal gives only **partial** involution. The two established determinants are **age** and **functioning testes**.

- Serum testosterone falls with age, but **intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) stay high**. DHT is **maintained, not elevated**; old "raised DHT" reports used cadaveric controls MCQ.
- Serum testosterone does not predict volume or PSA. Androgens act via growth factors rather than as direct mitogens, and they are anti-inflammatory.

ALGORITHM

1. Testosterone enters the stromal cell.
2. **Type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) converts it to DHT.
3. DHT acts on the stroma (autocrine) or diffuses to the epithelium (paracrine).
4. The DHT–AR complex binds DNA and switches on growth factors.
5. Growth factors drive epithelial survival and growth.

- ≈ **90%** of prostatic androgen is DHT of testicular origin; ≈10% is adrenal. DHT has higher AR affinity and forms a more stable complex.

Feature	Type 1	Type 2
Site	Skin, liver	Prostate **stromal fibroblasts**, some basal cells
5α-reductase deficiency	Normal	**Mutated**
Finasteride / dutasteride	Not inhibited / inhibited	Inhibited / inhibited
Role	Possibly cancer	Development and BPH growth

- Finasteride shrinks the gland as much as dual inhibition, so type 1 is unimportant in BPH.
- SRD5A2 lies on **chromosome 2p23** MCQ; its polymorphisms link inconsistently to BPH.

3.2 5α-reductase inhibitor (5-ARI) resistance

- 5-ARIs are the **only medical therapy that reduces volume**.
- Resistance is **not** failure to lower DHT but focal lack of luminal atrophy in nodules, or stromal hyperplasia, often linked to inflammation (NF-κB-driven AR variants, SRD5A2 silencing). The proposed definition is unchanged volume and no atrophy despite lower DHT.

3.3 Oestrogens

- In dogs, androgen plus oestradiol produces BPH; oestrogen induces AR and increases stroma. Oestrogen rises with age relative to testosterone, and adipose tissue aromatises.
- **Oestrogen receptor α (ERα) is stromal** and proliferative; **ERβ is epithelial** and antiproliferative. An ERβ agonist failed in a trial for LUTS.

3.4 Cell death; stromal–epithelial interaction

- Apoptosis after castration is active and probably mediated by transforming growth factor-β (TGF-β).
- Hypothesis: a **stromal defect removes a brake** on epithelial proliferation. Periurethral interstitial fibroblasts (carrying type 2 5α-reductase) increase in BPH and expand the TZ; **no drug targets them**.
- Chemokines such as interleukin-8 (IL-8) turn fibroblasts into collagen-producing myofibroblasts, and the resulting fibrosis contributes to LUTS.

3.5 Growth factors

Androgen permits; growth factors signal and are the likelier primary defect. The first prostatic growth factor found was basic fibroblast growth factor (FGF-2) MCQ.

Factor	Source and target	Role in BPH
FGF-2	Stroma (autocrine)	Raised 2–3-fold; induced by TGF-β and hypoxia
FGF-7 (keratinocyte growth factor, KGF)	Made by stroma; receptor **only on epithelium**	Leading stromal mediator of hormonal control
FGF-10	Smooth muscle to epithelium	Raised by androgens
TGF-β1	Stroma	**Inhibits epithelium** but is **mitogenic for fibroblasts**; BPH stroma shows a blunted response
Insulin-like growth factors (IGF-1, -2)	—	Transgenic IGF-1 causes enlargement; serum IGF-1 is unrelated to volume
Others	—	Epidermal growth factor (EGF), vascular endothelial growth factor (VEGF), FGF-17

- **Hypoxia hypothesis**: hypoxia induces HIF-1, which raises FGF-2 and FGF-7, so growing nodules become more hypoxic; vascular disease such as diabetes may add risk.

3.6 Inflammation

- **Activated T cells** secrete growth factors, and cytokines drive the stroma. Autoimmune disease increases BPH, whereas TNF antagonists are associated with less.
- **Inflammation correlates with enlargement and 5-ARI resistance**, but causality is unproven.

3.7 Metabolic syndrome, lifestyle

- Metabolic syndrome causes prostatic inflammation and LUTS via aromatisation, **hyperinsulinaemia driving sympathetic overactivity**, and low HDL.
- **Waist circumference** predicts prostatectomy and LUTS independently of BMI. BMI >35 versus <25 gives an odds ratio (OR) of **3.5** for a prostate ≥40 mL.

Factor	Direction
Diabetes, fasting glucose >110 mg/dL, obesity, central adiposity, hypertension (LUTS)	Increases risk
Physical activity; moderate alcohol	Lowers risk (alcohol lowers BPH, **not LUTS**)
Smoking, diet, vasectomy, sexual activity	No consistent effect (**vasectomy is not a risk factor**)
Jewish religion	More prostatectomy, not more BPH (access to care) MCQ

- **Depression** is strongly linked to LUTS, and LUTS severity correlates with sexual dysfunction.
- **Drugs that worsen LUTS**: α-sympathomimetic cold remedies, antidepressants, antihistamines and bronchodilators; anticholinergic and adrenergic drugs can precipitate retention.

3.8 Genetics and others

- In young men with large glands (>37 g), first-degree relatives have ≈ **4-fold** risk of surgery. Inheritance is **autosomal dominant**; ≈ **50%** of men operated on before 60 have heritable disease, versus ≈9% after 60.
- **Familial BPH** shows early onset, a **large gland**, normal androgens and a normal 5-ARI response. Only **dogs and humans** get BPH.

4 · Pathology

- **Every nodule arises in the TZ or the periurethral region**. The TZ lies outside the preprostatic sphincter; the periurethral zone lies inside it, with glands parallel to the urethra.
- The **middle lobe** is **periurethral**, because there is no TZ at the bladder neck MCQ. Patterns are middle, lateral, trilobar and posterior commissural (median bar).

> **KEY POINT**
>
> Only humans have a **capsule**, transmitting expansion onto the urethra; dogs get BPH but rarely obstruct. Transurethral incision of the prostate (TUIP) relieves obstruction without changing volume. Hence **size does not correlate with obstruction**.

Feature	Periurethral nodules	TZ nodules
Composition	Purely stromal (early), like **embryonic mesenchyme**	Glandular, tall epithelium
Origin	Interstitial fibroblasts	New duct branches forming a new ductal system

- **Two phases**: for ≈20 years nodule number rises; then glandular nodules grow.
- Small glands are mainly stroma and large glands mainly epithelial nodules. This variable stromal:epithelial ratio explains variable drug response, since α-blockers target smooth muscle and 5-ARIs target epithelium.

5 · Pathophysiology of obstruction

Many symptoms come from the bladder, so some persist after surgery.

ALGORITHM

1. Hyperplasia raises urethral resistance (static and dynamic components).
2. The detrusor compensates, at a cost to storage function.
3. Storage LUTS appear, with or without impaired emptying.
4. In a minority, the detrusor decompensates.

- The static component (bulk under the capsule, elastic ECM) is **not** reduced by α-blockade. The dynamic component is smooth muscle tone via **α1a** receptors (the most abundant subtype).
- Other regulators include endothelin and **phosphodiesterases 4 and 5 (PDE4/5)**, which is why PDE5 inhibitors help LUTS.
- Many LUTS arise from **obstruction-induced bladder change**. ≈ **One-third** of men keep voiding dysfunction (mostly storage) after relief.

- Storage changes are detrusor overactivity (DO) and lower compliance; emptying changes are reduced contractility and, in a minority, detrusor failure.

ALGORITHM

1. **Three-stage remodelling**:
2. Smooth muscle hypertrophy.
3. Compensation.
4. Decompensation (underactive bladder).

- Stretch, pressure and **hypoxia/ischaemia** switch the detrusor to a **secretory phenotype** that lays down ECM.
- **Trabeculation** reflects increased detrusor **collagen** and progresses to cellules, diverticula and failure; there is no accepted grading.

KEY POINT

Storage LUTS persist after transurethral resection of the prostate (TURP) because the collagen-rich, overactive detrusor does not reverse, so counsel men with DO before surgery. Acute urinary retention (AUR) is **not** the inevitable end: many men in AUR have an adequate detrusor and a precipitant.

6 · Natural history

Who progresses decides watchful waiting versus treatment.

- **Regression to the mean**: entry thresholds select men at a temporary extreme, so they seem to improve. It is part of every placebo effect.
- Within trials, IPSS, Qmax and volume barely correlate (restriction of range); only PSA and volume still correlate.
- ≈ **40%** improve without treatment (watchful waiting behaves like placebo); active drug gives about twice the placebo improvement.
- **Olmsted County**: IPSS rises ≈ **0.2–0.34 points/year**; Qmax falls ≈ **2.1%/year** (faster if age ≥70 or Qmax <10); volume grows ≈ **0.6 mL/year** (faster with higher PSA or volume). Volume >50 mL gives ≈3.5-fold odds of moderate–severe LUTS.
- A perceptible improvement is ≈ −2 IPSS points at low baseline versus −6 at high baseline.

Predictor of progression	Predicts
Age	Everything; AUR rises linearly from age 40 to 80+
PSA ≥1.4–1.5 ng/mL	Growth, symptoms and flow, AUR, surgery
Volume >30–40 mL	AUR (≈3-fold), surgery, new LUTS
IPSS ≥8 or high bother	AUR, surgery
Qmax <10–12 mL/s	AUR, flow decline, treatment
Raised post-void residual (PVR)	Treatment failure, AUR

- **MTOPS** (Medical Therapy of Prostatic Symptoms) placebo arm: progression **4.5 per 100 patient-years** (17% at 4 yr), mostly an **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**.
- **PLESS** (Proscar Long-term Efficacy and Safety Study) placebo arm: symptoms drifted back to baseline. Surgery was 10% at 4 yr, rising from ≈6% to ≈15% across PSA and volume

tertiles.

- **REDUCE** placebo arm: volume 40–80 mL and Qmax <10 each gave a hazard ratio of 1.67 for new LUTS.

> **KEY POINT**
>
> PSA is a cheap surrogate for volume, the main driver of AUR and surgery; models with >100 predictors barely beat PSA alone.

7 · Complications

These are the absolute indications for surgery.

> **KEY POINT**
>
> Death, renal failure, stones and infection are now rare. The common events are **AUR and surgery**, predicted by age, symptoms, low flow and a large gland (or high PSA).

7.1 Acute urinary retention

- Precipitants include infection, overdistension, fluid load, alcohol and bed rest; prostatic infarction is **not** proven.

	Spontaneous AUR	**Precipitated AUR**
Trigger	None	Surgery, anaesthesia, catheterisation, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	**75%**	**26%**

- AUR incidence is ≈ **4.5–6.8/1000 person-years** in the community, 18 in PLESS placebo and 6 in MTOPS. The hazard is constant over time.
- **Age and symptoms multiply risk**: in Olmsted, men in their 70s with IPSS >7 had 34.7/1000 person-years. Relative risks: IPSS ≥8 3.2; Qmax <12 3.9; volume >30 mL 3.0; PSA >1.4 several-fold.

7.2 Other complications

- **Renal**: "**silent prostatism**" is asymptomatic renal failure from BOO; MTOPS had none in >3000 men. †Catheterise first, watch for post-obstructive diuresis, let creatinine stabilise, then operate.
- **Bladder stones** are ≈ **8×** commoner because of stasis; renal stones are not.
- **Haematuria** reflects increased microvessel density; **finasteride** is first-line because it lowers VEGF. †Exclude cancer and stones first.
- **UTI** is rare (MTOPS 0.1/100 patient-years). **Incontinence** is overflow or urge (DO in up to half).
- **Surgery** is a soft end point and AUR a hard one. In Olmsted, moderate–severe symptoms, Qmax <12, volume >30 and PSA ≥1.4 each gave ≈4× treatment.

8 · Landmark trials

Trial	Design	Result	Take-home
VA Cooperative Study	TURP vs watchful waiting, moderate symptoms	Failure RR 0.48; 5-yr 10% vs 21%; ≈36% crossed over; delayed TURP smaller gains	Watchful waiting safe; delay may reduce benefit
PLESS	Finasteride vs placebo, 4 yr	Placebo returns to baseline; AUR/surgery rise with PSA/volume	PSA/volume predict progression
MTOPS	Placebo vs doxazosin vs finasteride vs combination	Placebo progression 4.5/100 patient-years, mostly IPSS ≥4; no renal failure	Defined "clinical progression"
Olmsted County Study	Population cohort, 40–79	IPSS +0.2–0.34/yr, Qmax −2.1%/yr, volume +0.6 mL/yr; AUR 6.8/1000 person-years	Best natural history; age, IPSS, Qmax, volume predict AUR

150 Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

1 · Definitions and the diagnostic problem

Not every man with urinary symptoms is obstructed; find the cause first.

- Benign prostatic hyperplasia (BPH) is **histological** proliferation in the **transition zone (TZ)**. It can lead to benign prostatic enlargement (BPE), benign prostatic obstruction (BPO, a form of bladder outlet obstruction, BOO) and lower urinary tract symptoms (LUTS).
- LUTS are multifactorial, including detrusor overactivity (DO) or underactivity (DUA), stricture and infection (UTI).

> **KEY POINT**
>
> Bothersome LUTS are not BPH-related BOO until assessed. Aims: **grade severity and bother** and **identify the cause**. Symptoms, flow, post-void residual (PVR), volume and cystoscopy cannot separate obstruction from bladder dysfunction; only the **pressure–flow study** can (gold standard, reserved for when it changes surgery).

Guidelines: American Urological Association (AUA) BPH (2021); European Association of Urology (EAU) non-neurogenic male LUTS (2022).

2 · History and symptom assessment

- **Comorbidities**: diabetes mellitus or insipidus, renal disease, heart failure, sleep apnoea, neurological disease, haematuria and prior lower-tract surgery.
- **Drugs**: loop diuretics, β-blockers, excess fluid and caffeine worsen storage; anticholinergics impair contractility.
- Assess **sexual function**. LUTS do **not** raise prostate cancer (PCa) risk.
- Storage symptoms (frequency, urgency, urge incontinence, nocturia) form **overactive bladder (OAB)**.

*See the paper answer **IPSS score** (Section 16, Ch 149).*

Also, from the condensed book (not in the paper answer):

- The AUA Symptom Index (AUASI) and International Prostate Symptom Score (IPSS) total **0–35**: mild **0–7**, moderate **8–19**, severe **20–35**.
- The minimum perceptible change is **3 points**; with a severe baseline, **percentage** reduction is more reliable.

3 · Examination and DRE

- Look for a palpable bladder, meatal stenosis and phimosis, and check **anal tone** and lower-limb neurology.
- **Digital rectal examination (DRE)** estimates volume and screens for PCa. It **underestimates prostate volume (PV) above 30 mL**; a flat gland suggests <30 mL.
- PV does not correlate with symptoms or obstruction, but the **30 mL** and **50 mL** cut-offs predict treatment response and progression.

4 · Frequency–volume charts and nocturia

- A frequency–volume chart (FVC) records times and volumes; a bladder diary adds intake and urgency. **Indication**: predominant **storage LUTS or nocturia**. **Duration**: **3 days**, since adherence falls by 7 days.
- **Nocturia** is waking ≥1 time to void. **Nocturnal polyuria** is night output **>20%** of 24-h output (young) or **>33%** (over 65).

ALGORITHM

1. **Nocturia**
2. Complete an FVC.
3. If †global polyuria (>40 mL/kg/24 h), look for diabetes or excess intake.
4. If nocturnal polyuria, look for heart failure, sleep apnoea, evening fluids or diuretic timing.
5. If reduced capacity, think of OAB or BOO.
6. Treat the cause. If nocturia persists, consider **sleep apnoea** (AUA).

Nocturnal polyuria: no fluid 2 h before bed, no evening alcohol or caffeine, retimed diuretics, with or without **desmopressin**. †It risks **hyponatraemia**, so check sodium at baseline, 1 week and 1 month.

5 · Laboratory tests

- **Urinalysis is mandatory** for UTI, microhaematuria, glycosuria and proteinuria. Gross haematuria needs exclusion of bladder cancer; **cytology** is for severe storage symptoms with dysuria, or smokers.
- **Prostate-specific antigen (PSA)** gives PCa risk, a volume surrogate and progression risk. Test only if life expectancy is **>10 years** and it changes management.
- PSA predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s). Baseline PSA also predicts acute urinary retention (AUR), surgery, and change in symptoms and maximum flow rate (Qmax).
- **5α-reductase inhibitors (5-ARIs) lower PSA by 40–50% by 12 months**, so take a baseline first. †Double the PSA after ≥6 months; any rise from the nadir needs evaluation.
- **Creatinine and eGFR** are checked (EAU) if renal impairment or retention is suspected.

6 · Instrumental investigations

PVR

- A PVR **<30 mL** is insignificant; persistently **>50 mL** is important; **200–300 mL** may mean DUA and a poorer response; **>300 mL** is significant chronic retention.
- High PVR means BOO **or DUA**; it predicts symptom deterioration (MTOPS >39 mL; ALTESS >93 mL), **not surgery**.

Uroflowmetry

- Record **Qmax**, voided volume and pattern. Repeat if voided volume is **<150 mL**.
- It **cannot distinguish BOO from DUA** (DUA in 9–48%). Qmax **10 mL/s** is more specific; **15 mL/s** is more sensitive.
- Optional at first; **recommended before active treatment** and for monitoring.

Urodynamics

SAQ Role of UDS before prostate surgery

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3a

Why consider UDS (urodynamic study)

- Symptoms, flow rate, PVR (post-void residual), prostate volume and cystoscopy **cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**: low Qmax (maximum flow rate) occurs in both. DUA is found in 9–48% of men with LUTS (lower urinary tract symptoms) studied.
- Surgery helps obstruction, not a weak bladder. **PFS (pressure–flow study) is the gold standard** for BOO.

Components

- **Filling cystometry:** sensation, compliance, **DO (detrusor overactivity)**.
- **PFS:** detrusor pressure at Qmax plotted against flow (ICS (International Continence Society) nomogram).

Index	Formula	Interpretation
BOOI (bladder outlet obstruction index)	PdetQmax − 2Qmax	>40 obstructed; 20–40 equivocal; <20 unobstructed
BCI (bladder contractility index)	PdetQmax + 5Qmax	>150 strong; 100–150 normal; <100 weak

- High pressure, low flow = BOO; low pressure, low flow = DUA.
- **Videourodynamics** adds anatomy (diverticula, reflux).

Indications before BPH (benign prostatic hyperplasia) surgery (EAU (European Association of Urology))

- Previous failed invasive treatment.
- Cannot void >150 mL (unreliable flow).
- PVR >300 mL.
- Age >80 years with mainly voiding LUTS.
- Age <50 years with mainly voiding LUTS.
- **Also:** suspected neurogenic bladder, prior pelvic surgery or radiation, Qmax >10 mL/s (AUA (American Urological Association): diagnostic uncertainty).

How results change management

- **Unobstructed or DUA:** poorer surgical results; counsel, consider CIC (clean intermittent catheterisation).
- **DO with BOO:** storage symptoms persist in about one-third after TURP (transurethral resection of the prostate); warn the patient.
- **Low compliance:** the only "absolute" urodynamic indication to relieve BOO (protects upper tracts).

Evidence and limitations

- Not routine. **UPSTREAM RCT (randomised controlled trial):** routine UDS did not change IPSS (International Prostate Symptom Score) or surgery rates.
- Invasive: UTI (urinary tract infection), discomfort, cost. Before **radical prostatectomy**, not routine.

Diagram to draw: ICS nomogram with obstructed, equivocal and unobstructed zones.

> **Key points to remember:**
> - PFS is the only test that separates BOO from DUA.
> - BOOI >40 obstructed; BCI <100 weak.
> - EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
> - UPSTREAM: routine UDS did not change outcomes.

Also, from the condensed book (not in the paper answer):

- Pressure–flow study (PFS) indications (EAU): failed invasive treatment; cannot void **>150 mL**; PVR **>300 mL**; age **>80** or **<50 years** with voiding LUTS; neurological disease or pelvic surgery. AUA: before surgery if **Qmax >10 mL/s**.
- DUA predicts long-term surgical failure.

Imaging and endoscopy

*See the paper answer **Methods to estimate prostate size** (Section 16, Ch 149).*

Also, from the condensed book (not in the paper answer):

- **Intravesical prostatic protrusion (IPP) ≥10 mm** is high grade; it predicts need for therapy and a **failed trial without catheter (TWOC)**. Bladder wall thickness and other non-invasive tests are accurate, but **none replaces PFS**.
- **Transrectal ultrasound (TRUS)** is the method of choice for PV, needed before a 5-ARI and **any BPH surgery**; a median lobe guides the choice of minimally invasive therapy.
- **Cystoscopy** neither diagnoses BOO nor decides treatment. Indications include haematuria, recurrent UTI, suspected stricture, and **excluding a median lobe before prostatic urethral lift (PUL)**.

7 · Evaluation algorithms

ALGORITHM

1. **EAU (men ≥40 yr)**
2. History (including sexual), symptom score, examination with DRE, urinalysis, PVR, and PSA if life expectancy >10 yr.
3. If DRE, PSA or urinalysis is abnormal, or neurological disease, follow the relevant guideline.
4. If not bothersome, no further work-up.
5. If PVR is significant, renal ultrasound with or without renal function.
6. If bothersome, FVC (storage or nocturia), then prostate ultrasound and uroflow if treating, then the medical algorithm.
7. Before surgery, endoscopy if it alters the choice; PFS for specific indications.

ALGORITHM

1. **AUA**
2. History, examination, IPSS, urinalysis, with or without PVR and flow.
3. If **PVR >300 mL**, manage as non-neurogenic chronic retention.
4. Counsel: behavioural, medical, procedural.
5. Reassess at **4–12 weeks**. If improved, follow up. If nocturia predominates, FVC and a drug. If incomplete, FVC, PVR and flow, urodynamics and cystoscopy, then the OAB pathway, surgery, or both.

8 · Watchful waiting (WW)

- **Indication**: mild–moderate uncomplicated LUTS, minimal bother. **Excluded**: gross haematuria, UTI, raised PSA or PVR, and neurological disease.
- WW is reassurance, education, lifestyle and **yearly** review; in the VA trial ≈a quarter crossed over because of **bother**.
- Lifestyle: fluids **1.5–2 L/day**, none 2 h before bed; less caffeine and evening alcohol; retime diuretics; **double voiding** and **urethral milking**.

> **KEY POINT**
>
> Treatment is driven by **bother and complicating features**, not by score or size alone.

9 · Medical therapy: choice

- Drugs are first line without an **absolute surgical indication** (refractory or recurrent AUR, recurrent UTI, renal insufficiency, stones, recurrent haematuria; see Ch 151).

LAQ BPH: medical management and surgical modalities

Paper answer · P2

Asked as:

- Describe medical management of BPH and role of various surgical modalities in its management. — *Summer 2015 Paper II Q1 (LAQ)*
- Medical treatment of BPH — *Practice Set 6 Paper II Q3f (SAQ)* (also answered in 2015-Summer Paper 2, Q1)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction, pathophysiology and rationale

- **BPH (benign prostatic hyperplasia)** is stromal and epithelial hyperplasia of the transition zone; it may cause BPO (benign prostatic obstruction) and **LUTS (lower urinary tract symptoms)**.
- Obstruction has a **static** component (gland bulk, DHT (dihydrotestosterone)-dependent / DHT-driven gland bulk) and a **dynamic** component (dynamic part: α1A-mediated smooth-muscle tone; static part: DHT-driven gland bulk). Drugs target each. The bladder responds with detrusor overactivity or later underactivity.

Evaluation (before treatment)

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35.

- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)**, frequency-volume chart, Qmax (maximum flow), PVR (post-void residual); PVR >300 mL is managed as chronic retention (PVR >300 mL = chronic retention).
- Measure prostate volume before surgery (AUA (American Urological Association)); pressure-flow study if unclear.

Medical management

Option / class	Doses	Key facts
Watchful waiting	—	Mild or non-bothersome LUTS; lifestyle advice
α1-blockers (first line)	Tamsulosin 0.4 mg; alfuzosin 10 mg; silodosin 8 mg; doxazosin 4–8 mg (titrate)	First line; act within days; no effect on gland size, AUR (acute urinary retention) or surgery (do not prevent AUR or surgery). Hypotension, ejaculatory dysfunction; silodosin causes most anejaculation. **IFIS (intraoperative floppy iris syndrome)**
5-ARIs (5α-reductase inhibitors)	Finasteride 5 mg; dutasteride 0.5 mg	For glands (prostate) >30–40 mL or PSA >1.5 ng/mL. Shrink the gland about 18–25% (another source: ~20%) (maximal at 6 months; act after 6 months), halve PSA. PLESS (Proscar Long-Term Efficacy and Safety Study): AUR or surgery down (↓) 51%. Low libido, ED (erectile dysfunction). Also BPH haematuria
Combination	—	For high-risk men. The α-blocker gives early relief; the 5-ARI prevents hard events
Antimuscarinic or β3-agonist (mirabegron, vibegron)	Add-on	Add for persisting storage symptoms; trials included only men with low PVR (PVR <150 mL)
Tadalafil 5 mg daily	PDE5 (phosphodiesterase-5) inhibitor	LUTS with or without (±) ED; not with nitrates
Desmopressin	—	Nocturnal polyuria (night urine >33% of 24-h output); check (monitor) sodium, especially ≥65 years

Combination therapy

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: progression fell (↓) 39% (doxazosin), 34% (finasteride) and **66% (combination)**; only finasteride or combination (finasteride-containing arms) reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination cut AUR or surgery by about 66% versus tamsulosin.
- Combine when **PV (prostate volume) >30–40 mL and PSA >1.5 ng/mL**; α-blocker may stop (withdrawable) after 6 months of combination (SMART (Symptoms Management After Reducing Therapy) trial).
- Tadalafil + finasteride fixed dose approved (FDA [Food and Drug Administration] 2021).

Side effects and important cautions

- **Side effects**: α-blockers cause dizziness, postural hypotension, ejaculatory dysfunction and **IFIS (intraoperative floppy iris syndrome)**, which persists after stopping, so avoid starting them before planned cataract surgery. 5-ARIs: low libido, ED, gynaecomastia.

- **IFIS:** billowing iris, prolapse, miosis; stopping the drug does not prevent it.
- 5-ARIs: double the PSA; not for cancer prevention (more high-grade tumours in PCPT [Prostate Cancer Prevention Trial] and REDUCE).
- **AUR**: catheterise (catheter), then TWOC (trial without catheter) on an α-blocker (ALFAUR (Alfuzosin in Acute Urinary Retention): success 62% vs 48%).
- **Also know:** phytotherapy (saw palmetto) not recommended by guidelines.

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency from BPO, failed medical therapy (surgery if refractory).

Modality	Size (AUA 2023)	Key feature
TURP (transurethral resection), monopolar or bipolar	30–80 mL	Reference standard; retrograde ejaculation ~two-thirds
TUIP (transurethral incision)	≤30 mL	Less retrograde ejaculation; more reoperation
PVP (photoselective vaporization)	≤80 mL	Safe when anticoagulation cannot stop; no histology
HoLEP/ThuLEP (laser enucleation)	Any size	Lowest retreatment, least bleeding
Simple prostatectomy (open or robotic)	>80 mL	Complete removal; diverticula or stones
Aquablation	30–80 mL (consider 80–150)	Heat-free; preserves ejaculation
PUL (prostatic urethral lift), Rezūm (water vapour)	30–80 mL (PUL: no median lobe)	Preserve sexual function; higher retreatment
iTind (temporary implant)	25–75 mL, no median lobe	Removed after a few days; modest efficacy
PAE (prostate artery embolisation)	Larger glands (AUA 2023: may be offered)	Local anaesthesia; less effective than TURP; radiation, non-target embolisation

- **TUR syndrome** (monopolar TURP): hypo-osmolar glycine or water is absorbed through open sinuses, causing dilutional hyponatraemia, fluid overload and confusion. Risk rises with gland >45 mL and resection >90 minutes. Treat with diuretics and, if severe, 3% saline, correcting slowly. **Bipolar TURP** uses saline and eliminates it.
- **Complications**: bleeding, stricture, bladder neck contracture, retreatment.
- **GOLIATH**: 180 W PVP non-inferior to TURP. **WATER (Waterjet Ablation Therapy for Endoscopic Resection)**: Aquablation non-inferior, with less retrograde ejaculation (10% vs 36%).
- **Also know:** TUVP (transurethral vaporization); TUMT (transurethral microwave therapy) dropped as legacy; Optilume paclitaxel-coated balloon (2023).

Follow-up

- Reassess at 4–12 weeks (α-blocker), 3–6 months (5-ARI) (another source: 6 months for 5-ARIs), then yearly; surgery if refractory.

Diagram to draw: Size-based surgical algorithm (≤30, 30–80, >80 mL); algorithm: LUTS → size, PSA, storage symptoms → drug choice.

Key points to remember:

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Also, from the condensed book (not in the paper answer):

- Antimuscarinics need PVR <200 mL; avoid oxybutynin in the elderly; hexane-extract Serenoa is an EAU option.

ALGORITHM

1. **EAU**
2. Not bothersome: WW.
3. Nocturnal polyuria: lifestyle with or without desmopressin.
4. Storage predominant: antimuscarinic or β3-agonist.
5. If **PV >40 mL and long-term treatment** intended: 5-ARI with or without an α1-blocker or phosphodiesterase-5 inhibitor (PDE5I).
6. Otherwise α1-blocker or PDE5I; if storage symptoms remain, add an antimuscarinic or β3-agonist. **AUA**: α-blocker (PDE5I if erectile dysfunction, ED) → **prostate >30 mL** → add 5-ARI → PDE5I trial → surgery.

KEY POINT

Reading trials: placebo improves scores and flow, so judge by a 3-point or percentage threshold; the goal includes **preventing progression**.

10 · α1-Blockers

SAQ Tamsulosin

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 12 Paper 2, Q3d

Introduction

- A **uroselective α1-adrenergic antagonist**, first-line drug for bothersome male LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia) (AUA [American Urological Association], EAU [European Association of Urology]).

Receptor basis

Subtype	Site	Effect of blockade
α1a (about 70% in prostate)	Prostate, bladder neck, prostatic urethra, iris dilator	Relieves dynamic obstruction; ejaculatory dysfunction; IFIS (intraoperative floppy iris syndrome)

Subtype	Site	Effect of blockade
α1b	Blood vessels	Hypotension (tamsulosin has 10-fold less affinity)
α1d	Detrusor, spinal afferents	Storage symptom relief

Mechanism

- Blocks noradrenaline at α1a receptors → relaxes prostatic and bladder-neck smooth muscle → lowers the **dynamic component** of BOO (bladder outlet obstruction). It does not shrink the prostate.

Pharmacology and dose

- **0.4 mg once daily** after the same meal (up to 0.8 mg); no titration.
- Modified release; hepatic metabolism (CYP3A4, CYP2D6); avoid strong CYP3A4 inhibitors (ketoconazole).
- Flow rises within 4–8 hours, symptoms by 1 week; 0.2 mg is not better than placebo.

Indications

1. **BPH/LUTS:** IPSS (International Prostate Symptom Score) falls about 30–40%; Qmax rises; storage and nocturia improve.
2. **Combination with 5ARI** (5α-reductase inhibitor) for prostate >30–40 mL: **CombAT** (dutasteride + tamsulosin) and MTOPS (combination cut progression by 66%).
3. **MET (medical expulsive therapy):** distal ureteric stones 5–10 mm; SUSPEND (2015) negative overall but included all sizes.
4. **AUR (acute urinary retention):** before trial without catheter.
5. **Also know:** ureteric stent symptoms; female functional outlet obstruction.

Adverse effects

- **Ejaculatory dysfunction** (dose-dependent; less than silodosin): retrograde ejaculation from bladder-neck relaxation, or anejaculation (emission failure).
- **IFIS:** floppy, billowing iris, iris prolapse, progressive miosis during cataract surgery; **highest risk with tamsulosin**. Stopping the drug does not remove the risk, so warn the ophthalmologist.
- Dizziness, rhinitis, headache; less orthostatic hypotension than non-selective agents. Caution: sulfonamide allergy.

Comparison

	Tamsulosin	Silodosin	Alfuzosin
Selectivity	α1a/α1d	Most α1a	Clinical only
Ejaculatory effect	Moderate	Highest	Lowest

Diagram to draw: prostate cross-section showing α1a receptors in stroma and bladder neck.

> **Key points to remember:**
>
> - 0.4 mg once daily; α1a/α1d selective; no titration.
> - IFIS highest with tamsulosin; persists after stopping.
> - MET benefit in distal stones 5–10 mm.

SAQ Silodosin

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "Silodosin" (Silodosin.) · P2 SAQ "Silodosin."

Definition and pharmacology

- **Uroselective α1-blocker**, the **most α1A-selective** agent (well over 100-fold vs α1B, or by another source several hundredfold over α1B; about 50-fold vs α1D), for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing vascular α1B gives **minimal hypotension** (few blood-pressure effects). Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Onset: flow improves within 2–6 hours, symptoms within 3–4 days. Metabolised by CYP3A4.

Mechanism of action

- **α1A receptors** (about 70% of prostatic α1 receptors) dominate in the prostate, bladder neck and prostatic urethra. Noradrenaline on them causes the **dynamic component** of BOO (bladder outlet obstruction).
- Blockade relaxes this smooth muscle.

Dose and contraindications

Item	Detail
Dose	**8 mg once daily with a meal**; no titration
Renal: CrCl (creatinine clearance) 30–50 mL/min	**4 mg daily**
Contraindicated	CrCl <30; severe liver (hepatic) impairment; strong CYP3A4 inhibitors (e.g. ketoconazole)
Caution	Other α-blockers (avoid), PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery

Indications and efficacy

Indication	Evidence
LUTS due to BPE (benign prostatic enlargement)	Japanese, American and European 12-week RCTs (randomised controlled trials): IPSS (International Prostate Symptom Score) and flow better than placebo; **non-inferior to tamsulosin 0.4 mg** (**Chapple 2011**, European trial), not better; effective in severe LUTS
MET (medical expulsive therapy)	Distal ureteric stones 5–10 mm; α-blockers reduce ureteric spasm; meta-analyses show higher, faster expulsion than tamsulosin; **SUSPEND (2015)** found no overall tamsulosin benefit
CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)	α-blocker-naive men with voiding symptoms; of recent onset

- **Greatest fall in BOOI (bladder outlet obstruction index)** of all α1-blockers (subgroup data).
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.
- **Also know:** before ureteroscopy (easier access); with propiverine for storage symptoms; nocturia (modest).

Adverse effects

- **Ejaculatory dysfunction / anejaculation (signature effect):** over a quarter of men in registration trials (about 14% vs 2% with tamsulosin head-to-head), yet few stop the drug; men with anejaculation had better responses. Mechanism: **emission failure** (vas and seminal vesicle α1A blockade) and bladder-neck relaxation. Reversible. Counsel young men.
- **Cardiovascular:** dizziness; orthostatic hypotension rare; no QT change; no excess hypotension in the elderly or with antihypertensives.
- Falls in older men: falls and fractures rise in men ≥66 on any α-blocker.
- **IFIS (intraoperative floppy iris syndrome):** not specifically studied but reported; class effect; warn the ophthalmologist before cataract surgery (stopping the drug does not remove risk).

Comparison with other α-blockers

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction	Efficacy in LUTS
Silodosin	α1A (highest); α1A ≫ α1D, α1B	8 mg OD	Lowest	**Highest**	Equal
Tamsulosin	α1A, α1D > α1B	0.4 mg OD	Low	Moderate	Equal
Alfuzosin	Clinically uroselective	10 mg OD	Low	Lowest	
Naftopidil	α1D	25–75 mg	Some	Less	
Doxazosin, terazosin	Non-selective; titrate	1–8 mg, 1–10 mg	Highest	Low	

Diagram to draw

- α1A in prostate and vas, α1B in vessels, α1D in bladder; prostate and bladder neck showing α1A receptor sites blocked by silodosin.

> **Key points to remember:**
> - Most α1A-selective α-blocker; 8 mg with a meal; no titration.
> - 4 mg if CrCl 30–50; avoid if <30.
> - Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
> - Anejaculation is the signature effect; least hypotension.

Also, from the condensed book (not in the paper answer):

- **No α1-blocker is clearly more effective**; they differ in adverse effects (AEs). Doxazosin cut progression 39% (MTOPS), **not AUR or surgery**.
- **ALLHAT**: the doxazosin arm stopped for heart failure, so never use it as sole antihypertensive; **never use two α-blockers**.

Class	Drug	Dose	Notes
Non-selective	Phenoxybenzamine	10 mg bd	α1+α2; abandoned
Short-acting	Prazosin, indoramin	bd	Not used for LUTS
Long-acting	Terazosin	5–10 mg od	Start **1 mg at bedtime**, titrate
	Doxazosin immediate release (IR)	2–8 mg od	Start 2 mg at bedtime, titrate

Class	Drug	Dose	Notes
	Doxazosin GITS (sustained release)	4–8 mg od	Start **4 mg**, **no titration**
Uroselective	Alfuzosin extended release (ER)	10 mg od	Clinically uroselective
	Tamsulosin	0.4 mg od (to 0.8)	α1a/α1d
	Silodosin	8 mg od	Most α1a-selective
	Naftopidil	25–75 mg/day	Only α1d-selective; Asia

KEY POINT

Titration: α1b blockade causes **first-dose hypotension** and syncope; a bedtime start and titration avoid it. Uroselective drugs are fixed-dose; doxazosin GITS starts at 4 mg without titration.

- They double the placebo symptom gain with **durable** benefit but do **not** shrink the gland; tamsulosin plateaus at **0.4 mg**.
- **AEs**: dizziness, asthenia, orthostatic hypotension and anejaculation. In the **elderly**, dizziness causes falls; severe hypotension is mostly in the **first 8 weeks**.
- **Sexual**: ED equals placebo. Ejaculatory dysfunction is mostly **anejaculation** (emission failure): silodosin (odds ratio ≈32) > tamsulosin (≈9) > alfuzosin.

KEY POINT

Intraoperative floppy iris syndrome (IFIS): **billowing iris, iris prolapse, progressive miosis**. α1a blockade causes dilator atrophy, so IFIS occurs **years after stopping**; stopping pre-op does **not** help. Highest with **tamsulosin**, then alfuzosin, doxazosin and naftopidil. †If cataract surgery is awaited, defer the α1-blocker.

Drug	Vascular	Ejaculatory	IFIS
Terazosin	High	Low	Reported
Doxazosin	High (GITS lower)	Low	Lower
Alfuzosin	Intermediate	Lowest uroselective	Intermediate
Tamsulosin	Low	Dose-dependent	**Highest**
Silodosin	Lowest	**Highest**	Not well studied

KEY POINT

One efficacy, three toxicities: **α1b blockade gives vascular effects** (terazosin, doxazosin); **α1a blockade affects ejaculation and causes IFIS** (silodosin, tamsulosin); **α1d blockade gives storage benefit** (naftopidil).

11 · 5α-Reductase inhibitors

The only drugs that prevent retention and surgery.

SAQ Dutasteride.

Paper answer · P2

Type: SAQ (10 marks)

Source: 2018-Summer Paper 2, Q3c

Pharmacology

- A **dual 5ARI (5α-reductase inhibitor)**: it blocks **type 1** (liver, skin) and **type 2** (prostate stroma, genital tissue) isoenzymes. Finasteride blocks type 2 only.
- **Mechanism:** blocks conversion of testosterone to DHT (dihydrotestosterone), the main prostatic androgen; serum DHT falls about **90%**. Epithelial atrophy shrinks the gland and relieves the **static** component of obstruction.
- **Dose: 0.5 mg once daily**; hepatic metabolism; **half-life about 5 weeks**.

Effects

- **Prostate volume falls about 25%** (maximum by 6 months).
- Symptoms and Qmax (maximum flow rate) improve modestly and slowly (**at least 6 months**).
- **Changes natural history:** pivotal trials: AUR (acute urinary retention) down 57%, BPH (benign prostatic hyperplasia) surgery down 48%.
- **PSA (prostate-specific antigen) halves** by 6–12 months: **double the PSA** to interpret it. A rise from the nadir on treatment needs evaluation.

Indications

- Moderate-to-severe LUTS (lower urinary tract symptoms) with an enlarged prostate: **>40 mL (EAU [European Association of Urology])** or PSA about >1.5 ng/mL; AUA (American Urological Association): >30 mL.
- **Combination with tamsulosin** for men at risk of progression (fixed-dose capsule available).
- Off-label: BPH haematuria (lowers prostatic microvessel density); before TURP (transurethral resection of the prostate) to reduce bleeding (inconsistent evidence).

Landmark trials

Trial	Result
CombAT (Combination of Avodart and Tamsulosin, 4 years; PV [prostate volume] ≥30 mL, PSA ≥1.5)	Combination best for symptoms and progression; dutasteride (and combination) beat tamsulosin for AUR and surgery
REDUCE (Reduction by Dutasteride of Prostate Cancer Events, 4 years)	Prostate cancer down about 23%, but slightly more Gleason 8–10; **not licensed for prevention**

Side effects

- **Sexual:** ED (erectile dysfunction), low libido, ejaculatory disorders; lessen with time.
- Gynaecomastia and breast tenderness; depression (FDA [Food and Drug Administration] warning).
- **No IFIS** (intraoperative floppy iris syndrome) and no hypotension, unlike α-blockers.

Precautions

- **Teratogenic:** pregnant women must not handle leaking capsules. **No blood donation for 6 months** after the last dose.

Diagram to draw: testosterone to DHT pathway with 5α-reductase types 1 and 2 blocked by dutasteride.

Key points to remember:

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

Also, from the condensed book (not in the paper answer):

- **Finasteride 5 mg** (type 2) lowers dihydrotestosterone (DHT) by †~70% and PV by 18–20%; in PLESS, AUR fell 57% and surgery 55%. It performs the same as dutasteride head to head.
- Indication: moderate–severe LUTS with PV >40 mL (EAU) or >30 mL (AUA).
- They relieve the **static component**. Benefit is **slow** (maximum shrinkage at **6 months**) and **volume dependent**, best at **>40 mL**.
- They are the **only class altering natural history**.
- PCPT and REDUCE showed fewer cancers but slightly more Gleason ≥8, so they are **not licensed for prevention**.
- **AEs**: ED, lower libido and lower ejaculate volume (≈2× placebo, waning after 1–2 yr), **depression** (FDA 2011) and rarely gynaecomastia. **Post-finasteride syndrome** is persistence of symptoms after stopping.

KEY POINT

5-ARIs treat the gland: slow, volume-dependent and modest for symptoms, but they prevent AUR and surgery, and they halve PSA (both a trap and a tool).

12 · Antimuscarinics

- They block detrusor **M2/M3** receptors, mainly on **afferent** pathways.
- **Indication**: predominant storage LUTS, **PVR <200 mL** and no prior AUR.
- Agents: darifenacin 7.5/15 mg, fesoterodine 4/8 mg, oxybutynin (IR 5 mg 3–4×/day; ER), propiverine, solifenacin 5/10 mg, tolterodine ER 4 mg and trospium.
- In BOO, PVR rises slightly and AUR equals placebo (trials excluded PVR ≥200).
- AEs: **dry mouth**, constipation and cognitive impairment. **Oxybutynin** has most brain penetration, so avoid it in the elderly; trospium least†.

KEY POINT

Antimuscarinics treat the bladder, so every recommendation is anchored to a PVR ceiling (200 mL).

13 · β3-Agonists

- β3 stimulation gives **storage relaxation without weakening voiding** (Qmax unchanged), with less dry mouth than antimuscarinics.
- **Mirabegron** starts at **25 mg** and rises to **50 mg**; in BOO, PVR rose only at 100 mg.
- AEs: nasopharyngitis, headache, blood pressure rise ≤1 mm Hg. **Contraindicated** in uncontrolled hypertension **≥180/110 mm Hg**; it is a moderate CYP2D6 inhibitor†.
- **Vibegron** 75 mg has no CYP2D6 inhibition; data are mainly in women.

14 · PDE5 inhibitors

- **Tadalafil 5 mg daily** is the only licensed agent (2.5 mg is ineffective). It lowers IPSS by ≈6 (placebo 3.6), **independent of ED**, and reduces nocturia. **Qmax is unchanged**.
- First choice for LUTS with ED (AUA). AEs: headache, back pain, dyspepsia, flushing. †Contraindicated with nitrates and riociguat.

15 · Combination therapy

- **MTOPS** (Medical Therapy of Prostatic Symptoms): progression = AUASI rise **≥4**, AUR, UTI or urosepsis, incontinence, or creatinine rise ≥1.5 mg/dL. Placebo 17%. Number needed to treat (NNT) for combination 8.4 (≈5 if PSA >4 and PV >40).
- **CombAT** (PV ≥30, PSA ≥1.5): AUR or surgery fell two-thirds versus tamsulosin but not significantly (NS) versus dutasteride.
- **Indication**: high risk, PV **>30–40 mL** and PSA **>1.5 ng/mL**. **SMART**: stop the α-blocker after 6 months.

MTOPS reduction vs placebo	Doxazosin	Finasteride	Combination
Overall progression	39%	34%	66%
≥4-point symptom rise	45%	30%	64%
AUR	NS	68%	81%
Invasive therapy	NS	64%	67%

KEY POINT

α-blockers prevent symptomatic progression only; 5-ARIs prevent AUR and surgery. The α-blocker gives fast dynamic relief; the 5-ARI gives slow static shrinkage.

Combination	Indication	Caution
α1-blocker + antimuscarinic	Persistent storage LUTS (tolterodine 4-arm; NEPTUNE)	PVR <150–200 mL only; AUR <1–3%
α1-blocker + β3-agonist	Persistent storage (MATCH)	PVR rise; limited data
α1-blocker + PDE5I	LUTS with or without ED; small gain	Hypotension; not recommended; start lowest dose
PDE5I + 5-ARI	Large prostate with or without ED	Single trial; FDA capsule 2021

16 · Phytotherapy

- Evidence is product-specific. Most **Serenoa repens** extracts show no benefit, but the **hexane extract (Permixon) 320 mg daily with food** reduced nocturia and raised Qmax and is an EAU option. AUA/EAU make **no general recommendation** for phytotherapy.

17 · Follow-up

- Worsening symptoms and rising PVR predict AUR.

ALGORITHM

1. Review at **4–12 weeks** (α-blocker response, AEs).
2. Review at **6 months** (5-ARI).
3. Then **annually**: score, flow, PVR; PSA on a 5-ARI if life expectancy >10 yr. WW: yearly.

18 · Acute urinary retention

SAQ Catheter free trial

Paper answer · P2

Type: SAQ (10 marks)

Source: 2016-Summer Paper 2, Q3e

Assumed meaning: trial without catheter (TWOC) after acute urinary retention.

Definition

- **TWOC (trial without catheter)**: catheter removal after decompression to test whether the patient voids adequately.
- **AUR (acute urinary retention)**: sudden, painful inability to void; commonest in older men with LUTS (lower urinary tract symptoms).

Types of AUR

Spontaneous	Precipitated
BPH (benign prostatic hyperplasia) progression	Anaesthesia, surgery, UTI (urinary tract infection), overdistension, anticholinergic or sympathomimetic drugs
TWOC fails more often; about 75% later need surgery	Better TWOC outcome; about a quarter need surgery

Initial management

- Catheterise; record drained volume; check creatinine; treat the trigger.
- Start an **α1-blocker** (alfuzosin or tamsulosin).

Rationale for α-blockers

- Sympathetic bladder-neck tone (dynamic component) adds to retention; α1-blockers relax it.
- **ALFAUR (Alfuzosin in Acute Urinary Retention)**: alfuzosin 10 mg for 3 days → TWOC success **62% vs 48%**.
- Meta-analysis: success relative risk about 1.5; fewer recurrences.

Procedure

- Remove the catheter after **2–3 days of α-blocker**, usually in the morning (no evidence favours short or long catheterisation).
- Encourage fluids; record voided volumes and **PVR (post-void residual)** by bladder scan.
- **Success**: comfortable voiding, low residual (unit threshold). **Failure**: recatheterise or start CISC (clean intermittent self-catheterisation).

Predictors of failure

- Advanced age; **drained volume >1 L**; spontaneous AUR; severe LUTS.
- Ultrasound: **IPP (intravesical prostatic protrusion) ≥10 mm**; thick bladder wall.

Results

- Success about **60%**; a second or third TWOC still succeeds in over a quarter.
- Recurrence after success is common, mostly within **6 months**; then offer surgery.

After TWOC

- **Success**: continue α-blocker; add a **5-ARI (5α-reductase inhibitor)** for large glands, because 5-ARIs, not α-blockers, prevent further AUR.
- **Failure**: TURP (transurethral resection of prostate) or other surgery; CISC or long-term catheter if unfit.
- **Also know:** immediate surgery has higher morbidity; TWOC also follows prostate surgery.

Diagram to draw: algorithm: AUR → catheter + α-blocker → TWOC at 2–3 days → success / failure.

> **Key points to remember:**
> - TWOC is first choice after AUR; success about 60%.
> - ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
> - Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

Also, from the condensed book (not in the paper answer):

- CombAT: AUR fell 68% versus tamsulosin but only 18% (NS) versus dutasteride; α-blockers do not prevent AUR (ALTESS, MTOPS).
- About two-thirds of AUR is spontaneous, with a worse TWOC outcome than precipitated AUR.

ALGORITHM

1. Catheterise (†suprapubic if urethral fails).
2. Start an **α1-blocker**.
3. Do a **TWOC at ~3–5 days**.
4. If he voids, continue (add a 5-ARI for a large gland).
5. If he fails, try up to **3 TWOCs**.
6. If AUR recurs or TWOC fails: PFS and surgery (see Ch 151).

- TWOC success is ≈ **60%**, and recurrence is mostly within **6 months**.
- Failure predictors: age, severe LUTS, drained volume **>1 L**, spontaneous AUR and high-grade IPP. 5-ARIs prevent AUR (PLESS, MTOPS).

> **KEY POINT**
>
> **The α-blocker rescues the TWOC** (dynamic component); **the 5-ARI prevents the next episode** (static component).

19 · Landmark trials

Trial	Design	Result	Take-home
VA transurethral resection (TURP) vs WW	Randomised trial, moderate LUTS	TURP better; most WW did not fail; crossover by bother	WW safe

Trial	Design	Result	Take-home
PLESS	Finasteride vs placebo, 4 yr, PV ~55 mL	AUR 7% to 3%, surgery 10% to 5%; combined fall 51%	5-ARI prevents AUR/surgery
Dutasteride phase III	2 yr, PV ≥30 mL	DHT fell 90%, PV 26%, AUR 57%, surgery 48%	≈ finasteride
MTOPS	Doxazosin/finasteride/combination/placebo, ≥4 yr	Progression fell 39/34/66%; AUR/surgery only with 5-ARI	Combination for high risk
CombAT	Dutasteride + tamsulosin, PV ≥30, PSA ≥1.5, 4 yr	AUR/surgery fell 66% vs tamsulosin	Combination for large gland
SMART	α-blocker withdrawal	No deterioration after 6 months	Stop α-blocker at 6 months
PCPT / REDUCE	5-ARI prevention	Fewer cancers, more high grade	Not licensed
ALFAUR	Alfuzosin vs placebo before TWOC	Success 62% vs 48%	α-blocker before TWOC

151 Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

1 · Scope and principles

- **American Urological Association (AUA) 2023 operations**: transurethral incision (TUIP), resection (TURP) and vaporisation (TUVP) of the prostate; photoselective vaporisation (PVP); prostatic urethral lift (PUL; UroLift); water vapour thermal therapy (WVTT; Rezūm); temporary implanted prostatic device (TIPD; iTind); robotic waterjet treatment (RWT; Aquablation); holmium or thulium laser enucleation (HoLEP/ThuLEP); simple prostatectomy; prostate artery embolisation (PAE).
- Transurethral microwave therapy (TUMT) was removed in 2023. Because the sham effect is large, new devices need sham-controlled trials.

> **KEY POINT**
>
> Volume predicts progression (acute urinary retention (AUR), surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** A gland >30 cc triples AUR risk (Olmsted County).

2 · Work-up and indications for surgery

- History, examination, IPSS and urinalysis at baseline (Statement (St) 1). International Prostate Symptom Score (IPSS) bands: **0–7 mild, 8–19 moderate, 20–35 severe**.
- Before surgery, post-void residual (PVR) is **required** (St 6). **Consider** size and shape by transrectal ultrasound (TRUS), abdominal ultrasound, cystoscopy, MRI or CT (St 5), uroflowmetry (St 7), and pressure–flow study if the diagnosis is uncertain (St 8).
- A raised prostate-specific antigen (PSA) needs biopsy **before** BPH surgery. Counsel on failure and retreatment (St 9).

AUA Statement 26 (surgery recommended)	† European Association of Urology (EAU): absolute vs relative
Renal insufficiency from BPH; refractory retention; recurrent urinary tract infections (UTIs); recurrent bladder stones or gross haematuria from BPH; lower urinary tract symptoms (LUTS) refractory to, or unwilling to take, medical therapy	**Absolute** (complicated BPH): refractory retention, upper-tract dilatation or renal impairment, recurrent UTI, bladder stones, recurrent haematuria refractory to 5α-reductase inhibitor (5-ARI). **Relative**: moderate–severe bothersome LUTS failing or declining medical therapy

- **Antibiotics (AUA 2019)**: one dose suffices for most transurethral cases. Cover gram-negatives and enterococci (1st/2nd-generation cephalosporin or co-trimoxazole), guided by antibiogram and urine culture. **Defer surgery if UTI is active and symptomatic**.

3 · Choosing the operation

Volume / situation	AUA 2023 options	Restrictions
<30 cc	TUIP, TURP, TUVP, PVP, HoLEP, ThuLEP, TIPD	TUIP only ≤30 cc; TIPD 25–75 cc, no obstructing middle lobe
30–80 cc	TURP, TUVP, PVP, HoLEP, ThuLEP, PUL, WVTT, RWT, TIPD	PUL: no obstructing median lobe; WVTT and RWT treat median lobes

Volume / situation	AUA 2023 options	Restrictions
>80–150 cc / >150 cc	HoLEP, ThuLEP, simple prostatectomy	TURP, PVP, TUVP stop at 80 cc
Higher bleeding risk	HoLEP, ThuLEP, PVP (St 43)	Otherwise hold anticoagulant/antiplatelet after liaising with prescriber
Preserve erection and ejaculation	PUL or WVTT (St 37)	Within 30–80 cc

> **KEY POINT**
>
> Only enucleation spans every size band, joined above 80 cc only by simple prostatectomy. Minimally invasive surgical therapies (MISTs) and RWT live in 30–80 cc, so volume and median lobe assessment come first.

4 · Monopolar TURP (m-TURP): the benchmark

Every newer operation is judged against TURP.

SAQ TURP instruments, techniques and monopolar vs bipolar

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "TURP instruments and techniques" (TURP instruments; TURP techniques) · P2 SAQ "Monopolar v/s Bipolar TURP."

Introduction

- **TURP (transurethral resection of the prostate):** an electrified (wire) loop removes transition-zone adenoma **from bladder neck to verumontanum, down to the surgical capsule**. The benchmark BPH (benign prostatic hyperplasia) operation; gold standard for glands up to 80 mL (AUA [American Urological Association] 2023); larger glands need enucleation or simple prostatectomy.

Resectoscope

Part	Description
Outer sheath	24–28 Fr; **insulated beak** prevents arcing to the urethra
Inner sheath	Creates **continuous flow**: inflow through the inner, outflow between sheaths (passive or suction). Lower bladder pressure, clear vision, no repeated emptying
Obturator	Blunt or visual; atraumatic passage
Working element	Moves the loop. **Passive:** spring holds loop inside sheath; cutting stroke by spring recoil. **Active:** surgeon's thumb pulls the extended loop in
Telescope	**Hopkins rod-lens**, usually **30°** (12° alternative)

Electrodes

- **Cutting loop:** tungsten (monopolar); **platinum–iridium (bipolar plasma)**. The plasma loop cuts at lower temperature and seals vessels.
- Roller ball or barrel (coagulation), vaporisation button, **Collins knife** (TUIP [transurethral incision of the prostate]).

Energy: generator and waveforms

- **Generator:** about 500 kHz, too fast to stimulate nerves.

- **Cut:** continuous waveform. **Coagulation:** interrupted, higher voltage. **Blend:** better haemostasis, cuts less well.

Monopolar versus bipolar: principles

- **Monopolar (m-TURP):** current runs from the loop through the patient to a grounding (dispersive) pad. Saline would disperse the current, so a **non-conducting, hypo-osmolar irrigant** (glycine 1.5%, sorbitol–mannitol, water) is needed.
- **Bipolar (b-TURP):** active and return poles are on the same instrument (the loop), so current stays local and **normal saline** can be used. The Plasma system turns saline into a plasma layer (orange glow) that cuts at lower temperature and seals vessels.
- **AUA:** no preference between monopolar and bipolar; choose by expertise.

Monopolar versus bipolar: comparison

Feature	Monopolar	Bipolar
Current path	Loop → patient → **dispersive pad**	Active and return poles on the loop; stays within the loop
Irrigant	**Non-conducting, hypo-osmolar:** glycine 1.5%, sorbitol, water	**Normal saline**
TUR (transurethral resection) syndrome	Yes; keep resection <90 min	Practically (virtually) eliminated
Resection time	Limited (speed over haemostasis)	No time pressure; large glands
Loop	Tungsten	Platinum–iridium (Plasma)
Bleeding, transfusion, clot retention	Higher	Lower
Catheter time, hospital stay	Longer	Shorter
Obturator reflex	Risk	Reduced
Efficacy: IPSS (International Prostate Symptom Score), QoL, PVR	Equivalent (Qmax marginally better with bipolar)	Equivalent
Stricture, BNC (bladder neck contracture), sexual function, retreatment	Similar	Similar

Imaging and irrigation

- **Light source:** xenon or LED (light-emitting diode) with fibreoptic cable; camera and monitor.
- **Irrigant** at body temperature; bag about **60 cm** high (each extra 10 cm more than doubles absorption); lowest height giving vision.

Accessories

- **Ellik evacuator or Toomey syringe** for chips.
- Van Buren sounds or Otis urethrotome: dilate meatus 2 Fr above sheath size (prevents stricture).
- **Three-way Foley catheter** (22–24 Fr) with catheter guide; continuous irrigation set.

Preparation

- Sterile urine; spinal or general anaesthesia; lithotomy with peroneal padding.
- **Inspect bladder first** (tumour, diverticula); identify ureteric orifices, verumontanum, external sphincter.

Resection techniques

- **Median lobe first**, then lateral lobes.
- **Modified Nesbit "encirclement":** start at 11–9 and 1–3 o'clock; expose but do not resect bladder neck fibres.
- **Floor-first:** channel at 5 or 7 o'clock to the capsule (sets depth early); widen laterally; lateral lobes "fall" into the fossa.
- **Anterior start:** 12 o'clock bladder neck, then quadrants down to the apex.
- **Chips:** long "canoe-like" chips with synchronised rocking.
- **Anterior tissue and apex last** in a bloodless field; leaving a little tissue is better than incontinence.
- Haemostasis: arterial bleeders must be fulgurated; venous ooze settles with catheter traction.
- Evacuate chips (Ellik evacuator); three-way catheter.
- **Also know:** TUVP (transurethral vaporisation, button electrode; ≤80 mL); TUIP (transurethral incision, <30 mL, no median lobe); bipolar enucleation; high bleeding risk: AUA prefers HoLEP (holmium laser enucleation) or PVP (photoselective vaporisation).

Troubleshooting a loop that will not cut

`Scope–generator connection → foot pedal → irrigant matches generator (glycine for monopolar, saline for bipolar) → patient grounding (pad contact)`

TUR syndrome (monopolar)

- Irrigant absorbed through open sinuses (venous pressure about 10 mmHg) → **dilutional hyponatraemia**: confusion, visual disturbance, nausea, bradycardia, hypertension, seizures; glycine also yields ammonia.
- **Risk factors:** gland >45 mL, resection >90 minutes, high irrigation bag.
- **Treatment:** stop surgery, check sodium; 3% saline 100 mL boluses for neurological symptoms; avoid over-rapid correction (central pontine myelinolysis).

Outcomes and evidence

- About 80% symptom improvement; low retreatment; retrograde ejaculation about two-thirds.
- Meta-analyses: monopolar and bipolar have equal efficacy up to 5 years; b-TURP safer (NNT about 50 to prevent one TUR syndrome, about 20 to prevent one clot retention).

Recent advances

- Bipolar plasma resection; chip-on-tip digital optics; laser enucleation scopes; single-use resectoscopes.
- **Also know:** TURis (sheath as return) caused strictures; bipolar enucleation and vaporisation (PlasmaButton).

Diagrams to draw

- Assembled resectoscope labelling sheaths, working element, loop and flow channels.
- Resection field with clock positions, bladder neck, verumontanum and sphincter.
- Monopolar circuit (loop → patient → pad) versus bipolar (loop to return electrode on the same resectoscope).

> **Key points to remember:**
>
> - Continuous-flow 24–28 Fr sheath, 30° telescope.
> - Tungsten loop (monopolar) vs platinum–iridium (bipolar).
> - Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
> - Bladder neck to verumontanum, depth to capsule.
> - Median lobe first; apex last.
> - Bipolar allows saline; TUR syndrome practically eliminated.
> - Equal efficacy; bipolar has less bleeding and shorter catheter time.
> - TUR syndrome risk: >45 mL gland, >90 minutes.

SAQ Measures to decrease complications of TURP

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 2, Q3b (TUR syndrome also answered in 2017-Winter Paper 3, Q3f)

Introduction

- TURP (transurethral resection of the prostate) is the benchmark; overall morbidity approaches one-fifth, mostly preventable.

Preoperative measures

- **Selection:** AUA (American Urological Association) 2023: TURP for glands up to 80 mL; larger glands need HoLEP (holmium laser enucleation) or simple prostatectomy.
- **Sterile urine:** treat UTI (urinary tract infection) first; single-dose prophylaxis covering gram-negatives and enterococci.
- **Anticoagulants:** stop after advice from the prescriber; if they cannot be stopped, choose HoLEP, ThuLEP or PVP (photoselective vaporisation).
- **Finasteride** for recurrent BPH (benign prostatic hyperplasia) bleeding.
- Spinal anaesthesia shows early TUR syndrome; pad the peroneal nerve.

Intraoperative measures

Complication	Prevention
TUR syndrome (hypo-osmolar fluid absorbed, dilutional hyponatraemia)	**Bipolar TURP with saline**; else irrigant ≤60 cm high, resection <90 min, avoid capsular perforation, check sodium in long resections
Bleeding	Systematic resection to capsule; control arterial bleeders as you go; bipolar "cut and seal"; catheter traction for venous ooze
Ureteric orifice injury	Identify orifices first; take a large median lobe down first
Perforation	Respect the capsule; visible fat means perforation

Complication	Prevention
Incontinence	**Stop at the verumontanum**; resect apex last in a bloodless field
Stricture	Calibrate or dilate meatus, lubricate generously, correct sheath size
Bladder neck contracture	Avoid over-resection and excess fulguration; TUIP (transurethral incision of the prostate) for small glands

Postoperative measures

- Large-bore three-way catheter; careful **continuous bladder irrigation** (blocked outflow distends the bladder and worsens bleeding).
- Even routine cases absorb 800–1000 mL of irrigant; check sodium if confused.
- TUR syndrome treatment: **3% saline 100 mL** boluses; raise sodium by no more than 12 mEq/L in 24 h (pontine myelinolysis).
- Stool softeners; no cycling for 4–6 weeks; counsel about **retrograde ejaculation** (about two-thirds).

Diagram to draw: resection field from bladder neck to verumontanum with the external sphincter marked.

> **Key points to remember:**
> - Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
> - Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
> - Verumontanum is the distal limit.

Also, from the condensed book (not in the paper answer):

- AUA: TURP for <80 cc (St 28), monopolar or bipolar by expertise (St 29). Monopolar irrigant is non-ionic (water, **1.5% glycine**†, sorbitol) and **hypo-osmolar**, which causes TUR syndrome.

ALGORITHM

1. Inspect the bladder. If tumour, resect it and defer TURP.
2. Identify orifices, neck, trigone, verumontanum (veru), sphincter; if orifices hidden, give indigo carmine or methylene blue.
3. Resect median lobe, then lateral lobes, then anterior tissue and apex last.
4. Evacuate chips, secure haemostasis, insert a large three-way catheter.

- **Anterior and apex last**, because the anterior fossa perforates easily and the sphincter lies more proximally anteriorly.
- **Arterial bleeding** is bright and must be fulgurated. **Venous bleeding** is darker and vanishes with a full bladder; once arteries are secured, use balloon traction.
- Even routine TURP absorbs **800–1000 mL** irrigant. The catheter comes out at 24–48 h.
- Warn of dysuria for weeks and eschar bleeding at **1–4 weeks**. Results last about 10 years.

5 · TURP complications

Missed TUR syndrome can kill.

SAQ TUR syndrome

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "TUR Syndrome" · P3 SAQ "TUR Syndrome."

Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorption of hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). Incidence low and falling; about 1–2% in large series.

Pathophysiology

- Prostatic venous pressure is about **10 mmHg**; irrigant at higher pressure enters open sinuses, or the periprostatic space after perforation.
- Even routine resection absorbs 800–1000 mL.
- Monopolar needs ion-free fluid: **glycine 1.5% (~200 mOsm/kg)**, sorbitol, mannitol or water (all hypo-osmolar; plasma ~290).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water moves into brain → **cerebral oedema**.
- **Glycine toxicity:** ammonia → encephalopathy; inhibitory retinal neurotransmitter → **transient blindness**.
- Sorbitol → hyperglycaemia; mannitol → osmotic diuresis.

Risk factors

- Gland **>45 mL**, resection **>90 minutes** (Veterans Affairs Cooperative Study), high irrigation pressure, capsular perforation, sterile water.

Clinical features

- Awake (spinal) patient: restlessness, confusion, **nausea, vomiting, visual disturbance**.
- Cardiovascular: **hypertension with bradycardia**, tachypnoea, later hypotension, ECG (electrocardiogram) changes.
- Severe: seizures, coma, herniation, death. **Na <120 mEq/L** is dangerous.

Prevention

- **Bipolar TURP or laser with saline** (eliminates it; NNT [number needed to treat] about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous-flow resectoscope; low-pressure irrigation.
- Limit resection time; stage huge glands; check sodium in long cases.
- Spinal anaesthesia for early detection; **1% ethanol-tagged irrigant** (breath test) monitors absorption.

Treatment

1. Stop surgery, secure haemostasis, insert catheter.
2. ABC (airway, breathing, circulation), oxygen, monitoring; send **sodium, osmolality**, glucose.
3. Fluid overload: **furosemide**; restrict fluids.

4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes, maximum 300 mL. **Target rise 4–6 mEq/L.**
5. Correction limit: **≤12 mEq/L in 24 h** for acute post-TURP hyponatraemia (8–10 mEq/L if symptoms start >48 h after surgery), avoiding **osmotic demyelination (central pontine myelinolysis)**.
6. Seizures: benzodiazepines; ICU (intensive care unit) care.

Other settings

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy) and hysteroscopy.

Diagram to draw

- Flowchart: absorption → overload + hyponatraemia → cerebral oedema.

Key points to remember:

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

Also, from the condensed book (not in the paper answer):

- Risk factors are a gland **>45 cc**, resection **>90 min**, perforation and irrigant height. Keep the bag at about **60 cm**, because prostatic venous pressure is only about **10 mm Hg**. Na^+ **<120 mEq/L** is severe.

ALGORITHM

1. Stop or expedite surgery, secure haemostasis, stop irrigation; send Na^+.
2. If overloaded, give a †loop diuretic.
3. If severe central nervous system signs: **3% saline 100 mL** (+2–3 mEq/L), repeat ×2 at 10 min (max 300 mL).
4. Target +4–6 mEq/L; ≤12 mEq/L in 24 h (≤8–10 if presentation >48 h). Monitor in high dependency.

KEY POINT

Acute post-TURP hyponatraemia is safer to correct than chronic: the brain has not yet shed its organic osmolytes, so osmotic demyelination is less likely. Bipolar saline TURP eliminates TUR syndrome.

- **Perforation** shows glistening fat. A limited extraperitoneal tear needs only a longer catheter, an extensive one needs drainage, and an intraperitoneal tear needs laparotomy. A persistent intra-operative erection responds to intracorporeal phenylephrine.

Late complication	Cause / features	Management
Bladder neck contracture (BNC)	Over-resection or over-fulguration (small glands); onset ~6 months, flow declines	Dilate, then incise over a wire into perivesical fat. Recurrent: add mitomycin C (10 mL of 0.4 mg/mL). Intractable: Y-V plasty

Late complication	Cause / features	Management
Stricture	Meatus or fossa navicularis	Dilate meatus 2 Fr above sheath; plenty of gel
Retrograde ejaculation	About **two-thirds**	Counsel; erection, orgasm, desire unaffected
Incontinence	Internal sphincter lost; apical over-resection causes total stress incontinence (SUI). Rare, usually transient	Veru is the distal limit; †pelvic floor exercises, then †sling or artificial sphincter
Bleeding	Early: incomplete haemostasis. Delayed (1–4 weeks): eschar	Early major: theatre. Delayed mild: fluids; †clot evacuation if retention
Erectile dysfunction (ED)	No conclusive causal evidence	—
Retreatment	Residual apical/anterior tissue, >80 cc, regrowth, BNC, stricture	Rises with size and pre-op retention

6 · Bipolar TURP (b-TURP)

*See the paper answer **TURP instruments, techniques and monopolar vs bipolar** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- It is equally effective and safer, so it may replace m-TURP as gold standard.

7 · TUIP

- TUIP releases capsule and bladder neck without removing adenoma. It is for **<30 cc only** (St 31) without a median lobe (relative contraindication; EAU the same). It suits a young man with a small gland who wants ejaculation.

ALGORITHM

1. Use a Collins knife, loop, or holmium/thulium laser.
2. Incise at 5 and/or 7 o'clock from distal to the orifice, through the bladder neck, to proximal to the veru.
3. Deepen to capsule (with or without fat); cauterise.
4. Catheter out and home on day 0–1.

- Compared with TURP it is shorter, bleeds less and relieves symptoms similarly, but gains less Qmax and needs **more reoperation**; retrograde ejaculation is rarer and BNC does not occur.

8 · Bipolar TUVP

- A broad electrode (PlasmaButton) vaporises and coagulates together, so haemostasis is excellent, but there is **no specimen**. Only **bipolar** is recommended (St 32: <30 and 30–80 cc); monopolar was abandoned for irritation, SUI and poor durability.
- Compared with TURP it needs a shorter catheter and fewer transfusions, but causes more dysuria, AUR and retreatment; retrograde ejaculation is as common.

> **KEY POINT**
>
> Vaporisation trades haemostasis for a coagulated, specimen-free bed. The coagulum causes dysuria and retention; with no anatomical plane, durability is weaker. PVP makes the same trade.

9 · Laser physics

SAQ Lasers in BPH, including green light

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P3 SAQ "Lasers in BPH, including HoLEP" (Various lasers in treatment of BPH; HOLEP) · P2 SAQ "LASERS in BPH." · P4 SAQ "Green light lasers"

HOLEP read as: holmium laser enucleation of the prostate for BPH (benign prostatic hyperplasia).

Principle

- LASER (light amplification by stimulated emission of radiation) turns light into local heat. The tissue effect depends on wavelength and **chromophore** (water or haemoglobin), which set **penetration depth**.
- About 60 °C coagulates tissue (delayed sloughing); ≥100 °C vaporises it (immediate removal). Coagulation lasers are obsolete.

Lasers used in BPH

Laser	Wavelength	Chromophore	Penetration	Procedure
Ho:YAG (holmium:yttrium-aluminium-garnet), pulsed	~2120 nm	Water	0.4 mm	**HoLEP** (holmium laser enucleation of the prostate)
Thulium:YAG (continuous) / **TFL** (thulium fibre laser)	2013 / 1940 nm	Water (TFL at absorption peak, about 4× holmium)	0.2 mm	ThuLEP (enucleation), ThuVEP (vapoenucleation), ThuVARP (vaporesection), vaporisation
KTP (potassium titanyl phosphate) / LBO (lithium triborate)	532 nm	Haemoglobin	0.8 mm	**PVP** (photoselective vaporisation), 120–180 W
Diode	940–1470 nm (various)	Water and haemoglobin (mixed)	0.5–5 mm	Vaporisation, enucleation
Nd:YAG (neodymium)	1064 nm	Weak	About 1 cm	Coagulation; abandoned (sloughing)

Enucleation versus vaporisation

Feature	Enucleation (HoLEP, ThuLEP)	Vaporisation (PVP)
Tissue	Whole transition zone to the capsule, then morcellated	Removed layer by layer
Histology	Yes	No
Gland size	**Any (size-independent)**	Up to 80 cc

Feature	Enucleation (HoLEP, ThuLEP)	Vaporisation (PVP)
Retreatment	Rare	Higher, especially >80 cc

HoLEP

- **HoLEP** removes the whole adenoma along the **surgical capsule**: the endoscopic equivalent of open simple prostatectomy.
- **Ho:YAG, 2120 nm**: water absorbs it; pulsed; **0.4 mm penetration**; coagulates vessels. Saline irrigation, so no TUR (transurethral resection) syndrome.
- **Indications: size-independent** (<30, 30–80, >80 mL), with or without median lobe; **anticoagulated or high bleeding risk** (AUA [American Urological Association]); alternative to open prostatectomy for large glands.
- **Steps:** 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr scope; **about 2.0 J × 40–50 Hz** for enucleation. 1. Bladder-neck incisions at 5 and 7 o'clock (median lobe) or one at 6 o'clock. 2. Perimontanal incision; **early apical release** protects the sphincter. 3. Develop the capsular plane to 12 o'clock; push lobes into the bladder (**en bloc, two- or three-lobe**). 4. Haemostasis with a defocused beam. 5. **Morcellation** with a full bladder; three-way catheter.
- **Limitation: steep learning curve** (about 50 cases).

Green light (KTP/LBO) lasers: physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam is passed through a **KTP or LBO crystal**. This doubles its frequency and halves the wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water, so vascular prostate absorbs it avidly and vaporisation is bloodless.
- Optical penetration **0.8 mm**, tissue effect 1–3 mm, thin (0.2 mm) coagulation rim. The result is bloodless **photoselective vaporisation**.

Green light platforms

System	Power	Note
KTP	80 W (later 100 W)	Popularised PVP (photoselective vaporisation of the prostate)
LBO HPS (High Performance System)	120 W	Faster vaporisation
LBO XPS	**180 W**	750 µm MoXy fibre: metal tip, fluid cooling

PVP technique

- 23–24 Fr cystoscope; side-firing fibre emitting at **70°**; saline irrigation.
- Open the bladder neck first, away from the orifices; sweep 15–30° in 1–2 seconds at ≤2 mm.
- Large bubbles mean efficient vaporisation. Pulsatile bleeding: coagulation mode (about 30 W).
- More than 5 kJ/cc predicts complete vaporisation.

Indications by guideline (AUA)

- HoLEP or ThuLEP for any size.
- PVP for prostates <30 cc and 30–80 cc (under 80 cc); not >80 cc (high retreatment).

- HoLEP, ThuLEP or PVP for men at **higher bleeding risk** (anticoagulated men, who can continue anticoagulants with PVP).

Evidence and outcomes

- **HoLEP: size-independent**; the only endoscopic procedure **superior to TURP** (transurethral resection of the prostate) for IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate), durable. Less bleeding, shorter catheter time and stay; longer operating time. Retreatment rare (whole transition zone removed).
- **ThuLEP:** outcomes equal to HoLEP.
- **GOLIATH RCT** (randomised controlled trial; 29 European centres, about 270 men, prostate ≤100 cc; 180 W PVP vs TURP): PVP **non-inferior** for IPSS, Qmax and complications at 2 years; shorter catheter time and stay; TURP faster. Retrograde ejaculation (about two-thirds) and reoperation equal. Retreatment high above 80 cc.

Complications

- Dysuria and storage symptoms (dysuria worst after PVP, the highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum after PVP.
- Transient storage symptoms and early / transient stress incontinence (SUI, stress urinary incontinence).
- Retrograde ejaculation in most (**about 75%**, three-quarters, after HoLEP).
- Bladder neck contracture (BNC) 0–3.2% (P2: 0–3%); stricture; capsular perforation (0.2–1% with PVP); ureteric orifice injury; **morcellator bladder injury**.
- No TUR syndrome (saline); transfusion very rare; erectile function unchanged; PVP leaves **no tissue for histology**.
- **Safety:** wavelength-specific eyewear for all; laser safety officer.

Recent advances

- **Moses pulse modulation**; ThuLEP (thulium enucleation); median-lobe-only HoLEP (antegrade ejaculation about 90%); same-day discharge.
- GreenLight enucleation; office and same-day catheter-free pathways.

Diagrams to draw

- HoLEP: sagittal prostate with enucleation plane along the capsule; adenoma enucleated to the capsule, pushed into the bladder, then morcellated.
- PVP: side-firing fibre at 70° vaporising a lateral lobe.

> **Key points to remember:**
>
> - Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm.
> - HoLEP is size-independent and the only endoscopic method superior to TURP; suits anticoagulated men.
> - Retrograde ejaculation about 75% after HoLEP.
> - Green light: 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration; 180 W XPS with MoXy fibre.
> - GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

Also, from the condensed book (not in the paper answer):

- Deeper penetration leaves more coagulum, hence Nd:YAG's sloughing and retention.

10 · PVP (GreenLight)

PVP suits anticoagulated men with glands under 80 cc.

*See the paper answer **Lasers in BPH, including green light** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- AUA St 33 recommends 120 W or 180 W PVP for <30 and 30–80 cc; anticoagulated men improve equally without more transfusion.

11 · HoLEP

Size-independent, and the only endoscopic operation beating TURP.

P4 P1 *HoLEP — size-independent (St 38/43), only endoscopic method superior to TURP; settings, steps, learning curve 20/50, BNC if <40 cc*

- It is durable to 18 years because the whole transition zone (TZ) goes, whereas TURP leaves a peripheral rim.

ALGORITHM

1. Incise the bladder neck at 5 and 7 o'clock (at 6 o'clock if no median lobe); carry to the veru and enucleate the median lobe.
2. Make a perimontanal inverted-U incision; open the apical plane.
3. Sweep anteriorly to 11–1 o'clock; incise the apical mucosal strip (no sphincter traction).
4. Develop the anterior plane to the neck ("Baleen" sign = bladder); release lobes into the bladder.
5. Haemostasis with a defocused beam; morcellate via nephroscope; 22 Fr three-way catheter, ~60 mL balloon.

- The smooth reflective capsule is the depth reference. If the morcellator engages mucosa, stop cutting and suction.
- **Transient SUI** is common early, in proportion to volume removed; early apical release reduces it.
- Retrograde ejaculation occurs in about **three-quarters**, but median-lobe-only HoLEP preserves it in ~90%. Erections are unchanged or improved and retreatment is rare.

12 · Thulium (ThuLEP) and variants

- Variants include ThuVARP (vaporesection), ThuVEP (vapo-incision) and ThuLEP (blunt dissection); steps match HoLEP.
- Function about equals HoLEP, with less bleeding than TURP (St 38 and 43). Retrograde ejaculation is ~four-fifths; erections are preserved.

	HoLEP	ThuLEP / ThuVEP
Learning curve	Steep (plateau ≥50)	Claimed shorter (~30)
Evidence	Largest; 18-year follow-up	Shorter follow-up

13 · MISTs

The ideal MIST is office-based, single session, needs no general anaesthesia, spares sexual function and ends medication. **Trade-off**: retreatment within 5 years exceeds surgery, and durability beyond 5 years is uncertain.

SAQ UroLift (prostatic urethral lift)

Paper answer · P4

Asked as:

- Urolift. — *Winter 2024 Paper IV Q3d (SAQ)*
- Urolift. — *Practice Set 6 Paper IV Q3b (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2024-Winter Paper 4, Q3d)

Read as PUL (prostatic urethral lift, UroLift) for BPH (benign prostatic hyperplasia).

Introduction

- A **non-ablative MIST (minimally invasive surgical therapy)**. Permanent implants pull (retract) the lateral lobes apart and open the prostatic urethra.
- **FDA (Food and Drug Administration) approval 2013**; extended to ≤100 mL in 2020.
- **Main advantage:** preserves erectile and **ejaculatory** function.

Implant and mechanism

- Spring-loaded implant with T-shaped ends:
- **Capsular tab:** nitinol.
- **Urethral end:** stainless steel (resists encrustation).
- **Connector:** polyethylene terephthalate suture under tension.
- Delivered through a **19-gauge needle** from a cystoscopic device.
- The gland is more compliant than the capsule, so tension compresses the lobe outward; later local atrophy and fibrosis. Implants are MRI (magnetic resonance imaging)-conditional (≤3 T).

Indications

- **AUA (American Urological Association): prostate 30–80 mL, no obstructing median lobe** (confirm by cystoscopy or ultrasound). EAU (European Association of Urology): <70 mL, no middle lobe, men wanting ejaculation preserved.
- Men wishing to stop medication.
- **Avoid:** obstructing median lobe (MedLift data non-randomised), large gland, active infection.

Technique

1. Local anaesthesia; rigid cystoscope.
2. Compress lateral lobe, fire needle, deploy implant; repeat on the other side.
3. **Usually 4–6 implants**, anterolateral, from bladder neck to apex, avoiding the neurovascular bundles.
4. **Keep implants >1.5 cm distal to the bladder neck** (encrustation).
5. Most go home without a catheter.

Evidence

	L.I.F.T. (2013)	BPH6 (2015)
Design	PUL vs sham, 2:1, 19 centres	PUL vs TURP (transurethral resection of prostate), 10 European centres
Result	IPSS (International Prostate Symptom Score) about halved, twice the sham effect; Qmax (maximum flow) +4 mL/s; no new sexual dysfunction	Composite end point met 52.3% vs 20.0%; antegrade ejaculation 100% vs 60.6%
Durability / retreatment	13.6% surgical retreatment at 5 years	13.6% vs 5.7% at 2 years; TURP better IPSS, Qmax, PVR (post-void residual)

- TURP wins on IPSS, Qmax and PVR (post-void residual).

Complications

- Transient dysuria, haematuria, urgency (<2 weeks); encrustation; rare pelvic haematoma; device malfunction (misfire).
- Lower major adverse events but **higher retreatment than TURP**.

Diagram to draw: axial prostate with implants pulling lateral lobes to the capsule.

> **Key points to remember:**
> - 30–80 mL, no median lobe; local anaesthesia.
> - Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
> - TURP wins (better) on IPSS, flow and durability.

Also, from the condensed book (not in the paper answer):

- AUA keeps 30–80 cc and has not adopted the FDA ≤100 cc label.
- PUL typically uses **4–6 implants**, placed anterolaterally between dorsal veins and neurovascular bundles.
- **WVTT (Rezūm)** is **convective**: radiofrequency steam is injected into the TZ, and condensation releases latent heat, causing 1.5–2 cm of necrosis confined by the capsule.
- Injections are given every 1 cm from neck to veru, with extra ones for a **median lobe**, under sedation or prostatic block.
- **Rezūm II** showed ~50% IPSS gain to 5 years, retreatment 4.4% and **no de novo ED**; reduced ejaculate settles by 12 months. (St 36/37).

> **KEY POINT**
>
> PUL moves tissue; WVTT ablates it. So PUL cannot treat a median lobe but needs no catheter and causes little dysuria, whereas WVTT treats median lobes at the cost of a catheter and weeks of irritation. Both preserve the bladder neck, and so antegrade ejaculation.

14 · Robotic waterjet (RWT, Aquablation)

- A non-thermal saline waterjet ablates tissue mapped on real-time biplanar TRUS. It **needs general or spinal anaesthesia, so it is not a MIST**. It suits 30–80 cc (St 39), treats

median lobes and gives no specimen.

- **Haemostasis is the weakness**: traction alone led to bleeding and transfusion, so **focal bladder-neck cautery** via resectoscope is now used.
- **WATER** showed non-inferiority to TURP to 5 years, with retrograde ejaculation 10% vs 36%.

15 · Temporary devices

- **iTind (TIPD)** is a nitinol device left **5–7 days** to cause ischaemic remodelling, then removed. St 41: 25–75 cc, no obstructing median lobe. In its sham trial the responder end point was met, but the absolute IPSS difference was not significant.
- **Optilume** (paclitaxel balloon) has single-arm data only and is not on the AUA list.

16 · Prostate artery embolisation (PAE)

No surgery, but less effective.

ALGORITHM

1. Femoral access under local anaesthesia; pelvic angiography with the catheter balloon as landmark.
2. Pass a 1.9–2.4 Fr microcatheter into the prostatic arteries (mostly from internal pudendal; often duplicated).
3. Inject microspheres to stasis; bilateral is better.

- **Contraindications**: underactive, acontractile or neurogenic bladder; stricture or BNC; large diverticula or stones; contrast allergy; chronic kidney disease.
- AUA St 40: by trained clinicians; no size limit; a role in refractory BPH haematuria. EAU: for men accepting results inferior to TURP.
- Compared with TURP, IPSS, Qmax and volume reduction are inferior but the stay is shorter; most men reintervene by 10 years.
- Complications include non-target embolisation, post-embolisation syndrome and AUR.

17 · Overall comparison

This is the counselling sheet.

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
m-/b-TURP	<80 cc	Spinal/general (GA); 1–2 d	Benchmark	Lost ~⅔	Bleeding, TUR syndrome (mono)	Low
TUIP	<30 cc	Day 0–1	Slightly less	Mostly kept	—	Higher
b-TUVP	<80 cc	Short catheter	Similar early, less durable	Lost	Dysuria, AUR	Higher
PVP	<80 cc; anticoagulated	Day 0–1	Non-inferior (GOLIATH)	Lost ~⅔	Dysuria, UTI	Higher than TURP

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
HoLEP/ThuLEP	Any; anticoagulated	GA/spinal; day 0–1	**Superior** (HoLEP)	Lost ~¾–⅘	Transient SUI, morcellator	Lowest
RWT	30–80 cc	GA/spinal	Non-inferior	Mostly kept	Bleeding	Low (short data)
PUL	30–80 cc, no median lobe	Local; often none	Less	**Preserved**	Encrustation	~14% at 5 y
WVTT	30–80 cc	Sedation; few days	Less	**Preserved**	Irritation	~4–8% at 5 y
iTind	25–75 cc, no median lobe	Local; device 5–7 d	Uncertain	**Preserved**	—	Unknown
PAE	Any	Local; day case	Inferior	Largely kept	Non-target embolisation	High

KEY POINT

The "gold standard" for a given man is the operation that meets his goals safely: a shared decision on symptoms, size and shape, bleeding risk, ejaculation wishes and surgeon expertise.

18 · Landmark trials

Trial	Design	Result	Take-home
L.I.F.T.	PUL vs sham, 2:1, 30–80 cc	IPSS ~halved; 13.6% surgical retreatment at 5 y	PUL works, preserves sexual function
BPH6	PUL vs TURP, ≤60 cc	Composite favoured PUL; TURP better IPSS/Qmax/PVR	"Superior" refers only to composite
Rezūm II	WVTT vs sham, 30–80 cc incl. median lobe	IPSS −11.2 vs −4.3; durable 5 y; no de novo ED	Convective MIST incl. median lobe
WATER	RWT vs TURP, double-blind	Non-inferior; better in 50–80 cc; less retrograde ejaculation	Aquablation for 30–80 cc
GOLIATH	180 W PVP vs TURP, ≤100 cc	Non-inferior to 2 y; shorter catheter and stay	PVP alternative to TURP
UK-ROPE	PAE vs TURP, matched registry	PAE inferior for IPSS/QoL; shorter stay	Lesser-efficacy option

152 Simple Prostatectomy: Open, Laparoscopic and Robotic

1 · Definition and surgical anatomy

- It enucleates only the benign prostatic hyperplasia (BPH) adenoma and is the gold standard for outlet obstruction with glands **>80 g**.
- The plane is the **surgical capsule (pseudocapsule)**: peripheral zone compressed by the enlarging transition zone (see Ch 148).
- **Urethral arteries enter posterolaterally just distal to the bladder neck**, so a figure-of-eight suture goes at **5 and 7 o'clock**.
- Eponyms MCQ : **Freyer** is suprapubic transvesical and **Millin (1945)** is retropubic.

2 · Indications

- Gland **>80–100 g** with bothersome lower urinary tract symptoms (LUTS) refractory to medical or endoscopic therapy.
- Retention, recurrent urinary tract infection (UTI), **bladder stones**, haematuria or obstructive nephropathy; a diverticulum needing surgery (relative); or inability to lie in lithotomy for transurethral resection (TURP).

3 · Pre-operative evaluation

- Check bleeding history, anticoagulants, prior pelvic surgery (which dictates the route), **urine culture** (treat first) and coagulation.

> **KEY POINT**
>
> Work-up asks three questions. Is the gland big enough (transrectal ultrasound)? Is there cancer (digital rectal examination, prostate-specific antigen, MRI)? Will the bladder empty (urodynamics)?

4 · Open techniques

Freyer opens the bladder; Millin opens the capsule.

SAQ Millin's prostatectomy

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Millin's prostatectomy" · P2 SAQ "Millin's prostatectomy"

Definition

- **Retropubic transcapsular (extravesical) simple prostatectomy** (Millin, 1945): adenoma enucleated through a transverse anterior capsulotomy **without opening the bladder**. Plane: the surgical capsule (compressed peripheral zone around the transition-zone adenoma).

Indications

- BPH (benign prostatic hyperplasia), gland **>80–100 g**, with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or upper-tract damage (obstructive complications), or when lithotomy is impossible.

- Prefer suprapubic (Freyer) for a large median lobe, diverticulum or bladder stones.
- EAU (European Association of Urology) and AUA (American Urological Association): simple prostatectomy or HoLEP (holmium laser enucleation) for large glands.

Work-up

- Exclude cancer: PSA (prostate-specific antigen), MRI (magnetic resonance imaging) ± biopsy; TRUS (transrectal ultrasound) volume; cystoscopy; urodynamics if bladder function is doubtful.

Steps

1. Lower midline or Pfannenstiel; extraperitoneal approach to the **space of Retzius**.
2. Control the dorsal venous complex: optional prophylactic ligation of DVC (dorsal venous complex) and pedicles (Walsh–Oesterling).
3. **Transverse capsulotomy ~2 cm distal to (below) the bladder neck** (a longitudinal cut may extend into the sphincter).
4. Find the adenoma–capsule plane with scissors; **finger enucleation**.
5. **Divide the apical urethra under vision** proximal to the sphincter, sparing it.
6. Trim the bladder neck lip; advance bladder neck mucosa into the fossa.
7. **Figure-of-eight sutures at 5 and 7 o'clock**, where prostatic arteries enter.
8. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin (retropubic)	Freyer (suprapubic)
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched by finger
Fossa haemostasis	Directly visible	Less visible
Stones, diverticulum, median lobe	Difficult	Ideal
Specific risk	(Dorsal) venous bleeding	Cystotomy leak

Complications

- Bleeding (transfusion <5%): control by catheter traction, Malament purse-string, O'Conor capsular plication.
- Urine leak, UTI (infection), ureteric injury, bladder neck contracture 2–6%, stricture, DVT (deep vein thrombosis) and thromboembolism.
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence very rare.

Versus TURP (transurethral resection of prostate)

- Complete removal, low (lower) retreatment, no TUR syndrome; but incision, longer stay.
- **Also know:** vesicocapsular variant.

Recent advances and modern alternatives

- HoLEP; laparoscopic, robotic multiport and single-port transvesical simple prostatectomy: less bleeding, shorter stay.

Diagram to draw

- Sagittal view of transverse capsulotomy and enucleation plane.

Key points to remember:

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

Also, from the condensed book (not in the paper answer):

- **Freyer** uses a transverse **cystotomy 1–2 cm proximal to the bladder neck**. It treats **median lobe, diverticula and stones**, but the apex is blind.
- **Millin**'s drawback is **dorsal venous complex (DVC) bleeding**.

KEY POINT

Every open variant: enter, find the pseudocapsular plane, divide the urethra at the apex **proximal to the verumontanum** without pulling on the sphincter, then 5/7 o'clock haemostasis with bladder neck mucosal advancement.

5 · Haemostasis

Technique	When / how
Pack fossa 5 min	First
5 and 7 o'clock figure-of-eight (2-0 chromic/polyglactin)	Every case; deep through proximal capsule
Bladder neck mucosal advancement ("trigonisation")	Haemostasis; prevents contracture
11 and 1 o'clock sutures	Anterior branches, occasionally
Walsh prophylactic control	Before Millin: ligate DVC and lateral pedicles, sparing neurovascular bundles (NVBs)
O'Conor plication	Diffuse bleeding: plicate the posterior capsule
Malament purse-string	Persistent bleeding: purse-string round the neck, tied suprapubically, partitions fossa from bladder

ALGORITHM

1. Catheter traction (50 mL balloon), with or without continuous bladder irrigation (CBI); exclude coagulopathy.
2. If bleeding continues, evacuate clot endoscopically and fulgurate.
3. If severe, reopen the fossa.

6 · Postoperative care (open)

- On day 2, if urine is clear, remove the Foley for a trial of void. If drain output is high, check **drain creatinine**.

7 · Laparoscopic and robotic

SAQ Laparoscopic Trans vesical Prostatectomy

Paper answer · P3

Type: SAQ (10 marks)
Source: 2025-Winter Paper 3, Q3d

Assumed meaning: laparoscopic transvesical (suprapubic) simple prostatectomy for large BPH (benign prostatic hyperplasia).

Introduction

- Laparoscopic version of the Freyer suprapubic operation: the adenoma is enucleated through a **cystotomy** in the **surgical capsule plane** (compressed peripheral zone). First LSP (laparoscopic simple prostatectomy): Mariano, 2002.

Indications

- Gland **>80–100 g** with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or obstructive nephropathy.
- **Transvesical route preferred** for a large intravesical median lobe, bladder stones or a diverticulum needing surgery.
- Work-up: PSA (prostate-specific antigen) ± MRI (magnetic resonance imaging) to exclude cancer; TRUS (transrectal ultrasound) volume; cystoscopy.

Steps

1. General anaesthesia, supine, Trendelenburg. **Ports:** 10-mm umbilical camera, two 10-mm working ports, 5-mm and 12-mm assistant ports. Transperitoneal or extraperitoneal (Retzius).
2. **Cystotomy** with stay sutures; identify ureteric orifices.
3. **Circumferential bladder-neck incision** (start 3 to 9 o'clock posteriorly), away from orifices.
4. Develop the **posterior plane first** to the apex, then the lateral lobes.
5. Divide anterior commissure; **transect urethra just proximal to the verumontanum** (protects sphincter).
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries enter posterolaterally).
7. **Bladder-neck mucosal advancement** to urethral mucosa covers the fossa and reduces bleeding.
8. 22 Fr three-way catheter; two-layer cystotomy closure.

Outcomes versus open

- Less blood loss, shorter catheter time and stay, less analgesia; **same functional results** (IPSS (International Prostate Symptom Score), flow rate, residual).
- Limitations: steep learning curve, poor ergonomics; **robotic simple prostatectomy** has largely replaced it. **Single-port transvesical robotic enucleation** (2.7 cm suprapubic incision, pneumovesicum) allows same-day discharge.

Complications

- Bleeding (transfusion rare), urine leak, ureteric orifice injury, bladder neck contracture (<1%; open 2–6%).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence rare.

Diagram to draw: sagittal view of cystotomy, bladder-neck incision and enucleation plane.

Key points to remember:

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

Also, from the condensed book (not in the paper answer):

- Mucosal advancement usually means **no CBI, drain or suprapubic tube**, and the catheter comes out on day 5–7.
- Avoid posterolateral cautery, because thermal NVB injury causes erectile dysfunction (ED).

ALGORITHM

1. Vertical cystotomy; identify orifices.
2. Incise the bladder neck 3→9 o'clock under the median lobe; develop the **posterior plane** to the apex, then the lobes.
3. Divide the anterior commissure, see the veru, and transect the urethra **just proximal to the veru**.
4. Place 5/7 barbed sutures; do **vesicourethral mucosal advancement**; close in two layers.

- **Single-port transvesical surgery** with **pneumovesicum 8 mm Hg** allows same-day discharge.

8 · Outcomes and comparisons

The route changes recovery, not function.

Comparison	Bottom line
Robotic vs open	Less bleeding, shorter catheter and stay, fewer complications; longer operation. Function **equal**
Single-port vs multiport	Shorter stay, less CBI and opioid; same complications
vs TURP	Lower retreatment, **no TUR syndrome**, treats stones and diverticula; but more bleeding, longer stay
†vs holmium laser enucleation (HoLEP)	†Equivalent outcomes; HoLEP bleeds less, but has a long learning curve and cannot treat diverticula

KEY POINT

Minimally invasive routes improve only perioperative morbidity; function is identical because enucleation is identical. Open transfusion is now **≤5%**.

9 · Complications

Complication	Cause / rate	Management
Extravasation	Incomplete closure; capsular tear	If drain creatinine equals urine, keep drain and Foley

Complication	Cause / rate	Management
Ureteric orifice injury	Neck incision or sutures	Stent
Bladder neck contracture	Open **2–6%** ; minimally invasive <1%	Dilation or Collings knife incision
Storage LUTS	Weeks–months	Anticholinergic or β3-agonist
Stress incontinence	Rare	Bulking (mild); artificial sphincter (severe)
Erectile dysfunction	**3–5%**	—
Retrograde ejaculation	**80–90%**	Counsel pre-op
UTI, stricture, venous thromboembolism (VTE)	VTE <1%	Prophylaxis

KEY POINT

The adenoma lies inside the pseudocapsule and proximal to the veru. Removal loses the bladder neck (retrograde ejaculation) but spares sphincter and NVBs, so incontinence and ED are rare.

★ Putting it together: the BPH pathway

A · Man with LUTS

ALGORITHM

1. **1. Assess:** International Prostate Symptom Score (IPSS: 0–7 mild, 8–19 moderate, 20–35 severe), rectal examination, urinalysis, PSA if it changes management; uroflow and post-void residual (PVR) before intervention. **2. If complications, operate:** refractory retention, recurrent UTI, stones, prostatic haematuria, renal impairment. **3. If mild**, watchful waiting. **4. If bothersome**, drugs (B); if they fail, a procedure (C).

B · Medical therapy

Drug	Use	Notes
α1-blocker	First line; works in days	Ejaculatory dysfunction; floppy iris
5α-reductase inhibitor (5-ARI)	Prostate >30–40 mL	Less retention and surgery; halves PSA
α-blocker and 5-ARI	Large gland, progression risk	Best (MTOPS, CombAT)
Add antimuscarinic or β3-agonist	Persistent storage LUTS	Check PVR first
Tadalafil 5 mg daily	LUTS with ED	No nitrates
Desmopressin	Nocturnal polyuria	Check sodium if ≥65

C · Choosing a procedure

Situation	Options
<30 mL	TUIP or TURP
30–80 mL	Bipolar TURP, PVP, HoLEP/ThuLEP, Aquablation; MISTs (urethral lift, water vapour)
>80 mL	HoLEP/ThuLEP; simple prostatectomy (stones/diverticulum)
Anticoagulation continued	HoLEP, ThuLEP, PVP
Preserve ejaculation	Urethral lift (no median lobe) or water vapour
Unfit	Long-term catheter or †prostatic artery embolisation

D · TURP complications

KEY POINT

TUR syndrome causes confusion and hyponatraemia. Stop, check sodium, give a diuretic if overloaded and 3% saline if severe; bipolar prevents it. Late problems are retrograde ejaculation (~two-thirds), stricture, bladder neck contracture and incontinence (see Ch 151).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ HoLEP

Paper answer · P1 · P4 · merged from 2

Asked as: P4 LAQ "HoLEP (holmium laser enucleation of the prostate)" (Holep; also SAQ: HOLEP.) · P1 SAQ "HoLEP in BPH"

Read as HoLEP (holmium laser enucleation of the prostate) for BPH (benign prostatic hyperplasia).

Introduction and principle

- **HoLEP** removes the whole transition-zone adenoma endoscopically **along the surgical capsule**. It is the endoscopic equivalent of open simple prostatectomy.
- **Evolution:** HoLAP (holmium laser ablation, 1994, slow) → HoLRP (resection, strips removed with a grasper) → **HoLEP (1998)**, made practical by the **tissue morcellator**.
- "Enucleation is enucleation": any energy giving a true **AEEP (anatomic endoscopic enucleation of the prostate)** gives durable results.

Laser physics

- **LASER:** light amplification by stimulated emission of radiation. Ho:YAG (holmium:yttrium–aluminium–garnet) = holmium atoms in a YAG (yttrium–aluminium–garnet) crystal.
- **Wavelength about 2120 nm**; chromophore **water** (absorbed by water); **pulsed**; optical penetration **0.4 mm**. It cuts by vaporisation with a thin coagulation zone; heat spreads 2–3 mm and coagulates small and medium vessels.
- Flexible quartz fibres; **normal saline** irrigation, so no TUR (transurethral resection) syndrome.

Laser	Wavelength	Chromophore	Penetration
Holmium:YAG	2120 nm	Water	0.4 mm
Thulium fibre	1940 nm	Water	0.2 mm
KTP (potassium-titanyl-phosphate, green light)	532 nm	Haemoglobin	0.8 mm
Nd:YAG	1064 nm	–	About 1 cm (sloughing)

MOSES (pulse modulation)

- **Moses effect:** the first part of each pulse boils a **vapour channel** in the water. Vapour absorbs the light far less, so the rest of the pulse passes through to the tissue ("parting the water").
- **Moses technology** splits each pulse to exploit this: more energy reaches tissue, and it **may shorten operating time**.

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without a median lobe (AUA [American Urological Association] Statement 38). Only enucleation spans every size band.
- **High bleeding risk or anticoagulated men** (AUA recommendation, Statement 43).
- Large glands otherwise needing open prostatectomy (alternative to open prostatectomy for large glands); retention; failed medical therapy.

Technique (steps)

- **Equipment:** 60–150 W Ho:YAG laser; **550 µm end-firing fibre**; 24–28 Fr continuous-flow laser resectoscope with 30° lens; saline.
- **Settings:** about 2.0 J × 40–50 Hz to enucleate; 1.0–2.0 J × 20 Hz to coagulate.
- **Three-lobe (bottom-up, early apical release):** 1. Median lobe: **bladder-neck incisions at 5 and 7 o'clock** to the capsule, joined proximal to the verumontanum; median lobe enucleated into the bladder. No median lobe: one 6 o'clock incision. 2. **Perimontanal inverted-U incision**; scope pushes laterally to open the lobe–capsule plane at the apex; **early apical release** to protect (spare) the sphincter. 3. Capsular plane developed anteriorly to 11–1 o'clock (round to 12 o'clock); **apical mucosal strip incised early** to avoid sphincter traction. 4. Anterior commissure divided; lateral lobes enucleated (detached) into the bladder (**en bloc, 2-lobe or 3-lobe**). 5. **Haemostasis** with a defocused beam.
- **Two-lobe:** 6 and 12 o'clock grooves; each half enucleated with its share of median lobe.
- **En bloc:** whole adenoma in one piece after circumferential apical release; usually paired with early apical release.
- **Morcellation:** nephroscope, two open inflow lines, **full bladder** and good haemostasis. Morcellators: Piranha (rotating), Versacut (oscillating). Fragments sent for histology.
- **Catheter:** 22 Fr three-way; same-day or next-day removal.

Outcomes and trials

- **Versus TURP (transurethral resection of the prostate):** RCT (randomised controlled trial) in glands <100 mL: better symptom score and PVR (post-void residual), less haemoglobin drop (less bleeding), shorter catheter time and stay, longer operation.
- **Meta-analyses:** better **IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable at **3–8 years**. **HoLEP is the only endoscopic procedure with efficacy superior to TURP.**
- **Versus open prostatectomy** for large glands (Naspro 2006; Kuntz 2008, 5 years): equal function, less morbidity, less blood loss, shorter catheter time and stay.
- **Durable; retreatment rare**: whole transition zone removed.
- **Anticoagulated men:** feasible; slightly longer stay; warfarin worse than DOACs (direct oral anticoagulants).
- **Guidelines:** EAU (European Association of Urology) and AUA list HoLEP (enucleation) as a first-line alternative to TURP and to open surgery for large (>80 mL) glands.

Complications

- **Intraoperative:** capsular perforation; ureteric orifice injury (glands >80 mL); **morcellator bladder injury** (stop and drain; intraperitoneal → repair); retained adenoma.
- **Early:** transient storage symptoms and dysuria; **transient (early) SUI (stress urinary incontinence)** from adenoma volume removed, falling over 6 months; persistent SUI rare.
- **Late: retrograde ejaculation about 75%**; BNC (bladder neck contracture) **0–3.2%** (small glands <40 mL, consider bladder-neck incision); meatal stricture; urethral stricture.
- Erectile function unchanged. Transfusion rare; conversion to TURP almost never.

Limitation: learning curve

- The main barrier is the **steep learning curve**: from 20 supervised cases; **efficiency plateaus after about 50 cases**. Mentoring and simulators help.

Recent advances

- High-power **Moses pulse modulation**; **ThuLEP** (thulium laser enucleation) and bipolar enucleation.
- **Ejaculation-preserving (ejaculation-sparing) and median-lobe-only HoLEP** (antegrade ejaculation about 90%).
- **Same-day discharge** and catheter removal.

Diagram to draw

- Sagittal prostate showing the enucleation plane along the surgical capsule, with 5 and 7 o'clock incisions.

> **Key points to remember:**
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
> - Size-independent; best (preferred) in anticoagulated men.
> - Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
> - Retrograde ejaculation about 75%; learning curve about 50 cases.

SAQ Prostatic stents

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3f (urethral stents also answered in Practice Set 4 Paper 4, Q3f and 2015-Summer Paper 4, Q3d)

Definition

- A device placed in the **prostatic urethra**, between bladder neck and verumontanum, to relieve BOO (bladder outlet obstruction) without resection. Mainly for men **unfit for surgery**.

Classification

Type	Examples	Features
Temporary, first generation	Urospiral, Prostakath (spirals)	Steel or gold-plated coil; not epithelialised
Temporary, second generation	**Memokath** (thermo-expandable nitinol); Allium (covered nitinol)	Expands with warm water, softens with cold water for removal
Temporary polyurethane	Intraurethral catheter, Spanner	Bulbar anchor; sphincter left free
Biodegradable	Polylactic acid stents	Dissolve; trial use
Permanent, epithelialising	**UroLume** (woven mesh)	Urothelium covers the mesh; withdrawn

Indications

- Retention or severe BOO from BPH (benign prostatic hyperplasia) in men unfit for surgery: avoids a long-term catheter.
- Drainage after thermal treatments; trial of de-obstruction.
- **Also know:** DSD (detrusor–sphincter dyssynergia), now abandoned; bulbar stricture.

Contraindications

- Poor detrusor contractility (do urodynamics if in doubt); untreated UTI (urinary tract infection); large median lobe; bladder stones; stricture.

Technique

- Measure prostatic urethral length cystoscopically. Place under local anaesthesia with endoscopic or fluoroscopic guidance. The stent must **end above the external sphincter**, or incontinence follows.

Complications

- **Migration** (commonest with temporary stents); **encrustation**; UTI; irritative symptoms, perineal pain, haematuria.
- **Hyperplastic ingrowth** and restenosis with permanent stents (difficult removal); incontinence if the sphincter is crossed.

Current status

- Most unfit men void after placement, but many stents are later removed.
- **EAU (European Association of Urology):** alternative to catheterisation in men unfit for surgery (weak). Not in the **AUA (American Urological Association) 2023** BPH list.
- **Recent: iTind**: temporary nitinol device left 5–7 days; it remodels the urethra by ischaemic pressure necrosis. AUA 2023 option for 25–75 mL without median lobe. **Optilume BPH** (paclitaxel balloon).

Diagram to draw: sagittal prostate with a Memokath from bladder neck to verumontanum, above the external sphincter.

> **Key points to remember:**
> - Temporary (Memokath) vs permanent (UroLume, withdrawn).
> - Need detrusor contractility; never cross the sphincter.
> - Migration, encrustation, ingrowth; iTind is the modern temporary device.

Rapid Revision Sheet · Section 16

Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate (LAQ)

- PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
- Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

IPSS score (SAQ)

- 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
- QoL 0–6 is separate; IPSS = AUA-SI + QoL.
- 3-point change is noticeable; IPSS does not diagnose BOO.

Methods to estimate prostate size (SAQ)

- Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
- DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
- Planimetry most precise; MRI most reproducible. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Role of UDS before prostate surgery (SAQ)

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

BPH: medical management and surgical modalities (LAQ)

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Tamsulosin (SAQ)

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.
- MET benefit in distal stones 5–10 mm.

Silodosin (SAQ)

- Most α1A-selective α-blocker; 8 mg with a meal; no titration.
- 4 mg if CrCl 30–50; avoid if <30.
- Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
- Anejaculation is the signature effect; least hypotension.

Dutasteride. (SAQ)

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

Catheter free trial (SAQ)

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

TURP instruments, techniques and monopolar vs bipolar (SAQ)

- Continuous-flow 24–28 Fr sheath, 30° telescope.
- Tungsten loop (monopolar) vs platinum–iridium (bipolar).
- Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
- Bladder neck to verumontanum, depth to capsule.
- Median lobe first; apex last.
- Bipolar allows saline; TUR syndrome practically eliminated.
- Equal efficacy; bipolar has less bleeding and shorter catheter time.
- TUR syndrome risk: >45 mL gland, >90 minutes.

Measures to decrease complications of TURP (SAQ)

- Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
- Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
- Verumontanum is the distal limit.

TUR syndrome (SAQ)

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

Lasers in BPH, including green light (SAQ)

- Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- HoLEP is size-independent and superior to TURP (only endoscopic method superior to TURP); suits anticoagulated men.
- Retrograde ejaculation about 75%.
- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

UroLift (prostatic urethral lift) (SAQ)

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
- TURP wins (better) on IPSS, flow and durability.

Millin's prostatectomy (SAQ)

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

Laparoscopic Trans vesical Prostatectomy (SAQ)

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

HoLEP (LAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
- Size-independent; best (preferred) in anticoagulated men.
- Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Prostatic stents (SAQ)

- Temporary (Memokath) vs permanent (UroLume, withdrawn).
- Need detrusor contractility; never cross the sphincter.
- Migration, encrustation, ingrowth; iTind is the modern temporary device.

SECTION 17

Prostate Cancer

153 Epidemiology, Aetiology and Prevention of Prostate Cancer

1 · Epidemiology — the headline

- It is the **second commonest** non-skin cancer and the **fifth** cause of cancer death in men; 2020 saw ≈ **1.4 million cases, 375,000 deaths**.
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest). US lifetime risk is ≈ **1 in 8**. 5-year survival is 97%, yet it is the second cause of cancer death in US men.
- **Incidence and mortality are decoupled.** Incidence follows screening and wealth, while mortality peaks where African ancestry meets poor access: the **Caribbean has the highest mortality in the world** MCQ.
- **Latent (autopsy) cancer** is ~5% under 30 y, 29% at 60–69 and **59% >79 y**, and higher in Black men at every age.

> **KEY POINT**
>
> In a prostate-specific antigen (PSA)-screened population, risk-factor studies may find determinants of being tested, so the preferred end point is **aggressive disease**. **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits falsely seem to raise incidence while lowering mortality.

2 · Established risk factors

Only three are unequivocal: **age, race (ancestry) and family history**.

- **Age**: mean age at diagnosis is 66 and it is rare under 40. Diagnosis at ≥70 carries a higher risk of cancer death.
- **Race**: Black men have the highest incidence (172.6/100,000 in the US) and **twice** the cancer-specific mortality of White men, presenting younger and more metastatic. Drivers are unequal access, comorbidity and **8q24** (and 17q21) alleles; with **equal access, outcome is equal** in non-metastatic disease.
- **Family history** is present in ~20%. Twin-study heritability is up to **58%**, so genes outweigh shared lifestyle.

Family history	Relative risk
Any first-degree relative (father or brother)	2–3×
Two first-degree relatives	~3.5×
Both father and brother	**9×**
Relative diagnosed <60 vs ≥60	~2.5× vs ~1.6×
Father/brother died of prostate cancer	2× risk of prostate cancer death

3 · Germline genetics

The key inherited genes are **DNA-repair genes**, which predict **aggressive** cancer.

Gene	Pathway	Association	Implication
BRCA2, BRCA1	Homologous recombination (HR)	BRCA2 familial, aggressive	Early screening; poly(ADP-ribose) polymerase (PARP) inhibitor; caution in active surveillance (AS)
ATM	HR	Metastatic disease	As for BRCA
PALB2, CHEK2	HR / DNA repair	Familial; PALB2 lethal	CHEK2: targeted screening
MLH1, MSH2, MSH6, PMS2 (**Lynch**)	Mismatch repair (MMR)	2–5% of cases	Checkpoint inhibitor; very sensitive to androgen deprivation therapy (ADT)
HOXB13	Prostate-specific homeobox	Hereditary prostate cancer	Targeted screening
8q24, 17q21	Genome-wide association study (GWAS) loci	Commoner in African ancestry	Explain part of the racial excess

- **Polygenic risk scores** predict any cancer, not aggressive cancer, so they are not yet practice-changing.

SAQ Indications of genomic testing in carcinoma prostate

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3e (PARP [poly-ADP-ribose polymerase] inhibitors also answered in 2024-Winter Paper 4, Q3f)

Types of test

Type	Sample	What it finds	Use
Germline	Blood or saliva	Inherited variants: BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13, MMR (mismatch repair: MLH1, MSH2, MSH6, PMS2)	Therapy, family testing, screening
Somatic	Tumour tissue (usually primary) or ctDNA (circulating tumour DNA)	HRR (homologous recombination repair) genes, CDK12, MSI-H (microsatellite instability-high)/dMMR (MMR deficiency)	Targeted therapy
Gene-expression classifiers	Biopsy or prostatectomy tissue	Decipher (22 RNA markers); Oncotype DX GPS (Genomic Prostate Score, 17 genes); Prolaris (cell-cycle genes)	Prognosis

Indications: germline testing

- **All metastatic** disease (AUA [American Urological Association], NCCN [National Comprehensive Cancer Network]): over 10% carry actionable DNA-repair mutations.
- **High- or very-high-risk** localised disease.
- **Intermediate risk with intraductal or cribriform histology** (consider).

- **Family history:** early-onset or metastatic prostate cancer (relative <60 years); BRCA-related (breast, ovarian, pancreatic) or Lynch-related (colorectal, endometrial, upper tract) cancers; known familial mutation.
- **Ashkenazi Jewish ancestry.**
- **Why:** PARP inhibitor eligibility; caution with active surveillance in BRCA2; carrier screening; cascade testing of relatives with counselling.

Indications: somatic testing

- **Metastatic disease, especially mCRPC (metastatic castration-resistant prostate cancer):** HRR alterations in up to 20–25%.
- **HRR (BRCA2 most)** → PARP inhibitors (olaparib, rucaparib, niraparib, talazoparib).
- **MSI-H/dMMR** (2–5%) → pembrolizumab (tumour-agnostic FDA [Food and Drug Administration] approval, 2017); then germline Lynch testing.
- Tissue preferred; ctDNA alternative, but **CHIP (clonal haematopoiesis of indeterminate potential)** gives false calls.

Indications: tissue classifiers

- **Low and favourable-intermediate risk:** active surveillance versus treatment; surveillance intensity. Never the sole basis.
- **Intermediate and high risk:** metastasis risk guides treatment intensity.
- **After prostatectomy:** high Decipher score may favour adjuvant or early salvage radiotherapy (retrospective evidence).
- **Limits:** tumour heterogeneity, retrospective data, cost.
- **Also know:** ConfirmMDx methylation before repeat biopsy; polygenic risk scores.

Diagram to draw: flowchart: metastatic → germline + somatic; localised high-risk or family history → germline; localised low or intermediate → tissue classifier.

> **Key points to remember:**
> - Germline: all metastatic, high-risk, family history, Ashkenazi.
> - Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
> - BRCA2: aggressive; avoid casual active surveillance.
> - Decipher, Oncotype GPS, Prolaris: prognosis only.

Also, from the condensed book (not in the paper answer):

- GWAS has found >200 single-nucleotide polymorphisms, each of small effect.

4 · Molecular aetiology and inflammation

- **Chronic inflammation** is in nearly every biopsy and may be causal; **proliferative inflammatory atrophy** is the proposed precursor, via high-grade prostatic intraepithelial neoplasia (HGPIN).
- **TMPRSS2-ERG** fusion puts the ERG oncogene under an androgen-regulated promoter. It is in ~half of Western tumours but uncommon in East Asians. See Ch 154

5 · Possible risk factors — body, metabolism, lifestyle

Most lifestyle factors act on **progression** rather than initiation; evidence is observational.

Factor	Total incidence	Aggressive / fatal
Height	~5% higher per 10 cm	More high-grade cancer and death
Obesity	Mixed; less low-grade	**Consistently raised**
Smoking	Not raised (smokers screen less)	**Mortality raised ~40%**; no excess after quitting ≥10 y
Metabolic syndrome, hypertension, cardiovascular disease (CVD)	Slightly raised	Metabolic syndrome: worse pathology
Diabetes	Lower (low testosterone, IGF-1 and PSA; metformin)	—
Vigorous activity	No clear effect	Less advanced disease
Alcohol	Unclear	Mortality raised at ≥45 g/day; binge drinking raises it

- Adiposity raises insulin, IGF-1, leptin, inflammation and oestrogen, which drives proliferation and blocks apoptosis.
- **Adult serum testosterone does not predict risk** MCQ.
- A **composite healthy lifestyle** means **30–50%** less lethal cancer; in the highest polygenic-risk quartile, lifetime lethal risk falls from 5.3% to 1.6%.
- **Diet**: World Cancer Research Fund 2018 found **no dietary factor with strong evidence**; dairy, calcium and lycopene were downgraded.
- **Vasectomy** is not a risk factor (detection bias), and the AUA says it need not be discussed.
- Transgender women on oestrogen have a median PSA ≈ **0.02 ng/mL** (50-fold lower), so cut-points must be lowered.

6 · Supplements

SAQ Antioxidants in urological malignancy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3a (prevention trials also in 2017-Summer Paper 4, Q3a)

Rationale

- ROS (reactive oxygen species) cause **oxidative DNA damage**; polyunsaturated fat and excess iron increase it. Antioxidants scavenge ROS, so diet studies suggested protection.
- Randomised trials of single high-dose supplements have been **null or harmful**.

Prostate cancer

Agent	Evidence	Result
SELECT (Selenium and Vitamin E Cancer Prevention Trial)	35,533 men; selenium 200 µg, vitamin E 400 IU, both, or placebo	Stopped early (median 5.46 years) for futility. No benefit. Vitamin E alone trended to more cancer (P = 0.06); extended follow-up showed a significant **increase (about 17%)**
ATBC (Alpha-Tocopherol, Beta-Carotene)	Low-dose vitamin E in heavy smokers	Secondary analysis: prostate cancer ↓32%, deaths ↓41%; beta-carotene ↑ lung cancer

Agent	Evidence	Result
Selenised yeast	Skin-cancer trial	Secondary fall in prostate cancer prompted SELECT
Lycopene (cooked tomato)	Meta-analysis: modestly lower risk, mainly advanced cancer	WCRF (World Cancer Research Fund) downgraded "probable" (2007) to "no conclusion" (2018); supplement trials mostly null
Green tea catechins	HGPIN (high-grade prostatic intraepithelial neoplasia)/ASAP trials	Large fall in progression to cancer (meta-analysis); dose impractical
Vitamin C, high-dose antioxidants	High doses act as pro-oxidants	Supplements above median → more advanced cancer

- **Also know:** vitamin D (VITAL, no effect); coffee (modestly lower advanced risk); pomegranate, soy and curcumin unproven.

Bladder cancer

- Supplemental beta-carotene, vitamins C and E and folate do not lower risk; SELECT showed no effect.
- Fruit and vegetable benefit disappears after adjusting for smoking. Smoking cessation is the real prevention.

Kidney cancer

- No antioxidant trial has shown benefit.

Why the trials failed

- Single nutrient at high dose; wrong chemical form; non-smoking populations; PSA-era indolent cancers; U-shaped dose effects.

Current position

- Antioxidant supplements are **not recommended** for cancer prevention. Advise a balanced diet, no smoking and normal weight.

Diagram to draw: ROS → DNA damage → cancer, with antioxidants blocking the first arrow.

> **Key points to remember:**
> - SELECT: no benefit; vitamin E alone increased prostate cancer.
> - ATBC benefit limited to smokers on low dose.
> - Lycopene downgraded to "no conclusion" (2018).

SAQ Prostate cancer prevention trials

Paper answer · P4

Type: SAQ (10 marks)
Source: 2017-Summer Paper 4, Q3a

Rationale

- Prostate cancer is common, slow and androgen-dependent. Men with 5α-reductase deficiency have rudimentary prostates. This supported **5ARI (5α-reductase inhibitor)** trials.

Landmark trials

	PCPT (2003)	REDUCE (2010)	SELECT (2009)
Agent	Finasteride 5 mg	Dutasteride 0.5 mg	Selenium 200 µg ± vitamin E 400 IU
Men	18,882; PSA (prostate-specific antigen) ≤3	PSA 2.5–10, prior negative biopsy	35,533
Design	7 years; end-of-study biopsy	Biopsy at 2 and 4 years	Stopped early for futility
All cancers	↓24.8%	↓22.8%	No benefit
High grade	Gleason 7–10 ↑ (6.4% vs 5.1%)	Gleason 8–10 ↑ (0.9% vs 0.6%)	Vitamin E alone ↑ risk

The high-grade controversy

- Proposed artefacts: better PSA performance and a smaller gland make cancer easier to find.
- The FDA (Food and Drug Administration) did **not accept** these. In 2011 it refused a prevention licence and added a **high-grade warning** to 5ARI labels.
- PCPT at 18.4 years: non-significant 25% fewer prostate cancer deaths. Real-world data show no mortality effect.
- **Conclusion:** 5ARIs reduce detection of low-grade cancer, not death.

Vitamins and other agents

- **ATBC** trial: low-dose vitamin E cut prostate cancer 32% in smokers (secondary end point). SELECT then failed.
- **VITAL** (vitamin D): no prostate effect. High-dose zinc, calcium and antioxidants may harm.
- **Aspirin/NSAIDs** (non-steroidal anti-inflammatory drugs): small effect. **Statins:** observational benefit only. **Metformin:** none.
- Diet: whole foods; avoid obesity.

Current position

- No agent is approved for prevention. 5ARIs may be discussed with men taking them for BPH (benign prostatic hyperplasia).

Diagram to draw: bar chart: all cancers ↓ vs high-grade ↑ in PCPT and REDUCE.

> **Key points to remember:**
> - PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
> - FDA warning; no approved preventive agent.
> - SELECT negative; vitamin E alone harmful.

Also, from the condensed book (not in the paper answer):

- Calcium supplements raise fatal cancer (relevant on ADT), and zinc ≥75 mg/day raises lethal cancer.
- No supplement benefits healthy men on a balanced diet.

KEY POINT

Vitamin E worked in the ATBC trial but failed in SELECT: ATBC used a tenfold lower dose in **heavy smokers** with little PSA screening (aggressive cancers); SELECT used 400 IU in non-smokers with widespread screening (indolent cancers). Different disease, different answer.

7 · Chemoprevention

*See the paper answer **Prostate cancer prevention trials** (Section 17, Ch 153).*

Also, from the condensed book (not in the paper answer):

- No preventive agent is FDA-approved. 5α-reductase inhibitors shrink the gland about 25%, so on a 5ARI you **double the PSA**. See Ch 154
- **Secondary prevention**: in the Prostate Cancer Lifestyle Trial (men on AS), plant-based diet, exercise and stress management meant **5% vs 27%** treated at 2 y.
- Activity (≥150 min/week moderate or ≥75 min vigorous) lowers cancer-specific mortality ~31%, while a Western diet and continued smoking raise cancer death.

KEY POINT

Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does, so lifestyle advice benefits both.

8 · PSA screening

LAQ PSA screening for prostate cancer: evidence, problems and solutions

Paper answer · P2

Asked as:

- What are the various problems in PSA screening — *Practice Set 8 Paper II Q1 (LAQ)*
- Screening of CA prostate. — *Practice Set 13 Paper II Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction (concept)

- Screening tests healthy, symptom-free (asymptomatic) men; its aim is lower prostate cancer mortality through earlier treatment (cure). The price is **overdiagnosis** (cancers that would never cause harm) and **overtreatment**.
- PSA (prostate-specific antigen) is a kallikrein (hK3) made by prostatic epithelium. It reaches blood when glandular architecture is disrupted. It is **organ-specific, not cancer-specific**.
- Since PSA testing began, (US) mortality has fallen by >40% (over 40%; SEER data), but at the price of overdiagnosis.

Tools

- **PSA:** biopsy threshold 4 ng/mL (many use 2.5–3; ERSPC used 3). Repeat a raised PSA before biopsy.
- **DRE (digital rectal examination):** little value alone; abnormal DRE raises risk.
- **Refinements:** PSA density ≥0.15; free PSA ≤25%; PHI (Prostate Health Index), 4Kscore, Stockholm3; urine tests (PCA3 [prostate cancer antigen 3], ExoDx, SelectMDx).

- **MRI (magnetic resonance imaging) before biopsy:** fewer biopsies and less low-grade overdiagnosis.

Evidence from screening trials

Trial	Design	Result
ERSPC (European Randomised Study of Screening for Prostate Cancer)	>182,000 men; PSA every 2–4 years vs none	**20% relative fall** in (prostate) cancer death at 16 years; 570 invited and 18 extra diagnoses per death prevented
PLCO (Prostate, Lung, Colorectal and Ovarian), USA	Annual PSA vs usual care	**No difference** at 17 years: control arm heavily PSA-tested (contamination), lower power
Göteborg-1 (Sweden)	Biennial PSA, men 50–64	44% fall in cancer death at 14 years (larger mortality reduction with biennial screening)
CAP (Cluster randomised trial of PSA testing for prostate cancer), UK	Single PSA invitation	No mortality difference at 10 years; only a small fall at 15 years
Göteborg-2 (2022)	MRI-targeted biopsy only	Halved overdiagnosis
- Corrected for non-attendance and contamination, ERSPC benefit is 27–31%; it appears only after about 10 years.		

Problems in PSA screening

1. Poor specificity (false positives)

- BPH (benign prostatic hyperplasia), prostatitis, retention, catheterisation, biopsy and TURP (transurethral resection of prostate) all raise PSA.
- At PSA 4–10 ng/mL with a normal DRE (digital rectal examination), only 17–32% of biopsies find cancer. Most men biopsied therefore have no cancer.

2. Poor sensitivity: no safe cut-off

- PCPT (Prostate Cancer Prevention Trial): risk is continuous; about 1 in 11 men biopsied at PSA 0–1 ng/mL has cancer.
- A high threshold misses curable cancer; a low threshold adds biopsies and overdiagnosis. Thresholds used: 4, 3 or 2.5 ng/mL.

3. Biological variability of PSA

- Rises with age, prostate volume (about 4% per mL) and Black race; physiological fluctuation; assay differences.
- Lowered by 5α-reductase inhibitors (about 50% at 12 months, so double the value), obesity, hypogonadism and statins.

4. Overdiagnosis (the central problem)

- Detection of cancers that would never have caused symptoms or death. Autopsy prevalence is high: 29% at 60–69 years and 59% after 79.
- Mostly Grade Group 1 cancers.

5. Overtreatment

- Overdiagnosed men treated radically suffer ED (erectile dysfunction), incontinence and bowel toxicity for no gain, especially above 65 years.

6. Harms of biopsy

- Pain, haematuria, haematospermia, retention and sepsis (transrectal route); anxiety from repeat biopsies.

7. Modest and delayed benefit

- Screening prevents only 20–30% of cancer deaths; no overall survival benefit has been shown. Men with life expectancy <10 years cannot benefit.

8. Biases in interpreting screening data

- **Lead-time bias**: earlier diagnosis lengthens apparent survival without delaying death.
- **Length-time bias**: screening preferentially finds slow-growing tumours.
- **Healthy-user bias**: health-conscious men seek testing, confounding risk-factor studies.

9. Unsettled policy

- Start age, stop age, interval and threshold are all debated; further large randomised trials are unlikely.
- Opportunistic screening is common and inequitable.

Guideline positions (guidelines)

Body	Recommendation
USPSTF (US Preventive Services Task Force) 2012	Grade D: against screening
USPSTF 2018 (current)	**Grade C, 55–69 years: individual shared decision; Grade D ≥70** (not ≥70; was Grade D in 2012)
AUA (American Urological Association)	Shared decision 55–69; every 2 years (2-yearly interval)
NCCN (National Comprehensive Cancer Network)	Baseline PSA and discussion at 45
EAU (European Association of Urology)	Risk-adapted early detection: informed men from 50; earlier start (45) with family history or African descent, 40 for BRCA2 carriers
- **No body recommends (advises) unselected population screening.**	
- **High risk:** Black men, father or brother diagnosed <65, BRCA2 and other DNA-repair mutations: start earlier.	
- Stop when life expectancy is under 10–15 years.	

Solutions: risk-adapted, MRI-first pathway (reducing harm)

```
Informed man, life expectancy >10–15 y → PSA (repeat if raised) → risk stratify: PSA density, free/total PSA, risk calculator, PHI / 4Kscore / Stockholm3 → MRI (magnetic resonance imaging) PI-RADS 1–2, low density → no biopsy, keep in follow-up PI-RADS ≥3 → targeted ± systematic biopsy (transperineal) → Grade Group 1 → active surveillance
```

- **Baseline PSA** predicts future risk (Malmö): low PSA at 40–60 allows longer intervals (intervals by baseline PSA).
- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI-targeted pathway found more significant cancer (38% vs 26%) and less insignificant cancer; 28% avoided biopsy.
- **Göteborg-2 (2022):** MRI with targeted biopsy in screening roughly halved overdiagnosis.
- **Active surveillance** of low-risk disease (Grade Group 1) uncouples overdiagnosis from overtreatment; MRI and biomarkers before biopsy.
- **Also know:** germline testing (BRCA2, ATM, HOXB13); polygenic risk scores; transperineal biopsy lowers sepsis.

Diagram to draw: iceberg of latent vs clinically significant cancer, plus the MRI-first flowchart (pathway: PSA → repeat → risk calculator or MRI → targeted biopsy → active surveillance or treatment).

> **Key points to remember:**
> - ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
> - Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
> - USPSTF 2018: Grade C at 55–69, Grade D ≥70.
> - Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Also, from the condensed book (not in the paper answer):
- Screening lowers cancer death ~**20–30%**, and the benefit builds with time.

9 · Landmark trials

Trial	Design	Result	Take-home
ERSPC (European screening study)	>182,000; PSA every 2–4 y vs none	Cancer death fell 16% (9 y), then ~20% (16 y); excess incidence fell 85% to 36%	Benefit is time-dependent
PLCO (US)	Annual PSA vs usual care	Control arm heavily screened	Contamination dilutes the effect
Göteborg-1	Swedish screening	Significantly fewer cancer deaths	Supports ERSPC
CAP (UK)	Single PSA invitation	No benefit at 10 y; tiny fall in death at 15 y	One-off screening is weak
PCPT	18,882, PSA ≤3; finasteride 5 mg 7 y	Cancer fell **24.8%** (18.4 vs 24.4%); Gleason 7–10 6.4 vs 5.1%; 18-y death fell 25% (not significant)	Fewer cancers, more high-grade
REDUCE	PSA 2.5–10, prior negative biopsy; dutasteride 0.5 mg 4 y	Cancer fell **22.8%**; Gleason 8–10 0.9 vs 0.6%	FDA warning
SELECT	35,533; selenium 200 µg with or without vitamin E 400 IU	Stopped for futility; vitamin E raised risk (P = 0.06)	No supplements

Trial	Design	Result	Take-home
VITAL	>25,000; vitamin D3 2000 IU 5 y	No prostate effect	Not preventive

154 Prostate Cancer Biomarkers

1 · Why biomarkers beyond PSA

- Prostate-specific antigen (PSA) is **organ-specific, not cancer-specific**, which causes negative biopsies and overdiagnosis of Grade Group (GG) 1. Guidelines advise **secondary testing** (imaging or biomarker) before biopsy; the target is **clinically significant cancer (GG ≥2)**. See Ch 153
- Settings are before initial or repeat biopsy, favourable-risk cancer (active surveillance (AS) vs treatment), and after radical prostatectomy (RP).

2 · PSA biology

- PSA is **hK3** (gene KLK3 on **19q13**), an androgen-regulated serine protease that **liquefies semen**. Half-life is **2–3 days**.
- It is made as **proPSA**, and **hK2 cleaves the leader** to give active PSA.
- Serum PSA rises when it leaks through a lost basal layer; per cell, cancer makes less PSA than benign tissue.

Serum form	% of PSA	Measured?
PSA–α1-antichymotrypsin (ACT)	60–90	Yes (predominant, inactive)
PSA–α2-macroglobulin	10–20	**No**: epitopes masked
PSA–α1-protease inhibitor	1–5	Yes
Free PSA	5–45	Yes (proPSA, benign PSA (BPSA), intact PSA)

Confounders

- After biopsy the rise lasts ≥**4 weeks**; ejaculation raises it for 24 h, so repeat after abstinence.
- **5α-reductase inhibitors (5ARIs)** lower PSA ~**50%** by 12 months, so **double** it.
- Age-specific upper limits (ng/mL): 40–49, 2.5; 50–59, 3.5; 60–69, 4.5; 70–79, 6.5.

3 · PSA derivatives

Derivative	Definition / values		Use
PSA density	PSA ÷ volume; cut-off (0.08–0.15)	**0.15**	After negative/equivocal magnetic resonance imaging (MRI); ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification
PSA velocity	Change per year		Minimal value; **never an isolated biopsy trigger**

Derivative	Definition / values	Use
% free PSA	Cancer PSA binds ACT, so **low %fPSA suggests cancer**; thresholds 15–25%	FDA: PSA 4–10, benign digital rectal examination (DRE); mainly **after negative biopsy**; unaffected by 5ARI

> **KEY POINT**
>
> %fPSA was validated against any cancer, not GG ≥2, so it overestimates significant risk.

- **[–2]proPSA** rises with Gleason and underpins the Prostate Health Index (PHI); **BPSA** tracks benign hyperplasia and has no prognostic value.

4 · Blood and urine tests (NCCN 2023 pre-biopsy)

Six tests estimate GG ≥2 risk; **thresholds are setting-specific** (higher for repeat biopsy).

Test	Components	Threshold / key fact
PHI	Blood: **(p2PSA ÷ fPSA) × √tPSA**	24–28.6 proposed; ~30% biopsies avoided
4Kscore	Blood: tPSA, fPSA, iPSA and **hK2**, with age, DRE, prior biopsy	**7.5%**: sensitivity 94%, negative predictive value (NPV) 95%; ≤7.5% = low long-term metastasis risk
IsoPSA	Blood: PSA isoform structure	Unaffected by 5ARI or α-blockers
SelectMDx	Post-DRE urine DLX1, HOXC6 mRNA with clinical data	NPV ~95%; needs prostate volume
ExoDx (EPI)	Urine exosomes: PCA3, ERG, SPDEF	**No DRE**; 15.6 initial, 20 repeat
MyProstateScore	Post-DRE urine PCA3, TMPRSS2:ERG and PSA	10 initial (sensitivity 97%), 15 repeat; validated in Black men

- **PCA3** is a long non-coding RNA. Score = (PCA3 ÷ KLK3 mRNA) × 1000 after an **attentive DRE**; thresholds 25/35. It is **independent of prostate size** and FDA-approved after a **previous negative biopsy**.
- **TMPRSS2:ERG** is **~100% specific** but in only ~half of tumours, so it is combined with PCA3.

> **KEY POINT**
>
> Urine tests follow a DRE because cancer RNA comes only from shed cells massaged into the urethra; ExoDx avoids this, as exosomes are shed spontaneously. Urinary markers are highly specific, but their coverage is limited, hence panels.

5 · Tissue-based biomarkers

SAQ Molecular staging of CA prostate

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3e (related: prognosis in 2019-Summer Paper 4, Q3e)

Definition and rationale

- Clinical staging (DRE (digital rectal examination), PSA (prostate-specific antigen), grade, imaging) misses occult spread and cannot fully predict behaviour.
- **Molecular staging** uses DNA, RNA or protein markers in tissue, blood or nodes to find hidden disease and predict outcome.

1. Detection of occult spread

- **RT-PCR (reverse transcription polymerase chain reaction)** for PSA or PSMA (prostate-specific membrane antigen) mRNA in blood, marrow or "negative" nodes: detects micrometastases, but inconsistent, so not used clinically.

- **CTCs (circulating tumour cells):** counts are prognostic in metastatic disease; **AR-V7** (splice variant lacking the ligand-binding domain) predicts resistance to ARPIs (androgen receptor pathway inhibitors).
- **ctDNA (circulating tumour DNA)** liquid biopsy: DNA-repair mutations; clonal haematopoiesis gives false calls.

2. Tissue genomic classifiers

Test	Basis	Predicts
Decipher	22-RNA classifier	Metastasis, cancer death; post-prostatectomy radiotherapy decision
Oncotype DX GPS (Genomic Prostate Score)	17 genes (12 cancer, 5 reference) by RT-PCR	Adverse pathology; AS (active surveillance) choice
Prolaris	31 cell-cycle progression genes	Progression, cancer death

- **Also know:** PORTOS (24-gene radiotherapy response score); ConfirmMDx (methylation, after negative biopsy).

3. Genetic markers

- **Germline:** BRCA2 (aggressive disease), BRCA1, ATM, CHEK2, Lynch genes, HOXB13 (risk, not aggressiveness); test in metastatic or high-risk disease.
- **Somatic:** PTEN loss, TMPRSS2:ERG fusion, homologous recombination repair defects (guide PARP (poly ADP-ribose polymerase) inhibitors).

4. Molecular imaging

- **PSMA PET (positron emission tomography):** more sensitive and specific than CT and bone scan for nodes and metastases (proPSMA), but cannot exclude micrometastases.

Clinical use

- Refines **AS vs treatment** and post-operative radiotherapy decisions; guides targeted therapy.

Limitations

- Mostly retrospective; **multifocality and heterogeneity** (the sampled focus may not be the worst); cost; added value still being tested (G-MAJOR trial).

Diagram to draw: clinical stage vs molecular stage (tissue, blood, nodes, imaging).

> **Key points to remember:**
> - Molecular staging finds occult disease and predicts behaviour beyond TNM.
> - Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
> - AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
> - Limitation: heterogeneity and mostly retrospective data.

Also, from the condensed book (not in the paper answer):

- **ConfirmMDx** measures GSTP1, APC and RASSF1 methylation in negative cores (NPV ~88–90%), using the **field ("halo") effect** around a missed tumour.
- **Decipher ≥0.4 post-RP** suggests benefit from radiotherapy.

6 · Germline testing

Up to 40–50% of risk is familial, mostly via **DNA-damage-repair** genes.

Gene	Risk	Phenotype
BRCA2	4.5× <65 y; ~8.6× overall	Younger, higher grade, shorter survival
BRCA1	1.8× <65 y	—
ATM, CHEK2	CHEK2 odds ratio ~2–3.4	ATM aggressive
Lynch (MSH2 etc.)	MSH2 ~10×	Also colorectal, upper tract
HOXB13 G84E	4.5×	**Not** more aggressive

Testing is for BRCA1/2 alone or a panel (BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM). Indications: see Ch 153.

7 · Clinical algorithm

ALGORITHM

1. **Raised PSA**: repeat it, excluding infection, instrumentation, ejaculation and 5ARI. Then do MRI and/or a biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density). If risk is high, biopsy.
2. **Negative biopsy, persistent suspicion**: MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx. If positive, targeted re-biopsy.
3. **Favourable-risk**: PSA density and Prolaris/Oncotype/Decipher decide AS vs treatment.
4. **After RP**: PSA should be undetectable. If ≥0.2 ng/mL and confirmed, this is biochemical recurrence; Decipher/PORTOS then guides adjuvant or salvage radiotherapy.

155 Prostate Biopsy: Techniques and Imaging

1 · Indications and contraindications

On transrectal ultrasound (TRUS) the peripheral (PZ) and central zones are homogeneous and echogenic; the transition zone (TZ) is heterogeneous. **39%** of cancers are isoechoic, so a normal grey scale never precludes biopsy.

SAQ TRUS and prostate biopsy

Paper answer · P1 · P2 · merged from 3

Asked as: P2 SAQ "Transrectal ultrasonography (TRUS)" (Transrectal ultrasonography; TRUS) · P2 SAQ "TRUS / prostate biopsy" (TRUS biopsy.; Prostate biopsy) · P1 SAQ "Transrectal ultrasound study of prostate"

Read as: transrectal ultrasonography (TRUS) of the prostate, and TRUS-guided prostate biopsy.

Physics, principle and technique of TRUS

- Piezoelectric crystals; higher frequency gives better resolution but less penetration (depth).
- The prostate lies directly in front of the rectum. Probes: **6–10 MHz** end-fire, side-fire or biplane.
- Left lateral position (or lithotomy for transperineal work), after DRE (digital rectal examination); transverse and sagittal planes, base to apex.

Normal sonographic anatomy

Zone (McNeal)	Glandular share	TRUS appearance	Cancer share
Peripheral	~70%	Homogeneous, echogenic (posterior)	~70%
Transition	5–10% (BPH [benign prostatic hyperplasia])	Heterogeneous, anterior	~20–25%
Central	~25%	Homogeneous, echogenic, at base	~5–10%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- Zones are not seen as such; TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone).
- **Corpora amylacea** mark the surgical capsule.
- Seminal vesicles about 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

Measurements

- **Volume (ellipsoid) = 0.52 (π/6) × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** (serial sections) for brachytherapy.
- **PSAD (PSA [prostate-specific antigen] density)** = PSA/volume; **>0.15 ng/mL/cc** is suspicious (suggests cancer).

Findings

- **Cancer:** classically a hypoechoic peripheral-zone lesion, but **39% are isoechoic**; false positives are BPH nodules, prostatitis and infarcts. Capsular bulge suggests extracapsular (extraprostatic) extension.
- **Cysts:** Müllerian (midline); ejaculatory duct cyst (just off midline); seminal vesicle cyst (lateral; renal agenesis).
- Abscess, calculi.
- After ADT (androgen deprivation therapy) or radiotherapy the gland shrinks and becomes diffusely hypoechoic.

Indications for TRUS

- **Guided biopsy:** 12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion.
- Volume before BPH surgery, brachytherapy or cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicles >1.5 cm wide, dilated ducts, midline cyst).
- Drainage of prostatic abscess or cysts; haematospermia.

Prostate biopsy: introduction

- TRUS-guided systematic 12-core biopsy has been the standard for diagnosing prostate cancer; about 30% are positive at first biopsy.

Prostate biopsy: indications and contraindications

- Raised PSA with risk factors; suspicious DRE; MRI PIRADS (Prostate Imaging Reporting and Data System) ≥3; suspected metastasis.
- **Repeat:** rising PSA; ASAP (atypical small acinar proliferation; 50% cancer risk); positive biomarkers (PHI [Prostate Health Index], 4Kscore, PCA3); active surveillance. Saturation biopsy only after negative biopsies.
- **Contraindications:** coagulopathy, acute UTI (urinary tract infection) or prostatitis, severe immunosuppression; no anus (use transperineal route).

MRI-first pathway

- **PROMIS (2017):** mpMRI (multiparametric MRI) sensitivity 93% vs 48% for TRUS biopsy.
- **PRECISION (2018):** MRI-targeted biopsy found more significant cancer (38% vs 26%) and avoided 28% of biopsies.
- Targeting: cognitive, MRI/US fusion or in-bore; 2–3 cores per target ± systematic cores.

Prostate biopsy: preparation

- Continue low-dose aspirin; stop warfarin and clopidogrel 5–7 days, DOACs (direct oral anticoagulants) 2–5 days before.
- **Transrectal: antibiotics for all** (contaminated route), ≤24 hours: fluoroquinolone, or cephalosporin ± aminoglycoside. Augmented or swab-directed if risk factors (antibiotics in last 6 months, diabetes, previous infection). Povidone-iodine rectal cleansing.
- Enema optional. **Periprostatic block:** 1–2% lidocaine, 5 mL each side at the junction of prostate base and seminal vesicle.

Prostate biopsy: technique

	Transrectal	Transperineal
Position	Left lateral	Lithotomy
Needle route	Rectal wall (contaminated)	Sterile perineal skin
Antibiotics	Mandatory	Skin-flora only or none
Anterior/apex	Poor	Good
Anaesthesia	Periprostatic block	Perineal and periapical triangle block

- Left lateral position; DRE; 6–10 MHz probe; measure volume (PSA density).
- 18-gauge spring-loaded needle; indent the capsule, withdraw slightly, then fire, so the core samples peripheral zone.
- **12 cores:** base, mid-gland and apex on each side, medial and lateral (sextant alone is inadequate; it missed lateral cancers). Add MRI-targeted cores (cognitive or fusion). TZ (transition zone) cores if gland >50 mL.
- **EAU (European Association of Urology) 2023 prefers transperineal biopsy.**

Complications (transrectal)

Complication	Rate
Haematuria	23–63%
Haematospermia	10–50%
Rectal bleeding	2–22%
Infection	0.1–7%; sepsis 0.3–3%
Retention	0.2–0.4%
Vasovagal reaction	1–5%

- Bleeding is commonest; sepsis is most serious.
- **Sepsis** from fluoroquinolone-resistant *E. coli*: carbapenem until cultures return; prevent with povidone-iodine rectal cleansing plus targeted or augmented prophylaxis.
- Rectal bleeding: probe pressure, balloon tamponade, endoscopy, embolisation.

Recent advances

- **Transperineal biopsy** under local anaesthesia: fewer infections, better apex and anterior sampling; preferred by EAU 2023.
- **MRI-first pathway** (PROMIS, PRECISION, above).
- Colour/power Doppler, contrast-enhanced TRUS, **elastography**, multiparametric ultrasound.
- **Micro-ultrasound (29 MHz; 70 µm resolution)** with PRI-MUS scoring, detection comparable to mpMRI.
- **Also know:** PSMA (prostate-specific membrane antigen) PET.

Diagrams to draw

- 12-core biopsy template on a transverse prostate outline.
- Transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

Key points to remember:

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
- MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
- 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
- Bleeding commonest; sepsis most serious.
- Transperineal now preferred.

Also, from the condensed book (not in the paper answer):

- **Repeat** biopsy follows rising prostate-specific antigen (PSA), atypical small acinar proliferation (ASAP; **≈50%** cancer risk), positive PCA3 or ConfirmMDx, suspicious MRI, or active surveillance (AS).
- National Comprehensive Cancer Network (NCCN): biopsy for abnormal digital rectal examination (DRE) with raised PSA, or repeat PSA >3 ng/mL with PI-RADS ≥3; PSA velocity alone, or abnormal DRE with normal PSA, is not enough. See Ch 154
- PSA density **>0.15 ng/mL/cc** suggests cancer. Volume = π/6 × transverse × anteroposterior × length.

2 · Pre-biopsy mpMRI

SAQ Multiparametric MRI, PI-RADS and fusion biopsy

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Prostate multiparametric MRI and endorectal coil MRI" (Multiparametric MRI : its components and interpretation for prostate cancer.; Endorectal coil MRI) · P4 SAQ "PI-RADS and MRI–ultrasound fusion biopsy"

Introduction

- **mpMRI (multiparametric magnetic resonance imaging):** anatomical plus functional sequences, usually at 3 T without endorectal coil. Reported by **PI-RADS v2.1 (Prostate Imaging Reporting and Data System, 2019)**.
- PI-RADS scores the likelihood of **clinically significant cancer** (Gleason ≥7, volume >0.5 cm^3 or EPE [extraprostatic extension]).

Components

Sequence	Principle	Cancer appearance
T2W (T2-weighted), triplanar	Anatomy; normal PZ (peripheral zone) bright	Low-signal lesion; "erased charcoal" in TZ (transition zone); shows EPE and SVI (seminal vesicle invasion)
DWI (diffusion-weighted imaging) with **ADC (apparent diffusion coefficient)** map; high b-value (≥1400)	Cellularity: cellular tumour restricts water movement	Bright on high-b DWI, **dark on ADC**; ADC falls as Gleason rises

Sequence	Principle	Cancer appearance
DCE (dynamic contrast-enhanced) T1 with gadolinium	Vascularity (angiogenesis): early enhancement and washout	Early focal enhancement (type 3 curve)

- **Biparametric MRI (T2W + DWI):** no contrast, cheaper, similar detection; poorer staging.
- **PI-QUAL (Prostate Imaging Quality):** image quality score (1–5) before reporting.

Interpretation: PI-RADS v2.1

Score	Clinically significant cancer
1	Highly unlikely
2	Unlikely
3	Equivocal
4	Likely; lesion <1.5 cm
5	Highly likely; ≥1.5 cm or definite EPE

Zone	Dominant sequence	Tie-breaker at score 3
Peripheral	**DWI**	DWI 3 + DCE positive → 4
Transition	**T2W**	T2W 3 + DWI 5 → 4; (v2.1) T2W 2 + DWI ≥4 → 3

- **DCE only upgrades PZ 3 to 4**; it never changes 4–5 or TZ scores.
- Cancer in about one-third of PI-RADS 3, over half of 4, 90% of 5.
- **PRECISE** score (1–5) tracks change during active surveillance.

Clinical use and evidence for MRI

- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI pathway with targeted biopsy of PI-RADS ≥3 detected more significant cancer (38% vs 26%); 28% avoided biopsy.
- EAU (European Association of Urology) and AUA (American Urological Association): MRI **before biopsy**; also staging, active surveillance, focal therapy planning.
- Limitations: low specificity (Cochrane 0.37); reader-dependent.

MRI–ultrasound fusion biopsy: principle

- Lesion contoured on MRI → software **co-registers it with live TRUS** in three dimensions; **elastic deformation** corrects probe distortion → needle aimed at the displayed target.
- Freehand electromagnetic or mechanical-arm tracking; transrectal or **transperineal** (fewer infections).

	Cognitive	Software fusion	In-bore MRI
Method	Aims from memory of MRI	Software overlay	Needle under MRI
Drawback	Operator-dependent; small lesion in large gland	Cost; registration error	Time, cost

- Meta-analysis (2022): no significant difference among the three; one RCT (randomised controlled trial) favoured fusion over cognitive.

Fusion biopsy: indications and evidence

- PI-RADS ≥3 lesion with raised PSA (prostate-specific antigen), first or repeat biopsy; active surveillance; focal therapy.
- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023:** 2–3 cores per target ± systematic cores.
- Targeting reduces overdiagnosis; MRI-invisible cancers keep systematic cores relevant.

Endorectal coil MRI: principle

- **ERC (endorectal coil):** a receive-only surface coil inside an inflatable balloon placed in the rectum against the prostate, used with an external pelvic phased-array coil.
- **"The currency of MRI is signal."** A coil close to the prostate gives higher **SNR (signal-to-noise ratio)**, allowing **higher spatial resolution**.

Endorectal coil MRI: technique

- Antispasmodic to reduce rectal motion artefact; balloon filled with air or perfluorocarbon (fluid reduces susceptibility artefact).
- Sequences as for mpMRI: **T2W** triplanar for anatomy, **DWI with ADC** map, **DCE**; ERC was traditionally needed for **MRSI (MR spectroscopic imaging)**.

Endorectal coil MRI: uses

- **Local staging:** extraprostatic extension and seminal vesicle invasion (both low T2 signal) before radical prostatectomy.
- **Local recurrence** after prostatectomy: recurrence isointense on T1, slightly hyperintense on T2; DCE raises accuracy at low PSA.
- **Also know:** focal therapy planning (ERC recommended at 1.5 T); seminal vesicles; female urethral diverticulum.

ERC versus no ERC

	With ERC	**Without ERC (3 T, pelvic coil)**
Signal and resolution	Higher	Adequate at 3 T
csPCa (clinically significant cancer) detection	GG (Grade Group) ≥3 equal; GG2 slightly better	GG ≥3 equal
Comfort, cost, time	Worse	Better
Gland	Deformed (fusion and radiotherapy planning harder)	Natural shape

ERC: disadvantages and contraindications

- Artefacts: **near-field signal flare** next to the coil (needs intensity correction), air susceptibility at the rectal wall, malposition.
- **Contraindications:** absent rectum (abdominoperineal resection), recent rectal surgery, anorectal fistula or inflammatory bowel disease, severe haemorrhoids or stricture, latex allergy.

ERC: current position

- At **3 T the ERC is largely historical**; PI-RADS v2.1 does not require it. **Consider it at 1.5 T** or with older scanners, or when body-coil quality is poor.

Diagrams to draw

- Prostate zones with a PZ lesion dark on ADC.
- MRI target on live TRUS with needle path.
- Sagittal pelvis with inflated endorectal coil against the posterior prostate.

Key points to remember:

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation; cognitive, fusion, in-bore: similar yield.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Also, from the condensed book (not in the paper answer):

- Multiparametric MRI is a **triage test**: with no lesion biopsy may be avoided. A lesion **>1.5 cm** or definite extraprostatic extension scores PI-RADS 5.

Feature	Peripheral zone	Transition zone
Dominant sequence	**DWI**	**T2WI**
Dominant score 3	Positive DCE upgrades to 4	DWI 5 upgrades to 4
v2.1 (2019)	—	T2WI 2 with DWI ≥4 becomes 3

KEY POINT

DCE never creates or downgrades PI-RADS 4–5 and never upgrades a TZ lesion. mpMRI misses a minority of significant cancers (mostly 3+4), so systematic cores remain relevant.

- Targeting is cognitive, software fusion or in-bore. Fusion-targeted cores are ≈3× likelier than systematic ones to find intermediate/high-risk disease.
- Take **2–4 cores per target** (AUA/SUO 2023: 2–3, with or without systematic cores) whenever biopsy is indicated and a PI-RADS ≥3 lesion exists.

3 · Systematic templates and technique

- **Extended 12-core** biopsy is the AUA standard, replacing sextant. Going from 6 to 12 cores raises detection, but 18–21 do not: **geometry, not core number**.
- Add TZ/anterior cores for repeat biopsy or gland >50 mL, and seminal vesicle cores for a palpable abnormality.
- Use an 18-gauge gun, and never reposition with the needle on the rectal wall.

4 · Transrectal vs transperineal (TP)

- TP passes clean perineal skin, causes no rectal bleeding and reaches apex and anterior gland (anterior-only cancers 6–20%).
- European Association of Urology 2023: TP is **preferred** (strong), with non-inferior detection. Give skin-flora cover or none; omitting antibiotics did not change the very low sepsis rate.

- Access is by brachytherapy grid, freehand guide (PrecisionPoint) or two 14-gauge cannulas.

5 · Antibiotic prophylaxis

- Every transrectal patient gets prophylaxis. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin with or without aminoglycoside; or 3rd-generation cephalosporin. Give ≥1 h before, for **≤24 h**.
- Single doses: levofloxacin 500 mg, gentamicin 5 mg/kg, amikacin 15 mg/kg, ceftriaxone 1–2 g. Aztreonam replaces aminoglycoside in renal impairment.
- The FDA 2016 fluoroquinolone warning covers tendon rupture, neuropathy, CNS effects and myasthenia.
- Infection risk factors are antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI, travel and rectal resistant E. coli. These men need augmented or swab-targeted prophylaxis, or TP.
- Endocarditis risk needs IV ampicillin (vancomycin if allergic) and gentamicin. An enema has no proven infection benefit.

6 · Anticoagulation and anaesthesia

Contraindications are coagulopathy, severe immunosuppression, and acute urinary tract infection (UTI) or prostatitis.

Agent	Action
Low-dose aspirin	Continue
Warfarin, clopidogrel	Stop 5–7 days; biopsy when INR <1.5
Direct oral anticoagulants (DOACs)	Stop 2–5 days
Mechanical valve	Bridge with heparin

- Transrectal: periprostatic block, 1–2% lidocaine 5 mL each side at the seminal vesicle–prostate junction (beats intrarectal gel).
- Transperineal: local to the skin, then 5 mL into the **periapical triangle** and 3 mL along the capsule.

7 · Complications and management

Complication	Rate	Management
Haematuria	23–63%	Clot retention (0.7%): catheter, washout
Haematospermia	10–50%	Lasts ≈1 month
Rectal bleeding	2–22%	Ladder (below)
Infection / UTI	0.1–7% / 2–6%	Oral or IV antibiotics
Sepsis	**0.3–3.1%**	Carbapenem until cultures
Vasovagal	1.4–5.3%	Trendelenburg, fluids
Retention	0.2–0.4%	Temporary catheter

ALGORITHM

1. **Post-biopsy fever**: assess for sepsis and take cultures.
2. Give **no empirical fluoroquinolone**, as E. coli is resistant. Start a carbapenem (or amikacin/3rd-generation cephalosporin by sensitivity) and resuscitate.

ALGORITHM

1. **Persistent rectal bleeding**: apply pressure.
2. If it continues, tamponade (sheath or Foley balloon). If still bleeding: endoscopy with adrenaline/sclerosant, then angioembolisation, then transrectal suture.

KEY POINT

Bleeding is the commonest complication overall; infection leads serious ones and admissions (1.0% rising to 4.1% in the resistance era). This explains universal prophylaxis and the move to the perineum.

8 · Repeat and saturation biopsy

SAQ Saturation biopsy of prostate

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Saturation biopsy in CA prostate" · P4 SAQ "Saturation biopsies of prostate"

Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often about 20–30), sampling the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone; the 12-core scheme (adds apical and far-lateral cores) is today's standard.
- Saturation reduces sampling error, mainly for **anterior and apical cancers** missed transrectally (anterior-only cancer 6–20%).

Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation.
- **Focal therapy** mapping (shows multifocality; TRUS [transrectal ultrasound] biopsy misses almost half of disease relevant to hemiablation).

Templates and technique

Template	Division	Cores
Original Barzell	Octants (anterior/posterior × apex/base, each side)	4–8 per octant
Modified Barzell	Apex and base of each side × 5 sectors	20

Template	Division	Cores
Ginsburg	Anterior, mid, posterior sectors per side	4 per sector

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine transrectally).

Results

- **Initial biopsy:** 12 to 18–21 cores did not raise detection.
- **Repeat setting:** early series gained; a later large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS biopsy 48%.

Complications

- More haematuria, haematospermia, pain and **urinary retention** than standard biopsy; anaesthesia and theatre time. Transperineal route has less infection.

Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** detected more significant cancer (38% vs 26%) and 28% avoided biopsy.
- **EAU (European Association of Urology) 2023** prefers the transperineal route.
- Still used for MRI-negative high suspicion and focal therapy planning (mapping).

Diagram to draw

- Transperineal grid over the prostate with sampling sectors.

> **Key points to remember:**
> - Saturation = >12 cores (18–21 or more); 5-mm template mapping.
> - No gain at initial biopsy; used after negatives and for focal therapy.
> - PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Also, from the condensed book (not in the paper answer):

- Micro-ultrasound (29 MHz) approaches mpMRI without contrast (emerging).

ALGORITHM

1. Cancer yield on successive biopsies in ERSPC: **22% → 10% → 5% → 4%**.

- Take a third or fourth biopsy only for high suspicion, using mpMRI with targeting, ConfirmMDx, anterior/TZ cores and the TP route.

9 · Landmark trials

Trial	Design	Result	Take-home
PROMIS (2017)	576: mpMRI, TRUS, 5-mm template mapping (reference)	Sensitivity 93% vs 48%; negative predictive value 89% vs 74%; 27% avoid biopsy	mpMRI better triage

Trial	Design	Result	Take-home
PRECISION (2018)	500 biopsy-naive: MRI with targeted biopsy vs 12-core TRUS	Significant cancer 38% vs 26%; 28% avoided biopsy	MRI pathway superior
MRI-FIRST (2019)	Paired systematic and targeted	Each alone missed cancers	Combine for maximal detection
PREVENT (2024)	TP without antibiotics vs transrectal targeted prophylaxis	Infection not significantly different; detection comparable	Both acceptable

10 · MCQ one-liners

MCQ ONE-LINERS

1. Absent seminal vesicle: ipsilateral renal agenesis in 79%.
2. Commonest false-positive hypoechoic lesion: BPH foci and prostatitis.
3. Müllerian cysts are midline; wolffian cysts are lateral.
4. Significant cancer in PROMIS: Grade Group ≥3 or core length ≥6 mm.

156 Pathology of Prostatic Neoplasia

1 · Precursor and intraductal lesions

P2 *PIN — definition, histology, precursor evidence, LGPIN not reported, isolated HGPIN needs no repeat biopsy, cribriform PIN reclassified*

- ~**20%** of high-grade prostatic intraepithelial neoplasia (HGPIN) carries the **TMPRSS2:ERG** fusion.
- **Intraductal carcinoma of the prostate (IDC-P)** is dense cribriform atypia with **basal cells retained**, usually **retrograde spread of high-grade invasive cancer**. It independently predicts early biochemical recurrence (BCR), and Grade Group (GG) 1 with IDC-P is upgraded at radical prostatectomy (RP) in **>80%**.

Lesion	Defining feature	Action on isolated finding
Low-grade PIN (LGPIN)	Mild atypia; irreproducible	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	No repeat biopsy
PINATYP	HGPIN with adjacent small atypical glands	Manage as atypical
Atypical intraductal proliferation (AIP)	Atypia beyond HGPIN, below IDC-P	Consider re-biopsy (if benign or GG1 elsewhere)
IDC-P	Dense cribriform, basal cells retained	Definitive therapy (or re-biopsy); **excludes active surveillance (AS)**

> **KEY POINT**
>
> HGPIN is a weak risk marker; IDC-P usually marks high-grade cancer already present. Traps: re-biopsy for HGPIN; surveillance for GG1 with IDC-P.

2 · Adenocarcinoma: location, spread, staging

- **Zones (McNeal)**: the peripheral zone (PZ) holds ~70% of cancers, including **85%** of T1c; they are posterolateral and palpable.
- The transition zone (TZ) holds ~20% of cancers, which are anterior, impalpable, under-sampled and lower grade. The central zone holds ~5%, rare but aggressive.
- Tumours are **multifocal in >85%**. With no true capsule, say "**extraprostatic extension**" (EPE); tumour exits via **perineural spaces**, posterolaterally.
- **Seminal vesicle invasion (SVI)** means tumour in the SV muscular wall, usually reached from the base through periseminal tissue.
- EPE is uncommon below **0.5 cm^3**; nodes and SVI are uncommon below **4 cm^3**.

Stage	Definition
T1a / T1b	Incidental at TUR of prostate (TURP)/enucleation; ≤5% / >5% of tissue
T1c	Non-palpable, found on needle biopsy
pT2	Organ-confined at RP
pT2+ (pT2x)	Positive margin at an intraprostatic incision or ill-defined boundary (apex)
pT3a	EPE; **includes microscopic bladder neck invasion**

Stage	Definition
pT3b	SVI
pT4	Other adjacent structures (rectum, levators, external sphincter, pelvic wall)

Trap: microscopic bladder neck invasion is pT3a; T1a/b depend on the 5% cut-off, not grade.

3 · Grading: Gleason and grade groups

SAQ Gleason's score and its significance

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 2, Q3f

Introduction

- **Gleason grading** (Donald Gleason, late 1960s) grades prostate adenocarcinoma by **glandular architecture at low magnification**, not cytology. Patterns run from 1 (most differentiated) to 5. Modified by ISUP (International Society of Urological Pathology) in **2005** and **2014**.

Patterns (modern)

Pattern	Architecture
1–2	Not used on biopsy now
3	Discrete, well-formed glands of any size
4	**Cribriform** (including glomeruloid), poorly formed or fused glands
5	Sheets, cords, single cells, solid nests, **comedonecrosis**

Scoring rules

- **Radical prostatectomy:** primary (commonest) + secondary pattern.
- **Biopsy (2005):** commonest + **highest** pattern, however small. Single pattern is doubled (3+3).
- A tertiary pattern 5 under 5% at prostatectomy is reported as a comment; it adds risk but does not change the grade group.
- Report **% pattern 4** in score 7; report cribriform and intraductal carcinoma (both adverse).

ISUP Grade Groups (2014; WHO [World Health Organization] 2016)

Grade group	Gleason score	5-year BCR (biochemical recurrence)-free after prostatectomy
1	3+3=6	96%
2	3+4=7	88%
3	4+3=7	63%
4	8 (4+4, 3+5, 5+3)	48%
5	9–10	26%

- Grade groups separate 3+4 from 4+3 and 8 from 9–10. Starting at "1" avoids patients misreading the lowest score, 6, as moderate cancer.

Significance

- **Most powerful biopsy predictor** of pathological stage, EPE (extraprostatic extension), nodal spread (0% in GG1, 4.3% in GG3), recurrence and death.
- **Risk groups** (D'Amico, NCCN [National Comprehensive Cancer Network]) combine grade, PSA (prostate-specific antigen) and stage.
- **Treatment choice:** GG1 → active surveillance; selected GG2 with little pattern 4 → surveillance; GG ≥4 → high risk, longer ADT (androgen deprivation therapy) with radiotherapy; also selects for brachytherapy and focal therapy.
- **Limitations:** biopsy undergrades about one-third of GG1 (sampling error); inter-observer variation; after ADT or radiotherapy, grading is unreliable.
- **Also know:** pure GG1 does not metastasise, so "is it cancer?" is debated.

Diagram to draw: Gleason's pattern diagram 3 → 4 → 5.

> **Key points to remember:**
> - Biopsy score = commonest + highest pattern.
> - 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
> - GG4 = score 8; GG5 = 9–10.

Also, from the condensed book (not in the paper answer):

- **Biopsy score = commonest plus highest pattern**. The lowest score in practice is **6**, which has **no nodal metastatic capacity**.

Pattern	Morphology			
3	Discrete well-formed glands			
4	**Cribriform (incl. glomeruloid)**, poorly formed or fused glands			
5	Sheets, cords, single cells, solid nests, **comedonecrosis**			
	GG (International Society of Urological Pathology (ISUP); World Health Organization 2016)	Gleason	Nodes at RP	5-yr BCR-free
1		3+3	0%	**96%**
2		3+4	0.6%	**88%**
3		4+3	4.3%	**63%**
4		4+4, 3+5, 5+3	—	**48%**
5		9–10	—	**26%**

> **KEY POINT**
>
> The steep drop is between GG2 and GG3: never lump them as "7". 3+5 and 5+3 are GG4; GG5 starts at 9. Pure GG1 cannot metastasise, so calling it "cancer" is debated.

- Report **% pattern 4** for GG2–3 (ISUP 2014). GG2 with limited pattern 4 may suit AS.
- **Tertiary pattern 5** (<5% at RP) raises BCR risk but **stays in its GG**, because nodal risk is not raised. On biopsy any higher pattern counts as secondary, however small.

Issue	ISUP 2019	Genitourinary Pathology Society (GUPS) 2019
% pattern 4 in GG2–3; cribriform	Report	Report
IDC-P with invasive cancer	**Incorporated into grade**	**Not graded**; comment
Targeted cores of one lesion	**Aggregate grade** (plus per-core)	Each core; aggregate optional

> **KEY POINT**
>
> Cribriform pattern and IDC-P mark aggressive disease and argue against AS even in GG2, so they must be reported. Aggregate grading of a targeted lesion avoids under- or over-grading.

Upgrading setting	Risk
Biopsy GG1, then RP	~⅓ upgraded
Biopsy GG2, then RP	~20% become GG3
GG1 with negative mpMRI (PI-RADS 1–2)	Still ~20% at 6 yr

Reclassification within 3 yr usually means missed disease rather than progression.

4 · Needle biopsy assessment

- Use **separate jars per site**. Pitfalls: **at the base, SV/central zone mimic HGPIN; at the apex, Cowper gland mimics cancer**.
- Report the **number of positive cores plus one other extent measure**, and report **perineural invasion** on biopsy (it raises EPE risk).
- Basal cell markers, **high-molecular-weight cytokeratin (HMWCK) and p63**, stain benign glands, so their **absence supports cancer**. **AMACR** is an adjunct only.
- **Adenosis**, in the TZ, is the chief mimic of low-grade cancer and carries **no increased cancer risk**.
- **Atypical small acinar proliferation (ASAP)**: cancer follows in ~**40%**. Get **expert review before re-biopsy**, then re-biopsy the atypical site.

> **KEY POINT**
>
> Sampling error is one-directional: a bad biopsy is believed, a good one is provisional, and an atypical one is re-examined before it is repeated.

5 · TUR and RP specimens

- At TUR, **grade each major nodule**. Nodes are uncommon with Gleason <8.
- **EPE**: focal vs nonfocal extent predicts progression. **SVI** carries **65%** 5-yr progression.
- **Margins** are positive in ~15% (robotic), but only ~50% progress. Report **extent (focal vs nonfocal, 3 mm) and grade at the margin**.
- Independent predictors of BCR are grade, EPE extent, margins, SVI and nodes. **Tumour volume is not independent**.

6 · Treatment effect

Treatment	Histological effect	Gleason score?
Hormone therapy	Grade looks artefactually higher	**No** where effect is present
Finasteride	No consistent change	Yes
Radiotherapy	Radiation atypia (up to 72 months)	Only if no treatment effect
Cryotherapy / high-intensity focused ultrasound (HIFU)	Infarction; residual cancer unaltered	**Yes**

After radiotherapy, cancer with treatment effect has a prognosis close to no cancer.

7 · Subtypes and other tumours

Subtype	Grading	Behaviour
Ductal (pure 0.4–0.8%; mixed ~5%)	**Pattern 4** (PIN-like 3; comedonecrosis 5)	Exophytic at the verumontanum; PSA may be normal; behaves like Gleason 8; **visceral recurrence at low PSA**
Mucinous	As usual	As acinar
Neuroendocrine differentiation (~half of acinar)	As usual	Not prognostic
Small cell (~50% mixed)	**Not graded**	Survival <1 yr
Squamous	—	**Osteolytic** metastases; hormone-unresponsive
Sarcomatoid	—	Dismal

- **Urothelial carcinoma in the prostate**: suburethral invasion is pT1 and stromal pT2, but **transmural spread from the bladder is pT4**. Urothelial cancer stains **GATA3**; prostate stains PSA and NKX3.1.
- Sarcomas are **rhabdomyosarcoma in children and leiomyosarcoma in adults**.

157 Diagnosis and Staging of Prostate Cancer

1 · Clinical presentation

- Early cancer is **asymptomatic**; symptoms imply advanced disease.
- **Locally advanced** disease causes voiding or ureteric obstruction and haematuria; **metastatic** disease causes bone pain, fracture, leg oedema and cord compression.

2 · Screening in brief

*See the paper answer **PSA screening for prostate cancer: evidence, problems and solutions** (Section 17, Ch 153).*

3 · DRE and PSA

- **Digital rectal examination (DRE)** has low screening value at PSA <3, but an abnormal DRE raises the risk of Grade Group (GG) 4–5 at every PSA level.
- **Prostate-specific antigen (PSA)** is organ-specific, not cancer-specific, and rises ≈4% per mL of volume.
- PSA is lowered by hypogonadism, obesity, statins and 5α-reductase inhibitors (5-ARIs). On a 5-ARI, **double PSA** after ≥12 months, or treat the nadir as the new baseline.
- Risk is **continuous**: the Prostate Cancer Prevention Trial (PCPT) found cancer in 15% of men with PSA ≤4.
- Thresholds are 4 or 2.5–3 (ERSPC used 3). Repeat a raised PSA before biopsy, and give no empirical antibiotics without urinary symptoms.

PSA (ng/mL)	Yield, normal DRE	Yield, suspicious DRE	High-grade (normal / suspicious)
0–1	8.8%	—	0.9%
4–10	17–32%	45–51%	4.1% / 11.7%
>10	43–65%	70–90%	19.4% / 50.5%

- Baseline PSA in the 40s predicts metastasis and death decades later (Malmö).
- Derivatives: PSA density ≥0.15; velocity >0.75 ng/mL/yr; free PSA ≤25% (at PSA 4–10). See Ch 154, 155

4 · Staging: TNM (AJCC 8th, 2017)

Pathological stage is more accurate and prognostic than clinical stage.

Category	Definition
T1a / T1b	Incidental at transurethral resection (TUR): ≤5% / >5% of tissue
T1c	Impalpable, found on needle biopsy
T2a / T2b / T2c	Palpable, confined: ≤½ of one side / >½ of one side / both sides
T3a / T3b	Extraprostatic extension (EPE) / seminal vesicle invasion (SVI)
T4	Fixed, or invading sphincter, rectum, bladder, levators or pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites with or without bone

AJCC prognostic groups add PSA and GG.

> **KEY POINT**
>
> T1 is defined by how the cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b; non-regional nodes are M1a; a biopsy-found impalpable cancer is T1c even if bilateral; clinical T is set by DRE, not imaging.

5 · Predicting extent and risk

- **PSA** cannot stage an individual. Organ-confined disease is ≈80% at PSA <4 and <50% above 10; nodes are positive in **75% at >50**.
- **Biopsy grade** is the most important biopsy finding, and tertiary pattern 5 on biopsy is recorded as the secondary pattern. See Ch 156

D'Amico	Stage	PSA	Gleason	10-yr disease-free after RP
Low (all criteria)	T1–T2a	≤10	≤6	83%
Intermediate (any one)	T2b	>10–20	7	46%
High (any one)	T2c	>20	8–10	29%

National Comprehensive Cancer Network (NCCN) group	Criteria
Very low	T1c, GG1, PSA <10, PSA density <0.15, <3 positive cores, ≤50% of any core
Low	T1–T2a, GG1, PSA <10
Favourable intermediate	One intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	≥2 intermediate factors, or GG3, or ≥50% cores positive
High	T3a, or GG4–5, or PSA >20
Very high	T3b–T4, primary pattern 5, >4 cores GG4–5, or ≥2 high-risk features

- CAPRA and **nomograms** predict better than groups for individuals; nodal nomograms decide pelvic lymph node dissection (PLND). Tissue genomic tests refine risk but await prospective validation.

6 · Staging imaging and nodes

Risk	Imaging
Low / intermediate, asymptomatic	No routine computed tomography (CT) or bone scan
High	Bone scan with CT or mpMRI; consider PSMA PET

- **Multiparametric MRI** is used for local staging (EPE, SVI), treatment planning and surveillance selection.
- **PSMA PET** (68Ga-PSMA-11, piflufolastat F-18) is FDA-approved for high-risk staging and is more sensitive and specific for nodes and metastases.
- **PLND** has no consistent oncological benefit; its value is **staging**. Do it when the nomogram nodal risk exceeds the guideline threshold.

KEY POINT

Imaging yield follows pre-test probability: in low-risk men most positive scans are false positives, leading to more tests, anxiety and wrongly denied cure.

7 · Landmark trials

Trial	Design	Result	Take-home
ERSPC	European screening randomised trial, from 1993	**20% relative reduction** in cancer death at 16 y	Saves lives, at the cost of overdiagnosis
PLCO	US screening randomised trial, from 1993	No mortality difference	Contaminated control arm
PCPT	Placebo arm biopsied regardless of PSA	15% cancer at PSA ≤4	No safe PSA level
proPSMA (2020)	PSMA PET-CT vs CT and bone scan, high-risk	Accuracy ≈92% vs 65%; less radiation	PSMA PET is the better first-line staging test

8 · MCQ one-liners

MCQ ONE-LINERS

1. Finasteride 1 mg lowers PSA as much as 5 mg.

158 Active Management Strategies for Localised Prostate Cancer

1 · The modern question: timing as well as modality

- In 15-year randomised data on favourable risk, surveillance gave **more metastasis but no difference in cancer mortality**.
- **Spectrum**: a man likely to die of competing risks gets **watchful waiting (WW)**; good longevity with unclear benefit means **active surveillance (AS)**; a man who needs therapy gets radical prostatectomy (RP), external-beam radiation therapy (EBRT), brachytherapy, or whole/partial-gland ablation.

> **KEY POINT**
>
> Cancer mortality is always weighed against competing mortality. A competing risk argues against treatment only when it will act before the cancer does.

2 · Host evaluation

- NCCN estimates life expectancy from Social Security Administration or WHO tables.
- In the Prostate Cancer Outcomes Study (PCOS), 10-year prostate cancer mortality was **3–18%** by risk, but other-cause mortality was **≥33%** with any comorbidity.
- The blunt **"10-year life expectancy" rule** undertreats older and overtreats younger men, so individualise.
- The **Charlson index** (1-year inpatient mortality) is reweighted as **PCCI** for 10-year mortality; NNT rises steeply with Charlson score.
- The automated **Care Assessment Needs (CAN) score** (Veterans Health Administration) best predicts 5-year survival MCQ.

Baseline function (IPSS, UCLA Prostate Cancer Index)

Baseline state	Favours	Reason
Obstructive lower urinary tract symptoms (LUTS), large gland	Surgery	RP relieves obstruction; brachytherapy increases retention
Severe irritative LUTS	Surgery	Worst after radiation
Gland >60 cm^3	Surgery (or neoadjuvant androgen-deprivation therapy, ADT)	Pubic arch interference; if >80 g, ADT (shrinks 25–40%) before radiotherapy (RT)
Chronic retention / neurogenic bladder	Surgery	RT causes bladder fibrosis; if RT is chosen, switch to intermittent self-catheterisation
Prior transurethral resection (TUR) defect	Surgery	Brachytherapy dosimetry is hard and incontinence increases. Do TURP before RT and delay EBRT **6–8 weeks**
Inflammatory bowel disease (IBD), bowel dysfunction, prior rectal surgery	Surgery	RT bowel toxicity

Baseline state	Favours	Reason
Poor sexual function	—	Recovery depends on age, baseline erection, PDE5 inhibitor use and vascular risk; poor baseline function means non-nerve-sparing surgery is used more.

Modality	Absolute contraindication	Relative contraindication
Radiotherapy	Previous radical pelvic irradiation	Bowel in field (prior surgery, hernia); uncontrolled diabetes; IBD; bowel dysfunction
Surgery	—	Previous pelvic RT; chronic steroids; severe cardiac/pulmonary/haematological comorbidity

3 · Risk stratification

NCCN group	Criteria	Staging imaging
Very low	**All**: T1c; grade group (GG) 1; prostate-specific antigen (PSA) <10; <3 positive cores, ≤50% each; PSA density <**0.15**	Not indicated
Low	**All**: T1–T2a; GG1; PSA <10	Not indicated
Favourable intermediate	Intermediate-risk factors (IRFs): T2b–c, GG2, PSA 10–20. **All**: 1 IRF; GG1–2; <50% cores positive	
Unfavourable intermediate	**Any**: 2–3 IRFs; GG3; ≥50% cores positive	Bone scan and CT/MRI, plus chest imaging; prostate-specific membrane antigen (PSMA) PET may replace these (NCCN 2023)
High	**Any**: T3a; GG4–5; PSA >20	
Very high	**Any**: T3b–T4; primary pattern 5; >4 cores GG4–5	

> **KEY POINT**
>
> Favourable groups need ALL criteria; adverse groups ANY one. PSA density appears only in very low risk; GG3 alone makes intermediate unfavourable. AUA/ASTRO/SUO use very low/low/intermediate/high without substratification.

- **Epstein criteria** (insignificant cancer): PSA density <0.15, Gleason ≤6, <3 positive cores, ≤50% of any core.
- Nomograms: **Partin** predicts pathological stage (accurate 72.4%); **Kattan** predicts relapse after RP; with **UCSF-CAPRA**, risk jumps at score ≥6.
- **proPSMA** (302 high-risk men): PSMA PET had accuracy **92% vs 65%** and sensitivity 85% vs 38% over conventional imaging, but a negative scan cannot exclude small nodes.
- Genomic tests are not routine (AUA 2022); use them only where they change an AS decision.

4 · Observation vs active surveillance

SAQ Expectant treatment of CA prostate

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3d

Rationale

- Overdiagnosis: autopsy prevalence of microfocal cancer roughly equals age as a percentage; lifetime risk of death is about 3%.
- **Grade group 1 (Gleason 6) has virtually no metastatic potential.** Deaths on observation come from higher grade missed at biopsy (about 30% upgraded at prostatectomy).

Two forms of expectant treatment

	AS (active surveillance)	**WW (watchful waiting)**
Patient	Fit for radical therapy, life expectancy ≥10 years	Limited life expectancy (age, comorbidity)
Intent	**Curative, deferred**	**Palliative**
Monitoring	PSA (prostate-specific antigen), DRE (digital rectal examination), MRI, biopsy	Infrequent PSA and review; no biopsy
Action	Radical treatment on reclassification	ADT (androgen deprivation therapy) or palliation for symptoms

Evidence

Trial	**Result**
SPCG-4 (pre-PSA)	Prostatectomy cut cancer death (19.6% vs 31.3%), mainly <65 years and higher grade
PIVOT	No significant mortality benefit; low-risk cancer death 2.7%
ProtecT (15 years)	Cancer death <3% in all arms; metastasis doubled on active monitoring (9.4% vs about 5%)

Selection for AS

- **Very low risk (Epstein):** grade group 1, ≤2 positive cores, no core >50%, PSA density <0.15.
- **Low risk:** grade group 1, PSA <10 ng/mL; all guidelines prefer AS.
- **Favourable intermediate risk** (small-volume grade group 2): selected cases.
- Caution: **BRCA2** carriers, high-volume disease.

AS protocol

- MRI and **confirmatory biopsy** within 1–3 years; PSA 6-monthly; DRE every 1–2 years; MRI every 18–24 months; biopsy every 3–5 years.
- **Triggers:** grade group ≥3 → treat; volume or PSA rise → MRI or biopsy first.

Outcomes

- Cancer mortality **0.5–5% at 15 years** by entry criteria (Johns Hopkins vs Toronto); about **25% treated by 5 years**.

Watchful waiting

- Switch AS to WW when life expectancy falls below about 5–10 years.

- **Also know:** genomic tests refine selection; 5ARI (5α-reductase inhibitor) has no proven role.

Diagram to draw: AS flowchart: diagnosis → MRI + confirmatory biopsy → surveillance → trigger → treatment.

Key points to remember:

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Also, from the condensed book (not in the paper answer):

- AS is **preferred for very low and most low risk** (AUA, ASTRO, SUO, NCCN); selected low-volume GG2 is acceptable.
- Limitations are **25–45%** occult high grade on systematic biopsy (mostly GG2), declining adherence and no agreed triggers. PSA density >0.15 predicts reclassification.

5 · Comparative oncological evidence

- Population studies favour RP (Swedish subdistribution HR ≈1.8; PCOS disease-specific HR 0.35, curves apart by 3 years), but this is largely **selection bias**.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RP vs WW; 695; pre-PSA; ~90% palpable	23.2 yr: cancer deaths 63 vs 99, relative risk (RR) **0.56**; absolute 11%; NNT **8**; most benefit <65 (RR 0.45), intermediate risk	Benefit in younger men with intermediate risk
PIVOT	RP vs observation; 731 (2000 planned); PSA era; older, comorbid	10 yr: no difference; benefit if PSA >10. 19.5 yr: all-cause ARR 5.5% and cancer 4.0% (both not significant); significant only in **intermediate risk** (ARR 14.5%). Bone metastases 4.7% vs 10.6%	No benefit in low risk; underpowered
ProtecT	1643 screen-detected, 50–69; monitoring vs RP vs RT with 3–6 mo ADT; 77% low risk	15-yr cancer mortality **2.7% vs 2.2% vs 2.9%** (P = 0.53); more metastasis with monitoring; 60% of monitored treated by 15 yr	AS safe in favourable risk; RP equals RT

KEY POINT

Competing mortality and low-risk enrolment dilute trial benefit, which appears only in younger intermediate-risk men with long follow-up.

7 · Functional outcomes compared

Cancer control is similar, so side effects usually decide.

Domain (vs AS; Chen, JAMA 2017)	RP	EBRT	Brachytherapy
Sexual	**Worst**; persists at 24 mo	Mild early decline; near AS by 24 mo	Mild early decline
Incontinence	**Worst**; persists at 24 mo	No change	No change
Obstructive/irritative	Worse at 3 mo, then **better than AS**	Worse at 3 mo; resolves by 12	**Worst acute**; resolves by 24 mo
Bowel	No change	**Worse**; persists at 24 mo	No change

- **PCOS** found differences at 2 and 5 years but **none at 15 years**. **ProtecT** found rising bloody stools after RT, with no difference in anxiety, depression or quality of life.
- In **CEASAR**, the erectile disadvantage of RP was confined to the top quartile of baseline function. In open vs robotic RP, the surgeon matters more than the approach.

> **KEY POINT**
>
> Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow decline. Curves converge by 15 years.

8 · Shared decision-making

- No treatment is clearly superior, so **patient preference carries unusual weight** (AUA, ASTRO, SUO).
- The **urologist's recommendation is the most influential factor**, so beware your own bias. **Longevity assessment may be the most important part of counselling**.

9 · Stepwise decision algorithm

ALGORITHM

1. **Treat or not?** Weigh disease and host factors. If longevity is short, then WW; if very low/low or selected favourable intermediate, then AS.
2. **Partial or whole gland?** Weigh focality, size, grade, potency, LUTS. If limited-focality intermediate risk, then focal ablation, ideally in a trial.
3. **Surgery, radiation or whole-gland ablation?** Compare outcomes, toxicities, baseline function and risk of secondary treatment.
4. **Delivery:** RP approach and node dissection, or EBRT vs brachytherapy, fractionation and ADT.

Risk / longevity	Usual recommendation
Limited longevity, low lethal potential	WW; ADT at symptoms
Very low	AS (best option)
Low	AS preferred for most; all therapies are options
Favourable intermediate	RP or RT; AS for select (low-volume GG2); focal therapy in trials
Unfavourable intermediate	RP, or RT with ADT; AS strongly discouraged

Risk / longevity	Usual recommendation
High / very high	RP, or RT with ADT; consider even with competing risks if the cancer threatens within 10 yr; no focal therapy

- **Surgery:** early incontinence and erectile dysfunction (ED) are the norm, so quote your own outcomes; LUTS may improve.
- **Radiation:** it spares short-term function but causes yearly decline; secondary rectal/bladder cancer at 15–20 years is controversial.
- **Whole-gland ablation:** long-term data are scant and fibrosis hampers salvage. Cryoablation almost always causes ED; HIFU preserves function better, but retreatment is **29–38%**.
- EBRT, brachytherapy, SBRT and protons give comparable 10-year cancer control.

159 Active Surveillance of Prostate Cancer

1 · Rationale: overdiagnosis

- **Latent reservoir**: autopsy cancer prevalence roughly equals age as a percentage. PSA testing and biopsy detect cancer in 15–25%, yet lifetime risk of cancer death is only **3%**; overdiagnosis is 87–94%.
- Active surveillance (AS) rests on **two premises**: progression takes a long time, and cure is still possible at the first sign of higher risk.

2 · AS versus watchful waiting

*See the paper answer **Expectant treatment of CA prostate** (Section 17, Ch 158).*

Also, from the condensed book (not in the paper answer):

- The switch to watchful waiting (WW) is typically near age 80.
- **Pitfall**: elderly men with unrecognised hypogonadism may progress (neuroendocrine dedifferentiation) without a PSA rise, so measure testosterone and get interval imaging.

> **KEY POINT**
>
> AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them.

3 · Biology of grade group 1

- **Pattern 3 is molecularly near-normal.** Patterns 4–5 show EGFR, HER2 and BCL2 changes and **PTEN deletion in 70–90%**; **MYC amplification with PTEN deletion** drives metastasis.
- **Metastatic potential of pure grade group (GG) 1 is effectively zero**, though occasional extraprostatic extension occurs.
- True grade progression is ~**1–2% per year**, mostly in high-volume Gleason 6, so follow-up must be long term.
- Deaths in low-risk series come from **occult higher grade** missed at biopsy, present in ~**30%** after systematic biopsy (85% of it 3+4).

Hazard	Misattribution of grade	True progression
Frequency	**Commonest (25–30%)**	1–2.3%/year
Usual result	Mostly GG2	Many 3+4 stay indolent
Remedy	MRI, targeted biopsy, predictors	Long-term surveillance

> **KEY POINT**
>
> The main hazard of AS is **sampling error at entry**, not biology over time. First job: get the entry grade right.

4 · Patient selection

Most GG1 should be on AS (AUA, EAU, ASCO, NICE and Cancer Care Ontario agree).

	Very low risk (Epstein)	Low risk
Definition	GG1; **≤2 positive cores; no core >50%; PSA density <0.15**	Any GG1, PSA <10
Progression to treatment	Lower	Higher
Metastasis	Exceptionally low	Exceptionally low

High-volume GG1 is still a candidate, but check it with multiparametric MRI (mpMRI) and targeted biopsy, re-biopsy or a genomic assay.

- **Toronto** is inclusive (all Gleason 6 and selected intermediate), with 15-yr cancer mortality of **5%** actuarial; **Johns Hopkins** is restrictive (Epstein very low), with **0.5%**. So **inclusion criteria set mortality, while intervention stays constant (~25% at 5 yr)**.
- In Toronto, baseline Gleason 7 had ≥20% metastasis at 15 yr (HR 3.75). In PRIAS (n=5302), cancer mortality was <1% at 10 yr, and cardiovascular disease was the commonest cause of death.
- **Favourable intermediate**: offer AS selectively for small-volume GG2. Only NICE lists AS and radical treatment as equal for intermediate risk.
- Exclude GG ≥3; intraductal or cribriform histology raises concern.
- **Black men** are AS candidates, perhaps with closer surveillance. **Family history** rarely justifies treatment, except metastatic disease <60 yr in a first-degree relative, which calls for germline testing.
- **BRCA2** carries the highest upgrading risk; consider definitive therapy even for low grade, or AS only with very close follow-up.
- Risk factors for higher grade are **PSA density >0.15**, Black race and biopsy volume; restricted diffusion on MRI is a powerful predictor.

5 · Monitoring protocol

ALGORITHM

1. At diagnosis, obtain mpMRI.
2. Do a **confirmation biopsy** (systematic + targeted) within 1–3 yr.
3. Check PSA 6-monthly, DRE every 1–2 yr and MRI ≤ every 18–24 months.
4. Repeat biopsy every 3–5 yr while stable.
5. If a trigger appears, then give definitive therapy; if life expectancy falls, then switch to WW.

- Confirmation biopsy may be omitted if the first biopsy was MRI-targeted. **Compliance is paramount**: compliers had better cancer mortality and metastasis-free survival.

Guideline	Intermediate risk	Schedule / MRI
Cancer Care Ontario	Treat; AS if selected	PSA 3–6-monthly; DRE yearly; biopsy within 6–12 mo then 3–5-yearly; MRI if discordant
AUA/ASTRO 2022	AS if selected	PSA 6–12-monthly; DRE 1–2-yearly; biopsy 1–4-yearly; MRI before initial or confirmation biopsy
NICE 2019	Radical treatment or AS	PSA 3–4-monthly in year 1, then 6-monthly; **MRI at enrolment**

Prompts definitive therapy	Prompts evaluation only
GG ≥3 (unless life expectancy significantly shortened)	More Gleason 6 volume: get MRI or a biomarker
Traditionally any Gleason ≥7 (now relaxed)	Small-volume GG2: may continue AS
	PSA rise alone: repeat MRI or biopsy

KEY POINT

Grade triggers treatment; volume and PSA trigger investigation. 10–15% leave AS by choice (anxiety) without a trigger.

6 · MRI in AS

- **At entry the evidence is strong**: mpMRI before confirmation biopsy, with **systematic and targeted cores**, upgrades about ⅓. NICE recommends it; NCCN lists it as an option.
- **Serial MRI alone is weak** (NPV ~0.8–0.9 for GG ≥2), so it must never be the only way to exclude progression. MRI-invisible cancers behave better.

7 · Biomarkers and genomics

- Decipher predicts upgrading in GG1, not GG2; Oncotype DX evidence is mixed and Prolaris has none specific to AS.
- **Bayesian problem**: with a 3% prior metastasis risk, a 90%-specific test labels 10% "significant", so test only higher-risk candidates (high volume, high PSA density, GG2).
- Use the tests to set surveillance intensity, **never as the sole basis for AS**; a favourable score does not override adverse features.

8 · Lifestyle, drugs and psychology

- **Stop smoking**, exercise vigorously (ERASE: lowered PSA), keep a healthy weight and eat a vegetable-based diet.
- Only vitamin D 1000–2000 IU/day is reasonable. **Selenium and vitamin E are contraindicated** (SELECT).
- 5α-reductase inhibitors are not used to prevent progression (REDEEM); metformin, apalutamide and enzalutamide are investigational.

9 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	695, pre-PSA, palpable; RP vs WW; 23 yr	Cancer death 19.6% vs 31.3% (relative risk 0.55)	Benefit at age <65 and GG ≥3; none in GG1–2
PIVOT	731 US veterans; RP vs observation; 18.6 yr	All-cause death 68% vs 73% (not significant); low-risk cancer death 2.7%	Modest benefit in intermediate risk
ProtecT	1643 PSA-detected; monitoring vs RP vs radiotherapy; 15 yr	Cancer mortality <3%, no difference; **metastases 9.4% vs 4.5% vs 5.0%**; 61% of monitoring arm treated	Active monitoring is not AS (no scheduled biopsy, >⅓ intermediate/high)

Trial	Design	Result	Take-home
ASIST	GG1 on AS; MRI-targeted + systematic vs systematic confirmation biopsy	Upgrading 33% vs 27% (not significant); ~50% less AS failure at 2 yr	MRI benefit shows over follow-up
REDEEM	Dutasteride vs placebo on AS	38% less "progression" (less GG1 volume), no upgrading difference	Cytoreduction, not disease modification

160 Open Radical Prostatectomy

1 · Principles

- Radical prostatectomy (RP) alone has been shown in a randomised controlled trial (RCT) to reduce metastasis and cancer death (**SPCG-4**). Most RPs are now robotic, without proven benefit.
- **Goals, in order**: cancer control, then avoiding technical complications, then continence, then potency.

> **KEY POINT**
>
> The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.

2 · Surgical anatomy

P1 P4 *Anatomy of prostate and neurovascular bundles; Prostate anatomy, newer surgical concepts and NVB preservation — fascial layers table, NVB of Walsh, inferior vesical urethral/capsular branches, accessory pudendals, Santorini plexus, nerve-sparing planes*

- Inferior vesical urethral branches enter at the bladder neck (**5 and 7 o'clock** pedicles); capsular branches mark the cavernous nerves.
- The pelvic plexus combines parasympathetic **S2–S4** and hypogastric sympathetic fibres. It lies beside the rectum, **5–11 cm** from the anal verge, centred on the seminal vesicle tip.
- The **striated sphincter** is a vertical tubular sheath, not a flat diaphragm, so apical dissection decides continence; it and the levator ani are both mainly **pudendal**.

> **KEY POINT**
>
> Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist.

3 · Selection and preoperative care

- **Harder, not contraindications**: prior pelvic surgery or radiotherapy, transurethral resection of the prostate (TURP), hernia mesh, large gland.
- **Defer surgery 6 weeks after biopsy and 12 weeks after TURP**, so haematoma settles.
- Inguinal hernia affects ~15% before and up to 20% after RP, so examine standing and repair it preperitoneally at RP.
- Give cephalosporin prophylaxis; the **fibreoptic headlight** is the most useful instrument.

4 · Retropubic RP — operative steps

ALGORITHM

1. Lower midline extraperitoneal incision; pelvic lymph node dissection (PLND).
2. Open endopelvic fascia; divide puboprostatic ligaments.
3. Control and divide the DVC, then the urethra and posterior sphincter.
4. Release the NVBs; divide pedicles, bladder neck and seminal vesicles.
5. Reconstruct the bladder neck and make the anastomosis.

- **PLND** is for intermediate/high risk, omitted for Grade Group 1–2 low-volume disease.
- **Endopelvic fascia**: open at the sidewall reflection, because too medial causes Santorini bleeding. Divide the **puboprostatic ligaments** but spare the **pubourethral** part, which anchors the sphincter.

DVC control

Feature	Stapler (preferred)	Suture ligation
Method	Vascular load; fire only if the catheter moves freely	3-0 figure-of-eight to pubic perichondrium, plus a back-bleeding stitch
Advantage	~200–400 mL less blood loss; less traction	Recreates puboprostatic support

- The **anterior apex is the commonest positive margin site**, so get absolute venous control before touching it.
- **Urethra**: transect close to the apex and place six 3-0 sutures through **mucosa and submucosa only**, because muscle bites delay continence.

Nerve sparing

- The **interfascial** plane (levator vs prostatic fascia) is correct; Walsh never uses intrafascial because of margin risk. Wide (extrafascial) excision is for induration, a fixed bundle or doubtful margin.
- Both NVBs are usually kept, because extracapsular extension rarely exceeds 1–2 mm. Decide in theatre: **if induration or a fixed bundle, excise that side**.
- **Technique**: release the prostate from the bundle, not the bundle from the prostate, starting **at the apex**. Use small clips and **never thermal energy**.
- **High anterior release** (before urethral division) means less traction and faster recovery.
- Avoid it if ≥2 **Tsuzuki criteria** are present: PSA >10, Gleason >6, >20% core involvement, >33% cores positive, perineural invasion, abnormal DRE.
- For **wide excision**, clamp **medial to lateral** on the rectum, since the reverse risks rectal injury.

Pedicles, bladder neck, reconstruction

- **Complete seminal vesicle excision is essential**; the pelvic plexus lies on their lateral surface.
- Close the bladder neck as a **tennis racquet** with mucosal eversion, and make a tension-free anastomosis; a peritoneal window lets haematoma decompress.

> **KEY POINT**
>
> Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB.

5 · Perineal RP (brief)

- Perineal RP reflects the fasciae off the prostate and spares the DVC, so it bleeds less, but PLND needs a separate approach.

6 · Intraoperative complications

Complication	Setting	Management
Haemorrhage (commonest, venous)	Endopelvic fascia, DVC	**Divide DVC completely and oversew** (best measure)
Obturator nerve	PLND	Reanastomose with fine non-absorbable suture
Rectal injury	Small prostate; apical/posterior plane	Two-layer repair and omentum; colostomy if prior radiotherapy
Ureteric injury	Dissecting in trigone	Reimplantation

- Typical loss is 300–1000 mL; partial DVC division bleeds more.
- **Rectal repair**: finish the prostatectomy, dilate the anus, close in 2 layers and place omentum between rectum and anastomosis to prevent a rectourethral fistula.

7 · Postoperative care

- Discharge on day 1–2. Give ketorolac only if creatinine <1.2 mg/dL and there is no haematuria.
- Keep the **catheter ~1 week**, because day-4 removal causes frequent retention; never remove it with significant extravasation.

8 · Postoperative complications

- Mortality is **0.2%**. **Delayed haemorrhage** needs **early exploration and evacuation**, which shortens stay.
- **Thromboembolism** is the major cause of death. It is highest with PLND and peaks at **14–28 days**. Use compression and early ambulation; treat with low-molecular-weight heparin (LMWH).
- **Bladder neck contracture** appears at ~3 months. Treat with dilatation, then cold-knife incision at 3, 6 and 9 o'clock with self-catheterisation and triamcinolone; reconstruction with an artificial sphincter is rarely needed.

*See the paper answer **Post-prostatectomy incontinence** (Section 8, Ch 123).*

Also, from the condensed book (not in the paper answer):

- Continence means **0–1 pad/24 h**. Prevent incontinence with a preserved apical sphincter, a precise tension-free mucosal anastomosis and a small supple neck.
- **Erectile dysfunction (ED)**: recovery is predicted by **age <65, preoperative potency, no diabetes and bilateral nerve sparing**, and continues up to 4 years. PDE5 inhibitor rehabilitation helps; nightly is not better than on-demand.

> **KEY POINT**
>
> Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs. Hence early re-exploration, clips near the bundles and a peritoneal window.

9 · Outcomes

Time	Continent	Potent (Walsh)
3 months	80%	38%
12 months	93%	73%
18–24 months	97%	86%

- These are single-expert figures; elsewhere only a third regained baseline potency by 2 years. RP improves lower urinary tract symptoms for up to 15 years.

10 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RCT, RP vs observation	Less metastasis and cancer death	RP cuts cancer death
Padma-Nathan	RCT, nightly sildenafil vs placebo	Better erections at 1 year	Rehabilitation works
REINVENT	RCT, vardenafil on-demand vs nightly	On-demand better	Daily dosing unproven

161 Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

1 · Background

SAQ Robotic radical prostatectomy

Paper answer · P1

Type: SAQ (10 marks)
Source: 2023-Winter Paper 1, Q3b

Introduction

- RARP (robot-assisted radical prostatectomy): Walsh's anatomical nerve-sparing principles done with **3D, 10× magnified vision and wristed instruments**. The robot removed the laparoscopic suturing barrier; da Vinci from 2000 (S, Si, Xi, SP).

Indications and contraindications

- Localised cancer (cT1–T2, selected cT3); life expectancy over 10 years.
- **Absolute:** uncorrectable bleeding diathesis; unfit for general anaesthesia.
- **Relative:** morbid obesity, prostate >80 g, median lobe, prior TURP (transurethral resection of the prostate), salvage surgery.

Set-up

- Low lithotomy, arms tucked, **steep Trendelenburg (22–25°)**, padding; pressure 12–15 mmHg; camera port 15–17 cm above pubis, three robotic and two assistant ports.

Steps (transperitoneal anterior approach)

1. Drop bladder; enter space of Retzius.
2. Incise endopelvic fascia; ligate DVC (dorsal venous complex).
3. Divide bladder neck (fat line, balloon traction); check for median lobe.
4. Seminal vesicles and vasa; incise Denonvilliers fascia.
5. Clip pedicles athermally; **interfascial nerve sparing** (extrafascial if high risk).
6. Apical dissection preserving urethral length.
7. Extended PLND (pelvic lymph node dissection) if indicated.
8. Posterior (Rocco) reconstruction; **Van Velthoven running anastomosis**.

Postoperative care

- ERAS (enhanced recovery after surgery); discharge within 24 h; catheter about 1 week (not before day 5: retention); pelvic floor training; penile rehabilitation with PDE5 (phosphodiesterase type 5) inhibitors.

Outcomes

- **Yaxley 2016** randomised trial: equal function and early margins versus open; less blood loss (300 versus 1400 mL) and shorter stay.
- Margins about 15% (9% if pT2); 1-year continence mid-90s% in expert hands.

Complications

- Transfusion 1–2%; bladder neck stricture <2% (open up to 16%); rectal injury; ureteric injury; lymphocele; positioning neuropathy, corneal abrasion; robot failure <0.5%.

Recent advances

- **Retzius-sparing** (earlier continence, more positive margins); Hood technique; single-port robot; same-day discharge.

Diagram to draw: port layout and fascial planes (intra-, inter-, extrafascial).

> **Key points to remember:**
> - Surgeon, not platform, decides outcome (Yaxley 2016).
> - Interfascial plane is the standard nerve-sparing plane.
> - Bladder neck stricture <2%; transfusion 1–2%.

Also, from the condensed book (not in the paper answer):

- Avoid relative contraindications (including neoadjuvant hormones) early in the learning curve.

> **KEY POINT**
>
> The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery.

2 · Patient selection

- The only absolute contraindications are uncorrectable bleeding diathesis and unfitness for general anaesthesia.
- After extraperitoneal mesh repair, use the **transperitoneal** route.
- **Obesity**: place longer trocars more cephalad and lower pneumoperitoneum to 10 mm Hg. A **large gland or median lobe** takes longer and bleeds more, but continence is comparable; **prior TURP** brings the ureteric orifices close to the bladder neck.
- Multiparametric MRI (mpMRI) for extracapsular extension (ECE) has specificity **0.88** and sensitivity **0.61**; it guides nerve sparing.

3 · Preparation and anaesthesia

- Consent for **conversion to open**; **general anaesthesia is mandatory**.
- **Positioning**: tuck and pad arms; use **no fixed shoulder rests** (brachial plexus injury) and avoid exaggerated hip flexion (femoral neuropraxia).
- CO2 pneumoperitoneum causes vagal response, oliguria and hypercarbia, so raise ventilation and lower pressure. Restrict fluids to 1.5–2 L, mostly after the anastomosis.
- **Steep Trendelenburg** causes facial, laryngeal and corneal oedema and raises intraocular pressure, so protect the eyes and consider delayed extubation if the face is very swollen.

4 · Access

- **Extraperitoneal** access gives a smaller space, protects bowel and needs less tilt (suits obesity and lung disease), but absorbs more CO2. **Transperitoneal** gives a larger space. Neither has a consistent advantage.
- Transperitoneal **anterior** (commonest; Retzius first) differs from **retrovesical** (seminal vesicles first, useful with a median lobe).
- Place the camera port **15–17 cm** from the symphysis; insufflate to 12–15 mm Hg.

5 · Operative steps (transperitoneal anterior RALRP)

ALGORITHM

1. Drop the bladder (Retzius); open the endopelvic fascia; ligate the DVC.
2. Divide the bladder neck; free the seminal vesicles and vasa.
3. Develop the posterior (Denonvilliers) plane; control pedicles and release the NVBs.
4. At the apex, divide the DVC and urethra; inspect the specimen.
5. Do pelvic lymph node dissection (PLND); entrap the specimen.
6. Posterior reconstruction, anastomosis, extraction.

- Open the endopelvic fascia with **no cautery**. Pneumoperitoneum tamponades DVC bleeding, so it can be sutured, stapled, or divided late and oversewn.

Bladder neck — no tactile feedback

- Divide anteriorly in the midline, lift the catheter, inspect for median lobe and orifices, then divide the posterior neck angled **45°** downward.
- After TURP the balloon falls into the defect, so division drifts too distal; a balloon off the midline suggests a median lobe. When in doubt, **divide more proximally**.

Posterior plane, pedicles, nerve sparing

P1 P4 *Anatomy of prostate and NVB; NVB preservation in RP — nerve-sparing planes, Denonvilliers excision for posterolateral ECE, injury mechanisms, apex = commonest margin*

- **Athermal (veil) release**: incise levator fascia anteromedially and sweep the NVB posterolaterally off the prostatic fascia, then Hem-o-lok the pedicle and divide sharply. The **intrafascial** plane is accepted in selected men.
- Pedicle clips near the anastomosis can migrate into the bladder, causing haematuria, infection and stones.

Apex and reconstruction

- Preserve urethral length; a small urethral rim on the apex lowers margins without harming continence.
- Reconstruct the bladder neck (tennis racquet) only for a large gland, median lobe or prior TURP, and protect the orifices.
- The **Rocco stitch** joins the Denonvilliers remnant and posterior bladder to the posterior rhabdosphincter. Its continence benefit is inconsistent; its agreed benefit is a **tension-free anastomosis**.
- The **Van Velthoven anastomosis** runs two arms anteriorly from 6 o'clock.

6 · Continence-preserving variants

*See the paper answer **Saturation biopsy of prostate** (Section 17, Ch 155).*

Also, from the condensed book (not in the paper answer):

- Retzius-sparing leaves the detrusor apron, puboprostatic ligaments and DVC untouched, on the idea that preserving anterior support beats reconstructing it.

7 · Pelvic lymph node dissection

P4 P1 *PLND templates; Lymphocele — templates (extended ~63%), staging not survival; lymphocele drain, then fenestration*

- **Indications**: may omit in low risk (nodes <5%). AUA 2018: unfavourable intermediate or high risk. EAU 2021: nomogram risk **>5%**. AUA/ASTRO 2022: risk-stratified by nomogram (Briganti and MSKCC nomograms predate PSMA PET).
- The standard template misses most drainage, so **if PLND is done, use the extended template**: proximal 2 cm above the common iliac bifurcation (ureter crossing), distal node of Cloquet, lateral **genitofemoral nerve**, medial bladder wall.
- **Clipping lymphatics** means fewer lymphoceles; node count is a poor quality surrogate.

PLND complications (~1 in 7)

- For the obturator nerve, remove a clip or repair a transection primarily. Minimise heat proximally, where the ureter is at risk.
- **Lymphocele is commonest** (**2–5%** significant). **Transperitoneal access is not protective**, because the bladder walls off the fossa; a **peritoneal interposition flap** reduces it. It classically presents with ipsilateral deep vein thrombosis (DVT).

8 · Postoperative care

- Use enhanced recovery, with same-day discharge for selected men; a drain is not routine.

Scenario	Catheter	Cystogram
Routine	~1 week, not before day 5	No
Extensive neck reconstruction / doubtful	≥10 days	Yes
Salvage	2–3 weeks	Strongly consider

- Removal at day 3–4 causes more retention. Extravasation delays but does not reduce long-term continence.
- **Penile rehabilitation**: PDE5 inhibitors help on-demand erections, but early rehabilitative use does not improve later response (3 RCTs). Use injections for non-responders.

9 · MIS-specific complications

Complication	Rate	Management
Transfusion	1–2%	Low, because pneumoperitoneum compresses veins
Bladder neck stricture	<2% (open up to 16%)	Dilatation or incision; rarely urethroplasty
Thromboembolism	~0.5%	Pneumatic compression; drugs only with multiple risk factors

Complication	Rate	Management
Robotic failure	<0.5%	Cancel or convert
Recognised rectal injury	Rare; higher in salvage	Multilayer repair and flap, no colostomy
Unrecognised rectal injury	Rare	Presents as rectovesical fistula; colostomy, later repair
Urine leak	Uncommon	Drain, longer catheter, cystogram

- **Ureteric injury** occurs behind the vas (posterior approach), at the posterior bladder neck after TURP or with a median lobe, and during proximal PLND.
- With DVT, image the pelvis for lymphocele or haematoma; with a significant lymphocele, do a duplex for occult DVT.

> **KEY POINT**
>
> Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated.

10 · Outcomes

- **Oncologic**: positive surgical margin (PSM) rates are equal open and robotic (~15%; ~9% organ-confined). PSM depends on tumour (Gleason ≥7, perineural invasion, stage) and surgeon volume, **not** on obesity, gland size or prior surgery.
- **Continence**: no-pad rates are in the mid-90s% at 1 year. Persistent mild leakage needs a sling or ProACT; severe (<2%) an artificial urinary sphincter.
- **Potency**: up to 90% at 1 year in expert hands; recovery continues to 5 years.
- **Salvage RALRP**: exclude metastases with PSMA PET and use full bowel prep and an air-bubble test. Continence is only 33–80%, and most men are impotent.

11 · Landmark trials

Trial	Design	Result	Take-home
Yaxley 2016	RCT, one open vs one robotic surgeon	Equal continence, potency, early oncology at 3 months; blood loss ~1400 vs 300 mL; shorter stay	Robot wins on recovery, not cancer or function
Retzius-sparing RCTs (Cochrane)	Single-centre RCTs	Earlier continence; equal at 12 months; more PSM	Use selectively

162 Radiation Therapy for Prostate Cancer

LAQ Radiotherapy and brachytherapy for CA prostate

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate?" · P2 SAQ "Radiotherapy and brachytherapy for CA prostate" (Radiotherapy for CA prostate.; Brachytherapy for CA prostate)

"Focal" is taken to mean localised (organ-confined and locally advanced) disease.

Radiobiology

- RT (radiotherapy) causes DNA double-strand breaks. Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum (~3 Gy). So larger fractions kill more tumour for the same late toxicity: the basis of **hypofractionation**. The low α/β ratio also favours high dose per fraction (HDR brachytherapy).

EBRT (external beam radiotherapy) techniques

Step	Advance
3D-CRT (three-dimensional conformal RT)	CT (computed tomography)-planned beams shaped to the prostate
IMRT (intensity-modulated RT)/VMAT (volumetric modulated arc therapy)	Inverse planning, steep dose gradients; less rectal toxicity; now standard; allows dose escalation
IGRT (image-guided RT)	Daily fiducials or cone-beam CT; margins 3–7 mm
MRI fusion, PSMA (prostate-specific membrane antigen) PET (positron emission tomography)	Better targeting; PSMA PET changes plans in up to half of high-risk men

EBRT fractionation schedules

Schedule	Dose per fraction	Example	Evidence
Conventional	1.8–2 Gy	78 Gy in 39	Dose escalation (RTOG 0126): better PSA control, no survival gain
Moderate hypofractionation (standard)	2.25–4 Gy	**60 Gy in 20**	**CHHiP** (non-inferior to 74 Gy in 37), **PROFIT**: non-inferior
SBRT (stereotactic body RT)	≥5 Gy	36.25 Gy in 5	**HYPO-RT-PC** non-inferior; **PACE-B** equal toxicity

Brachytherapy: definition and principle

- Brachytherapy (BT; Greek *brakhus*, short): radioactive sources placed **within the prostate**.
- **Steep dose fall-off** gives an intraprostatic dose far above what EBRT can safely give.

Brachytherapy: types

Feature	LDR (low dose rate), permanent seeds	HDR (high dose rate), temporary
Source	^{125}I (half-life 59.6 days), ^{103}Pd (17 days), ^{131}Cs (9.7 days)	**^{192}Ir** via remote afterloader
Monotherapy dose	^{125}I 145 Gy (P2: 140–160 Gy); ^{103}Pd 110–125 Gy	27 Gy in 2, 31.5 Gy in 3, 34–38 Gy in 4 fractions (single 19 Gy fraction fails)

Feature	LDR (low dose rate), permanent seeds	HDR (high dose rate), temporary
Monotherapy use	Low and favourable intermediate risk	
Dosimetry / quality	Post-implant CT at 14–30 days: **D90** (dose covering 90% of gland) **>90%** of prescription	Real-time planning; none afterwards

Brachytherapy: technique

- **TRUS (transrectal ultrasound)-guided transperineal** implant under anaesthesia: high lithotomy, stepper, template grid, needles, seeds deployed on withdrawal.
- **Peripheral loading** spares the urethra. Intraoperative planning gives best coverage.
- Quality: **D90 >90%** of prescription and **V100 >85%** of gland; urethral and rectal (V100 <1 cm^3) limits. Low D90 predicts failure.

Brachytherapy: indications

- **Monotherapy:** low and favourable intermediate risk (**RTOG 0232**: adding EBRT to implant gave no benefit in intermediate risk, more toxicity).
- **Boost** (EBRT 45 Gy + implant ± ADT [androgen deprivation therapy]) for unfavourable intermediate and high risk: **ASCENDE-RT:** LDR boost halved biochemical failure vs EBRT boost, with more grade 3 urinary toxicity (18.4% vs 5.2%).
- Also salvage and focal BT.

Brachytherapy: contraindications (relative)

- **IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s**, large median lobe; recent or large TURP (transurethral resection of the prostate) defect; gland >60 cm^3 (cytoreduce with ADT); pubic arch interference; prior pelvic RT; inflammatory bowel disease; cannot lie in lithotomy.

Brachytherapy: complications and follow-up

- Acute obstructive and irritative symptoms; retention (29% if IPSS >20, 2% if <10); α-blockers help.
- Chronic grade ≥2 urinary toxicity about 20%; urethral stricture; rectal complications 1–2%; rare rectourethral fistula; ED (erectile dysfunction).
- Nadir is deeper and later (about 2.5 years). **Benign PSA bounce** in about one-third: watch for 30 months before biopsy, do not salvage.

ADT with RT

- **Why:** ADT radiosensitises (less DNA repair) and causes apoptosis; RT up-regulates androgen-receptor signalling, which ADT blocks.

Risk group	ADT with EBRT
Low, favourable intermediate	None
Unfavourable intermediate	**4–6 months**
High / locally advanced	**18–36 months** (P2: 24–36 months) (EORTC 22863, PCS IV); add **abiraterone** if very high risk (STAMPEDE)

- **RT added to ADT** improves survival in locally advanced disease: **SPCG-7, NCIC PR.3/MRC PR07** (about 8–10% absolute).

After radical prostatectomy

- **Adjuvant RT** (pT3/positive margin): EORTC 22911, ARO 96-02 (biochemical control), **SWOG 8794** (metastasis-free and overall survival).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares over 60% of men RT.
- Salvage dose ≥64–66 Gy. **RTOG 9601:** 2 years bicalutamide improved overall survival. **SPPORT:** pelvic nodes + short ADT improve freedom from progression.

Advanced and metastatic disease

Setting	RT role	Evidence
Low-volume de novo mHSPC (metastatic hormone-sensitive)	**RT to prostate** + ADT	**STAMPEDE arm H**: overall survival HR (hazard ratio) 0.68 (3-year 73% → 81%); none in high volume
Oligometastases (≤3–5)	SBRT to all lesions (MDT, metastasis-directed therapy)	STOMP, **ORIOLE** (less progression), SABR-COMET
Painful bone metastases	Palliative EBRT, e.g. **8 Gy single fraction**; cord compression	Equal pain relief to multiple fractions
mCRPC (metastatic castration-resistant), bone-only symptomatic	**Radium-223** (α-emitter, calcium mimic, range <100 µm) 50 kBq/kg 4-weekly × 6	**ALSYMPCA**: overall survival 14.9 vs 11.3 months; fewer skeletal events
mCRPC, PSMA-positive, after taxane and ARPI (androgen-receptor pathway inhibitor)	**^{177}Lu-PSMA-617** (β-emitter radioligand)	**VISION**: survival 15.3 vs 11.3 months (HR 0.62); **TheraP**: better PSA response than cabazitaxel

- **Never combine radium-223 with abiraterone** (more deaths and fractures).
- **Also know:** protons (no proven gain over IMRT); strontium-89, samarium-153 (pain palliation); elective pelvic nodal RT (RTOG 94-13, GETUG-01: no survival gain).

Outcomes

- **ProtecT** (2016; 15-year 2023): equally low prostate-cancer mortality with RT, surgery or active monitoring; RT more bowel effects, surgery more incontinence and ED.

Response assessment

- **Phoenix failure:** PSA nadir + 2 ng/mL.
- Biopsy-proven local failure: salvage prostatectomy, brachytherapy, HIFU (high-intensity focused ultrasound) or cryotherapy.

Toxicity of RT

Type	Features and management
Acute (weeks)	Frequency, dysuria, diarrhoea; α-blockers
Late GI (gastrointestinal)	**Radiation proctitis** with bleeding at 6–18 months: sucralfate/steroid enemas, argon plasma coagulation, hyperbaric oxygen

Type	Features and management
Late GU (genitourinary)	Cystitis, haematuria, stricture
Sexual	ED, vascular; worse with ADT; PDE5 (phosphodiesterase type 5) inhibitors work in most
Rare	Second cancers, fistula

Recent advances

- **Hydrogel rectal spacer** (late rectal toxicity 2% vs 7%); MR-linac adaptive SBRT; **FLAME** focal boost to the MRI lesion; protons; PSMA radioligands moving earlier (PSMAfore).

Diagram to draw

- TRUS-guided perineal template brachytherapy (transperineal implant set-up).

Key points to remember:

- α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
- Phoenix: nadir + 2 ng/mL.
- LDR ^{125}I 145 Gy; HDR ^{192}Ir; D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

Also, from the condensed book (not in the paper answer):

- A single 19 Gy high-dose-rate fraction fails (§5).
- RADAR and RTOG 99-10 on ADT duration (§6).
- Salvage prostatectomy incontinence ~63% (§11).

1 · Radiobiology and fractionation

- Conventional fractionation gives **1.8–2 gray (Gy)** daily.
- In the **linear–quadratic model**, the **α/β ratio** sets a tissue's response to fraction size: low (2–4 Gy) in late-responding normal tissue, high (≥8 Gy) in most tumours.
- Prostate cancer has few cycling cells, so its α/β is **≈1.5 Gy**, **lower than the rectum (~3 Gy)**. Fractions >2 Gy therefore favour tumour kill over late toxicity, which is the rationale for hypofractionation.

Category	Dose/fraction	Typical course
Conventional	1.8–2 Gy	~8 wk (78 Gy in 39 fractions)
Moderate hypofractionation	2.25–4 Gy	~4 wk (**60 Gy/20**)
Ultrahypofractionation = stereotactic body RT (SBRT)	≥5 Gy	≤5–7 fractions (**36.25 Gy/5**)

2 · EBRT techniques

- MRI is fused with planning CT; PSMA PET changes management in up to half of higher-risk men (see Ch 152).

- **IMRT or VMAT** lowers gastrointestinal (GI) but not genitourinary (GU) toxicity versus 3D conformal RT, as the urethra lies inside the target.

Volume	Prostate application
Gross tumour volume (GTV)	Dominant intraprostatic lesion (DIL) on MRI
Clinical target volume (CTV)	Prostate, with or without proximal seminal vesicles and pelvic nodes
Planning target volume (PTV)	CTV plus a margin for motion and set-up: **3–7 mm** with image guidance

- **Image-guided RT (IGRT)** with daily gold fiducials or cone-beam CT is the NCCN standard; its smaller margins make large fractions safe.
- A **rectal spacer** (hydrogel or hyaluronic acid) is injected transperineally into Denonvilliers' fascia. Late rectal toxicity falls to **2% vs 7%** (hydrogel, 79.2 Gy).
- The MR-linac allows 2 mm margins, and protons have shown no randomised benefit over IMRT.

3 · Dose escalation

- Six randomised trials, including RTOG 0126, raised **biochemical control in all risk groups** with **no overall survival (OS) gain**, because salvage works and metastasis fell only slightly. Still, ≥~78 Gy is standard.
- **FLAME**: a focal boost to the MRI DIL improved biochemical progression-free survival (HR 0.45) without more toxicity.

4 · Hypofractionation

- **Moderate**: CHHiP 57 Gy/19 was not shown noninferior; PROFIT 60/20 was noninferior; RTOG 0415 70 Gy/28 was noninferior but more toxic late; HYPRO 64.6 Gy/19 was **not superior**.
- After prostatectomy, HYPORT 62.5 Gy/25 was noninferior to 66.6 Gy/37.
- **SBRT**: HYPO-RT-PC gave ~84% 5-yr failure-free survival, with more acute toxicity.
- **Dose ceiling**: there is no gain above **40 Gy/5**. Give it on alternate days with 3–5 mm margins.

5 · Brachytherapy (BT)

Urinary function decides eligibility.

	Low-dose-rate (LDR)	High-dose-rate (HDR)
Source	Permanent seeds (^{125}I, ^{103}Pd, ^{131}Cs)	Temporary ^{192}Ir (380 keV) in a remote afterloader
Technique	TRUS-guided transperineal template; **peripheral loading** spares urethra	Dose shaped by dwell times and positions; no post-implant dosimetry
Uses	Monotherapy for low/intermediate risk; boost	Boost (e.g. 15 Gy × 1 after 46 Gy); monotherapy 27 Gy/2, 34–38 Gy/4
Pitfall	Oedema, seed migration	**19 Gy × 1 fails**: 5-yr local failure 29% vs 3% (27 Gy/2)

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{125}I	**59.6 d**	140–160 Gy	108–110 Gy
^{103}Pd	17 d	110–125 Gy	90–100 Gy

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{131}Cs	9.7 d	100–115 Gy	80–85 Gy

- **Post-implant dosimetry** (CT at 14–30 days): **D90** >90% of prescription; **V100** >85%; urethral V150 <0.05 cm^3; rectal V100 <1 cm^3. A low D90 predicts biochemical failure. **MCQ** D90 is a dose, V100 a volume.
- **Relative contraindications**: **IPSS >20**; **Qmax <10 mL/s**; large median lobe; large TURP defect or TURP within 3–6 mo; prior pelvic RT; inflammatory bowel or connective tissue disease; **pubic arch interference**; inability to adopt lithotomy.
- A large gland shrinks ~30% with ADT. Retention risk is ≈29% if IPSS >20 vs ≈2% if <10, so start α-blockers before the implant.
- **With EBRT**: 45 Gy plus an implant at 60–70% of monotherapy dose. **RTOG 0232**: BT alone equalled EBRT plus BT in intermediate risk with less toxicity, so BT alone is standard.
- **ASCENDE-RT**: no OS difference; LDR boost grade 3 GU toxicity **18.4% vs 5.2%**.

> **KEY POINT**
>
> Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant.

6 · ADT with RT

- **Synergy**: ADT causes apoptosis, shrinks the target and lowers **Ku70**, impairing non-homologous end-joining repair (see Ch 164).

Risk group (NCCN)	With EBRT	With EBRT + BT	Key evidence
Low / favourable intermediate	None	—	RTOG 94-08: no benefit in low risk
Unfavourable intermediate	**4–6 mo**	—	RTOG 94-08, DFCI 95-096, TROG 96.01 (6 mo better than 3); RTOG 99-10: 9 mo no better than 4
High / very high	**24–36 mo** (≥18)	12–36 mo	EORTC 22863, RTOG 92-02, EORTC 22961 (36 mo better than 6); PCS IV: 18 mo = 36 mo

- Needed duration falls as RT dose rises: in **RADAR**, 18 mo beat 6 mo only without an HDR boost.
- **STAMPEDE**: adding 2 yr abiraterone in high-risk N0/N1 M0 improved metastasis-free survival and OS.

7 · Pelvic nodal RT

- RTOG 94-13 and GETUG-01 showed no durable benefit, mainly because a field border at L5/S1 misses the **common iliac nodes**; modern fields extend to **L4/L5**.
- **POP-RT** (node risk >20%, long ADT): whole-pelvis RT raised 5-yr biochemical failure-free survival (~95% vs ~81%), with no OS difference and more acute GI toxicity.

8 · Post-prostatectomy RT

- RADICALS-RT, GETUG-AFU 17 and RAVES, with their meta-analysis, make **early salvage preferred** over adjuvant RT, which may still suit very high risk (Gleason 8–10, pT3b) (see Ch 164).
- Dose is ≥64 Gy, and PSMA PET localises recurrence.

9 · Follow-up and biochemical failure

Do not mistake a benign bounce for failure.

- Benign glands persist, so PSA stays detectable. The nadir is ≤0.1 ng/mL after EBRT plus BT and higher after EBRT alone; it is reached at ~1.5 yr after EBRT and ~2.5 yr after BT.

	ASTRO 1996	**Phoenix 2005 (current)**
Failure	3 consecutive rises	PSA **≥ nadir + 2 ng/mL**
Date	Backdated to midway between nadir and first rise	When reached (or when salvage started)
Notes	Delays diagnosis ≥18 mo; invalid with ADT or BT	Best correlates with clinical relapse; valid with short ADT; fewer false calls from bounces

- Poor prognosis is signalled by failure <18 mo and especially **PSA doubling time <3 mo**, which predicts cancer death.
- A **bounce** (rise >0.2 ng/mL, then durable fall) occurs in ~1/3 after LDR, mostly in the first 3 yr, and predicts a better outcome.
- **Biopsy** is best at **30–36 mo**, since irradiated cells die slowly. Marked radiation effect is not graded and behaves as negative.

10 · Toxicity

Toxicity	**Features**	**Management / prevention**
Acute GU	Peaks ~2 mo, settles by 12 mo	α-blockers; catheter for retention
Late GU	Grade ≥2 ~20%; stricture; incontinence ~3%, more after TURP	IGRT halves late GU toxicity; peripheral loading
Late GI (proctitis)	Bleeding at 6–18 mo; IMRT 81 Gy grade ≥2 ~5%; BT 1–2%; colostomy ~0.3%	Steroid or sucralfate enemas, then **argon plasma coagulation** or hyperbaric oxygen, then colostomy
Erectile dysfunction	Vascular, over 24 mo; often due to ADT; best with BT alone (~76% potent at 5 yr)	PDE5 inhibitors work in ~¾
Second cancers	Small excess of bladder and rectal cancers	Investigate haematuria or rectal bleeding fully

11 · Salvage after local failure

- Salvage needs **biopsy-proven, local-only** recurrence. Options are salvage RP (incontinence ~63%), BT (preferred), SBRT re-irradiation, cryotherapy and HIFU.

12 · Landmark trials

Trial	Design	Result	Take-home
RTOG 0126	70.2 vs 79.2 Gy, low–intermediate	8-yr biochemical failure-free 65% vs 80%; OS same	Dose escalation improves control, not OS
CHHiP	74/37 vs 60/20 vs 57/19	60/20 noninferior	60 Gy/20 standard
HYPO-RT-PC	42.7/7 vs 78/39	Noninferior; equal late toxicity	Ultrahypofractionation valid
PACE-B	36.25/5 vs conventional/moderate	Similar 2-yr toxicity	SBRT safe
ASCENDE-RT	EBRT vs LDR boost (plus 12 mo ADT)	Biochemical failure HR 2.04 for EBRT boost; more GU toxicity with LDR	BT boost for high risk
TROG 96.01	RT with no, 3 or 6 mo ADT	6 mo improved OS; 3 mo did not	≥6 mo short-course
RTOG 94-08	66.6 Gy with or without 4 mo ADT, T1b–T2b	10-yr OS 57% vs 62%; intermediate risk only	ADT for intermediate, not low
EORTC 22863	RT with or without 3 yr goserelin, T3–T4	10-yr OS 39.8% vs 58.1%	Long ADT in high risk
EORTC 22961	6 vs 36 mo ADT	Short course not noninferior	Long ADT for high risk
PCS IV	18 vs 36 mo	No survival difference	18 mo acceptable
SPCG-7; NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	RT lowered cancer death and raised OS	Never ADT alone if fit
RTOG 94-13; GETUG-01	Pelvic vs prostate-only RT	No durable benefit	Field height matters
SPPORT	Salvage bed RT, with or without ADT and nodes	Pelvic RT plus ADT best	Treat nodes in salvage

163 Focal Therapy of Prostate Cancer

1 · Definition and rationale

- **Focal therapy (FT)** is guided ablation of an image-defined, biopsy-confirmed lesion with a **safety margin**. Sparing the neurovascular bundles (NVBs), sphincter and rectum preserves function.
- FT rests on mpMRI, targeted biopsy and the fact that **Gleason pattern 4/5** is lethal.

Index lesion hypothesis

- Cancer is multifocal in **50–91%** of radical prostatectomy (RP) specimens. The **index lesion (IL)**, the largest or highest-grade focus, is held to drive natural history and metastasis.
- **For**: the IL holds **80–90%** of tumour volume, and satellites are mostly <**0.5 mL** (Stamey "insignificant").
- **Against**: a satellite is the aggressive focus in 12–38%, and IL Gleason matches overall grade in only ~⅔.

> **KEY POINT**
>
> "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance.

- **FT and active surveillance (AS)** form one strategy: ablate grade 4 disease, watch the low-grade remainder, and retreat new foci.

2 · Patient selection

- The patient must **accept close surveillance**.
- **TRUS biopsy** misses almost half the disease relevant to hemiablation compared with **transperineal mapping biopsy (TMB, 5-mm grid)**.
- **mpMRI** detects high-grade and anterior tumours. In **PROMIS**, its sensitivity for clinically significant cancer (csPCa) was **93%** and specificity 41%, so MRI lesions need biopsy and MRI-negative areas still need systematic biopsy.
- mpMRI **underestimates volume**, so a margin is needed: **5 mm** ablates >95% of csPCa. About **90%** of outfield csPCa lies within **1 cm** of the target.

> **KEY POINT**
>
> Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland. Every consensus has kept all three.

Consensus criteria (2016–2019)

Variable	Consensus
Work-up	mpMRI plus targeted and systematic biopsy
Risk group	**Intermediate**; Gleason **3+4 = "sweet spot"**; 4+3 maximum; no cribriform or ductal histology
Size	≤1.5 mL (≤3 mL if not crossing midline); 20–25% of gland if crossing

Variable	Consensus
Residual untreated disease	Gleason 3+3 only (≤1 mm core length in 2016, tightened from 5 mm)
Life expectancy	Per guidelines; no fixed limit
Erectile dysfunction	**Not** a contraindication
Not suitable	High risk (no long-term data); low risk usually goes to AS

- The goal is now to treat and retreat so as to **delay or avoid radical therapy**. FT remains investigational, ideally within a trial.

3 · Ablation patterns and energies

- FT means **lesion, quadrant or hemiablation**; "hockey-stick" (subtotal) ablation is partial-gland ablation, not FT.

SAQ HIFU

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 2, Q3d

Definition and principle

- HIFU (high-intensity focused ultrasound) focuses ultrasound from a bowl-shaped transducer or lens onto a point. Energy is absorbed only where the beams converge, so tissue there is heated while the path is spared.
- **Intensity 1500–10,000 W/cm²** (diagnostic ultrasound causes negligible heating).
- **Mechanisms: thermal coagulative necrosis** (≥55 °C; prostate exceeds 100 °C within seconds) plus **cavitation** (bubbles collapse and damage cells).
- Each lesion is a **rice grain**; the target is "painted" with overlapping lesions. Ablated and viable tissue look alike on ultrasound, so gaps occur.

Devices and routes

- **Transrectal** probe that both monitors and treats (posterior zone).
- **TULSA (MRI [magnetic resonance imaging]-guided transurethral ultrasound ablation): MR thermometry** confirms lethal temperature; reaches the anterior gland.

Indications

Setting	Use
Focal therapy, prostate cancer	MRI-visible, unilateral, grade group 2–3 index lesion; hemiablation; based on the index-lesion hypothesis
Whole-gland ablation	Localised cancer; retreatment 29–38%
Salvage after EBRT (external beam radiotherapy)	Biopsy-proven local recurrence, **≥2 years after EBRT**
BPH (benign prostatic hyperplasia)	Approved
Kidney	Investigational (incomplete ablation)

Limitations

- **Anterior lesions** (energy lost, oedema displaces target), large gland, gland height and **calcification** (some advise TURP [transurethral resection of the prostate] first).

Outcomes

- **Focal HIFU:** continence 90–95%, potency 50–78%; infield positive biopsy 16.7–54% (csPCa [clinically significant prostate cancer] 0–21.8%).
- **HEAT registry** (1379 men): 7-year failure-free survival 69%.
- **Salvage HIFU:** biochemical progression-free survival about 48% at 3 years; continence 64–87.5%.

Complications

- Retention (about 8%), UTI (urinary tract infection) 6–18%, stricture (4%), ED (erectile dysfunction); rectourethral fistula rare (about 2% after salvage); osteitis pubis.

Follow-up

- PSA (prostate-specific antigen) falls only 50–80%; **mpMRI (multiparametric MRI) + targeted biopsy at 3–6 months**, then at 1–2 and 5 years. EAU (European Association of Urology): focal therapy within trials or registries.

Also know: histotripsy (>10,000 W/cm², mechanical liquefaction, visible bubble cloud).

Diagram to draw: transrectal transducer focusing beams on a rice-grain lesion in the posterior prostate.

> **Key points to remember:**
> - Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
> - Transrectal HIFU suits posterior lesions; TULSA for anterior.
> - Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

SAQ Cryotherapy for CA prostate

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "Cryotherapy for carcinoma prostate" · P2 SAQ "Biological effects of cryosurgery"

Principle

- Tissue destruction by extreme cold. **Argon** expanding in the probe tip cools by the **Joule–Thomson effect**; **helium** expanding warms (active thaw).
- Used for small renal masses and prostate cancer.
- Prostate cryotherapy is **transperineal, image-guided** freezing; suits **anterior** tumours. HIFU (high-intensity focused ultrasound) is the main alternative.
- Thermocouples guard sphincter and neurovascular bundles; a **urethral warmer** prevents sloughing.
- Generations: liquid nitrogen without imaging; TRUS (transrectal ultrasound) plus warmer; argon/helium 17-gauge probes; focal with MRI (magnetic resonance imaging) fusion.

Biological effects: mechanisms of cell death

Tissue dies during **both freezing and thawing**, by four mechanisms:

Mechanism	What happens
1. Direct cell injury	*Freeze:* extracellular water freezes first and becomes hyperosmolar; cells dehydrate and membranes distort. Rapid cooling forms **intracellular ice**, which ruptures organelles. *Thaw:* water rushes back, cells swell and burst; ice recrystallises into larger crystals
2. Vascular injury	Endothelial damage, platelet activation and **thrombosis** cause **vascular stasis** and ischaemic necrosis; the main delayed killing effect
3. Apoptosis	Sublethal cold at the periphery triggers programmed cell death
4. Immunomodulation	Necrotic cells release antigens and DAMPs (damage-associated molecular patterns); dendritic cells activate T cells against tumour

- **End result:** coagulative necrosis, then fibrous scar.

Factors deciding lethality

- **Temperature:** normal renal tissue dies at about −19.4 °C; tumour may need −40 to −50 °C. **Target ≤−40 °C.**
- **Fast freeze, slow thaw** maximise injury.
- **Double freeze–thaw cycle** enlarges the necrotic zone (standard of care); each freeze about 8–10 minutes.
- **Heat sink:** nearby large vessels warm tissue; more probes or longer freezing needed.

Zones of the ice ball

- Centre: complete necrosis. Edge: 0 °C and sublethal; below −20 °C only about 3 mm inside the edge. The lethal −40 °C isotherm lies well inside the visible ice ball.
- So the ice ball must extend **5–10 mm beyond the tumour**.

Indications in prostate cancer

1. **Salvage after radiotherapy** (main role): biopsy-proven local recurrence, no metastases, **PSA (prostate-specific antigen) <10 (ideally <4)**, doubling time ≥16 months.
2. **Primary whole-gland:** men unfit for surgery or radiotherapy.
3. **Focal or hemi-gland:** MRI-visible grade group 2–3 lesion; rests on the index-lesion hypothesis.

Technique

- TRUS-guided transperineal template: place cryoprobes, thermocouples and urethral warmer; real-time imaging shows the ice ball; freeze anterior to posterior until −40 °C is reached; helium thaw; repeat the cycle.
- **Also know:** passive versus active thaw (both used); insulated probe shafts reduce skin injury.

Outcomes

- Salvage: biochemical control **55–71% at 3–5 years** (Phoenix: nadir + 2).
- **Donnelly, Cancer 2010** (RCT, randomised controlled trial): **non-inferior** to external beam radiotherapy; worse sexual function.

Complications

	Primary / focal	Salvage
ED (erectile dysfunction)	**Almost always** (whole gland)	Very common

	Primary / focal	Salvage
Incontinence	Rare	Single digit
Rectourethral fistula	≤0.8% (focal)	1–3.4%
Retention	Commonest after focal	~4%

- **Also know:** urethral sloughing, stricture; focal ablation preserves erections better (COLD [Cryo On-Line Data] registry).

Comparison with other ablative energies

	Cryotherapy	HIFU	IRE (irreversible electroporation)
Energy	Freezing	Focused ultrasound heat (≥55 °C)	Electric pulses, non-thermal
Route	Transperineal	Transrectal	Transperineal
Best for	Anterior lesions	Posterior lesions	Lesions near nerves, vessels

Guidelines and advances

- **EAU (European Association of Urology):** primary or focal only in trials or registries. PSMA (prostate-specific membrane antigen) PET selects salvage patients.

Diagram to draw

- Prostate (or tumour) with probe and ice ball: −40 °C isotherm inside the 0 °C edge, and tumour margin.

> **Key points to remember:**
> - Argon freezes, helium thaws (Joule–Thomson); ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm, urethral warmer.
> - Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
> - Main role is salvage: PSA <10, doubling time ≥16 months.
> - Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

Also, from the condensed book (not in the paper answer):

- Focal cryotherapy 5-yr biochemical progression-free survival (BPFS) ~62–70%; COLD registry: partial equals whole-gland control with ~2× better erections.

Modality	Mechanism / route	Pros	Cons
TULSA (MRI-guided transurethral ultrasound ablation)	Transurethral transducer with MR thermometry	Anterior gland; control within 1.3 mm	In-bore MRI; csPCa on 12-month biopsy ~⅓
Focal laser	Interstitial fibre ≥60 °C; MR thermometry	Most predictable zone	Narrow margin, so **highest infield failure** (~40% at 6 yr)
Irreversible electroporation (IRE)	Non-thermal pulses >1600 V/cm open membrane pores	Spares connective tissue; good function	Needs deep anaesthesia; NVB and urethral injury on histology

Modality	Mechanism / route	Pros	Cons
Vascular-targeted photodynamic therapy (VTP)	IV padeliporfin (TOOKAD) 4 mg/kg activated by 753-nm light occludes vessels	**Only randomised evidence** (PCM301)	Gleason 6 only; perineal pain ~15%; light precautions
Radiofrequency	~100 °C	Fat limits conduction	Infield csPCa 20–30%
Focal brachytherapy	Low-dose-rate seeds or high-dose-rate (HDR) catheters	Mild early toxicity	HDR: stricture ≤8.2%, ED in half; biopsy only at 24 mo

KEY POINT

Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants all aim to make that edge lethal, so laser's precise but narrow zone gives the most infield recurrence.

4 · Outcomes

Modality	Infield positive biopsy	Outfield	Continence	Erections
Cryotherapy	7–39% overall (~¼ csPCa)		95–100%	47–100%
HIFU	17–54% (csPCa 0–22%)	7–43% (csPCa 2–8%)	90–95%	50–78%
Laser	11–50%	17–60%	95–100%	Minimal change
IRE	csPCa 1–10%	csPCa 5–16%	~All	Some decline
VTP	19–37%	17–19%	Preserved	Nil to mild fall

- HEAT (HIFU): 7-yr failure-free survival was **69%** (68% intermediate, 65% high risk).

5 · Complications

- The commonest is **urinary retention** (0–9%); others are infection, sloughing and stricture (~4–5% after HIFU or IRE).
- **Rectourethral fistula** occurs in ≤2% and is the most feared; some heal with a catheter for 4–6 weeks.

6 · Follow-up and failure

Follow-up relies on MRI and biopsy, not PSA.

- **PSA** falls only **50–80%**, and the Phoenix/ASTRO definitions are not validated after FT. A rise to nadir + **1.0** ng/mL at 12 months, or + 1.5 at 24–36 months, predicts csPCa.

Success (ICUD–SIU 2015)	Infield	Outfield
Short term	Eradication of treated focus	No csPCa — any outfield cancer = **selection failure**
Intermediate/long term	No csPCa	No csPCa (new cancer may be de novo)

ALGORITHM

1. At **3–6 months**: mpMRI and targeted biopsy (4–6 cores) of the treated zone.
2. At 1–2 and 5 years: mpMRI; if a new lesion appears, then targeted biopsy.
3. At 1–2 and 5 years: systematic biopsy of the untreated gland.

- **PI-FAB**: read contrast-enhanced (**DCE) images first**, as scar obscures T2 and diffusion. Score 1 (both negative) is monitored, score 2 (DCE positive only) is biopsied if PSA rises, and score 3 (both positive) is biopsied.

7 · Salvage

- After FT failure, options are **focal retreatment** or whole-gland RP or radiotherapy.

Focal salvage for radiorecurrent disease

	Cryotherapy	HIFU	IRE
Cancer control	5-yr BPFS ~50%	3-yr BPFS 48%	Progression-free survival 60%, metastasis-free survival 91% (48 months)
Continence	87–100%	64–87.5%	93%
Signature complication	Rectourethral fistula ~1.4%	Bladder-neck stricture, fistula ~2%; grade ≥3 ~¼	Urethral sludge ~9%

- Focal salvage controls cancer as well as whole-gland salvage, with better function.

8 · Landmark trials

Trial	Design	Result	Take-home
PCM301 (Azzouzi 2017)	415 low-risk men; padeliporfin VTP vs AS	Progression 28% vs 58%; conversion to radical therapy at 4 yr **24% vs 53%**; continence preserved	First level 1 evidence for FT — but Gleason 6 only, surrogate end points
PROMIS	mpMRI vs 5-mm TMB, 11 centres	csPCa sensitivity 93%, specificity 41%	MRI lesions need biopsy; systematic biopsy still needed
HEAT registry	Prospective focal HIFU, 1379 men	7-yr failure-free survival 69%	Largest FT dataset; includes high risk

KEY POINT

Open problems are leaving satellites, the cost of retreatment and adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy, whereas prostate FT relies on surveillance and retreatment.

164 Treatment of Locally Advanced Prostate Cancer

LAQ Staging and management of locally advanced (non-organ-confined) CA prostate

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Staging of prostate cancer and management of locally advanced (non-organ-confined) prostate cancer; complications of bilateral orchidectomy" (Describe staging of CA prostate and management of non-organ confirmed CA prostate; Desc. mgt optns for locally adv. CaP. Discuss complic. of b/l orchidectomy.; Describe management of LAPC) · P1 LAQ "Describe staging of CA prostate and management of nonorgan confined CA prostate"

Read as: staging of prostate cancer, and management of non-organ-confined (locally advanced and node-positive, non-metastatic) disease; describe the management options for locally advanced carcinoma prostate, and discuss the complications of bilateral orchidectomy. LAPC is read as locally advanced prostate cancer.

Introduction and definition

- **Locally advanced CaP (carcinoma prostate) / LAPC: cT3–4, N0 or N1 (N+), M0:** extracapsular extension (cT3a), seminal vesicle invasion (cT3b), fixed or invading adjacent organs (cT4), ± pelvic nodes.
- At least 10% of new prostate cancers are locally advanced; these men contribute a disproportionate share of cancer deaths.
- Non-organ-confined disease has a high chance of micrometastases. Any single treatment carries a high recurrence risk (single-modality treatment usually fails), so **multimodal (combined) therapy**, local therapy plus systemic ADT (androgen deprivation therapy), is the rule.
- In the PSA (prostate-specific antigen) era, most "high-risk" disease is defined by grade and PSA rather than palpation.
- Untreated, 15-year disease-specific survival is 57% for T3–T4 vs 81% for T1–T2 (Swedish cohort). Untreated, Gleason 8–10 disease carries high cancer mortality, so fit men need treatment.

Staging: TNM (tumour, node, metastasis) 2017 (AJCC [American Joint Committee on Cancer] 8th edition)

Stage	Definition
T1a / T1b	Not palpable; incidental at TURP (transurethral resection of prostate): ≤5% / >5% of resected tissue
T1c	Not palpable; found on needle biopsy (for raised PSA)
T2a / T2b / T2c	Palpable, confined: ≤ half of one lobe / > half of one lobe / both lobes
T3a	Extraprostatic extension (unilateral or bilateral), including microscopic bladder neck invasion
T3b	**Seminal vesicle invasion** (not T4)
T4	Fixed or invading external sphincter, rectum, bladder, levators, pelvic wall
N0 / N1	No nodes / regional (pelvic) nodes

Stage	Definition
M1a / M1b / M1c	Non-regional nodes / bone / other sites ± bone

- **Traps:** seminal vesicle invasion is T3b, not T4; non-regional nodes are M1a.
- **Prognostic stage groups** combine T, N, M, PSA and grade group: stage III = T3–4, PSA ≥20 ng/mL or grade group 5; IVA = N1; IVB = M1. pT2 is not subdivided.
- **Grade group (ISUP, International Society of Urological Pathology):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.

Risk groups

System	High risk	Very high / locally advanced
D'Amico	T2c, PSA >20 or Gleason 8–10 (any one)	–
NCCN (National Comprehensive Cancer Network)	cT3a, GG4–5 or PSA >20	cT3b–T4, primary Gleason pattern 5, ≥2 high-risk features, or >4 cores GG4–5
EAU (European Association of Urology)	PSA >20 ng/mL, ISUP 4–5 or cT2c	**Locally advanced: cT3–4 or cN+ (cN1), any PSA, any grade**

- Nomograms (CAPRA [Cancer of the Prostate Risk Assessment] score, Partin tables, Kattan nomograms) and genomic classifiers (Decipher) refine risk.

Staging work-up (evaluation)

- **DRE (digital rectal examination)** sets clinical T stage but under- and over-stages. Biopsy ISUP grade group.
- PSA predicts extent: organ-confined about 80% if PSA <4, under 50% if >10; nodes in about 20% if PSA >20 ng/mL and about 75% if >50 ng/mL.
- **mpMRI (multiparametric magnetic resonance imaging):** local extent; separates T2 from T3 (extraprostatic extension and seminal vesicle invasion; PI-RADS v2 reporting); best specificity in higher-risk men.
- **Bone scan + CT (computed tomography) or MRI** for high risk, PSA >20 or GG4–5.
- **PSMA (prostate-specific membrane antigen) PET-CT (positron emission tomography)** is more accurate; **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Assess life expectancy: a fit man in his 70s with high-grade disease will live long enough to die of it.

Management of locally advanced N0 (cT3–4 N0 M0)

A. Radiotherapy (EBRT, external-beam RT) + long-term ADT: the standard

- Dose-escalated IMRT (intensity-modulated RT) with image guidance, ≥78–81 Gy (P1: about 76–78 Gy), with or without pelvic node RT, with or without brachytherapy boost (**ASCENDE-RT:** LDR [low-dose-rate] boost halved biochemical failure, improving biochemical control; overall survival (OS) equal).
- **Why add ADT:** it shrinks the target and is synergistic with radiation: neoadjuvant, concurrent and adjuvant, **18–36 months** for high risk (NCCN; EAU 2–3 years).

Trial	Finding
EORTC 22863	3 years of goserelin with RT improved 5-year OS, 78% vs 62% (other sources: 79% vs 62%)
RTOG 85-31 / 92-02	Long adjuvant ADT better; RTOG 92-02: 28 vs 4 months ADT better on all endpoints except overall OS; OS gain in Gleason 8–10
TROG 96.01	6 months better than 3 months
SPCG-7; MRC PR07/NCIC PR.3	**RT is still needed with ADT:** adding RT to ADT reduced cancer death and gave an overall survival gain: SPCG-7 cancer death RR (relative risk) 0.44; MRC PR07 OS HR (hazard ratio) 0.77 (P1: HR 0.46). ADT alone is inferior
STAMPEDE (high-risk M0)	Adding 2 years of abiraterone: metastasis-free survival HR 0.53
RTOG 0521; GETUG 12	Docetaxel added modest benefit

- **Duration:** NCCN: ADT 18–36 months with EBRT (12–36 with brachytherapy boost); 18 vs 36 months gave similar OS.
- Pelvic nodal RT: RTOG 94-13 inconclusive; POP-RT (Tata Memorial) improved biochemical failure-free survival in high-risk N0.

B. Systemic intensification

- **STAMPEDE (abiraterone, M0):** for N1, or two of cT3–4, Gleason ≥8 or PSA ≥40 ng/mL, **abiraterone + prednisolone for 2 years** with ADT + RT gave metastasis-free survival HR 0.53 and OS HR 0.60. NCCN now lists it for very high risk.
- Docetaxel with RT and ADT (RTOG 0521, OS HR 0.69): not routine.

C. Radical prostatectomy (RP) + extended PLND (pelvic lymph node dissection, ePLND), within multimodal therapy: selected men

- For fit men, life expectancy >10 years, with a completely resectable tumour (mainly cT3a; selected cT3b, limited nodes). Clinical overstaging of cT3 occurs in 7–27%, so cT3 alone is no bar.
- **Advantages:** accurate pathological staging; removes the primary, the source of local morbidity (an untreated prostate causes retention, haematuria and ureteric obstruction that ADT does not prevent); RT kept for salvage.
- RP alone gives disease-free survival in at least half at 8–10 years; disease-specific survival 85–92% at 5 years and 79–82% at 10 years (10-year disease-specific survival about 80–90%), but biochemical recurrence is common, so multimodal care is planned.
- **Neoadjuvant ADT before RP:** clinical downstaging and fewer positive margins, but **no survival or PSA benefit**.
- **Adjuvant vs early salvage RT after RP:** EORTC 22911, SWOG 8794 (metastasis-free survival and OS improved) and ARO 96-02 favour adjuvant RT over observation. RADICALS-RT, RAVES and GETUG-AFU 17 favour **observation with early salvage RT** (at PSA rise) for most men with pT3 or a positive margin. Adjuvant RT is still considered for very high-risk pathology (pT3b, GG4–5 / Gleason 8–10, positive margins). Dose 64–66 Gy; adding short-term ADT to salvage RT improved progression-free survival (GETUG-AFU 16).

D. ADT alone

- Only for men unfit for or declining local therapy. Consider immediate ADT if symptomatic, PSA doubling time is short (<12 months) or PSA is very high (>50 ng/mL); otherwise

deferred ADT is acceptable. Early ADT delays progression; survival gain mainly in aggressive disease. In fit men it is inferior to ADT + RT.

- Options: bilateral orchidectomy, LHRH (luteinising hormone-releasing hormone) agonist or antagonist; bicalutamide 150 mg monotherapy spares sexual interest (harmful in localised disease); intermittent ADT.

E. Others

- Watchful waiting with deferred ADT in elderly comorbid men.
- Cryotherapy and HIFU (high-intensity focused ultrasound): not standard, investigational in high risk.
- **Also know:** TURP or stents for local obstruction; trial enrolment.

Node-positive disease (N1 M0)

Setting	Management
cN1 M0	RT to prostate and pelvic nodes + long-term ADT 2–3 years + abiraterone 2 years (STAMPEDE). RP + ePLND in selected men
pN1 (pN+) after RP	**Immediate ADT** (ECOG 7887: improved overall survival) **or ADT + adjuvant pelvic RT** (beats ADT alone); observation with undetectable PSA and ≤2 nodes

Metastatic (M1), in brief

- ADT plus an **ARPI (androgen receptor pathway inhibitor)** (abiraterone, enzalutamide, apalutamide) or a triplet with docetaxel; prostate RT for low-volume disease (see 2025-Winter Paper 3, Q3b).

Algorithm

```
Non-organ-confined, cT3–4 / cN1, M0 (PSMA PET or CT + bone scan) ├─ Fit, cT3a, resectable → RP + ePLND → early salvage RT ± ADT ├─ Most (cT3–4 or cN1) → EBRT + ADT 18–36 months │ very high risk / N1 → + abiraterone 2 years └─ Unfit / refuses → ADT alone (immediate if symptomatic, or deferred)
```

Complications and local sequelae

- ADT: hot flushes, osteoporosis, metabolic syndrome, cardiovascular risk, loss of libido; give bone protection.
- RT: proctitis, cystitis. RP: incontinence, erectile dysfunction.
- A prostate left in place can cause bladder outlet obstruction (channel TURP), ureteric obstruction (stent or nephrostomy) and haematuria (palliative RT).

Bilateral orchidectomy

- Removes about 95% of testosterone: **castrate level (<20 ng/dL) within 12 hours**. Cheap, no compliance issues, no flare, so useful in impending cord compression. Simple or subcapsular (all Leydig tissue must be removed).
- Versus LHRH agonists: fewer cardiac events, fractures and peripheral arterial disease reported.

Complications of bilateral orchidectomy

A. Surgical complications

- Scrotal haematoma (commonest), wound infection, chronic scrotal or groin pain.

- Psychological: empty scrotum and altered self-image. Prevent with subcapsular or epididymis-sparing technique, or a testicular prosthesis.
- Irreversible: intermittent ADT is impossible.

B. Systemic effects of castration

Effect	Details and management
Hot flushes	Commonest (50–80%); loss of hypothalamic oestrogen signalling; venlafaxine, gabapentin, megestrol or cyproterone if bothersome
Sexual dysfunction	Loss of libido, ED (erectile dysfunction), penile shortening; ED aids help selected men
Osteoporosis, fractures	Fracture risk up 20–45%; baseline DEXA (dual-energy X-ray absorptiometry), FRAX; calcium, vitamin D, exercise; **denosumab 60 mg 6-monthly** or zoledronic acid
Metabolic	Weight gain, fat gain, **sarcopenia**, insulin resistance, diabetes (HR about 1.34), metabolic syndrome
Cardiovascular	Coronary events, sudden cardiac death, thromboembolism
Anaemia	Normocytic; reversible after stopping
Others	Gynaecomastia (≤5%), fatigue, depression, cognitive change

- **Also know:** risk of castration resistance over time; check testosterone with rising PSA.

Follow-up

- PSA every 3–6 months; bone health, metabolic and cardiovascular monitoring on ADT.

Recent advances

- PSMA PET staging; trials adding apalutamide or enzalutamide (ATLAS, ENZARAD); SPCG-15 (RP vs RT + ADT).

Diagrams to draw

- TNM T-stage diagram / sagittal pelvis: T2 confined, T3a extracapsular, T3b seminal vesicle, T4 adjacent organs.
- Subcapsular orchidectomy.

> **Key points to remember:**
> - Seminal vesicle invasion is T3b; non-regional nodes are M1a.
> - Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
> - RT + long-term ADT (18–36 months; EAU 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
> - EORTC 22863 overall survival 79% vs 62%.
> - STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
> - Neoadjuvant ADT before RP: no survival gain.
> - Early salvage RT preferred over routine adjuvant RT for most men after RP.
> - Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

Also, from the condensed book (not in the paper answer):

- pN1: hormones plus RT beat hormones alone (OS 74% vs 55%) (§3).
- Bicalutamide 150 mg raised death in localised disease (HR 1.23) (§5).

1 · Definition and natural history

- The working definition is **T3–4, N0 or N1, M0**, and it calls for **multimodal therapy**.
- NCCN high risk is cT3a, PSA >20 or grade group (GG) 4–5. Very high risk is cT3b–T4, primary pattern 5, ≥2 high-risk features, or >4 cores of GG 4–5.
- Because of stage migration, high risk now mostly means **high grade**, not a palpable T3.
- **Grade drives death**: untreated Gleason 8–10 had 15-yr cancer mortality of 60–87% before the PSA era.

2 · Staging and risk assessment

- Clinical staging and imaging both **understage**.
- NCCN orders a bone scan and CT or MRI for cT3–T4, PSA >20, GG 4–5 or primary pattern 5; PSMA PET may substitute.
- In **proPSMA**, PSMA PET accuracy was **92% vs 65%**.

3 · Radical prostatectomy in high-risk disease

- **Clinical cT3 is overstaged in 7–27%**, and >⅓ of high-risk specimens are confined, so do not exclude men on clinical stage alone.
- NCCN advises RP with extended pelvic node dissection (**ePLND**) only if the tumour is **completely excisable**; otherwise give androgen deprivation (ADT, "AD" in trial reports) with radiotherapy (RT).
- In cT3, DSS after RP is **~80–90% at 10 yr**.

> **KEY POINT**
>
> Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37), higher than extracapsular extension (ECE), seminal vesicle invasion (SVI) or nodes, so remove all prostatic tissue regardless of stage.

Pathology after RP	5-yr PSA-free survival	Relative risk of progression
Organ-confined	95%	Reference
ECE	76%	2.2–2.7
SVI	37%	2.6
Node-positive	18%	3.3
Positive margin	—	**4.37** (highest)

Neoadjuvant AD before RP

- **Clinical downstaging occurs in 32–90%, but pathological in only 8–31%**. It has **no** effect on PSA recurrence, DSS or overall survival (OS).
- **CALGB 90203** (AD plus docetaxel before RP) missed its primary endpoint but improved MFS and OS.

Adjuvant and salvage RT after RP

Trial	Population / comparison	Result
EORTC 22911	pT3 and/or positive margin; immediate RT vs observation	Biochemical disease-free survival 74% vs 53%; **benefit only with positive margins**
SWOG 8794	Same	**MFS (HR 0.71) and OS (HR 0.72)** — the only OS-level evidence
ARO 96-02	pT3N0, undetectable PSA	Biochemical HR 0.53
RADICALS-RT, GETUG-AFU 17, RAVES	Adjuvant vs early salvage (64–66 Gy)	**Early salvage preferred**; 5-yr event-free 87–94% in both arms

- **Trap**: very high-risk men (Gleason 8–10, pT3b) were only ~20% of the timing trials, so adjuvant RT may still help them, as may extensive margins.
- Dose is **≥64 Gy**. **GETUG-AFU 16**: **3 months** of AD with salvage RT gave 10-yr progression-free survival 64% vs 49% (HR 0.54) (see Ch 162).

Adjuvant AD and pN1

- In **pT3N0**, adjuvant AD reduces progression but gives **no DSS or OS gain**.
- **ECOG 7887** (nodes found at RP): immediate AD improved OS, DSS and PFS.
- **AD plus RT beats AD alone** in pN1: 10-yr DSS 86% vs 70%, OS 74% vs 55%.

ALGORITHM

1. **pN1 after RP**
2. If favourable (few nodes, Gleason ≤8), then observe with PSA monitoring.
3. If adverse (>3 nodes, Gleason >8), then give AD plus RT (best OS).
4. Avoid AD alone in a fit man.

4 · RT + ADT

- The standard is **dose-escalated RT (>72, ideally ~78–81 Gy) with ADT**, often with pelvic RT (see Ch 162).
- ADT shrinks the target and is synergistic with RT.

Trial	Question	Verdict
RTOG 86-10	4 mo neoadjuvant/concurrent AD, bulky T2–T4	Disease-specific mortality fell from 36% to 23%; **OS unchanged**
TROG 96.01	RT with no, 3 or 6 mo NAD	6 mo lowered all-cause mortality (HR 0.63); 3 mo did not
RTOG 85-31	Indefinite adjuvant goserelin	Better control and MFS; OS gain only in Gleason 8–10
PCS IV	18 vs 36 mo	OS comparable; 18 mo better quality of life
RTOG 94-13	Pelvic vs prostate RT, and neoadjuvant vs adjuvant AD	Early PFS gain lost; whole pelvis with adjuvant AD had worst OS

- **Duration**: intermediate risk 4–6 mo; high risk (NCCN) **18–36 mo** with external beam RT (EBRT), 12–36 mo with a brachytherapy boost.
- **ASCENDE-RT**: a low-dose-rate boost halved biochemical recurrence vs an EBRT boost (HR 2.04), with equal OS.

- **Trap**: AD plus RT benefits often exclude OS. The OS gain is seen with long AD in high risk (EORTC 22863; RTOG 92-02 Gleason 8–10 OS 81% vs 71%).

Adding RT to ADT

- **SPCG-7**: cancer death relative risk **0.44**; overall mortality RR 0.68. **PR.3/PR07**: DSS HR 0.54, and 7-yr OS rose from 66% to 74%. So RT adds **8–10% OS**.

Systemic intensification

Trial	Added agent	Verdict
RTOG 0521	Docetaxel to long-term AD and RT	OS HR 0.69
GETUG 12	Docetaxel and estramustine to AD	Relapse-free 62% vs 50%
RTOG 9902	Paclitaxel, estramustine and etoposide	No benefit; thromboembolism
STAMPEDE (high-risk M0/N1)	**Abiraterone and prednisolone for 2 yr**, added to 3 yr AD and RT	**MFS HR 0.53, OS HR 0.60**

- STAMPEDE eligibility: **N1, or two of Gleason ≥8, cT3–4 and PSA ≥40**. **NCCN 2023 advises RT, ADT and 2 yr abiraterone for very high risk.**

5 · ADT alone and other options

- **MRC**: immediate rather than deferred AD gave fewer cancer deaths, mainly in M0.
- **Bicalutamide 150 mg** improved PFS in locally advanced disease but **raised death in localised disease** (HR 1.23), so never use it at low risk.
- AD alone does not stop local progression, and over half need a TURP.
- **RP vs RT with ADT**: there is no completed randomised trial; **SPCG-15** (cT3N0M0) is ongoing.

ALGORITHM

1. **Locally advanced / high-risk M0**
2. Stage with PSMA PET, or CT and bone scan.
3. If fit and resectable, then RP with ePLND. If pathology is adverse, then early salvage RT (adjuvant if pT3b, Gleason 8–10 or positive margins). If pN1, then observe or give AD plus RT.
4. Alternative: dose-escalated RT (with or without brachytherapy boost and pelvic nodes) plus 18–36 mo ADT.
5. If very high risk or N1, then add 2 yr abiraterone.
6. If unfit for local therapy, then ADT alone.

KEY POINT

Viva: "Why not ADT alone for T3 disease?" SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-7	ADT with or without RT, mostly T3	Cancer death RR 0.44	Add RT to ADT
NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	OS HR 0.77	Add RT to ADT
EORTC 22863	RT with or without 3 yr goserelin	5-yr OS 62% vs 79%	Long ADT in high risk
RTOG 92-02	4 vs 28 mo ADT	OS gain in Gleason 8–10	Prolong ADT for high grade
STAMPEDE (abiraterone)	ADT and RT, with or without 2 yr abiraterone	MFS HR 0.53, OS HR 0.60	Standard for very high risk/N1
SWOG 8794	Adjuvant RT vs observation	MFS and OS improved	Postoperative RT works
RADICALS-RT / RAVES / GETUG-AFU 17	Adjuvant vs early salvage	No difference	Early salvage for most
ECOG 7887	pN+ immediate vs delayed AD	OS improved	Treat pN+ early
proPSMA	PSMA PET vs conventional imaging	Accuracy 92% vs 65%	PSMA PET for staging

165 Management of Recurrent and Newly Metastatic Prostate Cancer

1 · Biochemical recurrence: concept and definitions

Biochemical recurrence (BCR) is a rise in prostate-specific antigen (PSA) after curative therapy, before clinical recurrence. Clinical recurrence is almost always preceded by it.

- After radical prostatectomy (RP), **¼–⅖** develop BCR within 10 years; only ≈half of higher-risk men with BCR fail clinically.
- At MSK, PSA **≥0.4 ng/mL followed by a further rise** best predicted metastasis.
- **Ultrasensitive PSA** is prognostic, not diagnostic; about half of early values fall on retesting, so repeat it.

Setting	Definition	Logic
Post-RP — AUA	PSA **≥0.2 ng/mL**	No gland remains, so any PSA is suspect
Post-RP — NCCN	Never undetectable, two rises, or PSA ≥0.1	Any of three triggers
Post-RP — EAU	No threshold; risk groups (below)	Kinetics plus grade
After external beam RT (EBRT) ± androgen deprivation therapy (ADT)	**Phoenix**: **nadir + 2 ng/mL**	Validated against metastasis and death
After low-dose-rate (LDR) brachytherapy	≥0.2 may suit better; **<0.2 at 4 years ≈ cure**	The very low nadir means Phoenix may flatter brachytherapy

EAU BCR risk group	After RP	After radiotherapy (RT)
Low risk (needs BOTH)	PSA doubling time (PSADT) **>1 year** AND grade group (GG) <4	Interval to failure **>18 months** AND GG <4
High risk (EITHER)	PSADT ≤1 year OR GG 4–5	Interval ≤18 months OR GG 4–5

> **KEY POINT**
>
> **PSA bounce** after RT is a transient rise of 0.1–0.5 ng/mL at a mean of ≈ **9 months**, followed by a fall. It is **favourable**, so monitor; never salvage on a bounce.

2 · Risk prediction and natural history

- A **positive margin** is the only modifiable BCR predictor.
- **CAPRA** is the pre-operative score and **CAPRA-S** the post-operative one MCQ. Only the MSK nomograms predict 15-year cancer-specific survival.
- **Pound**: untreated, a median **8 years** passes from BCR to metastasis, then **5 years** to death (P2-144).
- After Phoenix failure, metastasis follows at ≈5.4 years. Follow-up is lifelong, because ¼ recur after 5 years.

3 · Imaging at recurrence

P2 *65 yr male underwent RP... PSA has risen — BCR definitions, PSMA PET/mpMRI work-up, salvage RT dose/timing, RTOG 9601, GETUG-AFU 16, SPPORT, STOMP/ORIOLE, EMBARK, Pound prognosis*

- Bone scan and CT are rarely positive below PSA **10 ng/mL**.
- **Prostate-specific membrane antigen (PSMA) PET** is positive in ≈½ at PSA **<0.5**, and finds disease outside the salvage field in ≈1 in 5. Fluciclovine-guided salvage improved control (**EMPIRE-1**).
- Multiparametric MRI (mpMRI) is best for local recurrence. **Biopsy after RT** is mandatory before local salvage.

> **KEY POINT**
>
> Salvage RT works best at the lowest PSA, but imaging improves only as PSA rises, so risk features (grade, margins, kinetics) guide early action.

4 · Salvage after radical prostatectomy

- **Salvage RT (SRT)** is the only curative option (P2-144). Give **≥64–65 Gy** to the fossa.
- Delay costs control: ≈10% relapse-free survival is lost per 0.1 ng/mL in men with ≥2 adverse features.
- No visible lesion on mpMRI predicts SRT failure, because the disease is probably occult metastasis.

5 · Salvage after radiotherapy

Prerequisites are biopsy-proven local recurrence, no metastasis on imaging, excellent performance status, life expectancy ≥5–10 years and a shared decision. Every modality gives ≈**50–70%** control, so selection (local-only disease, low PSA, GG <4) matters more than modality.

Option	Selection / features	Signature morbidity
Salvage RP (open/robotic)	EAU: life expectancy ≥10 years, PSA **<10**, biopsy GG ≤3, N0M0 on imaging, initial cT1–T2.	Incontinence, bladder neck contracture, rectal injury, near-universal impotence
Salvage brachytherapy (LDR ≈ high-dose-rate)	Only after EBRT	Grade ≥3 urinary toxicity ≈1 in 6
Salvage stereotactic body RT (SBRT)	After EBRT	Low toxicity, modest control
Salvage cryotherapy	PSA <10 (ideally <4), PSADT **≥16 months**; a post-cryo nadir <0.2 predicts success	**Urethrorectal fistula** (1–3%), erectile dysfunction; incontinence single-digit with **urethral warming**
Salvage high-intensity focused ultrasound (HIFU)	≥2 years after EBRT	Fistula, osteitis pubis, stricture, incontinence

Recurrence after **focal therapy** is worked up like RT failure.

6 · Non-curative approach to BCR: observation vs ADT

- AUA: observe (or enrol in a trial) after maximal local therapy; give ADT only for higher metastatic risk (PSADT **<12 months** or high grade), preferably intermittent.
- EAU advises observing low-risk BCR, limited life expectancy, or men declining local therapy. Primary ADT for localised disease is **not** recommended.
- **Messing (ECOG)**: immediate ADT for node-positive (pN+) disease after RP improved overall survival (OS), but the trial was small and pre-dated modern imaging.
- Low-volume pN+ with undetectable PSA: **20–40%** are cured by surgery alone.
- Immediate ADT: TOAD OS hazard ratio (HR) 0.55; EORTC 30891 benefit only if PSA >50 or PSADT <12 months.
- **EMBARK** entry: PSADT **≤9 months**, with PSA ≥1 after RP or ≥2 above nadir after RT.

Intermittent vs continuous ADT

- **NCIC PR.7** (rising PSA after RT): intermittent was **non-inferior** (OS 8.8 vs 9.1 years).
- **SWOG 9346** (new metastatic): OS 5.1 vs 5.8 years, HR 1.10 (90% CI 0.99–1.23). This was **inconclusive**; intermittent may be worse here.

7 · Androgen deprivation therapy: biology and agents

Castration removes the ligand, not the receptor, so castration-resistant prostate cancer (CRPC) keeps androgen receptor (AR) signalling alive through AR amplification, ligand-binding-domain mutations, splice variants (**AR-V7**, poor prognosis) and tumour steroid synthesis. **Castrate testosterone: <50 ng/dL**.

SAQ GnRH (LHRH) analogues: agonists and antagonists

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "GnRH (LHRH) analogues: agonists and antagonists" (GNRH Analogue; LHRH agonist) · P2 SAQ "LHRH analogue"

Read as: GnRH (gonadotrophin-releasing hormone, also LHRH, luteinising hormone-releasing hormone) agonists and antagonists, chiefly in prostate cancer.

Physiology

- **LHRH (GnRH)** is a hypothalamic decapeptide released in **pulses**. Pulsatile GnRH drives pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone). LH → Leydig cells → testosterone (about 95% testicular; about 95% of total).
- **Continuous** exposure **desensitises and down-regulates** the pituitary receptor. This is the basis of agonist therapy.

GnRH (LHRH) agonists: drugs

Drug	Formulation
Leuprolide	Depot 1, 3, 6 monthly
Goserelin	SC (subcutaneous) implant 1 or 3 monthly
Triptorelin	IM (intramuscular) depot
Histrelin	Implant, 1 year
Buserelin	Nasal or SC

- Substitution at **position 6** of the decapeptide raises potency and half-life; depots last 1 month to 1 year.

Agonists: mechanism

`Agonist → initial LH surge (flare, testosterone up) → receptor down-regulation (1–2 weeks) → LH falls → castrate testosterone in 2–4 weeks`

- Survival **equals orchidectomy**.

Flare phenomenon

- LH rises up to **10-fold for 10–20 days** → bone pain, **spinal cord compression**, urinary retention.
- Dangerous with spinal or weight-bearing metastases.
- **Prevent with an antiandrogen** (e.g. bicalutamide, flutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → **antagonist** (degarelix) or orchidectomy instead.

GnRH antagonists

- Immediate competitive receptor block: LH falls 84% in 24 hours, **no flare**, FSH falls.
- **Degarelix** (SC monthly): castrate by day 3 in >95%; injection-site reactions.
- **Relugolix** (oral daily, FDA 2020). **HERO** vs leuprolide: sustained castration 97% vs 89%; faster recovery after stopping; **fewer MACE (major adverse cardiovascular events)** (about 3% vs 6%).
- Preferred for **impending cord compression** or severe bone pain when orchidectomy is unsuitable.
- **Also know:** abarelix restricted (severe allergy); PRONOUNCE (degarelix vs leuprolide, stopped early) showed no MACE difference.

Agonist vs antagonist

Agonist	Antagonist
Flare; needs antiandrogen	No flare; castrate by day 3 (degarelix)
FSH rises	FSH falls
Depot to yearly	Monthly SC or oral (relugolix; HERO)

Castrate level

- **<50 ng/dL** (traditional); **<20 ng/dL** is achievable (orchidectomy median about 15) and may be better (aim **<20 ng/dL**). Check testosterone when PSA (prostate-specific antigen) rises.

Indications in urology

- Metastatic hormone-sensitive prostate cancer (with ARPI [androgen receptor pathway inhibitor] or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- Recurrent prostate cancer: BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** intermittent ADT (androgen deprivation therapy) acceptable (PR.7); BPH (benign prostatic hyperplasia) volume reduction (not routine); central precocious puberty;

cryptorchidism (GnRH or hCG [human chorionic gonadotrophin]: low success, not first-line); endometriosis.

Adverse effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction).
- **Osteoporosis:** DEXA (dual-energy X-ray absorptiometry); calcium, vitamin D, denosumab or zoledronic acid.
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%), cognitive change.
- Slow testosterone recovery after stopping.

Diagram to draw

- Hypothalamic–pituitary–testis axis showing agonist and antagonist sites, and the flare.

> **Key points to remember:**
> - Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
> - Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
> - Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
> - Castrate <50 ng/dL; aim <20 ng/dL.

SAQ Maximum / combined androgen blockade

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Maximum (combined) androgen blockade and bicalutamide" (Maximum androgen blockade; Bicalutamide) · P4 SAQ "Combined Androgen Blockade."

Definition and rationale

- **MAB (maximum androgen blockade)**, also **CAB (combined androgen blockade)** or total androgen blockade: castration (orchidectomy or LHRH [luteinising hormone-releasing hormone] agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes about 90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) inside the prostate. The antiandrogen blocks these at the AR (androgen receptor).

Antiandrogens

Drug	Dose	Signature toxicity
Flutamide	250 mg three times daily	Diarrhoea, hepatotoxicity
Nilutamide	300 mg, then 150 mg daily	Delayed dark adaptation, pneumonitis
Bicalutamide	**50 mg daily** (with LHRH agent)	Best tolerated; gynaecomastia
Cyproterone (steroidal)	100 mg two to three times daily	Cardiovascular, hepatic; no CAB benefit, not used for CAB now

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

Evidence for MAB

- **Crawford (1989):** leuprolide + flutamide improved survival (partly flare control); **Eisenberger (1998):** orchidectomy + flutamide, **no** benefit.
- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, about 8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **about 2–3%** (borderline significance); cyproterone gave no benefit (slightly worse).
- Small gain, more cost and toxicity: **AUA (American Urological Association) advises against long-term CAB**.

Current uses of MAB

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) up to 10-fold, and testosterone, for 10–20 days, risking cord compression, pain or obstruction with bone metastases. Start antiandrogen **with or before** the agonist for **21–28 days**. Antagonists (degarelix, relugolix) or orchidectomy cause no flare.
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives a PSA (prostate-specific antigen) fall >50% in **15–30%** for 3.5–5 months (mutant AR turns the drug into an agonist).

Bicalutamide: introduction and mechanism

- Non-steroidal **AR antagonist**, called first- or second-generation. It competes with testosterone and DHT at the ligand-binding domain.
- Given alone, it blocks central feedback, so **LH and testosterone rise** (about 1.5 times); aromatisation to oestradiol causes gynaecomastia; libido is partly preserved.
- **Half-life about 6 days** (once daily); unaffected by age or renal impairment. Most potent and best tolerated of its class.

Uses of bicalutamide

Use	Dose	Evidence
CAB with LHRH agonist, metastatic	50 mg daily	Small survival gain (2–3%, PCTCG)
Flare prevention	50 mg, with or before the agonist, 2–4 weeks	Main use today
Monotherapy, locally advanced	150 mg daily	**EPC (Early Prostate Cancer) programme:** better progression-free survival; **harm in localised disease** (death HR [hazard ratio] 1.23)
With salvage radiotherapy	150 mg for 2 years	**RTOG 9601:** 12-year overall survival 76.3% vs 71.3%
CRPC (castration-resistant prostate cancer)	50 mg	Second-line step; largely replaced by newer agents

- 150 mg monotherapy roughly equals castration in locally advanced disease, with better sexual interest; not FDA (US Food and Drug Administration) approved alone, but used in some countries.
- **Also know:** stuttering priapism; feminising hormone therapy; studied in BPH (benign prostatic hyperplasia).

Adverse effects of bicalutamide

- **Gynaecomastia and breast pain** (two-thirds at 150 mg): prevent with breast radiotherapy (10–15 Gy single fraction) or tamoxifen 10 mg daily.
- **Hepatotoxicity** (monitor liver tests); rare **interstitial pneumonitis**; erectile dysfunction; fewer hot flushes than castration.

Modern intensification (replaces CAB in mHSPC, metastatic hormone-sensitive prostate cancer)

Trial	Addition to ADT (androgen deprivation therapy)	Result
LATITUDE, STAMPEDE (arm G)	Abiraterone + prednisone	OS (overall survival) HR 0.62 / 0.63
ENZAMET	Enzalutamide (vs ADT + older antiandrogen, a CAB control arm)	OS HR 0.67; enzalutamide beat CAB
ARCHES	Enzalutamide	Radiographic PFS (progression-free survival) HR 0.39
TITAN	Apalutamide	OS HR 0.67
PEACE-1, ARASENS	**Triplet**: + docetaxel + abiraterone or darolutamide	OS benefit (ARASENS HR 0.68)

- Docetaxel is also an option with ADT.
- **Comparison:** enzalutamide beat bicalutamide in CRPC (TERRAIN, STRIVE) and in **ENZAMET**.

Diagrams to draw

- Testicular and adrenal androgen sources with LHRH agonist and antiandrogen sites of action.
- Hypothalamus–pituitary–testis axis with bicalutamide blocking the AR.

Key points to remember:

- MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Method	Key facts
Bilateral orchiectomy	Castrate within hours. Fewer cardiac events and fractures than agonists with equal survival, so offer it for lifelong ADT
LHRH agonists / antagonists	See P4-103. HERO: castrate by day 4 in 56% vs 0%
Cyproterone acetate	Inferior to goserelin; cardiovascular events up to 10%; used 50–100 mg/day for hot flushes

Method	Key facts
AR-targeting agents (ARTAs): enzalutamide 160 mg, apalutamide 240 mg, darolutamide 600 mg twice daily	Seizures with enzalutamide; rash and hypothyroidism with apalutamide; darolutamide has the fewest effects
Abiraterone (selective CYP17), ketoconazole (non-selective)	Give abiraterone with prednisone to prevent ACTH-driven mineralocorticoid excess. Ketoconazole castrates within 4 h: an emergency option

KEY POINT

Flare matters with spinal or weight-bearing bone metastases: give an antiandrogen with the agonist for **21–28 days**. For **impending cord compression**, use an **antagonist or orchiectomy** (or ketoconazole). **Antiandrogen withdrawal**: 15–30% respond with bicalutamide and 0–10% with ARTAs, with no OS gain.

8 · ADT side effects and their management

Effect	Key facts	Management
Osteoporosis / fracture	Loss of peripheral oestrogen; fracture risk rises 20–45%	Baseline DEXA and FRAX. For all: calcium 1000–1200 mg and vitamin D 400–1000 IU/day, exercise, stop smoking. Give a drug if T-score is −1 to −2.5 with 10-year hip fracture risk **>3%** or major fracture **>20%** (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly or alendronate 70 mg weekly. Repeat DEXA at 1 year
Hot flushes (commonest, 50–80%)	Hypothalamic oestrogen loss	Oestrogens best; megestrol, cyproterone, venlafaxine, gabapentin. Clonidine is ineffective
Metabolic	Fat rises with sarcopenia; diabetes (agonist HR 1.44)	Exercise; monitor glucose and lipids
Cardiovascular	Agonists raise coronary events, sudden death and thromboembolism	Risk-factor control; orchiectomy or antagonist favoured
Anaemia	Normocytic; pre-ADT Hb <12 g/dL predicts shorter survival	Reversible ≈1 year after stopping
Gynaecomastia	≈½ with bicalutamide 150 mg or enzalutamide; ≤5% with castration	Prophylactic breast RT 10–15 Gy or tamoxifen

Follow-up on ADT

Check PSA and bloods 3–6-monthly. **If PSA rises, check testosterone first.** A rising PSA with castrate testosterone means **CRPC** (see Ch 166), so restage. The germline HSD3B1 (1245A>C) variant predicts ADT resistance MCQ.

9 · Metastatic hormone-sensitive prostate cancer (mHSPC)

LAQ Metastatic prostate cancer, including newly diagnosed and bone metastases

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Metastatic prostate cancer with bone metastases (retention and bone pain; bone mets; zoledronic acid)" (Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain; Management of bone mets in prostatic CA; also SAQ: Zolendronic acid) · P3 SAQ "Management of newly Metastatic prostate cancer"

Zolendronic acid: read as zoledronic acid.

Introduction

- Bone is the commonest metastatic site; visceral metastasis occurs in <20%.
- Aims: control the cancer, relieve pain, **prevent SREs (skeletal-related events)** and treat emergencies (cord compression, fracture).
- **Retention + bone pain case:** this is newly diagnosed (synchronous) **mHSPC (metastatic hormone-sensitive prostate cancer)** with two urgent problems: an obstructed outlet and painful bone metastases. Treat the emergencies first, start castration safely (no flare), then intensify systemic therapy.

Definitions

- **mHSPC:** no prior hormones. **Synchronous** (at diagnosis) or **metachronous** (after local therapy).
- **CHAARTED high volume:** ≥4 bone metastases with ≥1 outside the axial skeleton, or any visceral metastasis.
- **LATITUDE high risk:** ≥2 of Gleason ≥8, ≥3 bone lesions, visceral metastasis.

Pathophysiology of bone metastases

- Tumour cells lodge in axial marrow, adhering through integrins and collagen receptors.
- Metastases are mainly **osteoblastic (sclerotic)**, but resorption also rises. Tumour cytokines induce **RANKL (receptor activator of nuclear factor-κB ligand)**, which activates osteoclasts; resorbed bone releases growth factors that feed the tumour (**vicious cycle**). ADT (androgen deprivation therapy) adds osteoporosis through loss of oestrogen.
- **SREs:** pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone. Marrow replacement causes anaemia. Hypercalcaemia is rare and suggests neuroendocrine disease.

Evaluation (immediate assessment and work-up)

- **Clinical:** pain, neurological signs, performance status. **Ask and examine for cord compression:** back pain, leg weakness, sensory level, bowel symptoms. DRE (digital rectal examination).
- **Bloods:** PSA (prostate-specific antigen), ALP (alkaline phosphatase), calcium, creatinine, haemoglobin, testosterone.
- **Imaging:** ultrasound for hydronephrosis and residual; **bone scan and CT (computed tomography)** define volume; PSMA (prostate-specific membrane antigen) PET (positron emission tomography) finds more and is more sensitive, but trials used conventional imaging. **Plain X-ray or CT of painful weight-bearing sites** (lytic lesion, impending

fracture). **Whole-spine MRI (magnetic resonance imaging)** if any back pain or neurological sign.
- **Histology:** prostate biopsy (or TURP [transurethral resection of the prostate] chips).
- **Germline (and somatic) genetic testing** for all men with metastatic disease (>10% have DNA-repair mutations).
- **Volume (CHAARTED):** see definitions above.

Step 1: Relieve the retention

- **Urethral catheter** (suprapubic if urethral fails). Watch for post-obstructive diuresis.
- Persisting renal impairment with hydronephrosis means ureteric obstruction by tumour at the trigone: **nephrostomy or stent**.
- ADT shrinks the prostate, so try **TWOC (trial without catheter)** after a few months of ADT.
- If TWOC fails or symptoms persist: **channel (palliative) TURP**. ADT alone does not prevent local progression (older series: about a third needed TURP within 60 days of orchidectomy).

Step 2: Systemic treatment, hormone-sensitive (mHSPC): start ADT at once, without flare

- **ADT is the backbone, never alone** (orchidectomy, LHRH [luteinising hormone-releasing hormone] agonist or antagonist):

Option	Point
Bilateral orchidectomy	Castrate within hours; best for impending cord compression
GnRH (gonadotropin-releasing hormone) antagonist: degarelix 240 mg subcutaneously, then 80 mg monthly; oral relugolix	**No flare**; castrate by day 3 in >95% (degarelix); for impending cord compression
LHRH agonist + antiandrogen	Agonist raises LH (luteinising hormone) up to 10-fold, raising testosterone, for 10–20 days (**flare**). Give bicalutamide 50 mg daily with or before the agonist, for 21–28 days

- **Why avoid flare:** the testosterone surge can worsen bone pain, retention and cord compression in spinal metastases. Cover it with an antiandrogen for 21–28 days, or use an antagonist (degarelix) or orchidectomy.
- Target castrate testosterone **<50 ng/dL**. Antiandrogen alone is not recommended.

Intensified systemic therapy in mHSPC (doublets and triplets)

- ADT alone is no longer standard. Classify volume (CHAARTED) and risk (LATITUDE).

Add to ADT	Trial	Overall survival HR (hazard ratio)
Docetaxel 75 mg/m², 6 cycles	**CHAARTED** (benefit in high volume only); STAMPEDE	0.61; 0.78
Abiraterone 1000 mg + prednisone 5 mg	**LATITUDE**; STAMPEDE arm G	0.62; 0.63
Enzalutamide (ARPI, androgen receptor pathway inhibitor)	ARCHES (radiographic PFS [progression-free survival]); ENZAMET	0.67 (ENZAMET)

Add to ADT	Trial	Overall survival HR (hazard ratio)
Apalutamide (ARPI)	**TITAN**	0.67
Triplet: docetaxel 75 mg/m² × 6 + abiraterone	**PEACE-1** (synchronous)	0.75 (with docetaxel)
Triplet: docetaxel + darolutamide	**ARASENS**	0.68

- Hormonal agents work **regardless of volume**; docetaxel helps only high volume (GETUG-AFU 15 negative), so it is now given within a triplet. Triplets suit chemotherapy-fit men, especially high volume.
- *Guideline note:* the chapter lists ADT plus docetaxel as an option; current EAU (European Association of Urology) guidance gives docetaxel only within a triplet.

Local and metastasis-directed therapy in mHSPC

- **RT (radiotherapy) to the prostate in low-volume synchronous disease only:** STAMPEDE arm H, survival HR 0.68 (3-year 81% vs 73%); it also helps local control. HORRAD negative overall.
- Cytoreductive prostatectomy: trials only.
- **MDT (metastasis-directed therapy)** / **SBRT (stereotactic body RT)** for oligometastases (STOMP, ORIOLE) delays ADT; SBRT also for spinal re-irradiation.

Castration-resistant disease (mCRPC)

- ARPIs, docetaxel then cabazitaxel, PARP (poly-ADP-ribose polymerase) inhibitors for HRR (homologous recombination repair) mutations.
- **Radium-223** (alpha emitter, path <100 µm, spares marrow): for **symptomatic bone-only** disease (licensed only in castration-resistant disease); 50 kBq/kg every 4 weeks × 6. **ALSYMPCA:** overall survival 14.9 vs 11.3 months; delayed SREs. **Never with abiraterone** (more fractures and deaths).
- **^{177}Lu-PSMA-617** (beta emitter, PSMA-avid disease at any site): after ARPI and taxane; **VISION:** survival 15.3 vs 11.3 months.

Step 3: Bone pain

- **Analgesia** by the WHO (World Health Organization) analgesic ladder; neuropathic pain needs gabapentinoids or tricyclics; steroids for diffuse pain.
- **Focal pain: EBRT (external beam radiotherapy) single 8 Gy fraction.** Pain relief equals multiple fractions (30 Gy in 10 fractions, RTOG 9714); retreatment is commoner.
- X-ray or CT of painful weight-bearing sites to exclude lytic lesions and impending fracture; **orthopaedic fixation** then RT.
- **Diffuse pain:** steroids, wide-field RT; radiopharmaceuticals (strontium-89, samarium-153 relieve pain only; radium-223 adds survival, licensed only in castration-resistant disease).

Bone-targeted agents

Feature	Zoledronic acid	Denosumab
Mechanism / target	Bisphosphonate: inhibits osteoclast enzyme	Antibody against RANKL
SRE dose (mCRPC) / route	4 mg IV every 4 weeks, adjusted for creatinine clearance	120 mg SC (subcutaneous) every 4 weeks

Feature	Zoledronic acid	Denosumab
Renal adjustment	Yes	No
Evidence	Fewer SREs than placebo	Time to first SRE 20.7 vs 17.1 months vs zoledronate
Toxicity	Fever, myalgia, renal impairment, hypocalcaemia, ONJ	Hypocalcaemia, hypophosphataemia, ONJ 2–4%

- **Neither improves survival**; they protect the skeleton only.
- **In mHSPC, SRE doses are not indicated** (no benefit: CALGB 90202, STAMPEDE). SRE doses (zoledronic acid 4 mg or denosumab 120 mg 4-weekly) are for CRPC (castration-resistant prostate cancer). Use osteoporosis doses for ADT bone loss: denosumab 60 mg 6-monthly or zoledronic acid yearly, by DEXA (dual-energy X-ray absorptiometry) and FRAX.
- **ONJ (osteonecrosis of the jaw) prevention:** dental check before starting; calcium 1000 mg and vitamin D 800 IU daily.
- Give a bone agent with radium-223 or any ARPI.

Zoledronic acid (in detail)

- **Introduction:** a potent IV (intravenous) **nitrogen-containing bisphosphonate**.
- **Mechanism:**
- Binds hydroxyapatite at sites of active bone turnover and is taken up by **osteoclasts**.
- Inhibits **farnesyl pyrophosphate synthase** (mevalonate pathway), causing osteoclast dysfunction and apoptosis, so resorption falls.
- Breaks the "vicious cycle".
- Excreted unchanged by the kidneys.

Indication	Dose
SRE prevention, mCRPC (metastatic castration-resistant prostate cancer) and other bone metastases (RCC [renal cell carcinoma], bladder)	**4 mg IV over ≥15 min every 4 weeks**, adjusted for creatinine clearance
ADT bone loss (by DEXA and FRAX)	4–5 mg IV yearly
Hypercalcaemia of malignancy	4 mg IV with saline hydration

- 12-weekly dosing after the first doses is non-inferior to 4-weekly (CALGB 70604).
- **Not indicated at SRE doses in mHSPC:** no benefit (CALGB 90202, STAMPEDE).
- **Evidence: Zometa 039 trial (mCRPC):** fewer SREs than placebo (about 33% vs 44%). **Denosumab vs zoledronate (1904 men, mCRPC):** time to first SRE 20.7 vs 17.1 months. **Neither improves survival.**
- **Precautions and monitoring:** creatinine before every dose; avoid if creatinine clearance <30 mL/min. **Dental review before starting**; avoid invasive dental work during therapy. Calcium and vitamin D supplements.
- **Adverse effects: acute-phase reaction** (fever, myalgia) after the first infusion; nephrotoxicity, hypocalcaemia, hypophosphataemia; **ONJ**, atypical femoral fracture; ocular inflammation (rare).

Emergencies

Spinal cord compression: watch constantly (every focal back pain is cord compression until proven otherwise): `Back pain ± weakness, sensory level, sphincter change →`

dexamethasone 10 mg IV load, then 4–10 mg 6–12-hourly → urgent whole-spine MRI → decompressive surgery + RT (unstable spine, progression on RT, prior RT) or RT alone → orchidectomy or antagonist if hormone-naive (never agonist alone)

- Surgery plus RT beat RT alone for walking ability (**Patchell, 2005**).

Pathological or impending fracture: orthopaedic fixation then RT. **Mirels score** ≥9 (site, pain, lytic type, size) favours prophylactic fixation.

Algorithms

Retention + bone pain + metastases ├─ Catheter ± nephrostomy; cord signs? → dexamethasone → MRI → RT ± surgery ├─ Orchidectomy / antagonist / agonist + antiandrogen ├─ Analgesia + 8 Gy single-fraction RT; bone protection ├─ Add ARPI or triplet; prostate RT if low volume └─ TWOC after ADT → channel TURP if it fails

- Newly diagnosed mHSPC: split by volume and fitness to doublet or triplet ± prostate RT.

Follow-up

- PSA, ALP, liver tests, calcium, renal function and blood count every 3–6 months; testosterone if PSA rises; new scan if symptoms or PSA rise. Rising PSA with castrate testosterone = CRPC.
- PSA nadir ≤0.2 ng/mL at 7 months predicts long survival (SWOG 9346).
- PCWG3 (Prostate Cancer Working Group 3): bone scan progression needs ≥2 new lesions, confirmed later (early "flare" misleads).
- Manage ADT effects: bone loss (DEXA), diabetes, cardiovascular risk, hot flushes.
- **Also know:** stent or nephrostomy for obstruction; marrow-failure anaemia; hospice care.

Diagrams to draw

- The retention + bone pain flowchart above.
- Vicious cycle: tumour → RANKL → osteoclast → growth factors → tumour, with the sites blocked by denosumab and zoledronate.
- Algorithm splitting newly metastatic disease by volume and fitness to doublet or triplet ± prostate RT.

Key points to remember:

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: antiandrogen cover, antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief for focal pain; focal back pain = cord compression until MRI proves otherwise; dexamethasone and MRI.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance; no survival benefit.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

Also, from the condensed book (not in the paper answer):

- **SWOG 9346 PSA nadir at 7 months**: ≤0.2 gives median OS **75** months; 0.2–4 gives 44; >4 gives 13.
- Metachronous low volume does best (median OS ≈92 vs ≈43 months for synchronous high volume).
- Genes to test: germline BRCA1/2, ATM, PALB2, CHEK2, HOXB13, mismatch repair; somatic CDK12, MSI-H.

Definition	Criteria	Use
CHAARTED high volume	**≥4 bone metastases with ≥1 outside the axial skeleton**, OR any visceral metastasis (no grade criterion)	Docetaxel benefit
LATITUDE high risk	**≥2 of**: Gleason ≥8, ≥3 bone lesions, visceral	Abiraterone entry

- **ADT is the floor, never the ceiling**; long-term combined blockade is not recommended.
- Docetaxel is **75 mg/m² 3-weekly × 6** (meta-analysis HR 0.77).
- Observe only if asymptomatic with life expectancy <5 years. Cytoreductive prostatectomy is for trials only.
- **Metastasis-directed therapy (MDT)**: oligometastases are **≤3 sites** on PET, treated by SBRT or salvage node dissection.

10 · Landmark trials

Trial	Design	Result	Take-home
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Adjuvant vs early salvage RT (PSA 0.1–0.2)	Equivalent; ≥60% spared RT	Early salvage preferred
RTOG 0534 (SPPORT)	Bed vs bed plus 4–6 months ADT vs pelvis plus ADT	5-year freedom from progression 70.9 / 81.3 / **87.4%**	Add ADT and pelvic nodes (not OS)

Trial	Design	Result	Take-home
RTOG 9601	SRT ± 2 years bicalutamide 150 mg	12-year OS 76.3 vs 71.3%	OS benefit
GETUG-AFU 16	SRT ± 6 months goserelin	10-year BCR-free 64 vs 49% (HR 0.54); MFS HR 0.73; OS unchanged	Short ADT: no OS gain
EMBARK	High-risk BCR; leuprolide ± enzalutamide vs enzalutamide	MFS 87.3% (combination) / 80.0% (enzalutamide) / 71.4% (leuprolide); HR 0.42 and 0.63	Enzalutamide for high-risk non-metastatic HSPC
CHAARTED (ECOG 3805)	ADT ± docetaxel	OS 57.6 vs 44.0 months (HR 0.61); high volume **49.2 vs 32.2** months	High volume only
GETUG-AFU 15	ADT ± docetaxel	OS HR 0.88, not significant	Negative docetaxel trial
STAMPEDE (docetaxel; arm G abiraterone)	Multi-arm; metastatic and high-risk M0	Docetaxel OS HR 0.78; abiraterone OS HR 0.63, failure-free survival HR 0.29	Both work
LATITUDE	High-risk mHSPC; ADT ± abiraterone	OS HR 0.62; rPFS 33.0 vs 14.8 months	Abiraterone in high risk
ARCHES	ADT ± enzalutamide	rPFS HR 0.39	Volume-independent
ENZAMET	ADT plus enzalutamide vs ADT plus older antiandrogen	OS HR 0.67	Control was not placebo
TITAN	ADT ± apalutamide	rPFS HR 0.48; OS HR 0.67	Volume-independent
PEACE-1	Synchronous; 2×2, ± abiraterone and prostate RT	All: rPFS HR 0.54, OS HR 0.82; with docetaxel: rPFS 0.50, OS 0.75	Triplet with abiraterone
ARASENS	ADT and docetaxel, ± darolutamide	OS HR 0.68	Darolutamide in mHSPC only with docetaxel
STAMPEDE arm H	ADT ± prostate RT	Overall HR 0.92; low volume HR 0.68 (3-year OS 81 vs 73%)	RT to prostate in low volume

166 Management of Castration-Resistant Prostate Cancer

1 · Definition and biology

CRPC means progression (biochemical, radiographic or clinical) despite castrate testosterone **<50 ng/dL**. On androgen deprivation therapy (ADT) alone, metastatic overall survival (OS) was historically 24–36 months.

LAQ Castration-resistant prostate cancer (CRPC): definition and treatment

Paper answer · P2

Asked as:

- CRPC in Ca prostate. — *Summer 2020 Paper II Q3a (SAQ)*
- Discuss treatment of CRPC — *Practice Set 6 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2020-Summer Paper 2, Q3a)

Definition (EAU [European Association of Urology])

- CRPC (castration-resistant prostate cancer): castrate testosterone **<50 ng/dL (1.7 nmol/L)** plus either:
- **Biochemical progression:** three consecutive PSA (prostate-specific antigen) rises 1 week apart, two of them 50% above nadir, with PSA >2 ng/mL; or
- **Radiological progression:** ≥2 new bone lesions, or soft-tissue progression by RECIST (Response Evaluation Criteria in Solid Tumours).

Why it happens: the AR still drives growth

- Castration-resistant is **not** androgen-independent, so treatment first targets the AR (androgen receptor) more deeply.
- Mechanisms: AR overexpression and amplification; AR mutation (activated by other steroids); **splice variants lacking the ligand-binding domain**; adrenal and intratumoural androgen synthesis (CYP17 overexpressed); bypass pathways (PI3K/AKT, PTEN loss in ~40%).
- **Lineage plasticity:** neuroendocrine change with low PSA, visceral and lytic disease (visceral disease).

Evaluation

- Testosterone (exclude non-compliance), PSA, **PSADT (PSA doubling time)**; CT and bone scan; PSMA (prostate-specific membrane antigen) PET increasingly.
- Prognosis: performance status, pain, haemoglobin, ALP (alkaline phosphatase), LDH (lactate dehydrogenase), visceral disease.
- **Germline testing** in all metastatic disease; somatic sequencing of tumour tissue (germline and tumour testing) for HRR (homologous recombination repair) genes such as BRCA2, and MSI (microsatellite instability).
- Biopsy a metastasis if disease progresses fast with a low PSA (neuroendocrine change).

- PCWG3 (Prostate Cancer Clinical Trials Working Group 3): early bone-scan lesions may be **flare**; re-baseline at 12–16 weeks, then progression = ≥2 new lesions.

Clinical states model (PCWG3): `Localised → rising PSA → mCSPC (castration-sensitive) ↓ (on ADT) nmCRPC → mCRPC 1st line → 2nd line → 3rd line`

General principles of management

1. **Continue ADT (androgen deprivation therapy) lifelong**, including during chemotherapy; try antiandrogen withdrawal.
2. **Antiandrogen withdrawal** (e.g. bicalutamide): watch PSA 4–8 weeks; brief responses in a minority.
3. Choose by prior therapy, symptoms, visceral disease, genomics and comorbidity.
4. **Sequencing: switch mechanism**; a second ARPI rarely works (cross-resistance). Cross-resistance means a second ARPI (AR-pathway inhibitor) helps only 15–30%.

nmCRPC (non-metastatic)

- **PSADT ≤10 months:** add an ARPI. Each improved MFS (metastasis-free survival) and overall survival (improve metastasis-free and overall survival):
- Apalutamide 240 mg (**SPARTAN**: MFS HR 0.28); enzalutamide 160 mg (**PROSPER**); darolutamide 300 mg twice daily (**ARAMIS**: MFS 40.4 vs 18.4 months).
- Slower PSADT: continue ADT and monitor.

mCRPC: life-prolonging options

Class	Agent and dose	Key trial (overall survival)
ARPI (AR-pathway inhibitor)	**Abiraterone** 1000 mg fasting + prednisone	COU-AA-301: 14.8 vs 10.9 months; COU-AA-302 (pre-chemotherapy)
	Enzalutamide 160 mg	AFFIRM: 18.4 vs 13.6 months; PREVAIL
Taxane	**Docetaxel** 75 mg/m² 3-weekly + prednisone	TAX 327: 18.9 vs 16.4 months (mitoxantrone)
	Cabazitaxel 20–25 mg/m² after docetaxel	TROPIC: 15.1 vs 12.7; CARD: beats a second ARPI
Radioligand (mCRPC)	**Radium-223** 50 kBq/kg 4-weekly ×6 (symptomatic bone-only, no visceral)	ALSYMPCA: 14.9 vs 11.3 months
	^{177}Lu-PSMA-617 (PSMA-positive, after ARPI + taxane)	VISION: 15.3 vs 11.3 months
PARP (poly-ADP-ribose polymerase) inhibitor	Olaparib (HRR genes); rucaparib (BRCA)	PROfound; TRITON3
	Olaparib + abiraterone; niraparib + abiraterone; talazoparib + enzalutamide (first line)	PROpel; MAGNITUDE; TALAPRO-2
Immunotherapy	**Sipuleucel-T** (asymptomatic, no visceral disease)	IMPACT: 25.8 vs 21.7 months
	Pembrolizumab for MSI-high/dMMR (mismatch repair-deficient), 2–5%	Tumour-agnostic approval

- **Abiraterone** blocks CYP17, which also lowers cortisol. ACTH (adrenocorticotropic hormone) then rises and causes mineralocorticoid excess (hypokalaemia, hypertension, oedema), so prednisone is co-prescribed. Never combine with radium-223 (more deaths and fractures).
- **Enzalutamide:** seizures (~1%), fatigue, falls. Apalutamide: rash, hypothyroidism.
- **PARP inhibitors** cause synthetic lethality in HRR-deficient cells; benefit is greatest with BRCA2 and weak with ATM.

Sequencing by prior mCSPC therapy

Prior therapy	Next option
Docetaxel only	ARPI
ARPI only	Docetaxel, radium-223, sipuleucel-T
Both	Cabazitaxel, ^{177}Lu-PSMA, PARP inhibitor if HRR mutation

Neuroendocrine/aggressive variant

- Neuroendocrine variant: low PSA, liver or brain metastases, lytic bone lesions with hypercalcaemia. Treat with **platinum + etoposide** (or carboplatin + taxane) ± radiotherapy; survival <12 months.

Supportive and palliative care

- **Bone health:** denosumab 120 mg SC or zoledronic acid 4 mg IV (renal dose adjustment) every 4 weeks against skeletal events, with calcium and vitamin D. Denosumab delayed the first SRE (skeletal-related event) longer (20.7 vs 17.1 months); neither prolongs life. Risk: osteonecrosis of the jaw.
- Focal bone pain: external-beam radiotherapy (radiotherapy for bone pain).
- **Cord compression** needs high-dose dexamethasone → urgent MRI → radiotherapy ± decompression (Patchell trial: surgery + radiotherapy better).
- Ureteric obstruction: stent or nephrostomy. Anaemia; hospice care.

Recent advances

- PSMA-ligand therapy earlier (PSMAfore); AR degraders (ARV-110); AKT inhibition in PTEN loss; bipolar androgen therapy; antibody–drug conjugates.

Diagram to draw: clinical states: rising PSA → nmCRPC → mCRPC first, second, third line.

Diagram to draw: clinical states model with drugs placed at each state.

> **Key points to remember:**
> - Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
> - AR still drives CRPC.
> - nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
> - mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
> - Switch mechanism; never radium-223 with abiraterone.

Also, from the condensed book (not in the paper answer):

- A prostate-specific antigen (PSA) doubling time (PSADT) **<3 months** predicts a rapid course, so consider aggressive therapy.
- Bone pain or visceral-predominant disease calls for **upfront chemotherapy**.

- **Castration resistance is not androgen independence.** The androgen receptor (AR) still drives growth through amplification, mutations, the splice variant **AR-V7** and tumour steroid synthesis, so AR-pathway inhibitors (ARPIs) work.
- Once AR dependence is lost, use taxanes, radium-223, PSMA therapy, PARP inhibitors (PARPIs) or immunotherapy.

2 · Clinical states and progression criteria

The **Prostate Cancer Clinical Trials Working Group 3 (PCWG3)** clinical states model classifies men by prior therapy, from rising PSA through non-metastatic (nmCRPC) to metastatic CRPC (mCRPC).

- PCWG3 PSA progression is a rise of ≥25% and ≥2 ng/mL above nadir, confirmed ≥3 weeks later.

Modality	Pitfall	PCWG3 rule
PSA	Early rise with chemotherapy; flare with radium-223 and ^{177}Lu-PSMA	Confirm any rise
Bone scan	Early new lesions may be **flare** of healing disease	Rescan at **12–16 weeks** as the new baseline; progression needs **≥2 new lesions** confirmed by ≥2 more ("2 + 2")
CT	Soft tissue is little of the burden	RECIST; PFS preferred over response rate
PSMA PET	Apparent progression at 4 weeks resolves by 3 months in ≈13%	Criteria not defined

> **KEY POINT**
>
> Every marker (PSA, bone scan, PSMA PET) can flare early, so call progression only on confirmed, combined evidence.

- **Continue ADT indefinitely** during every later therapy.

3 · Germline and somatic testing

- **Germline testing** suits all metastatic disease, relevant family history (BRCA, Lynch, HOXB13) and Ashkenazi ancestry.
- **Somatic** sequencing usually uses the archival primary. In liquid biopsy, **clonal haematopoiesis (CHIP)** can give false DNA-repair calls.
- Somatic homologous recombination repair (HRR) mutations occur in **20–25%** of mCRPC. Mismatch repair deficiency (dMMR/MSI-H) occurs in **2–5%** and should prompt Lynch counselling.

4 · Non-metastatic CRPC (nmCRPC)

nmCRPC is a rising PSA on castration with no metastases on **conventional** imaging. PSA kinetics are the strongest predictor. **Add an ARPI (apalutamide, enzalutamide or**

darolutamide) to ADT when PSADT is ≤10 months. All three improve metastasis-free survival (MFS) and OS. Abiraterone is active but **not approved**.

5 · mCRPC: androgen-signalling inhibitors

Agent	Mechanism	Dose	Toxicity / notes
Abiraterone	Selective CYP17 inhibitor (17α-hydroxylase and 17,20-lyase)	1000 mg **fasting** with prednisone (10 mg/day in mCRPC)	Mineralocorticoid excess, hepatotoxicity; correct potassium first. Food raises absorption up to 10× (alternative: 250 mg with a low-fat meal)
Enzalutamide	Pure AR antagonist; blocks translocation and DNA binding	160 mg, no steroid	**Seizures (≈1%)**, fatigue, falls; CYP3A4/2C8 interactions
Apalutamide	Does not cross the blood–brain barrier (BBB)	240 mg	**Rash, hypothyroidism**, fractures
Darolutamide	Structurally distinct; does not cross the BBB	600 mg twice daily	Least CNS toxicity

KEY POINT

Prednisone with abiraterone: blocking CYP17 lowers cortisol, so ACTH rises and causes mineralocorticoid excess; prednisone suppresses ACTH. Enzalutamide needs no steroid. All ARPIs cause falls and fractures, so take care over 75.

6 · Cytotoxic chemotherapy

Cabazitaxel has poor P-glycoprotein affinity, so it works after docetaxel.

Agent	Dose / line	Signature toxicity
Docetaxel	**75 mg/m² 3-weekly** with prednisone 5 mg twice daily; first line	Neutropenia, **neuropathy**, oedema, hyperlacrimation, nail dystrophy
Cabazitaxel	**20 mg/m²** (25 if low neutropenic risk) 3-weekly with prednisone and G-CSF; after docetaxel	**Neutropenia** (grade 3–4 82%), **diarrhoea**
Mitoxantrone	12 mg/m² with prednisone; palliation	No survival benefit
Platinum (with etoposide or taxane)	Neuroendocrine variant; DNA-repair-deficient	—

7 · Radiopharmaceuticals

Feature	Radium-223	^{177}Lu-PSMA-617
Emitter	**Alpha** (path **<100 µm**, so marrow is spared); calcium mimetic	Beta; needs PSMA PET-positive disease
Eligible	**Symptomatic bone metastases, no visceral or bulky nodal disease**	PSMA-avid, any site, after ARPI and taxane
Schedule	50 kBq/kg IV 4-weekly × 6	6-weekly, up to 6 cycles
Toxicity	Diarrhoea, anaemia	Fatigue, **dry mouth**, anaemia

Feature	Radium-223	^{177}Lu-PSMA-617
Caution	**Never with abiraterone/prednisone**; bone agent with any ARPI	—

Strontium-89 and samarium-153 only palliate pain.

8 · PARP inhibitors

SAQ PARP inhibitors in Prostate cancer.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2024-Winter Paper 4, Q3f

Introduction

- **PARP (poly-ADP-ribose polymerase)** repairs single-strand DNA breaks. **PARPIs (PARP inhibitors):** olaparib, rucaparib, niraparib, talazoparib.

Mechanism: synthetic lethality

- **Catalytic inhibition** and **PARP trapping** on DNA leave single-strand breaks unrepaired.
- At replication these become **double-strand breaks**, normally repaired by **HRR (homologous recombination repair)**.
- Tumours lacking HRR (e.g. **BRCA2** loss) die; normal cells survive.
- **AR (androgen receptor)–PARP synergy:** AR maintains HRR genes, so AR blockade adds repair deficiency: basis for PARPI + ARPI (androgen receptor pathway inhibitor).

Who benefits: testing

- **HRR alterations in mCRPC (metastatic castration-resistant prostate cancer): up to 20–25%.** Genes: **BRCA2 (largest benefit)**, BRCA1, ATM (weak), PALB2, CHEK2, CDK12.
- **Germline testing** in all metastatic disease, strong family history or Ashkenazi ancestry; **somatic testing** on tissue or ctDNA (circulating tumour DNA), where clonal haematopoiesis can mislead.

Key trials

Trial	Setting	Result
PROfound	Olaparib vs enzalutamide/abiraterone after ARPI	BRCA/ATM cohort: imaging PFS (progression-free survival) **7.4 vs 3.6 months (HR [hazard ratio] 0.34)**; OS (overall survival) 19.1 vs 14.7 months in the final analysis (chapter quotes the interim 18.5 vs 15.1)
TRITON3	Rucaparib vs physician's choice	BRCA: PFS 11.2 vs 6.4 months (HR 0.50); no ATM benefit
PROpel	Olaparib + abiraterone, first-line mCRPC	OS 42.1 vs 34.7 months (P = 0.054, not significant); greatest in BRCA
MAGNITUDE	Niraparib + abiraterone, HRR-altered	BRCA rPFS (radiographic PFS) 19.5 vs 10.9 months (HR 0.55); no benefit without HRR alteration
TALAPRO-2	Talazoparib + enzalutamide, first line	All-comers rPFS not reached vs 21.9 months (HR 0.63); largest in BRCA

- **Also know:** TOPARP-A/B, TRITON2, GALAHAD (phase II).

Approvals (FDA [Food and Drug Administration])

- **Olaparib monotherapy:** HRR-mutated mCRPC after ARPI (2020).
- **Rucaparib:** BRCA-mutated mCRPC.
- **Olaparib or niraparib + abiraterone:** BRCA-mutated; **talazoparib + enzalutamide:** HRR-mutated (both 2023).

Toxicity

- **Anaemia** (commonest grade 3–4; monitor counts), fatigue, nausea; rare **MDS/AML (myelodysplastic syndrome/acute myeloid leukaemia)**, pulmonary embolism.

Recent advances

- **mCSPC (metastatic castration-sensitive prostate cancer)** trials: AMPLITUDE, TALAPRO-3; ATR inhibitors for ATM loss.

Diagram to draw: synthetic lethality: PARP block + BRCA loss → cell death.

> **Key points to remember:**
> - Synthetic lethality; BRCA2 gains most, ATM least.
> - PROfound: olaparib, PFS 7.4 vs 3.6 months.
> - Combinations: PROpel, MAGNITUDE, TALAPRO-2.

Also, from the condensed book (not in the paper answer):

- PARPIs are ineffective in unselected men; haemoglobin can fall suddenly, so watch blood counts.

9 · Immunotherapy

- **Sipuleucel-T**, an autologous dendritic-cell vaccine, was the first approved therapeutic cancer vaccine.
- It suits **asymptomatic/minimally symptomatic mCRPC with no visceral metastases or opioid pain**. It prolongs OS **without** PSA or PFS change.
- **Pembrolizumab** 200 mg IV 3-weekly is approved for **MSI-H/dMMR** or tumour mutational burden **≥10 mutations/megabase** after ≥1 line.

10 · Neuroendocrine/aggressive-variant CRPC

Feature	Usual mCRPC	Neuroendocrine variant
AR / PSA	Retained; PSA tracks disease	Lost; **PSA low despite progression**
Pattern	Osteoblastic bone ± nodes; visceral <20%	Bulky masses, liver, **osteolytic** lesions with hypercalcaemia, brain
Molecular	AR alterations	≥2 of PTEN loss, TP53 mutation, RB1 loss; low PSMA
Treatment	ARPI, taxane, radioligand, PARPI	**Platinum + etoposide** (or taxane + carboplatin) ± radiotherapy (RT)
Survival	Years	**<12 months**

Biopsy a metastasis if progression is rapid with a low PSA.

11 · Sequencing and cross-resistance

A second ARPI helps only **15–30%** because of cross-resistance, so **switch mechanism** (CARD). See the P2-140 sequencing table; add pembrolizumab for MSI-H/dMMR and a PARPI if HRR-mutated.

- Enzalutamide induces CYP3A4/2C9/2C19, so check anticoagulants and antiepileptics.
- AR-V7 in circulating tumour cells predicts resistance to abiraterone and enzalutamide, not to taxanes.

12 · Complications and palliative care

*See the paper answer **Metastatic prostate cancer, including newly diagnosed and bone metastases** (Section 17, Ch 165).*

Also, from the condensed book (not in the paper answer):

- Obstructive uropathy must be **relieved before chemotherapy**, but not every unilateral case needs drainage.

Every focal back pain in a man with bone metastases is cord compression until proven otherwise.

ALGORITHM

1. Back pain with or without deficit or sphincter change: suspect compression.
2. Give **high-dose IV dexamethasone** (10 mg IV, then 4–10 mg 6–12-hourly; 16–100 mg/day).
3. Arrange **urgent whole-spine MRI**.
4. Give RT with or without decompressive surgery.
5. Taper steroids over 2–3 weeks.

- Operate for progression during RT, unstable fracture, or recurrence after RT.
- Treat neuropathic pain with tricyclics or gabapentinoids, and start hospice care in time. Erythropoiesis-stimulating agents are not recommended for anaemia.

13 · Bone-targeted therapy

Two uses: ADT osteoporosis, and **prevention of skeletal-related events (SREs)** in CRPC bone metastases. There is **no survival effect**.

Feature	Zoledronic acid	Denosumab
Dose	**4 mg IV 4-weekly**, renal adjustment	**120 mg subcutaneous (SC) 4-weekly**; no renal adjustment
SRE	Better than placebo	**Better than zoledronic acid** (20.7 vs 17.1 months)
Toxicity	Flu-like, creatinine rise, hypocalcaemia, osteonecrosis of the jaw (ONJ)	Hypocalcaemia (grade ≥3 5%), ONJ **2–4%**

KEY POINT

ONJ follows dental extraction or poor teeth, so check the teeth first. Give calcium 1000 mg and vitamin D 800 IU daily with both agents. Bone agents are strongly recommended with radium-223.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTAN	nmCRPC, PSADT ≤10 mo; apalutamide	MFS 40.5 vs 16.2 mo (HR 0.28); OS HR 0.78	First nmCRPC drug
PROSPER	nmCRPC; enzalutamide	MFS improved; death reduced 27%	Enzalutamide for nmCRPC
ARAMIS	nmCRPC; darolutamide	MFS 40.4 vs 18.4 mo; OS HR 0.69	Least toxic
TAX 327	Docetaxel 3-weekly vs weekly vs mitoxantrone	OS 18.9 vs 17.3 vs 16.4 mo (HR 0.76)	First chemo to prolong survival
TROPIC	Post-docetaxel; cabazitaxel vs mitoxantrone	OS 15.1 vs 12.7 mo (HR 0.70)	Second-line chemo
PROSELICA / FIRSTANA	Cabazitaxel 20 vs 25 / vs docetaxel first line	20 non-inferior / not superior	20 mg/m²; docetaxel first
CARD	After docetaxel and 1 ARPI: cabazitaxel vs 2nd ARPI	OS 13.6 vs 11.0 mo (HR 0.64)	Switch mechanism
COU-AA-301	Post-docetaxel; abiraterone	OS 14.8 vs 10.9 mo (HR 0.65)	Post-chemo
COU-AA-302	Chemo-naive, asymptomatic	rPFS 16.5 vs 8.3 mo	Pre-chemo
AFFIRM	Post-docetaxel; enzalutamide	OS 18.4 vs 13.6 mo (HR 0.63)	Post-chemo
PREVAIL	Chemo-naive; enzalutamide	Death reduced 29%; radiographic progression reduced 81%	Pre-chemo
ALSYMPCA	Symptomatic bone-only; radium-223	OS 14.9 vs 11.3 mo (HR 0.70); SRE delayed	Alpha emitter prolongs life
VISION	Post-ARPI and taxane, PSMA-positive; ^{177}Lu-PSMA	OS 15.3 vs 11.3 mo (HR 0.62); rPFS HR 0.40	Approved 2022
TheraP	Phase II; ^{177}Lu-PSMA vs cabazitaxel	PSA response 65 vs 37%; fewer grade 3–4; OS equal	Lutetium ≥ cabazitaxel
IMPACT	Sipuleucel-T vs placebo	OS 25.8 vs 21.7 mo (HR 0.78); no PSA/PFS change	Vaccine approved 2010
PROfound	Post-ARPI; olaparib vs 2nd ARPI	BRCA/ATM rPFS 7.4 vs 3.6 mo (HR 0.34)	Olaparib, HRR panel
TRITON2/3	Rucaparib after ARPI	BRCA rPFS HR 0.50; no ATM benefit	BRCA only
PROpel / MAGNITUDE / TALAPRO-2	First-line PARPI with abiraterone / with enzalutamide	rPFS gains, largest in BRCA; PROpel OS 42.1 vs 34.7 mo (P = 0.054)	HRR/BRCA

Trial	Design	Result	Take-home
Denosumab vs zoledronic acid	Bone-metastatic CRPC	First SRE 20.7 vs 17.1 mo; OS equal	Denosumab for SRE

★ Putting it together

A · PSA to diagnosis

ALGORITHM

1. **Raised prostate-specific antigen (PSA) or abnormal DRE**: repeat it; exclude infection and instrumentation; double the PSA on 5α-reductase inhibitors (→ Ch 154).
2. **Multiparametric MRI (mpMRI) before biopsy**: PI-RADS (→ Ch 155).
3. **Biopsy** PI-RADS 3–5 targets plus systematic cores, preferably transperineal.
4. Assign **ISUP Grade Group (GG)** (→ Ch 156) and risk group (→ Ch 157). If unfavourable intermediate or worse, stage with PSMA PET (or CT and bone scan).

B · Localised disease: matching risk to treatment

NCCN favourable groups need ALL criteria, adverse groups ANY one.

Risk group	Defined by	Management
Very low	T1c, GG1, PSA <10, density <0.15, <3 positive cores (≤50% each)	Active surveillance (AS)
Low	T1–T2a, GG1, PSA <10	AS preferred
Favourable intermediate	One factor (T2b–c, GG2, PSA 10–20); <50% cores	Radical prostatectomy (RP) or radiotherapy (RT); AS if low-volume GG2
Unfavourable intermediate	2–3 factors, or GG3, or ≥50% cores	RP, or RT with 4–6 months ADT
High / very high	T3a, GG4–5 or PSA >20 / T3b–T4, primary pattern 5, ≥2 high-risk features	RP with extended node dissection, or RT with ADT 18–36 months; add abiraterone 2 years if N1 or ≥2 of Gleason ≥8, cT3–4, PSA ≥40 (STAMPEDE)

KEY POINT

SPCG-4, PIVOT and ProtecT show that radical treatment helps only younger men with ≥intermediate-risk disease. ProtecT: RP = RT for 15-year cancer survival.

C · Surgery vs radiotherapy vs focal therapy

	RP (Ch 160–161)	RT (Ch 162)	Focal (Ch 163)
Main toxicity	Incontinence, ED	Rectal/urinary irritation	Least; retreatment common
Advantages	Pathological staging	Suits comorbid men	Preserves function
Failure	PSA ≥0.2 ng/mL (confirmed)	Phoenix: nadir + 2 ng/mL	Undefined; biopsy-based
Salvage	Bed RT early at PSA 0.1–0.2 ± ADT	RP, cryotherapy, HIFU, brachytherapy	Repeat focal, RP, RT

D · Advanced disease

ALGORITHM

1. **Biochemical recurrence** (Ch 165): PSMA PET. After RP, salvage RT; after RT, local salvage if biopsy-proven. If high risk (PSADT ≤9 months), give ADT and enzalutamide (EMBARK).
2. **mHSPC**: never ADT alone; add an ARPI, or a docetaxel triplet if fit and high volume. If low volume, add prostate RT (STAMPEDE arm H). Test genes in all.
3. **nmCRPC** with PSADT ≤10 months: ARPI (SPARTAN, PROSPER, ARAMIS).
4. **mCRPC** (Ch 166): switch mechanism. Docetaxel → cabazitaxel (CARD); ^{177}Lu-PSMA (VISION); radium-223 for symptomatic bone-only disease (ALSYMPCA); PARP inhibitor if BRCA/HRR mutation (PROfound); pembrolizumab if MSI-H; protect bone.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 12 Paper 2, Q1 (biochemical recurrence also discussed in 2019-Summer Paper 4, Q2)

Assumed meaning: 6 months after RP (radical prostatectomy) the PSA (prostate-specific antigen) never became undetectable and is now ≥2 ng/mL, i.e. PSA persistence with early rise, suggesting residual or systemic disease.

Introduction

- After RP, PSA (half-life about 3 days) should become undetectable (<0.1 ng/mL) within about 6 weeks.
- **PSA persistence:** failure to become undetectable after RP. **BCR (biochemical recurrence):** a rise after an undetectable nadir.
- **Gleason 4+3 = grade group 3** (unfavourable intermediate risk).

Definitions of BCR after RP

Body	Definition
AUA (American Urological Association)	PSA ≥0.2 ng/mL
NCCN (National Comprehensive Cancer Network)	Failure to reach undetectable, two consecutive rises, or PSA ≥0.1 ng/mL
EAU (European Association of Urology) risk groups	**Low risk:** PSADT (PSA doubling time) >1 year **and** grade group <4. **High risk:** PSADT ≤1 year **or** grade group 4–5

Why is the PSA high? (sources)

1. **Residual local disease:** positive margin (the only modifiable factor), residual benign tissue (low, stable PSA).
2. **Occult nodal metastases** (limited or no lymph node dissection).
3. **Distant micrometastases:** likeliest with persistent PSA and a fast rise.

Evaluation

- **Repeat PSA** to confirm; calculate PSADT.
- **Review pathology:** pT stage, margin (length >1 mm, Gleason 4 at margin), seminal vesicles, nodes. **Also know:** CAPRA-S (Cancer of the Prostate Risk Assessment–Surgery) nomogram; Decipher genomic classifier.
- **PSMA (prostate-specific membrane antigen) PET/CT (positron emission tomography/computed tomography):** best test; detection about half below 0.5 ng/mL and nearly 100% above 2 ng/mL. Choline or fluciclovine if PSMA is unavailable.
- **mpMRI (multiparametric MRI) pelvis:** best for local recurrence in the prostatic fossa.

- CT and bone scan are rarely positive below PSA 10 ng/mL.

Treatment

```
Confirm PSA, PSADT, pathology → PSMA PET ± mpMRI ├ Negative or fossa only → salvage RT to bed + ADT ├ Pelvic nodes only → bed + pelvic node RT + ADT ├ Oligometastatic (≤3) → ADT ± metastasis-directed therapy └ Polymetastatic → treat as mHSPC: ADT + ARPI ± docetaxel
```

1. Salvage RT (radiotherapy): the only curative option

- Dose **≥64–66 Gy** to the bed (SAKK 09/10: 70 Gy gave no gain over 64 Gy); 40–45 Gy to pelvic nodes.
- Best at the lowest PSA: each 0.1 ng/mL costs about 2.6% recurrence-free survival.
- **RADICALS-RT, RAVES, GETUG-AFU 17 (ARTISTIC meta-analysis):** early salvage at PSA 0.1–0.2 equals adjuvant RT and spares ≥60% from RT. These apply to men who become undetectable, not to this patient.

2. Add ADT (androgen deprivation therapy) to salvage RT: essential here

Trial	Arms	Result
RTOG 9601 (Radiation Therapy Oncology Group)	RT ± 2 years bicalutamide 150 mg; PSA 0.2–4.0	12-year OS (overall survival) 76.3% vs 71.3%
GETUG-AFU 16	RT ± 6 months goserelin; PSA 0.2–2.0	10-year BCR-free 64% vs 49%; MFS (metastasis-free survival) better; OS unchanged
SPPORT (RTOG 0534)	Bed / bed + 4–6 months ADT / pelvis + ADT	5-year freedom from progression 70.9% / 81.3% / 87.4%
RADICALS-HD	None vs 6 vs 24 months ADT	24 months better MFS than 6 months
- At PSA ≥2 ng/mL with grade group 3, give whole-pelvis RT with **long-course (about 2 years) ADT**.		

3. Systemic therapy

- **PSMA-positive metastases:** ADT + ARPI (androgen receptor pathway inhibitor: abiraterone, enzalutamide, apalutamide) ± docetaxel (LATITUDE, ARCHES, PEACE-1, ARASENS).
- **Oligometastases (≤3):** MDT (metastasis-directed therapy) by SBRT (stereotactic body RT) delays ADT (**STOMP:** ADT-free survival 21 vs 13 months; **ORIOLE:** progression 19% vs 61%).
- **High-risk BCR not fit for salvage: EMBARK** (PSADT ≤9 months, PSA ≥1 ng/mL): enzalutamide + leuprolide, MFS HR (hazard ratio) 0.42.
- **Also know:** castrate testosterone <50 ng/dL; intermittent ADT acceptable (PR.7); observation only for EAU low-risk BCR or short life expectancy; salvage lymph node dissection is investigational.

Prognosis

- **Untreated BCR (Pound):** metastasis in about one-third, median 8 years; death a median 5 years later. Predictors: **BCR <2 years, PSADT <10 months, Gleason 8–10.**

- **This patient is poor-risk:** persistent PSA, high level within 6 months, likely short PSADT. Persistent PSA carries a higher risk of metastasis than late BCR.
- **Salvage RT:** one-third to one-half BCR-free at 4–5 years; poorer above PSA 0.5–1 ng/mL. Cancer-specific survival gain mainly with PSADT <6 months treated within 2 years.
- Lifelong PSA follow-up.

Diagram to draw: the treatment flowchart above, with PSA kinetics curve after RP.

> **Key points to remember:**
> - EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
> - PSMA PET first: almost 100% detection above PSA 2 ng/mL.
> - Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
> - Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

LAQ 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 14 Paper 2, Q1 (related: prognosis in 2019-Summer Paper 4, Q3e; staging of non-organ-confined disease in Practice Set 10 Paper 2, Q1)

"Mixed voiding symptoms" is read as mixed storage and voiding LUTS (lower urinary tract symptoms).

Introduction

- In a 48-year-old, a PSA (prostate-specific antigen) of 18 ng/mL is far above what BPH (benign prostatic hyperplasia) usually produces. **Prostate cancer must be excluded first**; the LUTS are then treated on their own merits.
- PSA is organ-specific, not cancer-specific: BPH, prostatitis, UTI (urinary tract infection), retention and instrumentation also raise it. At PSA >10 ng/mL, biopsy yield is 43–65% with a normal DRE (digital rectal examination) and 70–90% with a suspicious one.

History

- **LUTS:** IPSS (International Prostate Symptom Score) with bother score; storage versus voiding; retention, haematuria, recurrent UTI.
- **Red flags:** bone pain, weight loss, leg oedema, cord compression.
- Family history (prostate cancer <65 years; BRCA (breast cancer gene)-related cancers), recent catheter, fever or perineal pain (prostatitis), 5ARI (5α-reductase inhibitor) use, neurological disease, erectile function.

Examination

- **DRE:** size, nodule, induration, extension, fixity (clinical T stage). Palpable bladder, neurological and spinal examination.

Investigations

Test	Purpose
Urinalysis, culture	Exclude UTI or prostatitis; treat and repeat PSA later (no empirical antibiotics without symptoms)

Test	Purpose
Repeat PSA, free PSA	Confirm the rise; low free PSA favours cancer
Creatinine, ultrasound KUB (kidney, ureter, bladder)	Upper tracts, prostate volume (PSA density)
Uroflowmetry, PVR (post-void residual)	Baseline LUTS assessment
mpMRI (multiparametric MRI) before biopsy	PI-RADS (Prostate Imaging Reporting and Data System) score; guides targeted cores; local staging
Biopsy	Targeted (PI-RADS ≥3) **plus systematic** 12 cores; **transperineal** preferred (lower sepsis)

- At PSA 18, biopsy is needed **even if MRI is negative**. Blood or urine biomarkers (PHI (Prostate Health Index), 4Kscore, PCA3) matter mainly in the 2–10 ng/mL grey zone.

Staging if cancer is found

- Grade group, cores involved, PSA, T stage → risk group. PSA 10–20 makes him at least **intermediate risk** (D'Amico, NCCN (National Comprehensive Cancer Network)).
- **High risk:** bone scan with CT (computed tomography), or **PSMA PET (prostate-specific membrane antigen positron emission tomography)**. Organ-confined disease is <50% at PSA >10; nodes about 20% at PSA >20.
- **Germline testing** (BRCA2, mismatch repair genes) for high-risk or metastatic disease or strong family history.

Management of the cancer (life expectancy >25 years)

Situation	Management
Low or favourable intermediate risk	AS (active surveillance) is possible, but uncommon at PSA 18
Localised intermediate or high risk	**RP (radical prostatectomy)**, usually robotic, with extended PLND (pelvic lymph node dissection) when node risk exceeds the nomogram threshold; or **EBRT (external beam radiotherapy) + ADT (androgen deprivation therapy)**: 4–6 months intermediate, 18–36 months high risk
Locally advanced (T3–T4, N1)	Multimodal: RP ± adjuvant or early salvage RT, or EBRT + long-term ADT; abiraterone added for high-risk node-positive or high-risk N0 disease (STAMPEDE)
Metastatic	ADT + ARPI (androgen receptor pathway inhibitor) ± docetaxel; RT to the prostate for low volume

- **Why surgery suits him:** young, gives pathological staging, and **removes the obstructing gland**, curing the LUTS. SPCG-4: surgical benefit is concentrated in men <65.
- Brachytherapy is relatively contraindicated with severe LUTS, a large gland or prior TURP (transurethral resection of the prostate) (retention, incontinence).

If biopsy is negative

- PSA 18 still needs explaining (prostatitis, large gland, missed anterior tumour): repeat MRI-targeted transperineal biopsy; %free PSA, PCA3, ConfirmMDx; close PSA follow-up.

Management of LUTS

- **Behavioural measures** (fluid timing, less caffeine) and an **α-blocker** (e.g. tamsulosin) at once; it does not alter PSA.

- **Avoid starting a 5ARI until cancer is excluded:** it halves PSA and masks cancer; if used later, record a baseline PSA and double readings after 12 months.
- Persistent storage symptoms with low PVR: add an antimuscarinic or β3-agonist (mirabegron).
- **Surgery** (TURP, HoLEP (holmium laser enucleation of the prostate)) for retention, refractory symptoms or complications, only once cancer is excluded. In locally advanced cancer with retention, a channel TURP relieves obstruction.
- **Also know:** pressure-flow study before surgery at age <50 with mainly voiding LUTS (EAU (European Association of Urology)).

```
PSA 18 + LUTS → history, DRE, urine culture → repeat PSA → mpMRI → targeted + systematic biopsy (transperineal) Cancer → risk group + staging → RP / RT + ADT / systemic Benign → explain PSA, re-biopsy if needed → BPH pathway LUTS throughout: α-blocker; no 5ARI until cancer excluded
```

Diagram to draw: the flowchart above.

> **Key points to remember:**
> - PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
> - PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
> - Young fit man: RP treats both cancer and obstruction.
> - α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

LAQ Discuss management of organ confined CA prostate

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 15 Paper 2, Q2

Introduction

- Organ-confined prostate cancer is **cT1–T2 N0 M0**. Treatment must balance cancer mortality against **competing mortality**: life expectancy, comorbidity, baseline urinary, sexual and bowel function, and patient preference (shared decision-making).

Risk stratification (NCCN, National Comprehensive Cancer Network)

Risk group	Criteria
Very low	All of: T1c, GG (grade group) 1, PSA (prostate-specific antigen) <10 ng/mL, <3 positive cores with ≤50% cancer in each, PSA density <0.15
Low	All of: T1–T2a, GG1, PSA <10
Favourable intermediate	1 intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	2–3 intermediate factors, GG3, or ≥50% cores positive
High	Any of: T3a, GG4–5, PSA >20

- **Grade groups:** GG1 = 3+3; GG2 = 3+4; GG3 = 4+3; GG4 = 8; GG5 = 9–10.

Evaluation

- PSA, DRE (digital rectal examination), mpMRI (multiparametric MRI) with targeted and systematic biopsy.
- **Staging imaging** from unfavourable intermediate risk: bone scan + CT/MRI, or PSMA (prostate-specific membrane antigen) PET (positron emission tomography). **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Life expectancy (life tables, Charlson index); IPSS (International Prostate Symptom Score) and erectile function.
- **Also know:** Partin tables, Kattan nomogram, CAPRA (Cancer of the Prostate Risk Assessment) score; genomic tests (Decipher, Prolaris, Oncotype DX) for borderline surveillance decisions.

Options by risk

Risk	Preferred options
Very low / low	**Active surveillance (AS)**; watchful waiting if life expectancy short
Favourable intermediate	AS in selected men; RP (radical prostatectomy); EBRT (external-beam radiotherapy) or brachytherapy alone
Unfavourable intermediate	RP + PLND (pelvic lymph node dissection); EBRT + **4–6 months ADT** (androgen deprivation therapy)
High (organ-confined, e.g. GG4–5 or PSA >20)	RP + extended PLND (multimodal); EBRT + **18–36 months ADT**

1. Active surveillance vs watchful waiting

- **AS:** deferred treatment with **curative intent**. Monitor by PSA every 3–6 months, DRE, **confirmatory biopsy or MRI within 6–12 months**, then periodic MRI and biopsy. Trigger for treatment: grade progression (GG ≥2–3), increasing volume or patient choice.
- Entry criteria: Johns Hopkins (strict, Epstein), PRIAS (Prostate Cancer Research International Active Surveillance), Toronto (inclusive). About half are treated by 10–15 years.
- **Watchful waiting:** no curative intent; ADT for symptomatic progression. For limited life expectancy.

2. Radical prostatectomy

- Open retropubic or **robot-assisted** (similar outcomes in the randomised trial). Nerve sparing when there is no same-side extracapsular risk.
- **PLND** when nodal risk is significant on nomograms (extended template).
- **Complications:** incontinence, ED (erectile dysfunction), anastomotic stricture, rectal injury.
- **Advantages:** pathological staging; relieves obstruction; RT kept for salvage.

3. Radiotherapy

- **EBRT:** IMRT (intensity-modulated RT) with image guidance. Fractionation: conventional 78 Gy in 39; **moderate hypofractionation 60 Gy in 20** (CHHiP: non-inferior); **SBRT (stereotactic body RT) 36.25 Gy in 5** (PACE-B: similar toxicity).
- **Why add ADT:** synergistic cell kill and target shrinkage. **RTOG 94-08:** 4 months ADT improved OS in intermediate risk; **DFCI 95-096:** 6 months ADT improved 8-year OS (61% → 74%).

- **Brachytherapy:** LDR (low-dose-rate) seeds for low and favourable intermediate risk. Relative contraindications: gland >60 mL, high IPSS, recent TURP. HDR (high-dose-rate) or LDR **boost** with EBRT for higher risk (ASCENDE-RT: half the biochemical failure).
- **Toxicity:** proctitis, irritative urinary symptoms, late ED, second cancers (controversial).

4. Focal and whole-gland ablation

- HIFU (high-intensity focused ultrasound), cryotherapy, focal laser, IRE (irreversible electroporation), VTP (vascular-targeted photodynamic therapy).
- Based on the **index lesion hypothesis**. Best for MRI-visible, biopsy-confirmed, unifocal **intermediate-risk (GG2)** disease. Investigational; ideally in trials. VTP is the only modality with a randomised trial (versus AS in low risk).

Landmark trials

Trial	Comparison	Result
SPCG-4 (pre-PSA era)	RP vs watchful waiting	Cancer death RR (relative risk) 0.56 at 23 years; NNT 8; benefit <65 years
PIVOT	RP vs observation (PSA era)	No mortality difference; benefit only in intermediate risk
ProtecT	Monitoring vs RP vs RT + ADT	15-year cancer mortality 2.7%, 2.2%, 2.9% (no difference); more metastases with monitoring

- **ProtecT function:** RP worst for continence and sexual function; RT transient bowel effects.

Algorithm

```
Organ-confined CaP → risk group + life expectancy ├─ short life expectancy → watchful waiting ├─ very low / low → active surveillance ├─ favourable intermediate → AS (selected), RP, EBRT, brachytherapy └─ unfavourable intermediate / high → RP + PLND or EBRT + ADT (4–6 or 18–36 months)
```

Follow-up

- PSA every 6–12 months. **Biochemical recurrence:** after RP, PSA ≥0.2 ng/mL confirmed; after RT, **nadir + 2 ng/mL (Phoenix)**.

Recent advances

- PSMA PET staging; MRI-led surveillance; SBRT; genomic classifiers.

Diagram to draw: stepwise decision tree (treat or not → partial or whole gland → surgery or radiation).

> **Key points to remember:**
> - Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
> - AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
> - EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
> - Focal therapy rests on the index lesion hypothesis; still investigational.

LAQ Discuss advances in management of CA prostate

Paper answer · P4

Type: LAQ (25 marks)
Source: Practice Set 5 Paper 4, Q2 (related: 2025-Winter Paper 3, Q3b; 2024-Winter Paper 4, Q3f)

Introduction

- Three aims drive recent change: **find significant cancer and avoid diagnosing insignificant cancer**; **match local treatment to risk** with less morbidity; give **earlier, intensified and targeted** systemic therapy.

1. Screening and diagnosis

- **ERSPC (European Randomized Study of Screening for Prostate Cancer)** and Göteborg-1: PSA (prostate-specific antigen) screening lowers cancer-specific mortality, but PSA alone causes negative biopsies and overdiagnosis of GG (Grade Group) 1. Hence risk-stratified pathways.
- **Biomarkers** (reflex tests before biopsy):

Test	Sample	Content
PHI (Prostate Health Index)	Blood	(p2PSA ÷ free PSA) × √total PSA
4Kscore	Blood	Four kallikreins + clinical data; % risk of GG ≥2
STHLM3	Blood	4Kscore kallikreins + 232 risk SNPs (single-nucleotide polymorphisms) + clinical variables
PCA3 (prostate cancer antigen 3), SelectMDx	Urine after DRE (digital rectal examination)	RNA ratio (PCA3); DLX1 and HOXC6 (SelectMDx)
ExoDx IntelliScore	Urine, no DRE	Exosomal RNA of PCA3, ERG, SPDEF

- Tissue genomic classifiers (Decipher, Prolaris, Oncotype DX) refine risk for surveillance and adjuvant decisions.
- **MRI-first pathway:** mpMRI (multiparametric magnetic resonance imaging), scored by **PI-RADS (Prostate Imaging–Reporting and Data System)** 1–5, before biopsy.
- **PROMIS (2017):** mpMRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; NPV (negative predictive value) 89%; 27% could avoid biopsy.
- **PRECISION (2018):** MRI with targeted biopsy of PI-RADS ≥3 found more significant cancer (38% vs 26%); 28% avoided biopsy.
- So PI-RADS ≥3 lesions are biopsied, while PI-RADS 1–2 with low PSA density may safely avoid biopsy.
- **Transperineal biopsy** under local anaesthesia: near-zero sepsis; MRI–ultrasound fusion targeting. **Micro-ultrasound** (PRI-MUS scoring) matches mpMRI in registry data.
- **PSMA (prostate-specific membrane antigen) PET (positron emission tomography): proPSMA:** accuracy 92% vs 65% for CT (computed tomography) plus bone scan (sensitivity 85%, specificity 98%). Used for high-risk staging, biochemical recurrence and selecting radioligand therapy.

2. Localised disease

- **AS (active surveillance):** standard for GG1; extended to low-volume favourable intermediate risk (GG2), with MRI and PSA density. **ProtecT (15 years):** prostate cancer mortality about 2–3% in monitoring, surgery and radiotherapy arms; more metastasis on monitoring, no survival difference.

- **Robotic RP (radical prostatectomy): Retzius-sparing** (Bocciardi) gives earlier continence (1 week to 3 months; equal by 12 months); single-port robot; frozen-section-guided nerve sparing (NeuroSAFE).
- **Radiotherapy:**
- Moderate hypofractionation **60 Gy in 20** (CHHiP) is standard.
- **SBRT (stereotactic body radiotherapy) 36.25 Gy in 5:** PACE-B showed similar late toxicity at 2 years (chapter); its 5-year report (2024) showed non-inferior biochemical or clinical control. HYPO-RT-PC (42.7 Gy in 7) was also non-inferior.
- **FLAME:** MRI-lesion boost improved biochemical PFS (progression-free survival) (HR [hazard ratio] 0.45). **Hydrogel rectal spacer** reduces rectal toxicity.
- High-risk M0: **abiraterone added to ADT (androgen deprivation therapy) and RT (STAMPEDE):** MFS (metastasis-free survival) HR 0.53.
- **Focal therapy:** based on the **index lesion hypothesis** (the largest or highest-grade focus drives progression). Energies: HIFU (high-intensity focused ultrasound), cryotherapy, IRE (irreversible electroporation), focal laser, VTP (vascular-targeted photodynamic therapy, padeliporfin). **HEAT registry (1379 men):** failure-free survival 68% (intermediate risk); continence 90–95%. Still within trials or registries.

3. Recurrent and advanced disease

Setting	Advance	Evidence
High-risk BCR (biochemical recurrence)	PSMA-guided salvage; enzalutamide + leuprolide	**EMBARK:** MFS HR 0.42
mHSPC (metastatic hormone-sensitive)	ADT + ARPI (androgen receptor pathway inhibitor)	LATITUDE, TITAN, ENZAMET, ARCHES
	Triplet (ADT + docetaxel + ARPI)	**PEACE-1, ARASENS**
	Prostate RT, low volume	STAMPEDE arm H
	MDT (metastasis-directed therapy)	STOMP, ORIOLE
nmCRPC (non-metastatic castration-resistant)	Apalutamide, enzalutamide, darolutamide	SPARTAN, PROSPER, ARAMIS
HRR (homologous recombination repair)-mutated mCRPC	**PARP (poly-ADP-ribose polymerase) inhibitors**	**PROfound:** olaparib imaging PFS 7.4 vs 3.6 months; PROpel, MAGNITUDE, TALAPRO-2
PSMA-positive mCRPC	**^{177}Lu-PSMA-617**	**VISION:** OS (overall survival) 15.3 vs 11.3 months (HR 0.62); TheraP: PSA response 65% vs 37% vs cabazitaxel

- **ADT alone is no longer standard** for mHSPC; every fit man gets intensification.
- **Germline and somatic testing** for all metastatic disease; **pembrolizumab** for MSI-high (microsatellite instability-high) tumours.
- **Relugolix:** oral GnRH (gonadotropin-releasing hormone) antagonist; HERO showed better sustained castration than leuprolide.

- **Also know:** radium-223 (ALSYMPCA, OS HR 0.70); PSMAfore (lutetium before chemotherapy); AI in MRI reading and pathology.

Diagram to draw: MRI-first pathway: raised PSA → mpMRI → PI-RADS ≥3 targeted transperineal biopsy; PI-RADS 1–2 with low PSA density → surveillance.

> **Key points to remember:**
> - PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
> - proPSMA: PSMA PET accuracy 92% vs 65%.
> - ProtecT: no mortality difference at 15 years; AS is standard for GG1.
> - VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

SAQ Docetaxel

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 1, Q3b

Introduction

- Semisynthetic **taxane** cytotoxic. Since **TAX 327 (2004)** it is the standard first-line chemotherapy for mCRPC (metastatic castration-resistant prostate cancer).

Mechanism of action

- Binds **β-tubulin** and **stabilises microtubules** (GTP [guanosine triphosphate]-independent polymerisation). Cells arrest in **G2/M**.
- Induces apoptosis independent of TP53, through **BCL2 phosphorylation** and p27 induction.
- Prostate-specific: blocks microtubule transport of ligand-bound **AR (androgen receptor)** into the nucleus.

Dose and administration

- **75 mg/m² IV (intravenous) every 3 weeks with prednisone 5 mg twice daily**; 6 cycles in mHSPC (metastatic hormone-sensitive prostate cancer). Steroid premedication limits fluid retention and hypersensitivity. Hepatic metabolism: caution in liver dysfunction.

Uses in urology

Setting	Evidence
mCRPC	**TAX 327:** overall survival (OS) 18.9 vs 16.4 months versus mitoxantrone (HR [hazard ratio] 0.76); better pain and PSA (prostate-specific antigen) response. SWOG 9916 (with estramustine) also positive
mHSPC	**CHAARTED:** OS 57.6 vs 44.0 months (HR 0.61), benefit in **high volume**. STAMPEDE: OS HR 0.78. GETUG-AFU 15 not significant
Triplet	ADT (androgen deprivation therapy) + docetaxel + abiraterone (**PEACE-1**) or darolutamide (**ARASENS**) improves OS
High-risk localised	RTOG 0521: added to RT (radiotherapy) + long-term ADT, OS HR 0.69

Setting	Evidence
Bladder	Second-line single agent (response about 13%); **intravesical gemcitabine–docetaxel** for BCG (bacillus Calmette–Guérin)-unresponsive or BCG-naive high-grade NMIBC (non-muscle-invasive bladder cancer)
Penis	**TPF** (docetaxel, cisplatin, 5-fluorouracil)

CHAARTED high volume: ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

Toxicity

- **Neutropenia** and febrile neutropenia (dose-limiting), fatigue, **fluid retention** (peripheral oedema), **peripheral neuropathy**, hyperlacrimation, **nail dystrophy**, alopecia, liver dysfunction, hypersensitivity.
- 3-weekly: more neutropenia; weekly: more fatigue, diarrhoea, tearing.

Resistance and sequencing

- P-glycoprotein efflux pump mediates resistance. **Cabazitaxel** has poor P-glycoprotein affinity, so it works after docetaxel (**TROPIC**; **CARD** beat a second ARPI [androgen receptor pathway inhibitor]).
- **Also know:** failed combinations (bevacizumab, atrasentan).

Diagram to draw: microtubule stabilised by docetaxel causing G2/M arrest and blocking AR entry into the nucleus.

> **Key points to remember:**
> - 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
> - CHAARTED: benefit in high-volume mHSPC (HR 0.61).
> - Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.

SAQ PIN

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 2, Q3f

Assumed meaning: prostatic intraepithelial neoplasia.

Definition

- PIN (prostatic intraepithelial neoplasia): **architecturally benign** acini or ducts lined by **cytologically atypical** cells, with an intact or patchy **basal cell layer**.
- Graded **LGPIN (low-grade PIN)** and **HGPIN (high-grade PIN)**. LGPIN is not reported: it is irreproducible and carries no excess risk.

Histology

- HGPIN: nuclear enlargement, hyperchromasia, **prominent nucleoli**.
- Patterns: tufting, micropapillary, flat. **Cribriform lesions are now classed as IDC-P (intraductal carcinoma of the prostate) or AIP (atypical intraductal proliferation).**
- Immunohistochemistry: basal cells stain with **HMWCK (high-molecular-weight cytokeratin) and p63**; AMACR (α-methylacyl-CoA racemase) positive. Mainly in the peripheral zone.

Evidence that HGPIN is a precursor

- More and larger HGPIN foci in cancerous prostates; carcinoma seen budding from HGPIN.
- About 20% carry **TMPRSS2:ERG fusion** (about 50% of cancers). Extra genomic changes are needed for invasion.

Incidence

- About **4–5%** of biopsies (range 0–25%, due to observer variation).

Related lesions

Lesion	Meaning	Action on isolated finding
HGPIN	Weak risk marker	**No repeat biopsy**
PINATYP (HGPIN with adjacent atypical glands)	As "atypical"	Manage as ASAP (atypical small acinar proliferation)
ASAP	Cancer on re-biopsy about 40%	Repeat biopsy ± MRI (magnetic resonance imaging)
AIP	May herald IDC-P	Consider re-biopsy
IDC-P	Usually high-grade cancer present	Definitive treatment; excludes active surveillance

Management

- Isolated HGPIN, unifocal or multifocal: **no immediate repeat biopsy**. Decide further testing on PSA (prostate-specific antigen), DRE (digital rectal examination) and **mpMRI (multiparametric MRI)**.
- HGPIN on TURP (transurethral resection of the prostate) chips: no further sampling on current data.
- **Chemoprevention:** none proven; green tea showed reduced progression in small trials.
- *Guideline note:* EAU (European Association of Urology) guidelines still list extensive HGPIN (≥3 biopsy sites) as a reason for repeat biopsy; the chapter does not.

Diagram to draw: benign gland, HGPIN gland (atypical cells, intact basal layer) and cancer (no basal layer).

> **Key points to remember:**
> - Atypical cells, benign architecture, basal layer retained.
> - Isolated HGPIN needs no repeat biopsy; ASAP does.
> - Cribriform "PIN" is now IDC-P or AIP.

SAQ Imaging measures in CA prostate

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3e (mpMRI detail in 2024-Winter Paper 1, Q3d; PSMA PET in 2019-Summer Paper 4, Q3a)

1. Ultrasound

- **TRUS (transrectal ultrasound):** guides biopsy, measures volume (PSA [prostate-specific antigen] density). Cancer is a hypoechoic PZ (peripheral zone) lesion, but **39% are isoechoic**.

- **Also know:** Doppler, contrast and elastography (adjuncts); **micro-ultrasound** (29 MHz; PRI-MUS [Prostate Risk Identification using Micro-Ultrasound] score; close to MRI accuracy).

2. MRI: detection and local staging

- **mpMRI (multiparametric magnetic resonance imaging):**

Sequence	Shows	Cancer
T2-weighted	Anatomy; EPE (extraprostatic extension), SVI (seminal vesicle invasion)	Low signal
DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient)	Cellularity	Bright on DWI, dark on ADC
DCE (dynamic contrast-enhanced)	Vascularity	Early enhancement, washout

- **PI-RADS v2.1 (Prostate Imaging Reporting and Data System)** 1–5: DWI dominant in PZ, T2 in TZ (transition zone); DCE only upgrades PZ 3 to 4.
- **PROMIS** (sensitivity 93% vs 48% for TRUS biopsy) and **PRECISION** (more significant, less insignificant cancer): **MRI before biopsy**. Cochrane: sensitivity 0.91, specificity 0.37.
- **Also know:** biparametric MRI; **PI-QUAL** (Prostate Imaging Quality); **PRECISE** (scores change on surveillance); fusion biopsy.

3. Staging by risk group

- **Low or intermediate risk:** no routine CT (computed tomography) or bone scan.
- **High risk:** CT plus **^{99m}Tc-MDP (methylene diphosphonate) bone scan**, or **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**.
- **proPSMA:** accuracy 92% vs 65% for conventional imaging. Nodal sensitivity only 30–40%, specificity >90%: a negative scan does not exclude micrometastases.

4. Biochemical recurrence

- **PSMA PET** works at low PSA. **Calais:** 56% vs 26% for fluciclovine. **CONDOR:** management changed in 63.9%.
- Choline and fluciclovine PET are older; FDG (fluorodeoxyglucose) PET is poor except in neuroendocrine change.

5. Advanced disease

- PSMA PET selects men for **^{177}Lu-PSMA-617** (VISION); whole-body MRI. No proof yet that PSMA-guided treatment improves survival.

Diagram to draw: prostate zones with a PZ lesion dark on ADC; PSMA PET nodal uptake.

> **Key points to remember:**
> - MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
> - High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
> - PSMA nodal sensitivity 30–40%.

SAQ Methods to predict prognosis of patient of Cancer prostate

Paper answer · P4

Type: SAQ (10 marks)
Source: 2019-Summer Paper 4, Q3e

1. Clinical and pathological factors

- **PSA (prostate-specific antigen):** organ-confined about 80% at <4 but <50% at >10 ng/mL; density and velocity add information.
- **Clinical stage (DRE (digital rectal examination), TNM).**
- **Grade group (ISUP (International Society of Urological Pathology)) – strongest single factor:**

Grade group	Gleason	5-year BCR (biochemical recurrence)-free
1	≤6	96%
2	3+4	88%
3	4+3	63%
4	8	48%
5	9–10	26%

- Biopsy: % positive cores, **cribriform and intraductal** patterns, perineural invasion.
- Prostatectomy: margin (extent, grade), EPE (extraprostatic extension), seminal vesicle and node status.

2. Risk groups

D'Amico	Low	Intermediate	High
PSA (ng/mL)	≤10	10–20	>20
Gleason	≤6	7	8–10
Stage	T1–T2a	T2b	T2c

- **NCCN (National Comprehensive Cancer Network):** very low, low, favourable/unfavourable intermediate, high, very high.
- **CAPRA (Cancer of the Prostate Risk Assessment)** score 0–10 (age, PSA, Gleason, stage, % cores); **CAPRA-S** after surgery.

3. Nomograms

- **Partin tables:** organ-confined disease, EPE, seminal vesicle, node involvement.
- **MSKCC (Memorial Sloan Kettering) nomograms:** pre- and post-treatment BCR and 15-year cancer-specific survival.
- **Briganti nomogram:** risk of nodal invasion → decides PLND (pelvic lymph node dissection).

4. Imaging

- mpMRI (multiparametric MRI): PI-RADS (Prostate Imaging Reporting and Data System) score, tumour contact length, EPE.
- PSMA PET (prostate-specific membrane antigen PET) for metastatic burden.

5. Tissue biomarkers

Test	Basis	Use
Decipher	22-RNA classifier	Metastasis risk; post-op radiotherapy
Oncotype DX GPS	17 genes	Adverse pathology; active surveillance

Test	Basis	Use
Prolaris	Cell-cycle progression genes	Progression, cancer death

- Germline BRCA2 (BReast CAncer gene 2): aggressive disease.

6. After treatment

- **PSA doubling time (PSADT):** <3 months highest risk; Pound: time to BCR <2 years, PSADT <10 months, Gleason 8–10 predict metastasis.
- **EAU (European Association of Urology) BCR risk groups:** low risk = PSADT >1 year and grade group <4.
- PSA nadir on ADT (androgen deprivation therapy); **Phoenix** failure (nadir + 2) after radiotherapy.

Diagram to draw: flowchart from risk group to nomogram, biomarker and treatment decision.

> **Key points to remember:**
> - Grade group is the strongest predictor.
> - D'Amico, NCCN and CAPRA combine PSA, grade and stage.
> - Decipher, Oncotype, Prolaris add genomic risk.

Rapid Revision Sheet · Section 17

Indications of genomic testing in carcinoma prostate (SAQ)

- Germline: all metastatic, high-risk, family history, Ashkenazi.
- Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
- BRCA2: aggressive; avoid casual active surveillance.
- Decipher, Oncotype GPS, Prolaris: prognosis only.

Antioxidants in urological malignancy (SAQ)

- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

Prostate cancer prevention trials (SAQ)

- PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
- FDA warning; no approved preventive agent.
- SELECT negative; vitamin E alone harmful.

PSA screening for prostate cancer: evidence, problems and solutions (LAQ)

- ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
- Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
- USPSTF 2018: Grade C at 55–69, Grade D ≥70.
- Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Molecular staging of CA prostate (SAQ)

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

TRUS and prostate biopsy (SAQ)

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
- MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
- 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
- Bleeding commonest; sepsis most serious.
- Transperineal now preferred.

Multiparametric MRI, PI-RADS and fusion biopsy (SAQ)

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation.
- Cognitive, fusion, in-bore: similar yield.

- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Saturation biopsy of prostate (SAQ)

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Gleason's score and its significance (SAQ)

- Biopsy score = commonest + highest pattern.
- 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
- GG4 = score 8; GG5 = 9–10.

Expectant treatment of CA prostate (SAQ)

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Robotic radical prostatectomy (SAQ)

- Surgeon, not platform, decides outcome (Yaxley 2016).
- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

Radiotherapy and brachytherapy for CA prostate (LAQ)

- α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
- Phoenix: nadir + 2 ng/mL.
- LDR ^{125}I 145 Gy; HDR ^{192}Ir.
- D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

HIFU (SAQ)

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

Cryotherapy for CA prostate (SAQ)

- Argon freezes, helium thaws (Joule–Thomson); −40 °C, two cycles, urethral warmer.
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.
- Main role is salvage: PSA <10, doubling time ≥16 months.
- Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

Staging and management of locally advanced (non-organ-confined) CA prostate (LAQ)

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
- RT + long-term ADT (18–36 months; P1: 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
- RT + ADT: EORTC 22863 overall survival 79% vs 62%.
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Neoadjuvant ADT before RP: no survival gain.
- After RP: early salvage RT preferred over routine adjuvant RT for most men.
- Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

GnRH (LHRH) analogues: agonists and antagonists (SAQ)

- Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
- Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
- Castrate <50 ng/dL; aim <20 ng/dL.

Maximum / combined androgen blockade (SAQ)

- MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Metastatic prostate cancer, including newly diagnosed and bone metastases (LAQ)

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: cover the flare with antiandrogen, or use an antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief as longer courses for focal pain; dexamethasone and MRI for cord compression.
- Focal back pain = cord compression until MRI proves otherwise.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance.
- No survival benefit from bone agents.
- Dental check first (ONJ); calcium and vitamin D.

- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

Castration-resistant prostate cancer (CRPC): definition and treatment (LAQ)

- Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
- AR still drives CRPC.
- nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
- mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
- Switch mechanism; never radium-223 with abiraterone.

PARP inhibitors in Prostate cancer. (SAQ)

- Synthetic lethality; BRCA2 gains most, ATM least.
- PROfound: olaparib, PFS 7.4 vs 3.6 months.
- Combinations: PROpel, MAGNITUDE, TALAPRO-2.

65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis (LAQ)

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief. (LAQ)

- PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
- PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
- Young fit man: RP treats both cancer and obstruction.
- α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

Discuss management of organ confined CA prostate (LAQ)

- Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
- AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
- EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
- Focal therapy rests on the index lesion hypothesis; still investigational.

Discuss advances in management of CA prostate (LAQ)

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Docetaxel (SAQ)

- 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
- CHAARTED: benefit in high-volume mHSPC (HR 0.61).

- Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

PIN (SAQ)

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.

Imaging measures in CA prostate (SAQ)

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

Methods to predict prognosis of patient of Cancer prostate (SAQ)

- Grade group is the strongest predictor.
- D'Amico, NCCN and CAPRA combine PSA, grade and stage.
- Decipher, Oncotype, Prolaris add genomic risk.